Autodesk Robot Structural Analysis Professional 2015

Robot Object Model

© 2014 Autodesk, Inc. All Rights Reserved. Except as otherwise permitted by Autodesk, Inc., this publication, or parts thereof, may not be reproduced in any form, by any method, for any purpose. Certain materials included in this publication are reprinted with the permission of the copyright holder.

Disclaimer

THIS PUBLICATION AND THE INFORMATION CONTAINED HEREIN IS MADE AVAILABLE

BY AUTODESK, INC. “AS IS.” AUTODESK, INC. DISCLAIMS ALL WARRANTIES, EITHER

EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO ANY IMPLIED WARRANTIES

OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE REGARDING THESE MATERIALS.

Trademarks

The following are registered trademarks of Autodesk, Inc., in the USA and/or other countries:

Autodesk Robot Structural Analysis, Autodesk Concrete Building Structures, Spreadsheet Calculator, ATC, AutoCAD, Autodesk, Autodesk Inventor, Autodesk (logo), Buzzsaw,

Design Web Format, DWF, ViewCube, SteeringWheels, and Autodesk Revit.

All other brand names, product names or trademarks belong to their respective holders.

Third Party Software Program Credits

ACIS Copyright© 1989-2001 Spatial Corp. Portions Copyright© 2002 Autodesk, Inc.

Copyright© 1997 Microsoft Corporation. All rights reserved.

International CorrectSpell™ Spelling Correction System© 1995 by Lernout & Hauspie Speech

Products, N.V. All rights reserved.

InstallShield™ 3.0. Copyright© 1997 InstallShield Software Corporation. All rights reserved.

PANTONE® and other Pantone, Inc. trademarks are the property of Pantone, Inc.© Pantone, Inc.,2002.

Portions Copyright© 1991-1996 Arthur D. Applegate. All rights reserved.

Portions relating to JPEG © Copyright 1991-1998 Thomas G. Lane. All rights reserved. Portions of this software are based on the work of the Independent JPEG Group.

Portions relating to TIFF © Copyright 1997-1998 Sam Leffler. © Copyright 1991-1997 Silicon

Graphics, Inc. All rights reserved.

Government Use

Use, duplication, or disclosure by the U.S. Government is subject to restrictions as set forth in

FAR 12.212 (Commercial Computer Software-Restricted Rights) and DFAR 227.7202 (Rights in

Technical Data and Computer Software), as applicable.

Table of Contents

RobotOM Namespace 1

Robot Object Model 1

Data Server 1

Basic definitions 2

Querying mechanism 2

Labels - complex attributes of objects 17

IRobotPredefinedLabel 29

Uniform access to structure components 29

Structure components 37

Load cases 37

Load records 38

IRobotLoadRecordCommon 81

IRobotLoadRecordCommon Members 81

IRobotLoadRecordCommon Fields 81

IsAutoGenerated 81

Snow/wind loads 82

Generating 3D loads 82

Code combinations 148

Modal analysis parameters 189

Modal analysis parameters recognizing static forces 189

Seismic analysis parameters 209

IRobotSeismicResidualModeDefinitionType 294

IRobotSeismicResidualMode 294

IRobotSeismicResidualMode Members 295

IRobotSeismicResidualMode Fields 295

AugmentationFactor 295

DefinitionMethod 295

IsActive 296

LimitFrequency 296

IRobotSeismicAnalysis_EC8_General_Params 296

IRobotSeismicAnalysis_EC8_General_Params Members 296

IRobotSeismicAnalysis_EC8_General_Params Fields 297

Ag 297

B 298

BehaviorFactor 298

Direction 298

DirectionType 298

ExcitationDir 299

(c) 2014 Autodesk Inc. All rights reserved

Filter 299
ResidualMode 299
S 299
Spectrum 300
Tb 300
Tc 300
Td 300
IRobotSeismicAnalysis_EC8_SpectrumType 301
IRobotSeismicAnalysis_EC8_GroundType 301
IRobotSeismicAnalysis_EC8_Params 302
IRobotSeismicAnalysis_EC8_Params Members 302
IRobotSeismicAnalysis_EC8_Params Fields 302
Ag 303
BehaviorFactor 303
Direction 303
DirectionType 303
ExcitationDir 304
Filter 304
GroundType 304
ResidualMode 305
Spectrum 305
SpectrumType 305
Spectral analysis parameters 305
Non-linear analysis parameters 317
Moving loads 326
Buckling analysis parameters 359
Time history analysis parameters 364
IRobotTimeHistoryHHTParams 386
IRobotTimeHistoryHHTParams Members 386
IRobotTimeHistoryHHTParams Fields 386
Alpha 386
Beta 387
CoeffAlpha 387
Nonlinearity 387
NonlinearParams 388
IRobotTimeHistoryNewmarkAccelParams 388
IRobotTimeHistoryNewmarkAccelParams Members 388
IRobotTimeHistoryNewmarkAccelParams Fields 389
Alpha 389
Beta 389
Nonlinearity 389
NonlinearParams 390

(c) 2014 Autodesk Inc. All rights reserved

Harmonic analysis parameters 390
Push-over analysis parameters 391
FRF analysis parameters 395
Footfall analysis parameters 398
Parameters of simplified seismic analysis 447
IRobotSELFSeismicEngine 448
IRobotSELFSeismicEngine Members 448
IRobotSELFSeismicEngine Fields 448
GenerationParams 448
IRobotSELFSeismicEngine Methods 449
GenerateLoadCases 449
IRobotSELFSeismicGenerationParams 449
IRobotSELFSeismicGenerationParams Members 450
IRobotSELFSeismicGenerationParams Fields 450
CodeName 450
CodeNumber 450
ExcitationDir 451
ModalCaseParams 451
SeismicParams 451
TBaseMethod 451
IRobotSELFSeismicStructureParams 452
IRobotSELFSeismicStructureParams Members 452
IRobotSELFSeismicStructureParams Fields 452
BaseLevelCoordZ 452
BaseLevelDefMethod 453
BaseLevelStorey 453
TopLevelCoordZ 453
TopLevelDefMethod 454
TopLevelStorey 454
IRobotSELFSeismicLevelDefinitionMethod 454
IRobotSELFSeismicTBaseMethod 455
IRobotSELFSeismicAnalysisParams 455
IRobotSELFSeismicAnalysisParams Members 455
IRobotSELFSeismicAnalysisParams Fields 456
CodeName 456
CodeNumber 456
CodeParams 456
Eccentricities 457
StructureParams 457
TBaseMethod 457
IRobotSELFSeismicAnalysisParams Methods 457
GetExcitationDir 458

(c) 2014 Autodesk Inc. All rights reserved

IRobotSELFSeismic_ASCE_7_10_SiteClassType 458
IRobotSELFSeismic_ASCE_7_10_Params 458
IRobotSELFSeismic_ASCE_7_10_Params Members 459
IRobotSELFSeismic_ASCE_7_10_Params Fields 459
I 459
R 460
S1 460
SiteClass 460
Ss 460
StructureTypeX 461
StructureTypeY 461
TBaseValueX 461
TBaseValueY 461
TL 462
IRobotSELFSeismic_ASCE_7_10_StructureType 462
IRobotSELFSeismic_EC_8_Params 462
IRobotSELFSeismic_EC_8_Params Members 463
IRobotSELFSeismic_EC_8_Params Fields 463
Ag 463
Beta 464
Q 464
SiteClass 464
SpectrumType 465
StructureTypeX 465
StructureTypeY 465
IRobotSELFSeismic_EC_8_Params Methods 465
SiteClassEnvelopeCheck 466
SiteClassEnvelopeIsChecked 466
IRobotSELFSeismic_EC_8_StructureType 466
IRobotSELFSeismic_EC_8_SiteClass 467
IRobotSELFSeismic_EC_8_SpectrumType 467
IRobotSELFSeismic_AS_1170_4_Params 467
IRobotSELFSeismic_AS_1170_4_Params Members 468
IRobotSELFSeismic_AS_1170_4_Params Fields 468
Kp 468
Mi 469
Probability 469
SoilCategory 469
Sp 469
StructureTypeX 470
StructureTypeY 470
Z 470

(c) 2014 Autodesk Inc. All rights reserved

IRobotSELFSeismic_AS_1170_4_SoilCategoryType 470
IRobotSELFSeismic_AS_1170_4_StructureType 471
IRobotSELFSeismic_AS_1170_4_ProbabilityType 471
Wind loads simulation 472
IRobotWindLoadsSimulationEngine 472
IRobotWindLoadsSimulationEngine Members 472
IRobotWindLoadsSimulationEngine Fields 472
Params 473
IRobotWindLoadsSimulationEngine Methods 473
Generate 473
IRobotWindLoadsSimulationParams 473
IRobotWindLoadsSimulationParams Members 474
IRobotWindLoadsSimulationParams Fields 474
DeviationPercent 475
DirectionXNEnabled 475
DirectionXNYNEnabled 475
DirectionXNYPEnabled 475
DirectionXPEnabled 476
DirectionXPYNEnabled 476
DirectionXPYPEnabled 476
DirectionYNEnabled 476
DirectionYPEnabled 477
Elements 477
OpeningsClosed 477
TerrainLevel 477
Velocity 478
Nodes 478
Complex attributes of a node 478
Rigid links 479
Compatible nodes 488
Bars 538
Complex attributes of a bar 538
Cables 539
Elastic ground 542
IRobotUpliftDirection 545
IRobotUpliftSense 545
Sections 546
Releases 580
Geometrical imperfections 588
Non-linear hinges 592
Claddings 614
Complex attributes shared by objects of different types 659

(c) 2014 Autodesk Inc. All rights reserved

Material 659
Non-linear constraints 676
Objects 702
Complex attributes of an object 703
Linear releases 704
Panel calculation model 719
Finite elements 807
FE mesh generator 807
Grouping of objects 847
IRobotObjectStructuralType 850
Results 851
Calculation results for nodes 852
Calculation results for bars 869
Data structures containing complex calculation results 884
Calculation results for dynamic analyses 924
Calculation results for finite elements 935
IRobotResultStatusType 1033
IRobotCalculationResume 1034
IRobotCalculationResume Members 1034
IRobotCalculationResume Fields 1034
DiagonalStiffnessMatrixMax 1035
DiagonalStiffnessMatrixMin 1035
DiagonalStiffnessMatrixPrecision 1035
IRobotCalculationResume Methods 1035
GetEnergy 1036
GetEnergyPrecision 1036
Selections 1036
Edit operations 1048
Stories 1052
Quantity survey 1079
Project 1104
Project components 1105
Spread footing 1105
IRConcrFootingPierType 1128
RC Beam 1128
Continuous Footing 1168
MaterialType 1184
IRConcrContinuousFootingSupport Members 1185
IRConcrContinuousFootingSupport Fields 1185
IRConcrContinuousFootingSupport.MaterialType Field 1185
IRConcrContinuousFootingSupport.Name Field 1185
IRConcrContinuousFootingSupport.Width Field 1185

(c) 2014 Autodesk Inc. All rights reserved

RC Column 1185
IRConcrColumnBucklingModel 1202
IRConcrColumnBucklingModel Members 1203
IRConcrColumnBucklingModel Fields 1203
IsDirectionYOff 1204
IsDirectionZOff 1204
IsSwayY 1204
IsSwayZ 1204
IsTotalStructureHeight 1205
ky 1205
kz 1205
Ly 1205
Lz 1206
NumberOfStories 1206
TotalStructureHeight 1206
IRConcrColumnLoads 1207
IRConcrColumnLoads Members 1207
IRConcrColumnLoads Fields 1207
Count 1207
Item 1208
IRConcrColumnLoads Methods 1208
Add 1208
RemoveAll 1208
IRConcrColumnLoad 1209
IRConcrColumnLoad Members 1209
IRConcrColumnLoad Fields 1209
CaseName 1210
CaseType 1210
Fy 1210
Fz 1211
Gamma 1211
MnsY 1211
MnsZ 1211
MyA 1212
MyB 1212
MyC 1212
MzA 1212
MzB 1213
MzC 1213
N 1213
NdN 1213
IRConcrColumnLoadCaseType 1214

(c) 2014 Autodesk Inc. All rights reserved

RC Slab 1214
RC Deep Beam 1228
RC Wall 1232
Reinforcement 1245
Drawing 1266
Support of external file formats 1274
Job preferences 1281
Units and formats 1282
IRobotDatabaseType 1326
IRobotCalculationModelCoherence 1326
Backgrounds 1327
IRobotBackgroundLayers 1327
IRobotBackgroundLayers Members 1328
IRobotBackgroundLayers Fields 1328
Count 1328
IRobotBackgroundLayers Methods 1328
FindName 1329
Get 1329
IsImported 1329
SetImported 1330
IRobotBackgroundInsertParams 1330
IRobotBackgroundInsertParams Members 1330
IRobotBackgroundInsertParams Fields 1331
InsertionPoint 1331
Layers 1331
Mirror 1332
Plane 1332
PositionOnNormalAxis 1332
ReferencePoint 1333
RotationAngle 1333
ScalingFactor 1333
Units 1333
IRobotBackground 1334
IRobotBackground Members 1334
IRobotBackground Fields 1334
Color 1335
FilePath 1335
InsertParams 1335
IsOn 1336
Name 1336
Number 1336
VisibilityRange 1337

(c) 2014 Autodesk Inc. All rights reserved

IRobotBackground Methods 1337
Save 1337
IRobotBackgroundVisibilityRangeType 1337
IRobotBackgroundVisibilityRange 1338
IRobotBackgroundVisibilityRange Members 1338
IRobotBackgroundVisibilityRange Fields 1338
FromPos 1338
ToPos 1339
Type 1339
IRobotBackgroundServer 1339
IRobotBackgroundServer Members 1340
IRobotBackgroundServer Methods 1340
Create 1340
Get 1340
GetAllNumbers 1341
Remove 1341
RemoveAll 1341
IRobotActiveModelType 1360
Presentation of data 1360
Views and layouts 1360
Tables 1361
Graphical structure views 1379
Panel cuts 1381
Influence lines 1385
IRobotViewScreenCaptureResolution 1462
Views for RTF format files 1462
Screen captures 1465
Views for HTML format files 1467
IRobotDialogId 1476
Printouts 1476
Structural axes 1520
Application - main object of the model 1535
IRobotLicenseEntitlement 1547
IRobotLicenseEntitlementStatus 1547
IRobotCalculationsType 1548
Add-ins Manager 1548
Calculation module 1560
IRobotCalcEngineEvents 1591
IRobotCalcEngineEvents Members 1592
IRobotCalcEngineEvents Methods 1592
CalcMessage 1592
IRobotCalcMessageSource 1593

(c) 2014 Autodesk Inc. All rights reserved

IRobotCalcMessageSeverityLevel 1593
IRobotCalculationStatus 1593
IRobotCalculationMode 1594
Direct Analysis Method 1595
IRobotDAMCalcModule 1595
IRobotDAMCalcModule Members 1595
IRobotDAMCalcModule Fields 1596
IsActive 1596
Params 1596
IRobotDAMCalcModule Methods 1597
DeleteModel 1597
Run 1597
IRobotDAMAnalysisType 1597
IRobotDAMNotionalLoads 1598
IRobotDAMNotionalLoads Members 1598
IRobotDAMNotionalLoads Fields 1598
ActiveXN 1599
ActiveXP 1599
ActiveYN 1599
ActiveYP 1600
Coefficient 1600
GravityLoadCombEnabled 1600
LateralLoadCombEnabled 1601
LateralLoadCombType 1601
IRobotDAMLateralLoadCombType 1601
IRobotDAMReducedStiffness 1602
IRobotDAMReducedStiffness Members 1602
IRobotDAMReducedStiffness Fields 1603
RCBeamsValue 1603
RCColumnsAndWallsValue 1603
RCSlabsValue 1604
SteelMembersReductionType 1604
SteelMembersTauBValue 1604
SteelMembersValue 1605
IRobotDAMSteelMembersReductionType 1605
IRobotDAMParams 1605
IRobotDAMParams Members 1606
IRobotDAMParams Fields 1606
Analysis 1606
NotionalLoads 1607
ReducedStiffness 1607
IRobotDAMParams Methods 1607

(c) 2014 Autodesk Inc. All rights reserved

GetNLPDParams 1607
SetNLPDParams 1608
Update 1608
Global data types 1608
Geometric data types 1609
IRobotGeoPlane 1635
Concrete 1665
Supplementing the required reinforcement module with new codes 1665
IRBestForceDataIntegerValue 1699
IRBestForceDataSLSCombType 1700
IRBestCodeService2 1700
IRBestCodeService2 Members 1701
IRBestCodeService2 Fields 1701
RobotVersion 1701
ShowBucklingButton 1701
Plate and shell reinforcement 1702
IRConcrPlateCodeService2 1787
IRConcrPlateCodeService2 Members 1787
IRConcrPlateCodeService2 Fields 1787
RobotVersion 1787
ShowLongTermCracking 1788
Design of RC members 1788
Supplementing RC modules with new codes 1800
IRConcrCodeService 1800
IRConcrCodeService Members 1800
IRConcrCodeService Fields 1801
Beam 1801
Column 1801
IRConcrCodeService Methods 1801
IsConcrComponentServed 1801
IRConcrCodeColumn 1802
IRConcrCodeColumn Members 1802
IRConcrCodeColumn Methods 1802
IsCommandEnabled 1802
Verification 1803
IRConcrCodeColumnCommand 1803
IRConcrCodeReport 1803
IRConcrCodeReport Members 1804
IRConcrCodeReport Methods 1804
AddError 1804
AddMessage 1804
AddWarning 1805

(c) 2014 Autodesk Inc. All rights reserved

IRConcrCodeBeam 1805
IRConcrCodeBeam Members 1805
IRConcrCodeBeam Methods 1805
IsCommandEnabled 1806
Verification 1806
IRConcrCodeBeamCommand 1806
Steel and timber design 1807
EC3 Code 1808
CB71 Code 1853
Member section definition module 1937
Connection module 1938
Knee connection 1939
Column base connections 1970
Pinned column base 1971
Fixed column base 1979
Concrete column base 1989
Angle connection 2030
Tube connection 2042
Gusset plate connections 2059
Beam-principal beam connection 2123
External parameters 2180
Robot Kernel 2207
STR files analyzer 2218
Structure correction 2221

Index ff

(c) 2014 Autodesk Inc. All rights reserved

RobotOM Namespace

This is namespace RobotOM.

Robot Object Model

When modeling a structure, one uses a certain set of concepts that facilitate the process of defining and analyzing the structure. Here belong node, bar, load case and many others. While working in Robot, one encounters such concepts as project – represented by the RTD file, graphical viewers and tables – presenting different aspects of data describing a structure, as well as many other abstractions. In order to make it possible to use the above-mentioned concepts at the level of a programming language, they have been formalized and defined according to COM standards. Program members and data types used to define structures are described in Robot Object Model by means of appropriate interfaces. An interface is understood to mean a set of data, mutually linked in a logical manner, and a set of operations that can be performed on the data. The data are called attributes or members, and operations that can be performed are called functions or methods of the interface. The set of interface functions defines its functionality. Each interface has got a unique name. For instance, a node of a structure is represented by means of the IRobotNode interface. Among the data of the interface one finds, among others, three real numbers corresponding to node coordinates. In order to take advantage of the functionality of a given interface, one should create an object that realizes the interface. The manner of creating an object depends on the currently used programming language. In Visual Basic, for instance, a new object is created by means of the New operator or CreateObject function. Additionally, there is a certain interface type that describes the set of constant values or identifiers. Such interface represents enumeration type. The IRobotProjectType interface is a good example here. Identifiers referring to different types of projects available in Robot are its members. In order to use a member of such an interface, it suffices to provide its name. There is no necessity to create an object to realize it. Robot Object Model is a set of such interfaces. It is contained in the robotom.tlb file in the standard binary format, so that interfaces available in the model could be recognized by the compilers of programming languages and by the system software. The TLB standard extension comes from the Type Library name, as the interface is often, especially at the level of a programming language, identified with data type. Therefore, one can refer to an object realizing the interface called InterfaceName by calling it an object of the InterfaceName type.

Naming conventions

All names begin with a capital letter. If a name consists of more than one word, each name element begins with a capital letter. Additionally, an interface name begins with the IRobot prefix, but for interfaces associated with a specialized module the prefix coresponds to the module name (eg IRJoint for interfaces defining the connection module of Robot). The convention has been adopted to minimize the possibility of conflicting names in programming languages that cannot use many independent name spaces. The names of members of enumeration interface consist, in turn, of capital letters, and particular segments are separated by the underline sign ("_"). Additionally, each name begins with I_FLIN_, where FLIN refers to the first letters of interface name, with the prefix omitted. For instance, names of all elements of IRobotProjectType interface begin with the I_PT_ prefix (e.g. I_PT_FRAME_2D). Such convention facilitates the creation of unique names. The uniqueness of names of members of enumeration interfaces is necessary, for they are often interpreted in programming languages as global constants that should have unique names.

Units All functions of Robot Object Model use the standard SI units, independently of the current settings in Robot options. .

Data Server

All types of data used to define and model a structure (nodes, bars, load cases)) are managed by the Robot Object Model

(c) 2014 Autodesk Inc. All rights reserved

and the Data Server. The server also provides access to the results of calculations carried out for the defined structures. .

Interfaces

	Name	Description
	IRobotStructure (see page 1090)	RobotStructure represents an entire structure. Particular structure components are represented by appropriate interface components. .
	IRobotStructureCache (see page 1098)	Interface enabling buffered access to a structure. It enables adding quickly a whole component set to a structure. To add a whole data set to a structure, ApplyCache function of RobotStructure inerface should be called up. .
	IRobotStructureApplyInfo (see page 1101)	Interface providing access to information about structure components generated as a result of inserting a new data set to a structure. .
	IRobotStructureMergeData (see page 1102)	Interface defining data that may be merged into a structure by means of the Merge function.

I Basic definitions

Data Server manages a large amount of different types of data describing structures. In order to simplify and unify the manipulation of the data, certain basic abstractions have been defined. They allow one to classify the data. Each structure component is represented in the Data Server by means of an object of the appropriate type. All objects of the same type have the same set of attributes and the same functionality described with the set of operations that may be performed on the object. The set of attributes and operations is described formally by means of the appropriate interface. For instance, attributes and complete functionality of a structure node is described by the RobotNode interface. The object type determines univocally how it is possible to work with the object in question, and it defines the access interface for object attributes and functionalities. .

Enumerations

	Name	Description
	IRobotObjectType (see page 36)	RobotObjectType defines identifiers for all types of objects managed by the Data Server .

I.1 Querying mechanism

Data Server defines a simple querying mechanism that allows one to select only the relevant structure elements from among all the structure elements managed by the server. The selection process consists of two phases: determining the structure elements that are of interest for the designer, and taking them from the server and performing an operation on the elements. Selection defined by means of the object of RobotSelection type is used to determine the relevant sub-set of objects - structure components. Thus, a new selection is to be defined by creating a new object of RobotSelection type. By means of the operations available for an object of RobotSelection type, one may describe the selection of the relevant structure components. On the basis of the RobotSelection object, one may obtain a set of objects that meet the criteria determined in Selection. The mode of access to such a set is defined by means of the Collection described formally by the RobotCollection type. The collection simplifies the access to all its elements, orders and indexes them with numbers from 1 to the number of elements in the collection. Thus, it is easy to browse through the objects belonging to the collection. A collection created on the basis of the formerly defined selection allows one to get access to all the object that meet the selection criteria. .

Interfaces

	Name	Description
	IRobotSelection (see page 3)	RobotSelection allows one to define selections of particular structure components .

(c) 2014 Autodesk Inc. All rights reserved

	IRobotCollection (see page 9)	The RobotCollection type is defined to unify the access mode to the sub-set of objects of a given type. Each object belonging to a collection is linked with a number - the index of the object in the collection. In order to obtain an object, one should provide its index. Indexes assume values from 1 to Count (see page 10), where Count (see page 10) is the number of all objects in the collection. One should differentiate between the object index and the user-defined object number (e.g. node number). The user-defined number of an object is ascribed permanently to the object, unless the user changes... more (see page 9)
	IRobotMultiSelection (see page 11)	RobotMultiSelection interface describes a set of many selections of different types. Multi-selection may be useful to describe complex selections of objects of more than one type. .
	IRobotMultiCollection (see page 13)	The RobotMultiCollection groups many collections of objects of various types. .
	IRobotModeSelection (see page 15)	Interface describing selection of a mode for a modal load case. .

I.1.1 IRobotSelection

Class Hierarchy

RobotOM.IRobotSelection

C++

```
interface IRobotSelection : IDispatch;
```

C#

```
public interface IRobotSelection;
```

Visual Basic

```
Public Interface IRobotSelection
```

Description

RobotSelection allows one to define selections of particular structure components .

I.1.1.1 IRobotSelection Members

The following tables list the members exposed by IRobotSelection.

Public Fields

	Name	Description
	Count (see page 4)	Number of objects in the selection.
	Type (see page 4)	Types of objects described by selections (e.g. I_OT_NODE for node selection). The object type is applied when a selection is created and it may not be changed. .

Public Methods

	Name	Description
	Add (see page 5)	Adding the indicated selection to the current one. .
	AddOne (see page 5)	Function adds to the selection a single object with the user specified number.
	AddText (see page 6)	Function adds to the selection a list of objects described with the specified text.
	And (see page 6)	Operation of logical product with the selection provided as the argument .
	AndText (see page 6)	Function peforms the logical product operation of the selection and the object list described with the specified text. .
	Clear (see page 7)	Resetting selection - selection is empty after this operation .

(c) 2014 Autodesk Inc. All rights reserved

	Contains (see page 7)	Function returns the True value if the selection includes an object with the user-specified number.
	Exclude (see page 7)	Logical complement of the current selection and the selection indicated as the argument .
	ExcludeOne (see page 8)	Function deletes from the selection a single object with the user specified number.
	ExcludeText (see page 8)	Function deletes from the selection a list of objects described with the specified text.
	FromText (see page 8)	Robot allows one to define a selection of particular structure components by introducing a text - list describing the selection in the appropriate syntax. Such text may also be used as a parameter of the FromText method that creates on the basis of the text the relevant selection. A text provided as the parameter should describe the selection in the currently used interface language of Robot. .
	Get (see page 9)	Function returns the user number for the successive object included in the selection.
	ToText (see page 9)	Returns a selection description in the form of the appropriate text .

I.1.1.2 IRobotSelection Fields

The fields of the IRobotSelection class are listed here.

Public Fields

	Name	Description
	Count (see page 4)	Number of objects in the selection.
	Type (see page 4)	Types of objects described by selections (e.g. I_OT_NODE for node selection). The object type is applied when a selection is created and it may not be changed. .

I.1.1.2.1 Count

C++

```
HRESULT get_Count(long*);
```

C#

```
public long Count { get; }
```

Visual Basic

```
Public ReadOnly Count As long
```

Description

Number of objects in the selection.

Version

Available since version 8.2.

I.1.1.2.2 Type

C++

```
HRESULT get_Type(IRobotObjectType*);
```

C#

```
public IRobotObjectType Type { get; }
```

Visual Basic

```
Public ReadOnly Type As IRobotObjectType
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Types of objects described by selections (e.g. I_OT_NODE for node selection). The object type is applied when a selection is created and it may not be changed. .

I.1.1.3 IRobotSelection Methods

The methods of the IRobotSelection class are listed here.

Public Methods

	Name	Description
	Add (see page 5)	Adding the indicated selection to the current one. .
	AddOne (see page 5)	Function adds to the selection a single object with the user specified number.
	AddText (see page 6)	Function adds to the selection a list of objects described with the specified text.
	And (see page 6)	Operation of logical product with the selection provided as the argument .
	AndText (see page 6)	Function peforms the logical product operation of the selection and the object list described with the specified text. .
	Clear (see page 7)	Resetting selection - selection is empty after this operation .
	Contains (see page 7)	Function returns the True value if the selection includes an object with the user-specified number.
	Exclude (see page 7)	Logical complement of the current selection and the selection indicated as the argument .
	ExcludeOne (see page 8)	Function deletes from the selection a single object with the user specified number.
	ExcludeText (see page 8)	Function deletes from the selection a list of objects described with the specified text.
	FromText (see page 8)	Robot allows one to define a selection of particular structure components by introducing a text - list describing the selection in the appropriate syntax. Such text may also be used as a parameter of the FromText method that creates on the basis of the text the relevant selection. A text provided as the parameter should describe the selection in the currently used interface language of Robot. .
	Get (see page 9)	Function returns the user number for the successive object included in the selection.
	ToText (see page 9)	Returns a selection description in the form of the appropriate text .

I.1.1.3.1 Add

C++

```
HRESULT Add(IRobotSelection* _sel);
```

C#

```
public void Add(IRobotSelection _sel);
```

Visual Basic

```
Public Sub Add(ByRef _sel As IRobotSelection)
```

Description

Adding the indicated selection to the current one. .

I.1.1.3.2 AddOne

C++

```
HRESULT AddOne(long _obj_num);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public void AddOne(long _obj_num);
```

Visual Basic

```
Public Sub AddOne(_obj_num As long)
```

Description

Function adds to the selection a single object with the user specified number.

Version

Available since version 8.2.

I.1.1.3.3 AddText

C++

```
HRESULT AddText(BSTR _sel_text);
```

C#

```
public void AddText(String _sel_text);
```

Visual Basic

```
Public Sub AddText(_sel_text As String)
```

Description

Function adds to the selection a list of objects described with the specified text.

Version

Available since version 8.2.

I.1.1.3.4 And

C++

```
HRESULT And(IRobotSelection* _sel);
```

C#

```
public void And(IRobotSelection _sel);
```

Visual Basic

```
Public Sub And(ByRef _sel As IRobotSelection)
```

Description

Operation of logical product with the selection provided as the argument .

I.1.1.3.5 AndText

C++

```
HRESULT AndText(BSTR _sel_text);
```

C#

```
public void AndText(String _sel_text);
```

Visual Basic

```
Public Sub AndText(_sel_text As String)
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Function peforms the logical product operation of the selection and the object list described with the specified text. .

Version

Available since version 8.2.

I.1.1.3.6 Clear

C++

```
HRESULT Clear();
```

C#

```
public void Clear();
```

Visual Basic

```
Public Sub Clear()
```

Description

Resetting selection - selection is empty after this operation .

I.1.1.3.7 Contains

C++

```
HRESULT Contains(long _obj_num, VARIANT_BOOL* ret);
```

C#

```
public bool Contains(long _obj_num);
```

Visual Basic

```
Public Function Contains(_obj_num As long) As Boolean
```

Description

Function returns the True value if the selection includes an object with the user-specified number.

Version

Available since version 8.2.

I.1.1.3.8 Exclude

C++

```
HRESULT Exclude(IRobotSelection* _sel);
```

C#

```
public void Exclude(IRobotSelection _sel);
```

Visual Basic

```
Public Sub Exclude(ByRef _sel As IRobotSelection)
```

Description

Logical complement of the current selection and the selection indicated as the argument .

(c) 2014 Autodesk Inc. All rights reserved

I.1.1.3.9 ExcludeOne

C++

```
HRESULT ExcludeOne(long _obj_num);
```

C#

```
public void ExcludeOne(long _obj_num);
```

Visual Basic

```
Public Sub ExcludeOne(_obj_num As long)
```

Description

Function deletes from the selection a single object with the user specified number.

Version

Available since version 8.2.

I.1.1.3.10 ExcludeText

C++

```
HRESULT ExcludeText(BSTR _sel_text);
```

C#

```
public void ExcludeText(String _sel_text);
```

Visual Basic

```
Public Sub ExcludeText(_sel_text As String)
```

Description

Function deletes from the selection a list of objects described with the specified text.

Version

Available since version 8.2.

I.1.1.3.11 FromText

C++

```
HRESULT FromText(BSTR _sel_text);
```

C#

```
public void FromText(String _sel_text);
```

Visual Basic

```
Public Sub FromText(_sel_text As String)
```

Description

Robot allows one to define a selection of particular structure components by introducing a text - list describing the selection in the appropriate syntax. Such text may also be used as a parameter of the FromText method that creates on the basis of the text the relevant selection. A text provided as the parameter should describe the selection in the currently used interface language of Robot. .

(c) 2014 Autodesk Inc. All rights reserved

I.1.1.3.12 Get

C++

```
HRESULT Get(long _idx, long* ret);
```

C#

```
public long Get(long _idx);
```

Visual Basic

```
Public Function Get(_idx As long) As long
```

Description

Function returns the user number for the successive object included in the selection.

Version

Available since version 8.2.

I.1.1.3.13 ToText

C++

```
HRESULT ToText(BSTR* ret);
```

C#

```
public String ToText();
```

Visual Basic

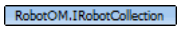
```
Public Function ToText() As String
```

Description

Returns a selection description in the form of the appropriate text .

I.1.2 IRobotCollection

Class Hierarchy



C++

```
interface IRobotCollection : IDispatch;
```

C#

```
public interface IRobotCollection;
```

Visual Basic

```
Public Interface IRobotCollection
```

Description

The RobotCollection type is defined to unify the access mode to the sub-set of objects of a given type. Each object belonging to a collection is linked with a number - the index of the object in the collection. In order to obtain an object, one should provide its index. Indexes assume values from 1 to Count (see page 10), where Count (see page 10) is the number of all objects in the collection. One should differentiate between the object index and the user-defined object number (e.g. node number). The user-defined number of an object is ascribed permanently to the object, unless the user changes the number. The index, in turn, corresponds to the position of the object in the given collection - the same object may have different indexes in different collections. Indexes are ascribed to objects in a given collection in such a way that the increasing index values correspond to the increasing values of user-defined numbers ascribed to the objects. For instance, if

(c) 2014 Autodesk Inc. All rights reserved

a collection consists of three objects with the user-defined numbers 24, 105 and 214, the objects obtain the following indexes in the same collection: 1, 2, 3. .

I.1.2.1 IRobotCollection Members

The following tables list the members exposed by IRobotCollection.

Public Fields

	Name	Description
	Count (see page 10)	Number of all objects in a collection (it is also the index of the last element of the collection) .

Public Methods

	Name	Description
	Get (see page 10)	The function returns an object with the indicated index in the collection. The type of returned object depends on the collection type - for a collection of nodes, the Get method returns an object of the RobotNode type. .

I.1.2.2 IRobotCollection Fields

The fields of the IRobotCollection class are listed here.

Public Fields

	Name	Description
	Count (see page 10)	Number of all objects in a collection (it is also the index of the last element of the collection) .

I.1.2.2.1 Count

C++

```
HRESULT get_Count(long*);
```

C#

```
public long Count { get; }
```

Visual Basic

```
Public ReadOnly Count As long
```

Description

Number of all objects in a collection (it is also the index of the last element of the collection) .

I.1.2.3 IRobotCollection Methods

The methods of the IRobotCollection class are listed here.

Public Methods

	Name	Description
	Get (see page 10)	The function returns an object with the indicated index in the collection. The type of returned object depends on the collection type - for a collection of nodes, the Get method returns an object of the RobotNode type. .

I.1.2.3.1 Get

C++

```
HRESULT Get(long _index, IDispatch* ret);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IDispatch Get(long _index);
```

Visual Basic

```
Public Function Get(_index As long) As IDispatch
```

Description

The function returns an object with the indicated index in the collection. The type of returned object depends on the collection type - for a collection of nodes, the Get method returns an object of the RobotNode type. .

I.1.3 IRobotMultiSelection

Class Hierarchy

C++

```
interface IRobotMultiSelection : IDispatch;
```

C#

```
public interface IRobotMultiSelection;
```

Visual Basic

```
Public Interface IRobotMultiSelection
```

Description

RobotMultiSelection interface describes a set of many selections of different types. Multi-selection may be useful to describe complex selections of objects of more than one type. .

I.1.3.1 IRobotMultiSelection Members

The following tables list the members exposed by IRobotMultiSelection.

Public Fields

	Name	Description
	CaseCmpnt (see page 12)	Selected number of a load case component (it refers only to code combinations); zero value indicates selection of all components (selection is not limited to a specific component).
	Modes (see page 12)	Mode selection is treated as a supplement to load case selection .

Public Methods

	Name	Description
	Exist (see page 13)	The function returns True (non-zero value) if the multi-selection contains the selection of objects of the indicated type. .
	Get (see page 13)	The function returns selection of objects of the indicated type .
	Set (see page 13)	The function sets the indicated selection as the selection of objects of a given type within the multi-selection .

I.1.3.2 IRobotMultiSelection Fields

The fields of the IRobotMultiSelection class are listed here.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	CaseCmpnt (see page 12)	Selected number of a load case component (it refers only to code combinations); zero value indicates selection of all components (selection is not limited to a specific component).
	Modes (see page 12)	Mode selection is treated as a supplement to load case selection .

I.1.3.2.1 CaseCmpnt

C++

```
HRESULT get_CaseCmpnt(long*);
HRESULT put_CaseCmpnt(long);
```

C#

```
public long CaseCmpnt { get; set; }
```

Visual Basic

```
Public CaseCmpnt As long
```

Description

Selected number of a load case component (it refers only to code combinations); zero value indicates selection of all components (selection is not limited to a specific component).

Version

Available since version 3.5.

I.1.3.2.2 Modes

C++

```
HRESULT get_Modes(IRobotModeSelection**);
```

C#

```
public IRobotModeSelection Modes { get; }
```

Visual Basic

```
Public ReadOnly Modes As IRobotModeSelection
```

Description

Mode selection is treated as a supplement to load case selection .

Version

Available since version 3.5.

I.1.3.3 IRobotMultiSelection Methods

The methods of the IRobotMultiSelection class are listed here.

Public Methods

	Name	Description
	Exist (see page 13)	The function returns True (non-zero value) if the multi-selection contains the selection of objects of the indicated type. .
	Get (see page 13)	The function returns selection of objects of the indicated type .
	Set (see page 13)	The function sets the indicated selection as the selection of objects of a given type within the multi-selection .

(c) 2014 Autodesk Inc. All rights reserved

I.1.3.3.1 Exist

C++

```
HRESULT Exist(IRobotObjectType _obj_type, VARIANT_BOOL* ret);
```

C#

```
public bool Exist(IRobotObjectType _obj_type);
```

Visual Basic

```
Public Function Exist(_obj_type As IRobotObjectType) As Boolean
```

Description

The function returns True (non-zero value) if the multi-selection contains the selection of objects of the indicated type. .

I.1.3.3.2 Get

C++

```
HRESULT Get(IRobotObjectType _obj_type, IRobotSelection** ret);
```

C#

```
public IRobotSelection Get(IRobotObjectType _obj_type);
```

Visual Basic

```
Public Function Get(_obj_type As IRobotObjectType) As IRobotSelection
```

Description

The function returns selection of objects of the indicated type .

I.1.3.3.3 Set

C++

```
HRESULT Set(IRobotObjectType _obj_type, IRobotSelection* _sel);
```

C#

```
public void Set(IRobotObjectType _obj_type, IRobotSelection _sel);
```

Visual Basic

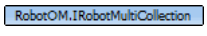
```
Public Sub Set(_obj_type As IRobotObjectType, ByRef _sel As IRobotSelection)
```

Description

The function sets the indicated selection as the selection of objects of a given type within the multi-selection .

I.1.4 IRobotMultiCollection

Class Hierarchy



C++

```
interface IRobotMultiCollection : IDispatch;
```

C#

```
public interface IRobotMultiCollection;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Interface IRobotMultiCollection
```

Description

The RobotMultiCollection groups many collections of objects of various types. .

I.1.4.1 IRobotMultiCollection Members

The following tables list the members exposed by IRobotMultiCollection.

Public Methods

	Name	Description
	Exist (see page 14)	The function returns True (non-zero value) if there is a collection of objects of the indicated type within the multi-collection .
	Get (see page 14)	The function returns a collection of objects of a similar type, contained in the given multi-collection. Before selecting a collection, one should check by means of the Exist (see page 14) function if it has been defined within the framework of the multi-collection. .
	Set (see page 15)	The function sets the indicated collection as the collection of objects of the indicated type within the multi-collection .

I.1.4.2 IRobotMultiCollection Methods

The methods of the IRobotMultiCollection class are listed here.

Public Methods

	Name	Description
	Exist (see page 14)	The function returns True (non-zero value) if there is a collection of objects of the indicated type within the multi-collection .
	Get (see page 14)	The function returns a collection of objects of a similar type, contained in the given multi-collection. Before selecting a collection, one should check by means of the Exist (see page 14) function if it has been defined within the framework of the multi-collection. .
	Set (see page 15)	The function sets the indicated collection as the collection of objects of the indicated type within the multi-collection .

I.1.4.2.1 Exist

C++

```
HRESULT Exist(IRobotObjectType _obj_type, VARIANT_BOOL* ret);
```

C#

```
public bool Exist(IRobotObjectType _obj_type);
```

Visual Basic

```
Public Function Exist(_obj_type As IRobotObjectType) As Boolean
```

Description

The function returns True (non-zero value) if there is a collection of objects of the indicated type within the multi-collection .

I.1.4.2.2 Get

C++

```
HRESULT Get(IRobotObjectType _obj_type, IRobotCollection** ret);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRobotCollection Get(IRobotObjectType _obj_type);
```

Visual Basic

```
Public Function Get(_obj_type As IRobotObjectType) As IRobotCollection
```

Description

The function returns a collection of objects of a similar type, contained in the given multi-collection. Before selecting a collection, one should check by means of the Exist (see page 14) function if it has been defined within the framework of the multi-collection. .

I.1.4.2.3 Set

C++

```
HRESULT Set(IRobotObjectType _obj_type, IRobotCollection* _collection);
```

C#

```
public void Set(IRobotObjectType _obj_type, IRobotCollection _collection);
```

Visual Basic

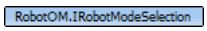
```
Public Sub Set(_obj_type As IRobotObjectType, ByRef _collection As IRobotCollection)
```

Description

The function sets the indicated collection as the collection of objects of the indicated type within the multi-collection .

I.1.5 IRobotModeSelection

Class Hierarchy



C++

```
interface IRobotModeSelection : IDispatch;
```

C#

```
public interface IRobotModeSelection;
```

Visual Basic

```
Public Interface IRobotModeSelection
```

Description

Interface describing selection of a mode for a modal load case. .

Version

Available since version 3.5.

I.1.5.1 IRobotModeSelection Members

The following tables list the members exposed by IRobotModeSelection.

Public Fields

	Name	Description
	Combination (see page 16)	Selected type of a combination which includes all load case modes .
	Mode (see page 16)	Mode number - its interpretation depends on a chosen type of mode selection .

(c) 2014 Autodesk Inc. All rights reserved

	Type (see page 16)	Type of mode selection.

I.1.5.2 IRobotModeSelection Fields

The fields of the IRobotModeSelection class are listed here.

Public Fields

	Name	Description
	Combination (see page 16)	Selected type of a combination which includes all load case modes .
	Mode (see page 16)	Mode number - its interpretation depends on a chosen type of mode selection .
	Type (see page 16)	Type of mode selection.

I.1.5.2.1 Combination

C++

```
HRESULT get_Combination(IRobotModeCombinationType*);
HRESULT put_Combination(IRobotModeCombinationType);
```

C#

```
public IRobotModeCombinationType Combination { get; set; }
```

Visual Basic

```
Public Combination As IRobotModeCombinationType
```

Description

Selected type of a combination which includes all load case modes .

Version

Available since version 3.5.

I.1.5.2.2 Mode

C++

```
HRESULT get_Mode(long*);
HRESULT put_Mode(long);
```

C#

```
public long Mode { get; set; }
```

Visual Basic

```
Public Mode As long
```

Description

Mode number - its interpretation depends on a chosen type of mode selection .

Version

Available since version 3.5.

I.1.5.2.3 Type

C++

```
HRESULT get_Type(IRobotModeSelectionType*);
HRESULT put_Type(IRobotModeSelectionType);
```

C#

```
public IRobotModeSelectionType Type { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Type As IRobotModeSelectionType
```

Description

Type of mode selection.

Version

Available since version 3.5.

I.2 Labels - complex attributes of objects

The notion of Label has been introduced to simplify the description of complex object attributes - structure components. A complex attribute is an attribute for the description of which a set of values is required, as opposed to a simple attribute described by one value. A support is an example of a complex attribute. In order to define a support, one should provide an entire set of values describing particular support parameters. On the other hand, the entire support is an attribute of a supported node. The set of all values describing particular quantities-values of a complex attribute becomes in a sense the value of this attribute. A label is used to name and identify such a complex value. It suffices to provide once the values of all attribute parameters (e.g. a support) and associate a name with this set (a character string), to make it possible to use the name only instead of the entire set of values. Thus, in order to apply an appropriate support to a node, it suffices to provide the name of a support (the label associated with the set of values describing all support parameters) and it is not necessary to reinforcement-introduce the values of particular support parameters. Since there are many different types of complex attributes, the notion of a complex attribute i.e. a label type (RobotLabelType) has been introduced. The name of a label is unique within the set of labels of the same type. .

Enumerations

	Name	Description
	IRobotLabelType (see page 17)	A set of Label types available in the Data Server is defined. .
	IRobotPredefinedLabel (see page 29)	Identifiers of labels automatically generated by program.

Interfaces

	Name	Description
	IRobotLabel (see page 18)	The RobotLabel interface describes the complex attribute referred to as Label in the Robot Object Model. The complex attribute is defined by an entire set of values and not by one value, as in the case of a simple attribute. The set of values is associated with a name - a character string. The set of values describing complex attribute parameters consists the data of the label, while the character string associated with the set consists the label name.
	IRobotLabelServer (see page 20)	All labels are managed (created, made accessible, deleted) by the label server. Irrespective of label type, a label must be defined by means of the server to be used in structure definition later on. .

I.2.1 IRobotLabelType

C++

```
enum IRobotLabelType;
```

C#

```
public enum IRobotLabelType;
```

Visual Basic

```
Public Enum IRobotLabelType
```

(c) 2014 Autodesk Inc. All rights reserved

Members

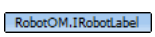
Members	Description
I_LT_NODE_RELEASE = 1	Label describing a node release .
I_LT_NODE_COMPATIBILITY = 2	Label describing a compatibility .
I_LT_BAR_SECTION = 3	Label describing a bar section .
I_LT_BAR_RELEASE = 4	Label describing a bar release .
I_LT_BAR_OFFSET = 5	Label describing a bar offset .
I_LT_BAR_CABLE = 6	Label describing a cable .
I_LT_BAREND_BRACKET = 7	Label describing a bracket at one of bar ends.
I_LT_PANEL_THICKNESS = 11	Label describing a plate thickness.
I_LT_PANEL_REINFORCEMENT = 12	Label describing a plate reinforcement.
I_LT_UNKNOWN = -1	Unknown label type Available since version 1.7.
I_LT_SUPPORT = 0	Label describing a support Available since version 2.0.
I_LT_MATERIAL = 8	Label describing a material Available since version 2.0.
I_LT_LINEAR_RELEASE = 13	Linear release between panels. Available since version 2.5.
I_LT_BAR_ELASTIC_GROUND = 14	Elastic ground (bars). Available since version 2.5.
I_LT_NODE_RIGID_LINK = 15	Rigid link. Available since version 2.5.
I_LT_MEMBER_TYPE = 16	Member type for steel/aluminum and timber design. Available since version 2.5.
I_LT_VEHICLE = 17	Vehicle definition. Available since version 3.
I_LT_SOLID_PROPERTIES = 18	Solid properties. Available since version 3.
I_LT_BAR_GEO_IMPERFECTIONS = 19	Geometrical imperfections. Available since version 3.
I_LT_BAR_NONLINEAR_HINGE = 20	Non-linear hinge. Available since version 3.
I_LT_CLADDING = 21	Available since version 3.5.
I_LT_PANEL_CALC_MODEL = 22	Label describing a panel calculation model. Available since version 9.7.
I_LT_MEMBER_REINFORCEMENT_PARAMS = 25	Available since version 12.

Description

A set of Label types available in the Data Server is defined. .

I.2.2 IRobotLabel

Class Hierarchy



C++

```
interface IRobotLabel : IDispatch;
```

C#

```
public interface IRobotLabel;
```

Visual Basic

```
Public Interface IRobotLabel
```

(c) 2014 Autodesk Inc. All rights reserved

Description

The RobotLabel interface describes the complex attribute referred to as Label in the Robot Object Model. The complex attribute is defined by an entire set of values and not by one value, as in the case of a simple attribute. The set of values is associated with a name - a character string. The set of values describing complex attribute parameters consists the data of the label, while the character string associated with the set consists the label name.

I.2.2.1 IRobotLabel Members

The following tables list the members exposed by IRobotLabel.

Public Fields

	Name	Description
	Data (see page 19)	Object containing the label data - set of values describing the complex attribute represented by the label. The object type depends on the label type. For instance, the data of a label describing a node support (i.e. labels of the I_LT_NODE_SUPPORT type) are represented by the object of the RobotNodeSupportData type. .
	Name (see page 20)	Name associated with the label - the name us unique among all the labels of the same type and therefore it may be used to identify the label. .
	Type (see page 20)	Label type .
	UniqueId (see page 20)	Unique label identifier Available since version 1.7.

I.2.2.2 IRobotLabel Fields

The fields of the IRobotLabel class are listed here.

Public Fields

	Name	Description
	Data (see page 19)	Object containing the label data - set of values describing the complex attribute represented by the label. The object type depends on the label type. For instance, the data of a label describing a node support (i.e. labels of the I_LT_NODE_SUPPORT type) are represented by the object of the RobotNodeSupportData type. .
	Name (see page 20)	Name associated with the label - the name us unique among all the labels of the same type and therefore it may be used to identify the label. .
	Type (see page 20)	Label type .
	UniqueId (see page 20)	Unique label identifier Available since version 1.7.

I.2.2.2.1 Data

C++

```
HRESULT get_Data(IDispatch*);
```

C#

```
public IDispatch Data { get; }
```

Visual Basic

```
Public ReadOnly Data As IDispatch
```

Description

Object containing the label data - set of values describing the complex attribute represented by the label. The object type depends on the label type. For instance, the data of a label describing a node support (i.e. labels of the I_LT_NODE_SUPPORT type) are represented by the object of the RobotNodeSupportData type. .

(c) 2014 Autodesk Inc. All rights reserved

I.2.2.2.2 Name

C++

```
HRESULT get_Name(BSTR*);
```

C#

```
public String Name { get; }
```

Visual Basic

```
Public ReadOnly Name As String
```

Description

Name associated with the label - the name us unique among all the labels of the same type and therefore it may be used to identify the label. .

I.2.2.2.3 Type

C++

```
HRESULT get_Type(IRobotLabelType*);
```

C#

```
public IRobotLabelType Type { get; }
```

Visual Basic

```
Public ReadOnly Type As IRobotLabelType
```

Description

Label type .

I.2.2.2.4 UniqueId

C++

```
HRESULT get_UniqueId(long*);
```

C#

```
public long UniqueId { get; }
```

Visual Basic

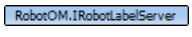
```
Public ReadOnly UniqueId As long
```

Description

Unique label identifier Available since version 1.7.

I.2.3 IRobotLabelServer

Class Hierarchy



C++

```
interface IRobotLabelServer : IDispatch;
```

C#

```
public interface IRobotLabelServer;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Interface IRobotLabelServer
```

Description

All labels are managed (created, made accessible, deleted) by the label server. Irrespective of label type, a label must be defined by means of the server to be used in structure definition later on. .

I.2.3.1 IRobotLabelServer Members

The following tables list the members exposed by IRobotLabelServer.

Public Methods

	Name	Description
	Create (see page 23)	The function creates and returns a new label of the defined type and with the defined name. if labels of the indicated type are stored in a database (as in the case foundation bar section type labels - I_LT_BAR_SECTION), then, the function Create looks through the currently available databases to fine a label with the defined name. If the label is not found in the database, all parameter values describing a label will be initialized with appropriate values from the database. If the user defines a label that is absent from the database, then, he should define appropriate values of... more (see page 23)
	CreateLike (see page 23)	Function creates and returns a new label of the indicated type with the indicated name. A data set defining the label will be copied from the label named the "_like_name" parameter.
	Delete (see page 24)	The function deletes a label of the indicatd type and name. If the label has been used as an attribute value of a structure component, then, the value of the attribute is set as undefined after deleting the label. For instance, if a bar had STEEL defined as its material, then, after deleting the STEEL label of the I_LT_BAR_MATERIAL type from the label server, the bar will have no material defined. .
	Exist (see page 24)	The function checks if there exist the label of the indicated type and name, i.e. if it has been saved in the structure. .
	FindWithId (see page 24)	Function finds and returns a label with the specified identifier. If such a label is not found, then an "empty reference" is returned (Nothing in the languages such as Visual Basic, NULL in the languages such as C++). Available since version 1.7.
	Get (see page 24)	The function returns the label of the indicated type and name. If such label already exists, running the function leads to critical error. .
	GetAll (see page 25)	The function returns all labels of all types that have been saved in the structure. .
	GetAvailableNames (see page 25)	Function returns the table with names of available labels of a given type. The available labels are those that may be assigned to structure elements by means of the SetLabel function. They include all the labels used earlier in the current structure and global labels - available in each structure.
	GetDefault (see page 25)	Function returns the name of a default label of the specified type.
	GetMany (see page 26)	The function returns a collection containing all label of the indicated type saved in the structure .
	GetPredefinedName (see page 26)	Function returns the current name of the given predefined label. .
	GetUniqueId (see page 26)	Function returns a unique identifier for the specified label.
	IsAvailable (see page 27)	Function returns a non-zero value (True) if the indicated label is available - it may be assigned to a structure element by means of the SetLabel function. .
	IsPredefinedName (see page 27)	Function returns True if the given type and labe name correspond with one of the predefined label. The identifier of the found predefined lable will be added to the initial parameter _ret_predef_lab.

(c) 2014 Autodesk Inc. All rights reserved

	IsUsed (see page 27)	Function checks if the label of the specified type with the specified name is currently used in th structure.
	SetDefault (see page 28)	Function sets a name of a default label of the specified type. Default labels are applied during automatic generation of structure elements e.g. as a result of edit operations performed on a structure.
	Store (see page 28)	Once a label is created and the values of its particular parameters are defined, one should save it in the structure so that it could be applied to appropriate structure component. If a label of the same type and name has already been saved in the structure, the values of all its parameters will be up-dated on the basis of the newly-defined values. .
	StoreWithName (see page 28)	Function saves the indicated label under the indicated name.

I.2.3.2 IRobotLabelServer Methods

The methods of the IRobotLabelServer class are listed here.

Public Methods

	Name	Description
	Create (see page 23)	The function creates and returns a new label of the defined type and with the defined name. if labels of the indicated type are stored in a database (as in the case foundation bar section type labels - I_LT_BAR_SECTION), then, the function Create looks through the currently available databases to fine a label with the defined name. If the label is not found in the database, all parameter values describing a label will be initialized with appropriate values from the database. If the user defines a label that is absent from the database, then, he should define appropriate values of... more (see page 23)
	CreateLike (see page 23)	Function creates and returns a new label of the indicated type with the indicated name. A data set defining the label will be copied from the label named the "_like_name" parameter.
	Delete (see page 24)	The function deletes a label of the indicatd type and name. If the label has been used as an attribute value of a structure component, then, the value of the attribute is set as undefined after deleting the label. For instance, if a bar had STEEL defined as its material, then, after deleting the STEEL label of the I_LT_BAR_MATERIAL type from the label server, the bar will have no material defined. .
	Exist (see page 24)	The function checks if there exist the label of the indicated type and name, i.e. if it has been saved in the structure. .
	FindWithId (see page 24)	Function finds and returns a label with the specified identifier. If such a label is not found, then an "empty reference" is returned (Nothing in the languages such as Visual Basic, NULL in the languages such as C++). Available since version 1.7.
	Get (see page 24)	The function returns the label of the indicated type and name. If such label already exists, running the function leads to critical error. .
	GetAll (see page 25)	The function returns all labels of all types that have been saved in the structure. .
	GetAvailableNames (see page 25)	Function returns the table with names of available labels of a given type. The available labels are those that may be assigned to structure elements by means of the SetLabel function. They include all the labels used earlier in the current structure and global labels - available in each structure.
	GetDefault (see page 25)	Function returns the name of a default label of the specified type.
	GetMany (see page 26)	The function returns a collection containing all label of the indicated type saved in the structure .
	GetPredefinedName (see page 26)	Function returns the current name of the given predefined label. .
	GetUniqueId (see page 26)	Function returns a unique identifier for the specified label.

(c) 2014 Autodesk Inc. All rights reserved

	IsAvailable (see page 27)	Function returns a non-zero value (True) if the indicated label is available - it may be assigned to a structure element by means of the SetLabel function. .
	IsPredefinedName (see page 27)	Function returns True if the given type and labe name correspond with one of the predefined label. The identifier of the found predefined lable will be added to the initial parameter _ret_predef_lab.
	IsUsed (see page 27)	Function checks if the label of the specified type with the specified name is currently used in th structure.
	SetDefault (see page 28)	Function sets a name of a default label of the specified type. Default labels are applied during automatic generation of structure elements e.g. as a result of edit operations performed on a structure.
	Store (see page 28)	Once a label is created and the values of its particular parameters are defined, one should save it in the structure so that it could be applied to appropriate structure component. If a label of the same type and name has already been saved in the structure, the values of all its parameters will be up-dated on the basis of the newly-defined values. .
	StoreWithName (see page 28)	Function saves the indicated label under the indicated name.

I.2.3.2.1 Create

C++

```
HRESULT Create(IRobotLabelType _label_type, BSTR _label_name, IRobotLabel** ret);
```

C#

```
public IRobotLabel Create(IRobotLabelType _label_type, String _label_name);
```

Visual Basic

```
Public Function Create(_label_type As IRobotLabelType, _label_name As String) As IRobotLabel
```

Description

The function creates and returns a new label of the defined type and with the defined name. if labels of the indicated type are stored in a database (as in the case foundation bar section type labels - I_LT_BAR_SECTION), then, the function Create looks through the currently available databases to fine a label with the defined name. If the label is not found in the database, all parameter values describing a label will be initialized with appropriate values from the database. If the user defines a label that is absent from the database, then, he should define appropriate values of all the parameters describing the complex attribute. .

I.2.3.2.2 CreateLike

C++

```
HRESULT CreateLike(IRobotLabelType _lab_type, BSTR _lab_name, BSTR _like_name,
IRobotLabel** ret);
```

C#

```
public IRobotLabel CreateLike(IRobotLabelType _lab_type, String _lab_name, String
_like_name);
```

Visual Basic

```
Public Function CreateLike(_lab_type As IRobotLabelType, _lab_name As String, _like_name As
String) As IRobotLabel
```

Description

Function creates and returns a new label of the indicated type with the indicated name. A data set defining the label will be copied from the label named the "_like_name" parameter.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.

I.2.3.2.3 Delete

C++

```
HRESULT Delete(IRobotLabelType _label_type, BSTR _label_name);
```

C#

```
public void Delete(IRobotLabelType _label_type, String _label_name);
```

Visual Basic

```
Public Sub Delete(_label_type As IRobotLabelType, _label_name As String)
```

Description

The function deletes a label of the indicatd type and name. If the label has been used as an attribute value of a structure component, then, the value of the attribute is set as undefined after deleting the label. For instance, if a bar had STEEL defined as its material, then, after deleting the STEEL label of the I_LT_BAR_MATERIAL type from the label server, the bar will have no material defined. .

I.2.3.2.4 Exist

C++

```
HRESULT Exist(IRobotLabelType _label_type, BSTR _label_name, VARIANT_BOOL* ret);
```

C#

```
public bool Exist(IRobotLabelType _label_type, String _label_name);
```

Visual Basic

```
Public Function Exist(_label_type As IRobotLabelType, _label_name As String) As Boolean
```

Description

The function checks if there exist the label of the indicated type and name, i.e. if it has been saved in the structure. .

I.2.3.2.5 FindWithId

C++

```
HRESULT FindWithId(long _unique_id, IRobotLabel** ret);
```

C#

```
public IRobotLabel FindWithId(long _unique_id);
```

Visual Basic

```
Public Function FindWithId(_unique_id As long) As IRobotLabel
```

Description

Function finds and returns a label with the specified identifier. If such a label is not found, then an "empty reference" is returned (Nothing in the languages such as Visual Basic, NULL in the languages such as C++). Available since version 1.7.

I.2.3.2.6 Get

C++

```
HRESULT Get(IRobotLabelType _label_type, BSTR _label_name, IRobotLabel** ret);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRobotLabel Get(IRobotLabelType _label_type, String _label_name);
```

Visual Basic

```
Public Function Get(_label_type As IRobotLabelType, _label_name As String) As IRobotLabel
```

Description

The function returns the label of the indicated type and name. If such label already exists, running the function leads to critical error. .

I.2.3.2.7 GetAll

C++

```
HRESULT GetAll(IRobotCollection** ret);
```

C#

```
public IRobotCollection GetAll();
```

Visual Basic

```
Public Function GetAll() As IRobotCollection
```

Description

The function returns all labels of all types that have been saved in the structure. .

I.2.3.2.8 GetAvailableNames

C++

```
HRESULT GetAvailableNames(IRobotLabelType _label_type, IRobotNamesArray** ret);
```

C#

```
public IRobotNamesArray GetAvailableNames(IRobotLabelType _label_type);
```

Visual Basic

```
Public Function GetAvailableNames(_label_type As IRobotLabelType) As IRobotNamesArray
```

Description

Function returns the table with names of available labels of a given type. The available labels are those that may be assigned to structure elements by means of the SetLabel function. They include all the labels used earlier in the current structure and global labels - available in each structure.

Version

Available since version 3.

I.2.3.2.9 GetDefault

C++

```
HRESULT GetDefault(IRobotLabelType _lab_type, BSTR* ret);
```

C#

```
public String GetDefault(IRobotLabelType _lab_type);
```

Visual Basic

```
Public Function GetDefault(_lab_type As IRobotLabelType) As String
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Function returns the name of a default label of the specified type.

Version

Available since version 3.5.

I.2.3.2.10 GetMany

C++

```
HRESULT GetMany(IRobotLabelType _label_type, IRobotCollection** ret);
```

C#

```
public IRobotCollection GetMany(IRobotLabelType _label_type);
```

Visual Basic

```
Public Function GetMany(_label_type As IRobotLabelType) As IRobotCollection
```

Description

The function returns a collection containing all label of the indicated type saved in the structure .

I.2.3.2.11 GetPredefinedName

C++

```
HRESULT GetPredefinedName(IRobotPredefinedLabel _predef_label, BSTR* ret);
```

C#

```
public String GetPredefinedName(IRobotPredefinedLabel _predef_label);
```

Visual Basic

```
Public Function GetPredefinedName(_predef_label As IRobotPredefinedLabel) As String
```

Description

Function returns the current name of the given predefined label. .

Version

Available since version 11.

I.2.3.2.12 GetUniqueId

C++

```
HRESULT GetUniqueId(IRobotLabelType _lab_type, BSTR _lab_name, long* ret);
```

C#

```
public long GetUniqueId(IRobotLabelType _lab_type, String _lab_name);
```

Visual Basic

```
Public Function GetUniqueId(_lab_type As IRobotLabelType, _lab_name As String) As long
```

Description

Function returns a unique identifier for the specified label.

Version

Available since version 8.2.

(c) 2014 Autodesk Inc. All rights reserved

I.2.3.2.13 IsAvailable

C++

```
HRESULT IsAvailable(IRobotLabelType _lab_type, BSTR _lab_name, VARIANT_BOOL* ret);
```

C#

```
public bool IsAvailable(IRobotLabelType _lab_type, String _lab_name);
```

Visual Basic

```
Public Function IsAvailable(_lab_type As IRobotLabelType, _lab_name As String) As Boolean
```

Description

Function returns a non-zero value (True) if the indicated label is available - it may be assigned to a structure element by means of the SetLabel function. .

Version

Available since version 3.

I.2.3.2.14 IsPredefinedName

C++

```
HRESULT IsPredefinedName(IRobotLabelType _lab_type, BSTR _lab_name, RobotPredefinedLabel*
_ret_predef_lab, VARIANT_BOOL* ret);
```

C#

```
public bool IsPredefinedName(IRobotLabelType _lab_type, String _lab_name,
RobotPredefinedLabel* _ret_predef_lab);
```

Visual Basic

```
Public Function IsPredefinedName(_lab_type As IRobotLabelType, _lab_name As String, ByRef
_ret_predef_lab As RobotPredefinedLabel*) As Boolean
```

Description

Function returns True if the given type and labe name correspond with one of the predefined label. The identifier of the found predefined lable will be added to the initial parameter _ret_predef_lab.

Version

Available since version 11.

I.2.3.2.15 IsUsed

C++

```
HRESULT IsUsed(IRobotLabelType _lab_type, BSTR _lab_name, VARIANT_BOOL* ret);
```

C#

```
public bool IsUsed(IRobotLabelType _lab_type, String _lab_name);
```

Visual Basic

```
Public Function IsUsed(_lab_type As IRobotLabelType, _lab_name As String) As Boolean
```

Description

Function checks if the label of the specified type with the specified name is currently used in th structure.

Version

Available since version 8.1.

(c) 2014 Autodesk Inc. All rights reserved

I.2.3.2.16 SetDefault

C++

```
HRESULT SetDefault(IRobotLabelType _lab_type, BSTR _lab_name, VARIANT_BOOL* ret);
```

C#

```
public bool SetDefault(IRobotLabelType _lab_type, String _lab_name);
```

Visual Basic

```
Public Function SetDefault(_lab_type As IRobotLabelType, _lab_name As String) As Boolean
```

Description

Function sets a name of a default label of the specified type. Default labels are applied during automatic generation of structure elements e.g. as a result of edit operations performed on a structure.

Version

Available since version 3.5.

I.2.3.2.17 Store

C++

```
HRESULT Store(IRobotLabel* _label);
```

C#

```
public void Store(IRobotLabel _label);
```

Visual Basic

```
Public Sub Store(ByRef _label As IRobotLabel)
```

Description

Once a label is created and the values of its particular parameters are defined, one should save it in the structure so that it could be applied to appropriate structure component. If a label of the same type and name has already been saved in the structure, the values of all its parameters will be up-dated on the basis of the newly-defined values. .

I.2.3.2.18 StoreWithName

C++

```
HRESULT StoreWithName(IRobotLabel* _label, BSTR _label_name);
```

C#

```
public void StoreWithName(IRobotLabel _label, String _label_name);
```

Visual Basic

```
Public Sub StoreWithName(ByRef _label As IRobotLabel, _label_name As String)
```

Description

Function saves the indicated label under the indicated name.

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

I.2.4 IRobotPredefinedLabel

C++

```
enum IRobotPredefinedLabel;
```

C#

```
public enum IRobotPredefinedLabel;
```

Visual Basic

```
Public Enum IRobotPredefinedLabel
```

Members

Members	Description
I_PL_MEMBER_TYPE_TIMBER_MEMBER = 1	Available since version 11.
I_PL_MEMBER_TYPE_TIMBER_COLUMN = 2	Available since version 11.
I_PL_MEMBER_TYPE_TIMBER_BEAM = 3	Available since version 11.
I_PL_MEMBER_TYPE_SIMPLE_BAR = 4	Available since version 11.
I_PL_MEMBER_TYPE_COLUMN = 5	Available since version 11.
I_PL_MEMBER_TYPE_BEAM = 6	Available since version 11.
I_PL_PANEL_CALC_MODEL_SHELL = 11	Available since version 11.
I_PL_PANEL_CALC_MODEL_SLAB_FLEXIBLE_DIAPHRAGM = 12	Available since version 11.
I_PL_PANEL_CALC_MODEL_SLAB_RIGID_DIAPHRAGM = 13	Available since version 11.
I_PL_PANEL_CALC_MODEL_CURTAIN_WALL = 14	Available since version 11.
I_PL_PANEL_CALC_MODEL_DECK_SLAB = 15	Available since version 11.
I_PL_PANEL_REINFORCEMENT_RC_FLOOR = 21	Available since version 11.
I_PL_PANEL_REINFORCEMENT_RC_WALL = 22	Available since version 11.
I_PL_PANEL_REINFORCEMENT_RC_SHELL = 23	Available since version 11.
I_PL_MEMBER_TYPE_RC_BEAM = 7	Available since version 11.
I_PL_MEMBER_TYPE_RC_COLUMN = 8	Available since version 11.
I_PL_MEMBER_REINFORCEMENT_PARAMS_STANDARD = 31	Available since version 12.
I_PL_PANEL_CALC_MODEL_SLAB_XY_DIAPHRAGM = 16	Available since version 12.

Description

Identifiers of labels automatically generated by program.

Version

Available since version 11.

I.3 Uniform access to structure components

In order to unify access mode to different structure components, the following two notions have been defined: Object (RobotDataObject) and Object Server (RobotDataObjectServer). An Object represents any structure component. The Object Server manages all objects of the same type. For instance, a structure node is represented in the Robot Object Model s an object of the RobotNode type, and all nodes are managed (created, made accessible, deleted) by the RobotNodeServer type. .

(c) 2014 Autodesk Inc. All rights reserved

Interfaces

	Name	Description
	IRobotDataObject (see page 30)	RobotDataObject defines a common interface for structure components. Each structure component has its own user-defined number that is unique among all the components of the same type. The user may use the number to identify objects. There is one uniform manner of ascribing and accessing complex attributes (labels) defined for all structure components. .
	IRobotDataObjectServer (see page 33)	All objects of the same type are managed by the same object server. RobotDataObjectServer defines a common interface for such servers. .

I.3.1 IRobotDataObject

Class Hierarchy

RobotOM.IRobotDataObject

C++

```
interface IRobotDataObject : IDispatch;
```

C#

```
public interface IRobotDataObject;
```

Visual Basic

```
Public Interface IRobotDataObject
```

Description

RobotDataObject defines a common interface for structure components. Each structure component has its own user-defined number that is unique among all the components of the same type. The user may use the number to identify objects. There is one uniform manner of ascribing and accessing complex attributes (labels) defined for all structure components. .

I.3.1.1 IRobotDataObject Members

The following tables list the members exposed by IRobotDataObject.

Public Fields

	Name	Description
	Number (see page 31)	Object number is assigned by the user; it is qunique among all the components of the same type. .

Public Methods

	Name	Description
	GetLabel (see page 31)	The function returns a label of the indicated type applied to the object. If the object does not have any label of the type, running the function leads to a critical error. .
	GetLabelName (see page 32)	The function returns the name of a label of the indicated type, applied to the object. If the object does not have any label of the type, running the function leads to a critical error. .
	GetLabels (see page 32)	The function returns a collection containing all labels (of different types) defined for a given object. .
	HasLabel (see page 32)	The function returns True (non-zero value) if a label of the type has already been defined for the object. .
	RemoveLabel (see page 32)	The function deletes a label of the indicated type. .
	SetLabel (see page 33)	The function applies the defined label to an object. At a given moment, an object may have only one label of a type applied (e.g. there may not be two supports defined for one node). If an object has a formerly applied label of the same type, it is replaced by the new one. .

(c) 2014 Autodesk Inc. All rights reserved

I.3.1.2 IRobotDataObject Fields

The fields of the IRobotDataObject class are listed here.

Public Fields

	Name	Description
	Number (see page 31)	Object number is assigned by the user; it is qunique among all the components of the same type. .

I.3.1.2.1 Number

C++

```
HRESULT get_Number(long*);
HRESULT put_Number(long);
```

C#

```
public long Number { get; set; }
```

Visual Basic

```
Public Number As long
```

Description

Object number is assigned by the user; it is qunique among all the components of the same type. .

I.3.1.3 IRobotDataObject Methods

The methods of the IRobotDataObject class are listed here.

Public Methods

	Name	Description
	GetLabel (see page 31)	The function returns a label of the indicated type applied to the object. If the object does not have any label of the type, running the function leads to a critical error. .
	GetLabelName (see page 32)	The function returns the name of a label of the indicated type, applied to the object. If the object does not have any label of the type, running the function leads to a critical error. .
	GetLabels (see page 32)	The function returns a collection containing all labels (of different types) defined for a given object. .
	HasLabel (see page 32)	The function returns True (non-zero value) if a label of the type has already been defined for the object. .
	RemoveLabel (see page 32)	The function deletes a label of the indicated type. .
	SetLabel (see page 33)	The function applies the defined label to an object. At a given moment, an object may have only one label of a type applied (e.g. there may not be two supports defined for one node). If an object has a formerly applied label of the same type, it is replaced by the new one. .

I.3.1.3.1 GetLabel

C++

```
HRESULT GetLabel(IRobotLabelType _label_type, IRobotLabel** ret);
```

C#

```
public IRobotLabel GetLabel(IRobotLabelType _label_type);
```

Visual Basic

```
Public Function GetLabel(_label_type As IRobotLabelType) As IRobotLabel
```

(c) 2014 Autodesk Inc. All rights reserved

Description

The function returns a label of the indicated type applied to the object. If the object does not have any label of the type, running the function leads to a critical error. .

I.3.1.3.2 GetLabelName

C++

```
HRESULT GetLabelName(IRobotLabelType _label_type, BSTR* ret);
```

C#

```
public String GetLabelName(IRobotLabelType _label_type);
```

Visual Basic

```
Public Function GetLabelName(_label_type As IRobotLabelType) As String
```

Description

The function returns the name of a label of the indicated type, applied to the object. If the object does not have any label of the type, running the function leads to a critical error. .

I.3.1.3.3 GetLabels

C++

```
HRESULT GetLabels(IRobotCollection** ret);
```

C#

```
public IRobotCollection GetLabels();
```

Visual Basic

```
Public Function GetLabels() As IRobotCollection
```

Description

The function returns a collection containing all labels (of different types) defined for a given object. .

I.3.1.3.4 HasLabel

C++

```
HRESULT HasLabel(IRobotLabelType _label_type, VARIANT_BOOL* ret);
```

C#

```
public bool HasLabel(IRobotLabelType _label_type);
```

Visual Basic

```
Public Function HasLabel(_label_type As IRobotLabelType) As Boolean
```

Description

The function returns True (non-zero value) if a label of the type has already been defined for the object. .

I.3.1.3.5 RemoveLabel

C++

```
HRESULT RemoveLabel(IRobotLabelType _label_type);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public void RemoveLabel(IRobotLabelType _label_type);
```

Visual Basic

```
Public Sub RemoveLabel(_label_type As IRobotLabelType)
```

Description

The function deletes a label of the indicated type. .

I.3.1.3.6 SetLabel

C++

```
HRESULT SetLabel(IRobotLabelType _label_type, BSTR _label_name);
```

C#

```
public void SetLabel(IRobotLabelType _label_type, String _label_name);
```

Visual Basic

```
Public Sub SetLabel(_label_type As IRobotLabelType, _label_name As String)
```

Description

The function applies the defined label to an object. At a given moment, an object may have only one label of a type applied (e.g. there may not be two supports defined for one node). If an object has a formerly applied label of the same type, it is replaced by the new one. .

I.3.2 IRobotDataObjectServer

Class Hierarchy

RobotOM.IRobotDataObjectServer

C++

```
interface IRobotDataObjectServer : IDispatch;
```

C#

```
public interface IRobotDataObjectServer;
```

Visual Basic

```
Public Interface IRobotDataObjectServer
```

Description

All objects of the same type are managed by the same object server. RobotDataObjectServer defines a common interface for such servers. .

I.3.2.1 IRobotDataObjectServer Members

The following tables list the members exposed by IRobotDataObjectServer.

Public Methods

	Name	Description
	Delete (see page 34)	The function deletes the object of the indicated number. .
	DeleteMany (see page 34)	The function deletes all objects that meet the criteria of the indicated selection. .
	Exist (see page 35)	The function returns True (non-zero value) if the object of the indicated name already exists. .

(c) 2014 Autodesk Inc. All rights reserved

	Get (see page 35)	The function returns an object with the indicated user-defined number. The type of the returned object agrees with the type of objects managed by the server. If the object of the indicated number does not exist, running the function leads to critical error. .
	GetAll (see page 35)	The function returns a collection containing all objects managed by the server. .
	GetMany (see page 36)	The function returns a collection of objects that meet the criteria of the indicated selection. .
	RemoveLabel (see page 36)	The function removes the labels of the indicated type from all objects that meet the criteria of the indicated selection. .
	SetLabel (see page 36)	The function applies a label (identified by the type and name) to all objects that meet the criteria of the indicated selection. .

I.3.2.2 IRobotDataObjectServer Methods

The methods of the IRobotDataObjectServer class are listed here.

Public Methods

	Name	Description
	Delete (see page 34)	The function deletes the object of the indicated number. .
	DeleteMany (see page 34)	The function deletes all objects that meet the criteria of the indicated selection. .
	Exist (see page 35)	The function returns True (non-zero value) if the object of the indicated name already exists. .
	Get (see page 35)	The function returns an object with the indicated user-defined number. The type of the returned object agrees with the type of objects managed by the server. If the object of the indicated number does not exist, running the function leads to critical error. .
	GetAll (see page 35)	The function returns a collection containing all objects managed by the server. .
	GetMany (see page 36)	The function returns a collection of objects that meet the criteria of the indicated selection. .
	RemoveLabel (see page 36)	The function removes the labels of the indicated type from all objects that meet the criteria of the indicated selection. .
	SetLabel (see page 36)	The function applies a label (identified by the type and name) to all objects that meet the criteria of the indicated selection. .

I.3.2.2.1 Delete

C++

```
HRESULT Delete(long _number);
```

C#

```
public void Delete(long _number);
```

Visual Basic

```
Public Sub Delete(_number As long)
```

Description

The function deletes the object of the indicated number. .

I.3.2.2.2 DeleteMany

C++

```
HRESULT DeleteMany(IRobotSelection* _selection);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public void DeleteMany(IRobotSelection _selection);
```

Visual Basic

```
Public Sub DeleteMany(ByRef _selection As IRobotSelection)
```

Description

The function deletes all objects that meet the criteria of the indicated selection. .

I.3.2.2.3 Exist

C++

```
HRESULT Exist(long _number, VARIANT_BOOL* ret);
```

C#

```
public bool Exist(long _number);
```

Visual Basic

```
Public Function Exist(_number As long) As Boolean
```

Description

The function returns True (non-zero value) if the object of the indicated name already exists. .

I.3.2.2.4 Get

C++

```
HRESULT Get(long _number, IRobotDataObject** ret);
```

C#

```
public IRobotDataObject Get(long _number);
```

Visual Basic

```
Public Function Get(_number As long) As IRobotDataObject
```

Description

The function returns an object with the indicated user-defined number. The type of the returned object agrees with the type of objects managed by the server. If the object of the indicated number does not exist, running the function leads to critical error. .

I.3.2.2.5 GetAll

C++

```
HRESULT GetAll(IRobotCollection** ret);
```

C#

```
public IRobotCollection GetAll();
```

Visual Basic

```
Public Function GetAll() As IRobotCollection
```

Description

The function returns a collection containing all objects managed by the server. .

(c) 2014 Autodesk Inc. All rights reserved

I.3.2.2.6 GetMany

C++

```
HRESULT GetMany(IRobotSelection* _selection, IRobotCollection** ret);
```

C#

```
public IRobotCollection GetMany(IRobotSelection _selection);
```

Visual Basic

```
Public Function GetMany(ByRef _selection As IRobotSelection) As IRobotCollection
```

Description

The function returns a collection of objects that meet the criteria of the indicated selection. .

I.3.2.2.7 RemoveLabel

C++

```
HRESULT RemoveLabel(IRobotSelection* _object_selection, IRobotLabelType _label_type);
```

C#

```
public void RemoveLabel(IRobotSelection _object_selection, IRobotLabelType _label_type);
```

Visual Basic

```
Public Sub RemoveLabel(ByRef _object_selection As IRobotSelection, _label_type As
IRobotLabelType)
```

Description

The function removes the labels of the indicated type from all objects that meet the criteria of the indicated selection. .

I.3.2.2.8 SetLabel

C++

```
HRESULT SetLabel(IRobotSelection* _object_selection, IRobotLabelType _label_type, BSTR
_label_name);
```

C#

```
public void SetLabel(IRobotSelection _object_selection, IRobotLabelType _label_type, String
_label_name);
```

Visual Basic

```
Public Sub SetLabel(ByRef _object_selection As IRobotSelection, _label_type As
IRobotLabelType, _label_name As String)
```

Description

The function applies a label (identified by the type and name) to all objects that meet the criteria of the indicated selection. .

I.4 IRobotObjectType

C++

```
enum IRobotObjectType;
```

C#

```
public enum IRobotObjectType;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Enum IRobotObjectType
```

Members

Members	Description
I_OT_NODE = 0	Node .
I_OT_BAR = 1	Bar .
I_OT_PANEL = 4	Panel.
I_OT_FINITE_ELEMENT = 5	Finite element.
I_OT_GEOMETRY = 6	Geometrical object .
I_OT_FAMILY = 3	Group.
I_OT_CASE = 2	Load case .
I_OT_OBJECT = -2	Shared auxiliary identifier for the following types of structure components: bar, slab, volumetric object and geometric object Available since version 2.0.
I_OT_UNDEFINED = -1	Auxiliary identifier denoting undefined or unknown object type Available since version 2.0.
I_OT_VOLUME = 7	Volumetric object. Available since version 3.

Description

RobotObjectType defines identifiers for all types of objects managed by the Data Server .

II Structure components

All structure components are represented by the objects of relevant data types. .

Enumerations

	Name	Description
	IRobotObjectStructuralType (see page 850)	Type of object as a structure component.

II.1 Load cases

Robot Object Model makes it possible to perform operations on load cases available in the Robot program. There are two basic types of load cases: a simple case and combinations. A simple case definition consists of a list of load records, while a combination contains a list of simple cases with the relevant coefficients. .

Enumerations

	Name	Description
	IRobotCaseNature (see page 410)	The set of case natures accepted by Robot is determined. .
	IRobotCaseAnalizeType (see page 410)	Robot Object Model defines a set of analysis types for load cases accepted by Robot. .
	IRobotCombinationType (see page 433)	Type of load case combinations available in Robot. .
	IRobotCaseType (see page 433)	Among load cases, one distinguishes three basic types: a simple case, a combination and a code combination. .
	IRobotLimitState (see page 435)	Limit state type.
	IRobotCaseAnalysisModesFilterType (see page 437)	Available methods of mode filtering.
	IRobotDynamicAnalysisDampingType (see page 446)	

(c) 2014 Autodesk Inc. All rights reserved

	IRobotCaseRelatedValueType (see page 446)	A set of identifiers for types of values that can be associated with the load case.

Interfaces

	Name	Description
	IRobotCase (see page 408)	The part of the load case interface that is shared by simple cases and combinations has been defined using RobotCase. .
	IRobotCaseServer (see page 411)	Server of load cases .
	IRobotSimpleCase (see page 420)	A simple load case is defined using the list of load records. .
	IRobotCaseCombination (see page 426)	A combination is defined by means of simple load cases with appropriate factors. .
	IRobotCaseFactorMngr (see page 431)	Interface granting access to the list of pairs (load case, coefficient). All pairs from the list are numbered (indexed) with numbers from 1 to the number of items (Count (see page 431)). .
	IRobotCaseFactor (see page 434)	Interface describing a pair (load case, coefficient) used in the definition of load case combination. .
	IRobotCaseCollection (see page 435)	Collection of load cases .
	IRobotCaseAnalysisModesFilter (see page 435)	Definition of modes taken into account during structure dynamic analysis. .
	IRobotDynamicAnalysisExcitationDirection (see page 438)	
	IRobotDynamicAnalysisDamping (see page 443)	Damping parameters for the dynamic analysis.

II.1.1 Load records

A load record is used to model loads applied to a structure .

Enumerations

	Name	Description
	IRobotLoadRecordType (see page 55)	Load record types available in the Robot Object Model .

Interfaces

	Name	Description
	IRobotLoadRecord (see page 56)	The load record is used to model a load applied to a structure. RobotLoadRecord is described by an interface shared by all load record types available in Robot. .
	IRobotLoadRecordMngr (see page 59)	Load record manager manages logical lists of inter-linked records (e.g. a list of records linked with the same load case). The records are numbered from 1 to the number of records in the list .
	IRobotLoadRecordLinear3D (see page 61)	Load record ot the I_LRT_LINEAR_3D type. .
	IRobotLoadRecordIn3Points (see page 63)	Load record for a load defined by three points .
	IRobotLoadRecordThermalIn3Points (see page 66)	Thermal load in three points. .
	IRobotLoadRecordInContour (see page 69)	Planar load on the contour. For a uniform load only the values for the first point of the load should be specified (I_ICRV_PX1, I_ICRV_PY1 and I_ICRV_PZ1) without providing the point coordinates. For a variable load all the load values and coordinates of the appropriate points should be specified applying the SetPoint (see page 72)() function. .

(c) 2014 Autodesk Inc. All rights reserved

	IRobotLoadRecordLinear (see page 73)	
	IRobotLoadRecord2 (see page 75)	Extended interface of the record that describes a load. .
	IRobotLoadRecordBarTrapezoidal (see page 77)	Trapezoidal load on bars.
	IRobotLoadRecordDead (see page 79)	Load record describing self-weight.
	IRobotLoadRecordCommon (see page 81)	The interface providing access to the extended set of common attributes for load records of different types.

II.1.1.1 Values describing load records

Each load record is defined by means of several numerical values. The significance of particular values depends on the load record type. A set if identifiers is defined to refer to particular values in the relevant load records. In order to define correctly a load record, the appropriate set of values (defined for this record type) should be recognized first. The value identifiers are used as parameters of the SetValue() and GetValue() functions for the load record. .

Enumerations

	Name	Description
	IRobotNodeForceRecordValues (see page 40)	Identifiers of the values describing load records of the following types: I_LRT_NODE_FORCE and I_LRT_NODE_FORCE_MASS. .
	IRobotNodeDisplacementRecordValues (see page 40)	The identifier of values for the records of the I_LRT_NODE_DISPLACEMENT type .
	IRobotNodeAuxiliaryRecordValues (see page 41)	Identifiers of values describing load records of the I_LRT_NODE_AUXILIARY type.
	IRobotBarDilatationRecordValues (see page 41)	Identifiers of the values describing load records of the following type: .
	IRobotBarForceConcentrateRecordValues (see page 41)	Identifier of the values describing load records of the following types: I_LRT_BAR_FORCE_CONCENTRATE, I_LRT_BAR_FORCE_CONCENTRATE_MASS .
	IRobotBarMomentDistributedRecordValues (see page 42)	Identifiers of the values describing load records of the following type: I_LRT_BAR_MOMENT_DISTRIBUTED .
	IRobotBarUniformRecordValues (see page 42)	Identifiers of the values describing load records of the following type: I_LRT_BAR_UNIFORM and I_LRT_BAR_UNIFORM_MASS .
	IRobotBarTrapezoidaleRecordValues (see page 43)	Identifiers of the values describing load records of the following type: I_LRT_BAR_TRAPEZOIDALE and I_LRT_BAR_TRAPEZOIDALE_MASS. .
	IRobotBarThermalRecordValues (see page 43)	Identifiers of the values describing load records of the following type: I_LRT_BAR_THERMAL .
	IRobotBarDeadRecordValues (see page 44)	Identifiers of the values describing load records of the following type: I_LRT_BAR_DEAD. .
	IRobotPointAuxiliaryRecordValues (see page 44)	Identifiers of the values describing load records of the following type: I_LRT_POINT_AUXILIARY. .
	IRobotIn3PointsRecordValues (see page 45)	Identifiers of the values describing load records of the following type: I_LRT_IN_3_POINTS. .
	IRobotPressureRecordValues (see page 45)	Identifiers of the values describing load records of the following type: I_LRT_PRESSURE. .
	IRobotLinearRecordValues (see page 46)	Identifiers of the values describing load records of the following type: I_LRT_LINEAR .
	IRobotMassActivationRecordValues (see page 46)	Identifiers of the values describing load records of the following type: I_LRT_MASS_ACTIVATION .
	IRobotLinear3DRecordValues (see page 47)	Identifiers of the values describing load records of the following type: I_LRT_LINEAR_3D .
	IRobotThermalIn3PointsRecordValues (see page 48)	Identifiers of the values describing load records of the following type: I_LRT_THERMAL_IN_3_POINTS (planar thermal load 3P). .

(c) 2014 Autodesk Inc. All rights reserved

	IRobotInContourRecordValues (see page 48)	Identifiers of the values describing load records of the following type: I_LRT_IN_CONTOUR (planar load on contour - uniform or variable) .
	IRobotUniformRecordValues (see page 49)	Identifiers of the values describing load records of the following type: I_LRT_UNIFORM .
	IRobotThermalRecordValues (see page 50)	Identifiers of the values describing load records of the following type: I_LRT_THERMAL (thermal load: uniform, linear, at three points) .
	IRobotNodeForceInPointRecordValues (see page 50)	Values that define a load resulting from a force acting at a point (load record type : I_LRT_NODE_FORCE_IN_POINT).
	IRobotLinearOnEdgesRecordValues (see page 51)	Parameters of linear load on edges.
	IRobotDeadRecordValues (see page 51)	
	IRobotSurfaceOnObjectRecordValues (see page 52)	Indentifiers of values for records of I_LRT_SURFACE_ON_OBJECT type.
	IRobotMobilePointForceRecordValues (see page 52)	Values describing a load record of the I_LRT_MOBILE_POINT_FORCE type. .
	IRobotMobileDistributedRecordValues (see page 53)	Values describing a load of the I_LRT_MOBILE_DISTRIBUTED type. .
	IRobotNodeVelocityRecordValues (see page 54)	Identifiers of values for the load records of the I_LRT_NODE_VELOCITY type.
	IRobotNodeAccelerationRecordValues (see page 54)	Identifiers of values for the load records of the I_LRT_NODE_ACCELERATION type.

II.1.1.1.1 IRobotNodeForceRecordValues

C++

```
enum IRobotNodeForceRecordValues;
```

C#

```
public enum IRobotNodeForceRecordValues;
```

Visual Basic

```
Public Enum IRobotNodeForceRecordValues
```

Description

Identifiers of the values describing load records of the following types: I_LRT_NODE_FORCE and I_LRT_NODE_FORCE_MASS. .

II.1.1.1.2 IRobotNodeDisplacementRecordValues

C++

```
enum IRobotNodeDisplacementRecordValues;
```

C#

```
public enum IRobotNodeDisplacementRecordValues;
```

Visual Basic

```
Public Enum IRobotNodeDisplacementRecordValues
```

Description

The identifier of values for the records of the I_LRT_NODE_DISPLACEMENT type .

(c) 2014 Autodesk Inc. All rights reserved

II.1.1.1.3 IRobotNodeAuxiliaryRecordValues

C++

```
enum IRobotNodeAuxiliaryRecordValues;
```

C#

```
public enum IRobotNodeAuxiliaryRecordValues;
```

Visual Basic

```
Public Enum IRobotNodeAuxiliaryRecordValues
```

Description

Identifiers of values describing load records of the I_LRT_NODE_AUXILIARY type.

II.1.1.1.4 IRobotBarDilatationRecordValues

C++

```
enum IRobotBarDilatationRecordValues;
```

C#

```
public enum IRobotBarDilatationRecordValues;
```

Visual Basic

```
Public Enum IRobotBarDilatationRecordValues
```

Description

Identifiers of the values describing load records of the following type: .

II.1.1.1.5 IRobotBarForceConcentrateRecordValues

C++

```
enum IRobotBarForceConcentrateRecordValues;
```

C#

```
public enum IRobotBarForceConcentrateRecordValues;
```

Visual Basic

```
Public Enum IRobotBarForceConcentrateRecordValues
```

Members

Members	Description
I_BFCRV_OFFSET_Y = 21	Offset value of the force application point (Y local direction). Available since version 3.
I_BFCRV_OFFSET_Z = 22	Offset value of the force application point (Z local direction). Available since version 3.
I_BFCRV_GENERATE_CALC_NODE = 12	1 - generation of calculation node 0 - no calculation node generated. Available since version 11.

Description

Identifier of the values describing load records of the following types: I_LRT_BAR_FORCE_CONCENTRATE, I_LRT_BAR_FORCE_CONCENTRATE_MASS .

(c) 2014 Autodesk Inc. All rights reserved

II.1.1.1.6 IRobotBarMomentDistributedRecordValues

C++

```
enum IRobotBarMomentDistributedRecordValues;
```

C#

```
public enum IRobotBarMomentDistributedRecordValues;
```

Visual Basic

```
Public Enum IRobotBarMomentDistributedRecordValues
```

Description

Identifiers of the values describing load records of the following type: I_LRT_BAR_MOMENT_DISTRIBUTED .

II.1.1.1.7 IRobotBarUniformRecordValues

C++

```
enum IRobotBarUniformRecordValues;
```

C#

```
public enum IRobotBarUniformRecordValues;
```

Visual Basic

```
Public Enum IRobotBarUniformRecordValues
```

Members

Members	Description
I_BURV_PX = 0	Linear pressure (or mass value) in X direction.
I_BURV_PY = 1	Linear pressure (or mass value) in Y direction.
I_BURV_PZ = 2	Linear pressure (or mass value) in Z direction.
I_BURV_ALPHA = 8	Alpha angle of the force vector (if additional vector rotation is defined).
I_BURV_BETA = 9	Beta angle of the force vector (if additional vector rotation is defined).
I_BURV_GAMMA = 10	Gamma angle of the force vector (if additional vector rotation is defined).
I_BURV_LOCAL = 11	Zero value if loads values refer to global coordinate system. Otherwise - to local.
I_BURV_PROJECTION = 12	Flag indicating whether load is to be projected (value different from zero) or not (zero).
I_BURV_RELATIVE = 13	Understanding of the distance values measured along the bar length: relative - different from zero, absolute - zero.
I_BURV_OFFSET_Y = 21	Offset value of the uniform load application point (Y local direction). Available since version 3.
I_BURV_OFFSET_Z = 22	Offset value of the uniform load application point (Z local direction). Available since version 3.

Description

Identifiers of the values describing load records of the following type: I_LRT_BAR_UNIFORM and I_LRT_BAR_UNIFORM_MASS .

(c) 2014 Autodesk Inc. All rights reserved

II.1.1.1.8 IRobotBarTrapezoidaleRecordValues

C++

```
enum IRobotBarTrapezoidaleRecordValues;
```

C#

```
public enum IRobotBarTrapezoidaleRecordValues;
```

Visual Basic

```
Public Enum IRobotBarTrapezoidaleRecordValues
```

Members

Members	Description
I_BTRV_PX2 = 0	A value of the X-component of the linear bar pressure in ending point.
I_BTRV_PY2 = 1	A value of the Y-component of the linear bar pressure in ending point.
I_BTRV_PZ2 = 2	A value of the Z-component of the linear bar pressure in ending point.
I_BTRV_PX1 = 3	A value of the X-component of the linear bar pressure in staring point.
I_BTRV_PY1 = 4	A value of the Y-component of the linear bar pressure in staring point.
I_BTRV_PZ1 = 5	A value of the Z-component of the linear bar pressure in staring point.
I_BTRV_X2 = 6	Distance of the ending point from the bar origin.
I_BTRV_X1 = 7	Distance of the starting point from the bar origin.
I_BTRV_ALPHA = 8	Alpha angle of the force vector (if additional vector rotation is defined).
I_BTRV_BETA = 9	Beta angle of the force vector (if additional vector rotation is defined).
I_BTRV_GAMMA = 10	Gamma angle of the force vector (if additional vector rotation is defined).
I_BTRV_PROJECTION = 12	Flag indicating whether the load is to be projected (value different from zero) or not (zero value).
I_BTRV_RELATIVE = 13	Understanding of the distance values measured along the bar length: relative - different from zero, absolute - zero.

Description

Identifiers of the values describing load records of the following type: I_LRT_BAR_TRAPEZOIDALE and I_LRT_BAR_TRAPEZOIDALE_MASS. .

II.1.1.1.9 IRobotBarThermalRecordValues

C++

```
enum IRobotBarThermalRecordValues;
```

C#

```
public enum IRobotBarThermalRecordValues;
```

Visual Basic

```
Public Enum IRobotBarThermalRecordValues
```

(c) 2014 Autodesk Inc. All rights reserved

Members

Members	Description
I_BTRV_TX = 0	Uniform temperature change (heating or cooling) of the bar.
I_BTRV_TY = 1	Difference in the temperature between the extreme cross section fibers along local y bar axis.
I_BTRV_TZ = 2	Difference in the temperature between the extreme cross section fibers along local z bar axis.

Description

Identifiers of the values describing load records of the following type: I_LRT_BAR_THERMAL .

II.1.1.1.10 IRobotBarDeadRecordValues

C++

```
enum IRobotBarDeadRecordValues;
```

C#

```
public enum IRobotBarDeadRecordValues;
```

Visual Basic

```
Public Enum IRobotBarDeadRecordValues
```

Members

Members	Description
I_BDRV_ENTIRE_STRUCTURE = 15	Flag indicating that load is applied to the whole structure. Available since version 3.

Description

Identifiers of the values describing load records of the following type: I_LRT_BAR_DEAD. .

II.1.1.1.11 IRobotPointAuxiliaryRecordValues

C++

```
enum IRobotPointAuxiliaryRecordValues;
```

C#

```
public enum IRobotPointAuxiliaryRecordValues;
```

Visual Basic

```
Public Enum IRobotPointAuxiliaryRecordValues
```

Members

Members	Description
I_PARV_FX = 0	X component value of the force load.
I_PARV_FY = 1	Y component value of the force load.
I_PARV_FZ = 2	Z component value of the force load.
I_PARV_CX = 3	X component value of the moment load.
I_PARV_CY = 4	Y component value of the moment load.
I_PARV_CZ = 5	Z component value of the moment load.
I_PARV_ALPHA = 8	Alpha angle of the force vector (if additional vector rotation is defined).
I_PARV_BETA = 9	Beta angle of the force vector (if additional vector rotation is defined).

(c) 2014 Autodesk Inc. All rights reserved

I_PARV_GAMMA = 10	Gamma angle of the force vector (if additional vector rotation is defined).
I_PARV_X = 11	X coordinate of a point in which force load is applied.
I_PARV_Y = 12	Y coordinate of a point in which force load is applied.
I_PARV_Z = 13	Z coordinate of a point in which force load is applied.

Description

Identifiers of the values describing load records of the following type: I_LRT_POINT_AUXILIARY. .

II.1.1.1.12 IRobotIn3PointsRecordValues

C++

```
enum IRobotIn3PointsRecordValues;
```

C#

```
public enum IRobotIn3PointsRecordValues;
```

Visual Basic

```
Public Enum IRobotIn3PointsRecordValues
```

Members

Members	Description
I_3PRV_PX1 = 0	A value of the pressure PX intensity acting in point 1.
I_3PRV_PY1 = 1	A value of the pressure PY intensity acting in point 2.
I_3PRV_PZ1 = 2	A value of the pressure PZ intensity acting in point 3.
I_3PRV_PX2 = 3	A value of the pressure PX intensity acting in point 2.
I_3PRV_PY2 = 4	A value of the pressure PY intensity acting in point 2.
I_3PRV_PZ2 = 5	A value of the pressure PZ intensity acting in point 3.
I_3PRV_PX3 = 6	A value of the pressure PX intensity acting in point 3.
I_3PRV_PY3 = 7	A value of the pressure PY intensity acting in point 2.
I_3PRV_PZ3 = 8	A value of the pressure PZ intensity acting in point 3.
I_3PRV_PROJECTION = 12	Projected/not projected load value (0/-1).
I_3PRV_N1 = 9	The first node number in which load pressure values are defined.
I_3PRV_N2 = 10	The second node number in which load pressure values are defined.
I_3PRV_N3 = 11	The third node number in which load pressure values are defined.
I_3PRV_LOCAL_SYSTEM = 13	Local/global loads.

Description

Identifiers of the values describing load records of the following type: I_LRT_IN_3_POINTS. .

II.1.1.1.13 IRobotPressureRecordValues

C++

```
enum IRobotPressureRecordValues;
```

C#

```
public enum IRobotPressureRecordValues;
```

Visual Basic

```
Public Enum IRobotPressureRecordValues
```

(c) 2014 Autodesk Inc. All rights reserved

Members

Members	Description
I_PRV_P = 0	Constant pressure value (independent from liquid).
I_PRV_RO = 1	Unit weight of liquid Available since version 2.0.
I_PRV_H = 2	Liquid level Available since version 2.0.
I_PRV_DIRECTION = 6	Direction of gravity for liquid (0 : -X, 1 : -Y, 2 : -Z, 3 : X, 4 : Y, 5 : Z).

Description

Identifiers of the values describing load records of the following type: I_LRT_PRESSURE. .

II.1.1.1.14 IRobotLinearRecordValues

C++

```
enum IRobotLinearRecordValues;
```

C#

```
public enum IRobotLinearRecordValues;
```

Visual Basic

```
Public Enum IRobotLinearRecordValues
```

Members

Members	Description
I_LRV_MX1 = 0	A value of the moment MX acting at point 1.
I_LRV_MX2 = 1	A value of the moment MX acting at point 2.
I_LRV_PZ1 = 2	A value of the force PZ acting at point 1.
I_LRV_PZ2 = 3	A value of the force PZ acting at point 2.
I_LRV_MY1 = 4	A value of the moment MY acting at point 1.
I_LRV_MY2 = 5	A value of the moment MY acting at point 2.
I_LRV_NODE_1 = 6	Node number defined at point 1.
I_LRV_NODE_2 = 7	Node number defined at point 1.
I_LRV_LOCAL = 11	Zero value if loads values refer to global coordinate system. Otherwise - to local.

Description

Identifiers of the values describing load records of the following type: I_LRT_LINEAR .

II.1.1.1.15 IRobotMassActivationRecordValues

C++

```
enum IRobotMassActivationRecordValues;
```

C#

```
public enum IRobotMassActivationRecordValues;
```

Visual Basic

```
Public Enum IRobotMassActivationRecordValues
```

Members

Members	Description
I_MARV_INPUT_DIR_X = 0	Flag to add/activate mass in global X-direction.

(c) 2014 Autodesk Inc. All rights reserved

I_MARV_INPUT_DIR_Y = 1	Flag to add/activate mass in global Y-direction.
I_MARV_INPUT_DIR_Z = 2	Flag to add/activate mass in global Z-direction.
I_MARV_FACTOR = 3	Load to mass conversion coefficient (standard = 1.0).
I_MARV_CASE_NUM = 4	Load case which is to be converted.
I_MARV_ACTIVATION_DIR = 6	Direction of mass conversion (load direction which components are to be converted to masses: 0 denotes X, 1 - Y, 2 - Z).
I_MARV_SIGN = 7	Conversion sign (direction of loads supplementary to ACTIVATION_DIR) +1 or -1.

Description

Identifiers of the values describing load records of the following type: I_LRT_MASS_ACTIVATION .

II.1.1.1.16 IRobotLinear3DRecordValues

C++

```
enum IRobotLinear3DRecordValues;
```

C#

```
public enum IRobotLinear3DRecordValues;
```

Visual Basic

```
Public Enum IRobotLinear3DRecordValues
```

Members

Members	Description
I_L3DRV_PX1 = 0	A value of the intensity of the lineraly changing load PX acting in point 1 (linear load is defined in 2 points).
I_L3DRV_PY1 = 1	A value of the intensity of the lineraly changing load PY acting in point 1 (linear load is defined in 2 points).
I_L3DRV_PZ1 = 2	A value of the intensity of the lineraly changing load PZ acting in point 1 (linear load is defined in 2 points).
I_L3DRV_MX1 = 3	A value of the intensity of the lineraly changing moment MX acting in point 1 (linear load is defined in 2 points).
I_L3DRV_MY1 = 4	A value of the intensity of the lineraly changing moment MY acting in point 1 (linear load is defined in 2 points).
I_L3DRV_MZ1 = 5	A value of the intensity of the lineraly changing moment MZ acting in point 1 (linear load is defined in 2 points).
I_L3DRV_PX2 = 6	A value of the intensity of the lineraly changing load PX acting in point 2 (linear load is defined in 2 points).
I_L3DRV_PY2 = 7	A value of the intensity of the lineraly changing load PY acting in point 2 (linear load is defined in 2 points).
I_L3DRV_PZ2 = 8	A value of the intensity of the lineraly changing load PZ acting in point 2 (linear load is defined in 2 points).
I_L3DRV_MX2 = 9	A value of the intensity of the lineraly changing moment MX acting in point 2 (linear load is defined in 2 points).
I_L3DRV_MY2 = 10	A value of the intensity of the lineraly changing moment MY acting in point 2 (linear load is defined in 2 points).
I_L3DRV_MZ2 = 11	A value of the intensity of the lineraly changing moment MZ acting in point 2 (linear load is defined in 2 points).
I_L3DRV_LOCAL = 13	Zero value if loads values refer to global coordinate system. Otherwise - to local.
I_L3DRV_GAMMA = 15	Rotation angle of the defined linear loads measured around the line of action.

(c) 2014 Autodesk Inc. All rights reserved

Description

Identifiers of the values describing load records of the following type: I_LRT_LINEAR_3D .

II.1.1.1.17 IRobotThermalIn3PointsRecordValues

C++

```
enum IRobotThermalIn3PointsRecordValues;
```

C#

```
public enum IRobotThermalIn3PointsRecordValues;
```

Visual Basic

```
Public Enum IRobotThermalIn3PointsRecordValues
```

Members

Members	Description
I_3PRV_TX1 = 0	Change of the middle surface temperature in the 1st point of surface for uniform or variable temperature change definition.
I_3PRV_TZ1 = 2	Change of the temperature gradient (along local z-axis) in the 1st point of surface for uniform or variable temperature change definition.
I_3PRV_TX2 = 3	Change of the middle surface temperature in the 2nd point of surface for uniform or variable temperature change definition.
I_3PRV_TZ2 = 5	Change of the temperature gradient (along local z-axis) in the 2nd point of surface for uniform or variable temperature change definition.
I_3PRV_TX3 = 6	Change of the middle surface temperature in the 3rd point of surface for uniform or variable temperature change definition.
I_3PRV_TZ3 = 8	Change of the temperature gradient (along local z-axis) in the 3rd point of surface for uniform or variable temperature change definition.

Description

Identifiers of the values describing load records of the following type: I_LRT_THERMAL_IN_3_POINTS (planar thermal load 3P). .

II.1.1.1.18 IRobotInContourRecordValues

C++

```
enum IRobotInContourRecordValues;
```

C#

```
public enum IRobotInContourRecordValues;
```

Visual Basic

```
Public Enum IRobotInContourRecordValues
```

Members

Members	Description
I_ICRV_PX1 = 0	Value of the X component of the pressure load in point 1 (for planar load defined in 3 points).
I_ICRV_PY1 = 1	Value of the Y component of the pressure load in point 1 (for planar load defined in 3 points).

(c) 2014 Autodesk Inc. All rights reserved

I_ICRV_PZ1 = 2	Value of the Z component of the pressure load in point 1 (for planar load defined in 3 points).
I_ICRV_PX2 = 3	Value of the X component of the pressure load in point 2 (for planar load defined in 3 points).
I_ICRV_PY2 = 4	Value of the Y component of the pressure load in point 2 (for planar load defined in 3 points).
I_ICRV_PZ2 = 5	Value of the Z component of the pressure load in point 2 (for planar load defined in 3 points).
I_ICRV_PX3 = 6	Value of the X component of the pressure load in point 3 (for planar load defined in 3 points).
I_ICRV_PY3 = 7	Value of the Y component of the pressure load in point 3 (for planar load defined in 3 points).
I_ICRV_PZ3 = 8	Value of the Z component of the pressure load in point 3 (for planar load defined in 3 points).
I_ICRV_PROJECTION = 12	Flag indicating whether the load is to be projected (value different from zero) or not (zero).
I_ICRV_NPOINTS = 13	Number of points defining a contour.
I_ICRV_LOCAL = 15	Variable describing coordinate system: local (the value not equal zero) or global (zero).

Description

Identifiers of the values describing load records of the following type: I_LRT_IN_CONTOUR (planar load on contour - uniform or variable) .

II.1.1.1.19 IRobotUniformRecordValues

C++

```
enum IRobotUniformRecordValues;
```

C#

```
public enum IRobotUniformRecordValues;
```

Visual Basic

```
Public Enum IRobotUniformRecordValues
```

Members

Members	Description
I_URV_PX = 0	Value of the X component of the pressure load acting on shell element.
I_URV_PY = 1	Value of the Y component of the pressure load acting on shell element.
I_URV_PZ = 2	Value of the Z component of the pressure load acting on shell element.
I_URV_LOCAL_SYSTEM = 11	Parameter assumes the value 0 (zero) when the load acts in the global system and the value different from zero when the load acts in the local system .
I_URV_PROJECTED = 12	Flag indicating whether the load is to be projected (value different from zero) or not (zero).

Description

Identifiers of the values describing load records of the following type: I_LRT_UNIFORM .

(c) 2014 Autodesk Inc. All rights reserved

II.1.1.1.20 IRobotThermalRecordValues

C++

```
enum IRobotThermalRecordValues;
```

C#

```
public enum IRobotThermalRecordValues;
```

Visual Basic

```
Public Enum IRobotThermalRecordValues
```

Members

Members	Description
I_TRV_T_1 = 0	Uniform temperature change (heating or cooling) of the bar.
I_TRV_T_2 = 3	Difference in the temperature between the extreme cross section fibers along local y bar axis.
I_TRV_T_3 = 6	Difference in the temperature between the extreme cross section fibers along local z bar axis.

Description

Identifiers of the values describing load records of the following type: I_LRT_THERMAL (thermal load: uniform, linear, at three points) .

II.1.1.1.21 IRobotNodeForceInPointRecordValues

C++

```
enum IRobotNodeForceInPointRecordValues;
```

C#

```
public enum IRobotNodeForceInPointRecordValues;
```

Visual Basic

```
Public Enum IRobotNodeForceInPointRecordValues
```

Members

Members	Description
I_NFIPRV_FX = 0	A value of concentrated force acting along the X direction Available since version 1.7.
I_NFIPRV_FY = 1	A value of concentrated force acting along the Y direction Available since version 1.7.
I_NFIPRV_FZ = 2	A value of concentrated force acting along the Z direction Available since version 1.7.
I_NFIPRV_MX = 3	A value of moment (about X axis) Available since version 1.7.
I_NFIPRV_MY = 4	A value of moment (about Y axis) Available since version 1.7.
I_NFIPRV_MZ = 5	A value of moment (about Z axis) Available since version 1.7.
I_NFIPRV_ALPHA = 8	Alpha angle of the force vector (if additional vector rotation defined).
I_NFIPRV_BETA = 9	Beta angle of the force vector (if additional vector rotation defined).
I_NFIPRV_GAMMA = 10	Gamma angle of the force vector (if additional vector rotation defined).
I_NFIPRV_POINT_X = 11	X coordinate of the point of load application Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

I_NFIPRV_POINT_Y = 12	Y coordinate of the point of load application Available since version 1.7.
I_NFIPRV_POINT_Z = 13	Z coordinate of the point of load application Available since version 1.7.

Description

Values that define a load resulting from a force acting at a point (load record type : I_LRT_NODE_FORCE_IN_POINT).

II.1.1.1.22 IRobotLinearOnEdgesRecordValues

C++

```
enum IRobotLinearOnEdgesRecordValues;
```

C#

```
public enum IRobotLinearOnEdgesRecordValues;
```

Visual Basic

```
Public Enum IRobotLinearOnEdgesRecordValues
```

Members

Members	Description
I_LOERV_PX = 0	Available since version 2.0.
I_LOERV_PY = 1	Available since version 2.0.
I_LOERV_PZ = 2	Available since version 2.0.
I_LOERV_MX = 3	Available since version 2.0.
I_LOERV_MY = 4	Available since version 2.0.
I_LOERV_MZ = 5	Available since version 2.0.
I_LOERV_GAMMA = 6	Available since version 2.0.
I_LOERV_LOCAL_SYSTEM = 11	Available since version 2.0.

Description

Parameters of linear load on edges.

II.1.1.1.23 IRobotDeadRecordValues

C++

```
enum IRobotDeadRecordValues;
```

C#

```
public enum IRobotDeadRecordValues;
```

Visual Basic

```
Public Enum IRobotDeadRecordValues
```

Members

Members	Description
I_DRV_X = 0	Weight multiplier for X-direction (if gravity load acts along this direction). Available since version 3.
I_DRV_Y = 1	Weight multiplier for Y-direction. Available since version 3.
I_DRV_Z = 2	Weight multiplier for Z-direction. Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

I_DRV_ENTIRE_STRUCTURE = 15	Flag which shows if weight of all structural elements is taken into account. Available since version 3.
I_DRV_COEFF = 3	Available since version 12.

Version

Available since version 3.

II.1.1.1.24 IRobotSurfaceOnObjectRecordValues

C++

```
enum IRobotSurfaceOnObjectRecordValues;
```

C#

```
public enum IRobotSurfaceOnObjectRecordValues;
```

Visual Basic

```
Public Enum IRobotSurfaceOnObjectRecordValues
```

Members

Members	Description
I_SOORV_PX = 0	Pressure value (on X-direction). Available since version 3.
I_SOORV_PY = 1	Pressure value (on Y-direction). Available since version 3.
I_SOORV_PZ = 2	Pressure value (on Z-direction). Available since version 3.
I_SOORV_LOCAL = 11	If equal zero - loads are defined in global coordinate system. Available since version 3.

Description

Indentifiers of values for records of I_LRT_SURFACE_ON_OBJECT type.

Version

Available since version 3.

II.1.1.1.25 IRobotMobilePointForceRecordValues

C++

```
enum IRobotMobilePointForceRecordValues;
```

C#

```
public enum IRobotMobilePointForceRecordValues;
```

Visual Basic

```
Public Enum IRobotMobilePointForceRecordValues
```

Members

Members	Description
I_MPFRV_X = 0	X coordinate of load application point. Available since version 3.
I_MPFRV_Y = 1	Y coordinate of load application point. Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

I_MPFRV_Z = 2	Z coordinate of load application point. Available since version 3.
I_MPFRV_FX = 3	Concentrated load value towards route local direction X. Available since version 3.
I_MPFRV_FY = 4	Concentrated load value towards route local direction Y. Available since version 3.
I_MPFRV_FZ = 5	Concentrated load value towards route local direction Z (downwards). Available since version 3.

Description

Values describing a load record of the I_LRT_MOBILE_POINT_FORCE type. .

Version

Available since version 3.

II.1.1.1.26 IRobotMobileDistributedRecordValues

C++

```
enum IRobotMobileDistributedRecordValues;
```

C#

```
public enum IRobotMobileDistributedRecordValues;
```

Visual Basic

```
Public Enum IRobotMobileDistributedRecordValues
```

Members

Members	Description
I_MDRV_X = 0	X coordinate of load application point. Available since version 3.
I_MDRV_Y = 1	Y coordinate of load application point. Available since version 3.
I_MDRV_Z = 2	Z coordinate of load application point. Available since version 3.
I_MDRV_PX = 3	Available since version 3.
I_MDRV_PY = 4	Available since version 3.
I_MDRV_PZ = 5	Surface load intensity towards route local direction Z (downwards). Available since version 3.
I_MDRV_DX = 6	Length of a segment/vehicle side along which the load is acting (along the vehicle axis). Available since version 3.
I_MDRV_DY = 7	Length of a segment/vehicle side along which the load is acting (perpendicular to the vehicle axis). Available since version 3.

Description

Values describing a load of the I_LRT_MOBILE_DISTRIBUTED type. .

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

II.1.1.1.27 IRobotNodeVelocityRecordValues

C++

```
enum IRobotNodeVelocityRecordValues;
```

C#

```
public enum IRobotNodeVelocityRecordValues;
```

Visual Basic

```
Public Enum IRobotNodeVelocityRecordValues
```

Members

Members	Description
I_NVRV_UX = 0	. Available since version 15.
I_NVRV_UY = 1	. Available since version 15.
I_NVRV_UZ = 2	. Available since version 15.
I_NVRV_ALPHA = 8	. Available since version 15.
I_NVRV_BETA = 9	. Available since version 15.
I_NVRV_GAMMA = 10	. Available since version 15.

Description

Identifiers of values for the load records of the I_LRT_NODE_VELOCITY type.

Version

Available since version 15.

II.1.1.1.28 IRobotNodeAccelerationRecordValues

C++

```
enum IRobotNodeAccelerationRecordValues;
```

C#

```
public enum IRobotNodeAccelerationRecordValues;
```

Visual Basic

```
Public Enum IRobotNodeAccelerationRecordValues
```

Members

Members	Description
I_NACRV_UX = 0	. Available since version 15.
I_NACRV_UY = 1	. Available since version 15.
I_NACRV_UZ = 2	. Available since version 15.
I_NACRV_ALPHA = 8	. Available since version 15.

(c) 2014 Autodesk Inc. All rights reserved

I_NACRV_BETA = 9	. Available since version 15.
I_NACRV_GAMMA = 10	. Available since version 15.

Description

Identifiers of values for the load records of the I_LRT_NODE_ACCELERATION type.

Version

Available since version 15.

II.1.1.2 IRobotLoadRecordType

C++

```
enum IRobotLoadRecordType;
```

C#

```
public enum IRobotLoadRecordType;
```

Visual Basic

```
Public Enum IRobotLoadRecordType
```

Members

Members	Description
I_LRT_NODE_FORCE = 0	Nodal force.
I_LRT_NODE_DISPLACEMENT = 1	Displacements of a support node.
I_LRT_BAR_DILATATION = 2	Shortening/extension of a bar.
I_LRT_BAR_FORCE_CONCENTRATED = 3	Concentrated force on the bar length.
I_LRT_BAR_MOMENT_DISTRIBUTED = 4	Distributed moment load on a bar.
I_LRT_BAR_UNIFORM = 5	Uniform linear load on a bar.
I_LRT_BAR_TRAPEZOIDALE = 6	Linear trapezoidal load on a bar.
I_LRT_BAR_THERMAL = 8	Thermal load on a bar.
I_LRT_LINEAR = 21	Linear load on FE, between two points (for slabs - 2D).
I_LRT_LINEAR_3D = 19	Linear load on FE, between two points (3D space).
I_LRT_POINT_AUXILIARY = 23	Concentrated load at a point.
I_LRT_IN_3_POINTS = 22	Planar load defined at three points.
I_LRT_PRESSURE = 24	Hydrostatic pressure.
I_LRT_THERMAL_IN_3_POINTS = 25	Planar thermal load defined at three points.
I_LRT_IN_CONTOUR = 28	Planar load defined on the contour (uniform or given at three points).
I_LRT_NODE_FORCE_MASS = 30	Concentrated nodal mass.
I_LRT_BAR_FORCE_CONCENTRATED_MASS = 33	Concentrated mass at a point on the bar length.
I_LRT_BAR_UNIFORM_MASS = 35	Uniform linear mass on a bar.
I_LRT_BAR_TRAPEZOIDALE_MASS = 36	Trapezoidal linear mass on a bar.
I_LRT_MASS_ACTIVATION = 39	Activation of a record for masses.
I_LRT_SPECTRUM_VALUE = 40	Spectrum value.
I_LRT_UNIFORM = 26	Uniform surface load.
I_LRT_THERMAL = 8	Planar thermal load.
I_LRT_NODE_FORCE_IN_POINT = 23	Concentrated force in the point Available since version 1.7.
I_LRT_LINEAR_ON_EDGES = 69	Linear load on edges Available since version 1.7.
I_LRT_DEAD = 7	Self-weight Available since version 2.0.

(c) 2014 Autodesk Inc. All rights reserved

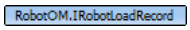
I_LRT_SURFACE_ON_OBJECT = 70	Surface load on object. Available since version 3.
I_LRT_MOBILE_POINT_FORCE = 53	Moving load - point force. Available since version 3.
I_LRT_MOBILE_DISTRIBUTED = 55	A moving distributed load: • linear, if value Dx or Dy equals zero • planar, if Dx and Dy assume non-zero values . Available since version 3.
I_LRT_NODE_VELOCITY = 10	. Available since version 15.
I_LRT_NODE_ACCELERATION = 11	. Available since version 15.

Description

Load record types available in the Robot Object Model .

II.1.1.3 IRobotLoadRecord

Class Hierarchy



C++

```
interface IRobotLoadRecord : IDispatch;
```

C#

```
public interface IRobotLoadRecord;
```

Visual Basic

```
Public Interface IRobotLoadRecord
```

Description

The load record is used to model a load applied to a structure. RobotLoadRecord is described by an interface shared by all load record types available in Robot. .

II.1.1.3.1 IRobotLoadRecord Members

The following tables list the members exposed by IRobotLoadRecord.

Public Fields

	Name	Description
	Description (see page 57)	Load record description .
	Objects (see page 57)	Selection of objects - structure components where a load is applied. The object type described by the selection corresponds to the type of objects to which the load may be applied. .
	ObjectType (see page 58)	Type (see page 58) of the object where a load is applied .
	Type (see page 58)	Load record type. .

(c) 2014 Autodesk Inc. All rights reserved

Public Methods

	Name	Description
	GetValue (see page 58)	The function returns the value of the indicated quantity describing a load The definition of a load record consists of a set of numbers - values describing the modeled load. The type and significance of particular values depend on the load record type and they are referred to by an appropriate identifier. .
	SetValue (see page 59)	The function sets the value of the indicated quantity describing a load. The definition of a load record consists of a set of numbers - values describing the modeled load. The type and significance of particular values depend on the load record type and they are referred to by an appropriate identifier. .

II.1.1.3.2 IRobotLoadRecord Fields

The fields of the IRobotLoadRecord class are listed here.

Public Fields

	Name	Description
	Description (see page 57)	Load record description .
	Objects (see page 57)	Selection of objects - structure components where a load is applied. The object type described by the selection corresponds to the type of objects to which the load may be applied. .
	ObjectType (see page 58)	Type (see page 58) of the object where a load is applied .
	Type (see page 58)	Load record type. .

II.1.1.3.2.1 Description

C++

```
HRESULT get_Description(BSTR*);
HRESULT put_Description(BSTR);
```

C#

```
public String Description { get; set; }
```

Visual Basic

```
Public Description As String
```

Description

Load record description .

II.1.1.3.2.2 Objects

C++

```
HRESULT get_Objects(IRobotSelection**);
```

C#

```
public IRobotSelection Objects { get; }
```

Visual Basic

```
Public ReadOnly Objects As IRobotSelection
```

Description

Selection of objects - structure components where a load is applied. The object type described by the selection corresponds to the type of objects to which the load may be applied. .

(c) 2014 Autodesk Inc. All rights reserved

II.1.1.3.2.3 ObjectType

C++

```
HRESULT get_ObjectType(IRobotObjectType*);
HRESULT put_ObjectType(IRobotObjectType);
```

C#

```
public IRobotObjectType ObjectType { get; set; }
```

Visual Basic

```
Public ObjectType As IRobotObjectType
```

Description

Type (see page 58) of the object where a load is applied .

II.1.1.3.2.4 Type

C++

```
HRESULT get_Type(IRobotLoadRecordType*);
```

C#

```
public IRobotLoadRecordType Type { get; }
```

Visual Basic

```
Public ReadOnly Type As IRobotLoadRecordType
```

Description

Load record type. .

II.1.1.3.3 IRobotLoadRecord Methods

The methods of the IRobotLoadRecord class are listed here.

Public Methods

	Name	Description
	GetValue (see page 58)	The function returns the value of the indicated quantity describing a load The definition of a load record consists of a set of numbers - values describing the modeled load. The type and significance of particular values depend on the load record type and they are referred to by an appropriate identifier. .
	SetValue (see page 59)	The function sets the value of the indicated quantity describing a load. The definition of a load record consists of a set of numbers - values describing the modeled load. The type and significance of particular values depend on the load record type and they are referred to by an appropriate identifier. .

II.1.1.3.3.1 GetValue

C++

```
HRESULT GetValue(short _value_id, double* ret);
```

C#

```
public double GetValue(short _value_id);
```

Visual Basic

```
Public Function GetValue(_value_id As short) As double
```

(c) 2014 Autodesk Inc. All rights reserved

Description

The function returns the value of the indicated quantity describing a load The definition of a load record consists of a set of numbers - values describing the modeled load. The type and significance of particular values depend on the load record type and they are referred to by an appropriate identifier. .

II.1.1.3.3.2 SetValue

C++

```
HRESULT SetValue(short _value_id, double _value);
```

C#

```
public void SetValue(short _value_id, double _value);
```

Visual Basic

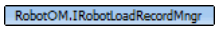
```
Public Sub SetValue(_value_id As short, _value As double)
```

Description

The function sets the value of the indicated quantity describing a load. The definition of a load record consists of a set of numbers - values describing the modeled load. The type and significance of particular values depend on the load record type and they are referred to by an appropriate identifier. .

II.1.1.4 IRobotLoadRecordMngr

Class Hierarchy



C++

```
interface IRobotLoadRecordMngr : IDispatch;
```

C#

```
public interface IRobotLoadRecordMngr;
```

Visual Basic

```
Public Interface IRobotLoadRecordMngr
```

Description

Load record manager manages logical lists of inter-linked records (e.g. a list of records linked with the same load case). The records are numbered from 1 to the number of records in the list .

II.1.1.4.1 IRobotLoadRecordMngr Members

The following tables list the members exposed by IRobotLoadRecordMngr.

Public Fields

	Name	Description
	Count (see page 60)	Number of currently managed load records .

Public Methods

	Name	Description
	Create (see page 60)	
	Delete (see page 61)	The function deletes the record with the indicated index. The list of all records is automatically re-numbered after this operation. .
	Get (see page 61)	The function returns a record with the indicated index on the list of the managed load records. .

(c) 2014 Autodesk Inc. All rights reserved

	New (see page 61)	The function creates a new load record of the indicated type. The newly-created record is added at the end of the managed list. The function gives the index of the crated record back. .

II.1.1.4.2 IRobotLoadRecordMngr Fields

The fields of the IRobotLoadRecordMngr class are listed here.

Public Fields

	Name	Description
	Count (see page 60)	Number of currently managed load records .

II.1.1.4.2.1 Count

C++

```
HRESULT get_Count(long*);
```

C#

```
public long Count { get; }
```

Visual Basic

```
Public ReadOnly Count As long
```

Description

Number of currently managed load records .

II.1.1.4.3 IRobotLoadRecordMngr Methods

The methods of the IRobotLoadRecordMngr class are listed here.

Public Methods

	Name	Description
	Create (see page 60)	
	Delete (see page 61)	The function deletes the record with the indicated index. The list of all records is automatically re-numbered after this operation. .
	Get (see page 61)	The function returns a record with the indicated index on the list of the managed load records. .
	New (see page 61)	The function creates a new load record of the indicated type. The newly-created record is added at the end of the managed list. The function gives the index of the crated record back. .

II.1.1.4.3.1 Create

C++

```
HRESULT Create(IRobotLoadRecordType _load_type, IRobotLoadRecord** ret);
```

C#

```
public IRobotLoadRecord Create(IRobotLoadRecordType _load_type);
```

Visual Basic

```
Public Function Create(_load_type As IRobotLoadRecordType) As IRobotLoadRecord
```

Version

Available since version 8.5.

(c) 2014 Autodesk Inc. All rights reserved

II.1.1.4.3.2 Delete

C++

```
HRESULT Delete(long _record_index);
```

C#

```
public void Delete(long _record_index);
```

Visual Basic

```
Public Sub Delete(_record_index As long)
```

Description

The function deletes the record with the indicated index. The list of all records is automatically re-numbered after this operation. .

II.1.1.4.3.3 Get

C++

```
HRESULT Get(long _record_index, IRobotLoadRecord** ret);
```

C#

```
public IRobotLoadRecord Get(long _record_index);
```

Visual Basic

```
Public Function Get(_record_index As long) As IRobotLoadRecord
```

Description

The function returns a record with the indicated index on the list of the managed load records. .

II.1.1.4.3.4 New

C++

```
HRESULT New(IRobotLoadRecordType _record_type, long* ret);
```

C#

```
public long New(IRobotLoadRecordType _record_type);
```

Visual Basic

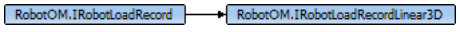
```
Public Function New(_record_type As IRobotLoadRecordType) As long
```

Description

The function creates a new load record of the indicated type. The newly-created record is added at the end of the managed list. The function gives the index of the crated record back. .

II.1.1.5 IRobotLoadRecordLinear3D

Class Hierarchy



C++

```
interface IRobotLoadRecordLinear3D : IRobotLoadRecord;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public interface IRobotLoadRecordLinear3D : IRobotLoadRecord;
```

Visual Basic

```
Public Interface IRobotLoadRecordLinear3D
```

Description

Load record ot the I_LRT_LINEAR_3D type. .

II.1.1.5.1 IRobotLoadRecordLinear3D Members

The following tables list the members exposed by IRobotLoadRecordLinear3D.

Public Fields

	Name	Description
	Description (see page 57)	Load record description .
	Objects (see page 57)	Selection of objects - structure components where a load is applied. The object type described by the selection corresponds to the type of objects to which the load may be applied. .
	ObjectType (see page 58)	Type (see page 58) of the object where a load is applied .
	Type (see page 58)	Load record type. .
	UniqueId (see page 62)	Unique identifier of a load record Available since version 1.7.

Public Methods

	Name	Description
	GetPoint (see page 63)	The function returns the coordinates of the indicated point of load application. .
	GetValue (see page 58)	The function returns the value of the indicated quantity describing a load The definition of a load record consists of a set of numbers - values describing the modeled load. The type and significance of particular values depend on the load record type and they are referred to by an appropriate identifier. .
	SetPoint (see page 63)	The function sets the coordinates of the indicated point of load application. .
	SetValue (see page 59)	The function sets the value of the indicated quantity describing a load. The definition of a load record consists of a set of numbers - values describing the modeled load. The type and significance of particular values depend on the load record type and they are referred to by an appropriate identifier. .

II.1.1.5.2 IRobotLoadRecordLinear3D Fields

The fields of the IRobotLoadRecordLinear3D class are listed here.

Public Fields

	Name	Description
	UniqueId (see page 62)	Unique identifier of a load record Available since version 1.7.

II.1.1.5.2.1 UniqueId

C++

```
HRESULT get_UniqueId(long*);
```

C#

```
public long UniqueId { get; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public ReadOnly UniqueId As long
```

Description

Unique identifier of a load record Available since version 1.7.

II.1.1.5.3 IRobotLoadRecordLinear3D Methods

The methods of the IRobotLoadRecordLinear3D class are listed here.

Public Methods

	Name	Description
	GetPoint (see page 63)	The function returns the coordinates of the indicated point of load application. .
	SetPoint (see page 63)	The function sets the coordinates of the indicated point of load application. .

II.1.1.5.3.1 GetPoint

C++

```
HRESULT GetPoint(int _which_point, double* _x, double* _y, double* _z);
```

C#

```
public void GetPoint(int _which_point, double* _x, double* _y, double* _z);
```

Visual Basic

```
Public Sub GetPoint(_which_point As int, ByRef _x As double*, ByRef _y As double*, ByRef _z
As double*)
```

Description

The function returns the coordinates of the indicated point of load application. .

II.1.1.5.3.2 SetPoint

C++

```
HRESULT SetPoint(int _which_point, double _x, double _y, double _z);
```

C#

```
public void SetPoint(int _which_point, double _x, double _y, double _z);
```

Visual Basic

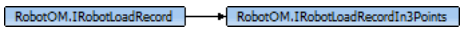
```
Public Sub SetPoint(_which_point As int, _x As double, _y As double, _z As double)
```

Description

The function sets the coordinates of the indicated point of load application. .

II.1.1.6 IRobotLoadRecordIn3Points

Class Hierarchy



C++

```
interface IRobotLoadRecordIn3Points : IRobotLoadRecord;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public interface IRobotLoadRecordIn3Points : IRobotLoadRecord;
```

Visual Basic

```
Public Interface IRobotLoadRecordIn3Points
```

Description

Load record for a load defined by three points .

II.1.1.6.1 IRobotLoadRecordIn3Points Members

The following tables list the members exposed by IRobotLoadRecordIn3Points.

Public Fields

	Name	Description
	Description (see page 57)	Load record description .
	Objects (see page 57)	Selection of objects - structure components where a load is applied. The object type described by the selection corresponds to the type of objects to which the load may be applied. .
	ObjectType (see page 58)	Type (see page 58) of the object where a load is applied .
	Type (see page 58)	Load record type. .
	UniqueId (see page 64)	Unique identifier of a load record Available since version 1.7.

Public Methods

	Name	Description
	GetGeoLimits (see page 65)	Function returns geometrical limits applied to loads. Available since version 2.0.
	GetPoint (see page 65)	The function returns coordinates of the indicated point of load application. .
	GetValue (see page 58)	The function returns the value of the indicated quantity describing a load The definition of a load record consists of a set of numbers - values describing the modeled load. The type and significance of particular values depend on the load record type and they are referred to by an appropriate identifier. .
	SetGeoLimits (see page 65)	Function sets geometrical limits for loads. Available since version 2.0.
	SetPoint (see page 66)	The function sets the coordinates of the indicated point of load application. .
	SetValue (see page 59)	The function sets the value of the indicated quantity describing a load. The definition of a load record consists of a set of numbers - values describing the modeled load. The type and significance of particular values depend on the load record type and they are referred to by an appropriate identifier. .

II.1.1.6.2 IRobotLoadRecordIn3Points Fields

The fields of the IRobotLoadRecordIn3Points class are listed here.

Public Fields

	Name	Description
	UniqueId (see page 64)	Unique identifier of a load record Available since version 1.7.

II.1.1.6.2.1 UniqueId

C++

```
HRESULT get_UniqueId(long*);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public long UniqueId { get; }
```

Visual Basic

```
Public ReadOnly UniqueId As long
```

Description

Unique identifier of a load record Available since version 1.7.

II.1.1.6.3 IRobotLoadRecordIn3Points Methods

The methods of the IRobotLoadRecordIn3Points class are listed here.

Public Methods

	Name	Description
	GetGeoLimits (see page 65)	Function returns geometrical limits applied to loads. Available since version 2.0.
	GetPoint (see page 65)	The function returns coordinates of the indicated point of load application. .
	SetGeoLimits (see page 65)	Function sets geometrical limits for loads. Available since version 2.0.
	SetPoint (see page 66)	The function sets the coordinates of the indicated point of load application. .

II.1.1.6.3.1 GetGeoLimits

C++

```
HRESULT GetGeoLimits(IRobotGeoLayer** ret);
```

C#

```
public IRobotGeoLayer GetGeoLimits();
```

Visual Basic

```
Public Function GetGeoLimits() As IRobotGeoLayer
```

Description

Function returns geometrical limits applied to loads. Available since version 2.0.

II.1.1.6.3.2 GetPoint

C++

```
HRESULT GetPoint(int _which_point, double* _x, double* _y, double* _z, VARIANT_BOOL* ret);
```

C#

```
public bool GetPoint(int _which_point, double* _x, double* _y, double* _z);
```

Visual Basic

```
Public Function GetPoint(_which_point As int, ByRef _x As double*, ByRef _y As double*,
ByRef _z As double*) As Boolean
```

Description

The function returns coordinates of the indicated point of load application. .

II.1.1.6.3.3 SetGeoLimits

C++

```
HRESULT SetGeoLimits(IRobotGeoLayer* _layer_def, VARIANT_BOOL* ret);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public bool SetGeoLimits(IRobotGeoLayer _layer_def);
```

Visual Basic

```
Public Function SetGeoLimits(ByRef _layer_def As IRobotGeoLayer) As Boolean
```

Description

Function sets geometrical limits for loads. Available since version 2.0.

II.1.1.6.3.4 SetPoint

C++

```
HRESULT SetPoint(int _which_point, double _x, double _y, double _z);
```

C#

```
public void SetPoint(int _which_point, double _x, double _y, double _z);
```

Visual Basic

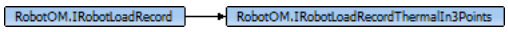
```
Public Sub SetPoint(_which_point As int, _x As double, _y As double, _z As double)
```

Description

The function sets the coordinates of the indicated point of load application. .

II.1.1.7 IRobotLoadRecordThermalIn3Points

Class Hierarchy



C++

```
interface IRobotLoadRecordThermalIn3Points : IRobotLoadRecord;
```

C#

```
public interface IRobotLoadRecordThermalIn3Points : IRobotLoadRecord;
```

Visual Basic

```
Public Interface IRobotLoadRecordThermalIn3Points
```

Description

Thermal load in three points. .

II.1.1.7.1 IRobotLoadRecordThermalIn3Points Members

The following tables list the members exposed by IRobotLoadRecordThermalIn3Points.

Public Fields

	Name	Description
	Description (see page 57)	Load record description .
	Objects (see page 57)	Selection of objects - structure components where a load is applied. The object type described by the selection corresponds to the type of objects to which the load may be applied. .
	ObjectType (see page 58)	Type (see page 58) of the object where a load is applied .
	Type (see page 58)	Load record type. .
	UniqueId (see page 67)	Unique identifier of a load record Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

Public Methods

	Name	Description
	GetGeoLimits (see page 68)	Function returns an object describing geometrical limitations of the load. If the load is not limited, then an empty reference to the object (NULL) will be returned. Available since version 2.0.
	GetPoint (see page 68)	The function sets the coordinates of the indicated point of load application. .
	GetValue (see page 58)	The function returns the value of the indicated quantity describing a load The definition of a load record consists of a set of numbers - values describing the modeled load. The type and significance of particular values depend on the load record type and they are referred to by an appropriate identifier. .
	SetGeoLimits (see page 68)	Function sets geometrical limits for loads. Available since version 2.0.
	SetPoint (see page 68)	The function sets the coordinates of the indicated point of load application. .
	SetValue (see page 59)	The function sets the value of the indicated quantity describing a load. The definition of a load record consists of a set of numbers - values describing the modeled load. The type and significance of particular values depend on the load record type and they are referred to by an appropriate identifier. .

II.1.1.7.2 IRobotLoadRecordThermalIn3Points Fields

The fields of the IRobotLoadRecordThermalIn3Points class are listed here.

Public Fields

	Name	Description
	UniqueId (see page 67)	Unique identifier of a load record Available since version 1.7.

II.1.1.7.2.1 UniqueId

C++

```
HRESULT get_UniqueId(long*);
```

C#

```
public long UniqueId { get; }
```

Visual Basic

```
Public ReadOnly UniqueId As long
```

Description

Unique identifier of a load record Available since version 1.7.

II.1.1.7.3 IRobotLoadRecordThermalIn3Points Methods

The methods of the IRobotLoadRecordThermalIn3Points class are listed here.

Public Methods

	Name	Description
	GetGeoLimits (see page 68)	Function returns an object describing geometrical limitations of the load. If the load is not limited, then an empty reference to the object (NULL) will be returned. Available since version 2.0.
	GetPoint (see page 68)	The function sets the coordinates of the indicated point of load application. .
	SetGeoLimits (see page 68)	Function sets geometrical limits for loads. Available since version 2.0.
	SetPoint (see page 68)	The function sets the coordinates of the indicated point of load application. .

(c) 2014 Autodesk Inc. All rights reserved

II.1.1.7.3.1 GetGeoLimits

C++

```
HRESULT GetGeoLimits(IRobotGeoLayer** ret);
```

C#

```
public IRobotGeoLayer GetGeoLimits();
```

Visual Basic

```
Public Function GetGeoLimits() As IRobotGeoLayer
```

Description

Function returns an object describing geometrical limitations of the load. If the load is not limited, then an empty reference to the object (NULL) will be returned. Available since version 2.0.

II.1.1.7.3.2 GetPoint

C++

```
HRESULT GetPoint(int _which_point, double* _x, double* _y, double* _z, VARIANT_BOOL* ret);
```

C#

```
public bool GetPoint(int _which_point, double* _x, double* _y, double* _z);
```

Visual Basic

```
Public Function GetPoint(_which_point As int, ByRef _x As double*, ByRef _y As double*,
ByRef _z As double*) As Boolean
```

Description

The function sets the coordinates of the indicated point of load application. .

II.1.1.7.3.3 SetGeoLimits

C++

```
HRESULT SetGeoLimits(IRobotGeoLayer* _layer_def, VARIANT_BOOL* ret);
```

C#

```
public bool SetGeoLimits(IRobotGeoLayer _layer_def);
```

Visual Basic

```
Public Function SetGeoLimits(ByRef _layer_def As IRobotGeoLayer) As Boolean
```

Description

Function sets geometrical limits for loads. Available since version 2.0.

II.1.1.7.3.4 SetPoint

C++

```
HRESULT SetPoint(int _which_point, double _x, double _y, double _z);
```

C#

```
public void SetPoint(int _which_point, double _x, double _y, double _z);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

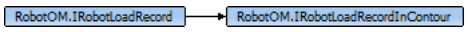
```
Public Sub SetPoint(_which_point As int, _x As double, _y As double, _z As double)
```

Description

The function sets the coordinates of the indicated point of load application. .

II.1.1.8 IRobotLoadRecordInContour

Class Hierarchy



C++

```
interface IRobotLoadRecordInContour : IRobotLoadRecord;
```

C#

```
public interface IRobotLoadRecordInContour : IRobotLoadRecord;
```

Visual Basic

```
Public Interface IRobotLoadRecordInContour
```

Description

Planar load on the contour. For a uniform load only the values for the first point of the load should be specified (I_ICRV_PX1, I_ICRV_PY1 and I_ICRV_PZ1) without providing the point coordinates. For a variable load all the load values and coordinates of the appropriate points should be specified applying the SetPoint (see page 72)() function.

.

II.1.1.8.1 IRobotLoadRecordInContour Members

The following tables list the members exposed by IRobotLoadRecordInContour.

Public Fields

	Name	Description
	Description (see page 57)	Load record description .
	Objects (see page 57)	Selection of objects - structure components where a load is applied. The object type described by the selection corresponds to the type of objects to which the load may be applied. .
	ObjectType (see page 58)	Type (see page 58) of the object where a load is applied .
	Type (see page 58)	Load record type. .
	UniqueId (see page 70)	Unique identifier of a load record Available since version 1.7.

Public Methods

	Name	Description
	GetContourPoint (see page 71)	The function gets the coordinates of the indicated point of a contour. .
	GetGeoLimits (see page 71)	Function returns an object describing geometrical limitations for a load. If they have not been defined, then an empty reference to the object will be returned (NULL). Available since version 2.0.
	GetPoint (see page 71)	The function gets the coordinates of a point. If the indicated point has not been defined, the function returns False (zero value). .
	GetValue (see page 58)	The function returns the value of the indicated quantity describing a load The definition of a load record consists of a set of numbers - values describing the modeled load. The type and significance of particular values depend on the load record type and they are referred to by an appropriate identifier. .
	GetVector (see page 71)	Function takes coordinates of a load projection vector. .

(c) 2014 Autodesk Inc. All rights reserved

	SetContourPoint (see page 72)	The function sets the coordinates of the indicated point of a contour. .
	SetGeoLimits (see page 72)	Function defines geometrical limitations for a load. If the limitations are defined successfully the function returns a value different from zero (True). Available since version 2.0.
	SetPoint (see page 72)	The function sets the coordinates of the indicated point of loading.Only for variable loads at three points. .
	SetValue (see page 59)	The function sets the value of the indicated quantity describing a load. The definition of a load record consists of a set of numbers - values describing the modeled load. The type and significance of particular values depend on the load record type and they are referred to by an appropriate identifier. .
	SetVector (see page 72)	Function sets vector of load projection.

II.1.1.8.2 IRobotLoadRecordInContour Fields

The fields of the IRobotLoadRecordInContour class are listed here.

Public Fields

	Name	Description
	UniqueId (see page 70)	Unique identifier of a load record Available since version 1.7.

II.1.1.8.2.1 UniqueId

C++

```
HRESULT get_UniqueId(long*);
```

C#

```
public long UniqueId { get; }
```

Visual Basic

```
Public ReadOnly UniqueId As long
```

Description

Unique identifier of a load record Available since version 1.7.

II.1.1.8.3 IRobotLoadRecordInContour Methods

The methods of the IRobotLoadRecordInContour class are listed here.

Public Methods

	Name	Description
	GetContourPoint (see page 71)	The function gets the coordinates of the indicated point of a contour. .
	GetGeoLimits (see page 71)	Function returns an object describing geometrical limitations for a load. If they have not been defined, then an empty reference to the object will be returned (NULL). Available since version 2.0.
	GetPoint (see page 71)	The function gets the coordinates of a point. If the indicated point has not been defined, the function returns False (zero value). .
	GetVector (see page 71)	Function takes coordinates of a load projection vector. .
	SetContourPoint (see page 72)	The function sets the coordinates of the indicated point of a contour. .
	SetGeoLimits (see page 72)	Function defines geometrical limitations for a load. If the limitations are defined successfully the function returns a value different from zero (True). Available since version 2.0.
	SetPoint (see page 72)	The function sets the coordinates of the indicated point of loading.Only for variable loads at three points. .
	SetVector (see page 72)	Function sets vector of load projection.

(c) 2014 Autodesk Inc. All rights reserved

II.1.1.8.3.1 GetContourPoint

C++

```
HRESULT GetContourPoint(int _which_point, double* _x, double* _y, double* _z);
```

C#

```
public void GetContourPoint(int _which_point, double* _x, double* _y, double* _z);
```

Visual Basic

```
Public Sub GetContourPoint(_which_point As int, ByRef _x As double*, ByRef _y As double*,
ByRef _z As double*)
```

Description

The function gets the coordinates of the indicated point of a contour. .

II.1.1.8.3.2 GetGeoLimits

C++

```
HRESULT GetGeoLimits(IRobotGeoLayer** ret);
```

C#

```
public IRobotGeoLayer GetGeoLimits();
```

Visual Basic

```
Public Function GetGeoLimits() As IRobotGeoLayer
```

Description

Function returns an object describing geometrical limitations for a load. If they have not been defined, then an empty reference to the object will be returned (NULL). Available since version 2.0.

II.1.1.8.3.3 GetPoint

C++

```
HRESULT GetPoint(int _which_point, double* _x, double* _y, double* _z, VARIANT_BOOL* ret);
```

C#

```
public bool GetPoint(int _which_point, double* _x, double* _y, double* _z);
```

Visual Basic

```
Public Function GetPoint(_which_point As int, ByRef _x As double*, ByRef _y As double*,
ByRef _z As double*) As Boolean
```

Description

The function gets the coordinates of a point. If the indicated point has not been defined, the function returns False (zero value). .

II.1.1.8.3.4 GetVector

C++

```
HRESULT GetVector(double* _vx, double* _vy, double* _vz);
```

C#

```
public void GetVector(double* _vx, double* _vy, double* _vz);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Sub GetVector(ByRef _vx As double*, ByRef _vy As double*, ByRef _vz As double*)
```

Description

Function takes coordinates of a load projection vector. .

II.1.1.8.3.5 SetContourPoint

C++

```
HRESULT SetContourPoint(int _which_point, double _x, double _y, double _z);
```

C#

```
public void SetContourPoint(int _which_point, double _x, double _y, double _z);
```

Visual Basic

```
Public Sub SetContourPoint(_which_point As int, _x As double, _y As double, _z As double)
```

Description

The function sets the coordinates of the indicated point of a contour. .

II.1.1.8.3.6 SetGeoLimits

C++

```
HRESULT SetGeoLimits(IRobotGeoLayer* _layer_def, VARIANT_BOOL* ret);
```

C#

```
public bool SetGeoLimits(IRobotGeoLayer _layer_def);
```

Visual Basic

```
Public Function SetGeoLimits(ByRef _layer_def As IRobotGeoLayer) As Boolean
```

Description

Function defines geometrical limitations for a load. If the limitations are defined successfully the function returns a value different from zero (True). Available since version 2.0.

II.1.1.8.3.7 SetPoint

C++

```
HRESULT SetPoint(int _which_point, double _x, double _y, double _z);
```

C#

```
public void SetPoint(int _which_point, double _x, double _y, double _z);
```

Visual Basic

```
Public Sub SetPoint(_which_point As int, _x As double, _y As double, _z As double)
```

Description

The function sets the coordinates of the indicated point of loading.Only for variable loads at three points. .

II.1.1.8.3.8 SetVector

C++

```
HRESULT SetVector(double _vx, double _vy, double _vz);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public void SetVector(double _vx, double _vy, double _vz);
```

Visual Basic

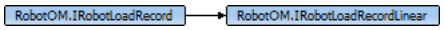
```
Public Sub SetVector(_vx As double, _vy As double, _vz As double)
```

Description

Function sets vector of load projection.

II.1.1.9 IRobotLoadRecordLinear

Class Hierarchy



C++

```
interface IRobotLoadRecordLinear : IRobotLoadRecord;
```

C#

```
public interface IRobotLoadRecordLinear : IRobotLoadRecord;
```

Visual Basic

```
Public Interface IRobotLoadRecordLinear
```

II.1.1.9.1 IRobotLoadRecordLinear Members

The following tables list the members exposed by IRobotLoadRecordLinear.

Public Fields

	Name	Description
	Description (see page 57)	Load record description .
	Objects (see page 57)	Selection of objects - structure components where a load is applied. The object type described by the selection corresponds to the type of objects to which the load may be applied. .
	ObjectType (see page 58)	Type (see page 58) of the object where a load is applied .
	Type (see page 58)	Load record type. .
	UniqueId (see page 74)	Unique identifier of a load record Available since version 1.7.

Public Methods

	Name	Description
	GetPoint (see page 74)	The function gets the coordinate of the indicated point. .
	GetValue (see page 58)	The function returns the value of the indicated quantity describing a load The definition of a load record consists of a set of numbers - values describing the modeled load. The type and significance of particular values depend on the load record type and they are referred to by an appropriate identifier. .
	SetPoint (see page 74)	The function sets the coordinate of the indicated point. .
	SetValue (see page 59)	The function sets the value of the indicated quantity describing a load. The definition of a load record consists of a set of numbers - values describing the modeled load. The type and significance of particular values depend on the load record type and they are referred to by an appropriate identifier. .

II.1.1.9.2 IRobotLoadRecordLinear Fields

The fields of the IRobotLoadRecordLinear class are listed here.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	UniqueId (see page 74)	Unique identifier of a load record Available since version 1.7.

II.1.1.9.2.1 UniqueId

C++

```
HRESULT get_UniqueId(long*);
```

C#

```
public long UniqueId { get; }
```

Visual Basic

```
Public ReadOnly UniqueId As long
```

Description

Unique identifier of a load record Available since version 1.7.

II.1.1.9.3 IRobotLoadRecordLinear Methods

The methods of the IRobotLoadRecordLinear class are listed here.

Public Methods

	Name	Description
	GetPoint (see page 74)	The function gets the coordinate of the indicated point. .
	SetPoint (see page 74)	The function sets the coordinate of the indicated point. .

II.1.1.9.3.1 GetPoint

C++

```
HRESULT GetPoint(int _which_point, double* _x, double* _y, double* _z);
```

C#

```
public void GetPoint(int _which_point, double* _x, double* _y, double* _z);
```

Visual Basic

```
Public Sub GetPoint(_which_point As int, ByRef _x As double*, ByRef _y As double*, ByRef _z
As double*)
```

Description

The function gets the coordinate of the indicated point. .

II.1.1.9.3.2 SetPoint

C++

```
HRESULT SetPoint(int _which_point, double _x, double _y, double _z);
```

C#

```
public void SetPoint(int _which_point, double _x, double _y, double _z);
```

Visual Basic

```
Public Sub SetPoint(_which_point As int, _x As double, _y As double, _z As double)
```

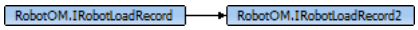
Description

The function sets the coordinate of the indicated point. .

(c) 2014 Autodesk Inc. All rights reserved

II.1.1.10 IRobotLoadRecord2

Class Hierarchy



C++

```
interface IRobotLoadRecord2 : IRobotLoadRecord;
```

C#

```
public interface IRobotLoadRecord2 : IRobotLoadRecord;
```

Visual Basic

```
Public Interface IRobotLoadRecord2
```

Description

Extended interface of the record that describes a load. .

II.1.1.10.1 IRobotLoadRecord2 Members

The following tables list the members exposed by IRobotLoadRecord2.

Public Fields

	Name	Description
	Description (see page 57)	Load record description .
	Objects (see page 57)	Selection of objects - structure components where a load is applied. The object type described by the selection corresponds to the type of objects to which the load may be applied. .
	ObjectType (see page 58)	Type (see page 58) of the object where a load is applied .
	Type (see page 58)	Load record type. .
	UniqueId (see page 76)	Unique identifier of a load record.

Public Methods

	Name	Description
	GetGeoLimits (see page 76)	Function returns an object describing a geometrical limitation defined for this load. If the limitation has not been defined, the function returns an empty reference. Available since version 2.0.
	GetValue (see page 58)	The function returns the value of the indicated quantity describing a load The definition of a load record consists of a set of numbers - values describing the modeled load. The type and significance of particular values depend on the load record type and they are referred to by an appropriate identifier. .
	SetGeoLimits (see page 76)	Function defines geometrical limitations for a load. It may happen to be the case that a definition method or load type does not allow defining geometrical limitations for it. The function returns the information if application of limitations has been successful. In order to delete geometrical limitations, an empty reference to the object should be provided as a parameter. Available since version 2.0.
	SetValue (see page 59)	The function sets the value of the indicated quantity describing a load. The definition of a load record consists of a set of numbers - values describing the modeled load. The type and significance of particular values depend on the load record type and they are referred to by an appropriate identifier. .

II.1.1.10.2 IRobotLoadRecord2 Fields

The fields of the IRobotLoadRecord2 class are listed here.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	UniqueId (see page 76)	Unique identifier of a load record.

II.1.1.10.2.1 UniqueId

C++

```
HRESULT get_UniqueId(long*);
```

C#

```
public long UniqueId { get; }
```

Visual Basic

```
Public ReadOnly UniqueId As long
```

Description

Unique identifier of a load record.

II.1.1.10.3 IRobotLoadRecord2 Methods

The methods of the IRobotLoadRecord2 class are listed here.

Public Methods

	Name	Description
	GetGeoLimits (see page 76)	Function returns an object describing a geometrical limitation defined for this load. If the limitation has not been defined, the function returns an empty reference. Available since version 2.0.
	SetGeoLimits (see page 76)	Function defines geometrical limitations for a load. It may happen to be the case that a definition method or load type does not allow defining geometrical limitations for it. The function returns the information if application of limitations has been successful. In order to delete geometrical limitations, an empty reference to the object should be provided as a parameter. Available since version 2.0.

II.1.1.10.3.1 GetGeoLimits

C++

```
HRESULT GetGeoLimits(IRobotGeoLayer** ret);
```

C#

```
public IRobotGeoLayer GetGeoLimits();
```

Visual Basic

```
Public Function GetGeoLimits() As IRobotGeoLayer
```

Description

Function returns an object describing a geometrical limitation defined for this load. If the limitation has not been defined, the function returns an empty reference. Available since version 2.0.

II.1.1.10.3.2 SetGeoLimits

C++

```
HRESULT SetGeoLimits(IRobotGeoLayer* _layer_def, VARIANT_BOOL* ret);
```

C#

```
public bool SetGeoLimits(IRobotGeoLayer _layer_def);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

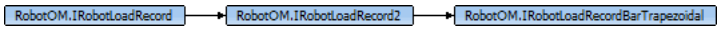
```
Public Function SetGeoLimits(ByRef _layer_def As IRobotGeoLayer) As Boolean
```

Description

Function defines geometrical limitations for a load. It may happen to be the case that a definition method or load type does not allow defining geometrical limitations for it. The function returns the information if application of limitations has been successful. In order to delete geometrical limitations, an empty reference to the object should be provided as a parameter. Available since version 2.0.

II.1.1.11 IRobotLoadRecordBarTrapezoidal

Class Hierarchy



C++

```
interface IRobotLoadRecordBarTrapezoidal : IRobotLoadRecord2;
```

C#

```
public interface IRobotLoadRecordBarTrapezoidal : IRobotLoadRecord2;
```

Visual Basic

```
Public Interface IRobotLoadRecordBarTrapezoidal
```

Description

Trapezoidal load on bars.

Version

Available since version 8.5.

II.1.1.11.1 IRobotLoadRecordBarTrapezoidal Members

The following tables list the members exposed by IRobotLoadRecordBarTrapezoidal.

Public Fields

	Name	Description
	Description (see page 57)	Load record description .
	Objects (see page 57)	Selection of objects - structure components where a load is applied. The object type described by the selection corresponds to the type of objects to which the load may be applied. .
	ObjectType (see page 58)	Type (see page 58) of the object where a load is applied .
	PointCount (see page 78)	Number of points defining the load.
	Type (see page 58)	Load record type. .
	UniqueId (see page 76)	Unique identifier of a load record.

Public Methods

	Name	Description
	GetGeoLimits (see page 76)	Function returns an object describing a geometrical limitation defined for this load. If the limitation has not been defined, the function returns an empty reference. Available since version 2.0.
	GetPoint (see page 78)	Function returns parameters of the load at the determined point.

(c) 2014 Autodesk Inc. All rights reserved

	GetValue (see page 58)	The function returns the value of the indicated quantity describing a load The definition of a load record consists of a set of numbers - values describing the modeled load. The type and significance of particular values depend on the load record type and they are referred to by an appropriate identifier. .
	SetGeoLimits (see page 76)	Function defines geometrical limitations for a load. It may happen to be the case that a definition method or load type does not allow defining geometrical limitations for it. The function returns the information if application of limitations has been successful. In order to delete geometrical limitations, an empty reference to the object should be provided as a parameter. Available since version 2.0.
	SetPoint (see page 79)	Function sets parameters of the load at the determined point.
	SetValue (see page 59)	The function sets the value of the indicated quantity describing a load. The definition of a load record consists of a set of numbers - values describing the modeled load. The type and significance of particular values depend on the load record type and they are referred to by an appropriate identifier. .

II.1.1.11.2 IRobotLoadRecordBarTrapezoidal Fields

The fields of the IRobotLoadRecordBarTrapezoidal class are listed here.

Public Fields

	Name	Description
	PointCount (see page 78)	Number of points defining the load.

II.1.1.11.2.1 PointCount

C++

```
HRESULT get_PointCount(long*);
HRESULT put_PointCount(long);
```

C#

```
public long PointCount { get; set; }
```

Visual Basic

```
Public PointCount As long
```

Description

Number of points defining the load.

Version

Available since version 8.5.

II.1.1.11.3 IRobotLoadRecordBarTrapezoidal Methods

The methods of the IRobotLoadRecordBarTrapezoidal class are listed here.

Public Methods

	Name	Description
	GetPoint (see page 78)	Function returns parameters of the load at the determined point.
	SetPoint (see page 79)	Function sets parameters of the load at the determined point.

II.1.1.11.3.1 GetPoint

C++

```
HRESULT GetPoint(long _pt_idx, double* _px, double* _py, double* _pz, double* _x);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public void GetPoint(long _pt_idx, double* _px, double* _py, double* _pz, double* _x);
```

Visual Basic

```
Public Sub GetPoint(_pt_idx As long, ByRef _px As double*, ByRef _py As double*, ByRef _pz
As double*, ByRef _x As double*)
```

Description

Function returns parameters of the load at the determined point.

Version

Available since version 8.5.

II.1.1.11.3.2 SetPoint

C++

```
HRESULT SetPoint(long _pt_idx, double _px, double _py, double _pz, double _x);
```

C#

```
public void SetPoint(long _pt_idx, double _px, double _py, double _pz, double _x);
```

Visual Basic

```
Public Sub SetPoint(_pt_idx As long, _px As double, _py As double, _pz As double, _x As
double)
```

Description

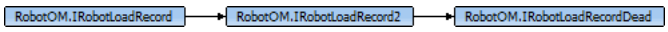
Function sets parameters of the load at the determined point.

Version

Available since version 8.5.

II.1.1.12 IRobotLoadRecordDead

Class Hierarchy



C++

```
interface IRobotLoadRecordDead : IRobotLoadRecord2;
```

C#

```
public interface IRobotLoadRecordDead : IRobotLoadRecord2;
```

Visual Basic

```
Public Interface IRobotLoadRecordDead
```

Description

Load record describing self-weight.

Version

Available since version 9.2.

II.1.1.12.1 IRobotLoadRecordDead Members

The following tables list the members exposed by IRobotLoadRecordDead.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	Description (see page 57)	Load record description .
	FiniteElems (see page 80)	Text describing the selection of finite elements to which the load is applied.
	Objects (see page 57)	Selection of objects - structure components where a load is applied. The object type described by the selection corresponds to the type of objects to which the load may be applied. .
	ObjectType (see page 58)	Type (see page 58) of the object where a load is applied .
	Type (see page 58)	Load record type. .
	UniqueId (see page 76)	Unique identifier of a load record.

Public Methods

	Name	Description
	GetGeoLimits (see page 76)	Function returns an object describing a geometrical limitation defined for this load. If the limitation has not been defined, the function returns an empty reference. Available since version 2.0.
	GetValue (see page 58)	The function returns the value of the indicated quantity describing a load The definition of a load record consists of a set of numbers - values describing the modeled load. The type and significance of particular values depend on the load record type and they are referred to by an appropriate identifier. .
	SetGeoLimits (see page 76)	Function defines geometrical limitations for a load. It may happen to be the case that a definition method or load type does not allow defining geometrical limitations for it. The function returns the information if application of limitations has been successful. In order to delete geometrical limitations, an empty reference to the object should be provided as a parameter. Available since version 2.0.
	SetValue (see page 59)	The function sets the value of the indicated quantity describing a load. The definition of a load record consists of a set of numbers - values describing the modeled load. The type and significance of particular values depend on the load record type and they are referred to by an appropriate identifier. .

II.1.1.12.2 IRobotLoadRecordDead Fields

The fields of the IRobotLoadRecordDead class are listed here.

Public Fields

	Name	Description
	FiniteElems (see page 80)	Text describing the selection of finite elements to which the load is applied.

II.1.1.12.2.1 FiniteElems

C++

```
HRESULT get_FiniteElems(BSTR*);
HRESULT put_FiniteElems(BSTR);
```

C#

```
public String FiniteElems { get; set; }
```

Visual Basic

```
Public FiniteElems As String
```

Description

Text describing the selection of finite elements to which the load is applied.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 9.2.

II.1.1.13 IRobotLoadRecordCommon

Class Hierarchy

C++

```
interface IRobotLoadRecordCommon : IDispatch;
```

C#

```
public interface IRobotLoadRecordCommon;
```

Visual Basic

```
Public Interface IRobotLoadRecordCommon
```

Description

The interface providing access to the extended set of common attributes for load records of different types.

Version

Available since version 12.

II.1.1.13.1 IRobotLoadRecordCommon Members

The following tables list the members exposed by IRobotLoadRecordCommon.

Public Fields

	Name	Description
	IsAutoGenerated (see page 81)	The flag indicating that a load record has been generated automatically.

II.1.1.13.2 IRobotLoadRecordCommon Fields

The fields of the IRobotLoadRecordCommon class are listed here.

Public Fields

	Name	Description
	IsAutoGenerated (see page 81)	The flag indicating that a load record has been generated automatically.

II.1.1.13.2.1 IsAutoGenerated

C++

```
HRESULT get_IsAutoGenerated(VARIANT_BOOL*);
```

C#

```
public bool IsAutoGenerated { get; }
```

Visual Basic

```
Public ReadOnly IsAutoGenerated As Boolean
```

Description

The flag indicating that a load record has been generated automatically.

Version

Available since version 12.

(c) 2014 Autodesk Inc. All rights reserved

II.1.2 Snow/wind loads

Enumerations

	Name	Description
	IRobotSWCodePLWindZone (see page 109)	
	IRobotSWCodePLWindSite (see page 109)	
	IRobotSWCodePLSnowZone (see page 110)	
	IRobotSWCodeFRWindSite (see page 129)	
	IRobotSWCodeFRWindType (see page 129)	
	IRobotSWCodeFRSurfaceType (see page 130)	
	IRobotSWCodeFRSnowType (see page 130)	
	IRobotSWCodePLWindPressDistribType (see page 130)	
	IRobotSWCodeECSiteType (see page 147)	
	IRobotSWCodeECGroundType (see page 147)	
	IRobotSWCodeECCdType (see page 148)	

Interfaces

	Name	Description
	IRobotSnowWindParams (see page 93)	Set of the non-code-specific parameters describing the manner of generating snow/wind loads. .
	IRobotSnowWindEngine (see page 96)	Generator of snow/wind loads.
	IRobotSWCodePLParams (see page 99)	Set of parameters for snow/wind loads for Polish code.
	IRobotSWCodeFRParams (see page 110)	Parameters of snow/wind load generation for the French code.
	IRobotSWCodeECParams (see page 131)	Parameters of snow/wind load generation for Eurocode. .

II.1.2.1 Generating 3D loads

Interfaces

	Name	Description
	IRobotSWStruct3D (see page 83)	Structure model defined for generation of 3D snow/wind loads. The model describes a 3D frame generated by copying n times a 2D frame in the third direction. All structure frames consist of the same number of elements. Corresponding elements in individual frames have the same indexes. If an element occurs only in gable frames, then the bar number (Bar) for the corresponding elements of internal frames is ignored.
	IRobotSWStruct3DElement (see page 85)	Element of a 3D structure model defined for generation of snow/wind loads.
	IRobotSWStruct3DFrame (see page 86)	Definition of a 2D frame, a component of a 3D frame, that models a structure for the needs of generation of snow/wind loads. The frame elements are indexed with values from the interval [1, ElemCount (see page 87)].

(c) 2014 Autodesk Inc. All rights reserved

	IRobotSWStruct3DGenParams (see page 88)	Parameters of generation of a 3D frame structure model for the needs of definition of moving loads.
	IRobotSWStruct3DPurlinGenParams (see page 91)	Definition of the method of generation of longitudinal bars (purlins) adjoining to the frame bar.

II.1.2.1.1 IRobotSWStruct3D

Class Hierarchy

RobotOM.IRobotSWStruct3D

C++

```
interface IRobotSWStruct3D : IDispatch;
```

C#

```
public interface IRobotSWStruct3D;
```

Visual Basic

```
Public Interface IRobotSWStruct3D
```

Description

Structure model defined for generation of 3D snow/wind loads. The model describes a 3D frame generated by copying n times a 2D frame in the third direction. All structure frames consist of the same number of elements. Corresponding elements in individual frames have the same indexes. If an element occurs only in gable frames, then the bar number (Bar) for the corresponding elements of internal frames is ignored.

II.1.2.1.1.1 IRobotSWStruct3D Members

The following tables list the members exposed by IRobotSWStruct3D.

Public Fields

	Name	Description
	FrameCount (see page 83)	Number of 2D frames forming a 3D frame.
	FrameElemCount (see page 84)	Number of elements of a single frame.

Public Methods

	Name	Description
	GetFrame (see page 84)	The function returns the frame with the specified index. The index of the first (front) frame equals 1, whereas the index of the last (rear) frame equals FrameCount (see page 83).
	SetFrame (see page 84)	The function defines the frame with the specified index. Frame indexes are values from the interval [1, FrameCount (see page 83)].

II.1.2.1.1.2 IRobotSWStruct3D Fields

The fields of the IRobotSWStruct3D class are listed here.

Public Fields

	Name	Description
	FrameCount (see page 83)	Number of 2D frames forming a 3D frame.
	FrameElemCount (see page 84)	Number of elements of a single frame.

II.1.2.1.1.2.1 FrameCount

C++

```
HRESULT get_FrameCount(long*);
HRESULT put_FrameCount(long);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public long FrameCount { get; set; }
```

Visual Basic

```
Public FrameCount As long
```

Description

Number of 2D frames forming a 3D frame.

II.1.2.1.1.2.2 FrameElemCount

C++

```
HRESULT get_FrameElemCount(long*);
```

C#

```
public long FrameElemCount { get; }
```

Visual Basic

```
Public ReadOnly FrameElemCount As long
```

Description

Number of elements of a single frame.

II.1.2.1.1.3 IRobotSWStruct3D Methods

The methods of the IRobotSWStruct3D class are listed here.

Public Methods

	Name	Description
	GetFrame (see page 84)	The function returns the frame with the specified index. The index of the first (front) frame equals 1, whereas the index of the last (rear) frame equals FrameCount (see page 83).
	SetFrame (see page 84)	The function defines the frame with the specified index. Frame indexes are values from the interval [1, FrameCount (see page 83)].

II.1.2.1.1.3.1 GetFrame

C++

```
HRESULT GetFrame(long _frame_idx, IRobotSWStruct3DFrame** ret);
```

C#

```
public IRobotSWStruct3DFrame GetFrame(long _frame_idx);
```

Visual Basic

```
Public Function GetFrame(_frame_idx As long) As IRobotSWStruct3DFrame
```

Description

The function returns the frame with the specified index. The index of the first (front) frame equals 1, whereas the index of the last (rear) frame equals FrameCount (see page 83).

II.1.2.1.1.3.2 SetFrame

C++

```
HRESULT SetFrame(long _frame_idx, IRobotSWStruct3DFrame* _frame);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public void SetFrame(long _frame_idx, IRobotSWStruct3DFrame _frame);
```

Visual Basic

```
Public Sub SetFrame(_frame_idx As long, ByRef _frame As IRobotSWStruct3DFrame)
```

Description

The function defines the frame with the specified index. Frame indexes are values from the interval [1, FrameCount (see page 83)].

II.1.2.1.2 IRobotSWStruct3DElement

Class Hierarchy

RobotOM.IRobotSWStruct3DElement

C++

```
interface IRobotSWStruct3DElement : IDispatch;
```

C#

```
public interface IRobotSWStruct3DElement;
```

Visual Basic

```
Public Interface IRobotSWStruct3DElement
```

Description

Element of a 3D structure model defined for generation of snow/wind loads.

II.1.2.1.2.1 IRobotSWStruct3DElement Members

The following tables list the members exposed by IRobotSWStruct3DElement.

Public Fields

	Name	Description
	Bar (see page 85)	Bar number.
	IsFacadeLoaded (see page 86)	Flag forcing load generation in the gable frame for a bar.
	IsFacadeOnly (see page 86)	Flag indicating that a bar occurs only in a gable frame (front or rear one).
	Purlins (see page 86)	Table with numbers of bars that define purlins adjoining to this bar.

II.1.2.1.2.2 IRobotSWStruct3DElement Fields

The fields of the IRobotSWStruct3DElement class are listed here.

Public Fields

	Name	Description
	Bar (see page 85)	Bar number.
	IsFacadeLoaded (see page 86)	Flag forcing load generation in the gable frame for a bar.
	IsFacadeOnly (see page 86)	Flag indicating that a bar occurs only in a gable frame (front or rear one).
	Purlins (see page 86)	Table with numbers of bars that define purlins adjoining to this bar.

II.1.2.1.2.2.1 Bar

C++

```
HRESULT get_Bar(long*);
HRESULT put_Bar(long);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public long Bar { get; set; }
```

Visual Basic

```
Public Bar As long
```

Description

Bar number.

II.1.2.1.2.2.2 IsFacadeLoaded

C++

```
HRESULT get_IsFacadeLoaded(VARIANT_BOOL*);
HRESULT put_IsFacadeLoaded(VARIANT_BOOL);
```

C#

```
public bool IsFacadeLoaded { get; set; }
```

Visual Basic

```
Public IsFacadeLoaded As Boolean
```

Description

Flag forcing load generation in the gable frame for a bar.

II.1.2.1.2.2.3 IsFacadeOnly

C++

```
HRESULT get_IsFacadeOnly(VARIANT_BOOL*);
HRESULT put_IsFacadeOnly(VARIANT_BOOL);
```

C#

```
public bool IsFacadeOnly { get; set; }
```

Visual Basic

```
Public IsFacadeOnly As Boolean
```

Description

Flag indicating that a bar occurs only in a gable frame (front or rear one).

II.1.2.1.2.2.4 Purlins

C++

```
HRESULT get_Purlins(IRobotNumbersArray**);
```

C#

```
public IRobotNumbersArray Purlins { get; }
```

Visual Basic

```
Public ReadOnly Purlins As IRobotNumbersArray
```

Description

Table with numbers of bars that define purlins adjoining to this bar.

II.1.2.1.3 IRobotSWStruct3DFrame

Class Hierarchy

RobotOM.IRobotSWStruct3DFrame

(c) 2014 Autodesk Inc. All rights reserved

C++

```
interface IRobotSWStruct3DFrame : IDispatch;
```

C#

```
public interface IRobotSWStruct3DFrame;
```

Visual Basic

```
Public Interface IRobotSWStruct3DFrame
```

Description

Definition of a 2D frame, a component of a 3D frame, that models a structure for the needs of generation of snow/wind loads. The frame elements are indexed with values from the interval [1, ElemCount (see page 87)].

II.1.2.1.3.1 IRobotSWStruct3DFrame Members

The following tables list the members exposed by IRobotSWStruct3DFrame.

Public Fields

	Name	Description
	ElemCount (see page 87)	Number of elements defining a frame.

Public Methods

	Name	Description
	GetElem (see page 88)	The function returns the frame element with the specified index.
	SetElem (see page 88)	The function positions the specified frame element.

II.1.2.1.3.2 IRobotSWStruct3DFrame Fields

The fields of the IRobotSWStruct3DFrame class are listed here.

Public Fields

	Name	Description
	ElemCount (see page 87)	Number of elements defining a frame.

II.1.2.1.3.2.1 ElemCount

C++

```
HRESULT get_ElemCount(long*);
```

C#

```
public long ElemCount { get; }
```

Visual Basic

```
Public ReadOnly ElemCount As long
```

Description

Number of elements defining a frame.

II.1.2.1.3.3 IRobotSWStruct3DFrame Methods

The methods of the IRobotSWStruct3DFrame class are listed here.

Public Methods

	Name	Description
	GetElem (see page 88)	The function returns the frame element with the specified index.

(c) 2014 Autodesk Inc. All rights reserved

	SetElem (see page 88)	The function positions the specified frame element.

II.1.2.1.3.3.1 GetElem

C++

```
HRESULT GetElem(long _elem_idx, IRobotSWStruct3DElement** ret);
```

C#

```
public IRobotSWStruct3DElement GetElem(long _elem_idx);
```

Visual Basic

```
Public Function GetElem(_elem_idx As long) As IRobotSWStruct3DElement
```

Description

The function returns the frame element with the specified index.

II.1.2.1.3.3.2 SetElem

C++

```
HRESULT SetElem(long _elem_idx, IRobotSWStruct3DElement* _elem);
```

C#

```
public void SetElem(long _elem_idx, IRobotSWStruct3DElement _elem);
```

Visual Basic

```
Public Sub SetElem(_elem_idx As long, ByRef _elem As IRobotSWStruct3DElement)
```

Description

The function positions the specified frame element.

II.1.2.1.4 IRobotSWStruct3DGenParams

Class Hierarchy

RobotOM.IRobotSWStruct3DGenParams

C++

```
interface IRobotSWStruct3DGenParams : IDispatch;
```

C#

```
public interface IRobotSWStruct3DGenParams;
```

Visual Basic

```
Public Interface IRobotSWStruct3DGenParams
```

Description

Parameters of generation of a 3D frame structure model for the needs of definition of moving loads.

II.1.2.1.4.1 IRobotSWStruct3DGenParams Members

The following tables list the members exposed by IRobotSWStruct3DGenParams.

Public Fields

	Name	Description
	Bars (see page 89)	Selection of bars forming the initial 2D frame.

(c) 2014 Autodesk Inc. All rights reserved

	FacadeLoadedBars (see page 89)	Selection of the gable frame bars for which generation of snow/wind loads will be forced.
	FacadeOnlyBars (see page 90)	Selection of bars to be copied only in gable walls.
	FrameCount (see page 90)	Ultimate number of 2D frames.
	Offsets (see page 90)	
	Spacings (see page 90)	Table including distances (in meters) between successive frames; If only one value (with the index 1) is defined in the table, then all frames will be spaced according to that value.

Public Methods

	Name	Description
	GetPurlins (see page 91)	The function returns parameters of generation of longitudinal bars for the indicated frame bar.
	SetPurlins (see page 91)	The function defines the method of generation of longitudinal bars for the indicated frame bar.

II.1.2.1.4.2 IRobotSWStruct3DGenParams Fields

The fields of the IRobotSWStruct3DGenParams class are listed here.

Public Fields

	Name	Description
	Bars (see page 89)	Selection of bars forming the initial 2D frame.
	FacadeLoadedBars (see page 89)	Selection of the gable frame bars for which generation of snow/wind loads will be forced.
	FacadeOnlyBars (see page 90)	Selection of bars to be copied only in gable walls.
	FrameCount (see page 90)	Ultimate number of 2D frames.
	Offsets (see page 90)	
	Spacings (see page 90)	Table including distances (in meters) between successive frames; If only one value (with the index 1) is defined in the table, then all frames will be spaced according to that value.

II.1.2.1.4.2.1 Bars

C++

```
HRESULT get_Bars(BSTR*);
HRESULT put_Bars(BSTR);
```

C#

```
public String Bars { get; set; }
```

Visual Basic

```
Public Bars As String
```

Description

Selection of bars forming the initial 2D frame.

II.1.2.1.4.2.2 FacadeLoadedBars

C++

```
HRESULT get_FacadeLoadedBars(BSTR*);
HRESULT put_FacadeLoadedBars(BSTR);
```

C#

```
public String FacadeLoadedBars { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public FacadeLoadedBars As String
```

Description

Selection of the gable frame bars for which generation of snow/wind loads will be forced.

II.1.2.1.4.2.3 FacadeOnlyBars

C++

```
HRESULT get_FacadeOnlyBars(BSTR*);
HRESULT put_FacadeOnlyBars(BSTR);
```

C#

```
public String FacadeOnlyBars { get; set; }
```

Visual Basic

```
Public FacadeOnlyBars As String
```

Description

Selection of bars to be copied only in gable walls.

II.1.2.1.4.2.4 FrameCount

C++

```
HRESULT get_FrameCount(long*);
HRESULT put_FrameCount(long);
```

C#

```
public long FrameCount { get; set; }
```

Visual Basic

```
Public FrameCount As long
```

Description

Ultimate number of 2D frames.

II.1.2.1.4.2.5 Offsets

C++

```
HRESULT get_Offsets(VARIANT_BOOL*);
HRESULT put_Offsets(VARIANT_BOOL);
```

C#

```
public bool Offsets { get; set; }
```

Visual Basic

```
Public Offsets As Boolean
```

II.1.2.1.4.2.6 Spacings

C++

```
HRESULT get_Spacings(IRobotValuesArray**);
```

C#

```
public IRobotValuesArray Spacings { get; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public ReadOnly Spacings As IRobotValuesArray
```

Description

Table including distances (in meters) between successive frames; If only one value (with the index 1) is defined in the table, then all frames will be spaced according to that value.

II.1.2.1.4.3 IRobotSWStruct3DGenParams Methods

The methods of the IRobotSWStruct3DGenParams class are listed here.

Public Methods

	Name	Description
	GetPurlins (see page 91)	The function returns parameters of generation of longitudinal bars for the indicated frame bar.
	SetPurlins (see page 91)	The function defines the method of generation of longitudinal bars for the indicated frame bar.

II.1.2.1.4.3.1 GetPurlins

C++

```
HRESULT GetPurlins(long _bar_num, IRobotSWStruct3DPurlinGenParams** ret);
```

C#

```
public IRobotSWStruct3DPurlinGenParams GetPurlins(long _bar_num);
```

Visual Basic

```
Public Function GetPurlins(_bar_num As long) As IRobotSWStruct3DPurlinGenParams
```

Description

The function returns parameters of generation of longitudinal bars for the indicated frame bar.

II.1.2.1.4.3.2 SetPurlins

C++

```
HRESULT SetPurlins(long _bar_num, IRobotSWStruct3DPurlinGenParams* _purlins_def);
```

C#

```
public void SetPurlins(long _bar_num, IRobotSWStruct3DPurlinGenParams _purlins_def);
```

Visual Basic

```
Public Sub SetPurlins(_bar_num As long, ByRef _purlins_def As
IRobotSWStruct3DPurlinGenParams)
```

Description

The function defines the method of generation of longitudinal bars for the indicated frame bar.

II.1.2.1.5 IRobotSWStruct3DPurlinGenParams

Class Hierarchy

RobotOM.IRobotSWStruct3DPurlinGenParams

C++

```
interface IRobotSWStruct3DPurlinGenParams : IDispatch;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public interface IRobotSWStruct3DPurlinGenParams;
```

Visual Basic

```
Public Interface IRobotSWStruct3DPurlinGenParams
```

Description

Definition of the method of generation of longitudinal bars (purlins) adjoining to the frame bar.

II.1.2.1.5.1 IRobotSWStruct3DPurlinGenParams Members

The following tables list the members exposed by IRobotSWStruct3DPurlinGenParams.

Public Fields

	Name	Description
	Locations (see page 92)	Table defining successive points at which purlins should adjoin to the bar.
	RelativeLocations (see page 92)	Flag indicating that location of successive purlins along the bar has been expressed in values relative to the beginning and length of the bar.
	SectionName (see page 93)	Name of the section that will be assigned to bars-purlins.

II.1.2.1.5.2 IRobotSWStruct3DPurlinGenParams Fields

The fields of the IRobotSWStruct3DPurlinGenParams class are listed here.

Public Fields

	Name	Description
	Locations (see page 92)	Table defining successive points at which purlins should adjoin to the bar.
	RelativeLocations (see page 92)	Flag indicating that location of successive purlins along the bar has been expressed in values relative to the beginning and length of the bar.
	SectionName (see page 93)	Name of the section that will be assigned to bars-purlins.

II.1.2.1.5.2.1 Locations

C++

```
HRESULT get_Locations(IRobotValuesArray**);
```

C#

```
public IRobotValuesArray Locations { get; }
```

Visual Basic

```
Public ReadOnly Locations As IRobotValuesArray
```

Description

Table defining successive points at which purlins should adjoin to the bar.

II.1.2.1.5.2.2 RelativeLocations

C++

```
HRESULT get_RelativeLocations(VARIANT_BOOL*);
HRESULT put_RelativeLocations(VARIANT_BOOL);
```

C#

```
public bool RelativeLocations { get; set; }
```

Visual Basic

```
Public RelativeLocations As Boolean
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Flag indicating that location of successive purlins along the bar has been expressed in values relative to the beginning and length of the bar.

II.1.2.1.5.2.3 SectionName

C++

```
HRESULT get_SectionName(BSTR*);
HRESULT put_SectionName(BSTR);
```

C#

```
public String SectionName { get; set; }
```

Visual Basic

```
Public SectionName As String
```

Description

Name of the section that will be assigned to bars-purlins.

II.1.2.2 IRobotSnowWindParams

Class Hierarchy

RobotOM.IRobotSnowWindParams

C++

```
interface IRobotSnowWindParams : IDispatch;
```

C#

```
public interface IRobotSnowWindParams;
```

Visual Basic

```
Public Interface IRobotSnowWindParams
```

Description

Set of the non-code-specific parameters describing the manner of generating snow/wind loads. .

II.1.2.2.1 IRobotSnowWindParams Members

The following tables list the members exposed by IRobotSnowWindParams.

Public Fields

	Name	Description
	BaseOnGround (see page 94)	Available since version 1.7.
	BaySpacing (see page 94)	Available since version 1.7.
	Envelope (see page 94)	Available since version 1.7.
	IsolatedRoofs (see page 95)	Available since version 1.7.
	IsSnow (see page 95)	Available since version 1.7.
	IsWind (see page 95)	Available since version 1.7.
	TotalDepth (see page 95)	Available since version 1.7.
	WithCavities (see page 96)	Available since version 1.7.
	WithParapets (see page 96)	Available since version 1.7.

II.1.2.2.2 IRobotSnowWindParams Fields

The fields of the IRobotSnowWindParams class are listed here.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	BaseOnGround (see page 94)	Available since version 1.7.
	BaySpacing (see page 94)	Available since version 1.7.
	Envelope (see page 94)	Available since version 1.7.
	IsolatedRoofs (see page 95)	Available since version 1.7.
	IsSnow (see page 95)	Available since version 1.7.
	IsWind (see page 95)	Available since version 1.7.
	TotalDepth (see page 95)	Available since version 1.7.
	WithCavities (see page 96)	Available since version 1.7.
	WithParapets (see page 96)	Available since version 1.7.

II.1.2.2.2.1 BaseOnGround

C++

```
HRESULT get_BaseOnGround(VARIANT_BOOL*);
HRESULT put_BaseOnGround(VARIANT_BOOL);
```

C#

```
public bool BaseOnGround { get; set; }
```

Visual Basic

```
Public BaseOnGround As Boolean
```

Description

Available since version 1.7.

II.1.2.2.2.2 BaySpacing

C++

```
HRESULT get_BaySpacing(double*);
HRESULT put_BaySpacing(double);
```

C#

```
public double BaySpacing { get; set; }
```

Visual Basic

```
Public BaySpacing As double
```

Description

Available since version 1.7.

II.1.2.2.2.3 Envelope

C++

```
HRESULT get_Envelope(BSTR*);
HRESULT put_Envelope(BSTR);
```

C#

```
public String Envelope { get; set; }
```

Visual Basic

```
Public Envelope As String
```

Description

Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

II.1.2.2.2.4 IsolatedRoofs

C++

```
HRESULT get_IsolatedRoofs(VARIANT_BOOL*);
HRESULT put_IsolatedRoofs(VARIANT_BOOL);
```

C#

```
public bool IsolatedRoofs { get; set; }
```

Visual Basic

```
Public IsolatedRoofs As Boolean
```

Description

Available since version 1.7.

II.1.2.2.2.5 IsSnow

C++

```
HRESULT get_IsSnow(VARIANT_BOOL*);
HRESULT put_IsSnow(VARIANT_BOOL);
```

C#

```
public bool IsSnow { get; set; }
```

Visual Basic

```
Public IsSnow As Boolean
```

Description

Available since version 1.7.

II.1.2.2.2.6 IsWind

C++

```
HRESULT get_IsWind(VARIANT_BOOL*);
HRESULT put_IsWind(VARIANT_BOOL);
```

C#

```
public bool IsWind { get; set; }
```

Visual Basic

```
Public IsWind As Boolean
```

Description

Available since version 1.7.

II.1.2.2.2.7 TotalDepth

C++

```
HRESULT get_TotalDepth(double*);
HRESULT put_TotalDepth(double);
```

C#

```
public double TotalDepth { get; set; }
```

Visual Basic

```
Public TotalDepth As double
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Available since version 1.7.

II.1.2.2.2.8 WithCavities

C++

```
HRESULT get_WithCavities(VARIANT_BOOL*);
HRESULT put_WithCavities(VARIANT_BOOL);
```

C#

```
public bool WithCavities { get; set; }
```

Visual Basic

```
Public WithCavities As Boolean
```

Description

Available since version 1.7.

II.1.2.2.2.9 WithParapets

C++

```
HRESULT get_WithParapets(VARIANT_BOOL*);
HRESULT put_WithParapets(VARIANT_BOOL);
```

C#

```
public bool WithParapets { get; set; }
```

Visual Basic

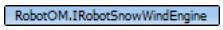
```
Public WithParapets As Boolean
```

Description

Available since version 1.7.

II.1.2.3 IRobotSnowWindEngine

Class Hierarchy



C++

```
interface IRobotSnowWindEngine : IDispatch;
```

C#

```
public interface IRobotSnowWindEngine;
```

Visual Basic

```
Public Interface IRobotSnowWindEngine
```

Description

Generator of snow/wind loads.

II.1.2.3.1 IRobotSnowWindEngine Members

The following tables list the members exposed by IRobotSnowWindEngine.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	CodeParams (see page 97)	Available since version 1.7.
	Params (see page 97)	Parameters of snow/wind generation (code independent) Available since version 1.7.

Public Methods

	Name	Description
	Generate (see page 98)	Function generates snow/wind loads on the basis of parameters set earlier. Function returns the collection of numbers of generated load cases. Available since version 1.7.
	Generate3D (see page 98)	The function generates snow/wind loads for the given 3D frame model. The function returns a collection of numbers of generated load cases.
	GenerateStruct3D (see page 98)	The function generates a 3D frame model based on defined parameters.
	ShowParamsDlg (see page 99)	Function opens a dialog for the currently selected snow and wind code. Function returns a value originally transferred through the dialog to the function responsible for dialog closeing (EndDialog).

II.1.2.3.2 IRobotSnowWindEngine Fields

The fields of the IRobotSnowWindEngine class are listed here.

Public Fields

	Name	Description
	CodeParams (see page 97)	Available since version 1.7.
	Params (see page 97)	Parameters of snow/wind generation (code independent) Available since version 1.7.

II.1.2.3.2.1 CodeParams

C++

```
HRESULT get_CodeParams(IDispatch*);
```

C#

```
public IDispatch CodeParams { get; }
```

Visual Basic

```
Public ReadOnly CodeParams As IDispatch
```

Description

Available since version 1.7.

II.1.2.3.2.2 Params

C++

```
HRESULT get_Params(IRobotSnowWindParams**);
```

C#

```
public IRobotSnowWindParams Params { get; }
```

Visual Basic

```
Public ReadOnly Params As IRobotSnowWindParams
```

Description

Parameters of snow/wind generation (code independent) Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

II.1.2.3.3 IRobotSnowWindEngine Methods

The methods of the IRobotSnowWindEngine class are listed here.

Public Methods

	Name	Description
	Generate (see page 98)	Function generates snow/wind loads on the basis of parameters set earlier. Function returns the collection of numbers of generated load cases. Available since version 1.7.
	Generate3D (see page 98)	The function generates snow/wind loads for the given 3D frame model. The function returns a collection of numbers of generated load cases.
	GenerateStruct3D (see page 98)	The function generates a 3D frame model based on defined parameters.
	ShowParamsDlg (see page 99)	Function opens a dialog for the currently selected snow and wind code. Function returns a value originally transferred through the dialog to the function responsible for dialog closeing (EndDialog).

II.1.2.3.3.1 Generate

C++

```
HRESULT Generate(IRobotNumbersCollection** ret);
```

C#

```
public IRobotNumbersCollection Generate();
```

Visual Basic

```
Public Function Generate() As IRobotNumbersCollection
```

Description

Function generates snow/wind loads on the basis of parameters set earlier. Function returns the collection of numbers of generated load cases. Available since version 1.7.

II.1.2.3.3.2 Generate3D

C++

```
HRESULT Generate3D(IRobotSWStruct3D* _struct_3d, IRobotNumbersCollection** ret);
```

C#

```
public IRobotNumbersCollection Generate3D(IRobotSWStruct3D _struct_3d);
```

Visual Basic

```
Public Function Generate3D(ByRef _struct_3d As IRobotSWStruct3D) As IRobotNumbersCollection
```

Description

The function generates snow/wind loads for the given 3D frame model. The function returns a collection of numbers of generated load cases.

II.1.2.3.3.3 GenerateStruct3D

C++

```
HRESULT GenerateStruct3D(IRobotSWStruct3DGenParams* _gen_params, IRobotSWStruct3D** ret);
```

C#

```
public IRobotSWStruct3D GenerateStruct3D(IRobotSWStruct3DGenParams _gen_params);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Function GenerateStruct3D(ByRef _gen_params As IRobotSWStruct3DGenParams) As
IRobotSWStruct3D
```

Description

The function generates a 3D frame model based on defined parameters.

II.1.2.3.3.4 ShowParamsDlg

C++

```
HRESULT ShowParamsDlg(long* ret);
```

C#

```
public long ShowParamsDlg();
```

Visual Basic

```
Public Function ShowParamsDlg() As long
```

Description

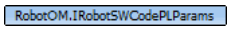
Function opens a dialog for the currently selected snow and wind code. Function returns a value originally transferred through the dialog to the function responsible for dialog closeing (EndDialog).

Version

Available since version 11.

II.1.2.4 IRobotSWCodePLParams

Class Hierarchy



C++

```
interface IRobotSWCodePLParams : IDispatch;
```

C#

```
public interface IRobotSWCodePLParams;
```

Visual Basic

```
Public Interface IRobotSWCodePLParams
```

Description

Set of parameters for snow/wind loads for Polish code.

II.1.2.4.1 IRobotSWCodePLParams Members

The following tables list the members exposed by IRobotSWCodePLParams.

Public Fields

	Name	Description
	Altitude (see page 101)	Altitude above sea level Available since version 1.7.
	IsolatedRoofs (see page 101)	Table containing numbers of recognized bars Available since version 1.7.
	NodalLoadsForAllBars (see page 102)	Flag indicating the method of modifying loads into nodal loads; If it is set as a value different from zero (True), then the loads for all the bars in the envelope will be modified. Available since version 1.7.
	NodalLoadsForBarsList (see page 102)	Selection list describing bars for which loads will be changed to nodal loads Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

	PermFront (see page 102)	Available since version 1.7.
	PermLeftSide (see page 102)	Available since version 1.7.
	PermRear (see page 103)	Available since version 1.7.
	PermRightSide (see page 103)	Available since version 1.7.
	ReferenceLevel (see page 103)	Available since version 1.7.
	SnowPressure (see page 103)	Available since version 1.7.
	SnowPressureAutomatic (see page 104)	Available since version 1.7.
	SnowRedistribution (see page 104)	Available since version 1.7.
	SnowZone (see page 104)	Available since version 1.7.
	StructureHeight (see page 104)	Structure height Available since version 1.7.
	WindDynamicAction (see page 105)	Available since version 1.7.
	WindDynamicDecrement (see page 105)	Available since version 1.7.
	WindDynamicPeriod (see page 105)	Available since version 1.7.
	WindMultipleRoofs (see page 106)	Available since version 1.7.
	WindPressure (see page 106)	Available since version 1.7.
	WindPressureAutomatic (see page 106)	Available since version 1.7.
	WindPressureDistribOnHeight (see page 106)	Distribution of wind pressure with respect to height Available since version 1.7.
	WindSite (see page 107)	Available since version 1.7.
	WindZone (see page 107)	Available since version 1.7.

Public Methods

	Name	Description
	IsolatedRoofGetLocation (see page 107)	Available since version 1.7.
	IsolatedRoofSetLocation (see page 108)	Available since version 1.7.
	SnowBarCoeffGet (see page 108)	Available since version 1.7.
	SnowBarCoeffSet (see page 108)	Available since version 1.7.
	WindBarCoeffGet (see page 109)	Function returns the coefficient value for a bar with the specified number. Coefficients are determined for the bars in the envelope. Available since version 1.7.
	WindBarCoeffSet (see page 109)	Function sets the coefficient value for a bar with the specified number. Coefficients can be determined only for the bars in the envelope. Available since version 1.7.

II.1.2.4.2 IRobotSWCodePLParams Fields

The fields of the IRobotSWCodePLParams class are listed here.

Public Fields

	Name	Description
	Altitude (see page 101)	Altitude above sea level Available since version 1.7.
	IsolatedRoofs (see page 101)	Table containing numbers of recognized bars Available since version 1.7.
	NodalLoadsForAllBars (see page 102)	Flag indicating the method of modifying loads into nodal loads; If it is set as a value different from zero (True), then the loads for all the bars in the envelope will be modified. Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

	NodalLoadsForBarsList (see page 102)	Selection list describing bars for which loads will be changed to nodal loads Available since version 1.7.
	PermFront (see page 102)	Available since version 1.7.
	PermLeftSide (see page 102)	Available since version 1.7.
	PermRear (see page 103)	Available since version 1.7.
	PermRightSide (see page 103)	Available since version 1.7.
	ReferenceLevel (see page 103)	Available since version 1.7.
	SnowPressure (see page 103)	Available since version 1.7.
	SnowPressureAutomatic (see page 104)	Available since version 1.7.
	SnowRedistribution (see page 104)	Available since version 1.7.
	SnowZone (see page 104)	Available since version 1.7.
	StructureHeight (see page 104)	Structure height Available since version 1.7.
	WindDynamicAction (see page 105)	Available since version 1.7.
	WindDynamicDecrement (see page 105)	Available since version 1.7.
	WindDynamicPeriod (see page 105)	Available since version 1.7.
	WindMultipleRoofs (see page 106)	Available since version 1.7.
	WindPressure (see page 106)	Available since version 1.7.
	WindPressureAutomatic (see page 106)	Available since version 1.7.
	WindPressureDistribOnHeight (see page 106)	Distribution of wind pressure with respect to height Available since version 1.7.
	WindSite (see page 107)	Available since version 1.7.
	WindZone (see page 107)	Available since version 1.7.

II.1.2.4.2.1 Altitude

C++

```
HRESULT get_Altitude(double*);
HRESULT put_Altitude(double);
```

C#

```
public double Altitude { get; set; }
```

Visual Basic

```
Public Altitude As double
```

Description

Altitude above sea level Available since version 1.7.

II.1.2.4.2.2 IsolatedRoofs

C++

```
HRESULT get_IsolatedRoofs(IRobotNumbersCollection**);
```

C#

```
public IRobotNumbersCollection IsolatedRoofs { get; }
```

Visual Basic

```
Public ReadOnly IsolatedRoofs As IRobotNumbersCollection
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Table containing numbers of recognized bars Available since version 1.7.

II.1.2.4.2.3 NodalLoadsForAllBars

C++

```
HRESULT get_NodalLoadsForAllBars(VARIANT_BOOL*);
HRESULT put_NodalLoadsForAllBars(VARIANT_BOOL);
```

C#

```
public bool NodalLoadsForAllBars { get; set; }
```

Visual Basic

```
Public NodalLoadsForAllBars As Boolean
```

Description

Flag indicating the method of modifying loads into nodal loads; If it is set as a value different from zero (True), then the loads for all the bars in the envelope will be modified. Available since version 1.7.

II.1.2.4.2.4 NodalLoadsForBarsList

C++

```
HRESULT get_NodalLoadsForBarsList(BSTR*);
HRESULT put_NodalLoadsForBarsList(BSTR);
```

C#

```
public String NodalLoadsForBarsList { get; set; }
```

Visual Basic

```
Public NodalLoadsForBarsList As String
```

Description

Selection list describing bars for which loads will be changed to nodal loads Available since version 1.7.

II.1.2.4.2.5 PermFront

C++

```
HRESULT get_PermFront(double*);
HRESULT put_PermFront(double);
```

C#

```
public double PermFront { get; set; }
```

Visual Basic

```
Public PermFront As double
```

Description

Available since version 1.7.

II.1.2.4.2.6 PermLeftSide

C++

```
HRESULT get_PermLeftSide(double*);
HRESULT put_PermLeftSide(double);
```

C#

```
public double PermLeftSide { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public PermLeftSide As double
```

Description

Available since version 1.7.

II.1.2.4.2.7 PermRear

C++

```
HRESULT get_PermRear(double*);
HRESULT put_PermRear(double);
```

C#

```
public double PermRear { get; set; }
```

Visual Basic

```
Public PermRear As double
```

Description

Available since version 1.7.

II.1.2.4.2.8 PermRightSide

C++

```
HRESULT get_PermRightSide(double*);
HRESULT put_PermRightSide(double);
```

C#

```
public double PermRightSide { get; set; }
```

Visual Basic

```
Public PermRightSide As double
```

Description

Available since version 1.7.

II.1.2.4.2.9 ReferenceLevel

C++

```
HRESULT get_ReferenceLevel(double*);
HRESULT put_ReferenceLevel(double);
```

C#

```
public double ReferenceLevel { get; set; }
```

Visual Basic

```
Public ReferenceLevel As double
```

Description

Available since version 1.7.

II.1.2.4.2.10 SnowPressure

C++

```
HRESULT get_SnowPressure(double*);
HRESULT put_SnowPressure(double);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double SnowPressure { get; set; }
```

Visual Basic

```
Public SnowPressure As double
```

Description

Available since version 1.7.

II.1.2.4.2.11 SnowPressureAutomatic

C++

```
HRESULT get_SnowPressureAutomatic(VARIANT_BOOL*);
HRESULT put_SnowPressureAutomatic(VARIANT_BOOL);
```

C#

```
public bool SnowPressureAutomatic { get; set; }
```

Visual Basic

```
Public SnowPressureAutomatic As Boolean
```

Description

Available since version 1.7.

II.1.2.4.2.12 SnowRedistribution

C++

```
HRESULT get_SnowRedistribution(VARIANT_BOOL*);
HRESULT put_SnowRedistribution(VARIANT_BOOL);
```

C#

```
public bool SnowRedistribution { get; set; }
```

Visual Basic

```
Public SnowRedistribution As Boolean
```

Description

Available since version 1.7.

II.1.2.4.2.13 SnowZone

C++

```
HRESULT get_SnowZone(IRobotSWCodePLSnowZone*);
HRESULT put_SnowZone(IRobotSWCodePLSnowZone);
```

C#

```
public IRobotSWCodePLSnowZone SnowZone { get; set; }
```

Visual Basic

```
Public SnowZone As IRobotSWCodePLSnowZone
```

Description

Available since version 1.7.

II.1.2.4.2.14 StructureHeight

C++

```
HRESULT get_StructureHeight(double*);
```

(c) 2014 Autodesk Inc. All rights reserved

```
HRESULT put_StructureHeight(double);
```

C#

```
public double StructureHeight { get; set; }
```

Visual Basic

```
Public StructureHeight As double
```

Description

Structure height Available since version 1.7.

II.1.2.4.2.15 WindDynamicAction

C++

```
HRESULT get_WindDynamicAction(VARIANT_BOOL*);
HRESULT put_WindDynamicAction(VARIANT_BOOL);
```

C#

```
public bool WindDynamicAction { get; set; }
```

Visual Basic

```
Public WindDynamicAction As Boolean
```

Description

Available since version 1.7.

II.1.2.4.2.16 WindDynamicDecrement

C++

```
HRESULT get_WindDynamicDecrement(double*);
HRESULT put_WindDynamicDecrement(double);
```

C#

```
public double WindDynamicDecrement { get; set; }
```

Visual Basic

```
Public WindDynamicDecrement As double
```

Description

Available since version 1.7.

II.1.2.4.2.17 WindDynamicPeriod

C++

```
HRESULT get_WindDynamicPeriod(double*);
HRESULT put_WindDynamicPeriod(double);
```

C#

```
public double WindDynamicPeriod { get; set; }
```

Visual Basic

```
Public WindDynamicPeriod As double
```

Description

Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

II.1.2.4.2.18 WindMultipleRoofs

C++

```
HRESULT get_WindMultipleRoofs(VARIANT_BOOL*);
HRESULT put_WindMultipleRoofs(VARIANT_BOOL);
```

C#

```
public bool WindMultipleRoofs { get; set; }
```

Visual Basic

```
Public WindMultipleRoofs As Boolean
```

Description

Available since version 1.7.

II.1.2.4.2.19 WindPressure

C++

```
HRESULT get_WindPressure(double*);
HRESULT put_WindPressure(double);
```

C#

```
public double WindPressure { get; set; }
```

Visual Basic

```
Public WindPressure As double
```

Description

Available since version 1.7.

II.1.2.4.2.20 WindPressureAutomatic

C++

```
HRESULT get_WindPressureAutomatic(VARIANT_BOOL*);
HRESULT put_WindPressureAutomatic(VARIANT_BOOL);
```

C#

```
public bool WindPressureAutomatic { get; set; }
```

Visual Basic

```
Public WindPressureAutomatic As Boolean
```

Description

Available since version 1.7.

II.1.2.4.2.21 WindPressureDistribOnHeight

C++

```
HRESULT get_WindPressureDistribOnHeight(IRobotSWCodePLWindPressDistribType*);
HRESULT put_WindPressureDistribOnHeight(IRobotSWCodePLWindPressDistribType);
```

C#

```
public IRobotSWCodePLWindPressDistribType WindPressureDistribOnHeight { get; set; }
```

Visual Basic

```
Public WindPressureDistribOnHeight As IRobotSWCodePLWindPressDistribType
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Distribution of wind pressure with respect to height Available since version 1.7.

II.1.2.4.2.22 WindSite

C++

```
HRESULT get_WindSite(IRobotSWCodePLWindSite*);
HRESULT put_WindSite(IRobotSWCodePLWindSite);
```

C#

```
public IRobotSWCodePLWindSite WindSite { get; set; }
```

Visual Basic

```
Public WindSite As IRobotSWCodePLWindSite
```

Description

Available since version 1.7.

II.1.2.4.2.23 WindZone

C++

```
HRESULT get_WindZone(IRobotSWCodePLWindZone*);
HRESULT put_WindZone(IRobotSWCodePLWindZone);
```

C#

```
public IRobotSWCodePLWindZone WindZone { get; set; }
```

Visual Basic

```
Public WindZone As IRobotSWCodePLWindZone
```

Description

Available since version 1.7.

II.1.2.4.3 IRobotSWCodePLParams Methods

The methods of the IRobotSWCodePLParams class are listed here.

Public Methods

	Name	Description
	IsolatedRoofGetLocation (see page 107)	Available since version 1.7.
	IsolatedRoofSetLocation (see page 108)	Available since version 1.7.
	SnowBarCoeffGet (see page 108)	Available since version 1.7.
	SnowBarCoeffSet (see page 108)	Available since version 1.7.
	WindBarCoeffGet (see page 109)	Function returns the coefficient value for a bar with the specified number. Coefficients are determined for the bars in the envelope. Available since version 1.7.
	WindBarCoeffSet (see page 109)	Function sets the coefficient value for a bar with the specified number. Coefficients can be determined only for the bars in the envelope. Available since version 1.7.

II.1.2.4.3.1 IsolatedRoofGetLocation

C++

```
HRESULT IsolatedRoofGetLocation(long _bar_num, VARIANT_BOOL _relative, double* ret);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double IsolatedRoofGetLocation(long _bar_num, bool _relative);
```

Visual Basic

```
Public Function IsolatedRoofGetLocation(_bar_num As long, _relative As Boolean) As double
```

Description

Available since version 1.7.

II.1.2.4.3.2 IsolatedRoofSetLocation

C++

```
HRESULT IsolatedRoofSetLocation(long _bar_num, double _loc, VARIANT_BOOL _relative);
```

C#

```
public void IsolatedRoofSetLocation(long _bar_num, double _loc, bool _relative);
```

Visual Basic

```
Public Sub IsolatedRoofSetLocation(_bar_num As long, _loc As double, _relative As Boolean)
```

Description

Available since version 1.7.

II.1.2.4.3.3 SnowBarCoeffGet

C++

```
HRESULT SnowBarCoeffGet(long _bar_num, double* ret);
```

C#

```
public double SnowBarCoeffGet(long _bar_num);
```

Visual Basic

```
Public Function SnowBarCoeffGet(_bar_num As long) As double
```

Description

Available since version 1.7.

II.1.2.4.3.4 SnowBarCoeffSet

C++

```
HRESULT SnowBarCoeffSet(long _bar_num, double _bar_coef);
```

C#

```
public void SnowBarCoeffSet(long _bar_num, double _bar_coef);
```

Visual Basic

```
Public Sub SnowBarCoeffSet(_bar_num As long, _bar_coef As double)
```

Description

Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

II.1.2.4.3.5 WindBarCoeffGet

C++

```
HRESULT WindBarCoeffGet(long _bar_num, double* ret);
```

C#

```
public double WindBarCoeffGet(long _bar_num);
```

Visual Basic

```
Public Function WindBarCoeffGet(_bar_num As long) As double
```

Description

Function returns the coefficient value for a bar with the specified number. Coefficients are determined for the bars in the envelope. Available since version 1.7.

II.1.2.4.3.6 WindBarCoeffSet

C++

```
HRESULT WindBarCoeffSet(long _bar_num, double _bar_coeff);
```

C#

```
public void WindBarCoeffSet(long _bar_num, double _bar_coeff);
```

Visual Basic

```
Public Sub WindBarCoeffSet(_bar_num As long, _bar_coeff As double)
```

Description

Function sets the coefficient value for a bar with the specified number. Coefficients can be determined only for the bars in the envelope. Available since version 1.7.

II.1.2.5 IRobotSWCodePLWindZone

C++

```
enum IRobotSWCodePLWindZone;
```

C#

```
public enum IRobotSWCodePLWindZone;
```

Visual Basic

```
Public Enum IRobotSWCodePLWindZone
```

Members

Members	Description
I_SWCPLWZ_I = 0	Available since version 1.7.
I_SWCPLWZ_II = 1	Available since version 1.7.
I_SWCPLWZ_IIA = 2	Available since version 1.7.
I_SWCPLWZ_IIB = 3	Available since version 1.7.
I_SWCPLWZ_III = 4	Available since version 1.7.

II.1.2.6 IRobotSWCodePLWindSite

C++

```
enum IRobotSWCodePLWindSite;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public enum IRobotSWCodePLWindSite;
```

Visual Basic

```
Public Enum IRobotSWCodePLWindSite
```

Members

Members	Description
I_SWCPLWS_A = 0	Available since version 1.7.
I_SWCPLWS_B = 1	Available since version 1.7.
I_SWCPLWS_C = 2	Available since version 1.7.

II.1.2.7 IRobotSWCodePLSnowZone

C++

```
enum IRobotSWCodePLSnowZone;
```

C#

```
public enum IRobotSWCodePLSnowZone;
```

Visual Basic

```
Public Enum IRobotSWCodePLSnowZone
```

Members

Members	Description
I_SWCPLSZ_I = 0	Available since version 1.7.
I_SWCPLSZ_II = 1	Available since version 1.7.
I_SWCPLSZ_III = 2	Available since version 1.7.
I_SWCPLSZ_IV = 3	Available since version 1.7.

II.1.2.8 IRobotSWCodeFRParams

Class Hierarchy

RobotOM.IRobotSWCodeFRParams

C++

```
interface IRobotSWCodeFRParams : IDispatch;
```

C#

```
public interface IRobotSWCodeFRParams;
```

Visual Basic

```
Public Interface IRobotSWCodeFRParams
```

Description

Parameters of snow/wind load generation for the French code.

II.1.2.8.1 IRobotSWCodeFRParams Members

The following tables list the members exposed by IRobotSWCodeFRParams.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	Altitude (see page 114)	Available since version 1.7.
	IsolatedRoofs (see page 114)	Available since version 1.7.
	NodalLoadsForAllBars (see page 114)	Available since version 1.7.
	NodalLoadsForBarsList (see page 115)	Available since version 1.7.
	OpenStructure (see page 115)	Available since version 1.7.
	PermDoorFront (see page 115)	Available since version 1.7.
	PermDoorFrontPresent (see page 115)	Available since version 1.7.
	PermDoorLeftSide (see page 116)	Available since version 1.7.
	PermDoorLeftSidePresent (see page 116)	Available since version 1.7.
	PermDoorRear (see page 116)	Available since version 1.7.
	PermDoorRearPresent (see page 116)	Available since version 1.7.
	PermDoorRightSide (see page 117)	Available since version 1.7.
	PermDoorRightSidePresent (see page 117)	Available since version 1.7.
	PermFront (see page 117)	Available since version 1.7.
	PermLeftSide (see page 117)	Available since version 1.7.
	PermRear (see page 118)	Available since version 1.7.
	PermRightSide (see page 118)	Available since version 1.7.
	ReferenceLevel (see page 118)	Available since version 1.7.
	RiseOfRoof (see page 119)	Available since version 1.7.
	RiseOfRoofAutomatic (see page 119)	Available since version 1.7.
	SnowGutterBars (see page 119)	Available since version 1.7.
	SnowIsWaterOutflow (see page 119)	Available since version 1.7.
	SnowObstacles (see page 120)	Obstacles - list of bars Available since version 1.7.
	SnowPressureExtreme (see page 120)	Available since version 1.7.
	SnowPressureExtremeAutomatic (see page 120)	Available since version 1.7.
	SnowPressureNormal (see page 120)	Available since version 1.7.
	SnowPressureNormalAutomatic (see page 121)	Available since version 1.7.
	SnowRedistribution (see page 121)	Available since version 1.7.
	SnowRegion (see page 121)	Symbol of snow region Available since version 1.7.
	SnowType (see page 121)	Available since version 1.7.
	SnowWaterOutflow (see page 122)	Water outflow Available since version 1.7.
	StructureHeight (see page 122)	Available since version 1.7.
	SurfaceLower (see page 122)	Available since version 1.7.
	SurfaceUpper (see page 122)	Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

	WindCoastalArea (see page 123)	Available since version 1.7.
	WindDeltaCoeff (see page 123)	Available since version 1.7.
	WindDeltaCoeffAutomatic (see page 123)	Available since version 1.7.
	WindDynamicAction (see page 124)	Available since version 1.7.
	WindDynamicActionCoeff (see page 124)	Available since version 1.7.
	WindDynamicActionCoeffAutomatic (see page 124)	Available since version 1.7.
	WindDynamicActionPeriod (see page 124)	Available since version 1.7.
	WindDynamicActionSteelStructure (see page 125)	Available since version 1.7.
	WindFacadeOffset (see page 125)	Available since version 1.7.
	WindMultipleRoof (see page 125)	Available since version 1.7.
	WindPressure (see page 125)	Available since version 1.7.
	WindPressureAutomatic (see page 126)	Available since version 1.7.
	WindPressureCeCiMinimum (see page 126)	Available since version 1.7.
	WindPressureVariable (see page 126)	Available since version 1.7.
	WindRegion (see page 126)	Name of wind region Available since version 1.7.
	WindSite (see page 127)	Available since version 1.7.
	WindType (see page 127)	Available since version 1.7.

Public Methods

	Name	Description
	IsolatedRoofGetLocation (see page 128)	Available since version 1.7.
	IsolatedRoofSetLocation (see page 128)	Available since version 1.7.
	SnowBarCoeffGet (see page 128)	Available since version 1.7.
	SnowBarCoeffSet (see page 128)	Available since version 1.7.
	WindBarCoeffGet (see page 129)	Available since version 1.7.
	WindBarCoeffSet (see page 129)	Available since version 1.7.

II.1.2.8.2 IRobotSWCodeFRParams Fields

The fields of the IRobotSWCodeFRParams class are listed here.

Public Fields

	Name	Description
	Altitude (see page 114)	Available since version 1.7.
	IsolatedRoofs (see page 114)	Available since version 1.7.
	NodalLoadsForAllBars (see page 114)	Available since version 1.7.
	NodalLoadsForBarsList (see page 115)	Available since version 1.7.
	OpenStructure (see page 115)	Available since version 1.7.
	PermDoorFront (see page 115)	Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

	PermDoorFrontPresent (see page 115)	Available since version 1.7.
	PermDoorLeftSide (see page 116)	Available since version 1.7.
	PermDoorLeftSidePresent (see page 116)	Available since version 1.7.
	PermDoorRear (see page 116)	Available since version 1.7.
	PermDoorRearPresent (see page 116)	Available since version 1.7.
	PermDoorRightSide (see page 117)	Available since version 1.7.
	PermDoorRightSidePresent (see page 117)	Available since version 1.7.
	PermFront (see page 117)	Available since version 1.7.
	PermLeftSide (see page 117)	Available since version 1.7.
	PermRear (see page 118)	Available since version 1.7.
	PermRightSide (see page 118)	Available since version 1.7.
	ReferenceLevel (see page 118)	Available since version 1.7.
	RiseOfRoof (see page 119)	Available since version 1.7.
	RiseOfRoofAutomatic (see page 119)	Available since version 1.7.
	SnowGutterBars (see page 119)	Available since version 1.7.
	SnowIsWaterOutflow (see page 119)	Available since version 1.7.
	SnowObstacles (see page 120)	Obstacles - list of bars Available since version 1.7.
	SnowPressureExtreme (see page 120)	Available since version 1.7.
	SnowPressureExtremeAutomatic (see page 120)	Available since version 1.7.
	SnowPressureNormal (see page 120)	Available since version 1.7.
	SnowPressureNormalAutomatic (see page 121)	Available since version 1.7.
	SnowRedistribution (see page 121)	Available since version 1.7.
	SnowRegion (see page 121)	Symbol of snow region Available since version 1.7.
	SnowType (see page 121)	Available since version 1.7.
	SnowWaterOutflow (see page 122)	Water outflow Available since version 1.7.
	StructureHeight (see page 122)	Available since version 1.7.
	SurfaceLower (see page 122)	Available since version 1.7.
	SurfaceUpper (see page 122)	Available since version 1.7.
	WindCoastalArea (see page 123)	Available since version 1.7.
	WindDeltaCoeff (see page 123)	Available since version 1.7.
	WindDeltaCoeffAutomatic (see page 123)	Available since version 1.7.
	WindDynamicAction (see page 124)	Available since version 1.7.
	WindDynamicActionCoeff (see page 124)	Available since version 1.7.
	WindDynamicActionCoeffAutomatic (see page 124)	Available since version 1.7.
	WindDynamicActionPeriod (see page 124)	Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

	WindDynamicActionSteelStructure (see page 125)	Available since version 1.7.
	WindFacadeOffset (see page 125)	Available since version 1.7.
	WindMultipleRoof (see page 125)	Available since version 1.7.
	WindPressure (see page 125)	Available since version 1.7.
	WindPressureAutomatic (see page 126)	Available since version 1.7.
	WindPressureCeCiMinimum (see page 126)	Available since version 1.7.
	WindPressureVariable (see page 126)	Available since version 1.7.
	WindRegion (see page 126)	Name of wind region Available since version 1.7.
	WindSite (see page 127)	Available since version 1.7.
	WindType (see page 127)	Available since version 1.7.

II.1.2.8.2.1 Altitude

C++

```
HRESULT get_Altitude(double*);
HRESULT put_Altitude(double);
```

C#

```
public double Altitude { get; set; }
```

Visual Basic

```
Public Altitude As double
```

Description

Available since version 1.7.

II.1.2.8.2.2 IsolatedRoofs

C++

```
HRESULT get_IsolatedRoofs(IRobotNumbersCollection**);
```

C#

```
public IRobotNumbersCollection IsolatedRoofs { get; }
```

Visual Basic

```
Public ReadOnly IsolatedRoofs As IRobotNumbersCollection
```

Description

Available since version 1.7.

II.1.2.8.2.3 NodalLoadsForAllBars

C++

```
HRESULT get_NodalLoadsForAllBars(VARIANT_BOOL*);
HRESULT put_NodalLoadsForAllBars(VARIANT_BOOL);
```

C#

```
public bool NodalLoadsForAllBars { get; set; }
```

Visual Basic

```
Public NodalLoadsForAllBars As Boolean
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Available since version 1.7.

II.1.2.8.2.4 NodalLoadsForBarsList

C++

```
HRESULT get_NodalLoadsForBarsList(BSTR*);
HRESULT put_NodalLoadsForBarsList(BSTR);
```

C#

```
public String NodalLoadsForBarsList { get; set; }
```

Visual Basic

```
Public NodalLoadsForBarsList As String
```

Description

Available since version 1.7.

II.1.2.8.2.5 OpenStructure

C++

```
HRESULT get_OpenStructure(VARIANT_BOOL*);
HRESULT put_OpenStructure(VARIANT_BOOL);
```

C#

```
public bool OpenStructure { get; set; }
```

Visual Basic

```
Public OpenStructure As Boolean
```

Description

Available since version 1.7.

II.1.2.8.2.6 PermDoorFront

C++

```
HRESULT get_PermDoorFront(double*);
HRESULT put_PermDoorFront(double);
```

C#

```
public double PermDoorFront { get; set; }
```

Visual Basic

```
Public PermDoorFront As double
```

Description

Available since version 1.7.

II.1.2.8.2.7 PermDoorFrontPresent

C++

```
HRESULT get_PermDoorFrontPresent(VARIANT_BOOL*);
HRESULT put_PermDoorFrontPresent(VARIANT_BOOL);
```

C#

```
public bool PermDoorFrontPresent { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public PermDoorFrontPresent As Boolean
```

Description

Available since version 1.7.

II.1.2.8.2.8 PermDoorLeftSide

C++

```
HRESULT get_PermDoorLeftSide(double*);
HRESULT put_PermDoorLeftSide(double);
```

C#

```
public double PermDoorLeftSide { get; set; }
```

Visual Basic

```
Public PermDoorLeftSide As double
```

Description

Available since version 1.7.

II.1.2.8.2.9 PermDoorLeftSidePresent

C++

```
HRESULT get_PermDoorLeftSidePresent(VARIANT_BOOL*);
HRESULT put_PermDoorLeftSidePresent(VARIANT_BOOL);
```

C#

```
public bool PermDoorLeftSidePresent { get; set; }
```

Visual Basic

```
Public PermDoorLeftSidePresent As Boolean
```

Description

Available since version 1.7.

II.1.2.8.2.10 PermDoorRear

C++

```
HRESULT get_PermDoorRear(double*);
HRESULT put_PermDoorRear(double);
```

C#

```
public double PermDoorRear { get; set; }
```

Visual Basic

```
Public PermDoorRear As double
```

Description

Available since version 1.7.

II.1.2.8.2.11 PermDoorRearPresent

C++

```
HRESULT get_PermDoorRearPresent(VARIANT_BOOL*);
HRESULT put_PermDoorRearPresent(VARIANT_BOOL);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public bool PermDoorRearPresent { get; set; }
```

Visual Basic

```
Public PermDoorRearPresent As Boolean
```

Description

Available since version 1.7.

II.1.2.8.2.12 PermDoorRightSide

C++

```
HRESULT get_PermDoorRightSide(double*);
HRESULT put_PermDoorRightSide(double);
```

C#

```
public double PermDoorRightSide { get; set; }
```

Visual Basic

```
Public PermDoorRightSide As double
```

Description

Available since version 1.7.

II.1.2.8.2.13 PermDoorRightSidePresent

C++

```
HRESULT get_PermDoorRightSidePresent(VARIANT_BOOL*);
HRESULT put_PermDoorRightSidePresent(VARIANT_BOOL);
```

C#

```
public bool PermDoorRightSidePresent { get; set; }
```

Visual Basic

```
Public PermDoorRightSidePresent As Boolean
```

Description

Available since version 1.7.

II.1.2.8.2.14 PermFront

C++

```
HRESULT get_PermFront(double*);
HRESULT put_PermFront(double);
```

C#

```
public double PermFront { get; set; }
```

Visual Basic

```
Public PermFront As double
```

Description

Available since version 1.7.

II.1.2.8.2.15 PermLeftSide

C++

```
HRESULT get_PermLeftSide(double*);
```

(c) 2014 Autodesk Inc. All rights reserved

```
HRESULT put_PermLeftSide(double);
```

C#

```
public double PermLeftSide { get; set; }
```

Visual Basic

```
Public PermLeftSide As double
```

Description

Available since version 1.7.

II.1.2.8.2.16 PermRear

C++

```
HRESULT get_PermRear(double*);
HRESULT put_PermRear(double);
```

C#

```
public double PermRear { get; set; }
```

Visual Basic

```
Public PermRear As double
```

Description

Available since version 1.7.

II.1.2.8.2.17 PermRightSide

C++

```
HRESULT get_PermRightSide(double*);
HRESULT put_PermRightSide(double);
```

C#

```
public double PermRightSide { get; set; }
```

Visual Basic

```
Public PermRightSide As double
```

Description

Available since version 1.7.

II.1.2.8.2.18 ReferenceLevel

C++

```
HRESULT get_ReferenceLevel(double*);
HRESULT put_ReferenceLevel(double);
```

C#

```
public double ReferenceLevel { get; set; }
```

Visual Basic

```
Public ReferenceLevel As double
```

Description

Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

II.1.2.8.2.19 RiseOfRoof

C++

```
HRESULT get_RiseOfRoof(double*);
HRESULT put_RiseOfRoof(double);
```

C#

```
public double RiseOfRoof { get; set; }
```

Visual Basic

```
Public RiseOfRoof As double
```

Description

Available since version 1.7.

II.1.2.8.2.20 RiseOfRoofAutomatic

C++

```
HRESULT get_RiseOfRoofAutomatic(VARIANT_BOOL*);
HRESULT put_RiseOfRoofAutomatic(VARIANT_BOOL);
```

C#

```
public bool RiseOfRoofAutomatic { get; set; }
```

Visual Basic

```
Public RiseOfRoofAutomatic As Boolean
```

Description

Available since version 1.7.

II.1.2.8.2.21 SnowGutterBars

C++

```
HRESULT get_SnowGutterBars(BSTR*);
HRESULT put_SnowGutterBars(BSTR);
```

C#

```
public String SnowGutterBars { get; set; }
```

Visual Basic

```
Public SnowGutterBars As String
```

Description

Available since version 1.7.

II.1.2.8.2.22 SnowIsWaterOutflow

C++

```
HRESULT get_SnowIsWaterOutflow(VARIANT_BOOL*);
HRESULT put_SnowIsWaterOutflow(VARIANT_BOOL);
```

C#

```
public bool SnowIsWaterOutflow { get; set; }
```

Visual Basic

```
Public SnowIsWaterOutflow As Boolean
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Available since version 1.7.

II.1.2.8.2.23 SnowObstacles

C++

```
HRESULT get_SnowObstacles(BSTR*);
HRESULT put_SnowObstacles(BSTR);
```

C#

```
public String SnowObstacles { get; set; }
```

Visual Basic

```
Public SnowObstacles As String
```

Description

Obstacles - list of bars Available since version 1.7.

II.1.2.8.2.24 SnowPressureExtreme

C++

```
HRESULT get_SnowPressureExtreme(double*);
HRESULT put_SnowPressureExtreme(double);
```

C#

```
public double SnowPressureExtreme { get; set; }
```

Visual Basic

```
Public SnowPressureExtreme As double
```

Description

Available since version 1.7.

II.1.2.8.2.25 SnowPressureExtremeAutomatic

C++

```
HRESULT get_SnowPressureExtremeAutomatic(VARIANT_BOOL*);
HRESULT put_SnowPressureExtremeAutomatic(VARIANT_BOOL);
```

C#

```
public bool SnowPressureExtremeAutomatic { get; set; }
```

Visual Basic

```
Public SnowPressureExtremeAutomatic As Boolean
```

Description

Available since version 1.7.

II.1.2.8.2.26 SnowPressureNormal

C++

```
HRESULT get_SnowPressureNormal(double*);
HRESULT put_SnowPressureNormal(double);
```

C#

```
public double SnowPressureNormal { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public SnowPressureNormal As double
```

Description

Available since version 1.7.

II.1.2.8.2.27 SnowPressureNormalAutomatic

C++

```
HRESULT get_SnowPressureNormalAutomatic(VARIANT_BOOL*);
HRESULT put_SnowPressureNormalAutomatic(VARIANT_BOOL);
```

C#

```
public bool SnowPressureNormalAutomatic { get; set; }
```

Visual Basic

```
Public SnowPressureNormalAutomatic As Boolean
```

Description

Available since version 1.7.

II.1.2.8.2.28 SnowRedistribution

C++

```
HRESULT get_SnowRedistribution(VARIANT_BOOL*);
HRESULT put_SnowRedistribution(VARIANT_BOOL);
```

C#

```
public bool SnowRedistribution { get; set; }
```

Visual Basic

```
Public SnowRedistribution As Boolean
```

Description

Available since version 1.7.

II.1.2.8.2.29 SnowRegion

C++

```
HRESULT get_SnowRegion(BSTR*);
HRESULT put_SnowRegion(BSTR);
```

C#

```
public String SnowRegion { get; set; }
```

Visual Basic

```
Public SnowRegion As String
```

Description

Symbol of snow region Available since version 1.7.

II.1.2.8.2.30 SnowType

C++

```
HRESULT get_SnowType(IRobotSWCodeFRSnowType*);
HRESULT put_SnowType(IRobotSWCodeFRSnowType);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRobotSWCodeFRSnowType SnowType { get; set; }
```

Visual Basic

```
Public SnowType As IRobotSWCodeFRSnowType
```

Description

Available since version 1.7.

II.1.2.8.2.31 SnowWaterOutflow

C++

```
HRESULT get_SnowWaterOutflow(double*);
HRESULT put_SnowWaterOutflow(double);
```

C#

```
public double SnowWaterOutflow { get; set; }
```

Visual Basic

```
Public SnowWaterOutflow As double
```

Description

Water outflow Available since version 1.7.

II.1.2.8.2.32 StructureHeight

C++

```
HRESULT get_StructureHeight(double*);
HRESULT put_StructureHeight(double);
```

C#

```
public double StructureHeight { get; set; }
```

Visual Basic

```
Public StructureHeight As double
```

Description

Available since version 1.7.

II.1.2.8.2.33 SurfaceLower

C++

```
HRESULT get_SurfaceLower(IRobotSWCodeFRSurfaceType*);
HRESULT put_SurfaceLower(IRobotSWCodeFRSurfaceType);
```

C#

```
public IRobotSWCodeFRSurfaceType SurfaceLower { get; set; }
```

Visual Basic

```
Public SurfaceLower As IRobotSWCodeFRSurfaceType
```

Description

Available since version 1.7.

II.1.2.8.2.34 SurfaceUpper

C++

```
HRESULT get_SurfaceUpper(IRobotSWCodeFRSurfaceType*);
```

(c) 2014 Autodesk Inc. All rights reserved

```
HRESULT put_SurfaceUpper(IRobotSWCodeFRSurfaceType);
```

C#

```
public IRobotSWCodeFRSurfaceType SurfaceUpper { get; set; }
```

Visual Basic

```
Public SurfaceUpper As IRobotSWCodeFRSurfaceType
```

Description

Available since version 1.7.

II.1.2.8.2.35 WindCoastalArea

C++

```
HRESULT get_WindCoastalArea(VARIANT_BOOL*);
HRESULT put_WindCoastalArea(VARIANT_BOOL);
```

C#

```
public bool WindCoastalArea { get; set; }
```

Visual Basic

```
Public WindCoastalArea As Boolean
```

Description

Available since version 1.7.

II.1.2.8.2.36 WindDeltaCoeff

C++

```
HRESULT get_WindDeltaCoeff(double*);
HRESULT put_WindDeltaCoeff(double);
```

C#

```
public double WindDeltaCoeff { get; set; }
```

Visual Basic

```
Public WindDeltaCoeff As double
```

Description

Available since version 1.7.

II.1.2.8.2.37 WindDeltaCoeffAutomatic

C++

```
HRESULT get_WindDeltaCoeffAutomatic(VARIANT_BOOL*);
HRESULT put_WindDeltaCoeffAutomatic(VARIANT_BOOL);
```

C#

```
public bool WindDeltaCoeffAutomatic { get; set; }
```

Visual Basic

```
Public WindDeltaCoeffAutomatic As Boolean
```

Description

Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

II.1.2.8.2.38 WindDynamicAction

C++

```
HRESULT get_WindDynamicAction(VARIANT_BOOL*);
HRESULT put_WindDynamicAction(VARIANT_BOOL);
```

C#

```
public bool WindDynamicAction { get; set; }
```

Visual Basic

```
Public WindDynamicAction As Boolean
```

Description

Available since version 1.7.

II.1.2.8.2.39 WindDynamicActionCoeff

C++

```
HRESULT get_WindDynamicActionCoeff(double*);
HRESULT put_WindDynamicActionCoeff(double);
```

C#

```
public double WindDynamicActionCoeff { get; set; }
```

Visual Basic

```
Public WindDynamicActionCoeff As double
```

Description

Available since version 1.7.

II.1.2.8.2.40 WindDynamicActionCoeffAutomatic

C++

```
HRESULT get_WindDynamicActionCoeffAutomatic(VARIANT_BOOL*);
HRESULT put_WindDynamicActionCoeffAutomatic(VARIANT_BOOL);
```

C#

```
public bool WindDynamicActionCoeffAutomatic { get; set; }
```

Visual Basic

```
Public WindDynamicActionCoeffAutomatic As Boolean
```

Description

Available since version 1.7.

II.1.2.8.2.41 WindDynamicActionPeriod

C++

```
HRESULT get_WindDynamicActionPeriod(double*);
HRESULT put_WindDynamicActionPeriod(double);
```

C#

```
public double WindDynamicActionPeriod { get; set; }
```

Visual Basic

```
Public WindDynamicActionPeriod As double
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Available since version 1.7.

II.1.2.8.2.42 WindDynamicActionSteelStructure

C++

```
HRESULT get_WindDynamicActionSteelStructure(VARIANT_BOOL*);
HRESULT put_WindDynamicActionSteelStructure(VARIANT_BOOL);
```

C#

```
public bool WindDynamicActionSteelStructure { get; set; }
```

Visual Basic

```
Public WindDynamicActionSteelStructure As Boolean
```

Description

Available since version 1.7.

II.1.2.8.2.43 WindFacadeOffset

C++

```
HRESULT get_WindFacadeOffset(VARIANT_BOOL*);
HRESULT put_WindFacadeOffset(VARIANT_BOOL);
```

C#

```
public bool WindFacadeOffset { get; set; }
```

Visual Basic

```
Public WindFacadeOffset As Boolean
```

Description

Available since version 1.7.

II.1.2.8.2.44 WindMultipleRoof

C++

```
HRESULT get_WindMultipleRoof(VARIANT_BOOL*);
HRESULT put_WindMultipleRoof(VARIANT_BOOL);
```

C#

```
public bool WindMultipleRoof { get; set; }
```

Visual Basic

```
Public WindMultipleRoof As Boolean
```

Description

Available since version 1.7.

II.1.2.8.2.45 WindPressure

C++

```
HRESULT get_WindPressure(double*);
HRESULT put_WindPressure(double);
```

C#

```
public double WindPressure { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public WindPressure As double
```

Description

Available since version 1.7.

II.1.2.8.2.46 WindPressureAutomatic

C++

```
HRESULT get_WindPressureAutomatic(VARIANT_BOOL*);
HRESULT put_WindPressureAutomatic(VARIANT_BOOL);
```

C#

```
public bool WindPressureAutomatic { get; set; }
```

Visual Basic

```
Public WindPressureAutomatic As Boolean
```

Description

Available since version 1.7.

II.1.2.8.2.47 WindPressureCeCiMinimum

C++

```
HRESULT get_WindPressureCeCiMinimum(VARIANT_BOOL*);
HRESULT put_WindPressureCeCiMinimum(VARIANT_BOOL);
```

C#

```
public bool WindPressureCeCiMinimum { get; set; }
```

Visual Basic

```
Public WindPressureCeCiMinimum As Boolean
```

Description

Available since version 1.7.

II.1.2.8.2.48 WindPressureVariable

C++

```
HRESULT get_WindPressureVariable(VARIANT_BOOL*);
HRESULT put_WindPressureVariable(VARIANT_BOOL);
```

C#

```
public bool WindPressureVariable { get; set; }
```

Visual Basic

```
Public WindPressureVariable As Boolean
```

Description

Available since version 1.7.

II.1.2.8.2.49 WindRegion

C++

```
HRESULT get_WindRegion(BSTR*);
HRESULT put_WindRegion(BSTR);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public String WindRegion { get; set; }
```

Visual Basic

```
Public WindRegion As String
```

Description

Name of wind region Available since version 1.7.

II.1.2.8.2.50 WindSite

C++

```
HRESULT get_WindSite(IRobotSWCodeFRWindSite*);
HRESULT put_WindSite(IRobotSWCodeFRWindSite);
```

C#

```
public IRobotSWCodeFRWindSite WindSite { get; set; }
```

Visual Basic

```
Public WindSite As IRobotSWCodeFRWindSite
```

Description

Available since version 1.7.

II.1.2.8.2.51 WindType

C++

```
HRESULT get_WindType(IRobotSWCodeFRWindType*);
HRESULT put_WindType(IRobotSWCodeFRWindType);
```

C#

```
public IRobotSWCodeFRWindType WindType { get; set; }
```

Visual Basic

```
Public WindType As IRobotSWCodeFRWindType
```

Description

Available since version 1.7.

II.1.2.8.3 IRobotSWCodeFRParams Methods

The methods of the IRobotSWCodeFRParams class are listed here.

Public Methods

	Name	Description
	IsolatedRoofGetLocation (see page 128)	Available since version 1.7.
	IsolatedRoofSetLocation (see page 128)	Available since version 1.7.
	SnowBarCoeffGet (see page 128)	Available since version 1.7.
	SnowBarCoeffSet (see page 128)	Available since version 1.7.
	WindBarCoeffGet (see page 129)	Available since version 1.7.
	WindBarCoeffSet (see page 129)	Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

II.1.2.8.3.1 IsolatedRoofGetLocation

C++

```
HRESULT IsolatedRoofGetLocation(long _bar_num, VARIANT_BOOL _relative, double* ret);
```

C#

```
public double IsolatedRoofGetLocation(long _bar_num, bool _relative);
```

Visual Basic

```
Public Function IsolatedRoofGetLocation(_bar_num As long, _relative As Boolean) As double
```

Description

Available since version 1.7.

II.1.2.8.3.2 IsolatedRoofSetLocation

C++

```
HRESULT IsolatedRoofSetLocation(long _bar_num, double _location, VARIANT_BOOL _relative);
```

C#

```
public void IsolatedRoofSetLocation(long _bar_num, double _location, bool _relative);
```

Visual Basic

```
Public Sub IsolatedRoofSetLocation(_bar_num As long, _location As double, _relative As
Boolean)
```

Description

Available since version 1.7.

II.1.2.8.3.3 SnowBarCoeffGet

C++

```
HRESULT SnowBarCoeffGet(long _bar_num, double* ret);
```

C#

```
public double SnowBarCoeffGet(long _bar_num);
```

Visual Basic

```
Public Function SnowBarCoeffGet(_bar_num As long) As double
```

Description

Available since version 1.7.

II.1.2.8.3.4 SnowBarCoeffSet

C++

```
HRESULT SnowBarCoeffSet(long _bar_num, double _coeff);
```

C#

```
public void SnowBarCoeffSet(long _bar_num, double _coeff);
```

Visual Basic

```
Public Sub SnowBarCoeffSet(_bar_num As long, _coeff As double)
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Available since version 1.7.

II.1.2.8.3.5 WindBarCoeffGet

C++

```
HRESULT WindBarCoeffGet(long _bar_num, double* ret);
```

C#

```
public double WindBarCoeffGet(long _bar_num);
```

Visual Basic

```
Public Function WindBarCoeffGet(_bar_num As long) As double
```

Description

Available since version 1.7.

II.1.2.8.3.6 WindBarCoeffSet

C++

```
HRESULT WindBarCoeffSet(long _bar_num, double _coeff);
```

C#

```
public void WindBarCoeffSet(long _bar_num, double _coeff);
```

Visual Basic

```
Public Sub WindBarCoeffSet(_bar_num As long, _coeff As double)
```

Description

Available since version 1.7.

II.1.2.9 IRobotSWCodeFRWindSite

C++

```
enum IRobotSWCodeFRWindSite;
```

C#

```
public enum IRobotSWCodeFRWindSite;
```

Visual Basic

```
Public Enum IRobotSWCodeFRWindSite
```

Members

Members	Description
I_SWCFRWS_OBSCURED = 0	Available since version 1.7.
I_SWCFRWS_NORMAL = 1	Available since version 1.7.
I_SWCFRWS_EXPOSED = 2	Available since version 1.7.

II.1.2.10 IRobotSWCodeFRWindType

C++

```
enum IRobotSWCodeFRWindType;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public enum IRobotSWCodeFRWindType;
```

Visual Basic

```
Public Enum IRobotSWCodeFRWindType
```

Members

Members	Description
I_SWCFRWT_NORMAL = 0	Available since version 1.7.
I_SWCFRWT_EXTREME = 1	Available since version 1.7.

II.1.2.11 IRobotSWCodeFRSurfaceType

C++

```
enum IRobotSWCodeFRSurfaceType;
```

C#

```
public enum IRobotSWCodeFRSurfaceType;
```

Visual Basic

```
Public Enum IRobotSWCodeFRSurfaceType
```

Members

Members	Description
I_SWCFRST_SMOOTH_OR_CORRUGATED = 0	Available since version 1.7.
I_SWCFRST_FOLDED_OR_CORRUGATED = 1	Available since version 1.7.
I_SWCFRST_RIBBED = 2	Available since version 1.7.

II.1.2.12 IRobotSWCodeFRSnowType

C++

```
enum IRobotSWCodeFRSnowType;
```

C#

```
public enum IRobotSWCodeFRSnowType;
```

Visual Basic

```
Public Enum IRobotSWCodeFRSnowType
```

Members

Members	Description
I_SWCFRST_NORMAL = 1	Available since version 1.7.
I_SWCFRST_ACCIDENTAL = 2	Available since version 1.7.
I_SWCFRST_NORMAL_AND_ACCIDENTAL = 0	Available since version 1.7.

II.1.2.13 IRobotSWCodePLWindPressDistribType

C++

```
enum IRobotSWCodePLWindPressDistribType;
```

C#

```
public enum IRobotSWCodePLWindPressDistribType;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

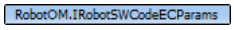
```
Public Enum IRobotSWCodePLWindPressDistribType
```

Members

Members	Description
I_SWCPLWPDT_CONSTANT = 0	Available since version 1.7.
I_SWCPLWPDT_VARIABLE = 1	Available since version 1.7.

II.1.2.14 IRobotSWCodeECParams

Class Hierarchy



C++

```
interface IRobotSWCodeECParams : IDispatch;
```

C#

```
public interface IRobotSWCodeECParams;
```

Visual Basic

```
Public Interface IRobotSWCodeECParams
```

Description

Parameters of snow/wind load generation for Eurocode. .

Version

Available since version 4.

II.1.2.14.1 IRobotSWCodeECParams Members

The following tables list the members exposed by IRobotSWCodeECParams.

Public Fields

	Name	Description
	Altitude (see page 134)	
	GlobalCDIR (see page 134)	
	GroundType (see page 134)	
	LeftWind2NordAngle (see page 135)	
	NodalLoadsForAllBars (see page 135)	
	NodalLoadsForBarsList (see page 135)	
	PermDoorFront (see page 135)	
	PermDoorFrontPresent (see page 136)	
	PermDoorLeftSide (see page 136)	
	PermDoorLeftSidePresent (see page 136)	
	PermDoorRear (see page 136)	
	PermDoorRearPresent (see page 137)	

(c) 2014 Autodesk Inc. All rights reserved

	PermDoorRightSide (see page 137)	
	PermDoorRightSidePresent (see page 137)	
	PermFront (see page 137)	
	PermLeftSide (see page 138)	
	PermRear (see page 138)	
	PermRightSide (see page 138)	
	ReferenceLevel (see page 138)	
	SnowGutterBars (see page 139)	
	SnowPressureExtreme (see page 139)	
	SnowPressureNormal (see page 139)	
	SnowRedistribution (see page 140)	
	StructureAge (see page 140)	
	StructureAgeCode (see page 140)	
	StructureHeight (see page 140)	
	StructureP (see page 141)	
	WindAutoCd (see page 141)	
	WindCALT (see page 141)	
	WindCd (see page 141)	
	WindCDIR (see page 142)	
	WindCdType (see page 142)	
	WindCt (see page 142)	
	WindCtAuto (see page 142)	
	WindCTEM (see page 143)	
	WindE (see page 143)	
	WindKT (see page 143)	
	WindPressureAutomatic (see page 143)	
	WindQref (see page 144)	
	WindQref0 (see page 144)	
	WindQref0p (see page 144)	
	WindSiteType (see page 145)	
	WindVref0 (see page 145)	
	WindZ0 (see page 145)	
	WindZMin (see page 145)	

Public Methods

	Name	Description
	SnowBarCoeffGet (see page 146)	
	SnowBarCoeffSet (see page 146)	
	WindBarCoeffGet (see page 146)	
	WindBarCoeffSet (see page 147)	

II.1.2.14.2 IRobotSWCodeECParams Fields

The fields of the IRobotSWCodeECParams class are listed here.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	Altitude (see page 134)	
	GlobalCDIR (see page 134)	
	GroundType (see page 134)	
	LeftWind2NordAngle (see page 135)	
	NodalLoadsForAllBars (see page 135)	
	NodalLoadsForBarsList (see page 135)	
	PermDoorFront (see page 135)	
	PermDoorFrontPresent (see page 136)	
	PermDoorLeftSide (see page 136)	
	PermDoorLeftSidePresent (see page 136)	
	PermDoorRear (see page 136)	
	PermDoorRearPresent (see page 137)	
	PermDoorRightSide (see page 137)	
	PermDoorRightSidePresent (see page 137)	
	PermFront (see page 137)	
	PermLeftSide (see page 138)	
	PermRear (see page 138)	
	PermRightSide (see page 138)	
	ReferenceLevel (see page 138)	
	SnowGutterBars (see page 139)	
	SnowPressureExtreme (see page 139)	
	SnowPressureNormal (see page 139)	
	SnowRedistribution (see page 140)	
	StructureAge (see page 140)	
	StructureAgeCode (see page 140)	
	StructureHeight (see page 140)	
	StructureP (see page 141)	
	WindAutoCd (see page 141)	
	WindCALT (see page 141)	
	WindCd (see page 141)	
	WindCDIR (see page 142)	
	WindCdType (see page 142)	
	WindCt (see page 142)	
	WindCtAuto (see page 142)	
	WindCTEM (see page 143)	
	WindE (see page 143)	
	WindKT (see page 143)	

(c) 2014 Autodesk Inc. All rights reserved

	WindPressureAutomatic (see page 143)	
	WindQref (see page 144)	
	WindQref0 (see page 144)	
	WindQref0p (see page 144)	
	WindSiteType (see page 145)	
	WindVref0 (see page 145)	
	WindZ0 (see page 145)	
	WindZMin (see page 145)	

II.1.2.14.2.1 Altitude

C++

```
HRESULT get_Altitude(double*);
HRESULT put_Altitude(double);
```

C#

```
public double Altitude { get; set; }
```

Visual Basic

```
Public Altitude As double
```

Version

Available since version 4.

II.1.2.14.2.2 GlobalCDIR

C++

```
HRESULT get_GlobalCDIR(VARIANT_BOOL*);
HRESULT put_GlobalCDIR(VARIANT_BOOL);
```

C#

```
public bool GlobalCDIR { get; set; }
```

Visual Basic

```
Public GlobalCDIR As Boolean
```

Version

Available since version 4.

II.1.2.14.2.3 GroundType

C++

```
HRESULT get_GroundType(IRobotSWCodeECGroundType*);
HRESULT put_GroundType(IRobotSWCodeECGroundType);
```

C#

```
public IRobotSWCodeECGroundType GroundType { get; set; }
```

Visual Basic

```
Public GroundType As IRobotSWCodeECGroundType
```

Version

Available since version 4.

(c) 2014 Autodesk Inc. All rights reserved

II.1.2.14.2.4 LeftWind2NordAngle

C++

```
HRESULT get_LeftWind2NordAngle(double*);
HRESULT put_LeftWind2NordAngle(double);
```

C#

```
public double LeftWind2NordAngle { get; set; }
```

Visual Basic

```
Public LeftWind2NordAngle As double
```

Version

Available since version 4.

II.1.2.14.2.5 NodalLoadsForAllBars

C++

```
HRESULT get_NodalLoadsForAllBars(VARIANT_BOOL*);
HRESULT put_NodalLoadsForAllBars(VARIANT_BOOL);
```

C#

```
public bool NodalLoadsForAllBars { get; set; }
```

Visual Basic

```
Public NodalLoadsForAllBars As Boolean
```

Version

Available since version 4.

II.1.2.14.2.6 NodalLoadsForBarsList

C++

```
HRESULT get_NodalLoadsForBarsList(BSTR*);
HRESULT put_NodalLoadsForBarsList(BSTR);
```

C#

```
public String NodalLoadsForBarsList { get; set; }
```

Visual Basic

```
Public NodalLoadsForBarsList As String
```

Version

Available since version 4.

II.1.2.14.2.7 PermDoorFront

C++

```
HRESULT get_PermDoorFront(double*);
HRESULT put_PermDoorFront(double);
```

C#

```
public double PermDoorFront { get; set; }
```

Visual Basic

```
Public PermDoorFront As double
```

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 4.

II.1.2.14.2.8 PermDoorFrontPresent

C++

```
HRESULT get_PermDoorFrontPresent(VARIANT_BOOL*);
HRESULT put_PermDoorFrontPresent(VARIANT_BOOL);
```

C#

```
public bool PermDoorFrontPresent { get; set; }
```

Visual Basic

```
Public PermDoorFrontPresent As Boolean
```

Version

Available since version 4.

II.1.2.14.2.9 PermDoorLeftSide

C++

```
HRESULT get_PermDoorLeftSide(double*);
HRESULT put_PermDoorLeftSide(double);
```

C#

```
public double PermDoorLeftSide { get; set; }
```

Visual Basic

```
Public PermDoorLeftSide As double
```

Version

Available since version 4.

II.1.2.14.2.10 PermDoorLeftSidePresent

C++

```
HRESULT get_PermDoorLeftSidePresent(VARIANT_BOOL*);
HRESULT put_PermDoorLeftSidePresent(VARIANT_BOOL);
```

C#

```
public bool PermDoorLeftSidePresent { get; set; }
```

Visual Basic

```
Public PermDoorLeftSidePresent As Boolean
```

Version

Available since version 4.

II.1.2.14.2.11 PermDoorRear

C++

```
HRESULT get_PermDoorRear(double*);
HRESULT put_PermDoorRear(double);
```

C#

```
public double PermDoorRear { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public PermDoorRear As double
```

Version

Available since version 4.

II.1.2.14.2.12 PermDoorRearPresent

C++

```
HRESULT get_PermDoorRearPresent(VARIANT_BOOL*);
HRESULT put_PermDoorRearPresent(VARIANT_BOOL);
```

C#

```
public bool PermDoorRearPresent { get; set; }
```

Visual Basic

```
Public PermDoorRearPresent As Boolean
```

Version

Available since version 4.

II.1.2.14.2.13 PermDoorRightSide

C++

```
HRESULT get_PermDoorRightSide(double*);
HRESULT put_PermDoorRightSide(double);
```

C#

```
public double PermDoorRightSide { get; set; }
```

Visual Basic

```
Public PermDoorRightSide As double
```

Version

Available since version 4.

II.1.2.14.2.14 PermDoorRightSidePresent

C++

```
HRESULT get_PermDoorRightSidePresent(VARIANT_BOOL*);
HRESULT put_PermDoorRightSidePresent(VARIANT_BOOL);
```

C#

```
public bool PermDoorRightSidePresent { get; set; }
```

Visual Basic

```
Public PermDoorRightSidePresent As Boolean
```

Version

Available since version 4.

II.1.2.14.2.15 PermFront

C++

```
HRESULT get_PermFront(double*);
HRESULT put_PermFront(double);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double PermFront { get; set; }
```

Visual Basic

```
Public PermFront As double
```

Version

Available since version 4.

II.1.2.14.2.16 PermLeftSide

C++

```
HRESULT get_PermLeftSide(double*);
HRESULT put_PermLeftSide(double);
```

C#

```
public double PermLeftSide { get; set; }
```

Visual Basic

```
Public PermLeftSide As double
```

Version

Available since version 4.

II.1.2.14.2.17 PermRear

C++

```
HRESULT get_PermRear(double*);
HRESULT put_PermRear(double);
```

C#

```
public double PermRear { get; set; }
```

Visual Basic

```
Public PermRear As double
```

Version

Available since version 4.

II.1.2.14.2.18 PermRightSide

C++

```
HRESULT get_PermRightSide(double*);
HRESULT put_PermRightSide(double);
```

C#

```
public double PermRightSide { get; set; }
```

Visual Basic

```
Public PermRightSide As double
```

Version

Available since version 4.

II.1.2.14.2.19 ReferenceLevel

C++

```
HRESULT get_ReferenceLevel(double*);
```

(c) 2014 Autodesk Inc. All rights reserved

```
HRESULT put_ReferenceLevel(double);
```

C#

```
public double ReferenceLevel { get; set; }
```

Visual Basic

```
Public ReferenceLevel As double
```

Version

Available since version 4.

II.1.2.14.2.20 SnowGutterBars

C++

```
HRESULT get_SnowGutterBars(BSTR*);
HRESULT put_SnowGutterBars(BSTR);
```

C#

```
public String SnowGutterBars { get; set; }
```

Visual Basic

```
Public SnowGutterBars As String
```

Version

Available since version 4.

II.1.2.14.2.21 SnowPressureExtreme

C++

```
HRESULT get_SnowPressureExtreme(double*);
HRESULT put_SnowPressureExtreme(double);
```

C#

```
public double SnowPressureExtreme { get; set; }
```

Visual Basic

```
Public SnowPressureExtreme As double
```

Version

Available since version 4.

II.1.2.14.2.22 SnowPressureNormal

C++

```
HRESULT get_SnowPressureNormal(double*);
HRESULT put_SnowPressureNormal(double);
```

C#

```
public double SnowPressureNormal { get; set; }
```

Visual Basic

```
Public SnowPressureNormal As double
```

Version

Available since version 4.

(c) 2014 Autodesk Inc. All rights reserved

II.1.2.14.2.23 SnowRedistribution

C++

```
HRESULT get_SnowRedistribution(VARIANT_BOOL*);
HRESULT put_SnowRedistribution(VARIANT_BOOL);
```

C#

```
public bool SnowRedistribution { get; set; }
```

Visual Basic

```
Public SnowRedistribution As Boolean
```

Version

Available since version 4.

II.1.2.14.2.24 StructureAge

C++

```
HRESULT get_StructureAge(double*);
HRESULT put_StructureAge(double);
```

C#

```
public double StructureAge { get; set; }
```

Visual Basic

```
Public StructureAge As double
```

Version

Available since version 4.

II.1.2.14.2.25 StructureAgeCode

C++

```
HRESULT get_StructureAgeCode(BSTR*);
HRESULT put_StructureAgeCode(BSTR);
```

C#

```
public String StructureAgeCode { get; set; }
```

Visual Basic

```
Public StructureAgeCode As String
```

Version

Available since version 4.

II.1.2.14.2.26 StructureHeight

C++

```
HRESULT get_StructureHeight(double*);
HRESULT put_StructureHeight(double);
```

C#

```
public double StructureHeight { get; set; }
```

Visual Basic

```
Public StructureHeight As double
```

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 4.

II.1.2.14.2.27 StructureP

C++

```
HRESULT get_StructureP(double*);
HRESULT put_StructureP(double);
```

C#

```
public double StructureP { get; set; }
```

Visual Basic

```
Public StructureP As double
```

Version

Available since version 4.

II.1.2.14.2.28 WindAutoCd

C++

```
HRESULT get_WindAutoCd(VARIANT_BOOL*);
HRESULT put_WindAutoCd(VARIANT_BOOL);
```

C#

```
public bool WindAutoCd { get; set; }
```

Visual Basic

```
Public WindAutoCd As Boolean
```

Version

Available since version 4.

II.1.2.14.2.29 WindCALT

C++

```
HRESULT get_WindCALT(double*);
HRESULT put_WindCALT(double);
```

C#

```
public double WindCALT { get; set; }
```

Visual Basic

```
Public WindCALT As double
```

Version

Available since version 4.

II.1.2.14.2.30 WindCd

C++

```
HRESULT get_WindCd(double*);
HRESULT put_WindCd(double);
```

C#

```
public double WindCd { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public WindCd As double
```

Version

Available since version 4.

II.1.2.14.2.31 WindCDIR

C++

```
HRESULT get_WindCDIR(double*);
HRESULT put_WindCDIR(double);
```

C#

```
public double WindCDIR { get; set; }
```

Visual Basic

```
Public WindCDIR As double
```

Version

Available since version 4.

II.1.2.14.2.32 WindCdType

C++

```
HRESULT get_WindCdType(IRobotSWCodeECCdType*);
HRESULT put_WindCdType(IRobotSWCodeECCdType);
```

C#

```
public IRobotSWCodeECCdType WindCdType { get; set; }
```

Visual Basic

```
Public WindCdType As IRobotSWCodeECCdType
```

Version

Available since version 4.

II.1.2.14.2.33 WindCt

C++

```
HRESULT get_WindCt(double*);
HRESULT put_WindCt(double);
```

C#

```
public double WindCt { get; set; }
```

Visual Basic

```
Public WindCt As double
```

Version

Available since version 4.

II.1.2.14.2.34 WindCtAuto

C++

```
HRESULT get_WindCtAuto(VARIANT_BOOL*);
HRESULT put_WindCtAuto(VARIANT_BOOL);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public bool WindCtAuto { get; set; }
```

Visual Basic

```
Public WindCtAuto As Boolean
```

Version

Available since version 4.

II.1.2.14.2.35 WindCTEM

C++

```
HRESULT get_WindCTEM(double*);
HRESULT put_WindCTEM(double);
```

C#

```
public double WindCTEM { get; set; }
```

Visual Basic

```
Public WindCTEM As double
```

Version

Available since version 4.

II.1.2.14.2.36 WindE

C++

```
HRESULT get_WindE(double*);
HRESULT put_WindE(double);
```

C#

```
public double WindE { get; set; }
```

Visual Basic

```
Public WindE As double
```

Version

Available since version 4.

II.1.2.14.2.37 WindKT

C++

```
HRESULT get_WindKT(double*);
HRESULT put_WindKT(double);
```

C#

```
public double WindKT { get; set; }
```

Visual Basic

```
Public WindKT As double
```

Version

Available since version 4.

II.1.2.14.2.38 WindPressureAutomatic

C++

```
HRESULT get_WindPressureAutomatic(VARIANT_BOOL*);
```

(c) 2014 Autodesk Inc. All rights reserved

```
HRESULT put_WindPressureAutomatic(VARIANT_BOOL);
```

C#

```
public bool WindPressureAutomatic { get; set; }
```

Visual Basic

```
Public WindPressureAutomatic As Boolean
```

Version

Available since version 4.

II.1.2.14.2.39 WindQref

C++

```
HRESULT get_WindQref(double*);
HRESULT put_WindQref(double);
```

C#

```
public double WindQref { get; set; }
```

Visual Basic

```
Public WindQref As double
```

Version

Available since version 4.

II.1.2.14.2.40 WindQref0

C++

```
HRESULT get_WindQref0(double*);
HRESULT put_WindQref0(double);
```

C#

```
public double WindQref0 { get; set; }
```

Visual Basic

```
Public WindQref0 As double
```

Version

Available since version 4.

II.1.2.14.2.41 WindQref0p

C++

```
HRESULT get_WindQref0p(double*);
HRESULT put_WindQref0p(double);
```

C#

```
public double WindQref0p { get; set; }
```

Visual Basic

```
Public WindQref0p As double
```

Version

Available since version 4.

(c) 2014 Autodesk Inc. All rights reserved

II.1.2.14.2.42 WindSiteType

C++

```
HRESULT get_WindSiteType(IRobotSWCodeECSiteType*);
HRESULT put_WindSiteType(IRobotSWCodeECSiteType);
```

C#

```
public IRobotSWCodeECSiteType WindSiteType { get; set; }
```

Visual Basic

```
Public WindSiteType As IRobotSWCodeECSiteType
```

Version

Available since version 4.

II.1.2.14.2.43 WindVref0

C++

```
HRESULT get_WindVref0(double*);
HRESULT put_WindVref0(double);
```

C#

```
public double WindVref0 { get; set; }
```

Visual Basic

```
Public WindVref0 As double
```

Version

Available since version 4.

II.1.2.14.2.44 WindZ0

C++

```
HRESULT get_WindZ0(double*);
HRESULT put_WindZ0(double);
```

C#

```
public double WindZ0 { get; set; }
```

Visual Basic

```
Public WindZ0 As double
```

Version

Available since version 4.

II.1.2.14.2.45 WindZMin

C++

```
HRESULT get_WindZMin(double*);
HRESULT put_WindZMin(double);
```

C#

```
public double WindZMin { get; set; }
```

Visual Basic

```
Public WindZMin As double
```

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 4.

II.1.2.14.3 IRobotSWCodeECParams Methods

The methods of the IRobotSWCodeECParams class are listed here.

Public Methods

	Name	Description
	SnowBarCoeffGet (see page 146)	
	SnowBarCoeffSet (see page 146)	
	WindBarCoeffGet (see page 146)	
	WindBarCoeffSet (see page 147)	

II.1.2.14.3.1 SnowBarCoeffGet

C++

```
HRESULT SnowBarCoeffGet(long _bar_num, double* ret);
```

C#

```
public double SnowBarCoeffGet(long _bar_num);
```

Visual Basic

```
Public Function SnowBarCoeffGet(_bar_num As long) As double
```

Version

Available since version 4.

II.1.2.14.3.2 SnowBarCoeffSet

C++

```
HRESULT SnowBarCoeffSet(long _bar_num, double _coeff);
```

C#

```
public void SnowBarCoeffSet(long _bar_num, double _coeff);
```

Visual Basic

```
Public Sub SnowBarCoeffSet(_bar_num As long, _coeff As double)
```

Version

Available since version 4.

II.1.2.14.3.3 WindBarCoeffGet

C++

```
HRESULT WindBarCoeffGet(long _bar_num, double* ret);
```

C#

```
public double WindBarCoeffGet(long _bar_num);
```

Visual Basic

```
Public Function WindBarCoeffGet(_bar_num As long) As double
```

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 4.

II.1.2.14.3.4 WindBarCoeffSet

C++

```
HRESULT WindBarCoeffSet(long _bar_num, double _coeff);
```

C#

```
public void WindBarCoeffSet(long _bar_num, double _coeff);
```

Visual Basic

```
Public Sub WindBarCoeffSet(_bar_num As long, _coeff As double)
```

Version

Available since version 4.

II.1.2.15 IRobotSWCodeECSiteType

C++

```
enum IRobotSWCodeECSiteType;
```

C#

```
public enum IRobotSWCodeECSiteType;
```

Visual Basic

```
Public Enum IRobotSWCodeECSiteType
```

Members

Members	Description
I_SWCECST_TYPE_I = 1	Available since version 4.
I_SWCECST_TYPE_II = 2	Available since version 4.
I_SWCECST_TYPE_III = 3	Available since version 4.
I_SWCECST_TYPE_IV = 4	Available since version 4.
I_SWCECST_TYPE_V = 5	Available since version 4.
I_SWCECST_TYPE_VI = 6	Available since version 4.

Version

Available since version 4.

II.1.2.16 IRobotSWCodeECGroundType

C++

```
enum IRobotSWCodeECGroundType;
```

C#

```
public enum IRobotSWCodeECGroundType;
```

Visual Basic

```
Public Enum IRobotSWCodeECGroundType
```

(c) 2014 Autodesk Inc. All rights reserved

Members

Members	Description
I_SWCECGT_TYPE_I = 1	Available since version 4.
I_SWCECGT_TYPE_II = 2	Available since version 4.
I_SWCECGT_TYPE_III = 3	Available since version 4.
I_SWCECGT_TYPE_IV = 4	Available since version 4.

Version

Available since version 4.

II.1.2.17 IRobotSWCodeECCdType

C++

```
enum IRobotSWCodeECCdType;
```

C#

```
public enum IRobotSWCodeECCdType;
```

Visual Basic

```
Public Enum IRobotSWCodeECCdType
```

Members

Members	Description
I_SWCECCT_TYPE_I = 1	Available since version 4.
I_SWCECCT_TYPE_II = 2	Available since version 4.
I_SWCECCT_TYPE_III = 3	Available since version 4.

Version

Available since version 4.

II.1.3 Code combinations

Enumerations

	Name	Description
	IRobotCodeCmbFlag (see page 157)	
	IRobotCodeCmbActionCoeffType (see page 163)	Coefficients used for describing actions. .
	IRobotCodeCmbCombPartType (see page 167)	
	IRobotCodeCmbCombType (see page 167)	
	IRobotCodeCmbDecidingValueType (see page 174)	
	IRobotCodeCmbOperator (see page 178)	Available operators used for defining relations between cases and case groups. .
	IRobotCodeCmbGenerationType (see page 188)	

Interfaces

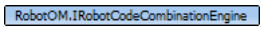
	Name	Description
	IRobotCodeCombinationEngine (see page 149)	

(c) 2014 Autodesk Inc. All rights reserved

	IRobotCodeCombination (see page 151)	Combination of load cases defined according to selected code.
	IRobotCodeCmbComponentMngr (see page 153)	Access interface to definitions of code combination components. .
	IRobotCodeCmbFactor (see page 154)	
	IRobotCodeCmbComponent (see page 155)	
	IRobotCodeCmbRegulations (see page 157)	Definition of the regulations describing the manner of generating code-defined loads. .
	IRobotCodeCmbActionServer (see page 160)	Definition of actions. .
	IRobotCodeCmbCombs (see page 164)	Combination definition.
	IRobotCodeCmbGenerationParams (see page 168)	Parameters of code combination generation.
	IRobotCodeCmbGroup (see page 175)	Definition of load case groups.
	IRobotCodeCmbGroupServer (see page 178)	Server managing the groups of cases.
	IRobotCodeCmbGroupRelation (see page 180)	Definition of relations between groups. Definition of relations between groups can be illustrated by means of the matrix whose elements are group numbers. All groups whose numbers are located in the same matrix row are in the same relation with each other (either AND or AND_OR). Whereas among the matrix rows the relation EXCLUSIVE_OR occurs. .
	IRobotCodeCmbGroupRelationServer (see page 185)	Server managing relations among groups. .
	IRobotCodeCmbActiveCaseInfo (see page 186)	

II.1.3.1 IRobotCodeCombinationEngine

Class Hierarchy



C++

```
interface IRobotCodeCombinationEngine : IDispatch;
```

C#

```
public interface IRobotCodeCombinationEngine;
```

Visual Basic

```
Public Interface IRobotCodeCombinationEngine
```

II.1.3.1.1 IRobotCodeCombinationEngine Members

The following tables list the members exposed by IRobotCodeCombinationEngine.

Public Fields

	Name	Description
	Params (see page 150)	Parameters of code combination generation Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

Public Methods

	Name	Description
	Generate (see page 150)	Function generates code-defined loads according to the selected code and current settings of generation parameters. Available since version 1.7.

II.1.3.1.2 IRobotCodeCombinationEngine Fields

The fields of the IRobotCodeCombinationEngine class are listed here.

Public Fields

	Name	Description
	Params (see page 150)	Parameters of code combination generation Available since version 1.7.

II.1.3.1.2.1 Params

C++

```
HRESULT get_Params(IRobotCodeCmbGenerationParams**);
```

C#

```
public IRobotCodeCmbGenerationParams Params { get; }
```

Visual Basic

```
Public ReadOnly Params As IRobotCodeCmbGenerationParams
```

Description

Parameters of code combination generation Available since version 1.7.

II.1.3.1.3 IRobotCodeCombinationEngine Methods

The methods of the IRobotCodeCombinationEngine class are listed here.

Public Methods

	Name	Description
	Generate (see page 150)	Function generates code-defined loads according to the selected code and current settings of generation parameters. Available since version 1.7.

II.1.3.1.3.1 Generate

C++

```
HRESULT Generate();
```

C#

```
public void Generate();
```

Visual Basic

```
Public Sub Generate()
```

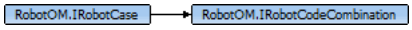
Description

Function generates code-defined loads according to the selected code and current settings of generation parameters. Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

II.1.3.2 IRobotCodeCombination

Class Hierarchy



C++

```
interface IRobotCodeCombination : IRobotCase;
```

C#

```
public interface IRobotCodeCombination : IRobotCase;
```

Visual Basic

```
Public Interface IRobotCodeCombination
```

Description

Combination of load cases defined according to selected code.

II.1.3.2.1 IRobotCodeCombination Members

The following tables list the members exposed by IRobotCodeCombination.

Public Fields

	Name	Description
	AnalizeType (see page 408)	Analysis type.
	CombinationType (see page 152)	Combination type Available since version 1.7.
	Components (see page 152)	Set of component combinations Available since version 1.7.
	Flag (see page 152)	Available since version 1.7.
	Name (see page 409)	User-defined case name .
	Nature (see page 409)	Case nature .
	Number (see page 409)	User-defined number linked with the load case .
	Type (see page 409)	Load case type - allows one to differentiate between a simple case and combination at the level of the shared part of the interface .
	UniqueId (see page 152)	Unique combination identifier Available since version 1.7.

Public Methods

	Name	Description
	FindByFlag (see page 153)	Function returns a user's number for the code combination linked with the combination that has a determined flag. If such a combination is not found, then zero value is returned. Available since version 1.7.

II.1.3.2.2 IRobotCodeCombination Fields

The fields of the IRobotCodeCombination class are listed here.

Public Fields

	Name	Description
	CombinationType (see page 152)	Combination type Available since version 1.7.
	Components (see page 152)	Set of component combinations Available since version 1.7.
	Flag (see page 152)	Available since version 1.7.
	UniqueId (see page 152)	Unique combination identifier Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

II.1.3.2.2.1 CombinationType

C++

```
HRESULT get_CombinationType(IRobotCombinationType*);
HRESULT put_CombinationType(IRobotCombinationType);
```

C#

```
public IRobotCombinationType CombinationType { get; set; }
```

Visual Basic

```
Public CombinationType As IRobotCombinationType
```

Description

Combination type Available since version 1.7.

II.1.3.2.2.2 Components

C++

```
HRESULT get_Components(IRobotCodeCmbComponentMngr**);
```

C#

```
public IRobotCodeCmbComponentMngr Components { get; }
```

Visual Basic

```
Public ReadOnly Components As IRobotCodeCmbComponentMngr
```

Description

Set of component combinations Available since version 1.7.

II.1.3.2.2.3 Flag

C++

```
HRESULT get_Flag(IRobotCodeCmbFlag*);
```

C#

```
public IRobotCodeCmbFlag Flag { get; }
```

Visual Basic

```
Public ReadOnly Flag As IRobotCodeCmbFlag
```

Description

Available since version 1.7.

II.1.3.2.2.4 UniqueId

C++

```
HRESULT get_UniqueId(long*);
```

C#

```
public long UniqueId { get; }
```

Visual Basic

```
Public ReadOnly UniqueId As long
```

Description

Unique combination identifier Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

II.1.3.2.3 IRobotCodeCombination Methods

The methods of the IRobotCodeCombination class are listed here.

Public Methods

	Name	Description
	FindByFlag (see page 153)	Function returns a user's number for the code combination linked with the combination that has a determined flag. If such a combination is not found, then zero value is returned. Available since version 1.7.

II.1.3.2.3.1 FindByFlag

C++

```
HRESULT FindByFlag(IRobotCodeCmbFlag _cmb_flag, long* ret);
```

C#

```
public long FindByFlag(IRobotCodeCmbFlag _cmb_flag);
```

Visual Basic

```
Public Function FindByFlag(_cmb_flag As IRobotCodeCmbFlag) As long
```

Description

Function returns a user's number for the code combination linked with the combination that has a determined flag. If such a combination is not found, then zero value is returned. Available since version 1.7.

II.1.3.3 IRobotCodeCmbComponentMngr

Class Hierarchy

RobotOM.IRobotCodeCmbComponentMngr

C++

```
interface IRobotCodeCmbComponentMngr : IDispatch;
```

C#

```
public interface IRobotCodeCmbComponentMngr;
```

Visual Basic

```
Public Interface IRobotCodeCmbComponentMngr
```

Description

Access interface to definitions of code combination components. .

II.1.3.3.1 IRobotCodeCmbComponentMngr Members

The following tables list the members exposed by IRobotCodeCmbComponentMngr.

Public Fields

	Name	Description
	Count (see page 154)	Number of combinations Available since version 1.7.

Public Methods

	Name	Description
	Get (see page 154)	Function returns a set of pairs (case number, coefficient) defining the selected combination. Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

II.1.3.3.2 IRobotCodeCmbComponentMngr Fields

The fields of the IRobotCodeCmbComponentMngr class are listed here.

Public Fields

	Name	Description
	Count (see page 154)	Number of combinations Available since version 1.7.

II.1.3.3.2.1 Count

C++

```
HRESULT get_Count(long*);
```

C#

```
public long Count { get; }
```

Visual Basic

```
Public ReadOnly Count As long
```

Description

Number of combinations Available since version 1.7.

II.1.3.3.3 IRobotCodeCmbComponentMngr Methods

The methods of the IRobotCodeCmbComponentMngr class are listed here.

Public Methods

	Name	Description
	Get (see page 154)	Function returns a set of pairs (case number, coefficient) defining the selected combination. Available since version 1.7.

II.1.3.3.3.1 Get

C++

```
HRESULT Get(long _cmb_idx, IRobotCodeCmbComponent** ret);
```

C#

```
public IRobotCodeCmbComponent Get(long _cmb_idx);
```

Visual Basic

```
Public Function Get(_cmb_idx As long) As IRobotCodeCmbComponent
```

Description

Function returns a set of pairs (case number, coefficient) defining the selected combination. Available since version 1.7.

II.1.3.4 IRobotCodeCmbFactor

Class Hierarchy

RobotOM.IRobotCodeCmbFactor

C++

```
interface IRobotCodeCmbFactor : IDispatch;
```

C#

```
public interface IRobotCodeCmbFactor;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Interface IRobotCodeCmbFactor
```

II.1.3.4.1 IRobotCodeCmbFactor Members

The following tables list the members exposed by IRobotCodeCmbFactor.

Public Fields

	Name	Description
	CaseNumber (see page 155)	Available since version 1.7.
	Factor (see page 155)	Available since version 1.7.

II.1.3.4.2 IRobotCodeCmbFactor Fields

The fields of the IRobotCodeCmbFactor class are listed here.

Public Fields

	Name	Description
	CaseNumber (see page 155)	Available since version 1.7.
	Factor (see page 155)	Available since version 1.7.

II.1.3.4.2.1 CaseNumber

C++

```
HRESULT get_CaseNumber(long*);
```

C#

```
public long CaseNumber { get; }
```

Visual Basic

```
Public ReadOnly CaseNumber As long
```

Description

Available since version 1.7.

II.1.3.4.2.2 Factor

C++

```
HRESULT get_Factor(double*);
```

C#

```
public double Factor { get; }
```

Visual Basic

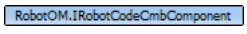
```
Public ReadOnly Factor As double
```

Description

Available since version 1.7.

II.1.3.5 IRobotCodeCmbComponent

Class Hierarchy



C++

```
interface IRobotCodeCmbComponent : IDispatch;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public interface IRobotCodeCmbComponent;
```

Visual Basic

```
Public Interface IRobotCodeCmbComponent
```

II.1.3.5.1 IRobotCodeCmbComponent Members

The following tables list the members exposed by IRobotCodeCmbComponent.

Public Fields

	Name	Description
	Count (see page 156)	Available since version 1.7.

Public Methods

	Name	Description
	Get (see page 156)	Available since version 1.7.

II.1.3.5.2 IRobotCodeCmbComponent Fields

The fields of the IRobotCodeCmbComponent class are listed here.

Public Fields

	Name	Description
	Count (see page 156)	Available since version 1.7.

II.1.3.5.2.1 Count

C++

```
HRESULT get_Count(long*);
```

C#

```
public long Count { get; }
```

Visual Basic

```
Public ReadOnly Count As long
```

Description

Available since version 1.7.

II.1.3.5.3 IRobotCodeCmbComponent Methods

The methods of the IRobotCodeCmbComponent class are listed here.

Public Methods

	Name	Description
	Get (see page 156)	Available since version 1.7.

II.1.3.5.3.1 Get

C++

```
HRESULT Get(long _idx, IRobotCodeCmbFactor** ret);
```

C#

```
public IRobotCodeCmbFactor Get(long _idx);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Function Get(_idx As long) As IRobotCodeCmbFactor
```

Description

Available since version 1.7.

II.1.3.6 IRobotCodeCmbFlag

C++

```
enum IRobotCodeCmbFlag;
```

C#

```
public enum IRobotCodeCmbFlag;
```

Visual Basic

```
Public Enum IRobotCodeCmbFlag
```

Members

Members	Description
I_CCF_MAIN = 3	Available since version 1.7.
I_CCF_MAX = 0	Available since version 1.7.
I_CCF_MIN = 1	Available since version 1.7.

II.1.3.7 IRobotCodeCmbRegulations

Class Hierarchy

RobotOM.IRobotCodeCmbRegulations

C++

```
interface IRobotCodeCmbRegulations : IDispatch;
```

C#

```
public interface IRobotCodeCmbRegulations;
```

Visual Basic

```
Public Interface IRobotCodeCmbRegulations
```

Description

Definition of the regulations describing the manner of generating code-defined loads. .

II.1.3.7.1 IRobotCodeCmbRegulations Members

The following tables list the members exposed by IRobotCodeCmbRegulations.

Public Fields

	Name	Description
	Actions (see page 158)	Actions described in the regulations Available since version 1.7.
	CodeName (see page 158)	Name of the regulations Available since version 1.7.
	Combinations (see page 158)	Combinations described in the regulations Available since version 1.7.
	IsEuroCode (see page 159)	Flag indicating if the regulations are compatible with the EUROCODE standard Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

	MaterialType (see page 159)	Material type; material type is recognized by its name - in case when a non-standard material name is used, its type will not be recognized correctly and the value designating "OTHER" unknown material type will be returned Available since version 1.7.
	Version (see page 159)	Text describing the version of the regulations Available since version 1.7.

II.1.3.7.2 IRobotCodeCmbRegulations Fields

The fields of the IRobotCodeCmbRegulations class are listed here.

Public Fields

	Name	Description
	Actions (see page 158)	Actions described in the regulations Available since version 1.7.
	CodeName (see page 158)	Name of the regulations Available since version 1.7.
	Combinations (see page 158)	Combinations described in the regulations Available since version 1.7.
	IsEuroCode (see page 159)	Flag indicating if the regulations are compatible with the EUROCODE standard Available since version 1.7.
	MaterialType (see page 159)	Material type; material type is recognized by its name - in case when a non-standard material name is used, its type will not be recognized correctly and the value designating "OTHER" unknown material type will be returned Available since version 1.7.
	Version (see page 159)	Text describing the version of the regulations Available since version 1.7.

II.1.3.7.2.1 Actions

C++

```
HRESULT get_Actions(IRobotCodeCmbActionServer**);
```

C#

```
public IRobotCodeCmbActionServer Actions { get; }
```

Visual Basic

```
Public ReadOnly Actions As IRobotCodeCmbActionServer
```

Description

Actions described in the regulations Available since version 1.7.

II.1.3.7.2.2 CodeName

C++

```
HRESULT get_CodeName(BSTR*);
HRESULT put_CodeName(BSTR);
```

C#

```
public String CodeName { get; set; }
```

Visual Basic

```
Public CodeName As String
```

Description

Name of the regulations Available since version 1.7.

II.1.3.7.2.3 Combinations

C++

```
HRESULT get_Combinations(IRobotCodeCmbCombs**);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRobotCodeCmbCombs Combinations { get; }
```

Visual Basic

```
Public ReadOnly Combinations As IRobotCodeCmbCombs
```

Description

Combinations described in the regulations Available since version 1.7.

II.1.3.7.2.4 IsEuroCode

C++

```
HRESULT get_IsEuroCode(VARIANT_BOOL*);
HRESULT put_IsEuroCode(VARIANT_BOOL);
```

C#

```
public bool IsEuroCode { get; set; }
```

Visual Basic

```
Public IsEuroCode As Boolean
```

Description

Flag indicating if the regulations are compatible with the EUROCODE standard Available since version 1.7.

II.1.3.7.2.5 MaterialType

C++

```
HRESULT get_MaterialType(IRobotMaterialType*);
HRESULT put_MaterialType(IRobotMaterialType);
```

C#

```
public IRobotMaterialType MaterialType { get; set; }
```

Visual Basic

```
Public MaterialType As IRobotMaterialType
```

Description

Material type; material type is recognized by its name - in case when a non-standard material name is used, its type will not be recognized correctly and the value designating "OTHER" unknown material type will be returned Available since version 1.7.

II.1.3.7.2.6 Version

C++

```
HRESULT get_Version(BSTR*);
HRESULT put_Version(BSTR);
```

C#

```
public String Version { get; set; }
```

Visual Basic

```
Public Version As String
```

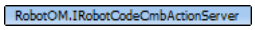
Description

Text describing the version of the regulations Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

II.1.3.8 IRobotCodeCmbActionServer

Class Hierarchy



C++

```
interface IRobotCodeCmbActionServer : IDispatch;
```

C#

```
public interface IRobotCodeCmbActionServer;
```

Visual Basic

```
Public Interface IRobotCodeCmbActionServer
```

Description

Definition of actions. .

II.1.3.8.1 IRobotCodeCmbActionServer Members

The following tables list the members exposed by IRobotCodeCmbActionServer.

Public Fields

	Name	Description
	Count (see page 161)	Number of defined actions Available since version 1.7.

Public Methods

	Name	Description
	GetCoeff (see page 161)	Function returns the value of a selected coefficient for a selected action. Available since version 1.7.
	GetName (see page 162)	Function returns the name of the indicated action. Available since version 1.7.
	GetNature (see page 162)	Function returns the nature of the selected action Available since version 1.7.
	New (see page 162)	Function defines a new action and returns its index which should be used while determining the values of individual coefficients. Available since version 1.7.
	Remove (see page 162)	Function deletes the indicated action definition. Available since version 1.7.
	SetCoeff (see page 163)	Function sets the value of a specified coefficient for a specified action. Available since version 1.7.
	SetName (see page 163)	Function sets (changes) the name of the indicated action. Available since version 1.7.
	SetNature (see page 163)	Function changes the nature of a selected action. Available since version 1.7.

II.1.3.8.2 IRobotCodeCmbActionServer Fields

The fields of the IRobotCodeCmbActionServer class are listed here.

Public Fields

	Name	Description
	Count (see page 161)	Number of defined actions Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

II.1.3.8.2.1 Count

C++

```
HRESULT get_Count(long*);
```

C#

```
public long Count { get; }
```

Visual Basic

```
Public ReadOnly Count As long
```

Description

Number of defined actions Available since version 1.7.

II.1.3.8.3 IRobotCodeCmbActionServer Methods

The methods of the IRobotCodeCmbActionServer class are listed here.

Public Methods

	Name	Description
	GetCoeff (see page 161)	Function returns the value of a selected coefficient for a selected action. Available since version 1.7.
	GetName (see page 162)	Function returns the name of the indicated action. Available since version 1.7.
	GetNature (see page 162)	Function returns the nature of the selected action Available since version 1.7.
	New (see page 162)	Function defines a new action and returns its index which should be used while determining the values of individual coefficients. Available since version 1.7.
	Remove (see page 162)	Function deletes the indicated action definition. Available since version 1.7.
	SetCoeff (see page 163)	Function sets the value of a specified coefficient for a specified action. Available since version 1.7.
	SetName (see page 163)	Function sets (changes) the name of the indicated action. Available since version 1.7.
	SetNature (see page 163)	Function changes the nature of a selected action. Available since version 1.7.

II.1.3.8.3.1 GetCoeff

C++

```
HRESULT GetCoeff(long _idx, IRobotCodeCmbActionCoeffType _coeff_type, double* ret);
```

C#

```
public double GetCoeff(long _idx, IRobotCodeCmbActionCoeffType _coeff_type);
```

Visual Basic

```
Public Function GetCoeff(_idx As long, _coeff_type As IRobotCodeCmbActionCoeffType) As
double
```

Description

Function returns the value of a selected coefficient for a selected action. Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

II.1.3.8.3.2 GetName

C++

```
HRESULT GetName(long _idx, BSTR* ret);
```

C#

```
public String GetName(long _idx);
```

Visual Basic

```
Public Function GetName(_idx As long) As String
```

Description

Function returns the name of the indicated action. Available since version 1.7.

II.1.3.8.3.3 GetNature

C++

```
HRESULT GetNature(long _idx, IRobotCaseNature* ret);
```

C#

```
public IRobotCaseNature GetNature(long _idx);
```

Visual Basic

```
Public Function GetNature(_idx As long) As IRobotCaseNature
```

Description

Function returns the nature of the selected action Available since version 1.7.

II.1.3.8.3.4 New

C++

```
HRESULT New(IRobotCaseNature _case_nature, BSTR _name, long* ret);
```

C#

```
public long New(IRobotCaseNature _case_nature, String _name);
```

Visual Basic

```
Public Function New(_case_nature As IRobotCaseNature, _name As String) As long
```

Description

Function defines a new action and returns its index which should be used while determining the values of individual coefficients. Available since version 1.7.

II.1.3.8.3.5 Remove

C++

```
HRESULT Remove(long _idx);
```

C#

```
public void Remove(long _idx);
```

Visual Basic

```
Public Sub Remove(_idx As long)
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Function deletes the indicated action definition. Available since version 1.7.

II.1.3.8.3.6 SetCoeff

C++

```
HRESULT SetCoeff(long _idx, IRobotCodeCmbActionCoeffType _coeff_type, double _coeff_val);
```

C#

```
public void SetCoeff(long _idx, IRobotCodeCmbActionCoeffType _coeff_type, double
_coeff_val);
```

Visual Basic

```
Public Sub SetCoeff(_idx As long, _coeff_type As IRobotCodeCmbActionCoeffType, _coeff_val
As double)
```

Description

Function sets the value of a specified coefficient for a specified action. Available since version 1.7.

II.1.3.8.3.7 SetName

C++

```
HRESULT SetName(long _idx, BSTR _name);
```

C#

```
public void SetName(long _idx, String _name);
```

Visual Basic

```
Public Sub SetName(_idx As long, _name As String)
```

Description

Function sets (changes) the name of the indicated action. Available since version 1.7.

II.1.3.8.3.8 SetNature

C++

```
HRESULT SetNature(long _idx, IRobotCaseNature _case_nat);
```

C#

```
public void SetNature(long _idx, IRobotCaseNature _case_nat);
```

Visual Basic

```
Public Sub SetNature(_idx As long, _case_nat As IRobotCaseNature)
```

Description

Function changes the nature of a selected action. Available since version 1.7.

II.1.3.9 IRobotCodeCmbActionCoeffType

C++

```
enum IRobotCodeCmbActionCoeffType;
```

C#

```
public enum IRobotCodeCmbActionCoeffType;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Enum IRobotCodeCmbActionCoeffType
```

Members

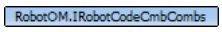
Members	Description
I_CCACT_GU_MAX = 0	Partial safety factor for ULS (maximum) Available since version 1.7.
I_CCACT_GU_MIN = 1	Partial safety factor for ULS (minimum) Available since version 1.7.
I_CCACT_GS = 2	Partial safety factor for SLS Available since version 1.7.
I_CCACT_GA = 3	Partial safety factor for accidental state Available since version 1.7.
I_CCACT_PSI0_1 = 4	Simultaneity coefficient for the first load Available since version 1.7.
I_CCACT_PSI0_2 = 5	Simultaneity coefficient for the second load Available since version 1.7.
I_CCACT_PSI0_3 = 6	Simultaneity coefficient for the third load Available since version 1.7.
I_CCACT_PSI0_N = 7	Simultaneity coefficient for the fourth load Available since version 1.7.
I_CCACT_PSI1 = 8	Factor for long-term loads Available since version 1.7.
I_CCACT_PSI2_1 = 9	Reduction factor for accidental combination Available since version 1.7.
I_CCACT_PSI2_N = 10	Reduction factor Available since version 1.7.
I_CCACT_PSIK = 11	Coefficient for second load and next loads Available since version 1.7.
I_CCACT_KSI_MAX = 12	Available since version 1.7.
I_CCACT_KSI_MIN = 13	Available since version 1.7.

Description

Coefficients used for describing actions. .

II.1.3.10 IRobotCodeCmbCombs

Class Hierarchy



C++

```
interface IRobotCodeCmbCombs : IDispatch;
```

C#

```
public interface IRobotCodeCmbCombs;
```

Visual Basic

```
Public Interface IRobotCodeCmbCombs
```

Description

Combination definition.

II.1.3.10.1 IRobotCodeCmbCombs Members

The following tables list the members exposed by IRobotCodeCmbCombs.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	Count (see page 165)	Number of combinations Available since version 1.7.

Public Methods

	Name	Description
	Get (see page 166)	Function returns the definition of selected combination. Available since version 1.7.
	New (see page 166)	Function creates a new combination and adds its definition at the end of the list. Index of the newly-created combination is returned. Available since version 1.7.
	Remove (see page 166)	Function deletes the definition of the selected combination. Available since version 1.7.
	Set (see page 166)	Function changes the definition of the selected combination. Available since version 1.7.

II.1.3.10.2 IRobotCodeCmbCombs Fields

The fields of the IRobotCodeCmbCombs class are listed here.

Public Fields

	Name	Description
	Count (see page 165)	Number of combinations Available since version 1.7.

II.1.3.10.2.1 Count

C++

```
HRESULT get_Count(long*);
```

C#

```
public long Count { get; }
```

Visual Basic

```
Public ReadOnly Count As long
```

Description

Number of combinations Available since version 1.7.

II.1.3.10.3 IRobotCodeCmbCombs Methods

The methods of the IRobotCodeCmbCombs class are listed here.

Public Methods

	Name	Description
	Get (see page 166)	Function returns the definition of selected combination. Available since version 1.7.
	New (see page 166)	Function creates a new combination and adds its definition at the end of the list. Index of the newly-created combination is returned. Available since version 1.7.
	Remove (see page 166)	Function deletes the definition of the selected combination. Available since version 1.7.
	Set (see page 166)	Function changes the definition of the selected combination. Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

II.1.3.10.3.1 Get

C++

```
HRESULT Get(long _idx, RobotLimitState* _limit_state, RobotCodeCmbCombType* _cmb_type, BSTR
_name, short* _dead_coef, short* _live_coef, short* _acc_coef, short* _seis_coef);
```

C#

```
public void Get(long _idx, RobotLimitState* _limit_state, RobotCodeCmbCombType* _cmb_type,
String _name, short* _dead_coef, short* _live_coef, short* _acc_coef, short* _seis_coef);
```

Visual Basic

```
Public Sub Get(_idx As long, ByRef _limit_state As RobotLimitState*, ByRef _cmb_type As
RobotCodeCmbCombType*, ByRef _name As String, ByRef _dead_coef As short*, ByRef _live_coef
As short*, ByRef _acc_coef As short*, ByRef _seis_coef As short*)
```

Description

Function returns the definition of selected combination. Available since version 1.7.

II.1.3.10.3.2 New

C++

```
HRESULT New(IRobotLimitState _lim_state, IRobotCodeCmbCombType _cmb_type, BSTR _name, short
_dead_coeff, short _live_coeff, short _acc_coeff, short _seis_coeff, long* ret);
```

C#

```
public long New(IRobotLimitState _lim_state, IRobotCodeCmbCombType _cmb_type, String _name,
short _dead_coeff, short _live_coeff, short _acc_coeff, short _seis_coeff);
```

Visual Basic

```
Public Function New(_lim_state As IRobotLimitState, _cmb_type As IRobotCodeCmbCombType,
_name As String, _dead_coeff As short, _live_coeff As short, _acc_coeff As short,
_seis_coeff As short) As long
```

Description

Function creates a new combination and adds its definition at the end of the list. Index of the newly-created combination is returned. Available since version 1.7.

II.1.3.10.3.3 Remove

C++

```
HRESULT Remove(long _idx);
```

C#

```
public void Remove(long _idx);
```

Visual Basic

```
Public Sub Remove(_idx As long)
```

Description

Function deletes the definition of the selected combination. Available since version 1.7.

II.1.3.10.3.4 Set

C++

```
HRESULT Set(long _idx, IRobotLimitState _limit_state, IRobotCodeCmbCombType _cmb_type, BSTR
_name, short _dead_coef, short _live_coef, short _acc_coef, short _seis_coef);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public void Set(long _idx, IRobotLimitState _limit_state, IRobotCodeCmbCombType _cmb_type,
String _name, short _dead_coef, short _live_coef, short _acc_coef, short _seis_coef);
```

Visual Basic

```
Public Sub Set(_idx As long, _limit_state As IRobotLimitState, _cmb_type As
IRobotCodeCmbCombType, _name As String, _dead_coef As short, _live_coef As short, _acc_coef
As short, _seis_coef As short)
```

Description

Function changes the definition of the selected combination. Available since version 1.7.

II.1.3.11 IRobotCodeCmbCombPartType

C++

```
enum IRobotCodeCmbCombPartType;
```

C#

```
public enum IRobotCodeCmbCombPartType;
```

Visual Basic

```
Public Enum IRobotCodeCmbCombPartType
```

Members

Members	Description
I_CCCPT_DEAD = 0	Available since version 1.7.
I_CCCPT_LIVE = 1	Available since version 1.7.
I_CCCPT_ACCIDENTAL = 2	Available since version 1.7.
I_CCCPT_SEISMIC = 3	Available since version 1.7.

II.1.3.12 IRobotCodeCmbCombType

C++

```
enum IRobotCodeCmbCombType;
```

C#

```
public enum IRobotCodeCmbCombType;
```

Visual Basic

```
Public Enum IRobotCodeCmbCombType
```

Members

Members	Description
I_CCCT_STANDARD = 1	Standard combination Available since version 1.7.
I_CCCT_SEISMIC = 2	Seismic combination Available since version 1.7.
I_CCCT_ACCIDENTAL = 3	Accidental combination Available since version 1.7.
I_CCCT_RARE = 4	Rare combination Available since version 1.7.
I_CCCT_FREQUENT = 5	Frequent combination Available since version 1.7.
I_CCCT_QUASI_PERM = 6	Quasi-permanent combination Available since version 1.7.
I_CCCT_USER = 7	User-defined combination Available since version 1.7.
I_CCCT_FUNDAMENTAL = 8	Basic combination Available since version 1.7.
I_CCCT_SIMPLIFIED = 9	Simplified combination Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

I_CCCT_EXTREMAL = 10	Extreme combination Available since version 1.7.

II.1.3.13 IRobotCodeCmbGenerationParams

Class Hierarchy

RobotOM.IRobotCodeCmbGenerationParams

C++

```
interface IRobotCodeCmbGenerationParams : IDispatch;
```

C#

```
public interface IRobotCodeCmbGenerationParams;
```

Visual Basic

```
Public Interface IRobotCodeCmbGenerationParams
```

Description

Parameters of code combination generation.

II.1.3.13.1 IRobotCodeCmbGenerationParams Members

The following tables list the members exposed by IRobotCodeCmbGenerationParams.

Public Fields

	Name	Description
	ActiveCases (see page 169)	Information about cases participating in a definition of code combinations (object collection of the RobotCodeCmbActiveCaseInfo type) Available since version 1.7.
	AllBars (see page 169)	Available since version 1.7.
	AllNodes (see page 170)	Available since version 1.7.
	BarSel (see page 170)	Bar selection (taken into account only for simplified generation when AllBars (see page 169) are set as zero - False) Available since version 1.7.
	ExtremalSnowFactor (see page 170)	Factor for extreme snow Available since version 1.7.
	GenType (see page 171)	Method of generation of code combinations.
	Groups (see page 171)	Groups Available since version 1.7.
	NodeSel (see page 171)	Node selection (taken into account only for simplified generation when AllNodes (see page 170) are set as zero - False) Available since version 1.7.
	PointsOnBar (see page 171)	Number of points along the bar length (it is taken into account only for simplified generation) Available since version 1.7.
	Regulations (see page 172)	Definition of the regulations Available since version 1.7.
	Relations (see page 172)	Relations between groups Available since version 1.7.

Public Methods

	Name	Description
	IsCombinationSelected (see page 173)	Function returns a value different from zero (True) if a specified template for generating combination is to be taken into account during generation. Available since version 1.7.
	IsCombinationTypeSelected (see page 173)	Function returns True if templates describing combination of the given type are used during combination generation.
	IsDecidingValueSelected (see page 173)	Function returns a value different from zero (True) if the specified decisive value has been selected. Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

	SelectCombination (see page 173)	Function enables selection of templates for calculating a combination among all the templates defined in the regulations. Available since version 1.7.
	SelectCombinationType (see page 174)	Function allows for a selection of templates describing combination of the given type from among all tamplates defined in the regulation.
	SelectDecidingValue (see page 174)	Function enables selection of decisive values. Available since version 1.7.

II.1.3.13.2 IRobotCodeCmbGenerationParams Fields

The fields of the IRobotCodeCmbGenerationParams class are listed here.

Public Fields

	Name	Description
	ActiveCases (see page 169)	Information about cases participating in a definition of code combinations (object collection of the RobotCodeCmbActiveCaseInfo type) Available since version 1.7.
	AllBars (see page 169)	Available since version 1.7.
	AllNodes (see page 170)	Available since version 1.7.
	BarSel (see page 170)	Bar selection (taken into account only for simplified generation when AllBars (see page 169) are set as zero - False) Available since version 1.7.
	ExtremalSnowFactor (see page 170)	Factor for extreme snow Available since version 1.7.
	GenType (see page 171)	Method of generation of code combinations.
	Groups (see page 171)	Groups Available since version 1.7.
	NodeSel (see page 171)	Node selection (taken into account only for simplified generation when AllNodes (see page 170) are set as zero - False) Available since version 1.7.
	PointsOnBar (see page 171)	Number of points along the bar length (it is taken into account only for simplified generation) Available since version 1.7.
	Regulations (see page 172)	Definition of the regulations Available since version 1.7.
	Relations (see page 172)	Relations between groups Available since version 1.7.

II.1.3.13.2.1 ActiveCases

C++

```
HRESULT get_ActiveCases(IRobotCollection**);
```

C#

```
public IRobotCollection ActiveCases { get; }
```

Visual Basic

```
Public ReadOnly ActiveCases As IRobotCollection
```

Description

Information about cases participating in a definition of code combinations (object collection of the RobotCodeCmbActiveCaseInfo type) Available since version 1.7.

II.1.3.13.2.2 AllBars

C++

```
HRESULT get_AllBars(VARIANT_BOOL*);
HRESULT put_AllBars(VARIANT_BOOL);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public bool AllBars { get; set; }
```

Visual Basic

```
Public AllBars As Boolean
```

Description

Available since version 1.7.

II.1.3.13.2.3 AllNodes

C++

```
HRESULT get_AllNodes(VARIANT_BOOL*);
HRESULT put_AllNodes(VARIANT_BOOL);
```

C#

```
public bool AllNodes { get; set; }
```

Visual Basic

```
Public AllNodes As Boolean
```

Description

Available since version 1.7.

II.1.3.13.2.4 BarSel

C++

```
HRESULT get_BarSel(BSTR*);
HRESULT put_BarSel(BSTR);
```

C#

```
public String BarSel { get; set; }
```

Visual Basic

```
Public BarSel As String
```

Description

Bar selection (taken into account only for simplified generation when AllBars (see page 169) are set as zero - False) Available since version 1.7.

II.1.3.13.2.5 ExtremalSnowFactor

C++

```
HRESULT get_ExtremalSnowFactor(double*);
HRESULT put_ExtremalSnowFactor(double);
```

C#

```
public double ExtremalSnowFactor { get; set; }
```

Visual Basic

```
Public ExtremalSnowFactor As double
```

Description

Factor for extreme snow Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

II.1.3.13.2.6 GenType

C++

```
HRESULT get_GenType(IRobotCodeCmbGenerationType*);
HRESULT put_GenType(IRobotCodeCmbGenerationType);
```

C#

```
public IRobotCodeCmbGenerationType GenType { get; set; }
```

Visual Basic

```
Public GenType As IRobotCodeCmbGenerationType
```

Description

Method of generation of code combinations.

Version

Available since version 8.2.

II.1.3.13.2.7 Groups

C++

```
HRESULT get_Groups(IRobotCodeCmbGroupServer**);
```

C#

```
public IRobotCodeCmbGroupServer Groups { get; }
```

Visual Basic

```
Public ReadOnly Groups As IRobotCodeCmbGroupServer
```

Description

Groups Available since version 1.7.

II.1.3.13.2.8 NodeSel

C++

```
HRESULT get_NodeSel(BSTR*);
HRESULT put_NodeSel(BSTR);
```

C#

```
public String NodeSel { get; set; }
```

Visual Basic

```
Public NodeSel As String
```

Description

Node selection (taken into account only for simplified generation when AllNodes (see page 170) are set as zero - False) Available since version 1.7.

II.1.3.13.2.9 PointsOnBar

C++

```
HRESULT get_PointsOnBar(short*);
HRESULT put_PointsOnBar(short);
```

C#

```
public short PointsOnBar { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public PointsOnBar As short
```

Description

Number of points along the bar length (it is taken into account only for simplified generation) Available since version 1.7.

II.1.3.13.2.10 Regulations

C++

```
HRESULT get_Regulations(IRobotCodeCmbRegulations**);
```

C#

```
public IRobotCodeCmbRegulations Regulations { get; }
```

Visual Basic

```
Public ReadOnly Regulations As IRobotCodeCmbRegulations
```

Description

Definition of the regulations Available since version 1.7.

II.1.3.13.2.11 Relations

C++

```
HRESULT get_Relations(IRobotCodeCmbGroupRelationServer**);
```

C#

```
public IRobotCodeCmbGroupRelationServer Relations { get; }
```

Visual Basic

```
Public ReadOnly Relations As IRobotCodeCmbGroupRelationServer
```

Description

Relations between groups Available since version 1.7.

II.1.3.13.3 IRobotCodeCmbGenerationParams Methods

The methods of the IRobotCodeCmbGenerationParams class are listed here.

Public Methods

	Name	Description
	IsCombinationSelected (see page 173)	Function returns a value different from zero (True) if a specified template for generating combination is to be taken into account during generation. Available since version 1.7.
	IsCombinationTypeSelected (see page 173)	Function returns True if templates describing combination of the given type are used during combination generation.
	IsDecidingValueSelected (see page 173)	Function returns a value different from zero (True) if the specified decisive value has been selected. Available since version 1.7.
	SelectCombination (see page 173)	Function enables selection of templates for calculating a combination among all the templates defined in the regulations. Available since version 1.7.
	SelectCombinationType (see page 174)	Function allows for a selection of templates describing combination of the given type from among all tamplates defined in the regulation.
	SelectDecidingValue (see page 174)	Function enables selection of decisive values. Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

II.1.3.13.3.1 IsCombinationSelected

C++

```
HRESULT IsCombinationSelected(long _cmb_idx, VARIANT_BOOL* ret);
```

C#

```
public bool IsCombinationSelected(long _cmb_idx);
```

Visual Basic

```
Public Function IsCombinationSelected(_cmb_idx As long) As Boolean
```

Description

Function returns a value different from zero (True) if a specified template for generating combination is to be taken into account during generation. Available since version 1.7.

II.1.3.13.3.2 IsCombinationTypeSelected

C++

```
HRESULT IsCombinationTypeSelected(IRobotCombinationType _cmb_type, VARIANT_BOOL* ret);
```

C#

```
public bool IsCombinationTypeSelected(IRobotCombinationType _cmb_type);
```

Visual Basic

```
Public Function IsCombinationTypeSelected(_cmb_type As IRobotCombinationType) As Boolean
```

Description

Function returns True if templates describing combination of the given type are used during combination generation.

Version

Available since version 11.

II.1.3.13.3.3 IsDecidingValueSelected

C++

```
HRESULT IsDecidingValueSelected(IRobotCodeCmbDecidingValueType _dvalue, VARIANT_BOOL* ret);
```

C#

```
public bool IsDecidingValueSelected(IRobotCodeCmbDecidingValueType _dvalue);
```

Visual Basic

```
Public Function IsDecidingValueSelected(_dvalue As IRobotCodeCmbDecidingValueType) As
Boolean
```

Description

Function returns a value different from zero (True) if the specified decisive value has been selected. Available since version 1.7.

II.1.3.13.3.4 SelectCombination

C++

```
HRESULT SelectCombination(long _cmb_idx, VARIANT_BOOL _select);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public void SelectCombination(long _cmb_idx, bool _select);
```

Visual Basic

```
Public Sub SelectCombination(_cmb_idx As long, _select As Boolean)
```

Description

Function enables selection of templates for calculating a combination among all the templates defined in the regulations. Available since version 1.7.

II.1.3.13.3.5 SelectCombinationType

C++

```
HRESULT SelectCombinationType(IRobotCombinationType _cmb_type, VARIANT_BOOL _select);
```

C#

```
public void SelectCombinationType(IRobotCombinationType _cmb_type, bool _select);
```

Visual Basic

```
Public Sub SelectCombinationType(_cmb_type As IRobotCombinationType, _select As Boolean)
```

Description

Function allows for a selection of templates describing combination of the given type from among all tamplates defined in the regulation.

Version

Available since version 11.

II.1.3.13.3.6 SelectDecidingValue

C++

```
HRESULT SelectDecidingValue(IRobotCodeCmbDecidingValueType _dvalue, VARIANT_BOOL _select);
```

C#

```
public void SelectDecidingValue(IRobotCodeCmbDecidingValueType _dvalue, bool _select);
```

Visual Basic

```
Public Sub SelectDecidingValue(_dvalue As IRobotCodeCmbDecidingValueType, _select As
Boolean)
```

Description

Function enables selection of decisive values. Available since version 1.7.

II.1.3.14 IRobotCodeCmbDecidingValueType

C++

```
enum IRobotCodeCmbDecidingValueType;
```

C#

```
public enum IRobotCodeCmbDecidingValueType;
```

Visual Basic

```
Public Enum IRobotCodeCmbDecidingValueType
```

(c) 2014 Autodesk Inc. All rights reserved

Members

Members	Description
I_CCDVT_FX = 1	Available since version 1.7.
I_CCDVT_FY = 2	Available since version 1.7.
I_CCDVT_FZ = 3	Available since version 1.7.
I_CCDVT_MX = 4	Available since version 1.7.
I_CCDVT_MY = 5	Available since version 1.7.
I_CCDVT_MZ = 6	Available since version 1.7.
I_CCDVT_UX = 7	Available since version 1.7.
I_CCDVT_UY = 8	Available since version 1.7.
I_CCDVT_UZ = 9	Available since version 1.7.
I_CCDVT_FXX_MAX = 10	Available since version 1.7.
I_CCDVT_MXX_MAX = 11	Available since version 1.7.
I_CCDVT_SXX_MAX = 12	Available since version 1.7.
I_CCDVT_REACTIONS = 35	Available since version 1.7.
I_CCDVT_SIGMA_X = 13	Available since version 1.7.
I_CCDVT_DEFLECTION = 14	Available since version 1.7.

II.1.3.15 IRobotCodeCmbGroup

Class Hierarchy

RobotOM.IRobotCodeCmbGroup

C++

```
interface IRobotCodeCmbGroup : IDispatch;
```

C#

```
public interface IRobotCodeCmbGroup;
```

Visual Basic

```
Public Interface IRobotCodeCmbGroup
```

Description

Definition of load case groups.

II.1.3.15.1 IRobotCodeCmbGroup Members

The following tables list the members exposed by IRobotCodeCmbGroup.

Public Fields

	Name	Description
	CaseCount (see page 176)	Number of load cases included in the group Available since version 1.7.
	Nature (see page 176)	Nature of cases included in the group Available since version 1.7.
	Operator (see page 176)	Operator determining relation between the group components Available since version 1.7.

Public Methods

	Name	Description
	Add (see page 177)	Function adds a case with the specified number to the group. If the case nature is incompatible with the group nature the function returns zero (False). Adding a case to this group means that it is simultaneously deleted from the group to which it has hitherto belonged. Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

	AddAll (see page 177)	Functions adds to the group all the load cases whose nature is compatible with the group nature. Simultaneously. it means that these cases will be deleted from the groups to which they have hitherto belonged. Available since version 1.7.
	IsFull (see page 178)	Function returns a value different from zero (True) if the natures of all the load cases in a group are compatible with the group nature. Available since version 1.7.

II.1.3.15.2 IRobotCodeCmbGroup Fields

The fields of the IRobotCodeCmbGroup class are listed here.

Public Fields

	Name	Description
	CaseCount (see page 176)	Number of load cases included in the group Available since version 1.7.
	Nature (see page 176)	Nature of cases included in the group Available since version 1.7.
	Operator (see page 176)	Operator determining relation between the group components Available since version 1.7.

II.1.3.15.2.1 CaseCount

C++

```
HRESULT get_CaseCount(long*);
```

C#

```
public long CaseCount { get; }
```

Visual Basic

```
Public ReadOnly CaseCount As long
```

Description

Number of load cases included in the group Available since version 1.7.

II.1.3.15.2.2 Nature

C++

```
HRESULT get_Nature(IRobotCaseNature*);
```

C#

```
public IRobotCaseNature Nature { get; }
```

Visual Basic

```
Public ReadOnly Nature As IRobotCaseNature
```

Description

Nature of cases included in the group Available since version 1.7.

II.1.3.15.2.3 Operator

C++

```
HRESULT get_Operator(IRobotCodeCmbOperator*);
HRESULT put_Operator(IRobotCodeCmbOperator);
```

C#

```
public IRobotCodeCmbOperator Operator { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Operator As IRobotCodeCmbOperator
```

Description

Operator determining relation between the group components Available since version 1.7.

II.1.3.15.3 IRobotCodeCmbGroup Methods

The methods of the IRobotCodeCmbGroup class are listed here.

Public Methods

	Name	Description
	Add (see page 177)	Function adds a case with the specified number to the group. If the case nature is incompatible with the group nature the function returns zero (False). Adding a case to this group means that it is simultaneously deleted from the group to which it has hitherto belonged. Available since version 1.7.
	AddAll (see page 177)	Functions adds to the group all the load cases whose nature is compatible with the group nature. Simultaneously. it means that these cases will be deleted from the groups to which they have hitherto belonged. Available since version 1.7.
	IsFull (see page 178)	Function returns a value different from zero (True) if the natures of all the load cases in a group are compatible with the group nature. Available since version 1.7.

II.1.3.15.3.1 Add

C++

```
HRESULT Add(long _case_num, VARIANT_BOOL* ret);
```

C#

```
public bool Add(long _case_num);
```

Visual Basic

```
Public Function Add(_case_num As long) As Boolean
```

Description

Function adds a case with the specified number to the group. If the case nature is incompatible with the group nature the function returns zero (False). Adding a case to this group means that it is simultaneously deleted from the group to which it has hitherto belonged. Available since version 1.7.

II.1.3.15.3.2 AddAll

C++

```
HRESULT AddAll();
```

C#

```
public void AddAll();
```

Visual Basic

```
Public Sub AddAll()
```

Description

Functions adds to the group all the load cases whose nature is compatible with the group nature. Simultaneously. it means that these cases will be deleted from the groups to which they have hitherto belonged. Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

II.1.3.15.3.3 IsFull

C++

```
HRESULT IsFull(VARIANT_BOOL* ret);
```

C#

```
public bool IsFull();
```

Visual Basic

```
Public Function IsFull() As Boolean
```

Description

Function returns a value different from zero (True) if the natures of all the load cases in a group are compatible with the group nature. Available since version 1.7.

II.1.3.16 IRobotCodeCmbOperator

C++

```
enum IRobotCodeCmbOperator;
```

C#

```
public enum IRobotCodeCmbOperator;
```

Visual Basic

```
Public Enum IRobotCodeCmbOperator
```

Members

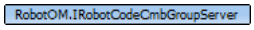
Members	Description
I_CCO_AND = 0	Operator: and Available since version 1.7.
I_CCO_EXCLUSIVE_OR = 1	Operator: exclusive or Available since version 1.7.
I_CCO_AND_OR = 2	Operator: and/or Available since version 1.7.

Description

Available operators used for defining relations between cases and case groups. .

II.1.3.17 IRobotCodeCmbGroupServer

Class Hierarchy



C++

```
interface IRobotCodeCmbGroupServer : IDispatch;
```

C#

```
public interface IRobotCodeCmbGroupServer;
```

Visual Basic

```
Public Interface IRobotCodeCmbGroupServer
```

Description

Server managing the groups of cases.

(c) 2014 Autodesk Inc. All rights reserved

II.1.3.17.1 IRobotCodeCmbGroupServer Members

The following tables list the members exposed by IRobotCodeCmbGroupServer.

Public Methods

	Name	Description
	FindByCase (see page 179)	Function finds the group which includes the specified load case and returns its number among the groups whose nature is compatible with the nature of that case. Available since version 1.7.
	Get (see page 179)	Function returns a group with the specified number among the groups of the defined nature. Groups are numbered within the nature from 1 to their number. Available since version 1.7.
	GetCount (see page 180)	Function returns the number of groups of the specified nature. Available since version 2.0.
	New (see page 180)	Function creates a new group with the defined parameters and returns its number. Groups are numbered within a given nature from 1 to Count. Available since version 1.7.

II.1.3.17.2 IRobotCodeCmbGroupServer Methods

The methods of the IRobotCodeCmbGroupServer class are listed here.

Public Methods

	Name	Description
	FindByCase (see page 179)	Function finds the group which includes the specified load case and returns its number among the groups whose nature is compatible with the nature of that case. Available since version 1.7.
	Get (see page 179)	Function returns a group with the specified number among the groups of the defined nature. Groups are numbered within the nature from 1 to their number. Available since version 1.7.
	GetCount (see page 180)	Function returns the number of groups of the specified nature. Available since version 2.0.
	New (see page 180)	Function creates a new group with the defined parameters and returns its number. Groups are numbered within a given nature from 1 to Count. Available since version 1.7.

II.1.3.17.2.1 FindByCase

C++

```
HRESULT FindByCase(long _case_num, long* ret);
```

C#

```
public long FindByCase(long _case_num);
```

Visual Basic

```
Public Function FindByCase(_case_num As long) As long
```

Description

Function finds the group which includes the specified load case and returns its number among the groups whose nature is compatible with the nature of that case. Available since version 1.7.

II.1.3.17.2.2 Get

C++

```
HRESULT Get(IRobotCaseNature _grp_nature, long _grp_number, IRobotCodeCmbGroup** ret);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRobotCodeCmbGroup Get(IRobotCaseNature _grp_nature, long _grp_number);
```

Visual Basic

```
Public Function Get(_grp_nature As IRobotCaseNature, _grp_number As long) As
IRobotCodeCmbGroup
```

Description

Function returns a group with the specified number among the groups of the defined nature. Groups are numbered within the nature from 1 to their number. Available since version 1.7.

II.1.3.17.2.3 GetCount

C++

```
HRESULT GetCount(IRobotCaseNature _grp_nature, long* ret);
```

C#

```
public long GetCount(IRobotCaseNature _grp_nature);
```

Visual Basic

```
Public Function GetCount(_grp_nature As IRobotCaseNature) As long
```

Description

Function returns the number of groups of the specified nature. Available since version 2.0.

II.1.3.17.2.4 New

C++

```
HRESULT New(IRobotCaseNature _grp_nature, IRobotCodeCmbOperator _oper, BSTR _case_list,
long* ret);
```

C#

```
public long New(IRobotCaseNature _grp_nature, IRobotCodeCmbOperator _oper, String
_case_list);
```

Visual Basic

```
Public Function New(_grp_nature As IRobotCaseNature, _oper As IRobotCodeCmbOperator,
_case_list As String) As long
```

Description

Function creates a new group with the defined parameters and returns its number. Groups are numbered within a given nature from 1 to Count. Available since version 1.7.

II.1.3.18 IRobotCodeCmbGroupRelation

Class Hierarchy

C++

```
interface IRobotCodeCmbGroupRelation : IDispatch;
```

C#

```
public interface IRobotCodeCmbGroupRelation;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Interface IRobotCodeCmbGroupRelation
```

Description

Definition of relations between groups. Definition of relations between groups can be illustrated by means of the matrix whose elements are group numbers. All groups whose numbers are located in the same matrix row are in the same relation with each other (either AND or AND_OR). Whereas among the matrix rows the relation EXCLUSIVE_OR occurs. .

II.1.3.18.1 IRobotCodeCmbGroupRelation Members

The following tables list the members exposed by IRobotCodeCmbGroupRelation.

Public Fields

	Name	Description
	Nature (see page 181)	Nature of groups defining relation Available since version 1.7.
	RowCount (see page 182)	Number of rows defining relation Available since version 1.7.

Public Methods

	Name	Description
	AddGroup (see page 182)	Function adds a specified number group to the indicated row (at the end of the row). Only the groups whose nature is compatible with the relation nature can be added to the relation. Available since version 1.7.
	Clear (see page 183)	Function deletes all the rows of the relation definition matrix. Available since version 1.7.
	GetGroup (see page 183)	Function returns the group number entered into the matrix into a specified position of the selected row. Available since version 1.7.
	GetGroupCount (see page 183)	Function returns the number of groups defining a selected matrix row. Matrix rows are numbered from 1 to RowCount (see page 182). Available since version 1.7.
	GetOperator (see page 184)	Function returns the operator for the indicated row of relation definition matrix. Available since version 1.7.
	NewRow (see page 184)	Function creates a new row of the relation definition matrix and returns its index. Available since version 1.7.
	RemoveRow (see page 184)	Function deletes a row with the indicated index from the relation definition matrix. The numbers of the remaining rows are updated automatically so that they constitute a sequence of numbers from 1 to RowCount (see page 182). Available since version 1.7.
	SetGroup (see page 184)	Function enters a specified number into the indicated position of the selected row. Only the groups whose nature is compatible with the relation nature can be saved in the relation. Available since version 1.7.
	SetOperator (see page 185)	Function sets the operator for the selected matrix row. Available since version 1.7.

II.1.3.18.2 IRobotCodeCmbGroupRelation Fields

The fields of the IRobotCodeCmbGroupRelation class are listed here.

Public Fields

	Name	Description
	Nature (see page 181)	Nature of groups defining relation Available since version 1.7.
	RowCount (see page 182)	Number of rows defining relation Available since version 1.7.

II.1.3.18.2.1 Nature

C++

```
HRESULT get_Nature(IRobotCaseNature*);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRobotCaseNature Nature { get; }
```

Visual Basic

```
Public ReadOnly Nature As IRobotCaseNature
```

Description

Nature of groups defining relation Available since version 1.7.

II.1.3.18.2.2 RowCount

C++

```
HRESULT get_RowCount(long*);
```

C#

```
public long RowCount { get; }
```

Visual Basic

```
Public ReadOnly RowCount As long
```

Description

Number of rows defining relation Available since version 1.7.

II.1.3.18.3 IRobotCodeCmbGroupRelation Methods

The methods of the IRobotCodeCmbGroupRelation class are listed here.

Public Methods

	Name	Description
	AddGroup (see page 182)	Function adds a specified number group to the indicated row (at the end of the row). Only the groups whose nature is compatible with the relation nature can be added to the relation. Available since version 1.7.
	Clear (see page 183)	Function deletes all the rows of the relation definition matrix. Available since version 1.7.
	GetGroup (see page 183)	Function returns the group number entered into the matrix into a specified position of the selected row. Available since version 1.7.
	GetGroupCount (see page 183)	Function returns the number of groups defining a selected matrix row. Matrix rows are numbered from 1 to RowCount (see page 182). Available since version 1.7.
	GetOperator (see page 184)	Function returns the operator for the indicated row of relation definition matrix. Available since version 1.7.
	NewRow (see page 184)	Function creates a new row of the relation definition matrix and returns its index. Available since version 1.7.
	RemoveRow (see page 184)	Function deletes a row with the indicated index from the relation definition matrix. The numbers of the remaining rows are updated automatically so that they constitute a sequence of numbers from 1 to RowCount (see page 182). Available since version 1.7.
	SetGroup (see page 184)	Function enters a specified number into the indicated position of the selected row. Only the groups whose nature is compatible with the relation nature can be saved in the relation. Available since version 1.7.
	SetOperator (see page 185)	Function sets the operator for the selected matrix row. Available since version 1.7.

II.1.3.18.3.1 AddGroup

C++

```
HRESULT AddGroup(long _row_idx, long _grp_number);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public void AddGroup(long _row_idx, long _grp_number);
```

Visual Basic

```
Public Sub AddGroup(_row_idx As long, _grp_number As long)
```

Description

Function adds a specified number group to the indicated row (at the end of the row). Only the groups whose nature is compatible with the relation nature can be added to the relation. Available since version 1.7.

II.1.3.18.3.2 Clear

C++

```
HRESULT Clear();
```

C#

```
public void Clear();
```

Visual Basic

```
Public Sub Clear()
```

Description

Function deletes all the rows of the relation definition matrix. Available since version 1.7.

II.1.3.18.3.3 GetGroup

C++

```
HRESULT GetGroup(long _row_idx, long _grp_idx, long* ret);
```

C#

```
public long GetGroup(long _row_idx, long _grp_idx);
```

Visual Basic

```
Public Function GetGroup(_row_idx As long, _grp_idx As long) As long
```

Description

Function returns the group number entered into the matrix into a specified position of the selected row. Available since version 1.7.

II.1.3.18.3.4 GetGroupCount

C++

```
HRESULT GetGroupCount(long _row_idx, long* ret);
```

C#

```
public long GetGroupCount(long _row_idx);
```

Visual Basic

```
Public Function GetGroupCount(_row_idx As long) As long
```

Description

Function returns the number of groups defining a selected matrix row. Matrix rows are numbered from 1 to RowCount (see page 182). Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

II.1.3.18.3.5 GetOperator

C++

```
HRESULT GetOperator(long _row_idx, IRobotCodeCmbOperator* ret);
```

C#

```
public IRobotCodeCmbOperator GetOperator(long _row_idx);
```

Visual Basic

```
Public Function GetOperator(_row_idx As long) As IRobotCodeCmbOperator
```

Description

Function returns the operator for the indicated row of relation definition matrix. Available since version 1.7.

II.1.3.18.3.6 NewRow

C++

```
HRESULT NewRow(IRobotCodeCmbOperator _oper_id, long* ret);
```

C#

```
public long NewRow(IRobotCodeCmbOperator _oper_id);
```

Visual Basic

```
Public Function NewRow(_oper_id As IRobotCodeCmbOperator) As long
```

Description

Function creates a new row of the relation definition matrix and returns its index. Available since version 1.7.

II.1.3.18.3.7 RemoveRow

C++

```
HRESULT RemoveRow(long _row_idx);
```

C#

```
public void RemoveRow(long _row_idx);
```

Visual Basic

```
Public Sub RemoveRow(_row_idx As long)
```

Description

Function deletes a row with the indicated index from the relation definition matrix. The numbers of the remaining rows are updated automatically so that they constitute a sequence of numbers from 1 to RowCount (see page 182). Available since version 1.7.

II.1.3.18.3.8 SetGroup

C++

```
HRESULT SetGroup(long _row_idx, long _grp_idx, long _grp_number);
```

C#

```
public void SetGroup(long _row_idx, long _grp_idx, long _grp_number);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Sub SetGroup(_row_idx As long, _grp_idx As long, _grp_number As long)
```

Description

Function enters a specified number into the indicated position of the selected row. Only the groups whose nature is compatible with the relation nature can be saved in the relation. Available since version 1.7.

II.1.3.18.3.9 SetOperator

C++

```
HRESULT SetOperator(long _row_idx, IRobotCodeCmbOperator _oper_id);
```

C#

```
public void SetOperator(long _row_idx, IRobotCodeCmbOperator _oper_id);
```

Visual Basic

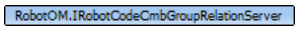
```
Public Sub SetOperator(_row_idx As long, _oper_id As IRobotCodeCmbOperator)
```

Description

Function sets the operator for the selected matrix row. Available since version 1.7.

II.1.3.19 IRobotCodeCmbGroupRelationServer

Class Hierarchy



C++

```
interface IRobotCodeCmbGroupRelationServer : IDispatch;
```

C#

```
public interface IRobotCodeCmbGroupRelationServer;
```

Visual Basic

```
Public Interface IRobotCodeCmbGroupRelationServer
```

Description

Server managing relations among groups. .

II.1.3.19.1 IRobotCodeCmbGroupRelationServer Members

The following tables list the members exposed by IRobotCodeCmbGroupRelationServer.

Public Methods

	Name	Description
	Get (see page 186)	Function returns the definition of the relation with the specified nature. Available since version 1.7.
	Set (see page 186)	Function saves a specified group relation definition. The previous group relation definition for the same nature will be replaced. Available since version 1.7.

II.1.3.19.2 IRobotCodeCmbGroupRelationServer Methods

The methods of the IRobotCodeCmbGroupRelationServer class are listed here.

(c) 2014 Autodesk Inc. All rights reserved

Public Methods

	Name	Description
	Get (see page 186)	Function returns the definition of the relation with the specified nature. Available since version 1.7.
	Set (see page 186)	Function saves a specified group relation definition. The previous group relation definition for the same nature will be replaced. Available since version 1.7.

II.1.3.19.2.1 Get

C++

```
HRESULT Get(IRobotCaseNature _rel_nature, IRobotCodeCmbGroupRelation** ret);
```

C#

```
public IRobotCodeCmbGroupRelation Get(IRobotCaseNature _rel_nature);
```

Visual Basic

```
Public Function Get(_rel_nature As IRobotCaseNature) As IRobotCodeCmbGroupRelation
```

Description

Function returns the definition of the relation with the specified nature. Available since version 1.7.

II.1.3.19.2.2 Set

C++

```
HRESULT Set(IRobotCodeCmbGroupRelation* _rel_def);
```

C#

```
public void Set(IRobotCodeCmbGroupRelation _rel_def);
```

Visual Basic

```
Public Sub Set(ByRef _rel_def As IRobotCodeCmbGroupRelation)
```

Description

Function saves a specified group relation definition. The previous group relation definition for the same nature will be replaced. Available since version 1.7.

II.1.3.20 IRobotCodeCmbActiveCaseInfo

Class Hierarchy

RobotOM.IRobotCodeCmbActiveCaseInfo

C++

```
interface IRobotCodeCmbActiveCaseInfo : IDispatch;
```

C#

```
public interface IRobotCodeCmbActiveCaseInfo;
```

Visual Basic

```
Public Interface IRobotCodeCmbActiveCaseInfo
```

II.1.3.20.1 IRobotCodeCmbActiveCaseInfo Members

The following tables list the members exposed by IRobotCodeCmbActiveCaseInfo.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	CaseNature (see page 187)	Nature of the load case Available since version 1.7.
	CaseNumber (see page 187)	Number of the load case Available since version 1.7.
	Coefficient (see page 187)	Factor Available since version 1.7.
	GroupNumber (see page 188)	Number of the group to which the case belongs Available since version 1.7.
	IsSelected (see page 188)	Flag indicating if the case will be taken into account while generating code combinations Available since version 1.7.

II.1.3.20.2 IRobotCodeCmbActiveCaseInfo Fields

The fields of the IRobotCodeCmbActiveCaseInfo class are listed here.

Public Fields

	Name	Description
	CaseNature (see page 187)	Nature of the load case Available since version 1.7.
	CaseNumber (see page 187)	Number of the load case Available since version 1.7.
	Coefficient (see page 187)	Factor Available since version 1.7.
	GroupNumber (see page 188)	Number of the group to which the case belongs Available since version 1.7.
	IsSelected (see page 188)	Flag indicating if the case will be taken into account while generating code combinations Available since version 1.7.

II.1.3.20.2.1 CaseNature

C++

```
HRESULT get_CaseNature(IRobotCaseNature*);
```

C#

```
public IRobotCaseNature CaseNature { get; }
```

Visual Basic

```
Public ReadOnly CaseNature As IRobotCaseNature
```

Description

Nature of the load case Available since version 1.7.

II.1.3.20.2.2 CaseNumber

C++

```
HRESULT get_CaseNumber(long*);
```

C#

```
public long CaseNumber { get; }
```

Visual Basic

```
Public ReadOnly CaseNumber As long
```

Description

Number of the load case Available since version 1.7.

II.1.3.20.2.3 Coefficient

C++

```
HRESULT get_Coefficient(double*);
```

(c) 2014 Autodesk Inc. All rights reserved

```
HRESULT put_Coefficient(double);
```

C#

```
public double Coefficient { get; set; }
```

Visual Basic

```
Public Coefficient As double
```

Description

Factor Available since version 1.7.

II.1.3.20.2.4 GroupNumber

C++

```
HRESULT get_GroupNumber(long*);
```

C#

```
public long GroupNumber { get; }
```

Visual Basic

```
Public ReadOnly GroupNumber As long
```

Description

Number of the group to which the case belongs Available since version 1.7.

II.1.3.20.2.5 IsSelected

C++

```
HRESULT get_IsSelected(VARIANT_BOOL*);
HRESULT put_IsSelected(VARIANT_BOOL);
```

C#

```
public bool IsSelected { get; set; }
```

Visual Basic

```
Public IsSelected As Boolean
```

Description

Flag indicating if the case will be taken into account while generating code combinations Available since version 1.7.

II.1.3.21 IRobotCodeCmbGenerationType

C++

```
enum IRobotCodeCmbGenerationType;
```

C#

```
public enum IRobotCodeCmbGenerationType;
```

Visual Basic

```
Public Enum IRobotCodeCmbGenerationType
```

Members

Members	Description
I_CCGT_NONE = 0	No generation of code combinations. Available since version 8.2.

(c) 2014 Autodesk Inc. All rights reserved

I_CCGT_SIMPLIFIED = 1	Generation of simplified code combinations. Available since version 8.2.
I_CCGT_FULL = 2	Generation of full code combinations. Available since version 8.2.
I_CCGT_MANUAL = 3	Generation of manual combinations. Available since version 11.

Version

Available since version 8.2.

II.1.4 Modal analysis parameters

Enumerations

	Name	Description
	IRobotModalAnalysisMode (see page 192)	Available types of modal analysis.
	IRobotModalAnalysisAlgorithm (see page 193)	Available methods of eigenproblem solution.
	IRobotModalAnalysisMassMatrixType (see page 193)	Available mass matrix types.
	IRobotModalAnalysisLimitType (see page 194)	Available quantities by means of which a limit value may be defined.

Interfaces

	Name	Description
	IRobotModalAnalysisParams (see page 194)	Parameters of modal analysis.
	IRobotModalAnalysisBase (see page 201)	Reduced base definition.
	IRobotModalAnalysisLimits (see page 203)	Definition of a limit range.
	IRobotModalAnalysisShifts (see page 205)	Shift definition.
	IRobotMassEccentricities (see page 206)	Definition of eccentricities.

II.1.4.1 Modal analysis parameters recognizing static forces

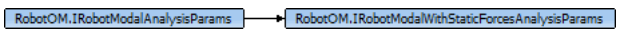
Available since version 2.5.

Interfaces

	Name	Description
	IRobotModalWithStaticForcesAnalysisParams (see page 189)	Modal analysis parameters with static forces taken into account.

II.1.4.1.1 IRobotModalWithStaticForcesAnalysisParams

Class Hierarchy



C++

```
interface IRobotModalWithStaticForcesAnalysisParams : IRobotModalAnalysisParams;
```

C#

```
public interface IRobotModalWithStaticForcesAnalysisParams : IRobotModalAnalysisParams;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Interface IRobotModalWithStaticForcesAnalysisParams
```

Description

Modal analysis parameters with static forces taken into account.

Version

Available since version 2.5.

II.1.4.1.1.1 IRobotModalWithStaticForcesAnalysisParams Members

The following tables list the members exposed by IRobotModalWithStaticForcesAnalysisParams.

Public Fields

	Name	Description
	Acceleration (see page 195)	Gravity value.
	Base (see page 196)	Reduced base definition.
	Damping (see page 196)	Damping value.
	DisregardDensity (see page 196)	Flag indicating if density of the structure element is to be taken into account during analysis .
	IncludeDampingInCalculations (see page 197)	Flag indicating if damping is to be taken into account in calculations.
	IterationsCount (see page 197)	Number of iterations.
	Limits (see page 197)	Limitation of the calculated mode number.
	MassEccentricities (see page 198)	Definition of eccentricities.
	MassMatrix (see page 198)	Mass matrix.
	MassParticipation (see page 198)	Definition of the value of mass participation percentage.
	Method (see page 199)	Method of eigenproblem solution
	Mode (see page 199)	Analysis mode.
	ModesCount (see page 199)	Number of modes.
	Nonlinear (see page 191)	Non-linear analysis.
	Shifts (see page 200)	Shift definition.
	SturmVerification (see page 200)	Switching on or off Sturm Check .
	Tolerance (see page 200)	Calculation tolerance value.

Public Methods

	Name	Description
	CreateFromStatic (see page 191)	Function initializes parameters of modal analysis with static forces taken into account based on the specified static load case. .
	GetStaticStateParams (see page 191)	Function returns parameters of non-linear analysis.
	SetStaticStateParams (see page 192)	Function sets parameters of non-linear analysis.

II.1.4.1.1.2 IRobotModalWithStaticForcesAnalysisParams Fields

The fields of the IRobotModalWithStaticForcesAnalysisParams class are listed here.

Public Fields

	Name	Description
	Nonlinear (see page 191)	Non-linear analysis.

(c) 2014 Autodesk Inc. All rights reserved

II.1.4.1.1.2.1 Nonlinear

C++

```
HRESULT get_Nonlinear(VARIANT_BOOL*);
HRESULT put_Nonlinear(VARIANT_BOOL);
```

C#

```
public bool Nonlinear { get; set; }
```

Visual Basic

```
Public Nonlinear As Boolean
```

Description

Non-linear analysis.

Version

Available since version 2.5.

II.1.4.1.1.3 IRobotModalWithStaticForcesAnalysisParams Methods

The methods of the IRobotModalWithStaticForcesAnalysisParams class are listed here.

Public Methods

	Name	Description
	CreateFromStatic (see page 191)	Function initializes parameters of modal analysis with static forces taken into account based on the specified static load case. .
	GetStaticStateParams (see page 191)	Function returns parameters of non-linear analysis.
	SetStaticStateParams (see page 192)	Function sets parameters of non-linear analysis.

II.1.4.1.1.3.1 CreateFromStatic

C++

```
HRESULT CreateFromStatic(long _static_case_num, VARIANT_BOOL* ret);
```

C#

```
public bool CreateFromStatic(long _static_case_num);
```

Visual Basic

```
Public Function CreateFromStatic(_static_case_num As long) As Boolean
```

Description

Function initializes parameters of modal analysis with static forces taken into account based on the specified static load case. .

Version

Available since version 2.5.

II.1.4.1.1.3.2 GetStaticStateParams

C++

```
HRESULT GetStaticStateParams(IRobotNonlinearAnalysisParams** ret);
```

C#

```
public IRobotNonlinearAnalysisParams GetStaticStateParams();
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Function GetStaticStateParams() As IRobotNonlinearAnalysisParams
```

Description

Function returns parameters of non-linear analysis.

Version

Available since version 2.5.

II.1.4.1.1.3.3 SetStaticStateParams

C++

```
HRESULT SetStaticStateParams(IRobotNonlinearAnalysisParams* _params);
```

C#

```
public void SetStaticStateParams(IRobotNonlinearAnalysisParams _params);
```

Visual Basic

```
Public Sub SetStaticStateParams(ByRef _params As IRobotNonlinearAnalysisParams)
```

Description

Function sets parameters of non-linear analysis.

Version

Available since version 2.5.

II.1.4.2 IRobotModalAnalysisMode

C++

```
enum IRobotModalAnalysisMode;
```

C#

```
public enum IRobotModalAnalysisMode;
```

Visual Basic

```
Public Enum IRobotModalAnalysisMode
```

Members

Members	Description
I_MAM_MODAL = 0	Modal analysis. Available since version 2.5.
I_MAM_SEISMIC = 1	Seismic analysis. Available since version 2.5.
I_MAM_SEISMIC_PSEUDO = 2	Pseudomodal seismic analysis. Available since version 2.5.

Description

Available types of modal analysis.

Version

Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

II.1.4.3 IRobotModalAnalysisAlgorithm

C++

```
enum IRobotModalAnalysisAlgorithm;
```

C#

```
public enum IRobotModalAnalysisAlgorithm;
```

Visual Basic

```
Public Enum IRobotModalAnalysisAlgorithm
```

Members

Members	Description
I_MAA_BLOCK_SUBSPACE_ITERATION = 0	Block subspace iteration. Available since version 2.5.
I_MAA_SUBSPACE_ITERATION = 1	Subspace iteration. Available since version 2.5.
I_MAA_LANCZOS = 2	Lanczos method. Available since version 2.5.
I_MAA_BASE_REDUCTION = 3	Base reduction method. Available since version 2.5.
I_MAA_PCG_RITZ = 4	Ritz method. Available since version 2.5.
I_MAA_PCG = 5	Preconditioned conjugate gradient method . Available since version 2.5.

Description

Available methods of eigenproblem solution.

Version

Available since version 2.5.

II.1.4.4 IRobotModalAnalysisMassMatrixType

C++

```
enum IRobotModalAnalysisMassMatrixType;
```

C#

```
public enum IRobotModalAnalysisMassMatrixType;
```

Visual Basic

```
Public Enum IRobotModalAnalysisMassMatrixType
```

Members

Members	Description
I_MAMMT_CONSISTENT = 0	Consistent mass matrix. Available since version 2.5.
I_MAMMT_LUMPED_WITH_ROTATIONS = 1	Lumped mass matrix with rotations. Available since version 2.5.
I_MAMMT_LUMPED = 2	Lumped mass matrix without rotations. Available since version 2.5.

Description

Available mass matrix types.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 2.5.

II.1.4.5 IRobotModalAnalysisLimitType

C++

```
enum IRobotModalAnalysisLimitType;
```

C#

```
public enum IRobotModalAnalysisLimitType;
```

Visual Basic

```
Public Enum IRobotModalAnalysisLimitType
```

Members

Members	Description
I_MALT_PULSATION = 0	Pulsation. Available since version 2.5.
I_MALT_FREQUENCY = 1	Frequency. Available since version 2.5.
I_MALT_PERIOD = 2	Period. Available since version 2.5.

Description

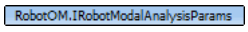
Available quantities by means of which a limit value may be defined.

Version

Available since version 2.5.

II.1.4.6 IRobotModalAnalysisParams

Class Hierarchy



C++

```
interface IRobotModalAnalysisParams : IDispatch;
```

C#

```
public interface IRobotModalAnalysisParams;
```

Visual Basic

```
Public Interface IRobotModalAnalysisParams
```

Description

Parameters of modal analysis.

Version

Available since version 2.5.

II.1.4.6.1 IRobotModalAnalysisParams Members

The following tables list the members exposed by IRobotModalAnalysisParams.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	Acceleration (see page 195)	Gravity value.
	Base (see page 196)	Reduced base definition.
	Damping (see page 196)	Damping value.
	DisregardDensity (see page 196)	Flag indicating if density of the structure element is to be taken into account during analysis .
	IncludeDampingInCalculations (see page 197)	Flag indicating if damping is to be taken into account in calculations.
	IterationsCount (see page 197)	Number of iterations.
	Limits (see page 197)	Limitation of the calculated mode number.
	MassEccentricities (see page 198)	Definition of eccentricities.
	MassMatrix (see page 198)	Mass matrix.
	MassParticipation (see page 198)	Definition of the value of mass participation percentage.
	Method (see page 199)	Method of eigenproblem solution
	Mode (see page 199)	Analysis mode.
	ModesCount (see page 199)	Number of modes.
	Shifts (see page 200)	Shift definition.
	SturmVerification (see page 200)	Switching on or off Sturm Check .
	Tolerance (see page 200)	Calculation tolerance value.

II.1.4.6.2 IRobotModalAnalysisParams Fields

The fields of the IRobotModalAnalysisParams class are listed here.

Public Fields

	Name	Description
	Acceleration (see page 195)	Gravity value.
	Base (see page 196)	Reduced base definition.
	Damping (see page 196)	Damping value.
	DisregardDensity (see page 196)	Flag indicating if density of the structure element is to be taken into account during analysis .
	IncludeDampingInCalculations (see page 197)	Flag indicating if damping is to be taken into account in calculations.
	IterationsCount (see page 197)	Number of iterations.
	Limits (see page 197)	Limitation of the calculated mode number.
	MassEccentricities (see page 198)	Definition of eccentricities.
	MassMatrix (see page 198)	Mass matrix.
	MassParticipation (see page 198)	Definition of the value of mass participation percentage.
	Method (see page 199)	Method of eigenproblem solution
	Mode (see page 199)	Analysis mode.
	ModesCount (see page 199)	Number of modes.
	Shifts (see page 200)	Shift definition.
	SturmVerification (see page 200)	Switching on or off Sturm Check .
	Tolerance (see page 200)	Calculation tolerance value.

II.1.4.6.2.1 Acceleration

C++

```
HRESULT get_Acceleration(double*);
```

(c) 2014 Autodesk Inc. All rights reserved

```
HRESULT put_Acceleration(double);
```

C#

```
public double Acceleration { get; set; }
```

Visual Basic

```
Public Acceleration As double
```

Description

Gravity value.

Version

Available since version 2.5.

II.1.4.6.2.2 Base

C++

```
HRESULT get_Base(IRobotModalAnalysisBase**);
```

C#

```
public IRobotModalAnalysisBase Base { get; }
```

Visual Basic

```
Public ReadOnly Base As IRobotModalAnalysisBase
```

Description

Reduced base definition.

Version

Available since version 2.5.

II.1.4.6.2.3 Damping

C++

```
HRESULT get_Damping(double*);
HRESULT put_Damping(double);
```

C#

```
public double Damping { get; set; }
```

Visual Basic

```
Public Damping As double
```

Description

Damping value.

Version

Available since version 2.5.

II.1.4.6.2.4 DisregardDensity

C++

```
HRESULT get_DisregardDensity(VARIANT_BOOL*);
HRESULT put_DisregardDensity(VARIANT_BOOL);
```

C#

```
public bool DisregardDensity { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public DisregardDensity As Boolean
```

Description

Flag indicating if density of the structure element is to be taken into account during analysis .

Version

Available since version 2.5.

II.1.4.6.2.5 IncludeDampingInCalculations

C++

```
HRESULT get_IncludeDampingInCalculations(VARIANT_BOOL*);
HRESULT put_IncludeDampingInCalculations(VARIANT_BOOL);
```

C#

```
public bool IncludeDampingInCalculations { get; set; }
```

Visual Basic

```
Public IncludeDampingInCalculations As Boolean
```

Description

Flag indicating if damping is to be taken into account in calculations.

Version

Available since version 2.5.

II.1.4.6.2.6 IterationsCount

C++

```
HRESULT get_IterationsCount(long*);
HRESULT put_IterationsCount(long);
```

C#

```
public long IterationsCount { get; set; }
```

Visual Basic

```
Public IterationsCount As long
```

Description

Number of iterations.

Version

Available since version 2.5.

II.1.4.6.2.7 Limits

C++

```
HRESULT get_Limits(IRobotModalAnalysisLimits**);
```

C#

```
public IRobotModalAnalysisLimits Limits { get; }
```

Visual Basic

```
Public ReadOnly Limits As IRobotModalAnalysisLimits
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Limitation of the calculated mode number.

Version

Available since version 2.5.

II.1.4.6.2.8 MassEccentricities

C++

```
HRESULT get_MassEccentricities(IRobotMassEccentricities**);
```

C#

```
public IRobotMassEccentricities MassEccentricities { get; }
```

Visual Basic

```
Public ReadOnly MassEccentricities As IRobotMassEccentricities
```

Description

Definition of eccentricities.

Version

Available since version 6.21.

II.1.4.6.2.9 MassMatrix

C++

```
HRESULT get_MassMatrix(IRobotModalAnalysisMassMatrixType*);
HRESULT put_MassMatrix(IRobotModalAnalysisMassMatrixType);
```

C#

```
public IRobotModalAnalysisMassMatrixType MassMatrix { get; set; }
```

Visual Basic

```
Public MassMatrix As IRobotModalAnalysisMassMatrixType
```

Description

Mass matrix.

Version

Available since version 2.5.

II.1.4.6.2.10 MassParticipation

C++

```
HRESULT get_MassParticipation(double*);
HRESULT put_MassParticipation(double);
```

C#

```
public double MassParticipation { get; set; }
```

Visual Basic

```
Public MassParticipation As double
```

Description

Definition of the value of mass participation percentage.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 2.5.

II.1.4.6.2.11 Method

C++

```
HRESULT get_Method(IRobotModalAnalysisAlgorithm*);
HRESULT put_Method(IRobotModalAnalysisAlgorithm);
```

C#

```
public IRobotModalAnalysisAlgorithm Method { get; set; }
```

Visual Basic

```
Public Method As IRobotModalAnalysisAlgorithm
```

Description

Method of eigenproblem solution

Notes

The selected algorithm of eigenproblem solution must be compatible with the active solver. .

Version

Available since version 2.5.

II.1.4.6.2.12 Mode

C++

```
HRESULT get_Mode(IRobotModalAnalysisMode*);
HRESULT put_Mode(IRobotModalAnalysisMode);
```

C#

```
public IRobotModalAnalysisMode Mode { get; set; }
```

Visual Basic

```
Public Mode As IRobotModalAnalysisMode
```

Description

Analysis mode.

Version

Available since version 2.5.

II.1.4.6.2.13 ModesCount

C++

```
HRESULT get_ModesCount(long*);
HRESULT put_ModesCount(long);
```

C#

```
public long ModesCount { get; set; }
```

Visual Basic

```
Public ModesCount As long
```

Description

Number of modes.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 2.5.

II.1.4.6.2.14 Shifts

C++

```
HRESULT get_Shifts(IRobotModalAnalysisShifts**);
```

C#

```
public IRobotModalAnalysisShifts Shifts { get; }
```

Visual Basic

```
Public ReadOnly Shifts As IRobotModalAnalysisShifts
```

Description

Shift definition.

Version

Available since version 2.5.

II.1.4.6.2.15 SturmVerification

C++

```
HRESULT get_SturmVerification(VARIANT_BOOL*);
HRESULT put_SturmVerification(VARIANT_BOOL);
```

C#

```
public bool SturmVerification { get; set; }
```

Visual Basic

```
Public SturmVerification As Boolean
```

Description

Switching on or off Sturm Check .

Version

Available since version 2.5.

II.1.4.6.2.16 Tolerance

C++

```
HRESULT get_Tolerance(double*);
HRESULT put_Tolerance(double);
```

C#

```
public double Tolerance { get; set; }
```

Visual Basic

```
Public Tolerance As double
```

Description

Calculation tolerance value.

Version

Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

II.1.4.7 IRobotModalAnalysisBase

Class Hierarchy

C++

```
interface IRobotModalAnalysisBase : IDispatch;
```

C#

```
public interface IRobotModalAnalysisBase;
```

Visual Basic

```
Public Interface IRobotModalAnalysisBase
```

Description

Reduced base definition.

Version

Available since version 2.5.

II.1.4.7.1 IRobotModalAnalysisBase Members

The following tables list the members exposed by IRobotModalAnalysisBase.

Public Methods

	Name	Description
	Add (see page 201)	Function generates a new element defining the base and returns its index. .
	Count (see page 202)	Function returns the number of elements defining the base. Elements defining the base are indexed from 1 to Count.
	Delete (see page 202)	Function deletes the indicated element defining the base.
	Get (see page 202)	Function returns the selected element defining the base. .

II.1.4.7.2 IRobotModalAnalysisBase Methods

The methods of the IRobotModalAnalysisBase class are listed here.

Public Methods

	Name	Description
	Add (see page 201)	Function generates a new element defining the base and returns its index. .
	Count (see page 202)	Function returns the number of elements defining the base. Elements defining the base are indexed from 1 to Count.
	Delete (see page 202)	Function deletes the indicated element defining the base.
	Get (see page 202)	Function returns the selected element defining the base. .

II.1.4.7.2.1 Add

C++

```
HRESULT Add(BSTR _nodes, VARIANT_BOOL _ux, VARIANT_BOOL _uy, VARIANT_BOOL _uz, long* ret);
```

C#

```
public long Add(String _nodes, bool _ux, bool _uy, bool _uz);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Function Add(_nodes As String, _ux As Boolean, _uy As Boolean, _uz As Boolean) As
long
```

Description

Function generates a new element defining the base and returns its index. .

Version

Available since version 2.5.

II.1.4.7.2.2 Count

C++

```
HRESULT Count(long* ret);
```

C#

```
public long Count();
```

Visual Basic

```
Public Function Count() As long
```

Description

Function returns the number of elements defining the base. Elements defining the base are indexed from 1 to Count.

Version

Available since version 2.5.

II.1.4.7.2.3 Delete

C++

```
HRESULT Delete(long _idx, VARIANT_BOOL* ret);
```

C#

```
public bool Delete(long _idx);
```

Visual Basic

```
Public Function Delete(_idx As long) As Boolean
```

Description

Function deletes the indicated element defining the base.

Version

Available since version 2.5.

II.1.4.7.2.4 Get

C++

```
HRESULT Get(long _idx, BSTR* _nodes, long* _ux, long* _uy, long* _uz);
```

C#

```
public void Get(long _idx, BSTR* _nodes, long* _ux, long* _uy, long* _uz);
```

Visual Basic

```
Public Sub Get(_idx As long, ByRef _nodes As BSTR*, ByRef _ux As long*, ByRef _uy As long*,
ByRef _uz As long*)
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Function returns the selected element defining the base. .

Version

Available since version 2.5.

II.1.4.8 IRobotModalAnalysisLimits

Class Hierarchy

C++

```
interface IRobotModalAnalysisLimits : IDispatch;
```

C#

```
public interface IRobotModalAnalysisLimits;
```

Visual Basic

```
Public Interface IRobotModalAnalysisLimits
```

Description

Definition of a limit range.

Version

Available since version 2.5.

II.1.4.8.1 IRobotModalAnalysisLimits Members

The following tables list the members exposed by IRobotModalAnalysisLimits.

Public Fields

	Name	Description
	FrequencyLimitValue (see page 204)	Limit frequency value.
	PeriodLimitValue (see page 204)	Limit period value.
	PulsationLimitValue (see page 204)	Limit pulsation value.

Public Methods

	Name	Description
	DefineLimits (see page 205)	Function defines the limit range based on the specified parameters. .

II.1.4.8.2 IRobotModalAnalysisLimits Fields

The fields of the IRobotModalAnalysisLimits class are listed here.

Public Fields

	Name	Description
	FrequencyLimitValue (see page 204)	Limit frequency value.
	PeriodLimitValue (see page 204)	Limit period value.
	PulsationLimitValue (see page 204)	Limit pulsation value.

(c) 2014 Autodesk Inc. All rights reserved

II.1.4.8.2.1 FrequencyLimitValue

C++

```
HRESULT get_FrequencyLimitValue(double*);
```

C#

```
public double FrequencyLimitValue { get; }
```

Visual Basic

```
Public ReadOnly FrequencyLimitValue As double
```

Description

Limit frequency value.

Version

Available since version 2.5.

II.1.4.8.2.2 PeriodLimitValue

C++

```
HRESULT get_PeriodLimitValue(double*);
```

C#

```
public double PeriodLimitValue { get; }
```

Visual Basic

```
Public ReadOnly PeriodLimitValue As double
```

Description

Limit period value.

Version

Available since version 2.5.

II.1.4.8.2.3 PulsationLimitValue

C++

```
HRESULT get_PulsationLimitValue(double*);
```

C#

```
public double PulsationLimitValue { get; }
```

Visual Basic

```
Public ReadOnly PulsationLimitValue As double
```

Description

Limit pulsation value.

Version

Available since version 2.5.

II.1.4.8.3 IRobotModalAnalysisLimits Methods

The methods of the IRobotModalAnalysisLimits class are listed here.

(c) 2014 Autodesk Inc. All rights reserved

Public Methods

	Name	Description
	DefineLimits (see page 205)	Function defines the limit range based on the specified parameters. .

II.1.4.8.3.1 DefineLimits

C++

```
HRESULT DefineLimits(IRobotModalAnalysisLimitType _limit_type, double _value);
```

C#

```
public void DefineLimits(IRobotModalAnalysisLimitType _limit_type, double _value);
```

Visual Basic

```
Public Sub DefineLimits(_limit_type As IRobotModalAnalysisLimitType, _value As double)
```

Description

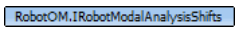
Function defines the limit range based on the specified parameters. .

Version

Available since version 2.5.

II.1.4.9 IRobotModalAnalysisShifts

Class Hierarchy



C++

```
interface IRobotModalAnalysisShifts : IDispatch;
```

C#

```
public interface IRobotModalAnalysisShifts;
```

Visual Basic

```
Public Interface IRobotModalAnalysisShifts
```

Description

Shift definition.

Version

Available since version 2.5.

II.1.4.9.1 IRobotModalAnalysisShifts Members

The following tables list the members exposed by IRobotModalAnalysisShifts.

Public Fields

	Name	Description
	IterationsCount (see page 206)	Number of iterations between shifts.

Public Methods

	Name	Description
	SetDefault (see page 206)	Function sets the default value for the number of iterations between shifts.

(c) 2014 Autodesk Inc. All rights reserved

II.1.4.9.2 IRobotModalAnalysisShifts Fields

The fields of the IRobotModalAnalysisShifts class are listed here.

Public Fields

	Name	Description
	IterationsCount (see page 206)	Number of iterations between shifts.

II.1.4.9.2.1 IterationsCount

C++

```
HRESULT get_IterationsCount(long*);
HRESULT put_IterationsCount(long);
```

C#

```
public long IterationsCount { get; set; }
```

Visual Basic

```
Public IterationsCount As long
```

Description

Number of iterations between shifts.

Version

Available since version 2.5.

II.1.4.9.3 IRobotModalAnalysisShifts Methods

The methods of the IRobotModalAnalysisShifts class are listed here.

Public Methods

	Name	Description
	SetDefault (see page 206)	Function sets the default value for the number of iterations between shifts.

II.1.4.9.3.1 SetDefault

C++

```
HRESULT SetDefault();
```

C#

```
public void SetDefault();
```

Visual Basic

```
Public Sub SetDefault()
```

Description

Function sets the default value for the number of iterations between shifts.

Version

Available since version 2.5.

II.1.4.10 IRobotMassEccentricities

Class Hierarchy

RobotOM.IRobotMassEccentricities

(c) 2014 Autodesk Inc. All rights reserved

C++

```
interface IRobotMassEccentricities : IDispatch;
```

C#

```
public interface IRobotMassEccentricities;
```

Visual Basic

```
Public Interface IRobotMassEccentricities
```

Description

Definition of eccentricities.

Version

Available since version 6.21.

II.1.4.10.1 IRobotMassEccentricities Members

The following tables list the members exposed by IRobotMassEccentricities.

Public Fields

	Name	Description
	IsDirX (see page 207)	Flag indicating if the value for the X direction is defined.
	IsDirY (see page 208)	Flag indicating if the value for the Y direction is defined.
	RelativeValues (see page 208)	Flag indicating if the specified values are relative or absolute.
	ValueDirX (see page 208)	Value for the X direction.
	ValueDirY (see page 209)	Value for the Y direction.

II.1.4.10.2 IRobotMassEccentricities Fields

The fields of the IRobotMassEccentricities class are listed here.

Public Fields

	Name	Description
	IsDirX (see page 207)	Flag indicating if the value for the X direction is defined.
	IsDirY (see page 208)	Flag indicating if the value for the Y direction is defined.
	RelativeValues (see page 208)	Flag indicating if the specified values are relative or absolute.
	ValueDirX (see page 208)	Value for the X direction.
	ValueDirY (see page 209)	Value for the Y direction.

II.1.4.10.2.1 IsDirX

C++

```
HRESULT get_IsDirX(VARIANT_BOOL*);
HRESULT put_IsDirX(VARIANT_BOOL);
```

C#

```
public bool IsDirX { get; set; }
```

Visual Basic

```
Public IsDirX As Boolean
```

Description

Flag indicating if the value for the X direction is defined.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 6.21.

II.1.4.10.2.2 IsDirY

C++

```
HRESULT get_IsDirY(VARIANT_BOOL*);
HRESULT put_IsDirY(VARIANT_BOOL);
```

C#

```
public bool IsDirY { get; set; }
```

Visual Basic

```
Public IsDirY As Boolean
```

Description

Flag indicating if the value for the Y direction is defined.

Version

Available since version 6.21.

II.1.4.10.2.3 RelativeValues

C++

```
HRESULT get_RelativeValues(VARIANT_BOOL*);
HRESULT put_RelativeValues(VARIANT_BOOL);
```

C#

```
public bool RelativeValues { get; set; }
```

Visual Basic

```
Public RelativeValues As Boolean
```

Description

Flag indicating if the specified values are relative or absolute.

Version

Available since version 6.21.

II.1.4.10.2.4 ValueDirX

C++

```
HRESULT get_ValueDirX(double*);
HRESULT put_ValueDirX(double);
```

C#

```
public double ValueDirX { get; set; }
```

Visual Basic

```
Public ValueDirX As double
```

Description

Value for the X direction.

Version

Available since version 6.21.

(c) 2014 Autodesk Inc. All rights reserved

II.1.4.10.2.5 ValueDirY

C++

```
HRESULT get_ValueDirY(double*);
HRESULT put_ValueDirY(double);
```

C#

```
public double ValueDirY { get; set; }
```

Visual Basic

```
Public ValueDirY As double
```

Description

Value for the Y direction.

Version

Available since version 6.21.

II.1.5 Seismic analysis parameters

Available since version 2.5.

Enumerations

	Name	Description
	IRobotSeismicAnalysis_AFPS_90_ZoneType (see page 253)	Available numbers of seismic zones.
	IRobotSeismicAnalysis_CIRSOC_103_ZoneType (see page 253)	Available numbers of seismic zones.
	IRobotSeismicAnalysis_P_100_92_ZoneType (see page 254)	Available numbers of seismic zones.
	IRobotSeismicAnalysis_PS_92_ZoneType (see page 254)	Available numbers of seismic zones.
	IRobotSeismicAnalysis_RPA_88_ZoneType (see page 255)	Available numbers of seismic zones.
	IRobotSeismicAnalysis_UBC_97_ZoneType (see page 255)	Available numbers of seismic zones.
	IRobotSeismicAnalysis_EAK_2000_ZoneType (see page 256)	Available numbers of seismic zones.
	IRobotSeismicAnalysis_TURKISH_23098_ZoneType (see page 256)	Available numbers of seismic zones.
	IRobotSeismicAnalysis_AFPS_90_StructureType (see page 256)	Available structure types.
	IRobotSeismicAnalysis_AFPS_90_SiteType (see page 257)	Available types of structure site.
	IRobotSeismicAnalysisSpectrumType (see page 257)	Available spectrum types. .
	IRobotSeismicAnalysis_CHINESE_StructureType (see page 258)	Available structure types.
	IRobotSeismicAnalysis_CHINESE_SiteType (see page 258)	Available types of structure site.
	IRobotSeismicAnalysis_CHINESE_IntensityType (see page 259)	Available intensity types.
	IRobotSeismicAnalysis_CHINESE_DesignType (see page 259)	Available design standards.
	IRobotSeismicAnalysis_CHINESE_EarthquakeType (see page 260)	Available types of earthquake distance.

(c) 2014 Autodesk Inc. All rights reserved

	IRobotSeismicAnalysis_CIRSOC_103_SoilType (see page 260)	Available ground types. .
	IRobotSeismicAnalysis_CIRSOC_103_StructureType (see page 260)	Available structure types.
	IRobotSeismicAnalysisDirectionType (see page 261)	Available direction types.
	IRobotSeismicAnalysis_DM_16_1_96_ProtectionCoeffType (see page 261)	Available values of seismic protection coefficients. .
	IRobotSeismicAnalysis_P_100_92_ImportanceClassType (see page 262)	Available importance classes. .
	IRobotSeismicAnalysis_PS_69_DampingType (see page 262)	Available damping types. .
	IRobotSeismicAnalysis_PS_69_SoilType (see page 263)	Available ground types. .
	IRobotSeismicAnalysis_PS_92_StructureType (see page 263)	Available structure types.
	IRobotSeismicAnalysis_PS_92_SiteType (see page 263)	Available site types.
	IRobotSeismicAnalysis_RPA_88_UsageType (see page 264)	Available usage types.
	IRobotSeismicAnalysis_RPA_88_CategoryType (see page 264)	Available categories.
	IRobotSeismicAnalysis_RPA_88_SoilType (see page 265)	Available soil types.
	IRobotSeismicAnalysis_TURKISH_23098_SoilType (see page 265)	Available soil types.
	IRobotSeismicAnalysis_UBC_97_SoilType (see page 266)	Available soil types.
	IRobotSeismicAnalysis_UBC_97_SourceType (see page 266)	Available seismic source types.
	IRobotSeismicAnalysis_IBC_2000_SiteClassType (see page 267)	Available soil types.
	IRobotSeismicAnalysis_EAK_2000_ImportanceFactorType (see page 267)	Available importance factors. .
	IRobotSeismicAnalysis_EAK_2000_GroundCategoryType (see page 268)	Available ground categories.
	IRobotSeismicAnalysis_RPS_2000_ZoneType (see page 269)	
	IRobotSeismicAnalysis_RPS_2000_SiteType (see page 269)	
	IRobotSeismicAnalysis_RPS_2000_StructureClass (see page 270)	
	IRobotSeismicAnalysis_RPA_2003_ZoneType (see page 272)	
	IRobotSeismicAnalysis_RPA_2003_UsageType (see page 273)	
	IRobotSeismicAnalysis_RPA_2003_SiteType (see page 273)	
	IRobotSeismicAnalysis_ITALY_ORDINANZA_SoilType (see page 278)	Soil type according to the Italian code Ordinanza 3274.
	IRobotSeismicAnalysis_ITALY_ORDINANZA_ZoneType (see page 278)	
	IRobotSeismicAnalysis_ITALY_ORDINANZA_Spectrum (see page 278)	
	IRobotSeismicAnalysis_ITALY_ORDINANZA_Direction (see page 278)	

(c) 2014 Autodesk Inc. All rights reserved

	IRobotSeismicAnalysis_IBC_2006_SiteClassType (see page 286)	Available soil types.
	IRobotSeismicAnalysis_PS_92_2008_ZoneType (see page 291)	Available numbers of seismic zones.
	IRobotSeismicAnalysis_PS_92_2008_StructureType (see page 292)	Available structure types.
	IRobotSeismicAnalysis_PS_92_2008_SiteType (see page 292)	Available site types.
	IRobotSeismicResidualModeDefinitionType (see page 294)	
	IRobotSeismicAnalysis_EC8_SpectrumType (see page 301)	
	IRobotSeismicAnalysis_EC8_GroundType (see page 301)	

Interfaces

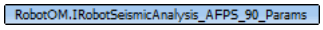
	Name	Description
	IRobotSeismicAnalysis_AFPS_90_Params (see page 212)	Seismic analysis parameters for the French AFPS code. .
	IRobotSeismicAnalysis_CHINESE_Params (see page 216)	Seismic analysis parameters for the Chinese code. .
	IRobotSeismicAnalysis_CIRSOC_103_Params (see page 220)	Seismic analysis parameters for the Argentinian code. .
	IRobotSeismicAnalysis_DM_16_1_96_Params (see page 223)	Seismic analysis parameters for the Italian DM 16.1.96 code. .
	IRobotSeismicAnalysis_P_100_92_Params (see page 225)	Seismic analysis parameters for the Romanian P100-92 code. .
	IRobotSeismicAnalysis_PS_69_Params (see page 228)	Seismic analysis parameters for the French PS69 code. .
	IRobotSeismicAnalysis_PS_92_Params (see page 231)	Seismic analysis parameters for the French PS 92 code. .
	IRobotSeismicAnalysis_RPA_88_Params (see page 236)	Seismic analysis parameters for the Algerian RPA 88 code. .
	IRobotSeismicAnalysis_TURKISH_23098_Params (see page 239)	Seismic analysis parameters for the Turkish code. .
	IRobotSeismicAnalysis_UBC_97_Params (see page 241)	Seismic analysis parameters for the American UBC 97 code. .
	IRobotSeismicAnalysis_IBC_2000_Params (see page 245)	Seismic analysis parameters for the IBC 2000 code issued in the USA. .
	IRobotSeismicAnalysis_EAK_2000_Params (see page 248)	Seismic analysis parameters for the Greek E.A.K. 2000 code. .
	IRobotSeismicAnalysis_PS_92_SiteEnvelope (see page 268)	
	IRobotSeismicAnalysis_RPS_2000_Params (see page 270)	Parameters of the Moroccan seismic code R.P.S. 2000.
	IRobotSeismicAnalysis_RPA_2003_Params (see page 273)	Parameters of the Algerian seismic code RPA 99 (2003). .
	IRobotSeismicAnalysis_ITALY_ORDINANZA_Params (see page 276)	Seismic analysis parameters for the Italian code Ordinanza 3274.
	IRobotSeismicAnalysis_P_100_2006_Params (see page 279)	Parameters of seismic analysis for Romanian code (P100-1-2006).
	IRobotSeismicAnalysis_IBC_2006_Params (see page 282)	Seismic analysis parameters for IBC 2006 released in the USA.
	IRobotSeismicAnalysis_PS_92_2008_Params (see page 286)	Seismic analysis parameters for the French code PS 92 - 2008.

(c) 2014 Autodesk Inc. All rights reserved

	IRobotSeismicAnalysis_PS_92_2008_SiteEnvelope (see page 293)	
	IRobotSeismicResidualMode (see page 294)	
	IRobotSeismicAnalysis_EC8_General_Params (see page 296)	.
	IRobotSeismicAnalysis_EC8_Params (see page 302)	.

II.1.5.1 IRobotSeismicAnalysis_AFPS_90_Params

Class Hierarchy



C++

```
interface IRobotSeismicAnalysis_AFPS_90_Params : IDispatch;
```

C#

```
public interface IRobotSeismicAnalysis_AFPS_90_Params;
```

Visual Basic

```
Public Interface IRobotSeismicAnalysis_AFPS_90_Params
```

Description

Seismic analysis parameters for the French AFPS code. .

Version

Available since version 2.5.

II.1.5.1.1 IRobotSeismicAnalysis_AFPS_90_Params Members

The following tables list the members exposed by IRobotSeismicAnalysis_AFPS_90_Params.

Public Fields

	Name	Description
	BehaviorFactor (see page 213)	Behavior factor.
	Direction (see page 213)	Seismic excitation direction.
	DirectionType (see page 213)	Direction (see page 213).
	Filter (see page 214)	Description of modes taken into account during structure dynamic analysis.
	ResidualMode (see page 214)	
	Site (see page 214)	Structure site type.
	SpectrumType (see page 215)	Spectrum type.
	StructureType (see page 215)	Structure type.
	Topography (see page 215)	Topography.
	ZoneType (see page 216)	Seismic zone number.

II.1.5.1.2 IRobotSeismicAnalysis_AFPS_90_Params Fields

The fields of the IRobotSeismicAnalysis_AFPS_90_Params class are listed here.

Public Fields

	Name	Description
	BehaviorFactor (see page 213)	Behavior factor.
	Direction (see page 213)	Seismic excitation direction.

(c) 2014 Autodesk Inc. All rights reserved

	DirectionType (see page 213)	Direction (see page 213).
	Filter (see page 214)	Description of modes taken into account during structure dynamic analysis.
	ResidualMode (see page 214)	
	Site (see page 214)	Structure site type.
	SpectrumType (see page 215)	Spectrum type.
	StructureType (see page 215)	Structure type.
	Topography (see page 215)	Topography.
	ZoneType (see page 216)	Seismic zone number.

II.1.5.1.2.1 BehaviorFactor

C++

```
HRESULT get_BehaviorFactor(double*);
HRESULT put_BehaviorFactor(double);
```

C#

```
public double BehaviorFactor { get; set; }
```

Visual Basic

```
Public BehaviorFactor As double
```

Description

Behavior factor.

Version

Available since version 2.5.

II.1.5.1.2.2 Direction

C++

```
HRESULT get_Direction(IRobotGeoPoint3D**);
HRESULT put_Direction(IRobotGeoPoint3D*);
```

C#

```
public IRobotGeoPoint3D Direction { get; set; }
```

Visual Basic

```
Public Direction As IRobotGeoPoint3D
```

Description

Seismic excitation direction.

Version

Available since version 2.5.

II.1.5.1.2.3 DirectionType

C++

```
HRESULT get_DirectionType(IRobotSeismicAnalysisDirectionType*);
HRESULT put_DirectionType(IRobotSeismicAnalysisDirectionType);
```

C#

```
public IRobotSeismicAnalysisDirectionType DirectionType { get; set; }
```

Visual Basic

```
Public DirectionType As IRobotSeismicAnalysisDirectionType
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Direction (see page 213).

Version

Available since version 2.5.

II.1.5.1.2.4 Filter

C++

```
HRESULT get_Filter(IRobotCaseAnalysisModesFilter**);
HRESULT put_Filter(IRobotCaseAnalysisModesFilter*);
```

C#

```
public IRobotCaseAnalysisModesFilter Filter { get; set; }
```

Visual Basic

```
Public Filter As IRobotCaseAnalysisModesFilter
```

Description

Description of modes taken into account during structure dynamic analysis.

Version

Available since version 2.5.

II.1.5.1.2.5 ResidualMode

C++

```
HRESULT get_ResidualMode(IRobotSeismicResidualMode**);
```

C#

```
public IRobotSeismicResidualMode ResidualMode { get; }
```

Visual Basic

```
Public ReadOnly ResidualMode As IRobotSeismicResidualMode
```

Version

Available since version 14.7.

II.1.5.1.2.6 Site

C++

```
HRESULT get_Site(IRobotSeismicAnalysis_AFPS_90_SiteType*);
HRESULT put_Site(IRobotSeismicAnalysis_AFPS_90_SiteType);
```

C#

```
public IRobotSeismicAnalysis_AFPS_90_SiteType Site { get; set; }
```

Visual Basic

```
Public Site As IRobotSeismicAnalysis_AFPS_90_SiteType
```

Description

Structure site type.

Version

Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

II.1.5.1.2.7 SpectrumType

C++

```
HRESULT get_SpectrumType(IRobotSeismicAnalysisSpectrumType*);
HRESULT put_SpectrumType(IRobotSeismicAnalysisSpectrumType);
```

C#

```
public IRobotSeismicAnalysisSpectrumType SpectrumType { get; set; }
```

Visual Basic

```
Public SpectrumType As IRobotSeismicAnalysisSpectrumType
```

Description

Spectrum type.

Version

Available since version 2.5.

II.1.5.1.2.8 StructureType

C++

```
HRESULT get_StructureType(IRobotSeismicAnalysis_AFPS_90_StructureType*);
HRESULT put_StructureType(IRobotSeismicAnalysis_AFPS_90_StructureType);
```

C#

```
public IRobotSeismicAnalysis_AFPS_90_StructureType StructureType { get; set; }
```

Visual Basic

```
Public StructureType As IRobotSeismicAnalysis_AFPS_90_StructureType
```

Description

Structure type.

Version

Available since version 2.5.

II.1.5.1.2.9 Topography

C++

```
HRESULT get_Topography(double*);
HRESULT put_Topography(double);
```

C#

```
public double Topography { get; set; }
```

Visual Basic

```
Public Topography As double
```

Description

Topography.

Version

Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

II.1.5.1.2.10 ZoneType

C++

```
HRESULT get_ZoneType(IRobotSeismicAnalysis_AFPS_90_ZoneType*);
HRESULT put_ZoneType(IRobotSeismicAnalysis_AFPS_90_ZoneType);
```

C#

```
public IRobotSeismicAnalysis_AFPS_90_ZoneType ZoneType { get; set; }
```

Visual Basic

```
Public ZoneType As IRobotSeismicAnalysis_AFPS_90_ZoneType
```

Description

Seismic zone number.

Version

Available since version 2.5.

II.1.5.2 IRobotSeismicAnalysis_CHINESE_Params

Class Hierarchy

RobotOM.IRobotSeismicAnalysis_CHINESE_Params

C++

```
interface IRobotSeismicAnalysis_CHINESE_Params : IDispatch;
```

C#

```
public interface IRobotSeismicAnalysis_CHINESE_Params;
```

Visual Basic

```
Public Interface IRobotSeismicAnalysis_CHINESE_Params
```

Description

Seismic analysis parameters for the Chinese code. .

Version

Available since version 2.5.

II.1.5.2.1 IRobotSeismicAnalysis_CHINESE_Params Members

The following tables list the members exposed by IRobotSeismicAnalysis_CHINESE_Params.

Public Fields

	Name	Description
	DesignStandard (see page 217)	Design standard.
	Direction (see page 217)	Seismic excitation direction.
	EarthquakeType (see page 218)	Earthquake distance.
	Factor (see page 218)	Factor.
	Filter (see page 218)	Description of modes taken into account during structure dynamic analysis.
	Intensity (see page 219)	Intensity.
	Site (see page 219)	Site.
	SiteTg (see page 219)	Site (see page 219).
	StructureType (see page 220)	Structure type.

(c) 2014 Autodesk Inc. All rights reserved

II.1.5.2.2 IRobotSeismicAnalysis_CHINESE_Params Fields

The fields of the IRobotSeismicAnalysis_CHINESE_Params class are listed here.

Public Fields

	Name	Description
	DesignStandard (see page 217)	Design standard.
	Direction (see page 217)	Seismic excitation direction.
	EarthquakeType (see page 218)	Earthquake distance.
	Factor (see page 218)	Factor.
	Filter (see page 218)	Description of modes taken into account during structure dynamic analysis.
	Intensity (see page 219)	Intensity.
	Site (see page 219)	Site.
	SiteTg (see page 219)	Site (see page 219).
	StructureType (see page 220)	Structure type.

II.1.5.2.2.1 DesignStandard

C++

```
HRESULT get_DesignStandard(IRobotSeismicAnalysis_CHINESE_DesignType*);
HRESULT put_DesignStandard(IRobotSeismicAnalysis_CHINESE_DesignType);
```

C#

```
public IRobotSeismicAnalysis_CHINESE_DesignType DesignStandard { get; set; }
```

Visual Basic

```
Public DesignStandard As IRobotSeismicAnalysis_CHINESE_DesignType
```

Description

Design standard.

Version

Available since version 2.5.

II.1.5.2.2.2 Direction

C++

```
HRESULT get_Direction(IRobotGeoPoint3D**);
HRESULT put_Direction(IRobotGeoPoint3D*);
```

C#

```
public IRobotGeoPoint3D Direction { get; set; }
```

Visual Basic

```
Public Direction As IRobotGeoPoint3D
```

Description

Seismic excitation direction.

Version

Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

II.1.5.2.2.3 EarthquakeType

C++

```
HRESULT get_EarthquakeType(IRobotSeismicAnalysis_CHINESE_EarthquakeType*);
HRESULT put_EarthquakeType(IRobotSeismicAnalysis_CHINESE_EarthquakeType);
```

C#

```
public IRobotSeismicAnalysis_CHINESE_EarthquakeType EarthquakeType { get; set; }
```

Visual Basic

```
Public EarthquakeType As IRobotSeismicAnalysis_CHINESE_EarthquakeType
```

Description

Earthquake distance.

Version

Available since version 2.5.

II.1.5.2.2.4 Factor

C++

```
HRESULT get_Factor(double*);
HRESULT put_Factor(double);
```

C#

```
public double Factor { get; set; }
```

Visual Basic

```
Public Factor As double
```

Description

Factor.

Version

Available since version 2.5.

II.1.5.2.2.5 Filter

C++

```
HRESULT get_Filter(IRobotCaseAnalysisModesFilter**);
HRESULT put_Filter(IRobotCaseAnalysisModesFilter*);
```

C#

```
public IRobotCaseAnalysisModesFilter Filter { get; set; }
```

Visual Basic

```
Public Filter As IRobotCaseAnalysisModesFilter
```

Description

Description of modes taken into account during structure dynamic analysis.

Version

Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

II.1.5.2.2.6 Intensity

C++

```
HRESULT get_Intensity(IRobotSeismicAnalysis_CHINESE_IntensityType*);
HRESULT put_Intensity(IRobotSeismicAnalysis_CHINESE_IntensityType);
```

C#

```
public IRobotSeismicAnalysis_CHINESE_IntensityType Intensity { get; set; }
```

Visual Basic

```
Public Intensity As IRobotSeismicAnalysis_CHINESE_IntensityType
```

Description

Intensity.

Version

Available since version 2.5.

II.1.5.2.2.7 Site

C++

```
HRESULT get_Site(IRobotSeismicAnalysis_CHINESE_SiteType*);
HRESULT put_Site(IRobotSeismicAnalysis_CHINESE_SiteType);
```

C#

```
public IRobotSeismicAnalysis_CHINESE_SiteType Site { get; set; }
```

Visual Basic

```
Public Site As IRobotSeismicAnalysis_CHINESE_SiteType
```

Description

Site.

Version

Available since version 2.5.

II.1.5.2.2.8 SiteTg

C++

```
HRESULT get_SiteTg(double*);
HRESULT put_SiteTg(double);
```

C#

```
public double SiteTg { get; set; }
```

Visual Basic

```
Public SiteTg As double
```

Description

Site (see page 219).

Version

Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

II.1.5.2.2.9 StructureType

C++

```
HRESULT get_StructureType(IRobotSeismicAnalysis_CHINESE_StructureType*);
HRESULT put_StructureType(IRobotSeismicAnalysis_CHINESE_StructureType);
```

C#

```
public IRobotSeismicAnalysis_CHINESE_StructureType StructureType { get; set; }
```

Visual Basic

```
Public StructureType As IRobotSeismicAnalysis_CHINESE_StructureType
```

Description

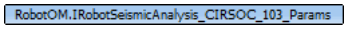
Structure type.

Version

Available since version 2.5.

II.1.5.3 IRobotSeismicAnalysis_CIRSOC_103_Params

Class Hierarchy



C++

```
interface IRobotSeismicAnalysis_CIRSOC_103_Params : IDispatch;
```

C#

```
public interface IRobotSeismicAnalysis_CIRSOC_103_Params;
```

Visual Basic

```
Public Interface IRobotSeismicAnalysis_CIRSOC_103_Params
```

Description

Seismic analysis parameters for the Argentinian code. .

Version

Available since version 2.5.

II.1.5.3.1 IRobotSeismicAnalysis_CIRSOC_103_Params Members

The following tables list the members exposed by IRobotSeismicAnalysis_CIRSOC_103_Params.

Public Fields

	Name	Description
	Direction (see page 221)	Seismic excitation direction.
	DirectionType (see page 221)	Direction (see page 221).
	Filter (see page 221)	Description of modes taken into account during structure dynamic analysis.
	PlasticityCoeff (see page 222)	Global plasticity coefficient.
	Soil (see page 222)	
	StructureType (see page 222)	Structure type.
	ZoneType (see page 223)	Seismic zone number.

II.1.5.3.2 IRobotSeismicAnalysis_CIRSOC_103_Params Fields

The fields of the IRobotSeismicAnalysis_CIRSOC_103_Params class are listed here.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	Direction (see page 221)	Seismic excitation direction.
	DirectionType (see page 221)	Direction (see page 221).
	Filter (see page 221)	Description of modes taken into account during structure dynamic analysis.
	PlasticityCoeff (see page 222)	Global plasticity coefficient.
	Soil (see page 222)	
	StructureType (see page 222)	Structure type.
	ZoneType (see page 223)	Seismic zone number.

II.1.5.3.2.1 Direction

C++

```
HRESULT get_Direction(IRobotGeoPoint3D**);
HRESULT put_Direction(IRobotGeoPoint3D*);
```

C#

```
public IRobotGeoPoint3D Direction { get; set; }
```

Visual Basic

```
Public Direction As IRobotGeoPoint3D
```

Description

Seismic excitation direction.

Version

Available since version 2.5.

II.1.5.3.2.2 DirectionType

C++

```
HRESULT get_DirectionType(IRobotSeismicAnalysisDirectionType*);
HRESULT put_DirectionType(IRobotSeismicAnalysisDirectionType);
```

C#

```
public IRobotSeismicAnalysisDirectionType DirectionType { get; set; }
```

Visual Basic

```
Public DirectionType As IRobotSeismicAnalysisDirectionType
```

Description

Direction (see page 221).

Version

Available since version 2.5.

II.1.5.3.2.3 Filter

C++

```
HRESULT get_Filter(IRobotCaseAnalysisModesFilter**);
HRESULT put_Filter(IRobotCaseAnalysisModesFilter*);
```

C#

```
public IRobotCaseAnalysisModesFilter Filter { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Filter As IRobotCaseAnalysisModesFilter
```

Description

Description of modes taken into account during structure dynamic analysis.

Version

Available since version 2.5.

II.1.5.3.2.4 PlasticityCoeff

C++

```
HRESULT get_PlasticityCoeff(double*);
HRESULT put_PlasticityCoeff(double);
```

C#

```
public double PlasticityCoeff { get; set; }
```

Visual Basic

```
Public PlasticityCoeff As double
```

Description

Global plasticity coefficient.

Version

Available since version 2.5.

II.1.5.3.2.5 Soil

C++

```
HRESULT get_Soil(RobotSeismicAnalysi_CIRSOC_103_SoilType*);
HRESULT put_Soil(RobotSeismicAnalysi_CIRSOC_103_SoilType);
```

C#

```
public RobotSeismicAnalysi_CIRSOC_103_SoilType Soil { get; set; }
```

Visual Basic

```
Public Soil As RobotSeismicAnalysi_CIRSOC_103_SoilType
```

Version

Available since version 2.5.

II.1.5.3.2.6 StructureType

C++

```
HRESULT get_StructureType(IRobotSeismicAnalysis_CIRSOC_103_StructureType*);
HRESULT put_StructureType(IRobotSeismicAnalysis_CIRSOC_103_StructureType);
```

C#

```
public IRobotSeismicAnalysis_CIRSOC_103_StructureType StructureType { get; set; }
```

Visual Basic

```
Public StructureType As IRobotSeismicAnalysis_CIRSOC_103_StructureType
```

Description

Structure type.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 2.5.

II.1.5.3.2.7 ZoneType

C++

```
HRESULT get_ZoneType(IRobotSeismicAnalysis_CIRSOC_103_ZoneType*);
HRESULT put_ZoneType(IRobotSeismicAnalysis_CIRSOC_103_ZoneType);
```

C#

```
public IRobotSeismicAnalysis_CIRSOC_103_ZoneType ZoneType { get; set; }
```

Visual Basic

```
Public ZoneType As IRobotSeismicAnalysis_CIRSOC_103_ZoneType
```

Description

Seismic zone number.

Version

Available since version 2.5.

II.1.5.4 IRobotSeismicAnalysis_DM_16_1_96_Params

Class Hierarchy

RobotOM.IRobotSeismicAnalysis_DM_16_1_96_Params

C++

```
interface IRobotSeismicAnalysis_DM_16_1_96_Params : IDispatch;
```

C#

```
public interface IRobotSeismicAnalysis_DM_16_1_96_Params;
```

Visual Basic

```
Public Interface IRobotSeismicAnalysis_DM_16_1_96_Params
```

Description

Seismic analysis parameters for the Italian DM 16.1.96 code. .

Version

Available since version 2.5.

II.1.5.4.1 IRobotSeismicAnalysis_DM_16_1_96_Params Members

The following tables list the members exposed by IRobotSeismicAnalysis_DM_16_1_96_Params.

Public Fields

	Name	Description
	Direction (see page 224)	Seismic excitation direction.
	Filter (see page 224)	Description of modes taken into account during structure dynamic analysis.
	SeismicCoeff (see page 224)	Seismic coefficient.
	SeismicProtectionCoeff (see page 225)	Seismic protection coefficient.

(c) 2014 Autodesk Inc. All rights reserved

II.1.5.4.2 IRobotSeismicAnalysis_DM_16_1_96_Params Fields

The fields of the IRobotSeismicAnalysis_DM_16_1_96_Params class are listed here.

Public Fields

	Name	Description
	Direction (see page 224)	Seismic excitation direction.
	Filter (see page 224)	Description of modes taken into account during structure dynamic analysis.
	SeismicCoeff (see page 224)	Seismic coefficient.
	SeismicProtectionCoeff (see page 225)	Seismic protection coefficient.

II.1.5.4.2.1 Direction

C++

```
HRESULT get_Direction(IRobotGeoPoint3D**);
HRESULT put_Direction(IRobotGeoPoint3D*);
```

C#

```
public IRobotGeoPoint3D Direction { get; set; }
```

Visual Basic

```
Public Direction As IRobotGeoPoint3D
```

Description

Seismic excitation direction.

Version

Available since version 2.5.

II.1.5.4.2.2 Filter

C++

```
HRESULT get_Filter(IRobotCaseAnalysisModesFilter**);
HRESULT put_Filter(IRobotCaseAnalysisModesFilter*);
```

C#

```
public IRobotCaseAnalysisModesFilter Filter { get; set; }
```

Visual Basic

```
Public Filter As IRobotCaseAnalysisModesFilter
```

Description

Description of modes taken into account during structure dynamic analysis.

Version

Available since version 2.5.

II.1.5.4.2.3 SeismicCoeff

C++

```
HRESULT get_SeismicCoeff(double*);
HRESULT put_SeismicCoeff(double);
```

C#

```
public double SeismicCoeff { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public SeismicCoeff As double
```

Description

Seismic coefficient.

Version

Available since version 2.5.

II.1.5.4.2.4 SeismicProtectionCoeff

C++

```
HRESULT get_SeismicProtectionCoeff(RobotSeismicAnalysisProtectionCoeffType*);
HRESULT put_SeismicProtectionCoeff(RobotSeismicAnalysisProtectionCoeffType);
```

C#

```
public RobotSeismicAnalysisProtectionCoeffType SeismicProtectionCoeff { get; set; }
```

Visual Basic

```
Public SeismicProtectionCoeff As RobotSeismicAnalysisProtectionCoeffType
```

Description

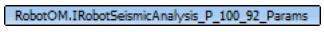
Seismic protection coefficient.

Version

Available since version 2.5.

II.1.5.5 IRobotSeismicAnalysis_P_100_92_Params

Class Hierarchy



C++

```
interface IRobotSeismicAnalysis_P_100_92_Params : IDispatch;
```

C#

```
public interface IRobotSeismicAnalysis_P_100_92_Params;
```

Visual Basic

```
Public Interface IRobotSeismicAnalysis_P_100_92_Params
```

Description

Seismic analysis parameters for the Romanian P100-92 code. .

Version

Available since version 2.5.

II.1.5.5.1 IRobotSeismicAnalysis_P_100_92_Params Members

The following tables list the members exposed by IRobotSeismicAnalysis_P_100_92_Params.

Public Fields

	Name	Description
	Direction (see page 226)	Seismic excitation direction.
	ExcitationDir (see page 226)	

(c) 2014 Autodesk Inc. All rights reserved

	Filter (see page 226)	Description of modes taken into account during structure dynamic analysis.
	ImportanceClass (see page 227)	Importance factor.
	Psi (see page 227)	
	Tc (see page 227)	
	ZoneType (see page 228)	Sesmic zone number.

II.1.5.5.2 IRobotSeismicAnalysis_P_100_92_Params Fields

The fields of the IRobotSeismicAnalysis_P_100_92_Params class are listed here.

Public Fields

	Name	Description
	Direction (see page 226)	Seismic excitation direction.
	ExcitationDir (see page 226)	
	Filter (see page 226)	Description of modes taken into account during structure dynamic analysis.
	ImportanceClass (see page 227)	Importance factor.
	Psi (see page 227)	
	Tc (see page 227)	
	ZoneType (see page 228)	Sesmic zone number.

II.1.5.5.2.1 Direction

C++

```
HRESULT get_Direction(IRobotGeoPoint3D**);
HRESULT put_Direction(IRobotGeoPoint3D*);
```

C#

```
public IRobotGeoPoint3D Direction { get; set; }
```

Visual Basic

```
Public Direction As IRobotGeoPoint3D
```

Description

Seismic excitation direction.

Version

Available since version 2.5.

II.1.5.5.2.2 ExcitationDir

C++

```
HRESULT get_ExcitationDir(IRobotDynamicAnalysisExcitationDirection**);
```

C#

```
public IRobotDynamicAnalysisExcitationDirection ExcitationDir { get; }
```

Visual Basic

```
Public ReadOnly ExcitationDir As IRobotDynamicAnalysisExcitationDirection
```

II.1.5.5.2.3 Filter

C++

```
HRESULT get_Filter(IRobotCaseAnalysisModesFilter**);
HRESULT put_Filter(IRobotCaseAnalysisModesFilter*);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRobotCaseAnalysisModesFilter Filter { get; set; }
```

Visual Basic

```
Public Filter As IRobotCaseAnalysisModesFilter
```

Description

Description of modes taken into account during structure dynamic analysis.

Version

Available since version 2.5.

II.1.5.5.2.4 ImportanceClass

C++

```
HRESULT get_ImportanceClass(IRobotSeismicAnalysis_P_100_92_ImportanceClassType*);
HRESULT put_ImportanceClass(IRobotSeismicAnalysis_P_100_92_ImportanceClassType);
```

C#

```
public IRobotSeismicAnalysis_P_100_92_ImportanceClassType ImportanceClass { get; set; }
```

Visual Basic

```
Public ImportanceClass As IRobotSeismicAnalysis_P_100_92_ImportanceClassType
```

Description

Importance factor.

Version

Available since version 2.5.

II.1.5.5.2.5 Psi

C++

```
HRESULT get_Psi(double*);
HRESULT put_Psi(double);
```

C#

```
public double Psi { get; set; }
```

Visual Basic

```
Public Psi As double
```

Version

Available since version 2.5.

II.1.5.5.2.6 Tc

C++

```
HRESULT get_Tc(double*);
HRESULT put_Tc(double);
```

C#

```
public double Tc { get; set; }
```

Visual Basic

```
Public Tc As double
```

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 2.5.

II.1.5.5.2.7 ZoneType

C++

```
HRESULT get_ZoneType(RobotSeismicAnalysis_P100_92_ZoneType*);
HRESULT put_ZoneType(RobotSeismicAnalysis_P100_92_ZoneType);
```

C#

```
public RobotSeismicAnalysis_P100_92_ZoneType ZoneType { get; set; }
```

Visual Basic

```
Public ZoneType As RobotSeismicAnalysis_P100_92_ZoneType
```

Description

Sesmic zone number.

Version

Available since version 2.5.

II.1.5.6 IRobotSeismicAnalysis_PS_69_Params

Class Hierarchy

RobotOM.IRobotSeismicAnalysis_PS_69_Params

C++

```
interface IRobotSeismicAnalysis_PS_69_Params : IDispatch;
```

C#

```
public interface IRobotSeismicAnalysis_PS_69_Params;
```

Visual Basic

```
Public Interface IRobotSeismicAnalysis_PS_69_Params
```

Description

Seismic analysis parameters for the French PS69 code. .

Version

Available since version 2.5.

II.1.5.6.1 IRobotSeismicAnalysis_PS_69_Params Members

The following tables list the members exposed by IRobotSeismicAnalysis_PS_69_Params.

Public Fields

	Name	Description
	Alpha (see page 229)	Seismic intensity factor.
	Damping (see page 229)	Damping.
	Delta (see page 229)	Foundation coefficient.
	Direction (see page 230)	Seismic excitation direction.
	Filter (see page 230)	Description of modes taken into account during structure dynamic analysis.
	Soil (see page 230)	Soil.

(c) 2014 Autodesk Inc. All rights reserved

II.1.5.6.2 IRobotSeismicAnalysis_PS_69_Params Fields

The fields of the IRobotSeismicAnalysis_PS_69_Params class are listed here.

Public Fields

	Name	Description
	Alpha (see page 229)	Seismic intensity factor.
	Damping (see page 229)	Damping.
	Delta (see page 229)	Foundation coefficient.
	Direction (see page 230)	Seismic excitation direction.
	Filter (see page 230)	Description of modes taken into account during structure dynamic analysis.
	Soil (see page 230)	Soil.

II.1.5.6.2.1 Alpha

C++

```
HRESULT get_Alpha(double*);
HRESULT put_Alpha(double);
```

C#

```
public double Alpha { get; set; }
```

Visual Basic

```
Public Alpha As double
```

Description

Seismic intensity factor.

Version

Available since version 2.5.

II.1.5.6.2.2 Damping

C++

```
HRESULT get_Damping(IRobotSeismicAnalysis_PS_69_DampingType*);
HRESULT put_Damping(IRobotSeismicAnalysis_PS_69_DampingType);
```

C#

```
public IRobotSeismicAnalysis_PS_69_DampingType Damping { get; set; }
```

Visual Basic

```
Public Damping As IRobotSeismicAnalysis_PS_69_DampingType
```

Description

Damping.

Version

Available since version 2.5.

II.1.5.6.2.3 Delta

C++

```
HRESULT get_Delta(double*);
HRESULT put_Delta(double);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double Delta { get; set; }
```

Visual Basic

```
Public Delta As double
```

Description

Foundation coefficient.

Version

Available since version 2.5.

II.1.5.6.2.4 Direction

C++

```
HRESULT get_Direction(IRobotGeoPoint3D**);
HRESULT put_Direction(IRobotGeoPoint3D*);
```

C#

```
public IRobotGeoPoint3D Direction { get; set; }
```

Visual Basic

```
Public Direction As IRobotGeoPoint3D
```

Description

Seismic excitation direction.

Version

Available since version 2.5.

II.1.5.6.2.5 Filter

C++

```
HRESULT get_Filter(IRobotCaseAnalysisModesFilter**);
HRESULT put_Filter(IRobotCaseAnalysisModesFilter*);
```

C#

```
public IRobotCaseAnalysisModesFilter Filter { get; set; }
```

Visual Basic

```
Public Filter As IRobotCaseAnalysisModesFilter
```

Description

Description of modes taken into account during structure dynamic analysis.

Version

Available since version 2.5.

II.1.5.6.2.6 Soil

C++

```
HRESULT get_Soil(IRobotSeismicAnalysis_PS_69_SoilType*);
HRESULT put_Soil(IRobotSeismicAnalysis_PS_69_SoilType);
```

C#

```
public IRobotSeismicAnalysis_PS_69_SoilType Soil { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Soil As IRobotSeismicAnalysis_PS_69_SoilType
```

Description

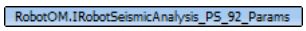
Soil.

Version

Available since version 2.5.

II.1.5.7 IRobotSeismicAnalysis_PS_92_Params

Class Hierarchy



C++

```
interface IRobotSeismicAnalysis_PS_92_Params : IDispatch;
```

C#

```
public interface IRobotSeismicAnalysis_PS_92_Params;
```

Visual Basic

```
Public Interface IRobotSeismicAnalysis_PS_92_Params
```

Description

Seismic analysis parameters for the French PS 92 code. .

Version

Available since version 2.5.

II.1.5.7.1 IRobotSeismicAnalysis_PS_92_Params Members

The following tables list the members exposed by IRobotSeismicAnalysis_PS_92_Params.

Public Fields

	Name	Description
	BehaviorFactor (see page 232)	Behavior factor.
	Direction (see page 232)	Seismic excitation direction.
	DirectionType (see page 233)	Direction (see page 232).
	ExcitationDir (see page 233)	Definition of excitation direction.
	Filter (see page 233)	Description of modes taken into account during structure dynamic analysis.
	ResidualMode (see page 233)	
	Site (see page 234)	Site.
	SiteEnvelope (see page 234)	
	SpectrumType (see page 234)	Spectrum.
	StructureType (see page 235)	Structure type.
	Topography (see page 235)	Topography.
	ZoneType (see page 235)	Seismic zone number.

II.1.5.7.2 IRobotSeismicAnalysis_PS_92_Params Fields

The fields of the IRobotSeismicAnalysis_PS_92_Params class are listed here.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	BehaviorFactor (see page 232)	Behavior factor.
	Direction (see page 232)	Seismic excitation direction.
	DirectionType (see page 233)	Direction (see page 232).
	ExcitationDir (see page 233)	Definition of excitation direction.
	Filter (see page 233)	Description of modes taken into account during structure dynamic analysis.
	ResidualMode (see page 233)	
	Site (see page 234)	Site.
	SiteEnvelope (see page 234)	
	SpectrumType (see page 234)	Spectrum.
	StructureType (see page 235)	Structure type.
	Topography (see page 235)	Topography.
	ZoneType (see page 235)	Seismic zone number.

II.1.5.7.2.1 BehaviorFactor

C++

```
HRESULT get_BehaviorFactor(double*);
HRESULT put_BehaviorFactor(double);
```

C#

```
public double BehaviorFactor { get; set; }
```

Visual Basic

```
Public BehaviorFactor As double
```

Description

Behavior factor.

Version

Available since version 2.5.

II.1.5.7.2.2 Direction

C++

```
HRESULT get_Direction(IRobotGeoPoint3D**);
HRESULT put_Direction(IRobotGeoPoint3D*);
```

C#

```
public IRobotGeoPoint3D Direction { get; set; }
```

Visual Basic

```
Public Direction As IRobotGeoPoint3D
```

Description

Seismic excitation direction.

Version

Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

II.1.5.7.2.3 DirectionType

C++

```
HRESULT get_DirectionType(IRobotSeismicAnalysisDirectionType*);
HRESULT put_DirectionType(IRobotSeismicAnalysisDirectionType);
```

C#

```
public IRobotSeismicAnalysisDirectionType DirectionType { get; set; }
```

Visual Basic

```
Public DirectionType As IRobotSeismicAnalysisDirectionType
```

Description

Direction (see page 232).

Version

Available since version 2.5.

II.1.5.7.2.4 ExcitationDir

C++

```
HRESULT get_ExcitationDir(IRobotDynamicAnalysisExcitationDirection**);
```

C#

```
public IRobotDynamicAnalysisExcitationDirection ExcitationDir { get; }
```

Visual Basic

```
Public ReadOnly ExcitationDir As IRobotDynamicAnalysisExcitationDirection
```

Description

Definition of excitation direction.

Version

Available since version 5.5.

II.1.5.7.2.5 Filter

C++

```
HRESULT get_Filter(IRobotCaseAnalysisModesFilter**);
HRESULT put_Filter(IRobotCaseAnalysisModesFilter*);
```

C#

```
public IRobotCaseAnalysisModesFilter Filter { get; set; }
```

Visual Basic

```
Public Filter As IRobotCaseAnalysisModesFilter
```

Description

Description of modes taken into account during structure dynamic analysis.

Version

Available since version 2.5.

II.1.5.7.2.6 ResidualMode

C++

```
HRESULT get_ResidualMode(IRobotSeismicResidualMode**);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRobotSeismicResidualMode ResidualMode { get; }
```

Visual Basic

```
Public ReadOnly ResidualMode As IRobotSeismicResidualMode
```

Version

Available since version 14.7.

II.1.5.7.2.7 Site

C++

```
HRESULT get_Site(IRobotSeismicAnalysis_PS_92_SiteType*);
HRESULT put_Site(IRobotSeismicAnalysis_PS_92_SiteType);
```

C#

```
public IRobotSeismicAnalysis_PS_92_SiteType Site { get; set; }
```

Visual Basic

```
Public Site As IRobotSeismicAnalysis_PS_92_SiteType
```

Description

Site.

Version

Available since version 2.5.

II.1.5.7.2.8 SiteEnvelope

C++

```
HRESULT get_SiteEnvelope(IRobotSeismicAnalysis_PS_92_SiteEnvelope**);
```

C#

```
public IRobotSeismicAnalysis_PS_92_SiteEnvelope SiteEnvelope { get; }
```

Visual Basic

```
Public ReadOnly SiteEnvelope As IRobotSeismicAnalysis_PS_92_SiteEnvelope
```

Version

Available since version 5.5.

II.1.5.7.2.9 SpectrumType

C++

```
HRESULT get_SpectrumType(IRobotSeismicAnalysisSpectrumType*);
HRESULT put_SpectrumType(IRobotSeismicAnalysisSpectrumType);
```

C#

```
public IRobotSeismicAnalysisSpectrumType SpectrumType { get; set; }
```

Visual Basic

```
Public SpectrumType As IRobotSeismicAnalysisSpectrumType
```

Description

Spectrum.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 2.5.

II.1.5.7.2.10 StructureType

C++

```
HRESULT get_StructureType(IRobotSeismicAnalysis_PS_92_StructureType*);
HRESULT put_StructureType(IRobotSeismicAnalysis_PS_92_StructureType);
```

C#

```
public IRobotSeismicAnalysis_PS_92_StructureType StructureType { get; set; }
```

Visual Basic

```
Public StructureType As IRobotSeismicAnalysis_PS_92_StructureType
```

Description

Structure type.

Version

Available since version 2.5.

II.1.5.7.2.11 Topography

C++

```
HRESULT get_Topography(double*);
HRESULT put_Topography(double);
```

C#

```
public double Topography { get; set; }
```

Visual Basic

```
Public Topography As double
```

Description

Topography.

Version

Available since version 2.5.

II.1.5.7.2.12 ZoneType

C++

```
HRESULT get_ZoneType(IRobotSeismicAnalysis_PS_92_ZoneType*);
HRESULT put_ZoneType(IRobotSeismicAnalysis_PS_92_ZoneType);
```

C#

```
public IRobotSeismicAnalysis_PS_92_ZoneType ZoneType { get; set; }
```

Visual Basic

```
Public ZoneType As IRobotSeismicAnalysis_PS_92_ZoneType
```

Description

Seismic zone number.

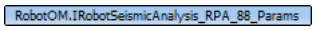
Version

Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

II.1.5.8 IRobotSeismicAnalysis_RPA_88_Params

Class Hierarchy



C++

```
interface IRobotSeismicAnalysis_RPA_88_Params : IDispatch;
```

C#

```
public interface IRobotSeismicAnalysis_RPA_88_Params;
```

Visual Basic

```
Public Interface IRobotSeismicAnalysis_RPA_88_Params
```

Description

Seismic analysis parameters for the Algerian RPA 88 code. .

Version

Available since version 2.5.

II.1.5.8.1 IRobotSeismicAnalysis_RPA_88_Params Members

The following tables list the members exposed by IRobotSeismicAnalysis_RPA_88_Params.

Public Fields

	Name	Description
	Category (see page 236)	Category.
	Direction (see page 237)	Seismic excitation direction.
	Filter (see page 237)	Description of modes taken into account during structure dynamic analysis.
	QualityFactor (see page 237)	Quality coefficient.
	Soil (see page 238)	Soil.
	Usage (see page 238)	Usage.
	ZoneType (see page 238)	Seismic zone number.

II.1.5.8.2 IRobotSeismicAnalysis_RPA_88_Params Fields

The fields of the IRobotSeismicAnalysis_RPA_88_Params class are listed here.

Public Fields

	Name	Description
	Category (see page 236)	Category.
	Direction (see page 237)	Seismic excitation direction.
	Filter (see page 237)	Description of modes taken into account during structure dynamic analysis.
	QualityFactor (see page 237)	Quality coefficient.
	Soil (see page 238)	Soil.
	Usage (see page 238)	Usage.
	ZoneType (see page 238)	Seismic zone number.

II.1.5.8.2.1 Category

C++

```
HRESULT get_Category(IRobotSeismicAnalysis_RPA_88_CategoryType*);
HRESULT put_Category(IRobotSeismicAnalysis_RPA_88_CategoryType);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRobotSeismicAnalysis_RPA_88_CategoryType Category { get; set; }
```

Visual Basic

```
Public Category As IRobotSeismicAnalysis_RPA_88_CategoryType
```

Description

Category.

Version

Available since version 2.5.

II.1.5.8.2.2 Direction

C++

```
HRESULT get_Direction(IRobotGeoPoint3D**);
HRESULT put_Direction(IRobotGeoPoint3D*);
```

C#

```
public IRobotGeoPoint3D Direction { get; set; }
```

Visual Basic

```
Public Direction As IRobotGeoPoint3D
```

Description

Seismic excitation direction.

Version

Available since version 2.5.

II.1.5.8.2.3 Filter

C++

```
HRESULT get_Filter(IRobotCaseAnalysisModesFilter**);
HRESULT put_Filter(IRobotCaseAnalysisModesFilter*);
```

C#

```
public IRobotCaseAnalysisModesFilter Filter { get; set; }
```

Visual Basic

```
Public Filter As IRobotCaseAnalysisModesFilter
```

Description

Description of modes taken into account during structure dynamic analysis.

Version

Available since version 2.5.

II.1.5.8.2.4 QualityFactor

C++

```
HRESULT get_QualityFactor(double*);
HRESULT put_QualityFactor(double);
```

C#

```
public double QualityFactor { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public QualityFactor As double
```

Description

Quality coefficient.

Version

Available since version 2.5.

II.1.5.8.2.5 Soil

C++

```
HRESULT get_Soil(IRobotSeismicAnalysis_RPA_88_SoilType*);
HRESULT put_Soil(IRobotSeismicAnalysis_RPA_88_SoilType);
```

C#

```
public IRobotSeismicAnalysis_RPA_88_SoilType Soil { get; set; }
```

Visual Basic

```
Public Soil As IRobotSeismicAnalysis_RPA_88_SoilType
```

Description

Soil.

Version

Available since version 2.5.

II.1.5.8.2.6 Usage

C++

```
HRESULT get_Usage(IRobotSeismicAnalysis_RPA_88_UsageType*);
HRESULT put_Usage(IRobotSeismicAnalysis_RPA_88_UsageType);
```

C#

```
public IRobotSeismicAnalysis_RPA_88_UsageType Usage { get; set; }
```

Visual Basic

```
Public Usage As IRobotSeismicAnalysis_RPA_88_UsageType
```

Description

Usage.

Version

Available since version 2.5.

II.1.5.8.2.7 ZoneType

C++

```
HRESULT get_ZoneType(IRobotSeismicAnalysis_RPA_88_ZoneType*);
HRESULT put_ZoneType(IRobotSeismicAnalysis_RPA_88_ZoneType);
```

C#

```
public IRobotSeismicAnalysis_RPA_88_ZoneType ZoneType { get; set; }
```

Visual Basic

```
Public ZoneType As IRobotSeismicAnalysis_RPA_88_ZoneType
```

(c) 2014 Autodesk Inc. All rights reserved

Description

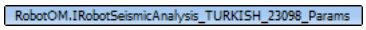
Seismic zone number.

Version

Available since version 2.5.

II.1.5.9 IRobotSeismicAnalysis_TURKISH_23098_Params

Class Hierarchy



C++

```
interface IRobotSeismicAnalysis_TURKISH_23098_Params : IDispatch;
```

C#

```
public interface IRobotSeismicAnalysis_TURKISH_23098_Params;
```

Visual Basic

```
Public Interface IRobotSeismicAnalysis_TURKISH_23098_Params
```

Description

Seismic analysis parameters for the Turkish code. .

Version

Available since version 2.5.

II.1.5.9.1 IRobotSeismicAnalysis_TURKISH_23098_Params Members

The following tables list the members exposed by IRobotSeismicAnalysis_TURKISH_23098_Params.

Public Fields

	Name	Description
	BehaviorFactor (see page 240)	Behavior factor.
	Direction (see page 240)	Seismic excitation direction.
	Filter (see page 240)	Description of modes taken into account during structure dynamic analysis.
	SoilType (see page 241)	Soil class.
	StructureImportance (see page 241)	Building importance coefficient.
	ZoneType (see page 241)	Seismic zone number.

II.1.5.9.2 IRobotSeismicAnalysis_TURKISH_23098_Params Fields

The fields of the IRobotSeismicAnalysis_TURKISH_23098_Params class are listed here.

Public Fields

	Name	Description
	BehaviorFactor (see page 240)	Behavior factor.
	Direction (see page 240)	Seismic excitation direction.
	Filter (see page 240)	Description of modes taken into account during structure dynamic analysis.
	SoilType (see page 241)	Soil class.
	StructureImportance (see page 241)	Building importance coefficient.

(c) 2014 Autodesk Inc. All rights reserved

	ZoneType (see page 241)	Seismic zone number.

II.1.5.9.2.1 BehaviorFactor

C++

```
HRESULT get_BehaviorFactor(double*);
HRESULT put_BehaviorFactor(double);
```

C#

```
public double BehaviorFactor { get; set; }
```

Visual Basic

```
Public BehaviorFactor As double
```

Description

Behavior factor.

Version

Available since version 2.5.

II.1.5.9.2.2 Direction

C++

```
HRESULT get_Direction(IRobotGeoPoint3D**);
HRESULT put_Direction(IRobotGeoPoint3D*);
```

C#

```
public IRobotGeoPoint3D Direction { get; set; }
```

Visual Basic

```
Public Direction As IRobotGeoPoint3D
```

Description

Seismic excitation direction.

Version

Available since version 2.5.

II.1.5.9.2.3 Filter

C++

```
HRESULT get_Filter(IRobotCaseAnalysisModesFilter**);
HRESULT put_Filter(IRobotCaseAnalysisModesFilter*);
```

C#

```
public IRobotCaseAnalysisModesFilter Filter { get; set; }
```

Visual Basic

```
Public Filter As IRobotCaseAnalysisModesFilter
```

Description

Description of modes taken into account during structure dynamic analysis.

Version

Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

II.1.5.9.2.4 SoilType

C++

```
HRESULT get_SoilType(IRobotSeismicAnalysis_TURKISH_23098_SoilType*);
HRESULT put_SoilType(IRobotSeismicAnalysis_TURKISH_23098_SoilType);
```

C#

```
public IRobotSeismicAnalysis_TURKISH_23098_SoilType SoilType { get; set; }
```

Visual Basic

```
Public SoilType As IRobotSeismicAnalysis_TURKISH_23098_SoilType
```

Description

Soil class.

Version

Available since version 2.5.

II.1.5.9.2.5 StructureImportance

C++

```
HRESULT get_StructureImportance(double*);
HRESULT put_StructureImportance(double);
```

C#

```
public double StructureImportance { get; set; }
```

Visual Basic

```
Public StructureImportance As double
```

Description

Building importance coefficient.

Version

Available since version 2.5.

II.1.5.9.2.6 ZoneType

C++

```
HRESULT get_ZoneType(IRobotSeismicAnalysis_TURKISH_23098_ZoneType*);
HRESULT put_ZoneType(IRobotSeismicAnalysis_TURKISH_23098_ZoneType);
```

C#

```
public IRobotSeismicAnalysis_TURKISH_23098_ZoneType ZoneType { get; set; }
```

Visual Basic

```
Public ZoneType As IRobotSeismicAnalysis_TURKISH_23098_ZoneType
```

Description

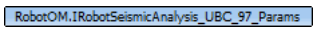
Seismic zone number.

Version

Available since version 2.5.

II.1.5.10 IRobotSeismicAnalysis_UBC_97_Params

Class Hierarchy



(c) 2014 Autodesk Inc. All rights reserved

C++

```
interface IRobotSeismicAnalysis_UBC_97_Params : IDispatch;
```

C#

```
public interface IRobotSeismicAnalysis_UBC_97_Params;
```

Visual Basic

```
Public Interface IRobotSeismicAnalysis_UBC_97_Params
```

Description

Seismic analysis parameters for the American UBC 97 code. .

Version

Available since version 2.5.

II.1.5.10.1 IRobotSeismicAnalysis_UBC_97_Params Members

The following tables list the members exposed by IRobotSeismicAnalysis_UBC_97_Params.

Public Fields

	Name	Description
	BehaviorFactor (see page 242)	Behavior factor.
	ClosestDistance (see page 243)	Minimum distance to a known seismic source (in km).
	Direction (see page 243)	Seismic excitation direction.
	ExcitationDir (see page 243)	
	Filter (see page 244)	Description of modes taken into account during structure dynamic analysis.
	I (see page 244)	Structure importance factor.
	Soil (see page 244)	Soil.
	Source (see page 245)	Seismic source type.
	ZoneType (see page 245)	Seismic zone number.

II.1.5.10.2 IRobotSeismicAnalysis_UBC_97_Params Fields

The fields of the IRobotSeismicAnalysis_UBC_97_Params class are listed here.

Public Fields

	Name	Description
	BehaviorFactor (see page 242)	Behavior factor.
	ClosestDistance (see page 243)	Minimum distance to a known seismic source (in km).
	Direction (see page 243)	Seismic excitation direction.
	ExcitationDir (see page 243)	
	Filter (see page 244)	Description of modes taken into account during structure dynamic analysis.
	I (see page 244)	Structure importance factor.
	Soil (see page 244)	Soil.
	Source (see page 245)	Seismic source type.
	ZoneType (see page 245)	Seismic zone number.

II.1.5.10.2.1 BehaviorFactor

C++

```
HRESULT get_BehaviorFactor(double*);
HRESULT put_BehaviorFactor(double);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double BehaviorFactor { get; set; }
```

Visual Basic

```
Public BehaviorFactor As double
```

Description

Behavior factor.

II.1.5.10.2.2 ClosestDistance

C++

```
HRESULT get_ClosestDistance(double*);
HRESULT put_ClosestDistance(double);
```

C#

```
public double ClosestDistance { get; set; }
```

Visual Basic

```
Public ClosestDistance As double
```

Description

Minimum distance to a known seismic source (in km).

Version

Available since version 2.5.

II.1.5.10.2.3 Direction

C++

```
HRESULT get_Direction(IRobotGeoPoint3D**);
HRESULT put_Direction(IRobotGeoPoint3D*);
```

C#

```
public IRobotGeoPoint3D Direction { get; set; }
```

Visual Basic

```
Public Direction As IRobotGeoPoint3D
```

Description

Seismic excitation direction.

Version

Available since version 2.5.

II.1.5.10.2.4 ExcitationDir

C++

```
HRESULT get_ExcitationDir(IRobotDynamicAnalysisExcitationDirection**);
```

C#

```
public IRobotDynamicAnalysisExcitationDirection ExcitationDir { get; }
```

Visual Basic

```
Public ReadOnly ExcitationDir As IRobotDynamicAnalysisExcitationDirection
```

(c) 2014 Autodesk Inc. All rights reserved

II.1.5.10.2.5 Filter

C++

```
HRESULT get_Filter(IRobotCaseAnalysisModesFilter**);
HRESULT put_Filter(IRobotCaseAnalysisModesFilter*);
```

C#

```
public IRobotCaseAnalysisModesFilter Filter { get; set; }
```

Visual Basic

```
Public Filter As IRobotCaseAnalysisModesFilter
```

Description

Description of modes taken into account during structure dynamic analysis.

Version

Available since version 2.5.

II.1.5.10.2.6 I

C++

```
HRESULT get_I(double*);
HRESULT put_I(double);
```

C#

```
public double I { get; set; }
```

Visual Basic

```
Public I As double
```

Description

Structure importance factor.

Version

Available since version 8.

II.1.5.10.2.7 Soil

C++

```
HRESULT get_Soil(IRobotSeismicAnalysis_UBC_97_SoilType*);
HRESULT put_Soil(IRobotSeismicAnalysis_UBC_97_SoilType);
```

C#

```
public IRobotSeismicAnalysis_UBC_97_SoilType Soil { get; set; }
```

Visual Basic

```
Public Soil As IRobotSeismicAnalysis_UBC_97_SoilType
```

Description

Soil.

Version

Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

II.1.5.10.2.8 Source

C++

```
HRESULT get_Source(IRobotSeismicAnalysis_UBC_97_SourceType*);
HRESULT put_Source(IRobotSeismicAnalysis_UBC_97_SourceType);
```

C#

```
public IRobotSeismicAnalysis_UBC_97_SourceType Source { get; set; }
```

Visual Basic

```
Public Source As IRobotSeismicAnalysis_UBC_97_SourceType
```

Description

Seismic source type.

Version

Available since version 2.5.

II.1.5.10.2.9 ZoneType

C++

```
HRESULT get_ZoneType(IRobotSeismicAnalysis_UBC_97_ZoneType*);
HRESULT put_ZoneType(IRobotSeismicAnalysis_UBC_97_ZoneType);
```

C#

```
public IRobotSeismicAnalysis_UBC_97_ZoneType ZoneType { get; set; }
```

Visual Basic

```
Public ZoneType As IRobotSeismicAnalysis_UBC_97_ZoneType
```

Description

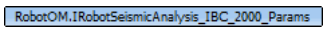
Seismic zone number.

Version

Available since version 2.5.

II.1.5.11 IRobotSeismicAnalysis_IBC_2000_Params

Class Hierarchy



C++

```
interface IRobotSeismicAnalysis_IBC_2000_Params : IDispatch;
```

C#

```
public interface IRobotSeismicAnalysis_IBC_2000_Params;
```

Visual Basic

```
Public Interface IRobotSeismicAnalysis_IBC_2000_Params
```

Description

Seismic analysis parameters for the IBC 2000 code issued in the USA. .

Version

Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

II.1.5.11.1 IRobotSeismicAnalysis_IBC_2000_Params Members

The following tables list the members exposed by IRobotSeismicAnalysis_IBC_2000_Params.

Public Fields

	Name	Description
	BehaviorFactor (see page 246)	Behaviour factor.
	Direction (see page 246)	Seismic excitation direction.
	ExcitationDir (see page 247)	
	Filter (see page 247)	Description of modes taken into account during structure dynamic analysis.
	Ie (see page 247)	Structure importance factor.
	S1 (see page 248)	Spectral acceleration for short periods.
	SiteClass (see page 248)	Soil.
	Ss (see page 248)	Spectral acceleration for 1-second period.

II.1.5.11.2 IRobotSeismicAnalysis_IBC_2000_Params Fields

The fields of the IRobotSeismicAnalysis_IBC_2000_Params class are listed here.

Public Fields

	Name	Description
	BehaviorFactor (see page 246)	Behaviour factor.
	Direction (see page 246)	Seismic excitation direction.
	ExcitationDir (see page 247)	
	Filter (see page 247)	Description of modes taken into account during structure dynamic analysis.
	Ie (see page 247)	Structure importance factor.
	S1 (see page 248)	Spectral acceleration for short periods.
	SiteClass (see page 248)	Soil.
	Ss (see page 248)	Spectral acceleration for 1-second period.

II.1.5.11.2.1 BehaviorFactor

C++

```
HRESULT get_BehaviorFactor(double*);
HRESULT put_BehaviorFactor(double);
```

C#

```
public double BehaviorFactor { get; set; }
```

Visual Basic

```
Public BehaviorFactor As double
```

Description

Behaviour factor.

Version

Available since version 8.

II.1.5.11.2.2 Direction

C++

```
HRESULT get_Direction(IRobotGeoPoint3D**);
HRESULT put_Direction(IRobotGeoPoint3D*);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRobotGeoPoint3D Direction { get; set; }
```

Visual Basic

```
Public Direction As IRobotGeoPoint3D
```

Description

Seismic excitation direction.

Version

Available since version 2.5.

II.1.5.11.2.3 ExcitationDir

C++

```
HRESULT get_ExcitationDir(IRobotDynamicAnalysisExcitationDirection**);
```

C#

```
public IRobotDynamicAnalysisExcitationDirection ExcitationDir { get; }
```

Visual Basic

```
Public ReadOnly ExcitationDir As IRobotDynamicAnalysisExcitationDirection
```

II.1.5.11.2.4 Filter

C++

```
HRESULT get_Filter(IRobotCaseAnalysisModesFilter**);
HRESULT put_Filter(IRobotCaseAnalysisModesFilter*);
```

C#

```
public IRobotCaseAnalysisModesFilter Filter { get; set; }
```

Visual Basic

```
Public Filter As IRobotCaseAnalysisModesFilter
```

Description

Description of modes taken into account during structure dynamic analysis.

Version

Available since version 2.5.

II.1.5.11.2.5 Ie

C++

```
HRESULT get_Ie(double*);
HRESULT put_Ie(double);
```

C#

```
public double Ie { get; set; }
```

Visual Basic

```
Public Ie As double
```

Description

Structure importance factor.

Version

Available since version 8.

(c) 2014 Autodesk Inc. All rights reserved

II.1.5.11.2.6 S1

C++

```
HRESULT get_S1(double*);
HRESULT put_S1(double);
```

C#

```
public double S1 { get; set; }
```

Visual Basic

```
Public S1 As double
```

Description

Spectral acceleration for short periods.

Version

Available since version 2.5.

II.1.5.11.2.7 SiteClass

C++

```
HRESULT get_SiteClass(IRobotSeismicAnalysis_IBC_2000_SiteClassType*);
HRESULT put_SiteClass(IRobotSeismicAnalysis_IBC_2000_SiteClassType);
```

C#

```
public IRobotSeismicAnalysis_IBC_2000_SiteClassType SiteClass { get; set; }
```

Visual Basic

```
Public SiteClass As IRobotSeismicAnalysis_IBC_2000_SiteClassType
```

Description

Soil.

Version

Available since version 2.5.

II.1.5.11.2.8 Ss

C++

```
HRESULT get_Ss(double*);
HRESULT put_Ss(double);
```

C#

```
public double Ss { get; set; }
```

Visual Basic

```
Public Ss As double
```

Description

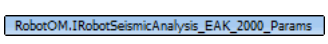
Spectral acceleration for 1-second period.

Version

Available since version 2.5.

II.1.5.12 IRobotSeismicAnalysis_EAK_2000_Params

Class Hierarchy



(c) 2014 Autodesk Inc. All rights reserved

C++

```
interface IRobotSeismicAnalysis_EAK_2000_Params : IDispatch;
```

C#

```
public interface IRobotSeismicAnalysis_EAK_2000_Params;
```

Visual Basic

```
Public Interface IRobotSeismicAnalysis_EAK_2000_Params
```

Description

Seismic analysis parameters for the Greek E.A.K. 2000 code. .

Version

Available since version 2.5.

II.1.5.12.1 IRobotSeismicAnalysis_EAK_2000_Params Members

The following tables list the members exposed by IRobotSeismicAnalysis_EAK_2000_Params.

Public Fields

	Name	Description
	BehaviorFactor (see page 250)	Behavior factor.
	Direction (see page 250)	Seismic excitation direction.
	DirectionType (see page 250)	Direction (see page 250).
	Filter (see page 251)	Description of modes taken into account during structure dynamic analysis.
	FoundationFactor (see page 251)	Foundation coefficient.
	GroundCategory (see page 251)	Ground category.
	ImportanceFactor (see page 252)	Structure importance.
	VerticalBehaviorFactor (see page 252)	Behavior coefficient for the vertical direction .
	VerticalFoundationFactor (see page 252)	Foundation coefficient for the vertical direction.
	ZoneType (see page 252)	Seismic zone number.

II.1.5.12.2 IRobotSeismicAnalysis_EAK_2000_Params Fields

The fields of the IRobotSeismicAnalysis_EAK_2000_Params class are listed here.

Public Fields

	Name	Description
	BehaviorFactor (see page 250)	Behavior factor.
	Direction (see page 250)	Seismic excitation direction.
	DirectionType (see page 250)	Direction (see page 250).
	Filter (see page 251)	Description of modes taken into account during structure dynamic analysis.
	FoundationFactor (see page 251)	Foundation coefficient.
	GroundCategory (see page 251)	Ground category.
	ImportanceFactor (see page 252)	Structure importance.
	VerticalBehaviorFactor (see page 252)	Behavior coefficient for the vertical direction .
	VerticalFoundationFactor (see page 252)	Foundation coefficient for the vertical direction.
	ZoneType (see page 252)	Seismic zone number.

(c) 2014 Autodesk Inc. All rights reserved

II.1.5.12.2.1 BehaviorFactor

C++

```
HRESULT get_BehaviorFactor(double*);
HRESULT put_BehaviorFactor(double);
```

C#

```
public double BehaviorFactor { get; set; }
```

Visual Basic

```
Public BehaviorFactor As double
```

Description

Behavior factor.

Version

Available since version 2.5.

II.1.5.12.2.2 Direction

C++

```
HRESULT get_Direction(IRobotGeoPoint3D**);
HRESULT put_Direction(IRobotGeoPoint3D*);
```

C#

```
public IRobotGeoPoint3D Direction { get; set; }
```

Visual Basic

```
Public Direction As IRobotGeoPoint3D
```

Description

Seismic excitation direction.

Version

Available since version 2.5.

II.1.5.12.2.3 DirectionType

C++

```
HRESULT get_DirectionType(IRobotSeismicAnalysisDirectionType*);
HRESULT put_DirectionType(IRobotSeismicAnalysisDirectionType);
```

C#

```
public IRobotSeismicAnalysisDirectionType DirectionType { get; set; }
```

Visual Basic

```
Public DirectionType As IRobotSeismicAnalysisDirectionType
```

Description

Direction (see page 250).

Version

Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

II.1.5.12.2.4 Filter

C++

```
HRESULT get_Filter(IRobotCaseAnalysisModesFilter**);
HRESULT put_Filter(IRobotCaseAnalysisModesFilter*);
```

C#

```
public IRobotCaseAnalysisModesFilter Filter { get; set; }
```

Visual Basic

```
Public Filter As IRobotCaseAnalysisModesFilter
```

Description

Description of modes taken into account during structure dynamic analysis.

Version

Available since version 2.5.

II.1.5.12.2.5 FoundationFactor

C++

```
HRESULT get_FoundationFactor(double*);
HRESULT put_FoundationFactor(double);
```

C#

```
public double FoundationFactor { get; set; }
```

Visual Basic

```
Public FoundationFactor As double
```

Description

Foundation coefficient.

Version

Available since version 2.5.

II.1.5.12.2.6 GroundCategory

C++

```
HRESULT get_GroundCategory(IRobotSeismicAnalysis_EAK_2000_GroundCategoryType*);
HRESULT put_GroundCategory(IRobotSeismicAnalysis_EAK_2000_GroundCategoryType);
```

C#

```
public IRobotSeismicAnalysis_EAK_2000_GroundCategoryType GroundCategory { get; set; }
```

Visual Basic

```
Public GroundCategory As IRobotSeismicAnalysis_EAK_2000_GroundCategoryType
```

Description

Ground category.

Version

Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

II.1.5.12.2.7 ImportanceFactor

C++

```
HRESULT get_ImportanceFactor(IRobotSeismicAnalysis_EAK_2000_ImportanceFactorType*);
HRESULT put_ImportanceFactor(IRobotSeismicAnalysis_EAK_2000_ImportanceFactorType);
```

C#

```
public IRobotSeismicAnalysis_EAK_2000_ImportanceFactorType ImportanceFactor { get; set; }
```

Visual Basic

```
Public ImportanceFactor As IRobotSeismicAnalysis_EAK_2000_ImportanceFactorType
```

Description

Structure importance.

Version

Available since version 2.5.

II.1.5.12.2.8 VerticalBehaviorFactor

C++

```
HRESULT get_VerticalBehaviorFactor(double*);
```

C#

```
public double VerticalBehaviorFactor { get; }
```

Visual Basic

```
Public ReadOnly VerticalBehaviorFactor As double
```

Description

Behavior coefficient for the vertical direction .

Version

Available since version 2.5.

II.1.5.12.2.9 VerticalFoundationFactor

C++

```
HRESULT get_VerticalFoundationFactor(double*);
```

C#

```
public double VerticalFoundationFactor { get; }
```

Visual Basic

```
Public ReadOnly VerticalFoundationFactor As double
```

Description

Foundation coefficient for the vertical direction.

Version

Available since version 2.5.

II.1.5.12.2.10 ZoneType

C++

```
HRESULT get_ZoneType(IRobotSeismicAnalysis_EAK_2000_ZoneType*);
HRESULT put_ZoneType(IRobotSeismicAnalysis_EAK_2000_ZoneType);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRobotSeismicAnalysis_EAK_2000_ZoneType ZoneType { get; set; }
```

Visual Basic

```
Public ZoneType As IRobotSeismicAnalysis_EAK_2000_ZoneType
```

Description

Seismic zone number.

Version

Available since version 2.5.

II.1.5.13 IRobotSeismicAnalysis_AFPS_90_ZoneType

C++

```
enum IRobotSeismicAnalysis_AFPS_90_ZoneType;
```

C#

```
public enum IRobotSeismicAnalysis_AFPS_90_ZoneType;
```

Visual Basic

```
Public Enum IRobotSeismicAnalysis_AFPS_90_ZoneType
```

Members

Members	Description
I_SAZT_AFPS_90_IA = 0	Available since version 2.5.
I_SAZT_AFPS_90_IB = 1	Available since version 2.5.
I_SAZT_AFPS_90_II = 2	Available since version 2.5.
I_SAZT_AFPS_90_III = 3	Available since version 2.5.

Description

Available numbers of seismic zones.

Version

Available since version 2.5.

II.1.5.14 IRobotSeismicAnalysis_CIRSOC_103_ZoneType

C++

```
enum IRobotSeismicAnalysis_CIRSOC_103_ZoneType;
```

C#

```
public enum IRobotSeismicAnalysis_CIRSOC_103_ZoneType;
```

Visual Basic

```
Public Enum IRobotSeismicAnalysis_CIRSOC_103_ZoneType
```

Members

Members	Description
I_SAZT_CIRSOC_103_0 = 0	Available since version 2.5.
I_SAZT_CIRSOC_103_1 = 1	Available since version 2.5.
I_SAZT_CIRSOC_103_2 = 2	Available since version 2.5.
I_SAZT_CIRSOC_103_3 = 3	Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

I_SAZT_CIRSOC_103_4 = 4	Available since version 2.5.

Description

Available numbers of seismic zones.

Version

Available since version 2.5.

II.1.5.15 IRobotSeismicAnalysis_P_100_92_ZoneType

C++

```
enum IRobotSeismicAnalysis_P_100_92_ZoneType;
```

C#

```
public enum IRobotSeismicAnalysis_P_100_92_ZoneType;
```

Visual Basic

```
Public Enum IRobotSeismicAnalysis_P_100_92_ZoneType
```

Members

Members	Description
I_SAZT_P_100_92_A = 0	Available since version 2.5.
I_SAZT_P_100_92_B = 1	Available since version 2.5.
I_SAZT_P_100_92_C = 2	Available since version 2.5.
I_SAZT_P_100_92_D = 3	Available since version 2.5.
I_SAZT_P_100_92_E = 4	Available since version 2.5.
I_SAZT_P_100_92_F = 5	Available since version 2.5.

Description

Available numbers of seismic zones.

Version

Available since version 2.5.

II.1.5.16 IRobotSeismicAnalysis_PS_92_ZoneType

C++

```
enum IRobotSeismicAnalysis_PS_92_ZoneType;
```

C#

```
public enum IRobotSeismicAnalysis_PS_92_ZoneType;
```

Visual Basic

```
Public Enum IRobotSeismicAnalysis_PS_92_ZoneType
```

Members

Members	Description
I_SAZT_PS_92_IA = 0	Available since version 2.5.
I_SAZT_PS_92_IB = 1	Available since version 2.5.
I_SAZT_PS_92_II = 2	Available since version 2.5.
I_SAZT_PS_92_III = 3	Available since version 2.5.

Description

Available numbers of seismic zones.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 2.5.

II.1.5.17 IRobotSeismicAnalysis_RPA_88_ZoneType

C++

```
enum IRobotSeismicAnalysis_RPA_88_ZoneType;
```

C#

```
public enum IRobotSeismicAnalysis_RPA_88_ZoneType;
```

Visual Basic

```
Public Enum IRobotSeismicAnalysis_RPA_88_ZoneType
```

Members

Members	Description
I_SAZT_RPA_88_I = 0	Available since version 2.5.
I_SAZT_RPA_88_II = 1	Available since version 2.5.
I_SAZT_RPA_88_III = 2	Available since version 2.5.

Description

Available numbers of seismic zones.

Version

Available since version 2.5.

II.1.5.18 IRobotSeismicAnalysis_UBC_97_ZoneType

C++

```
enum IRobotSeismicAnalysis_UBC_97_ZoneType;
```

C#

```
public enum IRobotSeismicAnalysis_UBC_97_ZoneType;
```

Visual Basic

```
Public Enum IRobotSeismicAnalysis_UBC_97_ZoneType
```

Members

Members	Description
I_SAZT_UBC_97_1 = 0	Available since version 2.5.
I_SAZT_UBC_97_2A = 1	Available since version 2.5.
I_SAZT_UBC_97_2B = 2	Available since version 2.5.
I_SAZT_UBC_97_3 = 3	Available since version 2.5.
I_SAZT_UBC_97_4 = 4	Available since version 2.5.

Description

Available numbers of seismic zones.

Version

Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

II.1.5.19 IRobotSeismicAnalysis_EAK_2000_ZoneType

C++

```
enum IRobotSeismicAnalysis_EAK_2000_ZoneType;
```

C#

```
public enum IRobotSeismicAnalysis_EAK_2000_ZoneType;
```

Visual Basic

```
Public Enum IRobotSeismicAnalysis_EAK_2000_ZoneType
```

Members

Members	Description
I_SAZT_EAK_2000_I = 0	Available since version 2.5.
I_SAZT_EAK_2000_II = 1	Available since version 2.5.
I_SAZT_EAK_2000_III = 2	Available since version 2.5.
I_SAZT_EAK_2000_IV = 3	Available since version 2.5.

Description

Available numbers of seismic zones.

Version

Available since version 2.5.

II.1.5.20 IRobotSeismicAnalysis_TURKISH_23098_ZoneType

C++

```
enum IRobotSeismicAnalysis_TURKISH_23098_ZoneType;
```

C#

```
public enum IRobotSeismicAnalysis_TURKISH_23098_ZoneType;
```

Visual Basic

```
Public Enum IRobotSeismicAnalysis_TURKISH_23098_ZoneType
```

Members

Members	Description
I_SAZT_TURKISH_23098_1 = 0	Available since version 2.5.
I_SAZT_TURKISH_23098_2 = 1	Available since version 2.5.
I_SAZT_TURKISH_23098_3 = 2	Available since version 2.5.
I_SAZT_TURKISH_23098_4 = 3	Available since version 2.5.

Description

Available numbers of seismic zones.

Version

Available since version 2.5.

II.1.5.21 IRobotSeismicAnalysis_AFPS_90_StructureType

C++

```
enum IRobotSeismicAnalysis_AFPS_90_StructureType;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public enum IRobotSeismicAnalysis_AFPS_90_StructureType;
```

Visual Basic

```
Public Enum IRobotSeismicAnalysis_AFPS_90_StructureType
```

Members

Members	Description
I_SAST_AFPS_90_A = 0	Available since version 2.5.
I_SAST_AFPS_90_B = 1	Available since version 2.5.
I_SAST_AFPS_90_C = 2	Available since version 2.5.

Description

Available structure types.

Version

Available since version 2.5.

II.1.5.22 IRobotSeismicAnalysis_AFPS_90_SiteType

C++

```
enum IRobotSeismicAnalysis_AFPS_90_SiteType;
```

C#

```
public enum IRobotSeismicAnalysis_AFPS_90_SiteType;
```

Visual Basic

```
Public Enum IRobotSeismicAnalysis_AFPS_90_SiteType
```

Members

Members	Description
I_SAST_AFPS_90_S0 = 0	Available since version 2.5.
I_SAST_AFPS_90_S1 = 1	Available since version 2.5.
I_SAST_AFPS_90_S2 = 2	Available since version 2.5.
I_SAST_AFPS_90_S3 = 3	Available since version 2.5.

Description

Available types of structure site.

Version

Available since version 2.5.

II.1.5.23 IRobotSeismicAnalysisSpectrumType

C++

```
enum IRobotSeismicAnalysisSpectrumType;
```

C#

```
public enum IRobotSeismicAnalysisSpectrumType;
```

Visual Basic

```
Public Enum IRobotSeismicAnalysisSpectrumType
```

(c) 2014 Autodesk Inc. All rights reserved

Members

Members	Description
I_SAST_DIMENSIONING = 0	Available since version 2.5.
I_SAST_ELASITC = 1	Available since version 2.5.

Description

Available spectrum types. .

Version

Available since version 2.5.

II.1.5.24 IRobotSeismicAnalysis_CHINESE_StructureType

C++

```
enum IRobotSeismicAnalysis_CHINESE_StructureType;
```

C#

```
public enum IRobotSeismicAnalysis_CHINESE_StructureType;
```

Visual Basic

```
Public Enum IRobotSeismicAnalysis_CHINESE_StructureType
```

Members

Members	Description
I_SAST_CHINESE_SPECIAL = 0	Available since version 2.5.
I_SAST_CHINESE_BUILDINGS = 1	Available since version 2.5.
I_SAST_CHINESE_BRIDGES = 2	Available since version 2.5.
I_SAST_CHINESE_HARBOR_BUILDINGS = 3	Available since version 2.5.

Description

Available structure types.

Version

Available since version 2.5.

II.1.5.25 IRobotSeismicAnalysis_CHINESE_SiteType

C++

```
enum IRobotSeismicAnalysis_CHINESE_SiteType;
```

C#

```
public enum IRobotSeismicAnalysis_CHINESE_SiteType;
```

Visual Basic

```
Public Enum IRobotSeismicAnalysis_CHINESE_SiteType
```

Members

Members	Description
I_SAST_CHINESE_I = 0	Available since version 2.5.
I_SAST_CHINESE_II = 1	Available since version 2.5.
I_SAST_CHINESE_III = 2	Available since version 2.5.
I_SAST_CHINESE_IV = 3	Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

Description

Available types of structure site.

Version

Available since version 2.5.

II.1.5.26 IRobotSeismicAnalysis_CHINESE_IntensityType

C++

```
enum IRobotSeismicAnalysis_CHINESE_IntensityType;
```

C#

```
public enum IRobotSeismicAnalysis_CHINESE_IntensityType;
```

Visual Basic

```
Public Enum IRobotSeismicAnalysis_CHINESE_IntensityType
```

Members

Members	Description
I_SAIT_CHINESE_6 = 0	Available since version 2.5.
I_SAIT_CHINESE_7 = 1	Available since version 2.5.
I_SAIT_CHINESE_8 = 2	Available since version 2.5.
I_SAIT_CHINESE_9 = 3	Available since version 2.5.

Description

Available intensity types.

Version

Available since version 2.5.

II.1.5.27 IRobotSeismicAnalysis_CHINESE_DesignType

C++

```
enum IRobotSeismicAnalysis_CHINESE_DesignType;
```

C#

```
public enum IRobotSeismicAnalysis_CHINESE_DesignType;
```

Visual Basic

```
Public Enum IRobotSeismicAnalysis_CHINESE_DesignType
```

Members

Members	Description
I_SADT_CHINESE_A = 0	Available since version 2.5.
I_SADT_CHINESE_B = 1	Available since version 2.5.

Description

Available design standards.

Version

Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

II.1.5.28 IRobotSeismicAnalysis_CHINESE_EarthquakeType

C++

```
enum IRobotSeismicAnalysis_CHINESE_EarthquakeType;
```

C#

```
public enum IRobotSeismicAnalysis_CHINESE_EarthquakeType;
```

Visual Basic

```
Public Enum IRobotSeismicAnalysis_CHINESE_EarthquakeType
```

Members

Members	Description
I_SAET_CHINESE_NEAR = 0	Available since version 2.5.
I_SAET_CHINESE_FAR = 1	Available since version 2.5.

Description

Available types of earthquake distance.

Version

Available since version 2.5.

II.1.5.29 IRobotSeismicAnalysis_CIRSOC_103_SoilType

C++

```
enum IRobotSeismicAnalysis_CIRSOC_103_SoilType;
```

C#

```
public enum IRobotSeismicAnalysis_CIRSOC_103_SoilType;
```

Visual Basic

```
Public Enum IRobotSeismicAnalysis_CIRSOC_103_SoilType
```

Members

Members	Description
I_SAST_CIRSOC_103_I = 0	Available since version 2.5.
I_SAST_CIRSOC_103_II = 1	Available since version 2.5.
I_SAST_CIRSOC_103_III = 2	Available since version 2.5.

Description

Available ground types. .

Version

Available since version 2.5.

II.1.5.30 IRobotSeismicAnalysis_CIRSOC_103_StructureType

C++

```
enum IRobotSeismicAnalysis_CIRSOC_103_StructureType;
```

C#

```
public enum IRobotSeismicAnalysis_CIRSOC_103_StructureType;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Enum IRobotSeismicAnalysis_CIRSOC_103_StructureType
```

Members

Members	Description
I_SAST_CIRSOC_103_A0 = 0	Available since version 2.5.
I_SAST_CIRSOC_103_A = 1	Available since version 2.5.
I_SAST_CIRSOC_103_B = 2	Available since version 2.5.

Description

Available structure types.

Version

Available since version 2.5.

II.1.5.31 IRobotSeismicAnalysisDirectionType

C++

```
enum IRobotSeismicAnalysisDirectionType;
```

C#

```
public enum IRobotSeismicAnalysisDirectionType;
```

Visual Basic

```
Public Enum IRobotSeismicAnalysisDirectionType
```

Members

Members	Description
I_SADT_HORIZONTAL = 0	Available since version 2.5.
I_SADT_VERTICAL = 1	Available since version 2.5.

Description

Available direction types.

Version

Available since version 2.5.

II.1.5.32 IRobotSeismicAnalysis_DM_16_1_96_ProtectionCoeffType

C++

```
enum IRobotSeismicAnalysis_DM_16_1_96_ProtectionCoeffType;
```

C#

```
public enum IRobotSeismicAnalysis_DM_16_1_96_ProtectionCoeffType;
```

Visual Basic

```
Public Enum IRobotSeismicAnalysis_DM_16_1_96_ProtectionCoeffType
```

Members

Members	Description
I_SAPCT_DM_16_1_96_1 = 0	Available since version 2.5.
I_SAPCT_DM_16_1_96_12 = 1	Available since version 2.5.
I_SAPCT_DM_16_1_96_14 = 2	Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

Description

Available values of seismic protection coefficients. .

Version

Available since version 2.5.

II.1.5.33 IRobotSeismicAnalysis_P_100_92_ImportanceClassType

C++

```
enum IRobotSeismicAnalysis_P_100_92_ImportanceClassType;
```

C#

```
public enum IRobotSeismicAnalysis_P_100_92_ImportanceClassType;
```

Visual Basic

```
Public Enum IRobotSeismicAnalysis_P_100_92_ImportanceClassType
```

Members

Members	Description
I_SAICT_100_92_I = 0	Available since version 2.5.
I_SAICT_100_92_II = 1	Available since version 2.5.
I_SAICT_100_92_III = 2	Available since version 2.5.
I_SAICT_100_92_IV = 3	Available since version 2.5.

Description

Available importance classes. .

Version

Available since version 2.5.

II.1.5.34 IRobotSeismicAnalysis_PS_69_DampingType

C++

```
enum IRobotSeismicAnalysis_PS_69_DampingType;
```

C#

```
public enum IRobotSeismicAnalysis_PS_69_DampingType;
```

Visual Basic

```
Public Enum IRobotSeismicAnalysis_PS_69_DampingType
```

Members

Members	Description
I_SADT_PS_69_NORMAL = 0	Available since version 2.5.
I_SADT_PS_69_AVERAGE = 1	Available since version 2.5.
I_SADT_PS_69_WEAK = 2	Available since version 2.5.

Description

Available damping types. .

Version

Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

II.1.5.35 IRobotSeismicAnalysis_PS_69_SoilType

C++

```
enum IRobotSeismicAnalysis_PS_69_SoilType;
```

C#

```
public enum IRobotSeismicAnalysis_PS_69_SoilType;
```

Visual Basic

```
Public Enum IRobotSeismicAnalysis_PS_69_SoilType
```

Members

Members	Description
I_SAST_PS_69_RIGID = 0	Available since version 2.5.
I_SAST_PS_69_FLEXIBLE = 1	Available since version 2.5.

Description

Available ground types. .

Version

Available since version 2.5.

II.1.5.36 IRobotSeismicAnalysis_PS_92_StructureType

C++

```
enum IRobotSeismicAnalysis_PS_92_StructureType;
```

C#

```
public enum IRobotSeismicAnalysis_PS_92_StructureType;
```

Visual Basic

```
Public Enum IRobotSeismicAnalysis_PS_92_StructureType
```

Members

Members	Description
I_SAST_PS_92_B = 0	Available since version 2.5.
I_SAST_PS_92_C = 1	Available since version 2.5.
I_SAST_PS_92_D = 2	Available since version 2.5.

Description

Available structure types.

Version

Available since version 2.5.

II.1.5.37 IRobotSeismicAnalysis_PS_92_SiteType

C++

```
enum IRobotSeismicAnalysis_PS_92_SiteType;
```

C#

```
public enum IRobotSeismicAnalysis_PS_92_SiteType;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Enum IRobotSeismicAnalysis_PS_92_SiteType
```

Members

Members	Description
I_SAST_PS_92_S0 = 0	Available since version 2.5.
I_SAST_PS_92_S1 = 1	Available since version 2.5.
I_SAST_PS_92_S2 = 2	Available since version 2.5.
I_SAST_PS_92_S3 = 3	Available since version 2.5.
I_SAST_PS_92_ENVELOPE = 4	Available since version 5.5.

Description

Available site types.

Version

Available since version 2.5.

II.1.5.38 IRobotSeismicAnalysis_RPA_88_UsageType

C++

```
enum IRobotSeismicAnalysis_RPA_88_UsageType;
```

C#

```
public enum IRobotSeismicAnalysis_RPA_88_UsageType;
```

Visual Basic

```
Public Enum IRobotSeismicAnalysis_RPA_88_UsageType
```

Members

Members	Description
I_SAUT_RPA_88_1 = 0	Available since version 2.5.
I_SAUT_RPA_88_2 = 1	Available since version 2.5.
I_SAUT_RPA_88_3 = 2	Available since version 2.5.

Description

Available usage types.

Version

Available since version 2.5.

II.1.5.39 IRobotSeismicAnalysis_RPA_88_CategoryType

C++

```
enum IRobotSeismicAnalysis_RPA_88_CategoryType;
```

C#

```
public enum IRobotSeismicAnalysis_RPA_88_CategoryType;
```

Visual Basic

```
Public Enum IRobotSeismicAnalysis_RPA_88_CategoryType
```

(c) 2014 Autodesk Inc. All rights reserved

Members

Members	Description
I_SACT_RPA_88_1 = 0	Available since version 2.5.
I_SACT_RPA_88_2 = 1	Available since version 2.5.
I_SACT_RPA_88_3 = 2	Available since version 2.5.
I_SACT_RPA_88_4 = 3	Available since version 2.5.
I_SACT_RPA_88_5 = 4	Available since version 2.5.
I_SACT_RPA_88_6 = 5	Available since version 2.5.
I_SACT_RPA_88_7 = 6	Available since version 2.5.
I_SACT_RPA_88_8 = 7	Available since version 2.5.

Description

Available categories.

Version

Available since version 2.5.

II.1.5.40 IRobotSeismicAnalysis_RPA_88_SoilType

C++

```
enum IRobotSeismicAnalysis_RPA_88_SoilType;
```

C#

```
public enum IRobotSeismicAnalysis_RPA_88_SoilType;
```

Visual Basic

```
Public Enum IRobotSeismicAnalysis_RPA_88_SoilType
```

Members

Members	Description
I_SAST_RPA_88_RIGID = 0	Available since version 2.5.
I_SAST_RPA_88_FLEXIBLE = 1	Available since version 2.5.

Description

Available soil types.

Version

Available since version 2.5.

II.1.5.41 IRobotSeismicAnalysis_TURKISH_23098_SoilType

C++

```
enum IRobotSeismicAnalysis_TURKISH_23098_SoilType;
```

C#

```
public enum IRobotSeismicAnalysis_TURKISH_23098_SoilType;
```

Visual Basic

```
Public Enum IRobotSeismicAnalysis_TURKISH_23098_SoilType
```

Members

Members	Description
I_SAST_TURKISH_23098_Z1 = 0	Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

I_SAST_TURKISH_23098_Z2 = 1	Available since version 2.5.
I_SAST_TURKISH_23098_Z3 = 2	Available since version 2.5.
I_SAST_TURKISH_23098_Z4 = 3	Available since version 2.5.

Description

Available soil types.

Version

Available since version 2.5.

II.1.5.42 IRobotSeismicAnalysis_UBC_97_SoilType

C++

```
enum IRobotSeismicAnalysis_UBC_97_SoilType;
```

C#

```
public enum IRobotSeismicAnalysis_UBC_97_SoilType;
```

Visual Basic

```
Public Enum IRobotSeismicAnalysis_UBC_97_SoilType
```

Members

Members	Description
I_SAST_UBC_97_Sa = 0	Available since version 2.5.
I_SAST_UBC_97_Sb = 1	Available since version 2.5.
I_SAST_UBC_97_Sc = 2	Available since version 2.5.
I_SAST_UBC_97_Sd = 3	Available since version 2.5.
I_SAST_UBC_97_Se = 4	Available since version 2.5.
I_SAST_UBC_97_Sf = 5	Available since version 2.5.

Description

Available soil types.

Version

Available since version 2.5.

II.1.5.43 IRobotSeismicAnalysis_UBC_97_SourceType

C++

```
enum IRobotSeismicAnalysis_UBC_97_SourceType;
```

C#

```
public enum IRobotSeismicAnalysis_UBC_97_SourceType;
```

Visual Basic

```
Public Enum IRobotSeismicAnalysis_UBC_97_SourceType
```

Members

Members	Description
I_SAST_UBC_97_A = 0	Available since version 2.5.
I_SAST_UBC_97_B = 1	Available since version 2.5.
I_SAST_UBC_97_C = 2	Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

Description

Available seismic source types.

Version

Available since version 2.5.

II.1.5.44 IRobotSeismicAnalysis_IBC_2000_SiteClassType

C++

```
enum IRobotSeismicAnalysis_IBC_2000_SiteClassType;
```

C#

```
public enum IRobotSeismicAnalysis_IBC_2000_SiteClassType;
```

Visual Basic

```
Public Enum IRobotSeismicAnalysis_IBC_2000_SiteClassType
```

Members

Members	Description
I_SASCT_IBC_2000_A = 0	Available since version 2.5.
I_SASCT_IBC_2000_B = 1	Available since version 2.5.
I_SASCT_IBC_2000_C = 2	Available since version 2.5.
I_SASCT_IBC_2000_D = 3	Available since version 2.5.
I_SASCT_IBC_2000_E = 4	Available since version 2.5.
I_SASCT_IBC_2000_F = 5	Available since version 2.5.

Description

Available soil types.

Version

Available since version 2.5.

II.1.5.45 IRobotSeismicAnalysis_EAK_2000_ImportanceFactorType

C++

```
enum IRobotSeismicAnalysis_EAK_2000_ImportanceFactorType;
```

C#

```
public enum IRobotSeismicAnalysis_EAK_2000_ImportanceFactorType;
```

Visual Basic

```
Public Enum IRobotSeismicAnalysis_EAK_2000_ImportanceFactorType
```

Members

Members	Description
I_SAIFT_EAK_2000_SIGMA1 = 0	Available since version 2.5.
I_SAIFT_EAK_2000_SIGMA2 = 1	Available since version 2.5.
I_SAIFT_EAK_2000_SIGMA3 = 2	Available since version 2.5.
I_SAIFT_EAK_2000_SIGMA4 = 3	Available since version 2.5.

Description

Available importance factors. .

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 2.5.

II.1.5.46 IRobotSeismicAnalysis_EAK_2000_GroundCategoryType

C++

```
enum IRobotSeismicAnalysis_EAK_2000_GroundCategoryType;
```

C#

```
public enum IRobotSeismicAnalysis_EAK_2000_GroundCategoryType;
```

Visual Basic

```
Public Enum IRobotSeismicAnalysis_EAK_2000_GroundCategoryType
```

Members

Members	Description
I_SAGCT_EAK_2000_ALPHA = 0	Available since version 2.5.
I_SAGCT_EAK_2000_BETA = 1	Available since version 2.5.
I_SAGCT_EAK_2000_GAMMA = 2	Available since version 2.5.
I_SAGCT_EAK_2000_DELTA = 3	Available since version 2.5.

Description

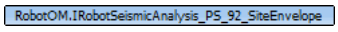
Available ground categories.

Version

Available since version 2.5.

II.1.5.47 IRobotSeismicAnalysis_PS_92_SiteEnvelope

Class Hierarchy



C++

```
interface IRobotSeismicAnalysis_PS_92_SiteEnvelope : IDispatch;
```

C#

```
public interface IRobotSeismicAnalysis_PS_92_SiteEnvelope;
```

Visual Basic

```
Public Interface IRobotSeismicAnalysis_PS_92_SiteEnvelope
```

Version

Available since version 5.5.

II.1.5.47.1 IRobotSeismicAnalysis_PS_92_SiteEnvelope Members

The following tables list the members exposed by IRobotSeismicAnalysis_PS_92_SiteEnvelope.

Public Methods

	Name	Description
	IsActive (see page 269)	Function returns the TRUE value if the indicated site type has been considered in the envelope. .
	SetActive (see page 269)	Function adds/deletes the indicated site type to/from the envelope. .

(c) 2014 Autodesk Inc. All rights reserved

II.1.5.47.2 IRobotSeismicAnalysis_PS_92_SiteEnvelope Methods

The methods of the IRobotSeismicAnalysis_PS_92_SiteEnvelope class are listed here.

Public Methods

	Name	Description
	IsActive (see page 269)	Function returns the TRUE value if the indicated site type has been considered in the envelope. .
	SetActive (see page 269)	Function adds/deletes the indicated site type to/from the envelope. .

II.1.5.47.2.1 IsActive

C++

```
HRESULT IsActive(IRobotSeismicAnalysis_PS_92_SiteType _site, VARIANT_BOOL* ret);
```

C#

```
public bool IsActive(IRobotSeismicAnalysis_PS_92_SiteType _site);
```

Visual Basic

```
Public Function IsActive(_site As IRobotSeismicAnalysis_PS_92_SiteType) As Boolean
```

Description

Function returns the TRUE value if the indicated site type has been considered in the envelope. .

II.1.5.47.2.2 SetActive

C++

```
HRESULT SetActive(IRobotSeismicAnalysis_PS_92_SiteType _site, VARIANT_BOOL _is_active = -1);
```

C#

```
public void SetActive(IRobotSeismicAnalysis_PS_92_SiteType _site, bool _is_active = -1);
```

Visual Basic

```
Public Sub SetActive(_site As IRobotSeismicAnalysis_PS_92_SiteType, Optional _is_active As
Boolean = -1)
```

Description

Function adds/deletes the indicated site type to/from the envelope. .

II.1.5.48 IRobotSeismicAnalysis_RPS_2000_ZoneType

C++

```
enum IRobotSeismicAnalysis_RPS_2000_ZoneType;
```

C#

```
public enum IRobotSeismicAnalysis_RPS_2000_ZoneType;
```

Visual Basic

```
Public Enum IRobotSeismicAnalysis_RPS_2000_ZoneType
```

II.1.5.49 IRobotSeismicAnalysis_RPS_2000_SiteType

C++

```
enum IRobotSeismicAnalysis_RPS_2000_SiteType;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public enum IRobotSeismicAnalysis_RPS_2000_SiteType;
```

Visual Basic

```
Public Enum IRobotSeismicAnalysis_RPS_2000_SiteType
```

II.1.5.50 IRobotSeismicAnalysis_RPS_2000_StructureClass

C++

```
enum IRobotSeismicAnalysis_RPS_2000_StructureClass;
```

C#

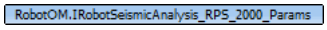
```
public enum IRobotSeismicAnalysis_RPS_2000_StructureClass;
```

Visual Basic

```
Public Enum IRobotSeismicAnalysis_RPS_2000_StructureClass
```

II.1.5.51 IRobotSeismicAnalysis_RPS_2000_Params

Class Hierarchy



C++

```
interface IRobotSeismicAnalysis_RPS_2000_Params : IDispatch;
```

C#

```
public interface IRobotSeismicAnalysis_RPS_2000_Params;
```

Visual Basic

```
Public Interface IRobotSeismicAnalysis_RPS_2000_Params
```

Description

Parameters of the Moroccan seismic code R.P.S. 2000.

II.1.5.51.1 IRobotSeismicAnalysis_RPS_2000_Params Members

The following tables list the members exposed by IRobotSeismicAnalysis_RPS_2000_Params.

Public Fields

	Name	Description
	BehaviorFactor (see page 271)	
	DirectionType (see page 271)	
	ExcitationDir (see page 271)	
	Filter (see page 271)	
	ResidualMode (see page 272)	
	Site (see page 272)	
	StructureClass (see page 272)	
	Zone (see page 272)	

II.1.5.51.2 IRobotSeismicAnalysis_RPS_2000_Params Fields

The fields of the IRobotSeismicAnalysis_RPS_2000_Params class are listed here.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	BehaviorFactor (see page 271)	
	DirectionType (see page 271)	
	ExcitationDir (see page 271)	
	Filter (see page 271)	
	ResidualMode (see page 272)	
	Site (see page 272)	
	StructureClass (see page 272)	
	Zone (see page 272)	

II.1.5.51.2.1 BehaviorFactor

C++

```
HRESULT get_BehaviorFactor(double*);
HRESULT put_BehaviorFactor(double);
```

C#

```
public double BehaviorFactor { get; set; }
```

Visual Basic

```
Public BehaviorFactor As double
```

II.1.5.51.2.2 DirectionType

C++

```
HRESULT get_DirectionType(IRobotSeismicAnalysisDirectionType*);
HRESULT put_DirectionType(IRobotSeismicAnalysisDirectionType);
```

C#

```
public IRobotSeismicAnalysisDirectionType DirectionType { get; set; }
```

Visual Basic

```
Public DirectionType As IRobotSeismicAnalysisDirectionType
```

II.1.5.51.2.3 ExcitationDir

C++

```
HRESULT get_ExcitationDir(IRobotDynamicAnalysisExcitationDirection**);
```

C#

```
public IRobotDynamicAnalysisExcitationDirection ExcitationDir { get; }
```

Visual Basic

```
Public ReadOnly ExcitationDir As IRobotDynamicAnalysisExcitationDirection
```

II.1.5.51.2.4 Filter

C++

```
HRESULT get_Filter(IRobotCaseAnalysisModesFilter**);
HRESULT put_Filter(IRobotCaseAnalysisModesFilter*);
```

C#

```
public IRobotCaseAnalysisModesFilter Filter { get; set; }
```

Visual Basic

```
Public Filter As IRobotCaseAnalysisModesFilter
```

(c) 2014 Autodesk Inc. All rights reserved

II.1.5.51.2.5 ResidualMode

C++

```
HRESULT get_ResidualMode(IRobotSeismicResidualMode**);
```

C#

```
public IRobotSeismicResidualMode ResidualMode { get; }
```

Visual Basic

```
Public ReadOnly ResidualMode As IRobotSeismicResidualMode
```

Version

Available since version 14.7.

II.1.5.51.2.6 Site

C++

```
HRESULT get_Site(IRobotSeismicAnalysis_RPS_2000_SiteType*);
HRESULT put_Site(IRobotSeismicAnalysis_RPS_2000_SiteType);
```

C#

```
public IRobotSeismicAnalysis_RPS_2000_SiteType Site { get; set; }
```

Visual Basic

```
Public Site As IRobotSeismicAnalysis_RPS_2000_SiteType
```

II.1.5.51.2.7 StructureClass

C++

```
HRESULT get_StructureClass(IRobotSeismicAnalysis_RPS_2000_StructureClass*);
HRESULT put_StructureClass(IRobotSeismicAnalysis_RPS_2000_StructureClass);
```

C#

```
public IRobotSeismicAnalysis_RPS_2000_StructureClass StructureClass { get; set; }
```

Visual Basic

```
Public StructureClass As IRobotSeismicAnalysis_RPS_2000_StructureClass
```

II.1.5.51.2.8 Zone

C++

```
HRESULT get_Zone(IRobotSeismicAnalysis_RPS_2000_ZoneType*);
HRESULT put_Zone(IRobotSeismicAnalysis_RPS_2000_ZoneType);
```

C#

```
public IRobotSeismicAnalysis_RPS_2000_ZoneType Zone { get; set; }
```

Visual Basic

```
Public Zone As IRobotSeismicAnalysis_RPS_2000_ZoneType
```

II.1.5.52 IRobotSeismicAnalysis_RPA_2003_ZoneType

C++

```
enum IRobotSeismicAnalysis_RPA_2003_ZoneType;
```

C#

```
public enum IRobotSeismicAnalysis_RPA_2003_ZoneType;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Enum IRobotSeismicAnalysis_RPA_2003_ZoneType
```

II.1.5.53 IRobotSeismicAnalysis_RPA_2003_UsageType

C++

```
enum IRobotSeismicAnalysis_RPA_2003_UsageType;
```

C#

```
public enum IRobotSeismicAnalysis_RPA_2003_UsageType;
```

Visual Basic

```
Public Enum IRobotSeismicAnalysis_RPA_2003_UsageType
```

II.1.5.54 IRobotSeismicAnalysis_RPA_2003_SiteType

C++

```
enum IRobotSeismicAnalysis_RPA_2003_SiteType;
```

C#

```
public enum IRobotSeismicAnalysis_RPA_2003_SiteType;
```

Visual Basic

```
Public Enum IRobotSeismicAnalysis_RPA_2003_SiteType
```

II.1.5.55 IRobotSeismicAnalysis_RPA_2003_Params

Class Hierarchy

RobotOM.IRobotSeismicAnalysis_RPA_2003_Params

C++

```
interface IRobotSeismicAnalysis_RPA_2003_Params : IDispatch;
```

C#

```
public interface IRobotSeismicAnalysis_RPA_2003_Params;
```

Visual Basic

```
Public Interface IRobotSeismicAnalysis_RPA_2003_Params
```

Description

Parameters of the Algerian seismic code RPA 99 (2003). .

II.1.5.55.1 IRobotSeismicAnalysis_RPA_2003_Params Members

The following tables list the members exposed by IRobotSeismicAnalysis_RPA_2003_Params.

Public Fields

	Name	Description
	BehaviorFactor (see page 274)	
	ExcitationDir (see page 274)	
	Filter (see page 274)	
	QualityCoef (see page 274)	
	ResidualMode (see page 275)	
	Site (see page 275)	

(c) 2014 Autodesk Inc. All rights reserved

	Usage (see page 275)	
	Zone (see page 275)	

II.1.5.55.2 IRobotSeismicAnalysis_RPA_2003_Params Fields

The fields of the IRobotSeismicAnalysis_RPA_2003_Params class are listed here.

Public Fields

	Name	Description
	BehaviorFactor (see page 274)	
	ExcitationDir (see page 274)	
	Filter (see page 274)	
	QualityCoef (see page 274)	
	ResidualMode (see page 275)	
	Site (see page 275)	
	Usage (see page 275)	
	Zone (see page 275)	

II.1.5.55.2.1 BehaviorFactor

C++

```
HRESULT get_BehaviorFactor(double*);
HRESULT put_BehaviorFactor(double);
```

C#

```
public double BehaviorFactor { get; set; }
```

Visual Basic

```
Public BehaviorFactor As double
```

II.1.5.55.2.2 ExcitationDir

C++

```
HRESULT get_ExcitationDir(IRobotDynamicAnalysisExcitationDirection**);
```

C#

```
public IRobotDynamicAnalysisExcitationDirection ExcitationDir { get; }
```

Visual Basic

```
Public ReadOnly ExcitationDir As IRobotDynamicAnalysisExcitationDirection
```

II.1.5.55.2.3 Filter

C++

```
HRESULT get_Filter(IRobotCaseAnalysisModesFilter**);
HRESULT put_Filter(IRobotCaseAnalysisModesFilter*);
```

C#

```
public IRobotCaseAnalysisModesFilter Filter { get; set; }
```

Visual Basic

```
Public Filter As IRobotCaseAnalysisModesFilter
```

II.1.5.55.2.4 QualityCoef

C++

```
HRESULT get_QualityCoef(double*);
HRESULT put_QualityCoef(double);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double QualityCoef { get; set; }
```

Visual Basic

```
Public QualityCoef As double
```

II.1.5.55.2.5 ResidualMode

C++

```
HRESULT get_ResidualMode(IRobotSeismicResidualMode**);
```

C#

```
public IRobotSeismicResidualMode ResidualMode { get; }
```

Visual Basic

```
Public ReadOnly ResidualMode As IRobotSeismicResidualMode
```

Version

Available since version 14.7.

II.1.5.55.2.6 Site

C++

```
HRESULT get_Site(IRobotSeismicAnalysis_RPA_2003_SiteType*);
HRESULT put_Site(IRobotSeismicAnalysis_RPA_2003_SiteType);
```

C#

```
public IRobotSeismicAnalysis_RPA_2003_SiteType Site { get; set; }
```

Visual Basic

```
Public Site As IRobotSeismicAnalysis_RPA_2003_SiteType
```

II.1.5.55.2.7 Usage

C++

```
HRESULT get_Usage(IRobotSeismicAnalysis_RPA_2003_UsageType*);
HRESULT put_Usage(IRobotSeismicAnalysis_RPA_2003_UsageType);
```

C#

```
public IRobotSeismicAnalysis_RPA_2003_UsageType Usage { get; set; }
```

Visual Basic

```
Public Usage As IRobotSeismicAnalysis_RPA_2003_UsageType
```

II.1.5.55.2.8 Zone

C++

```
HRESULT get_Zone(IRobotSeismicAnalysis_RPA_2003_ZoneType*);
HRESULT put_Zone(IRobotSeismicAnalysis_RPA_2003_ZoneType);
```

C#

```
public IRobotSeismicAnalysis_RPA_2003_ZoneType Zone { get; set; }
```

Visual Basic

```
Public Zone As IRobotSeismicAnalysis_RPA_2003_ZoneType
```

(c) 2014 Autodesk Inc. All rights reserved

II.1.5.56 IRobotSeismicAnalysis_ITALY_ORDINANZA_Params

Class Hierarchy

RobotOM.IRobotSeismicAnalysis_ITALY_ORDINANZA_Params

C++

```
interface IRobotSeismicAnalysis_ITALY_ORDINANZA_Params : IDispatch;
```

C#

```
public interface IRobotSeismicAnalysis_ITALY_ORDINANZA_Params;
```

Visual Basic

```
Public Interface IRobotSeismicAnalysis_ITALY_ORDINANZA_Params
```

Description

Seismic analysis parameters for the Italian code Ordinanza 3274.

II.1.5.56.1 IRobotSeismicAnalysis_ITALY_ORDINANZA_Params Members

The following tables list the members exposed by IRobotSeismicAnalysis_ITALY_ORDINANZA_Params.

Public Fields

	Name	Description
	Direction (see page 276)	
	ExcitationDir (see page 277)	
	FactorQ (see page 277)	
	Filter (see page 277)	
	Soil (see page 277)	
	Spectrum (see page 277)	
	Zone (see page 278)	

II.1.5.56.2 IRobotSeismicAnalysis_ITALY_ORDINANZA_Params Fields

The fields of the IRobotSeismicAnalysis_ITALY_ORDINANZA_Params class are listed here.

Public Fields

	Name	Description
	Direction (see page 276)	
	ExcitationDir (see page 277)	
	FactorQ (see page 277)	
	Filter (see page 277)	
	Soil (see page 277)	
	Spectrum (see page 277)	
	Zone (see page 278)	

II.1.5.56.2.1 Direction

C++

```
HRESULT get_Direction(IRobotSeismicAnalysis_ITALY_ORDINANZA_Direction*);
HRESULT put_Direction(IRobotSeismicAnalysis_ITALY_ORDINANZA_Direction);
```

C#

```
public IRobotSeismicAnalysis_ITALY_ORDINANZA_Direction Direction { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Direction As IRobotSeismicAnalysis_ITALY_ORDINANZA_Direction
```

II.1.5.56.2.2 ExcitationDir

C++

```
HRESULT get_ExcitationDir(IRobotDynamicAnalysisExcitationDirection**);
```

C#

```
public IRobotDynamicAnalysisExcitationDirection ExcitationDir { get; }
```

Visual Basic

```
Public ReadOnly ExcitationDir As IRobotDynamicAnalysisExcitationDirection
```

II.1.5.56.2.3 FactorQ

C++

```
HRESULT get_FactorQ(double*);
HRESULT put_FactorQ(double);
```

C#

```
public double FactorQ { get; set; }
```

Visual Basic

```
Public FactorQ As double
```

II.1.5.56.2.4 Filter

C++

```
HRESULT get_Filter(IRobotCaseAnalysisModesFilter**);
HRESULT put_Filter(IRobotCaseAnalysisModesFilter*);
```

C#

```
public IRobotCaseAnalysisModesFilter Filter { get; set; }
```

Visual Basic

```
Public Filter As IRobotCaseAnalysisModesFilter
```

II.1.5.56.2.5 Soil

C++

```
HRESULT get_Soil(IRobotSeismicAnalysis_ITALY_ORDINANZA_SoilType*);
HRESULT put_Soil(IRobotSeismicAnalysis_ITALY_ORDINANZA_SoilType);
```

C#

```
public IRobotSeismicAnalysis_ITALY_ORDINANZA_SoilType Soil { get; set; }
```

Visual Basic

```
Public Soil As IRobotSeismicAnalysis_ITALY_ORDINANZA_SoilType
```

II.1.5.56.2.6 Spectrum

C++

```
HRESULT get_Spectrum(IRobotSeismicAnalysis_ITALY_ORDINANZA_Spectrum*);
HRESULT put_Spectrum(IRobotSeismicAnalysis_ITALY_ORDINANZA_Spectrum);
```

C#

```
public IRobotSeismicAnalysis_ITALY_ORDINANZA_Spectrum Spectrum { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Spectrum As IRobotSeismicAnalysis_ITALY_ORDINANZA_Spectrum
```

II.1.5.56.2.7 Zone

C++

```
HRESULT get_Zone(IRobotSeismicAnalysis_ITALY_ORDINANZA_ZoneType*);
HRESULT put_Zone(IRobotSeismicAnalysis_ITALY_ORDINANZA_ZoneType);
```

C#

```
public IRobotSeismicAnalysis_ITALY_ORDINANZA_ZoneType Zone { get; set; }
```

Visual Basic

```
Public Zone As IRobotSeismicAnalysis_ITALY_ORDINANZA_ZoneType
```

II.1.5.57 IRobotSeismicAnalysis_ITALY_ORDINANZA_SoilType

C++

```
enum IRobotSeismicAnalysis_ITALY_ORDINANZA_SoilType;
```

C#

```
public enum IRobotSeismicAnalysis_ITALY_ORDINANZA_SoilType;
```

Visual Basic

```
Public Enum IRobotSeismicAnalysis_ITALY_ORDINANZA_SoilType
```

Description

Soil type according to the Italian code Ordinanza 3274.

II.1.5.58 IRobotSeismicAnalysis_ITALY_ORDINANZA_ZoneType

C++

```
enum IRobotSeismicAnalysis_ITALY_ORDINANZA_ZoneType;
```

C#

```
public enum IRobotSeismicAnalysis_ITALY_ORDINANZA_ZoneType;
```

Visual Basic

```
Public Enum IRobotSeismicAnalysis_ITALY_ORDINANZA_ZoneType
```

II.1.5.59 IRobotSeismicAnalysis_ITALY_ORDINANZA_Spectrum

C++

```
enum IRobotSeismicAnalysis_ITALY_ORDINANZA_Spectrum;
```

C#

```
public enum IRobotSeismicAnalysis_ITALY_ORDINANZA_Spectrum;
```

Visual Basic

```
Public Enum IRobotSeismicAnalysis_ITALY_ORDINANZA_Spectrum
```

II.1.5.60 IRobotSeismicAnalysis_ITALY_ORDINANZA_Direction

C++

```
enum IRobotSeismicAnalysis_ITALY_ORDINANZA_Direction;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

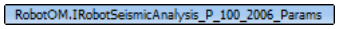
```
public enum IRobotSeismicAnalysis_ITALY_ORDINANZA_Direction;
```

Visual Basic

```
Public Enum IRobotSeismicAnalysis_ITALY_ORDINANZA_Direction
```

II.1.5.61 IRobotSeismicAnalysis_P_100_2006_Params

Class Hierarchy



C++

```
interface IRobotSeismicAnalysis_P_100_2006_Params : IDispatch;
```

C#

```
public interface IRobotSeismicAnalysis_P_100_2006_Params;
```

Visual Basic

```
Public Interface IRobotSeismicAnalysis_P_100_2006_Params
```

Description

Parameters of seismic analysis for Romanian code (P100-1-2006).

Version

Available since version 8.1.

II.1.5.61.1 IRobotSeismicAnalysis_P_100_2006_Params Members

The following tables list the members exposed by IRobotSeismicAnalysis_P_100_2006_Params.

Public Fields

	Name	Description
	Agg (see page 280)	
	B0 (see page 280)	
	BehaviorFactor (see page 280)	
	ExcitationDir (see page 280)	
	ImportanceFactor (see page 281)	
	SpectrumType (see page 281)	
	Tb (see page 281)	
	Tc (see page 281)	
	Td (see page 282)	

II.1.5.61.2 IRobotSeismicAnalysis_P_100_2006_Params Fields

The fields of the IRobotSeismicAnalysis_P_100_2006_Params class are listed here.

Public Fields

	Name	Description
	Agg (see page 280)	
	B0 (see page 280)	
	BehaviorFactor (see page 280)	
	ExcitationDir (see page 280)	
	ImportanceFactor (see page 281)	
	SpectrumType (see page 281)	

(c) 2014 Autodesk Inc. All rights reserved

	Tb (see page 281)	
	Tc (see page 281)	
	Td (see page 282)	

II.1.5.61.2.1 Agg

C++

```
HRESULT get_Agg(double*);
HRESULT put_Agg(double);
```

C#

```
public double Agg { get; set; }
```

Visual Basic

```
Public Agg As double
```

Version

Available since version 8.1.

II.1.5.61.2.2 B0

C++

```
HRESULT get_B0(double*);
HRESULT put_B0(double);
```

C#

```
public double B0 { get; set; }
```

Visual Basic

```
Public B0 As double
```

Version

Available since version 8.1.

II.1.5.61.2.3 BehaviorFactor

C++

```
HRESULT get_BehaviorFactor(double*);
HRESULT put_BehaviorFactor(double);
```

C#

```
public double BehaviorFactor { get; set; }
```

Visual Basic

```
Public BehaviorFactor As double
```

Version

Available since version 8.1.

II.1.5.61.2.4 ExcitationDir

C++

```
HRESULT get_ExcitationDir(IRobotDynamicAnalysisExcitationDirection**);
```

C#

```
public IRobotDynamicAnalysisExcitationDirection ExcitationDir { get; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public ReadOnly ExcitationDir As IRobotDynamicAnalysisExcitationDirection
```

Version

Available since version 8.1.

II.1.5.61.2.5 ImportanceFactor

C++

```
HRESULT get_ImportanceFactor(double*);
HRESULT put_ImportanceFactor(double);
```

C#

```
public double ImportanceFactor { get; set; }
```

Visual Basic

```
Public ImportanceFactor As double
```

Version

Available since version 8.1.

II.1.5.61.2.6 SpectrumType

C++

```
HRESULT get_SpectrumType(IRobotSeismicAnalysisSpectrumType*);
HRESULT put_SpectrumType(IRobotSeismicAnalysisSpectrumType);
```

C#

```
public IRobotSeismicAnalysisSpectrumType SpectrumType { get; set; }
```

Visual Basic

```
Public SpectrumType As IRobotSeismicAnalysisSpectrumType
```

Version

Available since version 8.1.

II.1.5.61.2.7 Tb

C++

```
HRESULT get_Tb(double*);
HRESULT put_Tb(double);
```

C#

```
public double Tb { get; set; }
```

Visual Basic

```
Public Tb As double
```

Version

Available since version 8.1.

II.1.5.61.2.8 Tc

C++

```
HRESULT get_Tc(double*);
HRESULT put_Tc(double);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double Tc { get; set; }
```

Visual Basic

```
Public Tc As double
```

Version

Available since version 8.1.

II.1.5.61.2.9 Td

C++

```
HRESULT get_Td(double*);
HRESULT put_Td(double);
```

C#

```
public double Td { get; set; }
```

Visual Basic

```
Public Td As double
```

Version

Available since version 8.1.

II.1.5.62 IRobotSeismicAnalysis_IBC_2006_Params

Class Hierarchy

RobotOM.IRobotSeismicAnalysis_IBC_2006_Params

C++

```
interface IRobotSeismicAnalysis_IBC_2006_Params : IDispatch;
```

C#

```
public interface IRobotSeismicAnalysis_IBC_2006_Params;
```

Visual Basic

```
Public Interface IRobotSeismicAnalysis_IBC_2006_Params
```

Description

Seismic analysis parameters for IBC 2006 released in the USA.

Version

Available since version 8.1.

II.1.5.62.1 IRobotSeismicAnalysis_IBC_2006_Params Members

The following tables list the members exposed by IRobotSeismicAnalysis_IBC_2006_Params.

Public Fields

	Name	Description
	BehaviorFactor (see page 283)	Behavior factor.
	Direction (see page 283)	Seismic excitation direction.
	ExcitationDir (see page 284)	
	Filter (see page 284)	Description of modes taken into account during the structure dynamic analysis.
	I (see page 284)	Structure importance factor.

(c) 2014 Autodesk Inc. All rights reserved

	S1 (see page 284)	Spectral acceleration for short periods.
	SiteClass (see page 285)	Soil.
	Ss (see page 285)	Spectral acceleration for the period 1s.
	TL (see page 285)	term period.

II.1.5.62.2 IRobotSeismicAnalysis_IBC_2006_Params Fields

The fields of the IRobotSeismicAnalysis_IBC_2006_Params class are listed here.

Public Fields

	Name	Description
	BehaviorFactor (see page 283)	Behavior factor.
	Direction (see page 283)	Seismic excitation direction.
	ExcitationDir (see page 284)	
	Filter (see page 284)	Description of modes taken into account during the structure dynamic analysis.
	I (see page 284)	Structure importance factor.
	S1 (see page 284)	Spectral acceleration for short periods.
	SiteClass (see page 285)	Soil.
	Ss (see page 285)	Spectral acceleration for the period 1s.
	TL (see page 285)	term period.

II.1.5.62.2.1 BehaviorFactor

C++

```
HRESULT get_BehaviorFactor(double*);
HRESULT put_BehaviorFactor(double);
```

C#

```
public double BehaviorFactor { get; set; }
```

Visual Basic

```
Public BehaviorFactor As double
```

Description

Behavior factor.

Version

Available since version 8.1.

II.1.5.62.2.2 Direction

C++

```
HRESULT get_Direction(IRobotGeoPoint3D**);
HRESULT put_Direction(IRobotGeoPoint3D*);
```

C#

```
public IRobotGeoPoint3D Direction { get; set; }
```

Visual Basic

```
Public Direction As IRobotGeoPoint3D
```

Description

Seismic excitation direction.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 8.1.

II.1.5.62.2.3 ExcitationDir

C++

```
HRESULT get_ExcitationDir(IRobotDynamicAnalysisExcitationDirection**);
```

C#

```
public IRobotDynamicAnalysisExcitationDirection ExcitationDir { get; }
```

Visual Basic

```
Public ReadOnly ExcitationDir As IRobotDynamicAnalysisExcitationDirection
```

Version

Available since version 8.1.

II.1.5.62.2.4 Filter

C++

```
HRESULT get_Filter(IRobotCaseAnalysisModesFilter**);
HRESULT put_Filter(IRobotCaseAnalysisModesFilter*);
```

C#

```
public IRobotCaseAnalysisModesFilter Filter { get; set; }
```

Visual Basic

```
Public Filter As IRobotCaseAnalysisModesFilter
```

Description

Description of modes taken into account during the structure dynamic analysis.

Version

Available since version 8.1.

II.1.5.62.2.5 I

C++

```
HRESULT get_I(double*);
HRESULT put_I(double);
```

C#

```
public double I { get; set; }
```

Visual Basic

```
Public I As double
```

Description

Structure importance factor.

Version

Available since version 8.1.

II.1.5.62.2.6 S1

C++

```
HRESULT get_S1(double*);
```

(c) 2014 Autodesk Inc. All rights reserved

```
HRESULT put_S1(double);
```

C#

```
public double S1 { get; set; }
```

Visual Basic

```
Public S1 As double
```

Description

Spectral acceleration for short periods.

Version

Available since version 8.1.

II.1.5.62.2.7 SiteClass

C++

```
HRESULT get_SiteClass(IRobotSeismicAnalysis_IBC_2006_SiteClassType*);
HRESULT put_SiteClass(IRobotSeismicAnalysis_IBC_2006_SiteClassType);
```

C#

```
public IRobotSeismicAnalysis_IBC_2006_SiteClassType SiteClass { get; set; }
```

Visual Basic

```
Public SiteClass As IRobotSeismicAnalysis_IBC_2006_SiteClassType
```

Description

Soil.

Version

Available since version 8.1.

II.1.5.62.2.8 Ss

C++

```
HRESULT get_Ss(double*);
HRESULT put_Ss(double);
```

C#

```
public double Ss { get; set; }
```

Visual Basic

```
Public Ss As double
```

Description

Spectral acceleration for the period 1s.

Version

Available since version 8.1.

II.1.5.62.2.9 TL

C++

```
HRESULT get_TL(double*);
HRESULT put_TL(double);
```

C#

```
public double TL { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public TL As double
```

Description

term period.

Version

Available since version 8.1.

II.1.5.63 IRobotSeismicAnalysis_IBC_2006_SiteClassType

C++

```
enum IRobotSeismicAnalysis_IBC_2006_SiteClassType;
```

C#

```
public enum IRobotSeismicAnalysis_IBC_2006_SiteClassType;
```

Visual Basic

```
Public Enum IRobotSeismicAnalysis_IBC_2006_SiteClassType
```

Members

Members	Description
I_SASCT_IBC_2006_A = 0	Available since version 8.1.
I_SASCT_IBC_2006_B = 1	Available since version 8.1.
I_SASCT_IBC_2006_C = 2	Available since version 8.1.
I_SASCT_IBC_2006_D = 3	Available since version 8.1.
I_SASCT_IBC_2006_E = 4	Available since version 8.1.
I_SASCT_IBC_2006_F = 5	Available since version 8.1.

Description

Available soil types.

Version

Available since version 8.1.

II.1.5.64 IRobotSeismicAnalysis_PS_92_2008_Params

Class Hierarchy

RobotOM.IRobotSeismicAnalysis_PS_92_2008_Params

C++

```
interface IRobotSeismicAnalysis_PS_92_2008_Params : IDispatch;
```

C#

```
public interface IRobotSeismicAnalysis_PS_92_2008_Params;
```

Visual Basic

```
Public Interface IRobotSeismicAnalysis_PS_92_2008_Params
```

Description

Seismic analysis parameters for the French code PS 92 - 2008.

Version

Available since version 8.7.

(c) 2014 Autodesk Inc. All rights reserved

II.1.5.64.1 IRobotSeismicAnalysis_PS_92_2008_Params Members

The following tables list the members exposed by IRobotSeismicAnalysis_PS_92_2008_Params.

Public Fields

	Name	Description
	Ag (see page 287)	Acceleration factor.
	BehaviorFactor (see page 288)	Behavior factor.
	Direction (see page 288)	Seismic excitation direction.
	DirectionType (see page 288)	Direction (see page 288).
	ExcitationDir (see page 289)	Definition of the excitation direction.
	Filter (see page 289)	Description of modes taken into account during the structure dynamic analysis.
	ResidualMode (see page 289)	
	Site (see page 289)	Site.
	SiteEnvelope (see page 290)	
	SpectrumType (see page 290)	Spectrum.
	StructureType (see page 290)	
	Topography (see page 291)	Topography.
	ZoneType (see page 291)	Seismic zone no.

II.1.5.64.2 IRobotSeismicAnalysis_PS_92_2008_Params Fields

The fields of the IRobotSeismicAnalysis_PS_92_2008_Params class are listed here.

Public Fields

	Name	Description
	Ag (see page 287)	Acceleration factor.
	BehaviorFactor (see page 288)	Behavior factor.
	Direction (see page 288)	Seismic excitation direction.
	DirectionType (see page 288)	Direction (see page 288).
	ExcitationDir (see page 289)	Definition of the excitation direction.
	Filter (see page 289)	Description of modes taken into account during the structure dynamic analysis.
	ResidualMode (see page 289)	
	Site (see page 289)	Site.
	SiteEnvelope (see page 290)	
	SpectrumType (see page 290)	Spectrum.
	StructureType (see page 290)	
	Topography (see page 291)	Topography.
	ZoneType (see page 291)	Seismic zone no.

II.1.5.64.2.1 Ag

C++

```
HRESULT get_Ag( double*);
HRESULT put_Ag( double);
```

C#

```
public  double Ag { get; set; }
```

Visual Basic

```
Public Ag As  double
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Acceleration factor.

Version

Available since version 9.

II.1.5.64.2.2 BehaviorFactor

C++

```
HRESULT get_BehaviorFactor(double*);
HRESULT put_BehaviorFactor(double);
```

C#

```
public double BehaviorFactor { get; set; }
```

Visual Basic

```
Public BehaviorFactor As double
```

Description

Behavior factor.

Version

Available since version 8.7.

II.1.5.64.2.3 Direction

C++

```
HRESULT get_Direction(IRobotGeoPoint3D**);
HRESULT put_Direction(IRobotGeoPoint3D*);
```

C#

```
public IRobotGeoPoint3D Direction { get; set; }
```

Visual Basic

```
Public Direction As IRobotGeoPoint3D
```

Description

Seismic excitation direction.

Version

Available since version 8.7.

II.1.5.64.2.4 DirectionType

C++

```
HRESULT get_DirectionType(IRobotSeismicAnalysisDirectionType*);
HRESULT put_DirectionType(IRobotSeismicAnalysisDirectionType);
```

C#

```
public IRobotSeismicAnalysisDirectionType DirectionType { get; set; }
```

Visual Basic

```
Public DirectionType As IRobotSeismicAnalysisDirectionType
```

Description

Direction (see page 288).

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 8.7.

II.1.5.64.2.5 ExcitationDir

C++

```
HRESULT get_ExcitationDir(IRobotDynamicAnalysisExcitationDirection**);
```

C#

```
public IRobotDynamicAnalysisExcitationDirection ExcitationDir { get; }
```

Visual Basic

```
Public ReadOnly ExcitationDir As IRobotDynamicAnalysisExcitationDirection
```

Description

Definition of the excitation direction.

Version

Available since version 8.7.

II.1.5.64.2.6 Filter

C++

```
HRESULT get_Filter(IRobotCaseAnalysisModesFilter**);
HRESULT put_Filter(IRobotCaseAnalysisModesFilter*);
```

C#

```
public IRobotCaseAnalysisModesFilter Filter { get; set; }
```

Visual Basic

```
Public Filter As IRobotCaseAnalysisModesFilter
```

Description

Description of modes taken into account during the structure dynamic analysis.

Version

Available since version 8.7.

II.1.5.64.2.7 ResidualMode

C++

```
HRESULT get_ResidualMode(IRobotSeismicResidualMode**);
```

C#

```
public IRobotSeismicResidualMode ResidualMode { get; }
```

Visual Basic

```
Public ReadOnly ResidualMode As IRobotSeismicResidualMode
```

Version

Available since version 14.7.

II.1.5.64.2.8 Site

C++

```
HRESULT get_Site(IRobotSeismicAnalysis_PS_92_2008_SiteType*);
HRESULT put_Site(IRobotSeismicAnalysis_PS_92_2008_SiteType);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRobotSeismicAnalysis_PS_92_2008_SiteType Site { get; set; }
```

Visual Basic

```
Public Site As IRobotSeismicAnalysis_PS_92_2008_SiteType
```

Description

Site.

Version

Available since version 8.7.

II.1.5.64.2.9 SiteEnvelope

C++

```
HRESULT get_SiteEnvelope(IRobotSeismicAnalysis_PS_92_2008_SiteEnvelope**);
```

C#

```
public IRobotSeismicAnalysis_PS_92_2008_SiteEnvelope SiteEnvelope { get; }
```

Visual Basic

```
Public ReadOnly SiteEnvelope As IRobotSeismicAnalysis_PS_92_2008_SiteEnvelope
```

Version

Available since version 8.7.

II.1.5.64.2.10 SpectrumType

C++

```
HRESULT get_SpectrumType(IRobotSeismicAnalysisSpectrumType*);
HRESULT put_SpectrumType(IRobotSeismicAnalysisSpectrumType);
```

C#

```
public IRobotSeismicAnalysisSpectrumType SpectrumType { get; set; }
```

Visual Basic

```
Public SpectrumType As IRobotSeismicAnalysisSpectrumType
```

Description

Spectrum.

Version

Available since version 8.7.

II.1.5.64.2.11 StructureType

C++

```
HRESULT get_StructureType(IRobotSeismicAnalysis_PS_92_2008_StructureType*);
HRESULT put_StructureType(IRobotSeismicAnalysis_PS_92_2008_StructureType);
```

C#

```
public IRobotSeismicAnalysis_PS_92_2008_StructureType StructureType { get; set; }
```

Visual Basic

```
Public StructureType As IRobotSeismicAnalysis_PS_92_2008_StructureType
```

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 8.7.

II.1.5.64.2.12 Topography

C++

```
HRESULT get_Topography(double*);
HRESULT put_Topography(double);
```

C#

```
public double Topography { get; set; }
```

Visual Basic

```
Public Topography As double
```

Description

Topography.

Version

Available since version 8.7.

II.1.5.64.2.13 ZoneType

C++

```
HRESULT get_ZoneType(IRobotSeismicAnalysis_PS_92_2008_ZoneType*);
HRESULT put_ZoneType(IRobotSeismicAnalysis_PS_92_2008_ZoneType);
```

C#

```
public IRobotSeismicAnalysis_PS_92_2008_ZoneType ZoneType { get; set; }
```

Visual Basic

```
Public ZoneType As IRobotSeismicAnalysis_PS_92_2008_ZoneType
```

Description

Seismic zone no.

Version

Available since version 8.7.

II.1.5.65 IRobotSeismicAnalysis_PS_92_2008_ZoneType

C++

```
enum IRobotSeismicAnalysis_PS_92_2008_ZoneType;
```

C#

```
public enum IRobotSeismicAnalysis_PS_92_2008_ZoneType;
```

Visual Basic

```
Public Enum IRobotSeismicAnalysis_PS_92_2008_ZoneType
```

Members

Members	Description
I_SAZT_PS_92_2008_2 = 0	Available since version 8.7.
I_SAZT_PS_92_2008_3 = 1	Available since version 8.7.
I_SAZT_PS_92_2008_4 = 2	Available since version 8.7.

(c) 2014 Autodesk Inc. All rights reserved

I_SAZT_PS_92_2008_5 = 4	Available since version 8.7.

Description

Available numbers of seismic zones.

Version

Available since version 8.7.

II.1.5.66 IRobotSeismicAnalysis_PS_92_2008_StructureType

C++

```
enum IRobotSeismicAnalysis_PS_92_2008_StructureType;
```

C#

```
public enum IRobotSeismicAnalysis_PS_92_2008_StructureType;
```

Visual Basic

```
Public Enum IRobotSeismicAnalysis_PS_92_2008_StructureType
```

Members

Members	Description
I_SAST_PS_92_2008_B = 0	Available since version 8.7.
I_SAST_PS_92_2008_C = 1	Available since version 8.7.
I_SAST_PS_92_2008_D = 2	Available since version 8.7.

Description

Available structure types.

Version

Available since version 8.7.

II.1.5.67 IRobotSeismicAnalysis_PS_92_2008_SiteType

C++

```
enum IRobotSeismicAnalysis_PS_92_2008_SiteType;
```

C#

```
public enum IRobotSeismicAnalysis_PS_92_2008_SiteType;
```

Visual Basic

```
Public Enum IRobotSeismicAnalysis_PS_92_2008_SiteType
```

Members

Members	Description
I_SAST_PS_92_2008_S0 = 0	Available since version 8.7.
I_SAST_PS_92_2008_S1 = 1	Available since version 8.7.
I_SAST_PS_92_2008_S2 = 2	Available since version 8.7.
I_SAST_PS_92_2008_S3 = 3	Available since version 8.7.
I_SAST_PS_92_2008_ENVELOPE = 4	Available since version 8.7.

Description

Available site types.

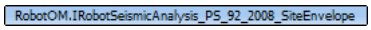
(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 8.7.

II.1.5.68 IRobotSeismicAnalysis_PS_92_2008_SiteEnvelope

Class Hierarchy



C++

```
interface IRobotSeismicAnalysis_PS_92_2008_SiteEnvelope : IDispatch;
```

C#

```
public interface IRobotSeismicAnalysis_PS_92_2008_SiteEnvelope;
```

Visual Basic

```
Public Interface IRobotSeismicAnalysis_PS_92_2008_SiteEnvelope
```

Version

Available since version 8.7.

II.1.5.68.1 IRobotSeismicAnalysis_PS_92_2008_SiteEnvelope Members

The following tables list the members exposed by IRobotSeismicAnalysis_PS_92_2008_SiteEnvelope.

Public Methods

	Name	Description
	IsActive (see page 293)	Function returns the TRUE value if the indicated site type has been included in the envelope.
	SetActive (see page 294)	Function adds/deletes an indicated site type to/from the envelope.

II.1.5.68.2 IRobotSeismicAnalysis_PS_92_2008_SiteEnvelope Methods

The methods of the IRobotSeismicAnalysis_PS_92_2008_SiteEnvelope class are listed here.

Public Methods

	Name	Description
	IsActive (see page 293)	Function returns the TRUE value if the indicated site type has been included in the envelope.
	SetActive (see page 294)	Function adds/deletes an indicated site type to/from the envelope.

II.1.5.68.2.1 IsActive

C++

```
HRESULT IsActive(IRobotSeismicAnalysis_PS_92_2008_SiteType _site, VARIANT_BOOL* ret);
```

C#

```
public bool IsActive(IRobotSeismicAnalysis_PS_92_2008_SiteType _site);
```

Visual Basic

```
Public Function IsActive(_site As IRobotSeismicAnalysis_PS_92_2008_SiteType) As Boolean
```

Description

Function returns the TRUE value if the indicated site type has been included in the envelope.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 8.7.

II.1.5.68.2.2 SetActive

C++

```
HRESULT SetActive(IRobotSeismicAnalysis_PS_92_2008_SiteType _site, VARIANT_BOOL _is_active
= -1);
```

C#

```
public void SetActive(IRobotSeismicAnalysis_PS_92_2008_SiteType _site, bool _is_active =
-1);
```

Visual Basic

```
Public Sub SetActive(_site As IRobotSeismicAnalysis_PS_92_2008_SiteType, Optional
_is_active As Boolean = -1)
```

Description

Function adds/deletes an indicated site type to/from the envelope.

Version

Available since version 8.7.

II.1.5.69 IRobotSeismicResidualModeDefinitionType

C++

```
enum IRobotSeismicResidualModeDefinitionType;
```

C#

```
public enum IRobotSeismicResidualModeDefinitionType;
```

Visual Basic

```
Public Enum IRobotSeismicResidualModeDefinitionType
```

Members

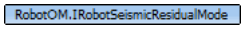
Members	Description
I_SRMDT_LAST_CALC_MODE = 0	Available since version 14.7.
I_SRMDT_LIMIT_FREQUENCY = 1	Available since version 14.7.

Version

Available since version 14.7.

II.1.5.70 IRobotSeismicResidualMode

Class Hierarchy



C++

```
interface IRobotSeismicResidualMode : IDispatch;
```

C#

```
public interface IRobotSeismicResidualMode;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Interface IRobotSeismicResidualMode
```

Version

Available since version 14.7.

II.1.5.70.1 IRobotSeismicResidualMode Members

The following tables list the members exposed by IRobotSeismicResidualMode.

Public Fields

	Name	Description
	AugmentationFactor (see page 295)	
	DefinitionMethod (see page 295)	
	IsActive (see page 296)	
	LimitFrequency (see page 296)	

II.1.5.70.2 IRobotSeismicResidualMode Fields

The fields of the IRobotSeismicResidualMode class are listed here.

Public Fields

	Name	Description
	AugmentationFactor (see page 295)	
	DefinitionMethod (see page 295)	
	IsActive (see page 296)	
	LimitFrequency (see page 296)	

II.1.5.70.2.1 AugmentationFactor

C++

```
HRESULT get_AugmentationFactor(double*);
HRESULT put_AugmentationFactor(double);
```

C#

```
public double AugmentationFactor { get; set; }
```

Visual Basic

```
Public AugmentationFactor As double
```

Version

Available since version 14.7.

II.1.5.70.2.2 DefinitionMethod

C++

```
HRESULT get_DefinitionMethod(IRobotSeismicResidualModeDefinitionType*);
HRESULT put_DefinitionMethod(IRobotSeismicResidualModeDefinitionType);
```

C#

```
public IRobotSeismicResidualModeDefinitionType DefinitionMethod { get; set; }
```

Visual Basic

```
Public DefinitionMethod As IRobotSeismicResidualModeDefinitionType
```

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 14.7.

II.1.5.70.2.3 IsActive

C++

```
HRESULT get_IsActive(VARIANT_BOOL*);
HRESULT put_IsActive(VARIANT_BOOL);
```

C#

```
public bool IsActive { get; set; }
```

Visual Basic

```
Public IsActive As Boolean
```

Version

Available since version 14.7.

II.1.5.70.2.4 LimitFrequency

C++

```
HRESULT get_LimitFrequency(double*);
HRESULT put_LimitFrequency(double);
```

C#

```
public double LimitFrequency { get; set; }
```

Visual Basic

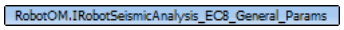
```
Public LimitFrequency As double
```

Version

Available since version 14.7.

II.1.5.71 IRobotSeismicAnalysis_EC8_General_Params

Class Hierarchy



C++

```
interface IRobotSeismicAnalysis_EC8_General_Params : IDispatch;
```

C#

```
public interface IRobotSeismicAnalysis_EC8_General_Params;
```

Visual Basic

```
Public Interface IRobotSeismicAnalysis_EC8_General_Params
```

Description

.

Version

Available since version 14.7.

II.1.5.71.1 IRobotSeismicAnalysis_EC8_General_Params Members

The following tables list the members exposed by IRobotSeismicAnalysis_EC8_General_Params.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	Ag (see page 297)	.
	B (see page 298)	
	BehaviorFactor (see page 298)	
	Direction (see page 298)	
	DirectionType (see page 298)	
	ExcitationDir (see page 299)	.
	Filter (see page 299)	
	ResidualMode (see page 299)	
	S (see page 299)	
	Spectrum (see page 300)	
	Tb (see page 300)	
	Tc (see page 300)	
	Td (see page 300)	

II.1.5.71.2 IRobotSeismicAnalysis_EC8_General_Params Fields

The fields of the IRobotSeismicAnalysis_EC8_General_Params class are listed here.

Public Fields

	Name	Description
	Ag (see page 297)	.
	B (see page 298)	
	BehaviorFactor (see page 298)	
	Direction (see page 298)	
	DirectionType (see page 298)	
	ExcitationDir (see page 299)	.
	Filter (see page 299)	
	ResidualMode (see page 299)	
	S (see page 299)	
	Spectrum (see page 300)	
	Tb (see page 300)	
	Tc (see page 300)	
	Td (see page 300)	

II.1.5.71.2.1 Ag

C++

```
HRESULT get_Ag(double*);
HRESULT put_Ag(double);
```

C#

```
public double Ag { get; set; }
```

Visual Basic

```
Public Ag As double
```

Description

.

Version

Available since version 14.7.

(c) 2014 Autodesk Inc. All rights reserved

II.1.5.71.2.2 B

C++

```
HRESULT get_B(double*);
HRESULT put_B(double);
```

C#

```
public double B { get; set; }
```

Visual Basic

```
Public B As double
```

Version

Available since version 14.7.

II.1.5.71.2.3 BehaviorFactor

C++

```
HRESULT get_BehaviorFactor(double*);
HRESULT put_BehaviorFactor(double);
```

C#

```
public double BehaviorFactor { get; set; }
```

Visual Basic

```
Public BehaviorFactor As double
```

Version

Available since version 14.7.

II.1.5.71.2.4 Direction

C++

```
HRESULT get_Direction(IRobotGeoPoint3D**);
HRESULT put_Direction(IRobotGeoPoint3D*);
```

C#

```
public IRobotGeoPoint3D Direction { get; set; }
```

Visual Basic

```
Public Direction As IRobotGeoPoint3D
```

Version

Available since version 14.7.

II.1.5.71.2.5 DirectionType

C++

```
HRESULT get_DirectionType(IRobotSeismicAnalysisDirectionType*);
HRESULT put_DirectionType(IRobotSeismicAnalysisDirectionType);
```

C#

```
public IRobotSeismicAnalysisDirectionType DirectionType { get; set; }
```

Visual Basic

```
Public DirectionType As IRobotSeismicAnalysisDirectionType
```

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 14.7.

II.1.5.71.2.6 ExcitationDir

C++

```
HRESULT get_ExcitationDir(IRobotDynamicAnalysisExcitationDirection**);
```

C#

```
public IRobotDynamicAnalysisExcitationDirection ExcitationDir { get; }
```

Visual Basic

```
Public ReadOnly ExcitationDir As IRobotDynamicAnalysisExcitationDirection
```

Description

.

Version

Available since version 14.7.

II.1.5.71.2.7 Filter

C++

```
HRESULT get_Filter(IRobotCaseAnalysisModesFilter**);
HRESULT put_Filter(IRobotCaseAnalysisModesFilter*);
```

C#

```
public IRobotCaseAnalysisModesFilter Filter { get; set; }
```

Visual Basic

```
Public Filter As IRobotCaseAnalysisModesFilter
```

Version

Available since version 14.7.

II.1.5.71.2.8 ResidualMode

C++

```
HRESULT get_ResidualMode(IRobotSeismicResidualMode**);
```

C#

```
public IRobotSeismicResidualMode ResidualMode { get; }
```

Visual Basic

```
Public ReadOnly ResidualMode As IRobotSeismicResidualMode
```

Version

Available since version 14.7.

II.1.5.71.2.9 S

C++

```
HRESULT get_S(double*);
HRESULT put_S(double);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double S { get; set; }
```

Visual Basic

```
Public S As double
```

Version

Available since version 14.7.

II.1.5.71.2.10 Spectrum

C++

```
HRESULT get_Spectrum(IRobotSeismicAnalysisSpectrumType*);
HRESULT put_Spectrum(IRobotSeismicAnalysisSpectrumType);
```

C#

```
public IRobotSeismicAnalysisSpectrumType Spectrum { get; set; }
```

Visual Basic

```
Public Spectrum As IRobotSeismicAnalysisSpectrumType
```

Version

Available since version 14.7.

II.1.5.71.2.11 Tb

C++

```
HRESULT get_Tb(double*);
HRESULT put_Tb(double);
```

C#

```
public double Tb { get; set; }
```

Visual Basic

```
Public Tb As double
```

Version

Available since version 14.7.

II.1.5.71.2.12 Tc

C++

```
HRESULT get_Tc(double*);
HRESULT put_Tc(double);
```

C#

```
public double Tc { get; set; }
```

Visual Basic

```
Public Tc As double
```

Version

Available since version 14.7.

II.1.5.71.2.13 Td

C++

```
HRESULT get_Td(double*);
```

(c) 2014 Autodesk Inc. All rights reserved

```
HRESULT put_Td(double);
```

C#

```
public double Td { get; set; }
```

Visual Basic

```
Public Td As double
```

Version

Available since version 14.7.

II.1.5.72 IRobotSeismicAnalysis_EC8_SpectrumType

C++

```
enum IRobotSeismicAnalysis_EC8_SpectrumType;
```

C#

```
public enum IRobotSeismicAnalysis_EC8_SpectrumType;
```

Visual Basic

```
Public Enum IRobotSeismicAnalysis_EC8_SpectrumType
```

Members

Members	Description
I_EC8_SAST_TYPE_1 = 0	Available since version 14.7.
I_EC8_SAST_TYPE_2 = 1	Available since version 14.7.

Version

Available since version 14.7.

II.1.5.73 IRobotSeismicAnalysis_EC8_GroundType

C++

```
enum IRobotSeismicAnalysis_EC8_GroundType;
```

C#

```
public enum IRobotSeismicAnalysis_EC8_GroundType;
```

Visual Basic

```
Public Enum IRobotSeismicAnalysis_EC8_GroundType
```

Members

Members	Description
I_EC8_SAGT_A = 0	Available since version 14.7.
I_EC8_SAGT_B = 1	Available since version 14.7.
I_EC8_SAGT_C = 2	Available since version 14.7.
I_EC8_SAGT_D = 3	Available since version 14.7.
I_EC8_SAGT_E = 4	Available since version 14.7.

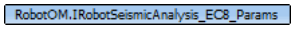
Version

Available since version 14.7.

(c) 2014 Autodesk Inc. All rights reserved

II.1.5.74 IRobotSeismicAnalysis_EC8_Params

Class Hierarchy



C++

```
interface IRobotSeismicAnalysis_EC8_Params : IDispatch;
```

C#

```
public interface IRobotSeismicAnalysis_EC8_Params;
```

Visual Basic

```
Public Interface IRobotSeismicAnalysis_EC8_Params
```

Description

.

Version

Available since version 14.7.

II.1.5.74.1 IRobotSeismicAnalysis_EC8_Params Members

The following tables list the members exposed by IRobotSeismicAnalysis_EC8_Params.

Public Fields

	Name	Description
	Ag (see page 303)	.
	BehaviorFactor (see page 303)	
	Direction (see page 303)	
	DirectionType (see page 303)	
	ExcitationDir (see page 304)	.
	Filter (see page 304)	
	GroundType (see page 304)	
	ResidualMode (see page 305)	
	Spectrum (see page 305)	
	SpectrumType (see page 305)	

II.1.5.74.2 IRobotSeismicAnalysis_EC8_Params Fields

The fields of the IRobotSeismicAnalysis_EC8_Params class are listed here.

Public Fields

	Name	Description
	Ag (see page 303)	.
	BehaviorFactor (see page 303)	
	Direction (see page 303)	
	DirectionType (see page 303)	
	ExcitationDir (see page 304)	.
	Filter (see page 304)	
	GroundType (see page 304)	
	ResidualMode (see page 305)	
	Spectrum (see page 305)	

(c) 2014 Autodesk Inc. All rights reserved

	SpectrumType (see page 305)	

II.1.5.74.2.1 Ag

C++

```
HRESULT get_Ag(double*);
HRESULT put_Ag(double);
```

C#

```
public double Ag { get; set; }
```

Visual Basic

```
Public Ag As double
```

Description

.

Version

Available since version 14.7.

II.1.5.74.2.2 BehaviorFactor

C++

```
HRESULT get_BehaviorFactor(double*);
HRESULT put_BehaviorFactor(double);
```

C#

```
public double BehaviorFactor { get; set; }
```

Visual Basic

```
Public BehaviorFactor As double
```

Version

Available since version 14.7.

II.1.5.74.2.3 Direction

C++

```
HRESULT get_Direction(IRobotGeoPoint3D**);
HRESULT put_Direction(IRobotGeoPoint3D*);
```

C#

```
public IRobotGeoPoint3D Direction { get; set; }
```

Visual Basic

```
Public Direction As IRobotGeoPoint3D
```

Version

Available since version 14.7.

II.1.5.74.2.4 DirectionType

C++

```
HRESULT get_DirectionType(IRobotSeismicAnalysisDirectionType*);
HRESULT put_DirectionType(IRobotSeismicAnalysisDirectionType);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRobotSeismicAnalysisDirectionType DirectionType { get; set; }
```

Visual Basic

```
Public DirectionType As IRobotSeismicAnalysisDirectionType
```

Version

Available since version 14.7.

II.1.5.74.2.5 ExcitationDir

C++

```
HRESULT get_ExcitationDir(IRobotDynamicAnalysisExcitationDirection**);
```

C#

```
public IRobotDynamicAnalysisExcitationDirection ExcitationDir { get; }
```

Visual Basic

```
Public ReadOnly ExcitationDir As IRobotDynamicAnalysisExcitationDirection
```

Description

.

Version

Available since version 14.7.

II.1.5.74.2.6 Filter

C++

```
HRESULT get_Filter(IRobotCaseAnalysisModesFilter**);
HRESULT put_Filter(IRobotCaseAnalysisModesFilter*);
```

C#

```
public IRobotCaseAnalysisModesFilter Filter { get; set; }
```

Visual Basic

```
Public Filter As IRobotCaseAnalysisModesFilter
```

Version

Available since version 14.7.

II.1.5.74.2.7 GroundType

C++

```
HRESULT get_GroundType(IRobotSeismicAnalysis_EC8_GroundType*);
HRESULT put_GroundType(IRobotSeismicAnalysis_EC8_GroundType);
```

C#

```
public IRobotSeismicAnalysis_EC8_GroundType GroundType { get; set; }
```

Visual Basic

```
Public GroundType As IRobotSeismicAnalysis_EC8_GroundType
```

Version

Available since version 14.7.

(c) 2014 Autodesk Inc. All rights reserved

II.1.5.74.2.8 ResidualMode

C++

```
HRESULT get_ResidualMode(IRobotSeismicResidualMode**);
```

C#

```
public IRobotSeismicResidualMode ResidualMode { get; }
```

Visual Basic

```
Public ReadOnly ResidualMode As IRobotSeismicResidualMode
```

Version

Available since version 14.7.

II.1.5.74.2.9 Spectrum

C++

```
HRESULT get_Spectrum(IRobotSeismicAnalysisSpectrumType*);
HRESULT put_Spectrum(IRobotSeismicAnalysisSpectrumType);
```

C#

```
public IRobotSeismicAnalysisSpectrumType Spectrum { get; set; }
```

Visual Basic

```
Public Spectrum As IRobotSeismicAnalysisSpectrumType
```

Version

Available since version 14.7.

II.1.5.74.2.10 SpectrumType

C++

```
HRESULT get_SpectrumType(IRobotSeismicAnalysis_EC8_SpectrumType*);
HRESULT put_SpectrumType(IRobotSeismicAnalysis_EC8_SpectrumType);
```

C#

```
public IRobotSeismicAnalysis_EC8_SpectrumType SpectrumType { get; set; }
```

Visual Basic

```
Public SpectrumType As IRobotSeismicAnalysis_EC8_SpectrumType
```

Version

Available since version 14.7.

II.1.6 Spectral analysis parameters

Available since version 2.5.

Enumerations

	Name	Description
	IRobotSpectralAnalysisAbscissaXAxisType (see page 316)	Available quantities on the abscissa axis. .
	IRobotSpectralAnalysisOrdinateYAxisType (see page 317)	Available quantities on the ordinate axis. .

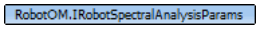
(c) 2014 Autodesk Inc. All rights reserved

Interfaces

	Name	Description
	IRobotSpectralAnalysisParams (see page 306)	Parameters of spectral analysis.
	IRobotSpectralAnalysisSpectrum (see page 308)	Spectrum definition.
	IRobotSpectralAnalysisPointsCollection (see page 313)	Collection of points defining spectrum. .

II.1.6.1 IRobotSpectralAnalysisParams

Class Hierarchy



C++

```
interface IRobotSpectralAnalysisParams : IDispatch;
```

C#

```
public interface IRobotSpectralAnalysisParams;
```

Visual Basic

```
Public Interface IRobotSpectralAnalysisParams
```

Description

Parameters of spectral analysis.

Version

Available since version 2.5.

II.1.6.1.1 IRobotSpectralAnalysisParams Members

The following tables list the members exposed by IRobotSpectralAnalysisParams.

Public Fields

	Name	Description
	Direction (see page 307)	Spectral excitation direction.
	ExcitationDir (see page 307)	
	Filter (see page 307)	Description of modes taken into account during structure dynamic analysis.
	Spectrum (see page 307)	Spectrum definition.

II.1.6.1.2 IRobotSpectralAnalysisParams Fields

The fields of the IRobotSpectralAnalysisParams class are listed here.

Public Fields

	Name	Description
	Direction (see page 307)	Spectral excitation direction.
	ExcitationDir (see page 307)	
	Filter (see page 307)	Description of modes taken into account during structure dynamic analysis.
	Spectrum (see page 307)	Spectrum definition.

(c) 2014 Autodesk Inc. All rights reserved

II.1.6.1.2.1 Direction

C++

```
HRESULT get_Direction(IRobotGeoPoint3D**);
HRESULT put_Direction(IRobotGeoPoint3D*);
```

C#

```
public IRobotGeoPoint3D Direction { get; set; }
```

Visual Basic

```
Public Direction As IRobotGeoPoint3D
```

Description

Spectral excitation direction.

Version

Available since version 2.5.

II.1.6.1.2.2 ExcitationDir

C++

```
HRESULT get_ExcitationDir(IRobotDynamicAnalysisExcitationDirection**);
```

C#

```
public IRobotDynamicAnalysisExcitationDirection ExcitationDir { get; }
```

Visual Basic

```
Public ReadOnly ExcitationDir As IRobotDynamicAnalysisExcitationDirection
```

II.1.6.1.2.3 Filter

C++

```
HRESULT get_Filter(IRobotCaseAnalysisModesFilter**);
HRESULT put_Filter(IRobotCaseAnalysisModesFilter*);
```

C#

```
public IRobotCaseAnalysisModesFilter Filter { get; set; }
```

Visual Basic

```
Public Filter As IRobotCaseAnalysisModesFilter
```

Description

Description of modes taken into account during structure dynamic analysis.

Version

Available since version 2.5.

II.1.6.1.2.4 Spectrum

C++

```
HRESULT get_Spectrum(IRobotSpectralAnalysisSpectrum**);
HRESULT put_Spectrum(IRobotSpectralAnalysisSpectrum*);
```

C#

```
public IRobotSpectralAnalysisSpectrum Spectrum { get; set; }
```

Visual Basic

```
Public Spectrum As IRobotSpectralAnalysisSpectrum
```

(c) 2014 Autodesk Inc. All rights reserved

Description

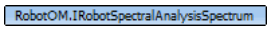
Spectrum definition.

Version

Available since version 2.5.

II.1.6.2 IRobotSpectralAnalysisSpectrum

Class Hierarchy



C++

```
interface IRobotSpectralAnalysisSpectrum : IDispatch;
```

C#

```
public interface IRobotSpectralAnalysisSpectrum;
```

Visual Basic

```
Public Interface IRobotSpectralAnalysisSpectrum
```

Description

Spectrum definition.

Version

Available since version 2.5.

II.1.6.2.1 IRobotSpectralAnalysisSpectrum Members

The following tables list the members exposed by IRobotSpectralAnalysisSpectrum.

Public Fields

	Name	Description
	AbscissaXAxis (see page 309)	Quantity on the abscissa axis .
	AbscissaXAxisLogarithmicScale (see page 309)	Logarithmic scale on the abscissa axis.
	Damping (see page 309)	Damping.
	Name (see page 310)	Spectrum name.
	OrdinateYAxis (see page 310)	Quantity on the ordinate axis.
	OrdinateYAxisLogarithmicScale (see page 310)	Logarithmic scale on the ordinate axis.
	Points (see page 311)	Points defining spectrum.

Public Methods

	Name	Description
	AddFromTimeHistory (see page 311)	To the collection of points defining spectrum the function adds points of the spectrum calculated based on the time-dependent function and specified parameters. .
	Average (see page 312)	Function calculates the spectrum by averaging the indicated spectra.
	LoadFromFile (see page 312)	Function reads the spectrum from a file. .
	SaveToFile (see page 312)	Function saves the spectrum to a file.

II.1.6.2.2 IRobotSpectralAnalysisSpectrum Fields

The fields of the IRobotSpectralAnalysisSpectrum class are listed here.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	AbscissaXAxis (see page 309)	Quantity on the abscissa axis .
	AbscissaXAxisLogarithmicScale (see page 309)	Logarithmic scale on the abscissa axis.
	Damping (see page 309)	Damping.
	Name (see page 310)	Spectrum name.
	OrdinateYAxis (see page 310)	Quantity on the ordinate axis.
	OrdinateYAxisLogarithmicScale (see page 310)	Logarithmic scale on the ordinate axis.
	Points (see page 311)	Points defining spectrum.

II.1.6.2.2.1 AbscissaXAxis

C++

```
HRESULT get_AbscissaXAxis(IRobotSpectralAnalysisAbscissaXAxisType*);
HRESULT put_AbscissaXAxis(IRobotSpectralAnalysisAbscissaXAxisType);
```

C#

```
public IRobotSpectralAnalysisAbscissaXAxisType AbscissaXAxis { get; set; }
```

Visual Basic

```
Public AbscissaXAxis As IRobotSpectralAnalysisAbscissaXAxisType
```

Description

Quantity on the abscissa axis .

Version

Available since version 2.5.

II.1.6.2.2.2 AbscissaXAxisLogarithmicScale

C++

```
HRESULT get_AbscissaXAxisLogarithmicScale(VARIANT_BOOL*);
HRESULT put_AbscissaXAxisLogarithmicScale(VARIANT_BOOL);
```

C#

```
public bool AbscissaXAxisLogarithmicScale { get; set; }
```

Visual Basic

```
Public AbscissaXAxisLogarithmicScale As Boolean
```

Description

Logarithmic scale on the abscissa axis.

Version

Available since version 2.5.

II.1.6.2.2.3 Damping

C++

```
HRESULT get_Damping(double*);
HRESULT put_Damping(double);
```

C#

```
public double Damping { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Damping As double
```

Description

Damping.

Version

Available since version 2.5.

II.1.6.2.2.4 Name

C++

```
HRESULT get_Name(BSTR*);
HRESULT put_Name(BSTR);
```

C#

```
public BSTR Name { get; set; }
```

Visual Basic

```
Public Name As BSTR
```

Description

Spectrum name.

Version

Available since version 2.5.

II.1.6.2.2.5 OrdinateYAxis

C++

```
HRESULT get_OrdinateYAxis(IRobotSpectralAnalysisOrdinateYAxisType*);
HRESULT put_OrdinateYAxis(IRobotSpectralAnalysisOrdinateYAxisType);
```

C#

```
public IRobotSpectralAnalysisOrdinateYAxisType OrdinateYAxis { get; set; }
```

Visual Basic

```
Public OrdinateYAxis As IRobotSpectralAnalysisOrdinateYAxisType
```

Description

Quantity on the ordinate axis.

Version

Available since version 2.5.

II.1.6.2.2.6 OrdinateYAxisLogarithmicScale

C++

```
HRESULT get_OrdinateYAxisLogarithmicScale(VARIANT_BOOL*);
HRESULT put_OrdinateYAxisLogarithmicScale(VARIANT_BOOL);
```

C#

```
public bool OrdinateYAxisLogarithmicScale { get; set; }
```

Visual Basic

```
Public OrdinateYAxisLogarithmicScale As Boolean
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Logarithmic scale on the ordinate axis.

Version

Available since version 2.5.

II.1.6.2.2.7 Points

C++

```
HRESULT get_Points(IRobotSpectralAnalysisPointsCollection**);
HRESULT put_Points(IRobotSpectralAnalysisPointsCollection*);
```

C#

```
public IRobotSpectralAnalysisPointsCollection Points { get; set; }
```

Visual Basic

```
Public Points As IRobotSpectralAnalysisPointsCollection
```

Description

Points defining spectrum.

Version

Available since version 2.5.

II.1.6.2.3 IRobotSpectralAnalysisSpectrum Methods

The methods of the IRobotSpectralAnalysisSpectrum class are listed here.

Public Methods

	Name	Description
	AddFromTimeHistory (see page 311)	To the collection of points defining spectrum the function adds points of the spectrum calculated based on the time-dependent function and specified parameters. .
	Average (see page 312)	Function calculates the spectrum by averaging the indicated spectra.
	LoadFromFile (see page 312)	Function reads the spectrum from a file. .
	SaveToFile (see page 312)	Function saves the spectrum to a file.

II.1.6.2.3.1 AddFromTimeHistory

C++

```
HRESULT AddFromTimeHistory(IRobotSpectralAnalysisPointsCollection* _time_history, double
_t_min, double _t_max, long _points_count, double _damping);
```

C#

```
public void AddFromTimeHistory(IRobotSpectralAnalysisPointsCollection _time_history, double
_t_min, double _t_max, long _points_count, double _damping);
```

Visual Basic

```
Public Sub AddFromTimeHistory(ByRef _time_history As
IRobotSpectralAnalysisPointsCollection, _t_min As double, _t_max As double, _points_count
As long, _damping As double)
```

Description

To the collection of points defining spectrum the function adds points of the spectrum calculated based on the time-dependent function and specified parameters. .

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 2.5.

II.1.6.2.3.2 Average

C++

```
HRESULT Average(IRobotSpectralAnalysisSpectrum* _spectrum_1,
IRobotSpectralAnalysisSpectrum* _spectrum_2, double _damping);
```

C#

```
public void Average(IRobotSpectralAnalysisSpectrum _spectrum_1,
IRobotSpectralAnalysisSpectrum _spectrum_2, double _damping);
```

Visual Basic

```
Public Sub Average(ByRef _spectrum_1 As IRobotSpectralAnalysisSpectrum, ByRef _spectrum_2
As IRobotSpectralAnalysisSpectrum, _damping As double)
```

Description

Function calculates the spectrum by averaging the indicated spectra.

Version

Available since version 2.5.

II.1.6.2.3.3 LoadFromFile

C++

```
HRESULT LoadFromFile(BSTR _file_path, BSTR _spectrum_name);
```

C#

```
public void LoadFromFile(BSTR _file_path, BSTR _spectrum_name);
```

Visual Basic

```
Public Sub LoadFromFile(_file_path As BSTR, _spectrum_name As BSTR)
```

Description

Function reads the spectrum from a file. .

Version

Available since version 2.5.

II.1.6.2.3.4 SaveToFile

C++

```
HRESULT SaveToFile(BSTR _file_path);
```

C#

```
public void SaveToFile(BSTR _file_path);
```

Visual Basic

```
Public Sub SaveToFile(_file_path As BSTR)
```

Description

Function saves the spectrum to a file.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 2.5.

II.1.6.3 IRobotSpectralAnalysisPointsCollection

Class Hierarchy

C++

```
interface IRobotSpectralAnalysisPointsCollection : IDispatch;
```

C#

```
public interface IRobotSpectralAnalysisPointsCollection;
```

Visual Basic

```
Public Interface IRobotSpectralAnalysisPointsCollection
```

Description

Collection of points defining spectrum. .

Version

Available since version 2.5.

II.1.6.3.1 IRobotSpectralAnalysisPointsCollection Members

The following tables list the members exposed by IRobotSpectralAnalysisPointsCollection.

Public Fields

	Name	Description
	Count (see page 313)	Number of points defining spectrum.

Public Methods

	Name	Description
	Add (see page 314)	Function adds a new point to the spectrum definition. .
	Clear (see page 314)	Function deletes all the points from the collection defining spectrum. .
	Get (see page 315)	Function returns coordinates of the selected point defining spectrum.
	LoadFromFile (see page 315)	Function reads out the point collection from a file. .
	Remove (see page 315)	Function deletes the point from the collection defining spectrum.
	SaveToFile (see page 316)	Function saves the point collection to a file. .
	Set (see page 316)	Function modifies the selected point defining the spectrum. .

II.1.6.3.2 IRobotSpectralAnalysisPointsCollection Fields

The fields of the IRobotSpectralAnalysisPointsCollection class are listed here.

Public Fields

	Name	Description
	Count (see page 313)	Number of points defining spectrum.

II.1.6.3.2.1 Count

C++

```
HRESULT get_Count(long*);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public long Count { get; }
```

Visual Basic

```
Public ReadOnly Count As long
```

Description

Number of points defining spectrum.

Version

Available since version 2.5.

II.1.6.3.3 IRobotSpectralAnalysisPointsCollection Methods

The methods of the IRobotSpectralAnalysisPointsCollection class are listed here.

Public Methods

	Name	Description
	Add (see page 314)	Function adds a new point to the spectrum definition. .
	Clear (see page 314)	Function deletes all the points from the collection defining spectrum. .
	Get (see page 315)	Function returns coordinates of the selected point defining spectrum.
	LoadFromFile (see page 315)	Function reads out the point collection from a file. .
	Remove (see page 315)	Function deletes the point from the collection defining spectrum.
	SaveToFile (see page 316)	Function saves the point collection to a file. .
	Set (see page 316)	Function modifies the selected point defining the spectrum. .

II.1.6.3.3.1 Add

C++

```
HRESULT Add(double _x, double _y);
```

C#

```
public void Add(double _x, double _y);
```

Visual Basic

```
Public Sub Add(_x As double, _y As double)
```

Description

Function adds a new point to the spectrum definition. .

Version

Available since version 2.5.

II.1.6.3.3.2 Clear

C++

```
HRESULT Clear();
```

C#

```
public void Clear();
```

Visual Basic

```
Public Sub Clear()
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Function deletes all the points from the collection defining spectrum. .

Version

Available since version 2.5.

II.1.6.3.3.3 Get

C++

```
HRESULT Get(long _pos, double* _x, double* _y);
```

C#

```
public void Get(long _pos, double* _x, double* _y);
```

Visual Basic

```
Public Sub Get(_pos As long, ByRef _x As double*, ByRef _y As double*)
```

Description

Function returns coordinates of the selected point defining spectrum.

Version

Available since version 2.5.

II.1.6.3.3.4 LoadFromFile

C++

```
HRESULT LoadFromFile(BSTR _file_path);
```

C#

```
public void LoadFromFile(BSTR _file_path);
```

Visual Basic

```
Public Sub LoadFromFile(_file_path As BSTR)
```

Description

Function reads out the point collection from a file. .

Version

Available since version 2.5.

II.1.6.3.3.5 Remove

C++

```
HRESULT Remove(long _pos);
```

C#

```
public void Remove(long _pos);
```

Visual Basic

```
Public Sub Remove(_pos As long)
```

Description

Function deletes the point from the collection defining spectrum.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 2.5.

II.1.6.3.3.6 SaveToFile

C++

```
HRESULT SaveToFile(BSTR _file_path);
```

C#

```
public void SaveToFile(BSTR _file_path);
```

Visual Basic

```
Public Sub SaveToFile(_file_path As BSTR)
```

Description

Function saves the point collection to a file. .

Version

Available since version 2.5.

II.1.6.3.3.7 Set

C++

```
HRESULT Set(long _pos, double _x, double _y);
```

C#

```
public void Set(long _pos, double _x, double _y);
```

Visual Basic

```
Public Sub Set(_pos As long, _x As double, _y As double)
```

Description

Function modifies the selected point defining the spectrum. .

Version

Available since version 2.5.

II.1.6.4 IRobotSpectralAnalysisAbscissaXAxisType

C++

```
enum IRobotSpectralAnalysisAbscissaXAxisType;
```

C#

```
public enum IRobotSpectralAnalysisAbscissaXAxisType;
```

Visual Basic

```
Public Enum IRobotSpectralAnalysisAbscissaXAxisType
```

Members

Members	Description
I_SAAXAT_PERIOD = 0	Period. Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

I_SAAXAT_PULSATION = 1	Pulsation. Available since version 2.5.
I_SAAXAT_FREQUENCY = 2	Frequency. Available since version 2.5.

Description

Available quantities on the abscissa axis. .

Version

Available since version 2.5.

II.1.6.5 IRobotSpectralAnalysisOrdinateYAxisType

C++

```
enum IRobotSpectralAnalysisOrdinateYAxisType;
```

C#

```
public enum IRobotSpectralAnalysisOrdinateYAxisType;
```

Visual Basic

```
Public Enum IRobotSpectralAnalysisOrdinateYAxisType
```

Members

Members	Description
I_SAOYAT_VELOCITY = 0	Velocity. Available since version 2.5.
I_SAOYAT_ACCELERATION = 1	Acceleration. Available since version 2.5.
I_SAOYAT_EXCITATION = 2	Displacement. Available since version 2.5.

Description

Available quantities on the ordinate axis. .

Version

Available since version 2.5.

II.1.7 Non-linear analysis parameters

Available since version 2.5.

Enumerations

	Name	Description
	IRobotNonlinearAnalysisAlgorithmType (see page 317)	Methods of non-linear analysis.

Interfaces

	Name	Description
	IRobotNonlinearAnalysisParams (see page 318)	Parameters of non-linear analysis.

II.1.7.1 IRobotNonlinearAnalysisAlgorithmType

C++

```
enum IRobotNonlinearAnalysisAlgorithmType;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public enum IRobotNonlinearAnalysisAlgorithmType;
```

Visual Basic

```
Public Enum IRobotNonlinearAnalysisAlgorithmType
```

Members

Members	Description
I_NAAT_INCREMENTAL_METHOD = 0	Incremental method of non-linear analysis. Available since version 2.5.
I_NAAT_DIRECT_ITERATION_METHOD = 1	Direct iteration method. Available since version 2.5.
I_NAAT_ARC_LENGTH_METHOD = 2	Available since version 2.5.
I_NAAT_PREDICTOR_CORRECTOR_METHOD = 3	Available since version 3.

Description

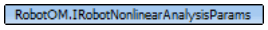
Methods of non-linear analysis.

Version

Available since version 2.5.

II.1.7.2 IRobotNonlinearAnalysisParams

Class Hierarchy



C++

```
interface IRobotNonlinearAnalysisParams : IDispatch;
```

C#

```
public interface IRobotNonlinearAnalysisParams;
```

Visual Basic

```
Public Interface IRobotNonlinearAnalysisParams
```

Description

Parameters of non-linear analysis.

Version

Available since version 2.5.

II.1.7.2.1 IRobotNonlinearAnalysisParams Members

The following tables list the members exposed by IRobotNonlinearAnalysisParams.

Public Fields

	Name	Description
	Algorithm (see page 320)	Method of non-linear analysis.
	DegreeOfFreedom (see page 320)	Definition of degree of freedom (for the "Arc-length" method) .
	DisplacementsRelativeCodeTolerance (see page 320)	Relative code tolerance for displacements.
	IncrementLengthReductionFactor (see page 321)	Increment length reduction factor.

(c) 2014 Autodesk Inc. All rights reserved

	IncrementLengthReductionNumber (see page 321)	Increment length reduction number .
	LineSearchMethodFactor (see page 321)	Control parameter for line-search method .
	LoadIncrementNumber (see page 322)	Number of load increments.
	MatrixUpdateAfterEachIteration (see page 322)	Matrix update after each iteration.
	MatrixUpdateAfterEachSubdivision (see page 322)	Matrix update after each subdivision.
	MaxDisplacement (see page 322)	Maximum displacement for the selected degree of freedom (for the "Arc-length" method) .
	MaximumIterationNumberForOneIncrement (see page 323)	Maximum iteration number for one increment.
	MaximumNumberOfBFGSCorrections (see page 323)	Maximum number of BFGS corrections.
	MaximumNumberOfLineSearches (see page 323)	Maximum number of line searches.
	MaxLoadFactor (see page 323)	Maximum load factor (for the "Arc-length" method) .
	NodeNumber (see page 324)	Node number (for the "Arc-length" method) .
	PDelta (see page 324)	Definition of P-delta analysis.
	ResidualForcesRelativeCodeTolerance (see page 324)	Relative code tolerance for residual forces.
	ResultListEachIteration (see page 325)	Relative code tolerance for residual forces.
	Stiff (see page 325)	(for "Arc-length" method).

Public Methods

	Name	Description
	GetSettingsFromPreferences (see page 325)	Function takes settings from preferences.
	SaveSettingsInPreferences (see page 326)	Function saves settings to preferences as default ones. .

II.1.7.2.2 IRobotNonlinearAnalysisParams Fields

The fields of the IRobotNonlinearAnalysisParams class are listed here.

Public Fields

	Name	Description
	Algorithm (see page 320)	Method of non-linear analysis.
	DegreeOfFreedom (see page 320)	Definition of degree of freedom (for the "Arc-length" method) .
	DisplacementsRelativeCodeTolerance (see page 320)	Relative code tolerance for displacements.
	IncrementLengthReductionFactor (see page 321)	Increment length reduction factor.
	IncrementLengthReductionNumber (see page 321)	Increment length reduction number .
	LineSearchMethodFactor (see page 321)	Control parameter for line-search method .
	LoadIncrementNumber (see page 322)	Number of load increments.
	MatrixUpdateAfterEachIteration (see page 322)	Matrix update after each iteration.

(c) 2014 Autodesk Inc. All rights reserved

	MatrixUpdateAfterEachSubdivision (see page 322)	Matrix update after each subdivision.
	MaxDisplacement (see page 322)	Maximum displacement for the selected degree of freedom (for the "Arc-length" method) .
	MaximumIterationNumberForOneIncrement (see page 323)	Maximum iteration number for one increment.
	MaximumNumberOfBFGSCorrections (see page 323)	Maximum number of BFGS corrections.
	MaximumNumberOfLineSearches (see page 323)	Maximum number of line searches.
	MaxLoadFactor (see page 323)	Maximum load factor (for the "Arc-length" method) .
	NodeNumber (see page 324)	Node number (for the "Arc-length" method) .
	PDelta (see page 324)	Definition of P-delta analysis.
	ResidualForcesRelativeCodeTolerance (see page 324)	Relative code tolerance for residual forces.
	ResultListEachIteration (see page 325)	Relative code tolerance for residual forces.
	Stiff (see page 325)	(for "Arc-length" method).

II.1.7.2.2.1 Algorithm

C++

```
HRESULT get_Algorithm(IRobotNonlinearAnalysisAlgorithmType*);
HRESULT put_Algorithm(IRobotNonlinearAnalysisAlgorithmType);
```

C#

```
public IRobotNonlinearAnalysisAlgorithmType Algorithm { get; set; }
```

Visual Basic

```
Public Algorithm As IRobotNonlinearAnalysisAlgorithmType
```

Description

Method of non-linear analysis.

II.1.7.2.2.2 DegreeOfFreedom

C++

```
HRESULT get_DegreeOfFreedom(IRobotDegreeOfFreedom*);
HRESULT put_DegreeOfFreedom(IRobotDegreeOfFreedom);
```

C#

```
public IRobotDegreeOfFreedom DegreeOfFreedom { get; set; }
```

Visual Basic

```
Public DegreeOfFreedom As IRobotDegreeOfFreedom
```

Description

Definition of degree of freedom (for the "Arc-length" method) .

Version

Available since version 2.5.

II.1.7.2.2.3 DisplacementsRelativeCodeTolerance

C++

```
HRESULT get_DisplacementsRelativeCodeTolerance(double*);
HRESULT put_DisplacementsRelativeCodeTolerance(double);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double DisplacementsRelativeCodeTolerance { get; set; }
```

Visual Basic

```
Public DisplacementsRelativeCodeTolerance As double
```

Description

Relative code tolerance for displacements.

Version

Available since version 2.5.

II.1.7.2.2.4 IncrementLengthReductionFactor

C++

```
HRESULT get_IncrementLengthReductionFactor(double*);
HRESULT put_IncrementLengthReductionFactor(double);
```

C#

```
public double IncrementLengthReductionFactor { get; set; }
```

Visual Basic

```
Public IncrementLengthReductionFactor As double
```

Description

Increment length reduction factor.

II.1.7.2.2.5 IncrementLengthReductionNumber

C++

```
HRESULT get_IncrementLengthReductionNumber(long*);
HRESULT put_IncrementLengthReductionNumber(long);
```

C#

```
public long IncrementLengthReductionNumber { get; set; }
```

Visual Basic

```
Public IncrementLengthReductionNumber As long
```

Description

Increment length reduction number .

II.1.7.2.2.6 LineSearchMethodFactor

C++

```
HRESULT get_LineSearchMethodFactor(double*);
HRESULT put_LineSearchMethodFactor(double);
```

C#

```
public double LineSearchMethodFactor { get; set; }
```

Visual Basic

```
Public LineSearchMethodFactor As double
```

Description

Control parameter for line-search method .

(c) 2014 Autodesk Inc. All rights reserved

II.1.7.2.2.7 LoadIncrementNumber

C++

```
HRESULT get_LoadIncrementNumber(long*);
HRESULT put_LoadIncrementNumber(long);
```

C#

```
public long LoadIncrementNumber { get; set; }
```

Visual Basic

```
Public LoadIncrementNumber As long
```

Description

Number of load increments.

II.1.7.2.2.8 MatrixUpdateAfterEachIteration

C++

```
HRESULT get_MatrixUpdateAfterEachIteration(VARIANT_BOOL*);
HRESULT put_MatrixUpdateAfterEachIteration(VARIANT_BOOL);
```

C#

```
public bool MatrixUpdateAfterEachIteration { get; set; }
```

Visual Basic

```
Public MatrixUpdateAfterEachIteration As Boolean
```

Description

Matrix update after each iteration.

II.1.7.2.2.9 MatrixUpdateAfterEachSubdivision

C++

```
HRESULT get_MatrixUpdateAfterEachSubdivision(VARIANT_BOOL*);
HRESULT put_MatrixUpdateAfterEachSubdivision(VARIANT_BOOL);
```

C#

```
public bool MatrixUpdateAfterEachSubdivision { get; set; }
```

Visual Basic

```
Public MatrixUpdateAfterEachSubdivision As Boolean
```

Description

Matrix update after each subdivision.

II.1.7.2.2.10 MaxDisplacement

C++

```
HRESULT get_MaxDisplacement(double*);
HRESULT put_MaxDisplacement(double);
```

C#

```
public double MaxDisplacement { get; set; }
```

Visual Basic

```
Public MaxDisplacement As double
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Maximum displacement for the selected degree of freedom (for the "Arc-length" method) .

Version

Available since version 2.5.

II.1.7.2.2.11 MaximumIterationNumberForOneIncrement

C++

```
HRESULT get_MaximumIterationNumberForOneIncrement(long*);
HRESULT put_MaximumIterationNumberForOneIncrement(long);
```

C#

```
public long MaximumIterationNumberForOneIncrement { get; set; }
```

Visual Basic

```
Public MaximumIterationNumberForOneIncrement As long
```

Description

Maximum iteration number for one increment.

II.1.7.2.2.12 MaximumNumberOfBFGSCorrections

C++

```
HRESULT get_MaximumNumberOfBFGSCorrections(long*);
HRESULT put_MaximumNumberOfBFGSCorrections(long);
```

C#

```
public long MaximumNumberOfBFGSCorrections { get; set; }
```

Visual Basic

```
Public MaximumNumberOfBFGSCorrections As long
```

Description

Maximum number of BFGS corrections.

II.1.7.2.2.13 MaximumNumberOfLineSearches

C++

```
HRESULT get_MaximumNumberOfLineSearches(long*);
HRESULT put_MaximumNumberOfLineSearches(long);
```

C#

```
public long MaximumNumberOfLineSearches { get; set; }
```

Visual Basic

```
Public MaximumNumberOfLineSearches As long
```

Description

Maximum number of line searches.

II.1.7.2.2.14 MaxLoadFactor

C++

```
HRESULT get_MaxLoadFactor(double*);
HRESULT put_MaxLoadFactor(double);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double MaxLoadFactor { get; set; }
```

Visual Basic

```
Public MaxLoadFactor As double
```

Description

Maximum load factor (for the "Arc-length" method) .

Version

Available since version 2.5.

II.1.7.2.2.15 NodeNumber

C++

```
HRESULT get_NodeNumber(long*);
HRESULT put_NodeNumber(long);
```

C#

```
public long NodeNumber { get; set; }
```

Visual Basic

```
Public NodeNumber As long
```

Description

Node number (for the "Arc-length" method) .

Version

Available since version 2.5.

II.1.7.2.2.16 PDelta

C++

```
HRESULT get_PDelta(VARIANT_BOOL*);
HRESULT put_PDelta(VARIANT_BOOL);
```

C#

```
public bool PDelta { get; set; }
```

Visual Basic

```
Public PDelta As Boolean
```

Description

Definition of P-delta analysis.

II.1.7.2.2.17 ResidualForcesRelativeCodeTolerance

C++

```
HRESULT get_ResidualForcesRelativeCodeTolerance(double*);
HRESULT put_ResidualForcesRelativeCodeTolerance(double);
```

C#

```
public double ResidualForcesRelativeCodeTolerance { get; set; }
```

Visual Basic

```
Public ResidualForcesRelativeCodeTolerance As double
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Relative code tolerance for residual forces.

II.1.7.2.2.18 ResultListEachIteration

C++

```
HRESULT get_ResultListEachIteration(VARIANT_BOOL*);
HRESULT put_ResultListEachIteration(VARIANT_BOOL);
```

C#

```
public bool ResultListEachIteration { get; set; }
```

Visual Basic

```
Public ResultListEachIteration As Boolean
```

Description

Relative code tolerance for residual forces.

II.1.7.2.2.19 Stiff

C++

```
HRESULT get_Stiff(VARIANT_BOOL*);
HRESULT put_Stiff(VARIANT_BOOL);
```

C#

```
public bool Stiff { get; set; }
```

Visual Basic

```
Public Stiff As Boolean
```

Description

(for "Arc-length" method).

Version

Available since version 2.5.

II.1.7.2.3 IRobotNonlinearAnalysisParams Methods

The methods of the IRobotNonlinearAnalysisParams class are listed here.

Public Methods

	Name	Description
	GetSettingsFromPreferences (see page 325)	Function takes settings from preferences.
	SaveSettingsInPreferences (see page 326)	Function saves settings to preferences as default ones. .

II.1.7.2.3.1 GetSettingsFromPreferences

C++

```
HRESULT GetSettingsFromPreferences();
```

C#

```
public void GetSettingsFromPreferences();
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Sub GetSettingsFromPreferences()
```

Description

Function takes settings from preferences.

Version

Available since version 2.5.

II.1.7.2.3.2 SaveSettingsInPreferences

C++

```
HRESULT SaveSettingsInPreferences();
```

C#

```
public void SaveSettingsInPreferences();
```

Visual Basic

```
Public Sub SaveSettingsInPreferences()
```

Description

Function saves settings to preferences as default ones. .

Version

Available since version 2.5.

II.1.8 Moving loads

Available since version 2.5.

Enumerations

	Name	Description
	IRobotMobileCaseFlag (see page 335)	Available types of a moving load case. .
	IRobotMobileCaseApplicationPlaneType (see page 343)	Type of application plane.
	IRobotVehicleLoadType (see page 352)	Load types.

Interfaces

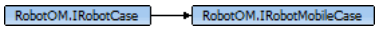
	Name	Description
	IRobotMobileCase (see page 327)	Moving load case.
	IRobotMobileCaseComponentMngr (see page 333)	Component list of moving load.
	IRobotMobileCaseComponent (see page 334)	Simple component of a moving load case. This component corresponds to the simple case generated at the next point of the route. .
	IRobotMobileCaseRoute (see page 336)	Definition of vehicle route.
	IRobotMobileCaseSegmentFactors (see page 340)	A set of coefficients assigned to a route segment, correcting vehicle loads. .
	IRobotVehicleData (see page 343)	Parameter set for vehicle definition.
	IRobotVehicleLoadMngr (see page 346)	List of loads that define a vehicle.

(c) 2014 Autodesk Inc. All rights reserved

	IRobotVehicleLoad (see page 348)	Record describing a single load associated with the vehicle. .
	IRobotVehicleDatabaseList (see page 352)	Interface describing the list of vehicle databases.
	IRobotVehicleDatabase (see page 356)	Interface for vehicle database support.

II.1.8.1 IRobotMobileCase

Class Hierarchy



C++

```
interface IRobotMobileCase : IRobotCase;
```

C#

```
public interface IRobotMobileCase : IRobotCase;
```

Visual Basic

```
Public Interface IRobotMobileCase
```

Description

Moving load case.

Version

Available since version 2.5.

II.1.8.1.1 IRobotMobileCase Members

The following tables list the members exposed by IRobotMobileCase.

Public Fields

	Name	Description
	AnalizeType (see page 408)	Analysis type.
	ApplicationPlaneBars (see page 328)	List of bars defining the plane to which the vehicle is applied (active parameter for ApplicationPlaneType (see page 329) = I_MCAPT_SELECTION).
	ApplicationPlaneType (see page 329)	Type (see page 409) of the plane to which the vehicle is applied.
	Components (see page 329)	List of moving load case components; Each component corresponds to a simple case generated due to applying the vehicle to the determined node on the route. .
	Flag (see page 329)	Flag determining type of a moving load case; Possible flag values are described by the IRobotMobileCaseFlag (see page 335) interface; In practice, this field is used for navigation between the base moving load case and its envelopes. .
	IsAuxiliary (see page 330)	
	Label (see page 330)	
	Name (see page 409)	User-defined case name .
	Nature (see page 409)	Case nature .
	NatureName (see page 330)	Name (see page 409) of nature.
	Number (see page 409)	User-defined number linked with the load case .
	Type (see page 409)	Load case type - allows one to differentiate between a simple case and combination at the level of the shared part of the interface .
	UniqueId (see page 330)	Unique identifier of a load case.

(c) 2014 Autodesk Inc. All rights reserved

	Vehicle (see page 331)	Name (see page 409) identifying the vehicle.

Public Methods

	Name	Description
	FindByFlag (see page 331)	Function returns number of the case of the determined type, associated with this moving load case. It allows navigation between the base moving load case and its envelopes. .
	GetRoute (see page 332)	Function takes the interface of route definition.
	SetNatureExt (see page 332)	Function assigns - to the case - the nature consistent with the indicated number of "action" described in the current regulations. .
	SetRoute (see page 332)	Function updates the route definition based on the indicated interface. .

II.1.8.1.2 IRobotMobileCase Fields

The fields of the IRobotMobileCase class are listed here.

Public Fields

	Name	Description
	ApplicationPlaneBars (see page 328)	List of bars defining the plane to which the vehicle is applied (active parameter for ApplicationPlaneType (see page 329) = I_MCAPT_SELECTION).
	ApplicationPlaneType (see page 329)	Type (see page 409) of the plane to which the vehicle is applied.
	Components (see page 329)	List of moving load case components; Each component corresponds to a simple case generated due to applying the vehicle to the determined node on the route. .
	Flag (see page 329)	Flag determining type of a moving load case; Possible flag values are described by the IRobotMobileCaseFlag (see page 335) interface; In practice, this field is used for navigation between the base moving load case and its envelopes. .
	IsAuxiliary (see page 330)	
	Label (see page 330)	
	NatureName (see page 330)	Name (see page 409) of nature.
	UniqueId (see page 330)	Unique identifier of a load case.
	Vehicle (see page 331)	Name (see page 409) identifying the vehicle.

II.1.8.1.2.1 ApplicationPlaneBars

C++

```
HRESULT get_ApplicationPlaneBars(BSTR*);
HRESULT put_ApplicationPlaneBars(BSTR);
```

C#

```
public String ApplicationPlaneBars { get; set; }
```

Visual Basic

```
Public ApplicationPlaneBars As String
```

Description

List of bars defining the plane to which the vehicle is applied (active parameter for ApplicationPlaneType (see page 329) = I_MCAPT_SELECTION).

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

II.1.8.1.2.2 ApplicationPlaneType

C++

```
HRESULT get_ApplicationPlaneType(IRobotMobileCaseApplicationPlaneType*);
HRESULT put_ApplicationPlaneType(IRobotMobileCaseApplicationPlaneType);
```

C#

```
public IRobotMobileCaseApplicationPlaneType ApplicationPlaneType { get; set; }
```

Visual Basic

```
Public ApplicationPlaneType As IRobotMobileCaseApplicationPlaneType
```

Description

Type (see page 409) of the plane to which the vehicle is applied.

Version

Available since version 3.

II.1.8.1.2.3 Components

C++

```
HRESULT get_Components(IRobotMobileCaseComponentMngr**);
```

C#

```
public IRobotMobileCaseComponentMngr Components { get; }
```

Visual Basic

```
Public ReadOnly Components As IRobotMobileCaseComponentMngr
```

Description

List of moving load case components; Each component corresponds to a simple case generated due to applying the vehicle to the determined node on the route. .

Version

Available since version 2.5.

II.1.8.1.2.4 Flag

C++

```
HRESULT get_Flag(IRobotMobileCaseFlag*);
```

C#

```
public IRobotMobileCaseFlag Flag { get; }
```

Visual Basic

```
Public ReadOnly Flag As IRobotMobileCaseFlag
```

Description

Flag determining type of a moving load case; Possible flag values are described by the IRobotMobileCaseFlag (see page 335) interface; In practice, this field is used for navigation between the base moving load case and its envelopes. .

Version

Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

II.1.8.1.2.5 IsAuxiliary

C++

```
HRESULT get_IsAuxiliary(VARIANT_BOOL*);
HRESULT put_IsAuxiliary(VARIANT_BOOL);
```

C#

```
public bool IsAuxiliary { get; set; }
```

Visual Basic

```
Public IsAuxiliary As Boolean
```

Version

Available since version 8.1.

II.1.8.1.2.6 Label

C++

```
HRESULT get_Label(BSTR*);
HRESULT put_Label(BSTR);
```

C#

```
public String Label { get; set; }
```

Visual Basic

```
Public Label As String
```

Version

Available since version 8.

II.1.8.1.2.7 NatureName

C++

```
HRESULT get_NatureName(BSTR*);
```

C#

```
public String NatureName { get; }
```

Visual Basic

```
Public ReadOnly NatureName As String
```

Description

Name (see page 409) of nature.

Version

Available since version 3.

II.1.8.1.2.8 UniqueId

C++

```
HRESULT get_UniqueId(long*);
```

C#

```
public long UniqueId { get; }
```

Visual Basic

```
Public ReadOnly UniqueId As long
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Unique identifier of a load case.

Version

Available since version 2.5.

II.1.8.1.2.9 Vehicle

C++

```
HRESULT get_Vehicle(BSTR*);
HRESULT put_Vehicle(BSTR);
```

C#

```
public String Vehicle { get; set; }
```

Visual Basic

```
Public Vehicle As String
```

Description

Name (see page 409) identifying the vehicle.

Version

Available since version 3.

II.1.8.1.3 IRobotMobileCase Methods

The methods of the IRobotMobileCase class are listed here.

Public Methods

	Name	Description
	FindByFlag (see page 331)	Function returns number of the case of the determined type, associated with this moving load case. It allows navigation between the base moving load case and its envelopes. .
	GetRoute (see page 332)	Function takes the interface of route definition.
	SetNatureExt (see page 332)	Function assigns - to the case - the nature consistent with the indicated number of "action" described in the current regulations. .
	SetRoute (see page 332)	Function updates the route definition based on the indicated interface. .

II.1.8.1.3.1 FindByFlag

C++

```
HRESULT FindByFlag(IRobotMobileCaseFlag _mc_flag, long* ret);
```

C#

```
public long FindByFlag(IRobotMobileCaseFlag _mc_flag);
```

Visual Basic

```
Public Function FindByFlag(_mc_flag As IRobotMobileCaseFlag) As long
```

Description

Function returns number of the case of the determined type, associated with this moving load case. It allows navigation between the base moving load case and its envelopes. .

Version

Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

II.1.8.1.3.2 GetRoute

C++

```
HRESULT GetRoute(IRobotMobileCaseRoute** ret);
```

C#

```
public IRobotMobileCaseRoute GetRoute();
```

Visual Basic

```
Public Function GetRoute() As IRobotMobileCaseRoute
```

Description

Function takes the interface of route definition.

Version

Available since version 3.

II.1.8.1.3.3 SetNatureExt

C++

```
HRESULT SetNatureExt(long _rgl_action);
```

C#

```
public void SetNatureExt(long _rgl_action);
```

Visual Basic

```
Public Sub SetNatureExt(_rgl_action As long)
```

Description

Function assigns - to the case - the nature consistent with the indicated number of "action" described in the current regulations. .

Version

Available since version 3.

II.1.8.1.3.4 SetRoute

C++

```
HRESULT SetRoute(IRobotMobileCaseRoute _route);
```

C#

```
public void SetRoute(IRobotMobileCaseRoute _route);
```

Visual Basic

```
Public Sub SetRoute(_route As IRobotMobileCaseRoute)
```

Description

Function updates the route definition based on the indicated interface. .

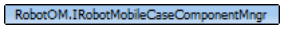
Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

II.1.8.2 IRobotMobileCaseComponentMngr

Class Hierarchy



C++

```
interface IRobotMobileCaseComponentMngr : IDispatch;
```

C#

```
public interface IRobotMobileCaseComponentMngr;
```

Visual Basic

```
Public Interface IRobotMobileCaseComponentMngr
```

Description

Component list of moving load.

Version

Available since version 2.5.

II.1.8.2.1 IRobotMobileCaseComponentMngr Members

The following tables list the members exposed by IRobotMobileCaseComponentMngr.

Public Fields

	Name	Description
	Count (see page 333)	Number of components on the list .

Public Methods

	Name	Description
	Get (see page 334)	Function returns the component with the specified index. Components of a moving load case are indexed from 1 to Count (see page 333). .

II.1.8.2.2 IRobotMobileCaseComponentMngr Fields

The fields of the IRobotMobileCaseComponentMngr class are listed here.

Public Fields

	Name	Description
	Count (see page 333)	Number of components on the list .

II.1.8.2.2.1 Count

C++

```
HRESULT get_Count(long*);
```

C#

```
public long Count { get; }
```

Visual Basic

```
Public ReadOnly Count As long
```

Description

Number of components on the list .

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 2.5.

II.1.8.2.3 IRobotMobileCaseComponentMngr Methods

The methods of the IRobotMobileCaseComponentMngr class are listed here.

Public Methods

	Name	Description
	Get (see page 334)	Function returns the component with the specified index. Components of a moving load case are indexed from 1 to Count (see page 333). .

II.1.8.2.3.1 Get

C++

```
HRESULT Get(long _cmpnt_idx, IRobotMobileCaseComponent** ret);
```

C#

```
public IRobotMobileCaseComponent Get(long _cmpnt_idx);
```

Visual Basic

```
Public Function Get(_cmpnt_idx As long) As IRobotMobileCaseComponent
```

Description

Function returns the component with the specified index. Components of a moving load case are indexed from 1 to Count (see page 333). .

Version

Available since version 2.5.

II.1.8.3 IRobotMobileCaseComponent

Class Hierarchy

RobotOM.IRobotMobileCaseComponent

C++

```
interface IRobotMobileCaseComponent : IDispatch;
```

C#

```
public interface IRobotMobileCaseComponent;
```

Visual Basic

```
Public Interface IRobotMobileCaseComponent
```

Description

Simple component of a moving load case. This component corresponds to the simple case generated at the next point of the route. .

Version

Available since version 2.5.

II.1.8.3.1 IRobotMobileCaseComponent Members

The following tables list the members exposed by IRobotMobileCaseComponent.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	Point (see page 335)	Number of the point on the route .
	Records (see page 335)	List of load records.

II.1.8.3.2 IRobotMobileCaseComponent Fields

The fields of the IRobotMobileCaseComponent class are listed here.

Public Fields

	Name	Description
	Point (see page 335)	Number of the point on the route .
	Records (see page 335)	List of load records.

II.1.8.3.2.1 Point

C++

```
HRESULT get_Point(long*);
```

C#

```
public long Point { get; }
```

Visual Basic

```
Public ReadOnly Point As long
```

Description

Number of the point on the route .

Version

Available since version 2.5.

II.1.8.3.2.2 Records

C++

```
HRESULT get_Records(IRobotLoadRecordMngr**);
```

C#

```
public IRobotLoadRecordMngr Records { get; }
```

Visual Basic

```
Public ReadOnly Records As IRobotLoadRecordMngr
```

Description

List of load records.

Version

Available since version 2.5.

II.1.8.4 IRobotMobileCaseFlag

C++

```
enum IRobotMobileCaseFlag;
```

C#

```
public enum IRobotMobileCaseFlag;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Enum IRobotMobileCaseFlag
```

Members

Members	Description
I_MCF_MAIN = 0	Base moving load case. Available since version 2.5.
I_MCF_MAX = 1	Top envelope. Available since version 2.5.
I_MCF_MIN = 2	Bottom envelope. Available since version 2.5.

Description

Available types of a moving load case. .

Version

Available since version 2.5.

II.1.8.5 IRobotMobileCaseRoute

Class Hierarchy

RobotOM.IRobotMobileCaseRoute

C++

```
interface IRobotMobileCaseRoute : IDispatch;
```

C#

```
public interface IRobotMobileCaseRoute;
```

Visual Basic

```
Public Interface IRobotMobileCaseRoute
```

Description

Definition of vehicle route.

Version

Available since version 3.

II.1.8.5.1 IRobotMobileCaseRoute Members

The following tables list the members exposed by IRobotMobileCaseRoute.

Public Fields

	Name	Description
	BeginingRouteLimit (see page 337)	Limit of vehicle position at the route beginning.
	EndRouteLimit (see page 337)	Limit of vehicle position at the route end.
	Geometry (see page 338)	Number of the object that determines route geometry .
	LoadDirection (see page 338)	Global direction of load application.
	Step (see page 338)	Step - the distance between two successive vehicle positions on a route, expressed in meters.
	Tolerance (see page 339)	Value determining conditions of generating the loads associated with the vehicle in the indicated structure .

(c) 2014 Autodesk Inc. All rights reserved

Public Methods

	Name	Description
	GetFactors (see page 339)	Function takes the vector of coefficients correcting loads for a route segment of the indicated number.
	SetFactors (see page 339)	Function updates the vector of coefficients correcting loads for a route segment of the indicated number and on the basis of the indicated interface .

II.1.8.5.2 IRobotMobileCaseRoute Fields

The fields of the IRobotMobileCaseRoute class are listed here.

Public Fields

	Name	Description
	BeginingRouteLimit (see page 337)	Limit of vehicle position at the route beginning.
	EndRouteLimit (see page 337)	Limit of vehicle position at the route end.
	Geometry (see page 338)	Number of the object that determines route geometry .
	LoadDirection (see page 338)	Global direction of load application.
	Step (see page 338)	Step - the distance between two successive vehicle positions on a route, expressed in meters.
	Tolerance (see page 339)	Value determining conditions of generating the loads associated with the vehicle in the indicated structure .

II.1.8.5.2.1 BeginingRouteLimit

C++

```
HRESULT get_BeginingRouteLimit(VARIANT_BOOL*);
HRESULT put_BeginingRouteLimit(VARIANT_BOOL);
```

C#

```
public bool BeginingRouteLimit { get; set; }
```

Visual Basic

```
Public BeginingRouteLimit As Boolean
```

Description

Limit of vehicle position at the route beginning.

Version

Available since version 3.

II.1.8.5.2.2 EndRouteLimit

C++

```
HRESULT get_EndRouteLimit(VARIANT_BOOL*);
HRESULT put_EndRouteLimit(VARIANT_BOOL);
```

C#

```
public bool EndRouteLimit { get; set; }
```

Visual Basic

```
Public EndRouteLimit As Boolean
```

Description

Limit of vehicle position at the route end.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.

II.1.8.5.2.3 Geometry

C++

```
HRESULT get_Geometry(long*);
HRESULT put_Geometry(long);
```

C#

```
public long Geometry { get; set; }
```

Visual Basic

```
Public Geometry As long
```

Description

Number of the object that determines route geometry .

Version

Available since version 3.

II.1.8.5.2.4 LoadDirection

C++

```
HRESULT get_LoadDirection(IRobotGeoPoint3D**);
```

C#

```
public IRobotGeoPoint3D LoadDirection { get; }
```

Visual Basic

```
Public ReadOnly LoadDirection As IRobotGeoPoint3D
```

Description

Global direction of load application.

Version

Available since version 3.

II.1.8.5.2.5 Step

C++

```
HRESULT get_Step( double*);
HRESULT put_Step( double);
```

C#

```
public  double Step { get; set; }
```

Visual Basic

```
Public Step As  double
```

Description

Step - the distance between two successive vehicle positions on a route, expressed in meters.

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

II.1.8.5.2.6 Tolerance

C++

```
HRESULT get_Tolerance(double*);
HRESULT put_Tolerance(double);
```

C#

```
public double Tolerance { get; set; }
```

Visual Basic

```
Public Tolerance As double
```

Description

Value determining conditions of generating the loads associated with the vehicle in the indicated structure .

Version

Available since version 3.

II.1.8.5.3 IRobotMobileCaseRoute Methods

The methods of the IRobotMobileCaseRoute class are listed here.

Public Methods

	Name	Description
	GetFactors (see page 339)	Function takes the vector of coefficients correcting loads for a route segment of the indicated number.
	SetFactors (see page 339)	Function updates the vector of coefficients correcting loads for a route segment of the indicated number and on the basis of the indicated interface .

II.1.8.5.3.1 GetFactors

C++

```
HRESULT GetFactors(long _segment, IRobotMobileCaseSegmentFactors** ret);
```

C#

```
public IRobotMobileCaseSegmentFactors GetFactors(long _segment);
```

Visual Basic

```
Public Function GetFactors(_segment As long) As IRobotMobileCaseSegmentFactors
```

Description

Function takes the vector of coefficients correcting loads for a route segment of the indicated number.

Version

Available since version 3.

II.1.8.5.3.2 SetFactors

C++

```
HRESULT SetFactors(long _segment, IRobotMobileCaseSegmentFactors _factors);
```

C#

```
public void SetFactors(long _segment, IRobotMobileCaseSegmentFactors _factors);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Sub SetFactors(_segment As long, _factors As IRobotMobileCaseSegmentFactors)
```

Description

Function updates the vector of coefficients correcting loads for a route segment of the indicated number and on the basis of the indicated interface .

Version

Available since version 3.

II.1.8.6 IRobotMobileCaseSegmentFactors

Class Hierarchy

RobotOM.IRobotMobileCaseSegmentFactors

C++

```
interface IRobotMobileCaseSegmentFactors : IDispatch;
```

C#

```
public interface IRobotMobileCaseSegmentFactors;
```

Visual Basic

```
Public Interface IRobotMobileCaseSegmentFactors
```

Description

A set of coefficients assigned to a route segment, correcting vehicle loads. .

Version

Available since version 3.

II.1.8.6.1 IRobotMobileCaseSegmentFactors Members

The following tables list the members exposed by IRobotMobileCaseSegmentFactors.

Public Fields

	Name	Description
	Gamma (see page 341)	Angle determining vehicle rotation about the vertical axis.
	HL (see page 341)	Scaling factor for horizontal transversal force H (from left side).
	HR (see page 341)	Scaling factor for horizontal transversal force H (from right side).
	LL (see page 342)	Scaling factor for horizontal longitudinal force L (from left side).
	LR (see page 342)	Scaling factor for horizontal longitudinal force L (from right side).
	VL (see page 342)	Scaling factor for vertical force V (from left side).
	VR (see page 343)	Scaling factor for vertical force V (from right side).

II.1.8.6.2 IRobotMobileCaseSegmentFactors Fields

The fields of the IRobotMobileCaseSegmentFactors class are listed here.

Public Fields

	Name	Description
	Gamma (see page 341)	Angle determining vehicle rotation about the vertical axis.
	HL (see page 341)	Scaling factor for horizontal transversal force H (from left side).
	HR (see page 341)	Scaling factor for horizontal transversal force H (from right side).
	LL (see page 342)	Scaling factor for horizontal longitudinal force L (from left side).

(c) 2014 Autodesk Inc. All rights reserved

	LR (see page 342)	Scaling factor for horizontal longitudinal force L (from right side).
	VL (see page 342)	Scaling factor for vertical force V (from left side).
	VR (see page 343)	Scaling factor for vertical force V (from right side).

II.1.8.6.2.1 Gamma

C++

```
HRESULT get_Gamma( double*);
HRESULT put_Gamma( double);
```

C#

```
public  double Gamma { get; set; }
```

Visual Basic

```
Public Gamma As  double
```

Description

Angle determining vehicle rotation about the vertical axis.

Version

Available since version 3.

II.1.8.6.2.2 HL

C++

```
HRESULT get_HL( double*);
HRESULT put_HL( double);
```

C#

```
public  double HL { get; set; }
```

Visual Basic

```
Public HL As  double
```

Description

Scaling factor for horizontal transversal force H (from left side).

Version

Available since version 3.

II.1.8.6.2.3 HR

C++

```
HRESULT get_HR( double*);
HRESULT put_HR( double);
```

C#

```
public  double HR { get; set; }
```

Visual Basic

```
Public HR As  double
```

Description

Scaling factor for horizontal transversal force H (from right side).

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

II.1.8.6.2.4 LL

C++

```
HRESULT get_LL( double*);
HRESULT put_LL( double);
```

C#

```
public  double LL { get; set; }
```

Visual Basic

```
Public LL As  double
```

Description

Scaling factor for horizontal longitudinal force L (from left side).

Version

Available since version 3.

II.1.8.6.2.5 LR

C++

```
HRESULT get_LR( double*);
HRESULT put_LR( double);
```

C#

```
public  double LR { get; set; }
```

Visual Basic

```
Public LR As  double
```

Description

Scaling factor for horizontal longitudinal force L (from right side).

Version

Available since version 3.

II.1.8.6.2.6 VL

C++

```
HRESULT get_VL( double*);
HRESULT put_VL( double);
```

C#

```
public  double VL { get; set; }
```

Visual Basic

```
Public VL As  double
```

Description

Scaling factor for vertical force V (from left side).

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

II.1.8.6.2.7 VR

C++

```
HRESULT get_VR( double*);
HRESULT put_VR( double);
```

C#

```
public  double VR { get; set; }
```

Visual Basic

```
Public VR As  double
```

Description

Scaling factor for vertical force V (from right side).

Version

Available since version 3.

II.1.8.7 IRobotMobileCaseApplicationPlaneType

C++

```
enum IRobotMobileCaseApplicationPlaneType;
```

C#

```
public enum IRobotMobileCaseApplicationPlaneType;
```

Visual Basic

```
Public Enum IRobotMobileCaseApplicationPlaneType
```

Members

Members	Description
I_MCAPT_AUTOMATIC = 1	Application - automatic. Available since version 3.
I_MCAPT_SELECTION = 2	Application - to selected bars. Available since version 3.

Description

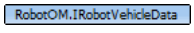
Type of application plane.

Version

Available since version 3.

II.1.8.8 IRobotVehicleData

Class Hierarchy



C++

```
interface IRobotVehicleData : IDispatch;
```

C#

```
public interface IRobotVehicleData;
```

Visual Basic

```
Public Interface IRobotVehicleData
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Parameter set for vehicle definition.

Version

Available since version 3.

II.1.8.8.1 IRobotVehicleData Members

The following tables list the members exposed by IRobotVehicleData.

Public Fields

	Name	Description
	b (see page 344)	Vehicle width.
	d1 (see page 344)	Distance between load and vehicle contour at the vehicle front .
	d2 (see page 345)	Distance between load and vehicle contour at the vehicle back.
	Loads (see page 345)	List of loads associated with the vehicle.

Public Methods

	Name	Description
	LoadFromDBase (see page 346)	Function reads vehicle parameters from the database.
	StoreToDBase (see page 346)	Function saves vehicle parameters to the database.

II.1.8.8.2 IRobotVehicleData Fields

The fields of the IRobotVehicleData class are listed here.

Public Fields

	Name	Description
	b (see page 344)	Vehicle width.
	d1 (see page 344)	Distance between load and vehicle contour at the vehicle front .
	d2 (see page 345)	Distance between load and vehicle contour at the vehicle back.
	Loads (see page 345)	List of loads associated with the vehicle.

II.1.8.8.2.1 b

C++

```
HRESULT get_b( double*);
HRESULT put_b( double);
```

C#

```
public  double b { get; set; }
```

Visual Basic

```
Public b As  double
```

Description

Vehicle width.

Version

Available since version 3.

II.1.8.8.2.2 d1

C++

```
HRESULT get_d1( double*);
HRESULT put_d1( double);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public  double d1 { get; set; }
```

Visual Basic

```
Public d1 As  double
```

Description

Distance between load and vehicle contour at the vehicle front .

Version

Available since version 3.

II.1.8.8.2.3 d2

C++

```
HRESULT get_d2( double*);
HRESULT put_d2( double);
```

C#

```
public  double d2 { get; set; }
```

Visual Basic

```
Public d2 As  double
```

Description

Distance between load and vehicle contour at the vehicle back.

Version

Available since version 3.

II.1.8.8.2.4 Loads

C++

```
HRESULT get_Loads(IRobotVehicleLoadMngr**);
```

C#

```
public IRobotVehicleLoadMngr Loads { get; }
```

Visual Basic

```
Public ReadOnly Loads As IRobotVehicleLoadMngr
```

Description

List of loads associated with the vehicle.

Version

Available since version 3.

II.1.8.8.3 IRobotVehicleData Methods

The methods of the IRobotVehicleData class are listed here.

Public Methods

	Name	Description
	LoadFromDBase (see page 346)	Function reads vehicle parameters from the database.
	StoreToDBase (see page 346)	Function saves vehicle parameters to the database.

(c) 2014 Autodesk Inc. All rights reserved

II.1.8.8.3.1 LoadFromDBase

C++

```
HRESULT LoadFromDBase(BSTR _vehicle_name, BSTR _db_name, VARIANT_BOOL* ret);
```

C#

```
public bool LoadFromDBase(String _vehicle_name, String _db_name);
```

Visual Basic

```
Public Function LoadFromDBase(_vehicle_name As String, _db_name As String) As Boolean
```

Description

Function reads vehicle parameters from the database.

Version

Available since version 3.

II.1.8.8.3.2 StoreToDBase

C++

```
HRESULT StoreToDBase(BSTR _vehicle_name, BSTR _db_name, VARIANT_BOOL* ret);
```

C#

```
public bool StoreToDBase(String _vehicle_name, String _db_name);
```

Visual Basic

```
Public Function StoreToDBase(_vehicle_name As String, _db_name As String) As Boolean
```

Description

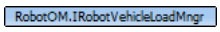
Function saves vehicle parameters to the database.

Version

Available since version 3.

II.1.8.9 IRobotVehicleLoadMngr

Class Hierarchy



C++

```
interface IRobotVehicleLoadMngr : IDispatch;
```

C#

```
public interface IRobotVehicleLoadMngr;
```

Visual Basic

```
Public Interface IRobotVehicleLoadMngr
```

Description

List of loads that define a vehicle.

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

II.1.8.9.1 IRobotVehicleLoadMngr Members

The following tables list the members exposed by IRobotVehicleLoadMngr.

Public Fields

	Name	Description
	Count (see page 347)	Number of records on the list.

Public Methods

	Name	Description
	Delete (see page 347)	Function removes a record from a list.
	Get (see page 348)	Function takes a record of the indicated index.
	New (see page 348)	Function adds a new load record at the end of the list.

II.1.8.9.2 IRobotVehicleLoadMngr Fields

The fields of the IRobotVehicleLoadMngr class are listed here.

Public Fields

	Name	Description
	Count (see page 347)	Number of records on the list.

II.1.8.9.2.1 Count

C++

```
HRESULT get_Count( long*);
```

C#

```
public  long Count { get; }
```

Visual Basic

```
Public ReadOnly Count As  long
```

Description

Number of records on the list.

Version

Available since version 3.

II.1.8.9.3 IRobotVehicleLoadMngr Methods

The methods of the IRobotVehicleLoadMngr class are listed here.

Public Methods

	Name	Description
	Delete (see page 347)	Function removes a record from a list.
	Get (see page 348)	Function takes a record of the indicated index.
	New (see page 348)	Function adds a new load record at the end of the list.

II.1.8.9.3.1 Delete

C++

```
HRESULT Delete(long _idx);
```

C#

```
public void Delete(long _idx);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Sub Delete(_idx As long)
```

Description

Function removes a record from a list.

Version

Available since version 3.

II.1.8.9.3.2 Get

C++

```
HRESULT Get(long _idx, IRobotVehicleLoad** ret);
```

C#

```
public IRobotVehicleLoad Get(long _idx);
```

Visual Basic

```
Public Function Get(_idx As long) As IRobotVehicleLoad
```

Description

Function takes a record of the indicated index.

Version

Available since version 3.

II.1.8.9.3.3 New

C++

```
HRESULT New(IRobotVehicleLoad** ret);
```

C#

```
public IRobotVehicleLoad New();
```

Visual Basic

```
Public Function New() As IRobotVehicleLoad
```

Description

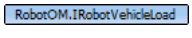
Function adds a new load record at the end of the list.

Version

Available since version 3.

II.1.8.10 IRobotVehicleLoad

Class Hierarchy



C++

```
interface IRobotVehicleLoad : IDispatch;
```

C#

```
public interface IRobotVehicleLoad;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Interface IRobotVehicleLoad
```

Description

Record describing a single load associated with the vehicle. .

Version

Available since version 3.

II.1.8.10.1 IRobotVehicleLoad Members

The following tables list the members exposed by IRobotVehicleLoad.

Public Fields

	Name	Description
	DX (see page 350)	For a linear load, it is the length of a segment on which the load is acting (along the vehicle axis); for a planar load, it is the length of a rectangle side on which the load is acting (along the vehicle axis); for concentrated force loads, the parameter is inactive.
	DY (see page 350)	For a linear load, it is the length of a segment on which the load is acting (perpendicular to the vehicle axis); for a planar load, it is the length of a rectangle side on which the load is acting (perpendicular to the vehicle axis); for concentrated force loads, the parameter is inactive .
	F (see page 350)	Depending on a load type, it is: value of a concentrated force, value of a linear load or value of a planar load .
	S (see page 351)	Depending on a load type, it is: value of force spacing, value of linear load spacing (only in the direction of Y axis) or value of planar load spacing (only in the direction of Y axis).
	Type (see page 351)	Load type.
	X (see page 351)	Coordinate value for the line along which the force is applied along the vehicle axis .

II.1.8.10.2 IRobotVehicleLoad Fields

The fields of the IRobotVehicleLoad class are listed here.

Public Fields

	Name	Description
	DX (see page 350)	For a linear load, it is the length of a segment on which the load is acting (along the vehicle axis); for a planar load, it is the length of a rectangle side on which the load is acting (along the vehicle axis); for concentrated force loads, the parameter is inactive.
	DY (see page 350)	For a linear load, it is the length of a segment on which the load is acting (perpendicular to the vehicle axis); for a planar load, it is the length of a rectangle side on which the load is acting (perpendicular to the vehicle axis); for concentrated force loads, the parameter is inactive .
	F (see page 350)	Depending on a load type, it is: value of a concentrated force, value of a linear load or value of a planar load .
	S (see page 351)	Depending on a load type, it is: value of force spacing, value of linear load spacing (only in the direction of Y axis) or value of planar load spacing (only in the direction of Y axis).
	Type (see page 351)	Load type.
	X (see page 351)	Coordinate value for the line along which the force is applied along the vehicle axis .

(c) 2014 Autodesk Inc. All rights reserved

II.1.8.10.2.1 DX

C++

```
HRESULT get_DX( double*);
HRESULT put_DX( double);
```

C#

```
public  double DX { get; set; }
```

Visual Basic

```
Public DX As  double
```

Description

For a linear load, it is the length of a segment on which the load is acting (along the vehicle axis); for a planar load, it is the length of a rectangle side on which the load is acting (along the vehicle axis); for concentrated force loads, the parameter is inactive.

Version

Available since version 3.

II.1.8.10.2.2 DY

C++

```
HRESULT get_DY( double*);
HRESULT put_DY( double);
```

C#

```
public  double DY { get; set; }
```

Visual Basic

```
Public DY As  double
```

Description

For a linear load, it is the length of a segment on which the load is acting (perpendicular to the vehicle axis); for a planar load, it is the length of a rectangle side on which the load is acting (perpendicular to the vehicle axis); for concentrated force loads, the parameter is inactive .

Version

Available since version 3.

II.1.8.10.2.3 F

C++

```
HRESULT get_F( double*);
HRESULT put_F( double);
```

C#

```
public  double F { get; set; }
```

Visual Basic

```
Public F As  double
```

Description

Depending on a load type, it is: value of a concentrated force, value of a linear load or value of a planar load .

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

II.1.8.10.2.4 S

C++

```
HRESULT get_S( double*);
HRESULT put_S( double);
```

C#

```
public  double S { get; set; }
```

Visual Basic

```
Public S As  double
```

Description

Depending on a load type, it is: value of force spacing, value of linear load spacing (only in the direction of Y axis) or value of planar load spacing (only in the direction of Y axis).

Version

Available since version 3.

II.1.8.10.2.5 Type

C++

```
HRESULT get_Type(IRobotVehicleLoadType*);
HRESULT put_Type(IRobotVehicleLoadType);
```

C#

```
public IRobotVehicleLoadType Type { get; set; }
```

Visual Basic

```
Public Type As IRobotVehicleLoadType
```

Description

Load type.

Version

Available since version 3.

II.1.8.10.2.6 X

C++

```
HRESULT get_X( double*);
HRESULT put_X( double);
```

C#

```
public  double X { get; set; }
```

Visual Basic

```
Public X As  double
```

Description

Coordinate value for the line along which the force is applied along the vehicle axis .

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

II.1.8.11 IRobotVehicleLoadType

C++

```
enum IRobotVehicleLoadType;
```

C#

```
public enum IRobotVehicleLoadType;
```

Visual Basic

```
Public Enum IRobotVehicleLoadType
```

Members

Members	Description
I_VLT_CONCENTRATED = 1	Concentrated force. Available since version 3.
I_VLT_LINEAR = 2	Linear load. Available since version 3.
I_VLT_SURFACE = 3	Surface load. Available since version 3.

Description

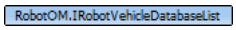
Load types.

Version

Available since version 3.

II.1.8.12 IRobotVehicleDatabaseList

Class Hierarchy



C++

```
interface IRobotVehicleDatabaseList : IDispatch;
```

C#

```
public interface IRobotVehicleDatabaseList;
```

Visual Basic

```
Public Interface IRobotVehicleDatabaseList
```

Description

Interface describing the list of vehicle databases.

Version

Available since version 3.

II.1.8.12.1 IRobotVehicleDatabaseList Members

The following tables list the members exposed by IRobotVehicleDatabaseList.

Public Fields

	Name	Description
	Count (see page 353)	Number of vehicle databases on the list.
	Default (see page 353)	Index of default vehicle database.

(c) 2014 Autodesk Inc. All rights reserved

Public Methods

	Name	Description
	Add (see page 354)	Function adds vehicle database of the indicated name at the end of the list. Status of the action is returned.
	AddFromFile (see page 354)	Function adds the vehicle database saved in the indicated file at the end of the list. Status of the operation execution is returned.
	Create (see page 355)	Function creates a new vehicle database.
	Find (see page 355)	Function searches the list seeking for the database of the indicated name. If there is no specified database on the list, zero value is returned, if there is, function returns the database index from the list. .
	Get (see page 355)	Function returns the name of the vehicle database located at the indicated position on the list.
	GetDatabase (see page 356)	Function returns the interface for the vehicle database with the index indicated on the list.
	Remove (see page 356)	Function deletes the database of the indicated index from the list. .

II.1.8.12.2 IRobotVehicleDatabaseList Fields

The fields of the IRobotVehicleDatabaseList class are listed here.

Public Fields

	Name	Description
	Count (see page 353)	Number of vehicle databases on the list.
	Default (see page 353)	Index of default vehicle database.

II.1.8.12.2.1 Count

C++

```
HRESULT get_Count(int*);
```

C#

```
public int Count { get; }
```

Visual Basic

```
Public ReadOnly Count As int
```

Description

Number of vehicle databases on the list.

Version

Available since version 3.

II.1.8.12.2.2 Default

C++

```
HRESULT get_Default(int*);
HRESULT put_Default(int);
```

C#

```
public int Default { get; set; }
```

Visual Basic

```
Public Default As int
```

Description

Index of default vehicle database.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.

II.1.8.12.3 IRobotVehicleDatabaseList Methods

The methods of the IRobotVehicleDatabaseList class are listed here.

Public Methods

	Name	Description
	Add (see page 354)	Function adds vehicle database of the indicated name at the end of the list. Status of the action is returned.
	AddFromFile (see page 354)	Function adds the vehicle database saved in the indicated file at the end of the list. Status of the operation execution is returned.
	Create (see page 355)	Function creates a new vehicle database.
	Find (see page 355)	Function searches the list seeking for the database of the indicated name. If there is no specified database on the list, zero value is returned, if there is, function returns the database index from the list. .
	Get (see page 355)	Function returns the name of the vehicle database located at the indicated position on the list.
	GetDatabase (see page 356)	Function returns the interface for the vehicle database with the index indicated on the list.
	Remove (see page 356)	Function deletes the database of the indicated index from the list. .

II.1.8.12.3.1 Add

C++

```
HRESULT Add(BSTR _db_name, VARIANT_BOOL* ret);
```

C#

```
public bool Add(String _db_name);
```

Visual Basic

```
Public Function Add(_db_name As String) As Boolean
```

Description

Function adds vehicle database of the indicated name at the end of the list. Status of the action is returned.

Version

Available since version 3.

II.1.8.12.3.2 AddFromFile

C++

```
HRESULT AddFromFile(BSTR _file_path, VARIANT_BOOL* ret);
```

C#

```
public bool AddFromFile(String _file_path);
```

Visual Basic

```
Public Function AddFromFile(_file_path As String) As Boolean
```

Description

Function adds the vehicle database saved in the indicated file at the end of the list. Status of the operation execution is returned.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.

II.1.8.12.3.3 Create

C++

```
HRESULT Create(BSTR _db_name, BSTR _long_db_name, BSTR _description);
```

C#

```
public void Create(String _db_name, String _long_db_name, String _description);
```

Visual Basic

```
Public Sub Create(_db_name As String, _long_db_name As String, _description As String)
```

Description

Function creates a new vehicle database.

Version

Available since version 3.

II.1.8.12.3.4 Find

C++

```
HRESULT Find(BSTR _db_name, int* ret);
```

C#

```
public int Find(String _db_name);
```

Visual Basic

```
Public Function Find(_db_name As String) As int
```

Description

Function searches the list seeking for the database of the indicated name. If there is no specified database on the list, zero value is returned, if there is, function returns the database index from the list. .

Version

Available since version 3.

II.1.8.12.3.5 Get

C++

```
HRESULT Get(int _idx, String* ret);
```

C#

```
public String Get(int _idx);
```

Visual Basic

```
Public Function Get(_idx As int) As String
```

Description

Function returns the name of the vehicle database located at the indicated position on the list.

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

II.1.8.12.3.6 GetDatabase

C++

```
HRESULT GetDatabase(int _idx, IRobotVehicleDatabase** ret);
```

C#

```
public IRobotVehicleDatabase GetDatabase(int _idx);
```

Visual Basic

```
Public Function GetDatabase(_idx As int) As IRobotVehicleDatabase
```

Description

Function returns the interface for the vehicle database with the index indicated on the list.

Version

Available since version 3.

II.1.8.12.3.7 Remove

C++

```
HRESULT Remove(int _idx);
```

C#

```
public void Remove(int _idx);
```

Visual Basic

```
Public Sub Remove(_idx As int)
```

Description

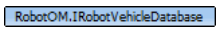
Function deletes the database of the indicated index from the list. .

Version

Available since version 3.

II.1.8.13 IRobotVehicleDatabase

Class Hierarchy



C++

```
interface IRobotVehicleDatabase : IDispatch;
```

C#

```
public interface IRobotVehicleDatabase;
```

Visual Basic

```
Public Interface IRobotVehicleDatabase
```

Description

Interface for vehicle database support.

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

II.1.8.13.1 IRobotVehicleDatabase Members

The following tables list the members exposed by IRobotVehicleDatabase.

Public Fields

	Name	Description
	Description (see page 357)	
	LongName (see page 357)	Additional description of a vehicle database.
	Name (see page 358)	Name identifying the vehicle database.

Public Methods

	Name	Description
	GetAll (see page 358)	Function returns a table containing names of all the vehicles from the database. .
	Load (see page 359)	Function reads vehicles from the vehicle database of the indicated name. If the specified database does not exist or an error occurs during read-out, zero value is returned. .
	LoadFromFile (see page 359)	Function reads vehicles from the database saved in the indicated file. If the specified path is incorrect or other reading error occurs, the function returns zero value. .

II.1.8.13.2 IRobotVehicleDatabase Fields

The fields of the IRobotVehicleDatabase class are listed here.

Public Fields

	Name	Description
	Description (see page 357)	
	LongName (see page 357)	Additional description of a vehicle database.
	Name (see page 358)	Name identifying the vehicle database.

II.1.8.13.2.1 Description

C++

```
HRESULT get_Description(BSTR*);
```

C#

```
public String Description { get; }
```

Visual Basic

```
Public ReadOnly Description As String
```

Version

Available since version 3.

II.1.8.13.2.2 LongName

C++

```
HRESULT get_LongName(BSTR*);
```

C#

```
public String LongName { get; }
```

Visual Basic

```
Public ReadOnly LongName As String
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Additional description of a vehicle database.

Version

Available since version 3.

II.1.8.13.2.3 Name

C++

```
HRESULT get_Name(BSTR*);
```

C#

```
public String Name { get; }
```

Visual Basic

```
Public ReadOnly Name As String
```

Description

Name identifying the vehicle database.

Version

Available since version 3.

II.1.8.13.3 IRobotVehicleDatabase Methods

The methods of the IRobotVehicleDatabase class are listed here.

Public Methods

	Name	Description
	GetAll (see page 358)	Function returns a table containing names of all the vehicles from the database. .
	Load (see page 359)	Function reads vehicles from the vehicle database of the indicated name. If the specified database does not exist or an error occurs during read-out, zero value is returned. .
	LoadFromFile (see page 359)	Function reads vehicles from the database saved in the indicated file. If the specified path is incorrect or other reading error occurs, the function returns zero value. .

II.1.8.13.3.1 GetAll

C++

```
HRESULT GetAll(IRobotNamesArray* ret);
```

C#

```
public IRobotNamesArray GetAll();
```

Visual Basic

```
Public Function GetAll() As IRobotNamesArray
```

Description

Function returns a table containing names of all the vehicles from the database. .

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

II.1.8.13.3.2 Load

C++

```
HRESULT Load(BSTR _db_name, VARIANT_BOOL* ret);
```

C#

```
public bool Load(String _db_name);
```

Visual Basic

```
Public Function Load(_db_name As String) As Boolean
```

Description

Function reads vehicles from the vehicle database of the indicated name. If the specified database does not exist or an error occurs during read-out, zero value is returned. .

Version

Available since version 3.

II.1.8.13.3.3 LoadFromFile

C++

```
HRESULT LoadFromFile(BSTR _file_path, VARIANT_BOOL* ret);
```

C#

```
public bool LoadFromFile(String _file_path);
```

Visual Basic

```
Public Function LoadFromFile(_file_path As String) As Boolean
```

Description

Function reads vehicles from the database saved in the indicated file. If the specified path is incorrect or other reading error occurs, the function returns zero value. .

Version

Available since version 3.

II.1.9 Buckling analysis parameters

Available since version 3.

Enumerations

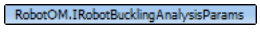
	Name	Description
	IRobotBucklingAnalysisMethod (see page 363)	Methods of eigenproblem solution for buckling analysis.

Interfaces

	Name	Description
	IRobotBucklingAnalysisParams (see page 359)	Buckling analysis parameters.

II.1.9.1 IRobotBucklingAnalysisParams

Class Hierarchy



(c) 2014 Autodesk Inc. All rights reserved

C++

```
interface IRobotBucklingAnalysisParams : IDispatch;
```

C#

```
public interface IRobotBucklingAnalysisParams;
```

Visual Basic

```
Public Interface IRobotBucklingAnalysisParams
```

Description

Buckling analysis parameters.

Version

Available since version 3.

II.1.9.1.1 IRobotBucklingAnalysisParams Members

The following tables list the members exposed by IRobotBucklingAnalysisParams.

Public Fields

	Name	Description
	Increment (see page 361)	Number of load increments.
	IsNonlinear (see page 361)	
	IterationsCount (see page 361)	Maximum number of iterations.
	Method (see page 361)	Method of eigenproblem solution.
	ModesCount (see page 362)	Number of buckling modes.
	NonlinearParams (see page 362)	Parameters of non-linear analysis (available if IsNonlinear (see page 361) flag is set).
	Shift (see page 362)	Shift value.
	SturmVerification (see page 363)	Flag switching on/off the algorithm that enables searching disregarded structure modes.
	Tolerance (see page 363)	Value of the error tolerance that is to be obtained during structure iteration analysis.

II.1.9.1.2 IRobotBucklingAnalysisParams Fields

The fields of the IRobotBucklingAnalysisParams class are listed here.

Public Fields

	Name	Description
	Increment (see page 361)	Number of load increments.
	IsNonlinear (see page 361)	
	IterationsCount (see page 361)	Maximum number of iterations.
	Method (see page 361)	Method of eigenproblem solution.
	ModesCount (see page 362)	Number of buckling modes.
	NonlinearParams (see page 362)	Parameters of non-linear analysis (available if IsNonlinear (see page 361) flag is set).
	Shift (see page 362)	Shift value.
	SturmVerification (see page 363)	Flag switching on/off the algorithm that enables searching disregarded structure modes.
	Tolerance (see page 363)	Value of the error tolerance that is to be obtained during structure iteration analysis.

(c) 2014 Autodesk Inc. All rights reserved

II.1.9.1.2.1 Increment

C++

```
HRESULT get_Increment(long*);
HRESULT put_Increment(long);
```

C#

```
public long Increment { get; set; }
```

Visual Basic

```
Public Increment As long
```

Description

Number of load increments.

Version

Available since version 3.

II.1.9.1.2.2 IsNonlinear

C++

```
HRESULT get_IsNonlinear(VARIANT_BOOL*);
```

C#

```
public bool IsNonlinear { get; }
```

Visual Basic

```
Public ReadOnly IsNonlinear As Boolean
```

Version

Available since version 3.

II.1.9.1.2.3 IterationsCount

C++

```
HRESULT get_IterationsCount(long*);
HRESULT put_IterationsCount(long);
```

C#

```
public long IterationsCount { get; set; }
```

Visual Basic

```
Public IterationsCount As long
```

Description

Maximum number of iterations.

Version

Available since version 3.

II.1.9.1.2.4 Method

C++

```
HRESULT get_Method(IRobotBucklingAnalysisMethod*);
HRESULT put_Method(IRobotBucklingAnalysisMethod);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRobotBucklingAnalysisMethod Method { get; set; }
```

Visual Basic

```
Public Method As IRobotBucklingAnalysisMethod
```

Description

Method of eigenproblem solution.

Version

Available since version 3.

II.1.9.1.2.5 ModesCount

C++

```
HRESULT get_ModesCount(long*);
HRESULT put_ModesCount(long);
```

C#

```
public long ModesCount { get; set; }
```

Visual Basic

```
Public ModesCount As long
```

Description

Number of buckling modes.

Version

Available since version 3.

II.1.9.1.2.6 NonlinearParams

C++

```
HRESULT get_NonlinearParams(IRobotNonlinearAnalysisParams**);
```

C#

```
public IRobotNonlinearAnalysisParams NonlinearParams { get; }
```

Visual Basic

```
Public ReadOnly NonlinearParams As IRobotNonlinearAnalysisParams
```

Description

Parameters of non-linear analysis (available if IsNonlinear (see page 361) flag is set).

Version

Available since version 3.

II.1.9.1.2.7 Shift

C++

```
HRESULT get_Shift(double*);
HRESULT put_Shift(double);
```

C#

```
public double Shift { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Shift As double
```

Description

Shift value.

Version

Available since version 3.

II.1.9.1.2.8 SturmVerification

C++

```
HRESULT get_SturmVerification(VARIANT_BOOL*);
HRESULT put_SturmVerification(VARIANT_BOOL);
```

C#

```
public bool SturmVerification { get; set; }
```

Visual Basic

```
Public SturmVerification As Boolean
```

Description

Flag switching on/off the algorithm that enables searching disregarded structure modes.

Version

Available since version 3.

II.1.9.1.2.9 Tolerance

C++

```
HRESULT get_Tolerance(double*);
HRESULT put_Tolerance(double);
```

C#

```
public double Tolerance { get; set; }
```

Visual Basic

```
Public Tolerance As double
```

Description

Value of the error tolerance that is to be obtained during structure iteration analysis.

Version

Available since version 3.

II.1.9.2 IRobotBucklingAnalysisMethod

C++

```
enum IRobotBucklingAnalysisMethod;
```

C#

```
public enum IRobotBucklingAnalysisMethod;
```

Visual Basic

```
Public Enum IRobotBucklingAnalysisMethod
```

(c) 2014 Autodesk Inc. All rights reserved

Members

Members	Description
I_BAM_SUBSPACE_ITERATION = 0	Subspace iteration. Available since version 3.
I_BAM_BLOCK_SUBSPACE_ITERATION = 1	Block subspace iteration. Available since version 3.

Description

Methods of eigenproblem solution for buckling analysis.

Version

Available since version 3.

II.1.10 Time history analysis parameters

Available since version 3.

Enumerations

	Name	Description
	IRobotTimeHistoryAnalysisMethod (see page 369)	Available methods of time history analysis. .

Interfaces

	Name	Description
	IRobotTimeHistoryAnalysisParams (see page 364)	Parameters of time history analysis.
	IRobotTimeHistoryNewmarkParams (see page 370)	Parameters of Newmark method for time history analysis. .
	IRobotTimeHistoryModalDecompositionParams (see page 371)	Parameters of modal decomposition method. It is a list of pairs [mode number, damping value]. For each mode, maximally one such a pair may exist. Additionally, the pairs are indexed with numbers from 1 to Count (see page 372). .
	IRobotTimeHistoryPointsCollection (see page 375)	Collection of points defining a time function for the time history analysis. .
	IRobotTimeHistoryFunctionList (see page 378)	List of defined time functions for time history analysis. .
	IRobotTimeHistoryNonlinearParams (see page 383)	Non-linear parameters of time history analysis.
	IRobotTimeHistoryHHTParams (see page 386)	Parameters of HHT (Hilber-Hughes-Taylor) method for time history analysis.
	IRobotTimeHistoryNewmarkAccelParams (see page 388)	Parameters of Newmark method (acceleration) for time history analysis.

II.1.10.1 IRobotTimeHistoryAnalysisParams

Class Hierarchy

RobotOM.IRobotTimeHistoryAnalysisParams

C++

```
interface IRobotTimeHistoryAnalysisParams : IDispatch;
```

C#

```
public interface IRobotTimeHistoryAnalysisParams;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Interface IRobotTimeHistoryAnalysisParams
```

Description

Parameters of time history analysis.

Version

Available since version 3.

II.1.10.1.1 IRobotTimeHistoryAnalysisParams Members

The following tables list the members exposed by IRobotTimeHistoryAnalysisParams.

Public Fields

	Name	Description
	Count (see page 365)	Number of defined time history analyses.
	Division (see page 366)	Number of time division for the result time step.
	End (see page 366)	Final value of the time variable for which the analysis is performed .
	InitialCase (see page 366)	Number of the load case for which results are to be taken as initial.
	Method (see page 367)	Time history analysis method.
	MethodParams (see page 367)	Parameters of selected method.
	TimeStep (see page 367)	Step of the time variable for which results are being saved.

Public Methods

	Name	Description
	Delete (see page 368)	Function deletes time history analysis definition of the index specified on the list.
	Find (see page 368)	Function returns index of the time history analysis definition for the specified static case (or mass). If for the indicated case the time history analysis has not been defined, function returns zero value. .
	Get (see page 369)	Function returns the time history analysis definition of the specified index. .
	Set (see page 369)	Function defines the time history analysis for the specified static case or mass.

II.1.10.1.2 IRobotTimeHistoryAnalysisParams Fields

The fields of the IRobotTimeHistoryAnalysisParams class are listed here.

Public Fields

	Name	Description
	Count (see page 365)	Number of defined time history analyses.
	Division (see page 366)	Number of time division for the result time step.
	End (see page 366)	Final value of the time variable for which the analysis is performed .
	InitialCase (see page 366)	Number of the load case for which results are to be taken as initial.
	Method (see page 367)	Time history analysis method.
	MethodParams (see page 367)	Parameters of selected method.
	TimeStep (see page 367)	Step of the time variable for which results are being saved.

II.1.10.1.2.1 Count

C++

```
HRESULT get_Count(long*);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public long Count { get; }
```

Visual Basic

```
Public ReadOnly Count As long
```

Description

Number of defined time history analyses.

Version

Available since version 3.

II.1.10.1.2.2 Division

C++

```
HRESULT get_Division(long*);
HRESULT put_Division(long);
```

C#

```
public long Division { get; set; }
```

Visual Basic

```
Public Division As long
```

Description

Number of time division for the result time step.

Version

Available since version 3.

II.1.10.1.2.3 End

C++

```
HRESULT get_End(double*);
HRESULT put_End(double);
```

C#

```
public double End { get; set; }
```

Visual Basic

```
Public End As double
```

Description

Final value of the time variable for which the analysis is performed .

Version

Available since version 3.

II.1.10.1.2.4 InitialCase

C++

```
HRESULT get_InitialCase(long*);
HRESULT put_InitialCase(long);
```

C#

```
public long InitialCase { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public InitialCase As long
```

Description

Number of the load case for which results are to be taken as initial.

Version

Available since version 11.

II.1.10.1.2.5 Method

C++

```
HRESULT get_Method(IRobotTimeHistoryAnalysisMethod*);
HRESULT put_Method(IRobotTimeHistoryAnalysisMethod);
```

C#

```
public IRobotTimeHistoryAnalysisMethod Method { get; set; }
```

Visual Basic

```
Public Method As IRobotTimeHistoryAnalysisMethod
```

Description

Time history analysis method.

Version

Available since version 3.

II.1.10.1.2.6 MethodParams

C++

```
HRESULT get_MethodParams(IDispatch*);
```

C#

```
public IDispatch MethodParams { get; }
```

Visual Basic

```
Public ReadOnly MethodParams As IDispatch
```

Description

Parameters of selected method.

Version

Available since version 3.

II.1.10.1.2.7 TimeStep

C++

```
HRESULT get_TimeStep(double*);
HRESULT put_TimeStep(double);
```

C#

```
public double TimeStep { get; set; }
```

Visual Basic

```
Public TimeStep As double
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Step of the time variable for which results are being saved.

Version

Available since version 3.

II.1.10.1.3 IRobotTimeHistoryAnalysisParams Methods

The methods of the IRobotTimeHistoryAnalysisParams class are listed here.

Public Methods

	Name	Description
	Delete (see page 368)	Function deletes time history analysis definition of the index specified on the list.
	Find (see page 368)	Function returns index of the time history analysis definition for the specified static case (or mass). If for the indicated case the time history analysis has not been defined, function returns zero value. .
	Get (see page 369)	Function returns the time history analysis definition of the specified index. .
	Set (see page 369)	Function defines the time history analysis for the specified static case or mass.

II.1.10.1.3.1 Delete

C++

```
HRESULT Delete(long _idx);
```

C#

```
public void Delete(long _idx);
```

Visual Basic

```
Public Sub Delete(_idx As long)
```

Description

Function deletes time history analysis definition of the index specified on the list.

Version

Available since version 3.

II.1.10.1.3.2 Find

C++

```
HRESULT Find(long _case_num, long* ret);
```

C#

```
public long Find(long _case_num);
```

Visual Basic

```
Public Function Find(_case_num As long) As long
```

Description

Function returns index of the time history analysis definition for the specified static case (or mass). If for the indicated case the time history analysis has not been defined, function returns zero value. .

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.

II.1.10.1.3.3 Get

C++

```
HRESULT Get(long _idx, long* _case_num, BSTR _time_fun, double* _coeff, double* _shift);
```

C#

```
public void Get(long _idx, long* _case_num, String _time_fun, double* _coeff, double*
_shift);
```

Visual Basic

```
Public Sub Get(_idx As long, ByRef _case_num As long*, ByRef _time_fun As String, ByRef
_coeff As double*, ByRef _shift As double*)
```

Description

Function returns the time history analysis definition of the specified index. .

Version

Available since version 3.

II.1.10.1.3.4 Set

C++

```
HRESULT Set(long _case_num, BSTR _time_fun, double _coeff, double _shift);
```

C#

```
public void Set(long _case_num, String _time_fun, double _coeff, double _shift);
```

Visual Basic

```
Public Sub Set(_case_num As long, _time_fun As String, _coeff As double, _shift As double)
```

Description

Function defines the time history analysis for the specified static case or mass.

Version

Available since version 3.

II.1.10.2 IRobotTimeHistoryAnalysisMethod

C++

```
enum IRobotTimeHistoryAnalysisMethod;
```

C#

```
public enum IRobotTimeHistoryAnalysisMethod;
```

Visual Basic

```
Public Enum IRobotTimeHistoryAnalysisMethod
```

Members

Members	Description
I_THAM_NEWMARK = 0	Available since version 3.
I_THAM_MODAL_DECOMPOSITION = 1	Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

I_THAM_HHT = 3	HHT (Hilber-Hughes-Taylor) method. Available since version 11.
I_THAM_NEWMARK_ACCELERATION = 4	Newmark method (acceleration). Available since version 11.

Description

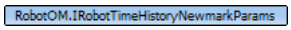
Available methods of time history analysis. .

Version

Available since version 3.

II.1.10.3 IRobotTimeHistoryNewmarkParams

Class Hierarchy



C++

```
interface IRobotTimeHistoryNewmarkParams : IDispatch;
```

C#

```
public interface IRobotTimeHistoryNewmarkParams;
```

Visual Basic

```
Public Interface IRobotTimeHistoryNewmarkParams
```

Description

Parameters of Newmark method for time history analysis. .

Version

Available since version 3.

II.1.10.3.1 IRobotTimeHistoryNewmarkParams Members

The following tables list the members exposed by IRobotTimeHistoryNewmarkParams.

Public Fields

	Name	Description
	Alpha (see page 370)	Damping parameter.
	Beta (see page 371)	Damping parameter.
	MassMatrixType (see page 371)	Stiffness matrix type.

II.1.10.3.2 IRobotTimeHistoryNewmarkParams Fields

The fields of the IRobotTimeHistoryNewmarkParams class are listed here.

Public Fields

	Name	Description
	Alpha (see page 370)	Damping parameter.
	Beta (see page 371)	Damping parameter.
	MassMatrixType (see page 371)	Stiffness matrix type.

II.1.10.3.2.1 Alpha

C++

```
HRESULT get_Alpha(double*);
HRESULT put_Alpha(double);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double Alpha { get; set; }
```

Visual Basic

```
Public Alpha As double
```

Description

Damping parameter.

Version

Available since version 3.

II.1.10.3.2.2 Beta

C++

```
HRESULT get_Beta(double*);
HRESULT put_Beta(double);
```

C#

```
public double Beta { get; set; }
```

Visual Basic

```
Public Beta As double
```

Description

Damping parameter.

Version

Available since version 3.

II.1.10.3.2.3 MassMatrixType

C++

```
HRESULT get_MassMatrixType(IRobotModalAnalysisMassMatrixType*);
HRESULT put_MassMatrixType(IRobotModalAnalysisMassMatrixType);
```

C#

```
public IRobotModalAnalysisMassMatrixType MassMatrixType { get; set; }
```

Visual Basic

```
Public MassMatrixType As IRobotModalAnalysisMassMatrixType
```

Description

Stiffness matrix type.

Version

Available since version 3.

II.1.10.4 IRobotTimeHistoryModalDecompositionParams

Class Hierarchy

RobotOM.IRobotTimeHistoryModalDecompositionParams

C++

```
interface IRobotTimeHistoryModalDecompositionParams : IDispatch;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public interface IRobotTimeHistoryModalDecompositionParams;
```

Visual Basic

```
Public Interface IRobotTimeHistoryModalDecompositionParams
```

Description

Parameters of modal decomposition method. It is a list of pairs [mode number, damping value]. For each mode, maximally one such a pair may exist. Additionally, the pairs are indexed with numbers from 1 to Count (see page 372). .

Version

Available since version 3.

II.1.10.4.1 IRobotTimeHistoryModalDecompositionParams Members

The following tables list the members exposed by IRobotTimeHistoryModalDecompositionParams.

Public Fields

	Name	Description
	Count (see page 372)	Number of defined mode pairs [mode number, damping value].

Public Methods

	Name	Description
	Delete (see page 373)	Function removes the selected list element.
	DeleteMode (see page 373)	Function deletes the pair containing the selected mode from the list.
	Get (see page 373)	Function returns the next pair from the list.
	GetDamping (see page 374)	Function returns a damping value for a selected mode.
	IsDefined (see page 374)	Function returns non-zero value (True), if for the specified mode damping has been defined. .
	SetDamping (see page 374)	Function sets a damping value for a given mode.

II.1.10.4.2 IRobotTimeHistoryModalDecompositionParams Fields

The fields of the IRobotTimeHistoryModalDecompositionParams class are listed here.

Public Fields

	Name	Description
	Count (see page 372)	Number of defined mode pairs [mode number, damping value].

II.1.10.4.2.1 Count

C++

```
HRESULT get_Count(long*);
```

C#

```
public long Count { get; }
```

Visual Basic

```
Public ReadOnly Count As long
```

Description

Number of defined mode pairs [mode number, damping value].

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

II.1.10.4.3 IRobotTimeHistoryModalDecompositionParams Methods

The methods of the IRobotTimeHistoryModalDecompositionParams class are listed here.

Public Methods

	Name	Description
	Delete (see page 373)	Function removes the selected list element.
	DeleteMode (see page 373)	Function deletes the pair containing the selected mode from the list.
	Get (see page 373)	Function returns the next pair from the list.
	GetDamping (see page 374)	Function returns a damping value for a selected mode.
	IsDefined (see page 374)	Function returns non-zero value (True), if for the specified mode damping has been defined. .
	SetDamping (see page 374)	Function sets a damping value for a given mode.

II.1.10.4.3.1 Delete

C++

```
HRESULT Delete(long _idx);
```

C#

```
public void Delete(long _idx);
```

Visual Basic

```
Public Sub Delete(_idx As long)
```

Description

Function removes the selected list element.

Version

Available since version 3.

II.1.10.4.3.2 DeleteMode

C++

```
HRESULT DeleteMode(long _mode);
```

C#

```
public void DeleteMode(long _mode);
```

Visual Basic

```
Public Sub DeleteMode(_mode As long)
```

Description

Function deletes the pair containing the selected mode from the list.

Version

Available since version 3.

II.1.10.4.3.3 Get

C++

```
HRESULT Get(long _idx, long* _mode, double* _damp);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public void Get(long _idx, long* _mode, double* _damp);
```

Visual Basic

```
Public Sub Get(_idx As long, ByRef _mode As long*, ByRef _damp As double*)
```

Description

Function returns the next pair from the list.

Version

Available since version 3.

II.1.10.4.3.4 GetDamping

C++

```
HRESULT GetDamping(long _mode, double* ret);
```

C#

```
public double GetDamping(long _mode);
```

Visual Basic

```
Public Function GetDamping(_mode As long) As double
```

Description

Function returns a damping value for a selected mode.

Version

Available since version 3.

II.1.10.4.3.5 IsDefined

C++

```
HRESULT IsDefined(long _mode, VARIANT_BOOL* ret);
```

C#

```
public bool IsDefined(long _mode);
```

Visual Basic

```
Public Function IsDefined(_mode As long) As Boolean
```

Description

Function returns non-zero value (True), if for the specified mode damping has been defined. .

Version

Available since version 3.

II.1.10.4.3.6 SetDamping

C++

```
HRESULT SetDamping(long _mode, double _damp);
```

C#

```
public void SetDamping(long _mode, double _damp);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Sub SetDamping(_mode As long, _damp As double)
```

Description

Function sets a damping value for a given mode.

Version

Available since version 3.

II.1.10.5 IRobotTimeHistoryPointsCollection

Class Hierarchy

RobotOM.IRobotTimeHistoryPointsCollection

C++

```
interface IRobotTimeHistoryPointsCollection : IDispatch;
```

C#

```
public interface IRobotTimeHistoryPointsCollection;
```

Visual Basic

```
Public Interface IRobotTimeHistoryPointsCollection
```

Description

Collection of points defining a time function for the time history analysis. .

Version

Available since version 3.

II.1.10.5.1 IRobotTimeHistoryPointsCollection Members

The following tables list the members exposed by IRobotTimeHistoryPointsCollection.

Public Fields

	Name	Description
	Count (see page 376)	Number of points in a collection.

Public Methods

	Name	Description
	Add (see page 376)	Function adds a new point of the (X, Y) coordinates. If a point of X coordinate has been defined earlier, then its value will be replaced by a new Y value. .
	Clear (see page 377)	Function removes all collection points.
	Delete (see page 377)	Function removes selected collection point.
	Find (see page 377)	Function returns index of the point defined for the X coordinate. If for the indicated X coordinate, Y value has not been defined, then function returns zero value (0). .
	Get (see page 378)	Function returns selected collection point.
	LoadFromFile (see page 378)	Function reads in the collection of points from the file and adds it to the points currently defined. .
	SaveToFile (see page 378)	Function saves the collection of points to a file. .

II.1.10.5.2 IRobotTimeHistoryPointsCollection Fields

The fields of the IRobotTimeHistoryPointsCollection class are listed here.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	Count (see page 376)	Number of points in a collection.

II.1.10.5.2.1 Count

C++

```
HRESULT get_Count(long*);
```

C#

```
public long Count { get; }
```

Visual Basic

```
Public ReadOnly Count As long
```

Description

Number of points in a collection.

Version

Available since version 3.

II.1.10.5.3 IRobotTimeHistoryPointsCollection Methods

The methods of the IRobotTimeHistoryPointsCollection class are listed here.

Public Methods

	Name	Description
	Add (see page 376)	Function adds a new point of the (X, Y) coordinates. If a point of X coordinate has been defined earlier, then its value will be replaced by a new Y value. .
	Clear (see page 377)	Function removes all collection points.
	Delete (see page 377)	Function removes selected collection point.
	Find (see page 377)	Function returns index of the point defined for the X coordinate. If for the indicated X coordinate, Y value has not been defined, then function returns zero value (0). .
	Get (see page 378)	Function returns selected collection point.
	LoadFromFile (see page 378)	Function reads in the collection of points from the file and adds it to the points currently defined. .
	SaveToFile (see page 378)	Function saves the collection of points to a file. .

II.1.10.5.3.1 Add

C++

```
HRESULT Add(double _x, double _y);
```

C#

```
public void Add(double _x, double _y);
```

Visual Basic

```
Public Sub Add(_x As double, _y As double)
```

Description

Function adds a new point of the (X, Y) coordinates. If a point of X coordinate has been defined earlier, then its value will be replaced by a new Y value. .

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.

II.1.10.5.3.2 Clear

C++

```
HRESULT Clear();
```

C#

```
public void Clear();
```

Visual Basic

```
Public Sub Clear()
```

Description

Function removes all collection points.

Version

Available since version 3.

II.1.10.5.3.3 Delete

C++

```
HRESULT Delete(long _idx);
```

C#

```
public void Delete(long _idx);
```

Visual Basic

```
Public Sub Delete(_idx As long)
```

Description

Function removes selected collection point.

Version

Available since version 3.

II.1.10.5.3.4 Find

C++

```
HRESULT Find(double _x, long* ret);
```

C#

```
public long Find(double _x);
```

Visual Basic

```
Public Function Find(_x As double) As long
```

Description

Function returns index of the point defined for the X coordinate. If for the indicated X coordinate, Y value has not been defined, then function returns zero value (0). .

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

II.1.10.5.3.5 Get

C++

```
HRESULT Get(long _idx, double* _x, double* _y);
```

C#

```
public void Get(long _idx, double* _x, double* _y);
```

Visual Basic

```
Public Sub Get(_idx As long, ByRef _x As double*, ByRef _y As double*)
```

Description

Function returns selected collection point.

Version

Available since version 3.

II.1.10.5.3.6 LoadFromFile

C++

```
HRESULT LoadFromFile(BSTR _file_path);
```

C#

```
public void LoadFromFile(String _file_path);
```

Visual Basic

```
Public Sub LoadFromFile(_file_path As String)
```

Description

Function reads in the collection of points from the file and adds it to the points currently defined. .

Version

Available since version 3.

II.1.10.5.3.7 SaveToFile

C++

```
HRESULT SaveToFile(BSTR _file_path);
```

C#

```
public void SaveToFile(String _file_path);
```

Visual Basic

```
Public Sub SaveToFile(_file_path As String)
```

Description

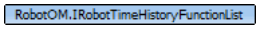
Function saves the collection of points to a file. .

Version

Available since version 3.

II.1.10.6 IRobotTimeHistoryFunctionList

Class Hierarchy



(c) 2014 Autodesk Inc. All rights reserved

C++

```
interface IRobotTimeHistoryFunctionList : IDispatch;
```

C#

```
public interface IRobotTimeHistoryFunctionList;
```

Visual Basic

```
Public Interface IRobotTimeHistoryFunctionList
```

Description

List of defined time functions for time history analysis. .

Version

Available since version 3.

II.1.10.6.1 IRobotTimeHistoryFunctionList Members

The following tables list the members exposed by IRobotTimeHistoryFunctionList.

Public Fields

	Name	Description
	Count (see page 379)	Number of defined (available) time functions.

Public Methods

	Name	Description
	AddFromFile (see page 380)	Function reads in the list of time functions from the indicated file. .
	Create (see page 380)	Function creates and returns a new (empty) collection of points defining a time function.
	CreateSum (see page 381)	Function creates and returns the collection of points as a sum of the indicated time functions.
	Delete (see page 381)	Function deletes the definition of time function of the specified index. .
	Find (see page 381)	Function returns index of the time function definition of the indicated name.
	Get (see page 382)	Function returns the collection of points defining a time function of the indicated index. .
	GetName (see page 382)	Function returns a name of the time function of the indicated index.
	SaveToFile (see page 382)	Function saves the list of time functions to the indicated file. .
	Store (see page 383)	Function saves the indicated collection of points as a definition of time function of the specified name. Index of the function definition from the list is returned.

II.1.10.6.2 IRobotTimeHistoryFunctionList Fields

The fields of the IRobotTimeHistoryFunctionList class are listed here.

Public Fields

	Name	Description
	Count (see page 379)	Number of defined (available) time functions.

II.1.10.6.2.1 Count

C++

```
HRESULT get_Count(long*);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public long Count { get; }
```

Visual Basic

```
Public ReadOnly Count As long
```

Description

Number of defined (available) time functions.

Version

Available since version 3.

II.1.10.6.3 IRobotTimeHistoryFunctionList Methods

The methods of the IRobotTimeHistoryFunctionList class are listed here.

Public Methods

	Name	Description
	AddFromFile (see page 380)	Function reads in the list of time functions from the indicated file. .
	Create (see page 380)	Function creates and returns a new (empty) collection of points defining a time function.
	CreateSum (see page 381)	Function creates and returns the collection of points as a sum of the indicated time functions.
	Delete (see page 381)	Function deletes the definition of time function of the specified index. .
	Find (see page 381)	Function returns index of the time function definition of the indicated name.
	Get (see page 382)	Function returns the collection of points defining a time function of the indicated index. .
	GetName (see page 382)	Function returns a name of the time function of the indicated index.
	SaveToFile (see page 382)	Function saves the list of time functions to the indicated file. .
	Store (see page 383)	Function saves the indicated collection of points as a definition of time function of the specified name. Index of the function definition from the list is returned.

II.1.10.6.3.1 AddFromFile

C++

```
HRESULT AddFromFile(BSTR _file_path);
```

C#

```
public void AddFromFile(String _file_path);
```

Visual Basic

```
Public Sub AddFromFile(_file_path As String)
```

Description

Function reads in the list of time functions from the indicated file. .

Version

Available since version 3.

II.1.10.6.3.2 Create

C++

```
HRESULT Create(IRobotTimeHistoryPointsCollection** ret);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRobotTimeHistoryPointsCollection Create();
```

Visual Basic

```
Public Function Create() As IRobotTimeHistoryPointsCollection
```

Description

Function creates and returns a new (empty) collection of points defining a time function.

Version

Available since version 3.

II.1.10.6.3.3 CreateSum

C++

```
HRESULT CreateSum(long _fun1_idx, double _fun1_coeff, long _fun2_idx, double _fun2_coeff,
IRobotTimeHistoryPointsCollection** ret);
```

C#

```
public IRobotTimeHistoryPointsCollection CreateSum(long _fun1_idx, double _fun1_coeff, long
_fun2_idx, double _fun2_coeff);
```

Visual Basic

```
Public Function CreateSum(_fun1_idx As long, _fun1_coeff As double, _fun2_idx As long,
_fun2_coeff As double) As IRobotTimeHistoryPointsCollection
```

Description

Function creates and returns the collection of points as a sum of the indicated time functions.

Version

Available since version 3.

II.1.10.6.3.4 Delete

C++

```
HRESULT Delete(long _idx);
```

C#

```
public void Delete(long _idx);
```

Visual Basic

```
Public Sub Delete(_idx As long)
```

Description

Function deletes the definition of time function of the specified index. .

Version

Available since version 3.

II.1.10.6.3.5 Find

C++

```
HRESULT Find(BSTR _function_name, long* ret);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public long Find(String _function_name);
```

Visual Basic

```
Public Function Find(_function_name As String) As long
```

Description

Function returns index of the time function definition of the indicated name.

Version

Available since version 3.

II.1.10.6.3.6 Get

C++

```
HRESULT Get(long _idx, IRobotTimeHistoryPointsCollection** ret);
```

C#

```
public IRobotTimeHistoryPointsCollection Get(long _idx);
```

Visual Basic

```
Public Function Get(_idx As long) As IRobotTimeHistoryPointsCollection
```

Description

Function returns the collection of points defining a time function of the indicated index. .

Version

Available since version 3.

II.1.10.6.3.7 GetName

C++

```
HRESULT GetName(long _idx, BSTR* ret);
```

C#

```
public String GetName(long _idx);
```

Visual Basic

```
Public Function GetName(_idx As long) As String
```

Description

Function returns a name of the time function of the indicated index.

Version

Available since version 3.

II.1.10.6.3.8 SaveToFile

C++

```
HRESULT SaveToFile(BSTR _file_path);
```

C#

```
public void SaveToFile(String _file_path);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Sub SaveToFile(_file_path As String)
```

Description

Function saves the list of time functions to the indicated file. .

Version

Available since version 3.

II.1.10.6.3.9 Store

C++

```
HRESULT Store(BSTR _function_name, IRobotTimeHistoryPointsCollection* _points, long* ret);
```

C#

```
public long Store(String _function_name, IRobotTimeHistoryPointsCollection _points);
```

Visual Basic

```
Public Function Store(_function_name As String, ByRef _points As
IRobotTimeHistoryPointsCollection) As long
```

Description

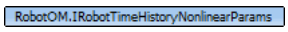
Function saves the indicated collection of points as a definition of time function of the specified name. Index of the function definition from the list is returned.

Version

Available since version 3.

II.1.10.7 IRobotTimeHistoryNonlinearParams

Class Hierarchy



C++

```
interface IRobotTimeHistoryNonlinearParams : IDispatch;
```

C#

```
public interface IRobotTimeHistoryNonlinearParams;
```

Visual Basic

```
Public Interface IRobotTimeHistoryNonlinearParams
```

Description

Non-linear parameters of time history analysis.

Version

Available since version 3.

II.1.10.7.1 IRobotTimeHistoryNonlinearParams Members

The following tables list the members exposed by IRobotTimeHistoryNonlinearParams.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	MatrixUpdateAfterEachIteration (see page 384)	Matrix update after each iteration.
	MatrixUpdateAfterEachSubdivision (see page 384)	Matrices are updated after each subdivision .
	MaximumIterationNumberForOneIncrement (see page 385)	Maximum number of iterations for an increment.
	PDelta (see page 385)	Definition of P-Delta analysis.
	ResidualForcesRelativeCodeTolerance (see page 385)	Relative norm tolerance for residual forces.

II.1.10.7.2 IRobotTimeHistoryNonlinearParams Fields

The fields of the IRobotTimeHistoryNonlinearParams class are listed here.

Public Fields

	Name	Description
	MatrixUpdateAfterEachIteration (see page 384)	Matrix update after each iteration.
	MatrixUpdateAfterEachSubdivision (see page 384)	Matrices are updated after each subdivision .
	MaximumIterationNumberForOneIncrement (see page 385)	Maximum number of iterations for an increment.
	PDelta (see page 385)	Definition of P-Delta analysis.
	ResidualForcesRelativeCodeTolerance (see page 385)	Relative norm tolerance for residual forces.

II.1.10.7.2.1 MatrixUpdateAfterEachIteration

C++

```
HRESULT get_MatrixUpdateAfterEachIteration(VARIANT_BOOL*);
HRESULT put_MatrixUpdateAfterEachIteration(VARIANT_BOOL);
```

C#

```
public bool MatrixUpdateAfterEachIteration { get; set; }
```

Visual Basic

```
Public MatrixUpdateAfterEachIteration As Boolean
```

Description

Matrix update after each iteration.

Version

Available since version 3.

II.1.10.7.2.2 MatrixUpdateAfterEachSubdivision

C++

```
HRESULT get_MatrixUpdateAfterEachSubdivision(VARIANT_BOOL*);
HRESULT put_MatrixUpdateAfterEachSubdivision(VARIANT_BOOL);
```

C#

```
public bool MatrixUpdateAfterEachSubdivision { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public MatrixUpdateAfterEachSubdivision As Boolean
```

Description

Matrices are updated after each subdivision .

Version

Available since version 3.

II.1.10.7.2.3 MaximumIterationNumberForOneIncrement

C++

```
HRESULT get_MaximumIterationNumberForOneIncrement(long*);
HRESULT put_MaximumIterationNumberForOneIncrement(long);
```

C#

```
public long MaximumIterationNumberForOneIncrement { get; set; }
```

Visual Basic

```
Public MaximumIterationNumberForOneIncrement As long
```

Description

Maximum number of iterations for an increment.

Version

Available since version 3.

II.1.10.7.2.4 PDelta

C++

```
HRESULT get_PDelta(VARIANT_BOOL*);
HRESULT put_PDelta(VARIANT_BOOL);
```

C#

```
public bool PDelta { get; set; }
```

Visual Basic

```
Public PDelta As Boolean
```

Description

Definition of P-Delta analysis.

Version

Available since version 3.

II.1.10.7.2.5 ResidualForcesRelativeCodeTolerance

C++

```
HRESULT get_ResidualForcesRelativeCodeTolerance(double*);
HRESULT put_ResidualForcesRelativeCodeTolerance(double);
```

C#

```
public double ResidualForcesRelativeCodeTolerance { get; set; }
```

Visual Basic

```
Public ResidualForcesRelativeCodeTolerance As double
```

(c) 2014 Autodesk Inc. All rights reserved

Description

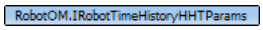
Relative norm tolerance for residual forces.

Version

Available since version 3.

II.1.10.8 IRobotTimeHistoryHHTParams

Class Hierarchy



C++

```
interface IRobotTimeHistoryHHTParams : IDispatch;
```

C#

```
public interface IRobotTimeHistoryHHTParams;
```

Visual Basic

```
Public Interface IRobotTimeHistoryHHTParams
```

Description

Parameters of HHT (Hilber-Hughes-Taylor) method for time history analysis.

Version

Available since version 11.

II.1.10.8.1 IRobotTimeHistoryHHTParams Members

The following tables list the members exposed by IRobotTimeHistoryHHTParams.

Public Fields

	Name	Description
	Alpha (see page 386)	Alpha parameter of Rayleigh damping.
	Beta (see page 387)	Rayleigh damping beta parameter.
	CoeffAlpha (see page 387)	Numerical damping.
	Nonlinearity (see page 387)	Flag which decides on considering the non-linear analysis.
	NonlinearParams (see page 388)	Non-linear parameters.

II.1.10.8.2 IRobotTimeHistoryHHTParams Fields

The fields of the IRobotTimeHistoryHHTParams class are listed here.

Public Fields

	Name	Description
	Alpha (see page 386)	Alpha parameter of Rayleigh damping.
	Beta (see page 387)	Rayleigh damping beta parameter.
	CoeffAlpha (see page 387)	Numerical damping.
	Nonlinearity (see page 387)	Flag which decides on considering the non-linear analysis.
	NonlinearParams (see page 388)	Non-linear parameters.

II.1.10.8.2.1 Alpha

C++

```
HRESULT get_Alpha(double*);
HRESULT put_Alpha(double);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double Alpha { get; set; }
```

Visual Basic

```
Public Alpha As double
```

Description

Alpha parameter of Rayleigh damping.

Version

Available since version 11.

II.1.10.8.2.2 Beta

C++

```
HRESULT get_Beta(double*);
HRESULT put_Beta(double);
```

C#

```
public double Beta { get; set; }
```

Visual Basic

```
Public Beta As double
```

Description

Rayleigh damping beta parameter.

Version

Available since version 11.

II.1.10.8.2.3 CoeffAlpha

C++

```
HRESULT get_CoeffAlpha(double*);
HRESULT put_CoeffAlpha(double);
```

C#

```
public double CoeffAlpha { get; set; }
```

Visual Basic

```
Public CoeffAlpha As double
```

Description

Numerical damping.

Version

Available since version 11.

II.1.10.8.2.4 Nonlinearity

C++

```
HRESULT get_Nonlinearity(VARIANT_BOOL*);
HRESULT put_Nonlinearity(VARIANT_BOOL);
```

C#

```
public bool Nonlinearity { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Nonlinearity As Boolean
```

Description

Flag which decides on considering the non-linear analysis.

Version

Available since version 11.

II.1.10.8.2.5 NonlinearParams

C++

```
HRESULT get_NonlinearParams(IRobotTimeHistoryNonlinearParams**);
```

C#

```
public IRobotTimeHistoryNonlinearParams NonlinearParams { get; }
```

Visual Basic

```
Public ReadOnly NonlinearParams As IRobotTimeHistoryNonlinearParams
```

Description

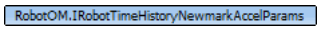
Non-linear parameters.

Version

Available since version 11.

II.1.10.9 IRobotTimeHistoryNewmarkAccelParams

Class Hierarchy



C++

```
interface IRobotTimeHistoryNewmarkAccelParams : IDispatch;
```

C#

```
public interface IRobotTimeHistoryNewmarkAccelParams;
```

Visual Basic

```
Public Interface IRobotTimeHistoryNewmarkAccelParams
```

Description

Parameters of Newmark method (acceleration) for time history analysis.

Version

Available since version 11.

II.1.10.9.1 IRobotTimeHistoryNewmarkAccelParams Members

The following tables list the members exposed by IRobotTimeHistoryNewmarkAccelParams.

Public Fields

	Name	Description
	Alpha (see page 389)	Alpha parameter of Rayleigh damping.
	Beta (see page 389)	Beta parameter of Rayleigh damping.
	Nonlinearity (see page 389)	Flag which decides on considering the non-linear analysis.

(c) 2014 Autodesk Inc. All rights reserved

	NonlinearParams (see page 390)	Parameters of non-linear analysis.

II.1.10.9.2 IRobotTimeHistoryNewmarkAccelParams Fields

The fields of the IRobotTimeHistoryNewmarkAccelParams class are listed here.

Public Fields

	Name	Description
	Alpha (see page 389)	Alpha parameter of Rayleigh damping.
	Beta (see page 389)	Beta parameter of Rayleigh damping.
	Nonlinearity (see page 389)	Flag which decides on considering the non-linear analysis.
	NonlinearParams (see page 390)	Parameters of non-linear analysis.

II.1.10.9.2.1 Alpha

C++

```
HRESULT get_Alpha(double*);
HRESULT put_Alpha(double);
```

C#

```
public double Alpha { get; set; }
```

Visual Basic

```
Public Alpha As double
```

Description

Alpha parameter of Rayleigh damping.

Version

Available since version 11.

II.1.10.9.2.2 Beta

C++

```
HRESULT get_Beta(double*);
HRESULT put_Beta(double);
```

C#

```
public double Beta { get; set; }
```

Visual Basic

```
Public Beta As double
```

Description

Beta parameter of Rayleigh damping.

Version

Available since version 11.

II.1.10.9.2.3 Nonlinearity

C++

```
HRESULT get_Nonlinearity(VARIANT_BOOL*);
HRESULT put_Nonlinearity(VARIANT_BOOL);
```

C#

```
public bool Nonlinearity { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Nonlinearity As Boolean
```

Description

Flag which decides on considering the non-linear analysis.

Version

Available since version 11.

II.1.10.9.2.4 NonlinearParams

C++

```
HRESULT get_NonlinearParams(IRobotTimeHistoryNonlinearParams**);
```

C#

```
public IRobotTimeHistoryNonlinearParams NonlinearParams { get; }
```

Visual Basic

```
Public ReadOnly NonlinearParams As IRobotTimeHistoryNonlinearParams
```

Description

Parameters of non-linear analysis.

Version

Available since version 11.

II.1.11 Harmonic analysis parameters

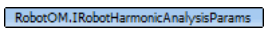
Available since version 3.

Interfaces

	Name	Description
	IRobotHarmonicAnalysisParams (see page 390)	

II.1.11.1 IRobotHarmonicAnalysisParams

Class Hierarchy



C++

```
interface IRobotHarmonicAnalysisParams : IDispatch;
```

C#

```
public interface IRobotHarmonicAnalysisParams;
```

Visual Basic

```
Public Interface IRobotHarmonicAnalysisParams
```

Version

Available since version 3.

II.1.11.1.1 IRobotHarmonicAnalysisParams Members

The following tables list the members exposed by IRobotHarmonicAnalysisParams.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	Excitation (see page 391)	Excitation.
	MassMatrix (see page 391)	Stiffness matrix type.

II.1.11.1.2 IRobotHarmonicAnalysisParams Fields

The fields of the IRobotHarmonicAnalysisParams class are listed here.

Public Fields

	Name	Description
	Excitation (see page 391)	Excitation.
	MassMatrix (see page 391)	Stiffness matrix type.

II.1.11.1.2.1 Excitation

C++

```
HRESULT get_Excitation(IRobotModalAnalysisLimits**);
```

C#

```
public IRobotModalAnalysisLimits Excitation { get; }
```

Visual Basic

```
Public ReadOnly Excitation As IRobotModalAnalysisLimits
```

Description

Excitation.

Version

Available since version 3.

II.1.11.1.2.2 MassMatrix

C++

```
HRESULT get_MassMatrix(IRobotModalAnalysisMassMatrixType*);
HRESULT put_MassMatrix(IRobotModalAnalysisMassMatrixType);
```

C#

```
public IRobotModalAnalysisMassMatrixType MassMatrix { get; set; }
```

Visual Basic

```
Public MassMatrix As IRobotModalAnalysisMassMatrixType
```

Description

Stiffness matrix type.

Version

Available since version 3.

II.1.12 Push-over analysis parameters

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

Enumerations

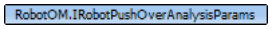
	Name	Description
	IRobotPushOverLoadDefinitionMethod (see page 395)	Available methods of load definition for push-over analysis.
	IRobotPushOverDirection (see page 395)	Available directions for Push-over analysis.

Interfaces

	Name	Description
	IRobotPushOverAnalysisParams (see page 392)	Parameters of Push-over analysis.

II.1.12.1 IRobotPushOverAnalysisParams

Class Hierarchy



C++

```
interface IRobotPushOverAnalysisParams : IDispatch;
```

C#

```
public interface IRobotPushOverAnalysisParams;
```

Visual Basic

```
Public Interface IRobotPushOverAnalysisParams
```

Description

Parameters of Push-over analysis.

Version

Available since version 3.

II.1.12.1.1 IRobotPushOverAnalysisParams Members

The following tables list the members exposed by IRobotPushOverAnalysisParams.

Public Fields

	Name	Description
	Direction (see page 393)	Direction.
	LoadDefinition (see page 393)	Method of load definition.
	MaxDisplacement (see page 393)	Maximum displacement.
	Node (see page 394)	Node number.
	Nonlinear (see page 394)	Flag switching on/off a non-linear analysis.
	NonlinearParams (see page 394)	Non-linear analysis parameters (considered only if the Nonlinear (see page 394) flag is set).

II.1.12.1.2 IRobotPushOverAnalysisParams Fields

The fields of the IRobotPushOverAnalysisParams class are listed here.

Public Fields

	Name	Description
	Direction (see page 393)	Direction.
	LoadDefinition (see page 393)	Method of load definition.

(c) 2014 Autodesk Inc. All rights reserved

	MaxDisplacement (see page 393)	Maximum displacement.
	Node (see page 394)	Node number.
	Nonlinear (see page 394)	Flag switching on/off a non-linear analysis.
	NonlinearParams (see page 394)	Non-linear analysis parameters (considered only if the Nonlinear (see page 394) flag is set).

II.1.12.1.2.1 Direction

C++

```
HRESULT get_Direction(IRobotPushOverDirection*);
HRESULT put_Direction(IRobotPushOverDirection);
```

C#

```
public IRobotPushOverDirection Direction { get; set; }
```

Visual Basic

```
Public Direction As IRobotPushOverDirection
```

Description

Direction.

Version

Available since version 3.

II.1.12.1.2.2 LoadDefinition

C++

```
HRESULT get_LoadDefinition(IRobotPushOverLoadDefinitionMethod*);
HRESULT put_LoadDefinition(IRobotPushOverLoadDefinitionMethod);
```

C#

```
public IRobotPushOverLoadDefinitionMethod LoadDefinition { get; set; }
```

Visual Basic

```
Public LoadDefinition As IRobotPushOverLoadDefinitionMethod
```

Description

Method of load definition.

Version

Available since version 3.

II.1.12.1.2.3 MaxDisplacement

C++

```
HRESULT get_MaxDisplacement(double*);
HRESULT put_MaxDisplacement(double);
```

C#

```
public double MaxDisplacement { get; set; }
```

Visual Basic

```
Public MaxDisplacement As double
```

Description

Maximum displacement.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.

II.1.12.1.2.4 Node

C++

```
HRESULT get_Node(long*);
HRESULT put_Node(long);
```

C#

```
public long Node { get; set; }
```

Visual Basic

```
Public Node As long
```

Description

Node number.

Version

Available since version 3.

II.1.12.1.2.5 Nonlinear

C++

```
HRESULT get_Nonlinear(VARIANT_BOOL*);
HRESULT put_Nonlinear(VARIANT_BOOL);
```

C#

```
public bool Nonlinear { get; set; }
```

Visual Basic

```
Public Nonlinear As Boolean
```

Description

Flag switching on/off a non-linear analysis.

Version

Available since version 3.

II.1.12.1.2.6 NonlinearParams

C++

```
HRESULT get_NonlinearParams(IRobotNonlinearAnalysisParams**);
```

C#

```
public IRobotNonlinearAnalysisParams NonlinearParams { get; }
```

Visual Basic

```
Public ReadOnly NonlinearParams As IRobotNonlinearAnalysisParams
```

Description

Non-linear analysis parameters (considered only if the Nonlinear (see page 394) flag is set).

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

II.1.12.2 IRobotPushOverLoadDefinitionMethod

C++

```
enum IRobotPushOverLoadDefinitionMethod;
```

C#

```
public enum IRobotPushOverLoadDefinitionMethod;
```

Visual Basic

```
Public Enum IRobotPushOverLoadDefinitionMethod
```

Members

Members	Description
I_POLDM_ACCELERATION = 0	Load definition based on the acceleration in the indicated direction. Available since version 3.
I_POLDM_USER_DEFINED = 2	Available since version 3.

Description

Available methods of load definition for push-over analysis.

Version

Available since version 3.

II.1.12.3 IRobotPushOverDirection

C++

```
enum IRobotPushOverDirection;
```

C#

```
public enum IRobotPushOverDirection;
```

Visual Basic

```
Public Enum IRobotPushOverDirection
```

Members

Members	Description
I_POD_UX_PLUS = 0	Available since version 3.
I_POD_UX_MINUS = 10	Available since version 3.
I_POD_UY_PLUS = 1	Available since version 3.
I_POD_UY_MINUS = 11	Available since version 3.
I_POD_UZ_PLUS = 2	Available since version 3.
I_POD_UZ_MINUS = 12	Available since version 3.

Description

Available directions for Push-over analysis.

Version

Available since version 3.

II.1.13 FRF analysis parameters

Available since version 9.7.

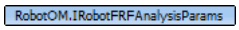
(c) 2014 Autodesk Inc. All rights reserved

Interfaces

	Name	Description
	IRobotFRFAnalysisParams (see page 396)	Parameters of FRF analysis.

II.1.13.1 IRobotFRFAnalysisParams

Class Hierarchy



C++

```
interface IRobotFRFAnalysisParams : IDispatch;
```

C#

```
public interface IRobotFRFAnalysisParams;
```

Visual Basic

```
Public Interface IRobotFRFAnalysisParams
```

Description

Parameters of FRF analysis.

Version

Available since version 9.7.

II.1.13.1.1 IRobotFRFAnalysisParams Members

The following tables list the members exposed by IRobotFRFAnalysisParams.

Public Fields

	Name	Description
	Damping (see page 397)	Damping parameters.
	FinalFrequency (see page 397)	Final frequency.
	FrequencyDivision (see page 397)	Division for frequency .
	IncludeEigenfrequencies (see page 397)	Flag indicating if eigenfrequencies should be included in calculations.
	InitialFrequency (see page 398)	Initial frequency.

II.1.13.1.2 IRobotFRFAnalysisParams Fields

The fields of the IRobotFRFAnalysisParams class are listed here.

Public Fields

	Name	Description
	Damping (see page 397)	Damping parameters.
	FinalFrequency (see page 397)	Final frequency.
	FrequencyDivision (see page 397)	Division for frequency .
	IncludeEigenfrequencies (see page 397)	Flag indicating if eigenfrequencies should be included in calculations.
	InitialFrequency (see page 398)	Initial frequency.

(c) 2014 Autodesk Inc. All rights reserved

II.1.13.1.2.1 Damping

C++

```
HRESULT get_Damping(IRobotDynamicAnalysisDamping**);
```

C#

```
public IRobotDynamicAnalysisDamping Damping { get; }
```

Visual Basic

```
Public ReadOnly Damping As IRobotDynamicAnalysisDamping
```

Description

Damping parameters.

Version

Available since version 9.7.

II.1.13.1.2.2 FinalFrequency

C++

```
HRESULT get_FinalFrequency(double*);
HRESULT put_FinalFrequency(double);
```

C#

```
public double FinalFrequency { get; set; }
```

Visual Basic

```
Public FinalFrequency As double
```

Description

Final frequency.

Version

Available since version 9.7.

II.1.13.1.2.3 FrequencyDivision

C++

```
HRESULT get_FrequencyDivision(long*);
HRESULT put_FrequencyDivision(long);
```

C#

```
public long FrequencyDivision { get; set; }
```

Visual Basic

```
Public FrequencyDivision As long
```

Description

Division for frequency .

Version

Available since version 9.7.

II.1.13.1.2.4 IncludeEigenfrequencies

C++

```
HRESULT get_IncludeEigenfrequencies(VARIANT_BOOL*);
```

(c) 2014 Autodesk Inc. All rights reserved

```
HRESULT put_IncludeEigenfrequencies(VARIANT_BOOL);
```

C#

```
public bool IncludeEigenfrequencies { get; set; }
```

Visual Basic

```
Public IncludeEigenfrequencies As Boolean
```

Description

Flag indicating if eigenfrequencies should be included in calculations.

Version

Available since version 9.7.

II.1.13.1.2.5 InitialFrequency

C++

```
HRESULT get_InitialFrequency(double*);
HRESULT put_InitialFrequency(double);
```

C#

```
public double InitialFrequency { get; set; }
```

Visual Basic

```
Public InitialFrequency As double
```

Description

Initial frequency.

Version

Available since version 9.7.

II.1.14 Footfall analysis parameters

Available since version 9.7.

Enumerations

	Name	Description
	IRobotFootfallAnalysisExcitationMethod (see page 398)	
	IRobotFootfallAnalysisExcitationForces (see page 399)	
	IRobotFootfallAnalysisNodeSelectionType (see page 401)	

Interfaces

	Name	Description
	IRobotFootfallAnalysisNodeSelection (see page 399)	
	IRobotFootfallAnalysisModalParams (see page 401)	Modal analysis parameters for the footfall analysis.
	IRobotFootfallAnalysisParams (see page 403)	Parameters of Footfall analysis.

II.1.14.1 IRobotFootfallAnalysisExcitationMethod

C++

```
enum IRobotFootfallAnalysisExcitationMethod;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public enum IRobotFootfallAnalysisExcitationMethod;
```

Visual Basic

```
Public Enum IRobotFootfallAnalysisExcitationMethod
```

Members

Members	Description
I_FAEM_SELF_EXCITATION = 1	Available since version 9.7.
I_FAEM_FULL_EXCITATION = 2	Available since version 9.7.

Version

Available since version 9.7.

II.1.14.2 IRobotFootfallAnalysisExcitationForces

C++

```
enum IRobotFootfallAnalysisExcitationForces;
```

C#

```
public enum IRobotFootfallAnalysisExcitationForces;
```

Visual Basic

```
Public Enum IRobotFootfallAnalysisExcitationForces
```

Members

Members	Description
I_FAEF_CONCRETE_CENTRE = 1	Available since version 9.7.
I_FAEF_SCI_P354 = 2	Available since version 9.7.

Version

Available since version 9.7.

II.1.14.3 IRobotFootfallAnalysisNodeSelection

Class Hierarchy

RobotOM.IRobotFootfallAnalysisNodeSelection

C++

```
interface IRobotFootfallAnalysisNodeSelection : IDispatch;
```

C#

```
public interface IRobotFootfallAnalysisNodeSelection;
```

Visual Basic

```
Public Interface IRobotFootfallAnalysisNodeSelection
```

Version

Available since version 9.7.

II.1.14.3.1 IRobotFootfallAnalysisNodeSelection Members

The following tables list the members exposed by IRobotFootfallAnalysisNodeSelection.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	SelectedNodes (see page 400)	
	SelectedPanels (see page 400)	
	Type (see page 400)	

II.1.14.3.2 IRobotFootfallAnalysisNodeSelection Fields

The fields of the IRobotFootfallAnalysisNodeSelection class are listed here.

Public Fields

	Name	Description
	SelectedNodes (see page 400)	
	SelectedPanels (see page 400)	
	Type (see page 400)	

II.1.14.3.2.1 SelectedNodes

C++

```
HRESULT get_SelectedNodes(IRobotSelection**);
```

C#

```
public IRobotSelection SelectedNodes { get; }
```

Visual Basic

```
Public ReadOnly SelectedNodes As IRobotSelection
```

Version

Available since version 9.7.

II.1.14.3.2.2 SelectedPanels

C++

```
HRESULT get_SelectedPanels(IRobotSelection**);
```

C#

```
public IRobotSelection SelectedPanels { get; }
```

Visual Basic

```
Public ReadOnly SelectedPanels As IRobotSelection
```

Version

Available since version 9.7.

II.1.14.3.2.3 Type

C++

```
HRESULT get_Type(IRobotFootfallAnalysisNodeSelectionType*);
HRESULT put_Type(IRobotFootfallAnalysisNodeSelectionType);
```

C#

```
public IRobotFootfallAnalysisNodeSelectionType Type { get; set; }
```

Visual Basic

```
Public Type As IRobotFootfallAnalysisNodeSelectionType
```

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 9.7.

II.1.14.4 IRobotFootfallAnalysisNodeSelectionType

C++

```
enum IRobotFootfallAnalysisNodeSelectionType;
```

C#

```
public enum IRobotFootfallAnalysisNodeSelectionType;
```

Visual Basic

```
Public Enum IRobotFootfallAnalysisNodeSelectionType
```

Members

Members	Description
I_FANST_ALL_NODES = 1	All nodes. Available since version 9.7.
I_FANST_SELECTED_NODES = 2	Selected nodes. Available since version 9.7.
I_FANST_NODES_BELONGING_TO_SELECTED_PANELS = 3	Nodes belonging to selected panels. Available since version 9.7.

Version

Available since version 9.7.

II.1.14.5 IRobotFootfallAnalysisModalParams

Class Hierarchy

RobotOM.IRobotFootfallAnalysisModalParams

C++

```
interface IRobotFootfallAnalysisModalParams : IDispatch;
```

C#

```
public interface IRobotFootfallAnalysisModalParams;
```

Visual Basic

```
Public Interface IRobotFootfallAnalysisModalParams
```

Description

Modal analysis parameters for the footfall analysis.

Version

Available since version 9.7.

II.1.14.5.1 IRobotFootfallAnalysisModalParams Members

The following tables list the members exposed by IRobotFootfallAnalysisModalParams.

Public Fields

	Name	Description
	FrequencyLimit (see page 402)	Frequency limit.
	IgnoreDensity (see page 402)	Flag indicating if density should be ignored.

(c) 2014 Autodesk Inc. All rights reserved

	IncludeMassForDirX (see page 403)	Flag indicating if masses should be considered on the X direction .
	IncludeMassForDirY (see page 403)	Flag indicating if masses should be considered on the Y direction.
	IncludeMassForDirZ (see page 403)	Flag indicating if masses should be considered on the Z direction.

II.1.14.5.2 IRobotFootfallAnalysisModalParams Fields

The fields of the IRobotFootfallAnalysisModalParams class are listed here.

Public Fields

	Name	Description
	FrequencyLimit (see page 402)	Frequency limit.
	IgnoreDensity (see page 402)	Flag indicating if density should be ignored.
	IncludeMassForDirX (see page 403)	Flag indicating if masses should be considered on the X direction .
	IncludeMassForDirY (see page 403)	Flag indicating if masses should be considered on the Y direction.
	IncludeMassForDirZ (see page 403)	Flag indicating if masses should be considered on the Z direction.

II.1.14.5.2.1 FrequencyLimit

C++

```
HRESULT get_FrequencyLimit(double*);
HRESULT put_FrequencyLimit(double);
```

C#

```
public double FrequencyLimit { get; set; }
```

Visual Basic

```
Public FrequencyLimit As double
```

Description

Frequency limit.

Version

Available since version 9.7.

II.1.14.5.2.2 IgnoreDensity

C++

```
HRESULT get_IgnoreDensity(VARIANT_BOOL*);
HRESULT put_IgnoreDensity(VARIANT_BOOL);
```

C#

```
public bool IgnoreDensity { get; set; }
```

Visual Basic

```
Public IgnoreDensity As Boolean
```

Description

Flag indicating if density should be ignored.

Version

Available since version 9.7.

(c) 2014 Autodesk Inc. All rights reserved

II.1.14.5.2.3 IncludeMassForDirX

C++

```
HRESULT get_IncludeMassForDirX(VARIANT_BOOL*);
HRESULT put_IncludeMassForDirX(VARIANT_BOOL);
```

C#

```
public bool IncludeMassForDirX { get; set; }
```

Visual Basic

```
Public IncludeMassForDirX As Boolean
```

Description

Flag indicating if masses should be considered on the X direction .

Version

Available since version 9.7.

II.1.14.5.2.4 IncludeMassForDirY

C++

```
HRESULT get_IncludeMassForDirY(VARIANT_BOOL*);
HRESULT put_IncludeMassForDirY(VARIANT_BOOL);
```

C#

```
public bool IncludeMassForDirY { get; set; }
```

Visual Basic

```
Public IncludeMassForDirY As Boolean
```

Description

Flag indicating if masses should be considered on the Y direction.

Version

Available since version 9.7.

II.1.14.5.2.5 IncludeMassForDirZ

C++

```
HRESULT get_IncludeMassForDirZ(VARIANT_BOOL*);
HRESULT put_IncludeMassForDirZ(VARIANT_BOOL);
```

C#

```
public bool IncludeMassForDirZ { get; set; }
```

Visual Basic

```
Public IncludeMassForDirZ As Boolean
```

Description

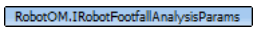
Flag indicating if masses should be considered on the Z direction.

Version

Available since version 9.7.

II.1.14.6 IRobotFootfallAnalysisParams

Class Hierarchy



(c) 2014 Autodesk Inc. All rights reserved

C++

```
interface IRobotFootfallAnalysisParams : IDispatch;
```

C#

```
public interface IRobotFootfallAnalysisParams;
```

Visual Basic

```
Public Interface IRobotFootfallAnalysisParams
```

Description

Parameters of Footfall analysis.

Version

Available since version 9.7.

II.1.14.6.1 IRobotFootfallAnalysisParams Members

The following tables list the members exposed by IRobotFootfallAnalysisParams.

Public Fields

	Name	Description
	Damping (see page 405)	Damping.
	ExcitationForces (see page 405)	
	ExcitationMethod (see page 405)	
	ExcitationNodes (see page 405)	Selection of excitation nodes.
	FootstepsNumber (see page 406)	Number of footsteps.
	MaxWalkingFrequency (see page 406)	Maximum walking frequency.
	MinWalkingFrequency (see page 406)	Minimum walking frequency.
	ModalParams (see page 407)	Parameters of modal analysis.
	ResponseNodes (see page 407)	Selection of response nodes.
	WalkersWeight (see page 407)	Walker's weight.

II.1.14.6.2 IRobotFootfallAnalysisParams Fields

The fields of the IRobotFootfallAnalysisParams class are listed here.

Public Fields

	Name	Description
	Damping (see page 405)	Damping.
	ExcitationForces (see page 405)	
	ExcitationMethod (see page 405)	
	ExcitationNodes (see page 405)	Selection of excitation nodes.
	FootstepsNumber (see page 406)	Number of footsteps.
	MaxWalkingFrequency (see page 406)	Maximum walking frequency.
	MinWalkingFrequency (see page 406)	Minimum walking frequency.
	ModalParams (see page 407)	Parameters of modal analysis.
	ResponseNodes (see page 407)	Selection of response nodes.
	WalkersWeight (see page 407)	Walker's weight.

(c) 2014 Autodesk Inc. All rights reserved

II.1.14.6.2.1 Damping

C++

```
HRESULT get_Damping(IRobotDynamicAnalysisDamping**);
```

C#

```
public IRobotDynamicAnalysisDamping Damping { get; }
```

Visual Basic

```
Public ReadOnly Damping As IRobotDynamicAnalysisDamping
```

Description

Damping.

Version

Available since version 9.7.

II.1.14.6.2.2 ExcitationForces

C++

```
HRESULT get_ExcitationForces(IRobotFootfallAnalysisExcitationForces*);
HRESULT put_ExcitationForces(IRobotFootfallAnalysisExcitationForces);
```

C#

```
public IRobotFootfallAnalysisExcitationForces ExcitationForces { get; set; }
```

Visual Basic

```
Public ExcitationForces As IRobotFootfallAnalysisExcitationForces
```

Version

Available since version 9.7.

II.1.14.6.2.3 ExcitationMethod

C++

```
HRESULT get_ExcitationMethod(IRobotFootfallAnalysisExcitationMethod*);
HRESULT put_ExcitationMethod(IRobotFootfallAnalysisExcitationMethod);
```

C#

```
public IRobotFootfallAnalysisExcitationMethod ExcitationMethod { get; set; }
```

Visual Basic

```
Public ExcitationMethod As IRobotFootfallAnalysisExcitationMethod
```

Version

Available since version 9.7.

II.1.14.6.2.4 ExcitationNodes

C++

```
HRESULT get_ExcitationNodes(IRobotFootfallAnalysisNodeSelection**);
```

C#

```
public IRobotFootfallAnalysisNodeSelection ExcitationNodes { get; }
```

Visual Basic

```
Public ReadOnly ExcitationNodes As IRobotFootfallAnalysisNodeSelection
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Selection of excitation nodes.

Version

Available since version 9.7.

II.1.14.6.2.5 FootstepsNumber

C++

```
HRESULT get_FootstepsNumber(long*);
HRESULT put_FootstepsNumber(long);
```

C#

```
public long FootstepsNumber { get; set; }
```

Visual Basic

```
Public FootstepsNumber As long
```

Description

Number of footsteps.

Version

Available since version 9.7.

II.1.14.6.2.6 MaxWalkingFrequency

C++

```
HRESULT get_MaxWalkingFrequency(double*);
HRESULT put_MaxWalkingFrequency(double);
```

C#

```
public double MaxWalkingFrequency { get; set; }
```

Visual Basic

```
Public MaxWalkingFrequency As double
```

Description

Maximum walking frequency.

Version

Available since version 9.7.

II.1.14.6.2.7 MinWalkingFrequency

C++

```
HRESULT get_MinWalkingFrequency(double*);
HRESULT put_MinWalkingFrequency(double);
```

C#

```
public double MinWalkingFrequency { get; set; }
```

Visual Basic

```
Public MinWalkingFrequency As double
```

Description

Minimum walking frequency.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 9.7.

II.1.14.6.2.8 ModalParams

C++

```
HRESULT get_ModalParams(IRobotFootfallAnalysisModalParams**);
```

C#

```
public IRobotFootfallAnalysisModalParams ModalParams { get; }
```

Visual Basic

```
Public ReadOnly ModalParams As IRobotFootfallAnalysisModalParams
```

Description

Parameters of modal analysis.

Version

Available since version 9.7.

II.1.14.6.2.9 ResponseNodes

C++

```
HRESULT get_ResponseNodes(IRobotFootfallAnalysisNodeSelection**);
```

C#

```
public IRobotFootfallAnalysisNodeSelection ResponseNodes { get; }
```

Visual Basic

```
Public ReadOnly ResponseNodes As IRobotFootfallAnalysisNodeSelection
```

Description

Selection of response nodes.

Version

Available since version 9.7.

II.1.14.6.2.10 WalkersWeight

C++

```
HRESULT get_WalkersWeight(double*);
HRESULT put_WalkersWeight(double);
```

C#

```
public double WalkersWeight { get; set; }
```

Visual Basic

```
Public WalkersWeight As double
```

Description

Walker's weight.

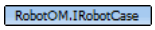
Version

Available since version 9.7.

(c) 2014 Autodesk Inc. All rights reserved

II.1.15 IRobotCase

Class Hierarchy



C++

```
interface IRobotCase : IDispatch;
```

C#

```
public interface IRobotCase;
```

Visual Basic

```
Public Interface IRobotCase
```

Description

The part of the load case interface that is shared by simple cases and combinations has been defined using RobotCase. .

II.1.15.1 IRobotCase Members

The following tables list the members exposed by IRobotCase.

Public Fields

	Name	Description
	AnalizeType (see page 408)	Analysis type.
	Name (see page 409)	User-defined case name .
	Nature (see page 409)	Case nature .
	Number (see page 409)	User-defined number linked with the load case .
	Type (see page 409)	Load case type - allows one to differentiate between a simple case and combination at the level of the shared part of the interface .

II.1.15.2 IRobotCase Fields

The fields of the IRobotCase class are listed here.

Public Fields

	Name	Description
	AnalizeType (see page 408)	Analysis type.
	Name (see page 409)	User-defined case name .
	Nature (see page 409)	Case nature .
	Number (see page 409)	User-defined number linked with the load case .
	Type (see page 409)	Load case type - allows one to differentiate between a simple case and combination at the level of the shared part of the interface .

II.1.15.2.1 AnalizeType

C++

```
HRESULT get_AnalizeType(IRobotCaseAnalizeType*);
HRESULT put_AnalizeType(IRobotCaseAnalizeType);
```

C#

```
public IRobotCaseAnalizeType AnalizeType { get; set; }
```

Visual Basic

```
Public AnalizeType As IRobotCaseAnalizeType
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Analysis type.

II.1.15.2.2 Name

C++

```
HRESULT get_Name(BSTR*);
HRESULT put_Name(BSTR);
```

C#

```
public String Name { get; set; }
```

Visual Basic

```
Public Name As String
```

Description

User-defined case name .

II.1.15.2.3 Nature

C++

```
HRESULT get_Nature(IRobotCaseNature*);
HRESULT put_Nature(IRobotCaseNature);
```

C#

```
public IRobotCaseNature Nature { get; set; }
```

Visual Basic

```
Public Nature As IRobotCaseNature
```

Description

Case nature .

II.1.15.2.4 Number

C++

```
HRESULT get_Number(long*);
HRESULT put_Number(long);
```

C#

```
public long Number { get; set; }
```

Visual Basic

```
Public Number As long
```

Description

User-defined number linked with the load case .

II.1.15.2.5 Type

C++

```
HRESULT get_Type(IRobotCaseType*);
```

C#

```
public IRobotCaseType Type { get; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public ReadOnly Type As IRobotCaseType
```

Description

Load case type - allows one to differentiate between a simple case and combination at the level of the shared part of the interface .

II.1.16 IRobotCaseNature

C++

```
enum IRobotCaseNature;
```

C#

```
public enum IRobotCaseNature;
```

Visual Basic

```
Public Enum IRobotCaseNature
```

Members

Members	Description
I_CN_PERMANENT = 0	Dead .
I_CN_TEMPERATURE = 4	Temperature.
I_CN_EXPLOATATION = 1	Live .
I_CN_WIND = 2	Wind.
I_CN_SNOW = 3	Snow.
I_CN_SEISMIC = 6	Seismic.
I_CN_ACCIDENTAL = 5	Available since version 1.7.

Description

The set of case natures accepted by Robot is determined. .

II.1.17 IRobotCaseAnalizeType

C++

```
enum IRobotCaseAnalizeType;
```

C#

```
public enum IRobotCaseAnalizeType;
```

Visual Basic

```
Public Enum IRobotCaseAnalizeType
```

Members

Members	Description
I_CAT_COMB_BUCKLING = -4	Buckling combination.
I_CAT_COMB_NONLINEAR_BUCKLING = -3	Non-linear buckling combination.
I_CAT_COMB_NONLINEAR = -1	Non-linear combination.
I_CAT_COMB = 0	Combination.
I_CAT_STATIC_LINEAR = 1	Static case.
I_CAT_STATIC_NONLINEAR = 2	Non-linear static case.
I_CAT_STATIC_LINEAR_AUXILIARY = 5	Auxiliary case.
I_CAT_DYNAMIC_MODAL = 11	Modal analysis.

(c) 2014 Autodesk Inc. All rights reserved

I_CAT_DYNAMIC_SPECTRAL = 12	Spectral analysis.
I_CAT_DYNAMIC_SEISMIC = 13	Seismic analysis.
I_CAT_DYNAMIC_HARMONIC = 14	Harmonic analysis.
I_CAT_MOBILE_MAIN = 30	Moving load analysis - main case.
I_CAT_COMB_CODE = -12	Code combination Available since version 1.7.
I_CAT_STATIC_BUCKLING = 4	Buckling analysis case Available since version 2.0.
I_CAT_STATIC_NONLINEAR_BUCKLING = 7	Non-linear buckling analysis case Available since version 2.0.
I_CAT_DYNAMIC_NONLINEAR_MODAL_WITH_STATIC_FORCE = 8	Non-linear modal analysis considering static forces Available since version 2.0.
I_CAT_TIME_HISTORY = 20	Time history analysis case. Available since version 3.
I_CAT_PUSH_OVER = 25	Push-over analysis. Available since version 3.
I_CAT_DYNAMIC_FRF = 15	Available since version 9.7.
I_CAT_DYNAMIC_FOOTFALL = 16	Available since version 9.7.
I_CAT_STATIC_ELF_SEISMIC = 9	Available since version 12.5.

Description

Robot Object Model defines a set of analysis types for load cases accepted by Robot. .

II.1.18 IRobotCaseServer

Class Hierarchy

RobotOM.IRobotCaseServer

C++

```
interface IRobotCaseServer : IDispatch;
```

C#

```
public interface IRobotCaseServer;
```

Visual Basic

```
Public Interface IRobotCaseServer
```

Description

Server of load cases .

II.1.18.1 IRobotCaseServer Members

The following tables list the members exposed by IRobotCaseServer.

Public Fields

	Name	Description
	CodeCmbEngine (see page 413)	Generator of code-dependent loads Available since version 1.7.
	FreeNumber (see page 413)	First available (free) user's number for a load case Available since version 1.7.
	QCmbTau (see page 413)	
	SELFSeismicEngine (see page 413)	Generator of loads for the simplified seismic analysis.
	SnowWindEngine (see page 414)	Generator of snow/wind loads Available since version 1.7.
	TimeHistoryFunctions (see page 414)	List of available time functions for time history analysis.

(c) 2014 Autodesk Inc. All rights reserved

	WindLoadsSimulationEngine (see page 414)	

Public Methods

	Name	Description
	BeginMultiOperation (see page 415)	Function enables speed-up of generation or modification of a whole group of load cases through appropriate optimization of some operations. After completing generation of load cases, EndMultiOperation (see page 417)() function should be called up to save changes in a structure.
	CreateCombination (see page 416)	The function creates and returns combinations with the defined parameters. .
	CreateMobile (see page 416)	Function creates moving load case.
	CreateSimple (see page 416)	The function creates and returns a simple load case with the defined parameters. .
	Delete (see page 417)	The function deletes from the structure model the load case with the indicated number .
	DeleteMany (see page 417)	The function deletes all the load cases that meet the selection criteria introduced as argument. .
	EndMultiOperation (see page 417)	Function updates definitions of load cases and combinations in a structure. It should be called up only if BeginMultiOperation (see page 415)() function has been called up earlier.
	Exist (see page 417)	The function checks if there exists a load case with the user-defined number.
	FindWithId (see page 418)	Function returns a load case number with the specified unique identifier. If the load case is not found, then the function returns zero value (0). Available since version 1.7.
	Get (see page 418)	The function returns a load case with the user-defined number. The type of the returned object depends on the real case type with the indicated number. If the indicated number defines a combination, the function returns an object of the RobotCaseCombination object type; otherwise, it will return RobotSimpleCase object type. If the case with the indicated number does not exist, running this function leads to critical error. .
	GetAll (see page 418)	The function returns a collection of all load cases defined for a given structure. .
	GetCmpntCount (see page 419)	Function returns a number of components of the specified load case. What a component means depends on a load case and analysis type; for example, for an FRF analysis case, components are successive steps.
	GetMany (see page 419)	The function returns a collection of load cases that meet the criterion of selection introduced as argument. .
	GetRelatedValue (see page 419)	Function returns a specified value type that is associated with the specified load case.
	GetUniqueId (see page 420)	Function returns a unique identifier for a load case with the user-specified number.
	IsAuxiliary (see page 420)	
	SetAuxiliary (see page 420)	

II.1.18.2 IRobotCaseServer Fields

The fields of the IRobotCaseServer class are listed here.

Public Fields

	Name	Description
	CodeCmbEngine (see page 413)	Generator of code-dependent loads Available since version 1.7.
	FreeNumber (see page 413)	First available (free) user's number for a load case Available since version 1.7.
	QCmbTau (see page 413)	

(c) 2014 Autodesk Inc. All rights reserved

	SELFSeismicEngine (see page 413)	Generator of loads for the simplified seismic analysis.
	SnowWindEngine (see page 414)	Generator of snow/wind loads Available since version 1.7.
	TimeHistoryFunctions (see page 414)	List of available time functions for time history analysis.
	WindLoadsSimulationEngine (see page 414)	

II.1.18.2.1 CodeCmbEngine

C++

```
HRESULT get_CodeCmbEngine(IRobotCodeCombinationEngine**);
```

C#

```
public IRobotCodeCombinationEngine CodeCmbEngine { get; }
```

Visual Basic

```
Public ReadOnly CodeCmbEngine As IRobotCodeCombinationEngine
```

Description

Generator of code-dependent loads Available since version 1.7.

II.1.18.2.2 FreeNumber

C++

```
HRESULT get_FreeNumber(long*);
```

C#

```
public long FreeNumber { get; }
```

Visual Basic

```
Public ReadOnly FreeNumber As long
```

Description

First available (free) user's number for a load case Available since version 1.7.

II.1.18.2.3 QCmbTau

C++

```
HRESULT get_QCmbTau(double*);
HRESULT put_QCmbTau(double);
```

C#

```
public double QCmbTau { get; set; }
```

Visual Basic

```
Public QCmbTau As double
```

II.1.18.2.4 SELFSeismicEngine

C++

```
HRESULT get_SELFSeismicEngine(IRobotSELFSeismicEngine**);
```

C#

```
public IRobotSELFSeismicEngine SELFSeismicEngine { get; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public ReadOnly SELFSeismicEngine As IRobotSELFSeismicEngine
```

Description

Generator of loads for the simplified seismic analysis.

Version

Available since version 12.5.

II.1.18.2.5 SnowWindEngine

C++

```
HRESULT get_SnowWindEngine(IRobotSnowWindEngine**);
```

C#

```
public IRobotSnowWindEngine SnowWindEngine { get; }
```

Visual Basic

```
Public ReadOnly SnowWindEngine As IRobotSnowWindEngine
```

Description

Generator of snow/wind loads Available since version 1.7.

II.1.18.2.6 TimeHistoryFunctions

C++

```
HRESULT get_TimeHistoryFunctions(IRobotTimeHistoryFunctionList**);
```

C#

```
public IRobotTimeHistoryFunctionList TimeHistoryFunctions { get; }
```

Visual Basic

```
Public ReadOnly TimeHistoryFunctions As IRobotTimeHistoryFunctionList
```

Description

List of available time functions for time history analysis.

Version

Available since version 3.

II.1.18.2.7 WindLoadsSimulationEngine

C++

```
HRESULT get_WindLoadsSimulationEngine(IRobotWindLoadsSimulationEngine**);
```

C#

```
public IRobotWindLoadsSimulationEngine WindLoadsSimulationEngine { get; }
```

Visual Basic

```
Public ReadOnly WindLoadsSimulationEngine As IRobotWindLoadsSimulationEngine
```

Version

Available since version 14.5.

II.1.18.3 IRobotCaseServer Methods

The methods of the IRobotCaseServer class are listed here.

(c) 2014 Autodesk Inc. All rights reserved

Public Methods

	Name	Description
	BeginMultiOperation (see page 415)	Function enables speed-up of generation or modification of a whole group of load cases through appropriate optimization of some operations. After completing generation of load cases, EndMultiOperation (see page 417)() function should be called up to save changes in a structure.
	CreateCombination (see page 416)	The function creates and returns combinations with the defined parameters. .
	CreateMobile (see page 416)	Function creates moving load case.
	CreateSimple (see page 416)	The function creates and returns a simple load case with the defined parameters. .
	Delete (see page 417)	The function deletes from the structure model the load case with the indicated number .
	DeleteMany (see page 417)	The function deletes all the load cases that meet the selection criteria introduced as argument. .
	EndMultiOperation (see page 417)	Function updates definitions of load cases and combinations in a structure. It should be called up only if BeginMultiOperation (see page 415)() function has been called up earlier.
	Exist (see page 417)	The function checks if there exists a load case with the user-defined number.
	FindWithId (see page 418)	Function returns a load case number with the specified unique identifier. If the load case is not found, then the function returns zero value (0). Available since version 1.7.
	Get (see page 418)	The function returns a load case with the user-defined number. The type of the returned object depends on the real case type with the indicated number. If the indicated number defines a combination, the function returns an object of the RobotCaseCombination object type; otherwise, it will return RobotSimpleCase object type. If the case with the indicated number does not exist, running this function leads to critical error. .
	GetAll (see page 418)	The function returns a collection of all load cases defined for a given structure. .
	GetCmpntCount (see page 419)	Function returns a number of components of the specified load case. What a component means depends on a load case and analysis type; for example, for an FRF analysis case, components are successive steps.
	GetMany (see page 419)	The function returns a collection of load cases that meet the criterion of selection introduced as argument. .
	GetRelatedValue (see page 419)	Function returns a specified value type that is associated with the specified load case.
	GetUniqueId (see page 420)	Function returns a unique identifier for a load case with the user-specified number.
	IsAuxiliary (see page 420)	
	SetAuxiliary (see page 420)	

II.1.18.3.1 BeginMultiOperation

C++

```
HRESULT BeginMultiOperation();
```

C#

```
public void BeginMultiOperation();
```

Visual Basic

```
Public Sub BeginMultiOperation()
```

Description

Function enables speed-up of generation or modification of a whole group of load cases through appropriate optimization of

(c) 2014 Autodesk Inc. All rights reserved

some operations. After completing generation of load cases, EndMultiOperation (see page 417)() function should be called up to save changes in a structure.

Version

Available since version 14.

II.1.18.3.2 CreateCombination

C++

```
HRESULT CreateCombination(long _num, BSTR _name, IRobotCombinationType _type,
IRobotCaseNature _nature, IRobotCaseAnalizeType _analize_type, IRobotCaseCombination** ret);
```

C#

```
public IRobotCaseCombination CreateCombination(long _num, String _name,
IRobotCombinationType _type, IRobotCaseNature _nature, IRobotCaseAnalizeType _analize_type);
```

Visual Basic

```
Public Function CreateCombination(_num As long, _name As String, _type As
IRobotCombinationType, _nature As IRobotCaseNature, _analize_type As IRobotCaseAnalizeType)
As IRobotCaseCombination
```

Description

The function creates and returns combinations with the defined parameters. .

II.1.18.3.3 CreateMobile

C++

```
HRESULT CreateMobile(long _number, BSTR _name, IRobotCaseNature _nature, IRobotMobileCase**
ret);
```

C#

```
public IRobotMobileCase CreateMobile(long _number, String _name, IRobotCaseNature _nature);
```

Visual Basic

```
Public Function CreateMobile(_number As long, _name As String, _nature As IRobotCaseNature)
As IRobotMobileCase
```

Description

Function creates moving load case.

Version

Available since version 3.

II.1.18.3.4 CreateSimple

C++

```
HRESULT CreateSimple(long _number, BSTR _name, IRobotCaseNature _nature,
IRobotCaseAnalizeType _analize_type, IRobotSimpleCase** ret);
```

C#

```
public IRobotSimpleCase CreateSimple(long _number, String _name, IRobotCaseNature _nature,
IRobotCaseAnalizeType _analize_type);
```

Visual Basic

```
Public Function CreateSimple(_number As long, _name As String, _nature As IRobotCaseNature,
_analize_type As IRobotCaseAnalizeType) As IRobotSimpleCase
```

(c) 2014 Autodesk Inc. All rights reserved

Description

The function creates and returns a simple load case with the defined parameters. .

II.1.18.3.5 Delete

C++

```
HRESULT Delete(long _number);
```

C#

```
public void Delete(long _number);
```

Visual Basic

```
Public Sub Delete(_number As long)
```

Description

The function deletes from the structure model the load case with the indicated number .

II.1.18.3.6 DeleteMany

C++

```
HRESULT DeleteMany(IRobotSelection* _case_selection);
```

C#

```
public void DeleteMany(IRobotSelection _case_selection);
```

Visual Basic

```
Public Sub DeleteMany(ByRef _case_selection As IRobotSelection)
```

Description

The function deletes all the load cases that meet the selection criteria introduced as argument. .

II.1.18.3.7 EndMultiOperation

C++

```
HRESULT EndMultiOperation();
```

C#

```
public void EndMultiOperation();
```

Visual Basic

```
Public Sub EndMultiOperation()
```

Description

Function updates definitions of load cases and combinations in a structure. It should be called up only if BeginMultiOperation (see page 415)() function has been called up earlier.

Version

Available since version 14.

II.1.18.3.8 Exist

C++

```
HRESULT Exist(long _number, VARIANT_BOOL* ret);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public bool Exist(long _number);
```

Visual Basic

```
Public Function Exist(_number As long) As Boolean
```

Description

The function checks if there exists a load case with the user-defined number.

II.1.18.3.9 FindWithId

C++

```
HRESULT FindWithId(long _unique_id, long* ret);
```

C#

```
public long FindWithId(long _unique_id);
```

Visual Basic

```
Public Function FindWithId(_unique_id As long) As long
```

Description

Function returns a load case number with the specified unique identifier. If the load case is not found, then the function returns zero value (0). Available since version 1.7.

II.1.18.3.10 Get

C++

```
HRESULT Get(long _number, IRobotCase** ret);
```

C#

```
public IRobotCase Get(long _number);
```

Visual Basic

```
Public Function Get(_number As long) As IRobotCase
```

Description

The function returns a load case with the user-defined number. The type of the returned object depends on the real case type with the indicated number. If the indicated number defines a combination, the function returns an object of the RobotCaseCombination object type; otherwise, it will return RobotSimpleCase object type. If the case with the indicated number does not exist, running this function leads to critical error. .

II.1.18.3.11 GetAll

C++

```
HRESULT GetAll(IRobotCaseCollection** ret);
```

C#

```
public IRobotCaseCollection GetAll();
```

Visual Basic

```
Public Function GetAll() As IRobotCaseCollection
```

Description

The function returns a collection of all load cases defined for a given structure. .

(c) 2014 Autodesk Inc. All rights reserved

II.1.18.3.12 GetCmpntCount

C++

```
HRESULT GetCmpntCount(long _case, long* ret);
```

C#

```
public long GetCmpntCount(long _case);
```

Visual Basic

```
Public Function GetCmpntCount(_case As long) As long
```

Description

Function returns a number of components of the specified load case. What a component means depends on a load case and analysis type; for example, for an FRF analysis case, components are successive steps.

Version

Available since version 9.7.

II.1.18.3.13 GetMany

C++

```
HRESULT GetMany(IRobotSelection* _case_selection, IRobotCaseCollection** ret);
```

C#

```
public IRobotCaseCollection GetMany(IRobotSelection _case_selection);
```

Visual Basic

```
Public Function GetMany(ByRef _case_selection As IRobotSelection) As IRobotCaseCollection
```

Description

The function returns a collection of load cases that meet the criterion of selection introduced as argument. .

II.1.18.3.14 GetRelatedValue

C++

```
HRESULT GetRelatedValue(IRobotCaseRelatedValueType _val_type, long _case, long _case_cmpnt
= 1, VARIANT* ret);
```

C#

```
public VARIANT GetRelatedValue(IRobotCaseRelatedValueType _val_type, long _case, long
_case_cmpnt = 1);
```

Visual Basic

```
Public Function GetRelatedValue(_val_type As IRobotCaseRelatedValueType, _case As long,
Optional _case_cmpnt As long = 1) As VARIANT
```

Description

Function returns a specified value type that is associated with the specified load case.

Version

Available since version 9.7.

(c) 2014 Autodesk Inc. All rights reserved

II.1.18.3.15 GetUniqueId

C++

```
HRESULT GetUniqueId(long _case_num, long* ret);
```

C#

```
public long GetUniqueId(long _case_num);
```

Visual Basic

```
Public Function GetUniqueId(_case_num As long) As long
```

Description

Function returns a unique identifier for a load case with the user-specified number.

Version

Available since version 8.2.

II.1.18.3.16 IsAuxiliary

C++

```
HRESULT IsAuxiliary(long _case_num, VARIANT_BOOL* ret);
```

C#

```
public bool IsAuxiliary(long _case_num);
```

Visual Basic

```
Public Function IsAuxiliary(_case_num As long) As Boolean
```

Version

Available since version 8.1.

II.1.18.3.17 SetAuxiliary

C++

```
HRESULT SetAuxiliary(long _case_num, VARIANT_BOOL _is_aux);
```

C#

```
public void SetAuxiliary(long _case_num, bool _is_aux);
```

Visual Basic

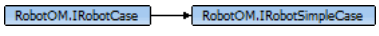
```
Public Sub SetAuxiliary(_case_num As long, _is_aux As Boolean)
```

Version

Available since version 8.1.

II.1.19 IRobotSimpleCase

Class Hierarchy



C++

```
interface IRobotSimpleCase : IRobotCase;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public interface IRobotSimpleCase : IRobotCase;
```

Visual Basic

```
Public Interface IRobotSimpleCase
```

Description

A simple load case is defined using the list of load records. .

II.1.19.1 IRobotSimpleCase Members

The following tables list the members exposed by IRobotSimpleCase.

Public Fields

	Name	Description
	AnalizeType (see page 408)	Analysis type.
	IsAuxiliary (see page 422)	
	Label (see page 422)	
	MainMode (see page 422)	Number (see page 409) of the main mode (0 denotes lack of the main mode) Available since version 1.7.
	ModesCount (see page 423)	Number (see page 409) of modes Available since version 1.7.
	Name (see page 409)	User-defined case name .
	Nature (see page 409)	Case nature .
	NatureName (see page 423)	Name (see page 409) of load case nature.
	Number (see page 409)	User-defined number linked with the load case .
	Records (see page 423)	List of load records associated with the case .
	TimeStepCount (see page 423)	Number (see page 409) of defined steps (it concerns only cases of time history analysis) .
	Type (see page 409)	Load case type - allows one to differentiate between a simple case and combination at the level of the shared part of the interface .
	UniqueId (see page 424)	Unique load case identifier Available since version 1.7.

Public Methods

	Name	Description
	GetAnalysisParams (see page 424)	Function returns the object describing analysis parameters defined for the load case. The type of the returned object depends on the analysis type that has been assigned to this load case (e.g. for modal cases the object of the RobotModalAnalysisParams type is returned).
	GetSeismicCode (see page 425)	Function returns the name of a seismic code for which a given load case has been defined. For cases whose analysis type is different from seismic analysis, the function returns an empty character string. .
	SetAnalysisParams (see page 425)	Function sets analysis parameters for the load case. If the specified parameters cannot be set for this case (e.g. due to incompatibility between the type of case analysis and type of the object defining parameters), then zero value (False) is returned. .
	SetNatureExt (see page 425)	Function assigns - to the case - the nature consistent with the indicated number of "action" described in the current regulations. .

II.1.19.2 IRobotSimpleCase Fields

The fields of the IRobotSimpleCase class are listed here.

Public Fields

	Name	Description
	IsAuxiliary (see page 422)	

(c) 2014 Autodesk Inc. All rights reserved

	Label (see page 422)	
	MainMode (see page 422)	Number (see page 409) of the main mode (0 denotes lack of the main mode) Available since version 1.7.
	ModesCount (see page 423)	Number (see page 409) of modes Available since version 1.7.
	NatureName (see page 423)	Name (see page 409) of load case nature.
	Records (see page 423)	List of load records associated with the case .
	TimeStepCount (see page 423)	Number (see page 409) of defined steps (it concerns only cases of time history analysis) .
	UniqueId (see page 424)	Unique load case identifier Available since version 1.7.

II.1.19.2.1 IsAuxiliary

C++

```
HRESULT get_IsAuxiliary(VARIANT_BOOL*);
HRESULT put_IsAuxiliary(VARIANT_BOOL);
```

C#

```
public bool IsAuxiliary { get; set; }
```

Visual Basic

```
Public IsAuxiliary As Boolean
```

Version

Available since version 8.1.

II.1.19.2.2 Label

C++

```
HRESULT get_Label(BSTR*);
HRESULT put_Label(BSTR);
```

C#

```
public String Label { get; set; }
```

Visual Basic

```
Public Label As String
```

Version

Available since version 8.

II.1.19.2.3 MainMode

C++

```
HRESULT get_MainMode(long*);
HRESULT put_MainMode(long);
```

C#

```
public long MainMode { get; set; }
```

Visual Basic

```
Public MainMode As long
```

Description

Number (see page 409) of the main mode (0 denotes lack of the main mode) Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

II.1.19.2.4 ModesCount

C++

```
HRESULT get_ModesCount(long*);
HRESULT put_ModesCount(long);
```

C#

```
public long ModesCount { get; set; }
```

Visual Basic

```
Public ModesCount As long
```

Description

Number (see page 409) of modes Available since version 1.7.

II.1.19.2.5 NatureName

C++

```
HRESULT get_NatureName(BSTR*);
```

C#

```
public String NatureName { get; }
```

Visual Basic

```
Public ReadOnly NatureName As String
```

Description

Name (see page 409) of load case nature.

Version

Available since version 3.

II.1.19.2.6 Records

C++

```
HRESULT get_Records(IRobotLoadRecordMngr**);
```

C#

```
public IRobotLoadRecordMngr Records { get; }
```

Visual Basic

```
Public ReadOnly Records As IRobotLoadRecordMngr
```

Description

List of load records associated with the case .

II.1.19.2.7 TimeStepCount

C++

```
HRESULT get_TimeStepCount(long*);
```

C#

```
public long TimeStepCount { get; }
```

Visual Basic

```
Public ReadOnly TimeStepCount As long
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Number (see page 409) of defined steps (it concerns only cases of time history analysis) .

Version

Available since version 3.

II.1.19.2.8 UniqueId

C++

```
HRESULT get_UniqueId(long*);
```

C#

```
public long UniqueId { get; }
```

Visual Basic

```
Public ReadOnly UniqueId As long
```

Description

Unique load case identifier Available since version 1.7.

II.1.19.3 IRobotSimpleCase Methods

The methods of the IRobotSimpleCase class are listed here.

Public Methods

	Name	Description
	GetAnalysisParams (see page 424)	Function returns the object describing analysis parameters defined for the load case. The type of the returned object depends on the analysis type that has been assigned to this load case (e.g. for modal cases the object of the RobotModalAnalysisParams type is returned).
	GetSeismicCode (see page 425)	Function returns the name of a seismic code for which a given load case has been defined. For cases whose analysis type is different from seismic analysis, the function returns an empty character string. .
	SetAnalysisParams (see page 425)	Function sets analysis parameters for the load case. If the specified parameters cannot be set for this case (e.g. due to incompatibility between the type of case analysis and type of the object defining parameters), then zero value (False) is returned. .
	SetNatureExt (see page 425)	Function assigns - to the case - the nature consistent with the indicated number of "action" described in the current regulations. .

II.1.19.3.1 GetAnalysisParams

C++

```
HRESULT GetAnalysisParams(IDispatch* ret);
```

C#

```
public IDispatch GetAnalysisParams();
```

Visual Basic

```
Public Function GetAnalysisParams() As IDispatch
```

Description

Function returns the object describing analysis parameters defined for the load case. The type of the returned object depends on the analysis type that has been assigned to this load case (e.g. for modal cases the object of the RobotModalAnalysisParams type is returned).

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 2.5.

II.1.19.3.2 GetSeismicCode

C++

```
HRESULT GetSeismicCode(BSTR* ret);
```

C#

```
public String GetSeismicCode();
```

Visual Basic

```
Public Function GetSeismicCode() As String
```

Description

Function returns the name of a seismic code for which a given load case has been defined. For cases whose analysis type is different from seismic analysis, the function returns an empty character string. .

Version

Available since version 2.5.

II.1.19.3.3 SetAnalysisParams

C++

```
HRESULT SetAnalysisParams(IDispatch _params, VARIANT_BOOL* ret);
```

C#

```
public bool SetAnalysisParams(IDispatch _params);
```

Visual Basic

```
Public Function SetAnalysisParams(_params As IDispatch) As Boolean
```

Description

Function sets analysis parameters for the load case. If the specified parameters cannot be set for this case (e.g. due to incompatibility between the type of case analysis and type of the object defining parameters), then zero value (False) is returned. .

Version

Available since version 2.5.

II.1.19.3.4 SetNatureExt

C++

```
HRESULT SetNatureExt(long _rgl_action);
```

C#

```
public void SetNatureExt(long _rgl_action);
```

Visual Basic

```
Public Sub SetNatureExt(_rgl_action As long)
```

Description

Function assigns - to the case - the nature consistent with the indicated number of "action" described in the current regulations. .

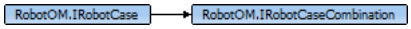
(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.

II.1.20 IRobotCaseCombination

Class Hierarchy



C++

```
interface IRobotCaseCombination : IRobotCase;
```

C#

```
public interface IRobotCaseCombination : IRobotCase;
```

Visual Basic

```
Public Interface IRobotCaseCombination
```

Description

A combination is defined by means of simple load cases with appropriate factors. .

II.1.20.1 IRobotCaseCombination Members

The following tables list the members exposed by IRobotCaseCombination.

Public Fields

	Name	Description
	AnalizeType (see page 408)	Analysis type.
	CaseFactors (see page 427)	List of pairs (simple case, factor) defining a combination .
	CombinationType (see page 427)	Combination type .
	IsAuxiliary (see page 427)	
	Label (see page 428)	
	Name (see page 409)	User-defined case name .
	Nature (see page 409)	Case nature .
	NatureName (see page 428)	Name (see page 409) of nature.
	Number (see page 409)	User-defined number linked with the load case .
	Quadratic (see page 428)	Indicator informing whether we are dealing with a quadratic combination .
	SeismicType (see page 429)	Seismic combination type .
	Signed (see page 429)	A flag indicating whether the combination is signed or not Available since version 1.7.
	Type (see page 409)	Load case type - allows one to differentiate between a simple case and combination at the level of the shared part of the interface .
	UniqueId (see page 429)	Unique identifier of load case combinations Available since version 1.7.

Public Methods

	Name	Description
	GetAnalysisParams (see page 430)	Function returns the object describing analysis parameters defined for the load case. The type of the returned object depends on the analysis type that has been assigned to this load case (e.g. for modal cases the object of the RobotModalAnalysisParams type is returned). .
	SetAnalysisParams (see page 430)	Function sets analysis parameters for the load case. If the parameters specified cannot be set for this case (e.g. due to incompatibility between the case analysis type and the type of the object defining parameters), then zero value (False) is returned. .

(c) 2014 Autodesk Inc. All rights reserved

	SetNatureExt (see page 430)	Function assigns - to the combination - the nature consistent with the indicated number of "action" described in the current regulations. .

II.1.20.2 IRobotCaseCombination Fields

The fields of the IRobotCaseCombination class are listed here.

Public Fields

	Name	Description
	CaseFactors (see page 427)	List of pairs (simple case, factor) defining a combination .
	CombinationType (see page 427)	Combination type .
	IsAuxiliary (see page 427)	
	Label (see page 428)	
	NatureName (see page 428)	Name (see page 409) of nature.
	Quadratic (see page 428)	Indicator informing whether we are dealing with a quadratic combination .
	SeismicType (see page 429)	Seismic combination type .
	Signed (see page 429)	A flag indicating whether the combination is signed or not Available since version 1.7.
	UniqueId (see page 429)	Unique identifier of load case combinations Available since version 1.7.

II.1.20.2.1 CaseFactors

C++

```
HRESULT get_CaseFactors(IRobotCaseFactorMngr**);
```

C#

```
public IRobotCaseFactorMngr CaseFactors { get; }
```

Visual Basic

```
Public ReadOnly CaseFactors As IRobotCaseFactorMngr
```

Description

List of pairs (simple case, factor) defining a combination .

II.1.20.2.2 CombinationType

C++

```
HRESULT get_CombinationType(IRobotCombinationType*);
HRESULT put_CombinationType(IRobotCombinationType);
```

C#

```
public IRobotCombinationType CombinationType { get; set; }
```

Visual Basic

```
Public CombinationType As IRobotCombinationType
```

Description

Combination type .

II.1.20.2.3 IsAuxiliary

C++

```
HRESULT get_IsAuxiliary(VARIANT_BOOL*);
HRESULT put_IsAuxiliary(VARIANT_BOOL);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public bool IsAuxiliary { get; set; }
```

Visual Basic

```
Public IsAuxiliary As Boolean
```

Version

Available since version 8.1.

II.1.20.2.4 Label

C++

```
HRESULT get_Label(BSTR*);
HRESULT put_Label(BSTR);
```

C#

```
public String Label { get; set; }
```

Visual Basic

```
Public Label As String
```

Version

Available since version 8.

II.1.20.2.5 NatureName

C++

```
HRESULT get_NatureName(BSTR*);
```

C#

```
public String NatureName { get; }
```

Visual Basic

```
Public ReadOnly NatureName As String
```

Description

Name (see page 409) of nature.

Version

Available since version 3.

II.1.20.2.6 Quadratic

C++

```
HRESULT get_Quadratic(VARIANT_BOOL*);
HRESULT put_Quadratic(VARIANT_BOOL);
```

C#

```
public bool Quadratic { get; set; }
```

Visual Basic

```
Public Quadratic As Boolean
```

Description

Indicator informing whether we are dealing with a quadratic combination .

(c) 2014 Autodesk Inc. All rights reserved

II.1.20.2.7 SeismicType

C++

```
HRESULT get_SeismicType(IRobotModeCombinationType*);
HRESULT put_SeismicType(IRobotModeCombinationType);
```

C#

```
public IRobotModeCombinationType SeismicType { get; set; }
```

Visual Basic

```
Public SeismicType As IRobotModeCombinationType
```

Description

Seismic combination type .

II.1.20.2.8 Signed

C++

```
HRESULT get_Signed(VARIANT_BOOL*);
HRESULT put_Signed(VARIANT_BOOL);
```

C#

```
public bool Signed { get; set; }
```

Visual Basic

```
Public Signed As Boolean
```

Description

A flag indicating whether the combination is signed or not Available since version 1.7.

II.1.20.2.9 UniqueId

C++

```
HRESULT get_UniqueId(long*);
```

C#

```
public long UniqueId { get; }
```

Visual Basic

```
Public ReadOnly UniqueId As long
```

Description

Unique identifier of load case combinations Available since version 1.7.

II.1.20.3 IRobotCaseCombination Methods

The methods of the IRobotCaseCombination class are listed here.

Public Methods

	Name	Description
	GetAnalysisParams (see page 430)	Function returns the object describing analysis parameters defined for the load case. The type of the returned object depends on the analysis type that has been assigned to this load case (e.g. for modal cases the object of the RobotModalAnalysisParams type is returned). .
	SetAnalysisParams (see page 430)	Function sets analysis parameters for the load case. If the parameters specified cannot be set for this case (e.g. due to incompatibility between the case analysis type and the type of the object defining parameters), then zero value (False) is returned. .

(c) 2014 Autodesk Inc. All rights reserved

	SetNatureExt (see page 430)	Function assigns - to the combination - the nature consistent with the indicated number of "action" described in the current regulations. .

II.1.20.3.1 GetAnalysisParams

C++

```
HRESULT GetAnalysisParams(IDispatch* ret);
```

C#

```
public IDispatch GetAnalysisParams();
```

Visual Basic

```
Public Function GetAnalysisParams() As IDispatch
```

Description

Function returns the object describing analysis parameters defined for the load case. The type of the returned object depends on the analysis type that has been assigned to this load case (e.g. for modal cases the object of the RobotModalAnalysisParams type is returned). .

Version

Available since version 2.5.

II.1.20.3.2 SetAnalysisParams

C++

```
HRESULT SetAnalysisParams(IDispatch _params, VARIANT_BOOL* ret);
```

C#

```
public bool SetAnalysisParams(IDispatch _params);
```

Visual Basic

```
Public Function SetAnalysisParams(_params As IDispatch) As Boolean
```

Description

Function sets analysis parameters for the load case. If the parameters specified cannot be set for this case (e.g. due to incompatibility between the case analysis type and the type of the object defining parameters), then zero value (False) is returned. .

Version

Available since version 2.5.

II.1.20.3.3 SetNatureExt

C++

```
HRESULT SetNatureExt(long _rgl_action);
```

C#

```
public void SetNatureExt(long _rgl_action);
```

Visual Basic

```
Public Sub SetNatureExt(_rgl_action As long)
```

Description

Function assigns - to the combination - the nature consistent with the indicated number of "action" described in the current regulations. .

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.

II.1.21 IRobotCaseFactorMngr

Class Hierarchy

C++

```
interface IRobotCaseFactorMngr : IDispatch;
```

C#

```
public interface IRobotCaseFactorMngr;
```

Visual Basic

```
Public Interface IRobotCaseFactorMngr
```

Description

Interface granting access to the list of pairs (load case, coefficient). All pairs from the list are numbered (indexed) with numbers from 1 to the number of items (Count (see page 431)). .

II.1.21.1 IRobotCaseFactorMngr Members

The following tables list the members exposed by IRobotCaseFactorMngr.

Public Fields

	Name	Description
	Count (see page 431)	Number of managed items .

Public Methods

	Name	Description
	Delete (see page 432)	The function deletes items with the defined index. .
	Get (see page 432)	The function returns the item with the indicated index. .
	New (see page 432)	The function creates a new pair (load case, coefficient) and adds to the list of managed items. It returns the index of the created pair. .

II.1.21.2 IRobotCaseFactorMngr Fields

The fields of the IRobotCaseFactorMngr class are listed here.

Public Fields

	Name	Description
	Count (see page 431)	Number of managed items .

II.1.21.2.1 Count

C++

```
HRESULT get_Count(long*);
```

C#

```
public long Count { get; }
```

Visual Basic

```
Public ReadOnly Count As long
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Number of managed items .

II.1.21.3 IRobotCaseFactorMngr Methods

The methods of the IRobotCaseFactorMngr class are listed here.

Public Methods

	Name	Description
	Delete (see page 432)	The function deletes items with the defined index. .
	Get (see page 432)	The function returns the item with the indicated index. .
	New (see page 432)	The function creates a new pair (load case, coefficient) and adds to the list of managed items. It returns the index of the created pair. .

II.1.21.3.1 Delete

C++

```
HRESULT Delete(long _index);
```

C#

```
public void Delete(long _index);
```

Visual Basic

```
Public Sub Delete(_index As long)
```

Description

The function deletes items with the defined index. .

II.1.21.3.2 Get

C++

```
HRESULT Get(long _index, IRobotCaseFactor** ret);
```

C#

```
public IRobotCaseFactor Get(long _index);
```

Visual Basic

```
Public Function Get(_index As long) As IRobotCaseFactor
```

Description

The function returns the item with the indicated index. .

II.1.21.3.3 New

C++

```
HRESULT New(long _case_number, double _factor, long* ret);
```

C#

```
public long New(long _case_number, double _factor);
```

Visual Basic

```
Public Function New(_case_number As long, _factor As double) As long
```

(c) 2014 Autodesk Inc. All rights reserved

Description

The function creates a new pair (load case, coefficient) and adds to the list of managed items. It returns the index of the created pair. .

II.1.22 IRobotCombinationType

C++

```
enum IRobotCombinationType;
```

C#

```
public enum IRobotCombinationType;
```

Visual Basic

```
Public Enum IRobotCombinationType
```

Members

Members	Description
I_CBT_SPC = 3	Special state.
I_CBT_ULS = 0	Ultimate limit state Available since version 2.0.
I_CBT_SLS = 1	Serviceability limit state Available since version 2.0.
I_CBT_ALS = 2	Accidental limit state Available since version 2.0.
I_CBT_SLS_EC_FRE = 4	Available since version 12.
I_CBT_SLS_EC_QPR = 5	Available since version 12.
I_CBT_SLS_EC_RAR = 6	Available since version 12.

Description

Type of load case combinations available in Robot. .

II.1.23 IRobotCaseType

C++

```
enum IRobotCaseType;
```

C#

```
public enum IRobotCaseType;
```

Visual Basic

```
Public Enum IRobotCaseType
```

Members

Members	Description
I_CT_SIMPLE = 0	Simple case .
I_CT_COMBINATION = 1	Combination .
I_CT_CODE_COMBINATION = 2	Code combination (defined according to a definition of the selected code) Available since version 1.7.
I_CT_MOBILE = 3	Moving load case. Available since version 2.5.

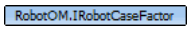
Description

Among load cases, one distinguishes three basic types: a simple case, a combination and a code combination. .

(c) 2014 Autodesk Inc. All rights reserved

II.1.24 IRobotCaseFactor

Class Hierarchy



C++

```
interface IRobotCaseFactor : IDispatch;
```

C#

```
public interface IRobotCaseFactor;
```

Visual Basic

```
Public Interface IRobotCaseFactor
```

Description

Interface describing a pair (load case, coefficient) used in the definition of load case combination. .

II.1.24.1 IRobotCaseFactor Members

The following tables list the members exposed by IRobotCaseFactor.

Public Fields

	Name	Description
	CaseNumber (see page 434)	Load case number .
	Factor (see page 434)	(multiplication) factor.

II.1.24.2 IRobotCaseFactor Fields

The fields of the IRobotCaseFactor class are listed here.

Public Fields

	Name	Description
	CaseNumber (see page 434)	Load case number .
	Factor (see page 434)	(multiplication) factor.

II.1.24.2.1 CaseNumber

C++

```
HRESULT get_CaseNumber(long*);
HRESULT put_CaseNumber(long);
```

C#

```
public long CaseNumber { get; set; }
```

Visual Basic

```
Public CaseNumber As long
```

Description

Load case number .

II.1.24.2.2 Factor

C++

```
HRESULT get_Factor(double*);
HRESULT put_Factor(double);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double Factor { get; set; }
```

Visual Basic

```
Public Factor As double
```

Description

(multiplication) factor.

II.1.25 IRobotCaseCollection

Class Hierarchy

C++

```
interface IRobotCaseCollection : IRobotCollection;
```

C#

```
public interface IRobotCaseCollection : IRobotCollection;
```

Visual Basic

```
Public Interface IRobotCaseCollection
```

Description

Collection of load cases .

II.1.26 IRobotLimitState

C++

```
enum IRobotLimitState;
```

C#

```
public enum IRobotLimitState;
```

Visual Basic

```
Public Enum IRobotLimitState
```

Members

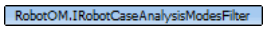
Members	Description
I_LS_ULS = 0	Ultimate limit state.
I_LS_SLS = 1	Serviceability limit state.
I_LS_ALS = 2	Accidental state.

Description

Limit state type.

II.1.27 IRobotCaseAnalysisModesFilter

Class Hierarchy



C++

```
interface IRobotCaseAnalysisModesFilter : IDispatch;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public interface IRobotCaseAnalysisModesFilter;
```

Visual Basic

```
Public Interface IRobotCaseAnalysisModesFilter
```

Description

Definition of modes taken into account during structure dynamic analysis. .

Version

Available since version 2.5.

II.1.27.1 IRobotCaseAnalysisModesFilter Members

The following tables list the members exposed by IRobotCaseAnalysisModesFilter.

Public Fields

	Name	Description
	MassPercentage (see page 436)	
	Modes (see page 436)	List of modes taken into account during structure dynamic analysis. .
	Type (see page 437)	Filtering type.

II.1.27.2 IRobotCaseAnalysisModesFilter Fields

The fields of the IRobotCaseAnalysisModesFilter class are listed here.

Public Fields

	Name	Description
	MassPercentage (see page 436)	
	Modes (see page 436)	List of modes taken into account during structure dynamic analysis. .
	Type (see page 437)	Filtering type.

II.1.27.2.1 MassPercentage

C++

```
HRESULT get_MassPercentage(double*);
HRESULT put_MassPercentage(double);
```

C#

```
public double MassPercentage { get; set; }
```

Visual Basic

```
Public MassPercentage As double
```

Version

Available since version 2.5.

II.1.27.2.2 Modes

C++

```
HRESULT get_Modes(BSTR*);
HRESULT put_Modes(BSTR);
```

C#

```
public BSTR Modes { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Modes As BSTR
```

Description

List of modes taken into account during structure dynamic analysis. .

Version

Available since version 2.5.

II.1.27.2.3 Type

C++

```
HRESULT get_Type(IRobotCaseAnalysisModesFilterType*);
HRESULT put_Type(IRobotCaseAnalysisModesFilterType);
```

C#

```
public IRobotCaseAnalysisModesFilterType Type { get; set; }
```

Visual Basic

```
Public Type As IRobotCaseAnalysisModesFilterType
```

Description

Filtering type.

Version

Available since version 2.5.

II.1.28 IRobotCaseAnalysisModesFilterType

C++

```
enum IRobotCaseAnalysisModesFilterType;
```

C#

```
public enum IRobotCaseAnalysisModesFilterType;
```

Visual Basic

```
Public Enum IRobotCaseAnalysisModesFilterType
```

Members

Members	Description
I_CAMFT_NON_ACTIVE = 0	Inactive filtering. Available since version 2.5.
I_CAMFT_MASS_PERCENTAGE = 1	During structure analysis only these modes will be taken into account whose value of the mass participation factor is greater than the specified one . Available since version 2.5.
I_CAMFT_LIST_OF_MODES = 2	During structure analysis only the modes specified by the user will be taken into account. Available since version 2.5.

Description

Available methods of mode filtering.

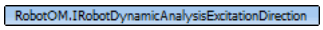
Version

Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

II.1.29 IRobotDynamicAnalysisExcitationDirection

Class Hierarchy



C++

```
interface IRobotDynamicAnalysisExcitationDirection : IDispatch;
```

C#

```
public interface IRobotDynamicAnalysisExcitationDirection;
```

Visual Basic

```
Public Interface IRobotDynamicAnalysisExcitationDirection
```

Version

Available since version 5.5.

II.1.29.1 IRobotDynamicAnalysisExcitationDirection Members

The following tables list the members exposed by IRobotDynamicAnalysisExcitationDirection.

Public Fields

	Name	Description
	CombType (see page 439)	
	Group1 (see page 439)	
	Group2 (see page 439)	
	Group3 (see page 440)	
	Lambda (see page 440)	
	Mi (see page 440)	
	NormalizedX (see page 440)	
	NormalizedY (see page 440)	
	NormalizedZ (see page 441)	
	QuadraticActive (see page 441)	
	QuadraticSigned (see page 441)	
	ResolutionActive (see page 441)	
	Rx (see page 441)	
	Ry (see page 442)	
	Rz (see page 442)	
	UseNormalized (see page 442)	
	X (see page 442)	
	Y (see page 442)	
	Z (see page 443)	

II.1.29.2 IRobotDynamicAnalysisExcitationDirection Fields

The fields of the IRobotDynamicAnalysisExcitationDirection class are listed here.

Public Fields

	Name	Description
	CombType (see page 439)	
	Group1 (see page 439)	
	Group2 (see page 439)	

(c) 2014 Autodesk Inc. All rights reserved

	Group3 (see page 440)	
	Lambda (see page 440)	
	Mi (see page 440)	
	NormalizedX (see page 440)	
	NormalizedY (see page 440)	
	NormalizedZ (see page 441)	
	QuadraticActive (see page 441)	
	QuadraticSigned (see page 441)	
	ResolutionActive (see page 441)	
	Rx (see page 441)	
	Ry (see page 442)	
	Rz (see page 442)	
	UseNormalized (see page 442)	
	X (see page 442)	
	Y (see page 442)	
	Z (see page 443)	

II.1.29.2.1 CombType

C++

```
HRESULT get_CombType(IRobotModeCombinationType*);
HRESULT put_CombType(IRobotModeCombinationType);
```

C#

```
public IRobotModeCombinationType CombType { get; set; }
```

Visual Basic

```
Public CombType As IRobotModeCombinationType
```

II.1.29.2.2 Group1

C++

```
HRESULT get_Group1(VARIANT_BOOL*);
HRESULT put_Group1(VARIANT_BOOL);
```

C#

```
public bool Group1 { get; set; }
```

Visual Basic

```
Public Group1 As Boolean
```

II.1.29.2.3 Group2

C++

```
HRESULT get_Group2(VARIANT_BOOL*);
HRESULT put_Group2(VARIANT_BOOL);
```

C#

```
public bool Group2 { get; set; }
```

Visual Basic

```
Public Group2 As Boolean
```

(c) 2014 Autodesk Inc. All rights reserved

II.1.29.2.4 Group3

C++

```
HRESULT get_Group3(VARIANT_BOOL*);
HRESULT put_Group3(VARIANT_BOOL);
```

C#

```
public bool Group3 { get; set; }
```

Visual Basic

```
Public Group3 As Boolean
```

II.1.29.2.5 Lambda

C++

```
HRESULT get_Lambda(double*);
HRESULT put_Lambda(double);
```

C#

```
public double Lambda { get; set; }
```

Visual Basic

```
Public Lambda As double
```

II.1.29.2.6 Mi

C++

```
HRESULT get_Mi(double*);
HRESULT put_Mi(double);
```

C#

```
public double Mi { get; set; }
```

Visual Basic

```
Public Mi As double
```

II.1.29.2.7 NormalizedX

C++

```
HRESULT get_NormalizedX(double*);
```

C#

```
public double NormalizedX { get; }
```

Visual Basic

```
Public ReadOnly NormalizedX As double
```

II.1.29.2.8 NormalizedY

C++

```
HRESULT get_NormalizedY(double*);
```

C#

```
public double NormalizedY { get; }
```

Visual Basic

```
Public ReadOnly NormalizedY As double
```

(c) 2014 Autodesk Inc. All rights reserved

II.1.29.2.9 NormalizedZ

C++

```
HRESULT get_NormalizedZ(double*);
```

C#

```
public double NormalizedZ { get; }
```

Visual Basic

```
Public ReadOnly NormalizedZ As double
```

II.1.29.2.10 QuadraticActive

C++

```
HRESULT get_QuadraticActive(VARIANT_BOOL*);
HRESULT put_QuadraticActive(VARIANT_BOOL);
```

C#

```
public bool QuadraticActive { get; set; }
```

Visual Basic

```
Public QuadraticActive As Boolean
```

II.1.29.2.11 QuadraticSigned

C++

```
HRESULT get_QuadraticSigned(VARIANT_BOOL*);
HRESULT put_QuadraticSigned(VARIANT_BOOL);
```

C#

```
public bool QuadraticSigned { get; set; }
```

Visual Basic

```
Public QuadraticSigned As Boolean
```

II.1.29.2.12 ResolutionActive

C++

```
HRESULT get_ResolutionActive(VARIANT_BOOL*);
HRESULT put_ResolutionActive(VARIANT_BOOL);
```

C#

```
public bool ResolutionActive { get; set; }
```

Visual Basic

```
Public ResolutionActive As Boolean
```

II.1.29.2.13 Rx

C++

```
HRESULT get_Rx(double*);
HRESULT put_Rx(double);
```

C#

```
public double Rx { get; set; }
```

Visual Basic

```
Public Rx As double
```

(c) 2014 Autodesk Inc. All rights reserved

II.1.29.2.14 Ry

C++

```
HRESULT get_Ry(double*);
HRESULT put_Ry(double);
```

C#

```
public double Ry { get; set; }
```

Visual Basic

```
Public Ry As double
```

II.1.29.2.15 Rz

C++

```
HRESULT get_Rz(double*);
HRESULT put_Rz(double);
```

C#

```
public double Rz { get; set; }
```

Visual Basic

```
Public Rz As double
```

II.1.29.2.16 UseNormalized

C++

```
HRESULT get_UseNormalized(VARIANT_BOOL*);
HRESULT put_UseNormalized(VARIANT_BOOL);
```

C#

```
public bool UseNormalized { get; set; }
```

Visual Basic

```
Public UseNormalized As Boolean
```

II.1.29.2.17 X

C++

```
HRESULT get_X(double*);
HRESULT put_X(double);
```

C#

```
public double X { get; set; }
```

Visual Basic

```
Public X As double
```

II.1.29.2.18 Y

C++

```
HRESULT get_Y(double*);
HRESULT put_Y(double);
```

C#

```
public double Y { get; set; }
```

Visual Basic

```
Public Y As double
```

(c) 2014 Autodesk Inc. All rights reserved

II.1.29.2.19 Z

C++

```
HRESULT get_Z(double*);
HRESULT put_Z(double);
```

C#

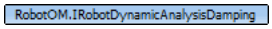
```
public double Z { get; set; }
```

Visual Basic

```
Public Z As double
```

II.1.30 IRobotDynamicAnalysisDamping

Class Hierarchy



C++

```
interface IRobotDynamicAnalysisDamping : IDispatch;
```

C#

```
public interface IRobotDynamicAnalysisDamping;
```

Visual Basic

```
Public Interface IRobotDynamicAnalysisDamping
```

Description

Damping parameters for the dynamic analysis.

Version

Available since version 9.7.

II.1.30.1 IRobotDynamicAnalysisDamping Members

The following tables list the members exposed by IRobotDynamicAnalysisDamping.

Public Fields

	Name	Description
	ConstValue (see page 444)	Constant damping value.
	Type (see page 444)	Damping type.

Public Methods

	Name	Description
	GetModeDamping (see page 445)	Function returns a damping value for the specified vibration mode.
	GetModes (see page 445)	Function returns a table with numbers of vibration modes with defined damping values.
	RemoveModeDamping (see page 445)	Function deletes a damping value for the specified vibration mode.
	SetModeDamping (see page 446)	Function sets a damping value for the specified vibration mode.

II.1.30.2 IRobotDynamicAnalysisDamping Fields

The fields of the IRobotDynamicAnalysisDamping class are listed here.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	ConstValue (see page 444)	Constant damping value.
	Type (see page 444)	Damping type.

II.1.30.2.1 ConstValue

C++

```
HRESULT get_ConstValue(double*);
HRESULT put_ConstValue(double);
```

C#

```
public double ConstValue { get; set; }
```

Visual Basic

```
Public ConstValue As double
```

Description

Constant damping value.

Version

Available since version 9.7.

II.1.30.2.2 Type

C++

```
HRESULT get_Type(IRobotDynamicAnalysisDampingType*);
HRESULT put_Type(IRobotDynamicAnalysisDampingType);
```

C#

```
public IRobotDynamicAnalysisDampingType Type { get; set; }
```

Visual Basic

```
Public Type As IRobotDynamicAnalysisDampingType
```

Description

Damping type.

Version

Available since version 9.7.

II.1.30.3 IRobotDynamicAnalysisDamping Methods

The methods of the IRobotDynamicAnalysisDamping class are listed here.

Public Methods

	Name	Description
	GetModeDamping (see page 445)	Function returns a damping value for the specified vibration mode.
	GetModes (see page 445)	Function returns a table with numbers of vibration modes with defined damping values.
	RemoveModeDamping (see page 445)	Function deletes a damping value for the specified vibration mode.
	SetModeDamping (see page 446)	Function sets a damping value for the specified vibration mode.

(c) 2014 Autodesk Inc. All rights reserved

II.1.30.3.1 GetModeDamping

C++

```
HRESULT GetModeDamping(long _mode, double* ret);
```

C#

```
public double GetModeDamping(long _mode);
```

Visual Basic

```
Public Function GetModeDamping(_mode As long) As double
```

Description

Function returns a damping value for the specified vibration mode.

Version

Available since version 9.7.

II.1.30.3.2 GetModes

C++

```
HRESULT GetModes(SAFEARRAY** ret);
```

C#

```
public Array GetModes();
```

Visual Basic

```
Public Function GetModes() As long)()
```

Description

Function returns a table with numbers of vibration modes with defined damping values.

Version

Available since version 9.7.

II.1.30.3.3 RemoveModeDamping

C++

```
HRESULT RemoveModeDamping(long _mode);
```

C#

```
public void RemoveModeDamping(long _mode);
```

Visual Basic

```
Public Sub RemoveModeDamping(_mode As long)
```

Description

Function deletes a damping value for the specified vibration mode.

Version

Available since version 9.7.

(c) 2014 Autodesk Inc. All rights reserved

II.1.30.3.4 SetModeDamping

C++

```
HRESULT SetModeDamping(long _mode, double _damp_value);
```

C#

```
public void SetModeDamping(long _mode, double _damp_value);
```

Visual Basic

```
Public Sub SetModeDamping(_mode As long, _damp_value As double)
```

Description

Function sets a damping value for the specified vibration mode.

Version

Available since version 9.7.

II.1.31 IRobotDynamicAnalysisDampingType

C++

```
enum IRobotDynamicAnalysisDampingType;
```

C#

```
public enum IRobotDynamicAnalysisDampingType;
```

Visual Basic

```
Public Enum IRobotDynamicAnalysisDampingType
```

Members

Members	Description
I_DADT_NONE = 0	Without damping. Available since version 9.7.
I_DADT_CONSTANT = 1	Constant damping. Available since version 9.7.
I_DADT_VARIABLE = 3	Damping that varies for each mode. Available since version 9.7.
I_DADT_MODAL = 2	Modal damping. Available since version 9.7.

Version

Available since version 9.7.

II.1.32 IRobotCaseRelatedValueType

C++

```
enum IRobotCaseRelatedValueType;
```

C#

```
public enum IRobotCaseRelatedValueType;
```

Visual Basic

```
Public Enum IRobotCaseRelatedValueType
```

(c) 2014 Autodesk Inc. All rights reserved

Members

Members	Description
I_CRVT_FRF_FREQUENCY = 1	Frequency for FRF analysis. Available since version 9.7.

Description

A set of identifiers for types of values that can be associated with the load case.

Version

Available since version 9.7.

II.1.33 Parameters of simplified seismic analysis

Parameters of seismic analysis (method of static equivalent lateral force).

Version

Available since version 12.5.

Enumerations

	Name	Description
	IRobotSELFSeismicLevelDefinitionMethod (see page 454)	
	IRobotSELFSeismicTBaseMethod (see page 455)	
	IRobotSELFSeismic_ASCE_7_10_SiteClassType (see page 458)	
	IRobotSELFSeismic_ASCE_7_10_StructureType (see page 462)	
	IRobotSELFSeismic_EC_8_StructureType (see page 466)	
	IRobotSELFSeismic_EC_8_SiteClass (see page 467)	
	IRobotSELFSeismic_EC_8_SpectrumType (see page 467)	
	IRobotSELFSeismic_AS_1170_4_SoilCategoryType (see page 470)	
	IRobotSELFSeismic_AS_1170_4_StructureType (see page 471)	
	IRobotSELFSeismic_AS_1170_4_ProbabilityType (see page 471)	

Interfaces

	Name	Description
	IRobotSELFSeismicEngine (see page 448)	Generator of load cases for the simplified seismic analysis.
	IRobotSELFSeismicGenerationParams (see page 449)	Parameters for generation of load cases.
	IRobotSELFSeismicStructureParams (see page 452)	
	IRobotSELFSeismicAnalysisParams (see page 455)	Parameters of the seismic analysis (method of static equivalent lateral force).
	IRobotSELFSeismic_ASCE_7_10_Params (see page 458)	Parameters of the ASCE 7-10 code for the simplified seismic analysis.
	IRobotSELFSeismic_EC_8_Params (see page 462)	Parameters of EC 8 for the simplified seismic analysis.

(c) 2014 Autodesk Inc. All rights reserved

	IRobotSELFSeismic_AS_1170_4_Params (see page 467)	Parameters of the AS 1170-4 code for the simplified seismic analysis.

II.1.33.1 IRobotSELFSeismicEngine

Class Hierarchy

RobotOM.IRobotSELFSeismicEngine

C++

```
interface IRobotSELFSeismicEngine : IDispatch;
```

C#

```
public interface IRobotSELFSeismicEngine;
```

Visual Basic

```
Public Interface IRobotSELFSeismicEngine
```

Description

Generator of load cases for the simplified seismic analysis.

Version

Available since version 12.5.

II.1.33.1.1 IRobotSELFSeismicEngine Members

The following tables list the members exposed by IRobotSELFSeismicEngine.

Public Fields

	Name	Description
	GenerationParams (see page 448)	Parameters for generation of load cases for the simplified seismic analysis.

Public Methods

	Name	Description
	GenerateLoadCases (see page 449)	The function generates new load cases for the simplified seismic analysis based on the parameters of generation set currently. The function returns the number of generated cases.

II.1.33.1.2 IRobotSELFSeismicEngine Fields

The fields of the IRobotSELFSeismicEngine class are listed here.

Public Fields

	Name	Description
	GenerationParams (see page 448)	Parameters for generation of load cases for the simplified seismic analysis.

II.1.33.1.2.1 GenerationParams

C++

```
HRESULT get_GenerationParams(IRobotSELFSeismicGenerationParams**);
```

C#

```
public IRobotSELFSeismicGenerationParams GenerationParams { get; }
```

Visual Basic

```
Public ReadOnly GenerationParams As IRobotSELFSeismicGenerationParams
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Parameters for generation of load cases for the simplified seismic analysis.

Version

Available since version 12.5.

II.1.33.1.3 IRobotSELFSeismicEngine Methods

The methods of the IRobotSELFSeismicEngine class are listed here.

Public Methods

	Name	Description
	GenerateLoadCases (see page 449)	The function generates new load cases for the simplified seismic analysis based on the parameters of generation set currently. The function returns the number of generated cases.

II.1.33.1.3.1 GenerateLoadCases

C++

```
HRESULT GenerateLoadCases(long* ret);
```

C#

```
public long GenerateLoadCases();
```

Visual Basic

```
Public Function GenerateLoadCases() As long
```

Description

The function generates new load cases for the simplified seismic analysis based on the parameters of generation set currently. The function returns the number of generated cases.

Version

Available since version 12.5.

II.1.33.2 IRobotSELFSeismicGenerationParams

Class Hierarchy

RobotOM.IRobotSELFSeismicGenerationParams

C++

```
interface IRobotSELFSeismicGenerationParams : IDispatch;
```

C#

```
public interface IRobotSELFSeismicGenerationParams;
```

Visual Basic

```
Public Interface IRobotSELFSeismicGenerationParams
```

Description

Parameters for generation of load cases.

Version

Available since version 12.5.

(c) 2014 Autodesk Inc. All rights reserved

II.1.33.2.1 IRobotSELFSeismicGenerationParams Members

The following tables list the members exposed by IRobotSELFSeismicGenerationParams.

Public Fields

	Name	Description
	CodeName (see page 450)	
	CodeNumber (see page 450)	
	ExcitationDir (see page 451)	
	ModalCaseParams (see page 451)	
	SeismicParams (see page 451)	
	TBaseMethod (see page 451)	

II.1.33.2.2 IRobotSELFSeismicGenerationParams Fields

The fields of the IRobotSELFSeismicGenerationParams class are listed here.

Public Fields

	Name	Description
	CodeName (see page 450)	
	CodeNumber (see page 450)	
	ExcitationDir (see page 451)	
	ModalCaseParams (see page 451)	
	SeismicParams (see page 451)	
	TBaseMethod (see page 451)	

II.1.33.2.2.1 CodeName

C++

```
HRESULT get_CodeName(BSTR*);
HRESULT put_CodeName(BSTR);
```

C#

```
public String CodeName { get; set; }
```

Visual Basic

```
Public CodeName As String
```

Version

Available since version 12.5.

II.1.33.2.2.2 CodeNumber

C++

```
HRESULT get_CodeNumber(long*);
HRESULT put_CodeNumber(long);
```

C#

```
public long CodeNumber { get; set; }
```

Visual Basic

```
Public CodeNumber As long
```

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 12.5.

II.1.33.2.2.3 ExcitationDir

C++

```
HRESULT get_ExcitationDir(IRobotDynamicAnalysisExcitationDirection**);
```

C#

```
public IRobotDynamicAnalysisExcitationDirection ExcitationDir { get; }
```

Visual Basic

```
Public ReadOnly ExcitationDir As IRobotDynamicAnalysisExcitationDirection
```

Version

Available since version 12.5.

II.1.33.2.2.4 ModalCaseParams

C++

```
HRESULT get_ModalCaseParams(IRobotModalAnalysisParams**);
```

C#

```
public IRobotModalAnalysisParams ModalCaseParams { get; }
```

Visual Basic

```
Public ReadOnly ModalCaseParams As IRobotModalAnalysisParams
```

Version

Available since version 12.5.

II.1.33.2.2.5 SeismicParams

C++

```
HRESULT get_SeismicParams(IRobotSELFSeismicAnalysisParams**);
```

C#

```
public IRobotSELFSeismicAnalysisParams SeismicParams { get; }
```

Visual Basic

```
Public ReadOnly SeismicParams As IRobotSELFSeismicAnalysisParams
```

Version

Available since version 12.5.

II.1.33.2.2.6 TBaseMethod

C++

```
HRESULT get_TBaseMethod(IRobotSELFSeismicTBaseMethod*);
HRESULT put_TBaseMethod(IRobotSELFSeismicTBaseMethod);
```

C#

```
public IRobotSELFSeismicTBaseMethod TBaseMethod { get; set; }
```

Visual Basic

```
Public TBaseMethod As IRobotSELFSeismicTBaseMethod
```

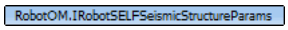
(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 12.5.

II.1.33.3 IRobotSELFSeismicStructureParams

Class Hierarchy



C++

```
interface IRobotSELFSeismicStructureParams : IDispatch;
```

C#

```
public interface IRobotSELFSeismicStructureParams;
```

Visual Basic

```
Public Interface IRobotSELFSeismicStructureParams
```

Version

Available since version 12.5.

II.1.33.3.1 IRobotSELFSeismicStructureParams Members

The following tables list the members exposed by IRobotSELFSeismicStructureParams.

Public Fields

	Name	Description
	BaseLevelCoordZ (see page 452)	
	BaseLevelDefMethod (see page 453)	
	BaseLevelStorey (see page 453)	Story no.
	TopLevelCoordZ (see page 453)	
	TopLevelDefMethod (see page 454)	
	TopLevelStorey (see page 454)	Story no.

II.1.33.3.2 IRobotSELFSeismicStructureParams Fields

The fields of the IRobotSELFSeismicStructureParams class are listed here.

Public Fields

	Name	Description
	BaseLevelCoordZ (see page 452)	
	BaseLevelDefMethod (see page 453)	
	BaseLevelStorey (see page 453)	Story no.
	TopLevelCoordZ (see page 453)	
	TopLevelDefMethod (see page 454)	
	TopLevelStorey (see page 454)	Story no.

II.1.33.3.2.1 BaseLevelCoordZ

C++

```
HRESULT get_BaseLevelCoordZ(double*);
HRESULT put_BaseLevelCoordZ(double);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double BaseLevelCoordZ { get; set; }
```

Visual Basic

```
Public BaseLevelCoordZ As double
```

Version

Available since version 12.5.

II.1.33.3.2.2 BaseLevelDefMethod

C++

```
HRESULT get_BaseLevelDefMethod(IRobotSELFSeismicLevelDefinitionMethod*);
HRESULT put_BaseLevelDefMethod(IRobotSELFSeismicLevelDefinitionMethod);
```

C#

```
public IRobotSELFSeismicLevelDefinitionMethod BaseLevelDefMethod { get; set; }
```

Visual Basic

```
Public BaseLevelDefMethod As IRobotSELFSeismicLevelDefinitionMethod
```

Version

Available since version 12.5.

II.1.33.3.2.3 BaseLevelStorey

C++

```
HRESULT get_BaseLevelStorey(long*);
HRESULT put_BaseLevelStorey(long);
```

C#

```
public long BaseLevelStorey { get; set; }
```

Visual Basic

```
Public BaseLevelStorey As long
```

Description

Story no.

Version

Available since version 12.5.

II.1.33.3.2.4 TopLevelCoordZ

C++

```
HRESULT get_TopLevelCoordZ(double*);
HRESULT put_TopLevelCoordZ(double);
```

C#

```
public double TopLevelCoordZ { get; set; }
```

Visual Basic

```
Public TopLevelCoordZ As double
```

Version

Available since version 12.5.

(c) 2014 Autodesk Inc. All rights reserved

II.1.33.3.2.5 TopLevelDefMethod

C++

```
HRESULT get_TopLevelDefMethod(IRobotSELFSeismicLevelDefinitionMethod*);
HRESULT put_TopLevelDefMethod(IRobotSELFSeismicLevelDefinitionMethod);
```

C#

```
public IRobotSELFSeismicLevelDefinitionMethod TopLevelDefMethod { get; set; }
```

Visual Basic

```
Public TopLevelDefMethod As IRobotSELFSeismicLevelDefinitionMethod
```

Version

Available since version 12.5.

II.1.33.3.2.6 TopLevelStorey

C++

```
HRESULT get_TopLevelStorey(long*);
HRESULT put_TopLevelStorey(long);
```

C#

```
public long TopLevelStorey { get; set; }
```

Visual Basic

```
Public TopLevelStorey As long
```

Description

Story no.

Version

Available since version 12.5.

II.1.33.4 IRobotSELFSeismicLevelDefinitionMethod

C++

```
enum IRobotSELFSeismicLevelDefinitionMethod;
```

C#

```
public enum IRobotSELFSeismicLevelDefinitionMethod;
```

Visual Basic

```
Public Enum IRobotSELFSeismicLevelDefinitionMethod
```

Members

Members	Description
I_SSLDM_AUTOMATIC = 0	Available since version 12.5.
I_SSLDM_BY_Z_COORDINATE = 1	Available since version 12.5.
I_SSLDM_BY_STOREY = 2	Available since version 12.5.

Version

Available since version 12.5.

(c) 2014 Autodesk Inc. All rights reserved

II.1.33.5 IRobotSELFSeismicTBaseMethod

C++

```
enum IRobotSELFSeismicTBaseMethod;
```

C#

```
public enum IRobotSELFSeismicTBaseMethod;
```

Visual Basic

```
Public Enum IRobotSELFSeismicTBaseMethod
```

Members

Members	Description
I_SSTBM_APPROXIMATE_BY_CODE = 0	Available since version 12.5.
I_SSTBM_USER_DEFINED = 1	Available since version 12.5.
I_SSTBM_PRECISE_MODAL = 2	Available since version 12.5.
I_SSTBM_PRECISE_MODAL_WITH_MASS = 3	Available since version 12.5.

Version

Available since version 12.5.

II.1.33.6 IRobotSELFSeismicAnalysisParams

Class Hierarchy

RobotOM.IRobotSELFSeismicAnalysisParams

C++

```
interface IRobotSELFSeismicAnalysisParams : IDispatch;
```

C#

```
public interface IRobotSELFSeismicAnalysisParams;
```

Visual Basic

```
Public Interface IRobotSELFSeismicAnalysisParams
```

Description

Parameters of the seismic analysis (method of static equivalent lateral force).

Version

Available since version 12.5.

II.1.33.6.1 IRobotSELFSeismicAnalysisParams Members

The following tables list the members exposed by IRobotSELFSeismicAnalysisParams.

Public Fields

	Name	Description
	CodeName (see page 456)	
	CodeNumber (see page 456)	
	CodeParams (see page 456)	Object defining code-specific parameters; the object type corresponds to a selected code, for example for the ASCE 7-10 code the object of RobotSELFSeismic_ASCE_7_10_Params class is available;.
	Eccentricities (see page 457)	
	StructureParams (see page 457)	

(c) 2014 Autodesk Inc. All rights reserved

	TBaseMethod (see page 457)	

Public Methods

	Name	Description
	GetExcitationDir (see page 458)	

II.1.33.6.2 IRobotSELFSeismicAnalysisParams Fields

The fields of the IRobotSELFSeismicAnalysisParams class are listed here.

Public Fields

	Name	Description
	CodeName (see page 456)	
	CodeNumber (see page 456)	
	CodeParams (see page 456)	Object defining code-specific parameters; the object type corresponds to a selected code, for example for the ASCE 7-10 code the object of RobotSELFSeismic_ASCE_7_10_Params class is available;.
	Eccentricities (see page 457)	
	StructureParams (see page 457)	
	TBaseMethod (see page 457)	

II.1.33.6.2.1 CodeName

C++

```
HRESULT get_CodeName(BSTR*);
```

C#

```
public String CodeName { get; }
```

Visual Basic

```
Public ReadOnly CodeName As String
```

Version

Available since version 12.5.

II.1.33.6.2.2 CodeNumber

C++

```
HRESULT get_CodeNumber(long*);
```

C#

```
public long CodeNumber { get; }
```

Visual Basic

```
Public ReadOnly CodeNumber As long
```

Version

Available since version 12.5.

II.1.33.6.2.3 CodeParams

C++

```
HRESULT get_CodeParams(IDispatch*);
```

C#

```
public IDispatch CodeParams { get; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public ReadOnly CodeParams As IDispatch
```

Description

Object defining code-specific parameters; the object type corresponds to a selected code, for example for the ASCE 7-10 code the object of RobotSELFSeismic_ASCE_7_10_Params class is available;.

Version

Available since version 12.5.

II.1.33.6.2.4 Eccentricities

C++

```
HRESULT get_Eccentricities(IRobotMassEccentricities**);
```

C#

```
public IRobotMassEccentricities Eccentricities { get; }
```

Visual Basic

```
Public ReadOnly Eccentricities As IRobotMassEccentricities
```

Version

Available since version 12.5.

II.1.33.6.2.5 StructureParams

C++

```
HRESULT get_StructureParams(IRobotSELFSeismicStructureParams**);
```

C#

```
public IRobotSELFSeismicStructureParams StructureParams { get; }
```

Visual Basic

```
Public ReadOnly StructureParams As IRobotSELFSeismicStructureParams
```

Version

Available since version 12.5.

II.1.33.6.2.6 TBaseMethod

C++

```
HRESULT get_TBaseMethod(IRobotSELFSeismicTBaseMethod*);
```

C#

```
public IRobotSELFSeismicTBaseMethod TBaseMethod { get; }
```

Visual Basic

```
Public ReadOnly TBaseMethod As IRobotSELFSeismicTBaseMethod
```

Version

Available since version 12.5.

II.1.33.6.3 IRobotSELFSeismicAnalysisParams Methods

The methods of the IRobotSELFSeismicAnalysisParams class are listed here.

(c) 2014 Autodesk Inc. All rights reserved

Public Methods

	Name	Description
	GetExcitationDir (see page 458)	

II.1.33.6.3.1 GetExcitationDir

C++

```
HRESULT GetExcitationDir(IRobotDynamicAnalysisExcitationDirection** ret);
```

C#

```
public IRobotDynamicAnalysisExcitationDirection GetExcitationDir();
```

Visual Basic

```
Public Function GetExcitationDir() As IRobotDynamicAnalysisExcitationDirection
```

Version

Available since version 12.5.

II.1.33.7 IRobotSELFSeismic_ASCE_7_10_SiteClassType

C++

```
enum IRobotSELFSeismic_ASCE_7_10_SiteClassType;
```

C#

```
public enum IRobotSELFSeismic_ASCE_7_10_SiteClassType;
```

Visual Basic

```
Public Enum IRobotSELFSeismic_ASCE_7_10_SiteClassType
```

Members

Members	Description
I_SSSCT_ASCE_7_10_A = 1	Available since version 12.5.
I_SSSCT_ASCE_7_10_B = 2	Available since version 12.5.
I_SSSCT_ASCE_7_10_C = 3	Available since version 12.5.
I_SSSCT_ASCE_7_10_D = 4	Available since version 12.5.
I_SSSCT_ASCE_7_10_E = 5	Available since version 12.5.
I_SSSCT_ASCE_7_10_F = 6	Available since version 12.5.

Version

Available since version 12.5.

II.1.33.8 IRobotSELFSeismic_ASCE_7_10_Params

Class Hierarchy

RobotOM.IRobotSELFSeismic_ASCE_7_10_Params

C++

```
interface IRobotSELFSeismic_ASCE_7_10_Params : IDispatch;
```

C#

```
public interface IRobotSELFSeismic_ASCE_7_10_Params;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Interface IRobotSELFSeismic_ASCE_7_10_Params
```

Description

Parameters of the ASCE 7-10 code for the simplified seismic analysis.

Version

Available since version 12.5.

II.1.33.8.1 IRobotSELFSeismic_ASCE_7_10_Params Members

The following tables list the members exposed by IRobotSELFSeismic_ASCE_7_10_Params.

Public Fields

	Name	Description
	I (see page 459)	
	R (see page 460)	
	S1 (see page 460)	
	SiteClass (see page 460)	
	Ss (see page 460)	
	StructureTypeX (see page 461)	
	StructureTypeY (see page 461)	
	TBaseValueX (see page 461)	
	TBaseValueY (see page 461)	
	TL (see page 462)	

II.1.33.8.2 IRobotSELFSeismic_ASCE_7_10_Params Fields

The fields of the IRobotSELFSeismic_ASCE_7_10_Params class are listed here.

Public Fields

	Name	Description
	I (see page 459)	
	R (see page 460)	
	S1 (see page 460)	
	SiteClass (see page 460)	
	Ss (see page 460)	
	StructureTypeX (see page 461)	
	StructureTypeY (see page 461)	
	TBaseValueX (see page 461)	
	TBaseValueY (see page 461)	
	TL (see page 462)	

II.1.33.8.2.1 I

C++

```
HRESULT get_I(double*);
HRESULT put_I(double);
```

C#

```
public double I { get; set; }
```

Visual Basic

```
Public I As double
```

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 12.5.

II.1.33.8.2.2 R

C++

```
HRESULT get_R(double*);
HRESULT put_R(double);
```

C#

```
public double R { get; set; }
```

Visual Basic

```
Public R As double
```

Version

Available since version 12.5.

II.1.33.8.2.3 S1

C++

```
HRESULT get_S1(double*);
HRESULT put_S1(double);
```

C#

```
public double S1 { get; set; }
```

Visual Basic

```
Public S1 As double
```

Version

Available since version 12.5.

II.1.33.8.2.4 SiteClass

C++

```
HRESULT get_SiteClass(IRobotSELFSeismic_ASCE_7_10_SiteClassType*);
HRESULT put_SiteClass(IRobotSELFSeismic_ASCE_7_10_SiteClassType);
```

C#

```
public IRobotSELFSeismic_ASCE_7_10_SiteClassType SiteClass { get; set; }
```

Visual Basic

```
Public SiteClass As IRobotSELFSeismic_ASCE_7_10_SiteClassType
```

Version

Available since version 12.5.

II.1.33.8.2.5 Ss

C++

```
HRESULT get_Ss(double*);
HRESULT put_Ss(double);
```

C#

```
public double Ss { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Ss As double
```

Version

Available since version 12.5.

II.1.33.8.2.6 StructureTypeX

C++

```
HRESULT get_StructureTypeX(IRobotSELFSeismic_ASCE_7_10_StructureType*);
HRESULT put_StructureTypeX(IRobotSELFSeismic_ASCE_7_10_StructureType);
```

C#

```
public IRobotSELFSeismic_ASCE_7_10_StructureType StructureTypeX { get; set; }
```

Visual Basic

```
Public StructureTypeX As IRobotSELFSeismic_ASCE_7_10_StructureType
```

Version

Available since version 12.5.

II.1.33.8.2.7 StructureTypeY

C++

```
HRESULT get_StructureTypeY(IRobotSELFSeismic_ASCE_7_10_StructureType*);
HRESULT put_StructureTypeY(IRobotSELFSeismic_ASCE_7_10_StructureType);
```

C#

```
public IRobotSELFSeismic_ASCE_7_10_StructureType StructureTypeY { get; set; }
```

Visual Basic

```
Public StructureTypeY As IRobotSELFSeismic_ASCE_7_10_StructureType
```

Version

Available since version 12.5.

II.1.33.8.2.8 TBaseValueX

C++

```
HRESULT get_TBaseValueX(double*);
HRESULT put_TBaseValueX(double);
```

C#

```
public double TBaseValueX { get; set; }
```

Visual Basic

```
Public TBaseValueX As double
```

Version

Available since version 12.5.

II.1.33.8.2.9 TBaseValueY

C++

```
HRESULT get_TBaseValueY(double*);
HRESULT put_TBaseValueY(double);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double TBaseValueY { get; set; }
```

Visual Basic

```
Public TBaseValueY As double
```

Version

Available since version 12.5.

II.1.33.8.2.10 TL

C++

```
HRESULT get_TL(double*);
HRESULT put_TL(double);
```

C#

```
public double TL { get; set; }
```

Visual Basic

```
Public TL As double
```

Version

Available since version 12.5.

II.1.33.9 IRobotSELFSeismic_ASCE_7_10_StructureType

C++

```
enum IRobotSELFSeismic_ASCE_7_10_StructureType;
```

C#

```
public enum IRobotSELFSeismic_ASCE_7_10_StructureType;
```

Visual Basic

```
Public Enum IRobotSELFSeismic_ASCE_7_10_StructureType
```

Members

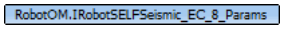
Members	Description
I_SSST_ASCE_7_10_STEEL_FRAMES = 1	Available since version 12.5.
I_SSST_ASCE_7_10_CONCRETE_FRAMES = 2	Available since version 12.5.
I_SSST_ASCE_7_10_BRACED_STEEL_FRAMES = 3	Available since version 12.5.
I_SSST_ASCE_7_10_OTHER_STRUCTURE_SYSTEM = 4	Available since version 12.5.

Version

Available since version 12.5.

II.1.33.10 IRobotSELFSeismic_EC_8_Params

Class Hierarchy



C++

```
interface IRobotSELFSeismic_EC_8_Params : IDispatch;
```

C#

```
public interface IRobotSELFSeismic_EC_8_Params;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Interface IRobotSELFSeismic_EC_8_Params
```

Description

Parameters of EC 8 for the simplified seismic analysis.

Version

Available since version 12.5.

II.1.33.10.1 IRobotSELFSeismic_EC_8_Params Members

The following tables list the members exposed by IRobotSELFSeismic_EC_8_Params.

Public Fields

	Name	Description
	Ag (see page 463)	
	Beta (see page 464)	
	Q (see page 464)	
	SiteClass (see page 464)	Ground type; to set the envelope for several ground types, you should set the sum of values to be included in the envelope.
	SpectrumType (see page 465)	
	StructureTypeX (see page 465)	
	StructureTypeY (see page 465)	

Public Methods

	Name	Description
	SiteClassEnvelopeCheck (see page 466)	
	SiteClassEnvelopeIsChecked (see page 466)	

II.1.33.10.2 IRobotSELFSeismic_EC_8_Params Fields

The fields of the IRobotSELFSeismic_EC_8_Params class are listed here.

Public Fields

	Name	Description
	Ag (see page 463)	
	Beta (see page 464)	
	Q (see page 464)	
	SiteClass (see page 464)	Ground type; to set the envelope for several ground types, you should set the sum of values to be included in the envelope.
	SpectrumType (see page 465)	
	StructureTypeX (see page 465)	
	StructureTypeY (see page 465)	

II.1.33.10.2.1 Ag

C++

```
HRESULT get_Ag(double*);
HRESULT put_Ag(double);
```

C#

```
public double Ag { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Ag As double
```

Version

Available since version 12.5.

II.1.33.10.2.2 Beta

C++

```
HRESULT get_Beta(double*);
HRESULT put_Beta(double);
```

C#

```
public double Beta { get; set; }
```

Visual Basic

```
Public Beta As double
```

Version

Available since version 12.5.

II.1.33.10.2.3 Q

C++

```
HRESULT get_Q(double*);
HRESULT put_Q(double);
```

C#

```
public double Q { get; set; }
```

Visual Basic

```
Public Q As double
```

Version

Available since version 12.5.

II.1.33.10.2.4 SiteClass

C++

```
HRESULT get_SiteClass(IRobotSELFSeismic_EC_8_SiteClass*);
HRESULT put_SiteClass(IRobotSELFSeismic_EC_8_SiteClass);
```

C#

```
public IRobotSELFSeismic_EC_8_SiteClass SiteClass { get; set; }
```

Visual Basic

```
Public SiteClass As IRobotSELFSeismic_EC_8_SiteClass
```

Description

Ground type; to set the envelope for several ground types, you should set the sum of values to be included in the envelope.

Version

Available since version 12.5.

(c) 2014 Autodesk Inc. All rights reserved

II.1.33.10.2.5 SpectrumType

C++

```
HRESULT get_SpectrumType(IRobotSELFSeismic_EC_8_SpectrumType*);
HRESULT put_SpectrumType(IRobotSELFSeismic_EC_8_SpectrumType);
```

C#

```
public IRobotSELFSeismic_EC_8_SpectrumType SpectrumType { get; set; }
```

Visual Basic

```
Public SpectrumType As IRobotSELFSeismic_EC_8_SpectrumType
```

Version

Available since version 12.5.

II.1.33.10.2.6 StructureTypeX

C++

```
HRESULT get_StructureTypeX(IRobotSELFSeismic_EC_8_StructureType*);
HRESULT put_StructureTypeX(IRobotSELFSeismic_EC_8_StructureType);
```

C#

```
public IRobotSELFSeismic_EC_8_StructureType StructureTypeX { get; set; }
```

Visual Basic

```
Public StructureTypeX As IRobotSELFSeismic_EC_8_StructureType
```

Version

Available since version 12.5.

II.1.33.10.2.7 StructureTypeY

C++

```
HRESULT get_StructureTypeY(IRobotSELFSeismic_EC_8_StructureType*);
HRESULT put_StructureTypeY(IRobotSELFSeismic_EC_8_StructureType);
```

C#

```
public IRobotSELFSeismic_EC_8_StructureType StructureTypeY { get; set; }
```

Visual Basic

```
Public StructureTypeY As IRobotSELFSeismic_EC_8_StructureType
```

Version

Available since version 12.5.

II.1.33.10.3 IRobotSELFSeismic_EC_8_Params Methods

The methods of the IRobotSELFSeismic_EC_8_Params class are listed here.

Public Methods

	Name	Description
	SiteClassEnvelopeCheck (see page 466)	
	SiteClassEnvelopeIsChecked (see page 466)	

(c) 2014 Autodesk Inc. All rights reserved

II.1.33.10.3.1 SiteClassEnvelopeCheck

C++

```
HRESULT SiteClassEnvelopeCheck(IRobotSELFSeismic_EC_8_SiteClass _siteClass, VARIANT_BOOL
_check);
```

C#

```
public void SiteClassEnvelopeCheck(IRobotSELFSeismic_EC_8_SiteClass _siteClass, bool
_check);
```

Visual Basic

```
Public Sub SiteClassEnvelopeCheck(_siteClass As IRobotSELFSeismic_EC_8_SiteClass, _check As
Boolean)
```

Version

Available since version 12.5.

II.1.33.10.3.2 SiteClassEnvelopeIsChecked

C++

```
HRESULT SiteClassEnvelopeIsChecked(IRobotSELFSeismic_EC_8_SiteClass _siteClass,
VARIANT_BOOL* ret);
```

C#

```
public bool SiteClassEnvelopeIsChecked(IRobotSELFSeismic_EC_8_SiteClass _siteClass);
```

Visual Basic

```
Public Function SiteClassEnvelopeIsChecked(_siteClass As IRobotSELFSeismic_EC_8_SiteClass)
As Boolean
```

Version

Available since version 12.5.

II.1.33.11 IRobotSELFSeismic_EC_8_StructureType

C++

```
enum IRobotSELFSeismic_EC_8_StructureType;
```

C#

```
public enum IRobotSELFSeismic_EC_8_StructureType;
```

Visual Basic

```
Public Enum IRobotSELFSeismic_EC_8_StructureType
```

Members

Members	Description
I_SSST_EC_8_STEEL_FRAMES = 1	Available since version 12.5.
I_SSST_EC_8_CONCRETE_FRAMES = 2	Available since version 12.5.
I_SSST_EC_8_BRACED_STEEL_FRAMES = 3	Available since version 12.5.
I_SSST_EC_8_OTHER_STRUCTURE_SYSTEM = 4	Available since version 12.5.

Version

Available since version 12.5.

(c) 2014 Autodesk Inc. All rights reserved

II.1.33.12 IRobotSELFSeismic_EC_8_SiteClass

C++

```
enum IRobotSELFSeismic_EC_8_SiteClass;
```

C#

```
public enum IRobotSELFSeismic_EC_8_SiteClass;
```

Visual Basic

```
Public Enum IRobotSELFSeismic_EC_8_SiteClass
```

Members

Members	Description
I_SSSC_EC_8_A = 1	Available since version 12.5.
I_SSSC_EC_8_B = 2	Available since version 12.5.
I_SSSC_EC_8_C = 4	Available since version 12.5.
I_SSSC_EC_8_D = 8	Available since version 12.5.
I_SSSC_EC_8_E = 16	Available since version 12.5.
I_SSSC_EC_8_ENVELOPE = 255	Available since version 12.5.

Version

Available since version 12.5.

II.1.33.13 IRobotSELFSeismic_EC_8_SpectrumType

C++

```
enum IRobotSELFSeismic_EC_8_SpectrumType;
```

C#

```
public enum IRobotSELFSeismic_EC_8_SpectrumType;
```

Visual Basic

```
Public Enum IRobotSELFSeismic_EC_8_SpectrumType
```

Members

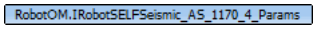
Members	Description
I_SSST_EC_8_Type1 = 1	Available since version 12.5.
I_SSST_EC_8_Type2 = 2	Available since version 12.5.

Version

Available since version 12.5.

II.1.33.14 IRobotSELFSeismic_AS_1170_4_Params

Class Hierarchy



C++

```
interface IRobotSELFSeismic_AS_1170_4_Params : IDispatch;
```

C#

```
public interface IRobotSELFSeismic_AS_1170_4_Params;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Interface IRobotSELFSeismic_AS_1170_4_Params
```

Description

Parameters of the AS 1170-4 code for the simplified seismic analysis.

Version

Available since version 12.5.

II.1.33.14.1 IRobotSELFSeismic_AS_1170_4_Params Members

The following tables list the members exposed by IRobotSELFSeismic_AS_1170_4_Params.

Public Fields

	Name	Description
	Kp (see page 468)	
	Mi (see page 469)	
	Probability (see page 469)	
	SoilCategory (see page 469)	
	Sp (see page 469)	
	StructureTypeX (see page 470)	
	StructureTypeY (see page 470)	
	Z (see page 470)	

II.1.33.14.2 IRobotSELFSeismic_AS_1170_4_Params Fields

The fields of the IRobotSELFSeismic_AS_1170_4_Params class are listed here.

Public Fields

	Name	Description
	Kp (see page 468)	
	Mi (see page 469)	
	Probability (see page 469)	
	SoilCategory (see page 469)	
	Sp (see page 469)	
	StructureTypeX (see page 470)	
	StructureTypeY (see page 470)	
	Z (see page 470)	

II.1.33.14.2.1 Kp

C++

```
HRESULT get_Kp( double*);
HRESULT put_Kp( double);
```

C#

```
public  double Kp { get; set; }
```

Visual Basic

```
Public Kp As  double
```

Version

Available since version 12.5.

(c) 2014 Autodesk Inc. All rights reserved

II.1.33.14.2.2 Mi

C++

```
HRESULT get_Mi( double*);
HRESULT put_Mi( double);
```

C#

```
public  double Mi { get; set; }
```

Visual Basic

```
Public Mi As  double
```

Version

Available since version 12.5.

II.1.33.14.2.3 Probability

C++

```
HRESULT get_Probability(IRobotSELFSeismic_AS_1170_4_ProbabilityType*);
HRESULT put_Probability(IRobotSELFSeismic_AS_1170_4_ProbabilityType);
```

C#

```
public IRobotSELFSeismic_AS_1170_4_ProbabilityType Probability { get; set; }
```

Visual Basic

```
Public Probability As IRobotSELFSeismic_AS_1170_4_ProbabilityType
```

Version

Available since version 12.5.

II.1.33.14.2.4 SoilCategory

C++

```
HRESULT get_SoilCategory(IRobotSELFSeismic_AS_1170_4_SoilCategoryType*);
HRESULT put_SoilCategory(IRobotSELFSeismic_AS_1170_4_SoilCategoryType);
```

C#

```
public IRobotSELFSeismic_AS_1170_4_SoilCategoryType SoilCategory { get; set; }
```

Visual Basic

```
Public SoilCategory As IRobotSELFSeismic_AS_1170_4_SoilCategoryType
```

Version

Available since version 12.5.

II.1.33.14.2.5 Sp

C++

```
HRESULT get_Sp(double*);
HRESULT put_Sp(double);
```

C#

```
public double Sp { get; set; }
```

Visual Basic

```
Public Sp As double
```

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 12.5.

II.1.33.14.2.6 StructureTypeX

C++

```
HRESULT get_StructureTypeX(IRobotSELFSeismic_AS_1170_4_StructureType*);
HRESULT put_StructureTypeX(IRobotSELFSeismic_AS_1170_4_StructureType);
```

C#

```
public IRobotSELFSeismic_AS_1170_4_StructureType StructureTypeX { get; set; }
```

Visual Basic

```
Public StructureTypeX As IRobotSELFSeismic_AS_1170_4_StructureType
```

Version

Available since version 12.5.

II.1.33.14.2.7 StructureTypeY

C++

```
HRESULT get_StructureTypeY(IRobotSELFSeismic_AS_1170_4_StructureType*);
HRESULT put_StructureTypeY(IRobotSELFSeismic_AS_1170_4_StructureType);
```

C#

```
public IRobotSELFSeismic_AS_1170_4_StructureType StructureTypeY { get; set; }
```

Visual Basic

```
Public StructureTypeY As IRobotSELFSeismic_AS_1170_4_StructureType
```

Version

Available since version 12.5.

II.1.33.14.2.8 Z

C++

```
HRESULT get_Z(double*);
HRESULT put_Z(double);
```

C#

```
public double Z { get; set; }
```

Visual Basic

```
Public Z As double
```

Version

Available since version 12.5.

II.1.33.15 IRobotSELFSeismic_AS_1170_4_SoilCategoryType

C++

```
enum IRobotSELFSeismic_AS_1170_4_SoilCategoryType;
```

C#

```
public enum IRobotSELFSeismic_AS_1170_4_SoilCategoryType;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Enum IRobotSELFSeismic_AS_1170_4_SoilCategoryType
```

Members

Members	Description
I_SSSC_AS_1170_4_A = 1	Available since version 12.5.
I_SSSC_AS_1170_4_B = 2	Available since version 12.5.
I_SSSC_AS_1170_4_C = 3	Available since version 12.5.
I_SSSC_AS_1170_4_D = 4	Available since version 12.5.
I_SSSC_AS_1170_4_E = 5	Available since version 12.5.

Version

Available since version 12.5.

II.1.33.16 IRobotSELFSeismic_AS_1170_4_StructureType

C++

```
enum IRobotSELFSeismic_AS_1170_4_StructureType;
```

C#

```
public enum IRobotSELFSeismic_AS_1170_4_StructureType;
```

Visual Basic

```
Public Enum IRobotSELFSeismic_AS_1170_4_StructureType
```

Members

Members	Description
I_SSST_AS_1170_4_STEEL_FRAMES = 1	Available since version 12.5.
I_SSST_AS_1170_4_CONCRETE_FRAMES = 2	Available since version 12.5.
I_SSST_AS_1170_4_BRACED_STEEL_FRAMES = 3	Available since version 12.5.
I_SSST_AS_1170_4_OTHER_STRUCTURES = 4	Available since version 12.5.

Version

Available since version 12.5.

II.1.33.17 IRobotSELFSeismic_AS_1170_4_ProbabilityType

C++

```
enum IRobotSELFSeismic_AS_1170_4_ProbabilityType;
```

C#

```
public enum IRobotSELFSeismic_AS_1170_4_ProbabilityType;
```

Visual Basic

```
Public Enum IRobotSELFSeismic_AS_1170_4_ProbabilityType
```

Members

Members	Description
I_SSPT_AS_1170_4_1_2500 = 1	Available since version 12.5.
I_SSPT_AS_1170_4_1_2000 = 2	Available since version 12.5.
I_SSPT_AS_1170_4_1_1500 = 3	Available since version 12.5.
I_SSPT_AS_1170_4_1_1000 = 4	Available since version 12.5.

(c) 2014 Autodesk Inc. All rights reserved

I_SSPT_AS_1170_4_1_800 = 5	Available since version 12.5.
I_SSPT_AS_1170_4_1_500 = 6	Available since version 12.5.
I_SSPT_AS_1170_4_1_250 = 7	Available since version 12.5.
I_SSPT_AS_1170_4_1_200 = 8	Available since version 12.5.
I_SSPT_AS_1170_4_1_100 = 9	Available since version 12.5.
I_SSPT_AS_1170_4_1_50 = 10	Available since version 12.5.
I_SSPT_AS_1170_4_1_25 = 11	Available since version 12.5.
I_SSPT_AS_1170_4_1_20 = 12	Available since version 12.5.

Version

Available since version 12.5.

II.1.34 Wind loads simulation

Available since version 14.5.

Interfaces

	Name	Description
	IRobotWindLoadsSimulationEngine (see page 472)	
	IRobotWindLoadsSimulationParams (see page 473)	

II.1.34.1 IRobotWindLoadsSimulationEngine

Class Hierarchy

RobotOM.IRobotWindLoadsSimulationEngine

C++

```
interface IRobotWindLoadsSimulationEngine : IDispatch;
```

C#

```
public interface IRobotWindLoadsSimulationEngine;
```

Visual Basic

```
Public Interface IRobotWindLoadsSimulationEngine
```

Version

Available since version 14.5.

II.1.34.1.1 IRobotWindLoadsSimulationEngine Members

The following tables list the members exposed by IRobotWindLoadsSimulationEngine.

Public Fields

	Name	Description
	Params (see page 473)	

Public Methods

	Name	Description
	Generate (see page 473)	

II.1.34.1.2 IRobotWindLoadsSimulationEngine Fields

The fields of the IRobotWindLoadsSimulationEngine class are listed here.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	Params (see page 473)	

II.1.34.1.2.1 Params

C++

```
HRESULT get_Params(IRobotWindLoadsSimulationParams**);
```

C#

```
public IRobotWindLoadsSimulationParams* Params { get; }
```

Visual Basic

```
Public ReadOnly Params As IRobotWindLoadsSimulationParams*
```

Version

Available since version 14.5.

II.1.34.1.3 IRobotWindLoadsSimulationEngine Methods

The methods of the IRobotWindLoadsSimulationEngine class are listed here.

Public Methods

	Name	Description
	Generate (see page 473)	

II.1.34.1.3.1 Generate

C++

```
HRESULT Generate(VARIANT_BOOL* ret);
```

C#

```
public bool Generate();
```

Visual Basic

```
Public Function Generate() As Boolean
```

Version

Available since version 14.5.

II.1.34.2 IRobotWindLoadsSimulationParams

Class Hierarchy

RobotOM.IRobotWindLoadsSimulationParams

C++

```
interface IRobotWindLoadsSimulationParams : IDispatch;
```

C#

```
public interface IRobotWindLoadsSimulationParams;
```

Visual Basic

```
Public Interface IRobotWindLoadsSimulationParams
```

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 14.5.

II.1.34.2.1 IRobotWindLoadsSimulationParams Members

The following tables list the members exposed by IRobotWindLoadsSimulationParams.

Public Fields

	Name	Description
	DeviationPercent (see page 475)	
	DirectionXNEnabled (see page 475)	
	DirectionXNYNEnabled (see page 475)	
	DirectionXNYPEnabled (see page 475)	
	DirectionXPEnabled (see page 476)	
	DirectionXPYNEnabled (see page 476)	
	DirectionXPYPEnabled (see page 476)	
	DirectionYNEnabled (see page 476)	
	DirectionYPEnabled (see page 477)	
	Elements (see page 477)	
	OpeningsClosed (see page 477)	
	TerrainLevel (see page 477)	
	Velocity (see page 478)	

II.1.34.2.2 IRobotWindLoadsSimulationParams Fields

The fields of the IRobotWindLoadsSimulationParams class are listed here.

Public Fields

	Name	Description
	DeviationPercent (see page 475)	
	DirectionXNEnabled (see page 475)	
	DirectionXNYNEnabled (see page 475)	
	DirectionXNYPEnabled (see page 475)	
	DirectionXPEnabled (see page 476)	
	DirectionXPYNEnabled (see page 476)	
	DirectionXPYPEnabled (see page 476)	
	DirectionYNEnabled (see page 476)	
	DirectionYPEnabled (see page 477)	
	Elements (see page 477)	

(c) 2014 Autodesk Inc. All rights reserved

	OpeningsClosed (see page 477)	
	TerrainLevel (see page 477)	
	Velocity (see page 478)	

II.1.34.2.2.1 DeviationPercent

C++

```
HRESULT get_DeviationPercent(double*);
HRESULT put_DeviationPercent(double);
```

C#

```
public double DeviationPercent { get; set; }
```

Visual Basic

```
Public DeviationPercent As double
```

Version

Available since version 14.5.

II.1.34.2.2.2 DirectionXNEnabled

C++

```
HRESULT get_DirectionXNEnabled(VARIANT_BOOL*);
HRESULT put_DirectionXNEnabled(VARIANT_BOOL);
```

C#

```
public bool DirectionXNEnabled { get; set; }
```

Visual Basic

```
Public DirectionXNEnabled As Boolean
```

Version

Available since version 14.5.

II.1.34.2.2.3 DirectionXNYNEnabled

C++

```
HRESULT get_DirectionXNYNEnabled(VARIANT_BOOL*);
HRESULT put_DirectionXNYNEnabled(VARIANT_BOOL);
```

C#

```
public bool DirectionXNYNEnabled { get; set; }
```

Visual Basic

```
Public DirectionXNYNEnabled As Boolean
```

Version

Available since version 14.5.

II.1.34.2.2.4 DirectionXNYPEnabled

C++

```
HRESULT get_DirectionXNYPEnabled(VARIANT_BOOL*);
HRESULT put_DirectionXNYPEnabled(VARIANT_BOOL);
```

C#

```
public bool DirectionXNYPEnabled { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public DirectionXNYPEnabled As Boolean
```

Version

Available since version 14.5.

II.1.34.2.2.5 DirectionXPEnabled

C++

```
HRESULT get_DirectionXPEnabled(VARIANT_BOOL*);
HRESULT put_DirectionXPEnabled(VARIANT_BOOL);
```

C#

```
public bool DirectionXPEnabled { get; set; }
```

Visual Basic

```
Public DirectionXPEnabled As Boolean
```

Version

Available since version 14.5.

II.1.34.2.2.6 DirectionXPYNEnabled

C++

```
HRESULT get_DirectionXPYNEnabled(VARIANT_BOOL*);
HRESULT put_DirectionXPYNEnabled(VARIANT_BOOL);
```

C#

```
public bool DirectionXPYNEnabled { get; set; }
```

Visual Basic

```
Public DirectionXPYNEnabled As Boolean
```

Version

Available since version 14.5.

II.1.34.2.2.7 DirectionXPYPEnabled

C++

```
HRESULT get_DirectionXPYPEnabled(VARIANT_BOOL*);
HRESULT put_DirectionXPYPEnabled(VARIANT_BOOL);
```

C#

```
public bool DirectionXPYPEnabled { get; set; }
```

Visual Basic

```
Public DirectionXPYPEnabled As Boolean
```

Version

Available since version 14.5.

II.1.34.2.2.8 DirectionYNEnabled

C++

```
HRESULT get_DirectionYNEnabled(VARIANT_BOOL*);
HRESULT put_DirectionYNEnabled(VARIANT_BOOL);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public bool DirectionYNEnabled { get; set; }
```

Visual Basic

```
Public DirectionYNEnabled As Boolean
```

Version

Available since version 14.5.

II.1.34.2.2.9 DirectionYPEnabled

C++

```
HRESULT get_DirectionYPEnabled(VARIANT_BOOL*);
HRESULT put_DirectionYPEnabled(VARIANT_BOOL);
```

C#

```
public bool DirectionYPEnabled { get; set; }
```

Visual Basic

```
Public DirectionYPEnabled As Boolean
```

Version

Available since version 14.5.

II.1.34.2.2.10 Elements

C++

```
HRESULT get_Elements(IRobotSelection **);
```

C#

```
public IRobotSelection * Elements { get; }
```

Visual Basic

```
Public ReadOnly Elements As IRobotSelection *
```

Version

Available since version 14.5.

II.1.34.2.2.11 OpeningsClosed

C++

```
HRESULT get_OpeningsClosed(VARIANT_BOOL*);
HRESULT put_OpeningsClosed(VARIANT_BOOL);
```

C#

```
public bool OpeningsClosed { get; set; }
```

Visual Basic

```
Public OpeningsClosed As Boolean
```

Version

Available since version 14.5.

II.1.34.2.2.12 TerrainLevel

C++

```
HRESULT get_TerrainLevel(double*);
HRESULT put_TerrainLevel(double);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double TerrainLevel { get; set; }
```

Visual Basic

```
Public TerrainLevel As double
```

Version

Available since version 14.5.

II.1.34.2.2.13 Velocity

C++

```
HRESULT get_Velocity( double*);
HRESULT put_Velocity( double);
```

C#

```
public  double Velocity { get; set; }
```

Visual Basic

```
Public Velocity As  double
```

Version

Available since version 14.5.

II.2 Nodes

A structure node is described by means of the RobotNode type. All the nodes are managed by the node server of the RobotNodeServer type. .

Interfaces

	Name	Description
	IRobotNode (see page 526)	Each structure node is represented in the Robot Object Model by means of the RobotNode interface. .
	IRobotNodeServer (see page 531)	Server of nodes manages all the structure nodes .

II.2.1 Complex attributes of a node

Robot Object Model provides access to structure node attributes by means of a uniform mechanism of labels. .

Enumerations

	Name	Description
	IRobotNodeSupportFixingDirection (see page 514)	Identifiers of directions that may be blocked out or released in a support. .
	IRobotNodeSupportOneDirectionFixingType (see page 514)	Identifiers of the manner of unilateral blocking of the indicated direction for supports.
	IRobotAdvancedSupportType (see page 517)	
	IRobotSupportEquivalentElasticityType (see page 522)	
	IRobotSupportColumnFixingType (see page 526)	

(c) 2014 Autodesk Inc. All rights reserved

Interfaces

	Name	Description
	IRobotNodeSupportData (see page 503)	Support parameters are described in the Robot Object Model by means of a set of values defined as RobotNodeSupportData. An object of this type contains data with the labels of the ILT_NODE_SUPPORT type. .
	IRobotEmitter (see page 515)	Emitter definition.
	IRobotAdvancedSupportData (see page 517)	
	IRobotSupportEquivalentElasticity (see page 520)	Definition of equivalent elasticity of a support.
	IRobotSupportEquivalentWallElasticity (see page 522)	
	IRobotSupportEquivalentColumnElasticity (see page 524)	

II.2.1.1 Rigid links

Available since version 2.5.

Interfaces

	Name	Description
	IRobotNodeRigidLinkData (see page 479)	Rigid link parameters for nodes.
	IRobotNodeRigidLinkDef (see page 482)	Definition of a node rigid link.
	IRobotNodeRigidLinkServer (see page 484)	Server allowing definition of rigid links for nodes. Moreover, the server provides access to the list of all the rigid links defined. .

II.2.1.1.1 IRobotNodeRigidLinkData

Class Hierarchy

RobotOM.IRobotNodeRigidLinkData

C++

```
interface IRobotNodeRigidLinkData : IDispatch;
```

C#

```
public interface IRobotNodeRigidLinkData;
```

Visual Basic

```
Public Interface IRobotNodeRigidLinkData
```

Description

Rigid link parameters for nodes.

Version

Available since version 2.5.

II.2.1.1.1.1 IRobotNodeRigidLinkData Members

The following tables list the members exposed by IRobotNodeRigidLinkData.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	RX (see page 480)	Blocked displacement in the RX direction.
	RY (see page 480)	Blocked displacement in the RY direction.
	RZ (see page 481)	Blocked displacement in the RZ direction.
	UX (see page 481)	Blocked displacement in the UX direction.
	UY (see page 481)	Blocked displacement in the UY direction.
	UZ (see page 482)	Blocked displacement in the UZ direction.

II.2.1.1.1.2 IRobotNodeRigidLinkData Fields

The fields of the IRobotNodeRigidLinkData class are listed here.

Public Fields

	Name	Description
	RX (see page 480)	Blocked displacement in the RX direction.
	RY (see page 480)	Blocked displacement in the RY direction.
	RZ (see page 481)	Blocked displacement in the RZ direction.
	UX (see page 481)	Blocked displacement in the UX direction.
	UY (see page 481)	Blocked displacement in the UY direction.
	UZ (see page 482)	Blocked displacement in the UZ direction.

II.2.1.1.1.2.1 RX

C++

```
HRESULT get_RX(VARIANT_BOOL*);
HRESULT put_RX(VARIANT_BOOL);
```

C#

```
public bool RX { get; set; }
```

Visual Basic

```
Public RX As Boolean
```

Description

Blocked displacement in the RX direction.

Version

Available since version 2.5.

II.2.1.1.1.2.2 RY

C++

```
HRESULT get_RY(VARIANT_BOOL*);
HRESULT put_RY(VARIANT_BOOL);
```

C#

```
public bool RY { get; set; }
```

Visual Basic

```
Public RY As Boolean
```

Description

Blocked displacement in the RY direction.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 2.5.

II.2.1.1.1.2.3 RZ

C++

```
HRESULT get_RZ(VARIANT_BOOL*);
HRESULT put_RZ(VARIANT_BOOL);
```

C#

```
public bool RZ { get; set; }
```

Visual Basic

```
Public RZ As Boolean
```

Description

Blocked displacement in the RZ direction.

Version

Available since version 2.5.

II.2.1.1.1.2.4 UX

C++

```
HRESULT get_UX(VARIANT_BOOL*);
HRESULT put_UX(VARIANT_BOOL);
```

C#

```
public bool UX { get; set; }
```

Visual Basic

```
Public UX As Boolean
```

Description

Blocked displacement in the UX direction.

Version

Available since version 2.5.

II.2.1.1.1.2.5 UY

C++

```
HRESULT get_UY(VARIANT_BOOL*);
HRESULT put_UY(VARIANT_BOOL);
```

C#

```
public bool UY { get; set; }
```

Visual Basic

```
Public UY As Boolean
```

Description

Blocked displacement in the UY direction.

Version

Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

II.2.1.1.1.2.6 UZ

C++

```
HRESULT get_UZ(VARIANT_BOOL*);
HRESULT put_UZ(VARIANT_BOOL);
```

C#

```
public bool UZ { get; set; }
```

Visual Basic

```
Public UZ As Boolean
```

Description

Blocked displacement in the UZ direction.

Version

Available since version 2.5.

II.2.1.1.2 IRobotNodeRigidLinkDef

Class Hierarchy

RobotOM.IRobotNodeRigidLinkDef

C++

```
interface IRobotNodeRigidLinkDef : IDispatch;
```

C#

```
public interface IRobotNodeRigidLinkDef;
```

Visual Basic

```
Public Interface IRobotNodeRigidLinkDef
```

Description

Definition of a node rigid link.

Version

Available since version 2.5.

II.2.1.1.2.1 IRobotNodeRigidLinkDef Members

The following tables list the members exposed by IRobotNodeRigidLinkDef.

Public Fields

	Name	Description
	LabelName (see page 483)	Name of the label storing data for the link .
	Master (see page 483)	Number of master node.
	Slaves (see page 483)	Selection of slave nodes.

II.2.1.1.2.2 IRobotNodeRigidLinkDef Fields

The fields of the IRobotNodeRigidLinkDef class are listed here.

Public Fields

	Name	Description
	LabelName (see page 483)	Name of the label storing data for the link .
	Master (see page 483)	Number of master node.

(c) 2014 Autodesk Inc. All rights reserved

	Slaves (see page 483)	Selection of slave nodes.

II.2.1.1.2.2.1 LabelName

C++

```
HRESULT get_LabelName(BSTR*);
HRESULT put_LabelName(BSTR);
```

C#

```
public String LabelName { get; set; }
```

Visual Basic

```
Public LabelName As String
```

Description

Name of the label storing data for the link .

Version

Available since version 2.5.

II.2.1.1.2.2.2 Master

C++

```
HRESULT get_Master(long*);
HRESULT put_Master(long);
```

C#

```
public long Master { get; set; }
```

Visual Basic

```
Public Master As long
```

Description

Number of master node.

Version

Available since version 2.5.

II.2.1.1.2.2.3 Slaves

C++

```
HRESULT get_Slaves(BSTR*);
HRESULT put_Slaves(BSTR);
```

C#

```
public String Slaves { get; set; }
```

Visual Basic

```
Public Slaves As String
```

Description

Selection of slave nodes.

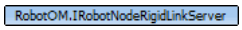
Version

Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

II.2.1.1.3 IRobotNodeRigidLinkServer

Class Hierarchy



C++

```
interface IRobotNodeRigidLinkServer : IDispatch;
```

C#

```
public interface IRobotNodeRigidLinkServer;
```

Visual Basic

```
Public Interface IRobotNodeRigidLinkServer
```

Description

Server allowing definition of rigid links for nodes. Moreover, the server provides access to the list of all the rigid links defined. .

Version

Available since version 2.5.

II.2.1.1.3.1 IRobotNodeRigidLinkServer Members

The following tables list the members exposed by IRobotNodeRigidLinkServer.

Public Fields

	Name	Description
	Count (see page 485)	Number of defined rigid links.

Public Methods

	Name	Description
	Find (see page 486)	Function locates the rigid link in which the specified node occurs (either as a master node or slave node) and returns index of the object describing this link. If no link has been defined for the determined node, then zero value is returned. .
	FindLabel (see page 486)	Function locates the rigid link described by the label of the specified name and returns first index of the object describing this link beginning with the determined start index. .
	FindMaster (see page 486)	Function locates a rigid link defined for the specified master node and returns index of the object describing this link. If no connection has been defined for the indicated node, then zero value is returned. .
	FindSlave (see page 487)	Function locates the rigid link, in which the node specified occurs as a slave node and returns index of the object describing this link. If no link has been defined for the determined node, then zero value is returned.
	Get (see page 487)	Function returns object defining the selected rigid link. By activating this function with successive indexes from 1 to Count (see page 485), the user may obtain information about all the rigid links defined. .
	GetLabel (see page 487)	Function returns label storing parameters of the selected rigid link.
	Remove (see page 488)	Function deletes the rigid link for which definition parameters are described by the object with the indicated index on the list of rigid links defined. .
	RemoveSlave (see page 488)	Function deletes the specified slave node from the definition of the link which includes the node.
	Set (see page 488)	Function defines a rigid link for the indicated nodes. .

(c) 2014 Autodesk Inc. All rights reserved

II.2.1.1.3.2 IRobotNodeRigidLinkServer Fields

The fields of the IRobotNodeRigidLinkServer class are listed here.

Public Fields

	Name	Description
	Count (see page 485)	Number of defined rigid links.

II.2.1.1.3.2.1 Count

C++

```
HRESULT get_Count(long*);
```

C#

```
public long Count { get; }
```

Visual Basic

```
Public ReadOnly Count As long
```

Description

Number of defined rigid links.

Version

Available since version 2.5.

II.2.1.1.3.3 IRobotNodeRigidLinkServer Methods

The methods of the IRobotNodeRigidLinkServer class are listed here.

Public Methods

	Name	Description
	Find (see page 486)	Function locates the rigid link in which the specified node occurs (either as a master node or slave node) and returns index of the object describing this link. If no link has been defined for the determined node, then zero value is returned. .
	FindLabel (see page 486)	Function locates the rigid link described by the label of the specified name and returns first index of the object describing this link beginning with the determined start index. .
	FindMaster (see page 486)	Function locates a rigid link defined for the specified master node and returns index of the object describing this link. If no connection has been defined for the indicated node, then zero value is returned. .
	FindSlave (see page 487)	Function locates the rigid link, in which the node specified occurs as a slave node and returns index of the object describing this link. If no link has been defined for the determined node, then zero value is returned.
	Get (see page 487)	Function returns object defining the selected rigid link. By activating this function with successive indexes from 1 to Count (see page 485), the user may obtain information about all the rigid links defined. .
	GetLabel (see page 487)	Function returns label storing parameters of the selected rigid link.
	Remove (see page 488)	Function deletes the rigid link for which definition parameters are described by the object with the indicated index on the list of rigid links defined. .
	RemoveSlave (see page 488)	Function deletes the specified slave node from the definition of the link which includes the node.
	Set (see page 488)	Function defines a rigid link for the indicated nodes. .

(c) 2014 Autodesk Inc. All rights reserved

II.2.1.1.3.3.1 Find

C++

```
HRESULT Find(long _node_num, long* ret);
```

C#

```
public long Find(long _node_num);
```

Visual Basic

```
Public Function Find(_node_num As long) As long
```

Description

Function locates the rigid link in which the specified node occurs (either as a master node or slave node) and returns index of the object describing this link. If no link has been defined for the determined node, then zero value is returned. .

Version

Available since version 2.5.

II.2.1.1.3.3.2 FindLabel

C++

```
HRESULT FindLabel(BSTR _label_name, long _start_def_idx = 1, long* ret);
```

C#

```
public long FindLabel(String _label_name, long _start_def_idx = 1);
```

Visual Basic

```
Public Function FindLabel(_label_name As String, Optional _start_def_idx As long = 1) As
long
```

Description

Function locates the rigid link described by the label of the specified name and returns first index of the object describing this link beginning with the determined start index. .

Version

Available since version 2.5.

II.2.1.1.3.3.3 FindMaster

C++

```
HRESULT FindMaster(long _master, long* ret);
```

C#

```
public long FindMaster(long _master);
```

Visual Basic

```
Public Function FindMaster(_master As long) As long
```

Description

Function locates a rigid link defined for the specified master node and returns index of the object describing this link. If no connection has been defined for the indicated node, then zero value is returned. .

Version

Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

II.2.1.1.3.3.4 FindSlave

C++

```
HRESULT FindSlave(long _slave_node, long* ret);
```

C#

```
public long FindSlave(long _slave_node);
```

Visual Basic

```
Public Function FindSlave(_slave_node As long) As long
```

Description

Function locates the rigid link, in which the node specified occurs as a slave node and returns index of the object describing this link. If no link has been defined for the determined node, then zero value is returned.

Version

Available since version 2.5.

II.2.1.1.3.3.5 Get

C++

```
HRESULT Get(long _def_idx, IRobotNodeRigidLinkDef** ret);
```

C#

```
public IRobotNodeRigidLinkDef Get(long _def_idx);
```

Visual Basic

```
Public Function Get(_def_idx As long) As IRobotNodeRigidLinkDef
```

Description

Function returns object defining the selected rigid link. By activating this function with successive indexes from 1 to Count (see page 485), the user may obtain information about all the rigid links defined. .

Version

Available since version 2.5.

II.2.1.1.3.3.6 GetLabel

C++

```
HRESULT GetLabel(long _def_idx, IRobotLabel** ret);
```

C#

```
public IRobotLabel GetLabel(long _def_idx);
```

Visual Basic

```
Public Function GetLabel(_def_idx As long) As IRobotLabel
```

Description

Function returns label storing parameters of the selected rigid link.

Version

Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

II.2.1.1.3.3.7 Remove

C++

```
HRESULT Remove(long _def_idx);
```

C#

```
public void Remove(long _def_idx);
```

Visual Basic

```
Public Sub Remove(_def_idx As long)
```

Description

Function deletes the rigid link for which definition parameters are described by the object with the indicated index on the list of rigid links defined. .

Version

Available since version 2.5.

II.2.1.1.3.3.8 RemoveSlave

C++

```
HRESULT RemoveSlave(long _slave);
```

C#

```
public void RemoveSlave(long _slave);
```

Visual Basic

```
Public Sub RemoveSlave(_slave As long)
```

Description

Function deletes the specified slave node from the definition of the link which includes the node.

Version

Available since version 2.5.

II.2.1.1.3.3.9 Set

C++

```
HRESULT Set(long _master, BSTR _slaves, BSTR _label_name, VARIANT_BOOL* ret);
```

C#

```
public bool Set(long _master, String _slaves, String _label_name);
```

Visual Basic

```
Public Function Set(_master As long, _slaves As String, _label_name As String) As Boolean
```

Description

Function defines a rigid link for the indicated nodes. .

Version

Available since version 2.5.

II.2.1.2 Compatible nodes

Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

Interfaces

	Name	Description
	IRobotNodeCompatibilityData (see page 489)	Compatible node.
	IRobotNodeCompatibilityDef (see page 497)	Definition of compatible node.
	IRobotNodeCompatibilityServer (see page 499)	Server allowing definition of compatible nodes.

II.2.1.2.1 IRobotNodeCompatibilityData

Class Hierarchy

RobotOM.IRobotNodeCompatibilityData

C++

```
interface IRobotNodeCompatibilityData : IDispatch;
```

C#

```
public interface IRobotNodeCompatibilityData;
```

Visual Basic

```
Public Interface IRobotNodeCompatibilityData
```

Description

Compatible node.

Version

Available since version 2.5.

II.2.1.2.1.1 IRobotNodeCompatibilityData Members

The following tables list the members exposed by IRobotNodeCompatibilityData.

Public Fields

	Name	Description
	Alpha (see page 490)	Value of the compatibility angle (Z axis).
	AX (see page 491)	Damping coeffcient (X axis).
	AY (see page 491)	Damping coeffcient (Y axis).
	AZ (see page 491)	Damping coeffcient (Z axis).
	Beta (see page 492)	Value of the compatibility angle (Y axis).
	BX (see page 492)	
	BY (see page 492)	
	BZ (see page 492)	
	Gamma (see page 493)	Value of the compatibility angle (X axis).
	HX (see page 493)	
	HY (see page 493)	
	HZ (see page 494)	
	KX (see page 494)	Elastic coefficient.
	KY (see page 494)	Elastic coefficient.
	KZ (see page 494)	Elastic coefficient.
	NonlinearModel (see page 495)	Non-linearity support.
	RX (see page 495)	Blocked direction (rotation about X axis).

(c) 2014 Autodesk Inc. All rights reserved

	RY (see page 495)	Blocked direction (rotation about Y axis).
	RZ (see page 496)	Blocked direction (rotation about Z axis).
	UX (see page 496)	Blocked direction (X axis).
	UY (see page 496)	Blocked direction (Y axis).
	UZ (see page 497)	Blocked direction (Z axis).

II.2.1.2.1.2 IRobotNodeCompatibilityData Fields

The fields of the IRobotNodeCompatibilityData class are listed here.

Public Fields

	Name	Description
	Alpha (see page 490)	Value of the compatibility angle (Z axis).
	AX (see page 491)	Damping coeffcient (X axis).
	AY (see page 491)	Damping coeffcient (Y axis).
	AZ (see page 491)	Damping coeffcient (Z axis).
	Beta (see page 492)	Value of the compatibility angle (Y axis).
	BX (see page 492)	
	BY (see page 492)	
	BZ (see page 492)	
	Gamma (see page 493)	Value of the compatibility angle (X axis).
	HX (see page 493)	
	HY (see page 493)	
	HZ (see page 494)	
	KX (see page 494)	Elastic coefficient.
	KY (see page 494)	Elastic coefficient.
	KZ (see page 494)	Elastic coefficient.
	NonlinearModel (see page 495)	Non-linearity support.
	RX (see page 495)	Blocked direction (rotation about X axis).
	RY (see page 495)	Blocked direction (rotation about Y axis).
	RZ (see page 496)	Blocked direction (rotation about Z axis).
	UX (see page 496)	Blocked direction (X axis).
	UY (see page 496)	Blocked direction (Y axis).
	UZ (see page 497)	Blocked direction (Z axis).

II.2.1.2.1.2.1 Alpha

C++

```
HRESULT get_Alpha(double*);
HRESULT put_Alpha(double);
```

C#

```
public double Alpha { get; set; }
```

Visual Basic

```
Public Alpha As double
```

Description

Value of the compatibility angle (Z axis).

Version

Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

II.2.1.2.1.2.2 AX

C++

```
HRESULT get_AX(double*);
HRESULT put_AX(double);
```

C#

```
public double AX { get; set; }
```

Visual Basic

```
Public AX As double
```

Description

Damping coeffcient (X axis).

Version

Available since version 2.5.

II.2.1.2.1.2.3 AY

C++

```
HRESULT get_AY(double*);
HRESULT put_AY(double);
```

C#

```
public double AY { get; set; }
```

Visual Basic

```
Public AY As double
```

Description

Damping coeffcient (Y axis).

Version

Available since version 2.5.

II.2.1.2.1.2.4 AZ

C++

```
HRESULT get_AZ(double*);
HRESULT put_AZ(double);
```

C#

```
public double AZ { get; set; }
```

Visual Basic

```
Public AZ As double
```

Description

Damping coeffcient (Z axis).

Version

Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

II.2.1.2.1.2.5 Beta

C++

```
HRESULT get_Beta(double*);
HRESULT put_Beta(double);
```

C#

```
public double Beta { get; set; }
```

Visual Basic

```
Public Beta As double
```

Description

Value of the compatibility angle (Y axis).

Version

Available since version 2.5.

II.2.1.2.1.2.6 BX

C++

```
HRESULT get_BX(double*);
HRESULT put_BX(double);
```

C#

```
public double BX { get; set; }
```

Visual Basic

```
Public BX As double
```

Version

Available since version 2.5.

II.2.1.2.1.2.7 BY

C++

```
HRESULT get_BY(double*);
HRESULT put_BY(double);
```

C#

```
public double BY { get; set; }
```

Visual Basic

```
Public BY As double
```

Version

Available since version 2.5.

II.2.1.2.1.2.8 BZ

C++

```
HRESULT get_BZ(double*);
HRESULT put_BZ(double);
```

C#

```
public double BZ { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public BZ As double
```

Version

Available since version 2.5.

II.2.1.2.1.2.9 Gamma

C++

```
HRESULT get_Gamma(double*);
HRESULT put_Gamma(double);
```

C#

```
public double Gamma { get; set; }
```

Visual Basic

```
Public Gamma As double
```

Description

Value of the compatibility angle (X axis).

Version

Available since version 2.5.

II.2.1.2.1.2.10 HX

C++

```
HRESULT get_HX(double*);
HRESULT put_HX(double);
```

C#

```
public double HX { get; set; }
```

Visual Basic

```
Public HX As double
```

Version

Available since version 2.5.

II.2.1.2.1.2.11 HY

C++

```
HRESULT get_HY(double*);
HRESULT put_HY(double);
```

C#

```
public double HY { get; set; }
```

Visual Basic

```
Public HY As double
```

Version

Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

II.2.1.2.1.2.12 HZ

C++

```
HRESULT get_HZ(double*);
HRESULT put_HZ(double);
```

C#

```
public double HZ { get; set; }
```

Visual Basic

```
Public HZ As double
```

Version

Available since version 2.5.

II.2.1.2.1.2.13 KX

C++

```
HRESULT get_KX(double*);
HRESULT put_KX(double);
```

C#

```
public double KX { get; set; }
```

Visual Basic

```
Public KX As double
```

Description

Elastic coefficient.

Version

Available since version 2.5.

II.2.1.2.1.2.14 KY

C++

```
HRESULT get_KY(double*);
HRESULT put_KY(double);
```

C#

```
public double KY { get; set; }
```

Visual Basic

```
Public KY As double
```

Description

Elastic coefficient.

Version

Available since version 2.5.

II.2.1.2.1.2.15 KZ

C++

```
HRESULT get_KZ(double*);
HRESULT put_KZ(double);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double KZ { get; set; }
```

Visual Basic

```
Public KZ As double
```

Description

Elastic coefficient.

Version

Available since version 2.5.

II.2.1.2.1.2.16 NonlinearModel

C++

```
HRESULT get_NonlinearModel(IRobotNonlinearLinkMngr**);
```

C#

```
public IRobotNonlinearLinkMngr NonlinearModel { get; }
```

Visual Basic

```
Public ReadOnly NonlinearModel As IRobotNonlinearLinkMngr
```

Description

Non-linearity support.

Version

Available since version 3.

II.2.1.2.1.2.17 RX

C++

```
HRESULT get_RX(VARIANT_BOOL*);
HRESULT put_RX(VARIANT_BOOL);
```

C#

```
public bool RX { get; set; }
```

Visual Basic

```
Public RX As Boolean
```

Description

Blocked direction (rotation about X axis).

Version

Available since version 2.5.

II.2.1.2.1.2.18 RY

C++

```
HRESULT get_RY(VARIANT_BOOL*);
HRESULT put_RY(VARIANT_BOOL);
```

C#

```
public bool RY { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public RY As Boolean
```

Description

Blocked direction (rotation about Y axis).

Version

Available since version 2.5.

II.2.1.2.1.2.19 RZ

C++

```
HRESULT get_RZ(VARIANT_BOOL*);
HRESULT put_RZ(VARIANT_BOOL);
```

C#

```
public bool RZ { get; set; }
```

Visual Basic

```
Public RZ As Boolean
```

Description

Blocked direction (rotation about Z axis).

Version

Available since version 2.5.

II.2.1.2.1.2.20 UX

C++

```
HRESULT get_UX(VARIANT_BOOL*);
HRESULT put_UX(VARIANT_BOOL);
```

C#

```
public bool UX { get; set; }
```

Visual Basic

```
Public UX As Boolean
```

Description

Blocked direction (X axis).

Version

Available since version 2.5.

II.2.1.2.1.2.21 UY

C++

```
HRESULT get_UY(VARIANT_BOOL*);
HRESULT put_UY(VARIANT_BOOL);
```

C#

```
public bool UY { get; set; }
```

Visual Basic

```
Public UY As Boolean
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Blocked direction (Y axis).

Version

Available since version 2.5.

II.2.1.2.1.2.22 UZ

C++

```
HRESULT get_UZ(VARIANT_BOOL*);
HRESULT put_UZ(VARIANT_BOOL);
```

C#

```
public bool UZ { get; set; }
```

Visual Basic

```
Public UZ As Boolean
```

Description

Blocked direction (Z axis).

Version

Available since version 2.5.

II.2.1.2.2 IRobotNodeCompatibilityDef

Class Hierarchy

RobotOM.IRobotNodeCompatibilityDef

C++

```
interface IRobotNodeCompatibilityDef : IDispatch;
```

C#

```
public interface IRobotNodeCompatibilityDef;
```

Visual Basic

```
Public Interface IRobotNodeCompatibilityDef
```

Description

Definition of compatible node.

Version

Available since version 2.5.

II.2.1.2.2.1 IRobotNodeCompatibilityDef Members

The following tables list the members exposed by IRobotNodeCompatibilityDef.

Public Fields

	Name	Description
	Bars (see page 498)	List of bars for which the compatibility model is defined .
	Compatible (see page 498)	Number of compatible node.
	LabelName (see page 498)	Name of the label storing data for a compatible node.
	Main (see page 499)	Number of master node.

(c) 2014 Autodesk Inc. All rights reserved

II.2.1.2.2.2 IRobotNodeCompatibilityDef Fields

The fields of the IRobotNodeCompatibilityDef class are listed here.

Public Fields

	Name	Description
	Bars (see page 498)	List of bars for which the compatibility model is defined .
	Compatible (see page 498)	Number of compatible node.
	LabelName (see page 498)	Name of the label storing data for a compatible node.
	Main (see page 499)	Number of master node.

II.2.1.2.2.2.1 Bars

C++

```
HRESULT get_Bars(String*);
HRESULT put_Bars(String);
```

C#

```
public String Bars { get; set; }
```

Visual Basic

```
Public Bars As String
```

Description

List of bars for which the compatibility model is defined .

Version

Available since version 2.5.

II.2.1.2.2.2.2 Compatible

C++

```
HRESULT get_Compatible(long*);
HRESULT put_Compatible(long);
```

C#

```
public long Compatible { get; set; }
```

Visual Basic

```
Public Compatible As long
```

Description

Number of compatible node.

Version

Available since version 2.5.

II.2.1.2.2.2.3 LabelName

C++

```
HRESULT get_LabelName(BSTR*);
HRESULT put_LabelName(BSTR);
```

C#

```
public String LabelName { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public LabelName As String
```

Description

Name of the label storing data for a compatible node.

Version

Available since version 2.5.

II.2.1.2.2.2.4 Main

C++

```
HRESULT get_Main(long*);
HRESULT put_Main(long);
```

C#

```
public long Main { get; set; }
```

Visual Basic

```
Public Main As long
```

Description

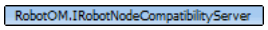
Number of master node.

Version

Available since version 2.5.

II.2.1.2.3 IRobotNodeCompatibilityServer

Class Hierarchy



C++

```
interface IRobotNodeCompatibilityServer : IDispatch;
```

C#

```
public interface IRobotNodeCompatibilityServer;
```

Visual Basic

```
Public Interface IRobotNodeCompatibilityServer
```

Description

Server allowing definition of compatible nodes.

Version

Available since version 2.5.

II.2.1.2.3.1 IRobotNodeCompatibilityServer Members

The following tables list the members exposed by IRobotNodeCompatibilityServer.

Public Fields

	Name	Description
	Count (see page 500)	Number of compatible nodes defined.

(c) 2014 Autodesk Inc. All rights reserved

Public Methods

	Name	Description
	Find (see page 501)	Function returns index of the first located compatible node definition in which the specified node occurs either as a master node or compatible node.
	FindCompatible (see page 501)	Function returns definition index of the compatible node of the indicated user number. .
	FindLabel (see page 501)	Function returns index of the first located compatible node definition with the specified name.
	FindMain (see page 502)	Function returns index of the first located definition of a compatible node in the specified master node beginning with the start index. .
	Get (see page 502)	Function returns object containing compatible node parameters. By activating this function with indexes from 1 to Count (see page 500), the user may obtain information about all the compatible nodes defined. .
	GetLabel (see page 502)	Function returns label describing parameters of the determined compatible node. .
	Remove (see page 503)	Function deletes the compatible node whose definition method describes the object of the specified index on the list of all compatible nodes. .
	Set (see page 503)	Function defines a compatible node for the selected bars at the indicated node. The selected bars must converge at the given node. .

II.2.1.2.3.2 IRobotNodeCompatibilityServer Fields

The fields of the IRobotNodeCompatibilityServer class are listed here.

Public Fields

	Name	Description
	Count (see page 500)	Number of compatible nodes defined.

II.2.1.2.3.2.1 Count

C++

```
HRESULT get_Count(long*);
HRESULT put_Count(long);
```

C#

```
public long Count { get; set; }
```

Visual Basic

```
Public Count As long
```

Description

Number of compatible nodes defined.

Version

Available since version 2.5.

II.2.1.2.3.3 IRobotNodeCompatibilityServer Methods

The methods of the IRobotNodeCompatibilityServer class are listed here.

Public Methods

	Name	Description
	Find (see page 501)	Function returns index of the first located compatible node definition in which the specified node occurs either as a master node or compatible node.

(c) 2014 Autodesk Inc. All rights reserved

	FindCompatible (see page 501)	Function returns definition index of the compatible node of the indicated user number. .
	FindLabel (see page 501)	Function returns index of the first located compatible node definition with the specified name.
	FindMain (see page 502)	Function returns index of the first located definition of a compatible node in the specified master node beginning with the start index. .
	Get (see page 502)	Function returns object containing compatible node parameters. By activating this function with indexes from 1 to Count (see page 500), the user may obtain information about all the compatible nodes defined. .
	GetLabel (see page 502)	Function returns label describing parameters of the determined compatible node. .
	Remove (see page 503)	Function deletes the compatible node whose definition method describes the object of the specified index on the list of all compatible nodes. .
	Set (see page 503)	Function defines a compatible node for the selected bars at the indicated node. The selected bars must converge at the given node. .

II.2.1.2.3.3.1 Find

C++

```
HRESULT Find(long _node, long _start_def_idx = 1, long* ret);
```

C#

```
public long Find(long _node, long _start_def_idx = 1);
```

Visual Basic

```
Public Function Find(_node As long, Optional _start_def_idx As long = 1) As long
```

Description

Function returns index of the first located compatible node definition in which the specified node occurs either as a master node or compatible node.

Version

Available since version 2.5.

II.2.1.2.3.3.2 FindCompatible

C++

```
HRESULT FindCompatible(long _compatible_node, long* ret);
```

C#

```
public long FindCompatible(long _compatible_node);
```

Visual Basic

```
Public Function FindCompatible(_compatible_node As long) As long
```

Description

Function returns definition index of the compatible node of the indicated user number. .

Version

Available since version 2.5.

II.2.1.2.3.3.3 FindLabel

C++

```
HRESULT FindLabel(BSTR _label_name, long _start_def_idx = 1, long* ret);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public long FindLabel(String _label_name, long _start_def_idx = 1);
```

Visual Basic

```
Public Function FindLabel(_label_name As String, Optional _start_def_idx As long = 1) As
long
```

Description

Function returns index of the first located compatible node definition with the specified name.

Version

Available since version 2.5.

II.2.1.2.3.3.4 FindMain

C++

```
HRESULT FindMain(long _main_node, long _start_def_idx = 1, long* ret);
```

C#

```
public long FindMain(long _main_node, long _start_def_idx = 1);
```

Visual Basic

```
Public Function FindMain(_main_node As long, Optional _start_def_idx As long = 1) As long
```

Description

Function returns index of the first located definition of a compatible node in the specified master node beginning with the start index. .

Version

Available since version 2.5.

II.2.1.2.3.3.5 Get

C++

```
HRESULT Get(long _def_idx, IRobotNodeCompatibilityDef** ret);
```

C#

```
public IRobotNodeCompatibilityDef Get(long _def_idx);
```

Visual Basic

```
Public Function Get(_def_idx As long) As IRobotNodeCompatibilityDef
```

Description

Function returns object containing compatible node parameters. By activating this function with indexes from 1 to Count (see page 500), the user may obtain information about all the compatible nodes defined. .

Version

Available since version 2.5.

II.2.1.2.3.3.6 GetLabel

C++

```
HRESULT GetLabel(long _def_idx, IRobotLabel** ret);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRobotLabel GetLabel(long _def_idx);
```

Visual Basic

```
Public Function GetLabel(_def_idx As long) As IRobotLabel
```

Description

Function returns label describing parameters of the determined compatible node. .

Version

Available since version 2.5.

II.2.1.2.3.3.7 Remove

C++

```
HRESULT Remove(long _def_idx);
```

C#

```
public void Remove(long _def_idx);
```

Visual Basic

```
Public Sub Remove(_def_idx As long)
```

Description

Function deletes the compatible node whose definition method describes the object of the specified index on the list of all compatible nodes. .

Version

Available since version 2.5.

II.2.1.2.3.3.8 Set

C++

```
HRESULT Set(long _node, BSTR _bar_list, BSTR _label_name, VARIANT_BOOL* ret);
```

C#

```
public bool Set(long _node, String _bar_list, String _label_name);
```

Visual Basic

```
Public Function Set(_node As long, _bar_list As String, _label_name As String) As Boolean
```

Description

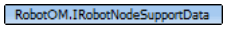
Function defines a compatible node for the selected bars at the indicated node. The selected bars must converge at the given node. .

Version

Available since version 2.5.

II.2.1.3 IRobotNodeSupportData

Class Hierarchy



(c) 2014 Autodesk Inc. All rights reserved

C++

```
interface IRobotNodeSupportData : IDispatch;
```

C#

```
public interface IRobotNodeSupportData;
```

Visual Basic

```
Public Interface IRobotNodeSupportData
```

Description

Support parameters are described in the Robot Object Model by means of a set of values defined as RobotNodeSupportData. An object of this type contains data with the labels of the ILT_NODE_SUPPORT type. .

II.2.1.3.1 IRobotNodeSupportData Members

The following tables list the members exposed by IRobotNodeSupportData.

Public Fields

	Name	Description
	Advanced (see page 506)	
	Alpha (see page 506)	Angle value in radians.
	AX (see page 506)	
	AY (see page 507)	
	AZ (see page 507)	
	Beta (see page 507)	Angle value in radians.
	BX (see page 507)	
	BY (see page 507)	
	BZ (see page 508)	
	ElasticLinear (see page 508)	Linear elasticity distribution; If the flag assumes value different from zero (True), then values of support elastic coefficients will be given per one running meter of length. Value of the elastic coefficient in a node is determined in the following manner: • for each node, the distance between centers of segments between successive nodes is determined • the determined distance is multiplied by the coefficient per one running meter defined by the user • the determined value is defined in the node .
	ElasticSurface (see page 508)	Surface elasticity distribution; If the flag assumes value different from zero (True), then value of support elasticity is given per surface unit and it is assigned to nodes in proportion to the area of elements adjoining each of the support nodes .
	Gamma (see page 509)	Angle value in radians.
	GlobalCoordSystem (see page 509)	Definition of a rigid support in the global / local system.
	HX (see page 509)	
	HY (see page 509)	
	HZ (see page 510)	
	KX (see page 510)	
	KY (see page 510)	
	KZ (see page 510)	
	NonlinearModel (see page 510)	Non-linearity support.

(c) 2014 Autodesk Inc. All rights reserved

	RX (see page 511)	
	RY (see page 511)	
	RZ (see page 511)	
	UX (see page 511)	
	UY (see page 512)	
	UZ (see page 512)	

Public Methods

	Name	Description
	GetAdvanced (see page 512)	Function returns support parameters applied during RC structure analysis. .
	GetOneDir (see page 513)	The function returns the manner of unilateral blocking of the indicated direction for support.
	IsFixed (see page 513)	The function checks if the given direction is blocked.
	SetAdvanced (see page 513)	Function defines support parameters applied during RC structure analysis. .
	SetFixed (see page 513)	The function allows one to block / un-block an indicated direction.
	SetOneDir (see page 514)	The function sets the manner of unilateral blocking of the indicated direction for the support.

II.2.1.3.2 IRobotNodeSupportData Fields

The fields of the IRobotNodeSupportData class are listed here.

Public Fields

	Name	Description
	Advanced (see page 506)	
	Alpha (see page 506)	Angle value in radians.
	AX (see page 506)	
	AY (see page 507)	
	AZ (see page 507)	
	Beta (see page 507)	Angle value in radians.
	BX (see page 507)	
	BY (see page 507)	
	BZ (see page 508)	
	ElasticLinear (see page 508)	Linear elasticity distribution; If the flag assumes value different from zero (True), then values of support elastic coefficients will be given per one running meter of length. Value of the elastic coefficient in a node is determined in the following manner: • for each node, the distance between centers of segments between successive nodes is determined • the determined distance is multiplied by the coefficient per one running meter defined by the user • the determined value is defined in the node .
	ElasticSurface (see page 508)	Surface elasticity distribution; If the flag assumes value different from zero (True), then value of support elasticity is given per surface unit and it is assigned to nodes in proportion to the area of elements adjoining each of the support nodes .
	Gamma (see page 509)	Angle value in radians.
	GlobalCoordSystem (see page 509)	Definition of a rigid support in the global / local system.
	HX (see page 509)	

(c) 2014 Autodesk Inc. All rights reserved

	HY (see page 509)	
	HZ (see page 510)	
	KX (see page 510)	
	KY (see page 510)	
	KZ (see page 510)	
	NonlinearModel (see page 510)	Non-linearity support.
	RX (see page 511)	
	RY (see page 511)	
	RZ (see page 511)	
	UX (see page 511)	
	UY (see page 512)	
	UZ (see page 512)	

II.2.1.3.2.1 Advanced

C++

```
HRESULT get_Advanced(IRobotAdvancedSupportData**);
```

C#

```
public IRobotAdvancedSupportData Advanced { get; }
```

Visual Basic

```
Public ReadOnly Advanced As IRobotAdvancedSupportData
```

Version

Available since version 8.5.

II.2.1.3.2.2 Alpha

C++

```
HRESULT get_Alpha(double*);
HRESULT put_Alpha(double);
```

C#

```
public double Alpha { get; set; }
```

Visual Basic

```
Public Alpha As double
```

Description

Angle value in radians.

II.2.1.3.2.3 AX

C++

```
HRESULT get_AX(double*);
HRESULT put_AX(double);
```

C#

```
public double AX { get; set; }
```

Visual Basic

```
Public AX As double
```

(c) 2014 Autodesk Inc. All rights reserved

II.2.1.3.2.4 AY

C++

```
HRESULT get_AY(double*);
HRESULT put_AY(double);
```

C#

```
public double AY { get; set; }
```

Visual Basic

```
Public AY As double
```

II.2.1.3.2.5 AZ

C++

```
HRESULT get_AZ(double*);
HRESULT put_AZ(double);
```

C#

```
public double AZ { get; set; }
```

Visual Basic

```
Public AZ As double
```

II.2.1.3.2.6 Beta

C++

```
HRESULT get_Beta(double*);
HRESULT put_Beta(double);
```

C#

```
public double Beta { get; set; }
```

Visual Basic

```
Public Beta As double
```

Description

Angle value in radians.

II.2.1.3.2.7 BX

C++

```
HRESULT get_BX(double*);
HRESULT put_BX(double);
```

C#

```
public double BX { get; set; }
```

Visual Basic

```
Public BX As double
```

II.2.1.3.2.8 BY

C++

```
HRESULT get_BY(double*);
HRESULT put_BY(double);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double BY { get; set; }
```

Visual Basic

```
Public BY As double
```

II.2.1.3.2.9 BZ

C++

```
HRESULT get_BZ(double*);
HRESULT put_BZ(double);
```

C#

```
public double BZ { get; set; }
```

Visual Basic

```
Public BZ As double
```

II.2.1.3.2.10 ElasticLinear

C++

```
HRESULT get_ElasticLinear(VARIANT_BOOL*);
HRESULT put_ElasticLinear(VARIANT_BOOL);
```

C#

```
public bool ElasticLinear { get; set; }
```

Visual Basic

```
Public ElasticLinear As Boolean
```

Description

Linear elasticity distribution; If the flag assumes value different from zero (True), then values of support elastic coefficients will be given per one running meter of length. Value of the elastic coefficient in a node is determined in the following manner:

- for each node, the distance between centers of segments between successive nodes is determined
- the determined distance is multiplied by the coefficient per one running meter defined by the user
- the determined value is defined in the node

.

Version

Available since version 2.5.

II.2.1.3.2.11 ElasticSurface

C++

```
HRESULT get_ElasticSurface(VARIANT_BOOL*);
HRESULT put_ElasticSurface(VARIANT_BOOL);
```

C#

```
public bool ElasticSurface { get; set; }
```

Visual Basic

```
Public ElasticSurface As Boolean
```

Description

Surface elasticity distribution; If the flag assumes value different from zero (True), then value of support elasticity is given per surface unit and it is assigned to nodes in proportion to the area of elements adjoining each of the support nodes .

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 2.5.

II.2.1.3.2.12 Gamma

C++

```
HRESULT get_Gamma(double*);
HRESULT put_Gamma(double);
```

C#

```
public double Gamma { get; set; }
```

Visual Basic

```
Public Gamma As double
```

Description

Angle value in radians.

II.2.1.3.2.13 GlobalCoordSystem

C++

```
HRESULT get_GlobalCoordSystem(VARIANT_BOOL*);
HRESULT put_GlobalCoordSystem(VARIANT_BOOL);
```

C#

```
public bool GlobalCoordSystem { get; set; }
```

Visual Basic

```
Public GlobalCoordSystem As Boolean
```

Description

Definition of a rigid support in the global / local system.

Version

Available since version 2.5.

II.2.1.3.2.14 HX

C++

```
HRESULT get_HX(double*);
HRESULT put_HX(double);
```

C#

```
public double HX { get; set; }
```

Visual Basic

```
Public HX As double
```

II.2.1.3.2.15 HY

C++

```
HRESULT get_HY(double*);
HRESULT put_HY(double);
```

C#

```
public double HY { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public HY As double
```

II.2.1.3.2.16 HZ

C++

```
HRESULT get_HZ(double*);
HRESULT put_HZ(double);
```

C#

```
public double HZ { get; set; }
```

Visual Basic

```
Public HZ As double
```

II.2.1.3.2.17 KX

C++

```
HRESULT get_KX(double*);
HRESULT put_KX(double);
```

C#

```
public double KX { get; set; }
```

Visual Basic

```
Public KX As double
```

II.2.1.3.2.18 KY

C++

```
HRESULT get_KY(double*);
HRESULT put_KY(double);
```

C#

```
public double KY { get; set; }
```

Visual Basic

```
Public KY As double
```

II.2.1.3.2.19 KZ

C++

```
HRESULT get_KZ(double*);
HRESULT put_KZ(double);
```

C#

```
public double KZ { get; set; }
```

Visual Basic

```
Public KZ As double
```

II.2.1.3.2.20 NonlinearModel

C++

```
HRESULT get_NonlinearModel(IRobotNonlinearLinkMngr**);
```

C#

```
public IRobotNonlinearLinkMngr NonlinearModel { get; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public ReadOnly NonlinearModel As IRobotNonlinearLinkMngr
```

Description

Non-linearity support.

Version

Available since version 3.

II.2.1.3.2.21 RX

C++

```
HRESULT get_RX(VARIANT_BOOL*);
HRESULT put_RX(VARIANT_BOOL);
```

C#

```
public bool RX { get; set; }
```

Visual Basic

```
Public RX As Boolean
```

II.2.1.3.2.22 RY

C++

```
HRESULT get_RY(VARIANT_BOOL*);
HRESULT put_RY(VARIANT_BOOL);
```

C#

```
public bool RY { get; set; }
```

Visual Basic

```
Public RY As Boolean
```

II.2.1.3.2.23 RZ

C++

```
HRESULT get_RZ(VARIANT_BOOL*);
HRESULT put_RZ(VARIANT_BOOL);
```

C#

```
public bool RZ { get; set; }
```

Visual Basic

```
Public RZ As Boolean
```

II.2.1.3.2.24 UX

C++

```
HRESULT get_UX(VARIANT_BOOL*);
HRESULT put_UX(VARIANT_BOOL);
```

C#

```
public bool UX { get; set; }
```

Visual Basic

```
Public UX As Boolean
```

(c) 2014 Autodesk Inc. All rights reserved

II.2.1.3.2.25 UY

C++

```
HRESULT get_UY(VARIANT_BOOL*);
HRESULT put_UY(VARIANT_BOOL);
```

C#

```
public bool UY { get; set; }
```

Visual Basic

```
Public UY As Boolean
```

II.2.1.3.2.26 UZ

C++

```
HRESULT get_UZ(VARIANT_BOOL*);
HRESULT put_UZ(VARIANT_BOOL);
```

C#

```
public bool UZ { get; set; }
```

Visual Basic

```
Public UZ As Boolean
```

II.2.1.3.3 IRobotNodeSupportData Methods

The methods of the IRobotNodeSupportData class are listed here.

Public Methods

	Name	Description
	GetAdvanced (see page 512)	Function returns support parameters applied during RC structure analysis. .
	GetOneDir (see page 513)	The function returns the manner of unilateral blocking of the indicated direction for support.
	IsFixed (see page 513)	The function checks if the given direction is blocked.
	SetAdvanced (see page 513)	Function defines support parameters applied during RC structure analysis. .
	SetFixed (see page 513)	The function allows one to block / un-block an indicated direction.
	SetOneDir (see page 514)	The function sets the manner of unilateral blocking of the indicated direction for the support.

II.2.1.3.3.1 GetAdvanced

C++

```
HRESULT GetAdvanced(double* _b, double* _h, IRobotAdvancedSupportType* ret);
```

C#

```
public IRobotAdvancedSupportType GetAdvanced(double* _b, double* _h);
```

Visual Basic

```
Public Function GetAdvanced(ByRef _b As double*, ByRef _h As double*) As
IRobotAdvancedSupportType
```

Description

Function returns support parameters applied during RC structure analysis. .

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 2.5.

II.2.1.3.3.2 GetOneDir

C++

```
HRESULT GetOneDir(IRobotNodeSupportFixingDirection _direction,
IRobotNodeSupportOneDirectionFixingType* ret);
```

C#

```
public IRobotNodeSupportOneDirectionFixingType GetOneDir(IRobotNodeSupportFixingDirection
_direction);
```

Visual Basic

```
Public Function GetOneDir(_direction As IRobotNodeSupportFixingDirection) As
IRobotNodeSupportOneDirectionFixingType
```

Description

The function returns the manner of unilateral blocking of the indicated direction for support.

II.2.1.3.3.3 IsFixed

C++

```
HRESULT IsFixed(IRobotNodeSupportFixingDirection _direction, VARIANT_BOOL* ret);
```

C#

```
public bool IsFixed(IRobotNodeSupportFixingDirection _direction);
```

Visual Basic

```
Public Function IsFixed(_direction As IRobotNodeSupportFixingDirection) As Boolean
```

Description

The function checks if the given direction is blocked.

II.2.1.3.3.4 SetAdvanced

C++

```
HRESULT SetAdvanced(IRobotAdvancedSupportType _type, double _b, double _h);
```

C#

```
public void SetAdvanced(IRobotAdvancedSupportType _type, double _b, double _h);
```

Visual Basic

```
Public Sub SetAdvanced(_type As IRobotAdvancedSupportType, _b As double, _h As double)
```

Description

Function defines support parameters applied during RC structure analysis. .

Version

Available since version 2.5.

II.2.1.3.3.5 SetFixed

C++

```
HRESULT SetFixed(IRobotNodeSupportFixingDirection _direction, VARIANT_BOOL _fixed);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public void SetFixed(IRobotNodeSupportFixingDirection _direction, bool _fixed);
```

Visual Basic

```
Public Sub SetFixed(_direction As IRobotNodeSupportFixingDirection, _fixed As Boolean)
```

Description

The function allows one to block / un-block an indicated direction.

II.2.1.3.3.6 SetOneDir

C++

```
HRESULT SetOneDir(IRobotNodeSupportFixingDirection _direction,
IRobotNodeSupportOneDirectionFixingType _how_blocked);
```

C#

```
public void SetOneDir(IRobotNodeSupportFixingDirection _direction,
IRobotNodeSupportOneDirectionFixingType _how_blocked);
```

Visual Basic

```
Public Sub SetOneDir(_direction As IRobotNodeSupportFixingDirection, _how_blocked As
IRobotNodeSupportOneDirectionFixingType)
```

Description

The function sets the manner of unilateral blocking of the indicated direction for the support.

II.2.1.4 IRobotNodeSupportFixingDirection

C++

```
enum IRobotNodeSupportFixingDirection;
```

C#

```
public enum IRobotNodeSupportFixingDirection;
```

Visual Basic

```
Public Enum IRobotNodeSupportFixingDirection
```

Description

Identifiers of directions that may be blocked out or released in a support. .

II.2.1.5 IRobotNodeSupportOneDirectionFixingType

C++

```
enum IRobotNodeSupportOneDirectionFixingType;
```

C#

```
public enum IRobotNodeSupportOneDirectionFixingType;
```

Visual Basic

```
Public Enum IRobotNodeSupportOneDirectionFixingType
```

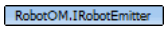
Description

Identifiers of the manner of unilateral blocking of the indicated direction for supports.

(c) 2014 Autodesk Inc. All rights reserved

II.2.1.6 IRobotEmitter

Class Hierarchy



C++

```
interface IRobotEmitter : IDispatch;
```

C#

```
public interface IRobotEmitter;
```

Visual Basic

```
Public Interface IRobotEmitter
```

Description

Emitter definition.

II.2.1.6.1 IRobotEmitter Members

The following tables list the members exposed by IRobotEmitter.

Public Fields

	Name	Description
	EstimatedElemNumber (see page 515)	Estimated element number in the sphere (diameter R1 (see page 516)) Available since version 1.7.
	H0 (see page 516)	Available since version 1.7.
	R1 (see page 516)	Available since version 1.7.
	R2 (see page 516)	Available since version 1.7.
	VariableMeshDensityIncrement (see page 516)	Available since version 1.7.

II.2.1.6.2 IRobotEmitter Fields

The fields of the IRobotEmitter class are listed here.

Public Fields

	Name	Description
	EstimatedElemNumber (see page 515)	Estimated element number in the sphere (diameter R1 (see page 516)) Available since version 1.7.
	H0 (see page 516)	Available since version 1.7.
	R1 (see page 516)	Available since version 1.7.
	R2 (see page 516)	Available since version 1.7.
	VariableMeshDensityIncrement (see page 516)	Available since version 1.7.

II.2.1.6.2.1 EstimatedElemNumber

C++

```
HRESULT get_EstimatedElemNumber(long*);
```

C#

```
public long EstimatedElemNumber { get; }
```

Visual Basic

```
Public ReadOnly EstimatedElemNumber As long
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Estimated element number in the sphere (diameter R1 (see page 516)) Available since version 1.7.

II.2.1.6.2.2 H0

C++

```
HRESULT get_H0(double*);
HRESULT put_H0(double);
```

C#

```
public double H0 { get; set; }
```

Visual Basic

```
Public H0 As double
```

Description

Available since version 1.7.

II.2.1.6.2.3 R1

C++

```
HRESULT get_R1(double*);
HRESULT put_R1(double);
```

C#

```
public double R1 { get; set; }
```

Visual Basic

```
Public R1 As double
```

Description

Available since version 1.7.

II.2.1.6.2.4 R2

C++

```
HRESULT get_R2(double*);
HRESULT put_R2(double);
```

C#

```
public double R2 { get; set; }
```

Visual Basic

```
Public R2 As double
```

Description

Available since version 1.7.

II.2.1.6.2.5 VariableMeshDensityIncrement

C++

```
HRESULT get_VariableMeshDensityIncrement(VARIANT_BOOL*);
HRESULT put_VariableMeshDensityIncrement(VARIANT_BOOL);
```

C#

```
public bool VariableMeshDensityIncrement { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public VariableMeshDensityIncrement As Boolean
```

Description

Available since version 1.7.

II.2.1.7 IRobotAdvancedSupportType

C++

```
enum IRobotAdvancedSupportType;
```

C#

```
public enum IRobotAdvancedSupportType;
```

Visual Basic

```
Public Enum IRobotAdvancedSupportType
```

Members

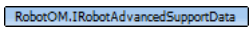
Members	Description
I_AST_NODAL = 1	Support defined in a node. Available since version 2.5.
I_AST_COLUMN_RECTANGULAR = 2	Rectangular column - it is required to define width b and height h of a column cross-section . Available since version 2.5.
I_AST_COLUMN_CIRCULAR = 3	Round column - it is required to define diameter b of a column cross-section . Available since version 2.5.
I_AST_WALL_MASONRY = 4	Masonry support defined by specifying the wall width b. Available since version 2.5.
I_AST_WALL_CONCRETE = 5	Concrete support defined by specifying the wall width b . Available since version 2.5.

Version

Available since version 2.5.

II.2.1.8 IRobotAdvancedSupportData

Class Hierarchy



C++

```
interface IRobotAdvancedSupportData : IDispatch;
```

C#

```
public interface IRobotAdvancedSupportData;
```

Visual Basic

```
Public Interface IRobotAdvancedSupportData
```

Version

Available since version 8.5.

(c) 2014 Autodesk Inc. All rights reserved

II.2.1.8.1 IRobotAdvancedSupportData Members

The following tables list the members exposed by IRobotAdvancedSupportData.

Public Fields

	Name	Description
	B (see page 518)	
	D (see page 518)	
	EquivalentElasticity (see page 519)	
	H (see page 519)	
	IsEquivalentElasticity (see page 519)	
	Type (see page 519)	Support type.

II.2.1.8.2 IRobotAdvancedSupportData Fields

The fields of the IRobotAdvancedSupportData class are listed here.

Public Fields

	Name	Description
	B (see page 518)	
	D (see page 518)	
	EquivalentElasticity (see page 519)	
	H (see page 519)	
	IsEquivalentElasticity (see page 519)	
	Type (see page 519)	Support type.

II.2.1.8.2.1 B

C++

```
HRESULT get_B(double*);
HRESULT put_B(double);
```

C#

```
public double B { get; set; }
```

Visual Basic

```
Public B As double
```

Version

Available since version 8.5.

II.2.1.8.2.2 D

C++

```
HRESULT get_D(double*);
HRESULT put_D(double);
```

C#

```
public double D { get; set; }
```

Visual Basic

```
Public D As double
```

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 8.5.

II.2.1.8.2.3 EquivalentElasticity

C++

```
HRESULT get_EquivalentElasticity(IRobotSupportEquivalentElasticity**);
```

C#

```
public IRobotSupportEquivalentElasticity EquivalentElasticity { get; }
```

Visual Basic

```
Public ReadOnly EquivalentElasticity As IRobotSupportEquivalentElasticity
```

Version

Available since version 8.5.

II.2.1.8.2.4 H

C++

```
HRESULT get_H(double*);
HRESULT put_H(double);
```

C#

```
public double H { get; set; }
```

Visual Basic

```
Public H As double
```

Version

Available since version 8.5.

II.2.1.8.2.5 IsEquivalentElasticity

C++

```
HRESULT get_IsEquivalentElasticity(VARIANT_BOOL*);
HRESULT put_IsEquivalentElasticity(VARIANT_BOOL);
```

C#

```
public bool IsEquivalentElasticity { get; set; }
```

Visual Basic

```
Public IsEquivalentElasticity As Boolean
```

Version

Available since version 8.5.

II.2.1.8.2.6 Type

C++

```
HRESULT get_Type(IRobotAdvancedSupportType*);
HRESULT put_Type(IRobotAdvancedSupportType);
```

C#

```
public IRobotAdvancedSupportType Type { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Type As IRobotAdvancedSupportType
```

Description

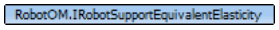
Support type.

Version

Available since version 8.5.

II.2.1.9 IRobotSupportEquivalentElasticity

Class Hierarchy



C++

```
interface IRobotSupportEquivalentElasticity : IDispatch;
```

C#

```
public interface IRobotSupportEquivalentElasticity;
```

Visual Basic

```
Public Interface IRobotSupportEquivalentElasticity
```

Description

Definition of equivalent elasticity of a support.

Version

Available since version 8.5.

II.2.1.9.1 IRobotSupportEquivalentElasticity Members

The following tables list the members exposed by IRobotSupportEquivalentElasticity.

Public Fields

	Name	Description
	E (see page 521)	Young's modulus value.
	PoissonRatio (see page 521)	Poisson's ratio value.
	Type (see page 521)	

Public Methods

	Name	Description
	SetMaterial (see page 522)	

II.2.1.9.2 IRobotSupportEquivalentElasticity Fields

The fields of the IRobotSupportEquivalentElasticity class are listed here.

Public Fields

	Name	Description
	E (see page 521)	Young's modulus value.
	PoissonRatio (see page 521)	Poisson's ratio value.
	Type (see page 521)	

(c) 2014 Autodesk Inc. All rights reserved

II.2.1.9.2.1 E

C++

```
HRESULT get_E(double*);
HRESULT put_E(double);
```

C#

```
public double E { get; set; }
```

Visual Basic

```
Public E As double
```

Description

Young's modulus value.

Version

Available since version 8.5.

II.2.1.9.2.2 PoissonRatio

C++

```
HRESULT get_PoissonRatio(double*);
HRESULT put_PoissonRatio(double);
```

C#

```
public double PoissonRatio { get; set; }
```

Visual Basic

```
Public PoissonRatio As double
```

Description

Poisson's ratio value.

Version

Available since version 8.5.

II.2.1.9.2.3 Type

C++

```
HRESULT get_Type(IRobotSupportEquivalentElasticityType*);
```

C#

```
public IRobotSupportEquivalentElasticityType Type { get; }
```

Visual Basic

```
Public ReadOnly Type As IRobotSupportEquivalentElasticityType
```

Version

Available since version 8.5.

II.2.1.9.3 IRobotSupportEquivalentElasticity Methods

The methods of the IRobotSupportEquivalentElasticity class are listed here.

Public Methods

	Name	Description
	SetMaterial (see page 522)	

(c) 2014 Autodesk Inc. All rights reserved

II.2.1.9.3.1 SetMaterial

C++

```
HRESULT SetMaterial(BSTR _material_name);
```

C#

```
public void SetMaterial(String _material_name);
```

Visual Basic

```
Public Sub SetMaterial(_material_name As String)
```

Version

Available since version 8.5.

II.2.1.10 IRobotSupportEquivalentElasticityType

C++

```
enum IRobotSupportEquivalentElasticityType;
```

C#

```
public enum IRobotSupportEquivalentElasticityType;
```

Visual Basic

```
Public Enum IRobotSupportEquivalentElasticityType
```

Members

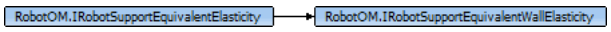
Members	Description
I_SEET_COLUMN = 1	Available since version 8.5.
I_SEET_WALL = 2	Available since version 8.5.

Version

Available since version 8.5.

II.2.1.11 IRobotSupportEquivalentWallElasticity

Class Hierarchy



C++

```
interface IRobotSupportEquivalentWallElasticity : IRobotSupportEquivalentElasticity;
```

C#

```
public interface IRobotSupportEquivalentWallElasticity : IRobotSupportEquivalentElasticity;
```

Visual Basic

```
Public Interface IRobotSupportEquivalentWallElasticity
```

Version

Available since version 8.5.

II.2.1.11.1 IRobotSupportEquivalentWallElasticity Members

The following tables list the members exposed by IRobotSupportEquivalentWallElasticity.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	E (see page 521)	Young's modulus value.
	L1 (see page 523)	
	L2 (see page 523)	
	PoissonRatio (see page 521)	Poisson's ratio value.
	ThroughTwoStories (see page 524)	
	Type (see page 521)	

Public Methods

	Name	Description
	SetMaterial (see page 522)	

II.2.1.11.2 IRobotSupportEquivalentWallElasticity Fields

The fields of the IRobotSupportEquivalentWallElasticity class are listed here.

Public Fields

	Name	Description
	L1 (see page 523)	
	L2 (see page 523)	
	ThroughTwoStories (see page 524)	

II.2.1.11.2.1 L1

C++

```
HRESULT get_L1(double*);
HRESULT put_L1(double);
```

C#

```
public double L1 { get; set; }
```

Visual Basic

```
Public L1 As double
```

Version

Available since version 8.5.

II.2.1.11.2.2 L2

C++

```
HRESULT get_L2(double*);
HRESULT put_L2(double);
```

C#

```
public double L2 { get; set; }
```

Visual Basic

```
Public L2 As double
```

Version

Available since version 8.5.

(c) 2014 Autodesk Inc. All rights reserved

II.2.1.11.2.3 ThroughTwoStories

C++

```
HRESULT get_ThroughTwoStories(VARIANT_BOOL*);
HRESULT put_ThroughTwoStories(VARIANT_BOOL);
```

C#

```
public bool ThroughTwoStories { get; set; }
```

Visual Basic

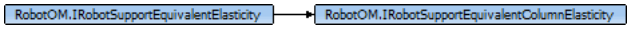
```
Public ThroughTwoStories As Boolean
```

Version

Available since version 8.5.

II.2.1.12 IRobotSupportEquivalentColumnElasticity

Class Hierarchy



C++

```
interface IRobotSupportEquivalentColumnElasticity : IRobotSupportEquivalentElasticity;
```

C#

```
public interface IRobotSupportEquivalentColumnElasticity :
IRobotSupportEquivalentElasticity;
```

Visual Basic

```
Public Interface IRobotSupportEquivalentColumnElasticity
```

Version

Available since version 8.5.

II.2.1.12.1 IRobotSupportEquivalentColumnElasticity Members

The following tables list the members exposed by IRobotSupportEquivalentColumnElasticity.

Public Fields

	Name	Description
	E (see page 521)	Young's modulus value.
	Fixing1 (see page 525)	
	Fixing2 (see page 525)	
	L1 (see page 525)	
	L2 (see page 526)	
	PoissonRatio (see page 521)	Poisson's ratio value.
	ThroughTwoStories (see page 526)	
	Type (see page 521)	

Public Methods

	Name	Description
	SetMaterial (see page 522)	

(c) 2014 Autodesk Inc. All rights reserved

II.2.1.12.2 IRobotSupportEquivalentColumnElasticity Fields

The fields of the IRobotSupportEquivalentColumnElasticity class are listed here.

Public Fields

	Name	Description
	Fixing1 (see page 525)	
	Fixing2 (see page 525)	
	L1 (see page 525)	
	L2 (see page 526)	
	ThroughTwoStories (see page 526)	

II.2.1.12.2.1 Fixing1

C++

```
HRESULT get_Fixing1(IRobotSupportColumnFixingType*);
HRESULT put_Fixing1(IRobotSupportColumnFixingType);
```

C#

```
public IRobotSupportColumnFixingType Fixing1 { get; set; }
```

Visual Basic

```
Public Fixing1 As IRobotSupportColumnFixingType
```

Version

Available since version 8.5.

II.2.1.12.2.2 Fixing2

C++

```
HRESULT get_Fixing2(IRobotSupportColumnFixingType*);
HRESULT put_Fixing2(IRobotSupportColumnFixingType);
```

C#

```
public IRobotSupportColumnFixingType Fixing2 { get; set; }
```

Visual Basic

```
Public Fixing2 As IRobotSupportColumnFixingType
```

Version

Available since version 8.5.

II.2.1.12.2.3 L1

C++

```
HRESULT get_L1(double*);
HRESULT put_L1(double);
```

C#

```
public double L1 { get; set; }
```

Visual Basic

```
Public L1 As double
```

Version

Available since version 8.5.

(c) 2014 Autodesk Inc. All rights reserved

II.2.1.12.2.4 L2

C++

```
HRESULT get_L2(double*);
HRESULT put_L2(double);
```

C#

```
public double L2 { get; set; }
```

Visual Basic

```
Public L2 As double
```

Version

Available since version 8.5.

II.2.1.12.2.5 ThroughTwoStories

C++

```
HRESULT get_ThroughTwoStories(VARIANT_BOOL*);
HRESULT put_ThroughTwoStories(VARIANT_BOOL);
```

C#

```
public bool ThroughTwoStories { get; set; }
```

Visual Basic

```
Public ThroughTwoStories As Boolean
```

Version

Available since version 8.5.

II.2.1.13 IRobotSupportColumnFixingType

C++

```
enum IRobotSupportColumnFixingType;
```

C#

```
public enum IRobotSupportColumnFixingType;
```

Visual Basic

```
Public Enum IRobotSupportColumnFixingType
```

Members

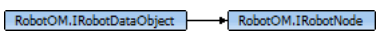
Members	Description
I_SCFT_FIXED = 1	Available since version 8.5.
I_SCFT_PINNED = 2	Available since version 8.5.
I_SCFT_ROLLER = 3	Available since version 8.5.

Version

Available since version 8.5.

II.2.2 IRobotNode

Class Hierarchy



(c) 2014 Autodesk Inc. All rights reserved

C++

```
interface IRobotNode : IRobotDataObject;
```

C#

```
public interface IRobotNode : IRobotDataObject;
```

Visual Basic

```
Public Interface IRobotNode
```

Description

Each structure node is represented in the Robot Object Model by means of the RobotNode interface. .

II.2.2.1 IRobotNode Members

The following tables list the members exposed by IRobotNode.

Public Fields

	Name	Description
	HasCalcSupport (see page 528)	Flag indicating that a support was created automatically at the node during generation of a structure model.
	HasEmitter (see page 528)	Flag indicating if emitter is assigned to the node Available since version 1.7.
	IsCalc (see page 528)	Flag indicating if the node has been created automatically due to generation of the calculation model.
	Number (see page 31)	Object number is assigned by the user; it is qunique among all the components of the same type. .
	UniqueId (see page 529)	Unique identifier of structure component Available since version 1.7.
	X (see page 529)	Coordinate x.
	Y (see page 529)	Coordinate y.
	Z (see page 529)	Coordinate z.

Public Methods

	Name	Description
	GetCalcSupport (see page 530)	Function makes available data defining a support created automatically at the node during generation of a structure model.
	GetEmitter (see page 530)	Function returns the emitter defined in the node. If the emitter has not been defined the function returns empty reference (NULL). Available since version 1.7.
	GetLabel (see page 31)	The function returns a label of the indicated type applied to the object. If the object does not have any label of the type, running the function leads to a critical error. .
	GetLabelName (see page 32)	The function returns the name of a label of the indicated type, applied to the object. If the object does not have any label of the type, running the function leads to a critical error. .
	GetLabels (see page 32)	The function returns a collection containing all labels (of different types) defined for a given object. .
	HasLabel (see page 32)	The function returns True (non-zero value) if a label of the type has already been defined for the object. .
	RemoveEmitter (see page 531)	Function deletes emitter from the node Available since version 1.7.
	RemoveLabel (see page 32)	The function deletes a label of the indicated type. .
	SetEmitter (see page 531)	Function defines emitter in the node Available since version 1.7.
	SetLabel (see page 33)	The function applies the defined label to an object. At a given moment, an object may have only one label of a type applied (e.g. there may not be two supports defined for one node). If an object has a formerly applied label of the same type, it is replaced by the new one. .

(c) 2014 Autodesk Inc. All rights reserved

II.2.2.2 IRobotNode Fields

The fields of the IRobotNode class are listed here.

Public Fields

	Name	Description
	HasCalcSupport (see page 528)	Flag indicating that a support was created automatically at the node during generation of a structure model.
	HasEmitter (see page 528)	Flag indicating if emitter is assigned to the node Available since version 1.7.
	IsCalc (see page 528)	Flag indicating if the node has been created automatically due to generation of the calculation model.
	UniqueId (see page 529)	Unique identifier of structure component Available since version 1.7.
	X (see page 529)	Coordinate x.
	Y (see page 529)	Coordinate y.
	Z (see page 529)	Coordinate z.

II.2.2.2.1 HasCalcSupport

C++

```
HRESULT get_HasCalcSupport(VARIANT_BOOL*);
```

C#

```
public bool HasCalcSupport { get; }
```

Visual Basic

```
Public ReadOnly HasCalcSupport As Boolean
```

Description

Flag indicating that a support was created automatically at the node during generation of a structure model.

Version

Available since version 8.

II.2.2.2.2 HasEmitter

C++

```
HRESULT get_HasEmitter(VARIANT_BOOL*);
```

C#

```
public bool HasEmitter { get; }
```

Visual Basic

```
Public ReadOnly HasEmitter As Boolean
```

Description

Flag indicating if emitter is assigned to the node Available since version 1.7.

II.2.2.2.3 IsCalc

C++

```
HRESULT get_IsCalc(VARIANT_BOOL*);
```

C#

```
public bool IsCalc { get; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public ReadOnly IsCalc As Boolean
```

Description

Flag indicating if the node has been created automatically due to generation of the calculation model.

Version

Available since version 2.5.

II.2.2.2.4 UniqueId

C++

```
HRESULT get_UniqueId(long*);
```

C#

```
public long UniqueId { get; }
```

Visual Basic

```
Public ReadOnly UniqueId As long
```

Description

Unique identifier of structure component Available since version 1.7.

II.2.2.2.5 X

C++

```
HRESULT get_X(double*);
HRESULT put_X(double);
```

C#

```
public double X { get; set; }
```

Visual Basic

```
Public X As double
```

Description

Coordinate x.

II.2.2.2.6 Y

C++

```
HRESULT get_Y(double*);
HRESULT put_Y(double);
```

C#

```
public double Y { get; set; }
```

Visual Basic

```
Public Y As double
```

Description

Coordinate y.

II.2.2.2.7 Z

C++

```
HRESULT get_Z(double*);
```

(c) 2014 Autodesk Inc. All rights reserved

```
HRESULT put_Z(double);
```

C#

```
public double Z { get; set; }
```

Visual Basic

```
Public Z As double
```

Description

Coordinate z.

II.2.2.3 IRobotNode Methods

The methods of the IRobotNode class are listed here.

Public Methods

	Name	Description
	GetCalcSupport (see page 530)	Function makes available data defining a support created automatically at the node during generation of a structure model.
	GetEmitter (see page 530)	Function returns the emitter defined in the node. If the emitter has not been defined the function returns empty reference (NULL). Available since version 1.7.
	RemoveEmitter (see page 531)	Function deletes emitter from the node Available since version 1.7.
	SetEmitter (see page 531)	Function defines emitter in the node Available since version 1.7.

II.2.2.3.1 GetCalcSupport

C++

```
HRESULT GetCalcSupport(IRobotNodeSupportData** ret);
```

C#

```
public IRobotNodeSupportData GetCalcSupport();
```

Visual Basic

```
Public Function GetCalcSupport() As IRobotNodeSupportData
```

Description

Function makes available data defining a support created automatically at the node during generation of a structure model.

Version

Available since version 8.

II.2.2.3.2 GetEmitter

C++

```
HRESULT GetEmitter(IRobotEmitter** ret);
```

C#

```
public IRobotEmitter GetEmitter();
```

Visual Basic

```
Public Function GetEmitter() As IRobotEmitter
```

Description

Function returns the emitter defined in the node. If the emitter has not been defined the function returns empty reference (NULL). Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

II.2.2.3.3 RemoveEmitter

C++

```
HRESULT RemoveEmitter();
```

C#

```
public void RemoveEmitter();
```

Visual Basic

```
Public Sub RemoveEmitter()
```

Description

Function deletes emitter from the node Available since version 1.7.

II.2.2.3.4 SetEmitter

C++

```
HRESULT SetEmitter(IRobotEmitter* _emit);
```

C#

```
public void SetEmitter(IRobotEmitter _emit);
```

Visual Basic

```
Public Sub SetEmitter(ByRef _emit As IRobotEmitter)
```

Description

Function defines emitter in the node Available since version 1.7.

II.2.3 IRobotNodeServer

Class Hierarchy

C++

```
interface IRobotNodeServer : IRobotDataObjectServer;
```

C#

```
public interface IRobotNodeServer : IRobotDataObjectServer;
```

Visual Basic

```
Public Interface IRobotNodeServer
```

Description

Server of nodes manages all the structure nodes .

II.2.3.1 IRobotNodeServer Members

The following tables list the members exposed by IRobotNodeServer.

Public Fields

	Name	Description
	CompatibleNodes (see page 533)	Compatible node server.
	FreeNumber (see page 533)	First available user number which can be assigned to the structure node Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

	NonlinearLinks (see page 533)	Non-linear connection models.
	RigidLinks (see page 533)	Rigid link server.

Public Methods

	Name	Description
	Create (see page 534)	The function creates a new node with the provided parameters. .
	Delete (see page 34)	The function deletes the object of the indicated number. .
	DeleteMany (see page 34)	The function deletes all objects that meet the criteria of the indicated selection. .
	Exist (see page 35)	The function returns True (non-zero value) if the object of the indicated name already exists. .
	FindWithId (see page 535)	Function returns a number of the node with a specified unique identifier. If such a node is not found, then, zero value is returned. Available since version 1.7.
	Get (see page 35)	The function returns an object with the indicated user-defined number. The type of the returned object agrees with the type of objects managed by the server. If the object of the indicated number does not exist, running the function leads to critical error. .
	GetAll (see page 35)	The function returns a collection containing all objects managed by the server. .
	GetCalcNodes (see page 535)	Function returns the collection of all nodes created automatically during generation of the calculation model. .
	GetCalcSupport (see page 535)	Function makes available definition of an automatic support at a specified node.
	GetConnectedBars (see page 535)	Function returns the table with numbers of bars adjoining the specified list of nodes. .
	GetMany (see page 36)	The function returns a collection of objects that meet the criteria of the indicated selection. .
	GetUniqueId (see page 536)	Function returns a unique identifier for the specified node.
	GetUserNodes (see page 536)	Function returns the collection of all user-defined nodes.
	HasCalcSupport (see page 536)	Function returns True value if a support was automatically created at the specified node during generation of a structure model.
	IsCalc (see page 537)	Function returns a non-zero value (True), if the node of the specified number has been created automatically during generation of the calculation model .
	RemoveEmitter (see page 537)	Function deletes emitter definition from nodes belonging to the specified selection. Available since version 1.7.
	RemoveLabel (see page 36)	The function removes the labels of the indicated type from all objects that meet the criteria of the indicated selection. .
	SetEmitter (see page 537)	Function defines a determined emitter on the specified node list. Available since version 1.7.
	SetLabel (see page 36)	The function applies a label (identified by the type and name) to all objects that meet the criteria of the indicated selection. .

II.2.3.2 IRobotNodeServer Fields

The fields of the IRobotNodeServer class are listed here.

Public Fields

	Name	Description
	CompatibleNodes (see page 533)	Compatible node server.
	FreeNumber (see page 533)	First available user number which can be assigned to the structure node Available since version 1.7.
	NonlinearLinks (see page 533)	Non-linear connection models.
	RigidLinks (see page 533)	Rigid link server.

(c) 2014 Autodesk Inc. All rights reserved

II.2.3.2.1 CompatibleNodes

C++

```
HRESULT get_CompatibleNodes(IRobotNodeCompatibilityServer**);
```

C#

```
public IRobotNodeCompatibilityServer CompatibleNodes { get; }
```

Visual Basic

```
Public ReadOnly CompatibleNodes As IRobotNodeCompatibilityServer
```

Description

Compatible node server.

Version

Available since version 2.5.

II.2.3.2.2 FreeNumber

C++

```
HRESULT get_FreeNumber(long*);
```

C#

```
public long FreeNumber { get; }
```

Visual Basic

```
Public ReadOnly FreeNumber As long
```

Description

First available user number which can be assigned to the structure node Available since version 1.7.

II.2.3.2.3 NonlinearLinks

C++

```
HRESULT get_NonlinearLinks(IRobotNonlinearLinkServer**);
```

C#

```
public IRobotNonlinearLinkServer NonlinearLinks { get; }
```

Visual Basic

```
Public ReadOnly NonlinearLinks As IRobotNonlinearLinkServer
```

Description

Non-linear connection models.

Version

Available since version 3.

II.2.3.2.4 RigidLinks

C++

```
HRESULT get_RigidLinks(IRobotNodeRigidLinkServer**);
```

C#

```
public IRobotNodeRigidLinkServer RigidLinks { get; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public ReadOnly RigidLinks As IRobotNodeRigidLinkServer
```

Description

Rigid link server.

Version

Available since version 2.5.

II.2.3.3 IRobotNodeServer Methods

The methods of the IRobotNodeServer class are listed here.

Public Methods

	Name	Description
	Create (see page 534)	The function creates a new node with the provided parameters. .
	FindWithId (see page 535)	Function returns a number of the node with a specified unique identifier. If such a node is not found, then, zero value is returned. Available since version 1.7.
	GetCalcNodes (see page 535)	Function returns the collection of all nodes created automatically during generation of the calculation model. .
	GetCalcSupport (see page 535)	Function makes available definition of an automatic support at a specified node.
	GetConnectedBars (see page 535)	Function returns the table with numbers of bars adjoining the specified list of nodes. .
	GetUniqueId (see page 536)	Function returns a unique identifier for the specified node.
	GetUserNodes (see page 536)	Function returns the collection of all user-defined nodes.
	HasCalcSupport (see page 536)	Function returns True value if a support was automatically created at the specified node during generation of a structure model.
	IsCalc (see page 537)	Function returns a non-zero value (True), if the node of the specified number has been created automatically during generation of the calculation model .
	RemoveEmitter (see page 537)	Function deletes emitter definition from nodes belonging to the specified selection. Available since version 1.7.
	SetEmitter (see page 537)	Function defines a determined emitter on the specified node list. Available since version 1.7.

II.2.3.3.1 Create

C++

```
HRESULT Create(long _node_number, double _x, double _y, double _z);
```

C#

```
public void Create(long _node_number, double _x, double _y, double _z);
```

Visual Basic

```
Public Sub Create(_node_number As long, _x As double, _y As double, _z As double)
```

Description

The function creates a new node with the provided parameters. .

II.2.3.3.2 FindWithId

C++

```
HRESULT FindWithId(long _unique_id, long* ret);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public long FindWithId(long _unique_id);
```

Visual Basic

```
Public Function FindWithId(_unique_id As long) As long
```

Description

Function returns a number of the node with a specified unique identifier. If such a node is not found, then, zero value is returned. Available since version 1.7.

II.2.3.3.3 GetCalcNodes

C++

```
HRESULT GetCalcNodes(IRobotCollection** ret);
```

C#

```
public IRobotCollection GetCalcNodes();
```

Visual Basic

```
Public Function GetCalcNodes() As IRobotCollection
```

Description

Function returns the collection of all nodes created automatically during generation of the calculation model. .

Version

Available since version 2.5.

II.2.3.3.4 GetCalcSupport

C++

```
HRESULT GetCalcSupport(long _node_number, IRobotNodeSupportData** ret);
```

C#

```
public IRobotNodeSupportData GetCalcSupport(long _node_number);
```

Visual Basic

```
Public Function GetCalcSupport(_node_number As long) As IRobotNodeSupportData
```

Description

Function makes available definition of an automatic support at a specified node.

Version

Available since version 8.

II.2.3.3.5 GetConnectedBars

C++

```
HRESULT GetConnectedBars(BSTR _node_sel, IRobotNumbersArray** ret);
```

C#

```
public IRobotNumbersArray GetConnectedBars(String _node_sel);
```

Visual Basic

```
Public Function GetConnectedBars(_node_sel As String) As IRobotNumbersArray
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Function returns the table with numbers of bars adjoining the specified list of nodes. .

Version

Available since version 3.5.

II.2.3.3.6 GetUniqueId

C++

```
HRESULT GetUniqueId(long _node_num, long* ret);
```

C#

```
public long GetUniqueId(long _node_num);
```

Visual Basic

```
Public Function GetUniqueId(_node_num As long) As long
```

Description

Function returns a unique identifier for the specified node.

Version

Available since version 8.2.

II.2.3.3.7 GetUserNodes

C++

```
HRESULT GetUserNodes(IRobotCollection** ret);
```

C#

```
public IRobotCollection GetUserNodes();
```

Visual Basic

```
Public Function GetUserNodes() As IRobotCollection
```

Description

Function returns the collection of all user-defined nodes.

Version

Available since version 2.5.

II.2.3.3.8 HasCalcSupport

C++

```
HRESULT HasCalcSupport(long _node_number, VARIANT_BOOL* ret);
```

C#

```
public bool HasCalcSupport(long _node_number);
```

Visual Basic

```
Public Function HasCalcSupport(_node_number As long) As Boolean
```

Description

Function returns True value if a support was automatically created at the specified node during generation of a structure model.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 8.

II.2.3.3.9 IsCalc

C++

```
HRESULT IsCalc(long _node_num, VARIANT_BOOL* ret);
```

C#

```
public bool IsCalc(long _node_num);
```

Visual Basic

```
Public Function IsCalc(_node_num As long) As Boolean
```

Description

Function returns a non-zero value (True), if the node of the specified number has been created automatically during generation of the calculation model .

Version

Available since version 2.5.

II.2.3.3.10 RemoveEmitter

C++

```
HRESULT RemoveEmitter(IRobotSelection* _node_sel);
```

C#

```
public void RemoveEmitter(IRobotSelection _node_sel);
```

Visual Basic

```
Public Sub RemoveEmitter(ByRef _node_sel As IRobotSelection)
```

Description

Function deletes emitter definition from nodes belonging to the specified selection. Available since version 1.7.

II.2.3.3.11 SetEmitter

C++

```
HRESULT SetEmitter(IRobotSelection* _node_sel, IRobotEmitter* _emit);
```

C#

```
public void SetEmitter(IRobotSelection _node_sel, IRobotEmitter _emit);
```

Visual Basic

```
Public Sub SetEmitter(ByRef _node_sel As IRobotSelection, ByRef _emit As IRobotEmitter)
```

Description

Function defines a determined emitter on the specified node list. Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

II.3 Bars

Enumerations

	Name	Description
	IRobotBarTensionCompression (see page 651)	

Interfaces

	Name	Description
	IRobotBar (see page 624)	Each bar in a structure is represented in the Object Model by a data object of the type: I_OT_BAR. The data and functionality related to the bar are available through time history RobotBar interface. The following complex bar attributes are defined by means of the labelling mechanism: Attribute Label type
	IRobotBarServer (see page 636)	Server of bars manages the bars defined in the structure .
	IRobotBarEnd (see page 648)	One can indicate a beginning and an end of each bar. The object of RobotBarEnd type, describing one of the bar ends, is created to make it possible to carry out operations affecting only the beginning or end of a bar (and not the entire bar).
	IRobotBarElement (see page 652)	Bar element.
	IRobotBarElementData (see page 654)	Bar (see page 655) element data.
	IRobotBarElementDataSet (see page 656)	Data set for bar elements.

II.3.1 Complex attributes of a bar

Complex attributes of a bar are defined by means of the labelling mechanism. .

Enumerations

	Name	Description
	IRobotBarEndBracketType (see page 621)	Available types of brackets. .
	IRobotBarEndBracketDataValue (see page 621)	A set of identifiers has been defined that determines particular parameters (attributes), describing a bracket at bar end. The identifier is used as a parameter of the function that gets or sets the value of the relevant bracket attribute. .
	IRobotBarOffsetAutoPosition (see page 623)	Available predefined position of an offset with respect to section. .
	IRobotBarOffsetMemberLength (see page 624)	

Interfaces

	Name	Description
	IRobotBarOffsetData (see page 617)	The structure describing an offset defined for a bar. .
	IRobotBarEndOffsetData (see page 619)	The structure describing a shift at one of bar ends.
	IRobotBarEndBracketData (see page 621)	The structure describing a bracket at bar end. .

II.3.1.1 Cables

Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

Enumerations

	Name	Description
	IRobotBarCableAssemblingParamType (see page 541)	Available types of assembling parameters.

Interfaces

	Name	Description
	IRobotBarCableData (see page 539)	Definition of cable parameters.

II.3.1.1.1 IRobotBarCableData

Class Hierarchy

RobotOM.IRobotBarCableData

C++

```
interface IRobotBarCableData : IDispatch;
```

C#

```
public interface IRobotBarCableData;
```

Visual Basic

```
Public Interface IRobotBarCableData
```

Description

Definition of cable parameters.

Version

Available since version 2.5.

II.3.1.1.1.1 IRobotBarCableData Members

The following tables list the members exposed by IRobotBarCableData.

Public Fields

	Name	Description
	AssemblingParam (see page 540)	Assembling parameter.
	AssemblingParamValue (see page 540)	Value of the assembling parameter.
	MaterialName (see page 540)	Name of the cable material.
	SectionAX (see page 541)	Cable section area.

II.3.1.1.1.2 IRobotBarCableData Fields

The fields of the IRobotBarCableData class are listed here.

Public Fields

	Name	Description
	AssemblingParam (see page 540)	Assembling parameter.
	AssemblingParamValue (see page 540)	Value of the assembling parameter.
	MaterialName (see page 540)	Name of the cable material.
	SectionAX (see page 541)	Cable section area.

(c) 2014 Autodesk Inc. All rights reserved

II.3.1.1.1.2.1 AssemblingParam

C++

```
HRESULT get_AssemblingParam(IRobotBarCableAssemblingParamType*);
HRESULT put_AssemblingParam(IRobotBarCableAssemblingParamType);
```

C#

```
public IRobotBarCableAssemblingParamType AssemblingParam { get; set; }
```

Visual Basic

```
Public AssemblingParam As IRobotBarCableAssemblingParamType
```

Description

Assembling parameter.

Version

Available since version 2.5.

II.3.1.1.1.2.2 AssemblingParamValue

C++

```
HRESULT get_AssemblingParamValue(double*);
HRESULT put_AssemblingParamValue(double);
```

C#

```
public double AssemblingParamValue { get; set; }
```

Visual Basic

```
Public AssemblingParamValue As double
```

Description

Value of the assembling parameter.

Version

Available since version 2.5.

II.3.1.1.1.2.3 MaterialName

C++

```
HRESULT get_MaterialName(BSTR*);
HRESULT put_MaterialName(BSTR);
```

C#

```
public BSTR MaterialName { get; set; }
```

Visual Basic

```
Public MaterialName As BSTR
```

Description

Name of the cable material.

Version

Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

II.3.1.1.1.2.4 SectionAX

C++

```
HRESULT get_SectionAX(double*);
HRESULT put_SectionAX(double);
```

C#

```
public double SectionAX { get; set; }
```

Visual Basic

```
Public SectionAX As double
```

Description

Cable section area.

Version

Available since version 2.5.

II.3.1.1.2 IRobotBarCableAssemblingParamType

C++

```
enum IRobotBarCableAssemblingParamType;
```

C#

```
public enum IRobotBarCableAssemblingParamType;
```

Visual Basic

```
Public Enum IRobotBarCableAssemblingParamType
```

Members

Members	Description
I_BCAPT_STRESSES_SIGMA = 1	Definition of normal stress as a cable parameter for the assembling load case. Available since version 2.5.
I_BCAPT_FORCE_FO = 2	Definition of tension force as a cable parameter for the assembling load case . Available since version 2.5.
I_BCAPT_LENGTH_L = 3	Definition of the unloaded cable length as a cable parameter for the assembling load case . Available since version 2.5.
I_BCAPT_ELONGATION_DL = 5	Definition of dilatation as a cable parameter for the assembling load case . Available since version 2.5.
I_BCAPT_ELONGATION_DL_RELATIVE = 4	Definition of relative dilatation as a cable parameter for the assembling load case. Available since version 2.5.

Description

Available types of assembling parameters.

Version

Available since version 2.5.

II.3.1.2 Elastic ground

Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

Enumerations

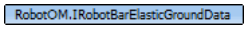
	Name	Description
	IRobotUpliftDirection (see page 545)	Available uplift directions.
	IRobotUpliftSense (see page 545)	Available uplift definitions in one direction.

Interfaces

	Name	Description
	IRobotBarElasticGroundData (see page 542)	Elastic ground parameters for bars. .

II.3.1.2.1 IRobotBarElasticGroundData

Class Hierarchy



C++

```
interface IRobotBarElasticGroundData : IDispatch;
```

C#

```
public interface IRobotBarElasticGroundData;
```

Visual Basic

```
Public Interface IRobotBarElasticGroundData
```

Description

Elastic ground parameters for bars. .

Version

Available since version 2.5.

II.3.1.2.1.1 IRobotBarElasticGroundData Members

The following tables list the members exposed by IRobotBarElasticGroundData.

Public Fields

	Name	Description
	HX (see page 543)	Eleastic ground coefficient HX in the RX direction of the global coordinate system.
	KY (see page 543)	Elastic ground coefficient KY in the UY direction of the global coordinate system .
	KZ (see page 544)	Eleastic ground coefficient KZ in the UZ direction of the global coordinate system.

Public Methods

	Name	Description
	GetOneDir (see page 544)	Function returns uplift definition for the given direction.
	SetOneDir (see page 544)	Function defines uplift for the given direction.

II.3.1.2.1.2 IRobotBarElasticGroundData Fields

The fields of the IRobotBarElasticGroundData class are listed here.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	HX (see page 543)	Eleastic ground coefficient HX in the RX direction of the global coordinate system.
	KY (see page 543)	Elastic ground coefficient KY in the UY direction of the global coordinate system .
	KZ (see page 544)	Eleastic ground coefficient KZ in the UZ direction of the global coordinate system.

II.3.1.2.1.2.1 HX

C++

```
HRESULT get_HX(double*);
HRESULT put_HX(double);
```

C#

```
public double HX { get; set; }
```

Visual Basic

```
Public HX As double
```

Description

Eleastic ground coefficient HX in the RX direction of the global coordinate system.

Version

Available since version 2.5.

II.3.1.2.1.2.2 KY

C++

```
HRESULT get_KY(double*);
HRESULT put_KY(double);
```

C#

```
public double KY { get; set; }
```

Visual Basic

```
Public KY As double
```

Description

Elastic ground coefficient KY in the UY direction of the global coordinate system .

Version

Available since version 2.5.

II.3.1.2.1.2.3 KZ

C++

```
HRESULT get_KZ(double*);
HRESULT put_KZ(double);
```

C#

```
public double KZ { get; set; }
```

Visual Basic

```
Public KZ As double
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Eleastic ground coefficient KZ in the UZ direction of the global coordinate system.

Version

Available since version 2.5.

II.3.1.2.1.3 IRobotBarElasticGroundData Methods

The methods of the IRobotBarElasticGroundData class are listed here.

Public Methods

	Name	Description
	GetOneDir (see page 544)	Function returns uplift definition for the given direction.
	SetOneDir (see page 544)	Function defines uplift for the given direction.

II.3.1.2.1.3.1 GetOneDir

C++

```
HRESULT GetOneDir(IRobotUpliftDirection _uplift_dir, IRobotUpliftSense* ret);
```

C#

```
public IRobotUpliftSense GetOneDir(IRobotUpliftDirection _uplift_dir);
```

Visual Basic

```
Public Function GetOneDir(_uplift_dir As IRobotUpliftDirection) As IRobotUpliftSense
```

Description

Function returns uplift definition for the given direction.

Version

Available since version 11.

II.3.1.2.1.3.2 SetOneDir

C++

```
HRESULT SetOneDir(IRobotUpliftDirection _uplift_dir, IRobotUpliftSense _uplift_type);
```

C#

```
public void SetOneDir(IRobotUpliftDirection _uplift_dir, IRobotUpliftSense _uplift_type);
```

Visual Basic

```
Public Sub SetOneDir(_uplift_dir As IRobotUpliftDirection, _uplift_type As
IRobotUpliftSense)
```

Description

Function defines uplift for the given direction.

Version

Available since version 11.

II.3.1.2.2 IRobotUpliftDirection

C++

```
enum IRobotUpliftDirection;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public enum IRobotUpliftDirection;
```

Visual Basic

```
Public Enum IRobotUpliftDirection
```

Members

Members	Description
I_UD_UY = 1	Uplift in the Y direction. Available since version 11.
I_UD_UZ = 2	Uplift in the Z direction. Available since version 11.

Description

Available uplift directions.

Version

Available since version 11.

II.3.1.2.3 IRobotUpliftSense

C++

```
enum IRobotUpliftSense;
```

C#

```
public enum IRobotUpliftSense;
```

Visual Basic

```
Public Enum IRobotUpliftSense
```

Members

Members	Description
I_US_NONE = 0	No uplift. Available since version 11.
I_US_MINUS = 1	Uplift in the negative direction. Available since version 11.
I_US_PLUS = 2	Uplift in the positive direction. Available since version 11.

Description

Available uplift definitions in one direction.

Version

Available since version 11.

II.3.1.3 Sections

Available since version 2.5.

Enumerations

	Name	Description
	IRobotBarSectionShapeType (see page 546)	A set of section shape types recognized by the program. .

(c) 2014 Autodesk Inc. All rights reserved

	IRobotBarSectionDataValue (see page 549)	A set of identifiers is defined to determine particular parameters (attributes) describing a bar section. Such an identifier is provided as the parameter of a function that gets or sets the value of the relevant section attribute. .
	IRobotBarSectionType (see page 550)	The set of section types recognized by the program has been defined. .
	IRobotBarSectionNonstdDataValue (see page 550)	A set of identifiers describing section parameters is defined for each type of non-standard user-defined section. .
	IRobotBarSectionConcreteDataValue (see page 551)	Characteristic parameters for RC sections. .
	IRobotBarSectionConcreteCutsPosition (see page 552)	Position of cuts on an RC section.
	IRobotBarSectionComponentShape (see page 576)	Shape types of sections being components of a compound section.
	IRobotBarSectionSpecialDataValue (see page 579)	A set of identifiers for parameters of special steel sections. Parameter names come from the section definition dialog box.

Interfaces

	Name	Description
	IRobotBarSectionData (see page 552)	Data defining a bar section. Object of the type is a Data component of each label of the ILT_BAR_SECTION type. .
	IRobotBarSectionNonstdData (see page 563)	Set of data used for describing the cross-section of a non-standard tapered section. .
	IRobotBarSectionConcreteData (see page 565)	Characteristic parameters for an RC member section. .
	IRobotBarSectionElasticParams (see page 569)	Elasto-plastic properties for sections. .
	IRobotBarSectionComplexData (see page 572)	Definition of compound section. .
	IRobotBarSectionSpecialData (see page 576)	Parameters of a special steel section.

II.3.1.3.1 IRobotBarSectionShapeType

C++

```
enum IRobotBarSectionShapeType;
```

C#

```
public enum IRobotBarSectionShapeType;
```

Visual Basic

```
Public Enum IRobotBarSectionShapeType
```

Members

Members	Description
I_BSST_USER_BOX = 91	Available since version 2.5.
I_BSST_USER_RECT = 92	Available since version 2.5.
I_BSST_USER_TUBE = 93	Available since version 2.5.
I_BSST_USER_I_BISYM = 94	Available since version 2.5.
I_BSST_USER_I_MONOSYM = 95	Available since version 2.5.
I_BSST_USER_T_SHAPE = 96	Available since version 2.5.
I_BSST_USER_C_SHAPE = 97	Available since version 2.5.
I_BSST_USER_CROSS = 201	Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

I_BSST_WOOD_RECT = 41	Timber rectangular section. Available since version 2.5.
I_BSST_WOOD_DRECT = 99	Timber double rectangular section. Available since version 2.5.
I_BSST_RECT_FILLED = 43	Steel rectangular solid section. Available since version 2.5.
I_BSST_CIRC_FILLED = 44	Steel round solid section. Available since version 2.5.
I_BSST_CONCR_COL_R = -108	Available since version 2.5.
I_BSST_CONCR_COL_T = -107	Available since version 2.5.
I_BSST_CONCR_COL_L = -106	Available since version 2.5.
I_BSST_CONCR_COL_Z = -105	Available since version 2.5.
I_BSST_CONCR_COL_P = -104	Available since version 2.5.
I_BSST_CONCR_COL_C = -103	Available since version 2.5.
I_BSST_CONCR_COL_CH = -102	Available since version 2.5.
I_BSST_CONCR_COL_CQ = -101	Available since version 2.5.
I_BSST_CONCR_BEAM_RECT = -3	RC rectangular beam . Available since version 2.5.
I_BSST_CONCR_BEAM_T = -2	Available since version 2.5.
I_BSST_CONCR_BEAM = -1	RC beam of any shape . Available since version 2.5.
I_BSST_COMP_2C_FACE = 1101	Two C-sections face to face. Available since version 4.
I_BSST_COMP_2C_BACK = 1102	Two C-sections back to back. Available since version 4.
I_BSST_COMP_2I = 1103	Two I-sections. Available since version 4.
I_BSST_COMP_CI = 1104	C-section and I-section. Available since version 4.
I_BSST_COMP_2LI = 1105	Two angles and one I-section. Available since version 4.
I_BSST_COMP_4L_FACE = 1106	Four angles face to face. Available since version 4.
I_BSST_COMP_4L_BACK = 1107	Four angles with legs back to back. Available since version 4.
I_BSST_COMP_2L_SHORT = 1108	Two angles with shorter legs back to back . Available since version 4.
I_BSST_COMP_2L_LONG = 1109	Two angles with longer legs back to back . Available since version 4.
I_BSST_COMP_2L_CROSS = 1110	Two angles - cross shape. Available since version 4.
I_BSST_CCL = 45	Semi-closed C-section.
I_BSST_URND = 46	Rounded C-section - rotated.
I_BSST_TRND = 47	Rectangular pipe.
I_BSST_CUAP = 48	C-sections set face to face, welded .
I_BSST_WOOD_CIRC = 100	Round timber section - solid.
I_BSST_USER_CIRC_FILLED = 101	User steel section - round, solid.
I_BSST_USER_POLYGONAL = 103	Pipe section - shaped like a regular polygon.
I_BSST_COLD_C_PLUS = 1010	Cold-formed C-section - semi-closed, with bends.

(c) 2014 Autodesk Inc. All rights reserved

I_BSST_COLD_SIGMA_SL = 1011	Cold-formed Sigma SL section.
I_BSST_COLD_SIGMA = 1012	Cold-formed Sigma section.
I_BSST_COLD_Z = 1013	Cold-formed Z-section.
I_BSST_COLD_L_LIPS = 1014	Cold-formed C-section - semi-closed.
I_BSST_COLD_Z_ROT = 1015	Cold-formed Z-section in the local - central coordinate system .
I_BSST_COMP_2L_FACE_SHORT = 1111	Two angles - shorter legs .
I_BSST_COMP_2L_FACE_LONG = 1112	Two angles - longer legs.
I_BSST_COMP_CI_BACK = 1113	C-section and I-section set back to back.
I_BSST_COMP_2C_FACE_WELD = 1201	Two C-sections set face to face, welded.
I_BSST_COMP_2C_BACK_WELD = 1202	Two C-sections set back to back, welded.
I_BSST_COMP_2I_WELD = 1203	Two welded I-sections.
I_BSST_COMP_CI_WELD = 1204	C-section and I-section set face to face - welded.
I_BSST_COMP_2LI_WELD = 1205	Two angles and I-section - welded.
I_BSST_COMP_4L_FACE_WELD = 1206	Four angles with legs set face to face - welded.
I_BSST_COMP_4L_BACK_WELD = 1207	Four angles set in the form of a cross - welded.
I_BSST_COMP_2L_SHORT_WELD = 1208	Two angles - shorter legs, welded.
I_BSST_COMP_2L_LONG_WELD = 1209	Two angles - longer legs, welded.
I_BSST_COMP_2L_CROSS_WELD = 1210	Two angles set in the form of a cross - welded .
I_BSST_SPEC_CORRUGATED_WEB = 104	Section with a corrugated web. Available since version 8.5.
I_BSST_SPEC_CASTELLATED_WEB_HEXAGONAL_OPENINGS = 50	Section with a castellated web (hexagonal openings) . Available since version 8.5.
I_BSST_SPEC_CASTELLATED_WEB_HEXAGONAL_OPENINGS_SHIFTED = 52	Section with a castellated web (hexagonal openings, with spacer plates). Available since version 8.5.
I_BSST_SPEC_CASTELLATED_WEB_ROUND_OPENINGS = 51	Section with a castellated web (round openings). Available since version 8.5.
I_BSST_CONCR_BEAM_I = -4	Available since version 8.7.
I_BSST_SPEC_SFB = 53	Available since version 8.7.
I_BSST_SPEC_IFBA = 54	Available since version 8.7.
I_BSST_SPEC_IFBB = 55	Available since version 8.7.
I_BSST_JOIST_K = 500	Standard joist of the K type. Available since version 9.
I_BSST_JOIST_LH = 501	Standard joist of the LH type. Available since version 9.
I_BSST_JOIST_KCS = 502	Joist of the KCS type in which a shear force is taken into account. Available since version 9.
I_BSST_JOIST_DLH = 503	Standard joist of the DLH type. Available since version 9.
I_BSST_USER_BOX_3 = 105	Available since version 14.5.
I_BSST_JOIST_G = 510	Available since version 14.5.
I_BSST_JOIST_VG = 511	Available since version 14.5.
I_BSST_JOIST_BG = 512	Available since version 14.5.
I_BSST_JOIST_SLH = 504	Available since version 14.7.

(c) 2014 Autodesk Inc. All rights reserved

Description

A set of section shape types recognized by the program. .

II.3.1.3.2 IRobotBarSectionDataValue

C++

```
enum IRobotBarSectionDataValue;
```

C#

```
public enum IRobotBarSectionDataValue;
```

Visual Basic

```
Public Enum IRobotBarSectionDataValue
```

Members

Members	Description
I_BSDV_SURFACE = 10	Painting area (perimeter).
I_BSDV_WEIGHT = 11	Nominal weight per length unit.
I_BSDV_D = 12	Section height - basic (maximum) vertical dimension of a section.
I_BSDV_BF = 13	Section width - basic (maximum) horizontal dimension of a section.
I_BSDV_TW = 14	Web thickness / vertical wall thickness.
I_BSDV_TF = 15	Flange thickness / horizontal wall thickness.
I_BSDV_RA = 16	Fillet radius.
I_BSDV_RI = 17	Fillet radius.
I_BSDV_S = 18	Spacing - distance between section elements (double angles).
I_BSDV_ZY = 19	Section plasticity modulus - bending around Y axis.
I_BSDV_ZZ = 20	Section plasticity modulus - bending around Z axis.
I_BSDV_WX = 21	Section modulus for calculation of torsional stresses.
I_BSDV_WY = 22	Section modulus for calculation of limit shear stresses along Y axis.
I_BSDV_WZ = 23	Section modulus for calculation of limit shear stresses along Z axis.
I_BSDV_GAMMA = 24	Angle between principal and main coordinate system axes.
I_BSDV_IOMEGA = 25	First moment of area.
I_BSDV_P1_LENGTH = 26	Length of plate 1 in the CROSS section type.
I_BSDV_P1_THICKNESS = 27	Thickness of plate 1 in the CROSS section type.
I_BSDV_P2_LENGTH = 28	Length of plate 2 in the CROSS section type.
I_BSDV_P2_THICKNESS = 29	Thickness of plate 2 in the CROSS section type.
I_BSDV_P3_LENGTH = 30	Length of plate 3 in the CROSS section type.
I_BSDV_P3_THICKNESS = 31	Thickness of plate 3 in the CROSS section type.
I_BSDV_P4_LENGTH = 32	Length of plate 4 in the CROSS section type.
I_BSDV_P4_THICKNESS = 33	Thickness of plate 4 in the CROSS section type.
I_BSDV_BF2 = 34	Second section width (compare I_BSDV_BF).
I_BSDV_TF2 = 35	Second flange thickness (compare I_BSDV_TF).
I_BSDV_DIM1 = 36	First dimension.
I_BSDV_DIM2 = 37	Second dimension.
I_BSDV_DIM3 = 38	Third dimension.
I_BSDV_ANGLE1 = 39	Additional angle.

(c) 2014 Autodesk Inc. All rights reserved

I_BSDV_ANGLE2 = 40	Additional angle.

Description

A set of identifiers is defined to determine particular parameters (attributes) describing a bar section. Such an identifier is provided as the parameter of a function that gets or sets the value of the relevant section attribute. .

II.3.1.3.3 IRobotBarSectionType

C++

```
enum IRobotBarSectionType;
```

C#

```
public enum IRobotBarSectionType;
```

Visual Basic

```
Public Enum IRobotBarSectionType
```

Members

Members	Description
I_BST_STANDARD = 0	Standard section (no additional list of data sets describing a tapered section); this section type is used for all sections except for non-standard user tapered sections and compound sections. .
I_BST_COMPLEX = 1000	Compound section. Available since version 4.
I_BST_SPECIAL = 1001	Special sections. Available since version 8.5.
I_BST_JOIST = 1002	Available since version 9.
I_BST_NS_BOX_3 = 20	Available since version 14.5.

Description

The set of section types recognized by the program has been defined. .

II.3.1.3.4 IRobotBarSectionNonstdDataValue

C++

```
enum IRobotBarSectionNonstdDataValue;
```

C#

```
public enum IRobotBarSectionNonstdDataValue;
```

Visual Basic

```
Public Enum IRobotBarSectionNonstdDataValue
```

Members

Members	Description
I_BSNDV_POLYGONAL_D_IS_INT = 3	Available since version 8.7.
I_BSNDV_BOX_3_H = 0	Available since version 14.5.
I_BSNDV_BOX_3_B = 1	Available since version 14.5.
I_BSNDV_BOX_3_TW = 2	Available since version 14.5.
I_BSNDV_BOX_3_TF = 3	Available since version 14.5.
I_BSNDV_BOX_3_B1 = 4	Available since version 14.5.
I_BSNDV_BOX_3_B2 = 5	Available since version 14.5.

(c) 2014 Autodesk Inc. All rights reserved

I_BSNDV_BOX_3_TF2 = 6	Available since version 14.5.

Description

A set of identifiers describing section parameters is defined for each type of non-standard user-defined section. .

II.3.1.3.5 IRobotBarSectionConcreteDataValue

C++

```
enum IRobotBarSectionConcreteDataValue;
```

C#

```
public enum IRobotBarSectionConcreteDataValue;
```

Visual Basic

```
Public Enum IRobotBarSectionConcreteDataValue
```

Members

Members	Description
I_BSCDV_COL_H = 0	Available since version 2.5.
I_BSCDV_COL_B = 1	Available since version 2.5.
I_BSCDV_COL_H1 = 2	Available since version 2.5.
I_BSCDV_COL_L1 = 3	Available since version 2.5.
I_BSCDV_COL_H2 = 4	Available since version 2.5.
I_BSCDV_COL_L2 = 5	Available since version 2.5.
I_BSCDV_COL_DE = 6	Available since version 2.5.
I_BSCDV_COL_N = 7	Available since version 2.5.
I_BSCDV_BEAM_H = 0	Available since version 2.5.
I_BSCDV_BEAM_B = 1	Available since version 2.5.
I_BSCDV_BEAM_PL = 2	Available since version 2.5.
I_BSCDV_BEAM_RL = 3	Available since version 2.5.
I_BSCDV_BEAM_EL1 = 4	Available since version 2.5.
I_BSCDV_BEAM_EL2 = 5	Available since version 2.5.
I_BSCDV_BEAM_PR = 6	Available since version 2.5.
I_BSCDV_BEAM_RR = 7	Available since version 2.5.
I_BSCDV_BEAM_ER1 = 8	Available since version 2.5.
I_BSCDV_BEAM_ER2 = 9	Available since version 2.5.
I_BSCDV_BEAM_HG1 = 10	Available since version 2.5.
I_BSCDV_BEAM_BG1 = 11	Available since version 2.5.
I_BSCDV_BEAM_HG2 = 12	Available since version 2.5.
I_BSCDV_BEAM_BG2 = 13	Available since version 2.5.
I_BSCDV_BEAM_T_H = 0	Available since version 2.5.
I_BSCDV_BEAM_T_B = 1	Available since version 2.5.
I_BSCDV_BEAM_T_HF = 100	Available since version 2.5.
I_BSCDV_BEAM_T_BF = 101	Available since version 2.5.
I_BSCDV_BEAM_RECT_H = 0	Available since version 2.5.
I_BSCDV_BEAM_RECT_B = 1	Available since version 2.5.
I_BSCDV_BEAM_I_B = 1	Available since version 8.7.
I_BSCDV_BEAM_I_H = 0	Available since version 8.7.
I_BSCDV_BEAM_I_B1 = 11	Available since version 8.7.
I_BSCDV_BEAM_I_B2 = 13	Available since version 8.7.

(c) 2014 Autodesk Inc. All rights reserved

I_BSCDV_BEAM_I_HF1 = 10	Available since version 8.7.
I_BSCDV_BEAM_I_HF2 = 12	Available since version 8.7.

Description

Characteristic parameters for RC sections. .

Version

Available since version 2.5.

II.3.1.3.6 IRobotBarSectionConcreteCutsPosition

C++

```
enum IRobotBarSectionConcreteCutsPosition;
```

C#

```
public enum IRobotBarSectionConcreteCutsPosition;
```

Visual Basic

```
Public Enum IRobotBarSectionConcreteCutsPosition
```

Members

Members	Description
I_BSCCP_NONE = 0	No cuts. Available since version 2.5.
I_BSCCP_LEFT = 1	Cuts on the left side. Available since version 2.5.
I_BSCCP_RIGHT = 2	Cuts on the right side. Available since version 2.5.

Description

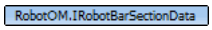
Position of cuts on an RC section.

Version

Available since version 2.5.

II.3.1.3.7 IRobotBarSectionData

Class Hierarchy



C++

```
interface IRobotBarSectionData : IDispatch;
```

C#

```
public interface IRobotBarSectionData;
```

Visual Basic

```
Public Interface IRobotBarSectionData
```

Description

Data defining a bar section. Object of the type is a Data component of each label of the ILT_BAR_SECTION type. .

II.3.1.3.7.1 IRobotBarSectionData Members

The following tables list the members exposed by IRobotBarSectionData.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	AliasCount (see page 555)	Number of additional section names.
	Concrete (see page 555)	RC section parameters.
	ElasticParams (see page 555)	Elasto-plastic properties.
	IsConcrete (see page 555)	Flag indicating if it is an RC section that is dealt with .
	IsJoist (see page 556)	
	IsSpecial (see page 556)	Flag indicating if the user is dealing with a special steel section.
	MaterialName (see page 556)	Name (see page 557) of the material associated with a bar section (material type must be consistent with the section type) .
	Members (see page 557)	Component enabling definition of a compound section; This component is active after previous setting the I_BST_COMPLEX value for the Type (see page 558) field (section type) and for the ShapeType (see page 558) field (shape type) - an appropriate shape available to compound sections. .
	Name (see page 557)	Section name .
	NonstdCount (see page 557)	Number of records describing parameters of non-standard tapered sections (for each section of the I_BST_STANDARD type, the value equals 0) .
	ShapeType (see page 558)	Section shape type .
	Special (see page 558)	Set of data describing a special steel section.
	Type (see page 558)	Section type .

Public Methods

	Name	Description
	CalcNonstdGeometry (see page 559)	The function calculates all the parameters determining section geometry on the basis of the non-standard parameters defined by the user. It should be run for a non-standard tapered section after defining all data sets describing sections in the points of variability. .
	CreateNonstd (see page 559)	The function creates and returns an object (data set) describing the section in a point indicated relatively along the bar (0 means bar beginning, 1 - bar end). The newly-created data structure is added at the end of objects describing a tapered section. The function is used only in the case of non-standard tapered sections. .
	DrawSymbol (see page 560)	Function draws a graphical symbol of a section in the specified area of the window.
	FindAlias (see page 560)	Function searches and returns the additional section name based on a given identifier.
	FindNonstd (see page 560)	The function finds an object describing the cross-section at a bar point indicated relative to its length. It returns the index of the object or zero, when the object is not found. .
	GetAlias (see page 561)	The function returns an additional section name.
	GetAliasEx (see page 561)	Function returns an additional section name and its identifier.
	GetNonstd (see page 561)	A definition of a tapered section consists of a list of objects describing a section in particular points of variability. An index from the range (1 to number of items in the list) is ascribed to each object from the list. The function returns an object with the indicated index. .
	GetValue (see page 562)	The function returns the value of the indicated section parameter. Each parameter describing a section has got its identifier. The set of possible identifiers is defined by the RobotBarSectionDataValue type. .
	LoadFromDBase (see page 562)	The function browses databases for all available sections of the indicated name and reads the data of the found section. If the section of the given name is found and properly read, the function returns the value True (non-zero value); otherwise, it returns False (zero). .

(c) 2014 Autodesk Inc. All rights reserved

	LoadFromDBase2 (see page 562)	Function searches a section of the specified name in the indicated section database, and next reads data of the section found. If a section of the specified name is found and correctly read, function returns the True value (different from zero), otherwise, it returns the False value (zero).
	RemoveNonstd (see page 563)	The function removes the positions with the indicated index from the list of objects (data sets) describing a tapered section. .
	SetValue (see page 563)	The function sets the value of the indicated section parameter. Each parameter describing a section has got its identifier. The set of possible identifiers is defined by the RobotBarSectionDataValue type. .

II.3.1.3.7.2 IRobotBarSectionData Fields

The fields of the IRobotBarSectionData class are listed here.

Public Fields

	Name	Description
	AliasCount (see page 555)	Number of additional section names.
	Concrete (see page 555)	RC section parameters.
	ElasticParams (see page 555)	Elasto-plastic properties.
	IsConcrete (see page 555)	Flag indicating if it is an RC section that is dealt with .
	IsJoist (see page 556)	
	IsSpecial (see page 556)	Flag indicating if the user is dealing with a special steel section.
	MaterialName (see page 556)	Name (see page 557) of the material associated with a bar section (material type must be consistent with the section type) .
	Members (see page 557)	Component enabling definition of a compound section; This component is active after previous setting the I_BST_COMPLEX value for the Type (see page 558) field (section type) and for the ShapeType (see page 558) field (shape type) - an appropriate shape available to compound sections. .
	Name (see page 557)	Section name .
	NonstdCount (see page 557)	Number of records describing parameters of non-standard tapered sections (for each section of the I_BST_STANDARD type, the value equals 0) .
	ShapeType (see page 558)	Section shape type .
	Special (see page 558)	Set of data describing a special steel section.
	Type (see page 558)	Section type .

II.3.1.3.7.2.1 AliasCount

C++

```
HRESULT get_AliasCount(long*);
```

C#

```
public long AliasCount { get; }
```

Visual Basic

```
Public ReadOnly AliasCount As long
```

Description

Number of additional section names.

Version

Available since version 6.1.

(c) 2014 Autodesk Inc. All rights reserved

II.3.1.3.7.2.2 Concrete

C++

```
HRESULT get_Concrete(IRobotBarSectionConcreteData**);
```

C#

```
public IRobotBarSectionConcreteData Concrete { get; }
```

Visual Basic

```
Public ReadOnly Concrete As IRobotBarSectionConcreteData
```

Description

RC section parameters.

Version

Available since version 2.5.

II.3.1.3.7.2.3 ElasticParams

C++

```
HRESULT get_ElasticParams(IRobotBarSectionElasticParams**);
```

C#

```
public IRobotBarSectionElasticParams ElasticParams { get; }
```

Visual Basic

```
Public ReadOnly ElasticParams As IRobotBarSectionElasticParams
```

Description

Elasto-plastic properties.

Version

Available since version 3.5.

II.3.1.3.7.2.4 IsConcrete

C++

```
HRESULT get_IsConcrete(VARIANT_BOOL*);
```

C#

```
public bool IsConcrete { get; }
```

Visual Basic

```
Public ReadOnly IsConcrete As Boolean
```

Description

Flag indicating if it is an RC section that is dealt with .

Version

Available since version 2.5.

II.3.1.3.7.2.5 IsJoist

C++

```
HRESULT get_IsJoist(VARIANT_BOOL*);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public bool IsJoist { get; }
```

Visual Basic

```
Public ReadOnly IsJoist As Boolean
```

Version

Available since version 9.

II.3.1.3.7.2.6 IsSpecial

C++

```
HRESULT get_IsSpecial(VARIANT_BOOL*);
```

C#

```
public bool IsSpecial { get; }
```

Visual Basic

```
Public ReadOnly IsSpecial As Boolean
```

Description

Flag indicating if the user is dealing with a special steel section.

Version

Available since version 8.7.

II.3.1.3.7.2.7 MaterialName

C++

```
HRESULT get_MaterialName(BSTR*);
HRESULT put_MaterialName(BSTR);
```

C#

```
public String MaterialName { get; set; }
```

Visual Basic

```
Public MaterialName As String
```

Description

Name (see page 557) of the material associated with a bar section (material type must be consistent with the section type) .

Version

Available since version 3.

II.3.1.3.7.2.8 Members

C++

```
HRESULT get_Members(IRobotBarSectionComplexData**);
```

C#

```
public IRobotBarSectionComplexData Members { get; }
```

Visual Basic

```
Public ReadOnly Members As IRobotBarSectionComplexData
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Component enabling definition of a compound section; This component is active after previous setting the I_BST_COMPLEX value for the Type (see page 558) field (section type) and for the ShapeType (see page 558) field (shape type) - an appropriate shape available to compound sections. .

Version

Available since version 4.

II.3.1.3.7.2.9 Name

C++

```
HRESULT get_Name(BSTR*);
HRESULT put_Name(BSTR);
```

C#

```
public String Name { get; set; }
```

Visual Basic

```
Public Name As String
```

Description

Section name .

II.3.1.3.7.2.10 NonstdCount

C++

```
HRESULT get_NonstdCount(long*);
```

C#

```
public long NonstdCount { get; }
```

Visual Basic

```
Public ReadOnly NonstdCount As long
```

Description

Number of records describing parameters of non-standard tapered sections (for each section of the I_BST_STANDARD type, the value equals 0) .

II.3.1.3.7.2.11 ShapeType

C++

```
HRESULT get_ShapeType(IRobotBarSectionShapeType*);
HRESULT put_ShapeType(IRobotBarSectionShapeType);
```

C#

```
public IRobotBarSectionShapeType ShapeType { get; set; }
```

Visual Basic

```
Public ShapeType As IRobotBarSectionShapeType
```

Description

Section shape type .

II.3.1.3.7.2.12 Special

C++

```
HRESULT get_Special(IRobotBarSectionSpecialData**);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRobotBarSectionSpecialData Special { get; }
```

Visual Basic

```
Public ReadOnly Special As IRobotBarSectionSpecialData
```

Description

Set of data describing a special steel section.

Version

Available since version 8.7.

II.3.1.3.7.2.13 Type

C++

```
HRESULT get_Type(IRobotBarSectionType*);
HRESULT put_Type(IRobotBarSectionType);
```

C#

```
public IRobotBarSectionType Type { get; set; }
```

Visual Basic

```
Public Type As IRobotBarSectionType
```

Description

Section type .

II.3.1.3.7.3 IRobotBarSectionData Methods

The methods of the IRobotBarSectionData class are listed here.

Public Methods

	Name	Description
	CalcNonstdGeometry (see page 559)	The function calculates all the parameters determining section geometry on the basis of the non-standard parameters defined by the user. It should be run for a non-standard tapered section after defining all data sets describing sections in the points of variability. .
	CreateNonstd (see page 559)	The function creates and returns an object (data set) describing the section in a point indicated relatively along the bar (0 means bar beginning, 1 - bar end). The newly-created data structure is added at the end of objects describing a tapered section. The function is used only in the case of non-standard tapered sections. .
	DrawSymbol (see page 560)	Function draws a graphical symbol of a section in the specified area of the window.
	FindAlias (see page 560)	Function searches and returns the additional section name based on a given identifier.
	FindNonstd (see page 560)	The function finds an object describing the cross-section at a bar point indicated relative to its length. It returns the index of the object or zero, when the object is not found. .
	GetAlias (see page 561)	The function returns an additional section name.
	GetAliasEx (see page 561)	Function returns an additional section name and its identifier.
	GetNonstd (see page 561)	A definition of a tapered section consists of a list of objects describing a section in particular points of variability. An index from the range (1 to number of items in the list) is ascribed to each object from the list. The function returns an object with the indicated index. .

(c) 2014 Autodesk Inc. All rights reserved

	GetValue (see page 562)	The function returns the value of the indicated section parameter. Each parameter describing a section has got its identifier. The set of possible identifiers is defined by the RobotBarSectionDataValue type. .
	LoadFromDBase (see page 562)	The function browses databases for all available sections of the indicated name and reads the data of the found section. If the section of the given name is found and properly read, the function returns the value True (non-zero value); otherwise, it returns False (zero). .
	LoadFromDBase2 (see page 562)	Function searches a section of the specified name in the indicated section database, and next reads data of the section found. If a section of the specified name is found and correctly read, function returns the True value (different from zero), otherwise, it returns the False value (zero).
	RemoveNonstd (see page 563)	The function removes the positions with the indicated index from the list of objects (data sets) describing a tapered section. .
	SetValue (see page 563)	The function sets the value of the indicated section parameter. Each parameter describing a section has got its identifier. The set of possible identifiers is defined by the RobotBarSectionDataValue type. .

II.3.1.3.7.3.1 CalcNonstdGeometry

C++

```
HRESULT CalcNonstdGeometry();
```

C#

```
public void CalcNonstdGeometry();
```

Visual Basic

```
Public Sub CalcNonstdGeometry()
```

Description

The function calculates all the parameters determining section geometry on the basis of the non-standard parameters defined by the user. It should be run for a non-standard tapered section after defining all data sets describing sections in the points of variability. .

II.3.1.3.7.3.2 CreateNonstd

C++

```
HRESULT CreateNonstd(double _relative_pos, IRobotBarSectionNonstdData** ret);
```

C#

```
public IRobotBarSectionNonstdData CreateNonstd(double _relative_pos);
```

Visual Basic

```
Public Function CreateNonstd(_relative_pos As double) As IRobotBarSectionNonstdData
```

Description

The function creates and returns an object (data set) describing the section in a point indicated relatively along the bar (0 means bar beginning, 1 - bar end). The newly-created data structure is added at the end of objects describing a tapered section. The function is used only in the case of non-standard tapered sections. .

II.3.1.3.7.3.3 DrawSymbol

C++

```
HRESULT DrawSymbol(long _window_handle, long _left, long _top, long _right, long _bottom);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public void DrawSymbol(long _window_handle, long _left, long _top, long _right, long
_bottom);
```

Visual Basic

```
Public Sub DrawSymbol(_window_handle As long, _left As long, _top As long, _right As long,
_bottom As long)
```

Description

Function draws a graphical symbol of a section in the specified area of the window.

Version

Available since version 7.5.

II.3.1.3.7.3.4 FindAlias

C++

```
HRESULT FindAlias(BSTR _alias_id, BSTR _alias, VARIANT_BOOL* ret);
```

C#

```
public bool FindAlias(String _alias_id, String _alias);
```

Visual Basic

```
Public Function FindAlias(_alias_id As String, ByRef _alias As String) As Boolean
```

Description

Function searches and returns the additional section name based on a given identifier.

Version

Available since version 7.5.

II.3.1.3.7.3.5 FindNonstd

C++

```
HRESULT FindNonstd(double _relative_pos, long* ret);
```

C#

```
public long FindNonstd(double _relative_pos);
```

Visual Basic

```
Public Function FindNonstd(_relative_pos As double) As long
```

Description

The function finds an object describing the cross-section at a bar point indicated relative to its length. It returns the index of the object or zero, when the object is not found. .

II.3.1.3.7.3.6 GetAlias

C++

```
HRESULT GetAlias(long _idx, BSTR* ret);
```

C#

```
public String GetAlias(long _idx);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Function GetAlias(_idx As long) As String
```

Description

The function returns an additional section name.

Version

Available since version 6.1.

II.3.1.3.7.3.7 GetAliasEx

C++

```
HRESULT GetAliasEx(long _idx, BSTR _alias_id, BSTR* ret);
```

C#

```
public String GetAliasEx(long _idx, String _alias_id);
```

Visual Basic

```
Public Function GetAliasEx(_idx As long, ByRef _alias_id As String) As String
```

Description

Function returns an additional section name and its identifier.

Version

Available since version 7.5.

II.3.1.3.7.3.8 GetNonstd

C++

```
HRESULT GetNonstd(long _index, IRobotBarSectionNonstdData** ret);
```

C#

```
public IRobotBarSectionNonstdData GetNonstd(long _index);
```

Visual Basic

```
Public Function GetNonstd(_index As long) As IRobotBarSectionNonstdData
```

Description

A definition of a tapered section consists of a list of objects describing a section in particular points of variability. An index from the range (1 to number of items in the list) is ascribed to each object from the list. The function returns an object with the indicated index. .

II.3.1.3.7.3.9 GetValue

C++

```
HRESULT GetValue(IRobotBarSectionDataValue _attribute, double* ret);
```

C#

```
public double GetValue(IRobotBarSectionDataValue _attribute);
```

Visual Basic

```
Public Function GetValue(_attribute As IRobotBarSectionDataValue) As double
```

(c) 2014 Autodesk Inc. All rights reserved

Description

The function returns the value of the indicated section parameter. Each parameter describing a section has got its identifier. The set of possible identifiers is defined by the RobotBarSectionDataValue type. .

II.3.1.3.7.3.10 LoadFromDBase

C++

```
HRESULT LoadFromDBase(BSTR _section_name, VARIANT_BOOL* ret);
```

C#

```
public bool LoadFromDBase(String _section_name);
```

Visual Basic

```
Public Function LoadFromDBase(_section_name As String) As Boolean
```

Description

The function browses databases for all available sections of the indicated name and reads the data of the found section. If the section of the given name is found and properly read, the function returns the value True (non-zero value); otherwise, it returns False (zero). .

II.3.1.3.7.3.11 LoadFromDBase2

C++

```
HRESULT LoadFromDBase2(BSTR _section_name, BSTR _dbase_name, VARIANT_BOOL* ret);
```

C#

```
public bool LoadFromDBase2(String _section_name, String _dbase_name);
```

Visual Basic

```
Public Function LoadFromDBase2(_section_name As String, _dbase_name As String) As Boolean
```

Description

Function searches a section of the specified name in the indicated section database, and next reads data of the section found. If a section of the specified name is found and correctly read, function returns the True value (different from zero), otherwise, it returns the False value (zero).

Version

Available since version 7.5.

II.3.1.3.7.3.12 RemoveNonstd

C++

```
HRESULT RemoveNonstd(long _index);
```

C#

```
public void RemoveNonstd(long _index);
```

Visual Basic

```
Public Sub RemoveNonstd(_index As long)
```

Description

The function removes the positions with the indicated index from the list of objects (data sets) describing a tapered section. .

(c) 2014 Autodesk Inc. All rights reserved

II.3.1.3.7.3.13 SetValue

C++

```
HRESULT SetValue(IRobotBarSectionDataValue _attribute, double _value);
```

C#

```
public void SetValue(IRobotBarSectionDataValue _attribute, double _value);
```

Visual Basic

```
Public Sub SetValue(_attribute As IRobotBarSectionDataValue, _value As double)
```

Description

The function sets the value of the indicated section parameter. Each parameter describing a section has got its identifier. The set of possible identifiers is defined by the RobotBarSectionDataValue type. .

II.3.1.3.8 IRobotBarSectionNonstdData

Class Hierarchy

RobotOM.IRobotBarSectionNonstdData

C++

```
interface IRobotBarSectionNonstdData : IDispatch;
```

C#

```
public interface IRobotBarSectionNonstdData;
```

Visual Basic

```
Public Interface IRobotBarSectionNonstdData
```

Description

Set of data used for describing the cross-section of a non-standard tapered section. .

II.3.1.3.8.1 IRobotBarSectionNonstdData Members

The following tables list the members exposed by IRobotBarSectionNonstdData.

Public Fields

	Name	Description
	Position (see page 564)	Relative position of a cross-section along a bar where a tapered section has been applied (0 - bar beginning, 1 - bar end) .

Public Methods

	Name	Description
	GetValue (see page 564)	The function returns tha value of the indicated section parameter. .
	SetValue (see page 564)	The function sets the value of the indicated parameter .

II.3.1.3.8.2 IRobotBarSectionNonstdData Fields

The fields of the IRobotBarSectionNonstdData class are listed here.

Public Fields

	Name	Description
	Position (see page 564)	Relative position of a cross-section along a bar where a tapered section has been applied (0 - bar beginning, 1 - bar end) .

(c) 2014 Autodesk Inc. All rights reserved

II.3.1.3.8.2.1 Position

C++

```
HRESULT get_Position(double*);
```

C#

```
public double Position { get; }
```

Visual Basic

```
Public ReadOnly Position As double
```

Description

Relative position of a cross-section along a bar where a tapered section has been applied (0 - bar beginning, 1 - bar end) .

II.3.1.3.8.3 IRobotBarSectionNonstdData Methods

The methods of the IRobotBarSectionNonstdData class are listed here.

Public Methods

	Name	Description
	GetValue (see page 564)	The function returns tha value of the indicated section parameter. .
	SetValue (see page 564)	The function sets the value of the indicated parameter .

II.3.1.3.8.3.1 GetValue

C++

```
HRESULT GetValue(IRobotBarSectionNonstdDataValue _attribute, double* ret);
```

C#

```
public double GetValue(IRobotBarSectionNonstdDataValue _attribute);
```

Visual Basic

```
Public Function GetValue(_attribute As IRobotBarSectionNonstdDataValue) As double
```

Description

The function returns tha value of the indicated section parameter. .

II.3.1.3.8.3.2 SetValue

C++

```
HRESULT SetValue(IRobotBarSectionNonstdDataValue _attribute, double _value);
```

C#

```
public void SetValue(IRobotBarSectionNonstdDataValue _attribute, double _value);
```

Visual Basic

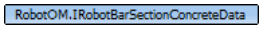
```
Public Sub SetValue(_attribute As IRobotBarSectionNonstdDataValue, _value As double)
```

Description

The function sets the value of the indicated parameter .

II.3.1.3.9 IRobotBarSectionConcreteData

Class Hierarchy



(c) 2014 Autodesk Inc. All rights reserved

C++

```
interface IRobotBarSectionConcreteData : IDispatch;
```

C#

```
public interface IRobotBarSectionConcreteData;
```

Visual Basic

```
Public Interface IRobotBarSectionConcreteData
```

Description

Characteristic parameters for an RC member section. .

Version

Available since version 2.5.

II.3.1.3.9.1 IRobotBarSectionConcreteData Members

The following tables list the members exposed by IRobotBarSectionConcreteData.

Public Fields

	Name	Description
	BeamCutsPosition (see page 566)	Position of cuts on a beam.
	IsBeam (see page 566)	
	IsColumn (see page 566)	

Public Methods

	Name	Description
	CalcGeometry (see page 567)	Function calculates section geometry based on the indicated characteristic values. .
	GetReduction (see page 567)	If coefficients of reduction of the moment of inertia were set earlier, function returns True value and specifies values of coefficients. Otherwise function returns False value.
	GetTapered (see page 568)	Function returns value different from zero (True), if a given section describes an RC beam with variable section width. At the same time, value of the variable section width is entered into the specified parameter. If it is not a section of variable width, then the function returns zero value (False). .
	GetValue (see page 568)	Function returns the value of the selected section parameter. .
	SetReduction (see page 568)	Function sets coefficients of reduction of the moment of inertia.
	SetTapered (see page 569)	Function defines variable section width for an RC beam.
	SetValue (see page 569)	Function sets the value of the selected section parameter. .

II.3.1.3.9.2 IRobotBarSectionConcreteData Fields

The fields of the IRobotBarSectionConcreteData class are listed here.

Public Fields

	Name	Description
	BeamCutsPosition (see page 566)	Position of cuts on a beam.
	IsBeam (see page 566)	
	IsColumn (see page 566)	

(c) 2014 Autodesk Inc. All rights reserved

II.3.1.3.9.2.1 BeamCutsPosition

C++

```
HRESULT get_BeamCutsPosition(IRobotBarSectionConcreteCutsPosition*);
HRESULT put_BeamCutsPosition(IRobotBarSectionConcreteCutsPosition);
```

C#

```
public IRobotBarSectionConcreteCutsPosition BeamCutsPosition { get; set; }
```

Visual Basic

```
Public BeamCutsPosition As IRobotBarSectionConcreteCutsPosition
```

Description

Position of cuts on a beam.

Version

Available since version 2.5.

II.3.1.3.9.2.2 IsBeam

C++

```
HRESULT get_IsBeam(VARIANT_BOOL*);
```

C#

```
public bool IsBeam { get; }
```

Visual Basic

```
Public ReadOnly IsBeam As Boolean
```

Version

Available since version 2.5.

II.3.1.3.9.2.3 IsColumn

C++

```
HRESULT get_IsColumn(VARIANT_BOOL*);
```

C#

```
public bool IsColumn { get; }
```

Visual Basic

```
Public ReadOnly IsColumn As Boolean
```

Version

Available since version 2.5.

II.3.1.3.9.3 IRobotBarSectionConcreteData Methods

The methods of the IRobotBarSectionConcreteData class are listed here.

Public Methods

	Name	Description
	CalcGeometry (see page 567)	Function calculates section geometry based on the indicated characteristic values. .
	GetReduction (see page 567)	If coefficients of reduction of the moment of inertia were set earlier, function returns True value and specifies values of coefficients. Otherwise function returns False value.

(c) 2014 Autodesk Inc. All rights reserved

	GetTapered (see page 568)	Function returns value different from zero (True), if a given section describes an RC beam with variable section width. At the same time, value of the variable section width is entered into the specified parameter. If it is not a section of variable width, then the function returns zero value (False). .
	GetValue (see page 568)	Function returns the value of the selected section parameter. .
	SetReduction (see page 568)	Function sets coefficients of reduction of the moment of inertia.
	SetTapered (see page 569)	Function defines variable section width for an RC beam.
	SetValue (see page 569)	Function sets the value of the selected section parameter. .

II.3.1.3.9.3.1 CalcGeometry

C++

```
HRESULT CalcGeometry();
```

C#

```
public void CalcGeometry();
```

Visual Basic

```
Public Sub CalcGeometry()
```

Description

Function calculates section geometry based on the indicated characteristic values. .

Version

Available since version 2.5.

II.3.1.3.9.3.2 GetReduction

C++

```
HRESULT GetReduction(double* _Ix_cf, double* _Iy_cf, double* _Iz_cf, VARIANT_BOOL* ret);
```

C#

```
public bool GetReduction(double* _Ix_cf, double* _Iy_cf, double* _Iz_cf);
```

Visual Basic

```
Public Function GetReduction(ByRef _Ix_cf As double*, ByRef _Iy_cf As double*, ByRef _Iz_cf
As double*) As Boolean
```

Description

If coefficients of reduction of the moment of inertia were set earlier, function returns True value and specifies values of coefficients. Otherwise function returns False value.

Version

Available since version 8.

II.3.1.3.9.3.3 GetTapered

C++

```
HRESULT GetTapered(double* _h2, VARIANT_BOOL* ret);
```

C#

```
public bool GetTapered(double* _h2);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Function GetTapered(ByRef _h2 As double*) As Boolean
```

Description

Function returns value different from zero (True), if a given section describes an RC beam with variable section width. At the same time, value of the variable section width is entered into the specified parameter. If it is not a section of variable width, then the function returns zero value (False). .

Version

Available since version 2.5.

II.3.1.3.9.3.4 GetValue

C++

```
HRESULT GetValue(IRobotBarSectionConcreteDataValue _param, double* ret);
```

C#

```
public double GetValue(IRobotBarSectionConcreteDataValue _param);
```

Visual Basic

```
Public Function GetValue(_param As IRobotBarSectionConcreteDataValue) As double
```

Description

Function returns the value of the selected section parameter. .

Version

Available since version 2.5.

II.3.1.3.9.3.5 SetReduction

C++

```
HRESULT SetReduction(VARIANT_BOOL _set, double _Ix_cf, double _Iy_cf, double _Iz_cf);
```

C#

```
public void SetReduction(bool _set, double _Ix_cf, double _Iy_cf, double _Iz_cf);
```

Visual Basic

```
Public Sub SetReduction(_set As Boolean, _Ix_cf As double, _Iy_cf As double, _Iz_cf As
double)
```

Description

Function sets coefficients of reduction of the moment of inertia.

Version

Available since version 8.

II.3.1.3.9.3.6 SetTapered

C++

```
HRESULT SetTapered(double _h2);
```

C#

```
public void SetTapered(double _h2);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Sub SetTapered(_h2 As double)
```

Description

Function defines variable section width for an RC beam.

Version

Available since version 2.5.

II.3.1.3.9.3.7 SetValue

C++

```
HRESULT SetValue(IRobotBarSectionConcreteDataValue _param, double _val);
```

C#

```
public void SetValue(IRobotBarSectionConcreteDataValue _param, double _val);
```

Visual Basic

```
Public Sub SetValue(_param As IRobotBarSectionConcreteDataValue, _val As double)
```

Description

Function sets the value of the selected section parameter. .

Version

Available since version 2.5.

II.3.1.3.10 IRobotBarSectionElasticParams

Class Hierarchy

RobotOM.IRobotBarSectionElasticParams

C++

```
interface IRobotBarSectionElasticParams : IDispatch;
```

C#

```
public interface IRobotBarSectionElasticParams;
```

Visual Basic

```
Public Interface IRobotBarSectionElasticParams
```

Description

Elasto-plastic properties for sections. .

Version

Available since version 3.5.

II.3.1.3.10.1 IRobotBarSectionElasticParams Members

The following tables list the members exposed by IRobotBarSectionElasticParams.

Public Fields

	Name	Description
	Active (see page 570)	Flag switching on/off definition of elasto-plastic properties .
	L1 (see page 571)	
	L2 (see page 571)	

(c) 2014 Autodesk Inc. All rights reserved

	MaterialModel (see page 571)	Material model .
	N (see page 571)	
	N1 (see page 572)	
	N2 (see page 572)	

II.3.1.3.10.2 IRobotBarSectionElasticParams Fields

The fields of the IRobotBarSectionElasticParams class are listed here.

Public Fields

	Name	Description
	Active (see page 570)	Flag switching on/off definition of elasto-plastic properties .
	L1 (see page 571)	
	L2 (see page 571)	
	MaterialModel (see page 571)	Material model .
	N (see page 571)	
	N1 (see page 572)	
	N2 (see page 572)	

II.3.1.3.10.2.1 Active

C++

```
HRESULT get_Active(VARIANT_BOOL*);
HRESULT put_Active(VARIANT_BOOL);
```

C#

```
public bool Active { get; set; }
```

Visual Basic

```
Public Active As Boolean
```

Description

Flag switching on/off definition of elasto-plastic properties .

Version

Available since version 3.5.

II.3.1.3.10.2.2 L1

C++

```
HRESULT get_L1(double*);
HRESULT put_L1(double);
```

C#

```
public double L1 { get; set; }
```

Visual Basic

```
Public L1 As double
```

Version

Available since version 3.5.

II.3.1.3.10.2.3 L2

C++

```
HRESULT get_L2(double*);
```

(c) 2014 Autodesk Inc. All rights reserved

```
HRESULT put_L2(double);
```

C#

```
public double L2 { get; set; }
```

Visual Basic

```
Public L2 As double
```

Version

Available since version 3.5.

II.3.1.3.10.2.4 MaterialModel

C++

```
HRESULT get_MaterialModel(IRobotMaterialElasticModel**);
```

C#

```
public IRobotMaterialElasticModel MaterialModel { get; }
```

Visual Basic

```
Public ReadOnly MaterialModel As IRobotMaterialElasticModel
```

Description

Material model .

Version

Available since version 3.5.

II.3.1.3.10.2.5 N

C++

```
HRESULT get_N(double*);
HRESULT put_N(double);
```

C#

```
public double N { get; set; }
```

Visual Basic

```
Public N As double
```

Version

Available since version 3.5.

II.3.1.3.10.2.6 N1

C++

```
HRESULT get_N1(double*);
HRESULT put_N1(double);
```

C#

```
public double N1 { get; set; }
```

Visual Basic

```
Public N1 As double
```

Version

Available since version 3.5.

(c) 2014 Autodesk Inc. All rights reserved

II.3.1.3.10.2.7 N2

C++

```
HRESULT get_N2(double*);
HRESULT put_N2(double);
```

C#

```
public double N2 { get; set; }
```

Visual Basic

```
Public N2 As double
```

Version

Available since version 3.5.

II.3.1.3.11 IRobotBarSectionComplexData

Class Hierarchy

RobotOM.IRobotBarSectionComplexData

C++

```
interface IRobotBarSectionComplexData : IDispatch;
```

C#

```
public interface IRobotBarSectionComplexData;
```

Visual Basic

```
Public Interface IRobotBarSectionComplexData
```

Description

Definition of compound section. .

Version

Available since version 4.

II.3.1.3.11.1 IRobotBarSectionComplexData Members

The following tables list the members exposed by IRobotBarSectionComplexData.

Public Fields

	Name	Description
	B (see page 573)	Spacing b.
	Count (see page 574)	Number of different simple sections composing a compound section; This number depends on the selected compound section shape - the ShapeType element of the IRobotBarSectionData (see page 552) interface. .
	D (see page 574)	Spacing d.

Public Methods

	Name	Description
	Get (see page 574)	Function returns section database name and section name for the indicated shape identifier. .
	GetShape (see page 575)	Function returns the shape identifier for a simple section being a component of a compound section.
	GetValue (see page 575)	Function returns value of the selected property for the indicated simple section. .

(c) 2014 Autodesk Inc. All rights reserved

	Set (see page 575)	Function sets section database name and section name for the indicated shape identifier.

II.3.1.3.11.2 IRobotBarSectionComplexData Fields

The fields of the IRobotBarSectionComplexData class are listed here.

Public Fields

	Name	Description
	B (see page 573)	Spacing b.
	Count (see page 574)	Number of different simple sections composing a compound section; This number depends on the selected compound section shape - the ShapeType element of the IRobotBarSectionData (see page 552) interface. .
	D (see page 574)	Spacing d.

II.3.1.3.11.2.1 B

C++

```
HRESULT get_B(double*);
HRESULT put_B(double);
```

C#

```
public double B { get; set; }
```

Visual Basic

```
Public B As double
```

Description

Spacing b.

Version

Available since version 4.

II.3.1.3.11.2.2 Count

C++

```
HRESULT get_Count(long*);
```

C#

```
public long Count { get; }
```

Visual Basic

```
Public ReadOnly Count As long
```

Description

Number of different simple sections composing a compound section; This number depends on the selected compound section shape - the ShapeType element of the IRobotBarSectionData (see page 552) interface. .

Version

Available since version 4.

II.3.1.3.11.2.3 D

C++

```
HRESULT get_D(double*);
HRESULT put_D(double);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double D { get; set; }
```

Visual Basic

```
Public D As double
```

Description

Spacing d.

Version

Available since version 4.

II.3.1.3.11.3 IRobotBarSectionComplexData Methods

The methods of the IRobotBarSectionComplexData class are listed here.

Public Methods

	Name	Description
	Get (see page 574)	Function returns section database name and section name for the indicated shape identifier. .
	GetShape (see page 575)	Function returns the shape identifier for a simple section being a component of a compound section.
	GetValue (see page 575)	Function returns value of the selected property for the indicated simple section. .
	Set (see page 575)	Function sets section database name and section name for the indicated shape identifier.

II.3.1.3.11.3.1 Get

C++

```
HRESULT Get(IRobotBarSectionComponentShape _shape, BSTR _database_name, BSTR _section_name,
VARIANT_BOOL* ret);
```

C#

```
public bool Get(IRobotBarSectionComponentShape _shape, String _database_name, String
_section_name);
```

Visual Basic

```
Public Function Get(_shape As IRobotBarSectionComponentShape, ByRef _database_name As
String, ByRef _section_name As String) As Boolean
```

Description

Function returns section database name and section name for the indicated shape identifier. .

Version

Available since version 4.

II.3.1.3.11.3.2 GetShape

C++

```
HRESULT GetShape(long _idx, IRobotBarSectionComponentShape* ret);
```

C#

```
public IRobotBarSectionComponentShape GetShape(long _idx);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Function GetShape(_idx As long) As IRobotBarSectionComponentShape
```

Description

Function returns the shape identifier for a simple section being a component of a compound section.

Version

Available since version 4.

II.3.1.3.11.3.3 GetValue

C++

```
HRESULT GetValue(IRobotBarSectionComponentShape _shape, IRobotBarSectionDataValue _val_id,
double* ret);
```

C#

```
public double GetValue(IRobotBarSectionComponentShape _shape, IRobotBarSectionDataValue
_val_id);
```

Visual Basic

```
Public Function GetValue(_shape As IRobotBarSectionComponentShape, _val_id As
IRobotBarSectionDataValue) As double
```

Description

Function returns value of the selected property for the indicated simple section. .

Version

Available since version 4.

II.3.1.3.11.3.4 Set

C++

```
HRESULT Set(IRobotBarSectionComponentShape _shape, BSTR _database_name, BSTR _section_name,
VARIANT_BOOL* ret);
```

C#

```
public bool Set(IRobotBarSectionComponentShape _shape, String _database_name, String
_section_name);
```

Visual Basic

```
Public Function Set(_shape As IRobotBarSectionComponentShape, _database_name As String,
_section_name As String) As Boolean
```

Description

Function sets section database name and section name for the indicated shape identifier.

Version

Available since version 4.

II.3.1.3.12 IRobotBarSectionComponentShape

C++

```
enum IRobotBarSectionComponentShape;
```

C#

```
public enum IRobotBarSectionComponentShape;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Enum IRobotBarSectionComponentShape
```

Members

Members	Description
I_BSCS_C = 1	C-section. Available since version 4.
I_BSCS_I = 2	I-section. Available since version 4.
I_BSCS_L = 3	Angle. Available since version 4.
I_BSCS_UNDEFINED = 0	Available since version 4.

Description

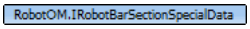
Shape types of sections being components of a compound section.

Version

Available since version 4.

II.3.1.3.13 IRobotBarSectionSpecialData

Class Hierarchy



C++

```
interface IRobotBarSectionSpecialData : IDispatch;
```

C#

```
public interface IRobotBarSectionSpecialData;
```

Visual Basic

```
Public Interface IRobotBarSectionSpecialData
```

Description

Parameters of a special steel section.

Version

Available since version 8.7.

II.3.1.3.13.1 IRobotBarSectionSpecialData Members

The following tables list the members exposed by IRobotBarSectionSpecialData.

Public Fields

	Name	Description
	DbFullName (see page 577)	Complete section database name.
	DbName (see page 578)	Section (see page 578) database name.
	Section (see page 578)	Section name.

Public Methods

	Name	Description
	GetValue (see page 578)	Function returns a value of the specified section parameter.
	SetValue (see page 579)	Function sets a value of the specified section parameter.

(c) 2014 Autodesk Inc. All rights reserved

II.3.1.3.13.2 IRobotBarSectionSpecialData Fields

The fields of the IRobotBarSectionSpecialData class are listed here.

Public Fields

	Name	Description
	DbFullName (see page 577)	Complete section database name.
	DbName (see page 578)	Section (see page 578) database name.
	Section (see page 578)	Section name.

II.3.1.3.13.2.1 DbFullName

C++

```
HRESULT get_DbFullName(BSTR*);
HRESULT put_DbFullName(BSTR);
```

C#

```
public String DbFullName { get; set; }
```

Visual Basic

```
Public DbFullName As String
```

Description

Complete section database name.

Version

Available since version 8.7.

II.3.1.3.13.2.2 DbName

C++

```
HRESULT get_DbName(BSTR*);
HRESULT put_DbName(BSTR);
```

C#

```
public String DbName { get; set; }
```

Visual Basic

```
Public DbName As String
```

Description

Section (see page 578) database name.

Version

Available since version 8.7.

II.3.1.3.13.2.3 Section

C++

```
HRESULT get_Section(BSTR*);
HRESULT put_Section(BSTR);
```

C#

```
public String Section { get; set; }
```

Visual Basic

```
Public Section As String
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Section name.

Version

Available since version 8.7.

II.3.1.3.13.3 IRobotBarSectionSpecialData Methods

The methods of the IRobotBarSectionSpecialData class are listed here.

Public Methods

	Name	Description
	GetValue (see page 578)	Function returns a value of the specified section parameter.
	SetValue (see page 579)	Function sets a value of the specified section parameter.

II.3.1.3.13.3.1 GetValue

C++

```
HRESULT GetValue(IRobotBarSectionSpecialDataValue _param_id, double* ret);
```

C#

```
public double GetValue(IRobotBarSectionSpecialDataValue _param_id);
```

Visual Basic

```
Public Function GetValue(_param_id As IRobotBarSectionSpecialDataValue) As double
```

Description

Function returns a value of the specified section parameter.

Version

Available since version 8.7.

II.3.1.3.13.3.2 SetValue

C++

```
HRESULT SetValue(IRobotBarSectionSpecialDataValue _param_id, double _param_value);
```

C#

```
public void SetValue(IRobotBarSectionSpecialDataValue _param_id, double _param_value);
```

Visual Basic

```
Public Sub SetValue(_param_id As IRobotBarSectionSpecialDataValue, _param_value As double)
```

Description

Function sets a value of the specified section parameter.

Version

Available since version 8.7.

II.3.1.3.14 IRobotBarSectionSpecialDataValue

C++

```
enum IRobotBarSectionSpecialDataValue;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public enum IRobotBarSectionSpecialDataValue;
```

Visual Basic

```
Public Enum IRobotBarSectionSpecialDataValue
```

Members

Members	Description
I_BSSDV_TW = 1	Available since version 8.7.
I_BSSDV_B1 = 2	Available since version 8.7.
I_BSSDV_TF1 = 3	Available since version 8.7.
I_BSSDV_H = 4	Available since version 8.7.
I_BSSDV_B2 = 5	Available since version 8.7.
I_BSSDV_TF2 = 6	Available since version 8.7.
I_BSSDV_D = 7	Available since version 8.7.
I_BSSDV_W = 8	Available since version 8.7.
I_BSSDV_C = 9	Available since version 8.7.
I_BSSDV_A = 10	Available since version 8.7.
I_BSSDV_HS = 11	Available since version 8.7.
I_BSSDV_BP = 12	Available since version 8.7.
I_BSSDV_TP = 13	Available since version 8.7.

Description

A set of identifiers for parameters of special steel sections. Parameter names come from the section definition dialog box.

Version

Available since version 8.7.

II.3.1.4 Releases

Available since version 2.5.

Enumerations

	Name	Description
	IRobotBarEndReleaseValue (see page 580)	Set of value that can be assumed by the parameters of a release at bar end. .

Interfaces

	Name	Description
	IRobotBarReleaseData (see page 581)	The set of data describing a release defined for a bar. The definition of the release consists of two sets of data - each one for each bar end. The data describing the entire release are stored as a data component of the labels of the following type: ILT_BAR_RELEASE. .
	IRobotBarEndReleaseData (see page 582)	Set of parameters defining a release at one bar end. The allowable values are determined by the RobotBarEndReleaseValue type. .

II.3.1.4.1 IRobotBarEndReleaseValue

C++

```
enum IRobotBarEndReleaseValue;
```

C#

```
public enum IRobotBarEndReleaseValue;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Enum IRobotBarEndReleaseValue
```

Members

Members	Description
I_BERV_NONE = 0	None .
I_BERV_ELASTIC = 4	Available since version 1.7.
I_BERV_ELASTIC_PLUS = 5	Available since version 1.7.
I_BERV_ELASTIC_MINUS = 6	Available since version 1.7.
I_BERV_NONLINEAR = 7	Non-linear release defined by the function. Available since version 9.
I_BERV_ELASTIC_REDUCED = 8	Elastic release defined by a partial rigidity coefficient. Available since version 9.
I_BERV_ELASTIC_REDUCED_PLUS = 9	Elastic release defined by a partial rigidity coefficient, unilateral plus. Available since version 9.
I_BERV_ELASTIC_REDUCED_MINUS = 10	Elastic release defined by a partial rigidity coefficient, unilateral minus. Available since version 9.

Description

Set of value that can be assumed by the parameters of a release at bar end. .

II.3.1.4.2 IRobotBarReleaseData

Class Hierarchy

RobotOM.IRobotBarReleaseData

C++

```
interface IRobotBarReleaseData : IDispatch;
```

C#

```
public interface IRobotBarReleaseData;
```

Visual Basic

```
Public Interface IRobotBarReleaseData
```

Description

The set of data describing a release defined for a bar. The definition of the release consists of two sets of data - each one for each bar end. The data describing the entire release are stored as a data component of the labels of the following type: ILT_BAR_RELEASE. .

II.3.1.4.2.1 IRobotBarReleaseData Members

The following tables list the members exposed by IRobotBarReleaseData.

Public Fields

	Name	Description
	EndNode (see page 581)	Data describing a release at bar end .
	StartNode (see page 582)	Data describing a release at bar beginning .

II.3.1.4.2.2 IRobotBarReleaseData Fields

The fields of the IRobotBarReleaseData class are listed here.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	EndNode (see page 581)	Data describing a release at bar end .
	StartNode (see page 582)	Data describing a release at bar beginning .

II.3.1.4.2.2.1 EndNode

C++

```
HRESULT get_EndNode(IRobotBarEndReleaseData**);
HRESULT put_EndNode(IRobotBarEndReleaseData*);
```

C#

```
public IRobotBarEndReleaseData EndNode { get; set; }
```

Visual Basic

```
Public EndNode As IRobotBarEndReleaseData
```

Description

Data describing a release at bar end .

II.3.1.4.2.2.2 StartNode

C++

```
HRESULT get_StartNode(IRobotBarEndReleaseData**);
HRESULT put_StartNode(IRobotBarEndReleaseData*);
```

C#

```
public IRobotBarEndReleaseData StartNode { get; set; }
```

Visual Basic

```
Public StartNode As IRobotBarEndReleaseData
```

Description

Data describing a release at bar beginning .

II.3.1.4.3 IRobotBarEndReleaseData

Class Hierarchy

RobotOM.IRobotBarEndReleaseData

C++

```
interface IRobotBarEndReleaseData : IDispatch;
```

C#

```
public interface IRobotBarEndReleaseData;
```

Visual Basic

```
Public Interface IRobotBarEndReleaseData
```

Description

Set of parameters defining a release at one bar end. The allowable values are determined by the RobotBarEndReleaseValue type. .

II.3.1.4.3.1 IRobotBarEndReleaseData Members

The following tables list the members exposed by IRobotBarEndReleaseData.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	AX (see page 583)	Damping coefficient (x axis direction).
	AY (see page 584)	Damping coefficient (y axis direction) .
	AZ (see page 584)	Damping coefficient (z axis direction) .
	BX (see page 584)	Bx damping coefficient .
	BY (see page 585)	By damping coefficient .
	BZ (see page 585)	Bz damping coefficient .
	HX (see page 585)	Available since version 1.7.
	HY (see page 585)	Available since version 1.7.
	HZ (see page 586)	Available since version 1.7.
	KX (see page 586)	Available since version 1.7.
	KY (see page 586)	Available since version 1.7.
	KZ (see page 586)	Available since version 1.7.
	NonlinearModel (see page 587)	Non-linearity support.
	RX (see page 587)	
	RY (see page 587)	
	RZ (see page 587)	
	UX (see page 588)	
	UY (see page 588)	
	UZ (see page 588)	

II.3.1.4.3.2 IRobotBarEndReleaseData Fields

The fields of the IRobotBarEndReleaseData class are listed here.

Public Fields

	Name	Description
	AX (see page 583)	Damping coefficient (x axis direction).
	AY (see page 584)	Damping coefficient (y axis direction) .
	AZ (see page 584)	Damping coefficient (z axis direction) .
	BX (see page 584)	Bx damping coefficient .
	BY (see page 585)	By damping coefficient .
	BZ (see page 585)	Bz damping coefficient .
	HX (see page 585)	Available since version 1.7.
	HY (see page 585)	Available since version 1.7.
	HZ (see page 586)	Available since version 1.7.
	KX (see page 586)	Available since version 1.7.
	KY (see page 586)	Available since version 1.7.
	KZ (see page 586)	Available since version 1.7.
	NonlinearModel (see page 587)	Non-linearity support.
	RX (see page 587)	
	RY (see page 587)	
	RZ (see page 587)	
	UX (see page 588)	
	UY (see page 588)	
	UZ (see page 588)	

(c) 2014 Autodesk Inc. All rights reserved

II.3.1.4.3.2.1 AX

C++

```
HRESULT get_AX( double*);
HRESULT put_AX( double);
```

C#

```
public  double AX { get; set; }
```

Visual Basic

```
Public AX As  double
```

Description

Damping coefficient (x axis direction).

Version

Available since version 2.5.

II.3.1.4.3.2.2 AY

C++

```
HRESULT get_AY( double*);
HRESULT put_AY( double);
```

C#

```
public  double AY { get; set; }
```

Visual Basic

```
Public AY As  double
```

Description

Damping coefficient (y axis direction) .

Version

Available since version 2.5.

II.3.1.4.3.2.3 AZ

C++

```
HRESULT get_AZ( double*);
HRESULT put_AZ( double);
```

C#

```
public  double AZ { get; set; }
```

Visual Basic

```
Public AZ As  double
```

Description

Damping coefficient (z axis direction) .

Version

Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

II.3.1.4.3.2.4 BX

C++

```
HRESULT get_BX( double*);
HRESULT put_BX( double);
```

C#

```
public  double BX { get; set; }
```

Visual Basic

```
Public BX As  double
```

Description

Bx damping coefficient .

Version

Available since version 2.5.

II.3.1.4.3.2.5 BY

C++

```
HRESULT get_BY( double*);
HRESULT put_BY( double);
```

C#

```
public  double BY { get; set; }
```

Visual Basic

```
Public BY As  double
```

Description

By damping coefficient .

Version

Available since version 2.5.

II.3.1.4.3.2.6 BZ

C++

```
HRESULT get_BZ( double*);
HRESULT put_BZ( double);
```

C#

```
public  double BZ { get; set; }
```

Visual Basic

```
Public BZ As  double
```

Description

Bz damping coefficient .

Version

Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

II.3.1.4.3.2.7 HX

C++

```
HRESULT get_HX(double*);
HRESULT put_HX(double);
```

C#

```
public double HX { get; set; }
```

Visual Basic

```
Public HX As double
```

Description

Available since version 1.7.

II.3.1.4.3.2.8 HY

C++

```
HRESULT get_HY(double*);
HRESULT put_HY(double);
```

C#

```
public double HY { get; set; }
```

Visual Basic

```
Public HY As double
```

Description

Available since version 1.7.

II.3.1.4.3.2.9 HZ

C++

```
HRESULT get_HZ(double*);
HRESULT put_HZ(double);
```

C#

```
public double HZ { get; set; }
```

Visual Basic

```
Public HZ As double
```

Description

Available since version 1.7.

II.3.1.4.3.2.10 KX

C++

```
HRESULT get_KX(double*);
HRESULT put_KX(double);
```

C#

```
public double KX { get; set; }
```

Visual Basic

```
Public KX As double
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Available since version 1.7.

II.3.1.4.3.2.11 KY

C++

```
HRESULT get_KY(double*);
HRESULT put_KY(double);
```

C#

```
public double KY { get; set; }
```

Visual Basic

```
Public KY As double
```

Description

Available since version 1.7.

II.3.1.4.3.2.12 KZ

C++

```
HRESULT get_KZ(double*);
HRESULT put_KZ(double);
```

C#

```
public double KZ { get; set; }
```

Visual Basic

```
Public KZ As double
```

Description

Available since version 1.7.

II.3.1.4.3.2.13 NonlinearModel

C++

```
HRESULT get_NonlinearModel(IRobotNonlinearLinkMngr**);
```

C#

```
public IRobotNonlinearLinkMngr NonlinearModel { get; }
```

Visual Basic

```
Public ReadOnly NonlinearModel As IRobotNonlinearLinkMngr
```

Description

Non-linearity support.

Version

Available since version 3.

II.3.1.4.3.2.14 RX

C++

```
HRESULT get_RX(IRobotBarEndReleaseValue*);
HRESULT put_RX(IRobotBarEndReleaseValue);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRobotBarEndReleaseValue RX { get; set; }
```

Visual Basic

```
Public RX As IRobotBarEndReleaseValue
```

II.3.1.4.3.2.15 RY

C++

```
HRESULT get_RY(IRobotBarEndReleaseValue*);
HRESULT put_RY(IRobotBarEndReleaseValue);
```

C#

```
public IRobotBarEndReleaseValue RY { get; set; }
```

Visual Basic

```
Public RY As IRobotBarEndReleaseValue
```

II.3.1.4.3.2.16 RZ

C++

```
HRESULT get_RZ(IRobotBarEndReleaseValue*);
HRESULT put_RZ(IRobotBarEndReleaseValue);
```

C#

```
public IRobotBarEndReleaseValue RZ { get; set; }
```

Visual Basic

```
Public RZ As IRobotBarEndReleaseValue
```

II.3.1.4.3.2.17 UX

C++

```
HRESULT get_UX(IRobotBarEndReleaseValue*);
HRESULT put_UX(IRobotBarEndReleaseValue);
```

C#

```
public IRobotBarEndReleaseValue UX { get; set; }
```

Visual Basic

```
Public UX As IRobotBarEndReleaseValue
```

II.3.1.4.3.2.18 UY

C++

```
HRESULT get_UY(IRobotBarEndReleaseValue*);
HRESULT put_UY(IRobotBarEndReleaseValue);
```

C#

```
public IRobotBarEndReleaseValue UY { get; set; }
```

Visual Basic

```
Public UY As IRobotBarEndReleaseValue
```

II.3.1.4.3.2.19 UZ

C++

```
HRESULT get_UZ(IRobotBarEndReleaseValue*);
HRESULT put_UZ(IRobotBarEndReleaseValue);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRobotBarEndReleaseValue UZ { get; set; }
```

Visual Basic

```
Public UZ As IRobotBarEndReleaseValue
```

II.3.1.5 Geometrical imperfections

Available since version 3.

Enumerations

	Name	Description
	IRobotBarGeoImperfectionsAxis (see page 592)	

Interfaces

	Name	Description
	IRobotBarGeoImperfectionsData (see page 589)	Data set describing bar geometrical imperfections. .

II.3.1.5.1 IRobotBarGeoImperfectionsData

Class Hierarchy

RobotOM.IRobotBarGeoImperfectionsData

C++

```
interface IRobotBarGeoImperfectionsData : IDispatch;
```

C#

```
public interface IRobotBarGeoImperfectionsData;
```

Visual Basic

```
Public Interface IRobotBarGeoImperfectionsData
```

Description

Data set describing bar geometrical imperfections. .

Version

Available since version 3.

II.3.1.5.1.1 IRobotBarGeoImperfectionsData Members

The following tables list the members exposed by IRobotBarGeoImperfectionsData.

Public Methods

	Name	Description
	GetBucklingCoeff (see page 590)	Function returns value of the buckling length coefficient for the selected direction. This value is relevant only if deflection is calulated automatically according to EC3.
	GetUser (see page 590)	Function returns the user-defined value of initial deflection for the indicated axis. .
	IsAutomatic (see page 590)	Function returns a non-zero value (True), if for a selected axis the initial deflection is calculated automatically according to EC3. .
	IsMinus (see page 591)	Function returns a non-zero value (True), if for the selected axis the negative deflection direction has been set. .
	SetAutomatic (see page 591)	Function allows determining - for a selected axis - if initial deflection is to be calculated automatically according to EC3, or not.

(c) 2014 Autodesk Inc. All rights reserved

	SetMinus (see page 591)	Function enables change of the deflection direction for the indicated axis.
	SetUser (see page 592)	Function sets the user-defined value of initial deflection for the indicated axis.

II.3.1.5.1.2 IRobotBarGeoImperfectionsData Methods

The methods of the IRobotBarGeoImperfectionsData class are listed here.

Public Methods

	Name	Description
	GetBucklingCoeff (see page 590)	Function returns value of the buckling length coefficient for the selected direction. This value is relevant only if deflection is calulated automatically according to EC3.
	GetUser (see page 590)	Function returns the user-defined value of initial deflection for the indicated axis. .
	IsAutomatic (see page 590)	Function returns a non-zero value (True), if for a selected axis the initial deflection is calculated automatically according to EC3. .
	IsMinus (see page 591)	Function returns a non-zero value (True), if for the selected axis the negative deflection direction has been set. .
	SetAutomatic (see page 591)	Function allows determining - for a selected axis - if initial deflection is to be calculated automatically according to EC3, or not.
	SetMinus (see page 591)	Function enables change of the deflection direction for the indicated axis.
	SetUser (see page 592)	Function sets the user-defined value of initial deflection for the indicated axis.

II.3.1.5.1.2.1 GetBucklingCoeff

C++

```
HRESULT GetBucklingCoeff(RobotBarGeoImperfectionAxis _axis, double* ret);
```

C#

```
public double GetBucklingCoeff(RobotBarGeoImperfectionAxis _axis);
```

Visual Basic

```
Public Function GetBucklingCoeff(_axis As RobotBarGeoImperfectionAxis) As double
```

Description

Function returns value of the buckling length coefficient for the selected direction. This value is relevant only if deflection is calulated automatically according to EC3.

Version

Available since version 3.

II.3.1.5.1.2.2 GetUser

C++

```
HRESULT GetUser(RobotBarGeoImperfectionAxis _axis, double* ret);
```

C#

```
public double GetUser(RobotBarGeoImperfectionAxis _axis);
```

Visual Basic

```
Public Function GetUser(_axis As RobotBarGeoImperfectionAxis) As double
```

Description

Function returns the user-defined value of initial deflection for the indicated axis. .

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.

II.3.1.5.1.2.3 IsAutomatic

C++

```
HRESULT IsAutomatic(RobotBarGeoImperfectionAxis _axis, VARIANT_BOOL* ret);
```

C#

```
public bool IsAutomatic(RobotBarGeoImperfectionAxis _axis);
```

Visual Basic

```
Public Function IsAutomatic(_axis As RobotBarGeoImperfectionAxis) As Boolean
```

Description

Function returns a non-zero value (True), if for a selected axis the initial deflection is calculated automatically according to EC3. .

Version

Available since version 3.

II.3.1.5.1.2.4 IsMinus

C++

```
HRESULT IsMinus(RobotBarGeoImperfectionAxis _axis, VARIANT_BOOL* ret);
```

C#

```
public bool IsMinus(RobotBarGeoImperfectionAxis _axis);
```

Visual Basic

```
Public Function IsMinus(_axis As RobotBarGeoImperfectionAxis) As Boolean
```

Description

Function returns a non-zero value (True), if for the selected axis the negative deflection direction has been set. .

Version

Available since version 3.

II.3.1.5.1.2.5 SetAutomatic

C++

```
HRESULT SetAutomatic(RobotBarGeoImperfectionAxis _axis, VARIANT_BOOL _automatic = True,
double _coeff = 1.0);
```

C#

```
public void SetAutomatic(RobotBarGeoImperfectionAxis _axis, bool _automatic = True, double
_coeff = 1.0);
```

Visual Basic

```
Public Sub SetAutomatic(_axis As RobotBarGeoImperfectionAxis, Optional _automatic As
Boolean = True, Optional _coeff As double = 1.0)
```

Description

Function allows determining - for a selected axis - if initial deflection is to be calculated automatically according to EC3, or not.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.

II.3.1.5.1.2.6 SetMinus

C++

```
HRESULT SetMinus(RobotBarGeoImperfectionAxis _axis, VARIANT_BOOL _minus = True);
```

C#

```
public void SetMinus(RobotBarGeoImperfectionAxis _axis, bool _minus = True);
```

Visual Basic

```
Public Sub SetMinus(_axis As RobotBarGeoImperfectionAxis, Optional _minus As Boolean = True)
```

Description

Function enables change of the deflection direction for the indicated axis.

Version

Available since version 3.

II.3.1.5.1.2.7 SetUser

C++

```
HRESULT SetUser(RobotBarGeoImperfectionAxis _axis, double _val);
```

C#

```
public void SetUser(RobotBarGeoImperfectionAxis _axis, double _val);
```

Visual Basic

```
Public Sub SetUser(_axis As RobotBarGeoImperfectionAxis, _val As double)
```

Description

Function sets the user-defined value of initial deflection for the indicated axis.

Version

Available since version 3.

II.3.1.5.2 IRobotBarGeoImperfectionsAxis

C++

```
enum IRobotBarGeoImperfectionsAxis;
```

C#

```
public enum IRobotBarGeoImperfectionsAxis;
```

Visual Basic

```
Public Enum IRobotBarGeoImperfectionsAxis
```

Members

Members	Description
I_BGIA_Z = 1	Local Z axis. Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

I_BGIA_Y = 0	Local Y axis. Available since version 3.

Version

Available since version 3.

II.3.1.6 Non-linear hinges

Available since version 3.

Enumerations

	Name	Description
	IRobotNonlinearHingeModelType (see page 603)	Types of non-linear hinge models.
	IRobotNonlinearHingeModelUnloadingType (see page 606)	Types of unloading methods.
	IRobotNonlinearHingeComponentType (see page 609)	Types of components characterizing the non-linear hinge.

Interfaces

	Name	Description
	IRobotNonlinearHingeModelServer (see page 593)	Server of non-linear hinge models. .
	IRobotNonlinearHingeModel (see page 595)	Definition of non-linear hinge model.
	IRobotNonlinearHingeModelAxisParams (see page 601)	Interface of parameters whose values depend on the semi-axis type.
	IRobotNonlinearHingeModelPoints (see page 604)	Interface for definition of curve points.
	IRobotNonlinearHingeData (see page 606)	Interface of non-linear hinge definition.
	IRobotNonlinearHingeDef (see page 609)	Interface describing a hinge defined in the structure. .
	IRobotNonlinearHingeServer (see page 612)	Server providing support for hinge definitions in a structure. .

II.3.1.6.1 IRobotNonlinearHingeModelServer

Class Hierarchy

RobotOM.IRobotNonlinearHingeModelServer

C++

```
interface IRobotNonlinearHingeModelServer : IDispatch;
```

C#

```
public interface IRobotNonlinearHingeModelServer;
```

Visual Basic

```
Public Interface IRobotNonlinearHingeModelServer
```

Description

Server of non-linear hinge models. .

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

II.3.1.6.1.1 IRobotNonlinearHingeModelServer Members

The following tables list the members exposed by IRobotNonlinearHingeModelServer.

Public Fields

	Name	Description
	Count (see page 594)	Number of available models.

Public Methods

	Name	Description
	Create (see page 594)	Function creates a new non-linear model of a hinge.
	Delete (see page 595)	Function deletes model of the indicated index.
	Find (see page 595)	Function returns the model index based on the specified name.
	Get (see page 595)	Function takes a non-linear hinge model of the indicated index. .

II.3.1.6.1.2 IRobotNonlinearHingeModelServer Fields

The fields of the IRobotNonlinearHingeModelServer class are listed here.

Public Fields

	Name	Description
	Count (see page 594)	Number of available models.

II.3.1.6.1.2.1 Count

C++

```
HRESULT get_Count( long*);
```

C#

```
public  long Count { get; }
```

Visual Basic

```
Public ReadOnly Count As  long
```

Description

Number of available models.

Version

Available since version 3.

II.3.1.6.1.3 IRobotNonlinearHingeModelServer Methods

The methods of the IRobotNonlinearHingeModelServer class are listed here.

Public Methods

	Name	Description
	Create (see page 594)	Function creates a new non-linear model of a hinge.
	Delete (see page 595)	Function deletes model of the indicated index.
	Find (see page 595)	Function returns the model index based on the specified name.
	Get (see page 595)	Function takes a non-linear hinge model of the indicated index. .

II.3.1.6.1.3.1 Create

C++

```
HRESULT Create(BSTR _name, IRobotNonlinearHingeModel** ret);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRobotNonlinearHingeModel Create(String _name);
```

Visual Basic

```
Public Function Create(_name As String) As IRobotNonlinearHingeModel
```

Description

Function creates a new non-linear model of a hinge.

Version

Available since version 3.

II.3.1.6.1.3.2 Delete

C++

```
HRESULT Delete(long _idx);
```

C#

```
public void Delete(long _idx);
```

Visual Basic

```
Public Sub Delete(_idx As long)
```

Description

Function deletes model of the indicated index.

Version

Available since version 3.

II.3.1.6.1.3.3 Find

C++

```
HRESULT Find(BSTR _name, long* ret);
```

C#

```
public long Find(String _name);
```

Visual Basic

```
Public Function Find(_name As String) As long
```

Description

Function returns the model index based on the specified name.

Version

Available since version 3.

II.3.1.6.1.3.4 Get

C++

```
HRESULT Get(long _idx, IRobotNonlinearHingeModel** ret);
```

C#

```
public IRobotNonlinearHingeModel Get(long _idx);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Function Get(_idx As long) As IRobotNonlinearHingeModel
```

Description

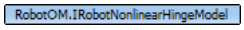
Function takes a non-linear hinge model of the indicated index. .

Version

Available since version 3.

II.3.1.6.2 IRobotNonlinearHingeModel

Class Hierarchy



C++

```
interface IRobotNonlinearHingeModel : IDispatch;
```

C#

```
public interface IRobotNonlinearHingeModel;
```

Visual Basic

```
Public Interface IRobotNonlinearHingeModel
```

Description

Definition of non-linear hinge model.

Version

Available since version 3.

II.3.1.6.2.1 IRobotNonlinearHingeModel Members

The following tables list the members exposed by IRobotNonlinearHingeModel.

Public Fields

	Name	Description
	LimitCoordX (see page 597)	Flag enforcing automatic estimation of a limit value for the quantity represented by abscissa; Example: for a model of the Force -Displacement type, it involves estimation of the force limit value. .
	LimitCoordXValue (see page 597)	Coefficient used for defining a limit value for the quantity represented by abscissa.
	LimitCoordY (see page 598)	Flag enforcing automatic estimation of a limit value for the quantity represented by ordinate; Example: for a model of the Force -Displacement type, it involves estimation of the displacement limit value. .
	LimitCoordYValue (see page 598)	Coefficient used for defining a limit value for the quantity represented by ordinate.
	MixedUnloadingValue (see page 598)	Value of the unloading factor for MIXED method.
	ModelType (see page 599)	Type of a non-linear hinge model.
	Name (see page 599)	Model name.
	Symetry (see page 599)	Curve symmetry .
	UnloadingMethod (see page 599)	Unloading method.

(c) 2014 Autodesk Inc. All rights reserved

Public Methods

	Name	Description
	GetAxisParams (see page 600)	Function takes the interface of the parameters characteristic of the indicated semi-axis. .
	GetPoints (see page 600)	Function takes the interface for definition of curve points. .
	SetAxisParams (see page 601)	Function updates parameters for the indicated semi-axis.
	SetPoints (see page 601)	Function updates a curve definition based on the specified interface. .

II.3.1.6.2.2 IRobotNonlinearHingeModel Fields

The fields of the IRobotNonlinearHingeModel class are listed here.

Public Fields

	Name	Description
	LimitCoordX (see page 597)	Flag enforcing automatic estimation of a limit value for the quantity represented by abscissa; Example: for a model of the Force -Displacement type, it involves estimation of the force limit value. .
	LimitCoordXValue (see page 597)	Coefficient used for defining a limit value for the quantity represented by abscissa.
	LimitCoordY (see page 598)	Flag enforcing automatic estimation of a limit value for the quantity represented by ordinate; Example: for a model of the Force -Displacement type, it involves estimation of the displacement limit value. .
	LimitCoordYValue (see page 598)	Coefficient used for defining a limit value for the quantity represented by ordinate.
	MixedUnloadingValue (see page 598)	Value of the unloading factor for MIXED method.
	ModelType (see page 599)	Type of a non-linear hinge model.
	Name (see page 599)	Model name.
	Symetry (see page 599)	Curve symmetry .
	UnloadingMethod (see page 599)	Unloading method.

II.3.1.6.2.2.1 LimitCoordX

C++

```
HRESULT get_LimitCoordX(VARIANT_BOOL*);
HRESULT put_LimitCoordX(VARIANT_BOOL);
```

C#

```
public bool LimitCoordX { get; set; }
```

Visual Basic

```
Public LimitCoordX As Boolean
```

Description

Flag enforcing automatic estimation of a limit value for the quantity represented by abscissa; Example: for a model of the Force -Displacement type, it involves estimation of the force limit value. .

Version

Available since version 3.

II.3.1.6.2.2.2 LimitCoordXValue

C++

```
HRESULT get_LimitCoordXValue(double*);
HRESULT put_LimitCoordXValue(double);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double LimitCoordXValue { get; set; }
```

Visual Basic

```
Public LimitCoordXValue As double
```

Description

Coefficient used for defining a limit value for the quantity represented by abscissa.

Version

Available since version 3.

II.3.1.6.2.2.3 LimitCoordY

C++

```
HRESULT get_LimitCoordY(VARIANT_BOOL*);
HRESULT put_LimitCoordY(VARIANT_BOOL);
```

C#

```
public bool LimitCoordY { get; set; }
```

Visual Basic

```
Public LimitCoordY As Boolean
```

Description

Flag enforcing automatic estimation of a limit value for the quantity represented by ordinate; Example: for a model of the Force -Displacement type, it involves estimation of the displacement limit value. .

Version

Available since version 3.

II.3.1.6.2.2.4 LimitCoordYValue

C++

```
HRESULT get_LimitCoordYValue(double*);
HRESULT put_LimitCoordYValue(double);
```

C#

```
public double LimitCoordYValue { get; set; }
```

Visual Basic

```
Public LimitCoordYValue As double
```

Description

Coefficient used for defining a limit value for the quantity represented by ordinate.

Version

Available since version 3.

II.3.1.6.2.2.5 MixedUnloadingValue

C++

```
HRESULT get_MixedUnloadingValue(double*);
HRESULT put_MixedUnloadingValue(double);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double MixedUnloadingValue { get; set; }
```

Visual Basic

```
Public MixedUnloadingValue As double
```

Description

Value of the unloading factor for MIXED method.

Version

Available since version 3.

II.3.1.6.2.2.6 ModelType

C++

```
HRESULT get_ModelType(IRobotNonlinearHingeModelType*);
HRESULT put_ModelType(IRobotNonlinearHingeModelType);
```

C#

```
public IRobotNonlinearHingeModelType ModelType { get; set; }
```

Visual Basic

```
Public ModelType As IRobotNonlinearHingeModelType
```

Description

Type of a non-linear hinge model.

Version

Available since version 3.

II.3.1.6.2.2.7 Name

C++

```
HRESULT get_Name(BSTR*);
HRESULT put_Name(BSTR);
```

C#

```
public String Name { get; set; }
```

Visual Basic

```
Public Name As String
```

Description

Model name.

Version

Available since version 3.

II.3.1.6.2.2.8 Symetry

C++

```
HRESULT get_Symetry(VARIANT_BOOL*);
```

C#

```
public bool Symetry { get; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public ReadOnly Symetry As Boolean
```

Description

Curve symmetry .

Version

Available since version 3.

II.3.1.6.2.2.9 UnloadingMethod

C++

```
HRESULT get_UnloadingMethod(IRobotNonlinearHingeModelUnloadingType*);
HRESULT put_UnloadingMethod(IRobotNonlinearHingeModelUnloadingType);
```

C#

```
public IRobotNonlinearHingeModelUnloadingType UnloadingMethod { get; set; }
```

Visual Basic

```
Public UnloadingMethod As IRobotNonlinearHingeModelUnloadingType
```

Description

Unloading method.

Version

Available since version 3.

II.3.1.6.2.3 IRobotNonlinearHingeModel Methods

The methods of the IRobotNonlinearHingeModel class are listed here.

Public Methods

	Name	Description
	GetAxisParams (see page 600)	Function takes the interface of the parameters characteristic of the indicated semi-axis. .
	GetPoints (see page 600)	Function takes the interface for definition of curve points. .
	SetAxisParams (see page 601)	Function updates parameters for the indicated semi-axis.
	SetPoints (see page 601)	Function updates a curve definition based on the specified interface. .

II.3.1.6.2.3.1 GetAxisParams

C++

```
HRESULT GetAxisParams(IRobotNonlinearLinkSemiAxisType _axis,
RobotNonlinearHingeModeAxisParams* ret);
```

C#

```
public RobotNonlinearHingeModeAxisParams GetAxisParams(IRobotNonlinearLinkSemiAxisType
_axis);
```

Visual Basic

```
Public Function GetAxisParams(_axis As IRobotNonlinearLinkSemiAxisType) As
RobotNonlinearHingeModeAxisParams
```

Description

Function takes the interface of the parameters characteristic of the indicated semi-axis. .

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.

II.3.1.6.2.3.2 GetPoints

C++

```
HRESULT GetPoints(IRobotNonlinearLinkSemiAxisType _axis, IRobotNonlinearHingeModelPoints**
ret);
```

C#

```
public IRobotNonlinearHingeModelPoints GetPoints(IRobotNonlinearLinkSemiAxisType _axis);
```

Visual Basic

```
Public Function GetPoints(_axis As IRobotNonlinearLinkSemiAxisType) As
IRobotNonlinearHingeModelPoints
```

Description

Function takes the interface for definition of curve points. .

Version

Available since version 3.

II.3.1.6.2.3.3 SetAxisParams

C++

```
HRESULT SetAxisParams(IRobotNonlinearLinkSemiAxisType _axis,
IRobotNonlinearHingeModelAxisParams* _params);
```

C#

```
public void SetAxisParams(IRobotNonlinearLinkSemiAxisType _axis,
IRobotNonlinearHingeModelAxisParams _params);
```

Visual Basic

```
Public Sub SetAxisParams(_axis As IRobotNonlinearLinkSemiAxisType, ByRef _params As
IRobotNonlinearHingeModelAxisParams)
```

Description

Function updates parameters for the indicated semi-axis.

Version

Available since version 3.

II.3.1.6.2.3.4 SetPoints

C++

```
HRESULT SetPoints(IRobotNonlinearLinkSemiAxisType _axis, IRobotNonlinearHingeModelPoints*
_points);
```

C#

```
public void SetPoints(IRobotNonlinearLinkSemiAxisType _axis,
IRobotNonlinearHingeModelPoints _points);
```

Visual Basic

```
Public Sub SetPoints(_axis As IRobotNonlinearLinkSemiAxisType, ByRef _points As
IRobotNonlinearHingeModelPoints)
```

(c) 2014 Autodesk Inc. All rights reserved

Description

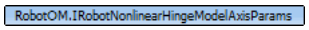
Function updates a curve definition based on the specified interface. .

Version

Available since version 3.

II.3.1.6.3 IRobotNonlinearHingeModelAxisParams

Class Hierarchy



C++

```
interface IRobotNonlinearHingeModelAxisParams : IDispatch;
```

C#

```
public interface IRobotNonlinearHingeModelAxisParams;
```

Visual Basic

```
Public Interface IRobotNonlinearHingeModelAxisParams
```

Description

Interface of parameters whose values depend on the semi-axis type.

Version

Available since version 3.

II.3.1.6.3.1 IRobotNonlinearHingeModelAxisParams Members

The following tables list the members exposed by IRobotNonlinearHingeModelAxisParams.

Public Fields

	Name	Description
	ImmediateOccupancy (see page 602)	Immediate occupancy.
	LifeSafety (see page 603)	Life safety.
	StructuralStability (see page 603)	Structural stability.

II.3.1.6.3.2 IRobotNonlinearHingeModelAxisParams Fields

The fields of the IRobotNonlinearHingeModelAxisParams class are listed here.

Public Fields

	Name	Description
	ImmediateOccupancy (see page 602)	Immediate occupancy.
	LifeSafety (see page 603)	Life safety.
	StructuralStability (see page 603)	Structural stability.

II.3.1.6.3.2.1 ImmediateOccupancy

C++

```
HRESULT get_ImmediateOccupancy(double*);
HRESULT put_ImmediateOccupancy(double);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double ImmediateOccupancy { get; set; }
```

Visual Basic

```
Public ImmediateOccupancy As double
```

Description

Immediate occupancy.

Version

Available since version 3.

II.3.1.6.3.2.2 LifeSafety

C++

```
HRESULT get_LifeSafety(double*);
HRESULT put_LifeSafety(double);
```

C#

```
public double LifeSafety { get; set; }
```

Visual Basic

```
Public LifeSafety As double
```

Description

Life safety.

Version

Available since version 3.

II.3.1.6.3.2.3 StructuralStability

C++

```
HRESULT get_StructuralStability(double*);
HRESULT put_StructuralStability(double);
```

C#

```
public double StructuralStability { get; set; }
```

Visual Basic

```
Public StructuralStability As double
```

Description

Structural stability.

Version

Available since version 3.

II.3.1.6.4 IRobotNonlinearHingeModelType

C++

```
enum IRobotNonlinearHingeModelType;
```

C#

```
public enum IRobotNonlinearHingeModelType;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Enum IRobotNonlinearHingeModelType
```

Members

Members	Description
I_NHMT_FORCE_DISPLACEMENT = 1	Force - displacement. Available since version 3.
I_NHMT_MOMENT_ROTATION = 2	Moment - rotation. Available since version 3.
I_NHMT_STRESS_PAIN = 3	Stress - strain. Available since version 3.

Description

Types of non-linear hinge models.

Version

Available since version 3.

II.3.1.6.5 IRobotNonlinearHingeModelPoints

Class Hierarchy

RobotOM.IRobotNonlinearHingeModelPoints

C++

```
interface IRobotNonlinearHingeModelPoints : IDispatch;
```

C#

```
public interface IRobotNonlinearHingeModelPoints;
```

Visual Basic

```
Public Interface IRobotNonlinearHingeModelPoints
```

Description

Interface for definition of curve points.

Version

Available since version 3.

II.3.1.6.5.1 IRobotNonlinearHingeModelPoints Members

The following tables list the members exposed by IRobotNonlinearHingeModelPoints.

Public Fields

	Name	Description
	Count (see page 605)	Number of points defining a curve.

Public Methods

	Name	Description
	Get (see page 605)	Function takes the interface for definition of a point of the indicated index. .
	Set (see page 605)	Function updates point definition based on the indicated interface.

II.3.1.6.5.2 IRobotNonlinearHingeModelPoints Fields

The fields of the IRobotNonlinearHingeModelPoints class are listed here.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	Count (see page 605)	Number of points defining a curve.

II.3.1.6.5.2.1 Count

C++

```
HRESULT get_Count( long*);
```

C#

```
public  long Count { get; }
```

Visual Basic

```
Public ReadOnly Count As  long
```

Description

Number of points defining a curve.

Version

Available since version 3.

II.3.1.6.5.3 IRobotNonlinearHingeModelPoints Methods

The methods of the IRobotNonlinearHingeModelPoints class are listed here.

Public Methods

	Name	Description
	Get (see page 605)	Function takes the interface for definition of a point of the indicated index. .
	Set (see page 605)	Function updates point definition based on the indicated interface.

II.3.1.6.5.3.1 Get

C++

```
HRESULT Get(long _idx, IRobotGeoPoint2D** ret);
```

C#

```
public IRobotGeoPoint2D Get(long _idx);
```

Visual Basic

```
Public Function Get(_idx As long) As IRobotGeoPoint2D
```

Description

Function takes the interface for definition of a point of the indicated index. .

Version

Available since version 3.

II.3.1.6.5.3.2 Set

C++

```
HRESULT Set(long _idx, IRobotGeoPoint2D* _point);
```

C#

```
public void Set(long _idx, IRobotGeoPoint2D _point);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Sub Set(_idx As long, ByRef _point As IRobotGeoPoint2D)
```

Description

Function updates point definition based on the indicated interface.

Version

Available since version 3.

II.3.1.6.6 IRobotNonlinearHingeModelUnloadingType

C++

```
enum IRobotNonlinearHingeModelUnloadingType;
```

C#

```
public enum IRobotNonlinearHingeModelUnloadingType;
```

Visual Basic

```
Public Enum IRobotNonlinearHingeModelUnloadingType
```

Members

Members	Description
I_NHMUT_ELASTIC = 1	Available since version 3.
I_NHMUT_PLASTIC = 2	Available since version 3.
I_NHMUT_DAMAGE = 3	Available since version 3.
I_NHMUT_MIXED = 4	Available since version 3.

Description

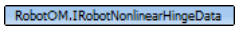
Types of unloading methods.

Version

Available since version 3.

II.3.1.6.7 IRobotNonlinearHingeData

Class Hierarchy



C++

```
interface IRobotNonlinearHingeData : IDispatch;
```

C#

```
public interface IRobotNonlinearHingeData;
```

Visual Basic

```
Public Interface IRobotNonlinearHingeData
```

Description

Interface of non-linear hinge definition.

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

II.3.1.6.7.1 IRobotNonlinearHingeData Members

The following tables list the members exposed by IRobotNonlinearHingeData.

Public Fields

	Name	Description
	NormalStress (see page 607)	Flag enforcing analysis of normal stresses in a bar section with the interactions between individual forces and moments considered .

Public Methods

	Name	Description
	GetModel (see page 608)	Function takes a model name for the indicated component. .
	IsActive (see page 608)	Function checks if the indicated component participates in a hinge definition. .
	Remove (see page 608)	Function excludes the indicated component from the hinge definition. .
	SetModel (see page 609)	Function ascribes a model of the indicated name to the component.

II.3.1.6.7.2 IRobotNonlinearHingeData Fields

The fields of the IRobotNonlinearHingeData class are listed here.

Public Fields

	Name	Description
	NormalStress (see page 607)	Flag enforcing analysis of normal stresses in a bar section with the interactions between individual forces and moments considered .

II.3.1.6.7.2.1 NormalStress

C++

```
HRESULT get_NormalStress(VARIANT_BOOL*);
HRESULT put_NormalStress(VARIANT_BOOL);
```

C#

```
public bool NormalStress { get; set; }
```

Visual Basic

```
Public NormalStress As Boolean
```

Description

Flag enforcing analysis of normal stresses in a bar section with the interactions between individual forces and moments considered .

Version

Available since version 3.

II.3.1.6.7.3 IRobotNonlinearHingeData Methods

The methods of the IRobotNonlinearHingeData class are listed here.

Public Methods

	Name	Description
	GetModel (see page 608)	Function takes a model name for the indicated component. .
	IsActive (see page 608)	Function checks if the indicated component participates in a hinge definition. .
	Remove (see page 608)	Function excludes the indicated component from the hinge definition. .
	SetModel (see page 609)	Function ascribes a model of the indicated name to the component.

(c) 2014 Autodesk Inc. All rights reserved

II.3.1.6.7.3.1 GetModel

C++

```
HRESULT GetModel(IRobotNonlinearHingeComponentType _cmpnt, BSTR* ret);
```

C#

```
public String GetModel(IRobotNonlinearHingeComponentType _cmpnt);
```

Visual Basic

```
Public Function GetModel(_cmpnt As IRobotNonlinearHingeComponentType) As String
```

Description

Function takes a model name for the indicated component. .

Version

Available since version 3.

II.3.1.6.7.3.2 IsActive

C++

```
HRESULT IsActive(IRobotNonlinearHingeComponentType _cmpnt, VARIANT_BOOL* ret);
```

C#

```
public bool IsActive(IRobotNonlinearHingeComponentType _cmpnt);
```

Visual Basic

```
Public Function IsActive(_cmpnt As IRobotNonlinearHingeComponentType) As Boolean
```

Description

Function checks if the indicated component participates in a hinge definition. .

Version

Available since version 3.

II.3.1.6.7.3.3 Remove

C++

```
HRESULT Remove(IRobotNonlinearHingeComponentType _cmpnt);
```

C#

```
public void Remove(IRobotNonlinearHingeComponentType _cmpnt);
```

Visual Basic

```
Public Sub Remove(_cmpnt As IRobotNonlinearHingeComponentType)
```

Description

Function excludes the indicated component from the hinge definition. .

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

II.3.1.6.7.3.4 SetModel

C++

```
HRESULT SetModel(IRobotNonlinearHingeComponentType _cmpnt, BSTR _model);
```

C#

```
public void SetModel(IRobotNonlinearHingeComponentType _cmpnt, String _model);
```

Visual Basic

```
Public Sub SetModel(_cmpnt As IRobotNonlinearHingeComponentType, _model As String)
```

Description

Function ascribes a model of the indicated name to the component.

Version

Available since version 3.

II.3.1.6.8 IRobotNonlinearHingeComponentType

C++

```
enum IRobotNonlinearHingeComponentType;
```

C#

```
public enum IRobotNonlinearHingeComponentType;
```

Visual Basic

```
Public Enum IRobotNonlinearHingeComponentType
```

Members

Members	Description
I_NHCT_FX = 1	Available since version 3.
I_NHCT_FY = 2	Available since version 3.
I_NHCT_FZ = 3	Available since version 3.
I_NHCT_MX = 4	Available since version 3.
I_NHCT_MY = 5	Available since version 3.
I_NHCT_MZ = 6	Available since version 3.
I_NHCT_SX = 7	Available since version 3.

Description

Types of components characterizing the non-linear hinge.

Version

Available since version 3.

II.3.1.6.9 IRobotNonlinearHingeDef

Class Hierarchy

RobotOM.IRobotNonlinearHingeDef

C++

```
interface IRobotNonlinearHingeDef : IDispatch;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public interface IRobotNonlinearHingeDef;
```

Visual Basic

```
Public Interface IRobotNonlinearHingeDef
```

Description

Interface describing a hinge defined in the structure. .

Version

Available since version 3.

II.3.1.6.9.1 IRobotNonlinearHingeDef Members

The following tables list the members exposed by IRobotNonlinearHingeDef.

Public Fields

	Name	Description
	Bar (see page 610)	Index of the bar on which a hinge is defined.
	LabelName (see page 611)	Name of the label storing hinge data.
	Offset (see page 611)	Coordinate determining the point on a bar where hinge is defined .
	Relative (see page 611)	Flag defining the manner of interpreting the Offset (see page 611) value: true value indicates that the offset coordinate is a relative value of the position on a bar, contained in the interval <0, 1>; false value indicates that the offset is treated as an absolute distance from the bar beginning - interval: <0, bar_length>.

II.3.1.6.9.2 IRobotNonlinearHingeDef Fields

The fields of the IRobotNonlinearHingeDef class are listed here.

Public Fields

	Name	Description
	Bar (see page 610)	Index of the bar on which a hinge is defined.
	LabelName (see page 611)	Name of the label storing hinge data.
	Offset (see page 611)	Coordinate determining the point on a bar where hinge is defined .
	Relative (see page 611)	Flag defining the manner of interpreting the Offset (see page 611) value: true value indicates that the offset coordinate is a relative value of the position on a bar, contained in the interval <0, 1>; false value indicates that the offset is treated as an absolute distance from the bar beginning - interval: <0, bar_length>.

II.3.1.6.9.2.1 Bar

C++

```
HRESULT get_Bar(long*);
HRESULT put_Bar(long);
```

C#

```
public long Bar { get; set; }
```

Visual Basic

```
Public Bar As long
```

Description

Index of the bar on which a hinge is defined.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.

II.3.1.6.9.2.2 LabelName

C++

```
HRESULT get_LabelName(BSTR*);
HRESULT put_LabelName(BSTR);
```

C#

```
public String LabelName { get; set; }
```

Visual Basic

```
Public LabelName As String
```

Description

Name of the label storing hinge data.

Version

Available since version 3.

II.3.1.6.9.2.3 Offset

C++

```
HRESULT get_Offset(long*);
HRESULT put_Offset(long);
```

C#

```
public long Offset { get; set; }
```

Visual Basic

```
Public Offset As long
```

Description

Coordinate determining the point on a bar where hinge is defined .

Version

Available since version 3.

II.3.1.6.9.2.4 Relative

C++

```
HRESULT get_Relative(VARIANT_BOOL*);
HRESULT put_Relative(VARIANT_BOOL);
```

C#

```
public bool Relative { get; set; }
```

Visual Basic

```
Public Relative As Boolean
```

Description

Flag defining the manner of interpreting the Offset (see page 611) value: true value indicates that the offset coordinate is a relative value of the position on a bar, contained in the interval <0, 1>; false value indicates that the offset is treated as an absolute distance from the bar beginning - interval: <0, bar_length>.

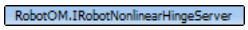
(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.

II.3.1.6.10 IRobotNonlinearHingeServer

Class Hierarchy



C++

```
interface IRobotNonlinearHingeServer : IDispatch;
```

C#

```
public interface IRobotNonlinearHingeServer;
```

Visual Basic

```
Public Interface IRobotNonlinearHingeServer
```

Description

Server providing support for hinge definitions in a structure. .

Version

Available since version 3.

II.3.1.6.10.1 IRobotNonlinearHingeServer Members

The following tables list the members exposed by IRobotNonlinearHingeServer.

Public Methods

	Name	Description
	Count (see page 613)	Function returns the number of hinges defined on the indicated bar (local mode) or defined in a whole structure (global mode). .
	Get (see page 613)	Function returns the interface of hinge definition in a structure. It enables work in two modes: in the local mode, by providing the index of a hinge on a bar and bar number and in the global mode, by providing the global index of a hinge in a structure. .
	Remove (see page 613)	Function deletes a definition of a non-linear hinge. It enables work in two modes: in the local mode, by providing the local index of hinge definition and bar number and in the global mode, by providing the global index of hinge definition. .
	Set (see page 614)	Function defines a hinge on the indicated bar with the indicated offset and returns the global (with respect to a whole structure) index of hinge definition. .

II.3.1.6.10.2 IRobotNonlinearHingeServer Methods

The methods of the IRobotNonlinearHingeServer class are listed here.

Public Methods

	Name	Description
	Count (see page 613)	Function returns the number of hinges defined on the indicated bar (local mode) or defined in a whole structure (global mode). .
	Get (see page 613)	Function returns the interface of hinge definition in a structure. It enables work in two modes: in the local mode, by providing the index of a hinge on a bar and bar number and in the global mode, by providing the global index of a hinge in a structure. .

(c) 2014 Autodesk Inc. All rights reserved

	Remove (see page 613)	Function deletes a definition of a non-linear hinge. It enables work in two modes: in the local mode, by providing the local index of hinge definition and bar number and in the global mode, by providing the global index of hinge definition. .
	Set (see page 614)	Function defines a hinge on the indicated bar with the indicated offset and returns the global (with respect to a whole structure) index of hinge definition. .

II.3.1.6.10.2.1 Count

C++

```
HRESULT Count(long _bar_number = 0, long* ret);
```

C#

```
public long Count(long _bar_number = 0);
```

Visual Basic

```
Public Function Count(Optional _bar_number As long = 0) As long
```

Description

Function returns the number of hinges defined on the indicated bar (local mode) or defined in a whole structure (global mode). .

Version

Available since version 3.

II.3.1.6.10.2.2 Get

C++

```
HRESULT Get(long _hinge_idx, long _bar_number = 0, IRobotNonlinearHingeDef** ret);
```

C#

```
public IRobotNonlinearHingeDef Get(long _hinge_idx, long _bar_number = 0);
```

Visual Basic

```
Public Function Get(_hinge_idx As long, Optional _bar_number As long = 0) As
IRobotNonlinearHingeDef
```

Description

Function returns the interface of hinge definition in a structure. It enables work in two modes: in the local mode, by providing the index of a hinge on a bar and bar number and in the global mode, by providing the global index of a hinge in a structure. .

Version

Available since version 3.

II.3.1.6.10.2.3 Remove

C++

```
HRESULT Remove(long _hinge_idx, long _bar_number = 0);
```

C#

```
public void Remove(long _hinge_idx, long _bar_number = 0);
```

Visual Basic

```
Public Sub Remove(_hinge_idx As long, Optional _bar_number As long = 0)
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Function deletes a definition of a non-linear hinge. It enables work in two modes: in the local mode, by providing the local index of hinge definition and bar number and in the global mode, by providing the global index of hinge definition. .

Version

Available since version 3.

II.3.1.6.10.2.4 Set

C++

```
HRESULT Set(long _bar_number, BSTR _label, double _offset, VARIANT_BOOL _relative, long*
ret);
```

C#

```
public long Set(long _bar_number, String _label, double _offset, bool _relative);
```

Visual Basic

```
Public Function Set(_bar_number As long, _label As String, _offset As double, _relative As
Boolean) As long
```

Description

Function defines a hinge on the indicated bar with the indicated offset and returns the global (with respect to a whole structure) index of hinge definition. .

Version

Available since version 3.

II.3.1.7 Claddings

Available since version 3.5.

Enumerations

	Name	Description
	IRobotCladdingType (see page 616)	Cladding type. .
	IRobotCladdingMethod (see page 616)	Method of load ditribution.

Interfaces

	Name	Description
	IRobotCladdingData (see page 614)	Cladding definition interface. .

II.3.1.7.1 IRobotCladdingData

Class Hierarchy

RobotOM.IRobotCladdingData

C++

```
interface IRobotCladdingData : IDispatch;
```

C#

```
public interface IRobotCladdingData;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Interface IRobotCladdingData
```

Description

Cladding definition interface. .

Version

Available since version 3.5.

II.3.1.7.1.1 IRobotCladdingData Members

The following tables list the members exposed by IRobotCladdingData.

Public Fields

	Name	Description
	Method (see page 615)	Method of load distribution.
	Type (see page 615)	Cladding type.

II.3.1.7.1.2 IRobotCladdingData Fields

The fields of the IRobotCladdingData class are listed here.

Public Fields

	Name	Description
	Method (see page 615)	Method of load distribution.
	Type (see page 615)	Cladding type.

II.3.1.7.1.2.1 Method

C++

```
HRESULT get_Method(IRobotCladdingMethod*);
HRESULT put_Method(IRobotCladdingMethod);
```

C#

```
public IRobotCladdingMethod Method { get; set; }
```

Visual Basic

```
Public Method As IRobotCladdingMethod
```

Description

Method of load distribution.

Version

Available since version 8.1.

II.3.1.7.1.2.2 Type

C++

```
HRESULT get_Type(IRobotCladdingType*);
HRESULT put_Type(IRobotCladdingType);
```

C#

```
public IRobotCladdingType Type { get; set; }
```

Visual Basic

```
Public Type As IRobotCladdingType
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Cladding type.

Version

Available since version 3.5.

II.3.1.7.2 IRobotCladdingType

C++

```
enum IRobotCladdingType;
```

C#

```
public enum IRobotCladdingType;
```

Visual Basic

```
Public Enum IRobotCladdingType
```

Members

Members	Description
I_CT_X = 0	Anisotropic parallel to local x axis . Available since version 3.5.
I_CT_Y = 1	Anisotropic perpendicular to local x axis . Available since version 3.5.
I_CT_XY = 2	Isotropic . Available since version 3.5.

Description

Cladding type. .

Version

Available since version 3.5.

II.3.1.7.3 IRobotCladdingMethod

C++

```
enum IRobotCladdingMethod;
```

C#

```
public enum IRobotCladdingMethod;
```

Visual Basic

```
Public Enum IRobotCladdingMethod
```

Members

Members	Description
I_CM_TRAPEZOIDAL = 0	Trapezoidal and triangular method. Available since version 8.1.

Description

Method of load ditribution.

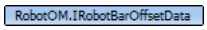
Version

Available since version 8.1.

(c) 2014 Autodesk Inc. All rights reserved

II.3.1.8 IRobotBarOffsetData

Class Hierarchy



C++

```
interface IRobotBarOffsetData : IDispatch;
```

C#

```
public interface IRobotBarOffsetData;
```

Visual Basic

```
Public Interface IRobotBarOffsetData
```

Description

The structure describing an offset defined for a bar. .

II.3.1.8.1 IRobotBarOffsetData Members

The following tables list the members exposed by IRobotBarOffsetData.

Public Fields

	Name	Description
	AxisOffset (see page 617)	
	CoordinateSystem (see page 618)	The coordinate system in which the offsets at the bar ends are defined.
	End (see page 618)	Offset at the end of a bar.
	ObjectNumber (see page 618)	
	Position (see page 618)	Position with respect to bar section (in local coordinate system) .
	Start (see page 619)	Offset at the beginning of a bar.

II.3.1.8.2 IRobotBarOffsetData Fields

The fields of the IRobotBarOffsetData class are listed here.

Public Fields

	Name	Description
	AxisOffset (see page 617)	
	CoordinateSystem (see page 618)	The coordinate system in which the offsets at the bar ends are defined.
	End (see page 618)	Offset at the end of a bar.
	ObjectNumber (see page 618)	
	Position (see page 618)	Position with respect to bar section (in local coordinate system) .
	Start (see page 619)	Offset at the beginning of a bar.

II.3.1.8.2.1 AxisOffset

C++

```
HRESULT get_AxisOffset(IRobotBarOffsetAutoPosition*);
HRESULT put_AxisOffset(IRobotBarOffsetAutoPosition);
```

C#

```
public IRobotBarOffsetAutoPosition AxisOffset { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public AxisOffset As IRobotBarOffsetAutoPosition
```

II.3.1.8.2.2 CoordinateSystem

C++

```
HRESULT get_CoordinateSystem(IRobotGeoCoordinateSystem*);
HRESULT put_CoordinateSystem(IRobotGeoCoordinateSystem);
```

C#

```
public IRobotGeoCoordinateSystem CoordinateSystem { get; set; }
```

Visual Basic

```
Public CoordinateSystem As IRobotGeoCoordinateSystem
```

Description

The coordinate system in which the offsets at the bar ends are defined.

II.3.1.8.2.3 End

C++

```
HRESULT get_End(IRobotBarEndOffsetData**);
```

C#

```
public IRobotBarEndOffsetData End { get; }
```

Visual Basic

```
Public ReadOnly End As IRobotBarEndOffsetData
```

Description

Offset at the end of a bar.

II.3.1.8.2.4 ObjectNumber

C++

```
HRESULT get_ObjectNumber(long*);
HRESULT put_ObjectNumber(long);
```

C#

```
public long ObjectNumber { get; set; }
```

Visual Basic

```
Public ObjectNumber As long
```

II.3.1.8.2.5 Position

C++

```
HRESULT get_Position(IRobotBarOffsetAutoPosition*);
HRESULT put_Position(IRobotBarOffsetAutoPosition);
```

C#

```
public IRobotBarOffsetAutoPosition Position { get; set; }
```

Visual Basic

```
Public Position As IRobotBarOffsetAutoPosition
```

Description

Position with respect to bar section (in local coordinate system) .

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.5.

II.3.1.8.2.6 Start

C++

```
HRESULT get_Start(IRobotBarEndOffsetData**);
```

C#

```
public IRobotBarEndOffsetData Start { get; }
```

Visual Basic

```
Public ReadOnly Start As IRobotBarEndOffsetData
```

Description

Offset at the beginning of a bar.

II.3.1.9 IRobotBarEndOffsetData

Class Hierarchy

RobotOM.IRobotBarEndOffsetData

C++

```
interface IRobotBarEndOffsetData : IDispatch;
```

C#

```
public interface IRobotBarEndOffsetData;
```

Visual Basic

```
Public Interface IRobotBarEndOffsetData
```

Description

The structure describing a shift at one of bar ends.

II.3.1.9.1 IRobotBarEndOffsetData Members

The following tables list the members exposed by IRobotBarEndOffsetData.

Public Fields

	Name	Description
	MemberLength (see page 620)	
	UX (see page 620)	
	UY (see page 620)	
	UZ (see page 620)	

II.3.1.9.2 IRobotBarEndOffsetData Fields

The fields of the IRobotBarEndOffsetData class are listed here.

Public Fields

	Name	Description
	MemberLength (see page 620)	
	UX (see page 620)	
	UY (see page 620)	

(c) 2014 Autodesk Inc. All rights reserved

	UZ (see page 620)	

II.3.1.9.2.1 MemberLength

C++

```
HRESULT get_MemberLength(IRobotBarOffsetMemberLength*);
HRESULT put_MemberLength(IRobotBarOffsetMemberLength);
```

C#

```
public IRobotBarOffsetMemberLength MemberLength { get; set; }
```

Visual Basic

```
Public MemberLength As IRobotBarOffsetMemberLength
```

II.3.1.9.2.2 UX

C++

```
HRESULT get_UX(double*);
HRESULT put_UX(double);
```

C#

```
public double UX { get; set; }
```

Visual Basic

```
Public UX As double
```

II.3.1.9.2.3 UY

C++

```
HRESULT get_UY(double*);
HRESULT put_UY(double);
```

C#

```
public double UY { get; set; }
```

Visual Basic

```
Public UY As double
```

II.3.1.9.2.4 UZ

C++

```
HRESULT get_UZ(double*);
HRESULT put_UZ(double);
```

C#

```
public double UZ { get; set; }
```

Visual Basic

```
Public UZ As double
```

II.3.1.10 IRobotBarEndBracketType

C++

```
enum IRobotBarEndBracketType;
```

C#

```
public enum IRobotBarEndBracketType;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Enum IRobotBarEndBracketType
```

Description

Available types of brackets. .

II.3.1.11 IRobotBarEndBracketDataValue

C++

```
enum IRobotBarEndBracketDataValue;
```

C#

```
public enum IRobotBarEndBracketDataValue;
```

Visual Basic

```
Public Enum IRobotBarEndBracketDataValue
```

Description

A set of identifiers has been defined that determines particular parameters (attributes), describing a bracket at bar end. The identifier is used as a parameter of the function that gets or sets the value of the relevant bracket attribute. .

II.3.1.12 IRobotBarEndBracketData

Class Hierarchy

RobotOM.IRobotBarEndBracketData

C++

```
interface IRobotBarEndBracketData : IDispatch;
```

C#

```
public interface IRobotBarEndBracketData;
```

Visual Basic

```
Public Interface IRobotBarEndBracketData
```

Description

The structure describing a bracket at bar end. .

II.3.1.12.1 IRobotBarEndBracketData Members

The following tables list the members exposed by IRobotBarEndBracketData.

Public Fields

	Name	Description
	Type (see page 622)	Bracket type.

Public Methods

	Name	Description
	GetValue (see page 622)	The function returns the value of the indicated bracket parameter. In order to find out if the returned value is of absolute or relative type, one should call the IsValueRelative (see page 623) function, providing the same identifier of bracket attribute. .

(c) 2014 Autodesk Inc. All rights reserved

	IsValueRelative (see page 623)	The function returns information about the character of the given bracket parameter. If zero is returned (False), it means that the indicated parameter is determined in absolute quantities, while a non-zero value (True) means that the bracket parameter is determined relative to the parameters of the bar section. .
	SetValue (see page 623)	The function sets the value of the indicated bracket parameter. .

II.3.1.12.2 IRobotBarEndBracketData Fields

The fields of the IRobotBarEndBracketData class are listed here.

Public Fields

	Name	Description
	Type (see page 622)	Bracket type.

II.3.1.12.2.1 Type

C++

```
HRESULT get_Type(IRobotBarEndBracketType*);
HRESULT put_Type(IRobotBarEndBracketType);
```

C#

```
public IRobotBarEndBracketType Type { get; set; }
```

Visual Basic

```
Public Type As IRobotBarEndBracketType
```

Description

Bracket type.

II.3.1.12.3 IRobotBarEndBracketData Methods

The methods of the IRobotBarEndBracketData class are listed here.

Public Methods

	Name	Description
	GetValue (see page 622)	The function returns the value of the indicated bracket parameter. In order to find out if the returned value is of absolute or relative type, one should call the IsValueRelative (see page 623) function, providing the same identifier of bracket attribute. .
	IsValueRelative (see page 623)	The function returns information about the character of the given bracket parameter. If zero is returned (False), it means that the indicated parameter is determined in absolute quantities, while a non-zero value (True) means that the bracket parameter is determined relative to the parameters of the bar section. .
	SetValue (see page 623)	The function sets the value of the indicated bracket parameter. .

II.3.1.12.3.1 GetValue

C++

```
HRESULT GetValue(IRobotBarEndBracketDataValue _param, double* ret);
```

C#

```
public double GetValue(IRobotBarEndBracketDataValue _param);
```

Visual Basic

```
Public Function GetValue(_param As IRobotBarEndBracketDataValue) As double
```

(c) 2014 Autodesk Inc. All rights reserved

Description

The function returns the value of the indicated bracket parameter. In order to find out if the returned value is of absolute or relative type, one should call the IsValueRelative (see page 623) function, providing the same identifier of bracket attribute. .

II.3.1.12.3.2 IsValueRelative

C++

```
HRESULT IsValueRelative(IRobotBarEndBracketDataValue _param, VARIANT_BOOL* ret);
```

C#

```
public bool IsValueRelative(IRobotBarEndBracketDataValue _param);
```

Visual Basic

```
Public Function IsValueRelative(_param As IRobotBarEndBracketDataValue) As Boolean
```

Description

The function returns information about the character of the given bracket parameter. If zero is returned (False), it means that the indicated parameter is determined in absolute quantities, while a non-zero value (True) means that the bracket parameter is determined relative to the parameters of the bar section. .

II.3.1.12.3.3 SetValue

C++

```
HRESULT SetValue(IRobotBarEndBracketDataValue _param, double _value, VARIANT_BOOL
_relative);
```

C#

```
public void SetValue(IRobotBarEndBracketDataValue _param, double _value, bool _relative);
```

Visual Basic

```
Public Sub SetValue(_param As IRobotBarEndBracketDataValue, _value As double, _relative As
Boolean)
```

Description

The function sets the value of the indicated bracket parameter. .

II.3.1.13 IRobotBarOffsetAutoPosition

C++

```
enum IRobotBarOffsetAutoPosition;
```

C#

```
public enum IRobotBarOffsetAutoPosition;
```

Visual Basic

```
Public Enum IRobotBarOffsetAutoPosition
```

Members

Members	Description
I_BOAP_0_0 = 0	Available since version 3.5.
I_BOAP_VPY_VPZ = 1	Available since version 3.5.
I_BOAP_0_VPZ = 2	Available since version 3.5.

(c) 2014 Autodesk Inc. All rights reserved

I_BOAP_VY_VPZ = 3	Available since version 3.5.
I_BOAP_VPY_0 = 4	Available since version 3.5.
I_BOAP_VY_0 = 5	Available since version 3.5.
I_BOAP_VPY_VZ = 6	Available since version 3.5.
I_BOAP_0_VZ = 7	Available since version 3.5.
I_BOAP_VY_VZ = 8	Available since version 3.5.

Description

Available predefined position of an offset with respect to section. .

Version

Available since version 3.5.

II.3.1.14 IRobotBarOffsetMemberLength

C++

```
enum IRobotBarOffsetMemberLength;
```

C#

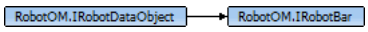
```
public enum IRobotBarOffsetMemberLength;
```

Visual Basic

```
Public Enum IRobotBarOffsetMemberLength
```

II.3.2 IRobotBar

Class Hierarchy



C++

```
interface IRobotBar : IRobotDataObject;
```

C#

```
public interface IRobotBar : IRobotDataObject;
```

Visual Basic

```
Public Interface IRobotBar
```

Description

Each bar in a structure is represented in the Object Model by a data object of the type: I_OT_BAR. The data and functionality related to the bar are available through time history RobotBar interface. The following complex bar attributes are defined by means of the labelling mechanism:

Attribute Label type

Section I_LT_BAR_SECTION Release I_LT_BAR_RELEASE Material I_LT_BAR_MATERIAL Offset I_LT_BAR_OFFSET .

II.3.2.1 IRobotBar Members

The following tables list the members exposed by IRobotBar.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	AnalyzeTTMethod (see page 627)	A flag indicating if the member is taken into consideration in the triangular / trapezoidal method.
	Elements (see page 627)	Collection containing one or more bar elements defined for the bar (the collection components are objects of the IRobotBarElement (see page 652) type).
	End (see page 627)	Bar end.
	EndNode (see page 628)	Number (see page 31) of the end bar node.
	Gamma (see page 628)	Gamma angle value defined for the bar. The gamma angle may be defined both for a bar and for a section. The resultant angle is a sum of both values. The value describes the angle for the bar. .
	InactiveBar (see page 628)	Flag enabling exclusion of a bar from the structure model - making it "inactive".
	IsSuperBar (see page 629)	Flag indicating that the bar is a superbar.
	Length (see page 629)	Bar length.
	Name (see page 629)	Bar name.
	NameTemplate (see page 629)	Bar name pattern.
	Number (see page 31)	Object number is assigned by the user; it is qunique among all the components of the same type. .
	ReversedOffset (see page 630)	Flag indicating if offset assigned to a bar is reversed with respect to bar orientation .
	ReversedRelease (see page 630)	Flag indicating if the member release is reversed in relation to member orientation.
	ReversedSection (see page 630)	Flag indicating if section assigned to the bar is inverted (section beginning is applied to the bar end).
	ShearForces (see page 631)	Flag indicating if shearing forces acting on a bar will be included in calculations of structure deformation.
	Start (see page 631)	Bar beginning .
	StartNode (see page 631)	Number (see page 31) of the initial bar node .
	StructuralType (see page 632)	Type of structure object.
	TensionCompression (see page 632)	Manner in which the bar carries stresses.
	TrussBar (see page 632)	Flag indicating if the bar is a truss bar.
	UniqueId (see page 633)	Unique identifier of structure component Available since version 1.7.

Public Methods

	Name	Description
	CalcGamma (see page 633)	Function calculates a Gamma (see page 628) angle value for a bar based on the coordinates (in the global system) of the local Z direction vector.
	ChangeOrientation (see page 634)	Function changes bar orientation to opposite.
	GetElemsData (see page 634)	Function fills out the specified object with data of elements belonging to the bar.
	GetLabel (see page 31)	The function returns a label of the indicated type applied to the object. If the object does not have any label of the type, running the function leads to a critical error. .
	GetLabelName (see page 32)	The function returns the name of a label of the indicated type, applied to the object. If the object does not have any label of the type, running the function leads to a critical error. .
	GetLabels (see page 32)	The function returns a collection containing all labels (of different types) defined for a given object. .
	GetLCS (see page 634)	Function returns Local Coordinate System for the bar.

(c) 2014 Autodesk Inc. All rights reserved

	GetSimpleBars (see page 635)	Function returns the selection of simple bars which are the superbar components. Prior to activating the function, the IsSuperBar (see page 629) flag should be checked. If the function is activated for a simple bar, it causes a critical error.
	HasLabel (see page 32)	The function returns True (non-zero value) if a label of the type has already been defined for the object. .
	RemoveLabel (see page 32)	The function deletes a label of the indicated type. .
	SetLabel (see page 33)	The function applies the defined label to an object. At a given moment, an object may have only one label of a type applied (e.g. there may not be two supports defined for one node). If an object has a formerly applied label of the same type, it is replaced by the new one. .
	SetOffset (see page 635)	Function assigns offset with a specified name to a bar, identically as the SetLabel (see page 33) function. In addition, it enables reversing the offset. .
	SetOrientation (see page 635)	Function adjusts bar orientation to the defined sense of the coordinate system axis.
	SetSection (see page 636)	Function assigns section of the indicated name to the bar, just as the SetLabel (see page 33) function does. Moreover, it allows inverting the section - applying section end to the bar beginning. .

II.3.2.2 IRobotBar Fields

The fields of the IRobotBar class are listed here.

Public Fields

	Name	Description
	AnalyzeTTMethod (see page 627)	A flag indicating if the member is taken into consideration in the triangular / trapezoidal method.
	Elements (see page 627)	Collection containing one or more bar elements defined for the bar (the collection components are objects of the IRobotBarElement (see page 652) type).
	End (see page 627)	Bar end.
	EndNode (see page 628)	Number (see page 31) of the end bar node.
	Gamma (see page 628)	Gamma angle value defined for the bar. The gamma angle may be defined both for a bar and for a section. The resultant angle is a sum of both values. The value describes the angle for the bar. .
	InactiveBar (see page 628)	Flag enabling exclusion of a bar from the structure model - making it "inactive".
	IsSuperBar (see page 629)	Flag indicating that the bar is a superbar.
	Length (see page 629)	Bar length.
	Name (see page 629)	Bar name.
	NameTemplate (see page 629)	Bar name pattern.
	ReversedOffset (see page 630)	Flag indicating if offset assigned to a bar is reversed with respect to bar orientation .
	ReversedRelease (see page 630)	Flag indicating if the member release is reversed in relation to member orientation.
	ReversedSection (see page 630)	Flag indicating if section assigned to the bar is inverted (section beginning is applied to the bar end).
	ShearForces (see page 631)	Flag indicating if shearing forces acting on a bar will be included in calculations of structure deformation.
	Start (see page 631)	Bar beginning .
	StartNode (see page 631)	Number (see page 31) of the initial bar node .
	StructuralType (see page 632)	Type of structure object.
	TensionCompression (see page 632)	Manner in which the bar carries stresses.
	TrussBar (see page 632)	Flag indicating if the bar is a truss bar.

(c) 2014 Autodesk Inc. All rights reserved

	UniqueId (see page 633)	Unique identifier of structure component Available since version 1.7.

II.3.2.2.1 AnalyzeTTMethod

C++

```
HRESULT get_AnalyzeTTMethod(VARIANT_BOOL*);
HRESULT put_AnalyzeTTMethod(VARIANT_BOOL);
```

C#

```
public bool AnalyzeTTMethod { get; set; }
```

Visual Basic

```
Public AnalyzeTTMethod As Boolean
```

Description

A flag indicating if the member is taken into consideration in the triangular / trapezoidal method.

Version

Available since version 11.

II.3.2.2.2 Elements

C++

```
HRESULT get_Elements(IRobotCollection**);
```

C#

```
public IRobotCollection Elements { get; }
```

Visual Basic

```
Public ReadOnly Elements As IRobotCollection
```

Description

Collection containing one or more bar elements defined for the bar (the collection components are objects of the IRobotBarElement (see page 652) type).

Version

Available since version 3.

II.3.2.2.3 End

C++

```
HRESULT get_End(IRobotBarEnd**);
```

C#

```
public IRobotBarEnd End { get; }
```

Visual Basic

```
Public ReadOnly End As IRobotBarEnd
```

Description

Bar end.

II.3.2.2.4 EndNode

C++

```
HRESULT get_EndNode(long*);
HRESULT put_EndNode(long);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public long EndNode { get; set; }
```

Visual Basic

```
Public EndNode As long
```

Description

Number (see page 31) of the end bar node.

II.3.2.2.5 Gamma

C++

```
HRESULT get_Gamma(double*);
HRESULT put_Gamma(double);
```

C#

```
public double Gamma { get; set; }
```

Visual Basic

```
Public Gamma As double
```

Description

Gamma angle value defined for the bar. The gamma angle may be defined both for a bar and for a section. The resultant angle is a sum of both values. The value describes the angle for the bar. .

II.3.2.2.6 InactiveBar

C++

```
HRESULT get_InactiveBar(VARIANT_BOOL*);
HRESULT put_InactiveBar(VARIANT_BOOL);
```

C#

```
public bool InactiveBar { get; set; }
```

Visual Basic

```
Public InactiveBar As Boolean
```

Description

Flag enabling exclusion of a bar from the structure model - making it "inactive".

Version

Available since version 4.5.

II.3.2.2.7 IsSuperBar

C++

```
HRESULT get_IsSuperBar(VARIANT_BOOL*);
```

C#

```
public bool IsSuperBar { get; }
```

Visual Basic

```
Public ReadOnly IsSuperBar As Boolean
```

Description

Flag indicating that the bar is a superbar.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.

II.3.2.2.8 Length

C++

```
HRESULT get_Length(double*);
```

C#

```
public double Length { get; }
```

Visual Basic

```
Public ReadOnly Length As double
```

Description

Bar length.

Version

Available since version 2.5.

II.3.2.2.9 Name

C++

```
HRESULT get_Name(BSTR*);
```

C#

```
public String Name { get; }
```

Visual Basic

```
Public ReadOnly Name As String
```

Description

Bar name.

Version

Available since version 7.5.

II.3.2.2.10 NameTemplate

C++

```
HRESULT get_NameTemplate(BSTR*);
HRESULT put_NameTemplate(BSTR);
```

C#

```
public String NameTemplate { get; set; }
```

Visual Basic

```
Public NameTemplate As String
```

Description

Bar name pattern.

Version

Available since version 7.5.

(c) 2014 Autodesk Inc. All rights reserved

II.3.2.2.11 ReversedOffset

C++

```
HRESULT get_ReversedOffset(VARIANT_BOOL*);
HRESULT put_ReversedOffset(VARIANT_BOOL);
```

C#

```
public bool ReversedOffset { get; set; }
```

Visual Basic

```
Public ReversedOffset As Boolean
```

Description

Flag indicating if offset assigned to a bar is reversed with respect to bar orientation .

Version

Available since version 3.5.

II.3.2.2.12 ReversedRelease

C++

```
HRESULT get_ReversedRelease(VARIANT_BOOL*);
HRESULT put_ReversedRelease(VARIANT_BOOL);
```

C#

```
public bool ReversedRelease { get; set; }
```

Visual Basic

```
Public ReversedRelease As Boolean
```

Description

Flag indicating if the member release is reversed in relation to member orientation.

Version

Available since version 11.

II.3.2.2.13 ReversedSection

C++

```
HRESULT get_ReversedSection(VARIANT_BOOL*);
HRESULT put_ReversedSection(VARIANT_BOOL);
```

C#

```
public bool ReversedSection { get; set; }
```

Visual Basic

```
Public ReversedSection As Boolean
```

Description

Flag indicating if section assigned to the bar is inverted (section beginning is applied to the bar end).

Version

Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

II.3.2.2.14 ShearForces

C++

```
HRESULT get_ShearForces(VARIANT_BOOL*);
HRESULT put_ShearForces(VARIANT_BOOL);
```

C#

```
public bool ShearForces { get; set; }
```

Visual Basic

```
Public ShearForces As Boolean
```

Description

Flag indicating if shearing forces acting on a bar will be included in calculations of structure deformation.

Version

Available since version 2.5.

II.3.2.2.15 Start

C++

```
HRESULT get_Start(IRobotBarEnd**);
```

C#

```
public IRobotBarEnd Start { get; }
```

Visual Basic

```
Public ReadOnly Start As IRobotBarEnd
```

Description

Bar beginning .

II.3.2.2.16 StartNode

C++

```
HRESULT get_StartNode(long*);
HRESULT put_StartNode(long);
```

C#

```
public long StartNode { get; set; }
```

Visual Basic

```
Public StartNode As long
```

Description

Number (see page 31) of the initial bar node .

II.3.2.2.17 StructuralType

C++

```
HRESULT get_StructuralType(IRobotObjectStructuralType*);
HRESULT put_StructuralType(IRobotObjectStructuralType);
```

C#

```
public IRobotObjectStructuralType StructuralType { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public StructuralType As IRobotObjectStructuralType
```

Description

Type of structure object.

Version

Available since version 9.7.

II.3.2.2.18 TensionCompression

C++

```
HRESULT get_TensionCompression(IRobotBarTensionCompression*);
HRESULT put_TensionCompression(IRobotBarTensionCompression);
```

C#

```
public IRobotBarTensionCompression TensionCompression { get; set; }
```

Visual Basic

```
Public TensionCompression As IRobotBarTensionCompression
```

Description

Manner in which the bar carries stresses.

Version

Available since version 2.5.

II.3.2.2.19 TrussBar

C++

```
HRESULT get_TrussBar(VARIANT_BOOL*);
HRESULT put_TrussBar(VARIANT_BOOL);
```

C#

```
public bool TrussBar { get; set; }
```

Visual Basic

```
Public TrussBar As Boolean
```

Description

Flag indicating if the bar is a truss bar.

Version

Available since version 2.5.

II.3.2.2.20 UniqueId

C++

```
HRESULT get_UniqueId(long*);
```

C#

```
public long UniqueId { get; }
```

Visual Basic

```
Public ReadOnly UniqueId As long
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Unique identifier of structure component Available since version 1.7.

II.3.2.3 IRobotBar Methods

The methods of the IRobotBar class are listed here.

Public Methods

	Name	Description
	CalcGamma (see page 633)	Function calculates a Gamma (see page 628) angle value for a bar based on the coordinates (in the global system) of the local Z direction vector.
	ChangeOrientation (see page 634)	Function changes bar orientation to opposite.
	GetElemsData (see page 634)	Function fills out the specified object with data of elements belonging to the bar.
	GetLCS (see page 634)	Function returns Local Coordinate System for the bar.
	GetSimpleBars (see page 635)	Function returns the selection of simple bars which are the superbar components. Prior to activating the function, the IsSuperBar (see page 629) flag should be checked. If the function is activated for a simple bar, it causes a critical error.
	SetOffset (see page 635)	Function assigns offset with a specified name to a bar, identically as the SetLabel (see page 33) function. In addition, it enables reversing the offset. .
	SetOrientation (see page 635)	Function adjusts bar orientation to the defined sense of the coordinate system axis.
	SetSection (see page 636)	Function assigns section of the indicated name to the bar, just as the SetLabel (see page 33) function does. Moreover, it allows inverting the section - applying section end to the bar beginning. .

II.3.2.3.1 CalcGamma

C++

```
HRESULT CalcGamma(double _local_Z_x, double _local_Z_y, double _local_Z_z, double* ret);
```

C#

```
public double CalcGamma(double _local_Z_x, double _local_Z_y, double _local_Z_z);
```

Visual Basic

```
Public Function CalcGamma(_local_Z_x As double, _local_Z_y As double, _local_Z_z As double)
As double
```

Description

Function calculates a Gamma (see page 628) angle value for a bar based on the coordinates (in the global system) of the local Z direction vector.

II.3.2.3.2 ChangeOrientation

C++

```
HRESULT ChangeOrientation();
```

C#

```
public void ChangeOrientation();
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Sub ChangeOrientation()
```

Description

Function changes bar orientation to opposite.

Version

Available since version 4.5.

II.3.2.3.3 GetElemsData

C++

```
HRESULT GetElemsData(IRobotBarElementDataSet* _ret_data);
```

C#

```
public void GetElemsData(IRobotBarElementDataSet _ret_data);
```

Visual Basic

```
Public Sub GetElemsData(ByRef _ret_data As IRobotBarElementDataSet)
```

Description

Function fills out the specified object with data of elements belonging to the bar.

Version

Available since version 9.

II.3.2.3.4 GetLCS

C++

```
HRESULT GetLCS(IRobotGeoPoint3D* _x, IRobotGeoPoint3D* _y, IRobotGeoPoint3D* _z);
```

C#

```
public void GetLCS(IRobotGeoPoint3D _x, IRobotGeoPoint3D _y, IRobotGeoPoint3D _z);
```

Visual Basic

```
Public Sub GetLCS(ByRef _x As IRobotGeoPoint3D, ByRef _y As IRobotGeoPoint3D, ByRef _z As
IRobotGeoPoint3D)
```

Description

Function returns Local Coordinate System for the bar.

Version

Available since version 14.7.

II.3.2.3.5 GetSimpleBars

C++

```
HRESULT GetSimpleBars(IRobotSelection** ret);
```

C#

```
public IRobotSelection GetSimpleBars();
```

Visual Basic

```
Public Function GetSimpleBars() As IRobotSelection
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Function returns the selection of simple bars which are the superbar components. Prior to activating the function, the IsSuperBar (see page 629) flag should be checked. If the function is activated for a simple bar, it causes a critical error.

Version

Available since version 3.

II.3.2.3.6 SetOffset

C++

```
HRESULT SetOffset(BSTR _offset_name, VARIANT_BOOL _reverse);
```

C#

```
public void SetOffset(String _offset_name, bool _reverse);
```

Visual Basic

```
Public Sub SetOffset(_offset_name As String, _reverse As Boolean)
```

Description

Function assigns offset with a specified name to a bar, identically as the SetLabel (see page 33) function. In addition, it enables reversing the offset. .

Version

Available since version 3.5.

II.3.2.3.7 SetOrientation

C++

```
HRESULT SetOrientation(IRobotGeoCoordinateAxisSense _dir);
```

C#

```
public void SetOrientation(IRobotGeoCoordinateAxisSense _dir);
```

Visual Basic

```
Public Sub SetOrientation(_dir As IRobotGeoCoordinateAxisSense)
```

Description

Function adjusts bar orientation to the defined sense of the coordinate system axis.

Version

Available since version 4.5.

II.3.2.3.8 SetSection

C++

```
HRESULT SetSection(BSTR _section_name, VARIANT_BOOL _reverse);
```

C#

```
public void SetSection(String _section_name, bool _reverse);
```

Visual Basic

```
Public Sub SetSection(_section_name As String, _reverse As Boolean)
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Function assigns section of the indicated name to the bar, just as the SetLabel (see page 33) function does. Moreover, it allows inverting the section - applying section end to the bar beginning. .

Version

Available since version 2.5.

II.3.3 IRobotBarServer

Class Hierarchy

C++

```
interface IRobotBarServer : IRobotDataObjectServer;
```

C#

```
public interface IRobotBarServer : IRobotDataObjectServer;
```

Visual Basic

```
Public Interface IRobotBarServer
```

Description

Server of bars manages the bars defined in the structure .

II.3.3.1 IRobotBarServer Members

The following tables list the members exposed by IRobotBarServer.

Public Fields

	Name	Description
	FreeNumber (see page 638)	First free user number available for bars Available since version 1.7.
	NonlinearHingeModels (see page 638)	Server of non-linear hinge models.
	NonlinearHinges (see page 639)	Non-linear hinge server.

Public Methods

	Name	Description
	BeginMultiOperation (see page 640)	Function enables speed-up of generation or modification of a whole group of bars through appropriate optimization of some operations. After completing generation of bars, EndMultiOperation (see page 642)() function should be called up to save changes in a structure.
	CalcGamma (see page 640)	Function calculates a Gamma angle value for a bar based on the coordinates (in the global system) of the local Z direction vector.
	ChangeOrientation (see page 641)	Function changes orientation of selected bars to opposite.
	Create (see page 641)	The function creates between the indicated nodes a bar with the defined number. .
	CreateSuperBar (see page 641)	Function generates a superbar with the user-defined number, composed of the indicated simple bars.
	Delete (see page 34)	The function deletes the object of the indicated number. .
	DeleteMany (see page 34)	The function deletes all objects that meet the criteria of the indicated selection. .
	EndMultiOperation (see page 642)	Function updates definitions of bars in a structure. It should be called up only if BeginMultiOperation (see page 640)() function has been called up earlier. .

(c) 2014 Autodesk Inc. All rights reserved

	Exist (see page 35)	The function returns True (non-zero value) if the object of the indicated name already exists. .
	FindWithId (see page 642)	Function returns a number of the bar with a specified unique identifier. If such a bar is not found, then, zero value is returned. Available since version 1.7.
	Get (see page 35)	The function returns an object with the indicated user-defined number. The type of the returned object agrees with the type of objects managed by the server. If the object of the indicated number does not exist, running the function leads to critical error. .
	GetAll (see page 35)	The function returns a collection containing all objects managed by the server. .
	GetAnalyzeTTMethodEnabled (see page 642)	Function returns selection of members taking into consideration in the triangular / trapezoid distribution method.
	GetElemsData (see page 642)	Function fills out the specified object with data of elements belonging to the indicated bars.
	GetLCS (see page 643)	Function returns Local Coordinate System for given bar.
	GetMany (see page 36)	The function returns a collection of objects that meet the criteria of the indicated selection. .
	GetName (see page 643)	Function returns a name of the specified bar.
	GetNameTemplate (see page 643)	Function returns a name pattern of the specified bar.
	GetStructuralType (see page 644)	Function returns a structure object type for the specified bar.
	GetUniqueId (see page 644)	Function returns a unique identifier for a bar with the user-specified number.
	IsInactive (see page 644)	Function checking if a given bar is assigned the "inactive" status.
	IsTrussBar (see page 645)	Function returns a value different from zero (True) if a given bar is a truss bar.
	RemoveLabel (see page 36)	The function removes the labels of the indicated type from all objects that meet the criteria of the indicated selection. .
	SetAnalyzeTTMethod (see page 645)	Function allows for adding/removing members to/from list of members taken into consideration in triangular/trapezoidal distribution method.
	SetInactive (see page 645)	Funtion setting the "inactive" status for the defined bar list.
	SetLabel (see page 36)	The function applies a label (identified by the type and name) to all objects that meet the criteria of the indicated selection. .
	SetNameTemplate (see page 646)	Function sets a name pattern for the specified bar.
	SetOrientation (see page 646)	Function adjusts orientation of selected bars to the defined sense of the coordinate system axis. .
	SetShearForces (see page 646)	For the indicated selection of bars, the function sets flag indicating if shearing forces are taken into account for them. .
	SetStructuralType (see page 647)	Function sets a structure object type for the specified bar list.
	SetTensionCompression (see page 647)	Function defines manner of carrying stresses for the specified bar selection. .
	SetTrussBar (see page 647)	Function enables definition of truss bars.
	Update (see page 647)	Function updates information on mutual relations between bars and nodes in a strucuture. It should be activated after generation of all structure bars and before definition of bar-node properties. .

II.3.3.2 IRobotBarServer Fields

The fields of the IRobotBarServer class are listed here.

Public Fields

	Name	Description
	FreeNumber (see page 638)	First free user number available for bars Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

	NonlinearHingeModels (see page 638)	Server of non-linear hinge models.
	NonlinearHinges (see page 639)	Non-linear hinge server.

II.3.3.2.1 FreeNumber

C++

```
HRESULT get_FreeNumber(long*);
```

C#

```
public long FreeNumber { get; }
```

Visual Basic

```
Public ReadOnly FreeNumber As long
```

Description

First free user number available for bars Available since version 1.7.

II.3.3.2.2 NonlinearHingeModels

C++

```
HRESULT get_NonlinearHingeModels(IRobotNonlinearHingeModelServer**);
```

C#

```
public IRobotNonlinearHingeModelServer NonlinearHingeModels { get; }
```

Visual Basic

```
Public ReadOnly NonlinearHingeModels As IRobotNonlinearHingeModelServer
```

Description

Server of non-linear hinge models.

Version

Available since version 3.

II.3.3.2.3 NonlinearHinges

C++

```
HRESULT get_NonlinearHinges(IRobotNonlinearHingeServer**);
```

C#

```
public IRobotNonlinearHingeServer NonlinearHinges { get; }
```

Visual Basic

```
Public ReadOnly NonlinearHinges As IRobotNonlinearHingeServer
```

Description

Non-linear hinge server.

Version

Available since version 3.

II.3.3.3 IRobotBarServer Methods

The methods of the IRobotBarServer class are listed here.

(c) 2014 Autodesk Inc. All rights reserved

Public Methods

	Name	Description
	BeginMultiOperation (see page 640)	Function enables speed-up of generation or modification of a whole group of bars through appropriate optimization of some operations. After completing generation of bars, EndMultiOperation (see page 642)() function should be called up to save changes in a structure.
	CalcGamma (see page 640)	Function calculates a Gamma angle value for a bar based on the coordinates (in the global system) of the local Z direction vector.
	ChangeOrientation (see page 641)	Function changes orientation of selected bars to opposite.
	Create (see page 641)	The function creates between the indicated nodes a bar with the defined number. .
	CreateSuperBar (see page 641)	Function generates a superbar with the user-defined number, composed of the indicated simple bars.
	EndMultiOperation (see page 642)	Function updates definitions of bars in a structure. It should be called up only if BeginMultiOperation (see page 640)() function has been called up earlier. .
	FindWithId (see page 642)	Function returns a number of the bar with a specified unique identifier. If such a bar is not found, then, zero value is returned. Available since version 1.7.
	GetAnalyzeTTMethodEnabled (see page 642)	Function returns selection of members taking into consideration in the triangular / trapezoid distribution method.
	GetElemsData (see page 642)	Function fills out the specified object with data of elements belonging to the indicated bars.
	GetLCS (see page 643)	Function returns Local Coordinate System for given bar.
	GetName (see page 643)	Function returns a name of the specified bar.
	GetNameTemplate (see page 643)	Function returns a name pattern of the specified bar.
	GetStructuralType (see page 644)	Function returns a structure object type for the specified bar.
	GetUniqueId (see page 644)	Function returns a unique identifier for a bar with the user-specified number.
	IsInactive (see page 644)	Function checking if a given bar is assigned the "inactive" status.
	IsTrussBar (see page 645)	Function returns a value different from zero (True) if a given bar is a truss bar.
	SetAnalyzeTTMethod (see page 645)	Function allows for adding/removing members to/from list of members taken into consideration in triangular/trapezoidal distribution method.
	SetInactive (see page 645)	Funtion setting the "inactive" status for the defined bar list.
	SetNameTemplate (see page 646)	Function sets a name pattern for the specified bar.
	SetOrientation (see page 646)	Function adjusts orientation of selected bars to the defined sense of the coordinate system axis. .
	SetShearForces (see page 646)	For the indicated selection of bars, the function sets flag indicating if shearing forces are taken into account for them. .
	SetStructuralType (see page 647)	Function sets a structure object type for the specified bar list.
	SetTensionCompression (see page 647)	Function defines manner of carrying stresses for the specified bar selection. .
	SetTrussBar (see page 647)	Function enables definition of truss bars.
	Update (see page 647)	Function updates information on mutual relations between bars and nodes in a strucuture. It should be activated after generation of all structure bars and before definition of bar-node properties. .

II.3.3.3.1 BeginMultiOperation

C++

```
HRESULT BeginMultiOperation();
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public void BeginMultiOperation();
```

Visual Basic

```
Public Sub BeginMultiOperation()
```

Description

Function enables speed-up of generation or modification of a whole group of bars through appropriate optimization of some operations. After completing generation of bars, EndMultiOperation (see page 642)() function should be called up to save changes in a structure.

Version

Available since version 14.7.

II.3.3.3.2 CalcGamma

C++

```
HRESULT CalcGamma(long _bar_num, double _local_Z_x, double _local_Z_y, double _local_Z_z,
double* ret);
```

C#

```
public double CalcGamma(long _bar_num, double _local_Z_x, double _local_Z_y, double
_local_Z_z);
```

Visual Basic

```
Public Function CalcGamma(_bar_num As long, _local_Z_x As double, _local_Z_y As double,
_local_Z_z As double) As double
```

Description

Function calculates a Gamma angle value for a bar based on the coordinates (in the global system) of the local Z direction vector.

II.3.3.3.3 ChangeOrientation

C++

```
HRESULT ChangeOrientation(BSTR _bars);
```

C#

```
public void ChangeOrientation(String _bars);
```

Visual Basic

```
Public Sub ChangeOrientation(_bars As String)
```

Description

Function changes orientation of selected bars to opposite.

Version

Available since version 4.5.

II.3.3.3.4 Create

C++

```
HRESULT Create(long _bar_num, long _start_node, long _end_node);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public void Create(long _bar_num, long _start_node, long _end_node);
```

Visual Basic

```
Public Sub Create(_bar_num As long, _start_node As long, _end_node As long)
```

Description

The function creates between the indicated nodes a bar with the defined number. .

II.3.3.3.5 CreateSuperBar

C++

```
HRESULT CreateSuperBar(long _number, BSTR _simple_bars);
```

C#

```
public void CreateSuperBar(long _number, String _simple_bars);
```

Visual Basic

```
Public Sub CreateSuperBar(_number As long, _simple_bars As String)
```

Description

Function generates a superbar with the user-defined number, composed of the indicated simple bars.

Version

Available since version 3.

II.3.3.3.6 EndMultiOperation

C++

```
HRESULT EndMultiOperation();
```

C#

```
public void EndMultiOperation();
```

Visual Basic

```
Public Sub EndMultiOperation()
```

Description

Function updates definitions of bars in a structure. It should be called up only if BeginMultiOperation (see page 640)() function has been called up earlier. .

Version

Available since version 14.7.

II.3.3.3.7 FindWithId

C++

```
HRESULT FindWithId(long _unique_id, long* ret);
```

C#

```
public long FindWithId(long _unique_id);
```

Visual Basic

```
Public Function FindWithId(_unique_id As long) As long
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Function returns a number of the bar with a specified unique identifier. If such a bar is not found, then, zero value is returned. Available since version 1.7.

II.3.3.3.8 GetAnalyzeTTMethodEnabled

C++

```
HRESULT GetAnalyzeTTMethodEnabled(IRobotSelection** ret);
```

C#

```
public IRobotSelection GetAnalyzeTTMethodEnabled();
```

Visual Basic

```
Public Function GetAnalyzeTTMethodEnabled() As IRobotSelection
```

Description

Function returns selection of members taking into consideration in the triangular / trapezoid distribution method.

Version

Available since version 11.

II.3.3.3.9 GetElemsData

C++

```
HRESULT GetElemsData(BSTR _bar_sel_txt, IRobotBarElementDataSet* _ret_data);
```

C#

```
public void GetElemsData(String _bar_sel_txt, IRobotBarElementDataSet _ret_data);
```

Visual Basic

```
Public Sub GetElemsData(_bar_sel_txt As String, ByRef _ret_data As IRobotBarElementDataSet)
```

Description

Function fills out the specified object with data of elements belonging to the indicated bars.

Version

Available since version 9.

II.3.3.3.10 GetLCS

C++

```
HRESULT GetLCS(long _bar_num, IRobotGeoPoint3D* _x, IRobotGeoPoint3D* _y, IRobotGeoPoint3D*
_z);
```

C#

```
public void GetLCS(long _bar_num, IRobotGeoPoint3D _x, IRobotGeoPoint3D _y,
IRobotGeoPoint3D _z);
```

Visual Basic

```
Public Sub GetLCS(_bar_num As long, ByRef _x As IRobotGeoPoint3D, ByRef _y As
IRobotGeoPoint3D, ByRef _z As IRobotGeoPoint3D)
```

Description

Function returns Local Coordinate System for given bar.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 14.7.

II.3.3.3.11 GetName

C++

```
HRESULT GetName(long _bar_num, BSTR* ret);
```

C#

```
public String GetName(long _bar_num);
```

Visual Basic

```
Public Function GetName(_bar_num As long) As String
```

Description

Function returns a name of the specified bar.

Version

Available since version 7.5.

II.3.3.3.12 GetNameTemplate

C++

```
HRESULT GetNameTemplate(long _bar_num, BSTR* ret);
```

C#

```
public String GetNameTemplate(long _bar_num);
```

Visual Basic

```
Public Function GetNameTemplate(_bar_num As long) As String
```

Description

Function returns a name pattern of the specified bar.

Version

Available since version 7.5.

II.3.3.3.13 GetStructuralType

C++

```
HRESULT GetStructuralType(long _bar, IRobotObjectStructuralType* ret);
```

C#

```
public IRobotObjectStructuralType GetStructuralType(long _bar);
```

Visual Basic

```
Public Function GetStructuralType(_bar As long) As IRobotObjectStructuralType
```

Description

Function returns a structure object type for the specified bar.

Version

Available since version 9.7.

(c) 2014 Autodesk Inc. All rights reserved

II.3.3.3.14 GetUniqueId

C++

```
HRESULT GetUniqueId(long _bar_num, long* ret);
```

C#

```
public long GetUniqueId(long _bar_num);
```

Visual Basic

```
Public Function GetUniqueId(_bar_num As long) As long
```

Description

Function returns a unique identifier for a bar with the user-specified number.

Version

Available since version 8.2.

II.3.3.3.15 IsInactive

C++

```
HRESULT IsInactive(long _bar_num, VARIANT_BOOL* ret);
```

C#

```
public bool IsInactive(long _bar_num);
```

Visual Basic

```
Public Function IsInactive(_bar_num As long) As Boolean
```

Description

Function checking if a given bar is assigned the "inactive" status.

Version

Available since version 4.5.

II.3.3.3.16 IsTrussBar

C++

```
HRESULT IsTrussBar(long _bar_num, VARIANT_BOOL* ret);
```

C#

```
public bool IsTrussBar(long _bar_num);
```

Visual Basic

```
Public Function IsTrussBar(_bar_num As long) As Boolean
```

Description

Function returns a value different from zero (True) if a given bar is a truss bar.

II.3.3.3.17 SetAnalyzeTTMethod

C++

```
HRESULT SetAnalyzeTTMethod(BSTR _bars, VARIANT_BOOL _analyze);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public void SetAnalyzeTTMethod(String _bars, bool _analyze);
```

Visual Basic

```
Public Sub SetAnalyzeTTMethod(_bars As String, _analyze As Boolean)
```

Description

Function allows for adding/removing members to/from list of members taken into consideration in triangular/trapezoidal distribution method.

Version

Available since version 11.

II.3.3.3.18 SetInactive

C++

```
HRESULT SetInactive(BSTR _bars, VARIANT_BOOL _inactive = true);
```

C#

```
public void SetInactive(String _bars, bool _inactive = true);
```

Visual Basic

```
Public Sub SetInactive(_bars As String, Optional _inactive As Boolean = true)
```

Description

Funtion setting the "inactive" status for the defined bar list.

Version

Available since version 4.5.

II.3.3.3.19 SetNameTemplate

C++

```
HRESULT SetNameTemplate(long _bar_num, BSTR _name_tmpl);
```

C#

```
public void SetNameTemplate(long _bar_num, String _name_tmpl);
```

Visual Basic

```
Public Sub SetNameTemplate(_bar_num As long, _name_tmpl As String)
```

Description

Function sets a name pattern for the specified bar.

Version

Available since version 7.5.

II.3.3.3.20 SetOrientation

C++

```
HRESULT SetOrientation(BSTR _bars, IRobotGeoCoordinateAxisSense _dir);
```

C#

```
public void SetOrientation(String _bars, IRobotGeoCoordinateAxisSense _dir);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Sub SetOrientation(_bars As String, _dir As IRobotGeoCoordinateAxisSense)
```

Description

Function adjusts orientation of selected bars to the defined sense of the coordinate system axis. .

Version

Available since version 4.5.

II.3.3.3.21 SetShearForces

C++

```
HRESULT SetShearForces(IRobotSelection* _bar_sel, VARIANT_BOOL _set = -1);
```

C#

```
public void SetShearForces(IRobotSelection _bar_sel, bool _set = -1);
```

Visual Basic

```
Public Sub SetShearForces(ByRef _bar_sel As IRobotSelection, Optional _set As Boolean = -1)
```

Description

For the indicated selection of bars, the function sets flag indicating if shearing forces are taken into account for them. .

Version

Available since version 2.5.

II.3.3.3.22 SetStructuralType

C++

```
HRESULT SetStructuralType(BSTR _bar_sel, IRobotObjectStructuralType _str_type);
```

C#

```
public void SetStructuralType(String _bar_sel, IRobotObjectStructuralType _str_type);
```

Visual Basic

```
Public Sub SetStructuralType(_bar_sel As String, _str_type As IRobotObjectStructuralType)
```

Description

Function sets a structure object type for the specified bar list.

Version

Available since version 9.7.

II.3.3.3.23 SetTensionCompression

C++

```
HRESULT SetTensionCompression(IRobotSelection* _bar_sel, IRobotBarTensionCompression _tc);
```

C#

```
public void SetTensionCompression(IRobotSelection _bar_sel, IRobotBarTensionCompression
_tc);
```

Visual Basic

```
Public Sub SetTensionCompression(ByRef _bar_sel As IRobotSelection, _tc As
```

(c) 2014 Autodesk Inc. All rights reserved

```
IRobotBarTensionCompression)
```

Description

Function defines manner of carrying stresses for the specified bar selection. .

Version

Available since version 2.5.

II.3.3.3.24 SetTrussBar

C++

```
HRESULT SetTrussBar(BSTR _bars, VARIANT_BOOL _truss_bar);
```

C#

```
public void SetTrussBar(String _bars, bool _truss_bar);
```

Visual Basic

```
Public Sub SetTrussBar(_bars As String, _truss_bar As Boolean)
```

Description

Function enables definition of truss bars.

II.3.3.3.25 Update

C++

```
HRESULT Update();
```

C#

```
public void Update();
```

Visual Basic

```
Public Sub Update()
```

Description

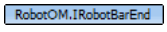
Function updates information on mutual relations between bars and nodes in a strucuture. It should be activated after generation of all structure bars and before definition of bar-node properties. .

Version

Available since version 3.5.

II.3.4 IRobotBarEnd

Class Hierarchy



C++

```
interface IRobotBarEnd : IDispatch;
```

C#

```
public interface IRobotBarEnd;
```

Visual Basic

```
Public Interface IRobotBarEnd
```

(c) 2014 Autodesk Inc. All rights reserved

Description

One can indicate a beginning and an end of each bar. The object of RobotBarEnd type, describing one of the bar ends, is created to make it possible to carry out operations affecting only the beginning or end of a bar (and not the entire bar).

II.3.4.1 IRobotBarEnd Members

The following tables list the members exposed by IRobotBarEnd.

Public Fields

	Name	Description
	Node (see page 649)	Node number (user-defined).

Public Methods

	Name	Description
	GetLabel (see page 649)	The function returns the label of the given type defined for the object.
	GetLabelName (see page 650)	The function returns the label of the given type, applied to the object.
	GetLabels (see page 650)	The function returns a collection containing all labels applied to the object. .
	GetOffsetValue (see page 650)	Function calculates and returns total values of offset at a bar end. .
	HasLabel (see page 650)	The function returns a non-zero value (True) if a label of the given type has been defined for the object.
	RemoveLabel (see page 651)	The function removes the label of the given type from the object. .
	SetLabel (see page 651)	The function applies a label of a given type to the object. .

II.3.4.2 IRobotBarEnd Fields

The fields of the IRobotBarEnd class are listed here.

Public Fields

	Name	Description
	Node (see page 649)	Node number (user-defined).

II.3.4.2.1 Node

C++

```
HRESULT get_Node(long*);
HRESULT put_Node(long);
```

C#

```
public long Node { get; set; }
```

Visual Basic

```
Public Node As long
```

Description

Node number (user-defined).

II.3.4.3 IRobotBarEnd Methods

The methods of the IRobotBarEnd class are listed here.

Public Methods

	Name	Description
	GetLabel (see page 649)	The function returns the label of the given type defined for the object.
	GetLabelName (see page 650)	The function returns the label of the given type, applied to the object.

(c) 2014 Autodesk Inc. All rights reserved

	GetLabels (see page 650)	The function returns a collection containing all labels applied to the object. .
	GetOffsetValue (see page 650)	Function calculates and returns total values of offset at a bar end. .
	HasLabel (see page 650)	The function returns a non-zero value (True) if a label of the given type has been defined for the object.
	RemoveLabel (see page 651)	The function removes the label of the given type from the object. .
	SetLabel (see page 651)	The function applies a label of a given type to the object. .

II.3.4.3.1 GetLabel

C++

```
HRESULT GetLabel(IRobotLabelType _lab_type, IRobotLabel** ret);
```

C#

```
public IRobotLabel GetLabel(IRobotLabelType _lab_type);
```

Visual Basic

```
Public Function GetLabel(_lab_type As IRobotLabelType) As IRobotLabel
```

Description

The function returns the label of the given type defined for the object.

II.3.4.3.2 GetLabelName

C++

```
HRESULT GetLabelName(IRobotLabelType _lab_type, BSTR* ret);
```

C#

```
public String GetLabelName(IRobotLabelType _lab_type);
```

Visual Basic

```
Public Function GetLabelName(_lab_type As IRobotLabelType) As String
```

Description

The function returns the label of the given type, applied to the object.

II.3.4.3.3 GetLabels

C++

```
HRESULT GetLabels(IRobotCollection** ret);
```

C#

```
public IRobotCollection GetLabels();
```

Visual Basic

```
Public Function GetLabels() As IRobotCollection
```

Description

The function returns a collection containing all labels applied to the object. .

II.3.4.3.4 GetOffsetValue

C++

```
HRESULT GetOffsetValue(IRobotBarEndOffsetData** ret);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRobotBarEndOffsetData GetOffsetValue();
```

Visual Basic

```
Public Function GetOffsetValue() As IRobotBarEndOffsetData
```

Description

Function calculates and returns total values of offset at a bar end. .

Version

Available since version 3.5.

II.3.4.3.5 HasLabel

C++

```
HRESULT HasLabel(IRobotLabelType _lab_type, VARIANT_BOOL* ret);
```

C#

```
public bool HasLabel(IRobotLabelType _lab_type);
```

Visual Basic

```
Public Function HasLabel(_lab_type As IRobotLabelType) As Boolean
```

Description

The function returns a non-zero value (True) if a label of the given type has been defined for the object.

II.3.4.3.6 RemoveLabel

C++

```
HRESULT RemoveLabel(IRobotLabelType _lab_type);
```

C#

```
public void RemoveLabel(IRobotLabelType _lab_type);
```

Visual Basic

```
Public Sub RemoveLabel(_lab_type As IRobotLabelType)
```

Description

The function removes the label of the given type from the object. .

II.3.4.3.7 SetLabel

C++

```
HRESULT SetLabel(IRobotLabelType _lab_type, BSTR _lab_name);
```

C#

```
public void SetLabel(IRobotLabelType _lab_type, String _lab_name);
```

Visual Basic

```
Public Sub SetLabel(_lab_type As IRobotLabelType, _lab_name As String)
```

Description

The function applies a label of a given type to the object. .

(c) 2014 Autodesk Inc. All rights reserved

II.3.5 IRobotBarTensionCompression

C++

```
enum IRobotBarTensionCompression;
```

C#

```
public enum IRobotBarTensionCompression;
```

Visual Basic

```
Public Enum IRobotBarTensionCompression
```

Members

Members	Description
I_BTC_STANDARD = 0	The bar carries both compressive and tensile stresses . Available since version 2.5.
I_BTC_TENSION_ONLY = 1	The bar carries only tensile stresses . Available since version 2.5.
I_BTC_COMPRESSION_ONLY = 2	The bar carries only compressive stresses. Available since version 2.5.

Version

Available since version 2.5.

II.3.6 IRobotBarElement

Class Hierarchy

RobotOM.IRobotBarElement

C++

```
interface IRobotBarElement : IDispatch;
```

C#

```
public interface IRobotBarElement;
```

Visual Basic

```
Public Interface IRobotBarElement
```

Description

Bar element.

Version

Available since version 3.

II.3.6.1 IRobotBarElement Members

The following tables list the members exposed by IRobotBarElement.

Public Fields

	Name	Description
	EndNode (see page 653)	Number (see page 653) of end node.
	Inactive (see page 653)	Flag enabling exclusion of a bar element from the structure calculation model.
	IsCalc (see page 653)	Flag indicating if the element has been created automatically during generation of the structure calculation model.

(c) 2014 Autodesk Inc. All rights reserved

	Number (see page 653)	
	StartNode (see page 654)	Number (see page 653) of origin node.

II.3.6.2 IRobotBarElement Fields

The fields of the IRobotBarElement class are listed here.

Public Fields

	Name	Description
	EndNode (see page 653)	Number (see page 653) of end node.
	Inactive (see page 653)	Flag enabling exclusion of a bar element from the structure calculation model.
	IsCalc (see page 653)	Flag indicating if the element has been created automatically during generation of the structure calculation model.
	Number (see page 653)	
	StartNode (see page 654)	Number (see page 653) of origin node.

II.3.6.2.1 EndNode

C++

```
HRESULT get_EndNode(long*);
```

C#

```
public long EndNode { get; }
```

Visual Basic

```
Public ReadOnly EndNode As long
```

Description

Number (see page 653) of end node.

Version

Available since version 3.

II.3.6.2.2 Inactive

C++

```
HRESULT get_Inactive(VARIANT_BOOL*);
HRESULT put_Inactive(VARIANT_BOOL);
```

C#

```
public bool Inactive { get; set; }
```

Visual Basic

```
Public Inactive As Boolean
```

Description

Flag enabling exclusion of a bar element from the structure calculation model.

Version

Available since version 4.5.

II.3.6.2.3 IsCalc

C++

```
HRESULT get_IsCalc(VARIANT_BOOL*);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public bool IsCalc { get; }
```

Visual Basic

```
Public ReadOnly IsCalc As Boolean
```

Description

Flag indicating if the element has been created automatically during generation of the structure calculation model.

Version

Available since version 3.

II.3.6.2.4 Number

C++

```
HRESULT get_Number(long*);
```

C#

```
public long Number { get; }
```

Visual Basic

```
Public ReadOnly Number As long
```

Version

Available since version 3.

II.3.6.2.5 StartNode

C++

```
HRESULT get_StartNode(long*);
```

C#

```
public long StartNode { get; }
```

Visual Basic

```
Public ReadOnly StartNode As long
```

Description

Number (see page 653) of origin node.

Version

Available since version 3.

II.3.7 IRobotBarElementData

Class Hierarchy

RobotOM.IRobotBarElementData

C++

```
interface IRobotBarElementData : IDispatch;
```

C#

```
public interface IRobotBarElementData;
```

Visual Basic

```
Public Interface IRobotBarElementData
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Bar (see page 655) element data.

Version

Available since version 9.

II.3.7.1 IRobotBarElementData Members

The following tables list the members exposed by IRobotBarElementData.

Public Fields

	Name	Description
	Bar (see page 655)	Number (see page 655) of a bar that the element belongs to.
	Number (see page 655)	Element no.

Public Methods

	Name	Description
	GetEndNode (see page 656)	Function returns a number and coordinates of the end node of the element.
	GetStartNode (see page 656)	Function returns a number and coordinates of the beginning node of the element.

II.3.7.2 IRobotBarElementData Fields

The fields of the IRobotBarElementData class are listed here.

Public Fields

	Name	Description
	Bar (see page 655)	Number (see page 655) of a bar that the element belongs to.
	Number (see page 655)	Element no.

II.3.7.2.1 Bar

C++

```
HRESULT get_Bar(long*);
```

C#

```
public long Bar { get; }
```

Visual Basic

```
Public ReadOnly Bar As long
```

Description

Number (see page 655) of a bar that the element belongs to.

Version

Available since version 9.

II.3.7.2.2 Number

C++

```
HRESULT get_Number(long*);
```

C#

```
public long Number { get; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public ReadOnly Number As long
```

Description

Element no.

Version

Available since version 9.

II.3.7.3 IRobotBarElementData Methods

The methods of the IRobotBarElementData class are listed here.

Public Methods

	Name	Description
	GetEndNode (see page 656)	Function returns a number and coordinates of the end node of the element.
	GetStartNode (see page 656)	Function returns a number and coordinates of the beginning node of the element.

II.3.7.3.1 GetEndNode

C++

```
HRESULT GetEndNode(double* _x, double* _y, double* _z, long* ret);
```

C#

```
public long GetEndNode(double* _x, double* _y, double* _z);
```

Visual Basic

```
Public Function GetEndNode(ByRef _x As double*, ByRef _y As double*, ByRef _z As double*)
As long
```

Description

Function returns a number and coordinates of the end node of the element.

Version

Available since version 9.

II.3.7.3.2 GetStartNode

C++

```
HRESULT GetStartNode(double* _x, double* _y, double* _z, long* ret);
```

C#

```
public long GetStartNode(double* _x, double* _y, double* _z);
```

Visual Basic

```
Public Function GetStartNode(ByRef _x As double*, ByRef _y As double*, ByRef _z As double*)
As long
```

Description

Function returns a number and coordinates of the beginning node of the element.

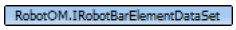
Version

Available since version 9.

(c) 2014 Autodesk Inc. All rights reserved

II.3.8 IRobotBarElementDataSet

Class Hierarchy



C++

```
interface IRobotBarElementDataSet : IDispatch;
```

C#

```
public interface IRobotBarElementDataSet;
```

Visual Basic

```
Public Interface IRobotBarElementDataSet
```

Description

Data set for bar elements.

Version

Available since version 9.

II.3.8.1 IRobotBarElementDataSet Members

The following tables list the members exposed by IRobotBarElementDataSet.

Public Fields

	Name	Description
	BarCount (see page 657)	Number of bars whose elements are included in the set.
	ElemCount (see page 658)	Number of elements in the set.

Public Methods

	Name	Description
	GetBarNumber (see page 658)	Function returns a user number for the bar with the specified index.
	GetElem (see page 658)	Function gives access to data of the element with the specified index.
	GetElemCountForBar (see page 659)	Function returns a number of elements for the bar with the specified index.
	GetElemForBar (see page 659)	Function returns data of the determined element for the specified bar.

II.3.8.2 IRobotBarElementDataSet Fields

The fields of the IRobotBarElementDataSet class are listed here.

Public Fields

	Name	Description
	BarCount (see page 657)	Number of bars whose elements are included in the set.
	ElemCount (see page 658)	Number of elements in the set.

II.3.8.2.1 BarCount

C++

```
HRESULT get_BarCount(long*);
```

C#

```
public long BarCount { get; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public ReadOnly BarCount As long
```

Description

Number of bars whose elements are included in the set.

Version

Available since version 9.

II.3.8.2.2 ElemCount

C++

```
HRESULT get_ElemCount(long*);
```

C#

```
public long ElemCount { get; }
```

Visual Basic

```
Public ReadOnly ElemCount As long
```

Description

Number of elements in the set.

Version

Available since version 9.

II.3.8.3 IRobotBarElementDataSet Methods

The methods of the IRobotBarElementDataSet class are listed here.

Public Methods

	Name	Description
	GetBarNumber (see page 658)	Function returns a user number for the bar with the specified index.
	GetElem (see page 658)	Function gives access to data of the element with the specified index.
	GetElemCountForBar (see page 659)	Function returns a number of elements for the bar with the specified index.
	GetElemForBar (see page 659)	Function returns data of the determined element for the specified bar.

II.3.8.3.1 GetBarNumber

C++

```
HRESULT GetBarNumber(long _bar_idx, long* ret);
```

C#

```
public long GetBarNumber(long _bar_idx);
```

Visual Basic

```
Public Function GetBarNumber(_bar_idx As long) As long
```

Description

Function returns a user number for the bar with the specified index.

Version

Available since version 9.

(c) 2014 Autodesk Inc. All rights reserved

II.3.8.3.2 GetElem

C++

```
HRESULT GetElem(long _elem_idx, IRobotBarElementData* _ret_data, long* ret);
```

C#

```
public long GetElem(long _elem_idx, IRobotBarElementData _ret_data);
```

Visual Basic

```
Public Function GetElem(_elem_idx As long, ByRef _ret_data As IRobotBarElementData) As long
```

Description

Function gives access to data of the element with the specified index.

Version

Available since version 9.

II.3.8.3.3 GetElemCountForBar

C++

```
HRESULT GetElemCountForBar(long _bar_idx, long* ret);
```

C#

```
public long GetElemCountForBar(long _bar_idx);
```

Visual Basic

```
Public Function GetElemCountForBar(_bar_idx As long) As long
```

Description

Function returns a number of elements for the bar with the specified index.

Version

Available since version 9.

II.3.8.3.4 GetElemForBar

C++

```
HRESULT GetElemForBar(long _bar_idx, long _elem_idx, IRobotBarElementData* _ret_data, long*
ret);
```

C#

```
public long GetElemForBar(long _bar_idx, long _elem_idx, IRobotBarElementData _ret_data);
```

Visual Basic

```
Public Function GetElemForBar(_bar_idx As long, _elem_idx As long, ByRef _ret_data As
IRobotBarElementData) As long
```

Description

Function returns data of the determined element for the specified bar.

Version

Available since version 9.

II.4 Complex attributes shared by objects of different types

(c) 2014 Autodesk Inc. All rights reserved

II.4.1 Material

Enumerations

	Name	Description
	IRobotMaterialType (see page 672)	Types of materials accepted by Robot. .
	IRobotMaterialTimberType (see page 672)	Types of timber materials defined by Robot. .
	IRobotMaterialModel (see page 673)	Available material models.
	IRobotMaterialElasticType (see page 673)	Available models of material elasticity.
	IRobotMaterialElasticUnloadingMethod (see page 676)	Available methods of material unloading.

Interfaces

	Name	Description
	IRobotMaterialData (see page 660)	Material parameters are described by means of the RobotMaterialdata interface. An object of the type contains data of each label (complex attribute) of the I_LT_BAR_MATERIAL type .
	IRobotMaterialElasticModel (see page 674)	Model (see page 675) of material elasticity. .

II.4.1.1 IRobotMaterialData

Class Hierarchy

RobotOM.IRobotMaterialData

C++

```
interface IRobotMaterialData : IDispatch;
```

C#

```
public interface IRobotMaterialData;
```

Visual Basic

```
Public Interface IRobotMaterialData
```

Description

Material parameters are described by means of the RobotMaterialdata interface. An object of the type contains data of each label (complex attribute) of the I_LT_BAR_MATERIAL type .

II.4.1.1.1 IRobotMaterialData Members

The following tables list the members exposed by IRobotMaterialData.

Public Fields

	Name	Description
	CB71_Category (see page 663)	Additional characteristics for CD71 code - 3 categories (I, II, II).
	CB71_Humidity (see page 663)	Additional characteristic for CB71 code - humidity in %.
	CB71_Nature (see page 663)	Additional characteristics for CD71 code - 2 natures (resinous / non-resinous).
	CB71_Retreat (see page 663)	Additional characteristic for CB71 code - retreat in %.
	CS (see page 664)	Reduction coefficient for shear limit stress (steel, aluminium) (not implemented).

(c) 2014 Autodesk Inc. All rights reserved

	Default (see page 664)	Default material (1 - yes, 0 - no).
	DumpCoef (see page 664)	Dumping coefficient .
	E (see page 664)	Young's modulus / axial Young modulus for timber.
	E_5 (see page 665)	Young's 5% module (timber) only for EC5 code.
	E_Trans (see page 665)	Young transverse module (timber) only for EC5 code.
	EC_Deformation (see page 665)	Shear deformation module.
	GMean (see page 665)	
	Kirchoff (see page 666)	Shear (transversal) module.
	LX (see page 666)	Thermal expansion coefficient.
	Name (see page 666)	Material name.
	NU (see page 666)	Poisson ratio .
	Nuance (see page 667)	Additional short material description.
	PN_Deformation (see page 667)	Shear deformation module for Polish code.
	PN_E_Additional (see page 667)	Additional Young module (timber) for Polish code (not implemented).
	PN_E_Trans (see page 667)	Young transverse module (timber) for Polish code (not implemented).
	RE (see page 668)	Yield point (steel, aluminum), compression resistance (concrete).
	RE_AxCompr (see page 668)	Axial compression resistance (timber) (not implemented).
	RE_AxTens (see page 668)	Axial tension resistance (timber).
	RE_Bending (see page 669)	Bending resistance (timber).
	RE_Shear (see page 669)	Shear resistance (timber).
	RE_TrCompr (see page 669)	Transverse compression resistance (timber).
	RE_TrTens (see page 669)	Transverse tension resistance (timber).
	RO (see page 670)	Material density (mass).
	RT (see page 670)	Tension resistance (steel).
	SecondName (see page 670)	Second name of the material, available only during editing material database under MS Access.
	Steel_Thermal (see page 670)	Thermal flag for steel materials. When it is on, it refers to special processing during production. .
	Timber_Type (see page 671)	Timber material type (glued multi-layered or normal) .
	Type (see page 671)	Material type.

Public Methods

	Name	Description
	LoadFromDBase (see page 671)	The function allows one to load the data on a material with the defined name to the material database. The returned logical value (True, False) indicates whether the information on the material with the defined name has been found in the material database or not. .
	SaveToDBase (see page 672)	Function saves material to the user database. .

II.4.1.1.2 IRobotMaterialData Fields

The fields of the IRobotMaterialData class are listed here.

Public Fields

	Name	Description
	CB71_Category (see page 663)	Additional characteristics for CD71 code - 3 categories (I, II, II).
	CB71_Humidity (see page 663)	Additional characteristic for CB71 code - humidity in %.
	CB71_Nature (see page 663)	Additional characteristics for CD71 code - 2 natures (resinous / non-resinous).
	CB71_Retreat (see page 663)	Additional characteristic for CB71 code - retreat in %.
	CS (see page 664)	Reduction coefficient for shear limit stress (steel, aluminium) (not implemented).

(c) 2014 Autodesk Inc. All rights reserved

	Default (see page 664)	Default material (1 - yes, 0 - no).
	DumpCoef (see page 664)	Dumping coefficient .
	E (see page 664)	Young's modulus / axial Young modulus for timber.
	E_5 (see page 665)	Young's 5% module (timber) only for EC5 code.
	E_Trans (see page 665)	Young transverse module (timber) only for EC5 code.
	EC_Deformation (see page 665)	Shear deformation module.
	GMean (see page 665)	
	Kirchoff (see page 666)	Shear (transversal) module.
	LX (see page 666)	Thermal expansion coefficient.
	Name (see page 666)	Material name.
	NU (see page 666)	Poisson ratio .
	Nuance (see page 667)	Additional short material description.
	PN_Deformation (see page 667)	Shear deformation module for Polish code.
	PN_E_Additional (see page 667)	Additional Young module (timber) for Polish code (not implemented).
	PN_E_Trans (see page 667)	Young transverse module (timber) for Polish code (not implemented).
	RE (see page 668)	Yield point (steel, aluminum), compression resistance (concrete).
	RE_AxCompr (see page 668)	Axial compression resistance (timber) (not implemented).
	RE_AxTens (see page 668)	Axial tension resistance (timber).
	RE_Bending (see page 669)	Bending resistance (timber).
	RE_Shear (see page 669)	Shear resistance (timber).
	RE_TrCompr (see page 669)	Transverse compression resistance (timber).
	RE_TrTens (see page 669)	Transverse tension resistance (timber).
	RO (see page 670)	Material density (mass).
	RT (see page 670)	Tension resistance (steel).
	SecondName (see page 670)	Second name of the material, available only during editing material database under MS Access.
	Steel_Thermal (see page 670)	Thermal flag for steel materials. When it is on, it refers to special processing during production. .
	Timber_Type (see page 671)	Timber material type (glued multi-layered or normal) .
	Type (see page 671)	Material type.

II.4.1.1.2.1 CB71_Category

C++

```
HRESULT get_CB71_Category(int*);
HRESULT put_CB71_Category(int);
```

C#

```
public int CB71_Category { get; set; }
```

Visual Basic

```
Public CB71_Category As int
```

Description

Additional characteristics for CD71 code - 3 categories (I, II, II).

II.4.1.1.2.2 CB71_Humidity

C++

```
HRESULT get_CB71_Humidity(double*);
HRESULT put_CB71_Humidity(double);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double CB71_Humidity { get; set; }
```

Visual Basic

```
Public CB71_Humidity As double
```

Description

Additional characteristic for CB71 code - humidity in %.

II.4.1.1.2.3 CB71_Nature

C++

```
HRESULT get_CB71_Nature(int*);
HRESULT put_CB71_Nature(int);
```

C#

```
public int CB71_Nature { get; set; }
```

Visual Basic

```
Public CB71_Nature As int
```

Description

Additional characteristics for CD71 code - 2 natures (resinous / non-resinous).

II.4.1.1.2.4 CB71_Retreat

C++

```
HRESULT get_CB71_Retreat(double*);
HRESULT put_CB71_Retreat(double);
```

C#

```
public double CB71_Retreat { get; set; }
```

Visual Basic

```
Public CB71_Retreat As double
```

Description

Additional characteristic for CB71 code - retreat in %.

II.4.1.1.2.5 CS

C++

```
HRESULT get_CS(double*);
HRESULT put_CS(double);
```

C#

```
public double CS { get; set; }
```

Visual Basic

```
Public CS As double
```

Description

Reduction coefficient for shear limit stress (steel, aluminium) (not implemented).

II.4.1.1.2.6 Default

C++

```
HRESULT get_Default(VARIANT_BOOL*);
```

(c) 2014 Autodesk Inc. All rights reserved

```
HRESULT put_Default(VARIANT_BOOL);
```

C#

```
public bool Default { get; set; }
```

Visual Basic

```
Public Default As Boolean
```

Description

Default material (1 - yes, 0 - no).

II.4.1.1.2.7 DumpCoef

C++

```
HRESULT get_DumpCoef(double*);
HRESULT put_DumpCoef(double);
```

C#

```
public double DumpCoef { get; set; }
```

Visual Basic

```
Public DumpCoef As double
```

Description

Dumping coefficient .

II.4.1.1.2.8 E

C++

```
HRESULT get_E(double*);
HRESULT put_E(double);
```

C#

```
public double E { get; set; }
```

Visual Basic

```
Public E As double
```

Description

Young's modulus / axial Young modulus for timber.

II.4.1.1.2.9 E_5

C++

```
HRESULT get_E_5(double*);
HRESULT put_E_5(double);
```

C#

```
public double E_5 { get; set; }
```

Visual Basic

```
Public E_5 As double
```

Description

Young's 5% module (timber) only for EC5 code.

(c) 2014 Autodesk Inc. All rights reserved

II.4.1.1.2.10 E_Trans

C++

```
HRESULT get_E_Trans(double*);
HRESULT put_E_Trans(double);
```

C#

```
public double E_Trans { get; set; }
```

Visual Basic

```
Public E_Trans As double
```

Description

Young transverse module (timber) only for EC5 code.

II.4.1.1.2.11 EC_Deformation

C++

```
HRESULT get_EC_Deformation(double*);
HRESULT put_EC_Deformation(double);
```

C#

```
public double EC_Deformation { get; set; }
```

Visual Basic

```
Public EC_Deformation As double
```

Description

Shear deformation module.

II.4.1.1.2.12 GMean

C++

```
HRESULT get_GMean(double*);
HRESULT put_GMean(double);
```

C#

```
public double GMean { get; set; }
```

Visual Basic

```
Public GMean As double
```

II.4.1.1.2.13 Kirchoff

C++

```
HRESULT get_Kirchoff(double*);
HRESULT put_Kirchoff(double);
```

C#

```
public double Kirchoff { get; set; }
```

Visual Basic

```
Public Kirchoff As double
```

Description

Shear (transversal) module.

(c) 2014 Autodesk Inc. All rights reserved

II.4.1.1.2.14 LX

C++

```
HRESULT get_LX(double*);
HRESULT put_LX(double);
```

C#

```
public double LX { get; set; }
```

Visual Basic

```
Public LX As double
```

Description

Thermal expansion coefficient.

II.4.1.1.2.15 Name

C++

```
HRESULT get_Name(BSTR*);
HRESULT put_Name(BSTR);
```

C#

```
public String Name { get; set; }
```

Visual Basic

```
Public Name As String
```

Description

Material name.

II.4.1.1.2.16 NU

C++

```
HRESULT get_NU(double*);
HRESULT put_NU(double);
```

C#

```
public double NU { get; set; }
```

Visual Basic

```
Public NU As double
```

Description

Poisson ratio .

II.4.1.1.2.17 Nuance

C++

```
HRESULT get_Nuance(BSTR*);
HRESULT put_Nuance(BSTR);
```

C#

```
public String Nuance { get; set; }
```

Visual Basic

```
Public Nuance As String
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Additional short material description.

II.4.1.1.2.18 PN_Deformation

C++

```
HRESULT get_PN_Deformation(double*);
HRESULT put_PN_Deformation(double);
```

C#

```
public double PN_Deformation { get; set; }
```

Visual Basic

```
Public PN_Deformation As double
```

Description

Shear deformation module for Polish code.

II.4.1.1.2.19 PN_E_Additional

C++

```
HRESULT get_PN_E_Additional(double*);
HRESULT put_PN_E_Additional(double);
```

C#

```
public double PN_E_Additional { get; set; }
```

Visual Basic

```
Public PN_E_Additional As double
```

Description

Additional Young module (timber) for Polish code (not implemented).

II.4.1.1.2.20 PN_E_Trans

C++

```
HRESULT get_PN_E_Trans(double*);
HRESULT put_PN_E_Trans(double);
```

C#

```
public double PN_E_Trans { get; set; }
```

Visual Basic

```
Public PN_E_Trans As double
```

Description

Young transverse module (timber) for Polish code (not implemented).

II.4.1.1.2.21 RE

C++

```
HRESULT get_RE(double*);
HRESULT put_RE(double);
```

C#

```
public double RE { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public RE As double
```

Description

Yield point (steel, aluminum), compression resistance (concrete).

II.4.1.1.2.22 RE_AxCompr

C++

```
HRESULT get_RE_AxCompr(double*);
HRESULT put_RE_AxCompr(double);
```

C#

```
public double RE_AxCompr { get; set; }
```

Visual Basic

```
Public RE_AxCompr As double
```

Description

Axial compression resistance (timber) (not implemented).

II.4.1.1.2.23 RE_AxTens

C++

```
HRESULT get_RE_AxTens(double*);
HRESULT put_RE_AxTens(double);
```

C#

```
public double RE_AxTens { get; set; }
```

Visual Basic

```
Public RE_AxTens As double
```

Description

Axial tension resistance (timber).

II.4.1.1.2.24 RE_Bending

C++

```
HRESULT get_RE_Bending(double*);
HRESULT put_RE_Bending(double);
```

C#

```
public double RE_Bending { get; set; }
```

Visual Basic

```
Public RE_Bending As double
```

Description

Bending resistance (timber).

II.4.1.1.2.25 RE_Shear

C++

```
HRESULT get_RE_Shear(double*);
HRESULT put_RE_Shear(double);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double RE_Shear { get; set; }
```

Visual Basic

```
Public RE_Shear As double
```

Description

Shear resistance (timber).

II.4.1.1.2.26 RE_TrCompr

C++

```
HRESULT get_RE_TrCompr(double*);
HRESULT put_RE_TrCompr(double);
```

C#

```
public double RE_TrCompr { get; set; }
```

Visual Basic

```
Public RE_TrCompr As double
```

Description

Transverse compression resistance (timber).

II.4.1.1.2.27 RE_TrTens

C++

```
HRESULT get_RE_TrTens(double*);
HRESULT put_RE_TrTens(double);
```

C#

```
public double RE_TrTens { get; set; }
```

Visual Basic

```
Public RE_TrTens As double
```

Description

Transverse tension resistance (timber).

II.4.1.1.2.28 RO

C++

```
HRESULT get_RO(double*);
HRESULT put_RO(double);
```

C#

```
public double RO { get; set; }
```

Visual Basic

```
Public RO As double
```

Description

Material density (mass).

II.4.1.1.2.29 RT

C++

```
HRESULT get_RT(double*);
```

(c) 2014 Autodesk Inc. All rights reserved

```
HRESULT put_RT(double);
```

C#

```
public double RT { get; set; }
```

Visual Basic

```
Public RT As double
```

Description

Tension resistance (steel).

II.4.1.1.2.30 SecondName

C++

```
HRESULT get_SecondName(BSTR*);
HRESULT put_SecondName(BSTR);
```

C#

```
public String SecondName { get; set; }
```

Visual Basic

```
Public SecondName As String
```

Description

Second name of the material, available only during editing material database under MS Access.

II.4.1.1.2.31 Steel_Thermal

C++

```
HRESULT get_Steel_Thermal(VARIANT_BOOL*);
HRESULT put_Steel_Thermal(VARIANT_BOOL);
```

C#

```
public bool Steel_Thermal { get; set; }
```

Visual Basic

```
Public Steel_Thermal As Boolean
```

Description

Thermal flag for steel materials. When it is on, it refers to special processing during production. .

II.4.1.1.2.32 Timber_Type

C++

```
HRESULT get_Timber_Type(IRobotMaterialTimberType*);
HRESULT put_Timber_Type(IRobotMaterialTimberType);
```

C#

```
public IRobotMaterialTimberType Timber_Type { get; set; }
```

Visual Basic

```
Public Timber_Type As IRobotMaterialTimberType
```

Description

Timber material type (glued multi-layered or normal) .

(c) 2014 Autodesk Inc. All rights reserved

II.4.1.1.2.33 Type

C++

```
HRESULT get_Type(IRobotMaterialType*);
HRESULT put_Type(IRobotMaterialType);
```

C#

```
public IRobotMaterialType Type { get; set; }
```

Visual Basic

```
Public Type As IRobotMaterialType
```

Description

Material type.

II.4.1.1.3 IRobotMaterialData Methods

The methods of the IRobotMaterialData class are listed here.

Public Methods

	Name	Description
	LoadFromDBase (see page 671)	The function allows one to load the data on a material with the defined name to the material database. The returned logical value (True, False) indicates whether the information on the material with the defined name has been found in the material database or not. .
	SaveToDBase (see page 672)	Function saves material to the user database. .

II.4.1.1.3.1 LoadFromDBase

C++

```
HRESULT LoadFromDBase(BSTR _material_name, VARIANT_BOOL* ret);
```

C#

```
public bool LoadFromDBase(String _material_name);
```

Visual Basic

```
Public Function LoadFromDBase(_material_name As String) As Boolean
```

Description

The function allows one to load the data on a material with the defined name to the material database. The returned logical value (True, False) indicates whether the information on the material with the defined name has been found in the material database or not. .

II.4.1.1.3.2 SaveToDBase

C++

```
HRESULT SaveToDBase();
```

C#

```
public void SaveToDBase();
```

Visual Basic

```
Public Sub SaveToDBase()
```

Description

Function saves material to the user database. .

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.

II.4.1.2 IRobotMaterialType

C++

```
enum IRobotMaterialType;
```

C#

```
public enum IRobotMaterialType;
```

Visual Basic

```
Public Enum IRobotMaterialType
```

Members

Members	Description
I_MT_STEEL = 1	Steel .
I_MT_ALUMINIUM = 3	Aluminum .
I_MT_TIMBER = 4	Timber .
I_MT_CONCRETE = 2	Concrete .
I_MT_OTHER = 5	Other .
I_MT_ALL = 0	Identifier denoting all material types Available since version 1.7.

Description

Types of materials accepted by Robot. .

II.4.1.3 IRobotMaterialTimberType

C++

```
enum IRobotMaterialTimberType;
```

C#

```
public enum IRobotMaterialTimberType;
```

Visual Basic

```
Public Enum IRobotMaterialTimberType
```

Members

Members	Description
I_MTT_NORMAL = 0	Normal timber .
I_MTT_GLUE_LAMINATED = 1	Multi-layered glued timber .
I_MTT_KERTO_S = 2	Available since version 1.7.
I_MTT_KERTO_Q = 3	Available since version 1.7.
I_MTT_KERTO_S_OLD = 4	Available since version 2.5.
I_MTT_KERTO_Q_OLD = 5	Available since version 2.5.

Description

Types of timber materials defined by Robot. .

(c) 2014 Autodesk Inc. All rights reserved

II.4.1.4 IRobotMaterialModel

C++

```
enum IRobotMaterialModel;
```

C#

```
public enum IRobotMaterialModel;
```

Visual Basic

```
Public Enum IRobotMaterialModel
```

Members

Members	Description
I_MM_ELASTIC = 0	Available since version 3.
I_MM_HUBER_MISES = 1	Available since version 3.
I_MM_DRUCKER_PRAGER = 2	Available since version 3.
I_MM_RANKIN = 3	Available since version 3.
I_MM_MOHR_COULOMB = 4	Available since version 3.

Description

Available material models.

Version

Available since version 3.

II.4.1.5 IRobotMaterialElasticType

C++

```
enum IRobotMaterialElasticType;
```

C#

```
public enum IRobotMaterialElasticType;
```

Visual Basic

```
Public Enum IRobotMaterialElasticType
```

Members

Members	Description
I_MET_PERFECTLY_PLASTIC = 0	Elastic-perfectly plastic model. Available since version 3.5.
I_MET_PLASTIC_WITH_HARDENING = 1	Elasto-plastic model with hardening . Available since version 3.5.

Description

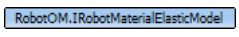
Available models of material elasticity.

Version

Available since version 3.5.

II.4.1.6 IRobotMaterialElasticModel

Class Hierarchy



(c) 2014 Autodesk Inc. All rights reserved

C++

```
interface IRobotMaterialElasticModel : IDispatch;
```

C#

```
public interface IRobotMaterialElasticModel;
```

Visual Basic

```
Public Interface IRobotMaterialElasticModel
```

Description

Model (see page 675) of material elasticity. .

Version

Available since version 3.5.

II.4.1.6.1 IRobotMaterialElasticModel Members

The following tables list the members exposed by IRobotMaterialElasticModel.

Public Fields

	Name	Description
	Coeff (see page 675)	Coefficient E1 / E for the elasto-plastic model with hardening .
	Model (see page 675)	
	UnloadingCoeff (see page 675)	Factor for the mixed unloading method .
	UnloadingMethod (see page 676)	Unloading method.

II.4.1.6.2 IRobotMaterialElasticModel Fields

The fields of the IRobotMaterialElasticModel class are listed here.

Public Fields

	Name	Description
	Coeff (see page 675)	Coefficient E1 / E for the elasto-plastic model with hardening .
	Model (see page 675)	
	UnloadingCoeff (see page 675)	Factor for the mixed unloading method .
	UnloadingMethod (see page 676)	Unloading method.

II.4.1.6.2.1 Coeff

C++

```
HRESULT get_Coeff(double*);
HRESULT put_Coeff(double);
```

C#

```
public double Coeff { get; set; }
```

Visual Basic

```
Public Coeff As double
```

Description

Coefficient E1 / E for the elasto-plastic model with hardening .

Version

Available since version 3.5.

(c) 2014 Autodesk Inc. All rights reserved

II.4.1.6.2.2 Model

C++

```
HRESULT get_Model(IRobotMaterialElasticType*);
HRESULT put_Model(IRobotMaterialElasticType);
```

C#

```
public IRobotMaterialElasticType Model { get; set; }
```

Visual Basic

```
Public Model As IRobotMaterialElasticType
```

Version

Available since version 3.5.

II.4.1.6.2.3 UnloadingCoeff

C++

```
HRESULT get_UnloadingCoeff(double*);
HRESULT put_UnloadingCoeff(double);
```

C#

```
public double UnloadingCoeff { get; set; }
```

Visual Basic

```
Public UnloadingCoeff As double
```

Description

Factor for the mixed unloading method .

Version

Available since version 3.5.

II.4.1.6.2.4 UnloadingMethod

C++

```
HRESULT get_UnloadingMethod(IRobotMaterialElasticUnloadingMethod*);
HRESULT put_UnloadingMethod(IRobotMaterialElasticUnloadingMethod);
```

C#

```
public IRobotMaterialElasticUnloadingMethod UnloadingMethod { get; set; }
```

Visual Basic

```
Public UnloadingMethod As IRobotMaterialElasticUnloadingMethod
```

Description

Unloading method.

Version

Available since version 3.5.

II.4.1.7 IRobotMaterialElasticUnloadingMethod

C++

```
enum IRobotMaterialElasticUnloadingMethod;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public enum IRobotMaterialElasticUnloadingMethod;
```

Visual Basic

```
Public Enum IRobotMaterialElasticUnloadingMethod
```

Members

Members	Description
I_MEUM_ELASTIC = 1	Elastic method of unloading . Available since version 3.5.
I_MEUM_PLASTIC = 2	Plastic method of unloading . Available since version 3.5.
I_MEUM_DAMAGE = 3	Damage method of unloading . Available since version 3.5.
I_MEUM_MIXED = 4	Mixed method of unloading . Available since version 3.5.

Description

Available methods of material unloading.

Version

Available since version 3.5.

II.4.2 Non-linear constraints

Non-linear connection models.

Version

Available since version 3.

Enumerations

	Name	Description
	IRobotNonlinearLinkCurveType (see page 687)	Available curve types:.
	IRobotNonlinearLinkModelType (see page 699)	Available relation types described by a non-linear model.
	IRobotNonlinearLinkSemiAxisType (see page 699)	Semi-axis types.

Interfaces

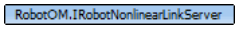
	Name	Description
	IRobotNonlinearLinkServer (see page 677)	Server of non-linear connection models. .
	IRobotNonlinearLink (see page 680)	Definintion of a non-linear connection model. .
	IRobotNonlinearLinkParams (see page 683)	Set of parameters describing any curve.
	IRobotNonlinearLinkParamsLinear (see page 684)	Interface specific to the LINEAR curve type. .
	IRobotNonlinearLinkParamsBLinear (see page 685)	Interface specific to the B_LINEAR curve type.
	IRobotNonlinearLinkParamsParabolic (see page 687)	Interface specific to the PARABOLIC and PARABOLIC_EC2 curve types.
	IRobotNonlinearLinkParamsPlastic (see page 689)	Interface specific to the PERFECTLY_PLASTIC curve type.

(c) 2014 Autodesk Inc. All rights reserved

	IRobotNonlinearLinkParamsPlasticHardening (see page 691)	Interface specific to the PLASTIC_HARDENING curve type.
	IRobotNonlinearLinkParamsGapHook (see page 693)	Interface specific to the GAP_HOOK curve type.
	IRobotNonlinearLinkParamsCustom (see page 694)	Interface specific to the CUSTOM curve type; it describes a curve composed of freely defined segments.
	IRobotNonlinearLinkParamsCustomSegment (see page 697)	Interface defining segment of the CUSTOM type curve.
	IRobotNonlinearLinkMngr (see page 700)	Support for relations of the non-linear model - direction type.

II.4.2.1 IRobotNonlinearLinkServer

Class Hierarchy



C++

```
interface IRobotNonlinearLinkServer : IDispatch;
```

C#

```
public interface IRobotNonlinearLinkServer;
```

Visual Basic

```
Public Interface IRobotNonlinearLinkServer
```

Description

Server of non-linear connection models. .

Version

Available since version 3.

II.4.2.1.1 IRobotNonlinearLinkServer Members

The following tables list the members exposed by IRobotNonlinearLinkServer.

Public Fields

	Name	Description
	Count (see page 678)	Number of available models.

Public Methods

	Name	Description
	Create (see page 678)	Function creates and returns a new non-linear connection model.
	Find (see page 679)	Function returns index of a model of the specified name. .
	Get (see page 679)	Function takes the interface for defining a model of the specified index. .
	Remove (see page 679)	Function deletes model of the specified index. .

II.4.2.1.2 IRobotNonlinearLinkServer Fields

The fields of the IRobotNonlinearLinkServer class are listed here.

Public Fields

	Name	Description
	Count (see page 678)	Number of available models.

(c) 2014 Autodesk Inc. All rights reserved

II.4.2.1.2.1 Count

C++

```
HRESULT get_Count( long*);
```

C#

```
public  long Count { get; }
```

Visual Basic

```
Public ReadOnly Count As  long
```

Description

Number of available models.

Version

Available since version 3.

II.4.2.1.3 IRobotNonlinearLinkServer Methods

The methods of the IRobotNonlinearLinkServer class are listed here.

Public Methods

	Name	Description
	Create (see page 678)	Function creates and returns a new non-linear connection model.
	Find (see page 679)	Function returns index of a model of the specified name. .
	Get (see page 679)	Function takes the interface for defining a model of the specified index. .
	Remove (see page 679)	Function deletes model of the specified index. .

II.4.2.1.3.1 Create

C++

```
HRESULT Create(BSTR _name, IRobotNonlinearLink** ret);
```

C#

```
public IRobotNonlinearLink Create(String _name);
```

Visual Basic

```
Public Function Create(_name As String) As IRobotNonlinearLink
```

Description

Function creates and returns a new non-linear connection model.

Version

Available since version 3.

II.4.2.1.3.2 Find

C++

```
HRESULT Find(BSTR _name, long* ret);
```

C#

```
public long Find(String _name);
```

Visual Basic

```
Public Function Find(_name As String) As long
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Function returns index of a model of the specified name. .

Version

Available since version 3.

II.4.2.1.3.3 Get

C++

```
HRESULT Get(long _idx, IRobotNonlinearLink** ret);
```

C#

```
public IRobotNonlinearLink Get(long _idx);
```

Visual Basic

```
Public Function Get(_idx As long) As IRobotNonlinearLink
```

Description

Function takes the interface for defining a model of the specified index. .

Version

Available since version 3.

II.4.2.1.3.4 Remove

C++

```
HRESULT Remove(long _idx);
```

C#

```
public void Remove(long _idx);
```

Visual Basic

```
Public Sub Remove(_idx As long)
```

Description

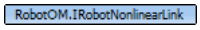
Function deletes model of the specified index. .

Version

Available since version 3.

II.4.2.2 IRobotNonlinearLink

Class Hierarchy



C++

```
interface IRobotNonlinearLink : IDispatch;
```

C#

```
public interface IRobotNonlinearLink;
```

Visual Basic

```
Public Interface IRobotNonlinearLink
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Definintion of a non-linear connection model. .

Version

Available since version 3.

II.4.2.2.1 IRobotNonlinearLink Members

The following tables list the members exposed by IRobotNonlinearLink.

Public Fields

	Name	Description
	ModelType (see page 681)	Type of the relationship described by the model.
	Name (see page 681)	Model name.
	Symetry (see page 681)	Curve symmetry.

Public Methods

	Name	Description
	GetCurveType (see page 682)	Function takes the curve type for the indicated semi-axis. .
	GetParams (see page 682)	Function takes a set of parameters describing a given curve. .
	SetCurveType (see page 683)	Function sets the curve type for the indicated semi-axis. .
	SetParams (see page 683)	Function updates the set of parameters describing a curve. .

II.4.2.2.2 IRobotNonlinearLink Fields

The fields of the IRobotNonlinearLink class are listed here.

Public Fields

	Name	Description
	ModelType (see page 681)	Type of the relationship described by the model.
	Name (see page 681)	Model name.
	Symetry (see page 681)	Curve symmetry.

II.4.2.2.2.1 ModelType

C++

```
HRESULT get_ModelType(IRobotNonlinearLinkModelType*);
HRESULT put_ModelType(IRobotNonlinearLinkModelType);
```

C#

```
public IRobotNonlinearLinkModelType ModelType { get; set; }
```

Visual Basic

```
Public ModelType As IRobotNonlinearLinkModelType
```

Description

Type of the relationship described by the model.

Version

Available since version 3.

II.4.2.2.2.2 Name

C++

```
HRESULT get_Name(BSTR*);
HRESULT put_Name(BSTR);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public String Name { get; set; }
```

Visual Basic

```
Public Name As String
```

Description

Model name.

Version

Available since version 3.

II.4.2.2.2.3 Symetry

C++

```
HRESULT get_Symetry(VARIANT_BOOL*);
```

C#

```
public bool Symetry { get; }
```

Visual Basic

```
Public ReadOnly Symetry As Boolean
```

Description

Curve symmetry.

Version

Available since version 3.

II.4.2.2.3 IRobotNonlinearLink Methods

The methods of the IRobotNonlinearLink class are listed here.

Public Methods

	Name	Description
	GetCurveType (see page 682)	Function takes the curve type for the indicated semi-axis. .
	GetParams (see page 682)	Function takes a set of parameters describing a given curve. .
	SetCurveType (see page 683)	Function sets the curve type for the indicated semi-axis. .
	SetParams (see page 683)	Function updates the set of parameters describing a curve. .

II.4.2.2.3.1 GetCurveType

C++

```
HRESULT GetCurveType(IRoborNonlinearLinkSemiAxisType _axis_type = I_NLSAT_ANY,
IRobotNonlinearLinkCurveType* ret);
```

C#

```
public IRobotNonlinearLinkCurveType GetCurveType(IRoborNonlinearLinkSemiAxisType _axis_type
= I_NLSAT_ANY);
```

Visual Basic

```
Public Function GetCurveType(Optional _axis_type As IRoborNonlinearLinkSemiAxisType =
I_NLSAT_ANY) As IRobotNonlinearLinkCurveType
```

Description

Function takes the curve type for the indicated semi-axis. .

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.

II.4.2.2.3.2 GetParams

C++

```
HRESULT GetParams(IRobotNonlinearLinkSemiAxisType _axis_type = I_NLSAT_ANY,
IRobotNonlinearLinkParams** ret);
```

C#

```
public IRobotNonlinearLinkParams GetParams(IRobotNonlinearLinkSemiAxisType _axis_type =
I_NLSAT_ANY);
```

Visual Basic

```
Public Function GetParams(Optional _axis_type As IRobotNonlinearLinkSemiAxisType =
I_NLSAT_ANY) As IRobotNonlinearLinkParams
```

Description

Function takes a set of parameters describing a given curve. .

Version

Available since version 3.

II.4.2.2.3.3 SetCurveType

C++

```
HRESULT SetCurveType(IRobotNonlinearLinkCurveType _curve_type,
IRobotNonlinearLinkSemiAxisType _axis_type = I_NLSAT_ANY);
```

C#

```
public void SetCurveType(IRobotNonlinearLinkCurveType _curve_type,
IRobotNonlinearLinkSemiAxisType _axis_type = I_NLSAT_ANY);
```

Visual Basic

```
Public Sub SetCurveType(_curve_type As IRobotNonlinearLinkCurveType, Optional _axis_type As
IRobotNonlinearLinkSemiAxisType = I_NLSAT_ANY)
```

Description

Function sets the curve type for the indicated semi-axis. .

Version

Available since version 3.

II.4.2.2.3.4 SetParams

C++

```
HRESULT SetParams(IRobotNonlinearLinkParams _params, IRobotNonlinearLinkSemiAxisType
_axis_type = I_NLSAT_ANY);
```

C#

```
public void SetParams(IRobotNonlinearLinkParams _params, IRobotNonlinearLinkSemiAxisType
_axis_type = I_NLSAT_ANY);
```

Visual Basic

```
Public Sub SetParams(_params As IRobotNonlinearLinkParams, Optional _axis_type As
IRobotNonlinearLinkSemiAxisType = I_NLSAT_ANY)
```

(c) 2014 Autodesk Inc. All rights reserved

Description

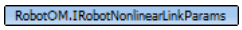
Function updates the set of parameters describing a curve. .

Version

Available since version 3.

II.4.2.3 IRobotNonlinearLinkParams

Class Hierarchy



C++

```
interface IRobotNonlinearLinkParams : IDispatch;
```

C#

```
public interface IRobotNonlinearLinkParams;
```

Visual Basic

```
Public Interface IRobotNonlinearLinkParams
```

Description

Set of parameters describing any curve.

Version

Available since version 3.

II.4.2.3.1 IRobotNonlinearLinkParams Members

The following tables list the members exposed by IRobotNonlinearLinkParams.

Public Fields

	Name	Description
	CurveType (see page 684)	Curve type.

II.4.2.3.2 IRobotNonlinearLinkParams Fields

The fields of the IRobotNonlinearLinkParams class are listed here.

Public Fields

	Name	Description
	CurveType (see page 684)	Curve type.

II.4.2.3.2.1 CurveType

C++

```
HRESULT get_CurveType(IRobotNonlinearLinkCurveType*);
```

C#

```
public IRobotNonlinearLinkCurveType CurveType { get; }
```

Visual Basic

```
Public ReadOnly CurveType As IRobotNonlinearLinkCurveType
```

Description

Curve type.

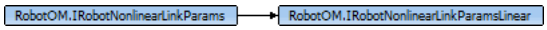
(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.

II.4.2.4 IRobotNonlinearLinkParamsLinear

Class Hierarchy



C++

```
interface IRobotNonlinearLinkParamsLinear : IRobotNonlinearLinkParams;
```

C#

```
public interface IRobotNonlinearLinkParamsLinear : IRobotNonlinearLinkParams;
```

Visual Basic

```
Public Interface IRobotNonlinearLinkParamsLinear
```

Description

Interface specific to the LINEAR curve type. .

Version

Available since version 3.

II.4.2.4.1 IRobotNonlinearLinkParamsLinear Members

The following tables list the members exposed by IRobotNonlinearLinkParamsLinear.

Public Fields

	Name	Description
	CurveType (see page 684)	Curve type.
	K (see page 685)	

II.4.2.4.2 IRobotNonlinearLinkParamsLinear Fields

The fields of the IRobotNonlinearLinkParamsLinear class are listed here.

Public Fields

	Name	Description
	K (see page 685)	

II.4.2.4.2.1 K

C++

```
HRESULT get_K( double*);
HRESULT put_K( double);
```

C#

```
public  double K { get; set; }
```

Visual Basic

```
Public K As  double
```

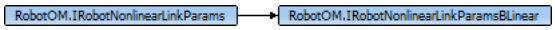
Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

II.4.2.5 IRobotNonlinearLinkParamsBLinear

Class Hierarchy



C++

```
interface IRobotNonlinearLinkParamsBLinear : IRobotNonlinearLinkParams;
```

C#

```
public interface IRobotNonlinearLinkParamsBLinear : IRobotNonlinearLinkParams;
```

Visual Basic

```
Public Interface IRobotNonlinearLinkParamsBLinear
```

Description

Interface specific to the B_LINEAR curve type.

Version

Available since version 3.

II.4.2.5.1 IRobotNonlinearLinkParamsBLinear Members

The following tables list the members exposed by IRobotNonlinearLinkParamsBLinear.

Public Fields

	Name	Description
	CurveType (see page 684)	Curve type.
	D1 (see page 686)	
	K1 (see page 686)	
	K2 (see page 687)	

II.4.2.5.2 IRobotNonlinearLinkParamsBLinear Fields

The fields of the IRobotNonlinearLinkParamsBLinear class are listed here.

Public Fields

	Name	Description
	D1 (see page 686)	
	K1 (see page 686)	
	K2 (see page 687)	

II.4.2.5.2.1 D1

C++

```
HRESULT get_D1( double*);
HRESULT put_D1( double);
```

C#

```
public  double D1 { get; set; }
```

Visual Basic

```
Public D1 As  double
```

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

II.4.2.5.2.2 K1

C++

```
HRESULT get_K1( double*);
HRESULT put_K1( double);
```

C#

```
public  double K1 { get; set; }
```

Visual Basic

```
Public K1 As  double
```

Version

Available since version 3.

II.4.2.5.2.3 K2

C++

```
HRESULT get_K2( double*);
HRESULT put_K2( double);
```

C#

```
public  double K2 { get; set; }
```

Visual Basic

```
Public K2 As  double
```

Version

Available since version 3.

II.4.2.6 IRobotNonlinearLinkCurveType

C++

```
enum IRobotNonlinearLinkCurveType;
```

C#

```
public enum IRobotNonlinearLinkCurveType;
```

Visual Basic

```
Public Enum IRobotNonlinearLinkCurveType
```

Members

Members	Description
I_NLCT_LINEAR = 1	Available since version 3.
I_NLCT_B_LINEAR = 2	Available since version 3.
I_NLCT_PARABOLIC = 3	Available since version 3.
I_NLCT_PARABOLIC_EC2 = 4	Available since version 3.
I_NLCT_PERFECTLY_PLASTIC = 5	Available since version 3.
I_NLCT_PLASTIC_WITH_HARDENING = 6	Available since version 3.
I_NLCT_GAP_HOOK = 7	Available since version 3.
I_NLCT_CUSTOM = 8	Available since version 3.

Description

Available curve types:.

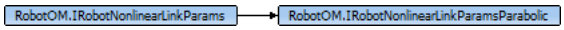
(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.

II.4.2.7 IRobotNonlinearLinkParamsParabolic

Class Hierarchy



C++

```
interface IRobotNonlinearLinkParamsParabolic : IRobotNonlinearLinkParams;
```

C#

```
public interface IRobotNonlinearLinkParamsParabolic : IRobotNonlinearLinkParams;
```

Visual Basic

```
Public Interface IRobotNonlinearLinkParamsParabolic
```

Description

Interface specific to the PARABOLIC and PARABOLIC_EC2 curve types.

Version

Available since version 3.

II.4.2.7.1 IRobotNonlinearLinkParamsParabolic Members

The following tables list the members exposed by IRobotNonlinearLinkParamsParabolic.

Public Fields

	Name	Description
	CurveType (see page 684)	Curve type.
	Dlim (see page 688)	
	Dmax (see page 688)	
	Flim (see page 689)	
	K (see page 689)	

II.4.2.7.2 IRobotNonlinearLinkParamsParabolic Fields

The fields of the IRobotNonlinearLinkParamsParabolic class are listed here.

Public Fields

	Name	Description
	Dlim (see page 688)	
	Dmax (see page 688)	
	Flim (see page 689)	
	K (see page 689)	

II.4.2.7.2.1 Dlim

C++

```
HRESULT get_Dlim( double*);
HRESULT put_Dlim( double);
```

C#

```
public  double Dlim { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Dlim As  double
```

Version

Available since version 3.

II.4.2.7.2.2 Dmax

C++

```
HRESULT get_Dmax( double*);
HRESULT put_Dmax( double);
```

C#

```
public  double Dmax { get; set; }
```

Visual Basic

```
Public Dmax As  double
```

Version

Available since version 3.

II.4.2.7.2.3 Flim

C++

```
HRESULT get_Flim( double*);
HRESULT put_Flim( double);
```

C#

```
public  double Flim { get; set; }
```

Visual Basic

```
Public Flim As  double
```

Version

Available since version 3.

II.4.2.7.2.4 K

C++

```
HRESULT get_K( double*);
HRESULT put_K( double);
```

C#

```
public  double K { get; set; }
```

Visual Basic

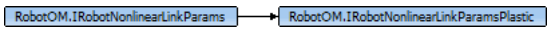
```
Public K As  double
```

Version

Available since version 3.

II.4.2.8 IRobotNonlinearLinkParamsPlastic

Class Hierarchy



(c) 2014 Autodesk Inc. All rights reserved

C++

```
interface IRobotNonlinearLinkParamsPlastic : IRobotNonlinearLinkParams;
```

C#

```
public interface IRobotNonlinearLinkParamsPlastic : IRobotNonlinearLinkParams;
```

Visual Basic

```
Public Interface IRobotNonlinearLinkParamsPlastic
```

Description

Interface specific to the PERFECTLY_PLASTIC curve type.

Version

Available since version 3.

II.4.2.8.1 IRobotNonlinearLinkParamsPlastic Members

The following tables list the members exposed by IRobotNonlinearLinkParamsPlastic.

Public Fields

	Name	Description
	CurveType (see page 684)	Curve type.
	Dlim (see page 690)	
	Flim (see page 690)	
	K (see page 691)	
	W (see page 691)	

II.4.2.8.2 IRobotNonlinearLinkParamsPlastic Fields

The fields of the IRobotNonlinearLinkParamsPlastic class are listed here.

Public Fields

	Name	Description
	Dlim (see page 690)	
	Flim (see page 690)	
	K (see page 691)	
	W (see page 691)	

II.4.2.8.2.1 Dlim

C++

```
HRESULT get_Dlim( double*);
HRESULT put_Dlim( double);
```

C#

```
public  double Dlim { get; set; }
```

Visual Basic

```
Public Dlim As  double
```

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

II.4.2.8.2.2 Flim

C++

```
HRESULT get_Flim( double*);
HRESULT put_Flim( double);
```

C#

```
public  double Flim { get; set; }
```

Visual Basic

```
Public Flim As  double
```

Version

Available since version 3.

II.4.2.8.2.3 K

C++

```
HRESULT get_K( double*);
HRESULT put_K( double);
```

C#

```
public  double K { get; set; }
```

Visual Basic

```
Public K As  double
```

Version

Available since version 3.

II.4.2.8.2.4 W

C++

```
HRESULT get_W( double*);
HRESULT put_W( double);
```

C#

```
public  double W { get; set; }
```

Visual Basic

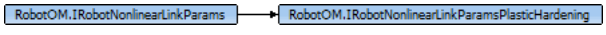
```
Public W As  double
```

Version

Available since version 3.

II.4.2.9 IRobotNonlinearLinkParamsPlasticHardening

Class Hierarchy



C++

```
interface IRobotNonlinearLinkParamsPlasticHardening : IRobotNonlinearLinkParams;
```

C#

```
public interface IRobotNonlinearLinkParamsPlasticHardening : IRobotNonlinearLinkParams;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Interface IRobotNonlinearLinkParamsPlasticHardening
```

Description

Interface specific to the PLASTIC_HARDENING curve type.

Version

Available since version 3.

II.4.2.9.1 IRobotNonlinearLinkParamsPlasticHardening Members

The following tables list the members exposed by IRobotNonlinearLinkParamsPlasticHardening.

Public Fields

	Name	Description
	CurveType (see page 684)	Curve type.
	Dlim (see page 692)	
	Flim (see page 692)	
	K0 (see page 692)	
	K1 (see page 693)	
	W (see page 693)	

II.4.2.9.2 IRobotNonlinearLinkParamsPlasticHardening Fields

The fields of the IRobotNonlinearLinkParamsPlasticHardening class are listed here.

Public Fields

	Name	Description
	Dlim (see page 692)	
	Flim (see page 692)	
	K0 (see page 692)	
	K1 (see page 693)	
	W (see page 693)	

II.4.2.9.2.1 Dlim

C++

```
HRESULT get_Dlim( double*);
HRESULT put_Dlim( double);
```

C#

```
public  double Dlim { get; set; }
```

Visual Basic

```
Public Dlim As  double
```

Version

Available since version 3.

II.4.2.9.2.2 Flim

C++

```
HRESULT get_Flim( double*);
HRESULT put_Flim( double);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public  double Flim { get; set; }
```

Visual Basic

```
Public Flim As  double
```

Version

Available since version 3.

II.4.2.9.2.3 K0

C++

```
HRESULT get_K0( double*);
HRESULT put_K0( double);
```

C#

```
public  double K0 { get; set; }
```

Visual Basic

```
Public K0 As  double
```

Version

Available since version 3.

II.4.2.9.2.4 K1

C++

```
HRESULT get_K1( double*);
HRESULT put_K1( double);
```

C#

```
public  double K1 { get; set; }
```

Visual Basic

```
Public K1 As  double
```

Version

Available since version 3.

II.4.2.9.2.5 W

C++

```
HRESULT get_W( double*);
HRESULT put_W( double);
```

C#

```
public  double W { get; set; }
```

Visual Basic

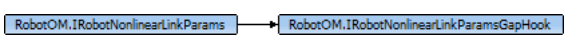
```
Public W As  double
```

Version

Available since version 3.

II.4.2.10 IRobotNonlinearLinkParamsGapHook

Class Hierarchy



(c) 2014 Autodesk Inc. All rights reserved

C++

```
interface IRobotNonlinearLinkParamsGapHook : IRobotNonlinearLinkParams;
```

C#

```
public interface IRobotNonlinearLinkParamsGapHook : IRobotNonlinearLinkParams;
```

Visual Basic

```
Public Interface IRobotNonlinearLinkParamsGapHook
```

Description

Interface specific to the GAP_HOOK curve type.

Version

Available since version 3.

II.4.2.10.1 IRobotNonlinearLinkParamsGapHook Members

The following tables list the members exposed by IRobotNonlinearLinkParamsGapHook.

Public Fields

	Name	Description
	CurveType (see page 684)	Curve type.
	D (see page 694)	
	K (see page 694)	

II.4.2.10.2 IRobotNonlinearLinkParamsGapHook Fields

The fields of the IRobotNonlinearLinkParamsGapHook class are listed here.

Public Fields

	Name	Description
	D (see page 694)	
	K (see page 694)	

II.4.2.10.2.1 D

C++

```
HRESULT get_D( double*);
HRESULT put_D( double);
```

C#

```
public  double D { get; set; }
```

Visual Basic

```
Public D As  double
```

Version

Available since version 3.

II.4.2.10.2.2 K

C++

```
HRESULT get_K( double*);
HRESULT put_K( double);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public  double K { get; set; }
```

Visual Basic

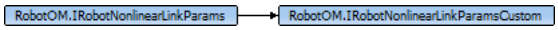
```
Public K As  double
```

Version

Available since version 3.

II.4.2.11 IRobotNonlinearLinkParamsCustom

Class Hierarchy



C++

```
interface IRobotNonlinearLinkParamsCustom : IRobotNonlinearLinkParams;
```

C#

```
public interface IRobotNonlinearLinkParamsCustom : IRobotNonlinearLinkParams;
```

Visual Basic

```
Public Interface IRobotNonlinearLinkParamsCustom
```

Description

Interface specific to the CUSTOM curve type; it describes a curve composed of freely defined segments.

Version

Available since version 3.

II.4.2.11.1 IRobotNonlinearLinkParamsCustom Members

The following tables list the members exposed by IRobotNonlinearLinkParamsCustom.

Public Fields

	Name	Description
	Count (see page 695)	Number of function segments.
	CurveType (see page 684)	Curve type.

Public Methods

	Name	Description
	Get (see page 696)	Function takes a set of parameters for defining a segment of a given index. .
	New (see page 696)	Function creates and returns a new curve segment. .
	Remove (see page 696)	Function deletes a segment of a given index. .
	Set (see page 697)	Function updates parameters describing a segment of a given index. .

II.4.2.11.2 IRobotNonlinearLinkParamsCustom Fields

The fields of the IRobotNonlinearLinkParamsCustom class are listed here.

Public Fields

	Name	Description
	Count (see page 695)	Number of function segments.

(c) 2014 Autodesk Inc. All rights reserved

II.4.2.11.2.1 Count

C++

```
HRESULT get_Count( long*);
```

C#

```
public  long Count { get; }
```

Visual Basic

```
Public ReadOnly Count As  long
```

Description

Number of function segments.

Version

Available since version 3.

II.4.2.11.3 IRobotNonlinearLinkParamsCustom Methods

The methods of the IRobotNonlinearLinkParamsCustom class are listed here.

Public Methods

	Name	Description
	Get (see page 696)	Function takes a set of parameters for defining a segment of a given index. .
	New (see page 696)	Function creates and returns a new curve segment. .
	Remove (see page 696)	Function deletes a segment of a given index. .
	Set (see page 697)	Function updates parameters describing a segment of a given index. .

II.4.2.11.3.1 Get

C++

```
HRESULT Get(long _idx, IRobotNonlinearLinkParamsCustomSegment** ret);
```

C#

```
public IRobotNonlinearLinkParamsCustomSegment Get(long _idx);
```

Visual Basic

```
Public Function Get(_idx As long) As IRobotNonlinearLinkParamsCustomSegment
```

Description

Function takes a set of parameters for defining a segment of a given index. .

Version

Available since version 3.

II.4.2.11.3.2 New

C++

```
HRESULT New(IRobotNonlinearLinkParamsCustomSegment** ret);
```

C#

```
public IRobotNonlinearLinkParamsCustomSegment New();
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Function New() As IRobotNonlinearLinkParamsCustomSegment
```

Description

Function creates and returns a new curve segment. .

Version

Available since version 3.

II.4.2.11.3.3 Remove

C++

```
HRESULT Remove(long _idx);
```

C#

```
public void Remove(long _idx);
```

Visual Basic

```
Public Sub Remove(_idx As long)
```

Description

Function deletes a segment of a given index. .

Version

Available since version 3.

II.4.2.11.3.4 Set

C++

```
HRESULT Set(long _idx, IRobotNonlinearLinkParamsCustomSegment* _segment);
```

C#

```
public void Set(long _idx, IRobotNonlinearLinkParamsCustomSegment _segment);
```

Visual Basic

```
Public Sub Set(_idx As long, ByRef _segment As IRobotNonlinearLinkParamsCustomSegment)
```

Description

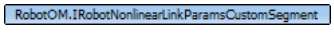
Function updates parameters describing a segment of a given index. .

Version

Available since version 3.

II.4.2.12 IRobotNonlinearLinkParamsCustomSegment

Class Hierarchy



C++

```
interface IRobotNonlinearLinkParamsCustomSegment : IDispatch;
```

C#

```
public interface IRobotNonlinearLinkParamsCustomSegment;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Interface IRobotNonlinearLinkParamsCustomSegment
```

Description

Interface defining segment of the CUSTOM type curve.

Version

Available since version 3.

II.4.2.12.1 IRobotNonlinearLinkParamsCustomSegment Members

The following tables list the members exposed by IRobotNonlinearLinkParamsCustomSegment.

Public Fields

	Name	Description
	Constant (see page 698)	Flag indicating if function expression is a constant number (y = a).
	Expression (see page 698)	Expression representing the functional relationship.
	OriginPoint (see page 698)	Segment beginning point.

II.4.2.12.2 IRobotNonlinearLinkParamsCustomSegment Fields

The fields of the IRobotNonlinearLinkParamsCustomSegment class are listed here.

Public Fields

	Name	Description
	Constant (see page 698)	Flag indicating if function expression is a constant number (y = a).
	Expression (see page 698)	Expression representing the functional relationship.
	OriginPoint (see page 698)	Segment beginning point.

II.4.2.12.2.1 Constant

C++

```
HRESULT get_Constant(VARIANT_BOOL*);
HRESULT put_Constant(VARIANT_BOOL);
```

C#

```
public bool Constant { get; set; }
```

Visual Basic

```
Public Constant As Boolean
```

Description

Flag indicating if function expression is a constant number (y = a).

Version

Available since version 3.

II.4.2.12.2.2 Expression

C++

```
HRESULT get_Expression(BSTR*);
HRESULT put_Expression(BSTR);
```

C#

```
public String Expression { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Expression As String
```

Description

Expression representing the functional relationship.

Version

Available since version 3.

II.4.2.12.2.3 OriginPoint

C++

```
HRESULT get_OriginPoint(double*);
HRESULT put_OriginPoint(double);
```

C#

```
public double OriginPoint { get; set; }
```

Visual Basic

```
Public OriginPoint As double
```

Description

Segment beginning point.

Version

Available since version 3.

II.4.2.13 IRobotNonlinearLinkModelType

C++

```
enum IRobotNonlinearLinkModelType;
```

C#

```
public enum IRobotNonlinearLinkModelType;
```

Visual Basic

```
Public Enum IRobotNonlinearLinkModelType
```

Members

Members	Description
I_NLMT_FORCE_DISPLACEMENT = 1	Force - displacement. Available since version 3.
I_NLMT_MOMENT_ROTATION = 2	Moment - rotation. Available since version 3.

Description

Available relation types described by a non-linear model.

Version

Available since version 3.

II.4.2.14 IRobotNonlinearLinkSemiAxisType

C++

```
enum IRobotNonlinearLinkSemiAxisType;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public enum IRobotNonlinearLinkSemiAxisType;
```

Visual Basic

```
Public Enum IRobotNonlinearLinkSemiAxisType
```

Members

Members	Description
I_NLSAT_POSITIVE = 1	Positive semi-axis. Available since version 3.
I_NLSAT_NEGATIVE = 2	Negative semi-axis.
I_NLSAT_ANY = 3	Arbitrary semi-axis. Available since version 3.

Description

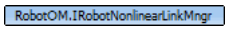
Semi-axis types.

Version

Available since version 3.

II.4.2.15 IRobotNonlinearLinkMngr

Class Hierarchy



C++

```
interface IRobotNonlinearLinkMngr : IDispatch;
```

C#

```
public interface IRobotNonlinearLinkMngr;
```

Visual Basic

```
Public Interface IRobotNonlinearLinkMngr
```

Description

Support for relations of the non-linear model - direction type.

Version

Available since version 3.

II.4.2.15.1 IRobotNonlinearLinkMngr Members

The following tables list the members exposed by IRobotNonlinearLinkMngr.

Public Methods

	Name	Description
	Delete (see page 701)	Function deletes a non-linear model from the indicated direction. .
	Get (see page 701)	Function returns the name of a non-linear model assigned to the specified direction.
	IsDefined (see page 701)	Function checks a non-zero value (True), if a non-linear model has been assigned to the indicated direction. .
	Set (see page 702)	Function determines a non-linear model for the specified drection. .

(c) 2014 Autodesk Inc. All rights reserved

II.4.2.15.2 IRobotNonlinearLinkMngr Methods

The methods of the IRobotNonlinearLinkMngr class are listed here.

Public Methods

	Name	Description
	Delete (see page 701)	Function deletes a non-linear model from the indicated direction. .
	Get (see page 701)	Function returns the name of a non-linear model assigned to the specified direction.
	IsDefined (see page 701)	Function checks a non-zero value (True), if a non-linear model has been assigned to the indicated direction. .
	Set (see page 702)	Function determines a non-linear model for the specified drection. .

II.4.2.15.2.1 Delete

C++

```
HRESULT Delete(IRobotDegreeOfFreedom _direction);
```

C#

```
public void Delete(IRobotDegreeOfFreedom _direction);
```

Visual Basic

```
Public Sub Delete(_direction As IRobotDegreeOfFreedom)
```

Description

Function deletes a non-linear model from the indicated direction. .

Version

Available since version 3.

II.4.2.15.2.2 Get

C++

```
HRESULT Get(IRobotDegreeOfFreedom _direction, BSTR* ret);
```

C#

```
public String Get(IRobotDegreeOfFreedom _direction);
```

Visual Basic

```
Public Function Get(_direction As IRobotDegreeOfFreedom) As String
```

Description

Function returns the name of a non-linear model assigned to the specified direction.

Version

Available since version 3.

II.4.2.15.2.3 IsDefined

C++

```
HRESULT IsDefined(IRobotDegreeOfFreedom _direction, VARIANT_BOOL* ret);
```

C#

```
public bool IsDefined(IRobotDegreeOfFreedom _direction);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Function IsDefined(_direction As IRobotDegreeOfFreedom) As Boolean
```

Description

Function checks a non-zero value (True), if a non-linear model has been assigned to the indicated direction. .

Version

Available since version 3.

II.4.2.15.2.4 Set

C++

```
HRESULT Set(IRobotDegreeOfFreedom _direction, BSTR _model_name);
```

C#

```
public void Set(IRobotDegreeOfFreedom _direction, String _model_name);
```

Visual Basic

```
Public Sub Set(_direction As IRobotDegreeOfFreedom, _model_name As String)
```

Description

Function determines a non-linear model for the specified drection. .

Version

Available since version 3.

II.5 Objects

Enumerations

	Name	Description
	IRobotObjModificationType (see page 744)	Set of identifiers has been defined for modifications which may be performed on specified objects - structure components. .
	IRobotObjOperationType (see page 744)	Set of identifiers has been defined for operations which may be performed on specified objects - structure components. .
	IRobotObjPartType (see page 779)	
	IRobotObjLocalXDirDefinitionType (see page 794)	Type of direction definition for the local X axis of the plate. .

Interfaces

	Name	Description
	IRobotObjModification (see page 745)	The basic interface describing modification of the initial object geometry .
	IRobotObjOperation (see page 747)	The basic interface describing transformation (translation, rotation, scaling, deformation) of the result of modification (the so called reference or image).
	IRobotObjOperationCollection (see page 748)	Collection of operations.
	IRobotObjModifExtrusion (see page 749)	The interface describing a modification of initial object geometry - extrude. .
	IRobotObjModifLathe (see page 750)	Interface describing modification of initial geometry of the object - revolve operation.
	IRobotObjModifPyramid (see page 751)	Interface describing modification of initial object geometry - extrude along polyline.

(c) 2014 Autodesk Inc. All rights reserved

	IRobotObjOperTranslation (see page 754)	The interface describing the operation of translation that transforms the result of modification of the initial object geometry. .
	IRobotObjOperScaling (see page 754)	The interface describing the operation of scaling, transforming the result of modification of object geometry .
	IRobotObjOperRotation (see page 756)	Interface describing the operation of rotation that transforms the result of modification of object geometry. .
	IRobotObjOperMeshing (see page 757)	The interface describing a deformation that transforms the result of modification of initial object geometry .
	IRobotObjEdge (see page 759)	Interface describing an object edge.
	IRobotObjAttributes (see page 763)	The interface describing additional attributes of an object or its part.
	IRobotObjPart (see page 768)	Interface describing an object component. .
	IRobotObjModificationCollection (see page 769)	Collection of modifications .
	IRobotObjEdgeCollection (see page 770)	Collection of edges.
	IRobotObjObject (see page 770)	
	IRobotObjObjectServer (see page 779)	Server of object that are structure components .
	IRobotObjPartMain (see page 794)	Main object component. It contains definition of the base geometry and geometric transformations which result in the geometry of the entire object. All object edges are also saved in the main component.
	IRobotObjPartReference (see page 799)	Object component definition which is the reference of the main component generated due to transformations. .
	IRobotObjPart2 (see page 801)	Extended definition of the object component. .
	IRobotObjMesh (see page 803)	Interface allowing management of the finite element mesh.

II.5.1 Complex attributes of an object

Enumerations

	Name	Description
	IRobotThicknessType (see page 724)	
	IRobotThicknessUpliftType (see page 724)	
	IRobotThicknessHomoType (see page 730)	
	IRobotThicknessOrthoType (see page 730)	
	IRobotThicknessOrthoDirType (see page 730)	
	IRobotThicknessMatrixValue (see page 740)	Physical interpretation of individual matrix values is described in the file of the Robot program help.

Interfaces

	Name	Description
	IRobotThicknessData (see page 722)	Thickness definition.
	IRobotThicknessHomoData (see page 725)	Definition of thickness parameters of the I_TT_HOMOGENEOUS type.
	IRobotThicknessOrthoData (see page 731)	
	IRobotThicknessMatrix (see page 739)	Object provides access to stiffness matrix values for orthotropic thicknesses.

(c) 2014 Autodesk Inc. All rights reserved

	IRobotSolidPropertiesData (see page 741)	Solid properties.

II.5.1.1 Linear releases

Available since version 2.5.

Enumerations

	Name	Description
	IRobotLinearReleaseDefinitionType (see page 708)	Available methods of release definition for one direction. .

Interfaces

	Name	Description
	IRobotLinearReleaseData (see page 704)	Linear release between panels.
	IRobotLinearReleaseServer (see page 708)	Server allowing definition of linear releases between panels. Moreover, it provides access to the list of objects describing all the releases defined. .
	IRobotLinearReleaseDef (see page 713)	Object (see page 715) describing definition parameters of the linear release between panels. The release is defined for the selected object component on the indicated edge. The edge may belong to object other than the released component.
	IRobotLinearReleaseDefList (see page 716)	List of definitions of linear releases.

II.5.1.1.1 IRobotLinearReleaseData

Class Hierarchy

RobotOM.IRobotLinearReleaseData

C++

```
interface IRobotLinearReleaseData : IDispatch;
```

C#

```
public interface IRobotLinearReleaseData;
```

Visual Basic

```
Public Interface IRobotLinearReleaseData
```

Description

Linear release between panels.

Version

Available since version 2.5.

II.5.1.1.1.1 IRobotLinearReleaseData Members

The following tables list the members exposed by IRobotLinearReleaseData.

Public Fields

	Name	Description
	HX (see page 705)	HX elastic coefficient for the elastic release in the RX (see page 706) direction.
	KX (see page 705)	KX elastic coefficient for the elastic release in the UX (see page 707) direction.

(c) 2014 Autodesk Inc. All rights reserved

	KY (see page 706)	KY elastic coefficient for the elastic release in the UY (see page 707) direction.
	KZ (see page 706)	KZ elastic coefficient for the elastic release in the UZ (see page 707) direction.
	RX (see page 706)	Definition of the release (RX direction).
	UX (see page 707)	Definition of the release (UX direction).
	UY (see page 707)	Definition of the release (UY direction).
	UZ (see page 707)	Definition of the release (UZ direction).

II.5.1.1.1.2 IRobotLinearReleaseData Fields

The fields of the IRobotLinearReleaseData class are listed here.

Public Fields

	Name	Description
	HX (see page 705)	HX elastic coefficient for the elastic release in the RX (see page 706) direction.
	KX (see page 705)	KX elastic coefficient for the elastic release in the UX (see page 707) direction.
	KY (see page 706)	KY elastic coefficient for the elastic release in the UY (see page 707) direction.
	KZ (see page 706)	KZ elastic coefficient for the elastic release in the UZ (see page 707) direction.
	RX (see page 706)	Definition of the release (RX direction).
	UX (see page 707)	Definition of the release (UX direction).
	UY (see page 707)	Definition of the release (UY direction).
	UZ (see page 707)	Definition of the release (UZ direction).

II.5.1.1.1.2.1 HX

C++

```
HRESULT get_HX(double*);
HRESULT put_HX(double);
```

C#

```
public double HX { get; set; }
```

Visual Basic

```
Public HX As double
```

Description

HX elastic coefficient for the elastic release in the RX (see page 706) direction.

Version

Available since version 2.5.

II.5.1.1.1.2.2 KX

C++

```
HRESULT get_KX(double*);
HRESULT put_KX(double);
```

C#

```
public double KX { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public KX As double
```

Description

KX elastic coefficient for the elastic release in the UX (see page 707) direction.

Version

Available since version 2.5.

II.5.1.1.1.2.3 KY

C++

```
HRESULT get_KY(double*);
HRESULT put_KY(double);
```

C#

```
public double KY { get; set; }
```

Visual Basic

```
Public KY As double
```

Description

KY elastic coefficient for the elastic release in the UY (see page 707) direction.

Version

Available since version 2.5.

II.5.1.1.1.2.4 KZ

C++

```
HRESULT get_KZ(double*);
HRESULT put_KZ(double);
```

C#

```
public double KZ { get; set; }
```

Visual Basic

```
Public KZ As double
```

Description

KZ elastic coefficient for the elastic release in the UZ (see page 707) direction.

Version

Available since version 2.5.

II.5.1.1.1.2.5 RX

C++

```
HRESULT get_RX(IRobotLinearReleaseDefinitionType*);
HRESULT put_RX(IRobotLinearReleaseDefinitionType);
```

C#

```
public IRobotLinearReleaseDefinitionType RX { get; set; }
```

Visual Basic

```
Public RX As IRobotLinearReleaseDefinitionType
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Definition of the release (RX direction).

Version

Available since version 2.5.

II.5.1.1.1.2.6 UX

C++

```
HRESULT get_UX(IRobotLinearReleaseDefinitionType*);
HRESULT put_UX(IRobotLinearReleaseDefinitionType);
```

C#

```
public IRobotLinearReleaseDefinitionType UX { get; set; }
```

Visual Basic

```
Public UX As IRobotLinearReleaseDefinitionType
```

Description

Definition of the release (UX direction).

Version

Available since version 2.5.

II.5.1.1.1.2.7 UY

C++

```
HRESULT get_UY(IRobotLinearReleaseDefinitionType*);
HRESULT put_UY(IRobotLinearReleaseDefinitionType);
```

C#

```
public IRobotLinearReleaseDefinitionType UY { get; set; }
```

Visual Basic

```
Public UY As IRobotLinearReleaseDefinitionType
```

Description

Definition of the release (UY direction).

Version

Available since version 2.5.

II.5.1.1.1.2.8 UZ

C++

```
HRESULT get_UZ(IRobotLinearReleaseDefinitionType*);
HRESULT put_UZ(IRobotLinearReleaseDefinitionType);
```

C#

```
public IRobotLinearReleaseDefinitionType UZ { get; set; }
```

Visual Basic

```
Public UZ As IRobotLinearReleaseDefinitionType
```

Description

Definition of the release (UZ direction).

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 2.5.

II.5.1.1.2 IRobotLinearReleaseDefinitionType

C++

```
enum IRobotLinearReleaseDefinitionType;
```

C#

```
public enum IRobotLinearReleaseDefinitionType;
```

Visual Basic

```
Public Enum IRobotLinearReleaseDefinitionType
```

Members

Members	Description
I_LRDT_NONE = 0	No release for the selected direction. Available since version 2.5.
I_LRDT_RELEASED = 1	Release definition for the selected direction. Available since version 2.5.
I_LRDT_MINUS = 2	Release definition in the direction opposite to the local system axis on the panel edge. Available since version 2.5.
I_LRDT_PLUS = 3	Release definition in the direction compatible with the local system axis on the panel edge . Available since version 2.5.

Description

Available methods of release definition for one direction. .

Version

Available since version 2.5.

II.5.1.1.3 IRobotLinearReleaseServer

Class Hierarchy

RobotOM.IRobotLinearReleaseServer

C++

```
interface IRobotLinearReleaseServer : IDispatch;
```

C#

```
public interface IRobotLinearReleaseServer;
```

Visual Basic

```
Public Interface IRobotLinearReleaseServer
```

Description

Server allowing definition of linear releases between panels. Moreover, it provides access to the list of objects describing all the releases defined. .

Version

Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

II.5.1.1.3.1 IRobotLinearReleaseServer Members

The following tables list the members exposed by IRobotLinearReleaseServer.

Public Fields

	Name	Description
	Count (see page 709)	Number of defined releases.

Public Methods

	Name	Description
	Find (see page 710)	Function returns index of the definition describing the release between the indicated objects. If no release has been defined between the indicated objects, then zero value is returned. .
	FindEdge (see page 711)	Function returns index of the first located linear release definition on the indicated edge. The list of the defined linear releases is searched beginning with the indicated start index. If the release is not located on the indicated edge, then zero value is returned. .
	FindLabel (see page 711)	Function returns index of the first located definition of the linear release with the indicated name. The list of the defined linear releases is searched beginning with the indicated start index. If the release is not located, then zero value is returned. .
	FindObject (see page 711)	Function returns index of the first located linear release definition for the specified object component. The list of the defined linear releases is searched beginning with the indicated start index. If the linear release is not located for the indicated object component, then zero value is returned. .
	Get (see page 712)	Function returns the objects containing parameters of the linear release definition. By activating this function with the successive indexes from 1 to Count (see page 709), the user may obtain information about all the defined linear releases between panels. .
	GetLabel (see page 712)	Function returns label describing parameters of the indicated linear release.
	Remove (see page 712)	Function deletes the linear release whose method of definition describes the object with the indicated index on the list of all the releases defined. .
	Set (see page 713)	Function defines a linear release for the selected object component on the indicated edge. The edge may belong to object other than the released component. Function returns value different from zero (True) if the release is defined. .
	SetMany (see page 713)	Function defines linear releases according to the specified list.

II.5.1.1.3.2 IRobotLinearReleaseServer Fields

The fields of the IRobotLinearReleaseServer class are listed here.

Public Fields

	Name	Description
	Count (see page 709)	Number of defined releases.

II.5.1.1.3.2.1 Count

C++

```
HRESULT get_Count(long*);
```

C#

```
public long Count { get; }
```

Visual Basic

```
Public ReadOnly Count As long
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Number of defined releases.

Version

Available since version 2.5.

II.5.1.1.3.3 IRobotLinearReleaseServer Methods

The methods of the IRobotLinearReleaseServer class are listed here.

Public Methods

	Name	Description
	Find (see page 710)	Function returns index of the definition describing the release between the indicated objects. If no release has been defined between the indicated objects, then zero value is returned. .
	FindEdge (see page 711)	Function returns index of the first located linear release definition on the indicated edge. The list of the defined linear releases is searched beginning with the indicated start index. If the release is not located on the indicated edge, then zero value is returned. .
	FindLabel (see page 711)	Function returns index of the first located definition of the linear release with the indicated name. The list of the defined linear releases is searched beginning with the indicated start index. If the release is not located, then zero value is returned. .
	FindObject (see page 711)	Function returns index of the first located linear release definition for the specified object component. The list of the defined linear releases is searched beginning with the indicated start index. If the linear release is not located for the indicated object component, then zero value is returned. .
	Get (see page 712)	Function returns the objects containing parameters of the linear release definition. By activating this function with the successive indexes from 1 to Count (see page 709), the user may obtain information about all the defined linear releases between panels. .
	GetLabel (see page 712)	Function returns label describing parameters of the indicated linear release.
	Remove (see page 712)	Function deletes the linear release whose method of definition describes the object with the indicated index on the list of all the releases defined. .
	Set (see page 713)	Function defines a linear release for the selected object component on the indicated edge. The edge may belong to object other than the released component. Function returns value different from zero (True) if the release is defined. .
	SetMany (see page 713)	Function defines linear releases according to the specified list.

II.5.1.1.3.3.1 Find

C++

```
HRESULT Find(long _edge_obj_num, long _edge_idx, long _obj_num, long _part_idx, long* ret);
```

C#

```
public long Find(long _edge_obj_num, long _edge_idx, long _obj_num, long _part_idx);
```

Visual Basic

```
Public Function Find(_edge_obj_num As long, _edge_idx As long, _obj_num As long, _part_idx
As long) As long
```

Description

Function returns index of the definition describing the release between the indicated objects. If no release has been defined between the indicated objects, then zero value is returned. .

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 2.5.

II.5.1.1.3.3.2 FindEdge

C++

```
HRESULT FindEdge(long _edge_obj, long _edge_idx, long _start_def_idx = 1, long* ret);
```

C#

```
public long FindEdge(long _edge_obj, long _edge_idx, long _start_def_idx = 1);
```

Visual Basic

```
Public Function FindEdge(_edge_obj As long, _edge_idx As long, Optional _start_def_idx As
long = 1) As long
```

Description

Function returns index of the first located linear release definition on the indicated edge. The list of the defined linear releases is searched beginning with the indicated start index. If the release is not located on the indicated edge, then zero value is returned. .

Version

Available since version 2.5.

II.5.1.1.3.3.3 FindLabel

C++

```
HRESULT FindLabel(BSTR _label_name, long _start_def_idx = 1, long* ret);
```

C#

```
public long FindLabel(String _label_name, long _start_def_idx = 1);
```

Visual Basic

```
Public Function FindLabel(_label_name As String, Optional _start_def_idx As long = 1) As
long
```

Description

Function returns index of the first located definition of the linear release with the indicated name. The list of the defined linear releases is searched beginning with the indicated start index. If the release is not located, then zero value is returned. .

Version

Available since version 2.5.

II.5.1.1.3.3.4 FindObject

C++

```
HRESULT FindObject(long _obj_num, long _part_idx, long _start_def_idx = 1, long* ret);
```

C#

```
public long FindObject(long _obj_num, long _part_idx, long _start_def_idx = 1);
```

Visual Basic

```
Public Function FindObject(_obj_num As long, _part_idx As long, Optional _start_def_idx As
long = 1) As long
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Function returns index of the first located linear release definition for the specified object component. The list of the defined linear releases is searched beginning with the indicated start index. If the linear release is not located for the indicated object component, then zero value is returned. .

Version

Available since version 2.5.

II.5.1.1.3.3.5 Get

C++

```
HRESULT Get(long _def_idx, IRobotLinearReleaseDef** ret);
```

C#

```
public IRobotLinearReleaseDef Get(long _def_idx);
```

Visual Basic

```
Public Function Get(_def_idx As long) As IRobotLinearReleaseDef
```

Description

Function returns the objects containing parameters of the linear release definition. By activating this function with the successive indexes from 1 to Count (see page 709), the user may obtain information about all the defined linear releases between panels. .

Version

Available since version 2.5.

II.5.1.1.3.3.6 GetLabel

C++

```
HRESULT GetLabel(long _def_idx, IRobotLabel** ret);
```

C#

```
public IRobotLabel GetLabel(long _def_idx);
```

Visual Basic

```
Public Function GetLabel(_def_idx As long) As IRobotLabel
```

Description

Function returns label describing parameters of the indicated linear release.

Version

Available since version 2.5.

II.5.1.1.3.3.7 Remove

C++

```
HRESULT Remove(long _def_idx);
```

C#

```
public void Remove(long _def_idx);
```

Visual Basic

```
Public Sub Remove(_def_idx As long)
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Function deletes the linear release whose method of definition describes the object with the indicated index on the list of all the releases defined. .

Version

Available since version 2.5.

II.5.1.1.3.3.8 Set

C++

```
HRESULT Set(long _edge_obj_num, long _edge_idx, long _obj_num, long _part_idx, BSTR
_label_name, VARIANT_BOOL* ret);
```

C#

```
public bool Set(long _edge_obj_num, long _edge_idx, long _obj_num, long _part_idx, String
_label_name);
```

Visual Basic

```
Public Function Set(_edge_obj_num As long, _edge_idx As long, _obj_num As long, _part_idx
As long, _label_name As String) As Boolean
```

Description

Function defines a linear release for the selected object component on the indicated edge. The edge may belong to object other than the released component. Function returns value different from zero (True) if the release is defined. .

Version

Available since version 2.5.

II.5.1.1.3.3.9 SetMany

C++

```
HRESULT SetMany(IRobotLinearReleaseDefList* _deflist);
```

C#

```
public void SetMany(IRobotLinearReleaseDefList _deflist);
```

Visual Basic

```
Public Sub SetMany(ByRef _deflist As IRobotLinearReleaseDefList)
```

Description

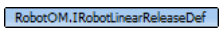
Function defines linear releases according to the specified list.

Version

Available since version 8.

II.5.1.1.4 IRobotLinearReleaseDef

Class Hierarchy



C++

```
interface IRobotLinearReleaseDef : IDispatch;
```

C#

```
public interface IRobotLinearReleaseDef;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Interface IRobotLinearReleaseDef
```

Description

Object (see page 715) describing definition parameters of the linear release between panels. The release is defined for the selected object component on the indicated edge. The edge may belong to object other than the released component.

Version

Available since version 2.5.

II.5.1.1.4.1 IRobotLinearReleaseDef Members

The following tables list the members exposed by IRobotLinearReleaseDef.

Public Fields

	Name	Description
	EdgeIdx (see page 714)	Index of the edge in the object.
	EdgeObject (see page 715)	Number of the object including the edge along which the release is defined.
	LabelName (see page 715)	Name of the label storing release parameters.
	Object (see page 715)	Number of the object for the component of which the release is defined .
	PartIdx (see page 716)	Index of the object component for which the release is defined .

II.5.1.1.4.2 IRobotLinearReleaseDef Fields

The fields of the IRobotLinearReleaseDef class are listed here.

Public Fields

	Name	Description
	EdgeIdx (see page 714)	Index of the edge in the object.
	EdgeObject (see page 715)	Number of the object including the edge along which the release is defined.
	LabelName (see page 715)	Name of the label storing release parameters.
	Object (see page 715)	Number of the object for the component of which the release is defined .
	PartIdx (see page 716)	Index of the object component for which the release is defined .

II.5.1.1.4.2.1 EdgeIdx

C++

```
HRESULT get_EdgeIdx(long*);
HRESULT put_EdgeIdx(long);
```

C#

```
public long EdgeIdx { get; set; }
```

Visual Basic

```
Public EdgeIdx As long
```

Description

Index of the edge in the object.

Version

Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

II.5.1.1.4.2.2 EdgeObject

C++

```
HRESULT get_EdgeObject(long*);
HRESULT put_EdgeObject(long);
```

C#

```
public long EdgeObject { get; set; }
```

Visual Basic

```
Public EdgeObject As long
```

Description

Number of the object including the edge along which the release is defined.

Version

Available since version 2.5.

II.5.1.1.4.2.3 LabelName

C++

```
HRESULT get_LabelName(BSTR*);
HRESULT put_LabelName(BSTR);
```

C#

```
public String LabelName { get; set; }
```

Visual Basic

```
Public LabelName As String
```

Description

Name of the label storing release parameters.

Version

Available since version 2.5.

II.5.1.1.4.2.4 Object

C++

```
HRESULT get_Object(long*);
HRESULT put_Object(long);
```

C#

```
public long Object { get; set; }
```

Visual Basic

```
Public Object As long
```

Description

Number of the object for the component of which the release is defined .

Version

Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

II.5.1.1.4.2.5 PartIdx

C++

```
HRESULT get_PartIdx(long*);
HRESULT put_PartIdx(long);
```

C#

```
public long PartIdx { get; set; }
```

Visual Basic

```
Public PartIdx As long
```

Description

Index of the object component for which the release is defined .

Version

Available since version 2.5.

II.5.1.1.5 IRobotLinearReleaseDefList

Class Hierarchy

RobotOM.IRobotLinearReleaseDefList

C++

```
interface IRobotLinearReleaseDefList : IDispatch;
```

C#

```
public interface IRobotLinearReleaseDefList;
```

Visual Basic

```
Public Interface IRobotLinearReleaseDefList
```

Description

List of definitions of linear releases.

Version

Available since version 8.

II.5.1.1.5.1 IRobotLinearReleaseDefList Members

The following tables list the members exposed by IRobotLinearReleaseDefList.

Public Fields

	Name	Description
	Count (see page 717)	

Public Methods

	Name	Description
	Add (see page 717)	Function adds a linear release definition to the list. Function returns an index of the definition added to the list.
	Clear (see page 718)	Function deletes all elements from the list.
	Get (see page 718)	Function makes available a linear release definition of the specified index.
	Remove (see page 718)	Function deletes a linear release definition of the specified index from the list.

(c) 2014 Autodesk Inc. All rights reserved

II.5.1.1.5.2 IRobotLinearReleaseDefList Fields

The fields of the IRobotLinearReleaseDefList class are listed here.

Public Fields

	Name	Description
	Count (see page 717)	

II.5.1.1.5.2.1 Count

C++

```
HRESULT get_Count(long*);
```

C#

```
public long Count { get; }
```

Visual Basic

```
Public ReadOnly Count As long
```

Version

Available since version 8.

II.5.1.1.5.3 IRobotLinearReleaseDefList Methods

The methods of the IRobotLinearReleaseDefList class are listed here.

Public Methods

	Name	Description
	Add (see page 717)	Function adds a linear release definition to the list. Function returns an index of the definition added to the list.
	Clear (see page 718)	Function deletes all elements from the list.
	Get (see page 718)	Function makes available a linear release definition of the specified index.
	Remove (see page 718)	Function deletes a linear release definition of the specified index from the list.

II.5.1.1.5.3.1 Add

C++

```
HRESULT Add(long _edge_obj_num, long _edge_idx, long _obj_num, long _part_idx, BSTR
_label_name, long* ret);
```

C#

```
public long Add(long _edge_obj_num, long _edge_idx, long _obj_num, long _part_idx, String
_label_name);
```

Visual Basic

```
Public Function Add(_edge_obj_num As long, _edge_idx As long, _obj_num As long, _part_idx
As long, _label_name As String) As long
```

Description

Function adds a linear release definition to the list. Function returns an index of the definition added to the list.

Version

Available since version 8.

(c) 2014 Autodesk Inc. All rights reserved

II.5.1.1.5.3.2 Clear

C++

```
HRESULT Clear();
```

C#

```
public void Clear();
```

Visual Basic

```
Public Sub Clear()
```

Description

Function deletes all elements from the list.

Version

Available since version 8.

II.5.1.1.5.3.3 Get

C++

```
HRESULT Get(long _idx, long* _edge_obj_num, long* _edge_idx, long* _obj_num, long*
_part_idx, BSTR _label_name, VARIANT_BOOL* ret);
```

C#

```
public bool Get(long _idx, long* _edge_obj_num, long* _edge_idx, long* _obj_num, long*
_part_idx, String _label_name);
```

Visual Basic

```
Public Function Get(_idx As long, ByRef _edge_obj_num As long*, ByRef _edge_idx As long*,
ByRef _obj_num As long*, ByRef _part_idx As long*, ByRef _label_name As String) As Boolean
```

Description

Function makes available a linear release definition of the specified index.

Version

Available since version 8.

II.5.1.1.5.3.4 Remove

C++

```
HRESULT Remove(long _idx);
```

C#

```
public void Remove(long _idx);
```

Visual Basic

```
Public Sub Remove(_idx As long)
```

Description

Function deletes a linear release definition of the specified index from the list.

Version

Available since version 8.

(c) 2014 Autodesk Inc. All rights reserved

II.5.1.2 Panel calculation model

Available since version 9.7.

Enumerations

	Name	Description
	IRobotPanelCalcModelFinElemType (see page 719)	
	IRobotPanelCalcModelLoadTransfer (see page 719)	
	IRobotPanelCalcModelDiaphragm (see page 720)	

Interfaces

	Name	Description
	IRobotPanelCalcModelData (see page 720)	Define a panel calculation model.

II.5.1.2.1 IRobotPanelCalcModelFinElemType

C++

```
enum IRobotPanelCalcModelFinElemType;
```

C#

```
public enum IRobotPanelCalcModelFinElemType;
```

Visual Basic

```
Public Enum IRobotPanelCalcModelFinElemType
```

Members

Members	Description
I_PCMFET_NONE = 0	Available since version 9.7.
I_PCMFET_SHELL = 1	Available since version 9.7.

Version

Available since version 9.7.

II.5.1.2.2 IRobotPanelCalcModelLoadTransfer

C++

```
enum IRobotPanelCalcModelLoadTransfer;
```

C#

```
public enum IRobotPanelCalcModelLoadTransfer;
```

Visual Basic

```
Public Enum IRobotPanelCalcModelLoadTransfer
```

Members

Members	Description
I_PCMLT_ANALYTICAL_FINITE_ELEMENTS = 1	Available since version 9.7.
I_PCMLT_SIMPLIFIED_TWO_WAY = 2	Available since version 9.7.
I_PCMLT_SIMPLIFIED_ONE_WAY = 3	Available since version 9.7.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 9.7.

II.5.1.2.3 IRobotPanelCalcModelDiaphragm

C++

```
enum IRobotPanelCalcModelDiaphragm;
```

C#

```
public enum IRobotPanelCalcModelDiaphragm;
```

Visual Basic

```
Public Enum IRobotPanelCalcModelDiaphragm
```

Members

Members	Description
I_PCMD_NONE = 0	Available since version 9.7.
I_PCMD_RIGID = 1	Available since version 9.7.
I_PCMD_FLEXIBLE = 2	Available since version 9.7.

Version

Available since version 9.7.

II.5.1.2.4 IRobotPanelCalcModelData

Class Hierarchy

C++

```
interface IRobotPanelCalcModelData : IDispatch;
```

C#

```
public interface IRobotPanelCalcModelData;
```

Visual Basic

```
Public Interface IRobotPanelCalcModelData
```

Description

Define a panel calculation model.

Version

Available since version 9.7.

II.5.1.2.4.1 IRobotPanelCalcModelData Members

The following tables list the members exposed by IRobotPanelCalcModelData.

Public Fields

	Name	Description
	Diaphragm (see page 721)	
	FinElemType (see page 721)	
	LoadTransfer (see page 721)	

(c) 2014 Autodesk Inc. All rights reserved

II.5.1.2.4.2 IRobotPanelCalcModelData Fields

The fields of the IRobotPanelCalcModelData class are listed here.

Public Fields

	Name	Description
	Diaphragm (see page 721)	
	FinElemType (see page 721)	
	LoadTransfer (see page 721)	

II.5.1.2.4.2.1 Diaphragm

C++

```
HRESULT get_Diaphragm(IRobotPanelCalcModelDiaphragm*);
HRESULT put_Diaphragm(IRobotPanelCalcModelDiaphragm);
```

C#

```
public IRobotPanelCalcModelDiaphragm Diaphragm { get; set; }
```

Visual Basic

```
Public Diaphragm As IRobotPanelCalcModelDiaphragm
```

Version

Available since version 9.7.

II.5.1.2.4.2.2 FinElemType

C++

```
HRESULT get_FinElemType(IRobotPanelCalcModelFinElemType*);
HRESULT put_FinElemType(IRobotPanelCalcModelFinElemType);
```

C#

```
public IRobotPanelCalcModelFinElemType FinElemType { get; set; }
```

Visual Basic

```
Public FinElemType As IRobotPanelCalcModelFinElemType
```

Version

Available since version 9.7.

II.5.1.2.4.2.3 LoadTransfer

C++

```
HRESULT get_LoadTransfer(IRobotPanelCalcModelLoadTransfer*);
HRESULT put_LoadTransfer(IRobotPanelCalcModelLoadTransfer);
```

C#

```
public IRobotPanelCalcModelLoadTransfer LoadTransfer { get; set; }
```

Visual Basic

```
Public LoadTransfer As IRobotPanelCalcModelLoadTransfer
```

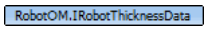
Version

Available since version 9.7.

(c) 2014 Autodesk Inc. All rights reserved

II.5.1.3 IRobotThicknessData

Class Hierarchy



C++

```
interface IRobotThicknessData : IDispatch;
```

C#

```
public interface IRobotThicknessData;
```

Visual Basic

```
Public Interface IRobotThicknessData
```

Description

Thickness definition.

II.5.1.3.1 IRobotThicknessData Members

The following tables list the members exposed by IRobotThicknessData.

Public Fields

	Name	Description
	Data (see page 723)	Set of data that are specific for the selected thickness type; parameters describing thickness of the I_TT_HOMOGENEOUS type are defined by means of RobotThicknessHomoData, whereas for the I_TT_ORTHOTROPIC type RobotThicknessOrthoData is defined Available since version 1.7.
	ElasticFoundation (see page 723)	Available since version 1.7.
	MaterialName (see page 723)	Name of the material connected with thickness Available since version 2.0.
	ThicknessType (see page 723)	Available since version 1.7.
	Uplift (see page 724)	Available since version 1.7.

II.5.1.3.2 IRobotThicknessData Fields

The fields of the IRobotThicknessData class are listed here.

Public Fields

	Name	Description
	Data (see page 723)	Set of data that are specific for the selected thickness type; parameters describing thickness of the I_TT_HOMOGENEOUS type are defined by means of RobotThicknessHomoData, whereas for the I_TT_ORTHOTROPIC type RobotThicknessOrthoData is defined Available since version 1.7.
	ElasticFoundation (see page 723)	Available since version 1.7.
	MaterialName (see page 723)	Name of the material connected with thickness Available since version 2.0.
	ThicknessType (see page 723)	Available since version 1.7.
	Uplift (see page 724)	Available since version 1.7.

II.5.1.3.2.1 Data

C++

```
HRESULT get_Data(Object*);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public Object Data { get; }
```

Visual Basic

```
Public ReadOnly Data As Object
```

Description

Set of data that are specific for the selected thickness type; parameters describing thickness of the I_TT_HOMOGENEOUS type are defined by means of RobotThicknessHomoData, whereas for the I_TT_ORTHOTROPIC type RobotThicknessOrthoData is defined Available since version 1.7.

II.5.1.3.2.2 ElasticFoundation

C++

```
HRESULT get_ElasticFoundation(double*);
HRESULT put_ElasticFoundation(double);
```

C#

```
public double ElasticFoundation { get; set; }
```

Visual Basic

```
Public ElasticFoundation As double
```

Description

Available since version 1.7.

II.5.1.3.2.3 MaterialName

C++

```
HRESULT get_MaterialName(BSTR*);
HRESULT put_MaterialName(BSTR);
```

C#

```
public String MaterialName { get; set; }
```

Visual Basic

```
Public MaterialName As String
```

Description

Name of the material connected with thickness Available since version 2.0.

II.5.1.3.2.4 ThicknessType

C++

```
HRESULT get_ThicknessType(IRobotThicknessType*);
HRESULT put_ThicknessType(IRobotThicknessType);
```

C#

```
public IRobotThicknessType ThicknessType { get; set; }
```

Visual Basic

```
Public ThicknessType As IRobotThicknessType
```

Description

Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

II.5.1.3.2.5 Uplift

C++

```
HRESULT get_Uplift(IRobotThicknessUpliftType*);
HRESULT put_Uplift(IRobotThicknessUpliftType);
```

C#

```
public IRobotThicknessUpliftType Uplift { get; set; }
```

Visual Basic

```
Public Uplift As IRobotThicknessUpliftType
```

Description

Available since version 1.7.

II.5.1.4 IRobotThicknessType

C++

```
enum IRobotThicknessType;
```

C#

```
public enum IRobotThicknessType;
```

Visual Basic

```
Public Enum IRobotThicknessType
```

Members

Members	Description
I_TT_HOMOGENEOUS = 0	Available since version 1.7.
I_TT_ORTHOTROPIC = 1	Available since version 1.7.

II.5.1.5 IRobotThicknessUpliftType

C++

```
enum IRobotThicknessUpliftType;
```

C#

```
public enum IRobotThicknessUpliftType;
```

Visual Basic

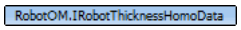
```
Public Enum IRobotThicknessUpliftType
```

Members

Members	Description
I_TUT_NONE = 0	Available since version 1.7.
I_TUT_MINUS = 1	Available since version 1.7.
I_TUT_PLUS = 2	Available since version 1.7.

II.5.1.6 IRobotThicknessHomoData

Class Hierarchy



(c) 2014 Autodesk Inc. All rights reserved

C++

```
interface IRobotThicknessHomoData : IDispatch;
```

C#

```
public interface IRobotThicknessHomoData;
```

Visual Basic

```
Public Interface IRobotThicknessHomoData
```

Description

Definition of thickness parameters of the I_TT_HOMOGENEOUS type.

II.5.1.6.1 IRobotThicknessHomoData Members

The following tables list the members exposed by IRobotThicknessHomoData.

Public Fields

	Name	Description
	Thick1 (see page 726)	Thickness value in the P1 point (set only for variable thickness) Available since version 1.7.
	Thick2 (see page 726)	Thickness value in the P2 point (set only for variable thickness) Available since version 1.7.
	Thick3 (see page 726)	Thickness value in the P3 point (set only for variable thickness) Available since version 1.7.
	ThickConst (see page 726)	Thickness value (if the thickness is uniform for the whole slab) Available since version 1.7.
	Type (see page 727)	Available since version 1.7.

Public Methods

	Name	Description
	GetP1 (see page 727)	Available since version 1.7.
	GetP2 (see page 728)	Available since version 1.7.
	GetP3 (see page 728)	Available since version 1.7.
	GetReduction (see page 728)	Function returns True value if a reduction coefficient of the moment of inertia has been set. The coefficient value will be entered to the specified parameter.
	SetP1 (see page 728)	Available since version 1.7.
	SetP2 (see page 729)	Available since version 1.7.
	SetP3 (see page 729)	Available since version 1.7.
	SetReduction (see page 729)	Function sets a coefficient of reduction of the moment of inertia.

II.5.1.6.2 IRobotThicknessHomoData Fields

The fields of the IRobotThicknessHomoData class are listed here.

Public Fields

	Name	Description
	Thick1 (see page 726)	Thickness value in the P1 point (set only for variable thickness) Available since version 1.7.
	Thick2 (see page 726)	Thickness value in the P2 point (set only for variable thickness) Available since version 1.7.
	Thick3 (see page 726)	Thickness value in the P3 point (set only for variable thickness) Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

	ThickConst (see page 726)	Thickness value (if the thickness is uniform for the whole slab) Available since version 1.7.
	Type (see page 727)	Available since version 1.7.

II.5.1.6.2.1 Thick1

C++

```
HRESULT get_Thick1(double*);
HRESULT put_Thick1(double);
```

C#

```
public double Thick1 { get; set; }
```

Visual Basic

```
Public Thick1 As double
```

Description

Thickness value in the P1 point (set only for variable thickness) Available since version 1.7.

II.5.1.6.2.2 Thick2

C++

```
HRESULT get_Thick2(double*);
HRESULT put_Thick2(double);
```

C#

```
public double Thick2 { get; set; }
```

Visual Basic

```
Public Thick2 As double
```

Description

Thickness value in the P2 point (set only for variable thickness) Available since version 1.7.

II.5.1.6.2.3 Thick3

C++

```
HRESULT get_Thick3(double*);
HRESULT put_Thick3(double);
```

C#

```
public double Thick3 { get; set; }
```

Visual Basic

```
Public Thick3 As double
```

Description

Thickness value in the P3 point (set only for variable thickness) Available since version 1.7.

II.5.1.6.2.4 ThickConst

C++

```
HRESULT get_ThickConst(double*);
HRESULT put_ThickConst(double);
```

C#

```
public double ThickConst { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public ThickConst As double
```

Description

Thickness value (if the thickness is uniform for the whole slab) Available since version 1.7.

II.5.1.6.2.5 Type

C++

```
HRESULT get_Type(IRobotThicknessHomoType*);
HRESULT put_Type(IRobotThicknessHomoType);
```

C#

```
public IRobotThicknessHomoType Type { get; set; }
```

Visual Basic

```
Public Type As IRobotThicknessHomoType
```

Description

Available since version 1.7.

II.5.1.6.3 IRobotThicknessHomoData Methods

The methods of the IRobotThicknessHomoData class are listed here.

Public Methods

	Name	Description
	GetP1 (see page 727)	Available since version 1.7.
	GetP2 (see page 728)	Available since version 1.7.
	GetP3 (see page 728)	Available since version 1.7.
	GetReduction (see page 728)	Function returns True value if a reduction coefficient of the moment of inertia has been set. The coefficient value will be entered to the specified parameter.
	SetP1 (see page 728)	Available since version 1.7.
	SetP2 (see page 729)	Available since version 1.7.
	SetP3 (see page 729)	Available since version 1.7.
	SetReduction (see page 729)	Function sets a coefficient of reduction of the moment of inertia.

II.5.1.6.3.1 GetP1

C++

```
HRESULT GetP1(double* _x, double* _y, double* _z);
```

C#

```
public void GetP1(double* _x, double* _y, double* _z);
```

Visual Basic

```
Public Sub GetP1(ByRef _x As double*, ByRef _y As double*, ByRef _z As double*)
```

Description

Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

II.5.1.6.3.2 GetP2

C++

```
HRESULT GetP2(double* _x, double* _y, double* _z);
```

C#

```
public void GetP2(double* _x, double* _y, double* _z);
```

Visual Basic

```
Public Sub GetP2(ByRef _x As double*, ByRef _y As double*, ByRef _z As double*)
```

Description

Available since version 1.7.

II.5.1.6.3.3 GetP3

C++

```
HRESULT GetP3(double* _x, double* _y, double* _z);
```

C#

```
public void GetP3(double* _x, double* _y, double* _z);
```

Visual Basic

```
Public Sub GetP3(ByRef _x As double*, ByRef _y As double*, ByRef _z As double*)
```

Description

Available since version 1.7.

II.5.1.6.3.4 GetReduction

C++

```
HRESULT GetReduction(double* _Ig_cf, VARIANT_BOOL* ret);
```

C#

```
public bool GetReduction(double* _Ig_cf);
```

Visual Basic

```
Public Function GetReduction(ByRef _Ig_cf As double*) As Boolean
```

Description

Function returns True value if a reduction coefficient of the moment of inertia has been set. The coefficient value will be entered to the specified parameter.

Version

Available since version 8.

II.5.1.6.3.5 SetP1

C++

```
HRESULT SetP1(double _x, double _y, double _z);
```

C#

```
public void SetP1(double _x, double _y, double _z);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Sub SetP1(_x As double, _y As double, _z As double)
```

Description

Available since version 1.7.

II.5.1.6.3.6 SetP2

C++

```
HRESULT SetP2(double _x, double _y, double _z);
```

C#

```
public void SetP2(double _x, double _y, double _z);
```

Visual Basic

```
Public Sub SetP2(_x As double, _y As double, _z As double)
```

Description

Available since version 1.7.

II.5.1.6.3.7 SetP3

C++

```
HRESULT SetP3(double _x, double _y, double _z);
```

C#

```
public void SetP3(double _x, double _y, double _z);
```

Visual Basic

```
Public Sub SetP3(_x As double, _y As double, _z As double)
```

Description

Available since version 1.7.

II.5.1.6.3.8 SetReduction

C++

```
HRESULT SetReduction(VARIANT_BOOL _set, double _Ig_cf);
```

C#

```
public void SetReduction(bool _set, double _Ig_cf);
```

Visual Basic

```
Public Sub SetReduction(_set As Boolean, _Ig_cf As double)
```

Description

Function sets a coefficient of reduction of the moment of inertia.

Version

Available since version 8.

(c) 2014 Autodesk Inc. All rights reserved

II.5.1.7 IRobotThicknessHomoType

C++

```
enum IRobotThicknessHomoType;
```

C#

```
public enum IRobotThicknessHomoType;
```

Visual Basic

```
Public Enum IRobotThicknessHomoType
```

Members

Members	Description
I_THT_CONSTANT = 0	Available since version 1.7.
I_THT_VARIABLE_ALONG_LINE = 1	Available since version 1.7.
I_THT_VARIABLE_ON_PLANE = 2	Available since version 1.7.

II.5.1.8 IRobotThicknessOrthoType

C++

```
enum IRobotThicknessOrthoType;
```

C#

```
public enum IRobotThicknessOrthoType;
```

Visual Basic

```
Public Enum IRobotThicknessOrthoType
```

Members

Members	Description
I_TOT_ONE_SIDED_UNIDIR_RIBS = 1	Available since version 1.7.
I_TOT_DOUBLE_SIDED_UNIDIR_RIBS = 2	Available since version 1.7.
I_TOT_ONE_SIDED_BIDIR_RIBS = 3	Available since version 1.7.
I_TOT_UNIDIR_BOX_FLOOR = 4	Available since version 1.7.
I_TOT_BIDIR_BOX_FLOOR = 5	Available since version 1.7.
I_TOT_GRILLAGE = 6	Available since version 1.7.
I_TOT_SLAB_ON_TRAPEZOID_PLATE = 7	Available since version 1.7.
I_TOT_USER = 8	Geometrical thickness parameters should be described by means of the appropriate matrices Available since version 1.7.
I_TOT_MATERIAL = 9	Material orthotropy Available since version 1.7.
I_TOT_HOLLOW_CORE_SLAB = 15	Available since version 11.

II.5.1.9 IRobotThicknessOrthoDirType

C++

```
enum IRobotThicknessOrthoDirType;
```

C#

```
public enum IRobotThicknessOrthoDirType;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Enum IRobotThicknessOrthoDirType
```

Members

Members	Description
I_TODT_DIR_X = 0	Available since version 1.7.
I_TODT_DIR_Y = 1	Available since version 1.7.
I_TODT_DIR_Z = 2	Available since version 1.7.
I_TODT_VECTOR = 3	Available since version 1.7.

II.5.1.10 IRobotThicknessOrthoData

Class Hierarchy

RobotOM.IRobotThicknessOrthoData

C++

```
interface IRobotThicknessOrthoData : IDispatch;
```

C#

```
public interface IRobotThicknessOrthoData;
```

Visual Basic

```
Public Interface IRobotThicknessOrthoData
```

II.5.1.10.1 IRobotThicknessOrthoData Members

The following tables list the members exposed by IRobotThicknessOrthoData.

Public Fields

	Name	Description
	A (see page 732)	Available since version 1.7.
	A1 (see page 733)	Available since version 1.7.
	A2 (see page 733)	Available since version 1.7.
	B (see page 733)	Available since version 1.7.
	B1 (see page 733)	Available since version 1.7.
	DirType (see page 734)	Available since version 1.7.
	DisregardBendStiffDirY (see page 734)	Disregard bending stiffness for Y direction.
	H (see page 734)	Available since version 1.7.
	H0 (see page 735)	Opening height.
	H1 (see page 735)	Available since version 1.7.
	H2 (see page 735)	Available since version 1.7.
	HA (see page 735)	Available since version 1.7.
	HB (see page 736)	Available since version 1.7.
	HC (see page 736)	Distance from lower edge to opening center.
	Matrix (see page 736)	Object providing access to stiffness matrix values Available since version 1.7.
	N1 (see page 736)	First stiffness coefficient for material orthotropy Available since version 1.7.
	N2 (see page 737)	Second stiffness coefficient for material orthotropy Available since version 1.7.
	Thick1 (see page 737)	Available since version 1.7.
	Thick2 (see page 737)	Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

	Thick3 (see page 738)	Available since version 1.7.
	Type (see page 738)	Available since version 1.7.

Public Methods

	Name	Description
	GetVector (see page 738)	Available since version 1.7.
	SetVector (see page 739)	Available since version 1.7.

II.5.1.10.2 IRobotThicknessOrthoData Fields

The fields of the IRobotThicknessOrthoData class are listed here.

Public Fields

	Name	Description
	A (see page 732)	Available since version 1.7.
	A1 (see page 733)	Available since version 1.7.
	A2 (see page 733)	Available since version 1.7.
	B (see page 733)	Available since version 1.7.
	B1 (see page 733)	Available since version 1.7.
	DirType (see page 734)	Available since version 1.7.
	DisregardBendStiffDirY (see page 734)	Disregard bending stiffness for Y direction.
	H (see page 734)	Available since version 1.7.
	H0 (see page 735)	Opening height.
	H1 (see page 735)	Available since version 1.7.
	H2 (see page 735)	Available since version 1.7.
	HA (see page 735)	Available since version 1.7.
	HB (see page 736)	Available since version 1.7.
	HC (see page 736)	Distance from lower edge to opening center.
	Matrix (see page 736)	Object providing access to stiffness matrix values Available since version 1.7.
	N1 (see page 736)	First stiffness coefficient for material orthotropy Available since version 1.7.
	N2 (see page 737)	Second stiffness coefficient for material orthotropy Available since version 1.7.
	Thick1 (see page 737)	Available since version 1.7.
	Thick2 (see page 737)	Available since version 1.7.
	Thick3 (see page 738)	Available since version 1.7.
	Type (see page 738)	Available since version 1.7.

II.5.1.10.2.1 A

C++

```
HRESULT get_A(double*);
HRESULT put_A(double);
```

C#

```
public double A { get; set; }
```

Visual Basic

```
Public A As double
```

Description

Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

II.5.1.10.2.2 A1

C++

```
HRESULT get_A1(double*);
HRESULT put_A1(double);
```

C#

```
public double A1 { get; set; }
```

Visual Basic

```
Public A1 As double
```

Description

Available since version 1.7.

II.5.1.10.2.3 A2

C++

```
HRESULT get_A2(double*);
HRESULT put_A2(double);
```

C#

```
public double A2 { get; set; }
```

Visual Basic

```
Public A2 As double
```

Description

Available since version 1.7.

II.5.1.10.2.4 B

C++

```
HRESULT get_B(double*);
HRESULT put_B(double);
```

C#

```
public double B { get; set; }
```

Visual Basic

```
Public B As double
```

Description

Available since version 1.7.

II.5.1.10.2.5 B1

C++

```
HRESULT get_B1(double*);
HRESULT put_B1(double);
```

C#

```
public double B1 { get; set; }
```

Visual Basic

```
Public B1 As double
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Available since version 1.7.

II.5.1.10.2.6 DirType

C++

```
HRESULT get_DirType(IRobotThicknessOrthoDirType*);
HRESULT put_DirType(IRobotThicknessOrthoDirType);
```

C#

```
public IRobotThicknessOrthoDirType DirType { get; set; }
```

Visual Basic

```
Public DirType As IRobotThicknessOrthoDirType
```

Description

Available since version 1.7.

II.5.1.10.2.7 DisregardBendStiffDirY

C++

```
HRESULT get_DisregardBendStiffDirY(VARIANT_BOOL*);
HRESULT put_DisregardBendStiffDirY(VARIANT_BOOL);
```

C#

```
public bool DisregardBendStiffDirY { get; set; }
```

Visual Basic

```
Public DisregardBendStiffDirY As Boolean
```

Description

Disregard bending stiffness for Y direction.

Version

Available since version 11.

II.5.1.10.2.8 H

C++

```
HRESULT get_H(double*);
HRESULT put_H(double);
```

C#

```
public double H { get; set; }
```

Visual Basic

```
Public H As double
```

Description

Available since version 1.7.

II.5.1.10.2.9 H0

C++

```
HRESULT get_H0(double*);
HRESULT put_H0(double);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double H0 { get; set; }
```

Visual Basic

```
Public H0 As double
```

Description

Opening height.

Version

Available since version 11.

II.5.1.10.2.10 H1

C++

```
HRESULT get_H1(double*);
HRESULT put_H1(double);
```

C#

```
public double H1 { get; set; }
```

Visual Basic

```
Public H1 As double
```

Description

Available since version 1.7.

II.5.1.10.2.11 H2

C++

```
HRESULT get_H2(double*);
HRESULT put_H2(double);
```

C#

```
public double H2 { get; set; }
```

Visual Basic

```
Public H2 As double
```

Description

Available since version 1.7.

II.5.1.10.2.12 HA

C++

```
HRESULT get_HA(double*);
HRESULT put_HA(double);
```

C#

```
public double HA { get; set; }
```

Visual Basic

```
Public HA As double
```

Description

Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

II.5.1.10.2.13 HB

C++

```
HRESULT get_HB(double*);
HRESULT put_HB(double);
```

C#

```
public double HB { get; set; }
```

Visual Basic

```
Public HB As double
```

Description

Available since version 1.7.

II.5.1.10.2.14 HC

C++

```
HRESULT get_HC(double*);
HRESULT put_HC(double);
```

C#

```
public double HC { get; set; }
```

Visual Basic

```
Public HC As double
```

Description

Distance from lower edge to opening center.

Version

Available since version 11.

II.5.1.10.2.15 Matrix

C++

```
HRESULT get_Matrix(IRobotThicknessMatrix**);
```

C#

```
public IRobotThicknessMatrix Matrix { get; }
```

Visual Basic

```
Public ReadOnly Matrix As IRobotThicknessMatrix
```

Description

Object providing access to stiffness matrix values Available since version 1.7.

II.5.1.10.2.16 N1

C++

```
HRESULT get_N1(double*);
HRESULT put_N1(double);
```

C#

```
public double N1 { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public N1 As double
```

Description

First stiffness coefficient for material orthotropy Available since version 1.7.

II.5.1.10.2.17 N2

C++

```
HRESULT get_N2(double*);
HRESULT put_N2(double);
```

C#

```
public double N2 { get; set; }
```

Visual Basic

```
Public N2 As double
```

Description

Second stiffness coefficient for material orthotropy Available since version 1.7.

II.5.1.10.2.18 Thick1

C++

```
HRESULT get_Thick1(double*);
HRESULT put_Thick1(double);
```

C#

```
public double Thick1 { get; set; }
```

Visual Basic

```
Public Thick1 As double
```

Description

Available since version 1.7.

II.5.1.10.2.19 Thick2

C++

```
HRESULT get_Thick2(double*);
HRESULT put_Thick2(double);
```

C#

```
public double Thick2 { get; set; }
```

Visual Basic

```
Public Thick2 As double
```

Description

Available since version 1.7.

II.5.1.10.2.20 Thick3

C++

```
HRESULT get_Thick3(double*);
HRESULT put_Thick3(double);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double Thick3 { get; set; }
```

Visual Basic

```
Public Thick3 As double
```

Description

Available since version 1.7.

II.5.1.10.2.21 Type

C++

```
HRESULT get_Type(IRobotThicknessOrthoType*);
HRESULT put_Type(IRobotThicknessOrthoType);
```

C#

```
public IRobotThicknessOrthoType Type { get; set; }
```

Visual Basic

```
Public Type As IRobotThicknessOrthoType
```

Description

Available since version 1.7.

II.5.1.10.3 IRobotThicknessOrthoData Methods

The methods of the IRobotThicknessOrthoData class are listed here.

Public Methods

	Name	Description
	GetVector (see page 738)	Available since version 1.7.
	SetVector (see page 739)	Available since version 1.7.

II.5.1.10.3.1 GetVector

C++

```
HRESULT GetVector(double* _x, double* _y, double* _z);
```

C#

```
public void GetVector(double* _x, double* _y, double* _z);
```

Visual Basic

```
Public Sub GetVector(ByRef _x As double*, ByRef _y As double*, ByRef _z As double*)
```

Description

Available since version 1.7.

II.5.1.10.3.2 SetVector

C++

```
HRESULT SetVector(double _x, double _y, double _z);
```

C#

```
public void SetVector(double _x, double _y, double _z);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

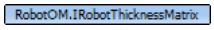
```
Public Sub SetVector(_x As double, _y As double, _z As double)
```

Description

Available since version 1.7.

II.5.1.11 IRobotThicknessMatrix

Class Hierarchy



C++

```
interface IRobotThicknessMatrix : IDispatch;
```

C#

```
public interface IRobotThicknessMatrix;
```

Visual Basic

```
Public Interface IRobotThicknessMatrix
```

Description

Object provides access to stiffness matrix values for orthotropic thicknesses.

II.5.1.11.1 IRobotThicknessMatrix Members

The following tables list the members exposed by IRobotThicknessMatrix.

Public Methods

	Name	Description
	GetValue (see page 740)	Function returns a value of the indicated matrix element. Available since version 1.7.
	SetValue (see page 740)	Function sets the value of the indicated matrix element. Available since version 1.7.

II.5.1.11.2 IRobotThicknessMatrix Methods

The methods of the IRobotThicknessMatrix class are listed here.

Public Methods

	Name	Description
	GetValue (see page 740)	Function returns a value of the indicated matrix element. Available since version 1.7.
	SetValue (see page 740)	Function sets the value of the indicated matrix element. Available since version 1.7.

II.5.1.11.2.1 GetValue

C++

```
HRESULT GetValue(IRobotThicknessMatrixValue _val_id, double* ret);
```

C#

```
public double GetValue(IRobotThicknessMatrixValue _val_id);
```

Visual Basic

```
Public Function GetValue(_val_id As IRobotThicknessMatrixValue) As double
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Function returns a value of the indicated matrix element. Available since version 1.7.

II.5.1.11.2.2 SetValue

C++

```
HRESULT SetValue(IRobotThicknessMatrixValue _val_id, double _val);
```

C#

```
public void SetValue(IRobotThicknessMatrixValue _val_id, double _val);
```

Visual Basic

```
Public Sub SetValue(_val_id As IRobotThicknessMatrixValue, _val As double)
```

Description

Function sets the value of the indicated matrix element. Available since version 1.7.

II.5.1.12 IRobotThicknessMatrixValue

C++

```
enum IRobotThicknessMatrixValue;
```

C#

```
public enum IRobotThicknessMatrixValue;
```

Visual Basic

```
Public Enum IRobotThicknessMatrixValue
```

Members

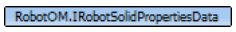
Members	Description
I_TMV_D_XXXX = 0	Defining this quantity makes sense only in the case of shell structures (it is ignored for plates) Available since version 1.7.
I_TMV_D_XXYY = 1	Defining this quantity makes sense only in the case of shell structures (it is ignored for plates) Available since version 1.7.
I_TMV_D_YYYY = 2	Defining this quantity makes sense only in the case of shell structures (it is ignored for plates) Available since version 1.7.
I_TMV_D_XYXY = 3	Defining this quantity makes sense only in the case of shell structures (it is ignored for plates) Available since version 1.7.
I_TMV_K_XXXX = 4	Available since version 1.7.
I_TMV_K_XXYY = 5	Available since version 1.7.
I_TMV_K_YYYY = 6	Available since version 1.7.
I_TMV_K_XYXY = 7	Available since version 1.7.
I_TMV_H_XX = 8	Available since version 1.7.
I_TMV_H_YY = 9	Available since version 1.7.

Description

Physical interpretation of individual matrix values is described in the file of the Robot program help.

II.5.1.13 IRobotSolidPropertiesData

Class Hierarchy



(c) 2014 Autodesk Inc. All rights reserved

C++

```
interface IRobotSolidPropertiesData : IDispatch;
```

C#

```
public interface IRobotSolidPropertiesData;
```

Visual Basic

```
Public Interface IRobotSolidPropertiesData
```

Description

Solid properties.

Version

Available since version 3.

II.5.1.13.1 IRobotSolidPropertiesData Members

The following tables list the members exposed by IRobotSolidPropertiesData.

Public Fields

	Name	Description
	DampCoef (see page 742)	Damping coefficient.
	E (see page 742)	Young's modulus.
	LX (see page 742)	Thermal expansion coefficient.
	MaterialModel (see page 743)	Material model.
	NU (see page 743)	Poisson ratio.
	RO (see page 743)	Density.

Public Methods

	Name	Description
	LoadFromDBase (see page 744)	Function reads data characteristic of a material of the indicated name from the material database. If the material of the indicated name is not found in the current material database, then zero value is returned (False). .

II.5.1.13.2 IRobotSolidPropertiesData Fields

The fields of the IRobotSolidPropertiesData class are listed here.

Public Fields

	Name	Description
	DampCoef (see page 742)	Damping coefficient.
	E (see page 742)	Young's modulus.
	LX (see page 742)	Thermal expansion coefficient.
	MaterialModel (see page 743)	Material model.
	NU (see page 743)	Poisson ratio.
	RO (see page 743)	Density.

II.5.1.13.2.1 DampCoef

C++

```
HRESULT get_DampCoef(double*);
HRESULT put_DampCoef(double);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double DampCoef { get; set; }
```

Visual Basic

```
Public DampCoef As double
```

Description

Damping coefficient.

Version

Available since version 3.

II.5.1.13.2.2 E

C++

```
HRESULT get_E(double*);
HRESULT put_E(double);
```

C#

```
public double E { get; set; }
```

Visual Basic

```
Public E As double
```

Description

Young's modulus.

Version

Available since version 3.

II.5.1.13.2.3 LX

C++

```
HRESULT get_LX(double*);
HRESULT put_LX(double);
```

C#

```
public double LX { get; set; }
```

Visual Basic

```
Public LX As double
```

Description

Thermal expansion coefficient.

Version

Available since version 3.

II.5.1.13.2.4 MaterialModel

C++

```
HRESULT get_MaterialModel(IRobotMaterialModel*);
HRESULT put_MaterialModel(IRobotMaterialModel);
```

C#

```
public IRobotMaterialModel MaterialModel { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public MaterialModel As IRobotMaterialModel
```

Description

Material model.

Version

Available since version 3.

II.5.1.13.2.5 NU

C++

```
HRESULT get_NU(double*);
HRESULT put_NU(double);
```

C#

```
public double NU { get; set; }
```

Visual Basic

```
Public NU As double
```

Description

Poisson ratio.

Version

Available since version 3.

II.5.1.13.2.6 RO

C++

```
HRESULT get_RO(double*);
HRESULT put_RO(double);
```

C#

```
public double RO { get; set; }
```

Visual Basic

```
Public RO As double
```

Description

Density.

Version

Available since version 3.

II.5.1.13.3 IRobotSolidPropertiesData Methods

The methods of the IRobotSolidPropertiesData class are listed here.

Public Methods

	Name	Description
	LoadFromDBase (see page 744)	Function reads data characteristic of a material of the indicated name from the material database. If the material of the indicated name is not found in the current material database, then zero value is returned (False). .

(c) 2014 Autodesk Inc. All rights reserved

II.5.1.13.3.1 LoadFromDBase

C++

```
HRESULT LoadFromDBase(BSTR _material_name, VARIANT_BOOL* ret);
```

C#

```
public bool LoadFromDBase(String _material_name);
```

Visual Basic

```
Public Function LoadFromDBase(_material_name As String) As Boolean
```

Description

Function reads data characteristic of a material of the indicated name from the material database. If the material of the indicated name is not found in the current material database, then zero value is returned (False). .

Version

Available since version 3.

II.5.2 IRobotObjModificationType

C++

```
enum IRobotObjModificationType;
```

C#

```
public enum IRobotObjModificationType;
```

Visual Basic

```
Public Enum IRobotObjModificationType
```

Members

Members	Description
I_OMT_NONE = 17	Undefined.
I_OMT_EXTRUSION = 7	Extrude.
I_OMT_LATHE = 8	Revolve.
I_OMT_PYRAMID = 9	Extrude along polyline.

Description

Set of identifiers has been defined for modifications which may be performed on specified objects - structure components. .

II.5.3 IRobotObjOperationType

C++

```
enum IRobotObjOperationType;
```

C#

```
public enum IRobotObjOperationType;
```

Visual Basic

```
Public Enum IRobotObjOperationType
```

Members

Members	Description
I_OOT_NONE = 16	Undefined.

(c) 2014 Autodesk Inc. All rights reserved

I_OOT_MESH = 1	Deformation.
I_OOT_TRANSLATE = 2	Translation.
I_OOT_SCALE = 3	Scaling.
I_OOT_ROTATE = 18	Rotation.

Description

Set of identifiers has been defined for operations which may be performed on specified objects - structure components. .

II.5.4 IRobotObjModification

Class Hierarchy

RobotOM.IRobotObjModification

C++

```
interface IRobotObjModification : IDispatch;
```

C#

```
public interface IRobotObjModification;
```

Visual Basic

```
Public Interface IRobotObjModification
```

Description

The basic interface describing modification of the initial object geometry .

II.5.4.1 IRobotObjModification Members

The following tables list the members exposed by IRobotObjModification.

Public Fields

	Name	Description
	Filled (see page 746)	The flag that indicates if the modification is filled with volumetric elements .
	NDiv (see page 746)	Number of divisions of lateral walls.
	Operations (see page 746)	Collection of operations describing transformation of the modification result (so called reference or image).
	Type (see page 747)	Modification type.

Public Methods

	Name	Description
	Add (see page 747)	The function adds a new operation.
	Clear (see page 747)	The function clears the table of operations.

II.5.4.2 IRobotObjModification Fields

The fields of the IRobotObjModification class are listed here.

Public Fields

	Name	Description
	Filled (see page 746)	The flag that indicates if the modification is filled with volumetric elements .
	NDiv (see page 746)	Number of divisions of lateral walls.
	Operations (see page 746)	Collection of operations describing transformation of the modification result (so called reference or image).

(c) 2014 Autodesk Inc. All rights reserved

	Type (see page 747)	Modification type.

II.5.4.2.1 Filled

C++

```
HRESULT get_Filled(VARIANT_BOOL*);
HRESULT put_Filled(VARIANT_BOOL);
```

C#

```
public bool Filled { get; set; }
```

Visual Basic

```
Public Filled As Boolean
```

Description

The flag that indicates if the modification is filled with volumetric elements .

II.5.4.2.2 NDiv

C++

```
HRESULT get_NDiv(long*);
HRESULT put_NDiv(long);
```

C#

```
public long NDiv { get; set; }
```

Visual Basic

```
Public NDiv As long
```

Description

Number of divisions of lateral walls.

II.5.4.2.3 Operations

C++

```
HRESULT get_Operations(IRobotObjOperationCollection**);
```

C#

```
public IRobotObjOperationCollection Operations { get; }
```

Visual Basic

```
Public ReadOnly Operations As IRobotObjOperationCollection
```

Description

Collection of operations describing transformation of the modification result (so called reference or image).

II.5.4.2.4 Type

C++

```
HRESULT get_Type(IRobotObjModificationType*);
```

C#

```
public IRobotObjModificationType Type { get; }
```

Visual Basic

```
Public ReadOnly Type As IRobotObjModificationType
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Modification type.

II.5.4.3 IRobotObjModification Methods

The methods of the IRobotObjModification class are listed here.

Public Methods

	Name	Description
	Add (see page 747)	The function adds a new operation.
	Clear (see page 747)	The function clears the table of operations.

II.5.4.3.1 Add

C++

```
HRESULT Add(IRobotObjOperation* _operation);
```

C#

```
public void Add(IRobotObjOperation _operation);
```

Visual Basic

```
Public Sub Add(ByRef _operation As IRobotObjOperation)
```

Description

The function adds a new operation.

II.5.4.3.2 Clear

C++

```
HRESULT Clear();
```

C#

```
public void Clear();
```

Visual Basic

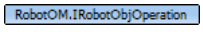
```
Public Sub Clear()
```

Description

The function clears the table of operations.

II.5.5 IRobotObjOperation

Class Hierarchy



C++

```
interface IRobotObjOperation : IDispatch;
```

C#

```
public interface IRobotObjOperation;
```

Visual Basic

```
Public Interface IRobotObjOperation
```

(c) 2014 Autodesk Inc. All rights reserved

Description

The basic interface describing transformation (translation, rotation, scaling, deformation) of the result of modification (the so called reference or image).

II.5.5.1 IRobotObjOperation Members

The following tables list the members exposed by IRobotObjOperation.

Public Fields

	Name	Description
	Type (see page 748)	Operation type.

II.5.5.2 IRobotObjOperation Fields

The fields of the IRobotObjOperation class are listed here.

Public Fields

	Name	Description
	Type (see page 748)	Operation type.

II.5.5.2.1 Type

C++

```
HRESULT get_Type(IRobotObjOperationType*);
```

C#

```
public IRobotObjOperationType Type { get; }
```

Visual Basic

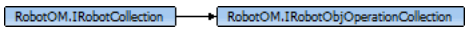
```
Public ReadOnly Type As IRobotObjOperationType
```

Description

Operation type.

II.5.6 IRobotObjOperationCollection

Class Hierarchy



C++

```
interface IRobotObjOperationCollection : IRobotCollection;
```

C#

```
public interface IRobotObjOperationCollection : IRobotCollection;
```

Visual Basic

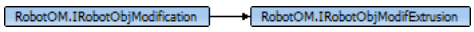
```
Public Interface IRobotObjOperationCollection
```

Description

Collection of operations.

II.5.7 IRobotObjModifExtrusion

Class Hierarchy



(c) 2014 Autodesk Inc. All rights reserved

C++

```
interface IRobotObjModifExtrusion : IRobotObjModification;
```

C#

```
public interface IRobotObjModifExtrusion : IRobotObjModification;
```

Visual Basic

```
Public Interface IRobotObjModifExtrusion
```

Description

The interface describing a modification of initial object geometry - extrude. .

II.5.7.1 IRobotObjModifExtrusion Members

The following tables list the members exposed by IRobotObjModifExtrusion.

Public Fields

	Name	Description
	Filled (see page 746)	The flag that indicates if the modification is filled with volumetric elements .
	NDiv (see page 746)	Number of divisions of lateral walls.
	Operations (see page 746)	Collection of operations describing transformation of the modification result (so called reference or image).
	Type (see page 747)	Modification type.
	Vector (see page 749)	Direction vector for extrusion.

Public Methods

	Name	Description
	Add (see page 747)	The function adds a new operation.
	Clear (see page 747)	The function clears the table of operations.

II.5.7.2 IRobotObjModifExtrusion Fields

The fields of the IRobotObjModifExtrusion class are listed here.

Public Fields

	Name	Description
	Vector (see page 749)	Direction vector for extrusion.

II.5.7.2.1 Vector

C++

```
HRESULT get_Vector(IRobotGeoPoint3D**);
HRESULT put_Vector(IRobotGeoPoint3D*);
```

C#

```
public IRobotGeoPoint3D Vector { get; set; }
```

Visual Basic

```
Public Vector As IRobotGeoPoint3D
```

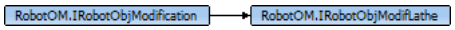
Description

Direction vector for extrusion.

(c) 2014 Autodesk Inc. All rights reserved

II.5.8 IRobotObjModifLathe

Class Hierarchy



C++

```
interface IRobotObjModifLathe : IRobotObjModification;
```

C#

```
public interface IRobotObjModifLathe : IRobotObjModification;
```

Visual Basic

```
Public Interface IRobotObjModifLathe
```

Description

Interface describing modification of initial geometry of the object - revolve operation.

II.5.8.1 IRobotObjModifLathe Members

The following tables list the members exposed by IRobotObjModifLathe.

Public Fields

	Name	Description
	Angle (see page 751)	Revolution angle.
	AxsP1 (see page 751)	The first point of revolution axis.
	AxsP2 (see page 751)	The second point of revolution axis.
	Filled (see page 746)	The flag that indicates if the modification is filled with volumetric elements .
	NDiv (see page 746)	Number of divisions of lateral walls.
	Operations (see page 746)	Collection of operations describing transformation of the modification result (so called reference or image).
	Type (see page 747)	Modification type.

Public Methods

	Name	Description
	Add (see page 747)	The function adds a new operation.
	Clear (see page 747)	The function clears the table of operations.

II.5.8.2 IRobotObjModifLathe Fields

The fields of the IRobotObjModifLathe class are listed here.

Public Fields

	Name	Description
	Angle (see page 751)	Revolution angle.
	AxsP1 (see page 751)	The first point of revolution axis.
	AxsP2 (see page 751)	The second point of revolution axis.

II.5.8.2.1 Angle

C++

```
HRESULT get_Angle(double*);
HRESULT put_Angle(double);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double Angle { get; set; }
```

Visual Basic

```
Public Angle As double
```

Description

Revolution angle.

II.5.8.2.2 AxsP1

C++

```
HRESULT get_AxsP1(IRobotGeoPoint3D**);
HRESULT put_AxsP1(IRobotGeoPoint3D*);
```

C#

```
public IRobotGeoPoint3D AxsP1 { get; set; }
```

Visual Basic

```
Public AxsP1 As IRobotGeoPoint3D
```

Description

The first point of revolution axis.

II.5.8.2.3 AxsP2

C++

```
HRESULT get_AxsP2(IRobotGeoPoint3D**);
HRESULT put_AxsP2(IRobotGeoPoint3D*);
```

C#

```
public IRobotGeoPoint3D AxsP2 { get; set; }
```

Visual Basic

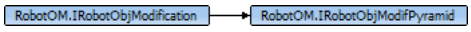
```
Public AxsP2 As IRobotGeoPoint3D
```

Description

The second point of revolution axis.

II.5.9 IRobotObjModifPyramid

Class Hierarchy



C++

```
interface IRobotObjModifPyramid : IRobotObjModification;
```

C#

```
public interface IRobotObjModifPyramid : IRobotObjModification;
```

Visual Basic

```
Public Interface IRobotObjModifPyramid
```

Description

Interface describing modification of initial object geometry - extrude along polyline.

(c) 2014 Autodesk Inc. All rights reserved

II.5.9.1 IRobotObjModifPyramid Members

The following tables list the members exposed by IRobotObjModifPyramid.

Public Fields

	Name	Description
	Factors (see page 752)	Direction (XYZ) scaling factors .
	Filled (see page 746)	The flag that indicates if the modification is filled with volumetric elements .
	NDiv (see page 746)	Number of divisions of lateral walls.
	Operations (see page 746)	Collection of operations describing transformation of the modification result (so called reference or image).
	Points (see page 753)	Polyline points.
	Type (see page 747)	Modification type.

Public Methods

	Name	Description
	Add (see page 747)	The function adds a new operation.
	AddPoint (see page 753)	Function adds a new point of the polyline with a scaling factor.
	Clear (see page 747)	The function clears the table of operations.
	ClearPoints (see page 753)	Available since version 1.7.

II.5.9.2 IRobotObjModifPyramid Fields

The fields of the IRobotObjModifPyramid class are listed here.

Public Fields

	Name	Description
	Factors (see page 752)	Direction (XYZ) scaling factors .
	Points (see page 753)	Polyline points.

II.5.9.2.1 Factors

C++

```
HRESULT get_Factors(IRobotGeoPoint3DCollection**);
```

C#

```
public IRobotGeoPoint3DCollection Factors { get; }
```

Visual Basic

```
Public ReadOnly Factors As IRobotGeoPoint3DCollection
```

Description

Direction (XYZ) scaling factors .

II.5.9.2.2 Points

C++

```
HRESULT get_Points(IRobotGeoPoint3DCollection**);
```

C#

```
public IRobotGeoPoint3DCollection Points { get; }
```

Visual Basic

```
Public ReadOnly Points As IRobotGeoPoint3DCollection
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Polyline points.

II.5.9.3 IRobotObjModifPyramid Methods

The methods of the IRobotObjModifPyramid class are listed here.

Public Methods

	Name	Description
	AddPoint (see page 753)	Function adds a new point of the polyline with a scaling factor.
	ClearPoints (see page 753)	Available since version 1.7.

II.5.9.3.1 AddPoint

C++

```
HRESULT AddPoint(IRobotGeoPoint3D* _point, IRobotGeoPoint3D* _factor);
```

C#

```
public void AddPoint(IRobotGeoPoint3D _point, IRobotGeoPoint3D _factor);
```

Visual Basic

```
Public Sub AddPoint(ByRef _point As IRobotGeoPoint3D, ByRef _factor As IRobotGeoPoint3D)
```

Description

Function adds a new point of the polyline with a scaling factor.

II.5.9.3.2 ClearPoints

C++

```
HRESULT ClearPoints();
```

C#

```
public void ClearPoints();
```

Visual Basic

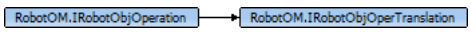
```
Public Sub ClearPoints()
```

Description

Available since version 1.7.

II.5.10 IRobotObjOperTranslation

Class Hierarchy



C++

```
interface IRobotObjOperTranslation : IRobotObjOperation;
```

C#

```
public interface IRobotObjOperTranslation : IRobotObjOperation;
```

Visual Basic

```
Public Interface IRobotObjOperTranslation
```

(c) 2014 Autodesk Inc. All rights reserved

Description

The interface describing the operation of translation that transforms the result of modification of the initial object geometry. .

II.5.10.1 IRobotObjOperTranslation Members

The following tables list the members exposed by IRobotObjOperTranslation.

Public Fields

	Name	Description
	Type (see page 748)	Operation type.
	Vector (see page 754)	Translation vector.

II.5.10.2 IRobotObjOperTranslation Fields

The fields of the IRobotObjOperTranslation class are listed here.

Public Fields

	Name	Description
	Vector (see page 754)	Translation vector.

II.5.10.2.1 Vector

C++

```
HRESULT get_Vector(IRobotGeoPoint3D**);
HRESULT put_Vector(IRobotGeoPoint3D*);
```

C#

```
public IRobotGeoPoint3D Vector { get; set; }
```

Visual Basic

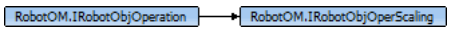
```
Public Vector As IRobotGeoPoint3D
```

Description

Translation vector.

II.5.11 IRobotObjOperScaling

Class Hierarchy



C++

```
interface IRobotObjOperScaling : IRobotObjOperation;
```

C#

```
public interface IRobotObjOperScaling : IRobotObjOperation;
```

Visual Basic

```
Public Interface IRobotObjOperScaling
```

Description

The interface describing the operation of scaling, transforming the result of modification of object geometry .

II.5.11.1 IRobotObjOperScaling Members

The following tables list the members exposed by IRobotObjOperScaling.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	Center (see page 755)	Scaling center.
	Factor (see page 755)	Scaling coefficient in the three main directions.
	Type (see page 748)	Operation type.

II.5.11.2 IRobotObjOperScaling Fields

The fields of the IRobotObjOperScaling class are listed here.

Public Fields

	Name	Description
	Center (see page 755)	Scaling center.
	Factor (see page 755)	Scaling coefficient in the three main directions.

II.5.11.2.1 Center

C++

```
HRESULT get_Center(IRobotGeoPoint3D**);
HRESULT put_Center(IRobotGeoPoint3D*);
```

C#

```
public IRobotGeoPoint3D Center { get; set; }
```

Visual Basic

```
Public Center As IRobotGeoPoint3D
```

Description

Scaling center.

II.5.11.2.2 Factor

C++

```
HRESULT get_Factor(IRobotGeoPoint3D**);
```

C#

```
public IRobotGeoPoint3D Factor { get; }
```

Visual Basic

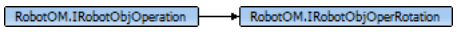
```
Public ReadOnly Factor As IRobotGeoPoint3D
```

Description

Scaling coefficient in the three main directions.

II.5.12 IRobotObjOperRotation

Class Hierarchy



C++

```
interface IRobotObjOperRotation : IRobotObjOperation;
```

C#

```
public interface IRobotObjOperRotation : IRobotObjOperation;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Interface IRobotObjOperRotation
```

Description

Interface describing the operation of rotation that transforms the result of modification of object geometry. .

II.5.12.1 IRobotObjOperRotation Members

The following tables list the members exposed by IRobotObjOperRotation.

Public Fields

	Name	Description
	Angle (see page 756)	Rotation angle.
	AxsP1 (see page 757)	The first point of rotation axis.
	AxsP2 (see page 757)	The second point of rotation axis.
	Type (see page 748)	Operation type.

II.5.12.2 IRobotObjOperRotation Fields

The fields of the IRobotObjOperRotation class are listed here.

Public Fields

	Name	Description
	Angle (see page 756)	Rotation angle.
	AxsP1 (see page 757)	The first point of rotation axis.
	AxsP2 (see page 757)	The second point of rotation axis.

II.5.12.2.1 Angle

C++

```
HRESULT get_Angle(double*);
HRESULT put_Angle(double);
```

C#

```
public double Angle { get; set; }
```

Visual Basic

```
Public Angle As double
```

Description

Rotation angle.

II.5.12.2.2 AxsP1

C++

```
HRESULT get_AxsP1(IRobotGeoPoint3D**);
HRESULT put_AxsP1(IRobotGeoPoint3D*);
```

C#

```
public IRobotGeoPoint3D AxsP1 { get; set; }
```

Visual Basic

```
Public AxsP1 As IRobotGeoPoint3D
```

Description

The first point of rotation axis.

(c) 2014 Autodesk Inc. All rights reserved

II.5.12.2.3 AxsP2

C++

```
HRESULT get_AxsP2(IRobotGeoPoint3D**);
HRESULT put_AxsP2(IRobotGeoPoint3D*);
```

C#

```
public IRobotGeoPoint3D AxsP2 { get; set; }
```

Visual Basic

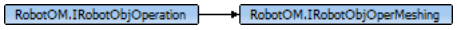
```
Public AxsP2 As IRobotGeoPoint3D
```

Description

The second point of rotation axis.

II.5.13 IRobotObjOperMeshing

Class Hierarchy



C++

```
interface IRobotObjOperMeshing : IRobotObjOperation;
```

C#

```
public interface IRobotObjOperMeshing : IRobotObjOperation;
```

Visual Basic

```
Public Interface IRobotObjOperMeshing
```

Description

The interface describing a deformation that transforms the result of modification of initial object geometry .

II.5.13.1 IRobotObjOperMeshing Members

The following tables list the members exposed by IRobotObjOperMeshing.

Public Fields

	Name	Description
	Points (see page 758)	Collection of indexes (numbers) of points that undergo displacement .
	Type (see page 748)	Operation type.
	Vectors (see page 758)	Collection of displacement vectors.

Public Methods

	Name	Description
	Add (see page 759)	The function adds a displacement vector for an indicated point that defines initial object geometry. The initial geometry of each object is defined through a set of points.
	Clear (see page 759)	The function clears the collection of vectors and point numbers.

II.5.13.2 IRobotObjOperMeshing Fields

The fields of the IRobotObjOperMeshing class are listed here.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	Points (see page 758)	Collection of indexes (numbers) of points that undergo displacement .
	Vectors (see page 758)	Collection of displacement vectors.

II.5.13.2.1 Points

C++

```
HRESULT get_Points(IRobotNumbersCollection**);
```

C#

```
public IRobotNumbersCollection Points { get; }
```

Visual Basic

```
Public ReadOnly Points As IRobotNumbersCollection
```

Description

Collection of indexes (numbers) of points that undergo displacement .

II.5.13.2.2 Vectors

C++

```
HRESULT get_Vectors(IRobotGeoPoint3DCollection**);
```

C#

```
public IRobotGeoPoint3DCollection Vectors { get; }
```

Visual Basic

```
Public ReadOnly Vectors As IRobotGeoPoint3DCollection
```

Description

Collection of displacement vectors.

II.5.13.3 IRobotObjOperMeshing Methods

The methods of the IRobotObjOperMeshing class are listed here.

Public Methods

	Name	Description
	Add (see page 759)	The function adds a displacement vector for an indicated point that defines initial object geometry. The initial geometry of each object is defined through a set of points.
	Clear (see page 759)	The function clears the collection of vectors and point numbers.

II.5.13.3.1 Add

C++

```
HRESULT Add(long _point, IRobotGeoPoint3D* _vector);
```

C#

```
public void Add(long _point, IRobotGeoPoint3D _vector);
```

Visual Basic

```
Public Sub Add(_point As long, ByRef _vector As IRobotGeoPoint3D)
```

(c) 2014 Autodesk Inc. All rights reserved

Description

The function adds a displacement vector for an indicated point that defines initial object geometry. The initial geometry of each object is defined through a set of points.

II.5.13.3.2 Clear

C++

```
HRESULT Clear();
```

C#

```
public void Clear();
```

Visual Basic

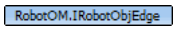
```
Public Sub Clear()
```

Description

The function clears the collection of vectors and point numbers.

II.5.14 IRobotObjEdge

Class Hierarchy



C++

```
interface IRobotObjEdge : IDispatch;
```

C#

```
public interface IRobotObjEdge;
```

Visual Basic

```
Public Interface IRobotObjEdge
```

Description

Interface describing an object edge.

II.5.14.1 IRobotObjEdge Members

The following tables list the members exposed by IRobotObjEdge.

Public Fields

	Name	Description
	Path (see page 760)	Edge geometry .

Public Methods

	Name	Description
	Divide (see page 761)	Fuction divides the edge at the indicated point. If the _relative flag is set as True, then the _x parameter determines a relative distance between the division point and edge beginning - a value contained in the interval (0, 1). Otherwise, _x determines a distance between the division point and edge beginning expressed in meters. .
	DivideByPlane (see page 761)	Function divides the edge by cutting it with the indicated plane of the defined coordinate on the normal axis.
	DivideN (see page 761)	Function divides the edge into N equal parts.
	GetLabel (see page 762)	

(c) 2014 Autodesk Inc. All rights reserved

	GetLabelName (see page 762)	
	GetLabels (see page 762)	
	HasLabel (see page 762)	
	RemoveLabel (see page 763)	
	SetLabel (see page 763)	

II.5.14.2 IRobotObjEdge Fields

The fields of the IRobotObjEdge class are listed here.

Public Fields

	Name	Description
	Path (see page 760)	Edge geometry .

II.5.14.2.1 Path

C++

```
HRESULT get_Path(IRobotGeoPoint3DCollection**);
```

C#

```
public IRobotGeoPoint3DCollection Path { get; }
```

Visual Basic

```
Public ReadOnly Path As IRobotGeoPoint3DCollection
```

Description

Edge geometry .

II.5.14.3 IRobotObjEdge Methods

The methods of the IRobotObjEdge class are listed here.

Public Methods

	Name	Description
	Divide (see page 761)	Fuction divides the edge at the indicated point. If the _relative flag is set as True, then the _x parameter determines a relative distance between the division point and edge beginning - a value contained in the interval (0, 1). Otherwise, _x determines a distance between the division point and edge beginning expressed in meters. .
	DivideByPlane (see page 761)	Function divides the edge by cutting it with the indicated plane of the defined coordinate on the normal axis.
	DivideN (see page 761)	Function divides the edge into N equal parts.
	GetLabel (see page 762)	
	GetLabelName (see page 762)	
	GetLabels (see page 762)	
	HasLabel (see page 762)	
	RemoveLabel (see page 763)	
	SetLabel (see page 763)	

II.5.14.3.1 Divide

C++

```
HRESULT Divide(double _x, VARIANT_BOOL _relative = false, VARIANT_BOOL _only_gen_node =
false);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public void Divide(double _x, bool _relative = false, bool _only_gen_node = false);
```

Visual Basic

```
Public Sub Divide(_x As double, Optional _relative As Boolean = false, Optional
_only_gen_node As Boolean = false)
```

Description

Fuction divides the edge at the indicated point. If the _relative flag is set as True, then the _x parameter determines a relative distance between the division point and edge beginning - a value contained in the interval (0, 1). Otherwise, _x determines a distance between the division point and edge beginning expressed in meters. .

Version

Available since version 4.5.

II.5.14.3.2 DivideByPlane

C++

```
HRESULT DivideByPlane(double _val, IRobotGeoCoordinateAxis _axis, VARIANT_BOOL
_only_gen_node = false);
```

C#

```
public void DivideByPlane(double _val, IRobotGeoCoordinateAxis _axis, bool _only_gen_node =
false);
```

Visual Basic

```
Public Sub DivideByPlane(_val As double, _axis As IRobotGeoCoordinateAxis, Optional
_only_gen_node As Boolean = false)
```

Description

Function divides the edge by cutting it with the indicated plane of the defined coordinate on the normal axis.

Version

Available since version 4.5.

II.5.14.3.3 DivideN

C++

```
HRESULT DivideN(long _n, VARIANT_BOOL _only_gen_nodes = false);
```

C#

```
public void DivideN(long _n, bool _only_gen_nodes = false);
```

Visual Basic

```
Public Sub DivideN(_n As long, Optional _only_gen_nodes As Boolean = false)
```

Description

Function divides the edge into N equal parts.

Version

Available since version 4.5.

(c) 2014 Autodesk Inc. All rights reserved

II.5.14.3.4 GetLabel

C++

```
HRESULT GetLabel(IRobotLabelType _lab_type, IRobotLabel** ret);
```

C#

```
public IRobotLabel GetLabel(IRobotLabelType _lab_type);
```

Visual Basic

```
Public Function GetLabel(_lab_type As IRobotLabelType) As IRobotLabel
```

II.5.14.3.5 GetLabelName

C++

```
HRESULT GetLabelName(IRobotLabelType _lab_type, BSTR* ret);
```

C#

```
public String GetLabelName(IRobotLabelType _lab_type);
```

Visual Basic

```
Public Function GetLabelName(_lab_type As IRobotLabelType) As String
```

II.5.14.3.6 GetLabels

C++

```
HRESULT GetLabels(IRobotCollection** ret);
```

C#

```
public IRobotCollection GetLabels();
```

Visual Basic

```
Public Function GetLabels() As IRobotCollection
```

II.5.14.3.7 HasLabel

C++

```
HRESULT HasLabel(IRobotLabelType _lab_type, VARIANT_BOOL* ret);
```

C#

```
public bool HasLabel(IRobotLabelType _lab_type);
```

Visual Basic

```
Public Function HasLabel(_lab_type As IRobotLabelType) As Boolean
```

II.5.14.3.8 RemoveLabel

C++

```
HRESULT RemoveLabel(IRobotLabelType _lab_type);
```

C#

```
public void RemoveLabel(IRobotLabelType _lab_type);
```

Visual Basic

```
Public Sub RemoveLabel(_lab_type As IRobotLabelType)
```

(c) 2014 Autodesk Inc. All rights reserved

II.5.14.3.9 SetLabel

C++

```
HRESULT SetLabel(IRobotLabelType _lab_type, BSTR _lab_name);
```

C#

```
public void SetLabel(IRobotLabelType _lab_type, String _lab_name);
```

Visual Basic

```
Public Sub SetLabel(_lab_type As IRobotLabelType, _lab_name As String)
```

II.5.15 IRobotObjAttributes

Class Hierarchy

C++

```
interface IRobotObjAttributes : IDispatch;
```

C#

```
public interface IRobotObjAttributes;
```

Visual Basic

```
Public Interface IRobotObjAttributes
```

Description

The interface describing additional attributes of an object or its part.

II.5.15.1 IRobotObjAttributes Members

The following tables list the members exposed by IRobotObjAttributes.

Public Fields

	Name	Description
	DirZ (see page 764)	The flag allowing change of the object component orientation to the opposite (change of the local Z axis sense) .
	Meshed (see page 764)	Flag determining if mesh is to be generated on an object component.

Public Methods

	Name	Description
	GetDirX (see page 765)	Function reads the local X axis direction. Read values are the coordinates of the X axis direction vector for definition in the Cartesian system or the coordinates of the system beginning for definition in the polar system. Additionally, the definition method is returned. Available since version 1.7.
	GetLabelName (see page 766)	
	GetLabels (see page 766)	
	GetLCS (see page 766)	Return Local Coordinate System of the panel.
	GetLCSDisplayPosition (see page 766)	
	HasLabel (see page 767)	
	RemoveLabel (see page 767)	

(c) 2014 Autodesk Inc. All rights reserved

	SetDirX (see page 767)	Function defines the local X axis direction in the defined coordinate system. Additional function parameters determine the coordinates of the X axis direction vector for definition in the Cartesian system or the coordinates of the system beginning for definition in the polar system. Available since version 1.7.
	SetLabel (see page 767)	

II.5.15.2 IRobotObjAttributes Fields

The fields of the IRobotObjAttributes class are listed here.

Public Fields

	Name	Description
	DirZ (see page 764)	The flag allowing change of the object component orientation to the opposite (change of the local Z axis sense) .
	Meshed (see page 764)	Flag determining if mesh is to be generated on an object component.

II.5.15.2.1 DirZ

C++

```
HRESULT get_DirZ(VARIANT_BOOL*);
HRESULT put_DirZ(VARIANT_BOOL);
```

C#

```
public bool DirZ { get; set; }
```

Visual Basic

```
Public DirZ As Boolean
```

Description

The flag allowing change of the object component orientation to the opposite (change of the local Z axis sense) .

II.5.15.2.2 Meshed

C++

```
HRESULT get_Meshed(VARIANT_BOOL*);
HRESULT put_Meshed(VARIANT_BOOL);
```

C#

```
public bool Meshed { get; set; }
```

Visual Basic

```
Public Meshed As Boolean
```

Description

Flag determining if mesh is to be generated on an object component.

II.5.15.3 IRobotObjAttributes Methods

The methods of the IRobotObjAttributes class are listed here.

Public Methods

	Name	Description
	GetDirX (see page 765)	Function reads the local X axis direction. Read values are the coordinates of the X axis direction vector for definition in the Cartesian system or the coordinates of the system beginning for definition in the polar system. Additionally, the definition method is returned. Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

	GetLabelName (see page 766)	
	GetLabels (see page 766)	
	GetLCS (see page 766)	Return Local Coordinate System of the panel.
	GetLCSDisplayPosition (see page 766)	
	HasLabel (see page 767)	
	RemoveLabel (see page 767)	
	SetDirX (see page 767)	Function defines the local X axis direction in the defined coordinate system. Additional function parameters determine the coordinates of the X axis direction vector for definition in the Cartesian system or the coordinates of the system beginning for definition in the polar system. Available since version 1.7.
	SetLabel (see page 767)	

II.5.15.3.1 GetDirX

C++

```
HRESULT GetDirX(double* _x, double* _y, double* _z, IRobotObjLocalXDirDefinitionType* ret);
```

C#

```
public IRobotObjLocalXDirDefinitionType GetDirX(double* _x, double* _y, double* _z);
```

Visual Basic

```
Public Function GetDirX(ByRef _x As double*, ByRef _y As double*, ByRef _z As double*) As
IRobotObjLocalXDirDefinitionType
```

Description

Function reads the local X axis direction. Read values are the coordinates of the X axis direction vector for definition in the Cartesian system or the coordinates of the system beginning for definition in the polar system. Additionally, the definition method is returned. Available since version 1.7.

II.5.15.3.2 GetLabelName

C++

```
HRESULT GetLabelName(IRobotLabelType _lab_type, BSTR* ret);
```

C#

```
public String GetLabelName(IRobotLabelType _lab_type);
```

Visual Basic

```
Public Function GetLabelName(_lab_type As IRobotLabelType) As String
```

II.5.15.3.3 GetLabels

C++

```
HRESULT GetLabels(IRobotCollection** ret);
```

C#

```
public IRobotCollection GetLabels();
```

Visual Basic

```
Public Function GetLabels() As IRobotCollection
```

(c) 2014 Autodesk Inc. All rights reserved

II.5.15.3.4 GetLCS

C++

```
HRESULT GetLCS(IRobotGeoPoint3D* _x, IRobotGeoPoint3D* _y, IRobotGeoPoint3D* _z);
```

C#

```
public void GetLCS(IRobotGeoPoint3D _x, IRobotGeoPoint3D _y, IRobotGeoPoint3D _z);
```

Visual Basic

```
Public Sub GetLCS(ByRef _x As IRobotGeoPoint3D, ByRef _y As IRobotGeoPoint3D, ByRef _z As
IRobotGeoPoint3D)
```

Description

Return Local Coordinate System of the panel.

Version

Available since version 14.7.

II.5.15.3.5 GetLCSDisplayPosition

C++

```
HRESULT GetLCSDisplayPosition(IRobotGeoPoint3D* _pt);
```

C#

```
public void GetLCSDisplayPosition(IRobotGeoPoint3D _pt);
```

Visual Basic

```
Public Sub GetLCSDisplayPosition(ByRef _pt As IRobotGeoPoint3D)
```

Version

Available since version 14.7.

II.5.15.3.6 HasLabel

C++

```
HRESULT HasLabel(IRobotLabelType _lab_type, VARIANT_BOOL* ret);
```

C#

```
public bool HasLabel(IRobotLabelType _lab_type);
```

Visual Basic

```
Public Function HasLabel(_lab_type As IRobotLabelType) As Boolean
```

II.5.15.3.7 RemoveLabel

C++

```
HRESULT RemoveLabel(IRobotLabelType _lab_type);
```

C#

```
public void RemoveLabel(IRobotLabelType _lab_type);
```

Visual Basic

```
Public Sub RemoveLabel(_lab_type As IRobotLabelType)
```

(c) 2014 Autodesk Inc. All rights reserved

II.5.15.3.8 SetDirX

C++

```
HRESULT SetDirX(IRobotObjLocalXDirDefinitionType _def_type, double _x, double _y, double
_z);
```

C#

```
public void SetDirX(IRobotObjLocalXDirDefinitionType _def_type, double _x, double _y,
double _z);
```

Visual Basic

```
Public Sub SetDirX(_def_type As IRobotObjLocalXDirDefinitionType, _x As double, _y As
double, _z As double)
```

Description

Function defines the local X axis direction in the defined coordinate system. Additional function parameters determine the coordinates of the X axis direction vector for definition in the Cartesian system or the coordinates of the system beginning for definition in the polar system. Available since version 1.7.

II.5.15.3.9 SetLabel

C++

```
HRESULT SetLabel(IRobotLabelType _lab_type, BSTR _lab_name);
```

C#

```
public void SetLabel(IRobotLabelType _lab_type, String _lab_name);
```

Visual Basic

```
Public Sub SetLabel(_lab_type As IRobotLabelType, _lab_name As String)
```

II.5.16 IRobotObjPart

Class Hierarchy

RobotOM.IRobotObjPart

C++

```
interface IRobotObjPart : IDispatch;
```

C#

```
public interface IRobotObjPart;
```

Visual Basic

```
Public Interface IRobotObjPart
```

Description

Interface describing an object component. .

II.5.16.1 IRobotObjPart Members

The following tables list the members exposed by IRobotObjPart.

Public Fields

	Name	Description
	Attribs (see page 768)	Attributes of the object component .

(c) 2014 Autodesk Inc. All rights reserved

	Type (see page 769)	Object component type Available since version 2.0.

Public Methods

	Name	Description
	GetGeometry (see page 769)	Function returns defintion of the object component geometry. Available since version 2.0.

II.5.16.2 IRobotObjPart Fields

The fields of the IRobotObjPart class are listed here.

Public Fields

	Name	Description
	Attribs (see page 768)	Attributes of the object component .
	Type (see page 769)	Object component type Available since version 2.0.

II.5.16.2.1 Attribs

C++

```
HRESULT get_Attribs(IRobotObjAttributes**);
```

C#

```
public IRobotObjAttributes Attribs { get; }
```

Visual Basic

```
Public ReadOnly Attribs As IRobotObjAttributes
```

Description

Attributes of the object component .

II.5.16.2.2 Type

C++

```
HRESULT get_Type(IRobotObjPartType*);
```

C#

```
public IRobotObjPartType Type { get; }
```

Visual Basic

```
Public ReadOnly Type As IRobotObjPartType
```

Description

Object component type Available since version 2.0.

II.5.16.3 IRobotObjPart Methods

The methods of the IRobotObjPart class are listed here.

Public Methods

	Name	Description
	GetGeometry (see page 769)	Function returns defintion of the object component geometry. Available since version 2.0.

II.5.16.3.1 GetGeometry

C++

```
HRESULT GetGeometry(IRobotGeoObject** ret);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRobotGeoObject GetGeometry();
```

Visual Basic

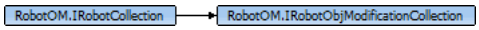
```
Public Function GetGeometry() As IRobotGeoObject
```

Description

Function returns defintion of the object component geometry. Available since version 2.0.

II.5.17 IRobotObjModificationCollection

Class Hierarchy



C++

```
interface IRobotObjModificationCollection : IRobotCollection;
```

C#

```
public interface IRobotObjModificationCollection : IRobotCollection;
```

Visual Basic

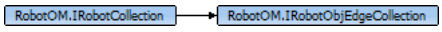
```
Public Interface IRobotObjModificationCollection
```

Description

Collection of modifications .

II.5.18 IRobotObjEdgeCollection

Class Hierarchy



C++

```
interface IRobotObjEdgeCollection : IRobotCollection;
```

C#

```
public interface IRobotObjEdgeCollection : IRobotCollection;
```

Visual Basic

```
Public Interface IRobotObjEdgeCollection
```

Description

Collection of edges.

II.5.19 IRobotObjObject

Class Hierarchy

C++

```
interface IRobotObjObject : IRobotDataObject;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public interface IRobotObjObject : IRobotDataObject;
```

Visual Basic

```
Public Interface IRobotObjObject
```

II.5.19.1 IRobotObjObject Members

The following tables list the members exposed by IRobotObjObject.

Public Fields

	Name	Description
	AnalyzeTTMethod (see page 772)	A flag indicating if the object is taken into consideration in the triangular / trapezoidal method.
	FiniteElems (see page 772)	List of all finite elements assigned to the object.
	Host (see page 772)	Number (see page 31) of the object which is a 'host' for this object.
	IsVolume (see page 773)	Flag (read-only) indicating if the object is a solid.
	Main (see page 773)	Main object component.
	Mesh (see page 773)	Component managing the finite element mesh for this object.
	Name (see page 774)	Object name.
	NameTemplate (see page 774)	Object name pattern.
	Nodes (see page 774)	List of nodes belonging to the object.
	Number (see page 31)	Object number is assigned by the user; it is qunique among all the components of the same type. .
	PartsCount (see page 774)	Number (see page 31) of object components .
	Reference (see page 775)	Main (see page 773) component reference Available since version 2.0.
	StructuralType (see page 775)	Type of structure object.
	UniqueId (see page 775)	Unique object identifier Available since version 1.7.

Public Methods

	Name	Description
	CalcArea (see page 776)	Function calculates the object area.
	CalcVol (see page 776)	Function calculates the solid volume. .
	GetFiniteElemsData (see page 777)	Function gives access to data of finite elements belonging to the object.
	GetHostedObjects (see page 777)	Function returns selection of objects for which this object is a host.
	GetLabel (see page 31)	The function returns a label of the indicated type applied to the object. If the object does not have any label of the type, running the function leads to a critical error. .
	GetLabelName (see page 32)	The function returns the name of a label of the indicated type, applied to the object. If the object does not have any label of the type, running the function leads to a critical error. .
	GetLabels (see page 32)	The function returns a collection containing all labels (of different types) defined for a given object. .
	GetPart (see page 777)	Function returns the indicated object component. Components are indexed from 1 to their number.
	GetPartType (see page 778)	The function returns the type of the indicated object part.
	HasLabel (see page 32)	The function returns True (non-zero value) if a label of the type has already been defined for the object. .
	Initialize (see page 778)	The function initializes or rebuilds the object on the basis of the base geometry and transformations defined in the main component. .
	RemoveLabel (see page 32)	The function deletes a label of the indicated type. .
	SetHostedObjects (see page 778)	Function sets the object as a host for objects described by the given selection.

(c) 2014 Autodesk Inc. All rights reserved

	SetLabel (see page 33)	The function applies the defined label to an object. At a given moment, an object may have only one label of a type applied (e.g. there may not be two supports defined for one node). If an object has a formerly applied label of the same type, it is replaced by the new one. .
	Update (see page 778)	The function updates definition of the object once its attributes are changed. .

II.5.19.2 IRobotObjObject Fields

The fields of the IRobotObjObject class are listed here.

Public Fields

	Name	Description
	AnalyzeTTMethod (see page 772)	A flag indicating if the object is taken into consideration in the triangular / trapezoidal method.
	FiniteElems (see page 772)	List of all finite elements assigned to the object.
	Host (see page 772)	Number (see page 31) of the object which is a 'host' for this object.
	IsVolume (see page 773)	Flag (read-only) indicating if the object is a solid.
	Main (see page 773)	Main object component.
	Mesh (see page 773)	Component managing the finite element mesh for this object.
	Name (see page 774)	Object name.
	NameTemplate (see page 774)	Object name pattern.
	Nodes (see page 774)	List of nodes belonging to the object.
	PartsCount (see page 774)	Number (see page 31) of object components .
	Reference (see page 775)	Main (see page 773) component reference Available since version 2.0.
	StructuralType (see page 775)	Type of structure object.
	UniqueId (see page 775)	Unique object identifier Available since version 1.7.

II.5.19.2.1 AnalyzeTTMethod

C++

```
HRESULT get_AnalyzeTTMethod(VARIANT_BOOL*);
HRESULT put_AnalyzeTTMethod(VARIANT_BOOL);
```

C#

```
public bool AnalyzeTTMethod { get; set; }
```

Visual Basic

```
Public AnalyzeTTMethod As Boolean
```

Description

A flag indicating if the object is taken into consideration in the triangular / trapezoidal method.

Version

Available since version 11.

II.5.19.2.2 FiniteElems

C++

```
HRESULT get_FiniteElems(BSTR*);
```

C#

```
public String FiniteElems { get; }
```

Visual Basic

```
Public ReadOnly FiniteElems As String
```

(c) 2014 Autodesk Inc. All rights reserved

Description

List of all finite elements assigned to the object.

Version

Available since version 2.5.

II.5.19.2.3 Host

C++

```
HRESULT get_Host(long*);
HRESULT put_Host(long);
```

C#

```
public long Host { get; set; }
```

Visual Basic

```
Public Host As long
```

Description

Number (see page 31) of the object which is a 'host' for this object.

Version

Available since version 11.

II.5.19.2.4 IsVolume

C++

```
HRESULT get_IsVolume(VARIANT_BOOL*);
```

C#

```
public bool IsVolume { get; }
```

Visual Basic

```
Public ReadOnly IsVolume As Boolean
```

Description

Flag (read-only) indicating if the object is a solid.

Version

Available since version 4.5.

II.5.19.2.5 Main

C++

```
HRESULT get_Main(IRobotObjPartMain**);
```

C#

```
public IRobotObjPartMain Main { get; }
```

Visual Basic

```
Public ReadOnly Main As IRobotObjPartMain
```

Description

Main object component.

(c) 2014 Autodesk Inc. All rights reserved

II.5.19.2.6 Mesh

C++

```
HRESULT get_Mesh(IRobotObjMesh**);
```

C#

```
public IRobotObjMesh Mesh { get; }
```

Visual Basic

```
Public ReadOnly Mesh As IRobotObjMesh
```

Description

Component managing the finite element mesh for this object.

Version

Available since version 3.

II.5.19.2.7 Name

C++

```
HRESULT get_Name(BSTR*);
```

C#

```
public String Name { get; }
```

Visual Basic

```
Public ReadOnly Name As String
```

Description

Object name.

Version

Available since version 7.5.

II.5.19.2.8 NameTemplate

C++

```
HRESULT get_NameTemplate(BSTR*);
HRESULT put_NameTemplate(BSTR);
```

C#

```
public String NameTemplate { get; set; }
```

Visual Basic

```
Public NameTemplate As String
```

Description

Object name pattern.

Version

Available since version 7.5.

II.5.19.2.9 Nodes

C++

```
HRESULT get_Nodes(BSTR*);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public String Nodes { get; }
```

Visual Basic

```
Public ReadOnly Nodes As String
```

Description

List of nodes belonging to the object.

Version

Available since version 4.

II.5.19.2.10 PartsCount

C++

```
HRESULT get_PartsCount(long*);
```

C#

```
public long PartsCount { get; }
```

Visual Basic

```
Public ReadOnly PartsCount As long
```

Description

Number (see page 31) of object components .

II.5.19.2.11 Reference

C++

```
HRESULT get_Reference(IRobotObjPartReference**);
```

C#

```
public IRobotObjPartReference Reference { get; }
```

Visual Basic

```
Public ReadOnly Reference As IRobotObjPartReference
```

Description

Main (see page 773) component reference Available since version 2.0.

II.5.19.2.12 StructuralType

C++

```
HRESULT get_StructuralType(IRobotObjectStructuralType*);
HRESULT put_StructuralType(IRobotObjectStructuralType);
```

C#

```
public IRobotObjectStructuralType StructuralType { get; set; }
```

Visual Basic

```
Public StructuralType As IRobotObjectStructuralType
```

Description

Type of structure object.

Version

Available since version 9.7.

(c) 2014 Autodesk Inc. All rights reserved

II.5.19.2.13 UniqueId

C++

```
HRESULT get_UniqueId(long*);
```

C#

```
public long UniqueId { get; }
```

Visual Basic

```
Public ReadOnly UniqueId As long
```

Description

Unique object identifier Available since version 1.7.

II.5.19.3 IRobotObjObject Methods

The methods of the IRobotObjObject class are listed here.

Public Methods

	Name	Description
	CalcArea (see page 776)	Function calculates the object area.
	CalcVol (see page 776)	Function calculates the solid volume. .
	GetFiniteElemsData (see page 777)	Function gives access to data of finite elements belonging to the object.
	GetHostedObjects (see page 777)	Function returns selection of objects for which this object is a host.
	GetPart (see page 777)	Function returns the indicated object component. Components are indexed from 1 to their number.
	GetPartType (see page 778)	The function returns the type of the indicated object part.
	Initialize (see page 778)	The function initializes or rebuilds the object on the basis of the base geometry and transformations defined in the main component. .
	SetHostedObjects (see page 778)	Function sets the object as a host for objects described by the given selection.
	Update (see page 778)	The function updates definition of the object once its attributes are changed. .

II.5.19.3.1 CalcArea

C++

```
HRESULT CalcArea(double* ret);
```

C#

```
public double CalcArea();
```

Visual Basic

```
Public Function CalcArea() As double
```

Description

Function calculates the object area.

Version

Available since version 4.5.

(c) 2014 Autodesk Inc. All rights reserved

II.5.19.3.2 CalcVol

C++

```
HRESULT CalcVol(double* ret);
```

C#

```
public double CalcVol();
```

Visual Basic

```
Public Function CalcVol() As double
```

Description

Function calculates the solid volume. .

Version

Available since version 4.5.

II.5.19.3.3 GetFiniteElemsData

C++

```
HRESULT GetFiniteElemsData(IRobotFiniteElementDataSet* _ret_data);
```

C#

```
public void GetFiniteElemsData(IRobotFiniteElementDataSet _ret_data);
```

Visual Basic

```
Public Sub GetFiniteElemsData(ByRef _ret_data As IRobotFiniteElementDataSet)
```

Description

Function gives access to data of finite elements belonging to the object.

Version

Available since version 9.

II.5.19.3.4 GetHostedObjects

C++

```
HRESULT GetHostedObjects(IRobotSelection** ret);
```

C#

```
public IRobotSelection GetHostedObjects();
```

Visual Basic

```
Public Function GetHostedObjects() As IRobotSelection
```

Description

Function returns selection of objects for which this object is a host.

Version

Available since version 11.

(c) 2014 Autodesk Inc. All rights reserved

II.5.19.3.5 GetPart

C++

```
HRESULT GetPart(long _part_num, IRobotObjPart** ret);
```

C#

```
public IRobotObjPart GetPart(long _part_num);
```

Visual Basic

```
Public Function GetPart(_part_num As long) As IRobotObjPart
```

Description

Function returns the indicated object component. Components are indexed from 1 to their number.

II.5.19.3.6 GetPartType

C++

```
HRESULT GetPartType(long _part_num, IRobotObjPartType* ret);
```

C#

```
public IRobotObjPartType GetPartType(long _part_num);
```

Visual Basic

```
Public Function GetPartType(_part_num As long) As IRobotObjPartType
```

Description

The function returns the type of the indicated object part.

II.5.19.3.7 Initialize

C++

```
HRESULT Initialize();
```

C#

```
public void Initialize();
```

Visual Basic

```
Public Sub Initialize()
```

Description

The function initializes or rebuilds the object on the basis of the base geometry and transformations defined in the main component. .

II.5.19.3.8 SetHostedObjects

C++

```
HRESULT SetHostedObjects(IRobotSelection* _obj_sel);
```

C#

```
public void SetHostedObjects(IRobotSelection _obj_sel);
```

Visual Basic

```
Public Sub SetHostedObjects(ByRef _obj_sel As IRobotSelection)
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Function sets the object as a host for objects described by the given selection.

Version

Available since version 11.

II.5.19.3.9 Update

C++

```
HRESULT Update();
```

C#

```
public void Update();
```

Visual Basic

```
Public Sub Update()
```

Description

The function updates definition of the object once its attributes are changed. .

II.5.20 IRobotObjPartType

C++

```
enum IRobotObjPartType;
```

C#

```
public enum IRobotObjPartType;
```

Visual Basic

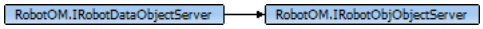
```
Public Enum IRobotObjPartType
```

Members

Members	Description
I_OPT_MAIN = 0	The main part of the object .
I_OPT_REFERENCE = 1	Reference - the image of the main part of the object after transformation.
I_OPT_SIDE = 2	Side - side object component.

II.5.21 IRobotObjObjectServer

Class Hierarchy



C++

```
interface IRobotObjObjectServer : IRobotDataObjectServer;
```

C#

```
public interface IRobotObjObjectServer : IRobotDataObjectServer;
```

Visual Basic

```
Public Interface IRobotObjObjectServer
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Server of object that are structure components .

II.5.21.1 IRobotObjObjectServer Members

The following tables list the members exposed by IRobotObjObjectServer.

Public Fields

	Name	Description
	AutoRecalcHoles (see page 781)	Flag switching on/off automatic detection of openings.
	FreeNumber (see page 782)	First free user number available for the object Available since version 1.7.
	LinearReleases (see page 782)	Server of linear releases defined for objects.
	Mesh (see page 782)	Component managing the finite element mesh on the level of the entire structure model (for all the objects defined in the structure) .

Public Methods

	Name	Description
	BeginMultiOperation (see page 784)	Function enables speed-up of generation or modification of a whole object group through appropriate optimization of some operations. After completing object generation, EndMultiOperation (see page 788)() function should be called up to update definitions of objects in a structure. .
	CalcArea (see page 784)	Function calculates area of the indicated object component. If the object component index equals 0, function returns area of the whole object. Attention: Before calling the function, mesh of surface finite elements should be generated. .
	CalcVol (see page 785)	Function calculates volume of the indicated solid or its component. If the object component index equals 0, function returns volume of the whole solid. Attention: Before calling the function, mesh of volumetric finite elements should be generated.
	Create (see page 785)	The function creates and returns a new structure object with the user-defined number .
	CreateArc (see page 785)	Function creates an arc based on the indicated points. .
	CreateCircle (see page 786)	Function creates a circle of the specified user-defined number based on three points. .
	CreateCone (see page 786)	Function creates a cone based on the indicated four points. First three points define the cone base and the fourth point defines position of the vertex. To create a truncated cone the top radius should be provided as a fifth defining point .
	CreateContour (see page 786)	Function creates a contour based on the specified points. .
	CreateCube (see page 787)	Function creates a cube based on the indicated four points - first three points (P1, P2, P3) define the cube base and the fourth point (PH) defines the height. The cube will be created in such a manner so that the PH point is connected by an edge with the P3 point. .
	CreateCylinder (see page 787)	Function creates a cylinder based on the indicated four points. First three points define the cylinder base and the fourth point (PH) defines its height. The cylinder will be created in such a manner so that the fourth point (PH) is a center of the second base of the cylinder. .
	CreateOnFiniteElems (see page 787)	Function creates one or more objects in such a manner so that they cover the indicated list of finite elements. Function returns the collection of the created object numbers. .
	CreatePolyline (see page 788)	Function creates a polyline of the specified user-defined number based on the indicated points. .
	CreateSolid (see page 788)	Function creates a solid based on the specified list of surface objects.
	Delete (see page 34)	The function deletes the object of the indicated number. .

(c) 2014 Autodesk Inc. All rights reserved

	DeleteMany (see page 34)	The function deletes all objects that meet the criteria of the indicated selection. .
	EndMultiOperation (see page 788)	Function updates definitions of objects in a structure. It should be called up only if BeginMultiOperation (see page 784)() function has been called up earlier. .
	Exist (see page 35)	The function returns True (non-zero value) if the object of the indicated name already exists. .
	FindWithId (see page 789)	Function returns a number of the object with the specified unique identifier. If the object is not found, then, zero value is returned (0). Available since version 1.7.
	Get (see page 35)	The function returns an object with the indicated user-defined number. The type of the returned object agrees with the type of objects managed by the server. If the object of the indicated number does not exist, running the function leads to critical error. .
	GetAll (see page 35)	The function returns a collection containing all objects managed by the server. .
	GetAnalyzeTTMethodEnabled (see page 789)	Function returns selection of objects taking into consideration in the triangular / trapezoidal distribution method.
	GetFiniteElemsData (see page 789)	Function returns finite element data for the determined object selection.
	GetHost (see page 790)	Function returns the number of host object for the given object.
	GetHostedObjects (see page 790)	Function returns a list of objects for which the given object is a host.
	GetMany (see page 36)	The function returns a collection of objects that meet the criteria of the indicated selection. .
	GetName (see page 790)	Function returns a name of the object of the given number.
	GetNameTemplate (see page 791)	Function returns the name pattern of an object of the specified number.
	GetStructuralType (see page 791)	Function returns the structure object type for an object with the specified number.
	GetUniqueId (see page 791)	Function returns a unique identifier for an object with the user-specified number.
	IsVolume (see page 792)	Function returns the True value, if the indicated object is a solid. .
	RemoveLabel (see page 36)	The function removes the labels of the indicated type from all objects that meet the criteria of the indicated selection. .
	SetAnalyzeTTMethod (see page 792)	Function allows for adding/removing objects to/from list of objects taken into consideration in triangular/trapezoidal ristribution method.
	SetHost (see page 792)	Function sets the number of host object for the given object.
	SetHostedObjects (see page 793)	Function sets a list of objects for which the given object is to be a host.
	SetLabel (see page 36)	The function applies a label (identified by the type and name) to all objects that meet the criteria of the indicated selection. .
	SetNameTemplate (see page 793)	Function sets a name pattern for the specified object.
	SetStructuralType (see page 793)	Function sets a structure object type for the specified object list.

II.5.21.2 IRobotObjObjectServer Fields

The fields of the IRobotObjObjectServer class are listed here.

Public Fields

	Name	Description
	AutoRecalcHoles (see page 781)	Flag switching on/off automatic detection of openings.
	FreeNumber (see page 782)	First free user number available for the object Available since version 1.7.
	LinearReleases (see page 782)	Server of linear releases defined for objects.
	Mesh (see page 782)	Component managing the finite element mesh on the level of the entire structure model (for all the objects defined in the structure) .

(c) 2014 Autodesk Inc. All rights reserved

II.5.21.2.1 AutoRecalcHoles

C++

```
HRESULT get_AutoRecalcHoles(VARIANT_BOOL*);
HRESULT put_AutoRecalcHoles(VARIANT_BOOL);
```

C#

```
public bool AutoRecalcHoles { get; set; }
```

Visual Basic

```
Public AutoRecalcHoles As Boolean
```

Description

Flag switching on/off automatic detection of openings.

Version

Available since version 3.5.

II.5.21.2.2 FreeNumber

C++

```
HRESULT get_FreeNumber(long*);
```

C#

```
public long FreeNumber { get; }
```

Visual Basic

```
Public ReadOnly FreeNumber As long
```

Description

First free user number available for the object Available since version 1.7.

II.5.21.2.3 LinearReleases

C++

```
HRESULT get_LinearReleases(IRobotLinearReleaseServer**);
```

C#

```
public IRobotLinearReleaseServer LinearReleases { get; }
```

Visual Basic

```
Public ReadOnly LinearReleases As IRobotLinearReleaseServer
```

Description

Server of linear releases defined for objects.

Version

Available since version 2.5.

II.5.21.2.4 Mesh

C++

```
HRESULT get_Mesh(IRobotObjMesh**);
```

C#

```
public IRobotObjMesh Mesh { get; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public ReadOnly Mesh As IRobotObjMesh
```

Description

Component managing the finite element mesh on the level of the entire structure model (for all the objects defined in the structure) .

Version

Available since version 3.

II.5.21.3 IRobotObjObjectServer Methods

The methods of the IRobotObjObjectServer class are listed here.

Public Methods

	Name	Description
	BeginMultiOperation (see page 784)	Function enables speed-up of generation or modification of a whole object group through appropriate optimization of some operations. After completing object generation, EndMultiOperation (see page 788)() function should be called up to update definitions of objects in a structure. .
	CalcArea (see page 784)	Function calculates area of the indicated object component. If the object component index equals 0, function returns area of the whole object. Attention: Before calling the function, mesh of surface finite elements should be generated. .
	CalcVol (see page 785)	Function calculates volume of the indicated solid or its component. If the object component index equals 0, function returns volume of the whole solid. Attention: Before calling the function, mesh of volumetric finite elements should be generated.
	Create (see page 785)	The function creates and returns a new structure object with the user-defined number .
	CreateArc (see page 785)	Function creates an arc based on the indicated points. .
	CreateCircle (see page 786)	Function creates a circle of the specified user-defined number based on three points. .
	CreateCone (see page 786)	Function creates a cone based on the indicated four points. First three points define the cone base and the fourth point defines position of the vertex. To create a truncated cone the top radius should be provided as a fifth defining point .
	CreateContour (see page 786)	Function creates a contour based on the specified points. .
	CreateCube (see page 787)	Function creates a cube based on the indicated four points - first three points (P1, P2, P3) define the cube base and the fourth point (PH) defines the height. The cube will be created in such a manner so that the PH point is connected by an edge with the P3 point. .
	CreateCylinder (see page 787)	Function creates a cylinder based on the indicated four points. First three points define the cylinder base and the fourth point (PH) defines its height. The cylinder will be created in such a manner so that the fourth point (PH) is a center of the second base of the cylinder. .
	CreateOnFiniteElems (see page 787)	Function creates one or more objects in such a manner so that they cover the indicated list of finite elements. Function returns the collection of the created object numbers. .
	CreatePolyline (see page 788)	Function creates a polyline of the specified user-defined number based on the indicated points. .
	CreateSolid (see page 788)	Function creates a solid based on the specified list of surface objects.
	EndMultiOperation (see page 788)	Function updates definitions of objects in a structure. It should be called up only if BeginMultiOperation (see page 784)() function has been called up earlier. .

(c) 2014 Autodesk Inc. All rights reserved

	FindWithId (see page 789)	Function returns a number of the object with the specified unique identifier. If the object is not found, then, zero value is returned (0). Available since version 1.7.
	GetAnalyzeTTMethodEnabled (see page 789)	Function returns selection of objects taking into consideration in the triangular / trapezoidal distribution method.
	GetFiniteElemsData (see page 789)	Function returns finite element data for the determined object selection.
	GetHost (see page 790)	Function returns the number of host object for the given object.
	GetHostedObjects (see page 790)	Function returns a list of objects for which the given object is a host.
	GetName (see page 790)	Function returns a name of the object of the given number.
	GetNameTemplate (see page 791)	Function returns the name pattern of an object of the specified number.
	GetStructuralType (see page 791)	Function returns the structure object type for an object with the specified number.
	GetUniqueId (see page 791)	Function returns a unique identifier for an object with the user-specified number.
	IsVolume (see page 792)	Function returns the True value, if the indicated object is a solid. .
	SetAnalyzeTTMethod (see page 792)	Function allows for adding/removing objects to/from list of objects taken into consideration in triangular/trapezoidal ristribution method.
	SetHost (see page 792)	Function sets the number of host object for the given object.
	SetHostedObjects (see page 793)	Function sets a list of objects for which the given object is to be a host.
	SetNameTemplate (see page 793)	Function sets a name pattern for the specified object.
	SetStructuralType (see page 793)	Function sets a structure object type for the specified object list.

II.5.21.3.1 BeginMultiOperation

C++

```
HRESULT BeginMultiOperation();
```

C#

```
public void BeginMultiOperation();
```

Visual Basic

```
Public Sub BeginMultiOperation()
```

Description

Function enables speed-up of generation or modification of a whole object group through appropriate optimization of some operations. After completing object generation, EndMultiOperation (see page 788)() function should be called up to update definitions of objects in a structure. .

Version

Available since version 3.5.

II.5.21.3.2 CalcArea

C++

```
HRESULT CalcArea(long _obj_num, long _part_idx = 0, double* ret);
```

C#

```
public double CalcArea(long _obj_num, long _part_idx = 0);
```

Visual Basic

```
Public Function CalcArea(_obj_num As long, Optional _part_idx As long = 0) As double
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Function calculates area of the indicated object component. If the object component index equals 0, function returns area of the whole object. Attention: Before calling the function, mesh of surface finite elements should be generated. .

Version

Available since version 4.5.

II.5.21.3.3 CalcVol

C++

```
HRESULT CalcVol(long _obj_num, long _part_idx = 0, double* ret);
```

C#

```
public double CalcVol(long _obj_num, long _part_idx = 0);
```

Visual Basic

```
Public Function CalcVol(_obj_num As long, Optional _part_idx As long = 0) As double
```

Description

Function calculates volume of the indicated solid or its component. If the object component index equals 0, function returns volume of the whole solid. Attention: Before calling the function, mesh of volumetric finite elements should be generated.

Version

Available since version 4.5.

II.5.21.3.4 Create

C++

```
HRESULT Create(long _num, IRobotObjObject** ret);
```

C#

```
public IRobotObjObject Create(long _num);
```

Visual Basic

```
Public Function Create(_num As long) As IRobotObjObject
```

Description

The function creates and returns a new structure object with the user-defined number .

II.5.21.3.5 CreateArc

C++

```
HRESULT CreateArc(long _number, IRobotPointsArray* _points, IRobotGeoArcDefinitionMethod
_creation_type = I_GADM_CENTER_BEGIN_END);
```

C#

```
public void CreateArc(long _number, IRobotPointsArray _points, IRobotGeoArcDefinitionMethod
_creation_type = I_GADM_CENTER_BEGIN_END);
```

Visual Basic

```
Public Sub CreateArc(_number As long, ByRef _points As IRobotPointsArray, Optional
_creation_type As IRobotGeoArcDefinitionMethod = I_GADM_CENTER_BEGIN_END)
```

Description

Function creates an arc based on the indicated points. .

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.

II.5.21.3.6 CreateCircle

C++

```
HRESULT CreateCircle(long _number, IRobotPointsArray* _points);
```

C#

```
public void CreateCircle(long _number, IRobotPointsArray _points);
```

Visual Basic

```
Public Sub CreateCircle(_number As long, ByRef _points As IRobotPointsArray)
```

Description

Function creates a circle of the specified user-defined number based on three points. .

Version

Available since version 3.

II.5.21.3.7 CreateCone

C++

```
HRESULT CreateCone(long _number, IRobotPointsArray* _points, VARIANT_BOOL _has_base, long
_divisions, long _sides, VARIANT_BOOL _is_volume = true, VARIANT_BOOL _has_top = true,
IRobotObjObject** ret);
```

C#

```
public IRobotObjObject CreateCone(long _number, IRobotPointsArray _points, bool _has_base,
long _divisions, long _sides, bool _is_volume = true, bool _has_top = true);
```

Visual Basic

```
Public Function CreateCone(_number As long, ByRef _points As IRobotPointsArray, _has_base
As Boolean, _divisions As long, _sides As long, Optional _is_volume As Boolean = true,
Optional _has_top As Boolean = true) As IRobotObjObject
```

Description

Function creates a cone based on the indicated four points. First three points define the cone base and the fourth point defines position of the vertex. To create a truncated cone the top radius should be provided as a fifth defining point .

Version

Available since version 3.

II.5.21.3.8 CreateContour

C++

```
HRESULT CreateContour(long _number, IRobotPointsArray* _points);
```

C#

```
public void CreateContour(long _number, IRobotPointsArray _points);
```

Visual Basic

```
Public Sub CreateContour(_number As long, ByRef _points As IRobotPointsArray)
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Function creates a contour based on the specified points. .

Version

Available since version 3.

II.5.21.3.9 CreateCube

C++

```
HRESULT CreateCube(long _number, IRobotPointsArray* _points, VARIANT_BOOL _has_base,
VARIANT_BOOL _has_top, long _divisions, VARIANT_BOOL _is_volume = true, IRobotObjObject**
ret);
```

C#

```
public IRobotObjObject CreateCube(long _number, IRobotPointsArray _points, bool _has_base,
bool _has_top, long _divisions, bool _is_volume = true);
```

Visual Basic

```
Public Function CreateCube(_number As long, ByRef _points As IRobotPointsArray, _has_base
As Boolean, _has_top As Boolean, _divisions As long, Optional _is_volume As Boolean = true)
As IRobotObjObject
```

Description

Function creates a cube based on the indicated four points - first three points (P1, P2, P3) define the cube base and the fourth point (PH) defines the height. The cube will be created in such a manner so that the PH point is connected by an edge with the P3 point. .

Version

Available since version 3.

II.5.21.3.10 CreateCylinder

C++

```
HRESULT CreateCylinder(long _number, IRobotPointsArray* _points, VARIANT_BOOL _has_base,
VARIANT_BOOL _has_top, long _divisions, long _sides, VARIANT_BOOL _is_volume = true,
IRobotObjObject** ret);
```

C#

```
public IRobotObjObject CreateCylinder(long _number, IRobotPointsArray _points, bool
_has_base, bool _has_top, long _divisions, long _sides, bool _is_volume = true);
```

Visual Basic

```
Public Function CreateCylinder(_number As long, ByRef _points As IRobotPointsArray,
_has_base As Boolean, _has_top As Boolean, _divisions As long, _sides As long, Optional
_is_volume As Boolean = true) As IRobotObjObject
```

Description

Function creates a cylinder based on the indicated four points. First three points define the cylinder base and the fourth point (PH) defines its height. The cylinder will be created in such a manner so that the fourth point (PH) is a center of the second base of the cylinder. .

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

II.5.21.3.11 CreateOnFiniteElems

C++

```
HRESULT CreateOnFiniteElems(BSTR _finite_elems, long _first_number,
IRobotNumbersCollection** ret);
```

C#

```
public IRobotNumbersCollection CreateOnFiniteElems(String _finite_elems, long
_first_number);
```

Visual Basic

```
Public Function CreateOnFiniteElems(_finite_elems As String, _first_number As long) As
IRobotNumbersCollection
```

Description

Function creates one or more objects in such a manner so that they cover the indicated list of finite elements. Function returns the collection of the created object numbers. .

Version

Available since version 3.

II.5.21.3.12 CreatePolyline

C++

```
HRESULT CreatePolyline(long _number, IRobotPointsArray* _points);
```

C#

```
public void CreatePolyline(long _number, IRobotPointsArray _points);
```

Visual Basic

```
Public Sub CreatePolyline(_number As long, ByRef _points As IRobotPointsArray)
```

Description

Function creates a polyline of the specified user-defined number based on the indicated points. .

Version

Available since version 3.

II.5.21.3.13 CreateSolid

C++

```
HRESULT CreateSolid(long _number, BSTR _objlist, IRobotObjObject** ret);
```

C#

```
public IRobotObjObject CreateSolid(long _number, String _objlist);
```

Visual Basic

```
Public Function CreateSolid(_number As long, _objlist As String) As IRobotObjObject
```

Description

Function creates a solid based on the specified list of surface objects.

Version

Available since version 8.2.

(c) 2014 Autodesk Inc. All rights reserved

II.5.21.3.14 EndMultiOperation

C++

```
HRESULT EndMultiOperation();
```

C#

```
public void EndMultiOperation();
```

Visual Basic

```
Public Sub EndMultiOperation()
```

Description

Function updates definitions of objects in a structure. It should be called up only if BeginMultiOperation (see page 784)() function has been called up earlier. .

Version

Available since version 3.5.

II.5.21.3.15 FindWithId

C++

```
HRESULT FindWithId(long _unique_id, long* ret);
```

C#

```
public long FindWithId(long _unique_id);
```

Visual Basic

```
Public Function FindWithId(_unique_id As long) As long
```

Description

Function returns a number of the object with the specified unique identifier. If the object is not found, then, zero value is returned (0). Available since version 1.7.

II.5.21.3.16 GetAnalyzeTTMethodEnabled

C++

```
HRESULT GetAnalyzeTTMethodEnabled(IRobotSelection** ret);
```

C#

```
public IRobotSelection GetAnalyzeTTMethodEnabled();
```

Visual Basic

```
Public Function GetAnalyzeTTMethodEnabled() As IRobotSelection
```

Description

Function returns selection of objects taking into consideration in the triangular / trapezoidal distribution method.

Version

Available since version 11.

II.5.21.3.17 GetFiniteElemsData

C++

```
HRESULT GetFiniteElemsData(BSTR _obj_sel_txt, IRobotFiniteElementDataSet* _ret_data);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public void GetFiniteElemsData(String _obj_sel_txt, IRobotFiniteElementDataSet _ret_data);
```

Visual Basic

```
Public Sub GetFiniteElemsData(_obj_sel_txt As String, ByRef _ret_data As
IRobotFiniteElementDataSet)
```

Description

Function returns finite element data for the determined object selection.

Version

Available since version 9.

II.5.21.3.18 GetHost

C++

```
HRESULT GetHost(long _obj_num, long* ret);
```

C#

```
public long GetHost(long _obj_num);
```

Visual Basic

```
Public Function GetHost(_obj_num As long) As long
```

Description

Function returns the number of host object for the given object.

Version

Available since version 11.

II.5.21.3.19 GetHostedObjects

C++

```
HRESULT GetHostedObjects(long _obj_num, IRobotSelection** ret);
```

C#

```
public IRobotSelection GetHostedObjects(long _obj_num);
```

Visual Basic

```
Public Function GetHostedObjects(_obj_num As long) As IRobotSelection
```

Description

Function returns a list of objects for which the given object is a host.

Version

Available since version 11.

II.5.21.3.20 GetName

C++

```
HRESULT GetName(long _obj_num, BSTR* ret);
```

C#

```
public String GetName(long _obj_num);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Function GetName(_obj_num As long) As String
```

Description

Function returns a name of the object of the given number.

Version

Available since version 7.5.

II.5.21.3.21 GetNameTemplate

C++

```
HRESULT GetNameTemplate(long _obj_num, BSTR* ret);
```

C#

```
public String GetNameTemplate(long _obj_num);
```

Visual Basic

```
Public Function GetNameTemplate(_obj_num As long) As String
```

Description

Function returns the name pattern of an object of the specified number.

Version

Available since version 7.5.

II.5.21.3.22 GetStructuralType

C++

```
HRESULT GetStructuralType(long _obj_num, IRobotObjectStructuralType* ret);
```

C#

```
public IRobotObjectStructuralType GetStructuralType(long _obj_num);
```

Visual Basic

```
Public Function GetStructuralType(_obj_num As long) As IRobotObjectStructuralType
```

Description

Function returns the structure object type for an object with the specified number.

Version

Available since version 9.7.

II.5.21.3.23 GetUniqueId

C++

```
HRESULT GetUniqueId(long _obj_num, long* ret);
```

C#

```
public long GetUniqueId(long _obj_num);
```

Visual Basic

```
Public Function GetUniqueId(_obj_num As long) As long
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Function returns a unique identifier for an object with the user-specified number.

Version

Available since version 8.2.

II.5.21.3.24 IsVolume

C++

```
HRESULT IsVolume(long _obj_num, VARIANT_BOOL* ret);
```

C#

```
public bool IsVolume(long _obj_num);
```

Visual Basic

```
Public Function IsVolume(_obj_num As long) As Boolean
```

Description

Function returns the True value, if the indicated object is a solid. .

Version

Available since version 4.5.

II.5.21.3.25 SetAnalyzeTTMethod

C++

```
HRESULT SetAnalyzeTTMethod(BSTR _objs, VARIANT_BOOL _analyze);
```

C#

```
public void SetAnalyzeTTMethod(String _objs, bool _analyze);
```

Visual Basic

```
Public Sub SetAnalyzeTTMethod(_objs As String, _analyze As Boolean)
```

Description

Function allows for adding/removing objects to/from list of objects taken into consideration in triangular/trapezoidal ristribution method.

Version

Available since version 11.

II.5.21.3.26 SetHost

C++

```
HRESULT SetHost(long _obj_num, long _host_num);
```

C#

```
public void SetHost(long _obj_num, long _host_num);
```

Visual Basic

```
Public Sub SetHost(_obj_num As long, _host_num As long)
```

Description

Function sets the number of host object for the given object.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 11.

II.5.21.3.27 SetHostedObjects

C++

```
HRESULT SetHostedObjects(long _obj_num, IRobotSelection* _hosted_objs);
```

C#

```
public void SetHostedObjects(long _obj_num, IRobotSelection _hosted_objs);
```

Visual Basic

```
Public Sub SetHostedObjects(_obj_num As long, ByRef _hosted_objs As IRobotSelection)
```

Description

Function sets a list of objects for which the given object is to be a host.

Version

Available since version 11.

II.5.21.3.28 SetNameTemplate

C++

```
HRESULT SetNameTemplate(long _obj_num, BSTR _name_tmpl);
```

C#

```
public void SetNameTemplate(long _obj_num, String _name_tmpl);
```

Visual Basic

```
Public Sub SetNameTemplate(_obj_num As long, _name_tmpl As String)
```

Description

Function sets a name pattern for the specified object.

Version

Available since version 7.5.

II.5.21.3.29 SetStructuralType

C++

```
HRESULT SetStructuralType(BSTR _obj_sel, IRobotObjectStructuralType _str_type);
```

C#

```
public void SetStructuralType(String _obj_sel, IRobotObjectStructuralType _str_type);
```

Visual Basic

```
Public Sub SetStructuralType(_obj_sel As String, _str_type As IRobotObjectStructuralType)
```

Description

Function sets a structure object type for the specified object list.

Version

Available since version 9.7.

(c) 2014 Autodesk Inc. All rights reserved

II.5.22 IRobotObjLocalXDirDefinitionType

C++

```
enum IRobotObjLocalXDirDefinitionType;
```

C#

```
public enum IRobotObjLocalXDirDefinitionType;
```

Visual Basic

```
Public Enum IRobotObjLocalXDirDefinitionType
```

Members

Members	Description
I_OLXDDT_UNDEFINED = 0	Local X axis direction is not defined (not significant) Available since version 1.7.
I_OLXDDT_CARTESIAN = 1	Definition of local X axis direction in the Cartesian coordinate system Available since version 1.7.
I_OLXDDT_POLAR = 2	Definition of local X axis direction in the polar coordinate system Available since version 1.7.

Description

Type of direction definition for the local X axis of the plate. .

II.5.23 IRobotObjPartMain

Class Hierarchy

C++

```
interface IRobotObjPartMain : IRobotObjPart;
```

C#

```
public interface IRobotObjPartMain : IRobotObjPart;
```

Visual Basic

```
Public Interface IRobotObjPartMain
```

Description

Main object component. It contains definition of the base geometry and geometric transformations which result in the geometry of the entire object. All object edges are also saved in the main component.

II.5.23.1 IRobotObjPartMain Members

The following tables list the members exposed by IRobotObjPartMain.

Public Fields

	Name	Description
	Attribs (see page 768)	Attributes of the object component .
	CurveDiv (see page 795)	Definition of object discretization Available since version 2.0. .
	DefPoints (see page 796)	Collection of points defining the base object geometry Available since version 2.0. .
	Edges (see page 796)	Collection of all object edges Available since version 2.0. .
	FiniteElems (see page 796)	List of finite elements assigned to the main object component.

(c) 2014 Autodesk Inc. All rights reserved

	Geometry (see page 796)	Definition of the base object geometry Available since version 2.0. .
	ModelPoints (see page 797)	Characteristic points used for contour discretization.
	Modifications (see page 797)	Collection of transformations of the base object geometry Available since version 2.0. .
	Nodes (see page 797)	
	Type (see page 769)	Object component type Available since version 2.0.

Public Methods

	Name	Description
	AddModification (see page 798)	Function adds the indicated transformation to the collection. Available since version 2.0. .
	CalcArea (see page 798)	Function calculates area of the object component. .
	CalcVol (see page 798)	Function calculates volume of the solid component. .
	ClearModifications (see page 799)	Function deletes all transformations from the collection of transformations. Available since version 2.0. .
	GetGeometry (see page 769)	Function returns defintion of the object component geometry. Available since version 2.0.

II.5.23.2 IRobotObjPartMain Fields

The fields of the IRobotObjPartMain class are listed here.

Public Fields

	Name	Description
	CurveDiv (see page 795)	Definition of object discretization Available since version 2.0. .
	DefPoints (see page 796)	Collection of points defining the base object geometry Available since version 2.0. .
	Edges (see page 796)	Collection of all object edges Available since version 2.0. .
	FiniteElems (see page 796)	List of finite elements assigned to the main object component.
	Geometry (see page 796)	Definition of the base object geometry Available since version 2.0. .
	ModelPoints (see page 797)	Characteristic points used for contour discretization.
	Modifications (see page 797)	Collection of transformations of the base object geometry Available since version 2.0. .
	Nodes (see page 797)	

II.5.23.2.1 CurveDiv

C++

```
HRESULT get_CurveDiv(IRobotGeoCurveDiv**);
HRESULT put_CurveDiv(IRobotGeoCurveDiv*);
```

C#

```
public IRobotGeoCurveDiv CurveDiv { get; set; }
```

Visual Basic

```
Public CurveDiv As IRobotGeoCurveDiv
```

Description

Definition of object discretization Available since version 2.0. .

II.5.23.2.2 DefPoints

C++

```
HRESULT get_DefPoints(IRobotGeoPoint3DCollection**);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRobotGeoPoint3DCollection DefPoints { get; }
```

Visual Basic

```
Public ReadOnly DefPoints As IRobotGeoPoint3DCollection
```

Description

Collection of points defining the base object geometry Available since version 2.0. .

II.5.23.2.3 Edges

C++

```
HRESULT get_Edges(IRobotObjEdgeCollection**);
```

C#

```
public IRobotObjEdgeCollection Edges { get; }
```

Visual Basic

```
Public ReadOnly Edges As IRobotObjEdgeCollection
```

Description

Collection of all object edges Available since version 2.0. .

II.5.23.2.4 FiniteElems

C++

```
HRESULT get_FiniteElems(BSTR*);
```

C#

```
public String FiniteElems { get; }
```

Visual Basic

```
Public ReadOnly FiniteElems As String
```

Description

List of finite elements assigned to the main object component.

Version

Available since version 2.5.

II.5.23.2.5 Geometry

C++

```
HRESULT get_Geometry(IRobotGeoObject**);
HRESULT put_Geometry(IRobotGeoObject*);
```

C#

```
public IRobotGeoObject Geometry { get; set; }
```

Visual Basic

```
Public Geometry As IRobotGeoObject
```

Description

Definition of the base object geometry Available since version 2.0. .

(c) 2014 Autodesk Inc. All rights reserved

II.5.23.2.6 ModelPoints

C++

```
HRESULT get_ModelPoints(IRobotGeoPoint3DCollection**);
```

C#

```
public IRobotGeoPoint3DCollection ModelPoints { get; }
```

Visual Basic

```
Public ReadOnly ModelPoints As IRobotGeoPoint3DCollection
```

Description

Characteristic points used for contour discretization.

Version

Available since version 3.

II.5.23.2.7 Modifications

C++

```
HRESULT get_Modifications(IRobotObjModificationCollection**);
```

C#

```
public IRobotObjModificationCollection Modifications { get; }
```

Visual Basic

```
Public ReadOnly Modifications As IRobotObjModificationCollection
```

Description

Collection of transformations of the base object geometry Available since version 2.0. .

II.5.23.2.8 Nodes

C++

```
HRESULT get_Nodes(BSTR*);
```

C#

```
public String Nodes { get; }
```

Visual Basic

```
Public ReadOnly Nodes As String
```

Version

Available since version 4.

II.5.23.3 IRobotObjPartMain Methods

The methods of the IRobotObjPartMain class are listed here.

Public Methods

	Name	Description
	AddModification (see page 798)	Function adds the indicated transformation to the collection. Available since version 2.0. .
	CalcArea (see page 798)	Function calculates area of the object component. .
	CalcVol (see page 798)	Function calculates volume of the solid component. .

(c) 2014 Autodesk Inc. All rights reserved

	ClearModifications (see page 799)	Function deletes all transformations from the collection of transformations. Available since version 2.0. .

II.5.23.3.1 AddModification

C++

```
HRESULT AddModification(IRobotObjModification* _mod_def);
```

C#

```
public void AddModification(IRobotObjModification _mod_def);
```

Visual Basic

```
Public Sub AddModification(ByRef _mod_def As IRobotObjModification)
```

Description

Function adds the indicated transformation to the collection. Available since version 2.0. .

II.5.23.3.2 CalcArea

C++

```
HRESULT CalcArea(double* ret);
```

C#

```
public double CalcArea();
```

Visual Basic

```
Public Function CalcArea() As double
```

Description

Function calculates area of the object component. .

Version

Available since version 4.5.

II.5.23.3.3 CalcVol

C++

```
HRESULT CalcVol(double* ret);
```

C#

```
public double CalcVol();
```

Visual Basic

```
Public Function CalcVol() As double
```

Description

Function calculates volume of the solid component. .

Version

Available since version 4.5.

II.5.23.3.4 ClearModifications

C++

```
HRESULT ClearModifications();
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public void ClearModifications();
```

Visual Basic

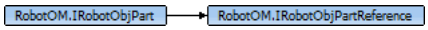
```
Public Sub ClearModifications()
```

Description

Function deletes all transformations from the collection of transformations. Available since version 2.0. .

II.5.24 IRobotObjPartReference

Class Hierarchy



C++

```
interface IRobotObjPartReference : IRobotObjPart;
```

C#

```
public interface IRobotObjPartReference : IRobotObjPart;
```

Visual Basic

```
Public Interface IRobotObjPartReference
```

Description

Object component definition which is the reference of the main component generated due to transformations. .

II.5.24.1 IRobotObjPartReference Members

The following tables list the members exposed by IRobotObjPartReference.

Public Fields

	Name	Description
	Attribs (see page 768)	Attributes of the object component .
	FiniteElems (see page 800)	List of finite elements assigned to this object component.
	Nodes (see page 800)	
	Type (see page 769)	Object component type Available since version 2.0.

Public Methods

	Name	Description
	CalcArea (see page 800)	Function calculates area of object component. .
	CalcVol (see page 801)	Function calculates volume of solid component. .
	GetGeometry (see page 769)	Function returns defintion of the object component geometry. Available since version 2.0.

II.5.24.2 IRobotObjPartReference Fields

The fields of the IRobotObjPartReference class are listed here.

Public Fields

	Name	Description
	FiniteElems (see page 800)	List of finite elements assigned to this object component.
	Nodes (see page 800)	

(c) 2014 Autodesk Inc. All rights reserved

II.5.24.2.1 FiniteElems

C++

```
HRESULT get_FiniteElems(BSTR*);
```

C#

```
public String FiniteElems { get; }
```

Visual Basic

```
Public ReadOnly FiniteElems As String
```

Description

List of finite elements assigned to this object component.

Version

Available since version 2.5.

II.5.24.2.2 Nodes

C++

```
HRESULT get_Nodes(BSTR*);
```

C#

```
public String Nodes { get; }
```

Visual Basic

```
Public ReadOnly Nodes As String
```

Version

Available since version 4.

II.5.24.3 IRobotObjPartReference Methods

The methods of the IRobotObjPartReference class are listed here.

Public Methods

	Name	Description
	CalcArea (see page 800)	Function calculates area of object component. .
	CalcVol (see page 801)	Function calculates volume of solid component. .

II.5.24.3.1 CalcArea

C++

```
HRESULT CalcArea(double* ret);
```

C#

```
public double CalcArea();
```

Visual Basic

```
Public Function CalcArea() As double
```

Description

Function calculates area of object component. .

Version

Available since version 4.5.

(c) 2014 Autodesk Inc. All rights reserved

II.5.24.3.2 CalcVol

C++

```
HRESULT CalcVol(double* ret);
```

C#

```
public double CalcVol();
```

Visual Basic

```
Public Function CalcVol() As double
```

Description

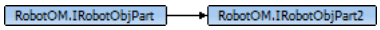
Function calculates volume of solid component. .

Version

Available since version 4.5.

II.5.25 IRobotObjPart2

Class Hierarchy



C++

```
interface IRobotObjPart2 : IRobotObjPart;
```

C#

```
public interface IRobotObjPart2 : IRobotObjPart;
```

Visual Basic

```
Public Interface IRobotObjPart2
```

Description

Extended definition of the object component. .

Version

Available since version 2.5.

II.5.25.1 IRobotObjPart2 Members

The following tables list the members exposed by IRobotObjPart2.

Public Fields

	Name	Description
	Attribs (see page 768)	Attributes of the object component .
	FiniteElems (see page 802)	Text containing the list of finite elements assigned to this object component .
	Nodes (see page 802)	
	Type (see page 769)	Object component type Available since version 2.0.

Public Methods

	Name	Description
	CalcArea (see page 803)	Function calculates area of the object component.
	CalcVol (see page 803)	Function calculates volume of the solid component.

(c) 2014 Autodesk Inc. All rights reserved

	GetGeometry (see page 769)	Function returns defintion of the object component geometry. Available since version 2.0.

II.5.25.2 IRobotObjPart2 Fields

The fields of the IRobotObjPart2 class are listed here.

Public Fields

	Name	Description
	FiniteElems (see page 802)	Text containing the list of finite elements assigned to this object component .
	Nodes (see page 802)	

II.5.25.2.1 FiniteElems

C++

```
HRESULT get_FiniteElems(BSTR*);
```

C#

```
public String FiniteElems { get; }
```

Visual Basic

```
Public ReadOnly FiniteElems As String
```

Description

Text containing the list of finite elements assigned to this object component .

Version

Available since version 2.5.

II.5.25.2.2 Nodes

C++

```
HRESULT get_Nodes(BSTR*);
```

C#

```
public String Nodes { get; }
```

Visual Basic

```
Public ReadOnly Nodes As String
```

Version

Available since version 4.

II.5.25.3 IRobotObjPart2 Methods

The methods of the IRobotObjPart2 class are listed here.

Public Methods

	Name	Description
	CalcArea (see page 803)	Function calculates area of the object component.
	CalcVol (see page 803)	Function calculates volume of the solid component.

II.5.25.3.1 CalcArea

C++

```
HRESULT CalcArea(double* ret);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double CalcArea();
```

Visual Basic

```
Public Function CalcArea() As double
```

Description

Function calculates area of the object component.

Version

Available since version 4.5.

II.5.25.3.2 CalcVol

C++

```
HRESULT CalcVol(double* ret);
```

C#

```
public double CalcVol();
```

Visual Basic

```
Public Function CalcVol() As double
```

Description

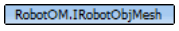
Function calculates volume of the solid component.

Version

Available since version 4.5.

II.5.26 IRobotObjMesh

Class Hierarchy



C++

```
interface IRobotObjMesh : IDispatch;
```

C#

```
public interface IRobotObjMesh;
```

Visual Basic

```
Public Interface IRobotObjMesh
```

Description

Interface allowing management of the finite element mesh.

Version

Available since version 3.

II.5.26.1 IRobotObjMesh Members

The following tables list the members exposed by IRobotObjMesh.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	Freeze (see page 804)	Flag enabling mesh freezing/unfreezing.
	IsGenerated (see page 805)	Flag indicating if the finite element mesh has been generated.
	Params (see page 805)	Parameters of mesh generation.

Public Methods

	Name	Description
	Generate (see page 805)	Function generates a finite element mesh.
	GetBasePoints (see page 806)	Function returns base points for a finite element mesh.
	GetQuality (see page 806)	Function checks a quality of a finite element mesh.
	Remove (see page 806)	Function removes a finite element mesh.
	SetBasePoints (see page 807)	Function sets base points for a finite element mesh.

II.5.26.2 IRobotObjMesh Fields

The fields of the IRobotObjMesh class are listed here.

Public Fields

	Name	Description
	Freeze (see page 804)	Flag enabling mesh freezing/unfreezing.
	IsGenerated (see page 805)	Flag indicating if the finite element mesh has been generated.
	Params (see page 805)	Parameters of mesh generation.

II.5.26.2.1 Freeze

C++

```
HRESULT get_Freeze(VARIANT_BOOL*);
HRESULT put_Freeze(VARIANT_BOOL);
```

C#

```
public bool Freeze { get; set; }
```

Visual Basic

```
Public Freeze As Boolean
```

Description

Flag enabling mesh freezing/unfreezing.

Version

Available since version 3.

II.5.26.2.2 IsGenerated

C++

```
HRESULT get_IsGenerated(VARIANT_BOOL*);
```

C#

```
public bool IsGenerated { get; }
```

Visual Basic

```
Public ReadOnly IsGenerated As Boolean
```

Description

Flag indicating if the finite element mesh has been generated.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.

II.5.26.2.3 Params

C++

```
HRESULT get_Params(IRobotMeshParams**);
```

C#

```
public IRobotMeshParams Params { get; }
```

Visual Basic

```
Public ReadOnly Params As IRobotMeshParams
```

Description

Parameters of mesh generation.

Version

Available since version 3.

II.5.26.3 IRobotObjMesh Methods

The methods of the IRobotObjMesh class are listed here.

Public Methods

	Name	Description
	Generate (see page 805)	Function generates a finite element mesh.
	GetBasePoints (see page 806)	Function returns base points for a finite element mesh.
	GetQuality (see page 806)	Function checks a quality of a finite element mesh.
	Remove (see page 806)	Function removes a finite element mesh.
	SetBasePoints (see page 807)	Function sets base points for a finite element mesh.

II.5.26.3.1 Generate

C++

```
HRESULT Generate();
```

C#

```
public void Generate();
```

Visual Basic

```
Public Sub Generate()
```

Description

Function generates a finite element mesh.

Version

Available since version 3.

II.5.26.3.2 GetBasePoints

C++

```
HRESULT GetBasePoints(IRobotNumbersArray** ret);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRobotNumbersArray GetBasePoints();
```

Visual Basic

```
Public Function GetBasePoints() As IRobotNumbersArray
```

Description

Function returns base points for a finite element mesh.

Version

Available since version 3.

II.5.26.3.3 GetQuality

C++

```
HRESULT GetQuality(double* _out_q1, double* _out_q2, BSTR _out_fes, double _in_precision =
0.5, BSTR _in_objs = "");
```

C#

```
public void GetQuality(double* _out_q1, double* _out_q2, String _out_fes, double
_in_precision = 0.5, String _in_objs = "");
```

Visual Basic

```
Public Sub GetQuality(ByRef _out_q1 As double*, ByRef _out_q2 As double*, ByRef _out_fes As
String, Optional _in_precision As double = 0.5, Optional _in_objs As String = "")
```

Description

Function checks a quality of a finite element mesh.

Version

Available since version 3.

II.5.26.3.4 Remove

C++

```
HRESULT Remove();
```

C#

```
public void Remove();
```

Visual Basic

```
Public Sub Remove()
```

Description

Function removes a finite element mesh.

Version

Available since version 3.

II.5.26.3.5 SetBasePoints

C++

```
HRESULT SetBasePoints(IRobotNumbersArray* _model_points);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public void SetBasePoints(IRobotNumbersArray _model_points);
```

Visual Basic

```
Public Sub SetBasePoints(ByRef _model_points As IRobotNumbersArray)
```

Description

Function sets base points for a finite element mesh.

Version

Available since version 3.

II.6 Finite elements

Available since version 2.5.

Enumerations

	Name	Description
	IRobotFiniteElementType (see page 835)	Available types of finite elements.

Interfaces

	Name	Description
	IRobotFiniteElement (see page 832)	Interface describing planar finite element. .
	IRobotFiniteElementNodes (see page 836)	Collection of nodes defining finite element.
	IRobotFiniteElementServer (see page 838)	Server of finite elements. .
	IRobotFiniteElementData (see page 842)	Interface gives access to finite element data.
	IRobotFiniteElementDataSet (see page 844)	Interface gives access to the finite element set.

II.6.1 FE mesh generator

Available since version 3.

Enumerations

	Name	Description
	IRobotMeshType (see page 808)	Types of generated finite element mesh (surface or volume).
	IRobotMeshForcingRatio (see page 809)	Forcing ratio of selected meshing parameter.
	IRobotMeshImplementDegree (see page 810)	Degrees of the selected method application while generating finite element mesh.
	IRobotMeshPanelDivType (see page 810)	Methods of contour division for the Coons method of finite element mesh generation.
	IRobotMeshSurfaceFEType (see page 811)	Finite element types applied while generating a surface element mesh. .
	IRobotMeshVolumetricFEType (see page 811)	Finite element types applied during generation of volumetric element mesh.
	IRobotMeshDelaunayType (see page 812)	Available variants of Delaunay's method.
	IRobotMeshMethodType (see page 812)	Acceptable meshing methods.

(c) 2014 Autodesk Inc. All rights reserved

	IRobotMeshGenerationType (see page 813)	Methods of finite element mesh generation.
	IRobotMeshAccessType (see page 831)	Availability of interfaces for mesh generation.
	IRobotMeshRefinementType (see page 831)	Methods of finite element mesh refinement.

Interfaces

	Name	Description
	IRobotMeshMethod (see page 813)	A set of parameters defining a mesh generation method and associated quantities. .
	IRobotMeshGeneration (see page 815)	Parameters of mesh generation.
	IRobotMeshCoonsParams (see page 817)	Parameters of Coons method.
	IRobotMeshSurfaceFiniteElems (see page 818)	Definition of surface finite elements applied during mesh generation. .
	IRobotMeshDelaunayParams (see page 820)	Parameters of Delaunay's method.
	IRobotMeshSurfaceParams (see page 824)	Parameters of the mesh generation using planar finite elements.
	IRobotMeshVolumeParams (see page 827)	Parameters of volumetric element mesh generation. .
	IRobotMeshParams (see page 829)	Parameters of the finite element mesh generation.

II.6.1.1 IRobotMeshType

C++

```
enum IRobotMeshType;
```

C#

```
public enum IRobotMeshType;
```

Visual Basic

```
Public Enum IRobotMeshType
```

Members

Members	Description
I_MT_FINE = 2	Fine meshing type. Available since version 3.
I_MT_COARSE = 0	Coarse meshing type. Available since version 3.
I_MT_NORMAL = 1	Normal meshing type. Available since version 3.
I_MT_USER = 3	User-defined meshing type. Available since version 3.
I_MT_WALLS_COARSE = 4	Available since version 12.
I_MT_WALLS_NORMAL = 5	Available since version 12.
I_MT_FLOORS_COARSE = 6	Available since version 12.
I_MT_FLOORS_NORMAL = 7	Available since version 12.

Description

Types of generated finite element mesh (surface or volume).

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.

II.6.1.2 IRobotMeshForcingRatio

C++

```
enum IRobotMeshForcingRatio;
```

C#

```
public enum IRobotMeshForcingRatio;
```

Visual Basic

```
Public Enum IRobotMeshForcingRatio
```

Members

Members	Description
I_MFR_NONE = 1	Forcing ratio - none. Available since version 3.
I_MFR_ANY = 2	Any forcing ratio. Available since version 3.
I_MFR_PROPOSED = 3	Proposed forcing ratio. Available since version 3.
I_MFR_RECOMMENDED = 4	Recommended forcing ratio. Available since version 3.
I_MFR_FORCED = 5	Forced forcing ratio. Available since version 3.

Description

Forcing ratio of selected meshing parameter.

Version

Available since version 3.

II.6.1.3 IRobotMeshImplementDegree

C++

```
enum IRobotMeshImplementDegree;
```

C#

```
public enum IRobotMeshImplementDegree;
```

Visual Basic

```
Public Enum IRobotMeshImplementDegree
```

Members

Members	Description
I_MIR_OFTEN = 1	Available since version 3.
I_MIR_RARELY = 2	Available since version 3.
I_MIR_NEVER = 3	Available since version 3.

Description

Degrees of the selected method application while generating finite element mesh.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.

II.6.1.4 IRobotMeshPanelDivType

C++

```
enum IRobotMeshPanelDivType;
```

C#

```
public enum IRobotMeshPanelDivType;
```

Visual Basic

```
Public Enum IRobotMeshPanelDivType
```

Members

Members	Description
I_MPDT_TRIANG_IN_TRIANG = 1	Division allowing generation of triangular finite elements in a selected triangular contour. Available since version 3.
I_MPDT_TRIANG_AND_SQUARE_IN_TRIANG = 2	Division allowing generation of triangular and square finite elements in a selected triangular contour . Available since version 3.
I_MPDT_TRIANG_AND_TRAPEZ_IN_TRIANG = 3	Division allowing generation of triangular and rhombic finite elements in a selected triangular contour . Available since version 3.
I_MPDT_SQUARE_IN_RECT = 4	Division allowing generation of square finite elements in a selected rectangular contour . Available since version 3.
I_MPDT_TRIANG_IN_RECT = 5	Division allowing generation of triangular finite elements in a selected rectangular contour . Available since version 3.

Description

Methods of contour division for the Coons method of finite element mesh generation.

Version

Available since version 3.

II.6.1.5 IRobotMeshSurfaceFEType

C++

```
enum IRobotMeshSurfaceFEType;
```

C#

```
public enum IRobotMeshSurfaceFEType;
```

Visual Basic

```
Public Enum IRobotMeshSurfaceFEType
```

Members

Members	Description
I_MSFET_3NODE_TRIANG = 1	3-node triangular surface finite elements . Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

I_MSFET_4NODE_QUADRIL = 2	4-node quadrilateral surface finite elements . Available since version 3.
I_MSFET_6NODE_TRIANG = 3	6-node triangular surface finite elements . Available since version 3.
I_MSFET_8NODE_QUADRIL = 4	8-node quadrilateral surface finite elements . Available since version 3.

Description

Finite element types applied while generating a surface element mesh. .

Version

Available since version 3.

II.6.1.6 IRobotMeshVolumetricFEType

C++

```
enum IRobotMeshVolumetricFEType;
```

C#

```
public enum IRobotMeshVolumetricFEType;
```

Visual Basic

```
Public Enum IRobotMeshVolumetricFEType
```

Members

Members	Description
I_MVFET_4NODE_TETRAHED = 1	4-node tetrahedral volumetric finite elements. Available since version 3.
I_MVFET_8NODE_HEXAHED = 2	8-node hexahedral volumetric finite elements. Available since version 3.

Description

Finite element types applied during generation of volumetric element mesh.

Version

Available since version 3.

II.6.1.7 IRobotMeshDelaunayType

C++

```
enum IRobotMeshDelaunayType;
```

C#

```
public enum IRobotMeshDelaunayType;
```

Visual Basic

```
Public Enum IRobotMeshDelaunayType
```

Members

Members	Description
I_MDT_DELAUNAY = 1	Standard Delaunay's method. Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

I_MDT_KANG = 2	Kang's method; selection of this variant indicates that the finite element mesh will be generated only on a contour near emitters according to the adopted method parameters (H0, Hmax, and Q). Available since version 3.
I_MDT_DELAUNAY_AND_KANG = 3	Delaunay and Kang method; selection of this variant indicates that near emitters the finite element mesh will be generated by means of Kang's method, whereas on the remaining contour - according to Delaunay's method. Available since version 3.

Description

Available variants of Delaunay's method.

Version

Available since version 3.

II.6.1.8 IRobotMeshMethodType

C++

```
enum IRobotMeshMethodType;
```

C#

```
public enum IRobotMeshMethodType;
```

Visual Basic

```
Public Enum IRobotMeshMethodType
```

Members

Members	Description
I_MMT_DELAUNAY = 1	Delaunay's method. Available since version 3.
I_MMT_COONS = 2	Coons method . Available since version 3.

Description

Acceptable meshing methods.

Version

Available since version 3.

II.6.1.9 IRobotMeshGenerationType

C++

```
enum IRobotMeshGenerationType;
```

C#

```
public enum IRobotMeshGenerationType;
```

Visual Basic

```
Public Enum IRobotMeshGenerationType
```

(c) 2014 Autodesk Inc. All rights reserved

Members

Members	Description
I_MGT_AUTOMATIC = 1	Automatic generation. Available since version 3.
I_MGT_USER = 2	Generation with user-defined parameters applied. Available since version 3.
I_MGT_ELEMENT_SIZE = 3	Available since version 4.5.

Description

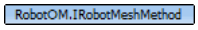
Methods of finite element mesh generation.

Version

Available since version 3.

II.6.1.10 IRobotMeshMethod

Class Hierarchy



C++

```
interface IRobotMeshMethod : IDispatch;
```

C#

```
public interface IRobotMeshMethod;
```

Visual Basic

```
Public Interface IRobotMeshMethod
```

Description

A set of parameters defining a mesh generation method and associated quantities. .

Version

Available since version 3.

II.6.1.10.1 IRobotMeshMethod Members

The following tables list the members exposed by IRobotMeshMethod.

Public Fields

	Name	Description
	ForcingRatio (see page 814)	Forcing ratio for the selected method.
	ImplementDegree (see page 814)	Degree of application for the selected method during mesh generation.
	Method (see page 814)	Meshing method.

II.6.1.10.2 IRobotMeshMethod Fields

The fields of the IRobotMeshMethod class are listed here.

Public Fields

	Name	Description
	ForcingRatio (see page 814)	Forcing ratio for the selected method.
	ImplementDegree (see page 814)	Degree of application for the selected method during mesh generation.
	Method (see page 814)	Meshing method.

(c) 2014 Autodesk Inc. All rights reserved

II.6.1.10.2.1 ForcingRatio

C++

```
HRESULT get_ForcingRatio(IRobotMeshForcingRatio*);
HRESULT put_ForcingRatio(IRobotMeshForcingRatio);
```

C#

```
public IRobotMeshForcingRatio ForcingRatio { get; set; }
```

Visual Basic

```
Public ForcingRatio As IRobotMeshForcingRatio
```

Description

Forcing ratio for the selected method.

Version

Available since version 3.

II.6.1.10.2.2 ImplementDegree

C++

```
HRESULT get_ImplementDegree(IRobotMeshImplementDegree*);
HRESULT put_ImplementDegree(IRobotMeshImplementDegree);
```

C#

```
public IRobotMeshImplementDegree ImplementDegree { get; set; }
```

Visual Basic

```
Public ImplementDegree As IRobotMeshImplementDegree
```

Description

Degree of application for the selected method during mesh generation.

Version

Available since version 3.

II.6.1.10.2.3 Method

C++

```
HRESULT get_Method(IRobotMeshMethodType*);
HRESULT put_Method(IRobotMeshMethodType);
```

C#

```
public IRobotMeshMethodType Method { get; set; }
```

Visual Basic

```
Public Method As IRobotMeshMethodType
```

Description

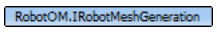
Meshing method.

Version

Available since version 3.

II.6.1.11 IRobotMeshGeneration

Class Hierarchy



(c) 2014 Autodesk Inc. All rights reserved

C++

```
interface IRobotMeshGeneration : IDispatch;
```

C#

```
public interface IRobotMeshGeneration;
```

Visual Basic

```
Public Interface IRobotMeshGeneration
```

Description

Parameters of mesh generation.

Version

Available since version 3.

II.6.1.11.1 IRobotMeshGeneration Members

The following tables list the members exposed by IRobotMeshGeneration.

Public Fields

	Name	Description
	Division1 (see page 816)	Division parameter determining the number of elements to be generated on the first contour edge (between first and second contour vertex).
	Division2 (see page 816)	Division parameter determining the number of elements to be generated on the second contour edge (between second and third contour vertex) .
	ElementSize (see page 816)	Expected length of the finite element edge (in meters) .
	Type (see page 817)	Meshing type.

II.6.1.11.2 IRobotMeshGeneration Fields

The fields of the IRobotMeshGeneration class are listed here.

Public Fields

	Name	Description
	Division1 (see page 816)	Division parameter determining the number of elements to be generated on the first contour edge (between first and second contour vertex).
	Division2 (see page 816)	Division parameter determining the number of elements to be generated on the second contour edge (between second and third contour vertex) .
	ElementSize (see page 816)	Expected length of the finite element edge (in meters) .
	Type (see page 817)	Meshing type.

II.6.1.11.2.1 Division1

C++

```
HRESULT get_Division1(double*);
HRESULT put_Division1(double);
```

C#

```
public double Division1 { get; set; }
```

Visual Basic

```
Public Division1 As double
```

Description

Division parameter determining the number of elements to be generated on the first contour edge (between first and second contour vertex).

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.

II.6.1.11.2.2 Division2

C++

```
HRESULT get_Division2(double*);
HRESULT put_Division2(double);
```

C#

```
public double Division2 { get; set; }
```

Visual Basic

```
Public Division2 As double
```

Description

Division parameter determining the number of elements to be generated on the second contour edge (between second and third contour vertex) .

Version

Available since version 3.

II.6.1.11.2.3 ElementSize

C++

```
HRESULT get_ElementSize(double*);
HRESULT put_ElementSize(double);
```

C#

```
public double ElementSize { get; set; }
```

Visual Basic

```
Public ElementSize As double
```

Description

Expected length of the finite element edge (in meters) .

Version

Available since version 4.5.

II.6.1.11.2.4 Type

C++

```
HRESULT get_Type(IRobotMeshGenerationType*);
HRESULT put_Type(IRobotMeshGenerationType);
```

C#

```
public IRobotMeshGenerationType Type { get; set; }
```

Visual Basic

```
Public Type As IRobotMeshGenerationType
```

Description

Meshing type.

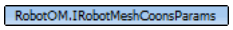
Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

II.6.1.12 IRobotMeshCoonsParams

Class Hierarchy



C++

```
interface IRobotMeshCoonsParams : IDispatch;
```

C#

```
public interface IRobotMeshCoonsParams;
```

Visual Basic

```
Public Interface IRobotMeshCoonsParams
```

Description

Parameters of Coons method.

Version

Available since version 3.

II.6.1.12.1 IRobotMeshCoonsParams Members

The following tables list the members exposed by IRobotMeshCoonsParams.

Public Fields

	Name	Description
	ForcingRatio (see page 818)	Forcing ratio for the selected method of contour division.
	PanelDivisionType (see page 818)	Manner of contour division for Coons method.

II.6.1.12.2 IRobotMeshCoonsParams Fields

The fields of the IRobotMeshCoonsParams class are listed here.

Public Fields

	Name	Description
	ForcingRatio (see page 818)	Forcing ratio for the selected method of contour division.
	PanelDivisionType (see page 818)	Manner of contour division for Coons method.

II.6.1.12.2.1 ForcingRatio

C++

```
HRESULT get_ForcingRatio(IRobotMeshForcingRatio*);
HRESULT put_ForcingRatio(IRobotMeshForcingRatio);
```

C#

```
public IRobotMeshForcingRatio ForcingRatio { get; set; }
```

Visual Basic

```
Public ForcingRatio As IRobotMeshForcingRatio
```

Description

Forcing ratio for the selected method of contour division.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.

II.6.1.12.2.2 PanelDivisionType

C++

```
HRESULT get_PanelDivisionType(IRobotMeshPanelDivType*);
HRESULT put_PanelDivisionType(IRobotMeshPanelDivType);
```

C#

```
public IRobotMeshPanelDivType PanelDivisionType { get; set; }
```

Visual Basic

```
Public PanelDivisionType As IRobotMeshPanelDivType
```

Description

Manner of contour division for Coons method.

Version

Available since version 3.

II.6.1.13 IRobotMeshSurfaceFiniteElems

Class Hierarchy

RobotOM.IRobotMeshSurfaceFiniteElems

C++

```
interface IRobotMeshSurfaceFiniteElems : IDispatch;
```

C#

```
public interface IRobotMeshSurfaceFiniteElems;
```

Visual Basic

```
Public Interface IRobotMeshSurfaceFiniteElems
```

Description

Definition of surface finite elements applied during mesh generation. .

Version

Available since version 3.

II.6.1.13.1 IRobotMeshSurfaceFiniteElems Members

The following tables list the members exposed by IRobotMeshSurfaceFiniteElems.

Public Fields

	Name	Description
	ConversionCoeff (see page 819)	Triangle-to-quadrilateral conversion coefficient .
	ForcingRatio (see page 819)	Forcing ratio for the selected type of surface finite elements.
	Type (see page 820)	Finite element types applied during generation of surface element mesh .

II.6.1.13.2 IRobotMeshSurfaceFiniteElems Fields

The fields of the IRobotMeshSurfaceFiniteElems class are listed here.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	ConversionCoeff (see page 819)	Triangle-to-quadrilateral conversion coefficient .
	ForcingRatio (see page 819)	Forcing ratio for the selected type of surface finite elements.
	Type (see page 820)	Finite element types applied during generation of surface element mesh .

II.6.1.13.2.1 ConversionCoeff

C++

```
HRESULT get_ConversionCoeff(double*);
HRESULT put_ConversionCoeff(double);
```

C#

```
public double ConversionCoeff { get; set; }
```

Visual Basic

```
Public ConversionCoeff As double
```

Description

Triangle-to-quadrilateral conversion coefficient .

Version

Available since version 3.

II.6.1.13.2.2 ForcingRatio

C++

```
HRESULT get_ForcingRatio(IRobotMeshForcingRatio*);
HRESULT put_ForcingRatio(IRobotMeshForcingRatio);
```

C#

```
public IRobotMeshForcingRatio ForcingRatio { get; set; }
```

Visual Basic

```
Public ForcingRatio As IRobotMeshForcingRatio
```

Description

Forcing ratio for the selected type of surface finite elements.

Version

Available since version 3.

II.6.1.13.2.3 Type

C++

```
HRESULT get_Type(IRobotMeshSurfaceFEType*);
HRESULT put_Type(IRobotMeshSurfaceFEType);
```

C#

```
public IRobotMeshSurfaceFEType Type { get; set; }
```

Visual Basic

```
Public Type As IRobotMeshSurfaceFEType
```

Description

Finite element types applied during generation of surface element mesh .

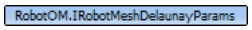
(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.

II.6.1.14 IRobotMeshDelaunayParams

Class Hierarchy



C++

```
interface IRobotMeshDelaunayParams : IDispatch;
```

C#

```
public interface IRobotMeshDelaunayParams;
```

Visual Basic

```
Public Interface IRobotMeshDelaunayParams
```

Description

Parameters of Delaunay's method.

Version

Available since version 3.

II.6.1.14.1 IRobotMeshDelaunayParams Members

The following tables list the members exposed by IRobotMeshDelaunayParams.

Public Fields

	Name	Description
	EmittersDefault (see page 821)	Default emitter.
	EmittersSmoothing (see page 822)	Smoothing of a finite element mesh.
	EmittersUser (see page 822)	User-defined emitter.
	H_max (see page 822)	Length of the penultimate wave before the process of increasing mesh elements is completed .
	H0 (see page 823)	First wave length.
	MeshDensity (see page 823)	Mesh density expressed by a value from the interval from 0 (fine mesh) to 1 (coarse mesh).
	NumberOfLevels (see page 823)	
	Q (see page 823)	Relation of lengths of two successive waves.
	RegularMesh (see page 824)	
	Type (see page 824)	Exact type of mesh generation method according to Delaunay's method.

II.6.1.14.2 IRobotMeshDelaunayParams Fields

The fields of the IRobotMeshDelaunayParams class are listed here.

Public Fields

	Name	Description
	EmittersDefault (see page 821)	Default emitter.
	EmittersSmoothing (see page 822)	Smoothing of a finite element mesh.
	EmittersUser (see page 822)	User-defined emitter.

(c) 2014 Autodesk Inc. All rights reserved

	H_max (see page 822)	Length of the penultimate wave before the process of increasing mesh elements is completed .
	H0 (see page 823)	First wave length.
	MeshDensity (see page 823)	Mesh density expressed by a value from the interval from 0 (fine mesh) to 1 (coarse mesh).
	NumberOfLevels (see page 823)	
	Q (see page 823)	Relation of lengths of two successive waves.
	RegularMesh (see page 824)	
	Type (see page 824)	Exact type of mesh generation method according to Delaunay's method.

II.6.1.14.2.1 EmittersDefault

C++

```
HRESULT get_EmittersDefault(VARIANT_BOOL*);
HRESULT put_EmittersDefault(VARIANT_BOOL);
```

C#

```
public bool EmittersDefault { get; set; }
```

Visual Basic

```
Public EmittersDefault As Boolean
```

Description

Default emitter.

Version

Available since version 3.

II.6.1.14.2.2 EmittersSmoothing

C++

```
HRESULT get_EmittersSmoothing(VARIANT_BOOL*);
HRESULT put_EmittersSmoothing(VARIANT_BOOL);
```

C#

```
public bool EmittersSmoothing { get; set; }
```

Visual Basic

```
Public EmittersSmoothing As Boolean
```

Description

Smoothing of a finite element mesh.

Version

Available since version 3.

II.6.1.14.2.3 EmittersUser

C++

```
HRESULT get_EmittersUser(VARIANT_BOOL*);
HRESULT put_EmittersUser(VARIANT_BOOL);
```

C#

```
public bool EmittersUser { get; set; }
```

Visual Basic

```
Public EmittersUser As Boolean
```

(c) 2014 Autodesk Inc. All rights reserved

Description

User-defined emitter.

Version

Available since version 3.

II.6.1.14.2.4 H_max

C++

```
HRESULT get_H_max(double*);
HRESULT put_H_max(double);
```

C#

```
public double H_max { get; set; }
```

Visual Basic

```
Public H_max As double
```

Description

Length of the penultimate wave before the process of increasing mesh elements is completed .

Version

Available since version 3.

II.6.1.14.2.5 H0

C++

```
HRESULT get_H0(double*);
HRESULT put_H0(double);
```

C#

```
public double H0 { get; set; }
```

Visual Basic

```
Public H0 As double
```

Description

First wave length.

Version

Available since version 3.

II.6.1.14.2.6 MeshDensity

C++

```
HRESULT get_MeshDensity(double*);
HRESULT put_MeshDensity(double);
```

C#

```
public double MeshDensity { get; set; }
```

Visual Basic

```
Public MeshDensity As double
```

Description

Mesh density expressed by a value from the interval from 0 (fine mesh) to 1 (coarse mesh).

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 13.

II.6.1.14.2.7 NumberOfLevels

C++

```
HRESULT get_NumberOfLevels(long*);
HRESULT put_NumberOfLevels(long);
```

C#

```
public long NumberOfLevels { get; set; }
```

Visual Basic

```
Public NumberOfLevels As long
```

Version

Available since version 13.

II.6.1.14.2.8 Q

C++

```
HRESULT get_Q(double*);
HRESULT put_Q(double);
```

C#

```
public double Q { get; set; }
```

Visual Basic

```
Public Q As double
```

Description

Relation of lengths of two successive waves.

Version

Available since version 3.

II.6.1.14.2.9 RegularMesh

C++

```
HRESULT get_RegularMesh(VARIANT_BOOL*);
HRESULT put_RegularMesh(VARIANT_BOOL);
```

C#

```
public bool RegularMesh { get; set; }
```

Visual Basic

```
Public RegularMesh As Boolean
```

Version

Available since version 9.5.

II.6.1.14.2.10 Type

C++

```
HRESULT get_Type(IRobotMeshDelaunayType*);
HRESULT put_Type(IRobotMeshDelaunayType);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRobotMeshDelaunayType Type { get; set; }
```

Visual Basic

```
Public Type As IRobotMeshDelaunayType
```

Description

Exact type of mesh generation method according to Delaunay's method.

Version

Available since version 3.

II.6.1.15 IRobotMeshSurfaceParams

Class Hierarchy

C++

```
interface IRobotMeshSurfaceParams : IDispatch;
```

C#

```
public interface IRobotMeshSurfaceParams;
```

Visual Basic

```
Public Interface IRobotMeshSurfaceParams
```

Description

Parameters of the mesh generation using planar finite elements.

Version

Available since version 3.

II.6.1.15.1 IRobotMeshSurfaceParams Members

The following tables list the members exposed by IRobotMeshSurfaceParams.

Public Fields

	Name	Description
	Coons (see page 825)	Parameters of Coons method.
	Delaunay (see page 826)	Parameters of Delaunay's method.
	FiniteElems (see page 826)	Definition of surface finite elements used during mesh generation.
	Generation (see page 826)	Parameters of mesh generation.
	Method (see page 826)	Definition of meshing method.

II.6.1.15.2 IRobotMeshSurfaceParams Fields

The fields of the IRobotMeshSurfaceParams class are listed here.

Public Fields

	Name	Description
	Coons (see page 825)	Parameters of Coons method.
	Delaunay (see page 826)	Parameters of Delaunay's method.
	FiniteElems (see page 826)	Definition of surface finite elements used during mesh generation.
	Generation (see page 826)	Parameters of mesh generation.
	Method (see page 826)	Definition of meshing method.

(c) 2014 Autodesk Inc. All rights reserved

II.6.1.15.2.1 Coons

C++

```
HRESULT get_Coons(IRobotMeshCoonsParams**);
```

C#

```
public IRobotMeshCoonsParams Coons { get; }
```

Visual Basic

```
Public ReadOnly Coons As IRobotMeshCoonsParams
```

Description

Parameters of Coons method.

Version

Available since version 3.

II.6.1.15.2.2 Delaunay

C++

```
HRESULT get_Delaunay(IRobotMeshDelaunayParams**);
```

C#

```
public IRobotMeshDelaunayParams Delaunay { get; }
```

Visual Basic

```
Public ReadOnly Delaunay As IRobotMeshDelaunayParams
```

Description

Parameters of Delaunay's method.

Version

Available since version 3.

II.6.1.15.2.3 FiniteElems

C++

```
HRESULT get_FiniteElems(IRobotMeshSurfaceFiniteElems**);
```

C#

```
public IRobotMeshSurfaceFiniteElems FiniteElems { get; }
```

Visual Basic

```
Public ReadOnly FiniteElems As IRobotMeshSurfaceFiniteElems
```

Description

Definition of surface finite elements used during mesh generation.

Version

Available since version 3.

II.6.1.15.2.4 Generation

C++

```
HRESULT get_Generation(IRobotMeshGeneration**);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRobotMeshGeneration Generation { get; }
```

Visual Basic

```
Public ReadOnly Generation As IRobotMeshGeneration
```

Description

Parameters of mesh generation.

Version

Available since version 3.

II.6.1.15.2.5 Method

C++

```
HRESULT get_Method(IRobotMeshMethod**);
```

C#

```
public IRobotMeshMethod Method { get; }
```

Visual Basic

```
Public ReadOnly Method As IRobotMeshMethod
```

Description

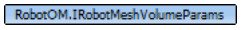
Definition of meshing method.

Version

Available since version 3.

II.6.1.16 IRobotMeshVolumeParams

Class Hierarchy



C++

```
interface IRobotMeshVolumeParams : IDispatch;
```

C#

```
public interface IRobotMeshVolumeParams;
```

Visual Basic

```
Public Interface IRobotMeshVolumeParams
```

Description

Parameters of volumetric element mesh generation. .

Version

Available since version 3.

II.6.1.16.1 IRobotMeshVolumeParams Members

The following tables list the members exposed by IRobotMeshVolumeParams.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	AdditionalSurfaceMeshing (see page 828)	Flag switching on additional mesh generation on a solid surface while generating a mesh of volumetric finite elements; if this option is switched on, it enables additional mesh generation on a solid (surface) contour which will affect the size of mesh elements within a solid .
	FiniteElemsType (see page 828)	Finite element type applied during generation of a volumetric element mesh.
	MeshDensity (see page 828)	Refinement degree for volumetric finite element mesh .

II.6.1.16.2 IRobotMeshVolumeParams Fields

The fields of the IRobotMeshVolumeParams class are listed here.

Public Fields

	Name	Description
	AdditionalSurfaceMeshing (see page 828)	Flag switching on additional mesh generation on a solid surface while generating a mesh of volumetric finite elements; if this option is switched on, it enables additional mesh generation on a solid (surface) contour which will affect the size of mesh elements within a solid .
	FiniteElemsType (see page 828)	Finite element type applied during generation of a volumetric element mesh.
	MeshDensity (see page 828)	Refinement degree for volumetric finite element mesh .

II.6.1.16.2.1 AdditionalSurfaceMeshing

C++

```
HRESULT get_AdditionalSurfaceMeshing(VARIANT_BOOL*);
HRESULT put_AdditionalSurfaceMeshing(VARIANT_BOOL);
```

C#

```
public bool AdditionalSurfaceMeshing { get; set; }
```

Visual Basic

```
Public AdditionalSurfaceMeshing As Boolean
```

Description

Flag switching on additional mesh generation on a solid surface while generating a mesh of volumetric finite elements; if this option is switched on, it enables additional mesh generation on a solid (surface) contour which will affect the size of mesh elements within a solid .

Version

Available since version 3.

II.6.1.16.2.2 FiniteElemsType

C++

```
HRESULT get_FiniteElemsType(IRobotMeshSurfaceFEType*);
HRESULT put_FiniteElemsType(IRobotMeshSurfaceFEType);
```

C#

```
public IRobotMeshSurfaceFEType FiniteElemsType { get; set; }
```

Visual Basic

```
Public FiniteElemsType As IRobotMeshSurfaceFEType
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Finite element type applied during generation of a volumetric element mesh.

Version

Available since version 3.

II.6.1.16.2.3 MeshDensity

C++

```
HRESULT get_MeshDensity(double*);
HRESULT put_MeshDensity(double);
```

C#

```
public double MeshDensity { get; set; }
```

Visual Basic

```
Public MeshDensity As double
```

Description

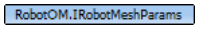
Refinement degree for volumetric finite element mesh .

Version

Available since version 3.

II.6.1.17 IRobotMeshParams

Class Hierarchy



C++

```
interface IRobotMeshParams : IDispatch;
```

C#

```
public interface IRobotMeshParams;
```

Visual Basic

```
Public Interface IRobotMeshParams
```

Description

Parameters of the finite element mesh generation.

Version

Available since version 3.

II.6.1.17.1 IRobotMeshParams Members

The following tables list the members exposed by IRobotMeshParams.

Public Fields

	Name	Description
	Flag (see page 829)	Flag of access to parameters of surface and volumetric mesh model.
	MeshType (see page 830)	Types of generated mesh for surface or volumetric finite elements.
	SurfaceParams (see page 830)	Set of attributes for generation of a surface element mesh.
	VolumeParams (see page 830)	Set of attributes for generation of a volumetric element mesh.

(c) 2014 Autodesk Inc. All rights reserved

II.6.1.17.2 IRobotMeshParams Fields

The fields of the IRobotMeshParams class are listed here.

Public Fields

	Name	Description
	Flag (see page 829)	Flag of access to parameters of surface and volumetric mesh model.
	MeshType (see page 830)	Types of generated mesh for surface or volumetric finite elements.
	SurfaceParams (see page 830)	Set of attributes for generation of a surface element mesh.
	VolumeParams (see page 830)	Set of attributes for generation of a volumetric element mesh.

II.6.1.17.2.1 Flag

C++

```
HRESULT get_Flag(IRobotMeshAccessType*);
```

C#

```
public IRobotMeshAccessType Flag { get; }
```

Visual Basic

```
Public ReadOnly Flag As IRobotMeshAccessType
```

Description

Flag of access to parameters of surface and volumetric mesh model.

Version

Available since version 3.

II.6.1.17.2.2 MeshType

C++

```
HRESULT get_MeshType(IRobotMeshType*);
HRESULT put_MeshType(IRobotMeshType);
```

C#

```
public IRobotMeshType MeshType { get; set; }
```

Visual Basic

```
Public MeshType As IRobotMeshType
```

Description

Types of generated mesh for surface or volumetric finite elements.

Version

Available since version 3.

II.6.1.17.2.3 SurfaceParams

C++

```
HRESULT get_SurfaceParams(IRobotMeshSurfaceParams**);
```

C#

```
public IRobotMeshSurfaceParams SurfaceParams { get; }
```

Visual Basic

```
Public ReadOnly SurfaceParams As IRobotMeshSurfaceParams
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Set of attributes for generation of a surface element mesh.

Version

Available since version 3.

II.6.1.17.2.4 VolumeParams

C++

```
HRESULT get_VolumeParams(IRobotMeshVolumeParams**);
```

C#

```
public IRobotMeshVolumeParams VolumeParams { get; }
```

Visual Basic

```
Public ReadOnly VolumeParams As IRobotMeshVolumeParams
```

Description

Set of attributes for generation of a volumetric element mesh.

Version

Available since version 3.

II.6.1.18 IRobotMeshAccessType

C++

```
enum IRobotMeshAccessType;
```

C#

```
public enum IRobotMeshAccessType;
```

Visual Basic

```
Public Enum IRobotMeshAccessType
```

Members

Members	Description
I_MAT_SURFACE_AND_VOLUMETRIC = 1	Flag indicating availability of the interface of surface mesh parameters and the interface of volumetric mesh parameters. Available since version 3.
I_MAT_SURFACE_ONLY = 2	Flag indicating availability only of the interface of surface mesh parameters . Available since version 3.
I_MAT_VOLUMETRIC_ONLY = 3	Flag indicating availability only of the interface of volumetric mesh parameters. Available since version 3.

Description

Availability of interfaces for mesh generation.

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

II.6.1.19 IRobotMeshRefinementType

C++

```
enum IRobotMeshRefinementType;
```

C#

```
public enum IRobotMeshRefinementType;
```

Visual Basic

```
Public Enum IRobotMeshRefinementType
```

Members

Members	Description
I_MRT_DOUBLE = 1	Each finite element edge is divided into two parts. Available since version 3.
I_MRT_TRIPLE = 2	Each finite element edge is divided into three parts. Available since version 3.
I_MRT_SIMPLE = 3	Finite element edges will not be divided. Available since version 3.

Description

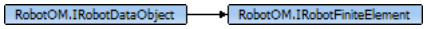
Methods of finite element mesh refinement.

Version

Available since version 3.

II.6.2 IRobotFiniteElement

Class Hierarchy



C++

```
interface IRobotFiniteElement : IRobotDataObject;
```

C#

```
public interface IRobotFiniteElement : IRobotDataObject;
```

Visual Basic

```
Public Interface IRobotFiniteElement
```

Description

Interface describing planar finite element. .

Version

Available since version 2.5.

II.6.2.1 IRobotFiniteElement Members

The following tables list the members exposed by IRobotFiniteElement.

Public Fields

	Name	Description
	FeType (see page 833)	Finite element type.
	Nodes (see page 833)	Nodes defining a finite element.

(c) 2014 Autodesk Inc. All rights reserved

	Number (see page 31)	Object number is assigned by the user; it is qunique among all the components of the same type. .
	ObjectNumber (see page 834)	Number (see page 31) of the object including the finite element.
	ObjectPartIdx (see page 834)	Index of the object component including the element.
	UniqueId (see page 834)	Unique element identifier .

Public Methods

	Name	Description
	CalcArea (see page 835)	Function calculates a finite element area.
	GetLabel (see page 31)	The function returns a label of the indicated type applied to the object. If the object does not have any label of the type, running the function leads to a critical error. .
	GetLabelName (see page 32)	The function returns the name of a label of the indicated type, applied to the object. If the object does not have any label of the type, running the function leads to a critical error. .
	GetLabels (see page 32)	The function returns a collection containing all labels (of different types) defined for a given object. .
	HasLabel (see page 32)	The function returns True (non-zero value) if a label of the type has already been defined for the object. .
	RemoveLabel (see page 32)	The function deletes a label of the indicated type. .
	SetLabel (see page 33)	The function applies the defined label to an object. At a given moment, an object may have only one label of a type applied (e.g. there may not be two supports defined for one node). If an object has a formerly applied label of the same type, it is replaced by the new one. .

II.6.2.2 IRobotFiniteElement Fields

The fields of the IRobotFiniteElement class are listed here.

Public Fields

	Name	Description
	FeType (see page 833)	Finite element type.
	Nodes (see page 833)	Nodes defining a finite element.
	ObjectNumber (see page 834)	Number (see page 31) of the object including the finite element.
	ObjectPartIdx (see page 834)	Index of the object component including the element.
	UniqueId (see page 834)	Unique element identifier .

II.6.2.2.1 FeType

C++

```
HRESULT get_FeType(IRobotFiniteElementType*);
HRESULT put_FeType(IRobotFiniteElementType);
```

C#

```
public IRobotFiniteElementType FeType { get; set; }
```

Visual Basic

```
Public FeType As IRobotFiniteElementType
```

Description

Finite element type.

Version

Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

II.6.2.2.2 Nodes

C++

```
HRESULT get_Nodes(IRobotFiniteElementNodes**);
```

C#

```
public IRobotFiniteElementNodes Nodes { get; }
```

Visual Basic

```
Public ReadOnly Nodes As IRobotFiniteElementNodes
```

Description

Nodes defining a finite element.

Version

Available since version 2.5.

II.6.2.2.3 ObjectNumber

C++

```
HRESULT get_ObjectNumber(long*);
```

C#

```
public long ObjectNumber { get; }
```

Visual Basic

```
Public ReadOnly ObjectNumber As long
```

Description

Number (see page 31) of the object including the finite element.

Version

Available since version 2.5.

II.6.2.2.4 ObjectPartIdx

C++

```
HRESULT get_ObjectPartIdx(long*);
```

C#

```
public long ObjectPartIdx { get; }
```

Visual Basic

```
Public ReadOnly ObjectPartIdx As long
```

Description

Index of the object component including the element.

Version

Available since version 2.5.

II.6.2.2.5 UniqueId

C++

```
HRESULT get_UniqueId(long*);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public long UniqueId { get; }
```

Visual Basic

```
Public ReadOnly UniqueId As long
```

Description

Unique element identifier .

Version

Available since version 2.5.

II.6.2.3 IRobotFiniteElement Methods

The methods of the IRobotFiniteElement class are listed here.

Public Methods

	Name	Description
	CalcArea (see page 835)	Function calculates a finite element area.

II.6.2.3.1 CalcArea

C++

```
HRESULT CalcArea(double* ret);
```

C#

```
public double CalcArea();
```

Visual Basic

```
Public Function CalcArea() As double
```

Description

Function calculates a finite element area.

Version

Available since version 7.5.

II.6.3 IRobotFiniteElementType

C++

```
enum IRobotFiniteElementType;
```

C#

```
public enum IRobotFiniteElementType;
```

Visual Basic

```
Public Enum IRobotFiniteElementType
```

Members

Members	Description
I_FET_T3 = 3	Three-node element. Available since version 2.5.
I_FET_Q4 = 4	Four-node element. Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

I_FET_T6 = 1	Six-node element. Available since version 2.5.
I_FET_Q8 = 2	Eight-node element. Available since version 2.5.
I_FET_VOL_B8 = 5	Available since version 7.5.
I_FET_VOL_T4 = 6	Available since version 7.5.
I_FET_VOL_W6 = 7	Available since version 7.5.
I_FET_VOL_B20 = 8	Available since version 7.5.
I_FET_VOL_T10 = 9	Available since version 7.5.
I_FET_VOL_W15 = 10	Available since version 7.5.

Description

Available types of finite elements.

Version

Available since version 2.5.

II.6.4 IRobotFiniteElementNodes

Class Hierarchy

RobotOM.IRobotFiniteElementNodes

C++

```
interface IRobotFiniteElementNodes : IDispatch;
```

C#

```
public interface IRobotFiniteElementNodes;
```

Visual Basic

```
Public Interface IRobotFiniteElementNodes
```

Description

Collection of nodes defining finite element.

Version

Available since version 2.5.

II.6.4.1 IRobotFiniteElementNodes Members

The following tables list the members exposed by IRobotFiniteElementNodes.

Public Fields

	Name	Description
	Count (see page 836)	Number of nodes in the collection (simultaneously, it is the index of the last node).

Public Methods

	Name	Description
	Get (see page 837)	Function returns number of the node with the specified index in the collection.
	GetAll (see page 837)	Function returns the table with numbers of nodes defining the finite element. .
	Set (see page 838)	Function inserts the specified node onto the indicated index in the collection of nodes defining finite element. .

(c) 2014 Autodesk Inc. All rights reserved

	SetAll (see page 838)	Function defines all nodes of the finite element.

II.6.4.2 IRobotFiniteElementNodes Fields

The fields of the IRobotFiniteElementNodes class are listed here.

Public Fields

	Name	Description
	Count (see page 836)	Number of nodes in the collection (simultaneously, it is the index of the last node).

II.6.4.2.1 Count

C++

```
HRESULT get_Count(long*);
```

C#

```
public long Count { get; }
```

Visual Basic

```
Public ReadOnly Count As long
```

Description

Number of nodes in the collection (simultaneously, it is the index of the last node).

Version

Available since version 2.5.

II.6.4.3 IRobotFiniteElementNodes Methods

The methods of the IRobotFiniteElementNodes class are listed here.

Public Methods

	Name	Description
	Get (see page 837)	Function returns number of the node with the specified index in the collection.
	GetAll (see page 837)	Function returns the table with numbers of nodes defining the finite element. .
	Set (see page 838)	Function inserts the specified node onto the indicated index in the collection of nodes defining finite element. .
	SetAll (see page 838)	Function defines all nodes of the finite element.

II.6.4.3.1 Get

C++

```
HRESULT Get(long _node_idx, long* ret);
```

C#

```
public long Get(long _node_idx);
```

Visual Basic

```
Public Function Get(_node_idx As long) As long
```

Description

Function returns number of the node with the specified index in the collection.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 2.5.

II.6.4.3.2 GetAll

C++

```
HRESULT GetAll(IRobotNumbersArray** ret);
```

C#

```
public IRobotNumbersArray GetAll();
```

Visual Basic

```
Public Function GetAll() As IRobotNumbersArray
```

Description

Function returns the table with numbers of nodes defining the finite element. .

Version

Available since version 2.5.

II.6.4.3.3 Set

C++

```
HRESULT Set(long _node_idx, long _node_num, VARIANT_BOOL* ret);
```

C#

```
public bool Set(long _node_idx, long _node_num);
```

Visual Basic

```
Public Function Set(_node_idx As long, _node_num As long) As Boolean
```

Description

Function inserts the specified node onto the indicated index in the collection of nodes defining finite element. .

Version

Available since version 2.5.

II.6.4.3.4 SetAll

C++

```
HRESULT SetAll(IRobotNumbersArray* _nodes_array);
```

C#

```
public void SetAll(IRobotNumbersArray _nodes_array);
```

Visual Basic

```
Public Sub SetAll(ByRef _nodes_array As IRobotNumbersArray)
```

Description

Function defines all nodes of the finite element.

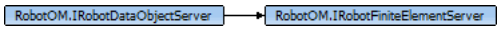
Version

Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

II.6.5 IRobotFiniteElementServer

Class Hierarchy



C++

```
interface IRobotFiniteElementServer : IRobotDataObjectServer;
```

C#

```
public interface IRobotFiniteElementServer : IRobotDataObjectServer;
```

Visual Basic

```
Public Interface IRobotFiniteElementServer
```

Description

Server of finite elements. .

Version

Available since version 2.5.

II.6.5.1 IRobotFiniteElementServer Members

The following tables list the members exposed by IRobotFiniteElementServer.

Public Fields

	Name	Description
	FreeNumber (see page 839)	First avaialble user-defined number that can be assigned to a finite element.

Public Methods

	Name	Description
	CalcArea (see page 840)	Function calculates an area of the finite element of the given number.
	Create (see page 840)	Function creates a new finite element defined on the indicated nodes. .
	Delete (see page 34)	The function deletes the object of the indicated number. .
	DeleteMany (see page 34)	The function deletes all objects that meet the criteria of the indicated selection. .
	Exist (see page 35)	The function returns True (non-zero value) if the object of the indicated name already exists. .
	Get (see page 35)	The function returns an object with the indicated user-defined number. The type of the returned object agrees with the type of objects managed by the server. If the object of the indicated number does not exist, running the function leads to critical error. .
	GetAll (see page 35)	The function returns a collection containing all objects managed by the server. .
	GetMany (see page 36)	The function returns a collection of objects that meet the criteria of the indicated selection. .
	MeshConcentrate (see page 841)	Function carries out mesh refinement for the indicated selection of finite elements. .
	MeshConsolidate (see page 841)	Function performs mesh consolidation for the indicated finite element list.
	RemoveLabel (see page 36)	The function removes the labels of the indicated type from all objects that meet the criteria of the indicated selection. .
	SetLabel (see page 36)	The function applies a label (identified by the type and name) to all objects that meet the criteria of the indicated selection. .

(c) 2014 Autodesk Inc. All rights reserved

	Update (see page 841)	Function reconstructs relations between finite elements and structure nodes. It should be activated, if the Create (see page 840) function with delayed model update, i.e., with parameter_ update_model set as False, has been used earlier. .

II.6.5.2 IRobotFiniteElementServer Fields

The fields of the IRobotFiniteElementServer class are listed here.

Public Fields

	Name	Description
	FreeNumber (see page 839)	First avaialble user-defined number that can be assigned to a finite element.

II.6.5.2.1 FreeNumber

C++

```
HRESULT get_FreeNumber(long*);
```

C#

```
public long FreeNumber { get; }
```

Visual Basic

```
Public ReadOnly FreeNumber As long
```

Description

First avaialble user-defined number that can be assigned to a finite element.

Version

Available since version 2.5.

II.6.5.3 IRobotFiniteElementServer Methods

The methods of the IRobotFiniteElementServer class are listed here.

Public Methods

	Name	Description
	CalcArea (see page 840)	Function calculates an area of the finite element of the given number.
	Create (see page 840)	Function creates a new finite element defined on the indicated nodes. .
	MeshConcentrate (see page 841)	Function carries out mesh refinement for the indicated selection of finite elements. .
	MeshConsolidate (see page 841)	Function performs mesh consolidation for the indicated finite element list.
	Update (see page 841)	Function reconstructs relations between finite elements and structure nodes. It should be activated, if the Create (see page 840) function with delayed model update, i.e., with parameter_ update_model set as False, has been used earlier. .

II.6.5.3.1 CalcArea

C++

```
HRESULT CalcArea(long _ele_num, double* ret);
```

C#

```
public double CalcArea(long _ele_num);
```

Visual Basic

```
Public Function CalcArea(_ele_num As long) As double
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Function calculates an area of the finite element of the given number.

Version

Available since version 7.5.

II.6.5.3.2 Create

C++

```
HRESULT Create(long _number, IRobotNumbersArray* _nodes, VARIANT_BOOL _update_model = True);
```

C#

```
public void Create(long _number, IRobotNumbersArray _nodes, bool _update_model = True);
```

Visual Basic

```
Public Sub Create(_number As long, ByRef _nodes As IRobotNumbersArray, Optional
_update_model As Boolean = True)
```

Description

Function creates a new finite element defined on the indicated nodes. .

Version

Available since version 3.

II.6.5.3.3 MeshConcentrate

C++

```
HRESULT MeshConcentrate(IRobotMeshRefinementType _type, IRobotSelection* _sel, VARIANT_BOOL
_auto_freeze = true);
```

C#

```
public void MeshConcentrate(IRobotMeshRefinementType _type, IRobotSelection _sel, bool
_auto_freeze = true);
```

Visual Basic

```
Public Sub MeshConcentrate(_type As IRobotMeshRefinementType, ByRef _sel As
IRobotSelection, Optional _auto_freeze As Boolean = true)
```

Description

Function carries out mesh refinement for the indicated selection of finite elements. .

Version

Available since version 3.

II.6.5.3.4 MeshConsolidate

C++

```
HRESULT MeshConsolidate(double _coeff, IRobotSelection* _sel, VARIANT_BOOL _auto_freeze =
true);
```

C#

```
public void MeshConsolidate(double _coeff, IRobotSelection _sel, bool _auto_freeze = true);
```

Visual Basic

```
Public Sub MeshConsolidate(_coeff As double, ByRef _sel As IRobotSelection, Optional
```

(c) 2014 Autodesk Inc. All rights reserved

```
_auto_freeze As Boolean = true)
```

Description

Function performs mesh consolidation for the indicated finite element list.

Version

Available since version 3.

II.6.5.3.5 Update

C++

```
HRESULT Update();
```

C#

```
public void Update();
```

Visual Basic

```
Public Sub Update()
```

Description

Function reconstructs relations between finite elements and structure nodes. It should be activated, if the Create (see page 840) function with delayed model update, i.e., with parameter_ update_model set as False, has been used earlier. .

Version

Available since version 3.

II.6.6 IRobotFiniteElementData

Class Hierarchy

RobotOM.IRobotFiniteElementData

C++

```
interface IRobotFiniteElementData : IDispatch;
```

C#

```
public interface IRobotFiniteElementData;
```

Visual Basic

```
Public Interface IRobotFiniteElementData
```

Description

Interface gives access to finite element data.

Version

Available since version 9.

II.6.6.1 IRobotFiniteElementData Members

The following tables list the members exposed by IRobotFiniteElementData.

Public Fields

	Name	Description
	NodeCount (see page 843)	Number (see page 843) of nodes defining the element.
	Number (see page 843)	Finite element number assigned by the user.
	Object (see page 843)	Number (see page 843) of an object that the finite element belongs to.

(c) 2014 Autodesk Inc. All rights reserved

Public Methods

	Name	Description
	GetNode (see page 844)	Function returns a number of the node with the specified index and the node coordinates.

II.6.6.2 IRobotFiniteElementData Fields

The fields of the IRobotFiniteElementData class are listed here.

Public Fields

	Name	Description
	NodeCount (see page 843)	Number (see page 843) of nodes defining the element.
	Number (see page 843)	Finite element number assigned by the user.
	Object (see page 843)	Number (see page 843) of an object that the finite element belongs to.

II.6.6.2.1 NodeCount

C++

```
HRESULT get_NodeCount(long*);
```

C#

```
public long NodeCount { get; }
```

Visual Basic

```
Public ReadOnly NodeCount As long
```

Description

Number (see page 843) of nodes defining the element.

Version

Available since version 9.

II.6.6.2.2 Number

C++

```
HRESULT get_Number(long*);
```

C#

```
public long Number { get; }
```

Visual Basic

```
Public ReadOnly Number As long
```

Description

Finite element number assigned by the user.

Version

Available since version 9.

II.6.6.2.3 Object

C++

```
HRESULT get_Object(long*);
```

C#

```
public long Object { get; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public ReadOnly Object As long
```

Description

Number (see page 843) of an object that the finite element belongs to.

Version

Available since version 9.

II.6.6.3 IRobotFiniteElementData Methods

The methods of the IRobotFiniteElementData class are listed here.

Public Methods

	Name	Description
	GetNode (see page 844)	Function returns a number of the node with the specified index and the node coordinates.

II.6.6.3.1 GetNode

C++

```
HRESULT GetNode(long _node_idx, double* _ret_x, double* _ret_y, double* _ret_z, long* ret);
```

C#

```
public long GetNode(long _node_idx, double* _ret_x, double* _ret_y, double* _ret_z);
```

Visual Basic

```
Public Function GetNode(_node_idx As long, ByRef _ret_x As double*, ByRef _ret_y As
double*, ByRef _ret_z As double*) As long
```

Description

Function returns a number of the node with the specified index and the node coordinates.

Version

Available since version 9.

II.6.7 IRobotFiniteElementDataSet

Class Hierarchy

RobotOM.IRobotFiniteElementDataSet

C++

```
interface IRobotFiniteElementDataSet : IDispatch;
```

C#

```
public interface IRobotFiniteElementDataSet;
```

Visual Basic

```
Public Interface IRobotFiniteElementDataSet
```

Description

Interface gives access to the finite element set.

Version

Available since version 9.

(c) 2014 Autodesk Inc. All rights reserved

II.6.7.1 IRobotFiniteElementDataSet Members

The following tables list the members exposed by IRobotFiniteElementDataSet.

Public Fields

	Name	Description
	ElemCount (see page 845)	Number of elements belonging to the set.
	ObjectCount (see page 845)	Number of objects whose elements are included in the set.

Public Methods

	Name	Description
	GetElem (see page 846)	Function returns a number of the element with the specified index and gives access to its data.
	GetElemCountForObject (see page 846)	Function returns a number of elements for the object with the specified index.
	GetElemForObject (see page 846)	Function returns data for the determined element of a given object.
	GetObjectNumber (see page 847)	Function returns a number of the object with the specified index.

II.6.7.2 IRobotFiniteElementDataSet Fields

The fields of the IRobotFiniteElementDataSet class are listed here.

Public Fields

	Name	Description
	ElemCount (see page 845)	Number of elements belonging to the set.
	ObjectCount (see page 845)	Number of objects whose elements are included in the set.

II.6.7.2.1 ElemCount

C++

```
HRESULT get_ElemCount(long*);
```

C#

```
public long ElemCount { get; }
```

Visual Basic

```
Public ReadOnly ElemCount As long
```

Description

Number of elements belonging to the set.

Version

Available since version 9.

II.6.7.2.2 ObjectCount

C++

```
HRESULT get_ObjectCount(long*);
```

C#

```
public long ObjectCount { get; }
```

Visual Basic

```
Public ReadOnly ObjectCount As long
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Number of objects whose elements are included in the set.

Version

Available since version 9.

II.6.7.3 IRobotFiniteElementDataSet Methods

The methods of the IRobotFiniteElementDataSet class are listed here.

Public Methods

	Name	Description
	GetElem (see page 846)	Function returns a number of the element with the specified index and gives access to its data.
	GetElemCountForObject (see page 846)	Function returns a number of elements for the object with the specified index.
	GetElemForObject (see page 846)	Function returns data for the determined element of a given object.
	GetObjectNumber (see page 847)	Function returns a number of the object with the specified index.

II.6.7.3.1 GetElem

C++

```
HRESULT GetElem(long _elem_idx, IRobotFiniteElementData* _elem_data, long* ret);
```

C#

```
public long GetElem(long _elem_idx, IRobotFiniteElementData _elem_data);
```

Visual Basic

```
Public Function GetElem(_elem_idx As long, ByRef _elem_data As IRobotFiniteElementData) As
long
```

Description

Function returns a number of the element with the specified index and gives access to its data.

Version

Available since version 9.

II.6.7.3.2 GetElemCountForObject

C++

```
HRESULT GetElemCountForObject(long _obj_idx, long* ret);
```

C#

```
public long GetElemCountForObject(long _obj_idx);
```

Visual Basic

```
Public Function GetElemCountForObject(_obj_idx As long) As long
```

Description

Function returns a number of elements for the object with the specified index.

Version

Available since version 9.

(c) 2014 Autodesk Inc. All rights reserved

II.6.7.3.3 GetElemForObject

C++

```
HRESULT GetElemForObject(long _obj_idx, long _elem_idx, IRobotFiniteElementData* _ret_data,
long* ret);
```

C#

```
public long GetElemForObject(long _obj_idx, long _elem_idx, IRobotFiniteElementData
_ret_data);
```

Visual Basic

```
Public Function GetElemForObject(_obj_idx As long, _elem_idx As long, ByRef _ret_data As
IRobotFiniteElementData) As long
```

Description

Function returns data for the determined element of a given object.

Version

Available since version 9.

II.6.7.3.4 GetObjectNumber

C++

```
HRESULT GetObjectNumber(long _obj_idx, long* ret);
```

C#

```
public long GetObjectNumber(long _obj_idx);
```

Visual Basic

```
Public Function GetObjectNumber(_obj_idx As long) As long
```

Description

Function returns a number of the object with the specified index.

Version

Available since version 9.

II.7 Grouping of objects

Available since version 7.5.

Interfaces

	Name	Description
	IRobotGroupObjectServer (see page 847)	Servers allows operations of grouping / ungrouping objects.

II.7.1 IRobotGroupObjectServer

Class Hierarchy

RobotOM.IRobotGroupObjectServer

C++

```
interface IRobotGroupObjectServer : IDispatch;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public interface IRobotGroupObjectServer;
```

Visual Basic

```
Public Interface IRobotGroupObjectServer
```

Description

Servers allows operations of grouping / ungrouping objects.

Version

Available since version 7.5.

II.7.1.1 IRobotGroupObjectServer Members

The following tables list the members exposed by IRobotGroupObjectServer.

Public Methods

	Name	Description
	Explode (see page 848)	Function ungroups a group of structure elements of the specified number.
	FindFirst (see page 849)	Function returns the number of the first object grouping structure elements. If no objects of this type have been defined, function returns zero value.
	FindNext (see page 849)	Function returns the number of the next object grouping structure elements. If no more grouping objects have been defined, function returns zero value.
	GetContents (see page 849)	Function returns selection lists of structure elements included in the specified "grouped" object.
	GroupGiven (see page 850)	Function groups (combines into one object) structure elements described by selection lists. Function returns the number of a created object.
	GroupSelected (see page 850)	Function groups (combines into one object) structure elements belonging to the current selection. Function returns the number of a created object.

II.7.1.2 IRobotGroupObjectServer Methods

The methods of the IRobotGroupObjectServer class are listed here.

Public Methods

	Name	Description
	Explode (see page 848)	Function ungroups a group of structure elements of the specified number.
	FindFirst (see page 849)	Function returns the number of the first object grouping structure elements. If no objects of this type have been defined, function returns zero value.
	FindNext (see page 849)	Function returns the number of the next object grouping structure elements. If no more grouping objects have been defined, function returns zero value.
	GetContents (see page 849)	Function returns selection lists of structure elements included in the specified "grouped" object.
	GroupGiven (see page 850)	Function groups (combines into one object) structure elements described by selection lists. Function returns the number of a created object.
	GroupSelected (see page 850)	Function groups (combines into one object) structure elements belonging to the current selection. Function returns the number of a created object.

II.7.1.2.1 Explode

C++

```
HRESULT Explode(long _group_obj);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public void Explode(long _group_obj);
```

Visual Basic

```
Public Sub Explode(_group_obj As long)
```

Description

Function ungroups a group of structure elements of the specified number.

Version

Available since version 7.5.

II.7.1.2.2 FindFirst

C++

```
HRESULT FindFirst(long* ret);
```

C#

```
public long FindFirst();
```

Visual Basic

```
Public Function FindFirst() As long
```

Description

Function returns the number of the first object grouping structure elements. If no objects of this type have been defined, function returns zero value.

Version

Available since version 7.5.

II.7.1.2.3 FindNext

C++

```
HRESULT FindNext(long _prev_group_obj, long* ret);
```

C#

```
public long FindNext(long _prev_group_obj);
```

Visual Basic

```
Public Function FindNext(_prev_group_obj As long) As long
```

Description

Function returns the number of the next object grouping structure elements. If no more grouping objects have been defined, function returns zero value.

Version

Available since version 7.5.

II.7.1.2.4 GetContents

C++

```
HRESULT GetContents(long _group_obj, BSTR _object_sel, BSTR _fe_sel, BSTR _node_sel);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public void GetContents(long _group_obj, String _object_sel, String _fe_sel, String
_node_sel);
```

Visual Basic

```
Public Sub GetContents(_group_obj As long, ByRef _object_sel As String, ByRef _fe_sel As
String, ByRef _node_sel As String)
```

Description

Function returns selection lists of structure elements included in the specified "grouped" object.

Version

Available since version 7.5.

II.7.1.2.5 GroupGiven

C++

```
HRESULT GroupGiven(BSTR _object_sel, BSTR _fe_sel, BSTR _node_sel, VARIANT_BOOL
_auto_include = True, long* ret);
```

C#

```
public long GroupGiven(String _object_sel, String _fe_sel, String _node_sel, bool
_auto_include = True);
```

Visual Basic

```
Public Function GroupGiven(_object_sel As String, _fe_sel As String, _node_sel As String,
Optional _auto_include As Boolean = True) As long
```

Description

Function groups (combines into one object) structure elements described by selection lists. Function returns the number of a created object.

Version

Available since version 7.5.

II.7.1.2.6 GroupSelected

C++

```
HRESULT GroupSelected(long* ret);
```

C#

```
public long GroupSelected();
```

Visual Basic

```
Public Function GroupSelected() As long
```

Description

Function groups (combines into one object) structure elements belonging to the current selection. Function returns the number of a created object.

Version

Available since version 7.5.

(c) 2014 Autodesk Inc. All rights reserved

II.8 IRobotObjectStructuralType

C++

```
enum IRobotObjectStructuralType;
```

C#

```
public enum IRobotObjectStructuralType;
```

Visual Basic

```
Public Enum IRobotObjectStructuralType
```

Members

Members	Description
I_OST_UNDEFINED = 0	Available since version 9.7.
I_OST_BEAM = 1	Available since version 9.7.
I_OST_COLUMN = 2	Available since version 9.7.
I_OST_SLAB = 4	Available since version 9.7.
I_OST_WALL = 5	Available since version 9.7.

Description

Type of object as a structure component.

Version

Available since version 9.7.

III Results

Enumerations

	Name	Description
	IRobotUniversalResultType (see page 1019)	Types of results made available by the RobotUniversalResultAccess object.
	IRobotResultParamType (see page 1025)	Set of parameters describing calculation results.
	IRobotResultQueryReturnType (see page 1033)	
	IRobotResultStatusType (see page 1033)	Type of data describing status of results.

Interfaces

	Name	Description
	IRobotResultServer (see page 1002)	The server makes available all types of results obtained by performing structure calculations. .
	IRobotUniversalResultAccess (see page 1007)	Universal access interface for calculation results. In order to obtain the relevant result the object should be first appropriately parametrized by determining what result type, for which load case and for which structure element is to be obtained, and then ResultValue (see page 1015) should be taken. .
	IRobotExtremeResultServer (see page 1017)	Server of extreme results.
	IRobotStructureValues (see page 1019)	Physical properties for the structure model.

(c) 2014 Autodesk Inc. All rights reserved

	IRobotResultRow (see page 1023)	Row of results calculated for determined values of a parameter set.
	IRobotResultRowSet (see page 1027)	Set of rows with calculation results.
	IRobotResultQueryParams (see page 1029)	Definition of a query to the result server.
	IRobotCalculationResume (see page 1034)	.

III.1 Calculation results for nodes

Interfaces

	Name	Description
	IRobotNodeResultServer (see page 852)	All types or results for nodes are made accessible by the appropriate server of the RobotNodeResultServer type. .
	IRobotReactionServer (see page 854)	The server providing access to values of reactions for nodes. .
	IRobotNodeDisplacementServer (see page 864)	The server provides access to all values of nodal displacements. .
	IRobotNodeBucklingServer (see page 866)	The server provides results of buckling analysis for nodes.
	IRobotPseudostaticForceServer (see page 867)	Server providing access to pseudostatic forces.

III.1.1 IRobotNodeResultServer

Class Hierarchy

RobotOM.IRobotNodeResultServer

C++

```
interface IRobotNodeResultServer : IDispatch;
```

C#

```
public interface IRobotNodeResultServer;
```

Visual Basic

```
Public Interface IRobotNodeResultServer
```

Description

All types or results for nodes are made accessible by the appropriate server of the RobotNodeResultServer type. .

III.1.1.1 IRobotNodeResultServer Members

The following tables list the members exposed by IRobotNodeResultServer.

Public Fields

	Name	Description
	Buckling (see page 853)	The server of results of buckling analysis for nodes.
	Displacements (see page 853)	Server provides access to nodal displacements .
	PseudostaticForces (see page 853)	Server providing access to values of pseudostatic forces Available since version 1.7.
	Reactions (see page 853)	Server provides access to reaction values in nodes. .

(c) 2014 Autodesk Inc. All rights reserved

III.1.1.2 IRobotNodeResultServer Fields

The fields of the IRobotNodeResultServer class are listed here.

Public Fields

	Name	Description
	Buckling (see page 853)	The server of results of buckling analysis for nodes.
	Displacements (see page 853)	Server provides access to nodal displacements .
	PseudostaticForces (see page 853)	Server providing access to values of pseudostatic forces Available since version 1.7.
	Reactions (see page 853)	Server provides access to reaction values in nodes. .

III.1.1.2.1 Buckling

C++

```
HRESULT get_Buckling(IRobotNodeBucklingServer**);
```

C#

```
public IRobotNodeBucklingServer Buckling { get; }
```

Visual Basic

```
Public ReadOnly Buckling As IRobotNodeBucklingServer
```

Description

The server of results of buckling analysis for nodes.

III.1.1.2.2 Displacements

C++

```
HRESULT get_Displacements(IRobotNodeDisplacementServer**);
```

C#

```
public IRobotNodeDisplacementServer Displacements { get; }
```

Visual Basic

```
Public ReadOnly Displacements As IRobotNodeDisplacementServer
```

Description

Server provides access to nodal displacements .

III.1.1.2.3 PseudostaticForces

C++

```
HRESULT get_PseudostaticForces(IRobotPseudostaticForceServer**);
```

C#

```
public IRobotPseudostaticForceServer PseudostaticForces { get; }
```

Visual Basic

```
Public ReadOnly PseudostaticForces As IRobotPseudostaticForceServer
```

Description

Server providing access to values of pseudostatic forces Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

III.1.1.2.4 Reactions

C++

```
HRESULT get_Reactions(IRobotReactionServer**);
```

C#

```
public IRobotReactionServer Reactions { get; }
```

Visual Basic

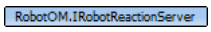
```
Public ReadOnly Reactions As IRobotReactionServer
```

Description

Server provides access to reaction values in nodes. .

III.1.2 IRobotReactionServer

Class Hierarchy



C++

```
interface IRobotReactionServer : IDispatch;
```

C#

```
public interface IRobotReactionServer;
```

Visual Basic

```
Public Interface IRobotReactionServer
```

Description

The server providing access to values of reactions for nodes. .

III.1.2.1 IRobotReactionServer Members

The following tables list the members exposed by IRobotReactionServer.

Public Methods

	Name	Description
	DDC (see page 856)	The function returns a structure containing Load Report reaction values for the indicated node and load case .
	DDCEx (see page 856)	Available since version 1.7.
	DDCLocal (see page 857)	The function returns a structure containing Load Report reaction values for the indicated node and load case .
	DDCLocalEx (see page 857)	Available since version 1.7.
	DDCSum (see page 857)	The function returns a structure containing values of reaction sums of the Load Report type for the indicated load case. .
	DDCSumEx (see page 857)	Available since version 1.7.
	DynCombDDC (see page 858)	Available since version 1.7.
	DynCombDDCLocal (see page 858)	Available since version 1.7.
	DynCombDDCSum (see page 858)	Available since version 1.7.
	DynCombLocal (see page 859)	Available since version 1.7.
	DynCombSum (see page 859)	Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

	DynCombSumForce (see page 859)	Available since version 1.7.
	DynCombValue (see page 859)	Available since version 1.7.
	DynDDC (see page 860)	Available since version 1.7.
	DynDDCLocal (see page 860)	Available since version 1.7.
	DynDDCSum (see page 860)	Available since version 1.7.
	DynLocal (see page 861)	Available since version 1.7.
	DynSum (see page 861)	Available since version 1.7.
	DynSumForce (see page 861)	Available since version 1.7.
	DynValue (see page 861)	Available since version 1.7.
	Local (see page 862)	The function returns a structure containing reaction values in the local coordinate system for the indicated node and load case. .
	LocalEx (see page 862)	Available since version 1.7.
	Sum (see page 862)	The function returns a structure containing values of reaction sums for the indicated load case .
	SumEx (see page 862)	Available since version 1.7.
	SumForce (see page 863)	The function returns a structure containing the value of sums of forces for the indicated load case. .
	SumForceEx (see page 863)	Available since version 1.7.
	Value (see page 863)	The function returns a structure containing the values of reactions for a given node and load case. .
	ValueEx (see page 863)	Function returns the structure containing reaction values for the specified node and load case. Available since version 1.7.

III.1.2.2 IRobotReactionServer Methods

The methods of the IRobotReactionServer class are listed here.

Public Methods

	Name	Description
	DDC (see page 856)	The function returns a structure containing Load Report reaction values for the indicated node and load case .
	DDCEx (see page 856)	Available since version 1.7.
	DDCLocal (see page 857)	The function returns a structure containing Load Report reaction values for the indicated node and load case .
	DDCLocalEx (see page 857)	Available since version 1.7.
	DDCSum (see page 857)	The function returns a structure containing values of reaction sums of the Load Report type for the indicated load case. .
	DDCSumEx (see page 857)	Available since version 1.7.
	DynCombDDC (see page 858)	Available since version 1.7.
	DynCombDDCLocal (see page 858)	Available since version 1.7.
	DynCombDDCSum (see page 858)	Available since version 1.7.
	DynCombLocal (see page 859)	Available since version 1.7.
	DynCombSum (see page 859)	Available since version 1.7.
	DynCombSumForce (see page 859)	Available since version 1.7.
	DynCombValue (see page 859)	Available since version 1.7.
	DynDDC (see page 860)	Available since version 1.7.
	DynDDCLocal (see page 860)	Available since version 1.7.
	DynDDCSum (see page 860)	Available since version 1.7.
	DynLocal (see page 861)	Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

	DynSum (see page 861)	Available since version 1.7.
	DynSumForce (see page 861)	Available since version 1.7.
	DynValue (see page 861)	Available since version 1.7.
	Local (see page 862)	The function returns a structure containing reaction values in the local coordinate system for the indicated node and load case. .
	LocalEx (see page 862)	Available since version 1.7.
	Sum (see page 862)	The function returns a structure containing values of reaction sums for the indicated load case .
	SumEx (see page 862)	Available since version 1.7.
	SumForce (see page 863)	The function returns a structure containing the value of sums of forces for the indicated load case. .
	SumForceEx (see page 863)	Available since version 1.7.
	Value (see page 863)	The function returns a structure containing the values of reactions for a given node and load case. .
	ValueEx (see page 863)	Function returns the structure containing reaction values for the specified node and load case. Available since version 1.7.

III.1.2.2.1 DDC

C++

```
HRESULT DDC(long _node, long _case_num, IRobotReactionData** ret);
```

C#

```
public IRobotReactionData DDC(long _node, long _case_num);
```

Visual Basic

```
Public Function DDC(_node As long, _case_num As long) As IRobotReactionData
```

Description

The function returns a structure containing Load Report reaction values for the indicated node and load case .

III.1.2.2.2 DDCEx

C++

```
HRESULT DDCEx(long _node, long _case, long _case_cmpnt = 1, IRobotReactionData** ret);
```

C#

```
public IRobotReactionData DDCEx(long _node, long _case, long _case_cmpnt = 1);
```

Visual Basic

```
Public Function DDCEx(_node As long, _case As long, Optional _case_cmpnt As long = 1) As
IRobotReactionData
```

Description

Available since version 1.7.

III.1.2.2.3 DDCLocal

C++

```
HRESULT DDCLocal(long _node, long _case_num, IRobotReactionData** ret);
```

C#

```
public IRobotReactionData DDCLocal(long _node, long _case_num);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Function DDCLocal(_node As long, _case_num As long) As IRobotReactionData
```

Description

The function returns a structure containing Load Report reaction values for the indicated node and load case .

III.1.2.2.4 DDCLocalEx

C++

```
HRESULT DDCLocalEx(long _node, long _case, long _case_cmpnt = 1, IRobotReactionData** ret);
```

C#

```
public IRobotReactionData DDCLocalEx(long _node, long _case, long _case_cmpnt = 1);
```

Visual Basic

```
Public Function DDCLocalEx(_node As long, _case As long, Optional _case_cmpnt As long = 1)
As IRobotReactionData
```

Description

Available since version 1.7.

III.1.2.2.5 DDCSum

C++

```
HRESULT DDCSum(long _case_num, IRobotReactionData** ret);
```

C#

```
public IRobotReactionData DDCSum(long _case_num);
```

Visual Basic

```
Public Function DDCSum(_case_num As long) As IRobotReactionData
```

Description

The function returns a structure containing values of reaction sums of the Load Report type for the indicated load case. .

III.1.2.2.6 DDCSumEx

C++

```
HRESULT DDCSumEx(long _case, long _case_cmpnt = 1, IRobotReactionData** ret);
```

C#

```
public IRobotReactionData DDCSumEx(long _case, long _case_cmpnt = 1);
```

Visual Basic

```
Public Function DDCSumEx(_case As long, Optional _case_cmpnt As long = 1) As
IRobotReactionData
```

Description

Available since version 1.7.

III.1.2.2.7 DynCombDDC

C++

```
HRESULT DynCombDDC(long _node, long _case, IRobotModeCombinationType _mode_cmb,
IRobotReactionData** ret);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRobotReactionData DynCombDDC(long _node, long _case, IRobotModeCombinationType
_mode_cmb);
```

Visual Basic

```
Public Function DynCombDDC(_node As long, _case As long, _mode_cmb As
IRobotModeCombinationType) As IRobotReactionData
```

Description

Available since version 1.7.

III.1.2.2.8 DynCombDDCLocal

C++

```
HRESULT DynCombDDCLocal(long _node, long _case, IRobotModeCombinationType _mode_cmb,
IRobotReactionData** ret);
```

C#

```
public IRobotReactionData DynCombDDCLocal(long _node, long _case, IRobotModeCombinationType
_mode_cmb);
```

Visual Basic

```
Public Function DynCombDDCLocal(_node As long, _case As long, _mode_cmb As
IRobotModeCombinationType) As IRobotReactionData
```

Description

Available since version 1.7.

III.1.2.2.9 DynCombDDCSum

C++

```
HRESULT DynCombDDCSum(long _case, IRobotModeCombinationType _mode_cmb, IRobotReactionData**
ret);
```

C#

```
public IRobotReactionData DynCombDDCSum(long _case, IRobotModeCombinationType _mode_cmb);
```

Visual Basic

```
Public Function DynCombDDCSum(_case As long, _mode_cmb As IRobotModeCombinationType) As
IRobotReactionData
```

Description

Available since version 1.7.

III.1.2.2.10 DynCombLocal

C++

```
HRESULT DynCombLocal(long _node, long _case, IRobotModeCombinationType _mode_cmb,
IRobotReactionData** ret);
```

C#

```
public IRobotReactionData DynCombLocal(long _node, long _case, IRobotModeCombinationType
_mode_cmb);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Function DynCombLocal(_node As long, _case As long, _mode_cmb As
IRobotModeCombinationType) As IRobotReactionData
```

Description

Available since version 1.7.

III.1.2.2.11 DynCombSum

C++

```
HRESULT DynCombSum(long _case, IRobotModeCombinationType _mode_cmb, IRobotReactionData**
ret);
```

C#

```
public IRobotReactionData DynCombSum(long _case, IRobotModeCombinationType _mode_cmb);
```

Visual Basic

```
Public Function DynCombSum(_case As long, _mode_cmb As IRobotModeCombinationType) As
IRobotReactionData
```

Description

Available since version 1.7.

III.1.2.2.12 DynCombSumForce

C++

```
HRESULT DynCombSumForce(long _case, IRobotModeCombinationType _mode_cmb,
IRobotReactionData** ret);
```

C#

```
public IRobotReactionData DynCombSumForce(long _case, IRobotModeCombinationType _mode_cmb);
```

Visual Basic

```
Public Function DynCombSumForce(_case As long, _mode_cmb As IRobotModeCombinationType) As
IRobotReactionData
```

Description

Available since version 1.7.

III.1.2.2.13 DynCombValue

C++

```
HRESULT DynCombValue(long _node, long _case, IRobotModeCombinationType _mode_cmb,
IRobotReactionData** ret);
```

C#

```
public IRobotReactionData DynCombValue(long _node, long _case, IRobotModeCombinationType
_mode_cmb);
```

Visual Basic

```
Public Function DynCombValue(_node As long, _case As long, _mode_cmb As
IRobotModeCombinationType) As IRobotReactionData
```

Description

Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

III.1.2.2.14 DynDDC

C++

```
HRESULT DynDDC(long _node, long _case, long _mode, IRobotReactionData** ret);
```

C#

```
public IRobotReactionData DynDDC(long _node, long _case, long _mode);
```

Visual Basic

```
Public Function DynDDC(_node As long, _case As long, _mode As long) As IRobotReactionData
```

Description

Available since version 1.7.

III.1.2.2.15 DynDDCLocal

C++

```
HRESULT DynDDCLocal(long _node, long _case, long _mode, IRobotReactionData** ret);
```

C#

```
public IRobotReactionData DynDDCLocal(long _node, long _case, long _mode);
```

Visual Basic

```
Public Function DynDDCLocal(_node As long, _case As long, _mode As long) As
IRobotReactionData
```

Description

Available since version 1.7.

III.1.2.2.16 DynDDCSum

C++

```
HRESULT DynDDCSum(long _case, long _mode, IRobotReactionData** ret);
```

C#

```
public IRobotReactionData DynDDCSum(long _case, long _mode);
```

Visual Basic

```
Public Function DynDDCSum(_case As long, _mode As long) As IRobotReactionData
```

Description

Available since version 1.7.

III.1.2.2.17 DynLocal

C++

```
HRESULT DynLocal(long _node, long _case, long _mode, IRobotReactionData** ret);
```

C#

```
public IRobotReactionData DynLocal(long _node, long _case, long _mode);
```

Visual Basic

```
Public Function DynLocal(_node As long, _case As long, _mode As long) As IRobotReactionData
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Available since version 1.7.

III.1.2.2.18 DynSum

C++

```
HRESULT DynSum(long _case, long _mode, IRobotReactionData** ret);
```

C#

```
public IRobotReactionData DynSum(long _case, long _mode);
```

Visual Basic

```
Public Function DynSum(_case As long, _mode As long) As IRobotReactionData
```

Description

Available since version 1.7.

III.1.2.2.19 DynSumForce

C++

```
HRESULT DynSumForce(long _case, long _mode, IRobotReactionData** ret);
```

C#

```
public IRobotReactionData DynSumForce(long _case, long _mode);
```

Visual Basic

```
Public Function DynSumForce(_case As long, _mode As long) As IRobotReactionData
```

Description

Available since version 1.7.

III.1.2.2.20 DynValue

C++

```
HRESULT DynValue(long _node, long _case, long _mode, IRobotReactionData** ret);
```

C#

```
public IRobotReactionData DynValue(long _node, long _case, long _mode);
```

Visual Basic

```
Public Function DynValue(_node As long, _case As long, _mode As long) As IRobotReactionData
```

Description

Available since version 1.7.

III.1.2.2.21 Local

C++

```
HRESULT Local(long _node, long _case_num, IRobotReactionData** ret);
```

C#

```
public IRobotReactionData Local(long _node, long _case_num);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Function Local(_node As long, _case_num As long) As IRobotReactionData
```

Description

The function returns a structure containing reaction values in the local coordinate system for the indicated node and load case. .

III.1.2.2.22 LocalEx

C++

```
HRESULT LocalEx(long _node, long _case, long _case_cmpnt = 1, IRobotReactionData** ret);
```

C#

```
public IRobotReactionData LocalEx(long _node, long _case, long _case_cmpnt = 1);
```

Visual Basic

```
Public Function LocalEx(_node As long, _case As long, Optional _case_cmpnt As long = 1) As
IRobotReactionData
```

Description

Available since version 1.7.

III.1.2.2.23 Sum

C++

```
HRESULT Sum(long _case_num, IRobotReactionData** ret);
```

C#

```
public IRobotReactionData Sum(long _case_num);
```

Visual Basic

```
Public Function Sum(_case_num As long) As IRobotReactionData
```

Description

The function returns a structure containing values of reaction sums for the indicated load case .

III.1.2.2.24 SumEx

C++

```
HRESULT SumEx(long _case, long _case_cmpnt = 1, IRobotReactionData** ret);
```

C#

```
public IRobotReactionData SumEx(long _case, long _case_cmpnt = 1);
```

Visual Basic

```
Public Function SumEx(_case As long, Optional _case_cmpnt As long = 1) As IRobotReactionData
```

Description

Available since version 1.7.

III.1.2.2.25 SumForce

C++

```
HRESULT SumForce(long _case_num, IRobotReactionData** ret);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRobotReactionData SumForce(long _case_num);
```

Visual Basic

```
Public Function SumForce(_case_num As long) As IRobotReactionData
```

Description

The function returns a structure containing the value of sums of forces for the indicated load case. .

III.1.2.2.26 SumForceEx

C++

```
HRESULT SumForceEx(long _case, long _case_cmpnt = 1, IRobotReactionData** ret);
```

C#

```
public IRobotReactionData SumForceEx(long _case, long _case_cmpnt = 1);
```

Visual Basic

```
Public Function SumForceEx(_case As long, Optional _case_cmpnt As long = 1) As
IRobotReactionData
```

Description

Available since version 1.7.

III.1.2.2.27 Value

C++

```
HRESULT Value(long _node, long _case_num, IRobotReactionData** ret);
```

C#

```
public IRobotReactionData Value(long _node, long _case_num);
```

Visual Basic

```
Public Function Value(_node As long, _case_num As long) As IRobotReactionData
```

Description

The function returns a structure containing the values of reactions for a given node and load case. .

III.1.2.2.28 ValueEx

C++

```
HRESULT ValueEx(long _node_num, long _case_num, long _case_cmpnt = 1, IRobotReactionData**
ret);
```

C#

```
public IRobotReactionData ValueEx(long _node_num, long _case_num, long _case_cmpnt = 1);
```

Visual Basic

```
Public Function ValueEx(_node_num As long, _case_num As long, Optional _case_cmpnt As long
= 1) As IRobotReactionData
```

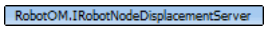
Description

Function returns the structure containing reaction values for the specified node and load case. Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

III.1.3 IRobotNodeDisplacementServer

Class Hierarchy



C++

```
interface IRobotNodeDisplacementServer : IDispatch;
```

C#

```
public interface IRobotNodeDisplacementServer;
```

Visual Basic

```
Public Interface IRobotNodeDisplacementServer
```

Description

The server provides access to all values of nodal displacements. .

III.1.3.1 IRobotNodeDisplacementServer Members

The following tables list the members exposed by IRobotNodeDisplacementServer.

Public Methods

	Name	Description
	DynCombValue (see page 865)	Function returns the data structure containing displacements for the specified node and defined mode combination of a selected load case. Available since version 1.7.
	DynValue (see page 865)	Function returns the data structure describing displacements in the defined node for a specified load case and specified mode. Available since version 1.7.
	Value (see page 865)	The function returns an object (set of data) describing displacements in the indicated node for the indicated load case. .
	ValueEx (see page 866)	Available since version 1.7.

III.1.3.2 IRobotNodeDisplacementServer Methods

The methods of the IRobotNodeDisplacementServer class are listed here.

Public Methods

	Name	Description
	DynCombValue (see page 865)	Function returns the data structure containing displacements for the specified node and defined mode combination of a selected load case. Available since version 1.7.
	DynValue (see page 865)	Function returns the data structure describing displacements in the defined node for a specified load case and specified mode. Available since version 1.7.
	Value (see page 865)	The function returns an object (set of data) describing displacements in the indicated node for the indicated load case. .
	ValueEx (see page 866)	Available since version 1.7.

III.1.3.2.1 DynCombValue

C++

```
HRESULT DynCombValue(long _node_num, long _case_num, IRobotModeCombinationType _mode_cmb,
IRobotDisplacementData** ret);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRobotDisplacementData DynCombValue(long _node_num, long _case_num,
IRobotModeCombinationType _mode_cmb);
```

Visual Basic

```
Public Function DynCombValue(_node_num As long, _case_num As long, _mode_cmb As
IRobotModeCombinationType) As IRobotDisplacementData
```

Description

Function returns the data structure containing displacements for the specified node and defined mode combination of a selected load case. Available since version 1.7.

III.1.3.2.2 DynValue

C++

```
HRESULT DynValue(long _node_num, long _case_num, long _mode, IRobotDisplacementData** ret);
```

C#

```
public IRobotDisplacementData DynValue(long _node_num, long _case_num, long _mode);
```

Visual Basic

```
Public Function DynValue(_node_num As long, _case_num As long, _mode As long) As
IRobotDisplacementData
```

Description

Function returns the data structure describing displacements in the defined node for a specified load case and specified mode. Available since version 1.7.

III.1.3.2.3 Value

C++

```
HRESULT Value(long _node_num, long _case_num, IRobotDisplacementData** ret);
```

C#

```
public IRobotDisplacementData Value(long _node_num, long _case_num);
```

Visual Basic

```
Public Function Value(_node_num As long, _case_num As long) As IRobotDisplacementData
```

Description

The function returns an object (set of data) describing displacements in the indicated node for the indicated load case. .

III.1.3.2.4 ValueEx

C++

```
HRESULT ValueEx(long _node, long _case, long _case_cmpnt, IRobotDisplacementData** ret);
```

C#

```
public IRobotDisplacementData ValueEx(long _node, long _case, long _case_cmpnt);
```

Visual Basic

```
Public Function ValueEx(_node As long, _case As long, _case_cmpnt As long) As
IRobotDisplacementData
```

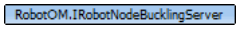
(c) 2014 Autodesk Inc. All rights reserved

Description

Available since version 1.7.

III.1.4 IRobotNodeBucklingServer

Class Hierarchy



C++

```
interface IRobotNodeBucklingServer : IDispatch;
```

C#

```
public interface IRobotNodeBucklingServer;
```

Visual Basic

```
Public Interface IRobotNodeBucklingServer
```

Description

The server provides results of buckling analysis for nodes.

III.1.4.1 IRobotNodeBucklingServer Members

The following tables list the members exposed by IRobotNodeBucklingServer.

Public Methods

	Name	Description
	EigenVector (see page 867)	The function returns the displacement vector for the indicated node, load case and mode.
	EigenVectorCmb (see page 867)	Available since version 1.7.

III.1.4.2 IRobotNodeBucklingServer Methods

The methods of the IRobotNodeBucklingServer class are listed here.

Public Methods

	Name	Description
	EigenVector (see page 867)	The function returns the displacement vector for the indicated node, load case and mode.
	EigenVectorCmb (see page 867)	Available since version 1.7.

III.1.4.2.1 EigenVector

C++

```
HRESULT EigenVector(long _node, long _case, long _mode, IRobotDisplacementData** ret);
```

C#

```
public IRobotDisplacementData EigenVector(long _node, long _case, long _mode);
```

Visual Basic

```
Public Function EigenVector(_node As long, _case As long, _mode As long) As
IRobotDisplacementData
```

Description

The function returns the displacement vector for the indicated node, load case and mode.

(c) 2014 Autodesk Inc. All rights reserved

III.1.4.2.2 EigenVectorCmb

C++

```
HRESULT EigenVectorCmb(long _node, long _case, IRobotModeCombinationType _mode_cmb,
IRobotDisplacementData** ret);
```

C#

```
public IRobotDisplacementData EigenVectorCmb(long _node, long _case,
IRobotModeCombinationType _mode_cmb);
```

Visual Basic

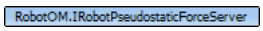
```
Public Function EigenVectorCmb(_node As long, _case As long, _mode_cmb As
IRobotModeCombinationType) As IRobotDdisplacementData
```

Description

Available since version 1.7.

III.1.5 IRobotPseudostaticForceServer

Class Hierarchy



C++

```
interface IRobotPseudostaticForceServer : IDispatch;
```

C#

```
public interface IRobotPseudostaticForceServer;
```

Visual Basic

```
Public Interface IRobotPseudostaticForceServer
```

Description

Server providing access to pseudostatic forces.

III.1.5.1 IRobotPseudostaticForceServer Members

The following tables list the members exposed by IRobotPseudostaticForceServer.

Public Methods

	Name	Description
	CombValue (see page 868)	Function returns pseudostatic force values in a specified node for a defined load case and mode. Available since version 1.7.
	Value (see page 868)	Function returns pseudostatic force values in a specified node for a defined load case and mode. Available since version 1.7.

III.1.5.2 IRobotPseudostaticForceServer Methods

The methods of the IRobotPseudostaticForceServer class are listed here.

Public Methods

	Name	Description
	CombValue (see page 868)	Function returns pseudostatic force values in a specified node for a defined load case and mode. Available since version 1.7.
	Value (see page 868)	Function returns pseudostatic force values in a specified node for a defined load case and mode. Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

III.1.5.2.1 CombValue

C++

```
HRESULT CombValue(long _node_num, long _case_num, IRobotModeCombinationType _mode_cmb,
IRobotForcesData** ret);
```

C#

```
public IRobotForcesData CombValue(long _node_num, long _case_num, IRobotModeCombinationType
_mode_cmb);
```

Visual Basic

```
Public Function CombValue(_node_num As long, _case_num As long, _mode_cmb As
IRobotModeCombinationType) As IRobotForcesData
```

Description

Function returns pseudostatic force values in a specified node for a defined load case and mode. Available since version 1.7.

III.1.5.2.2 Value

C++

```
HRESULT Value(long _node_num, long _case_num, long _mode_num, IRobotForcesData** ret);
```

C#

```
public IRobotForcesData Value(long _node_num, long _case_num, long _mode_num);
```

Visual Basic

```
Public Function Value(_node_num As long, _case_num As long, _mode_num As long) As
IRobotForcesData
```

Description

Function returns pseudostatic force values in a specified node for a defined load case and mode. Available since version 1.7.

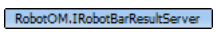
III.2 Calculation results for bars

Interfaces

	Name	Description
	IRobotBarResultServer (see page 869)	Server makes accessible all types of calculation results for bars. .
	IRobotBarForceServer (see page 872)	Server provides access to values of forces and moments for bars. .
	IRobotBarDeflectionServer (see page 875)	The server makes deflection values for bars available.
	IRobotBarStressServer (see page 879)	Server allows to get stress values for bars.
	IRobotBarBucklingServer (see page 881)	The server provides results of buckling analysis for bars.
	IRobotBarDisplacementServer (see page 882)	Server providing access to displacement results for bars. .

III.2.1 IRobotBarResultServer

Class Hierarchy



(c) 2014 Autodesk Inc. All rights reserved

C++

```
interface IRobotBarResultServer : IDispatch;
```

C#

```
public interface IRobotBarResultServer;
```

Visual Basic

```
Public Interface IRobotBarResultServer
```

Description

Server makes accessible all types of calculation results for bars. .

III.2.1.1 IRobotBarResultServer Members

The following tables list the members exposed by IRobotBarResultServer.

Public Fields

	Name	Description
	Buckling (see page 870)	Server of results of buckling analysis for bars.
	Deflections (see page 870)	Deflections.
	Displacements (see page 870)	Displacement server.
	Forces (see page 871)	Internal forces.
	Stresses (see page 871)	Stresses.

Public Methods

	Name	Description
	GeoImperfections (see page 871)	Function returns calculated deflection values in individual directions for the indicated bar.

III.2.1.2 IRobotBarResultServer Fields

The fields of the IRobotBarResultServer class are listed here.

Public Fields

	Name	Description
	Buckling (see page 870)	Server of results of buckling analysis for bars.
	Deflections (see page 870)	Deflections.
	Displacements (see page 870)	Displacement server.
	Forces (see page 871)	Internal forces.
	Stresses (see page 871)	Stresses.

III.2.1.2.1 Buckling

C++

```
HRESULT get_Buckling(IRobotBarBucklingServer**);
```

C#

```
public IRobotBarBucklingServer Buckling { get; }
```

Visual Basic

```
Public ReadOnly Buckling As IRobotBarBucklingServer
```

Description

Server of results of buckling analysis for bars.

(c) 2014 Autodesk Inc. All rights reserved

III.2.1.2.2 Deflections

C++

```
HRESULT get_Deflections(IRobotBarDeflectionServer**);
```

C#

```
public IRobotBarDeflectionServer Deflections { get; }
```

Visual Basic

```
Public ReadOnly Deflections As IRobotBarDeflectionServer
```

Description

Deflections.

III.2.1.2.3 Displacements

C++

```
HRESULT get_Displacements(IRobotBarDisplacementServer**);
```

C#

```
public IRobotBarDisplacementServer Displacements { get; }
```

Visual Basic

```
Public ReadOnly Displacements As IRobotBarDisplacementServer
```

Description

Displacement server.

Version

Available since version 2.5.

III.2.1.2.4 Forces

C++

```
HRESULT get_Forces(IRobotBarForceServer*);
```

C#

```
public IRobotBarForceServer Forces { get; }
```

Visual Basic

```
Public ReadOnly Forces As IRobotBarForceServer
```

Description

Internal forces.

III.2.1.2.5 Stresses

C++

```
HRESULT get_Stresses(IRobotBarStressServer**);
```

C#

```
public IRobotBarStressServer Stresses { get; }
```

Visual Basic

```
Public ReadOnly Stresses As IRobotBarStressServer
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Stresses.

III.2.1.3 IRobotBarResultServer Methods

The methods of the IRobotBarResultServer class are listed here.

Public Methods

	Name	Description
	GeoImperfections (see page 871)	Function returns calculated deflection values in individual directions for the indicated bar.

III.2.1.3.1 GeoImperfections

C++

```
HRESULT GeoImperfections(long _bar, IRobotBarDeflectionData** ret);
```

C#

```
public IRobotBarDeflectionData GeoImperfections(long _bar);
```

Visual Basic

```
Public Function GeoImperfections(_bar As long) As IRobotBarDeflectionData
```

Description

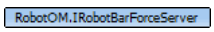
Function returns calculated deflection values in individual directions for the indicated bar.

Version

Available since version 3.

III.2.2 IRobotBarForceServer

Class Hierarchy



C++

```
interface IRobotBarForceServer : IDispatch;
```

C#

```
public interface IRobotBarForceServer;
```

Visual Basic

```
Public Interface IRobotBarForceServer
```

Description

Server provides access to values of forces and moments for bars. .

III.2.2.1 IRobotBarForceServer Members

The following tables list the members exposed by IRobotBarForceServer.

Public Methods

	Name	Description
	DynCombValue (see page 873)	Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

	DynCombValueByNPoints (see page 873)	Available since version 1.7.
	DynValue (see page 873)	Function returns the structure containing values of all forces and moments for a specified bar, mode of the load case and point on the bar. Available since version 1.7.
	DynValueByNPoints (see page 874)	Function returns the structure containing values of all forces and moments for a specified bar, mode of the load case and point on the bar. Available since version 1.7.
	Value (see page 874)	The function returns a structure containing the values of all forces and moments for the indicated bar, load case and point along a bar. .
	ValueByNPoints (see page 874)	The function returns a structure containing the values of all forces and moments for the indicated bar, load case at the indicated division point. .
	ValueByNPointsEx (see page 875)	Available since version 1.7.
	ValueEx (see page 875)	Available since version 1.7.

III.2.2.2 IRobotBarForceServer Methods

The methods of the IRobotBarForceServer class are listed here.

Public Methods

	Name	Description
	DynCombValue (see page 873)	Available since version 1.7.
	DynCombValueByNPoints (see page 873)	Available since version 1.7.
	DynValue (see page 873)	Function returns the structure containing values of all forces and moments for a specified bar, mode of the load case and point on the bar. Available since version 1.7.
	DynValueByNPoints (see page 874)	Function returns the structure containing values of all forces and moments for a specified bar, mode of the load case and point on the bar. Available since version 1.7.
	Value (see page 874)	The function returns a structure containing the values of all forces and moments for the indicated bar, load case and point along a bar. .
	ValueByNPoints (see page 874)	The function returns a structure containing the values of all forces and moments for the indicated bar, load case at the indicated division point. .
	ValueByNPointsEx (see page 875)	Available since version 1.7.
	ValueEx (see page 875)	Available since version 1.7.

III.2.2.2.1 DynCombValue

C++

```
HRESULT DynCombValue(long _bar_num, long _case_num, IRobotModeCombinationType _mode_cmb,
double _point, IRobotBarForceData** ret);
```

C#

```
public IRobotBarForceData DynCombValue(long _bar_num, long _case_num,
IRobotModeCombinationType _mode_cmb, double _point);
```

Visual Basic

```
Public Function DynCombValue(_bar_num As long, _case_num As long, _mode_cmb As
IRobotModeCombinationType, _point As double) As IRobotBarForceData
```

Description

Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

III.2.2.2.2 DynCombValueByNPoints

C++

```
HRESULT DynCombValueByNPoints(long _bar_num, long _case_num, IRobotModeCombinationType
_mode_cmb, int _no_points, int _point, IRobotBarForceData** ret);
```

C#

```
public IRobotBarForceData DynCombValueByNPoints(long _bar_num, long _case_num,
IRobotModeCombinationType _mode_cmb, int _no_points, int _point);
```

Visual Basic

```
Public Function DynCombValueByNPoints(_bar_num As long, _case_num As long, _mode_cmb As
IRobotModeCombinationType, _no_points As int, _point As int) As IRobotBarForceData
```

Description

Available since version 1.7.

III.2.2.2.3 DynValue

C++

```
HRESULT DynValue(long _bar_num, long _case_num, long _mode, double _point,
IRobotBarForceData** ret);
```

C#

```
public IRobotBarForceData DynValue(long _bar_num, long _case_num, long _mode, double
_point);
```

Visual Basic

```
Public Function DynValue(_bar_num As long, _case_num As long, _mode As long, _point As
double) As IRobotBarForceData
```

Description

Function returns the structure containing values of all forces and moments for a specified bar, mode of the load case and point on the bar. Available since version 1.7.

III.2.2.2.4 DynValueByNPoints

C++

```
HRESULT DynValueByNPoints(long _bar_num, long _case_num, long _mode, int _no_points, int
_point, IRobotBarForceData** ret);
```

C#

```
public IRobotBarForceData DynValueByNPoints(long _bar_num, long _case_num, long _mode, int
_no_points, int _point);
```

Visual Basic

```
Public Function DynValueByNPoints(_bar_num As long, _case_num As long, _mode As long,
_no_points As int, _point As int) As IRobotBarForceData
```

Description

Function returns the structure containing values of all forces and moments for a specified bar, mode of the load case and point on the bar. Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

III.2.2.2.5 Value

C++

```
HRESULT Value(long _bar_num, long _case_num, double _point, IRobotBarForceData** ret);
```

C#

```
public IRobotBarForceData Value(long _bar_num, long _case_num, double _point);
```

Visual Basic

```
Public Function Value(_bar_num As long, _case_num As long, _point As double) As
IRobotBarForceData
```

Description

The function returns a structure containing the values of all forces and moments for the indicated bar, load case and point along a bar. .

III.2.2.2.6 ValueByNPoints

C++

```
HRESULT ValueByNPoints(long _bar_num, long _case_num, int _no_points, int _point,
IRobotBarForceData* ret);
```

C#

```
public IRobotBarForceData ValueByNPoints(long _bar_num, long _case_num, int _no_points, int
_point);
```

Visual Basic

```
Public Function ValueByNPoints(_bar_num As long, _case_num As long, _no_points As int,
_point As int) As IRobotBarForceData
```

Description

The function returns a structure containing the values of all forces and moments for the indicated bar, load case at the indicated division point. .

III.2.2.2.7 ValueByNPointsEx

C++

```
HRESULT ValueByNPointsEx(long _bar_num, long _case_num, long _case_cmpnt, int _no_points,
int _point, IRobotBarForceData** ret);
```

C#

```
public IRobotBarForceData ValueByNPointsEx(long _bar_num, long _case_num, long _case_cmpnt,
int _no_points, int _point);
```

Visual Basic

```
Public Function ValueByNPointsEx(_bar_num As long, _case_num As long, _case_cmpnt As long,
_no_points As int, _point As int) As IRobotBarForceData
```

Description

Available since version 1.7.

III.2.2.2.8 ValueEx

C++

```
HRESULT ValueEx(long _bar_num, long _case_num, long _case_cmpnt, double _point,
IRobotBarForceData** ret);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRobotBarForceData ValueEx(long _bar_num, long _case_num, long _case_cmpnt, double
_point);
```

Visual Basic

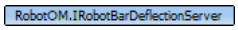
```
Public Function ValueEx(_bar_num As long, _case_num As long, _case_cmpnt As long, _point As
double) As IRobotBarForceData
```

Description

Available since version 1.7.

III.2.3 IRobotBarDeflectionServer

Class Hierarchy



C++

```
interface IRobotBarDeflectionServer : IDispatch;
```

C#

```
public interface IRobotBarDeflectionServer;
```

Visual Basic

```
Public Interface IRobotBarDeflectionServer
```

Description

The server makes deflection values for bars available.

III.2.3.1 IRobotBarDeflectionServer Members

The following tables list the members exposed by IRobotBarDeflectionServer.

Public Methods

	Name	Description
	DynCombMaxValue (see page 876)	The function makes available the maximum deflections for the square combination and appropriate positions on a bar.
	DynCombValue (see page 877)	The function makes available the deflection values for an indicated combination.
	DynMaxValue (see page 877)	The function makes maximum deflection values and appropriate positions on a bar in dynamics available.
	DynValue (see page 877)	The function gives back deflection values in dynamics.
	MaxValue (see page 878)	The function gives back maximum deflection values for a given bar, load case and appropriate positions on a bar.
	MaxValueEx (see page 878)	Available since version 1.7.
	Value (see page 878)	The function gives back a data stucture containing deflection values for a given bar, load case and a point on a bar. .
	ValueEx (see page 878)	Available since version 1.7.

III.2.3.2 IRobotBarDeflectionServer Methods

The methods of the IRobotBarDeflectionServer class are listed here.

(c) 2014 Autodesk Inc. All rights reserved

Public Methods

	Name	Description
	DynCombMaxValue (see page 876)	The function makes available the maximum deflections for the square combination and appropriate positions on a bar.
	DynCombValue (see page 877)	The function makes available the deflection values for an indicated combination.
	DynMaxValue (see page 877)	The function makes maximum deflection values and appropriate positions on a bar in dynamics available.
	DynValue (see page 877)	The function gives back deflection values in dynamics.
	MaxValue (see page 878)	The function gives back maximum deflection values for a given bar, load case and appropriate positions on a bar.
	MaxValueEx (see page 878)	Available since version 1.7.
	Value (see page 878)	The function gives back a data stucture containing deflection values for a given bar, load case and a point on a bar. .
	ValueEx (see page 878)	Available since version 1.7.

III.2.3.2.1 DynCombMaxValue

C++

```
HRESULT DynCombMaxValue(long _bar_num, long _case_num, IRobotModeCombinationType _cmb,
IRobotBarDeflectionData** ret);
```

C#

```
public IRobotBarDeflectionData DynCombMaxValue(long _bar_num, long _case_num,
IRobotModeCombinationType _cmb);
```

Visual Basic

```
Public Function DynCombMaxValue(_bar_num As long, _case_num As long, _cmb As
IRobotModeCombinationType) As IRobotBarDeflectionData
```

Description

The function makes available the maximum deflections for the square combination and appropriate positions on a bar.

III.2.3.2.2 DynCombValue

C++

```
HRESULT DynCombValue(long _bar_num, long _case_num, double _pos, IRobotModeCombinationType
_cmb, IRobotBarDeflectionData** ret);
```

C#

```
public IRobotBarDeflectionData DynCombValue(long _bar_num, long _case_num, double _pos,
IRobotModeCombinationType _cmb);
```

Visual Basic

```
Public Function DynCombValue(_bar_num As long, _case_num As long, _pos As double, _cmb As
IRobotModeCombinationType) As IRobotBarDeflectionData
```

Description

The function makes available the deflection values for an indicated combination.

III.2.3.2.3 DynMaxValue

C++

```
HRESULT DynMaxValue(long _bar_num, long _case_num, long _mode_num,
IRobotBarDeflectionData** ret);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRobotBarDeflectionData DynMaxValue(long _bar_num, long _case_num, long _mode_num);
```

Visual Basic

```
Public Function DynMaxValue(_bar_num As long, _case_num As long, _mode_num As long) As
IRobotBarDeflectionData
```

Description

The function makes maximum deflection values and appropriate positions on a bar in dynamics available.

III.2.3.2.4 DynValue

C++

```
HRESULT DynValue(long _bar_num, long _case_num, long _mode_num, double _pos,
IRobotBarDeflectionData** ret);
```

C#

```
public IRobotBarDeflectionData DynValue(long _bar_num, long _case_num, long _mode_num,
double _pos);
```

Visual Basic

```
Public Function DynValue(_bar_num As long, _case_num As long, _mode_num As long, _pos As
double) As IRobotBarDeflectionData
```

Description

The function gives back deflection values in dynamics.

III.2.3.2.5 MaxValue

C++

```
HRESULT MaxValue(long _bar_num, long _case_num, IRobotBarDeflectionData** ret);
```

C#

```
public IRobotBarDeflectionData MaxValue(long _bar_num, long _case_num);
```

Visual Basic

```
Public Function MaxValue(_bar_num As long, _case_num As long) As IRobotBarDeflectionData
```

Description

The function gives back maximum deflection values for a given bar, load case and appropriate positions on a bar.

III.2.3.2.6 MaxValueEx

C++

```
HRESULT MaxValueEx(long _bar_num, long _case_num, long _case_cmpnt,
IRobotBarDeflectionData** ret);
```

C#

```
public IRobotBarDeflectionData MaxValueEx(long _bar_num, long _case_num, long _case_cmpnt);
```

Visual Basic

```
Public Function MaxValueEx(_bar_num As long, _case_num As long, _case_cmpnt As long) As
IRobotBarDeflectionData
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Available since version 1.7.

III.2.3.2.7 Value

C++

```
HRESULT Value(long _bar_num, long _case_num, double _pos, IRobotBarDeflectionData** ret);
```

C#

```
public IRobotBarDeflectionData Value(long _bar_num, long _case_num, double _pos);
```

Visual Basic

```
Public Function Value(_bar_num As long, _case_num As long, _pos As double) As
IRobotBarDeflectionData
```

Description

The function gives back a data stucture containing deflection values for a given bar, load case and a point on a bar. .

III.2.3.2.8 ValueEx

C++

```
HRESULT ValueEx(long _bar_num, long _case_num, long _case_cmpnt, double _pos,
IRobotBarDeflectionData** ret);
```

C#

```
public IRobotBarDeflectionData ValueEx(long _bar_num, long _case_num, long _case_cmpnt,
double _pos);
```

Visual Basic

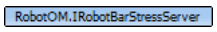
```
Public Function ValueEx(_bar_num As long, _case_num As long, _case_cmpnt As long, _pos As
double) As IRobotBarDeflectionData
```

Description

Available since version 1.7.

III.2.4 IRobotBarStressServer

Class Hierarchy



C++

```
interface IRobotBarStressServer : IDispatch;
```

C#

```
public interface IRobotBarStressServer;
```

Visual Basic

```
Public Interface IRobotBarStressServer
```

Description

Server allows to get stress values for bars.

III.2.4.1 IRobotBarStressServer Members

The following tables list the members exposed by IRobotBarStressServer.

(c) 2014 Autodesk Inc. All rights reserved

Public Methods

	Name	Description
	DynCombValue (see page 880)	Function returns stress values for a selected load case and specified mode combination, in the selected point on the bar. Available since version 1.7.
	DynValue (see page 880)	Function returns stress values for a selected load case and mode, in the selected point on the bar. Available since version 1.7.
	Value (see page 880)	The function makes the stress values for a given bar, load case and relative position on a bar available.
	ValueEx (see page 880)	Available since version 1.7.

III.2.4.2 IRobotBarStressServer Methods

The methods of the IRobotBarStressServer class are listed here.

Public Methods

	Name	Description
	DynCombValue (see page 880)	Function returns stress values for a selected load case and specified mode combination, in the selected point on the bar. Available since version 1.7.
	DynValue (see page 880)	Function returns stress values for a selected load case and mode, in the selected point on the bar. Available since version 1.7.
	Value (see page 880)	The function makes the stress values for a given bar, load case and relative position on a bar available.
	ValueEx (see page 880)	Available since version 1.7.

III.2.4.2.1 DynCombValue

C++

```
HRESULT DynCombValue(long _bar_num, long _case_num, IRobotModeCombinationType _mode_cmb,
double _pos, IRobotBarStressData** ret);
```

C#

```
public IRobotBarStressData DynCombValue(long _bar_num, long _case_num,
IRobotModeCombinationType _mode_cmb, double _pos);
```

Visual Basic

```
Public Function DynCombValue(_bar_num As long, _case_num As long, _mode_cmb As
IRobotModeCombinationType, _pos As double) As IRobotBarStressData
```

Description

Function returns stress values for a selected load case and specified mode combination, in the selected point on the bar. Available since version 1.7.

III.2.4.2.2 DynValue

C++

```
HRESULT DynValue(long _bar_num, long _case_num, long _mode_num, double _pos,
IRobotBarStressData** ret);
```

C#

```
public IRobotBarStressData DynValue(long _bar_num, long _case_num, long _mode_num, double
_pos);
```

Visual Basic

```
Public Function DynValue(_bar_num As long, _case_num As long, _mode_num As long, _pos As
```

(c) 2014 Autodesk Inc. All rights reserved

```
double) As IRobotBarStressData
```

Description

Function returns stress values for a selected load case and mode, in the selected point on the bar. Available since version 1.7.

III.2.4.2.3 Value

C++

```
HRESULT Value(long _bar_num, long _case_num, double _pos, IRobotBarStressData** ret);
```

C#

```
public IRobotBarStressData Value(long _bar_num, long _case_num, double _pos);
```

Visual Basic

```
Public Function Value(_bar_num As long, _case_num As long, _pos As double) As
IRobotBarStressData
```

Description

The function makes the stress values for a given bar, load case and relative position on a bar available.

III.2.4.2.4 ValueEx

C++

```
HRESULT ValueEx(long _bar_num, long _case_num, long _case_cmpnt, double _pos,
IRobotBarStressData** ret);
```

C#

```
public IRobotBarStressData ValueEx(long _bar_num, long _case_num, long _case_cmpnt, double
_pos);
```

Visual Basic

```
Public Function ValueEx(_bar_num As long, _case_num As long, _case_cmpnt As long, _pos As
double) As IRobotBarStressData
```

Description

Available since version 1.7.

III.2.5 IRobotBarBucklingServer

Class Hierarchy

C++

```
interface IRobotBarBucklingServer : IDispatch;
```

C#

```
public interface IRobotBarBucklingServer;
```

Visual Basic

```
Public Interface IRobotBarBucklingServer
```

Description

The server provides results of buckling analysis for bars.

(c) 2014 Autodesk Inc. All rights reserved

III.2.5.1 IRobotBarBucklingServer Members

The following tables list the members exposed by IRobotBarBucklingServer.

Public Methods

	Name	Description
	CriticalCoef (see page 881)	The function returns the value of the critical coefficient for the indicated load case and mode.
	EigenValue (see page 882)	The function returns eigenvalues for the indicated bar, load case and mode.

III.2.5.2 IRobotBarBucklingServer Methods

The methods of the IRobotBarBucklingServer class are listed here.

Public Methods

	Name	Description
	CriticalCoef (see page 881)	The function returns the value of the critical coefficient for the indicated load case and mode.
	EigenValue (see page 882)	The function returns eigenvalues for the indicated bar, load case and mode.

III.2.5.2.1 CriticalCoef

C++

```
HRESULT CriticalCoef(long _case_num, long _mode_num, double* ret);
```

C#

```
public double CriticalCoef(long _case_num, long _mode_num);
```

Visual Basic

```
Public Function CriticalCoef(_case_num As long, _mode_num As long) As double
```

Description

The function returns the value of the critical coefficient for the indicated load case and mode.

III.2.5.2.2 EigenValue

C++

```
HRESULT EigenValue(long _bar, long _case, long _mode, IRobotBarBucklingData** ret);
```

C#

```
public IRobotBarBucklingData EigenValue(long _bar, long _case, long _mode);
```

Visual Basic

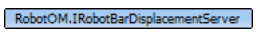
```
Public Function EigenValue(_bar As long, _case As long, _mode As long) As
IRobotBarBucklingData
```

Description

The function returns eigenvalues for the indicated bar, load case and mode.

III.2.6 IRobotBarDisplacementServer

Class Hierarchy



(c) 2014 Autodesk Inc. All rights reserved

C++

```
interface IRobotBarDisplacementServer : IDispatch;
```

C#

```
public interface IRobotBarDisplacementServer;
```

Visual Basic

```
Public Interface IRobotBarDisplacementServer
```

Description

Server providing access to displacement results for bars. .

Version

Available since version 2.5.

III.2.6.1 IRobotBarDisplacementServer Members

The following tables list the members exposed by IRobotBarDisplacementServer.

Public Methods

	Name	Description
	DynCombValue (see page 883)	Function returns the structure containing displacement values for the indicated bar, load case and mode combinations.
	DynValue (see page 883)	Function returns the structure containing displacement values for the indicated bar, load case and mode.
	Value (see page 884)	Function returns the structure containing displacement values for the indicated bar and load case. .

III.2.6.2 IRobotBarDisplacementServer Methods

The methods of the IRobotBarDisplacementServer class are listed here.

Public Methods

	Name	Description
	DynCombValue (see page 883)	Function returns the structure containing displacement values for the indicated bar, load case and mode combinations.
	DynValue (see page 883)	Function returns the structure containing displacement values for the indicated bar, load case and mode.
	Value (see page 884)	Function returns the structure containing displacement values for the indicated bar and load case. .

III.2.6.2.1 DynCombValue

C++

```
HRESULT DynCombValue(long _bar_num, double _pos, long _case_num, IRobotModeCombinationType
_mode_cmb, IRobotDisplacementData** ret);
```

C#

```
public IRobotDisplacementData DynCombValue(long _bar_num, double _pos, long _case_num,
IRobotModeCombinationType _mode_cmb);
```

Visual Basic

```
Public Function DynCombValue(_bar_num As long, _pos As double, _case_num As long, _mode_cmb
As IRobotModeCombinationType) As IRobotDisplacementData
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Function returns the structure containing displacement values for the indicated bar, load case and mode combinations.

Version

Available since version 2.5.

III.2.6.2.2 DynValue

C++

```
HRESULT DynValue(long _bar_num, double _pos, long _case_num, long _mode,
IRobotDisplacementData** ret);
```

C#

```
public IRobotDisplacementData DynValue(long _bar_num, double _pos, long _case_num, long
_mode);
```

Visual Basic

```
Public Function DynValue(_bar_num As long, _pos As double, _case_num As long, _mode As
long) As IRobotDisplacementData
```

Description

Function returns the structure containing displacement values for the indicated bar, load case and mode.

Version

Available since version 2.5.

III.2.6.2.3 Value

C++

```
HRESULT Value(long _bar_num, double _pos, long _case_num, long _case_cmpnt = 1,
IRobotDisplacementData** ret);
```

C#

```
public IRobotDisplacementData Value(long _bar_num, double _pos, long _case_num, long
_case_cmpnt = 1);
```

Visual Basic

```
Public Function Value(_bar_num As long, _pos As double, _case_num As long, Optional
_case_cmpnt As long = 1) As IRobotDisplacementData
```

Description

Function returns the structure containing displacement values for the indicated bar and load case. .

Version

Available since version 2.5.

III.3 Data structures containing complex calculation results

Enumerations

	Name	Description
	IRobotExtremeValueType (see page 915)	Set of results for which the extreme values are made accessible. .

(c) 2014 Autodesk Inc. All rights reserved

Interfaces

	Name	Description
	IRobotReactionData (see page 885)	The structure describing the value of reactions. .
	IRobotDisplacementData (see page 886)	Structure describing the value of displacements. .
	IRobotBarForceData (see page 888)	Structure describing values of forces, moments and elastic ground reaction for a bar. .
	IRobotBarStressData (see page 892)	A structure includes an information about stresses in a selected bar point for a selected load case.
	IRobotBarDeflectionData (see page 894)	A type of data describing deflections for the selected bar and case in a given bar point.
	IRobotBarBucklingData (see page 897)	The structure describing the eigenvalues for a single member, load case and mode. .
	IRobotEigenvalues (see page 899)	Data structure containing eigenvalues. .
	IRobotForcesData (see page 902)	Data structure containing force and moment values.
	IRobotTimeHistoryResults (see page 904)	Results for time history analysis.
	IRobotExtremeValue (see page 909)	Strucuture providing access to information about the extreme value. .
	IRobotExtremeParams (see page 913)	Strucuture containing parameters for which the extreme value is searched. .
	IRobotFRFResults (see page 918)	Set of results for FRF analysis.
	IRobotFootfallResults (see page 920)	Set of results for Footfall analysis.

III.3.1 IRobotReactionData

Class Hierarchy

RobotOM.IRobotReactionData

C++

```
interface IRobotReactionData : IDispatch;
```

C#

```
public interface IRobotReactionData;
```

Visual Basic

```
Public Interface IRobotReactionData
```

Description

The structure describing the value of reactions. .

III.3.1.1 IRobotReactionData Members

The following tables list the members exposed by IRobotReactionData.

Public Fields

	Name	Description
	FX (see page 885)	
	FY (see page 886)	
	FZ (see page 886)	

(c) 2014 Autodesk Inc. All rights reserved

	MX (see page 886)	
	MY (see page 886)	
	MZ (see page 886)	

III.3.1.2 IRobotReactionData Fields

The fields of the IRobotReactionData class are listed here.

Public Fields

	Name	Description
	FX (see page 885)	
	FY (see page 886)	
	FZ (see page 886)	
	MX (see page 886)	
	MY (see page 886)	
	MZ (see page 886)	

III.3.1.2.1 FX

C++

```
HRESULT get_FX(double*);
```

C#

```
public double FX { get; }
```

Visual Basic

```
Public ReadOnly FX As double
```

III.3.1.2.2 FY

C++

```
HRESULT get_FY(double*);
```

C#

```
public double FY { get; }
```

Visual Basic

```
Public ReadOnly FY As double
```

III.3.1.2.3 FZ

C++

```
HRESULT get_FZ(double*);
```

C#

```
public double FZ { get; }
```

Visual Basic

```
Public ReadOnly FZ As double
```

III.3.1.2.4 MX

C++

```
HRESULT get_MX(double*);
```

C#

```
public double MX { get; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public ReadOnly MX As double
```

III.3.1.2.5 MY

C++

```
HRESULT get_MY(double*);
```

C#

```
public double MY { get; }
```

Visual Basic

```
Public ReadOnly MY As double
```

III.3.1.2.6 MZ

C++

```
HRESULT get_MZ(double*);
```

C#

```
public double MZ { get; }
```

Visual Basic

```
Public ReadOnly MZ As double
```

III.3.2 IRobotDisplacementData

Class Hierarchy

C++

```
interface IRobotDisplacementData : IDispatch;
```

C#

```
public interface IRobotDisplacementData;
```

Visual Basic

```
Public Interface IRobotDisplacementData
```

Description

Structure describing the value of displacements. .

III.3.2.1 IRobotDisplacementData Members

The following tables list the members exposed by IRobotDisplacementData.

Public Fields

	Name	Description
	RX (see page 887)	
	RY (see page 887)	
	RZ (see page 888)	
	UX (see page 888)	
	UY (see page 888)	
	UZ (see page 888)	

(c) 2014 Autodesk Inc. All rights reserved

III.3.2.2 IRobotDisplacementData Fields

The fields of the IRobotDisplacementData class are listed here.

Public Fields

	Name	Description
	RX (see page 887)	
	RY (see page 887)	
	RZ (see page 888)	
	UX (see page 888)	
	UY (see page 888)	
	UZ (see page 888)	

III.3.2.2.1 RX

C++

```
HRESULT get_RX(double*);
```

C#

```
public double RX { get; }
```

Visual Basic

```
Public ReadOnly RX As double
```

III.3.2.2.2 RY

C++

```
HRESULT get_RY(double*);
```

C#

```
public double RY { get; }
```

Visual Basic

```
Public ReadOnly RY As double
```

III.3.2.2.3 RZ

C++

```
HRESULT get_RZ(double*);
```

C#

```
public double RZ { get; }
```

Visual Basic

```
Public ReadOnly RZ As double
```

III.3.2.2.4 UX

C++

```
HRESULT get_UX(double*);
```

C#

```
public double UX { get; }
```

Visual Basic

```
Public ReadOnly UX As double
```

(c) 2014 Autodesk Inc. All rights reserved

III.3.2.2.5 UY

C++

```
HRESULT get_UY(double*);
```

C#

```
public double UY { get; }
```

Visual Basic

```
Public ReadOnly UY As double
```

III.3.2.2.6 UZ

C++

```
HRESULT get_UZ(double*);
```

C#

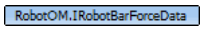
```
public double UZ { get; }
```

Visual Basic

```
Public ReadOnly UZ As double
```

III.3.3 IRobotBarForceData

Class Hierarchy



C++

```
interface IRobotBarForceData : IDispatch;
```

C#

```
public interface IRobotBarForceData;
```

Visual Basic

```
Public Interface IRobotBarForceData
```

Description

Structure describing values of forces, moments and elastic ground reaction for a bar. .

III.3.3.1 IRobotBarForceData Members

The following tables list the members exposed by IRobotBarForceData.

Public Fields

	Name	Description
	FX (see page 889)	
	FY (see page 890)	
	FZ (see page 890)	
	KY (see page 890)	
	KYAvailable (see page 890)	Flag indicating availability of the value of elastic ground reaction KY (see page 890).
	KZ (see page 891)	
	KZAvailable (see page 891)	Flag indicating availability of the value of elastic ground reaction KZ (see page 891).
	MX (see page 891)	

(c) 2014 Autodesk Inc. All rights reserved

	MY (see page 891)	
	MZ (see page 891)	

III.3.3.2 IRobotBarForceData Fields

The fields of the IRobotBarForceData class are listed here.

Public Fields

	Name	Description
	FX (see page 889)	
	FY (see page 890)	
	FZ (see page 890)	
	KY (see page 890)	
	KYAvailable (see page 890)	Flag indicating availability of the value of elastic ground reaction KY (see page 890).
	KZ (see page 891)	
	KZAvailable (see page 891)	Flag indicating availability of the value of elastic ground reaction KZ (see page 891).
	MX (see page 891)	
	MY (see page 891)	
	MZ (see page 891)	

III.3.3.2.1 FX

C++

```
HRESULT get_FX(double*);
```

C#

```
public double FX { get; }
```

Visual Basic

```
Public ReadOnly FX As double
```

III.3.3.2.2 FY

C++

```
HRESULT get_FY(double*);
```

C#

```
public double FY { get; }
```

Visual Basic

```
Public ReadOnly FY As double
```

III.3.3.2.3 FZ

C++

```
HRESULT get_FZ(double*);
```

C#

```
public double FZ { get; }
```

Visual Basic

```
Public ReadOnly FZ As double
```

(c) 2014 Autodesk Inc. All rights reserved

III.3.3.2.4 KY

C++

```
HRESULT get_KY(double*);
```

C#

```
public double KY { get; }
```

Visual Basic

```
Public ReadOnly KY As double
```

Version

Available since version 3.5.

III.3.3.2.5 KYAvailable

C++

```
HRESULT get_KYAvailable(VARIANT_BOOL*);
```

C#

```
public bool KYAvailable { get; }
```

Visual Basic

```
Public ReadOnly KYAvailable As Boolean
```

Description

Flag indicating availability of the value of elastic ground reaction KY (see page 890).

Version

Available since version 3.5.

III.3.3.2.6 KZ

C++

```
HRESULT get_KZ(double*);
```

C#

```
public double KZ { get; }
```

Visual Basic

```
Public ReadOnly KZ As double
```

Version

Available since version 3.5.

III.3.3.2.7 KZAvailable

C++

```
HRESULT get_KZAvailable(VARIANT_BOOL*);
```

C#

```
public bool KZAvailable { get; }
```

Visual Basic

```
Public ReadOnly KZAvailable As Boolean
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Flag indicating availability of the value of elastic ground reaction KZ (see page 891).

Version

Available since version 3.5.

III.3.3.2.8 MX

C++

```
HRESULT get_MX(double*);
```

C#

```
public double MX { get; }
```

Visual Basic

```
Public ReadOnly MX As double
```

III.3.3.2.9 MY

C++

```
HRESULT get_MY(double*);
```

C#

```
public double MY { get; }
```

Visual Basic

```
Public ReadOnly MY As double
```

III.3.3.2.10 MZ

C++

```
HRESULT get_MZ(double*);
```

C#

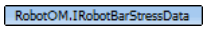
```
public double MZ { get; }
```

Visual Basic

```
Public ReadOnly MZ As double
```

III.3.4 IRobotBarStressData

Class Hierarchy



C++

```
interface IRobotBarStressData : IDispatch;
```

C#

```
public interface IRobotBarStressData;
```

Visual Basic

```
Public Interface IRobotBarStressData
```

Description

A structure includes an information about stresses in a selected bar point for a selected load case.

(c) 2014 Autodesk Inc. All rights reserved

III.3.4.1 IRobotBarStressData Members

The following tables list the members exposed by IRobotBarStressData.

Public Fields

	Name	Description
	FXSX (see page 893)	
	ShearY (see page 893)	
	ShearZ (see page 893)	
	Smax (see page 893)	
	SmaxMY (see page 893)	
	SmaxMZ (see page 894)	
	Smin (see page 894)	
	SminMY (see page 894)	
	SminMZ (see page 894)	
	Torsion (see page 894)	

III.3.4.2 IRobotBarStressData Fields

The fields of the IRobotBarStressData class are listed here.

Public Fields

	Name	Description
	FXSX (see page 893)	
	ShearY (see page 893)	
	ShearZ (see page 893)	
	Smax (see page 893)	
	SmaxMY (see page 893)	
	SmaxMZ (see page 894)	
	Smin (see page 894)	
	SminMY (see page 894)	
	SminMZ (see page 894)	
	Torsion (see page 894)	

III.3.4.2.1 FXSX

C++

```
HRESULT get_FXSX(double*);
```

C#

```
public double FXSX { get; }
```

Visual Basic

```
Public ReadOnly FXSX As double
```

III.3.4.2.2 ShearY

C++

```
HRESULT get_ShearY(double*);
```

C#

```
public double ShearY { get; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public ReadOnly ShearY As double
```

III.3.4.2.3 ShearZ

C++

```
HRESULT get_ShearZ(double*);
```

C#

```
public double ShearZ { get; }
```

Visual Basic

```
Public ReadOnly ShearZ As double
```

III.3.4.2.4 Smax

C++

```
HRESULT get_Smax(double*);
```

C#

```
public double Smax { get; }
```

Visual Basic

```
Public ReadOnly Smax As double
```

III.3.4.2.5 SmaxMY

C++

```
HRESULT get_SmaxMY(double*);
```

C#

```
public double SmaxMY { get; }
```

Visual Basic

```
Public ReadOnly SmaxMY As double
```

III.3.4.2.6 SmaxMZ

C++

```
HRESULT get_SmaxMZ(double*);
```

C#

```
public double SmaxMZ { get; }
```

Visual Basic

```
Public ReadOnly SmaxMZ As double
```

III.3.4.2.7 Smin

C++

```
HRESULT get_Smin(double*);
```

C#

```
public double Smin { get; }
```

Visual Basic

```
Public ReadOnly Smin As double
```

(c) 2014 Autodesk Inc. All rights reserved

III.3.4.2.8 SminMY

C++

```
HRESULT get_SminMY(double*);
```

C#

```
public double SminMY { get; }
```

Visual Basic

```
Public ReadOnly SminMY As double
```

III.3.4.2.9 SminMZ

C++

```
HRESULT get_SminMZ(double*);
```

C#

```
public double SminMZ { get; }
```

Visual Basic

```
Public ReadOnly SminMZ As double
```

III.3.4.2.10 Torsion

C++

```
HRESULT get_Torsion(double*);
```

C#

```
public double Torsion { get; }
```

Visual Basic

```
Public ReadOnly Torsion As double
```

III.3.5 IRobotBarDeflectionData

Class Hierarchy

C++

```
interface IRobotBarDeflectionData : IDispatch;
```

C#

```
public interface IRobotBarDeflectionData;
```

Visual Basic

```
Public Interface IRobotBarDeflectionData
```

Description

A type of data describing deflections for the selected bar and case in a given bar point.

III.3.5.1 IRobotBarDeflectionData Members

The following tables list the members exposed by IRobotBarDeflectionData.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	PosUX (see page 895)	A relative point position on a bar for which UX (see page 896) value was read .
	PosUY (see page 896)	A relative point position for which UY (see page 896) value was read.
	PosUZ (see page 896)	A relative point position for which UZ (see page 896) value was read.
	UX (see page 896)	
	UY (see page 896)	
	UZ (see page 896)	

III.3.5.2 IRobotBarDeflectionData Fields

The fields of the IRobotBarDeflectionData class are listed here.

Public Fields

	Name	Description
	PosUX (see page 895)	A relative point position on a bar for which UX (see page 896) value was read .
	PosUY (see page 896)	A relative point position for which UY (see page 896) value was read.
	PosUZ (see page 896)	A relative point position for which UZ (see page 896) value was read.
	UX (see page 896)	
	UY (see page 896)	
	UZ (see page 896)	

III.3.5.2.1 PosUX

C++

```
HRESULT get_PosUX(double*);
```

C#

```
public double PosUX { get; }
```

Visual Basic

```
Public ReadOnly PosUX As double
```

Description

A relative point position on a bar for which UX (see page 896) value was read .

III.3.5.2.2 PosUY

C++

```
HRESULT get_PosUY(double*);
```

C#

```
public double PosUY { get; }
```

Visual Basic

```
Public ReadOnly PosUY As double
```

Description

A relative point position for which UY (see page 896) value was read.

(c) 2014 Autodesk Inc. All rights reserved

III.3.5.2.3 PosUZ

C++

```
HRESULT get_PosUZ(double*);
```

C#

```
public double PosUZ { get; }
```

Visual Basic

```
Public ReadOnly PosUZ As double
```

Description

A relative point position for which UZ (see page 896) value was read.

III.3.5.2.4 UX

C++

```
HRESULT get_UX(double*);
```

C#

```
public double UX { get; }
```

Visual Basic

```
Public ReadOnly UX As double
```

III.3.5.2.5 UY

C++

```
HRESULT get_UY(double*);
```

C#

```
public double UY { get; }
```

Visual Basic

```
Public ReadOnly UY As double
```

III.3.5.2.6 UZ

C++

```
HRESULT get_UZ(double*);
```

C#

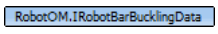
```
public double UZ { get; }
```

Visual Basic

```
Public ReadOnly UZ As double
```

III.3.6 IRobotBarBucklingData

Class Hierarchy



C++

```
interface IRobotBarBucklingData : IDispatch;
```

C#

```
public interface IRobotBarBucklingData;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Interface IRobotBarBucklingData
```

Description

The structure describing the eigenvalues for a single member, load case and mode. .

III.3.6.1 IRobotBarBucklingData Members

The following tables list the members exposed by IRobotBarBucklingData.

Public Fields

	Name	Description
	BuckLengthY (see page 898)	Buckling length Y.
	BuckLengthZ (see page 898)	Buckling length Z.
	CriticalCoef (see page 898)	Critical coefficient .
	CriticalForce (see page 898)	Critical force.
	SlendY (see page 899)	Slenderness Y.
	SlendZ (see page 899)	Slenderness Z.

III.3.6.2 IRobotBarBucklingData Fields

The fields of the IRobotBarBucklingData class are listed here.

Public Fields

	Name	Description
	BuckLengthY (see page 898)	Buckling length Y.
	BuckLengthZ (see page 898)	Buckling length Z.
	CriticalCoef (see page 898)	Critical coefficient .
	CriticalForce (see page 898)	Critical force.
	SlendY (see page 899)	Slenderness Y.
	SlendZ (see page 899)	Slenderness Z.

III.3.6.2.1 BuckLengthY

C++

```
HRESULT get_BuckLengthY(double*);
```

C#

```
public double BuckLengthY { get; }
```

Visual Basic

```
Public ReadOnly BuckLengthY As double
```

Description

Buckling length Y.

III.3.6.2.2 BuckLengthZ

C++

```
HRESULT get_BuckLengthZ(double*);
```

C#

```
public double BuckLengthZ { get; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public ReadOnly BuckLengthZ As double
```

Description

Buckling length Z.

III.3.6.2.3 CriticalCoef

C++

```
HRESULT get_CriticalCoef(double*);
```

C#

```
public double CriticalCoef { get; }
```

Visual Basic

```
Public ReadOnly CriticalCoef As double
```

Description

Critical coefficient .

III.3.6.2.4 CriticalForce

C++

```
HRESULT get_CriticalForce(double*);
```

C#

```
public double CriticalForce { get; }
```

Visual Basic

```
Public ReadOnly CriticalForce As double
```

Description

Critical force.

III.3.6.2.5 SlendY

C++

```
HRESULT get_SlendY(double*);
```

C#

```
public double SlendY { get; }
```

Visual Basic

```
Public ReadOnly SlendY As double
```

Description

Slenderness Y.

III.3.6.2.6 SlendZ

C++

```
HRESULT get_SlendZ(double*);
```

C#

```
public double SlendZ { get; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

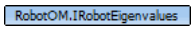
```
Public ReadOnly SlendZ As double
```

Description

Slenderness Z.

III.3.7 IRobotEigenvalues

Class Hierarchy



C++

```
interface IRobotEigenvalues : IDispatch;
```

C#

```
public interface IRobotEigenvalues;
```

Visual Basic

```
Public Interface IRobotEigenvalues
```

Description

Data structure containing eigenvalues. .

III.3.7.1 IRobotEigenvalues Members

The following tables list the members exposed by IRobotEigenvalues.

Public Fields

	Name	Description
	AvPartCoeff (see page 900)	Available since version 1.7.
	Damping (see page 900)	Available since version 1.7.
	EigenValue (see page 901)	Available since version 1.7.
	Energy (see page 901)	Available since version 1.7.
	Frequence (see page 901)	Available since version 1.7.
	Period (see page 901)	Available since version 1.7.
	Precision (see page 902)	Available since version 1.7.
	Pulsation (see page 902)	Available since version 1.7.

III.3.7.2 IRobotEigenvalues Fields

The fields of the IRobotEigenvalues class are listed here.

Public Fields

	Name	Description
	AvPartCoeff (see page 900)	Available since version 1.7.
	Damping (see page 900)	Available since version 1.7.
	EigenValue (see page 901)	Available since version 1.7.
	Energy (see page 901)	Available since version 1.7.
	Frequence (see page 901)	Available since version 1.7.
	Period (see page 901)	Available since version 1.7.
	Precision (see page 902)	Available since version 1.7.
	Pulsation (see page 902)	Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

III.3.7.2.1 AvPartCoeff

C++

```
HRESULT get_AvPartCoeff(double*);
```

C#

```
public double AvPartCoeff { get; }
```

Visual Basic

```
Public ReadOnly AvPartCoeff As double
```

Description

Available since version 1.7.

III.3.7.2.2 Damping

C++

```
HRESULT get_Damping(double*);
```

C#

```
public double Damping { get; }
```

Visual Basic

```
Public ReadOnly Damping As double
```

Description

Available since version 1.7.

III.3.7.2.3 EigenValue

C++

```
HRESULT get_EigenValue(double*);
```

C#

```
public double EigenValue { get; }
```

Visual Basic

```
Public ReadOnly EigenValue As double
```

Description

Available since version 1.7.

III.3.7.2.4 Energy

C++

```
HRESULT get_Energy(double*);
```

C#

```
public double Energy { get; }
```

Visual Basic

```
Public ReadOnly Energy As double
```

Description

Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

III.3.7.2.5 Frequence

C++

```
HRESULT get_Frequence(double*);
```

C#

```
public double Frequence { get; }
```

Visual Basic

```
Public ReadOnly Frequence As double
```

Description

Available since version 1.7.

III.3.7.2.6 Period

C++

```
HRESULT get_Period(double*);
```

C#

```
public double Period { get; }
```

Visual Basic

```
Public ReadOnly Period As double
```

Description

Available since version 1.7.

III.3.7.2.7 Precision

C++

```
HRESULT get_Precision(double*);
```

C#

```
public double Precision { get; }
```

Visual Basic

```
Public ReadOnly Precision As double
```

Description

Available since version 1.7.

III.3.7.2.8 Pulsation

C++

```
HRESULT get_Pulsation(double*);
```

C#

```
public double Pulsation { get; }
```

Visual Basic

```
Public ReadOnly Pulsation As double
```

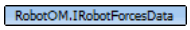
Description

Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

III.3.8 IRobotForcesData

Class Hierarchy



C++

```
interface IRobotForcesData : IDispatch;
```

C#

```
public interface IRobotForcesData;
```

Visual Basic

```
Public Interface IRobotForcesData
```

Description

Data structure containing force and moment values.

III.3.8.1 IRobotForcesData Members

The following tables list the members exposed by IRobotForcesData.

Public Fields

	Name	Description
	FX (see page 903)	Available since version 1.7.
	FY (see page 903)	Available since version 1.7.
	FZ (see page 903)	Available since version 1.7.
	MX (see page 904)	Available since version 1.7.
	MY (see page 904)	Available since version 1.7.
	MZ (see page 904)	Available since version 1.7.

III.3.8.2 IRobotForcesData Fields

The fields of the IRobotForcesData class are listed here.

Public Fields

	Name	Description
	FX (see page 903)	Available since version 1.7.
	FY (see page 903)	Available since version 1.7.
	FZ (see page 903)	Available since version 1.7.
	MX (see page 904)	Available since version 1.7.
	MY (see page 904)	Available since version 1.7.
	MZ (see page 904)	Available since version 1.7.

III.3.8.2.1 FX

C++

```
HRESULT get_FX(double*);
```

C#

```
public double FX { get; }
```

Visual Basic

```
Public ReadOnly FX As double
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Available since version 1.7.

III.3.8.2.2 FY

C++

```
HRESULT get_FY(double*);
```

C#

```
public double FY { get; }
```

Visual Basic

```
Public ReadOnly FY As double
```

Description

Available since version 1.7.

III.3.8.2.3 FZ

C++

```
HRESULT get_FZ(double*);
```

C#

```
public double FZ { get; }
```

Visual Basic

```
Public ReadOnly FZ As double
```

Description

Available since version 1.7.

III.3.8.2.4 MX

C++

```
HRESULT get_MX(double*);
```

C#

```
public double MX { get; }
```

Visual Basic

```
Public ReadOnly MX As double
```

Description

Available since version 1.7.

III.3.8.2.5 MY

C++

```
HRESULT get_MY(double*);
```

C#

```
public double MY { get; }
```

Visual Basic

```
Public ReadOnly MY As double
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Available since version 1.7.

III.3.8.2.6 MZ

C++

```
HRESULT get_MZ(double*);
```

C#

```
public double MZ { get; }
```

Visual Basic

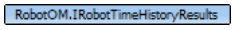
```
Public ReadOnly MZ As double
```

Description

Available since version 1.7.

III.3.9 IRobotTimeHistoryResults

Class Hierarchy



C++

```
interface IRobotTimeHistoryResults : IDispatch;
```

C#

```
public interface IRobotTimeHistoryResults;
```

Visual Basic

```
Public Interface IRobotTimeHistoryResults
```

Description

Results for time history analysis.

Version

Available since version 3.

III.3.9.1 IRobotTimeHistoryResults Members

The following tables list the members exposed by IRobotTimeHistoryResults.

Public Fields

	Name	Description
	ARX (see page 905)	Value of acceleration component ARX.
	ARY (see page 906)	Value of acceleration component ARY.
	ARZ (see page 906)	Value of acceleration component ARZ.
	AX (see page 906)	Value of acceleration component AX.
	AY (see page 907)	Value of acceleration component AY.
	AZ (see page 907)	Value of acceleration component AZ.
	Time (see page 907)	Time corresponding to the specified step.
	VRX (see page 908)	Value of velocity component VRX.
	VRY (see page 908)	Value of velocity component VRY.
	VRZ (see page 908)	Value of velocity component VRZ.

(c) 2014 Autodesk Inc. All rights reserved

	VX (see page 908)	Value of velocity component VX.
	VY (see page 909)	Value of velocity component VY.
	VZ (see page 909)	Value of velocity component VZ.

III.3.9.2 IRobotTimeHistoryResults Fields

The fields of the IRobotTimeHistoryResults class are listed here.

Public Fields

	Name	Description
	ARX (see page 905)	Value of acceleration component ARX.
	ARY (see page 906)	Value of acceleration component ARY.
	ARZ (see page 906)	Value of acceleration component ARZ.
	AX (see page 906)	Value of acceleration component AX.
	AY (see page 907)	Value of acceleration component AY.
	AZ (see page 907)	Value of acceleration component AZ.
	Time (see page 907)	Time corresponding to the specified step.
	VRX (see page 908)	Value of velocity component VRX.
	VRY (see page 908)	Value of velocity component VRY.
	VRZ (see page 908)	Value of velocity component VRZ.
	VX (see page 908)	Value of velocity component VX.
	VY (see page 909)	Value of velocity component VY.
	VZ (see page 909)	Value of velocity component VZ.

III.3.9.2.1 ARX

C++

```
HRESULT get_ARX( double*);
```

C#

```
public  double ARX { get; }
```

Visual Basic

```
Public ReadOnly ARX As  double
```

Description

Value of acceleration component ARX.

Version

Available since version 3.

III.3.9.2.2 ARY

C++

```
HRESULT get_ARY( double*);
```

C#

```
public  double ARY { get; }
```

Visual Basic

```
Public ReadOnly ARY As  double
```

Description

Value of acceleration component ARY.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.

III.3.9.2.3 ARZ

C++

```
HRESULT get_ARZ( double*);
```

C#

```
public  double ARZ { get; }
```

Visual Basic

```
Public ReadOnly ARZ As  double
```

Description

Value of acceleration component ARZ.

Version

Available since version 3.

III.3.9.2.4 AX

C++

```
HRESULT get_AX( double*);
```

C#

```
public  double AX { get; }
```

Visual Basic

```
Public ReadOnly AX As  double
```

Description

Value of acceleration component AX.

Version

Available since version 3.

III.3.9.2.5 AY

C++

```
HRESULT get_AY( double*);
```

C#

```
public  double AY { get; }
```

Visual Basic

```
Public ReadOnly AY As  double
```

Description

Value of acceleration component AY.

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

III.3.9.2.6 AZ

C++

```
HRESULT get_AZ( double*);
```

C#

```
public  double AZ { get; }
```

Visual Basic

```
Public ReadOnly AZ As  double
```

Description

Value of acceleration component AZ.

Version

Available since version 3.

III.3.9.2.7 Time

C++

```
HRESULT get_Time( double*);
```

C#

```
public  double Time { get; }
```

Visual Basic

```
Public ReadOnly Time As  double
```

Description

Time corresponding to the specified step.

Version

Available since version 3.

III.3.9.2.8 VRX

C++

```
HRESULT get_VRX( double*);
```

C#

```
public  double VRX { get; }
```

Visual Basic

```
Public ReadOnly VRX As  double
```

Description

Value of velocity component VRX.

Version

Available since version 3.

III.3.9.2.9 VRY

C++

```
HRESULT get_VRY( double*);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public  double VRY { get; }
```

Visual Basic

```
Public ReadOnly VRY As  double
```

Description

Value of velocity component VRY.

Version

Available since version 3.

III.3.9.2.10 VRZ

C++

```
HRESULT get_VRZ( double*);
```

C#

```
public  double VRZ { get; }
```

Visual Basic

```
Public ReadOnly VRZ As  double
```

Description

Value of velocity component VRZ.

Version

Available since version 3.

III.3.9.2.11 VX

C++

```
HRESULT get_VX( double*);
```

C#

```
public  double VX { get; }
```

Visual Basic

```
Public ReadOnly VX As  double
```

Description

Value of velocity component VX.

Version

Available since version 3.

III.3.9.2.12 VY

C++

```
HRESULT get_VY( double*);
```

C#

```
public  double VY { get; }
```

Visual Basic

```
Public ReadOnly VY As  double
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Value of velocity component VY.

Version

Available since version 3.

III.3.9.2.13 VZ

C++

```
HRESULT get_VZ( double*);
```

C#

```
public  double VZ { get; }
```

Visual Basic

```
Public ReadOnly VZ As  double
```

Description

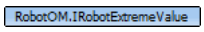
Value of velocity component VZ.

Version

Available since version 3.

III.3.10 IRobotExtremeValue

Class Hierarchy



C++

```
interface IRobotExtremeValue : IDispatch;
```

C#

```
public interface IRobotExtremeValue;
```

Visual Basic

```
Public Interface IRobotExtremeValue
```

Description

Strucuture providing access to information about the extreme value. .

Version

Available since version 3.5.

III.3.10.1 IRobotExtremeValue Members

The following tables list the members exposed by IRobotExtremeValue.

Public Fields

	Name	Description
	Bar (see page 911)	Number of the bar for which the extreme value has been obtained .
	Case (see page 911)	Number of the load case for which the specified extreme value has been obtained .

(c) 2014 Autodesk Inc. All rights reserved

	CaseCmpnt (see page 911)	Additional number indicating a load case component • for a code combination, it is a number of its component • for modal analysis, it is a mode number • for results of time history analysis, it is a step number.
	IsAvailable (see page 912)	Flag indicating if the extreme value has been taken correctly from the result server .
	ModeCmb (see page 912)	Designation of the mode combination for which the specified extreme value has been obtained (significant only when CaseCmpnt (see page 911) component equals zero) .
	Node (see page 912)	Number of the node in which the extreme value has been obtained .
	Position (see page 912)	Relative position on a bar in which the extreme value has been obtained .
	Value (see page 913)	Value.

III.3.10.2 IRobotExtremeValue Fields

The fields of the IRobotExtremeValue class are listed here.

Public Fields

	Name	Description
	Bar (see page 911)	Number of the bar for which the extreme value has been obtained .
	Case (see page 911)	Number of the load case for which the specified extreme value has been obtained .
	CaseCmpnt (see page 911)	Additional number indicating a load case component • for a code combination, it is a number of its component • for modal analysis, it is a mode number • for results of time history analysis, it is a step number.
	IsAvailable (see page 912)	Flag indicating if the extreme value has been taken correctly from the result server .
	ModeCmb (see page 912)	Designation of the mode combination for which the specified extreme value has been obtained (significant only when CaseCmpnt (see page 911) component equals zero) .
	Node (see page 912)	Number of the node in which the extreme value has been obtained .
	Position (see page 912)	Relative position on a bar in which the extreme value has been obtained .
	Value (see page 913)	Value.

III.3.10.2.1 Bar

C++

```
HRESULT get_Bar(long*);
```

C#

```
public long Bar { get; }
```

Visual Basic

```
Public ReadOnly Bar As long
```

Description

Number of the bar for which the extreme value has been obtained .

Version

Available since version 3.5.

(c) 2014 Autodesk Inc. All rights reserved

III.3.10.2.2 Case

C++

```
HRESULT get_Case(long*);
```

C#

```
public long Case { get; }
```

Visual Basic

```
Public ReadOnly Case As long
```

Description

Number of the load case for which the specified extreme value has been obtained .

Version

Available since version 3.5.

III.3.10.2.3 CaseCmpnt

C++

```
HRESULT get_CaseCmpnt(long*);
```

C#

```
public long CaseCmpnt { get; }
```

Visual Basic

```
Public ReadOnly CaseCmpnt As long
```

Description

Additional number indicating a load case component

- for a code combination, it is a number of its component
- for modal analysis, it is a mode number
- for results of time history analysis, it is a step number.

Version

Available since version 3.5.

III.3.10.2.4 IsAvailable

C++

```
HRESULT get_IsAvailable(VARIANT_BOOL*);
```

C#

```
public bool IsAvailable { get; }
```

Visual Basic

```
Public ReadOnly IsAvailable As Boolean
```

Description

Flag indicating if the extreme value has been taken correctly from the result server .

Version

Available since version 3.5.

(c) 2014 Autodesk Inc. All rights reserved

III.3.10.2.5 ModeCmb

C++

```
HRESULT get_ModeCmb(IRobotModeCombinationType*);
```

C#

```
public IRobotModeCombinationType ModeCmb { get; }
```

Visual Basic

```
Public ReadOnly ModeCmb As IRobotModeCombinationType
```

Description

Designation of the mode combination for which the specified extreme value has been obtained (significant only when CaseCmpnt (see page 911) component equals zero) .

Version

Available since version 3.5.

III.3.10.2.6 Node

C++

```
HRESULT get_Node(long*);
```

C#

```
public long Node { get; }
```

Visual Basic

```
Public ReadOnly Node As long
```

Description

Number of the node in which the extreme value has been obtained .

Version

Available since version 3.5.

III.3.10.2.7 Position

C++

```
HRESULT get_Position(double*);
```

C#

```
public double Position { get; }
```

Visual Basic

```
Public ReadOnly Position As double
```

Description

Relative position on a bar in which the extreme value has been obtained .

Version

Available since version 3.5.

III.3.10.2.8 Value

C++

```
HRESULT get_Value(double*);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double Value { get; }
```

Visual Basic

```
Public ReadOnly Value As double
```

Description

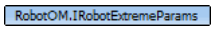
Value.

Version

Available since version 3.5.

III.3.11 IRobotExtremeParams

Class Hierarchy



C++

```
interface IRobotExtremeParams : IDispatch;
```

C#

```
public interface IRobotExtremeParams;
```

Visual Basic

```
Public Interface IRobotExtremeParams
```

Description

Strucuture containing parameters for which the extreme value is searched. .

Version

Available since version 3.5.

III.3.11.1 IRobotExtremeParams Members

The following tables list the members exposed by IRobotExtremeParams.

Public Fields

	Name	Description
	BarDivision (see page 914)	Number of bar divisions - significant only for bar results.
	Selection (see page 914)	Selection of structure components (nodes, bars, load cases) that should be considered while searching the extreme value.
	ValueType (see page 915)	Type of the result for which the extreme value is to be taken .

III.3.11.2 IRobotExtremeParams Fields

The fields of the IRobotExtremeParams class are listed here.

Public Fields

	Name	Description
	BarDivision (see page 914)	Number of bar divisions - significant only for bar results.
	Selection (see page 914)	Selection of structure components (nodes, bars, load cases) that should be considered while searching the extreme value.
	ValueType (see page 915)	Type of the result for which the extreme value is to be taken .

(c) 2014 Autodesk Inc. All rights reserved

III.3.11.2.1 BarDivision

C++

```
HRESULT get_BarDivision(long*);
HRESULT put_BarDivision(long);
```

C#

```
public long BarDivision { get; set; }
```

Visual Basic

```
Public BarDivision As long
```

Description

Number of bar divisions - significant only for bar results.

Version

Available since version 3.5.

III.3.11.2.2 Selection

C++

```
HRESULT get_Selection(IRobotMultiSelection**);
```

C#

```
public IRobotMultiSelection Selection { get; }
```

Visual Basic

```
Public ReadOnly Selection As IRobotMultiSelection
```

Description

Selection of structure components (nodes, bars, load cases) that should be considered while searching the extreme value.

Version

Available since version 3.5.

III.3.11.2.3 ValueType

C++

```
HRESULT get_ValueType(IRobotExtremeValueType*);
HRESULT put_ValueType(IRobotExtremeValueType);
```

C#

```
public IRobotExtremeValueType ValueType { get; set; }
```

Visual Basic

```
Public ValueType As IRobotExtremeValueType
```

Description

Type of the result for which the extreme value is to be taken .

Version

Available since version 3.5.

III.3.12 IRobotExtremeValueType

C++

```
enum IRobotExtremeValueType;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public enum IRobotExtremeValueType;
```

Visual Basic

```
Public Enum IRobotExtremeValueType
```

Members

Members	Description
I_EVT_REACTION_FX = 109	Available since version 3.5.
I_EVT_REACTION_FY = 110	Available since version 3.5.
I_EVT_REACTION_FZ = 111	Available since version 3.5.
I_EVT_REACTION_MX = 112	Available since version 3.5.
I_EVT_REACTION_MY = 113	Available since version 3.5.
I_EVT_REACTION_MZ = 114	Available since version 3.5.
I_EVT_DISPLACEMENT_NODE_UX = 141	Available since version 3.5.
I_EVT_DISPLACEMENT_NODE_UY = 142	Available since version 3.5.
I_EVT_DISPLACEMENT_NODE_UZ = 143	Available since version 3.5.
I_EVT_DISPLACEMENT_NODE_RX = 144	Available since version 3.5.
I_EVT_DISPLACEMENT_NODE_RY = 145	Available since version 3.5.
I_EVT_DISPLACEMENT_NODE_RZ = 146	Available since version 3.5.
I_EVT_DEFLECTION_UX = 153	Available since version 3.5.
I_EVT_DEFLECTION_UY = 154	Available since version 3.5.
I_EVT_DEFLECTION_UZ = 155	Available since version 3.5.
I_EVT_DISPLACEMENT_BAR_UX = 147	Available since version 3.5.
I_EVT_DISPLACEMENT_BAR_UY = 148	Available since version 3.5.
I_EVT_DISPLACEMENT_BAR_UZ = 149	Available since version 3.5.
I_EVT_DISPLACEMENT_BAR_RX = 150	Available since version 3.5.
I_EVT_DISPLACEMENT_BAR_RY = 151	Available since version 3.5.
I_EVT_DISPLACEMENT_BAR_RZ = 152	Available since version 3.5.
I_EVT_FORCE_BAR_FX = 165	Available since version 3.5.
I_EVT_FORCE_BAR_FY = 166	Available since version 3.5.
I_EVT_FORCE_BAR_FZ = 167	Available since version 3.5.
I_EVT_FORCE_BAR_MX = 168	Available since version 3.5.
I_EVT_FORCE_BAR_MY = 169	Available since version 3.5.
I_EVT_FORCE_BAR_MZ = 170	Available since version 3.5.
I_EVT_STRESS_BAR_SMAX = 171	Available since version 3.5.
I_EVT_STRESS_BAR_SMIN = 172	Available since version 3.5.
I_EVT_STRESS_BAR_SMAX_MY = 174	Available since version 3.5.
I_EVT_STRESS_BAR_SMAX_MZ = 175	Available since version 3.5.
I_EVT_STRESS_BAR_SMIN_MY = 176	Available since version 3.5.
I_EVT_STRESS_BAR_SMIN_MZ = 177	Available since version 3.5.
I_EVT_STRESS_BAR_FX_SX = 173	Available since version 3.5.
I_EVT_STRESS_BAR_TY = 179	Available since version 3.5.
I_EVT_STRESS_BAR_TZ = 180	Available since version 3.5.
I_EVT_STRESS_BAR_T = 182	Available since version 3.5.
I_EVT_PSEUDOSTATIC_FORCE_FX = 251	Available since version 3.5.
I_EVT_PSEUDOSTATIC_FORCE_FY = 252	Available since version 3.5.
I_EVT_PSEUDOSTATIC_FORCE_FZ = 253	Available since version 3.5.

(c) 2014 Autodesk Inc. All rights reserved

I_EVT_PSEUDOSTATIC_FORCE_MX = 254	Available since version 3.5.
I_EVT_PSEUDOSTATIC_FORCE_MY = 255	Available since version 3.5.
I_EVT_PSEUDOSTATIC_FORCE_MZ = 256	Available since version 3.5.
I_EVT_PSEUDOSTATIC_FORCE_GX = 1328	Available since version 3.5.
I_EVT_PSEUDOSTATIC_FORCE_GY = 1329	Available since version 3.5.
I_EVT_PSEUDOSTATIC_FORCE_GZ = 1330	Available since version 3.5.
I_EVT_PSEUDOSTATIC_FORCE_TX = 1331	Available since version 3.5.
I_EVT_PSEUDOSTATIC_FORCE_TY = 1332	Available since version 3.5.
I_EVT_PSEUDOSTATIC_FORCE_TZ = 1333	Available since version 3.5.
I_EVT_TIME_VX = 257	Available since version 3.5.
I_EVT_TIME_VY = 258	Available since version 3.5.
I_EVT_TIME_VZ = 259	Available since version 3.5.
I_EVT_TIME_VRX = 260	Available since version 3.5.
I_EVT_TIME_VRY = 261	Available since version 3.5.
I_EVT_TIME_VRZ = 262	Available since version 3.5.
I_EVT_TIME_AX = 263	Available since version 3.5.
I_EVT_TIME_AY = 264	Available since version 3.5.
I_EVT_TIME_AZ = 265	Available since version 3.5.
I_EVT_TIME_ARX = 266	Available since version 3.5.
I_EVT_TIME_ARY = 267	Available since version 3.5.
I_EVT_TIME_ARZ = 268	Available since version 3.5.
I_EVT_FRF_VX = 1783	VX velocity for FRF analysis. Available since version 9.7.
I_EVT_FRF_VY = 1784	VY velocity for FRF analysis. Available since version 9.7.
I_EVT_FRF_VZ = 1785	VZ velocity for FRF analysis. Available since version 9.7.
I_EVT_FOOTFALL_A = 1854	Overall maximum acceleration. Available since version 9.7.
I_EVT_FOOTFALL_RF_RESONANT = 1855	Maximum (resonant) response factor. Available since version 9.7.
I_EVT_FOOTFALL_RF_TRANSIENT = 1856	Maximum (transient) response factor. Available since version 9.7.
I_EVT_FOOTFALL_RF_OVERALL = 1857	Maximum (overall) response factor. Available since version 9.7.
I_EVT_FOOTFALL_VRMS = 1858	Available since version 9.7.
I_EVT_FOOTFALL_VRMQ = 1859	Available since version 9.7.
I_EVT_FOOTFALL_FREQUENCY = 1862	Available since version 9.7.
I_EVT_FOOTFALL_EXCITATION_NODE = 1861	Available since version 9.7.
I_EVT_MODAL_FREQUENCY = 404	Available since version 11.
I_EVT_MODAL_PERIOD = 405	Available since version 11.
I_EVT_MODAL_PULSATION = 406	Available since version 11.
I_EVT_MODAL_CURMASS_UX = 184	Available since version 11.
I_EVT_MODAL_CURMASS_UY = 185	Available since version 11.
I_EVT_MODAL_CURMASS_UZ = 186	Available since version 11.
I_EVT_MODAL_CURMASS_RX = 187	Available since version 11.
I_EVT_MODAL_CURMASS_RY = 188	Available since version 11.
I_EVT_MODAL_CURMASS_RZ = 189	Available since version 11.
I_EVT_MODAL_SUMMASS_UX = 190	Available since version 11.
I_EVT_MODAL_SUMMASS_UY = 191	Available since version 11.

(c) 2014 Autodesk Inc. All rights reserved

I_EVT_MODAL_SUMMASS_UZ = 192	Available since version 11.
I_EVT_MODAL_SUMMASS_RX = 193	Available since version 11.
I_EVT_MODAL_SUMMASS_RY = 194	Available since version 11.
I_EVT_MODAL_SUMMASS_RZ = 195	Available since version 11.
I_EVT_MODAL_TOTMASS_UX = 196	Available since version 11.
I_EVT_MODAL_TOTMASS_UY = 197	Available since version 11.
I_EVT_MODAL_TOTMASS_UZ = 198	Available since version 11.
I_EVT_MODAL_TOTMASS_RX = 199	Available since version 11.
I_EVT_MODAL_TOTMASS_RY = 200	Available since version 11.
I_EVT_MODAL_TOTMASS_RZ = 201	Available since version 11.
I_EVT_MODAL_PRECISION = 209	Available since version 11.
I_EVT_MODAL_DAMPING = 984	Available since version 11.
I_EVT_MODAL_ENERGY = 985	Available since version 11.
I_EVT_MODAL_AVERAGE_PARTICIPATION_COEFF = 986	Available since version 11.
I_EVT_MODAL_EIGENVALUE = 208	Available since version 11.
I_EVT_MODAL_EIGENVECTOR_UX = 228	Available since version 11.
I_EVT_MODAL_EIGENVECTOR_UY = 229	Available since version 11.
I_EVT_MODAL_EIGENVECTOR_UZ = 230	Available since version 11.
I_EVT_MODAL_EIGENVECTOR_RX = 231	Available since version 11.
I_EVT_MODAL_EIGENVECTOR_RY = 232	Available since version 11.
I_EVT_MODAL_EIGENVECTOR_RZ = 233	Available since version 11.
I_EVT_MODAL_EIGENVECTOR_UX_1 = 234	Available since version 11.
I_EVT_MODAL_EIGENVECTOR_UY_1 = 235	Available since version 11.
I_EVT_MODAL_EIGENVECTOR_UZ_1 = 236	Available since version 11.
I_EVT_MODAL_EIGENVECTOR_RX_1 = 237	Available since version 11.
I_EVT_MODAL_EIGENVECTOR_RY_1 = 238	Available since version 11.
I_EVT_MODAL_EIGENVECTOR_RZ_1 = 239	Available since version 11.
I_EVT_MODAL_PARTICIPATION_COEFF_UX = 202	Available since version 11.
I_EVT_MODAL_PARTICIPATION_COEFF_UY = 203	Available since version 11.
I_EVT_MODAL_PARTICIPATION_COEFF_UZ = 204	Available since version 11.
I_EVT_MODAL_PARTICIPATION_COEFF_RX = 205	Available since version 11.
I_EVT_MODAL_PARTICIPATION_COEFF_RY = 206	Available since version 11.
I_EVT_MODAL_PARTICIPATION_COEFF_RZ = 207	Available since version 11.
I_EVT_MODAL_MASSES_X = 1767	Available since version 11.
I_EVT_MODAL_MASSES_Y = 1768	Available since version 11.
I_EVT_MODAL_MASSES_Z = 1769	Available since version 11.
I_EVT_MODAL_NODE_MASSES_X = 1770	Available since version 11.
I_EVT_MODAL_NODE_MASSES_Y = 1771	Available since version 11.
I_EVT_MODAL_NODE_MASSES_Z = 1772	Available since version 11.

Description

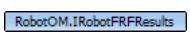
Set of results for which the extreme values are made accessible. .

Version

Available since version 3.5.

III.3.13 IRobotFRFResults

Class Hierarchy



(c) 2014 Autodesk Inc. All rights reserved

C++

```
interface IRobotFRFResults : IDispatch;
```

C#

```
public interface IRobotFRFResults;
```

Visual Basic

```
Public Interface IRobotFRFResults
```

Description

Set of results for FRF analysis.

Version

Available since version 9.7.

III.3.13.1 IRobotFRFResults Members

The following tables list the members exposed by IRobotFRFResults.

Public Fields

	Name	Description
	Frequency (see page 919)	A frequency value that corresponds to the specified step .
	VX (see page 919)	Value of VX velocity.
	VY (see page 919)	Value of VY velocity.
	VZ (see page 920)	Value of VZ velocity.

III.3.13.2 IRobotFRFResults Fields

The fields of the IRobotFRFResults class are listed here.

Public Fields

	Name	Description
	Frequency (see page 919)	A frequency value that corresponds to the specified step .
	VX (see page 919)	Value of VX velocity.
	VY (see page 919)	Value of VY velocity.
	VZ (see page 920)	Value of VZ velocity.

III.3.13.2.1 Frequency

C++

```
HRESULT get_Frequency(double*);
```

C#

```
public double Frequency { get; }
```

Visual Basic

```
Public ReadOnly Frequency As double
```

Description

A frequency value that corresponds to the specified step .

Version

Available since version 9.7.

(c) 2014 Autodesk Inc. All rights reserved

III.3.13.2.2 VX

C++

```
HRESULT get_VX(double*);
```

C#

```
public double VX { get; }
```

Visual Basic

```
Public ReadOnly VX As double
```

Description

Value of VX velocity.

Version

Available since version 9.7.

III.3.13.2.3 VY

C++

```
HRESULT get_VY(double*);
```

C#

```
public double VY { get; }
```

Visual Basic

```
Public ReadOnly VY As double
```

Description

Value of VY velocity.

Version

Available since version 9.7.

III.3.13.2.4 VZ

C++

```
HRESULT get_VZ(double*);
```

C#

```
public double VZ { get; }
```

Visual Basic

```
Public ReadOnly VZ As double
```

Description

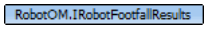
Value of VZ velocity.

Version

Available since version 9.7.

III.3.14 IRobotFootfallResults

Class Hierarchy



(c) 2014 Autodesk Inc. All rights reserved

C++

```
interface IRobotFootfallResults : IDispatch;
```

C#

```
public interface IRobotFootfallResults;
```

Visual Basic

```
Public Interface IRobotFootfallResults
```

Description

Set of results for Footfall analysis.

Version

Available since version 9.7.

III.3.14.1 IRobotFootfallResults Members

The following tables list the members exposed by IRobotFootfallResults.

Public Fields

	Name	Description
	A (see page 921)	Overall maximum acceleration.
	ExcitationNode (see page 921)	Excitation node.
	Frequency (see page 922)	Frequency.
	RF_Overall (see page 922)	Maximum (overall) response factor.
	RF_Resonant (see page 922)	Maximum (resonant) response factor.
	RF_Transient (see page 923)	Maximum (transient) response factor.
	VRMQ (see page 923)	VRMQ (Root Mean Quad Velocity).
	VRMS (see page 923)	VRMS (Root Mean Square Velocity).

III.3.14.2 IRobotFootfallResults Fields

The fields of the IRobotFootfallResults class are listed here.

Public Fields

	Name	Description
	A (see page 921)	Overall maximum acceleration.
	ExcitationNode (see page 921)	Excitation node.
	Frequency (see page 922)	Frequency.
	RF_Overall (see page 922)	Maximum (overall) response factor.
	RF_Resonant (see page 922)	Maximum (resonant) response factor.
	RF_Transient (see page 923)	Maximum (transient) response factor.
	VRMQ (see page 923)	VRMQ (Root Mean Quad Velocity).
	VRMS (see page 923)	VRMS (Root Mean Square Velocity).

III.3.14.2.1 A

C++

```
HRESULT get_A(double*);
```

C#

```
public double A { get; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public ReadOnly A As double
```

Description

Overall maximum acceleration.

Version

Available since version 9.7.

III.3.14.2.2 ExcitationNode

C++

```
HRESULT get_ExcitationNode(long*);
```

C#

```
public long ExcitationNode { get; }
```

Visual Basic

```
Public ReadOnly ExcitationNode As long
```

Description

Excitation node.

Version

Available since version 9.7.

III.3.14.2.3 Frequency

C++

```
HRESULT get_Frequency(double*);
```

C#

```
public double Frequency { get; }
```

Visual Basic

```
Public ReadOnly Frequency As double
```

Description

Frequency.

Version

Available since version 9.7.

III.3.14.2.4 RF_Overall

C++

```
HRESULT get_RF_Overall(double*);
```

C#

```
public double RF_Overall { get; }
```

Visual Basic

```
Public ReadOnly RF_Overall As double
```

Description

Maximum (overall) response factor.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 9.7.

III.3.14.2.5 RF_Resonant

C++

```
HRESULT get_RF_Resonant(double*);
```

C#

```
public double RF_Resonant { get; }
```

Visual Basic

```
Public ReadOnly RF_Resonant As double
```

Description

Maximum (resonant) response factor.

Version

Available since version 9.7.

III.3.14.2.6 RF_Transient

C++

```
HRESULT get_RF_Transient(double*);
```

C#

```
public double RF_Transient { get; }
```

Visual Basic

```
Public ReadOnly RF_Transient As double
```

Description

Maximum (transient) response factor.

Version

Available since version 9.7.

III.3.14.2.7 VRMQ

C++

```
HRESULT get_VRMQ(double*);
```

C#

```
public double VRMQ { get; }
```

Visual Basic

```
Public ReadOnly VRMQ As double
```

Description

VRMQ (Root Mean Quad Velocity).

Version

Available since version 9.7.

(c) 2014 Autodesk Inc. All rights reserved

III.3.14.2.8 VRMS

C++

```
HRESULT get_VRMS(double*);
```

C#

```
public double VRMS { get; }
```

Visual Basic

```
Public ReadOnly VRMS As double
```

Description

VRMS (Root Mean Square Velocity).

Version

Available since version 9.7.

III.4 Calculation results for dynamic analyses

Interfaces

	Name	Description
	IRobotAdvancedResultServer (see page 924)	
	IRobotEigenvaluesServer (see page 927)	Server providing access to eigenvalues.
	IRobotMassSumServer (see page 928)	
	IRobotSpectralCoefficients (see page 930)	Server provides access to spectral coefficient values. .
	IRobotEigenvectorsServer (see page 932)	Server providing access to eigenvectors. .
	IRobotTimeHistoryResultServer (see page 933)	Result server for time history analysis.
	IRobotFRFResultServer (see page 934)	Result server for FRF analysis.

III.4.1 IRobotAdvancedResultServer

Class Hierarchy

RobotOM.IRobotAdvancedResultServer

C++

```
interface IRobotAdvancedResultServer : IDispatch;
```

C#

```
public interface IRobotAdvancedResultServer;
```

Visual Basic

```
Public Interface IRobotAdvancedResultServer
```

III.4.1.1 IRobotAdvancedResultServer Members

The following tables list the members exposed by IRobotAdvancedResultServer.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	Eigenvalues (see page 925)	Available since version 1.7.
	Eigenvectors (see page 925)	Server providing access to eigenvector values Available since version 1.7.
	FRF (see page 925)	Result server for FRF analysis.
	MassSum (see page 926)	Server providing access to sum of masses Available since version 1.7.
	SpectralCoeffs (see page 926)	Server providing access to spectral coefficients Available since version 1.7.
	TimeHistory (see page 926)	Server providing access to results of time history analysis.

Public Methods

	Name	Description
	FootfallValue (see page 927)	Function returns a set of Footfall analysis results for the specified node and load case.

III.4.1.2 IRobotAdvancedResultServer Fields

The fields of the IRobotAdvancedResultServer class are listed here.

Public Fields

	Name	Description
	Eigenvalues (see page 925)	Available since version 1.7.
	Eigenvectors (see page 925)	Server providing access to eigenvector values Available since version 1.7.
	FRF (see page 925)	Result server for FRF analysis.
	MassSum (see page 926)	Server providing access to sum of masses Available since version 1.7.
	SpectralCoeffs (see page 926)	Server providing access to spectral coefficients Available since version 1.7.
	TimeHistory (see page 926)	Server providing access to results of time history analysis.

III.4.1.2.1 Eigenvalues

C++

```
HRESULT get_Eigenvalues(IRobotEigenvaluesServer**);
```

C#

```
public IRobotEigenvaluesServer Eigenvalues { get; }
```

Visual Basic

```
Public ReadOnly Eigenvalues As IRobotEigenvaluesServer
```

Description

Available since version 1.7.

III.4.1.2.2 Eigenvectors

C++

```
HRESULT get_Eigenvectors(IRobotEigenvectorsServer**);
```

C#

```
public IRobotEigenvectorsServer Eigenvectors { get; }
```

Visual Basic

```
Public ReadOnly Eigenvectors As IRobotEigenvectorsServer
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Server providing access to eigenvector values Available since version 1.7.

III.4.1.2.3 FRF

C++

```
HRESULT get_FRF(IRobotFRFResultServer**);
```

C#

```
public IRobotFRFResultServer FRF { get; }
```

Visual Basic

```
Public ReadOnly FRF As IRobotFRFResultServer
```

Description

Result server for FRF analysis.

Version

Available since version 9.7.

III.4.1.2.4 MassSum

C++

```
HRESULT get_MassSum(IRobotMassSumServer**);
```

C#

```
public IRobotMassSumServer MassSum { get; }
```

Visual Basic

```
Public ReadOnly MassSum As IRobotMassSumServer
```

Description

Server providing access to sum of masses Available since version 1.7.

III.4.1.2.5 SpectralCoeffs

C++

```
HRESULT get_SpectralCoeffs(IRobotSpectralCoefficients**);
```

C#

```
public IRobotSpectralCoefficients SpectralCoeffs { get; }
```

Visual Basic

```
Public ReadOnly SpectralCoeffs As IRobotSpectralCoefficients
```

Description

Server providing access to spectral coefficients Available since version 1.7.

III.4.1.2.6 TimeHistory

C++

```
HRESULT get_TimeHistory(IRobotTimeHistoryResultServer**);
```

C#

```
public IRobotTimeHistoryResultServer TimeHistory { get; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public ReadOnly TimeHistory As IRobotTimeHistoryResultServer
```

Description

Server providing access to results of time history analysis.

Version

Available since version 3.

III.4.1.3 IRobotAdvancedResultServer Methods

The methods of the IRobotAdvancedResultServer class are listed here.

Public Methods

	Name	Description
	FootfallValue (see page 927)	Function returns a set of Footfall analysis results for the specified node and load case.

III.4.1.3.1 FootfallValue

C++

```
HRESULT FootfallValue(long _node, long _case, IRobotFootfallResults** ret);
```

C#

```
public IRobotFootfallResults FootfallValue(long _node, long _case);
```

Visual Basic

```
Public Function FootfallValue(_node As long, _case As long) As IRobotFootfallResults
```

Description

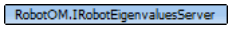
Function returns a set of Footfall analysis results for the specified node and load case.

Version

Available since version 9.7.

III.4.2 IRobotEigenvaluesServer

Class Hierarchy



C++

```
interface IRobotEigenvaluesServer : IDispatch;
```

C#

```
public interface IRobotEigenvaluesServer;
```

Visual Basic

```
Public Interface IRobotEigenvaluesServer
```

Description

Server providing access to eigenvalues.

III.4.2.1 IRobotEigenvaluesServer Members

The following tables list the members exposed by IRobotEigenvaluesServer.

(c) 2014 Autodesk Inc. All rights reserved

Public Methods

	Name	Description
	CombValue (see page 928)	Available since version 1.7.
	Value (see page 928)	Available since version 1.7.

III.4.2.2 IRobotEigenvaluesServer Methods

The methods of the IRobotEigenvaluesServer class are listed here.

Public Methods

	Name	Description
	CombValue (see page 928)	Available since version 1.7.
	Value (see page 928)	Available since version 1.7.

III.4.2.2.1 CombValue

C++

```
HRESULT CombValue(long _case_num, IRobotModeCombinationType _mode_cmb, IRobotEigenvalues**
ret);
```

C#

```
public IRobotEigenvalues CombValue(long _case_num, IRobotModeCombinationType _mode_cmb);
```

Visual Basic

```
Public Function CombValue(_case_num As long, _mode_cmb As IRobotModeCombinationType) As
IRobotEigenvalues
```

Description

Available since version 1.7.

III.4.2.2.2 Value

C++

```
HRESULT Value(long _case_num, long _mode_num, IRobotEigenvalues** ret);
```

C#

```
public IRobotEigenvalues Value(long _case_num, long _mode_num);
```

Visual Basic

```
Public Function Value(_case_num As long, _mode_num As long) As IRobotEigenvalues
```

Description

Available since version 1.7.

III.4.3 IRobotMassSumServer

Class Hierarchy

RobotOM.IRobotMassSumServer

C++

```
interface IRobotMassSumServer : IDispatch;
```

C#

```
public interface IRobotMassSumServer;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Interface IRobotMassSumServer
```

III.4.3.1 IRobotMassSumServer Members

The following tables list the members exposed by IRobotMassSumServer.

Public Methods

	Name	Description
	Current (see page 929)	Function returns values of participation masses expressed in percentages in the current mode for a specified case and mode, for individual directions. Available since version 1.7.
	PartCoeff (see page 929)	Function returns values of modal participation coefficients in a specified case for a defined mode, for individual directions. Available since version 1.7.
	Relative (see page 930)	Function returns values of participation mass sum expressed in percentages from mode 1 to the current mode for a specified case and mode, for individual directions. Available since version 1.7.
	Total (see page 930)	Function returns values of total participation mass sum in a specified case for a specified mode, for individual directions. Available since version 1.7.

III.4.3.2 IRobotMassSumServer Methods

The methods of the IRobotMassSumServer class are listed here.

Public Methods

	Name	Description
	Current (see page 929)	Function returns values of participation masses expressed in percentages in the current mode for a specified case and mode, for individual directions. Available since version 1.7.
	PartCoeff (see page 929)	Function returns values of modal participation coefficients in a specified case for a defined mode, for individual directions. Available since version 1.7.
	Relative (see page 930)	Function returns values of participation mass sum expressed in percentages from mode 1 to the current mode for a specified case and mode, for individual directions. Available since version 1.7.
	Total (see page 930)	Function returns values of total participation mass sum in a specified case for a specified mode, for individual directions. Available since version 1.7.

III.4.3.2.1 Current

C++

```
HRESULT Current(long _case_num, long _mode_num, IRobotDisplacementData** ret);
```

C#

```
public IRobotDisplacementData Current(long _case_num, long _mode_num);
```

Visual Basic

```
Public Function Current(_case_num As long, _mode_num As long) As IRobotDisplacementData
```

Description

Function returns values of participation masses expressed in percentages in the current mode for a specified case and mode, for individual directions. Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

III.4.3.2.2 PartCoeff

C++

```
HRESULT PartCoeff(long _case_num, long _mode_num, IRobotDisplacementData** ret);
```

C#

```
public IRobotDisplacementData PartCoeff(long _case_num, long _mode_num);
```

Visual Basic

```
Public Function PartCoeff(_case_num As long, _mode_num As long) As IRobotDisplacementData
```

Description

Function returns values of modal participation coefficients in a specified case for a defined mode, for individual directions. Available since version 1.7.

III.4.3.2.3 Relative

C++

```
HRESULT Relative(long _case_num, long _mode_num, IRobotDisplacementData** ret);
```

C#

```
public IRobotDisplacementData Relative(long _case_num, long _mode_num);
```

Visual Basic

```
Public Function Relative(_case_num As long, _mode_num As long) As IRobotDisplacementData
```

Description

Function returns values of participation mass sum expressed in percentages from mode 1 to the current mode for a specified case and mode, for individual directions. Available since version 1.7.

III.4.3.2.4 Total

C++

```
HRESULT Total(long _case_num, long _mode_num, IRobotDisplacementData** ret);
```

C#

```
public IRobotDisplacementData Total(long _case_num, long _mode_num);
```

Visual Basic

```
Public Function Total(_case_num As long, _mode_num As long) As IRobotDisplacementData
```

Description

Function returns values of total participation mass sum in a specified case for a specified mode, for individual directions. Available since version 1.7.

III.4.4 IRobotSpectralCoefficients

Class Hierarchy

C++

```
interface IRobotSpectralCoefficients : IDispatch;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public interface IRobotSpectralCoefficients;
```

Visual Basic

```
Public Interface IRobotSpectralCoefficients
```

Description

Server provides access to spectral coefficient values. .

III.4.4.1 IRobotSpectralCoefficients Members

The following tables list the members exposed by IRobotSpectralCoefficients.

Public Methods

	Name	Description
	ModeCoef (see page 931)	Function returns values of spectral mode coefficients in a specified case and mode for individual directions. Available since version 1.7.
	PartCoef (see page 931)	Function returns values of spectral participation coefficients in a specified case for a defined mode, for individual directions. Available since version 1.7.
	SpectrCoef (see page 932)	Function returns spectral coefficient value for a specified case and mode. Available since version 1.7.

III.4.4.2 IRobotSpectralCoefficients Methods

The methods of the IRobotSpectralCoefficients class are listed here.

Public Methods

	Name	Description
	ModeCoef (see page 931)	Function returns values of spectral mode coefficients in a specified case and mode for individual directions. Available since version 1.7.
	PartCoef (see page 931)	Function returns values of spectral participation coefficients in a specified case for a defined mode, for individual directions. Available since version 1.7.
	SpectrCoef (see page 932)	Function returns spectral coefficient value for a specified case and mode. Available since version 1.7.

III.4.4.2.1 ModeCoef

C++

```
HRESULT ModeCoef(long _case_num, long _mode_num, IRobotDisplacementData** ret);
```

C#

```
public IRobotDisplacementData ModeCoef(long _case_num, long _mode_num);
```

Visual Basic

```
Public Function ModeCoef(_case_num As long, _mode_num As long) As IRobotDisplacementData
```

Description

Function returns values of spectral mode coefficients in a specified case and mode for individual directions. Available since version 1.7.

III.4.4.2.2 PartCoef

C++

```
HRESULT PartCoef(long _case_num, long _mode_num, IRobotDisplacementData** ret);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRobotDisplacementData PartCoef(long _case_num, long _mode_num);
```

Visual Basic

```
Public Function PartCoef(_case_num As long, _mode_num As long) As IRobotDisplacementData
```

Description

Function returns values of spectral participation coefficients in a specified case for a defined mode, for individual directions. Available since version 1.7.

III.4.4.2.3 SpectrCoef

C++

```
HRESULT SpectrCoef(long _case_num, long _mode_num, double* ret);
```

C#

```
public double SpectrCoef(long _case_num, long _mode_num);
```

Visual Basic

```
Public Function SpectrCoef(_case_num As long, _mode_num As long) As double
```

Description

Function returns spectral coefficient value for a specified case and mode. Available since version 1.7.

III.4.5 IRobotEigenvectorsServer

Class Hierarchy

C++

```
interface IRobotEigenvectorsServer : IDispatch;
```

C#

```
public interface IRobotEigenvectorsServer;
```

Visual Basic

```
Public Interface IRobotEigenvectorsServer
```

Description

Server providing access to eigenvectors. .

III.4.5.1 IRobotEigenvectorsServer Members

The following tables list the members exposed by IRobotEigenvectorsServer.

Public Methods

	Name	Description
	CombValue (see page 933)	Function returns eigenvector for a specified node, load case and mode combination. Available since version 1.7.
	Value (see page 933)	Function returns eigenvector for a specified node, load case and mode. Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

III.4.5.2 IRobotEigenvectorsServer Methods

The methods of the IRobotEigenvectorsServer class are listed here.

Public Methods

	Name	Description
	CombValue (see page 933)	Function returns eigenvector for a specified node, load case and mode combination. Available since version 1.7.
	Value (see page 933)	Function returns eigenvector for a specified node, load case and mode. Available since version 1.7.

III.4.5.2.1 CombValue

C++

```
HRESULT CombValue(long _node_num, long _case_num, IRobotModeCombinationType _mode_cmb,
IRobotDisplacementData** ret);
```

C#

```
public IRobotDisplacementData CombValue(long _node_num, long _case_num,
IRobotModeCombinationType _mode_cmb);
```

Visual Basic

```
Public Function CombValue(_node_num As long, _case_num As long, _mode_cmb As
IRobotModeCombinationType) As IRobotDisplacementData
```

Description

Function returns eigenvector for a specified node, load case and mode combination. Available since version 1.7.

III.4.5.2.2 Value

C++

```
HRESULT Value(long _node_num, long _case_num, long _mode_num, IRobotDisplacementData** ret);
```

C#

```
public IRobotDisplacementData Value(long _node_num, long _case_num, long _mode_num);
```

Visual Basic

```
Public Function Value(_node_num As long, _case_num As long, _mode_num As long) As
IRobotDisplacementData
```

Description

Function returns eigenvector for a specified node, load case and mode. Available since version 1.7.

III.4.6 IRobotTimeHistoryResultServer

Class Hierarchy

RobotOM.IRobotTimeHistoryResultServer

C++

```
interface IRobotTimeHistoryResultServer : IDispatch;
```

C#

```
public interface IRobotTimeHistoryResultServer;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Interface IRobotTimeHistoryResultServer
```

Description

Result server for time history analysis.

Version

Available since version 3.

III.4.6.1 IRobotTimeHistoryResultServer Members

The following tables list the members exposed by IRobotTimeHistoryResultServer.

Public Methods

	Name	Description
	Value (see page 934)	Function returns a set of time history analysis results for a given node, case and step.

III.4.6.2 IRobotTimeHistoryResultServer Methods

The methods of the IRobotTimeHistoryResultServer class are listed here.

Public Methods

	Name	Description
	Value (see page 934)	Function returns a set of time history analysis results for a given node, case and step.

III.4.6.2.1 Value

C++

```
HRESULT Value(long _node, long _case, long _step, IRobotTimeHistoryResults** ret);
```

C#

```
public IRobotTimeHistoryResults Value(long _node, long _case, long _step);
```

Visual Basic

```
Public Function Value(_node As long, _case As long, _step As long) As
IRobotTimeHistoryResults
```

Description

Function returns a set of time history analysis results for a given node, case and step.

Version

Available since version 3.

III.4.7 IRobotFRFResultServer

Class Hierarchy

RobotOM.IRobotFRFResultServer

C++

```
interface IRobotFRFResultServer : IDispatch;
```

C#

```
public interface IRobotFRFResultServer;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Interface IRobotFRFResultServer
```

Description

Result server for FRF analysis.

Version

Available since version 9.7.

III.4.7.1 IRobotFRFResultServer Members

The following tables list the members exposed by IRobotFRFResultServer.

Public Methods

	Name	Description
	Value (see page 935)	Function returns a set of FRF analysis results for the specified node, load case and step.

III.4.7.2 IRobotFRFResultServer Methods

The methods of the IRobotFRFResultServer class are listed here.

Public Methods

	Name	Description
	Value (see page 935)	Function returns a set of FRF analysis results for the specified node, load case and step.

III.4.7.2.1 Value

C++

```
HRESULT Value(long _node, long _case, long _step, IRobotFRFResults** ret);
```

C#

```
public IRobotFRFResults Value(long _node, long _case, long _step);
```

Visual Basic

```
Public Function Value(_node As long, _case As long, _step As long) As IRobotFRFResults
```

Description

Function returns a set of FRF analysis results for the specified node, load case and step.

Version

Available since version 9.7.

III.5 Calculation results for finite elements

Available since version 2.5.

Enumerations

	Name	Description
	IRobotFeLayerType (see page 937)	Available designations of layers for which calculation results for planar finite elements may be obtained. .
	IRobotReinforceCalcMethods (see page 956)	Types of the reinforcement calculation hypotheses.
	IRobotFeResultReducedCutPosition (see page 977)	Available positions of a panel cut.

(c) 2014 Autodesk Inc. All rights reserved

	IRobotPanelCutDefinitionType (see page 984)	Types of panel cut definition. .
	IRobotFeResultType (see page 984)	Designation of the type of result calculated for finite elements.
	IRobotFeResultSmoothing (see page 986)	Available methods of averaging (smoothing) results.
	IRobotFeResultKind (see page 986)	

Interfaces

	Name	Description
	IRobotFeResultParams (see page 937)	Structure defining parameters necessary to take calculation results for finite elements. The type of returned values is strictly determined by the setting configuration of the fields: Panel (see page 941), Node (see page 941), Element (see page 940). The following variants are possible: 1. the Node (see page 941) node number is set globally averaged results at a node are returned 2. the Node (see page 941) node number and the Panel (see page 941) panel number are set results at a node averaged within a given panel are the results returned 3. the Node (see page 941) node number and the Element (see page 940) finite element number are set exact results at a node are returned 4. the Element (see page 940) finite element number is... more (see page 937)
	IRobotFeResultDetailed (see page 943)	Detailed result vector.
	IRobotFeResultServer (see page 950)	Result server for planar finite elements.
	IRobotFeResultPrincipal (see page 956)	Principal result vector.
	IRobotFeResultComplex (see page 963)	Complex result vector.
	IRobotFeResultReinforcement (see page 966)	Result vector for reinforcement.
	IRobotFeResultReduced (see page 971)	Reduced results for panels.
	IRobotPanelCutMngr (see page 978)	Manager of panel cuts. .
	IRobotPanelCut (see page 981)	Interface defining a cut through panel. .
	IRobotFeExtremeParams (see page 987)	Parameters of searching extreme results for finite elements. The list of nodes to be considered while searching extreme values is determined based on the settings in the fields NodeSel (see page 990), ElementSel (see page 989) and PanelSel (see page 991). If a list of nodes is set in the NodeSel (see page 990) field, then this list exactly will be considered in searching extreme values. If the NodeSel (see page 990) field is empty and the ElementSel (see page 989) field is set, then all the nodes belonging to the indicated finite elements will be considered. If the fields NodeSel (see page 990) and ElementSel (see page 989) are empty and the PanelSel (see page 991) field is set, then all the nodes belonging... more (see page 987)
	IRobotFeExtremeValue (see page 993)	Interface providing access to information about the extreme value. .

(c) 2014 Autodesk Inc. All rights reserved

	IRobotFeMultiResultType (see page 999)	Interface which enables indicating simultaneously - a great number of result types for finite elements.
	IRobotFeMultiExtremeValue (see page 1001)	Interface providing access to a great number of min./max. results for finite elements.

III.5.1 IRobotFeLayerType

C++

```
enum IRobotFeLayerType;
```

C#

```
public enum IRobotFeLayerType;
```

Visual Basic

```
Public Enum IRobotFeLayerType
```

Members

Members	Description
I_FLT_UPPER = 1	Top layer. Available since version 2.5.
I_FLT_MIDDLE = 2	Middle layer. Available since version 2.5.
I_FLT_LOWER = 3	Bottom layer. Available since version 2.5.
I_FLT_ARBITRARY = 4	Relative layer width defined arbitrarily. Available since version 2.5.

Description

Available designations of layers for which calculation results for planar finite elements may be obtained. .

Version

Available since version 2.5.

III.5.2 IRobotFeResultParams

Class Hierarchy

C++

```
interface IRobotFeResultParams : IDispatch;
```

C#

```
public interface IRobotFeResultParams;
```

Visual Basic

```
Public Interface IRobotFeResultParams
```

Description

Structure defining parameters necessary to take calculation results for finite elements. The type of returned values is strictly determined by the setting configuration of the fields: Panel (see page 941), Node (see page 941), Element (see page 940). The following variants are possible:

1. the Node (see page 941) node number is set

globally averaged results at a node are returned

(c) 2014 Autodesk Inc. All rights reserved

2. the Node (see page 941) node number and the Panel (see page 941) panel number are set

results at a node averaged within a given panel are the results returned

3. the Node (see page 941) node number and the Element (see page 940) finite element number are set

exact results at a node are returned

4. the Element (see page 940) finite element number is set

results at a center of the indicated element are the results returned

.

Version

Available since version 2.5.

III.5.2.1 IRobotFeResultParams Members

The following tables list the members exposed by IRobotFeResultParams.

Public Fields

	Name	Description
	CalcMethod (see page 939)	Type of reinforcement calculation method.
	Case (see page 939)	Load case number.
	CaseCmpnt (see page 939)	Component number for a complex component case or mode number for a dynamic case.
	Element (see page 940)	Finite element number.
	Layer (see page 940)	Layer.
	LayerArbitraryValue (see page 940)	Layer (see page 940) thickness.
	ModeCmb (see page 941)	Type of mode combination for a dynamic case .
	Node (see page 941)	Node number.
	Panel (see page 941)	Panel number.

Public Methods

	Name	Description
	GetDirX (see page 942)	Function takes the direction of the local X axis of the coordinate system in which the results will be presented. The values read are the coordinates of the direction vector of the X axis for definition in the Cartesian system or the coordinates of the central point for definition in the polar system. In case when values of all the arguments: x, y, z are set as zeros, then the system direction is compatible with the panel local system. In addition, the function returns a type of the system direction definition. .
	SetDirX (see page 942)	The function enables defining the X axis direction of the system in which the results will be presented. Activation of the function should take place only when the vector of detailed results is being taken. Function parameters determine the coordinates of the direction vector of the X axis in the Cartesian system or the coordinates of the central point in the polar system. In case when all the values: x, y, z are set as zeros, a direction compatible with the local system of the selected panel will be defined. .

III.5.2.2 IRobotFeResultParams Fields

The fields of the IRobotFeResultParams class are listed here.

Public Fields

	Name	Description
	CalcMethod (see page 939)	Type of reinforcement calculation method.
	Case (see page 939)	Load case number.

(c) 2014 Autodesk Inc. All rights reserved

	CaseCmpnt (see page 939)	Component number for a complex component case or mode number for a dynamic case.
	Element (see page 940)	Finite element number.
	Layer (see page 940)	Layer.
	LayerArbitraryValue (see page 940)	Layer (see page 940) thickness.
	ModeCmb (see page 941)	Type of mode combination for a dynamic case .
	Node (see page 941)	Node number.
	Panel (see page 941)	Panel number.

III.5.2.2.1 CalcMethod

C++

```
HRESULT get_CalcMethod(IRobotReinforceCalcMethods*);
HRESULT put_CalcMethod(IRobotReinforceCalcMethods);
```

C#

```
public IRobotReinforceCalcMethods CalcMethod { get; set; }
```

Visual Basic

```
Public CalcMethod As IRobotReinforceCalcMethods
```

Description

Type of reinforcement calculation method.

Version

Available since version 2.5.

III.5.2.2.2 Case

C++

```
HRESULT get_Case( long*);
HRESULT put_Case( long);
```

C#

```
public  long Case { get; set; }
```

Visual Basic

```
Public Case As  long
```

Description

Load case number.

Version

Available since version 2.5.

III.5.2.2.3 CaseCmpnt

C++

```
HRESULT get_CaseCmpnt( long*);
HRESULT put_CaseCmpnt( long);
```

C#

```
public  long CaseCmpnt { get; set; }
```

Visual Basic

```
Public CaseCmpnt As  long
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Component number for a complex component case or mode number for a dynamic case.

Version

Available since version 2.5.

III.5.2.2.4 Element

C++

```
HRESULT get_Element( long*);
HRESULT put_Element( long);
```

C#

```
public  long Element { get; set; }
```

Visual Basic

```
Public Element As  long
```

Description

Finite element number.

Version

Available since version 2.5.

III.5.2.2.5 Layer

C++

```
HRESULT get_Layer(IRobotFeLayerType*);
HRESULT put_Layer(IRobotFeLayerType);
```

C#

```
public IRobotFeLayerType Layer { get; set; }
```

Visual Basic

```
Public Layer As IRobotFeLayerType
```

Description

Layer.

Version

Available since version 2.5.

III.5.2.2.6 LayerArbitraryValue

C++

```
HRESULT get_LayerArbitraryValue( double*);
HRESULT put_LayerArbitraryValue( double);
```

C#

```
public  double LayerArbitraryValue { get; set; }
```

Visual Basic

```
Public LayerArbitraryValue As  double
```

Description

Layer (see page 940) thickness.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 2.5.

III.5.2.2.7 ModeCmb

C++

```
HRESULT get_ModeCmb(IRobotModeCombinationType*);
HRESULT put_ModeCmb(IRobotModeCombinationType);
```

C#

```
public IRobotModeCombinationType ModeCmb { get; set; }
```

Visual Basic

```
Public ModeCmb As IRobotModeCombinationType
```

Description

Type of mode combination for a dynamic case .

Version

Available since version 2.5.

III.5.2.2.8 Node

C++

```
HRESULT get_Node( long*);
HRESULT put_Node( long);
```

C#

```
public  long Node { get; set; }
```

Visual Basic

```
Public Node As  long
```

Description

Node number.

Version

Available since version 2.5.

III.5.2.2.9 Panel

C++

```
HRESULT get_Panel( long*);
HRESULT put_Panel( long);
```

C#

```
public  long Panel { get; set; }
```

Visual Basic

```
Public Panel As  long
```

Description

Panel number.

Version

Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

III.5.2.3 IRobotFeResultParams Methods

The methods of the IRobotFeResultParams class are listed here.

Public Methods

	Name	Description
	GetDirX (see page 942)	Function takes the direction of the local X axis of the coordinate system in which the results will be presented. The values read are the coordinates of the direction vector of the X axis for definition in the Cartesian system or the coordinates of the central point for definition in the polar system. In case when values of all the arguments: x, y, z are set as zeros, then the system direction is compatible with the panel local system. In addition, the function returns a type of the system direction definition. .
	SetDirX (see page 942)	The function enables defining the X axis direction of the system in which the results will be presented. Activation of the function should take place only when the vector of detailed results is being taken. Function parameters determine the coordinates of the direction vector of the X axis in the Cartesian system or the coordinates of the central point in the polar system. In case when all the values: x, y, z are set as zeros, a direction compatible with the local system of the selected panel will be defined. .

III.5.2.3.1 GetDirX

C++

```
HRESULT GetDirX(double* _x, double* _y, double* _z, IRobotObjLocalXDirDefinitionType* ret);
```

C#

```
public IRobotObjLocalXDirDefinitionType GetDirX(double* _x, double* _y, double* _z);
```

Visual Basic

```
Public Function GetDirX(ByRef _x As double*, ByRef _y As double*, ByRef _z As double*) As
IRobotObjLocalXDirDefinitionType
```

Description

Function takes the direction of the local X axis of the coordinate system in which the results will be presented. The values read are the coordinates of the direction vector of the X axis for definition in the Cartesian system or the coordinates of the central point for definition in the polar system. In case when values of all the arguments: x, y, z are set as zeros, then the system direction is compatible with the panel local system. In addition, the function returns a type of the system direction definition. .

Version

Available since version 2.5.

III.5.2.3.2 SetDirX

C++

```
HRESULT SetDirX(IRobotObjLocalXDirDefinitionType _def_type, double _x, double _y, double
_z);
```

C#

```
public void SetDirX(IRobotObjLocalXDirDefinitionType _def_type, double _x, double _y,
double _z);
```

Visual Basic

```
Public Sub SetDirX(_def_type As IRobotObjLocalXDirDefinitionType, _x As double, _y As
double, _z As double)
```

(c) 2014 Autodesk Inc. All rights reserved

Description

The function enables defining the X axis direction of the system in which the results will be presented. Activation of the function should take place only when the vector of detailed results is being taken. Function parameters determine the coordinates of the direction vector of the X axis in the Cartesian system or the coordinates of the central point in the polar system. In case when all the values: x, y, z are set as zeros, a direction compatible with the local system of the selected panel will be defined. .

Version

Available since version 2.5.

III.5.3 IRobotFeResultDetailed

Class Hierarchy

RobotOM.IRobotFeResultDetailed

C++

```
interface IRobotFeResultDetailed : IDispatch;
```

C#

```
public interface IRobotFeResultDetailed;
```

Visual Basic

```
Public Interface IRobotFeResultDetailed
```

Description

Detailed result vector.

Version

Available since version 2.5.

III.5.3.1 IRobotFeResultDetailed Members

The following tables list the members exposed by IRobotFeResultDetailed.

Public Fields

	Name	Description
	MXX (see page 944)	MXX moment value.
	MXY (see page 945)	MXY moment value .
	MYY (see page 945)	MYY moment value.
	NXX (see page 945)	NXX membrane force value.
	NXY (see page 946)	NXY membrane force value .
	NYY (see page 946)	NYY membrane force value .
	PNorm (see page 946)	PNorm soil reaction value.
	QXX (see page 946)	QXX shear force value.
	QYY (see page 947)	QYY shear force value.
	RNorm (see page 947)	RNorm rotation.
	RXX (see page 947)	RXX rotation.
	RYY (see page 948)	RYY rotation.
	SXX (see page 948)	SXX stress value.
	SXY (see page 948)	SXY stress value.
	SYY (see page 949)	SYY stress value.
	TXX (see page 949)	TXX shear stress value.

(c) 2014 Autodesk Inc. All rights reserved

	TYY (see page 949)	TYY shear stress value.
	UXX (see page 949)	UXX displacement.
	UYY (see page 950)	UYY displacement .
	WNorm (see page 950)	WNorm displacement .

III.5.3.2 IRobotFeResultDetailed Fields

The fields of the IRobotFeResultDetailed class are listed here.

Public Fields

	Name	Description
	MXX (see page 944)	MXX moment value.
	MXY (see page 945)	MXY moment value .
	MYY (see page 945)	MYY moment value.
	NXX (see page 945)	NXX membrane force value.
	NXY (see page 946)	NXY membrane force value .
	NYY (see page 946)	NYY membrane force value .
	PNorm (see page 946)	PNorm soil reaction value.
	QXX (see page 946)	QXX shear force value.
	QYY (see page 947)	QYY shear force value.
	RNorm (see page 947)	RNorm rotation.
	RXX (see page 947)	RXX rotation.
	RYY (see page 948)	RYY rotation.
	SXX (see page 948)	SXX stress value.
	SXY (see page 948)	SXY stress value.
	SYY (see page 949)	SYY stress value.
	TXX (see page 949)	TXX shear stress value.
	TYY (see page 949)	TYY shear stress value.
	UXX (see page 949)	UXX displacement.
	UYY (see page 950)	UYY displacement .
	WNorm (see page 950)	WNorm displacement .

III.5.3.2.1 MXX

C++

```
HRESULT get_MXX( double*);
```

C#

```
public  double MXX { get; }
```

Visual Basic

```
Public ReadOnly MXX As  double
```

Description

MXX moment value.

Version

Available since version 2.5.

III.5.3.2.2 MXY

C++

```
HRESULT get_MXY( double*);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public  double MXY { get; }
```

Visual Basic

```
Public ReadOnly MXY As  double
```

Description

MXY moment value .

Version

Available since version 2.5.

III.5.3.2.3 MYY

C++

```
HRESULT get_MYY( double*);
```

C#

```
public  double MYY { get; }
```

Visual Basic

```
Public ReadOnly MYY As  double
```

Description

MYY moment value.

Version

Available since version 2.5.

III.5.3.2.4 NXX

C++

```
HRESULT get_NXX( double*);
```

C#

```
public  double NXX { get; }
```

Visual Basic

```
Public ReadOnly NXX As  double
```

Description

NXX membrane force value.

Version

Available since version 2.5.

III.5.3.2.5 NXY

C++

```
HRESULT get_NXY( double*);
```

C#

```
public  double NXY { get; }
```

Visual Basic

```
Public ReadOnly NXY As  double
```

(c) 2014 Autodesk Inc. All rights reserved

Description

NXY membrane force value .

Version

Available since version 2.5.

III.5.3.2.6 NYY

C++

```
HRESULT get_NYY( double*);
```

C#

```
public  double NYY { get; }
```

Visual Basic

```
Public ReadOnly NYY As  double
```

Description

NYY membrane force value .

Version

Available since version 2.5.

III.5.3.2.7 PNorm

C++

```
HRESULT get_PNorm( double*);
```

C#

```
public  double PNorm { get; }
```

Visual Basic

```
Public ReadOnly PNorm As  double
```

Description

PNorm soil reaction value.

Version

Available since version 2.5.

III.5.3.2.8 QXX

C++

```
HRESULT get_QXX( double*);
```

C#

```
public  double QXX { get; }
```

Visual Basic

```
Public ReadOnly QXX As  double
```

Description

QXX shear force value.

Version

Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

III.5.3.2.9 QYY

C++

```
HRESULT get_QYY( double*);
```

C#

```
public  double QYY { get; }
```

Visual Basic

```
Public ReadOnly QYY As  double
```

Description

QYY shear force value.

Version

Available since version 2.5.

III.5.3.2.10 RNorm

C++

```
HRESULT get_RNorm( double*);
```

C#

```
public  double RNorm { get; }
```

Visual Basic

```
Public ReadOnly RNorm As  double
```

Description

RNorm rotation.

Version

Available since version 2.5.

III.5.3.2.11 RXX

C++

```
HRESULT get_RXX( double*);
```

C#

```
public  double RXX { get; }
```

Visual Basic

```
Public ReadOnly RXX As  double
```

Description

RXX rotation.

Version

Available since version 2.5.

III.5.3.2.12 RYY

C++

```
HRESULT get_RYY( double*);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public  double RYY { get; }
```

Visual Basic

```
Public ReadOnly RYY As  double
```

Description

RYY rotation.

Version

Available since version 2.5.

III.5.3.2.13 SXX

C++

```
HRESULT get_SXX( double*);
```

C#

```
public  double SXX { get; }
```

Visual Basic

```
Public ReadOnly SXX As  double
```

Description

SXX stress value.

Version

Available since version 2.5.

III.5.3.2.14 SXY

C++

```
HRESULT get_SXY( double*);
```

C#

```
public  double SXY { get; }
```

Visual Basic

```
Public ReadOnly SXY As  double
```

Description

SXY stress value.

Version

Available since version 2.5.

III.5.3.2.15 SYY

C++

```
HRESULT get_SYY( double*);
```

C#

```
public  double SYY { get; }
```

Visual Basic

```
Public ReadOnly SYY As  double
```

(c) 2014 Autodesk Inc. All rights reserved

Description

SYY stress value.

Version

Available since version 2.5.

III.5.3.2.16 TXX

C++

```
HRESULT get_TXX( double*);
```

C#

```
public  double TXX { get; }
```

Visual Basic

```
Public ReadOnly TXX As  double
```

Description

TXX shear stress value.

Version

Available since version 2.5.

III.5.3.2.17 TYY

C++

```
HRESULT get_TYY( double*);
```

C#

```
public  double TYY { get; }
```

Visual Basic

```
Public ReadOnly TYY As  double
```

Description

TYY shear stress value.

Version

Available since version 2.5.

III.5.3.2.18 UXX

C++

```
HRESULT get_UXX( double*);
```

C#

```
public  double UXX { get; }
```

Visual Basic

```
Public ReadOnly UXX As  double
```

Description

UXX displacement.

Version

Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

III.5.3.2.19 UYY

C++

```
HRESULT get_UYY( double*);
```

C#

```
public  double UYY { get; }
```

Visual Basic

```
Public ReadOnly UYY As  double
```

Description

UYY displacement .

Version

Available since version 2.5.

III.5.3.2.20 WNorm

C++

```
HRESULT get_WNorm( double*);
```

C#

```
public  double WNorm { get; }
```

Visual Basic

```
Public ReadOnly WNorm As  double
```

Description

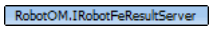
WNorm displacement .

Version

Available since version 2.5.

III.5.4 IRobotFeResultServer

Class Hierarchy



C++

```
interface IRobotFeResultServer : IDispatch;
```

C#

```
public interface IRobotFeResultServer;
```

Visual Basic

```
Public Interface IRobotFeResultServer
```

Description

Result server for planar finite elements.

Version

Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

III.5.4.1 IRobotFeResultServer Members

The following tables list the members exposed by IRobotFeResultServer.

Public Fields

	Name	Description
	PanelCuts (see page 952)	Manager of panel cuts .

Public Methods

	Name	Description
	Complex (see page 952)	Function returns a set of complex results corresponding to the parameters defined in the argument.
	Detailed (see page 953)	Function returns the detailed result vector corresponding to the parameters defined in the argument. .
	MaxValue (see page 953)	
	MinValue (see page 953)	
	MultiMaxValue (see page 954)	Function searches and returns the maximal values for all the specified types of results. Parameters of search for the maximal values are common for all the indicated types of results.
	MultiMinValue (see page 954)	Function searches and returns the minimal values for all the specified types of results. Parameters of search for the minimal values are common for all the indicated types of results.
	Principal (see page 954)	Function returns the principal result vector corresponding to the parameters defined in the argument. .
	Reduced (see page 955)	Function returns the vector of reduced results for panels corresponding to the specified parameters.
	ReducedEx (see page 955)	Function returns a vector of reduced results for panels, corresponding to the specified parameters.
	Reinforcement (see page 955)	Function returns a result vector for the reinforcement corresponding to the specified parameters. There are two variants of obtaining results by the appropriate setting of arguments: 1. first argument determines number of a panel, second argument determines node number 2. first argument determines number of a finite element, while second argument is set as zero.

III.5.4.2 IRobotFeResultServer Fields

The fields of the IRobotFeResultServer class are listed here.

Public Fields

	Name	Description
	PanelCuts (see page 952)	Manager of panel cuts .

III.5.4.2.1 PanelCuts

C++

```
HRESULT get_PanelCuts(IRobotPanelCutMngr**);
```

C#

```
public IRobotPanelCutMngr PanelCuts { get; }
```

Visual Basic

```
Public ReadOnly PanelCuts As IRobotPanelCutMngr
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Manager of panel cuts .

Version

Available since version 3.

III.5.4.3 IRobotFeResultServer Methods

The methods of the IRobotFeResultServer class are listed here.

Public Methods

	Name	Description
	Complex (see page 952)	Function returns a set of complex results corresponding to the parameters defined in the argument.
	Detailed (see page 953)	Function returns the detailed result vector corresponding to the parameters defined in the argument. .
	MaxValue (see page 953)	
	MinValue (see page 953)	
	MultiMaxValue (see page 954)	Function searches and returns the maximal values for all the specified types of results. Parameters of search for the maximal values are common for all the indicated types of results.
	MultiMinValue (see page 954)	Function searches and returns the minimal values for all the specified types of results. Parameters of search for the minimal values are common for all the indicated types of results.
	Principal (see page 954)	Function returns the principal result vector corresponding to the parameters defined in the argument. .
	Reduced (see page 955)	Function returns the vector of reduced results for panels corresponding to the specified parameters.
	ReducedEx (see page 955)	Function returns a vector of reduced results for panels, corresponding to the specified parameters.
	Reinforcement (see page 955)	Function returns a result vector for the reinforcement corresponding to the specified parameters. There are two variants of obtaining results by the appropriate setting of arguments: 1. first argument determines number of a panel, second argument determines node number 2. first argument determines number of a finite element, while second argument is set as zero.

III.5.4.3.1 Complex

C++

```
HRESULT Complex(IRobotFeResultParams* _param, IRobotFeResultComplex** ret);
```

C#

```
public IRobotFeResultComplex Complex(IRobotFeResultParams _param);
```

Visual Basic

```
Public Function Complex(ByRef _param As IRobotFeResultParams) As IRobotFeResultComplex
```

Description

Function returns a set of complex results corresponding to the parameters defined in the argument.

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

III.5.4.3.2 Detailed

C++

```
HRESULT Detailed(IRobotFeResultParams* _param, IRobotFeResultDetailed** ret);
```

C#

```
public IRobotFeResultDetailed Detailed(IRobotFeResultParams _param);
```

Visual Basic

```
Public Function Detailed(ByRef _param As IRobotFeResultParams) As IRobotFeResultDetailed
```

Description

Function returns the detailed result vector corresponding to the parameters defined in the argument. .

Version

Available since version 2.5.

III.5.4.3.3 MaxValue

C++

```
HRESULT MaxValue(IRobotFeExtremeParams* _params, IRobotFeExtremeValue** ret);
```

C#

```
public IRobotFeExtremeValue MaxValue(IRobotFeExtremeParams _params);
```

Visual Basic

```
Public Function MaxValue(ByRef _params As IRobotFeExtremeParams) As IRobotFeExtremeValue
```

Version

Available since version 4.

III.5.4.3.4 MinValue

C++

```
HRESULT MinValue(IRobotFeExtremeParams* _params, IRobotFeExtremeValue** ret);
```

C#

```
public IRobotFeExtremeValue MinValue(IRobotFeExtremeParams _params);
```

Visual Basic

```
Public Function MinValue(ByRef _params As IRobotFeExtremeParams) As IRobotFeExtremeValue
```

Version

Available since version 4.

III.5.4.3.5 MultiMaxValue

C++

```
HRESULT MultiMaxValue(IRobotFeExtremeParams* _params, IRobotFeMultiResultType* _result_ids,
IRobotFeMultiExtremeValue** ret);
```

C#

```
public IRobotFeMultiExtremeValue MultiMaxValue(IRobotFeExtremeParams _params,
IRobotFeMultiResultType _result_ids);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Function MultiMaxValue(ByRef _params As IRobotFeExtremeParams, ByRef _result_ids As
IRobotFeMultiResultType) As IRobotFeMultiExtremeValue
```

Description

Function searches and returns the maximal values for all the specified types of results. Parameters of search for the maximal values are common for all the indicated types of results.

Version

Available since version 4.1.

III.5.4.3.6 MultiMinValue

C++

```
HRESULT MultiMinValue(IRobotFeExtremeParams* _params, IRobotFeMultiResultType* _result_ids,
IRobotFeMultiExtremeValue** ret);
```

C#

```
public IRobotFeMultiExtremeValue MultiMinValue(IRobotFeExtremeParams _params,
IRobotFeMultiResultType _result_ids);
```

Visual Basic

```
Public Function MultiMinValue(ByRef _params As IRobotFeExtremeParams, ByRef _result_ids As
IRobotFeMultiResultType) As IRobotFeMultiExtremeValue
```

Description

Function searches and returns the minimal values for all the specified types of results. Parameters of search for the minimal values are common for all the indicated types of results.

Version

Available since version 4.1.

III.5.4.3.7 Principal

C++

```
HRESULT Principal(IRobotFeResultParams* _param, IRobotFeResultPrincipal** ret);
```

C#

```
public IRobotFeResultPrincipal Principal(IRobotFeResultParams _param);
```

Visual Basic

```
Public Function Principal(ByRef _param As IRobotFeResultParams) As IRobotFeResultPrincipal
```

Description

Function returns the principal result vector corresponding to the parameters defined in the argument. .

Version

Available since version 2.5.

III.5.4.3.8 Reduced

C++

```
HRESULT Reduced(long _panel, IRobotFeResultReducedCutPosition _cut, long _case,
IRobotFeResultReduced** ret);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRobotFeResultReduced Reduced(long _panel, IRobotFeResultReducedCutPosition _cut,
long _case);
```

Visual Basic

```
Public Function Reduced(_panel As long, _cut As IRobotFeResultReducedCutPosition, _case As
long) As IRobotFeResultReduced
```

Description

Function returns the vector of reduced results for panels corresponding to the specified parameters.

Version

Available since version 3.

III.5.4.3.9 ReducedEx

C++

```
HRESULT ReducedEx(long _panel, long _part_idx, IRobotFeResultReducedCutPosition _cut_pos,
long _case, long _case_cmpnt = 1, IRobotFeResultReduced** ret);
```

C#

```
public IRobotFeResultReduced ReducedEx(long _panel, long _part_idx,
IRobotFeResultReducedCutPosition _cut_pos, long _case, long _case_cmpnt = 1);
```

Visual Basic

```
Public Function ReducedEx(_panel As long, _part_idx As long, _cut_pos As
IRobotFeResultReducedCutPosition, _case As long, Optional _case_cmpnt As long = 1) As
IRobotFeResultReduced
```

Description

Function returns a vector of reduced results for panels, corresponding to the specified parameters.

Version

Available since version 5.5.

III.5.4.3.10 Reinforcement

C++

```
HRESULT Reinforcement(long _elem_or_panel, long _node = 0, IRobotFeResultReinforcement**
ret);
```

C#

```
public IRobotFeResultReinforcement Reinforcement(long _elem_or_panel, long _node = 0);
```

Visual Basic

```
Public Function Reinforcement(_elem_or_panel As long, Optional _node As long = 0) As
IRobotFeResultReinforcement
```

Description

Function returns a result vector for the reinforcement corresponding to the specified parameters. There are two variants of obtaining results by the appropriate setting of arguments:

1. first argument determines number of a panel, second argument determines node number
2. first argument determines number of a finite element, while second argument is set as zero.

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

III.5.5 IRobotReinforceCalcMethods

C++

```
enum IRobotReinforceCalcMethods;
```

C#

```
public enum IRobotReinforceCalcMethods;
```

Visual Basic

```
Public Enum IRobotReinforceCalcMethods
```

Members

Members	Description
I_RCM_NEN = 1	NEN hypothesis. Available since version 2.5.
I_RCM_WOOD_ARMER = 2	WOOD&ARMER hypothesis. Available since version 2.5.
I_RCM_ANALYTICAL = 3	Available since version 5.5.

Description

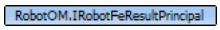
Types of the reinforcement calculation hypotheses.

Version

Available since version 2.5.

III.5.6 IRobotFeResultPrincipal

Class Hierarchy



C++

```
interface IRobotFeResultPrincipal : IDispatch;
```

C#

```
public interface IRobotFeResultPrincipal;
```

Visual Basic

```
Public Interface IRobotFeResultPrincipal
```

Description

Principal result vector.

Version

Available since version 2.5.

III.5.6.1 IRobotFeResultPrincipal Members

The following tables list the members exposed by IRobotFeResultPrincipal.

Public Fields

	Name	Description
	M1 (see page 958)	Value of M1 moment.
	M1_2 (see page 958)	Value of M(1-2) moment .
	M2 (see page 958)	Value of M2 moment .

(c) 2014 Autodesk Inc. All rights reserved

	MAL (see page 958)	Mal angle.
	N1 (see page 959)	Value of N1 membrane force .
	N1_2 (see page 959)	Value of N(1-2) membrane force .
	N2 (see page 959)	Value of N2 membrane force .
	NAL (see page 960)	Nal angle.
	Q1_2 (see page 960)	Value of Q(1-2) shear force .
	S1 (see page 960)	Value of s1 stress .
	S1_2 (see page 961)	Value of s(1-2) stress .
	S2 (see page 961)	Value of s2 stress .
	SAL (see page 961)	Sal angle.
	T1_2 (see page 961)	Value of T(1-2) shear stress .
	U (see page 962)	Value of U total displacement.
	UGX (see page 962)	Value of UGX global displacement.
	UGY (see page 962)	Value of UGY global displacement.
	UGZ (see page 963)	Value of UGZ global displacement.

III.5.6.2 IRobotFeResultPrincipal Fields

The fields of the IRobotFeResultPrincipal class are listed here.

Public Fields

	Name	Description
	M1 (see page 958)	Value of M1 moment.
	M1_2 (see page 958)	Value of M(1-2) moment .
	M2 (see page 958)	Value of M2 moment .
	MAL (see page 958)	Mal angle.
	N1 (see page 959)	Value of N1 membrane force .
	N1_2 (see page 959)	Value of N(1-2) membrane force .
	N2 (see page 959)	Value of N2 membrane force .
	NAL (see page 960)	Nal angle.
	Q1_2 (see page 960)	Value of Q(1-2) shear force .
	S1 (see page 960)	Value of s1 stress .
	S1_2 (see page 961)	Value of s(1-2) stress .
	S2 (see page 961)	Value of s2 stress .
	SAL (see page 961)	Sal angle.
	T1_2 (see page 961)	Value of T(1-2) shear stress .
	U (see page 962)	Value of U total displacement.
	UGX (see page 962)	Value of UGX global displacement.
	UGY (see page 962)	Value of UGY global displacement.
	UGZ (see page 963)	Value of UGZ global displacement.

III.5.6.2.1 M1

C++

```
HRESULT get_M1( double*);
```

C#

```
public  double M1 { get; }
```

Visual Basic

```
Public ReadOnly M1 As  double
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Value of M1 moment.

Version

Available since version 2.5.

III.5.6.2.2 M1_2

C++

```
HRESULT get_M1_2( double*);
```

C#

```
public  double M1_2 { get; }
```

Visual Basic

```
Public ReadOnly M1_2 As  double
```

Description

Value of M(1-2) moment .

Version

Available since version 2.5.

III.5.6.2.3 M2

C++

```
HRESULT get_M2( double*);
```

C#

```
public  double M2 { get; }
```

Visual Basic

```
Public ReadOnly M2 As  double
```

Description

Value of M2 moment .

Version

Available since version 2.5.

III.5.6.2.4 MAL

C++

```
HRESULT get_MAL( double*);
```

C#

```
public  double MAL { get; }
```

Visual Basic

```
Public ReadOnly MAL As  double
```

Description

Mal angle.

Version

Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

III.5.6.2.5 N1

C++

```
HRESULT get_N1( double*);
```

C#

```
public  double N1 { get; }
```

Visual Basic

```
Public ReadOnly N1 As  double
```

Description

Value of N1 membrane force .

Version

Available since version 2.5.

III.5.6.2.6 N1_2

C++

```
HRESULT get_N1_2( double*);
```

C#

```
public  double N1_2 { get; }
```

Visual Basic

```
Public ReadOnly N1_2 As  double
```

Description

Value of N(1-2) membrane force .

Version

Available since version 2.5.

III.5.6.2.7 N2

C++

```
HRESULT get_N2( double*);
```

C#

```
public  double N2 { get; }
```

Visual Basic

```
Public ReadOnly N2 As  double
```

Description

Value of N2 membrane force .

Version

Available since version 2.5.

III.5.6.2.8 NAL

C++

```
HRESULT get_NAL( double*);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public  double NAL { get; }
```

Visual Basic

```
Public ReadOnly NAL As  double
```

Description

Nal angle.

Version

Available since version 2.5.

III.5.6.2.9 Q1_2

C++

```
HRESULT get_Q1_2( double*);
```

C#

```
public  double Q1_2 { get; }
```

Visual Basic

```
Public ReadOnly Q1_2 As  double
```

Description

Value of Q(1-2) shear force .

Version

Available since version 2.5.

III.5.6.2.10 S1

C++

```
HRESULT get_S1( double*);
```

C#

```
public  double S1 { get; }
```

Visual Basic

```
Public ReadOnly S1 As  double
```

Description

Value of s1 stress .

Version

Available since version 2.5.

III.5.6.2.11 S1_2

C++

```
HRESULT get_S1_2( double*);
```

C#

```
public  double S1_2 { get; }
```

Visual Basic

```
Public ReadOnly S1_2 As  double
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Value of s(1-2) stress .

Version

Available since version 2.5.

III.5.6.2.12 S2

C++

```
HRESULT get_S2( double*);
```

C#

```
public  double S2 { get; }
```

Visual Basic

```
Public ReadOnly S2 As  double
```

Description

Value of s2 stress .

Version

Available since version 2.5.

III.5.6.2.13 SAL

C++

```
HRESULT get_SAL( double*);
```

C#

```
public  double SAL { get; }
```

Visual Basic

```
Public ReadOnly SAL As  double
```

Description

Sal angle.

Version

Available since version 2.5.

III.5.6.2.14 T1_2

C++

```
HRESULT get_T1_2( double*);
```

C#

```
public  double T1_2 { get; }
```

Visual Basic

```
Public ReadOnly T1_2 As  double
```

Description

Value of T(1-2) shear stress .

Version

Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

III.5.6.2.15 U

C++

```
HRESULT get_U( double*);
```

C#

```
public  double U { get; }
```

Visual Basic

```
Public ReadOnly U As  double
```

Description

Value of U total displacement.

Version

Available since version 2.5.

III.5.6.2.16 UGX

C++

```
HRESULT get_UGX( double*);
```

C#

```
public  double UGX { get; }
```

Visual Basic

```
Public ReadOnly UGX As  double
```

Description

Value of UGX global displacement.

Version

Available since version 2.5.

III.5.6.2.17 UGY

C++

```
HRESULT get_UGY( double*);
```

C#

```
public  double UGY { get; }
```

Visual Basic

```
Public ReadOnly UGY As  double
```

Description

Value of UGY global displacement.

Version

Available since version 2.5.

III.5.6.2.18 UGZ

C++

```
HRESULT get_UGZ( double*);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public  double UGZ { get; }
```

Visual Basic

```
Public ReadOnly UGZ As  double
```

Description

Value of UGZ global displacement.

Version

Available since version 2.5.

III.5.7 IRobotFeResultComplex

Class Hierarchy

C++

```
interface IRobotFeResultComplex : IDispatch;
```

C#

```
public interface IRobotFeResultComplex;
```

Visual Basic

```
Public Interface IRobotFeResultComplex
```

Description

Complex result vector.

Version

Available since version 3.

III.5.7.1 IRobotFeResultComplex Members

The following tables list the members exposed by IRobotFeResultComplex.

Public Fields

	Name	Description
	M_MISES (see page 964)	Value of reduced moment for surface finite elements.
	MXX_BOTTOM (see page 964)	Value of MXX designing moment of bottom reinforcement.
	MXX_TOP (see page 965)	Value of MXX designing moment of top reinforcement.
	MYY_BOTTOM (see page 965)	Value of MYY designing moment of bottom reinforcement.
	MYY_TOP (see page 965)	Value of MYY designing moment of top reinforcement.
	N_MISES (see page 966)	Value of reduced membrane force for surface finite elements .
	S_MISES (see page 966)	Reduced stress value for surface finite elements.

III.5.7.2 IRobotFeResultComplex Fields

The fields of the IRobotFeResultComplex class are listed here.

Public Fields

	Name	Description
	M_MISES (see page 964)	Value of reduced moment for surface finite elements.
	MXX_BOTTOM (see page 964)	Value of MXX designing moment of bottom reinforcement.
	MXX_TOP (see page 965)	Value of MXX designing moment of top reinforcement.

(c) 2014 Autodesk Inc. All rights reserved

	MYY_BOTTOM (see page 965)	Value of MYY designing moment of bottom reinforcement.
	MYY_TOP (see page 965)	Value of MYY designing moment of top reinforcement.
	N_MISES (see page 966)	Value of reduced membrane force for surface finite elements .
	S_MISES (see page 966)	Reduced stress value for surface finite elements.

III.5.7.2.1 M_MISES

C++

```
HRESULT get_M_MISES( double*);
```

C#

```
public  double M_MISES { get; }
```

Visual Basic

```
Public ReadOnly M_MISES As  double
```

Description

Value of reduced moment for surface finite elements.

Version

Available since version 3.

III.5.7.2.2 MXX_BOTTOM

C++

```
HRESULT get_MXX_BOTTOM( double*);
```

C#

```
public  double MXX_BOTTOM { get; }
```

Visual Basic

```
Public ReadOnly MXX_BOTTOM As  double
```

Description

Value of MXX designing moment of bottom reinforcement.

Version

Available since version 3.

III.5.7.2.3 MXX_TOP

C++

```
HRESULT get_MXX_TOP( double*);
```

C#

```
public  double MXX_TOP { get; }
```

Visual Basic

```
Public ReadOnly MXX_TOP As  double
```

Description

Value of MXX designing moment of top reinforcement.

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

III.5.7.2.4 MYY_BOTTOM

C++

```
HRESULT get_MYY_BOTTOM( double*);
```

C#

```
public  double MYY_BOTTOM { get; }
```

Visual Basic

```
Public ReadOnly MYY_BOTTOM As  double
```

Description

Value of MYY designing moment of bottom reinforcement.

Version

Available since version 3.

III.5.7.2.5 MYY_TOP

C++

```
HRESULT get_MYY_TOP( double*);
```

C#

```
public  double MYY_TOP { get; }
```

Visual Basic

```
Public ReadOnly MYY_TOP As  double
```

Description

Value of MYY designing moment of top reinforcement.

Version

Available since version 3.

III.5.7.2.6 N_MISES

C++

```
HRESULT get_N_MISES( double*);
```

C#

```
public  double N_MISES { get; }
```

Visual Basic

```
Public ReadOnly N_MISES As  double
```

Description

Value of reduced membrane force for surface finite elements .

Version

Available since version 3.

III.5.7.2.7 S_MISES

C++

```
HRESULT get_S_MISES( double*);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public  double S_MISES { get; }
```

Visual Basic

```
Public ReadOnly S_MISES As  double
```

Description

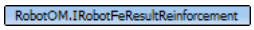
Reduced stress value for surface finite elements.

Version

Available since version 3.

III.5.8 IRobotFeResultReinforcement

Class Hierarchy



C++

```
interface IRobotFeResultReinforcement : IDispatch;
```

C#

```
public interface IRobotFeResultReinforcement;
```

Visual Basic

```
Public Interface IRobotFeResultReinforcement
```

Description

Result vector for reinforcement.

Version

Available since version 3.

III.5.8.1 IRobotFeResultReinforcement Members

The following tables list the members exposed by IRobotFeResultReinforcement.

Public Fields

	Name	Description
	A_MIN (see page 968)	Minimum reinforcement area for one layer and one reinforcement direction.
	AX (see page 968)	Cracking (X axis direction).
	AX_BOTTOM (see page 968)	Bottom reinforcement area (X axis).
	AX_TOP (see page 968)	Top reinforcement area (X axis).
	AY (see page 969)	Cracking (Y axis direction).
	AY_BOTTOM (see page 969)	Bottom reinforcement area (Y axis).
	AY_TOP (see page 969)	Top reinforcement area (Y axis).
	CalcError (see page 970)	
	E_AX_BOTTOM (see page 970)	Bottom reinforcement spacing (X axis).
	E_AX_TOP (see page 970)	Top reinforcement spacing (X axis).
	E_AY_BOTTOM (see page 971)	Bottom reinforcement spacing (Y axis).
	E_AY_TOP (see page 971)	Top reinforcement spacing (Y axis).
	F (see page 971)	Deflection value.

(c) 2014 Autodesk Inc. All rights reserved

III.5.8.2 IRobotFeResultReinforcement Fields

The fields of the IRobotFeResultReinforcement class are listed here.

Public Fields

	Name	Description
	A_MIN (see page 968)	Minimum reinforcement area for one layer and one reinforcement direction.
	AX (see page 968)	Cracking (X axis direction).
	AX_BOTTOM (see page 968)	Bottom reinforcement area (X axis).
	AX_TOP (see page 968)	Top reinforcement area (X axis).
	AY (see page 969)	Cracking (Y axis direction).
	AY_BOTTOM (see page 969)	Bottom reinforcement area (Y axis).
	AY_TOP (see page 969)	Top reinforcement area (Y axis).
	CalcError (see page 970)	
	E_AX_BOTTOM (see page 970)	Bottom reinforcement spacing (X axis).
	E_AX_TOP (see page 970)	Top reinforcement spacing (X axis).
	E_AY_BOTTOM (see page 971)	Bottom reinforcement spacing (Y axis).
	E_AY_TOP (see page 971)	Top reinforcement spacing (Y axis).
	F (see page 971)	Deflection value.

III.5.8.2.1 A_MIN

C++

```
HRESULT get_A_MIN( double*);
```

C#

```
public  double A_MIN { get; }
```

Visual Basic

```
Public ReadOnly A_MIN As  double
```

Description

Minimum reinforcement area for one layer and one reinforcement direction.

Version

Available since version 3.

III.5.8.2.2 AX

C++

```
HRESULT get_AX( double*);
```

C#

```
public  double AX { get; }
```

Visual Basic

```
Public ReadOnly AX As  double
```

Description

Cracking (X axis direction).

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

III.5.8.2.3 AX_BOTTOM

C++

```
HRESULT get_AX_BOTTOM( double*);
```

C#

```
public  double AX_BOTTOM { get; }
```

Visual Basic

```
Public ReadOnly AX_BOTTOM As  double
```

Description

Bottom reinforcement area (X axis).

Version

Available since version 3.

III.5.8.2.4 AX_TOP

C++

```
HRESULT get_AX_TOP( double*);
```

C#

```
public  double AX_TOP { get; }
```

Visual Basic

```
Public ReadOnly AX_TOP As  double
```

Description

Top reinforcement area (X axis).

Version

Available since version 3.

III.5.8.2.5 AY

C++

```
HRESULT get_AY( double*);
```

C#

```
public  double AY { get; }
```

Visual Basic

```
Public ReadOnly AY As  double
```

Description

Cracking (Y axis direction).

Version

Available since version 3.

III.5.8.2.6 AY_BOTTOM

C++

```
HRESULT get_AY_BOTTOM( double*);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public  double AY_BOTTOM { get; }
```

Visual Basic

```
Public ReadOnly AY_BOTTOM As  double
```

Description

Bottom reinforcement area (Y axis).

Version

Available since version 3.

III.5.8.2.7 AY_TOP

C++

```
HRESULT get_AY_TOP( double*);
```

C#

```
public  double AY_TOP { get; }
```

Visual Basic

```
Public ReadOnly AY_TOP As  double
```

Description

Top reinforcement area (Y axis).

Version

Available since version 3.

III.5.8.2.8 CalcError

C++

```
HRESULT get_CalcError(VARIANT_BOOL*);
```

C#

```
public bool CalcError { get; }
```

Visual Basic

```
Public ReadOnly CalcError As Boolean
```

Version

Available since version 3.

III.5.8.2.9 E_AX_BOTTOM

C++

```
HRESULT get_E_AX_BOTTOM( double*);
```

C#

```
public  double E_AX_BOTTOM { get; }
```

Visual Basic

```
Public ReadOnly E_AX_BOTTOM As  double
```

Description

Bottom reinforcement spacing (X axis).

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.

III.5.8.2.10 E_AX_TOP

C++

```
HRESULT get_E_AX_TOP( double*);
```

C#

```
public  double E_AX_TOP { get; }
```

Visual Basic

```
Public ReadOnly E_AX_TOP As  double
```

Description

Top reinforcement spacing (X axis).

Version

Available since version 3.

III.5.8.2.11 E_AY_BOTTOM

C++

```
HRESULT get_E_AY_BOTTOM( double*);
```

C#

```
public  double E_AY_BOTTOM { get; }
```

Visual Basic

```
Public ReadOnly E_AY_BOTTOM As  double
```

Description

Bottom reinforcement spacing (Y axis).

Version

Available since version 3.

III.5.8.2.12 E_AY_TOP

C++

```
HRESULT get_E_AY_TOP( double*);
```

C#

```
public  double E_AY_TOP { get; }
```

Visual Basic

```
Public ReadOnly E_AY_TOP As  double
```

Description

Top reinforcement spacing (Y axis).

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

III.5.8.2.13 F

C++

```
HRESULT get_F( double*);
```

C#

```
public  double F { get; }
```

Visual Basic

```
Public ReadOnly F As  double
```

Description

Deflection value.

Version

Available since version 3.

III.5.9 IRobotFeResultReduced

Class Hierarchy

C++

```
interface IRobotFeResultReduced : IDispatch;
```

C#

```
public interface IRobotFeResultReduced;
```

Visual Basic

```
Public Interface IRobotFeResultReduced
```

Description

Reduced results for panels.

Version

Available since version 3.

III.5.9.1 IRobotFeResultReduced Members

The following tables list the members exposed by IRobotFeResultReduced.

Public Fields

	Name	Description
	CutPos (see page 973)	Definition of the panel cut.
	Height (see page 973)	Panel height.
	Length (see page 973)	Length of panel cut.
	MY (see page 974)	MRy reduced moment outside the panel plane .
	MZ (see page 974)	MRz reduced moment in the panel plane.
	NodeLeftBottom (see page 974)	
	NodeLeftTop (see page 975)	
	NodeRightBottom (see page 975)	
	NodeRightTop (see page 975)	
	NX (see page 975)	NRx reduced force in the panel plane.
	SE (see page 976)	SRe normal stress in the middle layer.

(c) 2014 Autodesk Inc. All rights reserved

	SO (see page 976)	SRo normal stress in the middle layer.
	T (see page 976)	TR shear stress in the middle layer.
	TY (see page 977)	TRy reduced force in the panel plane.
	TZ (see page 977)	TRz shear force outside the panel plane.

III.5.9.2 IRobotFeResultReduced Fields

The fields of the IRobotFeResultReduced class are listed here.

Public Fields

	Name	Description
	CutPos (see page 973)	Definition of the panel cut.
	Height (see page 973)	Panel height.
	Length (see page 973)	Length of panel cut.
	MY (see page 974)	MRy reduced moment outside the panel plane .
	MZ (see page 974)	MRz reduced moment in the panel plane.
	NodeLeftBottom (see page 974)	
	NodeLeftTop (see page 975)	
	NodeRightBottom (see page 975)	
	NodeRightTop (see page 975)	
	NX (see page 975)	NRx reduced force in the panel plane.
	SE (see page 976)	SRe normal stress in the middle layer.
	SO (see page 976)	SRo normal stress in the middle layer.
	T (see page 976)	TR shear stress in the middle layer.
	TY (see page 977)	TRy reduced force in the panel plane.
	TZ (see page 977)	TRz shear force outside the panel plane.

III.5.9.2.1 CutPos

C++

```
HRESULT get_CutPos(IRobotFeResultReducedCutPosition*);
```

C#

```
public IRobotFeResultReducedCutPosition CutPos { get; }
```

Visual Basic

```
Public ReadOnly CutPos As IRobotFeResultReducedCutPosition
```

Description

Definition of the panel cut.

Version

Available since version 5.5.

III.5.9.2.2 Height

C++

```
HRESULT get_Height( double*);
```

C#

```
public  double Height { get; }
```

Visual Basic

```
Public ReadOnly Height As  double
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Panel height.

Version

Available since version 3.

III.5.9.2.3 Length

C++

```
HRESULT get_Length( double*);
```

C#

```
public  double Length { get; }
```

Visual Basic

```
Public ReadOnly Length As  double
```

Description

Length of panel cut.

Version

Available since version 3.

III.5.9.2.4 MY

C++

```
HRESULT get_MY( double*);
```

C#

```
public  double MY { get; }
```

Visual Basic

```
Public ReadOnly MY As  double
```

Description

MRy reduced moment outside the panel plane .

Version

Available since version 3.

III.5.9.2.5 MZ

C++

```
HRESULT get_MZ( double*);
```

C#

```
public  double MZ { get; }
```

Visual Basic

```
Public ReadOnly MZ As  double
```

Description

MRz reduced moment in the panel plane.

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

III.5.9.2.6 NodeLeftBottom

C++

```
HRESULT get_NodeLeftBottom(long*);
```

C#

```
public long NodeLeftBottom { get; }
```

Visual Basic

```
Public ReadOnly NodeLeftBottom As long
```

Version

Available since version 5.5.

III.5.9.2.7 NodeLeftTop

C++

```
HRESULT get_NodeLeftTop(long*);
```

C#

```
public long NodeLeftTop { get; }
```

Visual Basic

```
Public ReadOnly NodeLeftTop As long
```

Version

Available since version 5.5.

III.5.9.2.8 NodeRightBottom

C++

```
HRESULT get_NodeRightBottom(long*);
```

C#

```
public long NodeRightBottom { get; }
```

Visual Basic

```
Public ReadOnly NodeRightBottom As long
```

Version

Available since version 5.5.

III.5.9.2.9 NodeRightTop

C++

```
HRESULT get_NodeRightTop(long*);
```

C#

```
public long NodeRightTop { get; }
```

Visual Basic

```
Public ReadOnly NodeRightTop As long
```

Version

Available since version 5.5.

(c) 2014 Autodesk Inc. All rights reserved

III.5.9.2.10 NX

C++

```
HRESULT get_NX( double*);
```

C#

```
public  double NX { get; }
```

Visual Basic

```
Public ReadOnly NX As  double
```

Description

NRx reduced force in the panel plane.

Version

Available since version 3.

III.5.9.2.11 SE

C++

```
HRESULT get_SE( double*);
```

C#

```
public  double SE { get; }
```

Visual Basic

```
Public ReadOnly SE As  double
```

Description

SRe normal stress in the middle layer.

Version

Available since version 3.

III.5.9.2.12 SO

C++

```
HRESULT get_SO( double*);
```

C#

```
public  double SO { get; }
```

Visual Basic

```
Public ReadOnly SO As  double
```

Description

SRo normal stress in the middle layer.

Version

Available since version 3.

III.5.9.2.13 T

C++

```
HRESULT get_T( double*);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public  double T { get; }
```

Visual Basic

```
Public ReadOnly T As  double
```

Description

TR shear stress in the middle layer.

Version

Available since version 3.

III.5.9.2.14 TY

C++

```
HRESULT get_TY( double*);
```

C#

```
public  double TY { get; }
```

Visual Basic

```
Public ReadOnly TY As  double
```

Description

TRy reduced force in the panel plane.

Version

Available since version 3.

III.5.9.2.15 TZ

C++

```
HRESULT get_TZ( double*);
```

C#

```
public  double TZ { get; }
```

Visual Basic

```
Public ReadOnly TZ As  double
```

Description

TRz shear force outside the panel plane.

Version

Available since version 3.

III.5.10 IRobotFeResultReducedCutPosition

C++

```
enum IRobotFeResultReducedCutPosition;
```

C#

```
public enum IRobotFeResultReducedCutPosition;
```

Visual Basic

```
Public Enum IRobotFeResultReducedCutPosition
```

(c) 2014 Autodesk Inc. All rights reserved

Members

Members	Description
I_FRRCP_VERTICAL_LEFT = 4	Available since version 3.
I_FRRCP_HORIZONTAL_MIDDLE = 2	Available since version 3.
I_FRRCP_VERTICAL_RIGHT = 6	Available since version 3.
I_FRRCP_HORIZONTAL_TOP = 3	Available since version 3.
I_FRRCP_VERTICAL_MIDDLE = 5	Available since version 3.
I_FRRCP_HORIZONTAL_BOTTOM = 1	Available since version 3.

Description

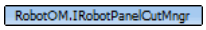
Available positions of a panel cut.

Version

Available since version 3.

III.5.11 IRobotPanelCutMngr

Class Hierarchy



C++

```
interface IRobotPanelCutMngr : IDispatch;
```

C#

```
public interface IRobotPanelCutMngr;
```

Visual Basic

```
Public Interface IRobotPanelCutMngr
```

Description

Manager of panel cuts. .

Version

Available since version 3.

III.5.11.1 IRobotPanelCutMngr Members

The following tables list the members exposed by IRobotPanelCutMngr.

Public Fields

	Name	Description
	Count (see page 978)	Number of available cuts.

Public Methods

	Name	Description
	Create (see page 979)	Function creates and returns a new cut.
	Find (see page 979)	Function returns index of the cut of the specified name. .
	Get (see page 980)	Function returns the cut of the indicated index. .
	GetName (see page 980)	Function returns name of the cut of the indicated index. .
	Remove (see page 980)	Function deletes the cut of the indicated index. .
	Store (see page 981)	Function saves the cut under the specified name. .

(c) 2014 Autodesk Inc. All rights reserved

III.5.11.2 IRobotPanelCutMngr Fields

The fields of the IRobotPanelCutMngr class are listed here.

Public Fields

	Name	Description
	Count (see page 978)	Number of available cuts.

III.5.11.2.1 Count

C++

```
HRESULT get_Count( long*);
```

C#

```
public  long Count { get; }
```

Visual Basic

```
Public ReadOnly Count As  long
```

Description

Number of available cuts.

Version

Available since version 3.

III.5.11.3 IRobotPanelCutMngr Methods

The methods of the IRobotPanelCutMngr class are listed here.

Public Methods

	Name	Description
	Create (see page 979)	Function creates and returns a new cut.
	Find (see page 979)	Function returns index of the cut of the specified name. .
	Get (see page 980)	Function returns the cut of the indicated index. .
	GetName (see page 980)	Function returns name of the cut of the indicated index. .
	Remove (see page 980)	Function deletes the cut of the indicated index. .
	Store (see page 981)	Function saves the cut under the specified name. .

III.5.11.3.1 Create

C++

```
HRESULT Create(IRobotPanelCut** ret);
```

C#

```
public IRobotPanelCut Create();
```

Visual Basic

```
Public Function Create() As IRobotPanelCut
```

Description

Function creates and returns a new cut.

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

III.5.11.3.2 Find

C++

```
HRESULT Find(BSTR _cut_name, long* ret);
```

C#

```
public long Find(String _cut_name);
```

Visual Basic

```
Public Function Find(_cut_name As String) As long
```

Description

Function returns index of the cut of the specified name. .

Version

Available since version 3.

III.5.11.3.3 Get

C++

```
HRESULT Get(long _idx, IRobotPanelCut** ret);
```

C#

```
public IRobotPanelCut Get(long _idx);
```

Visual Basic

```
Public Function Get(_idx As long) As IRobotPanelCut
```

Description

Function returns the cut of the indicated index. .

Version

Available since version 3.

III.5.11.3.4 GetName

C++

```
HRESULT GetName(long _idx, BSTR* ret);
```

C#

```
public String GetName(long _idx);
```

Visual Basic

```
Public Function GetName(_idx As long) As String
```

Description

Function returns name of the cut of the indicated index. .

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

III.5.11.3.5 Remove

C++

```
HRESULT Remove(long _idx);
```

C#

```
public void Remove(long _idx);
```

Visual Basic

```
Public Sub Remove(_idx As long)
```

Description

Function deletes the cut of the indicated index. .

Version

Available since version 3.

III.5.11.3.6 Store

C++

```
HRESULT Store(IRobotPanelCut* _cut, BSTR _name);
```

C#

```
public void Store(IRobotPanelCut _cut, String _name);
```

Visual Basic

```
Public Sub Store(ByRef _cut As IRobotPanelCut, _name As String)
```

Description

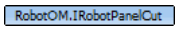
Function saves the cut under the specified name. .

Version

Available since version 3.

III.5.12 IRobotPanelCut

Class Hierarchy



C++

```
interface IRobotPanelCut : IDispatch;
```

C#

```
public interface IRobotPanelCut;
```

Visual Basic

```
Public Interface IRobotPanelCut
```

Description

Interface defining a cut through panel. .

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

III.5.12.1 IRobotPanelCut Members

The following tables list the members exposed by IRobotPanelCut.

Public Fields

	Name	Description
	Active (see page 982)	Flag indicating if a given cut is active.
	Color (see page 982)	Color applied while presenting a cut in graphical views .
	DefType (see page 982)	Type of a panel cut definition.
	Point1 (see page 983)	
	Point2 (see page 983)	
	Point3 (see page 983)	

III.5.12.2 IRobotPanelCut Fields

The fields of the IRobotPanelCut class are listed here.

Public Fields

	Name	Description
	Active (see page 982)	Flag indicating if a given cut is active.
	Color (see page 982)	Color applied while presenting a cut in graphical views .
	DefType (see page 982)	Type of a panel cut definition.
	Point1 (see page 983)	
	Point2 (see page 983)	
	Point3 (see page 983)	

III.5.12.2.1 Active

C++

```
HRESULT get_Active(VARIANT_BOOL*);
HRESULT put_Active(VARIANT_BOOL);
```

C#

```
public bool Active { get; set; }
```

Visual Basic

```
Public Active As Boolean
```

Description

Flag indicating if a given cut is active.

Version

Available since version 3.

III.5.12.2.2 Color

C++

```
HRESULT get_Color( long*);
HRESULT put_Color( long);
```

C#

```
public  long Color { get; set; }
```

Visual Basic

```
Public Color As  long
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Color applied while presenting a cut in graphical views .

Version

Available since version 3.

III.5.12.2.3 DefType

C++

```
HRESULT get_DefType(IRobotPanelCutDefinitionType*);
HRESULT put_DefType(IRobotPanelCutDefinitionType);
```

C#

```
public IRobotPanelCutDefinitionType DefType { get; set; }
```

Visual Basic

```
Public DefType As IRobotPanelCutDefinitionType
```

Description

Type of a panel cut definition.

Version

Available since version 3.

III.5.12.2.4 Point1

C++

```
HRESULT get_Point1(IRobotGeoPoint3D**);
```

C#

```
public IRobotGeoPoint3D Point1 { get; }
```

Visual Basic

```
Public ReadOnly Point1 As IRobotGeoPoint3D
```

Version

Available since version 3.

III.5.12.2.5 Point2

C++

```
HRESULT get_Point2(IRobotGeoPoint3D**);
```

C#

```
public IRobotGeoPoint3D Point2 { get; }
```

Visual Basic

```
Public ReadOnly Point2 As IRobotGeoPoint3D
```

Version

Available since version 3.

III.5.12.2.6 Point3

C++

```
HRESULT get_Point3(IRobotGeoPoint3D**);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRobotGeoPoint3D Point3 { get; }
```

Visual Basic

```
Public ReadOnly Point3 As IRobotGeoPoint3D
```

Version

Available since version 3.

III.5.13 IRobotPanelCutDefinitionType

C++

```
enum IRobotPanelCutDefinitionType;
```

C#

```
public enum IRobotPanelCutDefinitionType;
```

Visual Basic

```
Public Enum IRobotPanelCutDefinitionType
```

Members

Members	Description
I_PCDT_FULL_PLANE = 1	Unlimited cut plane, determined by two indicated points. Available since version 3.
I_PCDT_LIMITED_PLANE = 2	A cut plane defined by a segment determined by two indicated points and perpendicular to XY plane. Available since version 3.

Description

Types of panel cut definition. .

Version

Available since version 3.

III.5.14 IRobotFeResultType

C++

```
enum IRobotFeResultType;
```

C#

```
public enum IRobotFeResultType;
```

Visual Basic

```
Public Enum IRobotFeResultType
```

Members

Members	Description
I_FRT_DETAILED_SXX = 483	Available since version 4.
I_FRT_DETAILED_SYY = 484	Available since version 4.
I_FRT_DETAILED_SXY = 485	Available since version 4.
I_FRT_DETAILED_NXX = 492	Available since version 4.
I_FRT_DETAILED_NYY = 493	Available since version 4.
I_FRT_DETAILED_NXY = 494	Available since version 4.

(c) 2014 Autodesk Inc. All rights reserved

I_FRT_DETAILED_MXX = 501	Available since version 4.
I_FRT_DETAILED_MYY = 502	Available since version 4.
I_FRT_DETAILED_MXY = 503	Available since version 4.
I_FRT_DETAILED_TXX = 510	Available since version 4.
I_FRT_DETAILED_TYY = 511	Available since version 4.
I_FRT_DETAILED_QXX = 519	Available since version 4.
I_FRT_DETAILED_QYY = 520	Available since version 4.
I_FRT_DETAILED_UXX = 537	Available since version 4.
I_FRT_DETAILED_UYY = 538	Available since version 4.
I_FRT_DETAILED_WNORM = 531	Available since version 4.
I_FRT_DETAILED_RXX = 546	Available since version 4.
I_FRT_DETAILED_RYY = 547	Available since version 4.
I_FRT_DETAILED_RNORM = 549	Available since version 4.
I_FRT_DETAILED_PNORM = 558	Available since version 4.
I_FRT_PRINCIPAL_S1 = 487	Available since version 4.
I_FRT_PRINCIPAL_S2 = 488	Available since version 4.
I_FRT_PRINCIPAL_S1_2 = 490	Available since version 4.
I_FRT_PRINCIPAL_SAL = 489	Available since version 4.
I_FRT_PRINCIPAL_N1 = 496	Available since version 4.
I_FRT_PRINCIPAL_N2 = 497	Available since version 4.
I_FRT_PRINCIPAL_N1_2 = 499	Available since version 4.
I_FRT_PRINCIPAL_NAL = 498	Available since version 4.
I_FRT_PRINCIPAL_M1 = 505	Available since version 4.
I_FRT_PRINCIPAL_M2 = 506	Available since version 4.
I_FRT_PRINCIPAL_M1_2 = 508	Available since version 4.
I_FRT_PRINCIPAL_MAL = 507	Available since version 4.
I_FRT_PRINCIPAL_T1_2 = 517	Available since version 4.
I_FRT_PRINCIPAL_Q1_2 = 526	Available since version 4.
I_FRT_PRINCIPAL_UGX = 1302	Available since version 4.
I_FRT_PRINCIPAL_UGY = 1303	Available since version 4.
I_FRT_PRINCIPAL_UGZ = 1304	Available since version 4.
I_FRT_PRINCIPAL_U = 1305	Available since version 4.
I_FRT_COMPLEX_S_MISES = 491	Available since version 4.
I_FRT_COMPLEX_N_MISES = 500	Available since version 4.
I_FRT_COMPLEX_M_MISES = 509	Available since version 4.
I_FRT_COMPLEX_MXX_TOP_WA = 889	Available since version 4.
I_FRT_COMPLEX_MXX_BOTTOM_WA = 890	Available since version 4.
I_FRT_COMPLEX_MYY_TOP_WA = 891	Available since version 4.
I_FRT_COMPLEX_MYY_BOTTOM_WA = 892	Available since version 4.
I_FRT_COMPLEX_MXX_TOP_NEN = 893	Available since version 4.
I_FRT_COMPLEX_MXX_BOTTOM_NEN = 894	Available since version 4.
I_FRT_COMPLEX_MYY_TOP_NEN = 895	Available since version 4.
I_FRT_COMPLEX_MYY_BOTTOM_NEN = 896	Available since version 4.
I_FRT_REINF_AX_TOP = 0	Available since version 4.
I_FRT_REINF_AX_BOTTOM = 1	Available since version 4.
I_FRT_REINF_AY_TOP = 2	Available since version 4.
I_FRT_REINF_AY_BOTTOM = 3	Available since version 4.
I_FRT_REINF_E_AX_TOP = 4	Available since version 4.
I_FRT_REINF_E_AX_BOTTOM = 5	Available since version 4.

(c) 2014 Autodesk Inc. All rights reserved

I_FRT_REINF_E_AY_TOP = 6	Available since version 4.
I_FRT_REINF_E_AY_BOTTOM = 7	Available since version 4.
I_FRT_REINF_A_MIN = 8	Available since version 4.
I_FRT_REINF_AX = 9	Available since version 4.
I_FRT_REINF_AY = 10	Available since version 4.
I_FRT_REINF_F = 11	Available since version 4.
I_FRT_REDUCED_NX = 1217	Available since version 4.
I_FRT_REDUCED_MZ = 1218	Available since version 4.
I_FRT_REDUCED_TY = 1219	Available since version 4.
I_FRT_REDUCED_SO = 1220	Available since version 4.
I_FRT_REDUCED_SE = 1221	Available since version 4.
I_FRT_REDUCED_T = 1222	Available since version 4.
I_FRT_REDUCED_TZ = 1223	Available since version 4.
I_FRT_REDUCED_MY = 1224	Available since version 4.
I_FRT_REDUCED_LENGTH = 1225	Available since version 4.
I_FRT_REDUCED_HEIGHT = 1226	Available since version 4.

Description

Designation of the type of result calculated for finite elements.

Version

Available since version 4.

III.5.15 IRobotFeResultSmoothing

C++

```
enum IRobotFeResultSmoothing;
```

C#

```
public enum IRobotFeResultSmoothing;
```

Visual Basic

```
Public Enum IRobotFeResultSmoothing
```

Members

Members	Description
I_FRS_NO_SMOOTHING = 0	Exact results in nodes. Available since version 4.
I_FRS_GLOBAL_SMOOTHING = 1	Globally averaged results in nodes. Available since version 4.
I_FRS_SMOOTHING_WITHIN_A_PANEL = 2	Results in nodes - averaged within a panel. Available since version 4.
I_FRS_SMOOTHING_ACCORDING_TO_SELECTION = 3	Results in nodes averaged according to the selection of finite elements to which the nodes belong. Available since version 4.
I_FRS_IN_ELEMENT_CENTER = 4	Averaged results in finite element centers. Available since version 4.

Description

Available methods of averaging (smoothing) results.

Version

Available since version 4.

(c) 2014 Autodesk Inc. All rights reserved

III.5.16 IRobotFeResultKind

C++

```
enum IRobotFeResultKind;
```

C#

```
public enum IRobotFeResultKind;
```

Visual Basic

```
Public Enum IRobotFeResultKind
```

Members

Members	Description
I_FRK_PANEL_NODE = 0	Available since version 4.
I_FRK_ELEMENT_CENTER = 1	Available since version 4.
I_FRK_ELEMENT_NODE = 2	Available since version 4.

Version

Available since version 4.

III.5.17 IRobotFeExtremeParams

Class Hierarchy

RobotOM.IRobotFeExtremeParams

C++

```
interface IRobotFeExtremeParams : IDispatch;
```

C#

```
public interface IRobotFeExtremeParams;
```

Visual Basic

```
Public Interface IRobotFeExtremeParams
```

Description

Parameters of searching extreme results for finite elements.

The list of nodes to be considered while searching extreme values is determined based on the settings in the fields NodeSel (see page 990), ElementSel (see page 989) and PanelSel (see page 991). If a list of nodes is set in the NodeSel (see page 990) field, then this list exactly will be considered in searching extreme values. If the NodeSel (see page 990) field is empty and the ElementSel (see page 989) field is set, then all the nodes belonging to the indicated finite elements will be considered. If the fields NodeSel (see page 990) and ElementSel (see page 989) are empty and the PanelSel (see page 991) field is set, then all the nodes belonging to the indicated panels will be considered. If the fields NodeSel (see page 990), PanelSel (see page 991) and ElementSel (see page 989) are empty, then all the structure nodes will be considered.

For results in finite element centers, the list of elements to be considered is determined based on the ElementSel (see page 989) field. If the ElementSel (see page 989) list is empty and the PanelSel (see page 991) field is set, then all the finite elements belonging to the indicated panels will be considered. If the fields ElementSel (see page 989) and PanelSel (see page 991) are empty, then all the finite elements of the structure will be considered.

.

Version

Available since version 4.

(c) 2014 Autodesk Inc. All rights reserved

III.5.17.1 IRobotFeExtremeParams Members

The following tables list the members exposed by IRobotFeExtremeParams.

Public Fields

	Name	Description
	CaseCmpnt (see page 988)	Number of a load case component or mode number.
	CaseSel (see page 989)	List of load cases to be considered .
	ElementSel (see page 989)	List of finite elements to be considered.
	Layer (see page 989)	Layer.
	LayerArbitraryValue (see page 990)	Layer (see page 989) thickness.
	ModeCmb (see page 990)	Mode combination.
	NodeSel (see page 990)	List of nodes to be considered .
	PanelSel (see page 991)	List of panels to be considered.
	ReducedCutPos (see page 991)	
	ResultId (see page 991)	Designation of the result type.
	Smoothing (see page 992)	Method of result averaging.

Public Methods

	Name	Description
	GetDirX (see page 992)	
	Reset (see page 992)	
	SetDirX (see page 993)	

III.5.17.2 IRobotFeExtremeParams Fields

The fields of the IRobotFeExtremeParams class are listed here.

Public Fields

	Name	Description
	CaseCmpnt (see page 988)	Number of a load case component or mode number.
	CaseSel (see page 989)	List of load cases to be considered .
	ElementSel (see page 989)	List of finite elements to be considered.
	Layer (see page 989)	Layer.
	LayerArbitraryValue (see page 990)	Layer (see page 989) thickness.
	ModeCmb (see page 990)	Mode combination.
	NodeSel (see page 990)	List of nodes to be considered .
	PanelSel (see page 991)	List of panels to be considered.
	ReducedCutPos (see page 991)	
	ResultId (see page 991)	Designation of the result type.
	Smoothing (see page 992)	Method of result averaging.

III.5.17.2.1 CaseCmpnt

C++

```
HRESULT get_CaseCmpnt(long*);
HRESULT put_CaseCmpnt(long);
```

C#

```
public long CaseCmpnt { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public CaseCmpnt As long
```

Description

Number of a load case component or mode number.

Version

Available since version 4.1.

III.5.17.2.2 CaseSel

C++

```
HRESULT get_CaseSel(BSTR*);
HRESULT put_CaseSel(BSTR);
```

C#

```
public String CaseSel { get; set; }
```

Visual Basic

```
Public CaseSel As String
```

Description

List of load cases to be considered .

Version

Available since version 4.

III.5.17.2.3 ElementSel

C++

```
HRESULT get_ElementSel(BSTR*);
HRESULT put_ElementSel(BSTR);
```

C#

```
public String ElementSel { get; set; }
```

Visual Basic

```
Public ElementSel As String
```

Description

List of finite elements to be considered.

Version

Available since version 4.

III.5.17.2.4 Layer

C++

```
HRESULT get_Layer(IRobotFeLayerType*);
HRESULT put_Layer(IRobotFeLayerType);
```

C#

```
public IRobotFeLayerType Layer { get; set; }
```

Visual Basic

```
Public Layer As IRobotFeLayerType
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Layer.

Version

Available since version 4.

III.5.17.2.5 LayerArbitraryValue

C++

```
HRESULT get_LayerArbitraryValue(double*);
HRESULT put_LayerArbitraryValue(double);
```

C#

```
public double LayerArbitraryValue { get; set; }
```

Visual Basic

```
Public LayerArbitraryValue As double
```

Description

Layer (see page 989) thickness.

Version

Available since version 4.

III.5.17.2.6 ModeCmb

C++

```
HRESULT get_ModeCmb(IRobotModeCombinationType*);
HRESULT put_ModeCmb(IRobotModeCombinationType);
```

C#

```
public IRobotModeCombinationType ModeCmb { get; set; }
```

Visual Basic

```
Public ModeCmb As IRobotModeCombinationType
```

Description

Mode combination.

Version

Available since version 4.1.

III.5.17.2.7 NodeSel

C++

```
HRESULT get_NodeSel(BSTR*);
HRESULT put_NodeSel(BSTR);
```

C#

```
public String NodeSel { get; set; }
```

Visual Basic

```
Public NodeSel As String
```

Description

List of nodes to be considered .

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 4.

III.5.17.2.8 PanelSel

C++

```
HRESULT get_PanelSel(BSTR*);
HRESULT put_PanelSel(BSTR);
```

C#

```
public String PanelSel { get; set; }
```

Visual Basic

```
Public PanelSel As String
```

Description

List of panels to be considered.

Version

Available since version 4.

III.5.17.2.9 ReducedCutPos

C++

```
HRESULT get_ReducedCutPos(IRobotFeResultReducedCutPosition*);
HRESULT put_ReducedCutPos(IRobotFeResultReducedCutPosition);
```

C#

```
public IRobotFeResultReducedCutPosition ReducedCutPos { get; set; }
```

Visual Basic

```
Public ReducedCutPos As IRobotFeResultReducedCutPosition
```

Version

Available since version 4.

III.5.17.2.10 ResultId

C++

```
HRESULT get_ResultId(IRobotFeResultType*);
HRESULT put_ResultId(IRobotFeResultType);
```

C#

```
public IRobotFeResultType ResultId { get; set; }
```

Visual Basic

```
Public ResultId As IRobotFeResultType
```

Description

Designation of the result type.

Version

Available since version 4.

(c) 2014 Autodesk Inc. All rights reserved

III.5.17.2.11 Smoothing

C++

```
HRESULT get_Smoothing(IRobotFeResultSmoothing*);
HRESULT put_Smoothing(IRobotFeResultSmoothing);
```

C#

```
public IRobotFeResultSmoothing Smoothing { get; set; }
```

Visual Basic

```
Public Smoothing As IRobotFeResultSmoothing
```

Description

Method of result averaging.

Version

Available since version 4.

III.5.17.3 IRobotFeExtremeParams Methods

The methods of the IRobotFeExtremeParams class are listed here.

Public Methods

	Name	Description
	GetDirX (see page 992)	
	Reset (see page 992)	
	SetDirX (see page 993)	

III.5.17.3.1 GetDirX

C++

```
HRESULT GetDirX(double* _x, double* _y, double* _z, IRobotObjLocalXDirDefinitionType* ret);
```

C#

```
public IRobotObjLocalXDirDefinitionType GetDirX(double* _x, double* _y, double* _z);
```

Visual Basic

```
Public Function GetDirX(ByRef _x As double*, ByRef _y As double*, ByRef _z As double*) As
IRobotObjLocalXDirDefinitionType
```

Version

Available since version 4.

III.5.17.3.2 Reset

C++

```
HRESULT Reset();
```

C#

```
public void Reset();
```

Visual Basic

```
Public Sub Reset()
```

Version

Available since version 4.

(c) 2014 Autodesk Inc. All rights reserved

III.5.17.3.3 SetDirX

C++

```
HRESULT SetDirX(IRobotObjLocalXDirDefinitionType _def_type, double _x, double _y, double _z);
```

C#

```
public void SetDirX(IRobotObjLocalXDirDefinitionType _def_type, double _x, double _y, double _z);
```

Visual Basic

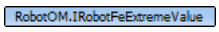
```
Public Sub SetDirX(_def_type As IRobotObjLocalXDirDefinitionType, _x As double, _y As double, _z As double)
```

Version

Available since version 4.

III.5.18 IRobotFeExtremeValue

Class Hierarchy



C++

```
interface IRobotFeExtremeValue : IDispatch;
```

C#

```
public interface IRobotFeExtremeValue;
```

Visual Basic

```
Public Interface IRobotFeExtremeValue
```

Description

Interface providing access to information about the extreme value. .

Version

Available since version 4.

III.5.18.1 IRobotFeExtremeValue Members

The following tables list the members exposed by IRobotFeExtremeValue.

Public Fields

	Name	Description
	Case (see page 995)	
	CaseCmpnt (see page 995)	Additional number indicating a load case component • for a code combination it is the number of its component • for a modal analysis case it is the number of a mode • for results of time history analysis it is the number of a step.
	Element (see page 995)	
	IsAvailable (see page 995)	Flag indicating if the extreme value has been taken correctly from the result server.
	Layer (see page 996)	

(c) 2014 Autodesk Inc. All rights reserved

	LayerArbitraryValue (see page 996)	
	ModeCmb (see page 996)	
	Node (see page 996)	
	Panel (see page 997)	
	ReducedCutPos (see page 997)	
	ResultId (see page 997)	
	Smoothing (see page 997)	
	Value (see page 998)	

Public Methods

	Name	Description
	GetDirX (see page 998)	Function takes the direction of the local X axis of the coordinate system, in which the specified extreme value has been obtained. The values read are the coordinates of the X axis direction vector for definition in the Cartesian system or the coordinates of the central point for definition in the polar system. If values of all the arguments: x, y, z are set as zeros, then the system direction is compatible with the panel local system. In addition, the function returns the type of the system direction definition. .

III.5.18.2 IRobotFeExtremeValue Fields

The fields of the IRobotFeExtremeValue class are listed here.

Public Fields

	Name	Description
	Case (see page 995)	
	CaseCmpnt (see page 995)	Additional number indicating a load case component • for a code combination it is the number of its component • for a modal analysis case it is the number of a mode • for results of time history analysis it is the number of a step.
	Element (see page 995)	
	IsAvailable (see page 995)	Flag indicating if the extreme value has been taken correctly from the result server.
	Layer (see page 996)	
	LayerArbitraryValue (see page 996)	
	ModeCmb (see page 996)	
	Node (see page 996)	
	Panel (see page 997)	
	ReducedCutPos (see page 997)	
	ResultId (see page 997)	
	Smoothing (see page 997)	
	Value (see page 998)	

III.5.18.2.1 Case

C++

```
HRESULT get_Case(long*);
```

C#

```
public long Case { get; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public ReadOnly Case As long
```

Version

Available since version 4.

III.5.18.2.2 CaseCmpnt

C++

```
HRESULT get_CaseCmpnt(long*);
```

C#

```
public long CaseCmpnt { get; }
```

Visual Basic

```
Public ReadOnly CaseCmpnt As long
```

Description

Additional number indicating a load case component

- for a code combination it is the number of its component
- for a modal analysis case it is the number of a mode
- for results of time history analysis it is the number of a step.

Version

Available since version 4.

III.5.18.2.3 Element

C++

```
HRESULT get_Element(long*);
```

C#

```
public long Element { get; }
```

Visual Basic

```
Public ReadOnly Element As long
```

Version

Available since version 4.

III.5.18.2.4 IsAvailable

C++

```
HRESULT get_IsAvailable(VARIANT_BOOL*);
```

C#

```
public bool IsAvailable { get; }
```

Visual Basic

```
Public ReadOnly IsAvailable As Boolean
```

Description

Flag indicating if the extreme value has been taken correctly from the result server.

Version

Available since version 4.

(c) 2014 Autodesk Inc. All rights reserved

III.5.18.2.5 Layer

C++

```
HRESULT get_Layer(IRobotFeLayerType*);
```

C#

```
public IRobotFeLayerType Layer { get; }
```

Visual Basic

```
Public ReadOnly Layer As IRobotFeLayerType
```

Version

Available since version 4.

III.5.18.2.6 LayerArbitraryValue

C++

```
HRESULT get_LayerArbitraryValue(double*);
```

C#

```
public double LayerArbitraryValue { get; }
```

Visual Basic

```
Public ReadOnly LayerArbitraryValue As double
```

Version

Available since version 4.

III.5.18.2.7 ModeCmb

C++

```
HRESULT get_ModeCmb(IRobotModeCombinationType*);
```

C#

```
public IRobotModeCombinationType ModeCmb { get; }
```

Visual Basic

```
Public ReadOnly ModeCmb As IRobotModeCombinationType
```

Version

Available since version 4.

III.5.18.2.8 Node

C++

```
HRESULT get_Node(long*);
```

C#

```
public long Node { get; }
```

Visual Basic

```
Public ReadOnly Node As long
```

Version

Available since version 4.

(c) 2014 Autodesk Inc. All rights reserved

III.5.18.2.9 Panel

C++

```
HRESULT get_Panel(long*);
```

C#

```
public long Panel { get; }
```

Visual Basic

```
Public ReadOnly Panel As long
```

Version

Available since version 4.

III.5.18.2.10 ReducedCutPos

C++

```
HRESULT get_ReducedCutPos(IRobotFeResultReducedCutPosition*);
```

C#

```
public IRobotFeResultReducedCutPosition ReducedCutPos { get; }
```

Visual Basic

```
Public ReadOnly ReducedCutPos As IRobotFeResultReducedCutPosition
```

Version

Available since version 4.

III.5.18.2.11 ResultId

C++

```
HRESULT get_ResultId(IRobotFeResultType*);
```

C#

```
public IRobotFeResultType ResultId { get; }
```

Visual Basic

```
Public ReadOnly ResultId As IRobotFeResultType
```

Version

Available since version 4.

III.5.18.2.12 Smoothing

C++

```
HRESULT get_Smoothing(IRobotFeResultSmoothing*);
```

C#

```
public IRobotFeResultSmoothing Smoothing { get; }
```

Visual Basic

```
Public ReadOnly Smoothing As IRobotFeResultSmoothing
```

Version

Available since version 4.

(c) 2014 Autodesk Inc. All rights reserved

III.5.18.2.13 Value

C++

```
HRESULT get_Value(double*);
```

C#

```
public double Value { get; }
```

Visual Basic

```
Public ReadOnly Value As double
```

Version

Available since version 4.

III.5.18.3 IRobotFeExtremeValue Methods

The methods of the IRobotFeExtremeValue class are listed here.

Public Methods

	Name	Description
	GetDirX (see page 998)	Function takes the direction of the local X axis of the coordinate system, in which the specified extreme value has been obtained. The values read are the coordinates of the X axis direction vector for definition in the Cartesian system or the coordinates of the central point for definition in the polar system. If values of all the arguments: x, y, z are set as zeros, then the system direction is compatible with the panel local system. In addition, the function returns the type of the system direction definition. .

III.5.18.3.1 GetDirX

C++

```
HRESULT GetDirX(double* _x, double* _y, double* _z, IRobotObjLocalXDirDefinitionType* ret);
```

C#

```
public IRobotObjLocalXDirDefinitionType GetDirX(double* _x, double* _y, double* _z);
```

Visual Basic

```
Public Function GetDirX(ByRef _x As double*, ByRef _y As double*, ByRef _z As double*) As
IRobotObjLocalXDirDefinitionType
```

Description

Function takes the direction of the local X axis of the coordinate system, in which the specified extreme value has been obtained. The values read are the coordinates of the X axis direction vector for definition in the Cartesian system or the coordinates of the central point for definition in the polar system. If values of all the arguments: x, y, z are set as zeros, then the system direction is compatible with the panel local system. In addition, the function returns the type of the system direction definition. .

Version

Available since version 4.

III.5.19 IRobotFeMultiResultType

Class Hierarchy

RobotOM.IRobotFeMultiResultType

(c) 2014 Autodesk Inc. All rights reserved

C++

```
interface IRobotFeMultiResultType : IDispatch;
```

C#

```
public interface IRobotFeMultiResultType;
```

Visual Basic

```
Public Interface IRobotFeMultiResultType
```

Description

Interface which enables indicating simultaneously - a great number of result types for finite elements.

Version

Available since version 4.1.

III.5.19.1 IRobotFeMultiResultType Members

The following tables list the members exposed by IRobotFeMultiResultType.

Public Fields

	Name	Description
	Count (see page 999)	

Public Methods

	Name	Description
	Add (see page 1000)	
	Get (see page 1000)	
	Remove (see page 1000)	

III.5.19.2 IRobotFeMultiResultType Fields

The fields of the IRobotFeMultiResultType class are listed here.

Public Fields

	Name	Description
	Count (see page 999)	

III.5.19.2.1 Count

C++

```
HRESULT get_Count(long*);
```

C#

```
public long Count { get; }
```

Visual Basic

```
Public ReadOnly Count As long
```

Version

Available since version 4.1.

III.5.19.3 IRobotFeMultiResultType Methods

The methods of the IRobotFeMultiResultType class are listed here.

(c) 2014 Autodesk Inc. All rights reserved

Public Methods

	Name	Description
	Add (see page 1000)	
	Get (see page 1000)	
	Remove (see page 1000)	

III.5.19.3.1 Add

C++

```
HRESULT Add(IRobotFeResultType _result_type);
```

C#

```
public void Add(IRobotFeResultType _result_type);
```

Visual Basic

```
Public Sub Add(_result_type As IRobotFeResultType)
```

Version

Available since version 4.1.

III.5.19.3.2 Get

C++

```
HRESULT Get(long _idx, IRobotFeResultType* ret);
```

C#

```
public IRobotFeResultType Get(long _idx);
```

Visual Basic

```
Public Function Get(_idx As long) As IRobotFeResultType
```

Version

Available since version 4.1.

III.5.19.3.3 Remove

C++

```
HRESULT Remove(IRobotFeResultType _result_type);
```

C#

```
public void Remove(IRobotFeResultType _result_type);
```

Visual Basic

```
Public Sub Remove(_result_type As IRobotFeResultType)
```

Version

Available since version 4.1.

III.5.20 IRobotFeMultiExtremeValue

Class Hierarchy

RobotOM.IRobotFeMultiExtremeValue

(c) 2014 Autodesk Inc. All rights reserved

C++

```
interface IRobotFeMultiExtremeValue : IDispatch;
```

C#

```
public interface IRobotFeMultiExtremeValue;
```

Visual Basic

```
Public Interface IRobotFeMultiExtremeValue
```

Description

Interface providing access to a great number of min./max. results for finite elements.

Version

Available since version 4.1.

III.5.20.1 IRobotFeMultiExtremeValue Members

The following tables list the members exposed by IRobotFeMultiExtremeValue.

Public Fields

	Name	Description
	Count (see page 1001)	

Public Methods

	Name	Description
	Get (see page 1002)	
	GetByResType (see page 1002)	Function returns the indicated min./max. result.

III.5.20.2 IRobotFeMultiExtremeValue Fields

The fields of the IRobotFeMultiExtremeValue class are listed here.

Public Fields

	Name	Description
	Count (see page 1001)	

III.5.20.2.1 Count

C++

```
HRESULT get_Count(long*);
```

C#

```
public long Count { get; }
```

Visual Basic

```
Public ReadOnly Count As long
```

Version

Available since version 4.1.

III.5.20.3 IRobotFeMultiExtremeValue Methods

The methods of the IRobotFeMultiExtremeValue class are listed here.

(c) 2014 Autodesk Inc. All rights reserved

Public Methods

	Name	Description
	Get (see page 1002)	
	GetByResType (see page 1002)	Function returns the indicated min./max. result.

III.5.20.3.1 Get

C++

```
HRESULT Get(long _idx, IRobotFeExtremeValue** ret);
```

C#

```
public IRobotFeExtremeValue Get(long _idx);
```

Visual Basic

```
Public Function Get(_idx As long) As IRobotFeExtremeValue
```

Version

Available since version 4.1.

III.5.20.3.2 GetByResType

C++

```
HRESULT GetByResType(IRobotFeResultType _result_type, IRobotFeExtremeValue** ret);
```

C#

```
public IRobotFeExtremeValue GetByResType(IRobotFeResultType _result_type);
```

Visual Basic

```
Public Function GetByResType(_result_type As IRobotFeResultType) As IRobotFeExtremeValue
```

Description

Function returns the indicated min./max. result.

Version

Available since version 4.1.

III.6 IRobotResultServer

Class Hierarchy

RobotOM.IRobotResultServer

C++

```
interface IRobotResultServer : IDispatch;
```

C#

```
public interface IRobotResultServer;
```

Visual Basic

```
Public Interface IRobotResultServer
```

Description

The server makes available all types of results obtained by performing structure calculations. .

(c) 2014 Autodesk Inc. All rights reserved

III.6.1 IRobotResultServer Members

The following tables list the members exposed by IRobotResultServer.

Public Fields

	Name	Description
	Advanced (see page 1004)	Available since version 1.7.
	Any (see page 1004)	Access to any calculation result .
	Bars (see page 1004)	Server of all results for bars .
	CalculationResume (see page 1004)	.
	Extremes (see page 1005)	Server providing access to some extreme values .
	FiniteElems (see page 1005)	Result server for finite elements .
	Nodes (see page 1005)	Server of all results for nodes .
	Status (see page 1006)	Results status.
	Storeys (see page 1006)	Results for stories.
	Total (see page 1006)	Physical properties of a whole structure.

Public Methods

	Name	Description
	Query (see page 1007)	Function fills out the specified set with results which correspond to the query parameters. Function returns information if all results have been got. To get all results, it is necessary to repeat calling up the function until the value I_RQRT_DONE value is returned.

III.6.2 IRobotResultServer Fields

The fields of the IRobotResultServer class are listed here.

Public Fields

	Name	Description
	Advanced (see page 1004)	Available since version 1.7.
	Any (see page 1004)	Access to any calculation result .
	Bars (see page 1004)	Server of all results for bars .
	CalculationResume (see page 1004)	.
	Extremes (see page 1005)	Server providing access to some extreme values .
	FiniteElems (see page 1005)	Result server for finite elements .
	Nodes (see page 1005)	Server of all results for nodes .
	Status (see page 1006)	Results status.
	Storeys (see page 1006)	Results for stories.
	Total (see page 1006)	Physical properties of a whole structure.

III.6.2.1 Advanced

C++

```
HRESULT get_Advanced(IRobotAdvancedResultServer**);
```

C#

```
public IRobotAdvancedResultServer Advanced { get; }
```

Visual Basic

```
Public ReadOnly Advanced As IRobotAdvancedResultServer
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Available since version 1.7.

III.6.2.2 Any

C++

```
HRESULT get_Any(IRobotUniversalResultAccess**);
```

C#

```
public IRobotUniversalResultAccess Any { get; }
```

Visual Basic

```
Public ReadOnly Any As IRobotUniversalResultAccess
```

Description

Access to any calculation result .

Version

Available since version 3.5.

III.6.2.3 Bars

C++

```
HRESULT get_Bars(IRobotBarResultServer**);
```

C#

```
public IRobotBarResultServer Bars { get; }
```

Visual Basic

```
Public ReadOnly Bars As IRobotBarResultServer
```

Description

Server of all results for bars .

III.6.2.4 CalculationResume

C++

```
HRESULT get_CalculationResume(IRobotCalculationResume**);
```

C#

```
public IRobotCalculationResume CalculationResume { get; }
```

Visual Basic

```
Public ReadOnly CalculationResume As IRobotCalculationResume
```

Description

.

Version

Available since version 15.

III.6.2.5 Extremes

C++

```
HRESULT get_Extremes(IRobotExtremeResultServer**);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRobotExtremeResultServer Extremes { get; }
```

Visual Basic

```
Public ReadOnly Extremes As IRobotExtremeResultServer
```

Description

Server providing access to some extreme values .

Version

Available since version 3.5.

III.6.2.6 FiniteElems

C++

```
HRESULT get_FiniteElems(IRobotFeResultServer**);
```

C#

```
public IRobotFeResultServer FiniteElems { get; }
```

Visual Basic

```
Public ReadOnly FiniteElems As IRobotFeResultServer
```

Description

Result server for finite elements .

Version

Available since version 2.5.

III.6.2.7 Nodes

C++

```
HRESULT get_Nodes(IRobotNodeResultServer**);
```

C#

```
public IRobotNodeResultServer Nodes { get; }
```

Visual Basic

```
Public ReadOnly Nodes As IRobotNodeResultServer
```

Description

Server of all results for nodes .

III.6.2.8 Status

C++

```
HRESULT get_Status(IRobotResultStatusType*);
```

C#

```
public IRobotResultStatusType Status { get; }
```

Visual Basic

```
Public ReadOnly Status As IRobotResultStatusType
```

Description

Results status.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 14.

III.6.2.9 Storeys

C++

```
HRESULT get_Storeys(IRobotStoreyResultServer**);
```

C#

```
public IRobotStoreyResultServer Storeys { get; }
```

Visual Basic

```
Public ReadOnly Storeys As IRobotStoreyResultServer
```

Description

Results for stories.

Version

Available since version 9.7.

III.6.2.10 Total

C++

```
HRESULT get_Total(IRobotStructureValues**);
```

C#

```
public IRobotStructureValues Total { get; }
```

Visual Basic

```
Public ReadOnly Total As IRobotStructureValues
```

Description

Physical properties of a whole structure.

Version

Available since version 7.5.

III.6.3 IRobotResultServer Methods

The methods of the IRobotResultServer class are listed here.

Public Methods

	Name	Description
	Query (see page 1007)	Function fills out the specified set with results which correspond to the query parameters. Function returns information if all results have been got. To get all results, it is necessary to repeat calling up the function until the value I_RQRT_DONE value is returned.

III.6.3.1 Query

C++

```
HRESULT Query(IRobotResultQueryParams* _params, IRobotResultRowSet* _ret_results,
IRobotResultQueryReturnType* ret);
```

C#

```
public IRobotResultQueryReturnType Query(IRobotResultQueryParams _params,
IRobotResultRowSet _ret_results);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Function Query(ByRef _params As IRobotResultQueryParams, ByRef _ret_results As
IRobotResultRowSet) As IRobotResultQueryReturnType
```

Description

Function fills out the specified set with results which correspond to the query parameters. Function returns information if all results have been got. To get all results, it is necessary to repeat calling up the function until the value I_RQRT_DONE value is returned.

Version

Available since version 8.2.

III.7 IRobotUniversalResultAccess

Class Hierarchy

C++

```
interface IRobotUniversalResultAccess : IDispatch;
```

C#

```
public interface IRobotUniversalResultAccess;
```

Visual Basic

```
Public Interface IRobotUniversalResultAccess
```

Description

Universal access interface for calculation results. In order to obtain the relevant result the object should be first appropriately parametrized by determining what result type, for which load case and for which structure element is to be obtained, and then ResultValue (see page 1015) should be taken. .

III.7.1 IRobotUniversalResultAccess Members

The following tables list the members exposed by IRobotUniversalResultAccess.

Public Fields

	Name	Description
	Available (see page 1009)	Flag indicating if the result for defined parameters can be obtained .
	Bar (see page 1010)	Bar number for which the result is to be taken .
	CalcPoint (see page 1010)	
	DivCount (see page 1010)	Number of bar division points (only for the results obtained in the defined point of bar division) .
	DivPoint (see page 1010)	Number of bar division point (only for the results obtained in the defined point of bar division) .
	Element (see page 1011)	Number of the finite element .
	Layer (see page 1011)	Layer.
	LayerArbitraryValue (see page 1011)	Layer (see page 1011) thickness.
	LinearSupports (see page 1012)	Linear supports.
	LoadCase (see page 1012)	Number of the load case for which the result is to be taken .
	LoadCaseCmpnt (see page 1012)	Number of the load case component (significant for cases including more than one component, e.g. code combination) .

(c) 2014 Autodesk Inc. All rights reserved

	Mode (see page 1013)	Number of the load case mode (significant for cases for which many modes are generated) .
	ModeCmb (see page 1013)	Type of mode combination .
	Node (see page 1013)	Node number for which the result is to be taken .
	Panel (see page 1013)	Panel number.
	ReducedCutPos (see page 1014)	Position of the panel cut (for reduced results).
	ReinforceCalcMethod (see page 1014)	Calculation method for reinforcement .
	RelativePoint (see page 1014)	Relative point on the bar (significant only for the results obtained in the defined bar point) .
	ResultId (see page 1015)	Identifier determining the result type which is to be taken .
	ResultType (see page 1015)	Result type.
	ResultValue (see page 1015)	Result value for defined parameters (provided that the Available (see page 1009) flag assumes value different from zero - True) .
	Storey (see page 1015)	Story name.

Public Methods

	Name	Description
	GetDirX (see page 1016)	Function takes direction of X axis of the local coordinate system in which results will be presented. The values read are the coordinates of the direction vector of X axis for definition in the Cartesian system or the coordinates of the central point for definition in the Polar system. If values of all the arguments: x, y, z are set as zeros, then the system direction is compatible with the local panel system. In addition, the function returns a type of system direction definition. .
	Reset (see page 1016)	Function sets as zero all the settings concerning the result type and the manner of its taking. .
	ResultValue3D (see page 1017)	Function returns a result in a form of a triple of numbers.
	SetDirX (see page 1017)	Function allows defining direction of X axis of the system in which results will be presented. The function should be activated only when the vector of detailed results is being taken. Function parameters determine coordinates of the direction vector of X axis in the Cartesian system or coordinates of the central point in the Polar system. If all the values: x, y, z are set as zeros, a direction compatible with the local system of the selected panel will be defined. .

III.7.2 IRobotUniversalResultAccess Fields

The fields of the IRobotUniversalResultAccess class are listed here.

Public Fields

	Name	Description
	Available (see page 1009)	Flag indicating if the result for defined parameters can be obtained .
	Bar (see page 1010)	Bar number for which the result is to be taken .
	CalcPoint (see page 1010)	
	DivCount (see page 1010)	Number of bar division points (only for the results obtained in the defined point of bar division) .
	DivPoint (see page 1010)	Number of bar division point (only for the results obtained in the defined point of bar division) .
	Element (see page 1011)	Number of the finite element .
	Layer (see page 1011)	Layer.
	LayerArbitraryValue (see page 1011)	Layer (see page 1011) thickness.
	LinearSupports (see page 1012)	Linear supports.
	LoadCase (see page 1012)	Number of the load case for which the result is to be taken .

(c) 2014 Autodesk Inc. All rights reserved

	LoadCaseCmpnt (see page 1012)	Number of the load case component (significant for cases including more than one component, e.g. code combination) .
	Mode (see page 1013)	Number of the load case mode (significant for cases for which many modes are generated) .
	ModeCmb (see page 1013)	Type of mode combination .
	Node (see page 1013)	Node number for which the result is to be taken .
	Panel (see page 1013)	Panel number.
	ReducedCutPos (see page 1014)	Position of the panel cut (for reduced results).
	ReinforceCalcMethod (see page 1014)	Calculation method for reinforcement .
	RelativePoint (see page 1014)	Relative point on the bar (significant only for the results obtained in the defined bar point) .
	ResultId (see page 1015)	Identifier determining the result type which is to be taken .
	ResultType (see page 1015)	Result type.
	ResultValue (see page 1015)	Result value for defined parameters (provided that the Available (see page 1009) flag assumes value different from zero - True) .
	Storey (see page 1015)	Story name.

III.7.2.1 Available

C++

```
HRESULT get_Available(VARIANT_BOOL*);
```

C#

```
public bool Available { get; }
```

Visual Basic

```
Public ReadOnly Available As Boolean
```

Description

Flag indicating if the result for defined parameters can be obtained .

III.7.2.2 Bar

C++

```
HRESULT get_Bar(long*);
HRESULT put_Bar(long);
```

C#

```
public long Bar { get; set; }
```

Visual Basic

```
Public Bar As long
```

Description

Bar number for which the result is to be taken .

III.7.2.3 CalcPoint

C++

```
HRESULT get_CalcPoint(long*);
HRESULT put_CalcPoint(long);
```

C#

```
public long CalcPoint { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public CalcPoint As long
```

Version

Available (see page 1009) since version 8.

III.7.2.4 DivCount

C++

```
HRESULT get_DivCount(long*);
HRESULT put_DivCount(long);
```

C#

```
public long DivCount { get; set; }
```

Visual Basic

```
Public DivCount As long
```

Description

Number of bar division points (only for the results obtained in the defined point of bar division) .

III.7.2.5 DivPoint

C++

```
HRESULT get_DivPoint(long*);
HRESULT put_DivPoint(long);
```

C#

```
public long DivPoint { get; set; }
```

Visual Basic

```
Public DivPoint As long
```

Description

Number of bar division point (only for the results obtained in the defined point of bar division) .

III.7.2.6 Element

C++

```
HRESULT get_Element(long*);
HRESULT put_Element(long);
```

C#

```
public long Element { get; set; }
```

Visual Basic

```
Public Element As long
```

Description

Number of the finite element .

Version

Available (see page 1009) since version 3.5.

(c) 2014 Autodesk Inc. All rights reserved

III.7.2.7 Layer

C++

```
HRESULT get_Layer(IRobotFeLayerType*);
HRESULT put_Layer(IRobotFeLayerType);
```

C#

```
public IRobotFeLayerType Layer { get; set; }
```

Visual Basic

```
Public Layer As IRobotFeLayerType
```

Description

Layer.

Version

Available (see page 1009) since version 3.5.

III.7.2.8 LayerArbitraryValue

C++

```
HRESULT get_LayerArbitraryValue(double*);
HRESULT put_LayerArbitraryValue(double);
```

C#

```
public double LayerArbitraryValue { get; set; }
```

Visual Basic

```
Public LayerArbitraryValue As double
```

Description

Layer (see page 1011) thickness.

Version

Available (see page 1009) since version 3.5.

III.7.2.9 LinearSupports

C++

```
HRESULT get_LinearSupports(VARIANT_BOOL*);
HRESULT put_LinearSupports(VARIANT_BOOL);
```

C#

```
public bool LinearSupports { get; set; }
```

Visual Basic

```
Public LinearSupports As Boolean
```

Description

Linear supports.

Version

Available (see page 1009) since version 8.1.

(c) 2014 Autodesk Inc. All rights reserved

III.7.2.10 LoadCase

C++

```
HRESULT get_LoadCase(long*);
HRESULT put_LoadCase(long);
```

C#

```
public long LoadCase { get; set; }
```

Visual Basic

```
Public LoadCase As long
```

Description

Number of the load case for which the result is to be taken .

III.7.2.11 LoadCaseCmpnt

C++

```
HRESULT get_LoadCaseCmpnt(long*);
HRESULT put_LoadCaseCmpnt(long);
```

C#

```
public long LoadCaseCmpnt { get; set; }
```

Visual Basic

```
Public LoadCaseCmpnt As long
```

Description

Number of the load case component (significant for cases including more than one component, e.g. code combination) .

III.7.2.12 Mode

C++

```
HRESULT get_Mode(long*);
HRESULT put_Mode(long);
```

C#

```
public long Mode { get; set; }
```

Visual Basic

```
Public Mode As long
```

Description

Number of the load case mode (significant for cases for which many modes are generated) .

III.7.2.13 ModeCmb

C++

```
HRESULT get_ModeCmb(IRobotModeCombinationType*);
HRESULT put_ModeCmb(IRobotModeCombinationType);
```

C#

```
public IRobotModeCombinationType ModeCmb { get; set; }
```

Visual Basic

```
Public ModeCmb As IRobotModeCombinationType
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Type of mode combination .

III.7.2.14 Node

C++

```
HRESULT get_Node(long*);
HRESULT put_Node(long);
```

C#

```
public long Node { get; set; }
```

Visual Basic

```
Public Node As long
```

Description

Node number for which the result is to be taken .

III.7.2.15 Panel

C++

```
HRESULT get_Panel(long*);
HRESULT put_Panel(long);
```

C#

```
public long Panel { get; set; }
```

Visual Basic

```
Public Panel As long
```

Description

Panel number.

Version

Available (see page 1009) since version 3.5.

III.7.2.16 ReducedCutPos

C++

```
HRESULT get_ReducedCutPos(IRobotFeResultReducedCutPosition*);
HRESULT put_ReducedCutPos(IRobotFeResultReducedCutPosition);
```

C#

```
public IRobotFeResultReducedCutPosition ReducedCutPos { get; set; }
```

Visual Basic

```
Public ReducedCutPos As IRobotFeResultReducedCutPosition
```

Description

Position of the panel cut (for reduced results).

Version

Available (see page 1009) since version 3.5.

(c) 2014 Autodesk Inc. All rights reserved

III.7.2.17 ReinforceCalcMethod

C++

```
HRESULT get_ReinforceCalcMethod(IRobotReinforceCalcMethods*);
HRESULT put_ReinforceCalcMethod(IRobotReinforceCalcMethods);
```

C#

```
public IRobotReinforceCalcMethods ReinforceCalcMethod { get; set; }
```

Visual Basic

```
Public ReinforceCalcMethod As IRobotReinforceCalcMethods
```

Description

Calculation method for reinforcement .

Version

Available (see page 1009) since version 3.5.

III.7.2.18 RelativePoint

C++

```
HRESULT get_RelativePoint(double*);
HRESULT put_RelativePoint(double);
```

C#

```
public double RelativePoint { get; set; }
```

Visual Basic

```
Public RelativePoint As double
```

Description

Relative point on the bar (significant only for the results obtained in the defined bar point) .

III.7.2.19 ResultId

C++

```
HRESULT get_ResultId(long*);
HRESULT put_ResultId(long);
```

C#

```
public long ResultId { get; set; }
```

Visual Basic

```
Public ResultId As long
```

Description

Identifier determining the result type which is to be taken .

III.7.2.20 ResultType

C++

```
HRESULT get_ResultType(IRobotUniversalResultType*);
```

C#

```
public IRobotUniversalResultType ResultType { get; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public ReadOnly ResultType As IRobotUniversalResultType
```

Description

Result type.

III.7.2.21 ResultValue

C++

```
HRESULT get_ResultValue(double*);
```

C#

```
public double ResultValue { get; }
```

Visual Basic

```
Public ReadOnly ResultValue As double
```

Description

Result value for defined parameters (provided that the Available (see page 1009) flag assumes value different from zero - True) .

III.7.2.22 Storey

C++

```
HRESULT get_Storey(BSTR*);
HRESULT put_Storey(BSTR);
```

C#

```
public String Storey { get; set; }
```

Visual Basic

```
Public Storey As String
```

Description

Story name.

III.7.3 IRobotUniversalResultAccess Methods

The methods of the IRobotUniversalResultAccess class are listed here.

Public Methods

	Name	Description
	GetDirX (see page 1016)	Function takes direction of X axis of the local coordinate system in which results will be presented. The values read are the coordinates of the direction vector of X axis for definition in the Cartesian system or the coordinates of the central point for definition in the Polar system. If values of all the arguments: x, y, z are set as zeros, then the system direction is compatible with the local panel system. In addition, the function returns a type of system direction definition. .
	Reset (see page 1016)	Function sets as zero all the settings concerning the result type and the manner of its taking. .
	ResultValue3D (see page 1017)	Function returns a result in a form of a triple of numbers.

(c) 2014 Autodesk Inc. All rights reserved

	SetDirX (see page 1017)	Function allows defining direction of X axis of the system in which results will be presented. The function should be activated only when the vector of detailed results is being taken. Function parameters determine coordinates of the direction vector of X axis in the Cartesian system or coordinates of the central point in the Polar system. If all the values: x, y, z are set as zeros, a direction compatible with the local system of the selected panel will be defined. .

III.7.3.1 GetDirX

C++

```
HRESULT GetDirX(double* _x, double* _y, double* _z, IRobotObjLocalXDirDefinitionType* ret);
```

C#

```
public IRobotObjLocalXDirDefinitionType GetDirX(double* _x, double* _y, double* _z);
```

Visual Basic

```
Public Function GetDirX(ByRef _x As double*, ByRef _y As double*, ByRef _z As double*) As
IRobotObjLocalXDirDefinitionType
```

Description

Function takes direction of X axis of the local coordinate system in which results will be presented. The values read are the coordinates of the direction vector of X axis for definition in the Cartesian system or the coordinates of the central point for definition in the Polar system. If values of all the arguments: x, y, z are set as zeros, then the system direction is compatible with the local panel system. In addition, the function returns a type of system direction definition. .

Version

Available (see page 1009) since version 3.5.

III.7.3.2 Reset

C++

```
HRESULT Reset();
```

C#

```
public void Reset();
```

Visual Basic

```
Public Sub Reset()
```

Description

Function sets as zero all the settings concerning the result type and the manner of its taking. .

III.7.3.3 ResultValue3D

C++

```
HRESULT ResultValue3D(double* _val1, double* _val2, double* _val3);
```

C#

```
public void ResultValue3D(double* _val1, double* _val2, double* _val3);
```

Visual Basic

```
Public Sub ResultValue3D(ByRef _val1 As double*, ByRef _val2 As double*, ByRef _val3 As
double*)
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Function returns a result in a form of a triple of numbers.

III.7.3.4 SetDirX

C++

```
HRESULT SetDirX(IRobotObjLocalXDirDefinitionType _def_type, double _x, double _y, double
_z);
```

C#

```
public void SetDirX(IRobotObjLocalXDirDefinitionType _def_type, double _x, double _y,
double _z);
```

Visual Basic

```
Public Sub SetDirX(_def_type As IRobotObjLocalXDirDefinitionType, _x As double, _y As
double, _z As double)
```

Description

Function allows defining direction of X axis of the system in which results will be presented. The function should be activated only when the vector of detailed results is being taken. Function parameters determine coordinates of the direction vector of X axis in the Cartesian system or coordinates of the central point in the Polar system. If all the values: x, y, z are set as zeros, a direction compatible with the local system of the selected panel will be defined. .

Version

Available (see page 1009) since version 3.5.

III.8 IRobotExtremeResultServer

Class Hierarchy

RobotOM.IRobotExtremeResultServer

C++

```
interface IRobotExtremeResultServer : IDispatch;
```

C#

```
public interface IRobotExtremeResultServer;
```

Visual Basic

```
Public Interface IRobotExtremeResultServer
```

Description

Server of extreme results.

Version

Available since version 3.5.

III.8.1 IRobotExtremeResultServer Members

The following tables list the members exposed by IRobotExtremeResultServer.

Public Methods

	Name	Description
	MaxValue (see page 1018)	Function returns the maximal value calculated for the parameters given. .
	MinValue (see page 1018)	Function returns the minimal value calculated for the parameters given. .

(c) 2014 Autodesk Inc. All rights reserved

III.8.2 IRobotExtremeResultServer Methods

The methods of the IRobotExtremeResultServer class are listed here.

Public Methods

	Name	Description
	MaxValue (see page 1018)	Function returns the maximal value calculated for the parameters given. .
	MinValue (see page 1018)	Function returns the minimal value calculated for the parameters given. .

III.8.2.1 MaxValue

C++

```
HRESULT MaxValue(IRobotExtremeParams* _params, IRobotExtremeValue** ret);
```

C#

```
public IRobotExtremeValue MaxValue(IRobotExtremeParams _params);
```

Visual Basic

```
Public Function MaxValue(ByRef _params As IRobotExtremeParams) As IRobotExtremeValue
```

Description

Function returns the maximal value calculated for the parameters given. .

Version

Available since version 3.5.

III.8.2.2 MinValue

C++

```
HRESULT MinValue(IRobotExtremeParams* _params, IRobotExtremeValue** ret);
```

C#

```
public IRobotExtremeValue MinValue(IRobotExtremeParams _params);
```

Visual Basic

```
Public Function MinValue(ByRef _params As IRobotExtremeParams) As IRobotExtremeValue
```

Description

Function returns the minimal value calculated for the parameters given. .

Version

Available since version 3.5.

III.9 IRobotUniversalResultType

C++

```
enum IRobotUniversalResultType;
```

C#

```
public enum IRobotUniversalResultType;
```

Visual Basic

```
Public Enum IRobotUniversalResultType
```

(c) 2014 Autodesk Inc. All rights reserved

Members

Members	Description
I_URT_VALUE = 1	Result has a form of a single value.
I_URT_VALUE_3D = 2	Result has a form of a triple of numbers.

Description

Types of results made available by the RobotUniversalResultAccess object.

III.10 IRobotStructureValues

Class Hierarchy

RobotOM.IRobotStructureValues

C++

```
interface IRobotStructureValues : IDispatch;
```

C#

```
public interface IRobotStructureValues;
```

Visual Basic

```
Public Interface IRobotStructureValues
```

Description

Physical properties for the structure model.

Version

Available since version 7.5.

III.10.1 IRobotStructureValues Members

The following tables list the members exposed by IRobotStructureValues.

Public Methods

	Name	Description
	GetEx2 (see page 1020)	Function returns additional eccentricity according to X axis for the given load case.
	GetEy2 (see page 1021)	Function returns additional eccentricity according to Y axis for the given load case.
	GetG (see page 1021)	Function returns coordinates of structure center of gravity for the given load case.
	GetIx (see page 1021)	Function returns mass moment of inertia according to X axis for the given load case.
	GetIy (see page 1022)	Function returns mass moment of inertia according to Y axis for the given load case.
	GetIz (see page 1022)	Function returns mass moment of inertia according to Z axis for the given load case.
	GetMass (see page 1022)	Function returns a mass of the whole structure for the given load case.
	GetT (see page 1023)	Function returns coordinates of structure center of torsion for the given load case.

III.10.2 IRobotStructureValues Methods

The methods of the IRobotStructureValues class are listed here.

(c) 2014 Autodesk Inc. All rights reserved

Public Methods

	Name	Description
	GetEx2 (see page 1020)	Function returns additional eccentricity according to X axis for the given load case.
	GetEy2 (see page 1021)	Function returns additional eccentricity according to Y axis for the given load case.
	GetG (see page 1021)	Function returns coordinates of structure center of gravity for the given load case.
	GetIx (see page 1021)	Function returns mass moment of inertia according to X axis for the given load case.
	GetIy (see page 1022)	Function returns mass moment of inertia according to Y axis for the given load case.
	GetIz (see page 1022)	Function returns mass moment of inertia according to Z axis for the given load case.
	GetMass (see page 1022)	Function returns a mass of the whole structure for the given load case.
	GetT (see page 1023)	Function returns coordinates of structure center of torsion for the given load case.

III.10.2.1 GetEx2

C++

```
HRESULT GetEx2(long _case_num, double* ret);
```

C#

```
public double GetEx2(long _case_num);
```

Visual Basic

```
Public Function GetEx2(_case_num As long) As double
```

Description

Function returns additional eccentricity according to X axis for the given load case.

Version

Available since version 7.5.

III.10.2.2 GetEy2

C++

```
HRESULT GetEy2(long _case_num, double* ret);
```

C#

```
public double GetEy2(long _case_num);
```

Visual Basic

```
Public Function GetEy2(_case_num As long) As double
```

Description

Function returns additional eccentricity according to Y axis for the given load case.

Version

Available since version 7.5.

(c) 2014 Autodesk Inc. All rights reserved

III.10.2.3 GetG

C++

```
HRESULT GetG(long _case_num, IRobotGeoPoint3D** ret);
```

C#

```
public IRobotGeoPoint3D GetG(long _case_num);
```

Visual Basic

```
Public Function GetG(_case_num As long) As IRobotGeoPoint3D
```

Description

Function returns coordinates of structure center of gravity for the given load case.

Version

Available since version 7.5.

III.10.2.4 GetIx

C++

```
HRESULT GetIx(long _case_num, double* ret);
```

C#

```
public double GetIx(long _case_num);
```

Visual Basic

```
Public Function GetIx(_case_num As long) As double
```

Description

Function returns mass moment of inertia according to X axis for the given load case.

Version

Available since version 7.5.

III.10.2.5 GetIy

C++

```
HRESULT GetIy(long _case_num, double* ret);
```

C#

```
public double GetIy(long _case_num);
```

Visual Basic

```
Public Function GetIy(_case_num As long) As double
```

Description

Function returns mass moment of inertia according to Y axis for the given load case.

Version

Available since version 7.5.

(c) 2014 Autodesk Inc. All rights reserved

III.10.2.6 GetIz

C++

```
HRESULT GetIz(long _case_num, double* ret);
```

C#

```
public double GetIz(long _case_num);
```

Visual Basic

```
Public Function GetIz(_case_num As long) As double
```

Description

Function returns mass moment of inertia according to Z axis for the given load case.

Version

Available since version 7.5.

III.10.2.7 GetMass

C++

```
HRESULT GetMass(long _case_num, double* ret);
```

C#

```
public double GetMass(long _case_num);
```

Visual Basic

```
Public Function GetMass(_case_num As long) As double
```

Description

Function returns a mass of the whole structure for the given load case.

Version

Available since version 7.5.

III.10.2.8 GetT

C++

```
HRESULT GetT(long _case_num, IRobotGeoPoint3D** ret);
```

C#

```
public IRobotGeoPoint3D GetT(long _case_num);
```

Visual Basic

```
Public Function GetT(_case_num As long) As IRobotGeoPoint3D
```

Description

Function returns coordinates of structure center of torsion for the given load case.

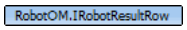
Version

Available since version 7.5.

(c) 2014 Autodesk Inc. All rights reserved

III.11 IRobotResultRow

Class Hierarchy



C++

```
interface IRobotResultRow : IDispatch;
```

C#

```
public interface IRobotResultRow;
```

Visual Basic

```
Public Interface IRobotResultRow
```

Description

Row of results calculated for determined values of a parameter set.

Version

Available since version 8.2.

III.11.1 IRobotResultRow Members

The following tables list the members exposed by IRobotResultRow.

Public Methods

	Name	Description
	GetParam (see page 1024)	Function returns values of determined parameters for which available results have been calculated.
	GetValue (see page 1024)	Function returns a value of the specified result.
	GetValueType (see page 1024)	Function returns a type of the value representing a specified result.
	IsAvailable (see page 1025)	Function checks if a specified result is available.

III.11.2 IRobotResultRow Methods

The methods of the IRobotResultRow class are listed here.

Public Methods

	Name	Description
	GetParam (see page 1024)	Function returns values of determined parameters for which available results have been calculated.
	GetValue (see page 1024)	Function returns a value of the specified result.
	GetValueType (see page 1024)	Function returns a type of the value representing a specified result.
	IsAvailable (see page 1025)	Function checks if a specified result is available.

III.11.2.1 GetParam

C++

```
HRESULT GetParam(IRobotResultParamType _result_param_id, VARIANT* ret);
```

C#

```
public VARIANT GetParam(IRobotResultParamType _result_param_id);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Function GetParam(_result_param_id As IRobotResultParamType) As VARIANT
```

Description

Function returns values of determined parameters for which available results have been calculated.

Version

Available since version 8.2.

III.11.2.2 GetValue

C++

```
HRESULT GetValue(long _result_id, VARIANT* ret);
```

C#

```
public VARIANT GetValue(long _result_id);
```

Visual Basic

```
Public Function GetValue(_result_id As long) As VARIANT
```

Description

Function returns a value of the specified result.

Version

Available since version 8.2.

III.11.2.3 GetValueType

C++

```
HRESULT GetValueType(long _result_id, IRobotUniversalResultType* ret);
```

C#

```
public IRobotUniversalResultType GetValueType(long _result_id);
```

Visual Basic

```
Public Function GetValueType(_result_id As long) As IRobotUniversalResultType
```

Description

Function returns a type of the value representing a specified result.

Version

Available since version 8.2.

III.11.2.4 IsAvailable

C++

```
HRESULT IsAvailable(long _result_id, VARIANT_BOOL* ret);
```

C#

```
public bool IsAvailable(long _result_id);
```

Visual Basic

```
Public Function IsAvailable(_result_id As long) As Boolean
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Function checks if a specified result is available.

Version

Available since version 8.2.

III.12 IRobotResultParamType

C++

```
enum IRobotResultParamType;
```

C#

```
public enum IRobotResultParamType;
```

Visual Basic

```
Public Enum IRobotResultParamType
```

Members

Members	Description
I_RPT_LOAD_CASE = 1	Load case no. Available since version 8.2.
I_RPT_LOAD_CASE_CMPNT = 2	Load case component no. Available since version 8.2.
I_RPT_NODE = 3	Node no. Available since version 8.2.
I_RPT_BAR = 4	Bar no. Available since version 8.2.
I_RPT_BAR_DIV_COUNT = 5	Number of bar division points. Available since version 8.2.
I_RPT_BAR_DIV_POINT = 6	Bar division point no. Available since version 8.2.
I_RPT_BAR_RELATIVE_POINT = 7	Relative point position on the bar. Available since version 8.2.
I_RPT_MODE = 8	Load case mode no. Available since version 8.2.
I_RPT_MODE_CMB = 9	Mode combination type (assumes values of the type IRobotModeCombinationType (see page 1040)). Available since version 8.2.
I_RPT_PANEL = 10	Panel no. Available since version 8.2.
I_RPT_ELEMENT = 11	Finite element no. Available since version 8.2.
I_RPT_LAYER = 12	Layer designation (assumes values of the type IRobotFeLayerType (see page 937)). Available since version 8.2.
I_RPT_LAYER_ARBITRARY_VALUE = 13	Layer thickness. Available since version 8.2.
I_RPT_REINFORCE_CALC_METHOD = 14	Calculation method for reinforcement (assumes values of the type IRobotReinforceCalcMethods (see page 956)). Available since version 8.2.

(c) 2014 Autodesk Inc. All rights reserved

I_RPT_REDUCED_CUT_POS = 15	Position of the cut through the panel for reduced results (assumes values of the type IRobotFeResultReducedCutPosition (see page 977)). Available since version 8.2.
I_RPT_STOREY = 16	Story name. Available since version 8.2.
I_RPT_CALC_POINT = 17	Available since version 8.2.
I_RPT_LINEAR_SUPPORTS = 18	Available since version 8.2.
I_RPT_DIR_X = 19	Direction of the X axis of the system in which results are presented - the table of three real numbers. Available since version 8.2.
I_RPT_PANEL_PART = 20	Panel component index. Available since version 8.2.
I_RPT_MAX_BUFFER_SIZE = 21	Maximal buffer size in bytes. Available since version 8.2.
I_RPT_MULTI_THREADS = 22	Flag indicating multithread access to results. Available since version 8.2.
I_RPT_BUFFER_CHUNK_SIZE = 23	Size of a single buffer fragment in bytes. Available since version 8.2.
I_RPT_THREAD_COUNT = 24	Number of threads which should be used for getting results. Available since version 8.2.
I_RPT_DIR_X_DEFTYPE = 25	Method of defining the X axis direction - value of the type IRobotObjLocalXDirDefinitionType (see page 794). Available since version 8.2.
I_RPT_SMOOTHING = 26	Method of result smoothing (values of the calculation type IRobotFeResultSmoothing (see page 986)). Available since version 8.2.
I_RPT_ITERATE_LOAD_CASES = 27	Parameter steering with getting the results; for results which do not depend on a load case the False value should be set. Available since version 8.7.
I_RPT_BAR_ELEMENT = 28	Available since version 12.
I_RPT_BAR_ELEMENT_DIV_COUNT = 29	Available since version 12.
I_RPT_BAR_ELEMENT_DIV_POINT = 30	Available since version 12.
I_RPT_RESULT_POINT_COORDINATES = 31	Set this on query to retrieve global coordinates of points where results are provided. Available since version 12.
I_RPT_BAR_ELEMENT_DIV_DISCONTINUITY = 32	Set this 1 (True) to offset division points on ends of an element in the case of discontinuity. The number of division points can be increased internally, so I_RPT_BAR_ELEMENT_DIV_COUNT should be read while reading results for a given element. Available since version 12.
I_RPT_BAR_ELEMENT_DIV_DISCONTINUITY_OFFSET = 33	Set a value for the relative offset used to shift the point of discontinuity, if you want it to be different from the default. Available since version 12.

Description

Set of parameters describing calculation results.

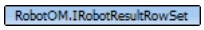
Version

Available since version 8.2.

(c) 2014 Autodesk Inc. All rights reserved

III.13 IRobotResultRowSet

Class Hierarchy



C++

```
interface IRobotResultRowSet : IDispatch;
```

C#

```
public interface IRobotResultRowSet;
```

Visual Basic

```
Public Interface IRobotResultRowSet
```

Description

Set of rows with calculation results.

Version

Available since version 8.2.

III.13.1 IRobotResultRowSet Members

The following tables list the members exposed by IRobotResultRowSet.

Public Fields

	Name	Description
	CurrentRow (see page 1028)	Current row of the result set.
	ResultIds (see page 1028)	Collection of result identifiers available in the set.

Public Methods

	Name	Description
	Clear (see page 1029)	
	MoveFirst (see page 1029)	Function sets the current row on the first position in the set. If the set is blank, the function returns the False value.
	MoveNext (see page 1029)	Function moves the current row to the next position in the set. If the current row has been set earlier on the last position, or the set is blank, the function returns the False value.

III.13.2 IRobotResultRowSet Fields

The fields of the IRobotResultRowSet class are listed here.

Public Fields

	Name	Description
	CurrentRow (see page 1028)	Current row of the result set.
	ResultIds (see page 1028)	Collection of result identifiers available in the set.

III.13.2.1 CurrentRow

C++

```
HRESULT get_CurrentRow(IRobotResultRow**);
```

C#

```
public IRobotResultRow CurrentRow { get; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public ReadOnly CurrentRow As IRobotResultRow
```

Description

Current row of the result set.

Version

Available since version 8.2.

III.13.2.2 ResultIds

C++

```
HRESULT get_ResultIds(IRobotNumbersCollection**);
```

C#

```
public IRobotNumbersCollection ResultIds { get; }
```

Visual Basic

```
Public ReadOnly ResultIds As IRobotNumbersCollection
```

Description

Collection of result identifiers available in the set.

Version

Available since version 8.2.

III.13.3 IRobotResultRowSet Methods

The methods of the IRobotResultRowSet class are listed here.

Public Methods

	Name	Description
	Clear (see page 1029)	
	MoveFirst (see page 1029)	Function sets the current row on the first position in the set. If the set is blank, the function returns the False value.
	MoveNext (see page 1029)	Function moves the current row to the next position in the set. If the current row has been set earlier on the last position, or the set is blank, the function returns the False value.

III.13.3.1 Clear

C++

```
HRESULT Clear();
```

C#

```
public void Clear();
```

Visual Basic

```
Public Sub Clear()
```

Version

Available since version 8.2.

(c) 2014 Autodesk Inc. All rights reserved

III.13.3.2 MoveFirst

C++

```
HRESULT MoveFirst(VARIANT_BOOL* ret);
```

C#

```
public bool MoveFirst();
```

Visual Basic

```
Public Function MoveFirst() As Boolean
```

Description

Function sets the current row on the first position in the set. If the set is blank, the function returns the False value.

Version

Available since version 8.2.

III.13.3.3 MoveNext

C++

```
HRESULT MoveNext(VARIANT_BOOL* ret);
```

C#

```
public bool MoveNext();
```

Visual Basic

```
Public Function MoveNext() As Boolean
```

Description

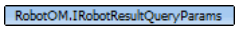
Function moves the current row to the next position in the set. If the current row has been set earlier on the last position, or the set is blank, the function returns the False value.

Version

Available since version 8.2.

III.14 IRobotResultQueryParams

Class Hierarchy



C++

```
interface IRobotResultQueryParams : IDispatch;
```

C#

```
public interface IRobotResultQueryParams;
```

Visual Basic

```
Public Interface IRobotResultQueryParams
```

Description

Definition of a query to the result server.

Version

Available since version 8.2.

(c) 2014 Autodesk Inc. All rights reserved

III.14.1 IRobotResultQueryParams Members

The following tables list the members exposed by IRobotResultQueryParams.

Public Fields

	Name	Description
	ResultIds (see page 1030)	Table of identifiers of results to be got.
	Selection (see page 1031)	Selection of structure components for which results should be got.

Public Methods

	Name	Description
	GetParam (see page 1031)	Function returns a value of the determined parameter.
	IsParamSet (see page 1032)	Function allows checking if a value has been set for a given parameter.
	Reset (see page 1032)	Function assigns zero values to all attributes of the query.
	SetParam (see page 1032)	Function sets a determined value of the specified parameter.

III.14.2 IRobotResultQueryParams Fields

The fields of the IRobotResultQueryParams class are listed here.

Public Fields

	Name	Description
	ResultIds (see page 1030)	Table of identifiers of results to be got.
	Selection (see page 1031)	Selection of structure components for which results should be got.

III.14.2.1 ResultIds

C++

```
HRESULT get_ResultIds(IRobotNumbersArray**);
```

C#

```
public IRobotNumbersArray ResultIds { get; }
```

Visual Basic

```
Public ReadOnly ResultIds As IRobotNumbersArray
```

Description

Table of identifiers of results to be got.

Version

Available since version 8.2.

III.14.2.2 Selection

C++

```
HRESULT get_Selection(IRobotMultiSelection**);
```

C#

```
public IRobotMultiSelection Selection { get; }
```

Visual Basic

```
Public ReadOnly Selection As IRobotMultiSelection
```

Description

Selection of structure components for which results should be got.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 8.2.

III.14.3 IRobotResultQueryParams Methods

The methods of the IRobotResultQueryParams class are listed here.

Public Methods

	Name	Description
	GetParam (see page 1031)	Function returns a value of the determined parameter.
	IsParamSet (see page 1032)	Function allows checking if a value has been set for a given parameter.
	Reset (see page 1032)	Function assigns zero values to all attributes of the query.
	SetParam (see page 1032)	Function sets a determined value of the specified parameter.

III.14.3.1 GetParam

C++

```
HRESULT GetParam(IRobotResultParamType _param_id, VARIANT* ret);
```

C#

```
public VARIANT GetParam(IRobotResultParamType _param_id);
```

Visual Basic

```
Public Function GetParam(_param_id As IRobotResultParamType) As VARIANT
```

Description

Function returns a value of the determined parameter.

Version

Available since version 8.2.

III.14.3.2 IsParamSet

C++

```
HRESULT IsParamSet(IRobotResultParamType _param_id, VARIANT_BOOL* ret);
```

C#

```
public bool IsParamSet(IRobotResultParamType _param_id);
```

Visual Basic

```
Public Function IsParamSet(_param_id As IRobotResultParamType) As Boolean
```

Description

Function allows checking if a value has been set for a given parameter.

Version

Available since version 8.2.

III.14.3.3 Reset

C++

```
HRESULT Reset();
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public void Reset();
```

Visual Basic

```
Public Sub Reset()
```

Description

Function assigns zero values to all attributes of the query.

Version

Available since version 8.2.

III.14.3.4 SetParam

C++

```
HRESULT SetParam(IRobotResultParamType _param_id, VARIANT _param_value);
```

C#

```
public void SetParam(IRobotResultParamType _param_id, VARIANT _param_value);
```

Visual Basic

```
Public Sub SetParam(_param_id As IRobotResultParamType, _param_value As VARIANT)
```

Description

Function sets a determined value of the specified parameter.

Version

Available since version 8.2.

III.15 IRobotResultQueryReturnType

C++

```
enum IRobotResultQueryReturnType;
```

C#

```
public enum IRobotResultQueryReturnType;
```

Visual Basic

```
Public Enum IRobotResultQueryReturnType
```

Members

Members	Description
I_RQRT_DONE = 1	Getting the results has been completed, all results have been got. Available since version 8.2.
I_RQRT_MORE_AVAILABLE = 2	Function that gets the results has given access only to a part of them, it is necessary to call up the function with the same parameters again to get the next part of results. Available since version 8.2.

Version

Available since version 8.2.

(c) 2014 Autodesk Inc. All rights reserved

III.16 IRobotResultStatusType

C++

```
enum IRobotResultStatusType;
```

C#

```
public enum IRobotResultStatusType;
```

Visual Basic

```
Public Enum IRobotResultStatusType
```

Members

Members	Description
I_RST_NONE = 0	No results. Available since version 14.
I_RST_AVAILABLE = 1	Results are available and up-to-date. Available since version 14.
I_RST_OUT_OF_DATE = 2	Results are out-of-date. Available since version 14.

Description

Type of data describing status of results.

Version

Available since version 14.

III.17 IRobotCalculationResume

Class Hierarchy

RobotOM.IRobotCalculationResume

C++

```
interface IRobotCalculationResume : IDispatch;
```

C#

```
public interface IRobotCalculationResume;
```

Visual Basic

```
Public Interface IRobotCalculationResume
```

Description

.

Version

Available since version 15.

III.17.1 IRobotCalculationResume Members

The following tables list the members exposed by IRobotCalculationResume.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	DiagonalStiffnessMatrixMax (see page 1035)	.
	DiagonalStiffnessMatrixMin (see page 1035)	.
	DiagonalStiffnessMatrixPrecision (see page 1035)	.

Public Methods

	Name	Description
	GetEnergy (see page 1036)	
	GetEnergyPrecision (see page 1036)	

III.17.2 IRobotCalculationResume Fields

The fields of the IRobotCalculationResume class are listed here.

Public Fields

	Name	Description
	DiagonalStiffnessMatrixMax (see page 1035)	.
	DiagonalStiffnessMatrixMin (see page 1035)	.
	DiagonalStiffnessMatrixPrecision (see page 1035)	.

III.17.2.1 DiagonalStiffnessMatrixMax

C++

```
HRESULT get_DiagonalStiffnessMatrixMax(double*);
```

C#

```
public double DiagonalStiffnessMatrixMax { get; }
```

Visual Basic

```
Public ReadOnly DiagonalStiffnessMatrixMax As double
```

Description

.

Version

Available since version 15.

III.17.2.2 DiagonalStiffnessMatrixMin

C++

```
HRESULT get_DiagonalStiffnessMatrixMin(double*);
```

C#

```
public double DiagonalStiffnessMatrixMin { get; }
```

Visual Basic

```
Public ReadOnly DiagonalStiffnessMatrixMin As double
```

(c) 2014 Autodesk Inc. All rights reserved

Description

.

Version

Available since version 15.

III.17.2.3 DiagonalStiffnessMatrixPrecision

C++

```
HRESULT get_DiagonalStiffnessMatrixPrecision(double*);
```

C#

```
public double DiagonalStiffnessMatrixPrecision { get; }
```

Visual Basic

```
Public ReadOnly DiagonalStiffnessMatrixPrecision As double
```

Description

.

Version

Available since version 15.

III.17.3 IRobotCalculationResume Methods

The methods of the IRobotCalculationResume class are listed here.

Public Methods

	Name	Description
	GetEnergy (see page 1036)	
	GetEnergyPrecision (see page 1036)	

III.17.3.1 GetEnergy

C++

```
HRESULT GetEnergy(long _load_case, double* ret);
```

C#

```
public double GetEnergy(long _load_case);
```

Visual Basic

```
Public Function GetEnergy(_load_case As long) As double
```

Version

Available since version 15.

III.17.3.2 GetEnergyPrecision

C++

```
HRESULT GetEnergyPrecision(long _load_case, double* ret);
```

C#

```
public double GetEnergyPrecision(long _load_case);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Function GetEnergyPrecision(_load_case As long) As double
```

Version

Available since version 15.

IV Selections

Enumerations

	Name	Description
	IRobotModeSelectionType (see page 1040)	A set of identifiers determining various manners of mode selection has been defined. .
	IRobotModeCombinationType (see page 1040)	A set of identifiers is defined to determine different types of combinations of modes. .
	IRobotPredefinedSelection (see page 1041)	

Interfaces

	Name	Description
	IRobotSelectionFactory (see page 1037)	Factory that creates objects representing selections. .
	IRobotGroup (see page 1041)	A group of structure components of a given type.
	IRobotGroupServer (see page 1045)	Object managing groups of structure elements.

IV.1 IRobotSelectionFactory

Class Hierarchy

RobotOM.IRobotSelectionFactory

C++

```
interface IRobotSelectionFactory : IDispatch;
```

C#

```
public interface IRobotSelectionFactory;
```

Visual Basic

```
Public Interface IRobotSelectionFactory
```

Description

Factory that creates objects representing selections. .

IV.1.1 IRobotSelectionFactory Members

The following tables list the members exposed by IRobotSelectionFactory.

Public Methods

	Name	Description
	Create (see page 1038)	The function creates and returns an object representing selections of structure components of an indicated type. The newly-created selection is empty. .
	CreateByLabel (see page 1038)	Function creates a selection of objects of the indicated type, which are assigned a label of the specified type and name. .

(c) 2014 Autodesk Inc. All rights reserved

	CreateByStorey (see page 1038)	Function creates a selection of objects of the specified type assigned to the story with the specified name.
	CreateFull (see page 1039)	Function creates selection containing all objects of the specified type. Available since version 1.7.
	CreateMulti (see page 1039)	The function creates and returns an empty object of the multi-selection type .
	CreatePredefined (see page 1039)	Available since version 1.7.
	Get (see page 1039)	The funcion gives the current selection of objects of a given type back.

IV.1.2 IRobotSelectionFactory Methods

The methods of the IRobotSelectionFactory class are listed here.

Public Methods

	Name	Description
	Create (see page 1038)	The function creates and returns an object representing selections of structure components of an indicated type. The newly-created selection is empty. .
	CreateByLabel (see page 1038)	Function creates a selection of objects of the indicated type, which are assigned a label of the specified type and name. .
	CreateByStorey (see page 1038)	Function creates a selection of objects of the specified type assigned to the story with the specified name.
	CreateFull (see page 1039)	Function creates selection containing all objects of the specified type. Available since version 1.7.
	CreateMulti (see page 1039)	The function creates and returns an empty object of the multi-selection type .
	CreatePredefined (see page 1039)	Available since version 1.7.
	Get (see page 1039)	The funcion gives the current selection of objects of a given type back.

IV.1.2.1 Create

C++

```
HRESULT Create(IRobotObjectType _object_type, IRobotSelection** ret);
```

C#

```
public IRobotSelection Create(IRobotObjectType _object_type);
```

Visual Basic

```
Public Function Create(_object_type As IRobotObjectType) As IRobotSelection
```

Description

The function creates and returns an object representing selections of structure components of an indicated type. The newly-created selection is empty. .

IV.1.2.2 CreateByLabel

C++

```
HRESULT CreateByLabel(IRobotObjectType _obj_type, IRobotLabelType _label_type, BSTR
_label_name, IRobotSelection** ret);
```

C#

```
public IRobotSelection CreateByLabel(IRobotObjectType _obj_type, IRobotLabelType
_label_type, String _label_name);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Function CreateByLabel(_obj_type As IRobotObjectType, _label_type As
IRobotLabelType, _label_name As String) As IRobotSelection
```

Description

Function creates a selection of objects of the indicated type, which are assigned a label of the specified type and name. .

Version

Available since version 3.

IV.1.2.3 CreateByStorey

C++

```
HRESULT CreateByStorey(IRobotObjectType _obj_type, BSTR _storey, IRobotSelection** ret);
```

C#

```
public IRobotSelection CreateByStorey(IRobotObjectType _obj_type, String _storey);
```

Visual Basic

```
Public Function CreateByStorey(_obj_type As IRobotObjectType, _storey As String) As
IRobotSelection
```

Description

Function creates a selection of objects of the specified type assigned to the story with the specified name.

Version

Available since version 9.5.

IV.1.2.4 CreateFull

C++

```
HRESULT CreateFull(IRobotObjectType _obj_type, IRobotSelection** ret);
```

C#

```
public IRobotSelection CreateFull(IRobotObjectType _obj_type);
```

Visual Basic

```
Public Function CreateFull(_obj_type As IRobotObjectType) As IRobotSelection
```

Description

Function creates selection containing all objects of the specified type. Available since version 1.7.

IV.1.2.5 CreateMulti

C++

```
HRESULT CreateMulti(IRobotMultiSelection** ret);
```

C#

```
public IRobotMultiSelection CreateMulti();
```

Visual Basic

```
Public Function CreateMulti() As IRobotMultiSelection
```

Description

The function creates and returns an empty object of the multi-selection type .

(c) 2014 Autodesk Inc. All rights reserved

IV.1.2.6 CreatePredefined

C++

```
HRESULT CreatePredefined(IRobotPredefinedSelection _predefined_sel, IRobotSelection** ret);
```

C#

```
public IRobotSelection CreatePredefined(IRobotPredefinedSelection _predefined_sel);
```

Visual Basic

```
Public Function CreatePredefined(_predefined_sel As IRobotPredefinedSelection) As
IRobotSelection
```

Description

Available since version 1.7.

IV.1.2.7 Get

C++

```
HRESULT Get(IRobotObjectType _obj_type, IRobotSelection** ret);
```

C#

```
public IRobotSelection Get(IRobotObjectType _obj_type);
```

Visual Basic

```
Public Function Get(_obj_type As IRobotObjectType) As IRobotSelection
```

Description

The funcion gives the current selection of objects of a given type back.

IV.2 IRobotModeSelectionType

C++

```
enum IRobotModeSelectionType;
```

C#

```
public enum IRobotModeSelectionType;
```

Visual Basic

```
Public Enum IRobotModeSelectionType
```

Members

Members	Description
I_MST_NONE = 1	Empty selection of modes.
I_MST_ALL = 2	Selection containing all modes .
I_MST_SINGLE = 3	Selection of one indicated mode .
I_MST_FIRST_N = 4	Selection of the first n modes .

Description

A set of identifiers determining various manners of mode selection has been defined. .

(c) 2014 Autodesk Inc. All rights reserved

IV.3 IRobotModeCombinationType

C++

```
enum IRobotModeCombinationType;
```

C#

```
public enum IRobotModeCombinationType;
```

Visual Basic

```
Public Enum IRobotModeCombinationType
```

Members

Members	Description
I_MCT_NONE = -1	No combinations .
I_MCT_SRSS = 0	SRSS type combination .
I_MCT_CQC = 1	CQC type combination .
I_MCT_10P = 2	10 percent.
I_MCT_2SM = 3	Double sum.
I_MCT_ALL = -2	Available since version 3.5.

Description

A set of identifiers is defined to determine different types of combinations of modes. .

IV.4 IRobotPredefinedSelection

C++

```
enum IRobotPredefinedSelection;
```

C#

```
public enum IRobotPredefinedSelection;
```

Visual Basic

```
Public Enum IRobotPredefinedSelection
```

Members

Members	Description
I_PS_CASE_SIMPLE_CASES = 2	Selection of all simple cases Available since version 1.7.
I_PS_CASE_COMBINATIONS = 3	Selection of all combination (but not code combinations) Available since version 1.7.
I_PS_CASE_CODE_COMBINATIONS = 4	Selection of all code combination Available since version 1.7.
I_PS_NODE_USER_NODES = 21	User-defined nodes. Available since version 2.5.
I_PS_NODE_CALC_NODES = 22	Nodes defined automatically during generation of a calculation model. Available since version 2.5.
I_PS_NODE_SUPPORTED = 23	Nodes at which supports have been defined. Available since version 2.5.
I_PS_BAR_INACTIVE = 31	Inactive bars. Available since version 4.5.

(c) 2014 Autodesk Inc. All rights reserved

IV.5 IRobotGroup

Class Hierarchy

RobotOM.IRobotGroup

C++

```
interface IRobotGroup : IDispatch;
```

C#

```
public interface IRobotGroup;
```

Visual Basic

```
Public Interface IRobotGroup
```

Description

A group of structure components of a given type.

Version

Available since version 3.

IV.5.1 IRobotGroup Members

The following tables list the members exposed by IRobotGroup.

Public Fields

	Name	Description
	Color (see page 1042)	Color assigned to a group.
	Name (see page 1042)	Group name.
	ObjectType (see page 1043)	Type of objects belonging to the group.
	SelList (see page 1043)	List describing the objects belonging to the group.

Public Methods

	Name	Description
	CreateCollection (see page 1044)	Function creates and returns the collection containing the objects belonging to the group. .
	CreateSelection (see page 1044)	Function creates and returns the selection describing the objects belonging to the group. .
	Read (see page 1044)	Function reads group definition from the structure model.
	Store (see page 1045)	Function saves the group definition in the structure model. .

IV.5.2 IRobotGroup Fields

The fields of the IRobotGroup class are listed here.

Public Fields

	Name	Description
	Color (see page 1042)	Color assigned to a group.
	Name (see page 1042)	Group name.
	ObjectType (see page 1043)	Type of objects belonging to the group.
	SelList (see page 1043)	List describing the objects belonging to the group.

(c) 2014 Autodesk Inc. All rights reserved

IV.5.2.1 Color

C++

```
HRESULT get_Color(long*);
HRESULT put_Color(long);
```

C#

```
public long Color { get; set; }
```

Visual Basic

```
Public Color As long
```

Description

Color assigned to a group.

Version

Available since version 3.

IV.5.2.2 Name

C++

```
HRESULT get_Name(BSTR*);
HRESULT put_Name(BSTR);
```

C#

```
public String Name { get; set; }
```

Visual Basic

```
Public Name As String
```

Description

Group name.

Version

Available since version 3.

IV.5.2.3 ObjectType

C++

```
HRESULT get_ObjectType(IRobotObjectType*);
HRESULT put_ObjectType(IRobotObjectType);
```

C#

```
public IRobotObjectType ObjectType { get; set; }
```

Visual Basic

```
Public ObjectType As IRobotObjectType
```

Description

Type of objects belonging to the group.

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

IV.5.2.4 SelList

C++

```
HRESULT get_SelList(BSTR*);
HRESULT put_SelList(BSTR);
```

C#

```
public String SelList { get; set; }
```

Visual Basic

```
Public SelList As String
```

Description

List describing the objects belonging to the group.

Version

Available since version 3.

IV.5.3 IRobotGroup Methods

The methods of the IRobotGroup class are listed here.

Public Methods

	Name	Description
	CreateCollection (see page 1044)	Function creates and returns the collection containing the objects belonging to the group. .
	CreateSelection (see page 1044)	Function creates and returns the selection describing the objects belonging to the group. .
	Read (see page 1044)	Function reads group definition from the structure model.
	Store (see page 1045)	Function saves the group definition in the structure model. .

IV.5.3.1 CreateCollection

C++

```
HRESULT CreateCollection(IRobotCollection** ret);
```

C#

```
public IRobotCollection CreateCollection();
```

Visual Basic

```
Public Function CreateCollection() As IRobotCollection
```

Description

Function creates and returns the collection containing the objects belonging to the group. .

Version

Available since version 3.

IV.5.3.2 CreateSelection

C++

```
HRESULT CreateSelection(IRobotSelection** ret);
```

C#

```
public IRobotSelection CreateSelection();
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Function CreateSelection() As IRobotSelection
```

Description

Function creates and returns the selection describing the objects belonging to the group. .

Version

Available since version 3.

IV.5.3.3 Read

C++

```
HRESULT Read();
```

C#

```
public void Read();
```

Visual Basic

```
Public Sub Read()
```

Description

Function reads group definition from the structure model.

Version

Available since version 3.

IV.5.3.4 Store

C++

```
HRESULT Store();
```

C#

```
public void Store();
```

Visual Basic

```
Public Sub Store()
```

Description

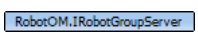
Function saves the group definition in the structure model. .

Version

Available since version 3.

IV.6 IRobotGroupServer

Class Hierarchy



C++

```
interface IRobotGroupServer : IDispatch;
```

C#

```
public interface IRobotGroupServer;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Interface IRobotGroupServer
```

Description

Object managing groups of structure elements.

Version

Available since version 3.

IV.6.1 IRobotGroupServer Members

The following tables list the members exposed by IRobotGroupServer.

Public Methods

	Name	Description
	Create (see page 1046)	Function creates a new group containing objects of the indicated type with the specified name. Function returns the index of the created group. .
	Delete (see page 1046)	Function deletes the group containing a given object type of the specified index.
	Find (see page 1047)	Function returns index of the group of the given type of objects. If such a group does not exist, then zero value is returned. .
	Get (see page 1047)	Function returns the group containing a given type of objects of the specified index. Groups are indexed from 1 to the number of groups containing a given type of objects.
	GetCount (see page 1047)	Function returns the number of groups containing a given type of objects defined in the structure. .

IV.6.2 IRobotGroupServer Methods

The methods of the IRobotGroupServer class are listed here.

Public Methods

	Name	Description
	Create (see page 1046)	Function creates a new group containing objects of the indicated type with the specified name. Function returns the index of the created group. .
	Delete (see page 1046)	Function deletes the group containing a given object type of the specified index.
	Find (see page 1047)	Function returns index of the group of the given type of objects. If such a group does not exist, then zero value is returned. .
	Get (see page 1047)	Function returns the group containing a given type of objects of the specified index. Groups are indexed from 1 to the number of groups containing a given type of objects.
	GetCount (see page 1047)	Function returns the number of groups containing a given type of objects defined in the structure. .

IV.6.2.1 Create

C++

```
HRESULT Create(IRobotObjectType _obj_type, BSTR _name, BSTR _sel_list = "", long _color =
0, long* ret);
```

C#

```
public long Create(IRobotObjectType _obj_type, String _name, String _sel_list = "", long
_color = 0);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Function Create(_obj_type As IRobotObjectType, _name As String, Optional _sel_list
As String = "", Optional _color As long = 0) As long
```

Description

Function creates a new group containing objects of the indicated type with the specified name. Function returns the index of the created group. .

Version

Available since version 3.

IV.6.2.2 Delete

C++

```
HRESULT Delete(IRobotObjectType _obj_type, long _idx);
```

C#

```
public void Delete(IRobotObjectType _obj_type, long _idx);
```

Visual Basic

```
Public Sub Delete(_obj_type As IRobotObjectType, _idx As long)
```

Description

Function deletes the group containing a given object type of the specified index.

Version

Available since version 3.

IV.6.2.3 Find

C++

```
HRESULT Find(IRobotObjectType _obj_type, BSTR _grp_name, long* ret);
```

C#

```
public long Find(IRobotObjectType _obj_type, String _grp_name);
```

Visual Basic

```
Public Function Find(_obj_type As IRobotObjectType, _grp_name As String) As long
```

Description

Function returns index of the group of the given type of objects. If such a group does not exist, then zero value is returned. .

Version

Available since version 3.

IV.6.2.4 Get

C++

```
HRESULT Get(IRobotObjectType _obj_type, long _idx, IRobotGroup** ret);
```

C#

```
public IRobotGroup Get(IRobotObjectType _obj_type, long _idx);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Function Get(_obj_type As IRobotObjectType, _idx As long) As IRobotGroup
```

Description

Function returns the group containing a given type of objects of the specified index. Groups are indexed from 1 to the number of groups containing a given type of objects.

Version

Available since version 3.

IV.6.2.5 GetCount

C++

```
HRESULT GetCount(IRobotObjectType _obj_type, long* ret);
```

C#

```
public long GetCount(IRobotObjectType _obj_type);
```

Visual Basic

```
Public Function GetCount(_obj_type As IRobotObjectType) As long
```

Description

Function returns the number of groups containing a given type of objects defined in the structure. .

Version

Available since version 3.

V Edit operations

Available since version 3.5.

Enumerations

	Name	Description
	IRobotTranslateOptions (see page 1048)	

Interfaces

	Name	Description
	IRobotStructureEditTools (see page 1048)	Interface grouping edit functions that operate on the entire structure or its part. .

V.1 IRobotTranslateOptions

C++

```
enum IRobotTranslateOptions;
```

C#

```
public enum IRobotTranslateOptions;
```

Visual Basic

```
Public Enum IRobotTranslateOptions
```

(c) 2014 Autodesk Inc. All rights reserved

Members

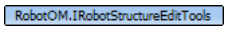
Members	Description
I_TO_COPY = 0	Available since version 3.5.
I_TO_COPY_WITH_DRAG = 1	Available since version 3.5.
I_TO_MOVE = 2	Available since version 3.5.

Version

Available since version 3.5.

V.2 IRobotStructureEditTools

Class Hierarchy



C++

```
interface IRobotStructureEditTools : IDispatch;
```

C#

```
public interface IRobotStructureEditTools;
```

Visual Basic

```
Public Interface IRobotStructureEditTools
```

Description

Interface grouping edit functions that operate on the entire structure or its part. .

Version

Available since version 3.5.

V.2.1 IRobotStructureEditTools Members

The following tables list the members exposed by IRobotStructureEditTools.

Public Methods

	Name	Description
	DivideBar (see page 1049)	Function divides the indicated bar based on the provided table with positions of division points. Function returns tables with numbers of bars generated as a result of the division operation. .
	SelMirror (see page 1050)	
	SelRotate (see page 1050)	Function performs the operation of rotation on the current selection. .
	SelScale (see page 1050)	Function performs the operation of scaling on the current selection. .
	SelTranslate (see page 1051)	Function performs the operation of translation on the current selection. .
	TranslateBar (see page 1051)	Function translates selected bar by the specified vector. Function returns a table with numbers of bars generated as a result of translation. .
	TranslateNode (see page 1052)	Function translates the selected node by the specified vector. Function returns a table with numbers of nodes generated as a result of translation. .

V.2.2 IRobotStructureEditTools Methods

The methods of the IRobotStructureEditTools class are listed here.

(c) 2014 Autodesk Inc. All rights reserved

Public Methods

	Name	Description
	DivideBar (see page 1049)	Function divides the indicated bar based on the provided table with positions of division points. Function returns tables with numbers of bars generated as a result of the division operation. .
	SelMirror (see page 1050)	
	SelRotate (see page 1050)	Function performs the operation of rotation on the current selection. .
	SelScale (see page 1050)	Function performs the operation of scaling on the current selection. .
	SelTranslate (see page 1051)	Function performs the operation of translation on the current selection. .
	TranslateBar (see page 1051)	Function translates selected bar by the specified vector. Function returns a table with numbers of bars generated as a result of translation. .
	TranslateNode (see page 1052)	Function translates the selected node by the specified vector. Function returns a table with numbers of nodes generated as a result of translation. .

V.2.2.1 DivideBar

C++

```
HRESULT DivideBar(long _bar_number, IRobotValuesArray* _div_points, VARIANT_BOOL _relative,
IRobotNumbersArray** ret);
```

C#

```
public IRobotNumbersArray DivideBar(long _bar_number, IRobotValuesArray _div_points, bool
_relative);
```

Visual Basic

```
Public Function DivideBar(_bar_number As long, ByRef _div_points As IRobotValuesArray,
_relative As Boolean) As IRobotNumbersArray
```

Description

Function divides the indicated bar based on the provided table with positions of division points. Function returns tables with numbers of bars generated as a result of the division operation. .

Version

Available since version 3.5.

V.2.2.2 SelMirror

C++

```
HRESULT SelMirror(double _p0x, double _p0y, double _p0z, double _p1x, double _p1y, double
_p1z, double _p2x, double _p2y, double _p2z, IRobotTranslateOptions _options, long
_repetitions);
```

C#

```
public void SelMirror(double _p0x, double _p0y, double _p0z, double _p1x, double _p1y,
double _p1z, double _p2x, double _p2y, double _p2z, IRobotTranslateOptions _options, long
_repetitions);
```

Visual Basic

```
Public Sub SelMirror(_p0x As double, _p0y As double, _p0z As double, _p1x As double, _p1y
As double, _p1z As double, _p2x As double, _p2y As double, _p2z As double, _options As
IRobotTranslateOptions, _repetitions As long)
```

Version

Available since version 4.

(c) 2014 Autodesk Inc. All rights reserved

V.2.2.3 SelRotate

C++

```
HRESULT SelRotate(double _p0x, double _p0y, double _p0z, double _p1x, double _p1y, double
_p1z, double _angle, IRobotTranslateOptions _options, long _repetitions);
```

C#

```
public void SelRotate(double _p0x, double _p0y, double _p0z, double _p1x, double _p1y,
double _p1z, double _angle, IRobotTranslateOptions _options, long _repetitions);
```

Visual Basic

```
Public Sub SelRotate(_p0x As double, _p0y As double, _p0z As double, _p1x As double, _p1y
As double, _p1z As double, _angle As double, _options As IRobotTranslateOptions,
_repetitions As long)
```

Description

Function performs the operation of rotation on the current selection. .

Version

Available since version 4.

V.2.2.4 SelScale

C++

```
HRESULT SelScale(double _p0x, double _p0y, double _p0z, double _factor,
IRobotTranslateOptions _options, long _repetitions);
```

C#

```
public void SelScale(double _p0x, double _p0y, double _p0z, double _factor,
IRobotTranslateOptions _options, long _repetitions);
```

Visual Basic

```
Public Sub SelScale(_p0x As double, _p0y As double, _p0z As double, _factor As double,
_options As IRobotTranslateOptions, _repetitions As long)
```

Description

Function performs the operation of scaling on the current selection. .

Version

Available since version 4.

V.2.2.5 SelTranslate

C++

```
HRESULT SelTranslate(double _dx, double _dy, double _dz, IRobotTranslateOptions _options,
long _repetitions);
```

C#

```
public void SelTranslate(double _dx, double _dy, double _dz, IRobotTranslateOptions
_options, long _repetitions);
```

Visual Basic

```
Public Sub SelTranslate(_dx As double, _dy As double, _dz As double, _options As
IRobotTranslateOptions, _repetitions As long)
```

Description

Function performs the operation of translation on the current selection. .

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 4.

V.2.2.6 TranslateBar

C++

```
HRESULT TranslateBar(long _bar_number, double _dx, double _dy, double _dz,
IRobotTranslateOptions _trans_option, long _repetitions = 1, IRobotNumbersArray** ret);
```

C#

```
public IRobotNumbersArray TranslateBar(long _bar_number, double _dx, double _dy, double
_dz, IRobotTranslateOptions _trans_option, long _repetitions = 1);
```

Visual Basic

```
Public Function TranslateBar(_bar_number As long, _dx As double, _dy As double, _dz As
double, _trans_option As IRobotTranslateOptions, Optional _repetitions As long = 1) As
IRobotNumbersArray
```

Description

Function translates selected bar by the specified vector. Function returns a table with numbers of bars generated as a result of translation. .

Version

Available since version 3.5.

V.2.2.7 TranslateNode

C++

```
HRESULT TranslateNode(long _node_number, double _dx, double _dy, double _dz,
IRobotTranslateOptions _trans_option, long _repetitions, IRobotNumbersArray** ret);
```

C#

```
public IRobotNumbersArray TranslateNode(long _node_number, double _dx, double _dy, double
_dz, IRobotTranslateOptions _trans_option, long _repetitions);
```

Visual Basic

```
Public Function TranslateNode(_node_number As long, _dx As double, _dy As double, _dz As
double, _trans_option As IRobotTranslateOptions, _repetitions As long) As IRobotNumbersArray
```

Description

Function translates the selected node by the specified vector. Function returns a table with numbers of nodes generated as a result of translation. .

Version

Available since version 3.5.

VI Stories

Interfaces

	Name	Description
	IRobotStorey (see page 1052)	Interface describing a story in a structure. .
	IRobotStoreyMngr (see page 1057)	Structure story manager.

(c) 2014 Autodesk Inc. All rights reserved

	IRobotStoreySelection (see page 1062)	Interface that enables selecting stories. Stories are identified by name or number. A story no. is simultaneously the index of this story in RobotStoreyMngr. The object can also be used for slecting levels. Adding a specified story to the selection adds the top level of this story. Additionally, you can add the building base level to the selection specifying 0 as a story no.
	IRobotStoreyValues (see page 1065)	Bar properties.
	IRobotStoreyReducedForces (see page 1069)	
	IRobotStoreyDdisplacements (see page 1074)	
	IRobotStoreyResultServer (see page 1077)	Server of results for stories.

VI.1 IRobotStorey

Class Hierarchy

RobotOM.IRobotStorey

C++

```
interface IRobotStorey : IDispatch;
```

C#

```
public interface IRobotStorey;
```

Visual Basic

```
Public Interface IRobotStorey
```

Description

Interface describing a story in a structure. .

VI.1.1 IRobotStorey Members

The following tables list the members exposed by IRobotStorey.

Public Fields

	Name	Description
	AutomaticSelection (see page 1053)	Flag indicating if selection of objects assigned to a story should be automatic .
	Color (see page 1054)	Color assigned to the story.
	Ex1 (see page 1054)	Accidental eccentricity along the X axis.
	Ey1 (see page 1054)	Accidental eccentricity along the Y axis.
	Height (see page 1055)	Story height.
	Lx (see page 1055)	Story dimension along the X axis.
	Ly (see page 1055)	Story dimension along the Y axis.
	Name (see page 1055)	Story name.
	Objects (see page 1056)	Selection of objects assigned to a story.
	TopLevel (see page 1056)	Top level of a story.

Public Methods

	Name	Description
	SetHeight (see page 1056)	Function changes the story height.
	SetTopLevel (see page 1057)	Function changes a position of the top level of a story.

(c) 2014 Autodesk Inc. All rights reserved

VI.1.2 IRobotStorey Fields

The fields of the IRobotStorey class are listed here.

Public Fields

	Name	Description
	AutomaticSelection (see page 1053)	Flag indicating if selection of objects assigned to a story should be automatic .
	Color (see page 1054)	Color assigned to the story.
	Ex1 (see page 1054)	Accidental eccentricity along the X axis.
	Ey1 (see page 1054)	Accidental eccentricity along the Y axis.
	Height (see page 1055)	Story height.
	Lx (see page 1055)	Story dimension along the X axis.
	Ly (see page 1055)	Story dimension along the Y axis.
	Name (see page 1055)	Story name.
	Objects (see page 1056)	Selection of objects assigned to a story.
	TopLevel (see page 1056)	Top level of a story.

VI.1.2.1 AutomaticSelection

C++

```
HRESULT get_AutomaticSelection(VARIANT_BOOL*);
HRESULT put_AutomaticSelection(VARIANT_BOOL);
```

C#

```
public bool AutomaticSelection { get; set; }
```

Visual Basic

```
Public AutomaticSelection As Boolean
```

Description

Flag indicating if selection of objects assigned to a story should be automatic .

VI.1.2.2 Color

C++

```
HRESULT get_Color(long*);
HRESULT put_Color(long);
```

C#

```
public long Color { get; set; }
```

Visual Basic

```
Public Color As long
```

Description

Color assigned to the story.

VI.1.2.3 Ex1

C++

```
HRESULT get_Ex1(double*);
```

C#

```
public double Ex1 { get; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public ReadOnly Ex1 As double
```

Description

Accidental eccentricity along the X axis.

Version

Available since version 9.7.

VI.1.2.4 Ey1

C++

```
HRESULT get_Ey1(double*);
```

C#

```
public double Ey1 { get; }
```

Visual Basic

```
Public ReadOnly Ey1 As double
```

Description

Accidental eccentricity along the Y axis.

Version

Available since version 9.7.

VI.1.2.5 Height

C++

```
HRESULT get_Height(double*);
HRESULT put_Height(double);
```

C#

```
public double Height { get; set; }
```

Visual Basic

```
Public Height As double
```

Description

Story height.

VI.1.2.6 Lx

C++

```
HRESULT get_Lx(double*);
```

C#

```
public double Lx { get; }
```

Visual Basic

```
Public ReadOnly Lx As double
```

Description

Story dimension along the X axis.

Version

Available since version 9.7.

(c) 2014 Autodesk Inc. All rights reserved

VI.1.2.7 Ly

C++

```
HRESULT get_Ly(double*);
```

C#

```
public double Ly { get; }
```

Visual Basic

```
Public ReadOnly Ly As double
```

Description

Story dimension along the Y axis.

Version

Available since version 9.7.

VI.1.2.8 Name

C++

```
HRESULT get_Name(BSTR*);
HRESULT put_Name(BSTR);
```

C#

```
public String Name { get; set; }
```

Visual Basic

```
Public Name As String
```

Description

Story name.

VI.1.2.9 Objects

C++

```
HRESULT get_Objects(BSTR*);
HRESULT put_Objects(BSTR);
```

C#

```
public String Objects { get; set; }
```

Visual Basic

```
Public Objects As String
```

Description

Selection of objects assigned to a story.

VI.1.2.10 TopLevel

C++

```
HRESULT get_TopLevel(double*);
HRESULT put_TopLevel(double);
```

C#

```
public double TopLevel { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public TopLevel As double
```

Description

Top level of a story.

Version

Available since version 9.5.

VI.1.3 IRobotStorey Methods

The methods of the IRobotStorey class are listed here.

Public Methods

	Name	Description
	SetHeight (see page 1056)	Function changes the story height.
	SetTopLevel (see page 1057)	Function changes a position of the top level of a story.

VI.1.3.1 SetHeight

C++

```
HRESULT SetHeight(double _height, VARIANT_BOOL _affect_geometry = -1);
```

C#

```
public void SetHeight(double _height, bool _affect_geometry = -1);
```

Visual Basic

```
Public Sub SetHeight(_height As double, Optional _affect_geometry As Boolean = -1)
```

Description

Function changes the story height.

Version

Available since version 9.5.

VI.1.3.2 SetTopLevel

C++

```
HRESULT SetTopLevel(double _top_level, VARIANT_BOOL _affect_geometry = -1);
```

C#

```
public void SetTopLevel(double _top_level, bool _affect_geometry = -1);
```

Visual Basic

```
Public Sub SetTopLevel(_top_level As double, Optional _affect_geometry As Boolean = -1)
```

Description

Function changes a position of the top level of a story.

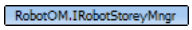
Version

Available since version 9.5.

(c) 2014 Autodesk Inc. All rights reserved

VI.2 IRobotStoreyMngr

Class Hierarchy



C++

```
interface IRobotStoreyMngr : IDispatch;
```

C#

```
public interface IRobotStoreyMngr;
```

Visual Basic

```
Public Interface IRobotStoreyMngr
```

Description

Structure story manager.

VI.2.1 IRobotStoreyMngr Members

The following tables list the members exposed by IRobotStoreyMngr.

Public Fields

	Name	Description
	BaseLevel (see page 1058)	Building base level.
	Count (see page 1059)	Number of stories defined in a structure.
	DisregardedObjects (see page 1059)	Selection of objects that should be disregarded in automatic recognition of structure elements belonging to individual stories.
	FilterStorey (see page 1059)	Name of the active filter story to which editing operations are limited; an empty character string indicates that a filter story is not set.

Public Methods

	Name	Description
	Create2 (see page 1060)	Function creates a new story with the specified parameters and returns its index. The created story is assigned a default automatic color and automatic object selection.
	Create2Ex (see page 1060)	Function creates a story with the specified parameters. Function returns the index of the created story.
	Delete (see page 1060)	Function deletes the story with the given index. .
	DeleteAll (see page 1061)	Function deletes all stories defined in a structure.
	Find (see page 1061)	Function searches for the story with the specified name and returns its index. If the story with the given name does not exist, function returns zero value. .
	Get (see page 1061)	Function returns definition of a story with the specified index. Stories are indexed with numbers from 1 to Count (see page 1059).

VI.2.2 IRobotStoreyMngr Fields

The fields of the IRobotStoreyMngr class are listed here.

Public Fields

	Name	Description
	BaseLevel (see page 1058)	Building base level.
	Count (see page 1059)	Number of stories defined in a structure.

(c) 2014 Autodesk Inc. All rights reserved

	DisregardedObjects (see page 1059)	Selection of objects that should be disregarded in automatic recognition of structure elements belonging to individual stories.
	FilterStorey (see page 1059)	Name of the active filter story to which editing operations are limited; an empty character string indicates that a filter story is not set.

VI.2.2.1 BaseLevel

C++

```
HRESULT get_BaseLevel(double*);
HRESULT put_BaseLevel(double);
```

C#

```
public double BaseLevel { get; set; }
```

Visual Basic

```
Public BaseLevel As double
```

Description

Building base level.

Version

Available since version 9.5.

VI.2.2.2 Count

C++

```
HRESULT get_Count(long*);
```

C#

```
public long Count { get; }
```

Visual Basic

```
Public ReadOnly Count As long
```

Description

Number of stories defined in a structure.

VI.2.2.3 DisregardedObjects

C++

```
HRESULT get_DisregardedObjects(BSTR*);
HRESULT put_DisregardedObjects(BSTR);
```

C#

```
public String DisregardedObjects { get; set; }
```

Visual Basic

```
Public DisregardedObjects As String
```

Description

Selection of objects that should be disregarded in automatic recognition of structure elements belonging to individual stories.

VI.2.2.4 FilterStorey

C++

```
HRESULT get_FilterStorey(BSTR*);
HRESULT put_FilterStorey(BSTR);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public String FilterStorey { get; set; }
```

Visual Basic

```
Public FilterStorey As String
```

Description

Name of the active filter story to which editing operations are limited; an empty character string indicates that a filter story is not set.

Version

Available since version 9.5.

VI.2.3 IRobotStoreyMngr Methods

The methods of the IRobotStoreyMngr class are listed here.

Public Methods

	Name	Description
	Create2 (see page 1060)	Function creates a new story with the specified parameters and returns its index. The created story is assigned a default automatic color and automatic object selection.
	Create2Ex (see page 1060)	Function creates a story with the specified parameters. Function returns the index of the created story.
	Delete (see page 1060)	Function deletes the story with the given index. .
	DeleteAll (see page 1061)	Function deletes all stories defined in a structure.
	Find (see page 1061)	Function searches for the story with the specified name and returns its index. If the story with the given name does not exist, function returns zero value. .
	Get (see page 1061)	Function returns definition of a story with the specified index. Stories are indexed with numbers from 1 to Count (see page 1059).

VI.2.3.1 Create2

C++

```
HRESULT Create2(BSTR _name, double _top_level, long* ret);
```

C#

```
public long Create2(String _name, double _top_level);
```

Visual Basic

```
Public Function Create2(_name As String, _top_level As double) As long
```

Description

Function creates a new story with the specified parameters and returns its index. The created story is assigned a default automatic color and automatic object selection.

Version

Available since version 9.5.

VI.2.3.2 Create2Ex

C++

```
HRESULT Create2Ex(BSTR _name, double _top_level, BSTR _objects = "", long _color = -1,
long* ret);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public long Create2Ex(String _name, double _top_level, String _objects = "", long _color =
-1);
```

Visual Basic

```
Public Function Create2Ex(_name As String, _top_level As double, Optional _objects As
String = "", Optional _color As long = -1) As long
```

Description

Function creates a story with the specified parameters. Function returns the index of the created story.

Version

Available since version 9.5.

VI.2.3.3 Delete

C++

```
HRESULT Delete(long _idx);
```

C#

```
public void Delete(long _idx);
```

Visual Basic

```
Public Sub Delete(_idx As long)
```

Description

Function deletes the story with the given index. .

VI.2.3.4 DeleteAll

C++

```
HRESULT DeleteAll();
```

C#

```
public void DeleteAll();
```

Visual Basic

```
Public Sub DeleteAll()
```

Description

Function deletes all stories defined in a structure.

VI.2.3.5 Find

C++

```
HRESULT Find(BSTR _story_name, long* ret);
```

C#

```
public long Find(String _story_name);
```

Visual Basic

```
Public Function Find(_story_name As String) As long
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Function searches for the story with the specified name and returns its index. If the story with the given name does not exist, function returns zero value. .

VI.2.3.6 Get

C++

```
HRESULT Get(long _idx, IRobotStorey** ret);
```

C#

```
public IRobotStorey Get(long _idx);
```

Visual Basic

```
Public Function Get(_idx As long) As IRobotStorey
```

Description

Function returns definition of a story with the specified index. Stories are indexed with numbers from 1 to Count (see page 1059).

VI.3 IRobotStoreySelection

Class Hierarchy

RobotOM.IRobotStoreySelection

C++

```
interface IRobotStoreySelection : IDispatch;
```

C#

```
public interface IRobotStoreySelection;
```

Visual Basic

```
Public Interface IRobotStoreySelection
```

Description

Interface that enables selecting stories. Stories are identified by name or number. A story no. is simultaneously the index of this story in RobotStoreyMngr. The object can also be used for slecting levels. Adding a specified story to the selection adds the top level of this story. Additionally, you can add the building base level to the selection specifying 0 as a story no.

Version

Available since version 9.7.

VI.3.1 IRobotStoreySelection Members

The following tables list the members exposed by IRobotStoreySelection.

Public Methods

	Name	Description
	Add (see page 1063)	Function adds to the selection a story with the specified number or name.
	AddAll (see page 1063)	Function adds all stories to the selection.
	Clear (see page 1063)	Function clears the selection.
	GetNames (see page 1064)	Function returns a table with names of stories that belong to the selection.
	GetNumbers (see page 1064)	Function returns a table with numbers of stories that belong to the selection.

(c) 2014 Autodesk Inc. All rights reserved

	Remove (see page 1064)	Function deletes from the selection a story with the specified number or name.

VI.3.2 IRobotStoreySelection Methods

The methods of the IRobotStoreySelection class are listed here.

Public Methods

	Name	Description
	Add (see page 1063)	Function adds to the selection a story with the specified number or name.
	AddAll (see page 1063)	Function adds all stories to the selection.
	Clear (see page 1063)	Function clears the selection.
	GetNames (see page 1064)	Function returns a table with names of stories that belong to the selection.
	GetNumbers (see page 1064)	Function returns a table with numbers of stories that belong to the selection.
	Remove (see page 1064)	Function deletes from the selection a story with the specified number or name.

VI.3.2.1 Add

C++

```
HRESULT Add(VARIANT _storey);
```

C#

```
public void Add(VARIANT _storey);
```

Visual Basic

```
Public Sub Add(_storey As VARIANT)
```

Description

Function adds to the selection a story with the specified number or name.

Version

Available since version 9.7.

VI.3.2.2 AddAll

C++

```
HRESULT AddAll();
```

C#

```
public void AddAll();
```

Visual Basic

```
Public Sub AddAll()
```

Description

Function adds all stories to the selection.

Version

Available since version 9.7.

(c) 2014 Autodesk Inc. All rights reserved

VI.3.2.3 Clear

C++

```
HRESULT Clear();
```

C#

```
public void Clear();
```

Visual Basic

```
Public Sub Clear()
```

Description

Function clears the selection.

Version

Available since version 9.7.

VI.3.2.4 GetNames

C++

```
HRESULT GetNames(SAFEARRAY** ret);
```

C#

```
public Array GetNames();
```

Visual Basic

```
Public Function GetNames() As string)()
```

Description

Function returns a table with names of stories that belong to the selection.

Version

Available since version 9.7.

VI.3.2.5 GetNumbers

C++

```
HRESULT GetNumbers(SAFEARRAY** ret);
```

C#

```
public Array GetNumbers();
```

Visual Basic

```
Public Function GetNumbers() As long)()
```

Description

Function returns a table with numbers of stories that belong to the selection.

Version

Available since version 9.7.

(c) 2014 Autodesk Inc. All rights reserved

VI.3.2.6 Remove

C++

```
HRESULT Remove(VARIANT _storey);
```

C#

```
public void Remove(VARIANT _storey);
```

Visual Basic

```
Public Sub Remove(_storey As VARIANT)
```

Description

Function deletes from the selection a story with the specified number or name.

Version

Available since version 9.7.

VI.4 IRobotStoreyValues

Class Hierarchy

RobotOM.IRobotStoreyValues

C++

```
interface IRobotStoreyValues : IDispatch;
```

C#

```
public interface IRobotStoreyValues;
```

Visual Basic

```
Public Interface IRobotStoreyValues
```

Description

Bar properties.

Version

Available since version 9.7.

VI.4.1 IRobotStoreyValues Members

The following tables list the members exposed by IRobotStoreyValues.

Public Fields

	Name	Description
	Ex0 (see page 1066)	Ex0 eccentricity - a distance between the center of rigidity and the center of gravity projected on the X axis.
	Ex1 (see page 1066)	Accidental eccentricity along the X axis.
	Ex2 (see page 1066)	Additional eccentricity along the X axis.
	Ey0 (see page 1067)	Ey0 eccentricity - a distance between the center of rigidity and the center of gravity projected on the Y axis.
	Ey1 (see page 1067)	Accidental eccentricity along the Y axis.
	Ey2 (see page 1067)	Additional eccentricity along the Y axis.
	G (see page 1068)	Center of gravity.
	Ix (see page 1068)	Moment of inertia (X axis).

(c) 2014 Autodesk Inc. All rights reserved

	Iy (see page 1068)	Moment of inertia (Y axis).
	Iz (see page 1068)	Moment of inertia (Z axis).
	Mass (see page 1069)	Mass.
	R (see page 1069)	Center of rigidity.

VI.4.2 IRobotStoreyValues Fields

The fields of the IRobotStoreyValues class are listed here.

Public Fields

	Name	Description
	Ex0 (see page 1066)	Ex0 eccentricity - a distance between the center of rigidity and the center of gravity projected on the X axis.
	Ex1 (see page 1066)	Accidental eccentricity along the X axis.
	Ex2 (see page 1066)	Additional eccentricity along the X axis.
	Ey0 (see page 1067)	Ey0 eccentricity - a distance between the center of rigidity and the center of gravity projected on the Y axis.
	Ey1 (see page 1067)	Accidental eccentricity along the Y axis.
	Ey2 (see page 1067)	Additional eccentricity along the Y axis.
	G (see page 1068)	Center of gravity.
	Ix (see page 1068)	Moment of inertia (X axis).
	Iy (see page 1068)	Moment of inertia (Y axis).
	Iz (see page 1068)	Moment of inertia (Z axis).
	Mass (see page 1069)	Mass.
	R (see page 1069)	Center of rigidity.

VI.4.2.1 Ex0

C++

```
HRESULT get_Ex0(double*);
```

C#

```
public double Ex0 { get; }
```

Visual Basic

```
Public ReadOnly Ex0 As double
```

Description

Ex0 eccentricity - a distance between the center of rigidity and the center of gravity projected on the X axis.

Version

Available since version 9.7.

VI.4.2.2 Ex1

C++

```
HRESULT get_Ex1(double*);
```

C#

```
public double Ex1 { get; }
```

Visual Basic

```
Public ReadOnly Ex1 As double
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Accidental eccentricity along the X axis.

Version

Available since version 9.7.

VI.4.2.3 Ex2

C++

```
HRESULT get_Ex2(double*);
```

C#

```
public double Ex2 { get; }
```

Visual Basic

```
Public ReadOnly Ex2 As double
```

Description

Additional eccentricity along the X axis.

Version

Available since version 9.7.

VI.4.2.4 Ey0

C++

```
HRESULT get_Ey0(double*);
```

C#

```
public double Ey0 { get; }
```

Visual Basic

```
Public ReadOnly Ey0 As double
```

Description

Ey0 eccentricity - a distance between the center of rigidity and the center of gravity projected on the Y axis.

Version

Available since version 9.7.

VI.4.2.5 Ey1

C++

```
HRESULT get_Ey1(double*);
```

C#

```
public double Ey1 { get; }
```

Visual Basic

```
Public ReadOnly Ey1 As double
```

Description

Accidental eccentricity along the Y axis.

Version

Available since version 9.7.

(c) 2014 Autodesk Inc. All rights reserved

VI.4.2.6 Ey2

C++

```
HRESULT get_Ey2(double*);
```

C#

```
public double Ey2 { get; }
```

Visual Basic

```
Public ReadOnly Ey2 As double
```

Description

Additional eccentricity along the Y axis.

Version

Available since version 9.7.

VI.4.2.7 G

C++

```
HRESULT get_G(IRobotGeoPoint3D**);
```

C#

```
public IRobotGeoPoint3D G { get; }
```

Visual Basic

```
Public ReadOnly G As IRobotGeoPoint3D
```

Description

Center of gravity.

Version

Available since version 9.7.

VI.4.2.8 Ix

C++

```
HRESULT get_Ix(double*);
```

C#

```
public double Ix { get; }
```

Visual Basic

```
Public ReadOnly Ix As double
```

Description

Moment of inertia (X axis).

Version

Available since version 9.7.

VI.4.2.9 Iy

C++

```
HRESULT get_Iy(double*);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double Iy { get; }
```

Visual Basic

```
Public ReadOnly Iy As double
```

Description

Moment of inertia (Y axis).

Version

Available since version 9.7.

VI.4.2.10 Iz

C++

```
HRESULT get_Iz(double*);
```

C#

```
public double Iz { get; }
```

Visual Basic

```
Public ReadOnly Iz As double
```

Description

Moment of inertia (Z axis).

Version

Available since version 9.7.

VI.4.2.11 Mass

C++

```
HRESULT get_Mass(double*);
```

C#

```
public double Mass { get; }
```

Visual Basic

```
Public ReadOnly Mass As double
```

Description

Mass.

Version

Available since version 9.7.

VI.4.2.12 R

C++

```
HRESULT get_R(IRobotGeoPoint3D**);
```

C#

```
public IRobotGeoPoint3D R { get; }
```

Visual Basic

```
Public ReadOnly R As IRobotGeoPoint3D
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Center of rigidity.

Version

Available since version 9.7.

VI.5 IRobotStoreyReducedForces

Class Hierarchy

C++

```
interface IRobotStoreyReducedForces : IDispatch;
```

C#

```
public interface IRobotStoreyReducedForces;
```

Visual Basic

```
Public Interface IRobotStoreyReducedForces
```

Version

Available since version 9.7.

VI.5.1 IRobotStoreyReducedForces Members

The following tables list the members exposed by IRobotStoreyReducedForces.

Public Fields

	Name	Description
	FX (see page 1071)	
	FX_ToColumns (see page 1071)	
	FX_ToWalls (see page 1071)	
	FY (see page 1071)	
	FY_ToColumns (see page 1072)	
	FY_ToWalls (see page 1072)	
	FZ (see page 1072)	
	FZ_ToColumns (see page 1072)	
	FZ_ToWalls (see page 1073)	
	MX (see page 1073)	
	MY (see page 1073)	
	MZ (see page 1073)	

VI.5.2 IRobotStoreyReducedForces Fields

The fields of the IRobotStoreyReducedForces class are listed here.

Public Fields

	Name	Description
	FX (see page 1071)	
	FX_ToColumns (see page 1071)	
	FX_ToWalls (see page 1071)	
	FY (see page 1071)	

(c) 2014 Autodesk Inc. All rights reserved

	FY_ToColumns (see page 1072)	
	FY_ToWalls (see page 1072)	
	FZ (see page 1072)	
	FZ_ToColumns (see page 1072)	
	FZ_ToWalls (see page 1073)	
	MX (see page 1073)	
	MY (see page 1073)	
	MZ (see page 1073)	

VI.5.2.1 FX

C++

```
HRESULT get_FX(double*);
```

C#

```
public double FX { get; }
```

Visual Basic

```
Public ReadOnly FX As double
```

Version

Available since version 9.7.

VI.5.2.2 FX_ToColumns

C++

```
HRESULT get_FX_ToColumns(double*);
```

C#

```
public double FX_ToColumns { get; }
```

Visual Basic

```
Public ReadOnly FX_ToColumns As double
```

Version

Available since version 9.7.

VI.5.2.3 FX_ToWalls

C++

```
HRESULT get_FX_ToWalls(double*);
```

C#

```
public double FX_ToWalls { get; }
```

Visual Basic

```
Public ReadOnly FX_ToWalls As double
```

Version

Available since version 9.7.

VI.5.2.4 FY

C++

```
HRESULT get_FY(double*);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double FY { get; }
```

Visual Basic

```
Public ReadOnly FY As double
```

Version

Available since version 9.7.

VI.5.2.5 FY_ToColumns

C++

```
HRESULT get_FY_ToColumns(double*);
```

C#

```
public double FY_ToColumns { get; }
```

Visual Basic

```
Public ReadOnly FY_ToColumns As double
```

Version

Available since version 9.7.

VI.5.2.6 FY_ToWalls

C++

```
HRESULT get_FY_ToWalls(double*);
```

C#

```
public double FY_ToWalls { get; }
```

Visual Basic

```
Public ReadOnly FY_ToWalls As double
```

Version

Available since version 9.7.

VI.5.2.7 FZ

C++

```
HRESULT get_FZ(double*);
```

C#

```
public double FZ { get; }
```

Visual Basic

```
Public ReadOnly FZ As double
```

Version

Available since version 9.7.

VI.5.2.8 FZ_ToColumns

C++

```
HRESULT get_FZ_ToColumns(double*);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double FZ_ToColumns { get; }
```

Visual Basic

```
Public ReadOnly FZ_ToColumns As double
```

Version

Available since version 9.7.

VI.5.2.9 FZ_ToWalls

C++

```
HRESULT get_FZ_ToWalls(double*);
```

C#

```
public double FZ_ToWalls { get; }
```

Visual Basic

```
Public ReadOnly FZ_ToWalls As double
```

Version

Available since version 9.7.

VI.5.2.10 MX

C++

```
HRESULT get_MX(double*);
```

C#

```
public double MX { get; }
```

Visual Basic

```
Public ReadOnly MX As double
```

Version

Available since version 9.7.

VI.5.2.11 MY

C++

```
HRESULT get_MY(double*);
```

C#

```
public double MY { get; }
```

Visual Basic

```
Public ReadOnly MY As double
```

Version

Available since version 9.7.

VI.5.2.12 MZ

C++

```
HRESULT get_MZ(double*);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double MZ { get; }
```

Visual Basic

```
Public ReadOnly MZ As double
```

Version

Available since version 9.7.

VI.6 IRobotStoreyDisplacements

Class Hierarchy

C++

```
interface IRobotStoreyDisplacements : IDispatch;
```

C#

```
public interface IRobotStoreyDisplacements;
```

Visual Basic

```
Public Interface IRobotStoreyDisplacements
```

Version

Available since version 9.7.

VI.6.1 IRobotStoreyDisplacements Members

The following tables list the members exposed by IRobotStoreyDisplacements.

Public Fields

	Name	Description
	DrUX (see page 1075)	
	DrUY (see page 1075)	
	MaxUX (see page 1075)	
	MaxUY (see page 1075)	
	MinUX (see page 1076)	
	MinUY (see page 1076)	
	NodeMaxUX (see page 1076)	
	NodeMaxUY (see page 1076)	
	NodeMinUX (see page 1077)	
	NodeMinUY (see page 1077)	

VI.6.2 IRobotStoreyDisplacements Fields

The fields of the IRobotStoreyDisplacements class are listed here.

Public Fields

	Name	Description
	DrUX (see page 1075)	
	DrUY (see page 1075)	
	MaxUX (see page 1075)	
	MaxUY (see page 1075)	

(c) 2014 Autodesk Inc. All rights reserved

	MinUX (see page 1076)	
	MinUY (see page 1076)	
	NodeMaxUX (see page 1076)	
	NodeMaxUY (see page 1076)	
	NodeMinUX (see page 1077)	
	NodeMinUY (see page 1077)	

VI.6.2.1 DrUX

C++

```
HRESULT get_DrUX(double*);
```

C#

```
public double DrUX { get; }
```

Visual Basic

```
Public ReadOnly DrUX As double
```

Version

Available since version 9.7.

VI.6.2.2 DrUY

C++

```
HRESULT get_DrUY(double*);
```

C#

```
public double DrUY { get; }
```

Visual Basic

```
Public ReadOnly DrUY As double
```

Version

Available since version 9.7.

VI.6.2.3 MaxUX

C++

```
HRESULT get_MaxUX(double*);
```

C#

```
public double MaxUX { get; }
```

Visual Basic

```
Public ReadOnly MaxUX As double
```

Version

Available since version 9.7.

VI.6.2.4 MaxUY

C++

```
HRESULT get_MaxUY(double*);
```

C#

```
public double MaxUY { get; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public ReadOnly MaxUY As double
```

Version

Available since version 9.7.

VI.6.2.5 MinUX

C++

```
HRESULT get_MinUX(double*);
```

C#

```
public double MinUX { get; }
```

Visual Basic

```
Public ReadOnly MinUX As double
```

Version

Available since version 9.7.

VI.6.2.6 MinUY

C++

```
HRESULT get_MinUY(double*);
```

C#

```
public double MinUY { get; }
```

Visual Basic

```
Public ReadOnly MinUY As double
```

Version

Available since version 9.7.

VI.6.2.7 NodeMaxUX

C++

```
HRESULT get_NodeMaxUX(long*);
```

C#

```
public long NodeMaxUX { get; }
```

Visual Basic

```
Public ReadOnly NodeMaxUX As long
```

Version

Available since version 9.7.

VI.6.2.8 NodeMaxUY

C++

```
HRESULT get_NodeMaxUY(long*);
```

C#

```
public long NodeMaxUY { get; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public ReadOnly NodeMaxUY As long
```

Version

Available since version 9.7.

VI.6.2.9 NodeMinUX

C++

```
HRESULT get_NodeMinUX(long*);
```

C#

```
public long NodeMinUX { get; }
```

Visual Basic

```
Public ReadOnly NodeMinUX As long
```

Version

Available since version 9.7.

VI.6.2.10 NodeMinUY

C++

```
HRESULT get_NodeMinUY(long*);
```

C#

```
public long NodeMinUY { get; }
```

Visual Basic

```
Public ReadOnly NodeMinUY As long
```

Version

Available since version 9.7.

VI.7 IRobotStoreyResultServer

Class Hierarchy

C++

```
interface IRobotStoreyResultServer : IDispatch;
```

C#

```
public interface IRobotStoreyResultServer;
```

Visual Basic

```
Public Interface IRobotStoreyResultServer
```

Description

Server of results for stories.

Version

Available since version 9.7.

(c) 2014 Autodesk Inc. All rights reserved

VI.7.1 IRobotStoreyResultServer Members

The following tables list the members exposed by IRobotStoreyResultServer.

Public Methods

	Name	Description
	Displacements (see page 1078)	Function returns values of displacements for the specified story and load case.
	ReducedForces (see page 1078)	Function returns a set of reduced forces for the specified story and load case.
	Values (see page 1079)	Function returns a set of values for the specified story and load case.

VI.7.2 IRobotStoreyResultServer Methods

The methods of the IRobotStoreyResultServer class are listed here.

Public Methods

	Name	Description
	Displacements (see page 1078)	Function returns values of displacements for the specified story and load case.
	ReducedForces (see page 1078)	Function returns a set of reduced forces for the specified story and load case.
	Values (see page 1079)	Function returns a set of values for the specified story and load case.

VI.7.2.1 Displacements

C++

```
HRESULT Displacements(long _storey, long _case, IRobotStoreyDisplacements** ret);
```

C#

```
public IRobotStoreyDisplacements Displacements(long _storey, long _case);
```

Visual Basic

```
Public Function Displacements(_storey As long, _case As long) As IRobotStoreyDisplacements
```

Description

Function returns values of displacements for the specified story and load case.

Version

Available since version 9.7.

VI.7.2.2 ReducedForces

C++

```
HRESULT ReducedForces(long _storey, long _case, IRobotStoreyReducedForces** ret);
```

C#

```
public IRobotStoreyReducedForces ReducedForces(long _storey, long _case);
```

Visual Basic

```
Public Function ReducedForces(_storey As long, _case As long) As IRobotStoreyReducedForces
```

Description

Function returns a set of reduced forces for the specified story and load case.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 9.7.

VI.7.2.3 Values

C++

```
HRESULT Values(long _storey, long _case, IRobotStoreyValues** ret);
```

C#

```
public IRobotStoreyValues Values(long _storey, long _case);
```

Visual Basic

```
Public Function Values(_storey As long, _case As long) As IRobotStoreyValues
```

Description

Function returns a set of values for the specified story and load case.

Version

Available since version 9.7.

VII Quantity survey

Available since version 7.5.

Interfaces

	Name	Description
	IRobotMaterialQuantitySurvey (see page 1079)	
	IRobotBarSectionQuantitySurvey (see page 1082)	
	IRobotThicknessQuantitySurvey (see page 1085)	
	IRobotStructureQuantitySurvey (see page 1088)	Quantity survey of a structure.

VII.1 IRobotMaterialQuantitySurvey

Class Hierarchy

RobotOM.IRobotMaterialQuantitySurvey

C++

```
interface IRobotMaterialQuantitySurvey : IDispatch;
```

C#

```
public interface IRobotMaterialQuantitySurvey;
```

Visual Basic

```
Public Interface IRobotMaterialQuantitySurvey
```

Version

Available since version 7.5.

(c) 2014 Autodesk Inc. All rights reserved

VII.1.1 IRobotMaterialQuantitySurvey Members

The following tables list the members exposed by IRobotMaterialQuantitySurvey.

Public Fields

	Name	Description
	Count (see page 1080)	Number of all different materials used in a structure.

Public Methods

	Name	Description
	GetName (see page 1081)	Function returns a material name.
	GetType (see page 1081)	Function returns a type of the given material.
	GetVolume (see page 1081)	Function returns the total volume of all structure components of the specified type and of the specified material.
	GetWeight (see page 1082)	Function returns the total weight of all structure components of the specified type and of the specified material.

VII.1.2 IRobotMaterialQuantitySurvey Fields

The fields of the IRobotMaterialQuantitySurvey class are listed here.

Public Fields

	Name	Description
	Count (see page 1080)	Number of all different materials used in a structure.

VII.1.2.1 Count

C++

```
HRESULT get_Count(long*);
```

C#

```
public long Count { get; }
```

Visual Basic

```
Public ReadOnly Count As long
```

Description

Number of all different materials used in a structure.

Version

Available since version 7.5.

VII.1.3 IRobotMaterialQuantitySurvey Methods

The methods of the IRobotMaterialQuantitySurvey class are listed here.

Public Methods

	Name	Description
	GetName (see page 1081)	Function returns a material name.
	GetType (see page 1081)	Function returns a type of the given material.
	GetVolume (see page 1081)	Function returns the total volume of all structure components of the specified type and of the specified material.
	GetWeight (see page 1082)	Function returns the total weight of all structure components of the specified type and of the specified material.

(c) 2014 Autodesk Inc. All rights reserved

VII.1.3.1 GetName

C++

```
HRESULT GetName(long _idx, BSTR* ret);
```

C#

```
public String GetName(long _idx);
```

Visual Basic

```
Public Function GetName(_idx As long) As String
```

Description

Function returns a material name.

Version

Available since version 7.5.

VII.1.3.2 GetType

C++

```
HRESULT GetType(VARIANT _idx_or_name, IRobotMaterialType* ret);
```

C#

```
public IRobotMaterialType GetType(VARIANT _idx_or_name);
```

Visual Basic

```
Public Function GetType(_idx_or_name As VARIANT) As IRobotMaterialType
```

Description

Function returns a type of the given material.

Version

Available since version 7.5.

VII.1.3.3 GetVolume

C++

```
HRESULT GetVolume(IRobotObjectType _obj_type, VARIANT _idx_or_name, double* ret);
```

C#

```
public double GetVolume(IRobotObjectType _obj_type, VARIANT _idx_or_name);
```

Visual Basic

```
Public Function GetVolume(_obj_type As IRobotObjectType, _idx_or_name As VARIANT) As double
```

Description

Function returns the total volume of all structure components of the specified type and of the specified material.

Version

Available since version 7.5.

(c) 2014 Autodesk Inc. All rights reserved

VII.1.3.4 GetWeight

C++

```
HRESULT GetWeight(IRobotObjectType _obj_type, VARIANT _idx_or_name, double* ret);
```

C#

```
public double GetWeight(IRobotObjectType _obj_type, VARIANT _idx_or_name);
```

Visual Basic

```
Public Function GetWeight(_obj_type As IRobotObjectType, _idx_or_name As VARIANT) As double
```

Description

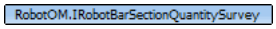
Function returns the total weight of all structure components of the specified type and of the specified material.

Version

Available since version 7.5.

VII.2 IRobotBarSectionQuantitySurvey

Class Hierarchy



C++

```
interface IRobotBarSectionQuantitySurvey : IDispatch;
```

C#

```
public interface IRobotBarSectionQuantitySurvey;
```

Visual Basic

```
Public Interface IRobotBarSectionQuantitySurvey
```

Version

Available since version 7.5.

VII.2.1 IRobotBarSectionQuantitySurvey Members

The following tables list the members exposed by IRobotBarSectionQuantitySurvey.

Public Fields

	Name	Description
	Count (see page 1083)	Number of different bar sections used in a structure.

Public Methods

	Name	Description
	GetLength (see page 1083)	Function returns the total length of all structure members of a selected section.
	GetName (see page 1084)	Function returns the name of a section of the given index.
	GetPaintArea (see page 1084)	Function returns the total painting area of all structure members of a selected section.
	GetUnitWeight (see page 1084)	Function returns a unit weight of a selected section.
	GetVolume (see page 1085)	Function returns the total volume of all structure members of a selected section.

(c) 2014 Autodesk Inc. All rights reserved

	GetWeight (see page 1085)	Function returns the total weight of all structure members of a selected section.

VII.2.2 IRobotBarSectionQuantitySurvey Fields

The fields of the IRobotBarSectionQuantitySurvey class are listed here.

Public Fields

	Name	Description
	Count (see page 1083)	Number of different bar sections used in a structure.

VII.2.2.1 Count

C++

```
HRESULT get_Count(long*);
```

C#

```
public long Count { get; }
```

Visual Basic

```
Public ReadOnly Count As long
```

Description

Number of different bar sections used in a structure.

Version

Available since version 7.5.

VII.2.3 IRobotBarSectionQuantitySurvey Methods

The methods of the IRobotBarSectionQuantitySurvey class are listed here.

Public Methods

	Name	Description
	GetLength (see page 1083)	Function returns the total length of all structure members of a selected section.
	GetName (see page 1084)	Function returns the name of a section of the given index.
	GetPaintArea (see page 1084)	Function returns the total painting area of all structure members of a selected section.
	GetUnitWeight (see page 1084)	Function returns a unit weight of a selected section.
	GetVolume (see page 1085)	Function returns the total volume of all structure members of a selected section.
	GetWeight (see page 1085)	Function returns the total weight of all structure members of a selected section.

VII.2.3.1 GetLength

C++

```
HRESULT GetLength(VARIANT _idx_or_name, double* ret);
```

C#

```
public double GetLength(VARIANT _idx_or_name);
```

Visual Basic

```
Public Function GetLength(_idx_or_name As VARIANT) As double
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Function returns the total length of all structure members of a selected section.

Version

Available since version 7.5.

VII.2.3.2 GetName

C++

```
HRESULT GetName(long _idx, BSTR* ret);
```

C#

```
public String GetName(long _idx);
```

Visual Basic

```
Public Function GetName(_idx As long) As String
```

Description

Function returns the name of a section of the given index.

Version

Available since version 7.5.

VII.2.3.3 GetPaintArea

C++

```
HRESULT GetPaintArea(VARIANT _idx_or_name, double* ret);
```

C#

```
public double GetPaintArea(VARIANT _idx_or_name);
```

Visual Basic

```
Public Function GetPaintArea(_idx_or_name As VARIANT) As double
```

Description

Function returns the total painting area of all structure members of a selected section.

Version

Available since version 7.5.

VII.2.3.4 GetUnitWeight

C++

```
HRESULT GetUnitWeight(VARIANT _idx_or_name, double* ret);
```

C#

```
public double GetUnitWeight(VARIANT _idx_or_name);
```

Visual Basic

```
Public Function GetUnitWeight(_idx_or_name As VARIANT) As double
```

Description

Function returns a unit weight of a selected section.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 7.5.

VII.2.3.5 GetVolume

C++

```
HRESULT GetVolume(VARIANT _idx_or_name, double* ret);
```

C#

```
public double GetVolume(VARIANT _idx_or_name);
```

Visual Basic

```
Public Function GetVolume(_idx_or_name As VARIANT) As double
```

Description

Function returns the total volume of all structure members of a selected section.

Version

Available since version 7.5.

VII.2.3.6 GetWeight

C++

```
HRESULT GetWeight(VARIANT _idx_or_name, double* ret);
```

C#

```
public double GetWeight(VARIANT _idx_or_name);
```

Visual Basic

```
Public Function GetWeight(_idx_or_name As VARIANT) As double
```

Description

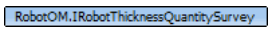
Function returns the total weight of all structure members of a selected section.

Version

Available since version 7.5.

VII.3 IRobotThicknessQuantitySurvey

Class Hierarchy



C++

```
interface IRobotThicknessQuantitySurvey : IDispatch;
```

C#

```
public interface IRobotThicknessQuantitySurvey;
```

Visual Basic

```
Public Interface IRobotThicknessQuantitySurvey
```

Version

Available since version 7.5.

(c) 2014 Autodesk Inc. All rights reserved

VII.3.1 IRobotThicknessQuantitySurvey Members

The following tables list the members exposed by IRobotThicknessQuantitySurvey.

Public Fields

	Name	Description
	Count (see page 1086)	Number of different panel thicknesses used in a structure.

Public Methods

	Name	Description
	GetArea (see page 1087)	Function returns the total area of all structure panels of a selected thickness.
	GetName (see page 1087)	Function returns a thickness name of the given index.
	GetUnitWeight (see page 1087)	Function returns a unit weight for a selected thickness.
	GetVolume (see page 1088)	Function returns the total volume of all structure panels of a selected thickness.
	GetWeight (see page 1088)	Function returns the total weight of all structure panels of a selected thickness.

VII.3.2 IRobotThicknessQuantitySurvey Fields

The fields of the IRobotThicknessQuantitySurvey class are listed here.

Public Fields

	Name	Description
	Count (see page 1086)	Number of different panel thicknesses used in a structure.

VII.3.2.1 Count

C++

```
HRESULT get_Count(long*);
```

C#

```
public long Count { get; }
```

Visual Basic

```
Public ReadOnly Count As long
```

Description

Number of different panel thicknesses used in a structure.

Version

Available since version 7.5.

VII.3.3 IRobotThicknessQuantitySurvey Methods

The methods of the IRobotThicknessQuantitySurvey class are listed here.

Public Methods

	Name	Description
	GetArea (see page 1087)	Function returns the total area of all structure panels of a selected thickness.
	GetName (see page 1087)	Function returns a thickness name of the given index.
	GetUnitWeight (see page 1087)	Function returns a unit weight for a selected thickness.
	GetVolume (see page 1088)	Function returns the total volume of all structure panels of a selected thickness.

(c) 2014 Autodesk Inc. All rights reserved

	GetWeight (see page 1088)	Function returns the total weight of all structure panels of a selected thickness.

VII.3.3.1 GetArea

C++

```
HRESULT GetArea(VARIANT _idx_or_name, double* ret);
```

C#

```
public double GetArea(VARIANT _idx_or_name);
```

Visual Basic

```
Public Function GetArea(_idx_or_name As VARIANT) As double
```

Description

Function returns the total area of all structure panels of a selected thickness.

Version

Available since version 7.5.

VII.3.3.2 GetName

C++

```
HRESULT GetName(long _idx, BSTR* ret);
```

C#

```
public String GetName(long _idx);
```

Visual Basic

```
Public Function GetName(_idx As long) As String
```

Description

Function returns a thickness name of the given index.

Version

Available since version 7.5.

VII.3.3.3 GetUnitWeight

C++

```
HRESULT GetUnitWeight(VARIANT _idx_or_name, double* ret);
```

C#

```
public double GetUnitWeight(VARIANT _idx_or_name);
```

Visual Basic

```
Public Function GetUnitWeight(_idx_or_name As VARIANT) As double
```

Description

Function returns a unit weight for a selected thickness.

Version

Available since version 7.5.

(c) 2014 Autodesk Inc. All rights reserved

VII.3.3.4 GetVolume

C++

```
HRESULT GetVolume(VARIANT _idx_or_name, double* ret);
```

C#

```
public double GetVolume(VARIANT _idx_or_name);
```

Visual Basic

```
Public Function GetVolume(_idx_or_name As VARIANT) As double
```

Description

Function returns the total volume of all structure panels of a selected thickness.

Version

Available since version 7.5.

VII.3.3.5 GetWeight

C++

```
HRESULT GetWeight(VARIANT _idx_or_name, double* ret);
```

C#

```
public double GetWeight(VARIANT _idx_or_name);
```

Visual Basic

```
Public Function GetWeight(_idx_or_name As VARIANT) As double
```

Description

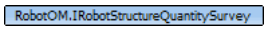
Function returns the total weight of all structure panels of a selected thickness.

Version

Available since version 7.5.

VII.4 IRobotStructureQuantitySurvey

Class Hierarchy



C++

```
interface IRobotStructureQuantitySurvey : IDispatch;
```

C#

```
public interface IRobotStructureQuantitySurvey;
```

Visual Basic

```
Public Interface IRobotStructureQuantitySurvey
```

Description

Quantity survey of a structure.

Version

Available since version 7.5.

(c) 2014 Autodesk Inc. All rights reserved

VII.4.1 IRobotStructureQuantitySurvey Members

The following tables list the members exposed by IRobotStructureQuantitySurvey.

Public Fields

	Name	Description
	BarSections (see page 1089)	
	Materials (see page 1089)	
	PanelThickness (see page 1089)	

VII.4.2 IRobotStructureQuantitySurvey Fields

The fields of the IRobotStructureQuantitySurvey class are listed here.

Public Fields

	Name	Description
	BarSections (see page 1089)	
	Materials (see page 1089)	
	PanelThickness (see page 1089)	

VII.4.2.1 BarSections

C++

```
HRESULT get_BarSections(IRobotBarSectionQuantitySurvey**);
```

C#

```
public IRobotBarSectionQuantitySurvey BarSections { get; }
```

Visual Basic

```
Public ReadOnly BarSections As IRobotBarSectionQuantitySurvey
```

Version

Available since version 7.5.

VII.4.2.2 Materials

C++

```
HRESULT get_Materials(IRobotMaterialQuantitySurvey**);
```

C#

```
public IRobotMaterialQuantitySurvey Materials { get; }
```

Visual Basic

```
Public ReadOnly Materials As IRobotMaterialQuantitySurvey
```

Version

Available since version 7.5.

VII.4.2.3 PanelThickness

C++

```
HRESULT get_PanelThickness(IRobotThicknessQuantitySurvey**);
```

C#

```
public IRobotThicknessQuantitySurvey PanelThickness { get; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

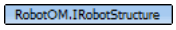
```
Public ReadOnly PanelThickness As IRobotThicknessQuantitySurvey
```

Version

Available since version 7.5.

VIII IRobotStructure

Class Hierarchy



C++

```
interface IRobotStructure : IDispatch;
```

C#

```
public interface IRobotStructure;
```

Visual Basic

```
Public Interface IRobotStructure
```

Description

RobotStructure represents an entire structure. Particular structure components are represented by appropriate interface components. .

VIII.1 IRobotStructure Members

The following tables list the members exposed by IRobotStructure.

Public Fields

	Name	Description
	Bars (see page 1092)	Structure bar server .
	Cases (see page 1092)	Load case server .
	Edit (see page 1092)	Set of auxiliary edit functions .
	ExtParams (see page 1092)	External parameter server.
	FiniteElems (see page 1093)	Finite element server.
	GroupObjects (see page 1093)	Server for grouping of structure elements.
	Groups (see page 1093)	Server managing groups of structure components.
	IsCalcModelGenerated (see page 1094)	A flag indicating if the computational model for the structure has been generated and if it is available.
	Labels (see page 1094)	Server of complex attributes, such as node supports, bar sections and others. .
	Nodes (see page 1094)	Structure node server .
	Objects (see page 1094)	Object server .
	QuantitySurvey (see page 1095)	Quantity survey.
	Results (see page 1095)	Server of the currently available results .
	ResultsFreeze (see page 1095)	
	Selections (see page 1095)	Server managing selections (selection factory) .
	Storeys (see page 1096)	Stories defined in a structure.
	Type (see page 1096)	Structure type.

(c) 2014 Autodesk Inc. All rights reserved

Public Methods

	Name	Description
	ApplyCache (see page 1097)	Function inserts data from a specified buffer to a structure and returns information about new structure components generated in this way. .
	Clear (see page 1097)	Function deletes all elements of a structure model.
	CreateCache (see page 1097)	Function creates and returns an empty buffer to which new structure components can be added. To insert a whole component set to a structure, ApplyCache (see page 1097) function should be called up. .
	ExportXml (see page 1098)	Function exports selected structure components and calculation results to the indicated xml file.
	Merge (see page 1098)	Function attaches a given data set to the curent structure.

VIII.2 IRobotStructure Fields

The fields of the IRobotStructure class are listed here.

Public Fields

	Name	Description
	Bars (see page 1092)	Structure bar server .
	Cases (see page 1092)	Load case server .
	Edit (see page 1092)	Set of auxiliary edit functions .
	ExtParams (see page 1092)	External parameter server.
	FiniteElems (see page 1093)	Finite element server.
	GroupObjects (see page 1093)	Server for grouping of structure elements.
	Groups (see page 1093)	Server managing groups of structure components.
	IsCalcModelGenerated (see page 1094)	A flag indicating if the computational model for the structure has been generated and if it is available.
	Labels (see page 1094)	Server of complex attributes, such as node supports, bar sections and others. .
	Nodes (see page 1094)	Structure node server .
	Objects (see page 1094)	Object server .
	QuantitySurvey (see page 1095)	Quantity survey.
	Results (see page 1095)	Server of the currently available results .
	ResultsFreeze (see page 1095)	
	Selections (see page 1095)	Server managing selections (selection factory) .
	Storeys (see page 1096)	Stories defined in a structure.
	Type (see page 1096)	Structure type.

VIII.2.1 Bars

C++

```
HRESULT get_Bars(IRobotBarServer**);
```

C#

```
public IRobotBarServer Bars { get; }
```

Visual Basic

```
Public ReadOnly Bars As IRobotBarServer
```

Description

Structure bar server .

(c) 2014 Autodesk Inc. All rights reserved

VIII.2.2 Cases

C++

```
HRESULT get_Cases(IRobotCaseServer**);
```

C#

```
public IRobotCaseServer Cases { get; }
```

Visual Basic

```
Public ReadOnly Cases As IRobotCaseServer
```

Description

Load case server .

VIII.2.3 Edit

C++

```
HRESULT get_Edit(IRobotStructureEditTools**);
```

C#

```
public IRobotStructureEditTools Edit { get; }
```

Visual Basic

```
Public ReadOnly Edit As IRobotStructureEditTools
```

Description

Set of auxiliary edit functions .

Version

Available since version 3.5.

VIII.2.4 ExtParams

C++

```
HRESULT get_ExtParams(IRobotParamServer**);
```

C#

```
public IRobotParamServer ExtParams { get; }
```

Visual Basic

```
Public ReadOnly ExtParams As IRobotParamServer
```

Description

External parameter server.

Version

Available since version 8.2.

VIII.2.5 FiniteElems

C++

```
HRESULT get_FiniteElems(IRobotFiniteElementServer**);
```

C#

```
public IRobotFiniteElementServer FiniteElems { get; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public ReadOnly FiniteElems As IRobotFiniteElementServer
```

Description

Finite element server.

Version

Available since version 2.5.

VIII.2.6 GroupObjects

C++

```
HRESULT get_GroupObjects(IRobotGroupObjectServer**);
```

C#

```
public IRobotGroupObjectServer GroupObjects { get; }
```

Visual Basic

```
Public ReadOnly GroupObjects As IRobotGroupObjectServer
```

Description

Server for grouping of structure elements.

Version

Available since version 7.5.

VIII.2.7 Groups

C++

```
HRESULT get_Groups(IRobotGroupServer**);
```

C#

```
public IRobotGroupServer Groups { get; }
```

Visual Basic

```
Public ReadOnly Groups As IRobotGroupServer
```

Description

Server managing groups of structure components.

Version

Available since version 3.

VIII.2.8 IsCalcModelGenerated

C++

```
HRESULT get_IsCalcModelGenerated(VARIANT_BOOL*);
```

C#

```
public bool IsCalcModelGenerated { get; }
```

Visual Basic

```
Public ReadOnly IsCalcModelGenerated As Boolean
```

Description

A flag indicating if the computational model for the structure has been generated and if it is available.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 13.4.

VIII.2.9 Labels

C++

```
HRESULT get_Labels(IRobotLabelServer**);
```

C#

```
public IRobotLabelServer Labels { get; }
```

Visual Basic

```
Public ReadOnly Labels As IRobotLabelServer
```

Description

Server of complex attributes, such as node supports, bar sections and others. .

VIII.2.10 Nodes

C++

```
HRESULT get_Nodes(IRobotNodeServer**);
```

C#

```
public IRobotNodeServer Nodes { get; }
```

Visual Basic

```
Public ReadOnly Nodes As IRobotNodeServer
```

Description

Structure node server .

VIII.2.11 Objects

C++

```
HRESULT get_Objects(IRobotObjObjectServer**);
```

C#

```
public IRobotObjObjectServer Objects { get; }
```

Visual Basic

```
Public ReadOnly Objects As IRobotObjObjectServer
```

Description

Object server .

VIII.2.12 QuantitySurvey

C++

```
HRESULT get_QuantitySurvey(IRobotStructureQuantitySurvey**);
```

C#

```
public IRobotStructureQuantitySurvey QuantitySurvey { get; }
```

Visual Basic

```
Public ReadOnly QuantitySurvey As IRobotStructureQuantitySurvey
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Quantity survey.

Version

Available since version 7.5.

VIII.2.13 Results

C++

```
HRESULT get_Results(IRobotResultServer**);
```

C#

```
public IRobotResultServer Results { get; }
```

Visual Basic

```
Public ReadOnly Results As IRobotResultServer
```

Description

Server of the currently available results .

VIII.2.14 ResultsFreeze

C++

```
HRESULT get_ResultsFreeze(VARIANT_BOOL*);
HRESULT put_ResultsFreeze(VARIANT_BOOL);
```

C#

```
public bool ResultsFreeze { get; set; }
```

Visual Basic

```
Public ResultsFreeze As Boolean
```

Version

Available since version 4.

VIII.2.15 Selections

C++

```
HRESULT get_Selections(IRobotSelectionFactory**);
```

C#

```
public IRobotSelectionFactory Selections { get; }
```

Visual Basic

```
Public ReadOnly Selections As IRobotSelectionFactory
```

Description

Server managing selections (selection factory) .

VIII.2.16 Storeys

C++

```
HRESULT get_Storeys(IRobotStoreyMngr**);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRobotStoreyMngr Storeys { get; }
```

Visual Basic

```
Public ReadOnly Storeys As IRobotStoreyMngr
```

Description

Stories defined in a structure.

VIII.2.17 Type

C++

```
HRESULT get_Type(IRobotProjectType*);
```

C#

```
public IRobotProjectType Type { get; }
```

Visual Basic

```
Public ReadOnly Type As IRobotProjectType
```

Description

Structure type.

Version

Available since version 8.

VIII.3 IRobotStructure Methods

The methods of the IRobotStructure class are listed here.

Public Methods

	Name	Description
	ApplyCache (see page 1097)	Function inserts data from a specified buffer to a structure and returns information about new structure components generated in this way. .
	Clear (see page 1097)	Function deletes all elements of a structure model.
	CreateCache (see page 1097)	Function creates and returns an empty buffer to which new structure components can be added. To insert a whole component set to a structure, ApplyCache (see page 1097) function should be called up. .
	ExportXml (see page 1098)	Function exports selected structure components and calculation results to the indicated xml file.
	Merge (see page 1098)	Function attaches a given data set to the curent structure.

VIII.3.1 ApplyCache

C++

```
HRESULT ApplyCache(IRobotStructureCache* _struct_cache, IRobotStructureApplyInfo** ret);
```

C#

```
public IRobotStructureApplyInfo ApplyCache(IRobotStructureCache _struct_cache);
```

Visual Basic

```
Public Function ApplyCache(ByRef _struct_cache As IRobotStructureCache) As
IRobotStructureApplyInfo
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Function inserts data from a specified buffer to a structure and returns information about new structure components generated in this way. .

Version

Available since version 3.5.

VIII.3.2 Clear

C++

```
HRESULT Clear();
```

C#

```
public void Clear();
```

Visual Basic

```
Public Sub Clear()
```

Description

Function deletes all elements of a structure model.

VIII.3.3 CreateCache

C++

```
HRESULT CreateCache(IRobotStructureCache** ret);
```

C#

```
public IRobotStructureCache CreateCache();
```

Visual Basic

```
Public Function CreateCache() As IRobotStructureCache
```

Description

Function creates and returns an empty buffer to which new structure components can be added. To insert a whole component set to a structure, ApplyCache (see page 1097) function should be called up. .

Version

Available since version 3.5.

VIII.3.4 ExportXml

C++

```
HRESULT ExportXml(BSTR _input_xml, BSTR _output_xml);
```

C#

```
public void ExportXml(String _input_xml, String _output_xml);
```

Visual Basic

```
Public Sub ExportXml(_input_xml As String, _output_xml As String)
```

Description

Function exports selected structure components and calculation results to the indicated xml file.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 4.1.

VIII.3.5 Merge

C++

```
HRESULT Merge(IRobotStructureMergeData* _data, IUnknown _params);
```

C#

```
public void Merge(IRobotStructureMergeData _data, IUnknown _params);
```

Visual Basic

```
Public Sub Merge(ByRef _data As IRobotStructureMergeData, _params As IUnknown)
```

Description

Function attaches a given data set to the curent structure.

Version

Available since version 8.

IX IRobotStructureCache

Class Hierarchy

RobotOM.IRobotStructureCache

C++

```
interface IRobotStructureCache : IDispatch;
```

C#

```
public interface IRobotStructureCache;
```

Visual Basic

```
Public Interface IRobotStructureCache
```

Description

Interface enabling buffered access to a structure. It enables adding quickly a whole component set to a structure. To add a whole data set to a structure, ApplyCache function of RobotStructure inerface should be called up. .

Version

Available since version 3.5.

IX.1 IRobotStructureCache Members

The following tables list the members exposed by IRobotStructureCache.

Public Methods

	Name	Description
	AddBar (see page 1099)	Function adds a new bar to the buffer.
	AddNode (see page 1100)	Function adds a new node to the buffer.
	EnsureNodeExist (see page 1100)	Function checks if a node exists at the position of defined coordinates. If yes, then function returns its number. If no, then function creates a new node and returns its number.

(c) 2014 Autodesk Inc. All rights reserved

	SetBarLabel (see page 1100)	Function enables assigning a specified label to an earlier-generated bar.
	SetBarName (see page 1101)	Function sets a name schema for the specified bar.

IX.2 IRobotStructureCache Methods

The methods of the IRobotStructureCache class are listed here.

Public Methods

	Name	Description
	AddBar (see page 1099)	Function adds a new bar to the buffer.
	AddNode (see page 1100)	Function adds a new node to the buffer.
	EnsureNodeExist (see page 1100)	Function checks if a node exists at the position of defined coordinates. If yes, then function returns its number. If no, then function creates a new node and returns its number.
	SetBarLabel (see page 1100)	Function enables assigning a specified label to an earlier-generated bar.
	SetBarName (see page 1101)	Function sets a name schema for the specified bar.

IX.2.1 AddBar

C++

```
HRESULT AddBar(long _number, long _start_node, long _end_node, BSTR _sect_name, BSTR
_mater_name, double _gamma = 0);
```

C#

```
public void AddBar(long _number, long _start_node, long _end_node, String _sect_name,
String _mater_name, double _gamma = 0);
```

Visual Basic

```
Public Sub AddBar(_number As long, _start_node As long, _end_node As long, _sect_name As
String, _mater_name As String, Optional _gamma As double = 0)
```

Description

Function adds a new bar to the buffer.

Version

Available since version 3.5.

IX.2.2 AddNode

C++

```
HRESULT AddNode(long _number, double _x, double _y, double _z);
```

C#

```
public void AddNode(long _number, double _x, double _y, double _z);
```

Visual Basic

```
Public Sub AddNode(_number As long, _x As double, _y As double, _z As double)
```

Description

Function adds a new node to the buffer.

Version

Available since version 3.5.

(c) 2014 Autodesk Inc. All rights reserved

IX.2.3 EnsureNodeExist

C++

```
HRESULT EnsureNodeExist(double _x, double _y, double _z, long* ret);
```

C#

```
public long EnsureNodeExist(double _x, double _y, double _z);
```

Visual Basic

```
Public Function EnsureNodeExist(_x As double, _y As double, _z As double) As long
```

Description

Function checks if a node exists at the position of defined coordinates. If yes, then function returns its number. If no, then function creates a new node and returns its number.

IX.2.4 SetBarLabel

C++

```
HRESULT SetBarLabel(long _bar, IRobotLabelType _lab_type, BSTR _lab_name, long _ext_par =
0);
```

C#

```
public void SetBarLabel(long _bar, IRobotLabelType _lab_type, String _lab_name, long
_ext_par = 0);
```

Visual Basic

```
Public Sub SetBarLabel(_bar As long, _lab_type As IRobotLabelType, _lab_name As String,
Optional _ext_par As long = 0)
```

Description

Function enables assigning a specified label to an earlier-generated bar.

Version

Available since version 4.1.

IX.2.5 SetBarName

C++

```
HRESULT SetBarName(long _bar_num, BSTR _name_tmpl);
```

C#

```
public void SetBarName(long _bar_num, String _name_tmpl);
```

Visual Basic

```
Public Sub SetBarName(_bar_num As long, _name_tmpl As String)
```

Description

Function sets a name schema for the specified bar.

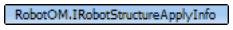
Version

Available since version 8.2.

(c) 2014 Autodesk Inc. All rights reserved

X IRobotStructureApplyInfo

Class Hierarchy



C++

```
interface IRobotStructureApplyInfo : IDispatch;
```

C#

```
public interface IRobotStructureApplyInfo;
```

Visual Basic

```
Public Interface IRobotStructureApplyInfo
```

Description

Interface providing access to information about structure components generated as a result of inserting a new data set to a structure. .

Version

Available since version 3.5.

X.1 IRobotStructureApplyInfo Members

The following tables list the members exposed by IRobotStructureApplyInfo.

Public Fields

	Name	Description
	Bars (see page 1102)	Table with bar numbers.
	Nodes (see page 1102)	Table with node numbers.

X.2 IRobotStructureApplyInfo Fields

The fields of the IRobotStructureApplyInfo class are listed here.

Public Fields

	Name	Description
	Bars (see page 1102)	Table with bar numbers.
	Nodes (see page 1102)	Table with node numbers.

X.2.1 Bars

C++

```
HRESULT get_Bars(IRobotNumbersArray**);
```

C#

```
public IRobotNumbersArray Bars { get; }
```

Visual Basic

```
Public ReadOnly Bars As IRobotNumbersArray
```

Description

Table with bar numbers.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.5.

X.2.2 Nodes

C++

```
HRESULT get_Nodes(IRobotNumbersArray**);
```

C#

```
public IRobotNumbersArray Nodes { get; }
```

Visual Basic

```
Public ReadOnly Nodes As IRobotNumbersArray
```

Description

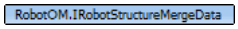
Table with node numbers.

Version

Available since version 3.5.

XI IRobotStructureMergeData

Class Hierarchy



C++

```
interface IRobotStructureMergeData : IDispatch;
```

C#

```
public interface IRobotStructureMergeData;
```

Visual Basic

```
Public Interface IRobotStructureMergeData
```

Description

Interface defining data that may be merged into a structure by means of the Merge function.

Version

Available since version 8.

XI.1 IRobotStructureMergeData Members

The following tables list the members exposed by IRobotStructureMergeData.

Public Fields

	Name	Description
	Structure (see page 1103)	Auxiliary structure.

Public Methods

	Name	Description
	CreateStructure (see page 1104)	Function creates an empty auxiliary structure of the specified type.
	LoadStructure (see page 1104)	Function loads an auxiliary structure from the specified RTD file.

(c) 2014 Autodesk Inc. All rights reserved

XI.2 IRobotStructureMergeData Fields

The fields of the IRobotStructureMergeData class are listed here.

Public Fields

	Name	Description
	Structure (see page 1103)	Auxiliary structure.

XI.2.1 Structure

C++

```
HRESULT get_Structure(IRobotStructure**);
```

C#

```
public IRobotStructure Structure { get; }
```

Visual Basic

```
Public ReadOnly Structure As IRobotStructure
```

Description

Auxiliary structure.

Version

Available since version 8.

XI.3 IRobotStructureMergeData Methods

The methods of the IRobotStructureMergeData class are listed here.

Public Methods

	Name	Description
	CreateStructure (see page 1104)	Function creates an empty auxiliary structure of the specified type.
	LoadStructure (see page 1104)	Function loads an auxiliary structure from the specified RTD file.

XI.3.1 CreateStructure

C++

```
HRESULT CreateStructure(IRobotProjectType _type, VARIANT_BOOL* ret);
```

C#

```
public bool CreateStructure(IRobotProjectType _type);
```

Visual Basic

```
Public Function CreateStructure(_type As IRobotProjectType) As Boolean
```

Description

Function creates an empty auxiliary structure of the specified type.

Version

Available since version 8.

(c) 2014 Autodesk Inc. All rights reserved

XI.3.2 LoadStructure

C++

```
HRESULT LoadStructure(BSTR _rtd_fpath, VARIANT_BOOL* ret);
```

C#

```
public bool LoadStructure(String _rtd_fpath);
```

Visual Basic

```
Public Function LoadStructure(_rtd_fpath As String) As Boolean
```

Description

Function loads an auxiliary structure from the specified RTD file.

Version

Available since version 8.

Project

Robot Object Model allows one to manage an entire project (task) saved to an RTD-format file. .

Enumerations

	Name	Description
	IRobotProjectType (see page 1342)	A set of identifiers is defined to refer to structure types recognized by Robot.
	IRobotProjectSaveFormat (see page 1358)	Available formats for project saving.
	IRobotActiveModelType (see page 1360)	.

Interfaces

	Name	Description
	IRobotProject (see page 1342)	RobotProject interface allows for managing all the data saved in an RTD-format file. .
	IRobotFileInsertParams (see page 1355)	The proces of inserting a structure, defined in an external file of a non-RTD format, is carried out according to parameters determining the insertion mode. The RobotInsertParams interface provides access to the parameters determining the manner of inserting the structure into the current project. .
	IRobotProjectEvents (see page 1358)	Events generated by the RobotProject object.

I Project components

A project may contain many components of different types. They are stored in a tree-like hierarchical structure and managed by an appropriate component manager. .

Enumerations

	Name	Description
	IRobotProjectComponentType (see page 1269)	A set of identifiers is defined to describe different types of components that may be saved in a project. .

(c) 2014 Autodesk Inc. All rights reserved

Interfaces

	Name	Description
	IRobotProjectComponent (see page 1269)	Common interface granting access to any project component. .
	IRobotProjectComponentMngr (see page 1271)	Manager of project components manages tree-like structures containing all components defined in a project. Project components are divided into different levels. The manager makes available a list of all levels that are indexed with numbers from 1 to the number of levels. Additionally, each level has its name. Project components may be taken by levels. It is possible to take from the project all the components of an indicated type from the level of the indicated name. .

I.1 Spread footing

A set of interfaces allowing access to a component of the 'spread footing' project. .

Version

Available since version 3.

Enumerations

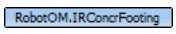
	Name	Description
	IRConcrFootingDimType (see page 1118)	A set of values identifying individual spread footing dimensions. .
	IRConcrFootingShapeType (see page 1118)	A set of values defining the spread footing shape. .
	IRConcrFootingType (see page 1119)	A set of values defining the foundation type. .
	IRConcrFootResultType (see page 1127)	
	IRConcrFootingPierType (see page 1128)	

Interfaces

	Name	Description
	IRConcrFooting (see page 1106)	Interface representing the spread footing.
	IRConcrFootingGeometry (see page 1115)	Interface representing the spread footing geometry.
	IRConcrFootingGround (see page 1119)	Interface representing soils and reference level of the footing.
	IRConcrFootingLoads (see page 1121)	Interface representing spread footing loads and backfill loads. .
	IRConcrFootingResults (see page 1121)	Interface representing calculation results for the spread footing. .
	IRConcrFootingCalculationOptions (see page 1123)	
	IRConcrFootingPattern (see page 1125)	

I.1.1 IRConcrFooting

Class Hierarchy



C++

```
interface IRConcrFooting : IDispatch;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public interface IRConcrFooting;
```

Visual Basic

```
Public Interface IRConcrFooting
```

Description

Interface representing the spread footing.

Version

Available since version 3.

I.1.1.1 IRConcrFooting Members

The following tables list the members exposed by IRConcrFooting.

Public Fields

	Name	Description
	CalculationOptions (see page 1108)	
	Concrete (see page 1108)	
	FootRotation (see page 1108)	
	Geometry (see page 1108)	Footing geometry.
	Ground (see page 1109)	Soils under the spread footing.
	IsActive (see page 1109)	
	IsSelected (see page 1109)	
	Loads (see page 1110)	Spread footing loads and backfill loads .
	Name (see page 1110)	
	NumberOfElements (see page 1110)	
	Pattern (see page 1110)	
	Reinforcement (see page 1111)	
	Results (see page 1111)	Analysis results for spread footing.
	Steel (see page 1111)	
	StructureUserNo (see page 1111)	
	StructureUserNoCount (see page 1112)	
	UniqueId (see page 1112)	

Public Methods

	Name	Description
	Activate (see page 1113)	
	Calculate (see page 1113)	
	CreateCalculationNoteRtf (see page 1113)	
	CreateFromNodes (see page 1113)	
	CreateReinforcement (see page 1114)	
	DisplayDrawing (see page 1114)	
	Save (see page 1114)	
	Verify (see page 1114)	

(c) 2014 Autodesk Inc. All rights reserved

I.1.1.2 IRConcrFooting Fields

The fields of the IRConcrFooting class are listed here.

Public Fields

	Name	Description
	CalculationOptions (see page 1108)	
	Concrete (see page 1108)	
	FootRotation (see page 1108)	
	Geometry (see page 1108)	Footing geometry.
	Ground (see page 1109)	Soils under the spread footing.
	IsActive (see page 1109)	
	IsSelected (see page 1109)	
	Loads (see page 1110)	Spread footing loads and backfill loads .
	Name (see page 1110)	
	NumberOfElements (see page 1110)	
	Pattern (see page 1110)	
	Reinforcement (see page 1111)	
	Results (see page 1111)	Analysis results for spread footing.
	Steel (see page 1111)	
	StructureUserNo (see page 1111)	
	StructureUserNoCount (see page 1112)	
	UniqueId (see page 1112)	

I.1.1.2.1 CalculationOptions

C++

```
HRESULT get_CalculationOptions(IRConcrFootingCalculationOptions**);
```

C#

```
public IRConcrFootingCalculationOptions CalculationOptions { get; }
```

Visual Basic

```
Public ReadOnly CalculationOptions As IRConcrFootingCalculationOptions
```

Version

Available since version 10.

I.1.1.2.2 Concrete

C++

```
HRESULT get_Concrete(IRConcrConcrete**);
```

C#

```
public IRConcrConcrete Concrete { get; }
```

Visual Basic

```
Public ReadOnly Concrete As IRConcrConcrete
```

Version

Available since version 10.

(c) 2014 Autodesk Inc. All rights reserved

I.1.1.2.3 FootRotation

C++

```
HRESULT get_FootRotation(double*);
HRESULT put_FootRotation(double);
```

C#

```
public double FootRotation { get; set; }
```

Visual Basic

```
Public FootRotation As double
```

Version

Available since version 11.

I.1.1.2.4 Geometry

C++

```
HRESULT get_Geometry(IRConcrFootingGeometry**);
```

C#

```
public IRConcrFootingGeometry Geometry { get; }
```

Visual Basic

```
Public ReadOnly Geometry As IRConcrFootingGeometry
```

Description

Footing geometry.

Version

Available since version 3.

I.1.1.2.5 Ground

C++

```
HRESULT get_Ground(IRConcrFootingGround**);
```

C#

```
public IRConcrFootingGround Ground { get; }
```

Visual Basic

```
Public ReadOnly Ground As IRConcrFootingGround
```

Description

Soils under the spread footing.

Version

Available since version 3.

I.1.1.2.6 IsActive

C++

```
HRESULT get_IsActive(VARIANT_BOOL*);
```

C#

```
public bool IsActive { get; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public ReadOnly IsActive As Boolean
```

Version

Available since version 10.

I.1.1.2.7 IsSelected

C++

```
HRESULT get_IsSelected(VARIANT_BOOL*);
HRESULT put_IsSelected(VARIANT_BOOL);
```

C#

```
public bool IsSelected { get; set; }
```

Visual Basic

```
Public IsSelected As Boolean
```

Version

Available since version 12.

I.1.1.2.8 Loads

C++

```
HRESULT get_Loads(IRConcrFootingLoads**);
```

C#

```
public IRConcrFootingLoads Loads { get; }
```

Visual Basic

```
Public ReadOnly Loads As IRConcrFootingLoads
```

Description

Spread footing loads and backfill loads .

Version

Available since version 3.

I.1.1.2.9 Name

C++

```
HRESULT get_Name(BSTR*);
```

C#

```
public String Name { get; }
```

Visual Basic

```
Public ReadOnly Name As String
```

Version

Available since version 10.

I.1.1.2.10 NumberOfElements

C++

```
HRESULT get_NumberOfElements( long*);
HRESULT put_NumberOfElements( long);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public  long NumberOfElements { get; set; }
```

Visual Basic

```
Public NumberOfElements As  long
```

Version

Available since version 11.

I.1.1.2.11 Pattern

C++

```
HRESULT get_Pattern(IRConcrFootingPattern**);
```

C#

```
public IRConcrFootingPattern Pattern { get; }
```

Visual Basic

```
Public ReadOnly Pattern As IRConcrFootingPattern
```

Version

Available since version 10.

I.1.1.2.12 Reinforcement

C++

```
HRESULT get_Reinforcement(IRConcrReinforcement**);
```

C#

```
public IRConcrReinforcement Reinforcement { get; }
```

Visual Basic

```
Public ReadOnly Reinforcement As IRConcrReinforcement
```

Version

Available since version 10.

I.1.1.2.13 Results

C++

```
HRESULT get_Results(IRConcrFootingResults**);
```

C#

```
public IRConcrFootingResults Results { get; }
```

Visual Basic

```
Public ReadOnly Results As IRConcrFootingResults
```

Description

Analysis results for spread footing.

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

I.1.1.2.14 Steel

C++

```
HRESULT get_Steel(IRConcrSteel**);
```

C#

```
public IRConcrSteel Steel { get; }
```

Visual Basic

```
Public ReadOnly Steel As IRConcrSteel
```

Version

Available since version 10.

I.1.1.2.15 StructureUserNo

C++

```
HRESULT get_StructureUserNo(long*);
HRESULT put_StructureUserNo(long);
```

C#

```
public long StructureUserNo { get; set; }
```

Visual Basic

```
Public StructureUserNo As long
```

Version

Available since version 10.

I.1.1.2.16 StructureUserNoCount

C++

```
HRESULT get_StructureUserNoCount(short*);
HRESULT put_StructureUserNoCount(short);
```

C#

```
public short StructureUserNoCount { get; set; }
```

Visual Basic

```
Public StructureUserNoCount As short
```

Version

Available since version 10.

I.1.1.2.17 UniqueId

C++

```
HRESULT get_UniqueId(long*);
```

C#

```
public long UniqueId { get; }
```

Visual Basic

```
Public ReadOnly UniqueId As long
```

Version

Available since version 10.

(c) 2014 Autodesk Inc. All rights reserved

I.1.1.3 IRConcrFooting Methods

The methods of the IRConcrFooting class are listed here.

Public Methods

	Name	Description
	Activate (see page 1113)	
	Calculate (see page 1113)	
	CreateCalculationNoteRtf (see page 1113)	
	CreateFromNodes (see page 1113)	
	CreateReinforcement (see page 1114)	
	DisplayDrawing (see page 1114)	
	Save (see page 1114)	
	Verify (see page 1114)	

I.1.1.3.1 Activate

C++

```
HRESULT Activate();
```

C#

```
public void Activate();
```

Visual Basic

```
Public Sub Activate()
```

Version

Available since version 10.

I.1.1.3.2 Calculate

C++

```
HRESULT Calculate(VARIANT_BOOL _createReinforcement, VARIANT_BOOL* ret);
```

C#

```
public bool Calculate(bool _createReinforcement);
```

Visual Basic

```
Public Function Calculate(_createReinforcement As Boolean) As Boolean
```

Version

Available since version 10.

I.1.1.3.3 CreateCalculationNoteRtf

C++

```
HRESULT CreateCalculationNoteRtf(BSTR _fileName, VARIANT_BOOL* ret);
```

C#

```
public bool CreateCalculationNoteRtf(String _fileName);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Function CreateCalculationNoteRtf(_fileName As String) As Boolean
```

Version

Available since version 10.

I.1.1.3.4 CreateFromNodes

C++

```
HRESULT CreateFromNodes(BSTR _selText, VARIANT_BOOL* ret);
```

C#

```
public bool CreateFromNodes(String _selText);
```

Visual Basic

```
Public Function CreateFromNodes(_selText As String) As Boolean
```

Version

Available since version 10.

I.1.1.3.5 CreateReinforcement

C++

```
HRESULT CreateReinforcement();
```

C#

```
public void CreateReinforcement();
```

Visual Basic

```
Public Sub CreateReinforcement()
```

Version

Available since version 10.

I.1.1.3.6 DisplayDrawing

C++

```
HRESULT DisplayDrawing();
```

C#

```
public void DisplayDrawing();
```

Visual Basic

```
Public Sub DisplayDrawing()
```

Version

Available since version 10.

I.1.1.3.7 Save

C++

```
HRESULT Save();
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public void Save();
```

Visual Basic

```
Public Sub Save()
```

Version

Available since version 10.

I.1.1.3.8 Verify

C++

```
HRESULT Verify(VARIANT_BOOL _showErrors, long* ret);
```

C#

```
public long Verify(bool _showErrors);
```

Visual Basic

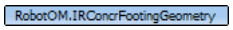
```
Public Function Verify(_showErrors As Boolean) As long
```

Version

Available since version 10.

I.1.2 IRConcrFootingGeometry

Class Hierarchy



C++

```
interface IRConcrFootingGeometry : IDispatch;
```

C#

```
public interface IRConcrFootingGeometry;
```

Visual Basic

```
Public Interface IRConcrFootingGeometry
```

Description

Interface representing the spread footing geometry.

Version

Available since version 3.

I.1.2.1 IRConcrFootingGeometry Members

The following tables list the members exposed by IRConcrFootingGeometry.

Public Fields

	Name	Description
	ConcreteVolume (see page 1116)	
	PierType (see page 1116)	
	Shape (see page 1116)	Shape of the spread footing section.
	ShutteringArea (see page 1116)	

(c) 2014 Autodesk Inc. All rights reserved

	Type (see page 1117)	Foundation type.

Public Methods

	Name	Description
	GetDim (see page 1117)	Function returns the value of a relevant spread footing dimension. .
	SetDim (see page 1117)	Function sets the value of a relevant spread footing dimension. .

I.1.2.2 IRConcrFootingGeometry Fields

The fields of the IRConcrFootingGeometry class are listed here.

Public Fields

	Name	Description
	ConcreteVolume (see page 1116)	
	PierType (see page 1116)	
	Shape (see page 1116)	Shape of the spread footing section.
	ShutteringArea (see page 1116)	
	Type (see page 1117)	Foundation type.

I.1.2.2.1 ConcreteVolume

C++

```
HRESULT get_ConcreteVolume(double*);
```

C#

```
public double ConcreteVolume { get; }
```

Visual Basic

```
Public ReadOnly ConcreteVolume As double
```

Version

Available since version 10.

I.1.2.2.2 PierType

C++

```
HRESULT get_PierType(IRConcrFootingPierType*);
HRESULT put_PierType(IRConcrFootingPierType);
```

C#

```
public IRConcrFootingPierType PierType { get; set; }
```

Visual Basic

```
Public PierType As IRConcrFootingPierType
```

Version

Available since version 11.

I.1.2.2.3 Shape

C++

```
HRESULT get_Shape(IRConcrFootingShapeType*);
HRESULT put_Shape(IRConcrFootingShapeType);
```

C#

```
public IRConcrFootingShapeType Shape { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Shape As IRConcrFootingShapeType
```

Description

Shape of the spread footing section.

Version

Available since version 3.

I.1.2.2.4 ShutteringArea

C++

```
HRESULT get_ShutteringArea(double*);
```

C#

```
public double ShutteringArea { get; }
```

Visual Basic

```
Public ReadOnly ShutteringArea As double
```

Version

Available since version 10.

I.1.2.2.5 Type

C++

```
HRESULT get_Type(IRConcrFootingType*);
HRESULT put_Type(IRConcrFootingType);
```

C#

```
public IRConcrFootingType Type { get; set; }
```

Visual Basic

```
Public Type As IRConcrFootingType
```

Description

Foundation type.

Version

Available since version 3.

I.1.2.3 IRConcrFootingGeometry Methods

The methods of the IRConcrFootingGeometry class are listed here.

Public Methods

	Name	Description
	GetDim (see page 1117)	Function returns the value of a relevant spread footing dimension. .
	SetDim (see page 1117)	Function sets the value of a relevant spread footing dimension. .

I.1.2.3.1 GetDim

C++

```
HRESULT GetDim(IRConcrFootingDimType _dimType, double* ret);
```

C#

```
public double GetDim(IRConcrFootingDimType _dimType);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Function GetDim(_dimType As IRConcrFootingDimType) As double
```

Description

Function returns the value of a relevant spread footing dimension. .

Version

Available since version 3.

I.1.2.3.2 SetDim

C++

```
HRESULT SetDim(IRConcrFootingDimType _dimType, double _val);
```

C#

```
public void SetDim(IRConcrFootingDimType _dimType, double _val);
```

Visual Basic

```
Public Sub SetDim(_dimType As IRConcrFootingDimType, _val As double)
```

Description

Function sets the value of a relevant spread footing dimension. .

Version

Available since version 3.

I.1.3 IRConcrFootingDimType

C++

```
enum IRConcrFootingDimType;
```

C#

```
public enum IRConcrFootingDimType;
```

Visual Basic

```
Public Enum IRConcrFootingDimType
```

Members

Members	Description
I_CFDT_FOOT_DIM_A = 1	Available since version 3.
I_CFDT_FOOT_DIM_B = 2	Available since version 3.
I_CFDT_FOOT_DIM_H1 = 3	Available since version 3.
I_CFDT_FOOT_DIM_H2 = 4	Available since version 3.
I_CFDT_FOOT_DIM_H3 = 5	Available since version 3.
I_CFDT_FOOT_DIM_H4 = 6	Available since version 3.
I_CFDT_FOOT_DIM_EX = 7	Available since version 3.
I_CFDT_FOOT_DIM_EY = 8	Available since version 3.
I_CFDT_FOOT_DIM_AP = 9	Available since version 3.
I_CFDT_FOOT_DIM_BP = 10	Available since version 3.
I_CFDT_FOOT_DIM_L = 11	Available since version 3.
I_CFDT_FOOT_DIM_COL_A = 12	Available since version 3.
I_CFDT_FOOT_DIM_COL_B = 13	Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

Description

A set of values identifying individual spread footing dimensions. .

Version

Available since version 3.

I.1.4 IRConcrFootingShapeType

C++

```
enum IRConcrFootingShapeType;
```

C#

```
public enum IRConcrFootingShapeType;
```

Visual Basic

```
Public Enum IRConcrFootingShapeType
```

Members

Members	Description
I_CFST_FOOT_SHAPE_RECT = 1	Spread footing of a rectangular section. Available since version 3.
I_CFST_FOOT_SHAPE_TRAPEZOIDAL = 2	Spread footing of a trapezoidal section. Available since version 3.
I_CFST_FOOT_SHAPE_TWO_CL = 3	Spread footing under two columns. Available since version 11.

Description

A set of values defining the spread footing shape. .

Version

Available since version 3.

I.1.5 IRConcrFootingType

C++

```
enum IRConcrFootingType;
```

C#

```
public enum IRConcrFootingType;
```

Visual Basic

```
Public Enum IRConcrFootingType
```

Members

Members	Description
I_CFT_FOOT_TYPE_FOOTING = 1	Column footing. Available since version 3.
I_CFT_FOOT_TYPE_WALL = 2	Continuous footing. Available since version 3.

Description

A set of values defining the foundation type. .

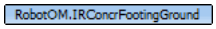
(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.

I.1.6 IRConcrFootingGround

Class Hierarchy



C++

```
interface IRConcrFootingGround : IDispatch;
```

C#

```
public interface IRConcrFootingGround;
```

Visual Basic

```
Public Interface IRConcrFootingGround
```

Description

Interface representing soils and reference level of the footing.

Version

Available since version 3.

I.1.6.1 IRConcrFootingGround Members

The following tables list the members exposed by IRConcrFootingGround.

Public Fields

	Name	Description
	ColumnPierLevel (see page 1120)	Column pier level.
	SoilLevel (see page 1120)	Reference level of the footing .

I.1.6.2 IRConcrFootingGround Fields

The fields of the IRConcrFootingGround class are listed here.

Public Fields

	Name	Description
	ColumnPierLevel (see page 1120)	Column pier level.
	SoilLevel (see page 1120)	Reference level of the footing .

I.1.6.2.1 ColumnPierLevel

C++

```
HRESULT get_ColumnPierLevel( double*);
HRESULT put_ColumnPierLevel( double);
```

C#

```
public  double ColumnPierLevel { get; set; }
```

Visual Basic

```
Public ColumnPierLevel As  double
```

Description

Column pier level.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.

I.1.6.2.2 SoilLevel

C++

```
HRESULT get_SoilLevel( double*);
HRESULT put_SoilLevel( double);
```

C#

```
public  double SoilLevel { get; set; }
```

Visual Basic

```
Public SoilLevel As  double
```

Description

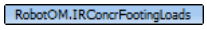
Reference level of the footing .

Version

Available since version 3.

I.1.7 IRConcrFootingLoads

Class Hierarchy



C++

```
interface IRConcrFootingLoads : IDispatch;
```

C#

```
public interface IRConcrFootingLoads;
```

Visual Basic

```
Public Interface IRConcrFootingLoads
```

Description

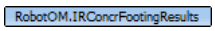
Interface representing spread footing loads and backfill loads. .

Version

Available since version 3.

I.1.8 IRConcrFootingResults

Class Hierarchy



C++

```
interface IRConcrFootingResults : IDispatch;
```

C#

```
public interface IRConcrFootingResults;
```

Visual Basic

```
Public Interface IRConcrFootingResults
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Interface representing calculation results for the spread footing. .

Version

Available since version 3.

I.1.8.1 IRConcrFootingResults Members

The following tables list the members exposed by IRConcrFootingResults.

Public Fields

	Name	Description
	Ax (see page 1122)	Required reinforcement area (X direction).
	Ay (see page 1122)	Required reinforcement area (Y direction).
	Val (see page 1122)	

I.1.8.2 IRConcrFootingResults Fields

The fields of the IRConcrFootingResults class are listed here.

Public Fields

	Name	Description
	Ax (see page 1122)	Required reinforcement area (X direction).
	Ay (see page 1122)	Required reinforcement area (Y direction).
	Val (see page 1122)	

I.1.8.2.1 Ax

C++

```
HRESULT get_Ax( double*);
HRESULT put_Ax( double);
```

C#

```
public  double Ax { get; set; }
```

Visual Basic

```
Public Ax As  double
```

Description

Required reinforcement area (X direction).

Version

Available since version 3.

I.1.8.2.2 Ay

C++

```
HRESULT get_Ay( double*);
HRESULT put_Ay( double);
```

C#

```
public  double Ay { get; set; }
```

Visual Basic

```
Public Ay As  double
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Required reinforcement area (Y direction).

Version

Available since version 3.

I.1.8.2.3 Val

C++

```
HRESULT get_Val(double*);
HRESULT put_Val(double);
```

C#

```
public double Val { get; set; }
```

Visual Basic

```
Public Val As double
```

Version

Available since version 10.

I.1.9 IRConcrFootingCalculationOptions

Class Hierarchy

RobotOM.IRConcrFootingCalculationOptions

C++

```
interface IRConcrFootingCalculationOptions : IDispatch;
```

C#

```
public interface IRConcrFootingCalculationOptions;
```

Visual Basic

```
Public Interface IRConcrFootingCalculationOptions
```

Version

Available since version 10.

I.1.9.1 IRConcrFootingCalculationOptions Members

The following tables list the members exposed by IRConcrFootingCalculationOptions.

Public Fields

	Name	Description
	Cover (see page 1124)	
	NominalCover (see page 1124)	
	NominalPierCover (see page 1124)	
	PierCover (see page 1124)	

I.1.9.2 IRConcrFootingCalculationOptions Fields

The fields of the IRConcrFootingCalculationOptions class are listed here.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	Cover (see page 1124)	
	NominalCover (see page 1124)	
	NominalPierCover (see page 1124)	
	PierCover (see page 1124)	

I.1.9.2.1 Cover

C++

```
HRESULT get_Cover(double*);
HRESULT put_Cover(double);
```

C#

```
public double Cover { get; set; }
```

Visual Basic

```
Public Cover As double
```

Version

Available since version 10.

I.1.9.2.2 NominalCover

C++

```
HRESULT get_NominalCover(double*);
```

C#

```
public double NominalCover { get; }
```

Visual Basic

```
Public ReadOnly NominalCover As double
```

Version

Available since version 11.

I.1.9.2.3 NominalPierCover

C++

```
HRESULT get_NominalPierCover( double*);
```

C#

```
public  double NominalPierCover { get; }
```

Visual Basic

```
Public ReadOnly NominalPierCover As  double
```

Version

Available since version 11.

I.1.9.2.4 PierCover

C++

```
HRESULT get_PierCover(double*);
HRESULT put_PierCover(double);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double PierCover { get; set; }
```

Visual Basic

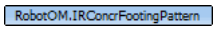
```
Public PierCover As double
```

Version

Available since version 11.

I.1.10 IRConcrFootingPattern

Class Hierarchy



C++

```
interface IRConcrFootingPattern : IDispatch;
```

C#

```
public interface IRConcrFootingPattern;
```

Visual Basic

```
Public Interface IRConcrFootingPattern
```

Version

Available since version 10.

I.1.10.1 IRConcrFootingPattern Members

The following tables list the members exposed by IRConcrFootingPattern.

Public Fields

	Name	Description
	Diameter1 (see page 1126)	
	Diameter2 (see page 1126)	
	Spacing1Max (see page 1126)	
	Spacing1Min (see page 1126)	
	Spacing2Max (see page 1127)	
	Spacing2Min (see page 1127)	

I.1.10.2 IRConcrFootingPattern Fields

The fields of the IRConcrFootingPattern class are listed here.

Public Fields

	Name	Description
	Diameter1 (see page 1126)	
	Diameter2 (see page 1126)	
	Spacing1Max (see page 1126)	
	Spacing1Min (see page 1126)	
	Spacing2Max (see page 1127)	
	Spacing2Min (see page 1127)	

(c) 2014 Autodesk Inc. All rights reserved

I.1.10.2.1 Diameter1

C++

```
HRESULT get_Diameter1(double*);
HRESULT put_Diameter1(double);
```

C#

```
public double Diameter1 { get; set; }
```

Visual Basic

```
Public Diameter1 As double
```

Version

Available since version 10.

I.1.10.2.2 Diameter2

C++

```
HRESULT get_Diameter2(double*);
HRESULT put_Diameter2(double);
```

C#

```
public double Diameter2 { get; set; }
```

Visual Basic

```
Public Diameter2 As double
```

Version

Available since version 10.

I.1.10.2.3 Spacing1Max

C++

```
HRESULT get_Spacing1Max(double*);
HRESULT put_Spacing1Max(double);
```

C#

```
public double Spacing1Max { get; set; }
```

Visual Basic

```
Public Spacing1Max As double
```

Version

Available since version 10.

I.1.10.2.4 Spacing1Min

C++

```
HRESULT get_Spacing1Min(double*);
HRESULT put_Spacing1Min(double);
```

C#

```
public double Spacing1Min { get; set; }
```

Visual Basic

```
Public Spacing1Min As double
```

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 10.

I.1.10.2.5 Spacing2Max

C++

```
HRESULT get_Spacing2Max(double*);
HRESULT put_Spacing2Max(double);
```

C#

```
public double Spacing2Max { get; set; }
```

Visual Basic

```
Public Spacing2Max As double
```

Version

Available since version 10.

I.1.10.2.6 Spacing2Min

C++

```
HRESULT get_Spacing2Min(double*);
HRESULT put_Spacing2Min(double);
```

C#

```
public double Spacing2Min { get; set; }
```

Visual Basic

```
Public Spacing2Min As double
```

Version

Available since version 10.

I.1.11 IRConcrFootResultType

C++

```
enum IRConcrFootResultType;
```

C#

```
public enum IRConcrFootResultType;
```

Visual Basic

```
Public Enum IRConcrFootResultType
```

Members

Members	Description
I_FRT_FOOT_RES_AX = 1	Available since version 10.
I_FRT_FOOT_RES_AY = 2	Available since version 10.

Version

Available since version 10.

(c) 2014 Autodesk Inc. All rights reserved

I.1.12 IRConcrFootingPierType

C++

```
enum IRConcrFootingPierType;
```

C#

```
public enum IRConcrFootingPierType;
```

Visual Basic

```
Public Enum IRConcrFootingPierType
```

Members

Members	Description
I_CFT_FOOT_PIER_TYPE_PLAIN = 1	Available since version 11.
I_CFT_FOOT_PIER_TYPE_DOWELED = 2	Available since version 11.
I_CFT_FOOT_PIER_TYPE_BOLTED = 3	Available since version 11.
I_CFT_FOOT_PIER_TYPE_HINGE_1 = 4	Available since version 11.
I_CFT_FOOT_PIER_TYPE_HINGE_2 = 5	Available since version 11.
I_CFT_FOOT_PIER_TYPE_SOCKETED = 6	Available since version 11.

Version

Available since version 11.

I.2 RC Beam

Available since version 10.

Enumerations

	Name	Description
	IRConcrBeamLinearLoadType (see page 1164)	
	IRConcrBeamLoadNatureType (see page 1165)	
	IRConcrBeamPointLoadType (see page 1165)	
	IRConcrBeamSectionType (see page 1165)	
	IRConcrBeamSectionDimType (see page 1166)	
	IRConcrBeamCrackingType (see page 1167)	
	IRConcrBeamSupportType (see page 1167)	
	IRConcrBeamSupportMaterialType (see page 1168)	

Interfaces

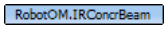
	Name	Description
	IRConcrBeam (see page 1129)	
	IRConcrBeamCalcOptions (see page 1138)	
	IRConcrBeamGeometry (see page 1141)	

(c) 2014 Autodesk Inc. All rights reserved

	IRConcrBeamImportOptions (see page 1145)	
	IRConcrBeamLinearLoad (see page 1147)	
	IRConcrBeamPatternOptions (see page 1152)	
	IRConcrBeamPointLoad (see page 1153)	
	IRConcrBeamSegment (see page 1155)	
	IRConcrBeamSpan (see page 1156)	
	IRConcrBeamSpanNumbers (see page 1159)	
	IRConcrBeamStoryOptions (see page 1161)	
	IRConcrBeamSupport (see page 1162)	

I.2.1 IRConcrBeam

Class Hierarchy



C++

```
interface IRConcrBeam : IDispatch;
```

C#

```
public interface IRConcrBeam;
```

Visual Basic

```
Public Interface IRConcrBeam
```

Version

Available since version 10.

I.2.1.1 IRConcrBeam Members

The following tables list the members exposed by IRConcrBeam.

Public Fields

	Name	Description
	CalculationOptions (see page 1131)	
	Concrete (see page 1131)	
	Geometry (see page 1131)	
	ImportOptions (see page 1132)	
	IsActive (see page 1132)	
	IsSelected (see page 1132)	
	LinearLoad (see page 1132)	
	LinearLoadsCount (see page 1133)	
	Name (see page 1133)	

(c) 2014 Autodesk Inc. All rights reserved

	NumberOfElements (see page 1133)	
	PatternOptions (see page 1133)	
	PointLoad (see page 1134)	
	PointLoadsCount (see page 1134)	
	Reinforcement (see page 1134)	
	Steel (see page 1134)	
	StoryOptions (see page 1135)	
	StructureUserNo (see page 1135)	
	StructureUserNoCount (see page 1135)	
	UniqueId (see page 1135)	

Public Methods

	Name	Description
	Activate (see page 1136)	
	Calculate (see page 1136)	
	CreateCalculationNoteRtf (see page 1137)	
	CreateFromBars (see page 1137)	
	GenerateSpliceBars (see page 1137)	
	Save (see page 1137)	
	Verify (see page 1138)	
	VerifySpan (see page 1138)	

I.2.1.2 IRConcrBeam Fields

The fields of the IRConcrBeam class are listed here.

Public Fields

	Name	Description
	CalculationOptions (see page 1131)	
	Concrete (see page 1131)	
	Geometry (see page 1131)	
	ImportOptions (see page 1132)	
	IsActive (see page 1132)	
	IsSelected (see page 1132)	
	LinearLoad (see page 1132)	
	LinearLoadsCount (see page 1133)	
	Name (see page 1133)	
	NumberOfElements (see page 1133)	
	PatternOptions (see page 1133)	
	PointLoad (see page 1134)	
	PointLoadsCount (see page 1134)	
	Reinforcement (see page 1134)	
	Steel (see page 1134)	
	StoryOptions (see page 1135)	

(c) 2014 Autodesk Inc. All rights reserved

	StructureUserNo (see page 1135)	
	StructureUserNoCount (see page 1135)	
	UniqueId (see page 1135)	

I.2.1.2.1 CalculationOptions

C++

```
HRESULT get_CalculationOptions(IRConcrBeamCalcOptions**);
```

C#

```
public IRConcrBeamCalcOptions CalculationOptions { get; }
```

Visual Basic

```
Public ReadOnly CalculationOptions As IRConcrBeamCalcOptions
```

Version

Available since version 10.

I.2.1.2.2 Concrete

C++

```
HRESULT get_Concrete(IRConcrConcrete**);
```

C#

```
public IRConcrConcrete Concrete { get; }
```

Visual Basic

```
Public ReadOnly Concrete As IRConcrConcrete
```

Version

Available since version 10.

I.2.1.2.3 Geometry

C++

```
HRESULT get_Geometry(IRConcrBeamGeometry**);
```

C#

```
public IRConcrBeamGeometry Geometry { get; }
```

Visual Basic

```
Public ReadOnly Geometry As IRConcrBeamGeometry
```

Version

Available since version 10.

I.2.1.2.4 ImportOptions

C++

```
HRESULT get_ImportOptions(IRConcrBeamImportOptions**);
```

C#

```
public IRConcrBeamImportOptions ImportOptions { get; }
```

Visual Basic

```
Public ReadOnly ImportOptions As IRConcrBeamImportOptions
```

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 10.

I.2.1.2.5 IsActive

C++

```
HRESULT get_IsActive(VARIANT_BOOL*);
```

C#

```
public bool IsActive { get; }
```

Visual Basic

```
Public ReadOnly IsActive As Boolean
```

Version

Available since version 10.

I.2.1.2.6 IsSelected

C++

```
HRESULT get_IsSelected(VARIANT_BOOL*);
HRESULT put_IsSelected(VARIANT_BOOL);
```

C#

```
public bool IsSelected { get; set; }
```

Visual Basic

```
Public IsSelected As Boolean
```

Version

Available since version 12.

I.2.1.2.7 LinearLoad

C++

```
HRESULT get_LinearLoad(IRConcrBeamLinearLoad**);
```

C#

```
public IRConcrBeamLinearLoad LinearLoad { get; }
```

Visual Basic

```
Public ReadOnly LinearLoad As IRConcrBeamLinearLoad
```

Version

Available since version 10.

I.2.1.2.8 LinearLoadsCount

C++

```
HRESULT get_LinearLoadsCount(short*);
HRESULT put_LinearLoadsCount(short);
```

C#

```
public short LinearLoadsCount { get; set; }
```

Visual Basic

```
Public LinearLoadsCount As short
```

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 10.

I.2.1.2.9 Name

C++

```
HRESULT get_Name(BSTR*);
```

C#

```
public String Name { get; }
```

Visual Basic

```
Public ReadOnly Name As String
```

Version

Available since version 10.

I.2.1.2.10 NumberOfElements

C++

```
HRESULT get_NumberOfElements(long*);
HRESULT put_NumberOfElements(long);
```

C#

```
public long NumberOfElements { get; set; }
```

Visual Basic

```
Public NumberOfElements As long
```

Version

Available since version 12.

I.2.1.2.11 PatternOptions

C++

```
HRESULT get_PatternOptions(IRConcrBeamPatternOptions**);
```

C#

```
public IRConcrBeamPatternOptions PatternOptions { get; }
```

Visual Basic

```
Public ReadOnly PatternOptions As IRConcrBeamPatternOptions
```

Version

Available since version 10.

I.2.1.2.12 PointLoad

C++

```
HRESULT get_PointLoad(IRConcrBeamPointLoad**);
```

C#

```
public IRConcrBeamPointLoad PointLoad { get; }
```

Visual Basic

```
Public ReadOnly PointLoad As IRConcrBeamPointLoad
```

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 10.

I.2.1.2.13 PointLoadsCount

C++

```
HRESULT get_PointLoadsCount(short*);
HRESULT put_PointLoadsCount(short);
```

C#

```
public short PointLoadsCount { get; set; }
```

Visual Basic

```
Public PointLoadsCount As short
```

Version

Available since version 10.

I.2.1.2.14 Reinforcement

C++

```
HRESULT get_Reinforcement(IRConcrReinforcement**);
```

C#

```
public IRConcrReinforcement Reinforcement { get; }
```

Visual Basic

```
Public ReadOnly Reinforcement As IRConcrReinforcement
```

Version

Available since version 10.

I.2.1.2.15 Steel

C++

```
HRESULT get_Steel(IRConcrSteel**);
```

C#

```
public IRConcrSteel Steel { get; }
```

Visual Basic

```
Public ReadOnly Steel As IRConcrSteel
```

Version

Available since version 10.

I.2.1.2.16 StoryOptions

C++

```
HRESULT get_StoryOptions(IRConcrBeamStoryOptions**);
```

C#

```
public IRConcrBeamStoryOptions StoryOptions { get; }
```

Visual Basic

```
Public ReadOnly StoryOptions As IRConcrBeamStoryOptions
```

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 10.

I.2.1.2.17 StructureUserNo

C++

```
HRESULT get_StructureUserNo(long*);
HRESULT put_StructureUserNo(long);
```

C#

```
public long StructureUserNo { get; set; }
```

Visual Basic

```
Public StructureUserNo As long
```

Version

Available since version 10.

I.2.1.2.18 StructureUserNoCount

C++

```
HRESULT get_StructureUserNoCount(short*);
HRESULT put_StructureUserNoCount(short);
```

C#

```
public short StructureUserNoCount { get; set; }
```

Visual Basic

```
Public StructureUserNoCount As short
```

Version

Available since version 10.

I.2.1.2.19 UniqueId

C++

```
HRESULT get_UniqueId(long*);
```

C#

```
public long UniqueId { get; }
```

Visual Basic

```
Public ReadOnly UniqueId As long
```

Version

Available since version 10.

I.2.1.3 IRConcrBeam Methods

The methods of the IRConcrBeam class are listed here.

Public Methods

	Name	Description
	Activate (see page 1136)	
	Calculate (see page 1136)	

(c) 2014 Autodesk Inc. All rights reserved

	CreateCalculationNoteRtf (see page 1137)	
	CreateFromBars (see page 1137)	
	GenerateSpliceBars (see page 1137)	
	Save (see page 1137)	
	Verify (see page 1138)	
	VerifySpan (see page 1138)	

I.2.1.3.1 Activate

C++

```
HRESULT Activate();
```

C#

```
public void Activate();
```

Visual Basic

```
Public Sub Activate()
```

Version

Available since version 10.

I.2.1.3.2 Calculate

C++

```
HRESULT Calculate(VARIANT_BOOL _createReinforcement, VARIANT_BOOL* ret);
```

C#

```
public bool Calculate(bool _createReinforcement);
```

Visual Basic

```
Public Function Calculate(_createReinforcement As Boolean) As Boolean
```

Version

Available since version 10.

I.2.1.3.3 CreateCalculationNoteRtf

C++

```
HRESULT CreateCalculationNoteRtf(BSTR _fileName, VARIANT_BOOL* ret);
```

C#

```
public bool CreateCalculationNoteRtf(String _fileName);
```

Visual Basic

```
Public Function CreateCalculationNoteRtf(_fileName As String) As Boolean
```

Version

Available since version 10.

(c) 2014 Autodesk Inc. All rights reserved

I.2.1.3.4 CreateFromBars

C++

```
HRESULT CreateFromBars(VARIANT_BOOL* ret);
```

C#

```
public bool CreateFromBars();
```

Visual Basic

```
Public Function CreateFromBars() As Boolean
```

Version

Available since version 10.

I.2.1.3.5 GenerateSpliceBars

C++

```
HRESULT GenerateSpliceBars(VARIANT_BOOL* ret);
```

C#

```
public bool GenerateSpliceBars();
```

Visual Basic

```
Public Function GenerateSpliceBars() As Boolean
```

Version

Available since version 10.

I.2.1.3.6 Save

C++

```
HRESULT Save();
```

C#

```
public void Save();
```

Visual Basic

```
Public Sub Save()
```

Version

Available since version 10.

I.2.1.3.7 Verify

C++

```
HRESULT Verify(VARIANT_BOOL _showErrors, VARIANT_BOOL* ret);
```

C#

```
public bool Verify(bool _showErrors);
```

Visual Basic

```
Public Function Verify(_showErrors As Boolean) As Boolean
```

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 10.

I.2.1.3.8 VerifySpan

C++

```
HRESULT VerifySpan(VARIANT_BOOL _showErrors, short _spanNo, long* ret);
```

C#

```
public long VerifySpan(bool _showErrors, short _spanNo);
```

Visual Basic

```
Public Function VerifySpan(_showErrors As Boolean, _spanNo As short) As long
```

Version

Available since version 10.

I.2.2 IRConcrBeamCalcOptions

Class Hierarchy

RobotOM.IRConcrBeamCalcOptions

C++

```
interface IRConcrBeamCalcOptions : IDispatch;
```

C#

```
public interface IRConcrBeamCalcOptions;
```

Visual Basic

```
Public Interface IRConcrBeamCalcOptions
```

Version

Available since version 10.

I.2.2.1 IRConcrBeamCalcOptions Members

The following tables list the members exposed by IRConcrBeamCalcOptions.

Public Fields

	Name	Description
	CalcSpanLengthInAxis (see page 1139)	
	CoverBottom (see page 1140)	
	CoverBottomFixed (see page 1140)	
	CoverSide (see page 1140)	
	CoverSideFixed (see page 1140)	
	CoverTop (see page 1141)	
	CoverTopFixed (see page 1141)	

I.2.2.2 IRConcrBeamCalcOptions Fields

The fields of the IRConcrBeamCalcOptions class are listed here.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	CalcSpanLengthInAxis (see page 1139)	
	CoverBottom (see page 1140)	
	CoverBottomFixed (see page 1140)	
	CoverSide (see page 1140)	
	CoverSideFixed (see page 1140)	
	CoverTop (see page 1141)	
	CoverTopFixed (see page 1141)	

I.2.2.2.1 CalcSpanLengthInAxis

C++

```
HRESULT get_CalcSpanLengthInAxis(VARIANT_BOOL*);
HRESULT put_CalcSpanLengthInAxis(VARIANT_BOOL);
```

C#

```
public bool CalcSpanLengthInAxis { get; set; }
```

Visual Basic

```
Public CalcSpanLengthInAxis As Boolean
```

Version

Available since version 10.

I.2.2.2.2 CoverBottom

C++

```
HRESULT get_CoverBottom(double*);
HRESULT put_CoverBottom(double);
```

C#

```
public double CoverBottom { get; set; }
```

Visual Basic

```
Public CoverBottom As double
```

Version

Available since version 10.

I.2.2.2.3 CoverBottomFixed

C++

```
HRESULT get_CoverBottomFixed(VARIANT_BOOL*);
HRESULT put_CoverBottomFixed(VARIANT_BOOL);
```

C#

```
public bool CoverBottomFixed { get; set; }
```

Visual Basic

```
Public CoverBottomFixed As Boolean
```

Version

Available since version 10.

(c) 2014 Autodesk Inc. All rights reserved

I.2.2.2.4 CoverSide

C++

```
HRESULT get_CoverSide(double*);
HRESULT put_CoverSide(double);
```

C#

```
public double CoverSide { get; set; }
```

Visual Basic

```
Public CoverSide As double
```

Version

Available since version 10.

I.2.2.2.5 CoverSideFixed

C++

```
HRESULT get_CoverSideFixed(VARIANT_BOOL*);
HRESULT put_CoverSideFixed(VARIANT_BOOL);
```

C#

```
public bool CoverSideFixed { get; set; }
```

Visual Basic

```
Public CoverSideFixed As Boolean
```

Version

Available since version 10.

I.2.2.2.6 CoverTop

C++

```
HRESULT get_CoverTop(double*);
HRESULT put_CoverTop(double);
```

C#

```
public double CoverTop { get; set; }
```

Visual Basic

```
Public CoverTop As double
```

Version

Available since version 10.

I.2.2.2.7 CoverTopFixed

C++

```
HRESULT get_CoverTopFixed(VARIANT_BOOL*);
HRESULT put_CoverTopFixed(VARIANT_BOOL);
```

C#

```
public bool CoverTopFixed { get; set; }
```

Visual Basic

```
Public CoverTopFixed As Boolean
```

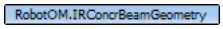
(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 10.

I.2.3 IRConcrBeamGeometry

Class Hierarchy



C++

```
interface IRConcrBeamGeometry : IDispatch;
```

C#

```
public interface IRConcrBeamGeometry;
```

Visual Basic

```
Public Interface IRConcrBeamGeometry
```

Version

Available since version 10.

I.2.3.1 IRConcrBeamGeometry Members

The following tables list the members exposed by IRConcrBeamGeometry.

Public Fields

	Name	Description
	AutoNameSpans (see page 1142)	
	AutoNameSupports (see page 1143)	
	ConcreteVolume (see page 1143)	
	LeftCantilever (see page 1143)	
	RightCantilever (see page 1143)	
	ShutteringArea (see page 1144)	
	Span (see page 1144)	
	SpanNumber (see page 1144)	

Public Methods

	Name	Description
	UpdateInternalGeometryData (see page 1145)	

I.2.3.2 IRConcrBeamGeometry Fields

The fields of the IRConcrBeamGeometry class are listed here.

Public Fields

	Name	Description
	AutoNameSpans (see page 1142)	
	AutoNameSupports (see page 1143)	
	ConcreteVolume (see page 1143)	
	LeftCantilever (see page 1143)	
	RightCantilever (see page 1143)	

(c) 2014 Autodesk Inc. All rights reserved

	ShutteringArea (see page 1144)	
	Span (see page 1144)	
	SpanNumber (see page 1144)	

I.2.3.2.1 AutoNameSpans

C++

```
HRESULT get_AutoNameSpans(VARIANT_BOOL*);
HRESULT put_AutoNameSpans(VARIANT_BOOL);
```

C#

```
public bool AutoNameSpans { get; set; }
```

Visual Basic

```
Public AutoNameSpans As Boolean
```

Version

Available since version 10.

I.2.3.2.2 AutoNameSupports

C++

```
HRESULT get_AutoNameSupports(VARIANT_BOOL*);
HRESULT put_AutoNameSupports(VARIANT_BOOL);
```

C#

```
public bool AutoNameSupports { get; set; }
```

Visual Basic

```
Public AutoNameSupports As Boolean
```

Version

Available since version 10.

I.2.3.2.3 ConcreteVolume

C++

```
HRESULT get_ConcreteVolume(double*);
```

C#

```
public double ConcreteVolume { get; }
```

Visual Basic

```
Public ReadOnly ConcreteVolume As double
```

Version

Available since version 10.

I.2.3.2.4 LeftCantilever

C++

```
HRESULT get_LeftCantilever(VARIANT_BOOL*);
HRESULT put_LeftCantilever(VARIANT_BOOL);
```

C#

```
public bool LeftCantilever { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public LeftCantilever As Boolean
```

Version

Available since version 10.

I.2.3.2.5 RightCantilever

C++

```
HRESULT get_RightCantilever(VARIANT_BOOL*);
HRESULT put_RightCantilever(VARIANT_BOOL);
```

C#

```
public bool RightCantilever { get; set; }
```

Visual Basic

```
Public RightCantilever As Boolean
```

Version

Available since version 10.

I.2.3.2.6 ShutteringArea

C++

```
HRESULT get_ShutteringArea(double*);
```

C#

```
public double ShutteringArea { get; }
```

Visual Basic

```
Public ReadOnly ShutteringArea As double
```

Version

Available since version 10.

I.2.3.2.7 Span

C++

```
HRESULT get_Span(IRConcrBeamSpan**);
```

C#

```
public IRConcrBeamSpan Span { get; }
```

Visual Basic

```
Public ReadOnly Span As IRConcrBeamSpan
```

Version

Available since version 10.

I.2.3.2.8 SpanNumber

C++

```
HRESULT get_SpanNumber(short*);
HRESULT put_SpanNumber(short);
```

C#

```
public short SpanNumber { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public SpanNumber As short
```

Version

Available since version 10.

I.2.3.3 IRConcrBeamGeometry Methods

The methods of the IRConcrBeamGeometry class are listed here.

Public Methods

	Name	Description
	UpdateInternalGeometryData (see page 1145)	

I.2.3.3.1 UpdateInternalGeometryData

C++

```
HRESULT UpdateInternalGeometryData();
```

C#

```
public void UpdateInternalGeometryData();
```

Visual Basic

```
Public Sub UpdateInternalGeometryData()
```

Version

Available since version 10.

I.2.4 IRConcrBeamImportOptions

Class Hierarchy

C++

```
interface IRConcrBeamImportOptions : IDispatch;
```

C#

```
public interface IRConcrBeamImportOptions;
```

Visual Basic

```
Public Interface IRConcrBeamImportOptions
```

Version

Available since version 10.

I.2.4.1 IRConcrBeamImportOptions Members

The following tables list the members exposed by IRConcrBeamImportOptions.

Public Fields

	Name	Description
	GroupByGeometry (see page 1146)	
	GroupByLevel (see page 1146)	

(c) 2014 Autodesk Inc. All rights reserved

	ImportCombinations (see page 1146)	
	LiveLongCoeff (see page 1147)	
	RunCalcAuto (see page 1147)	
	ShowDialog (see page 1147)	

I.2.4.2 IRConcrBeamImportOptions Fields

The fields of the IRConcrBeamImportOptions class are listed here.

Public Fields

	Name	Description
	GroupByGeometry (see page 1146)	
	GroupByLevel (see page 1146)	
	ImportCombinations (see page 1146)	
	LiveLongCoeff (see page 1147)	
	RunCalcAuto (see page 1147)	
	ShowDialog (see page 1147)	

I.2.4.2.1 GroupByGeometry

C++

```
HRESULT get_GroupByGeometry(VARIANT_BOOL*);
HRESULT put_GroupByGeometry(VARIANT_BOOL);
```

C#

```
public bool GroupByGeometry { get; set; }
```

Visual Basic

```
Public GroupByGeometry As Boolean
```

Version

Available since version 10.

I.2.4.2.2 GroupByLevel

C++

```
HRESULT get_GroupByLevel(VARIANT_BOOL*);
HRESULT put_GroupByLevel(VARIANT_BOOL);
```

C#

```
public bool GroupByLevel { get; set; }
```

Visual Basic

```
Public GroupByLevel As Boolean
```

Version

Available since version 10.

I.2.4.2.3 ImportCombinations

C++

```
HRESULT get_ImportCombinations(VARIANT_BOOL*);
HRESULT put_ImportCombinations(VARIANT_BOOL);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public bool ImportCombinations { get; set; }
```

Visual Basic

```
Public ImportCombinations As Boolean
```

Version

Available since version 10.

I.2.4.2.4 LiveLongCoeff

C++

```
HRESULT get_LiveLongCoeff(double*);
HRESULT put_LiveLongCoeff(double);
```

C#

```
public double LiveLongCoeff { get; set; }
```

Visual Basic

```
Public LiveLongCoeff As double
```

Version

Available since version 10.

I.2.4.2.5 RunCalcAuto

C++

```
HRESULT get_RunCalcAuto(VARIANT_BOOL*);
HRESULT put_RunCalcAuto(VARIANT_BOOL);
```

C#

```
public bool RunCalcAuto { get; set; }
```

Visual Basic

```
Public RunCalcAuto As Boolean
```

Version

Available since version 10.

I.2.4.2.6 ShowDialog

C++

```
HRESULT get_ShowDialog(VARIANT_BOOL*);
HRESULT put_ShowDialog(VARIANT_BOOL);
```

C#

```
public bool ShowDialog { get; set; }
```

Visual Basic

```
Public ShowDialog As Boolean
```

Version

Available since version 10.

(c) 2014 Autodesk Inc. All rights reserved

I.2.5 IRConcrBeamLinearLoad

Class Hierarchy

RobotOM.IRConcrBeamLinearLoad

C++

```
interface IRConcrBeamLinearLoad : IDispatch;
```

C#

```
public interface IRConcrBeamLinearLoad;
```

Visual Basic

```
Public Interface IRConcrBeamLinearLoad
```

Version

Available since version 10.

I.2.5.1 IRConcrBeamLinearLoad Members

The following tables list the members exposed by IRConcrBeamLinearLoad.

Public Fields

	Name	Description
	Case (see page 1148)	
	Nature (see page 1149)	
	RelativeCoordinates (see page 1149)	
	Spans (see page 1149)	
	Type (see page 1150)	
	Value1 (see page 1150)	
	Value2 (see page 1150)	
	Value3 (see page 1150)	
	X1 (see page 1151)	
	X2 (see page 1151)	
	X3 (see page 1151)	
	X4 (see page 1151)	

I.2.5.2 IRConcrBeamLinearLoad Fields

The fields of the IRConcrBeamLinearLoad class are listed here.

Public Fields

	Name	Description
	Case (see page 1148)	
	Nature (see page 1149)	
	RelativeCoordinates (see page 1149)	
	Spans (see page 1149)	
	Type (see page 1150)	
	Value1 (see page 1150)	
	Value2 (see page 1150)	
	Value3 (see page 1150)	

(c) 2014 Autodesk Inc. All rights reserved

	X1 (see page 1151)	
	X2 (see page 1151)	
	X3 (see page 1151)	
	X4 (see page 1151)	

I.2.5.2.1 Case

C++

```
HRESULT get_Case(short*);
HRESULT put_Case(short);
```

C#

```
public short Case { get; set; }
```

Visual Basic

```
Public Case As short
```

Version

Available since version 10.

I.2.5.2.2 Nature

C++

```
HRESULT get_Nature(IRConcrBeamLoadNatureType*);
HRESULT put_Nature(IRConcrBeamLoadNatureType);
```

C#

```
public IRConcrBeamLoadNatureType Nature { get; set; }
```

Visual Basic

```
Public Nature As IRConcrBeamLoadNatureType
```

Version

Available since version 10.

I.2.5.2.3 RelativeCoordinates

C++

```
HRESULT get_RelativeCoordinates(VARIANT_BOOL*);
HRESULT put_RelativeCoordinates(VARIANT_BOOL);
```

C#

```
public bool RelativeCoordinates { get; set; }
```

Visual Basic

```
Public RelativeCoordinates As Boolean
```

Version

Available since version 10.

I.2.5.2.4 Spans

C++

```
HRESULT get_Spans(IRConcrBeamSpanNumbers**);
```

C#

```
public IRConcrBeamSpanNumbers Spans { get; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public ReadOnly Spans As IRConcrBeamSpanNumbers
```

Version

Available since version 10.

I.2.5.2.5 Type

C++

```
HRESULT get_Type(IRConcrBeamLinearLoadType*);
HRESULT put_Type(IRConcrBeamLinearLoadType);
```

C#

```
public IRConcrBeamLinearLoadType Type { get; set; }
```

Visual Basic

```
Public Type As IRConcrBeamLinearLoadType
```

Version

Available since version 10.

I.2.5.2.6 Value1

C++

```
HRESULT get_Value1(double*);
HRESULT put_Value1(double);
```

C#

```
public double Value1 { get; set; }
```

Visual Basic

```
Public Value1 As double
```

Version

Available since version 10.

I.2.5.2.7 Value2

C++

```
HRESULT get_Value2(double*);
HRESULT put_Value2(double);
```

C#

```
public double Value2 { get; set; }
```

Visual Basic

```
Public Value2 As double
```

Version

Available since version 10.

I.2.5.2.8 Value3

C++

```
HRESULT get_Value3(double*);
HRESULT put_Value3(double);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double Value3 { get; set; }
```

Visual Basic

```
Public Value3 As double
```

Version

Available since version 10.

I.2.5.2.9 X1

C++

```
HRESULT get_X1(double*);
HRESULT put_X1(double);
```

C#

```
public double X1 { get; set; }
```

Visual Basic

```
Public X1 As double
```

Version

Available since version 10.

I.2.5.2.10 X2

C++

```
HRESULT get_X2(double*);
HRESULT put_X2(double);
```

C#

```
public double X2 { get; set; }
```

Visual Basic

```
Public X2 As double
```

Version

Available since version 10.

I.2.5.2.11 X3

C++

```
HRESULT get_X3(double*);
HRESULT put_X3(double);
```

C#

```
public double X3 { get; set; }
```

Visual Basic

```
Public X3 As double
```

Version

Available since version 10.

I.2.5.2.12 X4

C++

```
HRESULT get_X4(double*);
```

(c) 2014 Autodesk Inc. All rights reserved

```
HRESULT put_X4(double);
```

C#

```
public double X4 { get; set; }
```

Visual Basic

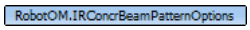
```
Public X4 As double
```

Version

Available since version 10.

I.2.6 IRConcrBeamPatternOptions

Class Hierarchy



C++

```
interface IRConcrBeamPatternOptions : IDispatch;
```

C#

```
public interface IRConcrBeamPatternOptions;
```

Visual Basic

```
Public Interface IRConcrBeamPatternOptions
```

Version

Available since version 10.

I.2.6.1 IRConcrBeamPatternOptions Members

The following tables list the members exposed by IRConcrBeamPatternOptions.

Public Fields

	Name	Description
	SpanBySpan (see page 1152)	

I.2.6.2 IRConcrBeamPatternOptions Fields

The fields of the IRConcrBeamPatternOptions class are listed here.

Public Fields

	Name	Description
	SpanBySpan (see page 1152)	

I.2.6.2.1 SpanBySpan

C++

```
HRESULT get_SpanBySpan(VARIANT_BOOL*);
HRESULT put_SpanBySpan(VARIANT_BOOL);
```

C#

```
public bool SpanBySpan { get; set; }
```

Visual Basic

```
Public SpanBySpan As Boolean
```

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 10.

I.2.7 IRConcrBeamPointLoad

Class Hierarchy

C++

```
interface IRConcrBeamPointLoad : IDispatch;
```

C#

```
public interface IRConcrBeamPointLoad;
```

Visual Basic

```
Public Interface IRConcrBeamPointLoad
```

Version

Available since version 10.

I.2.7.1 IRConcrBeamPointLoad Members

The following tables list the members exposed by IRConcrBeamPointLoad.

Public Fields

	Name	Description
	Case (see page 1153)	
	Nature (see page 1154)	
	RelativeCoordinates (see page 1154)	
	Spans (see page 1154)	
	Type (see page 1154)	
	Value (see page 1155)	
	X1 (see page 1155)	

I.2.7.2 IRConcrBeamPointLoad Fields

The fields of the IRConcrBeamPointLoad class are listed here.

Public Fields

	Name	Description
	Case (see page 1153)	
	Nature (see page 1154)	
	RelativeCoordinates (see page 1154)	
	Spans (see page 1154)	
	Type (see page 1154)	
	Value (see page 1155)	
	X1 (see page 1155)	

I.2.7.2.1 Case

C++

```
HRESULT get_Case(short*);
```

(c) 2014 Autodesk Inc. All rights reserved

```
HRESULT put_Case(short);
```

C#

```
public short Case { get; set; }
```

Visual Basic

```
Public Case As short
```

Version

Available since version 10.

I.2.7.2.2 Nature

C++

```
HRESULT get_Nature(IRConcrBeamLoadNatureType*);
HRESULT put_Nature(IRConcrBeamLoadNatureType);
```

C#

```
public IRConcrBeamLoadNatureType Nature { get; set; }
```

Visual Basic

```
Public Nature As IRConcrBeamLoadNatureType
```

Version

Available since version 10.

I.2.7.2.3 RelativeCoordinates

C++

```
HRESULT get_RelativeCoordinates(VARIANT_BOOL*);
HRESULT put_RelativeCoordinates(VARIANT_BOOL);
```

C#

```
public bool RelativeCoordinates { get; set; }
```

Visual Basic

```
Public RelativeCoordinates As Boolean
```

Version

Available since version 10.

I.2.7.2.4 Spans

C++

```
HRESULT get_Spans(IRConcrBeamSpanNumbers**);
```

C#

```
public IRConcrBeamSpanNumbers Spans { get; }
```

Visual Basic

```
Public ReadOnly Spans As IRConcrBeamSpanNumbers
```

Version

Available since version 10.

(c) 2014 Autodesk Inc. All rights reserved

I.2.7.2.5 Type

C++

```
HRESULT get_Type(IRConcrBeamPointLoadType*);
HRESULT put_Type(IRConcrBeamPointLoadType);
```

C#

```
public IRConcrBeamPointLoadType Type { get; set; }
```

Visual Basic

```
Public Type As IRConcrBeamPointLoadType
```

Version

Available since version 10.

I.2.7.2.6 Value

C++

```
HRESULT get_Value(double*);
HRESULT put_Value(double);
```

C#

```
public double Value { get; set; }
```

Visual Basic

```
Public Value As double
```

Version

Available since version 10.

I.2.7.2.7 X1

C++

```
HRESULT get_X1(double*);
HRESULT put_X1(double);
```

C#

```
public double X1 { get; set; }
```

Visual Basic

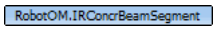
```
Public X1 As double
```

Version

Available since version 10.

I.2.8 IRConcrBeamSegment

Class Hierarchy



C++

```
interface IRConcrBeamSegment : IDispatch;
```

C#

```
public interface IRConcrBeamSegment;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Interface IRConcrBeamSegment
```

Version

Available since version 10.

I.2.8.1 IRConcrBeamSegment Members

The following tables list the members exposed by IRConcrBeamSegment.

Public Fields

	Name	Description
	Dim (see page 1156)	
	SectionType (see page 1156)	

I.2.8.2 IRConcrBeamSegment Fields

The fields of the IRConcrBeamSegment class are listed here.

Public Fields

	Name	Description
	Dim (see page 1156)	
	SectionType (see page 1156)	

I.2.8.2.1 Dim

C++

```
HRESULT get_Dim(double*);
HRESULT put_Dim(double);
```

C#

```
public double Dim { get; set; }
```

Visual Basic

```
Public Dim As double
```

Version

Available since version 10.

I.2.8.2.2 SectionType

C++

```
HRESULT get_SectionType(IRConcrBeamSectionType*);
HRESULT put_SectionType(IRConcrBeamSectionType);
```

C#

```
public IRConcrBeamSectionType SectionType { get; set; }
```

Visual Basic

```
Public SectionType As IRConcrBeamSectionType
```

Version

Available since version 10.

(c) 2014 Autodesk Inc. All rights reserved

I.2.9 IRConcrBeamSpan

Class Hierarchy

RobotOM.IRConcrBeamSpan

C++

```
interface IRConcrBeamSpan : IDispatch;
```

C#

```
public interface IRConcrBeamSpan;
```

Visual Basic

```
Public Interface IRConcrBeamSpan
```

Version

Available since version 10.

I.2.9.1 IRConcrBeamSpan Members

The following tables list the members exposed by IRConcrBeamSpan.

Public Fields

	Name	Description
	CalculationLength (see page 1157)	
	HasOpenings (see page 1158)	
	LeftSupport (see page 1158)	
	Length (see page 1158)	
	Name (see page 1158)	
	RightSupport (see page 1159)	
	Segment (see page 1159)	
	SegmentNumber (see page 1159)	

I.2.9.2 IRConcrBeamSpan Fields

The fields of the IRConcrBeamSpan class are listed here.

Public Fields

	Name	Description
	CalculationLength (see page 1157)	
	HasOpenings (see page 1158)	
	LeftSupport (see page 1158)	
	Length (see page 1158)	
	Name (see page 1158)	
	RightSupport (see page 1159)	
	Segment (see page 1159)	
	SegmentNumber (see page 1159)	

I.2.9.2.1 CalculationLength

C++

```
HRESULT get_CalculationLength(double*);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double CalculationLength { get; }
```

Visual Basic

```
Public ReadOnly CalculationLength As double
```

Version

Available since version 10.

I.2.9.2.2 HasOpenings

C++

```
HRESULT get_HasOpenings(VARIANT_BOOL*);
```

C#

```
public bool HasOpenings { get; }
```

Visual Basic

```
Public ReadOnly HasOpenings As Boolean
```

Version

Available since version 10.

I.2.9.2.3 LeftSupport

C++

```
HRESULT get_LeftSupport(IRConcrBeamSupport**);
```

C#

```
public IRConcrBeamSupport LeftSupport { get; }
```

Visual Basic

```
Public ReadOnly LeftSupport As IRConcrBeamSupport
```

Version

Available since version 10.

I.2.9.2.4 Length

C++

```
HRESULT get_Length(double*);
HRESULT put_Length(double);
```

C#

```
public double Length { get; set; }
```

Visual Basic

```
Public Length As double
```

Version

Available since version 10.

I.2.9.2.5 Name

C++

```
HRESULT get_Name(BSTR*);
HRESULT put_Name(BSTR);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public String Name { get; set; }
```

Visual Basic

```
Public Name As String
```

Version

Available since version 10.

I.2.9.2.6 RightSupport

C++

```
HRESULT get_RightSupport(IRConcrBeamSupport**);
```

C#

```
public IRConcrBeamSupport RightSupport { get; }
```

Visual Basic

```
Public ReadOnly RightSupport As IRConcrBeamSupport
```

Version

Available since version 10.

I.2.9.2.7 Segment

C++

```
HRESULT get_Segment(IRConcrBeamSegment**);
```

C#

```
public IRConcrBeamSegment Segment { get; }
```

Visual Basic

```
Public ReadOnly Segment As IRConcrBeamSegment
```

Version

Available since version 10.

I.2.9.2.8 SegmentNumber

C++

```
HRESULT get_SegmentNumber(short*);
HRESULT put_SegmentNumber(short);
```

C#

```
public short SegmentNumber { get; set; }
```

Visual Basic

```
Public SegmentNumber As short
```

Version

Available since version 10.

I.2.10 IRConcrBeamSpanNumbers

Class Hierarchy

(c) 2014 Autodesk Inc. All rights reserved

C++

```
interface IRConcrBeamSpanNumbers : IDispatch;
```

C#

```
public interface IRConcrBeamSpanNumbers;
```

Visual Basic

```
Public Interface IRConcrBeamSpanNumbers
```

Version

Available since version 10.

I.2.10.1 IRConcrBeamSpanNumbers Members

The following tables list the members exposed by IRConcrBeamSpanNumbers.

Public Fields

	Name	Description
	Count (see page 1160)	
	Item (see page 1160)	

Public Methods

	Name	Description
	Add (see page 1161)	
	RemoveAll (see page 1161)	

I.2.10.2 IRConcrBeamSpanNumbers Fields

The fields of the IRConcrBeamSpanNumbers class are listed here.

Public Fields

	Name	Description
	Count (see page 1160)	
	Item (see page 1160)	

I.2.10.2.1 Count

C++

```
HRESULT get_Count(long*);
```

C#

```
public long Count { get; }
```

Visual Basic

```
Public ReadOnly Count As long
```

Version

Available since version 10.

I.2.10.2.2 Item

C++

```
HRESULT get_Item(short*);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public short Item { get; }
```

Visual Basic

```
Public ReadOnly Item As short
```

Version

Available since version 10.

I.2.10.3 IRConcrBeamSpanNumbers Methods

The methods of the IRConcrBeamSpanNumbers class are listed here.

Public Methods

	Name	Description
	Add (see page 1161)	
	RemoveAll (see page 1161)	

I.2.10.3.1 Add

C++

```
HRESULT Add(short _newVal);
```

C#

```
public void Add(short _newVal);
```

Visual Basic

```
Public Sub Add(_newVal As short)
```

Version

Available since version 10.

I.2.10.3.2 RemoveAll

C++

```
HRESULT RemoveAll();
```

C#

```
public void RemoveAll();
```

Visual Basic

```
Public Sub RemoveAll()
```

Version

Available since version 10.

I.2.11 IRConcrBeamStoryOptions

Class Hierarchy

RobotOM.IRConcrBeamStoryOptions

C++

```
interface IRConcrBeamStoryOptions : IDispatch;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public interface IRConcrBeamStoryOptions;
```

Visual Basic

```
Public Interface IRConcrBeamStoryOptions
```

Version

Available since version 10.

I.2.11.1 IRConcrBeamStoryOptions Members

The following tables list the members exposed by IRConcrBeamStoryOptions.

Public Fields

	Name	Description
	Cracking (see page 1162)	

I.2.11.2 IRConcrBeamStoryOptions Fields

The fields of the IRConcrBeamStoryOptions class are listed here.

Public Fields

	Name	Description
	Cracking (see page 1162)	

I.2.11.2.1 Cracking

C++

```
HRESULT get_Cracking(IRConcrBeamCrackingType*);
HRESULT put_Cracking(IRConcrBeamCrackingType);
```

C#

```
public IRConcrBeamCrackingType Cracking { get; set; }
```

Visual Basic

```
Public Cracking As IRConcrBeamCrackingType
```

Version

Available since version 10.

I.2.12 IRConcrBeamSupport

Class Hierarchy

RobotOM.IRConcrBeamSupport

C++

```
interface IRConcrBeamSupport : IDispatch;
```

C#

```
public interface IRConcrBeamSupport;
```

Visual Basic

```
Public Interface IRConcrBeamSupport
```

Version

Available since version 10.

(c) 2014 Autodesk Inc. All rights reserved

I.2.12.1 IRConcrBeamSupport Members

The following tables list the members exposed by IRConcrBeamSupport.

Public Fields

	Name	Description
	MaterialType (see page 1163)	
	Name (see page 1163)	
	Type (see page 1164)	
	Width (see page 1164)	

I.2.12.2 IRConcrBeamSupport Fields

The fields of the IRConcrBeamSupport class are listed here.

Public Fields

	Name	Description
	MaterialType (see page 1163)	
	Name (see page 1163)	
	Type (see page 1164)	
	Width (see page 1164)	

I.2.12.2.1 MaterialType

C++

```
HRESULT get_MaterialType(IRConcrBeamSupportMaterialType*);
HRESULT put_MaterialType(IRConcrBeamSupportMaterialType);
```

C#

```
public IRConcrBeamSupportMaterialType MaterialType { get; set; }
```

Visual Basic

```
Public MaterialType As IRConcrBeamSupportMaterialType
```

Version

Available since version 10.

I.2.12.2.2 Name

C++

```
HRESULT get_Name(BSTR*);
HRESULT put_Name(BSTR);
```

C#

```
public String Name { get; set; }
```

Visual Basic

```
Public Name As String
```

Version

Available since version 10.

I.2.12.2.3 Type

C++

```
HRESULT get_Type(IRConcrBeamSupportType*);
HRESULT put_Type(IRConcrBeamSupportType);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRConcrBeamSupportType Type { get; set; }
```

Visual Basic

```
Public Type As IRConcrBeamSupportType
```

Version

Available since version 10.

I.2.12.2.4 Width

C++

```
HRESULT get_Width(double*);
HRESULT put_Width(double);
```

C#

```
public double Width { get; set; }
```

Visual Basic

```
Public Width As double
```

Version

Available since version 10.

I.2.13 IRConcrBeamLinearLoadType

C++

```
enum IRConcrBeamLinearLoadType;
```

C#

```
public enum IRConcrBeamLinearLoadType;
```

Visual Basic

```
Public Enum IRConcrBeamLinearLoadType
```

Members

Members	Description
I_LLT_BEAM_LOAD_SELF_WEIGHT = 1	Available since version 10.
I_LLT_BEAM_LOAD_UNIFORM = 2	Available since version 10.
I_LLT_BEAM_LOAD_CONTINUOUS = 3	Available since version 10.
I_LLT_BEAM_LOAD_TRAPEZOIDAL1 = 4	Available since version 10.
I_LLT_BEAM_LOAD_TRAPEZOIDAL2 = 5	Available since version 10.
I_LLT_BEAM_LOAD_TRAPEZOIDAL3 = 6	Available since version 10.

Version

Available since version 10.

I.2.14 IRConcrBeamLoadNatureType

C++

```
enum IRConcrBeamLoadNatureType;
```

C#

```
public enum IRConcrBeamLoadNatureType;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Enum IRConcrBeamLoadNatureType
```

Members

Members	Description
I_BLN_BEAM_NATURE_DEAD = 1	Available since version 10.
I_BLN_BEAM_NATURE_LIVE = 2	Available since version 10.
I_BLN_BEAM_NATURE_WIND = 3	Available since version 10.
I_BLN_BEAM_NATURE_SNOW = 4	Available since version 10.
I_BLN_BEAM_NATURE_SEISMIC = 5	Available since version 10.

Version

Available since version 10.

I.2.15 IRConcrBeamPointLoadType

C++

```
enum IRConcrBeamPointLoadType;
```

C#

```
public enum IRConcrBeamPointLoadType;
```

Visual Basic

```
Public Enum IRConcrBeamPointLoadType
```

Members

Members	Description
I_PLT_BEAM_LOAD_FORCE = 1	Available since version 10.
I_PLT_BEAM_LOAD_MOMENT = 2	Available since version 10.
I_PLT_BEAM_LOAD_AXIAL_FORCE = 3	Available since version 10.
I_PLT_BEAM_LOAD_TORSION_MOMENT = 4	Available since version 10.
I_PLT_BEAM_LOAD_UNIFORM_TORSION_MOMENT = 5	Available since version 10.

Version

Available since version 10.

I.2.16 IRConcrBeamSectionType

C++

```
enum IRConcrBeamSectionType;
```

C#

```
public enum IRConcrBeamSectionType;
```

Visual Basic

```
Public Enum IRConcrBeamSectionType
```

Members

Members	Description
I_CBS_BEAM_SECTION_NG_NP = 1	Available since version 10.
I_CBS_BEAM_SECTION_NG_LP = 2	Available since version 10.

(c) 2014 Autodesk Inc. All rights reserved

I_CBS_BEAM_SECTION_NG_RP = 3	Available since version 10.
I_CBS_BEAM_SECTION_NG_BP = 4	Available since version 10.
I_CBS_BEAM_SECTION_LG_NP = 5	Available since version 10.
I_CBS_BEAM_SECTION_LG_LP = 6	Available since version 10.
I_CBS_BEAM_SECTION_LG_RP = 7	Available since version 10.
I_CBS_BEAM_SECTION_LG_BP = 8	Available since version 10.
I_CBS_BEAM_SECTION_RG_NP = 9	Available since version 10.
I_CBS_BEAM_SECTION_RG_LP = 10	Available since version 10.
I_CBS_BEAM_SECTION_RG_RP = 11	Available since version 10.
I_CBS_BEAM_SECTION_RG_BP = 12	Available since version 10.

Version

Available since version 10.

I.2.17 IRConcrBeamSectionDimType

C++

```
enum IRConcrBeamSectionDimType;
```

C#

```
public enum IRConcrBeamSectionDimType;
```

Visual Basic

```
Public Enum IRConcrBeamSectionDimType
```

Members

Members	Description
I_BSD_BEAM_SECTION_DIM_L = 1	Available since version 10.
I_BSD_BEAM_SECTION_DIM_H1 = 2	Available since version 10.
I_BSD_BEAM_SECTION_DIM_H2 = 3	Available since version 10.
I_BSD_BEAM_SECTION_DIM_OFFY1 = 4	Available since version 10.
I_BSD_BEAM_SECTION_DIM_OFFY2 = 5	Available since version 10.
I_BSD_BEAM_SECTION_DIM_OFFZ1 = 6	Available since version 10.
I_BSD_BEAM_SECTION_DIM_OFFZ2 = 7	Available since version 10.
I_BSD_BEAM_SECTION_DIM_B = 8	Available since version 10.
I_BSD_BEAM_SECTION_DIM_EL1 = 9	Available since version 10.
I_BSD_BEAM_SECTION_DIM_EL2 = 10	Available since version 10.
I_BSD_BEAM_SECTION_DIM_PL = 11	Available since version 10.
I_BSD_BEAM_SECTION_DIM_RL = 12	Available since version 10.
I_BSD_BEAM_SECTION_DIM_HG1 = 13	Available since version 10.
I_BSD_BEAM_SECTION_DIM_BG1 = 14	Available since version 10.
I_BSD_BEAM_SECTION_DIM_HG2 = 15	Available since version 10.
I_BSD_BEAM_SECTION_DIM_BG2 = 16	Available since version 10.
I_BSD_BEAM_SECTION_DIM_ER1 = 17	Available since version 10.
I_BSD_BEAM_SECTION_DIM_ER2 = 18	Available since version 10.
I_BSD_BEAM_SECTION_DIM_PR = 19	Available since version 10.
I_BSD_BEAM_SECTION_DIM_RR = 20	Available since version 10.
I_BSD_BEAM_SECTION_DIM_HG3 = 21	Available since version 10.
I_BSD_BEAM_SECTION_DIM_BG3 = 22	Available since version 10.
I_BSD_BEAM_SECTION_DIM_HG4 = 23	Available since version 10.

(c) 2014 Autodesk Inc. All rights reserved

I_BSD_BEAM_SECTION_DIM_BG4 = 24	Available since version 10.
I_BSD_BEAM_SECTION_DIM_PL_CALC = 25	Available since version 10.
I_BSD_BEAM_SECTION_DIM_PR_CALC = 26	Available since version 10.
I_BSD_BEAM_SECTION_DIM_CL = 27	Available since version 10.
I_BSD_BEAM_SECTION_DIM_CR = 28	Available since version 10.

Version

Available since version 10.

I.2.18 IRConcrBeamCrackingType

C++

```
enum IRConcrBeamCrackingType;
```

C#

```
public enum IRConcrBeamCrackingType;
```

Visual Basic

```
Public Enum IRConcrBeamCrackingType
```

Members

Members	Description
I_CBC_BEAM_CRACKING_UNDEFINED = 1	Available since version 10.
I_CBC_BEAM_CRACKING_PERMISSIBLE = 2	Available since version 10.
I_CBC_BEAM_CRACKING_LIMITED = 3	Available since version 10.
I_CBC_BEAM_CRACKING_NOT_PERMISSIBLE = 4	Available since version 10.

Version

Available since version 10.

I.2.19 IRConcrBeamSupportType

C++

```
enum IRConcrBeamSupportType;
```

C#

```
public enum IRConcrBeamSupportType;
```

Visual Basic

```
Public Enum IRConcrBeamSupportType
```

Members

Members	Description
IBST_BEAM_SUPPORT_PINNED = 1	Available since version 10.
IBST_BEAM_SUPPORT_FIXED = 2	Available since version 10.
IBST_BEAM_SUPPORT_SWAY = 3	Available since version 10.

Version

Available since version 10.

(c) 2014 Autodesk Inc. All rights reserved

I.2.20 IRConcrBeamSupportMaterialType

C++

```
enum IRConcrBeamSupportMaterialType;
```

C#

```
public enum IRConcrBeamSupportMaterialType;
```

Visual Basic

```
Public Enum IRConcrBeamSupportMaterialType
```

Members

Members	Description
I_BSM_BEAM_SUPPORT_CONCRETE = 1	Available since version 10.
I_BSM_BEAM_SUPPORT_MASONRY = 2	Available since version 10.

Version

Available since version 10.

I.3 Continuous Footing

Available since version 10.

Enumerations

	Name	Description
	IRConcrContinuousFootingSectionType (see page 1183)	
	IRConcrContinuousFootingSectionDimType (see page 1183)	
	IRConcrContinuousFootingSupportMaterialType (see page 1184)	

Interfaces

	Name	Description
	IRConcrContinuousFooting (see page 1168)	
	IRConcrContinuousFootingGeometry (see page 1175)	
	IRConcrContinuousFootingSegment (see page 1179)	
	IRConcrContinuousFootingSpan (see page 1180)	
	IRConcrContinuousFootingSupport (see page 1184)	.MaterialType (see page 1185)

I.3.1 IRConcrContinuousFooting

Class Hierarchy

RobotOM.IRConcrContinuousFooting

C++

```
interface IRConcrContinuousFooting : IDispatch;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public interface IRConcrContinuousFooting;
```

Visual Basic

```
Public Interface IRConcrContinuousFooting
```

Version

Available since version 10.

I.3.1.1 IRConcrContinuousFooting Members

The following tables list the members exposed by IRConcrContinuousFooting.

Public Fields

	Name	Description
	CalculationOptions (see page 1170)	
	Concrete (see page 1170)	
	Geometry (see page 1171)	
	ImportOptions (see page 1171)	
	IsActive (see page 1171)	
	IsSelected (see page 1171)	
	Name (see page 1172)	
	NumberOfElements (see page 1172)	
	PatternOptions (see page 1172)	
	Reinforcement (see page 1172)	
	Steel (see page 1173)	
	StructureUserNo (see page 1173)	
	StructureUserNoCount (see page 1173)	
	UniqueId (see page 1173)	

Public Methods

	Name	Description
	Activate (see page 1174)	
	Calculate (see page 1174)	
	CreateCalculationNoteRtf (see page 1174)	
	Save (see page 1175)	
	Verify (see page 1175)	

I.3.1.2 IRConcrContinuousFooting Fields

The fields of the IRConcrContinuousFooting class are listed here.

Public Fields

	Name	Description
	CalculationOptions (see page 1170)	
	Concrete (see page 1170)	
	Geometry (see page 1171)	
	ImportOptions (see page 1171)	

(c) 2014 Autodesk Inc. All rights reserved

	IsActive (see page 1171)	
	IsSelected (see page 1171)	
	Name (see page 1172)	
	NumberOfElements (see page 1172)	
	PatternOptions (see page 1172)	
	Reinforcement (see page 1172)	
	Steel (see page 1173)	
	StructureUserNo (see page 1173)	
	StructureUserNoCount (see page 1173)	
	UniqueId (see page 1173)	

I.3.1.2.1 CalculationOptions

C++

```
HRESULT get_CalculationOptions(IRConcrBeamCalcOptions**);
```

C#

```
public IRConcrBeamCalcOptions CalculationOptions { get; }
```

Visual Basic

```
Public ReadOnly CalculationOptions As IRConcrBeamCalcOptions
```

Version

Available since version 10.

I.3.1.2.2 Concrete

C++

```
HRESULT get_Concrete(IRConcrConcrete**);
```

C#

```
public IRConcrConcrete Concrete { get; }
```

Visual Basic

```
Public ReadOnly Concrete As IRConcrConcrete
```

Version

Available since version 10.

I.3.1.2.3 Geometry

C++

```
HRESULT get_Geometry(IRConcrContinuousFootingGeometry**);
```

C#

```
public IRConcrContinuousFootingGeometry Geometry { get; }
```

Visual Basic

```
Public ReadOnly Geometry As IRConcrContinuousFootingGeometry
```

Version

Available since version 10.

(c) 2014 Autodesk Inc. All rights reserved

I.3.1.2.4 ImportOptions

C++

```
HRESULT get_ImportOptions(IRConcrBeamImportOptions**);
```

C#

```
public IRConcrBeamImportOptions ImportOptions { get; }
```

Visual Basic

```
Public ReadOnly ImportOptions As IRConcrBeamImportOptions
```

Version

Available since version 10.

I.3.1.2.5 IsActive

C++

```
HRESULT get_IsActive(VARIANT_BOOL*);
```

C#

```
public bool IsActive { get; }
```

Visual Basic

```
Public ReadOnly IsActive As Boolean
```

Version

Available since version 10.

I.3.1.2.6 IsSelected

C++

```
HRESULT get_IsSelected(VARIANT_BOOL*);
HRESULT put_IsSelected(VARIANT_BOOL);
```

C#

```
public bool IsSelected { get; set; }
```

Visual Basic

```
Public IsSelected As Boolean
```

Version

Available since version 12.

I.3.1.2.7 Name

C++

```
HRESULT get_Name(BSTR*);
```

C#

```
public String Name { get; }
```

Visual Basic

```
Public ReadOnly Name As String
```

Version

Available since version 10.

(c) 2014 Autodesk Inc. All rights reserved

I.3.1.2.8 NumberOfElements

C++

```
HRESULT get_NumberOfElements(long*);
HRESULT put_NumberOfElements(long);
```

C#

```
public long NumberOfElements { get; set; }
```

Visual Basic

```
Public NumberOfElements As long
```

Version

Available since version 12.

I.3.1.2.9 PatternOptions

C++

```
HRESULT get_PatternOptions(IRConcrBeamPatternOptions**);
```

C#

```
public IRConcrBeamPatternOptions PatternOptions { get; }
```

Visual Basic

```
Public ReadOnly PatternOptions As IRConcrBeamPatternOptions
```

Version

Available since version 10.

I.3.1.2.10 Reinforcement

C++

```
HRESULT get_Reinforcement(IRConcrReinforcement**);
```

C#

```
public IRConcrReinforcement Reinforcement { get; }
```

Visual Basic

```
Public ReadOnly Reinforcement As IRConcrReinforcement
```

Version

Available since version 10.

I.3.1.2.11 Steel

C++

```
HRESULT get_Steel(IRConcrSteel**);
```

C#

```
public IRConcrSteel Steel { get; }
```

Visual Basic

```
Public ReadOnly Steel As IRConcrSteel
```

Version

Available since version 10.

(c) 2014 Autodesk Inc. All rights reserved

I.3.1.2.12 StructureUserNo

C++

```
HRESULT get_StructureUserNo(long*);
HRESULT put_StructureUserNo(long);
```

C#

```
public long StructureUserNo { get; set; }
```

Visual Basic

```
Public StructureUserNo As long
```

Version

Available since version 10.

I.3.1.2.13 StructureUserNoCount

C++

```
HRESULT get_StructureUserNoCount(short*);
HRESULT put_StructureUserNoCount(short);
```

C#

```
public short StructureUserNoCount { get; set; }
```

Visual Basic

```
Public StructureUserNoCount As short
```

Version

Available since version 10.

I.3.1.2.14 UniqueId

C++

```
HRESULT get_UniqueId(long*);
```

C#

```
public long UniqueId { get; }
```

Visual Basic

```
Public ReadOnly UniqueId As long
```

Version

Available since version 10.

I.3.1.3 IRConcrContinuousFooting Methods

The methods of the IRConcrContinuousFooting class are listed here.

Public Methods

	Name	Description
	Activate (see page 1174)	
	Calculate (see page 1174)	
	CreateCalculationNoteRtf (see page 1174)	
	Save (see page 1175)	
	Verify (see page 1175)	

(c) 2014 Autodesk Inc. All rights reserved

I.3.1.3.1 Activate

C++

```
HRESULT Activate();
```

C#

```
public void Activate();
```

Visual Basic

```
Public Sub Activate()
```

Version

Available since version 10.

I.3.1.3.2 Calculate

C++

```
HRESULT Calculate(VARIANT_BOOL _createReinforcement, VARIANT_BOOL* ret);
```

C#

```
public bool Calculate(bool _createReinforcement);
```

Visual Basic

```
Public Function Calculate(_createReinforcement As Boolean) As Boolean
```

Version

Available since version 10.

I.3.1.3.3 CreateCalculationNoteRtf

C++

```
HRESULT CreateCalculationNoteRtf(BSTR _fileName, VARIANT_BOOL* ret);
```

C#

```
public bool CreateCalculationNoteRtf(String _fileName);
```

Visual Basic

```
Public Function CreateCalculationNoteRtf(_fileName As String) As Boolean
```

Version

Available since version 10.

I.3.1.3.4 Save

C++

```
HRESULT Save();
```

C#

```
public void Save();
```

Visual Basic

```
Public Sub Save()
```

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 10.

I.3.1.3.5 Verify

C++

```
HRESULT Verify(VARIANT_BOOL _showErrors, long* ret);
```

C#

```
public long Verify(bool _showErrors);
```

Visual Basic

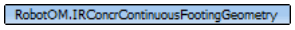
```
Public Function Verify(_showErrors As Boolean) As long
```

Version

Available since version 10.

I.3.2 IRConcrContinuousFootingGeometry

Class Hierarchy



C++

```
interface IRConcrContinuousFootingGeometry : IDispatch;
```

C#

```
public interface IRConcrContinuousFootingGeometry;
```

Visual Basic

```
Public Interface IRConcrContinuousFootingGeometry
```

Version

Available since version 10.

I.3.2.1 IRConcrContinuousFootingGeometry Members

The following tables list the members exposed by IRConcrContinuousFootingGeometry.

Public Fields

	Name	Description
	AutoNameSpans (see page 1176)	
	AutoNameSupports (see page 1177)	
	ConcreteVolume (see page 1177)	
	LeftCantilever (see page 1177)	
	RightCantilever (see page 1177)	
	ShutteringArea (see page 1178)	
	Span (see page 1178)	
	SpanNumber (see page 1178)	

(c) 2014 Autodesk Inc. All rights reserved

Public Methods

	Name	Description
	UpdateInternalGeometryData (see page 1179)	

I.3.2.2 IRConcrContinuousFootingGeometry Fields

The fields of the IRConcrContinuousFootingGeometry class are listed here.

Public Fields

	Name	Description
	AutoNameSpans (see page 1176)	
	AutoNameSupports (see page 1177)	
	ConcreteVolume (see page 1177)	
	LeftCantilever (see page 1177)	
	RightCantilever (see page 1177)	
	ShutteringArea (see page 1178)	
	Span (see page 1178)	
	SpanNumber (see page 1178)	

I.3.2.2.1 AutoNameSpans

C++

```
HRESULT get_AutoNameSpans(VARIANT_BOOL*);
HRESULT put_AutoNameSpans(VARIANT_BOOL);
```

C#

```
public bool AutoNameSpans { get; set; }
```

Visual Basic

```
Public AutoNameSpans As Boolean
```

Version

Available since version 10.

I.3.2.2.2 AutoNameSupports

C++

```
HRESULT get_AutoNameSupports(VARIANT_BOOL*);
HRESULT put_AutoNameSupports(VARIANT_BOOL);
```

C#

```
public bool AutoNameSupports { get; set; }
```

Visual Basic

```
Public AutoNameSupports As Boolean
```

Version

Available since version 10.

I.3.2.2.3 ConcreteVolume

C++

```
HRESULT get_ConcreteVolume(double*);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double ConcreteVolume { get; }
```

Visual Basic

```
Public ReadOnly ConcreteVolume As double
```

Version

Available since version 10.

I.3.2.2.4 LeftCantilever

C++

```
HRESULT get_LeftCantilever(VARIANT_BOOL*);
HRESULT put_LeftCantilever(VARIANT_BOOL);
```

C#

```
public bool LeftCantilever { get; set; }
```

Visual Basic

```
Public LeftCantilever As Boolean
```

Version

Available since version 10.

I.3.2.2.5 RightCantilever

C++

```
HRESULT get_RightCantilever(VARIANT_BOOL*);
HRESULT put_RightCantilever(VARIANT_BOOL);
```

C#

```
public bool RightCantilever { get; set; }
```

Visual Basic

```
Public RightCantilever As Boolean
```

Version

Available since version 10.

I.3.2.2.6 ShutteringArea

C++

```
HRESULT get_ShutteringArea(double*);
```

C#

```
public double ShutteringArea { get; }
```

Visual Basic

```
Public ReadOnly ShutteringArea As double
```

Version

Available since version 10.

I.3.2.2.7 Span

C++

```
HRESULT get_Span(IRConcrContinuousFootingSpan**);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRConcrContinuousFootingSpan Span { get; }
```

Visual Basic

```
Public ReadOnly Span As IRConcrContinuousFootingSpan
```

Version

Available since version 10.

I.3.2.2.8 SpanNumber

C++

```
HRESULT get_SpanNumber(short*);
HRESULT put_SpanNumber(short);
```

C#

```
public short SpanNumber { get; set; }
```

Visual Basic

```
Public SpanNumber As short
```

Version

Available since version 10.

I.3.2.3 IRConcrContinuousFootingGeometry Methods

The methods of the IRConcrContinuousFootingGeometry class are listed here.

Public Methods

	Name	Description
	UpdateInternalGeometryData (see page 1179)	

I.3.2.3.1 UpdateInternalGeometryData

C++

```
HRESULT UpdateInternalGeometryData();
```

C#

```
public void UpdateInternalGeometryData();
```

Visual Basic

```
Public Sub UpdateInternalGeometryData()
```

Version

Available since version 10.

I.3.3 IRConcrContinuousFootingSegment

Class Hierarchy

C++

```
interface IRConcrContinuousFootingSegment : IDispatch;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public interface IRConcrContinuousFootingSegment;
```

Visual Basic

```
Public Interface IRConcrContinuousFootingSegment
```

Version

Available since version 10.

I.3.3.1 IRConcrContinuousFootingSegment Members

The following tables list the members exposed by IRConcrContinuousFootingSegment.

Public Fields

	Name	Description
	Dim (see page 1179)	
	SectionType (see page 1180)	

I.3.3.2 IRConcrContinuousFootingSegment Fields

The fields of the IRConcrContinuousFootingSegment class are listed here.

Public Fields

	Name	Description
	Dim (see page 1179)	
	SectionType (see page 1180)	

I.3.3.2.1 Dim

C++

```
HRESULT get_Dim(double*);
HRESULT put_Dim(double);
```

C#

```
public double Dim { get; set; }
```

Visual Basic

```
Public Dim As double
```

Version

Available since version 10.

I.3.3.2.2 SectionType

C++

```
HRESULT get_SectionType(IRConcrContinuousFootingSectionType*);
HRESULT put_SectionType(IRConcrContinuousFootingSectionType);
```

C#

```
public IRConcrContinuousFootingSectionType SectionType { get; set; }
```

Visual Basic

```
Public SectionType As IRConcrContinuousFootingSectionType
```

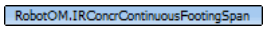
Version

Available since version 10.

(c) 2014 Autodesk Inc. All rights reserved

I.3.4 IRConcrContinuousFootingSpan

Class Hierarchy



C++

```
interface IRConcrContinuousFootingSpan : IDispatch;
```

C#

```
public interface IRConcrContinuousFootingSpan;
```

Visual Basic

```
Public Interface IRConcrContinuousFootingSpan
```

Version

Available since version 10.

I.3.4.1 IRConcrContinuousFootingSpan Members

The following tables list the members exposed by IRConcrContinuousFootingSpan.

Public Fields

	Name	Description
	CalculationLength (see page 1181)	
	LeftSupport (see page 1181)	
	Length (see page 1182)	
	Name (see page 1182)	
	RightSupport (see page 1182)	
	Segment (see page 1182)	
	SegmentNumber (see page 1183)	

I.3.4.2 IRConcrContinuousFootingSpan Fields

The fields of the IRConcrContinuousFootingSpan class are listed here.

Public Fields

	Name	Description
	CalculationLength (see page 1181)	
	LeftSupport (see page 1181)	
	Length (see page 1182)	
	Name (see page 1182)	
	RightSupport (see page 1182)	
	Segment (see page 1182)	
	SegmentNumber (see page 1183)	

I.3.4.2.1 CalculationLength

C++

```
HRESULT get_CalculationLength(double*);
```

C#

```
public double CalculationLength { get; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public ReadOnly CalculationLength As double
```

Version

Available since version 10.

I.3.4.2.2 LeftSupport

C++

```
HRESULT get_LeftSupport(IRConcrContinuousFootingSupport
**);
```

C#

```
public IRConcrContinuousFootingSupport
 LeftSupport { get; }
```

Visual Basic

```
Public ReadOnly LeftSupport As IRConcrContinuousFootingSupport
```

Version

Available since version 10.

I.3.4.2.3 Length

C++

```
HRESULT get_Length(double*);
HRESULT put_Length(double);
```

C#

```
public double Length { get; set; }
```

Visual Basic

```
Public Length As double
```

Version

Available since version 10.

I.3.4.2.4 Name

C++

```
HRESULT get_Name(BSTR*);
HRESULT put_Name(BSTR);
```

C#

```
public String Name { get; set; }
```

Visual Basic

```
Public Name As String
```

Version

Available since version 10.

I.3.4.2.5 RightSupport

C++

```
HRESULT get_RightSupport(IRConcrContinuousFootingSupport
**);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRConcrContinuousFootingSupport
 RightSupport { get; }
```

Visual Basic

```
Public ReadOnly RightSupport As IRConcrContinuousFootingSupport
```

Version

Available since version 10.

I.3.4.2.6 Segment

C++

```
HRESULT get_Segment(IRConcrContinuousFootingSegment**);
```

C#

```
public IRConcrContinuousFootingSegment Segment { get; }
```

Visual Basic

```
Public ReadOnly Segment As IRConcrContinuousFootingSegment
```

Version

Available since version 10.

I.3.4.2.7 SegmentNumber

C++

```
HRESULT get_SegmentNumber(short*);
HRESULT put_SegmentNumber(short);
```

C#

```
public short SegmentNumber { get; set; }
```

Visual Basic

```
Public SegmentNumber As short
```

Version

Available since version 10.

I.3.5 IRConcrContinuousFootingSectionType

C++

```
enum IRConcrContinuousFootingSectionType;
```

C#

```
public enum IRConcrContinuousFootingSectionType;
```

Visual Basic

```
Public Enum IRConcrContinuousFootingSectionType
```

Members

Members	Description
I_CBS_CFOOT_SECTION_NP = 1	Available since version 10.
I_CBS_CFOOT_SECTION_LP = 2	Available since version 10.
I_CBS_CFOOT_SECTION_RP = 3	Available since version 10.

(c) 2014 Autodesk Inc. All rights reserved

I_CBS_CFOOT_SECTION_BP = 4	Available since version 10.

Version

Available since version 10.

I.3.6 IRConcrContinuousFootingSectionDimType

C++

```
enum IRConcrContinuousFootingSectionDimType;
```

C#

```
public enum IRConcrContinuousFootingSectionDimType;
```

Visual Basic

```
Public Enum IRConcrContinuousFootingSectionDimType
```

Members

Members	Description
I_BSD_CFOOT_SECTION_DIM_L = 1	Available since version 10.
I_BSD_CFOOT_SECTION_DIM_H1 = 2	Available since version 10.
I_BSD_CFOOT_SECTION_DIM_H2 = 3	Available since version 10.
I_BSD_CFOOT_SECTION_DIM_B = 4	Available since version 10.
I_BSD_CFOOT_SECTION_DIM_EL1 = 5	Available since version 10.
I_BSD_CFOOT_SECTION_DIM_ER1 = 6	Available since version 10.
I_BSD_CFOOT_SECTION_DIM_PL = 7	Available since version 10.
I_BSD_CFOOT_SECTION_DIM_RL = 8	Available since version 10.
I_BSD_CFOOT_SECTION_DIM_PR = 9	Available since version 10.
I_BSD_CFOOT_SECTION_DIM_RR = 10	Available since version 10.

Version

Available since version 10.

I.3.7 IRConcrContinuousFootingSupportMaterialType

C++

```
enum IRConcrContinuousFootingSupportMaterialType;
```

C#

```
public enum IRConcrContinuousFootingSupportMaterialType;
```

Visual Basic

```
Public Enum IRConcrContinuousFootingSupportMaterialType
```

Members

Members	Description
I_BSM_CFOOT_SUPPORT_CONCRETE = 1	Available since version 10.
I_BSM_CFOOT_SUPPORT_MASONRY = 2	Available since version 10.

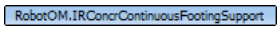
Version

Available since version 10.

(c) 2014 Autodesk Inc. All rights reserved

I.3.8 MaterialType

Class Hierarchy



C++

```
HRESULT get_MaterialType(IRConcrContinuousFootingSupportMaterialType*);
HRESULT put_MaterialType(IRConcrContinuousFootingSupportMaterialType);
```

C#

```
public IRConcrContinuousFootingSupportMaterialType MaterialType { get; set; }
```

Visual Basic

```
Public MaterialType As IRConcrContinuousFootingSupportMaterialType
```

Description

.MaterialType (see page 1185)

Version

Available since version 10.

I.3.8.1 IRConcrContinuousFootingSupport Members

The following tables list the members exposed by IRConcrContinuousFootingSupport.

Public Fields

	Name	Description
	MaterialType (see page 1185)	This is MaterialType, a member of class IRConcrContinuousFootingSupport.
	Name (see page 1185)	This is Name, a member of class IRConcrContinuousFootingSupport.
	Width (see page 1185)	This is Width, a member of class IRConcrContinuousFootingSupport.

I.3.8.2 IRConcrContinuousFootingSupport Fields

The fields of the IRConcrContinuousFootingSupport class are listed here.

Public Fields

	Name	Description
	MaterialType (see page 1185)	This is MaterialType, a member of class IRConcrContinuousFootingSupport.
	Name (see page 1185)	This is Name, a member of class IRConcrContinuousFootingSupport.
	Width (see page 1185)	This is Width, a member of class IRConcrContinuousFootingSupport.

I.3.8.2.1 IRConcrContinuousFootingSupport.MaterialType Field

C#

```
IRConcrContinuousFootingSupportMaterialType MaterialType;
```

Description

This is MaterialType, a member of class IRConcrContinuousFootingSupport.

I.3.8.2.2 IRConcrContinuousFootingSupport.Name Field

C#

```
String Name;
```

(c) 2014 Autodesk Inc. All rights reserved

Description

This is Name, a member of class IRConcrContinuousFootingSupport.

I.3.8.2.3 IRConcrContinuousFootingSupport.Width Field

C#

```
double Width;
```

Description

This is Width, a member of class IRConcrContinuousFootingSupport.

I.4 RC Column

Available since version 10.

Enumerations

	Name	Description
	IRConcrColumnDimensionType (see page 1201)	
	IRConcrColumnSectionType (see page 1202)	
	IRConcrColumnLoadCaseType (see page 1214)	

Interfaces

	Name	Description
	IRConcrColumn (see page 1186)	
	IRConcrColumnCalcOptions (see page 1195)	
	IRConcrColumnGeometry (see page 1196)	
	IRConcrColumnImportOptions (see page 1198)	
	IRConcrColumnPatternOptions (see page 1200)	
	IRConcrColumnBucklingModel (see page 1202)	
	IRConcrColumnLoads (see page 1207)	
	IRConcrColumnLoad (see page 1209)	

I.4.1 IRConcrColumn

Class Hierarchy

RobotOM.IRConcrColumn

C++

```
interface IRConcrColumn : IDispatch;
```

C#

```
public interface IRConcrColumn;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Interface IRConcrColumn
```

Version

Available since version 10.

I.4.1.1 IRConcrColumn Members

The following tables list the members exposed by IRConcrColumn.

Public Fields

	Name	Description
	BucklingModel (see page 1188)	
	CalculationOptions (see page 1188)	
	Concrete (see page 1188)	
	Geometry (see page 1189)	
	HasUpperColumn (see page 1189)	
	ImportOptions (see page 1189)	
	IsActive (see page 1189)	
	IsSelected (see page 1190)	
	Loads (see page 1190)	
	Name (see page 1190)	
	NumberOfElements (see page 1190)	
	PatternOptions (see page 1191)	
	Reinforcement (see page 1191)	
	Steel (see page 1191)	
	StructureUserNo (see page 1191)	
	StructureUserNoCount (see page 1192)	
	UniqueId (see page 1192)	
	UpperColumn (see page 1192)	
	UpperColumnDY (see page 1192)	
	UpperColumnDZ (see page 1193)	

Public Methods

	Name	Description
	Activate (see page 1193)	
	Calculate (see page 1194)	
	CreateCalculationNoteRtf (see page 1194)	
	CreateFromBars (see page 1194)	
	Save (see page 1194)	
	Verify (see page 1195)	

I.4.1.2 IRConcrColumn Fields

The fields of the IRConcrColumn class are listed here.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	BucklingModel (see page 1188)	
	CalculationOptions (see page 1188)	
	Concrete (see page 1188)	
	Geometry (see page 1189)	
	HasUpperColumn (see page 1189)	
	ImportOptions (see page 1189)	
	IsActive (see page 1189)	
	IsSelected (see page 1190)	
	Loads (see page 1190)	
	Name (see page 1190)	
	NumberOfElements (see page 1190)	
	PatternOptions (see page 1191)	
	Reinforcement (see page 1191)	
	Steel (see page 1191)	
	StructureUserNo (see page 1191)	
	StructureUserNoCount (see page 1192)	
	UniqueId (see page 1192)	
	UpperColumn (see page 1192)	
	UpperColumnDY (see page 1192)	
	UpperColumnDZ (see page 1193)	

I.4.1.2.1 BucklingModel

C++

```
HRESULT get_BucklingModel(IRConcrColumnBucklingModel**);
```

C#

```
public IRConcrColumnBucklingModel BucklingModel { get; }
```

Visual Basic

```
Public ReadOnly BucklingModel As IRConcrColumnBucklingModel
```

Version

Available since version 11.

I.4.1.2.2 CalculationOptions

C++

```
HRESULT get_CalculationOptions(IRConcrColumnCalcOptions**);
```

C#

```
public IRConcrColumnCalcOptions CalculationOptions { get; }
```

Visual Basic

```
Public ReadOnly CalculationOptions As IRConcrColumnCalcOptions
```

Version

Available since version 10.

(c) 2014 Autodesk Inc. All rights reserved

I.4.1.2.3 Concrete

C++

```
HRESULT get_Concrete(IRConcrConcrete**);
```

C#

```
public IRConcrConcrete Concrete { get; }
```

Visual Basic

```
Public ReadOnly Concrete As IRConcrConcrete
```

Version

Available since version 10.

I.4.1.2.4 Geometry

C++

```
HRESULT get_Geometry(IRConcrColumnGeometry**);
```

C#

```
public IRConcrColumnGeometry Geometry { get; }
```

Visual Basic

```
Public ReadOnly Geometry As IRConcrColumnGeometry
```

Version

Available since version 10.

I.4.1.2.5 HasUpperColumn

C++

```
HRESULT get_HasUpperColumn(VARIANT_BOOL*);
```

C#

```
public bool HasUpperColumn { get; }
```

Visual Basic

```
Public ReadOnly HasUpperColumn As Boolean
```

Version

Available since version 12.

I.4.1.2.6 ImportOptions

C++

```
HRESULT get_ImportOptions(IRConcrColumnImportOptions**);
```

C#

```
public IRConcrColumnImportOptions ImportOptions { get; }
```

Visual Basic

```
Public ReadOnly ImportOptions As IRConcrColumnImportOptions
```

Version

Available since version 10.

(c) 2014 Autodesk Inc. All rights reserved

I.4.1.2.7 IsActive

C++

```
HRESULT get_IsActive(VARIANT_BOOL*);
```

C#

```
public bool IsActive { get; }
```

Visual Basic

```
Public ReadOnly IsActive As Boolean
```

Version

Available since version 10.

I.4.1.2.8 IsSelected

C++

```
HRESULT get_IsSelected(VARIANT_BOOL*);
HRESULT put_IsSelected(VARIANT_BOOL);
```

C#

```
public bool IsSelected { get; set; }
```

Visual Basic

```
Public IsSelected As Boolean
```

Version

Available since version 12.

I.4.1.2.9 Loads

C++

```
HRESULT get_Loads(IRConcrColumnLoads**);
```

C#

```
public IRConcrColumnLoads Loads { get; }
```

Visual Basic

```
Public ReadOnly Loads As IRConcrColumnLoads
```

Version

Available since version 11.

I.4.1.2.10 Name

C++

```
HRESULT get_Name(BSTR*);
```

C#

```
public String Name { get; }
```

Visual Basic

```
Public ReadOnly Name As String
```

Version

Available since version 10.

(c) 2014 Autodesk Inc. All rights reserved

I.4.1.2.11 NumberOfElements

C++

```
HRESULT get_NumberOfElements( long*);
HRESULT put_NumberOfElements( long);
```

C#

```
public  long NumberOfElements { get; set; }
```

Visual Basic

```
Public NumberOfElements As  long
```

Version

Available since version 10.

I.4.1.2.12 PatternOptions

C++

```
HRESULT get_PatternOptions(IRConcrColumnPatternOptions**);
```

C#

```
public IRConcrColumnPatternOptions PatternOptions { get; }
```

Visual Basic

```
Public ReadOnly PatternOptions As IRConcrColumnPatternOptions
```

Version

Available since version 10.

I.4.1.2.13 Reinforcement

C++

```
HRESULT get_Reinforcement(IRConcrReinforcement**);
```

C#

```
public IRConcrReinforcement Reinforcement { get; }
```

Visual Basic

```
Public ReadOnly Reinforcement As IRConcrReinforcement
```

Version

Available since version 10.

I.4.1.2.14 Steel

C++

```
HRESULT get_Steel(IRConcrSteel**);
```

C#

```
public IRConcrSteel Steel { get; }
```

Visual Basic

```
Public ReadOnly Steel As IRConcrSteel
```

Version

Available since version 10.

(c) 2014 Autodesk Inc. All rights reserved

I.4.1.2.15 StructureUserNo

C++

```
HRESULT get_StructureUserNo(long*);
HRESULT put_StructureUserNo(long);
```

C#

```
public long StructureUserNo { get; set; }
```

Visual Basic

```
Public StructureUserNo As long
```

Version

Available since version 10.

I.4.1.2.16 StructureUserNoCount

C++

```
HRESULT get_StructureUserNoCount(short*);
HRESULT put_StructureUserNoCount(short);
```

C#

```
public short StructureUserNoCount { get; set; }
```

Visual Basic

```
Public StructureUserNoCount As short
```

Version

Available since version 10.

I.4.1.2.17 UniqueId

C++

```
HRESULT get_UniqueId(long*);
```

C#

```
public long UniqueId { get; }
```

Visual Basic

```
Public ReadOnly UniqueId As long
```

Version

Available since version 10.

I.4.1.2.18 UpperColumn

C++

```
HRESULT get_UpperColumn(IRConcrColumn**);
HRESULT put_UpperColumn(IRConcrColumn*);
```

C#

```
public IRConcrColumn UpperColumn { get; set; }
```

Visual Basic

```
Public UpperColumn As IRConcrColumn
```

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 12.

I.4.1.2.19 UpperColumnDY

C++

```
HRESULT get_UpperColumnDY( double*);
HRESULT put_UpperColumnDY( double);
```

C#

```
public  double UpperColumnDY { get; set; }
```

Visual Basic

```
Public UpperColumnDY As  double
```

Version

Available since version 12.

I.4.1.2.20 UpperColumnDZ

C++

```
HRESULT get_UpperColumnDZ( double*);
HRESULT put_UpperColumnDZ( double);
```

C#

```
public  double UpperColumnDZ { get; set; }
```

Visual Basic

```
Public UpperColumnDZ As  double
```

Version

Available since version 12.

I.4.1.3 IRConcrColumn Methods

The methods of the IRConcrColumn class are listed here.

Public Methods

	Name	Description
	Activate (see page 1193)	
	Calculate (see page 1194)	
	CreateCalculationNoteRtf (see page 1194)	
	CreateFromBars (see page 1194)	
	Save (see page 1194)	
	Verify (see page 1195)	

I.4.1.3.1 Activate

C++

```
HRESULT Activate();
```

C#

```
public void Activate();
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Sub Activate()
```

Version

Available since version 10.

I.4.1.3.2 Calculate

C++

```
HRESULT Calculate(VARIANT_BOOL _createReinforcement, VARIANT_BOOL* ret);
```

C#

```
public bool Calculate(bool _createReinforcement);
```

Visual Basic

```
Public Function Calculate(_createReinforcement As Boolean) As Boolean
```

Version

Available since version 10.

I.4.1.3.3 CreateCalculationNoteRtf

C++

```
HRESULT CreateCalculationNoteRtf(BSTR _fileName, VARIANT_BOOL* ret);
```

C#

```
public bool CreateCalculationNoteRtf(String _fileName);
```

Visual Basic

```
Public Function CreateCalculationNoteRtf(_fileName As String) As Boolean
```

Version

Available since version 10.

I.4.1.3.4 CreateFromBars

C++

```
HRESULT CreateFromBars(BSTR _selText, VARIANT_BOOL* ret);
```

C#

```
public bool CreateFromBars(String _selText);
```

Visual Basic

```
Public Function CreateFromBars(_selText As String) As Boolean
```

Version

Available since version 10.

I.4.1.3.5 Save

C++

```
HRESULT Save();
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public void Save();
```

Visual Basic

```
Public Sub Save()
```

Version

Available since version 10.

I.4.1.3.6 Verify

C++

```
HRESULT Verify(VARIANT_BOOL _showErrors, long* ret);
```

C#

```
public long Verify(bool _showErrors);
```

Visual Basic

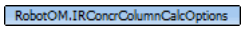
```
Public Function Verify(_showErrors As Boolean) As long
```

Version

Available since version 10.

I.4.2 IRConcrColumnCalcOptions

Class Hierarchy



C++

```
interface IRConcrColumnCalcOptions : IDispatch;
```

C#

```
public interface IRConcrColumnCalcOptions;
```

Visual Basic

```
Public Interface IRConcrColumnCalcOptions
```

Version

Available since version 10.

I.4.2.1 IRConcrColumnCalcOptions Members

The following tables list the members exposed by IRConcrColumnCalcOptions.

Public Fields

	Name	Description
	Cover (see page 1195)	

I.4.2.2 IRConcrColumnCalcOptions Fields

The fields of the IRConcrColumnCalcOptions class are listed here.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	Cover (see page 1195)	

I.4.2.2.1 Cover

C++

```
HRESULT get_Cover(double*);
HRESULT put_Cover(double);
```

C#

```
public double Cover { get; set; }
```

Visual Basic

```
Public Cover As double
```

Version

Available since version 10.

I.4.3 IRConcrColumnGeometry

Class Hierarchy

RobotOM.IRConcrColumnGeometry

C++

```
interface IRConcrColumnGeometry : IDispatch;
```

C#

```
public interface IRConcrColumnGeometry;
```

Visual Basic

```
Public Interface IRConcrColumnGeometry
```

Version

Available since version 10.

I.4.3.1 IRConcrColumnGeometry Members

The following tables list the members exposed by IRConcrColumnGeometry.

Public Fields

	Name	Description
	ConcreteVolume (see page 1197)	
	Dim (see page 1197)	
	DimIsFixed (see page 1197)	
	Section (see page 1197)	
	SectionName (see page 1198)	
	ShutteringArea (see page 1198)	

I.4.3.2 IRConcrColumnGeometry Fields

The fields of the IRConcrColumnGeometry class are listed here.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	ConcreteVolume (see page 1197)	
	Dim (see page 1197)	
	DimIsFixed (see page 1197)	
	Section (see page 1197)	
	SectionName (see page 1198)	
	ShutteringArea (see page 1198)	

I.4.3.2.1 ConcreteVolume

C++

```
HRESULT get_ConcreteVolume(double*);
```

C#

```
public double ConcreteVolume { get; }
```

Visual Basic

```
Public ReadOnly ConcreteVolume As double
```

Version

Available since version 10.

I.4.3.2.2 Dim

C++

```
HRESULT get_Dim(double*);
HRESULT put_Dim(double);
```

C#

```
public double Dim { get; set; }
```

Visual Basic

```
Public Dim As double
```

Version

Available since version 10.

I.4.3.2.3 DimIsFixed

C++

```
HRESULT get_DimIsFixed(VARIANT_BOOL*);
HRESULT put_DimIsFixed(VARIANT_BOOL);
```

C#

```
public bool DimIsFixed { get; set; }
```

Visual Basic

```
Public DimIsFixed As Boolean
```

Version

Available since version 11.

(c) 2014 Autodesk Inc. All rights reserved

I.4.3.2.4 Section

C++

```
HRESULT get_Section(IRConcrColumnSectionType*);
HRESULT put_Section(IRConcrColumnSectionType);
```

C#

```
public IRConcrColumnSectionType Section { get; set; }
```

Visual Basic

```
Public Section As IRConcrColumnSectionType
```

Version

Available since version 10.

I.4.3.2.5 SectionName

C++

```
HRESULT get_SectionName(BSTR*);
HRESULT put_SectionName(BSTR);
```

C#

```
public String SectionName { get; set; }
```

Visual Basic

```
Public SectionName As String
```

Version

Available since version 10.

I.4.3.2.6 ShutteringArea

C++

```
HRESULT get_ShutteringArea(double*);
```

C#

```
public double ShutteringArea { get; }
```

Visual Basic

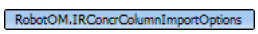
```
Public ReadOnly ShutteringArea As double
```

Version

Available since version 10.

I.4.4 IRConcrColumnImportOptions

Class Hierarchy



C++

```
interface IRConcrColumnImportOptions : IDispatch;
```

C#

```
public interface IRConcrColumnImportOptions;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Interface IRConcrColumnImportOptions
```

Version

Available since version 10.

I.4.4.1 IRConcrColumnImportOptions Members

The following tables list the members exposed by IRConcrColumnImportOptions.

Public Fields

	Name	Description
	GroupByGeometry (see page 1199)	
	GroupByLevel (see page 1199)	
	ImportCombinations (see page 1200)	
	RunCalcAuto (see page 1200)	
	ShowDialog (see page 1200)	

I.4.4.2 IRConcrColumnImportOptions Fields

The fields of the IRConcrColumnImportOptions class are listed here.

Public Fields

	Name	Description
	GroupByGeometry (see page 1199)	
	GroupByLevel (see page 1199)	
	ImportCombinations (see page 1200)	
	RunCalcAuto (see page 1200)	
	ShowDialog (see page 1200)	

I.4.4.2.1 GroupByGeometry

C++

```
HRESULT get_GroupByGeometry(VARIANT_BOOL*);
HRESULT put_GroupByGeometry(VARIANT_BOOL);
```

C#

```
public bool GroupByGeometry { get; set; }
```

Visual Basic

```
Public GroupByGeometry As Boolean
```

Version

Available since version 10.

I.4.4.2.2 GroupByLevel

C++

```
HRESULT get_GroupByLevel(VARIANT_BOOL*);
HRESULT put_GroupByLevel(VARIANT_BOOL);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public bool GroupByLevel { get; set; }
```

Visual Basic

```
Public GroupByLevel As Boolean
```

Version

Available since version 10.

I.4.4.2.3 ImportCombinations

C++

```
HRESULT get_ImportCombinations(VARIANT_BOOL*);
HRESULT put_ImportCombinations(VARIANT_BOOL);
```

C#

```
public bool ImportCombinations { get; set; }
```

Visual Basic

```
Public ImportCombinations As Boolean
```

Version

Available since version 10.

I.4.4.2.4 RunCalcAuto

C++

```
HRESULT get_RunCalcAuto(VARIANT_BOOL*);
HRESULT put_RunCalcAuto(VARIANT_BOOL);
```

C#

```
public bool RunCalcAuto { get; set; }
```

Visual Basic

```
Public RunCalcAuto As Boolean
```

Version

Available since version 10.

I.4.4.2.5 ShowDialog

C++

```
HRESULT get_ShowDialog(VARIANT_BOOL*);
HRESULT put_ShowDialog(VARIANT_BOOL);
```

C#

```
public bool ShowDialog { get; set; }
```

Visual Basic

```
Public ShowDialog As Boolean
```

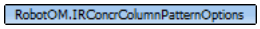
Version

Available since version 10.

(c) 2014 Autodesk Inc. All rights reserved

I.4.5 IRConcrColumnPatternOptions

Class Hierarchy



C++

```
interface IRConcrColumnPatternOptions : IDispatch;
```

C#

```
public interface IRConcrColumnPatternOptions;
```

Visual Basic

```
Public Interface IRConcrColumnPatternOptions
```

Version

Available since version 10.

I.4.5.1 IRConcrColumnPatternOptions Members

The following tables list the members exposed by IRConcrColumnPatternOptions.

Public Fields

	Name	Description
	TiesToBeam (see page 1201)	

I.4.5.2 IRConcrColumnPatternOptions Fields

The fields of the IRConcrColumnPatternOptions class are listed here.

Public Fields

	Name	Description
	TiesToBeam (see page 1201)	

I.4.5.2.1 TiesToBeam

C++

```
HRESULT get_TiesToBeam(VARIANT_BOOL*);
HRESULT put_TiesToBeam(VARIANT_BOOL);
```

C#

```
public bool TiesToBeam { get; set; }
```

Visual Basic

```
Public TiesToBeam As Boolean
```

Version

Available since version 10.

I.4.6 IRConcrColumnDimensionType

C++

```
enum IRConcrColumnDimensionType;
```

C#

```
public enum IRConcrColumnDimensionType;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Enum IRConcrColumnDimensionType
```

Members

Members	Description
I_CCD_COLUMN_DIM_HSP = 1	Available since version 10.
I_CCD_COLUMN_DIM_HSD = 2	Available since version 10.
I_CCD_COLUMN_DIM_EPD = 3	Available since version 10.
I_CCD_COLUMN_DIM_A = 4	Available since version 10.
I_CCD_COLUMN_DIM_B = 5	Available since version 10.
I_CCD_COLUMN_DIM_C = 6	Available since version 10.
I_CCD_COLUMN_DIM_H1 = 7	Available since version 10.
I_CCD_COLUMN_DIM_L1 = 8	Available since version 10.
I_CCD_COLUMN_DIM_H2 = 9	Available since version 10.
I_CCD_COLUMN_DIM_L2 = 10	Available since version 10.
I_CCD_COLUMN_DIM_N = 11	Available since version 10.
I_CCD_COLUMN_DIM_DE = 12	Available since version 10.
I_CCD_COLUMN_DIM_HSP_X = 1	Available since version 10.
I_CCD_COLUMN_DIM_HSP_Y = 13	Available since version 10.

Version

Available since version 10.

I.4.7 IRConcrColumnSectionType

C++

```
enum IRConcrColumnSectionType;
```

C#

```
public enum IRConcrColumnSectionType;
```

Visual Basic

```
Public Enum IRConcrColumnSectionType
```

Members

Members	Description
I_CCT_COLUMN_SEC_R = 1	Available since version 10.
I_CCT_COLUMN_SEC_L = 2	Available since version 10.
I_CCT_COLUMN_SEC_TC = 3	Available since version 10.
I_CCT_COLUMN_SEC_Z = 4	Available since version 10.
I_CCT_COLUMN_SEC_P = 5	Available since version 10.
I_CCT_COLUMN_SEC_C36 = 6	Available since version 10.
I_CCT_COLUMN_SEC_C18 = 7	Available since version 10.
I_CCT_COLUMN_SEC_C90 = 8	Available since version 10.

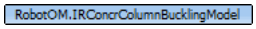
Version

Available since version 10.

(c) 2014 Autodesk Inc. All rights reserved

I.4.8 IRConcrColumnBucklingModel

Class Hierarchy



C++

```
interface IRConcrColumnBucklingModel : IDispatch;
```

C#

```
public interface IRConcrColumnBucklingModel;
```

Visual Basic

```
Public Interface IRConcrColumnBucklingModel
```

Version

Available since version 11.

I.4.8.1 IRConcrColumnBucklingModel Members

The following tables list the members exposed by IRConcrColumnBucklingModel.

Public Fields

	Name	Description
	IsDirectionYOff (see page 1204)	
	IsDirectionZOff (see page 1204)	
	IsSwayY (see page 1204)	
	IsSwayZ (see page 1204)	
	IsTotalStructureHeight (see page 1205)	
	ky (see page 1205)	
	kz (see page 1205)	
	Ly (see page 1205)	
	Lz (see page 1206)	
	NumberOfStories (see page 1206)	
	TotalStructureHeight (see page 1206)	

I.4.8.2 IRConcrColumnBucklingModel Fields

The fields of the IRConcrColumnBucklingModel class are listed here.

Public Fields

	Name	Description
	IsDirectionYOff (see page 1204)	
	IsDirectionZOff (see page 1204)	
	IsSwayY (see page 1204)	
	IsSwayZ (see page 1204)	
	IsTotalStructureHeight (see page 1205)	
	ky (see page 1205)	
	kz (see page 1205)	
	Ly (see page 1205)	

(c) 2014 Autodesk Inc. All rights reserved

	Lz (see page 1206)	
	NumberOfStories (see page 1206)	
	TotalStructureHeight (see page 1206)	

I.4.8.2.1 IsDirectionYOff

C++

```
HRESULT get_IsDirectionYOff(VARIANT_BOOL*);
HRESULT put_IsDirectionYOff(VARIANT_BOOL);
```

C#

```
public bool IsDirectionYOff { get; set; }
```

Visual Basic

```
Public IsDirectionYOff As Boolean
```

Version

Available since version 11.

I.4.8.2.2 IsDirectionZOff

C++

```
HRESULT get_IsDirectionZOff(VARIANT_BOOL*);
HRESULT put_IsDirectionZOff(VARIANT_BOOL);
```

C#

```
public bool IsDirectionZOff { get; set; }
```

Visual Basic

```
Public IsDirectionZOff As Boolean
```

Version

Available since version 11.

I.4.8.2.3 IsSwayY

C++

```
HRESULT get_IsSwayY(VARIANT_BOOL*);
HRESULT put_IsSwayY(VARIANT_BOOL);
```

C#

```
public bool IsSwayY { get; set; }
```

Visual Basic

```
Public IsSwayY As Boolean
```

Version

Available since version 11.

I.4.8.2.4 IsSwayZ

C++

```
HRESULT get_IsSwayZ(VARIANT_BOOL*);
HRESULT put_IsSwayZ(VARIANT_BOOL);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public bool IsSwayZ { get; set; }
```

Visual Basic

```
Public IsSwayZ As Boolean
```

Version

Available since version 11.

I.4.8.2.5 IsTotalStructureHeight

C++

```
HRESULT get_IsTotalStructureHeight(VARIANT_BOOL*);
HRESULT put_IsTotalStructureHeight(VARIANT_BOOL);
```

C#

```
public bool IsTotalStructureHeight { get; set; }
```

Visual Basic

```
Public IsTotalStructureHeight As Boolean
```

Version

Available since version 11.

I.4.8.2.6 ky

C++

```
HRESULT get_ky(double*);
HRESULT put_ky(double);
```

C#

```
public double ky { get; set; }
```

Visual Basic

```
Public ky As double
```

Version

Available since version 11.

I.4.8.2.7 kz

C++

```
HRESULT get_kz(double*);
HRESULT put_kz(double);
```

C#

```
public double kz { get; set; }
```

Visual Basic

```
Public kz As double
```

Version

Available since version 11.

I.4.8.2.8 Ly

C++

```
HRESULT get_Ly(double*);
```

(c) 2014 Autodesk Inc. All rights reserved

```
HRESULT put_Ly(double);
```

C#

```
public double Ly { get; set; }
```

Visual Basic

```
Public Ly As double
```

Version

Available since version 11.

I.4.8.2.9 Lz

C++

```
HRESULT get_Lz(double*);
HRESULT put_Lz(double);
```

C#

```
public double Lz { get; set; }
```

Visual Basic

```
Public Lz As double
```

Version

Available since version 11.

I.4.8.2.10 NumberOfStories

C++

```
HRESULT get_NumberOfStories( long*);
HRESULT put_NumberOfStories( long);
```

C#

```
public  long NumberOfStories { get; set; }
```

Visual Basic

```
Public NumberOfStories As  long
```

Version

Available since version 11.

I.4.8.2.11 TotalStructureHeight

C++

```
HRESULT get_TotalStructureHeight( double*);
HRESULT put_TotalStructureHeight( double);
```

C#

```
public  double TotalStructureHeight { get; set; }
```

Visual Basic

```
Public TotalStructureHeight As  double
```

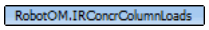
Version

Available since version 11.

(c) 2014 Autodesk Inc. All rights reserved

I.4.9 IRConcrColumnLoads

Class Hierarchy



C++

```
interface IRConcrColumnLoads : IDispatch;
```

C#

```
public interface IRConcrColumnLoads;
```

Visual Basic

```
Public Interface IRConcrColumnLoads
```

Version

Available since version 11.

I.4.9.1 IRConcrColumnLoads Members

The following tables list the members exposed by IRConcrColumnLoads.

Public Fields

	Name	Description
	Count (see page 1207)	
	Item (see page 1208)	

Public Methods

	Name	Description
	Add (see page 1208)	
	RemoveAll (see page 1208)	

I.4.9.2 IRConcrColumnLoads Fields

The fields of the IRConcrColumnLoads class are listed here.

Public Fields

	Name	Description
	Count (see page 1207)	
	Item (see page 1208)	

I.4.9.2.1 Count

C++

```
HRESULT get_Count(long*);
```

C#

```
public long Count { get; }
```

Visual Basic

```
Public ReadOnly Count As long
```

Version

Available since version 11.

(c) 2014 Autodesk Inc. All rights reserved

I.4.9.2.2 Item

C++

```
HRESULT get_Item(IRConcrColumnLoad**);
```

C#

```
public IRConcrColumnLoad Item { get; }
```

Visual Basic

```
Public ReadOnly Item As IRConcrColumnLoad
```

Version

Available since version 11.

I.4.9.3 IRConcrColumnLoads Methods

The methods of the IRConcrColumnLoads class are listed here.

Public Methods

	Name	Description
	Add (see page 1208)	
	RemoveAll (see page 1208)	

I.4.9.3.1 Add

C++

```
HRESULT Add(IRConcrColumnLoad** ret);
```

C#

```
public IRConcrColumnLoad Add();
```

Visual Basic

```
Public Function Add() As IRConcrColumnLoad
```

Version

Available since version 11.

I.4.9.3.2 RemoveAll

C++

```
HRESULT RemoveAll();
```

C#

```
public void RemoveAll();
```

Visual Basic

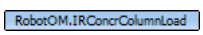
```
Public Sub RemoveAll()
```

Version

Available since version 11.

I.4.10 IRConcrColumnLoad

Class Hierarchy



(c) 2014 Autodesk Inc. All rights reserved

C++

```
interface IRConcrColumnLoad : IDispatch;
```

C#

```
public interface IRConcrColumnLoad;
```

Visual Basic

```
Public Interface IRConcrColumnLoad
```

Version

Available since version 11.

I.4.10.1 IRConcrColumnLoad Members

The following tables list the members exposed by IRConcrColumnLoad.

Public Fields

	Name	Description
	CaseName (see page 1210)	
	CaseType (see page 1210)	
	Fy (see page 1210)	
	Fz (see page 1211)	
	Gamma (see page 1211)	
	MnsY (see page 1211)	
	MnsZ (see page 1211)	
	MyA (see page 1212)	
	MyB (see page 1212)	
	MyC (see page 1212)	
	MzA (see page 1212)	
	MzB (see page 1213)	
	MzC (see page 1213)	
	N (see page 1213)	
	NdN (see page 1213)	

I.4.10.2 IRConcrColumnLoad Fields

The fields of the IRConcrColumnLoad class are listed here.

Public Fields

	Name	Description
	CaseName (see page 1210)	
	CaseType (see page 1210)	
	Fy (see page 1210)	
	Fz (see page 1211)	
	Gamma (see page 1211)	
	MnsY (see page 1211)	
	MnsZ (see page 1211)	
	MyA (see page 1212)	
	MyB (see page 1212)	
	MyC (see page 1212)	
	MzA (see page 1212)	
	MzB (see page 1213)	

(c) 2014 Autodesk Inc. All rights reserved

	MzC (see page 1213)	
	N (see page 1213)	
	NdN (see page 1213)	

I.4.10.2.1 CaseName

C++

```
HRESULT get_CaseName(BSTR*);
HRESULT put_CaseName(BSTR);
```

C#

```
public String CaseName { get; set; }
```

Visual Basic

```
Public CaseName As String
```

Version

Available since version 11.

I.4.10.2.2 CaseType

C++

```
HRESULT get_CaseType(IRConcrColumnLoadCaseType*);
HRESULT put_CaseType(IRConcrColumnLoadCaseType);
```

C#

```
public IRConcrColumnLoadCaseType CaseType { get; set; }
```

Visual Basic

```
Public CaseType As IRConcrColumnLoadCaseType
```

Version

Available since version 11.

I.4.10.2.3 Fy

C++

```
HRESULT get_Fy(double*);
HRESULT put_Fy(double);
```

C#

```
public double Fy { get; set; }
```

Visual Basic

```
Public Fy As double
```

Version

Available since version 11.

I.4.10.2.4 Fz

C++

```
HRESULT get_Fz(double*);
HRESULT put_Fz(double);
```

C#

```
public double Fz { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Fz As double
```

Version

Available since version 11.

I.4.10.2.5 Gamma

C++

```
HRESULT get_Gamma(double*);
HRESULT put_Gamma(double);
```

C#

```
public double Gamma { get; set; }
```

Visual Basic

```
Public Gamma As double
```

Version

Available since version 11.

I.4.10.2.6 MnsY

C++

```
HRESULT get_MnsY(double*);
HRESULT put_MnsY(double);
```

C#

```
public double MnsY { get; set; }
```

Visual Basic

```
Public MnsY As double
```

Version

Available since version 11.

I.4.10.2.7 MnsZ

C++

```
HRESULT get_MnsZ(double*);
HRESULT put_MnsZ(double);
```

C#

```
public double MnsZ { get; set; }
```

Visual Basic

```
Public MnsZ As double
```

Version

Available since version 11.

I.4.10.2.8 MyA

C++

```
HRESULT get_MyA(double*);
HRESULT put_MyA(double);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double MyA { get; set; }
```

Visual Basic

```
Public MyA As double
```

Version

Available since version 11.

I.4.10.2.9 MyB

C++

```
HRESULT get_MyB(double*);
HRESULT put_MyB(double);
```

C#

```
public double MyB { get; set; }
```

Visual Basic

```
Public MyB As double
```

Version

Available since version 11.

I.4.10.2.10 MyC

C++

```
HRESULT get_MyC(double*);
HRESULT put_MyC(double);
```

C#

```
public double MyC { get; set; }
```

Visual Basic

```
Public MyC As double
```

Version

Available since version 11.

I.4.10.2.11 MzA

C++

```
HRESULT get_MzA(double*);
HRESULT put_MzA(double);
```

C#

```
public double MzA { get; set; }
```

Visual Basic

```
Public MzA As double
```

Version

Available since version 11.

I.4.10.2.12 MzB

C++

```
HRESULT get_MzB(double*);
```

(c) 2014 Autodesk Inc. All rights reserved

```
HRESULT put_MzB(double);
```

C#

```
public double MzB { get; set; }
```

Visual Basic

```
Public MzB As double
```

Version

Available since version 11.

I.4.10.2.13 MzC

C++

```
HRESULT get_MzC(double*);
HRESULT put_MzC(double);
```

C#

```
public double MzC { get; set; }
```

Visual Basic

```
Public MzC As double
```

Version

Available since version 11.

I.4.10.2.14 N

C++

```
HRESULT get_N(double*);
HRESULT put_N(double);
```

C#

```
public double N { get; set; }
```

Visual Basic

```
Public N As double
```

Version

Available since version 11.

I.4.10.2.15 NdN

C++

```
HRESULT get_NdN(double*);
HRESULT put_NdN(double);
```

C#

```
public double NdN { get; set; }
```

Visual Basic

```
Public NdN As double
```

Version

Available since version 11.

(c) 2014 Autodesk Inc. All rights reserved

I.4.11 IRConcrColumnLoadCaseType

C++

```
enum IRConcrColumnLoadCaseType;
```

C#

```
public enum IRConcrColumnLoadCaseType;
```

Visual Basic

```
Public Enum IRConcrColumnLoadCaseType
```

Members

Members	Description
I_RCL_LOAD_CASE_PERMANENT = 1	Available since version 11.
I_RCL_LOAD_CASE_EXPLOITATION = 2	Available since version 11.
I_RCL_LOAD_CASE_ACCIDENTAL = 3	Available since version 11.
I_RCL_LOAD_CASE_WIND = 4	Available since version 11.
I_RCL_LOAD_CASE_DESIGN = 5	Available since version 11.
I_RCL_LOAD_CASE_DESIGN_ACC = 6	Available since version 11.
I_RCL_LOAD_CASE_TEMPERATURE = 7	Available since version 11.
I_RCL_LOAD_CASE_SNOW = 8	Available since version 11.
I_RCL_LOAD_CASE_TEST = 9	Available since version 11.
I_RCL_LOAD_CASE_SEISMIC = 10	Available since version 11.

Version

Available since version 11.

I.5 RC Slab

Available since version 10.

Enumerations

	Name	Description
	IRConcrSlabRnfType (see page 1226)	
	IRConcrSlabRnfSegmentType (see page 1227)	
	IRConcrSlabSupportType (see page 1227)	

Interfaces

	Name	Description
	IRConcrSlab (see page 1215)	
	IRConcrSlabCalculationOptions (see page 1220)	
	IRConcrSlabGeometry (see page 1221)	
	IRConcrSlabPatternOptions (see page 1222)	
	IRConcrSlabWizard (see page 1223)	

(c) 2014 Autodesk Inc. All rights reserved

I.5.1 IRConcrSlab

Class Hierarchy

RobotOM.IRConcrSlab

C++

```
interface IRConcrSlab : IDispatch;
```

C#

```
public interface IRConcrSlab;
```

Visual Basic

```
Public Interface IRConcrSlab
```

Version

Available since version 10.

I.5.1.1 IRConcrSlab Members

The following tables list the members exposed by IRConcrSlab.

Public Fields

	Name	Description
	CalculationOptions (see page 1216)	
	Concrete (see page 1216)	
	Geometry (see page 1216)	
	IsSelected (see page 1217)	
	Name (see page 1217)	
	PatternOptions (see page 1217)	
	Reinforcement (see page 1217)	
	Steel (see page 1218)	
	UniqueId (see page 1218)	

Public Methods

	Name	Description
	Activate (see page 1218)	
	CreateCalculationNoteRtf (see page 1219)	
	CreateFromObjects (see page 1219)	
	Save (see page 1219)	
	StartWizard (see page 1220)	

I.5.1.2 IRConcrSlab Fields

The fields of the IRConcrSlab class are listed here.

Public Fields

	Name	Description
	CalculationOptions (see page 1216)	
	Concrete (see page 1216)	
	Geometry (see page 1216)	

(c) 2014 Autodesk Inc. All rights reserved

	IsSelected (see page 1217)	
	Name (see page 1217)	
	PatternOptions (see page 1217)	
	Reinforcement (see page 1217)	
	Steel (see page 1218)	
	UniqueId (see page 1218)	

I.5.1.2.1 CalculationOptions

C++

```
HRESULT get_CalculationOptions(IRConcrSlabCalculationOptions**);
```

C#

```
public IRConcrSlabCalculationOptions CalculationOptions { get; }
```

Visual Basic

```
Public ReadOnly CalculationOptions As IRConcrSlabCalculationOptions
```

Version

Available since version 10.

I.5.1.2.2 Concrete

C++

```
HRESULT get_Concrete(IRConcrConcrete**);
```

C#

```
public IRConcrConcrete Concrete { get; }
```

Visual Basic

```
Public ReadOnly Concrete As IRConcrConcrete
```

Version

Available since version 10.

I.5.1.2.3 Geometry

C++

```
HRESULT get_Geometry(IRConcrSlabGeometry**);
```

C#

```
public IRConcrSlabGeometry Geometry { get; }
```

Visual Basic

```
Public ReadOnly Geometry As IRConcrSlabGeometry
```

Version

Available since version 10.

I.5.1.2.4 IsSelected

C++

```
HRESULT get_IsSelected(VARIANT_BOOL*);
HRESULT put_IsSelected(VARIANT_BOOL);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public bool IsSelected { get; set; }
```

Visual Basic

```
Public IsSelected As Boolean
```

Version

Available since version 12.

I.5.1.2.5 Name

C++

```
HRESULT get_Name(BSTR*);
```

C#

```
public String Name { get; }
```

Visual Basic

```
Public ReadOnly Name As String
```

Version

Available since version 10.

I.5.1.2.6 PatternOptions

C++

```
HRESULT get_PatternOptions(IRConcrSlabPatternOptions**);
```

C#

```
public IRConcrSlabPatternOptions PatternOptions { get; }
```

Visual Basic

```
Public ReadOnly PatternOptions As IRConcrSlabPatternOptions
```

Version

Available since version 10.

I.5.1.2.7 Reinforcement

C++

```
HRESULT get_Reinforcement(IRConcrReinforcement**);
```

C#

```
public IRConcrReinforcement Reinforcement { get; }
```

Visual Basic

```
Public ReadOnly Reinforcement As IRConcrReinforcement
```

Version

Available since version 10.

I.5.1.2.8 Steel

C++

```
HRESULT get_Steel(IRConcrSteel**);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRConcrSteel Steel { get; }
```

Visual Basic

```
Public ReadOnly Steel As IRConcrSteel
```

Version

Available since version 10.

I.5.1.2.9 UniqueId

C++

```
HRESULT get_UniqueId(long*);
```

C#

```
public long UniqueId { get; }
```

Visual Basic

```
Public ReadOnly UniqueId As long
```

Version

Available since version 10.

I.5.1.3 IRConcrSlab Methods

The methods of the IRConcrSlab class are listed here.

Public Methods

	Name	Description
	Activate (see page 1218)	
	CreateCalculationNoteRtf (see page 1219)	
	CreateFromObjects (see page 1219)	
	Save (see page 1219)	
	StartWizard (see page 1220)	

I.5.1.3.1 Activate

C++

```
HRESULT Activate();
```

C#

```
public void Activate();
```

Visual Basic

```
Public Sub Activate()
```

Version

Available since version 10.

I.5.1.3.2 CreateCalculationNoteRtf

C++

```
HRESULT CreateCalculationNoteRtf(BSTR _fileName, VARIANT_BOOL* ret);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public bool CreateCalculationNoteRtf(String _fileName);
```

Visual Basic

```
Public Function CreateCalculationNoteRtf(_fileName As String) As Boolean
```

Version

Available since version 10.

I.5.1.3.3 CreateFromObjects

C++

```
HRESULT CreateFromObjects(BSTR _selText, VARIANT_BOOL* ret);
```

C#

```
public bool CreateFromObjects(String _selText);
```

Visual Basic

```
Public Function CreateFromObjects(_selText As String) As Boolean
```

Version

Available since version 10.

I.5.1.3.4 Save

C++

```
HRESULT Save();
```

C#

```
public void Save();
```

Visual Basic

```
Public Sub Save()
```

Version

Available since version 10.

I.5.1.3.5 StartWizard

C++

```
HRESULT StartWizard(IRConcrSlabWizard** ret);
```

C#

```
public IRConcrSlabWizard StartWizard();
```

Visual Basic

```
Public Function StartWizard() As IRConcrSlabWizard
```

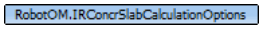
Version

Available since version 10.

(c) 2014 Autodesk Inc. All rights reserved

I.5.2 IRConcrSlabCalculationOptions

Class Hierarchy



C++

```
interface IRConcrSlabCalculationOptions : IDispatch;
```

C#

```
public interface IRConcrSlabCalculationOptions;
```

Visual Basic

```
Public Interface IRConcrSlabCalculationOptions
```

Version

Available since version 10.

I.5.2.1 IRConcrSlabCalculationOptions Members

The following tables list the members exposed by IRConcrSlabCalculationOptions.

Public Fields

	Name	Description
	Cover (see page 1220)	
	CoverLateral (see page 1221)	

I.5.2.2 IRConcrSlabCalculationOptions Fields

The fields of the IRConcrSlabCalculationOptions class are listed here.

Public Fields

	Name	Description
	Cover (see page 1220)	
	CoverLateral (see page 1221)	

I.5.2.2.1 Cover

C++

```
HRESULT get_Cover(double*);
HRESULT put_Cover(double);
```

C#

```
public double Cover { get; set; }
```

Visual Basic

```
Public Cover As double
```

Version

Available since version 10.

I.5.2.2.2 CoverLateral

C++

```
HRESULT get_CoverLateral(double*);
HRESULT put_CoverLateral(double);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double CoverLateral { get; set; }
```

Visual Basic

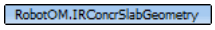
```
Public CoverLateral As double
```

Version

Available since version 10.

I.5.3 IRConcrSlabGeometry

Class Hierarchy



C++

```
interface IRConcrSlabGeometry : IDispatch;
```

C#

```
public interface IRConcrSlabGeometry;
```

Visual Basic

```
Public Interface IRConcrSlabGeometry
```

Version

Available since version 10.

I.5.3.1 IRConcrSlabGeometry Members

The following tables list the members exposed by IRConcrSlabGeometry.

Public Fields

	Name	Description
	ConcreteVolume (see page 1222)	
	ShutteringArea (see page 1222)	

I.5.3.2 IRConcrSlabGeometry Fields

The fields of the IRConcrSlabGeometry class are listed here.

Public Fields

	Name	Description
	ConcreteVolume (see page 1222)	
	ShutteringArea (see page 1222)	

I.5.3.2.1 ConcreteVolume

C++

```
HRESULT get_ConcreteVolume(double*);
```

C#

```
public double ConcreteVolume { get; }
```

Visual Basic

```
Public ReadOnly ConcreteVolume As double
```

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 10.

I.5.3.2.2 ShutteringArea

C++

```
HRESULT get_ShutteringArea(double*);
```

C#

```
public double ShutteringArea { get; }
```

Visual Basic

```
Public ReadOnly ShutteringArea As double
```

Version

Available since version 10.

I.5.4 IRConcrSlabPatternOptions

Class Hierarchy

RobotOM.IRConcrSlabPatternOptions

C++

```
interface IRConcrSlabPatternOptions : IDispatch;
```

C#

```
public interface IRConcrSlabPatternOptions;
```

Visual Basic

```
Public Interface IRConcrSlabPatternOptions
```

Version

Available since version 10.

I.5.4.1 IRConcrSlabPatternOptions Members

The following tables list the members exposed by IRConcrSlabPatternOptions.

Public Fields

	Name	Description
	ReinforcementSegment (see page 1223)	
	ReinforcementType (see page 1223)	

I.5.4.2 IRConcrSlabPatternOptions Fields

The fields of the IRConcrSlabPatternOptions class are listed here.

Public Fields

	Name	Description
	ReinforcementSegment (see page 1223)	
	ReinforcementType (see page 1223)	

(c) 2014 Autodesk Inc. All rights reserved

I.5.4.2.1 ReinforcementSegment

C++

```
HRESULT get_ReinforcementSegment(IRConcrSlabRnfSegmentType*);
HRESULT put_ReinforcementSegment(IRConcrSlabRnfSegmentType);
```

C#

```
public IRConcrSlabRnfSegmentType ReinforcementSegment { get; set; }
```

Visual Basic

```
Public ReinforcementSegment As IRConcrSlabRnfSegmentType
```

Version

Available since version 10.

I.5.4.2.2 ReinforcementType

C++

```
HRESULT get_ReinforcementType(IRConcrSlabRnfType*);
HRESULT put_ReinforcementType(IRConcrSlabRnfType);
```

C#

```
public IRConcrSlabRnfType ReinforcementType { get; set; }
```

Visual Basic

```
Public ReinforcementType As IRConcrSlabRnfType
```

Version

Available since version 10.

I.5.5 IRConcrSlabWizard

Class Hierarchy

RobotOM.IRConcrSlabWizard

C++

```
interface IRConcrSlabWizard : IDispatch;
```

C#

```
public interface IRConcrSlabWizard;
```

Visual Basic

```
Public Interface IRConcrSlabWizard
```

Version

Available since version 10.

I.5.5.1 IRConcrSlabWizard Members

The following tables list the members exposed by IRConcrSlabWizard.

Public Fields

	Name	Description
	Thickness (see page 1224)	

(c) 2014 Autodesk Inc. All rights reserved

Public Methods

	Name	Description
	AddColumnSupport (see page 1225)	
	AddOpening (see page 1225)	
	AddOpeningNode (see page 1225)	
	AddSlabNode (see page 1226)	
	AddWallSupport (see page 1226)	
	Finish (see page 1226)	

I.5.5.2 IRConcrSlabWizard Fields

The fields of the IRConcrSlabWizard class are listed here.

Public Fields

	Name	Description
	Thickness (see page 1224)	

I.5.5.2.1 Thickness

C++

```
HRESULT get_Thickness(double*);
HRESULT put_Thickness(double);
```

C#

```
public double Thickness { get; set; }
```

Visual Basic

```
Public Thickness As double
```

Version

Available since version 10.

I.5.5.3 IRConcrSlabWizard Methods

The methods of the IRConcrSlabWizard class are listed here.

Public Methods

	Name	Description
	AddColumnSupport (see page 1225)	
	AddOpening (see page 1225)	
	AddOpeningNode (see page 1225)	
	AddSlabNode (see page 1226)	
	AddWallSupport (see page 1226)	
	Finish (see page 1226)	

I.5.5.3.1 AddColumnSupport

C++

```
HRESULT AddColumnSupport(double _x, double _y, double _b, double _h, double _alpha);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public void AddColumnSupport(double _x, double _y, double _b, double _h, double _alpha);
```

Visual Basic

```
Public Sub AddColumnSupport(_x As double, _y As double, _b As double, _h As double, _alpha
As double)
```

Version

Available since version 10.

I.5.5.3.2 AddOpening

C++

```
HRESULT AddOpening();
```

C#

```
public void AddOpening();
```

Visual Basic

```
Public Sub AddOpening()
```

Version

Available since version 10.

I.5.5.3.3 AddOpeningNode

C++

```
HRESULT AddOpeningNode(double _x, double _y);
```

C#

```
public void AddOpeningNode(double _x, double _y);
```

Visual Basic

```
Public Sub AddOpeningNode(_x As double, _y As double)
```

Version

Available since version 10.

I.5.5.3.4 AddSlabNode

C++

```
HRESULT AddSlabNode(double _x, double _y);
```

C#

```
public void AddSlabNode(double _x, double _y);
```

Visual Basic

```
Public Sub AddSlabNode(_x As double, _y As double)
```

Version

Available since version 10.

(c) 2014 Autodesk Inc. All rights reserved

I.5.5.3.5 AddWallSupport

C++

```
HRESULT AddWallSupport(double _x1, double _y1, double _x2, double _y2, double _thickness,
IRConcrSlabSupportType _type);
```

C#

```
public void AddWallSupport(double _x1, double _y1, double _x2, double _y2, double
_thickness, IRConcrSlabSupportType _type);
```

Visual Basic

```
Public Sub AddWallSupport(_x1 As double, _y1 As double, _x2 As double, _y2 As double,
_thickness As double, _type As IRConcrSlabSupportType)
```

Version

Available since version 10.

I.5.5.3.6 Finish

C++

```
HRESULT Finish(VARIANT_BOOL* ret);
```

C#

```
public bool Finish();
```

Visual Basic

```
Public Function Finish() As Boolean
```

Version

Available since version 10.

I.5.6 IRConcrSlabRnfType

C++

```
enum IRConcrSlabRnfType;
```

C#

```
public enum IRConcrSlabRnfType;
```

Visual Basic

```
Public Enum IRConcrSlabRnfType
```

Members

Members	Description
I_CSR_SLAB_RNF_BAR_ONLY = 1	Available since version 10.
I_CSR_SLAB_RNF_NET_ONLY = 2	Available since version 10.
I_CSR_SLAB_RNF_BAR_AND_NET = 3	Available since version 10.

Version

Available since version 10.

(c) 2014 Autodesk Inc. All rights reserved

I.5.7 IRConcrSlabRnfSegmentType

C++

```
enum IRConcrSlabRnfSegmentType;
```

C#

```
public enum IRConcrSlabRnfSegmentType;
```

Visual Basic

```
Public Enum IRConcrSlabRnfSegmentType
```

Members

Members	Description
I_SRS_SLAB_SEG_SINGLE = 1	Available since version 10.
I_SRS_SLAB_SEG_ENTIRE = 2	Available since version 10.

Version

Available since version 10.

I.5.8 IRConcrSlabSupportType

C++

```
enum IRConcrSlabSupportType;
```

C#

```
public enum IRConcrSlabSupportType;
```

Visual Basic

```
Public Enum IRConcrSlabSupportType
```

Members

Members	Description
I_CSS_SLAB_SUPPORT_PINNED = 1	Available since version 10.
I_CSS_SLAB_SUPPORT_FIXED = 2	Available since version 10.

Version

Available since version 10.

I.6 RC Deep Beam

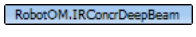
Available since version 10.

Interfaces

	Name	Description
	IRConcrDeepBeam (see page 1228)	
	IRConcrDeepBeamGeometry (see page 1232)	

I.6.1 IRConcrDeepBeam

Class Hierarchy



(c) 2014 Autodesk Inc. All rights reserved

C++

```
interface IRConcrDeepBeam : IDispatch;
```

C#

```
public interface IRConcrDeepBeam;
```

Visual Basic

```
Public Interface IRConcrDeepBeam
```

Version

Available since version 10.

I.6.1.1 IRConcrDeepBeam Members

The following tables list the members exposed by IRConcrDeepBeam.

Public Fields

	Name	Description
	Concrete (see page 1229)	
	Geometry (see page 1229)	
	IsActive (see page 1229)	
	IsSelected (see page 1230)	
	Name (see page 1230)	
	Reinforcement (see page 1230)	
	Steel (see page 1230)	
	UniqueId (see page 1231)	

Public Methods

	Name	Description
	Activate (see page 1231)	
	CreateCalculationNoteRtf (see page 1231)	
	CreateFromObjects (see page 1232)	
	Save (see page 1232)	

I.6.1.2 IRConcrDeepBeam Fields

The fields of the IRConcrDeepBeam class are listed here.

Public Fields

	Name	Description
	Concrete (see page 1229)	
	Geometry (see page 1229)	
	IsActive (see page 1229)	
	IsSelected (see page 1230)	
	Name (see page 1230)	
	Reinforcement (see page 1230)	
	Steel (see page 1230)	
	UniqueId (see page 1231)	

(c) 2014 Autodesk Inc. All rights reserved

I.6.1.2.1 Concrete

C++

```
HRESULT get_Concrete(IRConcrConcrete**);
```

C#

```
public IRConcrConcrete Concrete { get; }
```

Visual Basic

```
Public ReadOnly Concrete As IRConcrConcrete
```

Version

Available since version 10.

I.6.1.2.2 Geometry

C++

```
HRESULT get_Geometry(IRConcrDeepBeamGeometry**);
```

C#

```
public IRConcrDeepBeamGeometry Geometry { get; }
```

Visual Basic

```
Public ReadOnly Geometry As IRConcrDeepBeamGeometry
```

Version

Available since version 10.

I.6.1.2.3 IsActive

C++

```
HRESULT get_IsActive(VARIANT_BOOL*);
```

C#

```
public bool IsActive { get; }
```

Visual Basic

```
Public ReadOnly IsActive As Boolean
```

Version

Available since version 10.

I.6.1.2.4 IsSelected

C++

```
HRESULT get_IsSelected(VARIANT_BOOL*);
HRESULT put_IsSelected(VARIANT_BOOL);
```

C#

```
public bool IsSelected { get; set; }
```

Visual Basic

```
Public IsSelected As Boolean
```

Version

Available since version 12.

(c) 2014 Autodesk Inc. All rights reserved

I.6.1.2.5 Name

C++

```
HRESULT get_Name(BSTR*);
```

C#

```
public String Name { get; }
```

Visual Basic

```
Public ReadOnly Name As String
```

Version

Available since version 10.

I.6.1.2.6 Reinforcement

C++

```
HRESULT get_Reinforcement(IRConcrReinforcement**);
```

C#

```
public IRConcrReinforcement Reinforcement { get; }
```

Visual Basic

```
Public ReadOnly Reinforcement As IRConcrReinforcement
```

Version

Available since version 10.

I.6.1.2.7 Steel

C++

```
HRESULT get_Steel(IRConcrSteel**);
```

C#

```
public IRConcrSteel Steel { get; }
```

Visual Basic

```
Public ReadOnly Steel As IRConcrSteel
```

Version

Available since version 10.

I.6.1.2.8 UniqueId

C++

```
HRESULT get_UniqueId(long*);
```

C#

```
public long UniqueId { get; }
```

Visual Basic

```
Public ReadOnly UniqueId As long
```

Version

Available since version 10.

(c) 2014 Autodesk Inc. All rights reserved

I.6.1.3 IRConcrDeepBeam Methods

The methods of the IRConcrDeepBeam class are listed here.

Public Methods

	Name	Description
	Activate (see page 1231)	
	CreateCalculationNoteRtf (see page 1231)	
	CreateFromObjects (see page 1232)	
	Save (see page 1232)	

I.6.1.3.1 Activate

C++

```
HRESULT Activate();
```

C#

```
public void Activate();
```

Visual Basic

```
Public Sub Activate()
```

Version

Available since version 10.

I.6.1.3.2 CreateCalculationNoteRtf

C++

```
HRESULT CreateCalculationNoteRtf(BSTR _fileName, VARIANT_BOOL* ret);
```

C#

```
public bool CreateCalculationNoteRtf(String _fileName);
```

Visual Basic

```
Public Function CreateCalculationNoteRtf(_fileName As String) As Boolean
```

Version

Available since version 10.

I.6.1.3.3 CreateFromObjects

C++

```
HRESULT CreateFromObjects(BSTR _selText, VARIANT_BOOL* ret);
```

C#

```
public bool CreateFromObjects(String _selText);
```

Visual Basic

```
Public Function CreateFromObjects(_selText As String) As Boolean
```

Version

Available since version 10.

(c) 2014 Autodesk Inc. All rights reserved

I.6.1.3.4 Save

C++

```
HRESULT Save();
```

C#

```
public void Save();
```

Visual Basic

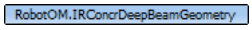
```
Public Sub Save()
```

Version

Available since version 10.

I.6.2 IRConcrDeepBeamGeometry

Class Hierarchy



C++

```
interface IRConcrDeepBeamGeometry : IDispatch;
```

C#

```
public interface IRConcrDeepBeamGeometry;
```

Visual Basic

```
Public Interface IRConcrDeepBeamGeometry
```

Version

Available since version 10.

I.7 RC Wall

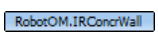
Available since version 10.

Interfaces

	Name	Description
	IRConcrWall (see page 1233)	
	IRConcrWallGeometry (see page 1238)	
	IRConcrWallOpening (see page 1241)	
	IRConcrWallOpenings (see page 1243)	

I.7.1 IRConcrWall

Class Hierarchy



C++

```
interface IRConcrWall : IDispatch;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public interface IRConcrWall;
```

Visual Basic

```
Public Interface IRConcrWall
```

Version

Available since version 10.

I.7.1.1 IRConcrWall Members

The following tables list the members exposed by IRConcrWall.

Public Fields

	Name	Description
	Concrete (see page 1234)	
	Geometry (see page 1234)	
	IsSelected (see page 1234)	
	Name (see page 1235)	
	Reinforcement (see page 1235)	
	Steel (see page 1235)	
	UniqueId (see page 1235)	

Public Methods

	Name	Description
	Activate (see page 1236)	
	Calculate (see page 1236)	
	CreateCalculationNoteRtf (see page 1236)	
	CreateFromObjects (see page 1237)	
	Save (see page 1237)	
	Verify (see page 1237)	

I.7.1.2 IRConcrWall Fields

The fields of the IRConcrWall class are listed here.

Public Fields

	Name	Description
	Concrete (see page 1234)	
	Geometry (see page 1234)	
	IsSelected (see page 1234)	
	Name (see page 1235)	
	Reinforcement (see page 1235)	
	Steel (see page 1235)	
	UniqueId (see page 1235)	

I.7.1.2.1 Concrete

C++

```
HRESULT get_Concrete(IRConcrConcrete**);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRConcrConcrete Concrete { get; }
```

Visual Basic

```
Public ReadOnly Concrete As IRConcrConcrete
```

Version

Available since version 10.

I.7.1.2.2 Geometry

C++

```
HRESULT get_Geometry(IRConcrWallGeometry**);
```

C#

```
public IRConcrWallGeometry Geometry { get; }
```

Visual Basic

```
Public ReadOnly Geometry As IRConcrWallGeometry
```

Version

Available since version 10.

I.7.1.2.3 IsSelected

C++

```
HRESULT get_IsSelected(VARIANT_BOOL*);
HRESULT put_IsSelected(VARIANT_BOOL);
```

C#

```
public bool IsSelected { get; set; }
```

Visual Basic

```
Public IsSelected As Boolean
```

Version

Available since version 12.

I.7.1.2.4 Name

C++

```
HRESULT get_Name(BSTR*);
```

C#

```
public String Name { get; }
```

Visual Basic

```
Public ReadOnly Name As String
```

Version

Available since version 10.

I.7.1.2.5 Reinforcement

C++

```
HRESULT get_Reinforcement(IRConcrReinforcement**);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRConcrReinforcement Reinforcement { get; }
```

Visual Basic

```
Public ReadOnly Reinforcement As IRConcrReinforcement
```

Version

Available since version 10.

I.7.1.2.6 Steel

C++

```
HRESULT get_Steel(IRConcrSteel**);
```

C#

```
public IRConcrSteel Steel { get; }
```

Visual Basic

```
Public ReadOnly Steel As IRConcrSteel
```

Version

Available since version 10.

I.7.1.2.7 UniqueId

C++

```
HRESULT get_UniqueId(long*);
```

C#

```
public long UniqueId { get; }
```

Visual Basic

```
Public ReadOnly UniqueId As long
```

Version

Available since version 10.

I.7.1.3 IRConcrWall Methods

The methods of the IRConcrWall class are listed here.

Public Methods

	Name	Description
	Activate (see page 1236)	
	Calculate (see page 1236)	
	CreateCalculationNoteRtf (see page 1236)	
	CreateFromObjects (see page 1237)	
	Save (see page 1237)	
	Verify (see page 1237)	

I.7.1.3.1 Activate

C++

```
HRESULT Activate();
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public void Activate();
```

Visual Basic

```
Public Sub Activate()
```

Version

Available since version 10.

I.7.1.3.2 Calculate

C++

```
HRESULT Calculate(VARIANT_BOOL _createReinforcement, VARIANT_BOOL* ret);
```

C#

```
public bool Calculate(bool _createReinforcement);
```

Visual Basic

```
Public Function Calculate(_createReinforcement As Boolean) As Boolean
```

Version

Available since version 10.

I.7.1.3.3 CreateCalculationNoteRtf

C++

```
HRESULT CreateCalculationNoteRtf(BSTR _fileName, VARIANT_BOOL* ret);
```

C#

```
public bool CreateCalculationNoteRtf(String _fileName);
```

Visual Basic

```
Public Function CreateCalculationNoteRtf(_fileName As String) As Boolean
```

Version

Available since version 10.

I.7.1.3.4 CreateFromObjects

C++

```
HRESULT CreateFromObjects(BSTR _selTaxt, VARIANT_BOOL* ret);
```

C#

```
public bool CreateFromObjects(String _selTaxt);
```

Visual Basic

```
Public Function CreateFromObjects(_selTaxt As String) As Boolean
```

Version

Available since version 10.

(c) 2014 Autodesk Inc. All rights reserved

I.7.1.3.5 Save

C++

```
HRESULT Save();
```

C#

```
public void Save();
```

Visual Basic

```
Public Sub Save()
```

Version

Available since version 10.

I.7.1.3.6 Verify

C++

```
HRESULT Verify(VARIANT_BOOL _showErrors, long* ret);
```

C#

```
public long Verify(bool _showErrors);
```

Visual Basic

```
Public Function Verify(_showErrors As Boolean) As long
```

Version

Available since version 10.

I.7.2 IRConcrWallGeometry

Class Hierarchy

RobotOM.IRConcrWallGeometry

C++

```
interface IRConcrWallGeometry : IDispatch;
```

C#

```
public interface IRConcrWallGeometry;
```

Visual Basic

```
Public Interface IRConcrWallGeometry
```

Version

Available since version 10.

I.7.2.1 IRConcrWallGeometry Members

The following tables list the members exposed by IRConcrWallGeometry.

Public Fields

	Name	Description
	Height (see page 1239)	
	LeftBoundaryLength (see page 1239)	

(c) 2014 Autodesk Inc. All rights reserved

	LeftBoundaryThickness (see page 1239)	
	Length (see page 1239)	
	Name (see page 1240)	
	Openings (see page 1240)	
	RightBoundaryLength (see page 1240)	
	RightBoundaryThickness (see page 1240)	
	Thickness (see page 1241)	

I.7.2.2 IRConcrWallGeometry Fields

The fields of the IRConcrWallGeometry class are listed here.

Public Fields

	Name	Description
	Height (see page 1239)	
	LeftBoundaryLength (see page 1239)	
	LeftBoundaryThickness (see page 1239)	
	Length (see page 1239)	
	Name (see page 1240)	
	Openings (see page 1240)	
	RightBoundaryLength (see page 1240)	
	RightBoundaryThickness (see page 1240)	
	Thickness (see page 1241)	

I.7.2.2.1 Height

C++

```
HRESULT get_Height(double*);
HRESULT put_Height(double);
```

C#

```
public double Height { get; set; }
```

Visual Basic

```
Public Height As double
```

Version

Available since version 10.

I.7.2.2.2 LeftBoundaryLength

C++

```
HRESULT get_LeftBoundaryLength(double*);
HRESULT put_LeftBoundaryLength(double);
```

C#

```
public double LeftBoundaryLength { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public LeftBoundaryLength As double
```

Version

Available since version 10.

I.7.2.2.3 LeftBoundaryThickness

C++

```
HRESULT get_LeftBoundaryThickness(double*);
HRESULT put_LeftBoundaryThickness(double);
```

C#

```
public double LeftBoundaryThickness { get; set; }
```

Visual Basic

```
Public LeftBoundaryThickness As double
```

Version

Available since version 10.

I.7.2.2.4 Length

C++

```
HRESULT get_Length(double*);
HRESULT put_Length(double);
```

C#

```
public double Length { get; set; }
```

Visual Basic

```
Public Length As double
```

Version

Available since version 10.

I.7.2.2.5 Name

C++

```
HRESULT get_Name(BSTR*);
HRESULT put_Name(BSTR);
```

C#

```
public String Name { get; set; }
```

Visual Basic

```
Public Name As String
```

Version

Available since version 10.

I.7.2.2.6 Openings

C++

```
HRESULT get_Openings(IRConcrWallOpenings**);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRConcrWallOpenings Openings { get; }
```

Visual Basic

```
Public ReadOnly Openings As IRConcrWallOpenings
```

Version

Available since version 10.

I.7.2.2.7 RightBoundaryLength

C++

```
HRESULT get_RightBoundaryLength(double*);
HRESULT put_RightBoundaryLength(double);
```

C#

```
public double RightBoundaryLength { get; set; }
```

Visual Basic

```
Public RightBoundaryLength As double
```

Version

Available since version 10.

I.7.2.2.8 RightBoundaryThickness

C++

```
HRESULT get_RightBoundaryThickness(double*);
HRESULT put_RightBoundaryThickness(double);
```

C#

```
public double RightBoundaryThickness { get; set; }
```

Visual Basic

```
Public RightBoundaryThickness As double
```

Version

Available since version 10.

I.7.2.2.9 Thickness

C++

```
HRESULT get_Thickness(double*);
HRESULT put_Thickness(double);
```

C#

```
public double Thickness { get; set; }
```

Visual Basic

```
Public Thickness As double
```

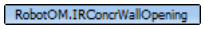
Version

Available since version 10.

(c) 2014 Autodesk Inc. All rights reserved

I.7.3 IRConcrWallOpening

Class Hierarchy



C++

```
interface IRConcrWallOpening : IDispatch;
```

C#

```
public interface IRConcrWallOpening;
```

Visual Basic

```
Public Interface IRConcrWallOpening
```

Version

Available since version 10.

I.7.3.1 IRConcrWallOpening Members

The following tables list the members exposed by IRConcrWallOpening.

Public Fields

	Name	Description
	Lx (see page 1242)	
	Lz (see page 1242)	
	Name (see page 1242)	
	PosX (see page 1243)	
	PosZ (see page 1243)	

I.7.3.2 IRConcrWallOpening Fields

The fields of the IRConcrWallOpening class are listed here.

Public Fields

	Name	Description
	Lx (see page 1242)	
	Lz (see page 1242)	
	Name (see page 1242)	
	PosX (see page 1243)	
	PosZ (see page 1243)	

I.7.3.2.1 Lx

C++

```
HRESULT get_Lx(double*);
HRESULT put_Lx(double);
```

C#

```
public double Lx { get; set; }
```

Visual Basic

```
Public Lx As double
```

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 10.

I.7.3.2.2 Lz

C++

```
HRESULT get_Lz(double*);
HRESULT put_Lz(double);
```

C#

```
public double Lz { get; set; }
```

Visual Basic

```
Public Lz As double
```

Version

Available since version 10.

I.7.3.2.3 Name

C++

```
HRESULT get_Name(BSTR*);
HRESULT put_Name(BSTR);
```

C#

```
public String Name { get; set; }
```

Visual Basic

```
Public Name As String
```

Version

Available since version 10.

I.7.3.2.4 PosX

C++

```
HRESULT get_PosX(double*);
HRESULT put_PosX(double);
```

C#

```
public double PosX { get; set; }
```

Visual Basic

```
Public PosX As double
```

Version

Available since version 10.

I.7.3.2.5 PosZ

C++

```
HRESULT get_PosZ(double*);
HRESULT put_PosZ(double);
```

C#

```
public double PosZ { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

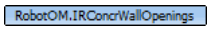
```
Public PosZ As double
```

Version

Available since version 10.

I.7.4 IRConcrWallOpenings

Class Hierarchy



C++

```
interface IRConcrWallOpenings : IDispatch;
```

C#

```
public interface IRConcrWallOpenings;
```

Visual Basic

```
Public Interface IRConcrWallOpenings
```

Version

Available since version 10.

I.7.4.1 IRConcrWallOpenings Members

The following tables list the members exposed by IRConcrWallOpenings.

Public Fields

	Name	Description
	Count (see page 1244)	
	Item (see page 1244)	

Public Methods

	Name	Description
	Add (see page 1244)	

I.7.4.2 IRConcrWallOpenings Fields

The fields of the IRConcrWallOpenings class are listed here.

Public Fields

	Name	Description
	Count (see page 1244)	
	Item (see page 1244)	

I.7.4.2.1 Count

C++

```
HRESULT get_Count(long*);
```

C#

```
public long Count { get; }
```

Visual Basic

```
Public ReadOnly Count As long
```

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 10.

I.7.4.2.2 Item

C++

```
HRESULT get_Item(IRConcrWallOpening**);
```

C#

```
public IRConcrWallOpening Item { get; }
```

Visual Basic

```
Public ReadOnly Item As IRConcrWallOpening
```

Version

Available since version 10.

I.7.4.3 IRConcrWallOpenings Methods

The methods of the IRConcrWallOpenings class are listed here.

Public Methods

	Name	Description
	Add (see page 1244)	

I.7.4.3.1 Add

C++

```
HRESULT Add(IRConcrWallOpening** ret);
```

C#

```
public IRConcrWallOpening Add();
```

Visual Basic

```
Public Function Add() As IRConcrWallOpening
```

Version

Available since version 10.

I.8 Reinforcement

Available since version 10.

Enumerations

	Name	Description
	IRConcrBarType (see page 1257)	
	IRConcrBarSubtype (see page 1258)	
	IRConcrSteelType (see page 1260)	
	IRConcrSteelSurfaceType (see page 1264)	

(c) 2014 Autodesk Inc. All rights reserved

Interfaces

	Name	Description
	IRConcrReinforcement (see page 1245)	
	IRConcrReinforcingBar (see page 1249)	
	IRConcrReinforcingBars (see page 1255)	
	IRConcrSteelNames (see page 1258)	
	IRConcrSteel (see page 1260)	
	IRConcrConcrete (see page 1263)	
	IRConcrSteelStrengths (see page 1264)	
	IRConcrBarDiameters (see page 1264)	

I.8.1 IRConcrReinforcement

Class Hierarchy

RobotOM.IRConcrReinforcement

C++

```
interface IRConcrReinforcement : IDispatch;
```

C#

```
public interface IRConcrReinforcement;
```

Visual Basic

```
Public Interface IRConcrReinforcement
```

Version

Available since version 10.

I.8.1.1 IRConcrReinforcement Members

The following tables list the members exposed by IRConcrReinforcement.

Public Fields

	Name	Description
	Bars (see page 1246)	
	Comment (see page 1247)	
	GroupGlobally (see page 1247)	
	GroupingMethod (see page 1247)	
	IsFrozen (see page 1247)	

Public Methods

	Name	Description
	Freeze (see page 1248)	
	LockRefresh (see page 1248)	
	RemoveAll (see page 1248)	

(c) 2014 Autodesk Inc. All rights reserved

	RemoveParametric (see page 1249)	

I.8.1.2 IRConcrReinforcement Fields

The fields of the IRConcrReinforcement class are listed here.

Public Fields

	Name	Description
	Bars (see page 1246)	
	Comment (see page 1247)	
	GroupGlobally (see page 1247)	
	GroupingMethod (see page 1247)	
	IsFrozen (see page 1247)	

I.8.1.2.1 Bars

C++

```
HRESULT get_Bars(IRConcrReinforcingBars**);
```

C#

```
public IRConcrReinforcingBars Bars { get; }
```

Visual Basic

```
Public ReadOnly Bars As IRConcrReinforcingBars
```

Version

Available since version 10.

I.8.1.2.2 Comment

C++

```
HRESULT get_Comment(BSTR*);
HRESULT put_Comment(BSTR);
```

C#

```
public String Comment { get; set; }
```

Visual Basic

```
Public Comment As String
```

Version

Available since version 10.

I.8.1.2.3 GroupGlobally

C++

```
HRESULT get_GroupGlobally(VARIANT_BOOL*);
HRESULT put_GroupGlobally(VARIANT_BOOL);
```

C#

```
public bool GroupGlobally { get; set; }
```

Visual Basic

```
Public GroupGlobally As Boolean
```

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 10.

I.8.1.2.4 GroupingMethod

C++

```
HRESULT get_GroupingMethod(long*);
HRESULT put_GroupingMethod(long);
```

C#

```
public long GroupingMethod { get; set; }
```

Visual Basic

```
Public GroupingMethod As long
```

Version

Available since version 10.

I.8.1.2.5 IsFrozen

C++

```
HRESULT get_IsFrozen(VARIANT_BOOL*);
```

C#

```
public bool IsFrozen { get; }
```

Visual Basic

```
Public ReadOnly IsFrozen As Boolean
```

Version

Available since version 11.

I.8.1.3 IRConcrReinforcement Methods

The methods of the IRConcrReinforcement class are listed here.

Public Methods

	Name	Description
	Freeze (see page 1248)	
	LockRefresh (see page 1248)	
	RemoveAll (see page 1248)	
	RemoveParametric (see page 1249)	

I.8.1.3.1 Freeze

C++

```
HRESULT Freeze(VARIANT_BOOL _freeze, VARIANT_BOOL* ret);
```

C#

```
public bool Freeze(bool _freeze);
```

Visual Basic

```
Public Function Freeze(_freeze As Boolean) As Boolean
```

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 10.

I.8.1.3.2 LockRefresh

C++

```
HRESULT LockRefresh(VARIANT_BOOL _lock, VARIANT_BOOL* ret);
```

C#

```
public bool LockRefresh(bool _lock);
```

Visual Basic

```
Public Function LockRefresh(_lock As Boolean) As Boolean
```

Version

Available since version 10.

I.8.1.3.3 RemoveAll

C++

```
HRESULT RemoveAll();
```

C#

```
public void RemoveAll();
```

Visual Basic

```
Public Sub RemoveAll()
```

Version

Available since version 10.

I.8.1.3.4 RemoveParametric

C++

```
HRESULT RemoveParametric();
```

C#

```
public void RemoveParametric();
```

Visual Basic

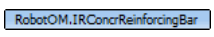
```
Public Sub RemoveParametric()
```

Version

Available since version 11.

I.8.2 IRConcrReinforcingBar

Class Hierarchy



C++

```
interface IRConcrReinforcingBar : IDispatch;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public interface IRConcrReinforcingBar;
```

Visual Basic

```
Public Interface IRConcrReinforcingBar
```

Version

Available since version 10.

I.8.2.1 IRConcrReinforcingBar Members

The following tables list the members exposed by IRConcrReinforcingBar.

Public Fields

	Name	Description
	DirectionX (see page 1250)	
	DirectionY (see page 1251)	
	DirectionZ (see page 1251)	
	GroupNumber (see page 1251)	
	IsParametric (see page 1251)	
	Position (see page 1252)	
	ROBarEditorId (see page 1252)	
	Span (see page 1252)	
	SteelType (see page 1252)	
	Subtype (see page 1253)	
	Type (see page 1253)	
	Weight (see page 1253)	

Public Methods

	Name	Description
	CreateCopy (see page 1254)	
	GetROBarEditor (see page 1254)	
	SetBarShape (see page 1254)	
	SetDirections (see page 1255)	
	SetROBarEditor (see page 1255)	

I.8.2.2 IRConcrReinforcingBar Fields

The fields of the IRConcrReinforcingBar class are listed here.

Public Fields

	Name	Description
	DirectionX (see page 1250)	
	DirectionY (see page 1251)	
	DirectionZ (see page 1251)	
	GroupNumber (see page 1251)	
	IsParametric (see page 1251)	
	Position (see page 1252)	
	ROBarEditorId (see page 1252)	
	Span (see page 1252)	
	SteelType (see page 1252)	
	Subtype (see page 1253)	

(c) 2014 Autodesk Inc. All rights reserved

	Type (see page 1253)	
	Weight (see page 1253)	

I.8.2.2.1 DirectionX

C++

```
HRESULT get_DirectionX(IRobotGeoPoint3D**);
```

C#

```
public IRobotGeoPoint3D DirectionX { get; }
```

Visual Basic

```
Public ReadOnly DirectionX As IRobotGeoPoint3D
```

Version

Available since version 10.

I.8.2.2.2 DirectionY

C++

```
HRESULT get_DirectionY(IRobotGeoPoint3D**);
```

C#

```
public IRobotGeoPoint3D DirectionY { get; }
```

Visual Basic

```
Public ReadOnly DirectionY As IRobotGeoPoint3D
```

Version

Available since version 10.

I.8.2.2.3 DirectionZ

C++

```
HRESULT get_DirectionZ(IRobotGeoPoint3D**);
```

C#

```
public IRobotGeoPoint3D DirectionZ { get; }
```

Visual Basic

```
Public ReadOnly DirectionZ As IRobotGeoPoint3D
```

Version

Available since version 10.

I.8.2.2.4 GroupNumber

C++

```
HRESULT get_GroupNumber( int*);
HRESULT put_GroupNumber( int);
```

C#

```
public  int GroupNumber { get; set; }
```

Visual Basic

```
Public GroupNumber As  int
```

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 10.

I.8.2.2.5 IsParametric

C++

```
HRESULT get_IsParametric(VARIANT_BOOL*);
HRESULT put_IsParametric(VARIANT_BOOL);
```

C#

```
public bool IsParametric { get; set; }
```

Visual Basic

```
Public IsParametric As Boolean
```

Version

Available since version 11.

I.8.2.2.6 Position

C++

```
HRESULT get_Position(IRobotGeoPoint3D**);
```

C#

```
public IRobotGeoPoint3D Position { get; }
```

Visual Basic

```
Public ReadOnly Position As IRobotGeoPoint3D
```

Version

Available since version 10.

I.8.2.2.7 ROBarEditorId

C++

```
HRESULT get_ROBarEditorId(long*);
```

C#

```
public long ROBarEditorId { get; }
```

Visual Basic

```
Public ReadOnly ROBarEditorId As long
```

Version

Available since version 10.

I.8.2.2.8 Span

C++

```
HRESULT get_Span(short*);
HRESULT put_Span(short);
```

C#

```
public short Span { get; set; }
```

Visual Basic

```
Public Span As short
```

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 10.

I.8.2.2.9 SteelType

C++

```
HRESULT get_SteelType(IRConcrSteelType*);
HRESULT put_SteelType(IRConcrSteelType);
```

C#

```
public IRConcrSteelType SteelType { get; set; }
```

Visual Basic

```
Public SteelType As IRConcrSteelType
```

Version

Available since version 10.

I.8.2.2.10 Subtype

C++

```
HRESULT get_Subtype(IRConcrBarSubtype*);
```

C#

```
public IRConcrBarSubtype Subtype { get; }
```

Visual Basic

```
Public ReadOnly Subtype As IRConcrBarSubtype
```

Version

Available since version 10.

I.8.2.2.11 Type

C++

```
HRESULT get_Type(IRConcrBarType*);
```

C#

```
public IRConcrBarType Type { get; }
```

Visual Basic

```
Public ReadOnly Type As IRConcrBarType
```

Version

Available since version 10.

I.8.2.2.12 Weight

C++

```
HRESULT get_Weight( double*);
```

C#

```
public  double Weight { get; }
```

Visual Basic

```
Public ReadOnly Weight As  double
```

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 10.

I.8.2.3 IRConcrReinforcingBar Methods

The methods of the IRConcrReinforcingBar class are listed here.

Public Methods

	Name	Description
	CreateCopy (see page 1254)	
	GetROBarEditor (see page 1254)	
	SetBarShape (see page 1254)	
	SetDirections (see page 1255)	
	SetROBarEditor (see page 1255)	

I.8.2.3.1 CreateCopy

C++

```
HRESULT CreateCopy(double _offX, double _offY, double _offZ, long _number, VARIANT_BOOL*
ret);
```

C#

```
public bool CreateCopy(double _offX, double _offY, double _offZ, long _number);
```

Visual Basic

```
Public Function CreateCopy(_offX As double, _offY As double, _offZ As double, _number As
long) As Boolean
```

Version

Available since version 10.

I.8.2.3.2 GetROBarEditor

C++

```
HRESULT GetROBarEditor(VARIANT_BOOL _readOnly, IDispatch** ret);
```

C#

```
public IDispatch* GetROBarEditor(bool _readOnly);
```

Visual Basic

```
Public Function GetROBarEditor(_readOnly As Boolean) As IDispatch*
```

Version

Available since version 10.

I.8.2.3.3 SetBarShape

C++

```
HRESULT SetBarShape(VARIANT_BOOL* ret);
```

C#

```
public bool SetBarShape();
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Function SetBarShape() As Boolean
```

Version

Available since version 10.

I.8.2.3.4 SetDirections

C++

```
HRESULT SetDirections(double _Xx, double _Xy, double _Xz, double _Yx, double _Yy, double
_Yz, VARIANT_BOOL* ret);
```

C#

```
public bool SetDirections(double _Xx, double _Xy, double _Xz, double _Yx, double _Yy,
double _Yz);
```

Visual Basic

```
Public Function SetDirections(_Xx As double, _Xy As double, _Xz As double, _Yx As double,
_Yy As double, _Yz As double) As Boolean
```

Version

Available since version 10.

I.8.2.3.5 SetROBarEditor

C++

```
HRESULT SetROBarEditor(IDispatch* _editor);
```

C#

```
public void SetROBarEditor(IDispatch* _editor);
```

Visual Basic

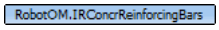
```
Public Sub SetROBarEditor(ByRef _editor As IDispatch*)
```

Version

Available since version 10.

I.8.3 IRConcrReinforcingBars

Class Hierarchy



C++

```
interface IRConcrReinforcingBars : IDispatch;
```

C#

```
public interface IRConcrReinforcingBars;
```

Visual Basic

```
Public Interface IRConcrReinforcingBars
```

Version

Available since version 10.

(c) 2014 Autodesk Inc. All rights reserved

I.8.3.1 IRConcrReinforcingBars Members

The following tables list the members exposed by IRConcrReinforcingBars.

Public Fields

	Name	Description
	Count (see page 1256)	
	Item (see page 1256)	

Public Methods

	Name	Description
	Add (see page 1257)	

I.8.3.2 IRConcrReinforcingBars Fields

The fields of the IRConcrReinforcingBars class are listed here.

Public Fields

	Name	Description
	Count (see page 1256)	
	Item (see page 1256)	

I.8.3.2.1 Count

C++

```
HRESULT get_Count(long*);
```

C#

```
public long Count { get; }
```

Visual Basic

```
Public ReadOnly Count As long
```

Version

Available since version 10.

I.8.3.2.2 Item

C++

```
HRESULT get_Item(IRConcrReinforcingBar**);
```

C#

```
public IRConcrReinforcingBar Item { get; }
```

Visual Basic

```
Public ReadOnly Item As IRConcrReinforcingBar
```

Version

Available since version 10.

I.8.3.3 IRConcrReinforcingBars Methods

The methods of the IRConcrReinforcingBars class are listed here.

(c) 2014 Autodesk Inc. All rights reserved

Public Methods

	Name	Description
	Add (see page 1257)	

I.8.3.3.1 Add

C++

```
HRESULT Add(BSTR _shapeCode, IRConcrBarType _type, IRConcrBarSubType _subType, double
_diameter, double _bendDiam = 0, IRConcrReinforcingBar** ret);
```

C#

```
public IRConcrReinforcingBar Add(String _shapeCode, IRConcrBarType _type, IRConcrBarSubType
_subType, double _diameter, double _bendDiam = 0);
```

Visual Basic

```
Public Function Add(_shapeCode As String, _type As IRConcrBarType, _subType As
IRConcrBarSubType, _diameter As double, Optional _bendDiam As double = 0) As
IRConcrReinforcingBar
```

Version

Available since version 10.

I.8.4 IRConcrBarType

C++

```
enum IRConcrBarType;
```

C#

```
public enum IRConcrBarType;
```

Visual Basic

```
Public Enum IRConcrBarType
```

Members

Members	Description
I_CBT_BAR_TYPE_UNDEFINED = 0	Available since version 10.
I_CBT_BAR_TYPE_LON_MAIN = 1	Available since version 10.
I_CBT_BAR_TYPE_TRA_MAIN = 2	Available since version 10.
I_CBT_BAR_TYPE_LON_HOLE = 4	Available since version 10.
I_CBT_BAR_TYPE_TRA_HOLE = 8	Available since version 10.
I_CBT_BAR_TYPE_LON_SURF = 16	Available since version 10.
I_CBT_BAR_TYPE_TRA_SURF = 32	Available since version 10.
I_CBT_BAR_TYPE_LON_CONS = 64	Available since version 10.
I_CBT_BAR_TYPE_TRA_CONS = 128	Available since version 10.
I_CBT_BAR_TYPE_LON_CORN = 256	Available since version 10.
I_CBT_BAR_TYPE_TRA_CORN = 512	Available since version 10.
I_CBT_BAR_TYPE_LON_JOIN = 1024	Available since version 10.
I_CBT_BAR_TYPE_TRA_JOIN = 2048	Available since version 10.
I_CBT_BAR_TYPE_LON_SPEC = 4096	Available since version 10.
I_CBT_BAR_TYPE_TRA_SPEC = 8192	Available since version 10.
I_CBT_BAR_TYPE_LON_DIST = 32768	Available since version 10.
I_CBT_BAR_TYPE_LON_BOAT = 65536	Available since version 10.

(c) 2014 Autodesk Inc. All rights reserved

I_CBT_BAR_TYPE_LON_TORS = 131072	Available since version 10.
I_CBT_BAR_TYPE_LON_MAIN_FIRST = 262144	Available since version 10.
I_CBT_BAR_TYPE_LON_JOIN_STRUT = 524288	Available since version 10.
I_CBT_BAR_TYPE_LON_FLAN = 1048576	Available since version 10.

Version

Available since version 10.

I.8.5 IRConcrBarSubtype

C++

```
enum IRConcrBarSubtype;
```

C#

```
public enum IRConcrBarSubtype;
```

Visual Basic

```
Public Enum IRConcrBarSubtype
```

Members

Members	Description
I_CBS_BAR_SUBTYPE_ANY = 0	Available since version 10.
I_CBS_BAR_SUBTYPE_UP = 1	Available since version 10.
I_CBS_BAR_SUBTYPE_UP_RIGHT = 2	Available since version 10.
I_CBS_BAR_SUBTYPE_DOWN = 4	Available since version 10.
I_CBS_BAR_SUBTYPE_INC_LEFT = 8	Available since version 10.
I_CBS_BAR_SUBTYPE_INC_RIGHT = 16	Available since version 10.
I_CBS_BAR_SUBTYPE_STRAIGHT = 32	Available since version 10.
I_CBS_BAR_SUBTYPE_INC_45 = 64	Available since version 10.
I_CBS_BAR_SUBTYPE_DIR_VER = 128	Available since version 10.
I_CBS_BAR_SUBTYPE_DIR_HOR = 256	Available since version 10.
I_CBS_BAR_SUBTYPE_EXT = 512	Available since version 10.
I_CBS_BAR_SUBTYPE_INT = 1024	Available since version 10.
I_CBS_BAR_SUBTYPE_SHA = 2048	Available since version 10.
I_CBS_BAR_SUBTYPE_PIN = 4096	Available since version 10.
I_CBS_BAR_SUBTYPE_JOIN = 8192	Available since version 10.
I_CBS_BAR_SUBTYPE_SPEC = 16384	Available since version 10.
I_CBS_BAR_SUBTYPE_UBAR = 32768	Available since version 10.
I_CBS_BAR_SUBTYPE_DIST = 65536	Available since version 10.
I_CBS_BAR_SUBTYPE_WALL = 131072	Available since version 10.
I_CBS_BAR_SUBTYPE_FOOT = 262144	Available since version 10.
I_CBS_BAR_SUBTYPE_TABLE = 524288	Available since version 10.
I_CBS_BAR_SUBTYPE_GROIN = 1048576	Available since version 10.
I_CBS_BAR_SUBTYPE_FRONT = 2097152	Available since version 10.
I_CBS_BAR_SUBTYPE_BACK = 4194304	Available since version 10.
I_CBS_BAR_SUBTYPE_LEFT = 8388608	Available since version 10.
I_CBS_BAR_SUBTYPE_RIGHT = 16777216	Available since version 10.

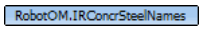
Version

Available since version 10.

(c) 2014 Autodesk Inc. All rights reserved

I.8.6 IRConcrSteelNames

Class Hierarchy



C++

```
interface IRConcrSteelNames : IDispatch;
```

C#

```
public interface IRConcrSteelNames;
```

Visual Basic

```
Public Interface IRConcrSteelNames
```

Version

Available since version 10.

I.8.6.1 IRConcrSteelNames Members

The following tables list the members exposed by IRConcrSteelNames.

Public Fields

	Name	Description
	Count (see page 1259)	
	Item (see page 1259)	

I.8.6.2 IRConcrSteelNames Fields

The fields of the IRConcrSteelNames class are listed here.

Public Fields

	Name	Description
	Count (see page 1259)	
	Item (see page 1259)	

I.8.6.2.1 Count

C++

```
HRESULT get_Count(long*);
```

C#

```
public long Count { get; }
```

Visual Basic

```
Public ReadOnly Count As long
```

Version

Available since version 10.

I.8.6.2.2 Item

C++

```
HRESULT get_Item(BSTR*);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public String Item { get; }
```

Visual Basic

```
Public ReadOnly Item As String
```

Version

Available since version 10.

I.8.7 IRConcrSteelType

C++

```
enum IRConcrSteelType;
```

C#

```
public enum IRConcrSteelType;
```

Visual Basic

```
Public Enum IRConcrSteelType
```

Members

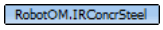
Members	Description
I_CST_STEEL_TYPE_LONG = 1	Available since version 10.
I_CST_STEEL_TYPE_TRAN = 2	Available since version 10.
I_CST_STEEL_TYPE_WIRE = 3	Available since version 10.
I_CST_STEEL_TYPE_SPEC = 4	Available since version 10.

Version

Available since version 10.

I.8.8 IRConcrSteel

Class Hierarchy



C++

```
interface IRConcrSteel : IDispatch;
```

C#

```
public interface IRConcrSteel;
```

Visual Basic

```
Public Interface IRConcrSteel
```

Version

Available since version 10.

I.8.8.1 IRConcrSteel Members

The following tables list the members exposed by IRConcrSteel.

Public Fields

	Name	Description
	AvailableNames (see page 1261)	

(c) 2014 Autodesk Inc. All rights reserved

	AvailableStrengths (see page 1261)	
	BarDiameters (see page 1262)	
	BarsDatabasePath (see page 1262)	Available since version 10.
	CharacteristicStrength (see page 1262)	
	SelectedName (see page 1262)	
	SurfaceType (see page 1262)	

I.8.8.2 IRConcrSteel Fields

The fields of the IRConcrSteel class are listed here.

Public Fields

	Name	Description
	AvailableNames (see page 1261)	
	AvailableStrengths (see page 1261)	
	BarDiameters (see page 1262)	
	BarsDatabasePath (see page 1262)	Available since version 10.
	CharacteristicStrength (see page 1262)	
	SelectedName (see page 1262)	
	SurfaceType (see page 1262)	

I.8.8.2.1 AvailableNames

C++

```
HRESULT get_AvailableNames(IRConcrSteelNames**);
```

C#

```
public IRConcrSteelNames AvailableNames { get; }
```

Visual Basic

```
Public ReadOnly AvailableNames As IRConcrSteelNames
```

Version

Available since version 10.

I.8.8.2.2 AvailableStrengths

C++

```
HRESULT get_AvailableStrengths(IRConcrSteelStrengths**);
```

C#

```
public IRConcrSteelStrengths AvailableStrengths { get; }
```

Visual Basic

```
Public ReadOnly AvailableStrengths As IRConcrSteelStrengths
```

Version

Available since version 10.

(c) 2014 Autodesk Inc. All rights reserved

I.8.8.2.3 BarDiameters

C++

```
HRESULT get_BarDiameters(IRConcrBarDiameters**);
```

C#

```
public IRConcrBarDiameters BarDiameters { get; }
```

Visual Basic

```
Public ReadOnly BarDiameters As IRConcrBarDiameters
```

Version

Available since version 10.

I.8.8.2.4 BarsDatabasePath

C#

```
void BarsDatabasePath;
```

Description

Available since version 10.

I.8.8.2.5 CharacteristicStrength

C++

```
HRESULT get_CharacteristicStrength(double*);
HRESULT put_CharacteristicStrength(double);
```

C#

```
public double CharacteristicStrength { get; set; }
```

Visual Basic

```
Public CharacteristicStrength As double
```

Version

Available since version 10.

I.8.8.2.6 SelectedName

C++

```
HRESULT get_SelectedName(BSTR*);
HRESULT put_SelectedName(BSTR);
```

C#

```
public String SelectedName { get; set; }
```

Visual Basic

```
Public SelectedName As String
```

Version

Available since version 10.

I.8.8.2.7 SurfaceType

C++

```
HRESULT get_SurfaceType(IRConcrSteelSurfaceType*);
HRESULT put_SurfaceType(IRConcrSteelSurfaceType);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRConcrSteelSurfaceType SurfaceType { get; set; }
```

Visual Basic

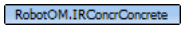
```
Public SurfaceType As IRConcrSteelSurfaceType
```

Version

Available since version 10.

I.8.9 IRConcrConcrete

Class Hierarchy



C++

```
interface IRConcrConcrete : IDispatch;
```

C#

```
public interface IRConcrConcrete;
```

Visual Basic

```
Public Interface IRConcrConcrete
```

Version

Available since version 10.

I.8.9.1 IRConcrConcrete Members

The following tables list the members exposed by IRConcrConcrete.

Public Fields

	Name	Description
	CharacteristicStrength (see page 1263)	

I.8.9.2 IRConcrConcrete Fields

The fields of the IRConcrConcrete class are listed here.

Public Fields

	Name	Description
	CharacteristicStrength (see page 1263)	

I.8.9.2.1 CharacteristicStrength

C++

```
HRESULT get_CharacteristicStrength(double*);
HRESULT put_CharacteristicStrength(double);
```

C#

```
public double CharacteristicStrength { get; set; }
```

Visual Basic

```
Public CharacteristicStrength As double
```

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 10.

I.8.10 IRConcrSteelSurfaceType

C++

```
enum IRConcrSteelSurfaceType;
```

C#

```
public enum IRConcrSteelSurfaceType;
```

Visual Basic

```
Public Enum IRConcrSteelSurfaceType
```

Members

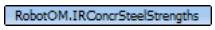
Members	Description
I_SST_SURFACE_RIBBED = 1	Available since version 10.
I_SST_SURFACE_PLAIN = 2	Available since version 10.

Version

Available since version 10.

I.8.11 IRConcrSteelStrengths

Class Hierarchy



C++

```
interface IRConcrSteelStrengths : IDispatch;
```

C#

```
public interface IRConcrSteelStrengths;
```

Visual Basic

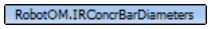
```
Public Interface IRConcrSteelStrengths
```

Version

Available since version 10.

I.8.12 IRConcrBarDiameters

Class Hierarchy



C++

```
interface IRConcrBarDiameters : IDispatch;
```

C#

```
public interface IRConcrBarDiameters;
```

Visual Basic

```
Public Interface IRConcrBarDiameters
```

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 10.

I.8.12.1 IRConcrBarDiameters Members

The following tables list the members exposed by IRConcrBarDiameters.

Public Fields

	Name	Description
	Count (see page 1265)	
	Item (see page 1265)	

Public Methods

	Name	Description
	Add (see page 1266)	
	RemoveAll (see page 1266)	

I.8.12.2 IRConcrBarDiameters Fields

The fields of the IRConcrBarDiameters class are listed here.

Public Fields

	Name	Description
	Count (see page 1265)	
	Item (see page 1265)	

I.8.12.2.1 Count

C++

```
HRESULT get_Count(long*);
```

C#

```
public long Count { get; }
```

Visual Basic

```
Public ReadOnly Count As long
```

Version

Available since version 10.

I.8.12.2.2 Item

C++

```
HRESULT get_Item(double*);
```

C#

```
public double Item { get; }
```

Visual Basic

```
Public ReadOnly Item As double
```

Version

Available since version 10.

(c) 2014 Autodesk Inc. All rights reserved

I.8.12.3 IRConcrBarDiameters Methods

The methods of the IRConcrBarDiameters class are listed here.

Public Methods

	Name	Description
	Add (see page 1266)	
	RemoveAll (see page 1266)	

I.8.12.3.1 Add

C++

```
HRESULT Add(double _newDiam);
```

C#

```
public void Add(double _newDiam);
```

Visual Basic

```
Public Sub Add(_newDiam As double)
```

Version

Available since version 10.

I.8.12.3.2 RemoveAll

C++

```
HRESULT RemoveAll();
```

C#

```
public void RemoveAll();
```

Visual Basic

```
Public Sub RemoveAll()
```

Version

Available since version 10.

I.9 Drawing

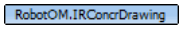
Available since version 10.

Interfaces

	Name	Description
	IRConcrDrawing (see page 1266)	
	IRConcrDrawingBarGroup (see page 1268)	

I.9.1 IRConcrDrawing

Class Hierarchy



C++

```
interface IRConcrDrawing : IDispatch;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public interface IRConcrDrawing;
```

Visual Basic

```
Public Interface IRConcrDrawing
```

Version

Available since version 10.

I.9.1.1 IRConcrDrawing Members

The following tables list the members exposed by IRConcrDrawing.

Public Fields

	Name	Description
	Group (see page 1267)	
	GroupCount (see page 1267)	

I.9.1.2 IRConcrDrawing Fields

The fields of the IRConcrDrawing class are listed here.

Public Fields

	Name	Description
	Group (see page 1267)	
	GroupCount (see page 1267)	

I.9.1.2.1 Group

C++

```
HRESULT get_Group(IRConcrDrawingBarGroup**);
```

C#

```
public IRConcrDrawingBarGroup Group { get; }
```

Visual Basic

```
Public ReadOnly Group As IRConcrDrawingBarGroup
```

Version

Available since version 10.

I.9.1.2.2 GroupCount

C++

```
HRESULT get_GroupCount(long*);
```

C#

```
public long GroupCount { get; }
```

Visual Basic

```
Public ReadOnly GroupCount As long
```

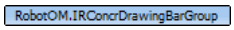
Version

Available since version 10.

(c) 2014 Autodesk Inc. All rights reserved

I.9.2 IRConcrDrawingBarGroup

Class Hierarchy



C++

```
interface IRConcrDrawingBarGroup : IDispatch;
```

C#

```
public interface IRConcrDrawingBarGroup;
```

Visual Basic

```
Public Interface IRConcrDrawingBarGroup
```

Version

Available since version 10.

I.9.2.1 IRConcrDrawingBarGroup Members

The following tables list the members exposed by IRConcrDrawingBarGroup.

Public Fields

	Name	Description
	Bar (see page 1268)	
	BarCount (see page 1269)	

I.9.2.2 IRConcrDrawingBarGroup Fields

The fields of the IRConcrDrawingBarGroup class are listed here.

Public Fields

	Name	Description
	Bar (see page 1268)	
	BarCount (see page 1269)	

I.9.2.2.1 Bar

C++

```
HRESULT get_Bar(IRConcrReinforcingBar**);
```

C#

```
public IRConcrReinforcingBar Bar { get; }
```

Visual Basic

```
Public ReadOnly Bar As IRConcrReinforcingBar
```

Version

Available since version 10.

I.9.2.2.2 BarCount

C++

```
HRESULT get_BarCount(long*);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public long BarCount { get; }
```

Visual Basic

```
Public ReadOnly BarCount As long
```

Version

Available since version 10.

I.10 IRobotProjectComponentType

C++

```
enum IRobotProjectComponentType;
```

C#

```
public enum IRobotProjectComponentType;
```

Visual Basic

```
Public Enum IRobotProjectComponentType
```

Members

Members	Description
I_PCT_BEAM = 3	Concrete beam .
I_PCT_COLUMN = 4	Concrete column .
I_PCT_FOOT = 5	Concrete footing .
I_PCT_JOINT = 6	Steel connection .
I_PCT_DRAWING = 7	Plotter drawing .
I_PCT_BWALL = 8	Concrete wall .
I_PCT_RETAINING_WALL = 10	Retaining wall. Available since version 3.
I_PCT_SLAB = 9	Available since version 3.
I_PCT_CONTINUOUS_FOOT = 11	Available since version 3.

Description

A set of identifiers is defined to describe different types of components that may be saved in a project. .

I.11 IRobotProjectComponent

Class Hierarchy

C++

```
interface IRobotProjectComponent : IDispatch;
```

C#

```
public interface IRobotProjectComponent;
```

Visual Basic

```
Public Interface IRobotProjectComponent
```

Description

Common interface granting access to any project component. .

(c) 2014 Autodesk Inc. All rights reserved

I.11.1 IRobotProjectComponent Members

The following tables list the members exposed by IRobotProjectComponent.

Public Fields

	Name	Description
	Data (see page 1270)	Object providing access to data on a project component; the type of the object depends on the component type .
	Name (see page 1270)	Component name .
	Type (see page 1271)	Project component type .

I.11.2 IRobotProjectComponent Fields

The fields of the IRobotProjectComponent class are listed here.

Public Fields

	Name	Description
	Data (see page 1270)	Object providing access to data on a project component; the type of the object depends on the component type .
	Name (see page 1270)	Component name .
	Type (see page 1271)	Project component type .

I.11.2.1 Data

C++

```
HRESULT get_Data(IDispatch*);
HRESULT put_Data(IDispatch);
```

C#

```
public IDispatch Data { get; set; }
```

Visual Basic

```
Public Data As IDispatch
```

Description

Object providing access to data on a project component; the type of the object depends on the component type .

I.11.2.2 Name

C++

```
HRESULT get_Name(BSTR*);
HRESULT put_Name(BSTR);
```

C#

```
public String Name { get; set; }
```

Visual Basic

```
Public Name As String
```

Description

Component name .

I.11.2.3 Type

C++

```
HRESULT get_Type(IRobotProjectComponentType*);
HRESULT put_Type(IRobotProjectComponentType);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRobotProjectComponentType Type { get; set; }
```

Visual Basic

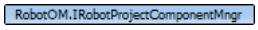
```
Public Type As IRobotProjectComponentType
```

Description

Project component type .

I.12 IRobotProjectComponentMngr

Class Hierarchy



C++

```
interface IRobotProjectComponentMngr : IDispatch;
```

C#

```
public interface IRobotProjectComponentMngr;
```

Visual Basic

```
Public Interface IRobotProjectComponentMngr
```

Description

Manager of project components manages tree-like structures containing all components defined in a project. Project components are divided into different levels. The manager makes available a list of all levels that are indexed with numbers from 1 to the number of levels. Additionally, each level has its name. Project components may be taken by levels. It is possible to take from the project all the components of an indicated type from the level of the indicated name. .

I.12.1 IRobotProjectComponentMngr Members

The following tables list the members exposed by IRobotProjectComponentMngr.

Public Fields

	Name	Description
	LevelCount (see page 1272)	Number of levels in the structure containing all project components. .
	StdLevelName (see page 1272)	Name of standard level Available since version 2.0.

Public Methods

	Name	Description
	Create (see page 1273)	Function creates a project component of the defined type and locate it on a specified level. Available since version 2.0.
	Get (see page 1273)	The function returns a collection of components of the indicated type from the level of the indicated name. Collection components are objects of the RobotProjectComponent type. .
	GetLevelName (see page 1273)	The function returns the name of the level with the indicated index. Levels are indexed with numbers from 1 to LevelCount (see page 1272). .

I.12.2 IRobotProjectComponentMngr Fields

The fields of the IRobotProjectComponentMngr class are listed here.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	LevelCount (see page 1272)	Number of levels in the structure containing all project components. .
	StdLevelName (see page 1272)	Name of standard level Available since version 2.0.

I.12.2.1 LevelCount

C++

```
HRESULT get_LevelCount(long*);
HRESULT put_LevelCount(long);
```

C#

```
public long LevelCount { get; set; }
```

Visual Basic

```
Public LevelCount As long
```

Description

Number of levels in the structure containing all project components. .

I.12.2.2 StdLevelName

C++

```
HRESULT get_StdLevelName(BSTR*);
```

C#

```
public String StdLevelName { get; }
```

Visual Basic

```
Public ReadOnly StdLevelName As String
```

Description

Name of standard level Available since version 2.0.

I.12.3 IRobotProjectComponentMngr Methods

The methods of the IRobotProjectComponentMngr class are listed here.

Public Methods

	Name	Description
	Create (see page 1273)	Function creates a project component of the defined type and locate it on a specified level. Available since version 2.0.
	Get (see page 1273)	The function returns a collection of components of the indicated type from the level of the indicated name. Collection components are objects of the RobotProjectComponent type. .
	GetLevelName (see page 1273)	The function returns the name of the level with the indicated index. Levels are indexed with numbers from 1 to LevelCount (see page 1272). .

I.12.3.1 Create

C++

```
HRESULT Create(IRobotProjectComponentType _cmpnt_type, BSTR _cmpnt_name, BSTR _level_name,
IRobotProjectComponent** ret);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRobotProjectComponent Create(IRobotProjectComponentType _cmpnt_type, String
_cmpnt_name, String _level_name);
```

Visual Basic

```
Public Function Create(_cmpnt_type As IRobotProjectComponentType, _cmpnt_name As String,
_level_name As String) As IRobotProjectComponent
```

Description

Function creates a project component of the defined type and locate it on a specified level. Available since version 2.0.

I.12.3.2 Get

C++

```
HRESULT Get(IRobotProjectComponentType _cmpnt_type, BSTR _level_name, IRobotCollection**
ret);
```

C#

```
public IRobotCollection Get(IRobotProjectComponentType _cmpnt_type, String _level_name);
```

Visual Basic

```
Public Function Get(_cmpnt_type As IRobotProjectComponentType, _level_name As String) As
IRobotCollection
```

Description

The function returns a collection of components of the indicated type from the level of the indicated name. Collection components are objects of the RobotProjectComponent type. .

I.12.3.3 GetLevelName

C++

```
HRESULT GetLevelName(long _level_idx, BSTR* ret);
```

C#

```
public String GetLevelName(long _level_idx);
```

Visual Basic

```
Public Function GetLevelName(_level_idx As long) As String
```

Description

The function returns the name of the level with the indicated index. Levels are indexed with numbers from 1 to LevelCount (see page 1272). .

II Support of external file formats

A structure and an entire project is saved to the file (RTD format). It is, however, possible to open and save structures defined in other external file formats. .

Enumerations

	Name	Description
	IRobotExternalFileFormat (see page 1274)	A set of identifiers is defined to describe different external file formats accepted by Robot. .

(c) 2014 Autodesk Inc. All rights reserved

	IRobotSTRParamType (see page 1281)	The Robot text file allows one to define variables of different types. Depending on variable type, they may assume values of different character. A set of identifiers is defined to refer to different types of variables available in STR files. .

Interfaces

	Name	Description
	IRobotSTRParams (see page 1274)	Interface granting access to the set of parameters (variables) defined by the STR format files. .
	IRobotSTRParameter (see page 1276)	The interface describing the variable of text file in STR format. .

II.1 IRobotExternalFileFormat

C++

```
enum IRobotExternalFileFormat;
```

C#

```
public enum IRobotExternalFileFormat;
```

Visual Basic

```
Public Enum IRobotExternalFileFormat
```

Members

Members	Description
I_EFF_STR = 1	Text file STR format .
I_EFF_DXF = 2	DXF format .

Description

A set of identifiers is defined to describe different external file formats accepted by Robot. .

II.2 IRobotSTRParams

Class Hierarchy

RobotOM.IRobotSTRParams

C++

```
interface IRobotSTRParams : IDispatch;
```

C#

```
public interface IRobotSTRParams;
```

Visual Basic

```
Public Interface IRobotSTRParams
```

Description

Interface granting access to the set of parameters (variables) defined by the STR format files. .

II.2.1 IRobotSTRParams Members

The following tables list the members exposed by IRobotSTRParams.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	Count (see page 1275)	Number of managed parameters .

Public Methods

	Name	Description
	FindParameter (see page 1276)	The function returns the index associated with the parameter of the indicated name .
	GetParameter (see page 1276)	The function returns a structure describing a parameter with the indicated index .

II.2.2 IRobotSTRParams Fields

The fields of the IRobotSTRParams class are listed here.

Public Fields

	Name	Description
	Count (see page 1275)	Number of managed parameters .

II.2.2.1 Count

C++

```
HRESULT get_Count(long*);
```

C#

```
public long Count { get; }
```

Visual Basic

```
Public ReadOnly Count As long
```

Description

Number of managed parameters .

II.2.3 IRobotSTRParams Methods

The methods of the IRobotSTRParams class are listed here.

Public Methods

	Name	Description
	FindParameter (see page 1276)	The function returns the index associated with the parameter of the indicated name .
	GetParameter (see page 1276)	The function returns a structure describing a parameter with the indicated index .

II.2.3.1 FindParameter

C++

```
HRESULT FindParameter(BSTR _param_name, long* ret);
```

C#

```
public long FindParameter(String _param_name);
```

Visual Basic

```
Public Function FindParameter(_param_name As String) As long
```

(c) 2014 Autodesk Inc. All rights reserved

Description

The function returns the index associated with the parameter of the indicated name .

II.2.3.2 GetParameter

C++

```
HRESULT GetParameter(long _idx, IRobotSTRParameter** ret);
```

C#

```
public IRobotSTRParameter GetParameter(long _idx);
```

Visual Basic

```
Public Function GetParameter(_idx As long) As IRobotSTRParameter
```

Description

The function returns a structure describing a parameter with the indicated index .

II.3 IRobotSTRParameter

Class Hierarchy

RobotOM.IRobotSTRParameter

C++

```
interface IRobotSTRParameter : IDispatch;
```

C#

```
public interface IRobotSTRParameter;
```

Visual Basic

```
Public Interface IRobotSTRParameter
```

Description

The interface describing the variable of text file in STR format. .

II.3.1 IRobotSTRParameter Members

The following tables list the members exposed by IRobotSTRParameter.

Public Fields

	Name	Description
	Description (see page 1277)	Variable description .
	DoubleVal_1 (see page 1278)	Value of the variable of the I_STR_PT_DOUBLE type. It is also the first out of three values of the variable of the I_STR_PT_DOUBLE_3 type. .
	DoubleVal_2 (see page 1278)	The second out of three variables of the I_STR_PT_DOUBLE_3 type .
	DoubleVal_3 (see page 1278)	The third out of three variables of the I_STR_PT_DOUBLE_3 type .
	FilePathVal (see page 1279)	Value of the variable of the I_STR_PT_GROUP type representing the access path to the file with the appropriate bitmap. .
	IntegerVal (see page 1279)	Value of the variable of the I_STR_PT_INTEGER type .
	IsActive (see page 1279)	Flag indicating if the variable is active .
	Name (see page 1279)	Variable name .
	SelectionVal (see page 1280)	Value of the variable of the I_STR_PT_SELECTION type describing a selection by means of text .

(c) 2014 Autodesk Inc. All rights reserved

	TextList (see page 1280)	String of characters containing all possible values (separated by commas) of the variable of the I_STR_PT_TEXT_LIST type. .
	TextVal (see page 1280)	Value of the variable of the I_STR_PT_TEXT type. It is also the value currently selected from the list for the variable of the I_STR_PT_TEXT_LIST type. .
	Type (see page 1280)	Variable type .

II.3.2 IRobotSTRParameter Fields

The fields of the IRobotSTRParameter class are listed here.

Public Fields

	Name	Description
	Description (see page 1277)	Variable description .
	DoubleVal_1 (see page 1278)	Value of the variable of the I_STR_PT_DOUBLE type. It is also the first out of three values of the variable of the I_STR_PT_DOUBLE_3 type. .
	DoubleVal_2 (see page 1278)	The second out of three variables of the I_STR_PT_DOUBLE_3 type .
	DoubleVal_3 (see page 1278)	The third out of three variables of the I_STR_PT_DOUBLE_3 type .
	FilePathVal (see page 1279)	Value of the variable of the I_STR_PT_GROUP type representing the access path to the file with the appropriate bitmap. .
	IntegerVal (see page 1279)	Value of the variable of the I_STR_PT_INTEGER type .
	IsActive (see page 1279)	Flag indicating if the variable is active .
	Name (see page 1279)	Variable name .
	SelectionVal (see page 1280)	Value of the variable of the I_STR_PT_SELECTION type describing a selection by means of text .
	TextList (see page 1280)	String of characters containing all possible values (separated by commas) of the variable of the I_STR_PT_TEXT_LIST type. .
	TextVal (see page 1280)	Value of the variable of the I_STR_PT_TEXT type. It is also the value currently selected from the list for the variable of the I_STR_PT_TEXT_LIST type. .
	Type (see page 1280)	Variable type .

II.3.2.1 Description

C++

```
HRESULT get_Description(BSTR*);
```

C#

```
public String Description { get; }
```

Visual Basic

```
Public ReadOnly Description As String
```

Description

Variable description .

II.3.2.2 DoubleVal_1

C++

```
HRESULT get_DoubleVal_1(double*);
HRESULT put_DoubleVal_1(double);
```

C#

```
public double DoubleVal_1 { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public DoubleVal_1 As double
```

Description

Value of the variable of the I_STR_PT_DOUBLE type. It is also the first out of three values of the variable of the I_STR_PT_DOUBLE_3 type. .

II.3.2.3 DoubleVal_2

C++

```
HRESULT get_DoubleVal_2(double*);
HRESULT put_DoubleVal_2(double);
```

C#

```
public double DoubleVal_2 { get; set; }
```

Visual Basic

```
Public DoubleVal_2 As double
```

Description

The second out of three variables of the I_STR_PT_DOUBLE_3 type .

II.3.2.4 DoubleVal_3

C++

```
HRESULT get_DoubleVal_3(double*);
HRESULT put_DoubleVal_3(double);
```

C#

```
public double DoubleVal_3 { get; set; }
```

Visual Basic

```
Public DoubleVal_3 As double
```

Description

The third out of three variables of the I_STR_PT_DOUBLE_3 type .

II.3.2.5 FilePathVal

C++

```
HRESULT get_FilePathVal(BSTR*);
HRESULT put_FilePathVal(BSTR);
```

C#

```
public String FilePathVal { get; set; }
```

Visual Basic

```
Public FilePathVal As String
```

Description

Value of the variable of the I_STR_PT_GROUP type representing the access path to the file with the appropriate bitmap. .

II.3.2.6 IntegerVal

C++

```
HRESULT get_IntegerVal(long*);
HRESULT put_IntegerVal(long);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public long IntegerVal { get; set; }
```

Visual Basic

```
Public IntegerVal As long
```

Description

Value of the variable of the I_STR_PT_INTEGER type .

II.3.2.7 IsActive

C++

```
HRESULT get_IsActive(VARIANT_BOOL*);
```

C#

```
public bool IsActive { get; }
```

Visual Basic

```
Public ReadOnly IsActive As Boolean
```

Description

Flag indicating if the variable is active .

II.3.2.8 Name

C++

```
HRESULT get_Name(BSTR*);
```

C#

```
public String Name { get; }
```

Visual Basic

```
Public ReadOnly Name As String
```

Description

Variable name .

II.3.2.9 SelectionVal

C++

```
HRESULT get_SelectionVal(BSTR*);
HRESULT put_SelectionVal(BSTR);
```

C#

```
public String SelectionVal { get; set; }
```

Visual Basic

```
Public SelectionVal As String
```

Description

Value of the variable of the I_STR_PT_SELECTION type describing a selection by means of text .

II.3.2.10 TextList

C++

```
HRESULT get_TextList(BSTR*);
HRESULT put_TextList(BSTR);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public String TextList { get; set; }
```

Visual Basic

```
Public TextList As String
```

Description

String of characters containing all possible values (separated by commas) of the variable of the I_STR_PT_TEXT_LIST type. .

II.3.2.11 TextVal

C++

```
HRESULT get_TextVal(BSTR*);
HRESULT put_TextVal(BSTR);
```

C#

```
public String TextVal { get; set; }
```

Visual Basic

```
Public TextVal As String
```

Description

Value of the variable of the I_STR_PT_TEXT type. It is also the value currently selected from the list for the variable of the I_STR_PT_TEXT_LIST type. .

II.3.2.12 Type

C++

```
HRESULT get_Type(IRobotSTRParamType*);
```

C#

```
public IRobotSTRParamType Type { get; }
```

Visual Basic

```
Public ReadOnly Type As IRobotSTRParamType
```

Description

Variable type .

II.4 IRobotSTRParamType

C++

```
enum IRobotSTRParamType;
```

C#

```
public enum IRobotSTRParamType;
```

Visual Basic

```
Public Enum IRobotSTRParamType
```

Members

Members	Description
I_STR_PT_INTEGER = 1	Variable assuming the value in the form of an integer .
I_STR_PT_DOUBLE = 2	Variable assuming the value in the form of a real number .

(c) 2014 Autodesk Inc. All rights reserved

I_STR_PT_TEXT = 3	Variable assuming the value in the form of a text .
I_STR_PT_SELECTION = 4	Variable assuming the value in the form of a text describing selection .
I_STR_PT_DOUBLE_3 = 5	Variable assuming the value described by means of three real numbers .
I_STR_PT_TEXT_LIST = 6	Variable assuming the value in the form of a specially formated text. It consists of fragments separated by commas, defining the list of possible text values, and the (repeated) currently-selected fragment. .
I_STR_PT_SECTION = 7	Variable containing the section name (as a text) .
I_STR_PT_GROUP = 8	Variable representing a group of varibles. Its value consists of two elements: group name (stored in the TextVal variable), and the path to the file containing the bitmap associated with the group (FilePathVal) .

Description

The Robot text file allows one to define variables of different types. Depending on variable type, they may assume values of different character. A set of identifiers is defined to refer to different types of variables available in STR files. .

III Job preferences

Enumerations

	Name	Description
	IRobotCodeType (see page 1312)	Set of identifiers determining the types of codes that may be set in the Robot program.
	IRobotDatabaseType (see page 1326)	Available database types.
	IRobotCalculationModelCoherence (see page 1326)	

Interfaces

	Name	Description
	IRobotProjectPreferences (see page 1296)	Interface describing the settings for the current project (job).
	IRobotSectionDatabaseList (see page 1305)	The interface describing a list of section databases .
	IRobotMaterialDatabase (see page 1308)	The interface providing access to material database .
	IRobotSectionDatabase (see page 1312)	The interface providing access to section database.
	IRobotProjectPreferencesEvents (see page 1315)	Events related to the change of settings for the current project.
	IRobotEurocodeSteelDesignFactors (see page 1316)	Partial factors for the steel design according to Eurocode.
	IRobotEurocodeSteelConnectionFactors (see page 1319)	Partial factors for the steel connection design according to Eurocode.
	IRobotEurocodeFactors (see page 1324)	Partial factors for Eurocodes.

III.1 Units and formats

Available since version 3.5.

(c) 2014 Autodesk Inc. All rights reserved

Enumerations

	Name	Description
	IRobotUnitType (see page 1282)	Unit types.
	IRobotUnitEditionType (see page 1285)	Types of editable units. .

Interfaces

	Name	Description
	IRobotUnitData (see page 1283)	Unit definition interface.
	IRobotUnitMngr (see page 1286)	Interface of unit support management. .
	IRobotUnitComplexData (see page 1290)	Interface of composed unit definition. A composed unit is a unit being a function of other two units. For example, unit of the moment is a product of force unit and length unit. .
	IRobotUnitEditionServer (see page 1291)	Interface enabling unit edition. .
	IRobotUnitEditionData (see page 1294)	Interface of unit edit definition. An edit definition is a definition which determines a base unit and a proportionality factor. .
	IRobotUnitMngrEvents (see page 1295)	Events (notifications) generated by the unit manager.

III.1.1 IRobotUnitType

C++

```
enum IRobotUnitType;
```

C#

```
public enum IRobotUnitType;
```

Visual Basic

```
Public Enum IRobotUnitType
```

Members

Members	Description
I_UT_STRUCTURE_DIMENSION = 1	Available since version 3.5.
I_UT_SECTION_DIMENSION = 2	Available since version 3.5.
I_UT_SECTION_PROPERTIES = 3	Available since version 3.5.
I_UT_STEEL_CONECTIONS = 4	Available since version 3.5.
I_UT_DIAMETER_RC_BASE = 5	Available since version 3.5.
I_UT_REINFORCEMENT_AREAS = 6	Available since version 3.5.
I_UT_FORCE = 7	Available since version 3.5.
I_UT_MOMENT = 8	Available since version 3.5.
I_UT_STRESS = 9	Available since version 3.5.
I_UT_DISPLACEMENT = 10	Available since version 3.5.
I_UT_ANGLE_ROTATION_DATA = 11	Available since version 3.5.
I_UT_ANGLE_ROTATION_RESULT = 12	Available since version 3.5.
I_UT_TEMPERATURE = 13	Available since version 3.5.
I_UT_WEIGHT = 14	Available since version 3.5.
I_UT_MASS = 15	Available since version 3.5.
I_UT_DIMENSIONLESS_QUALITY = 16	Available since version 3.5.
I_UT_RULER = 17	Available since version 3.5.

(c) 2014 Autodesk Inc. All rights reserved

Description

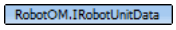
Unit types.

Version

Available since version 3.5.

III.1.2 IRobotUnitData

Class Hierarchy



C++

```
interface IRobotUnitData : IDispatch;
```

C#

```
public interface IRobotUnitData;
```

Visual Basic

```
Public Interface IRobotUnitData
```

Description

Unit definition interface.

Version

Available since version 3.5.

III.1.2.1 IRobotUnitData Members

The following tables list the members exposed by IRobotUnitData.

Public Fields

	Name	Description
	E (see page 1284)	Type (see page 1285) of unit presentation: true value - exponential presentation, false value - decimal presentation .
	Name (see page 1284)	Unit name.
	Precision (see page 1285)	Precision of unit presentation - number of decimal places.
	Type (see page 1285)	Unit type.

III.1.2.2 IRobotUnitData Fields

The fields of the IRobotUnitData class are listed here.

Public Fields

	Name	Description
	E (see page 1284)	Type (see page 1285) of unit presentation: true value - exponential presentation, false value - decimal presentation .
	Name (see page 1284)	Unit name.
	Precision (see page 1285)	Precision of unit presentation - number of decimal places.
	Type (see page 1285)	Unit type.

III.1.2.2.1 E

C++

```
HRESULT get_E(VARIANT_BOOL*);
HRESULT put_E(VARIANT_BOOL);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public bool E { get; set; }
```

Visual Basic

```
Public E As Boolean
```

Description

Type (see page 1285) of unit presentation: true value - exponential presentation, false value - decimal presentation .

Version

Available since version 3.5.

III.1.2.2.2 Name

C++

```
HRESULT get_Name(BSTR*);
HRESULT put_Name(BSTR);
```

C#

```
public String Name { get; set; }
```

Visual Basic

```
Public Name As String
```

Description

Unit name.

Version

Available since version 3.5.

III.1.2.2.3 Precision

C++

```
HRESULT get_Precision(long*);
HRESULT put_Precision(long);
```

C#

```
public long Precision { get; set; }
```

Visual Basic

```
Public Precision As long
```

Description

Precision of unit presentation - number of decimal places.

Version

Available since version 3.5.

III.1.2.2.4 Type

C++

```
HRESULT get_Type(IRobotUnitType*);
```

C#

```
public IRobotUnitType Type { get; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public ReadOnly Type As IRobotUnitType
```

Description

Unit type.

Version

Available since version 3.5.

III.1.3 IRobotUnitEditionType

C++

```
enum IRobotUnitEditionType;
```

C#

```
public enum IRobotUnitEditionType;
```

Visual Basic

```
Public Enum IRobotUnitEditionType
```

Members

Members	Description
I_UMT_LENGTH = 0	Available since version 3.5.
I_UMT_FORCE = 1	Available since version 3.5.
I_UMT_MASS = 2	Available since version 3.5.

Description

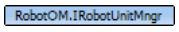
Types of editable units. .

Version

Available since version 3.5.

III.1.4 IRobotUnitMngr

Class Hierarchy



C++

```
interface IRobotUnitMngr : IDispatch;
```

C#

```
public interface IRobotUnitMngr;
```

Visual Basic

```
Public Interface IRobotUnitMngr
```

Description

Interface of unit support management. .

Version

Available since version 3.5.

(c) 2014 Autodesk Inc. All rights reserved

III.1.4.1 IRobotUnitMngr Members

The following tables list the members exposed by IRobotUnitMngr.

Public Fields

	Name	Description
	UnitEdition (see page 1287)	Edition server.
	UseMetricAsDefault (see page 1287)	Flage indicating if the software uses metric units by default.

Public Methods

	Name	Description
	Count (see page 1288)	Function returns the number of available units of the specified type. .
	Get (see page 1288)	Function takes definition of the active unit for a given type.
	GetCoeff (see page 1288)	The function returns the coefficient of conversion of the currently-selected unit of a given type to the base unit.
	GetCoeff2 (see page 1289)	The function returns the conversion factor for a unit of a given type and specified index to the base unit.
	GetName (see page 1289)	Function takes name of the unit of the specified type with the determined index. .
	Refresh (see page 1289)	Function forces refreshing current information about units in the whole application.
	Set (see page 1289)	Function determines active unit of the specified type based on the provided definition. .

III.1.4.2 IRobotUnitMngr Fields

The fields of the IRobotUnitMngr class are listed here.

Public Fields

	Name	Description
	UnitEdition (see page 1287)	Edition server.
	UseMetricAsDefault (see page 1287)	Flage indicating if the software uses metric units by default.

III.1.4.2.1 UnitEdition

C++

```
HRESULT get_UnitEdition(IRobotUnitEditionServer**);
```

C#

```
public IRobotUnitEditionServer UnitEdition { get; }
```

Visual Basic

```
Public ReadOnly UnitEdition As IRobotUnitEditionServer
```

Description

Edition server.

Version

Available since version 3.5.

III.1.4.2.2 UseMetricAsDefault

C++

```
HRESULT get_UseMetricAsDefault(VARIANT_BOOL*);
HRESULT put_UseMetricAsDefault(VARIANT_BOOL);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public bool UseMetricAsDefault { get; set; }
```

Visual Basic

```
Public UseMetricAsDefault As Boolean
```

Description

Flage indicating if the software uses metric units by default.

Version

Available since version 9.7.

III.1.4.3 IRobotUnitMngr Methods

The methods of the IRobotUnitMngr class are listed here.

Public Methods

	Name	Description
	Count (see page 1288)	Function returns the number of available units of the specified type. .
	Get (see page 1288)	Function takes definition of the active unit for a given type.
	GetCoeff (see page 1288)	The function returns the coefficient of conversion of the currently-selected unit of a given type to the base unit.
	GetCoeff2 (see page 1289)	The function returns the conversion factor for a unit of a given type and specified index to the base unit.
	GetName (see page 1289)	Function takes name of the unit of the specified type with the determined index. .
	Refresh (see page 1289)	Function forces refreshing current information about units in the whole application.
	Set (see page 1289)	Function determines active unit of the specified type based on the provided definition. .

III.1.4.3.1 Count

C++

```
HRESULT Count(IRobotUnitType _type, long* ret);
```

C#

```
public long Count(IRobotUnitType _type);
```

Visual Basic

```
Public Function Count(_type As IRobotUnitType) As long
```

Description

Function returns the number of available units of the specified type. .

Version

Available since version 3.5.

III.1.4.3.2 Get

C++

```
HRESULT Get(IRobotUnitType _type, IRobotUnitData** ret);
```

C#

```
public IRobotUnitData Get(IRobotUnitType _type);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Function Get(_type As IRobotUnitType) As IRobotUnitData
```

Description

Function takes definition of the active unit for a given type.

Version

Available since version 3.5.

III.1.4.3.3 GetCoeff

C++

```
HRESULT GetCoeff(IRobotUnitType _unit_type, double* ret);
```

C#

```
public double GetCoeff(IRobotUnitType _unit_type);
```

Visual Basic

```
Public Function GetCoeff(_unit_type As IRobotUnitType) As double
```

Description

The function returns the coefficient of conversion of the currently-selected unit of a given type to the base unit.

III.1.4.3.4 GetCoeff2

C++

```
HRESULT GetCoeff2(IRobotUnitType _unit_type, long _unit_idx, double* ret);
```

C#

```
public double GetCoeff2(IRobotUnitType _unit_type, long _unit_idx);
```

Visual Basic

```
Public Function GetCoeff2(_unit_type As IRobotUnitType, _unit_idx As long) As double
```

Description

The function returns the conversion factor for a unit of a given type and specified index to the base unit.

Version

Available since version 12.

III.1.4.3.5 GetName

C++

```
HRESULT GetName(IRobotUnitType _type, long _idx);
```

C#

```
public void GetName(IRobotUnitType _type, long _idx);
```

Visual Basic

```
Public Sub GetName(_type As IRobotUnitType, _idx As long)
```

Description

Function takes name of the unit of the specified type with the determined index. .

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.5.

III.1.4.3.6 Refresh

C++

```
HRESULT Refresh();
```

C#

```
public void Refresh();
```

Visual Basic

```
Public Sub Refresh()
```

Description

Function forces refreshing current information about units in the whole application.

III.1.4.3.7 Set

C++

```
HRESULT Set(IRobotUnitType _type, IRobotUnitData _data);
```

C#

```
public void Set(IRobotUnitType _type, IRobotUnitData _data);
```

Visual Basic

```
Public Sub Set(_type As IRobotUnitType, _data As IRobotUnitData)
```

Description

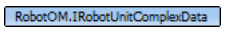
Function determines active unit of the specified type based on the provided definition. .

Version

Available since version 3.5.

III.1.5 IRobotUnitComplexData

Class Hierarchy



C++

```
interface IRobotUnitComplexData : IDispatch;
```

C#

```
public interface IRobotUnitComplexData;
```

Visual Basic

```
Public Interface IRobotUnitComplexData
```

Description

Interface of composed unit definition. A composed unit is a unit being a function of other two units. For example, unit of the moment is a product of force unit and length unit. .

Version

Available since version 3.5.

(c) 2014 Autodesk Inc. All rights reserved

III.1.5.1 IRobotUnitComplexData Members

The following tables list the members exposed by IRobotUnitComplexData.

Public Fields

	Name	Description
	Name2 (see page 1290)	Name of the second component unit. .

III.1.5.2 IRobotUnitComplexData Fields

The fields of the IRobotUnitComplexData class are listed here.

Public Fields

	Name	Description
	Name2 (see page 1290)	Name of the second component unit. .

III.1.5.2.1 Name2

C++

```
HRESULT get_Name2(BSTR*);
HRESULT put_Name2(BSTR);
```

C#

```
public String Name2 { get; set; }
```

Visual Basic

```
Public Name2 As String
```

Description

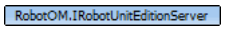
Name of the second component unit. .

Version

Available since version 3.5.

III.1.6 IRobotUnitEditionServer

Class Hierarchy



C++

```
interface IRobotUnitEditionServer : IDispatch;
```

C#

```
public interface IRobotUnitEditionServer;
```

Visual Basic

```
Public Interface IRobotUnitEditionServer
```

Description

Interface enabling unit edition. .

Version

Available since version 3.5.

(c) 2014 Autodesk Inc. All rights reserved

III.1.6.1 IRobotUnitEditionServer Members

The following tables list the members exposed by IRobotUnitEditionServer.

Public Methods

	Name	Description
	Count (see page 1292)	Function returns the number of units of the specified type. .
	Delete (see page 1292)	Function deletes a unit of the specified type with the determined index. .
	Find (see page 1292)	Function returns index of the unit of the specified type with the determined name. .
	Get (see page 1293)	Function takes the edit definition of unit of the specified type with the specified index. .
	New (see page 1293)	Function generates a new edit unit based on given parameters. .
	Set (see page 1293)	Function updates unit definition. .

III.1.6.2 IRobotUnitEditionServer Methods

The methods of the IRobotUnitEditionServer class are listed here.

Public Methods

	Name	Description
	Count (see page 1292)	Function returns the number of units of the specified type. .
	Delete (see page 1292)	Function deletes a unit of the specified type with the determined index. .
	Find (see page 1292)	Function returns index of the unit of the specified type with the determined name. .
	Get (see page 1293)	Function takes the edit definition of unit of the specified type with the specified index. .
	New (see page 1293)	Function generates a new edit unit based on given parameters. .
	Set (see page 1293)	Function updates unit definition. .

III.1.6.2.1 Count

C++

```
HRESULT Count(IRobotUnitEditionType _type, long* ret);
```

C#

```
public long Count(IRobotUnitEditionType _type);
```

Visual Basic

```
Public Function Count(_type As IRobotUnitEditionType) As long
```

Description

Function returns the number of units of the specified type. .

Version

Available since version 3.5.

III.1.6.2.2 Delete

C++

```
HRESULT Delete(IRobotUnitEditionType _type, long _idx);
```

C#

```
public void Delete(IRobotUnitEditionType _type, long _idx);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Sub Delete(_type As IRobotUnitEditionType, _idx As long)
```

Description

Function deletes a unit of the specified type with the determined index. .

Version

Available since version 3.5.

III.1.6.2.3 Find

C++

```
HRESULT Find(IRobotUnitEditionType _type, BSTR _name, long* ret);
```

C#

```
public long Find(IRobotUnitEditionType _type, String _name);
```

Visual Basic

```
Public Function Find(_type As IRobotUnitEditionType, _name As String) As long
```

Description

Function returns index of the unit of the specified type with the determined name. .

Version

Available since version 3.5.

III.1.6.2.4 Get

C++

```
HRESULT Get(IRobotUnitEditionType _type, long _idx, IRobotUnitEditionData** ret);
```

C#

```
public IRobotUnitEditionData Get(IRobotUnitEditionType _type, long _idx);
```

Visual Basic

```
Public Function Get(_type As IRobotUnitEditionType, _idx As long) As IRobotUnitEditionData
```

Description

Function takes the edit definition of unit of the specified type with the specified index. .

Version

Available since version 3.5.

III.1.6.2.5 New

C++

```
HRESULT New(IRobotUnitEditionType _type, name _unit, double _coeff, long* ret);
```

C#

```
public long New(IRobotUnitEditionType _type, name _unit, double _coeff);
```

Visual Basic

```
Public Function New(_type As IRobotUnitEditionType, _unit As name, _coeff As double) As long
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Function generates a new edit unit based on given parameters. .

Version

Available since version 3.5.

III.1.6.2.6 Set

C++

```
HRESULT Set(IRobotUnitEditionData _data);
```

C#

```
public void Set(IRobotUnitEditionData _data);
```

Visual Basic

```
Public Sub Set(_data As IRobotUnitEditionData)
```

Description

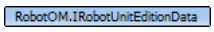
Function updates unit definition. .

Version

Available since version 3.5.

III.1.7 IRobotUnitEditionData

Class Hierarchy



C++

```
interface IRobotUnitEditionData : IDispatch;
```

C#

```
public interface IRobotUnitEditionData;
```

Visual Basic

```
Public Interface IRobotUnitEditionData
```

Description

Interface of unit edit definition. An edit definition is a definition which determines a base unit and a proportionality factor. .

Version

Available since version 3.5.

III.1.7.1 IRobotUnitEditionData Members

The following tables list the members exposed by IRobotUnitEditionData.

Public Fields

	Name	Description
	Coefficient (see page 1294)	Number determining the ratio of defined unit to base unit.
	Type (see page 1295)	Type of edit unit.
	Unit (see page 1295)	Name of base unit .

(c) 2014 Autodesk Inc. All rights reserved

III.1.7.2 IRobotUnitEditionData Fields

The fields of the IRobotUnitEditionData class are listed here.

Public Fields

	Name	Description
	Coefficient (see page 1294)	Number determining the ratio of defined unit to base unit.
	Type (see page 1295)	Type of edit unit.
	Unit (see page 1295)	Name of base unit .

III.1.7.2.1 Coefficient

C++

```
HRESULT get_Coefficient(double*);
HRESULT put_Coefficient(double);
```

C#

```
public double Coefficient { get; set; }
```

Visual Basic

```
Public Coefficient As double
```

Description

Number determining the ratio of defined unit to base unit.

Version

Available since version 3.5.

III.1.7.2.2 Type

C++

```
HRESULT get_Type(IRobotUnitEditionType*);
```

C#

```
public IRobotUnitEditionType Type { get; }
```

Visual Basic

```
Public ReadOnly Type As IRobotUnitEditionType
```

Description

Type of edit unit.

Version

Available since version 3.5.

III.1.7.2.3 Unit

C++

```
HRESULT get_Unit(BSTR*);
HRESULT put_Unit(BSTR);
```

C#

```
public String Unit { get; set; }
```

Visual Basic

```
Public Unit As String
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Name of base unit .

Version

Available since version 3.5.

III.1.8 IRobotUnitMngrEvents

Class Hierarchy

RobotOM.IRobotUnitMngrEvents

C++

```
interface IRobotUnitMngrEvents : IDispatch;
```

C#

```
public interface IRobotUnitMngrEvents;
```

Visual Basic

```
Public Interface IRobotUnitMngrEvents
```

Description

Events (notifications) generated by the unit manager.

III.1.8.1 IRobotUnitMngrEvents Members

The following tables list the members exposed by IRobotUnitMngrEvents.

Public Methods

	Name	Description
	UnitsChanged (see page 1296)	Event generated after any change in the settings of units.

III.1.8.2 IRobotUnitMngrEvents Methods

The methods of the IRobotUnitMngrEvents class are listed here.

Public Methods

	Name	Description
	UnitsChanged (see page 1296)	Event generated after any change in the settings of units.

III.1.8.2.1 UnitsChanged

C++

```
HRESULT UnitsChanged();
```

C#

```
public void UnitsChanged();
```

Visual Basic

```
Public Sub UnitsChanged()
```

Description

Event generated after any change in the settings of units.

(c) 2014 Autodesk Inc. All rights reserved

III.2 IRobotProjectPreferences

Class Hierarchy

RobotOM.IRobotProjectPreferences

C++

```
interface IRobotProjectPreferences : IDispatch;
```

C#

```
public interface IRobotProjectPreferences;
```

Visual Basic

```
Public Interface IRobotProjectPreferences
```

Description

Interface describing the settings for the current project (job).

III.2.1 IRobotProjectPreferences Members

The following tables list the members exposed by IRobotProjectPreferences.

Public Fields

	Name	Description
	CalcModelCoherence (see page 1298)	
	EurocodeFactors (see page 1298)	Partial factors for Eurocode.
	KinematicConstraints (see page 1299)	
	Materials (see page 1299)	Active material database .
	MeshAutoAdjust (see page 1299)	Automatic adjustment of finite element mesh.
	MeshAutoAdjustIterationCount (see page 1299)	
	MeshParams (see page 1300)	Meshing parameters.
	MeshParamsFloors (see page 1300)	Mesh generation parameters for slabs.
	MeshParamsWalls (see page 1300)	Mesh generation parameters for walls.
	SectionsActive (see page 1301)	The list of active (selected) section databases .
	SectionsFound (see page 1301)	The list of available (found) names for the section database .
	Units (see page 1301)	Object managing units and formats.
	VehiclesActive (see page 1301)	List of names of active (selected) vehicle databases.
	VehiclesFound (see page 1302)	List of names of available (found) vehicle databases.

Public Methods

	Name	Description
	GetActiveCode (see page 1302)	The function returns the name of the active code of the indicated type.
	GetActiveCodeNumber (see page 1303)	Function returns the numerical identifier of an active code of a given type.
	GetCurrentDatabase (see page 1303)	Function returns a name of the current database of the given type. .
	Save (see page 1303)	Function forces saving changes in the project settings. .
	SetActiveCode (see page 1304)	The function sets the active code of the indicated type. If selection of an indicated code is not possible, then, zero value is returned (False). .

(c) 2014 Autodesk Inc. All rights reserved

	SetActiveCodeNumber (see page 1304)	Function sets an active code of a given type based on the specified numerical identifier of the code.
	SetCurrentDatabase (see page 1304)	Function sets the given database as a current database of the given type.

III.2.2 IRobotProjectPreferences Fields

The fields of the IRobotProjectPreferences class are listed here.

Public Fields

	Name	Description
	CalcModelCoherence (see page 1298)	
	EurocodeFactors (see page 1298)	Partial factors for Eurocode.
	KinematicConstraints (see page 1299)	
	Materials (see page 1299)	Active material database .
	MeshAutoAdjust (see page 1299)	Automatic adjustment of finite element mesh.
	MeshAutoAdjustIterationCount (see page 1299)	
	MeshParams (see page 1300)	Meshing parameters.
	MeshParamsFloors (see page 1300)	Mesh generation parameters for slabs.
	MeshParamsWalls (see page 1300)	Mesh generation parameters for walls.
	SectionsActive (see page 1301)	The list of active (selected) section databases .
	SectionsFound (see page 1301)	The list of available (found) names for the section database .
	Units (see page 1301)	Object managing units and formats.
	VehiclesActive (see page 1301)	List of names of active (selected) vehicle databases.
	VehiclesFound (see page 1302)	List of names of available (found) vehicle databases.

III.2.2.1 CalcModelCoherence

C++

```
HRESULT get_CalcModelCoherence(IRobotCalculationModelCoherence*);
HRESULT put_CalcModelCoherence(IRobotCalculationModelCoherence);
```

C#

```
public IRobotCalculationModelCoherence CalcModelCoherence { get; set; }
```

Visual Basic

```
Public CalcModelCoherence As IRobotCalculationModelCoherence
```

Version

Available since version 12.

III.2.2.2 EurocodeFactors

C++

```
HRESULT get_EurocodeFactors(IRobotEurocodeFactors**);
```

C#

```
public IRobotEurocodeFactors EurocodeFactors { get; }
```

Visual Basic

```
Public ReadOnly EurocodeFactors As IRobotEurocodeFactors
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Partial factors for Eurocode.

Version

Available since version 9.7.

III.2.2.3 KinematicConstraints

C++

```
HRESULT get_KinematicConstraints(VARIANT_BOOL*);
HRESULT put_KinematicConstraints(VARIANT_BOOL);
```

C#

```
public bool KinematicConstraints { get; set; }
```

Visual Basic

```
Public KinematicConstraints As Boolean
```

Version

Available since version 12.

III.2.2.4 Materials

C++

```
HRESULT get_Materials(IRobotMaterialDatabase**);
```

C#

```
public IRobotMaterialDatabase Materials { get; }
```

Visual Basic

```
Public ReadOnly Materials As IRobotMaterialDatabase
```

Description

Active material database .

III.2.2.5 MeshAutoAdjust

C++

```
HRESULT get_MeshAutoAdjust(VARIANT_BOOL*);
HRESULT put_MeshAutoAdjust(VARIANT_BOOL);
```

C#

```
public bool MeshAutoAdjust { get; set; }
```

Visual Basic

```
Public MeshAutoAdjust As Boolean
```

Description

Automatic adjustment of finite element mesh.

Version

Available since version 4.5.

III.2.2.6 MeshAutoAdjustIterationCount

C++

```
HRESULT get_MeshAutoAdjustIterationCount(long*);
```

(c) 2014 Autodesk Inc. All rights reserved

```
HRESULT put_MeshAutoAdjustIterationCount(long);
```

C#

```
public long MeshAutoAdjustIterationCount { get; set; }
```

Visual Basic

```
Public MeshAutoAdjustIterationCount As long
```

Version

Available since version 13.4.

III.2.2.7 MeshParams

C++

```
HRESULT get_MeshParams(IRobotMeshParams**);
```

C#

```
public IRobotMeshParams MeshParams { get; }
```

Visual Basic

```
Public ReadOnly MeshParams As IRobotMeshParams
```

Description

Meshing parameters.

Version

Available since version 3.

III.2.2.8 MeshParamsFloors

C++

```
HRESULT get_MeshParamsFloors(IRobotMeshParams**);
```

C#

```
public IRobotMeshParams MeshParamsFloors { get; }
```

Visual Basic

```
Public ReadOnly MeshParamsFloors As IRobotMeshParams
```

Description

Mesh generation parameters for slabs.

Version

Available since version 12.

III.2.2.9 MeshParamsWalls

C++

```
HRESULT get_MeshParamsWalls(IRobotMeshParams**);
```

C#

```
public IRobotMeshParams MeshParamsWalls { get; }
```

Visual Basic

```
Public ReadOnly MeshParamsWalls As IRobotMeshParams
```

Description

Mesh generation parameters for walls.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 12.

III.2.2.10 SectionsActive

C++

```
HRESULT get_SectionsActive(IRobotSectionDatabaseList**);
```

C#

```
public IRobotSectionDatabaseList SectionsActive { get; }
```

Visual Basic

```
Public ReadOnly SectionsActive As IRobotSectionDatabaseList
```

Description

The list of active (selected) section databases .

III.2.2.11 SectionsFound

C++

```
HRESULT get_SectionsFound(IRobotSectionDatabaseList**);
```

C#

```
public IRobotSectionDatabaseList SectionsFound { get; }
```

Visual Basic

```
Public ReadOnly SectionsFound As IRobotSectionDatabaseList
```

Description

The list of available (found) names for the section database .

III.2.2.12 Units

C++

```
HRESULT get_Units(IRobotUnitMngr**);
```

C#

```
public IRobotUnitMngr Units { get; }
```

Visual Basic

```
Public ReadOnly Units As IRobotUnitMngr
```

Description

Object managing units and formats.

Version

Available since version 3.

III.2.2.13 VehiclesActive

C++

```
HRESULT get_VehiclesActive(IRobotVehicleDatabaseList**);
```

C#

```
public IRobotVehicleDatabaseList VehiclesActive { get; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public ReadOnly VehiclesActive As IRobotVehicleDatabaseList
```

Description

List of names of active (selected) vehicle databases.

Version

Available since version 3.

III.2.2.14 VehiclesFound

C++

```
HRESULT get_VehiclesFound(IRobotVehicleDatabaseList**);
```

C#

```
public IRobotVehicleDatabaseList VehiclesFound { get; }
```

Visual Basic

```
Public ReadOnly VehiclesFound As IRobotVehicleDatabaseList
```

Description

List of names of available (found) vehicle databases.

Version

Available since version 3.

III.2.3 IRobotProjectPreferences Methods

The methods of the IRobotProjectPreferences class are listed here.

Public Methods

	Name	Description
	GetActiveCode (see page 1302)	The function returns the name of the active code of the indicated type.
	GetActiveCodeNumber (see page 1303)	Function returns the numerical identifier of an active code of a given type.
	GetCurrentDatabase (see page 1303)	Function returns a name of the current database of the given type. .
	Save (see page 1303)	Function forces saving changes in the project settings. .
	SetActiveCode (see page 1304)	The function sets the active code of the indicated type. If selection of an indicated code is not possible, then, zero value is returned (False). .
	SetActiveCodeNumber (see page 1304)	Function sets an active code of a given type based on the specified numerical identifier of the code.
	SetCurrentDatabase (see page 1304)	Function sets the given database as a current database of the given type.

III.2.3.1 GetActiveCode

C++

```
HRESULT GetActiveCode(IRobotCodeType _code_type, BSTR* ret);
```

C#

```
public String GetActiveCode(IRobotCodeType _code_type);
```

Visual Basic

```
Public Function GetActiveCode(_code_type As IRobotCodeType) As String
```

(c) 2014 Autodesk Inc. All rights reserved

Description

The function returns the name of the active code of the indicated type.

III.2.3.2 GetActiveCodeNumber

C++

```
HRESULT GetActiveCodeNumber(IRobotCodeType _code_type, long* ret);
```

C#

```
public long GetActiveCodeNumber(IRobotCodeType _code_type);
```

Visual Basic

```
Public Function GetActiveCodeNumber(_code_type As IRobotCodeType) As long
```

Description

Function returns the numerical identifier of an active code of a given type.

Version

Available since version 6.5.

III.2.3.3 GetCurrentDatabase

C++

```
HRESULT GetCurrentDatabase(IRobotDatabaseType _db_type, BSTR* ret);
```

C#

```
public String GetCurrentDatabase(IRobotDatabaseType _db_type);
```

Visual Basic

```
Public Function GetCurrentDatabase(_db_type As IRobotDatabaseType) As String
```

Description

Function returns a name of the current database of the given type. .

Version

Available since version 11.

III.2.3.4 Save

C++

```
HRESULT Save();
```

C#

```
public void Save();
```

Visual Basic

```
Public Sub Save()
```

Description

Function forces saving changes in the project settings. .

Version

Available since version 4.5.

(c) 2014 Autodesk Inc. All rights reserved

III.2.3.5 SetActiveCode

C++

```
HRESULT SetActiveCode(IRobotCodeType _code_type, BSTR _code_name, VARIANT_BOOL* ret);
```

C#

```
public bool SetActiveCode(IRobotCodeType _code_type, String _code_name);
```

Visual Basic

```
Public Function SetActiveCode(_code_type As IRobotCodeType, _code_name As String) As Boolean
```

Description

The function sets the active code of the indicated type. If selection of an indicated code is not possible, then, zero value is returned (False). .

III.2.3.6 SetActiveCodeNumber

C++

```
HRESULT SetActiveCodeNumber(IRobotCodeType _code_type, long _code_number, VARIANT_BOOL*
ret);
```

C#

```
public bool SetActiveCodeNumber(IRobotCodeType _code_type, long _code_number);
```

Visual Basic

```
Public Function SetActiveCodeNumber(_code_type As IRobotCodeType, _code_number As long) As
Boolean
```

Description

Function sets an active code of a given type based on the specified numerical identifier of the code.

Version

Available since version 6.5.

III.2.3.7 SetCurrentDatabase

C++

```
HRESULT SetCurrentDatabase(IRobotDatabaseType _db_type, BSTR _db_name, VARIANT_BOOL* ret);
```

C#

```
public bool SetCurrentDatabase(IRobotDatabaseType _db_type, String _db_name);
```

Visual Basic

```
Public Function SetCurrentDatabase(_db_type As IRobotDatabaseType, _db_name As String) As
Boolean
```

Description

Function sets the given database as a current database of the given type.

Version

Available since version 11.

(c) 2014 Autodesk Inc. All rights reserved

III.3 IRobotSectionDatabaseList

Class Hierarchy

RobotOM.IRobotSectionDatabaseList

C++

```
interface IRobotSectionDatabaseList : IDispatch;
```

C#

```
public interface IRobotSectionDatabaseList;
```

Visual Basic

```
Public Interface IRobotSectionDatabaseList
```

Description

The interface describing a list of section databases .

III.3.1 IRobotSectionDatabaseList Members

The following tables list the members exposed by IRobotSectionDatabaseList.

Public Fields

	Name	Description
	Count (see page 1306)	Number of section databases in the list.

Public Methods

	Name	Description
	Add (see page 1306)	The function adds a section database with a given name at the end of the list. It returns the message about carrying out the operation successfully.
	AddFromFile (see page 1307)	The function adds a section database defined in the indicated file (.mdb) at the end of the list. It returns the message about carrying out the operation successfully.
	ChangeIndex (see page 1307)	The function imposes a new position in the list for the selected database. By means of the function one may define a required order of section databases in the list. The order is important while searching through the databases for a section with a given name. .
	Find (see page 1307)	The function returns the list index of the database with the indicated name. If the required database is absent from the list, the function returns zero. .
	Get (see page 1307)	The function returns the name of section database saved in the list under the indicated index. The list elements are indexed from 1 to Count (see page 1306).
	GetDatabase (see page 1308)	The function returns the section database whose name is saved in the list in the indicated position.
	Remove (see page 1308)	The function deletees a database from the list. .

III.3.2 IRobotSectionDatabaseList Fields

The fields of the IRobotSectionDatabaseList class are listed here.

Public Fields

	Name	Description
	Count (see page 1306)	Number of section databases in the list.

(c) 2014 Autodesk Inc. All rights reserved

III.3.2.1 Count

C++

```
HRESULT get_Count(int*);
```

C#

```
public int Count { get; }
```

Visual Basic

```
Public ReadOnly Count As int
```

Description

Number of section databases in the list.

III.3.3 IRobotSectionDatabaseList Methods

The methods of the IRobotSectionDatabaseList class are listed here.

Public Methods

	Name	Description
	Add (see page 1306)	The function adds a section database with a given name at the end of the list. It returns the message about carrying out the operation successfully.
	AddFromFile (see page 1307)	The function adds a section database defined in the indicated file (.mdb) at the end of the list. It returns the message about carrying out the operation successfully.
	ChangeIndex (see page 1307)	The function imposes a new position in the list for the selected database. By means of the function one may define a required order of section databases in the list. The order is important while searching through the databases for a section with a given name. .
	Find (see page 1307)	The function returns the list index of the database with the indicated name. If the required database is absent from the list, the function returns zero. .
	Get (see page 1307)	The function returns the name of section database saved in the list under the indicated index. The list elements are indexed from 1 to Count (see page 1306).
	GetDatabase (see page 1308)	The function returns the section database whose name is saved in the list in the indicated position.
	Remove (see page 1308)	The function deletees a database from the list. .

III.3.3.1 Add

C++

```
HRESULT Add(BSTR _db_name, VARIANT_BOOL* ret);
```

C#

```
public bool Add(String _db_name);
```

Visual Basic

```
Public Function Add(_db_name As String) As Boolean
```

Description

The function adds a section database with a given name at the end of the list. It returns the message about carrying out the operation successfully.

(c) 2014 Autodesk Inc. All rights reserved

III.3.3.2 AddFromFile

C++

```
HRESULT AddFromFile(BSTR _file_path, VARIANT_BOOL* ret);
```

C#

```
public bool AddFromFile(String _file_path);
```

Visual Basic

```
Public Function AddFromFile(_file_path As String) As Boolean
```

Description

The function adds a section database defined in the indicated file (.mdb) at the end of the list. It returns the message about carrying out the operation successfully.

III.3.3.3 ChangeIndex

C++

```
HRESULT ChangeIndex(int _cur_idx, int _new_idx);
```

C#

```
public void ChangeIndex(int _cur_idx, int _new_idx);
```

Visual Basic

```
Public Sub ChangeIndex(_cur_idx As int, _new_idx As int)
```

Description

The function imposes a new position in the list for the selected database. By means of the function one may define a required order of section databases in the list. The order is important while searching through the databases for a section with a given name. .

III.3.3.4 Find

C++

```
HRESULT Find(BSTR _db_name, int* ret);
```

C#

```
public int Find(String _db_name);
```

Visual Basic

```
Public Function Find(_db_name As String) As int
```

Description

The function returns the list index of the database with the indicated name. If the required database is absent from the list, the function returns zero. .

III.3.3.5 Get

C++

```
HRESULT Get(int _idx, BSTR* ret);
```

C#

```
public String Get(int _idx);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Function Get(_idx As int) As String
```

Description

The function returns the name of section database saved in the list under the indicated index. The list elements are indexed from 1 to Count (see page 1306).

III.3.3.6 GetDatabase

C++

```
HRESULT GetDatabase(int _idx, IRobotSectionDatabase** ret);
```

C#

```
public IRobotSectionDatabase GetDatabase(int _idx);
```

Visual Basic

```
Public Function GetDatabase(_idx As int) As IRobotSectionDatabase
```

Description

The function returns the section database whose name is saved in the list in the indicated position.

III.3.3.7 Remove

C++

```
HRESULT Remove(int _idx);
```

C#

```
public void Remove(int _idx);
```

Visual Basic

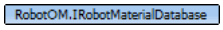
```
Public Sub Remove(_idx As int)
```

Description

The function deletees a database from the list. .

III.4 IRobotMaterialDatabase

Class Hierarchy



C++

```
interface IRobotMaterialDatabase : IDispatch;
```

C#

```
public interface IRobotMaterialDatabase;
```

Visual Basic

```
Public Interface IRobotMaterialDatabase
```

Description

The interface providing access to material database .

(c) 2014 Autodesk Inc. All rights reserved

III.4.1 IRobotMaterialDatabase Members

The following tables list the members exposed by IRobotMaterialDatabase.

Public Fields

	Name	Description
	Name (see page 1309)	Material database name .

Public Methods

	Name	Description
	Get (see page 1310)	The function returns a table of names of materials of the indicated type that are defined in the database .
	GetAll (see page 1310)	The function returns a table of names of all materials defined in the database .
	GetDefault (see page 1310)	Function returns the name of a selected type of default material. .
	Load (see page 1311)	The function reads materials from the database with the indicated name. If there occurs an error during the operation of reading from the database or the database with the name does not exist, the function returns zero (False).
	LoadFromFile (see page 1311)	The function reads the materials from the indicated file. If there occurs an error during the operation of reading from the database, the function returns zero (False). .
	SetDefault (see page 1311)	Function sets the name of a selected type of default material. .

III.4.2 IRobotMaterialDatabase Fields

The fields of the IRobotMaterialDatabase class are listed here.

Public Fields

	Name	Description
	Name (see page 1309)	Material database name .

III.4.2.1 Name

C++

```
HRESULT get_Name(BSTR*);
```

C#

```
public String Name { get; }
```

Visual Basic

```
Public ReadOnly Name As String
```

Description

Material database name .

III.4.3 IRobotMaterialDatabase Methods

The methods of the IRobotMaterialDatabase class are listed here.

Public Methods

	Name	Description
	Get (see page 1310)	The function returns a table of names of materials of the indicated type that are defined in the database .
	GetAll (see page 1310)	The function returns a table of names of all materials defined in the database .

(c) 2014 Autodesk Inc. All rights reserved

	GetDefault (see page 1310)	Function returns the name of a selected type of default material. .
	Load (see page 1311)	The function reads materials from the database with the indicated name. If there occurs an error during the operation of reading from the database or the database with the name does not exist, the function returns zero (False).
	LoadFromFile (see page 1311)	The function reads the materials from the indicated file. If there occurs an error during the operation of reading from the database, the function returns zero (False). .
	SetDefault (see page 1311)	Function sets the name of a selected type of default material. .

III.4.3.1 Get

C++

```
HRESULT Get(IRobotMaterialType _mater_type, IRobotNamesArray** ret);
```

C#

```
public IRobotNamesArray Get(IRobotMaterialType _mater_type);
```

Visual Basic

```
Public Function Get(_mater_type As IRobotMaterialType) As IRobotNamesArray
```

Description

The function returns a table of names of materials of the indicated type that are defined in the database .

III.4.3.2 GetAll

C++

```
HRESULT GetAll(IRobotNamesArray** ret);
```

C#

```
public IRobotNamesArray GetAll();
```

Visual Basic

```
Public Function GetAll() As IRobotNamesArray
```

Description

The function returns a table of names of all materials defined in the database .

III.4.3.3 GetDefault

C++

```
HRESULT GetDefault(IRobotMaterialType _mater_type, BSTR* ret);
```

C#

```
public String GetDefault(IRobotMaterialType _mater_type);
```

Visual Basic

```
Public Function GetDefault(_mater_type As IRobotMaterialType) As String
```

Description

Function returns the name of a selected type of default material. .

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

III.4.3.4 Load

C++

```
HRESULT Load(BSTR _dbase_name, VARIANT_BOOL* ret);
```

C#

```
public bool Load(String _dbase_name);
```

Visual Basic

```
Public Function Load(_dbase_name As String) As Boolean
```

Description

The function reads materials from the database with the indicated name. If there occurs an error during the operation of reading from the database or the database with the name does not exist, the function returns zero (False).

III.4.3.5 LoadFromFile

C++

```
HRESULT LoadFromFile(BSTR _file_path, VARIANT_BOOL* ret);
```

C#

```
public bool LoadFromFile(String _file_path);
```

Visual Basic

```
Public Function LoadFromFile(_file_path As String) As Boolean
```

Description

The function reads the materials from the indicated file. If there occurs an error during the operation of reading from the database, the function returns zero (False). .

III.4.3.6 SetDefault

C++

```
HRESULT SetDefault(IRobotMaterialType _mater_type, BSTR _mater_name);
```

C#

```
public void SetDefault(IRobotMaterialType _mater_type, String _mater_name);
```

Visual Basic

```
Public Sub SetDefault(_mater_type As IRobotMaterialType, _mater_name As String)
```

Description

Function sets the name of a selected type of default material. .

Version

Available since version 3.

III.5 IRobotCodeType

C++

```
enum IRobotCodeType;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public enum IRobotCodeType;
```

Visual Basic

```
Public Enum IRobotCodeType
```

Members

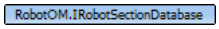
Members	Description
I_CT_STEEL_STRUCTURES = 0	The code for steel structures .
I_CT_STEEL_CONNECTIONS = 1	The code for steel connections .
I_CT_TIMBER_STRUCTURES = 2	The code for timber structures .
I_CT_RC_REAL_REINF = 3	The code for real reinforcement .
I_CT_RC_THEORETICAL_REINF = 4	The code for theoretical reinforcement .
I_CT_FOUNDATIONS_DESIGN = 5	Code for spread footing design.
I_CT_CODE_COMBINATIONS = 9	Code combinations.
I_CT_SNOW_WIND_LOADS = 6	Code for snow/wind loads.
I_CT_SEISMIC_LOADS = 7	Code for seismic loads.
I_CT_MOVING_LOADS = 8	Code for moving loads.

Description

Set of identifiers determining the types of codes that may be set in the Robot program.

III.6 IRobotSectionDatabase

Class Hierarchy



C++

```
interface IRobotSectionDatabase : IDispatch;
```

C#

```
public interface IRobotSectionDatabase;
```

Visual Basic

```
Public Interface IRobotSectionDatabase
```

Description

The interface providing access to section database.

III.6.1 IRobotSectionDatabase Members

The following tables list the members exposed by IRobotSectionDatabase.

Public Fields

	Name	Description
	Description (see page 1313)	Section database description.
	FullName (see page 1313)	Full name of section database.
	Name (see page 1314)	Section database name .

(c) 2014 Autodesk Inc. All rights reserved

Public Methods

	Name	Description
	GetAll (see page 1314)	The function returns a table containing names of all sections described in the database.
	Load (see page 1314)	The function reads sections from the section database with the indicated name. If such section database does not exist or there occurs an error during the reading process, the function returns zero. .
	LoadFromFile (see page 1315)	The function reads sections from the database saved in the indicated file. If there occurred an error during the reading process, the function returns zero.

III.6.2 IRobotSectionDatabase Fields

The fields of the IRobotSectionDatabase class are listed here.

Public Fields

	Name	Description
	Description (see page 1313)	Section database description.
	FullName (see page 1313)	Full name of section database.
	Name (see page 1314)	Section database name .

III.6.2.1 Description

C++

```
HRESULT get_Description(BSTR*);
```

C#

```
public String Description { get; }
```

Visual Basic

```
Public ReadOnly Description As String
```

Description

Section database description.

Version

Available since version 7.5.

III.6.2.2 FullName

C++

```
HRESULT get_FullName(BSTR*);
```

C#

```
public String FullName { get; }
```

Visual Basic

```
Public ReadOnly FullName As String
```

Description

Full name of section database.

Version

Available since version 4.

(c) 2014 Autodesk Inc. All rights reserved

III.6.2.3 Name

C++

```
HRESULT get_Name(BSTR*);
```

C#

```
public String Name { get; }
```

Visual Basic

```
Public ReadOnly Name As String
```

Description

Section database name .

III.6.3 IRobotSectionDatabase Methods

The methods of the IRobotSectionDatabase class are listed here.

Public Methods

	Name	Description
	GetAll (see page 1314)	The function returns a table containing names of all sections described in the database.
	Load (see page 1314)	The function reads sections from the section database with the indicated name. If such section database does not exist or there occurs an error during the reading process, the function returns zero. .
	LoadFromFile (see page 1315)	The function reads sections from the database saved in the indicated file. If there occurred an error during the reading process, the function returns zero.

III.6.3.1 GetAll

C++

```
HRESULT GetAll(IRobotNamesArray** ret);
```

C#

```
public IRobotNamesArray GetAll();
```

Visual Basic

```
Public Function GetAll() As IRobotNamesArray
```

Description

The function returns a table containing names of all sections described in the database.

III.6.3.2 Load

C++

```
HRESULT Load(BSTR _db_name, VARIANT_BOOL* ret);
```

C#

```
public bool Load(String _db_name);
```

Visual Basic

```
Public Function Load(_db_name As String) As Boolean
```

(c) 2014 Autodesk Inc. All rights reserved

Description

The function reads sections from the section database with the indicated name. If such section database does not exist or there occurs an error during the reading process, the function returns zero. .

III.6.3.3 LoadFromFile

C++

```
HRESULT LoadFromFile(BSTR _file_path, VARIANT_BOOL* ret);
```

C#

```
public bool LoadFromFile(String _file_path);
```

Visual Basic

```
Public Function LoadFromFile(_file_path As String) As Boolean
```

Description

The function reads sections from the database saved in the indicated file. If there occurred an error during the reading process, the function returns zero.

III.7 IRobotProjectPreferencesEvents

Class Hierarchy

RobotOM.IRobotProjectPreferencesEvents

C++

```
interface IRobotProjectPreferencesEvents : IDispatch;
```

C#

```
public interface IRobotProjectPreferencesEvents;
```

Visual Basic

```
Public Interface IRobotProjectPreferencesEvents
```

Description

Events related to the change of settings for the current project.

III.7.1 IRobotProjectPreferencesEvents Members

The following tables list the members exposed by IRobotProjectPreferencesEvents.

Public Methods

	Name	Description
	OnDialogOK (see page 1316)	Event generated after accepting changes made in the project settings dialog box by the user.

III.7.2 IRobotProjectPreferencesEvents Methods

The methods of the IRobotProjectPreferencesEvents class are listed here.

Public Methods

	Name	Description
	OnDialogOK (see page 1316)	Event generated after accepting changes made in the project settings dialog box by the user.

(c) 2014 Autodesk Inc. All rights reserved

III.7.2.1 OnDialogOK

C++

```
HRESULT OnDialogOK();
```

C#

```
public void OnDialogOK();
```

Visual Basic

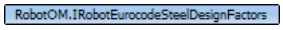
```
Public Sub OnDialogOK()
```

Description

Event generated after accepting changes made in the project settings dialog box by the user.

III.8 IRobotEurocodeSteelDesignFactors

Class Hierarchy



C++

```
interface IRobotEurocodeSteelDesignFactors : IDispatch;
```

C#

```
public interface IRobotEurocodeSteelDesignFactors;
```

Visual Basic

```
Public Interface IRobotEurocodeSteelDesignFactors
```

Description

Partial factors for the steel design according to Eurocode.

Version

Available since version 9.7.

III.8.1 IRobotEurocodeSteelDesignFactors Members

The following tables list the members exposed by IRobotEurocodeSteelDesignFactors.

Public Fields

	Name	Description
	Gamma0 (see page 1317)	Resistance of elements and sections gM0.
	Gamma1 (see page 1317)	Resistance of elements and sections gM1.
	Gamma2 (see page 1318)	Resistance of elements and sections gM2.
	GammaFire (see page 1318)	Safety factor for fire conditions.

Public Methods

	Name	Description
	LoadFromCode (see page 1318)	Function loads values of coefficients from the code with the specified name.
	LoadFromCodeNumber (see page 1319)	Function loads values of coefficients from the specified code.

(c) 2014 Autodesk Inc. All rights reserved

III.8.2 IRobotEurocodeSteelDesignFactors Fields

The fields of the IRobotEurocodeSteelDesignFactors class are listed here.

Public Fields

	Name	Description
	Gamma0 (see page 1317)	Resistance of elements and sections gM0.
	Gamma1 (see page 1317)	Resistance of elements and sections gM1.
	Gamma2 (see page 1318)	Resistance of elements and sections gM2.
	GammaFire (see page 1318)	Safety factor for fire conditions.

III.8.2.1 Gamma0

C++

```
HRESULT get_Gamma0(double*);
HRESULT put_Gamma0(double);
```

C#

```
public double Gamma0 { get; set; }
```

Visual Basic

```
Public Gamma0 As double
```

Description

Resistance of elements and sections gM0.

Version

Available since version 9.7.

III.8.2.2 Gamma1

C++

```
HRESULT get_Gamma1(double*);
HRESULT put_Gamma1(double);
```

C#

```
public double Gamma1 { get; set; }
```

Visual Basic

```
Public Gamma1 As double
```

Description

Resistance of elements and sections gM1.

Version

Available since version 9.7.

III.8.2.3 Gamma2

C++

```
HRESULT get_Gamma2(double*);
HRESULT put_Gamma2(double);
```

C#

```
public double Gamma2 { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Gamma2 As double
```

Description

Resistance of elements and sections gM2.

Version

Available since version 9.7.

III.8.2.4 GammaFire

C++

```
HRESULT get_GammaFire(double*);
HRESULT put_GammaFire(double);
```

C#

```
public double GammaFire { get; set; }
```

Visual Basic

```
Public GammaFire As double
```

Description

Safety factor for fire conditions.

Version

Available since version 9.7.

III.8.3 IRobotEurocodeSteelDesignFactors Methods

The methods of the IRobotEurocodeSteelDesignFactors class are listed here.

Public Methods

	Name	Description
	LoadFromCode (see page 1318)	Function loads values of coefficients from the code with the specified name.
	LoadFromCodeNumber (see page 1319)	Function loads values of coefficients from the specified code.

III.8.3.1 LoadFromCode

C++

```
HRESULT LoadFromCode(BSTR _code_name, VARIANT_BOOL* ret);
```

C#

```
public bool LoadFromCode(String _code_name);
```

Visual Basic

```
Public Function LoadFromCode(_code_name As String) As Boolean
```

Description

Function loads values of coefficients from the code with the specified name.

Version

Available since version 9.7.

(c) 2014 Autodesk Inc. All rights reserved

III.8.3.2 LoadFromCodeNumber

C++

```
HRESULT LoadFromCodeNumber(long _code_number, VARIANT_BOOL* ret);
```

C#

```
public bool LoadFromCodeNumber(long _code_number);
```

Visual Basic

```
Public Function LoadFromCodeNumber(_code_number As long) As Boolean
```

Description

Function loads values of coefficients from the specified code.

Version

Available since version 9.7.

III.9 IRobotEurocodeSteelConnectionFactors

Class Hierarchy

RobotOM.IRobotEurocodeSteelConnectionFactors

C++

```
interface IRobotEurocodeSteelConnectionFactors : IDispatch;
```

C#

```
public interface IRobotEurocodeSteelConnectionFactors;
```

Visual Basic

```
Public Interface IRobotEurocodeSteelConnectionFactors
```

Description

Partial factors for the steel connection design according to Eurocode.

Version

Available since version 9.7.

III.9.1 IRobotEurocodeSteelConnectionFactors Members

The following tables list the members exposed by IRobotEurocodeSteelConnectionFactors.

Public Fields

	Name	Description
	Gamma0 (see page 1320)	Resistance of elements and sections gM0.
	Gamma1 (see page 1321)	Resistance of elements and sections gM1.
	Gamma2 (see page 1321)	Resistance of fasteners and plates in bearing gM2.
	Gamma3 (see page 1321)	Slip resistance - at ultimate limit state gM3.
	Gamma3Ser (see page 1322)	Slip resistance - at serviceability limit state gM3ser.
	Gamma4 (see page 1322)	Bearing resistance of an injection bolt gM4.
	Gamma5 (see page 1322)	Resistance of joints in hollow section lattice girder gM5.
	Gamma6 (see page 1323)	Resistance of pins at serviceability limit state gM6.
	Gamma7 (see page 1323)	Preload of high strength bolts gM7.

(c) 2014 Autodesk Inc. All rights reserved

	GammaC (see page 1323)	Concrete capacity gC .

Public Methods

	Name	Description
	LoadFromCode (see page 1324)	Function loads values of coefficients from the code with the specified name.
	LoadFromCodeNumber (see page 1324)	Function loads values of coefficients from the specified code.

III.9.2 IRobotEurocodeSteelConnectionFactors Fields

The fields of the IRobotEurocodeSteelConnectionFactors class are listed here.

Public Fields

	Name	Description
	Gamma0 (see page 1320)	Resistance of elements and sections gM0.
	Gamma1 (see page 1321)	Resistance of elements and sections gM1.
	Gamma2 (see page 1321)	Resistance of fasteners and plates in bearing gM2.
	Gamma3 (see page 1321)	Slip resistance - at ultimate limit state gM3.
	Gamma3Ser (see page 1322)	Slip resistance - at serviceability limit state gM3ser.
	Gamma4 (see page 1322)	Bearing resistance of an injection bolt gM4.
	Gamma5 (see page 1322)	Resistance of joints in hollow section lattice girder gM5.
	Gamma6 (see page 1323)	Resistance of pins at serviceability limit state gM6.
	Gamma7 (see page 1323)	Preload of high strength bolts gM7.
	GammaC (see page 1323)	Concrete capacity gC .

III.9.2.1 Gamma0

C++

```
HRESULT get_Gamma0(double*);
HRESULT put_Gamma0(double);
```

C#

```
public double Gamma0 { get; set; }
```

Visual Basic

```
Public Gamma0 As double
```

Description

Resistance of elements and sections gM0.

Version

Available since version 9.7.

III.9.2.2 Gamma1

C++

```
HRESULT get_Gamma1(double*);
HRESULT put_Gamma1(double);
```

C#

```
public double Gamma1 { get; set; }
```

Visual Basic

```
Public Gamma1 As double
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Resistance of elements and sections gM1.

Version

Available since version 9.7.

III.9.2.3 Gamma2

C++

```
HRESULT get_Gamma2(double*);
HRESULT put_Gamma2(double);
```

C#

```
public double Gamma2 { get; set; }
```

Visual Basic

```
Public Gamma2 As double
```

Description

Resistance of fasteners and plates in bearing gM2.

Version

Available since version 9.7.

III.9.2.4 Gamma3

C++

```
HRESULT get_Gamma3(double*);
HRESULT put_Gamma3(double);
```

C#

```
public double Gamma3 { get; set; }
```

Visual Basic

```
Public Gamma3 As double
```

Description

Slip resistance - at ultimate limit state gM3.

Version

Available since version 9.7.

III.9.2.5 Gamma3Ser

C++

```
HRESULT get_Gamma3Ser(double*);
HRESULT put_Gamma3Ser(double);
```

C#

```
public double Gamma3Ser { get; set; }
```

Visual Basic

```
Public Gamma3Ser As double
```

Description

Slip resistance - at serviceability limit state gM3ser.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 9.7.

III.9.2.6 Gamma4

C++

```
HRESULT get_Gamma4(double*);
HRESULT put_Gamma4(double);
```

C#

```
public double Gamma4 { get; set; }
```

Visual Basic

```
Public Gamma4 As double
```

Description

Bearing resistance of an injection bolt gM4.

Version

Available since version 9.7.

III.9.2.7 Gamma5

C++

```
HRESULT get_Gamma5(double*);
HRESULT put_Gamma5(double);
```

C#

```
public double Gamma5 { get; set; }
```

Visual Basic

```
Public Gamma5 As double
```

Description

Resistance of joints in hollow section lattice girder gM5.

Version

Available since version 9.7.

III.9.2.8 Gamma6

C++

```
HRESULT get_Gamma6(double*);
HRESULT put_Gamma6(double);
```

C#

```
public double Gamma6 { get; set; }
```

Visual Basic

```
Public Gamma6 As double
```

Description

Resistance of pins at serviceability limit state gM6.

Version

Available since version 9.7.

(c) 2014 Autodesk Inc. All rights reserved

III.9.2.9 Gamma7

C++

```
HRESULT get_Gamma7(double*);
HRESULT put_Gamma7(double);
```

C#

```
public double Gamma7 { get; set; }
```

Visual Basic

```
Public Gamma7 As double
```

Description

Preload of high strength bolts gM7.

Version

Available since version 9.7.

III.9.2.10 GammaC

C++

```
HRESULT get_GammaC(double*);
HRESULT put_GammaC(double);
```

C#

```
public double GammaC { get; set; }
```

Visual Basic

```
Public GammaC As double
```

Description

Concrete capacity gC .

Version

Available since version 9.7.

III.9.3 IRobotEurocodeSteelConnectionFactors Methods

The methods of the IRobotEurocodeSteelConnectionFactors class are listed here.

Public Methods

	Name	Description
	LoadFromCode (see page 1324)	Function loads values of coefficients from the code with the specified name.
	LoadFromCodeNumber (see page 1324)	Function loads values of coefficients from the specified code.

III.9.3.1 LoadFromCode

C++

```
HRESULT LoadFromCode(BSTR _code_name, VARIANT_BOOL* ret);
```

C#

```
public bool LoadFromCode(String _code_name);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Function LoadFromCode(_code_name As String) As Boolean
```

Description

Function loads values of coefficients from the code with the specified name.

Version

Available since version 9.7.

III.9.3.2 LoadFromCodeNumber

C++

```
HRESULT LoadFromCodeNumber(long _code_number, VARIANT_BOOL* ret);
```

C#

```
public bool LoadFromCodeNumber(long _code_number);
```

Visual Basic

```
Public Function LoadFromCodeNumber(_code_number As long) As Boolean
```

Description

Function loads values of coefficients from the specified code.

Version

Available since version 9.7.

III.10 IRobotEurocodeFactors

Class Hierarchy

RobotOM.IRobotEurocodeFactors

C++

```
interface IRobotEurocodeFactors : IDispatch;
```

C#

```
public interface IRobotEurocodeFactors;
```

Visual Basic

```
Public Interface IRobotEurocodeFactors
```

Description

Partial factors for Eurocodes.

Version

Available since version 9.7.

III.10.1 IRobotEurocodeFactors Members

The following tables list the members exposed by IRobotEurocodeFactors.

Public Fields

	Name	Description
	SteelConnections (see page 1325)	Partial factors for steel connections according to Eurocode.

(c) 2014 Autodesk Inc. All rights reserved

	SteelDesign (see page 1325)	Partial factors for the steel design according to Eurocode.

III.10.2 IRobotEurocodeFactors Fields

The fields of the IRobotEurocodeFactors class are listed here.

Public Fields

	Name	Description
	SteelConnections (see page 1325)	Partial factors for steel connections according to Eurocode.
	SteelDesign (see page 1325)	Partial factors for the steel design according to Eurocode.

III.10.2.1 SteelConnections

C++

```
HRESULT get_SteelConnections(IRobotEurocodeSteelConnectionFactors**);
```

C#

```
public IRobotEurocodeSteelConnectionFactors SteelConnections { get; }
```

Visual Basic

```
Public ReadOnly SteelConnections As IRobotEurocodeSteelConnectionFactors
```

Description

Partial factors for steel connections according to Eurocode.

Version

Available since version 9.7.

III.10.2.2 SteelDesign

C++

```
HRESULT get_SteelDesign(IRobotEurocodeSteelDesignFactors**);
```

C#

```
public IRobotEurocodeSteelDesignFactors SteelDesign { get; }
```

Visual Basic

```
Public ReadOnly SteelDesign As IRobotEurocodeSteelDesignFactors
```

Description

Partial factors for the steel design according to Eurocode.

Version

Available since version 9.7.

III.11 IRobotDatabaseType

C++

```
enum IRobotDatabaseType;
```

C#

```
public enum IRobotDatabaseType;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Enum IRobotDatabaseType
```

Members

Members	Description
I_DT_SECTIONS = 1	Database of steel and timber sections. Available since version 11.
I_DT_VEHICLE_LOADS = 2	Vehicle database. Available since version 11.
I_DT_STANDARD_LOADS = 3	Database of loads. Available since version 11.
I_DT_BUILDING_SOILS = 4	Soil database. Available since version 11.
I_DT_BOLTS = 5	Bolt database. Available since version 11.
I_DT_ANCHOR_BOLTS = 6	Anchor bolt database. Available since version 11.
I_DT_REINFORCING_BARS = 7	Database of reinforcing bars. Available since version 11.
I_DT_WIRE_FABRICS = 8	Database of wire fabrics. Available since version 11.

Description

Available database types.

Version

Available since version 11.

III.12 IRobotCalculationModelCoherence

C++

```
enum IRobotCalculationModelCoherence;
```

C#

```
public enum IRobotCalculationModelCoherence;
```

Visual Basic

```
Public Enum IRobotCalculationModelCoherence
```

Members

Members	Description
I_CMC_COHERENT_FE_MESH = 1	Available since version 12.
I_CMC_USE_OF_KINEMATIC_CONSTRAINTS = 2	Available since version 12.

Version

Available since version 12.

IV Backgrounds

Available since version 11.

(c) 2014 Autodesk Inc. All rights reserved

Enumerations

	Name	Description
	IRobotBackgroundVisibilityRangeType (see page 1337)	Available types of background visibility range.

Interfaces

	Name	Description
	IRobotBackgroundLayers (see page 1327)	Background layers.
	IRobotBackgroundInsertParams (see page 1330)	Parameters of locating a background in the project.
	IRobotBackground (see page 1334)	
	IRobotBackgroundVisibilityRange (see page 1338)	
	IRobotBackgroundServer (see page 1339)	Object managing backgrounds in the project.

IV.1 IRobotBackgroundLayers

Class Hierarchy

C++

```
interface IRobotBackgroundLayers : IDispatch;
```

C#

```
public interface IRobotBackgroundLayers;
```

Visual Basic

```
Public Interface IRobotBackgroundLayers
```

Description

Background layers.

Version

Available since version 11.

IV.1.1 IRobotBackgroundLayers Members

The following tables list the members exposed by IRobotBackgroundLayers.

Public Fields

	Name	Description
	Count (see page 1328)	Number of layers.

Public Methods

	Name	Description
	FindName (see page 1329)	Function returns index of the layer of the given name.
	Get (see page 1329)	Function makes data for a layer of the given index available.
	IsImported (see page 1329)	Function returns True if a layer with the given index is imported with the background to the project.
	SetImported (see page 1330)	Function allows for changing the status for the layer of the given index.

(c) 2014 Autodesk Inc. All rights reserved

IV.1.2 IRobotBackgroundLayers Fields

The fields of the IRobotBackgroundLayers class are listed here.

Public Fields

	Name	Description
	Count (see page 1328)	Number of layers.

IV.1.2.1 Count

C++

```
HRESULT get_Count(long*);
```

C#

```
public long Count { get; }
```

Visual Basic

```
Public ReadOnly Count As long
```

Description

Number of layers.

Version

Available since version 11.

IV.1.3 IRobotBackgroundLayers Methods

The methods of the IRobotBackgroundLayers class are listed here.

Public Methods

	Name	Description
	FindName (see page 1329)	Function returns index of the layer of the given name.
	Get (see page 1329)	Function makes data for a layer of the given index available.
	IsImported (see page 1329)	Function returns True if a layer with the given index is imported with the background to the project.
	SetImported (see page 1330)	Function allows for changing the status for the layer of the given index.

IV.1.3.1 FindName

C++

```
HRESULT FindName(BSTR _layer_name, long* ret);
```

C#

```
public long FindName(String _layer_name);
```

Visual Basic

```
Public Function FindName(_layer_name As String) As long
```

Description

Function returns index of the layer of the given name.

Version

Available since version 11.

(c) 2014 Autodesk Inc. All rights reserved

IV.1.3.2 Get

C++

```
HRESULT Get(long _layer_idx, BSTR _layer_name, bool* _is_imported);
```

C#

```
public void Get(long _layer_idx, String _layer_name, bool* _is_imported);
```

Visual Basic

```
Public Sub Get(_layer_idx As long, ByRef _layer_name As String, ByRef _is_imported As bool*)
```

Description

Function makes data for a layer of the given index available.

Version

Available since version 11.

IV.1.3.3 IsImported

C++

```
HRESULT IsImported(long _layer_idx, VARIANT_BOOL* ret);
```

C#

```
public bool IsImported(long _layer_idx);
```

Visual Basic

```
Public Function IsImported(_layer_idx As long) As Boolean
```

Description

Function returns True if a layer with the given index is imported with the background to the project.

Version

Available since version 11.

IV.1.3.4 SetImported

C++

```
HRESULT SetImported(long _layer_idx, VARIANT_BOOL _is_imported);
```

C#

```
public void SetImported(long _layer_idx, bool _is_imported);
```

Visual Basic

```
Public Sub SetImported(_layer_idx As long, _is_imported As Boolean)
```

Description

Function allows for changing the status for the layer of the given index.

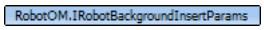
Version

Available since version 11.

(c) 2014 Autodesk Inc. All rights reserved

IV.2 IRobotBackgroundInsertParams

Class Hierarchy



C++

```
interface IRobotBackgroundInsertParams : IDispatch;
```

C#

```
public interface IRobotBackgroundInsertParams;
```

Visual Basic

```
Public Interface IRobotBackgroundInsertParams
```

Description

Parameters of locating a background in the project.

Version

Available since version 11.

IV.2.1 IRobotBackgroundInsertParams Members

The following tables list the members exposed by IRobotBackgroundInsertParams.

Public Fields

	Name	Description
	InsertionPoint (see page 1331)	A point on the plane in which the characteristic point of the background will be located.
	Layers (see page 1331)	Background layers.
	Mirror (see page 1332)	Background mirror.
	Plane (see page 1332)	Plane of background insertion.
	PositionOnNormalAxis (see page 1332)	Location of the selected plane on the axis perpendicular to it.
	ReferencePoint (see page 1333)	Characteristic point on the background.
	RotationAngle (see page 1333)	
	ScalingFactor (see page 1333)	Scale factor.
	Units (see page 1333)	

IV.2.2 IRobotBackgroundInsertParams Fields

The fields of the IRobotBackgroundInsertParams class are listed here.

Public Fields

	Name	Description
	InsertionPoint (see page 1331)	A point on the plane in which the characteristic point of the background will be located.
	Layers (see page 1331)	Background layers.
	Mirror (see page 1332)	Background mirror.
	Plane (see page 1332)	Plane of background insertion.
	PositionOnNormalAxis (see page 1332)	Location of the selected plane on the axis perpendicular to it.
	ReferencePoint (see page 1333)	Characteristic point on the background.
	RotationAngle (see page 1333)	

(c) 2014 Autodesk Inc. All rights reserved

	ScalingFactor (see page 1333)	Scale factor.
	Units (see page 1333)	

IV.2.2.1 InsertionPoint

C++

```
HRESULT get_InsertionPoint(IRobotGeoPoint2D**);
```

C#

```
public IRobotGeoPoint2D InsertionPoint { get; }
```

Visual Basic

```
Public ReadOnly InsertionPoint As IRobotGeoPoint2D
```

Description

A point on the plane in which the characteristic point of the background will be located.

Version

Available since version 11.

IV.2.2.2 Layers

C++

```
HRESULT get_Layers(IRobotBackgroundLayers**);
```

C#

```
public IRobotBackgroundLayers Layers { get; }
```

Visual Basic

```
Public ReadOnly Layers As IRobotBackgroundLayers
```

Description

Background layers.

Version

Available since version 11.

IV.2.2.3 Mirror

C++

```
HRESULT get_Mirror(VARIANT_BOOL*);
HRESULT put_Mirror(VARIANT_BOOL);
```

C#

```
public bool Mirror { get; set; }
```

Visual Basic

```
Public Mirror As Boolean
```

Description

Background mirror.

Version

Available since version 11.

(c) 2014 Autodesk Inc. All rights reserved

IV.2.2.4 Plane

C++

```
HRESULT get_Plane(IRobotGeoPlane*);
HRESULT put_Plane(IRobotGeoPlane);
```

C#

```
public IRobotGeoPlane Plane { get; set; }
```

Visual Basic

```
Public Plane As IRobotGeoPlane
```

Description

Plane of background insertion.

Version

Available since version 11.

IV.2.2.5 PositionOnNormalAxis

C++

```
HRESULT get_PositionOnNormalAxis(double*);
HRESULT put_PositionOnNormalAxis(double);
```

C#

```
public double PositionOnNormalAxis { get; set; }
```

Visual Basic

```
Public PositionOnNormalAxis As double
```

Description

Location of the selected plane on the axis perpendicular to it.

Version

Available since version 11.

IV.2.2.6 ReferencePoint

C++

```
HRESULT get_ReferencePoint(IRobotGeoPoint2D**);
```

C#

```
public IRobotGeoPoint2D ReferencePoint { get; }
```

Visual Basic

```
Public ReadOnly ReferencePoint As IRobotGeoPoint2D
```

Description

Characteristic point on the background.

Version

Available since version 11.

IV.2.2.7 RotationAngle

C++

```
HRESULT get_RotationAngle(double*);
```

(c) 2014 Autodesk Inc. All rights reserved

```
HRESULT put_RotationAngle(double);
```

C#

```
public double RotationAngle { get; set; }
```

Visual Basic

```
Public RotationAngle As double
```

Version

Available since version 11.

IV.2.2.8 ScalingFactor

C++

```
HRESULT get_ScalingFactor(double*);
HRESULT put_ScalingFactor(double);
```

C#

```
public double ScalingFactor { get; set; }
```

Visual Basic

```
Public ScalingFactor As double
```

Description

Scale factor.

Version

Available since version 11.

IV.2.2.9 Units

C++

```
HRESULT get_Units(BSTR*);
HRESULT put_Units(BSTR);
```

C#

```
public String Units { get; set; }
```

Visual Basic

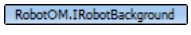
```
Public Units As String
```

Version

Available since version 11.

IV.3 IRobotBackground

Class Hierarchy



C++

```
interface IRobotBackground : IDispatch;
```

C#

```
public interface IRobotBackground;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Interface IRobotBackground
```

Version

Available since version 11.

IV.3.1 IRobotBackground Members

The following tables list the members exposed by IRobotBackground.

Public Fields

	Name	Description
	Color (see page 1335)	
	FilePath (see page 1335)	Path to a file with definition of background imported to project as a link.
	InsertParams (see page 1335)	Parameters of placing a background in the project.
	IsOn (see page 1336)	Flag switching on/off background displaying.
	Name (see page 1336)	
	Number (see page 1336)	Unique number identifying background.
	VisibilityRange (see page 1337)	Visibility range.

Public Methods

	Name	Description
	Save (see page 1337)	Function saves changed background definition to the project.

IV.3.2 IRobotBackground Fields

The fields of the IRobotBackground class are listed here.

Public Fields

	Name	Description
	Color (see page 1335)	
	FilePath (see page 1335)	Path to a file with definition of background imported to project as a link.
	InsertParams (see page 1335)	Parameters of placing a background in the project.
	IsOn (see page 1336)	Flag switching on/off background displaying.
	Name (see page 1336)	
	Number (see page 1336)	Unique number identifying background.
	VisibilityRange (see page 1337)	Visibility range.

IV.3.2.1 Color

C++

```
HRESULT get_Color(long*);
HRESULT put_Color(long);
```

C#

```
public long Color { get; set; }
```

Visual Basic

```
Public Color As long
```

Version

Available since version 11.

(c) 2014 Autodesk Inc. All rights reserved

IV.3.2.2 FilePath

C++

```
HRESULT get_FilePath(BSTR*);
HRESULT put_FilePath(BSTR);
```

C#

```
public String FilePath { get; set; }
```

Visual Basic

```
Public FilePath As String
```

Description

Path to a file with definition of background imported to project as a link.

Version

Available since version 11.

IV.3.2.3 InsertParams

C++

```
HRESULT get_InsertParams(IRobotBackgroundInsertParams**);
```

C#

```
public IRobotBackgroundInsertParams InsertParams { get; }
```

Visual Basic

```
Public ReadOnly InsertParams As IRobotBackgroundInsertParams
```

Description

Parameters of placing a background in the project.

Version

Available since version 11.

IV.3.2.4 IsOn

C++

```
HRESULT get_IsOn(VARIANT_BOOL*);
HRESULT put_IsOn(VARIANT_BOOL);
```

C#

```
public bool IsOn { get; set; }
```

Visual Basic

```
Public IsOn As Boolean
```

Description

Flag switching on/off background displaying.

Version

Available since version 11.

IV.3.2.5 Name

C++

```
HRESULT get_Name(BSTR*);
```

(c) 2014 Autodesk Inc. All rights reserved

```
HRESULT put_Name(BSTR);
```

C#

```
public String Name { get; set; }
```

Visual Basic

```
Public Name As String
```

Version

Available since version 11.

IV.3.2.6 Number

C++

```
HRESULT get_Number(long*);
```

C#

```
public long Number { get; }
```

Visual Basic

```
Public ReadOnly Number As long
```

Description

Unique number identifying background.

Version

Available since version 11.

IV.3.2.7 VisibilityRange

C++

```
HRESULT get_VisibilityRange(IRobotBackgroundVisibilityRange**);
```

C#

```
public IRobotBackgroundVisibilityRange VisibilityRange { get; }
```

Visual Basic

```
Public ReadOnly VisibilityRange As IRobotBackgroundVisibilityRange
```

Description

Visibility range.

Version

Available since version 11.

IV.3.3 IRobotBackground Methods

The methods of the IRobotBackground class are listed here.

Public Methods

	Name	Description
	Save (see page 1337)	Function saves changed background definition to the project.

IV.3.3.1 Save

C++

```
HRESULT Save(VARIANT_BOOL* ret);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public bool Save();
```

Visual Basic

```
Public Function Save() As Boolean
```

Description

Function saves changed background definition to the project.

Version

Available since version 11.

IV.4 IRobotBackgroundVisibilityRangeType

C++

```
enum IRobotBackgroundVisibilityRangeType;
```

C#

```
public enum IRobotBackgroundVisibilityRangeType;
```

Visual Basic

```
Public Enum IRobotBackgroundVisibilityRangeType
```

Members

Members	Description
I_BVRT_AUTOMATIC = 1	Available since version 11.
I_BVRT_USER_DEFINED = 2	Available since version 11.
I_BVRT_UNLIMITED = 3	Available since version 11.

Description

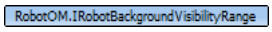
Available types of background visibility range.

Version

Available since version 11.

IV.5 IRobotBackgroundVisibilityRange

Class Hierarchy



C++

```
interface IRobotBackgroundVisibilityRange : IDispatch;
```

C#

```
public interface IRobotBackgroundVisibilityRange;
```

Visual Basic

```
Public Interface IRobotBackgroundVisibilityRange
```

Version

Available since version 11.

(c) 2014 Autodesk Inc. All rights reserved

IV.5.1 IRobotBackgroundVisibilityRange Members

The following tables list the members exposed by IRobotBackgroundVisibilityRange.

Public Fields

	Name	Description
	FromPos (see page 1338)	
	ToPos (see page 1339)	
	Type (see page 1339)	

IV.5.2 IRobotBackgroundVisibilityRange Fields

The fields of the IRobotBackgroundVisibilityRange class are listed here.

Public Fields

	Name	Description
	FromPos (see page 1338)	
	ToPos (see page 1339)	
	Type (see page 1339)	

IV.5.2.1 FromPos

C++

```
HRESULT get_FromPos(double*);
HRESULT put_FromPos(double);
```

C#

```
public double FromPos { get; set; }
```

Visual Basic

```
Public FromPos As double
```

Version

Available since version 11.

IV.5.2.2 ToPos

C++

```
HRESULT get_ToPos(double*);
HRESULT put_ToPos(double);
```

C#

```
public double ToPos { get; set; }
```

Visual Basic

```
Public ToPos As double
```

Version

Available since version 11.

IV.5.2.3 Type

C++

```
HRESULT get_Type(IRobotBackgroundVisibilityRangeType*);
HRESULT put_Type(IRobotBackgroundVisibilityRangeType);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRobotBackgroundVisibilityRangeType Type { get; set; }
```

Visual Basic

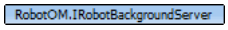
```
Public Type As IRobotBackgroundVisibilityRangeType
```

Version

Available since version 11.

IV.6 IRobotBackgroundServer

Class Hierarchy



C++

```
interface IRobotBackgroundServer : IDispatch;
```

C#

```
public interface IRobotBackgroundServer;
```

Visual Basic

```
Public Interface IRobotBackgroundServer
```

Description

Object managing backgrounds in the project.

Version

Available since version 11.

IV.6.1 IRobotBackgroundServer Members

The following tables list the members exposed by IRobotBackgroundServer.

Public Methods

	Name	Description
	Create (see page 1340)	Function creates an object defining background on the basis of the selected file.
	Get (see page 1340)	Function returns definition of the background of the given number.
	GetAllNumbers (see page 1341)	Function returns table containing numbers of all backgrounds defined in the project.
	Remove (see page 1341)	Function returns the background of the given number.
	RemoveAll (see page 1341)	Function deletes all backgrounds from project.

IV.6.2 IRobotBackgroundServer Methods

The methods of the IRobotBackgroundServer class are listed here.

Public Methods

	Name	Description
	Create (see page 1340)	Function creates an object defining background on the basis of the selected file.
	Get (see page 1340)	Function returns definition of the background of the given number.
	GetAllNumbers (see page 1341)	Function returns table containing numbers of all backgrounds defined in the project.

(c) 2014 Autodesk Inc. All rights reserved

	Remove (see page 1341)	Function returns the background of the given number.
	RemoveAll (see page 1341)	Function deletes all backgrounds from project.

IV.6.2.1 Create

C++

```
HRESULT Create(BSTR _file_path, VARIANT_BOOL _as_link, IRobotBackground** ret);
```

C#

```
public IRobotBackground Create(String _file_path, bool _as_link);
```

Visual Basic

```
Public Function Create(_file_path As String, _as_link As Boolean) As IRobotBackground
```

Description

Function creates an object defining background on the basis of the selected file.

Version

Available since version 11.

IV.6.2.2 Get

C++

```
HRESULT Get(long _number, IRobotBackground** ret);
```

C#

```
public IRobotBackground Get(long _number);
```

Visual Basic

```
Public Function Get(_number As long) As IRobotBackground
```

Description

Function returns definition of the background of the given number.

Version

Available since version 11.

IV.6.2.3 GetAllNumbers

C++

```
HRESULT GetAllNumbers(SAFEARRAY** ret);
```

C#

```
public Array GetAllNumbers();
```

Visual Basic

```
Public Function GetAllNumbers() As long)()
```

Description

Function returns table containing numbers of all backgrounds defined in the project.

Version

Available since version 11.

(c) 2014 Autodesk Inc. All rights reserved

IV.6.2.4 Remove

C++

```
HRESULT Remove(long _number);
```

C#

```
public void Remove(long _number);
```

Visual Basic

```
Public Sub Remove(_number As long)
```

Description

Function returns the background of the given number.

Version

Available since version 11.

IV.6.2.5 RemoveAll

C++

```
HRESULT RemoveAll();
```

C#

```
public void RemoveAll();
```

Visual Basic

```
Public Sub RemoveAll()
```

Description

Function deletes all backgrounds from project.

Version

Available since version 11.

V IRobotProjectType

C++

```
enum IRobotProjectType;
```

C#

```
public enum IRobotProjectType;
```

Visual Basic

```
Public Enum IRobotProjectType
```

Members

Members	Description
I_PT_CONCRETE_DEEP_BEAM = 18	Deep beam. Available since version 2.5.
I_PT_BUILDING = 19	Available since version 9.5.

Description

A set of identifiers is defined to refer to structure types recognized by Robot.

(c) 2014 Autodesk Inc. All rights reserved

VI IRobotProject

Class Hierarchy

RobotOM.IRobotProject

C++

```
interface IRobotProject : IDispatch;
```

C#

```
public interface IRobotProject;
```

Visual Basic

```
Public Interface IRobotProject
```

Description

RobotProject interface allows for managing all the data saved in an RTD-format file. .

VI.1 IRobotProject Members

The following tables list the members exposed by IRobotProject.

Public Fields

	Name	Description
	ActiveModel (see page 1345)	.
	AxisMngr (see page 1345)	
	Backgrounds (see page 1345)	Backgrounds.
	CalcEngine (see page 1346)	Object representing the calculation module of Robot .
	ComponentMngr (see page 1346)	Manager of all project components .
	ConcrReinfEngine (see page 1346)	Calculation module for the concrete reinforcement.
	Connections (see page 1346)	Server of connections defined in the project. .
	DimServer (see page 1347)	Design and verification server for steel and timber members.
	ExtFileName (see page 1347)	The path to the last external file (read by means of the OpenExtFile (see page 1353) function) in a non-RTD format.
	ExtFileParams (see page 1347)	Set of parameters opened by means of ReadExtFileParams (see page 1353)() function from an external-format file .
	FileInsertParams (see page 1348)	Collection of parameter related to the manner of inserting into the project the currently opened external file .
	FileName (see page 1348)	Path to the file where the project is saved .
	IsActive (see page 1348)	A flag indicating if the project is active or not - opened from a file or defined by the New (see page 1352)() function. It can happen that in the Robot program there is no project opened. In such a situation the IsActive flag equals zero (False).
	Name (see page 1348)	Project name .
	Preferences (see page 1349)	The object describing job parameters .
	PrintEngine (see page 1349)	Module that prints project documentation .
	Structure (see page 1349)	The object representing a structure defined in the project .
	Type (see page 1349)	Project type (type of structure defined in the project) .
	UniqueId (see page 1350)	Unique string of characters enabling project identification (RTD file) independently of the file name and its location on disk, which may change while working with the project; (available since version 1.7).

(c) 2014 Autodesk Inc. All rights reserved

	ViewMngr (see page 1350)	Manager controlling all views related to the project .

Public Methods

	Name	Description
	Close (see page 1351)	The function closes the current project without saving changes. .
	InsertExtFile (see page 1351)	The function inserts a structure defined in the external format (non-RTD format) file to the current project. Values of parameters describing definition and method of structure insertion may be read out (by means of the ReadExtFileParams (see page 1353) function) and changed earlier. If the structure is inserted correctly to the current project, then the function returns value different from zero (True).
	New (see page 1352)	The function creates a new project of the indicated type. .
	NewFromTemplate (see page 1352)	Function creates a new document on the basis of a template defined by a specified file. Available since version 2.0.
	Open (see page 1352)	The function opens an existing project from the indicated file in RTD format. .
	OpenExtFile (see page 1353)	The function opens a file in a format different from RTD. When opening the file, the program uses the current values of parameters read earlier from the file by means of the ReadExtFileParams (see page 1353) function. The parameters are divided into two sets: FileInsertParams (see page 1348)- parameters describing the manner of inserting the structure defined in the external file to the current project, and ExtFileParams (see page 1347) - set of variables defined in the external file to provide a parametrized structure description. The function returns a non-zero value (True) if the file is opened correctly and inserted into the project. It returns a zero value... more (see page 1353)
	ReadExtFileParams (see page 1353)	The function reads variables from the indicated external file in a non-RTD format. The values of the variables will be used during the process of opening the file by means of the OpenExtFile (see page 1353) function. Once the variables are read by means of the function, two sets of parameters are available: FileInsertParams (see page 1348) - parameters describing the manner of inserting the structure defined in the external file to the current project, and ExtFileParams (see page 1347) - set of variables defined in the external file to provide a parametrized structure description. The value of each parameter from each set may be modified and the modification... more (see page 1353)
	Save (see page 1353)	The function saves the current project to the file whose full access path is described by the FileName (see page 1348) variable. .
	SaveAs (see page 1354)	The function saves the current project to the file accessed by the indicated path. .
	SaveAsExtFile (see page 1354)	The function saves the current project to the file in the indicated external format (other than RTD). .
	SaveToFormat (see page 1354)	Function saves the project to a file of the selected format. .

VI.2 IRobotProject Fields

The fields of the IRobotProject class are listed here.

Public Fields

	Name	Description
	ActiveModel (see page 1345)	.
	AxisMngr (see page 1345)	
	Backgrounds (see page 1345)	Backgrounds.
	CalcEngine (see page 1346)	Object representing the calculation module of Robot .
	ComponentMngr (see page 1346)	Manager of all project components .
	ConcrReinfEngine (see page 1346)	Calculation module for the concrete reinforcement.

(c) 2014 Autodesk Inc. All rights reserved

	Connections (see page 1346)	Server of connections defined in the project. .
	DimServer (see page 1347)	Design and verification server for steel and timber members.
	ExtFileName (see page 1347)	The path to the last external file (read by means of the OpenExtFile (see page 1353) function) in a non-RTD format.
	ExtFileParams (see page 1347)	Set of parameters opened by means of ReadExtFileParams (see page 1353)() function from an external-format file .
	FileInsertParams (see page 1348)	Collection of parameter related to the manner of inserting into the project the currently opened external file .
	FileName (see page 1348)	Path to the file where the project is saved .
	IsActive (see page 1348)	A flag indicating if the project is active or not - opened from a file or defined by the New (see page 1352)() function. It can happen that in the Robot program there is no project opened. In such a situation the IsActive flag equals zero (False).
	Name (see page 1348)	Project name .
	Preferences (see page 1349)	The object describing job parameters .
	PrintEngine (see page 1349)	Module that prints project documentation .
	Structure (see page 1349)	The object representing a structure defined in the project .
	Type (see page 1349)	Project type (type of structure defined in the project) .
	UniqueId (see page 1350)	Unique string of characters enabling project identification (RTD file) independently of the file name and its location on disk, which may change while working with the project; (available since version 1.7).
	ViewMngr (see page 1350)	Manager controlling all views related to the project .

VI.2.1 ActiveModel

C++

```
HRESULT get_ActiveModel(IRobotActiveModelType*);
HRESULT put_ActiveModel(IRobotActiveModelType);
```

C#

```
public IRobotActiveModelType ActiveModel { get; set; }
```

Visual Basic

```
Public ActiveModel As IRobotActiveModelType
```

Description

.

Version

Available since version 15.

VI.2.2 AxisMngr

C++

```
HRESULT get_AxisMngr(IRobotStructuralAxisGridMngr**);
```

C#

```
public IRobotStructuralAxisGridMngr AxisMngr { get; }
```

Visual Basic

```
Public ReadOnly AxisMngr As IRobotStructuralAxisGridMngr
```

VI.2.3 Backgrounds

C++

```
HRESULT get_Backgrounds(IRobotBackgroundServer**);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRobotBackgroundServer Backgrounds { get; }
```

Visual Basic

```
Public ReadOnly Backgrounds As IRobotBackgroundServer
```

Description

Backgrounds.

Version

Available since version 11.

VI.2.4 CalcEngine

C++

```
HRESULT get_CalcEngine(IRobotCalcEngine**);
```

C#

```
public IRobotCalcEngine CalcEngine { get; }
```

Visual Basic

```
Public ReadOnly CalcEngine As IRobotCalcEngine
```

Description

Object representing the calculation module of Robot .

VI.2.5 ComponentMngr

C++

```
HRESULT get_ComponentMngr(IRobotProjectComponentMngr**);
```

C#

```
public IRobotProjectComponentMngr ComponentMngr { get; }
```

Visual Basic

```
Public ReadOnly ComponentMngr As IRobotProjectComponentMngr
```

Description

Manager of all project components .

VI.2.6 ConcrReinfEngine

C++

```
HRESULT get_ConcrReinfEngine(IRConcrCalcEngine**);
```

C#

```
public IRConcrCalcEngine ConcrReinfEngine { get; }
```

Visual Basic

```
Public ReadOnly ConcrReinfEngine As IRConcrCalcEngine
```

Description

Calculation module for the concrete reinforcement.

Version

Available since version 5.5.

(c) 2014 Autodesk Inc. All rights reserved

VI.2.7 Connections

C++

```
HRESULT get_Connections(IRJointConnectionServer**);
```

C#

```
public IRJointConnectionServer Connections { get; }
```

Visual Basic

```
Public ReadOnly Connections As IRJointConnectionServer
```

Description

Server of connections defined in the project. .

VI.2.8 DimServer

C++

```
HRESULT get_DimServer(IRDimServer**);
```

C#

```
public IRDimServer DimServer { get; }
```

Visual Basic

```
Public ReadOnly DimServer As IRDimServer
```

Description

Design and verification server for steel and timber members.

Version

Available since version 2.5.

VI.2.9 ExtFileName

C++

```
HRESULT get_ExtFileName(BSTR*);
```

C#

```
public String ExtFileName { get; }
```

Visual Basic

```
Public ReadOnly ExtFileName As String
```

Description

The path to the last external file (read by means of the OpenExtFile (see page 1353) function) in a non-RTD format.

VI.2.10 ExtFileParams

C++

```
HRESULT get_ExtFileParams(IDispatch*);
```

C#

```
public IDispatch ExtFileParams { get; }
```

Visual Basic

```
Public ReadOnly ExtFileParams As IDispatch
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Set of parameters opened by means of ReadExtFileParams (see page 1353)() function from an external-format file .

VI.2.11 FileInsertParams

C++

```
HRESULT get_FileInsertParams(IRobotFileInsertParams**);
```

C#

```
public IRobotFileInsertParams FileInsertParams { get; }
```

Visual Basic

```
Public ReadOnly FileInsertParams As IRobotFileInsertParams
```

Description

Collection of parameter related to the manner of inserting into the project the currently opened external file .

VI.2.12 FileName

C++

```
HRESULT get_FileName(BSTR*);
```

C#

```
public String FileName { get; }
```

Visual Basic

```
Public ReadOnly FileName As String
```

Description

Path to the file where the project is saved .

VI.2.13 IsActive

C++

```
HRESULT get_IsActive(VARIANT_BOOL*);
```

C#

```
public bool IsActive { get; }
```

Visual Basic

```
Public ReadOnly IsActive As Boolean
```

Description

A flag indicating if the project is active or not - opened from a file or defined by the New (see page 1352)() function. It can happen that in the Robot program there is no project opened. In such a situation the IsActive flag equals zero (False).

VI.2.14 Name

C++

```
HRESULT get_Name(BSTR*);
```

C#

```
public String Name { get; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public ReadOnly Name As String
```

Description

Project name .

VI.2.15 Preferences

C++

```
HRESULT get_Preferences(IRobotProjectPreferences**);
```

C#

```
public IRobotProjectPreferences Preferences { get; }
```

Visual Basic

```
Public ReadOnly Preferences As IRobotProjectPreferences
```

Description

The object describing job parameters .

VI.2.16 PrintEngine

C++

```
HRESULT get_PrintEngine(IRobotPrintEngine**);
```

C#

```
public IRobotPrintEngine PrintEngine { get; }
```

Visual Basic

```
Public ReadOnly PrintEngine As IRobotPrintEngine
```

Description

Module that prints project documentation .

VI.2.17 Structure

C++

```
HRESULT get_Structure(IRobotStructure**);
```

C#

```
public IRobotStructure Structure { get; }
```

Visual Basic

```
Public ReadOnly Structure As IRobotStructure
```

Description

The object representing a structure defined in the project .

VI.2.18 Type

C++

```
HRESULT get_Type(IRobotProjectType*);
```

C#

```
public IRobotProjectType Type { get; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public ReadOnly Type As IRobotProjectType
```

Description

Project type (type of structure defined in the project) .

VI.2.19 UniqueId

C++

```
HRESULT get_UniqueId(BSTR*);
```

C#

```
public String UniqueId { get; }
```

Visual Basic

```
Public ReadOnly UniqueId As String
```

Description

Unique string of characters enabling project identification (RTD file) independently of the file name and its location on disk, which may change while working with the project; (available since version 1.7).

VI.2.20 ViewMngr

C++

```
HRESULT get_ViewMngr(IRobotViewMngr**);
```

C#

```
public IRobotViewMngr ViewMngr { get; }
```

Visual Basic

```
Public ReadOnly ViewMngr As IRobotViewMngr
```

Description

Manager controlling all views related to the project .

VI.3 IRobotProject Methods

The methods of the IRobotProject class are listed here.

Public Methods

	Name	Description
	Close (see page 1351)	The function closes the current project without saving changes. .
	InsertExtFile (see page 1351)	The function inserts a structure defined in the external format (non-RTD format) file to the current project. Values of parameters describing definition and method of structure insertion may be read out (by means of the ReadExtFileParams (see page 1353) function) and changed earlier. If the structure is inserted correctly to the current project, then the function returns value different from zero (True).
	New (see page 1352)	The function creates a new project of the indicated type. .
	NewFromTemplate (see page 1352)	Function creates a new document on the basis of a template defined by a specified file. Available since version 2.0.
	Open (see page 1352)	The function opens an existing project from the indicated file in RTD format. .

(c) 2014 Autodesk Inc. All rights reserved

	OpenExtFile (see page 1353)	The function opens a file in a format different from RTD. When opening the file, the program uses the current values of parameters read earlier from the file by means of the ReadExtFileParams (see page 1353) function. The parameters are divided into two sets: FileInsertParams (see page 1348)- parameters describing the manner of inserting the structure defined in the external file to the current project, and ExtFileParams (see page 1347) - set of variables defined in the external file to provide a parametrized structure description. The function returns a non-zero value (True) if the file is opened correctly and inserted into the project. It returns a zero value... more (see page 1353)
	ReadExtFileParams (see page 1353)	The function reads variables from the indicated external file in a non-RTD format. The values of the variables will be used during the process of opening the file by means of the OpenExtFile (see page 1353) function. Once the variables are read by means of the function, two sets of parameters are available: FileInsertParams (see page 1348) - parameters describing the manner of inserting the structure defined in the external file to the current project, and ExtFileParams (see page 1347) - set of variables defined in the external file to provide a parametrized structure description. The value of each parameter from each set may be modified and the modification... more (see page 1353)
	Save (see page 1353)	The function saves the current project to the file whose full access path is described by the FileName (see page 1348) variable. .
	SaveAs (see page 1354)	The function saves the current project to the file accessed by the indicated path. .
	SaveAsExtFile (see page 1354)	The function saves the current project to the file in the indicated external format (other than RTD). .
	SaveToFormat (see page 1354)	Function saves the project to a file of the selected format. .

VI.3.1 Close

C++

```
HRESULT Close();
```

C#

```
public void Close();
```

Visual Basic

```
Public Sub Close()
```

Description

The function closes the current project without saving changes. .

VI.3.2 InsertExtFile

C++

```
HRESULT InsertExtFile(BSTR _file_path, IRobotExternalFileFormat _file_format, VARIANT_BOOL
_ignore_warnings, VARIANT_BOOL _only_geometry, VARIANT_BOOL* ret);
```

C#

```
public bool InsertExtFile(String _file_path, IRobotExternalFileFormat _file_format, bool
_ignore_warnings, bool _only_geometry);
```

Visual Basic

```
Public Function InsertExtFile(_file_path As String, _file_format As
IRobotExternalFileFormat, _ignore_warnings As Boolean, _only_geometry As Boolean) As Boolean
```

Description

The function inserts a structure defined in the external format (non-RTD format) file to the current project. Values of

(c) 2014 Autodesk Inc. All rights reserved

parameters describing definition and method of structure insertion may be read out (by means of the ReadExtFileParams (see page 1353) function) and changed earlier. If the structure is inserted correctly to the current project, then the function returns value different from zero (True).

Version

Available since version 2.5.

VI.3.3 New

C++

```
HRESULT New(IRobotProjectType _prj_type);
```

C#

```
public void New(IRobotProjectType _prj_type);
```

Visual Basic

```
Public Sub New(_prj_type As IRobotProjectType)
```

Description

The function creates a new project of the indicated type. .

VI.3.4 NewFromTemplate

C++

```
HRESULT NewFromTemplate(BSTR _tmpl_file);
```

C#

```
public void NewFromTemplate(String _tmpl_file);
```

Visual Basic

```
Public Sub NewFromTemplate(_tmpl_file As String)
```

Description

Function creates a new document on the basis of a template defined by a specified file. Available since version 2.0.

VI.3.5 Open

C++

```
HRESULT Open(BSTR _file_path);
```

C#

```
public void Open(String _file_path);
```

Visual Basic

```
Public Sub Open(_file_path As String)
```

Description

The function opens an existing project from the indicated file in RTD format. .

VI.3.6 OpenExtFile

C++

```
HRESULT OpenExtFile(BSTR _file_path, IRobotExternalFileFormat _format, VARIANT_BOOL
_ignore_warnings, VARIANT_BOOL* ret);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public bool OpenExtFile(String _file_path, IRobotExternalFileFormat _format, bool
_ignore_warnings);
```

Visual Basic

```
Public Function OpenExtFile(_file_path As String, _format As IRobotExternalFileFormat,
_ignore_warnings As Boolean) As Boolean
```

Description

The function opens a file in a format different from RTD. When opening the file, the program uses the current values of parameters read earlier from the file by means of the ReadExtFileParams (see page 1353) function. The parameters are divided into two sets: FileInsertParams (see page 1348)- parameters describing the manner of inserting the structure defined in the external file to the current project, and ExtFileParams (see page 1347) - set of variables defined in the external file to provide a parametrized structure description. The function returns a non-zero value (True) if the file is opened correctly and inserted into the project. It returns a zero value (False) in the case of errors during file opening. .

VI.3.7 ReadExtFileParams

C++

```
HRESULT ReadExtFileParams(BSTR _file_path, IRobotExternalFileFormat _file_format,
VARIANT_BOOL* ret);
```

C#

```
public bool ReadExtFileParams(String _file_path, IRobotExternalFileFormat _file_format);
```

Visual Basic

```
Public Function ReadExtFileParams(_file_path As String, _file_format As
IRobotExternalFileFormat) As Boolean
```

Description

The function reads variables from the indicated external file in a non-RTD format. The values of the variables will be used during the process of opening the file by means of the OpenExtFile (see page 1353) function. Once the variables are read by means of the function, two sets of parameters are available: FileInsertParams (see page 1348) - parameters describing the manner of inserting the structure defined in the external file to the current project, and ExtFileParams (see page 1347) - set of variables defined in the external file to provide a parametrized structure description. The value of each parameter from each set may be modified and the modification will be recognized during opening the file by means of the OpenExtFile (see page 1353) function. The function returns zero (False) in the case of errors during analysis of the indicated external file. .

VI.3.8 Save

C++

```
HRESULT Save();
```

C#

```
public void Save();
```

Visual Basic

```
Public Sub Save()
```

Description

The function saves the current project to the file whose full access path is described by the FileName (see page 1348) variable. .

(c) 2014 Autodesk Inc. All rights reserved

VI.3.9 SaveAs

C++

```
HRESULT SaveAs(BSTR _file_path);
```

C#

```
public void SaveAs(String _file_path);
```

Visual Basic

```
Public Sub SaveAs(_file_path As String)
```

Description

The function saves the current project to the file accessed by the indicated path. .

VI.3.10 SaveAsExtFile

C++

```
HRESULT SaveAsExtFile(BSTR _file_path, IRobotExternalFileFormat _format);
```

C#

```
public void SaveAsExtFile(String _file_path, IRobotExternalFileFormat _format);
```

Visual Basic

```
Public Sub SaveAsExtFile(_file_path As String, _format As IRobotExternalFileFormat)
```

Description

The function saves the current project to the file in the indicated external format (other than RTD). .

VI.3.11 SaveToFormat

C++

```
HRESULT SaveToFormat(IRobotProjectSaveFormat _file_format, BSTR _file_path, VARIANT_BOOL*
ret);
```

C#

```
public bool SaveToFormat(IRobotProjectSaveFormat _file_format, String _file_path);
```

Visual Basic

```
Public Function SaveToFormat(_file_format As IRobotProjectSaveFormat, _file_path As String)
As Boolean
```

Description

Function saves the project to a file of the selected format. .

Version

Available since version 3.

VII IRobotFileInsertParams

Class Hierarchy

RobotOM.IRobotFileInsertParams

(c) 2014 Autodesk Inc. All rights reserved

C++

```
interface IRobotFileInsertParams : IDispatch;
```

C#

```
public interface IRobotFileInsertParams;
```

Visual Basic

```
Public Interface IRobotFileInsertParams
```

Description

The proces of inserting a structure, defined in an external file of a non-RTD format, is carried out according to parameters determining the insertion mode. The RobotInsertParams interface provides access to the parameters determining the manner of inserting the structure into the current project. .

VII.1 IRobotFileInsertParams Members

The following tables list the members exposed by IRobotFileInsertParams.

Public Fields

	Name	Description
	AsObject (see page 1356)	The flag indicating if the structure being inserted is to be treated as an object .
	ReferenceNode (see page 1356)	Node number - structure insertion point .
	ScaleFactor (see page 1356)	Scaling coefficient .

Public Methods

	Name	Description
	GetInsertionPoint (see page 1357)	The function sets the coordinates of the insertion point. .
	GetRotation (see page 1357)	The function provides access to the rotation angle defined for the inserted structure. .
	SetInsertionPoint (see page 1357)	The function sets the coordinates of the insertion point. .
	SetRotation (see page 1357)	The function allows one to determine the angle of the iserted structure. .

VII.2 IRobotFileInsertParams Fields

The fields of the IRobotFileInsertParams class are listed here.

Public Fields

	Name	Description
	AsObject (see page 1356)	The flag indicating if the structure being inserted is to be treated as an object .
	ReferenceNode (see page 1356)	Node number - structure insertion point .
	ScaleFactor (see page 1356)	Scaling coefficient .

VII.2.1 AsObject

C++

```
HRESULT get_AsObject(VARIANT_BOOL*);
HRESULT put_AsObject(VARIANT_BOOL);
```

C#

```
public bool AsObject { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public AsObject As Boolean
```

Description

The flag indicating if the structure being inserted is to be treated as an object .

VII.2.2 ReferenceNode

C++

```
HRESULT get_ReferenceNode(long*);
HRESULT put_ReferenceNode(long);
```

C#

```
public long ReferenceNode { get; set; }
```

Visual Basic

```
Public ReferenceNode As long
```

Description

Node number - structure insertion point .

VII.2.3 ScaleFactor

C++

```
HRESULT get_ScaleFactor(double*);
HRESULT put_ScaleFactor(double);
```

C#

```
public double ScaleFactor { get; set; }
```

Visual Basic

```
Public ScaleFactor As double
```

Description

Scaling coefficient .

VII.3 IRobotFileInsertParams Methods

The methods of the IRobotFileInsertParams class are listed here.

Public Methods

	Name	Description
	GetInsertionPoint (see page 1357)	The function sets the coordinates of the insertion point. .
	GetRotation (see page 1357)	The function provides access to the rotation angle defined for the inserted structure. .
	SetInsertionPoint (see page 1357)	The function sets the coordinates of the insertion point. .
	SetRotation (see page 1357)	The function allows one to determine the angle of the iserted structure. .

VII.3.1 GetInsertionPoint

C++

```
HRESULT GetInsertionPoint(double* _x, double* _y, double* _z);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public void GetInsertionPoint(double* _x, double* _y, double* _z);
```

Visual Basic

```
Public Sub GetInsertionPoint(ByRef _x As double*, ByRef _y As double*, ByRef _z As double*)
```

Description

The function sets the coordinates of the insertion point. .

VII.3.2 GetRotation

C++

```
HRESULT GetRotation(double* _alpha, double* _beta, double* _gamma);
```

C#

```
public void GetRotation(double* _alpha, double* _beta, double* _gamma);
```

Visual Basic

```
Public Sub GetRotation(ByRef _alpha As double*, ByRef _beta As double*, ByRef _gamma As
double*)
```

Description

The function provides access to the rotation angle defined for the inserted structure. .

VII.3.3 SetInsertionPoint

C++

```
HRESULT SetInsertionPoint(double _x, double _y, double _z);
```

C#

```
public void SetInsertionPoint(double _x, double _y, double _z);
```

Visual Basic

```
Public Sub SetInsertionPoint(_x As double, _y As double, _z As double)
```

Description

The function sets the coordinates of the insertion point. .

VII.3.4 SetRotation

C++

```
HRESULT SetRotation(double _alpha, double _beta, double _gamma);
```

C#

```
public void SetRotation(double _alpha, double _beta, double _gamma);
```

Visual Basic

```
Public Sub SetRotation(_alpha As double, _beta As double, _gamma As double)
```

Description

The function allows one to determine the angle of the iserted structure. .

(c) 2014 Autodesk Inc. All rights reserved

VIII IRobotProjectSaveFormat

C++

```
enum IRobotProjectSaveFormat;
```

C#

```
public enum IRobotProjectSaveFormat;
```

Visual Basic

```
Public Enum IRobotProjectSaveFormat
```

Members

Members	Description
I_PSF_RTD = 0	Available since version 3.
I_PSF_RTD_NORESULTS = 1	Available since version 3.
I_PSF_STR = 2	Available since version 3.
I_PSF_DXF = 3	Available since version 3.
I_PSF_DXF_V14 = 4	Available since version 3.
I_PSF_DWG = 5	Available since version 3.
I_PSF_ANF = 6	Available since version 3.
I_PSF_WRL = 7	Available since version 3.
I_PSF_S = 8	Available since version 3.
I_PSF_SAT = 9	Available since version 3.
I_PSF_STP_CIM = 10	Available since version 3.
I_PSF_STP_DSTV = 11	Available since version 3.

Description

Available formats for project saving.

Version

Available since version 3.

IX IRobotProjectEvents

Class Hierarchy

RobotOM.IRobotProjectEvents

C++

```
interface IRobotProjectEvents : IDispatch;
```

C#

```
public interface IRobotProjectEvents;
```

Visual Basic

```
Public Interface IRobotProjectEvents
```

Description

Events generated by the RobotProject object.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 7.5.

IX.1 IRobotProjectEvents Members

The following tables list the members exposed by IRobotProjectEvents.

Public Methods

	Name	Description
	OnClose (see page 1359)	Event generated directly before closing the document.
	OnSave (see page 1359)	Event generated directly before saving the project to the file of the specified path.

IX.2 IRobotProjectEvents Methods

The methods of the IRobotProjectEvents class are listed here.

Public Methods

	Name	Description
	OnClose (see page 1359)	Event generated directly before closing the document.
	OnSave (see page 1359)	Event generated directly before saving the project to the file of the specified path.

IX.2.1 OnClose

C++

```
HRESULT OnClose(VARIANT_BOOL _is_modified);
```

C#

```
public void OnClose(bool _is_modified);
```

Visual Basic

```
Public Sub OnClose(_is_modified As Boolean)
```

Description

Event generated directly before closing the document.

Version

Available since version 7.5.

IX.2.2 OnSave

C++

```
HRESULT OnSave(BSTR _file_path, VARIANT_BOOL _is_modified);
```

C#

```
public void OnSave(String _file_path, bool _is_modified);
```

Visual Basic

```
Public Sub OnSave(_file_path As String, _is_modified As Boolean)
```

Description

Event generated directly before saving the project to the file of the specified path.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 7.5.

X IRobotActiveModelType

C++

```
enum IRobotActiveModelType;
```

C#

```
public enum IRobotActiveModelType;
```

Visual Basic

```
Public Enum IRobotActiveModelType
```

Members

Members	Description
I_AMT_MAIN = 0	. Available since version 15.
I_AMT_DAM = 1	. Available since version 15.

Description

.

Version

Available since version 15.

Presentation of data

I Views and layouts

Enumerations

	Name	Description
	IRobotLayoutId (see page 1475)	Set of identifiers for the layouts defined in the Robot program. .
	IRobotDialogId (see page 1476)	Identifiers of available dialogs.

Interfaces

	Name	Description
	IRobotViewMngr (see page 1469)	View manager controls all the views and layouts available in Robot. .

I.1 Tables

Enumerations

	Name	Description
	IRobotTableType (see page 1361)	One can differentiate a number of table types presenting different types of data. Each table type has got its identifier. .

(c) 2014 Autodesk Inc. All rights reserved

	IRobotTableDataType (see page 1361)	A set of identifiers is defined to refer to types of data presented in tables. .
	IRobotTableConfigFlag (see page 1362)	Tables may be configured in many ways. A set of flags is defined to help one configure tables. .
	IRobotTableConfigValue (see page 1363)	Set of values used to configure tables. .
	IRobotTableScreenCaptureUpdateType (see page 1379)	

Interfaces

	Name	Description
	IRobotTableConfig (see page 1363)	Each table may be configured by means of a configuring object.
	IRobotTable (see page 1364)	Interface representing tables. .
	IRobotTableFrame (see page 1373)	Interface representing the viewer composed of many tables. Each table is a separate tab defined within the same main frame.
	IRobotTableScreenCaptureParams (see page 1376)	Parameters of the screen capture for a table.

I.1.1 IRobotTableType

C++

```
enum IRobotTableType;
```

C#

```
public enum IRobotTableType;
```

Visual Basic

```
Public Enum IRobotTableType
```

Description

One can differentiate a number of table types presenting different types of data. Each table type has got its identifier. .

I.1.2 IRobotTableDataType

C++

```
enum IRobotTableDataType;
```

C#

```
public enum IRobotTableDataType;
```

Visual Basic

```
Public Enum IRobotTableDataType
```

Description

A set of identifiers is defined to refer to types of data presented in tables. .

I.1.3 IRobotTableConfigFlag

C++

```
enum IRobotTableConfigFlag;
```

C#

```
public enum IRobotTableConfigFlag;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Enum IRobotTableConfigFlag
```

Members

Members	Description
I_TCF_DETAILS = 1	Detailed information Available since version 1.7.
I_TCF_GLOBAL_COORDINATES = 2	Global coordinates Available since version 1.7.
I_TCF_LOCAL_COORDINATES = 3	Local coordinates Available since version 1.7.
I_TCF_COEXISTENT_VALUES = 4	Available since version 1.7.
I_TCF_EXTREME_COMBINATIONS = 5	Available since version 1.7.
I_TCF_PROP_STD_TAPERED_SECTION = 6	Available since version 1.7.
I_TCF_COMB_DEFINITIONS = 7	Available since version 1.7.
I_TCF_COMB_CODE_COMPONENTS = 8	Available since version 1.7.
I_TCF_COMB_MOVING_LOADS = 9	Available since version 1.7.
I_TCF_COMB_TIME_HISTORY_COMPONENTS = 10	Available since version 1.7.
I_TCF_REAC_VALUES = 11	Available since version 1.7.
I_TCF_REAC_APPLIED_FORCES_SUM = 12	Available since version 1.7.
I_TCF_REAC_EQUILIBRIUM_PRECISION = 13	Available since version 1.7.
I_TCF_REAC_RESIDUUM = 14	Available since version 1.7.
I_TCF_BAR_DEFL_DEFLECTIONS = 15	Available since version 1.7.
I_TCF_BAR_DEFL_MAX_DEFLECTIONS = 16	Available since version 1.7.
I_TCF_BAR_DEFL_DISPLACEMENTS = 17	Available since version 1.7.
I_TCF_FE_DIR_X_AXIS = 18	Available since version 1.7.
I_TCF_FE_DIR_Y_AXIS = 19	Available since version 1.7.
I_TCF_FE_DIR_Z_AXIS = 20	Available since version 1.7.
I_TCF_FE_LAYER_UPPER = 21	Available since version 1.7.
I_TCF_FE_LAYER_MIDDLE = 22	Available since version 1.7.
I_TCF_FE_LAYER_LOWER = 23	Available since version 1.7.
I_TCF_FE_LAYER_MAX = 24	Available since version 1.7.
I_TCF_FE_LAYER_MIN = 25	Available since version 1.7.
I_TCF_FE_LAYER_MAX_ABSOLUTE = 26	Available since version 1.7.
I_TCF_FE_IN_NODES = 27	Available since version 1.7.
I_TCF_FE_IN_ELEMENT_CENTERS = 28	Available since version 1.7.
I_TCF_REINF_IN_NODES = 29	Available since version 1.7.
I_TCF_REINF_IN_ELEMENT_CENTERS = 30	Available since version 1.7.
I_TCF_REAC_DDC = 31	Available since version 1.7.
I_TCF_LOCALS = 32	Available since version 1.7.
I_TCF_FE_IN_ELENODE = 33	Available since version 1.7.
I_TCF_ORDER_BY_CASE = 34	Available since version 1.7.
I_TCF_CASE_INFO = 35	Available since version 1.7.
I_TCF_GLOBAL_ANALYSIS_RESULTS_NPOINTS = 36	Available since version 3.
I_TCF_GLOBAL_ANALYSIS_RESULTS_RELATIVE = 37	Available since version 3.

Description

Tables may be configured in many ways. A set of flags is defined to help one configure tables. .

(c) 2014 Autodesk Inc. All rights reserved

I.1.4 IRobotTableConfigValue

C++

```
enum IRobotTableConfigValue;
```

C#

```
public enum IRobotTableConfigValue;
```

Visual Basic

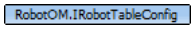
```
Public Enum IRobotTableConfigValue
```

Description

Set of values used to configure tables. .

I.1.5 IRobotTableConfig

Class Hierarchy



C++

```
interface IRobotTableConfig : IDispatch;
```

C#

```
public interface IRobotTableConfig;
```

Visual Basic

```
Public Interface IRobotTableConfig
```

Description

Each table may be configured by means of a configuring object.

I.1.5.1 IRobotTableConfig Members

The following tables list the members exposed by IRobotTableConfig.

Public Methods

	Name	Description
	SetFlag (see page 1364)	The function switches {on/off} the indicated flag configuring the table. .
	SetValue (see page 1364)	The function sets the value of the indicated variable configuring the table. .

I.1.5.2 IRobotTableConfig Methods

The methods of the IRobotTableConfig class are listed here.

Public Methods

	Name	Description
	SetFlag (see page 1364)	The function switches {on/off} the indicated flag configuring the table. .
	SetValue (see page 1364)	The function sets the value of the indicated variable configuring the table. .

I.1.5.2.1 SetFlag

C++

```
HRESULT SetFlag(IRobotTableConfigFlag _flag, VARIANT_BOOL _set);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public void SetFlag(IRobotTableConfigFlag _flag, bool _set);
```

Visual Basic

```
Public Sub SetFlag(_flag As IRobotTableConfigFlag, _set As Boolean)
```

Description

The function switches {on/off} the indicated flag configuring the table. .

I.1.5.2.2 SetValue

C++

```
HRESULT SetValue(IRobotTableConfigValue _value_id, double _val);
```

C#

```
public void SetValue(IRobotTableConfigValue _value_id, double _val);
```

Visual Basic

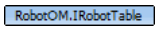
```
Public Sub SetValue(_value_id As IRobotTableConfigValue, _val As double)
```

Description

The function sets the value of the indicated variable configuring the table. .

I.1.6 IRobotTable

Class Hierarchy



C++

```
interface IRobotTable : IDispatch;
```

C#

```
public interface IRobotTable;
```

Visual Basic

```
Public Interface IRobotTable
```

Description

Interface representing tables. .

I.1.6.1 IRobotTable Members

The following tables list the members exposed by IRobotTable.

Public Fields

	Name	Description
	ColCount (see page 1366)	Number of table columns.
	Configuration (see page 1366)	Object controlling table configuration .
	Printable (see page 1366)	Object representing tables; it may be added to the printout and printed .
	RowCount (see page 1367)	Number of table rows.
	Title (see page 1367)	Table title .
	UserControl (see page 1367)	Flag that enables giving/taking back table navigation to/from the interactive program user .

(c) 2014 Autodesk Inc. All rights reserved

	Visible (see page 1367)	Variable determining if the table is to be displayed or not .
	Window (see page 1368)	

Public Methods

	Name	Description
	AddColumn (see page 1369)	The function adds to a table a new column that will present data of the indicated type .
	GetColWidth (see page 1369)	Function returns a width of the column with the specified index.
	GetDataType (see page 1369)	Function returns a type of data presented in the column with the specified index.
	GetRowHeight (see page 1370)	Function returns a height of the row with the specified index.
	MakeScreenCapture (see page 1370)	Function saves a screen capture with the specified parameters.
	Select (see page 1370)	The function sets a filter defined by the indicated selection on the table .
	SelectAllFromStore (see page 1370)	The function sets a filter on the table, defined by means of all selections formerly saved in the table by means of the StoreSelection (see page 1373)() function. .
	SelectFromStore (see page 1371)	The function sets a filter on the table. The filter is defined by means of the selection saved earlier by the StoreSelection (see page 1373)() function.
	SelectMode (see page 1371)	The function sets a filter on the table, described by the indicated mode selection. .
	SelectModeFromStore (see page 1371)	The function sets a filter on the table, defined by means of mode selection formerly saved in the table by means of the StoreModeSelection (see page 1372)(). .
	SetColWidth (see page 1372)	Function sets a width of the column with the specified index.
	SetRowHeight (see page 1372)	Function sets a height of the row with the specified index.
	StoreModeSelection (see page 1372)	The function saves the indicated mode selection to the table. .
	StoreSelection (see page 1373)	The function saves the indicated selection in the table without modifying its filter. One may save one selection of each type of objects - structure components - in the table. .

I.1.6.2 IRobotTable Fields

The fields of the IRobotTable class are listed here.

Public Fields

	Name	Description
	ColCount (see page 1366)	Number of table columns.
	Configuration (see page 1366)	Object controlling table configuration .
	Printable (see page 1366)	Object representing tables; it may be added to the printout and printed .
	RowCount (see page 1367)	Number of table rows.
	Title (see page 1367)	Table title .
	UserControl (see page 1367)	Flag that enables giving/taking back table navigation to/from the interactive program user .
	Visible (see page 1367)	Variable determining if the table is to be displayed or not .
	Window (see page 1368)	

I.1.6.2.1 ColCount

C++

```
HRESULT get_ColCount(long*);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public long ColCount { get; }
```

Visual Basic

```
Public ReadOnly ColCount As long
```

Description

Number of table columns.

Version

Available since version 8.2.

I.1.6.2.2 Configuration

C++

```
HRESULT get_Configuration(IRobotTableConfig**);
```

C#

```
public IRobotTableConfig Configuration { get; }
```

Visual Basic

```
Public ReadOnly Configuration As IRobotTableConfig
```

Description

Object controlling table configuration .

I.1.6.2.3 Printable

C++

```
HRESULT get_Printable(IRobotPrintable**);
```

C#

```
public IRobotPrintable Printable { get; }
```

Visual Basic

```
Public ReadOnly Printable As IRobotPrintable
```

Description

Object representing tables; it may be added to the printout and printed .

I.1.6.2.4 RowCount

C++

```
HRESULT get_RowCount(long*);
```

C#

```
public long RowCount { get; }
```

Visual Basic

```
Public ReadOnly RowCount As long
```

Description

Number of table rows.

Version

Available since version 8.2.

(c) 2014 Autodesk Inc. All rights reserved

I.1.6.2.5 Title

C++

```
HRESULT get_Title(BSTR*);
HRESULT put_Title(BSTR);
```

C#

```
public String Title { get; set; }
```

Visual Basic

```
Public Title As String
```

Description

Table title .

I.1.6.2.6 UserControl

C++

```
HRESULT get_UserControl(VARIANT_BOOL*);
HRESULT put_UserControl(VARIANT_BOOL);
```

C#

```
public bool UserControl { get; set; }
```

Visual Basic

```
Public UserControl As Boolean
```

Description

Flag that enables giving/taking back table navigation to/from the interactive program user .

Version

Available since version 3.

I.1.6.2.7 Visible

C++

```
HRESULT get_Visible(VARIANT_BOOL*);
HRESULT put_Visible(VARIANT_BOOL);
```

C#

```
public bool Visible { get; set; }
```

Visual Basic

```
Public Visible As Boolean
```

Description

Variable determining if the table is to be displayed or not .

I.1.6.2.8 Window

C++

```
HRESULT get_Window(IRobotWindow**);
```

C#

```
public IRobotWindow Window { get; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public ReadOnly Window As IRobotWindow
```

Version

Available since version 3.

I.1.6.3 IRobotTable Methods

The methods of the IRobotTable class are listed here.

Public Methods

	Name	Description
	AddColumn (see page 1369)	The function adds to a table a new column that will present data of the indicated type .
	GetColWidth (see page 1369)	Function returns a width of the column with the specified index.
	GetDataType (see page 1369)	Function returns a type of data presented in the column with the specified index.
	GetRowHeight (see page 1370)	Function returns a height of the row with the specified index.
	MakeScreenCapture (see page 1370)	Function saves a screen capture with the specified parameters.
	Select (see page 1370)	The function sets a filter defined by the indicated selection on the table .
	SelectAllFromStore (see page 1370)	The function sets a filter on the table, defined by means of all selections formerly saved in the table by means of the StoreSelection (see page 1373)() function. .
	SelectFromStore (see page 1371)	The function sets a filter on the table. The filter is defined by means of the selection saved earlier by the StoreSelection (see page 1373)() function.
	SelectMode (see page 1371)	The function sets a filter on the table, described by the indicated mode selection. .
	SelectModeFromStore (see page 1371)	The function sets a filter on the table, defined by means of mode selection formerly saved in the table by means of the StoreModeSelection (see page 1372)(). .
	SetColWidth (see page 1372)	Function sets a width of the column with the specified index.
	SetRowHeight (see page 1372)	Function sets a height of the row with the specified index.
	StoreModeSelection (see page 1372)	The function saves the indicated mode selection to the table. .
	StoreSelection (see page 1373)	The function saves the indicated selection in the table without modifying its filter. One may save one selection of each type of objects - structure components - in the table. .

I.1.6.3.1 AddColumn

C++

```
HRESULT AddColumn(IRobotTableDataType _data_type);
```

C#

```
public void AddColumn(IRobotTableDataType _data_type);
```

Visual Basic

```
Public Sub AddColumn(_data_type As IRobotTableDataType)
```

Description

The function adds to a table a new column that will present data of the indicated type .

(c) 2014 Autodesk Inc. All rights reserved

I.1.6.3.2 GetColWidth

C++

```
HRESULT GetColWidth(long _col_idx, long* ret);
```

C#

```
public long GetColWidth(long _col_idx);
```

Visual Basic

```
Public Function GetColWidth(_col_idx As long) As long
```

Description

Function returns a width of the column with the specified index.

Version

Available since version 8.2.

I.1.6.3.3 GetDataType

C++

```
HRESULT GetDataType(long _col_idx, IRobotTableDataType* ret);
```

C#

```
public IRobotTableDataType GetDataType(long _col_idx);
```

Visual Basic

```
Public Function GetDataType(_col_idx As long) As IRobotTableDataType
```

Description

Function returns a type of data presented in the column with the specified index.

Version

Available since version 8.2.

I.1.6.3.4 GetRowHeight

C++

```
HRESULT GetRowHeight(long _row_idx, long* ret);
```

C#

```
public long GetRowHeight(long _row_idx);
```

Visual Basic

```
Public Function GetRowHeight(_row_idx As long) As long
```

Description

Function returns a height of the row with the specified index.

Version

Available since version 8.2.

(c) 2014 Autodesk Inc. All rights reserved

I.1.6.3.5 MakeScreenCapture

C++

```
HRESULT MakeScreenCapture(IRobotTableScreenCaptureParams* _sc_params);
```

C#

```
public void MakeScreenCapture(IRobotTableScreenCaptureParams _sc_params);
```

Visual Basic

```
Public Sub MakeScreenCapture(ByRef _sc_params As IRobotTableScreenCaptureParams)
```

Description

Function saves a screen capture with the specified parameters.

Version

Available since version 8.2.

I.1.6.3.6 Select

C++

```
HRESULT Select(IRobotObjectType _sel_type, BSTR _sel_text);
```

C#

```
public void Select(IRobotObjectType _sel_type, String _sel_text);
```

Visual Basic

```
Public Sub Select(_sel_type As IRobotObjectType, _sel_text As String)
```

Description

The function sets a filter defined by the indicated selection on the table .

I.1.6.3.7 SelectAllFromStore

C++

```
HRESULT SelectAllFromStore();
```

C#

```
public void SelectAllFromStore();
```

Visual Basic

```
Public Sub SelectAllFromStore()
```

Description

The function sets a filter on the table, defined by means of all selections formerly saved in the table by means of the StoreSelection (see page 1373)() function. .

I.1.6.3.8 SelectFromStore

C++

```
HRESULT SelectFromStore(IRobotObjectType _sel_type);
```

C#

```
public void SelectFromStore(IRobotObjectType _sel_type);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Sub SelectFromStore(_sel_type As IRobotObjectType)
```

Description

The function sets a filter on the table. The filter is defined by means of the selection saved earlier by the StoreSelection (see page 1373)() function.

I.1.6.3.9 SelectMode

C++

```
HRESULT SelectMode(IRobotModeSelectionType _type, BSTR _sel_text, IRobotModeCombinationType
_comb);
```

C#

```
public void SelectMode(IRobotModeSelectionType _type, String _sel_text,
IRobotModeCombinationType _comb);
```

Visual Basic

```
Public Sub SelectMode(_type As IRobotModeSelectionType, _sel_text As String, _comb As
IRobotModeCombinationType)
```

Description

The function sets a filter on the table, described by the indicated mode selection. .

I.1.6.3.10 SelectModeFromStore

C++

```
HRESULT SelectModeFromStore();
```

C#

```
public void SelectModeFromStore();
```

Visual Basic

```
Public Sub SelectModeFromStore()
```

Description

The function sets a filter on the table, defined by means of mode selection formerly saved in the table by means of the StoreModeSelection (see page 1372)(). .

I.1.6.3.11 SetColWidth

C++

```
HRESULT SetColWidth(long _col_idx, long _col_width);
```

C#

```
public void SetColWidth(long _col_idx, long _col_width);
```

Visual Basic

```
Public Sub SetColWidth(_col_idx As long, _col_width As long)
```

Description

Function sets a width of the column with the specified index.

Version

Available since version 8.2.

(c) 2014 Autodesk Inc. All rights reserved

I.1.6.3.12 SetRowHeight

C++

```
HRESULT SetRowHeight(long _row_idx, long _row_height);
```

C#

```
public void SetRowHeight(long _row_idx, long _row_height);
```

Visual Basic

```
Public Sub SetRowHeight(_row_idx As long, _row_height As long)
```

Description

Function sets a height of the row with the specified index.

Version

Available since version 8.2.

I.1.6.3.13 StoreModeSelection

C++

```
HRESULT StoreModeSelection(IRobotModeSelectionType _type, BSTR _sel_text,
IRobotModeCombinationType _comb);
```

C#

```
public void StoreModeSelection(IRobotModeSelectionType _type, String _sel_text,
IRobotModeCombinationType _comb);
```

Visual Basic

```
Public Sub StoreModeSelection(_type As IRobotModeSelectionType, _sel_text As String, _comb
As IRobotModeCombinationType)
```

Description

The function saves the indicated mode selection to the table. .

I.1.6.3.14 StoreSelection

C++

```
HRESULT StoreSelection(IRobotObjectType _sel_type, BSTR _sel_txt);
```

C#

```
public void StoreSelection(IRobotObjectType _sel_type, String _sel_txt);
```

Visual Basic

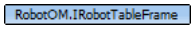
```
Public Sub StoreSelection(_sel_type As IRobotObjectType, _sel_txt As String)
```

Description

The function saves the indicated selection in the table without modifying its filter. One may save one selection of each type of objects - structure components - in the table. .

I.1.7 IRobotTableFrame

Class Hierarchy



(c) 2014 Autodesk Inc. All rights reserved

C++

```
interface IRobotTableFrame : IDispatch;
```

C#

```
public interface IRobotTableFrame;
```

Visual Basic

```
Public Interface IRobotTableFrame
```

Description

Interface representing the viewer composed of many tables. Each table is a separate tab defined within the same main frame.

Version

Available since version 3.

I.1.7.1 IRobotTableFrame Members

The following tables list the members exposed by IRobotTableFrame.

Public Fields

	Name	Description
	Count (see page 1374)	Number of tables defined in the same main frame.
	Current (see page 1374)	Index of active table.
	Window (see page 1374)	

Public Methods

	Name	Description
	Get (see page 1375)	Function returns the table of the specified index.
	GetName (see page 1375)	Function returns the name of the tab on which the table of the indicated index is displayed. .
	SetName (see page 1375)	Function sets the name for the tab which presents the table of the indicated index.

I.1.7.2 IRobotTableFrame Fields

The fields of the IRobotTableFrame class are listed here.

Public Fields

	Name	Description
	Count (see page 1374)	Number of tables defined in the same main frame.
	Current (see page 1374)	Index of active table.
	Window (see page 1374)	

I.1.7.2.1 Count

C++

```
HRESULT get_Count(long*);
```

C#

```
public long Count { get; }
```

Visual Basic

```
Public ReadOnly Count As long
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Number of tables defined in the same main frame.

Version

Available since version 3.

I.1.7.2.2 Current

C++

```
HRESULT get_Current(long*);
HRESULT put_Current(long);
```

C#

```
public long Current { get; set; }
```

Visual Basic

```
Public Current As long
```

Description

Index of active table.

Version

Available since version 3.

I.1.7.2.3 Window

C++

```
HRESULT get_Window(IRobotWindow**);
```

C#

```
public IRobotWindow Window { get; }
```

Visual Basic

```
Public ReadOnly Window As IRobotWindow
```

Version

Available since version 3.

I.1.7.3 IRobotTableFrame Methods

The methods of the IRobotTableFrame class are listed here.

Public Methods

	Name	Description
	Get (see page 1375)	Function returns the table of the specified index.
	GetName (see page 1375)	Function returns the name of the tab on which the table of the indicated index is displayed. .
	SetName (see page 1375)	Function sets the name for the tab which presents the table of the indicated index.

I.1.7.3.1 Get

C++

```
HRESULT Get(long _idx, IRobotTable** ret);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRobotTable Get(long _idx);
```

Visual Basic

```
Public Function Get(_idx As long) As IRobotTable
```

Description

Function returns the table of the specified index.

Version

Available since version 3.

I.1.7.3.2 GetName

C++

```
HRESULT GetName(long _table_idx, BSTR* ret);
```

C#

```
public String GetName(long _table_idx);
```

Visual Basic

```
Public Function GetName(_table_idx As long) As String
```

Description

Function returns the name of the tab on which the table of the indicated index is displayed. .

Version

Available since version 3.

I.1.7.3.3 SetName

C++

```
HRESULT SetName(long _table_idx, BSTR _name);
```

C#

```
public void SetName(long _table_idx, String _name);
```

Visual Basic

```
Public Sub SetName(_table_idx As long, _name As String)
```

Description

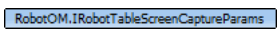
Function sets the name for the tab which presents the table of the indicated index.

Version

Available since version 3.

I.1.8 IRobotTableScreenCaptureParams

Class Hierarchy



C++

```
interface IRobotTableScreenCaptureParams : IDispatch;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public interface IRobotTableScreenCaptureParams;
```

Visual Basic

```
Public Interface IRobotTableScreenCaptureParams
```

Description

Parameters of the screen capture for a table.

Version

Available since version 8.2.

I.1.8.1 IRobotTableScreenCaptureParams Members

The following tables list the members exposed by IRobotTableScreenCaptureParams.

Public Fields

	Name	Description
	AddInfoTab (see page 1377)	
	AdjustColumnWidth (see page 1377)	The flag indicating if width of the table columns should be adjusted automatically to width of the html page.
	Comment (see page 1377)	Comment text to be added to the screen capture.
	IncludeDateAndTime (see page 1378)	The flag indicating if the current date and time should be added automatically to the screen capture.
	Name (see page 1378)	
	UpdateType (see page 1378)	

I.1.8.2 IRobotTableScreenCaptureParams Fields

The fields of the IRobotTableScreenCaptureParams class are listed here.

Public Fields

	Name	Description
	AddInfoTab (see page 1377)	
	AdjustColumnWidth (see page 1377)	The flag indicating if width of the table columns should be adjusted automatically to width of the html page.
	Comment (see page 1377)	Comment text to be added to the screen capture.
	IncludeDateAndTime (see page 1378)	The flag indicating if the current date and time should be added automatically to the screen capture.
	Name (see page 1378)	
	UpdateType (see page 1378)	

I.1.8.2.1 AddInfoTab

C++

```
HRESULT get_AddInfoTab(VARIANT_BOOL*);
HRESULT put_AddInfoTab(VARIANT_BOOL);
```

C#

```
public bool AddInfoTab { get; set; }
```

Visual Basic

```
Public AddInfoTab As Boolean
```

Version

Available since version 8.2.

(c) 2014 Autodesk Inc. All rights reserved

I.1.8.2.2 AdjustColumnWidth

C++

```
HRESULT get_AdjustColumnWidth(VARIANT_BOOL*);
HRESULT put_AdjustColumnWidth(VARIANT_BOOL);
```

C#

```
public bool AdjustColumnWidth { get; set; }
```

Visual Basic

```
Public AdjustColumnWidth As Boolean
```

Description

The flag indicating if width of the table columns should be adjusted automatically to width of the html page.

Version

Available since version 12.

I.1.8.2.3 Comment

C++

```
HRESULT get_Comment(BSTR*);
HRESULT put_Comment(BSTR);
```

C#

```
public String Comment { get; set; }
```

Visual Basic

```
Public Comment As String
```

Description

Comment text to be added to the screen capture.

Version

Available since version 12.

I.1.8.2.4 IncludeDateAndTime

C++

```
HRESULT get_IncludeDateAndTime(VARIANT_BOOL*);
HRESULT put_IncludeDateAndTime(VARIANT_BOOL);
```

C#

```
public bool IncludeDateAndTime { get; set; }
```

Visual Basic

```
Public IncludeDateAndTime As Boolean
```

Description

The flag indicating if the current date and time should be added automatically to the screen capture.

Version

Available since version 12.

(c) 2014 Autodesk Inc. All rights reserved

I.1.8.2.5 Name

C++

```
HRESULT get_Name(BSTR*);
HRESULT put_Name(BSTR);
```

C#

```
public String Name { get; set; }
```

Visual Basic

```
Public Name As String
```

Version

Available since version 8.2.

I.1.8.2.6 UpdateType

C++

```
HRESULT get_UpdateType(IRobotTableScreenCaptureUpdateType*);
HRESULT put_UpdateType(IRobotTableScreenCaptureUpdateType);
```

C#

```
public IRobotTableScreenCaptureUpdateType UpdateType { get; set; }
```

Visual Basic

```
Public UpdateType As IRobotTableScreenCaptureUpdateType
```

Version

Available since version 8.2.

I.1.9 IRobotTableScreenCaptureUpdateType

C++

```
enum IRobotTableScreenCaptureUpdateType;
```

C#

```
public enum IRobotTableScreenCaptureUpdateType;
```

Visual Basic

```
Public Enum IRobotTableScreenCaptureUpdateType
```

Members

Members	Description
I_TSCUT_UPDATED_UPON_PRINTING = 1	Available since version 8.2.
I_TSCUT_CURRENT_VIEW = 2	Available since version 8.2.
I_TSCUT_COPY_TO_CLIPBOARD = 4	The screen capture will be saved only to the clipboard. Available since version 12.

Version

Available since version 8.2.

(c) 2014 Autodesk Inc. All rights reserved

I.2 Graphical structure views

Enumerations

	Name	Description
	IRobotViewType (see page 1395)	Set of available types of graphical views of the structure that can be created by means of the view manager. .
	IRobotViewProjection (see page 1405)	The set of identifiers determining different types of projections to be presented in graphical views has been defined. .
	IRobotViewVisibilityStatusType (see page 1406)	Available types of visibility status for a graphical viewer.
	IRobotViewVisibilityStatusValue (see page 1407)	Accessible values for particular types of visibility status.
	IRobotViewScaleType (see page 1412)	Available types of a scale defined for a graphical viewer.
	IRobotViewDetailedAnalysisTableTab (see page 1415)	Identifiers of the tabs in the table being a part of the view presenting detailed analysis results. Available since version 2.0.
	IRobotViewDisplayAttributes (see page 1418)	Set of attributes whose visibility may be switched on/off for graphical views. .
	IRobotViewHiddenLinesDisplayType (see page 1421)	Possible types of displaying hidden lines.
	IRobotViewDiagramResultType (see page 1426)	Set of result types presented by means of diagrams on bars.
	IRobotViewDiagramDescriptionType (see page 1427)	Available methods of locating descriptions for diagrams.
	IRobotViewDiagramFillingType (see page 1428)	Available methods of diagram filling.
	IRobotViewDiagramSignDifferType (see page 1428)	Available methods of distinguishing positive and negative values on diagrams.
	IRobotViewBarMapResultType (see page 1430)	Set of result types presented by means of maps on bars.
	IRobotViewDetailedAnalysisResultType (see page 1432)	Result types presented by detailed analysis view. .
	IRobotViewFeMapResultType (see page 1436)	Result types presented by means of maps on finite elements. .
	IRobotViewFeMapLocalSystemType (see page 1443)	
	IRobotViewFeMapCrossPresentationType (see page 1444)	Manner of presenting crosses in the map view for surface finite elements. .
	IRobotViewFeMapLayerType (see page 1444)	
	IRobotViewFeMapSmoothingType (see page 1444)	
	IRobotViewDiagramPositionType (see page 1447)	Available diagram positions with respect to the panel plane.
	IRobotViewReinforcementResultType (see page 1448)	Available reinforcement results presented in the garphical view. .
	IRobotViewGlobalAnalysisResultsType (see page 1452)	Types of result display.
	IRobotViewGlobalAnalysisParamsType (see page 1453)	Parameter types.
	IRobotViewScreenCaptureUpdateType (see page 1459)	Available methods of updating the contents of a screen capture for the graphical view.
	IRobotViewDiagramValueType (see page 1461)	Types of values displayed on diagrams.

(c) 2014 Autodesk Inc. All rights reserved

	IRobotViewScreenCaptureResolution (see page 1462)	

Interfaces

	Name	Description
	IRobotView (see page 1396)	Interface representing the graphical structure view. .
	IRobotViewDetailedAnalysis (see page 1412)	View presenting results of the detailed analysis. Available since version 2.0.
	IRobotViewDisplayParams (see page 1416)	Parameters of display for graphical viewers.
	IRobotViewDiagramParams (see page 1422)	Interface controlling the manner of displaying diagrams on bars. .
	IRobotViewDiagrams (see page 1429)	A viewer dedicated to result presentation.
	IRobotViewBarMaps (see page 1429)	View presenting maps on bars with a scale. .
	IRobotViewFeMaps (see page 1429)	View presenting maps on finite elements with a scale. .
	IRobotViewBarMapParams (see page 1430)	Parameters describing the manner of displaying maps on bars.
	IRobotViewDetailedAnalysisParams (see page 1433)	Parameters of detailed analysis view.
	IRobotViewFeMapParams (see page 1438)	Parameters of presenting maps on finite elements. .
	IRobotView2 (see page 1445)	Extended graphical view interface. .
	IRobotViewGlobalAnalysisParams (see page 1449)	Interface controlling display of global analysis results. .
	IRobotViewGlobalAnalysisResultsParams (see page 1450)	Interface controlling result display.
	IRobotViewGlobalAnalysis (see page 1453)	View presenting global analysis results. .
	IRobotViewScreenCaptureParams (see page 1456)	Parameters of a screen capture for the graphical view.
	IRobotView3 (see page 1459)	

I.2.1 Panel cuts

Available since version 3.

Interfaces

	Name	Description
	IRobotViewPanelCuts (see page 1381)	View presenting cuts through panels.
	IRobotViewPanelCutParams (see page 1381)	Parameters navigating panel cut display.

I.2.1.1 IRobotViewPanelCuts

Class Hierarchy

C++

```
interface IRobotViewPanelCuts : IRobotView3;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public interface IRobotViewPanelCuts : IRobotView3;
```

Visual Basic

```
Public Interface IRobotViewPanelCuts
```

Description

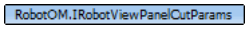
View presenting cuts through panels.

Version

Available since version 3.

I.2.1.2 IRobotViewPanelCutParams

Class Hierarchy



C++

```
interface IRobotViewPanelCutParams : IDispatch;
```

C#

```
public interface IRobotViewPanelCutParams;
```

Visual Basic

```
Public Interface IRobotViewPanelCutParams
```

Description

Parameters navigating panel cut display.

Version

Available since version 3.

I.2.1.2.1 IRobotViewPanelCutParams Members

The following tables list the members exposed by IRobotViewPanelCutParams.

Public Fields

	Name	Description
	CurrentReinforcementResult (see page 1382)	Currently displayed result type for reinforcement.
	CurrentResult (see page 1383)	Currently displayed result type.
	Descritpions (see page 1383)	Method of description display.
	Filling (see page 1383)	Method of diagram filling.
	IntegralValue (see page 1384)	Activation of a label containing the value of integral for the selected component along the cutting line length.
	Layer (see page 1384)	Layer.
	LayerArbritraryValue (see page 1384)	Layer (see page 1384) thickness.
	Position (see page 1385)	Type of diagram position with respect to the panel plane.
	Smoothing (see page 1385)	Type of diagram smoothing.

I.2.1.2.2 IRobotViewPanelCutParams Fields

The fields of the IRobotViewPanelCutParams class are listed here.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	CurrentReinforcementResult (see page 1382)	Currently displayed result type for reinforcement.
	CurrentResult (see page 1383)	Currently displayed result type.
	Descritpions (see page 1383)	Method of description display.
	Filling (see page 1383)	Method of diagram filling.
	IntegralValue (see page 1384)	Activation of a label containing the value of integral for the selected component along the cutting line length.
	Layer (see page 1384)	Layer.
	LayerArbritraryValue (see page 1384)	Layer (see page 1384) thickness.
	Position (see page 1385)	Type of diagram position with respect to the panel plane.
	Smoothing (see page 1385)	Type of diagram smoothing.

I.2.1.2.2.1 CurrentReinforcementResult

C++

```
HRESULT get_CurrentReinforcementResult(IRobotViewReinforcementResultType*);
HRESULT put_CurrentReinforcementResult(IRobotViewReinforcementResultType);
```

C#

```
public IRobotViewReinforcementResultType CurrentReinforcementResult { get; set; }
```

Visual Basic

```
Public CurrentReinforcementResult As IRobotViewReinforcementResultType
```

Description

Currently displayed result type for reinforcement.

Version

Available since version 3.

I.2.1.2.2.2 CurrentResult

C++

```
HRESULT get_CurrentResult(IRobotViewFeMapResultType*);
HRESULT put_CurrentResult(IRobotViewFeMapResultType);
```

C#

```
public IRobotViewFeMapResultType CurrentResult { get; set; }
```

Visual Basic

```
Public CurrentResult As IRobotViewFeMapResultType
```

Description

Currently displayed result type.

Version

Available since version 3.

I.2.1.2.2.3 Descritpions

C++

```
HRESULT get_Descritpions(IRobotViewDiagramDescriptionType*);
HRESULT put_Descritpions(IRobotViewDiagramDescriptionType);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRobotViewDiagramDescriptionType Descritpions { get; set; }
```

Visual Basic

```
Public Descritpions As IRobotViewDiagramDescriptionType
```

Description

Method of description display.

Version

Available since version 3.

I.2.1.2.2.4 Filling

C++

```
HRESULT get_Filling(IRobotViewDiagramFillingType*);
HRESULT put_Filling(IRobotViewDiagramFillingType);
```

C#

```
public IRobotViewDiagramFillingType Filling { get; set; }
```

Visual Basic

```
Public Filling As IRobotViewDiagramFillingType
```

Description

Method of diagram filling.

Version

Available since version 3.

I.2.1.2.2.5 IntegralValue

C++

```
HRESULT get_IntegralValue(VARIANT_BOOL*);
HRESULT put_IntegralValue(VARIANT_BOOL);
```

C#

```
public bool IntegralValue { get; set; }
```

Visual Basic

```
Public IntegralValue As Boolean
```

Description

Activation of a label containing the value of integral for the selected component along the cutting line length.

Version

Available since version 3.

I.2.1.2.2.6 Layer

C++

```
HRESULT get_Layer(IRobotViewFeMapLayerType*);
HRESULT put_Layer(IRobotViewFeMapLayerType);
```

C#

```
public IRobotViewFeMapLayerType Layer { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Layer As IRobotViewFeMapLayerType
```

Description

Layer.

Version

Available since version 3.

I.2.1.2.2.7 LayerArbritraryValue

C++

```
HRESULT get_LayerArbritraryValue(double*);
HRESULT put_LayerArbritraryValue(double);
```

C#

```
public double LayerArbritraryValue { get; set; }
```

Visual Basic

```
Public LayerArbritraryValue As double
```

Description

Layer (see page 1384) thickness.

Version

Available since version 3.

I.2.1.2.2.8 Position

C++

```
HRESULT get_Position(IRobotViewDiagramPositionType*);
HRESULT put_Position(IRobotViewDiagramPositionType);
```

C#

```
public IRobotViewDiagramPositionType Position { get; set; }
```

Visual Basic

```
Public Position As IRobotViewDiagramPositionType
```

Description

Type of diagram position with respect to the panel plane.

Version

Available since version 3.

I.2.1.2.2.9 Smoothing

C++

```
HRESULT get_Smoothing(IRobotViewFeMapSmoothingType*);
HRESULT put_Smoothing(IRobotViewFeMapSmoothingType);
```

C#

```
public IRobotViewFeMapSmoothingType Smoothing { get; set; }
```

Visual Basic

```
Public Smoothing As IRobotViewFeMapSmoothingType
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Type of diagram smoothing.

Version

Available since version 3.

I.2.2 Influence lines

Available since version 3.5.

Enumerations

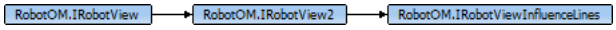
	Name	Description
	IRobotViewInfluenceLinesResultType (see page 1393)	Available result types for the influence line view. .
	IRobotViewInfluenceLinesLocalSystemType (see page 1394)	Available types of local coordinate systems. .
	IRobotViewInfluenceLinesLayerType (see page 1395)	Types of layers.

Interfaces

	Name	Description
	IRobotViewInfluenceLines (see page 1386)	Graphical view presenting an influence line.
	IRobotViewInfluenceLinesParams (see page 1388)	Parameters of an influence line view. .

I.2.2.1 IRobotViewInfluenceLines

Class Hierarchy



C++

```
interface IRobotViewInfluenceLines : IRobotView2;
```

C#

```
public interface IRobotViewInfluenceLines : IRobotView2;
```

Visual Basic

```
Public Interface IRobotViewInfluenceLines
```

Description

Graphical view presenting an influence line.

Version

Available since version 3.5.

I.2.2.1.1 IRobotViewInfluenceLines Members

The following tables list the members exposed by IRobotViewInfluenceLines.

Public Fields

	Name	Description
	DiagramMagnification (see page 1398)	Diagram enlarging (scale: 1 to 10).
	ParamsBarMap (see page 1398)	Parameters of displaying maps an bars Available since version 2.0.

(c) 2014 Autodesk Inc. All rights reserved

	ParamsDiagram (see page 1398)	Parameters of displaying diagrams on the view Available since version 2.0.
	ParamsDisplay (see page 1398)	Display parameters (a component enabling control of individual attribute display) Available since version 2.0.
	ParamsFeMap (see page 1399)	Display parameters of maps presenting results for surface finite elements Available since version 2.0.
	ParamsInfluenceLines (see page 1387)	View parameters. Available since version 3.5.
	ParamsPanelCut (see page 1446)	Parameters of displaying panel cuts.
	Printable (see page 1399)	Object representing the view that may be attached to the printout and printed .
	Projection (see page 1399)	Current projection .
	Selection (see page 1399)	Selection of structure components presented in the view .
	Title (see page 1400)	View title .
	Type (see page 1400)	View type .
	UserControl (see page 1447)	Flag that enables giving/taking back graphical view navigation to/from the interactive program user .
	Visible (see page 1400)	View display flag .
	Window (see page 1447)	

Public Methods

	Name	Description
	CopyToClipboard (see page 1401)	The function inserts the view into the clipboard.
	GetRotationPoint (see page 1401)	The function gives back a rotation center during a view rotation.
	GetScale (see page 1402)	The function gives a value of the given scale back. A set of available scales describes RobotViewScaleType type.
	GetSize (see page 1402)	The function returns the size of the view in mm. .
	GetWorkPoint (see page 1402)	The function gives a work point back (on the XY plane it gives Z level) .
	GetZoom (see page 1402)	The function returns the current zoom - coordinates of view sides in the current coordinate system .
	IsLocal (see page 1403)	The function gives the status of a local coordinate system back.
	MakeScreenCapture (see page 1388)	
	Redraw (see page 1403)	The function draws the view. .
	Rotate (see page 1403)	The function rotates structure view with respect to the indicated global coordinate axis, with the rotation point recognized.
	SetGlobal (see page 1403)	The function sets a global coordinate system.
	SetLocal (see page 1404)	The function sets a local coordinate system. If the given points do not describe a local plane in the unambiguous way the function gives a zero value (False) back.
	SetRotationPoint (see page 1404)	The funcion sets the rotation center during the view rotation.
	SetScale (see page 1404)	The function assigns a given value to the scale.
	SetSize (see page 1404)	The function sets the view size in mm .
	SetWorkPoint (see page 1405)	The function sets a working point (for the XY plane Z level is set). .
	SetZoom (see page 1405)	The function sets the current zoom - coordinates of view sides in the current coordinate system .

I.2.2.1.2 IRobotViewInfluenceLines Fields

The fields of the IRobotViewInfluenceLines class are listed here.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	ParamsInfluenceLines (see page 1387)	View parameters. Available since version 3.5.

I.2.2.1.2.1 ParamsInfluenceLines

C#

```
void ParamsInfluenceLines;
```

Description

View parameters.

Available since version 3.5.

I.2.2.1.3 IRobotViewInfluenceLines Methods

The methods of the IRobotViewInfluenceLines class are listed here.

Public Methods

	Name	Description
	MakeScreenCapture (see page 1388)	

I.2.2.1.3.1 MakeScreenCapture

C++

```
HRESULT MakeScreenCapture(IRobotViewScreenCaptureParams* _sc_params);
```

C#

```
public void MakeScreenCapture(IRobotViewScreenCaptureParams _sc_params);
```

Visual Basic

```
Public Sub MakeScreenCapture(ByRef _sc_params As IRobotViewScreenCaptureParams)
```

I.2.2.2 IRobotViewInfluenceLinesParams

Class Hierarchy

RobotOM.IRobotViewInfluenceLinesParams

C++

```
interface IRobotViewInfluenceLinesParams : IDispatch;
```

C#

```
public interface IRobotViewInfluenceLinesParams;
```

Visual Basic

```
Public Interface IRobotViewInfluenceLinesParams
```

Description

Parameters of an influence line view. .

Version

Available since version 3.5.

(c) 2014 Autodesk Inc. All rights reserved

I.2.2.2.1 IRobotViewInfluenceLinesParams Members

The following tables list the members exposed by IRobotViewInfluenceLinesParams.

Public Fields

	Name	Description
	Direction (see page 1389)	Type of the local reference system.
	DirectionData (see page 1389)	Vector defining the local reference system; used in appropriate system types (CARTESIAN_ALONG_VECTOR and POLAR_IN_POINT).
	DirectionNode (see page 1390)	Node number for definition of the POLAR_IN_POINT local reference system .
	Element (see page 1390)	Number of element/bar for which an influence line will be generated .
	Layer (see page 1390)	Layer type.
	LayerArbitraryVal (see page 1391)	Value of the ARBITRARY type layer.
	Position (see page 1391)	Relative coordinate of the point of element/bar for which an influence line will be generated .
	RangeFrom (see page 1391)	Initial number of the moving load position with which generation of an influence line will start .
	RangeTo (see page 1392)	Final number of the moving load position (associated with RangeFrom (see page 1391)).

Public Methods

	Name	Description
	IsOn (see page 1392)	Function checks if a given result type is currently displayed. .
	Set (see page 1392)	Function manages display of results of the specified type. .

I.2.2.2.2 IRobotViewInfluenceLinesParams Fields

The fields of the IRobotViewInfluenceLinesParams class are listed here.

Public Fields

	Name	Description
	Direction (see page 1389)	Type of the local reference system.
	DirectionData (see page 1389)	Vector defining the local reference system; used in appropriate system types (CARTESIAN_ALONG_VECTOR and POLAR_IN_POINT).
	DirectionNode (see page 1390)	Node number for definition of the POLAR_IN_POINT local reference system .
	Element (see page 1390)	Number of element/bar for which an influence line will be generated .
	Layer (see page 1390)	Layer type.
	LayerArbitraryVal (see page 1391)	Value of the ARBITRARY type layer.
	Position (see page 1391)	Relative coordinate of the point of element/bar for which an influence line will be generated .
	RangeFrom (see page 1391)	Initial number of the moving load position with which generation of an influence line will start .
	RangeTo (see page 1392)	Final number of the moving load position (associated with RangeFrom (see page 1391)).

I.2.2.2.2.1 Direction

C++

```
HRESULT get_Direction(IRobotViewInfluenceLinesLocalSystemType*);
HRESULT put_Direction(IRobotViewInfluenceLinesLocalSystemType);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRobotViewInfluenceLinesLocalSystemType Direction { get; set; }
```

Visual Basic

```
Public Direction As IRobotViewInfluenceLinesLocalSystemType
```

Description

Type of the local reference system.

Version

Available since version 3.5.

I.2.2.2.2.2 DirectionData

C++

```
HRESULT get_DirectionData(IRobotGeoPoint3D**);
HRESULT put_DirectionData(IRobotGeoPoint3D*);
```

C#

```
public IRobotGeoPoint3D DirectionData { get; set; }
```

Visual Basic

```
Public DirectionData As IRobotGeoPoint3D
```

Description

Vector defining the local reference system; used in appropriate system types (CARTESIAN_ALONG_VECTOR and POLAR_IN_POINT).

Version

Available since version 3.5.

I.2.2.2.2.3 DirectionNode

C++

```
HRESULT get_DirectionNode(long*);
HRESULT put_DirectionNode(long);
```

C#

```
public long DirectionNode { get; set; }
```

Visual Basic

```
Public DirectionNode As long
```

Description

Node number for definition of the POLAR_IN_POINT local reference system .

Version

Available since version 3.5.

I.2.2.2.2.4 Element

C++

```
HRESULT get_Element(long*);
HRESULT put_Element(long);
```

C#

```
public long Element { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Element As long
```

Description

Number of element/bar for which an influence line will be generated .

Version

Available since version 3.5.

I.2.2.2.2.5 Layer

C++

```
HRESULT get_Layer(IRobotViewInfluenceLinesLayerType*);
HRESULT put_Layer(IRobotViewInfluenceLinesLayerType);
```

C#

```
public IRobotViewInfluenceLinesLayerType Layer { get; set; }
```

Visual Basic

```
Public Layer As IRobotViewInfluenceLinesLayerType
```

Description

Layer type.

Version

Available since version 3.5.

I.2.2.2.2.6 LayerArbitraryVal

C++

```
HRESULT get_LayerArbitraryVal(double*);
HRESULT put_LayerArbitraryVal(double);
```

C#

```
public double LayerArbitraryVal { get; set; }
```

Visual Basic

```
Public LayerArbitraryVal As double
```

Description

Value of the ARBITRARY type layer.

Version

Available since version 3.5.

I.2.2.2.2.7 Position

C++

```
HRESULT get_Position(double*);
HRESULT put_Position(double);
```

C#

```
public double Position { get; set; }
```

Visual Basic

```
Public Position As double
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Relative coordinate of the point of element/bar for which an influence line will be generated .

Version

Available since version 3.5.

I.2.2.2.2.8 RangeFrom

C++

```
HRESULT get_RangeFrom(long*);
HRESULT put_RangeFrom(long);
```

C#

```
public long RangeFrom { get; set; }
```

Visual Basic

```
Public RangeFrom As long
```

Description

Initial number of the moving load position with which generation of an influence line will start .

Version

Available since version 3.5.

I.2.2.2.2.9 RangeTo

C++

```
HRESULT get_RangeTo(long*);
HRESULT put_RangeTo(long);
```

C#

```
public long RangeTo { get; set; }
```

Visual Basic

```
Public RangeTo As long
```

Description

Final number of the moving load position (associated with RangeFrom (see page 1391)).

Version

Available since version 3.5.

I.2.2.2.3 IRobotViewInfluenceLinesParams Methods

The methods of the IRobotViewInfluenceLinesParams class are listed here.

Public Methods

	Name	Description
	IsOn (see page 1392)	Function checks if a given result type is currently displayed. .
	Set (see page 1392)	Function manages display of results of the specified type. .

I.2.2.2.3.1 IsOn

C++

```
HRESULT IsOn(IRobotViewInfluenceLinesResultTypes _type, VARIANT_BOOL* ret);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public bool IsOn(IRobotViewInfluenceLinesResultTypes _type);
```

Visual Basic

```
Public Function IsOn(_type As IRobotViewInfluenceLinesResultTypes) As Boolean
```

Description

Function checks if a given result type is currently displayed. .

Version

Available since version 3.5.

I.2.2.2.3.2 Set

C++

```
HRESULT Set(IRobotViewInfluenceLinesResultTypes _type, VARIANT_BOOL _val);
```

C#

```
public void Set(IRobotViewInfluenceLinesResultTypes _type, bool _val);
```

Visual Basic

```
Public Sub Set(_type As IRobotViewInfluenceLinesResultTypes, _val As Boolean)
```

Description

Function manages display of results of the specified type. .

Version

Available since version 3.5.

I.2.2.3 IRobotViewInfluenceLinesResultType

C++

```
enum IRobotViewInfluenceLinesResultType;
```

C#

```
public enum IRobotViewInfluenceLinesResultType;
```

Visual Basic

```
Public Enum IRobotViewInfluenceLinesResultType
```

Members

Members	Description
I_VILRT_NTM_FX = 1	Available since version 3.5.
I_VILRT_NTM_FY = 2	Available since version 3.5.
I_VILRT_NTM_FZ = 3	Available since version 3.5.
I_VILRT_NTM_MX = 4	Available since version 3.5.
I_VILRT_NTM_MY = 5	Available since version 3.5.
I_VILRT_NTM_MZ = 6	Available since version 3.5.
I_VILRT_NTM_KY = 7	Available since version 3.5.
I_VILRT_NTM_KZ = 8	Available since version 3.5.
I_VILRT_NTM_UX = 9	Available since version 3.5.
I_VILRT_NTM_UZ = 10	Available since version 3.5.
I_VILRT_NODES_UX = 11	Available since version 3.5.

(c) 2014 Autodesk Inc. All rights reserved

I_VILRT_NODES_UY = 12	Available since version 3.5.
I_VILRT_NODES_UZ = 13	Available since version 3.5.
I_VILRT_NODES_RX = 14	Available since version 3.5.
I_VILRT_NODES_RY = 15	Available since version 3.5.
I_VILRT_NODES_RZ = 16	Available since version 3.5.
I_VILRT_NODES_FX = 17	Available since version 3.5.
I_VILRT_NODES_FY = 18	Available since version 3.5.
I_VILRT_NODES_FZ = 19	Available since version 3.5.
I_VILRT_NODES_MX = 20	Available since version 3.5.
I_VILRT_NODES_MY = 21	Available since version 3.5.
I_VILRT_NODES_MZ = 22	Available since version 3.5.
I_VILRT_DETAILED_STRESS_XX = 23	Available since version 3.5.
I_VILRT_DETAILED_STRESS_YY = 24	Available since version 3.5.
I_VILRT_DETAILED_STRESS_XY = 25	Available since version 3.5.
I_VILRT_DETAILED_STRESS_Z = 26	Available since version 3.5.
I_VILRT_DETAILED_MEMBRANE_FORCE_XX = 27	Available since version 3.5.
I_VILRT_DETAILED_MEMBRANE_FORCE_YY = 28	Available since version 3.5.
I_VILRT_DETAILED_MEMBRANE_FORCE_XY = 29	Available since version 3.5.
I_VILRT_DETAILED_MOMENTS_XX = 30	Available since version 3.5.
I_VILRT_DETAILED_MOMENTS_YY = 31	Available since version 3.5.
I_VILRT_DETAILED_MOMENTS_XY = 32	Available since version 3.5.
I_VILRT_DETAILED_SHEAR_STRESS_YY = 34	Available since version 3.5.
I_VILRT_DETAILED_SHEAR_FORCE_XX = 35	Available since version 3.5.
I_VILRT_DETAILED_SHEAR_FORCE_YY = 36	Available since version 3.5.
I_VILRT_DETAILED_DISPLACEMENT_XX = 37	Available since version 3.5.
I_VILRT_DETAILED_DISPLACEMENT_YY = 38	Available since version 3.5.
I_VILRT_DETAILED_DISPLACEMENT_Z = 39	Available since version 3.5.
I_VILRT_DETAILED_ROTATION_XX = 40	Available since version 3.5.
I_VILRT_DETAILED_ROTATION_YY = 41	Available since version 3.5.
I_VILRT_DETAILED_ROTATION_Z = 42	Available since version 3.5.
I_VILRT_DETAILED_SOIL_REACTION_Z = 43	Available since version 3.5.
I_VILRT_EXTREME_STRESS_1 = 44	Available since version 3.5.
I_VILRT_EXTREME_STRESS_2 = 45	Available since version 3.5.
I_VILRT_EXTREME_STRESS_1_2 = 46	Available since version 3.5.
I_VILRT_EXTREME_STRESS_ANGLE = 47	Available since version 3.5.
I_VILRT_EXTREME_MEMBRANE_FORCES_1 = 48	Available since version 3.5.
I_VILRT_EXTREME_MEMBRANE_FORCES_2 = 49	Available since version 3.5.
I_VILRT_EXTREME_MEMBRANE_FORCES_1_2 = 50	Available since version 3.5.
I_VILRT_EXTREME_MEMBRANE_FORCES_ANGLE = 51	Available since version 3.5.
I_VILRT_EXTREME_MOMENTS_1 = 52	Available since version 3.5.
I_VILRT_EXTREME_MOMENTS_2 = 53	Available since version 3.5.
I_VILRT_EXTREME_MOMENTS_1_2 = 54	Available since version 3.5.
I_VILRT_EXTREME_MOMENTS_ANGLE = 55	Available since version 3.5.
I_VILRT_EXTREME_SHEAR_STRESS_1_2 = 56	Available since version 3.5.
I_VILRT_EXTREME_SHEAR_FORCE_1_2 = 57	Available since version 3.5.
I_VILRT_COMPLEX_STRESS = 58	Available since version 3.5.
I_VILRT_COMPLEX_MEMBRANE_FORCE = 59	Available since version 3.5.
I_VILRT_COMPLEX_MOMENT = 60	Available since version 3.5.

(c) 2014 Autodesk Inc. All rights reserved

Description

Available result types for the influence line view. .

Version

Available since version 3.5.

I.2.2.4 IRobotViewInfluenceLinesLocalSystemType

C++

```
enum IRobotViewInfluenceLinesLocalSystemType;
```

C#

```
public enum IRobotViewInfluenceLinesLocalSystemType;
```

Visual Basic

```
Public Enum IRobotViewInfluenceLinesLocalSystemType
```

Members

Members	Description
I_VILLST_CARTESIAN_ALONG_X = 0	Available since version 3.5.
I_VILLST_CARTESIAN_ALONG_Z = 2	Available since version 3.5.
I_VILLST_CARTESIAN_ALONG_VECTOR = 3	Available since version 3.5.
I_VILLST_POLAR_IN_NODE = 4	Available since version 3.5.
I_VILLST_POLAR_IN_POINT = 5	Available since version 3.5.
I_VILLST_AUTOMATIC = 6	Available since version 3.5.

Description

Available types of local coordinate systems. .

Version

Available since version 3.5.

I.2.2.5 IRobotViewInfluenceLinesLayerType

C++

```
enum IRobotViewInfluenceLinesLayerType;
```

C#

```
public enum IRobotViewInfluenceLinesLayerType;
```

Visual Basic

```
Public Enum IRobotViewInfluenceLinesLayerType
```

Members

Members	Description
I_VILLT_UPPER = 0	Available since version 3.5.
I_VILLT_MIDDLE = 1	Available since version 3.5.
I_VILLT_LOWER = 2	Available since version 3.5.
I_VILLT_ARBITRARY = 3	Available since version 3.5.

Description

Types of layers.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.5.

I.2.3 IRobotViewType

C++

```
enum IRobotViewType;
```

C#

```
public enum IRobotViewType;
```

Visual Basic

```
Public Enum IRobotViewType
```

Members

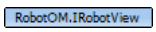
Members	Description
I_VT_STANDARD = 1	Standard structure view .
I_VT_DETAILED_ANALYSIS = 2	A view presenting detailed analysis results Available since version 1.7.
I_VT_DIAGRAMS = 3	Available since version 2.0.
I_VT_MAPS_ON_BARS = 4	Available since version 2.0.
I_VT_MAPS_ON_FINITE_ELEMENTS = 5	Available since version 2.0.
I_VT_PANEL_CUTS = 6	Available since version 3.
I_VT_GLOBAL_ANALYSIS = 7	Available since version 3.
I_VT_INFLUENCE_LINES = 8	Available since version 3.5.

Description

Set of available types of graphical views of the structure that can be created by means of the view manager. .

I.2.4 IRobotView

Class Hierarchy



C++

```
interface IRobotView : IDispatch;
```

C#

```
public interface IRobotView;
```

Visual Basic

```
Public Interface IRobotView
```

Description

Interface representing the graphical structure view. .

I.2.4.1 IRobotView Members

The following tables list the members exposed by IRobotView.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	DiagramMagnification (see page 1398)	Diagram enlarging (scale: 1 to 10).
	ParamsBarMap (see page 1398)	Parameters of displaying maps an bars Available since version 2.0.
	ParamsDiagram (see page 1398)	Parameters of displaying diagrams on the view Available since version 2.0.
	ParamsDisplay (see page 1398)	Display parameters (a component enabling control of individual attribute display) Available since version 2.0.
	ParamsFeMap (see page 1399)	Display parameters of maps presenting results for surface finite elements Available since version 2.0.
	Printable (see page 1399)	Object representing the view that may be attached to the printout and printed .
	Projection (see page 1399)	Current projection .
	Selection (see page 1399)	Selection of structure components presented in the view .
	Title (see page 1400)	View title .
	Type (see page 1400)	View type .
	Visible (see page 1400)	View display flag .

Public Methods

	Name	Description
	CopyToClipboard (see page 1401)	The function inserts the view into the clipboard.
	GetRotationPoint (see page 1401)	The function gives back a rotation center during a view rotation.
	GetScale (see page 1402)	The function gives a value of the given scale back. A set of available scales describes RobotViewScaleType type.
	GetSize (see page 1402)	The function returns the size of the view in mm. .
	GetWorkPoint (see page 1402)	The function gives a work point back (on the XY plane it gives Z level) .
	GetZoom (see page 1402)	The function returns the current zoom - coordinates of view sides in the current coordinate system .
	IsLocal (see page 1403)	The function gives the status of a local coordinate system back.
	Redraw (see page 1403)	The function draws the view. .
	Rotate (see page 1403)	The function rotates structure view with respect to the indicated global coordinate axis, with the rotation point recognized.
	SetGlobal (see page 1403)	The function sets a global coordinate system.
	SetLocal (see page 1404)	The function sets a local coordinate system. If the given points do not describe a local plane in the unambiguous way the function gives a zero value (False) back.
	SetRotationPoint (see page 1404)	The funcion sets the rotation center during the view rotation.
	SetScale (see page 1404)	The function assigns a given value to the scale.
	SetSize (see page 1404)	The function sets the view size in mm .
	SetWorkPoint (see page 1405)	The function sets a working point (for the XY plane Z level is set). .
	SetZoom (see page 1405)	The function sets the current zoom - coordinates of view sides in the current coordinate system .

I.2.4.2 IRobotView Fields

The fields of the IRobotView class are listed here.

Public Fields

	Name	Description
	DiagramMagnification (see page 1398)	Diagram enlarging (scale: 1 to 10).

(c) 2014 Autodesk Inc. All rights reserved

	ParamsBarMap (see page 1398)	Parameters of displaying maps an bars Available since version 2.0.
	ParamsDiagram (see page 1398)	Parameters of displaying diagrams on the view Available since version 2.0.
	ParamsDisplay (see page 1398)	Display parameters (a component enabling control of individual attribute display) Available since version 2.0.
	ParamsFeMap (see page 1399)	Display parameters of maps presenting results for surface finite elements Available since version 2.0.
	Printable (see page 1399)	Object representing the view that may be attached to the printout and printed .
	Projection (see page 1399)	Current projection .
	Selection (see page 1399)	Selection of structure components presented in the view .
	Title (see page 1400)	View title .
	Type (see page 1400)	View type .
	Visible (see page 1400)	View display flag .

I.2.4.2.1 DiagramMagnification

C++

```
HRESULT get_DiagramMagnification(int*);
HRESULT put_DiagramMagnification(int);
```

C#

```
public int DiagramMagnification { get; set; }
```

Visual Basic

```
Public DiagramMagnification As int
```

Description

Diagram enlarging (scale: 1 to 10).

I.2.4.2.2 ParamsBarMap

C++

```
HRESULT get_ParamsBarMap(IRobotViewBarMapParams**);
```

C#

```
public IRobotViewBarMapParams ParamsBarMap { get; }
```

Visual Basic

```
Public ReadOnly ParamsBarMap As IRobotViewBarMapParams
```

Description

Parameters of displaying maps an bars Available since version 2.0.

I.2.4.2.3 ParamsDiagram

C++

```
HRESULT get_ParamsDiagram(IRobotViewDiagramParams**);
```

C#

```
public IRobotViewDiagramParams ParamsDiagram { get; }
```

Visual Basic

```
Public ReadOnly ParamsDiagram As IRobotViewDiagramParams
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Parameters of displaying diagrams on the view Available since version 2.0.

I.2.4.2.4 ParamsDisplay

C++

```
HRESULT get_ParamsDisplay(IRobotViewDisplayParams**);
```

C#

```
public IRobotViewDisplayParams ParamsDisplay { get; }
```

Visual Basic

```
Public ReadOnly ParamsDisplay As IRobotViewDisplayParams
```

Description

Display parameters (a component enabling control of individual attribute display) Available since version 2.0.

I.2.4.2.5 ParamsFeMap

C++

```
HRESULT get_ParamsFeMap(IRobotViewFeMapParams**);
```

C#

```
public IRobotViewFeMapParams ParamsFeMap { get; }
```

Visual Basic

```
Public ReadOnly ParamsFeMap As IRobotViewFeMapParams
```

Description

Display parameters of maps presenting results for surface finite elements Available since version 2.0.

I.2.4.2.6 Printable

C++

```
HRESULT get_Printable(IRobotPrintable**);
```

C#

```
public IRobotPrintable Printable { get; }
```

Visual Basic

```
Public ReadOnly Printable As IRobotPrintable
```

Description

Object representing the view that may be attached to the printout and printed .

I.2.4.2.7 Projection

C++

```
HRESULT get_Projection(IRobotViewProjection*);
HRESULT put_Projection(IRobotViewProjection);
```

C#

```
public IRobotViewProjection Projection { get; set; }
```

Visual Basic

```
Public Projection As IRobotViewProjection
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Current projection .

I.2.4.2.8 Selection

C++

```
HRESULT get_Selection(IRobotMultiSelection**);
```

C#

```
public IRobotMultiSelection Selection { get; }
```

Visual Basic

```
Public ReadOnly Selection As IRobotMultiSelection
```

Description

Selection of structure components presented in the view .

I.2.4.2.9 Title

C++

```
HRESULT get_Title(BSTR*);
HRESULT put_Title(BSTR);
```

C#

```
public String Title { get; set; }
```

Visual Basic

```
Public Title As String
```

Description

View title .

I.2.4.2.10 Type

C++

```
HRESULT get_Type(IRobotViewType*);
```

C#

```
public IRobotViewType Type { get; }
```

Visual Basic

```
Public ReadOnly Type As IRobotViewType
```

Description

View type .

I.2.4.2.11 Visible

C++

```
HRESULT get_Visible(VARIANT_BOOL*);
HRESULT put_Visible(VARIANT_BOOL);
```

C#

```
public bool Visible { get; set; }
```

Visual Basic

```
Public Visible As Boolean
```

(c) 2014 Autodesk Inc. All rights reserved

Description

View display flag .

I.2.4.3 IRobotView Methods

The methods of the IRobotView class are listed here.

Public Methods

	Name	Description
	CopyToClipboard (see page 1401)	The function inserts the view into the clipboard.
	GetRotationPoint (see page 1401)	The function gives back a rotation center during a view rotation.
	GetScale (see page 1402)	The function gives a value of the given scale back. A set of available scales describes RobotViewScaleType type.
	GetSize (see page 1402)	The function returns the size of the view in mm. .
	GetWorkPoint (see page 1402)	The function gives a work point back (on the XY plane it gives Z level) .
	GetZoom (see page 1402)	The function returns the current zoom - coordinates of view sides in the current coordinate system .
	IsLocal (see page 1403)	The function gives the status of a local coordinate system back.
	Redraw (see page 1403)	The function draws the view. .
	Rotate (see page 1403)	The function rotates structure view with respect to the indicated global coordinate axis, with the rotation point recognized.
	SetGlobal (see page 1403)	The function sets a global coordinate system.
	SetLocal (see page 1404)	The function sets a local coordinate system. If the given points do not describe a local plane in the unambiguous way the function gives a zero value (False) back.
	SetRotationPoint (see page 1404)	The funcion sets the rotation center during the view rotation.
	SetScale (see page 1404)	The function assigns a given value to the scale.
	SetSize (see page 1404)	The function sets the view size in mm .
	SetWorkPoint (see page 1405)	The function sets a working point (for the XY plane Z level is set). .
	SetZoom (see page 1405)	The function sets the current zoom - coordinates of view sides in the current coordinate system .

I.2.4.3.1 CopyToClipboard

C++

```
HRESULT CopyToClipboard();
```

C#

```
public void CopyToClipboard();
```

Visual Basic

```
Public Sub CopyToClipboard()
```

Description

The function inserts the view into the clipboard.

I.2.4.3.2 GetRotationPoint

C++

```
HRESULT GetRotationPoint(IRobotGeoPoint3D** ret);
```

C#

```
public IRobotGeoPoint3D GetRotationPoint();
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Function GetRotationPoint() As IRobotGeoPoint3D
```

Description

The function gives back a rotation center during a view rotation.

I.2.4.3.3 GetScale

C++

```
HRESULT GetScale(int _scale_id, double* ret);
```

C#

```
public double GetScale(int _scale_id);
```

Visual Basic

```
Public Function GetScale(_scale_id As int) As double
```

Description

The function gives a value of the given scale back. A set of available scales describes RobotViewScaleType type.

I.2.4.3.4 GetSize

C++

```
HRESULT GetSize(double* _width_in_mm, double* _height_in_mm);
```

C#

```
public void GetSize(double* _width_in_mm, double* _height_in_mm);
```

Visual Basic

```
Public Sub GetSize(ByRef _width_in_mm As double*, ByRef _height_in_mm As double*)
```

Description

The function returns the size of the view in mm. .

I.2.4.3.5 GetWorkPoint

C++

```
HRESULT GetWorkPoint(IRobotGeoPoint3D** ret);
```

C#

```
public IRobotGeoPoint3D GetWorkPoint();
```

Visual Basic

```
Public Function GetWorkPoint() As IRobotGeoPoint3D
```

Description

The function gives a work point back (on the XY plane it gives Z level) .

I.2.4.3.6 GetZoom

C++

```
HRESULT GetZoom(double* _left, double* _top, double* _right, double* _bottom);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public void GetZoom(double* _left, double* _top, double* _right, double* _bottom);
```

Visual Basic

```
Public Sub GetZoom(ByRef _left As double*, ByRef _top As double*, ByRef _right As double*,
ByRef _bottom As double*)
```

Description

The function returns the current zoom - coordinates of view sides in the current coordinate system .

I.2.4.3.7 IsLocal

C++

```
HRESULT IsLocal(VARIANT_BOOL* ret);
```

C#

```
public bool IsLocal();
```

Visual Basic

```
Public Function IsLocal() As Boolean
```

Description

The function gives the status of a local coordinate system back.

I.2.4.3.8 Redraw

C++

```
HRESULT Redraw(VARIANT_BOOL _force_init_zoom);
```

C#

```
public void Redraw(bool _force_init_zoom);
```

Visual Basic

```
Public Sub Redraw(_force_init_zoom As Boolean)
```

Description

The function draws the view. .

I.2.4.3.9 Rotate

C++

```
HRESULT Rotate(IRobotGeoCoordinateAxis _axis, double _angle);
```

C#

```
public void Rotate(IRobotGeoCoordinateAxis _axis, double _angle);
```

Visual Basic

```
Public Sub Rotate(_axis As IRobotGeoCoordinateAxis, _angle As double)
```

Description

The function rotates structure view with respect to the indicated global coordinate axis, with the rotation point recognized.

(c) 2014 Autodesk Inc. All rights reserved

I.2.4.3.10 SetGlobal

C++

```
HRESULT SetGlobal();
```

C#

```
public void SetGlobal();
```

Visual Basic

```
Public Sub SetGlobal()
```

Description

The function sets a global coordinate system.

I.2.4.3.11 SetLocal

C++

```
HRESULT SetLocal(IRobotGeoPoint3D* _p1, IRobotGeoPoint3D* _p2, IRobotGeoPoint3D* _p3,
VARIANT_BOOL* ret);
```

C#

```
public bool SetLocal(IRobotGeoPoint3D _p1, IRobotGeoPoint3D _p2, IRobotGeoPoint3D _p3);
```

Visual Basic

```
Public Function SetLocal(ByRef _p1 As IRobotGeoPoint3D, ByRef _p2 As IRobotGeoPoint3D,
ByRef _p3 As IRobotGeoPoint3D) As Boolean
```

Description

The function sets a local coordinate system. If the given points do not describe a local plane in the unambiguous way the function gives a zero value (False) back.

I.2.4.3.12 SetRotationPoint

C++

```
HRESULT SetRotationPoint(IRobotGeoPoint3D* _rot_point);
```

C#

```
public void SetRotationPoint(IRobotGeoPoint3D _rot_point);
```

Visual Basic

```
Public Sub SetRotationPoint(ByRef _rot_point As IRobotGeoPoint3D)
```

Description

The funcion sets the rotation center during the view rotation.

I.2.4.3.13 SetScale

C++

```
HRESULT SetScale(int _scale_id, double _val);
```

C#

```
public void SetScale(int _scale_id, double _val);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Sub SetScale(_scale_id As int, _val As double)
```

Description

The function assigns a given value to the scale.

I.2.4.3.14 SetSize

C++

```
HRESULT SetSize(double _width_in_mm, double _height_in_mm);
```

C#

```
public void SetSize(double _width_in_mm, double _height_in_mm);
```

Visual Basic

```
Public Sub SetSize(_width_in_mm As double, _height_in_mm As double)
```

Description

The function sets the view size in mm .

I.2.4.3.15 SetWorkPoint

C++

```
HRESULT SetWorkPoint(IRobotGeoPoint3D* _point);
```

C#

```
public void SetWorkPoint(IRobotGeoPoint3D _point);
```

Visual Basic

```
Public Sub SetWorkPoint(ByRef _point As IRobotGeoPoint3D)
```

Description

The function sets a working point (for the XY plane Z level is set). .

I.2.4.3.16 SetZoom

C++

```
HRESULT SetZoom(double _left, double _top, double _right, double _bottom);
```

C#

```
public void SetZoom(double _left, double _top, double _right, double _bottom);
```

Visual Basic

```
Public Sub SetZoom(_left As double, _top As double, _right As double, _bottom As double)
```

Description

The function sets the current zoom - coordinates of view sides in the current coordinate system .

I.2.5 IRobotViewProjection

C++

```
enum IRobotViewProjection;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public enum IRobotViewProjection;
```

Visual Basic

```
Public Enum IRobotViewProjection
```

Members

Members	Description
I_VP_XY_3D = 4	Deep projection (XY plane).
I_VP_YZ_3D = 5	Deep projection (YZ plane).
I_VP_XZ_3D = 6	Deep projection (XZ plane).

Description

The set of identifiers determining different types of projections to be presented in graphical views has been defined. .

I.2.6 IRobotViewVisibilityStatusType

C++

```
enum IRobotViewVisibilityStatusType;
```

C#

```
public enum IRobotViewVisibilityStatusType;
```

Visual Basic

```
Public Enum IRobotViewVisibilityStatusType
```

Members

Members	Description
I_VVST_STRUCTURE_3D = 0	3D structure.
I_VVST_SECTIONS = 1	Sections.
I_VVST_ANIMATION = 2	Animation.
I_VVST_SUPERPOS = 3	Case superposition.
I_VVST_MODE = 4	Modes.
I_VVST_BUCKLING = 5	Buckling.
I_VVST_PLOT = 6	A plotter drawing.
I_VVST_SCALE = 7	Scale.
I_VVST_FILL = 8	A method of area filling.
I_VVST_ACTIVE_CASE = 9	A number of active case.
I_VVST_MOBILE = 10	Settings for moving loads.
I_VVST_ACTIVE_MODE = 11	A number of active mode.
I_VVST_ACTIVE_PHASE = 12	An active structure phase.
I_VVST_ACTIVE_QCMB = 13	Square combination mode.
I_VVST_STRUCTURE = 14	Structure.
I_VVST_NODES = 15	Node numbers.
I_VVST_ELE = 16	Element numbers.
I_VVST_NOT_SELECTED = 17	Objects not selected.
I_VVST_EXPLODE = 18	Explode.
I_VVST_MTC = 19	Materials, compression, tension.
I_VVST_SECTION_NAMES = 20	Section names.
I_VVST_SECTION_SYMB = 21	Section symbols.
I_VVST_SECTION_DRAW = 22	Section drawings.

(c) 2014 Autodesk Inc. All rights reserved

I_VVST_SECTION_SURF = 23	Section area.
I_VVST_LOCAL = 24	Local systems.
I_VVST_THICKNESS = 25	Thicknesses.
I_VVST_SUPPORTS = 26	Supports.
I_VVST_RELEASES = 27	Releases.
I_VVST_RIGID_LINKS = 28	Rigid links.
I_VVST_LOADS = 29	Loads.
I_VVST_NODE_LOADS = 30	Nodal loads.
I_VVST_BAR_LOADS = 31	Bar loads.
I_VVST_DEFORMATION = 32	Deformed structure.
I_VVST_FORCES = 33	Forces.
I_VVST_EXTREMES = 34	Extreme forces.
I_VVST_REACTIONS = 41	Residual reactions and vectors.
I_VVST_FE = 42	Panels and finite elements.
I_VVST_STRESSES = 43	Stress curvatures.
I_VVST_FE_CUTS = 44	Cuts through finite elements.
I_VVST_FE_DIRECTION = 45	A direction of result calculations in finite elements.
I_VVST_FE_LAY = 46	A layer of stress calculations in finite elements.
I_VVST_REINF_DIR = 47	Main direction of reinforcement.
I_VVST_BAR_MAPS = 48	Maps on bars.
I_VVST_TITLE = 49	Title.
I_VVST_OFFSETS = 50	Offsets.
I_VVST_COMPATIBILITIES = 51	Compatibilities.
I_VVST_ENVELOPES = 52	Envelopes.
I_VVST_ANALYSIS = 53	Analysis.
I_VVST_DIM_LINES = 54	Dimension lines.
I_VVST_REINF_CROSSES = 55	Reinforcement crosses.
I_VVST_STRESSES_GLOBAL = 56	Available since version 1.7.
I_VVST_DESCRIPTIONS = 57	Available since version 2.0.
I_VVST_GRID = 58	Available since version 2.0.
I_VVST_RULER = 59	Available since version 2.0.
_VVST_STRUCTURAL_AXIS = 60	Available since version 2.0.
I_VVST_MAPS = 61	Available since version 2.0.
I_VVST_MAPS_LAYER = 62	Available since version 2.0.
I_VVST_MAPS_DEFORMATION = 63	Available since version 2.0.

Description

Available types of visibility status for a graphical viewer.

I.2.7 IRobotViewVisibilityStatusValue

C++

```
enum IRobotViewVisibilityStatusValue;
```

C#

```
public enum IRobotViewVisibilityStatusValue;
```

Visual Basic

```
Public Enum IRobotViewVisibilityStatusValue
```

(c) 2014 Autodesk Inc. All rights reserved

Members

Members	Description
I_VVSV_MTC_MATERIAL = 0x01	Material.
I_VVSV_MTC_COMPRESSION = 0x02	Compression.
I_VVSV_MTC_TENSION = 0x04	Tension.
I_VVSV_MTC_CABLE = 0x08	Cables.
I_VVSV_STRUCTURE_ON = 0x01	Structure geometry.
I_VVSV_STRUCTURE_HDL = 0x04	Hidden lines.
I_VVSV_STRUCTURE_SHD = 0x10	Shading.
I_VVSV_STRUCTURE_HDF = 0x40	Colorful sides, black edges.
I_VVSV_STRUCTURE_HDR = 0x80	Shading with edge correction. .
I_VVSV_SUPPORTS_SYMB = 0x01	Support symbols.
I_VVSV_SUPPORTS_COD = 0x02	Support codes.
I_VVSV_SUPPORTS_LBL = 0x04	Labels for all visible attributes (instead of codes).
I_VVSV_ANALYSIS_DET = 0x01	Detailed analysis.
I_VVSV_ANALYSIS_GLO = 0x02	Global analysis.
I_VVSV_ANALYSIS_INF = 0x08	Analysis of influence lines .
I_VVSV_ENVELOPES_MIN = 0x01	Minimum envelope.
I_VVSV_ENVELOPES_MAX = 0x02	Maximum envelope.
I_VVSV_ENVELOPES_CMP = 0x04	All envelope components.
I_VVSV_ENVELOPES_ONE = 0x08	Single components in complex cases.
I_VVSV_ENVELOPES_HST = 0x10	Time history analysis case.
I_VVSV_ENVELOPES_CUT = 0x20	The analysis in cuts.
I_VVSV_ENVELOPES_SEU = 0x40	The analysis of a single bar.
I_VVSV_SECTION_DRAW_SHAPE = 0x01	Section shapes.
I_VVSV_SECTION_DRAW_COLOR = 0x10	Section colors.
I_VVSV_SECTION_DRAW_TOP_BOTTOM = 0x20	Top and bottom of elements in colors.
I_VVSV_SECTION_DRAW_GROUP = 0x40	Group colors.
I_VVSV_SECTION_SYMB_SYMB = 0x1	Section symbols.
I_VVSV_SECTION_SYMB_ELA = 0x2	Elastic ground for bars.
I_VVSV_FILL_FILL = 0x1	Map filling.
I_VVSV_FILL_AVG = 0x2	Averaging values in finite elements.
I_VVSV_FILL_VAL = 0x4	Values on maps.
I_VVSV_FILL_ISO = 0x8	Isolines.
I_VVSV_FILL_BOR = 0x10	Contour.
I_VVSV_FILL_TAN = 0x20	Cuts on normal plane.
I_VVSV_FILL_DEF = 0x40	Deformation.
I_VVSV_FILL_SMOOTH_NON = 0	No smoothing.
I_VVSV_FILL_SMOOTH_GLO = 0x100	Global smoothing.
I_VVSV_FILL_SMOOTH_LOC = 0x200	Smoothing within panels.
I_VVSV_FILL_SMOOTH_SEL = 0x300	Smoothing on selection.
I_VVSV_RIGID_LINKS_RIG = 0x1	Rigid links.
I_VVSV_RIGID_LINKS_SYM = 0x2	Rigid links - symbols.
I_VVSV_RIGID_LINKS_DET = 0x4	Rigid links - details.
I_VVSV_RIGID_LINKS_SUR = 0x8	Rigid links - areas.
I_VVSV_FE_CON = 0x1	Contours.
I_VVSV_FE_COI = 0x2	Inside part of contours .
I_VVSV_FE_COL = 0x4	Thickness colors.

(c) 2014 Autodesk Inc. All rights reserved

I_VVSV_FE_COE = 0x8	Contour components.
I_VVSV_FE_MES = 0x10	Mesh.
I_VVSV_FE_EMI = 0x20	Emitters.
I_VVSV_FE_RES = 0x40	Results.
I_VVSV_FE_DSC = 0x80	Description of panels.
I_VVSV_FE_DSO = 0x100	Description of objects.
I_VVSV_FE_EDG = 0x200	Description of edges.
I_VVSV_FE_ADV = 0x400	Detailed description of objects.
I_VVSV_ELE_BAR_NUM = 0x1	Bar numbers.
I_VVSV_ELE_CAL_NUM = 0x2	Calculation element numbers.
I_VVSV_ELE_FE_NUM = 0x4	Finite element numbers.
I_VVSV_ELE_FE_VIS = 0x8	Finite elements.
I_VVSV_MOBILE_ROUTE = 0x1	Route.
I_VVSV_MOBILE_CAR = 0x2	Vehicle.
I_VVSV_MOBILE_ELE = 0x4	Elements.
I_VVSV_MOBILE_LOADS = 0x8	Loads.
I_VVSV_STRUCTURE_HDY = 0x100	Quick shading of faces Available since version 1.7.
I_VVSV_FILL_SMOOTH_CAR = 0x400	Available since version 1.7.
I_VVSV_FORCES_FX = 0x0001	Available since version 1.7.
I_VVSV_FORCES_FY = 0x0002	Available since version 1.7.
I_VVSV_FORCES_FZ = 0x0004	Available since version 1.7.
I_VVSV_FORCES_FXC = 0x0008	Available since version 1.7.
I_VVSV_FORCES_MX = 0x0010	Available since version 1.7.
I_VVSV_FORCES_MY = 0x0020	Available since version 1.7.
I_VVSV_FORCES_MZ = 0x0040	Available since version 1.7.
I_VVSV_FORCES_UX = 0x0100	Available since version 1.7.
I_VVSV_FORCES_UY = 0x0200	Available since version 1.7.
I_VVSV_FORCES_UZ = 0x0400	Available since version 1.7.
I_VVSV_FORCES_DFL = 0x1000	Available since version 1.7.
I_VVSV_STRESSES_NORMALMIN = 0x0001	Available since version 1.7.
I_VVSV_STRESSES_NORMALMAX = 0x0002	Available since version 1.7.
I_VVSV_STRESSES_FLEXMIN_MY = 0x0004	Available since version 1.7.
I_VVSV_STRESSES_FLEXMIN_MZ = 0x0008	Available since version 1.7.
I_VVSV_STRESSES_FLEXMAX_MY = 0x0010	Available since version 1.7.
I_VVSV_STRESSES_FLEXMAX_MZ = 0x0020	Available since version 1.7.
I_VVSV_STRESSES_AXIAL = 0x0040	Available since version 1.7.
I_VVSV_STRESSES_SHEAR_Y = 0x0080	Available since version 1.7.
I_VVSV_STRESSES_SHEAR_Z = 0x0100	Available since version 1.7.
I_VVSV_STRESSES_TORSION = 0x0200	Available since version 1.7.
I_VVSV_DEFORMATION_EXACT = 0x0001	Available since version 1.7.
I_VVSV_DEFORMATION_STD = 0x0002	Available since version 1.7.
I_VVSV_REACTIONS_F = 0x01	Available since version 1.7.
I_VVSV_REACTIONS_M = 0x02	Available since version 1.7.
I_VVSV_REACTIONS_RESID_F = 0x04	Available since version 1.7.
I_VVSV_REACTIONS_RESID_M = 0x08	Available since version 1.7.
I_VVSV_REACTIONS_VAL = 0x10	Available since version 1.7.
I_VVSV_REACTIONS_ROT = 0x20	Available since version 1.7.
I_VVSV_REACTIONS_DIV = 0x40	Available since version 1.7.
I_VVSV_LOADS_SYMBOL = 0x01	Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

I_VVSV_LOADS_VALUE = 0x02	Available since version 1.7.
I_VVSV_STRESSES_GLOBAL_USER_MIN = 0x001	Available since version 1.7.
I_VVSV_STRESSES_GLOBAL_USER_MAX = 0x002	Available since version 1.7.
I_VVSV_STRESSES_GLOBAL_NORMAL_MIN = 0x004	Available since version 1.7.
I_VVSV_STRESSES_GLOBAL_NORMAL_MAX = 0x008	Available since version 1.7.
I_VVSV_STRESSES_GLOBAL_TAU_MIN = 0x010	Available since version 1.7.
I_VVSV_STRESSES_GLOBAL_TAU_MAX = 0x020	Available since version 1.7.
I_VVSV_STRESSES_GLOBAL_MISES_MIN = 0x040	Available since version 1.7.
I_VVSV_STRESSES_GLOBAL_MISES_MAX = 0x080	Available since version 1.7.
I_VVSV_BAR_MAPS_FX = 165	Available since version 1.7.
I_VVSV_BAR_MAPS_FY = 166	Available since version 1.7.
I_VVSV_BAR_MAPS_FZ = 167	Available since version 1.7.
I_VVSV_BAR_MAPS_MX = 168	Available since version 1.7.
I_VVSV_BAR_MAPS_MY = 169	Available since version 1.7.
I_VVSV_BAR_MAPS_MZ = 170	Available since version 1.7.
I_VVSV_BAR_MAPS_STRESS_S_MAX = 171	Available since version 1.7.
I_VVSV_BAR_MAPS_STRESS_S_MIN = 172	Available since version 1.7.
I_VVSV_BAR_MAPS_STRESS_FX_AX = 173	Available since version 1.7.
I_VVSV_BAR_MAPS_STRESS_S_MAX_MY = 174	Available since version 1.7.
I_VVSV_BAR_MAPS_STRESS_S_MAX_MZ = 175	Available since version 1.7.
I_VVSV_BAR_MAPS_STRESS_S_MIN_MY = 176	Available since version 1.7.
I_VVSV_BAR_MAPS_STRESS_S_MIN_MZ = 177	Available since version 1.7.
I_VVSV_BAR_MAPS_SHEAR_STRESS_TY = 179	Available since version 1.7.
I_VVSV_BAR_MAPS_SHEAR_STRESS_TZ = 180	Available since version 1.7.
I_VVSV_BAR_MAPS_SHEAR_STRESS_T = 181	Available since version 1.7.
I_VVSV_BAR_MAPS_DESIGN_RATIO = 796	Available since version 1.7.
I_VVSV_BAR_MAPS_DESIGN_MEMB_LENGTH = 271	Available since version 1.7.
I_VVSV_BAR_MAPS_DESIGN_SLEND_LAY = 831	Available since version 1.7.
I_VVSV_BAR_MAPS_DESIGN_SLEND_LAZ = 832	Available since version 1.7.
I_VVSV_DESCRIPTIONS_NONE = 0	Available since version 2.0.
I_VVSV_DESCRIPTIONS_HORIZONTAL = 0x01	Available since version 2.0.
I_VVSV_DESCRIPTIONS_VERTICAL = 0x02	Available since version 2.0.
I_VVSV_REINFORCEMENT_A1 = 0x0001	Available since version 2.0.
I_VVSV_REINFORCEMENT_A1R = 0x0010	Available since version 2.0.
I_VVSV_REINFORCEMENT_A2 = 0x0002	Available since version 2.0.
I_VVSV_REINFORCEMENT_A2R = 0x0020	Available since version 2.0.
I_VVSV_REINFORCEMENT_PRC = 0x0004	Available since version 2.0.
I_VVSV_REINFORCEMENT_PRCR = 0x0008	Available since version 2.0.
I_VVSV_REINFORCEMENT_DISTRIB = 0x0040	Available since version 2.0.
I_VVSV_REINFORCEMENT_DISTRIBR = 0x0080	Available since version 2.0.
I_VVSV_REINFORCEMENT_A1BARS = 0x0100	Available since version 2.0.
I_VVSV_REINFORCEMENT_A2BARS = 0x0200	Available since version 2.0.
I_VVSV_MAPS_DETAILED_STRESSES_XX = 483	Available since version 2.0.
I_VVSV_MAPS_DETAILED_STRESSES_YY = 484	Available since version 2.0.
I_VVSV_MAPS_DETAILED_STRESSES_XY = 485	Available since version 2.0.
I_VVSV_MAPS_DETAILED_STRESSES_Z = 486	Available since version 2.0.
I_VVSV_MAPS_DETAILED_MEMB_FORCES_XX = 492	Available since version 2.0.
I_VVSV_MAPS_DETAILED_MEMB_FORCES_YY = 493	Available since version 2.0.
I_VVSV_MAPS_DETAILED_MEMB_FORCES_XY = 494	Available since version 2.0.

(c) 2014 Autodesk Inc. All rights reserved

I_VVSV_MAPS_DETAILED_MOMENTS_XX = 501	Available since version 2.0.
I_VVSV_MAPS_DETAILED_MOMENTS_YY = 502	Available since version 2.0.
I_VVSV_MAPS_DETAILED_MOMENTS_XY = 503	Available since version 2.0.
I_VVSV_MAPS_DETAILED_SHEAR_STRESSES_XX = 510	Available since version 2.0.
I_VVSV_MAPS_DETAILED_SHEAR_STRESSES_YY = 511	Available since version 2.0.
I_VVSV_MAPS_DETAILED_SHEAR_FORCES_XX = 519	Available since version 2.0.
I_VVSV_MAPS_DETAILED_SHEAR_FORCES_YY = 520	Available since version 2.0.
I_VVSV_MAPS_DETAILED_DISPLACEMENTS_XX = 537	Available since version 2.0.
I_VVSV_MAPS_DETAILED_DISPLACEMENTS_YY = 538	Available since version 2.0.
I_VVSV_MAPS_DETAILED_DISPLACEMENTS_Z = 531	Available since version 2.0.
I_VVSV_MAPS_DETAILED_ROTATION_XX = 546	Available since version 2.0.
I_VVSV_MAPS_DETAILED_ROTATION_YY = 547	Available since version 2.0.
I_VVSV_MAPS_DETAILED_ROTATION_Z = 549	Available since version 2.0.
I_VVSV_MAPS_DETAILED_SOIL_REACTIONS = 558	Available since version 2.0.
I_VVSV_MAPS_PRINCIPAL_STRESSES_1 = 487	Available since version 2.0.
I_VVSV_MAPS_PRINCIPAL_STRESSES_2 = 488	Available since version 2.0.
I_VVSV_MAPS_PRINCIPAL_STRESSES_1_2 = 490	Available since version 2.0.
I_VVSV_MAPS_PRINCIPAL_STRESSES_ANGLE = 489	Available since version 2.0.
I_VVSV_MAPS_PRINCIPAL_MEMB_FORCES_1 = 496	Available since version 2.0.
I_VVSV_MAPS_PRINCIPAL_MEMB_FORCES_2 = 497	Available since version 2.0.
I_VVSV_MAPS_PRINCIPAL_MEMB_FORCES_1_2 = 499	Available since version 2.0.
I_VVSV_MAPS_PRINCIPAL_MEMB_FORCES_ANGLE = 498	Available since version 2.0.
I_VVSV_MAPS_PRINCIPAL_MOMENTS_1 = 505	Available since version 2.0.
I_VVSV_MAPS_PRINCIPAL_MOMENTS_2 = 506	Available since version 2.0.
I_VVSV_MAPS_PRINCIPAL_MOMENTS_ANGLE = 507	Available since version 2.0.
I_VVSV_MAPS_PRINCIPAL_MOMENTS_1_2 = 508	Available since version 2.0.
I_VVSV_MAPS_PRINCIPAL_SHEAR_STRESSES = 517	Available since version 2.0.
I_VVSV_MAPS_PRINCIPAL_SHEAR_FORCES = 526	Available since version 2.0.
I_VVSV_MAPS_PRINCIPAL_GLOBAL_DISP_X = 1302	Available since version 2.0.
I_VVSV_MAPS_PRINCIPAL_GLOBAL_DISP_Y = 1303	Available since version 2.0.
I_VVSV_MAPS_PRINCIPAL_GLOBAL_DISP_Z = 1304	Available since version 2.0.
I_VVSV_MAPS_PRINCIPAL_TOTAL_DISP = 1305	Available since version 2.0.
I_VVSV_MAPS_COMPLEX_STRESSES = 491	Available since version 2.0.
I_VVSV_MAPS_COMPLEX_MEMB_FORCES = 500	Available since version 2.0.
I_VVSV_MAPS_COMPLEX_MOMENTS = 509	Available since version 2.0.
I_VVSV_MAPS_COMPLEX_REINFORCEMENT_TOP_MXX = 889	Available since version 2.0.
I_VVSV_MAPS_COMPLEX_REINFORCEMENT_TOP_MYY = 891	Available since version 2.0.
I_VVSV_MAPS_COMPLEX_REINFORCEMENT_BOTTOM_MXX = 890	Available since version 2.0.
I_VVSV_MAPS_COMPLEX_REINFORCEMENT_BOTTOM_MYY = 892	Available since version 2.0.
I_VVSV_MAPS_CROSS_S = 1404	Available since version 2.0.
I_VVSV_MAPS_CROSS_N = 1405	Available since version 2.0.
I_VVSV_MAPS_CROSS_M = 1406	Available since version 2.0.
I_VVSV_MAPS_LAYER_UPPER = 0	Available since version 2.0.
I_VVSV_MAPS_LAYER_MIDDLE = 1	Available since version 2.0.
I_VVSV_MAPS_LAYER_LOWER = 2	Available since version 2.0.

(c) 2014 Autodesk Inc. All rights reserved

I_VVSV_MAPS_LAYER_MAXIMUM = 3	Available since version 2.0.
I_VVSV_MAPS_LAYER_MINIMUM = 4	Available since version 2.0.
I_VVSV_MAPS_LAYER_ABS_MINIMUM = 5	Available since version 2.0.
I_VVSV_MAPS_LAYER_ARBITRARY = 6	Available since version 2.0.
I_VVSV_MAPS_ISOLINES = 0x0001	Available since version 2.0.
I_VVSV_MAPS_DESCRIPTION = 0x0002	Available since version 2.0.
I_VVSV_MAPS_SMOOTH_NON = 0x0004	Available since version 2.0.
I_VVSV_MAPS_SMOOTH_GLO = 0x0008	Available since version 2.0.
I_VVSV_MAPS_SMOOTH_LOC = 0x0010	Available since version 2.0.
I_VVSV_MAPS_SMOOTH_SEL = 0x0020	Available since version 2.0.
I_VVSV_MAPS_SMOOTH_CAR = 0x0030	Available since version 2.0.
I_VVSV_FORCES_BAR_REACT_KY = 0x0080	Available since version 3.
I_VVSV_FORCES_BAR_REACT_KZ = 0x0800	Available since version 3.
I_VVSV_FILL_FE = 0x80	Finite element mesh. Available since version 4.1.

Description

Accessible values for particular types of visibility status.

I.2.8 IRobotViewScaleType

C++

```
enum IRobotViewScaleType;
```

C#

```
public enum IRobotViewScaleType;
```

Visual Basic

```
Public Enum IRobotViewScaleType
```

Members

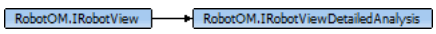
Members	Description
I_VST_FX = 0	A scale for FX force.
I_VST_FY = 1	A scale for FY force .
I_VST_FZ = 2	A scale for FZ force .
I_VST_MX = 3	A scale for MX .
I_VST_MY = 4	A scale for MY .
I_VST_MZ = 5	A scale for MZ .
I_VST_DEFORM = 6	A scale for deformation.
I_VST_SIG = 7	A scale for axial stresses.
I_VST_TAU = 8	A scale for shear stresses .

Description

Available types of a scale defined for a graphical viewer.

I.2.9 IRobotViewDetailedAnalysis

Class Hierarchy



(c) 2014 Autodesk Inc. All rights reserved

C++

```
interface IRobotViewDetailedAnalysis : IRobotView;
```

C#

```
public interface IRobotViewDetailedAnalysis : IRobotView;
```

Visual Basic

```
Public Interface IRobotViewDetailedAnalysis
```

Description

View presenting results of the detailed analysis. Available since version 2.0.

I.2.9.1 IRobotViewDetailedAnalysis Members

The following tables list the members exposed by IRobotViewDetailedAnalysis.

Public Fields

	Name	Description
	CurrentTableTab (see page 1414)	Identifier of the active tab in the tabular part of the view.
	DiagramMagnification (see page 1398)	Diagram enlarging (scale: 1 to 10).
	ParamsBarMap (see page 1398)	Parameters of displaying maps an bars Available since version 2.0.
	ParamsDetailed (see page 1414)	Parameters of displaying detailed analysis results Available since version 2.0.
	ParamsDiagram (see page 1398)	Parameters of displaying diagrams on the view Available since version 2.0.
	ParamsDisplay (see page 1398)	Display parameters (a component enabling control of individual attribute display) Available since version 2.0.
	ParamsFeMap (see page 1399)	Display parameters of maps presenting results for surface finite elements Available since version 2.0.
	Printable (see page 1399)	Object representing the view that may be attached to the printout and printed .
	Projection (see page 1399)	Current projection .
	Selection (see page 1399)	Selection of structure components presented in the view .
	Title (see page 1400)	View title .
	Type (see page 1400)	View type .
	UserControl (see page 1414)	Flag that enables giving/taking back graphical view navigation to/from the interactive program user .
	Visible (see page 1400)	View display flag .
	Window (see page 1415)	

Public Methods

	Name	Description
	CopyToClipboard (see page 1401)	The function inserts the view into the clipboard.
	GetRotationPoint (see page 1401)	The function gives back a rotation center during a view rotation.
	GetScale (see page 1402)	The function gives a value of the given scale back. A set of available scales describes RobotViewScaleType type.
	GetSize (see page 1402)	The function returns the size of the view in mm. .
	GetWorkPoint (see page 1402)	The function gives a work point back (on the XY plane it gives Z level) .
	GetZoom (see page 1402)	The function returns the current zoom - coordinates of view sides in the current coordinate system .
	IsLocal (see page 1403)	The function gives the status of a local coordinate system back.

(c) 2014 Autodesk Inc. All rights reserved

	MakeScreenCapture (see page 1415)	
	Redraw (see page 1403)	The function draws the view. .
	Rotate (see page 1403)	The function rotates structure view with respect to the indicated global coordinate axis, with the rotation point recognized.
	SetGlobal (see page 1403)	The function sets a global coordinate system.
	SetLocal (see page 1404)	The function sets a local coordinate system. If the given points do not describe a local plane in the unambiguous way the function gives a zero value (False) back.
	SetRotationPoint (see page 1404)	The funcion sets the rotation center during the view rotation.
	SetScale (see page 1404)	The function assigns a given value to the scale.
	SetSize (see page 1404)	The function sets the view size in mm .
	SetWorkPoint (see page 1405)	The function sets a working point (for the XY plane Z level is set). .
	SetZoom (see page 1405)	The function sets the current zoom - coordinates of view sides in the current coordinate system .

I.2.9.2 IRobotViewDetailedAnalysis Fields

The fields of the IRobotViewDetailedAnalysis class are listed here.

Public Fields

	Name	Description
	CurrentTableTab (see page 1414)	Identifier of the active tab in the tabular part of the view.
	ParamsDetailed (see page 1414)	Parameters of displaying detailed analysis results Available since version 2.0.
	UserControl (see page 1414)	Flag that enables giving/taking back graphical view navigation to/from the interactive program user .
	Window (see page 1415)	

I.2.9.2.1 CurrentTableTab

C++

```
HRESULT get_CurrentTableTab(IRobotViewDetailedAnalysisTableTab*);
HRESULT put_CurrentTableTab(IRobotViewDetailedAnalysisTableTab);
```

C#

```
public IRobotViewDetailedAnalysisTableTab CurrentTableTab { get; set; }
```

Visual Basic

```
Public CurrentTableTab As IRobotViewDetailedAnalysisTableTab
```

Description

Identifier of the active tab in the tabular part of the view.

I.2.9.2.2 ParamsDetailed

C++

```
HRESULT get_ParamsDetailed(IRobotViewDetailedAnalysisParams**);
```

C#

```
public IRobotViewDetailedAnalysisParams ParamsDetailed { get; }
```

Visual Basic

```
Public ReadOnly ParamsDetailed As IRobotViewDetailedAnalysisParams
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Parameters of displaying detailed analysis results Available since version 2.0.

I.2.9.2.3 UserControl

C++

```
HRESULT get_UserControl(VARIANT_BOOL*);
HRESULT put_UserControl(VARIANT_BOOL);
```

C#

```
public bool UserControl { get; set; }
```

Visual Basic

```
Public UserControl As Boolean
```

Description

Flag that enables giving/taking back graphical view navigation to/from the interactive program user .

Version

Available since version 3.

I.2.9.2.4 Window

C++

```
HRESULT get_Window(IRobotWindow**);
```

C#

```
public IRobotWindow Window { get; }
```

Visual Basic

```
Public ReadOnly Window As IRobotWindow
```

Version

Available since version 3.

I.2.9.3 IRobotViewDetailedAnalysis Methods

The methods of the IRobotViewDetailedAnalysis class are listed here.

Public Methods

	Name	Description
	MakeScreenCapture (see page 1415)	

I.2.9.3.1 MakeScreenCapture

C++

```
HRESULT MakeScreenCapture(IRobotViewScreenCaptureParams* _sc_params);
```

C#

```
public void MakeScreenCapture(IRobotViewScreenCaptureParams _sc_params);
```

Visual Basic

```
Public Sub MakeScreenCapture(ByRef _sc_params As IRobotViewScreenCaptureParams)
```

(c) 2014 Autodesk Inc. All rights reserved

I.2.10 IRobotViewDetailedAnalysisTableTab

C++

```
enum IRobotViewDetailedAnalysisTableTab;
```

C#

```
public enum IRobotViewDetailedAnalysisTableTab;
```

Visual Basic

```
Public Enum IRobotViewDetailedAnalysisTableTab
```

Members

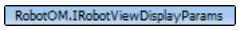
Members	Description
I_VDATT_LOCAL_EXTREMES = 1	Tab presenting local extreme values.
I_VDATT_GLOBAL_EXTREMES = 2	Tab presenting global extreme values.

Description

Identifiers of the tabs in the table being a part of the view presenting detailed analysis results. Available since version 2.0.

I.2.11 IRobotViewDisplayParams

Class Hierarchy



C++

```
interface IRobotViewDisplayParams : IDispatch;
```

C#

```
public interface IRobotViewDisplayParams;
```

Visual Basic

```
Public Interface IRobotViewDisplayParams
```

Description

Parameters of display for graphical viewers.

I.2.11.1 IRobotViewDisplayParams Members

The following tables list the members exposed by IRobotViewDisplayParams.

Public Fields

	Name	Description
	HiddenLines (see page 1417)	Type of displaying hidden lines Available since version 2.0.
	SymbolSize (see page 1417)	Symbol size (from 1 to 10) Available since version 2.0.

Public Methods

	Name	Description
	IsOn (see page 1417)	Function returns a value different from zero (True) if a selected attribute is switched on. Available since version 2.0.
	Set (see page 1418)	Function switches on/off a selected visibility attribute. Available since version 2.0.

(c) 2014 Autodesk Inc. All rights reserved

I.2.11.2 IRobotViewDisplayParams Fields

The fields of the IRobotViewDisplayParams class are listed here.

Public Fields

	Name	Description
	HiddenLines (see page 1417)	Type of displaying hidden lines Available since version 2.0.
	SymbolSize (see page 1417)	Symbol size (from 1 to 10) Available since version 2.0.

I.2.11.2.1 HiddenLines

C++

```
HRESULT get_HiddenLines(IRobotViewHiddenLinesDisplayType*);
HRESULT put_HiddenLines(IRobotViewHiddenLinesDisplayType);
```

C#

```
public IRobotViewHiddenLinesDisplayType HiddenLines { get; set; }
```

Visual Basic

```
Public HiddenLines As IRobotViewHiddenLinesDisplayType
```

Description

Type of displaying hidden lines Available since version 2.0.

I.2.11.2.2 SymbolSize

C++

```
HRESULT get_SymbolSize(short*);
HRESULT put_SymbolSize(short);
```

C#

```
public short SymbolSize { get; set; }
```

Visual Basic

```
Public SymbolSize As short
```

Description

Symbol size (from 1 to 10) Available since version 2.0.

I.2.11.3 IRobotViewDisplayParams Methods

The methods of the IRobotViewDisplayParams class are listed here.

Public Methods

	Name	Description
	IsOn (see page 1417)	Function returns a value different from zero (True) if a selected attribute is switched on. Available since version 2.0.
	Set (see page 1418)	Function switches on/off a selected visibility attribute. Available since version 2.0.

I.2.11.3.1 IsOn

C++

```
HRESULT IsOn(IRobotViewDisplayAttributes _v_attrib, VARIANT_BOOL* ret);
```

C#

```
public bool IsOn(IRobotViewDisplayAttributes _v_attrib);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Function IsOn(_v_attrib As IRobotViewDisplayAttributes) As Boolean
```

Description

Function returns a value different from zero (True) if a selected attribute is switched on. Available since version 2.0.

I.2.11.3.2 Set

C++

```
HRESULT Set(IRobotViewDisplayAttributes _v_attrib, VARIANT_BOOL _on_off);
```

C#

```
public void Set(IRobotViewDisplayAttributes _v_attrib, bool _on_off);
```

Visual Basic

```
Public Sub Set(_v_attrib As IRobotViewDisplayAttributes, _on_off As Boolean)
```

Description

Function switches on/off a selected visibility attribute. Available since version 2.0.

I.2.12 IRobotViewDisplayAttributes

C++

```
enum IRobotViewDisplayAttributes;
```

C#

```
public enum IRobotViewDisplayAttributes;
```

Visual Basic

```
Public Enum IRobotViewDisplayAttributes
```

Members

Members	Description
I_VDA_STRUCTURE_STRUCTURE = 1	Available since version 2.0.
I_VDA_STRUCTURE_NODE_NUMBERS = 2	Available since version 2.0.
I_VDA_STRUCTURE_BAR_NUMBERS = 3	Available since version 2.0.
I_VDA_STRUCTURE_SUPPORT_SYMBOLS = 4	Available since version 2.0.
I_VDA_STRUCTURE_SUPPORT_CODES = 5	Available since version 2.0.
I_VDA_STRUCTURE_ONLY_FOR_SELECTED_OBJECTS = 6	Available since version 2.0.
I_VDA_STRUCTURE_GROUP_COLORS = 8	Available since version 2.0.
I_VDA_STRUCTURE_ATTRIBUTE_LABELS = 9	Available since version 2.0.
I_VDA_SECTIONS_NAME = 11	Available since version 2.0.
I_VDA_SECTIONS_COLORS = 12	Available since version 2.0.
I_VDA_SECTIONS_SYMBOLS = 13	Available since version 2.0.
I_VDA_SECTIONS_SHAPE = 14	Available since version 2.0.
I_VDA_SECTIONS_SURFACE = 15	Available since version 2.0.
I_VDA_SECTIONS_MATERIAL = 16	Available since version 2.0.
I_VDA_LOADS_VALUES = 18	Available since version 2.0.
I_VDA_LOADS_MOVING_ROUTE = 19	Available since version 2.0.
I_VDA_LOADS_MOVING_VEHICLE = 20	Available since version 2.0.

(c) 2014 Autodesk Inc. All rights reserved

I_VDA_LOADS_MOVING_ELEMENTS = 21	Available since version 2.0.
I_VDA_ADVANCED_RELEASE_SYMBOLS = 23	Available since version 2.0.
I_VDA_ADVANCED_RELEASE_CODES = 24	Available since version 2.0.
I_VDA_ADVANCED_OFFSETS = 25	Available since version 2.0.
I_VDA_ADVANCED_COMPATIBLE_NODES = 26	Available since version 2.0.
I_VDA_ADVANCED_RIGID_LINKS = 27	Available since version 2.0.
I_VDA_ADVANCED_RIGID_LINK_SURFACE = 28	Available since version 2.0.
I_VDA_ADVANCED_ELASTIC_FOUNDATION = 29	Available since version 2.0.
I_VDA_ADVANCED_CABLES = 30	Available since version 2.0.
I_VDA_ADVANCED_TENSION_COMPRESSION = 31	Available since version 2.0.
I_VDA_ADVANCED_STEEL_CONNECTIONS = 32	Available since version 2.0.
I_VDA_FE_PANEL_CONTOURS = 33	Available since version 2.0.
I_VDA_FE_PANEL_INTERIOR = 34	Available since version 2.0.
I_VDA_FE_THICKNESS = 35	Available since version 2.0.
I_VDA_FE_PANEL_NUMBERS = 36	Available since version 2.0.
I_VDA_FE_PANEL_COMPLEX_DESC = 37	Available since version 2.0.
I_VDA_FE_COLOR_LEGEND = 38	Available since version 2.0.
I_VDA_FE_CONTOUR_COMPONENTS = 39	Available since version 2.0.
I_VDA_FE_MESH_PREVIEW = 40	Available since version 2.0.
I_VDA_FE_EDGE_NUMBERS = 41	Available since version 2.0.
I_VDA_FE_FINITE_ELEMENTS = 42	Available since version 2.0.
I_VDA_FE_FINITE_ELEMENT_NUMBERS = 43	Available since version 2.0.
I_VDA_FE_EMITTERS = 44	Available since version 2.0.
I_VDA_OTHER_STRUCTURAL_AXIS = 45	Available since version 2.0.
I_VDA_OTHER_DIMENSION_LINES = 46	Available since version 2.0.
I_VDA_OTHER_GRID = 47	Available since version 2.0.
I_VDA_OTHER_RULER = 48	Available since version 2.0.
I_VDA_OTHER_HIDE_NODES = 49	Available since version 2.0.
I_VDA_OTHER_HIDE_INACTIVE = 51	Available since version 2.0.
I_VDA_OTHER_CALC_ELEM_NUMBERS = 52	Available since version 2.0.
I_VDA_OTHER_CALC_POINT_NUMBERS = 53	Available since version 2.0.
I_VDA_OTHER_OBJECTS_OUT_OF_PLANE = 54	Available since version 2.0.
I_VDA_SECTIONS_MEMBER_TYPE_NAME = 55	Bar type - name. Available since version 3.5.
I_VDA_SECTIONS_MEMBER_TYPE_LEGEND = 56	Bar type - color legend. Available since version 3.5.
I_VDA_ADVANCED_GEOIMPERFECTIONS = 57	Geometrical imperfections. Available since version 3.5.
I_VDA_ADVANCED_GEOIMPERFECTIONS_NAME = 58	Geometrical imperfections - name. Available since version 3.5.
I_VDA_ADVANCED_NONLINEAR_HINGES = 59	Non-linear hinges. Available since version 3.5.
I_VDA_STRUCTURE_SUPPORT_SHAPES = 60	Support shapes. Available since version 9.7.
I_VDA_STRUCTURE_EXPLODE_BARS = 61	Available since version 9.7.
I_VDA_STRUCTURE_EXPLODE_FE = 62	Available since version 9.7.
I_VDA_STRUCTURE_LOCAL_SYSTEM_BARS = 63	Available since version 9.7.
I_VDA_STRUCTURE_LOCAL_SYSTEM_PANELS = 64	Available since version 9.7.
I_VDA_STRUCTURE_LOCAL_SYSTEM_FE = 65	Available since version 9.7.
I_VDA_STRUCTURE_BAR_NAMES = 66	Available since version 9.7.

(c) 2014 Autodesk Inc. All rights reserved

I_VDA_SECTIONS_CODE_GROUPS = 67	Available since version 9.7.
I_VDA_LOADS_SYMBOLS_CONCENTRATED = 68	Symbols of concentrated loads. Available since version 9.7.
I_VDA_LOADS_SYMBOLS_LINEAR = 69	Symbols of linear loads. Available since version 9.7.
I_VDA_LOADS_SYMBOLS_PLANAR = 70	Symbols of planar loads. Available since version 9.7.
I_VDA_LOADS_SYMBOLS_UNIFORM_SIZE = 71	Uniform size of load symbols. Available since version 9.7.
I_VDA_LOADS_AUTOMATICALLY = 72	Automatically-generated forces. Available since version 9.7.
I_VDA_LOADS_DISTRIBUTION_REGIONS = 73	Regions of load distribution. Available since version 9.7.
I_VDA_STRUCTURE_SUPPORT_DETAILED_SYMBOLS = 74	Detailed support symbols. Available since version 9.7.
I_VDA_STRUCTURE_SUPPORT_DIRECTION = 75	Symbol of support direction. Available since version 9.7.
I_VDA_ADVANCED_STORIES_COLORS = 76	Color designation of stories. Available since version 9.7.
I_VDA_ADVANCED_STORIES_RESULTS = 77	Results for stories. Available since version 9.7.
I_VDA_ADVANCED_RC_BEAM = 78	RC beams. Available since version 9.7.
I_VDA_ADVANCED_RC_COLUMN = 79	RC columns. Available since version 9.7.
I_VDA_ADVANCED_RC_FOUNDATION = 80	RC foundations. Available since version 9.7.
I_VDA_ADVANCED_RC_CONTINOUS_FOOTING = 81	RC continuous footings. Available since version 9.7.
I_VDA_ADVANCED_RC_WALL = 82	RC walls. Available since version 9.7.
I_VDA_ADVANCED_RC_DEEP_BEAM = 83	RC deep beams. Available since version 9.7.
I_VDA_ADVANCED_STEEL_CONNECTION_NUMBERS = 84	Numbers of steel connections. Available since version 9.7.
I_VDA_ADVANCED_STEEL_CONNECTION_NAMES = 85	Names of steel connections. Available since version 9.7.
I_VDA_ADVANCED_STEEL_CONNECTION_TYPES = 86	Types of steel connections. Available since version 9.7.
I_VDA_FE_CLADDING_INTERIOR = 87	Filling of claddings. Available since version 9.7.
I_VDA_FE_FE_INTERIOR = 88	Filling of FEs. Available since version 9.7.
I_VDA_FE_PANEL_THICKNESSES = 89	Panel thickness. Available since version 9.7.
I_VDA_FE_PANEL_REINFORCEMENT = 90	Panel reinforcement. Available since version 9.7.
I_VDA_FE_PANEL_NAMES = 91	Panel names. Available since version 9.7.
I_VDA_FE_LOAD_DIRECTION = 93	Direction of load distribution. Available since version 9.7.

(c) 2014 Autodesk Inc. All rights reserved

I_VDA_FE_CHARACTERISTIC_POINTS = 94	Characteristic points. Available since version 9.7.
I_VDA_FE_AUXILIARY_OBJECTS = 95	Auxiliary objects. Available since version 9.7.
I_VDA_FE_CLADDING_COLORS = 96	Colors of claddings. Available since version 9.7.
I_VDA_OTHER_STRUCTURAL_AXIS_DESCRIPTION = 97	Labels of structure axes. Available since version 9.7.
I_VDA_VIEWOGL_HIDE_INVISIBLE_LINES = 98	Hide invisible lines. Available since version 9.7.
I_VDA_VIEWOGL_COLOR_FACES = 99	Color faces. Available since version 9.7.
I_VDA_VIEWOGL_LIGHT = 100	Light. Available since version 9.7.
I_VDA_VIEWOGL_LIGHT_ON_MAPS = 101	Light on maps. Available since version 9.7.
I_VDA_VIEWOGL_TRANSLUCENT = 102	Translucent faces. Available since version 9.7.
I_VDA_VIEWOGL_BLACK_EDGES = 103	Black edges. Available since version 9.7.
I_VDA_VIEWOGL_ANTYALIASING = 104	Antialiasing. Available since version 9.7.
I_VDA_VIEWOGL_HIDE_XY_CUTTING_PLANES = 105	Hide XY cutting plane. Available since version 9.7.
I_VDA_VIEWOGL_HIDE_XZ_CUTTING_PLANES = 106	Hide XZ cutting plane. Available since version 9.7.
I_VDA_VIEWOGL_HIDE_YZ_CUTTING_PLANES = 107	Hide YZ cutting plane. Available since version 9.7.
I_VDA_VIEWOGL_DRAW_OUT_OF_SCREEN = 108	Draw objects positioned out of screen. Available since version 9.7.
I_VDA_VIEWOGL_REDRAW = 109	Full redrawing after modification. Available since version 9.7.
I_VDA_VIEWOGL_DETAILS = 110	Draw details during structure viewing. Available since version 9.7.
I_VDA_VIEW_HIDDEN_LINES = 111	Hidden lines. Available since version 9.7.
I_VDA_VIEW_COLOR = 112	Colored sides. Available since version 9.7.
I_VDA_VIEW_SHADING = 113	Shading. Available since version 9.7.
I_VDA_VIEW_SHADING_EDGES = 114	Shading/edges. Available since version 9.7.
I_VDA_VIEW_QUICK_SHADING = 115	Quick shading of faces. Available since version 9.7.
I_VDA_VIEW_NONE = 116	Available since version 9.7.
I_VDA_LOADS_PRESSURE_MAP = 117	Available since version 14.5.

Description

Set of attributes whose visibility may be switched on/off for graphical views. .

(c) 2014 Autodesk Inc. All rights reserved

I.2.13 IRobotViewHiddenLinesDisplayType

C++

```
enum IRobotViewHiddenLinesDisplayType;
```

C#

```
public enum IRobotViewHiddenLinesDisplayType;
```

Visual Basic

```
Public Enum IRobotViewHiddenLinesDisplayType
```

Members

Members	Description
I_VHLDT_HIDDEN_LINES = 0x04	Available since version 2.0.
I_VHLDT_SHADING = 0x10	Available since version 2.0.
I_VHLDT_COLOR_SIDES_EDGES = 0x40	Available since version 2.0.
I_VHLDT_SHADING_EDGES = 0x80	Available since version 2.0.
I_VHLDT_QUICK_SHADING_OF_FACES = 0x100	Available since version 2.0.
I_VHLDT_NONE = 0	Available since version 2.0.

Description

Possible types of displaying hidden lines.

I.2.14 IRobotViewDiagramParams

Class Hierarchy

RobotOM.IRobotViewDiagramParams

C++

```
interface IRobotViewDiagramParams : IDispatch;
```

C#

```
public interface IRobotViewDiagramParams;
```

Visual Basic

```
Public Interface IRobotViewDiagramParams
```

Description

Interface controlling the manner of displaying diagrams on bars. .

I.2.14.1 IRobotViewDiagramParams Members

The following tables list the members exposed by IRobotViewDiagramParams.

Public Fields

	Name	Description
	Descriptions (see page 1423)	The manner of locating the diagram description Available since version 2.0.
	Filling (see page 1423)	Type of diagram filling Available since version 2.0.
	PositiveNegative (see page 1424)	Manner of differentiating the values of different signs on the diagrams Available since version 2.0.
	ReactionsInLocalSystem (see page 1424)	Available since version 2.0.

(c) 2014 Autodesk Inc. All rights reserved

	Values (see page 1424)	

Public Methods

	Name	Description
	GetColor (see page 1425)	Function returns color which is used while drawing a diagram of a specified result type. Available since version 2.0.
	GetScale (see page 1425)	Function returns scale value for a specified result diagram for 1 cm. Available since version 2.0.
	IsOn (see page 1425)	Function returns value different from zero (True) if a specified result type is switched on (i.e. it is displayed in the view). Available since version 2.0.
	Set (see page 1426)	Function switches on/off diagram display for a specified result type. Available since version 2.0.
	SetColor (see page 1426)	Function sets color which is to be used while drawing a diagram of a specified result type. Available since version 2.0.
	SetScale (see page 1426)	Function sets a diagram scale for 1 cm for a specified result type. If the same scale is valid for several associated results it is set for all of them. If scale definition for a specified result type makes no sense, then scale value is ignored. Available since version 2.0.

I.2.14.2 IRobotViewDiagramParams Fields

The fields of the IRobotViewDiagramParams class are listed here.

Public Fields

	Name	Description
	Descriptions (see page 1423)	The manner of locating the diagram description Available since version 2.0.
	Filling (see page 1423)	Type of diagram filling Available since version 2.0.
	PositiveNegative (see page 1424)	Manner of differentiating the values of different signs on the diagrams Available since version 2.0.
	ReactionsInLocalSystem (see page 1424)	Available since version 2.0.
	Values (see page 1424)	

I.2.14.2.1 Descriptions

C++

```
HRESULT get_Descriptions(IRobotViewDiagramDescriptionType*);
HRESULT put_Descriptions(IRobotViewDiagramDescriptionType);
```

C#

```
public IRobotViewDiagramDescriptionType Descriptions { get; set; }
```

Visual Basic

```
Public Descriptions As IRobotViewDiagramDescriptionType
```

Description

The manner of locating the diagram description Available since version 2.0.

I.2.14.2.2 Filling

C++

```
HRESULT get_Filling(IRobotViewDiagramFillingType*);
HRESULT put_Filling(IRobotViewDiagramFillingType);
```

C#

```
public IRobotViewDiagramFillingType Filling { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Filling As IRobotViewDiagramFillingType
```

Description

Type of diagram filling Available since version 2.0.

I.2.14.2.3 PositiveNegative

C++

```
HRESULT get_PositiveNegative(IRobotViewDiagramSignDifferType*);
HRESULT put_PositiveNegative(IRobotViewDiagramSignDifferType);
```

C#

```
public IRobotViewDiagramSignDifferType PositiveNegative { get; set; }
```

Visual Basic

```
Public PositiveNegative As IRobotViewDiagramSignDifferType
```

Description

Manner of differentiating the values of different signs on the diagrams Available since version 2.0.

I.2.14.2.4 ReactionsInLocalSystem

C++

```
HRESULT get_ReactionsInLocalSystem(VARIANT_BOOL*);
HRESULT put_ReactionsInLocalSystem(VARIANT_BOOL);
```

C#

```
public bool ReactionsInLocalSystem { get; set; }
```

Visual Basic

```
Public ReactionsInLocalSystem As Boolean
```

Description

Available since version 2.0.

I.2.14.2.5 Values

C++

```
HRESULT get_Values(IRobotViewDiagramValueType*);
HRESULT put_Values(IRobotViewDiagramValueType);
```

C#

```
public IRobotViewDiagramValueType Values { get; set; }
```

Visual Basic

```
Public Values As IRobotViewDiagramValueType
```

Version

Available since version 8.1.

I.2.14.3 IRobotViewDiagramParams Methods

The methods of the IRobotViewDiagramParams class are listed here.

(c) 2014 Autodesk Inc. All rights reserved

Public Methods

	Name	Description
	GetColor (see page 1425)	Function returns color which is used while drawing a diagram of a specified result type. Available since version 2.0.
	GetScale (see page 1425)	Function returns scale value for a specified result diagram for 1 cm. Available since version 2.0.
	IsOn (see page 1425)	Function returns value different from zero (True) if a specified result type is switched on (i.e. it is displayed in the view). Available since version 2.0.
	Set (see page 1426)	Function switches on/off diagram display for a specified result type. Available since version 2.0.
	SetColor (see page 1426)	Function sets color which is to be used while drawing a diagram of a specified result type. Available since version 2.0.
	SetScale (see page 1426)	Function sets a diagram scale for 1 cm for a specified result type. If the same scale is valid for several associated results it is set for all of them. If scale definition for a specified result type makes no sense, then scale value is ignored. Available since version 2.0.

I.2.14.3.1 GetColor

C++

```
HRESULT GetColor(IRobotViewDiagramResultType _res_type, long* ret);
```

C#

```
public long GetColor(IRobotViewDiagramResultType _res_type);
```

Visual Basic

```
Public Function GetColor(_res_type As IRobotViewDiagramResultType) As long
```

Description

Function returns color which is used while drawing a diagram of a specified result type. Available since version 2.0.

I.2.14.3.2 GetScale

C++

```
HRESULT GetScale(IRobotViewDiagramResultType _res_type, double* ret);
```

C#

```
public double GetScale(IRobotViewDiagramResultType _res_type);
```

Visual Basic

```
Public Function GetScale(_res_type As IRobotViewDiagramResultType) As double
```

Description

Function returns scale value for a specified result diagram for 1 cm. Available since version 2.0.

I.2.14.3.3 IsOn

C++

```
HRESULT IsOn(IRobotViewDiagramResultType _res_type, VARIANT_BOOL* ret);
```

C#

```
public bool IsOn(IRobotViewDiagramResultType _res_type);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Function IsOn(_res_type As IRobotViewDiagramResultType) As Boolean
```

Description

Function returns value different from zero (True) if a specified result type is switched on (i.e. it is displayed in the view). Available since version 2.0.

I.2.14.3.4 Set

C++

```
HRESULT Set(IRobotViewDiagramResultType _res_type, VARIANT_BOOL _on_off);
```

C#

```
public void Set(IRobotViewDiagramResultType _res_type, bool _on_off);
```

Visual Basic

```
Public Sub Set(_res_type As IRobotViewDiagramResultType, _on_off As Boolean)
```

Description

Function switches on/off diagram display for a specified result type. Available since version 2.0.

I.2.14.3.5 SetColor

C++

```
HRESULT SetColor(IRobotViewDiagramResultType _res_type, long _color);
```

C#

```
public void SetColor(IRobotViewDiagramResultType _res_type, long _color);
```

Visual Basic

```
Public Sub SetColor(_res_type As IRobotViewDiagramResultType, _color As long)
```

Description

Function sets color which is to be used while drawing a diagram of a specified result type. Available since version 2.0.

I.2.14.3.6 SetScale

C++

```
HRESULT SetScale(IRobotViewDiagramResultType _res_type, double _scale_for_1_cm);
```

C#

```
public void SetScale(IRobotViewDiagramResultType _res_type, double _scale_for_1_cm);
```

Visual Basic

```
Public Sub SetScale(_res_type As IRobotViewDiagramResultType, _scale_for_1_cm As double)
```

Description

Function sets a diagram scale for 1 cm for a specified result type. If the same scale is valid for several associated results it is set for all of them. If scale definition for a specified result type makes no sense, then scale value is ignored. Available since version 2.0.

(c) 2014 Autodesk Inc. All rights reserved

I.2.15 IRobotViewDiagramResultType

C++

```
enum IRobotViewDiagramResultType;
```

C#

```
public enum IRobotViewDiagramResultType;
```

Visual Basic

```
Public Enum IRobotViewDiagramResultType
```

Members

Members	Description
I_VDRT_NTM_FX = 1	Available since version 2.0.
I_VDRT_NTM_FY = 2	Available since version 2.0.
I_VDRT_NTM_FZ = 3	Available since version 2.0.
I_VDRT_NTM_MX = 4	Available since version 2.0.
I_VDRT_NTM_MY = 5	Available since version 2.0.
I_VDRT_NTM_MZ = 6	Available since version 2.0.
I_VDRT_DEFORMATION_DEFORMATION = 7	Available since version 2.0.
I_VDRT_DEFORMATION_EXACT = 8	Available since version 2.0.
I_VDRT_STRESS_S_MAX = 9	Available since version 2.0.
I_VDRT_STRESS_S_MIN = 10	Available since version 2.0.
I_VDRT_STRESS_BENDING_S_MAX_MY = 11	Available since version 2.0.
I_VDRT_STRESS_BENDING_S_MAX_MZ = 12	Available since version 2.0.
I_VDRT_STRESS_BENDING_S_MIN_MY = 13	Available since version 2.0.
I_VDRT_STRESS_BENDING_S_MIN_MZ = 14	Available since version 2.0.
I_VDRT_STRESS_AXIAL_FX_AX = 15	Available since version 2.0.
I_VDRT_STRESS_SHEAR_TY = 16	Available since version 2.0.
I_VDRT_STRESS_SHEAR_TZ = 17	Available since version 2.0.
I_VDRT_STRESS_TORSION = 18	Available since version 2.0.
I_VDRT_REACTION_FORCES = 19	Available since version 2.0.
I_VDRT_REACTION_MOMENTS = 20	Available since version 2.0.
I_VDRT_REACTION_RESIDUAL_FORCES = 21	Available since version 2.0.
I_VDRT_REACTION_RESIDUAL_MOMENTS = 22	Available since version 2.0.
I_VDRT_REINFORCE_TOP = 23	Available since version 2.0.
I_VDRT_REINFORCE_TOP_REAL = 24	Available since version 2.0.
I_VDRT_REINFORCE_BOTTOM = 25	Available since version 2.0.
I_VDRT_REINFORCE_BOTTOM_REAL = 26	Available since version 2.0.
I_VDRT_REINFORCE_RATIO = 27	Available since version 2.0.
I_VDRT_REINFORCE_RATIO_REAL = 28	Available since version 2.0.
I_VDRT_REINFORCE_STIRRUP_SPACING = 29	Available since version 2.0.
I_VDRT_REINFORCE_STIRRUP_SPACING_REAL = 30	Available since version 2.0.
I_VDRT_REINFORCE_NUMBER_OF_UPPER_BARS = 31	Available since version 2.0.
I_VDRT_REINFORCE_NUMBER_OF_LOWER_BARS = 32	Available since version 2.0.
I_VDRT_NTM_KY = 33	Coefficient of elastic ground reaction in Y direction. Available since version 3.
I_VDRT_NTM_KZ = 34	Coefficient of elastic ground reaction in Z direction. Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

Description

Set of result types presented by means of diagrams on bars.

I.2.16 IRobotViewDiagramDescriptionType

C++

```
enum IRobotViewDiagramDescriptionType;
```

C#

```
public enum IRobotViewDiagramDescriptionType;
```

Visual Basic

```
Public Enum IRobotViewDiagramDescriptionType
```

Members

Members	Description
I_VDDT_NONE = 0	Available since version 2.0.
I_VDDT_LABELS = 1	Available since version 8.1.
I_VDDT_TEXT = 2	Available since version 8.1.

Description

Available methods of locating descriptions for diagrams.

I.2.17 IRobotViewDiagramFillingType

C++

```
enum IRobotViewDiagramFillingType;
```

C#

```
public enum IRobotViewDiagramFillingType;
```

Visual Basic

```
Public Enum IRobotViewDiagramFillingType
```

Members

Members	Description
I_VDFT_FENCE = 0	Available since version 2.0.
I_VDFT_FILLED = 1	Available since version 2.0.

Description

Available methods of diagram filling.

I.2.18 IRobotViewDiagramSignDifferType

C++

```
enum IRobotViewDiagramSignDifferType;
```

C#

```
public enum IRobotViewDiagramSignDifferType;
```

Visual Basic

```
Public Enum IRobotViewDiagramSignDifferType
```

(c) 2014 Autodesk Inc. All rights reserved

Members

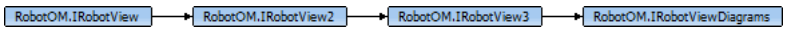
Members	Description
I_VDSDT_UNDIFFERENTIATED = 0	Available since version 2.0.
I_VDSDT_DIFFERENTIATED = 1	Available since version 2.0.

Description

Available methods of distinguishing positive and negative values on diagrams.

I.2.19 IRobotViewDiagrams

Class Hierarchy



C++

```
interface IRobotViewDiagrams : IRobotView3;
```

C#

```
public interface IRobotViewDiagrams : IRobotView3;
```

Visual Basic

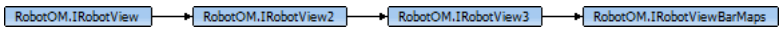
```
Public Interface IRobotViewDiagrams
```

Description

A viewer dedicated to result presentation.

I.2.20 IRobotViewBarMaps

Class Hierarchy



C++

```
interface IRobotViewBarMaps : IRobotView3;
```

C#

```
public interface IRobotViewBarMaps : IRobotView3;
```

Visual Basic

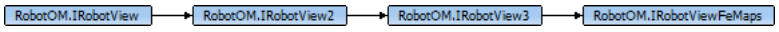
```
Public Interface IRobotViewBarMaps
```

Description

View presenting maps on bars with a scale. .

I.2.21 IRobotViewFeMaps

Class Hierarchy



C++

```
interface IRobotViewFeMaps : IRobotView3;
```

C#

```
public interface IRobotViewFeMaps : IRobotView3;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Interface IRobotViewFeMaps
```

Description

View presenting maps on finite elements with a scale. .

I.2.22 IRobotViewBarMapResultType

C++

```
enum IRobotViewBarMapResultType;
```

C#

```
public enum IRobotViewBarMapResultType;
```

Visual Basic

```
Public Enum IRobotViewBarMapResultType
```

Members

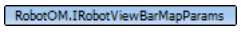
Members	Description
I_VBMRT_NTM_FX = 165	Available since version 2.0.
I_VBMRT_NTM_FY = 166	Available since version 2.0.
I_VBMRT_NTM_FZ = 167	Available since version 2.0.
I_VBMRT_NTM_MX = 168	Available since version 2.0.
I_VBMRT_NTM_MY = 169	Available since version 2.0.
I_VBMRT_NTM_MZ = 170	Available since version 2.0.
I_VBMRT_STRESS_S_MAX = 171	Available since version 2.0.
I_VBMRT_STRESS_S_MIN = 172	Available since version 2.0.
I_VBMRT_STRESS_S_MAX_MY = 174	Available since version 2.0.
I_VBMRT_STRESS_S_MIN_MY = 176	Available since version 2.0.
I_VBMRT_STRESS_S_MAX_MZ = 175	Available since version 2.0.
I_VBMRT_STRESS_S_MIN_MZ = 177	Available since version 2.0.
I_VBMRT_STRESS_FX_AX = 173	Available since version 2.0.
I_VBMRT_STRESS_SHEAR_TY = 179	Available since version 2.0.
I_VBMRT_STRESS_SHEAR_TZ = 180	Available since version 2.0.
I_VBMRT_STRESS_TORSION_T = 182	Available since version 2.0.
I_VBMRT_DESIGN_RATIO = 796	Available since version 2.0.
I_VBMRT_DESIGN_MEMBER_LENGTH = 271	Available since version 2.0.
I_VBMRT_DESIGN_SLEND_LAY = 831	Available since version 2.0.
I_VBMRT_DESIGN_SLEND_LAZ = 832	Available since version 2.0.
I_VBMRT_NOTHING = 0	Available since version 2.0.

Description

Set of result types presented by means of maps on bars.

I.2.23 IRobotViewBarMapParams

Class Hierarchy



C++

```
interface IRobotViewBarMapParams : IDispatch;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public interface IRobotViewBarMapParams;
```

Visual Basic

```
Public Interface IRobotViewBarMapParams
```

Description

Parameters describing the manner of displaying maps on bars.

I.2.23.1 IRobotViewBarMapParams Members

The following tables list the members exposed by IRobotViewBarMapParams.

Public Fields

	Name	Description
	CurrentResult (see page 1431)	Result type for which a map on bars is currently displayed.
	Descriptions (see page 1431)	Manner of displaying map description Available since version 2.0.
	MapThicknessCoeff (see page 1432)	Thickness coefficient of a map drawing line in relation to the thickness of a bar drawing line Available since version 2.0.
	StructureDeformation (see page 1432)	Flag switching on/off structure deformation drawing Available since version 2.0.

I.2.23.2 IRobotViewBarMapParams Fields

The fields of the IRobotViewBarMapParams class are listed here.

Public Fields

	Name	Description
	CurrentResult (see page 1431)	Result type for which a map on bars is currently displayed.
	Descriptions (see page 1431)	Manner of displaying map description Available since version 2.0.
	MapThicknessCoeff (see page 1432)	Thickness coefficient of a map drawing line in relation to the thickness of a bar drawing line Available since version 2.0.
	StructureDeformation (see page 1432)	Flag switching on/off structure deformation drawing Available since version 2.0.

I.2.23.2.1 CurrentResult

C++

```
HRESULT get_CurrentResult(IRobotViewBarMapResultType*);
HRESULT put_CurrentResult(IRobotViewBarMapResultType);
```

C#

```
public IRobotViewBarMapResultType CurrentResult { get; set; }
```

Visual Basic

```
Public CurrentResult As IRobotViewBarMapResultType
```

Description

Result type for which a map on bars is currently displayed.

I.2.23.2.2 Descriptions

C++

```
HRESULT get_Descriptions(IRobotViewDiagramDescriptionType*);
HRESULT put_Descriptions(IRobotViewDiagramDescriptionType);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRobotViewDiagramDescriptionType Descriptions { get; set; }
```

Visual Basic

```
Public Descriptions As IRobotViewDiagramDescriptionType
```

Description

Manner of displaying map description Available since version 2.0.

I.2.23.2.3 MapThicknessCoeff

C++

```
HRESULT get_MapThicknessCoeff(double*);
HRESULT put_MapThicknessCoeff(double);
```

C#

```
public double MapThicknessCoeff { get; set; }
```

Visual Basic

```
Public MapThicknessCoeff As double
```

Description

Thickness coefficient of a map drawing line in relation to the thickness of a bar drawing line Available since version 2.0.

I.2.23.2.4 StructureDeformation

C++

```
HRESULT get_StructureDeformation(VARIANT_BOOL*);
HRESULT put_StructureDeformation(VARIANT_BOOL);
```

C#

```
public bool StructureDeformation { get; set; }
```

Visual Basic

```
Public StructureDeformation As Boolean
```

Description

Flag switching on/off structure deformation drawing Available since version 2.0.

I.2.24 IRobotViewDetailedAnalysisResultType

C++

```
enum IRobotViewDetailedAnalysisResultType;
```

C#

```
public enum IRobotViewDetailedAnalysisResultType;
```

Visual Basic

```
Public Enum IRobotViewDetailedAnalysisResultType
```

Members

Members	Description
I_VDART_NTM_FX = 1	Available since version 2.0.
I_VDART_NTM_FY = 2	Available since version 2.0.
I_VDART_NTM_FZ = 3	Available since version 2.0.

(c) 2014 Autodesk Inc. All rights reserved

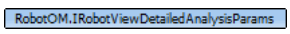
I_VDART_NTM_MX = 4	Available since version 2.0.
I_VDART_NTM_MY = 5	Available since version 2.0.
I_VDART_NTM_MZ = 6	Available since version 2.0.
I_VDART_NTM_UX = 7	Available since version 2.0.
I_VDART_NTM_UY = 8	Available since version 2.0.
I_VDART_NTM_UZ = 9	Available since version 2.0.
I_VDART_STRESS_S_MAX = 10	Available since version 2.0.
I_VDART_STRESS_S_MIN = 11	Available since version 2.0.
I_VDART_STRESS_BENDING_S_MAX_MY = 12	Available since version 2.0.
I_VDART_STRESS_BENDING_S_MIN_MY = 13	Available since version 2.0.
I_VDART_STRESS_BENDING_S_MAX_MZ = 14	Available since version 2.0.
I_VDART_STRESS_BENDING_S_MIN_MZ = 15	Available since version 2.0.
I_VDART_STRESS_AXIAL_FX_AX = 16	Available since version 2.0.
I_VDART_STRESS_SHEAR_TY = 17	Available since version 2.0.
I_VDART_STRESS_SHEAR_TZ = 18	Available since version 2.0.
I_VDART_STRESS_TORSION_T = 19	Available since version 2.0.
I_VDART_REINFORCE_TOP = 20	Available since version 2.0.
I_VDART_REINFORCE_TOP_REAL = 21	Available since version 2.0.
I_VDART_REINFORCE_BOTTOM = 22	Available since version 2.0.
I_VDART_REINFORCE_BOTTOM_REAL = 23	Available since version 2.0.
I_VDART_REINFORCE_RATIO = 24	Available since version 2.0.
I_VDART_REINFORCE_RATIO_REAL = 25	Available since version 2.0.
I_VDART_REINFORCE_STIRRUP_SPACING = 26	Available since version 2.0.
I_VDART_REINFORCE_STIRRUP_SPACING_REAL = 27	Available since version 2.0.
I_VDART_REINFORCE_NUMBER_OF_UPPER_BARS = 28	Available since version 2.0.
I_VDART_REINFORCE_NUMBER_OF_LOWER_BARS = 29	Available since version 2.0.

Description

Result types presented by detailed analysis view. .

I.2.25 IRobotViewDetailedAnalysisParams

Class Hierarchy



C++

```
interface IRobotViewDetailedAnalysisParams : IDispatch;
```

C#

```
public interface IRobotViewDetailedAnalysisParams;
```

Visual Basic

```
Public Interface IRobotViewDetailedAnalysisParams
```

Description

Parameters of detailed analysis view.

I.2.25.1 IRobotViewDetailedAnalysisParams Members

The following tables list the members exposed by IRobotViewDetailedAnalysisParams.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	Descriptions (see page 1434)	Manner of displaying diagram description Available since version 2.0.
	Filling (see page 1434)	Type of diagram filling Available since version 2.0.
	PositiveNegative (see page 1435)	Manner of presenting values of different signs on the diagram Available since version 2.0.
	ReinforceShowTheoreticAndRealVals (see page 1435)	Available since version 2.0.

Public Methods

	Name	Description
	GetColor (see page 1435)	Function returns color assigned to a specified result type. Available since version 2.0.
	IsOn (see page 1436)	Function returns value different from zero (True) if a selected result type is displayed by the detailed analysis view. Available since version 2.0.
	Set (see page 1436)	Function switches on/off display of a defined result type. Available since version 2.0.
	SetColor (see page 1436)	Function sets color which is to be used while drawing a diagram for a specified result type. Available since version 2.0.

I.2.25.2 IRobotViewDetailedAnalysisParams Fields

The fields of the IRobotViewDetailedAnalysisParams class are listed here.

Public Fields

	Name	Description
	Descriptions (see page 1434)	Manner of displaying diagram description Available since version 2.0.
	Filling (see page 1434)	Type of diagram filling Available since version 2.0.
	PositiveNegative (see page 1435)	Manner of presenting values of different signs on the diagram Available since version 2.0.
	ReinforceShowTheoreticAndRealVals (see page 1435)	Available since version 2.0.

I.2.25.2.1 Descriptions

C++

```
HRESULT get_Descriptions(IRobotViewDiagramDescriptionType*);
HRESULT put_Descriptions(IRobotViewDiagramDescriptionType);
```

C#

```
public IRobotViewDiagramDescriptionType Descriptions { get; set; }
```

Visual Basic

```
Public Descriptions As IRobotViewDiagramDescriptionType
```

Description

Manner of displaying diagram description Available since version 2.0.

I.2.25.2.2 Filling

C++

```
HRESULT get_Filling(IRobotViewDiagramFillingType*);
HRESULT put_Filling(IRobotViewDiagramFillingType);
```

C#

```
public IRobotViewDiagramFillingType Filling { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Filling As IRobotViewDiagramFillingType
```

Description

Type of diagram filling Available since version 2.0.

I.2.25.2.3 PositiveNegative

C++

```
HRESULT get_PositiveNegative(IRobotViewDiagramSignDifferType*);
HRESULT put_PositiveNegative(IRobotViewDiagramSignDifferType);
```

C#

```
public IRobotViewDiagramSignDifferType PositiveNegative { get; set; }
```

Visual Basic

```
Public PositiveNegative As IRobotViewDiagramSignDifferType
```

Description

Manner of presenting values of different signs on the diagram Available since version 2.0.

I.2.25.2.4 ReinforceShowTheoreticAndRealVals

C++

```
HRESULT get_ReinforceShowTheoreticAndRealVals(VARIANT_BOOL*);
HRESULT put_ReinforceShowTheoreticAndRealVals(VARIANT_BOOL);
```

C#

```
public bool ReinforceShowTheoreticAndRealVals { get; set; }
```

Visual Basic

```
Public ReinforceShowTheoreticAndRealVals As Boolean
```

Description

Available since version 2.0.

I.2.25.3 IRobotViewDetailedAnalysisParams Methods

The methods of the IRobotViewDetailedAnalysisParams class are listed here.

Public Methods

	Name	Description
	GetColor (see page 1435)	Function returns color assigned to a specified result type. Available since version 2.0.
	IsOn (see page 1436)	Function returns value different from zero (True) if a selected result type is displayed by the detailed analysis view. Available since version 2.0.
	Set (see page 1436)	Function switches on/off display of a defined result type. Available since version 2.0.
	SetColor (see page 1436)	Function sets color which is to be used while drawing a diagram for a specified result type. Available since version 2.0.

I.2.25.3.1 GetColor

C++

```
HRESULT GetColor(IRobotViewDetailedAnalysisResultType _res_type, long* ret);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public long GetColor(IRobotViewDetailedAnalysisResultType _res_type);
```

Visual Basic

```
Public Function GetColor(_res_type As IRobotViewDetailedAnalysisResultType) As long
```

Description

Function returns color assigned to a specified result type. Available since version 2.0.

I.2.25.3.2 IsOn

C++

```
HRESULT IsOn(IRobotViewDetailedAnalysisResultType _res_type, VARIANT_BOOL* ret);
```

C#

```
public bool IsOn(IRobotViewDetailedAnalysisResultType _res_type);
```

Visual Basic

```
Public Function IsOn(_res_type As IRobotViewDetailedAnalysisResultType) As Boolean
```

Description

Function returns value different from zero (True) if a selected result type is displayed by the detailed analysis view. Available since version 2.0.

I.2.25.3.3 Set

C++

```
HRESULT Set(IRobotViewDetailedAnalysisResultType _res_type, VARIANT_BOOL _on_off);
```

C#

```
public void Set(IRobotViewDetailedAnalysisResultType _res_type, bool _on_off);
```

Visual Basic

```
Public Sub Set(_res_type As IRobotViewDetailedAnalysisResultType, _on_off As Boolean)
```

Description

Function switches on/off display of a defined result type. Available since version 2.0.

I.2.25.3.4 SetColor

C++

```
HRESULT SetColor(IRobotViewDetailedAnalysisResultType _res_type, long _color);
```

C#

```
public void SetColor(IRobotViewDetailedAnalysisResultType _res_type, long _color);
```

Visual Basic

```
Public Sub SetColor(_res_type As IRobotViewDetailedAnalysisResultType, _color As long)
```

Description

Function sets color which is to be used while drawing a diagram for a specified result type. Available since version 2.0.

(c) 2014 Autodesk Inc. All rights reserved

I.2.26 IRobotViewFeMapResultType

C++

```
enum IRobotViewFeMapResultType;
```

C#

```
public enum IRobotViewFeMapResultType;
```

Visual Basic

```
Public Enum IRobotViewFeMapResultType
```

Members

Members	Description
I_VFMRT_DETAILED_STRESS_XX = 483	Available since version 2.0.
I_VFMRT_DETAILED_STRESS_YY = 484	Available since version 2.0.
I_VFMRT_DETAILED_STRESS_XY = 485	Available since version 2.0.
I_VFMRT_DETAILED_STRESS_Z = 486	Available since version 2.0.
I_VFMRT_DETAILED_MEMBRANE_FORCE_XX = 492	Available since version 2.0.
I_VFMRT_DETAILED_MEMBRANE_FORCE_YY = 493	Available since version 2.0.
I_VFMRT_DETAILED_MEMBRANE_FORCE_XY = 494	Available since version 2.0.
I_VFMRT_DETAILED_MOMENT_XX = 501	Available since version 2.0.
I_VFMRT_DETAILED_MOMENT_YY = 502	Available since version 2.0.
I_VFMRT_DETAILED_MOMENT_XY = 503	Available since version 2.0.
I_VFMRT_DETAILED_SHEAR_STRESS_XX = 510	Available since version 2.0.
I_VFMRT_DETAILED_SHEAR_STRESS_YY = 511	Available since version 2.0.
I_VFMRT_DETAILED_SHEAR_FORCE_XX = 519	Available since version 2.0.
I_VFMRT_DETAILED_SHEAR_FORCE_YY = 520	Available since version 2.0.
I_VFMRT_DETAILED_DISPLACEMENT_XX = 537	Available since version 2.0.
I_VFMRT_DETAILED_DISPLACEMENT_YY = 538	Available since version 2.0.
I_VFMRT_DETAILED_ROTATION_XX = 546	Available since version 2.0.
I_VFMRT_DETAILED_ROTATION_YY = 547	Available since version 2.0.
I_VFMRT_DETAILED_ROTATION_Z = 549	Available since version 2.0.
I_VFMRT_DETAILED_SOIL_REACTION_Z = 558	Available since version 2.0.
I_VFMRT_PRINCIPAL_STRESS_1 = 487	Available since version 2.0.
I_VFMRT_PRINCIPAL_STRESS_2 = 488	Available since version 2.0.
I_VFMRT_PRINCIPAL_STRESS_1_2 = 490	Available since version 2.0.
I_VFMRT_PRINCIPAL_STRESS_ANGLE = 489	Available since version 2.0.
I_VFMRT_PRINCIPAL_MEMBRANE_FORCE_1 = 496	Available since version 2.0.
I_VFMRT_PRINCIPAL_MEMBRANE_FORCE_2 = 497	Available since version 2.0.
I_VFMRT_PRINCIPAL_MEMBRANE_FORCE_1_2 = 499	Available since version 2.0.
I_VFMRT_PRINCIPAL_MEMBRANE_FORCE_ANGLE = 498	Available since version 2.0.
I_VFMRT_PRINCIPAL_MOMENT_1 = 505	Available since version 2.0.
I_VFMRT_PRINCIPAL_MOMENT_2 = 506	Available since version 2.0.
I_VFMRT_PRINCIPAL_MOMENT_1_2 = 508	Available since version 2.0.
I_VFMRT_PRINCIPAL_MOMENT_ANGLE = 507	Available since version 2.0.
I_VFMRT_PRINCIPAL_SHEAR_STRESS_1_2 = 517	Available since version 2.0.
I_VFMRT_PRINCIPAL_SHEAR_FORCE_1_2 = 526	Available since version 2.0.
I_VFMRT_GLOBAL_DISPLACEMENT_X = 1302	Available since version 2.0.
I_VFMRT_GLOBAL_DISPLACEMENT_Y = 1303	Available since version 2.0.

(c) 2014 Autodesk Inc. All rights reserved

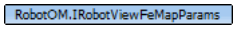
I_VFMRT_GLOBAL_DISPLACEMENT_Z = 1304	Available since version 2.0.
I_VFMRT_TOTAL_DISPLACEMENTS = 313058	Available since version 2.0.
I_VFMRT_COMPLEX_STRESSES = 491	Available since version 2.0.
I_VFMRT_COMPLEX_MEMBRANE_FORCES = 500	Available since version 2.0.
I_VFMRT_COMPLEX_MOMENTS = 509	Available since version 2.0.
I_VFMRT_COMPLEX_REINFORCE_TOP_MXX = 889	Available since version 2.0.
I_VFMRT_COMPLEX_REINFORCE_TOP_MYY = 891	Available since version 2.0.
I_VFMRT_COMPLEX_REINFORCE_BOTTOM_MXX = 890	Available since version 2.0.
I_VFMRT_COMPLEX_REINFORCE_BOTTOM_MYY = 892	Available since version 2.0.

Description

Result types presented by means of maps on finite elements. .

I.2.27 IRobotViewFeMapParams

Class Hierarchy



C++

```
interface IRobotViewFeMapParams : IDispatch;
```

C#

```
public interface IRobotViewFeMapParams;
```

Visual Basic

```
Public Interface IRobotViewFeMapParams
```

Description

Parameters of presenting maps on finite elements. .

I.2.27.1 IRobotViewFeMapParams Members

The following tables list the members exposed by IRobotViewFeMapParams.

Public Fields

	Name	Description
	CrossPresentation (see page 1439)	Available since version 2.0.
	CurrentResult (see page 1440)	Currently displayed result type Available since version 2.0.
	DeformationActive (see page 1440)	Available since version 2.0.
	DeformationConstScale (see page 1440)	Available since version 2.0.
	Direction (see page 1440)	Available since version 2.0.
	DirectionData (see page 1441)	Additional data describing direction; depending on the selected method of determining direction, it contains coordinates of the point or vector defining the direction (for the polar system defined in a node, the DirectionNode (see page 1441) variable denoting a node number should be applied) Available since version 2.0.
	DirectionNode (see page 1441)	Number of the node determining direction in case when direction is defined in the polar system by indicating a structure node Available since version 2.0.
	Isolines (see page 1441)	Available since version 2.0.
	Layer (see page 1441)	Available since version 2.0.

(c) 2014 Autodesk Inc. All rights reserved

	LayerArbitraryVal (see page 1442)	Available since version 2.0.
	MapScale (see page 1442)	Available since version 2.0.
	Smoothing (see page 1442)	Available since version 2.0.
	WithDescription (see page 1443)	Available since version 2.0.

Public Methods

	Name	Description
	SetDirectionData (see page 1443)	

I.2.27.2 IRobotViewFeMapParams Fields

The fields of the IRobotViewFeMapParams class are listed here.

Public Fields

	Name	Description
	CrossPresentation (see page 1439)	Available since version 2.0.
	CurrentResult (see page 1440)	Currently displayed result type Available since version 2.0.
	DeformationActive (see page 1440)	Available since version 2.0.
	DeformationConstScale (see page 1440)	Available since version 2.0.
	Direction (see page 1440)	Available since version 2.0.
	DirectionData (see page 1441)	Additional data describing direction; depending on the selected method of determining direction, it contains coordinates of the point or vector defining the direction (for the polar system defined in a node, the DirectionNode (see page 1441) variable denoting a node number should be applied) Available since version 2.0.
	DirectionNode (see page 1441)	Number of the node determining direction in case when direction is defined in the polar system by indicating a structure node Available since version 2.0.
	Isolines (see page 1441)	Available since version 2.0.
	Layer (see page 1441)	Available since version 2.0.
	LayerArbitraryVal (see page 1442)	Available since version 2.0.
	MapScale (see page 1442)	Available since version 2.0.
	Smoothing (see page 1442)	Available since version 2.0.
	WithDescription (see page 1443)	Available since version 2.0.

I.2.27.2.1 CrossPresentation

C++

```
HRESULT get_CrossPresentation(IRobotViewFeMapCrossPresentationType*);
HRESULT put_CrossPresentation(IRobotViewFeMapCrossPresentationType);
```

C#

```
public IRobotViewFeMapCrossPresentationType CrossPresentation { get; set; }
```

Visual Basic

```
Public CrossPresentation As IRobotViewFeMapCrossPresentationType
```

Description

Available since version 2.0.

(c) 2014 Autodesk Inc. All rights reserved

I.2.27.2.2 CurrentResult

C++

```
HRESULT get_CurrentResult(IRobotViewFeMapResultType*);
HRESULT put_CurrentResult(IRobotViewFeMapResultType);
```

C#

```
public IRobotViewFeMapResultType CurrentResult { get; set; }
```

Visual Basic

```
Public CurrentResult As IRobotViewFeMapResultType
```

Description

Currently displayed result type Available since version 2.0.

I.2.27.2.3 DeformationActive

C++

```
HRESULT get_DeformationActive(VARIANT_BOOL*);
HRESULT put_DeformationActive(VARIANT_BOOL);
```

C#

```
public bool DeformationActive { get; set; }
```

Visual Basic

```
Public DeformationActive As Boolean
```

Description

Available since version 2.0.

I.2.27.2.4 DeformationConstScale

C++

```
HRESULT get_DeformationConstScale(VARIANT_BOOL*);
HRESULT put_DeformationConstScale(VARIANT_BOOL);
```

C#

```
public bool DeformationConstScale { get; set; }
```

Visual Basic

```
Public DeformationConstScale As Boolean
```

Description

Available since version 2.0.

I.2.27.2.5 Direction

C++

```
HRESULT get_Direction(IRobotViewFeMapLocalSystemType*);
HRESULT put_Direction(IRobotViewFeMapLocalSystemType);
```

C#

```
public IRobotViewFeMapLocalSystemType Direction { get; set; }
```

Visual Basic

```
Public Direction As IRobotViewFeMapLocalSystemType
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Available since version 2.0.

I.2.27.2.6 DirectionData

C++

```
HRESULT get_DirectionData(IRobotGeoPoint3D**);
```

C#

```
public IRobotGeoPoint3D DirectionData { get; }
```

Visual Basic

```
Public ReadOnly DirectionData As IRobotGeoPoint3D
```

Description

Additional data describing direction; depending on the selected method of determining direction, it contains coordinates of the point or vector defining the direction (for the polar system defined in a node, the DirectionNode (see page 1441) variable denoting a node number should be applied) Available since version 2.0.

I.2.27.2.7 DirectionNode

C++

```
HRESULT get_DirectionNode(long*);
HRESULT put_DirectionNode(long);
```

C#

```
public long DirectionNode { get; set; }
```

Visual Basic

```
Public DirectionNode As long
```

Description

Number of the node determining direction in case when direction is defined in the polar system by indicating a structure node Available since version 2.0.

I.2.27.2.8 Isolines

C++

```
HRESULT get_Isolines(VARIANT_BOOL*);
HRESULT put_Isolines(VARIANT_BOOL);
```

C#

```
public bool Isolines { get; set; }
```

Visual Basic

```
Public Isolines As Boolean
```

Description

Available since version 2.0.

I.2.27.2.9 Layer

C++

```
HRESULT get_Layer(IRobotViewFeMapLayerType*);
HRESULT put_Layer(IRobotViewFeMapLayerType);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRobotViewFeMapLayerType Layer { get; set; }
```

Visual Basic

```
Public Layer As IRobotViewFeMapLayerType
```

Description

Available since version 2.0.

I.2.27.2.10 LayerArbitraryVal

C++

```
HRESULT get_LayerArbitraryVal(double*);
HRESULT put_LayerArbitraryVal(double);
```

C#

```
public double LayerArbitraryVal { get; set; }
```

Visual Basic

```
Public LayerArbitraryVal As double
```

Description

Available since version 2.0.

I.2.27.2.11 MapScale

C++

```
HRESULT get_MapScale(double*);
HRESULT put_MapScale(double);
```

C#

```
public double MapScale { get; set; }
```

Visual Basic

```
Public MapScale As double
```

Description

Available since version 2.0.

I.2.27.2.12 Smoothing

C++

```
HRESULT get_Smoothing(IRobotViewFeMapSmoothingType*);
HRESULT put_Smoothing(IRobotViewFeMapSmoothingType);
```

C#

```
public IRobotViewFeMapSmoothingType Smoothing { get; set; }
```

Visual Basic

```
Public Smoothing As IRobotViewFeMapSmoothingType
```

Description

Available since version 2.0.

I.2.27.2.13 WithDescription

C++

```
HRESULT get_WithDescription(VARIANT_BOOL*);
```

(c) 2014 Autodesk Inc. All rights reserved

```
HRESULT put_WithDescription(VARIANT_BOOL);
```

C#

```
public bool WithDescription { get; set; }
```

Visual Basic

```
Public WithDescription As Boolean
```

Description

Available since version 2.0.

I.2.27.3 IRobotViewFeMapParams Methods

The methods of the IRobotViewFeMapParams class are listed here.

Public Methods

	Name	Description
	SetDirectionData (see page 1443)	

I.2.27.3.1 SetDirectionData

C++

```
HRESULT SetDirectionData(IRobotGeoPoint3D* _dir_data);
```

C#

```
public void SetDirectionData(IRobotGeoPoint3D _dir_data);
```

Visual Basic

```
Public Sub SetDirectionData(ByRef _dir_data As IRobotGeoPoint3D)
```

Version

Available since version 3.5.

I.2.28 IRobotViewFeMapLocalSystemType

C++

```
enum IRobotViewFeMapLocalSystemType;
```

C#

```
public enum IRobotViewFeMapLocalSystemType;
```

Visual Basic

```
Public Enum IRobotViewFeMapLocalSystemType
```

Members

Members	Description
I_VFMLST_AUTOMATIC = 0	Available since version 2.0.
I_VFMLST_CARTESIAN_ALONG_X = 1	Available since version 2.0.
I_VFMLST_CARTESIAN_ALONG_Y = 2	Available since version 2.0.
I_VFMLST_CARTESIAN_ALONG_Z = 3	Available since version 2.0.
I_VFMLST_CARTESIAN_ALONG_VECTOR = 4	Available since version 2.0.
I_VFMLST_POLAR_IN_NODE = 5	Available since version 2.0.
I_VFMLST_POLAR_IN_POINT = 6	Available since version 2.0.

(c) 2014 Autodesk Inc. All rights reserved

I.2.29 IRobotViewFeMapCrossPresentationType

C++

```
enum IRobotViewFeMapCrossPresentationType;
```

C#

```
public enum IRobotViewFeMapCrossPresentationType;
```

Visual Basic

```
Public Enum IRobotViewFeMapCrossPresentationType
```

Members

Members	Description
I_VFMCPT_NONE = 0	Available since version 2.0.
I_VFMCPT_STRESSES = 1404	Available since version 2.0.
I_VFMCPT_NORMAL_FORCES = 1405	Available since version 2.0.
I_VFMCPT_MOMENTS = 1406	Available since version 2.0.

Description

Manner of presenting crosses in the map view for surface finite elements. .

I.2.30 IRobotViewFeMapLayerType

C++

```
enum IRobotViewFeMapLayerType;
```

C#

```
public enum IRobotViewFeMapLayerType;
```

Visual Basic

```
Public Enum IRobotViewFeMapLayerType
```

Members

Members	Description
I_VFMLT_UPPER = 0	Available since version 2.0.
I_VFMLT_MIDDLE = 1	Available since version 2.0.
I_VFMLT_LOWER = 2	Available since version 2.0.
I_VFMLT_MAXIMUM = 3	Available since version 2.0.
I_VFMLT_MINIMUM = 4	Available since version 2.0.
I_VFMLT_ABSOLUTE_MAXIMUM = 5	Available since version 2.0.
I_VFMLT_ARBITRARY = 6	Available since version 2.0.

I.2.31 IRobotViewFeMapSmoothingType

C++

```
enum IRobotViewFeMapSmoothingType;
```

C#

```
public enum IRobotViewFeMapSmoothingType;
```

Visual Basic

```
Public Enum IRobotViewFeMapSmoothingType
```

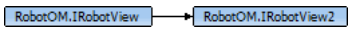
(c) 2014 Autodesk Inc. All rights reserved

Members

Members	Description
I_VFMST_NO_SMOOTHING = 0	Available since version 2.0.
I_VFMST_GLOBAL_SMOOTHING = 1	Available since version 2.0.
I_VFMST_SMOOTHING_WITH_PANEL = 2	Available since version 2.0.
I_VFMST_SMOOTHING_ACCORDING_TO_SELECTION = 3	Available since version 2.0.

I.2.32 IRobotView2

Class Hierarchy



C++

```
interface IRobotView2 : IRobotView;
```

C#

```
public interface IRobotView2 : IRobotView;
```

Visual Basic

```
Public Interface IRobotView2
```

Description

Extended graphical view interface. .

Version

Available since version 3.

I.2.32.1 IRobotView2 Members

The following tables list the members exposed by IRobotView2.

Public Fields

	Name	Description
	DiagramMagnification (see page 1398)	Diagram enlarging (scale: 1 to 10).
	ParamsBarMap (see page 1398)	Parameters of displaying maps an bars Available since version 2.0.
	ParamsDiagram (see page 1398)	Parameters of displaying diagrams on the view Available since version 2.0.
	ParamsDisplay (see page 1398)	Display parameters (a component enabling control of individual attribute display) Available since version 2.0.
	ParamsFeMap (see page 1399)	Display parameters of maps presenting results for surface finite elements Available since version 2.0.
	ParamsPanelCut (see page 1446)	Parameters of displaying panel cuts.
	Printable (see page 1399)	Object representing the view that may be attached to the printout and printed .
	Projection (see page 1399)	Current projection .
	Selection (see page 1399)	Selection of structure components presented in the view .
	Title (see page 1400)	View title .
	Type (see page 1400)	View type .
	UserControl (see page 1447)	Flag that enables giving/taking back graphical view navigation to/from the interactive program user .
	Visible (see page 1400)	View display flag .

(c) 2014 Autodesk Inc. All rights reserved

	Window (see page 1447)	

Public Methods

	Name	Description
	CopyToClipboard (see page 1401)	The function inserts the view into the clipboard.
	GetRotationPoint (see page 1401)	The function gives back a rotation center during a view rotation.
	GetScale (see page 1402)	The function gives a value of the given scale back. A set of available scales describes RobotViewScaleType type.
	GetSize (see page 1402)	The function returns the size of the view in mm. .
	GetWorkPoint (see page 1402)	The function gives a work point back (on the XY plane it gives Z level) .
	GetZoom (see page 1402)	The function returns the current zoom - coordinates of view sides in the current coordinate system .
	IsLocal (see page 1403)	The function gives the status of a local coordinate system back.
	Redraw (see page 1403)	The function draws the view. .
	Rotate (see page 1403)	The function rotates structure view with respect to the indicated global coordinate axis, with the rotation point recognized.
	SetGlobal (see page 1403)	The function sets a global coordinate system.
	SetLocal (see page 1404)	The function sets a local coordinate system. If the given points do not describe a local plane in the unambiguous way the function gives a zero value (False) back.
	SetRotationPoint (see page 1404)	The funcion sets the rotation center during the view rotation.
	SetScale (see page 1404)	The function assigns a given value to the scale.
	SetSize (see page 1404)	The function sets the view size in mm .
	SetWorkPoint (see page 1405)	The function sets a working point (for the XY plane Z level is set). .
	SetZoom (see page 1405)	The function sets the current zoom - coordinates of view sides in the current coordinate system .

I.2.32.2 IRobotView2 Fields

The fields of the IRobotView2 class are listed here.

Public Fields

	Name	Description
	ParamsPanelCut (see page 1446)	Parameters of displaying panel cuts.
	UserControl (see page 1447)	Flag that enables giving/taking back graphical view navigation to/from the interactive program user .
	Window (see page 1447)	

I.2.32.2.1 ParamsPanelCut

C++

```
HRESULT get_ParamsPanelCut(IRobotViewPanelCutParams**);
```

C#

```
public IRobotViewPanelCutParams ParamsPanelCut { get; }
```

Visual Basic

```
Public ReadOnly ParamsPanelCut As IRobotViewPanelCutParams
```

Description

Parameters of displaying panel cuts.

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

I.2.32.2.2 UserControl

C++

```
HRESULT get_UserControl(VARIANT_BOOL*);
HRESULT put_UserControl(VARIANT_BOOL);
```

C#

```
public bool UserControl { get; set; }
```

Visual Basic

```
Public UserControl As Boolean
```

Description

Flag that enables giving/taking back graphical view navigation to/from the interactive program user .

Version

Available since version 3.

I.2.32.2.3 Window

C++

```
HRESULT get_Window(IRobotWindow**);
```

C#

```
public IRobotWindow Window { get; }
```

Visual Basic

```
Public ReadOnly Window As IRobotWindow
```

Version

Available since version 3.

I.2.33 IRobotViewDiagramPositionType

C++

```
enum IRobotViewDiagramPositionType;
```

C#

```
public enum IRobotViewDiagramPositionType;
```

Visual Basic

```
Public Enum IRobotViewDiagramPositionType
```

Members

Members	Description
I_VDPT_NORMAL = 1	Diagram perpendicular to the panel plane. Available since version 3.
I_VDPT_IN_PLANE = 2	Diagram in the panel plane. Available since version 3.

Description

Available diagram positions with respect to the panel plane.

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

I.2.34 IRobotViewReinforcementResultType

C++

```
enum IRobotViewReinforcementResultType;
```

C#

```
public enum IRobotViewReinforcementResultType;
```

Visual Basic

```
Public Enum IRobotViewReinforcementResultType
```

Members

Members	Description
I_VRRT_AX_TOP = 577	Top reinforcement area (X direction). Available since version 3.
I_VRRT_AX_BOTTOM = 579	Bottom reinforcement area (X direction). Available since version 3.
I_VRRT_AY_TOP = 578	Top reinforcement area (Y direction). Available since version 3.
I_VRRT_AY_BOTTOM = 580	Bottom reinforcement area (Y direction). Available since version 3.
I_VRRT_E_AX_TOP = 699	Top reinforcement spacing (X direction). Available since version 3.
I_VRRT_E_AX_BOTTOM = 701	Bottom reinforcement spacing (X direction). Available since version 3.
I_VRRT_E_AY_TOP = 700	Top reinforcement spacing (Y direction). Available since version 3.
I_VRRT_E_AY_BOTTOM = 702	Bottom reinforcement spacing (Y direction). Available since version 3.
I_VRRT_A_MIN = 1119	Minimal reinforcement area for one layer and one reinforcement direction. Available since version 3.
I_VRRT_AX = 1116	Cracking (X axis direction). Available since version 3.
I_VRRT_AY = 1117	Cracking (Y axis direction). Available since version 3.
I_VRRT_F = 1118	Deflection value. Available since version 3.

Description

Available reinforcement results presented in the garphical view. .

Version

Available since version 3.

I.2.35 IRobotViewGlobalAnalysisParams

Class Hierarchy

RobotOM.IRobotViewGlobalAnalysisParams

C++

```
interface IRobotViewGlobalAnalysisParams : IDispatch;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public interface IRobotViewGlobalAnalysisParams;
```

Visual Basic

```
Public Interface IRobotViewGlobalAnalysisParams
```

Description

Interface controlling display of global analysis results. .

Version

Available since version 3.

I.2.35.1 IRobotViewGlobalAnalysisParams Members

The following tables list the members exposed by IRobotViewGlobalAnalysisParams.

Public Fields

	Name	Description
	Results (see page 1449)	Method of result display .

Public Methods

	Name	Description
	IsOn (see page 1450)	Function checks status of the indicated parameter. .
	Set (see page 1450)	Function sets status for the indicated parameter. .

I.2.35.2 IRobotViewGlobalAnalysisParams Fields

The fields of the IRobotViewGlobalAnalysisParams class are listed here.

Public Fields

	Name	Description
	Results (see page 1449)	Method of result display .

I.2.35.2.1 Results

C++

```
HRESULT get_Results(IRobotViewGlobalAnalysisResultsParams**);
```

C#

```
public IRobotViewGlobalAnalysisResultsParams Results { get; }
```

Visual Basic

```
Public ReadOnly Results As IRobotViewGlobalAnalysisResultsParams
```

Description

Method of result display .

Version

Available since version 3.

I.2.35.3 IRobotViewGlobalAnalysisParams Methods

The methods of the IRobotViewGlobalAnalysisParams class are listed here.

(c) 2014 Autodesk Inc. All rights reserved

Public Methods

	Name	Description
	IsOn (see page 1450)	Function checks status of the indicated parameter. .
	Set (see page 1450)	Function sets status for the indicated parameter. .

I.2.35.3.1 IsOn

C++

```
HRESULT IsOn(IRobotViewGlobalAnalysisParamsType _type, VARIANT_BOOL* ret);
```

C#

```
public bool IsOn(IRobotViewGlobalAnalysisParamsType _type);
```

Visual Basic

```
Public Function IsOn(_type As IRobotViewGlobalAnalysisParamsType) As Boolean
```

Description

Function checks status of the indicated parameter. .

Version

Available since version 3.

I.2.35.3.2 Set

C++

```
HRESULT Set(IRobotViewGlobalAnalysisParamsType _type, VARIANT_BOOL _status);
```

C#

```
public void Set(IRobotViewGlobalAnalysisParamsType _type, bool _status);
```

Visual Basic

```
Public Sub Set(_type As IRobotViewGlobalAnalysisParamsType, _status As Boolean)
```

Description

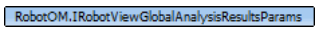
Function sets status for the indicated parameter. .

Version

Available since version 3.

I.2.36 IRobotViewGlobalAnalysisResultsParams

Class Hierarchy



C++

```
interface IRobotViewGlobalAnalysisResultsParams : IDispatch;
```

C#

```
public interface IRobotViewGlobalAnalysisResultsParams;
```

Visual Basic

```
Public Interface IRobotViewGlobalAnalysisResultsParams
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Interface controlling result display.

Version

Available since version 3.

I.2.36.1 IRobotViewGlobalAnalysisResultsParams Members

The following tables list the members exposed by IRobotViewGlobalAnalysisResultsParams.

Public Fields

	Name	Description
	NPointsValue (see page 1451)	Number of points along the bar length.
	RelativeValue (see page 1452)	Relative point coordinate.
	Type (see page 1452)	Method of result presentation.

I.2.36.2 IRobotViewGlobalAnalysisResultsParams Fields

The fields of the IRobotViewGlobalAnalysisResultsParams class are listed here.

Public Fields

	Name	Description
	NPointsValue (see page 1451)	Number of points along the bar length.
	RelativeValue (see page 1452)	Relative point coordinate.
	Type (see page 1452)	Method of result presentation.

I.2.36.2.1 NPointsValue

C++

```
HRESULT get_NPointsValue(long*);
HRESULT put_NPointsValue(long);
```

C#

```
public long NPointsValue { get; set; }
```

Visual Basic

```
Public NPointsValue As long
```

Description

Number of points along the bar length.

Version

Available since version 3.

I.2.36.2.2 RelativeValue

C++

```
HRESULT get_RelativeValue(double*);
HRESULT put_RelativeValue(double);
```

C#

```
public double RelativeValue { get; set; }
```

Visual Basic

```
Public RelativeValue As double
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Relative point coordinate.

Version

Available since version 3.

I.2.36.2.3 Type

C++

```
HRESULT get_Type(IRobotViewGlobalAnalysisResultsType*);
HRESULT put_Type(IRobotViewGlobalAnalysisResultsType);
```

C#

```
public IRobotViewGlobalAnalysisResultsType Type { get; set; }
```

Visual Basic

```
Public Type As IRobotViewGlobalAnalysisResultsType
```

Description

Method of result presentation.

Version

Available since version 3.

I.2.37 IRobotViewGlobalAnalysisResultsType

C++

```
enum IRobotViewGlobalAnalysisResultsType;
```

C#

```
public enum IRobotViewGlobalAnalysisResultsType;
```

Visual Basic

```
Public Enum IRobotViewGlobalAnalysisResultsType
```

Members

Members	Description
I_VGART_N_POINTS = 1	Available since version 3.
I_VGART_RELATIVE = 2	Available since version 3.

Description

Types of result display.

Version

Available since version 3.

I.2.38 IRobotViewGlobalAnalysisParamsType

C++

```
enum IRobotViewGlobalAnalysisParamsType;
```

C#

```
public enum IRobotViewGlobalAnalysisParamsType;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Enum IRobotViewGlobalAnalysisParamsType
```

Members

Members	Description
I_VGAPT_FORCE_FX = 1	Available since version 3.
I_VGAPT_FORCE_FY = 2	Available since version 3.
I_VGAPT_FORCE_FZ = 3	Available since version 3.
I_VGAPT_FORCE_MX = 4	Available since version 3.
I_VGAPT_FORCE_MY = 5	Available since version 3.
I_VGAPT_FORCE_MZ = 6	Available since version 3.
I_VGAPT_FORCE_KX = 7	Available since version 3.
I_VGAPT_FORCE_KY = 8	Available since version 3.
I_VGAPT_STRESS_NORMAL = 9	Available since version 3.
I_VGAPT_STRESS_EXTREME = 10	Available since version 3.
I_VGAPT_STRESS_BENDING_SMY = 11	Available since version 3.
I_VGAPT_STRESS_BENDING_SMZ = 12	Available since version 3.
I_VGAPT_STRESS_AXIAL_SXFX = 13	Available since version 3.
I_VGAPT_STRESS_SHEAR = 14	Available since version 3.
I_VGAPT_STRESS_TORSION = 15	Available since version 3.
I_VGAPT_DESIGN_RATIO = 16	Available since version 3.
I_VGAPT_DESIGN_MEMBER_LENGTH = 17	Available since version 3.
I_VGAPT_DESIGN_SLENDERNESS_LAY = 18	Available since version 3.
I_VGAPT_DESIGN_SLENDERNESS_LAZ = 19	Available since version 3.

Description

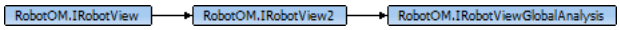
Parameter types.

Version

Available since version 3.

I.2.39 IRobotViewGlobalAnalysis

Class Hierarchy



C++

```
interface IRobotViewGlobalAnalysis : IRobotView2;
```

C#

```
public interface IRobotViewGlobalAnalysis : IRobotView2;
```

Visual Basic

```
Public Interface IRobotViewGlobalAnalysis
```

Description

View presenting global analysis results. .

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

I.2.39.1 IRobotViewGlobalAnalysis Members

The following tables list the members exposed by IRobotViewGlobalAnalysis.

Public Fields

	Name	Description
	DiagramMagnification (see page 1398)	Diagram enlarging (scale: 1 to 10).
	ParamsBarMap (see page 1398)	Parameters of displaying maps an bars Available since version 2.0.
	ParamsDiagram (see page 1398)	Parameters of displaying diagrams on the view Available since version 2.0.
	ParamsDisplay (see page 1398)	Display parameters (a component enabling control of individual attribute display) Available since version 2.0.
	ParamsFeMap (see page 1399)	Display parameters of maps presenting results for surface finite elements Available since version 2.0.
	ParamsGlobalAnalysis (see page 1455)	Display parameters for global analysis results.
	ParamsPanelCut (see page 1446)	Parameters of displaying panel cuts.
	Printable (see page 1399)	Object representing the view that may be attached to the printout and printed .
	Projection (see page 1399)	Current projection .
	Selection (see page 1399)	Selection of structure components presented in the view .
	Title (see page 1400)	View title .
	Type (see page 1400)	View type .
	UserControl (see page 1447)	Flag that enables giving/taking back graphical view navigation to/from the interactive program user .
	Visible (see page 1400)	View display flag .
	Window (see page 1447)	

Public Methods

	Name	Description
	CopyToClipboard (see page 1401)	The function inserts the view into the clipboard.
	GetRotationPoint (see page 1401)	The function gives back a rotation center during a view rotation.
	GetScale (see page 1402)	The function gives a value of the given scale back. A set of available scales describes RobotViewScaleType type.
	GetSize (see page 1402)	The function returns the size of the view in mm. .
	GetWorkPoint (see page 1402)	The function gives a work point back (on the XY plane it gives Z level) .
	GetZoom (see page 1402)	The function returns the current zoom - coordinates of view sides in the current coordinate system .
	IsLocal (see page 1403)	The function gives the status of a local coordinate system back.
	MakeScreenCapture (see page 1455)	
	Redraw (see page 1403)	The function draws the view. .
	Rotate (see page 1403)	The function rotates structure view with respect to the indicated global coordinate axis, with the rotation point recognized.
	SetGlobal (see page 1403)	The function sets a global coordinate system.
	SetLocal (see page 1404)	The function sets a local coordinate system. If the given points do not describe a local plane in the unambiguous way the function gives a zero value (False) back.
	SetRotationPoint (see page 1404)	The funcion sets the rotation center during the view rotation.
	SetScale (see page 1404)	The function assigns a given value to the scale.
	SetSize (see page 1404)	The function sets the view size in mm .

(c) 2014 Autodesk Inc. All rights reserved

	SetWorkPoint (see page 1405)	The function sets a working point (for the XY plane Z level is set). .
	SetZoom (see page 1405)	The function sets the current zoom - coordinates of view sides in the current coordinate system .

I.2.39.2 IRobotViewGlobalAnalysis Fields

The fields of the IRobotViewGlobalAnalysis class are listed here.

Public Fields

	Name	Description
	ParamsGlobalAnalysis (see page 1455)	Display parameters for global analysis results.

I.2.39.2.1 ParamsGlobalAnalysis

C++

```
HRESULT get_ParamsGlobalAnalysis(IRobotViewGlobalAnalysisParams**);
```

C#

```
public IRobotViewGlobalAnalysisParams ParamsGlobalAnalysis { get; }
```

Visual Basic

```
Public ReadOnly ParamsGlobalAnalysis As IRobotViewGlobalAnalysisParams
```

Description

Display parameters for global analysis results.

Version

Available since version 3.

I.2.39.3 IRobotViewGlobalAnalysis Methods

The methods of the IRobotViewGlobalAnalysis class are listed here.

Public Methods

	Name	Description
	MakeScreenCapture (see page 1455)	

I.2.39.3.1 MakeScreenCapture

C++

```
HRESULT MakeScreenCapture(IRobotViewScreenCaptureParams* _sc_params);
```

C#

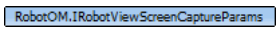
```
public void MakeScreenCapture(IRobotViewScreenCaptureParams _sc_params);
```

Visual Basic

```
Public Sub MakeScreenCapture(ByRef _sc_params As IRobotViewScreenCaptureParams)
```

I.2.40 IRobotViewScreenCaptureParams

Class Hierarchy



C++

```
interface IRobotViewScreenCaptureParams : IDispatch;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public interface IRobotViewScreenCaptureParams;
```

Visual Basic

```
Public Interface IRobotViewScreenCaptureParams
```

Description

Parameters of a screen capture for the graphical view.

I.2.40.1 IRobotViewScreenCaptureParams Members

The following tables list the members exposed by IRobotViewScreenCaptureParams.

Public Fields

	Name	Description
	Comment (see page 1457)	Comment text to be added to the screen capture.
	IncludeDateAndTime (see page 1457)	The flag indicating if the current date and time should be added automatically to the screen capture.
	Name (see page 1457)	Screen capture name.
	Orientation (see page 1458)	Screen capture orientation (landscape or portait).
	Resolution (see page 1458)	
	ScaleAutomatic (see page 1458)	True value denotes selection of the automatic scale, while False value indicates selection of the user scale defined by ScaleValue (see page 1459) value.
	ScaleValue (see page 1459)	User scale value - distance expressed in meters (m) which corresponds to one centimeter (1 cm) in the graphical view.
	UpdateType (see page 1459)	

I.2.40.2 IRobotViewScreenCaptureParams Fields

The fields of the IRobotViewScreenCaptureParams class are listed here.

Public Fields

	Name	Description
	Comment (see page 1457)	Comment text to be added to the screen capture.
	IncludeDateAndTime (see page 1457)	The flag indicating if the current date and time should be added automatically to the screen capture.
	Name (see page 1457)	Screen capture name.
	Orientation (see page 1458)	Screen capture orientation (landscape or portait).
	Resolution (see page 1458)	
	ScaleAutomatic (see page 1458)	True value denotes selection of the automatic scale, while False value indicates selection of the user scale defined by ScaleValue (see page 1459) value.
	ScaleValue (see page 1459)	User scale value - distance expressed in meters (m) which corresponds to one centimeter (1 cm) in the graphical view.
	UpdateType (see page 1459)	

I.2.40.2.1 Comment

C++

```
HRESULT get_Comment(BSTR*);
HRESULT put_Comment(BSTR);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public String Comment { get; set; }
```

Visual Basic

```
Public Comment As String
```

Description

Comment text to be added to the screen capture.

Version

Available since version 12.

I.2.40.2.2 IncludeDateAndTime

C++

```
HRESULT get_IncludeDateAndTime(VARIANT_BOOL*);
HRESULT put_IncludeDateAndTime(VARIANT_BOOL);
```

C#

```
public bool IncludeDateAndTime { get; set; }
```

Visual Basic

```
Public IncludeDateAndTime As Boolean
```

Description

The flag indicating if the current date and time should be added automatically to the screen capture.

Version

Available since version 12.

I.2.40.2.3 Name

C++

```
HRESULT get_Name(BSTR*);
HRESULT put_Name(BSTR);
```

C#

```
public String Name { get; set; }
```

Visual Basic

```
Public Name As String
```

Description

Screen capture name.

I.2.40.2.4 Orientation

C++

```
HRESULT get_Orientation(IRobotPageSetupOrientation*);
HRESULT put_Orientation(IRobotPageSetupOrientation);
```

C#

```
public IRobotPageSetupOrientation Orientation { get; set; }
```

Visual Basic

```
Public Orientation As IRobotPageSetupOrientation
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Screen capture orientation (landscape or portait).

I.2.40.2.5 Resolution

C++

```
HRESULT get_Resolution(IRobotViewScreenCaptureResolution*);
HRESULT put_Resolution(IRobotViewScreenCaptureResolution);
```

C#

```
public IRobotViewScreenCaptureResolution Resolution { get; set; }
```

Visual Basic

```
Public Resolution As IRobotViewScreenCaptureResolution
```

Version

Available since version 12.

I.2.40.2.6 ScaleAutomatic

C++

```
HRESULT get_ScaleAutomatic(VARIANT_BOOL*);
HRESULT put_ScaleAutomatic(VARIANT_BOOL);
```

C#

```
public bool ScaleAutomatic { get; set; }
```

Visual Basic

```
Public ScaleAutomatic As Boolean
```

Description

True value denotes selection of the automatic scale, while False value indicates selection of the user scale defined by ScaleValue (see page 1459) value.

I.2.40.2.7 ScaleValue

C++

```
HRESULT get_ScaleValue(double*);
HRESULT put_ScaleValue(double);
```

C#

```
public double ScaleValue { get; set; }
```

Visual Basic

```
Public ScaleValue As double
```

Description

User scale value - distance expressed in meters (m) which corresponds to one centimeter (1 cm) in the graphical view.

I.2.40.2.8 UpdateType

C++

```
HRESULT get_UpdateType(IRobotViewScreenCaptureUpdateType*);
HRESULT put_UpdateType(IRobotViewScreenCaptureUpdateType);
```

C#

```
public IRobotViewScreenCaptureUpdateType UpdateType { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public UpdateType As IRobotViewScreenCaptureUpdateType
```

I.2.41 IRobotViewScreenCaptureUpdateType

C++

```
enum IRobotViewScreenCaptureUpdateType;
```

C#

```
public enum IRobotViewScreenCaptureUpdateType;
```

Visual Basic

```
Public Enum IRobotViewScreenCaptureUpdateType
```

Members

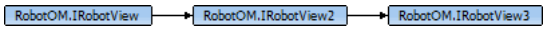
Members	Description
I_SCUT_UPDATED_UPON_PRINTING = 1	The view contents will be updated automatically while printing.
I_SCUT_CURRENT_VIEW = 2	The view will be printed in the form it had while making a screen capture - the view contents will not be updated.
I_SCUT_UPDATED_WHOLE_STRUCTURE = 3	The view contents will be updated while printing in such a way so that the view presents the whole current structure.
I_SCUT_COPY_TO_CLIPBOARD = 4	The screen capture will be saved only to the clipboard. Available since version 12.

Description

Available methods of updating the contents of a screen capture for the graphical view.

I.2.42 IRobotView3

Class Hierarchy



C++

```
interface IRobotView3 : IRobotView2;
```

C#

```
public interface IRobotView3 : IRobotView2;
```

Visual Basic

```
Public Interface IRobotView3
```

I.2.42.1 IRobotView3 Members

The following tables list the members exposed by IRobotView3.

Public Fields

	Name	Description
	DiagramMagnification (see page 1398)	Diagram enlarging (scale: 1 to 10).
	ParamsBarMap (see page 1398)	Parameters of displaying maps an bars Available since version 2.0.
	ParamsDiagram (see page 1398)	Parameters of displaying diagrams on the view Available since version 2.0.
	ParamsDisplay (see page 1398)	Display parameters (a component enabling control of individual attribute display) Available since version 2.0.

(c) 2014 Autodesk Inc. All rights reserved

	ParamsFeMap (see page 1399)	Display parameters of maps presenting results for surface finite elements Available since version 2.0.
	ParamsPanelCut (see page 1446)	Parameters of displaying panel cuts.
	Printable (see page 1399)	Object representing the view that may be attached to the printout and printed .
	Projection (see page 1399)	Current projection .
	Selection (see page 1399)	Selection of structure components presented in the view .
	Title (see page 1400)	View title .
	Type (see page 1400)	View type .
	UserControl (see page 1447)	Flag that enables giving/taking back graphical view navigation to/from the interactive program user .
	Visible (see page 1400)	View display flag .
	Window (see page 1447)	

Public Methods

	Name	Description
	CopyToClipboard (see page 1401)	The function inserts the view into the clipboard.
	GetRotationPoint (see page 1401)	The function gives back a rotation center during a view rotation.
	GetScale (see page 1402)	The function gives a value of the given scale back. A set of available scales describes RobotViewScaleType type.
	GetSize (see page 1402)	The function returns the size of the view in mm. .
	GetWorkPoint (see page 1402)	The function gives a work point back (on the XY plane it gives Z level) .
	GetZoom (see page 1402)	The function returns the current zoom - coordinates of view sides in the current coordinate system .
	IsLocal (see page 1403)	The function gives the status of a local coordinate system back.
	MakeScreenCapture (see page 1461)	Function makes a screen capture of a graphical view.
	Redraw (see page 1403)	The function draws the view. .
	Rotate (see page 1403)	The function rotates structure view with respect to the indicated global coordinate axis, with the rotation point recognized.
	SetGlobal (see page 1403)	The function sets a global coordinate system.
	SetLocal (see page 1404)	The function sets a local coordinate system. If the given points do not describe a local plane in the unambiguous way the function gives a zero value (False) back.
	SetRotationPoint (see page 1404)	The funcion sets the rotation center during the view rotation.
	SetScale (see page 1404)	The function assigns a given value to the scale.
	SetSize (see page 1404)	The function sets the view size in mm .
	SetWorkPoint (see page 1405)	The function sets a working point (for the XY plane Z level is set). .
	SetZoom (see page 1405)	The function sets the current zoom - coordinates of view sides in the current coordinate system .

I.2.42.2 IRobotView3 Methods

The methods of the IRobotView3 class are listed here.

Public Methods

	Name	Description
	MakeScreenCapture (see page 1461)	Function makes a screen capture of a graphical view.

(c) 2014 Autodesk Inc. All rights reserved

I.2.42.2.1 MakeScreenCapture

C++

```
HRESULT MakeScreenCapture(IRobotViewScreenCaptureParams* _sc_params);
```

C#

```
public void MakeScreenCapture(IRobotViewScreenCaptureParams _sc_params);
```

Visual Basic

```
Public Sub MakeScreenCapture(ByRef _sc_params As IRobotViewScreenCaptureParams)
```

Description

Function makes a screen capture of a graphical view.

I.2.43 IRobotViewDiagramValueType

C++

```
enum IRobotViewDiagramValueType;
```

C#

```
public enum IRobotViewDiagramValueType;
```

Visual Basic

```
Public Enum IRobotViewDiagramValueType
```

Members

Members	Description
I_VDVT_ALL = 0	Available since version 8.1.
I_VDVT_LOCAL_EXTREMES = 1	Available since version 8.1.
I_VDVT_GLOBAL_EXTREMES = 2	Available since version 8.1.

Description

Types of values displayed on diagrams.

Version

Available since version 8.1.

I.2.44 IRobotViewScreenCaptureResolution

C++

```
enum IRobotViewScreenCaptureResolution;
```

C#

```
public enum IRobotViewScreenCaptureResolution;
```

Visual Basic

```
Public Enum IRobotViewScreenCaptureResolution
```

Members

Members	Description
I_VSCR_DEFAULT = 0	Available since version 12.
I_VSCR_2048 = 1	Available since version 12.
I_VSCR_3072 = 2	Available since version 12.

(c) 2014 Autodesk Inc. All rights reserved

I_VSCR_4096 = 3	Available since version 12.

Version

Available since version 12.

I.3 Views for RTF format files

Interfaces

	Name	Description
	IRobotRtfView (see page 1462)	Files in RTF format are frequently used to present data or results. Robot enriched the format with the possibility of introducing into an RTF file the variables defined in the program. A variable is written to file in the form of its name, marked with the "@" sign at both ends. The view RobotRtfView automatically extends variables to give them their current value.

I.3.1 IRobotRtfView

Class Hierarchy

RobotOM.IRobotRtfView

C++

```
interface IRobotRtfView : IDispatch;
```

C#

```
public interface IRobotRtfView;
```

Visual Basic

```
Public Interface IRobotRtfView
```

Description

Files in RTF format are frequently used to present data or results. Robot enriched the format with the possibility of introducing into an RTF file the variables defined in the program. A variable is written to file in the form of its name, marked with the "@" sign at both ends. The view RobotRtfView automatically extends variables to give them their current value.

I.3.1.1 IRobotRtfView Members

The following tables list the members exposed by IRobotRtfView.

Public Fields

	Name	Description
	Printable (see page 1463)	Object representing a view, that may be added to the print-out and printed.

Public Methods

	Name	Description
	AppendFromFile (see page 1464)	The function pastes the contents of the indicated view at the end of the view. .
	Evaluate (see page 1464)	The function extends all the variables in the text and changes their names to their current values. There is no corresponding undo operation. .
	LoadFromFile (see page 1464)	The function reads the indicated file in RTF format. .
	SaveToFile (see page 1464)	The function saves the content of the view to the indicated file. If the "as_template" parameter is not equal zero (set to True), the view contents will be written as a template (variable names and not their values will be saved to file).

(c) 2014 Autodesk Inc. All rights reserved

I.3.1.2 IRobotRtfView Fields

The fields of the IRobotRtfView class are listed here.

Public Fields

	Name	Description
	Printable (see page 1463)	Object representing a view, that may be added to the print-out and printed.

I.3.1.2.1 Printable

C++

```
HRESULT get_Printable(IRobotPrintable**);
```

C#

```
public IRobotPrintable Printable { get; }
```

Visual Basic

```
Public ReadOnly Printable As IRobotPrintable
```

Description

Object representing a view, that may be added to the print-out and printed.

I.3.1.3 IRobotRtfView Methods

The methods of the IRobotRtfView class are listed here.

Public Methods

	Name	Description
	AppendFromFile (see page 1464)	The function pastes the contents of the indicated view at the end of the view. .
	Evaluate (see page 1464)	The function extends all the variables in the text and changes their names to their current values. There is no corresponding undo operation. .
	LoadFromFile (see page 1464)	The function reads the indicated file in RTF format. .
	SaveToFile (see page 1464)	The function saves the content of the view to the indicated file. If the "as_template" parameter is not equal zero (set to True), the view contents will be written as a template (variable names and not their values will be saved to file).

I.3.1.3.1 AppendFromFile

C++

```
HRESULT AppendFromFile(BSTR _file_path, VARIANT_BOOL* ret);
```

C#

```
public bool AppendFromFile(String _file_path);
```

Visual Basic

```
Public Function AppendFromFile(_file_path As String) As Boolean
```

Description

The function pastes the contents of the indicated view at the end of the view. .

(c) 2014 Autodesk Inc. All rights reserved

I.3.1.3.2 Evaluate

C++

```
HRESULT Evaluate();
```

C#

```
public void Evaluate();
```

Visual Basic

```
Public Sub Evaluate()
```

Description

The function extends all the variables in the text and changes their names to their current values. There is no corresponding undo operation. .

I.3.1.3.3 LoadFromFile

C++

```
HRESULT LoadFromFile(BSTR _file_path, VARIANT_BOOL* ret);
```

C#

```
public bool LoadFromFile(String _file_path);
```

Visual Basic

```
Public Function LoadFromFile(_file_path As String) As Boolean
```

Description

The function reads the indicated file in RTF format. .

I.3.1.3.4 SaveToFile

C++

```
HRESULT SaveToFile(BSTR _file_path, VARIANT_BOOL _as_template = false, VARIANT_BOOL* ret);
```

C#

```
public bool SaveToFile(String _file_path, bool _as_template = false);
```

Visual Basic

```
Public Function SaveToFile(_file_path As String, Optional _as_template As Boolean = false)
As Boolean
```

Description

The function saves the content of the view to the indicated file. If the "as_template" parameter is not equal zero (set to True), the view contents will be written as a template (variable names and not their values will be saved to file).

I.4 Screen captures

The Robot program allows saving the current state (settings and/or contents) of a selected view into the archive to be used later at the stage of creating documentation. It is performed by making a screen capture of a selected view. The screen captures are identified by their names. .

(c) 2014 Autodesk Inc. All rights reserved

Interfaces

	Name	Description
	IRobotScreenCaptureMngr (see page 1465)	An object manages screen captures defined in the current project. .

I.4.1 IRobotScreenCaptureMngr

Class Hierarchy

RobotOM.IRobotScreenCaptureMngr

C++

```
interface IRobotScreenCaptureMngr : IDispatch;
```

C#

```
public interface IRobotScreenCaptureMngr;
```

Visual Basic

```
Public Interface IRobotScreenCaptureMngr
```

Description

An object manages screen captures defined in the current project. .

I.4.1.1 IRobotScreenCaptureMngr Members

The following tables list the members exposed by IRobotScreenCaptureMngr.

Public Fields

	Name	Description
	Count (see page 1466)	Number of defined screen captures Available since version 1.7.

Public Methods

	Name	Description
	Find (see page 1466)	Function returns index of a screen capture with a specified name. If screen capture with this name does not exist, then zero value is returned. Available since version 1.7.
	Get (see page 1467)	Function returns name of a screen capture with a specified index. Screen captures are indexed from 1 to Count (see page 1466). Available since version 1.7.
	Remove (see page 1467)	Function deletes the archive containing a screen capture with a specified name from the project. Available since version 1.7.

I.4.1.2 IRobotScreenCaptureMngr Fields

The fields of the IRobotScreenCaptureMngr class are listed here.

Public Fields

	Name	Description
	Count (see page 1466)	Number of defined screen captures Available since version 1.7.

I.4.1.2.1 Count

C++

```
HRESULT get_Count(long*);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public long Count { get; }
```

Visual Basic

```
Public ReadOnly Count As long
```

Description

Number of defined screen captures Available since version 1.7.

I.4.1.3 IRobotScreenCaptureMngr Methods

The methods of the IRobotScreenCaptureMngr class are listed here.

Public Methods

	Name	Description
	Find (see page 1466)	Function returns index of a screen capture with a specified name. If screen capture with this name does not exist, then zero value is returned. Available since version 1.7.
	Get (see page 1467)	Function returns name of a screen capture with a specified index. Screen captures are indexed from 1 to Count (see page 1466). Available since version 1.7.
	Remove (see page 1467)	Function deletes the archive containing a screen capture with a specified name from the project. Available since version 1.7.

I.4.1.3.1 Find

C++

```
HRESULT Find(BSTR _sc_name, long* ret);
```

C#

```
public long Find(String _sc_name);
```

Visual Basic

```
Public Function Find(_sc_name As String) As long
```

Description

Function returns index of a screen capture with a specified name. If screen capture with this name does not exist, then zero value is returned. Available since version 1.7.

I.4.1.3.2 Get

C++

```
HRESULT Get(long _idx, BSTR* ret);
```

C#

```
public String Get(long _idx);
```

Visual Basic

```
Public Function Get(_idx As long) As String
```

Description

Function returns name of a screen capture with a specified index. Screen captures are indexed from 1 to Count (see page 1466). Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

I.4.1.3.3 Remove

C++

```
HRESULT Remove(BSTR _sc_name);
```

C#

```
public void Remove(String _sc_name);
```

Visual Basic

```
Public Sub Remove(_sc_name As String)
```

Description

Function deletes the archive containing a screen capture with a specified name from the project. Available since version 1.7.

I.5 Views for HTML format files

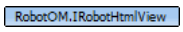
Available since version 7.5.

Interfaces

	Name	Description
	IRobotHtmlView (see page 1467)	

I.5.1 IRobotHtmlView

Class Hierarchy



C++

```
interface IRobotHtmlView : IDispatch;
```

C#

```
public interface IRobotHtmlView;
```

Visual Basic

```
Public Interface IRobotHtmlView
```

Version

Available since version 7.5.

I.5.1.1 IRobotHtmlView Members

The following tables list the members exposed by IRobotHtmlView.

Public Fields

	Name	Description
	Printable (see page 1468)	Object representing a view that may be attached to the printout and printed.

Public Methods

	Name	Description
	LoadFromFile (see page 1469)	Function loads an HTML format file.
	SaveToFile (see page 1469)	Function saves the view contents to the specified HTML format file.

(c) 2014 Autodesk Inc. All rights reserved

I.5.1.2 IRobotHtmlView Fields

The fields of the IRobotHtmlView class are listed here.

Public Fields

	Name	Description
	Printable (see page 1468)	Object representing a view that may be attached to the printout and printed.

I.5.1.2.1 Printable

C++

```
HRESULT get_Printable(IRobotPrintable**);
```

C#

```
public IRobotPrintable Printable { get; }
```

Visual Basic

```
Public ReadOnly Printable As IRobotPrintable
```

Description

Object representing a view that may be attached to the printout and printed.

Version

Available since version 7.5.

I.5.1.3 IRobotHtmlView Methods

The methods of the IRobotHtmlView class are listed here.

Public Methods

	Name	Description
	LoadFromFile (see page 1469)	Function loads an HTML format file.
	SaveToFile (see page 1469)	Function saves the view contents to the specified HTML format file.

I.5.1.3.1 LoadFromFile

C++

```
HRESULT LoadFromFile(BSTR _file_path, VARIANT_BOOL* ret);
```

C#

```
public bool LoadFromFile(String _file_path);
```

Visual Basic

```
Public Function LoadFromFile(_file_path As String) As Boolean
```

Description

Function loads an HTML format file.

Version

Available since version 7.5.

I.5.1.3.2 SaveToFile

C++

```
HRESULT SaveToFile(BSTR _file_path, VARIANT_BOOL* ret);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public bool SaveToFile(String _file_path);
```

Visual Basic

```
Public Function SaveToFile(_file_path As String) As Boolean
```

Description

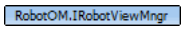
Function saves the view contents to the specified HTML format file.

Version

Available since version 7.5.

I.6 IRobotViewMngr

Class Hierarchy



C++

```
interface IRobotViewMngr : IDispatch;
```

C#

```
public interface IRobotViewMngr;
```

Visual Basic

```
Public Interface IRobotViewMngr
```

Description

View manager controls all the views and layouts available in Robot. .

I.6.1 IRobotViewMngr Members

The following tables list the members exposed by IRobotViewMngr.

Public Fields

	Name	Description
	CurrentLayout (see page 1471)	Identifier of active layout.
	NewViewAsTab (see page 1471)	Flag indicating if a new view is created as a tab.
	RecycleViews (see page 1471)	
	TableCount (see page 1471)	Number of opened tables.
	ViewCount (see page 1472)	Number of opened graphical views.

Public Methods

	Name	Description
	CreateFromSc (see page 1472)	Function creates and returns a view on the basis of a screen capture with the specified name. Type of the created view depends on the screen capture contents. If structure view manager is unable to make any of the defined view interfaces available, then, IRobotPrintable (see page 1477) interface is returned. If a screen capture with the specified name does not exist or if during its reading fro the archive an error occurs, then, empty reference will be returned (zero from the level of C language, Nothing from the level of Visual Basic). Available since version 1.7.
	CreateRtfView (see page 1473)	The function creates a view for files in RTF format.
	CreateTable (see page 1473)	The function creates and returns a table of the indicated type that will present the appropriate type of data. .

(c) 2014 Autodesk Inc. All rights reserved

	CreateView (see page 1473)	The function creates and returns the graphical view of the indicated type. .
	GetTable (see page 1474)	Function returns the interface for the table of the specified index. .
	GetView (see page 1474)	Function returns the interaface for the graphical view of the indicated index. .
	Refresh (see page 1474)	The function results in refreshing all the displayed views. .
	ShowDialog (see page 1475)	Function displays a dialog of the given identifier. For modal dialogs the function returns a code transferred originally as a result of the dialog operation. For non-modal dialogs the function returns a True/False status informing whether the dialog displaying failed or not. .

I.6.2 IRobotViewMngr Fields

The fields of the IRobotViewMngr class are listed here.

Public Fields

	Name	Description
	CurrentLayout (see page 1471)	Identifier of active layout.
	NewViewAsTab (see page 1471)	Flag indicating if a new view is created as a tab.
	RecycleViews (see page 1471)	
	TableCount (see page 1471)	Number of opened tables.
	ViewCount (see page 1472)	Number of opened graphical views.

I.6.2.1 CurrentLayout

C++

```
HRESULT get_CurrentLayout(IRobotLayoutId*);
HRESULT put_CurrentLayout(IRobotLayoutId);
```

C#

```
public IRobotLayoutId CurrentLayout { get; set; }
```

Visual Basic

```
Public CurrentLayout As IRobotLayoutId
```

Description

Identifier of active layout.

Version

Available since version 3.

I.6.2.2 NewViewAsTab

C++

```
HRESULT get_NewViewAsTab(VARIANT_BOOL*);
HRESULT put_NewViewAsTab(VARIANT_BOOL);
```

C#

```
public bool NewViewAsTab { get; set; }
```

Visual Basic

```
Public NewViewAsTab As Boolean
```

Description

Flag indicating if a new view is created as a tab.

Version

Available since version 9.7.

(c) 2014 Autodesk Inc. All rights reserved

I.6.2.3 RecycleViews

C++

```
HRESULT get_RecycleViews(VARIANT_BOOL*);
HRESULT put_RecycleViews(VARIANT_BOOL);
```

C#

```
public bool RecycleViews { get; set; }
```

Visual Basic

```
Public RecycleViews As Boolean
```

I.6.2.4 TableCount

C++

```
HRESULT get_TableCount(long*);
```

C#

```
public long TableCount { get; }
```

Visual Basic

```
Public ReadOnly TableCount As long
```

Description

Number of opened tables.

Version

Available since version 3.

I.6.2.5 ViewCount

C++

```
HRESULT get_ViewCount(long*);
```

C#

```
public long ViewCount { get; }
```

Visual Basic

```
Public ReadOnly ViewCount As long
```

Description

Number of opened graphical views.

Version

Available since version 3.

I.6.3 IRobotViewMngr Methods

The methods of the IRobotViewMngr class are listed here.

(c) 2014 Autodesk Inc. All rights reserved

Public Methods

	Name	Description
	CreateFromSc (see page 1472)	Function creates and returns a view on the basis of a screen capture with the specified name. Type of the created view depends on the screen capture contents. If structure view manager is unable to make any of the defined view interfaces available, then, IRobotPrintable (see page 1477) interface is returned. If a screen capture with the specified name does not exist or if during its reading fro the archive an error occurs, then, empty reference will be returned (zero from the level of C language, Nothing from the level of Visual Basic). Available since version 1.7.
	CreateRtfView (see page 1473)	The function creates a view for files in RTF format.
	CreateTable (see page 1473)	The function creates and returns a table of the indicated type that will present the appropriate type of data. .
	CreateView (see page 1473)	The function creates and returns the graphical view of the indicated type. .
	GetTable (see page 1474)	Function returns the interface for the table of the specified index. .
	GetView (see page 1474)	Function returns the interaface for the graphical view of the indicated index. .
	Refresh (see page 1474)	The function results in refreshing all the displayed views. .
	ShowDialog (see page 1475)	Function displays a dialog of the given identifier. For modal dialogs the function returns a code transferred originally as a result of the dialog operation. For non-modal dialogs the function returns a True/False status informing whether the dialog displaying failed or not. .

I.6.3.1 CreateFromSc

C++

```
HRESULT CreateFromSc(BSTR _sc_name, IDispatch* ret);
```

C#

```
public IDispatch CreateFromSc(String _sc_name);
```

Visual Basic

```
Public Function CreateFromSc(_sc_name As String) As IDispatch
```

Description

Function creates and returns a view on the basis of a screen capture with the specified name. Type of the created view depends on the screen capture contents. If structure view manager is unable to make any of the defined view interfaces available, then, IRobotPrintable (see page 1477) interface is returned. If a screen capture with the specified name does not exist or if during its reading fro the archive an error occurs, then, empty reference will be returned (zero from the level of C language, Nothing from the level of Visual Basic). Available since version 1.7.

I.6.3.2 CreateRtfView

C++

```
HRESULT CreateRtfView(IRobotRtfView** ret);
```

C#

```
public IRobotRtfView CreateRtfView();
```

Visual Basic

```
Public Function CreateRtfView() As IRobotRtfView
```

Description

The function creates a view for files in RTF format.

(c) 2014 Autodesk Inc. All rights reserved

I.6.3.3 CreateTable

C++

```
HRESULT CreateTable(IRobotTableType _tab_type, IRobotTableDataType _data_type,
IRobotTable** ret);
```

C#

```
public IRobotTable CreateTable(IRobotTableType _tab_type, IRobotTableDataType _data_type);
```

Visual Basic

```
Public Function CreateTable(_tab_type As IRobotTableType, _data_type As
IRobotTableDataType) As IRobotTable
```

Description

The function creates and returns a table of the indicated type that will present the appropriate type of data. .

I.6.3.4 CreateView

C++

```
HRESULT CreateView(IRobotViewType _view_type, IRobotView** ret);
```

C#

```
public IRobotView CreateView(IRobotViewType _view_type);
```

Visual Basic

```
Public Function CreateView(_view_type As IRobotViewType) As IRobotView
```

Description

The function creates and returns the graphical view of the indicated type. .

I.6.3.5 GetTable

C++

```
HRESULT GetTable(long _table_idx, IRobotTableFrame** ret);
```

C#

```
public IRobotTableFrame GetTable(long _table_idx);
```

Visual Basic

```
Public Function GetTable(_table_idx As long) As IRobotTableFrame
```

Description

Function returns the interface for the table of the specified index. .

Version

Available since version 3.

I.6.3.6 GetView

C++

```
HRESULT GetView(long _view_idx, IRobotView2** ret);
```

C#

```
public IRobotView2 GetView(long _view_idx);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Function GetView(_view_idx As long) As IRobotView2
```

Description

Function returns the interaface for the graphical view of the indicated index. .

Version

Available since version 3.

I.6.3.7 Refresh

C++

```
HRESULT Refresh();
```

C#

```
public void Refresh();
```

Visual Basic

```
Public Sub Refresh()
```

Description

The function results in refreshing all the displayed views. .

I.6.3.8 ShowDialog

C++

```
HRESULT ShowDialog(IRobotDialogId _dlg_id, long _parent_wnd = 0, VARIANT_BOOL _modal =
True, VARIANT* ret);
```

C#

```
public VARIANT ShowDialog(IRobotDialogId _dlg_id, long _parent_wnd = 0, bool _modal = True);
```

Visual Basic

```
Public Function ShowDialog(_dlg_id As IRobotDialogId, Optional _parent_wnd As long = 0,
Optional _modal As Boolean = True) As VARIANT
```

Description

Function displays a dialog of the given identifier. For modal dialogs the function returns a code transferred originally as a result of the dialog operation. For non-modal dialogs the function returns a True/False status informing whether the dialog displaying failed or not. .

Version

Available since version 11.

I.7 IRobotLayoutId

C++

```
enum IRobotLayoutId;
```

C#

```
public enum IRobotLayoutId;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Enum IRobotLayoutId
```

Members

Members	Description
I_LI_MODEL_GEOMETRY = 1	Available since version 3.
I_LI_MODEL_NODES = 2	Available since version 3.
I_LI_MODEL_BARS = 3	Available since version 3.
I_LI_MODEL_SUPPORTS = 4	Available since version 3.
I_LI_MODEL_PROPERTIES = 5	Available since version 3.
I_LI_MODEL_LOADS = 6	Available since version 3.
I_LI_RC_COLUMN_DEFINITION = 11	Available since version 3.
I_LI_RC_BEAM_DEFINITION = 21	Available since version 3.
I_LI_RC_BEAM_WALL_DEFINITION = 31	Available since version 3.
I_LI_FOOTING_DEFINITION = 41	Available since version 3.
I_LI_DESIGN_CONNECTIONS = 51	Available since version 3.
I_LI_DESIGN_STEEL_ALUMINUM = 52	Available since version 3.
I_LI_DESIGN_WOOD = 53	Available since version 3.
I_LI_DESIGN_RC_MEMBERS = 54	Available since version 3.
I_LI_TOOL_SECTION_DEFINITION = 61	Available since version 3.
I_LI_TOOL_TEXT_FILE = 62	Available since version 3.
I_LI_TOOL_FINAL_DRAWING = 63	Available since version 3.
I_LI_FOUNDATION_DEFINITION = 71	Available since version 3.
I_LI_RESULTS_DIAGRAMS = 81	Available since version 3.
I_LI_RESULTS_DETAILED = 82	Available since version 3.
I_LI_RESULTS_STRESS_BARS = 83	Available since version 3.
I_LI_RESULTS_STRESS_STRUCTURE = 84	Available since version 3.
I_LI_RESULTS_MAPS = 85	Available since version 3.
I_LI_RC_SLAB_REQUIRED_REINF = 91	Available since version 3.
I_LI_RC_SLAB_PROVIDED_REINF = 92	Available since version 3.
I_LI_RC_SLAB_REINFORCEMENT = 93	Available since version 3.

Description

Set of identifiers for the layouts defined in the Robot program. .

Version

Available since version 3.

I.8 IRobotDialogId

C++

```
enum IRobotDialogId;
```

C#

```
public enum IRobotDialogId;
```

Visual Basic

```
Public Enum IRobotDialogId
```

(c) 2014 Autodesk Inc. All rights reserved

Members

Members	Description
I_DI_BAR_SECTIONS = 1	Section dialog. Available since version 11.
I_DI_SUPPORTS = 2	Support dialog. Available since version 11.
I_DI_PANEL_THICKNESS = 3	Panel thickness dialog. Available since version 11.

Description

Identifiers of available dialogs.

Version

Available since version 11.

II Printouts

Enumerations

	Name	Description
	IRobotVariablePredefinedId (see page 1491)	Set of identifiers for standard (previously prepared for the user) variables. .
	IRobotOutputFileFormat (see page 1502)	Available file formats to which a printout may be directed. .
	IRobotPageSetupOrientation (see page 1514)	Page setup.
	IRobotPageSetupFrameType (see page 1515)	Types of frames used for decoration of individual page elements. .
	IRobotPageSetupTocLocation (see page 1517)	
	IRobotReportItemType (see page 1517)	Available types of printout components.
	IRobotExternalPreviewFormat (see page 1518)	External format of a print preview.

Interfaces

	Name	Description
	IRobotPrintable (see page 1477)	Interface of any data object that is to be printed by the printing module of Robot. .
	IRobotPrintEngine (see page 1480)	Interface representing the Robot printing module. .
	IRobotVariableMngr (see page 1487)	Manager of variables (parameters) used in printout templates.
	IRobotPageSetup (see page 1491)	Interface allows reading and modifying current page settings and enables template management. The Robot program allows definition of many templates storing page settings, which are differentiated by name. At the given moment one of the templates is set as current (default). .
	IRobotReportTemplateMngr (see page 1502)	There is a possibility to save a previously composed printout as a template, which may be used many times afterwards in the same or different project. Printout templates are recognized by names. All templates available in the current project are managed by the template manager. .
	IRobotReportStdElementRtf (see page 1505)	Standard (predefined) printout element based on an RTF format template file.

(c) 2014 Autodesk Inc. All rights reserved

	IRobotReportItemList (see page 1507)	List of printout components.
	IRobotReportItem (see page 1509)	Interface that describes a printout element.
	IRobotPageSetupTableOfContents (see page 1515)	Interface that describes the table of contents.
	IRobotVariableMngrExtension (see page 1518)	Interface of external extension for the manager of variables (parameters) used in printouts.

II.1 IRobotPrintable

Class Hierarchy

RobotOM.IRobotPrintable

C++

```
interface IRobotPrintable : IDispatch;
```

C#

```
public interface IRobotPrintable;
```

Visual Basic

```
Public Interface IRobotPrintable
```

Description

Interface of any data object that is to be printed by the printing module of Robot. .

II.1.1 IRobotPrintable Members

The following tables list the members exposed by IRobotPrintable.

Public Fields

	Name	Description
	Comment (see page 1478)	Text of the comment to be added to the object on the printout.
	IncludeDateTime (see page 1478)	The flag indicating if the current date and time is added to the object on the printout.
	StartFromNewPage (see page 1479)	Falg indicating whether an object should be printed from a new page .
	Title (see page 1479)	Title.

Public Methods

	Name	Description
	SaveToFile (see page 1479)	Function exports object contents to the indicated file with the specified format. Available since version 1.7.

II.1.2 IRobotPrintable Fields

The fields of the IRobotPrintable class are listed here.

Public Fields

	Name	Description
	Comment (see page 1478)	Text of the comment to be added to the object on the printout.
	IncludeDateTime (see page 1478)	The flag indicating if the current date and time is added to the object on the printout.

(c) 2014 Autodesk Inc. All rights reserved

	StartFromNewPage (see page 1479)	Falg indicating whether an object should be printed from a new page .
	Title (see page 1479)	Title.

II.1.2.1 Comment

C++

```
HRESULT get_Comment(BSTR*);
HRESULT put_Comment(BSTR);
```

C#

```
public String Comment { get; set; }
```

Visual Basic

```
Public Comment As String
```

Description

Text of the comment to be added to the object on the printout.

Version

Available since version 12.

II.1.2.2 IncludeDateTime

C++

```
HRESULT get_IncludeDateTime(VARIANT_BOOL*);
HRESULT put_IncludeDateTime(VARIANT_BOOL);
```

C#

```
public bool IncludeDateTime { get; set; }
```

Visual Basic

```
Public IncludeDateTime As Boolean
```

Description

The flag indicating if the current date and time is added to the object on the printout.

Version

Available since version 12.

II.1.2.3 StartFromNewPage

C#

```
void StartFromNewPage;
```

Description

Falg indicating whether an object should be printed from a new page .

II.1.2.4 Title

C++

```
HRESULT get_Title(BSTR*);
HRESULT put_Title(BSTR);
```

C#

```
public String Title { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Title As String
```

Description

Title.

II.1.3 IRobotPrintable Methods

The methods of the IRobotPrintable class are listed here.

Public Methods

	Name	Description
	SaveToFile (see page 1479)	Function exports object contents to the indicated file with the specified format. Available since version 1.7.

II.1.3.1 SaveToFile

C++

```
HRESULT SaveToFile(BSTR _file_path, IRobotOutputFileFormat _file_format, VARIANT_BOOL* ret);
```

C#

```
public bool SaveToFile(String _file_path, IRobotOutputFileFormat _file_format);
```

Visual Basic

```
Public Function SaveToFile(_file_path As String, _file_format As IRobotOutputFileFormat) As
Boolean
```

Description

Function exports object contents to the indicated file with the specified format. Available since version 1.7.

II.2 IRobotPrintEngine

Class Hierarchy

RobotOM.IRobotPrintEngine

C++

```
interface IRobotPrintEngine : IDispatch;
```

C#

```
public interface IRobotPrintEngine;
```

Visual Basic

```
Public Interface IRobotPrintEngine
```

Description

Interface representing the Robot printing module. .

II.2.1 IRobotPrintEngine Members

The following tables list the members exposed by IRobotPrintEngine.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	IsWhilePreview (see page 1481)	Flag indicating that the application is currently working in the "pint preview" mode.
	OrganizerItems (see page 1481)	List of printout components defined in the Printout Composition dialog box.
	PageSetup (see page 1482)	Page setup.
	ReportTemplates (see page 1482)	List of available printout templates Available since version 1.7.
	ScreenCaptures (see page 1482)	List of available screen captures Available since version 1.7.

Public Methods

	Name	Description
	AddScToReport (see page 1483)	Function adds screen capture with a specified name at the end of the current printout. If a screen capture with a specified name is not found, zero value is returned (False). Available since version 1.7.
	AddTemplateToReport (see page 1483)	Function adds contents of a printout template with a specified name at the end of the printout. If a template with this name is not found, then function returns zero value (False). Available since version 1.7.
	AddToReport (see page 1484)	The function adds the indicated object at the end of the currently compoesed printout. .
	ClosePreview (see page 1484)	Function forces closing a print preview.
	CreateReportFromOrganizer (see page 1484)	Function creates a printout composed of elements currently defined in the Printout Composition dialog box.
	ExternalPreviewReport (see page 1485)	Function launches an external program and exports the current printout to it.
	PreviewReport (see page 1485)	The function runs the print preview of the current printout composition .
	PrintReport (see page 1485)	The function results in sending the current printout composition to the printer. .
	RemoveFromReport (see page 1485)	The function deletes the indicated object from the current printout. .
	RemoveScFromReport (see page 1486)	Function deletes screen capture with a specified name from the current printout. First screen capture with this name occurring on the printout component list is deleted. Available since version 1.7.
	ResetReport (see page 1486)	The function deletes all objects from the current printout. .
	SaveReportToFile (see page 1486)	Function exports printout to the specified file with the defined format. Available since version 1.7.
	SaveReportToOrganizer (see page 1486)	Function saves the current printout as a current list of printout elements presented in the Printout Composition dialog box.
	SaveReportToTemplate (see page 1487)	Function saves printout to the template with a specified name. If a template with this name already exists its contents will be replaced. Available since version 1.7.

II.2.2 IRobotPrintEngine Fields

The fields of the IRobotPrintEngine class are listed here.

Public Fields

	Name	Description
	IsWhilePreview (see page 1481)	Flag indicating that the application is currently working in the "pint preview" mode.
	OrganizerItems (see page 1481)	List of printout components defined in the Printout Composition dialog box.
	PageSetup (see page 1482)	Page setup.

(c) 2014 Autodesk Inc. All rights reserved

	ReportTemplates (see page 1482)	List of available printout templates Available since version 1.7.
	ScreenCaptures (see page 1482)	List of available screen captures Available since version 1.7.

II.2.2.1 IsWhilePreview

C++

```
HRESULT get_IsWhilePreview(VARIANT_BOOL*);
```

C#

```
public bool IsWhilePreview { get; }
```

Visual Basic

```
Public ReadOnly IsWhilePreview As Boolean
```

Description

Flag indicating that the application is currently working in the "pint preview" mode.

II.2.2.2 OrganizerItems

C++

```
HRESULT get_OrganizerItems(IRobotReportItemList**);
```

C#

```
public IRobotReportItemList OrganizerItems { get; }
```

Visual Basic

```
Public ReadOnly OrganizerItems As IRobotReportItemList
```

Description

List of printout components defined in the Printout Composition dialog box.

Version

Available since version 5.5.

II.2.2.3 PageSetup

C++

```
HRESULT get_PageSetup(IRobotPageSetup**);
```

C#

```
public IRobotPageSetup PageSetup { get; }
```

Visual Basic

```
Public ReadOnly PageSetup As IRobotPageSetup
```

Description

Page setup.

II.2.2.4 ReportTemplates

C++

```
HRESULT get_ReportTemplates(IRobotReportTemplateMngr**);
```

C#

```
public IRobotReportTemplateMngr ReportTemplates { get; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public ReadOnly ReportTemplates As IRobotReportTemplateMngr
```

Description

List of available printout templates Available since version 1.7.

II.2.2.5 ScreenCaptures

C++

```
HRESULT get_ScreenCaptures(IRobotScreenCaptureMngr**);
```

C#

```
public IRobotScreenCaptureMngr ScreenCaptures { get; }
```

Visual Basic

```
Public ReadOnly ScreenCaptures As IRobotScreenCaptureMngr
```

Description

List of available screen captures Available since version 1.7.

II.2.3 IRobotPrintEngine Methods

The methods of the IRobotPrintEngine class are listed here.

Public Methods

	Name	Description
	AddScToReport (see page 1483)	Function adds screen capture with a specified name at the end of the current printout. If a screen capture with a specified name is not found, zero value is returned (False). Available since version 1.7.
	AddTemplateToReport (see page 1483)	Function adds contents of a printout template with a specified name at the end of the printout. If a template with this name is not found, then function returns zero value (False). Available since version 1.7.
	AddToReport (see page 1484)	The function adds the indicated object at the end of the currently compoesed printout. .
	ClosePreview (see page 1484)	Function forces closing a print preview.
	CreateReportFromOrganizer (see page 1484)	Function creates a printout composed of elements currently defined in the Printout Composition dialog box.
	ExternalPreviewReport (see page 1485)	Function launches an external program and exports the current printout to it.
	PreviewReport (see page 1485)	The function runs the print preview of the current printout composition .
	PrintReport (see page 1485)	The function results in sending the current printout composition to the printer. .
	RemoveFromReport (see page 1485)	The function deletes the indicated object from the current printout. .
	RemoveScFromReport (see page 1486)	Function deletes screen capture with a specified name from the current printout. First screen capture with this name occurring on the printout component list is deleted. Available since version 1.7.
	ResetReport (see page 1486)	The function deletes all objects from the current printout. .
	SaveReportToFile (see page 1486)	Function exports printout to the specified file with the defined format. Available since version 1.7.
	SaveReportToOrganizer (see page 1486)	Function saves the current printout as a current list of printout elements presented in the Printout Composition dialog box.
	SaveReportToTemplate (see page 1487)	Function saves printout to the template with a specified name. If a template with this name already exists its contents will be replaced. Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

II.2.3.1 AddScToReport

C++

```
HRESULT AddScToReport(BSTR _sc_name, VARIANT_BOOL* ret);
```

C#

```
public bool AddScToReport(String _sc_name);
```

Visual Basic

```
Public Function AddScToReport(_sc_name As String) As Boolean
```

Description

Function adds screen capture with a specified name at the end of the current printout. If a screen capture with a specified name is not found, zero value is returned (False). Available since version 1.7.

II.2.3.2 AddTemplateToReport

C++

```
HRESULT AddTemplateToReport(BSTR _tmpl_name, VARIANT_BOOL* ret);
```

C#

```
public bool AddTemplateToReport(String _tmpl_name);
```

Visual Basic

```
Public Function AddTemplateToReport(_tmpl_name As String) As Boolean
```

Description

Function adds contents of a printout template with a specified name at the end of the printout. If a template with this name is not found, then function returns zero value (False). Available since version 1.7.

II.2.3.3 AddToReport

C++

```
HRESULT AddToReport(IRobotPrintable* _obj);
```

C#

```
public void AddToReport(IRobotPrintable _obj);
```

Visual Basic

```
Public Sub AddToReport(ByRef _obj As IRobotPrintable)
```

Description

The function adds the indicated object at the end of the currently compoesed printout. .

II.2.3.4 ClosePreview

C++

```
HRESULT ClosePreview();
```

C#

```
public void ClosePreview();
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Sub ClosePreview()
```

Description

Function forces closing a print preview.

II.2.3.5 CreateReportFromOrganizer

C++

```
HRESULT CreateReportFromOrganizer();
```

C#

```
public void CreateReportFromOrganizer();
```

Visual Basic

```
Public Sub CreateReportFromOrganizer()
```

Description

Function creates a printout composed of elements currently defined in the Printout Composition dialog box.

Version

Available since version 6.4.

II.2.3.6 ExternalPreviewReport

C++

```
HRESULT ExternalPreviewReport(IRobotExternalPreviewFormat _format);
```

C#

```
public void ExternalPreviewReport(IRobotExternalPreviewFormat _format);
```

Visual Basic

```
Public Sub ExternalPreviewReport(_format As IRobotExternalPreviewFormat)
```

Description

Function launches an external program and exports the current printout to it.

Version

Available since version 6.4.

II.2.3.7 PreviewReport

C++

```
HRESULT PreviewReport();
```

C#

```
public void PreviewReport();
```

Visual Basic

```
Public Sub PreviewReport()
```

Description

The function runs the print preview of the current printout composition .

(c) 2014 Autodesk Inc. All rights reserved

II.2.3.8 PrintReport

C++

```
HRESULT PrintReport();
```

C#

```
public void PrintReport();
```

Visual Basic

```
Public Sub PrintReport()
```

Description

The function results in sending the current printout composition to the printer. .

II.2.3.9 RemoveFromReport

C++

```
HRESULT RemoveFromReport(IRobotPrintable* _obj);
```

C#

```
public void RemoveFromReport(IRobotPrintable _obj);
```

Visual Basic

```
Public Sub RemoveFromReport(ByRef _obj As IRobotPrintable)
```

Description

The function deletes the indicated object from the current printout. .

II.2.3.10 RemoveScFromReport

C++

```
HRESULT RemoveScFromReport(BSTR _sc_name);
```

C#

```
public void RemoveScFromReport(String _sc_name);
```

Visual Basic

```
Public Sub RemoveScFromReport(_sc_name As String)
```

Description

Function deletes screen capture with a specified name from the current printout. First screen capture with this name occurring on the printout component list is deleted. Available since version 1.7.

II.2.3.11 ResetReport

C++

```
HRESULT ResetReport();
```

C#

```
public void ResetReport();
```

Visual Basic

```
Public Sub ResetReport()
```

(c) 2014 Autodesk Inc. All rights reserved

Description

The function deletes all objects from the current printout. .

II.2.3.12 SaveReportToFile

C++

```
HRESULT SaveReportToFile(BSTR _file_path, IRobotOutputFileFormat _file_format,
VARIANT_BOOL* ret);
```

C#

```
public bool SaveReportToFile(String _file_path, IRobotOutputFileFormat _file_format);
```

Visual Basic

```
Public Function SaveReportToFile(_file_path As String, _file_format As
IRobotOutputFileFormat) As Boolean
```

Description

Function exports printout to the specified file with the defined format. Available since version 1.7.

II.2.3.13 SaveReportToOrganizer

C++

```
HRESULT SaveReportToOrganizer();
```

C#

```
public void SaveReportToOrganizer();
```

Visual Basic

```
Public Sub SaveReportToOrganizer()
```

Description

Function saves the current printout as a current list of printout elements presented in the Printout Composition dialog box.

Version

Available since version 6.4.

II.2.3.14 SaveReportToTemplate

C++

```
HRESULT SaveReportToTemplate(BSTR _tmpl_name);
```

C#

```
public void SaveReportToTemplate(String _tmpl_name);
```

Visual Basic

```
Public Sub SaveReportToTemplate(_tmpl_name As String)
```

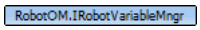
Description

Function saves printout to the template with a specified name. If a template with this name already exists its contents will be replaced. Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

II.3 IRobotVariableMngr

Class Hierarchy



C++

```
interface IRobotVariableMngr : IDispatch;
```

C#

```
public interface IRobotVariableMngr;
```

Visual Basic

```
Public Interface IRobotVariableMngr
```

Description

Manager of variables (parameters) used in printout templates.

II.3.1 IRobotVariableMngr Members

The following tables list the members exposed by IRobotVariableMngr.

Public Methods

	Name	Description
	AddExtension (see page 1488)	Function enables extending the manager with an external object providing values for variables.
	Delete (see page 1489)	The function deletes a variable with the given name. .
	Exist (see page 1489)	The function checks if there exists a variable with the indicated name .
	GetPredefinedValue (see page 1489)	Function returns the value of the given standard variable.
	GetValue (see page 1489)	The function returns the value of the indicated variable. If the variable with the indicated name does not exist, the function returns an empty string.
	RemoveExtension (see page 1490)	Function deletes a specified object from the extension list.
	SetPredefinedValue (see page 1490)	Function sets value of a specified standard variable. Some variables are read-only. For such a variable, values cannot be set. Then, this function returns zero (False). .
	SetValue (see page 1490)	The function ascribes a defined value to the variable with the given name. If the variable does not exist, it will be defined. If it does exist, its value will be changed. .

II.3.2 IRobotVariableMngr Methods

The methods of the IRobotVariableMngr class are listed here.

Public Methods

	Name	Description
	AddExtension (see page 1488)	Function enables extending the manager with an external object providing values for variables.
	Delete (see page 1489)	The function deletes a variable with the given name. .
	Exist (see page 1489)	The function checks if there exists a variable with the indicated name .
	GetPredefinedValue (see page 1489)	Function returns the value of the given standard variable.

(c) 2014 Autodesk Inc. All rights reserved

	GetValue (see page 1489)	The function returns the value of the indicated variable. If the variable with the indicated name does not exist, the function returns an empty string.
	RemoveExtension (see page 1490)	Function deletes a specified object from the extension list.
	SetPredefinedValue (see page 1490)	Function sets value of a specified standard variable. Some variables are read-only. For such a variable, values cannot be set. Then, this function returns zero (False). .
	SetValue (see page 1490)	The function ascribes a defined value to the variable with the given name. If the variable does not exist, it will be defined. If it does exist, its value will be changed. .

II.3.2.1 AddExtension

C++

```
HRESULT AddExtension(IRobotVariableMngrExtension* _ext_obj);
```

C#

```
public void AddExtension(IRobotVariableMngrExtension _ext_obj);
```

Visual Basic

```
Public Sub AddExtension(ByRef _ext_obj As IRobotVariableMngrExtension)
```

Description

Function enables extending the manager with an external object providing values for variables.

II.3.2.2 Delete

C++

```
HRESULT Delete(BSTR _var_name);
```

C#

```
public void Delete(String _var_name);
```

Visual Basic

```
Public Sub Delete(_var_name As String)
```

Description

The function deletes a variable with the given name. .

II.3.2.3 Exist

C++

```
HRESULT Exist(BSTR _var_name, VARIANT_BOOL* ret);
```

C#

```
public bool Exist(String _var_name);
```

Visual Basic

```
Public Function Exist(_var_name As String) As Boolean
```

Description

The function checks if there exists a variable with the indicated name .

(c) 2014 Autodesk Inc. All rights reserved

II.3.2.4 GetPredefinedValue

C++

```
HRESULT GetPredefinedValue(IRobotVariablePredefinedId _var_id, BSTR* ret);
```

C#

```
public String GetPredefinedValue(IRobotVariablePredefinedId _var_id);
```

Visual Basic

```
Public Function GetPredefinedValue(_var_id As IRobotVariablePredefinedId) As String
```

Description

Function returns the value of the given standard variable.

II.3.2.5 GetValue

C++

```
HRESULT GetValue(BSTR _var_name, BSTR* ret);
```

C#

```
public String GetValue(String _var_name);
```

Visual Basic

```
Public Function GetValue(_var_name As String) As String
```

Description

The function returns the value of the indicated variable. If the variable with the indicated name does not exist, the function returns an empty string.

II.3.2.6 RemoveExtension

C++

```
HRESULT RemoveExtension(IRobotVariableMngrExtension* _ext_obj);
```

C#

```
public void RemoveExtension(IRobotVariableMngrExtension _ext_obj);
```

Visual Basic

```
Public Sub RemoveExtension(ByRef _ext_obj As IRobotVariableMngrExtension)
```

Description

Function deletes a specified object from the extension list.

II.3.2.7 SetPredefinedValue

C++

```
HRESULT SetPredefinedValue(IRobotVariablePredefinedId _var_id, BSTR _var_value,
VARIANT_BOOL* ret);
```

C#

```
public bool SetPredefinedValue(IRobotVariablePredefinedId _var_id, String _var_value);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Function SetPredefinedValue(_var_id As IRobotVariablePredefinedId, _var_value As String) As Boolean
```

Description

Function sets value of a specified standard variable. Some variables are read-only. For such a variable, values cannot be set. Then, this function returns zero (False). .

II.3.2.8 SetValue

C++

```
HRESULT SetValue(BSTR _var_name, BSTR _var_value);
```

C#

```
public void SetValue(String _var_name, String _var_value);
```

Visual Basic

```
Public Sub SetValue(_var_name As String, _var_value As String)
```

Description

The function ascribes a defined value to the variable with the given name. If the variable does not exist, it will be defined. If it does exist, its value will be changed. .

II.4 IRobotVariablePredefinedId

C++

```
enum IRobotVariablePredefinedId;
```

C#

```
public enum IRobotVariablePredefinedId;
```

Visual Basic

```
Public Enum IRobotVariablePredefinedId
```

Members

Members	Description
I_VPI_REPORT_PAGE_NUMBER = 0	Number of a print-out page (initial or current during printing).
I_VPI_DATE = 1	Current date.
I_VPI_TIME = 2	Current time.
I_VPI_ROBOT_APP = 3	Program name .
I_VPI_ROBOT_VER = 4	Program version.
I_VPI_ROBOT_PROVIDER = 5	Name of the program supplier (dealer).
I_VPI_USER_NAME = 6	User's name .
I_VPI_USER_ADDRESS = 7	User's address.
I_VPI_PROJECT_NAME = 8	Project name .
I_VPI_PROJECT_FILE = 9	Name of the file containing the project.
I_VPI_PROJECT_DIRECTORY = 10	Folder where the file with the project is stored.
I_VPI_PROJECT_SIZE = 11	Size of the file with the project.
I_VPI_PROJECT_CREATED = 12	Date when the project file was created.
I_VPI_PROJECT_MODIFIED = 13	Date of the last modification of the project file.
I_VPI_PROJECT_VERSION = 14	Project version.

(c) 2014 Autodesk Inc. All rights reserved

Description

Set of identifiers for standard (previously prepared for the user) variables. .

II.5 IRobotPageSetup

Class Hierarchy

RobotOM.IRobotPageSetup

C++

```
interface IRobotPageSetup : IDispatch;
```

C#

```
public interface IRobotPageSetup;
```

Visual Basic

```
Public Interface IRobotPageSetup
```

Description

Interface allows reading and modifying current page settings and enables template management. The Robot program allows definition of many templates storing page settings, which are differentiated by name. At the given moment one of the templates is set as current (default). .

II.5.1 IRobotPageSetup Members

The following tables list the members exposed by IRobotPageSetup.

Public Fields

	Name	Description
	Footer (see page 1493)	Footer on each printout page .
	FromEdgeFooter (see page 1494)	Distance of footer from the bottom edge of paper in millimeters.
	FromEdgeHeader (see page 1494)	Distance of header from the top edge of paper in millimeters.
	Gutter (see page 1494)	Gutter in millimeters.
	Header (see page 1495)	Header on each printout page.
	IsCurrent (see page 1495)	Flag that enables setting page parameters as the current ones .
	MarginBottom (see page 1495)	Bottom margin in millimeters.
	MarginLeft (see page 1496)	Left margin in millimeters.
	MarginRight (see page 1496)	Right margin in millimeters.
	MarginTop (see page 1496)	Top margin in millimeters.
	PageOrientation (see page 1497)	Page setup.
	PaperHeight (see page 1497)	Paper height in millimeters.
	PaperSize (see page 1497)	Identifier of paper size.
	PaperWidth (see page 1498)	Paper width in millimeters.
	StartPageNumber (see page 1498)	Initial page number Attention: Initial page number is not stored in a template. Setting a value of the initial page number means the same as setting a value of the variable named PAGE_NUMBER by means of the IRobotVariableMngr (see page 1487) interface.
	TemplateCount (see page 1498)	Number of the remembered templates with page settings .
	TemplateName (see page 1499)	Template name.
	TextFrame (see page 1499)	Page contents frame.
	TitlePage (see page 1499)	Printout title page .

(c) 2014 Autodesk Inc. All rights reserved

	Toc (see page 1499)	Parameters of the table of contents.
	Variables (see page 1500)	Manager of variables (parameters).

Public Methods

	Name	Description
	GetTemplateName (see page 1500)	Function returns the name of a template with the specified index. Templates are indexed with values from 1 to TemplateCount (see page 1498). .
	Load (see page 1501)	Function reads page settings from the specified template. .
	LoadCurrent (see page 1501)	Function reads current page settings. .
	Save (see page 1501)	Function saves current page settings. .
	SaveAs (see page 1502)	Function saves current page settings as a template with the specified name. .

II.5.2 IRobotPageSetup Fields

The fields of the IRobotPageSetup class are listed here.

Public Fields

	Name	Description
	Footer (see page 1493)	Footer on each printout page .
	FromEdgeFooter (see page 1494)	Distance of footer from the bottom edge of paper in millimeters.
	FromEdgeHeader (see page 1494)	Distance of header from the top edge of paper in millimeters.
	Gutter (see page 1494)	Gutter in millimeters.
	Header (see page 1495)	Header on each printout page.
	IsCurrent (see page 1495)	Flag that enables setting page parameters as the current ones .
	MarginBottom (see page 1495)	Bottom margin in millimeters.
	MarginLeft (see page 1496)	Left margin in millimeters.
	MarginRight (see page 1496)	Right margin in millimeters.
	MarginTop (see page 1496)	Top margin in millimeters.
	PageOrientation (see page 1497)	Page setup.
	PaperHeight (see page 1497)	Paper height in millimeters.
	PaperSize (see page 1497)	Identifier of paper size.
	PaperWidth (see page 1498)	Paper width in millimeters.
	StartPageNumber (see page 1498)	Initial page number Attention: Initial page number is not stored in a template. Setting a value of the initial page number means the same as setting a value of the variable named PAGE_NUMBER by means of the IRobotVariableMngr (see page 1487) interface.
	TemplateCount (see page 1498)	Number of the remembered templates with page settings .
	TemplateName (see page 1499)	Template name.
	TextFrame (see page 1499)	Page contents frame.
	TitlePage (see page 1499)	Printout title page .
	Toc (see page 1499)	Parameters of the table of contents.
	Variables (see page 1500)	Manager of variables (parameters).

II.5.2.1 Footer

C++

```
HRESULT get_Footer(IRobotReportStdElementRtf**);
```

C#

```
public IRobotReportStdElementRtf Footer { get; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public ReadOnly Footer As IRobotReportStdElementRtf
```

Description

Footer on each printout page .

Version

Available since version 2.5.

II.5.2.2 FromEdgeFooter

C++

```
HRESULT get_FromEdgeFooter(double*);
HRESULT put_FromEdgeFooter(double);
```

C#

```
public double FromEdgeFooter { get; set; }
```

Visual Basic

```
Public FromEdgeFooter As double
```

Description

Distance of footer from the bottom edge of paper in millimeters.

Version

Available since version 5.5.

II.5.2.3 FromEdgeHeader

C++

```
HRESULT get_FromEdgeHeader(double*);
HRESULT put_FromEdgeHeader(double);
```

C#

```
public double FromEdgeHeader { get; set; }
```

Visual Basic

```
Public FromEdgeHeader As double
```

Description

Distance of header from the top edge of paper in millimeters.

Version

Available since version 5.5.

II.5.2.4 Gutter

C++

```
HRESULT get_Gutter(double*);
HRESULT put_Gutter(double);
```

C#

```
public double Gutter { get; set; }
```

Visual Basic

```
Public Gutter As double
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Gutter in millimeters.

Version

Available since version 5.5.

II.5.2.5 Header

C++

```
HRESULT get_Header(IRobotReportStdElementRtf**);
```

C#

```
public IRobotReportStdElementRtf Header { get; }
```

Visual Basic

```
Public ReadOnly Header As IRobotReportStdElementRtf
```

Description

Header on each printout page.

Version

Available since version 2.5.

II.5.2.6 IsCurrent

C++

```
HRESULT get_IsCurrent(VARIANT_BOOL*);
HRESULT put_IsCurrent(VARIANT_BOOL);
```

C#

```
public bool IsCurrent { get; set; }
```

Visual Basic

```
Public IsCurrent As Boolean
```

Description

Flag that enables setting page parameters as the current ones .

Version

Available since version 5.5.

II.5.2.7 MarginBottom

C++

```
HRESULT get_MarginBottom(double*);
HRESULT put_MarginBottom(double);
```

C#

```
public double MarginBottom { get; set; }
```

Visual Basic

```
Public MarginBottom As double
```

Description

Bottom margin in millimeters.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 5.5.

II.5.2.8 MarginLeft

C++

```
HRESULT get_MarginLeft(double*);
HRESULT put_MarginLeft(double);
```

C#

```
public double MarginLeft { get; set; }
```

Visual Basic

```
Public MarginLeft As double
```

Description

Left margin in millimeters.

Version

Available since version 5.5.

II.5.2.9 MarginRight

C++

```
HRESULT get_MarginRight(double*);
HRESULT put_MarginRight(double);
```

C#

```
public double MarginRight { get; set; }
```

Visual Basic

```
Public MarginRight As double
```

Description

Right margin in millimeters.

Version

Available since version 5.5.

II.5.2.10 MarginTop

C++

```
HRESULT get_MarginTop(double*);
HRESULT put_MarginTop(double);
```

C#

```
public double MarginTop { get; set; }
```

Visual Basic

```
Public MarginTop As double
```

Description

Top margin in millimeters.

Version

Available since version 5.5.

(c) 2014 Autodesk Inc. All rights reserved

II.5.2.11 PageOrientation

C++

```
HRESULT get_PageOrientation(IRobotPageSetupOrientation*);
HRESULT put_PageOrientation(IRobotPageSetupOrientation);
```

C#

```
public IRobotPageSetupOrientation PageOrientation { get; set; }
```

Visual Basic

```
Public PageOrientation As IRobotPageSetupOrientation
```

Description

Page setup.

Version

Available since version 5.5.

II.5.2.12 PaperHeight

C++

```
HRESULT get_PaperHeight(long*);
HRESULT put_PaperHeight(long);
```

C#

```
public long PaperHeight { get; set; }
```

Visual Basic

```
Public PaperHeight As long
```

Description

Paper height in millimeters.

Version

Available since version 5.5.

II.5.2.13 PaperSize

C++

```
HRESULT get_PaperSize(long*);
HRESULT put_PaperSize(long);
```

C#

```
public long PaperSize { get; set; }
```

Visual Basic

```
Public PaperSize As long
```

Description

Identifier of paper size.

Version

Available since version 5.5.

(c) 2014 Autodesk Inc. All rights reserved

II.5.2.14 PaperWidth

C++

```
HRESULT get_PaperWidth(long*);
HRESULT put_PaperWidth(long);
```

C#

```
public long PaperWidth { get; set; }
```

Visual Basic

```
Public PaperWidth As long
```

Description

Paper width in millimeters.

Version

Available since version 5.5.

II.5.2.15 StartPageNumber

C++

```
HRESULT get_StartPageNumber(long*);
HRESULT put_StartPageNumber(long);
```

C#

```
public long StartPageNumber { get; set; }
```

Visual Basic

```
Public StartPageNumber As long
```

Description

Initial page number Attention: Initial page number is not stored in a template. Setting a value of the initial page number means the same as setting a value of the variable named PAGE_NUMBER by means of the IRobotVariableMngr (see page 1487) interface.

Version

Available since version 5.5.

II.5.2.16 TemplateCount

C++

```
HRESULT get_TemplateCount(long*);
```

C#

```
public long TemplateCount { get; }
```

Visual Basic

```
Public ReadOnly TemplateCount As long
```

Description

Number of the remembered templates with page settings .

Version

Available since version 5.5.

(c) 2014 Autodesk Inc. All rights reserved

II.5.2.17 TemplateName

C++

```
HRESULT get_TemplateName(BSTR*);
```

C#

```
public String TemplateName { get; }
```

Visual Basic

```
Public ReadOnly TemplateName As String
```

Description

Template name.

Version

Available since version 5.5.

II.5.2.18 TextFrame

C++

```
HRESULT get_TextFrame(IRobotPageSetupFrameType*);
HRESULT put_TextFrame(IRobotPageSetupFrameType);
```

C#

```
public IRobotPageSetupFrameType TextFrame { get; set; }
```

Visual Basic

```
Public TextFrame As IRobotPageSetupFrameType
```

Description

Page contents frame.

Version

Available since version 5.5.

II.5.2.19 TitlePage

C++

```
HRESULT get_TitlePage(IRobotReportStdElementRtf**);
```

C#

```
public IRobotReportStdElementRtf TitlePage { get; }
```

Visual Basic

```
Public ReadOnly TitlePage As IRobotReportStdElementRtf
```

Description

Printout title page .

Version

Available since version 2.5.

II.5.2.20 Toc

C++

```
HRESULT get_Toc(IRobotPageSetupTableOfContents**);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRobotPageSetupTableOfContents Toc { get; }
```

Visual Basic

```
Public ReadOnly Toc As IRobotPageSetupTableOfContents
```

Description

Parameters of the table of contents.

Version

Available since version 5.5.

II.5.2.21 Variables

C++

```
HRESULT get_Variables(IRobotVariableMngr**);
```

C#

```
public IRobotVariableMngr Variables { get; }
```

Visual Basic

```
Public ReadOnly Variables As IRobotVariableMngr
```

Description

Manager of variables (parameters).

II.5.3 IRobotPageSetup Methods

The methods of the IRobotPageSetup class are listed here.

Public Methods

	Name	Description
	GetTemplateName (see page 1500)	Function returns the name of a template with the specified index. Templates are indexed with values from 1 to TemplateCount (see page 1498). .
	Load (see page 1501)	Function reads page settings from the specified template. .
	LoadCurrent (see page 1501)	Function reads current page settings. .
	Save (see page 1501)	Function saves current page settings. .
	SaveAs (see page 1502)	Function saves current page settings as a template with the specified name. .

II.5.3.1 GetTemplateName

C++

```
HRESULT GetTemplateName(long _idx, BSTR* ret);
```

C#

```
public String GetTemplateName(long _idx);
```

Visual Basic

```
Public Function GetTemplateName(_idx As long) As String
```

Description

Function returns the name of a template with the specified index. Templates are indexed with values from 1 to TemplateCount (see page 1498). .

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 5.5.

II.5.3.2 Load

C++

```
HRESULT Load(BSTR _template_name, VARIANT_BOOL* ret);
```

C#

```
public bool Load(String _template_name);
```

Visual Basic

```
Public Function Load(_template_name As String) As Boolean
```

Description

Function reads page settings from the specified template. .

Version

Available since version 5.5.

II.5.3.3 LoadCurrent

C++

```
HRESULT LoadCurrent(VARIANT_BOOL* ret);
```

C#

```
public bool LoadCurrent();
```

Visual Basic

```
Public Function LoadCurrent() As Boolean
```

Description

Function reads current page settings. .

Version

Available since version 5.5.

II.5.3.4 Save

C++

```
HRESULT Save(VARIANT_BOOL* ret);
```

C#

```
public bool Save();
```

Visual Basic

```
Public Function Save() As Boolean
```

Description

Function saves current page settings. .

Version

Available since version 5.5.

(c) 2014 Autodesk Inc. All rights reserved

II.5.3.5 SaveAs

C++

```
HRESULT SaveAs(BSTR _template_name, VARIANT_BOOL* ret);
```

C#

```
public bool SaveAs(String _template_name);
```

Visual Basic

```
Public Function SaveAs(_template_name As String) As Boolean
```

Description

Function saves current page settings as a template with the specified name. .

Version

Available since version 5.5.

II.6 IRobotOutputFileFormat

C++

```
enum IRobotOutputFileFormat;
```

C#

```
public enum IRobotOutputFileFormat;
```

Visual Basic

```
Public Enum IRobotOutputFileFormat
```

Members

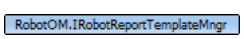
Members	Description
I_OFF_TEXT = 1	Available since version 1.7.
I_OFF_RTF = 2	Available since version 1.7.
I_OFF_RTF_JPEG = 3	RTF format file with the inserted JPEG format graphics. Available since version 5.5.
I_OFF_HTML = 4	HTML format file. Available since version 7.5.
I_OFF_PNG = 5	PNG format file. Available since version 12.

Description

Available file formats to which a printout may be directed. .

II.7 IRobotReportTemplateMngr

Class Hierarchy



C++

```
interface IRobotReportTemplateMngr : IDispatch;
```

C#

```
public interface IRobotReportTemplateMngr;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Interface IRobotReportTemplateMngr
```

Description

There is a possibility to save a previously composed printout as a template, which may be used many times afterwards in the same or different project. Printout templates are recognized by names. All templates available in the current project are managed by the template manager. .

II.7.1 IRobotReportTemplateMngr Members

The following tables list the members exposed by IRobotReportTemplateMngr.

Public Fields

	Name	Description
	Count (see page 1503)	Number of available printout templates Available since version 1.7.

Public Methods

	Name	Description
	Find (see page 1504)	Function finds a printout template with the specified name and returns its index. If the printout template is not found, then zero value is returned. Available since version 1.7.
	Get (see page 1504)	Function returns name of a template with a specified index. Printout templates are indexed from 1 to Count (see page 1503). Available since version 1.7.
	Remove (see page 1504)	Function deletes a printout template with the specified name. Available since version 1.7.

II.7.2 IRobotReportTemplateMngr Fields

The fields of the IRobotReportTemplateMngr class are listed here.

Public Fields

	Name	Description
	Count (see page 1503)	Number of available printout templates Available since version 1.7.

II.7.2.1 Count

C++

```
HRESULT get_Count(long*);
```

C#

```
public long Count { get; }
```

Visual Basic

```
Public ReadOnly Count As long
```

Description

Number of available printout templates Available since version 1.7.

II.7.3 IRobotReportTemplateMngr Methods

The methods of the IRobotReportTemplateMngr class are listed here.

(c) 2014 Autodesk Inc. All rights reserved

Public Methods

	Name	Description
	Find (see page 1504)	Function finds a printout template with the specified name and returns its index. If the printout template is not found, then zero value is returned. Available since version 1.7.
	Get (see page 1504)	Function returns name of a template with a specified index. Printout templates are indexed from 1 to Count (see page 1503). Available since version 1.7.
	Remove (see page 1504)	Function deletes a printout template with the specified name. Available since version 1.7.

II.7.3.1 Find

C++

```
HRESULT Find(BSTR _tmpl_name, long* ret);
```

C#

```
public long Find(String _tmpl_name);
```

Visual Basic

```
Public Function Find(_tmpl_name As String) As long
```

Description

Function finds a printout template with the specified name and returns its index. If the printout template is not found, then zero value is returned. Available since version 1.7.

II.7.3.2 Get

C++

```
HRESULT Get(long _idx, BSTR* ret);
```

C#

```
public String Get(long _idx);
```

Visual Basic

```
Public Function Get(_idx As long) As String
```

Description

Function returns name of a template with a specified index. Printout templates are indexed from 1 to Count (see page 1503). Available since version 1.7.

II.7.3.3 Remove

C++

```
HRESULT Remove(BSTR _tmpl_name);
```

C#

```
public void Remove(String _tmpl_name);
```

Visual Basic

```
Public Sub Remove(_tmpl_name As String)
```

Description

Function deletes a printout template with the specified name. Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

II.8 IRobotReportStdElementRtf

Class Hierarchy

RobotOM.IRobotReportStdElementRtf

C++

```
interface IRobotReportStdElementRtf : IDispatch;
```

C#

```
public interface IRobotReportStdElementRtf;
```

Visual Basic

```
Public Interface IRobotReportStdElementRtf
```

Description

Standard (predefined) printout element based on an RTF format template file.

Version

Available since version 2.5.

II.8.1 IRobotReportStdElementRtf Members

The following tables list the members exposed by IRobotReportStdElementRtf.

Public Fields

	Name	Description
	Active (see page 1505)	Flag indicating if the element is active (i.e. it will be added to the printout).
	Frame (see page 1506)	Frame type.

Public Methods

	Name	Description
	LoadFromFile (see page 1506)	Function reads in the printout element contents from the indicated file. .
	RestoreDefaults (see page 1507)	Function restores the standard contents of a printout element. .
	SaveToFile (see page 1507)	Function saves an element to a file of selected format.

II.8.2 IRobotReportStdElementRtf Fields

The fields of the IRobotReportStdElementRtf class are listed here.

Public Fields

	Name	Description
	Active (see page 1505)	Flag indicating if the element is active (i.e. it will be added to the printout).
	Frame (see page 1506)	Frame type.

II.8.2.1 Active

C++

```
HRESULT get_Active(VARIANT_BOOL*);
HRESULT put_Active(VARIANT_BOOL);
```

C#

```
public bool Active { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Active As Boolean
```

Description

Flag indicating if the element is active (i.e. it will be added to the printout).

Version

Available since version 2.5.

II.8.2.2 Frame

C++

```
HRESULT get_Frame(IRobotPageSetupFrameType*);
HRESULT put_Frame(IRobotPageSetupFrameType);
```

C#

```
public IRobotPageSetupFrameType Frame { get; set; }
```

Visual Basic

```
Public Frame As IRobotPageSetupFrameType
```

Description

Frame type.

Version

Available since version 5.5.

II.8.3 IRobotReportStdElementRtf Methods

The methods of the IRobotReportStdElementRtf class are listed here.

Public Methods

	Name	Description
	LoadFromFile (see page 1506)	Function reads in the printout element contents from the indicated file. .
	RestoreDefaults (see page 1507)	Function restores the standard contents of a printout element. .
	SaveToFile (see page 1507)	Function saves an element to a file of selected format.

II.8.3.1 LoadFromFile

C++

```
HRESULT LoadFromFile(BSTR _file_path, VARIANT_BOOL* ret);
```

C#

```
public bool LoadFromFile(String _file_path);
```

Visual Basic

```
Public Function LoadFromFile(_file_path As String) As Boolean
```

Description

Function reads in the printout element contents from the indicated file. .

Version

Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

II.8.3.2 RestoreDefaults

C++

```
HRESULT RestoreDefaults();
```

C#

```
public void RestoreDefaults();
```

Visual Basic

```
Public Sub RestoreDefaults()
```

Description

Function restores the standard contents of a printout element. .

Version

Available since version 2.5.

II.8.3.3 SaveToFile

C++

```
HRESULT SaveToFile(BSTR _fpath, IRobotOutputFileFormat _format, VARIANT_BOOL* ret);
```

C#

```
public bool SaveToFile(String _fpath, IRobotOutputFileFormat _format);
```

Visual Basic

```
Public Function SaveToFile(_fpath As String, _format As IRobotOutputFileFormat) As Boolean
```

Description

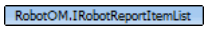
Function saves an element to a file of selected format.

Version

Available since version 5.5.

II.9 IRobotReportItemList

Class Hierarchy



C++

```
interface IRobotReportItemList : IDispatch;
```

C#

```
public interface IRobotReportItemList;
```

Visual Basic

```
Public Interface IRobotReportItemList
```

Description

List of printout components.

Version

Available since version 5.5.

(c) 2014 Autodesk Inc. All rights reserved

II.9.1 IRobotReportItemList Members

The following tables list the members exposed by IRobotReportItemList.

Public Fields

	Name	Description
	Count (see page 1508)	Number of elements on the list.

Public Methods

	Name	Description
	Get (see page 1508)	Function returns a printout component with the specified index. Printout components are indexed from 1 to Count (see page 1508). .

II.9.2 IRobotReportItemList Fields

The fields of the IRobotReportItemList class are listed here.

Public Fields

	Name	Description
	Count (see page 1508)	Number of elements on the list.

II.9.2.1 Count

C++

```
HRESULT get_Count(long*);
```

C#

```
public long Count { get; }
```

Visual Basic

```
Public ReadOnly Count As long
```

Description

Number of elements on the list.

Version

Available since version 5.5.

II.9.3 IRobotReportItemList Methods

The methods of the IRobotReportItemList class are listed here.

Public Methods

	Name	Description
	Get (see page 1508)	Function returns a printout component with the specified index. Printout components are indexed from 1 to Count (see page 1508). .

II.9.3.1 Get

C++

```
HRESULT Get(long _idx, IRobotReportItem** ret);
```

C#

```
public IRobotReportItem Get(long _idx);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Function Get(_idx As long) As IRobotReportItem
```

Description

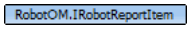
Function returns a printout component with the specified index. Printout components are indexed from 1 to Count (see page 1508). .

Version

Available since version 5.5.

II.10 IRobotReportItem

Class Hierarchy



C++

```
interface IRobotReportItem : IDispatch;
```

C#

```
public interface IRobotReportItem;
```

Visual Basic

```
Public Interface IRobotReportItem
```

Description

Interface that describes a printout element.

Version

Available since version 5.5.

II.10.1 IRobotReportItem Members

The following tables list the members exposed by IRobotReportItem.

Public Fields

	Name	Description
	ExcludeFromReport (see page 1510)	
	HasNoteAfter (see page 1510)	Flag indicating that at a note has been inserted after the printout component .
	HasNoteBefore (see page 1511)	Flag indicating that a note has been inserted before the printout component .
	IncludeDateTime (see page 1511)	
	NoteAfter (see page 1511)	Note inserted after the printout component .
	NoteBefore (see page 1512)	Note inserted before the printout component.
	StartFromNewPage (see page 1512)	Flag that indicates if an element should be printed on a new page.
	Title (see page 1512)	Element title in RTF format.
	TitleText (see page 1513)	Element title.
	Type (see page 1513)	Type of printout element.

(c) 2014 Autodesk Inc. All rights reserved

Public Methods

	Name	Description
	CreateView (see page 1513)	Fuction creates and returns a view corresponding to the printout component. A type of the created view depends on the contents of the printout component. If access to any of the defined interfaces for views is impossible, then the IRobotPrintable (see page 1477) interface is returned. In case an error occurs, the empty reference is returned (zero for the C language, Nothing for Visual Basic). Function makes the interface accessible for the object reperesented by the printout component of I_RIT_VIEW type. .
	GetPageTemplate (see page 1514)	Function returns the name of a page template for the printout component of I_RIT_PAGE_TEMPLATE type. .

II.10.2 IRobotReportItem Fields

The fields of the IRobotReportItem class are listed here.

Public Fields

	Name	Description
	ExcludeFromReport (see page 1510)	
	HasNoteAfter (see page 1510)	Flag indicating that at a note has been inserted after the printout component .
	HasNoteBefore (see page 1511)	Flag indicating that a note has been inserted before the printout component .
	IncludeDateTime (see page 1511)	
	NoteAfter (see page 1511)	Note inserted after the printout component .
	NoteBefore (see page 1512)	Note inserted before the printout component.
	StartFromNewPage (see page 1512)	Flag that indicates if an element should be printed on a new page.
	Title (see page 1512)	Element title in RTF format.
	TitleText (see page 1513)	Element title.
	Type (see page 1513)	Type of printout element.

II.10.2.1 ExcludeFromReport

C++

```
HRESULT get_ExcludeFromReport(VARIANT_BOOL*);
HRESULT put_ExcludeFromReport(VARIANT_BOOL);
```

C#

```
public bool ExcludeFromReport { get; set; }
```

Visual Basic

```
Public ExcludeFromReport As Boolean
```

Version

Available since version 14.5.

II.10.2.2 HasNoteAfter

C++

```
HRESULT get_HasNoteAfter(VARIANT_BOOL*);
```

C#

```
public bool HasNoteAfter { get; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public ReadOnly HasNoteAfter As Boolean
```

Description

Flag indicating that at a note has been inserted after the printout component .

Version

Available since version 5.5.

II.10.2.3 HasNoteBefore

C++

```
HRESULT get_HasNoteBefore(VARIANT_BOOL*);
```

C#

```
public bool HasNoteBefore { get; }
```

Visual Basic

```
Public ReadOnly HasNoteBefore As Boolean
```

Description

Flag indicating that a note has been inserted before the printout component .

Version

Available since version 5.5.

II.10.2.4 IncludeDateTime

C++

```
HRESULT get_IncludeDateTime(VARIANT_BOOL*);
```

C#

```
public bool IncludeDateTime { get; }
```

Visual Basic

```
Public ReadOnly IncludeDateTime As Boolean
```

Version

Available since version 12.

II.10.2.5 NoteAfter

C++

```
HRESULT get_NoteAfter(IRobotRtfView**);
```

C#

```
public IRobotRtfView NoteAfter { get; }
```

Visual Basic

```
Public ReadOnly NoteAfter As IRobotRtfView
```

Description

Note inserted after the printout component .

Version

Available since version 5.5.

(c) 2014 Autodesk Inc. All rights reserved

II.10.2.6 NoteBefore

C++

```
HRESULT get_NoteBefore(IRobotRtfView**);
```

C#

```
public IRobotRtfView NoteBefore { get; }
```

Visual Basic

```
Public ReadOnly NoteBefore As IRobotRtfView
```

Description

Note inserted before the printout component.

Version

Available since version 5.5.

II.10.2.7 StartFromNewPage

C++

```
HRESULT get_StartFromNewPage(VARIANT_BOOL*);
HRESULT put_StartFromNewPage(VARIANT_BOOL);
```

C#

```
public bool StartFromNewPage { get; set; }
```

Visual Basic

```
Public StartFromNewPage As Boolean
```

Description

Flag that indicates if an element should be printed on a new page.

Version

Available since version 5.5.

II.10.2.8 Title

C++

```
HRESULT get_Title(IRobotRtfView**);
```

C#

```
public IRobotRtfView Title { get; }
```

Visual Basic

```
Public ReadOnly Title As IRobotRtfView
```

Description

Element title in RTF format.

Version

Available since version 5.5.

II.10.2.9 TitleText

C++

```
HRESULT get_TitleText(BSTR*);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public String TitleText { get; }
```

Visual Basic

```
Public ReadOnly TitleText As String
```

Description

Element title.

Version

Available since version 5.5.

II.10.2.10 Type

C++

```
HRESULT get_Type(IRobotReportItemType*);
```

C#

```
public IRobotReportItemType Type { get; }
```

Visual Basic

```
Public ReadOnly Type As IRobotReportItemType
```

Description

Type of printout element.

Version

Available since version 5.5.

II.10.3 IRobotReportItem Methods

The methods of the IRobotReportItem class are listed here.

Public Methods

	Name	Description
	CreateView (see page 1513)	Fuction creates and returns a view corresponding to the printout component. A type of the created view depends on the contents of the printout component. If access to any of the defined interfaces for views is impossible, then the IRobotPrintable (see page 1477) interface is returned. In case an error occurs, the empty reference is returned (zero for the C language, Nothing for Visual Basic). Function makes the interface accessible for the object reperesented by the printout component of I_RIT_VIEW type. .
	GetPageTemplate (see page 1514)	Function returns the name of a page template for the printout component of I_RIT_PAGE_TEMPLATE type. .

II.10.3.1 CreateView

C++

```
HRESULT CreateView(IDispatch** ret);
```

C#

```
public IDispatch* CreateView();
```

Visual Basic

```
Public Function CreateView() As IDispatch*
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Fuction creates and returns a view corresponding to the printout component. A type of the created view depends on the contents of the printout component. If access to any of the defined interfaces for views is impossible, then the IRobotPrintable (see page 1477) interface is returned. In case an error occurs, the empty reference is returned (zero for the C language, Nothing for Visual Basic). Function makes the interface accessible for the object reperesented by the printout component of I_RIT_VIEW type. .

Version

Available since version 5.5.

II.10.3.2 GetPageTemplate

C++

```
HRESULT GetPageTemplate(BSTR* ret);
```

C#

```
public String GetPageTemplate();
```

Visual Basic

```
Public Function GetPageTemplate() As String
```

Description

Function returns the name of a page template for the printout component of I_RIT_PAGE_TEMPLATE type. .

Version

Available since version 5.5.

II.11 IRobotPageSetupOrientation

C++

```
enum IRobotPageSetupOrientation;
```

C#

```
public enum IRobotPageSetupOrientation;
```

Visual Basic

```
Public Enum IRobotPageSetupOrientation
```

Members

Members	Description
I_PSO_PORTRAIT = 1	Available since version 5.5.
I_PSO_LANDSCAPE = 2	Available since version 5.5.

Description

Page setup.

Version

Available since version 5.5.

II.12 IRobotPageSetupFrameType

C++

```
enum IRobotPageSetupFrameType;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public enum IRobotPageSetupFrameType;
```

Visual Basic

```
Public Enum IRobotPageSetupFrameType
```

Members

Members	Description
I_PSFT_NONE = 0	No frame. Available since version 5.5.
I_PSFT_FRAME = 1	Frame. Available since version 5.5.
I_PSFT_SEPARATION_LINE = 2	Separation line. Available since version 5.5.

Description

Types of frames used for decoration of individual page elements. .

Version

Available since version 5.5.

II.13 IRobotPageSetupTableOfContents

Class Hierarchy

RobotOM.IRobotPageSetupTableOfContents

C++

```
interface IRobotPageSetupTableOfContents : IDispatch;
```

C#

```
public interface IRobotPageSetupTableOfContents;
```

Visual Basic

```
Public Interface IRobotPageSetupTableOfContents
```

Description

Interface that describes the table of contents.

Version

Available since version 5.5.

II.13.1 IRobotPageSetupTableOfContents Members

The following tables list the members exposed by IRobotPageSetupTableOfContents.

Public Fields

	Name	Description
	Active (see page 1516)	Flag that switches on/off the table of contents.
	IncludeTitle (see page 1516)	
	Location (see page 1516)	

(c) 2014 Autodesk Inc. All rights reserved

II.13.2 IRobotPageSetupTableOfContents Fields

The fields of the IRobotPageSetupTableOfContents class are listed here.

Public Fields

	Name	Description
	Active (see page 1516)	Flag that switches on/off the table of contents.
	IncludeTitle (see page 1516)	
	Location (see page 1516)	

II.13.2.1 Active

C++

```
HRESULT get_Active(VARIANT_BOOL*);
HRESULT put_Active(VARIANT_BOOL);
```

C#

```
public bool Active { get; set; }
```

Visual Basic

```
Public Active As Boolean
```

Description

Flag that switches on/off the table of contents.

Version

Available since version 5.5.

II.13.2.2 IncludeTitle

C++

```
HRESULT get_IncludeTitle(VARIANT_BOOL*);
HRESULT put_IncludeTitle(VARIANT_BOOL);
```

C#

```
public bool IncludeTitle { get; set; }
```

Visual Basic

```
Public IncludeTitle As Boolean
```

Version

Available since version 5.5.

II.13.2.3 Location

C++

```
HRESULT get_Location(IRobotPageSetupTocLocation*);
HRESULT put_Location(IRobotPageSetupTocLocation);
```

C#

```
public IRobotPageSetupTocLocation Location { get; set; }
```

Visual Basic

```
Public Location As IRobotPageSetupTocLocation
```

Version

Available since version 5.5.

(c) 2014 Autodesk Inc. All rights reserved

II.14 IRobotPageSetupTocLocation

C++

```
enum IRobotPageSetupTocLocation;
```

C#

```
public enum IRobotPageSetupTocLocation;
```

Visual Basic

```
Public Enum IRobotPageSetupTocLocation
```

Members

Members	Description
I_PSTL_BEGINNING = 1	At the beginning. Available since version 5.5.
I_PSTL_END = 2	At the end. Available since version 5.5.

Version

Available since version 5.5.

II.15 IRobotReportItemType

C++

```
enum IRobotReportItemType;
```

C#

```
public enum IRobotReportItemType;
```

Visual Basic

```
Public Enum IRobotReportItemType
```

Members

Members	Description
I_RIT_BLANK_PAGE = 1	Printout component that represents a blank page. Available since version 5.5.
I_RIT_PAGE_TEMPLATE = 2	Printout component that enforces change of the page template . Available since version 5.5.
I_RIT_VIEW = 3	Printout component representing a view or other object that may be printed. Available since version 5.5.

Description

Available types of printout components.

Version

Available since version 5.5.

(c) 2014 Autodesk Inc. All rights reserved

II.16 IRobotExternalPreviewFormat

C++

```
enum IRobotExternalPreviewFormat;
```

C#

```
public enum IRobotExternalPreviewFormat;
```

Visual Basic

```
Public Enum IRobotExternalPreviewFormat
```

Members

Members	Description
EPF_HTML = 1	Print preview in the html file browser. Available since version 6.4.
EPF_MS_OFFICE = 2	Presentation of a printout in the Word program of the Microsoft Office package; it requires installing the Microsoft Office package. Available since version 6.4.
EPF_OPEN_OFFICE = 3	Presentation of a printout in the text editor of the Open Office package; it requires installing the Open Office package. Available since version 6.4.

Description

External format of a print preview.

Version

Available since version 6.4.

II.17 IRobotVariableMngrExtension

Class Hierarchy

RobotOM.IRobotVariableMngrExtension

C++

```
interface IRobotVariableMngrExtension : IDispatch;
```

C#

```
public interface IRobotVariableMngrExtension;
```

Visual Basic

```
Public Interface IRobotVariableMngrExtension
```

Description

Interface of external extension for the manager of variables (parameters) used in printouts.

II.17.1 IRobotVariableMngrExtension Members

The following tables list the members exposed by IRobotVariableMngrExtension.

(c) 2014 Autodesk Inc. All rights reserved

Public Methods

	Name	Description
	GetIndexedValue (see page 1519)	Function returns a value of the variable indexed for a given value of an index. If the object does not support the variable of a given name, then function returns FALSE value (zero).
	GetValue (see page 1519)	Function fills out a value of the variable of a given name. If the object does not support this variable, then function returns FALSE value (zero).

II.17.2 IRobotVariableMngrExtension Methods

The methods of the IRobotVariableMngrExtension class are listed here.

Public Methods

	Name	Description
	GetIndexedValue (see page 1519)	Function returns a value of the variable indexed for a given value of an index. If the object does not support the variable of a given name, then function returns FALSE value (zero).
	GetValue (see page 1519)	Function fills out a value of the variable of a given name. If the object does not support this variable, then function returns FALSE value (zero).

II.17.2.1 GetIndexedValue

C++

```
HRESULT GetIndexedValue(BSTR _var_name, long _cur_index, long _max_index, BSTR _var_value,
VARIANT_BOOL* ret);
```

C#

```
public bool GetIndexedValue(String _var_name, long _cur_index, long _max_index, String
_var_value);
```

Visual Basic

```
Public Function GetIndexedValue(_var_name As String, _cur_index As long, _max_index As
long, ByRef _var_value As String) As Boolean
```

Description

Function returns a value of the variable indexed for a given value of an index. If the object does not support the variable of a given name, then function returns FALSE value (zero).

II.17.2.2 GetValue

C++

```
HRESULT GetValue(BSTR _var_name, BSTR _var_value, VARIANT_BOOL* ret);
```

C#

```
public bool GetValue(String _var_name, String _var_value);
```

Visual Basic

```
Public Function GetValue(_var_name As String, ByRef _var_value As String) As Boolean
```

Description

Function fills out a value of the variable of a given name. If the object does not support this variable, then function returns FALSE value (zero).

(c) 2014 Autodesk Inc. All rights reserved

III Structural axes

Enumerations

	Name	Description
	IRobotStructuralAxisLabelType (see page 1520)	Available methods of structure axis description.
	IRobotStructuralAxisGridType (see page 1525)	Available types of structure axis grids.

Interfaces

	Name	Description
	IRobotStructuralAxisSequenceList (see page 1520)	List of structure axis sequences.
	IRobotStructuralAxisGrid (see page 1526)	Structure axis grid.
	IRobotStructuralAxisGridCartesian (see page 1527)	Cartesian grid of structure axes.
	IRobotStructuralAxisGridMngr (see page 1531)	Manager of structure axis grids.

III.1 IRobotStructuralAxisLabelType

C++

```
enum IRobotStructuralAxisLabelType;
```

C#

```
public enum IRobotStructuralAxisLabelType;
```

Visual Basic

```
Public Enum IRobotStructuralAxisLabelType
```

Members

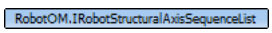
Members	Description
I_SALT_123 = 1	Description with successive numbers.
I_SALT_ABC = 2	Description with successive alphabet letters.
I_SALT_VALUE = 3	Description with values.
I_SALT_DEFINE = 4	Description with a defined character string.
I_SALT_VARIOUS = 5	Application of different methods of axis description.

Description

Available methods of structure axis description.

III.2 IRobotStructuralAxisSequenceList

Class Hierarchy



C++

```
interface IRobotStructuralAxisSequenceList : IDispatch;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public interface IRobotStructuralAxisSequenceList;
```

Visual Basic

```
Public Interface IRobotStructuralAxisSequenceList
```

Description

List of structure axis sequences.

III.2.1 IRobotStructuralAxisSequenceList Members

The following tables list the members exposed by IRobotStructuralAxisSequenceList.

Public Fields

	Name	Description
	AxisCount (see page 1522)	Number of defined structure axes.
	SequenceCount (see page 1522)	Number of defined structure axis sequences.
	StartPosition (see page 1522)	Beginning point of the list.

Public Methods

	Name	Description
	AddSequence (see page 1523)	Function inserts a new structure axis sequence. While creating the first sequence, the start axis will also be created at the position StartPosition (see page 1522).
	Clear (see page 1523)	Function deletes all structure axes.
	DeleteSequence (see page 1523)	Function deletes the structure axis sequence of a given index.
	FindAxisByPos (see page 1524)	Function returns an index of the axis of a given position. If no structure axis has been defined at the given position, then function returns zero value.
	GetAxis (see page 1524)	Function returns data for the structure axis of a given index.
	GetSequence (see page 1524)	Function returns data for the sequence of a given index.
	SetAxisLabel (see page 1524)	Function changes a description for the structure axis of a given index.
	SetLabelFormat (see page 1525)	Function sets the current method of structure axis description.
	SingleOutAxis (see page 1525)	Function steers with singling out of a structure axis in graphical views.

III.2.2 IRobotStructuralAxisSequenceList Fields

The fields of the IRobotStructuralAxisSequenceList class are listed here.

Public Fields

	Name	Description
	AxisCount (see page 1522)	Number of defined structure axes.
	SequenceCount (see page 1522)	Number of defined structure axis sequences.
	StartPosition (see page 1522)	Beginning point of the list.

III.2.2.1 AxisCount

C++

```
HRESULT get_AxisCount(long*);
```

C#

```
public long AxisCount { get; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public ReadOnly AxisCount As long
```

Description

Number of defined structure axes.

III.2.2.2 SequenceCount

C++

```
HRESULT get_SequenceCount(long*);
```

C#

```
public long SequenceCount { get; }
```

Visual Basic

```
Public ReadOnly SequenceCount As long
```

Description

Number of defined structure axis sequences.

III.2.2.3 StartPosition

C++

```
HRESULT get_StartPosition(double*);
HRESULT put_StartPosition(double);
```

C#

```
public double StartPosition { get; set; }
```

Visual Basic

```
Public StartPosition As double
```

Description

Beginning point of the list.

III.2.3 IRobotStructuralAxisSequenceList Methods

The methods of the IRobotStructuralAxisSequenceList class are listed here.

Public Methods

	Name	Description
	AddSequence (see page 1523)	Function inserts a new structure axis sequence. While creating the first sequence, the start axis will also be created at the position StartPosition (see page 1522).
	Clear (see page 1523)	Function deletes all structure axes.
	DeleteSequence (see page 1523)	Function deletes the structure axis sequence of a given index.
	FindAxisByPos (see page 1524)	Function returns an index of the axis of a given position. If no structure axis has been defined at the given position, then function returns zero value.
	GetAxis (see page 1524)	Function returns data for the structure axis of a given index.
	GetSequence (see page 1524)	Function returns data for the sequence of a given index.
	SetAxisLabel (see page 1524)	Function changes a description for the structure axis of a given index.
	SetLabelFormat (see page 1525)	Function sets the current method of structure axis description.
	SingleOutAxis (see page 1525)	Function steers with singling out of a structure axis in graphical views.

(c) 2014 Autodesk Inc. All rights reserved

III.2.3.1 AddSequence

C++

```
HRESULT AddSequence(double _distance, long _repeat_number);
```

C#

```
public void AddSequence(double _distance, long _repeat_number);
```

Visual Basic

```
Public Sub AddSequence(_distance As double, _repeat_number As long)
```

Description

Function inserts a new structure axis sequence. While creating the first sequence, the start axis will also be created at the position StartPosition (see page 1522).

III.2.3.2 Clear

C++

```
HRESULT Clear();
```

C#

```
public void Clear();
```

Visual Basic

```
Public Sub Clear()
```

Description

Function deletes all structure axes.

III.2.3.3 DeleteSequence

C++

```
HRESULT DeleteSequence(long _idx);
```

C#

```
public void DeleteSequence(long _idx);
```

Visual Basic

```
Public Sub DeleteSequence(_idx As long)
```

Description

Function deletes the structure axis sequence of a given index.

III.2.3.4 FindAxisByPos

C++

```
HRESULT FindAxisByPos(double _pos, long* ret);
```

C#

```
public long FindAxisByPos(double _pos);
```

Visual Basic

```
Public Function FindAxisByPos(_pos As double) As long
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Function returns an index of the axis of a given position. If no structure axis has been defined at the given position, then function returns zero value.

III.2.3.5 GetAxis

C++

```
HRESULT GetAxis(long _idx, BSTR _label, double* _position, bool* _singleout, VARIANT_BOOL*
ret);
```

C#

```
public bool GetAxis(long _idx, String _label, double* _position, bool* _singleout);
```

Visual Basic

```
Public Function GetAxis(_idx As long, ByRef _label As String, ByRef _position As double*,
ByRef _singleout As bool*) As Boolean
```

Description

Function returns data for the structure axis of a given index.

III.2.3.6 GetSequence

C++

```
HRESULT GetSequence(long _idx, double* _distance, long* _repeat_number);
```

C#

```
public void GetSequence(long _idx, double* _distance, long* _repeat_number);
```

Visual Basic

```
Public Sub GetSequence(_idx As long, ByRef _distance As double*, ByRef _repeat_number As
long*)
```

Description

Function returns data for the sequence of a given index.

III.2.3.7 SetAxisLabel

C++

```
HRESULT SetAxisLabel(long _idx, BSTR _label);
```

C#

```
public void SetAxisLabel(long _idx, String _label);
```

Visual Basic

```
Public Sub SetAxisLabel(_idx As long, _label As String)
```

Description

Function changes a description for the structure axis of a given index.

III.2.3.8 SetLabelFormat

C++

```
HRESULT SetLabelFormat(IRobotStructuralAxisLabelType _label_type, BSTR _label_def = "",
VARIANT_BOOL _reversed = false);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public void SetLabelFormat(IRobotStructuralAxisLabelType _label_type, String _label_def =
"", bool _reversed = false);
```

Visual Basic

```
Public Sub SetLabelFormat(_label_type As IRobotStructuralAxisLabelType, Optional _label_def
As String = "", Optional _reversed As Boolean = false)
```

Description

Function sets the current method of structure axis description.

III.2.3.9 SingleOutAxis

C++

```
HRESULT SingleOutAxis(long _idx, VARIANT_BOOL _singleout = true);
```

C#

```
public void SingleOutAxis(long _idx, bool _singleout = true);
```

Visual Basic

```
Public Sub SingleOutAxis(_idx As long, Optional _singleout As Boolean = true)
```

Description

Function steers with singling out of a structure axis in graphical views.

III.3 IRobotStructuralAxisGridType

C++

```
enum IRobotStructuralAxisGridType;
```

C#

```
public enum IRobotStructuralAxisGridType;
```

Visual Basic

```
Public Enum IRobotStructuralAxisGridType
```

Members

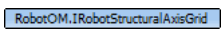
Members	Description
I_SAGT_CARTESIAN = 1	Cartesian grid.
I_SAGT_CYLINDRICAL = 2	Cylindrical grid.

Description

Available types of structure axis grids.

III.4 IRobotStructuralAxisGrid

Class Hierarchy



C++

```
interface IRobotStructuralAxisGrid : IDispatch;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public interface IRobotStructuralAxisGrid;
```

Visual Basic

```
Public Interface IRobotStructuralAxisGrid
```

Description

Structure axis grid.

III.4.1 IRobotStructuralAxisGrid Members

The following tables list the members exposed by IRobotStructuralAxisGrid.

Public Fields

	Name	Description
	Name (see page 1526)	Grid name.
	Type (see page 1527)	Grid type.

Public Methods

	Name	Description
	Save (see page 1527)	Function saves the grid definition.

III.4.2 IRobotStructuralAxisGrid Fields

The fields of the IRobotStructuralAxisGrid class are listed here.

Public Fields

	Name	Description
	Name (see page 1526)	Grid name.
	Type (see page 1527)	Grid type.

III.4.2.1 Name

C++

```
HRESULT get_Name(BSTR*);
HRESULT put_Name(BSTR);
```

C#

```
public String Name { get; set; }
```

Visual Basic

```
Public Name As String
```

Description

Grid name.

III.4.2.2 Type

C++

```
HRESULT get_Type(IRobotStructuralAxisGridType*);
```

C#

```
public IRobotStructuralAxisGridType Type { get; }
```

Visual Basic

```
Public ReadOnly Type As IRobotStructuralAxisGridType
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Grid type.

III.4.3 IRobotStructuralAxisGrid Methods

The methods of the IRobotStructuralAxisGrid class are listed here.

Public Methods

	Name	Description
	Save (see page 1527)	Function saves the grid definition.

III.4.3.1 Save

C++

```
HRESULT Save();
```

C#

```
public void Save();
```

Visual Basic

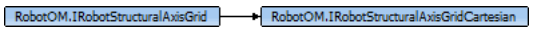
```
Public Sub Save()
```

Description

Function saves the grid definition.

III.5 IRobotStructuralAxisGridCartesian

Class Hierarchy



C++

```
interface IRobotStructuralAxisGridCartesian : IRobotStructuralAxisGrid;
```

C#

```
public interface IRobotStructuralAxisGridCartesian : IRobotStructuralAxisGrid;
```

Visual Basic

```
Public Interface IRobotStructuralAxisGridCartesian
```

Description

Cartesian grid of structure axes.

III.5.1 IRobotStructuralAxisGridCartesian Members

The following tables list the members exposed by IRobotStructuralAxisGridCartesian.

Public Fields

	Name	Description
	IncludeStoreysInZ (see page 1528)	Flag indicating that the sequence of structure axes defined on the OZ axis, should take stories into account automatically.
	Name (see page 1526)	Grid name.
	RotationAngle (see page 1529)	Angle of grid rotation.
	RotationAxis (see page 1529)	Axis of grid rotation.

(c) 2014 Autodesk Inc. All rights reserved

	StoreysInZ (see page 1529)	Selection of stories included on the OZ axis, if the IncludeStoreysInZ (see page 1528) flag has a True value.
	Type (see page 1527)	Grid type.
	X (see page 1530)	Sequence of structure axes defined on the OX axis.
	Y (see page 1530)	Sequence of structure axes defined on the OY axis.
	Z (see page 1530)	Sequence of structure axes defined on the OZ axis.

Public Methods

	Name	Description
	GetRelativeToPoint (see page 1531)	Function returns information about the point with respect to which the grid coordinates were defined. If (0,0,0) is the reference point, the function returns the False value.
	Save (see page 1527)	Function saves the grid definition.
	SetRelativeToPoint (see page 1531)	Function specifies coordinates of the point with respect to which the grid coordinates were defined.

III.5.2 IRobotStructuralAxisGridCartesian Fields

The fields of the IRobotStructuralAxisGridCartesian class are listed here.

Public Fields

	Name	Description
	IncludeStoreysInZ (see page 1528)	Flag indicating that the sequence of structure axes defined on the OZ axis, should take stories into account automatically.
	RotationAngle (see page 1529)	Angle of grid rotation.
	RotationAxis (see page 1529)	Axis of grid rotation.
	StoreysInZ (see page 1529)	Selection of stories included on the OZ axis, if the IncludeStoreysInZ (see page 1528) flag has a True value.
	X (see page 1530)	Sequence of structure axes defined on the OX axis.
	Y (see page 1530)	Sequence of structure axes defined on the OY axis.
	Z (see page 1530)	Sequence of structure axes defined on the OZ axis.

III.5.2.1 IncludeStoreysInZ

C++

```
HRESULT get_IncludeStoreysInZ(VARIANT_BOOL*);
HRESULT put_IncludeStoreysInZ(VARIANT_BOOL);
```

C#

```
public bool IncludeStoreysInZ { get; set; }
```

Visual Basic

```
Public IncludeStoreysInZ As Boolean
```

Description

Flag indicating that the sequence of structure axes defined on the OZ axis, should take stories into account automatically.

Version

Available since version 9.5.

III.5.2.2 RotationAngle

C++

```
HRESULT get_RotationAngle(double*);
HRESULT put_RotationAngle(double);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double RotationAngle { get; set; }
```

Visual Basic

```
Public RotationAngle As double
```

Description

Angle of grid rotation.

III.5.2.3 RotationAxis

C++

```
HRESULT get_RotationAxis(IRobotGeoCoordinateAxis*);
HRESULT put_RotationAxis(IRobotGeoCoordinateAxis);
```

C#

```
public IRobotGeoCoordinateAxis RotationAxis { get; set; }
```

Visual Basic

```
Public RotationAxis As IRobotGeoCoordinateAxis
```

Description

Axis of grid rotation.

III.5.2.4 StoreysInZ

C++

```
HRESULT get_StoreysInZ(IRobotStoreySelection**);
```

C#

```
public IRobotStoreySelection StoreysInZ { get; }
```

Visual Basic

```
Public ReadOnly StoreysInZ As IRobotStoreySelection
```

Description

Selection of stories included on the OZ axis, if the IncludeStoreysInZ (see page 1528) flag has a True value.

Version

Available since version 9.7.

III.5.2.5 X

C++

```
HRESULT get_X(IRobotStructuralAxisSequenceList**);
```

C#

```
public IRobotStructuralAxisSequenceList X { get; }
```

Visual Basic

```
Public ReadOnly X As IRobotStructuralAxisSequenceList
```

Description

Sequence of structure axes defined on the OX axis.

(c) 2014 Autodesk Inc. All rights reserved

III.5.2.6 Y

C++

```
HRESULT get_Y(IRobotStructuralAxisSequenceList**);
```

C#

```
public IRobotStructuralAxisSequenceList Y { get; }
```

Visual Basic

```
Public ReadOnly Y As IRobotStructuralAxisSequenceList
```

Description

Sequence of structure axes defined on the OY axis.

III.5.2.7 Z

C++

```
HRESULT get_Z(IRobotStructuralAxisSequenceList**);
```

C#

```
public IRobotStructuralAxisSequenceList Z { get; }
```

Visual Basic

```
Public ReadOnly Z As IRobotStructuralAxisSequenceList
```

Description

Sequence of structure axes defined on the OZ axis.

III.5.3 IRobotStructuralAxisGridCartesian Methods

The methods of the IRobotStructuralAxisGridCartesian class are listed here.

Public Methods

	Name	Description
	GetRelativeToPoint (see page 1531)	Function returns information about the point with respect to which the grid coordinates were defined. If (0,0,0) is the reference point, the function returns the False value.
	SetRelativeToPoint (see page 1531)	Function specifies coordinates of the point with respect to which the grid coordinates were defined.

III.5.3.1 GetRelativeToPoint

C++

```
HRESULT GetRelativeToPoint(double* _x, double* _y, double* _z, VARIANT_BOOL* ret);
```

C#

```
public bool GetRelativeToPoint(double* _x, double* _y, double* _z);
```

Visual Basic

```
Public Function GetRelativeToPoint(ByRef _x As double*, ByRef _y As double*, ByRef _z As
double*) As Boolean
```

Description

Function returns information about the point with respect to which the grid coordinates were defined. If (0,0,0) is the reference point, the function returns the False value.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 9.2.

III.5.3.2 SetRelativeToPoint

C++

```
HRESULT SetRelativeToPoint(VARIANT_BOOL _set, double _x = 0, double _y = 0, double _z = 0);
```

C#

```
public void SetRelativeToPoint(bool _set, double _x = 0, double _y = 0, double _z = 0);
```

Visual Basic

```
Public Sub SetRelativeToPoint(_set As Boolean, Optional _x As double = 0, Optional _y As
double = 0, Optional _z As double = 0)
```

Description

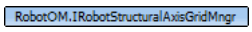
Function specifies coordinates of the point with respect to which the grid coordinates were defined.

Version

Available since version 9.2.

III.6 IRobotStructuralAxisGridMngr

Class Hierarchy



C++

```
interface IRobotStructuralAxisGridMngr : IDispatch;
```

C#

```
public interface IRobotStructuralAxisGridMngr;
```

Visual Basic

```
Public Interface IRobotStructuralAxisGridMngr
```

Description

Manager of structure axis grids.

III.6.1 IRobotStructuralAxisGridMngr Members

The following tables list the members exposed by IRobotStructuralAxisGridMngr.

Public Fields

	Name	Description
	Count (see page 1532)	Number of defined grids.

Public Methods

	Name	Description
	Activate (see page 1533)	Function activates/deactivates the grid of a given index.
	Clear (see page 1533)	Function deletes all grids.
	Create (see page 1533)	Function creates and returns a new blank grid of a defined type.
	Delete (see page 1533)	Function deletes the grid of a given index.

(c) 2014 Autodesk Inc. All rights reserved

	FindByName (see page 1534)	Function returns an index of the grid of a given name or a zero value if the grid is not found.
	Get (see page 1534)	Function makes available the grid of a given index.
	GetByName (see page 1534)	Function makes available the grid of a given name.
	IsActive (see page 1535)	Function returns activity state of the grid of a given index.

III.6.2 IRobotStructuralAxisGridMngr Fields

The fields of the IRobotStructuralAxisGridMngr class are listed here.

Public Fields

	Name	Description
	Count (see page 1532)	Number of defined grids.

III.6.2.1 Count

C++

```
HRESULT get_Count(long*);
```

C#

```
public long Count { get; }
```

Visual Basic

```
Public ReadOnly Count As long
```

Description

Number of defined grids.

III.6.3 IRobotStructuralAxisGridMngr Methods

The methods of the IRobotStructuralAxisGridMngr class are listed here.

Public Methods

	Name	Description
	Activate (see page 1533)	Function activates/deactivates the grid of a given index.
	Clear (see page 1533)	Function deletes all grids.
	Create (see page 1533)	Function creates and returns a new blank grid of a defined type.
	Delete (see page 1533)	Function deletes the grid of a given index.
	FindByName (see page 1534)	Function returns an index of the grid of a given name or a zero value if the grid is not found.
	Get (see page 1534)	Function makes available the grid of a given index.
	GetByName (see page 1534)	Function makes available the grid of a given name.
	IsActive (see page 1535)	Function returns activity state of the grid of a given index.

III.6.3.1 Activate

C++

```
HRESULT Activate(long _idx, VARIANT_BOOL _activate = true);
```

C#

```
public void Activate(long _idx, bool _activate = true);
```

Visual Basic

```
Public Sub Activate(_idx As long, Optional _activate As Boolean = true)
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Function activates/deactivates the grid of a given index.

III.6.3.2 Clear

C++

```
HRESULT Clear();
```

C#

```
public void Clear();
```

Visual Basic

```
Public Sub Clear()
```

Description

Function deletes all grids.

III.6.3.3 Create

C++

```
HRESULT Create(IRobotStructuralAxisGridType _type, BSTR _name, IRobotStructuralAxisGrid**
ret);
```

C#

```
public IRobotStructuralAxisGrid Create(IRobotStructuralAxisGridType _type, String _name);
```

Visual Basic

```
Public Function Create(_type As IRobotStructuralAxisGridType, _name As String) As
IRobotStructuralAxisGrid
```

Description

Function creates and returns a new blank grid of a defined type.

III.6.3.4 Delete

C++

```
HRESULT Delete(long _idx);
```

C#

```
public void Delete(long _idx);
```

Visual Basic

```
Public Sub Delete(_idx As long)
```

Description

Function deletes the grid of a given index.

III.6.3.5 FindByName

C++

```
HRESULT FindByName(BSTR _name, long* ret);
```

C#

```
public long FindByName(String _name);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Function FindByName(_name As String) As long
```

Description

Function returns an index of the grid of a given name or a zero value if the grid is not found.

III.6.3.6 Get

C++

```
HRESULT Get(long _idx, IRobotStructuralAxisGrid** ret);
```

C#

```
public IRobotStructuralAxisGrid Get(long _idx);
```

Visual Basic

```
Public Function Get(_idx As long) As IRobotStructuralAxisGrid
```

Description

Function makes available the grid of a given index.

III.6.3.7 GetByName

C++

```
HRESULT GetByName(BSTR _name, IRobotStructuralAxisGrid** ret);
```

C#

```
public IRobotStructuralAxisGrid GetByName(String _name);
```

Visual Basic

```
Public Function GetByName(_name As String) As IRobotStructuralAxisGrid
```

Description

Function makes available the grid of a given name.

III.6.3.8 IsActive

C++

```
HRESULT IsActive(long _idx, VARIANT_BOOL* ret);
```

C#

```
public bool IsActive(long _idx);
```

Visual Basic

```
Public Function IsActive(_idx As long) As Boolean
```

Description

Function returns activity state of the grid of a given index.

(c) 2014 Autodesk Inc. All rights reserved

Application - main object of the model

Enumerations

	Name	Description
	IRobotDirectory (see page 1544)	The set of identifiers representing folders used by Robot program.
	IRobotLanguage (see page 1544)	
	IRobotQuitOption (see page 1545)	Set of identifiers describing the manner of exiting the program. .
	IRobotDirectoryExtension (see page 1545)	Set of identifiers representing extensions for Robot program folders. .
	IRobotLicenseEntitlement (see page 1547)	
	IRobotLicenseEntitlementStatus (see page 1547)	
	IRobotCalculationsType (see page 1548)	

Interfaces

	Name	Description
	IRobotApplication (see page 1535)	Interface representing the entire application is the main object of the Robot Object Model. .
	IRobotPreferences (see page 1541)	Interface describing the current settings for the entire application. .
	IRobotPreferencesEvents (see page 1546)	Events related to the change of settings for the whole application.

I IRobotApplication

Class Hierarchy

RobotOM.IRobotApplication

C++

```
interface IRobotApplication : IDispatch;
```

C#

```
public interface IRobotApplication;
```

Visual Basic

```
Public Interface IRobotApplication
```

Description

Interface representing the entire application is the main object of the Robot Object Model. .

I.1 IRobotApplication Members

The following tables list the members exposed by IRobotApplication.

Public Fields

	Name	Description
	CmpntFactory (see page 1537)	

(c) 2014 Autodesk Inc. All rights reserved

	Interactive (see page 1537)	Flag indicating if interactive work with the program is expected .
	Is360 (see page 1537)	
	Kernel (see page 1538)	Component enabling access to the object of the Robot application by means of the IRobotKernel (see page 2208) interface.
	Preferences (see page 1538)	The object describing application parameters. .
	ProgramVersion (see page 1538)	Designation of the Robot program version .
	Project (see page 1538)	Current project .
	UserControl (see page 1539)	Flag used for giving/taking back program navigation to/from the interactive user .
	Version (see page 1539)	Indicator of Robot Object Model saved in the following format [Version].[Extended] (e.g. 1.02) .
	Visible (see page 1539)	Flag of the main Robot window visibility .
	Window (see page 1540)	Main window of Robot program Available since version 1.7.

Public Methods

	Name	Description
	LicenseCheckEntitlement (see page 1540)	The function returns status of a given entitlement.
	Quit (see page 1540)	Function closes the program. Available since version 1.7.

I.2 IRobotApplication Fields

The fields of the IRobotApplication class are listed here.

Public Fields

	Name	Description
	CmpntFactory (see page 1537)	
	Interactive (see page 1537)	Flag indicating if interactive work with the program is expected .
	Is360 (see page 1537)	
	Kernel (see page 1538)	Component enabling access to the object of the Robot application by means of the IRobotKernel (see page 2208) interface.
	Preferences (see page 1538)	The object describing application parameters. .
	ProgramVersion (see page 1538)	Designation of the Robot program version .
	Project (see page 1538)	Current project .
	UserControl (see page 1539)	Flag used for giving/taking back program navigation to/from the interactive user .
	Version (see page 1539)	Indicator of Robot Object Model saved in the following format [Version].[Extended] (e.g. 1.02) .
	Visible (see page 1539)	Flag of the main Robot window visibility .
	Window (see page 1540)	Main window of Robot program Available since version 1.7.

I.2.1 CmpntFactory

C++

```
HRESULT get_CmpntFactory(IRobotComponentFactory**);
```

C#

```
public IRobotComponentFactory CmpntFactory { get; }
```

Visual Basic

```
Public ReadOnly CmpntFactory As IRobotComponentFactory
```

Version

Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

I.2.2 Interactive

C++

```
HRESULT get_Interactive(VARIANT_BOOL*);
HRESULT put_Interactive(VARIANT_BOOL);
```

C#

```
public bool Interactive { get; set; }
```

Visual Basic

```
Public Interactive As Boolean
```

Description

Flag indicating if interactive work with the program is expected .

I.2.3 Is360

C++

```
HRESULT get_Is360(VARIANT_BOOL*);
```

C#

```
public bool Is360 { get; }
```

Visual Basic

```
Public ReadOnly Is360 As Boolean
```

Version

Available since version 13.4.

I.2.4 Kernel

C++

```
HRESULT get_Kernel(IRobotKernel**);
```

C#

```
public IRobotKernel Kernel { get; }
```

Visual Basic

```
Public ReadOnly Kernel As IRobotKernel
```

Description

Component enabling access to the object of the Robot application by means of the IRobotKernel (see page 2208) interface.

Version

Available since version 2.5.

I.2.5 Preferences

C++

```
HRESULT get_Preferences(IRobotPreferences**);
```

C#

```
public IRobotPreferences Preferences { get; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public ReadOnly Preferences As IRobotPreferences
```

Description

The object describing application parameters. .

I.2.6 ProgramVersion

C++

```
HRESULT get_ProgramVersion(BSTR*);
```

C#

```
public String ProgramVersion { get; }
```

Visual Basic

```
Public ReadOnly ProgramVersion As String
```

Description

Designation of the Robot program version .

Version

Available since version 3.

I.2.7 Project

C++

```
HRESULT get_Project(IRobotProject**);
```

C#

```
public IRobotProject Project { get; }
```

Visual Basic

```
Public ReadOnly Project As IRobotProject
```

Description

Current project .

I.2.8 UserControl

C++

```
HRESULT get_UserControl(VARIANT_BOOL*);
HRESULT put_UserControl(VARIANT_BOOL);
```

C#

```
public bool UserControl { get; set; }
```

Visual Basic

```
Public UserControl As Boolean
```

Description

Flag used for giving/taking back program navigation to/from the interactive user .

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

I.2.9 Version

C++

```
HRESULT get_Version(double*);
```

C#

```
public double Version { get; }
```

Visual Basic

```
Public ReadOnly Version As double
```

Description

Indicator of Robot Object Model saved in the following format [Version].[Extended] (e.g. 1.02) .

I.2.10 Visible

C++

```
HRESULT get_Visible(VARIANT_BOOL*);
HRESULT put_Visible(VARIANT_BOOL);
```

C#

```
public bool Visible { get; set; }
```

Visual Basic

```
Public Visible As Boolean
```

Description

Flag of the main Robot window visibility .

I.2.11 Window

C++

```
HRESULT get_Window(IRobotWindow**);
```

C#

```
public IRobotWindow Window { get; }
```

Visual Basic

```
Public ReadOnly Window As IRobotWindow
```

Description

Main window of Robot program Available since version 1.7.

I.3 IRobotApplication Methods

The methods of the IRobotApplication class are listed here.

Public Methods

	Name	Description
	LicenseCheckEntitlement (see page 1540)	The function returns status of a given entitlement.
	Quit (see page 1540)	Function closes the program. Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

I.3.1 LicenseCheckEntitlement

C++

```
HRESULT LicenseCheckEntitlement(IRobotLicenseEntitlement _entitlement,
IRobotLicenseEntitlementStatus* ret);
```

C#

```
public IRobotLicenseEntitlementStatus LicenseCheckEntitlement(IRobotLicenseEntitlement
_entitlement);
```

Visual Basic

```
Public Function LicenseCheckEntitlement(_entitlement As IRobotLicenseEntitlement) As
IRobotLicenseEntitlementStatus
```

Description

The function returns status of a given entitlement.

Version

Available since version 13.4.

I.3.2 Quit

C++

```
HRESULT Quit(IRobotQuitOption _quit_option);
```

C#

```
public void Quit(IRobotQuitOption _quit_option);
```

Visual Basic

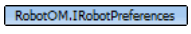
```
Public Sub Quit(_quit_option As IRobotQuitOption)
```

Description

Function closes the program. Available since version 1.7.

II IRobotPreferences

Class Hierarchy



C++

```
interface IRobotPreferences : IDispatch;
```

C#

```
public interface IRobotPreferences;
```

Visual Basic

```
Public Interface IRobotPreferences
```

Description

Interface describing the current settings for the entire application. .

(c) 2014 Autodesk Inc. All rights reserved

II.1 IRobotPreferences Members

The following tables list the members exposed by IRobotPreferences.

Public Fields

	Name	Description
	CalculationsType (see page 1542)	
	CloudCalculationsEnabled (see page 1542)	
	Multiprocessing (see page 1542)	Flag activating parallel processing - multiprocessing.
	OpenGL (see page 1542)	Flag that enables switching on/off OpenGL graphics.

Public Methods

	Name	Description
	GetDirectory (see page 1543)	The function returns full path to the desired Robot folder. .
	GetLanguage (see page 1543)	Function returns the current setting of the specified language. Code (number) of the appropriate country is returned (e.g. 33 for France).

II.2 IRobotPreferences Fields

The fields of the IRobotPreferences class are listed here.

Public Fields

	Name	Description
	CalculationsType (see page 1542)	
	CloudCalculationsEnabled (see page 1542)	
	Multiprocessing (see page 1542)	Flag activating parallel processing - multiprocessing.
	OpenGL (see page 1542)	Flag that enables switching on/off OpenGL graphics.

II.2.1 CalculationsType

C++

```
HRESULT get_CalculationsType(IRobotCalculationsType*);
HRESULT put_CalculationsType(IRobotCalculationsType);
```

C#

```
public IRobotCalculationsType CalculationsType { get; set; }
```

Visual Basic

```
Public CalculationsType As IRobotCalculationsType
```

Version

Available since version 13.4.

II.2.2 CloudCalculationsEnabled

C++

```
HRESULT get_CloudCalculationsEnabled(VARIANT_BOOL*);
HRESULT put_CloudCalculationsEnabled(VARIANT_BOOL);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public bool CloudCalculationsEnabled { get; set; }
```

Visual Basic

```
Public CloudCalculationsEnabled As Boolean
```

Version

Available since version 13.4.

II.2.3 Multiprocessing

C++

```
HRESULT get_Multiprocessing(VARIANT_BOOL*);
HRESULT put_Multiprocessing(VARIANT_BOOL);
```

C#

```
public bool Multiprocessing { get; set; }
```

Visual Basic

```
Public Multiprocessing As Boolean
```

Description

Flag activating parallel processing - multiprocessing.

Version

Available since version 8.5.

II.2.4 OpenGL

C++

```
HRESULT get_OpenGL(VARIANT_BOOL*);
HRESULT put_OpenGL(VARIANT_BOOL);
```

C#

```
public bool OpenGL { get; set; }
```

Visual Basic

```
Public OpenGL As Boolean
```

Description

Flag that enables switching on/off OpenGL graphics.

Version

Available since version 5.5.

II.3 IRobotPreferences Methods

The methods of the IRobotPreferences class are listed here.

Public Methods

	Name	Description
	GetDirectory (see page 1543)	The function returns full path to the desired Robot folder. .
	GetLanguage (see page 1543)	Function returns the current setting of the specified language. Code (number) of the appropriate country is returned (e.g. 33 for France).

(c) 2014 Autodesk Inc. All rights reserved

II.3.1 GetDirectory

C++

```
HRESULT GetDirectory(IRobotDirectory _robot_dir, BSTR* ret);
```

C#

```
public String GetDirectory(IRobotDirectory _robot_dir);
```

Visual Basic

```
Public Function GetDirectory(_robot_dir As IRobotDirectory) As String
```

Description

The function returns full path to the desired Robot folder. .

II.3.2 GetLanguage

C++

```
HRESULT GetLanguage(IRobotLanguage _lang_id, int* ret);
```

C#

```
public int GetLanguage(IRobotLanguage _lang_id);
```

Visual Basic

```
Public Function GetLanguage(_lang_id As IRobotLanguage) As int
```

Description

Function returns the current setting of the specified language. Code (number) of the appropriate country is returned (e.g. 33 for France).

III IRobotDirectory

C++

```
enum IRobotDirectory;
```

C#

```
public enum IRobotDirectory;
```

Visual Basic

```
Public Enum IRobotDirectory
```

Members

Members	Description
I_D_EXE = 1	Folder with executable files.
I_D_RES = 2	Folder with resources.
I_D_CONF = 3	Folder with configuration files.
I_D_USER_CONF = 4	Folder with user's configuration files .
I_D_TEMPLATE = 5	Folder with template files .
I_D_HELP = 6	Folder with help files.
I_D_USER_OUTPUT = 7	User's target (start-up) folder .
I_D_MAIN = 8	The main Robot folder .
I_D_USER_MAIN = 9	Main user folder.

(c) 2014 Autodesk Inc. All rights reserved

I_D_USER_PROJECTS = 10	Project folder.
I_D_USER_TEMPLATE = 11	Available since version 3.

Description

The set of identifiers representing folders used by Robot program.

IV IRobotLanguage

C++

```
enum IRobotLanguage;
```

C#

```
public enum IRobotLanguage;
```

Visual Basic

```
Public Enum IRobotLanguage
```

Members

Members	Description
I_L_WORK = 1	Interface language.
I_L_PRINTOUT = 2	Printout language.
I_L_REGIONAL = 3	Regional language.

V IRobotQuitOption

C++

```
enum IRobotQuitOption;
```

C#

```
public enum IRobotQuitOption;
```

Visual Basic

```
Public Enum IRobotQuitOption
```

Members

Members	Description
I_QO_DISCARD_CHANGES = 0	Changes made in the project will not be saved Available since version 1.7.
I_QO_PROMPT_TO_SAVE_CHANGES = 1	If changes that require saving to file have been made in the project, the program will display a message asking if the changes are to be saved Available since version 1.7.
I_QO_SAVE_CHANGES = 2	Changes will be saved automatically Available since version 1.7.

Description

Set of identifiers describing the manner of exiting the program. .

(c) 2014 Autodesk Inc. All rights reserved

VI IRobotDirectoryExtension

C++

```
enum IRobotDirectoryExtension;
```

C#

```
public enum IRobotDirectoryExtension;
```

Visual Basic

```
Public Enum IRobotDirectoryExtension
```

Members

Members	Description
I_DE_TEMPLATE = 51	Extension for the TEMPLATE folder. Available since version 3.5.
I_DE_USER_CONF = 52	Available since version 3.5.
I_DE_CONF = 53	Available since version 3.5.
I_DE_USER_TEMPLATE = 54	Available since version 3.5.

Description

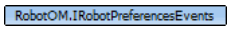
Set of identifiers representing extensions for Robot program folders. .

Version

Available since version 3.5.

VII IRobotPreferencesEvents

Class Hierarchy



C++

```
interface IRobotPreferencesEvents : IDispatch;
```

C#

```
public interface IRobotPreferencesEvents;
```

Visual Basic

```
Public Interface IRobotPreferencesEvents
```

Description

Events related to the change of settings for the whole application.

VII.1 IRobotPreferencesEvents Members

The following tables list the members exposed by IRobotPreferencesEvents.

Public Methods

	Name	Description
	OnDialogOK (see page 1546)	Event generated after accepting changes made in the program settings dialog box by the user.

(c) 2014 Autodesk Inc. All rights reserved

VII.2 IRobotPreferencesEvents Methods

The methods of the IRobotPreferencesEvents class are listed here.

Public Methods

	Name	Description
	OnDialogOK (see page 1546)	Event generated after accepting changes made in the program settings dialog box by the user.

VII.2.1 OnDialogOK

C++

```
HRESULT OnDialogOK();
```

C#

```
public void OnDialogOK();
```

Visual Basic

```
Public Sub OnDialogOK()
```

Description

Event generated after accepting changes made in the program settings dialog box by the user.

VIII IRobotLicenseEntitlement

C++

```
enum IRobotLicenseEntitlement;
```

C#

```
public enum IRobotLicenseEntitlement;
```

Visual Basic

```
Public Enum IRobotLicenseEntitlement
```

Members

Members	Description
I_LE_LOCAL_SOLVE = 0	Entitlement enabling to perform calculations on the local machine. Available since version 13.4.
I_LE_CLOUD_SOLVE = 1	Entitlement enabling to perform calculations in the calculation cloud. Available since version 13.4.

Version

Available since version 13.4.

IX IRobotLicenseEntitlementStatus

C++

```
enum IRobotLicenseEntitlementStatus;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public enum IRobotLicenseEntitlementStatus;
```

Visual Basic

```
Public Enum IRobotLicenseEntitlementStatus
```

Members

Members	Description
I_LES_ENTITLED = 0	Available since version 13.4.
I_LES_NOT_ENTITLED = 1	No entitlement. Available since version 13.4.
I_LES_UNKNOWN = 2	It is impossible to determine the possessed entitlements because there are problems with a network connection. Available since version 13.4.
I_LES_NOT_SIGNED_IN = 3	It is impossible to determine the possessed entitlements because the user is not logged in. Available since version 14.

Version

Available since version 13.4.

X IRobotCalculationsType

C++

```
enum IRobotCalculationsType;
```

C#

```
public enum IRobotCalculationsType;
```

Visual Basic

```
Public Enum IRobotCalculationsType
```

Members

Members	Description
I_CT_CALCULATE_LOCALLY = 0	Calculations will be performed locally on the user machine. Available since version 13.4.
I_CT_CALCULATE_IN_CLOUD = 1	Calculations will be performed in the calculation cloud. Available since version 13.4.

Version

Available since version 13.4.

Add-ins Manager

Interfaces

	Name	Description
	IRobotAddIn (see page 1548)	Definition of the Robot program extension. This interface must implement each component that is integrated with the Robot program and that extends its functionality. .

(c) 2014 Autodesk Inc. All rights reserved

	IRobotAddInMngr (see page 1551)	Add-ins Manager keeps tracks to all add-ins currently registered in Robot.
	IRobotCmdList (see page 1552)	List of information about commands.
	IRobotCmdInfo (see page 1554)	Information about the command and the menu option linked with it.
	IRobotAddInRegistrar (see page 1556)	Information about each Robot program extension must be saved as an appropriate position in the system register. RobotAddInRegistrar facilitates the process of extension registration.

I IRobotAddIn

Class Hierarchy

RobotOM.IRobotAddIn

C++

```
interface IRobotAddIn : IDispatch;
```

C#

```
public interface IRobotAddIn;
```

Visual Basic

```
Public Interface IRobotAddIn
```

Description

Definition of the Robot program extension. This interface must implement each component that is integrated with the Robot program and that extends its functionality. .

I.1 IRobotAddIn Members

The following tables list the members exposed by IRobotAddIn.

Public Methods

	Name	Description
	Connect (see page 1550)	Function wil be called up by the Robot application when the extension is connected to the program: for the first time - upon user's request or during first activation of the program after installing the extension; for the next time - while activating the Robot program again. The extension should remember the access interface to the Robot application (RobotApplication) so that the extension can use its functionality. The extension, while being connected, is ascribed its identifier which should be remembered since it will be needed during subsequent communication with the Robot program. While connecting the extension to the application for... more (see page 1550)
	Disconnect (see page 1550)	This function will be called up before the extension is disconnected with the Robot program (e.g. while exiting the application or upon user's request who wants no longer to use the extension). If there is any reason for the extension not to be disconnected with the program at the given moment, the function should return zero value (False). .
	DoCommand (see page 1551)	Function will be called up by the Robot program if a user selects the menu option linked with the command supported by this extension. .
	GetExpectedVersion (see page 1551)	Function should return the number of the RobotOS model version which is used by the extension. It will enable warning a user in a situation when the extension that is being installed by the user expects a newer Robot program version than the one currently used by the user. .

(c) 2014 Autodesk Inc. All rights reserved

	InstallCommands (see page 1551)	Function will be called up by the application to obtain information about the commands that are made available by the extension. List specified by the parameter should be filled out with the information about all the commands defined by the extension. If the operation is performed successfully, the function should return a value different from zero (True). .

I.2 IRobotAddIn Methods

The methods of the IRobotAddIn class are listed here.

Public Methods

	Name	Description
	Connect (see page 1550)	Function wil be called up by the Robot application when the extension is connected to the program: for the first time - upon user's request or during first activation of the program after installing the extension; for the next time - while activating the Robot program again. The extension should remember the access interface to the Robot application (RobotApplication) so that the extension can use its functionality. The extension, while being connected, is ascribed its identifier which should be remembered since it will be needed during subsequent communication with the Robot program. While connecting the extension to the application for... more (see page 1550)
	Disconnect (see page 1550)	This function will be called up before the extension is disconnected with the Robot program (e.g. while exiting the application or upon user's request who wants no longer to use the extension). If there is any reason for the extension not to be disconnected with the program at the given moment, the function should return zero value (False). .
	DoCommand (see page 1551)	Function will be called up by the Robot program if a user selects the menu option linked with the command supported by this extension. .
	GetExpectedVersion (see page 1551)	Function should return the number of the RobotOS model version which is used by the extension. It will enable warning a user in a situation when the extension that is being installed by the user expects a newer Robot program version than the one currently used by the user. .
	InstallCommands (see page 1551)	Function will be called up by the application to obtain information about the commands that are made available by the extension. List specified by the parameter should be filled out with the information about all the commands defined by the extension. If the operation is performed successfully, the function should return a value different from zero (True). .

I.2.1 Connect

C++

```
HRESULT Connect(IRobotApplication _robot_app, long _add_in_id, VARIANT_BOOL _first_time,
VARIANT_BOOL* ret);
```

C#

```
public bool Connect(IRobotApplication _robot_app, long _add_in_id, bool _first_time);
```

Visual Basic

```
Public Function Connect(_robot_app As IRobotApplication, _add_in_id As long, _first_time As
Boolean) As Boolean
```

Description

Function wil be called up by the Robot application when the extension is connected to the program: for the first time - upon user's request or during first activation of the program after installing the extension; for the next time - while activating the Robot program again. The extension should remember the access interface to the Robot application (RobotApplication) so

(c) 2014 Autodesk Inc. All rights reserved

that the extension can use its functionality. The extension, while being connected, is ascribed its identifier which should be remembered since it will be needed during subsequent communication with the Robot program. While connecting the extension to the application for the first time it is possible to perform additional initializing operations. If all is performed correctly, then function should return a value different from zero (True).

I.2.2 Disconnect

C++

```
HRESULT Disconnect(VARIANT_BOOL* ret);
```

C#

```
public bool Disconnect();
```

Visual Basic

```
Public Function Disconnect() As Boolean
```

Description

This function will be called up before the extension is disconnected with the Robot program (e.g. while exiting the application or upon user's request who wants no longer to use the extension). If there is any reason for the extension not to be disconnected with the program at the given moment, the function should return zero value (False). .

I.2.3 DoCommand

C++

```
HRESULT DoCommand(long _cmd_id);
```

C#

```
public void DoCommand(long _cmd_id);
```

Visual Basic

```
Public Sub DoCommand(_cmd_id As long)
```

Description

Function will be called up by the Robot program if a user selects the menu option linked with the command supported by this extension. .

I.2.4 GetExpectedVersion

C++

```
HRESULT GetExpectedVersion(double* ret);
```

C#

```
public double GetExpectedVersion();
```

Visual Basic

```
Public Function GetExpectedVersion() As double
```

Description

Function should return the number of the RobotOS model version which is used by the extension. It will enable warning a user in a situation when the extension that is being installed by the user expects a newer Robot program version than the one currently used by the user. .

(c) 2014 Autodesk Inc. All rights reserved

I.2.5 InstallCommands

C++

```
HRESULT InstallCommands(IRobotCmdList* _cmd_list, VARIANT_BOOL* ret);
```

C#

```
public bool InstallCommands(IRobotCmdList _cmd_list);
```

Visual Basic

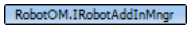
```
Public Function InstallCommands(ByRef _cmd_list As IRobotCmdList) As Boolean
```

Description

Function will be called up by the application to obtain information about the commands that are made available by the extension. List specified by the parameter should be filled out with the information about all the commands defined by the extension. If the operation is performed successfully, the function should return a value different from zero (True). .

II IRobotAddInMngr

Class Hierarchy



C++

```
interface IRobotAddInMngr : IDispatch;
```

C#

```
public interface IRobotAddInMngr;
```

Visual Basic

```
Public Interface IRobotAddInMngr
```

Description

Add-ins Manager keeps tracks to all add-ins currently registered in Robot.

II.1 IRobotAddInMngr Members

The following tables list the members exposed by IRobotAddInMngr.

Public Methods

	Name	Description
	InstallCommand (see page 1552)	

II.2 IRobotAddInMngr Methods

The methods of the IRobotAddInMngr class are listed here.

Public Methods

	Name	Description
	InstallCommand (see page 1552)	

(c) 2014 Autodesk Inc. All rights reserved

II.2.1 InstallCommand

C++

```
HRESULT InstallCommand(long _add_in_id, long _cmd_id, BSTR _cmd_name, VARIANT_BOOL* ret);
```

C#

```
public bool InstallCommand(long _add_in_id, long _cmd_id, String _cmd_name);
```

Visual Basic

```
Public Function InstallCommand(_add_in_id As long, _cmd_id As long, _cmd_name As String) As
Boolean
```

III IRobotCmdList

Class Hierarchy

RobotOM.IRobotCmdList

C++

```
interface IRobotCmdList : IDispatch;
```

C#

```
public interface IRobotCmdList;
```

Visual Basic

```
Public Interface IRobotCmdList
```

Description

List of information about commands.

III.1 IRobotCmdList Members

The following tables list the members exposed by IRobotCmdList.

Public Fields

	Name	Description
	Count (see page 1553)	Number of elements of the list Available since version 1.7.

Public Methods

	Name	Description
	Get (see page 1554)	Function returns the specified list element. List elements are indexed from 1 to Count (see page 1553). Available since version 1.7.
	New (see page 1554)	Function creates a new list element and adds it at the end. The index of the element on the list is returned. Available since version 1.7.

III.2 IRobotCmdList Fields

The fields of the IRobotCmdList class are listed here.

Public Fields

	Name	Description
	Count (see page 1553)	Number of elements of the list Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

III.2.1 Count

C++

```
HRESULT get_Count(long*);
```

C#

```
public long Count { get; }
```

Visual Basic

```
Public ReadOnly Count As long
```

Description

Number of elements of the list Available since version 1.7.

III.3 IRobotCmdList Methods

The methods of the IRobotCmdList class are listed here.

Public Methods

	Name	Description
	Get (see page 1554)	Function returns the specified list element. List elements are indexed from 1 to Count (see page 1553). Available since version 1.7.
	New (see page 1554)	Function creates a new list element and adds it at the end. The index of the element on the list is returned. Available since version 1.7.

III.3.1 Get

C++

```
HRESULT Get(long _idx, IRobotCmdInfo** ret);
```

C#

```
public IRobotCmdInfo Get(long _idx);
```

Visual Basic

```
Public Function Get(_idx As long) As IRobotCmdInfo
```

Description

Function returns the specified list element. List elements are indexed from 1 to Count (see page 1553). Available since version 1.7.

III.3.2 New

C++

```
HRESULT New(long _cmd_id, BSTR _cmd_name, long* ret);
```

C#

```
public long New(long _cmd_id, String _cmd_name);
```

Visual Basic

```
Public Function New(_cmd_id As long, _cmd_name As String) As long
```

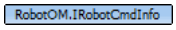
Description

Function creates a new list element and adds it at the end. The index of the element on the list is returned. Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

IV IRobotCmdInfo

Class Hierarchy



C++

```
interface IRobotCmdInfo : IDispatch;
```

C#

```
public interface IRobotCmdInfo;
```

Visual Basic

```
Public Interface IRobotCmdInfo
```

Description

Information about the command and the menu option linked with it.

IV.1 IRobotCmdInfo Members

The following tables list the members exposed by IRobotCmdInfo.

Public Fields

	Name	Description
	Id (see page 1555)	Command identifier Available since version 1.7.
	MenuChecked (see page 1555)	Indicator of marking the menu option linked with the command 0 - not marked 1 - marked 2 - intermediate state Available since version 1.7.
	MenuEnabled (see page 1556)	Flag indicating if the menu option linked with the command is available Available since version 1.7.
	Name (see page 1556)	Name of the menu option linked with the command Available since version 1.7.

IV.2 IRobotCmdInfo Fields

The fields of the IRobotCmdInfo class are listed here.

Public Fields

	Name	Description
	Id (see page 1555)	Command identifier Available since version 1.7.
	MenuChecked (see page 1555)	Indicator of marking the menu option linked with the command 0 - not marked 1 - marked 2 - intermediate state Available since version 1.7.
	MenuEnabled (see page 1556)	Flag indicating if the menu option linked with the command is available Available since version 1.7.
	Name (see page 1556)	Name of the menu option linked with the command Available since version 1.7.

IV.2.1 Id

C++

```
HRESULT get_Id(long*);
HRESULT put_Id(long);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public long Id { get; set; }
```

Visual Basic

```
Public Id As long
```

Description

Command identifier Available since version 1.7.

IV.2.2 MenuChecked

C++

```
HRESULT get_MenuChecked(long*);
HRESULT put_MenuChecked(long);
```

C#

```
public long MenuChecked { get; set; }
```

Visual Basic

```
Public MenuChecked As long
```

Description

Indicator of marking the menu option linked with the command 0 - not marked 1 - marked 2 - intermediate state

Available since version 1.7.

IV.2.3 MenuEnabled

C++

```
HRESULT get_MenuEnabled(VARIANT_BOOL*);
HRESULT put_MenuEnabled(VARIANT_BOOL);
```

C#

```
public bool MenuEnabled { get; set; }
```

Visual Basic

```
Public MenuEnabled As Boolean
```

Description

Flag indicating if the menu option linked with the command is available Available since version 1.7.

IV.2.4 Name

C++

```
HRESULT get_Name(BSTR*);
HRESULT put_Name(BSTR);
```

C#

```
public String Name { get; set; }
```

Visual Basic

```
Public Name As String
```

Description

Name of the menu option linked with the command Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

V IRobotAddInRegistrar

Class Hierarchy

RobotOM.IRobotAddInRegistrar

C++

```
interface IRobotAddInRegistrar : IDispatch;
```

C#

```
public interface IRobotAddInRegistrar;
```

Visual Basic

```
Public Interface IRobotAddInRegistrar
```

Description

Information about each Robot program extension must be saved as an appropriate position in the system register. RobotAddInRegistrar facilitates the process of extension registration.

V.1 IRobotAddInRegistrar Members

The following tables list the members exposed by IRobotAddInRegistrar.

Public Fields

	Name	Description
	Guid (see page 1557)	GUID identifier of the main extension component (presented interchangeably with ProgId) Available since version 1.7.
	ProductName (see page 1558)	Extension name Available since version 1.7.
	ProgId (see page 1558)	Identifier of the main extension component (presented interchangeably with Guid (see page 1557)) Available since version 1.7.
	ProviderName (see page 1558)	Name of the extension supplier Available since version 1.7.

Public Methods

	Name	Description
	InstallMenu (see page 1559)	Function adds menu with the specified name to all Robot program views. The commands specified in the list will are the menu options. They will be supported by the extension whose ProgId or Guid (see page 1557) have been set earlier. If identifier of the extension supporting commands from the list is not specified, the function returns zero value (False). Available since version 1.7.
	Register (see page 1559)	Function saves information about the extension to the register. If registration is performed successfully, a value different from zero is returned (True). Registration failure may result from lack of settings concerning suppler name or product as well as identifier (Guid (see page 1557) or ProgId). Available since version 1.7.
	Unregister (see page 1559)	Function deletes information about the extension from the register. Available since version 1.7.

V.2 IRobotAddInRegistrar Fields

The fields of the IRobotAddInRegistrar class are listed here.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	Guid (see page 1557)	GUID identifier of the main extension component (presented interchangeably with ProgId) Available since version 1.7.
	ProductName (see page 1558)	Extension name Available since version 1.7.
	ProgId (see page 1558)	Identifier of the main extension component (presented interchangeably with Guid (see page 1557)) Available since version 1.7.
	ProviderName (see page 1558)	Name of the extension supplier Available since version 1.7.

V.2.1 Guid

C++

```
HRESULT get_Guid(BSTR*);
HRESULT put_Guid(BSTR);
```

C#

```
public String Guid { get; set; }
```

Visual Basic

```
Public Guid As String
```

Description

GUID identifier of the main extension component (presented interchangeably with ProgId) Available since version 1.7.

V.2.2 ProductName

C++

```
HRESULT get_ProductName(BSTR*);
HRESULT put_ProductName(BSTR);
```

C#

```
public String ProductName { get; set; }
```

Visual Basic

```
Public ProductName As String
```

Description

Extension name Available since version 1.7.

V.2.3 ProgId

C++

```
HRESULT get_ProgId(BSTR*);
HRESULT put_ProgId(BSTR);
```

C#

```
public String ProgId { get; set; }
```

Visual Basic

```
Public ProgId As String
```

Description

Identifier of the main extension component (presented interchangeably with Guid (see page 1557)) Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

V.2.4 ProviderName

C++

```
HRESULT get_ProviderName(BSTR*);
HRESULT put_ProviderName(BSTR);
```

C#

```
public String ProviderName { get; set; }
```

Visual Basic

```
Public ProviderName As String
```

Description

Name of the extension supplier Available since version 1.7.

V.3 IRobotAddInRegistrar Methods

The methods of the IRobotAddInRegistrar class are listed here.

Public Methods

	Name	Description
	InstallMenu (see page 1559)	Function adds menu with the specified name to all Robot program views. The commands specified in the list will are the menu options. They will be supported by the extension whose ProgId or Guid (see page 1557) have been set earlier. If identifier of the extension supporting commands from the list is not specified, the function returns zero value (False). Available since version 1.7.
	Register (see page 1559)	Function saves information about the extension to the register. If registration is performed successfully, a value different from zero is returned (True). Registration failure may result from lack of settings concerning suppler name or product as well as identifier (Guid (see page 1557) or ProgId). Available since version 1.7.
	Unregister (see page 1559)	Function deletes information about the extension from the register. Available since version 1.7.

V.3.1 InstallMenu

C++

```
HRESULT InstallMenu(BSTR _menu_name, IRobotCmdList* _options, VARIANT_BOOL* ret);
```

C#

```
public bool InstallMenu(String _menu_name, IRobotCmdList _options);
```

Visual Basic

```
Public Function InstallMenu(_menu_name As String, ByRef _options As IRobotCmdList) As
Boolean
```

Description

Function adds menu with the specified name to all Robot program views. The commands specified in the list will are the menu options. They will be supported by the extension whose ProgId or Guid (see page 1557) have been set earlier. If identifier of the extension supporting commands from the list is not specified, the function returns zero value (False). Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

V.3.2 Register

C++

```
HRESULT Register(VARIANT_BOOL* ret);
```

C#

```
public bool Register();
```

Visual Basic

```
Public Function Register() As Boolean
```

Description

Function saves information about the extension to the register. If registration is performed successfully, a value different from zero is returned (True). Registration failure may result from lack of settings concerning suppler name or product as well as identifier (Guid (see page 1557) or Progld). Available since version 1.7.

V.3.3 Unregister

C++

```
HRESULT Unregister();
```

C#

```
public void Unregister();
```

Visual Basic

```
Public Sub Unregister()
```

Description

Function deletes information about the extension from the register. Available since version 1.7.

Calculation module

Enumerations

	Name	Description
	IRobotEquationSolvingMethod (see page 1567)	Available methods of equation set solution.
	IRobotStructureAutoVerificationType (see page 1572)	Available types of automatic structure verification.
	IRobotSparseMSolverMethod (see page 1573)	
	IRobotIterativePrediconditionerType (see page 1574)	
	IRobotIterativeSolverMethod (see page 1578)	
	IRobotIterativeSolverMemoryUsage (see page 1578)	
	IRobotStructureAnalysisModalParticipationCoeff (see page 1579)	Available methods of participation coefficient calculation.
	IRobotSeismicResultsPanelDirection (see page 1584)	
	IRobotCalcMessageSource (see page 1593)	

(c) 2014 Autodesk Inc. All rights reserved

	IRobotCalcMessageSeverityLevel (see page 1593)	
	IRobotCalculationStatus (see page 1593)	
	IRobotCalculationMode (see page 1594)	Available methods of performing calculations.

Interfaces

	Name	Description
	IRobotCalcEngine (see page 1561)	Interface representing the calculation module of Robot program. .
	IRobotStructureAnalysisParams (see page 1567)	Parameters of structure analysis.
	IRobotSparseMSolverParams (see page 1572)	
	IRobotIterativeSolverParams (see page 1574)	
	IRobotSeismicResultsSaveParams (see page 1579)	Parameters of saving quadratic combination results for seismic analysis in a project.
	IRobotModelGenerationParams (see page 1584)	Generation parameters for structure calculation model.
	IRobotBucklingDeformationParams (see page 1589)	Parameters of buckling deformation.
	IRobotCalcEngineEvents (see page 1591)	Events generated by the RobotCalcEngine calculation module.

I IRobotCalcEngine

Class Hierarchy

RobotOM.IRobotCalcEngine

C++

```
interface IRobotCalcEngine : IDispatch;
```

C#

```
public interface IRobotCalcEngine;
```

Visual Basic

```
Public Interface IRobotCalcEngine
```

Description

Interface representing the calculation module of Robot program. .

I.1 IRobotCalcEngine Members

The following tables list the members exposed by IRobotCalcEngine.

Public Fields

	Name	Description
	AnalysisParams (see page 1562)	Parameters of structure analysis.
	AutoFreezeResults (see page 1562)	Flag enforcing automatic freezing of results after calculations .
	AutoGenerateModel (see page 1563)	Flag steering automatic model generation that directly precedes structure calculations.

(c) 2014 Autodesk Inc. All rights reserved

	BucklingDeformation (see page 1563)	Buckling deformation.
	DAM (see page 1563)	.
	GenerationParams (see page 1564)	Parameters of calculation model generation.
	SaveResultsInExternalFile (see page 1564)	Flag enforcing result saving in the external file *.RT_.
	SeismicResultsSaveParams (see page 1564)	Saving parameters for seismic analysis results .
	StatusWindowParent (see page 1565)	Handle to parent window for the calculation status window .
	UseStatusWindow (see page 1565)	Flag that steers the status window display while performing time-consuming calculations .

Public Methods

	Name	Description
	Calculate (see page 1566)	The function recalculates the entire structure. If there appeared an error during calculations, the function returns zero value (False), otherwise it returns a non-zero value (True) .
	CalculateEx (see page 1566)	The functions performs structure analysis. If the calculations succeeded and results are available the status I_CS_COMPLETED is returned. Otherwise, the appropriate status with info about reasons of error(s) is returned.
	GenerateModel (see page 1566)	Function generates a structure calculation model. .
	StopCalculation (see page 1567)	Function causes interrupting the currently performed calculations.

I.2 IRobotCalcEngine Fields

The fields of the IRobotCalcEngine class are listed here.

Public Fields

	Name	Description
	AnalysisParams (see page 1562)	Parameters of structure analysis.
	AutoFreezeResults (see page 1562)	Flag enforcing automatic freezing of results after calculations .
	AutoGenerateModel (see page 1563)	Flag steering automatic model generation that directly precedes structure calculations.
	BucklingDeformation (see page 1563)	Buckling deformation.
	DAM (see page 1563)	.
	GenerationParams (see page 1564)	Parameters of calculation model generation.
	SaveResultsInExternalFile (see page 1564)	Flag enforcing result saving in the external file *.RT_.
	SeismicResultsSaveParams (see page 1564)	Saving parameters for seismic analysis results .
	StatusWindowParent (see page 1565)	Handle to parent window for the calculation status window .
	UseStatusWindow (see page 1565)	Flag that steers the status window display while performing time-consuming calculations .

I.2.1 AnalysisParams

C++

```
HRESULT get_AnalysisParams(IRobotStructureAnalysisParams**);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRobotStructureAnalysisParams AnalysisParams { get; }
```

Visual Basic

```
Public ReadOnly AnalysisParams As IRobotStructureAnalysisParams
```

Description

Parameters of structure analysis.

Version

Available since version 3.

I.2.2 AutoFreezeResults

C++

```
HRESULT get_AutoFreezeResults(VARIANT_BOOL*);
HRESULT put_AutoFreezeResults(VARIANT_BOOL);
```

C#

```
public bool AutoFreezeResults { get; set; }
```

Visual Basic

```
Public AutoFreezeResults As Boolean
```

Description

Flag enforcing automatic freezing of results after calculations .

Version

Available since version 3.

I.2.3 AutoGenerateModel

C++

```
HRESULT get_AutoGenerateModel(VARIANT_BOOL*);
HRESULT put_AutoGenerateModel(VARIANT_BOOL);
```

C#

```
public bool AutoGenerateModel { get; set; }
```

Visual Basic

```
Public AutoGenerateModel As Boolean
```

Description

Flag steering automatic model generation that directly precedes structure calculations.

Version

Available since version 2.5.

I.2.4 BucklingDeformation

C++

```
HRESULT get_BucklingDeformation(IRobotBucklingDeformationParams**);
```

C#

```
public IRobotBucklingDeformationParams BucklingDeformation { get; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public ReadOnly BucklingDeformation As IRobotBucklingDeformationParams
```

Description

Buckling deformation.

Version

Available since version 3.

I.2.5 DAM

C++

```
HRESULT get_DAM(IRobotDAMCalcModule**);
```

C#

```
public IRobotDAMCalcModule DAM { get; }
```

Visual Basic

```
Public ReadOnly DAM As IRobotDAMCalcModule
```

Description

.

Version

Available since version 15.

I.2.6 GenerationParams

C++

```
HRESULT get_GenerationParams(IRobotModelGenerationParams**);
```

C#

```
public IRobotModelGenerationParams GenerationParams { get; }
```

Visual Basic

```
Public ReadOnly GenerationParams As IRobotModelGenerationParams
```

Description

Parameters of calculation model generation.

Version

Available since version 3.

I.2.7 SaveResultsInExternalFile

C++

```
HRESULT get_SaveResultsInExternalFile(VARIANT_BOOL*);
HRESULT put_SaveResultsInExternalFile(VARIANT_BOOL);
```

C#

```
public bool SaveResultsInExternalFile { get; set; }
```

Visual Basic

```
Public SaveResultsInExternalFile As Boolean
```

Description

Flag enforcing result saving in the external file *.RT_.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.

I.2.8 SeismicResultsSaveParams

C++

```
HRESULT get_SeismicResultsSaveParams(IRobotSeismicResultsSaveParams**);
```

C#

```
public IRobotSeismicResultsSaveParams SeismicResultsSaveParams { get; }
```

Visual Basic

```
Public ReadOnly SeismicResultsSaveParams As IRobotSeismicResultsSaveParams
```

Description

Saving parameters for seismic analysis results .

Version

Available since version 3.

I.2.9 StatusWindowParent

C++

```
HRESULT get_StatusWindowParent(long*);
HRESULT put_StatusWindowParent(long);
```

C#

```
public long StatusWindowParent { get; set; }
```

Visual Basic

```
Public StatusWindowParent As long
```

Description

Handle to parent window for the calculation status window .

Version

Available since version 5.5.

I.2.10 UseStatusWindow

C++

```
HRESULT get_UseStatusWindow(VARIANT_BOOL*);
HRESULT put_UseStatusWindow(VARIANT_BOOL);
```

C#

```
public bool UseStatusWindow { get; set; }
```

Visual Basic

```
Public UseStatusWindow As Boolean
```

Description

Flag that steers the status window display while performing time-consuming calculations .

Version

Available since version 5.5.

(c) 2014 Autodesk Inc. All rights reserved

I.3 IRobotCalcEngine Methods

The methods of the IRobotCalcEngine class are listed here.

Public Methods

	Name	Description
	Calculate (see page 1566)	The function recalculates the entire structure. If there appeared an error during calculations, the function returns zero value (False), otherwise it returns a non-zero value (True) .
	CalculateEx (see page 1566)	The functions performs structure analysis. If the calculations succeeded and results are available the status I_CS_COMPLETED is returned. Otherwise, the appropriate status with info about reasons of error(s) is returned.
	GenerateModel (see page 1566)	Function generates a structure calculation model. .
	StopCalculation (see page 1567)	Function causes interrupting the currently performed calculations.

I.3.1 Calculate

C++

```
HRESULT Calculate(VARIANT_BOOL* ret);
```

C#

```
public bool Calculate();
```

Visual Basic

```
Public Function Calculate() As Boolean
```

Description

The function recalculates the entire structure. If there appeared an error during calculations, the function returns zero value (False), otherwise it returns a non-zero value (True) .

I.3.2 CalculateEx

C++

```
HRESULT CalculateEx(IRobotCalculationMode _calc_mode, IRobotCalculationStatus* ret);
```

C#

```
public IRobotCalculationStatus CalculateEx(IRobotCalculationMode _calc_mode);
```

Visual Basic

```
Public Function CalculateEx(_calc_mode As IRobotCalculationMode) As IRobotCalculationStatus
```

Description

The functions performs structure analysis. If the calculations succeeded and results are available the status I_CS_COMPLETED is returned. Otherwise, the appropriate status with info about reasons of error(s) is returned.

Version

Available since version 13.4.

I.3.3 GenerateModel

C++

```
HRESULT GenerateModel();
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public void GenerateModel();
```

Visual Basic

```
Public Sub GenerateModel()
```

Description

Function generates a structure calculation model. .

Version

Available since version 3.5.

I.3.4 StopCalculation

C++

```
HRESULT StopCalculation();
```

C#

```
public void StopCalculation();
```

Visual Basic

```
Public Sub StopCalculation()
```

Description

Function causes interrupting the currently performed calculations.

Version

Available since version 13.4.

II IRobotEquationSolvingMethod

C++

```
enum IRobotEquationSolvingMethod;
```

C#

```
public enum IRobotEquationSolvingMethod;
```

Visual Basic

```
Public Enum IRobotEquationSolvingMethod
```

Members

Members	Description
I_ESM_FRONTAL = 0	Available since version 3.
I_ESM_SKYLINE = 1	Available since version 3.
I_ESM_SPARSE = 4	Available since version 3.
I_ESM_SPARSE_M = 5	Available since version 3.
I_ESM_ITERATIVE = 2	Available since version 3.
I_ESM_AUTO = 3	Available since version 3.
I_ESM_MULTI_THREADED = 7	Available since version 12.

Description

Available methods of equation set solution.

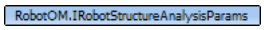
(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.

III IRobotStructureAnalysisParams

Class Hierarchy



C++

```
interface IRobotStructureAnalysisParams : IDispatch;
```

C#

```
public interface IRobotStructureAnalysisParams;
```

Visual Basic

```
Public Interface IRobotStructureAnalysisParams
```

Description

Parameters of structure analysis.

Version

Available since version 3.

III.1 IRobotStructureAnalysisParams Members

The following tables list the members exposed by IRobotStructureAnalysisParams.

Public Fields

	Name	Description
	AutoBarMerging (see page 1569)	Automatic bar merging during import of geometry.
	AutoVerification (see page 1569)	Selected type of automatic structure verification.
	DSCAlgorithm (see page 1569)	DSC algorithm.
	EquationSolvingMethod (see page 1570)	Selected method of equation set solution .
	FictitiousRigidityCoeff (see page 1570)	Fictitious rigidity coefficient.
	IgnoreWarnings (see page 1570)	
	IterativeParams (see page 1571)	
	ModalParticipationCoeff (see page 1571)	Method of participation coefficient calculation.
	RLINKElements (see page 1571)	
	SparseMParams (see page 1571)	

III.2 IRobotStructureAnalysisParams Fields

The fields of the IRobotStructureAnalysisParams class are listed here.

Public Fields

	Name	Description
	AutoBarMerging (see page 1569)	Automatic bar merging during import of geometry.
	AutoVerification (see page 1569)	Selected type of automatic structure verification.

(c) 2014 Autodesk Inc. All rights reserved

	DSCAlgorithm (see page 1569)	DSC algorithm.
	EquationSolvingMethod (see page 1570)	Selected method of equation set solution .
	FictitiousRigidityCoeff (see page 1570)	Fictitious rigidity coefficient.
	IgnoreWarnings (see page 1570)	
	IterativeParams (see page 1571)	
	ModalParticipationCoeff (see page 1571)	Method of participation coefficient calculation.
	RLINKElements (see page 1571)	
	SparseMParams (see page 1571)	

III.2.1 AutoBarMerging

C++

```
HRESULT get_AutoBarMerging(VARIANT_BOOL*);
HRESULT put_AutoBarMerging(VARIANT_BOOL);
```

C#

```
public bool AutoBarMerging { get; set; }
```

Visual Basic

```
Public AutoBarMerging As Boolean
```

Description

Automatic bar merging during import of geometry.

Version

Available since version 3.

III.2.2 AutoVerification

C++

```
HRESULT get_AutoVerification(IRobotStructureAutoVerificationType*);
HRESULT put_AutoVerification(IRobotStructureAutoVerificationType);
```

C#

```
public IRobotStructureAutoVerificationType AutoVerification { get; set; }
```

Visual Basic

```
Public AutoVerification As IRobotStructureAutoVerificationType
```

Description

Selected type of automatic structure verification.

Version

Available since version 3.

III.2.3 DSCAlgorithm

C++

```
HRESULT get_DSCAlgorithm(VARIANT_BOOL*);
HRESULT put_DSCAlgorithm(VARIANT_BOOL);
```

C#

```
public bool DSCAlgorithm { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public DSCAlgorithm As Boolean
```

Description

DSC algorithm.

Version

Available since version 3.

III.2.4 EquationSolvingMethod

C++

```
HRESULT get_EquationSolvingMethod(IRobotEquationSolvingMethod*);
HRESULT put_EquationSolvingMethod(IRobotEquationSolvingMethod);
```

C#

```
public IRobotEquationSolvingMethod EquationSolvingMethod { get; set; }
```

Visual Basic

```
Public EquationSolvingMethod As IRobotEquationSolvingMethod
```

Description

Selected method of equation set solution .

Version

Available since version 3.

III.2.5 FictitiousRigidityCoeff

C++

```
HRESULT get_FictitiousRigidityCoeff(double*);
HRESULT put_FictitiousRigidityCoeff(double);
```

C#

```
public double FictitiousRigidityCoeff { get; set; }
```

Visual Basic

```
Public FictitiousRigidityCoeff As double
```

Description

Fictitious rigidity coefficient.

Version

Available since version 3.

III.2.6 IgnoreWarnings

C++

```
HRESULT get_IgnoreWarnings(VARIANT_BOOL*);
HRESULT put_IgnoreWarnings(VARIANT_BOOL);
```

C#

```
public bool IgnoreWarnings { get; set; }
```

Visual Basic

```
Public IgnoreWarnings As Boolean
```

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.

III.2.7 IterativeParams

C++

```
HRESULT get_IterativeParams(IRobotIterativeSolverParams**);
```

C#

```
public IRobotIterativeSolverParams IterativeParams { get; }
```

Visual Basic

```
Public ReadOnly IterativeParams As IRobotIterativeSolverParams
```

Version

Available since version 3.

III.2.8 ModalParticipationCoeff

C++

```
HRESULT get_ModalParticipationCoeff(IRobotStructureAnalysisModalParticipationCoeff*);
HRESULT put_ModalParticipationCoeff(IRobotStructureAnalysisModalParticipationCoeff);
```

C#

```
public IRobotStructureAnalysisModalParticipationCoeff ModalParticipationCoeff { get; set; }
```

Visual Basic

```
Public ModalParticipationCoeff As IRobotStructureAnalysisModalParticipationCoeff
```

Description

Method of participation coefficient calculation.

Version

Available since version 3.

III.2.9 RLINKElements

C++

```
HRESULT get_RLINKElements(VARIANT_BOOL*);
HRESULT put_RLINKElements(VARIANT_BOOL);
```

C#

```
public bool RLINKElements { get; set; }
```

Visual Basic

```
Public RLINKElements As Boolean
```

Version

Available since version 11.

III.2.10 SparseMParams

C++

```
HRESULT get_SparseMParams(IRobotSparseMSolverParams**);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRobotSparseMSolverParams SparseMParams { get; }
```

Visual Basic

```
Public ReadOnly SparseMParams As IRobotSparseMSolverParams
```

Version

Available since version 3.

IV IRobotStructureAutoVerificationType

C++

```
enum IRobotStructureAutoVerificationType;
```

C#

```
public enum IRobotStructureAutoVerificationType;
```

Visual Basic

```
Public Enum IRobotStructureAutoVerificationType
```

Members

Members	Description
I_SAVT_NONE = 0	Available since version 3.
I_SAVT_ERRORS_ONLY = 1	Available since version 3.
I_SAVT_ERRORS_AND_WARNINGS = 2	Available since version 3.

Description

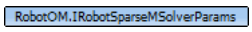
Available types of automatic structure verification.

Version

Available since version 3.

V IRobotSparseMSolverParams

Class Hierarchy



C++

```
interface IRobotSparseMSolverParams : IDispatch;
```

C#

```
public interface IRobotSparseMSolverParams;
```

Visual Basic

```
Public Interface IRobotSparseMSolverParams
```

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

V.1 IRobotSparseMSolverParams Members

The following tables list the members exposed by IRobotSparseMSolverParams.

Public Fields

	Name	Description
	Method (see page 1573)	

V.2 IRobotSparseMSolverParams Fields

The fields of the IRobotSparseMSolverParams class are listed here.

Public Fields

	Name	Description
	Method (see page 1573)	

V.2.1 Method

C++

```
HRESULT get_Method(IRobotSparseMSolverMethod*);
HRESULT put_Method(IRobotSparseMSolverMethod);
```

C#

```
public IRobotSparseMSolverMethod Method { get; set; }
```

Visual Basic

```
Public Method As IRobotSparseMSolverMethod
```

Version

Available since version 3.

VI IRobotSparseMSolverMethod

C++

```
enum IRobotSparseMSolverMethod;
```

C#

```
public enum IRobotSparseMSolverMethod;
```

Visual Basic

```
Public Enum IRobotSparseMSolverMethod
```

Members

Members	Description
I_SMSM_NDM = 3	Nested dissection method. Available since version 3.
I_SMSM_MDA = 4	Minimal degrees algorithm (MDA). Available since version 3.

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

VII IRobotIterativePrediconditionerType

C++

```
enum IRobotIterativePrediconditionerType;
```

C#

```
public enum IRobotIterativePrediconditionerType;
```

Visual Basic

```
Public Enum IRobotIterativePrediconditionerType
```

Members

Members	Description
I_IPT_DIAGONAL = 0	Available since version 3.
I_IPT_GAUSS = 1	Available since version 3.
I_IPT_CHOLESKY = 2	Available since version 3.
I_IPT_ICCF = 3	Available since version 3.

Version

Available since version 3.

VIII IRobotIterativeSolverParams

Class Hierarchy

RobotOM.IRobotIterativeSolverParams

C++

```
interface IRobotIterativeSolverParams : IDispatch;
```

C#

```
public interface IRobotIterativeSolverParams;
```

Visual Basic

```
Public Interface IRobotIterativeSolverParams
```

Version

Available since version 3.

VIII.1 IRobotIterativeSolverParams Members

The following tables list the members exposed by IRobotIterativeSolverParams.

Public Fields

	Name	Description
	AggregationLevelsCount (see page 1575)	Number of aggregation levels.
	AnalyseDiagonale (see page 1575)	Diagonal analysis.
	CalcKMatrix (see page 1576)	Calculation of the K matrix.

(c) 2014 Autodesk Inc. All rights reserved

	InternalIterationsCount (see page 1576)	Number of internal iterations.
	MemoryUsage (see page 1576)	
	Method (see page 1577)	
	Multilevel (see page 1577)	
	PrediconditionerType (see page 1577)	
	Tolerance (see page 1578)	Tolerance definition.

VIII.2 IRobotIterativeSolverParams Fields

The fields of the IRobotIterativeSolverParams class are listed here.

Public Fields

	Name	Description
	AggregationLevelsCount (see page 1575)	Number of aggregation levels.
	AnalyseDiagonale (see page 1575)	Diagonal analysis.
	CalcKMatrix (see page 1576)	Calculation of the K matrix.
	InternalIterationsCount (see page 1576)	Number of internal iterations.
	MemoryUsage (see page 1576)	
	Method (see page 1577)	
	Multilevel (see page 1577)	
	PrediconditionerType (see page 1577)	
	Tolerance (see page 1578)	Tolerance definition.

VIII.2.1 AggregationLevelsCount

C++

```
HRESULT get_AggregationLevelsCount(long*);
HRESULT put_AggregationLevelsCount(long);
```

C#

```
public long AggregationLevelsCount { get; set; }
```

Visual Basic

```
Public AggregationLevelsCount As long
```

Description

Number of aggregation levels.

Version

Available since version 3.

VIII.2.2 AnalyseDiagonale

C++

```
HRESULT get_AnalyseDiagonale(VARIANT_BOOL*);
HRESULT put_AnalyseDiagonale(VARIANT_BOOL);
```

C#

```
public bool AnalyseDiagonale { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public AnalyseDiagonale As Boolean
```

Description

Diagonal analysis.

Version

Available since version 3.

VIII.2.3 CalcKMatrix

C++

```
HRESULT get_CalcKMatrix(VARIANT_BOOL*);
HRESULT put_CalcKMatrix(VARIANT_BOOL);
```

C#

```
public bool CalcKMatrix { get; set; }
```

Visual Basic

```
Public CalcKMatrix As Boolean
```

Description

Calculation of the K matrix.

Version

Available since version 3.

VIII.2.4 InternalIterationsCount

C++

```
HRESULT get_InternalIterationsCount(long*);
HRESULT put_InternalIterationsCount(long);
```

C#

```
public long InternalIterationsCount { get; set; }
```

Visual Basic

```
Public InternalIterationsCount As long
```

Description

Number of internal iterations.

Version

Available since version 3.

VIII.2.5 MemoryUsage

C++

```
HRESULT get_MemoryUsage(IRobotIterativeSolverMemoryUsage*);
HRESULT put_MemoryUsage(IRobotIterativeSolverMemoryUsage);
```

C#

```
public IRobotIterativeSolverMemoryUsage MemoryUsage { get; set; }
```

Visual Basic

```
Public MemoryUsage As IRobotIterativeSolverMemoryUsage
```

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.

VIII.2.6 Method

C++

```
HRESULT get_Method(IRobotIterativeSolverMethod*);
HRESULT put_Method(IRobotIterativeSolverMethod);
```

C#

```
public IRobotIterativeSolverMethod Method { get; set; }
```

Visual Basic

```
Public Method As IRobotIterativeSolverMethod
```

Version

Available since version 3.

VIII.2.7 Multilevel

C++

```
HRESULT get_Multilevel(VARIANT_BOOL*);
HRESULT put_Multilevel(VARIANT_BOOL);
```

C#

```
public bool Multilevel { get; set; }
```

Visual Basic

```
Public Multilevel As Boolean
```

Version

Available since version 3.

VIII.2.8 PrediconditionerType

C++

```
HRESULT get_PrediconditionerType(IRobotIterativePrediconditionerType*);
HRESULT put_PrediconditionerType(IRobotIterativePrediconditionerType);
```

C#

```
public IRobotIterativePrediconditionerType PrediconditionerType { get; set; }
```

Visual Basic

```
Public PrediconditionerType As IRobotIterativePrediconditionerType
```

Version

Available since version 3.

VIII.2.9 Tolerance

C++

```
HRESULT get_Tolerance(double*);
HRESULT put_Tolerance(double);
```

C#

```
public double Tolerance { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Tolerance As double
```

Description

Tolerance definition.

Version

Available since version 3.

IX IRobotIterativeSolverMethod

C++

```
enum IRobotIterativeSolverMethod;
```

C#

```
public enum IRobotIterativeSolverMethod;
```

Visual Basic

```
Public Enum IRobotIterativeSolverMethod
```

Members

Members	Description
I_ISM_0 = 0	Available since version 3.
I_ISM_1 = 1	Available since version 3.
I_ISM_2 = 2	Available since version 3.

Version

Available since version 3.

X IRobotIterativeSolverMemoryUsage

C++

```
enum IRobotIterativeSolverMemoryUsage;
```

C#

```
public enum IRobotIterativeSolverMemoryUsage;
```

Visual Basic

```
Public Enum IRobotIterativeSolverMemoryUsage
```

Members

Members	Description
I_ISMU_MIN = 0	Available since version 3.
I_ISMU_1_2 = 2	Available since version 3.
I_ISMU_1_4 = 1	Available since version 3.
I_ISMU_MAX = 3	Available since version 3.

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

XI IRobotStructureAnalysisModalParticipationCoeff

C++

```
enum IRobotStructureAnalysisModalParticipationCoeff;
```

C#

```
public enum IRobotStructureAnalysisModalParticipationCoeff;
```

Visual Basic

```
Public Enum IRobotStructureAnalysisModalParticipationCoeff
```

Members

Members	Description
I_SAMPC_SUM_ABSOLUTE_VALUES = 0	Sum of absolute values. Available since version 3.
I_SAMPC_SQUARE_ROOT_OF_SUM_SQUARES = 1	Square root of sum of squares. Available since version 3.

Description

Available methods of participation coefficient calculation.

Version

Available since version 3.

XII IRobotSeismicResultsSaveParams

Class Hierarchy

RobotOM.IRobotSeismicResultsSaveParams

C++

```
interface IRobotSeismicResultsSaveParams : IDispatch;
```

C#

```
public interface IRobotSeismicResultsSaveParams;
```

Visual Basic

```
Public Interface IRobotSeismicResultsSaveParams
```

Description

Parameters of saving quadratic combination results for seismic analysis in a project.

Version

Available since version 3.

XII.1 IRobotSeismicResultsSaveParams Members

The following tables list the members exposed by IRobotSeismicResultsSaveParams.

Public Fields

	Name	Description
	Displacements (see page 1581)	Flag enforcing saving of displacements .

(c) 2014 Autodesk Inc. All rights reserved

	Forces (see page 1581)	Flag enforcing saving of forces.
	LocalDisplacements (see page 1581)	Local displacements.
	NMQ (see page 1581)	Flag enforcing saving of N membrane forces, M moments and Q shear forces.
	OnlyQuadraticCombs (see page 1582)	Result save only for quadratic combinations.
	PanelsDir (see page 1582)	Direction for panels.
	PointNumber (see page 1582)	Definition of a number of points along the bar length.
	Reactions (see page 1583)	Flag enforcing saving of reactions.
	Reduced (see page 1583)	Flag enforcing saving of reduced results .
	Save (see page 1583)	Flag indicating if results are saved in the project.
	Stresses (see page 1584)	Flag enforcing saving of stresses .

XII.2 IRobotSeismicResultsSaveParams Fields

The fields of the IRobotSeismicResultsSaveParams class are listed here.

Public Fields

	Name	Description
	Displacements (see page 1581)	Flag enforcing saving of displacements .
	Forces (see page 1581)	Flag enforcing saving of forces.
	LocalDisplacements (see page 1581)	Local displacements.
	NMQ (see page 1581)	Flag enforcing saving of N membrane forces, M moments and Q shear forces.
	OnlyQuadraticCombs (see page 1582)	Result save only for quadratic combinations.
	PanelsDir (see page 1582)	Direction for panels.
	PointNumber (see page 1582)	Definition of a number of points along the bar length.
	Reactions (see page 1583)	Flag enforcing saving of reactions.
	Reduced (see page 1583)	Flag enforcing saving of reduced results .
	Save (see page 1583)	Flag indicating if results are saved in the project.
	Stresses (see page 1584)	Flag enforcing saving of stresses .

XII.2.1 Displacements

C++

```
HRESULT get_Displacements(VARIANT_BOOL*);
HRESULT put_Displacements(VARIANT_BOOL);
```

C#

```
public bool Displacements { get; set; }
```

Visual Basic

```
Public Displacements As Boolean
```

Description

Flag enforcing saving of displacements .

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

XII.2.2 Forces

C++

```
HRESULT get_Forces(VARIANT_BOOL*);
HRESULT put_Forces(VARIANT_BOOL);
```

C#

```
public bool Forces { get; set; }
```

Visual Basic

```
Public Forces As Boolean
```

Description

Flag enforcing saving of forces.

Version

Available since version 3.

XII.2.3 LocalDisplacements

C++

```
HRESULT get_LocalDisplacements(VARIANT_BOOL*);
HRESULT put_LocalDisplacements(VARIANT_BOOL);
```

C#

```
public bool LocalDisplacements { get; set; }
```

Visual Basic

```
Public LocalDisplacements As Boolean
```

Description

Local displacements.

XII.2.4 NMQ

C++

```
HRESULT get_NMQ(VARIANT_BOOL*);
HRESULT put_NMQ(VARIANT_BOOL);
```

C#

```
public bool NMQ { get; set; }
```

Visual Basic

```
Public NMQ As Boolean
```

Description

Flag enforcing saving of N membrane forces, M moments and Q shear forces.

Version

Available since version 3.

XII.2.5 OnlyQuadraticCombs

C++

```
HRESULT get_OnlyQuadraticCombs(VARIANT_BOOL*);
HRESULT put_OnlyQuadraticCombs(VARIANT_BOOL);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public bool OnlyQuadraticCombs { get; set; }
```

Visual Basic

```
Public OnlyQuadraticCombs As Boolean
```

Description

Result save only for quadratic combinations.

Version

Available since version 5.6.

XII.2.6 PanelsDir

C++

```
HRESULT get_PanelsDir(IRobotSeismicResultsPanelDirection*);
HRESULT put_PanelsDir(IRobotSeismicResultsPanelDirection);
```

C#

```
public IRobotSeismicResultsPanelDirection PanelsDir { get; set; }
```

Visual Basic

```
Public PanelsDir As IRobotSeismicResultsPanelDirection
```

Description

Direction for panels.

Version

Available since version 3.

XII.2.7 PointNumber

C++

```
HRESULT get_PointNumber(long*);
HRESULT put_PointNumber(long);
```

C#

```
public long PointNumber { get; set; }
```

Visual Basic

```
Public PointNumber As long
```

Description

Definition of a number of points along the bar length.

Version

Available since version 3.

XII.2.8 Reactions

C++

```
HRESULT get_Reactions(VARIANT_BOOL*);
HRESULT put_Reactions(VARIANT_BOOL);
```

C#

```
public bool Reactions { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Reactions As Boolean
```

Description

Flag enforcing saving of reactions.

Version

Available since version 3.

XII.2.9 Reduced

C++

```
HRESULT get_Reduced(VARIANT_BOOL*);
HRESULT put_Reduced(VARIANT_BOOL);
```

C#

```
public bool Reduced { get; set; }
```

Visual Basic

```
Public Reduced As Boolean
```

Description

Flag enforcing saving of reduced results .

Version

Available since version 3.

XII.2.10 Save

C++

```
HRESULT get_Save(VARIANT_BOOL*);
HRESULT put_Save(VARIANT_BOOL);
```

C#

```
public bool Save { get; set; }
```

Visual Basic

```
Public Save As Boolean
```

Description

Flag indicating if results are saved in the project.

Version

Available since version 3.

XII.2.11 Stresses

C++

```
HRESULT get_Stresses(VARIANT_BOOL*);
HRESULT put_Stresses(VARIANT_BOOL);
```

C#

```
public bool Stresses { get; set; }
```

Visual Basic

```
Public Stresses As Boolean
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Flag enforcing saving of stresses .

Version

Available since version 3.

XIII IRobotSeismicResultsPanelDirection

C++

```
enum IRobotSeismicResultsPanelDirection;
```

C#

```
public enum IRobotSeismicResultsPanelDirection;
```

Visual Basic

```
Public Enum IRobotSeismicResultsPanelDirection
```

Members

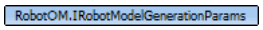
Members	Description
I_SRPD_AUTOMATIC = 0	Available since version 3.
I_SRPD_DIR_X = 1	Available since version 3.
I_SRPD_DIR_Y = 2	Available since version 3.
I_SRPD_DIR_Z = 3	Available since version 3.

Version

Available since version 3.

XIV IRobotModelGenerationParams

Class Hierarchy



C++

```
interface IRobotModelGenerationParams : IDispatch;
```

C#

```
public interface IRobotModelGenerationParams;
```

Visual Basic

```
Public Interface IRobotModelGenerationParams
```

Description

Generation parameters for structure calculation model.

Version

Available since version 3.

XIV.1 IRobotModelGenerationParams Members

The following tables list the members exposed by IRobotModelGenerationParams.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	AssemblingCase (see page 1586)	Assembling case number.
	GenerateNodes_BarsAndFiniteElems (see page 1586)	Node generation at intersections of bars and finite elements.
	GenerateNodes_DiagonalBars (see page 1586)	Node generation at intersections of inclined bars.
	GenerateNodes_VertHorizBars (see page 1587)	Node generation at intersections of vertical and horizontal bars.
	MaxElementLength (see page 1587)	Maximum length of calculation element.
	NeglectedBars (see page 1587)	List of bars to be disregarded during generation of the calculation model.
	NeglectedGeoObjects (see page 1588)	List of geometrical objects to be disregarded during generation of the calculation model.
	ToleranceAutomatic (see page 1588)	Enforcing the automatic value for tolerance.
	ToleranceValue (see page 1588)	Tolerance value.

XIV.2 IRobotModelGenerationParams Fields

The fields of the IRobotModelGenerationParams class are listed here.

Public Fields

	Name	Description
	AssemblingCase (see page 1586)	Assembling case number.
	GenerateNodes_BarsAndFiniteElems (see page 1586)	Node generation at intersections of bars and finite elements.
	GenerateNodes_DiagonalBars (see page 1586)	Node generation at intersections of inclined bars.
	GenerateNodes_VertHorizBars (see page 1587)	Node generation at intersections of vertical and horizontal bars.
	MaxElementLength (see page 1587)	Maximum length of calculation element.
	NeglectedBars (see page 1587)	List of bars to be disregarded during generation of the calculation model.
	NeglectedGeoObjects (see page 1588)	List of geometrical objects to be disregarded during generation of the calculation model.
	ToleranceAutomatic (see page 1588)	Enforcing the automatic value for tolerance.
	ToleranceValue (see page 1588)	Tolerance value.

XIV.2.1 AssemblingCase

C++

```
HRESULT get_AssemblingCase(long*);
HRESULT put_AssemblingCase(long);
```

C#

```
public long AssemblingCase { get; set; }
```

Visual Basic

```
Public AssemblingCase As long
```

Description

Assembling case number.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.

XIV.2.2 GenerateNodes_BarsAndFiniteElems

C++

```
HRESULT get_GenerateNodes_BarsAndFiniteElems(VARIANT_BOOL*);
HRESULT put_GenerateNodes_BarsAndFiniteElems(VARIANT_BOOL);
```

C#

```
public bool GenerateNodes_BarsAndFiniteElems { get; set; }
```

Visual Basic

```
Public GenerateNodes_BarsAndFiniteElems As Boolean
```

Description

Node generation at intersections of bars and finite elements.

Version

Available since version 3.

XIV.2.3 GenerateNodes_DiagonalBars

C++

```
HRESULT get_GenerateNodes_DiagonalBars(VARIANT_BOOL*);
HRESULT put_GenerateNodes_DiagonalBars(VARIANT_BOOL);
```

C#

```
public bool GenerateNodes_DiagonalBars { get; set; }
```

Visual Basic

```
Public GenerateNodes_DiagonalBars As Boolean
```

Description

Node generation at intersections of inclined bars.

Version

Available since version 3.

XIV.2.4 GenerateNodes_VertHorizBars

C++

```
HRESULT get_GenerateNodes_VertHorizBars(VARIANT_BOOL*);
HRESULT put_GenerateNodes_VertHorizBars(VARIANT_BOOL);
```

C#

```
public bool GenerateNodes_VertHorizBars { get; set; }
```

Visual Basic

```
Public GenerateNodes_VertHorizBars As Boolean
```

Description

Node generation at intersections of vertical and horizontal bars.

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

XIV.2.5 MaxElementLength

C++

```
HRESULT get_MaxElementLength(double*);
HRESULT put_MaxElementLength(double);
```

C#

```
public double MaxElementLength { get; set; }
```

Visual Basic

```
Public MaxElementLength As double
```

Description

Maximum length of calculation element.

Version

Available since version 3.

XIV.2.6 NeglectedBars

C++

```
HRESULT get_NeglectedBars(BSTR*);
HRESULT put_NeglectedBars(BSTR);
```

C#

```
public String NeglectedBars { get; set; }
```

Visual Basic

```
Public NeglectedBars As String
```

Description

List of bars to be disregarded during generation of the calculation model.

Version

Available since version 3.

XIV.2.7 NeglectedGeoObjects

C++

```
HRESULT get_NeglectedGeoObjects(BSTR*);
HRESULT put_NeglectedGeoObjects(BSTR);
```

C#

```
public String NeglectedGeoObjects { get; set; }
```

Visual Basic

```
Public NeglectedGeoObjects As String
```

Description

List of geometrical objects to be disregarded during generation of the calculation model.

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

XIV.2.8 ToleranceAutomatic

C++

```
HRESULT get_ToleranceAutomatic(VARIANT_BOOL*);
HRESULT put_ToleranceAutomatic(VARIANT_BOOL);
```

C#

```
public bool ToleranceAutomatic { get; set; }
```

Visual Basic

```
Public ToleranceAutomatic As Boolean
```

Description

Enforcing the automatic value for tolerance.

Version

Available since version 3.

XIV.2.9 ToleranceValue

C++

```
HRESULT get_ToleranceValue(double*);
HRESULT put_ToleranceValue(double);
```

C#

```
public double ToleranceValue { get; set; }
```

Visual Basic

```
Public ToleranceValue As double
```

Description

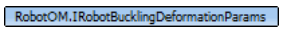
Tolerance value.

Version

Available since version 3.

XV IRobotBucklingDeformationParams

Class Hierarchy



C++

```
interface IRobotBucklingDeformationParams : IDispatch;
```

C#

```
public interface IRobotBucklingDeformationParams;
```

Visual Basic

```
Public Interface IRobotBucklingDeformationParams
```

Description

Parameters of buckling deformation.

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

XV.1 IRobotBucklingDeformationParams Members

The following tables list the members exposed by IRobotBucklingDeformationParams.

Public Fields

	Name	Description
	CaseNumber (see page 1590)	Number of selected buckling analysis case (zero value indicates that no case has been selected).
	MaxDisplacement (see page 1590)	Maximum displacement.
	OmitCaseForDeformations (see page 1590)	Flag indicating if a selected buckling case is to be disregarded for a structure including deformations .

Public Methods

	Name	Description
	GetModeCoeff (see page 1591)	Function returns coefficient for the indicated mode.
	SetModeCoeff (see page 1591)	Function defines coefficient for the indicated mode. .

XV.2 IRobotBucklingDeformationParams Fields

The fields of the IRobotBucklingDeformationParams class are listed here.

Public Fields

	Name	Description
	CaseNumber (see page 1590)	Number of selected buckling analysis case (zero value indicates that no case has been selected).
	MaxDisplacement (see page 1590)	Maximum displacement.
	OmitCaseForDeformations (see page 1590)	Flag indicating if a selected buckling case is to be disregarded for a structure including deformations .

XV.2.1 CaseNumber

C++

```
HRESULT get_CaseNumber(long*);
HRESULT put_CaseNumber(long);
```

C#

```
public long CaseNumber { get; set; }
```

Visual Basic

```
Public CaseNumber As long
```

Description

Number of selected buckling analysis case (zero value indicates that no case has been selected).

Version

Available since version 3.

XV.2.2 MaxDisplacement

C++

```
HRESULT get_MaxDisplacement(double*);
HRESULT put_MaxDisplacement(double);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double MaxDisplacement { get; set; }
```

Visual Basic

```
Public MaxDisplacement As double
```

Description

Maximum displacement.

Version

Available since version 3.

XV.2.3 OmitCaseForDeformations

C++

```
HRESULT get_OmitCaseForDeformations(VARIANT_BOOL*);
HRESULT put_OmitCaseForDeformations(VARIANT_BOOL);
```

C#

```
public bool OmitCaseForDeformations { get; set; }
```

Visual Basic

```
Public OmitCaseForDeformations As Boolean
```

Description

Flag indicating if a selected buckling case is to be disregarded for a structure including deformations .

Version

Available since version 3.

XV.3 IRobotBucklingDeformationParams Methods

The methods of the IRobotBucklingDeformationParams class are listed here.

Public Methods

	Name	Description
	GetModeCoeff (see page 1591)	Function returns coefficient for the indicated mode.
	SetModeCoeff (see page 1591)	Function defines coefficient for the indicated mode. .

XV.3.1 GetModeCoeff

C++

```
HRESULT GetModeCoeff(long _mode, double* ret);
```

C#

```
public double GetModeCoeff(long _mode);
```

Visual Basic

```
Public Function GetModeCoeff(_mode As long) As double
```

Description

Function returns coefficient for the indicated mode.

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

XV.3.2 SetModeCoeff

C++

```
HRESULT SetModeCoeff(long _mode, double _coeff);
```

C#

```
public void SetModeCoeff(long _mode, double _coeff);
```

Visual Basic

```
Public Sub SetModeCoeff(_mode As long, _coeff As double)
```

Description

Function defines coefficient for the indicated mode. .

Version

Available since version 3.

XVI IRobotCalcEngineEvents

Class Hierarchy

RobotOM.IRobotCalcEngineEvents

C++

```
interface IRobotCalcEngineEvents : IDispatch;
```

C#

```
public interface IRobotCalcEngineEvents;
```

Visual Basic

```
Public Interface IRobotCalcEngineEvents
```

Description

Events generated by the RobotCalcEngine calculation module.

Version

Available since version 13.4.

XVI.1 IRobotCalcEngineEvents Members

The following tables list the members exposed by IRobotCalcEngineEvents.

Public Methods

	Name	Description
	CalcMessage (see page 1592)	Function makes the messages from verification of computational model and structure analysis available.

XVI.2 IRobotCalcEngineEvents Methods

The methods of the IRobotCalcEngineEvents class are listed here.

(c) 2014 Autodesk Inc. All rights reserved

Public Methods

	Name	Description
	CalcMessage (see page 1592)	Function makes the messages from verification of computational model and structure analysis available.

XVI.2.1 CalcMessage

C++

```
HRESULT CalcMessage(IRobotCalcMessageSource _msgSource, RobotCalcMessageSeverity
_msgSeverity, long _msgId, BSTR _msgText, IRobotObjectType _objType, BSTR _objSel);
```

C#

```
public void CalcMessage(IRobotCalcMessageSource _msgSource, RobotCalcMessageSeverity
_msgSeverity, long _msgId, String _msgText, IRobotObjectType _objType, String _objSel);
```

Visual Basic

```
Public Sub CalcMessage(_msgSource As IRobotCalcMessageSource, _msgSeverity As
RobotCalcMessageSeverity, _msgId As long, _msgText As String, _objType As IRobotObjectType,
_objSel As String)
```

Description

Function makes the messages from verification of computational model and structure analysis available.

Version

Available since version 13.4.

XVII IRobotCalcMessageSource

C++

```
enum IRobotCalcMessageSource;
```

C#

```
public enum IRobotCalcMessageSource;
```

Visual Basic

```
Public Enum IRobotCalcMessageSource
```

Members

Members	Description
I_CMS_VERIFICATION = 3	Verification of the computational model correctness. Available since version 13.4.
I_CMS_CALCULATIONS = 1	Structure analysis. Available since version 13.4.

Version

Available since version 13.4.

XVIII IRobotCalcMessageSeverityLevel

C++

```
enum IRobotCalcMessageSeverityLevel;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public enum IRobotCalcMessageSeverityLevel;
```

Visual Basic

```
Public Enum IRobotCalcMessageSeverityLevel
```

Members

Members	Description
I_CMSL_WARNING = 2	Available since version 13.4.
I_CMSL_ERROR = 3	Available since version 13.4.
I_CMSL_NOTE = 1	Available since version 13.4.

Version

Available since version 13.4.

XIX IRobotCalculationStatus

C++

```
enum IRobotCalculationStatus;
```

C#

```
public enum IRobotCalculationStatus;
```

Visual Basic

```
Public Enum IRobotCalculationStatus
```

Members

Members	Description
I_CS_COMPLETED = 0	Calculations succeeded; results are available. Available since version 13.4.
I_CS_FAILED_VERIFICATION = 1	Calculations have been stopped because of errors or warnings that occured during the verification stage; no results. Available since version 13.4.
I_CS_FAILED_CALCULATION = 2	Calculations failed; no reults. Available since version 13.4.
I_CS_CANCELLED_BY_USER = 3	Calculations were interrupted by the user; no results. Available since version 13.4.
I_CS_FAILED_GENERATION = 4	An error has occured during generation of computational model; no results. Available since version 13.4.
I_CS_IN_PROGRESS = -1	Calculations in progress. Available since version 13.4.
I_CS_FAILED_NO_ENTITLEMENT = 5	No entitlements to perform calculations. Available since version 13.4.

Version

Available since version 13.4.

(c) 2014 Autodesk Inc. All rights reserved

XX IRobotCalculationMode

C++

```
enum IRobotCalculationMode;
```

C#

```
public enum IRobotCalculationMode;
```

Visual Basic

```
Public Enum IRobotCalculationMode
```

Members

Members	Description
I_CM_LOCAL = 0	Calculations performed in the synchronous way on the local machine. Available since version 13.4.
I_CM_LOCAL_ASYNCHRONOUS = 1	Calculations performed in the asynchronous way on the local machine. Available since version 13.4.
I_CM_CLOUD = 2	Calculations performed in the synchronous way in the calculation cloud. Available since version 13.4.
I_CM_CLOUD_ASYNCHRONOUS = 3	Calculations performed in the asynchronous way in the calculation cloud. Available since version 13.4.

Description

Available methods of performing calculations.

Version

Available since version 13.4.

XXI Direct Analysis Method

Available since version 15.

Enumerations

	Name	Description
	IRobotDAMAnalysisType (see page 1597)	.
	IRobotDAMLateralLoadCombType (see page 1601)	.
	IRobotDAMSteelMembersReductionType (see page 1605)	.

Interfaces

	Name	Description
	IRobotDAMCalcModule (see page 1595)	Calculation module for Direct Analysis Method.
	IRobotDAMNotionalLoads (see page 1598)	.
	IRobotDAMReducedStiffness (see page 1602)	.

(c) 2014 Autodesk Inc. All rights reserved

	IRobotDAMParams (see page 1605)	.

XXI.1 IRobotDAMCalcModule

Class Hierarchy

C++

```
interface IRobotDAMCalcModule : IDispatch;
```

C#

```
public interface IRobotDAMCalcModule;
```

Visual Basic

```
Public Interface IRobotDAMCalcModule
```

Description

Calculation module for Direct Analysis Method.

Version

Available since version 15.

XXI.1.1 IRobotDAMCalcModule Members

The following tables list the members exposed by IRobotDAMCalcModule.

Public Fields

	Name	Description
	IsActive (see page 1596)	.
	Params (see page 1596)	.

Public Methods

	Name	Description
	DeleteModel (see page 1597)	
	Run (see page 1597)	

XXI.1.2 IRobotDAMCalcModule Fields

The fields of the IRobotDAMCalcModule class are listed here.

Public Fields

	Name	Description
	IsActive (see page 1596)	.
	Params (see page 1596)	.

XXI.1.2.1 IsActive

C++

```
HRESULT get_IsActive(VARIANT_BOOL*);
HRESULT put_IsActive(VARIANT_BOOL);
```

C#

```
public bool IsActive { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public IsActive As Boolean
```

Description

.

Version

Available since version 15.

XXI.1.2.2 Params

C++

```
HRESULT get_Params(IRobotDAMParams**);
```

C#

```
public IRobotDAMParams Params { get; }
```

Visual Basic

```
Public ReadOnly Params As IRobotDAMParams
```

Description

.

Version

Available since version 15.

XXI.1.3 IRobotDAMCalcModule Methods

The methods of the IRobotDAMCalcModule class are listed here.

Public Methods

	Name	Description
	DeleteModel (see page 1597)	
	Run (see page 1597)	

XXI.1.3.1 DeleteModel

C++

```
HRESULT DeleteModel();
```

C#

```
public void DeleteModel();
```

Visual Basic

```
Public Sub DeleteModel()
```

Version

Available since version 15.

XXI.1.3.2 Run

C++

```
HRESULT Run(VARIANT_BOOL* ret);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public bool Run();
```

Visual Basic

```
Public Function Run() As Boolean
```

Version

Available since version 15.

XXI.2 IRobotDAMAnalysisType

C++

```
enum IRobotDAMAnalysisType;
```

C#

```
public enum IRobotDAMAnalysisType;
```

Visual Basic

```
Public Enum IRobotDAMAnalysisType
```

Members

Members	Description
I_DAT_ANSI_AISC_360_10_LRFD = 0	. Available since version 15.
I_DAT_ANSI_AISC_360_10_ASD = 1	. Available since version 15.
I_DAT_USER_DEFINED = 2	. Available since version 15.

Description

.

Version

Available since version 15.

XXI.3 IRobotDAMNotionalLoads

Class Hierarchy

RobotOM.IRobotDAMNotionalLoads

C++

```
interface IRobotDAMNotionalLoads : IDispatch;
```

C#

```
public interface IRobotDAMNotionalLoads;
```

Visual Basic

```
Public Interface IRobotDAMNotionalLoads
```

Description

.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 15.

XXI.3.1 IRobotDAMNotionalLoads Members

The following tables list the members exposed by IRobotDAMNotionalLoads.

Public Fields

	Name	Description
	ActiveXN (see page 1599)	.
	ActiveXP (see page 1599)	.
	ActiveYN (see page 1599)	.
	ActiveYP (see page 1600)	.
	Coefficient (see page 1600)	.
	GravityLoadCombEnabled (see page 1600)	.
	LateralLoadCombEnabled (see page 1601)	.
	LateralLoadCombType (see page 1601)	.

XXI.3.2 IRobotDAMNotionalLoads Fields

The fields of the IRobotDAMNotionalLoads class are listed here.

Public Fields

	Name	Description
	ActiveXN (see page 1599)	.
	ActiveXP (see page 1599)	.
	ActiveYN (see page 1599)	.
	ActiveYP (see page 1600)	.
	Coefficient (see page 1600)	.
	GravityLoadCombEnabled (see page 1600)	.
	LateralLoadCombEnabled (see page 1601)	.
	LateralLoadCombType (see page 1601)	.

XXI.3.2.1 ActiveXN

C++

```
HRESULT get_ActiveXN(VARIANT_BOOL*);
HRESULT put_ActiveXN(VARIANT_BOOL);
```

C#

```
public bool ActiveXN { get; set; }
```

Visual Basic

```
Public ActiveXN As Boolean
```

Description

.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 15.

XXI.3.2.2 ActiveXP

C++

```
HRESULT get_ActiveXP(VARIANT_BOOL*);
HRESULT put_ActiveXP(VARIANT_BOOL);
```

C#

```
public bool ActiveXP { get; set; }
```

Visual Basic

```
Public ActiveXP As Boolean
```

Description

.

Version

Available since version 15.

XXI.3.2.3 ActiveYN

C++

```
HRESULT get_ActiveYN(VARIANT_BOOL*);
HRESULT put_ActiveYN(VARIANT_BOOL);
```

C#

```
public bool ActiveYN { get; set; }
```

Visual Basic

```
Public ActiveYN As Boolean
```

Description

.

Version

Available since version 15.

XXI.3.2.4 ActiveYP

C++

```
HRESULT get_ActiveYP(VARIANT_BOOL*);
HRESULT put_ActiveYP(VARIANT_BOOL);
```

C#

```
public bool ActiveYP { get; set; }
```

Visual Basic

```
Public ActiveYP As Boolean
```

Description

.

Version

Available since version 15.

(c) 2014 Autodesk Inc. All rights reserved

XXI.3.2.5 Coefficient

C++

```
HRESULT get_Coefficient(double*);
HRESULT put_Coefficient(double);
```

C#

```
public double Coefficient { get; set; }
```

Visual Basic

```
Public Coefficient As double
```

Description

.

Version

Available since version 15.

XXI.3.2.6 GravityLoadCombEnabled

C++

```
HRESULT get_GravityLoadCombEnabled(VARIANT_BOOL*);
HRESULT put_GravityLoadCombEnabled(VARIANT_BOOL);
```

C#

```
public bool GravityLoadCombEnabled { get; set; }
```

Visual Basic

```
Public GravityLoadCombEnabled As Boolean
```

Description

.

Version

Available since version 15.

XXI.3.2.7 LateralLoadCombEnabled

C++

```
HRESULT get_LateralLoadCombEnabled(VARIANT_BOOL*);
HRESULT put_LateralLoadCombEnabled(VARIANT_BOOL);
```

C#

```
public bool LateralLoadCombEnabled { get; set; }
```

Visual Basic

```
Public LateralLoadCombEnabled As Boolean
```

Description

.

Version

Available since version 15.

(c) 2014 Autodesk Inc. All rights reserved

XXI.3.2.8 LateralLoadCombType

C++

```
HRESULT get_LateralLoadCombType(IRobotDAMLateralLoadCombType*);
HRESULT put_LateralLoadCombType(IRobotDAMLateralLoadCombType);
```

C#

```
public IRobotDAMLateralLoadCombType LateralLoadCombType { get; set; }
```

Visual Basic

```
Public LateralLoadCombType As IRobotDAMLateralLoadCombType
```

Description

.

Version

Available since version 15.

XXI.4 IRobotDAMLateralLoadCombType

C++

```
enum IRobotDAMLateralLoadCombType;
```

C#

```
public enum IRobotDAMLateralLoadCombType;
```

Visual Basic

```
Public Enum IRobotDAMLateralLoadCombType
```

Members

Members	Description
I_DLLCT_LATERAL_LOAD_DIR = 0	. Available since version 15.
I_DLLCT_ONE_DIR_ONLY = 1	. Available since version 15.
I_DLLCT_TWO_DIR_INDEPENDENTLY = 2	. Available since version 15.

Description

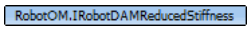
.

Version

Available since version 15.

XXI.5 IRobotDAMReducedStiffness

Class Hierarchy



C++

```
interface IRobotDAMReducedStiffness : IDispatch;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public interface IRobotDAMReducedStiffness;
```

Visual Basic

```
Public Interface IRobotDAMReducedStiffness
```

Description

.

Version

Available since version 15.

XXI.5.1 IRobotDAMReducedStiffness Members

The following tables list the members exposed by IRobotDAMReducedStiffness.

Public Fields

	Name	Description
	RCBeamsValue (see page 1603)	.
	RCColumnsAndWallsValue (see page 1603)	.
	RCSlabsValue (see page 1604)	.
	SteelMembersReductionType (see page 1604)	.
	SteelMembersTauBValue (see page 1604)	.
	SteelMembersValue (see page 1605)	.

XXI.5.2 IRobotDAMReducedStiffness Fields

The fields of the IRobotDAMReducedStiffness class are listed here.

Public Fields

	Name	Description
	RCBeamsValue (see page 1603)	.
	RCColumnsAndWallsValue (see page 1603)	.
	RCSlabsValue (see page 1604)	.
	SteelMembersReductionType (see page 1604)	.
	SteelMembersTauBValue (see page 1604)	.
	SteelMembersValue (see page 1605)	.

XXI.5.2.1 RCBeamsValue

C++

```
HRESULT get_RCBeamsValue(double*);
HRESULT put_RCBeamsValue(double);
```

C#

```
public double RCBeamsValue { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public RCBeamsValue As double
```

Description

.

Version

Available since version 15.

XXI.5.2.2 RCColumnsAndWallsValue

C++

```
HRESULT get_RCColumnsAndWallsValue(double*);
HRESULT put_RCColumnsAndWallsValue(double);
```

C#

```
public double RCColumnsAndWallsValue { get; set; }
```

Visual Basic

```
Public RCColumnsAndWallsValue As double
```

Description

.

Version

Available since version 15.

XXI.5.2.3 RCSlabsValue

C++

```
HRESULT get_RCSlabsValue(double*);
HRESULT put_RCSlabsValue(double);
```

C#

```
public double RCSlabsValue { get; set; }
```

Visual Basic

```
Public RCSlabsValue As double
```

Description

.

Version

Available since version 15.

XXI.5.2.4 SteelMembersReductionType

C++

```
HRESULT get_SteelMembersReductionType(IRobotDAMSteelMembersReductionType*);
HRESULT put_SteelMembersReductionType(IRobotDAMSteelMembersReductionType);
```

C#

```
public IRobotDAMSteelMembersReductionType SteelMembersReductionType { get; set; }
```

Visual Basic

```
Public SteelMembersReductionType As IRobotDAMSteelMembersReductionType
```

(c) 2014 Autodesk Inc. All rights reserved

Description

.

Version

Available since version 15.

XXI.5.2.5 SteelMembersTauBValue

C++

```
HRESULT get_SteelMembersTauBValue(double*);
HRESULT put_SteelMembersTauBValue(double);
```

C#

```
public double SteelMembersTauBValue { get; set; }
```

Visual Basic

```
Public SteelMembersTauBValue As double
```

Description

.

Version

Available since version 15.

XXI.5.2.6 SteelMembersValue

C++

```
HRESULT get_SteelMembersValue(double*);
HRESULT put_SteelMembersValue(double);
```

C#

```
public double SteelMembersValue { get; set; }
```

Visual Basic

```
Public SteelMembersValue As double
```

Description

.

Version

Available since version 15.

XXI.6 IRobotDAMSteelMembersReductionType

C++

```
enum IRobotDAMSteelMembersReductionType;
```

C#

```
public enum IRobotDAMSteelMembersReductionType;
```

Visual Basic

```
Public Enum IRobotDAMSteelMembersReductionType
```

(c) 2014 Autodesk Inc. All rights reserved

Members

Members	Description
I_DSMRT_FACTOR_0_8 = 0	. Available since version 15.
I_DSMRT_FACTOR_0_8_AND_TAU_B = 1	. Available since version 15.
I_DSMRT_FACTOR_0_8_AND_NL_INCREASE = 2	. Available since version 15.

Description

.

Version

Available since version 15.

XXI.7 IRobotDAMParams

Class Hierarchy

RobotOM.IRobotDAMParams

C++

```
interface IRobotDAMParams : IDispatch;
```

C#

```
public interface IRobotDAMParams;
```

Visual Basic

```
Public Interface IRobotDAMParams
```

Description

.

Version

Available since version 15.

XXI.7.1 IRobotDAMParams Members

The following tables list the members exposed by IRobotDAMParams.

Public Fields

	Name	Description
	Analysis (see page 1606)	.
	NotionalLoads (see page 1607)	.
	ReducedStiffness (see page 1607)	.

Public Methods

	Name	Description
	GetNLPDParams (see page 1607)	
	SetNLPDParams (see page 1608)	
	Update (see page 1608)	

(c) 2014 Autodesk Inc. All rights reserved

XXI.7.2 IRobotDAMParams Fields

The fields of the IRobotDAMParams class are listed here.

Public Fields

	Name	Description
	Analysis (see page 1606)	.
	NotionalLoads (see page 1607)	.
	ReducedStiffness (see page 1607)	.

XXI.7.2.1 Analysis

C++

```
HRESULT get_Analysis(IRobotDAMAnalysisType*);
HRESULT put_Analysis(IRobotDAMAnalysisType);
```

C#

```
public IRobotDAMAnalysisType Analysis { get; set; }
```

Visual Basic

```
Public Analysis As IRobotDAMAnalysisType
```

Description

.

Version

Available since version 15.

XXI.7.2.2 NotionalLoads

C++

```
HRESULT get_NotionalLoads(IRobotDAMNotionalLoads**);
```

C#

```
public IRobotDAMNotionalLoads NotionalLoads { get; }
```

Visual Basic

```
Public ReadOnly NotionalLoads As IRobotDAMNotionalLoads
```

Description

.

Version

Available since version 15.

XXI.7.2.3 ReducedStiffness

C++

```
HRESULT get_ReducedStiffness(IRobotDAMReducedStiffness**);
```

C#

```
public IRobotDAMReducedStiffness ReducedStiffness { get; }
```

Visual Basic

```
Public ReadOnly ReducedStiffness As IRobotDAMReducedStiffness
```

(c) 2014 Autodesk Inc. All rights reserved

Description

.

Version

Available since version 15.

XXI.7.3 IRobotDAMParams Methods

The methods of the IRobotDAMParams class are listed here.

Public Methods

	Name	Description
	GetNLPDParams (see page 1607)	
	SetNLPDParams (see page 1608)	
	Update (see page 1608)	

XXI.7.3.1 GetNLPDParams

C++

```
HRESULT GetNLPDParams(IRobotNonlinearAnalysisParams** ret);
```

C#

```
public IRobotNonlinearAnalysisParams* GetNLPDParams();
```

Visual Basic

```
Public Function GetNLPDParams() As IRobotNonlinearAnalysisParams*
```

Version

Available since version 15.

XXI.7.3.2 SetNLPDParams

C++

```
HRESULT SetNLPDParams(IRobotNonlinearAnalysisParams* ret);
```

C#

```
public IRobotNonlinearAnalysisParams SetNLPDParams();
```

Visual Basic

```
Public Function SetNLPDParams() As IRobotNonlinearAnalysisParams
```

Version

Available since version 15.

XXI.7.3.3 Update

C++

```
HRESULT Update();
```

C#

```
public void Update();
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Sub Update()
```

Version

Available since version 15.

Global data types

Enumerations

	Name	Description
	IRobotWindowState (see page 1653)	Available window states.
	IRobotComponentType (see page 1656)	Available component types that can be created by means of the component factory.
	IRobotDegreeOfFreedom (see page 1657)	Available degrees of freedom.

Interfaces

	Name	Description
	IRobotValuesArray (see page 1636)	Interface representing a one-dimensional table of real numbers.
	IRobotNamesArray (see page 1638)	The interface describing a one-dimensional table of names represented by character strings.
	IRobotNumbersArray (see page 1641)	The interface representing a table of integers. Table elements are indexed from 1 to Count (see page 1641). .
	IRobotObjectsArray (see page 1643)	The interface providing access to the table of objects .
	IRobotNumbersCollection (see page 1645)	Collection of numbers (integers).
	IRobotCodeRegistrar (see page 1646)	In order to simplify the process of registering external codes extending the Robot program, RobotCodeRegistrar has been defined.
	IRobotWindow (see page 1650)	General interface describing a window in the Robot program. It describes both the main application window and windows which comprise individual views. .
	IRobotProtectionInfo (see page 1653)	Object provides access to information about protection of the Robot program and its extensions.
	IRobotComponentFactory (see page 1654)	Auxiliary component factory. It should be applied instead of the standard New operator for creation of "free" components in case when the application is to cooperate with both the Robot program and Kernel, depending on the user selection. .
	IRobotPointsArray (see page 1658)	Table of points in the 3D space. .
	IRobotSerializable (see page 1660)	Interface of the object whose data can be saved in/read from the stream.
	IRobotNumbersDictionary (see page 1662)	Dictionary of pairs (integer, integer).

(c) 2014 Autodesk Inc. All rights reserved

Geometric data types

Enumerations

	Name	Description
	IRobotGeoCoordinateAxis (see page 1614)	Coordinate axes.
	IRobotGeoCoordinateSystem (see page 1615)	Types of coordinate systems.
	IRobotGeoSegmentType (see page 1615)	Types of segments that may compose complex geometrical objects.
	IRobotGeoObjectType (see page 1616)	Types of geometrical objects.
	IRobotGeoArcDefinitionMethod (see page 1634)	Available methods of arc definition.
	IRobotGeoCoordinateAxisSense (see page 1635)	Senses of the coordinate system axes.
	IRobotGeoPlane (see page 1635)	

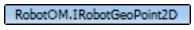
Interfaces

	Name	Description
	IRobotGeoPoint2D (see page 1610)	The interface describing a point on a plane.
	IRobotGeoPoint3D (see page 1612)	The interface describing a point in space.
	IRobotGeoCurveDiv (see page 1616)	The data type describing object discretization.
	IRobotGeoSegment (see page 1618)	Auxiliary interface for the definition of polylines and contours whose fragments may take the form of arcs or roundings.
	IRobotGeoSegmentLine (see page 1619)	Auxiliary interface for defining polylines and contours - a segment with or without a rounding.
	IRobotGeoSegmentArc (see page 1621)	Auxiliary interface for polyline and contour definition - arc.
	IRobotGeoObject (see page 1623)	The basic interface describing a geometrical object .
	IRobotGeoPolyline (see page 1624)	Interface describing a polyline.
	IRobotGeoSegmentCollection (see page 1626)	Collection of segments of geometrical objects.
	IRobotGeoContour (see page 1626)	Interface describing a contour. .
	IRobotGeoArc (see page 1628)	Interface describing an arc.
	IRobotGeoCircle (see page 1630)	Interface describing a circle.
	IRobotGeoPoint3DCollection (see page 1631)	Collection of points.
	IRobotGeoLayer (see page 1632)	Definition of a layer as a space sector. Layer is defined by specifying a cutting plane (by means of three points: P1 (see page 1633), P2 (see page 1633) and P3 (see page 1634)) and the fourth PDir (see page 1634) point determining a half-space direction. Additionally, layer thickness may be determined by specifying the Thickness (see page 1634) value. If the layer thickness is not specified, the layer describes the entire half-space.

(c) 2014 Autodesk Inc. All rights reserved

I.1 IRobotGeoPoint2D

Class Hierarchy



C++

```
interface IRobotGeoPoint2D : IDispatch;
```

C#

```
public interface IRobotGeoPoint2D;
```

Visual Basic

```
Public Interface IRobotGeoPoint2D
```

Description

The interface describing a point on a plane.

I.1.1 IRobotGeoPoint2D Members

The following tables list the members exposed by IRobotGeoPoint2D.

Public Fields

	Name	Description
	X (see page 1611)	X coordinate.
	Y (see page 1611)	Y coordinate.

Public Methods

	Name	Description
	Get (see page 1612)	The function takes all point coordinates.
	Set (see page 1612)	The function sets all point coordinates.

I.1.2 IRobotGeoPoint2D Fields

The fields of the IRobotGeoPoint2D class are listed here.

Public Fields

	Name	Description
	X (see page 1611)	X coordinate.
	Y (see page 1611)	Y coordinate.

I.1.2.1 X

C++

```
HRESULT get_X(double*);
HRESULT put_X(double);
```

C#

```
public double X { get; set; }
```

Visual Basic

```
Public X As double
```

Description

X coordinate.

(c) 2014 Autodesk Inc. All rights reserved

I.1.2.2 Y

C++

```
HRESULT get_Y(double*);
HRESULT put_Y(double);
```

C#

```
public double Y { get; set; }
```

Visual Basic

```
Public Y As double
```

Description

Y coordinate.

I.1.3 IRobotGeoPoint2D Methods

The methods of the IRobotGeoPoint2D class are listed here.

Public Methods

	Name	Description
	Get (see page 1612)	The function takes all point coordinates.
	Set (see page 1612)	The function sets all point coordinates.

I.1.3.1 Get

C++

```
HRESULT Get(double* _x, double* _y);
```

C#

```
public void Get(double* _x, double* _y);
```

Visual Basic

```
Public Sub Get(ByRef _x As double*, ByRef _y As double*)
```

Description

The function takes all point coordinates.

I.1.3.2 Set

C++

```
HRESULT Set(double _x, double _y);
```

C#

```
public void Set(double _x, double _y);
```

Visual Basic

```
Public Sub Set(_x As double, _y As double)
```

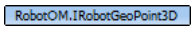
Description

The function sets all point coordinates.

(c) 2014 Autodesk Inc. All rights reserved

I.2 IRobotGeoPoint3D

Class Hierarchy



C++

```
interface IRobotGeoPoint3D : IDispatch;
```

C#

```
public interface IRobotGeoPoint3D;
```

Visual Basic

```
Public Interface IRobotGeoPoint3D
```

Description

The interface describing a point in space.

I.2.1 IRobotGeoPoint3D Members

The following tables list the members exposed by IRobotGeoPoint3D.

Public Fields

	Name	Description
	X (see page 1613)	X coordinate .
	Y (see page 1613)	Y coordinate.
	Z (see page 1613)	Z coordinate.

Public Methods

	Name	Description
	Get (see page 1614)	The function gives back all point coordinates.
	Set (see page 1614)	The function sets all point coordinates. .

I.2.2 IRobotGeoPoint3D Fields

The fields of the IRobotGeoPoint3D class are listed here.

Public Fields

	Name	Description
	X (see page 1613)	X coordinate .
	Y (see page 1613)	Y coordinate.
	Z (see page 1613)	Z coordinate.

I.2.2.1 X

C++

```
HRESULT get_X(double*);
HRESULT put_X(double);
```

C#

```
public double X { get; set; }
```

Visual Basic

```
Public X As double
```

(c) 2014 Autodesk Inc. All rights reserved

Description

X coordinate .

I.2.2.2 Y

C++

```
HRESULT get_Y(double*);
HRESULT put_Y(double);
```

C#

```
public double Y { get; set; }
```

Visual Basic

```
Public Y As double
```

Description

Y coordinate.

I.2.2.3 Z

C++

```
HRESULT get_Z(double*);
HRESULT put_Z(double);
```

C#

```
public double Z { get; set; }
```

Visual Basic

```
Public Z As double
```

Description

Z coordinate.

I.2.3 IRobotGeoPoint3D Methods

The methods of the IRobotGeoPoint3D class are listed here.

Public Methods

	Name	Description
	Get (see page 1614)	The function gives back all point coordinates.
	Set (see page 1614)	The function sets all point coordinates. .

I.2.3.1 Get

C++

```
HRESULT Get(double* _x, double* _y, double* _z);
```

C#

```
public void Get(double* _x, double* _y, double* _z);
```

Visual Basic

```
Public Sub Get(ByRef _x As double*, ByRef _y As double*, ByRef _z As double*)
```

Description

The function gives back all point coordinates.

(c) 2014 Autodesk Inc. All rights reserved

I.2.3.2 Set

C++

```
HRESULT Set(double _x, double _y, double _z);
```

C#

```
public void Set(double _x, double _y, double _z);
```

Visual Basic

```
Public Sub Set(_x As double, _y As double, _z As double)
```

Description

The function sets all point coordinates. .

I.3 IRobotGeoCoordinateAxis

C++

```
enum IRobotGeoCoordinateAxis;
```

C#

```
public enum IRobotGeoCoordinateAxis;
```

Visual Basic

```
Public Enum IRobotGeoCoordinateAxis
```

Members

Members	Description
I_GCA_OX = 1	Axis OX.
I_GCA_OY = 2	Axis OY.
I_GCA_OZ = 3	Axis OZ.
I_GCA_NONE = 0	No coordinate axis is indicated. Available since version 7.

Description

Coordinate axes.

I.4 IRobotGeoCoordinateSystem

C++

```
enum IRobotGeoCoordinateSystem;
```

C#

```
public enum IRobotGeoCoordinateSystem;
```

Visual Basic

```
Public Enum IRobotGeoCoordinateSystem
```

Members

Members	Description
I_GCS_GLOBAL = 0	In global system.
I_GCS_LOCAL_AFTER_OFFSET = 1	Local coordinate system of a bar in the position considering translation.

(c) 2014 Autodesk Inc. All rights reserved

I_GCS_LOCAL_ORIGINAL = 2	Local coordinate system of a bar.

Description

Types of coordinate systems.

I.5 IRobotGeoSegmentType

C++

```
enum IRobotGeoSegmentType;
```

C#

```
public enum IRobotGeoSegmentType;
```

Visual Basic

```
Public Enum IRobotGeoSegmentType
```

Members

Members	Description
I_GST_NONE = 0	Undefined.
I_GST_LINE = 1	Line (with a rounding).
I_GST_ARC = 2	Arc.

Description

Types of segments that may compose complex geometrical objects.

I.6 IRobotGeoObjectType

C++

```
enum IRobotGeoObjectType;
```

C#

```
public enum IRobotGeoObjectType;
```

Visual Basic

```
Public Enum IRobotGeoObjectType
```

Members

Members	Description
I_GOT_NONE = 0	Undefined.
I_GOT_POLYLINE = 1	Polyline.
I_GOT_CONTOUR = 2	Contour.
I_GOT_ARC = 4	Arc.
I_GOT_CIRCLE = 7	Circle.
I_GOT_INTERSECTION = 100	Overlap of objects .

Description

Types of geometrical objects.

I.7 IRobotGeoCurveDiv

Class Hierarchy

(c) 2014 Autodesk Inc. All rights reserved

C++

```
interface IRobotGeoCurveDiv : IDispatch;
```

C#

```
public interface IRobotGeoCurveDiv;
```

Visual Basic

```
Public Interface IRobotGeoCurveDiv
```

Description

The data type describing object discretization.

I.7.1 IRobotGeoCurveDiv Members

The following tables list the members exposed by IRobotGeoCurveDiv.

Public Fields

	Name	Description
	IsAnalytical (see page 1617)	Arc defined analytically (arc discretization options are not available).
	Length (see page 1617)	Segment length.
	Mode (see page 1617)	0 - fixed number of divisions 1 - number of divisions per 360 degrees 2 - segment length.
	N (see page 1618)	Number of divisions.

I.7.2 IRobotGeoCurveDiv Fields

The fields of the IRobotGeoCurveDiv class are listed here.

Public Fields

	Name	Description
	IsAnalytical (see page 1617)	Arc defined analytically (arc discretization options are not available).
	Length (see page 1617)	Segment length.
	Mode (see page 1617)	0 - fixed number of divisions 1 - number of divisions per 360 degrees 2 - segment length.
	N (see page 1618)	Number of divisions.

I.7.2.1 IsAnalytical

C++

```
HRESULT get_IsAnalytical(VARIANT_BOOL*);
HRESULT put_IsAnalytical(VARIANT_BOOL);
```

C#

```
public bool IsAnalytical { get; set; }
```

Visual Basic

```
Public IsAnalytical As Boolean
```

Description

Arc defined analytically (arc discretization options are not available).

Version

Available since version 8.2.

(c) 2014 Autodesk Inc. All rights reserved

I.7.2.2 Length

C++

```
HRESULT get_Length(double*);
HRESULT put_Length(double);
```

C#

```
public double Length { get; set; }
```

Visual Basic

```
Public Length As double
```

Description

Segment length.

Version

Available since version 8.2.

I.7.2.3 Mode

C++

```
HRESULT get_Mode(long*);
HRESULT put_Mode(long);
```

C#

```
public long Mode { get; set; }
```

Visual Basic

```
Public Mode As long
```

Description

0 - fixed number of divisions 1 - number of divisions per 360 degrees 2 - segment length.

I.7.2.4 N

C++

```
HRESULT get_N(long*);
HRESULT put_N(long);
```

C#

```
public long N { get; set; }
```

Visual Basic

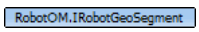
```
Public N As long
```

Description

Number of divisions.

I.8 IRobotGeoSegment

Class Hierarchy



C++

```
interface IRobotGeoSegment : IDispatch;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public interface IRobotGeoSegment;
```

Visual Basic

```
Public Interface IRobotGeoSegment
```

Description

Auxiliary interface for the definition of polylines and contours whose fragments may take the form of arcs or roundings.

I.8.1 IRobotGeoSegment Members

The following tables list the members exposed by IRobotGeoSegment.

Public Fields

	Name	Description
	P1 (see page 1619)	Initial point of a segment (end point is taken from the next segment).
	Type (see page 1619)	Segment type.

I.8.2 IRobotGeoSegment Fields

The fields of the IRobotGeoSegment class are listed here.

Public Fields

	Name	Description
	P1 (see page 1619)	Initial point of a segment (end point is taken from the next segment).
	Type (see page 1619)	Segment type.

I.8.2.1 P1

C++

```
HRESULT get_P1(IRobotGeoPoint3D**);
HRESULT put_P1(IRobotGeoPoint3D*);
```

C#

```
public IRobotGeoPoint3D P1 { get; set; }
```

Visual Basic

```
Public P1 As IRobotGeoPoint3D
```

Description

Initial point of a segment (end point is taken from the next segment).

I.8.2.2 Type

C++

```
HRESULT get_Type(IRobotGeoSegmentType*);
```

C#

```
public IRobotGeoSegmentType Type { get; }
```

Visual Basic

```
Public ReadOnly Type As IRobotGeoSegmentType
```

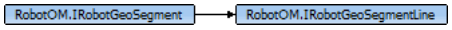
Description

Segment type.

(c) 2014 Autodesk Inc. All rights reserved

I.9 IRobotGeoSegmentLine

Class Hierarchy



C++

```
interface IRobotGeoSegmentLine : IRobotGeoSegment;
```

C#

```
public interface IRobotGeoSegmentLine : IRobotGeoSegment;
```

Visual Basic

```
Public Interface IRobotGeoSegmentLine
```

Description

Auxiliary interface for defining polylines and contours - a segment with or without a rounding.

I.9.1 IRobotGeoSegmentLine Members

The following tables list the members exposed by IRobotGeoSegmentLine.

Public Fields

	Name	Description
	Div (see page 1620)	Rounding discretization.
	NDiv (see page 1620)	Number of divisions - discretization of the segment.
	P1 (see page 1619)	Initial point of a segment (end point is taken from the next segment).
	Radius (see page 1620)	Rounding radius.
	Round (see page 1621)	Rounded or not.
	Type (see page 1619)	Segment type.

I.9.2 IRobotGeoSegmentLine Fields

The fields of the IRobotGeoSegmentLine class are listed here.

Public Fields

	Name	Description
	Div (see page 1620)	Rounding discretization.
	NDiv (see page 1620)	Number of divisions - discretization of the segment.
	Radius (see page 1620)	Rounding radius.
	Round (see page 1621)	Rounded or not.

I.9.2.1 Div

C++

```
HRESULT get_Div(IRobotGeoCurveDiv**);
HRESULT put_Div(IRobotGeoCurveDiv*);
```

C#

```
public IRobotGeoCurveDiv Div { get; set; }
```

Visual Basic

```
Public Div As IRobotGeoCurveDiv
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Rounding discretization.

I.9.2.2 NDiv

C++

```
HRESULT get_NDiv(long*);
HRESULT put_NDiv(long);
```

C#

```
public long NDiv { get; set; }
```

Visual Basic

```
Public NDiv As long
```

Description

Number of divisions - discretization of the segment.

I.9.2.3 Radius

C++

```
HRESULT get_Radius(double*);
HRESULT put_Radius(double);
```

C#

```
public double Radius { get; set; }
```

Visual Basic

```
Public Radius As double
```

Description

Rounding radius.

I.9.2.4 Round

C++

```
HRESULT get_Round(VARIANT_BOOL*);
HRESULT put_Round(VARIANT_BOOL);
```

C#

```
public bool Round { get; set; }
```

Visual Basic

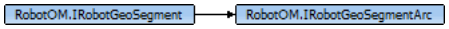
```
Public Round As Boolean
```

Description

Rounded or not.

I.10 IRobotGeoSegmentArc

Class Hierarchy



C++

```
interface IRobotGeoSegmentArc : IRobotGeoSegment;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public interface IRobotGeoSegmentArc : IRobotGeoSegment;
```

Visual Basic

```
Public Interface IRobotGeoSegmentArc
```

Description

Auxiliary interface for polyline and contour definition - arc.

I.10.1 IRobotGeoSegmentArc Members

The following tables list the members exposed by IRobotGeoSegmentArc.

Public Fields

	Name	Description
	Div (see page 1622)	Arc discretization.
	P1 (see page 1619)	Initial point of a segment (end point is taken from the next segment).
	P2 (see page 1622)	Point on an arc.
	Type (see page 1619)	Segment type.

Public Methods

	Name	Description
	Set (see page 1622)	The function stets the initial arc point P1 (see page 1619) and a point along the arc P2 (see page 1622).

I.10.2 IRobotGeoSegmentArc Fields

The fields of the IRobotGeoSegmentArc class are listed here.

Public Fields

	Name	Description
	Div (see page 1622)	Arc discretization.
	P2 (see page 1622)	Point on an arc.

I.10.2.1 Div

C++

```
HRESULT get_Div(IRobotGeoCurveDiv**);
HRESULT put_Div(IRobotGeoCurveDiv*);
```

C#

```
public IRobotGeoCurveDiv Div { get; set; }
```

Visual Basic

```
Public Div As IRobotGeoCurveDiv
```

Description

Arc discretization.

I.10.2.2 P2

C++

```
HRESULT get_P2(IRobotGeoPoint3D**);
HRESULT put_P2(IRobotGeoPoint3D*);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRobotGeoPoint3D P2 { get; set; }
```

Visual Basic

```
Public P2 As IRobotGeoPoint3D
```

Description

Point on an arc.

I.10.3 IRobotGeoSegmentArc Methods

The methods of the IRobotGeoSegmentArc class are listed here.

Public Methods

	Name	Description
	Set (see page 1622)	The function stets the initial arc point P1 (see page 1619) and a point along the arc P2 (see page 1622).

I.10.3.1 Set

C++

```
HRESULT Set(IRobotGeoPoint3D* _p1, IRobotGeoPoint3D* _p2);
```

C#

```
public void Set(IRobotGeoPoint3D _p1, IRobotGeoPoint3D _p2);
```

Visual Basic

```
Public Sub Set(ByRef _p1 As IRobotGeoPoint3D, ByRef _p2 As IRobotGeoPoint3D)
```

Description

The function stets the initial arc point P1 (see page 1619) and a point along the arc P2 (see page 1622).

I.11 IRobotGeoObject

Class Hierarchy

RobotOM.IRobotGeoObject

C++

```
interface IRobotGeoObject : IDispatch;
```

C#

```
public interface IRobotGeoObject;
```

Visual Basic

```
Public Interface IRobotGeoObject
```

Description

The basic interface describing a geometrical object .

I.11.1 IRobotGeoObject Members

The following tables list the members exposed by IRobotGeoObject.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	Type (see page 1623)	Object type.

Public Methods

	Name	Description
	Initialize (see page 1624)	The function initializes internal data on the basis of the data defining the object. .

I.11.2 IRobotGeoObject Fields

The fields of the IRobotGeoObject class are listed here.

Public Fields

	Name	Description
	Type (see page 1623)	Object type.

I.11.2.1 Type

C++

```
HRESULT get_Type(IRobotGeoObjectType*);
```

C#

```
public IRobotGeoObjectType Type { get; }
```

Visual Basic

```
Public ReadOnly Type As IRobotGeoObjectType
```

Description

Object type.

I.11.3 IRobotGeoObject Methods

The methods of the IRobotGeoObject class are listed here.

Public Methods

	Name	Description
	Initialize (see page 1624)	The function initializes internal data on the basis of the data defining the object. .

I.11.3.1 Initialize

C++

```
HRESULT Initialize();
```

C#

```
public void Initialize();
```

Visual Basic

```
Public Sub Initialize()
```

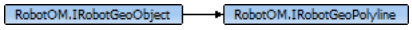
Description

The function initializes internal data on the basis of the data defining the object. .

(c) 2014 Autodesk Inc. All rights reserved

I.12 IRobotGeoPolyline

Class Hierarchy



C++

```
interface IRobotGeoPolyline : IRobotGeoObject;
```

C#

```
public interface IRobotGeoPolyline : IRobotGeoObject;
```

Visual Basic

```
Public Interface IRobotGeoPolyline
```

Description

Interface describing a polyline.

I.12.1 IRobotGeoPolyline Members

The following tables list the members exposed by IRobotGeoPolyline.

Public Fields

	Name	Description
	Segments (see page 1625)	Collection of segments describing a polyline.
	Type (see page 1623)	Object type.

Public Methods

	Name	Description
	Add (see page 1625)	The function adds a new segment to the polyline. .
	Clear (see page 1626)	The function clears a collection of segments of a polyline.
	Initialize (see page 1624)	The function initializes internal data on the basis of the data defining the object. .

I.12.2 IRobotGeoPolyline Fields

The fields of the IRobotGeoPolyline class are listed here.

Public Fields

	Name	Description
	Segments (see page 1625)	Collection of segments describing a polyline.

I.12.2.1 Segments

C++

```
HRESULT get_Segments(IRobotGeoSegmentCollection**);
```

C#

```
public IRobotGeoSegmentCollection Segments { get; }
```

Visual Basic

```
Public ReadOnly Segments As IRobotGeoSegmentCollection
```

Description

Collection of segments describing a polyline.

(c) 2014 Autodesk Inc. All rights reserved

I.12.3 IRobotGeoPolyline Methods

The methods of the IRobotGeoPolyline class are listed here.

Public Methods

	Name	Description
	Add (see page 1625)	The function adds a new segment to the polyline. .
	Clear (see page 1626)	The function clears a collection of segments of a polyline.

I.12.3.1 Add

C++

```
HRESULT Add(IRobotGeoSegment* _segment);
```

C#

```
public void Add(IRobotGeoSegment _segment);
```

Visual Basic

```
Public Sub Add(ByRef _segment As IRobotGeoSegment)
```

Description

The function adds a new segment to the polyline. .

I.12.3.2 Clear

C++

```
HRESULT Clear();
```

C#

```
public void Clear();
```

Visual Basic

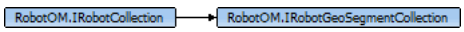
```
Public Sub Clear()
```

Description

The function clears a collection of segments of a polyline.

I.13 IRobotGeoSegmentCollection

Class Hierarchy



C++

```
interface IRobotGeoSegmentCollection : IRobotCollection;
```

C#

```
public interface IRobotGeoSegmentCollection : IRobotCollection;
```

Visual Basic

```
Public Interface IRobotGeoSegmentCollection
```

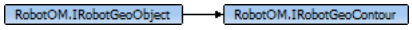
Description

Collection of segments of geometrical objects.

(c) 2014 Autodesk Inc. All rights reserved

I.14 IRobotGeoContour

Class Hierarchy



C++

```
interface IRobotGeoContour : IRobotGeoObject;
```

C#

```
public interface IRobotGeoContour : IRobotGeoObject;
```

Visual Basic

```
Public Interface IRobotGeoContour
```

Description

Interface describing a contour. .

I.14.1 IRobotGeoContour Members

The following tables list the members exposed by IRobotGeoContour.

Public Fields

	Name	Description
	Segments (see page 1627)	Collection of segments describing a contour (the contour is closed automatically; one does not have to define the last segment).
	Type (see page 1623)	Object type.

Public Methods

	Name	Description
	Add (see page 1627)	The function adds a new segment. The segment may be a line or an arc.
	Clear (see page 1628)	The function clears a segment collection .
	Initialize (see page 1624)	The function initializes internal data on the basis of the data defining the object. .

I.14.2 IRobotGeoContour Fields

The fields of the IRobotGeoContour class are listed here.

Public Fields

	Name	Description
	Segments (see page 1627)	Collection of segments describing a contour (the contour is closed automatically; one does not have to define the last segment).

I.14.2.1 Segments

C++

```
HRESULT get_Segments(IRobotGeoSegmentCollection**);
```

C#

```
public IRobotGeoSegmentCollection Segments { get; }
```

Visual Basic

```
Public ReadOnly Segments As IRobotGeoSegmentCollection
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Collection of segments describing a contour (the contour is closed automatically; one does not have to define the last segment).

I.14.3 IRobotGeoContour Methods

The methods of the IRobotGeoContour class are listed here.

Public Methods

	Name	Description
	Add (see page 1627)	The function adds a new segment. The segment may be a line or an arc.
	Clear (see page 1628)	The function clears a segment collection .

I.14.3.1 Add

C++

```
HRESULT Add(IRobotGeoSegment* _segment);
```

C#

```
public void Add(IRobotGeoSegment _segment);
```

Visual Basic

```
Public Sub Add(ByRef _segment As IRobotGeoSegment)
```

Description

The function adds a new segment. The segment may be a line or an arc.

I.14.3.2 Clear

C++

```
HRESULT Clear();
```

C#

```
public void Clear();
```

Visual Basic

```
Public Sub Clear()
```

Description

The function clears a segment collection .

I.15 IRobotGeoArc

Class Hierarchy

C++

```
interface IRobotGeoArc : IRobotGeoObject;
```

C#

```
public interface IRobotGeoArc : IRobotGeoObject;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Interface IRobotGeoArc
```

Description

Interface describing an arc.

I.15.1 IRobotGeoArc Members

The following tables list the members exposed by IRobotGeoArc.

Public Fields

	Name	Description
	P1 (see page 1629)	Initial point of the arc.
	P2 (see page 1629)	Final point of the arc.
	P3 (see page 1629)	A point along the arc.
	Type (see page 1623)	Object type.

Public Methods

	Name	Description
	Initialize (see page 1624)	The function initializes internal data on the basis of the data defining the object. .

I.15.2 IRobotGeoArc Fields

The fields of the IRobotGeoArc class are listed here.

Public Fields

	Name	Description
	P1 (see page 1629)	Initial point of the arc.
	P2 (see page 1629)	Final point of the arc.
	P3 (see page 1629)	A point along the arc.

I.15.2.1 P1

C++

```
HRESULT get_P1(IRobotGeoPoint3D**);
HRESULT put_P1(IRobotGeoPoint3D*);
```

C#

```
public IRobotGeoPoint3D P1 { get; set; }
```

Visual Basic

```
Public P1 As IRobotGeoPoint3D
```

Description

Initial point of the arc.

I.15.2.2 P2

C++

```
HRESULT get_P2(IRobotGeoPoint3D**);
HRESULT put_P2(IRobotGeoPoint3D*);
```

C#

```
public IRobotGeoPoint3D P2 { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public P2 As IRobotGeoPoint3D
```

Description

Final point of the arc.

I.15.2.3 P3

C++

```
HRESULT get_P3(IRobotGeoPoint3D**);
HRESULT put_P3(IRobotGeoPoint3D*);
```

C#

```
public IRobotGeoPoint3D P3 { get; set; }
```

Visual Basic

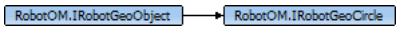
```
Public P3 As IRobotGeoPoint3D
```

Description

A point along the arc.

I.16 IRobotGeoCircle

Class Hierarchy



C++

```
interface IRobotGeoCircle : IRobotGeoObject;
```

C#

```
public interface IRobotGeoCircle : IRobotGeoObject;
```

Visual Basic

```
Public Interface IRobotGeoCircle
```

Description

Interface describing a circle.

I.16.1 IRobotGeoCircle Members

The following tables list the members exposed by IRobotGeoCircle.

Public Fields

	Name	Description
	P1 (see page 1630)	A point belonging to a circle.
	P2 (see page 1631)	A point belonging to a circle.
	P3 (see page 1631)	A point belonging to a circle.
	PC (see page 1631)	The circle center.
	Type (see page 1623)	Object type.

Public Methods

	Name	Description
	Initialize (see page 1624)	The function initializes internal data on the basis of the data defining the object. .

(c) 2014 Autodesk Inc. All rights reserved

I.16.2 IRobotGeoCircle Fields

The fields of the IRobotGeoCircle class are listed here.

Public Fields

	Name	Description
	P1 (see page 1630)	A point belonging to a circle.
	P2 (see page 1631)	A point belonging to a circle.
	P3 (see page 1631)	A point belonging to a circle.
	PC (see page 1631)	The circle center.

I.16.2.1 P1

C++

```
HRESULT get_P1(IRobotGeoPoint3D**);
HRESULT put_P1(IRobotGeoPoint3D*);
```

C#

```
public IRobotGeoPoint3D P1 { get; set; }
```

Visual Basic

```
Public P1 As IRobotGeoPoint3D
```

Description

A point belonging to a circle.

I.16.2.2 P2

C++

```
HRESULT get_P2(IRobotGeoPoint3D**);
HRESULT put_P2(IRobotGeoPoint3D*);
```

C#

```
public IRobotGeoPoint3D P2 { get; set; }
```

Visual Basic

```
Public P2 As IRobotGeoPoint3D
```

Description

A point belonging to a circle.

I.16.2.3 P3

C++

```
HRESULT get_P3(IRobotGeoPoint3D**);
HRESULT put_P3(IRobotGeoPoint3D*);
```

C#

```
public IRobotGeoPoint3D P3 { get; set; }
```

Visual Basic

```
Public P3 As IRobotGeoPoint3D
```

Description

A point belonging to a circle.

(c) 2014 Autodesk Inc. All rights reserved

I.16.2.4 PC

C++

```
HRESULT get_PC(IRobotGeoPoint3D**);
```

C#

```
public IRobotGeoPoint3D PC { get; }
```

Visual Basic

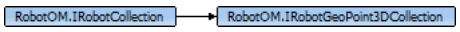
```
Public ReadOnly PC As IRobotGeoPoint3D
```

Description

The circle center.

I.17 IRobotGeoPoint3DCollection

Class Hierarchy



C++

```
interface IRobotGeoPoint3DCollection : IRobotCollection;
```

C#

```
public interface IRobotGeoPoint3DCollection : IRobotCollection;
```

Visual Basic

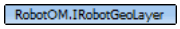
```
Public Interface IRobotGeoPoint3DCollection
```

Description

Collection of points.

I.18 IRobotGeoLayer

Class Hierarchy



C++

```
interface IRobotGeoLayer : IDispatch;
```

C#

```
public interface IRobotGeoLayer;
```

Visual Basic

```
Public Interface IRobotGeoLayer
```

Description

Definition of a layer as a space sector. Layer is defined by specifying a cutting plane (by means of three points: P1 (see page 1633), P2 (see page 1633) and P3 (see page 1634)) and the fourth PDir (see page 1634) point determining a half-space direction. Additionally, layer thickness may be determined by specifying the Thickness (see page 1634) value. If the layer thickness is not specified, the layer describes the entire half-space.

(c) 2014 Autodesk Inc. All rights reserved

I.18.1 IRobotGeoLayer Members

The following tables list the members exposed by IRobotGeoLayer.

Public Fields

	Name	Description
	IsThickDefined (see page 1633)	Auxiliary flag indicating if the layer thickness has been defined Available since version 2.0.
	P1 (see page 1633)	First point (P1) defining the cutting plane Available since version 2.0.
	P2 (see page 1633)	Second point (P2) defining the cutting plane Available since version 2.0.
	P3 (see page 1634)	Third point (P3) defining the cutting plane Available since version 2.0.
	PDir (see page 1634)	Point defining direction (i.e. it indicates one of the half-spaces determined by the cutting plane) Available since version 2.0.
	Thickness (see page 1634)	Layer thickness Available since version 2.0.

I.18.2 IRobotGeoLayer Fields

The fields of the IRobotGeoLayer class are listed here.

Public Fields

	Name	Description
	IsThickDefined (see page 1633)	Auxiliary flag indicating if the layer thickness has been defined Available since version 2.0.
	P1 (see page 1633)	First point (P1) defining the cutting plane Available since version 2.0.
	P2 (see page 1633)	Second point (P2) defining the cutting plane Available since version 2.0.
	P3 (see page 1634)	Third point (P3) defining the cutting plane Available since version 2.0.
	PDir (see page 1634)	Point defining direction (i.e. it indicates one of the half-spaces determined by the cutting plane) Available since version 2.0.
	Thickness (see page 1634)	Layer thickness Available since version 2.0.

I.18.2.1 IsThickDefined

C++

```
HRESULT get_IsThickDefined(VARIANT_BOOL*);
HRESULT put_IsThickDefined(VARIANT_BOOL);
```

C#

```
public bool IsThickDefined { get; set; }
```

Visual Basic

```
Public IsThickDefined As Boolean
```

Description

Auxiliary flag indicating if the layer thickness has been defined Available since version 2.0.

I.18.2.2 P1

C++

```
HRESULT get_P1(IRobotGeoPoint3D**);
```

C#

```
public IRobotGeoPoint3D P1 { get; }
```

Visual Basic

```
Public ReadOnly P1 As IRobotGeoPoint3D
```

(c) 2014 Autodesk Inc. All rights reserved

Description

First point (P1) defining the cutting plane Available since version 2.0.

I.18.2.3 P2

C++

```
HRESULT get_P2(IRobotGeoPoint3D**);
```

C#

```
public IRobotGeoPoint3D P2 { get; }
```

Visual Basic

```
Public ReadOnly P2 As IRobotGeoPoint3D
```

Description

Second point (P2) defining the cutting plane Available since version 2.0.

I.18.2.4 P3

C++

```
HRESULT get_P3(IRobotGeoPoint3D**);
```

C#

```
public IRobotGeoPoint3D P3 { get; }
```

Visual Basic

```
Public ReadOnly P3 As IRobotGeoPoint3D
```

Description

Third point (P3) defining the cutting plane Available since version 2.0.

I.18.2.5 PDir

C++

```
HRESULT get_PDir(IRobotGeoPoint3D**);
```

C#

```
public IRobotGeoPoint3D PDir { get; }
```

Visual Basic

```
Public ReadOnly PDir As IRobotGeoPoint3D
```

Description

Point defining direction (i.e. it indicates one of the half-spaces determined by the cutting plane) Available since version 2.0.

I.18.2.6 Thickness

C++

```
HRESULT get_Thickness(double*);
HRESULT put_Thickness(double);
```

C#

```
public double Thickness { get; set; }
```

Visual Basic

```
Public Thickness As double
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Layer thickness Available since version 2.0.

I.19 IRobotGeoArcDefinitionMethod

C++

```
enum IRobotGeoArcDefinitionMethod;
```

C#

```
public enum IRobotGeoArcDefinitionMethod;
```

Visual Basic

```
Public Enum IRobotGeoArcDefinitionMethod
```

Members

Members	Description
I_GADM_CENTER_BEGIN_END = 0	Available since version 3.
I_GADM_BEGIN_MIDDLE_END = 1	Available since version 3.

Description

Available methods of arc definition.

Version

Available since version 3.

I.20 IRobotGeoCoordinateAxisSense

C++

```
enum IRobotGeoCoordinateAxisSense;
```

C#

```
public enum IRobotGeoCoordinateAxisSense;
```

Visual Basic

```
Public Enum IRobotGeoCoordinateAxisSense
```

Members

Members	Description
I_GCAS_OX_PLUS = 1	Available since version 4.5.
I_GCAS_OY_PLUS = 2	Available since version 4.5.
I_GCAS_OZ_PLUS = 3	Available since version 4.5.
I_GCAS_OX_MINUS = -1	Available since version 4.5.
I_GCAS_OY_MINUS = -2	Available since version 4.5.
I_GCAS_OZ_MINUS = -3	Available since version 4.5.

Description

Senses of the coordinate system axes.

Version

Available since version 4.5.

(c) 2014 Autodesk Inc. All rights reserved

I.21 IRobotGeoPlane

C++

```
enum IRobotGeoPlane;
```

C#

```
public enum IRobotGeoPlane;
```

Visual Basic

```
Public Enum IRobotGeoPlane
```

Members

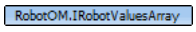
Members	Description
I_GP_XY = 0	Available since version 11.
I_GP_XZ = 1	Available since version 11.
I_GP_YZ = 2	Available since version 11.

Version

Available since version 11.

II IRobotValuesArray

Class Hierarchy



C++

```
interface IRobotValuesArray : IDispatch;
```

C#

```
public interface IRobotValuesArray;
```

Visual Basic

```
Public Interface IRobotValuesArray
```

Description

Interface representing a one-dimensional table of real numbers.

II.1 IRobotValuesArray Members

The following tables list the members exposed by IRobotValuesArray.

Public Fields

	Name	Description
	Count (see page 1637)	Number of elements in the table (at the same time, it is the index of the last element in the table).

Public Methods

	Name	Description
	Get (see page 1637)	The function returns the value of table element with the given index. Table elements are indexed from 1 to Count (see page 1637).

(c) 2014 Autodesk Inc. All rights reserved

	Set (see page 1637)	The function sets the value of table element with the given index. Table elements are indexed from 1 to Count (see page 1637).
	SetSize (see page 1638)	The function sets the table size. Table elements are indexed from 1 to table size.

II.2 IRobotValuesArray Fields

The fields of the IRobotValuesArray class are listed here.

Public Fields

	Name	Description
	Count (see page 1637)	Number of elements in the table (at the same time, it is the index of the last element in the table).

II.2.1 Count

C++

```
HRESULT get_Count(long*);
```

C#

```
public long Count { get; }
```

Visual Basic

```
Public ReadOnly Count As long
```

Description

Number of elements in the table (at the same time, it is the index of the last element in the table).

II.3 IRobotValuesArray Methods

The methods of the IRobotValuesArray class are listed here.

Public Methods

	Name	Description
	Get (see page 1637)	The function returns the value of table element with the given index. Table elements are indexed from 1 to Count (see page 1637).
	Set (see page 1637)	The function sets the value of table element with the given index. Table elements are indexed from 1 to Count (see page 1637).
	SetSize (see page 1638)	The function sets the table size. Table elements are indexed from 1 to table size.

II.3.1 Get

C++

```
HRESULT Get(long _idx, double* ret);
```

C#

```
public double Get(long _idx);
```

Visual Basic

```
Public Function Get(_idx As long) As double
```

Description

The function returns the value of table element with the given index. Table elements are indexed from 1 to Count (see page 1637).

(c) 2014 Autodesk Inc. All rights reserved

II.3.2 Set

C++

```
HRESULT Set(long _idx, double _val);
```

C#

```
public void Set(long _idx, double _val);
```

Visual Basic

```
Public Sub Set(_idx As long, _val As double)
```

Description

The function sets the value of table element with the given index. Table elements are indexed from 1 to Count (see page 1637).

II.3.3 SetSize

C++

```
HRESULT SetSize(long _count);
```

C#

```
public void SetSize(long _count);
```

Visual Basic

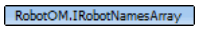
```
Public Sub SetSize(_count As long)
```

Description

The function sets the table size. Table elements are indexed from 1 to table size.

III IRobotNamesArray

Class Hierarchy



C++

```
interface IRobotNamesArray : IDispatch;
```

C#

```
public interface IRobotNamesArray;
```

Visual Basic

```
Public Interface IRobotNamesArray
```

Description

The interface describing a one-dimensional table of names represented by character strings.

III.1 IRobotNamesArray Members

The following tables list the members exposed by IRobotNamesArray.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	Count (see page 1639)	Number of elements in the table (it is also the index of the last element in the table).

Public Methods

	Name	Description
	Find (see page 1639)	Function returns index of the name specified in the table. Table elements are searched beginning with the determined start index. If the specified name is not found, function returns value -1. .
	Get (see page 1640)	The function returns table element with the given index. Table element are indexed from 1 to Count (see page 1639). .
	Set (see page 1640)	The function sets the value of the given table element. The table elements are indexed from 1 to Count (see page 1639).
	SetSize (see page 1640)	The function sets a new table size. Table elements are indexed from 1 to Count (see page 1639). .

III.2 IRobotNamesArray Fields

The fields of the IRobotNamesArray class are listed here.

Public Fields

	Name	Description
	Count (see page 1639)	Number of elements in the table (it is also the index of the last element in the table).

III.2.1 Count

C++

```
HRESULT get_Count(long*);
```

C#

```
public long Count { get; }
```

Visual Basic

```
Public ReadOnly Count As long
```

Description

Number of elements in the table (it is also the index of the last element in the table).

III.3 IRobotNamesArray Methods

The methods of the IRobotNamesArray class are listed here.

Public Methods

	Name	Description
	Find (see page 1639)	Function returns index of the name specified in the table. Table elements are searched beginning with the determined start index. If the specified name is not found, function returns value -1. .
	Get (see page 1640)	The function returns table element with the given index. Table element are indexed from 1 to Count (see page 1639). .
	Set (see page 1640)	The function sets the value of the given table element. The table elements are indexed from 1 to Count (see page 1639).
	SetSize (see page 1640)	The function sets a new table size. Table elements are indexed from 1 to Count (see page 1639). .

(c) 2014 Autodesk Inc. All rights reserved

III.3.1 Find

C++

```
HRESULT Find(BSTR _to_find, long _start_idx, long* ret);
```

C#

```
public long Find(String _to_find, long _start_idx);
```

Visual Basic

```
Public Function Find(_to_find As String, _start_idx As long) As long
```

Description

Function returns index of the name specified in the table. Table elements are searched beginning with the determined start index. If the specified name is not found, function returns value -1. .

Version

Available since version 3.5.

III.3.2 Get

C++

```
HRESULT Get(long _idx, BSTR* ret);
```

C#

```
public String Get(long _idx);
```

Visual Basic

```
Public Function Get(_idx As long) As String
```

Description

The function returns table element with the given index. Table element are indexed from 1 to Count (see page 1639). .

III.3.3 Set

C++

```
HRESULT Set(long _idx, BSTR _text);
```

C#

```
public void Set(long _idx, String _text);
```

Visual Basic

```
Public Sub Set(_idx As long, _text As String)
```

Description

The function sets the value of the given table element. The table elements are indexed from 1 to Count (see page 1639).

III.3.4 SetSize

C++

```
HRESULT SetSize(long _count);
```

C#

```
public void SetSize(long _count);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Sub SetSize(_count As long)
```

Description

The function sets a new table size. Table elements are indexed from 1 to Count (see page 1639). .

IV IRobotNumbersArray

Class Hierarchy

RobotOM.IRobotNumbersArray

C++

```
interface IRobotNumbersArray : IDispatch;
```

C#

```
public interface IRobotNumbersArray;
```

Visual Basic

```
Public Interface IRobotNumbersArray
```

Description

The interface representing a table of integers. Table elements are indexed from 1 to Count (see page 1641). .

IV.1 IRobotNumbersArray Members

The following tables list the members exposed by IRobotNumbersArray.

Public Fields

	Name	Description
	Count (see page 1641)	Table size (at the same time, this is the index of the last element in the table) .

Public Methods

	Name	Description
	Get (see page 1642)	The function returns a table element with the indicated index. Table elements are indexed from 1 to Count (see page 1641). .
	Set (see page 1642)	The function sets the value of table element with the given index. Table elements are indexed from 1 to Count (see page 1641).
	SetSize (see page 1642)	The function sets a new table size. Table elements will be indexed from 1 to table size. .

IV.2 IRobotNumbersArray Fields

The fields of the IRobotNumbersArray class are listed here.

Public Fields

	Name	Description
	Count (see page 1641)	Table size (at the same time, this is the index of the last element in the table) .

(c) 2014 Autodesk Inc. All rights reserved

IV.2.1 Count

C++

```
HRESULT get_Count(long*);
```

C#

```
public long Count { get; }
```

Visual Basic

```
Public ReadOnly Count As long
```

Description

Table size (at the same time, this is the index of the last element in the table) .

IV.3 IRobotNumbersArray Methods

The methods of the IRobotNumbersArray class are listed here.

Public Methods

	Name	Description
	Get (see page 1642)	The function returns a table element with the indicated index. Table elements are indexed from 1 to Count (see page 1641). .
	Set (see page 1642)	The function sets the value of table element with the given index. Table elements are indexed from 1 to Count (see page 1641).
	SetSize (see page 1642)	The function sets a new table size. Table elements will be indexed from 1 to table size. .

IV.3.1 Get

C++

```
HRESULT Get(long _idx, long* ret);
```

C#

```
public long Get(long _idx);
```

Visual Basic

```
Public Function Get(_idx As long) As long
```

Description

The function returns a table element with the indicated index. Table elements are indexed from 1 to Count (see page 1641). .

IV.3.2 Set

C++

```
HRESULT Set(long _idx, long _val);
```

C#

```
public void Set(long _idx, long _val);
```

Visual Basic

```
Public Sub Set(_idx As long, _val As long)
```

(c) 2014 Autodesk Inc. All rights reserved

Description

The function sets the value of table element with the given index. Table elements are indexed from 1 to Count (see page 1641).

IV.3.3 SetSize

C++

```
HRESULT SetSize(long _count);
```

C#

```
public void SetSize(long _count);
```

Visual Basic

```
Public Sub SetSize(_count As long)
```

Description

The function sets a new table size. Table elements will be indexed from 1 to table size. .

V IRobotObjectsArray

Class Hierarchy

RobotOM.IRobotObjectsArray

C++

```
interface IRobotObjectsArray : IDispatch;
```

C#

```
public interface IRobotObjectsArray;
```

Visual Basic

```
Public Interface IRobotObjectsArray
```

Description

The interface providing access to the table of objects .

V.1 IRobotObjectsArray Members

The following tables list the members exposed by IRobotObjectsArray.

Public Fields

	Name	Description
	Count (see page 1644)	Number of objects in the table .

Public Methods

	Name	Description
	Get (see page 1644)	The function returns the indicated table element. .
	Set (see page 1644)	The function sets the indicated table element. .
	SetSize (see page 1645)	The function sets the table size. Table elements are indexed from 1 to table size. The table size is also the number of its elements. .

(c) 2014 Autodesk Inc. All rights reserved

V.2 IRobotObjectsArray Fields

The fields of the IRobotObjectsArray class are listed here.

Public Fields

	Name	Description
	Count (see page 1644)	Number of objects in the table .

V.2.1 Count

C++

```
HRESULT get_Count(long*);
```

C#

```
public long Count { get; }
```

Visual Basic

```
Public ReadOnly Count As long
```

Description

Number of objects in the table .

V.3 IRobotObjectsArray Methods

The methods of the IRobotObjectsArray class are listed here.

Public Methods

	Name	Description
	Get (see page 1644)	The function returns the indicated table element. .
	Set (see page 1644)	The function sets the indicated table element. .
	SetSize (see page 1645)	The function sets the table size. Table elements are indexed from 1 to table size. The table size is also the number of its elements. .

V.3.1 Get

C++

```
HRESULT Get(long _idx, IDispatch* ret);
```

C#

```
public IDispatch Get(long _idx);
```

Visual Basic

```
Public Function Get(_idx As long) As IDispatch
```

Description

The function returns the indicated table element. .

V.3.2 Set

C++

```
HRESULT Set(long _idx, IDispatch _object);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public void Set(long _idx, IDispatch _object);
```

Visual Basic

```
Public Sub Set(_idx As long, _object As IDispatch)
```

Description

The function sets the indicated table element. .

V.3.3 SetSize

C++

```
HRESULT SetSize(long _count);
```

C#

```
public void SetSize(long _count);
```

Visual Basic

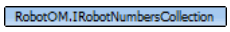
```
Public Sub SetSize(_count As long)
```

Description

The function sets the table size. Table elements are indexed from 1 to table size. The table size is also the number of its elements. .

VI IRobotNumbersCollection

Class Hierarchy



C++

```
interface IRobotNumbersCollection : IDispatch;
```

C#

```
public interface IRobotNumbersCollection;
```

Visual Basic

```
Public Interface IRobotNumbersCollection
```

Description

Collection of numbers (integers).

VI.1 IRobotNumbersCollection Members

The following tables list the members exposed by IRobotNumbersCollection.

Public Fields

	Name	Description
	Count (see page 1646)	Number of elements in the collection .

(c) 2014 Autodesk Inc. All rights reserved

Public Methods

	Name	Description
	Get (see page 1646)	The function returns the required collection element. The elements are indexed from 1 to Count (see page 1646).

VI.2 IRobotNumbersCollection Fields

The fields of the IRobotNumbersCollection class are listed here.

Public Fields

	Name	Description
	Count (see page 1646)	Number of elements in the collection .

VI.2.1 Count

C++

```
HRESULT get_Count(long*);
```

C#

```
public long Count { get; }
```

Visual Basic

```
Public ReadOnly Count As long
```

Description

Number of elements in the collection .

VI.3 IRobotNumbersCollection Methods

The methods of the IRobotNumbersCollection class are listed here.

Public Methods

	Name	Description
	Get (see page 1646)	The function returns the required collection element. The elements are indexed from 1 to Count (see page 1646).

VI.3.1 Get

C++

```
HRESULT Get(long _idx, long* ret);
```

C#

```
public long Get(long _idx);
```

Visual Basic

```
Public Function Get(_idx As long) As long
```

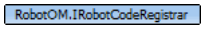
Description

The function returns the required collection element. The elements are indexed from 1 to Count (see page 1646).

(c) 2014 Autodesk Inc. All rights reserved

VII IRobotCodeRegistrar

Class Hierarchy



C++

```
interface IRobotCodeRegistrar : IDispatch;
```

C#

```
public interface IRobotCodeRegistrar;
```

Visual Basic

```
Public Interface IRobotCodeRegistrar
```

Description

In order to simplify the process of registering external codes extending the Robot program, RobotCodeRegistrar has been defined.

VII.1 IRobotCodeRegistrar Members

The following tables list the members exposed by IRobotCodeRegistrar.

Public Fields

	Name	Description
	CodeName (see page 1647)	Code name .
	CodeType (see page 1648)	Code type.
	Country (see page 1648)	Name of a country where the code is implemented.
	Guid (see page 1648)	The text containing CLSID of the main external code implementation component (the field may be filled alternatively from ProgId (see page 1649)).
	Manufacturer (see page 1649)	Name of the producer/dealer of the given code implementation.
	ProgId (see page 1649)	The text containing ProgID of the main external code implementation component (the field may be set alternatively with the Guid (see page 1648) field) .

Public Methods

	Name	Description
	Register (see page 1649)	The function registers code implementation on the basis of the set values. .
	Unregister (see page 1650)	The function unregisters the code implementation on the basis of the set values.

VII.2 IRobotCodeRegistrar Fields

The fields of the IRobotCodeRegistrar class are listed here.

Public Fields

	Name	Description
	CodeName (see page 1647)	Code name .
	CodeType (see page 1648)	Code type.
	Country (see page 1648)	Name of a country where the code is implemented.

(c) 2014 Autodesk Inc. All rights reserved

	Guid (see page 1648)	The text containing CLSID of the main external code implementation component (the field may be filled alternatively from ProgId (see page 1649)).
	Manufacturer (see page 1649)	Name of the producer/dealer of the given code implementation.
	ProgId (see page 1649)	The text containing ProgID of the main external code implementation component (the field may be set alternatively with the Guid (see page 1648) field) .

VII.2.1 CodeName

C++

```
HRESULT get_CodeName(BSTR*);
HRESULT put_CodeName(BSTR);
```

C#

```
public String CodeName { get; set; }
```

Visual Basic

```
Public CodeName As String
```

Description

Code name .

VII.2.2 CodeType

C++

```
HRESULT get_CodeType(IRobotCodeType*);
HRESULT put_CodeType(IRobotCodeType);
```

C#

```
public IRobotCodeType CodeType { get; set; }
```

Visual Basic

```
Public CodeType As IRobotCodeType
```

Description

Code type.

VII.2.3 Country

C++

```
HRESULT get_Country(BSTR*);
HRESULT put_Country(BSTR);
```

C#

```
public String Country { get; set; }
```

Visual Basic

```
Public Country As String
```

Description

Name of a country where the code is implemented.

Version

Available since version 12.

(c) 2014 Autodesk Inc. All rights reserved

VII.2.4 Guid

C++

```
HRESULT get_Guid(BSTR*);
HRESULT put_Guid(BSTR);
```

C#

```
public String Guid { get; set; }
```

Visual Basic

```
Public Guid As String
```

Description

The text containing CLSID of the main external code implementation component (the field may be filled alternatively from ProgId (see page 1649)).

VII.2.5 Manufacturer

C++

```
HRESULT get_Manufacturer(BSTR*);
HRESULT put_Manufacturer(BSTR);
```

C#

```
public String Manufacturer { get; set; }
```

Visual Basic

```
Public Manufacturer As String
```

Description

Name of the producer/dealer of the given code implementation.

VII.2.6 ProgId

C++

```
HRESULT get_ProgId(BSTR*);
HRESULT put_ProgId(BSTR);
```

C#

```
public String ProgId { get; set; }
```

Visual Basic

```
Public ProgId As String
```

Description

The text containing ProgID of the main external code implementation component (the field may be set alternatively with the Guid (see page 1648) field) .

VII.3 IRobotCodeRegistrar Methods

The methods of the IRobotCodeRegistrar class are listed here.

Public Methods

	Name	Description
	Register (see page 1649)	The function registers code implementation on the basis of the set values. .

(c) 2014 Autodesk Inc. All rights reserved

	Unregister (see page 1650)	The function unregisters the code implementation on the basis of the set values.

VII.3.1 Register

C++

```
HRESULT Register();
```

C#

```
public void Register();
```

Visual Basic

```
Public Sub Register()
```

Description

The function registers code implementation on the basis of the set values. .

VII.3.2 Unregister

C++

```
HRESULT Unregister();
```

C#

```
public void Unregister();
```

Visual Basic

```
Public Sub Unregister()
```

Description

The function unregisters the code implementation on the basis of the set values.

VIII IRobotWindow

Class Hierarchy

RobotOM.IRobotWindow

C++

```
interface IRobotWindow : IDispatch;
```

C#

```
public interface IRobotWindow;
```

Visual Basic

```
Public Interface IRobotWindow
```

Description

General interface describing a window in the Robot program. It describes both the main application window and windows which comprise individual views. .

VIII.1 IRobotWindow Members

The following tables list the members exposed by IRobotWindow.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	Caption (see page 1651)	
	Handle (see page 1651)	System window handle Available since version 1.7.
	IsActive (see page 1651)	Indicator of window activity Available since version 1.7.
	State (see page 1652)	State of the window Available since version 1.7.

Public Methods

	Name	Description
	Activate (see page 1652)	Function results in activating the window. Available since version 1.7.
	SendMessage (see page 1652)	The function allows to send a message to the window.

VIII.2 IRobotWindow Fields

The fields of the IRobotWindow class are listed here.

Public Fields

	Name	Description
	Caption (see page 1651)	
	Handle (see page 1651)	System window handle Available since version 1.7.
	IsActive (see page 1651)	Indicator of window activity Available since version 1.7.
	State (see page 1652)	State of the window Available since version 1.7.

VIII.2.1 Caption

C++

```
HRESULT get_Caption(BSTR*);
HRESULT put_Caption(BSTR);
```

C#

```
public String Caption { get; set; }
```

Visual Basic

```
Public Caption As String
```

Version

Available since version 3.

VIII.2.2 Handle

C++

```
HRESULT get_Handle(long*);
```

C#

```
public long Handle { get; }
```

Visual Basic

```
Public ReadOnly Handle As long
```

Description

System window handle Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

VIII.2.3 IsActive

C++

```
HRESULT get_IsActive(VARIANT_BOOL*);
```

C#

```
public bool IsActive { get; }
```

Visual Basic

```
Public ReadOnly IsActive As Boolean
```

Description

Indicator of window activity Available since version 1.7.

VIII.2.4 State

C++

```
HRESULT get_State(IRobotWindowState*);
HRESULT put_State(IRobotWindowState);
```

C#

```
public IRobotWindowState State { get; set; }
```

Visual Basic

```
Public State As IRobotWindowState
```

Description

State of the window Available since version 1.7.

VIII.3 IRobotWindow Methods

The methods of the IRobotWindow class are listed here.

Public Methods

	Name	Description
	Activate (see page 1652)	Function results in activating the window. Available since version 1.7.
	SendMessage (see page 1652)	The function allows to send a message to the window.

VIII.3.1 Activate

C++

```
HRESULT Activate();
```

C#

```
public void Activate();
```

Visual Basic

```
Public Sub Activate()
```

Description

Function results in activating the window. Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

VIII.3.2 SendMessage

C++

```
HRESULT SendMessage(long _msg_id, long _w_param, long _l_param);
```

C#

```
public void SendMessage(long _msg_id, long _w_param, long _l_param);
```

Visual Basic

```
Public Sub SendMessage(_msg_id As long, _w_param As long, _l_param As long)
```

Description

The function allows to send a message to the window.

Version

Available since version 3.

IX IRobotWindowState

C++

```
enum IRobotWindowState;
```

C#

```
public enum IRobotWindowState;
```

Visual Basic

```
Public Enum IRobotWindowState
```

Members

Members	Description
I_WS_NORMAL = 0	Available since version 1.7.
I_WS_MAXIMIZE = 1	Available since version 1.7.
I_WS_MINIMIZE = 2	Available since version 1.7.

Description

Available window states.

X IRobotProtectionInfo

Class Hierarchy

C++

```
interface IRobotProtectionInfo : IDispatch;
```

C#

```
public interface IRobotProtectionInfo;
```

Visual Basic

```
Public Interface IRobotProtectionInfo
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Object provides access to information about protection of the Robot program and its extensions.

X.1 IRobotProtectionInfo Members

The following tables list the members exposed by IRobotProtectionInfo.

Public Methods

	Name	Description
	IsExtensionEnabled (see page 1654)	Function returns value different from zero (True), if the external module (application) - Robot program extension - has been activated in the Robot program protection system. Available since version 2.0.

X.2 IRobotProtectionInfo Methods

The methods of the IRobotProtectionInfo class are listed here.

Public Methods

	Name	Description
	IsExtensionEnabled (see page 1654)	Function returns value different from zero (True), if the external module (application) - Robot program extension - has been activated in the Robot program protection system. Available since version 2.0.

X.2.1 IsExtensionEnabled

C++

```
HRESULT IsExtensionEnabled(long _ext_num, VARIANT_BOOL* ret);
```

C#

```
public bool IsExtensionEnabled(long _ext_num);
```

Visual Basic

```
Public Function IsExtensionEnabled(_ext_num As long) As Boolean
```

Description

Function returns value different from zero (True), if the external module (application) - Robot program extension - has been activated in the Robot program protection system. Available since version 2.0.

XI IRobotComponentFactory

Class Hierarchy

RobotOM.IRobotComponentFactory

C++

```
interface IRobotComponentFactory : IDispatch;
```

C#

```
public interface IRobotComponentFactory;
```

Visual Basic

```
Public Interface IRobotComponentFactory
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Auxiliary component factory. It should be applied instead of the standard New operator for creation of "free" components in case when the application is to cooperate with both the Robot program and Kernel, depending on the user selection. .

Version

Available since version 2.5.

XI.1 IRobotComponentFactory Members

The following tables list the members exposed by IRobotComponentFactory.

Public Methods

	Name	Description
	Create (see page 1655)	Function creates a new component of the indicated type.
	CreateExt (see page 1655)	Function creates a new component supported by the indicated extension of the Robot program kernel.

XI.2 IRobotComponentFactory Methods

The methods of the IRobotComponentFactory class are listed here.

Public Methods

	Name	Description
	Create (see page 1655)	Function creates a new component of the indicated type.
	CreateExt (see page 1655)	Function creates a new component supported by the indicated extension of the Robot program kernel.

XI.2.1 Create

C++

```
HRESULT Create(IRobotComponentType _cmpnt_type, IDispatch* ret);
```

C#

```
public IDispatch Create(IRobotComponentType _cmpnt_type);
```

Visual Basic

```
Public Function Create(_cmpnt_type As IRobotComponentType) As IDispatch
```

Description

Function creates a new component of the indicated type.

Version

Available since version 2.5.

XI.2.2 CreateExt

C++

```
HRESULT CreateExt(BSTR _extmdl_name, long _cmpnt_type, IDispatch* ret);
```

C#

```
public IDispatch CreateExt(String _extmdl_name, long _cmpnt_type);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Function CreateExt(_extmdl_name As String, _cmpnt_type As long) As IDispatch
```

Description

Function creates a new component supported by the indicated extension of the Robot program kernel.

Version

Available since version 2.5.

XII IRobotComponentType

C++

```
enum IRobotComponentType;
```

C#

```
public enum IRobotComponentType;
```

Visual Basic

```
Public Enum IRobotComponentType
```

Members

Members	Description
I_CT_VALUES_ARRAY = 1	Available since version 2.5.
I_CT_NUMBERS_ARRAY = 2	Available since version 2.5.
I_CT_NAMES_ARRAY = 3	Available since version 2.5.
I_CT_OBJECTS_ARRAY = 4	Available since version 2.5.
I_CT_GEO_POINT_2D = 5	Available since version 2.5.
I_CT_GEO_POINT_3D = 6	Available since version 2.5.
I_CT_GEO_CURVE_DIV = 7	Available since version 2.5.
I_CT_GEO_SEGMENT = 8	Available since version 2.5.
I_CT_GEO_SEGMENT_LINE = 9	Available since version 2.5.
I_CT_GEO_SEGMENT_ARC = 10	Available since version 2.5.
I_CT_GEO_OBJECT = 11	Available since version 2.5.
I_CT_GEO_POLYLINE = 12	Available since version 2.5.
I_CT_GEO_CONTOUR = 13	Available since version 2.5.
I_CT_GEO_ARC = 14	Available since version 2.5.
I_CT_GEO_CIRCLE = 15	Available since version 2.5.
I_CT_GEO_SEGMENT_COLLECTION = 16	Available since version 2.5.
I_CT_GEO_POINT_3D_COLLECTION = 17	Available since version 2.5.
I_CT_GEO_LAYER = 18	Available since version 2.5.
I_CT_EMITTER = 19	Available since version 2.5.
I_CT_JOINT_LOAD = 201	Available since version 2.5.
I_CT_JOINT_KNEE_LOAD = 202	Available since version 2.5.
I_CT_JOINT_ANGLE_LOAD = 203	Available since version 2.5.
I_CT_JOINT_TUBE_LOAD = 204	Available since version 2.5.
I_CT_JOINT_PINNED_LOAD = 205	Available since version 2.5.
I_CT_JOINT_FIXED_LOAD = 206	Available since version 2.5.
I_CT_JOINT_GUSSET_SIMPLE_LOAD = 207	Available since version 2.5.
I_CT_JOINT_GUSSET_CROSS_LOAD = 208	Available since version 2.5.
I_CT_JOINT_GUSSET_FLANGE_LOAD = 209	Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

I_CT_MODIF_EXTRUSION = 29	Available since version 2.5.
I_CT_MODIF_LATHE = 30	Available since version 2.5.
I_CT_MODIF_PYRAMID = 31	Available since version 2.5.
I_CT_OPER_TRANSLATION = 32	Available since version 2.5.
I_CT_OPER_SCALING = 33	Available since version 2.5.
I_CT_OPER_ROTATION = 34	Available since version 2.5.
I_CT_OPER_MESHING = 35	Available since version 2.5.
I_CT_SPECTRAL_ANALYSIS_POINTS_COLLECTION = 36	Available since version 2.5.
I_CT_SPECTRAL_ANALYSIS_SPECTRUM = 37	Available since version 2.5.
I_CT_CASE_ANALYSIS_MODES_FILTER = 38	Available since version 2.5.
I_CT_FE_RESULT_PARAMS = 39	Object steering the parameters of getting results for finite elements. Available since version 2.5.
I_CT_RTF_VIEW = 40	View for RTF files. Available since version 2.5.
I_CT_POINTS_ARRAY = 41	Table of points in the 3D space (IRobotPointsArray (see page 1658)). Available since version 3.
I_CT_EXTREME_PARAMS = 42	Parameters of taking extreme values . Available since version 3.5.
I_CT_STRUCTURE_GEO_ANALYSER = 43	Structure geometry analyzer. Available since version 3.5.
I_CT_FE_EXTREME_PARAMS = 44	Available since version 4.
I_CT_FE_MULTI_RESULT_TYPE = 45	Available since version 4.1.
I_CT_SW_STRUCT3D_GEN_PARAMS = 46	Parameters of generation of a 3D structure model for snow/wind loads (IRobotSWStruct3DGenParams (see page 88) object type).
I_CT_SW_STRUCT3D_PURLIN_GEN_PARAMS = 47	IRobotSWStruct3DPurlinGenParams (see page 91) object type.
I_CT_SW_STRUCT3D = 48	3D structure model for generation of snow/wind loads (IRobotSWStruct3D (see page 83) object type).
I_CT_SW_STRUCT3D_FRAME = 49	IRobotSWStruct3DFrame (see page 86) object type.
I_CT_SW_STRUCT3D_ELEMENT = 50	IRobotSWStruct3DFrame (see page 86) object type.
I_CT_HTML_VIEW = 52	View for the HTML format files. Available since version 7.5.
I_CT_STRUCTURE_MERGE_DATA = 53	Available since version 8.
I_CT_NUMBERS_DICTIONARY = 54	Available since version 8.1.
I_CT_TABLE_SCREEN_CAPTURE_PARAMS = 55	Available since version 8.2.
I_CT_PARAM_DEF = 56	Definition of an external parameter. Available since version 8.2.
I_CT_RESULT_QUERY_PARAMS = 57	Parameters of a query to the result server. Available since version 8.2.

Description

Available component types that can be created by means of the component factory.

Version

Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

XIII IRobotDegreeOfFreedom

C++

```
enum IRobotDegreeOfFreedom;
```

C#

```
public enum IRobotDegreeOfFreedom;
```

Visual Basic

```
Public Enum IRobotDegreeOfFreedom
```

Members

Members	Description
I_DOF_UX = 0	Available since version 2.5.
I_DOF_UY = 1	Available since version 2.5.
I_DOF_UZ = 2	Available since version 2.5.
I_DOF_RX = 3	Available since version 2.5.
I_DOF_RY = 4	Available since version 2.5.
I_DOF_RZ = 5	Available since version 2.5.

Description

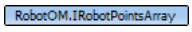
Available degrees of freedom.

Version

Available since version 2.5.

XIV IRobotPointsArray

Class Hierarchy



C++

```
interface IRobotPointsArray : IDispatch;
```

C#

```
public interface IRobotPointsArray;
```

Visual Basic

```
Public Interface IRobotPointsArray
```

Description

Table of points in the 3D space. .

Version

Available since version 3.

XIV.1 IRobotPointsArray Members

The following tables list the members exposed by IRobotPointsArray.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	Count (see page 1659)	Number of elements (at the same time, it is the index of the last table element).

Public Methods

	Name	Description
	Get (see page 1659)	Function takes point coordinates saved in the table under the indicated index. Table elements are indexed from 1 to Count (see page 1659). .
	Set (see page 1660)	Function saves the specified point coordinates under the indicated index to the table. Table elements are indexed from 1 to Count (see page 1659).
	SetSize (see page 1660)	Function sets a table size.

XIV.2 IRobotPointsArray Fields

The fields of the IRobotPointsArray class are listed here.

Public Fields

	Name	Description
	Count (see page 1659)	Number of elements (at the same time, it is the index of the last table element).

XIV.2.1 Count

C++

```
HRESULT get_Count(long*);
```

C#

```
public long Count { get; }
```

Visual Basic

```
Public ReadOnly Count As long
```

Description

Number of elements (at the same time, it is the index of the last table element).

Version

Available since version 3.

XIV.3 IRobotPointsArray Methods

The methods of the IRobotPointsArray class are listed here.

Public Methods

	Name	Description
	Get (see page 1659)	Function takes point coordinates saved in the table under the indicated index. Table elements are indexed from 1 to Count (see page 1659). .
	Set (see page 1660)	Function saves the specified point coordinates under the indicated index to the table. Table elements are indexed from 1 to Count (see page 1659).
	SetSize (see page 1660)	Function sets a table size.

(c) 2014 Autodesk Inc. All rights reserved

XIV.3.1 Get

C++

```
HRESULT Get(long _idx, double* _x, double* _y, double* _z);
```

C#

```
public void Get(long _idx, double* _x, double* _y, double* _z);
```

Visual Basic

```
Public Sub Get(_idx As long, ByRef _x As double*, ByRef _y As double*, ByRef _z As double*)
```

Description

Function takes point coordinates saved in the table under the indicated index. Table elements are indexed from 1 to Count (see page 1659). .

Version

Available since version 3.

XIV.3.2 Set

C++

```
HRESULT Set(long _idx, double _x, double _y, double _z);
```

C#

```
public void Set(long _idx, double _x, double _y, double _z);
```

Visual Basic

```
Public Sub Set(_idx As long, _x As double, _y As double, _z As double)
```

Description

Function saves the specified point coordinates under the indicated index to the table. Table elements are indexed from 1 to Count (see page 1659).

Version

Available since version 3.

XIV.3.3 SetSize

C++

```
HRESULT SetSize(long _size);
```

C#

```
public void SetSize(long _size);
```

Visual Basic

```
Public Sub SetSize(_size As long)
```

Description

Function sets a table size.

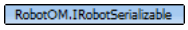
Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

XV IRobotSerializable

Class Hierarchy



C++

```
interface IRobotSerializable : IDispatch;
```

C#

```
public interface IRobotSerializable;
```

Visual Basic

```
Public Interface IRobotSerializable
```

Description

Interface of the object whose data can be saved in/read from the stream.

Version

Available since version 8.

XV.1 IRobotSerializable Members

The following tables list the members exposed by IRobotSerializable.

Public Methods

	Name	Description
	Read (see page 1661)	Function reads object data from the specified stream.
	Write (see page 1661)	Function saves object data in the specified stream.

XV.2 IRobotSerializable Methods

The methods of the IRobotSerializable class are listed here.

Public Methods

	Name	Description
	Read (see page 1661)	Function reads object data from the specified stream.
	Write (see page 1661)	Function saves object data in the specified stream.

XV.2.1 Read

C++

```
HRESULT Read(IUnknown _source_stream, VARIANT_BOOL* ret);
```

C#

```
public bool Read(IUnknown _source_stream);
```

Visual Basic

```
Public Function Read(_source_stream As IUnknown) As Boolean
```

Description

Function reads object data from the specified stream.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 8.

XV.2.2 Write

C++

```
HRESULT Write(IUnknown _dest_stream, VARIANT_BOOL* ret);
```

C#

```
public bool Write(IUnknown _dest_stream);
```

Visual Basic

```
Public Function Write(_dest_stream As IUnknown) As Boolean
```

Description

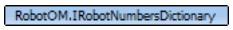
Function saves object data in the specified stream.

Version

Available since version 8.

XVI IRobotNumbersDictionary

Class Hierarchy



C++

```
interface IRobotNumbersDictionary : IDispatch;
```

C#

```
public interface IRobotNumbersDictionary;
```

Visual Basic

```
Public Interface IRobotNumbersDictionary
```

Description

Dictionary of pairs (integer, integer).

Version

Available since version 8.1.

XVI.1 IRobotNumbersDictionary Members

The following tables list the members exposed by IRobotNumbersDictionary.

Public Fields

	Name	Description
	Count (see page 1663)	Number of pairs defined in the dictionary.

Public Methods

	Name	Description
	Delete (see page 1663)	Function deletes a pair with the specified key.

(c) 2014 Autodesk Inc. All rights reserved

	Find (see page 1663)	Function returns an index of the pair with the specified key. If the pair is not found, function returns zero value.
	FindGet (see page 1664)	Function returns an index of the pair with the specified key and a value assigned to the key. If the pair is not found, function returns zero value.
	Get (see page 1664)	Function returns a pair of the specified index.
	Set (see page 1664)	Function adds a new pair to the dictionary. If a pair with the specified key already exists in the dictionary, its value will be replaced.

XVI.2 IRobotNumbersDictionary Fields

The fields of the IRobotNumbersDictionary class are listed here.

Public Fields

	Name	Description
	Count (see page 1663)	Number of pairs defined in the dictionary.

XVI.2.1 Count

C++

```
HRESULT get_Count(long*);
```

C#

```
public long Count { get; }
```

Visual Basic

```
Public ReadOnly Count As long
```

Description

Number of pairs defined in the dictionary.

Version

Available since version 8.1.

XVI.3 IRobotNumbersDictionary Methods

The methods of the IRobotNumbersDictionary class are listed here.

Public Methods

	Name	Description
	Delete (see page 1663)	Function deletes a pair with the specified key.
	Find (see page 1663)	Function returns an index of the pair with the specified key. If the pair is not found, function returns zero value.
	FindGet (see page 1664)	Function returns an index of the pair with the specified key and a value assigned to the key. If the pair is not found, function returns zero value.
	Get (see page 1664)	Function returns a pair of the specified index.
	Set (see page 1664)	Function adds a new pair to the dictionary. If a pair with the specified key already exists in the dictionary, its value will be replaced.

XVI.3.1 Delete

C++

```
HRESULT Delete(long _key);
```

C#

```
public void Delete(long _key);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Sub Delete(_key As long)
```

Description

Function deletes a pair with the specified key.

Version

Available since version 8.1.

XVI.3.2 Find

C++

```
HRESULT Find(long _key, long* ret);
```

C#

```
public long Find(long _key);
```

Visual Basic

```
Public Function Find(_key As long) As long
```

Description

Function returns an index of the pair with the specified key. If the pair is not found, function returns zero value.

Version

Available since version 8.1.

XVI.3.3 FindGet

C++

```
HRESULT FindGet(long _key, long* _value, long* ret);
```

C#

```
public long FindGet(long _key, long* _value);
```

Visual Basic

```
Public Function FindGet(_key As long, ByRef _value As long*) As long
```

Description

Function returns an index of the pair with the specified key and a value assigned to the key. If the pair is not found, function returns zero value.

Version

Available since version 8.1.

XVI.3.4 Get

C++

```
HRESULT Get(long _idx, long* _key, long* _value);
```

C#

```
public void Get(long _idx, long* _key, long* _value);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Sub Get(_idx As long, ByRef _key As long*, ByRef _value As long*)
```

Description

Function returns a pair of the specified index.

Version

Available since version 8.1.

XVI.3.5 Set

C++

```
HRESULT Set(long _key, long _value);
```

C#

```
public void Set(long _key, long _value);
```

Visual Basic

```
Public Sub Set(_key As long, _value As long)
```

Description

Function adds a new pair to the dictionary. If a pair with the specified key already exists in the dictionary, its value will be replaced.

Version

Available since version 8.1.

Concrete

Enumerations

	Name	Description
	IRConcrBarSectionGeometryType (see page 1798)	Types of geometry of an RC member section .

Interfaces

	Name	Description
	IRConcrBarSectionData (see page 1794)	A (see page 1795) simplified interface describing an RC member section.
	IRConcrCalcEngine (see page 1798)	RC calculation module.

I Supplementing the required reinforcement module with new codes

Enumerations

	Name	Description
	IRBestCalcParamsDataDoubleValue (see page 1678)	A set of parameter identifiers of real number type for the RBestCalcParamsData interface. The values of the set may be used as arguments defining the parameter type for the following functions of RBestCalcParamsData interface: SetDouble and GetDouble. .

(c) 2014 Autodesk Inc. All rights reserved

	IRBestCalcParamsDataIntegerValue (see page 1679)	An identifier set for integer type parameters for the RBestCalcPlaramsData interface.
	IRBestCalcParamsDataStringValue (see page 1680)	The identifier set for text values transferred for the RBestCalcParamsData interface.
	IRBestMemberDataDoubleValue (see page 1683)	A set of identifiers of real number parameters for the RBestMemberData interface.
	IRBestMemberDataIntegerValue (see page 1684)	An identifier set of integer type parameters for the BestMemberData interface. .
	IRBestResultsDoubleValue (see page 1685)	A set of identifiers of real-number parameters for the RBestResults interface. .
	IRBestResultsIntegerValue (see page 1686)	A set of identifiers of real-number parameters for the interface RBestResults.
	IRBestCalcErrors (see page 1686)	The set of identifiers describing errors in calculations of theoretical reinforcement. If there occurred an error during calculations, the information about this fact should be written in the object of the RBestResults type, transferred as a parameter of the GetResults function of the RBestCodeCalcEngine interface. To write the information, one should use the SetInteger function that introduces integral values into an object. As an argument determining the type of integral value, one should provide the relevant identifier of error type, while as the error value, one should provide the relevant value from the set RBestCalcErrors. .
	IRBestCalcWarnings (see page 1687)	Warning bit masks for calculations of theoretical reinforcement. .
	IRBestResultsStringValue (see page 1688)	Set of identifiers of text parameters for the interface RBestResults.
	IRBestDimParamsIntegerValue (see page 1690)	Available identifiers of parameters of an integer type for the RBestDimParams interface.
	IRBestMemberDataStringValue (see page 1694)	The set of identifiers of text parameters for the RBestMemberData interface. .
	IRBestMemberType (see page 1695)	A set of identifiers for a type of RC member or plate.
	IRBestCodeCalculationType (see page 1696)	
	IRBestCalculationType (see page 1698)	
	IRBestDimParamsDoubleValue (see page 1698)	
	IRBestBendType (see page 1698)	
	IRBestDirection (see page 1698)	
	IRBestLevel (see page 1699)	
	IRBestForceDataDoubleValue (see page 1699)	
	IRBestCoordSystem (see page 1699)	
	IRBestForceDataIntegerValue (see page 1699)	
	IRBestForceDataSLSCombType (see page 1700)	Available since version 11.0.

Interfaces

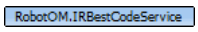
	Name	Description
	IRBestCodeService (see page 1667)	RBestCodeService makes code-dependent definitions and calculations for theoretical reinforcement of RC members available.
	IRBestCalcParamsDlg (see page 1669)	The interface describes the dialog box that allows the parameter definition for theoretical reinforcement calculations for RC members. .

(c) 2014 Autodesk Inc. All rights reserved

	IRBestMemberDlg (see page 1670)	The interface describes the dialog box that makes a code-dependent definition of a RC member type possible.
	IRBestParamSet (see page 1671)	The interface describes a set of parameters that can be divided into two categories: values with a well defined meaning independent from code and these ones that meaning is connected with the given code. All values are available by the function set of the type Get/Set. The identifier of the taken or set quantity is their parameter. Code-dependent parameters have identifiers beginning from USER_VALUE upwards.
	IRBestCalcParamsData (see page 1677)	The interface defines calculation parameters of theoretical reinforcement for an RC member. Available parameters are described by the following types: RBestCalcParamsDataDoubleValue, RBestCalcParamsDataIntegerValue and RBestCalcParamsDataStringValue.
	IRBestCalcParamsDataList (see page 1680)	An object list of the BestCalcParamsData type. Each list element is associated with a label name. All the names are saved in the LabNames (see page 1681) table.
	IRBestMemberData (see page 1683)	The interface describes a data set for a type of RC member. Accessible parameters are described by the following types: RBestMemberDataDoubleValue, RBestMemberDataIntegerValue and RBestMemberDataStringValue. .
	IRBestResults (see page 1684)	The interface describes calculation results of theoretical reinforcement for single section of RC member. Available values are described by the types: RBestResultsDoubleValue, RBestResultsIntegerValue and RBestResultsStringValue.
	IRBestForceData (see page 1688)	A set of forces acting in a bar section. .
	IRBestDimParams (see page 1690)	A set of additional values parametrizing code (design) calculations. The RBestDimParamsIntegerValue type describes the set of available parameters. .
	IRBestCodeCalcEngine (see page 1691)	The interface describes the object performing code calculations of theoretical reinforcement for RC members.
	IRBestPlateCalcParamsDlg (see page 1695)	
	IRBestCodeServiceExt (see page 1696)	Available since version 2.0.
	IRBestCodeService2 (see page 1700)	

I.1 IRBestCodeService

Class Hierarchy



C++

```
interface IRBestCodeService : IDispatch;
```

C#

```
public interface IRBestCodeService;
```

Visual Basic

```
Public Interface IRBestCodeService
```

Description

RBestCodeService makes code-dependent definitions and calculations for theoretical reinforcement of RC members available.

(c) 2014 Autodesk Inc. All rights reserved

I.1.1 IRBestCodeService Members

The following tables list the members exposed by IRBestCodeService.

Public Fields

	Name	Description
	CalcEngine (see page 1668)	The object performing code calculations.
	CalcParamsDlg (see page 1668)	A dialog box that allows to edit calculation parameters.
	MemberDlg (see page 1668)	A dialog box that allows to edit RC member parameters .

I.1.2 IRBestCodeService Fields

The fields of the IRBestCodeService class are listed here.

Public Fields

	Name	Description
	CalcEngine (see page 1668)	The object performing code calculations.
	CalcParamsDlg (see page 1668)	A dialog box that allows to edit calculation parameters.
	MemberDlg (see page 1668)	A dialog box that allows to edit RC member parameters .

I.1.2.1 CalcEngine

C++

```
HRESULT get_CalcEngine(IRBestCodeCalcEngine**);
```

C#

```
public IRBestCodeCalcEngine CalcEngine { get; }
```

Visual Basic

```
Public ReadOnly CalcEngine As IRBestCodeCalcEngine
```

Description

The object performing code calculations.

I.1.2.2 CalcParamsDlg

C++

```
HRESULT get_CalcParamsDlg(IRBestCalcParamsDlg**);
```

C#

```
public IRBestCalcParamsDlg CalcParamsDlg { get; }
```

Visual Basic

```
Public ReadOnly CalcParamsDlg As IRBestCalcParamsDlg
```

Description

A dialog box that allows to edit calculation parameters.

I.1.2.3 MemberDlg

C++

```
HRESULT get_MemberDlg(IRBestMemberDlg**);
```

C#

```
public IRBestMemberDlg MemberDlg { get; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public ReadOnly MemberDlg As IRBestMemberDlg
```

Description

A dialog box that allows to edit RC member parameters .

I.2 IRBestCalcParamsDlg

Class Hierarchy

C++

```
interface IRBestCalcParamsDlg : IDispatch;
```

C#

```
public interface IRBestCalcParamsDlg;
```

Visual Basic

```
Public Interface IRBestCalcParamsDlg
```

Description

The interface describes the dialog box that allows the parameter definition for theoretical reinforcement calculations for RC members. .

I.2.1 IRBestCalcParamsDlg Members

The following tables list the members exposed by IRBestCalcParamsDlg.

Public Methods

	Name	Description
	DoModal (see page 1669)	The function creates and displays a new modal dialog box that allows to edit the list of objects describing calculation parameters.
	SetStandard (see page 1670)	The function stets standard (default) parameter values for the object. .

I.2.2 IRBestCalcParamsDlg Methods

The methods of the IRBestCalcParamsDlg class are listed here.

Public Methods

	Name	Description
	DoModal (see page 1669)	The function creates and displays a new modal dialog box that allows to edit the list of objects describing calculation parameters.
	SetStandard (see page 1670)	The function stets standard (default) parameter values for the object. .

I.2.2.1 DoModal

C++

```
HRESULT DoModal(IRBestCalcParamsDataList* _param_list);
```

C#

```
public void DoModal(IRBestCalcParamsDataList _param_list);
```

Visual Basic

```
Public Sub DoModal(ByRef _param_list As IRBestCalcParamsDataList)
```

(c) 2014 Autodesk Inc. All rights reserved

Description

The function creates and displays a new modal dialog box that allows to edit the list of objects describing calculation parameters.

I.2.2.2 SetStandard

C++

```
HRESULT SetStandard(IRBestCalcParamsData* _params);
```

C#

```
public void SetStandard(IRBestCalcParamsData _params);
```

Visual Basic

```
Public Sub SetStandard(ByRef _params As IRBestCalcParamsData)
```

Description

The function stets standard (default) parameter values for the object. .

I.3 IRBestMemberDlg

Class Hierarchy

RobotOM.IRBestMemberDlg

C++

```
interface IRBestMemberDlg : IDispatch;
```

C#

```
public interface IRBestMemberDlg;
```

Visual Basic

```
Public Interface IRBestMemberDlg
```

Description

The interface describes the dialog box that makes a code-dependent definition of a RC member type possible.

I.3.1 IRBestMemberDlg Members

The following tables list the members exposed by IRBestMemberDlg.

Public Methods

	Name	Description
	DoModal (see page 1671)	The function creates and displays a modal dialog box that allows to edit the bar code parameters saved in a given object.
	SetStandard (see page 1671)	The function sets standard (default) values in the indicated object for the code-dependent parameters.

I.3.2 IRBestMemberDlg Methods

The methods of the IRBestMemberDlg class are listed here.

Public Methods

	Name	Description
	DoModal (see page 1671)	The function creates and displays a modal dialog box that allows to edit the bar code parameters saved in a given object.

(c) 2014 Autodesk Inc. All rights reserved

	SetStandard (see page 1671)	The function sets standard (default) values in the indicated object for the code-dependent parameters.

I.3.2.1 DoModal

C++

```
HRESULT DoModal(IRBestMemberData* _params);
```

C#

```
public void DoModal(IRBestMemberData _params);
```

Visual Basic

```
Public Sub DoModal(ByRef _params As IRBestMemberData)
```

Description

The function creates and displays a modal dialog box that allows to edit the bar code parameters saved in a given object.

I.3.2.2 SetStandard

C++

```
HRESULT SetStandard(IRBestMemberData* _params);
```

C#

```
public void SetStandard(IRBestMemberData _params);
```

Visual Basic

```
Public Sub SetStandard(ByRef _params As IRBestMemberData)
```

Description

The function sets standard (default) values in the indicated object for the code-dependent parameters.

I.4 IRBestParamSet

Class Hierarchy

RobotOM.IRBestParamSet

C++

```
interface IRBestParamSet : IDispatch;
```

C#

```
public interface IRBestParamSet;
```

Visual Basic

```
Public Interface IRBestParamSet
```

Description

The interface describes a set of parameters that can be divided into two categories: values with a well defined meaning independent from code and these ones that meaning is connected with the given code. All values are available by the function set of the type Get/Set. The identifier of the taken or set quantity is their parameter. Code-dependent parameters have identifiers beginning from USER_VALUE upwards.

I.4.1 IRBestParamSet Members

The following tables list the members exposed by IRBestParamSet.

(c) 2014 Autodesk Inc. All rights reserved

Public Methods

	Name	Description
	Clear (see page 1673)	The function deletes all settings of parameters carried out by means of the group of functions Set*. .
	ClearDouble (see page 1673)	The function deletes the setting of the real-number type parameter with the given identifier.
	ClearInteger (see page 1674)	The function deletes the setting of the integer type parameter with the given identifier.
	ClearString (see page 1674)	The function deletes the setting of the text type parameter with the given identifier.
	GetDouble (see page 1674)	The function takes a real number, which is a parameter value with a given identifier. The function returns zero (False) if the parameter value with the indicated identifier was not set.
	GetInteger (see page 1674)	The function takes an integer number, which is a parameter value with a given identifier. It gives back zero (False) if the value of the parameter with the indicated identifier was not set.
	GetString (see page 1675)	The function takes a text (character string) which is a parameter value with the given identifier. It gives zero (False) back if the value of the parameter with the indicated identifier was not set. .
	IsValidDouble (see page 1675)	The function checks if the indicated real number is a correct value of the parameter with the indicated identifier. It returns zero (False) if the value is not correct. .
	IsValidInteger (see page 1675)	The function checks if the indicated integer is a correct value of the parameter with the indicated identifier. It returns zero (False) if the value is not correct. .
	IsValidString (see page 1675)	The function checks if the indicated text (character string) is a correct value of the parameter with the indicated identifier. It returns zero (False) if the value is not correct. .
	SetDouble (see page 1676)	The function sets a value of the given parameter if it is correct (according to IsValidDouble (see page 1675)). If the parameter value is incorrect, the function returns zero (False).
	SetInteger (see page 1676)	The function sets a value of the given parameter if it is correct (according to IsValidInteger (see page 1675)). If the parameter value is incorrect, the function returns zero (False).
	SetString (see page 1676)	The function sets the value of the given parameter, if correct (according to IsValidString (see page 1675)). If the parameter value is not correct, the function returns zero (False). .

I.4.2 IRBestParamSet Methods

The methods of the IRBestParamSet class are listed here.

Public Methods

	Name	Description
	Clear (see page 1673)	The function deletes all settings of parameters carried out by means of the group of functions Set*. .
	ClearDouble (see page 1673)	The function deletes the setting of the real-number type parameter with the given identifier.
	ClearInteger (see page 1674)	The function deletes the setting of the integer type parameter with the given identifier.
	ClearString (see page 1674)	The function deletes the setting of the text type parameter with the given identifier.
	GetDouble (see page 1674)	The function takes a real number, which is a parameter value with a given identifier. The function returns zero (False) if the parameter value with the indicated identifier was not set.

(c) 2014 Autodesk Inc. All rights reserved

	GetInteger (see page 1674)	The function takes an integer number, which is a parameter value with a given identifier. It gives back zero (False) if the value of the parameter with the indicated identifier was not set.
	GetString (see page 1675)	The function takes a text (character string) which is a parameter value with the given identifier. It gives zero (False) back if the value of the parameter with the indicated identifier was not set. .
	IsValidDouble (see page 1675)	The function checks if the indicated real number is a correct value of the parameter with the indicated identifier. It returns zero (False) if the value is not correct. .
	IsValidInteger (see page 1675)	The function checks if the indicated integer is a correct value of the parameter with the indicated identifier. It returns zero (False) if the value is not correct. .
	IsValidString (see page 1675)	The function checks if the indicated text (character string) is a correct value of the parameter with the indicated identifier. It returns zero (False) if the value is not correct. .
	SetDouble (see page 1676)	The function sets a value of the given parameter if it is correct (according to IsValidDouble (see page 1675)). If the parameter value is incorrect, the function returns zero (False).
	SetInteger (see page 1676)	The function sets a value of the given parameter if it is correct (according to IsValidInteger (see page 1675)). If the parameter value is incorrect, the function returns zero (False).
	SetString (see page 1676)	The function sets the value of the given parameter, if correct (according to IsValidString (see page 1675)). If the parameter value is not correct, the function returns zero (False). .

I.4.2.1 Clear

C++

```
HRESULT Clear();
```

C#

```
public void Clear();
```

Visual Basic

```
Public Sub Clear()
```

Description

The function deletes all settings of parameters carried out by means of the group of functions Set*. .

I.4.2.2 ClearDouble

C++

```
HRESULT ClearDouble(int _param_id);
```

C#

```
public void ClearDouble(int _param_id);
```

Visual Basic

```
Public Sub ClearDouble(_param_id As int)
```

Description

The function deletes the setting of the real-number type parameter with the given identifier.

(c) 2014 Autodesk Inc. All rights reserved

I.4.2.3 ClearInteger

C++

```
HRESULT ClearInteger(int _param_id);
```

C#

```
public void ClearInteger(int _param_id);
```

Visual Basic

```
Public Sub ClearInteger(_param_id As int)
```

Description

The function deletes the setting of the integer type parameter with the given identifier.

I.4.2.4 ClearString

C++

```
HRESULT ClearString(int _param_id);
```

C#

```
public void ClearString(int _param_id);
```

Visual Basic

```
Public Sub ClearString(_param_id As int)
```

Description

The function deletes the setting of the text type parameter with the given identifier.

I.4.2.5 GetDouble

C++

```
HRESULT GetDouble(int _param_id, double* _value, VARIANT_BOOL* ret);
```

C#

```
public bool GetDouble(int _param_id, double* _value);
```

Visual Basic

```
Public Function GetDouble(_param_id As int, ByRef _value As double*) As Boolean
```

Description

The function takes a real number, which is a parameter value with a given identifier. The function returns zero (False) if the parameter value with the indicated identifier was not set.

I.4.2.6 GetInteger

C++

```
HRESULT GetInteger(int _param_id, int* _value, VARIANT_BOOL* ret);
```

C#

```
public bool GetInteger(int _param_id, int* _value);
```

Visual Basic

```
Public Function GetInteger(_param_id As int, ByRef _value As int*) As Boolean
```

(c) 2014 Autodesk Inc. All rights reserved

Description

The function takes an integer number, which is a parameter value with a given identifier. It gives back zero (False) if the value of the parameter with the indicated identifier was not set.

I.4.2.7 GetString

C++

```
HRESULT GetString(int _param_id, BSTR* _value, VARIANT_BOOL* ret);
```

C#

```
public bool GetString(int _param_id, BSTR* _value);
```

Visual Basic

```
Public Function GetString(_param_id As int, ByRef _value As BSTR*) As Boolean
```

Description

The function takes a text (character string) which is a parameter value with the given identifier. It gives zero (False) back if the value of the parameter with the indicated identifier was not set. .

I.4.2.8 IsValidDouble

C++

```
HRESULT IsValidDouble(int _param_id, double _value, VARIANT_BOOL* ret);
```

C#

```
public bool IsValidDouble(int _param_id, double _value);
```

Visual Basic

```
Public Function IsValidDouble(_param_id As int, _value As double) As Boolean
```

Description

The function checks if the indicated real number is a correct value of the parameter with the indicated identifier. It returns zero (False) if the value is not correct. .

I.4.2.9 IsValidInteger

C++

```
HRESULT IsValidInteger(int _param_id, int _value, VARIANT_BOOL* ret);
```

C#

```
public bool IsValidInteger(int _param_id, int _value);
```

Visual Basic

```
Public Function IsValidInteger(_param_id As int, _value As int) As Boolean
```

Description

The function checks if the indicated integer is a correct value of the parameter with the indicated identifier. It returns zero (False) if the value is not correct. .

I.4.2.10 IsValidString

C++

```
HRESULT IsValidString(int _param_id, BSTR _value, VARIANT_BOOL* ret);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public bool IsValidString(int _param_id, String _value);
```

Visual Basic

```
Public Function IsValidString(_param_id As int, _value As String) As Boolean
```

Description

The function checks if the indicated text (character string) is a correct value of the parameter with the indicated identifier. It returns zero (False) if the value is not correct. .

I.4.2.11 SetDouble

C++

```
HRESULT SetDouble(int _param_id, double _value, VARIANT_BOOL* ret);
```

C#

```
public bool SetDouble(int _param_id, double _value);
```

Visual Basic

```
Public Function SetDouble(_param_id As int, _value As double) As Boolean
```

Description

The function sets a value of the given parameter if it is correct (according to IsValidDouble (see page 1675)). If the parameter value is incorrect, the function returns zero (False).

I.4.2.12 SetInteger

C++

```
HRESULT SetInteger(int _param_id, int _value, VARIANT_BOOL* ret);
```

C#

```
public bool SetInteger(int _param_id, int _value);
```

Visual Basic

```
Public Function SetInteger(_param_id As int, _value As int) As Boolean
```

Description

The function sets a value of the given parameter if it is correct (according to IsValidInteger (see page 1675)). If the parameter value is incorrect, the function returns zero (False).

I.4.2.13 SetString

C++

```
HRESULT SetString(int _param_id, BSTR _value, VARIANT_BOOL* ret);
```

C#

```
public bool SetString(int _param_id, BSTR _value);
```

Visual Basic

```
Public Function SetString(_param_id As int, _value As BSTR) As Boolean
```

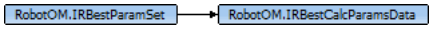
Description

The function sets the value of the given parameter, if correct (according to IsValidString (see page 1675)). If the parameter value is not correct, the function returns zero (False). .

(c) 2014 Autodesk Inc. All rights reserved

I.5 IRBestCalcParamsData

Class Hierarchy



C++

```
interface IRBestCalcParamsData : IRBestParamSet;
```

C#

```
public interface IRBestCalcParamsData : IRBestParamSet;
```

Visual Basic

```
Public Interface IRBestCalcParamsData
```

Description

The interface defines calculation parameters of theoretical reinforcement for an RC member. Available parameters are described by the following types: RBestCalcParamsDataDoubleValue, RBestCalcParamsDataIntegerValue and RBestCalcParamsDataStringValue.

I.5.1 IRBestCalcParamsData Members

The following tables list the members exposed by IRBestCalcParamsData.

Public Fields

	Name	Description
	ModularityList (see page 1678)	A list of modularity values.

Public Methods

	Name	Description
	Clear (see page 1673)	The function deletes all settings of parameters carried out by means of the group of functions Set*. .
	ClearDouble (see page 1673)	The function deletes the setting of the real-number type parameter with the given identifier.
	ClearInteger (see page 1674)	The function deletes the setting of the integer type parameter with the given identifier.
	ClearString (see page 1674)	The function deletes the setting of the text type parameter with the given identifier.
	GetDouble (see page 1674)	The function takes a real number, which is a parameter value with a given identifier. The function returns zero (False) if the parameter value with the indicated identifier was not set.
	GetInteger (see page 1674)	The function takes an integer number, which is a parameter value with a given identifier. It gives back zero (False) if the value of the parameter with the indicated identifier was not set.
	GetString (see page 1675)	The function takes a text (character string) which is a parameter value with the given identifier. It gives zero (False) back if the value of the parameter with the indicated identifier was not set. .
	IsValidDouble (see page 1675)	The function checks if the indicated real number is a correct value of the parameter with the indicated identifier. It returns zero (False) if the value is not correct. .
	IsValidInteger (see page 1675)	The function checks if the indicated integer is a correct value of the parameter with the indicated identifier. It returns zero (False) if the value is not correct. .
	IsValidString (see page 1675)	The function checks if the indicated text (character string) is a correct value of the parameter with the indicated identifier. It returns zero (False) if the value is not correct. .

(c) 2014 Autodesk Inc. All rights reserved

	SetDouble (see page 1676)	The function sets a value of the given parameter if it is correct (according to IsValidDouble (see page 1675)). If the parameter value is incorrect, the function returns zero (False).
	SetInteger (see page 1676)	The function sets a value of the given parameter if it is correct (according to IsValidInteger (see page 1675)). If the parameter value is incorrect, the function returns zero (False).
	SetString (see page 1676)	The function sets the value of the given parameter, if correct (according to IsValidString (see page 1675)). If the parameter value is not correct, the function returns zero (False). .

I.5.2 IRBestCalcParamsData Fields

The fields of the IRBestCalcParamsData class are listed here.

Public Fields

	Name	Description
	ModularityList (see page 1678)	A list of modularity values.

I.5.2.1 ModularityList

C++

```
HRESULT get_ModularityList(IRobotValuesArray**);
```

C#

```
public IRobotValuesArray ModularityList { get; }
```

Visual Basic

```
Public ReadOnly ModularityList As IRobotValuesArray
```

Description

A list of modularity values.

I.6 IRBestCalcParamsDataDoubleValue

C++

```
enum IRBestCalcParamsDataDoubleValue;
```

C#

```
public enum IRBestCalcParamsDataDoubleValue;
```

Visual Basic

```
Public Enum IRBestCalcParamsDataDoubleValue
```

Members

Members	Description
I_BCPDDV_LONG_TENSION_BAR_DIAM = 0	Diameter of top reinforcing bars.
I_BCPDDV_LONG_COMPRESSION_BAR_DIAM = 1	Diameter of bottom reinforcing bars.
I_BCPDDV_TRAN_BAR_DIAM = 2	Diameter of transversal reinforcement.
I_BCPDDV_TRAN_INCLINATION = 3	A transversal reinforcement inclination angle (in radians).
I_BCPDDV_SAND_CONTENT = 4	Sand percentage [0..1] (for lightweight concrete).
I_BCPDDV_LONG_STEEL_FE = 5	Steel resistance of longitudinal reinforcement.
I_BCPDDV_TRAN_STEEL_FE = 6	Steel resistance of transversal reinforcement .
I_BCPDDV_CONCRETE_FC = 7	Concrete resistance.
I_BCPDDV_COVER_SIZE = 8	Cover (calculated to bar axis).

(c) 2014 Autodesk Inc. All rights reserved

I_BCPDDV_CONCRETE_CREEP = 9	Concrete creep coefficient.
I_BCPDDV_REDISTRIBUTION = 10	Redistribution coefficient.
I_BCPDDV_MAX_BREAK = 11	Maximum cracking width.
I_BCPDDV_USER_VALUE = 1000	The smallest identifier reserved for the user-defined values.
I_BCPDDV_MAX_DEFLECTION = 12	Available since version 2.0.
I_BCPDDV_MAX_CRACK = 13	Available since version 2.0.
I_BCPDDV_MAIN_RNF_COORD_X = 14	Available since version 2.0.
I_BCPDDV_MAIN_RNF_COORD_Y = 15	Available since version 2.0.
I_BCPDDV_MAIN_RNF_COORD_Z = 16	Available since version 2.0.
I_BCPDDV_FLT2F = 17	Available since version 2.0.

Description

A set of parameter identifiers of real number type for the RBestCalcParamsData interface. The values of the set may be used as arguments defining the parameter type for the following functions of RBestCalcParamsData interface: SetDouble and GetDouble. .

I.7 IRBestCalcParamsDataIntegerValue

C++

```
enum IRBestCalcParamsDataIntegerValue;
```

C#

```
public enum IRBestCalcParamsDataIntegerValue;
```

Visual Basic

```
Public Enum IRBestCalcParamsDataIntegerValue
```

Members

Members	Description
I_BCPDIV_SEISMICS = 0	Seismic risk.
I_BCPDIV_CRACKING = 1	An exposure type .
I_BCPDIV_TRAN_TYPE = 2	Type of transversal reinforcement (0 - stirrups, 1 - spiral).
I_BCPDIV_LEGS_NUM = 3	Number of legs in stirrups.
I_BCPDIV_CONCRETE_LIGHT = 4	Lightweight concrete used (0 - no, 1 - yes).
I_BCPDIV_TRAN_ZONES_NUM = 5	Number of transversal reinforcement sections in final results (refers to beams) .
I_BCPDIV_TRAN_ZONES_OPTIMALIZATION = 6	Optimization of spacing of transversal reinforcement sections (0 - switched off, 1 - switched on).
I_BCPDIV_LONG_STEEL_BY_CLASS = 7	Steel resistance of longitudinal reinforcement defined on the basis of steel grade (0 - no, 1 - yes) .
I_BCPDIV_TRAN_STEEL_BY_CLASS = 8	Steel resistance of transversal reinforcement defined on the basis of steel grade (0 - no, 1 - yes).
I_BCPDIV_CONCRETE_BY_CLASS = 9	Concrete resistance defined on the basis of a concrete class (0 - no, 1 - yes).
I_BCPDIV_CONCRETE_BY_CONSTRUCTION = 10	Concrete resistance taken from the structure (0 - no, 1 - yes).
I_BCPDIV_LONG_BAR_DIAM_THE_SAME = 11	Equal bar diameters of a top and bottom longitudinal reinforcement (0 - no, 1 - yes).
I_BCPDIV_USER_VALUE = 1000	The smallest identifier reserved for the user-defined values .
I_BCPDIV_CALC_DEFLECTION = 12	Available since version 2.0.
I_BCPDIV_CALC_CRACK = 13	Available since version 2.0.
I_BCPDIV_ADJUST_DEFLECTION = 14	Available since version 2.0.
I_BCPDIV_FLT2F = 15	Available since version 2.0.

(c) 2014 Autodesk Inc. All rights reserved

I_BCPDIV_MAIN_RNF_COORD_SYSTEM = 16	Available since version 2.0.

Description

An identifier set for integer type parameters for the RBestCalcPlaramsData interface.

I.8 IRBestCalcParamsDataStringValue

C++

```
enum IRBestCalcParamsDataStringValue;
```

C#

```
public enum IRBestCalcParamsDataStringValue;
```

Visual Basic

```
Public Enum IRBestCalcParamsDataStringValue
```

Members

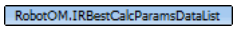
Members	Description
I_BCPDSV_NAME = 0	Name of parameter set.
I_BCPDSV_LONG_REINF_CLASS = 1	Name of steel grade of longitudinal reinforcement.
I_BCPDSV_TRAN_REINF_CLASS = 2	Name of steel grade of transversal reinforcement.
I_BCPDSV_CONCRETE_CLASS = 3	Name of concrete class.
I_BCPDSV_USER_VALUE = 1000	The smallest identifier reserved for the user-defined values .

Description

The identifier set for text values transferred for the RBestCalcParamsData interface.

I.9 IRBestCalcParamsDataList

Class Hierarchy



C++

```
interface IRBestCalcParamsDataList : IDispatch;
```

C#

```
public interface IRBestCalcParamsDataList;
```

Visual Basic

```
Public Interface IRBestCalcParamsDataList
```

Description

An object list of the BestCalcParamsData type. Each list element is associated with a label name. All the names are saved in the LabNames (see page 1681) table.

I.9.1 IRBestCalcParamsDataList Members

The following tables list the members exposed by IRBestCalcParamsDataList.

Public Fields

	Name	Description
	LabNames (see page 1681)	Table with label names associated with particular sets of parameters .

(c) 2014 Autodesk Inc. All rights reserved

	Selected (see page 1681)	Name of the label associated with the indicated element from the list (if the name is empty, no element from the list is indicated).

Public Methods

	Name	Description
	Create (see page 1682)	The function creates and returns a new empty set of parameters.
	Delete (see page 1682)	The function deletes from the list an element associated with the given label .
	Get (see page 1682)	The function gives a list element associated with the given label .
	Store (see page 1683)	The function writes the defined set of parameters as the data for the label of the defined name. Definition of a new name results in the creation of a new label associated with the defined set of parameters. If an existing name is selected, the data associated with it will be overwritten. .

I.9.2 IRBestCalcParamsDataList Fields

The fields of the IRBestCalcParamsDataList class are listed here.

Public Fields

	Name	Description
	LabNames (see page 1681)	Table with label names associated with particular sets of parameters .
	Selected (see page 1681)	Name of the label associated with the indicated element from the list (if the name is empty, no element from the list is indicated).

I.9.2.1 LabNames

C++

```
HRESULT get_LabNames(IRobotNamesArray**);
```

C#

```
public IRobotNamesArray LabNames { get; }
```

Visual Basic

```
Public ReadOnly LabNames As IRobotNamesArray
```

Description

Table with label names associated with particular sets of parameters .

I.9.2.2 Selected

C++

```
HRESULT get_Selected(BSTR*);
HRESULT put_Selected(BSTR);
```

C#

```
public String Selected { get; set; }
```

Visual Basic

```
Public Selected As String
```

Description

Name of the label associated with the indicated element from the list (if the name is empty, no element from the list is indicated).

I.9.3 IRBestCalcParamsDataList Methods

The methods of the IRBestCalcParamsDataList class are listed here.

(c) 2014 Autodesk Inc. All rights reserved

Public Methods

	Name	Description
	Create (see page 1682)	The function creates and returns a new empty set of parameters.
	Delete (see page 1682)	The function deletes from the list an element associated with the given label .
	Get (see page 1682)	The function gives a list element associated with the given label .
	Store (see page 1683)	The function writes the defined set of parameters as the data for the label of the defined name. Definition of a new name results in the creation of a new label associated with the defined set of parameters. If an existing name is selected, the data associated with it will be overwritten. .

I.9.3.1 Create

C++

```
HRESULT Create(IRBestCalcParamsData** ret);
```

C#

```
public IRBestCalcParamsData Create();
```

Visual Basic

```
Public Function Create() As IRBestCalcParamsData
```

Description

The function creates and returns a new empty set of parameters.

I.9.3.2 Delete

C++

```
HRESULT Delete(BSTR _lab_name, VARIANT_BOOL* ret);
```

C#

```
public bool Delete(String _lab_name);
```

Visual Basic

```
Public Function Delete(_lab_name As String) As Boolean
```

Description

The function deletes from the list an element associated with the given label .

I.9.3.3 Get

C++

```
HRESULT Get(BSTR _lab_name, IRBestCalcParamsData** ret);
```

C#

```
public IRBestCalcParamsData Get(String _lab_name);
```

Visual Basic

```
Public Function Get(_lab_name As String) As IRBestCalcParamsData
```

Description

The function gives a list element associated with the given label .

(c) 2014 Autodesk Inc. All rights reserved

I.9.3.4 Store

C++

```
HRESULT Store(BSTR _lab_name, IRBestCalcParamsData* _params, VARIANT_BOOL* ret);
```

C#

```
public bool Store(String _lab_name, IRBestCalcParamsData _params);
```

Visual Basic

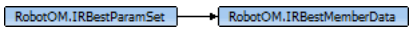
```
Public Function Store(_lab_name As String, ByRef _params As IRBestCalcParamsData) As Boolean
```

Description

The function writes the defined set of parameters as the data for the label of the defined name. Definition of a new name results in the creation of a new label associated with the defined set of parameters. If an existing name is selected, the data associated with it will be overwritten. .

I.10 IRBestMemberData

Class Hierarchy



C++

```
interface IRBestMemberData : IRBestParamSet;
```

C#

```
public interface IRBestMemberData : IRBestParamSet;
```

Visual Basic

```
Public Interface IRBestMemberData
```

Description

The interface describes a data set for a type of RC member. Accessible parameters are described by the following types: RBestMemberDataDoubleValue, RBestMemberDataIntegerValue and RBestMemberDataStringValue. .

I.11 IRBestMemberDataDoubleValue

C++

```
enum IRBestMemberDataDoubleValue;
```

C#

```
public enum IRBestMemberDataDoubleValue;
```

Visual Basic

```
Public Enum IRBestMemberDataDoubleValue
```

Members

Members	Description
I_BMDDV_LENGTH_Y = 0	Length in Y direction.
I_BMDDV_LENGTH_Z = 1	Length in Z direction.
I_BMDDV_BUCK_COEF_Y = 2	Buckling coefficient Y.
I_BMDDV_BUCK_COEF_Z = 3	Buckling coefficient Z.

(c) 2014 Autodesk Inc. All rights reserved

I_BMDDV_SUPPORT_WIDTH_1 = 4	Length of the first support (calculated to axis).
I_BMDDV_SUPPORT_WIDTH_2 = 5	Length of the second support (calculated to axis).
I_BMDDV_USER_VALUE = 1000	The smallest identifier reserved for the user-defined values.

Description

A set of identifiers of real number parameters for the RBestMemberData interface.

I.12 IRBestMemberDataIntegerValue

C++

```
enum IRBestMemberDataIntegerValue;
```

C#

```
public enum IRBestMemberDataIntegerValue;
```

Visual Basic

```
Public Enum IRBestMemberDataIntegerValue
```

Members

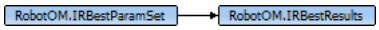
Members	Description
I_BMDIV_CALC_AS_SLENDER_Y = 1	Calculations of slender columns in direction Y (0 - no, 1 - yes).
I_BMDIV_SWAY_Y = 3	Sway structure in direction Y (0 - no, 1 - yes).
I_BMDIV_SWAY_Z = 4	Sway structure in direction Z (0 - no, 1 - yes).
I_BMDIV_USER_VALUE = 1000	The smallest identifier reserved for a user-defined value.
I_BMDIV_TYPE = 0	Assumes one of the values of the calculable type IRBestMemberType (see page 1695).
I_BMDIV_CALC_AS_SLENDER_Z = 2	Calculations of slender columns in direction Z (0 - no, 1 - yes).

Description

An identifier set of integer type parameters for the BestMemberData interface. .

I.13 IRBestResults

Class Hierarchy



C++

```
interface IRBestResults : IRBestParamSet;
```

C#

```
public interface IRBestResults : IRBestParamSet;
```

Visual Basic

```
Public Interface IRBestResults
```

Description

The interface describes calculation results of theoretical reinforcement for single section of RC member. Available values are described by the types: RBestResultsDoubleValue, RBestResultsIntegerValue and RBestResultsStringValue.

(c) 2014 Autodesk Inc. All rights reserved

I.14 IRBestResultsDoubleValue

C++

```
enum IRBestResultsDoubleValue;
```

C#

```
public enum IRBestResultsDoubleValue;
```

Visual Basic

```
Public Enum IRBestResultsDoubleValue
```

Members

Members	Description
I_BRDV_AS_TOP = 0	Top reinforcement area.
I_BRDV_AS_BOTTOM = 1	Bottom reinforcement area.
I_BRDV_AS_LEFT = 2	Left reinforcement area.
I_BRDV_AS_RIGHT = 3	Right reinforcement area.
I_BRDV_STIRR_SPACE = 9	Stirrup spacing.
I_BRDV_STIRR_SPACE_MOD = 10	Stirrup spacing with modularity taken into consideration.
I_BRDV_LAMBDA_Y = 13	Lambda coefficient for Y direction.
I_BRDV_LAMBDA_Z = 14	Lambda coefficient for Z direction.
I_BRDV_CRACK_WIDTH = 15	Cracking width.
I_BRDV_USER_VALUE = 1000	The smallest identifier reserved for user-defined values. .
I_BRDV_AS_MIN = 4	The minimum code-defined reinforcement area.
I_BRDV_AS_MAX = 5	The maximum code-defined reinforcement area.
I_BRDV_RO = 6	Reinforcement density.
I_BRDV_RO_MIN = 7	Minimum code-defined reinforcement density.
I_BRDV_RO_MAX = 8	Maximum code-defined reinforcement density.
I_BRDV_STIRR_SPACE_MIN = 11	Minimum code-defined stirrup spacing.
I_BRDV_STIRR_SPACE_MAX = 12	Maximum code-defined stirrup spacing.
I_BRDV_RIGIDITY = 16	Available since version 2.0.
I_BRDV_BAR_SPACING = 17	Available since version 2.0.
I_BRDV_AS_PROVIDED_TOP_MY = 18	Provided top reinforcement (My). Available since version 11.
I_BRDV_AS_PROVIDED_TOP_MZ = 19	Provided top reinforcement (Mz). Available since version 11.
I_BRDV_AS_PROVIDED_BOTTOM_MY = 20	Provided bottom reinforcement (My). Available since version 11.
I_BRDV_AS_PROVIDED_BOTTOM_MZ = 21	Provided bottom reinforcement (Mz). Available since version 11.
I_BRDV_CRACK_WIDTH_LT_TOP = 22	Available since version 12.5.
I_BRDV_CRACK_WIDTH_LT_BOTTOM = 23	Available since version 12.5.
I_BRDV_CRACK_WIDTH_ST_TOP = 24	Available since version 12.5.
I_BRDV_CRACK_WIDTH_ST_BOTTOM = 25	Available since version 12.5.
I_BRDV_STIFFNESS_TOP = 26	Available since version 12.5.
I_BRDV_STIFFNESS_BOTTOM = 27	Available since version 12.5.
I_BRDV_AS_MIN_TOP = 28	Available since version 12.5.
I_BRDV_AS_MIN_BOTTOM = 29	Available since version 12.5.

(c) 2014 Autodesk Inc. All rights reserved

Description

A set of identifiers of real-number parameters for the RBestResults interface. .

I.15 IRBestResultsIntegerValue

C++

```
enum IRBestResultsIntegerValue;
```

C#

```
public enum IRBestResultsIntegerValue;
```

Visual Basic

```
Public Enum IRBestResultsIntegerValue
```

Members

Members	Description
I_BRIV_BEND_CASE_IDX = 0	Index of the designing load case in the force vector for the calculations carried out for bending.
I_BRIV_BEND_ERROR_NUM = 3	The error number (RBestCalcErrors determines available error numbers) for calculations for bending.
I_BRIV_SHEAR_CASE_IDX = 1	Index of the designing load case in the force vector for the calculations carried out for shearing.
I_BRIV_TORSION_CASE_IDX = 2	Index of the designing load case in the force vector for the calculations carried out for torsion.
I_BRIV_USER_VALUE = 1000	The smallest identifier reserved for user-defined values .
I_BRIV_BEND_WARNING_NUM = 6	Warning bit mask for calculations carried out for bending (the bit values of the available warnings are determined by the interface RBestCalcWarnings).
I_BRIV_SHEAR_WARNING_NUM = 7	Warning bit mask for calculations carried out for shearing.
I_BRIV_TORSION_WARNING_NUM = 8	Warning bit mask for calculations carried out for torsion.
I_BRIV_SHEAR_ERROR_NUM = 4	Error number for calculations carried out for shearing.
I_BRIV_TORSION_ERROR_NUM = 5	Error number for calculations carried out for torsion.

Description

A set of identifiers of real-number parameters for the interface RBestResults.

I.16 IRBestCalcErrors

C++

```
enum IRBestCalcErrors;
```

C#

```
public enum IRBestCalcErrors;
```

Visual Basic

```
Public Enum IRBestCalcErrors
```

Members

Members	Description
I_BCE_NO_ERRORS = 0	Correct calculations.
I_BCE_NO_CALC = 1	No calculations.
I_BCE_SECTION_TOO_SMALL = 2	Section values are too small for a defined moment.

(c) 2014 Autodesk Inc. All rights reserved

I_BCE_SECTION_SHEAR = 3	Section values are too small for a given shear force.
I_BCE_BCKL_COLUMN = 4	Axial force reaches a critical value .
I_BCE_USER_ERROR = -1	An error (defined meaning).

Description

The set of identifiers describing errors in calculations of theoretical reinforcement. If there occurred an error during calculations, the information about this fact should be written in the object of the RBestResults type, transferred as a parameter of the GetResults function of the RBestCodeCalcEngine interface. To write the information, one should use the SetInteger function that introduces integral values into an object. As an argument determining the type of integral value, one should provide the relevant identifier of error type, while as the error value, one should provide the relevant value from the set RBestCalcErrors. .

I.17 IRBestCalcWarnings

C++

```
enum IRBestCalcWarnings;
```

C#

```
public enum IRBestCalcWarnings;
```

Visual Basic

```
Public Enum IRBestCalcWarnings
```

Members

Members	Description
I_BCW_NO_WARNINGS = 0	No warnings.
I_BCW_RO_MIN = 0x1	A reinforcement ratio too low.
I_BCW_RO_MAX = 0x2	A reinforcement ratio too high.
I_BCW_SPACE_BY_CODE = 0x4	Stirrup spacing calculated from a maximum stirrup spacing given in the code.
I_BCW_LAMBDA_MAX = 0x8	Slenderness exceeds a maximum value given in the code.
I_BCW_BREAK_MAX = 0x10	Cracking width exceeds a maximum value given in the code.
I_BCW_USER_WARNING = 0x20	Warning (defined meaning).
I_BCW_STEEL_EXC_RAR = 0x40	Admissible steel stress is exceeded (rare combination). Available since version 12.
I_BCW_CONCRETE_EXC_RAR = 0x80	Admissible concrete stress is exceeded (rare combination). Available since version 12.
I_BCW_CONCRETE_EXC_QPR = 0x100	Admissible concrete stress is exceeded (quasi-permanent combination). Available since version 12.
I_BCW_COT_THETA_CHANGED = 0x200	CotTHETA value is changed. Available since version 12.

Description

Warning bit masks for calculations of theoretical reinforcement. .

I.18 IRBestResultsStringValue

C++

```
enum IRBestResultsStringValue;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public enum IRBestResultsStringValue;
```

Visual Basic

```
Public Enum IRBestResultsStringValue
```

Members

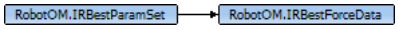
Members	Description
I_BRSV_BEND_ERROR_NAME = 0	A message text about a definable error in calculations for bending.
I_BRSV_BEND_WARNING_NAME = 3	A message text about a definable warning in calculations for bending.
I_BRSV_SHEAR_WARNING_NAME = 4	A message text about a definable warning in calculations for shearing.
I_BRSV_TORSION_WARNING_NAME = 5	A message text about a definable warning in calculations for torsion.
I_BRSV_SHEAR_ERROR_NAME = 1	A message text about a definable error in calculations for shearing.
I_BRSV_TORSION_ERROR_NAME = 2	A message text about a definable error in calculations for torsion.
I_BRSV_USER_VALUE = 1000	The smallest identifier reserved for a user-defined value .

Description

Set of identifiers of text parameters for the interface RBestResults.

I.19 IRBestForceData

Class Hierarchy



C++

```
interface IRBestForceData : IRBestParamSet;
```

C#

```
public interface IRBestForceData : IRBestParamSet;
```

Visual Basic

```
Public Interface IRBestForceData
```

Description

A set of forces acting in a bar section. .

I.19.1 IRBestForceData Members

The following tables list the members exposed by IRBestForceData.

Public Fields

	Name	Description
	LimitState (see page 1690)	Type of analyzed limit state.
	Values (see page 1690)	Values of forces acting in a bar section.

(c) 2014 Autodesk Inc. All rights reserved

Public Methods

	Name	Description
	Clear (see page 1673)	The function deletes all settings of parameters carried out by means of the group of functions Set*. .
	ClearDouble (see page 1673)	The function deletes the setting of the real-number type parameter with the given identifier.
	ClearInteger (see page 1674)	The function deletes the setting of the integer type parameter with the given identifier.
	ClearString (see page 1674)	The function deletes the setting of the text type parameter with the given identifier.
	GetDouble (see page 1674)	The function takes a real number, which is a parameter value with a given identifier. The function returns zero (False) if the parameter value with the indicated identifier was not set.
	GetInteger (see page 1674)	The function takes an integer number, which is a parameter value with a given identifier. It gives back zero (False) if the value of the parameter with the indicated identifier was not set.
	GetString (see page 1675)	The function takes a text (character string) which is a parameter value with the given identifier. It gives zero (False) back if the value of the parameter with the indicated identifier was not set. .
	IsValidDouble (see page 1675)	The function checks if the indicated real number is a correct value of the parameter with the indicated identifier. It returns zero (False) if the value is not correct. .
	IsValidInteger (see page 1675)	The function checks if the indicated integer is a correct value of the parameter with the indicated identifier. It returns zero (False) if the value is not correct. .
	IsValidString (see page 1675)	The function checks if the indicated text (character string) is a correct value of the parameter with the indicated identifier. It returns zero (False) if the value is not correct. .
	SetDouble (see page 1676)	The function sets a value of the given parameter if it is correct (according to IsValidDouble (see page 1675)). If the parameter value is incorrect, the function returns zero (False).
	SetInteger (see page 1676)	The function sets a value of the given parameter if it is correct (according to IsValidInteger (see page 1675)). If the parameter value is incorrect, the function returns zero (False).
	SetString (see page 1676)	The function sets the value of the given parameter, if correct (according to IsValidString (see page 1675)). If the parameter value is not correct, the function returns zero (False). .

I.19.2 IRBestForceData Fields

The fields of the IRBestForceData class are listed here.

Public Fields

	Name	Description
	LimitState (see page 1690)	Type of analyzed limit state.
	Values (see page 1690)	Values of forces acting in a bar section.

I.19.2.1 LimitState

C++

```
HRESULT get_LimitState(IRobotLimitState*);
```

C#

```
public IRobotLimitState LimitState { get; }
```

Visual Basic

```
Public ReadOnly LimitState As IRobotLimitState
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Type of analyzed limit state.

I.19.2.2 Values

C++

```
HRESULT get_Values(IRobotBarForceData**);
```

C#

```
public IRobotBarForceData Values { get; }
```

Visual Basic

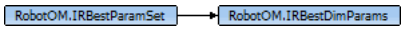
```
Public ReadOnly Values As IRobotBarForceData
```

Description

Values of forces acting in a bar section.

I.20 IRBestDimParams

Class Hierarchy



C++

```
interface IRBestDimParams : IRBestParamSet;
```

C#

```
public interface IRBestDimParams : IRBestParamSet;
```

Visual Basic

```
Public Interface IRBestDimParams
```

Description

A set of additional values parametrizing code (design) calculations. The RBestDimParamsIntegerValue type describes the set of available parameters. .

I.21 IRBestDimParamsIntegerValue

C++

```
enum IRBestDimParamsIntegerValue;
```

C#

```
public enum IRBestDimParamsIntegerValue;
```

Visual Basic

```
Public Enum IRBestDimParamsIntegerValue
```

Members

Members	Description
I_BDPIV_BEND_SIMPLE = 0	Calculations for simple bending (1 - on, 0 - off).
I_BDPIV_BEND_UNIAXIAL = 1	Calculations for offset compression and bending (1 - on, 0 - off).
I_BDPIV_BEND_BIAXIAL = 2	Calculations for offset biaxial compression (1 - on, 0 - off).

(c) 2014 Autodesk Inc. All rights reserved

I_BDPIV_SHEAR = 3	Calculations for shearing (1 - on, 0 - off).
I_BDPIV_TORSION = 4	Calculations for torsion (1 - on, 0 - off).
I_BDPIV_USER_VALUE = 1000	The smallest identifier reserved for the user-defined values .
I_BDPIV_CALCULATION_TYPE = 7	Available since version 2.0.
I_BDPIV_DIR_Y = 5	Available since version 2.0.
I_BDPIV_DIR_Z = 6	Available since version 2.0.
I_BDPIV_DIRECTION = 8	Available since version 2.0.
I_BDPIV_BEND = 9	Available since version 2.0.
I_BDPIV_LEVEL = 10	Available since version 2.0.

Description

Available identifiers of parameters of an integer type for the RBestDimParams interface.

I.22 IRBestCodeCalcEngine

Class Hierarchy

C++

```
interface IRBestCodeCalcEngine : IDispatch;
```

C#

```
public interface IRBestCodeCalcEngine;
```

Visual Basic

```
Public Interface IRBestCodeCalcEngine
```

Description

The interface describes the object performing code calculations of theoretical reinforcement for RC members.

I.22.1 IRBestCodeCalcEngine Members

The following tables list the members exposed by IRBestCodeCalcEngine.

Public Methods

	Name	Description
	Calculate (see page 1692)	The function performs calculations on a base of set parameters.
	GetResults (see page 1692)	The function returns the results of recently performed calculations to the indicated object.
	SetForces (see page 1693)	The function parametrizes the calculation process in the case of short beams and columns. It provides information about forces acting in a section (forces), as well as the position of the section along the member (pos [0 .. member length]).
	SetForcesArray (see page 1693)	
	SetForcesSlender (see page 1693)	The function parametrizes the calculation process in the case of slender columns. It provides information about forces acting in column cross-sections: at column ends (forces_begin, forces_end) and at the intermediate cross-section (forces_middle).
	SetGeometry (see page 1694)	The function parametrizes the calculation process. It provides information about the geometry of a bar section.
	SetParams (see page 1694)	The function parametrizes the calculation process. It transfers information concerning parameters for calculation of theoretical reinforcement (calc_params) as well as RC member parameters (memb_def).

(c) 2014 Autodesk Inc. All rights reserved

I.22.2 IRBestCodeCalcEngine Methods

The methods of the IRBestCodeCalcEngine class are listed here.

Public Methods

	Name	Description
	Calculate (see page 1692)	The function performs calculations on a base of set parameters.
	GetResults (see page 1692)	The function returns the results of recently performed calculations to the indicated object.
	SetForces (see page 1693)	The function parametrizes the calculation process in the case of short beams and columns. It provides information about forces acting in a section (forces), as well as the position of the section along the member (pos [0 .. member length]).
	SetForcesArray (see page 1693)	
	SetForcesSlender (see page 1693)	The function parametrizes the calculation process in the case of slender columns. It provides information about forces acting in column cross-sections: at column ends (forces_begin, forces_end) and at the intermediate cross-section (forces_middle).
	SetGeometry (see page 1694)	The function parametrizes the calculation process. It provides information about the geometry of a bar section.
	SetParams (see page 1694)	The function parametrizes the calculation process. It transfers information concerning parameters for calculation of theoretical reinforcement (calc_params) as well as RC member parameters (memb_def).

I.22.2.1 Calculate

C++

```
HRESULT Calculate(IRBestDimParams* _dim_params);
```

C#

```
public void Calculate(IRBestDimParams _dim_params);
```

Visual Basic

```
Public Sub Calculate(ByRef _dim_params As IRBestDimParams)
```

Description

The function performs calculations on a base of set parameters.

I.22.2.2 GetResults

C++

```
HRESULT GetResults(IRBestResults* _results);
```

C#

```
public void GetResults(IRBestResults _results);
```

Visual Basic

```
Public Sub GetResults(ByRef _results As IRBestResults)
```

Description

The function returns the results of recently performed calculations to the indicated object.

(c) 2014 Autodesk Inc. All rights reserved

I.22.2.3 SetForces

C++

```
HRESULT SetForces(IRobotObjectsArray* _forces, double _pos);
```

C#

```
public void SetForces(IRobotObjectsArray _forces, double _pos);
```

Visual Basic

```
Public Sub SetForces(ByRef _forces As IRobotObjectsArray, _pos As double)
```

Description

The function parametrizes the calculation process in the case of short beams and columns. It provides information about forces acting in a section (forces), as well as the position of the section along the member (pos [0 .. member length]).

I.22.2.4 SetForcesArray

C++

```
HRESULT SetForcesArray(SAFEARRAY* _forces, SAFEARRAY* _params, double _pos);
```

C#

```
public void SetForcesArray(Array _forces, Array _params, double _pos);
```

Visual Basic

```
Public Sub SetForcesArray(ByRef _forces As double)(), ByRef _params As int)(), _pos As
double)
```

Version

Available since version 12.

I.22.2.5 SetForcesSlender

C++

```
HRESULT SetForcesSlender(IRobotObjectsArray* _forces_begin, IRobotObjectsArray*
_forces_midle, IRobotObjectsArray* _forces_end);
```

C#

```
public void SetForcesSlender(IRobotObjectsArray _forces_begin, IRobotObjectsArray
_forces_midle, IRobotObjectsArray _forces_end);
```

Visual Basic

```
Public Sub SetForcesSlender(ByRef _forces_begin As IRobotObjectsArray, ByRef _forces_midle
As IRobotObjectsArray, ByRef _forces_end As IRobotObjectsArray)
```

Description

The function parametrizes the calculation process in the case of slender columns. It provides information about forces acting in column cross-sections: at column ends (forces_begin, forces_end) and at the intermediate cross-section (forces_middle).

I.22.2.6 SetGeometry

C++

```
HRESULT SetGeometry(IRConcrBarSectionData* _sect_geo);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public void SetGeometry(IRConcrBarSectionData _sect_geo);
```

Visual Basic

```
Public Sub SetGeometry(ByRef _sect_geo As IRConcrBarSectionData)
```

Description

The function parametrizes the calculation process. It provides information about the geometry of a bar section.

I.22.2.7 SetParams

C++

```
HRESULT SetParams(IRBestCalcParamsData* _calc_params, IRBestMemberData* _memb_def);
```

C#

```
public void SetParams(IRBestCalcParamsData _calc_params, IRBestMemberData _memb_def);
```

Visual Basic

```
Public Sub SetParams(ByRef _calc_params As IRBestCalcParamsData, ByRef _memb_def As
IRBestMemberData)
```

Description

The function parametrizes the calculation process. It transfers information concerning parameters for calculation of theoretical reinforcement (calc_params) as well as RC member parameters (memb_def).

I.23 IRBestMemberDataStringValue

C++

```
enum IRBestMemberDataStringValue;
```

C#

```
public enum IRBestMemberDataStringValue;
```

Visual Basic

```
Public Enum IRBestMemberDataStringValue
```

Members

Members	Description
I_BMDSV_USER_VALUE = 1000	The smallest identifier reserved for the user .

Description

The set of identifiers of text parameters for the RBestMemberData interface. .

I.24 IRBestMemberType

C++

```
enum IRBestMemberType;
```

C#

```
public enum IRBestMemberType;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Enum IRBestMemberType
```

Description

A set of identifiers for a type of RC member or plate.

I.25 IRBestPlateCalcParamsDlg

Class Hierarchy

RobotOM.IRBestPlateCalcParamsDlg

C++

```
interface IRBestPlateCalcParamsDlg : IDispatch;
```

C#

```
public interface IRBestPlateCalcParamsDlg;
```

Visual Basic

```
Public Interface IRBestPlateCalcParamsDlg
```

I.25.1 IRBestPlateCalcParamsDlg Members

The following tables list the members exposed by IRBestPlateCalcParamsDlg.

Public Methods

	Name	Description
	DoModal (see page 1695)	Available since version 1.7.
	SetStandard (see page 1696)	Available since version 1.7.

I.25.2 IRBestPlateCalcParamsDlg Methods

The methods of the IRBestPlateCalcParamsDlg class are listed here.

Public Methods

	Name	Description
	DoModal (see page 1695)	Available since version 1.7.
	SetStandard (see page 1696)	Available since version 1.7.

I.25.2.1 DoModal

C++

```
HRESULT DoModal(IRBestCalcParamsData* _params, BSTR _label_name);
```

C#

```
public void DoModal(IRBestCalcParamsData _params, String _label_name);
```

Visual Basic

```
Public Sub DoModal(ByRef _params As IRBestCalcParamsData, _label_name As String)
```

Description

Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

I.25.2.2 SetStandard

C++

```
HRESULT SetStandard(IRBestCalcParamsData* _params);
```

C#

```
public void SetStandard(IRBestCalcParamsData _params);
```

Visual Basic

```
Public Sub SetStandard(ByRef _params As IRBestCalcParamsData)
```

Description

Available since version 1.7.

I.26 IRBestCodeCalculationType

C++

```
enum IRBestCodeCalculationType;
```

C#

```
public enum IRBestCodeCalculationType;
```

Visual Basic

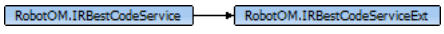
```
Public Enum IRBestCodeCalculationType
```

Members

Members	Description
I_BCCT_PLATE = 0	Available since version 1.7.
I_BCCT_MEMBER = 1	Available since version 1.7.

I.27 IRBestCodeServiceExt

Class Hierarchy



C++

```
interface IRBestCodeServiceExt : IRBestCodeService;
```

C#

```
public interface IRBestCodeServiceExt : IRBestCodeService;
```

Visual Basic

```
Public Interface IRBestCodeServiceExt
```

Description

Available since version 2.0.

I.27.1 IRBestCodeServiceExt Members

The following tables list the members exposed by IRBestCodeServiceExt.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	CalcEngine (see page 1668)	The object performing code calculations.
	CalcParamsDlg (see page 1668)	A dialog box that allows to edit calculation parameters.
	MemberDlg (see page 1668)	A dialog box that allows to edit RC member parameters .

Public Methods

	Name	Description
	GetCalcParamsDlg (see page 1697)	Function returns the interface to the appropriate dialog box for calculation parameter definition. Available since version 2.0.
	IsServed (see page 1697)	Available since version 2.0.

I.27.2 IRBestCodeServiceExt Methods

The methods of the IRBestCodeServiceExt class are listed here.

Public Methods

	Name	Description
	GetCalcParamsDlg (see page 1697)	Function returns the interface to the appropriate dialog box for calculation parameter definition. Available since version 2.0.
	IsServed (see page 1697)	Available since version 2.0.

I.27.2.1 GetCalcParamsDlg

C++

```
HRESULT GetCalcParamsDlg(IRBestCodeCalculationType _calc_type, IUnknown* ret);
```

C#

```
public IUnknown GetCalcParamsDlg(IRBestCodeCalculationType _calc_type);
```

Visual Basic

```
Public Function GetCalcParamsDlg(_calc_type As IRBestCodeCalculationType) As IUnknown
```

Description

Function returns the interface to the appropriate dialog box for calculation parameter definition. Available since version 2.0.

I.27.2.2 IsServed

C++

```
HRESULT IsServed(IRBestCodeCalculationType _calc_type, VARIANT_BOOL* ret);
```

C#

```
public bool IsServed(IRBestCodeCalculationType _calc_type);
```

Visual Basic

```
Public Function IsServed(_calc_type As IRBestCodeCalculationType) As Boolean
```

Description

Available since version 2.0.

(c) 2014 Autodesk Inc. All rights reserved

I.28 IRBestCalculationType

C++

```
enum IRBestCalculationType;
```

C#

```
public enum IRBestCalculationType;
```

Visual Basic

```
Public Enum IRBestCalculationType
```

I.29 IRBestDimParamsDoubleValue

C++

```
enum IRBestDimParamsDoubleValue;
```

C#

```
public enum IRBestDimParamsDoubleValue;
```

Visual Basic

```
Public Enum IRBestDimParamsDoubleValue
```

Members

Members	Description
I_BDPDV_CONCRETE_FC = 11	Available since version 2.0.

I.30 IRBestBendType

C++

```
enum IRBestBendType;
```

C#

```
public enum IRBestBendType;
```

Visual Basic

```
Public Enum IRBestBendType
```

I.31 IRBestDirection

C++

```
enum IRBestDirection;
```

C#

```
public enum IRBestDirection;
```

Visual Basic

```
Public Enum IRBestDirection
```

(c) 2014 Autodesk Inc. All rights reserved

I.32 IRBestLevel

C++

```
enum IRBestLevel;
```

C#

```
public enum IRBestLevel;
```

Visual Basic

```
Public Enum IRBestLevel
```

Members

Members	Description
I_BL_DOWN = 1	Available since version 2.0.

I.33 IRBestForceDataDoubleValue

C++

```
enum IRBestForceDataDoubleValue;
```

C#

```
public enum IRBestForceDataDoubleValue;
```

Visual Basic

```
Public Enum IRBestForceDataDoubleValue
```

Members

Members	Description
I_BFDDV_COEF_LONG = 0	Available since version 2.0.
I_BFDDV_COEF_LONG2TOT = 1	Available since version 12.5.

I.34 IRBestCoordSystem

C++

```
enum IRBestCoordSystem;
```

C#

```
public enum IRBestCoordSystem;
```

Visual Basic

```
Public Enum IRBestCoordSystem
```

Members

Members	Description
I_BCS_CARTESIAN = 1	Available since version 2.0.
I_BCS_POLAR = 2	Available since version 2.0.

I.35 IRBestForceDataIntegerValue

C++

```
enum IRBestForceDataIntegerValue;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public enum IRBestForceDataIntegerValue;
```

Visual Basic

```
Public Enum IRBestForceDataIntegerValue
```

Members

Members	Description
I_BFDIV_SLS_COMB_TYPE = 0	Available from version 11.0. Available since version 11.

Version

Available since version 11.

I.36 IRBestForceDataSLSCombType

C++

```
enum IRBestForceDataSLSCombType;
```

C#

```
public enum IRBestForceDataSLSCombType;
```

Visual Basic

```
Public Enum IRBestForceDataSLSCombType
```

Members

Members	Description
I_BFDSLS_COMB_TYPE_NONE = 0	Available since version 11.
I_BFDSLS_COMB_TYPE_RAR = 1	Available since version 11.
I_BFDSLS_COMB_TYPE_FRE = 2	Available since version 11.
I_BFDSLS_COMB_TYPE_QPR = 3	Available since version 11.

Description

Available since version 11.0.

Version

Available since version 11.

I.37 IRBestCodeService2

Class Hierarchy

C++

```
interface IRBestCodeService2 : IDispatch;
```

C#

```
public interface IRBestCodeService2;
```

Visual Basic

```
Public Interface IRBestCodeService2
```

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 12.5.

I.37.1 IRBestCodeService2 Members

The following tables list the members exposed by IRBestCodeService2.

Public Fields

	Name	Description
	RobotVersion (see page 1701)	
	ShowBucklingButton (see page 1701)	

I.37.2 IRBestCodeService2 Fields

The fields of the IRBestCodeService2 class are listed here.

Public Fields

	Name	Description
	RobotVersion (see page 1701)	
	ShowBucklingButton (see page 1701)	

I.37.2.1 RobotVersion

C++

```
HRESULT get_RobotVersion( BSTR*);
```

C#

```
public  BSTR RobotVersion { get; }
```

Visual Basic

```
Public ReadOnly RobotVersion As  BSTR
```

Version

Available since version 12.5.

I.37.2.2 ShowBucklingButton

C++

```
HRESULT get_ShowBucklingButton(VARIANT_BOOL*);
```

C#

```
public bool ShowBucklingButton { get; }
```

Visual Basic

```
Public ReadOnly ShowBucklingButton As Boolean
```

Version

Available since version 12.5.

(c) 2014 Autodesk Inc. All rights reserved

II Plate and shell reinforcement

Enumerations

	Name	Description
	IRConcrReinforceDirection (see page 1709)	Available reinforcement directions for plates/shells.
	IRConcr_PN_SteelGrades (see page 1721)	Reinforcing steel grades for Polish codes: PN-84 and PN-99.
	IRConcr_PN84_ConcreteGrades (see page 1721)	Concrete class for Polish code PN-84.
	IRConcr_PN84_ExposureRatings (see page 1722)	Types of exposure rating defined for the Polish code PN-84. .
	IRConcr_PN99_ExposureRatings (see page 1722)	Types of exposure rating defined for the Polish code PN-99. .
	IRConcr_PN99_ConcreteGrades (see page 1730)	Concrete classes for Polish code PN-99.
	IRConcr_BAEL_CrackingType (see page 1739)	
	IRConcr_BAEL_EnvironmentType (see page 1739)	
	IRConcr_BAEL_WaterLevel (see page 1739)	
	IRConcr_BAEL_ConcreteGrades (see page 1740)	
	IRConcr_BAEL_SteelGrades (see page 1740)	
	IRConcr_PN84_HumidityType (see page 1740)	Types of humidity defined for the PN-84 code .
	IRConcr_ACI318_LoadActionPeriodType (see page 1748)	Available duration times of variable long-term loads. .
	IRConcr_ACI318_SteelGrades (see page 1748)	
	IRConcr_ACI318_BarDim (see page 1750)	Available bar diameters for the code ACI 318/99.
	IRConcr_ACI318_MetricBarDIm (see page 1751)	Available bar diameters for the code ACI 318/99 metric.
	IRConcrReinforceBarType (see page 1751)	Types (designations) of reinforcing bars singled out during reinforcement definition. .
	IRConcr_BS8110_ExposureRatings (see page 1758)	Exposure types defined for the code BS 8110. .
	IRConcr_BS8110_ConcreteAge (see page 1758)	Predicted values of concrete age.
	IRConcr_BS8110_ConcreteGrades (see page 1759)	Concrete classes according to the code BS 8110. .
	IRConcr_BS8110_PartialSafetyFactors (see page 1760)	Safety factor types.
	IRConcrReinforceCalcType (see page 1765)	
	IRConcr_SNIP_SteelGrades (see page 1771)	
	IRConcr_SNIP_ConcreteGrades (see page 1771)	
	IRConcr_SNIP_ConcreteTypes (see page 1771)	

(c) 2014 Autodesk Inc. All rights reserved

	IRConcr_SNIP_CuringMethods (see page 1771)	
	IRConcr_SNIP_Exposure (see page 1774)	
	IRConcr_EC2_ConcreteGrades (see page 1774)	Concrete classes according to Eurocode 2.
	IRConcr_EC2_ITALIAN_SteelGrades (see page 1774)	
	IRConcr_EC2_ExposureRatings (see page 1774)	Exposure types according to Eurocode 2.
	IRConcr_EC2_NAD (see page 1779)	
	IRConcrMinimumReinforcementType (see page 1779)	
	IRConcr_IS2000_EnvironmentType (see page 1780)	Exposure types defined for the IS 456:2000 code.

Interfaces

	Name	Description
	IRConcrReinforceData (see page 1704)	Definition of plate and shell reinforcement type includes parameters that are code-independent.
	IRConcrConcreteParams (see page 1709)	Data characterizing concrete. .
	IRConcrSteelParams (see page 1712)	Set of parameters describing reinforcing steel properties.
	IRConcr_PN84_ReinforceData (see page 1715)	Parameters of plate and shell reinforcement for the PN-84 code. Steel grades are described by the identifier set RConr_PN_SteeGrades, concrete classes by RConcr_PN84_ConcreteGrades, whereas types of exposure rating RConcr_PN84_ExposureRatings.
	IRConcr_PN99_ReinforceData (see page 1723)	Reinforcement parameters for the Polish code PN-99. Steel grades are described by the identifier set RConr_PN_SteeGrades, concrete classes by RConcr_PN99_ConcreteGrades, whereas types of exposure rating by RConcr_PN99_ExposureRatings. .
	IRConcr_BAEL_ReinforceData (see page 1731)	Parameters of plate and shell reinforcement according to the BAEL code (91 and 99). Steel grades are determined by the identifiers from the set RConcr_BAEL_SteelGrades whereas concrete classes by RConcr_BAEL_ConcreteGrades. .
	IRConcr_ACI318_ReinforceData (see page 1741)	Interface of reinforcement parameters for the codes: "ACI 318/99" and "ACI 318/99 metric". .
	IRConcr_ACI318_ConcreteParams (see page 1749)	Parameters describing concrete for the code ACI 318/99. .
	IRConcr_BS8110_ReinforceData (see page 1752)	Interface of reinforcement parameters for the code BS8110.
	IRConcrSlabRequiredReinfCalcParams (see page 1760)	Calculation parameters for the required (theoretical) reinforcement for plates and shells. .
	IRConcrSlabRequiredReinfEngine (see page 1763)	Calculation module for required (theoretical) reinforcement of plates and shells. .
	IRConcrReinforceData2 (see page 1765)	
	IRConcrReinforceDataMain (see page 1767)	
	IRConcr_SNIP_ReinforceData (see page 1769)	Interface of plate/shell reinforcement parameters for the code "SNIP 2.03.01-84".
	IRConcr_SNIP_ConcreteParams (see page 1772)	
	IRConcr_EC2_ReinforceData (see page 1775)	

(c) 2014 Autodesk Inc. All rights reserved

	IRConcr_IS2000_ReinforceData (see page 1780)	Reinforcement parameters for the Indian code IS 456:2000.
	IRConcrPlateCodeService2 (see page 1787)	

II.1 IRConcrReinforceData

Class Hierarchy

RobotOM.IRConcrReinforceData

C++

```
interface IRConcrReinforceData : IDispatch;
```

C#

```
public interface IRConcrReinforceData;
```

Visual Basic

```
Public Interface IRConcrReinforceData
```

Description

Definition of plate and shell reinforcement type includes parameters that are code-independent.

II.1.1 IRConcrReinforceData Members

The following tables list the members exposed by IRConcrReinforceData.

Public Fields

	Name	Description
	BarDim_D1_Bot (see page 1705)	Diameter of reinforcing bars of bottom main reinforcement Available since version 1.7.
	BarDim_D1_Up (see page 1705)	Diameter of reinforcing bars of top main reinforcement Available since version 1.7.
	BarDim_D2_Bot (see page 1706)	Reinforcing bar diameter of bottom reinforcement in the direction perpendicular to the main reinforcement Available since version 1.7.
	BarDim_D2_Up (see page 1706)	Reinforcing bar diameter of top reinforcement in the direction perpendicular to the main reinforcement Available since version 1.7.
	CodeName (see page 1706)	Code name Available since version 1.7.
	Concrete (see page 1706)	Available since version 1.7.
	Cover_Bot (see page 1707)	Cover - bottom reinforcement Available since version 1.7.
	Cover_Up (see page 1707)	Cover - top reinforcement Available since version 1.7.
	ReinforcingSteel (see page 1707)	Available since version 1.7.

Public Methods

	Name	Description
	GetMainDirection (see page 1708)	Function returns the main reiforcement direction. Interpretation of returned coordinate values depends on the manner of determinig direction (direction vector coordinates, if the direction is determined by means of the vector, ALONG_VECTOR, or the point coordinates in the polar system - POLAR_POINT). Available since version 1.7.
	SetMainDirection (see page 1708)	Function defines the main reiforcement direction. Parameter interpretation depends on the manner of determinig direction (direction vector coordinates, if the direction is determined by means of the vector, ALONG_VECTOR, or the point coordinates in the polar system - POLAR_POINT, in the remaining cases the coordinate values are ignored). Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

II.1.2 IRConcrReinforceData Fields

The fields of the IRConcrReinforceData class are listed here.

Public Fields

	Name	Description
	BarDim_D1_Bot (see page 1705)	Diameter of reinforcing bars of bottom main reinforcement Available since version 1.7.
	BarDim_D1_Up (see page 1705)	Diameter of reinforcing bars of top main reinforcement Available since version 1.7.
	BarDim_D2_Bot (see page 1706)	Reinforcing bar diameter of bottom reinforcement in the direction perpendicular to the main reinforcement Available since version 1.7.
	BarDim_D2_Up (see page 1706)	Reinforcing bar diameter of top reinforcement in the direction perpendicular to the main reinforcement Available since version 1.7.
	CodeName (see page 1706)	Code name Available since version 1.7.
	Concrete (see page 1706)	Available since version 1.7.
	Cover_Bot (see page 1707)	Cover - bottom reinforcement Available since version 1.7.
	Cover_Up (see page 1707)	Cover - top reinforcement Available since version 1.7.
	ReinforcingSteel (see page 1707)	Available since version 1.7.

II.1.2.1 BarDim_D1_Bot

C++

```
HRESULT get_BarDim_D1_Bot(double*);
HRESULT put_BarDim_D1_Bot(double);
```

C#

```
public double BarDim_D1_Bot { get; set; }
```

Visual Basic

```
Public BarDim_D1_Bot As double
```

Description

Diameter of reinforcing bars of bottom main reinforcement Available since version 1.7.

II.1.2.2 BarDim_D1_Up

C++

```
HRESULT get_BarDim_D1_Up(double*);
HRESULT put_BarDim_D1_Up(double);
```

C#

```
public double BarDim_D1_Up { get; set; }
```

Visual Basic

```
Public BarDim_D1_Up As double
```

Description

Diameter of reinforcing bars of top main reinforcement Available since version 1.7.

II.1.2.3 BarDim_D2_Bot

C++

```
HRESULT get_BarDim_D2_Bot(double*);
HRESULT put_BarDim_D2_Bot(double);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double BarDim_D2_Bot { get; set; }
```

Visual Basic

```
Public BarDim_D2_Bot As double
```

Description

Reinforcing bar diameter of bottom reinforcement in the direction perpendicular to the main reinforcement Available since version 1.7.

II.1.2.4 BarDim_D2_Up

C++

```
HRESULT get_BarDim_D2_Up(double*);
HRESULT put_BarDim_D2_Up(double);
```

C#

```
public double BarDim_D2_Up { get; set; }
```

Visual Basic

```
Public BarDim_D2_Up As double
```

Description

Reinforcing bar diameter of top reinforcement in the direction perpendicular to the main reinforcement Available since version 1.7.

II.1.2.5 CodeName

C++

```
HRESULT get_CodeName(BSTR*);
```

C#

```
public String CodeName { get; }
```

Visual Basic

```
Public ReadOnly CodeName As String
```

Description

Code name Available since version 1.7.

II.1.2.6 Concrete

C++

```
HRESULT get_Concrete(IRConcrConcreteParams**);
```

C#

```
public IRConcrConcreteParams Concrete { get; }
```

Visual Basic

```
Public ReadOnly Concrete As IRConcrConcreteParams
```

Description

Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

II.1.2.7 Cover_Bot

C++

```
HRESULT get_Cover_Bot(double*);
HRESULT put_Cover_Bot(double);
```

C#

```
public double Cover_Bot { get; set; }
```

Visual Basic

```
Public Cover_Bot As double
```

Description

Cover - bottom reinforcement Available since version 1.7.

II.1.2.8 Cover_Up

C++

```
HRESULT get_Cover_Up(double*);
HRESULT put_Cover_Up(double);
```

C#

```
public double Cover_Up { get; set; }
```

Visual Basic

```
Public Cover_Up As double
```

Description

Cover - top reinforcement Available since version 1.7.

II.1.2.9 ReinforcingSteel

C++

```
HRESULT get_ReinforcingSteel(IRConcrSteelParams**);
```

C#

```
public IRConcrSteelParams ReinforcingSteel { get; }
```

Visual Basic

```
Public ReadOnly ReinforcingSteel As IRConcrSteelParams
```

Description

Available since version 1.7.

II.1.3 IRConcrReinforceData Methods

The methods of the IRConcrReinforceData class are listed here.

Public Methods

	Name	Description
	GetMainDirection (see page 1708)	Function returns the main reiforcement direction. Interpretation of returned coordinate values depends on the manner of determinig direction (direction vector coordinates, if the direction is determined by means of the vector, ALONG_VECTOR, or the point coordinates in the polar system - POLAR_POINT). Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

	SetMainDirection (see page 1708)	Function defines the main reiforcement direction. Parameter interpretation depends on the manner of determinig direction (direction vector coordinates, if the direction is determined by means of the vector, ALONG_VECTOR, or the point coordinates in the polar system - POLAR_POINT, in the remaining cases the coordinate values are ignored). Available since version 1.7.

II.1.3.1 GetMainDirection

C++

```
HRESULT GetMainDirection(double* _x, double* _y, double* _z, IRConcrReinforceDirection*
ret);
```

C#

```
public IRConcrReinforceDirection GetMainDirection(double* _x, double* _y, double* _z);
```

Visual Basic

```
Public Function GetMainDirection(ByRef _x As double*, ByRef _y As double*, ByRef _z As
double*) As IRConcrReinforceDirection
```

Description

Function returns the main reiforcement direction. Interpretation of returned coordinate values depends on the manner of determinig direction (direction vector coordinates, if the direction is determined by means of the vector, ALONG_VECTOR, or the point coordinates in the polar system - POLAR_POINT). Available since version 1.7.

II.1.3.2 SetMainDirection

C++

```
HRESULT SetMainDirection(IRConcrReinforceDirection _dir_type, double _x, double _y, double
_z);
```

C#

```
public void SetMainDirection(IRConcrReinforceDirection _dir_type, double _x, double _y,
double _z);
```

Visual Basic

```
Public Sub SetMainDirection(_dir_type As IRConcrReinforceDirection, _x As double, _y As
double, _z As double)
```

Description

Function defines the main reiforcement direction. Parameter interpretation depends on the manner of determinig direction (direction vector coordinates, if the direction is determined by means of the vector, ALONG_VECTOR, or the point coordinates in the polar system - POLAR_POINT, in the remaining cases the coordinate values are ignored). Available since version 1.7.

II.2 IRConcrReinforceDirection

C++

```
enum IRConcrReinforceDirection;
```

C#

```
public enum IRConcrReinforceDirection;
```

Visual Basic

```
Public Enum IRConcrReinforceDirection
```

(c) 2014 Autodesk Inc. All rights reserved

Members

Members	Description
I_CRD_ALONG_X = 0	Available since version 1.7.
I_CRD_ALONG_Y = 1	Available since version 1.7.
I_CRD_ALONG_Z = 2	Available since version 1.7.
I_CRD_CARTESIAN_ALONG_VECTOR = 3	Direction determined in the Cartesian system by means of the specified vector Available since version 1.7.
I_CRD_POLAR_POINT = 4	Direction determined in the polar system by means of the specified point Available since version 1.7.
I_CRD_AUTOMATIC = 5	Automatic direction compatible to local X axis of the panel. Available since version 7.5.

Description

Available reinforcement directions for plates/shells.

II.3 IRConcrConcreteParams

Class Hierarchy

RobotOM.IRConcrConcreteParams

C++

```
interface IRConcrConcreteParams : IDispatch;
```

C#

```
public interface IRConcrConcreteParams;
```

Visual Basic

```
Public Interface IRConcrConcreteParams
```

Description

Data characterizing concrete. .

II.3.1 IRConcrConcreteParams Members

The following tables list the members exposed by IRConcrConcreteParams.

Public Fields

	Name	Description
	Ec (see page 1710)	Elasticity modulus for concrete; parameter assumes value that equals -1 when the value of the appropriate material assigned to a panel is taken for calculations Available since version 1.7.
	Fc (see page 1711)	Characteristic compressive strength; parameter assumes value that equals -1 when the value of the appropriate material assigned to a panel is taken for calculations Available since version 1.7.
	Fc_calc (see page 1711)	Design compressive strength (Fc_calc = Fc (see page 1711) / gammaC, where gammaC denotes safety factor); parameter assumes value that equals -1 when the value of the appropriate material assigned to a panel is taken for calculations Available since version 1.7.
	Fcj (see page 1711)	Value that is used only in the French code and denotes Fc (see page 1711) (characteristic compressive strength); parameter assumes value that equals -1 when the value of the appropriate material assigned to a panel is taken for calculations Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

	Ft (see page 1712)	Characteristic tensile strength; parameter assumes value that equals -1 when the value of the appropriate material assigned to a panel is taken for calculations Available since version 1.7.
	Grade (see page 1712)	Identifier denoting a predefined concrete class (according to the code it should be compared with the appropriate set of defined identifiers) Available since version 1.7.

II.3.2 IRConcrConcreteParams Fields

The fields of the IRConcrConcreteParams class are listed here.

Public Fields

	Name	Description
	Ec (see page 1710)	Elasticity modulus for concrete; parameter assumes value that equals -1 when the value of the appropriate material assigned to a panel is taken for calculations Available since version 1.7.
	Fc (see page 1711)	Characteristic compressive strength; parameter assumes value that equals -1 when the value of the appropriate material assigned to a panel is taken for calculations Available since version 1.7.
	Fc_calc (see page 1711)	Design compressive strength (Fc_calc = Fc (see page 1711) / gammaC, where gammaC denotes safety factor); parameter assumes value that equals -1 when the value of the appropriate material assigned to a panel is taken for calculations Available since version 1.7.
	Fcj (see page 1711)	Value that is used only in the French code and denotes Fc (see page 1711) (characteristic compressive strength); parameter assumes value that equals -1 when the value of the appropriate material assigned to a panel is taken for calculations Available since version 1.7.
	Ft (see page 1712)	Characteristic tensile strength; parameter assumes value that equals -1 when the value of the appropriate material assigned to a panel is taken for calculations Available since version 1.7.
	Grade (see page 1712)	Identifier denoting a predefined concrete class (according to the code it should be compared with the appropriate set of defined identifiers) Available since version 1.7.

II.3.2.1 Ec

C++

```
HRESULT get_Ec(double*);
HRESULT put_Ec(double);
```

C#

```
public double Ec { get; set; }
```

Visual Basic

```
Public Ec As double
```

Description

Elasticity modulus for concrete; parameter assumes value that equals -1 when the value of the appropriate material assigned to a panel is taken for calculations

Available since version 1.7.

II.3.2.2 Fc

C++

```
HRESULT get_Fc(double*);
HRESULT put_Fc(double);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double Fc { get; set; }
```

Visual Basic

```
Public Fc As double
```

Description

Characteristic compressive strength; parameter assumes value that equals -1 when the value of the appropriate material assigned to a panel is taken for calculations Available since version 1.7.

II.3.2.3 Fc_calc

C++

```
HRESULT get_Fc_calc(double*);
HRESULT put_Fc_calc(double);
```

C#

```
public double Fc_calc { get; set; }
```

Visual Basic

```
Public Fc_calc As double
```

Description

Design compressive strength (Fc_calc = Fc (see page 1711) / gammaC, where gammaC denotes safety factor); parameter assumes value that equals -1 when the value of the appropriate material assigned to a panel is taken for calculations Available since version 1.7.

II.3.2.4 Fcj

C++

```
HRESULT get_Fcj(double*);
HRESULT put_Fcj(double);
```

C#

```
public double Fcj { get; set; }
```

Visual Basic

```
Public Fcj As double
```

Description

Value that is used only in the French code and denotes Fc (see page 1711) (characteristic compressive strength); parameter assumes value that equals -1 when the value of the appropriate material assigned to a panel is taken for calculations Available since version 1.7.

II.3.2.5 Ft

C++

```
HRESULT get_Ft(double*);
HRESULT put_Ft(double);
```

C#

```
public double Ft { get; set; }
```

Visual Basic

```
Public Ft As double
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Characteristic tensile strength; parameter assumes value that equals -1 when the value of the appropriate material assigned to a panel is taken for calculations Available since version 1.7.

II.3.2.6 Grade

C++

```
HRESULT get_Grade(short*);
HRESULT put_Grade(short);
```

C#

```
public short Grade { get; set; }
```

Visual Basic

```
Public Grade As short
```

Description

Identifier denoting a predefined concrete class (according to the code it should be compared with the appropriate set of defined identifiers) Available since version 1.7.

II.4 IRConcrSteelParams

Class Hierarchy

RobotOM.IRConcrSteelParams

C++

```
interface IRConcrSteelParams : IDispatch;
```

C#

```
public interface IRConcrSteelParams;
```

Visual Basic

```
Public Interface IRConcrSteelParams
```

Description

Set of parameters describing reinforcing steel properties.

II.4.1 IRConcrSteelParams Members

The following tables list the members exposed by IRConcrSteelParams.

Public Fields

	Name	Description
	Ey (see page 1713)	Value of limit elastic strain Available since version 1.7.
	Ft (see page 1713)	Characteristic tensile strength Available since version 1.7.
	Ft_calc (see page 1714)	Calculation tensile strength (Ft_calc = Ft (see page 1713) / gammaS, where gammaS is a safety factor) Available since version 1.7.
	Fy (see page 1714)	Characteristic yield point Available since version 1.7.
	Fy_calc (see page 1714)	Calculation yield point(Fy_calc = Fy (see page 1714) / gammaS, where gammaS is a safety factor) Available since version 1.7.
	Grade (see page 1715)	Identifier denoting a predefined steel grade (according to code it should be compared with the appropriate set of defined identifiers) Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

II.4.2 IRConcrSteelParams Fields

The fields of the IRConcrSteelParams class are listed here.

Public Fields

	Name	Description
	Ey (see page 1713)	Value of limit elastic strain Available since version 1.7.
	Ft (see page 1713)	Characteristic tensile strength Available since version 1.7.
	Ft_calc (see page 1714)	Calculation tensile strength (Ft_calc = Ft (see page 1713) / gammaS, where gammaS is a safety factor) Available since version 1.7.
	Fy (see page 1714)	Characteristic yield point Available since version 1.7.
	Fy_calc (see page 1714)	Calculation yield point(Fy_calc = Fy (see page 1714) / gammaS, where gammaS is a safety factor) Available since version 1.7.
	Grade (see page 1715)	Identifier denoting a predefined steel grade (according to code it should be compared with the appropriate set of defined identifiers) Available since version 1.7.

II.4.2.1 Ey

C++

```
HRESULT get_Ey(double*);
HRESULT put_Ey(double);
```

C#

```
public double Ey { get; set; }
```

Visual Basic

```
Public Ey As double
```

Description

Value of limit elastic strain Available since version 1.7.

II.4.2.2 Ft

C++

```
HRESULT get_Ft(double*);
HRESULT put_Ft(double);
```

C#

```
public double Ft { get; set; }
```

Visual Basic

```
Public Ft As double
```

Description

Characteristic tensile strength Available since version 1.7.

II.4.2.3 Ft_calc

C++

```
HRESULT get_Ft_calc(double*);
HRESULT put_Ft_calc(double);
```

C#

```
public double Ft_calc { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Ft_calc As double
```

Description

Calculation tensile strength (Ft_calc = Ft (see page 1713) / gammaS, where gammaS is a safety factor) Available since version 1.7.

II.4.2.4 Fy

C++

```
HRESULT get_Fy(double*);
HRESULT put_Fy(double);
```

C#

```
public double Fy { get; set; }
```

Visual Basic

```
Public Fy As double
```

Description

Characteristic yield point Available since version 1.7.

II.4.2.5 Fy_calc

C++

```
HRESULT get_Fy_calc(double*);
HRESULT put_Fy_calc(double);
```

C#

```
public double Fy_calc { get; set; }
```

Visual Basic

```
Public Fy_calc As double
```

Description

Calculation yield point(Fy_calc = Fy (see page 1714) / gammaS, where gammaS is a safety factor) Available since version 1.7.

II.4.2.6 Grade

C++

```
HRESULT get_Grade(short*);
HRESULT put_Grade(short);
```

C#

```
public short Grade { get; set; }
```

Visual Basic

```
Public Grade As short
```

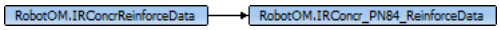
Description

Identifier denoting a predefined steel grade (according to code it should be compared with the appropriate set of defined identifiers) Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

II.5 IRConcr_PN84_ReinforceData

Class Hierarchy



C++

```
interface IRConcr_PN84_ReinforceData : IRConcrReinforceData;
```

C#

```
public interface IRConcr_PN84_ReinforceData : IRConcrReinforceData;
```

Visual Basic

```
Public Interface IRConcr_PN84_ReinforceData
```

Description

Parameters of plate and shell reinforcement for the PN-84 code. Steel grades are described by the identifier set RConr_PN_SteeGrades, concrete classes by RConcr_PN84_ConcreteGrades, whereas types of exposure rating RConcr_PN84_ExposureRatings.

II.5.1 IRConcr_PN84_ReinforceData Members

The following tables list the members exposed by IRConcr_PN84_ReinforceData.

Public Fields

	Name	Description
	BarDim_D1_Bot (see page 1705)	Diameter of reinforcing bars of bottom main reinforcement Available since version 1.7.
	BarDim_D1_Up (see page 1705)	Diameter of reinforcing bars of top main reinforcement Available since version 1.7.
	BarDim_D2_Bot (see page 1706)	Reinforcing bar diameter of bottom reinforcement in the direction perpendicular to the main reinforcement Available since version 1.7.
	BarDim_D2_Up (see page 1706)	Reinforcing bar diameter of top reinforcement in the direction perpendicular to the main reinforcement Available since version 1.7.
	CheckConstructionStages (see page 1717)	Structure check in the phases of structure construction Available since version 1.7.
	CodeName (see page 1706)	Code name Available since version 1.7.
	Concrete (see page 1706)	Available since version 1.7.
	Cover_Bot (see page 1707)	Cover - bottom reinforcement Available since version 1.7.
	Cover_Up (see page 1707)	Cover - top reinforcement Available since version 1.7.
	LongActionAbove100 (see page 1717)	term action of the temperature over 100 degrees Available since version 1.7.
	Main (see page 1718)	
	ReinforcingSteel (see page 1707)	Available since version 1.7.
	SingleShortLoad (see page 1718)	Single-time short-term load Available since version 1.7.
	SLS_ConcreteAge (see page 1718)	Concrete (see page 1706) age (days) Available since version 1.7.
	SLS_Cracking (see page 1718)	Flag indicating if cracking width is to be calculated Available since version 1.7.
	SLS_CrackingReinfCorrect (see page 1719)	Flag switching on automatic correction of cracking width by increasing reinforcement area Available since version 1.7.
	SLS_CreepingCoef (see page 1719)	Concrete (see page 1706) creeping coefficient Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

	SLS_Deflection (see page 1719)	Flag indicating if a deflection value for RC slab taking cracking into account is to be calculated Available since version 1.7.
	SLS_DeflectionReinfCorrect (see page 1719)	Flag switching on automatic correction of deflections by increasing reinforcement area Available since version 1.7.
	SLS_EnvHumidity (see page 1720)	Type of environment humidity Available since version 1.7.
	SLS_Exposure (see page 1720)	Exposure rating Available since version 1.7.
	SLS_LoadsRatio (see page 1720)	term loads to total loads ratio Available since version 1.7.
	SLS_MaxCracking (see page 1720)	Admissible cracking width Available since version 1.7.
	SLS_MaxDeflection (see page 1721)	Admissible deflection value Available since version 1.7.

Public Methods

	Name	Description
	GetMainDirection (see page 1708)	Function returns the main reiforcement direction. Interpretation of returned coordinate values depends on the manner of determinig direction (direction vector coordinates, if the direction is determined by means of the vector, ALONG_VECTOR, or the point coordinates in the polar system - POLAR_POINT). Available since version 1.7.
	SetMainDirection (see page 1708)	Function defines the main reiforcement direction. Parameter interpretation depends on the manner of determinig direction (direction vector coordinates, if the direction is determined by means of the vector, ALONG_VECTOR, or the point coordinates in the polar system - POLAR_POINT, in the remaining cases the coordinate values are ignored). Available since version 1.7.

II.5.2 IRConcr_PN84_ReinforceData Fields

The fields of the IRConcr_PN84_ReinforceData class are listed here.

Public Fields

	Name	Description
	CheckConstructionStages (see page 1717)	Structure check in the phases of structure construction Available since version 1.7.
	LongActionAbove100 (see page 1717)	term action of the temperature over 100 degrees Available since version 1.7.
	Main (see page 1718)	
	SingleShortLoad (see page 1718)	Single-time short-term load Available since version 1.7.
	SLS_ConcreteAge (see page 1718)	Concrete (see page 1706) age (days) Available since version 1.7.
	SLS_Cracking (see page 1718)	Flag indicating if cracking width is to be calculated Available since version 1.7.
	SLS_CrackingReinfCorrect (see page 1719)	Flag switching on automatic correction of cracking width by increasing reinforcement area Available since version 1.7.
	SLS_CreepingCoef (see page 1719)	Concrete (see page 1706) creeping coefficient Available since version 1.7.
	SLS_Deflection (see page 1719)	Flag indicating if a deflection value for RC slab taking cracking into account is to be calculated Available since version 1.7.
	SLS_DeflectionReinfCorrect (see page 1719)	Flag switching on automatic correction of deflections by increasing reinforcement area Available since version 1.7.
	SLS_EnvHumidity (see page 1720)	Type of environment humidity Available since version 1.7.
	SLS_Exposure (see page 1720)	Exposure rating Available since version 1.7.
	SLS_LoadsRatio (see page 1720)	term loads to total loads ratio Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

	SLS_MaxCracking (see page 1720)	Admissible cracking width Available since version 1.7.
	SLS_MaxDeflection (see page 1721)	Admissible deflection value Available since version 1.7.

II.5.2.1 CheckConstructionStages

C++

```
HRESULT get_CheckConstructionStages(VARIANT_BOOL*);
HRESULT put_CheckConstructionStages(VARIANT_BOOL);
```

C#

```
public bool CheckConstructionStages { get; set; }
```

Visual Basic

```
Public CheckConstructionStages As Boolean
```

Description

Structure check in the phases of structure construction Available since version 1.7.

II.5.2.2 LongActionAbove100

C++

```
HRESULT get_LongActionAbove100(VARIANT_BOOL*);
HRESULT put_LongActionAbove100(VARIANT_BOOL);
```

C#

```
public bool LongActionAbove100 { get; set; }
```

Visual Basic

```
Public LongActionAbove100 As Boolean
```

Description

term action of the temperature over 100 degrees Available since version 1.7.

II.5.2.3 Main

C++

```
HRESULT get_Main(IRConcrReinforceDataMain**);
```

C#

```
public IRConcrReinforceDataMain Main { get; }
```

Visual Basic

```
Public ReadOnly Main As IRConcrReinforceDataMain
```

II.5.2.4 SingleShortLoad

C++

```
HRESULT get_SingleShortLoad(VARIANT_BOOL*);
HRESULT put_SingleShortLoad(VARIANT_BOOL);
```

C#

```
public bool SingleShortLoad { get; set; }
```

Visual Basic

```
Public SingleShortLoad As Boolean
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Single-time short-term load Available since version 1.7.

II.5.2.5 SLS_ConcreteAge

C++

```
HRESULT get_SLS_ConcreteAge(short*);
HRESULT put_SLS_ConcreteAge(short);
```

C#

```
public short SLS_ConcreteAge { get; set; }
```

Visual Basic

```
Public SLS_ConcreteAge As short
```

Description

Concrete (see page 1706) age (days) Available since version 1.7.

II.5.2.6 SLS_Cracking

C++

```
HRESULT get_SLS_Cracking(VARIANT_BOOL*);
HRESULT put_SLS_Cracking(VARIANT_BOOL);
```

C#

```
public bool SLS_Cracking { get; set; }
```

Visual Basic

```
Public SLS_Cracking As Boolean
```

Description

Flag indicating if cracking width is to be calculated Available since version 1.7.

II.5.2.7 SLS_CrackingReinfCorrect

C++

```
HRESULT get_SLS_CrackingReinfCorrect(VARIANT_BOOL*);
HRESULT put_SLS_CrackingReinfCorrect(VARIANT_BOOL);
```

C#

```
public bool SLS_CrackingReinfCorrect { get; set; }
```

Visual Basic

```
Public SLS_CrackingReinfCorrect As Boolean
```

Description

Flag switching on automatic correction of cracking width by increasing reinforcement area Available since version 1.7.

II.5.2.8 SLS_CreepingCoef

C++

```
HRESULT get_SLS_CreepingCoef(double*);
HRESULT put_SLS_CreepingCoef(double);
```

C#

```
public double SLS_CreepingCoef { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public SLS_CreepingCoef As double
```

Description

Concrete (see page 1706) creeping coefficient Available since version 1.7.

II.5.2.9 SLS_Deflection

C++

```
HRESULT get_SLS_Deflection(VARIANT_BOOL*);
HRESULT put_SLS_Deflection(VARIANT_BOOL);
```

C#

```
public bool SLS_Deflection { get; set; }
```

Visual Basic

```
Public SLS_Deflection As Boolean
```

Description

Flag indicating if a deflection value for RC slab taking cracking into account is to be calculated Available since version 1.7.

II.5.2.10 SLS_DeflectionReinfCorrect

C++

```
HRESULT get_SLS_DeflectionReinfCorrect(VARIANT_BOOL*);
HRESULT put_SLS_DeflectionReinfCorrect(VARIANT_BOOL);
```

C#

```
public bool SLS_DeflectionReinfCorrect { get; set; }
```

Visual Basic

```
Public SLS_DeflectionReinfCorrect As Boolean
```

Description

Flag switching on automatic correction of deflections by increasing reinforcement area Available since version 1.7.

II.5.2.11 SLS_EnvHumidity

C++

```
HRESULT get_SLS_EnvHumidity(IRConcr_PN84_HumidityType*);
HRESULT put_SLS_EnvHumidity(IRConcr_PN84_HumidityType);
```

C#

```
public IRConcr_PN84_HumidityType SLS_EnvHumidity { get; set; }
```

Visual Basic

```
Public SLS_EnvHumidity As IRConcr_PN84_HumidityType
```

Description

Type of environment humidity Available since version 1.7.

II.5.2.12 SLS_Exposure

C++

```
HRESULT get_SLS_Exposure(short*);
HRESULT put_SLS_Exposure(short);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public short SLS_Exposure { get; set; }
```

Visual Basic

```
Public SLS_Exposure As short
```

Description

Exposure rating Available since version 1.7.

II.5.2.13 SLS_LoadsRatio

C++

```
HRESULT get_SLS_LoadsRatio(double*);
HRESULT put_SLS_LoadsRatio(double);
```

C#

```
public double SLS_LoadsRatio { get; set; }
```

Visual Basic

```
Public SLS_LoadsRatio As double
```

Description

term loads to total loads ratio Available since version 1.7.

II.5.2.14 SLS_MaxCracking

C++

```
HRESULT get_SLS_MaxCracking(double*);
HRESULT put_SLS_MaxCracking(double);
```

C#

```
public double SLS_MaxCracking { get; set; }
```

Visual Basic

```
Public SLS_MaxCracking As double
```

Description

Admissible cracking width Available since version 1.7.

II.5.2.15 SLS_MaxDeflection

C++

```
HRESULT get_SLS_MaxDeflection(double*);
HRESULT put_SLS_MaxDeflection(double);
```

C#

```
public double SLS_MaxDeflection { get; set; }
```

Visual Basic

```
Public SLS_MaxDeflection As double
```

Description

Admissible deflection value Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

II.6 IRConcr_PN_SteelGrades

C++

```
enum IRConcr_PN_SteelGrades;
```

C#

```
public enum IRConcr_PN_SteelGrades;
```

Visual Basic

```
Public Enum IRConcr_PN_SteelGrades
```

Members

Members	Description
I_CPNSG_A_0 = 2	Available since version 1.7.
I_CPNSG_A_I = 3	Available since version 1.7.
I_CPNSG_A_II = 4	Available since version 1.7.
I_CPNSG_A_III = 5	Available since version 1.7.
I_CPNSG_A_IIIN = 6	Available since version 1.7.

Description

Reinforcing steel grades for Polish codes: PN-84 and PN-99.

II.7 IRConcr_PN84_ConcreteGrades

C++

```
enum IRConcr_PN84_ConcreteGrades;
```

C#

```
public enum IRConcr_PN84_ConcreteGrades;
```

Visual Basic

```
Public Enum IRConcr_PN84_ConcreteGrades
```

Members

Members	Description
I_CPN84CG_ANOTHER = -1	Other concrete type Available since version 1.7.
I_CPN84CG_B10 = 0	Available since version 1.7.
I_CPN84CG_B12_5 = 1	Available since version 1.7.
I_CPN84CG_B15 = 2	Available since version 1.7.
I_CPN84CG_B17_5 = 3	Available since version 1.7.
I_CPN84CG_B20 = 4	Available since version 1.7.
I_CPN84CG_B25 = 5	Available since version 1.7.
I_CPN84CG_B30 = 6	Available since version 1.7.
I_CPN84CG_B35 = 7	Available since version 1.7.
I_CPN84CG_B40 = 8	Available since version 1.7.
I_CPN84CG_B50 = 9	Available since version 1.7.
I_CPN84CG_B55 = 10	Available since version 1.7.
I_CPN84CG_B60 = 11	Available since version 1.7.
I_CPN84CG_B65 = 12	Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

Description

Concrete class for Polish code PN-84.

II.8 IRConcr_PN84_ExposureRatings

C++

```
enum IRConcr_PN84_ExposureRatings;
```

C#

```
public enum IRConcr_PN84_ExposureRatings;
```

Visual Basic

```
Public Enum IRConcr_PN84_ExposureRatings
```

Members

Members	Description
I_CPN84ER_MILD = 0	Available since version 1.7.
I_CPN84ER_MODERATE = 1	Available since version 1.7.
I_CPN84ER_SEVERE_OR_LEAKPROOF = 2	Available since version 1.7.

Description

Types of exposure rating defined for the Polish code PN-84. .

II.9 IRConcr_PN99_ExposureRatings

C++

```
enum IRConcr_PN99_ExposureRatings;
```

C#

```
public enum IRConcr_PN99_ExposureRatings;
```

Visual Basic

```
Public Enum IRConcr_PN99_ExposureRatings
```

Members

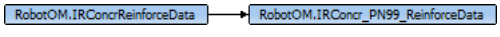
Members	Description
I_CPN99ER_X0 = 0	Available since version 8.5.
I_CPN99ER_XC1_XC2_XC3_XC4 = 1	Available since version 8.5.
I_CPN99ER_XF1_XF3 = 2	Available since version 8.5.
I_CPN99ER_XD1_XD2_XD3 = 3	Available since version 8.5.
I_CPN99ER_XS1_XS2_XS3 = 4	Available since version 8.5.
I_CPN99ER_XF2_XF4 = 5	Available since version 8.5.
I_CPN99ER_XA1_XA2_XA3 = 6	Available since version 8.5.

Description

Types of exposure rating defined for the Polish code PN-99. .

II.10 IRConcr_PN99_ReinforceData

Class Hierarchy



(c) 2014 Autodesk Inc. All rights reserved

C++

```
interface IRConcr_PN99_ReinforceData : IRConcrReinforceData;
```

C#

```
public interface IRConcr_PN99_ReinforceData : IRConcrReinforceData;
```

Visual Basic

```
Public Interface IRConcr_PN99_ReinforceData
```

Description

Reinforcement parameters for the Polish code PN-99. Steel grades are described by the identifier set RConr_PN_SteeGrades, concrete classes by RConcr_PN99_ConcreteGrades, whereas types of exposure rating by RConcr_PN99_ExposureRatings. .

II.10.1 IRConcr_PN99_ReinforceData Members

The following tables list the members exposed by IRConcr_PN99_ReinforceData.

Public Fields

	Name	Description
	BarDim_D1_Bot (see page 1705)	Diameter of reinforcing bars of bottom main reinforcement Available since version 1.7.
	BarDim_D1_Up (see page 1705)	Diameter of reinforcing bars of top main reinforcement Available since version 1.7.
	BarDim_D2_Bot (see page 1706)	Reinforcing bar diameter of bottom reinforcement in the direction perpendicular to the main reinforcement Available since version 1.7.
	BarDim_D2_Up (see page 1706)	Reinforcing bar diameter of top reinforcement in the direction perpendicular to the main reinforcement Available since version 1.7.
	CodeName (see page 1706)	Code name Available since version 1.7.
	Concrete (see page 1706)	Available since version 1.7.
	Cover_Bot (see page 1707)	Cover - bottom reinforcement Available since version 1.7.
	Cover_Up (see page 1707)	Cover - top reinforcement Available since version 1.7.
	IsSpecialStructure (see page 1725)	Structure of special importance.
	Main (see page 1726)	
	ReinforcingSteel (see page 1707)	Available since version 1.7.
	SLS_Concrete (see page 1726)	Concrete (see page 1706) age (years) Available since version 1.7.
	SLS_ConcreteAge (see page 1726)	Concrete (see page 1706) age expressed in days (it is significant to which range among the ranges determined by the code the day number belongs) Available since version 1.7.
	SLS_Cracking (see page 1726)	Flag indicating if cracking width is to be calculated Available since version 1.7.
	SLS_CrackingReinfCorrect (see page 1727)	Flag switching on automatic correction of cracking width by increasing area of reinforcement Available since version 1.7.
	SLS_CreepingCoef (see page 1727)	
	SLS_CreepingCoefValue (see page 1727)	
	SLS_Deflection (see page 1727)	Flag indicating if a deflection value for RC slab taking cracking into account is to be calculated Available since version 1.7.
	SLS_DeflectionReinfCorrect (see page 1728)	Flag switching on automatic correction of deflections by increasing reinforcement area Available since version 1.7.
	SLS_EnvHumidityVal (see page 1728)	Environment humidity value (in %) Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

	SLS_ExposureBot (see page 1728)	Environment class for the bottom area.
	SLS_ExposureTop (see page 1728)	Environment class for the top area.
	SLS_LoadRatio (see page 1729)	
	SLS_MaxCrackingBot (see page 1729)	Maximum cracking for the bottom area.
	SLS_MaxCrackingBotEnabled (see page 1729)	
	SLS_MaxCrackingTop (see page 1729)	Maximum cracking for the top area.
	SLS_MaxCrackingTopEnabled (see page 1730)	
	SLS_MaxDeflection (see page 1730)	Admissible deflection value Available since version 1.7.

Public Methods

	Name	Description
	GetMainDirection (see page 1708)	Function returns the main reiforcement direction. Interpretation of returned coordinate values depends on the manner of determinig direction (direction vector coordinates, if the direction is determined by means of the vector, ALONG_VECTOR, or the point coordinates in the polar system - POLAR_POINT). Available since version 1.7.
	SetMainDirection (see page 1708)	Function defines the main reiforcement direction. Parameter interpretation depends on the manner of determinig direction (direction vector coordinates, if the direction is determined by means of the vector, ALONG_VECTOR, or the point coordinates in the polar system - POLAR_POINT, in the remaining cases the coordinate values are ignored). Available since version 1.7.

II.10.2 IRConcr_PN99_ReinforceData Fields

The fields of the IRConcr_PN99_ReinforceData class are listed here.

Public Fields

	Name	Description
	IsSpecialStructure (see page 1725)	Structure of special importance.
	Main (see page 1726)	
	SLS_Concrete (see page 1726)	Concrete (see page 1706) age (years) Available since version 1.7.
	SLS_ConcreteAge (see page 1726)	Concrete (see page 1706) age expressed in days (it is significant to which range among the ranges determined by the code the day number belongs) Available since version 1.7.
	SLS_Cracking (see page 1726)	Flag indicating if cracking width is to be calculated Available since version 1.7.
	SLS_CrackingReinfCorrect (see page 1727)	Flag switching on automatic correction of cracking width by increasing area of reinforcement Available since version 1.7.
	SLS_CreepingCoef (see page 1727)	
	SLS_CreepingCoefValue (see page 1727)	
	SLS_Deflection (see page 1727)	Flag indicating if a deflection value for RC slab taking cracking into account is to be calculated Available since version 1.7.
	SLS_DeflectionReinfCorrect (see page 1728)	Flag switching on automatic correction of deflections by increasing reinforcement area Available since version 1.7.
	SLS_EnvHumidityVal (see page 1728)	Environment humidity value (in %) Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

	SLS_ExposureBot (see page 1728)	Environment class for the bottom area.
	SLS_ExposureTop (see page 1728)	Environment class for the top area.
	SLS_LoadRatio (see page 1729)	
	SLS_MaxCrackingBot (see page 1729)	Maximum cracking for the bottom area.
	SLS_MaxCrackingBotEnabled (see page 1729)	
	SLS_MaxCrackingTop (see page 1729)	Maximum cracking for the top area.
	SLS_MaxCrackingTopEnabled (see page 1730)	
	SLS_MaxDeflection (see page 1730)	Admissible deflection value Available since version 1.7.

II.10.2.1 IsSpecialStructure

C++

```
HRESULT get_IsSpecialStructure(VARIANT_BOOL*);
HRESULT put_IsSpecialStructure(VARIANT_BOOL);
```

C#

```
public bool IsSpecialStructure { get; set; }
```

Visual Basic

```
Public IsSpecialStructure As Boolean
```

Description

Structure of special importance.

Version

Available since version 8.5.

II.10.2.2 Main

C++

```
HRESULT get_Main(IRConcrReinforceDataMain**);
```

C#

```
public IRConcrReinforceDataMain Main { get; }
```

Visual Basic

```
Public ReadOnly Main As IRConcrReinforceDataMain
```

II.10.2.3 SLS_Concrete

C++

```
HRESULT get_SLS_Concrete(double*);
HRESULT put_SLS_Concrete(double);
```

C#

```
public double SLS_Concrete { get; set; }
```

Visual Basic

```
Public SLS_Concrete As double
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Concrete (see page 1706) age (years) Available since version 1.7.

II.10.2.4 SLS_ConcreteAge

C++

```
HRESULT get_SLS_ConcreteAge(short*);
HRESULT put_SLS_ConcreteAge(short);
```

C#

```
public short SLS_ConcreteAge { get; set; }
```

Visual Basic

```
Public SLS_ConcreteAge As short
```

Description

Concrete (see page 1706) age expressed in days (it is significant to which range among the ranges determined by the code the day number belongs) Available since version 1.7.

II.10.2.5 SLS_Cracking

C++

```
HRESULT get_SLS_Cracking(VARIANT_BOOL*);
HRESULT put_SLS_Cracking(VARIANT_BOOL);
```

C#

```
public bool SLS_Cracking { get; set; }
```

Visual Basic

```
Public SLS_Cracking As Boolean
```

Description

Flag indicating if cracking width is to be calculated Available since version 1.7.

II.10.2.6 SLS_CrackingReinfCorrect

C++

```
HRESULT get_SLS_CrackingReinfCorrect(VARIANT_BOOL*);
HRESULT put_SLS_CrackingReinfCorrect(VARIANT_BOOL);
```

C#

```
public bool SLS_CrackingReinfCorrect { get; set; }
```

Visual Basic

```
Public SLS_CrackingReinfCorrect As Boolean
```

Description

Flag switching on automatic correction of cracking width by increasing area of reinforcement Available since version 1.7.

II.10.2.7 SLS_CreepingCoef

C++

```
HRESULT get_SLS_CreepingCoef(VARIANT_BOOL*);
HRESULT put_SLS_CreepingCoef(VARIANT_BOOL);
```

C#

```
public bool SLS_CreepingCoef { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public SLS_CreepingCoef As Boolean
```

II.10.2.8 SLS_CreepingCoefValue

C++

```
HRESULT get_SLS_CreepingCoefValue(double*);
HRESULT put_SLS_CreepingCoefValue(double);
```

C#

```
public double SLS_CreepingCoefValue { get; set; }
```

Visual Basic

```
Public SLS_CreepingCoefValue As double
```

II.10.2.9 SLS_Deflection

C++

```
HRESULT get_SLS_Deflection(VARIANT_BOOL*);
HRESULT put_SLS_Deflection(VARIANT_BOOL);
```

C#

```
public bool SLS_Deflection { get; set; }
```

Visual Basic

```
Public SLS_Deflection As Boolean
```

Description

Flag indicating if a deflection value for RC slab taking cracking into account is to be calculated Available since version 1.7.

II.10.2.10 SLS_DeflectionReinfCorrect

C++

```
HRESULT get_SLS_DeflectionReinfCorrect(VARIANT_BOOL*);
HRESULT put_SLS_DeflectionReinfCorrect(VARIANT_BOOL);
```

C#

```
public bool SLS_DeflectionReinfCorrect { get; set; }
```

Visual Basic

```
Public SLS_DeflectionReinfCorrect As Boolean
```

Description

Flag switching on automatic correction of deflections by increasing reinforcement area Available since version 1.7.

II.10.2.11 SLS_EnvHumidityVal

C++

```
HRESULT get_SLS_EnvHumidityVal(double*);
HRESULT put_SLS_EnvHumidityVal(double);
```

C#

```
public double SLS_EnvHumidityVal { get; set; }
```

Visual Basic

```
Public SLS_EnvHumidityVal As double
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Environment humidity value (in %) Available since version 1.7.

II.10.2.12 SLS_ExposureBot

C++

```
HRESULT get_SLS_ExposureBot(short*);
HRESULT put_SLS_ExposureBot(short);
```

C#

```
public short SLS_ExposureBot { get; set; }
```

Visual Basic

```
Public SLS_ExposureBot As short
```

Description

Environment class for the bottom area.

Version

Available since version 8.5.

II.10.2.13 SLS_ExposureTop

C++

```
HRESULT get_SLS_ExposureTop(short*);
HRESULT put_SLS_ExposureTop(short);
```

C#

```
public short SLS_ExposureTop { get; set; }
```

Visual Basic

```
Public SLS_ExposureTop As short
```

Description

Environment class for the top area.

Version

Available since version 8.5.

II.10.2.14 SLS_LoadRatio

C++

```
HRESULT get_SLS_LoadRatio(double*);
HRESULT put_SLS_LoadRatio(double);
```

C#

```
public double SLS_LoadRatio { get; set; }
```

Visual Basic

```
Public SLS_LoadRatio As double
```

II.10.2.15 SLS_MaxCrackingBot

C++

```
HRESULT get_SLS_MaxCrackingBot(double*);
HRESULT put_SLS_MaxCrackingBot(double);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double SLS_MaxCrackingBot { get; set; }
```

Visual Basic

```
Public SLS_MaxCrackingBot As double
```

Description

Maximum cracking for the bottom area.

Version

Available since version 8.5.

II.10.2.16 SLS_MaxCrackingBotEnabled

C++

```
HRESULT get_SLS_MaxCrackingBotEnabled(VARIANT_BOOL*);
HRESULT put_SLS_MaxCrackingBotEnabled(VARIANT_BOOL);
```

C#

```
public bool SLS_MaxCrackingBotEnabled { get; set; }
```

Visual Basic

```
Public SLS_MaxCrackingBotEnabled As Boolean
```

Version

Available since version 8.5.

II.10.2.17 SLS_MaxCrackingTop

C++

```
HRESULT get_SLS_MaxCrackingTop(double*);
HRESULT put_SLS_MaxCrackingTop(double);
```

C#

```
public double SLS_MaxCrackingTop { get; set; }
```

Visual Basic

```
Public SLS_MaxCrackingTop As double
```

Description

Maximum cracking for the top area.

Version

Available since version 8.5.

II.10.2.18 SLS_MaxCrackingTopEnabled

C++

```
HRESULT get_SLS_MaxCrackingTopEnabled(VARIANT_BOOL*);
HRESULT put_SLS_MaxCrackingTopEnabled(VARIANT_BOOL);
```

C#

```
public bool SLS_MaxCrackingTopEnabled { get; set; }
```

Visual Basic

```
Public SLS_MaxCrackingTopEnabled As Boolean
```

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 8.5.

II.10.2.19 SLS_MaxDeflection

C++

```
HRESULT get_SLS_MaxDeflection(double*);
HRESULT put_SLS_MaxDeflection(double);
```

C#

```
public double SLS_MaxDeflection { get; set; }
```

Visual Basic

```
Public SLS_MaxDeflection As double
```

Description

Admissible deflection value Available since version 1.7.

II.11 IRConcr_PN99_ConcreteGrades

C++

```
enum IRConcr_PN99_ConcreteGrades;
```

C#

```
public enum IRConcr_PN99_ConcreteGrades;
```

Visual Basic

```
Public Enum IRConcr_PN99_ConcreteGrades
```

Members

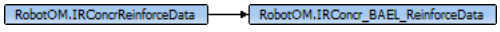
Members	Description
I_CPN99CG_ANOTHER = -1	Other concrete class Available since version 1.7.
I_CPN99CG_B15 = 0	Available since version 1.7.
I_CPN99CG_B20 = 1	Available since version 1.7.
I_CPN99CG_B25 = 2	Available since version 1.7.
I_CPN99CG_B30 = 3	Available since version 1.7.
I_CPN99CG_B37 = 4	Available since version 1.7.
I_CPN99CG_B45 = 5	Available since version 1.7.
I_CPN99CG_B50 = 6	Available since version 1.7.
I_CPN99CG_B55 = 7	Available since version 1.7.
I_CPN99CG_B60 = 8	Available since version 1.7.
I_CPN99CG_B65 = 9	Available since version 1.7.
I_CPN99CG_B70 = 10	Available since version 1.7.

Description

Concrete classes for Polish code PN-99.

II.12 IRConcr_BAEL_ReinforceData

Class Hierarchy



(c) 2014 Autodesk Inc. All rights reserved

C++

```
interface IRConcr_BAEL_ReinforceData : IRConcrReinforceData;
```

C#

```
public interface IRConcr_BAEL_ReinforceData : IRConcrReinforceData;
```

Visual Basic

```
Public Interface IRConcr_BAEL_ReinforceData
```

Description

Parameters of plate and shell reinforcement according to the BAEL code (91 and 99). Steel grades are determined by the identifiers from the set RConcr_BAEL_SteelGrades whereas concrete classes by RConcr_BAEL_ConcreteGrades. .

II.12.1 IRConcr_BAEL_ReinforceData Members

The following tables list the members exposed by IRConcr_BAEL_ReinforceData.

Public Fields

	Name	Description
	BarDim_D1_Bot (see page 1705)	Diameter of reinforcing bars of bottom main reinforcement Available since version 1.7.
	BarDim_D1_Up (see page 1705)	Diameter of reinforcing bars of top main reinforcement Available since version 1.7.
	BarDim_D2_Bot (see page 1706)	Reinforcing bar diameter of bottom reinforcement in the direction perpendicular to the main reinforcement Available since version 1.7.
	BarDim_D2_Up (see page 1706)	Reinforcing bar diameter of top reinforcement in the direction perpendicular to the main reinforcement Available since version 1.7.
	CodeName (see page 1706)	Code name Available since version 1.7.
	Concrete (see page 1706)	Available since version 1.7.
	Cover_Bot (see page 1707)	Cover - bottom reinforcement Available since version 1.7.
	Cover_Up (see page 1707)	Cover - top reinforcement Available since version 1.7.
	CrackAlpha (see page 1733)	Additional parameter resulting from the requirements of DTU 14-1 or Fascicule 74 Available since version 1.7.
	CrackAlphaBot (see page 1734)	
	CrackAlphaTop (see page 1734)	
	CrackEnv (see page 1734)	Additional parameter resulting from the requirements of DTU 14-1 or Fascicule 74 Available since version 1.7.
	CrackEnvBot (see page 1734)	
	CrackEnvTop (see page 1735)	
	CrackExtraParams (see page 1735)	Flag indicating if additional parameters resulting from the requirements of DTU 14-1 (for limited cracking) or Fascicule 74 (for not permissible cracking) are defined Available since version 2.0.
	CrackExtraParamsBot (see page 1735)	
	CrackExtraParamsTop (see page 1736)	
	Cracking (see page 1736)	Available since version 1.7.
	CrackingBot (see page 1736)	
	CrackingTop (see page 1736)	
	CrackWaterLevel (see page 1737)	Additional parameter resulting from the requirements of DTU 14-1 Available since version 1.7.
	CrackWaterLevelBot (see page 1737)	

(c) 2014 Autodesk Inc. All rights reserved

	CrackWaterLevelTop (see page 1737)	
	Main (see page 1737)	
	ReinforcingSteel (see page 1707)	Available since version 1.7.
	SLS_Deflection (see page 1738)	Available since version 1.7.
	SLS_DeflectionReinfCorrect (see page 1738)	Available since version 1.7.
	SLS_MaxDeflection (see page 1738)	Available since version 1.7.
	SteelSymbol (see page 1738)	Available since version 1.7.

Public Methods

	Name	Description
	GetMainDirection (see page 1708)	Function returns the main reiforcement direction. Interpretation of returned coordinate values depends on the manner of determinig direction (direction vector coordinates, if the direction is determined by means of the vector, ALONG_VECTOR, or the point coordinates in the polar system - POLAR_POINT). Available since version 1.7.
	SetMainDirection (see page 1708)	Function defines the main reiforcement direction. Parameter interpretation depends on the manner of determinig direction (direction vector coordinates, if the direction is determined by means of the vector, ALONG_VECTOR, or the point coordinates in the polar system - POLAR_POINT, in the remaining cases the coordinate values are ignored). Available since version 1.7.

II.12.2 IRConcr_BAEL_ReinforceData Fields

The fields of the IRConcr_BAEL_ReinforceData class are listed here.

Public Fields

	Name	Description
	CrackAlpha (see page 1733)	Additional parameter resulting from the requirements of DTU 14-1 or Fascicule 74 Available since version 1.7.
	CrackAlphaBot (see page 1734)	
	CrackAlphaTop (see page 1734)	
	CrackEnv (see page 1734)	Additional parameter resulting from the requirements of DTU 14-1 or Fascicule 74 Available since version 1.7.
	CrackEnvBot (see page 1734)	
	CrackEnvTop (see page 1735)	
	CrackExtraParams (see page 1735)	Flag indicating if additional parameters resulting from the requirements of DTU 14-1 (for limited cracking) or Fascicule 74 (for not permissible cracking) are defined Available since version 2.0.
	CrackExtraParamsBot (see page 1735)	
	CrackExtraParamsTop (see page 1736)	
	Cracking (see page 1736)	Available since version 1.7.
	CrackingBot (see page 1736)	
	CrackingTop (see page 1736)	
	CrackWaterLevel (see page 1737)	Additional parameter resulting from the requirements of DTU 14-1 Available since version 1.7.
	CrackWaterLevelBot (see page 1737)	
	CrackWaterLevelTop (see page 1737)	

(c) 2014 Autodesk Inc. All rights reserved

	Main (see page 1737)	
	SLS_Deflection (see page 1738)	Available since version 1.7.
	SLS_DeflectionReinfCorrect (see page 1738)	Available since version 1.7.
	SLS_MaxDeflection (see page 1738)	Available since version 1.7.
	SteelSymbol (see page 1738)	Available since version 1.7.

II.12.2.1 CrackAlpha

C++

```
HRESULT get_CrackAlpha(double*);
HRESULT put_CrackAlpha(double);
```

C#

```
public double CrackAlpha { get; set; }
```

Visual Basic

```
Public CrackAlpha As double
```

Description

Additional parameter resulting from the requirements of DTU 14-1 or Fascicule 74 Available since version 1.7.

II.12.2.2 CrackAlphaBot

C++

```
HRESULT get_CrackAlphaBot(double*);
HRESULT put_CrackAlphaBot(double);
```

C#

```
public double CrackAlphaBot { get; set; }
```

Visual Basic

```
Public CrackAlphaBot As double
```

Version

Available since version 8.

II.12.2.3 CrackAlphaTop

C++

```
HRESULT get_CrackAlphaTop(double*);
HRESULT put_CrackAlphaTop(double);
```

C#

```
public double CrackAlphaTop { get; set; }
```

Visual Basic

```
Public CrackAlphaTop As double
```

Version

Available since version 8.

II.12.2.4 CrackEnv

C++

```
HRESULT get_CrackEnv(IRConcr_BAEL_EnvironmentType*);
```

(c) 2014 Autodesk Inc. All rights reserved

```
HRESULT put_CrackEnv(IRConcr_BAEL_EnvironmentType);
```

C#

```
public IRConcr_BAEL_EnvironmentType CrackEnv { get; set; }
```

Visual Basic

```
Public CrackEnv As IRConcr_BAEL_EnvironmentType
```

Description

Additional parameter resulting from the requirements of DTU 14-1 or Fascicule 74 Available since version 1.7.

II.12.2.5 CrackEnvBot

C++

```
HRESULT get_CrackEnvBot(IRConcr_BAEL_EnvironmentType*);
HRESULT put_CrackEnvBot(IRConcr_BAEL_EnvironmentType);
```

C#

```
public IRConcr_BAEL_EnvironmentType CrackEnvBot { get; set; }
```

Visual Basic

```
Public CrackEnvBot As IRConcr_BAEL_EnvironmentType
```

Version

Available since version 8.

II.12.2.6 CrackEnvTop

C++

```
HRESULT get_CrackEnvTop(IRConcr_BAEL_EnvironmentType*);
HRESULT put_CrackEnvTop(IRConcr_BAEL_EnvironmentType);
```

C#

```
public IRConcr_BAEL_EnvironmentType CrackEnvTop { get; set; }
```

Visual Basic

```
Public CrackEnvTop As IRConcr_BAEL_EnvironmentType
```

Version

Available since version 8.

II.12.2.7 CrackExtraParams

C++

```
HRESULT get_CrackExtraParams(VARIANT_BOOL*);
HRESULT put_CrackExtraParams(VARIANT_BOOL);
```

C#

```
public bool CrackExtraParams { get; set; }
```

Visual Basic

```
Public CrackExtraParams As Boolean
```

Description

Flag indicating if additional parameters resulting from the requirements of DTU 14-1 (for limited cracking) or Fascicule 74 (for not permissible cracking) are defined Available since version 2.0.

(c) 2014 Autodesk Inc. All rights reserved

II.12.2.8 CrackExtraParamsBot

C++

```
HRESULT get_CrackExtraParamsBot(VARIANT_BOOL*);
HRESULT put_CrackExtraParamsBot(VARIANT_BOOL);
```

C#

```
public bool CrackExtraParamsBot { get; set; }
```

Visual Basic

```
Public CrackExtraParamsBot As Boolean
```

Version

Available since version 8.

II.12.2.9 CrackExtraParamsTop

C++

```
HRESULT get_CrackExtraParamsTop(VARIANT_BOOL*);
HRESULT put_CrackExtraParamsTop(VARIANT_BOOL);
```

C#

```
public bool CrackExtraParamsTop { get; set; }
```

Visual Basic

```
Public CrackExtraParamsTop As Boolean
```

Version

Available since version 8.

II.12.2.10 Cracking

C++

```
HRESULT get_Cracking(IRConcr_BAEL_CrackingType*);
HRESULT put_Cracking(IRConcr_BAEL_CrackingType);
```

C#

```
public IRConcr_BAEL_CrackingType Cracking { get; set; }
```

Visual Basic

```
Public Cracking As IRConcr_BAEL_CrackingType
```

Description

Available since version 1.7.

II.12.2.11 CrackingBot

C++

```
HRESULT get_CrackingBot(IRConcr_BAEL_CrackingType*);
HRESULT put_CrackingBot(IRConcr_BAEL_CrackingType);
```

C#

```
public IRConcr_BAEL_CrackingType CrackingBot { get; set; }
```

Visual Basic

```
Public CrackingBot As IRConcr_BAEL_CrackingType
```

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 8.

II.12.2.12 CrackingTop

C++

```
HRESULT get_CrackingTop(IRConcr_BAEL_CrackingType*);
HRESULT put_CrackingTop(IRConcr_BAEL_CrackingType);
```

C#

```
public IRConcr_BAEL_CrackingType CrackingTop { get; set; }
```

Visual Basic

```
Public CrackingTop As IRConcr_BAEL_CrackingType
```

Version

Available since version 8.

II.12.2.13 CrackWaterLevel

C++

```
HRESULT get_CrackWaterLevel(IRConcr_BAEL_WaterLevel*);
HRESULT put_CrackWaterLevel(IRConcr_BAEL_WaterLevel);
```

C#

```
public IRConcr_BAEL_WaterLevel CrackWaterLevel { get; set; }
```

Visual Basic

```
Public CrackWaterLevel As IRConcr_BAEL_WaterLevel
```

Description

Additional parameter resulting from the requirements of DTU 14-1 Available since version 1.7.

II.12.2.14 CrackWaterLevelBot

C++

```
HRESULT get_CrackWaterLevelBot(IRConcr_BAEL_WaterLevel*);
HRESULT put_CrackWaterLevelBot(IRConcr_BAEL_WaterLevel);
```

C#

```
public IRConcr_BAEL_WaterLevel CrackWaterLevelBot { get; set; }
```

Visual Basic

```
Public CrackWaterLevelBot As IRConcr_BAEL_WaterLevel
```

Version

Available since version 8.

II.12.2.15 CrackWaterLevelTop

C++

```
HRESULT get_CrackWaterLevelTop(IRConcr_BAEL_WaterLevel*);
HRESULT put_CrackWaterLevelTop(IRConcr_BAEL_WaterLevel);
```

C#

```
public IRConcr_BAEL_WaterLevel CrackWaterLevelTop { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public CrackWaterLevelTop As IRConcr_BAEL_WaterLevel
```

Version

Available since version 8.

II.12.2.16 Main

C++

```
HRESULT get_Main(IRConcrReinforceDataMain**);
```

C#

```
public IRConcrReinforceDataMain Main { get; }
```

Visual Basic

```
Public ReadOnly Main As IRConcrReinforceDataMain
```

II.12.2.17 SLS_Deflection

C++

```
HRESULT get_SLS_Deflection(VARIANT_BOOL*);
HRESULT put_SLS_Deflection(VARIANT_BOOL);
```

C#

```
public bool SLS_Deflection { get; set; }
```

Visual Basic

```
Public SLS_Deflection As Boolean
```

Description

Available since version 1.7.

II.12.2.18 SLS_DeflectionReinfCorrect

C++

```
HRESULT get_SLS_DeflectionReinfCorrect(VARIANT_BOOL*);
HRESULT put_SLS_DeflectionReinfCorrect(VARIANT_BOOL);
```

C#

```
public bool SLS_DeflectionReinfCorrect { get; set; }
```

Visual Basic

```
Public SLS_DeflectionReinfCorrect As Boolean
```

Description

Available since version 1.7.

II.12.2.19 SLS_MaxDeflection

C++

```
HRESULT get_SLS_MaxDeflection(double*);
HRESULT put_SLS_MaxDeflection(double);
```

C#

```
public double SLS_MaxDeflection { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public SLS_MaxDeflection As double
```

Description

Available since version 1.7.

II.12.2.20 SteelSymbol

C++

```
HRESULT get_SteelSymbol(BSTR*);
HRESULT put_SteelSymbol(BSTR);
```

C#

```
public String SteelSymbol { get; set; }
```

Visual Basic

```
Public SteelSymbol As String
```

Description

Available since version 1.7.

II.13 IRConcr_BAEL_CrackingType

C++

```
enum IRConcr_BAEL_CrackingType;
```

C#

```
public enum IRConcr_BAEL_CrackingType;
```

Visual Basic

```
Public Enum IRConcr_BAEL_CrackingType
```

Members

Members	Description
I_CBAELCT_OTHER = 3	Available since version 1.7.
I_CBAELCT_NOT_SEVERE = 0	Available since version 1.7.
I_CBAELCT_SEVERE = 1	Available since version 1.7.
I_CBAELCT_VERY_SEVERE = 2	Available since version 1.7.

II.14 IRConcr_BAEL_EnvironmentType

C++

```
enum IRConcr_BAEL_EnvironmentType;
```

C#

```
public enum IRConcr_BAEL_EnvironmentType;
```

Visual Basic

```
Public Enum IRConcr_BAEL_EnvironmentType
```

Members

Members	Description
I_CBAELET_AGGRESSIVE_WATER = 1	Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

I_CBAELET_NORMAL_WATER = 0	Available since version 1.7.

II.15 IRConcr_BAEL_WaterLevel

C++

```
enum IRConcr_BAEL_WaterLevel;
```

C#

```
public enum IRConcr_BAEL_WaterLevel;
```

Visual Basic

```
Public Enum IRConcr_BAEL_WaterLevel
```

Members

Members	Description
I_CBAELWL_LOW = 0	Available since version 1.7.
I_CBAELWL_HIGH = 1	Available since version 1.7.
I_CBAELWL_EXCEPTIONAL = 2	Available since version 1.7.

II.16 IRConcr_BAEL_ConcreteGrades

C++

```
enum IRConcr_BAEL_ConcreteGrades;
```

C#

```
public enum IRConcr_BAEL_ConcreteGrades;
```

Visual Basic

```
Public Enum IRConcr_BAEL_ConcreteGrades
```

Members

Members	Description
I_CBAELCG_OTHER = -1	Available since version 1.7.
I_CBAELCG_C12_15 = 0	Available since version 1.7.
I_CBAELCG_C16_20 = 1	Available since version 1.7.
I_CBAELCG_C20_25 = 2	Available since version 1.7.
I_CBAELCG_C25_30 = 3	Available since version 1.7.
I_CBAELCG_C30_37 = 4	Available since version 1.7.
I_CBAELCG_C35_45 = 5	Available since version 1.7.
I_CBAELCG_C40_50 = 6	Available since version 1.7.
I_CBAELCG_C45_55 = 7	Available since version 1.7.
I_CBAELCG_C50_60 = 8	Available since version 1.7.

II.17 IRConcr_BAEL_SteelGrades

C++

```
enum IRConcr_BAEL_SteelGrades;
```

C#

```
public enum IRConcr_BAEL_SteelGrades;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Enum IRConcr_BAEL_SteelGrades
```

Members

Members	Description
I_CBAELSG_PLAIN = 0	Available since version 1.7.
I_CBAELSG_DEFORMED = 1	Available since version 1.7.

II.18 IRConcr_PN84_HumidityType

C++

```
enum IRConcr_PN84_HumidityType;
```

C#

```
public enum IRConcr_PN84_HumidityType;
```

Visual Basic

```
Public Enum IRConcr_PN84_HumidityType
```

Members

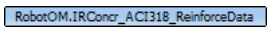
Members	Description
I_CPN84HT_40 = 0	Up to 40 percent Available since version 2.0.
I_CPN84HT_40_75 = 1	From 40 to 75 percent Available since version 2.0.
I_CPN84HT_75 = 2	Over 75 percent Available since version 2.0.
I_CPN84HT_WATER = 3	Water Available since version 2.0.

Description

Types of humidity defined for the PN-84 code .

II.19 IRConcr_ACI318_ReinforceData

Class Hierarchy



C++

```
interface IRConcr_ACI318_ReinforceData : IDispatch;
```

C#

```
public interface IRConcr_ACI318_ReinforceData;
```

Visual Basic

```
Public Interface IRConcr_ACI318_ReinforceData
```

Description

Interface of reinforcement parameters for the codes: "ACI 318/99" and "ACI 318/99 metric". .

Version

Available since version 3.5.

II.19.1 IRConcr_ACI318_ReinforceData Members

The following tables list the members exposed by IRConcr_ACI318_ReinforceData.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	IsMetric (see page 1743)	Flag informing about the code variant: true value for "ACI 318/99 metric", false value for "ACI 318/99". .
	Main (see page 1743)	
	SLS_Cracking (see page 1743)	Flag determining calculation of cracking width.
	SLS_CreepingCoef (see page 1743)	
	SLS_CreepingCoefValue (see page 1744)	
	SLS_Deflection (see page 1744)	Flag determining calculation of a deflection value.
	SLS_DeflectionReinfCorrection (see page 1744)	Flag managing automatic correction of deflections by increasing the area of reinforcement .
	SLS_Factor (see page 1744)	Flag indicating if the concrete creep coefficient should be consiered in reinforcement definition .
	SLS_FactorValue (see page 1745)	Value of concrete creep coefficient.
	SLS_LoadActionPeriod (see page 1745)	Duration of variable long-term loads.
	SLS_LoadRatio (see page 1745)	Ratio of long-term loads to total loads.
	SLS_MaxCracking (see page 1746)	Admissible value of cracking width.
	SLS_MaxDeflection (see page 1746)	Admissible value of deflection.

Public Methods

	Name	Description
	GetBarDim (see page 1747)	Function takes diameter of the specified bar. .
	GetMetricBarDim (see page 1747)	Equivalent of the GetBarDim (see page 1747) function for the metric variant of the code. .
	SetBarDim (see page 1747)	Function sets a diameter of the specified bar.
	SetMetricBarDim (see page 1748)	Equivalent of the SetBarDim (see page 1747) function for the metric variant of the code. .

II.19.2 IRConcr_ACI318_ReinforceData Fields

The fields of the IRConcr_ACI318_ReinforceData class are listed here.

Public Fields

	Name	Description
	IsMetric (see page 1743)	Flag informing about the code variant: true value for "ACI 318/99 metric", false value for "ACI 318/99". .
	Main (see page 1743)	
	SLS_Cracking (see page 1743)	Flag determining calculation of cracking width.
	SLS_CreepingCoef (see page 1743)	
	SLS_CreepingCoefValue (see page 1744)	
	SLS_Deflection (see page 1744)	Flag determining calculation of a deflection value.
	SLS_DeflectionReinfCorrection (see page 1744)	Flag managing automatic correction of deflections by increasing the area of reinforcement .
	SLS_Factor (see page 1744)	Flag indicating if the concrete creep coefficient should be consiered in reinforcement definition .

(c) 2014 Autodesk Inc. All rights reserved

	SLS_FactorValue (see page 1745)	Value of concrete creep coefficient.
	SLS_LoadActionPeriod (see page 1745)	Duration of variable long-term loads.
	SLS_LoadRatio (see page 1745)	Ratio of long-term loads to total loads.
	SLS_MaxCracking (see page 1746)	Admissible value of cracking width.
	SLS_MaxDeflection (see page 1746)	Admissible value of deflection.

II.19.2.1 IsMetric

C++

```
HRESULT get_IsMetric(VARIANT_BOOL*);
```

C#

```
public bool IsMetric { get; }
```

Visual Basic

```
Public ReadOnly IsMetric As Boolean
```

Description

Flag informing about the code variant: true value for "ACI 318/99 metric", false value for "ACI 318/99". .

Version

Available since version 3.5.

II.19.2.2 Main

C++

```
HRESULT get_Main(IRConcrReinforceDataMain**);
```

C#

```
public IRConcrReinforceDataMain Main { get; }
```

Visual Basic

```
Public ReadOnly Main As IRConcrReinforceDataMain
```

II.19.2.3 SLS_Cracking

C++

```
HRESULT get_SLS_Cracking(VARIANT_BOOL*);
HRESULT put_SLS_Cracking(VARIANT_BOOL);
```

C#

```
public bool SLS_Cracking { get; set; }
```

Visual Basic

```
Public SLS_Cracking As Boolean
```

Description

Flag determining calculation of cracking width.

Version

Available since version 3.5.

(c) 2014 Autodesk Inc. All rights reserved

II.19.2.4 SLS_CreepingCoef

C++

```
HRESULT get_SLS_CreepingCoef(VARIANT_BOOL*);
HRESULT put_SLS_CreepingCoef(VARIANT_BOOL);
```

C#

```
public bool SLS_CreepingCoef { get; set; }
```

Visual Basic

```
Public SLS_CreepingCoef As Boolean
```

II.19.2.5 SLS_CreepingCoefValue

C++

```
HRESULT get_SLS_CreepingCoefValue(double*);
HRESULT put_SLS_CreepingCoefValue(double);
```

C#

```
public double SLS_CreepingCoefValue { get; set; }
```

Visual Basic

```
Public SLS_CreepingCoefValue As double
```

II.19.2.6 SLS_Deflection

C++

```
HRESULT get_SLS_Deflection(VARIANT_BOOL*);
HRESULT put_SLS_Deflection(VARIANT_BOOL);
```

C#

```
public bool SLS_Deflection { get; set; }
```

Visual Basic

```
Public SLS_Deflection As Boolean
```

Description

Flag determining calculation of a deflection value.

Version

Available since version 3.5.

II.19.2.7 SLS_DeflectionReinfCorrection

C++

```
HRESULT get_SLS_DeflectionReinfCorrection(VARIANT_BOOL*);
HRESULT put_SLS_DeflectionReinfCorrection(VARIANT_BOOL);
```

C#

```
public bool SLS_DeflectionReinfCorrection { get; set; }
```

Visual Basic

```
Public SLS_DeflectionReinfCorrection As Boolean
```

Description

Flag managing automatic correction of deflections by increasing the area of reinforcement .

Version

Available since version 3.5.

(c) 2014 Autodesk Inc. All rights reserved

II.19.2.8 SLS_Factor

C++

```
HRESULT get_SLS_Factor(VARIANT_BOOL*);
HRESULT put_SLS_Factor(VARIANT_BOOL);
```

C#

```
public bool SLS_Factor { get; set; }
```

Visual Basic

```
Public SLS_Factor As Boolean
```

Description

Flag indicating if the concrete creep coefficient should be consiered in reinforcement definition .

Version

Available since version 3.5.

II.19.2.9 SLS_FactorValue

C++

```
HRESULT get_SLS_FactorValue(double*);
HRESULT put_SLS_FactorValue(double);
```

C#

```
public double SLS_FactorValue { get; set; }
```

Visual Basic

```
Public SLS_FactorValue As double
```

Description

Value of concrete creep coefficient.

Version

Available since version 3.5.

II.19.2.10 SLS_LoadActionPeriod

C++

```
HRESULT get_SLS_LoadActionPeriod(IRConcr_ACI318_LoadActionPeriodType*);
HRESULT put_SLS_LoadActionPeriod(IRConcr_ACI318_LoadActionPeriodType);
```

C#

```
public IRConcr_ACI318_LoadActionPeriodType SLS_LoadActionPeriod { get; set; }
```

Visual Basic

```
Public SLS_LoadActionPeriod As IRConcr_ACI318_LoadActionPeriodType
```

Description

Duration of variable long-term loads.

Version

Available since version 3.5.

(c) 2014 Autodesk Inc. All rights reserved

II.19.2.11 SLS_LoadRatio

C++

```
HRESULT get_SLS_LoadRatio(double*);
HRESULT put_SLS_LoadRatio(double);
```

C#

```
public double SLS_LoadRatio { get; set; }
```

Visual Basic

```
Public SLS_LoadRatio As double
```

Description

Ratio of long-term loads to total loads.

Version

Available since version 3.5.

II.19.2.12 SLS_MaxCracking

C++

```
HRESULT get_SLS_MaxCracking(double*);
HRESULT put_SLS_MaxCracking(double);
```

C#

```
public double SLS_MaxCracking { get; set; }
```

Visual Basic

```
Public SLS_MaxCracking As double
```

Description

Admissible value of cracking width.

Version

Available since version 3.5.

II.19.2.13 SLS_MaxDeflection

C++

```
HRESULT get_SLS_MaxDeflection(double*);
HRESULT put_SLS_MaxDeflection(double);
```

C#

```
public double SLS_MaxDeflection { get; set; }
```

Visual Basic

```
Public SLS_MaxDeflection As double
```

Description

Admissible value of deflection.

Version

Available since version 3.5.

II.19.3 IRConcr_ACI318_ReinforceData Methods

The methods of the IRConcr_ACI318_ReinforceData class are listed here.

(c) 2014 Autodesk Inc. All rights reserved

Public Methods

	Name	Description
	GetBarDim (see page 1747)	Function takes diameter of the specified bar. .
	GetMetricBarDim (see page 1747)	Equivalent of the GetBarDim (see page 1747) function for the metric variant of the code. .
	SetBarDim (see page 1747)	Function sets a diameter of the specified bar.
	SetMetricBarDim (see page 1748)	Equivalent of the SetBarDim (see page 1747) function for the metric variant of the code. .

II.19.3.1 GetBarDim

C++

```
HRESULT GetBarDim(IRConcrReinforceBarType _type, IRConcr_ACI318_BarDim* ret);
```

C#

```
public IRConcr_ACI318_BarDim GetBarDim(IRConcrReinforceBarType _type);
```

Visual Basic

```
Public Function GetBarDim(_type As IRConcrReinforceBarType) As IRConcr_ACI318_BarDim
```

Description

Function takes diameter of the specified bar. .

Version

Available since version 3.5.

II.19.3.2 GetMetricBarDim

C++

```
HRESULT GetMetricBarDim(IRConncrReinforceBarType _bar_type, IRConncr_ACI318_MetricBarDim*
ret);
```

C#

```
public IRConncr_ACI318_MetricBarDim GetMetricBarDim(IRConncrReinforceBarType _bar_type);
```

Visual Basic

```
Public Function GetMetricBarDim(_bar_type As IRConncrReinforceBarType) As
IRConncr_ACI318_MetricBarDim
```

Description

Equivalent of the GetBarDim (see page 1747) function for the metric variant of the code. .

Version

Available since version 3.5.

II.19.3.3 SetBarDim

C++

```
HRESULT SetBarDim(IRConcrReinforceBarType _type, IRConcr_ACI318_BarDim _val);
```

C#

```
public void SetBarDim(IRConcrReinforceBarType _type, IRConcr_ACI318_BarDim _val);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Sub SetBarDim(_type As IRConcrReinforceBarType, _val As IRConcr_ACI318_BarDim)
```

Description

Function sets a diameter of the specified bar.

Version

Available since version 3.5.

II.19.3.4 SetMetricBarDim

C++

```
HRESULT SetMetricBarDim(IRConncrReinforceBarType _type, IRConncr_ACI318_MetricBarDim _val);
```

C#

```
public void SetMetricBarDim(IRConncrReinforceBarType _type, IRConncr_ACI318_MetricBarDim
_val);
```

Visual Basic

```
Public Sub SetMetricBarDim(_type As IRConncrReinforceBarType, _val As
IRConncr_ACI318_MetricBarDim)
```

Description

Equivalent of the SetBarDim (see page 1747) function for the metric variant of the code. .

Version

Available since version 3.5.

II.20 IRConcr_ACI318_LoadActionPeriodType

C++

```
enum IRConcr_ACI318_LoadActionPeriodType;
```

C#

```
public enum IRConcr_ACI318_LoadActionPeriodType;
```

Visual Basic

```
Public Enum IRConcr_ACI318_LoadActionPeriodType
```

Members

Members	Description
I_ACI318_LAPT_3 = 0	Available since version 3.5.
I_ACI318_LAPT_6 = 1	Available since version 3.5.
I_ACI318_LAPT_12 = 2	Available since version 3.5.
I_ACI318_LAPT_60 = 3	Available since version 3.5.

Description

Available duration times of variable long-term loads. .

Version

Available since version 3.5.

(c) 2014 Autodesk Inc. All rights reserved

II.21 IRConcr_ACI318_SteelGrades

C++

```
enum IRConcr_ACI318_SteelGrades;
```

C#

```
public enum IRConcr_ACI318_SteelGrades;
```

Visual Basic

```
Public Enum IRConcr_ACI318_SteelGrades
```

Members

Members	Description
I_ACI318_ST_40 = 0	Available since version 3.5.
I_ACI318_ST_50 = 1	Available since version 3.5.
I_ACI318_ST_60 = 2	Available since version 3.5.
I_ACI318_ST_75 = 3	Available since version 3.5.

Version

Available since version 3.5.

II.22 IRConcr_ACI318_ConcreteParams

Class Hierarchy

C++

```
interface IRConcr_ACI318_ConcreteParams : IRConcrConcreteParams;
```

C#

```
public interface IRConcr_ACI318_ConcreteParams : IRConcrConcreteParams;
```

Visual Basic

```
Public Interface IRConcr_ACI318_ConcreteParams
```

Description

Parameters describing concrete for the code ACI 318/99. .

Version

Available since version 3.5.

II.22.1 IRConcr_ACI318_ConcreteParams Members

The following tables list the members exposed by IRConcr_ACI318_ConcreteParams.

Public Fields

	Name	Description
	AsInStructure (see page 1750)	Flag indicating if reinforcement definition considers compression resistance of the material currently addded to the panel .
	Ec (see page 1710)	Elasticity modulus for concrete; parameter assumes value that equals -1 when the value of the appropriate material assigned to a panel is taken for calculations Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

	Fc (see page 1711)	Characteristic compressive strength; parameter assumes value that equals -1 when the value of the appropriate material assigned to a panel is taken for calculations Available since version 1.7.
	Fc_calc (see page 1711)	Design compressive strength (Fc_calc = Fc (see page 1711) / gammaC, where gammaC denotes safety factor); parameter assumes value that equals -1 when the value of the appropriate material assigned to a panel is taken for calculations Available since version 1.7.
	Fcj (see page 1711)	Value that is used only in the French code and denotes Fc (see page 1711) (characteristic compressive strength); parameter assumes value that equals -1 when the value of the appropriate material assigned to a panel is taken for calculations Available since version 1.7.
	Ft (see page 1712)	Characteristic tensile strength; parameter assumes value that equals -1 when the value of the appropriate material assigned to a panel is taken for calculations Available since version 1.7.
	Grade (see page 1712)	Identifier denoting a predefined concrete class (according to the code it should be compared with the appropriate set of defined identifiers) Available since version 1.7.

II.22.2 IRConcr_ACI318_ConcreteParams Fields

The fields of the IRConcr_ACI318_ConcreteParams class are listed here.

Public Fields

	Name	Description
	AsInStructure (see page 1750)	Flag indicating if reinforcement definition considers compression resistance of the material currently addded to the panel .

II.22.2.1 AsInStructure

C++

```
HRESULT get_AsInStructure(VARIANT_BOOL*);
HRESULT put_AsInStructure(VARIANT_BOOL);
```

C#

```
public bool AsInStructure { get; set; }
```

Visual Basic

```
Public AsInStructure As Boolean
```

Description

Flag indicating if reinforcement definition considers compression resistance of the material currently addded to the panel .

Version

Available since version 3.5.

II.23 IRConcr_ACI318_BarDim

C++

```
enum IRConcr_ACI318_BarDim;
```

C#

```
public enum IRConcr_ACI318_BarDim;
```

Visual Basic

```
Public Enum IRConcr_ACI318_BarDim
```

(c) 2014 Autodesk Inc. All rights reserved

Members

Members	Description
I_ACI318_BR_3 = 0	Available since version 3.5.
I_ACI318_BR_5 = 2	Available since version 3.5.
I_ACI318_BR_6 = 3	Available since version 3.5.
I_ACI318_BR_7 = 4	Available since version 3.5.
I_ACI318_BR_8 = 5	Available since version 3.5.
I_ACI318_BR_9 = 6	Available since version 3.5.
I_ACI318_BR_10 = 7	Available since version 3.5.
I_ACI318_BR_11 = 8	Available since version 3.5.
I_ACI318_BR_14 = 9	Available since version 3.5.
I_ACI318_BR_18 = 10	Available since version 3.5.

Description

Available bar diameters for the code ACI 318/99.

Version

Available since version 3.5.

II.24 IRConcr_ACI318_MetricBarDIm

C++

```
enum IRConcr_ACI318_MetricBarDIm;
```

C#

```
public enum IRConcr_ACI318_MetricBarDIm;
```

Visual Basic

```
Public Enum IRConcr_ACI318_MetricBarDIm
```

Members

Members	Description
I_ACI318_MBD_5 = 0	Available since version 3.5.
I_ACI318_MBD_6 = 1	Available since version 3.5.
I_ACI318_MBD_8 = 2	Available since version 3.5.
I_ACI318_MBD_10 = 3	Available since version 3.5.
I_ACI318_MBD_12 = 4	Available since version 3.5.
I_ACI318_MBD_14 = 5	Available since version 3.5.
I_ACI318_MBD_16 = 6	Available since version 3.5.
I_ACI318_MBD_20 = 7	Available since version 3.5.
I_ACI318_MBD_25 = 8	Available since version 3.5.
I_ACI318_MBD_32 = 9	Available since version 3.5.
I_ACI318_MBD_40 = 10	Available since version 3.5.

Description

Available bar diameters for the code ACI 318/99 metric.

Version

Available since version 3.5.

(c) 2014 Autodesk Inc. All rights reserved

II.25 IRConcrReinforceBarType

C++

```
enum IRConcrReinforceBarType;
```

C#

```
public enum IRConcrReinforceBarType;
```

Visual Basic

```
Public Enum IRConcrReinforceBarType
```

Members

Members	Description
I_RBT_D1_TOP = 0	Available since version 3.5.
I_RBT_D1_BOTTOM = 1	Available since version 3.5.
I_RBT_D2_TOP = 2	Available since version 3.5.
I_RBT_D2_BOTTOM = 3	Available since version 3.5.

Description

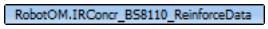
Types (designations) of reinforcing bars singled out during reinforcement definition. .

Version

Available since version 3.5.

II.26 IRConcr_BS8110_ReinforceData

Class Hierarchy



C++

```
interface IRConcr_BS8110_ReinforceData : IDispatch;
```

C#

```
public interface IRConcr_BS8110_ReinforceData;
```

Visual Basic

```
Public Interface IRConcr_BS8110_ReinforceData
```

Description

Interface of reinforcement parameters for the code BS8110.

Version

Available since version 3.5.

II.26.1 IRConcr_BS8110_ReinforceData Members

The following tables list the members exposed by IRConcr_BS8110_ReinforceData.

Public Fields

	Name	Description
	Main (see page 1753)	

(c) 2014 Autodesk Inc. All rights reserved

	PartialSafetyFactors (see page 1754)	Safety factor.
	SLS_ConcreteAge (see page 1754)	Concrete age (days) .
	SLS_Cracking (see page 1754)	Flag determinig calculation of cracking width.
	SLS_CrackingReinfCorection (see page 1755)	Flag managing automatic correction of cracking width by increasing reinforcement area.
	SLS_CreepingCoef (see page 1755)	Flag indicating if concrete creep coefficient should be considered in reinforcement definition.
	SLS_CreepingCoefValue (see page 1755)	Concrete creep coefficient.
	SLS_Deflection (see page 1756)	Flag determinig calculation of a deflection value.
	SLS_DeflectionReinfCorection (see page 1756)	Flag managing automatic correction of deflections by increasing reinforcement area.
	SLS_EnvHumidity (see page 1756)	Humidity (%).
	SLS_Exposure (see page 1757)	Exposure.
	SLS_LoadRatio (see page 1757)	Ratio of long-term loads to total loads.
	SLS_MaxCracking (see page 1757)	Admissible value of cracking width.
	SLS_MaxDeflection (see page 1758)	Admissible deflection value.

II.26.2 IRConcr_BS8110_ReinforceData Fields

The fields of the IRConcr_BS8110_ReinforceData class are listed here.

Public Fields

	Name	Description
	Main (see page 1753)	
	PartialSafetyFactors (see page 1754)	Safety factor.
	SLS_ConcreteAge (see page 1754)	Concrete age (days) .
	SLS_Cracking (see page 1754)	Flag determinig calculation of cracking width.
	SLS_CrackingReinfCorection (see page 1755)	Flag managing automatic correction of cracking width by increasing reinforcement area.
	SLS_CreepingCoef (see page 1755)	Flag indicating if concrete creep coefficient should be considered in reinforcement definition.
	SLS_CreepingCoefValue (see page 1755)	Concrete creep coefficient.
	SLS_Deflection (see page 1756)	Flag determinig calculation of a deflection value.
	SLS_DeflectionReinfCorection (see page 1756)	Flag managing automatic correction of deflections by increasing reinforcement area.
	SLS_EnvHumidity (see page 1756)	Humidity (%).
	SLS_Exposure (see page 1757)	Exposure.
	SLS_LoadRatio (see page 1757)	Ratio of long-term loads to total loads.
	SLS_MaxCracking (see page 1757)	Admissible value of cracking width.
	SLS_MaxDeflection (see page 1758)	Admissible deflection value.

(c) 2014 Autodesk Inc. All rights reserved

II.26.2.1 Main

C++

```
HRESULT get_Main(IRConcrReinforceDataMain**);
```

C#

```
public IRConcrReinforceDataMain Main { get; }
```

Visual Basic

```
Public ReadOnly Main As IRConcrReinforceDataMain
```

II.26.2.2 PartialSafetyFactors

C++

```
HRESULT get_PartialSafetyFactors(IRConcr_BS8110_PartialSafetyFactors*);
HRESULT put_PartialSafetyFactors(IRConcr_BS8110_PartialSafetyFactors);
```

C#

```
public IRConcr_BS8110_PartialSafetyFactors PartialSafetyFactors { get; set; }
```

Visual Basic

```
Public PartialSafetyFactors As IRConcr_BS8110_PartialSafetyFactors
```

Description

Safety factor.

Version

Available since version 3.5.

II.26.2.3 SLS_ConcreteAge

C++

```
HRESULT get_SLS_ConcreteAge(IRConcr_BS8110_ConcreteAge*);
HRESULT put_SLS_ConcreteAge(IRConcr_BS8110_ConcreteAge);
```

C#

```
public IRConcr_BS8110_ConcreteAge SLS_ConcreteAge { get; set; }
```

Visual Basic

```
Public SLS_ConcreteAge As IRConcr_BS8110_ConcreteAge
```

Description

Concrete age (days) .

Version

Available since version 3.5.

II.26.2.4 SLS_Cracking

C++

```
HRESULT get_SLS_Cracking(VARIANT_BOOL*);
HRESULT put_SLS_Cracking(VARIANT_BOOL);
```

C#

```
public bool SLS_Cracking { get; set; }
```

Visual Basic

```
Public SLS_Cracking As Boolean
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Flag determinig calculation of cracking width.

Version

Available since version 3.5.

II.26.2.5 SLS_CrackingReinfCorection

C++

```
HRESULT get_SLS_CrackingReinfCorection(VARIANT_BOOL*);
HRESULT put_SLS_CrackingReinfCorection(VARIANT_BOOL);
```

C#

```
public bool SLS_CrackingReinfCorection { get; set; }
```

Visual Basic

```
Public SLS_CrackingReinfCorection As Boolean
```

Description

Flag managing automatic correction of cracking width by increasing reinforcement area.

Version

Available since version 3.5.

II.26.2.6 SLS_CreepingCoef

C++

```
HRESULT get_SLS_CreepingCoef(VARIANT_BOOL*);
HRESULT put_SLS_CreepingCoef(VARIANT_BOOL);
```

C#

```
public bool SLS_CreepingCoef { get; set; }
```

Visual Basic

```
Public SLS_CreepingCoef As Boolean
```

Description

Flag indicating if concrete creep coefficient should be considered in reinforcement definition.

Version

Available since version 3.5.

II.26.2.7 SLS_CreepingCoefValue

C++

```
HRESULT get_SLS_CreepingCoefValue(double*);
HRESULT put_SLS_CreepingCoefValue(double);
```

C#

```
public double SLS_CreepingCoefValue { get; set; }
```

Visual Basic

```
Public SLS_CreepingCoefValue As double
```

Description

Concrete creep coefficient.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.5.

II.26.2.8 SLS_Deflection

C++

```
HRESULT get_SLS_Deflection(VARIANT_BOOL*);
HRESULT put_SLS_Deflection(VARIANT_BOOL);
```

C#

```
public bool SLS_Deflection { get; set; }
```

Visual Basic

```
Public SLS_Deflection As Boolean
```

Description

Flag determinig calculation of a deflection value.

Version

Available since version 3.5.

II.26.2.9 SLS_DeflectionReinfCorection

C++

```
HRESULT get_SLS_DeflectionReinfCorection(VARIANT_BOOL*);
HRESULT put_SLS_DeflectionReinfCorection(VARIANT_BOOL);
```

C#

```
public bool SLS_DeflectionReinfCorection { get; set; }
```

Visual Basic

```
Public SLS_DeflectionReinfCorection As Boolean
```

Description

Flag managing automatic correction of deflections by increasing reinforcement area.

Version

Available since version 3.5.

II.26.2.10 SLS_EnvHumidity

C++

```
HRESULT get_SLS_EnvHumidity(short*);
HRESULT put_SLS_EnvHumidity(short);
```

C#

```
public short SLS_EnvHumidity { get; set; }
```

Visual Basic

```
Public SLS_EnvHumidity As short
```

Description

Humidity (%).

Version

Available since version 3.5.

(c) 2014 Autodesk Inc. All rights reserved

II.26.2.11 SLS_Exposure

C++

```
HRESULT get_SLS_Exposure(IRConcr_BS8110_ExposureRatings*);
HRESULT put_SLS_Exposure(IRConcr_BS8110_ExposureRatings);
```

C#

```
public IRConcr_BS8110_ExposureRatings SLS_Exposure { get; set; }
```

Visual Basic

```
Public SLS_Exposure As IRConcr_BS8110_ExposureRatings
```

Description

Exposure.

Version

Available since version 3.5.

II.26.2.12 SLS_LoadRatio

C++

```
HRESULT get_SLS_LoadRatio(double*);
HRESULT put_SLS_LoadRatio(double);
```

C#

```
public double SLS_LoadRatio { get; set; }
```

Visual Basic

```
Public SLS_LoadRatio As double
```

Description

Ratio of long-term loads to total loads.

Version

Available since version 3.5.

II.26.2.13 SLS_MaxCracking

C++

```
HRESULT get_SLS_MaxCracking(double*);
HRESULT put_SLS_MaxCracking(double);
```

C#

```
public double SLS_MaxCracking { get; set; }
```

Visual Basic

```
Public SLS_MaxCracking As double
```

Description

Admissible value of cracking width.

Version

Available since version 3.5.

(c) 2014 Autodesk Inc. All rights reserved

II.26.2.14 SLS_MaxDeflection

C++

```
HRESULT get_SLS_MaxDeflection(double*);
HRESULT put_SLS_MaxDeflection(double);
```

C#

```
public double SLS_MaxDeflection { get; set; }
```

Visual Basic

```
Public SLS_MaxDeflection As double
```

Description

Admissible deflection value.

Version

Available since version 3.5.

II.27 IRConcr_BS8110_ExposureRatings

C++

```
enum IRConcr_BS8110_ExposureRatings;
```

C#

```
public enum IRConcr_BS8110_ExposureRatings;
```

Visual Basic

```
Public Enum IRConcr_BS8110_ExposureRatings
```

Members

Members	Description
I_BS8110_ER_MILD = 0	Available since version 3.5.
I_BS8110_ER_MODERATE = 1	Available since version 3.5.
I_BS8110_ER_SEVERE = 2	Available since version 3.5.
I_BS8110_ER_VERY_SEVERE = 3	Available since version 3.5.
I_BS8110_ER_MOST_SEVERE = 4	Available since version 3.5.

Description

Exposure types defined for the code BS 8110. .

Version

Available since version 3.5.

II.28 IRConcr_BS8110_ConcreteAge

C++

```
enum IRConcr_BS8110_ConcreteAge;
```

C#

```
public enum IRConcr_BS8110_ConcreteAge;
```

Visual Basic

```
Public Enum IRConcr_BS8110_ConcreteAge
```

(c) 2014 Autodesk Inc. All rights reserved

Members

Members	Description
I_BS8110_CA_1 = 0	1 day. Available since version 3.5.
I_BS8110_CA_3 = 1	3 days. Available since version 3.5.
I_BS8110_CA_7 = 2	7 days. Available since version 3.5.
I_BS8110_CA_28 = 3	28 days. Available since version 3.5.
I_BS8110_CA_90 = 4	90 days. Available since version 3.5.
I_BS8110_CA_365 = 5	365 days. Available since version 3.5.

Description

Predicted values of concrete age.

Version

Available since version 3.5.

II.29 IRConcr_BS8110_ConcreteGrades

C++

```
enum IRConcr_BS8110_ConcreteGrades;
```

C#

```
public enum IRConcr_BS8110_ConcreteGrades;
```

Visual Basic

```
Public Enum IRConcr_BS8110_ConcreteGrades
```

Members

Members	Description
I_BS8110_CG_AUTO = 0	Available since version 3.5.
I_BS8110_CG_C12_15 = 1	Available since version 3.5.
I_BS8110_CG_C16_20 = 2	Available since version 3.5.
I_BS8110_CG_C20_25 = 3	Available since version 3.5.
I_BS8110_CG_C25_30 = 4	Available since version 3.5.
I_BS8110_CG_C30_37 = 5	Available since version 3.5.
I_BS8110_CG_C35_45 = 6	Available since version 3.5.
I_BS8110_CG_C40_50 = 7	Available since version 3.5.
I_BS8110_CG_C45_55 = 8	Available since version 3.5.
I_BS8110_CG_C50_60 = 9	Available since version 3.5.

Description

Concrete classes according to the code BS 8110. .

Version

Available since version 3.5.

(c) 2014 Autodesk Inc. All rights reserved

II.30 IRConcr_BS8110_PartialSafetyFactors

C++

```
enum IRConcr_BS8110_PartialSafetyFactors;
```

C#

```
public enum IRConcr_BS8110_PartialSafetyFactors;
```

Visual Basic

```
Public Enum IRConcr_BS8110_PartialSafetyFactors
```

Members

Members	Description
I_BS8110_PSF_1985 = 0	Available since version 3.5.
I_BS8110_PSF_1997 = 1	Available since version 3.5.

Description

Safety factor types.

Version

Available since version 3.5.

II.31 IRConcrSlabRequiredReinfCalcParams

Class Hierarchy

RobotOM.IRConcrSlabRequiredReinfCalcParams

C++

```
interface IRConcrSlabRequiredReinfCalcParams : IDispatch;
```

C#

```
public interface IRConcrSlabRequiredReinfCalcParams;
```

Visual Basic

```
Public Interface IRConcrSlabRequiredReinfCalcParams
```

Description

Calculation parameters for the required (theoretical) reinforcement for plates and shells. .

Version

Available since version 5.5.

II.31.1 IRConcrSlabRequiredReinfCalcParams Members

The following tables list the members exposed by IRConcrSlabRequiredReinfCalcParams.

Public Fields

	Name	Description
	CasesACC (see page 1761)	
	CasesSLS (see page 1761)	
	CasesULS (see page 1762)	
	DisplayErrors (see page 1762)	

(c) 2014 Autodesk Inc. All rights reserved

	ForcesReduction (see page 1762)	
	GloballyAvgDesginForces (see page 1762)	
	Method (see page 1763)	Calculation method.
	Panels (see page 1763)	Selection of objects for which calculations will be performed.

II.31.2 IRConcrSlabRequiredReinfCalcParams Fields

The fields of the IRConcrSlabRequiredReinfCalcParams class are listed here.

Public Fields

	Name	Description
	CasesACC (see page 1761)	
	CasesSLS (see page 1761)	
	CasesULS (see page 1762)	
	DisplayErrors (see page 1762)	
	ForcesReduction (see page 1762)	
	GloballyAvgDesginForces (see page 1762)	
	Method (see page 1763)	Calculation method.
	Panels (see page 1763)	Selection of objects for which calculations will be performed.

II.31.2.1 CasesACC

C++

```
HRESULT get_CasesACC(IRobotSelection**);
```

C#

```
public IRobotSelection CasesACC { get; }
```

Visual Basic

```
Public ReadOnly CasesACC As IRobotSelection
```

Version

Available since version 5.5.

II.31.2.2 CasesSLS

C++

```
HRESULT get_CasesSLS(IRobotSelection**);
```

C#

```
public IRobotSelection CasesSLS { get; }
```

Visual Basic

```
Public ReadOnly CasesSLS As IRobotSelection
```

Version

Available since version 5.5.

II.31.2.3 CasesULS

C++

```
HRESULT get_CasesULS(IRobotSelection**);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRobotSelection CasesULS { get; }
```

Visual Basic

```
Public ReadOnly CasesULS As IRobotSelection
```

Version

Available since version 5.5.

II.31.2.4 DisplayErrors

C++

```
HRESULT get_DisplayErrors(VARIANT_BOOL*);
HRESULT put_DisplayErrors(VARIANT_BOOL);
```

C#

```
public bool DisplayErrors { get; set; }
```

Visual Basic

```
Public DisplayErrors As Boolean
```

Version

Available since version 5.5.

II.31.2.5 ForcesReduction

C++

```
HRESULT get_ForcesReduction(VARIANT_BOOL*);
HRESULT put_ForcesReduction(VARIANT_BOOL);
```

C#

```
public bool ForcesReduction { get; set; }
```

Visual Basic

```
Public ForcesReduction As Boolean
```

Version

Available since version 5.5.

II.31.2.6 GloballyAvgDesginForces

C++

```
HRESULT get_GloballyAvgDesginForces(VARIANT_BOOL*);
HRESULT put_GloballyAvgDesginForces(VARIANT_BOOL);
```

C#

```
public bool GloballyAvgDesginForces { get; set; }
```

Visual Basic

```
Public GloballyAvgDesginForces As Boolean
```

Version

Available since version 5.5.

II.31.2.7 Method

C++

```
HRESULT get_Method(IRobotReinforceCalcMethods*);
```

(c) 2014 Autodesk Inc. All rights reserved

```
HRESULT put_Method(IRobotReinforceCalcMethods);
```

C#

```
public IRobotReinforceCalcMethods Method { get; set; }
```

Visual Basic

```
Public Method As IRobotReinforceCalcMethods
```

Description

Calculation method.

Version

Available since version 5.5.

II.31.2.8 Panels

C++

```
HRESULT get_Panels(IRobotSelection**);
```

C#

```
public IRobotSelection Panels { get; }
```

Visual Basic

```
Public ReadOnly Panels As IRobotSelection
```

Description

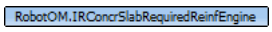
Selection of objects for which calculations will be performed.

Version

Available since version 5.5.

II.32 IRConcrSlabRequiredReinfEngine

Class Hierarchy



C++

```
interface IRConcrSlabRequiredReinfEngine : IDispatch;
```

C#

```
public interface IRConcrSlabRequiredReinfEngine;
```

Visual Basic

```
Public Interface IRConcrSlabRequiredReinfEngine
```

Description

Calculation module for required (theoretical) reinforcement of plates and shells. .

Version

Available since version 5.5.

II.32.1 IRConcrSlabRequiredReinfEngine Members

The following tables list the members exposed by IRConcrSlabRequiredReinfEngine.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	Params (see page 1764)	Calculation parameters.

Public Methods

	Name	Description
	Calculate (see page 1765)	Function performs calculations of the required (theoretical) reinforcement of plates and shells according to the currently-set parameters. .
	GetCalculatedPanels (see page 1765)	Function returns selection of slabs for which results of required reinforcement calculations are available.

II.32.2 IRConcrSlabRequiredReinfEngine Fields

The fields of the IRConcrSlabRequiredReinfEngine class are listed here.

Public Fields

	Name	Description
	Params (see page 1764)	Calculation parameters.

II.32.2.1 Params

C++

```
HRESULT get_Params(IRConcrSlabRequiredReinfCalcParams**);
```

C#

```
public IRConcrSlabRequiredReinfCalcParams Params { get; }
```

Visual Basic

```
Public ReadOnly Params As IRConcrSlabRequiredReinfCalcParams
```

Description

Calculation parameters.

Version

Available since version 5.5.

II.32.3 IRConcrSlabRequiredReinfEngine Methods

The methods of the IRConcrSlabRequiredReinfEngine class are listed here.

Public Methods

	Name	Description
	Calculate (see page 1765)	Function performs calculations of the required (theoretical) reinforcement of plates and shells according to the currently-set parameters. .
	GetCalculatedPanels (see page 1765)	Function returns selection of slabs for which results of required reinforcement calculations are available.

II.32.3.1 Calculate

C++

```
HRESULT Calculate(VARIANT_BOOL* ret);
```

C#

```
public bool Calculate();
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Function Calculate() As Boolean
```

Description

Function performs calculations of the required (theoretical) reinforcement of plates and shells according to the currently-set parameters. .

Version

Available since version 5.5.

II.32.3.2 GetCalculatedPanels

C++

```
HRESULT GetCalculatedPanels(IRobotSelection** ret);
```

C#

```
public IRobotSelection GetCalculatedPanels();
```

Visual Basic

```
Public Function GetCalculatedPanels() As IRobotSelection
```

Description

Function returns selection of slabs for which results of required reinforcement calculations are available.

Version

Available since version 11.

II.33 IRConcrReinforceCalcType

C++

```
enum IRConcrReinforceCalcType;
```

C#

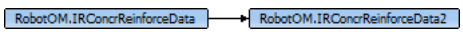
```
public enum IRConcrReinforceCalcType;
```

Visual Basic

```
Public Enum IRConcrReinforceCalcType
```

II.34 IRConcrReinforceData2

Class Hierarchy



C++

```
interface IRConcrReinforceData2 : IRConcrReinforceData;
```

C#

```
public interface IRConcrReinforceData2 : IRConcrReinforceData;
```

Visual Basic

```
Public Interface IRConcrReinforceData2
```

(c) 2014 Autodesk Inc. All rights reserved

II.34.1 IRConcrReinforceData2 Members

The following tables list the members exposed by IRConcrReinforceData2.

Public Fields

	Name	Description
	BarDim_D1_Bot (see page 1705)	Diameter of reinforcing bars of bottom main reinforcement Available since version 1.7.
	BarDim_D1_Up (see page 1705)	Diameter of reinforcing bars of top main reinforcement Available since version 1.7.
	BarDim_D2_Bot (see page 1706)	Reinforcing bar diameter of bottom reinforcement in the direction perpendicular to the main reinforcement Available since version 1.7.
	BarDim_D2_Up (see page 1706)	Reinforcing bar diameter of top reinforcement in the direction perpendicular to the main reinforcement Available since version 1.7.
	CodeName (see page 1706)	Code name Available since version 1.7.
	Concrete (see page 1706)	Available since version 1.7.
	Cover_Bot (see page 1707)	Cover - bottom reinforcement Available since version 1.7.
	Cover_Up (see page 1707)	Cover - top reinforcement Available since version 1.7.
	Main (see page 1766)	
	ReinforcingSteel (see page 1707)	Available since version 1.7.

Public Methods

	Name	Description
	GetMainDirection (see page 1708)	Function returns the main reiforcement direction. Interpretation of returned coordinate values depends on the manner of determinig direction (direction vector coordinates, if the direction is determined by means of the vector, ALONG_VECTOR, or the point coordinates in the polar system - POLAR_POINT). Available since version 1.7.
	SetMainDirection (see page 1708)	Function defines the main reiforcement direction. Parameter interpretation depends on the manner of determinig direction (direction vector coordinates, if the direction is determined by means of the vector, ALONG_VECTOR, or the point coordinates in the polar system - POLAR_POINT, in the remaining cases the coordinate values are ignored). Available since version 1.7.

II.34.2 IRConcrReinforceData2 Fields

The fields of the IRConcrReinforceData2 class are listed here.

Public Fields

	Name	Description
	Main (see page 1766)	

II.34.2.1 Main

C++

```
HRESULT get_Main(IRConcrReinforceDataMain**);
```

C#

```
public IRConcrReinforceDataMain Main { get; }
```

Visual Basic

```
Public ReadOnly Main As IRConcrReinforceDataMain
```

(c) 2014 Autodesk Inc. All rights reserved

II.35 IRConcrReinforceDataMain

Class Hierarchy

C++

```
interface IRConcrReinforceDataMain : IRConcrReinforceData;
```

C#

```
public interface IRConcrReinforceDataMain : IRConcrReinforceData;
```

Visual Basic

```
Public Interface IRConcrReinforceDataMain
```

II.35.1 IRConcrReinforceDataMain Members

The following tables list the members exposed by IRConcrReinforceDataMain.

Public Fields

	Name	Description
	BarDim_D1_Bot (see page 1705)	Diameter of reinforcing bars of bottom main reinforcement Available since version 1.7.
	BarDim_D1_Up (see page 1705)	Diameter of reinforcing bars of top main reinforcement Available since version 1.7.
	BarDim_D2_Bot (see page 1706)	Reinforcing bar diameter of bottom reinforcement in the direction perpendicular to the main reinforcement Available since version 1.7.
	BarDim_D2_Up (see page 1706)	Reinforcing bar diameter of top reinforcement in the direction perpendicular to the main reinforcement Available since version 1.7.
	CodeName (see page 1706)	Code name Available since version 1.7.
	Concrete (see page 1706)	Available since version 1.7.
	Cover_Bot (see page 1707)	Cover - bottom reinforcement Available since version 1.7.
	Cover_Up (see page 1707)	Cover - top reinforcement Available since version 1.7.
	MembraneReinfInOneLayer (see page 1768)	Membrane reinforcement in one layer.
	MinimumReinf (see page 1768)	
	ReinfCalcType (see page 1768)	
	ReinforcingSteel (see page 1707)	Available since version 1.7.
	UnidirReinf (see page 1769)	Unidirectional reinforcement.

Public Methods

	Name	Description
	GetMainDirection (see page 1708)	Function returns the main reiforcement direction. Interpretation of returned coordinate values depends on the manner of determinig direction (direction vector coordinates, if the direction is determined by means of the vector, ALONG_VECTOR, or the point coordinates in the polar system - POLAR_POINT). Available since version 1.7.
	SetMainDirection (see page 1708)	Function defines the main reiforcement direction. Parameter interpretation depends on the manner of determinig direction (direction vector coordinates, if the direction is determined by means of the vector, ALONG_VECTOR, or the point coordinates in the polar system - POLAR_POINT, in the remaining cases the coordinate values are ignored). Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

II.35.2 IRConcrReinforceDataMain Fields

The fields of the IRConcrReinforceDataMain class are listed here.

Public Fields

	Name	Description
	MembraneReinfInOneLayer (see page 1768)	Membrane reinforcement in one layer.
	MinimumReinf (see page 1768)	
	ReinfCalcType (see page 1768)	
	UnidirReinf (see page 1769)	Unidirectional reinforcement.

II.35.2.1 MembraneReinfInOneLayer

C++

```
HRESULT get_MembraneReinfInOneLayer(VARIANT_BOOL*);
HRESULT put_MembraneReinfInOneLayer(VARIANT_BOOL);
```

C#

```
public bool MembraneReinfInOneLayer { get; set; }
```

Visual Basic

```
Public MembraneReinfInOneLayer As Boolean
```

Description

Membrane reinforcement in one layer.

II.35.2.2 MinimumReinf

C++

```
HRESULT get_MinimumReinf(IRConcrMinimumReinforcementType*);
HRESULT put_MinimumReinf(IRConcrMinimumReinforcementType);
```

C#

```
public IRConcrMinimumReinforcementType MinimumReinf { get; set; }
```

Visual Basic

```
Public MinimumReinf As IRConcrMinimumReinforcementType
```

Version

Available since version 8.5.

II.35.2.3 ReinfCalcType

C++

```
HRESULT get_ReinfCalcType(IRConcrReinforceCalcType*);
HRESULT put_ReinfCalcType(IRConcrReinforceCalcType);
```

C#

```
public IRConcrReinforceCalcType ReinfCalcType { get; set; }
```

Visual Basic

```
Public ReinfCalcType As IRConcrReinforceCalcType
```

(c) 2014 Autodesk Inc. All rights reserved

II.35.2.4 UnidirReinf

C++

```
HRESULT get_UnidirReinf(VARIANT_BOOL*);
HRESULT put_UnidirReinf(VARIANT_BOOL);
```

C#

```
public bool UnidirReinf { get; set; }
```

Visual Basic

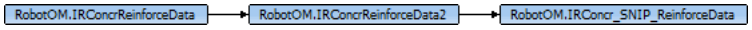
```
Public UnidirReinf As Boolean
```

Description

Unidirectional reinforcement.

II.36 IRConcr_SNIP_ReinforceData

Class Hierarchy



C++

```
interface IRConcr_SNIP_ReinforceData : IRConcrReinforceData2;
```

C#

```
public interface IRConcr_SNIP_ReinforceData : IRConcrReinforceData2;
```

Visual Basic

```
Public Interface IRConcr_SNIP_ReinforceData
```

Description

Interface of plate/shell reinforcement parameters for the code "SNIP 2.03.01-84".

II.36.1 IRConcr_SNIP_ReinforceData Members

The following tables list the members exposed by IRConcr_SNIP_ReinforceData.

Public Fields

	Name	Description
	BarDim_D1_Bot (see page 1705)	Diameter of reinforcing bars of bottom main reinforcement Available since version 1.7.
	BarDim_D1_Up (see page 1705)	Diameter of reinforcing bars of top main reinforcement Available since version 1.7.
	BarDim_D2_Bot (see page 1706)	Reinforcing bar diameter of bottom reinforcement in the direction perpendicular to the main reinforcement Available since version 1.7.
	BarDim_D2_Up (see page 1706)	Reinforcing bar diameter of top reinforcement in the direction perpendicular to the main reinforcement Available since version 1.7.
	CodeName (see page 1706)	Code name Available since version 1.7.
	Concrete (see page 1706)	Available since version 1.7.
	Cover_Bot (see page 1707)	Cover - bottom reinforcement Available since version 1.7.
	Cover_Up (see page 1707)	Cover - top reinforcement Available since version 1.7.
	Main (see page 1766)	
	ReinforcingSteel (see page 1707)	Available since version 1.7.
	SLS_Cracking (see page 1770)	

(c) 2014 Autodesk Inc. All rights reserved

	SLS_CrackingReinfCorrection (see page 1770)	
	SLS_Exposure (see page 1771)	

Public Methods

	Name	Description
	GetMainDirection (see page 1708)	Function returns the main reiforcement direction. Interpretation of returned coordinate values depends on the manner of determinig direction (direction vector coordinates, if the direction is determined by means of the vector, ALONG_VECTOR, or the point coordinates in the polar system - POLAR_POINT). Available since version 1.7.
	SetMainDirection (see page 1708)	Function defines the main reiforcement direction. Parameter interpretation depends on the manner of determinig direction (direction vector coordinates, if the direction is determined by means of the vector, ALONG_VECTOR, or the point coordinates in the polar system - POLAR_POINT, in the remaining cases the coordinate values are ignored). Available since version 1.7.

II.36.2 IRConcr_SNIP_ReinforceData Fields

The fields of the IRConcr_SNIP_ReinforceData class are listed here.

Public Fields

	Name	Description
	SLS_Cracking (see page 1770)	
	SLS_CrackingReinfCorrection (see page 1770)	
	SLS_Exposure (see page 1771)	

II.36.2.1 SLS_Cracking

C++

```
HRESULT get_SLS_Cracking(VARIANT_BOOL*);
HRESULT put_SLS_Cracking(VARIANT_BOOL);
```

C#

```
public bool SLS_Cracking { get; set; }
```

Visual Basic

```
Public SLS_Cracking As Boolean
```

II.36.2.2 SLS_CrackingReinfCorrection

C++

```
HRESULT get_SLS_CrackingReinfCorrection(VARIANT_BOOL*);
HRESULT put_SLS_CrackingReinfCorrection(VARIANT_BOOL);
```

C#

```
public bool SLS_CrackingReinfCorrection { get; set; }
```

Visual Basic

```
Public SLS_CrackingReinfCorrection As Boolean
```

II.36.2.3 SLS_Exposure

C++

```
HRESULT get_SLS_Exposure(IRConcr_SNIP_Exposure*);
HRESULT put_SLS_Exposure(IRConcr_SNIP_Exposure);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRConcr_SNIP_Exposure SLS_Exposure { get; set; }
```

Visual Basic

```
Public SLS_Exposure As IRConcr_SNIP_Exposure
```

II.37 IRConcr_SNIP_SteelGrades

C++

```
enum IRConcr_SNIP_SteelGrades;
```

C#

```
public enum IRConcr_SNIP_SteelGrades;
```

Visual Basic

```
Public Enum IRConcr_SNIP_SteelGrades
```

II.38 IRConcr_SNIP_ConcreteGrades

C++

```
enum IRConcr_SNIP_ConcreteGrades;
```

C#

```
public enum IRConcr_SNIP_ConcreteGrades;
```

Visual Basic

```
Public Enum IRConcr_SNIP_ConcreteGrades
```

II.39 IRConcr_SNIP_ConcreteTypes

C++

```
enum IRConcr_SNIP_ConcreteTypes;
```

C#

```
public enum IRConcr_SNIP_ConcreteTypes;
```

Visual Basic

```
Public Enum IRConcr_SNIP_ConcreteTypes
```

II.40 IRConcr_SNIP_CuringMethods

C++

```
enum IRConcr_SNIP_CuringMethods;
```

C#

```
public enum IRConcr_SNIP_CuringMethods;
```

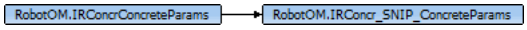
Visual Basic

```
Public Enum IRConcr_SNIP_CuringMethods
```

(c) 2014 Autodesk Inc. All rights reserved

II.41 IRConcr_SNIP_ConcreteParams

Class Hierarchy



C++

```
interface IRConcr_SNIP_ConcreteParams : IRConcrConcreteParams;
```

C#

```
public interface IRConcr_SNIP_ConcreteParams : IRConcrConcreteParams;
```

Visual Basic

```
Public Interface IRConcr_SNIP_ConcreteParams
```

II.41.1 IRConcr_SNIP_ConcreteParams Members

The following tables list the members exposed by IRConcr_SNIP_ConcreteParams.

Public Fields

	Name	Description
	ConcretingInLayers (see page 1773)	
	CuringMethod (see page 1773)	
	Ec (see page 1710)	Elasticity modulus for concrete; parameter assumes value that equals -1 when the value of the appropriate material assigned to a panel is taken for calculations Available since version 1.7.
	Fc (see page 1711)	Characteristic compressive strength; parameter assumes value that equals -1 when the value of the appropriate material assigned to a panel is taken for calculations Available since version 1.7.
	Fc_calc (see page 1711)	Design compressive strength (Fc_calc = Fc (see page 1711) / gammaC, where gammaC denotes safety factor); parameter assumes value that equals -1 when the value of the appropriate material assigned to a panel is taken for calculations Available since version 1.7.
	Fcj (see page 1711)	Value that is used only in the French code and denotes Fc (see page 1711) (characteristic compressive strength); parameter assumes value that equals -1 when the value of the appropriate material assigned to a panel is taken for calculations Available since version 1.7.
	Ft (see page 1712)	Characteristic tensile strength; parameter assumes value that equals -1 when the value of the appropriate material assigned to a panel is taken for calculations Available since version 1.7.
	Grade (see page 1712)	Identifier denoting a predefined concrete class (according to the code it should be compared with the appropriate set of defined identifiers) Available since version 1.7.
	HighHumidity (see page 1773)	
	Type (see page 1773)	

II.41.2 IRConcr_SNIP_ConcreteParams Fields

The fields of the IRConcr_SNIP_ConcreteParams class are listed here.

Public Fields

	Name	Description
	ConcretingInLayers (see page 1773)	

(c) 2014 Autodesk Inc. All rights reserved

	CuringMethod (see page 1773)	
	HighHumidity (see page 1773)	
	Type (see page 1773)	

II.41.2.1 ConcretingInLayers

C++

```
HRESULT get_ConcretingInLayers(VARIANT_BOOL*);
HRESULT put_ConcretingInLayers(VARIANT_BOOL);
```

C#

```
public bool ConcretingInLayers { get; set; }
```

Visual Basic

```
Public ConcretingInLayers As Boolean
```

II.41.2.2 CuringMethod

C++

```
HRESULT get_CuringMethod(IRConcr_SNIP_CuringMethods*);
HRESULT put_CuringMethod(IRConcr_SNIP_CuringMethods);
```

C#

```
public IRConcr_SNIP_CuringMethods CuringMethod { get; set; }
```

Visual Basic

```
Public CuringMethod As IRConcr_SNIP_CuringMethods
```

II.41.2.3 HighHumidity

C++

```
HRESULT get_HighHumidity(VARIANT_BOOL*);
HRESULT put_HighHumidity(VARIANT_BOOL);
```

C#

```
public bool HighHumidity { get; set; }
```

Visual Basic

```
Public HighHumidity As Boolean
```

II.41.2.4 Type

C++

```
HRESULT get_Type(IRConcr_SNIP_ConcreteTypes*);
HRESULT put_Type(IRConcr_SNIP_ConcreteTypes);
```

C#

```
public IRConcr_SNIP_ConcreteTypes Type { get; set; }
```

Visual Basic

```
Public Type As IRConcr_SNIP_ConcreteTypes
```

II.42 IRConcr_SNIP_Exposure

C++

```
enum IRConcr_SNIP_Exposure;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public enum IRConcr_SNIP_Exposure;
```

Visual Basic

```
Public Enum IRConcr_SNIP_Exposure
```

II.43 IRConcr_EC2_ConcreteGrades

C++

```
enum IRConcr_EC2_ConcreteGrades;
```

C#

```
public enum IRConcr_EC2_ConcreteGrades;
```

Visual Basic

```
Public Enum IRConcr_EC2_ConcreteGrades
```

Members

Members	Description
I_CEC2CG_AUTOMATIC = -1	Material defined for a panel will be used.

Description

Concrete classes according to Eurocode 2.

II.44 IRConcr_EC2_ITALIAN_SteelGrades

C++

```
enum IRConcr_EC2_ITALIAN_SteelGrades;
```

C#

```
public enum IRConcr_EC2_ITALIAN_SteelGrades;
```

Visual Basic

```
Public Enum IRConcr_EC2_ITALIAN_SteelGrades
```

II.45 IRConcr_EC2_ExposureRatings

C++

```
enum IRConcr_EC2_ExposureRatings;
```

C#

```
public enum IRConcr_EC2_ExposureRatings;
```

Visual Basic

```
Public Enum IRConcr_EC2_ExposureRatings
```

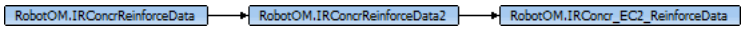
Description

Exposure types according to Eurocode 2.

(c) 2014 Autodesk Inc. All rights reserved

II.46 IRConcr_EC2_ReinforceData

Class Hierarchy



C++

```
interface IRConcr_EC2_ReinforceData : IRConcrReinforceData2;
```

C#

```
public interface IRConcr_EC2_ReinforceData : IRConcrReinforceData2;
```

Visual Basic

```
Public Interface IRConcr_EC2_ReinforceData
```

II.46.1 IRConcr_EC2_ReinforceData Members

The following tables list the members exposed by IRConcr_EC2_ReinforceData.

Public Fields

	Name	Description
	BarDim_D1_Bot (see page 1705)	Diameter of reinforcing bars of bottom main reinforcement Available since version 1.7.
	BarDim_D1_Up (see page 1705)	Diameter of reinforcing bars of top main reinforcement Available since version 1.7.
	BarDim_D2_Bot (see page 1706)	Reinforcing bar diameter of bottom reinforcement in the direction perpendicular to the main reinforcement Available since version 1.7.
	BarDim_D2_Up (see page 1706)	Reinforcing bar diameter of top reinforcement in the direction perpendicular to the main reinforcement Available since version 1.7.
	CodeName (see page 1706)	Code name Available since version 1.7.
	Concrete (see page 1706)	Available since version 1.7.
	Cover_Bot (see page 1707)	Cover - bottom reinforcement Available since version 1.7.
	Cover_Up (see page 1707)	Cover - top reinforcement Available since version 1.7.
	Main (see page 1766)	
	NAD (see page 1777)	
	ReinforcingSteel (see page 1707)	Available since version 1.7.
	SLS_ConcreteAge (see page 1777)	
	SLS_Cracking (see page 1777)	
	SLS_CrackingReinfCorrect (see page 1777)	
	SLS_CreepingCoef (see page 1777)	
	SLS_CreepingCoefValue (see page 1778)	
	SLS_Deflection (see page 1778)	
	SLS_DeflectionReinfCorrect (see page 1778)	
	SLS_EnvHumidityVal (see page 1778)	
	SLS_Exposure (see page 1778)	
	SLS_MaxCracking (see page 1779)	

(c) 2014 Autodesk Inc. All rights reserved

	SLS_MaxDeflection (see page 1779)	

Public Methods

	Name	Description
	GetMainDirection (see page 1708)	Function returns the main reiforcement direction. Interpretation of returned coordinate values depends on the manner of determinig direction (direction vector coordinates, if the direction is determined by means of the vector, ALONG_VECTOR, or the point coordinates in the polar system - POLAR_POINT). Available since version 1.7.
	SetMainDirection (see page 1708)	Function defines the main reiforcement direction. Parameter interpretation depends on the manner of determinig direction (direction vector coordinates, if the direction is determined by means of the vector, ALONG_VECTOR, or the point coordinates in the polar system - POLAR_POINT, in the remaining cases the coordinate values are ignored). Available since version 1.7.

II.46.2 IRConcr_EC2_ReinforceData Fields

The fields of the IRConcr_EC2_ReinforceData class are listed here.

Public Fields

	Name	Description
	NAD (see page 1777)	
	SLS_ConcreteAge (see page 1777)	
	SLS_Cracking (see page 1777)	
	SLS_CrackingReinfCorrect (see page 1777)	
	SLS_CreepingCoef (see page 1777)	
	SLS_CreepingCoefValue (see page 1778)	
	SLS_Deflection (see page 1778)	
	SLS_DeflectionReinfCorrect (see page 1778)	
	SLS_EnvHumidityVal (see page 1778)	
	SLS_Exposure (see page 1778)	
	SLS_MaxCracking (see page 1779)	
	SLS_MaxDeflection (see page 1779)	

II.46.2.1 NAD

C++

```
HRESULT get_NAD(IRConcr_EC2_NAD*);
HRESULT put_NAD(IRConcr_EC2_NAD);
```

C#

```
public IRConcr_EC2_NAD NAD { get; set; }
```

Visual Basic

```
Public NAD As IRConcr_EC2_NAD
```

(c) 2014 Autodesk Inc. All rights reserved

II.46.2.2 SLS_ConcreteAge

C++

```
HRESULT get_SLS_ConcreteAge(short*);
HRESULT put_SLS_ConcreteAge(short);
```

C#

```
public short SLS_ConcreteAge { get; set; }
```

Visual Basic

```
Public SLS_ConcreteAge As short
```

II.46.2.3 SLS_Cracking

C++

```
HRESULT get_SLS_Cracking(VARIANT_BOOL*);
HRESULT put_SLS_Cracking(VARIANT_BOOL);
```

C#

```
public bool SLS_Cracking { get; set; }
```

Visual Basic

```
Public SLS_Cracking As Boolean
```

II.46.2.4 SLS_CrackingReinfCorrect

C++

```
HRESULT get_SLS_CrackingReinfCorrect(VARIANT_BOOL*);
HRESULT put_SLS_CrackingReinfCorrect(VARIANT_BOOL);
```

C#

```
public bool SLS_CrackingReinfCorrect { get; set; }
```

Visual Basic

```
Public SLS_CrackingReinfCorrect As Boolean
```

II.46.2.5 SLS_CreepingCoef

C++

```
HRESULT get_SLS_CreepingCoef(VARIANT_BOOL*);
HRESULT put_SLS_CreepingCoef(VARIANT_BOOL);
```

C#

```
public bool SLS_CreepingCoef { get; set; }
```

Visual Basic

```
Public SLS_CreepingCoef As Boolean
```

II.46.2.6 SLS_CreepingCoefValue

C++

```
HRESULT get_SLS_CreepingCoefValue(double*);
HRESULT put_SLS_CreepingCoefValue(double);
```

C#

```
public double SLS_CreepingCoefValue { get; set; }
```

Visual Basic

```
Public SLS_CreepingCoefValue As double
```

(c) 2014 Autodesk Inc. All rights reserved

II.46.2.7 SLS_Deflection

C++

```
HRESULT get_SLS_Deflection(VARIANT_BOOL*);
HRESULT put_SLS_Deflection(VARIANT_BOOL);
```

C#

```
public bool SLS_Deflection { get; set; }
```

Visual Basic

```
Public SLS_Deflection As Boolean
```

II.46.2.8 SLS_DeflectionReinfCorrect

C++

```
HRESULT get_SLS_DeflectionReinfCorrect(VARIANT_BOOL*);
HRESULT put_SLS_DeflectionReinfCorrect(VARIANT_BOOL);
```

C#

```
public bool SLS_DeflectionReinfCorrect { get; set; }
```

Visual Basic

```
Public SLS_DeflectionReinfCorrect As Boolean
```

II.46.2.9 SLS_EnvHumidityVal

C++

```
HRESULT get_SLS_EnvHumidityVal(double*);
HRESULT put_SLS_EnvHumidityVal(double);
```

C#

```
public double SLS_EnvHumidityVal { get; set; }
```

Visual Basic

```
Public SLS_EnvHumidityVal As double
```

II.46.2.10 SLS_Exposure

C++

```
HRESULT get_SLS_Exposure(IRConcr_EC2_ExposureRatings*);
HRESULT put_SLS_Exposure(IRConcr_EC2_ExposureRatings);
```

C#

```
public IRConcr_EC2_ExposureRatings SLS_Exposure { get; set; }
```

Visual Basic

```
Public SLS_Exposure As IRConcr_EC2_ExposureRatings
```

II.46.2.11 SLS_MaxCracking

C++

```
HRESULT get_SLS_MaxCracking(double*);
HRESULT put_SLS_MaxCracking(double);
```

C#

```
public double SLS_MaxCracking { get; set; }
```

Visual Basic

```
Public SLS_MaxCracking As double
```

(c) 2014 Autodesk Inc. All rights reserved

II.46.2.12 SLS_MaxDeflection

C++

```
HRESULT get_SLS_MaxDeflection(double*);
HRESULT put_SLS_MaxDeflection(double);
```

C#

```
public double SLS_MaxDeflection { get; set; }
```

Visual Basic

```
Public SLS_MaxDeflection As double
```

II.47 IRConcr_EC2_NAD

C++

```
enum IRConcr_EC2_NAD;
```

C#

```
public enum IRConcr_EC2_NAD;
```

Visual Basic

```
Public Enum IRConcr_EC2_NAD
```

II.48 IRConcrMinimumReinforcementType

C++

```
enum IRConcrMinimumReinforcementType;
```

C#

```
public enum IRConcrMinimumReinforcementType;
```

Visual Basic

```
Public Enum IRConcrMinimumReinforcementType
```

Members

Members	Description
I_CMRT_NONE = 0	Available since version 8.5.
I_CMRT_FOR_NONZERO_AS = 1	Available since version 8.5.
I_CMRT_FOR_WHOLE_PANEL = 2	Available since version 8.5.

Version

Available since version 8.5.

II.49 IRConcr_IS2000_EnvironmentType

C++

```
enum IRConcr_IS2000_EnvironmentType;
```

C#

```
public enum IRConcr_IS2000_EnvironmentType;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Enum IRConcr_IS2000_EnvironmentType
```

Members

Members	Description
I_IS2000_ER_MILD = 0	Available since version 9.5.
I_IS2000_ER_MODERATE = 1	Available since version 9.5.
I_IS2000_ER_SEVERE = 2	Available since version 9.5.
I_IS2000_ER_VERY_SEVERE = 3	Available since version 9.5.
I_IS2000_ER_MOST_SEVERE = 4	Available since version 9.5.

Description

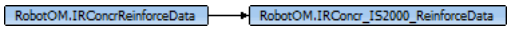
Exposure types defined for the IS 456:2000 code.

Version

Available since version 9.5.

II.50 IRConcr_IS2000_ReinforceData

Class Hierarchy



C++

```
interface IRConcr_IS2000_ReinforceData : IRConcrReinforceData;
```

C#

```
public interface IRConcr_IS2000_ReinforceData : IRConcrReinforceData;
```

Visual Basic

```
Public Interface IRConcr_IS2000_ReinforceData
```

Description

Reinforcement parameters for the Indian code IS 456:2000.

Version

Available since version 9.5.

II.50.1 IRConcr_IS2000_ReinforceData Members

The following tables list the members exposed by IRConcr_IS2000_ReinforceData.

Public Fields

	Name	Description
	BarDim_D1_Bot (see page 1705)	Diameter of reinforcing bars of bottom main reinforcement Available since version 1.7.
	BarDim_D1_Up (see page 1705)	Diameter of reinforcing bars of top main reinforcement Available since version 1.7.
	BarDim_D2_Bot (see page 1706)	Reinforcing bar diameter of bottom reinforcement in the direction perpendicular to the main reinforcement Available since version 1.7.
	BarDim_D2_Up (see page 1706)	Reinforcing bar diameter of top reinforcement in the direction perpendicular to the main reinforcement Available since version 1.7.
	CodeName (see page 1706)	Code name Available since version 1.7.
	Concrete (see page 1706)	Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

	Cover_Bot (see page 1707)	Cover - bottom reinforcement Available since version 1.7.
	Cover_Up (see page 1707)	Cover - top reinforcement Available since version 1.7.
	Main (see page 1782)	
	ReinforcingSteel (see page 1707)	Available since version 1.7.
	SLS_ConcreteAge (see page 1783)	
	SLS_Cracking (see page 1783)	
	SLS_CrackingReinfCorrect (see page 1783)	
	SLS_CreepingCoeff (see page 1783)	
	SLS_CreepingCoeffValue (see page 1784)	
	SLS_Deflection (see page 1784)	
	SLS_DeflectionReinfCorrect (see page 1784)	
	SLS_ExposureBot (see page 1784)	
	SLS_ExposureTop (see page 1785)	
	SLS_LoadRatio (see page 1785)	
	SLS_MaxCrackingBot (see page 1785)	
	SLS_MaxCrackingBotEnabled (see page 1785)	
	SLS_MaxCrackingTop (see page 1786)	
	SLS_MaxCrackingTopEnabled (see page 1786)	
	SLS_MaxDeflection (see page 1786)	

Public Methods

	Name	Description
	GetMainDirection (see page 1708)	Function returns the main reiforcement direction. Interpretation of returned coordinate values depends on the manner of determinig direction (direction vector coordinates, if the direction is determined by means of the vector, ALONG_VECTOR, or the point coordinates in the polar system - POLAR_POINT). Available since version 1.7.
	SetMainDirection (see page 1708)	Function defines the main reiforcement direction. Parameter interpretation depends on the manner of determinig direction (direction vector coordinates, if the direction is determined by means of the vector, ALONG_VECTOR, or the point coordinates in the polar system - POLAR_POINT, in the remaining cases the coordinate values are ignored). Available since version 1.7.

II.50.2 IRConcr_IS2000_ReinforceData Fields

The fields of the IRConcr_IS2000_ReinforceData class are listed here.

Public Fields

	Name	Description
	Main (see page 1782)	
	SLS_ConcreteAge (see page 1783)	
	SLS_Cracking (see page 1783)	

(c) 2014 Autodesk Inc. All rights reserved

	SLS_CrackingReinfCorrect (see page 1783)	
	SLS_CreepingCoeff (see page 1783)	
	SLS_CreepingCoeffValue (see page 1784)	
	SLS_Deflection (see page 1784)	
	SLS_DeflectionReinfCorrect (see page 1784)	
	SLS_ExposureBot (see page 1784)	
	SLS_ExposureTop (see page 1785)	
	SLS_LoadRatio (see page 1785)	
	SLS_MaxCrackingBot (see page 1785)	
	SLS_MaxCrackingBotEnabled (see page 1785)	
	SLS_MaxCrackingTop (see page 1786)	
	SLS_MaxCrackingTopEnabled (see page 1786)	
	SLS_MaxDeflection (see page 1786)	

II.50.2.1 Main

C++

```
HRESULT get_Main(IRConcrReinforceDataMain**);
```

C#

```
public IRConcrReinforceDataMain Main { get; }
```

Visual Basic

```
Public ReadOnly Main As IRConcrReinforceDataMain
```

Version

Available since version 9.5.

II.50.2.2 SLS_ConcreteAge

C++

```
HRESULT get_SLS_ConcreteAge(short*);
HRESULT put_SLS_ConcreteAge(short);
```

C#

```
public short SLS_ConcreteAge { get; set; }
```

Visual Basic

```
Public SLS_ConcreteAge As short
```

Version

Available since version 9.5.

(c) 2014 Autodesk Inc. All rights reserved

II.50.2.3 SLS_Cracking

C++

```
HRESULT get_SLS_Cracking(VARIANT_BOOL*);
HRESULT put_SLS_Cracking(VARIANT_BOOL);
```

C#

```
public bool SLS_Cracking { get; set; }
```

Visual Basic

```
Public SLS_Cracking As Boolean
```

Version

Available since version 9.5.

II.50.2.4 SLS_CrackingReinfCorrect

C++

```
HRESULT get_SLS_CrackingReinfCorrect(VARIANT_BOOL*);
HRESULT put_SLS_CrackingReinfCorrect(VARIANT_BOOL);
```

C#

```
public bool SLS_CrackingReinfCorrect { get; set; }
```

Visual Basic

```
Public SLS_CrackingReinfCorrect As Boolean
```

Version

Available since version 9.5.

II.50.2.5 SLS_CreepingCoeff

C++

```
HRESULT get_SLS_CreepingCoeff(VARIANT_BOOL*);
HRESULT put_SLS_CreepingCoeff(VARIANT_BOOL);
```

C#

```
public bool SLS_CreepingCoeff { get; set; }
```

Visual Basic

```
Public SLS_CreepingCoeff As Boolean
```

Version

Available since version 9.5.

II.50.2.6 SLS_CreepingCoeffValue

C++

```
HRESULT get_SLS_CreepingCoeffValue(double*);
HRESULT put_SLS_CreepingCoeffValue(double);
```

C#

```
public double SLS_CreepingCoeffValue { get; set; }
```

Visual Basic

```
Public SLS_CreepingCoeffValue As double
```

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 9.5.

II.50.2.7 SLS_Deflection

C++

```
HRESULT get_SLS_Deflection(VARIANT_BOOL*);
HRESULT put_SLS_Deflection(VARIANT_BOOL);
```

C#

```
public bool SLS_Deflection { get; set; }
```

Visual Basic

```
Public SLS_Deflection As Boolean
```

Version

Available since version 9.5.

II.50.2.8 SLS_DeflectionReinfCorrect

C++

```
HRESULT get_SLS_DeflectionReinfCorrect(VARIANT_BOOL*);
HRESULT put_SLS_DeflectionReinfCorrect(VARIANT_BOOL);
```

C#

```
public bool SLS_DeflectionReinfCorrect { get; set; }
```

Visual Basic

```
Public SLS_DeflectionReinfCorrect As Boolean
```

Version

Available since version 9.5.

II.50.2.9 SLS_ExposureBot

C++

```
HRESULT get_SLS_ExposureBot(short*);
HRESULT put_SLS_ExposureBot(short);
```

C#

```
public short SLS_ExposureBot { get; set; }
```

Visual Basic

```
Public SLS_ExposureBot As short
```

Version

Available since version 9.5.

II.50.2.10 SLS_ExposureTop

C++

```
HRESULT get_SLS_ExposureTop(short*);
HRESULT put_SLS_ExposureTop(short);
```

C#

```
public short SLS_ExposureTop { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public SLS_ExposureTop As short
```

Version

Available since version 9.5.

II.50.2.11 SLS_LoadRatio

C++

```
HRESULT get_SLS_LoadRatio(double*);
HRESULT put_SLS_LoadRatio(double);
```

C#

```
public double SLS_LoadRatio { get; set; }
```

Visual Basic

```
Public SLS_LoadRatio As double
```

Version

Available since version 9.5.

II.50.2.12 SLS_MaxCrackingBot

C++

```
HRESULT get_SLS_MaxCrackingBot(double*);
HRESULT put_SLS_MaxCrackingBot(double);
```

C#

```
public double SLS_MaxCrackingBot { get; set; }
```

Visual Basic

```
Public SLS_MaxCrackingBot As double
```

Version

Available since version 9.5.

II.50.2.13 SLS_MaxCrackingBotEnabled

C++

```
HRESULT get_SLS_MaxCrackingBotEnabled(VARIANT_BOOL*);
HRESULT put_SLS_MaxCrackingBotEnabled(VARIANT_BOOL);
```

C#

```
public bool SLS_MaxCrackingBotEnabled { get; set; }
```

Visual Basic

```
Public SLS_MaxCrackingBotEnabled As Boolean
```

Version

Available since version 9.5.

II.50.2.14 SLS_MaxCrackingTop

C++

```
HRESULT get_SLS_MaxCrackingTop(double*);
HRESULT put_SLS_MaxCrackingTop(double);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double SLS_MaxCrackingTop { get; set; }
```

Visual Basic

```
Public SLS_MaxCrackingTop As double
```

Version

Available since version 9.5.

II.50.2.15 SLS_MaxCrackingTopEnabled

C++

```
HRESULT get_SLS_MaxCrackingTopEnabled(VARIANT_BOOL*);
HRESULT put_SLS_MaxCrackingTopEnabled(VARIANT_BOOL);
```

C#

```
public bool SLS_MaxCrackingTopEnabled { get; set; }
```

Visual Basic

```
Public SLS_MaxCrackingTopEnabled As Boolean
```

Version

Available since version 9.5.

II.50.2.16 SLS_MaxDeflection

C++

```
HRESULT get_SLS_MaxDeflection(double*);
HRESULT put_SLS_MaxDeflection(double);
```

C#

```
public double SLS_MaxDeflection { get; set; }
```

Visual Basic

```
Public SLS_MaxDeflection As double
```

Version

Available since version 9.5.

II.51 IRConcrPlateCodeService2

Class Hierarchy

C++

```
interface IRConcrPlateCodeService2 : IDispatch;
```

C#

```
public interface IRConcrPlateCodeService2;
```

Visual Basic

```
Public Interface IRConcrPlateCodeService2
```

Version

Available since version 12.5.

(c) 2014 Autodesk Inc. All rights reserved

II.51.1 IRConcrPlateCodeService2 Members

The following tables list the members exposed by IRConcrPlateCodeService2.

Public Fields

	Name	Description
	RobotVersion (see page 1787)	
	ShowLongTermCracking (see page 1788)	

II.51.2 IRConcrPlateCodeService2 Fields

The fields of the IRConcrPlateCodeService2 class are listed here.

Public Fields

	Name	Description
	RobotVersion (see page 1787)	
	ShowLongTermCracking (see page 1788)	

II.51.2.1 RobotVersion

C++

```
HRESULT get_RobotVersion(BSTR*);
```

C#

```
public String RobotVersion { get; }
```

Visual Basic

```
Public ReadOnly RobotVersion As String
```

Version

Available since version 12.5.

II.51.2.2 ShowLongTermCracking

C++

```
HRESULT get_ShowLongTermCracking(VARIANT_BOOL*);
```

C#

```
public bool ShowLongTermCracking { get; }
```

Visual Basic

```
Public ReadOnly ShowLongTermCracking As Boolean
```

Version

Available since version 12.5.

III Design of RC members

Available since version 7.5.

(c) 2014 Autodesk Inc. All rights reserved

Enumerations

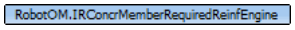
	Name	Description
	IRConcrBeamCalcPointDefinitionType (see page 1794)	Available methods of definition of calculation points for a beam.

Interfaces

	Name	Description
	IRConcrMemberRequiredReinfEngine (see page 1788)	Calculation module for the design of required reinforcement of RC members.
	IRConcrMemberRequiredReinfCalcParams (see page 1790)	Calculation parameters for the design of required reinforcement of RC members.

III.1 IRConcrMemberRequiredReinfEngine

Class Hierarchy



C++

```
interface IRConcrMemberRequiredReinfEngine : IDispatch;
```

C#

```
public interface IRConcrMemberRequiredReinfEngine;
```

Visual Basic

```
Public Interface IRConcrMemberRequiredReinfEngine
```

Description

Calculation module for the design of required reinforcement of RC members.

Version

Available since version 7.5.

III.1.1 IRConcrMemberRequiredReinfEngine Members

The following tables list the members exposed by IRConcrMemberRequiredReinfEngine.

Public Fields

	Name	Description
	Params (see page 1789)	Calculation parameters.

Public Methods

	Name	Description
	Calculate (see page 1789)	Function performs calculations of the required reinforcement of RC members according to the currently-set parameters.
	GetCalculatedMembers (see page 1790)	Function returns selection of members for which results of required reinforcement calculations are available.

III.1.2 IRConcrMemberRequiredReinfEngine Fields

The fields of the IRConcrMemberRequiredReinfEngine class are listed here.

Public Fields

	Name	Description
	Params (see page 1789)	Calculation parameters.

(c) 2014 Autodesk Inc. All rights reserved

III.1.2.1 Params

C++

```
HRESULT get_Params(IRConcrMemberRequiredReinfCalcParams**);
```

C#

```
public IRConcrMemberRequiredReinfCalcParams Params { get; }
```

Visual Basic

```
Public ReadOnly Params As IRConcrMemberRequiredReinfCalcParams
```

Description

Calculation parameters.

Version

Available since version 7.5.

III.1.3 IRConcrMemberRequiredReinfEngine Methods

The methods of the IRConcrMemberRequiredReinfEngine class are listed here.

Public Methods

	Name	Description
	Calculate (see page 1789)	Function performs calculations of the required reinforcement of RC members according to the currently-set parameters.
	GetCalculatedMembers (see page 1790)	Function returns selection of members for which results of required reinforcement calculations are available.

III.1.3.1 Calculate

C++

```
HRESULT Calculate(VARIANT_BOOL* ret);
```

C#

```
public bool Calculate();
```

Visual Basic

```
Public Function Calculate() As Boolean
```

Description

Function performs calculations of the required reinforcement of RC members according to the currently-set parameters.

Version

Available since version 7.5.

III.1.3.2 GetCalculatedMembers

C++

```
HRESULT GetCalculatedMembers(IRobotSelection** ret);
```

C#

```
public IRobotSelection GetCalculatedMembers();
```

Visual Basic

```
Public Function GetCalculatedMembers() As IRobotSelection
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Function returns selection of members for which results of required reinforcement calculations are available.

Version

Available since version 11.

III.2 IRConcrMemberRequiredReinfCalcParams

Class Hierarchy

RobotOM.IRConcrMemberRequiredReinfCalcParams

C++

```
interface IRConcrMemberRequiredReinfCalcParams : IDispatch;
```

C#

```
public interface IRConcrMemberRequiredReinfCalcParams;
```

Visual Basic

```
Public Interface IRConcrMemberRequiredReinfCalcParams
```

Description

Calculation parameters for the design of required reinforcement of RC members.

Version

Available since version 7.5.

III.2.1 IRConcrMemberRequiredReinfCalcParams Members

The following tables list the members exposed by IRConcrMemberRequiredReinfCalcParams.

Public Fields

	Name	Description
	BeamPointsType (see page 1791)	Method of definition of division points for a beam.
	BeamPointsValue (see page 1792)	Value determining points of the beam division.
	CasesALS (see page 1792)	
	CasesSLS (see page 1792)	
	CasesULS (see page 1792)	
	CombALS (see page 1793)	
	CombSLS (see page 1793)	
	CombULS (see page 1793)	
	Members (see page 1793)	Bar selection.

III.2.2 IRConcrMemberRequiredReinfCalcParams Fields

The fields of the IRConcrMemberRequiredReinfCalcParams class are listed here.

Public Fields

	Name	Description
	BeamPointsType (see page 1791)	Method of definition of division points for a beam.
	BeamPointsValue (see page 1792)	Value determining points of the beam division.
	CasesALS (see page 1792)	

(c) 2014 Autodesk Inc. All rights reserved

	CasesSLS (see page 1792)	
	CasesULS (see page 1792)	
	CombALS (see page 1793)	
	CombSLS (see page 1793)	
	CombULS (see page 1793)	
	Members (see page 1793)	Bar selection.

III.2.2.1 BeamPointsType

C++

```
HRESULT get_BeamPointsType(IRConcrBeamCalcPointDefinitionType*);
HRESULT put_BeamPointsType(IRConcrBeamCalcPointDefinitionType);
```

C#

```
public IRConcrBeamCalcPointDefinitionType BeamPointsType { get; set; }
```

Visual Basic

```
Public BeamPointsType As IRConcrBeamCalcPointDefinitionType
```

Description

Method of definition of division points for a beam.

Version

Available since version 7.5.

III.2.2.2 BeamPointsValue

C++

```
HRESULT get_BeamPointsValue(double*);
HRESULT put_BeamPointsValue(double);
```

C#

```
public double BeamPointsValue { get; set; }
```

Visual Basic

```
Public BeamPointsValue As double
```

Description

Value determining points of the beam division.

Version

Available since version 7.5.

III.2.2.3 CasesALS

C++

```
HRESULT get_CasesALS(IRobotSelection**);
```

C#

```
public IRobotSelection CasesALS { get; }
```

Visual Basic

```
Public ReadOnly CasesALS As IRobotSelection
```

Version

Available since version 7.5.

(c) 2014 Autodesk Inc. All rights reserved

III.2.2.4 CasesSLS

C++

```
HRESULT get_CasesSLS(IRobotSelection**);
```

C#

```
public IRobotSelection CasesSLS { get; }
```

Visual Basic

```
Public ReadOnly CasesSLS As IRobotSelection
```

Version

Available since version 7.5.

III.2.2.5 CasesULS

C++

```
HRESULT get_CasesULS(IRobotSelection**);
```

C#

```
public IRobotSelection CasesULS { get; }
```

Visual Basic

```
Public ReadOnly CasesULS As IRobotSelection
```

Version

Available since version 7.5.

III.2.2.6 CombALS

C++

```
HRESULT get_CombALS(VARIANT_BOOL*);
HRESULT put_CombALS(VARIANT_BOOL);
```

C#

```
public bool CombALS { get; set; }
```

Visual Basic

```
Public CombALS As Boolean
```

Version

Available since version 7.5.

III.2.2.7 CombSLS

C++

```
HRESULT get_CombSLS(VARIANT_BOOL*);
HRESULT put_CombSLS(VARIANT_BOOL);
```

C#

```
public bool CombSLS { get; set; }
```

Visual Basic

```
Public CombSLS As Boolean
```

Version

Available since version 7.5.

(c) 2014 Autodesk Inc. All rights reserved

III.2.2.8 CombULS

C++

```
HRESULT get_CombULS(VARIANT_BOOL*);
HRESULT put_CombULS(VARIANT_BOOL);
```

C#

```
public bool CombULS { get; set; }
```

Visual Basic

```
Public CombULS As Boolean
```

Version

Available since version 7.5.

III.2.2.9 Members

C++

```
HRESULT get_Members(IRobotSelection**);
```

C#

```
public IRobotSelection Members { get; }
```

Visual Basic

```
Public ReadOnly Members As IRobotSelection
```

Description

Bar selection.

Version

Available since version 7.5.

III.3 IRConcrBeamCalcPointDefinitionType

C++

```
enum IRConcrBeamCalcPointDefinitionType;
```

C#

```
public enum IRConcrBeamCalcPointDefinitionType;
```

Visual Basic

```
Public Enum IRConcrBeamCalcPointDefinitionType
```

Members

Members	Description
I_CBCPDT_POINT_COUNT = 0	Number of points for beam. Available since version 7.5.
I_CBCPDT_POINT_SPACING = 1	Distance between successive division points expressed in meters. Available since version 7.5.

Description

Available methods of definition of calculation points for a beam.

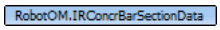
(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 7.5.

IV IRConcrBarSectionData

Class Hierarchy



C++

```
interface IRConcrBarSectionData : IDispatch;
```

C#

```
public interface IRConcrBarSectionData;
```

Visual Basic

```
Public Interface IRConcrBarSectionData
```

Description

A (see page 1795) simplified interface describing an RC member section.

IV.1 IRConcrBarSectionData Members

The following tables list the members exposed by IRConcrBarSectionData.

Public Fields

	Name	Description
	A (see page 1795)	
	Ac (see page 1796)	
	Ap (see page 1796)	
	B (see page 1796)	
	Bf (see page 1796)	
	Bfb (see page 1796)	
	H (see page 1797)	
	Hf (see page 1797)	
	Hfb (see page 1797)	
	N (see page 1797)	
	Per (see page 1797)	
	Type (see page 1798)	Section geometry type.

IV.2 IRConcrBarSectionData Fields

The fields of the IRConcrBarSectionData class are listed here.

Public Fields

	Name	Description
	A (see page 1795)	
	Ac (see page 1796)	
	Ap (see page 1796)	
	B (see page 1796)	
	Bf (see page 1796)	

(c) 2014 Autodesk Inc. All rights reserved

	Bfb (see page 1796)	
	H (see page 1797)	
	Hf (see page 1797)	
	Hfb (see page 1797)	
	N (see page 1797)	
	Per (see page 1797)	
	Type (see page 1798)	Section geometry type.

IV.2.1 A

C++

```
HRESULT get_A(double*);
HRESULT put_A(double);
```

C#

```
public double A { get; set; }
```

Visual Basic

```
Public A As double
```

IV.2.2 Ac

C++

```
HRESULT get_Ac(double*);
HRESULT put_Ac(double);
```

C#

```
public double Ac { get; set; }
```

Visual Basic

```
Public Ac As double
```

IV.2.3 Ap

C++

```
HRESULT get_Ap(double*);
HRESULT put_Ap(double);
```

C#

```
public double Ap { get; set; }
```

Visual Basic

```
Public Ap As double
```

IV.2.4 B

C++

```
HRESULT get_B(double*);
HRESULT put_B(double);
```

C#

```
public double B { get; set; }
```

Visual Basic

```
Public B As double
```

(c) 2014 Autodesk Inc. All rights reserved

IV.2.5 Bf

C++

```
HRESULT get_Bf(double*);
HRESULT put_Bf(double);
```

C#

```
public double Bf { get; set; }
```

Visual Basic

```
Public Bf As double
```

IV.2.6 Bfb

C++

```
HRESULT get_Bfb(double*);
HRESULT put_Bfb(double);
```

C#

```
public double Bfb { get; set; }
```

Visual Basic

```
Public Bfb As double
```

IV.2.7 H

C++

```
HRESULT get_H(double*);
HRESULT put_H(double);
```

C#

```
public double H { get; set; }
```

Visual Basic

```
Public H As double
```

IV.2.8 Hf

C++

```
HRESULT get_Hf(double*);
HRESULT put_Hf(double);
```

C#

```
public double Hf { get; set; }
```

Visual Basic

```
Public Hf As double
```

IV.2.9 Hfb

C++

```
HRESULT get_Hfb(double*);
HRESULT put_Hfb(double);
```

C#

```
public double Hfb { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Hfb As double
```

IV.2.10 N

C++

```
HRESULT get_N(int*);
HRESULT put_N(int);
```

C#

```
public int N { get; set; }
```

Visual Basic

```
Public N As int
```

IV.2.11 Per

C++

```
HRESULT get_Per(double*);
HRESULT put_Per(double);
```

C#

```
public double Per { get; set; }
```

Visual Basic

```
Public Per As double
```

IV.2.12 Type

C++

```
HRESULT get_Type(IRConcrBarSectionGeometryType*);
HRESULT put_Type(IRConcrBarSectionGeometryType);
```

C#

```
public IRConcrBarSectionGeometryType Type { get; set; }
```

Visual Basic

```
Public Type As IRConcrBarSectionGeometryType
```

Description

Section geometry type.

V IRConcrBarSectionGeometryType

C++

```
enum IRConcrBarSectionGeometryType;
```

C#

```
public enum IRConcrBarSectionGeometryType;
```

Visual Basic

```
Public Enum IRConcrBarSectionGeometryType
```

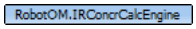
Description

Types of geometry of an RC member section .

(c) 2014 Autodesk Inc. All rights reserved

VI IRConcrCalcEngine

Class Hierarchy



C++

```
interface IRConcrCalcEngine : IDispatch;
```

C#

```
public interface IRConcrCalcEngine;
```

Visual Basic

```
Public Interface IRConcrCalcEngine
```

Description

RC calculation module.

Version

Available since version 5.5.

VI.1 IRConcrCalcEngine Members

The following tables list the members exposed by IRConcrCalcEngine.

Public Fields

	Name	Description
	MemberRequiredReinf (see page 1799)	Module calculating the required reinforcement for RC members.
	SlabRequiredReinf (see page 1799)	Module for calculation of the required (theoretical) reinforcement for plates and shells .

VI.2 IRConcrCalcEngine Fields

The fields of the IRConcrCalcEngine class are listed here.

Public Fields

	Name	Description
	MemberRequiredReinf (see page 1799)	Module calculating the required reinforcement for RC members.
	SlabRequiredReinf (see page 1799)	Module for calculation of the required (theoretical) reinforcement for plates and shells .

VI.2.1 MemberRequiredReinf

C++

```
HRESULT get_MemberRequiredReinf(IRConcrMemberRequiredReinfEngine**);
```

C#

```
public IRConcrMemberRequiredReinfEngine MemberRequiredReinf { get; }
```

Visual Basic

```
Public ReadOnly MemberRequiredReinf As IRConcrMemberRequiredReinfEngine
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Module calculating the required reinforcement for RC members.

Version

Available since version 7.5.

VI.2.2 SlabRequiredReinf

C++

```
HRESULT get_SlabRequiredReinf(IRConcrSlabRequiredReinfEngine**);
```

C#

```
public IRConcrSlabRequiredReinfEngine SlabRequiredReinf { get; }
```

Visual Basic

```
Public ReadOnly SlabRequiredReinf As IRConcrSlabRequiredReinfEngine
```

Description

Module for calculation of the required (theoretical) reinforcement for plates and shells .

Version

Available since version 5.5.

VII Supplementing RC modules with new codes

Available since version 12.

Enumerations

	Name	Description
	IRConcrCodeColumnCommand (see page 1803)	
	IRConcrCodeBeamCommand (see page 1806)	

Interfaces

	Name	Description
	IRConcrCodeService (see page 1800)	RConcrCodeService allows code-specific definitions and calculations for provided reinforcement of RC elements. Available from 12.0 version.
	IRConcrCodeColumn (see page 1802)	
	IRConcrCodeReport (see page 1803)	
	IRConcrCodeBeam (see page 1805)	

VII.1 IRConcrCodeService

Class Hierarchy

RobotOM.IRConcrCodeService

C++

```
interface IRConcrCodeService : IDispatch;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public interface IRConcrCodeService;
```

Visual Basic

```
Public Interface IRConcrCodeService
```

Description

RConcrCodeService allows code-specific definitions and calculations for provided reinforcement of RC elements. Available from 12.0 version.

Version

Available since version 12.

VII.1.1 IRConcrCodeService Members

The following tables list the members exposed by IRConcrCodeService.

Public Fields

	Name	Description
	Beam (see page 1801)	
	Column (see page 1801)	

Public Methods

	Name	Description
	IsConcrComponentServed (see page 1801)	

VII.1.2 IRConcrCodeService Fields

The fields of the IRConcrCodeService class are listed here.

Public Fields

	Name	Description
	Beam (see page 1801)	
	Column (see page 1801)	

VII.1.2.1 Beam

C++

```
HRESULT get_Beam(IRConcrCodeBeam**);
```

C#

```
public IRConcrCodeBeam Beam { get; }
```

Visual Basic

```
Public ReadOnly Beam As IRConcrCodeBeam
```

Version

Available since version 12.

VII.1.2.2 Column

C++

```
HRESULT get_Column(IRConcrCodeColumn**);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRConcrCodeColumn Column { get; }
```

Visual Basic

```
Public ReadOnly Column As IRConcrCodeColumn
```

Version

Available since version 12.

VII.1.3 IRConcrCodeService Methods

The methods of the IRConcrCodeService class are listed here.

Public Methods

	Name	Description
	IsConcrComponentServed (see page 1801)	

VII.1.3.1 IsConcrComponentServed

C++

```
HRESULT IsConcrComponentServed(IRobotProjectComponentType _module, VARIANT_BOOL* ret);
```

C#

```
public bool IsConcrComponentServed(IRobotProjectComponentType _module);
```

Visual Basic

```
Public Function IsConcrComponentServed(_module As IRobotProjectComponentType) As Boolean
```

Version

Available since version 12.

VII.2 IRConcrCodeColumn

Class Hierarchy

RobotOM.IRConcrCodeColumn

C++

```
interface IRConcrCodeColumn : IDispatch;
```

C#

```
public interface IRConcrCodeColumn;
```

Visual Basic

```
Public Interface IRConcrCodeColumn
```

Version

Available since version 12.

VII.2.1 IRConcrCodeColumn Members

The following tables list the members exposed by IRConcrCodeColumn.

(c) 2014 Autodesk Inc. All rights reserved

Public Methods

	Name	Description
	IsCommandEnabled (see page 1802)	
	Verification (see page 1803)	

VII.2.2 IRConcrCodeColumn Methods

The methods of the IRConcrCodeColumn class are listed here.

Public Methods

	Name	Description
	IsCommandEnabled (see page 1802)	
	Verification (see page 1803)	

VII.2.2.1 IsCommandEnabled

C++

```
HRESULT IsCommandEnabled(IRConcrCodeColumnCommand _command, VARIANT_BOOL* ret);
```

C#

```
public bool IsCommandEnabled(IRConcrCodeColumnCommand _command);
```

Visual Basic

```
Public Function IsCommandEnabled(_command As IRConcrCodeColumnCommand) As Boolean
```

Version

Available since version 12.

VII.2.2.2 Verification

C++

```
HRESULT Verification(IRConcrColumn* _column, IRConcrCodeReport* _report, VARIANT_BOOL* ret);
```

C#

```
public bool Verification(IRConcrColumn _column, IRConcrCodeReport _report);
```

Visual Basic

```
Public Function Verification(ByRef _column As IRConcrColumn, ByRef _report As
IRConcrCodeReport) As Boolean
```

Version

Available since version 12.

VII.3 IRConcrCodeColumnCommand

C++

```
enum IRConcrCodeColumnCommand;
```

C#

```
public enum IRConcrCodeColumnCommand;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Enum IRConcrCodeColumnCommand
```

Members

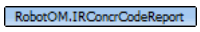
Members	Description
I_CCCC_CL_VERIFICATION = 1	Available since version 12.

Version

Available since version 12.

VII.4 IRConcrCodeReport

Class Hierarchy



C++

```
interface IRConcrCodeReport : IDispatch;
```

C#

```
public interface IRConcrCodeReport;
```

Visual Basic

```
Public Interface IRConcrCodeReport
```

Version

Available since version 12.

VII.4.1 IRConcrCodeReport Members

The following tables list the members exposed by IRConcrCodeReport.

Public Methods

	Name	Description
	AddError (see page 1804)	
	AddMessage (see page 1804)	
	AddWarning (see page 1805)	

VII.4.2 IRConcrCodeReport Methods

The methods of the IRConcrCodeReport class are listed here.

Public Methods

	Name	Description
	AddError (see page 1804)	
	AddMessage (see page 1804)	
	AddWarning (see page 1805)	

VII.4.2.1 AddError

C++

```
HRESULT AddError(BSTR _text);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public void AddError(String _text);
```

Visual Basic

```
Public Sub AddError(_text As String)
```

Version

Available since version 12.

VII.4.2.2 AddMessage

C++

```
HRESULT AddMessage(BSTR _message);
```

C#

```
public void AddMessage(String _message);
```

Visual Basic

```
Public Sub AddMessage(_message As String)
```

Version

Available since version 12.

VII.4.2.3 AddWarning

C++

```
HRESULT AddWarning(BSTR _warning);
```

C#

```
public void AddWarning(String _warning);
```

Visual Basic

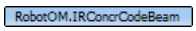
```
Public Sub AddWarning(_warning As String)
```

Version

Available since version 12.

VII.5 IRConcrCodeBeam

Class Hierarchy



C++

```
interface IRConcrCodeBeam : IDispatch;
```

C#

```
public interface IRConcrCodeBeam;
```

Visual Basic

```
Public Interface IRConcrCodeBeam
```

Version

Available since version 12.

(c) 2014 Autodesk Inc. All rights reserved

VII.5.1 IRConcrCodeBeam Members

The following tables list the members exposed by IRConcrCodeBeam.

Public Methods

	Name	Description
	IsCommandEnabled (see page 1806)	
	Verification (see page 1806)	

VII.5.2 IRConcrCodeBeam Methods

The methods of the IRConcrCodeBeam class are listed here.

Public Methods

	Name	Description
	IsCommandEnabled (see page 1806)	
	Verification (see page 1806)	

VII.5.2.1 IsCommandEnabled

C++

```
HRESULT IsCommandEnabled(IRConcrCodeBeamCommand _command, VARIANT_BOOL* ret);
```

C#

```
public bool IsCommandEnabled(IRConcrCodeBeamCommand _command);
```

Visual Basic

```
Public Function IsCommandEnabled(_command As IRConcrCodeBeamCommand) As Boolean
```

Version

Available since version 12.

VII.5.2.2 Verification

C++

```
HRESULT Verification(IRConcrBeam* _beam, IRConcrCodeReport* _report, VARIANT_BOOL* ret);
```

C#

```
public bool Verification(IRConcrBeam _beam, IRConcrCodeReport _report);
```

Visual Basic

```
Public Function Verification(ByRef _beam As IRConcrBeam, ByRef _report As
IRConcrCodeReport) As Boolean
```

Version

Available since version 12.

VII.6 IRConcrCodeBeamCommand

C++

```
enum IRConcrCodeBeamCommand;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public enum IRConcrCodeBeamCommand;
```

Visual Basic

```
Public Enum IRConcrCodeBeamCommand
```

Members

Members	Description
I_CCBC_BM_VERIFICATION = 1	Available since version 12.

Version

Available since version 12.

Steel and timber design

Enumerations

	Name	Description
	IRDimStreamType (see page 1889)	
	IRDimMembDefType (see page 1892)	
	IRDimMembDefMatType (see page 1893)	
	IRDimMembDefLengthDataType (see page 1893)	
	IRDimMembDefBucklingDataType (see page 1893)	
	IRDimMembDefDispDataType (see page 1893)	
	IRDimMembDefDeflDataType (see page 1894)	
	IRDimEffDefParamType (see page 1898)	
	IRDimEffDefDirType (see page 1898)	
	IRDimEffDefIntPsType (see page 1898)	
	IRDimMatDefType (see page 1908)	
	IRDimMatDefValType (see page 1908)	
	IRDimMatDefLongExValType (see page 1908)	
	IRDimMatDefDblExValType (see page 1908)	
	IRDimProfDefType (see page 1912)	
	IRDimProfDefItemType (see page 1912)	
	IRDimProfDefValType (see page 1912)	

(c) 2014 Autodesk Inc. All rights reserved

	IRDimCalcStateFlagType (see page 1915)	
	IRDimCalcStateParamType (see page 1915)	
	IRDimCalcStateParamValue (see page 1915)	
	IRDimCalcStateValueType (see page 1916)	
	IRDimMembCalcRetValue (see page 1919)	
	IRDimMembCalcBuckType (see page 1919)	
	IRDimMembResTableLineType (see page 1920)	
	IRDimMembResTableComp (see page 1920)	
	IRDimUnitType (see page 1935)	

Interfaces

	Name	Description
	IRDimStream (see page 1889)	
	IRDimMembDef (see page 1894)	
	IRDimEffDef (see page 1899)	
	IRDimMatDef (see page 1908)	
	IRDimProfDef (see page 1913)	
	IRDimCalcState (see page 1916)	
	IRDimMembSrv (see page 1918)	
	IRDimMembRes (see page 1920)	
	IRDimMembCalc (see page 1928)	
	IRDimCodeService (see page 1932)	
	IRDimClient (see page 1934)	
	IRDimUnits (see page 1935)	
	IRDimServer (see page 1937)	Server responsible for design and verification of steel and timber members.

EC3 Code

Available since version 3.5.

Enumerations

	Name	Description
	IRDimLoadTypeEC3 (see page 1817)	
	IRDimLaterBuckTypeEC3 (see page 1817)	
	IRDimLoadLevelEC3 (see page 1817)	
	IRDimLatBuckCoeffDiagramEC3 (see page 1818)	
	IRDimBuckDiagramEC3 (see page 1818)	

(c) 2014 Autodesk Inc. All rights reserved

	IRDimYieldStrengthTypeEC3 (see page 1852)	

Interfaces

	Name	Description
	IRDimMembParamsEC3 (see page 1809)	
	IRDimCodeResEC3 (see page 1819)	

I.1 IRDimMembParamsEC3

Class Hierarchy

RobotOM.IRDimMembParamsEC3

C++

```
interface IRDimMembParamsEC3 : IDispatch;
```

C#

```
public interface IRDimMembParamsEC3;
```

Visual Basic

```
Public Interface IRDimMembParamsEC3
```

Version

Available since version 3.5.

I.1.1 IRDimMembParamsEC3 Members

The following tables list the members exposed by IRDimMembParamsEC3.

Public Fields

	Name	Description
	BuckLengthCoeffY (see page 1811)	
	BuckLengthCoeffZ (see page 1811)	
	BucklingDiagramY (see page 1811)	
	BucklingDiagramZ (see page 1812)	
	HotRolledPipes (see page 1812)	
	LatBuckType (see page 1812)	
	LatCoeffLowerFlange (see page 1812)	
	LatCoeffLowerFlangeValue (see page 1813)	
	LatCoeffUpperFlange (see page 1813)	
	LatCoeffUpperFlangeValue (see page 1813)	
	LoadLevel (see page 1813)	
	LoadLevelValue (see page 1814)	
	LoadTypeY (see page 1814)	

(c) 2014 Autodesk Inc. All rights reserved

	LoadTypeZ (see page 1814)	
	MaterCoeffGamma0 (see page 1815)	
	MaterCoeffGamma1 (see page 1815)	
	RelLimitDeflUy (see page 1815)	
	RelLimitdeflUz (see page 1815)	
	TensAreaNetGros (see page 1816)	
	TubeControl (see page 1816)	
	YieldStrengthType (see page 1816)	
	YieldStrengthValue (see page 1816)	

I.1.2 IRDimMembParamsEC3 Fields

The fields of the IRDimMembParamsEC3 class are listed here.

Public Fields

	Name	Description
	BuckLengthCoeffY (see page 1811)	
	BuckLengthCoeffZ (see page 1811)	
	BucklingDiagramY (see page 1811)	
	BucklingDiagramZ (see page 1812)	
	HotRolledPipes (see page 1812)	
	LatBuckType (see page 1812)	
	LatCoeffLowerFlange (see page 1812)	
	LatCoeffLowerFlangeValue (see page 1813)	
	LatCoeffUpperFlange (see page 1813)	
	LatCoeffUpperFlangeValue (see page 1813)	
	LoadLevel (see page 1813)	
	LoadLevelValue (see page 1814)	
	LoadTypeY (see page 1814)	
	LoadTypeZ (see page 1814)	
	MaterCoeffGamma0 (see page 1815)	
	MaterCoeffGamma1 (see page 1815)	
	RelLimitDeflUy (see page 1815)	
	RelLimitdeflUz (see page 1815)	
	TensAreaNetGros (see page 1816)	
	TubeControl (see page 1816)	
	YieldStrengthType (see page 1816)	

(c) 2014 Autodesk Inc. All rights reserved

	YieldStrengthValue (see page 1816)	

I.1.2.1 BuckLengthCoeffY

C++

```
HRESULT get_BuckLengthCoeffY(double*);
HRESULT put_BuckLengthCoeffY(double);
```

C#

```
public double BuckLengthCoeffY { get; set; }
```

Visual Basic

```
Public BuckLengthCoeffY As double
```

Version

Available since version 3.

I.1.2.2 BuckLengthCoeffZ

C++

```
HRESULT get_BuckLengthCoeffZ(double*);
HRESULT put_BuckLengthCoeffZ(double);
```

C#

```
public double BuckLengthCoeffZ { get; set; }
```

Visual Basic

```
Public BuckLengthCoeffZ As double
```

Version

Available since version 3.

I.1.2.3 BucklingDiagramY

C++

```
HRESULT get_BucklingDiagramY(IRDimBuckDiagramEC3*);
HRESULT put_BucklingDiagramY(IRDimBuckDiagramEC3);
```

C#

```
public IRDimBuckDiagramEC3 BucklingDiagramY { get; set; }
```

Visual Basic

```
Public BucklingDiagramY As IRDimBuckDiagramEC3
```

Version

Available since version 3.

I.1.2.4 BucklingDiagramZ

C++

```
HRESULT get_BucklingDiagramZ(IRDimBuckDiagramEC3*);
HRESULT put_BucklingDiagramZ(IRDimBuckDiagramEC3);
```

C#

```
public IRDimBuckDiagramEC3 BucklingDiagramZ { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public BucklingDiagramZ As IRDimBuckDiagramEC3
```

Version

Available since version 3.

I.1.2.5 HotRolledPipes

C++

```
HRESULT get_HotRolledPipes(VARIANT_BOOL*);
HRESULT put_HotRolledPipes(VARIANT_BOOL);
```

C#

```
public bool HotRolledPipes { get; set; }
```

Visual Basic

```
Public HotRolledPipes As Boolean
```

Version

Available since version 3.5.

I.1.2.6 LatBuckType

C++

```
HRESULT get_LatBuckType(IRDimLaterBuckTypeEC3*);
HRESULT put_LatBuckType(IRDimLaterBuckTypeEC3);
```

C#

```
public IRDimLaterBuckTypeEC3 LatBuckType { get; set; }
```

Visual Basic

```
Public LatBuckType As IRDimLaterBuckTypeEC3
```

Version

Available since version 3.

I.1.2.7 LatCoeffLowerFlange

C++

```
HRESULT get_LatCoeffLowerFlange(IRDimLatBuckCoeffDiagramEC3*);
HRESULT put_LatCoeffLowerFlange(IRDimLatBuckCoeffDiagramEC3);
```

C#

```
public IRDimLatBuckCoeffDiagramEC3 LatCoeffLowerFlange { get; set; }
```

Visual Basic

```
Public LatCoeffLowerFlange As IRDimLatBuckCoeffDiagramEC3
```

Version

Available since version 3.

I.1.2.8 LatCoeffLowerFlangeValue

C++

```
HRESULT get_LatCoeffLowerFlangeValue(double*);
HRESULT put_LatCoeffLowerFlangeValue(double);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double LatCoeffLowerFlangeValue { get; set; }
```

Visual Basic

```
Public LatCoeffLowerFlangeValue As double
```

Version

Available since version 3.

I.1.2.9 LatCoeffUpperFlange

C++

```
HRESULT get_LatCoeffUpperFlange(IRDimLatBuckCoeffDiagramEC3*);
HRESULT put_LatCoeffUpperFlange(IRDimLatBuckCoeffDiagramEC3);
```

C#

```
public IRDimLatBuckCoeffDiagramEC3 LatCoeffUpperFlange { get; set; }
```

Visual Basic

```
Public LatCoeffUpperFlange As IRDimLatBuckCoeffDiagramEC3
```

Version

Available since version 3.

I.1.2.10 LatCoeffUpperFlangeValue

C++

```
HRESULT get_LatCoeffUpperFlangeValue(double*);
HRESULT put_LatCoeffUpperFlangeValue(double);
```

C#

```
public double LatCoeffUpperFlangeValue { get; set; }
```

Visual Basic

```
Public LatCoeffUpperFlangeValue As double
```

Version

Available since version 3.

I.1.2.11 LoadLevel

C++

```
HRESULT get_LoadLevel(IRDimLoadLevelEC3*);
HRESULT put_LoadLevel(IRDimLoadLevelEC3);
```

C#

```
public IRDimLoadLevelEC3 LoadLevel { get; set; }
```

Visual Basic

```
Public LoadLevel As IRDimLoadLevelEC3
```

Version

Available since version 3.

I.1.2.12 LoadLevelValue

C++

```
HRESULT get_LoadLevelValue(double*);
```

(c) 2014 Autodesk Inc. All rights reserved

```
HRESULT put_LoadLevelValue(double);
```

C#

```
public double LoadLevelValue { get; set; }
```

Visual Basic

```
Public LoadLevelValue As double
```

Version

Available since version 3.

I.1.2.13 LoadTypeY

C++

```
HRESULT get_LoadTypeY(IRDimLoadTypeEC3*);
HRESULT put_LoadTypeY(IRDimLoadTypeEC3);
```

C#

```
public IRDimLoadTypeEC3 LoadTypeY { get; set; }
```

Visual Basic

```
Public LoadTypeY As IRDimLoadTypeEC3
```

Version

Available since version 3.

I.1.2.14 LoadTypeZ

C++

```
HRESULT get_LoadTypeZ(IRDimLoadTypeEC3*);
HRESULT put_LoadTypeZ(IRDimLoadTypeEC3);
```

C#

```
public IRDimLoadTypeEC3 LoadTypeZ { get; set; }
```

Visual Basic

```
Public LoadTypeZ As IRDimLoadTypeEC3
```

Version

Available since version 3.

I.1.2.15 MaterCoeffGamma0

C++

```
HRESULT get_MaterCoeffGamma0(double*);
HRESULT put_MaterCoeffGamma0(double);
```

C#

```
public double MaterCoeffGamma0 { get; set; }
```

Visual Basic

```
Public MaterCoeffGamma0 As double
```

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

I.1.2.16 MaterCoeffGamma1

C++

```
HRESULT get_MaterCoeffGamma1(double*);
HRESULT put_MaterCoeffGamma1(double);
```

C#

```
public double MaterCoeffGamma1 { get; set; }
```

Visual Basic

```
Public MaterCoeffGamma1 As double
```

Version

Available since version 3.

I.1.2.17 RelLimitDeflUy

C++

```
HRESULT get_RelLimitDeflUy(double*);
HRESULT put_RelLimitDeflUy(double);
```

C#

```
public double RelLimitDeflUy { get; set; }
```

Visual Basic

```
Public RelLimitDeflUy As double
```

Version

Available since version 3.

I.1.2.18 RelLimitdeflUz

C++

```
HRESULT get_RelLimitdeflUz(double*);
HRESULT put_RelLimitdeflUz(double);
```

C#

```
public double RelLimitdeflUz { get; set; }
```

Visual Basic

```
Public RelLimitdeflUz As double
```

Version

Available since version 3.

I.1.2.19 TensAreaNetGros

C++

```
HRESULT get_TensAreaNetGros(double*);
HRESULT put_TensAreaNetGros(double);
```

C#

```
public double TensAreaNetGros { get; set; }
```

Visual Basic

```
Public TensAreaNetGros As double
```

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.

I.1.2.20 TubeControl

C++

```
HRESULT get_TubeControl(VARIANT_BOOL*);
HRESULT put_TubeControl(VARIANT_BOOL);
```

C#

```
public bool TubeControl { get; set; }
```

Visual Basic

```
Public TubeControl As Boolean
```

Version

Available since version 3.

I.1.2.21 YieldStrengthType

C++

```
HRESULT get_YieldStrengthType(IRDimYieldStrengthTypeEC3*);
HRESULT put_YieldStrengthType(IRDimYieldStrengthTypeEC3);
```

C#

```
public IRDimYieldStrengthTypeEC3 YieldStrengthType { get; set; }
```

Visual Basic

```
Public YieldStrengthType As IRDimYieldStrengthTypeEC3
```

Version

Available since version 3.5.

I.1.2.22 YieldStrengthValue

C++

```
HRESULT get_YieldStrengthValue(double*);
HRESULT put_YieldStrengthValue(double);
```

C#

```
public double YieldStrengthValue { get; set; }
```

Visual Basic

```
Public YieldStrengthValue As double
```

Version

Available since version 3.

I.2 IRDimLoadTypeEC3

C++

```
enum IRDimLoadTypeEC3;
```

C#

```
public enum IRDimLoadTypeEC3;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Enum IRDimLoadTypeEC3
```

Members

Members	Description
I_DLT_EC3_UNIFORM_LOAD = 0	Available since version 3.
I_DLT_EC3_UNIFORM_MOMENT = 1	Available since version 3.
I_DLT_EC3_CONCENTRATED_FORCE = 2	Available since version 3.
I_DLT_EC3_MOMENTS_AT_ENDS = 3	Available since version 3.

Version

Available since version 3.5.

I.3 IRDimLaterBuckTypeEC3

C++

```
enum IRDimLaterBuckTypeEC3;
```

C#

```
public enum IRDimLaterBuckTypeEC3;
```

Visual Basic

```
Public Enum IRDimLaterBuckTypeEC3
```

Members

Members	Description
I_DLBT_EC3_SYMMETR_LOADED = 0	Available since version 3.
I_DLBT_EC3_CANTILEVER = 1	Available since version 3.
I_DLBT_EC3_NO = 2	Available since version 3.

Version

Available since version 3.5.

I.4 IRDimLoadLevelEC3

C++

```
enum IRDimLoadLevelEC3;
```

C#

```
public enum IRDimLoadLevelEC3;
```

Visual Basic

```
Public Enum IRDimLoadLevelEC3
```

Members

Members	Description
I_DLL_EC3_UPP_EDGE_LOADED = 0	Available since version 3.
I_DLL_EC3_UPP_SECT_PAR_LOADED = 1	Available since version 3.
I_DLL_EC3_CENTER_LOADED = 2	Available since version 3.
I_DLL_EC3_LOW_SEC_PART_LOADED = 3	Available since version 3.
I_DLL_EC3_LOW_EDGE_LOADED = 4	Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.5.

I.5 IRDimLatBuckCoeffDiagramEC3

C++

```
enum IRDimLatBuckCoeffDiagramEC3;
```

C#

```
public enum IRDimLatBuckCoeffDiagramEC3;
```

Visual Basic

```
Public Enum IRDimLatBuckCoeffDiagramEC3
```

Members

Members	Description
I_DLBCD_EC3_CANTILEVER_2_0 = 0	Available since version 3.
I_DLBCD_EC3_INTERNAL_BRACINGS = 5	Available since version 3.
I_DLBCD_EC3_NO = 4	Available since version 3.
I_DLBCD_EC3_PINNED_PINNED_1_0 = 1	Available since version 3.
I_DLBCD_EC3_STIFF_STIFF_0_5 = 2	Available since version 3.
I_DLBCD_EC3_USER_DEFINED = 3	Available since version 3.

Version

Available since version 3.5.

I.6 IRDimBuckDiagramEC3

C++

```
enum IRDimBuckDiagramEC3;
```

C#

```
public enum IRDimBuckDiagramEC3;
```

Visual Basic

```
Public Enum IRDimBuckDiagramEC3
```

Members

Members	Description
I_DBD_EC3_AUTO = 12	Available since version 3.
I_DBD_EC3_CANTILEVER_2_0 = 3	Available since version 3.
I_DBD_EC3_INTERN_ADJBAR_6 = 8	Available since version 3.
I_DBD_EC3_INTERNAL_BRACINGS = 13	Available since version 3.
I_DBD_EC3_NO = 11	Available since version 3.
I_DBD_EC3_PINNED_ADJBAR_1 = 4	Available since version 3.
I_DBD_EC3_PINNED_ADJBAR_3 = 6	Available since version 3.
I_DBD_EC3_PINNED_PINNED_1_0 = 0	Available since version 3.
I_DBD_EC3_PINNED_STIFF_0_7 = 2	Available since version 3.
I_DBD_EC3_STIFF_ADJBAR_1 = 5	Available since version 3.
I_DBD_EC3_STIFF_ADJBAR_3 = 7	Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

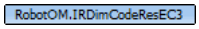
I_DBD_EC3_STIFF_STIFF_0_5 = 1	Available since version 3.
I_DBD_EC3_TRUSS_CHORD_0_9 = 9	Available since version 3.
I_DBD_EC3_TRUSS_DIAGONAL_0_8 = 10	Available since version 3.
I_DBD_EC3_USER_DEFINED = -1	Available since version 3.

Version

Available since version 3.5.

I.7 IRDimCodeResEC3

Class Hierarchy



C++

```
interface IRDimCodeResEC3 : IDispatch;
```

C#

```
public interface IRDimCodeResEC3;
```

Visual Basic

```
Public Interface IRDimCodeResEC3
```

Version

Available since version 3.5.

I.7.1 IRDimCodeResEC3 Members

The following tables list the members exposed by IRDimCodeResEC3.

Public Fields

	Name	Description
	AxForceExcentrEny (see page 1827)	
	AxForcExcentrEnz (see page 1827)	
	BendParamBetaMlty (see page 1827)	
	BendParamBetaMltz (see page 1828)	
	BendParamBetaMz (see page 1828)	
	BendParamBetMy (see page 1828)	
	BuckCoeffMinXi (see page 1828)	
	BuckCoeffXy (see page 1829)	
	BuckCoeffXz (see page 1829)	
	BuckCurveCoeffAlfy (see page 1829)	
	BuckCurveCoeffAlfz (see page 1829)	
	BuckCurveNumbY (see page 1830)	
	BuckCurveNumbZ (see page 1830)	

(c) 2014 Autodesk Inc. All rights reserved

	BuckParamFiy (see page 1830)	
	BuckParamFiz (see page 1830)	
	BuckParamKy (see page 1831)	
	BuckParamKz (see page 1831)	
	BuckRelSlendLaby (see page 1831)	
	BuckRelSlendLabz (see page 1831)	
	BuckSLendLamy (see page 1832)	
	BuckStrenNbyrd (see page 1832)	
	BuckStrenNbzrd (see page 1832)	
	BuclSlendLamz (see page 1832)	
	ClassOfSect (see page 1833)	
	ClassOfSectElem1 (see page 1833)	
	ClassOfSectElem2 (see page 1833)	
	ClassOfSectElem3 (see page 1833)	
	ClassOfSectElem4 (see page 1834)	
	CompParamBetaA (see page 1834)	
	CritMomenMcr (see page 1834)	
	EffectiveMomenMeff (see page 1834)	
	EffSectAreaSeff (see page 1835)	
	EffSectModulWyeff (see page 1835)	
	EffSectModulWzeff (see page 1835)	
	ElastMomStrenMelyrd (see page 1835)	
	ElastMomStrenMelzrd (see page 1836)	
	ElastSectModulWyel (see page 1836)	
	ElastSectModulWzel (see page 1836)	
	FlangeAreaAf (see page 1836)	
	FlangeAreaAw (see page 1837)	
	InteractParamAlfa (see page 1837)	
	InteractParamBeta (see page 1837)	
	InteractParamMilt (see page 1837)	
	InteractParamMiy (see page 1838)	
	InteractParamMiz (see page 1838)	
	IsBuckY (see page 1838)	
	IsBuckZ (see page 1838)	
	LaodLevel (see page 1839)	

(c) 2014 Autodesk Inc. All rights reserved

	LatBuckCoeffXlt (see page 1839)	
	LatBuckLengthLd (see page 1839)	
	LatBuckMomStrenmbrd (see page 1839)	
	LatBuckParamBetaW (see page 1840)	
	LatBuckParamC1 (see page 1840)	
	LatBuckParamC2 (see page 1840)	
	LatBuckParamFilt (see page 1840)	
	LatBuckParamKlt (see page 1841)	
	LatBuckParamLd (see page 1841)	
	LatBuckSlendLamlt (see page 1841)	
	LowFlanSlend1 (see page 1841)	
	LowFlanSlend2 (see page 1842)	
	MaEffRatio (see page 1842)	
	MaterCoeffGamma0 (see page 1842)	
	MaterCoeffGamma1 (see page 1842)	
	MaterCoeffGamma2 (see page 1843)	
	MaterialCapacityFy (see page 1843)	
	MaxBuckSlend (see page 1843)	
	MomStrengthMpzEuro (see page 1843)	
	OverallBuckStrenNbrd (see page 1844)	
	PartEffRatio1 (see page 1844)	
	PartEffRatio2 (see page 1844)	
	PartEffRatio3 (see page 1844)	
	PartEffRatio4 (see page 1845)	
	PartEffRatio5 (see page 1845)	
	PlastAxForcStrenNplrd (see page 1845)	
	PlastCompStrenNcrd (see page 1845)	
	PlastMomStrenMplyrd (see page 1846)	
	PlastMomStrenMplzrd (see page 1846)	
	PlastSectModulWypl (see page 1846)	
	PlastSectModulWzpl (see page 1846)	
	PlastTensStrenNtrd (see page 1847)	

(c) 2014 Autodesk Inc. All rights reserved

	ReducMomStrenMcyrd (see page 1847)	
	ReducMomStrenMczrd (see page 1847)	
	ReducMomStrenMnyrd (see page 1847)	
	ReducMomStrenMnzrd (see page 1848)	
	ReducMomStrenMvyrd (see page 1848)	
	ReducMomStrenMvzrd (see page 1848)	
	ShearStrenTplyrd (see page 1848)	
	ShearStrenTplzrd (see page 1849)	
	StrsComp (see page 1849)	
	StrsLftEdgeMZ (see page 1849)	
	StrsLowEdgeMY (see page 1849)	
	StrsRgtEdgeMZ (see page 1850)	
	StrsShearY (see page 1850)	
	StrsShearZ (see page 1850)	
	StrsTens (see page 1850)	
	StrsUprEdgeMY (see page 1851)	
	TensStrenNurd (see page 1851)	
	TorsMomInertIt (see page 1851)	
	UppFlanSlend (see page 1851)	
	WarpingConstantIw (see page 1852)	
	WebSlend (see page 1852)	
	WebSlend1 (see page 1852)	

I.7.2 IRDimCodeResEC3 Fields

The fields of the IRDimCodeResEC3 class are listed here.

Public Fields

	Name	Description
	AxForceExcentrEny (see page 1827)	
	AxForcExcentrEnz (see page 1827)	
	BendParamBetaMlty (see page 1827)	
	BendParamBetaMltz (see page 1828)	
	BendParamBetaMz (see page 1828)	
	BendParamBetMy (see page 1828)	
	BuckCoeffMinXi (see page 1828)	
	BuckCoeffXy (see page 1829)	
	BuckCoeffXz (see page 1829)	

(c) 2014 Autodesk Inc. All rights reserved

	BuckCurveCoeffAlfy (see page 1829)	
	BuckCurveCoeffAlfz (see page 1829)	
	BuckCurveNumbY (see page 1830)	
	BuckCurveNumbZ (see page 1830)	
	BuckParamFiy (see page 1830)	
	BuckParamFiz (see page 1830)	
	BuckParamKy (see page 1831)	
	BuckParamKz (see page 1831)	
	BuckRelSlendLaby (see page 1831)	
	BuckRelSlendLabz (see page 1831)	
	BuckSLendLamy (see page 1832)	
	BuckStrenNbyrd (see page 1832)	
	BuckStrenNbzrd (see page 1832)	
	BuclSlendLamz (see page 1832)	
	ClassOfSect (see page 1833)	
	ClassOfSectElem1 (see page 1833)	
	ClassOfSectElem2 (see page 1833)	
	ClassOfSectElem3 (see page 1833)	
	ClassOfSectElem4 (see page 1834)	
	CompParamBetaA (see page 1834)	
	CritMomenMcr (see page 1834)	
	EffectiveMomenMeff (see page 1834)	
	EffSectAreaSeff (see page 1835)	
	EffSectModulWyeff (see page 1835)	
	EffSectModulWzeff (see page 1835)	
	ElastMomStrenMelyrd (see page 1835)	
	ElastMomStrenMelzrd (see page 1836)	
	ElastSectModulWyel (see page 1836)	
	ElastSectModulWzel (see page 1836)	
	FlangeAreaAf (see page 1836)	
	FlangeAreaAw (see page 1837)	
	InteractParamAlfa (see page 1837)	
	InteractParamBeta (see page 1837)	

(c) 2014 Autodesk Inc. All rights reserved

	InteractParamMilt (see page 1837)	
	InteractParamMiy (see page 1838)	
	InteractParamMiz (see page 1838)	
	IsBuckY (see page 1838)	
	IsBuckZ (see page 1838)	
	LaodLevel (see page 1839)	
	LatBuckCoeffXlt (see page 1839)	
	LatBuckLengthLd (see page 1839)	
	LatBuckMomStrenmbrd (see page 1839)	
	LatBuckParamBetaW (see page 1840)	
	LatBuckParamC1 (see page 1840)	
	LatBuckParamC2 (see page 1840)	
	LatBuckParamFilt (see page 1840)	
	LatBuckParamKlt (see page 1841)	
	LatBuckParamLd (see page 1841)	
	LatBuckSlendLamlt (see page 1841)	
	LowFlanSlend1 (see page 1841)	
	LowFlanSlend2 (see page 1842)	
	MaEffRatio (see page 1842)	
	MaterCoeffGamma0 (see page 1842)	
	MaterCoeffGamma1 (see page 1842)	
	MaterCoeffGamma2 (see page 1843)	
	MaterialCapacityFy (see page 1843)	
	MaxBuckSlend (see page 1843)	
	MomStrengthMpzEuro (see page 1843)	
	OverallBuckStrenNbrd (see page 1844)	
	PartEffRatio1 (see page 1844)	
	PartEffRatio2 (see page 1844)	
	PartEffRatio3 (see page 1844)	
	PartEffRatio4 (see page 1845)	
	PartEffRatio5 (see page 1845)	
	PlastAxForcStrenNplrd (see page 1845)	
	PlastCompStrenNcrd (see page 1845)	

(c) 2014 Autodesk Inc. All rights reserved

	PlastMomStrenMplyrd (see page 1846)	
	PlastMomStrenMplzrd (see page 1846)	
	PlastSectModulWypl (see page 1846)	
	PlastSectModulWzpl (see page 1846)	
	PlastTensStrenNtrd (see page 1847)	
	ReducMomStrenMcyrd (see page 1847)	
	ReducMomStrenMczrd (see page 1847)	
	ReducMomStrenMnyrd (see page 1847)	
	ReducMomStrenMnzrd (see page 1848)	
	ReducMomStrenMvyrd (see page 1848)	
	ReducMomStrenMvzrd (see page 1848)	
	ShearStrenTplyrd (see page 1848)	
	ShearStrenTplzrd (see page 1849)	
	StrsComp (see page 1849)	
	StrsLftEdgeMZ (see page 1849)	
	StrsLowEdgeMY (see page 1849)	
	StrsRgtEdgeMZ (see page 1850)	
	StrsShearY (see page 1850)	
	StrsShearZ (see page 1850)	
	StrsTens (see page 1850)	
	StrsUprEdgeMY (see page 1851)	
	TensStrenNurd (see page 1851)	
	TorsMomInertIt (see page 1851)	
	UppFlanSlend (see page 1851)	
	WarpingConstantIw (see page 1852)	
	WebSlend (see page 1852)	
	WebSlend1 (see page 1852)	

I.7.2.1 AxForceExcentrEny

C++

```
HRESULT get_AxForceExcentrEny(double*);
```

C#

```
public double AxForceExcentrEny { get; }
```

Visual Basic

```
Public ReadOnly AxForceExcentrEny As double
```

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

I.7.2.2 AxForcExcentrEnz

C++

```
HRESULT get_AxForcExcentrEnz(double*);
```

C#

```
public double AxForcExcentrEnz { get; }
```

Visual Basic

```
Public ReadOnly AxForcExcentrEnz As double
```

Version

Available since version 3.

I.7.2.3 BendParamBetaMlty

C++

```
HRESULT get_BendParamBetaMlty(double*);
```

C#

```
public double BendParamBetaMlty { get; }
```

Visual Basic

```
Public ReadOnly BendParamBetaMlty As double
```

Version

Available since version 3.

I.7.2.4 BendParamBetaMltz

C++

```
HRESULT get_BendParamBetaMltz(double*);
```

C#

```
public double BendParamBetaMltz { get; }
```

Visual Basic

```
Public ReadOnly BendParamBetaMltz As double
```

Version

Available since version 3.

I.7.2.5 BendParamBetaMz

C++

```
HRESULT get_BendParamBetaMz(double*);
```

C#

```
public double BendParamBetaMz { get; }
```

Visual Basic

```
Public ReadOnly BendParamBetaMz As double
```

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

I.7.2.6 BendParamBetMy

C++

```
HRESULT get_BendParamBetMy(double*);
```

C#

```
public double BendParamBetMy { get; }
```

Visual Basic

```
Public ReadOnly BendParamBetMy As double
```

Version

Available since version 3.

I.7.2.7 BuckCoeffMinXi

C++

```
HRESULT get_BuckCoeffMinXi(double*);
```

C#

```
public double BuckCoeffMinXi { get; }
```

Visual Basic

```
Public ReadOnly BuckCoeffMinXi As double
```

Version

Available since version 3.

I.7.2.8 BuckCoeffXy

C++

```
HRESULT get_BuckCoeffXy(double*);
```

C#

```
public double BuckCoeffXy { get; }
```

Visual Basic

```
Public ReadOnly BuckCoeffXy As double
```

Version

Available since version 3.

I.7.2.9 BuckCoeffXz

C++

```
HRESULT get_BuckCoeffXz(double*);
```

C#

```
public double BuckCoeffXz { get; }
```

Visual Basic

```
Public ReadOnly BuckCoeffXz As double
```

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

I.7.2.10 BuckCurveCoeffAlfy

C++

```
HRESULT get_BuckCurveCoeffAlfy(double*);
```

C#

```
public double BuckCurveCoeffAlfy { get; }
```

Visual Basic

```
Public ReadOnly BuckCurveCoeffAlfy As double
```

Version

Available since version 3.

I.7.2.11 BuckCurveCoeffAlfz

C++

```
HRESULT get_BuckCurveCoeffAlfz(double*);
```

C#

```
public double BuckCurveCoeffAlfz { get; }
```

Visual Basic

```
Public ReadOnly BuckCurveCoeffAlfz As double
```

Version

Available since version 3.

I.7.2.12 BuckCurveNumbY

C++

```
HRESULT get_BuckCurveNumbY(double*);
```

C#

```
public double BuckCurveNumbY { get; }
```

Visual Basic

```
Public ReadOnly BuckCurveNumbY As double
```

Version

Available since version 3.

I.7.2.13 BuckCurveNumbZ

C++

```
HRESULT get_BuckCurveNumbZ(double*);
```

C#

```
public double BuckCurveNumbZ { get; }
```

Visual Basic

```
Public ReadOnly BuckCurveNumbZ As double
```

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

I.7.2.14 BuckParamFiy

C++

```
HRESULT get_BuckParamFiy(double*);
```

C#

```
public double BuckParamFiy { get; }
```

Visual Basic

```
Public ReadOnly BuckParamFiy As double
```

Version

Available since version 3.

I.7.2.15 BuckParamFiz

C++

```
HRESULT get_BuckParamFiz(double*);
```

C#

```
public double BuckParamFiz { get; }
```

Visual Basic

```
Public ReadOnly BuckParamFiz As double
```

Version

Available since version 3.

I.7.2.16 BuckParamKy

C++

```
HRESULT get_BuckParamKy(double*);
```

C#

```
public double BuckParamKy { get; }
```

Visual Basic

```
Public ReadOnly BuckParamKy As double
```

Version

Available since version 3.

I.7.2.17 BuckParamKz

C++

```
HRESULT get_BuckParamKz(double*);
```

C#

```
public double BuckParamKz { get; }
```

Visual Basic

```
Public ReadOnly BuckParamKz As double
```

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

I.7.2.18 BuckRelSlendLaby

C++

```
HRESULT get_BuckRelSlendLaby(double*);
```

C#

```
public double BuckRelSlendLaby { get; }
```

Visual Basic

```
Public ReadOnly BuckRelSlendLaby As double
```

Version

Available since version 3.

I.7.2.19 BuckRelSlendLabz

C++

```
HRESULT get_BuckRelSlendLabz(double*);
```

C#

```
public double BuckRelSlendLabz { get; }
```

Visual Basic

```
Public ReadOnly BuckRelSlendLabz As double
```

Version

Available since version 3.

I.7.2.20 BuckSLendLamy

C++

```
HRESULT get_BuckSLendLamy(double*);
```

C#

```
public double BuckSLendLamy { get; }
```

Visual Basic

```
Public ReadOnly BuckSLendLamy As double
```

Version

Available since version 3.

I.7.2.21 BuckStrenNbyrd

C++

```
HRESULT get_BuckStrenNbyrd(double*);
```

C#

```
public double BuckStrenNbyrd { get; }
```

Visual Basic

```
Public ReadOnly BuckStrenNbyrd As double
```

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

I.7.2.22 BuckStrenNbzrd

C++

```
HRESULT get_BuckStrenNbzrd(double*);
```

C#

```
public double BuckStrenNbzrd { get; }
```

Visual Basic

```
Public ReadOnly BuckStrenNbzrd As double
```

Version

Available since version 3.

I.7.2.23 BuclSlendLamz

C++

```
HRESULT get_BuclSlendLamz(double*);
```

C#

```
public double BuclSlendLamz { get; }
```

Visual Basic

```
Public ReadOnly BuclSlendLamz As double
```

Version

Available since version 3.

I.7.2.24 ClassOfSect

C++

```
HRESULT get_ClassOfSect(long*);
```

C#

```
public long ClassOfSect { get; }
```

Visual Basic

```
Public ReadOnly ClassOfSect As long
```

Version

Available since version 3.

I.7.2.25 ClassOfSectElem1

C++

```
HRESULT get_ClassOfSectElem1(long*);
```

C#

```
public long ClassOfSectElem1 { get; }
```

Visual Basic

```
Public ReadOnly ClassOfSectElem1 As long
```

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

I.7.2.26 ClassOfSectElem2

C++

```
HRESULT get_ClassOfSectElem2(long*);
```

C#

```
public long ClassOfSectElem2 { get; }
```

Visual Basic

```
Public ReadOnly ClassOfSectElem2 As long
```

Version

Available since version 3.

I.7.2.27 ClassOfSectElem3

C++

```
HRESULT get_ClassOfSectElem3(long*);
```

C#

```
public long ClassOfSectElem3 { get; }
```

Visual Basic

```
Public ReadOnly ClassOfSectElem3 As long
```

Version

Available since version 3.

I.7.2.28 ClassOfSectElem4

C++

```
HRESULT get_ClassOfSectElem4(long*);
```

C#

```
public long ClassOfSectElem4 { get; }
```

Visual Basic

```
Public ReadOnly ClassOfSectElem4 As long
```

Version

Available since version 3.

I.7.2.29 CompParamBetaA

C++

```
HRESULT get_CompParamBetaA(double*);
```

C#

```
public double CompParamBetaA { get; }
```

Visual Basic

```
Public ReadOnly CompParamBetaA As double
```

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

I.7.2.30 CritMomenMcr

C++

```
HRESULT get_CritMomenMcr(double*);
```

C#

```
public double CritMomenMcr { get; }
```

Visual Basic

```
Public ReadOnly CritMomenMcr As double
```

Version

Available since version 3.

I.7.2.31 EffectiveMomenMeff

C++

```
HRESULT get_EffectiveMomenMeff(double*);
```

C#

```
public double EffectiveMomenMeff { get; }
```

Visual Basic

```
Public ReadOnly EffectiveMomenMeff As double
```

Version

Available since version 3.

I.7.2.32 EffSectAreaSeff

C++

```
HRESULT get_EffSectAreaSeff(double*);
```

C#

```
public double EffSectAreaSeff { get; }
```

Visual Basic

```
Public ReadOnly EffSectAreaSeff As double
```

Version

Available since version 3.

I.7.2.33 EffSectModulWyeff

C++

```
HRESULT get_EffSectModulWyeff(double*);
```

C#

```
public double EffSectModulWyeff { get; }
```

Visual Basic

```
Public ReadOnly EffSectModulWyeff As double
```

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

I.7.2.34 EffSectModulWzeff

C++

```
HRESULT get_EffSectModulWzeff(double*);
```

C#

```
public double EffSectModulWzeff { get; }
```

Visual Basic

```
Public ReadOnly EffSectModulWzeff As double
```

Version

Available since version 3.

I.7.2.35 ElastMomStrenMelyrd

C++

```
HRESULT get_ElastMomStrenMelyrd(double*);
```

C#

```
public double ElastMomStrenMelyrd { get; }
```

Visual Basic

```
Public ReadOnly ElastMomStrenMelyrd As double
```

Version

Available since version 3.

I.7.2.36 ElastMomStrenMelzrd

C++

```
HRESULT get_ElastMomStrenMelzrd(double*);
```

C#

```
public double ElastMomStrenMelzrd { get; }
```

Visual Basic

```
Public ReadOnly ElastMomStrenMelzrd As double
```

Version

Available since version 3.

I.7.2.37 ElastSectModulWyel

C++

```
HRESULT get_ElastSectModulWyel(double*);
```

C#

```
public double ElastSectModulWyel { get; }
```

Visual Basic

```
Public ReadOnly ElastSectModulWyel As double
```

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

I.7.2.38 ElastSectModulWzel

C++

```
HRESULT get_ElastSectModulWzel(double*);
```

C#

```
public double ElastSectModulWzel { get; }
```

Visual Basic

```
Public ReadOnly ElastSectModulWzel As double
```

Version

Available since version 3.

I.7.2.39 FlangeAreaAf

C++

```
HRESULT get_FlangeAreaAf(double*);
```

C#

```
public double FlangeAreaAf { get; }
```

Visual Basic

```
Public ReadOnly FlangeAreaAf As double
```

Version

Available since version 3.

I.7.2.40 FlangeAreaAw

C++

```
HRESULT get_FlangeAreaAw(double*);
```

C#

```
public double FlangeAreaAw { get; }
```

Visual Basic

```
Public ReadOnly FlangeAreaAw As double
```

Version

Available since version 3.

I.7.2.41 InteractParamAlfa

C++

```
HRESULT get_InteractParamAlfa(double*);
```

C#

```
public double InteractParamAlfa { get; }
```

Visual Basic

```
Public ReadOnly InteractParamAlfa As double
```

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

I.7.2.42 InteractParamBeta

C++

```
HRESULT get_InteractParamBeta(double*);
```

C#

```
public double InteractParamBeta { get; }
```

Visual Basic

```
Public ReadOnly InteractParamBeta As double
```

Version

Available since version 3.

I.7.2.43 InteractParamMilt

C++

```
HRESULT get_InteractParamMilt(double*);
```

C#

```
public double InteractParamMilt { get; }
```

Visual Basic

```
Public ReadOnly InteractParamMilt As double
```

Version

Available since version 3.

I.7.2.44 InteractParamMiy

C++

```
HRESULT get_InteractParamMiy(double*);
```

C#

```
public double InteractParamMiy { get; }
```

Visual Basic

```
Public ReadOnly InteractParamMiy As double
```

Version

Available since version 3.

I.7.2.45 InteractParamMiz

C++

```
HRESULT get_InteractParamMiz(double*);
```

C#

```
public double InteractParamMiz { get; }
```

Visual Basic

```
Public ReadOnly InteractParamMiz As double
```

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

I.7.2.46 IsBuckY

C++

```
HRESULT get_IsBuckY(long*);
```

C#

```
public long IsBuckY { get; }
```

Visual Basic

```
Public ReadOnly IsBuckY As long
```

Version

Available since version 3.

I.7.2.47 IsBuckZ

C++

```
HRESULT get_IsBuckZ(long*);
```

C#

```
public long IsBuckZ { get; }
```

Visual Basic

```
Public ReadOnly IsBuckZ As long
```

Version

Available since version 3.

I.7.2.48 LaodLevel

C++

```
HRESULT get_LaodLevel(double*);
```

C#

```
public double LaodLevel { get; }
```

Visual Basic

```
Public ReadOnly LaodLevel As double
```

Version

Available since version 3.

I.7.2.49 LatBuckCoeffXlt

C++

```
HRESULT get_LatBuckCoeffXlt(double*);
```

C#

```
public double LatBuckCoeffXlt { get; }
```

Visual Basic

```
Public ReadOnly LatBuckCoeffXlt As double
```

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

I.7.2.50 LatBuckLengthLd

C++

```
HRESULT get_LatBuckLengthLd(double*);
```

C#

```
public double LatBuckLengthLd { get; }
```

Visual Basic

```
Public ReadOnly LatBuckLengthLd As double
```

Version

Available since version 3.

I.7.2.51 LatBuckMomStrenmbrd

C++

```
HRESULT get_LatBuckMomStrenmbrd(double*);
```

C#

```
public double LatBuckMomStrenmbrd { get; }
```

Visual Basic

```
Public ReadOnly LatBuckMomStrenmbrd As double
```

Version

Available since version 3.

I.7.2.52 LatBuckParamBetaW

C++

```
HRESULT get_LatBuckParamBetaW(double*);
```

C#

```
public double LatBuckParamBetaW { get; }
```

Visual Basic

```
Public ReadOnly LatBuckParamBetaW As double
```

Version

Available since version 3.

I.7.2.53 LatBuckParamC1

C++

```
HRESULT get_LatBuckParamC1(double*);
```

C#

```
public double LatBuckParamC1 { get; }
```

Visual Basic

```
Public ReadOnly LatBuckParamC1 As double
```

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

I.7.2.54 LatBuckParamC2

C++

```
HRESULT get_LatBuckParamC2(double*);
```

C#

```
public double LatBuckParamC2 { get; }
```

Visual Basic

```
Public ReadOnly LatBuckParamC2 As double
```

Version

Available since version 3.

I.7.2.55 LatBuckParamFilt

C++

```
HRESULT get_LatBuckParamFilt(double*);
```

C#

```
public double LatBuckParamFilt { get; }
```

Visual Basic

```
Public ReadOnly LatBuckParamFilt As double
```

Version

Available since version 3.

I.7.2.56 LatBuckParamKlt

C++

```
HRESULT get_LatBuckParamKlt(double*);
```

C#

```
public double LatBuckParamKlt { get; }
```

Visual Basic

```
Public ReadOnly LatBuckParamKlt As double
```

Version

Available since version 3.

I.7.2.57 LatBuckParamLd

C++

```
HRESULT get_LatBuckParamLd(long*);
```

C#

```
public long LatBuckParamLd { get; }
```

Visual Basic

```
Public ReadOnly LatBuckParamLd As long
```

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

I.7.2.58 LatBuckSlendLamlt

C++

```
HRESULT get_LatBuckSlendLamlt(double*);
```

C#

```
public double LatBuckSlendLamlt { get; }
```

Visual Basic

```
Public ReadOnly LatBuckSlendLamlt As double
```

Version

Available since version 3.

I.7.2.59 LowFlanSlend1

C++

```
HRESULT get_LowFlanSlend1(double*);
```

C#

```
public double LowFlanSlend1 { get; }
```

Visual Basic

```
Public ReadOnly LowFlanSlend1 As double
```

Version

Available since version 3.

I.7.2.60 LowFlanSlend2

C++

```
HRESULT get_LowFlanSlend2(double*);
```

C#

```
public double LowFlanSlend2 { get; }
```

Visual Basic

```
Public ReadOnly LowFlanSlend2 As double
```

Version

Available since version 3.

I.7.2.61 MaEffRatio

C++

```
HRESULT get_MaEffRatio(double*);
```

C#

```
public double MaEffRatio { get; }
```

Visual Basic

```
Public ReadOnly MaEffRatio As double
```

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

I.7.2.62 MaterCoeffGamma0

C++

```
HRESULT get_MaterCoeffGamma0(double*);
```

C#

```
public double MaterCoeffGamma0 { get; }
```

Visual Basic

```
Public ReadOnly MaterCoeffGamma0 As double
```

Version

Available since version 3.

I.7.2.63 MaterCoeffGamma1

C++

```
HRESULT get_MaterCoeffGamma1(double*);
```

C#

```
public double MaterCoeffGamma1 { get; }
```

Visual Basic

```
Public ReadOnly MaterCoeffGamma1 As double
```

Version

Available since version 3.

I.7.2.64 MaterCoeffGamma2

C++

```
HRESULT get_MaterCoeffGamma2(double*);
```

C#

```
public double MaterCoeffGamma2 { get; }
```

Visual Basic

```
Public ReadOnly MaterCoeffGamma2 As double
```

Version

Available since version 3.

I.7.2.65 MaterialCapacityFy

C++

```
HRESULT get_MaterialCapacityFy(double*);
```

C#

```
public double MaterialCapacityFy { get; }
```

Visual Basic

```
Public ReadOnly MaterialCapacityFy As double
```

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

I.7.2.66 MaxBuckSlend

C++

```
HRESULT get_MaxBuckSlend(double*);
```

C#

```
public double MaxBuckSlend { get; }
```

Visual Basic

```
Public ReadOnly MaxBuckSlend As double
```

Version

Available since version 3.

I.7.2.67 MomStrengthMpzEuro

C++

```
HRESULT get_MomStrengthMpzEuro(double*);
```

C#

```
public double MomStrengthMpzEuro { get; }
```

Visual Basic

```
Public ReadOnly MomStrengthMpzEuro As double
```

Version

Available since version 3.

I.7.2.68 OverallBuckStrenNbrd

C++

```
HRESULT get_OverallBuckStrenNbrd(double*);
```

C#

```
public double OverallBuckStrenNbrd { get; }
```

Visual Basic

```
Public ReadOnly OverallBuckStrenNbrd As double
```

Version

Available since version 3.

I.7.2.69 PartEffRatio1

C++

```
HRESULT get_PartEffRatio1(double*);
```

C#

```
public double PartEffRatio1 { get; }
```

Visual Basic

```
Public ReadOnly PartEffRatio1 As double
```

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

I.7.2.70 PartEffRatio2

C++

```
HRESULT get_PartEffRatio2(double*);
```

C#

```
public double PartEffRatio2 { get; }
```

Visual Basic

```
Public ReadOnly PartEffRatio2 As double
```

Version

Available since version 3.

I.7.2.71 PartEffRatio3

C++

```
HRESULT get_PartEffRatio3(double*);
```

C#

```
public double PartEffRatio3 { get; }
```

Visual Basic

```
Public ReadOnly PartEffRatio3 As double
```

Version

Available since version 3.

I.7.2.72 PartEffRatio4

C++

```
HRESULT get_PartEffRatio4(double*);
```

C#

```
public double PartEffRatio4 { get; }
```

Visual Basic

```
Public ReadOnly PartEffRatio4 As double
```

Version

Available since version 3.

I.7.2.73 PartEffRatio5

C++

```
HRESULT get_PartEffRatio5(double*);
```

C#

```
public double PartEffRatio5 { get; }
```

Visual Basic

```
Public ReadOnly PartEffRatio5 As double
```

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

I.7.2.74 PlastAxForcStrenNplrd

C++

```
HRESULT get_PlastAxForcStrenNplrd(double*);
```

C#

```
public double PlastAxForcStrenNplrd { get; }
```

Visual Basic

```
Public ReadOnly PlastAxForcStrenNplrd As double
```

Version

Available since version 3.

I.7.2.75 PlastCompStrenNcrd

C++

```
HRESULT get_PlastCompStrenNcrd(double*);
```

C#

```
public double PlastCompStrenNcrd { get; }
```

Visual Basic

```
Public ReadOnly PlastCompStrenNcrd As double
```

Version

Available since version 3.

I.7.2.76 PlastMomStrenMplyrd

C++

```
HRESULT get_PlastMomStrenMplyrd(double*);
```

C#

```
public double PlastMomStrenMplyrd { get; }
```

Visual Basic

```
Public ReadOnly PlastMomStrenMplyrd As double
```

Version

Available since version 3.

I.7.2.77 PlastMomStrenMplzrd

C++

```
HRESULT get_PlastMomStrenMplzrd(double*);
```

C#

```
public double PlastMomStrenMplzrd { get; }
```

Visual Basic

```
Public ReadOnly PlastMomStrenMplzrd As double
```

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

I.7.2.78 PlastSectModulWypl

C++

```
HRESULT get_PlastSectModulWypl(double*);
```

C#

```
public double PlastSectModulWypl { get; }
```

Visual Basic

```
Public ReadOnly PlastSectModulWypl As double
```

Version

Available since version 3.

I.7.2.79 PlastSectModulWzpl

C++

```
HRESULT get_PlastSectModulWzpl(double*);
```

C#

```
public double PlastSectModulWzpl { get; }
```

Visual Basic

```
Public ReadOnly PlastSectModulWzpl As double
```

Version

Available since version 3.

I.7.2.80 PlastTensStrenNtrd

C++

```
HRESULT get_PlastTensStrenNtrd(double*);
```

C#

```
public double PlastTensStrenNtrd { get; }
```

Visual Basic

```
Public ReadOnly PlastTensStrenNtrd As double
```

Version

Available since version 3.

I.7.2.81 ReducMomStrenMcyrd

C++

```
HRESULT get_ReducMomStrenMcyrd(double*);
```

C#

```
public double ReducMomStrenMcyrd { get; }
```

Visual Basic

```
Public ReadOnly ReducMomStrenMcyrd As double
```

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

I.7.2.82 ReducMomStrenMczrd

C++

```
HRESULT get_ReducMomStrenMczrd(double*);
```

C#

```
public double ReducMomStrenMczrd { get; }
```

Visual Basic

```
Public ReadOnly ReducMomStrenMczrd As double
```

Version

Available since version 3.

I.7.2.83 ReducMomStrenMnyrd

C++

```
HRESULT get_ReducMomStrenMnyrd(double*);
```

C#

```
public double ReducMomStrenMnyrd { get; }
```

Visual Basic

```
Public ReadOnly ReducMomStrenMnyrd As double
```

Version

Available since version 3.

I.7.2.84 ReducMomStrenMnzrd

C++

```
HRESULT get_ReducMomStrenMnzrd(double*);
```

C#

```
public double ReducMomStrenMnzrd { get; }
```

Visual Basic

```
Public ReadOnly ReducMomStrenMnzrd As double
```

Version

Available since version 3.

I.7.2.85 ReducMomStrenMvyrd

C++

```
HRESULT get_ReducMomStrenMvyrd(double*);
```

C#

```
public double ReducMomStrenMvyrd { get; }
```

Visual Basic

```
Public ReadOnly ReducMomStrenMvyrd As double
```

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

I.7.2.86 ReducMomStrenMvzrd

C++

```
HRESULT get_ReducMomStrenMvzrd(double*);
```

C#

```
public double ReducMomStrenMvzrd { get; }
```

Visual Basic

```
Public ReadOnly ReducMomStrenMvzrd As double
```

Version

Available since version 3.

I.7.2.87 ShearStrenTplyrd

C++

```
HRESULT get_ShearStrenTplyrd(double*);
```

C#

```
public double ShearStrenTplyrd { get; }
```

Visual Basic

```
Public ReadOnly ShearStrenTplyrd As double
```

Version

Available since version 3.

I.7.2.88 ShearStrenTplzrd

C++

```
HRESULT get_ShearStrenTplzrd(double*);
```

C#

```
public double ShearStrenTplzrd { get; }
```

Visual Basic

```
Public ReadOnly ShearStrenTplzrd As double
```

Version

Available since version 3.

I.7.2.89 StrsComp

C++

```
HRESULT get_StrsComp(double*);
```

C#

```
public double StrsComp { get; }
```

Visual Basic

```
Public ReadOnly StrsComp As double
```

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

I.7.2.90 StrsLftEdgeMZ

C++

```
HRESULT get_StrsLftEdgeMZ(double*);
```

C#

```
public double StrsLftEdgeMZ { get; }
```

Visual Basic

```
Public ReadOnly StrsLftEdgeMZ As double
```

Version

Available since version 3.

I.7.2.91 StrsLowEdgeMY

C++

```
HRESULT get_StrsLowEdgeMY(double*);
```

C#

```
public double StrsLowEdgeMY { get; }
```

Visual Basic

```
Public ReadOnly StrsLowEdgeMY As double
```

Version

Available since version 3.

I.7.2.92 StrsRgtEdgeMZ

C++

```
HRESULT get_StrsRgtEdgeMZ(double*);
```

C#

```
public double StrsRgtEdgeMZ { get; }
```

Visual Basic

```
Public ReadOnly StrsRgtEdgeMZ As double
```

Version

Available since version 3.

I.7.2.93 StrsShearY

C++

```
HRESULT get_StrsShearY(double*);
```

C#

```
public double StrsShearY { get; }
```

Visual Basic

```
Public ReadOnly StrsShearY As double
```

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

I.7.2.94 StrsShearZ

C++

```
HRESULT get_StrsShearZ(double*);
```

C#

```
public double StrsShearZ { get; }
```

Visual Basic

```
Public ReadOnly StrsShearZ As double
```

Version

Available since version 3.

I.7.2.95 StrsTens

C++

```
HRESULT get_StrsTens(double*);
```

C#

```
public double StrsTens { get; }
```

Visual Basic

```
Public ReadOnly StrsTens As double
```

Version

Available since version 3.

I.7.2.96 StrsUprEdgeMY

C++

```
HRESULT get_StrsUprEdgeMY(double*);
```

C#

```
public double StrsUprEdgeMY { get; }
```

Visual Basic

```
Public ReadOnly StrsUprEdgeMY As double
```

Version

Available since version 3.

I.7.2.97 TensStrenNurd

C++

```
HRESULT get_TensStrenNurd(double*);
```

C#

```
public double TensStrenNurd { get; }
```

Visual Basic

```
Public ReadOnly TensStrenNurd As double
```

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

I.7.2.98 TorsMomInertIt

C++

```
HRESULT get_TorsMomInertIt(double*);
```

C#

```
public double TorsMomInertIt { get; }
```

Visual Basic

```
Public ReadOnly TorsMomInertIt As double
```

Version

Available since version 3.

I.7.2.99 UppFlanSlend

C++

```
HRESULT get_UppFlanSlend(double*);
```

C#

```
public double UppFlanSlend { get; }
```

Visual Basic

```
Public ReadOnly UppFlanSlend As double
```

Version

Available since version 3.

I.7.2.100 WarpingConstantIw

C++

```
HRESULT get_WarpingConstantIw(double*);
```

C#

```
public double WarpingConstantIw { get; }
```

Visual Basic

```
Public ReadOnly WarpingConstantIw As double
```

Version

Available since version 3.

I.7.2.101 WebSlend

C++

```
HRESULT get_WebSlend(double*);
```

C#

```
public double WebSlend { get; }
```

Visual Basic

```
Public ReadOnly WebSlend As double
```

Version

Available since version 3.

(c) 2014 Autodesk Inc. All rights reserved

I.7.2.102 WebSlend1

C++

```
HRESULT get_WebSlend1(double*);
```

C#

```
public double WebSlend1 { get; }
```

Visual Basic

```
Public ReadOnly WebSlend1 As double
```

Version

Available since version 3.

I.8 IRDimYieldStrengthTypeEC3

C++

```
enum IRDimYieldStrengthTypeEC3;
```

C#

```
public enum IRDimYieldStrengthTypeEC3;
```

Visual Basic

```
Public Enum IRDimYieldStrengthTypeEC3
```

Members

Members	Description
I_DYST_EC3_BASIC = 0	Available since version 3.5.
I_DYST_EC3_AVERAGE = 1	Available since version 3.5.

Version

Available since version 3.5.

II CB71 Code

Available since version 3.5.

Enumerations

	Name	Description
	IRDimBuckDiagramCB71 (see page 1887)	
	IRDimLaterBuckTypeCB71 (see page 1887)	
	IRDimLoadLevelCB71 (see page 1888)	
	IRDimLoadTypeCB71 (see page 1888)	
	IRDimFireMemberPositionsCB71 (see page 1888)	
	IRDimFireProtectionTimesCB71 (see page 1889)	

(c) 2014 Autodesk Inc. All rights reserved

Interfaces

	Name	Description
	IRDimMembParamsCB71 (see page 1853)	
	IRDimCodeResCB71 (see page 1863)	

II.1 IRDimMembParamsCB71

Class Hierarchy

RobotOM.IRDimMembParamsCB71

C++

```
interface IRDimMembParamsCB71 : IDispatch;
```

C#

```
public interface IRDimMembParamsCB71;
```

Visual Basic

```
Public Interface IRDimMembParamsCB71
```

Version

Available since version 3.5.

II.1.1 IRDimMembParamsCB71 Members

The following tables list the members exposed by IRDimMembParamsCB71.

Public Fields

	Name	Description
	ArcBeamCheck (see page 1856)	
	ArcRadius (see page 1856)	
	BuckComposBeamCoefCey (see page 1856)	
	BuckComposBeamCoefCez (see page 1857)	
	BuckIsComposedY (see page 1857)	
	BuckIsComposedZ (see page 1857)	
	BuckLengthCoeffZ (see page 1857)	
	BucklingDiagramY (see page 1858)	
	BucklingDiagramZ (see page 1858)	
	BucklLengthCoeffY (see page 1858)	
	FireContinProtForComposBeam (see page 1858)	
	FireLftSideProt (see page 1859)	
	FireLowSideProt (see page 1859)	

(c) 2014 Autodesk Inc. All rights reserved

	FireMembPosition (see page 1859)	
	FireProtectionTime (see page 1859)	
	FireRequiredResist (see page 1860)	
	FireRgtSideProt (see page 1860)	
	FireUprSideProt (see page 1860)	
	ForceDistFromBeginDistX (see page 1861)	
	HumChangeDeltaH (see page 1861)	
	InflDeflCoefTheta (see page 1861)	
	LatBuckType (see page 1861)	
	LatCoeffLowerFlangeValue (see page 1862)	
	LatCoeffUpperFlangeValue (see page 1862)	
	LoadLevel (see page 1862)	
	LoadTypeY (see page 1862)	
	TensAreaNetGros (see page 1863)	
	TubeControl (see page 1863)	

II.1.2 IRDimMembParamsCB71 Fields

The fields of the IRDimMembParamsCB71 class are listed here.

Public Fields

	Name	Description
	ArcBeamCheck (see page 1856)	
	ArcRadius (see page 1856)	
	BuckComposBeamCoefCey (see page 1856)	
	BuckComposBeamCoefCez (see page 1857)	
	BuckIsComposedY (see page 1857)	
	BuckIsComposedZ (see page 1857)	
	BuckLengthCoeffZ (see page 1857)	
	BucklingDiagramY (see page 1858)	
	BucklingDiagramZ (see page 1858)	
	BucklLengthCoeffY (see page 1858)	
	FireContinProtForComposBeam (see page 1858)	
	FireLftSideProt (see page 1859)	
	FireLowSideProt (see page 1859)	
	FireMembPosition (see page 1859)	

(c) 2014 Autodesk Inc. All rights reserved

	FireProtectionTime (see page 1859)	
	FireRequiredResist (see page 1860)	
	FireRgtSideProt (see page 1860)	
	FireUprSideProt (see page 1860)	
	ForceDistFromBeginDistX (see page 1861)	
	HumChangeDeltaH (see page 1861)	
	InflDeflCoefTheta (see page 1861)	
	LatBuckType (see page 1861)	
	LatCoeffLowerFlangeValue (see page 1862)	
	LatCoeffUpperFlangeValue (see page 1862)	
	LoadLevel (see page 1862)	
	LoadTypeY (see page 1862)	
	TensAreaNetGros (see page 1863)	
	TubeControl (see page 1863)	

II.1.2.1 ArcBeamCheck

C++

```
HRESULT get_ArcBeamCheck(VARIANT_BOOL*);
HRESULT put_ArcBeamCheck(VARIANT_BOOL);
```

C#

```
public bool ArcBeamCheck { get; set; }
```

Visual Basic

```
Public ArcBeamCheck As Boolean
```

Version

Available since version 3.5.

II.1.2.2 ArcRadius

C++

```
HRESULT get_ArcRadius(double*);
HRESULT put_ArcRadius(double);
```

C#

```
public double ArcRadius { get; set; }
```

Visual Basic

```
Public ArcRadius As double
```

Version

Available since version 3.5.

(c) 2014 Autodesk Inc. All rights reserved

II.1.2.3 BuckComposBeamCoefCey

C++

```
HRESULT get_BuckComposBeamCoefCey(double*);
HRESULT put_BuckComposBeamCoefCey(double);
```

C#

```
public double BuckComposBeamCoefCey { get; set; }
```

Visual Basic

```
Public BuckComposBeamCoefCey As double
```

Version

Available since version 3.5.

II.1.2.4 BuckComposBeamCoefCez

C++

```
HRESULT get_BuckComposBeamCoefCez(double*);
HRESULT put_BuckComposBeamCoefCez(double);
```

C#

```
public double BuckComposBeamCoefCez { get; set; }
```

Visual Basic

```
Public BuckComposBeamCoefCez As double
```

Version

Available since version 3.5.

II.1.2.5 BuckIsComposedY

C++

```
HRESULT get_BuckIsComposedY(VARIANT_BOOL*);
HRESULT put_BuckIsComposedY(VARIANT_BOOL);
```

C#

```
public bool BuckIsComposedY { get; set; }
```

Visual Basic

```
Public BuckIsComposedY As Boolean
```

Version

Available since version 3.5.

II.1.2.6 BuckIsComposedZ

C++

```
HRESULT get_BuckIsComposedZ(VARIANT_BOOL*);
HRESULT put_BuckIsComposedZ(VARIANT_BOOL);
```

C#

```
public bool BuckIsComposedZ { get; set; }
```

Visual Basic

```
Public BuckIsComposedZ As Boolean
```

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.5.

II.1.2.7 BuckLengthCoeffZ

C++

```
HRESULT get_BuckLengthCoeffZ(double*);
HRESULT put_BuckLengthCoeffZ(double);
```

C#

```
public double BuckLengthCoeffZ { get; set; }
```

Visual Basic

```
Public BuckLengthCoeffZ As double
```

Version

Available since version 3.5.

II.1.2.8 BucklingDiagramY

C++

```
HRESULT get_BucklingDiagramY(IRDimBuckDiagramCB71*);
HRESULT put_BucklingDiagramY(IRDimBuckDiagramCB71);
```

C#

```
public IRDimBuckDiagramCB71 BucklingDiagramY { get; set; }
```

Visual Basic

```
Public BucklingDiagramY As IRDimBuckDiagramCB71
```

Version

Available since version 3.5.

II.1.2.9 BucklingDiagramZ

C++

```
HRESULT get_BucklingDiagramZ(IRDimBuckDiagramCB71*);
HRESULT put_BucklingDiagramZ(IRDimBuckDiagramCB71);
```

C#

```
public IRDimBuckDiagramCB71 BucklingDiagramZ { get; set; }
```

Visual Basic

```
Public BucklingDiagramZ As IRDimBuckDiagramCB71
```

Version

Available since version 3.5.

II.1.2.10 BucklLengthCoeffY

C++

```
HRESULT get_BucklLengthCoeffY(double*);
HRESULT put_BucklLengthCoeffY(double);
```

C#

```
public double BucklLengthCoeffY { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public BucklLengthCoeffY As double
```

Version

Available since version 3.5.

II.1.2.11 FireContinProtForComposBeam

C++

```
HRESULT get_FireContinProtForComposBeam(VARIANT_BOOL*);
HRESULT put_FireContinProtForComposBeam(VARIANT_BOOL);
```

C#

```
public bool FireContinProtForComposBeam { get; set; }
```

Visual Basic

```
Public FireContinProtForComposBeam As Boolean
```

Version

Available since version 3.5.

II.1.2.12 FireLftSideProt

C++

```
HRESULT get_FireLftSideProt(VARIANT_BOOL*);
HRESULT put_FireLftSideProt(VARIANT_BOOL);
```

C#

```
public bool FireLftSideProt { get; set; }
```

Visual Basic

```
Public FireLftSideProt As Boolean
```

Version

Available since version 3.5.

II.1.2.13 FireLowSideProt

C++

```
HRESULT get_FireLowSideProt(VARIANT_BOOL*);
HRESULT put_FireLowSideProt(VARIANT_BOOL);
```

C#

```
public bool FireLowSideProt { get; set; }
```

Visual Basic

```
Public FireLowSideProt As Boolean
```

Version

Available since version 3.5.

II.1.2.14 FireMembPosition

C++

```
HRESULT get_FireMembPosition(IRDimFireMemberPositionsCB71*);
HRESULT put_FireMembPosition(IRDimFireMemberPositionsCB71);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRDimFireMemberPositionsCB71 FireMembPosition { get; set; }
```

Visual Basic

```
Public FireMembPosition As IRDimFireMemberPositionsCB71
```

Version

Available since version 3.5.

II.1.2.15 FireProtectionTime

C++

```
HRESULT get_FireProtectionTime(IRDimFireProtectionTimesCB71*);
HRESULT put_FireProtectionTime(IRDimFireProtectionTimesCB71);
```

C#

```
public IRDimFireProtectionTimesCB71 FireProtectionTime { get; set; }
```

Visual Basic

```
Public FireProtectionTime As IRDimFireProtectionTimesCB71
```

Version

Available since version 3.5.

II.1.2.16 FireRequiredResist

C++

```
HRESULT get_FireRequiredResist(double*);
HRESULT put_FireRequiredResist(double);
```

C#

```
public double FireRequiredResist { get; set; }
```

Visual Basic

```
Public FireRequiredResist As double
```

Version

Available since version 3.5.

II.1.2.17 FireRgtSideProt

C++

```
HRESULT get_FireRgtSideProt(VARIANT_BOOL*);
HRESULT put_FireRgtSideProt(VARIANT_BOOL);
```

C#

```
public bool FireRgtSideProt { get; set; }
```

Visual Basic

```
Public FireRgtSideProt As Boolean
```

Version

Available since version 3.5.

II.1.2.18 FireUprSideProt

C++

```
HRESULT get_FireUprSideProt(VARIANT_BOOL*);
```

(c) 2014 Autodesk Inc. All rights reserved

```
HRESULT put_FireUprSideProt(VARIANT_BOOL);
```

C#

```
public bool FireUprSideProt { get; set; }
```

Visual Basic

```
Public FireUprSideProt As Boolean
```

Version

Available since version 3.5.

II.1.2.19 ForceDistFromBeginDistX

C++

```
HRESULT get_ForceDistFromBeginDistX(double*);
HRESULT put_ForceDistFromBeginDistX(double);
```

C#

```
public double ForceDistFromBeginDistX { get; set; }
```

Visual Basic

```
Public ForceDistFromBeginDistX As double
```

Version

Available since version 3.5.

II.1.2.20 HumChangeDeltaH

C++

```
HRESULT get_HumChangeDeltaH(double*);
HRESULT put_HumChangeDeltaH(double);
```

C#

```
public double HumChangeDeltaH { get; set; }
```

Visual Basic

```
Public HumChangeDeltaH As double
```

Version

Available since version 3.5.

II.1.2.21 InflDeflCoefTheta

C++

```
HRESULT get_InflDeflCoefTheta(double*);
HRESULT put_InflDeflCoefTheta(double);
```

C#

```
public double InflDeflCoefTheta { get; set; }
```

Visual Basic

```
Public InflDeflCoefTheta As double
```

Version

Available since version 3.5.

(c) 2014 Autodesk Inc. All rights reserved

II.1.2.22 LatBuckType

C++

```
HRESULT get_LatBuckType(IRDimLaterBuckTypeCB71*);
HRESULT put_LatBuckType(IRDimLaterBuckTypeCB71);
```

C#

```
public IRDimLaterBuckTypeCB71 LatBuckType { get; set; }
```

Visual Basic

```
Public LatBuckType As IRDimLaterBuckTypeCB71
```

Version

Available since version 3.5.

II.1.2.23 LatCoeffLowerFlangeValue

C++

```
HRESULT get_LatCoeffLowerFlangeValue(double*);
HRESULT put_LatCoeffLowerFlangeValue(double);
```

C#

```
public double LatCoeffLowerFlangeValue { get; set; }
```

Visual Basic

```
Public LatCoeffLowerFlangeValue As double
```

Version

Available since version 3.5.

II.1.2.24 LatCoeffUpperFlangeValue

C++

```
HRESULT get_LatCoeffUpperFlangeValue(double*);
HRESULT put_LatCoeffUpperFlangeValue(double);
```

C#

```
public double LatCoeffUpperFlangeValue { get; set; }
```

Visual Basic

```
Public LatCoeffUpperFlangeValue As double
```

Version

Available since version 3.5.

II.1.2.25 LoadLevel

C++

```
HRESULT get_LoadLevel(IRDimLoadLevelCB71*);
HRESULT put_LoadLevel(IRDimLoadLevelCB71);
```

C#

```
public IRDimLoadLevelCB71 LoadLevel { get; set; }
```

Visual Basic

```
Public LoadLevel As IRDimLoadLevelCB71
```

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.5.

II.1.2.26 LoadTypeY

C++

```
HRESULT get_LoadTypeY(IRDimLoadTypeCB71*);
HRESULT put_LoadTypeY(IRDimLoadTypeCB71);
```

C#

```
public IRDimLoadTypeCB71 LoadTypeY { get; set; }
```

Visual Basic

```
Public LoadTypeY As IRDimLoadTypeCB71
```

Version

Available since version 3.5.

II.1.2.27 TensAreaNetGros

C++

```
HRESULT get_TensAreaNetGros(double*);
HRESULT put_TensAreaNetGros(double);
```

C#

```
public double TensAreaNetGros { get; set; }
```

Visual Basic

```
Public TensAreaNetGros As double
```

Version

Available since version 3.5.

II.1.2.28 TubeControl

C++

```
HRESULT get_TubeControl(VARIANT_BOOL*);
HRESULT put_TubeControl(VARIANT_BOOL);
```

C#

```
public bool TubeControl { get; set; }
```

Visual Basic

```
Public TubeControl As Boolean
```

Version

Available since version 3.5.

II.2 IRDimCodeResCB71

Class Hierarchy

C++

```
interface IRDimCodeResCB71 : IDispatch;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public interface IRDimCodeResCB71;
```

Visual Basic

```
Public Interface IRDimCodeResCB71
```

Version

Available since version 3.5.

II.2.1 IRDimCodeResCB71 Members

The following tables list the members exposed by IRDimCodeResCB71.

Public Fields

	Name	Description
	BadWidthBfUnderFire (see page 1869)	
	BuckCoefKMax (see page 1869)	
	BuckCoefKy (see page 1869)	
	BuckCoefKz (see page 1870)	
	BuckLengthComposBeamLey (see page 1870)	
	BuckLengthComposBeamLez (see page 1870)	
	BuckLengthLfy (see page 1870)	
	BuckLengthLfz (see page 1871)	
	BuckSlendComposBeamLaym (see page 1871)	
	BuckSlendComposBeamLazm (see page 1871)	
	BuckSLendLay (see page 1871)	
	BuckSlendLaz (see page 1872)	
	CoefDependOnAngle (see page 1872)	
	CoefDependOnHeight (see page 1872)	Available since version 3.5.
	CoefDependOnHumAndBend (see page 1872)	
	CoefDependOnHumAndComp (see page 1872)	
	CoefDependOnWidth (see page 1873)	
	EfficiencyRatio (see page 1873)	
	FireHightInMidSpanH (see page 1873)	
	FireInerMomInMidSpanIy (see page 1873)	
	FireInerMomInMidSpanIz (see page 1874)	
	FireInerMomIx (see page 1874)	
	FireInerMomIy (see page 1874)	
	FireInerMomIz (see page 1874)	
	FireLftEdgeDistVpy (see page 1875)	

(c) 2014 Autodesk Inc. All rights reserved

	FireLowEdgeDistVpz (see page 1875)	
	FireProtectCoefK2 (see page 1875)	
	FireProtection (see page 1875)	
	FireProtectionPosition (see page 1876)	
	FireRedCoefK1b (see page 1876)	
	FireRedCoefK1h (see page 1876)	
	FireResistance (see page 1876)	
	FireRgtEdgeDistVy (see page 1877)	
	FireSecAreaInMidSpanS (see page 1877)	
	FireSecAreaS (see page 1877)	
	FireSeModulWy (see page 1877)	
	FireSeModulWZ (see page 1878)	
	FireShearAreaSY (see page 1878)	
	FireShearAreaSZ (see page 1878)	
	FireUprEdgeDistVz (see page 1878)	
	FireWidthAtMembEndB (see page 1879)	
	FireWidthAtMembEndH (see page 1879)	
	FireWidthB (see page 1879)	
	FireWidthH (see page 1879)	
	FireWidthInMidSpanB (see page 1880)	
	IsBuckY (see page 1880)	
	IsBuckZ (see page 1880)	
	LatBuckCoefKinst (see page 1880)	
	LatBuckLengthCoef (see page 1881)	
	LatBuckLengthLd (see page 1881)	
	LatBuckRelSlendLam (see page 1881)	
	LatBuckStrsCrit (see page 1881)	
	MatAxCompResist (see page 1882)	
	MatAxTensResist (see page 1882)	
	MatBendResist (see page 1882)	
	MatShearResist (see page 1882)	
	MatTrCompResist (see page 1883)	
	MatTrTensResist (see page 1883)	
	StrsBend (see page 1883)	
	StrsBendInCompEdgeMY (see page 1883)	

(c) 2014 Autodesk Inc. All rights reserved

	StrsBendInCompEdgeMZ (see page 1884)	
	StrsBendInCurveBeam (see page 1884)	
	StrsComp (see page 1884)	
	StrsFinal (see page 1884)	
	StrsLftEdgeMZ (see page 1885)	
	StrsLowEdgeMY (see page 1885)	
	StrsRgtEdgeMZ (see page 1885)	
	StrsShearY (see page 1885)	
	StrsShearZ (see page 1886)	
	StrsTens (see page 1886)	
	StrsUprEdgeMY (see page 1886)	
	TimberType (see page 1886)	

II.2.2 IRDimCodeResCB71 Fields

The fields of the IRDimCodeResCB71 class are listed here.

Public Fields

	Name	Description
	BadWidthBfUnderFire (see page 1869)	
	BuckCoefKMax (see page 1869)	
	BuckCoefKy (see page 1869)	
	BuckCoefKz (see page 1870)	
	BuckLengthComposBeamLey (see page 1870)	
	BuckLengthComposBeamLez (see page 1870)	
	BuckLengthLfy (see page 1870)	
	BuckLengthLfz (see page 1871)	
	BuckSlendComposBeamLaym (see page 1871)	
	BuckSlendComposBeamLazm (see page 1871)	
	BuckSLendLay (see page 1871)	
	BuckSlendLaz (see page 1872)	
	CoefDependOnAngle (see page 1872)	
	CoefDependOnHeight (see page 1872)	Available since version 3.5.
	CoefDependOnHumAndBend (see page 1872)	
	CoefDependOnHumAndComp (see page 1872)	
	CoefDependOnWidth (see page 1873)	
	EfficiencyRatio (see page 1873)	
	FireHightInMidSpanH (see page 1873)	
	FireInerMomInMidSpanIy (see page 1873)	

(c) 2014 Autodesk Inc. All rights reserved

	FireInerMomInMidSpanIz (see page 1874)	
	FireInerMomIx (see page 1874)	
	FireInerMomIy (see page 1874)	
	FireInerMomIz (see page 1874)	
	FireLftEdgeDistVpy (see page 1875)	
	FireLowEdgeDistVpz (see page 1875)	
	FireProtectCoefK2 (see page 1875)	
	FireProtection (see page 1875)	
	FireProtectionPosition (see page 1876)	
	FireRedCoefK1b (see page 1876)	
	FireRedCoefK1h (see page 1876)	
	FireResistance (see page 1876)	
	FireRgtEdgeDistVy (see page 1877)	
	FireSecAreaInMidSpanS (see page 1877)	
	FireSecAreaS (see page 1877)	
	FireSeModulWy (see page 1877)	
	FireSeModulWZ (see page 1878)	
	FireShearAreaSY (see page 1878)	
	FireShearAreaSZ (see page 1878)	
	FireUprEdgeDistVz (see page 1878)	
	FireWidthAtMembEndB (see page 1879)	
	FireWidthAtMembEndH (see page 1879)	
	FireWidthB (see page 1879)	
	FireWidthH (see page 1879)	
	FireWidthInMidSpanB (see page 1880)	
	IsBuckY (see page 1880)	
	IsBuckZ (see page 1880)	
	LatBuckCoefKinst (see page 1880)	
	LatBuckLengthCoef (see page 1881)	
	LatBuckLengthLd (see page 1881)	
	LatBuckRelSlendLam (see page 1881)	
	LatBuckStrsCrit (see page 1881)	
	MatAxCompResist (see page 1882)	
	MatAxTensResist (see page 1882)	
	MatBendResist (see page 1882)	

(c) 2014 Autodesk Inc. All rights reserved

	MatShearResist (see page 1882)	
	MatTrCompResist (see page 1883)	
	MatTrTensResist (see page 1883)	
	StrsBend (see page 1883)	
	StrsBendInCompEdgeMY (see page 1883)	
	StrsBendInCompEdgeMZ (see page 1884)	
	StrsBendInCurveBeam (see page 1884)	
	StrsComp (see page 1884)	
	StrsFinal (see page 1884)	
	StrsLftEdgeMZ (see page 1885)	
	StrsLowEdgeMY (see page 1885)	
	StrsRgtEdgeMZ (see page 1885)	
	StrsShearY (see page 1885)	
	StrsShearZ (see page 1886)	
	StrsTens (see page 1886)	
	StrsUprEdgeMY (see page 1886)	
	TimberType (see page 1886)	

II.2.2.1 BadWidthBfUnderFire

C++

```
HRESULT get_BadWidthBfUnderFire(VARIANT_BOOL*);
```

C#

```
public bool BadWidthBfUnderFire { get; }
```

Visual Basic

```
Public ReadOnly BadWidthBfUnderFire As Boolean
```

Version

Available since version 3.5.

II.2.2.2 BuckCoefKMax

C++

```
HRESULT get_BuckCoefKMax(double*);
```

C#

```
public double BuckCoefKMax { get; }
```

Visual Basic

```
Public ReadOnly BuckCoefKMax As double
```

Version

Available since version 3.5.

II.2.2.3 BuckCoefKy

C++

```
HRESULT get_BuckCoefKy(double*);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double BuckCoefKy { get; }
```

Visual Basic

```
Public ReadOnly BuckCoefKy As double
```

Version

Available since version 3.5.

II.2.2.4 BuckCoefKz

C++

```
HRESULT get_BuckCoefKz(double*);
```

C#

```
public double BuckCoefKz { get; }
```

Visual Basic

```
Public ReadOnly BuckCoefKz As double
```

Version

Available since version 3.5.

II.2.2.5 BuckLengthComposBeamLey

C++

```
HRESULT get_BuckLengthComposBeamLey(double*);
```

C#

```
public double BuckLengthComposBeamLey { get; }
```

Visual Basic

```
Public ReadOnly BuckLengthComposBeamLey As double
```

Version

Available since version 3.5.

II.2.2.6 BuckLengthComposBeamLez

C++

```
HRESULT get_BuckLengthComposBeamLez(double*);
```

C#

```
public double BuckLengthComposBeamLez { get; }
```

Visual Basic

```
Public ReadOnly BuckLengthComposBeamLez As double
```

Version

Available since version 3.5.

II.2.2.7 BuckLengthLfy

C++

```
HRESULT get_BuckLengthLfy(double*);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double BuckLengthLfy { get; }
```

Visual Basic

```
Public ReadOnly BuckLengthLfy As double
```

Version

Available since version 3.5.

II.2.2.8 BuckLengthLfz

C++

```
HRESULT get_BuckLengthLfz(double*);
```

C#

```
public double BuckLengthLfz { get; }
```

Visual Basic

```
Public ReadOnly BuckLengthLfz As double
```

Version

Available since version 3.5.

II.2.2.9 BuckSlendComposBeamLaym

C++

```
HRESULT get_BuckSlendComposBeamLaym(double*);
```

C#

```
public double BuckSlendComposBeamLaym { get; }
```

Visual Basic

```
Public ReadOnly BuckSlendComposBeamLaym As double
```

Version

Available since version 3.5.

II.2.2.10 BuckSlendComposBeamLazm

C++

```
HRESULT get_BuckSlendComposBeamLazm(double*);
```

C#

```
public double BuckSlendComposBeamLazm { get; }
```

Visual Basic

```
Public ReadOnly BuckSlendComposBeamLazm As double
```

Version

Available since version 3.5.

II.2.2.11 BuckSLendLay

C++

```
HRESULT get_BuckSLendLay(double*);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double BuckSLendLay { get; }
```

Visual Basic

```
Public ReadOnly BuckSLendLay As double
```

Version

Available since version 3.5.

II.2.2.12 BuckSlendLaz

C++

```
HRESULT get_BuckSlendLaz(double*);
```

C#

```
public double BuckSlendLaz { get; }
```

Visual Basic

```
Public ReadOnly BuckSlendLaz As double
```

Version

Available since version 3.5.

II.2.2.13 CoefDependOnAngle

C++

```
HRESULT get_CoefDependOnAngle(double*);
```

C#

```
public double CoefDependOnAngle { get; }
```

Visual Basic

```
Public ReadOnly CoefDependOnAngle As double
```

Version

Available since version 3.5.

II.2.2.14 CoefDependOnHeight

C#

```
void CoefDependOnHeight;
```

Description

Available since version 3.5.

II.2.2.15 CoefDependOnHumAndBend

C++

```
HRESULT get_CoefDependOnHumAndBend(double*);
```

C#

```
public double CoefDependOnHumAndBend { get; }
```

Visual Basic

```
Public ReadOnly CoefDependOnHumAndBend As double
```

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.5.

II.2.2.16 CoefDependOnHumAndComp

C++

```
HRESULT get_CoefDependOnHumAndComp(double*);
```

C#

```
public double CoefDependOnHumAndComp { get; }
```

Visual Basic

```
Public ReadOnly CoefDependOnHumAndComp As double
```

Version

Available since version 3.5.

II.2.2.17 CoefDependOnWidth

C++

```
HRESULT get_CoefDependOnWidth(double*);
```

C#

```
public double CoefDependOnWidth { get; }
```

Visual Basic

```
Public ReadOnly CoefDependOnWidth As double
```

Version

Available since version 3.5.

II.2.2.18 EfficiencyRatio

C++

```
HRESULT get_EfficiencyRatio(double*);
```

C#

```
public double EfficiencyRatio { get; }
```

Visual Basic

```
Public ReadOnly EfficiencyRatio As double
```

Version

Available since version 3.5.

II.2.2.19 FireHightInMidSpanH

C++

```
HRESULT get_FireHightInMidSpanH(double*);
```

C#

```
public double FireHightInMidSpanH { get; }
```

Visual Basic

```
Public ReadOnly FireHightInMidSpanH As double
```

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.5.

II.2.2.20 FireInerMomInMidSpanIy

C++

```
HRESULT get_FireInerMomInMidSpanIy(double*);
```

C#

```
public double FireInerMomInMidSpanIy { get; }
```

Visual Basic

```
Public ReadOnly FireInerMomInMidSpanIy As double
```

Version

Available since version 3.5.

II.2.2.21 FireInerMomInMidSpanIz

C++

```
HRESULT get_FireInerMomInMidSpanIz(double*);
```

C#

```
public double FireInerMomInMidSpanIz { get; }
```

Visual Basic

```
Public ReadOnly FireInerMomInMidSpanIz As double
```

Version

Available since version 3.5.

II.2.2.22 FireInerMomIx

C++

```
HRESULT get_FireInerMomIx(double*);
```

C#

```
public double FireInerMomIx { get; }
```

Visual Basic

```
Public ReadOnly FireInerMomIx As double
```

Version

Available since version 3.5.

II.2.2.23 FireInerMomIy

C++

```
HRESULT get_FireInerMomIy(double*);
```

C#

```
public double FireInerMomIy { get; }
```

Visual Basic

```
Public ReadOnly FireInerMomIy As double
```

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.5.

II.2.2.24 FireInerMomIz

C++

```
HRESULT get_FireInerMomIz(double*);
```

C#

```
public double FireInerMomIz { get; }
```

Visual Basic

```
Public ReadOnly FireInerMomIz As double
```

Version

Available since version 3.5.

II.2.2.25 FireLftEdgeDistVpy

C++

```
HRESULT get_FireLftEdgeDistVpy(double*);
```

C#

```
public double FireLftEdgeDistVpy { get; }
```

Visual Basic

```
Public ReadOnly FireLftEdgeDistVpy As double
```

Version

Available since version 3.5.

II.2.2.26 FireLowEdgeDistVpz

C++

```
HRESULT get_FireLowEdgeDistVpz(double*);
```

C#

```
public double FireLowEdgeDistVpz { get; }
```

Visual Basic

```
Public ReadOnly FireLowEdgeDistVpz As double
```

Version

Available since version 3.5.

II.2.2.27 FireProtectCoefK2

C++

```
HRESULT get_FireProtectCoefK2(double*);
```

C#

```
public double FireProtectCoefK2 { get; }
```

Visual Basic

```
Public ReadOnly FireProtectCoefK2 As double
```

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.5.

II.2.2.28 FireProtection

C++

```
HRESULT get_FireProtection(short*);
```

C#

```
public short FireProtection { get; }
```

Visual Basic

```
Public ReadOnly FireProtection As short
```

Version

Available since version 3.5.

II.2.2.29 FireProtectionPosition

C++

```
HRESULT get_FireProtectionPosition(short*);
```

C#

```
public short FireProtectionPosition { get; }
```

Visual Basic

```
Public ReadOnly FireProtectionPosition As short
```

Version

Available since version 3.5.

II.2.2.30 FireRedCoefK1b

C++

```
HRESULT get_FireRedCoefK1b(double*);
```

C#

```
public double FireRedCoefK1b { get; }
```

Visual Basic

```
Public ReadOnly FireRedCoefK1b As double
```

Version

Available since version 3.5.

II.2.2.31 FireRedCoefK1h

C++

```
HRESULT get_FireRedCoefK1h(double*);
```

C#

```
public double FireRedCoefK1h { get; }
```

Visual Basic

```
Public ReadOnly FireRedCoefK1h As double
```

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.5.

II.2.2.32 FireResistance

C++

```
HRESULT get_FireResistance(double*);
```

C#

```
public double FireResistance { get; }
```

Visual Basic

```
Public ReadOnly FireResistance As double
```

Version

Available since version 3.5.

II.2.2.33 FireRgtEdgeDistVy

C++

```
HRESULT get_FireRgtEdgeDistVy(double*);
```

C#

```
public double FireRgtEdgeDistVy { get; }
```

Visual Basic

```
Public ReadOnly FireRgtEdgeDistVy As double
```

Version

Available since version 3.5.

II.2.2.34 FireSecAreaInMidSpanS

C++

```
HRESULT get_FireSecAreaInMidSpanS(double*);
```

C#

```
public double FireSecAreaInMidSpanS { get; }
```

Visual Basic

```
Public ReadOnly FireSecAreaInMidSpanS As double
```

Version

Available since version 3.5.

II.2.2.35 FireSecAreaS

C++

```
HRESULT get_FireSecAreaS(double*);
```

C#

```
public double FireSecAreaS { get; }
```

Visual Basic

```
Public ReadOnly FireSecAreaS As double
```

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.5.

II.2.2.36 FireSeModulWy

C++

```
HRESULT get_FireSeModulWy(double*);
```

C#

```
public double FireSeModulWy { get; }
```

Visual Basic

```
Public ReadOnly FireSeModulWy As double
```

Version

Available since version 3.5.

II.2.2.37 FireSeModulWZ

C++

```
HRESULT get_FireSeModulWZ(double*);
```

C#

```
public double FireSeModulWZ { get; }
```

Visual Basic

```
Public ReadOnly FireSeModulWZ As double
```

Version

Available since version 3.5.

II.2.2.38 FireShearAreaSY

C++

```
HRESULT get_FireShearAreaSY(double*);
```

C#

```
public double FireShearAreaSY { get; }
```

Visual Basic

```
Public ReadOnly FireShearAreaSY As double
```

Version

Available since version 3.5.

II.2.2.39 FireShearAreaSZ

C++

```
HRESULT get_FireShearAreaSZ(double*);
```

C#

```
public double FireShearAreaSZ { get; }
```

Visual Basic

```
Public ReadOnly FireShearAreaSZ As double
```

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.5.

II.2.2.40 FireUprEdgeDistVz

C++

```
HRESULT get_FireUprEdgeDistVz(double*);
```

C#

```
public double FireUprEdgeDistVz { get; }
```

Visual Basic

```
Public ReadOnly FireUprEdgeDistVz As double
```

Version

Available since version 3.5.

II.2.2.41 FireWidthAtMembEndB

C++

```
HRESULT get_FireWidthAtMembEndB(double*);
```

C#

```
public double FireWidthAtMembEndB { get; }
```

Visual Basic

```
Public ReadOnly FireWidthAtMembEndB As double
```

Version

Available since version 3.5.

II.2.2.42 FireWidthAtMembEndH

C++

```
HRESULT get_FireWidthAtMembEndH(double*);
```

C#

```
public double FireWidthAtMembEndH { get; }
```

Visual Basic

```
Public ReadOnly FireWidthAtMembEndH As double
```

Version

Available since version 3.5.

II.2.2.43 FireWidthB

C++

```
HRESULT get_FireWidthB(double*);
```

C#

```
public double FireWidthB { get; }
```

Visual Basic

```
Public ReadOnly FireWidthB As double
```

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.5.

II.2.2.44 FireWidthH

C++

```
HRESULT get_FireWidthH(double*);
```

C#

```
public double FireWidthH { get; }
```

Visual Basic

```
Public ReadOnly FireWidthH As double
```

Version

Available since version 3.5.

II.2.2.45 FireWidthInMidSpanB

C++

```
HRESULT get_FireWidthInMidSpanB(double*);
```

C#

```
public double FireWidthInMidSpanB { get; }
```

Visual Basic

```
Public ReadOnly FireWidthInMidSpanB As double
```

Version

Available since version 3.5.

II.2.2.46 IsBuckY

C++

```
HRESULT get_IsBuckY(short*);
```

C#

```
public short IsBuckY { get; }
```

Visual Basic

```
Public ReadOnly IsBuckY As short
```

Version

Available since version 3.5.

II.2.2.47 IsBuckZ

C++

```
HRESULT get_IsBuckZ(short*);
```

C#

```
public short IsBuckZ { get; }
```

Visual Basic

```
Public ReadOnly IsBuckZ As short
```

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.5.

II.2.2.48 LatBuckCoefKinst

C++

```
HRESULT get_LatBuckCoefKinst(double*);
```

C#

```
public double LatBuckCoefKinst { get; }
```

Visual Basic

```
Public ReadOnly LatBuckCoefKinst As double
```

Version

Available since version 3.5.

II.2.2.49 LatBuckLengthCoef

C++

```
HRESULT get_LatBuckLengthCoef(double*);
```

C#

```
public double LatBuckLengthCoef { get; }
```

Visual Basic

```
Public ReadOnly LatBuckLengthCoef As double
```

Version

Available since version 3.5.

II.2.2.50 LatBuckLengthLd

C++

```
HRESULT get_LatBuckLengthLd(double*);
```

C#

```
public double LatBuckLengthLd { get; }
```

Visual Basic

```
Public ReadOnly LatBuckLengthLd As double
```

Version

Available since version 3.5.

II.2.2.51 LatBuckRelSlendLam

C++

```
HRESULT get_LatBuckRelSlendLam(double*);
```

C#

```
public double LatBuckRelSlendLam { get; }
```

Visual Basic

```
Public ReadOnly LatBuckRelSlendLam As double
```

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.5.

II.2.2.52 LatBuckStrsCrit

C++

```
HRESULT get_LatBuckStrsCrit(double*);
```

C#

```
public double LatBuckStrsCrit { get; }
```

Visual Basic

```
Public ReadOnly LatBuckStrsCrit As double
```

Version

Available since version 3.5.

II.2.2.53 MatAxCompResist

C++

```
HRESULT get_MatAxCompResist(double*);
```

C#

```
public double MatAxCompResist { get; }
```

Visual Basic

```
Public ReadOnly MatAxCompResist As double
```

Version

Available since version 3.5.

II.2.2.54 MatAxTensResist

C++

```
HRESULT get_MatAxTensResist(double*);
```

C#

```
public double MatAxTensResist { get; }
```

Visual Basic

```
Public ReadOnly MatAxTensResist As double
```

Version

Available since version 3.5.

II.2.2.55 MatBendResist

C++

```
HRESULT get_MatBendResist(double*);
```

C#

```
public double MatBendResist { get; }
```

Visual Basic

```
Public ReadOnly MatBendResist As double
```

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.5.

II.2.2.56 MatShearResist

C++

```
HRESULT get_MatShearResist(double*);
```

C#

```
public double MatShearResist { get; }
```

Visual Basic

```
Public ReadOnly MatShearResist As double
```

Version

Available since version 3.5.

II.2.2.57 MatTrCompResist

C++

```
HRESULT get_MatTrCompResist(double*);
```

C#

```
public double MatTrCompResist { get; }
```

Visual Basic

```
Public ReadOnly MatTrCompResist As double
```

Version

Available since version 3.5.

II.2.2.58 MatTrTensResist

C++

```
HRESULT get_MatTrTensResist(double*);
```

C#

```
public double MatTrTensResist { get; }
```

Visual Basic

```
Public ReadOnly MatTrTensResist As double
```

Version

Available since version 3.5.

II.2.2.59 StrsBend

C++

```
HRESULT get_StrsBend(double*);
```

C#

```
public double StrsBend { get; }
```

Visual Basic

```
Public ReadOnly StrsBend As double
```

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.5.

II.2.2.60 StrsBendInCompEdgeMY

C++

```
HRESULT get_StrsBendInCompEdgeMY(double*);
```

C#

```
public double StrsBendInCompEdgeMY { get; }
```

Visual Basic

```
Public ReadOnly StrsBendInCompEdgeMY As double
```

Version

Available since version 3.5.

II.2.2.61 StrsBendInCompEdgeMZ

C++

```
HRESULT get_StrsBendInCompEdgeMZ(double*);
```

C#

```
public double StrsBendInCompEdgeMZ { get; }
```

Visual Basic

```
Public ReadOnly StrsBendInCompEdgeMZ As double
```

Version

Available since version 3.5.

II.2.2.62 StrsBendInCurveBeam

C++

```
HRESULT get_StrsBendInCurveBeam(double*);
```

C#

```
public double StrsBendInCurveBeam { get; }
```

Visual Basic

```
Public ReadOnly StrsBendInCurveBeam As double
```

Version

Available since version 3.5.

II.2.2.63 StrsComp

C++

```
HRESULT get_StrsComp(double*);
```

C#

```
public double StrsComp { get; }
```

Visual Basic

```
Public ReadOnly StrsComp As double
```

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.5.

II.2.2.64 StrsFinal

C++

```
HRESULT get_StrsFinal(double*);
```

C#

```
public double StrsFinal { get; }
```

Visual Basic

```
Public ReadOnly StrsFinal As double
```

Version

Available since version 3.5.

II.2.2.65 StrsLftEdgeMZ

C++

```
HRESULT get_StrsLftEdgeMZ(double*);
```

C#

```
public double StrsLftEdgeMZ { get; }
```

Visual Basic

```
Public ReadOnly StrsLftEdgeMZ As double
```

II.2.2.66 StrsLowEdgeMY

C++

```
HRESULT get_StrsLowEdgeMY(double*);
```

C#

```
public double StrsLowEdgeMY { get; }
```

Visual Basic

```
Public ReadOnly StrsLowEdgeMY As double
```

Version

Available since version 3.5.

II.2.2.67 StrsRgtEdgeMZ

C++

```
HRESULT get_StrsRgtEdgeMZ(double*);
```

C#

```
public double StrsRgtEdgeMZ { get; }
```

Visual Basic

```
Public ReadOnly StrsRgtEdgeMZ As double
```

Version

Available since version 3.5.

(c) 2014 Autodesk Inc. All rights reserved

II.2.2.68 StrsShearY

C++

```
HRESULT get_StrsShearY(double*);
```

C#

```
public double StrsShearY { get; }
```

Visual Basic

```
Public ReadOnly StrsShearY As double
```

Version

Available since version 3.5.

II.2.2.69 StrsShearZ

C++

```
HRESULT get_StrsShearZ(double*);
```

C#

```
public double StrsShearZ { get; }
```

Visual Basic

```
Public ReadOnly StrsShearZ As double
```

Version

Available since version 3.5.

II.2.2.70 StrsTens

C++

```
HRESULT get_StrsTens(double*);
```

C#

```
public double StrsTens { get; }
```

Visual Basic

```
Public ReadOnly StrsTens As double
```

Version

Available since version 3.5.

II.2.2.71 StrsUprEdgeMY

C++

```
HRESULT get_StrsUprEdgeMY(double*);
```

C#

```
public double StrsUprEdgeMY { get; }
```

Visual Basic

```
Public ReadOnly StrsUprEdgeMY As double
```

Version

Available since version 3.5.

(c) 2014 Autodesk Inc. All rights reserved

II.2.2.72 TimberType

C++

```
HRESULT get_TimberType(short*);
```

C#

```
public short TimberType { get; }
```

Visual Basic

```
Public ReadOnly TimberType As short
```

Version

Available since version 3.5.

II.3 IRDimBuckDiagramCB71

C++

```
enum IRDimBuckDiagramCB71;
```

C#

```
public enum IRDimBuckDiagramCB71;
```

Visual Basic

```
Public Enum IRDimBuckDiagramCB71
```

Members

Members	Description
I_BD_CB71_PINNED_PINNED_1_0 = 0	Available since version 3.5.
I_BD_CB71_STIFF_STIFF_0_65 = 1	Available since version 3.5.
I_BD_CB71_STIFF_PINNED_0_8 = 2	Available since version 3.5.
I_BD_CB71_CANTILEVER = 3	Available since version 3.5.
I_BD_CB71_NONE = 4	Available since version 3.5.
I_BD_CB71_INTERNAL_BRACINGS = 5	Available since version 3.5.

Version

Available since version 3.5.

II.4 IRDimLaterBuckTypeCB71

C++

```
enum IRDimLaterBuckTypeCB71;
```

C#

```
public enum IRDimLaterBuckTypeCB71;
```

Visual Basic

```
Public Enum IRDimLaterBuckTypeCB71
```

Members

Members	Description
I_LBT_CB71_PINNED_SUPPORTS = 0	Available since version 3.5.
I_LBT_CB71_FIXED_SUPPORTS = 1	Available since version 3.5.

(c) 2014 Autodesk Inc. All rights reserved

I_LBT_CB71_CANTILEVER = 2	Available since version 3.5.
I_LBT_CB71_NONE = 3	Available since version 3.5.

Version

Available since version 3.5.

II.5 IRDimLoadLevelCB71

C++

```
enum IRDimLoadLevelCB71;
```

C#

```
public enum IRDimLoadLevelCB71;
```

Visual Basic

```
Public Enum IRDimLoadLevelCB71
```

Members

Members	Description
I_LL_CB71_UP = 0	Available since version 3.5.
I_LL_CB71_MIDDLE = 1	Available since version 3.5.
I_LL_CB71_BOTTOM = 2	Available since version 3.5.

Version

Available since version 3.5.

II.6 IRDimLoadTypeCB71

C++

```
enum IRDimLoadTypeCB71;
```

C#

```
public enum IRDimLoadTypeCB71;
```

Visual Basic

```
Public Enum IRDimLoadTypeCB71
```

Members

Members	Description
I_DLT_CB71_MOMENT_AT_END = 0	Available since version 3.5.
I_DLT_CB71_UNIFORM_LOAD = 1	Available since version 3.5.
I_DLT_CB71_FORCE_AT_DISTANCE = 2	Available since version 3.5.

Version

Available since version 3.5.

II.7 IRDimFireMemberPositionsCB71

C++

```
enum IRDimFireMemberPositionsCB71;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public enum IRDimFireMemberPositionsCB71;
```

Visual Basic

```
Public Enum IRDimFireMemberPositionsCB71
```

Members

Members	Description
I_FMP_CB71_AUTOMATIC = 0	Available since version 3.5.
I_FMP_CB71_VERTICAL = 1	Available since version 3.5.
I_FMP_CB71_HORIZONTAL = 2	Available since version 3.5.

Version

Available since version 3.5.

II.8 IRDimFireProtectionTimesCB71

C++

```
enum IRDimFireProtectionTimesCB71;
```

C#

```
public enum IRDimFireProtectionTimesCB71;
```

Visual Basic

```
Public Enum IRDimFireProtectionTimesCB71
```

Members

Members	Description
I_FPT_CB71_LESS = 0	< 1/4. Available since version 3.5.
I_FPT_CB71_MORE = 1	> 1/4. Available since version 3.5.

Version

Available since version 3.5.

III IRDimStreamType

C++

```
enum IRDimStreamType;
```

C#

```
public enum IRDimStreamType;
```

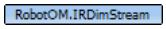
Visual Basic

```
Public Enum IRDimStreamType
```

(c) 2014 Autodesk Inc. All rights reserved

IV IRDimStream

Class Hierarchy



C++

```
interface IRDimStream : IDispatch;
```

C#

```
public interface IRDimStream;
```

Visual Basic

```
Public Interface IRDimStream
```

IV.1 IRDimStream Members

The following tables list the members exposed by IRDimStream.

Public Methods

	Name	Description
	Clear (see page 1890)	
	ReadDouble (see page 1891)	
	ReadLong (see page 1891)	
	ReadText (see page 1891)	
	SeekSet (see page 1891)	
	Size (see page 1891)	
	WriteDouble (see page 1892)	
	WriteLong (see page 1892)	
	WriteText (see page 1892)	

IV.2 IRDimStream Methods

The methods of the IRDimStream class are listed here.

Public Methods

	Name	Description
	Clear (see page 1890)	
	ReadDouble (see page 1891)	
	ReadLong (see page 1891)	
	ReadText (see page 1891)	
	SeekSet (see page 1891)	
	Size (see page 1891)	
	WriteDouble (see page 1892)	
	WriteLong (see page 1892)	
	WriteText (see page 1892)	

IV.2.1 Clear

C++

```
HRESULT Clear();
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public void Clear();
```

Visual Basic

```
Public Sub Clear()
```

IV.2.2 ReadDouble

C++

```
HRESULT ReadDouble(double* ret);
```

C#

```
public double ReadDouble();
```

Visual Basic

```
Public Function ReadDouble() As double
```

IV.2.3 ReadLong

C++

```
HRESULT ReadLong(long* ret);
```

C#

```
public long ReadLong();
```

Visual Basic

```
Public Function ReadLong() As long
```

IV.2.4 ReadText

C++

```
HRESULT ReadText(BSTR* ret);
```

C#

```
public String ReadText();
```

Visual Basic

```
Public Function ReadText() As String
```

IV.2.5 SeekSet

C++

```
HRESULT SeekSet(IRDimStreamType _type, long _pos);
```

C#

```
public void SeekSet(IRDimStreamType _type, long _pos);
```

Visual Basic

```
Public Sub SeekSet(_type As IRDimStreamType, _pos As long)
```

IV.2.6 Size

C++

```
HRESULT Size(IRDimStreamType _type, long* ret);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public long Size(IRDimStreamType _type);
```

Visual Basic

```
Public Function Size(_type As IRDimStreamType) As long
```

IV.2.7 WriteDouble

C++

```
HRESULT WriteDouble(double _val);
```

C#

```
public void WriteDouble(double _val);
```

Visual Basic

```
Public Sub WriteDouble(_val As double)
```

IV.2.8 WriteLong

C++

```
HRESULT WriteLong(long _val);
```

C#

```
public void WriteLong(long _val);
```

Visual Basic

```
Public Sub WriteLong(_val As long)
```

IV.2.9 WriteText

C++

```
HRESULT WriteText(BSTR _val);
```

C#

```
public void WriteText(String _val);
```

Visual Basic

```
Public Sub WriteText(_val As String)
```

V IRDimMembDefType

C++

```
enum IRDimMembDefType;
```

C#

```
public enum IRDimMembDefType;
```

Visual Basic

```
Public Enum IRDimMembDefType
```

(c) 2014 Autodesk Inc. All rights reserved

VI IRDimMembDefMatType

C++

```
enum IRDimMembDefMatType;
```

C#

```
public enum IRDimMembDefMatType;
```

Visual Basic

```
Public Enum IRDimMembDefMatType
```

Members

Members	Description
I_DMDMT_CONCRETE = 3	Available since version 7.5.

VII IRDimMembDefLengthDataType

C++

```
enum IRDimMembDefLengthDataType;
```

C#

```
public enum IRDimMembDefLengthDataType;
```

Visual Basic

```
Public Enum IRDimMembDefLengthDataType
```

VIII IRDimMembDefBucklingDataType

C++

```
enum IRDimMembDefBucklingDataType;
```

C#

```
public enum IRDimMembDefBucklingDataType;
```

Visual Basic

```
Public Enum IRDimMembDefBucklingDataType
```

IX IRDimMembDefDispDataType

C++

```
enum IRDimMembDefDispDataType;
```

C#

```
public enum IRDimMembDefDispDataType;
```

Visual Basic

```
Public Enum IRDimMembDefDispDataType
```

(c) 2014 Autodesk Inc. All rights reserved

X IRDimMembDefDeflDataType

C++

```
enum IRDimMembDefDeflDataType;
```

C#

```
public enum IRDimMembDefDeflDataType;
```

Visual Basic

```
Public Enum IRDimMembDefDeflDataType
```

XI IRDimMembDef

Class Hierarchy

RobotOM.IRDimMembDef

C++

```
interface IRDimMembDef : IDispatch;
```

C#

```
public interface IRDimMembDef;
```

Visual Basic

```
Public Interface IRDimMembDef
```

XI.1 IRDimMembDef Members

The following tables list the members exposed by IRDimMembDef.

Public Fields

	Name	Description
	ClientID (see page 1895)	
	Length (see page 1895)	
	MatType (see page 1895)	
	Name (see page 1895)	
	Type (see page 1896)	

Public Methods

	Name	Description
	DeflYZRelLimit (see page 1896)	
	DisplXYRelLimit (see page 1896)	
	IsBuckCoefConst (see page 1897)	
	IsDeflectionYZ (see page 1897)	
	IsDisplacementXY (see page 1897)	
	LengthYZUV (see page 1897)	
	Retrieve (see page 1897)	
	Store (see page 1898)	

(c) 2014 Autodesk Inc. All rights reserved

XI.2 IRDimMembDef Fields

The fields of the IRDimMembDef class are listed here.

Public Fields

	Name	Description
	ClientID (see page 1895)	
	Length (see page 1895)	
	MatType (see page 1895)	
	Name (see page 1895)	
	Type (see page 1896)	

XI.2.1 ClientID

C++

```
HRESULT get_ClientID(BSTR*);
```

C#

```
public String ClientID { get; }
```

Visual Basic

```
Public ReadOnly ClientID As String
```

XI.2.2 Length

C++

```
HRESULT get_Length(double*);
HRESULT put_Length(double);
```

C#

```
public double Length { get; set; }
```

Visual Basic

```
Public Length As double
```

XI.2.3 MatType

C++

```
HRESULT get_MatType(IRDimMembDefMatType*);
```

C#

```
public IRDimMembDefMatType MatType { get; }
```

Visual Basic

```
Public ReadOnly MatType As IRDimMembDefMatType
```

XI.2.4 Name

C++

```
HRESULT get_Name(BSTR*);
```

C#

```
public String Name { get; }
```

Visual Basic

```
Public ReadOnly Name As String
```

(c) 2014 Autodesk Inc. All rights reserved

XI.2.5 Type

C++

```
HRESULT get_Type(IRDimMembDefType*);
HRESULT put_Type(IRDimMembDefType);
```

C#

```
public IRDimMembDefType Type { get; set; }
```

Visual Basic

```
Public Type As IRDimMembDefType
```

XI.3 IRDimMembDef Methods

The methods of the IRDimMembDef class are listed here.

Public Methods

	Name	Description
	DeflYZRelLimit (see page 1896)	
	DisplXYRelLimit (see page 1896)	
	IsBuckCoefConst (see page 1897)	
	IsDeflectionYZ (see page 1897)	
	IsDisplacementXY (see page 1897)	
	LengthYZUV (see page 1897)	
	Retrieve (see page 1897)	
	Store (see page 1898)	

XI.3.1 DeflYZRelLimit

C++

```
HRESULT DeflYZRelLimit(IRDimMembDefDeflDataType _type, double* ret);
```

C#

```
public double DeflYZRelLimit(IRDimMembDefDeflDataType _type);
```

Visual Basic

```
Public Function DeflYZRelLimit(_type As IRDimMembDefDeflDataType) As double
```

XI.3.2 DisplXYRelLimit

C++

```
HRESULT DisplXYRelLimit(IRDimMembDefDispDataType _type, double* ret);
```

C#

```
public double DisplXYRelLimit(IRDimMembDefDispDataType _type);
```

Visual Basic

```
Public Function DisplXYRelLimit(_type As IRDimMembDefDispDataType) As double
```

(c) 2014 Autodesk Inc. All rights reserved

XI.3.3 IsBuckCoefConst

C++

```
HRESULT IsBuckCoefConst(IRDimMembDefBucklingDataType _type, VARIANT_BOOL* ret);
```

C#

```
public bool IsBuckCoefConst(IRDimMembDefBucklingDataType _type);
```

Visual Basic

```
Public Function IsBuckCoefConst(_type As IRDimMembDefBucklingDataType) As Boolean
```

XI.3.4 IsDeflectionYZ

C++

```
HRESULT IsDeflectionYZ(IRDimMembDefDeflDataType _type, VARIANT_BOOL* ret);
```

C#

```
public bool IsDeflectionYZ(IRDimMembDefDeflDataType _type);
```

Visual Basic

```
Public Function IsDeflectionYZ(_type As IRDimMembDefDeflDataType) As Boolean
```

XI.3.5 IsDisplacementXY

C++

```
HRESULT IsDisplacementXY(IRDimMembDefDispDataType _type, VARIANT_BOOL* ret);
```

C#

```
public bool IsDisplacementXY(IRDimMembDefDispDataType _type);
```

Visual Basic

```
Public Function IsDisplacementXY(_type As IRDimMembDefDispDataType) As Boolean
```

XI.3.6 LengthYZUV

C++

```
HRESULT LengthYZUV(IRDimMembDefLengthDataType _type, double* ret);
```

C#

```
public double LengthYZUV(IRDimMembDefLengthDataType _type);
```

Visual Basic

```
Public Function LengthYZUV(_type As IRDimMembDefLengthDataType) As double
```

XI.3.7 Retrieve

C++

```
HRESULT Retrieve(IRDimStream* _str);
```

C#

```
public void Retrieve(IRDimStream _str);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Sub Retrieve(ByRef _str As IRDimStream)
```

XI.3.8 Store

C++

```
HRESULT Store(IRDimStream* _str);
```

C#

```
public void Store(IRDimStream _str);
```

Visual Basic

```
Public Sub Store(ByRef _str As IRDimStream)
```

XII IRDimEffDefParamType

C++

```
enum IRDimEffDefParamType;
```

C#

```
public enum IRDimEffDefParamType;
```

Visual Basic

```
Public Enum IRDimEffDefParamType
```

XIII IRDimEffDefDirType

C++

```
enum IRDimEffDefDirType;
```

C#

```
public enum IRDimEffDefDirType;
```

Visual Basic

```
Public Enum IRDimEffDefDirType
```

XIV IRDimEffDefIntPsType

C++

```
enum IRDimEffDefIntPsType;
```

C#

```
public enum IRDimEffDefIntPsType;
```

Visual Basic

```
Public Enum IRDimEffDefIntPsType
```

(c) 2014 Autodesk Inc. All rights reserved

XV IRDimEffDef

Class Hierarchy

RobotOM.IRDimEffDef

C++

```
interface IRDimEffDef : IDispatch;
```

C#

```
public interface IRDimEffDef;
```

Visual Basic

```
Public Interface IRDimEffDef
```

XV.1 IRDimEffDef Members

The following tables list the members exposed by IRDimEffDef.

Public Fields

	Name	Description
	MX (see page 1900)	
	MY (see page 1900)	
	MZ (see page 1901)	
	N (see page 1901)	
	QY (see page 1901)	
	QZ (see page 1901)	

Public Methods

	Name	Description
	Clear (see page 1902)	
	Read_IntPsEff_M_1P4L (see page 1903)	
	Read_IntPsEff_M_3P4L (see page 1903)	
	Read_IntPsEff_M_MID (see page 1903)	
	Read_IntPsEff_M1 (see page 1903)	
	Read_IntPsEff_M12 (see page 1903)	
	Read_IntPsEff_M2 (see page 1904)	
	Read_IntPsEff_MN_MAX (see page 1904)	
	Read_IntPsEff_MP_MAX (see page 1904)	
	Read_M_1P4L (see page 1904)	
	Read_M_3P4L (see page 1904)	
	Read_M_MID (see page 1905)	
	Read_M1 (see page 1905)	
	Read_M12 (see page 1905)	

(c) 2014 Autodesk Inc. All rights reserved

	Read_M2 (see page 1905)	
	Read_MN_MAX (see page 1905)	
	Read_MP_MAX (see page 1906)	
	ReadParam (see page 1906)	
	WriteForces (see page 1906)	
	WriteIntPsEffSet1 (see page 1906)	
	WriteIntPsEffSet2 (see page 1907)	
	WriteParam (see page 1907)	
	WriteValuesSet1 (see page 1907)	
	WriteValuesSet2 (see page 1907)	

XV.2 IRDimEffDef Fields

The fields of the IRDimEffDef class are listed here.

Public Fields

	Name	Description
	MX (see page 1900)	
	MY (see page 1900)	
	MZ (see page 1901)	
	N (see page 1901)	
	QY (see page 1901)	
	QZ (see page 1901)	

XV.2.1 MX

C++

```
HRESULT get_MX(double*);
```

C#

```
public double MX { get; }
```

Visual Basic

```
Public ReadOnly MX As double
```

XV.2.2 MY

C++

```
HRESULT get_MY(double*);
```

C#

```
public double MY { get; }
```

Visual Basic

```
Public ReadOnly MY As double
```

XV.2.3 MZ

C++

```
HRESULT get_MZ(double*);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double MZ { get; }
```

Visual Basic

```
Public ReadOnly MZ As double
```

XV.2.4 N

C++

```
HRESULT get_N(double*);
```

C#

```
public double N { get; }
```

Visual Basic

```
Public ReadOnly N As double
```

XV.2.5 QY

C++

```
HRESULT get_QY(double*);
```

C#

```
public double QY { get; }
```

Visual Basic

```
Public ReadOnly QY As double
```

XV.2.6 QZ

C++

```
HRESULT get_QZ(double*);
```

C#

```
public double QZ { get; }
```

Visual Basic

```
Public ReadOnly QZ As double
```

XV.3 IRDimEffDef Methods

The methods of the IRDimEffDef class are listed here.

Public Methods

	Name	Description
	Clear (see page 1902)	
	Read_IntPsEff_M_1P4L (see page 1903)	
	Read_IntPsEff_M_3P4L (see page 1903)	
	Read_IntPsEff_M_MID (see page 1903)	
	Read_IntPsEff_M1 (see page 1903)	
	Read_IntPsEff_M12 (see page 1903)	

(c) 2014 Autodesk Inc. All rights reserved

	Read_IntPsEff_M2 (see page 1904)	
	Read_IntPsEff_MN_MAX (see page 1904)	
	Read_IntPsEff_MP_MAX (see page 1904)	
	Read_M_1P4L (see page 1904)	
	Read_M_3P4L (see page 1904)	
	Read_M_MID (see page 1905)	
	Read_M1 (see page 1905)	
	Read_M12 (see page 1905)	
	Read_M2 (see page 1905)	
	Read_MN_MAX (see page 1905)	
	Read_MP_MAX (see page 1906)	
	ReadParam (see page 1906)	
	WriteForces (see page 1906)	
	WriteIntPsEffSet1 (see page 1906)	
	WriteIntPsEffSet2 (see page 1907)	
	WriteParam (see page 1907)	
	WriteValuesSet1 (see page 1907)	
	WriteValuesSet2 (see page 1907)	

XV.3.1 Clear

C++

```
HRESULT Clear();
```

C#

```
public void Clear();
```

Visual Basic

```
Public Sub Clear()
```

XV.3.2 Read_IntPsEff_M_1P4L

C++

```
HRESULT Read_IntPsEff_M_1P4L(IRDimEffDefIntPsType _type, double* ret);
```

C#

```
public double Read_IntPsEff_M_1P4L(IRDimEffDefIntPsType _type);
```

Visual Basic

```
Public Function Read_IntPsEff_M_1P4L(_type As IRDimEffDefIntPsType) As double
```

XV.3.3 Read_IntPsEff_M_3P4L

C++

```
HRESULT Read_IntPsEff_M_3P4L(IRDimEffDefIntPsType _type, double* ret);
```

C#

```
public double Read_IntPsEff_M_3P4L(IRDimEffDefIntPsType _type);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Function Read_IntPsEff_M_3P4L(_type As IRDimEffDefIntPsType) As double
```

XV.3.4 Read_IntPsEff_M_MID

C++

```
HRESULT Read_IntPsEff_M_MID(IRDimEffDefIntPsType _type, double* ret);
```

C#

```
public double Read_IntPsEff_M_MID(IRDimEffDefIntPsType _type);
```

Visual Basic

```
Public Function Read_IntPsEff_M_MID(_type As IRDimEffDefIntPsType) As double
```

XV.3.5 Read_IntPsEff_M1

C++

```
HRESULT Read_IntPsEff_M1(IRDimEffDefIntPsType _type, double* ret);
```

C#

```
public double Read_IntPsEff_M1(IRDimEffDefIntPsType _type);
```

Visual Basic

```
Public Function Read_IntPsEff_M1(_type As IRDimEffDefIntPsType) As double
```

XV.3.6 Read_IntPsEff_M12

C++

```
HRESULT Read_IntPsEff_M12(IRDimEffDefIntPsType _type, double* ret);
```

C#

```
public double Read_IntPsEff_M12(IRDimEffDefIntPsType _type);
```

Visual Basic

```
Public Function Read_IntPsEff_M12(_type As IRDimEffDefIntPsType) As double
```

XV.3.7 Read_IntPsEff_M2

C++

```
HRESULT Read_IntPsEff_M2(IRDimEffDefIntPsType _type, double* ret);
```

C#

```
public double Read_IntPsEff_M2(IRDimEffDefIntPsType _type);
```

Visual Basic

```
Public Function Read_IntPsEff_M2(_type As IRDimEffDefIntPsType) As double
```

XV.3.8 Read_IntPsEff_MN_MAX

C++

```
HRESULT Read_IntPsEff_MN_MAX(IRDimEffDefIntPsType _type, double* ret);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double Read_IntPsEff_MN_MAX(IRDimEffDefIntPsType _type);
```

Visual Basic

```
Public Function Read_IntPsEff_MN_MAX(_type As IRDimEffDefIntPsType) As double
```

XV.3.9 Read_IntPsEff_MP_MAX

C++

```
HRESULT Read_IntPsEff_MP_MAX(IRDimEffDefIntPsType _type, double* ret);
```

C#

```
public double Read_IntPsEff_MP_MAX(IRDimEffDefIntPsType _type);
```

Visual Basic

```
Public Function Read_IntPsEff_MP_MAX(_type As IRDimEffDefIntPsType) As double
```

XV.3.10 Read_M_1P4L

C++

```
HRESULT Read_M_1P4L(IRDimEffDefDirType _type, double* ret);
```

C#

```
public double Read_M_1P4L(IRDimEffDefDirType _type);
```

Visual Basic

```
Public Function Read_M_1P4L(_type As IRDimEffDefDirType) As double
```

XV.3.11 Read_M_3P4L

C++

```
HRESULT Read_M_3P4L(IRDimEffDefDirType _type, double* ret);
```

C#

```
public double Read_M_3P4L(IRDimEffDefDirType _type);
```

Visual Basic

```
Public Function Read_M_3P4L(_type As IRDimEffDefDirType) As double
```

XV.3.12 Read_M_MID

C++

```
HRESULT Read_M_MID(IRDimEffDefDirType _type, double* ret);
```

C#

```
public double Read_M_MID(IRDimEffDefDirType _type);
```

Visual Basic

```
Public Function Read_M_MID(_type As IRDimEffDefDirType) As double
```

XV.3.13 Read_M1

C++

```
HRESULT Read_M1(IRDimEffDefDirType _type, double* ret);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double Read_M1(IRDimEffDefDirType _type);
```

Visual Basic

```
Public Function Read_M1(_type As IRDimEffDefDirType) As double
```

XV.3.14 Read_M12

C++

```
HRESULT Read_M12(IRDimEffDefDirType _type, double* ret);
```

C#

```
public double Read_M12(IRDimEffDefDirType _type);
```

Visual Basic

```
Public Function Read_M12(_type As IRDimEffDefDirType) As double
```

XV.3.15 Read_M2

C++

```
HRESULT Read_M2(IRDimEffDefDirType _type, double* ret);
```

C#

```
public double Read_M2(IRDimEffDefDirType _type);
```

Visual Basic

```
Public Function Read_M2(_type As IRDimEffDefDirType) As double
```

XV.3.16 Read_MN_MAX

C++

```
HRESULT Read_MN_MAX(IRDimEffDefDirType _type, double* ret);
```

C#

```
public double Read_MN_MAX(IRDimEffDefDirType _type);
```

Visual Basic

```
Public Function Read_MN_MAX(_type As IRDimEffDefDirType) As double
```

XV.3.17 Read_MP_MAX

C++

```
HRESULT Read_MP_MAX(IRDimEffDefDirType _type, double* ret);
```

C#

```
public double Read_MP_MAX(IRDimEffDefDirType _type);
```

Visual Basic

```
Public Function Read_MP_MAX(_type As IRDimEffDefDirType) As double
```

(c) 2014 Autodesk Inc. All rights reserved

XV.3.18 ReadParam

C++

```
HRESULT ReadParam(IRDimEffDefParamType _type, long* ret);
```

C#

```
public long ReadParam(IRDimEffDefParamType _type);
```

Visual Basic

```
Public Function ReadParam(_type As IRDimEffDefParamType) As long
```

XV.3.19 WriteForces

C++

```
HRESULT WriteForces(double _N, double _QY, double _QZ, double _MX, double _MY, double _MZ);
```

C#

```
public void WriteForces(double _N, double _QY, double _QZ, double _MX, double _MY, double
_MZ);
```

Visual Basic

```
Public Sub WriteForces(_N As double, _QY As double, _QZ As double, _MX As double, _MY As
double, _MZ As double)
```

XV.3.20 WriteIntPsEffSet1

C++

```
HRESULT WriteIntPsEffSet1(IRDimEffDefIntPsType _type, double _M1, double _M2, double _M12);
```

C#

```
public void WriteIntPsEffSet1(IRDimEffDefIntPsType _type, double _M1, double _M2, double
_M12);
```

Visual Basic

```
Public Sub WriteIntPsEffSet1(_type As IRDimEffDefIntPsType, _M1 As double, _M2 As double,
_M12 As double)
```

XV.3.21 WriteIntPsEffSet2

C++

```
HRESULT WriteIntPsEffSet2(IRDimEffDefIntPsType _type, double _MP_MAX, double _MN_MAX,
double _M_MID, double _M_1P4L, double _M_3P4L);
```

C#

```
public void WriteIntPsEffSet2(IRDimEffDefIntPsType _type, double _MP_MAX, double _MN_MAX,
double _M_MID, double _M_1P4L, double _M_3P4L);
```

Visual Basic

```
Public Sub WriteIntPsEffSet2(_type As IRDimEffDefIntPsType, _MP_MAX As double, _MN_MAX As
double, _M_MID As double, _M_1P4L As double, _M_3P4L As double)
```

XV.3.22 WriteParam

C++

```
HRESULT WriteParam(IRDimEffDefParamType _type, long _val);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public void WriteParam(IRDimEffDefParamType _type, long _val);
```

Visual Basic

```
Public Sub WriteParam(_type As IRDimEffDefParamType, _val As long)
```

XV.3.23 WriteValuesSet1

C++

```
HRESULT WriteValuesSet1(IRDimEffDefDirType _type, double _M1, double _M2, double _M12);
```

C#

```
public void WriteValuesSet1(IRDimEffDefDirType _type, double _M1, double _M2, double _M12);
```

Visual Basic

```
Public Sub WriteValuesSet1(_type As IRDimEffDefDirType, _M1 As double, _M2 As double, _M12
As double)
```

XV.3.24 WriteValuesSet2

C++

```
HRESULT WriteValuesSet2(IRDimEffDefDirType _type, double _MP_MAX, double _MN_MAX, double
_M_MID, double _M_1P4L, double _M_3P4L);
```

C#

```
public void WriteValuesSet2(IRDimEffDefDirType _type, double _MP_MAX, double _MN_MAX,
double _M_MID, double _M_1P4L, double _M_3P4L);
```

Visual Basic

```
Public Sub WriteValuesSet2(_type As IRDimEffDefDirType, _MP_MAX As double, _MN_MAX As
double, _M_MID As double, _M_1P4L As double, _M_3P4L As double)
```

XVI IRDimMatDefType

C++

```
enum IRDimMatDefType;
```

C#

```
public enum IRDimMatDefType;
```

Visual Basic

```
Public Enum IRDimMatDefType
```

XVII IRDimMatDefValType

C++

```
enum IRDimMatDefValType;
```

C#

```
public enum IRDimMatDefValType;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Enum IRDimMatDefValType
```

XVIII IRDimMatDefLongExValType

C++

```
enum IRDimMatDefLongExValType;
```

C#

```
public enum IRDimMatDefLongExValType;
```

Visual Basic

```
Public Enum IRDimMatDefLongExValType
```

XIX IRDimMatDefDblExValType

C++

```
enum IRDimMatDefDblExValType;
```

C#

```
public enum IRDimMatDefDblExValType;
```

Visual Basic

```
Public Enum IRDimMatDefDblExValType
```

XX IRDimMatDef

Class Hierarchy

RobotOM.IRDimMatDef

C++

```
interface IRDimMatDef : IDispatch;
```

C#

```
public interface IRDimMatDef;
```

Visual Basic

```
Public Interface IRDimMatDef
```

XX.1 IRDimMatDef Members

The following tables list the members exposed by IRDimMatDef.

Public Fields

	Name	Description
	Name (see page 1909)	
	SecondName (see page 1910)	
	Type (see page 1910)	

(c) 2014 Autodesk Inc. All rights reserved

Public Methods

	Name	Description
	ReadDoubleExtraValue (see page 1910)	
	ReadLongExtraValue (see page 1911)	
	ReadValue (see page 1911)	
	SetNames (see page 1911)	
	WriteDoubleExtraValue (see page 1911)	
	WriteLongExtraValue (see page 1911)	
	WriteValue (see page 1912)	

XX.2 IRDimMatDef Fields

The fields of the IRDimMatDef class are listed here.

Public Fields

	Name	Description
	Name (see page 1909)	
	SecondName (see page 1910)	
	Type (see page 1910)	

XX.2.1 Name

C++

```
HRESULT get_Name(BSTR*);
```

C#

```
public String Name { get; }
```

Visual Basic

```
Public ReadOnly Name As String
```

XX.2.2 SecondName

C++

```
HRESULT get_SecondName(BSTR*);
```

C#

```
public String SecondName { get; }
```

Visual Basic

```
Public ReadOnly SecondName As String
```

XX.2.3 Type

C++

```
HRESULT get_Type(IRDimMatDefType*);
HRESULT put_Type(IRDimMatDefType);
```

C#

```
public IRDimMatDefType Type { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Type As IRDimMatDefType
```

XX.3 IRDimMatDef Methods

The methods of the IRDimMatDef class are listed here.

Public Methods

	Name	Description
	ReadDoubleExtraValue (see page 1910)	
	ReadLongExtraValue (see page 1911)	
	ReadValue (see page 1911)	
	SetNames (see page 1911)	
	WriteDoubleExtraValue (see page 1911)	
	WriteLongExtraValue (see page 1911)	
	WriteValue (see page 1912)	

XX.3.1 ReadDoubleExtraValue

C++

```
HRESULT ReadDoubleExtraValue(IRDimMatDefDblExValType _type, double* ret);
```

C#

```
public double ReadDoubleExtraValue(IRDimMatDefDblExValType _type);
```

Visual Basic

```
Public Function ReadDoubleExtraValue(_type As IRDimMatDefDblExValType) As double
```

XX.3.2 ReadLongExtraValue

C++

```
HRESULT ReadLongExtraValue(IRDimMatDefLongExValType _type, long* ret);
```

C#

```
public long ReadLongExtraValue(IRDimMatDefLongExValType _type);
```

Visual Basic

```
Public Function ReadLongExtraValue(_type As IRDimMatDefLongExValType) As long
```

XX.3.3 ReadValue

C++

```
HRESULT ReadValue(IRDimMatDefValType _type, double* ret);
```

C#

```
public double ReadValue(IRDimMatDefValType _type);
```

Visual Basic

```
Public Function ReadValue(_type As IRDimMatDefValType) As double
```

(c) 2014 Autodesk Inc. All rights reserved

XX.3.4 SetNames

C++

```
HRESULT SetNames(BSTR _name, BSTR _second_name);
```

C#

```
public void SetNames(String _name, String _second_name);
```

Visual Basic

```
Public Sub SetNames(_name As String, _second_name As String)
```

XX.3.5 WriteDoubleExtraValue

C++

```
HRESULT WriteDoubleExtraValue(IRDimMatDefDblExValType _type, double _val);
```

C#

```
public void WriteDoubleExtraValue(IRDimMatDefDblExValType _type, double _val);
```

Visual Basic

```
Public Sub WriteDoubleExtraValue(_type As IRDimMatDefDblExValType, _val As double)
```

XX.3.6 WriteLongExtraValue

C++

```
HRESULT WriteLongExtraValue(IRDimMatDefLongExValType _type, long _val);
```

C#

```
public void WriteLongExtraValue(IRDimMatDefLongExValType _type, long _val);
```

Visual Basic

```
Public Sub WriteLongExtraValue(_type As IRDimMatDefLongExValType, _val As long)
```

XX.3.7 WriteValue

C++

```
HRESULT WriteValue(IRDimMatDefValType _type, double _val);
```

C#

```
public void WriteValue(IRDimMatDefValType _type, double _val);
```

Visual Basic

```
Public Sub WriteValue(_type As IRDimMatDefValType, _val As double)
```

XXI IRDimProfDefType

C++

```
enum IRDimProfDefType;
```

C#

```
public enum IRDimProfDefType;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Enum IRDimProfDefType
```

XXII IRDimProfDefItemType

C++

```
enum IRDimProfDefItemType;
```

C#

```
public enum IRDimProfDefItemType;
```

Visual Basic

```
Public Enum IRDimProfDefItemType
```

XXIII IRDimProfDefValType

C++

```
enum IRDimProfDefValType;
```

C#

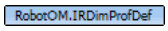
```
public enum IRDimProfDefValType;
```

Visual Basic

```
Public Enum IRDimProfDefValType
```

XXIV IRDimProfDef

Class Hierarchy



C++

```
interface IRDimProfDef : IDispatch;
```

C#

```
public interface IRDimProfDef;
```

Visual Basic

```
Public Interface IRDimProfDef
```

XXIV.1 IRDimProfDef Members

The following tables list the members exposed by IRDimProfDef.

Public Fields

	Name	Description
	IsVar (see page 1913)	
	Name (see page 1914)	
	Type (see page 1914)	

(c) 2014 Autodesk Inc. All rights reserved

Public Methods

	Name	Description
	Clear (see page 1914)	
	ReadValue (see page 1914)	
	WriteValue (see page 1915)	

XXIV.2 IRDimProfDef Fields

The fields of the IRDimProfDef class are listed here.

Public Fields

	Name	Description
	IsVar (see page 1913)	
	Name (see page 1914)	
	Type (see page 1914)	

XXIV.2.1 IsVar

C++

```
HRESULT get_IsVar(VARIANT_BOOL*);
```

C#

```
public bool IsVar { get; }
```

Visual Basic

```
Public ReadOnly IsVar As Boolean
```

XXIV.2.2 Name

C++

```
HRESULT get_Name(BSTR*);
HRESULT put_Name(BSTR);
```

C#

```
public String Name { get; set; }
```

Visual Basic

```
Public Name As String
```

XXIV.2.3 Type

C++

```
HRESULT get_Type(IRDimProfDefType*);
HRESULT put_Type(IRDimProfDefType);
```

C#

```
public IRDimProfDefType Type { get; set; }
```

Visual Basic

```
Public Type As IRDimProfDefType
```

XXIV.3 IRDimProfDef Methods

The methods of the IRDimProfDef class are listed here.

(c) 2014 Autodesk Inc. All rights reserved

Public Methods

	Name	Description
	Clear (see page 1914)	
	ReadValue (see page 1914)	
	WriteValue (see page 1915)	

XXIV.3.1 Clear

C++

```
HRESULT Clear();
```

C#

```
public void Clear();
```

Visual Basic

```
Public Sub Clear()
```

XXIV.3.2 ReadValue

C++

```
HRESULT ReadValue(IRDimProfDefItemType _item, IRDimProfDefValType _type, double* ret);
```

C#

```
public double ReadValue(IRDimProfDefItemType _item, IRDimProfDefValType _type);
```

Visual Basic

```
Public Function ReadValue(_item As IRDimProfDefItemType, _type As IRDimProfDefValType) As
double
```

XXIV.3.3 WriteValue

C++

```
HRESULT WriteValue(IRDimProfDefItemType _item, IRDimProfDefValType _type, double _val);
```

C#

```
public void WriteValue(IRDimProfDefItemType _item, IRDimProfDefValType _type, double _val);
```

Visual Basic

```
Public Sub WriteValue(_item As IRDimProfDefItemType, _type As IRDimProfDefValType, _val As
double)
```

XXV IRDimCalcStateFlagType

C++

```
enum IRDimCalcStateFlagType;
```

C#

```
public enum IRDimCalcStateFlagType;
```

Visual Basic

```
Public Enum IRDimCalcStateFlagType
```

(c) 2014 Autodesk Inc. All rights reserved

XXVI IRDimCalcStateParamType

C++

```
enum IRDimCalcStateParamType;
```

C#

```
public enum IRDimCalcStateParamType;
```

Visual Basic

```
Public Enum IRDimCalcStateParamType
```

XXVII IRDimCalcStateParamValue

C++

```
enum IRDimCalcStateParamValue;
```

C#

```
public enum IRDimCalcStateParamValue;
```

Visual Basic

```
Public Enum IRDimCalcStateParamValue
```

XXVIII IRDimCalcStateValueType

C++

```
enum IRDimCalcStateValueType;
```

C#

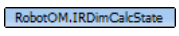
```
public enum IRDimCalcStateValueType;
```

Visual Basic

```
Public Enum IRDimCalcStateValueType
```

XXIX IRDimCalcState

Class Hierarchy



C++

```
interface IRDimCalcState : IDispatch;
```

C#

```
public interface IRDimCalcState;
```

Visual Basic

```
Public Interface IRDimCalcState
```

(c) 2014 Autodesk Inc. All rights reserved

XXIX.1 IRDimCalcState Members

The following tables list the members exposed by IRDimCalcState.

Public Methods

	Name	Description
	GetParam (see page 1917)	
	GetParamValue (see page 1917)	
	IsFlagSet (see page 1917)	
	SetFlag (see page 1917)	
	SetParam (see page 1918)	
	SetParamValue (see page 1918)	

XXIX.2 IRDimCalcState Methods

The methods of the IRDimCalcState class are listed here.

Public Methods

	Name	Description
	GetParam (see page 1917)	
	GetParamValue (see page 1917)	
	IsFlagSet (see page 1917)	
	SetFlag (see page 1917)	
	SetParam (see page 1918)	
	SetParamValue (see page 1918)	

XXIX.2.1 GetParam

C++

```
HRESULT GetParam(IRDimCalcStateParamType _type, IRDimCalcStateParamValue* ret);
```

C#

```
public IRDimCalcStateParamValue GetParam(IRDimCalcStateParamType _type);
```

Visual Basic

```
Public Function GetParam(_type As IRDimCalcStateParamType) As IRDimCalcStateParamValue
```

XXIX.2.2 GetParamValue

C++

```
HRESULT GetParamValue(IRDimCalcStateValueType _type, double* ret);
```

C#

```
public double GetParamValue(IRDimCalcStateValueType _type);
```

Visual Basic

```
Public Function GetParamValue(_type As IRDimCalcStateValueType) As double
```

XXIX.2.3 IsFlagSet

C++

```
HRESULT IsFlagSet(IRDimCalcStateFlagType _type, VARIANT_BOOL* ret);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public bool IsFlagSet(IRDimCalcStateFlagType _type);
```

Visual Basic

```
Public Function IsFlagSet(_type As IRDimCalcStateFlagType) As Boolean
```

XXIX.2.4 SetFlag

C++

```
HRESULT SetFlag(IRDimCalcStateFlagType _type, VARIANT_BOOL _val);
```

C#

```
public void SetFlag(IRDimCalcStateFlagType _type, bool _val);
```

Visual Basic

```
Public Sub SetFlag(_type As IRDimCalcStateFlagType, _val As Boolean)
```

XXIX.2.5 SetParam

C++

```
HRESULT SetParam(IRDimCalcStateParamType _type, IRDimCalcStateParamValue _val);
```

C#

```
public void SetParam(IRDimCalcStateParamType _type, IRDimCalcStateParamValue _val);
```

Visual Basic

```
Public Sub SetParam(_type As IRDimCalcStateParamType, _val As IRDimCalcStateParamValue)
```

XXIX.2.6 SetParamValue

C++

```
HRESULT SetParamValue(IRDimCalcStateValueType _type, double _val);
```

C#

```
public void SetParamValue(IRDimCalcStateValueType _type, double _val);
```

Visual Basic

```
Public Sub SetParamValue(_type As IRDimCalcStateValueType, _val As double)
```

XXX IRDimMembSrv

Class Hierarchy

RobotOM.IRDimMembSrv

C++

```
interface IRDimMembSrv : IDispatch;
```

C#

```
public interface IRDimMembSrv;
```

Visual Basic

```
Public Interface IRDimMembSrv
```

(c) 2014 Autodesk Inc. All rights reserved

XXX.1 IRDimMembSrv Members

The following tables list the members exposed by IRDimMembSrv.

Public Methods

	Name	Description
	CheckLabelName (see page 1919)	
	Save (see page 1919)	

XXX.2 IRDimMembSrv Methods

The methods of the IRDimMembSrv class are listed here.

Public Methods

	Name	Description
	CheckLabelName (see page 1919)	
	Save (see page 1919)	

XXX.2.1 CheckLabelName

C++

```
HRESULT CheckLabelName(BSTR _name, bool* _is_defined, bool* _can_be_saved);
```

C#

```
public void CheckLabelName(String _name, bool* _is_defined, bool* _can_be_saved);
```

Visual Basic

```
Public Sub CheckLabelName(_name As String, ByRef _is_defined As bool*, ByRef _can_be_saved
As bool*)
```

XXX.2.2 Save

C++

```
HRESULT Save(IRDimMembDef* _val);
```

C#

```
public void Save(IRDimMembDef _val);
```

Visual Basic

```
Public Sub Save(ByRef _val As IRDimMembDef)
```

XXXI IRDimMembCalcRetValue

C++

```
enum IRDimMembCalcRetValue;
```

C#

```
public enum IRDimMembCalcRetValue;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Enum IRDimMembCalcRetValue
```

XXXII IRDimMembCalcBuckType

C++

```
enum IRDimMembCalcBuckType;
```

C#

```
public enum IRDimMembCalcBuckType;
```

Visual Basic

```
Public Enum IRDimMembCalcBuckType
```

XXXIII IRDimMembResTableLineType

C++

```
enum IRDimMembResTableLineType;
```

C#

```
public enum IRDimMembResTableLineType;
```

Visual Basic

```
Public Enum IRDimMembResTableLineType
```

XXXIV IRDimMembResTableComp

C++

```
enum IRDimMembResTableComp;
```

C#

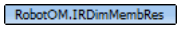
```
public enum IRDimMembResTableComp;
```

Visual Basic

```
Public Enum IRDimMembResTableComp
```

XXXV IRDimMembRes

Class Hierarchy



C++

```
interface IRDimMembRes : IDispatch;
```

C#

```
public interface IRDimMembRes;
```

Visual Basic

```
Public Interface IRDimMembRes
```

(c) 2014 Autodesk Inc. All rights reserved

XXXV.1 IRDimMembRes Members

The following tables list the members exposed by IRDimMembRes.

Public Fields

	Name	Description
	EffRatio (see page 1922)	
	Language (see page 1922)	
	Ratio (see page 1922)	
	ResOfCalc (see page 1922)	
	RtfFileName (see page 1922)	
	SlendY (see page 1923)	
	SlendZ (see page 1923)	
	Units (see page 1923)	

Public Methods

	Name	Description
	BlockCount (see page 1924)	
	CreateResWnd (see page 1924)	
	GetEffDefAccess (see page 1924)	
	GetLineComponent (see page 1925)	
	GetLineType (see page 1925)	
	GetMatDefAccess (see page 1925)	
	GetMaxLineNo (see page 1925)	
	GetMembDefAccess (see page 1926)	
	GetProfDefAccess (see page 1926)	
	IsLineActive (see page 1926)	
	IsStatement (see page 1926)	
	RecognizedPQ (see page 1926)	
	RefreshUnits (see page 1927)	
	ReplaceMark (see page 1927)	
	Retrieve (see page 1927)	
	SetEffDef (see page 1927)	
	SetMatDef (see page 1927)	
	SetMembDef (see page 1928)	
	SetProfDef (see page 1928)	

XXXV.2 IRDimMembRes Fields

The fields of the IRDimMembRes class are listed here.

Public Fields

	Name	Description
	EffRatio (see page 1922)	
	Language (see page 1922)	
	Ratio (see page 1922)	
	ResOfCalc (see page 1922)	

(c) 2014 Autodesk Inc. All rights reserved

	RtfFileName (see page 1922)	
	SlendY (see page 1923)	
	SlendZ (see page 1923)	
	Units (see page 1923)	

XXXV.2.1 EffRatio

C++

```
HRESULT get_EffRatio(double*);
```

C#

```
public double EffRatio { get; }
```

Visual Basic

```
Public ReadOnly EffRatio As double
```

XXXV.2.2 Language

C++

```
HRESULT get_Language(long*);
HRESULT put_Language(long);
```

C#

```
public long Language { get; set; }
```

Visual Basic

```
Public Language As long
```

XXXV.2.3 Ratio

C++

```
HRESULT get_Ratio(double*);
```

C#

```
public double Ratio { get; }
```

Visual Basic

```
Public ReadOnly Ratio As double
```

XXXV.2.4 ResOfCalc

C++

```
HRESULT get_ResOfCalc(IRDimMembCalcRetValue*);
```

C#

```
public IRDimMembCalcRetValue ResOfCalc { get; }
```

Visual Basic

```
Public ReadOnly ResOfCalc As IRDimMembCalcRetValue
```

XXXV.2.5 RtfFileName

C++

```
HRESULT get_RtfFileName(BSTR*);
```

C#

```
public String RtfFileName { get; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public ReadOnly RtfFileName As String
```

XXXV.2.6 SlendY

C++

```
HRESULT get_SlendY(double*);
```

C#

```
public double SlendY { get; }
```

Visual Basic

```
Public ReadOnly SlendY As double
```

XXXV.2.7 SlendZ

C++

```
HRESULT get_SlendZ(double*);
```

C#

```
public double SlendZ { get; }
```

Visual Basic

```
Public ReadOnly SlendZ As double
```

XXXV.2.8 Units

C++

```
HRESULT get_Units(IRDimUnits**);
HRESULT put_Units(IRDimUnits*);
```

C#

```
public IRDimUnits Units { get; set; }
```

Visual Basic

```
Public Units As IRDimUnits
```

XXXV.3 IRDimMembRes Methods

The methods of the IRDimMembRes class are listed here.

Public Methods

	Name	Description
	BlockCount (see page 1924)	
	CreateResWnd (see page 1924)	
	GetEffDefAccess (see page 1924)	
	GetLineComponent (see page 1925)	
	GetLineType (see page 1925)	
	GetMatDefAccess (see page 1925)	
	GetMaxLineNo (see page 1925)	
	GetMembDefAccess (see page 1926)	
	GetProfDefAccess (see page 1926)	

(c) 2014 Autodesk Inc. All rights reserved

	IsLineActive (see page 1926)	
	IsStatement (see page 1926)	
	RecognizedPQ (see page 1926)	
	RefreshUnits (see page 1927)	
	ReplaceMark (see page 1927)	
	Retrieve (see page 1927)	
	SetEffDef (see page 1927)	
	SetMatDef (see page 1927)	
	SetMembDef (see page 1928)	
	SetProfDef (see page 1928)	

XXXV.3.1 BlockCount

C++

```
HRESULT BlockCount(BSTR _name, long* ret);
```

C#

```
public long BlockCount(String _name);
```

Visual Basic

```
Public Function BlockCount(_name As String) As long
```

XXXV.3.2 CreateResWnd

C++

```
HRESULT CreateResWnd(long _parent_hwnd, long* ret);
```

C#

```
public long CreateResWnd(long _parent_hwnd);
```

Visual Basic

```
Public Function CreateResWnd(_parent_hwnd As long) As long
```

XXXV.3.3 GetEffDefAccess

C++

```
HRESULT GetEffDefAccess(IRDimEffDef** ret);
```

C#

```
public IRDimEffDef GetEffDefAccess();
```

Visual Basic

```
Public Function GetEffDefAccess() As IRDimEffDef
```

XXXV.3.4 GetLineComponent

C++

```
HRESULT GetLineComponent(long _line_no, IRDimMembResTableComp _cmpnt_no, BSTR* ret);
```

C#

```
public String GetLineComponent(long _line_no, IRDimMembResTableComp _cmpnt_no);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Function GetLineComponent(_line_no As long, _cmpnt_no As IRDimMembResTableComp) As
String
```

XXXV.3.5 GetLineType

C++

```
HRESULT GetLineType(long _line_no, IRDimMembResTableLineType* ret);
```

C#

```
public IRDimMembResTableLineType GetLineType(long _line_no);
```

Visual Basic

```
Public Function GetLineType(_line_no As long) As IRDimMembResTableLineType
```

XXXV.3.6 GetMatDefAccess

C++

```
HRESULT GetMatDefAccess(IRDimMatDef** ret);
```

C#

```
public IRDimMatDef GetMatDefAccess();
```

Visual Basic

```
Public Function GetMatDefAccess() As IRDimMatDef
```

XXXV.3.7 GetMaxLineNo

C++

```
HRESULT GetMaxLineNo(long* ret);
```

C#

```
public long GetMaxLineNo();
```

Visual Basic

```
Public Function GetMaxLineNo() As long
```

XXXV.3.8 GetMembDefAccess

C++

```
HRESULT GetMembDefAccess(IRDimMembDef** ret);
```

C#

```
public IRDimMembDef GetMembDefAccess();
```

Visual Basic

```
Public Function GetMembDefAccess() As IRDimMembDef
```

XXXV.3.9 GetProfDefAccess

C++

```
HRESULT GetProfDefAccess(IRDimProfDef** ret);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRDimProfDef GetProfDefAccess();
```

Visual Basic

```
Public Function GetProfDefAccess() As IRDimProfDef
```

XXXV.3.10 IsLineActive

C++

```
HRESULT IsLineActive(long _line_no, VARIANT_BOOL* ret);
```

C#

```
public bool IsLineActive(long _line_no);
```

Visual Basic

```
Public Function IsLineActive(_line_no As long) As Boolean
```

XXXV.3.11 IsStatement

C++

```
HRESULT IsStatement(BSTR _name, VARIANT_BOOL* ret);
```

C#

```
public bool IsStatement(String _name);
```

Visual Basic

```
Public Function IsStatement(_name As String) As Boolean
```

XXXV.3.12 RecognizedPQ

C++

```
HRESULT RecognizedPQ(BSTR _name, BSTR _vstr, BSTR _ustr, VARIANT_BOOL* ret);
```

C#

```
public bool RecognizedPQ(String _name, String _vstr, String _ustr);
```

Visual Basic

```
Public Function RecognizedPQ(_name As String, ByRef _vstr As String, ByRef _ustr As String)
As Boolean
```

XXXV.3.13 RefreshUnits

C++

```
HRESULT RefreshUnits(VARIANT_BOOL _are_robot_units);
```

C#

```
public void RefreshUnits(bool _are_robot_units);
```

Visual Basic

```
Public Sub RefreshUnits(_are_robot_units As Boolean)
```

(c) 2014 Autodesk Inc. All rights reserved

XXXV.3.14 ReplaceMark

C++

```
HRESULT ReplaceMark(BSTR _mark, BSTR _mark_out, VARIANT_BOOL* ret);
```

C#

```
public bool ReplaceMark(String _mark, String _mark_out);
```

Visual Basic

```
Public Function ReplaceMark(_mark As String, ByRef _mark_out As String) As Boolean
```

XXXV.3.15 Retrieve

C++

```
HRESULT Retrieve(IRDimStream* _str);
```

C#

```
public void Retrieve(IRDimStream _str);
```

Visual Basic

```
Public Sub Retrieve(ByRef _str As IRDimStream)
```

XXXV.3.16 SetEffDef

C++

```
HRESULT SetEffDef(IRDimEffDef* _eff_def);
```

C#

```
public void SetEffDef(IRDimEffDef _eff_def);
```

Visual Basic

```
Public Sub SetEffDef(ByRef _eff_def As IRDimEffDef)
```

XXXV.3.17 SetMatDef

C++

```
HRESULT SetMatDef(IRDimMatDef* _mat_def);
```

C#

```
public void SetMatDef(IRDimMatDef _mat_def);
```

Visual Basic

```
Public Sub SetMatDef(ByRef _mat_def As IRDimMatDef)
```

XXXV.3.18 SetMembDef

C++

```
HRESULT SetMembDef(IRDimMembDef* _memb_def);
```

C#

```
public void SetMembDef(IRDimMembDef _memb_def);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Sub SetMembDef(ByRef _memb_def As IRDimMembDef)
```

XXXV.3.19 SetProfDef

C++

```
HRESULT SetProfDef(IRDimProfDef* _prof_def);
```

C#

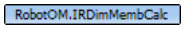
```
public void SetProfDef(IRDimProfDef _prof_def);
```

Visual Basic

```
Public Sub SetProfDef(ByRef _prof_def As IRDimProfDef)
```

XXXVI IRDimMembCalc

Class Hierarchy



C++

```
interface IRDimMembCalc : IDispatch;
```

C#

```
public interface IRDimMembCalc;
```

Visual Basic

```
Public Interface IRDimMembCalc
```

XXXVI.1 IRDimMembCalc Members

The following tables list the members exposed by IRDimMembCalc.

Public Fields

	Name	Description
	IsExtraMomentsSet1 (see page 1929)	
	IsExtraMomentsSet2 (see page 1929)	
	IsIntPointsMomentsSet1 (see page 1930)	
	IsIntPointsMomentsSet2 (see page 1930)	

Public Methods

	Name	Description
	CalculBuckling (see page 1930)	
	CalculMember (see page 1931)	
	GetRatio (see page 1931)	
	GetResultsInterface (see page 1931)	
	SetCalcState (see page 1931)	
	SetEffDef (see page 1931)	

(c) 2014 Autodesk Inc. All rights reserved

	SetMatDef (see page 1932)	
	SetMembDef (see page 1932)	
	SetProfDef (see page 1932)	

XXXVI.2 IRDimMembCalc Fields

The fields of the IRDimMembCalc class are listed here.

Public Fields

	Name	Description
	IsExtraMomentsSet1 (see page 1929)	
	IsExtraMomentsSet2 (see page 1929)	
	IsIntPointsMomentsSet1 (see page 1930)	
	IsIntPointsMomentsSet2 (see page 1930)	

XXXVI.2.1 IsExtraMomentsSet1

C++

```
HRESULT get_IsExtraMomentsSet1(VARIANT_BOOL*);
```

C#

```
public bool IsExtraMomentsSet1 { get; }
```

Visual Basic

```
Public ReadOnly IsExtraMomentsSet1 As Boolean
```

XXXVI.2.2 IsExtraMomentsSet2

C++

```
HRESULT get_IsExtraMomentsSet2(VARIANT_BOOL*);
```

C#

```
public bool IsExtraMomentsSet2 { get; }
```

Visual Basic

```
Public ReadOnly IsExtraMomentsSet2 As Boolean
```

XXXVI.2.3 IsIntPointsMomentsSet1

C++

```
HRESULT get_IsIntPointsMomentsSet1(VARIANT_BOOL*);
```

C#

```
public bool IsIntPointsMomentsSet1 { get; }
```

Visual Basic

```
Public ReadOnly IsIntPointsMomentsSet1 As Boolean
```

XXXVI.2.4 IsIntPointsMomentsSet2

C++

```
HRESULT get_IsIntPointsMomentsSet2(VARIANT_BOOL*);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public bool IsIntPointsMomentsSet2 { get; }
```

Visual Basic

```
Public ReadOnly IsIntPointsMomentsSet2 As Boolean
```

XXXVI.3 IRDimMembCalc Methods

The methods of the IRDimMembCalc class are listed here.

Public Methods

	Name	Description
	CalculBuckling (see page 1930)	
	CalculMember (see page 1931)	
	GetRatio (see page 1931)	
	GetResultsInterface (see page 1931)	
	SetCalcState (see page 1931)	
	SetEffDef (see page 1931)	
	SetMatDef (see page 1932)	
	SetMembDef (see page 1932)	
	SetProfDef (see page 1932)	

XXXVI.3.1 CalculBuckling

C++

```
HRESULT CalculBuckling();
```

C#

```
public void CalculBuckling();
```

Visual Basic

```
Public Sub CalculBuckling()
```

XXXVI.3.2 CalculMember

C++

```
HRESULT CalculMember(IRDimMembCalcRetValue* ret);
```

C#

```
public IRDimMembCalcRetValue CalculMember();
```

Visual Basic

```
Public Function CalculMember() As IRDimMembCalcRetValue
```

XXXVI.3.3 GetRatio

C++

```
HRESULT GetRatio(double* ret);
```

C#

```
public double GetRatio();
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Function GetRatio() As double
```

XXXVI.3.4 GetResultsInterface

C++

```
HRESULT GetResultsInterface(IRDimMembRes** ret);
```

C#

```
public IRDimMembRes GetResultsInterface();
```

Visual Basic

```
Public Function GetResultsInterface() As IRDimMembRes
```

XXXVI.3.5 SetCalcState

C++

```
HRESULT SetCalcState(IRDimCalcState* _st);
```

C#

```
public void SetCalcState(IRDimCalcState _st);
```

Visual Basic

```
Public Sub SetCalcState(ByRef _st As IRDimCalcState)
```

XXXVI.3.6 SetEffDef

C++

```
HRESULT SetEffDef(IRDimEffDef* _eff_data);
```

C#

```
public void SetEffDef(IRDimEffDef _eff_data);
```

Visual Basic

```
Public Sub SetEffDef(ByRef _eff_data As IRDimEffDef)
```

XXXVI.3.7 SetMatDef

C++

```
HRESULT SetMatDef(IRDimMatDef* _mat_def);
```

C#

```
public void SetMatDef(IRDimMatDef _mat_def);
```

Visual Basic

```
Public Sub SetMatDef(ByRef _mat_def As IRDimMatDef)
```

XXXVI.3.8 SetMembDef

C++

```
HRESULT SetMembDef(IRDimMembDef* _memb_def);
```

C#

```
public void SetMembDef(IRDimMembDef _memb_def);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Sub SetMembDef(ByRef _memb_def As IRDimMembDef)
```

XXXVI.3.9 SetProfDef

C++

```
HRESULT SetProfDef(IRDimProfDef* _prof_def);
```

C#

```
public void SetProfDef(IRDimProfDef _prof_def);
```

Visual Basic

```
Public Sub SetProfDef(ByRef _prof_def As IRDimProfDef)
```

XXXVII IRDimCodeService

Class Hierarchy

RobotOM.IRDimCodeService

C++

```
interface IRDimCodeService : IDispatch;
```

C#

```
public interface IRDimCodeService;
```

Visual Basic

```
Public Interface IRDimCodeService
```

XXXVII.1 IRDimCodeService Members

The following tables list the members exposed by IRDimCodeService.

Public Methods

	Name	Description
	EditMembDef (see page 1933)	
	GetDefaultMembDef (see page 1933)	
	GetMembCalc (see page 1934)	

XXXVII.2 IRDimCodeService Methods

The methods of the IRDimCodeService class are listed here.

Public Methods

	Name	Description
	EditMembDef (see page 1933)	
	GetDefaultMembDef (see page 1933)	
	GetMembCalc (see page 1934)	

(c) 2014 Autodesk Inc. All rights reserved

XXXVII.2.1 EditMembDef

C++

```
HRESULT EditMembDef(long _language, VARIANT_BOOL _inModalWindow, long _parentHWND,
IRDimMembDef* _val);
```

C#

```
public void EditMembDef(long _language, bool _inModalWindow, long _parentHWND, IRDimMembDef
_val);
```

Visual Basic

```
Public Sub EditMembDef(_language As long, _inModalWindow As Boolean, _parentHWND As long,
ByRef _val As IRDimMembDef)
```

XXXVII.2.2 GetDefaultMembDef

C++

```
HRESULT GetDefaultMembDef(IRDimMembDefType _type, IRDimMembDef** ret);
```

C#

```
public IRDimMembDef GetDefaultMembDef(IRDimMembDefType _type);
```

Visual Basic

```
Public Function GetDefaultMembDef(_type As IRDimMembDefType) As IRDimMembDef
```

XXXVII.2.3 GetMembCalc

C++

```
HRESULT GetMembCalc(IRDimMembCalc** ret);
```

C#

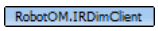
```
public IRDimMembCalc GetMembCalc();
```

Visual Basic

```
Public Function GetMembCalc() As IRDimMembCalc
```

XXXVIII IRDimClient

Class Hierarchy



C++

```
interface IRDimClient : IDispatch;
```

C#

```
public interface IRDimClient;
```

Visual Basic

```
Public Interface IRDimClient
```

XXXVIII.1 IRDimClient Members

The following tables list the members exposed by IRDimClient.

(c) 2014 Autodesk Inc. All rights reserved

Public Methods

	Name	Description
	GetRDimCodeService (see page 1934)	

XXXVIII.2 IRDimClient Methods

The methods of the IRDimClient class are listed here.

Public Methods

	Name	Description
	GetRDimCodeService (see page 1934)	

XXXVIII.2.1 GetRDimCodeService

C++

```
HRESULT GetRDimCodeService(IRDimCodeService** ret);
```

C#

```
public IRDimCodeService GetRDimCodeService();
```

Visual Basic

```
Public Function GetRDimCodeService() As IRDimCodeService
```

XXXIX IRDimUnitType

C++

```
enum IRDimUnitType;
```

C#

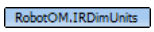
```
public enum IRDimUnitType;
```

Visual Basic

```
Public Enum IRDimUnitType
```

XL IRDimUnits

Class Hierarchy



C++

```
interface IRDimUnits : IDispatch;
```

C#

```
public interface IRDimUnits;
```

Visual Basic

```
Public Interface IRDimUnits
```

(c) 2014 Autodesk Inc. All rights reserved

XL.1 IRDimUnits Members

The following tables list the members exposed by IRDimUnits.

Public Fields

	Name	Description
	AreRobotUnits (see page 1936)	

Public Methods

	Name	Description
	Format (see page 1936)	
	ReadToUserCoef (see page 1936)	
	ReadUserName (see page 1937)	
	Refresh (see page 1937)	

XL.2 IRDimUnits Fields

The fields of the IRDimUnits class are listed here.

Public Fields

	Name	Description
	AreRobotUnits (see page 1936)	

XL.2.1 AreRobotUnits

C++

```
HRESULT get_AreRobotUnits(VARIANT_BOOL*);
```

C#

```
public bool AreRobotUnits { get; }
```

Visual Basic

```
Public ReadOnly AreRobotUnits As Boolean
```

XL.3 IRDimUnits Methods

The methods of the IRDimUnits class are listed here.

Public Methods

	Name	Description
	Format (see page 1936)	
	ReadToUserCoef (see page 1936)	
	ReadUserName (see page 1937)	
	Refresh (see page 1937)	

XL.3.1 Format

C++

```
HRESULT Format(IRDimUnitType _unit_type, double _val, BSTR* ret);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public String Format(IRDimUnitType _unit_type, double _val);
```

Visual Basic

```
Public Function Format(_unit_type As IRDimUnitType, _val As double) As String
```

XL.3.2 ReadToUserCoef

C++

```
HRESULT ReadToUserCoef(IRDimUnitType _unit_type, double* ret);
```

C#

```
public double ReadToUserCoef(IRDimUnitType _unit_type);
```

Visual Basic

```
Public Function ReadToUserCoef(_unit_type As IRDimUnitType) As double
```

XL.3.3 ReadUserName

C++

```
HRESULT ReadUserName(IRDimUnitType _unit_type, BSTR* ret);
```

C#

```
public String ReadUserName(IRDimUnitType _unit_type);
```

Visual Basic

```
Public Function ReadUserName(_unit_type As IRDimUnitType) As String
```

XL.3.4 Refresh

C++

```
HRESULT Refresh(VARIANT_BOOL _are_robot_units);
```

C#

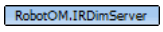
```
public void Refresh(bool _are_robot_units);
```

Visual Basic

```
Public Sub Refresh(_are_robot_units As Boolean)
```

XLI IRDimServer

Class Hierarchy



C++

```
interface IRDimServer : IDispatch;
```

C#

```
public interface IRDimServer;
```

Visual Basic

```
Public Interface IRDimServer
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Server responsible for design and verification of steel and timber members.

Version

Available since version 2.5.

Member section definition module

Names of interfaces from the module of bar section definition start with the RSect prefix.

Version

Available since version 3.

Connection module

Names of interfaces defined for the connection module begin with the RJoint prefix.

Notes

All values described by interfaces of a connection module are given in millimeters.

Enumerations

	Name	Description
	IRJointConnectionDefType (see page 2156)	A method of connection definition.
	IRJointConnectionType (see page 2156)	Types of connections available in a connections module of Robot.
	IRJointWebFlangeRelativePos (see page 2159)	Possible types of a beam - column mutual position. .
	IRJointBoltType (see page 2169)	Available bolt types.
	IRJointWebType (see page 2169)	Types of reinforcing plates.
	IRJointLoadType (see page 2179)	
	IRJointAngleUnit (see page 2179)	
	IRJointExtType (see page 2180)	
	IRJointComponentType (see page 2180)	

Interfaces

	Name	Description
	IRJointConnectionInfo (see page 2157)	Information about connection. .
	IRJointConnection (see page 2159)	A data type describing common features for all connections that can be defined in Robot.
	IRJointWeld (see page 2162)	Description of a weld.
	IRJointPlate (see page 2163)	Plate description.
	IRJointBolts (see page 2166)	Definition of a group of bolts with common parameters.
	IRJointConnectionServer (see page 2169)	The connection server administers all connection defined in the project. Each connection is associated with a number. The numbering of connections may be discontinuous.
	IRJointProfile (see page 2174)	Structure describing a single connection component. .

(c) 2014 Autodesk Inc. All rights reserved

	IRJointLoad (see page 2178)	Interface describing a load applied to the connection. .

I Knee connection

Enumerations

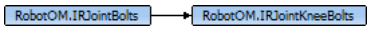
	Name	Description
	IRJointKneeType (see page 1950)	Accessible types of a knee connection.
	IRJointKneeFixType (see page 1950)	Available types of a knee connection as for joint type.
	IRJointKneeReinfType (see page 1951)	Available bracket types for a knee connection.
	IRJointKneeWebStiffType (see page 1965)	Bracket types of a column web for a knee connection. .
	IRJointKneeDiagonalStiffType (see page 1965)	Available slanting bracket types for a column web.
	IRJointKneeMaterials (see page 1969)	

Interfaces

	Name	Description
	IRJointKneeBolts (see page 1939)	Definition of a bolt group for a knee connection.
	IRJointKneeBracket (see page 1943)	Knee bracket.
	IRJointKneeWebPlate (see page 1947)	Description of a plate stiffening a column web.
	IRJointKnee (see page 1951)	Knee connection.
	IRJointKneeDiagonalStiff (see page 1965)	An inclined stiffener of a column.
	IRJointKneeStiffColumn (see page 1967)	Column bracket in a knee connection. .
	IRJointKneeLoad (see page 1968)	Interface describing a load applied to a knee connection. .

I.1 IRJointKneeBolts

Class Hierarchy



C++

```
interface IRJointKneeBolts : IRJointBolts;
```

C#

```
public interface IRJointKneeBolts : IRJointBolts;
```

Visual Basic

```
Public Interface IRJointKneeBolts
```

Description

Definition of a bolt group for a knee connection.

(c) 2014 Autodesk Inc. All rights reserved

I.1.1 IRJointKneeBolts Members

The following tables list the members exposed by IRJointKneeBolts.

Public Fields

	Name	Description
	Area (see page 2167)	Bolt section area.
	ClassName (see page 1940)	Bolt class.
	Cols (see page 1941)	Number of bolts in a column.
	Diameter (see page 2167)	Bolt diameter.
	DiameterName (see page 1941)	Bolt diameter.
	EqualSpac (see page 1941)	Bolts are even-spaced if EqualSpac equals 1.
	Friction (see page 2168)	Bolt friction.
	Height1 (see page 1942)	Distance h1.
	Rows (see page 1942)	Number of bolts in a row.
	SpacingH (see page 1942)	Distances between bolts in the horizontal direction.
	SpacingV (see page 1943)	Distances between bolts in the vertical direction.
	Symmetry (see page 1943)	Bolts are centered with respect to the plate if Symmetry equals 1.

I.1.2 IRJointKneeBolts Fields

The fields of the IRJointKneeBolts class are listed here.

Public Fields

	Name	Description
	ClassName (see page 1940)	Bolt class.
	Cols (see page 1941)	Number of bolts in a column.
	DiameterName (see page 1941)	Bolt diameter.
	EqualSpac (see page 1941)	Bolts are even-spaced if EqualSpac equals 1.
	Height1 (see page 1942)	Distance h1.
	Rows (see page 1942)	Number of bolts in a row.
	SpacingH (see page 1942)	Distances between bolts in the horizontal direction.
	SpacingV (see page 1943)	Distances between bolts in the vertical direction.
	Symmetry (see page 1943)	Bolts are centered with respect to the plate if Symmetry equals 1.

I.1.2.1 ClassName

C++

```
HRESULT get_ClassName(BSTR*);
HRESULT put_ClassName(BSTR);
```

C#

```
public BSTR ClassName { get; set; }
```

Visual Basic

```
Public ClassName As BSTR
```

Description

Bolt class.

Version

Available since version 9.5.

(c) 2014 Autodesk Inc. All rights reserved

I.1.2.2 Cols

C++

```
HRESULT get_Cols(int*);
HRESULT put_Cols(int);
```

C#

```
public int Cols { get; set; }
```

Visual Basic

```
Public Cols As int
```

Description

Number of bolts in a column.

Version

Available since version 9.5.

I.1.2.3 DiameterName

C++

```
HRESULT get_DiameterName(BSTR*);
HRESULT put_DiameterName(BSTR);
```

C#

```
public BSTR DiameterName { get; set; }
```

Visual Basic

```
Public DiameterName As BSTR
```

Description

Bolt diameter.

Version

Available since version 9.5.

I.1.2.4 EqualSpac

C++

```
HRESULT get_EqualSpac(int*);
HRESULT put_EqualSpac(int);
```

C#

```
public int EqualSpac { get; set; }
```

Visual Basic

```
Public EqualSpac As int
```

Description

Bolts are even-spaced if EqualSpac equals 1.

Version

Available since version 9.5.

(c) 2014 Autodesk Inc. All rights reserved

I.1.2.5 Height1

C++

```
HRESULT get_Height1(double*);
HRESULT put_Height1(double);
```

C#

```
public double Height1 { get; set; }
```

Visual Basic

```
Public Height1 As double
```

Description

Distance h1.

Version

Available since version 9.5.

I.1.2.6 Rows

C++

```
HRESULT get_Rows(int*);
HRESULT put_Rows(int);
```

C#

```
public int Rows { get; set; }
```

Visual Basic

```
Public Rows As int
```

Description

Number of bolts in a row.

Version

Available since version 9.5.

I.1.2.7 SpacingH

C++

```
HRESULT get_SpacingH(IRobotValuesArray**);
```

C#

```
public IRobotValuesArray SpacingH { get; }
```

Visual Basic

```
Public ReadOnly SpacingH As IRobotValuesArray
```

Description

Distances between bolts in the horizontal direction.

I.1.2.8 SpacingV

C++

```
HRESULT get_SpacingV(IRobotValuesArray**);
```

C#

```
public IRobotValuesArray SpacingV { get; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public ReadOnly SpacingV As IRobotValuesArray
```

Description

Distances between bolts in the vertical direction.

I.1.2.9 Symmetry

C++

```
HRESULT get_Symmetry(int*);
HRESULT put_Symmetry(int);
```

C#

```
public int Symmetry { get; set; }
```

Visual Basic

```
Public Symmetry As int
```

Description

Bolts are centered with respect to the plate if Symmetry equals 1.

Version

Available since version 9.5.

I.2 IRJointKneeBracket

Class Hierarchy

RobotOM.IRJointKneeBracket

C++

```
interface IRJointKneeBracket : IDispatch;
```

C#

```
public interface IRJointKneeBracket;
```

Visual Basic

```
Public Interface IRJointKneeBracket
```

Description

Knee bracket.

I.2.1 IRJointKneeBracket Members

The following tables list the members exposed by IRJointKneeBracket.

Public Fields

	Name	Description
	Angle (see page 1944)	Inclination angle of reinforcement.
	Exist (see page 1944)	Information about a bracket existence.
	Height (see page 1945)	Bracket height.
	Length (see page 1945)	Bracket length.
	Material (see page 1945)	Name of the bracket material .
	ThickFlange (see page 1946)	Thickness of bracket flange .
	ThickWeb (see page 1946)	Thickness of bracket web.

(c) 2014 Autodesk Inc. All rights reserved

	Width (see page 1946)	Bracket width.

Public Methods

	Name	Description
	AngleExt (see page 1947)	Angle (see page 1944) of bracket inclination, depending on the parameter - in degrees or radians.

I.2.2 IRJointKneeBracket Fields

The fields of the IRJointKneeBracket class are listed here.

Public Fields

	Name	Description
	Angle (see page 1944)	Inclination angle of reinforcement.
	Exist (see page 1944)	Information about a bracket existence.
	Height (see page 1945)	Bracket height.
	Length (see page 1945)	Bracket length.
	Material (see page 1945)	Name of the bracket material .
	ThickFlange (see page 1946)	Thickness of bracket flange .
	ThickWeb (see page 1946)	Thickness of bracket web.
	Width (see page 1946)	Bracket width.

I.2.2.1 Angle

C++

```
HRESULT get_Angle(double*);
HRESULT put_Angle(double);
```

C#

```
public double Angle { get; set; }
```

Visual Basic

```
Public Angle As double
```

Description

Inclination angle of reinforcement.

I.2.2.2 Exist

C++

```
HRESULT get_Exist(VARIANT_BOOL*);
HRESULT put_Exist(VARIANT_BOOL);
```

C#

```
public bool Exist { get; set; }
```

Visual Basic

```
Public Exist As Boolean
```

Description

Information about a bracket existence.

I.2.2.3 Height

C++

```
HRESULT get_Height(double*);
```

(c) 2014 Autodesk Inc. All rights reserved

```
HRESULT put_Height(double);
```

C#

```
public double Height { get; set; }
```

Visual Basic

```
Public Height As double
```

Description

Bracket height.

I.2.2.4 Length

C++

```
HRESULT get_Length(double*);
HRESULT put_Length(double);
```

C#

```
public double Length { get; set; }
```

Visual Basic

```
Public Length As double
```

Description

Bracket length.

I.2.2.5 Material

C++

```
HRESULT get_Material(BSTR*);
HRESULT put_Material(BSTR);
```

C#

```
public BSTR Material { get; set; }
```

Visual Basic

```
Public Material As BSTR
```

Description

Name of the bracket material .

Version

Available since version 9.5.

I.2.2.6 ThickFlange

C++

```
HRESULT get_ThickFlange(double*);
HRESULT put_ThickFlange(double);
```

C#

```
public double ThickFlange { get; set; }
```

Visual Basic

```
Public ThickFlange As double
```

Description

Thickness of bracket flange .

(c) 2014 Autodesk Inc. All rights reserved

I.2.2.7 ThickWeb

C++

```
HRESULT get_ThickWeb(double*);
HRESULT put_ThickWeb(double);
```

C#

```
public double ThickWeb { get; set; }
```

Visual Basic

```
Public ThickWeb As double
```

Description

Thickness of bracket web.

I.2.2.8 Width

C++

```
HRESULT get_Width(double*);
HRESULT put_Width(double);
```

C#

```
public double Width { get; set; }
```

Visual Basic

```
Public Width As double
```

Description

Bracket width.

I.2.3 IRJointKneeBracket Methods

The methods of the IRJointKneeBracket class are listed here.

Public Methods

	Name	Description
	AngleExt (see page 1947)	Angle (see page 1944) of bracket inclination, depending on the parameter - in degrees or radians.

I.2.3.1 AngleExt

C++

```
HRESULT AngleExt(IRJointAngleUnit _unit, double* ret);
```

C#

```
public double AngleExt(IRJointAngleUnit _unit);
```

Visual Basic

```
Public Function AngleExt(_unit As IRJointAngleUnit) As double
```

Description

Angle (see page 1944) of bracket inclination, depending on the parameter - in degrees or radians.

Version

Available since version 9.5.

(c) 2014 Autodesk Inc. All rights reserved

I.3 IRJointKneeWebPlate

Class Hierarchy

C++

```
interface IRJointKneeWebPlate : IDispatch;
```

C#

```
public interface IRJointKneeWebPlate;
```

Visual Basic

```
Public Interface IRJointKneeWebPlate
```

Description

Description of a plate stiffening a column web.

I.3.1 IRJointKneeWebPlate Members

The following tables list the members exposed by IRJointKneeWebPlate.

Public Fields

	Name	Description
	Bolts (see page 1948)	Bolt group.
	Height (see page 1948)	Available since version 1.7.
	Material (see page 1948)	Name of the component material.
	Thick (see page 1949)	Available since version 1.7.
	Type (see page 1949)	
	WeldLong (see page 1949)	A longitudinal weld.
	WeldTran (see page 1949)	A transversal weld.
	Width (see page 1950)	Available since version 1.7.

I.3.2 IRJointKneeWebPlate Fields

The fields of the IRJointKneeWebPlate class are listed here.

Public Fields

	Name	Description
	Bolts (see page 1948)	Bolt group.
	Height (see page 1948)	Available since version 1.7.
	Material (see page 1948)	Name of the component material.
	Thick (see page 1949)	Available since version 1.7.
	Type (see page 1949)	
	WeldLong (see page 1949)	A longitudinal weld.
	WeldTran (see page 1949)	A transversal weld.
	Width (see page 1950)	Available since version 1.7.

I.3.2.1 Bolts

C++

```
HRESULT get_Bolts(IRJointKneeBolts**);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRJointKneeBolts Bolts { get; }
```

Visual Basic

```
Public ReadOnly Bolts As IRJointKneeBolts
```

Description

Bolt group.

I.3.2.2 Height

C++

```
HRESULT get_Height(double*);
HRESULT put_Height(double);
```

C#

```
public double Height { get; set; }
```

Visual Basic

```
Public Height As double
```

Description

Available since version 1.7.

I.3.2.3 Material

C++

```
HRESULT get_Material(BSTR*);
HRESULT put_Material(BSTR);
```

C#

```
public BSTR Material { get; set; }
```

Visual Basic

```
Public Material As BSTR
```

Description

Name of the component material.

Version

Available since version 9.5.

I.3.2.4 Thick

C++

```
HRESULT get_Thick(double*);
HRESULT put_Thick(double);
```

C#

```
public double Thick { get; set; }
```

Visual Basic

```
Public Thick As double
```

Description

Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

I.3.2.5 Type

C++

```
HRESULT get_Type(IRJointWebType*);
HRESULT put_Type(IRJointWebType);
```

C#

```
public IRJointWebType Type { get; set; }
```

Visual Basic

```
Public Type As IRJointWebType
```

I.3.2.6 WeldLong

C++

```
HRESULT get_WeldLong(IRJointWeld**);
```

C#

```
public IRJointWeld WeldLong { get; }
```

Visual Basic

```
Public ReadOnly WeldLong As IRJointWeld
```

Description

A longitudinal weld.

I.3.2.7 WeldTran

C++

```
HRESULT get_WeldTran(IRJointWeld**);
```

C#

```
public IRJointWeld WeldTran { get; }
```

Visual Basic

```
Public ReadOnly WeldTran As IRJointWeld
```

Description

A transversal weld.

I.3.2.8 Width

C++

```
HRESULT get_Width(double*);
HRESULT put_Width(double);
```

C#

```
public double Width { get; set; }
```

Visual Basic

```
Public Width As double
```

Description

Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

I.4 IRJointKneeType

C++

```
enum IRJointKneeType;
```

C#

```
public enum IRJointKneeType;
```

Visual Basic

```
Public Enum IRJointKneeType
```

Members

Members	Description
I_JKT_BEAM2BEAM = 0	Beam-to-beam connection.
I_JKT_BEAM2COLUMN = 1	A frame knee connection.
I_JKT_BEAM2COLUMN_CONTINUE = 2	A beam-to-column connection.

Description

Accessible types of a knee connection.

I.5 IRJointKneeFixType

C++

```
enum IRJointKneeFixType;
```

C#

```
public enum IRJointKneeFixType;
```

Visual Basic

```
Public Enum IRJointKneeFixType
```

Description

Available types of a knee connection as for joint type.

I.6 IRJointKneeReinfType

C++

```
enum IRJointKneeReinfType;
```

C#

```
public enum IRJointKneeReinfType;
```

Visual Basic

```
Public Enum IRJointKneeReinfType
```

Description

Available bracket types for a knee connection.

(c) 2014 Autodesk Inc. All rights reserved

I.7 IRJointKnee

Class Hierarchy

C++

```
interface IRJointKnee : IRJointConnection;
```

C#

```
public interface IRJointKnee : IRJointConnection;
```

Visual Basic

```
Public Interface IRJointKnee
```

Description

Knee connection.

I.7.1 IRJointKnee Members

The following tables list the members exposed by IRJointKnee.

Public Fields

	Name	Description
	Beam (see page 1954)	Beam section .
	Bolts (see page 1954)	A group of connecting bolts.
	BracketLow (see page 1955)	Lower bracket of a beam.
	BracketUp (see page 1955)	Upper bracket of a beam.
	Column (see page 1955)	Column section .
	DefComponentMaterial (see page 1955)	Name of the default material of components.
	FixType (see page 1956)	Out-of-date parameter - retained to ensure compatibility with the previous version.
	FlanPlateLower (see page 1956)	Label of the lower bracket of the column flange.
	FlanPlatePosLower (see page 1956)	Lower bracket of the column flange - ebd parameter.
	FlanPlatePosUpper (see page 1957)	Upper bracket of the column flange - ebu parameter.
	FlanPlateUpper (see page 1957)	Label of the upper bracket of the column flange.
	IsBolted (see page 1957)	The connection is bolted if IsBolted equals 1, otherwise, the connection is welded.
	IsDefComponentMaterialSet (see page 1958)	All components are assigned the same material if IsDefComponentMaterialSet equals 1.
	KneeType (see page 1958)	A type of a knee connection.
	Material (see page 1958)	Available since version ARSA 2010.
	MaterPlates (see page 1958)	
	Plate (see page 1959)	Main plate.
	PlatePosition (see page 1959)	Distance between the top edge of the slab and the top edge of the adjoining element.
	PlatePositionLow (see page 1959)	Distance between bottom plate edge and the bottom edge of the adjoining element .
	StiffDiag (see page 1959)	Inclined column bracket.
	StiffLow (see page 1960)	Bottom column bracing.

(c) 2014 Autodesk Inc. All rights reserved

	StiffTypeLow (see page 1960)	A type of lower bracket.
	StiffTypeUp (see page 1960)	A type of upper bracket.
	StiffUp (see page 1960)	Top column bracing.
	TensionPlateLow (see page 1961)	Bottom plate reinforcing the beam.
	TensionPlateUp (see page 1961)	Top plate reinforcing the beam.
	Type (see page 2160)	A connection type.
	VStiffLow (see page 1961)	A bottom, vertical bracing of a beam.
	VStiffUp (see page 1961)	A top, vertical bracing of a beam.
	WebBeamStiffenerVLower (see page 1962)	Label of the lower vertical stiffener of a beam.
	WebBeamStiffenerVUpper (see page 1962)	Label of the upper vertical stiffener of a beam.
	WebColumnStiffenerHLower (see page 1962)	Label of the lower horizontal stiffener of a column.
	WebColumnStiffenerHUpper (see page 1963)	Label of the upper horizontal stiffener of a column.
	WebPlate (see page 1963)	A plate reinforcing a column web.
	WebStiffType (see page 1963)	A bracket type of a column web.
	WeldBracketDownFlange (see page 1963)	Thickness of the weld connecting the flange of lower bracket with the column.
	WeldBracketUpFlange (see page 1964)	Thickness of the weld connecting the flange of upper bracket with the column .
	WeldFlange (see page 1964)	
	WeldStiff (see page 1964)	
	WeldWeb (see page 1964)	
	WFRelPos (see page 2161)	A type of relative position of a beam and a column.

Public Methods

	Name	Description
	GetFromRobot (see page 2161)	The function gets connection parameters connection parameters from Robot and fills an object with the data.
	SetToRobot (see page 2161)	The function sets and saves connection parameter in Robot.

I.7.2 IRJointKnee Fields

The fields of the IRJointKnee class are listed here.

Public Fields

	Name	Description
	Beam (see page 1954)	Beam section .
	Bolts (see page 1954)	A group of connecting bolts.
	BracketLow (see page 1955)	Lower bracket of a beam.
	BracketUp (see page 1955)	Upper bracket of a beam.
	Column (see page 1955)	Column section .
	DefComponentMaterial (see page 1955)	Name of the default material of components.
	FixType (see page 1956)	Out-of-date parameter - retained to ensure compatibility with the previous version.
	FlanPlateLower (see page 1956)	Label of the lower bracket of the column flange.
	FlanPlatePosLower (see page 1956)	Lower bracket of the column flange - ebd parameter.

(c) 2014 Autodesk Inc. All rights reserved

	FlanPlatePosUpper (see page 1957)	Upper bracket of the column flange - ebu parameter.
	FlanPlateUpper (see page 1957)	Label of the upper bracket of the column flange.
	IsBolted (see page 1957)	The connection is bolted if IsBolted equals 1, otherwise, the connection is welded.
	IsDefComponentMaterialSet (see page 1958)	All components are assigned the same material if IsDefComponentMaterialSet equals 1.
	KneeType (see page 1958)	A type of a knee connection.
	Material (see page 1958)	Available since version ARSA 2010.
	MaterPlates (see page 1958)	
	Plate (see page 1959)	Main plate.
	PlatePosition (see page 1959)	Distance between the top edge of the slab and the top edge of the adjoining element.
	PlatePositionLow (see page 1959)	Distance between bottom plate edge and the bottom edge of the adjoining element .
	StiffDiag (see page 1959)	Inclined column bracket.
	StiffLow (see page 1960)	Bottom column bracing.
	StiffTypeLow (see page 1960)	A type of lower bracket.
	StiffTypeUp (see page 1960)	A type of upper bracket.
	StiffUp (see page 1960)	Top column bracing.
	TensionPlateLow (see page 1961)	Bottom plate reinforcing the beam.
	TensionPlateUp (see page 1961)	Top plate reinforcing the beam.
	VStiffLow (see page 1961)	A bottom, vertical bracing of a beam.
	VStiffUp (see page 1961)	A top, vertical bracing of a beam.
	WebBeamStiffenerVLower (see page 1962)	Label of the lower vertical stiffener of a beam.
	WebBeamStiffenerVUpper (see page 1962)	Label of the upper vertical stiffener of a beam.
	WebColumnStiffenerHLower (see page 1962)	Label of the lower horizontal stiffener of a column.
	WebColumnStiffenerHUpper (see page 1963)	Label of the upper horizontal stiffener of a column.
	WebPlate (see page 1963)	A plate reinforcing a column web.
	WebStiffType (see page 1963)	A bracket type of a column web.
	WeldBracketDownFlange (see page 1963)	Thickness of the weld connecting the flange of lower bracket with the column.
	WeldBracketUpFlange (see page 1964)	Thickness of the weld connecting the flange of upper bracket with the column .
	WeldFlange (see page 1964)	
	WeldStiff (see page 1964)	
	WeldWeb (see page 1964)	

I.7.2.1 Beam

C++

```
HRESULT get_Beam(IRJointProfile**);
```

C#

```
public IRJointProfile Beam { get; }
```

Visual Basic

```
Public ReadOnly Beam As IRJointProfile
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Beam section .

I.7.2.2 Bolts

C++

```
HRESULT get_Bolts(IRJointKneeBolts**);
```

C#

```
public IRJointKneeBolts Bolts { get; }
```

Visual Basic

```
Public ReadOnly Bolts As IRJointKneeBolts
```

Description

A group of connecting bolts.

I.7.2.3 BracketLow

C++

```
HRESULT get_BracketLow(IRJointKneeBracket**);
```

C#

```
public IRJointKneeBracket BracketLow { get; }
```

Visual Basic

```
Public ReadOnly BracketLow As IRJointKneeBracket
```

Description

Lower bracket of a beam.

I.7.2.4 BracketUp

C++

```
HRESULT get_BracketUp(IRJointKneeBracket**);
```

C#

```
public IRJointKneeBracket BracketUp { get; }
```

Visual Basic

```
Public ReadOnly BracketUp As IRJointKneeBracket
```

Description

Upper bracket of a beam.

I.7.2.5 Column

C++

```
HRESULT get_Column(IRJointProfile**);
```

C#

```
public IRJointProfile Column { get; }
```

Visual Basic

```
Public ReadOnly Column As IRJointProfile
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Column section .

I.7.2.6 DefComponentMaterial

C++

```
HRESULT get_DefComponentMaterial( BSTR*);
HRESULT put_DefComponentMaterial( BSTR);
```

C#

```
public  BSTR DefComponentMaterial { get; set; }
```

Visual Basic

```
Public DefComponentMaterial As  BSTR
```

Description

Name of the default material of components.

Version

Available since version 9.5.

I.7.2.7 FixType

C++

```
HRESULT get_FixType(IRJointKneeFixType*);
HRESULT put_FixType(IRJointKneeFixType);
```

C#

```
public IRJointKneeFixType FixType { get; set; }
```

Visual Basic

```
Public FixType As IRJointKneeFixType
```

Description

Out-of-date parameter - retained to ensure compatibility with the previous version.

I.7.2.8 FlanPlateLower

C++

```
HRESULT get_FlanPlateLower(IRJointPlate**);
```

C#

```
public IRJointPlate FlanPlateLower { get; }
```

Visual Basic

```
Public ReadOnly FlanPlateLower As IRJointPlate
```

Description

Label of the lower bracket of the column flange.

Version

Available since version 9.5.

I.7.2.9 FlanPlatePosLower

C++

```
HRESULT get_FlanPlatePosLower(double*);
```

(c) 2014 Autodesk Inc. All rights reserved

```
HRESULT put_FlanPlatePosLower(double);
```

C#

```
public double FlanPlatePosLower { get; set; }
```

Visual Basic

```
Public FlanPlatePosLower As double
```

Description

Lower bracket of the column flange - ebd parameter.

Version

Available since version 9.5.

I.7.2.10 FlanPlatePosUpper

C++

```
HRESULT get_FlanPlatePosUpper(double*);
HRESULT put_FlanPlatePosUpper(double);
```

C#

```
public double FlanPlatePosUpper { get; set; }
```

Visual Basic

```
Public FlanPlatePosUpper As double
```

Description

Upper bracket of the column flange - ebu parameter.

Version

Available since version 9.5.

I.7.2.11 FlanPlateUpper

C++

```
HRESULT get_FlanPlateUpper(IRJointPlate**);
```

C#

```
public IRJointPlate FlanPlateUpper { get; }
```

Visual Basic

```
Public ReadOnly FlanPlateUpper As IRJointPlate
```

Description

Label of the upper bracket of the column flange.

Version

Available since version 9.5.

I.7.2.12 IsBolted

C++

```
HRESULT get_IsBolted(int*);
HRESULT put_IsBolted(int);
```

C#

```
public int IsBolted { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public IsBolted As int
```

Description

The connection is bolted if IsBolted equals 1, otherwise, the connection is welded.

Version

Available since version 9.5.

I.7.2.13 IsDefComponentMaterialSet

C++

```
HRESULT get_IsDefComponentMaterialSet(int*);
HRESULT put_IsDefComponentMaterialSet(int);
```

C#

```
public int IsDefComponentMaterialSet { get; set; }
```

Visual Basic

```
Public IsDefComponentMaterialSet As int
```

Description

All components are assigned the same material if IsDefComponentMaterialSet equals 1.

Version

Available since version 9.5.

I.7.2.14 KneeType

C++

```
HRESULT get_KneeType(IRJointKneeType*);
HRESULT put_KneeType(IRJointKneeType);
```

C#

```
public IRJointKneeType KneeType { get; set; }
```

Visual Basic

```
Public KneeType As IRJointKneeType
```

Description

A type of a knee connection.

I.7.2.15 Material

C++

```
HRESULT get_Material(IRJointKneeMaterials*);
HRESULT put_Material(IRJointKneeMaterials);
```

C#

```
public IRJointKneeMaterials Material { get; set; }
```

Visual Basic

```
Public Material As IRJointKneeMaterials
```

Description

Available since version ARSA 2010.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 10.

I.7.2.16 MaterPlates

C++

```
HRESULT get_MaterPlates(IRobotMaterialData**);
```

C#

```
public IRobotMaterialData MaterPlates { get; }
```

Visual Basic

```
Public ReadOnly MaterPlates As IRobotMaterialData
```

I.7.2.17 Plate

C++

```
HRESULT get_Plate(IRJointPlate**);
```

C#

```
public IRJointPlate Plate { get; }
```

Visual Basic

```
Public ReadOnly Plate As IRJointPlate
```

Description

Main plate.

I.7.2.18 PlatePosition

C++

```
HRESULT get_PlatePosition(double*);
HRESULT put_PlatePosition(double);
```

C#

```
public double PlatePosition { get; set; }
```

Visual Basic

```
Public PlatePosition As double
```

Description

Distance between the top edge of the slab and the top edge of the adjoining element.

Version

Available since version 9.5.

I.7.2.19 PlatePositionLow

C++

```
HRESULT get_PlatePositionLow(double*);
HRESULT put_PlatePositionLow(double);
```

C#

```
public double PlatePositionLow { get; set; }
```

Visual Basic

```
Public PlatePositionLow As double
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Distance between bottom plate edge and the bottom edge of the adjoining element .

I.7.2.20 StiffDiag

C++

```
HRESULT get_StiffDiag(IRJointKneeDiagonalStiff**);
```

C#

```
public IRJointKneeDiagonalStiff StiffDiag { get; }
```

Visual Basic

```
Public ReadOnly StiffDiag As IRJointKneeDiagonalStiff
```

Description

Inclined column bracket.

I.7.2.21 StiffLow

C++

```
HRESULT get_StiffLow(IRJointKneeStiffColumn**);
```

C#

```
public IRJointKneeStiffColumn StiffLow { get; }
```

Visual Basic

```
Public ReadOnly StiffLow As IRJointKneeStiffColumn
```

Description

Bottom column bracing.

I.7.2.22 StiffTypeLow

C++

```
HRESULT get_StiffTypeLow(IRJointKneeReinfType*);
HRESULT put_StiffTypeLow(IRJointKneeReinfType);
```

C#

```
public IRJointKneeReinfType StiffTypeLow { get; set; }
```

Visual Basic

```
Public StiffTypeLow As IRJointKneeReinfType
```

Description

A type of lower bracket.

I.7.2.23 StiffTypeUp

C++

```
HRESULT get_StiffTypeUp(IRJointKneeReinfType*);
HRESULT put_StiffTypeUp(IRJointKneeReinfType);
```

C#

```
public IRJointKneeReinfType StiffTypeUp { get; set; }
```

Visual Basic

```
Public StiffTypeUp As IRJointKneeReinfType
```

(c) 2014 Autodesk Inc. All rights reserved

Description

A type of upper bracket.

I.7.2.24 StiffUp

C++

```
HRESULT get_StiffUp(IRJointKneeStiffColumn**);
```

C#

```
public IRJointKneeStiffColumn StiffUp { get; }
```

Visual Basic

```
Public ReadOnly StiffUp As IRJointKneeStiffColumn
```

Description

Top column bracing.

I.7.2.25 TensionPlateLow

C++

```
HRESULT get_TensionPlateLow(IRJointPlate**);
```

C#

```
public IRJointPlate TensionPlateLow { get; }
```

Visual Basic

```
Public ReadOnly TensionPlateLow As IRJointPlate
```

Description

Bottom plate reinforcing the beam.

I.7.2.26 TensionPlateUp

C++

```
HRESULT get_TensionPlateUp(IRJointPlate**);
```

C#

```
public IRJointPlate TensionPlateUp { get; }
```

Visual Basic

```
Public ReadOnly TensionPlateUp As IRJointPlate
```

Description

Top plate reinforcing the beam.

I.7.2.27 VStiffLow

C++

```
HRESULT get_VStiffLow(IRJointPlate**);
```

C#

```
public IRJointPlate VStiffLow { get; }
```

Visual Basic

```
Public ReadOnly VStiffLow As IRJointPlate
```

(c) 2014 Autodesk Inc. All rights reserved

Description

A bottom, vertical bracing of a beam.

I.7.2.28 VStiffUp

C++

```
HRESULT get_VStiffUp(IRJointPlate**);
```

C#

```
public IRJointPlate VStiffUp { get; }
```

Visual Basic

```
Public ReadOnly VStiffUp As IRJointPlate
```

Description

A top, vertical bracing of a beam.

I.7.2.29 WebBeamStiffenerVLower

C++

```
HRESULT get_WebBeamStiffenerVLower(IRJointPlate**);
```

C#

```
public IRJointPlate WebBeamStiffenerVLower { get; }
```

Visual Basic

```
Public ReadOnly WebBeamStiffenerVLower As IRJointPlate
```

Description

Label of the lower vertical stiffener of a beam.

Version

Available since version 9.5.

I.7.2.30 WebBeamStiffenerVUpper

C++

```
HRESULT get_WebBeamStiffenerVUpper(IRJointPlate**);
```

C#

```
public IRJointPlate WebBeamStiffenerVUpper { get; }
```

Visual Basic

```
Public ReadOnly WebBeamStiffenerVUpper As IRJointPlate
```

Description

Label of the upper vertical stiffener of a beam.

Version

Available since version 9.5.

I.7.2.31 WebColumnStiffenerHLower

C++

```
HRESULT get_WebColumnStiffenerHLower(IRJointPlate**);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRJointPlate WebColumnStiffenerHLower { get; }
```

Visual Basic

```
Public ReadOnly WebColumnStiffenerHLower As IRJointPlate
```

Description

Label of the lower horizontal stiffener of a column.

Version

Available since version 9.5.

I.7.2.32 WebColumnStiffenerHUpper

C++

```
HRESULT get_WebColumnStiffenerHUpper(IRJointPlate**);
```

C#

```
public IRJointPlate WebColumnStiffenerHUpper { get; }
```

Visual Basic

```
Public ReadOnly WebColumnStiffenerHUpper As IRJointPlate
```

Description

Label of the upper horizontal stiffener of a column.

Version

Available since version 9.5.

I.7.2.33 WebPlate

C++

```
HRESULT get_WebPlate(IRJointKneeWebPlate**);
```

C#

```
public IRJointKneeWebPlate WebPlate { get; }
```

Visual Basic

```
Public ReadOnly WebPlate As IRJointKneeWebPlate
```

Description

A plate reinforcing a column web.

I.7.2.34 WebStiffType

C++

```
HRESULT get_WebStiffType(IRJointKneeWebStiffType*);
HRESULT put_WebStiffType(IRJointKneeWebStiffType);
```

C#

```
public IRJointKneeWebStiffType WebStiffType { get; set; }
```

Visual Basic

```
Public WebStiffType As IRJointKneeWebStiffType
```

Description

A bracket type of a column web.

(c) 2014 Autodesk Inc. All rights reserved

I.7.2.35 WeldBracketDownFlange

C++

```
HRESULT get_WeldBracketDownFlange(IRJointWeld**);
```

C#

```
public IRJointWeld WeldBracketDownFlange { get; }
```

Visual Basic

```
Public ReadOnly WeldBracketDownFlange As IRJointWeld
```

Description

Thickness of the weld connecting the flange of lower bracket with the column.

I.7.2.36 WeldBracketUpFlange

C++

```
HRESULT get_WeldBracketUpFlange(IRJointWeld**);
```

C#

```
public IRJointWeld WeldBracketUpFlange { get; }
```

Visual Basic

```
Public ReadOnly WeldBracketUpFlange As IRJointWeld
```

Description

Thickness of the weld connecting the flange of upper bracket with the column .

Version

Available since version 9.5.

I.7.2.37 WeldFlange

C++

```
HRESULT get_WeldFlange(IRJointWeld**);
```

C#

```
public IRJointWeld WeldFlange { get; }
```

Visual Basic

```
Public ReadOnly WeldFlange As IRJointWeld
```

I.7.2.38 WeldStiff

C++

```
HRESULT get_WeldStiff(IRJointWeld**);
```

C#

```
public IRJointWeld WeldStiff { get; }
```

Visual Basic

```
Public ReadOnly WeldStiff As IRJointWeld
```

I.7.2.39 WeldWeb

C++

```
HRESULT get_WeldWeb(IRJointWeld**);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRJointWeld WeldWeb { get; }
```

Visual Basic

```
Public ReadOnly WeldWeb As IRJointWeld
```

I.8 IRJointKneeWebStiffType

C++

```
enum IRJointKneeWebStiffType;
```

C#

```
public enum IRJointKneeWebStiffType;
```

Visual Basic

```
Public Enum IRJointKneeWebStiffType
```

Description

Bracket types of a column web for a knee connection. .

I.9 IRJointKneeDiagonalStiffType

C++

```
enum IRJointKneeDiagonalStiffType;
```

C#

```
public enum IRJointKneeDiagonalStiffType;
```

Visual Basic

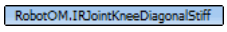
```
Public Enum IRJointKneeDiagonalStiffType
```

Description

Available slanting bracket types for a column web.

I.10 IRJointKneeDiagonalStiff

Class Hierarchy



C++

```
interface IRJointKneeDiagonalStiff : IDispatch;
```

C#

```
public interface IRJointKneeDiagonalStiff;
```

Visual Basic

```
Public Interface IRJointKneeDiagonalStiff
```

Description

An inclined stiffener of a column.

(c) 2014 Autodesk Inc. All rights reserved

I.10.1 IRJointKneeDiagonalStiff Members

The following tables list the members exposed by IRJointKneeDiagonalStiff.

Public Fields

	Name	Description
	Exist (see page 1966)	
	Material (see page 1966)	Name of the component material.
	Thick (see page 1966)	Thickness.
	Type (see page 1967)	A stiffener type.

I.10.2 IRJointKneeDiagonalStiff Fields

The fields of the IRJointKneeDiagonalStiff class are listed here.

Public Fields

	Name	Description
	Exist (see page 1966)	
	Material (see page 1966)	Name of the component material.
	Thick (see page 1966)	Thickness.
	Type (see page 1967)	A stiffener type.

I.10.2.1 Exist

C++

```
HRESULT get_Exist(VARIANT_BOOL*);
```

C#

```
public bool Exist { get; }
```

Visual Basic

```
Public ReadOnly Exist As Boolean
```

I.10.2.2 Material

C++

```
HRESULT get_Material(BSTR*);
HRESULT put_Material(BSTR);
```

C#

```
public BSTR Material { get; set; }
```

Visual Basic

```
Public Material As BSTR
```

Description

Name of the component material.

Version

Available since version 9.5.

I.10.2.3 Thick

C++

```
HRESULT get_Thick(double*);
HRESULT put_Thick(double);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double Thick { get; set; }
```

Visual Basic

```
Public Thick As double
```

Description

Thickness.

I.10.2.4 Type

C++

```
HRESULT get_Type(IRJointKneeDiagonalStiffType*);
HRESULT put_Type(IRJointKneeDiagonalStiffType);
```

C#

```
public IRJointKneeDiagonalStiffType Type { get; set; }
```

Visual Basic

```
Public Type As IRJointKneeDiagonalStiffType
```

Description

A stiffener type.

I.11 IRJointKneeStiffColumn

Class Hierarchy

RobotOM.IRJointKneeStiffColumn

C++

```
interface IRJointKneeStiffColumn : IDispatch;
```

C#

```
public interface IRJointKneeStiffColumn;
```

Visual Basic

```
Public Interface IRJointKneeStiffColumn
```

Description

Column bracket in a knee connection. .

I.11.1 IRJointKneeStiffColumn Members

The following tables list the members exposed by IRJointKneeStiffColumn.

Public Fields

	Name	Description
	Thick (see page 1968)	Thickness.

I.11.2 IRJointKneeStiffColumn Fields

The fields of the IRJointKneeStiffColumn class are listed here.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	Thick (see page 1968)	Thickness.

I.11.2.1 Thick

C++

```
HRESULT get_Thick(double*);
HRESULT put_Thick(double);
```

C#

```
public double Thick { get; set; }
```

Visual Basic

```
Public Thick As double
```

Description

Thickness.

I.12 IRJointKneeLoad

Class Hierarchy

C++

```
interface IRJointKneeLoad : IRJointLoad;
```

C#

```
public interface IRJointKneeLoad : IRJointLoad;
```

Visual Basic

```
Public Interface IRJointKneeLoad
```

Description

Interface describing a load applied to a knee connection. .

I.12.1 IRJointKneeLoad Members

The following tables list the members exposed by IRJointKneeLoad.

Public Fields

	Name	Description
	Cases (see page 2178)	Selection of load cases, defined for the structure. .
	M (see page 1969)	
	N (see page 1969)	
	Q (see page 1969)	
	Type (see page 2179)	Load type (defined manually or taken from Robot).

I.12.2 IRJointKneeLoad Fields

The fields of the IRJointKneeLoad class are listed here.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	M (see page 1969)	
	N (see page 1969)	
	Q (see page 1969)	

I.12.2.1 M

C++

```
HRESULT get_M(double*);
HRESULT put_M(double);
```

C#

```
public double M { get; set; }
```

Visual Basic

```
Public M As double
```

I.12.2.2 N

C++

```
HRESULT get_N(double*);
HRESULT put_N(double);
```

C#

```
public double N { get; set; }
```

Visual Basic

```
Public N As double
```

I.12.2.3 Q

C++

```
HRESULT get_Q(double*);
HRESULT put_Q(double);
```

C#

```
public double Q { get; set; }
```

Visual Basic

```
Public Q As double
```

I.13 IRJointKneeMaterials

C++

```
enum IRJointKneeMaterials;
```

C#

```
public enum IRJointKneeMaterials;
```

Visual Basic

```
Public Enum IRJointKneeMaterials
```

Members

Members	Description
I_JKM_PLATE = 0	Available since version 9.7.

(c) 2014 Autodesk Inc. All rights reserved

I_JKM_PLATE_LEFT = 1	Available since version 9.7.
I_JKM_FLAN_PLATE_UPPER = 2	Available since version 9.7.
I_JKM_FLAN_PLATE_LOWER = 3	Available since version 9.7.
I_JKM_WEB_COLUMN_STIFF_H_UPPER = 4	Available since version 9.7.
I_JKM_WEB_COLUMN_STIFF_H_LOWER = 5	Available since version 9.7.
I_JKM_WEB_BEAM_STIFF_V_UPPER = 6	Available since version 9.7.
I_JKM_WEB_BEAM_STIFF_V_LOWER = 7	Available since version 9.7.
I_JKM_TENSION_PLATE_UPPER = 8	Available since version 9.7.
I_JKM_TENSION_PLATE_LOWER = 9	Available since version 9.7.

Version

Available since version 9.7.

II Column base connections

Enumerations

	Name	Description
	IRJointAnchorType (see page 2003)	Available anchorage types.
	IRJointFootPlateType (see page 2009)	Available connection types - plate/angle.
	IRJointWedgeType (see page 2013)	Available wedge types.
	IRJointAnchorPlateType (see page 2019)	Types of plates near anchor.
	IRJointFootStiffType (see page 2029)	

Interfaces

	Name	Description
	IRJointAnchorBolt (see page 2001)	Anchorage bolt.
	IRJointAnchorPlate (see page 2002)	Plate of an anchor bolt.
	IRJointAnchor (see page 2004)	Anchorage definition.
	IRJointFootBolts (see page 2006)	Bolt group of identical parameters for pinned column base. .
	IRJointFootPlate (see page 2009)	
	IRJointWedge (see page 2011)	
	IRJointBearingPlate (see page 2013)	
	IRJointFootWelds (see page 2014)	Description of connection element welds.
	IRJointColumnBracket (see page 2016)	Column stiffener.
	IRJointFootStiffenerVert (see page 2019)	
	IRJointFootStiffenerHoriz (see page 2021)	
	IRJointFootMaterials (see page 2022)	
	IRJointColumnSquare (see page 2024)	

(c) 2014 Autodesk Inc. All rights reserved

	IRJointFootStiffenerSimple (see page 2025)	
	IRJointFootStiffenerComplex (see page 2027)	

II.1 Pinned column base

Enumerations

	Name	Description
	IRJointPinnedColumnBaseStiffType (see page 1971)	

Interfaces

	Name	Description
	IRJointPinnedColumnBase (see page 1971)	Pinned column base.
	IRJointPinnedLoad (see page 1977)	

II.1.1 IRJointPinnedColumnBaseStiffType

C++

```
enum IRJointPinnedColumnBaseStiffType;
```

C#

```
public enum IRJointPinnedColumnBaseStiffType;
```

Visual Basic

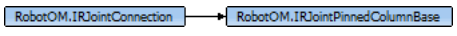
```
Public Enum IRJointPinnedColumnBaseStiffType
```

Members

Members	Description
I_JPCBST_LRDIAG = 0	Available since version 1.7.
I_JPCBST_LDIAG = 1	Available since version 1.7.
I_JPCBST_RDIAG = 2	Available since version 1.7.
I_JPCBST_LRI = 3	Available since version 1.7.
I_JPCBST_LI = 4	Available since version 1.7.
I_JPCBST_RI = 5	Available since version 1.7.
I_JPCBST_NONE = 6	Available since version 1.7.

II.1.2 IRJointPinnedColumnBase

Class Hierarchy



C++

```
interface IRJointPinnedColumnBase : IRJointConnection;
```

C#

```
public interface IRJointPinnedColumnBase : IRJointConnection;
```

Visual Basic

```
Public Interface IRJointPinnedColumnBase
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Pinned column base.

II.1.2.1 IRJointPinnedColumnBase Members

The following tables list the members exposed by IRJointPinnedColumnBase.

Public Fields

	Name	Description
	Anchor (see page 1973)	Anchorage Available since version 1.7.
	Base (see page 1973)	Base Available since version 1.7.
	BasePlateMaterial (see page 1973)	
	Bearing (see page 1974)	Bearing Available since version 1.7.
	BearingPlate (see page 1974)	Bearing (see page 1974) plate Available since version 1.7.
	Bolts (see page 1974)	Group of connecting bolts Available since version 1.7.
	FootPlate (see page 1974)	Plate Available since version 1.7.
	Materials (see page 1975)	Available since version 1.7.
	NodeNumber (see page 1975)	
	Profile (see page 1975)	Column section Available since version 1.7.
	Square (see page 1975)	Available since version 1.7.
	StiffHoriz (see page 1976)	Available since version 1.7.
	StiffType (see page 1976)	Available since version 1.7.
	StiffVert (see page 1976)	Available since version 1.7.
	Type (see page 2160)	A connection type.
	Washer (see page 1976)	Washer Available since version 1.7.
	Wedge (see page 1977)	Wedge Available since version 1.7.
	Welds (see page 1977)	Welds in a connection Available since version 1.7.
	WFRelPos (see page 2161)	A type of relative position of a beam and a column.

Public Methods

	Name	Description
	GetFromRobot (see page 2161)	The function gets connection parameters connection parameters from Robot and fills an object with the data.
	SetToRobot (see page 2161)	The function sets and saves connection parameter in Robot.

II.1.2.2 IRJointPinnedColumnBase Fields

The fields of the IRJointPinnedColumnBase class are listed here.

Public Fields

	Name	Description
	Anchor (see page 1973)	Anchorage Available since version 1.7.
	Base (see page 1973)	Base Available since version 1.7.
	BasePlateMaterial (see page 1973)	
	Bearing (see page 1974)	Bearing Available since version 1.7.
	BearingPlate (see page 1974)	Bearing (see page 1974) plate Available since version 1.7.
	Bolts (see page 1974)	Group of connecting bolts Available since version 1.7.
	FootPlate (see page 1974)	Plate Available since version 1.7.
	Materials (see page 1975)	Available since version 1.7.
	NodeNumber (see page 1975)	

(c) 2014 Autodesk Inc. All rights reserved

	Profile (see page 1975)	Column section Available since version 1.7.
	Square (see page 1975)	Available since version 1.7.
	StiffHoriz (see page 1976)	Available since version 1.7.
	StiffType (see page 1976)	Available since version 1.7.
	StiffVert (see page 1976)	Available since version 1.7.
	Washer (see page 1976)	Washer Available since version 1.7.
	Wedge (see page 1977)	Wedge Available since version 1.7.
	Welds (see page 1977)	Welds in a connection Available since version 1.7.

II.1.2.2.1 Anchor

C++

```
HRESULT get_Anchor(IRJointAnchor**);
```

C#

```
public IRJointAnchor Anchor { get; }
```

Visual Basic

```
Public ReadOnly Anchor As IRJointAnchor
```

Description

Anchorage Available since version 1.7.

II.1.2.2.2 Base

C++

```
HRESULT get_Base(IRJointPlate**);
```

C#

```
public IRJointPlate Base { get; }
```

Visual Basic

```
Public ReadOnly Base As IRJointPlate
```

Description

Base Available since version 1.7.

II.1.2.2.3 BasePlateMaterial

C++

```
HRESULT get_BasePlateMaterial(IRobotMaterialData**);
```

C#

```
public IRobotMaterialData BasePlateMaterial { get; }
```

Visual Basic

```
Public ReadOnly BasePlateMaterial As IRobotMaterialData
```

Version

Available since version 4.5.

II.1.2.2.4 Bearing

C++

```
HRESULT get_Bearing(IRJointPlate**);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRJointPlate Bearing { get; }
```

Visual Basic

```
Public ReadOnly Bearing As IRJointPlate
```

Description

Bearing Available since version 1.7.

II.1.2.2.5 BearingPlate

C++

```
HRESULT get_BearingPlate(IRJointBearingPlate**);
```

C#

```
public IRJointBearingPlate BearingPlate { get; }
```

Visual Basic

```
Public ReadOnly BearingPlate As IRJointBearingPlate
```

Description

Bearing (see page 1974) plate Available since version 1.7.

II.1.2.2.6 Bolts

C++

```
HRESULT get_Bolts(IRJointFootBolts**);
```

C#

```
public IRJointFootBolts Bolts { get; }
```

Visual Basic

```
Public ReadOnly Bolts As IRJointFootBolts
```

Description

Group of connecting bolts Available since version 1.7.

II.1.2.2.7 FootPlate

C++

```
HRESULT get_FootPlate(IRJointFootPlate**);
```

C#

```
public IRJointFootPlate FootPlate { get; }
```

Visual Basic

```
Public ReadOnly FootPlate As IRJointFootPlate
```

Description

Plate Available since version 1.7.

II.1.2.2.8 Materials

C++

```
HRESULT get_Materials(IRJointFootMaterials**);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRJointFootMaterials Materials { get; }
```

Visual Basic

```
Public ReadOnly Materials As IRJointFootMaterials
```

Description

Available since version 1.7.

II.1.2.2.9 NodeNumber

C++

```
HRESULT get_NodeNumber(long*);
HRESULT put_NodeNumber(long);
```

C#

```
public long NodeNumber { get; set; }
```

Visual Basic

```
Public NodeNumber As long
```

Version

Available since version 4.5.

II.1.2.2.10 Profile

C++

```
HRESULT get_Profile(IRJointProfile**);
```

C#

```
public IRJointProfile Profile { get; }
```

Visual Basic

```
Public ReadOnly Profile As IRJointProfile
```

Description

Column section Available since version 1.7.

II.1.2.2.11 Square

C++

```
HRESULT get_Square(IRJointColumnSquare**);
```

C#

```
public IRJointColumnSquare Square { get; }
```

Visual Basic

```
Public ReadOnly Square As IRJointColumnSquare
```

Description

Available since version 1.7.

II.1.2.2.12 StiffHoriz

C++

```
HRESULT get_StiffHoriz(IRJointFootStiffenerHoriz**);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRJointFootStiffenerHoriz StiffHoriz { get; }
```

Visual Basic

```
Public ReadOnly StiffHoriz As IRJointFootStiffenerHoriz
```

Description

Available since version 1.7.

II.1.2.2.13 StiffType

C++

```
HRESULT get_StiffType(IRJointPinnedColumnBaseStiffType*);
HRESULT put_StiffType(IRJointPinnedColumnBaseStiffType);
```

C#

```
public IRJointPinnedColumnBaseStiffType StiffType { get; set; }
```

Visual Basic

```
Public StiffType As IRJointPinnedColumnBaseStiffType
```

Description

Available since version 1.7.

II.1.2.2.14 StiffVert

C++

```
HRESULT get_StiffVert(IRJointFootStiffenerVert**);
```

C#

```
public IRJointFootStiffenerVert StiffVert { get; }
```

Visual Basic

```
Public ReadOnly StiffVert As IRJointFootStiffenerVert
```

Description

Available since version 1.7.

II.1.2.2.15 Washer

C++

```
HRESULT get_Washer(IRJointPlate**);
```

C#

```
public IRJointPlate Washer { get; }
```

Visual Basic

```
Public ReadOnly Washer As IRJointPlate
```

Description

Washer Available since version 1.7.

II.1.2.2.16 Wedge

C++

```
HRESULT get_Wedge(IRJointWedge**);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRJointWedge Wedge { get; }
```

Visual Basic

```
Public ReadOnly Wedge As IRJointWedge
```

Description

Wedge Available since version 1.7.

II.1.2.2.17 Welds

C++

```
HRESULT get_Welds(IRJointFootWelds**);
```

C#

```
public IRJointFootWelds Welds { get; }
```

Visual Basic

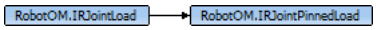
```
Public ReadOnly Welds As IRJointFootWelds
```

Description

Welds in a connection Available since version 1.7.

II.1.3 IRJointPinnedLoad

Class Hierarchy



C++

```
interface IRJointPinnedLoad : IRJointLoad;
```

C#

```
public interface IRJointPinnedLoad : IRJointLoad;
```

Visual Basic

```
Public Interface IRJointPinnedLoad
```

II.1.3.1 IRJointPinnedLoad Members

The following tables list the members exposed by IRJointPinnedLoad.

Public Fields

	Name	Description
	Cases (see page 2178)	Selection of load cases, defined for the structure. .
	Nc (see page 1978)	Available since version 1.7.
	Nt (see page 1978)	Available since version 1.7.
	NTy (see page 1978)	Available since version 1.7.
	NTz (see page 1979)	Available since version 1.7.
	Ty (see page 1979)	Available since version 1.7.
	Type (see page 2179)	Load type (defined manually or taken from Robot).
	Tz (see page 1979)	Available since version 1.7.

II.1.3.2 IRJointPinnedLoad Fields

The fields of the IRJointPinnedLoad class are listed here.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	Nc (see page 1978)	Available since version 1.7.
	Nt (see page 1978)	Available since version 1.7.
	NTy (see page 1978)	Available since version 1.7.
	NTz (see page 1979)	Available since version 1.7.
	Ty (see page 1979)	Available since version 1.7.
	Tz (see page 1979)	Available since version 1.7.

II.1.3.2.1 Nc

C++

```
HRESULT get_Nc(double*);
HRESULT put_Nc(double);
```

C#

```
public double Nc { get; set; }
```

Visual Basic

```
Public Nc As double
```

Description

Available since version 1.7.

II.1.3.2.2 Nt

C++

```
HRESULT get_Nt(double*);
HRESULT put_Nt(double);
```

C#

```
public double Nt { get; set; }
```

Visual Basic

```
Public Nt As double
```

Description

Available since version 1.7.

II.1.3.2.3 NTy

C++

```
HRESULT get_NTy(double*);
HRESULT put_NTy(double);
```

C#

```
public double NTy { get; set; }
```

Visual Basic

```
Public NTy As double
```

Description

Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

II.1.3.2.4 NTz

C++

```
HRESULT get_NTz(double*);
HRESULT put_NTz(double);
```

C#

```
public double NTz { get; set; }
```

Visual Basic

```
Public NTz As double
```

Description

Available since version 1.7.

II.1.3.2.5 Ty

C++

```
HRESULT get_Ty(double*);
HRESULT put_Ty(double);
```

C#

```
public double Ty { get; set; }
```

Visual Basic

```
Public Ty As double
```

Description

Available since version 1.7.

II.1.3.2.6 Tz

C++

```
HRESULT get_Tz(double*);
HRESULT put_Tz(double);
```

C#

```
public double Tz { get; set; }
```

Visual Basic

```
Public Tz As double
```

Description

Available since version 1.7.

II.2 Fixed column base

Enumerations

	Name	Description
	IRJointFixedColumnBaseStiffType (see page 1988)	

Interfaces

	Name	Description
	IRJointFixedLoad (see page 1980)	

(c) 2014 Autodesk Inc. All rights reserved

	IRJointFixedColumnBase (see page 1982)	
	IRJointFixedFootWelds (see page 1987)	

II.2.1 IRJointFixedLoad

Class Hierarchy

C++

```
interface IRJointFixedLoad : IRJointLoad;
```

C#

```
public interface IRJointFixedLoad : IRJointLoad;
```

Visual Basic

```
Public Interface IRJointFixedLoad
```

II.2.1.1 IRJointFixedLoad Members

The following tables list the members exposed by IRJointFixedLoad.

Public Fields

	Name	Description
	Cases (see page 2178)	Selection of load cases, defined for the structure. .
	Fx (see page 1980)	Available since version 1.7.
	Fy (see page 1981)	Available since version 1.7.
	Fz (see page 1981)	Available since version 1.7.
	My (see page 1981)	Available since version 1.7.
	Mz (see page 1981)	
	Type (see page 2179)	Load type (defined manually or taken from Robot).

II.2.1.2 IRJointFixedLoad Fields

The fields of the IRJointFixedLoad class are listed here.

Public Fields

	Name	Description
	Fx (see page 1980)	Available since version 1.7.
	Fy (see page 1981)	Available since version 1.7.
	Fz (see page 1981)	Available since version 1.7.
	My (see page 1981)	Available since version 1.7.
	Mz (see page 1981)	

II.2.1.2.1 Fx

C++

```
HRESULT get_Fx(double*);
HRESULT put_Fx(double);
```

C#

```
public double Fx { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Fx As double
```

Description

Available since version 1.7.

II.2.1.2.2 Fy

C++

```
HRESULT get_Fy(double*);
HRESULT put_Fy(double);
```

C#

```
public double Fy { get; set; }
```

Visual Basic

```
Public Fy As double
```

Description

Available since version 1.7.

II.2.1.2.3 Fz

C++

```
HRESULT get_Fz(double*);
HRESULT put_Fz(double);
```

C#

```
public double Fz { get; set; }
```

Visual Basic

```
Public Fz As double
```

Description

Available since version 1.7.

II.2.1.2.4 My

C++

```
HRESULT get_My(double*);
HRESULT put_My(double);
```

C#

```
public double My { get; set; }
```

Visual Basic

```
Public My As double
```

Description

Available since version 1.7.

II.2.1.2.5 Mz

C++

```
HRESULT get_Mz( double*);
HRESULT put_Mz( double);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public  double Mz { get; set; }
```

Visual Basic

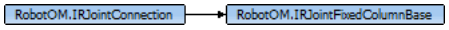
```
Public Mz As  double
```

Version

Available since version 7.5.

II.2.2 IRJointFixedColumnBase

Class Hierarchy



C++

```
interface IRJointFixedColumnBase : IRJointConnection;
```

C#

```
public interface IRJointFixedColumnBase : IRJointConnection;
```

Visual Basic

```
Public Interface IRJointFixedColumnBase
```

II.2.2.1 IRJointFixedColumnBase Members

The following tables list the members exposed by IRJointFixedColumnBase.

Public Fields

	Name	Description
	Anchor (see page 1983)	Available since version 1.7.
	Base (see page 1983)	Available since version 1.7.
	BasePlateMaterial (see page 1984)	
	Bolts (see page 1984)	Available since version 1.7.
	ComplexStiff (see page 1984)	Available since version 1.7.
	FootPlate (see page 1984)	Available since version 1.7.
	Materials (see page 1985)	Available since version 1.7.
	NodeNumber (see page 1985)	
	Profile (see page 1985)	Available since version 1.7.
	SimpleStiff (see page 1985)	Available since version 1.7.
	StiffType (see page 1986)	Available since version 1.7.
	Type (see page 2160)	A connection type.
	Washer (see page 1986)	Available since version 1.7.
	Wedge (see page 1986)	Available since version 1.7.
	Welds (see page 1986)	Available since version 1.7.
	WFRelPos (see page 2161)	A type of relative position of a beam and a column.

Public Methods

	Name	Description
	GetFromRobot (see page 2161)	The function gets connection parameters connection parameters from Robot and fills an object with the data.
	SetToRobot (see page 2161)	The function sets and saves connection parameter in Robot.

(c) 2014 Autodesk Inc. All rights reserved

II.2.2.2 IRJointFixedColumnBase Fields

The fields of the IRJointFixedColumnBase class are listed here.

Public Fields

	Name	Description
	Anchor (see page 1983)	Available since version 1.7.
	Base (see page 1983)	Available since version 1.7.
	BasePlateMaterial (see page 1984)	
	Bolts (see page 1984)	Available since version 1.7.
	ComplexStiff (see page 1984)	Available since version 1.7.
	FootPlate (see page 1984)	Available since version 1.7.
	Materials (see page 1985)	Available since version 1.7.
	NodeNumber (see page 1985)	
	Profile (see page 1985)	Available since version 1.7.
	SimpleStiff (see page 1985)	Available since version 1.7.
	StiffType (see page 1986)	Available since version 1.7.
	Washer (see page 1986)	Available since version 1.7.
	Wedge (see page 1986)	Available since version 1.7.
	Welds (see page 1986)	Available since version 1.7.

II.2.2.2.1 Anchor

C++

```
HRESULT get_Anchor(IRJointAnchor**);
```

C#

```
public IRJointAnchor Anchor { get; }
```

Visual Basic

```
Public ReadOnly Anchor As IRJointAnchor
```

Description

Available since version 1.7.

II.2.2.2.2 Base

C++

```
HRESULT get_Base(IRJointPlate**);
```

C#

```
public IRJointPlate Base { get; }
```

Visual Basic

```
Public ReadOnly Base As IRJointPlate
```

Description

Available since version 1.7.

II.2.2.2.3 BasePlateMaterial

C++

```
HRESULT get_BasePlateMaterial(IRobotMaterialData**);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRobotMaterialData BasePlateMaterial { get; }
```

Visual Basic

```
Public ReadOnly BasePlateMaterial As IRobotMaterialData
```

Version

Available since version 4.5.

II.2.2.2.4 Bolts

C++

```
HRESULT get_Bolts(IRJointFootBolts**);
```

C#

```
public IRJointFootBolts Bolts { get; }
```

Visual Basic

```
Public ReadOnly Bolts As IRJointFootBolts
```

Description

Available since version 1.7.

II.2.2.2.5 ComplexStiff

C++

```
HRESULT get_ComplexStiff(IRJointFootStiffenerComplex**);
```

C#

```
public IRJointFootStiffenerComplex ComplexStiff { get; }
```

Visual Basic

```
Public ReadOnly ComplexStiff As IRJointFootStiffenerComplex
```

Description

Available since version 1.7.

II.2.2.2.6 FootPlate

C++

```
HRESULT get_FootPlate(IRJointFootPlate**);
```

C#

```
public IRJointFootPlate FootPlate { get; }
```

Visual Basic

```
Public ReadOnly FootPlate As IRJointFootPlate
```

Description

Available since version 1.7.

II.2.2.2.7 Materials

C++

```
HRESULT get_Materials(IRJointFootMaterials**);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRJointFootMaterials Materials { get; }
```

Visual Basic

```
Public ReadOnly Materials As IRJointFootMaterials
```

Description

Available since version 1.7.

II.2.2.2.8 NodeNumber

C++

```
HRESULT get_NodeNumber(long*);
HRESULT put_NodeNumber(long);
```

C#

```
public long NodeNumber { get; set; }
```

Visual Basic

```
Public NodeNumber As long
```

Version

Available since version 4.5.

II.2.2.2.9 Profile

C++

```
HRESULT get_Profile(IRJointProfile**);
```

C#

```
public IRJointProfile Profile { get; }
```

Visual Basic

```
Public ReadOnly Profile As IRJointProfile
```

Description

Available since version 1.7.

II.2.2.2.10 SimpleStiff

C++

```
HRESULT get_SimpleStiff(IRJointFootStiffenerSimple**);
```

C#

```
public IRJointFootStiffenerSimple SimpleStiff { get; }
```

Visual Basic

```
Public ReadOnly SimpleStiff As IRJointFootStiffenerSimple
```

Description

Available since version 1.7.

II.2.2.2.11 StiffType

C++

```
HRESULT get_StiffType(IRJointFixedColumnBaseStiffType*);
HRESULT put_StiffType(IRJointFixedColumnBaseStiffType);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRJointFixedColumnBaseStiffType StiffType { get; set; }
```

Visual Basic

```
Public StiffType As IRJointFixedColumnBaseStiffType
```

Description

Available since version 1.7.

II.2.2.2.12 Washer

C++

```
HRESULT get_Washer(IRJointPlate**);
```

C#

```
public IRJointPlate Washer { get; }
```

Visual Basic

```
Public ReadOnly Washer As IRJointPlate
```

Description

Available since version 1.7.

II.2.2.2.13 Wedge

C++

```
HRESULT get_Wedge(IRJointWedge**);
```

C#

```
public IRJointWedge Wedge { get; }
```

Visual Basic

```
Public ReadOnly Wedge As IRJointWedge
```

Description

Available since version 1.7.

II.2.2.2.14 Welds

C++

```
HRESULT get_Welds(IRJointFixedFootWelds**);
```

C#

```
public IRJointFixedFootWelds Welds { get; }
```

Visual Basic

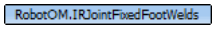
```
Public ReadOnly Welds As IRJointFixedFootWelds
```

Description

Available since version 1.7.

II.2.3 IRJointFixedFootWelds

Class Hierarchy



(c) 2014 Autodesk Inc. All rights reserved

C++

```
interface IRJointFixedFootWelds : IDispatch;
```

C#

```
public interface IRJointFixedFootWelds;
```

Visual Basic

```
Public Interface IRJointFixedFootWelds
```

II.2.3.1 IRJointFixedFootWelds Members

The following tables list the members exposed by IRJointFixedFootWelds.

Public Fields

	Name	Description
	FootPlate (see page 1987)	Available since version 1.7.
	Stiff (see page 1988)	Available since version 1.7.
	Washer (see page 1988)	Available since version 1.7.
	Wedge (see page 1988)	Available since version 1.7.

II.2.3.2 IRJointFixedFootWelds Fields

The fields of the IRJointFixedFootWelds class are listed here.

Public Fields

	Name	Description
	FootPlate (see page 1987)	Available since version 1.7.
	Stiff (see page 1988)	Available since version 1.7.
	Washer (see page 1988)	Available since version 1.7.
	Wedge (see page 1988)	Available since version 1.7.

II.2.3.2.1 FootPlate

C++

```
HRESULT get_FootPlate(IRJointWeld**);
```

C#

```
public IRJointWeld FootPlate { get; }
```

Visual Basic

```
Public ReadOnly FootPlate As IRJointWeld
```

Description

Available since version 1.7.

II.2.3.2.2 Stiff

C++

```
HRESULT get_Stiff(IRJointWeld**);
```

C#

```
public IRJointWeld Stiff { get; }
```

Visual Basic

```
Public ReadOnly Stiff As IRJointWeld
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Available since version 1.7.

II.2.3.2.3 Washer

C++

```
HRESULT get_Washer(IRJointWeld**);
```

C#

```
public IRJointWeld Washer { get; }
```

Visual Basic

```
Public ReadOnly Washer As IRJointWeld
```

Description

Available since version 1.7.

II.2.3.2.4 Wedge

C++

```
HRESULT get_Wedge(IRJointWeld**);
```

C#

```
public IRJointWeld Wedge { get; }
```

Visual Basic

```
Public ReadOnly Wedge As IRJointWeld
```

Description

Available since version 1.7.

II.2.4 IRJointFixedColumnBaseStiffType

C++

```
enum IRJointFixedColumnBaseStiffType;
```

C#

```
public enum IRJointFixedColumnBaseStiffType;
```

Visual Basic

```
Public Enum IRJointFixedColumnBaseStiffType
```

Members

Members	Description
I_JFCBST_NONE = 0	Available since version 1.7.
I_JFCBST_SIMPLE = 1	Available since version 1.7.
I_JFCBST_COMPLEX = 2	Available since version 1.7.

II.3 Concrete column base

Enumerations

	Name	Description
	IRJointSpreadFootingType (see page 1993)	

(c) 2014 Autodesk Inc. All rights reserved

	IRJointColumnBasePlateCalcModel (see page 1998)	

Interfaces

	Name	Description
	IRJointConcreteMaterials (see page 1989)	
	IRJointConcreteColumn (see page 1990)	
	IRJointConcreteColumnFoundation (see page 1994)	
	IRJointConcreteColumnLoad (see page 1998)	

II.3.1 IRJointConcreteMaterials

Class Hierarchy

RobotOM.IRJointConcreteMaterials

C++

```
interface IRJointConcreteMaterials : IDispatch;
```

C#

```
public interface IRJointConcreteMaterials;
```

Visual Basic

```
Public Interface IRJointConcreteMaterials
```

II.3.1.1 IRJointConcreteMaterials Members

The following tables list the members exposed by IRJointConcreteMaterials.

Public Fields

	Name	Description
	CoeffBA (see page 1990)	Available since version 1.7.
	Dosage (see page 1990)	Available since version 1.7.
	Sigma (see page 1990)	Available since version 1.7.

II.3.1.2 IRJointConcreteMaterials Fields

The fields of the IRJointConcreteMaterials class are listed here.

Public Fields

	Name	Description
	CoeffBA (see page 1990)	Available since version 1.7.
	Dosage (see page 1990)	Available since version 1.7.
	Sigma (see page 1990)	Available since version 1.7.

II.3.1.2.1 CoeffBA

C++

```
HRESULT get_CoeffBA(double*);
HRESULT put_CoeffBA(double);
```

C#

```
public double CoeffBA { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public CoeffBA As double
```

Description

Available since version 1.7.

II.3.1.2.2 Dosage

C++

```
HRESULT get_Dosage(double*);
HRESULT put_Dosage(double);
```

C#

```
public double Dosage { get; set; }
```

Visual Basic

```
Public Dosage As double
```

Description

Available since version 1.7.

II.3.1.2.3 Sigma

C++

```
HRESULT get_Sigma(double*);
HRESULT put_Sigma(double);
```

C#

```
public double Sigma { get; set; }
```

Visual Basic

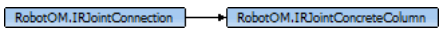
```
Public Sigma As double
```

Description

Available since version 1.7.

II.3.2 IRJointConcreteColumn

Class Hierarchy



C++

```
interface IRJointConcreteColumn : IRJointConnection;
```

C#

```
public interface IRJointConcreteColumn : IRJointConnection;
```

Visual Basic

```
Public Interface IRJointConcreteColumn
```

II.3.2.1 IRJointConcreteColumn Members

The following tables list the members exposed by IRJointConcreteColumn.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	Base (see page 1992)	
	CalcModel (see page 1992)	
	Depth (see page 1992)	
	Foundation (see page 1992)	
	Materials (see page 1993)	
	Profile (see page 1993)	
	SpreadFootingType (see page 1993)	
	Type (see page 2160)	A connection type.
	WFRelPos (see page 2161)	A type of relative position of a beam and a column.

Public Methods

	Name	Description
	GetFromRobot (see page 2161)	The function gets connection parameters connection parameters from Robot and fills an object with the data.
	SetToRobot (see page 2161)	The function sets and saves connection parameter in Robot.

II.3.2.2 IRJointConcreteColumn Fields

The fields of the IRJointConcreteColumn class are listed here.

Public Fields

	Name	Description
	Base (see page 1992)	
	CalcModel (see page 1992)	
	Depth (see page 1992)	
	Foundation (see page 1992)	
	Materials (see page 1993)	
	Profile (see page 1993)	
	SpreadFootingType (see page 1993)	

II.3.2.2.1 Base

C++

```
HRESULT get_Base(IRJointPlate**);
```

C#

```
public IRJointPlate Base { get; }
```

Visual Basic

```
Public ReadOnly Base As IRJointPlate
```

Version

Available since version 7.5.

II.3.2.2.2 CalcModel

C++

```
HRESULT get_CalcModel(IRJointColumnBasePlateCalcModel*);
HRESULT put_CalcModel(IRJointColumnBasePlateCalcModel);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRJointColumnBasePlateCalcModel CalcModel { get; set; }
```

Visual Basic

```
Public CalcModel As IRJointColumnBasePlateCalcModel
```

Version

Available since version 7.5.

II.3.2.2.3 Depth

C++

```
HRESULT get_Depth(double*);
HRESULT put_Depth(double);
```

C#

```
public double Depth { get; set; }
```

Visual Basic

```
Public Depth As double
```

II.3.2.2.4 Foundation

C++

```
HRESULT get_Foundation(IRJointConcreteColumnFoundation**);
```

C#

```
public IRJointConcreteColumnFoundation Foundation { get; }
```

Visual Basic

```
Public ReadOnly Foundation As IRJointConcreteColumnFoundation
```

Version

Available since version 7.5.

II.3.2.2.5 Materials

C++

```
HRESULT get_Materials(IRJointConcreteMaterials**);
```

C#

```
public IRJointConcreteMaterials Materials { get; }
```

Visual Basic

```
Public ReadOnly Materials As IRJointConcreteMaterials
```

Version

Available since version 7.5.

II.3.2.2.6 Profile

C++

```
HRESULT get_Profile(IRJointProfile**);
```

C#

```
public IRJointProfile Profile { get; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public ReadOnly Profile As IRJointProfile
```

II.3.2.2.7 SpreadFootingType

C++

```
HRESULT get_SpreadFootingType(IRJointSpreadFootingType*);
HRESULT put_SpreadFootingType(IRJointSpreadFootingType);
```

C#

```
public IRJointSpreadFootingType SpreadFootingType { get; set; }
```

Visual Basic

```
Public SpreadFootingType As IRJointSpreadFootingType
```

II.3.3 IRJointSpreadFootingType

C++

```
enum IRJointSpreadFootingType;
```

C#

```
public enum IRJointSpreadFootingType;
```

Visual Basic

```
Public Enum IRJointSpreadFootingType
```

Members

Members	Description
I_JSFT_RECT_CROSS_SECT = 1	Available since version 7.5.
I_JSFT_ABRUPT_CHANGES_HEIGHT = 2	Available since version 7.5.
I_JSFT_TAPERED = 3	Available since version 7.5.

Version

Available since version 7.5.

II.3.4 IRJointConcreteColumnFoundation

Class Hierarchy

RobotOM.IRJointConcreteColumnFoundation

C++

```
interface IRJointConcreteColumnFoundation : IDispatch;
```

C#

```
public interface IRJointConcreteColumnFoundation;
```

Visual Basic

```
Public Interface IRJointConcreteColumnFoundation
```

Version

Available since version 7.5.

II.3.4.1 IRJointConcreteColumnFoundation Members

The following tables list the members exposed by IRJointConcreteColumnFoundation.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	par_a1 (see page 1995)	
	par_b (see page 1995)	
	par_b1 (see page 1995)	
	par_b2 (see page 1995)	
	par_h (see page 1996)	
	par_h1 (see page 1996)	
	par_h2 (see page 1996)	
	par_l (see page 1996)	
	par_l1 (see page 1997)	
	par_l2 (see page 1997)	
	par_o1 (see page 1997)	
	par_o2 (see page 1997)	

II.3.4.2 IRJointConcreteColumnFoundation Fields

The fields of the IRJointConcreteColumnFoundation class are listed here.

Public Fields

	Name	Description
	par_a1 (see page 1995)	
	par_b (see page 1995)	
	par_b1 (see page 1995)	
	par_b2 (see page 1995)	
	par_h (see page 1996)	
	par_h1 (see page 1996)	
	par_h2 (see page 1996)	
	par_l (see page 1996)	
	par_l1 (see page 1997)	
	par_l2 (see page 1997)	
	par_o1 (see page 1997)	
	par_o2 (see page 1997)	

II.3.4.2.1 par_a1

C++

```
HRESULT get_par_a1(double*);
HRESULT put_par_a1(double);
```

C#

```
public double par_a1 { get; set; }
```

Visual Basic

```
Public par_a1 As double
```

Version

Available since version 7.5.

II.3.4.2.2 par_b

C++

```
HRESULT get_par_b(double*);
```

(c) 2014 Autodesk Inc. All rights reserved

```
HRESULT put_par_b(double);
```

C#

```
public double par_b { get; set; }
```

Visual Basic

```
Public par_b As double
```

Version

Available since version 7.5.

II.3.4.2.3 par_b1

C++

```
HRESULT get_par_b1(double*);
HRESULT put_par_b1(double);
```

C#

```
public double par_b1 { get; set; }
```

Visual Basic

```
Public par_b1 As double
```

Version

Available since version 7.5.

II.3.4.2.4 par_b2

C++

```
HRESULT get_par_b2(double*);
HRESULT put_par_b2(double);
```

C#

```
public double par_b2 { get; set; }
```

Visual Basic

```
Public par_b2 As double
```

Version

Available since version 7.5.

II.3.4.2.5 par_h

C++

```
HRESULT get_par_h(double*);
HRESULT put_par_h(double);
```

C#

```
public double par_h { get; set; }
```

Visual Basic

```
Public par_h As double
```

Version

Available since version 7.5.

(c) 2014 Autodesk Inc. All rights reserved

II.3.4.2.6 par_h1

C++

```
HRESULT get_par_h1(double*);
HRESULT put_par_h1(double);
```

C#

```
public double par_h1 { get; set; }
```

Visual Basic

```
Public par_h1 As double
```

Version

Available since version 7.5.

II.3.4.2.7 par_h2

C++

```
HRESULT get_par_h2(double*);
HRESULT put_par_h2(double);
```

C#

```
public double par_h2 { get; set; }
```

Visual Basic

```
Public par_h2 As double
```

Version

Available since version 7.5.

II.3.4.2.8 par_l

C++

```
HRESULT get_par_l(double*);
HRESULT put_par_l(double);
```

C#

```
public double par_l { get; set; }
```

Visual Basic

```
Public par_l As double
```

Version

Available since version 7.5.

II.3.4.2.9 par_l1

C++

```
HRESULT get_par_l1(double*);
HRESULT put_par_l1(double);
```

C#

```
public double par_l1 { get; set; }
```

Visual Basic

```
Public par_l1 As double
```

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 7.5.

II.3.4.2.10 par_l2

C++

```
HRESULT get_par_l2(double*);
HRESULT put_par_l2(double);
```

C#

```
public double par_l2 { get; set; }
```

Visual Basic

```
Public par_l2 As double
```

Version

Available since version 7.5.

II.3.4.2.11 par_o1

C++

```
HRESULT get_par_o1(double*);
HRESULT put_par_o1(double);
```

C#

```
public double par_o1 { get; set; }
```

Visual Basic

```
Public par_o1 As double
```

Version

Available since version 7.5.

II.3.4.2.12 par_o2

C++

```
HRESULT get_par_o2(double*);
HRESULT put_par_o2(double);
```

C#

```
public double par_o2 { get; set; }
```

Visual Basic

```
Public par_o2 As double
```

Version

Available since version 7.5.

II.3.5 IRJointColumnBasePlateCalcModel

C++

```
enum IRJointColumnBasePlateCalcModel;
```

C#

```
public enum IRJointColumnBasePlateCalcModel;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Enum IRJointColumnBasePlateCalcModel
```

Members

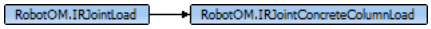
Members	Description
I_JCBPCM_PLASTIC = 0	Available since version 7.5.
I_JCBPCM_ELASTIC = 1	Available since version 7.5.

Version

Available since version 7.5.

II.3.6 IRJointConcreteColumnLoad

Class Hierarchy



C++

```
interface IRJointConcreteColumnLoad : IRJointLoad;
```

C#

```
public interface IRJointConcreteColumnLoad : IRJointLoad;
```

Visual Basic

```
Public Interface IRJointConcreteColumnLoad
```

Version

Available since version 7.5.

II.3.6.1 IRJointConcreteColumnLoad Members

The following tables list the members exposed by IRJointConcreteColumnLoad.

Public Fields

	Name	Description
	Cases (see page 2178)	Selection of load cases, defined for the structure. .
	Fx (see page 1999)	
	FxAssemb (see page 1999)	
	Fy (see page 2000)	
	Fz (see page 2000)	
	My (see page 2000)	
	Mz (see page 2000)	
	Type (see page 2179)	Load type (defined manually or taken from Robot).

II.3.6.2 IRJointConcreteColumnLoad Fields

The fields of the IRJointConcreteColumnLoad class are listed here.

Public Fields

	Name	Description
	Fx (see page 1999)	
	FxAssemb (see page 1999)	
	Fy (see page 2000)	
	Fz (see page 2000)	

(c) 2014 Autodesk Inc. All rights reserved

	My (see page 2000)	
	Mz (see page 2000)	

II.3.6.2.1 Fx

C++

```
HRESULT get_Fx(double*);
HRESULT put_Fx(double);
```

C#

```
public double Fx { get; set; }
```

Visual Basic

```
Public Fx As double
```

Version

Available since version 7.5.

II.3.6.2.2 FxAssemb

C++

```
HRESULT get_FxAssemb(double*);
HRESULT put_FxAssemb(double);
```

C#

```
public double FxAssemb { get; set; }
```

Visual Basic

```
Public FxAssemb As double
```

Version

Available since version 7.5.

II.3.6.2.3 Fy

C++

```
HRESULT get_Fy(double*);
HRESULT put_Fy(double);
```

C#

```
public double Fy { get; set; }
```

Visual Basic

```
Public Fy As double
```

Version

Available since version 7.5.

II.3.6.2.4 Fz

C++

```
HRESULT get_Fz(double*);
HRESULT put_Fz(double);
```

C#

```
public double Fz { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Fz As double
```

Version

Available since version 7.5.

II.3.6.2.5 My

C++

```
HRESULT get_My(double*);
HRESULT put_My(double);
```

C#

```
public double My { get; set; }
```

Visual Basic

```
Public My As double
```

Version

Available since version 7.5.

II.3.6.2.6 Mz

C++

```
HRESULT get_Mz(double*);
HRESULT put_Mz(double);
```

C#

```
public double Mz { get; set; }
```

Visual Basic

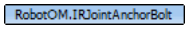
```
Public Mz As double
```

Version

Available since version 7.5.

II.4 IRJointAnchorBolt

Class Hierarchy



C++

```
interface IRJointAnchorBolt : IDispatch;
```

C#

```
public interface IRJointAnchorBolt;
```

Visual Basic

```
Public Interface IRJointAnchorBolt
```

Description

Anchorage bolt.

(c) 2014 Autodesk Inc. All rights reserved

II.4.1 IRJointAnchorBolt Members

The following tables list the members exposed by IRJointAnchorBolt.

Public Fields

	Name	Description
	Length1 (see page 2001)	Available since version 1.7.
	Length2 (see page 2002)	Available since version 1.7.
	Length3 (see page 2002)	Available since version 1.7.
	Length4 (see page 2002)	Available since version 1.7.

II.4.2 IRJointAnchorBolt Fields

The fields of the IRJointAnchorBolt class are listed here.

Public Fields

	Name	Description
	Length1 (see page 2001)	Available since version 1.7.
	Length2 (see page 2002)	Available since version 1.7.
	Length3 (see page 2002)	Available since version 1.7.
	Length4 (see page 2002)	Available since version 1.7.

II.4.2.1 Length1

C++

```
HRESULT get_Length1(double*);
HRESULT put_Length1(double);
```

C#

```
public double Length1 { get; set; }
```

Visual Basic

```
Public Length1 As double
```

Description

Available since version 1.7.

II.4.2.2 Length2

C++

```
HRESULT get_Length2(double*);
HRESULT put_Length2(double);
```

C#

```
public double Length2 { get; set; }
```

Visual Basic

```
Public Length2 As double
```

Description

Available since version 1.7.

II.4.2.3 Length3

C++

```
HRESULT get_Length3(double*);
```

(c) 2014 Autodesk Inc. All rights reserved

```
HRESULT put_Length3(double);
```

C#

```
public double Length3 { get; set; }
```

Visual Basic

```
Public Length3 As double
```

Description

Available since version 1.7.

II.4.2.4 Length4

C++

```
HRESULT get_Length4(double*);
HRESULT put_Length4(double);
```

C#

```
public double Length4 { get; set; }
```

Visual Basic

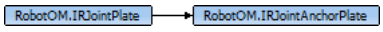
```
Public Length4 As double
```

Description

Available since version 1.7.

II.5 IRJointAnchorPlate

Class Hierarchy



C++

```
interface IRJointAnchorPlate : IRJointPlate;
```

C#

```
public interface IRJointAnchorPlate : IRJointPlate;
```

Visual Basic

```
Public Interface IRJointAnchorPlate
```

Description

Plate of an anchor bolt.

II.5.1 IRJointAnchorPlate Members

The following tables list the members exposed by IRJointAnchorPlate.

Public Fields

	Name	Description
	Exist (see page 2164)	Information about a plate existance .
	Length (see page 2164)	Plate length .
	Material (see page 2164)	Name of the component material.
	Thick (see page 2165)	Plate thickness.
	Type (see page 2003)	Shape of a plate near anchor Available since version 1.7.
	Width (see page 2165)	Plate width.

(c) 2014 Autodesk Inc. All rights reserved

Public Methods

	Name	Description
	AngleExt (see page 2165)	Angle of bracket inclination, depending on the parameter - in degrees or radians.

II.5.2 IRJointAnchorPlate Fields

The fields of the IRJointAnchorPlate class are listed here.

Public Fields

	Name	Description
	Type (see page 2003)	Shape of a plate near anchor Available since version 1.7.

II.5.2.1 Type

C++

```
HRESULT get_Type(IRJointAnchorPlateType*);
HRESULT put_Type(IRJointAnchorPlateType);
```

C#

```
public IRJointAnchorPlateType Type { get; set; }
```

Visual Basic

```
Public Type As IRJointAnchorPlateType
```

Description

Shape of a plate near anchor Available since version 1.7.

II.6 IRJointAnchorType

C++

```
enum IRJointAnchorType;
```

C#

```
public enum IRJointAnchorType;
```

Visual Basic

```
Public Enum IRJointAnchorType
```

Members

Members	Description
I_JAT_ROD = 0	Available since version 1.7.
I_JAT_SIMPLE = 1	Available since version 1.7.
I_JAT_PIN = 2	Available since version 1.7.
I_JAT_PLATE = 3	Available since version 1.7.

Description

Available anchorage types.

II.7 IRJointAnchor

Class Hierarchy

RobotOM.IRJointAnchor

(c) 2014 Autodesk Inc. All rights reserved

C++

```
interface IRJointAnchor : IDispatch;
```

C#

```
public interface IRJointAnchor;
```

Visual Basic

```
Public Interface IRJointAnchor
```

Description

Anchorage definition.

II.7.1 IRJointAnchor Members

The following tables list the members exposed by IRJointAnchor.

Public Fields

	Name	Description
	AnchorPlate (see page 2005)	Plate of an anchor bolt Available since version 1.7.
	Bolts (see page 2005)	Arrangement and type of anchor bolts Available since version 1.7.
	Tige (see page 2005)	Parameters of anchor bolt Available since version 1.7.
	Type (see page 2005)	Anchorage type Available since version 1.7.

II.7.2 IRJointAnchor Fields

The fields of the IRJointAnchor class are listed here.

Public Fields

	Name	Description
	AnchorPlate (see page 2005)	Plate of an anchor bolt Available since version 1.7.
	Bolts (see page 2005)	Arrangement and type of anchor bolts Available since version 1.7.
	Tige (see page 2005)	Parameters of anchor bolt Available since version 1.7.
	Type (see page 2005)	Anchorage type Available since version 1.7.

II.7.2.1 AnchorPlate

C++

```
HRESULT get_AnchorPlate(IRJointAnchorPlate**);
```

C#

```
public IRJointAnchorPlate AnchorPlate { get; }
```

Visual Basic

```
Public ReadOnly AnchorPlate As IRJointAnchorPlate
```

Description

Plate of an anchor bolt Available since version 1.7.

II.7.2.2 Bolts

C++

```
HRESULT get_Bolts(IRJointFootBolts**);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRJointFootBolts Bolts { get; }
```

Visual Basic

```
Public ReadOnly Bolts As IRJointFootBolts
```

Description

Arrangement and type of anchor bolts Available since version 1.7.

II.7.2.3 Tige

C++

```
HRESULT get_Tige(IRJointAnchorBolt**);
```

C#

```
public IRJointAnchorBolt Tige { get; }
```

Visual Basic

```
Public ReadOnly Tige As IRJointAnchorBolt
```

Description

Parameters of anchor bolt Available since version 1.7.

II.7.2.4 Type

C++

```
HRESULT get_Type(IRJointAnchorType*);
HRESULT put_Type(IRJointAnchorType);
```

C#

```
public IRJointAnchorType Type { get; set; }
```

Visual Basic

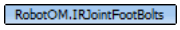
```
Public Type As IRJointAnchorType
```

Description

Anchorage type Available since version 1.7.

II.8 IRJointFootBolts

Class Hierarchy



C++

```
interface IRJointFootBolts : IDispatch;
```

C#

```
public interface IRJointFootBolts;
```

Visual Basic

```
Public Interface IRJointFootBolts
```

Description

Bolt group of identical parameters for pinned column base. .

(c) 2014 Autodesk Inc. All rights reserved

II.8.1 IRJointFootBolts Members

The following tables list the members exposed by IRJointFootBolts.

Public Fields

	Name	Description
	Area (see page 2007)	Available since version 1.7.
	ClassName (see page 2007)	Name of the bolt class Available since version 1.7.
	Diameter (see page 2007)	Bolt diameter Available since version 1.7.
	DiameterName (see page 2007)	Available since version 1.7.
	Distance (see page 2008)	Available since version 1.7.
	Friction (see page 2008)	Available since version 1.7.
	Rows (see page 2008)	Available since version 1.7.
	SpacingH (see page 2008)	Available since version 1.7.
	SpacingV (see page 2009)	Available since version 1.7.

II.8.2 IRJointFootBolts Fields

The fields of the IRJointFootBolts class are listed here.

Public Fields

	Name	Description
	Area (see page 2007)	Available since version 1.7.
	ClassName (see page 2007)	Name of the bolt class Available since version 1.7.
	Diameter (see page 2007)	Bolt diameter Available since version 1.7.
	DiameterName (see page 2007)	Available since version 1.7.
	Distance (see page 2008)	Available since version 1.7.
	Friction (see page 2008)	Available since version 1.7.
	Rows (see page 2008)	Available since version 1.7.
	SpacingH (see page 2008)	Available since version 1.7.
	SpacingV (see page 2009)	Available since version 1.7.

II.8.2.1 Area

C++

```
HRESULT get_Area(double*);
```

C#

```
public double Area { get; }
```

Visual Basic

```
Public ReadOnly Area As double
```

Description

Available since version 1.7.

II.8.2.2 ClassName

C++

```
HRESULT get_ClassName(BSTR*);
HRESULT put_ClassName(BSTR);
```

C#

```
public String ClassName { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public ClassName As String
```

Description

Name of the bolt class Available since version 1.7.

II.8.2.3 Diameter

C++

```
HRESULT get_Diameter(long*);
```

C#

```
public long Diameter { get; }
```

Visual Basic

```
Public ReadOnly Diameter As long
```

Description

Bolt diameter Available since version 1.7.

II.8.2.4 DiameterName

C++

```
HRESULT get_DiameterName(BSTR*);
HRESULT put_DiameterName(BSTR);
```

C#

```
public String DiameterName { get; set; }
```

Visual Basic

```
Public DiameterName As String
```

Description

Available since version 1.7.

II.8.2.5 Distance

C++

```
HRESULT get_Distance(double*);
HRESULT put_Distance(double);
```

C#

```
public double Distance { get; set; }
```

Visual Basic

```
Public Distance As double
```

Description

Available since version 1.7.

II.8.2.6 Friction

C++

```
HRESULT get_Friction(double*);
HRESULT put_Friction(double);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double Friction { get; set; }
```

Visual Basic

```
Public Friction As double
```

Description

Available since version 1.7.

II.8.2.7 Rows

C++

```
HRESULT get_Rows(long*);
HRESULT put_Rows(long);
```

C#

```
public long Rows { get; set; }
```

Visual Basic

```
Public Rows As long
```

Description

Available since version 1.7.

II.8.2.8 SpacingH

C++

```
HRESULT get_SpacingH(double*);
HRESULT put_SpacingH(double);
```

C#

```
public double SpacingH { get; set; }
```

Visual Basic

```
Public SpacingH As double
```

Description

Available since version 1.7.

II.8.2.9 SpacingV

C++

```
HRESULT get_SpacingV(double*);
HRESULT put_SpacingV(double);
```

C#

```
public double SpacingV { get; set; }
```

Visual Basic

```
Public SpacingV As double
```

Description

Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

II.9 IRJointFootPlateType

C++

```
enum IRJointFootPlateType;
```

C#

```
public enum IRJointFootPlateType;
```

Visual Basic

```
Public Enum IRJointFootPlateType
```

Members

Members	Description
I_JFPT_NONE = 0	Available since version 1.7.
I_JFPT_WELD = 1	Available since version 1.7.
I_JFPT_ANGLE = 2	Available since version 1.7.

Description

Available connection types - plate/angle.

II.10 IRJointFootPlate

Class Hierarchy

C++

```
interface IRJointFootPlate : IRJointPlate;
```

C#

```
public interface IRJointFootPlate : IRJointPlate;
```

Visual Basic

```
Public Interface IRJointFootPlate
```

II.10.1 IRJointFootPlate Members

The following tables list the members exposed by IRJointFootPlate.

Public Fields

	Name	Description
	Diameter (see page 2010)	Opening diameters in a plate for bolts Available since version 1.7.
	Exist (see page 2164)	Information about a plate existance .
	Length (see page 2164)	Plate length .
	Material (see page 2164)	Name of the component material.
	Thick (see page 2165)	Plate thickness.
	Type (see page 2010)	Connection type - plate/angle Available since version 1.7.
	Width (see page 2165)	Plate width.

Public Methods

	Name	Description
	AngleExt (see page 2165)	Angle of bracket inclination, depending on the parameter - in degrees or radians.

(c) 2014 Autodesk Inc. All rights reserved

II.10.2 IRJointFootPlate Fields

The fields of the IRJointFootPlate class are listed here.

Public Fields

	Name	Description
	Diameter (see page 2010)	Opening diameters in a plate for bolts Available since version 1.7.
	Type (see page 2010)	Connection type - plate/angle Available since version 1.7.

II.10.2.1 Diameter

C++

```
HRESULT get_Diameter(double*);
HRESULT put_Diameter(double);
```

C#

```
public double Diameter { get; set; }
```

Visual Basic

```
Public Diameter As double
```

Description

Opening diameters in a plate for bolts Available since version 1.7.

II.10.2.2 Type

C++

```
HRESULT get_Type(IRJointFootPlateType*);
HRESULT put_Type(IRJointFootPlateType);
```

C#

```
public IRJointFootPlateType Type { get; set; }
```

Visual Basic

```
Public Type As IRJointFootPlateType
```

Description

Connection type - plate/angle Available since version 1.7.

II.11 IRJointWedge

Class Hierarchy

RobotOM.IRJointWedge

C++

```
interface IRJointWedge : IDispatch;
```

C#

```
public interface IRJointWedge;
```

Visual Basic

```
Public Interface IRJointWedge
```

II.11.1 IRJointWedge Members

The following tables list the members exposed by IRJointWedge.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	Length (see page 2011)	Available since version 1.7.
	Profile (see page 2012)	Wedge section Available since version 1.7.
	Thick (see page 2012)	Available since version 1.7.
	Type (see page 2012)	Wedge type Available since version 1.7.
	Width (see page 2012)	Available since version 1.7.
	XTypeMaterial (see page 2013)	Available since version 1.7.

II.11.2 IRJointWedge Fields

The fields of the IRJointWedge class are listed here.

Public Fields

	Name	Description
	Length (see page 2011)	Available since version 1.7.
	Profile (see page 2012)	Wedge section Available since version 1.7.
	Thick (see page 2012)	Available since version 1.7.
	Type (see page 2012)	Wedge type Available since version 1.7.
	Width (see page 2012)	Available since version 1.7.
	XTypeMaterial (see page 2013)	Available since version 1.7.

II.11.2.1 Length

C++

```
HRESULT get_Length(double*);
HRESULT put_Length(double);
```

C#

```
public double Length { get; set; }
```

Visual Basic

```
Public Length As double
```

Description

Available since version 1.7.

II.11.2.2 Profile

C++

```
HRESULT get_Profile(IRJointProfile**);
```

C#

```
public IRJointProfile Profile { get; }
```

Visual Basic

```
Public ReadOnly Profile As IRJointProfile
```

Description

Wedge section Available since version 1.7.

II.11.2.3 Thick

C++

```
HRESULT get_Thick(double*);
```

(c) 2014 Autodesk Inc. All rights reserved

```
HRESULT put_Thick(double);
```

C#

```
public double Thick { get; set; }
```

Visual Basic

```
Public Thick As double
```

Description

Available since version 1.7.

II.11.2.4 Type

C++

```
HRESULT get_Type(IRJointWedgeType*);
HRESULT put_Type(IRJointWedgeType);
```

C#

```
public IRJointWedgeType Type { get; set; }
```

Visual Basic

```
Public Type As IRJointWedgeType
```

Description

Wedge type Available since version 1.7.

II.11.2.5 Width

C++

```
HRESULT get_Width(double*);
HRESULT put_Width(double);
```

C#

```
public double Width { get; set; }
```

Visual Basic

```
Public Width As double
```

Description

Available since version 1.7.

II.11.2.6 XTypeMaterial

C++

```
HRESULT get_XTypeMaterial(BSTR*);
HRESULT put_XTypeMaterial(BSTR);
```

C#

```
public String XTypeMaterial { get; set; }
```

Visual Basic

```
Public XTypeMaterial As String
```

Description

Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

II.12 IRJointWedgeType

C++

```
enum IRJointWedgeType;
```

C#

```
public enum IRJointWedgeType;
```

Visual Basic

```
Public Enum IRJointWedgeType
```

Members

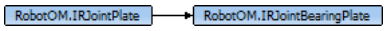
Members	Description
I_JWT_NONE = 0	Available since version 1.7.
I_JWT_I = 1	Available since version 1.7.
I_JWT_L = 2	Available since version 1.7.
I_JWT_T = 3	Available since version 1.7.
I_JWT_X = 4	Available since version 1.7.

Description

Available wedge types.

II.13 IRJointBearingPlate

Class Hierarchy



C++

```
interface IRJointBearingPlate : IRJointPlate;
```

C#

```
public interface IRJointBearingPlate : IRJointPlate;
```

Visual Basic

```
Public Interface IRJointBearingPlate
```

II.13.1 IRJointBearingPlate Members

The following tables list the members exposed by IRJointBearingPlate.

Public Fields

	Name	Description
	Exist (see page 2164)	Information about a plate existance .
	Length (see page 2164)	Plate length .
	Material (see page 2164)	Name of the component material.
	Thick (see page 2165)	Plate thickness.
	ThickBearingBar (see page 2014)	Thickness of limit element Available since version 1.7.
	Width (see page 2165)	Plate width.

(c) 2014 Autodesk Inc. All rights reserved

Public Methods

	Name	Description
	AngleExt (see page 2165)	Angle of bracket inclination, depending on the parameter - in degrees or radians.

II.13.2 IRJointBearingPlate Fields

The fields of the IRJointBearingPlate class are listed here.

Public Fields

	Name	Description
	ThickBearingBar (see page 2014)	Thickness of limit element Available since version 1.7.

II.13.2.1 ThickBearingBar

C++

```
HRESULT get_ThickBearingBar(double*);
HRESULT put_ThickBearingBar(double);
```

C#

```
public double ThickBearingBar { get; set; }
```

Visual Basic

```
Public ThickBearingBar As double
```

Description

Thickness of limit element Available since version 1.7.

II.14 IRJointFootWelds

Class Hierarchy

RobotOM.IRJointFootWelds

C++

```
interface IRJointFootWelds : IDispatch;
```

C#

```
public interface IRJointFootWelds;
```

Visual Basic

```
Public Interface IRJointFootWelds
```

Description

Description of connection element welds.

II.14.1 IRJointFootWelds Members

The following tables list the members exposed by IRJointFootWelds.

Public Fields

	Name	Description
	Bearing (see page 2015)	Available since version 1.7.
	FootPlate (see page 2015)	Available since version 1.7.
	Stiff (see page 2016)	Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

	Washer (see page 2016)	Available since version 1.7.
	Wedge (see page 2016)	Available since version 1.7.

II.14.2 IRJointFootWelds Fields

The fields of the IRJointFootWelds class are listed here.

Public Fields

	Name	Description
	Bearing (see page 2015)	Available since version 1.7.
	FootPlate (see page 2015)	Available since version 1.7.
	Stiff (see page 2016)	Available since version 1.7.
	Washer (see page 2016)	Available since version 1.7.
	Wedge (see page 2016)	Available since version 1.7.

II.14.2.1 Bearing

C++

```
HRESULT get_Bearing(IRJointWeld**);
```

C#

```
public IRJointWeld Bearing { get; }
```

Visual Basic

```
Public ReadOnly Bearing As IRJointWeld
```

Description

Available since version 1.7.

II.14.2.2 FootPlate

C++

```
HRESULT get_FootPlate(IRJointWeld**);
```

C#

```
public IRJointWeld FootPlate { get; }
```

Visual Basic

```
Public ReadOnly FootPlate As IRJointWeld
```

Description

Available since version 1.7.

II.14.2.3 Stiff

C++

```
HRESULT get_Stiff(IRJointWeld**);
```

C#

```
public IRJointWeld Stiff { get; }
```

Visual Basic

```
Public ReadOnly Stiff As IRJointWeld
```

Description

Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

II.14.2.4 Washer

C++

```
HRESULT get_Washer(IRJointWeld**);
```

C#

```
public IRJointWeld Washer { get; }
```

Visual Basic

```
Public ReadOnly Washer As IRJointWeld
```

Description

Available since version 1.7.

II.14.2.5 Wedge

C++

```
HRESULT get_Wedge(IRJointWeld**);
```

C#

```
public IRJointWeld Wedge { get; }
```

Visual Basic

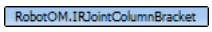
```
Public ReadOnly Wedge As IRJointWeld
```

Description

Available since version 1.7.

II.15 IRJointColumnBracket

Class Hierarchy



C++

```
interface IRJointColumnBracket : IDispatch;
```

C#

```
public interface IRJointColumnBracket;
```

Visual Basic

```
Public Interface IRJointColumnBracket
```

Description

Column stiffener.

II.15.1 IRJointColumnBracket Members

The following tables list the members exposed by IRJointColumnBracket.

Public Fields

	Name	Description
	Height (see page 2017)	Available since version 1.7.
	Length (see page 2018)	Available since version 1.7.
	Thick (see page 2018)	Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

	VThick (see page 2018)	Available since version 1.7.
	Width (see page 2018)	Available since version 1.7.

Public Methods

	Name	Description
	Exist (see page 2019)	Function returns information about the stiffener existence. Available since version 1.7.

II.15.2 IRJointColumnBracket Fields

The fields of the IRJointColumnBracket class are listed here.

Public Fields

	Name	Description
	Height (see page 2017)	Available since version 1.7.
	Length (see page 2018)	Available since version 1.7.
	Thick (see page 2018)	Available since version 1.7.
	VThick (see page 2018)	Available since version 1.7.
	Width (see page 2018)	Available since version 1.7.

II.15.2.1 Height

C++

```
HRESULT get_Height(double*);
HRESULT put_Height(double);
```

C#

```
public double Height { get; set; }
```

Visual Basic

```
Public Height As double
```

Description

Available since version 1.7.

II.15.2.2 Length

C++

```
HRESULT get_Length(double*);
HRESULT put_Length(double);
```

C#

```
public double Length { get; set; }
```

Visual Basic

```
Public Length As double
```

Description

Available since version 1.7.

II.15.2.3 Thick

C++

```
HRESULT get_Thick(double*);
HRESULT put_Thick(double);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double Thick { get; set; }
```

Visual Basic

```
Public Thick As double
```

Description

Available since version 1.7.

II.15.2.4 VThick

C++

```
HRESULT get_VThick(double*);
HRESULT put_VThick(double);
```

C#

```
public double VThick { get; set; }
```

Visual Basic

```
Public VThick As double
```

Description

Available since version 1.7.

II.15.2.5 Width

C++

```
HRESULT get_Width(double*);
HRESULT put_Width(double);
```

C#

```
public double Width { get; set; }
```

Visual Basic

```
Public Width As double
```

Description

Available since version 1.7.

II.15.3 IRJointColumnBracket Methods

The methods of the IRJointColumnBracket class are listed here.

Public Methods

	Name	Description
	Exist (see page 2019)	Function returns information about the stiffener existence. Available since version 1.7.

II.15.3.1 Exist

C++

```
HRESULT Exist(VARIANT_BOOL* ret);
```

C#

```
public bool Exist();
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Function Exist() As Boolean
```

Description

Function returns information about the stiffener existence. Available since version 1.7.

II.16 IRJointAnchorPlateType

C++

```
enum IRJointAnchorPlateType;
```

C#

```
public enum IRJointAnchorPlateType;
```

Visual Basic

```
Public Enum IRJointAnchorPlateType
```

Members

Members	Description
I_JAPT_RECT = 0	Available since version 1.7.
I_JAPT_TUBE = 1	Available since version 1.7.

Description

Types of plates near anchor.

II.17 IRJointFootStiffenerVert

Class Hierarchy

RobotOM.IRJointFootStiffenerVert

C++

```
interface IRJointFootStiffenerVert : IDispatch;
```

C#

```
public interface IRJointFootStiffenerVert;
```

Visual Basic

```
Public Interface IRJointFootStiffenerVert
```

II.17.1 IRJointFootStiffenerVert Members

The following tables list the members exposed by IRJointFootStiffenerVert.

Public Fields

	Name	Description
	Length (see page 2020)	Available since version 1.7.
	Thick (see page 2020)	Available since version 1.7.
	WidthSpacing (see page 2021)	Available since version 1.7.

II.17.2 IRJointFootStiffenerVert Fields

The fields of the IRJointFootStiffenerVert class are listed here.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	Length (see page 2020)	Available since version 1.7.
	Thick (see page 2020)	Available since version 1.7.
	WidthSpacing (see page 2021)	Available since version 1.7.

II.17.2.1 Length

C++

```
HRESULT get_Length(double*);
HRESULT put_Length(double);
```

C#

```
public double Length { get; set; }
```

Visual Basic

```
Public Length As double
```

Description

Available since version 1.7.

II.17.2.2 Thick

C++

```
HRESULT get_Thick(double*);
HRESULT put_Thick(double);
```

C#

```
public double Thick { get; set; }
```

Visual Basic

```
Public Thick As double
```

Description

Available since version 1.7.

II.17.2.3 WidthSpacing

C++

```
HRESULT get_WidthSpacing(double*);
HRESULT put_WidthSpacing(double);
```

C#

```
public double WidthSpacing { get; set; }
```

Visual Basic

```
Public WidthSpacing As double
```

Description

Available since version 1.7.

II.18 IRJointFootStiffenerHoriz

Class Hierarchy

RobotOM.IRJointFootStiffenerHoriz

(c) 2014 Autodesk Inc. All rights reserved

C++

```
interface IRJointFootStiffenerHoriz : IDispatch;
```

C#

```
public interface IRJointFootStiffenerHoriz;
```

Visual Basic

```
Public Interface IRJointFootStiffenerHoriz
```

II.18.1 IRJointFootStiffenerHoriz Members

The following tables list the members exposed by IRJointFootStiffenerHoriz.

Public Fields

	Name	Description
	Thick (see page 2021)	Available since version 1.7.

II.18.2 IRJointFootStiffenerHoriz Fields

The fields of the IRJointFootStiffenerHoriz class are listed here.

Public Fields

	Name	Description
	Thick (see page 2021)	Available since version 1.7.

II.18.2.1 Thick

C++

```
HRESULT get_Thick(double*);
HRESULT put_Thick(double);
```

C#

```
public double Thick { get; set; }
```

Visual Basic

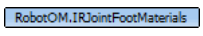
```
Public Thick As double
```

Description

Available since version 1.7.

II.19 IRJointFootMaterials

Class Hierarchy



C++

```
interface IRJointFootMaterials : IDispatch;
```

C#

```
public interface IRJointFootMaterials;
```

Visual Basic

```
Public Interface IRJointFootMaterials
```

(c) 2014 Autodesk Inc. All rights reserved

II.19.1 IRJointFootMaterials Members

The following tables list the members exposed by IRJointFootMaterials.

Public Fields

	Name	Description
	CementAmount (see page 2023)	Available since version 1.7.
	ConcrClass (see page 2023)	Available since version 1.7.
	ConcrSteelCoeff (see page 2023)	Available since version 1.7.
	PlateSigma (see page 2023)	Available since version 1.7.
	PlateYoung (see page 2024)	Available since version 1.7.

II.19.2 IRJointFootMaterials Fields

The fields of the IRJointFootMaterials class are listed here.

Public Fields

	Name	Description
	CementAmount (see page 2023)	Available since version 1.7.
	ConcrClass (see page 2023)	Available since version 1.7.
	ConcrSteelCoeff (see page 2023)	Available since version 1.7.
	PlateSigma (see page 2023)	Available since version 1.7.
	PlateYoung (see page 2024)	Available since version 1.7.

II.19.2.1 CementAmount

C++

```
HRESULT get_CementAmount(double*);
HRESULT put_CementAmount(double);
```

C#

```
public double CementAmount { get; set; }
```

Visual Basic

```
Public CementAmount As double
```

Description

Available since version 1.7.

II.19.2.2 ConcrClass

C++

```
HRESULT get_ConcrClass(double*);
HRESULT put_ConcrClass(double);
```

C#

```
public double ConcrClass { get; set; }
```

Visual Basic

```
Public ConcrClass As double
```

Description

Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

II.19.2.3 ConcrSteelCoeff

C++

```
HRESULT get_ConcrSteelCoeff(double*);
HRESULT put_ConcrSteelCoeff(double);
```

C#

```
public double ConcrSteelCoeff { get; set; }
```

Visual Basic

```
Public ConcrSteelCoeff As double
```

Description

Available since version 1.7.

II.19.2.4 PlateSigma

C++

```
HRESULT get_PlateSigma(double*);
HRESULT put_PlateSigma(double);
```

C#

```
public double PlateSigma { get; set; }
```

Visual Basic

```
Public PlateSigma As double
```

Description

Available since version 1.7.

II.19.2.5 PlateYoung

C++

```
HRESULT get_PlateYoung(double*);
HRESULT put_PlateYoung(double);
```

C#

```
public double PlateYoung { get; set; }
```

Visual Basic

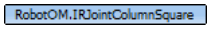
```
Public PlateYoung As double
```

Description

Available since version 1.7.

II.20 IRJointColumnSquare

Class Hierarchy



C++

```
interface IRJointColumnSquare : IDispatch;
```

C#

```
public interface IRJointColumnSquare;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Interface IRJointColumnSquare
```

II.20.1 IRJointColumnSquare Members

The following tables list the members exposed by IRJointColumnSquare.

Public Fields

	Name	Description
	Diameter (see page 2025)	Available since version 1.7.
	Length (see page 2025)	Available since version 1.7.
	Profile (see page 2025)	Available since version 1.7.

II.20.2 IRJointColumnSquare Fields

The fields of the IRJointColumnSquare class are listed here.

Public Fields

	Name	Description
	Diameter (see page 2025)	Available since version 1.7.
	Length (see page 2025)	Available since version 1.7.
	Profile (see page 2025)	Available since version 1.7.

II.20.2.1 Diameter

C++

```
HRESULT get_Diameter(double*);
HRESULT put_Diameter(double);
```

C#

```
public double Diameter { get; set; }
```

Visual Basic

```
Public Diameter As double
```

Description

Available since version 1.7.

II.20.2.2 Length

C++

```
HRESULT get_Length(double*);
HRESULT put_Length(double);
```

C#

```
public double Length { get; set; }
```

Visual Basic

```
Public Length As double
```

Description

Available since version 1.7.

II.20.2.3 Profile

C++

```
HRESULT get_Profile(IRJointProfile**);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRJointProfile Profile { get; }
```

Visual Basic

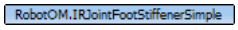
```
Public ReadOnly Profile As IRJointProfile
```

Description

Available since version 1.7.

II.21 IRJointFootStiffenerSimple

Class Hierarchy



C++

```
interface IRJointFootStiffenerSimple : IDispatch;
```

C#

```
public interface IRJointFootStiffenerSimple;
```

Visual Basic

```
Public Interface IRJointFootStiffenerSimple
```

II.21.1 IRJointFootStiffenerSimple Members

The following tables list the members exposed by IRJointFootStiffenerSimple.

Public Fields

	Name	Description
	Height (see page 2026)	Available since version 1.7.
	Length (see page 2026)	Available since version 1.7.
	Thick (see page 2027)	Available since version 1.7.
	Type (see page 2027)	Available since version 1.7.
	Width (see page 2027)	Available since version 1.7.

II.21.2 IRJointFootStiffenerSimple Fields

The fields of the IRJointFootStiffenerSimple class are listed here.

Public Fields

	Name	Description
	Height (see page 2026)	Available since version 1.7.
	Length (see page 2026)	Available since version 1.7.
	Thick (see page 2027)	Available since version 1.7.
	Type (see page 2027)	Available since version 1.7.
	Width (see page 2027)	Available since version 1.7.

II.21.2.1 Height

C++

```
HRESULT get_Height(double*);
HRESULT put_Height(double);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double Height { get; set; }
```

Visual Basic

```
Public Height As double
```

Description

Available since version 1.7.

II.21.2.2 Length

C++

```
HRESULT get_Length(double*);
HRESULT put_Length(double);
```

C#

```
public double Length { get; set; }
```

Visual Basic

```
Public Length As double
```

Description

Available since version 1.7.

II.21.2.3 Thick

C++

```
HRESULT get_Thick(double*);
HRESULT put_Thick(double);
```

C#

```
public double Thick { get; set; }
```

Visual Basic

```
Public Thick As double
```

Description

Available since version 1.7.

II.21.2.4 Type

C++

```
HRESULT get_Type(IRJointFootStiffType*);
HRESULT put_Type(IRJointFootStiffType);
```

C#

```
public IRJointFootStiffType Type { get; set; }
```

Visual Basic

```
Public Type As IRJointFootStiffType
```

Description

Available since version 1.7.

II.21.2.5 Width

C++

```
HRESULT get_Width(double*);
```

(c) 2014 Autodesk Inc. All rights reserved

```
HRESULT put_Width(double);
```

C#

```
public double Width { get; set; }
```

Visual Basic

```
Public Width As double
```

Description

Available since version 1.7.

II.22 IRJointFootStiffenerComplex

Class Hierarchy

RobotOM.IRJointFootStiffenerComplex

C++

```
interface IRJointFootStiffenerComplex : IDispatch;
```

C#

```
public interface IRJointFootStiffenerComplex;
```

Visual Basic

```
Public Interface IRJointFootStiffenerComplex
```

II.22.1 IRJointFootStiffenerComplex Members

The following tables list the members exposed by IRJointFootStiffenerComplex.

Public Fields

	Name	Description
	Height (see page 2028)	Available since version 1.7.
	Length (see page 2028)	Available since version 1.7.
	ThickPlateHor (see page 2029)	Available since version 1.7.
	ThickStiff (see page 2029)	Available since version 1.7.

II.22.2 IRJointFootStiffenerComplex Fields

The fields of the IRJointFootStiffenerComplex class are listed here.

Public Fields

	Name	Description
	Height (see page 2028)	Available since version 1.7.
	Length (see page 2028)	Available since version 1.7.
	ThickPlateHor (see page 2029)	Available since version 1.7.
	ThickStiff (see page 2029)	Available since version 1.7.

II.22.2.1 Height

C++

```
HRESULT get_Height(double*);
HRESULT put_Height(double);
```

C#

```
public double Height { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Height As double
```

Description

Available since version 1.7.

II.22.2.2 Length

C++

```
HRESULT get_Length(double*);
HRESULT put_Length(double);
```

C#

```
public double Length { get; set; }
```

Visual Basic

```
Public Length As double
```

Description

Available since version 1.7.

II.22.2.3 ThickPlateHor

C++

```
HRESULT get_ThickPlateHor(double*);
HRESULT put_ThickPlateHor(double);
```

C#

```
public double ThickPlateHor { get; set; }
```

Visual Basic

```
Public ThickPlateHor As double
```

Description

Available since version 1.7.

II.22.2.4 ThickStiff

C++

```
HRESULT get_ThickStiff(double*);
HRESULT put_ThickStiff(double);
```

C#

```
public double ThickStiff { get; set; }
```

Visual Basic

```
Public ThickStiff As double
```

Description

Available since version 1.7.

II.23 IRJointFootStiffType

C++

```
enum IRJointFootStiffType;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public enum IRJointFootStiffType;
```

Visual Basic

```
Public Enum IRJointFootStiffType
```

Members

Members	Description
I_JFST_1 = 1	Available since version 1.7.
I_JFST_2 = 2	Available since version 1.7.
I_JFST_3 = 3	Available since version 1.7.
I_JFST_4 = 4	Available since version 1.7.
I_JFST_5 = 5	Available since version 1.7.

III Angle connection

Enumerations

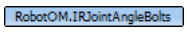
	Name	Description
	IRJointAngleType (see page 2035)	
	IRJointAngleProfilePosition (see page 2040)	

Interfaces

	Name	Description
	IRJointAngleBolts (see page 2030)	
	IRJointAngle (see page 2033)	
	IRJointBeamCut (see page 2036)	Definition of beam cut-outs. .
	IRJointWithAngles (see page 2037)	Angle (see page 2039) connection.
	IRJointAngleLoad (see page 2041)	

III.1 IRJointAngleBolts

Class Hierarchy



C++

```
interface IRJointAngleBolts : IDispatch;
```

C#

```
public interface IRJointAngleBolts;
```

Visual Basic

```
Public Interface IRJointAngleBolts
```

III.1.1 IRJointAngleBolts Members

The following tables list the members exposed by IRJointAngleBolts.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	Area (see page 2031)	Available since version 1.7.
	ClassName (see page 2031)	Available since version 1.7.
	Diameter (see page 2031)	Available since version 1.7.
	DiameterName (see page 2032)	Available since version 1.7.
	DistFromUpperEdge (see page 2032)	Available since version 1.7.
	DistFromWeb (see page 2032)	Available since version 1.7.
	Friction (see page 2033)	Available since version 1.7.
	Rows (see page 2033)	Available since version 1.7.
	Spacing (see page 2033)	Available since version 1.7.

III.1.2 IRJointAngleBolts Fields

The fields of the IRJointAngleBolts class are listed here.

Public Fields

	Name	Description
	Area (see page 2031)	Available since version 1.7.
	ClassName (see page 2031)	Available since version 1.7.
	Diameter (see page 2031)	Available since version 1.7.
	DiameterName (see page 2032)	Available since version 1.7.
	DistFromUpperEdge (see page 2032)	Available since version 1.7.
	DistFromWeb (see page 2032)	Available since version 1.7.
	Friction (see page 2033)	Available since version 1.7.
	Rows (see page 2033)	Available since version 1.7.
	Spacing (see page 2033)	Available since version 1.7.

III.1.2.1 Area

C++

```
HRESULT get_Area(double*);
```

C#

```
public double Area { get; }
```

Visual Basic

```
Public ReadOnly Area As double
```

Description

Available since version 1.7.

III.1.2.2 ClassName

C++

```
HRESULT get_ClassName(BSTR*);
HRESULT put_ClassName(BSTR);
```

C#

```
public String ClassName { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public ClassName As String
```

Description

Available since version 1.7.

III.1.2.3 Diameter

C++

```
HRESULT get_Diameter(long*);
```

C#

```
public long Diameter { get; }
```

Visual Basic

```
Public ReadOnly Diameter As long
```

Description

Available since version 1.7.

III.1.2.4 DiameterName

C++

```
HRESULT get_DiameterName(BSTR*);
HRESULT put_DiameterName(BSTR);
```

C#

```
public String DiameterName { get; set; }
```

Visual Basic

```
Public DiameterName As String
```

Description

Available since version 1.7.

III.1.2.5 DistFromUpperEdge

C++

```
HRESULT get_DistFromUpperEdge(double*);
HRESULT put_DistFromUpperEdge(double);
```

C#

```
public double DistFromUpperEdge { get; set; }
```

Visual Basic

```
Public DistFromUpperEdge As double
```

Description

Available since version 1.7.

III.1.2.6 DistFromWeb

C++

```
HRESULT get_DistFromWeb(double*);
HRESULT put_DistFromWeb(double);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double DistFromWeb { get; set; }
```

Visual Basic

```
Public DistFromWeb As double
```

Description

Available since version 1.7.

III.1.2.7 Friction

C++

```
HRESULT get_Friction(double*);
HRESULT put_Friction(double);
```

C#

```
public double Friction { get; set; }
```

Visual Basic

```
Public Friction As double
```

Description

Available since version 1.7.

III.1.2.8 Rows

C++

```
HRESULT get_Rows(long*);
HRESULT put_Rows(long);
```

C#

```
public long Rows { get; set; }
```

Visual Basic

```
Public Rows As long
```

Description

Available since version 1.7.

III.1.2.9 Spacing

C++

```
HRESULT get_Spacing(double*);
HRESULT put_Spacing(double);
```

C#

```
public double Spacing { get; set; }
```

Visual Basic

```
Public Spacing As double
```

Description

Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

III.2 IRJointAngle

Class Hierarchy

RobotOM.IRJointAngle

C++

```
interface IRJointAngle : IDispatch;
```

C#

```
public interface IRJointAngle;
```

Visual Basic

```
Public Interface IRJointAngle
```

III.2.1 IRJointAngle Members

The following tables list the members exposed by IRJointAngle.

Public Fields

	Name	Description
	DistFromUpperBeamEdge (see page 2034)	Distance from the top beam edge Available since version 1.7.
	Length (see page 2034)	Angle height Available since version 1.7.
	Profile (see page 2035)	Available since version 1.7.
	ProfilePosition (see page 2035)	Available since version 1.7.
	Type (see page 2035)	Available since version 1.7.

III.2.2 IRJointAngle Fields

The fields of the IRJointAngle class are listed here.

Public Fields

	Name	Description
	DistFromUpperBeamEdge (see page 2034)	Distance from the top beam edge Available since version 1.7.
	Length (see page 2034)	Angle height Available since version 1.7.
	Profile (see page 2035)	Available since version 1.7.
	ProfilePosition (see page 2035)	Available since version 1.7.
	Type (see page 2035)	Available since version 1.7.

III.2.2.1 DistFromUpperBeamEdge

C++

```
HRESULT get_DistFromUpperBeamEdge(double*);
HRESULT put_DistFromUpperBeamEdge(double);
```

C#

```
public double DistFromUpperBeamEdge { get; set; }
```

Visual Basic

```
Public DistFromUpperBeamEdge As double
```

Description

Distance from the top beam edge Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

III.2.2.2 Length

C++

```
HRESULT get_Length(double*);
HRESULT put_Length(double);
```

C#

```
public double Length { get; set; }
```

Visual Basic

```
Public Length As double
```

Description

Angle height Available since version 1.7.

III.2.2.3 Profile

C++

```
HRESULT get_Profile(IRJointProfile**);
```

C#

```
public IRJointProfile Profile { get; }
```

Visual Basic

```
Public ReadOnly Profile As IRJointProfile
```

Description

Available since version 1.7.

III.2.2.4 ProfilePosition

C++

```
HRESULT get_ProfilePosition(IRJointAngleProfilePosition*);
HRESULT put_ProfilePosition(IRJointAngleProfilePosition);
```

C#

```
public IRJointAngleProfilePosition ProfilePosition { get; set; }
```

Visual Basic

```
Public ProfilePosition As IRJointAngleProfilePosition
```

Description

Available since version 1.7.

III.2.2.5 Type

C++

```
HRESULT get_Type(IRJointAngleType*);
HRESULT put_Type(IRJointAngleType);
```

C#

```
public IRJointAngleType Type { get; set; }
```

Visual Basic

```
Public Type As IRJointAngleType
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Available since version 1.7.

III.3 IRJointAngleType

C++

```
enum IRJointAngleType;
```

C#

```
public enum IRJointAngleType;
```

Visual Basic

```
Public Enum IRJointAngleType
```

Members

Members	Description
I_JST_ALONG_BEARER = 0	Available since version 1.7.
I_JST_ALONG_SECOND = 1	Available since version 1.7.

III.4 IRJointBeamCut

Class Hierarchy

RobotOM.IRJointBeamCut

C++

```
interface IRJointBeamCut : IDispatch;
```

C#

```
public interface IRJointBeamCut;
```

Visual Basic

```
Public Interface IRJointBeamCut
```

Description

Definition of beam cut-outs. .

III.4.1 IRJointBeamCut Members

The following tables list the members exposed by IRJointBeamCut.

Public Fields

	Name	Description
	HeightLower (see page 2037)	Length (see page 2037) of a beam cut-out (bottom) Available since version 1.7.
	HeightUpper (see page 2037)	Length (see page 2037) of a beam cut-out (top) Available since version 1.7.
	Length (see page 2037)	Cut-out length Available since version 1.7.

III.4.2 IRJointBeamCut Fields

The fields of the IRJointBeamCut class are listed here.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	HeightLower (see page 2037)	Length (see page 2037) of a beam cut-out (bottom) Available since version 1.7.
	HeightUpper (see page 2037)	Length (see page 2037) of a beam cut-out (top) Available since version 1.7.
	Length (see page 2037)	Cut-out length Available since version 1.7.

III.4.2.1 HeightLower

C++

```
HRESULT get_HeightLower(double*);
HRESULT put_HeightLower(double);
```

C#

```
public double HeightLower { get; set; }
```

Visual Basic

```
Public HeightLower As double
```

Description

Length (see page 2037) of a beam cut-out (bottom) Available since version 1.7.

III.4.2.2 HeightUpper

C++

```
HRESULT get_HeightUpper(double*);
HRESULT put_HeightUpper(double);
```

C#

```
public double HeightUpper { get; set; }
```

Visual Basic

```
Public HeightUpper As double
```

Description

Length (see page 2037) of a beam cut-out (top) Available since version 1.7.

III.4.2.3 Length

C++

```
HRESULT get_Length(double*);
HRESULT put_Length(double);
```

C#

```
public double Length { get; set; }
```

Visual Basic

```
Public Length As double
```

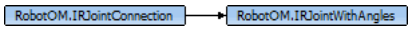
Description

Cut-out length Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

III.5 IRJointWithAngles

Class Hierarchy



C++

```
interface IRJointWithAngles : IRJointConnection;
```

C#

```
public interface IRJointWithAngles : IRJointConnection;
```

Visual Basic

```
Public Interface IRJointWithAngles
```

Description

Angle (see page 2039) connection.

III.5.1 IRJointWithAngles Members

The following tables list the members exposed by IRJointWithAngles.

Public Fields

	Name	Description
	Angle (see page 2039)	Angle Available since version 1.7.
	BeamBolts (see page 2039)	Bolt group on the beam side Available since version 1.7.
	BeamCut (see page 2039)	Beam cut-out Available since version 1.7.
	BeamProfile (see page 2039)	Beam section Available since version 1.7.
	ColumnBolts (see page 2040)	Bolt group on the column side Available since version 1.7.
	ColumnProfile (see page 2040)	Column section Available since version 1.7.
	Distance (see page 2040)	Distance between a column and beam Available since version 1.7.
	Type (see page 2160)	A connection type.
	WFRelPos (see page 2161)	A type of relative position of a beam and a column.

Public Methods

	Name	Description
	GetFromRobot (see page 2161)	The function gets connection parameters connection parameters from Robot and fills an object with the data.
	SetToRobot (see page 2161)	The function sets and saves connection parameter in Robot.

III.5.2 IRJointWithAngles Fields

The fields of the IRJointWithAngles class are listed here.

Public Fields

	Name	Description
	Angle (see page 2039)	Angle Available since version 1.7.
	BeamBolts (see page 2039)	Bolt group on the beam side Available since version 1.7.
	BeamCut (see page 2039)	Beam cut-out Available since version 1.7.
	BeamProfile (see page 2039)	Beam section Available since version 1.7.
	ColumnBolts (see page 2040)	Bolt group on the column side Available since version 1.7.
	ColumnProfile (see page 2040)	Column section Available since version 1.7.
	Distance (see page 2040)	Distance between a column and beam Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

III.5.2.1 Angle

C++

```
HRESULT get_Angle(IRJointAngle**);
```

C#

```
public IRJointAngle Angle { get; }
```

Visual Basic

```
Public ReadOnly Angle As IRJointAngle
```

Description

Angle Available since version 1.7.

III.5.2.2 BeamBolts

C++

```
HRESULT get_BeamBolts(IRJointAngleBolts**);
```

C#

```
public IRJointAngleBolts BeamBolts { get; }
```

Visual Basic

```
Public ReadOnly BeamBolts As IRJointAngleBolts
```

Description

Bolt group on the beam side Available since version 1.7.

III.5.2.3 BeamCut

C++

```
HRESULT get_BeamCut(IRJointBeamCut**);
```

C#

```
public IRJointBeamCut BeamCut { get; }
```

Visual Basic

```
Public ReadOnly BeamCut As IRJointBeamCut
```

Description

Beam cut-out Available since version 1.7.

III.5.2.4 BeamProfile

C++

```
HRESULT get_BeamProfile(IRJointProfile**);
```

C#

```
public IRJointProfile BeamProfile { get; }
```

Visual Basic

```
Public ReadOnly BeamProfile As IRJointProfile
```

Description

Beam section Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

III.5.2.5 ColumnBolts

C++

```
HRESULT get_ColumnBolts(IRJointAngleBolts**);
```

C#

```
public IRJointAngleBolts ColumnBolts { get; }
```

Visual Basic

```
Public ReadOnly ColumnBolts As IRJointAngleBolts
```

Description

Bolt group on the column side Available since version 1.7.

III.5.2.6 ColumnProfile

C++

```
HRESULT get_ColumnProfile(IRJointProfile**);
```

C#

```
public IRJointProfile ColumnProfile { get; }
```

Visual Basic

```
Public ReadOnly ColumnProfile As IRJointProfile
```

Description

Column section Available since version 1.7.

III.5.2.7 Distance

C++

```
HRESULT get_Distance(double*);
HRESULT put_Distance(double);
```

C#

```
public double Distance { get; set; }
```

Visual Basic

```
Public Distance As double
```

Description

Distance between a column and beam Available since version 1.7.

III.6 IRJointAngleProfilePosition

C++

```
enum IRJointAngleProfilePosition;
```

C#

```
public enum IRJointAngleProfilePosition;
```

Visual Basic

```
Public Enum IRJointAngleProfilePosition
```

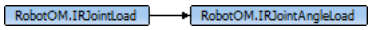
(c) 2014 Autodesk Inc. All rights reserved

Members

Members	Description
I_JAPP_LONG_FLANGE_TO_COLUMN = 0	Available since version 1.7.
I_JAPP_SHORT_FLANGE_TO_COLUMN = 1	Available since version 1.7.

III.7 IRJointAngleLoad

Class Hierarchy



C++

```
interface IRJointAngleLoad : IRJointLoad;
```

C#

```
public interface IRJointAngleLoad : IRJointLoad;
```

Visual Basic

```
Public Interface IRJointAngleLoad
```

III.7.1 IRJointAngleLoad Members

The following tables list the members exposed by IRJointAngleLoad.

Public Fields

	Name	Description
	Cases (see page 2178)	Selection of load cases, defined for the structure. .
	Fz (see page 2041)	Available since version 1.7.
	Type (see page 2179)	Load type (defined manually or taken from Robot).

III.7.2 IRJointAngleLoad Fields

The fields of the IRJointAngleLoad class are listed here.

Public Fields

	Name	Description
	Fz (see page 2041)	Available since version 1.7.

III.7.2.1 Fz

C++

```
HRESULT get_Fz(double*);
HRESULT put_Fz(double);
```

C#

```
public double Fz { get; set; }
```

Visual Basic

```
Public Fz As double
```

Description

Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

IV Tube connection

Enumerations

	Name	Description
	IRJointTubeType (see page 2046)	
	IRJointTubeProfileType (see page 2047)	

Interfaces

	Name	Description
	IRJointTube (see page 2042)	
	IRJointTubeFlangeProfile (see page 2047)	
	IRJointTubeDiagProfile (see page 2050)	
	IRJointTubePostProfile (see page 2053)	
	IRJointTubeLoad (see page 2055)	

IV.1 IRJointTube

Class Hierarchy

C++

```
interface IRJointTube : IRJointConnection;
```

C#

```
public interface IRJointTube : IRJointConnection;
```

Visual Basic

```
Public Interface IRJointTube
```

IV.1.1 IRJointTube Members

The following tables list the members exposed by IRJointTube.

Public Fields

	Name	Description
	DiagLeftLower (see page 2043)	Available since version 1.7.
	DiagLeftUpper (see page 2044)	Section of a first diagonal Available since version 1.7.
	DiagRightUpper (see page 2044)	Available since version 1.7.
	Flange (see page 2044)	Chord section Available since version 1.7.
	PostUpper (see page 2044)	Available since version 1.7.
	StiffHBracketMaterial (see page 2045)	Available since version ARSA 2010.
	StiffHoriz (see page 2045)	Available since version 1.7.
	StiffLateral (see page 2045)	Available since version 1.7.
	TubeType (see page 2045)	Type (see page 2160) of tube connection Available since version 1.7.
	Type (see page 2160)	A connection type.
	WeldDiag (see page 2046)	Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

	WeldStiff (see page 2046)	Available since version 1.7.
	WFRelPos (see page 2161)	A type of relative position of a beam and a column.

Public Methods

	Name	Description
	GetFromRobot (see page 2161)	The function gets connection parameters connection parameters from Robot and fills an object with the data.
	SetToRobot (see page 2161)	The function sets and saves connection parameter in Robot.

IV.1.2 IRJointTube Fields

The fields of the IRJointTube class are listed here.

Public Fields

	Name	Description
	DiagLeftLower (see page 2043)	Available since version 1.7.
	DiagLeftUpper (see page 2044)	Section of a first diagonal Available since version 1.7.
	DiagRightUpper (see page 2044)	Available since version 1.7.
	Flange (see page 2044)	Chord section Available since version 1.7.
	PostUpper (see page 2044)	Available since version 1.7.
	StiffHBracketMaterial (see page 2045)	Available since version ARSA 2010.
	StiffHoriz (see page 2045)	Available since version 1.7.
	StiffLateral (see page 2045)	Available since version 1.7.
	TubeType (see page 2045)	Type (see page 2160) of tube connection Available since version 1.7.
	WeldDiag (see page 2046)	Available since version 1.7.
	WeldStiff (see page 2046)	Available since version 1.7.

IV.1.2.1 DiagLeftLower

C++

```
HRESULT get_DiagLeftLower(IRJointTubeDiagProfile**);
```

C#

```
public IRJointTubeDiagProfile DiagLeftLower { get; }
```

Visual Basic

```
Public ReadOnly DiagLeftLower As IRJointTubeDiagProfile
```

Description

Available since version 1.7.

IV.1.2.2 DiagLeftUpper

C++

```
HRESULT get_DiagLeftUpper(IRJointTubeDiagProfile**);
```

C#

```
public IRJointTubeDiagProfile DiagLeftUpper { get; }
```

Visual Basic

```
Public ReadOnly DiagLeftUpper As IRJointTubeDiagProfile
```

Description

Section of a first diagonal Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

IV.1.2.3 DiagRightUpper

C++

```
HRESULT get_DiagRightUpper(IRJointTubeDiagProfile**);
```

C#

```
public IRJointTubeDiagProfile DiagRightUpper { get; }
```

Visual Basic

```
Public ReadOnly DiagRightUpper As IRJointTubeDiagProfile
```

Description

Available since version 1.7.

IV.1.2.4 Flange

C++

```
HRESULT get_Flange(IRJointTubeFlangeProfile**);
```

C#

```
public IRJointTubeFlangeProfile Flange { get; }
```

Visual Basic

```
Public ReadOnly Flange As IRJointTubeFlangeProfile
```

Description

Chord section Available since version 1.7.

IV.1.2.5 PostUpper

C++

```
HRESULT get_PostUpper(IRJointTubePostProfile**);
```

C#

```
public IRJointTubePostProfile PostUpper { get; }
```

Visual Basic

```
Public ReadOnly PostUpper As IRJointTubePostProfile
```

Description

Available since version 1.7.

IV.1.2.6 StiffHBracketMaterial

C++

```
HRESULT get_StiffHBracketMaterial( BSTR*);
HRESULT put_StiffHBracketMaterial( BSTR);
```

C#

```
public  BSTR StiffHBracketMaterial { get; set; }
```

Visual Basic

```
Public StiffHBracketMaterial As  BSTR
```

Description

Available since version ARSA 2010.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 10.

IV.1.2.7 StiffHoriz

C++

```
HRESULT get_StiffHoriz(IRJointPlate**);
```

C#

```
public IRJointPlate StiffHoriz { get; }
```

Visual Basic

```
Public ReadOnly StiffHoriz As IRJointPlate
```

Description

Available since version 1.7.

IV.1.2.8 StiffLateral

C++

```
HRESULT get_StiffLateral(IRJointPlate**);
```

C#

```
public IRJointPlate StiffLateral { get; }
```

Visual Basic

```
Public ReadOnly StiffLateral As IRJointPlate
```

Description

Available since version 1.7.

IV.1.2.9 TubeType

C++

```
HRESULT get_TubeType(IRJointTubeType*);
HRESULT put_TubeType(IRJointTubeType);
```

C#

```
public IRJointTubeType TubeType { get; set; }
```

Visual Basic

```
Public TubeType As IRJointTubeType
```

Description

Type (see page 2160) of tube connection Available since version 1.7.

IV.1.2.10 WeldDiag

C++

```
HRESULT get_WeldDiag(IRJointWeld**);
```

C#

```
public IRJointWeld WeldDiag { get; }
```

Visual Basic

```
Public ReadOnly WeldDiag As IRJointWeld
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Available since version 1.7.

IV.1.2.11 WeldStiff

C++

```
HRESULT get_WeldStiff(IRJointWeld**);
```

C#

```
public IRJointWeld WeldStiff { get; }
```

Visual Basic

```
Public ReadOnly WeldStiff As IRJointWeld
```

Description

Available since version 1.7.

IV.2 IRJointTubeType

C++

```
enum IRJointTubeType;
```

C#

```
public enum IRJointTubeType;
```

Visual Basic

```
Public Enum IRJointTubeType
```

Members

Members	Description
I_JTT_T = 0	Available since version 1.7.
I_JTT_Y = 1	Available since version 1.7.
I_JTT_X = 2	Available since version 1.7.
I_JTT_N = 3	Available since version 1.7.
I_JTT_K = 4	Available since version 1.7.
I_JTT_KT = 5	Available since version 1.7.
I_JTT_NONE = -1	Available since version 1.7.

IV.3 IRJointTubeProfileType

C++

```
enum IRJointTubeProfileType;
```

C#

```
public enum IRJointTubeProfileType;
```

Visual Basic

```
Public Enum IRJointTubeProfileType
```

Members

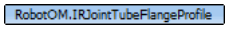
Members	Description
I_JTPT_RECT = 0	Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

I_JTPT_TUBE = 1	Available since version 1.7.
I_JTPT_RECT_RECT = 2	Available since version 1.7.
I_JTPT_RECT_TUBE = 3	Available since version 1.7.
I_JTPT_TUBE_TUBE = 4	Available since version 1.7.
I_JTPT_UNKNOWN = -1	Available since version 1.7.

IV.4 IRJointTubeFlangeProfile

Class Hierarchy



C++

```
interface IRJointTubeFlangeProfile : IDispatch;
```

C#

```
public interface IRJointTubeFlangeProfile;
```

Visual Basic

```
Public Interface IRJointTubeFlangeProfile
```

IV.4.1 IRJointTubeFlangeProfile Members

The following tables list the members exposed by IRJointTubeFlangeProfile.

Public Fields

	Name	Description
	BarNumber (see page 2048)	Bar number.
	Excentr (see page 2048)	Available since version 1.7.
	Exist (see page 2048)	
	Material (see page 2049)	Available since version 1.7.
	Profile (see page 2049)	Available since version ARSA 2010.
	Section (see page 2049)	Available since version 1.7.

IV.4.2 IRJointTubeFlangeProfile Fields

The fields of the IRJointTubeFlangeProfile class are listed here.

Public Fields

	Name	Description
	BarNumber (see page 2048)	Bar number.
	Excentr (see page 2048)	Available since version 1.7.
	Exist (see page 2048)	
	Material (see page 2049)	Available since version 1.7.
	Profile (see page 2049)	Available since version ARSA 2010.
	Section (see page 2049)	Available since version 1.7.

IV.4.2.1 BarNumber

C++

```
HRESULT get_BarNumber(long*);
HRESULT put_BarNumber(long);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public long BarNumber { get; set; }
```

Visual Basic

```
Public BarNumber As long
```

Description

Bar number.

Version

Available since version 3.5.

IV.4.2.2 Excentr

C++

```
HRESULT get_Excentr(double*);
HRESULT put_Excentr(double);
```

C#

```
public double Excentr { get; set; }
```

Visual Basic

```
Public Excentr As double
```

Description

Available since version 1.7.

IV.4.2.3 Exist

C++

```
HRESULT get_Exist(VARIANT_BOOL*);
```

C#

```
public bool Exist { get; }
```

Visual Basic

```
Public ReadOnly Exist As Boolean
```

Version

Available since version 4.5.

IV.4.2.4 Material

C++

```
HRESULT get_Material(BSTR*);
HRESULT put_Material(BSTR);
```

C#

```
public String Material { get; set; }
```

Visual Basic

```
Public Material As String
```

Description

Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

IV.4.2.5 Profile

C++

```
HRESULT get_Profile(IRJointProfile**);
```

C#

```
public IRJointProfile Profile { get; }
```

Visual Basic

```
Public ReadOnly Profile As IRJointProfile
```

Description

Available since version ARSA 2010.

Version

Available since version 10.

IV.4.2.6 Section

C++

```
HRESULT get_Section(BSTR*);
HRESULT put_Section(BSTR);
```

C#

```
public String Section { get; set; }
```

Visual Basic

```
Public Section As String
```

Description

Available since version 1.7.

IV.5 IRJointTubeDiagProfile

Class Hierarchy

RobotOM.IRJointTubeDiagProfile

C++

```
interface IRJointTubeDiagProfile : IDispatch;
```

C#

```
public interface IRJointTubeDiagProfile;
```

Visual Basic

```
Public Interface IRJointTubeDiagProfile
```

IV.5.1 IRJointTubeDiagProfile Members

The following tables list the members exposed by IRJointTubeDiagProfile.

Public Fields

	Name	Description
	Angle (see page 2050)	Available since version 1.7.
	BarNumber (see page 2051)	Bar number.
	Exist (see page 2051)	

(c) 2014 Autodesk Inc. All rights reserved

	Gap (see page 2051)	Available since version 1.7.
	Length (see page 2052)	Available since version 1.7.
	Material (see page 2052)	Available since version 1.7.
	Overlap (see page 2052)	Available since version 1.7.
	Profile (see page 2052)	Available from version ARSA 2010.
	Section (see page 2053)	Available since version 1.7.

IV.5.2 IRJointTubeDiagProfile Fields

The fields of the IRJointTubeDiagProfile class are listed here.

Public Fields

	Name	Description
	Angle (see page 2050)	Available since version 1.7.
	BarNumber (see page 2051)	Bar number.
	Exist (see page 2051)	
	Gap (see page 2051)	Available since version 1.7.
	Length (see page 2052)	Available since version 1.7.
	Material (see page 2052)	Available since version 1.7.
	Overlap (see page 2052)	Available since version 1.7.
	Profile (see page 2052)	Available from version ARSA 2010.
	Section (see page 2053)	Available since version 1.7.

IV.5.2.1 Angle

C++

```
HRESULT get_Angle(double*);
HRESULT put_Angle(double);
```

C#

```
public double Angle { get; set; }
```

Visual Basic

```
Public Angle As double
```

Description

Available since version 1.7.

IV.5.2.2 BarNumber

C++

```
HRESULT get_BarNumber(long*);
HRESULT put_BarNumber(long);
```

C#

```
public long BarNumber { get; set; }
```

Visual Basic

```
Public BarNumber As long
```

Description

Bar number.

Version

Available since version 3.5.

(c) 2014 Autodesk Inc. All rights reserved

IV.5.2.3 Exist

C++

```
HRESULT get_Exist(VARIANT_BOOL*);
```

C#

```
public bool Exist { get; }
```

Visual Basic

```
Public ReadOnly Exist As Boolean
```

Version

Available since version 4.5.

IV.5.2.4 Gap

C++

```
HRESULT get_Gap(double*);
HRESULT put_Gap(double);
```

C#

```
public double Gap { get; set; }
```

Visual Basic

```
Public Gap As double
```

Description

Available since version 1.7.

IV.5.2.5 Length

C++

```
HRESULT get_Length(double*);
HRESULT put_Length(double);
```

C#

```
public double Length { get; set; }
```

Visual Basic

```
Public Length As double
```

Description

Available since version 1.7.

IV.5.2.6 Material

C++

```
HRESULT get_Material(BSTR*);
HRESULT put_Material(BSTR);
```

C#

```
public String Material { get; set; }
```

Visual Basic

```
Public Material As String
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Available since version 1.7.

IV.5.2.7 Overlap

C++

```
HRESULT get_Overlap(double*);
HRESULT put_Overlap(double);
```

C#

```
public double Overlap { get; set; }
```

Visual Basic

```
Public Overlap As double
```

Description

Available since version 1.7.

IV.5.2.8 Profile

C++

```
HRESULT get_Profile(IRJointProfile**);
```

C#

```
public IRJointProfile Profile { get; }
```

Visual Basic

```
Public ReadOnly Profile As IRJointProfile
```

Description

Available from version ARSA 2010.

Version

Available since version 10.

IV.5.2.9 Section

C++

```
HRESULT get_Section(BSTR*);
HRESULT put_Section(BSTR);
```

C#

```
public String Section { get; set; }
```

Visual Basic

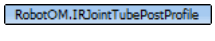
```
Public Section As String
```

Description

Available since version 1.7.

IV.6 IRJointTubePostProfile

Class Hierarchy



(c) 2014 Autodesk Inc. All rights reserved

C++

```
interface IRJointTubePostProfile : IDispatch;
```

C#

```
public interface IRJointTubePostProfile;
```

Visual Basic

```
Public Interface IRJointTubePostProfile
```

IV.6.1 IRJointTubePostProfile Members

The following tables list the members exposed by IRJointTubePostProfile.

Public Fields

	Name	Description
	BarNumber (see page 2054)	
	Exist (see page 2054)	
	Length (see page 2054)	Available since version 1.7.
	Material (see page 2055)	Available since version 1.7.
	Profile (see page 2055)	Available since version ARSA 2010.
	Section (see page 2055)	Available since version 1.7.

IV.6.2 IRJointTubePostProfile Fields

The fields of the IRJointTubePostProfile class are listed here.

Public Fields

	Name	Description
	BarNumber (see page 2054)	
	Exist (see page 2054)	
	Length (see page 2054)	Available since version 1.7.
	Material (see page 2055)	Available since version 1.7.
	Profile (see page 2055)	Available since version ARSA 2010.
	Section (see page 2055)	Available since version 1.7.

IV.6.2.1 BarNumber

C++

```
HRESULT get_BarNumber(long*);
HRESULT put_BarNumber(long);
```

C#

```
public long BarNumber { get; set; }
```

Visual Basic

```
Public BarNumber As long
```

Version

Available since version 4.5.

IV.6.2.2 Exist

C++

```
HRESULT get_Exist(VARIANT_BOOL*);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public bool Exist { get; }
```

Visual Basic

```
Public ReadOnly Exist As Boolean
```

Version

Available since version 4.5.

IV.6.2.3 Length

C++

```
HRESULT get_Length(double*);
HRESULT put_Length(double);
```

C#

```
public double Length { get; set; }
```

Visual Basic

```
Public Length As double
```

Description

Available since version 1.7.

IV.6.2.4 Material

C++

```
HRESULT get_Material(BSTR*);
HRESULT put_Material(BSTR);
```

C#

```
public String Material { get; set; }
```

Visual Basic

```
Public Material As String
```

Description

Available since version 1.7.

IV.6.2.5 Profile

C++

```
HRESULT get_Profile(IRJointProfile**);
```

C#

```
public IRJointProfile Profile { get; }
```

Visual Basic

```
Public ReadOnly Profile As IRJointProfile
```

Description

Available since version ARSA 2010.

Version

Available since version 10.

(c) 2014 Autodesk Inc. All rights reserved

IV.6.2.6 Section

C++

```
HRESULT get_Section(BSTR*);
HRESULT put_Section(BSTR);
```

C#

```
public String Section { get; set; }
```

Visual Basic

```
Public Section As String
```

Description

Available since version 1.7.

IV.7 IRJointTubeLoad

Class Hierarchy

C++

```
interface IRJointTubeLoad : IRJointLoad;
```

C#

```
public interface IRJointTubeLoad : IRJointLoad;
```

Visual Basic

```
Public Interface IRJointTubeLoad
```

IV.7.1 IRJointTubeLoad Members

The following tables list the members exposed by IRJointTubeLoad.

Public Fields

	Name	Description
	Cases (see page 2178)	Selection of load cases, defined for the structure. .
	DiagLeftLowerM (see page 2057)	Available since version 1.7.
	DiagLeftLowerN (see page 2057)	Available since version 1.7.
	DiagLeftUpperM (see page 2057)	Available since version 1.7.
	DiagLeftUpperN (see page 2057)	Available since version 1.7.
	DiagRightUpperM (see page 2058)	Available since version 1.7.
	DiagRightUpperN (see page 2058)	Available since version 1.7.
	FlangeM (see page 2058)	Available since version 1.7.
	FlangeN (see page 2058)	Available since version 1.7.
	FlangeQ (see page 2059)	Available since version 1.7.
	PostUpperN (see page 2059)	Available since version 1.7.
	Type (see page 2179)	Load type (defined manually or taken from Robot).

IV.7.2 IRJointTubeLoad Fields

The fields of the IRJointTubeLoad class are listed here.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	DiagLeftLowerM (see page 2057)	Available since version 1.7.
	DiagLeftLowerN (see page 2057)	Available since version 1.7.
	DiagLeftUpperM (see page 2057)	Available since version 1.7.
	DiagLeftUpperN (see page 2057)	Available since version 1.7.
	DiagRightUpperM (see page 2058)	Available since version 1.7.
	DiagRightUpperN (see page 2058)	Available since version 1.7.
	FlangeM (see page 2058)	Available since version 1.7.
	FlangeN (see page 2058)	Available since version 1.7.
	FlangeQ (see page 2059)	Available since version 1.7.
	PostUpperN (see page 2059)	Available since version 1.7.

IV.7.2.1 DiagLeftLowerM

C++

```
HRESULT get_DiagLeftLowerM(double*);
HRESULT put_DiagLeftLowerM(double);
```

C#

```
public double DiagLeftLowerM { get; set; }
```

Visual Basic

```
Public DiagLeftLowerM As double
```

Description

Available since version 1.7.

IV.7.2.2 DiagLeftLowerN

C++

```
HRESULT get_DiagLeftLowerN(double*);
HRESULT put_DiagLeftLowerN(double);
```

C#

```
public double DiagLeftLowerN { get; set; }
```

Visual Basic

```
Public DiagLeftLowerN As double
```

Description

Available since version 1.7.

IV.7.2.3 DiagLeftUpperM

C++

```
HRESULT get_DiagLeftUpperM(double*);
HRESULT put_DiagLeftUpperM(double);
```

C#

```
public double DiagLeftUpperM { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public DiagLeftUpperM As double
```

Description

Available since version 1.7.

IV.7.2.4 DiagLeftUpperN

C++

```
HRESULT get_DiagLeftUpperN(double*);
HRESULT put_DiagLeftUpperN(double);
```

C#

```
public double DiagLeftUpperN { get; set; }
```

Visual Basic

```
Public DiagLeftUpperN As double
```

Description

Available since version 1.7.

IV.7.2.5 DiagRightUpperM

C++

```
HRESULT get_DiagRightUpperM(double*);
HRESULT put_DiagRightUpperM(double);
```

C#

```
public double DiagRightUpperM { get; set; }
```

Visual Basic

```
Public DiagRightUpperM As double
```

Description

Available since version 1.7.

IV.7.2.6 DiagRightUpperN

C++

```
HRESULT get_DiagRightUpperN(double*);
HRESULT put_DiagRightUpperN(double);
```

C#

```
public double DiagRightUpperN { get; set; }
```

Visual Basic

```
Public DiagRightUpperN As double
```

Description

Available since version 1.7.

IV.7.2.7 FlangeM

C++

```
HRESULT get_FlangeM(double*);
HRESULT put_FlangeM(double);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double FlangeM { get; set; }
```

Visual Basic

```
Public FlangeM As double
```

Description

Available since version 1.7.

IV.7.2.8 FlangeN

C++

```
HRESULT get_FlangeN(double*);
HRESULT put_FlangeN(double);
```

C#

```
public double FlangeN { get; set; }
```

Visual Basic

```
Public FlangeN As double
```

Description

Available since version 1.7.

IV.7.2.9 FlangeQ

C++

```
HRESULT get_FlangeQ(double*);
HRESULT put_FlangeQ(double);
```

C#

```
public double FlangeQ { get; set; }
```

Visual Basic

```
Public FlangeQ As double
```

Description

Available since version 1.7.

IV.7.2.10 PostUpperN

C++

```
HRESULT get_PostUpperN(double*);
HRESULT put_PostUpperN(double);
```

C#

```
public double PostUpperN { get; set; }
```

Visual Basic

```
Public PostUpperN As double
```

Description

Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

V Gusset plate connections

Enumerations

	Name	Description
	IRJointGussetFixType (see page 2068)	
	IRJointGussetSimpleProfilePosition (see page 2068)	
	IRJointGussetCornersType (see page 2078)	
	IRJointGussetCrossProfileCutting (see page 2087)	
	IRJointGussetFlangeProfileCutting (see page 2102)	
	IRJointGussetDiagonalePositionType (see page 2122)	
	IRJointGussetFlangePlateRegularType (see page 2123)	

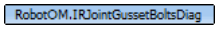
Interfaces

	Name	Description
	IRJointGussetBoltsDiag (see page 2060)	
	IRJointGussetWeldsDiag (see page 2063)	
	IRJointGussetDiagonale (see page 2065)	
	IRJointGussetSimple (see page 2068)	
	IRJointGussetSimplePlate (see page 2072)	
	IRJointGussetSimpleAttachment (see page 2078)	
	IRJointGussetSimpleAttachBolts (see page 2080)	
	IRJointGussetSimpleAttachBoltsHorizontal (see page 2082)	
	IRJointGussetSimpleAttachBoltsVertical (see page 2084)	
	IRJointGussetSimpleAttachWelds (see page 2085)	
	IRJointGussetCrossPlate (see page 2087)	
	IRJointGussetCross (see page 2096)	
	IRJointGussetFlangePlate (see page 2102)	
	IRJointGussetFlange (see page 2112)	
	IRJointGussetSimpleLoad (see page 2117)	
	IRJointGussetCrossLoad (see page 2118)	
	IRJointGussetFlangeLoad (see page 2120)	

(c) 2014 Autodesk Inc. All rights reserved

V.1 IRJointGussetBoltsDiag

Class Hierarchy



C++

```
interface IRJointGussetBoltsDiag : IDispatch;
```

C#

```
public interface IRJointGussetBoltsDiag;
```

Visual Basic

```
Public Interface IRJointGussetBoltsDiag
```

V.1.1 IRJointGussetBoltsDiag Members

The following tables list the members exposed by IRJointGussetBoltsDiag.

Public Fields

	Name	Description
	BoltAxisShift (see page 2061)	Available since version 1.7.
	ClassName (see page 2061)	Available since version 1.7.
	Diameter (see page 2062)	Available since version 1.7.
	DistanceH1 (see page 2062)	Available since version 1.7.
	Friction (see page 2062)	Available since version 1.7.
	Rows (see page 2062)	Available since version 1.7.
	Spacing (see page 2063)	Available since version 1.7.

V.1.2 IRJointGussetBoltsDiag Fields

The fields of the IRJointGussetBoltsDiag class are listed here.

Public Fields

	Name	Description
	BoltAxisShift (see page 2061)	Available since version 1.7.
	ClassName (see page 2061)	Available since version 1.7.
	Diameter (see page 2062)	Available since version 1.7.
	DistanceH1 (see page 2062)	Available since version 1.7.
	Friction (see page 2062)	Available since version 1.7.
	Rows (see page 2062)	Available since version 1.7.
	Spacing (see page 2063)	Available since version 1.7.

V.1.2.1 BoltAxisShift

C++

```
HRESULT get_BoltAxisShift(double*);
HRESULT put_BoltAxisShift(double);
```

C#

```
public double BoltAxisShift { get; set; }
```

Visual Basic

```
Public BoltAxisShift As double
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Available since version 1.7.

V.1.2.2 ClassName

C++

```
HRESULT get_ClassName(BSTR*);
HRESULT put_ClassName(BSTR);
```

C#

```
public String ClassName { get; set; }
```

Visual Basic

```
Public ClassName As String
```

Description

Available since version 1.7.

V.1.2.3 Diameter

C++

```
HRESULT get_Diameter(double*);
HRESULT put_Diameter(double);
```

C#

```
public double Diameter { get; set; }
```

Visual Basic

```
Public Diameter As double
```

Description

Available since version 1.7.

V.1.2.4 DistanceH1

C++

```
HRESULT get_DistanceH1(double*);
HRESULT put_DistanceH1(double);
```

C#

```
public double DistanceH1 { get; set; }
```

Visual Basic

```
Public DistanceH1 As double
```

Description

Available since version 1.7.

V.1.2.5 Friction

C++

```
HRESULT get_Friction(double*);
HRESULT put_Friction(double);
```

C#

```
public double Friction { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Friction As double
```

Description

Available since version 1.7.

V.1.2.6 Rows

C++

```
HRESULT get_Rows(long*);
HRESULT put_Rows(long);
```

C#

```
public long Rows { get; set; }
```

Visual Basic

```
Public Rows As long
```

Description

Available since version 1.7.

V.1.2.7 Spacing

C++

```
HRESULT get_Spacing(IRobotValuesArray**);
```

C#

```
public IRobotValuesArray Spacing { get; }
```

Visual Basic

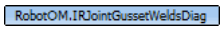
```
Public ReadOnly Spacing As IRobotValuesArray
```

Description

Available since version 1.7.

V.2 IRJointGussetWeldsDiag

Class Hierarchy



C++

```
interface IRJointGussetWeldsDiag : IDispatch;
```

C#

```
public interface IRJointGussetWeldsDiag;
```

Visual Basic

```
Public Interface IRJointGussetWeldsDiag
```

V.2.1 IRJointGussetWeldsDiag Members

The following tables list the members exposed by IRJointGussetWeldsDiag.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	Length1 (see page 2064)	Available since version ARSA 2010.
	Length2 (see page 2064)	Available since version ARSA 2010.
	ThickCorner (see page 2064)	Available since version 1.7.
	ThickFlange (see page 2065)	Available since version 1.7.

V.2.2 IRJointGussetWeldsDiag Fields

The fields of the IRJointGussetWeldsDiag class are listed here.

Public Fields

	Name	Description
	Length1 (see page 2064)	Available since version ARSA 2010.
	Length2 (see page 2064)	Available since version ARSA 2010.
	ThickCorner (see page 2064)	Available since version 1.7.
	ThickFlange (see page 2065)	Available since version 1.7.

V.2.2.1 Length1

C++

```
HRESULT get_Length1( double*);
HRESULT put_Length1( double);
```

C#

```
public  double Length1 { get; set; }
```

Visual Basic

```
Public Length1 As  double
```

Description

Available since version ARSA 2010.

Version

Available since version 10.

V.2.2.2 Length2

C++

```
HRESULT get_Length2( double*);
HRESULT put_Length2( double);
```

C#

```
public  double Length2 { get; set; }
```

Visual Basic

```
Public Length2 As  double
```

Description

Available since version ARSA 2010.

Version

Available since version 10.

(c) 2014 Autodesk Inc. All rights reserved

V.2.2.3 ThickCorner

C++

```
HRESULT get_ThickCorner(double*);
HRESULT put_ThickCorner(double);
```

C#

```
public double ThickCorner { get; set; }
```

Visual Basic

```
Public ThickCorner As double
```

Description

Available since version 1.7.

V.2.2.4 ThickFlange

C++

```
HRESULT get_ThickFlange(double*);
HRESULT put_ThickFlange(double);
```

C#

```
public double ThickFlange { get; set; }
```

Visual Basic

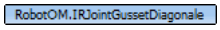
```
Public ThickFlange As double
```

Description

Available since version 1.7.

V.3 IRJointGussetDiagonale

Class Hierarchy



C++

```
interface IRJointGussetDiagonale : IDispatch;
```

C#

```
public interface IRJointGussetDiagonale;
```

Visual Basic

```
Public Interface IRJointGussetDiagonale
```

V.3.1 IRJointGussetDiagonale Members

The following tables list the members exposed by IRJointGussetDiagonale.

Public Fields

	Name	Description
	BarNumber (see page 2066)	
	BoltsDiag (see page 2066)	Available since version 1.7.
	DistanceEC (see page 2066)	Available since version 1.7.
	Exist (see page 2067)	
	Position (see page 2067)	Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

	Profile (see page 2067)	Available since version 1.7.
	WeldsDiag (see page 2067)	Available since version 1.7.

V.3.2 IRJointGussetDiagonale Fields

The fields of the IRJointGussetDiagonale class are listed here.

Public Fields

	Name	Description
	BarNumber (see page 2066)	
	BoltsDiag (see page 2066)	Available since version 1.7.
	DistanceEC (see page 2066)	Available since version 1.7.
	Exist (see page 2067)	
	Position (see page 2067)	Available since version 1.7.
	Profile (see page 2067)	Available since version 1.7.
	WeldsDiag (see page 2067)	Available since version 1.7.

V.3.2.1 BarNumber

C++

```
HRESULT get_BarNumber(long*);
HRESULT put_BarNumber(long);
```

C#

```
public long BarNumber { get; set; }
```

Visual Basic

```
Public BarNumber As long
```

Version

Available since version 4.5.

V.3.2.2 BoltsDiag

C++

```
HRESULT get_BoltsDiag(IRJointGussetBoltsDiag**);
```

C#

```
public IRJointGussetBoltsDiag BoltsDiag { get; }
```

Visual Basic

```
Public ReadOnly BoltsDiag As IRJointGussetBoltsDiag
```

Description

Available since version 1.7.

V.3.2.3 DistanceEC

C++

```
HRESULT get_DistanceEC(double*);
HRESULT put_DistanceEC(double);
```

C#

```
public double DistanceEC { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public DistanceEC As double
```

Description

Available since version 1.7.

V.3.2.4 Exist

C++

```
HRESULT get_Exist(VARIANT_BOOL*);
HRESULT put_Exist(VARIANT_BOOL);
```

C#

```
public bool Exist { get; set; }
```

Visual Basic

```
Public Exist As Boolean
```

Version

Available since version 4.5.

V.3.2.5 Position

C++

```
HRESULT get_Position(IRJointGussetDiagonalePositionType*);
HRESULT put_Position(IRJointGussetDiagonalePositionType);
```

C#

```
public IRJointGussetDiagonalePositionType Position { get; set; }
```

Visual Basic

```
Public Position As IRJointGussetDiagonalePositionType
```

Description

Available since version 1.7.

V.3.2.6 Profile

C++

```
HRESULT get_Profile(IRJointProfile**);
```

C#

```
public IRJointProfile Profile { get; }
```

Visual Basic

```
Public ReadOnly Profile As IRJointProfile
```

Description

Available since version 1.7.

V.3.2.7 WeldsDiag

C++

```
HRESULT get_WeldsDiag(IRJointGussetWeldsDiag**);
```

C#

```
public IRJointGussetWeldsDiag WeldsDiag { get; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public ReadOnly WeldsDiag As IRJointGussetWeldsDiag
```

Description

Available since version 1.7.

V.4 IRJointGussetFixType

C++

```
enum IRJointGussetFixType;
```

C#

```
public enum IRJointGussetFixType;
```

Visual Basic

```
Public Enum IRJointGussetFixType
```

Members

Members	Description
I_JGFT_BOLTED = 0	Available since version 1.7.
I_JGFT_WELDED = 1	Available since version 1.7.

V.5 IRJointGussetSimpleProfilePosition

C++

```
enum IRJointGussetSimpleProfilePosition;
```

C#

```
public enum IRJointGussetSimpleProfilePosition;
```

Visual Basic

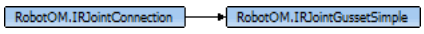
```
Public Enum IRJointGussetSimpleProfilePosition
```

Members

Members	Description
I_JGSPP_VERTICAL = 0	Available since version 1.7.
I_JGSPP_DIAGONAL = 1	Available since version 1.7.

V.6 IRJointGussetSimple

Class Hierarchy



C++

```
interface IRJointGussetSimple : IRJointConnection;
```

C#

```
public interface IRJointGussetSimple : IRJointConnection;
```

Visual Basic

```
Public Interface IRJointGussetSimple
```

(c) 2014 Autodesk Inc. All rights reserved

V.6.1 IRJointGussetSimple Members

The following tables list the members exposed by IRJointGussetSimple.

Public Fields

	Name	Description
	Attachment (see page 2069)	Available since version 1.7.
	BoltsClassVector (see page 2070)	Available since version ARSA 2010.
	BoltsDimensionNamesVector (see page 2070)	Available since version ARSA 2010.
	BoltsDimensionsVector (see page 2070)	Available since version ARSA 2010.
	Diagonale (see page 2071)	Available since version 1.7.
	FixType (see page 2071)	Available since version 1.7.
	GussetPlate (see page 2071)	Available since version 1.7.
	ProfilePosition (see page 2071)	Available since version 1.7.
	Type (see page 2160)	A connection type.
	WFRelPos (see page 2161)	A type of relative position of a beam and a column.

Public Methods

	Name	Description
	GetFromRobot (see page 2161)	The function gets connection parameters connection parameters from Robot and fills an object with the data.
	SetToRobot (see page 2161)	The function sets and saves connection parameter in Robot.

V.6.2 IRJointGussetSimple Fields

The fields of the IRJointGussetSimple class are listed here.

Public Fields

	Name	Description
	Attachment (see page 2069)	Available since version 1.7.
	BoltsClassVector (see page 2070)	Available since version ARSA 2010.
	BoltsDimensionNamesVector (see page 2070)	Available since version ARSA 2010.
	BoltsDimensionsVector (see page 2070)	Available since version ARSA 2010.
	Diagonale (see page 2071)	Available since version 1.7.
	FixType (see page 2071)	Available since version 1.7.
	GussetPlate (see page 2071)	Available since version 1.7.
	ProfilePosition (see page 2071)	Available since version 1.7.

V.6.2.1 Attachment

C++

```
HRESULT get_Attachment(IRJointGussetSimpleAttachment**);
```

C#

```
public IRJointGussetSimpleAttachment Attachment { get; }
```

Visual Basic

```
Public ReadOnly Attachment As IRJointGussetSimpleAttachment
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Available since version 1.7.

V.6.2.2 BoltsClassVector

C++

```
HRESULT get_BoltsClassVector(IRobotNamesArray**);
```

C#

```
public IRobotNamesArray BoltsClassVector { get; }
```

Visual Basic

```
Public ReadOnly BoltsClassVector As IRobotNamesArray
```

Description

Available since version ARSA 2010.

Version

Available since version 10.

V.6.2.3 BoltsDimensionNamesVector

C++

```
HRESULT get_BoltsDimensionNamesVector(IRobotNamesArray**);
```

C#

```
public IRobotNamesArray BoltsDimensionNamesVector { get; }
```

Visual Basic

```
Public ReadOnly BoltsDimensionNamesVector As IRobotNamesArray
```

Description

Available since version ARSA 2010.

Version

Available since version 10.

V.6.2.4 BoltsDimensionsVector

C++

```
HRESULT get_BoltsDimensionsVector(IRobotValuesArray**);
```

C#

```
public IRobotValuesArray BoltsDimensionsVector { get; }
```

Visual Basic

```
Public ReadOnly BoltsDimensionsVector As IRobotValuesArray
```

Description

Available since version ARSA 2010.

Version

Available since version 10.

(c) 2014 Autodesk Inc. All rights reserved

V.6.2.5 Diagonale

C++

```
HRESULT get_Diagonale(IRJointGussetDiagonale**);
```

C#

```
public IRJointGussetDiagonale Diagonale { get; }
```

Visual Basic

```
Public ReadOnly Diagonale As IRJointGussetDiagonale
```

Description

Available since version 1.7.

V.6.2.6 FixType

C++

```
HRESULT get_FixType(IRJointGussetFixType*);
HRESULT put_FixType(IRJointGussetFixType);
```

C#

```
public IRJointGussetFixType FixType { get; set; }
```

Visual Basic

```
Public FixType As IRJointGussetFixType
```

Description

Available since version 1.7.

V.6.2.7 GussetPlate

C++

```
HRESULT get_GussetPlate(IRJointGussetSimplePlate**);
```

C#

```
public IRJointGussetSimplePlate GussetPlate { get; }
```

Visual Basic

```
Public ReadOnly GussetPlate As IRJointGussetSimplePlate
```

Description

Available since version 1.7.

V.6.2.8 ProfilePosition

C++

```
HRESULT get_ProfilePosition(IRJointGussetSimpleProfilePosition*);
HRESULT put_ProfilePosition(IRJointGussetSimpleProfilePosition);
```

C#

```
public IRJointGussetSimpleProfilePosition ProfilePosition { get; set; }
```

Visual Basic

```
Public ProfilePosition As IRJointGussetSimpleProfilePosition
```

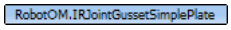
Description

Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

V.7 IRJointGussetSimplePlate

Class Hierarchy



C++

```
interface IRJointGussetSimplePlate : IDispatch;
```

C#

```
public interface IRJointGussetSimplePlate;
```

Visual Basic

```
Public Interface IRJointGussetSimplePlate
```

V.7.1 IRJointGussetSimplePlate Members

The following tables list the members exposed by IRJointGussetSimplePlate.

Public Fields

	Name	Description
	CornersType (see page 2073)	Available since version 1.7.
	DistanceD (see page 2073)	Available since version 1.7.
	DistanceEH (see page 2073)	Available since version 1.7.
	DistanceEV (see page 2074)	Available since version 1.7.
	H1 (see page 2074)	Available since version ARSA 2010.
	H2 (see page 2074)	Available since version ARSA 2010.
	H3 (see page 2075)	Available since version ARSA 2010.
	H4 (see page 2075)	Available since version ARSA 2010.
	Height (see page 2075)	Available since version 1.7.
	Material (see page 2076)	Available since version 1.7.
	Thick (see page 2076)	Available since version 1.7.
	V1 (see page 2076)	Available since version ARSA 2010.
	V2 (see page 2076)	Available since version ARSA 2010.
	V3 (see page 2077)	Available since version ARSA 2010.
	V4 (see page 2077)	Available since version ARSA 2010.
	Width (see page 2077)	Available since version 1.7.

V.7.2 IRJointGussetSimplePlate Fields

The fields of the IRJointGussetSimplePlate class are listed here.

Public Fields

	Name	Description
	CornersType (see page 2073)	Available since version 1.7.
	DistanceD (see page 2073)	Available since version 1.7.
	DistanceEH (see page 2073)	Available since version 1.7.
	DistanceEV (see page 2074)	Available since version 1.7.
	H1 (see page 2074)	Available since version ARSA 2010.
	H2 (see page 2074)	Available since version ARSA 2010.
	H3 (see page 2075)	Available since version ARSA 2010.
	H4 (see page 2075)	Available since version ARSA 2010.

(c) 2014 Autodesk Inc. All rights reserved

	Height (see page 2075)	Available since version 1.7.
	Material (see page 2076)	Available since version 1.7.
	Thick (see page 2076)	Available since version 1.7.
	V1 (see page 2076)	Available since version ARSA 2010.
	V2 (see page 2076)	Available since version ARSA 2010.
	V3 (see page 2077)	Available since version ARSA 2010.
	V4 (see page 2077)	Available since version ARSA 2010.
	Width (see page 2077)	Available since version 1.7.

V.7.2.1 CornersType

C++

```
HRESULT get_CornersType(IRJointGussetCornersType*);
HRESULT put_CornersType(IRJointGussetCornersType);
```

C#

```
public IRJointGussetCornersType CornersType { get; set; }
```

Visual Basic

```
Public CornersType As IRJointGussetCornersType
```

Description

Available since version 1.7.

V.7.2.2 DistanceD

C++

```
HRESULT get_DistanceD(double*);
HRESULT put_DistanceD(double);
```

C#

```
public double DistanceD { get; set; }
```

Visual Basic

```
Public DistanceD As double
```

Description

Available since version 1.7.

V.7.2.3 DistanceEH

C++

```
HRESULT get_DistanceEH(double*);
HRESULT put_DistanceEH(double);
```

C#

```
public double DistanceEH { get; set; }
```

Visual Basic

```
Public DistanceEH As double
```

Description

Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

V.7.2.4 DistanceEV

C++

```
HRESULT get_DistanceEV(double*);
HRESULT put_DistanceEV(double);
```

C#

```
public double DistanceEV { get; set; }
```

Visual Basic

```
Public DistanceEV As double
```

Description

Available since version 1.7.

V.7.2.5 H1

C++

```
HRESULT get_H1(double*);
HRESULT put_H1(double);
```

C#

```
public double H1 { get; set; }
```

Visual Basic

```
Public H1 As double
```

Description

Available since version ARSA 2010.

Version

Available since version 10.

V.7.2.6 H2

C++

```
HRESULT get_H2(double*);
HRESULT put_H2(double);
```

C#

```
public double H2 { get; set; }
```

Visual Basic

```
Public H2 As double
```

Description

Available since version ARSA 2010.

Version

Available since version 10.

V.7.2.7 H3

C++

```
HRESULT get_H3(double*);
HRESULT put_H3(double);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double H3 { get; set; }
```

Visual Basic

```
Public H3 As double
```

Description

Available since version ARSA 2010.

Version

Available since version 10.

V.7.2.8 H4

C++

```
HRESULT get_H4(double*);
HRESULT put_H4(double);
```

C#

```
public double H4 { get; set; }
```

Visual Basic

```
Public H4 As double
```

Description

Available since version ARSA 2010.

Version

Available since version 10.

V.7.2.9 Height

C++

```
HRESULT get_Height(double*);
HRESULT put_Height(double);
```

C#

```
public double Height { get; set; }
```

Visual Basic

```
Public Height As double
```

Description

Available since version 1.7.

V.7.2.10 Material

C++

```
HRESULT get_Material(BSTR*);
HRESULT put_Material(BSTR);
```

C#

```
public String Material { get; set; }
```

Visual Basic

```
Public Material As String
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Available since version 1.7.

V.7.2.11 Thick

C++

```
HRESULT get_Thick(double*);
HRESULT put_Thick(double);
```

C#

```
public double Thick { get; set; }
```

Visual Basic

```
Public Thick As double
```

Description

Available since version 1.7.

V.7.2.12 V1

C++

```
HRESULT get_V1(double*);
HRESULT put_V1(double);
```

C#

```
public double V1 { get; set; }
```

Visual Basic

```
Public V1 As double
```

Description

Available since version ARSA 2010.

Version

Available since version 10.

V.7.2.13 V2

C++

```
HRESULT get_V2(double*);
HRESULT put_V2(double);
```

C#

```
public double V2 { get; set; }
```

Visual Basic

```
Public V2 As double
```

Description

Available since version ARSA 2010.

Version

Available since version 10.

(c) 2014 Autodesk Inc. All rights reserved

V.7.2.14 V3

C++

```
HRESULT get_V3(double*);
HRESULT put_V3(double);
```

C#

```
public double V3 { get; set; }
```

Visual Basic

```
Public V3 As double
```

Description

Available since version ARSA 2010.

Version

Available since version 10.

V.7.2.15 V4

C++

```
HRESULT get_V4(double*);
HRESULT put_V4(double);
```

C#

```
public double V4 { get; set; }
```

Visual Basic

```
Public V4 As double
```

Description

Available since version ARSA 2010.

Version

Available since version 10.

V.7.2.16 Width

C++

```
HRESULT get_Width(double*);
HRESULT put_Width(double);
```

C#

```
public double Width { get; set; }
```

Visual Basic

```
Public Width As double
```

Description

Available since version 1.7.

V.8 IRJointGussetCornersType

C++

```
enum IRJointGussetCornersType;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public enum IRJointGussetCornersType;
```

Visual Basic

```
Public Enum IRJointGussetCornersType
```

Members

Members	Description
I_JGCT_ACUTE = 0	Available since version 1.7.
I_JGCT_CUT_IN = 1	Available since version 1.7.
I_JGCT_CUT_OUT = 2	Available since version 1.7.

V.9 IRJointGussetSimpleAttachment

Class Hierarchy

RobotOM.IRJointGussetSimpleAttachment

C++

```
interface IRJointGussetSimpleAttachment : IDispatch;
```

C#

```
public interface IRJointGussetSimpleAttachment;
```

Visual Basic

```
Public Interface IRJointGussetSimpleAttachment
```

V.9.1 IRJointGussetSimpleAttachment Members

The following tables list the members exposed by IRJointGussetSimpleAttachment.

Public Fields

	Name	Description
	BoltsAttach (see page 2079)	Available since version 1.7.
	FixType (see page 2079)	Available since version 1.7.
	WeldsAttach (see page 2079)	Available since version 1.7.
	WeldType (see page 2080)	Available since version ARSA 2010.

V.9.2 IRJointGussetSimpleAttachment Fields

The fields of the IRJointGussetSimpleAttachment class are listed here.

Public Fields

	Name	Description
	BoltsAttach (see page 2079)	Available since version 1.7.
	FixType (see page 2079)	Available since version 1.7.
	WeldsAttach (see page 2079)	Available since version 1.7.
	WeldType (see page 2080)	Available since version ARSA 2010.

V.9.2.1 BoltsAttach

C++

```
HRESULT get_BoltsAttach(IRJointGussetSimpleAttachBolts**);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRJointGussetSimpleAttachBolts BoltsAttach { get; }
```

Visual Basic

```
Public ReadOnly BoltsAttach As IRJointGussetSimpleAttachBolts
```

Description

Available since version 1.7.

V.9.2.2 FixType

C++

```
HRESULT get_FixType(IRJointGussetFixType*);
HRESULT put_FixType(IRJointGussetFixType);
```

C#

```
public IRJointGussetFixType FixType { get; set; }
```

Visual Basic

```
Public FixType As IRJointGussetFixType
```

Description

Available since version 1.7.

V.9.2.3 WeldsAttach

C++

```
HRESULT get_WeldsAttach(IRJointGussetSimpleAttachWelds**);
```

C#

```
public IRJointGussetSimpleAttachWelds WeldsAttach { get; }
```

Visual Basic

```
Public ReadOnly WeldsAttach As IRJointGussetSimpleAttachWelds
```

Description

Available since version 1.7.

V.9.2.4 WeldType

C++

```
HRESULT get_WeldType(RJointWeldType*);
HRESULT put_WeldType(RJointWeldType);
```

C#

```
public RJointWeldType WeldType { get; set; }
```

Visual Basic

```
Public WeldType As RJointWeldType
```

Description

Available since version ARSA 2010.

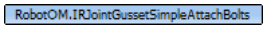
Version

Available since version 10.

(c) 2014 Autodesk Inc. All rights reserved

V.10 IRJointGussetSimpleAttachBolts

Class Hierarchy



C++

```
interface IRJointGussetSimpleAttachBolts : IDispatch;
```

C#

```
public interface IRJointGussetSimpleAttachBolts;
```

Visual Basic

```
Public Interface IRJointGussetSimpleAttachBolts
```

V.10.1 IRJointGussetSimpleAttachBolts Members

The following tables list the members exposed by IRJointGussetSimpleAttachBolts.

Public Fields

	Name	Description
	ClassName (see page 2081)	Available since version 1.7.
	Diameter (see page 2081)	Available since version 1.7.
	Friction (see page 2081)	Available since version 1.7.
	Horizontal (see page 2081)	Available since version 1.7.
	Vertical (see page 2082)	Available since version 1.7.

V.10.2 IRJointGussetSimpleAttachBolts Fields

The fields of the IRJointGussetSimpleAttachBolts class are listed here.

Public Fields

	Name	Description
	ClassName (see page 2081)	Available since version 1.7.
	Diameter (see page 2081)	Available since version 1.7.
	Friction (see page 2081)	Available since version 1.7.
	Horizontal (see page 2081)	Available since version 1.7.
	Vertical (see page 2082)	Available since version 1.7.

V.10.2.1 ClassName

C++

```
HRESULT get_ClassName(BSTR*);
HRESULT put_ClassName(BSTR);
```

C#

```
public String ClassName { get; set; }
```

Visual Basic

```
Public ClassName As String
```

Description

Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

V.10.2.2 Diameter

C++

```
HRESULT get_Diameter(double*);
HRESULT put_Diameter(double);
```

C#

```
public double Diameter { get; set; }
```

Visual Basic

```
Public Diameter As double
```

Description

Available since version 1.7.

V.10.2.3 Friction

C++

```
HRESULT get_Friction(double*);
HRESULT put_Friction(double);
```

C#

```
public double Friction { get; set; }
```

Visual Basic

```
Public Friction As double
```

Description

Available since version 1.7.

V.10.2.4 Horizontal

C++

```
HRESULT get_Horizontal(IRJointGussetSimpleAttachBoltsHorizontal**);
```

C#

```
public IRJointGussetSimpleAttachBoltsHorizontal Horizontal { get; }
```

Visual Basic

```
Public ReadOnly Horizontal As IRJointGussetSimpleAttachBoltsHorizontal
```

Description

Available since version 1.7.

V.10.2.5 Vertical

C++

```
HRESULT get_Vertical(IRJointGussetSimpleAttachBoltsVertical**);
```

C#

```
public IRJointGussetSimpleAttachBoltsVertical Vertical { get; }
```

Visual Basic

```
Public ReadOnly Vertical As IRJointGussetSimpleAttachBoltsVertical
```

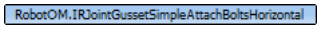
Description

Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

V.11 IRJointGussetSimpleAttachBoltsHorizontal

Class Hierarchy



C++

```
interface IRJointGussetSimpleAttachBoltsHorizontal : IDispatch;
```

C#

```
public interface IRJointGussetSimpleAttachBoltsHorizontal;
```

Visual Basic

```
Public Interface IRJointGussetSimpleAttachBoltsHorizontal
```

V.11.1 IRJointGussetSimpleAttachBoltsHorizontal Members

The following tables list the members exposed by IRJointGussetSimpleAttachBoltsHorizontal.

Public Fields

	Name	Description
	DistanceEB (see page 2083)	Available since version 1.7.
	DistanceH1 (see page 2083)	Available since version 1.7.
	Rows (see page 2083)	Available since version 1.7.
	Spacing (see page 2083)	Available since version 1.7.

V.11.2 IRJointGussetSimpleAttachBoltsHorizontal Fields

The fields of the IRJointGussetSimpleAttachBoltsHorizontal class are listed here.

Public Fields

	Name	Description
	DistanceEB (see page 2083)	Available since version 1.7.
	DistanceH1 (see page 2083)	Available since version 1.7.
	Rows (see page 2083)	Available since version 1.7.
	Spacing (see page 2083)	Available since version 1.7.

V.11.2.1 DistanceEB

C++

```
HRESULT get_DistanceEB(double*);
HRESULT put_DistanceEB(double);
```

C#

```
public double DistanceEB { get; set; }
```

Visual Basic

```
Public DistanceEB As double
```

Description

Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

V.11.2.2 DistanceH1

C++

```
HRESULT get_DistanceH1(double*);
HRESULT put_DistanceH1(double);
```

C#

```
public double DistanceH1 { get; set; }
```

Visual Basic

```
Public DistanceH1 As double
```

Description

Available since version 1.7.

V.11.2.3 Rows

C++

```
HRESULT get_Rows(long*);
HRESULT put_Rows(long);
```

C#

```
public long Rows { get; set; }
```

Visual Basic

```
Public Rows As long
```

Description

Available since version 1.7.

V.11.2.4 Spacing

C++

```
HRESULT get_Spacing(IRobotValuesArray**);
```

C#

```
public IRobotValuesArray Spacing { get; }
```

Visual Basic

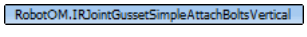
```
Public ReadOnly Spacing As IRobotValuesArray
```

Description

Available since version 1.7.

V.12 IRJointGussetSimpleAttachBoltsVertical

Class Hierarchy



C++

```
interface IRJointGussetSimpleAttachBoltsVertical : IDispatch;
```

C#

```
public interface IRJointGussetSimpleAttachBoltsVertical;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Interface IRJointGussetSimpleAttachBoltsVertical
```

V.12.1 IRJointGussetSimpleAttachBoltsVertical Members

The following tables list the members exposed by IRJointGussetSimpleAttachBoltsVertical.

Public Fields

	Name	Description
	DistanceEB (see page 2084)	Available since version 1.7.
	DistanceH1 (see page 2085)	Available since version 1.7.
	Rows (see page 2085)	Available since version 1.7.
	Spacing (see page 2085)	Available since version 1.7.

V.12.2 IRJointGussetSimpleAttachBoltsVertical Fields

The fields of the IRJointGussetSimpleAttachBoltsVertical class are listed here.

Public Fields

	Name	Description
	DistanceEB (see page 2084)	Available since version 1.7.
	DistanceH1 (see page 2085)	Available since version 1.7.
	Rows (see page 2085)	Available since version 1.7.
	Spacing (see page 2085)	Available since version 1.7.

V.12.2.1 DistanceEB

C++

```
HRESULT get_DistanceEB(double*);
HRESULT put_DistanceEB(double);
```

C#

```
public double DistanceEB { get; set; }
```

Visual Basic

```
Public DistanceEB As double
```

Description

Available since version 1.7.

V.12.2.2 DistanceH1

C++

```
HRESULT get_DistanceH1(double*);
HRESULT put_DistanceH1(double);
```

C#

```
public double DistanceH1 { get; set; }
```

Visual Basic

```
Public DistanceH1 As double
```

Description

Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

V.12.2.3 Rows

C++

```
HRESULT get_Rows(long*);
HRESULT put_Rows(long);
```

C#

```
public long Rows { get; set; }
```

Visual Basic

```
Public Rows As long
```

Description

Available since version 1.7.

V.12.2.4 Spacing

C++

```
HRESULT get_Spacing(IRobotValuesArray**);
```

C#

```
public IRobotValuesArray Spacing { get; }
```

Visual Basic

```
Public ReadOnly Spacing As IRobotValuesArray
```

Description

Available since version 1.7.

V.13 IRJointGussetSimpleAttachWelds

Class Hierarchy

RobotOM.IRJointGussetSimpleAttachWelds

C++

```
interface IRJointGussetSimpleAttachWelds : IDispatch;
```

C#

```
public interface IRJointGussetSimpleAttachWelds;
```

Visual Basic

```
Public Interface IRJointGussetSimpleAttachWelds
```

V.13.1 IRJointGussetSimpleAttachWelds Members

The following tables list the members exposed by IRJointGussetSimpleAttachWelds.

Public Fields

	Name	Description
	ThickEdgeA (see page 2086)	Available since version 1.7.
	ThickEdgeB (see page 2086)	Available since version 1.7.

V.13.2 IRJointGussetSimpleAttachWelds Fields

The fields of the IRJointGussetSimpleAttachWelds class are listed here.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	ThickEdgeA (see page 2086)	Available since version 1.7.
	ThickEdgeB (see page 2086)	Available since version 1.7.

V.13.2.1 ThickEdgeA

C++

```
HRESULT get_ThickEdgeA(double*);
HRESULT put_ThickEdgeA(double);
```

C#

```
public double ThickEdgeA { get; set; }
```

Visual Basic

```
Public ThickEdgeA As double
```

Description

Available since version 1.7.

V.13.2.2 ThickEdgeB

C++

```
HRESULT get_ThickEdgeB(double*);
HRESULT put_ThickEdgeB(double);
```

C#

```
public double ThickEdgeB { get; set; }
```

Visual Basic

```
Public ThickEdgeB As double
```

Description

Available since version 1.7.

V.14 IRJointGussetCrossProfileCutting

C++

```
enum IRJointGussetCrossProfileCutting;
```

C#

```
public enum IRJointGussetCrossProfileCutting;
```

Visual Basic

```
Public Enum IRJointGussetCrossProfileCutting
```

Members

Members	Description
I_JGCPC_CONTINUE_LEFTUPPER_RIGHTLOWER = 0	Available since version 1.7.
I_JGCPC_CONTINUE_LEFTLOWER_RIGHTUPPER = 1	Available since version 1.7.
I_JGCPC_ALL_CUTTED = 2	Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

V.15 IRJointGussetCrossPlate

Class Hierarchy

RobotOM.IRJointGussetCrossPlate

C++

```
interface IRJointGussetCrossPlate : IDispatch;
```

C#

```
public interface IRJointGussetCrossPlate;
```

Visual Basic

```
Public Interface IRJointGussetCrossPlate
```

V.15.1 IRJointGussetCrossPlate Members

The following tables list the members exposed by IRJointGussetCrossPlate.

Public Fields

	Name	Description
	B1 (see page 2089)	Available since version 1.7.
	B2 (see page 2089)	Available since version 1.7.
	B3 (see page 2089)	Available since version 1.7.
	B4 (see page 2089)	Available since version 1.7.
	CornersType (see page 2090)	Available since version 1.7.
	CutH1 (see page 2090)	Available since version ARSA 2010.
	CutH2 (see page 2090)	Available since version ARSA 2010.
	CutH3 (see page 2091)	Available since version ARSA 2010.
	CutH4 (see page 2091)	Available since version ARSA 2010.
	CutV1 (see page 2091)	Available since version ARSA 2010.
	CutV2 (see page 2092)	Available since version ARSA 2010.
	CutV3 (see page 2092)	Available since version ARSA 2010.
	CutV4 (see page 2092)	Available since version ARSA 2010.
	H1 (see page 2093)	Available since version 1.7.
	H2 (see page 2093)	Available since version 1.7.
	H3 (see page 2093)	Available since version 1.7.
	H4 (see page 2093)	Available since version 1.7.
	Height (see page 2094)	Available since version ARSA 2010.
	HOffset (see page 2094)	Available since version ARSA 2010.
	HorizontalOffset (see page 2094)	Available since version ARSA 2010.
	Length (see page 2095)	Available since version ARSA 2010.
	Material (see page 2095)	Available since version 1.7.
	Thick (see page 2095)	Available since version 1.7.
	VOffset (see page 2095)	Available since version ARSA 2010.

V.15.2 IRJointGussetCrossPlate Fields

The fields of the IRJointGussetCrossPlate class are listed here.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	B1 (see page 2089)	Available since version 1.7.
	B2 (see page 2089)	Available since version 1.7.
	B3 (see page 2089)	Available since version 1.7.
	B4 (see page 2089)	Available since version 1.7.
	CornersType (see page 2090)	Available since version 1.7.
	CutH1 (see page 2090)	Available since version ARSA 2010.
	CutH2 (see page 2090)	Available since version ARSA 2010.
	CutH3 (see page 2091)	Available since version ARSA 2010.
	CutH4 (see page 2091)	Available since version ARSA 2010.
	CutV1 (see page 2091)	Available since version ARSA 2010.
	CutV2 (see page 2092)	Available since version ARSA 2010.
	CutV3 (see page 2092)	Available since version ARSA 2010.
	CutV4 (see page 2092)	Available since version ARSA 2010.
	H1 (see page 2093)	Available since version 1.7.
	H2 (see page 2093)	Available since version 1.7.
	H3 (see page 2093)	Available since version 1.7.
	H4 (see page 2093)	Available since version 1.7.
	Height (see page 2094)	Available since version ARSA 2010.
	HOffset (see page 2094)	Available since version ARSA 2010.
	HorizontalOffset (see page 2094)	Available since version ARSA 2010.
	Length (see page 2095)	Available since version ARSA 2010.
	Material (see page 2095)	Available since version 1.7.
	Thick (see page 2095)	Available since version 1.7.
	VOffset (see page 2095)	Available since version ARSA 2010.

V.15.2.1 B1

C++

```
HRESULT get_B1(double*);
HRESULT put_B1(double);
```

C#

```
public double B1 { get; set; }
```

Visual Basic

```
Public B1 As double
```

Description

Available since version 1.7.

V.15.2.2 B2

C++

```
HRESULT get_B2(double*);
HRESULT put_B2(double);
```

C#

```
public double B2 { get; set; }
```

Visual Basic

```
Public B2 As double
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Available since version 1.7.

V.15.2.3 B3

C++

```
HRESULT get_B3(double*);
HRESULT put_B3(double);
```

C#

```
public double B3 { get; set; }
```

Visual Basic

```
Public B3 As double
```

Description

Available since version 1.7.

V.15.2.4 B4

C++

```
HRESULT get_B4(double*);
HRESULT put_B4(double);
```

C#

```
public double B4 { get; set; }
```

Visual Basic

```
Public B4 As double
```

Description

Available since version 1.7.

V.15.2.5 CornersType

C++

```
HRESULT get_CornersType(IRJointGussetCornersType*);
HRESULT put_CornersType(IRJointGussetCornersType);
```

C#

```
public IRJointGussetCornersType CornersType { get; set; }
```

Visual Basic

```
Public CornersType As IRJointGussetCornersType
```

Description

Available since version 1.7.

V.15.2.6 CutH1

C++

```
HRESULT get_CutH1(double*);
HRESULT put_CutH1(double);
```

C#

```
public double CutH1 { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public CutH1 As double
```

Description

Available since version ARSA 2010.

Version

Available since version 10.

V.15.2.7 CutH2

C++

```
HRESULT get_CutH2(double*);
HRESULT put_CutH2(double);
```

C#

```
public double CutH2 { get; set; }
```

Visual Basic

```
Public CutH2 As double
```

Description

Available since version ARSA 2010.

Version

Available since version 10.

V.15.2.8 CutH3

C++

```
HRESULT get_CutH3(double*);
HRESULT put_CutH3(double);
```

C#

```
public double CutH3 { get; set; }
```

Visual Basic

```
Public CutH3 As double
```

Description

Available since version ARSA 2010.

Version

Available since version 10.

V.15.2.9 CutH4

C++

```
HRESULT get_CutH4(double*);
HRESULT put_CutH4(double);
```

C#

```
public double CutH4 { get; set; }
```

Visual Basic

```
Public CutH4 As double
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Available since version ARSA 2010.

Version

Available since version 10.

V.15.2.10 CutV1

C++

```
HRESULT get_CutV1(double*);
HRESULT put_CutV1(double);
```

C#

```
public double CutV1 { get; set; }
```

Visual Basic

```
Public CutV1 As double
```

Description

Available since version ARSA 2010.

Version

Available since version 10.

V.15.2.11 CutV2

C++

```
HRESULT get_CutV2(double*);
HRESULT put_CutV2(double);
```

C#

```
public double CutV2 { get; set; }
```

Visual Basic

```
Public CutV2 As double
```

Description

Available since version ARSA 2010.

Version

Available since version 10.

V.15.2.12 CutV3

C++

```
HRESULT get_CutV3(double*);
HRESULT put_CutV3(double);
```

C#

```
public double CutV3 { get; set; }
```

Visual Basic

```
Public CutV3 As double
```

Description

Available since version ARSA 2010.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 10.

V.15.2.13 CutV4

C++

```
HRESULT get_CutV4(double*);
HRESULT put_CutV4(double);
```

C#

```
public double CutV4 { get; set; }
```

Visual Basic

```
Public CutV4 As double
```

Description

Available since version ARSA 2010.

Version

Available since version 10.

V.15.2.14 H1

C++

```
HRESULT get_H1(double*);
HRESULT put_H1(double);
```

C#

```
public double H1 { get; set; }
```

Visual Basic

```
Public H1 As double
```

Description

Available since version 1.7.

V.15.2.15 H2

C++

```
HRESULT get_H2(double*);
HRESULT put_H2(double);
```

C#

```
public double H2 { get; set; }
```

Visual Basic

```
Public H2 As double
```

Description

Available since version 1.7.

V.15.2.16 H3

C++

```
HRESULT get_H3(double*);
HRESULT put_H3(double);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double H3 { get; set; }
```

Visual Basic

```
Public H3 As double
```

Description

Available since version 1.7.

V.15.2.17 H4

C++

```
HRESULT get_H4(double*);
HRESULT put_H4(double);
```

C#

```
public double H4 { get; set; }
```

Visual Basic

```
Public H4 As double
```

Description

Available since version 1.7.

V.15.2.18 Height

C++

```
HRESULT get_Height(double*);
HRESULT put_Height(double);
```

C#

```
public double Height { get; set; }
```

Visual Basic

```
Public Height As double
```

Description

Available since version ARSA 2010.

Version

Available since version 10.

V.15.2.19 HOffset

C++

```
HRESULT get_HOffset(double*);
HRESULT put_HOffset(double);
```

C#

```
public double HOffset { get; set; }
```

Visual Basic

```
Public HOffset As double
```

Description

Available since version ARSA 2010.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 10.

V.15.2.20 HorizontalOffset

C++

```
HRESULT get_HorizontalOffset(double*);
HRESULT put_HorizontalOffset(double);
```

C#

```
public double HorizontalOffset { get; set; }
```

Visual Basic

```
Public HorizontalOffset As double
```

Description

Available since version ARSA 2010.

Version

Available since version 10.

V.15.2.21 Length

C++

```
HRESULT get_Length(double*);
HRESULT put_Length(double);
```

C#

```
public double Length { get; set; }
```

Visual Basic

```
Public Length As double
```

Description

Available since version ARSA 2010.

Version

Available since version 10.

V.15.2.22 Material

C++

```
HRESULT get_Material(BSTR*);
HRESULT put_Material(BSTR);
```

C#

```
public String Material { get; set; }
```

Visual Basic

```
Public Material As String
```

Description

Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

V.15.2.23 Thick

C++

```
HRESULT get_Thick(double*);
HRESULT put_Thick(double);
```

C#

```
public double Thick { get; set; }
```

Visual Basic

```
Public Thick As double
```

Description

Available since version 1.7.

V.15.2.24 VOffset

C++

```
HRESULT get_VOffset(double*);
HRESULT put_VOffset(double);
```

C#

```
public double VOffset { get; set; }
```

Visual Basic

```
Public VOffset As double
```

Description

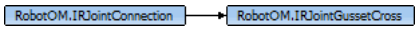
Available since version ARSA 2010.

Version

Available since version 10.

V.16 IRJointGussetCross

Class Hierarchy



C++

```
interface IRJointGussetCross : IRJointConnection;
```

C#

```
public interface IRJointGussetCross : IRJointConnection;
```

Visual Basic

```
Public Interface IRJointGussetCross
```

V.16.1 IRJointGussetCross Members

The following tables list the members exposed by IRJointGussetCross.

Public Fields

	Name	Description
	BoltsClassVector (see page 2097)	Available since ARSA 2010.

(c) 2014 Autodesk Inc. All rights reserved

	BoltsDimensionNamesVector (see page 2098)	Available since ARSA 2010.
	BoltsDimensionsVector (see page 2098)	Available since ARSA 2010.
	DiagLeftLower (see page 2098)	Available since version 1.7.
	DiagLeftUpper (see page 2099)	Available since version 1.7.
	DiagRightLower (see page 2099)	Available since version 1.7.
	DiagRightUpper (see page 2099)	Available since version 1.7.
	FixType_LeftL_RightU (see page 2099)	Available since version 1.7.
	FixType_LeftLower (see page 2100)	Available since ARSA 2010.
	FixType_LeftU_RightL (see page 2100)	Available since version 1.7.
	FixType_LeftUpper (see page 2100)	Available since ARSA 2010.
	FixType_RightLower (see page 2101)	Available since version ARSA 2010.
	FixType_RightUpper (see page 2101)	Available since ARSA 2010.
	GussetPlate (see page 2101)	Available since version 1.7.
	ProfileCutting (see page 2101)	Available since version 1.7.
	Type (see page 2160)	A connection type.
	WFRelPos (see page 2161)	A type of relative position of a beam and a column.

Public Methods

	Name	Description
	GetFromRobot (see page 2161)	The function gets connection parameters connection parameters from Robot and fills an object with the data.
	SetToRobot (see page 2161)	The function sets and saves connection parameter in Robot.

V.16.2 IRJointGussetCross Fields

The fields of the IRJointGussetCross class are listed here.

Public Fields

	Name	Description
	BoltsClassVector (see page 2097)	Available since ARSA 2010.
	BoltsDimensionNamesVector (see page 2098)	Available since ARSA 2010.
	BoltsDimensionsVector (see page 2098)	Available since ARSA 2010.
	DiagLeftLower (see page 2098)	Available since version 1.7.
	DiagLeftUpper (see page 2099)	Available since version 1.7.
	DiagRightLower (see page 2099)	Available since version 1.7.
	DiagRightUpper (see page 2099)	Available since version 1.7.
	FixType_LeftL_RightU (see page 2099)	Available since version 1.7.
	FixType_LeftLower (see page 2100)	Available since ARSA 2010.
	FixType_LeftU_RightL (see page 2100)	Available since version 1.7.
	FixType_LeftUpper (see page 2100)	Available since ARSA 2010.

(c) 2014 Autodesk Inc. All rights reserved

	FixType_RightLower (see page 2101)	Available since version ARSA 2010.
	FixType_RightUpper (see page 2101)	Available since ARSA 2010.
	GussetPlate (see page 2101)	Available since version 1.7.
	ProfileCutting (see page 2101)	Available since version 1.7.

V.16.2.1 BoltsClassVector

C++

```
HRESULT get_BoltsClassVector(IRobotNamesArray**);
```

C#

```
public IRobotNamesArray BoltsClassVector { get; }
```

Visual Basic

```
Public ReadOnly BoltsClassVector As IRobotNamesArray
```

Description

Available since ARSA 2010.

Version

Available since version 10.

V.16.2.2 BoltsDimensionNamesVector

C++

```
HRESULT get_BoltsDimensionNamesVector(IRobotNamesArray**);
```

C#

```
public IRobotNamesArray BoltsDimensionNamesVector { get; }
```

Visual Basic

```
Public ReadOnly BoltsDimensionNamesVector As IRobotNamesArray
```

Description

Available since ARSA 2010.

Version

Available since version 10.

V.16.2.3 BoltsDimensionsVector

C++

```
HRESULT get_BoltsDimensionsVector(IRobotValuesArray**);
```

C#

```
public IRobotValuesArray BoltsDimensionsVector { get; }
```

Visual Basic

```
Public ReadOnly BoltsDimensionsVector As IRobotValuesArray
```

Description

Available since ARSA 2010.

Version

Available since version 10.

(c) 2014 Autodesk Inc. All rights reserved

V.16.2.4 DiagLeftLower

C++

```
HRESULT get_DiagLeftLower(IRJointGussetDiagonale**);
```

C#

```
public IRJointGussetDiagonale DiagLeftLower { get; }
```

Visual Basic

```
Public ReadOnly DiagLeftLower As IRJointGussetDiagonale
```

Description

Available since version 1.7.

V.16.2.5 DiagLeftUpper

C++

```
HRESULT get_DiagLeftUpper(IRJointGussetDiagonale**);
```

C#

```
public IRJointGussetDiagonale DiagLeftUpper { get; }
```

Visual Basic

```
Public ReadOnly DiagLeftUpper As IRJointGussetDiagonale
```

Description

Available since version 1.7.

V.16.2.6 DiagRightLower

C++

```
HRESULT get_DiagRightLower(IRJointGussetDiagonale**);
```

C#

```
public IRJointGussetDiagonale DiagRightLower { get; }
```

Visual Basic

```
Public ReadOnly DiagRightLower As IRJointGussetDiagonale
```

Description

Available since version 1.7.

V.16.2.7 DiagRightUpper

C++

```
HRESULT get_DiagRightUpper(IRJointGussetDiagonale**);
```

C#

```
public IRJointGussetDiagonale DiagRightUpper { get; }
```

Visual Basic

```
Public ReadOnly DiagRightUpper As IRJointGussetDiagonale
```

Description

Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

V.16.2.8 FixType_LeftL_RightU

C++

```
HRESULT get_FixType_LeftL_RightU(IRJointGussetFixType*);
HRESULT put_FixType_LeftL_RightU(IRJointGussetFixType);
```

C#

```
public IRJointGussetFixType FixType_LeftL_RightU { get; set; }
```

Visual Basic

```
Public FixType_LeftL_RightU As IRJointGussetFixType
```

Description

Available since version 1.7.

V.16.2.9 FixType_LeftLower

C++

```
HRESULT get_FixType_LeftLower(IRJointGussetFixType*);
HRESULT put_FixType_LeftLower(IRJointGussetFixType);
```

C#

```
public IRJointGussetFixType FixType_LeftLower { get; set; }
```

Visual Basic

```
Public FixType_LeftLower As IRJointGussetFixType
```

Description

Available since ARSA 2010.

Version

Available since version 10.

V.16.2.10 FixType_LeftU_RightL

C++

```
HRESULT get_FixType_LeftU_RightL(IRJointGussetFixType*);
HRESULT put_FixType_LeftU_RightL(IRJointGussetFixType);
```

C#

```
public IRJointGussetFixType FixType_LeftU_RightL { get; set; }
```

Visual Basic

```
Public FixType_LeftU_RightL As IRJointGussetFixType
```

Description

Available since version 1.7.

V.16.2.11 FixType_LeftUpper

C++

```
HRESULT get_FixType_LeftUpper(IRJointGussetFixType*);
HRESULT put_FixType_LeftUpper(IRJointGussetFixType);
```

C#

```
public IRJointGussetFixType FixType_LeftUpper { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public FixType_LeftUpper As IRJointGussetFixType
```

Description

Available since ARSA 2010.

Version

Available since version 10.

V.16.2.12 FixType_RightLower

C++

```
HRESULT get_FixType_RightLower(IRJointGussetFixType*);
HRESULT put_FixType_RightLower(IRJointGussetFixType);
```

C#

```
public IRJointGussetFixType FixType_RightLower { get; set; }
```

Visual Basic

```
Public FixType_RightLower As IRJointGussetFixType
```

Description

Available since version ARSA 2010.

Version

Available since version 10.

V.16.2.13 FixType_RightUpper

C++

```
HRESULT get_FixType_RightUpper(IRJointGussetFixType*);
HRESULT put_FixType_RightUpper(IRJointGussetFixType);
```

C#

```
public IRJointGussetFixType FixType_RightUpper { get; set; }
```

Visual Basic

```
Public FixType_RightUpper As IRJointGussetFixType
```

Description

Available since ARSA 2010.

Version

Available since version 10.

V.16.2.14 GussetPlate

C++

```
HRESULT get_GussetPlate(IRJointGussetCrossPlate**);
```

C#

```
public IRJointGussetCrossPlate GussetPlate { get; }
```

Visual Basic

```
Public ReadOnly GussetPlate As IRJointGussetCrossPlate
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Available since version 1.7.

V.16.2.15 ProfileCutting

C++

```
HRESULT get_ProfileCutting(IRJointGussetCrossProfileCutting*);
HRESULT put_ProfileCutting(IRJointGussetCrossProfileCutting);
```

C#

```
public IRJointGussetCrossProfileCutting ProfileCutting { get; set; }
```

Visual Basic

```
Public ProfileCutting As IRJointGussetCrossProfileCutting
```

Description

Available since version 1.7.

V.17 IRJointGussetFlangeProfileCutting

C++

```
enum IRJointGussetFlangeProfileCutting;
```

C#

```
public enum IRJointGussetFlangeProfileCutting;
```

Visual Basic

```
Public Enum IRJointGussetFlangeProfileCutting
```

Members

Members	Description
I_JGFPC_CONTINUOUS = 0	Available since version 1.7.
I_JGFPC_CUTTED = 1	Available since version 1.7.

V.18 IRJointGussetFlangePlate

Class Hierarchy

RobotOM.IRJointGussetFlangePlate

C++

```
interface IRJointGussetFlangePlate : IDispatch;
```

C#

```
public interface IRJointGussetFlangePlate;
```

Visual Basic

```
Public Interface IRJointGussetFlangePlate
```

V.18.1 IRJointGussetFlangePlate Members

The following tables list the members exposed by IRJointGussetFlangePlate.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	B1 (see page 2104)	Available since version 1.7.
	B2 (see page 2104)	Available since version 1.7.
	B3 (see page 2104)	Available since version 1.7.
	B4 (see page 2104)	Available since version 1.7.
	CornersType (see page 2105)	Available since version 1.7.
	CutH1 (see page 2105)	Available since ARSA 2010.
	CutH2 (see page 2105)	Available since ARSA 2010.
	CutH3 (see page 2106)	Available since ARSA 2010.
	CutH4 (see page 2106)	Available since ARSA 2010.
	CutV1 (see page 2106)	Available since version ARSA 2010.
	CutV2 (see page 2107)	Available since ARSA 2010.
	CutV3 (see page 2107)	Available since ARSA 2010.
	CutV4 (see page 2107)	Available since ARSA 2010.
	EH (see page 2108)	Available since version 1.7.
	EV (see page 2108)	Available since version 1.7.
	Geight (see page 2108)	Available since version ARSA 2010.
	H1 (see page 2108)	Available since version 1.7.
	H2 (see page 2109)	Available since version 1.7.
	H3 (see page 2109)	Available since version 1.7.
	H4 (see page 2109)	Available since version 1.7.
	HOffset (see page 2109)	Available since version ARSA 2010.
	Length (see page 2110)	Available since version ARSA 2010.
	Material (see page 2110)	Available since version 1.7.
	NewPlateDefinition (see page 2110)	Available since version ARSA 2010.
	Thick (see page 2111)	Available since version 1.7.
	Type (see page 2111)	Available since version 1.7.
	VOffset (see page 2111)	Available since version ARSA 2010.

V.18.2 IRJointGussetFlangePlate Fields

The fields of the IRJointGussetFlangePlate class are listed here.

Public Fields

	Name	Description
	B1 (see page 2104)	Available since version 1.7.
	B2 (see page 2104)	Available since version 1.7.
	B3 (see page 2104)	Available since version 1.7.
	B4 (see page 2104)	Available since version 1.7.
	CornersType (see page 2105)	Available since version 1.7.
	CutH1 (see page 2105)	Available since ARSA 2010.
	CutH2 (see page 2105)	Available since ARSA 2010.
	CutH3 (see page 2106)	Available since ARSA 2010.
	CutH4 (see page 2106)	Available since ARSA 2010.
	CutV1 (see page 2106)	Available since version ARSA 2010.
	CutV2 (see page 2107)	Available since ARSA 2010.
	CutV3 (see page 2107)	Available since ARSA 2010.
	CutV4 (see page 2107)	Available since ARSA 2010.

(c) 2014 Autodesk Inc. All rights reserved

	EH (see page 2108)	Available since version 1.7.
	EV (see page 2108)	Available since version 1.7.
	Geight (see page 2108)	Available since version ARSA 2010.
	H1 (see page 2108)	Available since version 1.7.
	H2 (see page 2109)	Available since version 1.7.
	H3 (see page 2109)	Available since version 1.7.
	H4 (see page 2109)	Available since version 1.7.
	HOffset (see page 2109)	Available since version ARSA 2010.
	Length (see page 2110)	Available since version ARSA 2010.
	Material (see page 2110)	Available since version 1.7.
	NewPlateDefinition (see page 2110)	Available since version ARSA 2010.
	Thick (see page 2111)	Available since version 1.7.
	Type (see page 2111)	Available since version 1.7.
	VOffset (see page 2111)	Available since version ARSA 2010.

V.18.2.1 B1

C++

```
HRESULT get_B1(double*);
HRESULT put_B1(double);
```

C#

```
public double B1 { get; set; }
```

Visual Basic

```
Public B1 As double
```

Description

Available since version 1.7.

V.18.2.2 B2

C++

```
HRESULT get_B2(double*);
HRESULT put_B2(double);
```

C#

```
public double B2 { get; set; }
```

Visual Basic

```
Public B2 As double
```

Description

Available since version 1.7.

V.18.2.3 B3

C++

```
HRESULT get_B3(double*);
HRESULT put_B3(double);
```

C#

```
public double B3 { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public B3 As double
```

Description

Available since version 1.7.

V.18.2.4 B4

C++

```
HRESULT get_B4(double*);
HRESULT put_B4(double);
```

C#

```
public double B4 { get; set; }
```

Visual Basic

```
Public B4 As double
```

Description

Available since version 1.7.

V.18.2.5 CornersType

C++

```
HRESULT get_CornersType(IRJointGussetCornersType*);
HRESULT put_CornersType(IRJointGussetCornersType);
```

C#

```
public IRJointGussetCornersType CornersType { get; set; }
```

Visual Basic

```
Public CornersType As IRJointGussetCornersType
```

Description

Available since version 1.7.

V.18.2.6 CutH1

C++

```
HRESULT get_CutH1(double*);
HRESULT put_CutH1(double);
```

C#

```
public double CutH1 { get; set; }
```

Visual Basic

```
Public CutH1 As double
```

Description

Available since ARSA 2010.

Version

Available since version 10.

(c) 2014 Autodesk Inc. All rights reserved

V.18.2.7 CutH2

C++

```
HRESULT get_CutH2(double*);
HRESULT put_CutH2(double);
```

C#

```
public double CutH2 { get; set; }
```

Visual Basic

```
Public CutH2 As double
```

Description

Available since ARSA 2010.

Version

Available since version 10.

V.18.2.8 CutH3

C++

```
HRESULT get_CutH3(double*);
HRESULT put_CutH3(double);
```

C#

```
public double CutH3 { get; set; }
```

Visual Basic

```
Public CutH3 As double
```

Description

Available since ARSA 2010.

Version

Available since version 10.

V.18.2.9 CutH4

C++

```
HRESULT get_CutH4(double*);
HRESULT put_CutH4(double);
```

C#

```
public double CutH4 { get; set; }
```

Visual Basic

```
Public CutH4 As double
```

Description

Available since ARSA 2010.

Version

Available since version 10.

(c) 2014 Autodesk Inc. All rights reserved

V.18.2.10 CutV1

C++

```
HRESULT get_CutV1(double*);
HRESULT put_CutV1(double);
```

C#

```
public double CutV1 { get; set; }
```

Visual Basic

```
Public CutV1 As double
```

Description

Available since version ARSA 2010.

Version

Available since version 10.

V.18.2.11 CutV2

C++

```
HRESULT get_CutV2(double*);
HRESULT put_CutV2(double);
```

C#

```
public double CutV2 { get; set; }
```

Visual Basic

```
Public CutV2 As double
```

Description

Available since ARSA 2010.

Version

Available since version 10.

V.18.2.12 CutV3

C++

```
HRESULT get_CutV3(double*);
HRESULT put_CutV3(double);
```

C#

```
public double CutV3 { get; set; }
```

Visual Basic

```
Public CutV3 As double
```

Description

Available since ARSA 2010.

Version

Available since version 10.

(c) 2014 Autodesk Inc. All rights reserved

V.18.2.13 CutV4

C++

```
HRESULT get_CutV4(double*);
HRESULT put_CutV4(double);
```

C#

```
public double CutV4 { get; set; }
```

Visual Basic

```
Public CutV4 As double
```

Description

Available since ARSA 2010.

Version

Available since version 10.

V.18.2.14 EH

C++

```
HRESULT get_EH(double*);
HRESULT put_EH(double);
```

C#

```
public double EH { get; set; }
```

Visual Basic

```
Public EH As double
```

Description

Available since version 1.7.

V.18.2.15 EV

C++

```
HRESULT get_EV(double*);
HRESULT put_EV(double);
```

C#

```
public double EV { get; set; }
```

Visual Basic

```
Public EV As double
```

Description

Available since version 1.7.

V.18.2.16 Geight

C++

```
HRESULT get_Geight(double*);
HRESULT put_Geight(double);
```

C#

```
public double Geight { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Geight As double
```

Description

Available since version ARSA 2010.

Version

Available since version 10.

V.18.2.17 H1

C++

```
HRESULT get_H1(double*);
HRESULT put_H1(double);
```

C#

```
public double H1 { get; set; }
```

Visual Basic

```
Public H1 As double
```

Description

Available since version 1.7.

V.18.2.18 H2

C++

```
HRESULT get_H2(double*);
HRESULT put_H2(double);
```

C#

```
public double H2 { get; set; }
```

Visual Basic

```
Public H2 As double
```

Description

Available since version 1.7.

V.18.2.19 H3

C++

```
HRESULT get_H3(double*);
HRESULT put_H3(double);
```

C#

```
public double H3 { get; set; }
```

Visual Basic

```
Public H3 As double
```

Description

Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

V.18.2.20 H4

C++

```
HRESULT get_H4(double*);
HRESULT put_H4(double);
```

C#

```
public double H4 { get; set; }
```

Visual Basic

```
Public H4 As double
```

Description

Available since version 1.7.

V.18.2.21 HOffset

C++

```
HRESULT get_HOffset(double*);
HRESULT put_HOffset(double);
```

C#

```
public double HOffset { get; set; }
```

Visual Basic

```
Public HOffset As double
```

Description

Available since version ARSA 2010.

Version

Available since version 10.

V.18.2.22 Length

C++

```
HRESULT get_Length(double*);
HRESULT put_Length(double);
```

C#

```
public double Length { get; set; }
```

Visual Basic

```
Public Length As double
```

Description

Available since version ARSA 2010.

Version

Available since version 10.

V.18.2.23 Material

C++

```
HRESULT get_Material(BSTR*);
HRESULT put_Material(BSTR);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public String Material { get; set; }
```

Visual Basic

```
Public Material As String
```

Description

Available since version 1.7.

V.18.2.24 NewPlateDefinition

C++

```
HRESULT get_NewPlateDefinition(VARIANT_BOOL*);
HRESULT put_NewPlateDefinition(VARIANT_BOOL);
```

C#

```
public bool NewPlateDefinition { get; set; }
```

Visual Basic

```
Public NewPlateDefinition As Boolean
```

Description

Available since version ARSA 2010.

Version

Available since version 10.

V.18.2.25 Thick

C++

```
HRESULT get_Thick(double*);
HRESULT put_Thick(double);
```

C#

```
public double Thick { get; set; }
```

Visual Basic

```
Public Thick As double
```

Description

Available since version 1.7.

V.18.2.26 Type

C++

```
HRESULT get_Type(IRJointGussetFlangePlateRegularType*);
HRESULT put_Type(IRJointGussetFlangePlateRegularType);
```

C#

```
public IRJointGussetFlangePlateRegularType Type { get; set; }
```

Visual Basic

```
Public Type As IRJointGussetFlangePlateRegularType
```

Description

Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

V.18.2.27 VOffset

C++

```
HRESULT get_VOffset(double*);
HRESULT put_VOffset(double);
```

C#

```
public double VOffset { get; set; }
```

Visual Basic

```
Public VOffset As double
```

Description

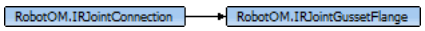
Available since version ARSA 2010.

Version

Available since version 10.

V.19 IRJointGussetFlange

Class Hierarchy



C++

```
interface IRJointGussetFlange : IRJointConnection;
```

C#

```
public interface IRJointGussetFlange : IRJointConnection;
```

Visual Basic

```
Public Interface IRJointGussetFlange
```

V.19.1 IRJointGussetFlange Members

The following tables list the members exposed by IRJointGussetFlange.

Public Fields

	Name	Description
	BoltsClassVector (see page 2113)	Available since version ARSA 2010.
	BoltsDimensionNamesVector (see page 2114)	Available since ARSA 2010.
	BoltsDimensionsVector (see page 2114)	Available since version ARSA 2010.
	DiagonalLeft (see page 2114)	Available since version 1.7.
	DiagonalRight (see page 2115)	Available since version 1.7.
	DiagonalUp (see page 2115)	Available since version 1.7.
	FixType_DiagonalLeft (see page 2115)	Available since version 1.7.
	FixType_DiagonalRight (see page 2115)	Available since version 1.7.
	FixType_DiagonalUp (see page 2116)	Available since version 1.7.
	FixType_FlangeLeft (see page 2116)	Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

	FixType_FlangeRight (see page 2116)	Available since version 1.7.
	FlangeLeft (see page 2116)	Available since version 1.7.
	FlangeRight (see page 2117)	Available since version 1.7.
	GussetPlate (see page 2117)	Available since version 1.7.
	ProfileCutting (see page 2117)	Available since version 1.7.
	Type (see page 2160)	A connection type.
	WFRelPos (see page 2161)	A type of relative position of a beam and a column.

Public Methods

	Name	Description
	GetFromRobot (see page 2161)	The function gets connection parameters connection parameters from Robot and fills an object with the data.
	SetToRobot (see page 2161)	The function sets and saves connection parameter in Robot.

V.19.2 IRJointGussetFlange Fields

The fields of the IRJointGussetFlange class are listed here.

Public Fields

	Name	Description
	BoltsClassVector (see page 2113)	Available since version ARSA 2010.
	BoltsDimensionNamesVector (see page 2114)	Available since ARSA 2010.
	BoltsDimensionsVector (see page 2114)	Available since version ARSA 2010.
	DiagonalLeft (see page 2114)	Available since version 1.7.
	DiagonalRight (see page 2115)	Available since version 1.7.
	DiagonalUp (see page 2115)	Available since version 1.7.
	FixType_DiagonalLeft (see page 2115)	Available since version 1.7.
	FixType_DiagonalRight (see page 2115)	Available since version 1.7.
	FixType_DiagonalUp (see page 2116)	Available since version 1.7.
	FixType_FlangeLeft (see page 2116)	Available since version 1.7.
	FixType_FlangeRight (see page 2116)	Available since version 1.7.
	FlangeLeft (see page 2116)	Available since version 1.7.
	FlangeRight (see page 2117)	Available since version 1.7.
	GussetPlate (see page 2117)	Available since version 1.7.
	ProfileCutting (see page 2117)	Available since version 1.7.

V.19.2.1 BoltsClassVector

C++

```
HRESULT get_BoltsClassVector(IRobotNamesArray**);
```

C#

```
public IRobotNamesArray BoltsClassVector { get; }
```

Visual Basic

```
Public ReadOnly BoltsClassVector As IRobotNamesArray
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Available since version ARSA 2010.

Version

Available since version 10.

V.19.2.2 BoltsDimensionNamesVector

C++

```
HRESULT get_BoltsDimensionNamesVector(IRobotNamesArray**);
```

C#

```
public IRobotNamesArray BoltsDimensionNamesVector { get; }
```

Visual Basic

```
Public ReadOnly BoltsDimensionNamesVector As IRobotNamesArray
```

Description

Available since ARSA 2010.

Version

Available since version 10.

V.19.2.3 BoltsDimensionsVector

C++

```
HRESULT get_BoltsDimensionsVector(IRobotValuesArray**);
```

C#

```
public IRobotValuesArray BoltsDimensionsVector { get; }
```

Visual Basic

```
Public ReadOnly BoltsDimensionsVector As IRobotValuesArray
```

Description

Available since version ARSA 2010.

Version

Available since version 10.

V.19.2.4 DiagonalLeft

C++

```
HRESULT get_DiagonalLeft(IRJointGussetDiagonale**);
```

C#

```
public IRJointGussetDiagonale DiagonalLeft { get; }
```

Visual Basic

```
Public ReadOnly DiagonalLeft As IRJointGussetDiagonale
```

Description

Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

V.19.2.5 DiagonalRight

C++

```
HRESULT get_DiagonalRight(IRJointGussetDiagonale**);
```

C#

```
public IRJointGussetDiagonale DiagonalRight { get; }
```

Visual Basic

```
Public ReadOnly DiagonalRight As IRJointGussetDiagonale
```

Description

Available since version 1.7.

V.19.2.6 DiagonalUp

C++

```
HRESULT get_DiagonalUp(IRJointGussetDiagonale**);
```

C#

```
public IRJointGussetDiagonale DiagonalUp { get; }
```

Visual Basic

```
Public ReadOnly DiagonalUp As IRJointGussetDiagonale
```

Description

Available since version 1.7.

V.19.2.7 FixType_DiagonalLeft

C++

```
HRESULT get_FixType_DiagonalLeft(IRJointGussetFixType*);
HRESULT put_FixType_DiagonalLeft(IRJointGussetFixType);
```

C#

```
public IRJointGussetFixType FixType_DiagonalLeft { get; set; }
```

Visual Basic

```
Public FixType_DiagonalLeft As IRJointGussetFixType
```

Description

Available since version 1.7.

V.19.2.8 FixType_DiagonalRight

C++

```
HRESULT get_FixType_DiagonalRight(IRJointGussetFixType*);
HRESULT put_FixType_DiagonalRight(IRJointGussetFixType);
```

C#

```
public IRJointGussetFixType FixType_DiagonalRight { get; set; }
```

Visual Basic

```
Public FixType_DiagonalRight As IRJointGussetFixType
```

Description

Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

V.19.2.9 FixType_DiagonalUp

C++

```
HRESULT get_FixType_DiagonalUp(IRJointGussetFixType*);
HRESULT put_FixType_DiagonalUp(IRJointGussetFixType);
```

C#

```
public IRJointGussetFixType FixType_DiagonalUp { get; set; }
```

Visual Basic

```
Public FixType_DiagonalUp As IRJointGussetFixType
```

Description

Available since version 1.7.

V.19.2.10 FixType_FlangeLeft

C++

```
HRESULT get_FixType_FlangeLeft(IRJointGussetFixType*);
HRESULT put_FixType_FlangeLeft(IRJointGussetFixType);
```

C#

```
public IRJointGussetFixType FixType_FlangeLeft { get; set; }
```

Visual Basic

```
Public FixType_FlangeLeft As IRJointGussetFixType
```

Description

Available since version 1.7.

V.19.2.11 FixType_FlangeRight

C++

```
HRESULT get_FixType_FlangeRight(IRJointGussetFixType*);
HRESULT put_FixType_FlangeRight(IRJointGussetFixType);
```

C#

```
public IRJointGussetFixType FixType_FlangeRight { get; set; }
```

Visual Basic

```
Public FixType_FlangeRight As IRJointGussetFixType
```

Description

Available since version 1.7.

V.19.2.12 FlangeLeft

C++

```
HRESULT get_FlangeLeft(IRJointGussetDiagonale**);
```

C#

```
public IRJointGussetDiagonale FlangeLeft { get; }
```

Visual Basic

```
Public ReadOnly FlangeLeft As IRJointGussetDiagonale
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Available since version 1.7.

V.19.2.13 FlangeRight

C++

```
HRESULT get_FlangeRight(IRJointGussetDiagonale**);
```

C#

```
public IRJointGussetDiagonale FlangeRight { get; }
```

Visual Basic

```
Public ReadOnly FlangeRight As IRJointGussetDiagonale
```

Description

Available since version 1.7.

V.19.2.14 GussetPlate

C++

```
HRESULT get_GussetPlate(IRJointGussetFlangePlate**);
```

C#

```
public IRJointGussetFlangePlate GussetPlate { get; }
```

Visual Basic

```
Public ReadOnly GussetPlate As IRJointGussetFlangePlate
```

Description

Available since version 1.7.

V.19.2.15 ProfileCutting

C++

```
HRESULT get_ProfileCutting(IRJointGussetFlangeProfileCutting*);
HRESULT put_ProfileCutting(IRJointGussetFlangeProfileCutting);
```

C#

```
public IRJointGussetFlangeProfileCutting ProfileCutting { get; set; }
```

Visual Basic

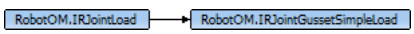
```
Public ProfileCutting As IRJointGussetFlangeProfileCutting
```

Description

Available since version 1.7.

V.20 IRJointGussetSimpleLoad

Class Hierarchy



C++

```
interface IRJointGussetSimpleLoad : IRJointLoad;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public interface IRJointGussetSimpleLoad : IRJointLoad;
```

Visual Basic

```
Public Interface IRJointGussetSimpleLoad
```

V.20.1 IRJointGussetSimpleLoad Members

The following tables list the members exposed by IRJointGussetSimpleLoad.

Public Fields

	Name	Description
	Cases (see page 2178)	Selection of load cases, defined for the structure. .
	Fx (see page 2118)	Available since version 1.7.
	Type (see page 2179)	Load type (defined manually or taken from Robot).

V.20.2 IRJointGussetSimpleLoad Fields

The fields of the IRJointGussetSimpleLoad class are listed here.

Public Fields

	Name	Description
	Fx (see page 2118)	Available since version 1.7.

V.20.2.1 Fx

C++

```
HRESULT get_Fx(double*);
HRESULT put_Fx(double);
```

C#

```
public double Fx { get; set; }
```

Visual Basic

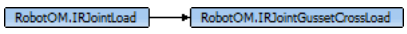
```
Public Fx As double
```

Description

Available since version 1.7.

V.21 IRJointGussetCrossLoad

Class Hierarchy



C++

```
interface IRJointGussetCrossLoad : IRJointLoad;
```

C#

```
public interface IRJointGussetCrossLoad : IRJointLoad;
```

Visual Basic

```
Public Interface IRJointGussetCrossLoad
```

(c) 2014 Autodesk Inc. All rights reserved

V.21.1 IRJointGussetCrossLoad Members

The following tables list the members exposed by IRJointGussetCrossLoad.

Public Fields

	Name	Description
	Cases (see page 2178)	Selection of load cases, defined for the structure. .
	Fx_LeftLower (see page 2119)	Available since version 1.7.
	Fx_LeftUpper (see page 2119)	Available since version 1.7.
	Fx_RightLower (see page 2120)	Available since version 1.7.
	Fx_RightUpper (see page 2120)	Available since version 1.7.
	Type (see page 2179)	Load type (defined manually or taken from Robot).

V.21.2 IRJointGussetCrossLoad Fields

The fields of the IRJointGussetCrossLoad class are listed here.

Public Fields

	Name	Description
	Fx_LeftLower (see page 2119)	Available since version 1.7.
	Fx_LeftUpper (see page 2119)	Available since version 1.7.
	Fx_RightLower (see page 2120)	Available since version 1.7.
	Fx_RightUpper (see page 2120)	Available since version 1.7.

V.21.2.1 Fx_LeftLower

C++

```
HRESULT get_Fx_LeftLower(double*);
HRESULT put_Fx_LeftLower(double);
```

C#

```
public double Fx_LeftLower { get; set; }
```

Visual Basic

```
Public Fx_LeftLower As double
```

Description

Available since version 1.7.

V.21.2.2 Fx_LeftUpper

C++

```
HRESULT get_Fx_LeftUpper(double*);
HRESULT put_Fx_LeftUpper(double);
```

C#

```
public double Fx_LeftUpper { get; set; }
```

Visual Basic

```
Public Fx_LeftUpper As double
```

Description

Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

V.21.2.3 Fx_RightLower

C++

```
HRESULT get_Fx_RightLower(double*);
HRESULT put_Fx_RightLower(double);
```

C#

```
public double Fx_RightLower { get; set; }
```

Visual Basic

```
Public Fx_RightLower As double
```

Description

Available since version 1.7.

V.21.2.4 Fx_RightUpper

C++

```
HRESULT get_Fx_RightUpper(double*);
HRESULT put_Fx_RightUpper(double);
```

C#

```
public double Fx_RightUpper { get; set; }
```

Visual Basic

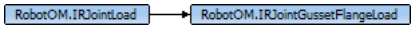
```
Public Fx_RightUpper As double
```

Description

Available since version 1.7.

V.22 IRJointGussetFlangeLoad

Class Hierarchy



C++

```
interface IRJointGussetFlangeLoad : IRJointLoad;
```

C#

```
public interface IRJointGussetFlangeLoad : IRJointLoad;
```

Visual Basic

```
Public Interface IRJointGussetFlangeLoad
```

V.22.1 IRJointGussetFlangeLoad Members

The following tables list the members exposed by IRJointGussetFlangeLoad.

Public Fields

	Name	Description
	Cases (see page 2178)	Selection of load cases, defined for the structure. .
	Fx_FlangeLeft (see page 2121)	Available since version 1.7.
	Fx_FlangeRight (see page 2121)	Available since version 1.7.
	Fx_LeftUpper (see page 2122)	Available since version 1.7.
	Fx_RightUpper (see page 2122)	Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

	Fx_Upper (see page 2122)	Available since version 1.7.
	Type (see page 2179)	Load type (defined manually or taken from Robot).

V.22.2 IRJointGussetFlangeLoad Fields

The fields of the IRJointGussetFlangeLoad class are listed here.

Public Fields

	Name	Description
	Fx_FlangeLeft (see page 2121)	Available since version 1.7.
	Fx_FlangeRight (see page 2121)	Available since version 1.7.
	Fx_LeftUpper (see page 2122)	Available since version 1.7.
	Fx_RightUpper (see page 2122)	Available since version 1.7.
	Fx_Upper (see page 2122)	Available since version 1.7.

V.22.2.1 Fx_FlangeLeft

C++

```
HRESULT get_Fx_FlangeLeft(double*);
HRESULT put_Fx_FlangeLeft(double);
```

C#

```
public double Fx_FlangeLeft { get; set; }
```

Visual Basic

```
Public Fx_FlangeLeft As double
```

Description

Available since version 1.7.

V.22.2.2 Fx_FlangeRight

C++

```
HRESULT get_Fx_FlangeRight(double*);
HRESULT put_Fx_FlangeRight(double);
```

C#

```
public double Fx_FlangeRight { get; set; }
```

Visual Basic

```
Public Fx_FlangeRight As double
```

Description

Available since version 1.7.

V.22.2.3 Fx_LeftUpper

C++

```
HRESULT get_Fx_LeftUpper(double*);
HRESULT put_Fx_LeftUpper(double);
```

C#

```
public double Fx_LeftUpper { get; set; }
```

Visual Basic

```
Public Fx_LeftUpper As double
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Available since version 1.7.

V.22.2.4 Fx_RightUpper

C++

```
HRESULT get_Fx_RightUpper(double*);
HRESULT put_Fx_RightUpper(double);
```

C#

```
public double Fx_RightUpper { get; set; }
```

Visual Basic

```
Public Fx_RightUpper As double
```

Description

Available since version 1.7.

V.22.2.5 Fx_Upper

C++

```
HRESULT get_Fx_Upper(double*);
HRESULT put_Fx_Upper(double);
```

C#

```
public double Fx_Upper { get; set; }
```

Visual Basic

```
Public Fx_Upper As double
```

Description

Available since version 1.7.

V.23 IRJointGussetDiagonalePositionType

C++

```
enum IRJointGussetDiagonalePositionType;
```

C#

```
public enum IRJointGussetDiagonalePositionType;
```

Visual Basic

```
Public Enum IRJointGussetDiagonalePositionType
```

Members

Members	Description
I_JGDPT_LONG_FLANGE = 0	Available since version 1.7.
I_JGDPT_SHORT_FLANGE = 1	Available since version 1.7.
I_JGDPT_LONG_FLANGE_INV = 2	Available since version 1.7.
I_JGDPT_SHORT_FLANGE_INV = 3	Available since version 1.7.

(c) 2014 Autodesk Inc. All rights reserved

V.24 IRJointGussetFlangePlateRegularType

C++

```
enum IRJointGussetFlangePlateRegularType;
```

C#

```
public enum IRJointGussetFlangePlateRegularType;
```

Visual Basic

```
Public Enum IRJointGussetFlangePlateRegularType
```

Members

Members	Description
I_JGFPRT_REGULAR = 0	Available since version 1.7.
I_JGFPRT_IRREGULAR = 1	Available since version 1.7.

VI Beam-principal beam connection

Available since version 7.5.

Enumerations

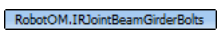
	Name	Description
	IRJointConnectorsType (see page 2152)	
	IRJointElementType (see page 2152)	

Interfaces

	Name	Description
	IRJointBeamGirderBolts (see page 2124)	Bolt description for stiffener and plate.
	IRJointBeamGirderStiffener (see page 2128)	Stiffener description.
	IRJointBeamGirderPlate (see page 2132)	Plate description.
	IRJointBeamGirderSeat (see page 2136)	Description of seat profile.
	IRJointBeamGirder (see page 2139)	Beam-principal beam connection.
	IRJointBeamGirderLoad (see page 2146)	
	IRJointBeamGirderSeatBolts (see page 2149)	
	IRJointBeamGirderAngle (see page 2153)	
	IRJointBeamGirderBeam (see page 2155)	

VI.1 IRJointBeamGirderBolts

Class Hierarchy



(c) 2014 Autodesk Inc. All rights reserved

C++

```
interface IRJointBeamGirderBolts : IDispatch;
```

C#

```
public interface IRJointBeamGirderBolts;
```

Visual Basic

```
Public Interface IRJointBeamGirderBolts
```

Description

Bolt description for stiffener and plate.

Version

Available since version 7.5.

VI.1.1 IRJointBeamGirderBolts Members

The following tables list the members exposed by IRJointBeamGirderBolts.

Public Fields

	Name	Description
	Area (see page 2125)	
	ClassName (see page 2125)	Bolt class.
	Cols (see page 2126)	
	Diameter (see page 2126)	Bolt diameter.
	DiameterName (see page 2126)	
	DistFromUpperBeamEdge (see page 2126)	Distance of 1st bolt row to upper beam edge.
	DistFromVertBeamEdge (see page 2127)	Distance of 1st bolt column to vertical beam edge.
	Friction (see page 2127)	
	Rows (see page 2127)	Number of bolt rows.
	SpacingH (see page 2128)	Horizontal spacing.
	SpacingV (see page 2128)	Vertical spacing.

VI.1.2 IRJointBeamGirderBolts Fields

The fields of the IRJointBeamGirderBolts class are listed here.

Public Fields

	Name	Description
	Area (see page 2125)	
	ClassName (see page 2125)	Bolt class.
	Cols (see page 2126)	
	Diameter (see page 2126)	Bolt diameter.
	DiameterName (see page 2126)	
	DistFromUpperBeamEdge (see page 2126)	Distance of 1st bolt row to upper beam edge.
	DistFromVertBeamEdge (see page 2127)	Distance of 1st bolt column to vertical beam edge.
	Friction (see page 2127)	
	Rows (see page 2127)	Number of bolt rows.
	SpacingH (see page 2128)	Horizontal spacing.

(c) 2014 Autodesk Inc. All rights reserved

	SpacingV (see page 2128)	Vertical spacing.

VI.1.2.1 Area

C++

```
HRESULT get_Area( double*);
```

C#

```
public  double Area { get; }
```

Visual Basic

```
Public ReadOnly Area As  double
```

Version

Available since version 7.5.

VI.1.2.2 ClassName

C++

```
HRESULT get_ClassName(BSTR*);
HRESULT put_ClassName(BSTR);
```

C#

```
public String ClassName { get; set; }
```

Visual Basic

```
Public ClassName As String
```

Description

Bolt class.

Version

Available since version 7.5.

VI.1.2.3 Cols

C++

```
HRESULT get_Cols( long*);
HRESULT put_Cols( long);
```

C#

```
public  long Cols { get; set; }
```

Visual Basic

```
Public Cols As  long
```

Version

Available since version 7.5.

VI.1.2.4 Diameter

C++

```
HRESULT get_Diameter( long*);
```

C#

```
public  long Diameter { get; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public ReadOnly Diameter As  long
```

Description

Bolt diameter.

Version

Available since version 7.5.

VI.1.2.5 DiameterName

C++

```
HRESULT get_DiameterName(BSTR*);
HRESULT put_DiameterName(BSTR);
```

C#

```
public String DiameterName { get; set; }
```

Visual Basic

```
Public DiameterName As String
```

Version

Available since version 7.5.

VI.1.2.6 DistFromUpperBeamEdge

C++

```
HRESULT get_DistFromUpperBeamEdge( double*);
HRESULT put_DistFromUpperBeamEdge( double);
```

C#

```
public  double DistFromUpperBeamEdge { get; set; }
```

Visual Basic

```
Public DistFromUpperBeamEdge As  double
```

Description

Distance of 1st bolt row to upper beam edge.

Version

Available since version 7.5.

VI.1.2.7 DistFromVertBeamEdge

C++

```
HRESULT get_DistFromVertBeamEdge(double*);
HRESULT put_DistFromVertBeamEdge(double);
```

C#

```
public double DistFromVertBeamEdge { get; set; }
```

Visual Basic

```
Public DistFromVertBeamEdge As double
```

Description

Distance of 1st bolt column to vertical beam edge.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 7.5.

VI.1.2.8 Friction

C++

```
HRESULT get_Friction( double*);
HRESULT put_Friction( double);
```

C#

```
public  double Friction { get; set; }
```

Visual Basic

```
Public Friction As  double
```

Version

Available since version 7.5.

VI.1.2.9 Rows

C++

```
HRESULT get_Rows( long*);
HRESULT put_Rows( long);
```

C#

```
public  long Rows { get; set; }
```

Visual Basic

```
Public Rows As  long
```

Description

Number of bolt rows.

Version

Available since version 7.5.

VI.1.2.10 SpacingH

C++

```
HRESULT get_SpacingH( double*);
HRESULT put_SpacingH( double);
```

C#

```
public  double SpacingH { get; set; }
```

Visual Basic

```
Public SpacingH As  double
```

Description

Horizontal spacing.

Version

Available since version 7.5.

(c) 2014 Autodesk Inc. All rights reserved

VI.1.2.11 SpacingV

C++

```
HRESULT get_SpacingV( double*);
HRESULT put_SpacingV( double);
```

C#

```
public  double SpacingV { get; set; }
```

Visual Basic

```
Public SpacingV As  double
```

Description

Vertical spacing.

Version

Available since version 7.5.

VI.2 IRJointBeamGirderStiffener

Class Hierarchy

C++

```
interface IRJointBeamGirderStiffener : IDispatch;
```

C#

```
public interface IRJointBeamGirderStiffener;
```

Visual Basic

```
Public Interface IRJointBeamGirderStiffener
```

Description

Stiffener description.

Version

Available since version 7.5.

VI.2.1 IRJointBeamGirderStiffener Members

The following tables list the members exposed by IRJointBeamGirderStiffener.

Public Fields

	Name	Description
	BoltsBeam (see page 2129)	Bolts for stiffener-beam connection.
	ConnectorsType (see page 2130)	Connector type for stiffener-beam connection.
	CutHeightDown (see page 2130)	
	CutHeightUp (see page 2130)	
	CutLength (see page 2131)	Length (see page 2131) of stiffener cut-off.
	Length (see page 2131)	Stiffener length.
	Material (see page 2131)	Stiffener material.
	Thick (see page 2132)	Stiffener thickness.
	WeldsBeam (see page 2132)	Welds for stiffener-beam connection.

(c) 2014 Autodesk Inc. All rights reserved

VI.2.2 IRJointBeamGirderStiffener Fields

The fields of the IRJointBeamGirderStiffener class are listed here.

Public Fields

	Name	Description
	BoltsBeam (see page 2129)	Bolts for stiffener-beam connection.
	ConnectorsType (see page 2130)	Connector type for stiffener-beam connection.
	CutHeightDown (see page 2130)	
	CutHeightUp (see page 2130)	
	CutLength (see page 2131)	Length (see page 2131) of stiffener cut-off.
	Length (see page 2131)	Stiffener length.
	Material (see page 2131)	Stiffener material.
	Thick (see page 2132)	Stiffener thickness.
	WeldsBeam (see page 2132)	Welds for stiffener-beam connection.

VI.2.2.1 BoltsBeam

C++

```
HRESULT get_BoltsBeam(IRJointBeamGirderBolts**);
```

C#

```
public IRJointBeamGirderBolts BoltsBeam { get; }
```

Visual Basic

```
Public ReadOnly BoltsBeam As IRJointBeamGirderBolts
```

Description

Bolts for stiffener-beam connection.

Version

Available since version 7.5.

VI.2.2.2 ConnectorsType

C++

```
HRESULT get_ConnectorsType(IRJointConnectorsType*);
HRESULT put_ConnectorsType(IRJointConnectorsType);
```

C#

```
public IRJointConnectorsType ConnectorsType { get; set; }
```

Visual Basic

```
Public ConnectorsType As IRJointConnectorsType
```

Description

Connector type for stiffener-beam connection.

Version

Available since version 7.5.

VI.2.2.3 CutHeightDown

C++

```
HRESULT get_CutHeightDown(double*);
HRESULT put_CutHeightDown(double);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public double CutHeightDown { get; set; }
```

Visual Basic

```
Public CutHeightDown As double
```

Version

Available since version 7.5.

VI.2.2.4 CutHeightUp

C++

```
HRESULT get_CutHeightUp(double*);
HRESULT put_CutHeightUp(double);
```

C#

```
public double CutHeightUp { get; set; }
```

Visual Basic

```
Public CutHeightUp As double
```

Version

Available since version 7.5.

VI.2.2.5 CutLength

C++

```
HRESULT get_CutLength(double*);
HRESULT put_CutLength(double);
```

C#

```
public double CutLength { get; set; }
```

Visual Basic

```
Public CutLength As double
```

Description

Length (see page 2131) of stiffener cut-off.

Version

Available since version 7.5.

VI.2.2.6 Length

C++

```
HRESULT get_Length(double*);
HRESULT put_Length(double);
```

C#

```
public double Length { get; set; }
```

Visual Basic

```
Public Length As double
```

Description

Stiffener length.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 7.5.

VI.2.2.7 Material

C++

```
HRESULT get_Material(BSTR*);
HRESULT put_Material(BSTR);
```

C#

```
public String Material { get; set; }
```

Visual Basic

```
Public Material As String
```

Description

Stiffener material.

Version

Available since version 7.5.

VI.2.2.8 Thick

C++

```
HRESULT get_Thick(double*);
HRESULT put_Thick(double);
```

C#

```
public double Thick { get; set; }
```

Visual Basic

```
Public Thick As double
```

Description

Stiffener thickness.

Version

Available since version 7.5.

VI.2.2.9 WeldsBeam

C++

```
HRESULT get_WeldsBeam(IRJointWeld**);
```

C#

```
public IRJointWeld WeldsBeam { get; }
```

Visual Basic

```
Public ReadOnly WeldsBeam As IRJointWeld
```

Description

Welds for stiffener-beam connection.

Version

Available since version 7.5.

(c) 2014 Autodesk Inc. All rights reserved

VI.3 IRJointBeamGirderPlate

Class Hierarchy

RobotOM.IRJointBeamGirderPlate

C++

```
interface IRJointBeamGirderPlate : IDispatch;
```

C#

```
public interface IRJointBeamGirderPlate;
```

Visual Basic

```
Public Interface IRJointBeamGirderPlate
```

Description

Plate description.

Version

Available since version 7.5.

VI.3.1 IRJointBeamGirderPlate Members

The following tables list the members exposed by IRJointBeamGirderPlate.

Public Fields

	Name	Description
	BoltsBeam (see page 2133)	Bolts connecting plate with beam.
	ConnectorsType (see page 2134)	Connector type for the plate-beam connection.
	DistFromUpperBeamEdge (see page 2134)	Distance of upper plate edge to upper beam edge.
	Length (see page 2134)	Plate length.
	Material (see page 2135)	Material name for plate.
	Thick (see page 2135)	Plate thickness.
	WeldsBeam (see page 2135)	Welds connecting plate with beam.
	WeldsGirder (see page 2135)	Welds connecting principal beam flange with plate.
	Width (see page 2136)	Plate width.

VI.3.2 IRJointBeamGirderPlate Fields

The fields of the IRJointBeamGirderPlate class are listed here.

Public Fields

	Name	Description
	BoltsBeam (see page 2133)	Bolts connecting plate with beam.
	ConnectorsType (see page 2134)	Connector type for the plate-beam connection.
	DistFromUpperBeamEdge (see page 2134)	Distance of upper plate edge to upper beam edge.
	Length (see page 2134)	Plate length.
	Material (see page 2135)	Material name for plate.
	Thick (see page 2135)	Plate thickness.
	WeldsBeam (see page 2135)	Welds connecting plate with beam.
	WeldsGirder (see page 2135)	Welds connecting principal beam flange with plate.

(c) 2014 Autodesk Inc. All rights reserved

	Width (see page 2136)	Plate width.

VI.3.2.1 BoltsBeam

C++

```
HRESULT get_BoltsBeam(IRJointBeamGirderBolts**);
```

C#

```
public IRJointBeamGirderBolts BoltsBeam { get; }
```

Visual Basic

```
Public ReadOnly BoltsBeam As IRJointBeamGirderBolts
```

Description

Bolts connecting plate with beam.

Version

Available since version 7.5.

VI.3.2.2 ConnectorsType

C++

```
HRESULT get_ConnectorsType(IRJointConnectorsType*);
HRESULT put_ConnectorsType(IRJointConnectorsType);
```

C#

```
public IRJointConnectorsType ConnectorsType { get; set; }
```

Visual Basic

```
Public ConnectorsType As IRJointConnectorsType
```

Description

Connector type for the plate-beam connection.

Version

Available since version 7.5.

VI.3.2.3 DistFromUpperBeamEdge

C++

```
HRESULT get_DistFromUpperBeamEdge( double*);
HRESULT put_DistFromUpperBeamEdge( double);
```

C#

```
public  double DistFromUpperBeamEdge { get; set; }
```

Visual Basic

```
Public DistFromUpperBeamEdge As  double
```

Description

Distance of upper plate edge to upper beam edge.

Version

Available since version 7.5.

(c) 2014 Autodesk Inc. All rights reserved

VI.3.2.4 Length

C++

```
HRESULT get_Length( double*);
HRESULT put_Length( double);
```

C#

```
public  double Length { get; set; }
```

Visual Basic

```
Public Length As  double
```

Description

Plate length.

Version

Available since version 7.5.

VI.3.2.5 Material

C++

```
HRESULT get_Material(BSTR*);
HRESULT put_Material(BSTR);
```

C#

```
public String Material { get; set; }
```

Visual Basic

```
Public Material As String
```

Description

Material name for plate.

Version

Available since version 7.5.

VI.3.2.6 Thick

C++

```
HRESULT get_Thick( double*);
HRESULT put_Thick( double);
```

C#

```
public  double Thick { get; set; }
```

Visual Basic

```
Public Thick As  double
```

Description

Plate thickness.

Version

Available since version 7.5.

(c) 2014 Autodesk Inc. All rights reserved

VI.3.2.7 WeldsBeam

C++

```
HRESULT get_WeldsBeam(IRJointWeld**);
```

C#

```
public IRJointWeld WeldsBeam { get; }
```

Visual Basic

```
Public ReadOnly WeldsBeam As IRJointWeld
```

Description

Welds connecting plate with beam.

Version

Available since version 7.5.

VI.3.2.8 WeldsGirder

C++

```
HRESULT get_WeldsGirder(IRJointWeld**);
```

C#

```
public IRJointWeld WeldsGirder { get; }
```

Visual Basic

```
Public ReadOnly WeldsGirder As IRJointWeld
```

Description

Welds connecting principal beam flange with plate.

Version

Available since version 7.5.

VI.3.2.9 Width

C++

```
HRESULT get_Width( double*);
HRESULT put_Width( double);
```

C#

```
public  double Width { get; set; }
```

Visual Basic

```
Public Width As  double
```

Description

Plate width.

Version

Available since version 7.5.

VI.4 IRJointBeamGirderSeat

Class Hierarchy

RobotOM.IRJointBeamGirderSeat

(c) 2014 Autodesk Inc. All rights reserved

C++

```
interface IRJointBeamGirderSeat : IDispatch;
```

C#

```
public interface IRJointBeamGirderSeat;
```

Visual Basic

```
Public Interface IRJointBeamGirderSeat
```

Description

Description of seat profile.

Version

Available since version 7.5.

VI.4.1 IRJointBeamGirderSeat Members

The following tables list the members exposed by IRJointBeamGirderSeat.

Public Fields

	Name	Description
	BoltsBeam (see page 2137)	Bolts connecting seat profile and beam flange.
	BoltsGirder (see page 2137)	Bolts connecting seat profile and principal beam web.
	ConnectorsToWebType (see page 2138)	Connector type for seat profile-principal beam web connection.
	Length (see page 2138)	Length of seat profile.
	Material (see page 2138)	Name of seat profile material.
	Section (see page 2139)	Section name.
	WeldsGirder (see page 2139)	Welds connecting seat profile and principal beam web.

VI.4.2 IRJointBeamGirderSeat Fields

The fields of the IRJointBeamGirderSeat class are listed here.

Public Fields

	Name	Description
	BoltsBeam (see page 2137)	Bolts connecting seat profile and beam flange.
	BoltsGirder (see page 2137)	Bolts connecting seat profile and principal beam web.
	ConnectorsToWebType (see page 2138)	Connector type for seat profile-principal beam web connection.
	Length (see page 2138)	Length of seat profile.
	Material (see page 2138)	Name of seat profile material.
	Section (see page 2139)	Section name.
	WeldsGirder (see page 2139)	Welds connecting seat profile and principal beam web.

VI.4.2.1 BoltsBeam

C++

```
HRESULT get_BoltsBeam(IRJointBeamGirderSeatBolts**);
```

C#

```
public IRJointBeamGirderSeatBolts BoltsBeam { get; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public ReadOnly BoltsBeam As IRJointBeamGirderSeatBolts
```

Description

Bolts connecting seat profile and beam flange.

Version

Available since version 7.5.

VI.4.2.2 BoltsGirder

C++

```
HRESULT get_BoltsGirder(IRJointBeamGirderSeatBolts**);
```

C#

```
public IRJointBeamGirderSeatBolts BoltsGirder { get; }
```

Visual Basic

```
Public ReadOnly BoltsGirder As IRJointBeamGirderSeatBolts
```

Description

Bolts connecting seat profile and principal beam web.

Version

Available since version 7.5.

VI.4.2.3 ConnectorsToWebType

C++

```
HRESULT get_ConnectorsToWebType(IRJointConnectorsType*);
HRESULT put_ConnectorsToWebType(IRJointConnectorsType);
```

C#

```
public IRJointConnectorsType ConnectorsToWebType { get; set; }
```

Visual Basic

```
Public ConnectorsToWebType As IRJointConnectorsType
```

Description

Connector type for seat profile-principal beam web connection.

Version

Available since version 7.5.

VI.4.2.4 Length

C++

```
HRESULT get_Length(double*);
HRESULT put_Length(double);
```

C#

```
public double Length { get; set; }
```

Visual Basic

```
Public Length As double
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Length of seat profile.

Version

Available since version 7.5.

VI.4.2.5 Material

C++

```
HRESULT get_Material(BSTR*);
HRESULT put_Material(BSTR);
```

C#

```
public String Material { get; set; }
```

Visual Basic

```
Public Material As String
```

Description

Name of seat profile material.

Version

Available since version 7.5.

VI.4.2.6 Section

C++

```
HRESULT get_Section(BSTR*);
HRESULT put_Section(BSTR);
```

C#

```
public String Section { get; set; }
```

Visual Basic

```
Public Section As String
```

Description

Section name.

Version

Available since version 7.5.

VI.4.2.7 WeldsGirder

C++

```
HRESULT get_WeldsGirder(IRJointWeld**);
```

C#

```
public IRJointWeld WeldsGirder { get; }
```

Visual Basic

```
Public ReadOnly WeldsGirder As IRJointWeld
```

Description

Welds connecting seat profile and principal beam web.

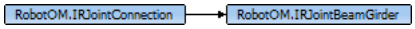
(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 7.5.

VI.5 IRJointBeamGirder

Class Hierarchy



C++

```
interface IRJointBeamGirder : IRJointConnection;
```

C#

```
public interface IRJointBeamGirder : IRJointConnection;
```

Visual Basic

```
Public Interface IRJointBeamGirder
```

Description

Beam-principal beam connection.

Version

Available since version 7.5.

VI.5.1 IRJointBeamGirder Members

The following tables list the members exposed by IRJointBeamGirder.

Public Fields

	Name	Description
	AngleLeft (see page 2141)	
	AngleRight (see page 2141)	
	BeamLeft (see page 2142)	Right beam section.
	BeamRight (see page 2142)	Section of principal beam.
	DistGirderFlangeToBeamFlangeLeft (see page 2142)	
	DistGirderFlangeToBeamFlangeRight (see page 2143)	
	DistGirderToBeamLeft (see page 2143)	
	DistGirderToBeamRight (see page 2143)	
	ElementType (see page 2143)	
	Girder (see page 2144)	Section of principal beam.
	PlateLeft (see page 2144)	
	PlateRight (see page 2144)	
	SeatLeftDown (see page 2144)	
	SeatLeftUp (see page 2145)	
	SeatRightDown (see page 2145)	
	SeatRightUp (see page 2145)	
	StiffenerLeft (see page 2145)	
	StiffenerRight (see page 2146)	
	Type (see page 2160)	A connection type.

(c) 2014 Autodesk Inc. All rights reserved

	WFRelPos (see page 2161)	A type of relative position of a beam and a column.

Public Methods

	Name	Description
	GetFromRobot (see page 2161)	The function gets connection parameters connection parameters from Robot and fills an object with the data.
	SetToRobot (see page 2161)	The function sets and saves connection parameter in Robot.

VI.5.2 IRJointBeamGirder Fields

The fields of the IRJointBeamGirder class are listed here.

Public Fields

	Name	Description
	AngleLeft (see page 2141)	
	AngleRight (see page 2141)	
	BeamLeft (see page 2142)	Right beam section.
	BeamRight (see page 2142)	Section of principal beam.
	DistGirderFlangeToBeamFlangeLeft (see page 2142)	
	DistGirderFlangeToBeamFlangeRight (see page 2143)	
	DistGirderToBeamLeft (see page 2143)	
	DistGirderToBeamRight (see page 2143)	
	ElementType (see page 2143)	
	Girder (see page 2144)	Section of principal beam.
	PlateLeft (see page 2144)	
	PlateRight (see page 2144)	
	SeatLeftDown (see page 2144)	
	SeatLeftUp (see page 2145)	
	SeatRightDown (see page 2145)	
	SeatRightUp (see page 2145)	
	StiffenerLeft (see page 2145)	
	StiffenerRight (see page 2146)	

VI.5.2.1 AngleLeft

C++

```
HRESULT get_AngleLeft(IRJointBeamGirderAngle**);
```

C#

```
public IRJointBeamGirderAngle AngleLeft { get; }
```

Visual Basic

```
Public ReadOnly AngleLeft As IRJointBeamGirderAngle
```

Version

Available since version 7.5.

(c) 2014 Autodesk Inc. All rights reserved

VI.5.2.2 AngleRight

C++

```
HRESULT get_AngleRight(IRJointBeamGirderAngle**);
```

C#

```
public IRJointBeamGirderAngle AngleRight { get; }
```

Visual Basic

```
Public ReadOnly AngleRight As IRJointBeamGirderAngle
```

Version

Available since version 7.5.

VI.5.2.3 BeamLeft

C++

```
HRESULT get_BeamLeft(IRJointBeamGirderBeam**);
```

C#

```
public IRJointBeamGirderBeam BeamLeft { get; }
```

Visual Basic

```
Public ReadOnly BeamLeft As IRJointBeamGirderBeam
```

Description

Right beam section.

Version

Available since version 7.5.

VI.5.2.4 BeamRight

C++

```
HRESULT get_BeamRight(IRJointBeamGirderBeam**);
```

C#

```
public IRJointBeamGirderBeam BeamRight { get; }
```

Visual Basic

```
Public ReadOnly BeamRight As IRJointBeamGirderBeam
```

Description

Section of principal beam.

Version

Available since version 7.5.

VI.5.2.5 DistGirderFlangeToBeamFlangeLeft

C++

```
HRESULT get_DistGirderFlangeToBeamFlangeLeft(double*);
HRESULT put_DistGirderFlangeToBeamFlangeLeft(double);
```

C#

```
public double DistGirderFlangeToBeamFlangeLeft { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public DistGirderFlangeToBeamFlangeLeft As double
```

Version

Available since version 7.5.

VI.5.2.6 DistGirderFlangeToBeamFlangeRight

C++

```
HRESULT get_DistGirderFlangeToBeamFlangeRight(double*);
HRESULT put_DistGirderFlangeToBeamFlangeRight(double);
```

C#

```
public double DistGirderFlangeToBeamFlangeRight { get; set; }
```

Visual Basic

```
Public DistGirderFlangeToBeamFlangeRight As double
```

Version

Available since version 7.5.

VI.5.2.7 DistGirderToBeamLeft

C++

```
HRESULT get_DistGirderToBeamLeft(double*);
HRESULT put_DistGirderToBeamLeft(double);
```

C#

```
public double DistGirderToBeamLeft { get; set; }
```

Visual Basic

```
Public DistGirderToBeamLeft As double
```

Version

Available since version 7.5.

VI.5.2.8 DistGirderToBeamRight

C++

```
HRESULT get_DistGirderToBeamRight(double*);
HRESULT put_DistGirderToBeamRight(double);
```

C#

```
public double DistGirderToBeamRight { get; set; }
```

Visual Basic

```
Public DistGirderToBeamRight As double
```

Version

Available since version 7.5.

VI.5.2.9 ElementType

C++

```
HRESULT get_ElementType(IRJointElementType*);
HRESULT put_ElementType(IRJointElementType);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public IRJointElementType ElementType { get; set; }
```

Visual Basic

```
Public ElementType As IRJointElementType
```

Version

Available since version 7.5.

VI.5.2.10 Girder

C++

```
HRESULT get_Girder(IRJointProfile**);
```

C#

```
public IRJointProfile Girder { get; }
```

Visual Basic

```
Public ReadOnly Girder As IRJointProfile
```

Description

Section of principal beam.

Version

Available since version 7.5.

VI.5.2.11 PlateLeft

C++

```
HRESULT get_PlateLeft(IRJointBeamGirderPlate**);
```

C#

```
public IRJointBeamGirderPlate PlateLeft { get; }
```

Visual Basic

```
Public ReadOnly PlateLeft As IRJointBeamGirderPlate
```

Version

Available since version 7.5.

VI.5.2.12 PlateRight

C++

```
HRESULT get_PlateRight(IRJointBeamGirderPlate**);
```

C#

```
public IRJointBeamGirderPlate PlateRight { get; }
```

Visual Basic

```
Public ReadOnly PlateRight As IRJointBeamGirderPlate
```

Version

Available since version 7.5.

(c) 2014 Autodesk Inc. All rights reserved

VI.5.2.13 SeatLeftDown

C++

```
HRESULT get_SeatLeftDown(IRJointBeamGirderSeat**);
```

C#

```
public IRJointBeamGirderSeat SeatLeftDown { get; }
```

Visual Basic

```
Public ReadOnly SeatLeftDown As IRJointBeamGirderSeat
```

Version

Available since version 7.5.

VI.5.2.14 SeatLeftUp

C++

```
HRESULT get_SeatLeftUp(IRJointBeamGirderSeat**);
```

C#

```
public IRJointBeamGirderSeat SeatLeftUp { get; }
```

Visual Basic

```
Public ReadOnly SeatLeftUp As IRJointBeamGirderSeat
```

Version

Available since version 7.5.

VI.5.2.15 SeatRightDown

C++

```
HRESULT get_SeatRightDown(IRJointBeamGirderSeat**);
```

C#

```
public IRJointBeamGirderSeat SeatRightDown { get; }
```

Visual Basic

```
Public ReadOnly SeatRightDown As IRJointBeamGirderSeat
```

Version

Available since version 7.5.

VI.5.2.16 SeatRightUp

C++

```
HRESULT get_SeatRightUp(IRJointBeamGirderSeat**);
```

C#

```
public IRJointBeamGirderSeat SeatRightUp { get; }
```

Visual Basic

```
Public ReadOnly SeatRightUp As IRJointBeamGirderSeat
```

Version

Available since version 7.5.

(c) 2014 Autodesk Inc. All rights reserved

VI.5.2.17 StiffenerLeft

C++

```
HRESULT get_StiffenerLeft(IRJointBeamGirderStiffener**);
```

C#

```
public IRJointBeamGirderStiffener StiffenerLeft { get; }
```

Visual Basic

```
Public ReadOnly StiffenerLeft As IRJointBeamGirderStiffener
```

Version

Available since version 7.5.

VI.5.2.18 StiffenerRight

C++

```
HRESULT get_StiffenerRight(IRJointBeamGirderStiffener**);
```

C#

```
public IRJointBeamGirderStiffener StiffenerRight { get; }
```

Visual Basic

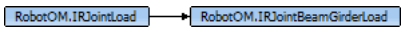
```
Public ReadOnly StiffenerRight As IRJointBeamGirderStiffener
```

Version

Available since version 7.5.

VI.6 IRJointBeamGirderLoad

Class Hierarchy



C++

```
interface IRJointBeamGirderLoad : IRJointLoad;
```

C#

```
public interface IRJointBeamGirderLoad : IRJointLoad;
```

Visual Basic

```
Public Interface IRJointBeamGirderLoad
```

Version

Available since version 7.5.

VI.6.1 IRJointBeamGirderLoad Members

The following tables list the members exposed by IRJointBeamGirderLoad.

Public Fields

	Name	Description
	Cases (see page 2178)	Selection of load cases, defined for the structure. .
	LFx (see page 2147)	Axial force (left side).
	LFz (see page 2147)	Shear force (left side).

(c) 2014 Autodesk Inc. All rights reserved

	LMy (see page 2147)	Bending moment (left side).
	RFx (see page 2148)	Axial force (right side).
	RFz (see page 2148)	Shear force (right side).
	RMy (see page 2148)	Bending moment (right side).
	Type (see page 2179)	Load type (defined manually or taken from Robot).

VI.6.2 IRJointBeamGirderLoad Fields

The fields of the IRJointBeamGirderLoad class are listed here.

Public Fields

	Name	Description
	LFx (see page 2147)	Axial force (left side).
	LFz (see page 2147)	Shear force (left side).
	LMy (see page 2147)	Bending moment (left side).
	RFx (see page 2148)	Axial force (right side).
	RFz (see page 2148)	Shear force (right side).
	RMy (see page 2148)	Bending moment (right side).

VI.6.2.1 LFx

C++

```
HRESULT get_LFx( double*);
HRESULT put_LFx( double);
```

C#

```
public  double LFx { get; set; }
```

Visual Basic

```
Public LFx As  double
```

Description

Axial force (left side).

Version

Available since version 7.5.

VI.6.2.2 LFz

C++

```
HRESULT get_LFz( double*);
HRESULT put_LFz( double);
```

C#

```
public  double LFz { get; set; }
```

Visual Basic

```
Public LFz As  double
```

Description

Shear force (left side).

Version

Available since version 7.5.

(c) 2014 Autodesk Inc. All rights reserved

VI.6.2.3 LMy

C++

```
HRESULT get_LMy( double*);
HRESULT put_LMy( double);
```

C#

```
public  double LMy { get; set; }
```

Visual Basic

```
Public LMy As  double
```

Description

Bending moment (left side).

Version

Available since version 7.5.

VI.6.2.4 RFx

C++

```
HRESULT get_RFx( double*);
HRESULT put_RFx( double);
```

C#

```
public  double RFx { get; set; }
```

Visual Basic

```
Public RFx As  double
```

Description

Axial force (right side).

Version

Available since version 7.5.

VI.6.2.5 RFz

C++

```
HRESULT get_RFz( double*);
HRESULT put_RFz( double);
```

C#

```
public  double RFz { get; set; }
```

Visual Basic

```
Public RFz As  double
```

Description

Shear force (right side).

Version

Available since version 7.5.

(c) 2014 Autodesk Inc. All rights reserved

VI.6.2.6 RMy

C++

```
HRESULT get_RMy( double*);
HRESULT put_RMy( double);
```

C#

```
public  double RMy { get; set; }
```

Visual Basic

```
Public RMy As  double
```

Description

Bending moment (right side).

Version

Available since version 7.5.

VI.7 IRJointBeamGirderSeatBolts

Class Hierarchy

RobotOM.IRJointBeamGirderSeatBolts

C++

```
interface IRJointBeamGirderSeatBolts : IDispatch;
```

C#

```
public interface IRJointBeamGirderSeatBolts;
```

Visual Basic

```
Public Interface IRJointBeamGirderSeatBolts
```

Version

Available since version 7.5.

VI.7.1 IRJointBeamGirderSeatBolts Members

The following tables list the members exposed by IRJointBeamGirderSeatBolts.

Public Fields

	Name	Description
	Area (see page 2150)	
	ClassName (see page 2150)	
	Cols (see page 2150)	
	Diameter (see page 2151)	
	DiameterName (see page 2151)	
	DistFromPerpendicularArmEdge (see page 2151)	Distance of bolt row to edge of perpendicular leg of seat profile.
	DistFromVertEdge (see page 2151)	Distance of extreme bolt to vertical edge of seat profile.
	Friction (see page 2152)	

(c) 2014 Autodesk Inc. All rights reserved

VI.7.2 IRJointBeamGirderSeatBolts Fields

The fields of the IRJointBeamGirderSeatBolts class are listed here.

Public Fields

	Name	Description
	Area (see page 2150)	
	ClassName (see page 2150)	
	Cols (see page 2150)	
	Diameter (see page 2151)	
	DiameterName (see page 2151)	
	DistFromPerpendicularArmEdge (see page 2151)	Distance of bolt row to edge of perpendicular leg of seat profile.
	DistFromVertEdge (see page 2151)	Distance of extreme bolt to vertical edge of seat profile.
	Friction (see page 2152)	

VI.7.2.1 Area

C++

```
HRESULT get_Area( double*);
```

C#

```
public  double Area { get; }
```

Visual Basic

```
Public ReadOnly Area As  double
```

Version

Available since version 7.5.

VI.7.2.2 ClassName

C++

```
HRESULT get_ClassName(BSTR*);
HRESULT put_ClassName(BSTR);
```

C#

```
public String ClassName { get; set; }
```

Visual Basic

```
Public ClassName As String
```

Version

Available since version 7.5.

VI.7.2.3 Cols

C++

```
HRESULT get_Cols( long*);
HRESULT put_Cols( long);
```

C#

```
public  long Cols { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Cols As  long
```

Version

Available since version 7.5.

VI.7.2.4 Diameter

C++

```
HRESULT get_Diameter( long*);
```

C#

```
public  long Diameter { get; }
```

Visual Basic

```
Public ReadOnly Diameter As  long
```

Version

Available since version 7.5.

VI.7.2.5 DiameterName

C++

```
HRESULT get_DiameterName(BSTR*);
HRESULT put_DiameterName(BSTR);
```

C#

```
public String DiameterName { get; set; }
```

Visual Basic

```
Public DiameterName As String
```

Version

Available since version 7.5.

VI.7.2.6 DistFromPerpendicularArmEdge

C++

```
HRESULT get_DistFromPerpendicularArmEdge( double*);
HRESULT put_DistFromPerpendicularArmEdge( double);
```

C#

```
public  double DistFromPerpendicularArmEdge { get; set; }
```

Visual Basic

```
Public DistFromPerpendicularArmEdge As  double
```

Description

Distance of bolt row to edge of perpendicular leg of seat profile.

Version

Available since version 7.5.

VI.7.2.7 DistFromVertEdge

C++

```
HRESULT get_DistFromVertEdge( double*);
```

(c) 2014 Autodesk Inc. All rights reserved

```
HRESULT put_DistFromVertEdge( double);
```

C#

```
public  double DistFromVertEdge { get; set; }
```

Visual Basic

```
Public DistFromVertEdge As  double
```

Description

Distance of extreme bolt to vertical edge of seat profile.

Version

Available since version 7.5.

VI.7.2.8 Friction

C++

```
HRESULT get_Friction( double*);
HRESULT put_Friction( double);
```

C#

```
public  double Friction { get; set; }
```

Visual Basic

```
Public Friction As  double
```

Version

Available since version 7.5.

VI.8 IRJointConnectorsType

C++

```
enum IRJointConnectorsType;
```

C#

```
public enum IRJointConnectorsType;
```

Visual Basic

```
Public Enum IRJointConnectorsType
```

Members

Members	Description
I_JCT_BOLTS = 0	Available since version 7.5.
I_JCT_WELDS = 1	Available since version 7.5.

Version

Available since version 7.5.

VI.9 IRJointElementType

C++

```
enum IRJointElementType;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public enum IRJointElementType;
```

Visual Basic

```
Public Enum IRJointElementType
```

Members

Members	Description
I_JET_R_ANGLES_WEB = 0	Available since version 7.5.
I_JET_R_ANGLES_FLANGE = 1	Available since version 7.5.
I_JET_R_STIFF_SHORT = 2	Available since version 7.5.
I_JET_R_STIFF_LONG = 3	Available since version 7.5.
I_JET_R_SEATS = 4	Available since version 7.5.
I_JET_R_SEATS_ANGLES = 5	Available since version 7.5.
I_JET_R_PLATES = 6	Available since version 7.5.
I_JET_L_ANGLES_R_ANGLES_WEB = 7	Available since version 7.5.
I_JET_L_ANGLES_R_ANGLES_FLANGE = 8	Available since version 7.5.
I_JET_L_STIFF_SHORT_R_STIFF_SHORT = 9	Available since version 7.5.
I_JET_L_STIFF_SHORT_R_STIFF_LONG = 10	Available since version 7.5.
I_JET_L_STIFF_LONG_R_STIFF_SHORT = 11	Available since version 7.5.
I_JET_L_STIFF_LONG_R_STIFF_LONG = 12	Available since version 7.5.
I_JET_L_SEATS_R_SEATS = 13	Available since version 7.5.
I_JET_L_SEATS_R_SEATS_ANGLES = 14	Available since version 7.5.
I_JET_L_SEATS_ANGLES_R_SEATS = 15	Available since version 7.5.
I_JET_L_SEATS_ANGLES_R_SEATS_ANGLES = 16	Available since version 7.5.
I_JET_L_PLATES_R_PLATES = 17	Available since version 7.5.
I_JET_L_PLATES_R_ANGLES = 18	Available since version 7.5.
I_JET_L_ANGLES_R_PLATES = 19	Available since version 7.5.

Version

Available since version 7.5.

VI.10 IRJointBeamGirderAngle

Class Hierarchy

RobotOM.IRJointBeamGirderAngle

C++

```
interface IRJointBeamGirderAngle : IDispatch;
```

C#

```
public interface IRJointBeamGirderAngle;
```

Visual Basic

```
Public Interface IRJointBeamGirderAngle
```

Version

Available since version 7.5.

(c) 2014 Autodesk Inc. All rights reserved

VI.10.1 IRJointBeamGirderAngle Members

The following tables list the members exposed by IRJointBeamGirderAngle.

Public Fields

	Name	Description
	BoltsBeam (see page 2154)	
	BoltsGirder (see page 2154)	
	Element (see page 2154)	

VI.10.2 IRJointBeamGirderAngle Fields

The fields of the IRJointBeamGirderAngle class are listed here.

Public Fields

	Name	Description
	BoltsBeam (see page 2154)	
	BoltsGirder (see page 2154)	
	Element (see page 2154)	

VI.10.2.1 BoltsBeam

C++

```
HRESULT get_BoltsBeam(IRJointAngleBolts**);
```

C#

```
public IRJointAngleBolts BoltsBeam { get; }
```

Visual Basic

```
Public ReadOnly BoltsBeam As IRJointAngleBolts
```

Version

Available since version 7.5.

VI.10.2.2 BoltsGirder

C++

```
HRESULT get_BoltsGirder(IRJointAngleBolts**);
```

C#

```
public IRJointAngleBolts BoltsGirder { get; }
```

Visual Basic

```
Public ReadOnly BoltsGirder As IRJointAngleBolts
```

Version

Available since version 7.5.

VI.10.2.3 Element

C++

```
HRESULT get_Element(IRJointAngle**);
```

C#

```
public IRJointAngle Element { get; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public ReadOnly Element As IRJointAngle
```

Version

Available since version 7.5.

VI.11 IRJointBeamGirderBeam

Class Hierarchy

C++

```
interface IRJointBeamGirderBeam : IDispatch;
```

C#

```
public interface IRJointBeamGirderBeam;
```

Visual Basic

```
Public Interface IRJointBeamGirderBeam
```

Version

Available since version 7.5.

VI.11.1 IRJointBeamGirderBeam Members

The following tables list the members exposed by IRJointBeamGirderBeam.

Public Fields

	Name	Description
	CutEnd (see page 2155)	
	Profile (see page 2155)	

VI.11.2 IRJointBeamGirderBeam Fields

The fields of the IRJointBeamGirderBeam class are listed here.

Public Fields

	Name	Description
	CutEnd (see page 2155)	
	Profile (see page 2155)	

VI.11.2.1 CutEnd

C++

```
HRESULT get_CutEnd(IRJointBeamCut**);
```

C#

```
public IRJointBeamCut CutEnd { get; }
```

Visual Basic

```
Public ReadOnly CutEnd As IRJointBeamCut
```

Version

Available since version 7.5.

(c) 2014 Autodesk Inc. All rights reserved

VI.11.2.2 Profile

C++

```
HRESULT get_Profile(IRJointProfile**);
```

C#

```
public IRJointProfile Profile { get; }
```

Visual Basic

```
Public ReadOnly Profile As IRJointProfile
```

Version

Available since version 7.5.

VII IRJointConnectionDefType

C++

```
enum IRJointConnectionDefType;
```

C#

```
public enum IRJointConnectionDefType;
```

Visual Basic

```
Public Enum IRJointConnectionDefType
```

Members

Members	Description
I_JCDT_IN_STRUCTURE = 0	A connection defined in a structure.
I_JCDT_STANDALONE = 1	A connection defined independently on a structure.

Description

A method of connection definition.

VIII IRJointConnectionType

C++

```
enum IRJointConnectionType;
```

C#

```
public enum IRJointConnectionType;
```

Visual Basic

```
Public Enum IRJointConnectionType
```

Members

Members	Description
I_JCT_KNEE_BOLTED = 0	A bolted knee connection.
I_JCT_KNEE_WELDED = 1	A welded knee connection .
I_JCT_ANGLE = 2	Angle connection.
I_JCT_TRNS = 3	Transversal connection of beams (not available).
I_JCT_COL_PINNED = 5	A pinned column base.

(c) 2014 Autodesk Inc. All rights reserved

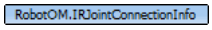
I_JCT_COL_FIXED = 6	A fixed column base.
I_JCT_COL_CONCR = 7	A concrete column base.
I_JCT_TUBE = 8	Tube connection.
I_JCT_GUSSET_SIMPLE = 9	Available since version 1.7.
I_JCT_GUSSET_CROSS = 10	Available since version 1.7.
I_JCT_GUSSET_FLANGE = 11	Available since version 1.7.
I_JCT_BEAM_GIRDER = 12	Available since version 7.5.

Description

Types of connections available in a connections module of Robot.

IX IRJointConnectionInfo

Class Hierarchy



C++

```
interface IRJointConnectionInfo : IDispatch;
```

C#

```
public interface IRJointConnectionInfo;
```

Visual Basic

```
Public Interface IRJointConnectionInfo
```

Description

Information about connection. .

IX.1 IRJointConnectionInfo Members

The following tables list the members exposed by IRJointConnectionInfo.

Public Fields

	Name	Description
	DefType (see page 2158)	A method of connection definition.
	Elements (see page 2158)	A selection of elements creating a connection.
	Node (see page 2158)	Number (see page 2158) of a connection node.
	Number (see page 2158)	Connection number in the Robot connection server.
	Type (see page 2159)	Connection type.
	UniqueId (see page 2159)	Unique connection identifier Available since version 1.7.

IX.2 IRJointConnectionInfo Fields

The fields of the IRJointConnectionInfo class are listed here.

Public Fields

	Name	Description
	DefType (see page 2158)	A method of connection definition.
	Elements (see page 2158)	A selection of elements creating a connection.
	Node (see page 2158)	Number (see page 2158) of a connection node.
	Number (see page 2158)	Connection number in the Robot connection server.

(c) 2014 Autodesk Inc. All rights reserved

	Type (see page 2159)	Connection type.
	UniqueId (see page 2159)	Unique connection identifier Available since version 1.7.

IX.2.1 DefType

C++

```
HRESULT get_DefType(IRJointConnectionDefType*);
HRESULT put_DefType(IRJointConnectionDefType);
```

C#

```
public IRJointConnectionDefType DefType { get; set; }
```

Visual Basic

```
Public DefType As IRJointConnectionDefType
```

Description

A method of connection definition.

IX.2.2 Elements

C++

```
HRESULT get_Elements(IRobotSelection**);
```

C#

```
public IRobotSelection Elements { get; }
```

Visual Basic

```
Public ReadOnly Elements As IRobotSelection
```

Description

A selection of elements creating a connection.

IX.2.3 Node

C++

```
HRESULT get_Node( long*);
HRESULT put_Node( long);
```

C#

```
public  long Node { get; set; }
```

Visual Basic

```
Public Node As  long
```

Description

Number (see page 2158) of a connection node.

IX.2.4 Number

C++

```
HRESULT get_Number(long*);
HRESULT put_Number(long);
```

C#

```
public long Number { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Number As long
```

Description

Connection number in the Robot connection server.

IX.2.5 Type

C++

```
HRESULT get_Type(IRJointConnectionType*);
HRESULT put_Type(IRJointConnectionType);
```

C#

```
public IRJointConnectionType Type { get; set; }
```

Visual Basic

```
Public Type As IRJointConnectionType
```

Description

Connection type.

IX.2.6 UniqueId

C++

```
HRESULT get_UniqueId(long*);
```

C#

```
public long UniqueId { get; }
```

Visual Basic

```
Public ReadOnly UniqueId As long
```

Description

Unique connection identifier Available since version 1.7.

X IRJointWebFlangeRelativePos

C++

```
enum IRJointWebFlangeRelativePos;
```

C#

```
public enum IRJointWebFlangeRelativePos;
```

Visual Basic

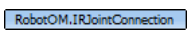
```
Public Enum IRJointWebFlangeRelativePos
```

Description

Possible types of a beam - column mutual position. .

XI IRJointConnection

Class Hierarchy



(c) 2014 Autodesk Inc. All rights reserved

C++

```
interface IRJointConnection : IDispatch;
```

C#

```
public interface IRJointConnection;
```

Visual Basic

```
Public Interface IRJointConnection
```

Description

A data type describing common features for all connections that can be defined in Robot.

XI.1 IRJointConnection Members

The following tables list the members exposed by IRJointConnection.

Public Fields

	Name	Description
	Type (see page 2160)	A connection type.
	WFRelPos (see page 2161)	A type of relative position of a beam and a column.

Public Methods

	Name	Description
	GetFromRobot (see page 2161)	The function gets connection parameters connection parameters from Robot and fills an object with the data.
	SetToRobot (see page 2161)	The function sets and saves connection parameter in Robot.

XI.2 IRJointConnection Fields

The fields of the IRJointConnection class are listed here.

Public Fields

	Name	Description
	Type (see page 2160)	A connection type.
	WFRelPos (see page 2161)	A type of relative position of a beam and a column.

XI.2.1 Type

C++

```
HRESULT get_Type(IRJointConnectionType*);
```

C#

```
public IRJointConnectionType Type { get; }
```

Visual Basic

```
Public ReadOnly Type As IRJointConnectionType
```

Description

A connection type.

XI.2.2 WFRelPos

C++

```
HRESULT get_WFRelPos(IRJointWebFlangeRelativePos*);
```

(c) 2014 Autodesk Inc. All rights reserved

```
HRESULT put_WFRelPos(IRJointWebFlangeRelativePos);
```

C#

```
public IRJointWebFlangeRelativePos WFRelPos { get; set; }
```

Visual Basic

```
Public WFRelPos As IRJointWebFlangeRelativePos
```

Description

A type of relative position of a beam and a column.

XI.3 IRJointConnection Methods

The methods of the IRJointConnection class are listed here.

Public Methods

	Name	Description
	GetFromRobot (see page 2161)	The function gets connection parameters connection parameters from Robot and fills an object with the data.
	SetToRobot (see page 2161)	The function sets and saves connection parameter in Robot.

XI.3.1 GetFromRobot

C++

```
HRESULT GetFromRobot(IRJointConnectionInfo* _info);
```

C#

```
public void GetFromRobot(IRJointConnectionInfo _info);
```

Visual Basic

```
Public Sub GetFromRobot(ByRef _info As IRJointConnectionInfo)
```

Description

The function gets connection parameters connection parameters from Robot and fills an object with the data.

XI.3.2 SetToRobot

C++

```
HRESULT SetToRobot(IRJointConnectionInfo* _info);
```

C#

```
public void SetToRobot(IRJointConnectionInfo _info);
```

Visual Basic

```
Public Sub SetToRobot(ByRef _info As IRJointConnectionInfo)
```

Description

The function sets and saves connection parameter in Robot.

XII IRJointWeld

Class Hierarchy

RobotOM.IRJointWeld

(c) 2014 Autodesk Inc. All rights reserved

C++

```
interface IRJointWeld : IDispatch;
```

C#

```
public interface IRJointWeld;
```

Visual Basic

```
Public Interface IRJointWeld
```

Description

Description of a weld.

XII.1 IRJointWeld Members

The following tables list the members exposed by IRJointWeld.

Public Fields

	Name	Description
	Strength (see page 2162)	Weld resistance.
	Thick (see page 2163)	Weld thickness.

XII.2 IRJointWeld Fields

The fields of the IRJointWeld class are listed here.

Public Fields

	Name	Description
	Strength (see page 2162)	Weld resistance.
	Thick (see page 2163)	Weld thickness.

XII.2.1 Strength

C++

```
HRESULT get_Strength(double*);
HRESULT put_Strength(double);
```

C#

```
public double Strength { get; set; }
```

Visual Basic

```
Public Strength As double
```

Description

Weld resistance.

XII.2.2 Thick

C++

```
HRESULT get_Thick(double*);
HRESULT put_Thick(double);
```

C#

```
public double Thick { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

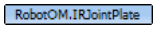
```
Public Thick As double
```

Description

Weld thickness.

XIII IRJointPlate

Class Hierarchy



C++

```
interface IRJointPlate : IDispatch;
```

C#

```
public interface IRJointPlate;
```

Visual Basic

```
Public Interface IRJointPlate
```

Description

Plate description.

XIII.1 IRJointPlate Members

The following tables list the members exposed by IRJointPlate.

Public Fields

	Name	Description
	Exist (see page 2164)	Information about a plate existance .
	Length (see page 2164)	Plate length .
	Material (see page 2164)	Name of the component material.
	Thick (see page 2165)	Plate thickness.
	Width (see page 2165)	Plate width.

Public Methods

	Name	Description
	AngleExt (see page 2165)	Angle of bracket inclination, depending on the parameter - in degrees or radians.

XIII.2 IRJointPlate Fields

The fields of the IRJointPlate class are listed here.

Public Fields

	Name	Description
	Exist (see page 2164)	Information about a plate existance .
	Length (see page 2164)	Plate length .
	Material (see page 2164)	Name of the component material.
	Thick (see page 2165)	Plate thickness.
	Width (see page 2165)	Plate width.

(c) 2014 Autodesk Inc. All rights reserved

XIII.2.1 Exist

C++

```
HRESULT get_Exist(VARIANT_BOOL*);
```

C#

```
public bool Exist { get; }
```

Visual Basic

```
Public ReadOnly Exist As Boolean
```

Description

Information about a plate existance .

XIII.2.2 Length

C++

```
HRESULT get_Length(double*);
HRESULT put_Length(double);
```

C#

```
public double Length { get; set; }
```

Visual Basic

```
Public Length As double
```

Description

Plate length .

XIII.2.3 Material

C++

```
HRESULT get_Material(BSTR*);
HRESULT put_Material(BSTR);
```

C#

```
public BSTR Material { get; set; }
```

Visual Basic

```
Public Material As BSTR
```

Description

Name of the component material.

Version

Available since version 9.5.

XIII.2.4 Thick

C++

```
HRESULT get_Thick(double*);
HRESULT put_Thick(double);
```

C#

```
public double Thick { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Thick As double
```

Description

Plate thickness.

XIII.2.5 Width

C++

```
HRESULT get_Width(double*);
HRESULT put_Width(double);
```

C#

```
public double Width { get; set; }
```

Visual Basic

```
Public Width As double
```

Description

Plate width.

XIII.3 IRJointPlate Methods

The methods of the IRJointPlate class are listed here.

Public Methods

	Name	Description
	AngleExt (see page 2165)	Angle of bracket inclination, depending on the parameter - in degrees or radians.

XIII.3.1 AngleExt

C++

```
HRESULT AngleExt(double* ret);
```

C#

```
public double AngleExt();
```

Visual Basic

```
Public Function AngleExt() As double
```

Description

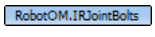
Angle of bracket inclination, depending on the parameter - in degrees or radians.

Version

Available since version 9.5.

XIV IRJointBolts

Class Hierarchy



C++

```
interface IRJointBolts : IDispatch;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public interface IRJointBolts;
```

Visual Basic

```
Public Interface IRJointBolts
```

Description

Definition of a group of bolts with common parameters.

XIV.1 IRJointBolts Members

The following tables list the members exposed by IRJointBolts.

Public Fields

	Name	Description
	Area (see page 2167)	Bolt section area.
	ClassName (see page 2167)	Bolt class.
	Cols (see page 2167)	Number of columns.
	Diameter (see page 2167)	Bolt diameter.
	DiameterName (see page 2168)	Bolt diameter name.
	Friction (see page 2168)	Bolt friction.
	Height1 (see page 2168)	Distance of first bolt (e.g. from the top edge of a plate for knee connections) .
	Rows (see page 2168)	Number of rows.

XIV.2 IRJointBolts Fields

The fields of the IRJointBolts class are listed here.

Public Fields

	Name	Description
	Area (see page 2167)	Bolt section area.
	ClassName (see page 2167)	Bolt class.
	Cols (see page 2167)	Number of columns.
	Diameter (see page 2167)	Bolt diameter.
	DiameterName (see page 2168)	Bolt diameter name.
	Friction (see page 2168)	Bolt friction.
	Height1 (see page 2168)	Distance of first bolt (e.g. from the top edge of a plate for knee connections) .
	Rows (see page 2168)	Number of rows.

XIV.2.1 Area

C++

```
HRESULT get_Area(double*);
```

C#

```
public double Area { get; }
```

Visual Basic

```
Public ReadOnly Area As double
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Bolt section area.

XIV.2.2 ClassName

C++

```
HRESULT get_ClassName(BSTR*);
HRESULT put_ClassName(BSTR);
```

C#

```
public String ClassName { get; set; }
```

Visual Basic

```
Public ClassName As String
```

Description

Bolt class.

XIV.2.3 Cols

C++

```
HRESULT get_Cols(int*);
HRESULT put_Cols(int);
```

C#

```
public int Cols { get; set; }
```

Visual Basic

```
Public Cols As int
```

Description

Number of columns.

XIV.2.4 Diameter

C++

```
HRESULT get_Diameter(double*);
```

C#

```
public double Diameter { get; }
```

Visual Basic

```
Public ReadOnly Diameter As double
```

Description

Bolt diameter.

XIV.2.5 DiameterName

C++

```
HRESULT get_DiameterName(BSTR*);
HRESULT put_DiameterName(BSTR);
```

C#

```
public String DiameterName { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public DiameterName As String
```

Description

Bolt diameter name.

XIV.2.6 Friction

C++

```
HRESULT get_Friction(double*);
HRESULT put_Friction(double);
```

C#

```
public double Friction { get; set; }
```

Visual Basic

```
Public Friction As double
```

Description

Bolt friction.

XIV.2.7 Height1

C++

```
HRESULT get_Height1(double*);
HRESULT put_Height1(double);
```

C#

```
public double Height1 { get; set; }
```

Visual Basic

```
Public Height1 As double
```

Description

Distance of first bolt (e.g. from the top edge of a plate for knee connections) .

XIV.2.8 Rows

C++

```
HRESULT get_Rows(int*);
HRESULT put_Rows(int);
```

C#

```
public int Rows { get; set; }
```

Visual Basic

```
Public Rows As int
```

Description

Number of rows.

XV IRJointBoltType

C++

```
enum IRJointBoltType;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public enum IRJointBoltType;
```

Visual Basic

```
Public Enum IRJointBoltType
```

Description

Available bolt types.

XVI IRJointWebType

C++

```
enum IRJointWebType;
```

C#

```
public enum IRJointWebType;
```

Visual Basic

```
Public Enum IRJointWebType
```

Description

Types of reinforcing plates.

XVII IRJointConnectionServer

Class Hierarchy

RobotOM.IRJointConnectionServer

C++

```
interface IRJointConnectionServer : IDispatch;
```

C#

```
public interface IRJointConnectionServer;
```

Visual Basic

```
Public Interface IRJointConnectionServer
```

Description

The connection server administers all connection defined in the project. Each connection is associated with a number. The numbering of connections may be discontinuous.

XVII.1 IRJointConnectionServer Members

The following tables list the members exposed by IRJointConnectionServer.

Public Fields

	Name	Description
	Count (see page 2170)	Number of defined connections .

(c) 2014 Autodesk Inc. All rights reserved

Public Methods

	Name	Description
	Calculate (see page 2171)	The function carries out verification of the connection with the given number for the defined loads and returns the verification results.
	CalculateNote (see page 2171)	The function carries out calculations for the indicated connection as the function Calculate (see page 2171)(). Additionally, it generates a calculation note with calculation results.
	Create (see page 2172)	The function creates and returns an object representing a new connection of the indicated type.
	CreateInfo (see page 2172)	The function creates and returns a new object representing the information about a connection.
	Delete (see page 2172)	The function removes the connection with the given number. .
	Exist (see page 2173)	The function checks if there is a connection with the indicated number. .
	Find (see page 2173)	The function finds a connection defined for the indicated selection of bars and returns its number. If there is no such connection, it returns the value 0.
	FindWithId (see page 2173)	Function returns number of a connection with the specified unique identifier. If the connection is not found, zero value is returned. Available since version 1.7.
	Get (see page 2173)	The function returns the connection with the given number. .
	GetAllNumbers (see page 2174)	The function returns a table containing numbers of all defined connections.
	GetInfo (see page 2174)	The function returns an object describing the connection with the given number.

XVII.2 IRJointConnectionServer Fields

The fields of the IRJointConnectionServer class are listed here.

Public Fields

	Name	Description
	Count (see page 2170)	Number of defined connections .

XVII.2.1 Count

C++

```
HRESULT get_Count(long*);
```

C#

```
public long Count { get; }
```

Visual Basic

```
Public ReadOnly Count As long
```

Description

Number of defined connections .

XVII.3 IRJointConnectionServer Methods

The methods of the IRJointConnectionServer class are listed here.

(c) 2014 Autodesk Inc. All rights reserved

Public Methods

	Name	Description
	Calculate (see page 2171)	The function carries out verification of the connection with the given number for the defined loads and returns the verification results.
	CalculateNote (see page 2171)	The function carries out calculations for the indicated connection as the function Calculate (see page 2171)(). Additionally, it generates a calculation note with calculation results.
	Create (see page 2172)	The function creates and returns an object representing a new connection of the indicated type.
	CreateInfo (see page 2172)	The function creates and returns a new object representing the information about a connection.
	Delete (see page 2172)	The function removes the connection with the given number. .
	Exist (see page 2173)	The function checks if there is a connection with the indicated number. .
	Find (see page 2173)	The function finds a connection defined for the indicated selection of bars and returns its number. If there is no such connection, it returns the value 0.
	FindWithId (see page 2173)	Function returns number of a connection with the specified unique identifier. If the connection is not found, zero value is returned. Available since version 1.7.
	Get (see page 2173)	The function returns the connection with the given number. .
	GetAllNumbers (see page 2174)	The function returns a table containing numbers of all defined connections.
	GetInfo (see page 2174)	The function returns an object describing the connection with the given number.

XVII.3.1 Calculate

C++

```
HRESULT Calculate(long _num, IRJointLoad* _joint_loads, double* ret);
```

C#

```
public double Calculate(long _num, IRJointLoad _joint_loads);
```

Visual Basic

```
Public Function Calculate(_num As long, ByRef _joint_loads As IRJointLoad) As double
```

Description

The function carries out verification of the connection with the given number for the defined loads and returns the verification results.

XVII.3.2 CalculateNote

C++

```
HRESULT CalculateNote(long _num, IRJointLoad* _joint_loads, BSTR _file_path, double* ret);
```

C#

```
public double CalculateNote(long _num, IRJointLoad _joint_loads, String _file_path);
```

Visual Basic

```
Public Function CalculateNote(_num As long, ByRef _joint_loads As IRJointLoad, _file_path
As String) As double
```

Description

The function carries out calculations for the indicated connection as the function Calculate (see page 2171)(). Additionally,

(c) 2014 Autodesk Inc. All rights reserved

it generates a calculation note with calculation results.

XVII.3.3 Create

C++

```
HRESULT Create(IRJointConnectionType _joint_type, IRJointConnection** ret);
```

C#

```
public IRJointConnection Create(IRJointConnectionType _joint_type);
```

Visual Basic

```
Public Function Create(_joint_type As IRJointConnectionType) As IRJointConnection
```

Description

The function creates and returns an object representing a new connection of the indicated type.

XVII.3.4 CreateInfo

C++

```
HRESULT CreateInfo(IRJointConnectionInfo** ret);
```

C#

```
public IRJointConnectionInfo CreateInfo();
```

Visual Basic

```
Public Function CreateInfo() As IRJointConnectionInfo
```

Description

The function creates and returns a new object representing the information about a connection.

XVII.3.5 Delete

C++

```
HRESULT Delete(long _num);
```

C#

```
public void Delete(long _num);
```

Visual Basic

```
Public Sub Delete(_num As long)
```

Description

The function removes the connection with the given number. .

XVII.3.6 Exist

C++

```
HRESULT Exist(long _num, VARIANT_BOOL* ret);
```

C#

```
public bool Exist(long _num);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Function Exist(_num As long) As Boolean
```

Description

The function checks if there is a connection with the indicated number. .

XVII.3.7 Find

C++

```
HRESULT Find(BSTR _sel_text, long* ret);
```

C#

```
public long Find(String _sel_text);
```

Visual Basic

```
Public Function Find(_sel_text As String) As long
```

Description

The function finds a connection defined for the indicated selection of bars and returns its number. If there is no such connection, it returns the value 0.

XVII.3.8 FindWithId

C++

```
HRESULT FindWithId(long _unique_id, long* ret);
```

C#

```
public long FindWithId(long _unique_id);
```

Visual Basic

```
Public Function FindWithId(_unique_id As long) As long
```

Description

Function returns number of a connection with the specified unique identifier. If the connection is not found, zero value is returned. Available since version 1.7.

XVII.3.9 Get

C++

```
HRESULT Get(long _num, IRJointConnection** ret);
```

C#

```
public IRJointConnection Get(long _num);
```

Visual Basic

```
Public Function Get(_num As long) As IRJointConnection
```

Description

The function returns the connection with the given number. .

(c) 2014 Autodesk Inc. All rights reserved

XVII.3.10 GetAllNumbers

C++

```
HRESULT GetAllNumbers(IRobotNumbersArray** ret);
```

C#

```
public IRobotNumbersArray GetAllNumbers();
```

Visual Basic

```
Public Function GetAllNumbers() As IRobotNumbersArray
```

Description

The function returns a table containing numbers of all defined connections.

XVII.3.11 GetInfo

C++

```
HRESULT GetInfo(long _num, IRJointConnectionInfo** ret);
```

C#

```
public IRJointConnectionInfo GetInfo(long _num);
```

Visual Basic

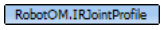
```
Public Function GetInfo(_num As long) As IRJointConnectionInfo
```

Description

The function returns an object describing the connection with the given number.

XVIII IRJointProfile

Class Hierarchy



C++

```
interface IRJointProfile : IDispatch;
```

C#

```
public interface IRJointProfile;
```

Visual Basic

```
Public Interface IRJointProfile
```

Description

Structure describing a single connection component. .

XVIII.1 IRJointProfile Members

The following tables list the members exposed by IRJointProfile.

(c) 2014 Autodesk Inc. All rights reserved

Public Fields

	Name	Description
	Angle (see page 2175)	Connection angle.
	BarNumber (see page 2175)	Bar number.
	Material (see page 2176)	Material name .
	Section (see page 2176)	Section name .

Public Methods

	Name	Description
	AngleExt (see page 2177)	Angle (see page 2175) of section inclination, depending on the parameter - in degrees or radians.
	Edit (see page 2177)	Function gets a section type through the user interface.
	EditComponent (see page 2177)	Function takes component type through user interface.

XVIII.2 IRJointProfile Fields

The fields of the IRJointProfile class are listed here.

Public Fields

	Name	Description
	Angle (see page 2175)	Connection angle.
	BarNumber (see page 2175)	Bar number.
	Material (see page 2176)	Material name .
	Section (see page 2176)	Section name .

XVIII.2.1 Angle

C++

```
HRESULT get_Angle(double*);
HRESULT put_Angle(double);
```

C#

```
public double Angle { get; set; }
```

Visual Basic

```
Public Angle As double
```

Description

Connection angle.

XVIII.2.2 BarNumber

C++

```
HRESULT get_BarNumber(long*);
HRESULT put_BarNumber(long);
```

C#

```
public long BarNumber { get; set; }
```

Visual Basic

```
Public BarNumber As long
```

Description

Bar number.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.5.

XVIII.2.3 Material

C++

```
HRESULT get_Material(BSTR*);
HRESULT put_Material(BSTR);
```

C#

```
public String Material { get; set; }
```

Visual Basic

```
Public Material As String
```

Description

Material name .

XVIII.2.4 Section

C++

```
HRESULT get_Section(BSTR*);
HRESULT put_Section(BSTR);
```

C#

```
public String Section { get; set; }
```

Visual Basic

```
Public Section As String
```

Description

Section name .

XVIII.3 IRJointProfile Methods

The methods of the IRJointProfile class are listed here.

Public Methods

	Name	Description
	AngleExt (see page 2177)	Angle (see page 2175) of section inclination, depending on the parameter - in degrees or radians.
	Edit (see page 2177)	Function gets a section type through the user interface.
	EditComponent (see page 2177)	Function takes component type through user interface.

XVIII.3.1 AngleExt

C++

```
HRESULT AngleExt(IRJointAngleUnit _unit, double* ret);
```

C#

```
public double AngleExt(IRJointAngleUnit _unit);
```

Visual Basic

```
Public Function AngleExt(_unit As IRJointAngleUnit) As double
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Angle (see page 2175) of section inclination, depending on the parameter - in degrees or radians.

Version

Available since version 9.5.

XVIII.3.2 Edit

C++

```
HRESULT Edit(IRJointConnectionType _cType, void* ret);
```

C#

```
public void Edit(IRJointConnectionType _cType);
```

Visual Basic

```
Public Function Edit(_cType As IRJointConnectionType) As void
```

Description

Function gets a section type through the user interface.

Version

Available since version 9.5.

XVIII.3.3 EditComponent

C++

```
HRESULT EditComponent(IRJointComponentType _cType, void* ret);
```

C#

```
public void EditComponent(IRJointComponentType _cType);
```

Visual Basic

```
Public Function EditComponent(_cType As IRJointComponentType) As void
```

Description

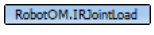
Function takes component type through user interface.

Version

Available since version 10.

XIX IRJointLoad

Class Hierarchy



C++

```
interface IRJointLoad : IDispatch;
```

C#

```
public interface IRJointLoad;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Interface IRJointLoad
```

Description

Interface describing a load applied to the connection. .

XIX.1 IRJointLoad Members

The following tables list the members exposed by IRJointLoad.

Public Fields

	Name	Description
	Cases (see page 2178)	Selection of load cases, defined for the structure. .
	Type (see page 2179)	Load type (defined manually or taken from Robot).

XIX.2 IRJointLoad Fields

The fields of the IRJointLoad class are listed here.

Public Fields

	Name	Description
	Cases (see page 2178)	Selection of load cases, defined for the structure. .
	Type (see page 2179)	Load type (defined manually or taken from Robot).

XIX.2.1 Cases

C++

```
HRESULT get_Cases(IRobotSelection**);
```

C#

```
public IRobotSelection Cases { get; }
```

Visual Basic

```
Public ReadOnly Cases As IRobotSelection
```

Description

Selection of load cases, defined for the structure. .

XIX.2.2 Type

C++

```
HRESULT get_Type(IRJointLoadType*);
HRESULT put_Type(IRJointLoadType);
```

C#

```
public IRJointLoadType Type { get; set; }
```

Visual Basic

```
Public Type As IRJointLoadType
```

Description

Load type (defined manually or taken from Robot).

(c) 2014 Autodesk Inc. All rights reserved

XX IRJointLoadType

C++

```
enum IRJointLoadType;
```

C#

```
public enum IRJointLoadType;
```

Visual Basic

```
Public Enum IRJointLoadType
```

Members

Members	Description
I_JLT_ROBOT_FORCES = 1	Load taken from structure calculations .
I_JLT_MANUAL_FORCES = 2	Loads defined manually for the connection.

XXI IRJointAngleUnit

C++

```
enum IRJointAngleUnit;
```

C#

```
public enum IRJointAngleUnit;
```

Visual Basic

```
Public Enum IRJointAngleUnit
```

Members

Members	Description
I_JAU_RADIANS = 0	Available since version 9.5.
I_JAU_DEGREES = 1	Available since version 9.5.

Version

Available since version 9.5.

XXII IRJointExtType

C++

```
enum IRJointExtType;
```

C#

```
public enum IRJointExtType;
```

Visual Basic

```
Public Enum IRJointExtType
```

Members

Members	Description
I_EXT_NONE = 0	Available since version 9.5.

(c) 2014 Autodesk Inc. All rights reserved

I_EXT_ANGLE2BEAM = 1	Available since version 9.5.

Version

Available since version 9.5.

XXIII IRJointComponentType

C++

```
enum IRJointComponentType;
```

C#

```
public enum IRJointComponentType;
```

Visual Basic

```
Public Enum IRJointComponentType
```

Members

Members	Description
I_JCMT_BEAM_MAIN = 0	Available since version 10.
I_JCMT_BEAM_ANGLES = 1	Available since version 10.
I_JCMT_BEAM_BRACKETS = 2	Available since version 10.

Version

Available since version 10.

External parameters

It is possible to define arbitrary parameters and assign them to inividual structure elements and to the entire project.

Version

Available since version 8.2.

Enumerations

	Name	Description
	IRobotParamValueType (see page 2202)	Available types of external parameter values.

Interfaces

	Name	Description
	IRobotParamServer (see page 2181)	Interface for access to parameters of structure elements of all types. Structure elements are indicated by means of unique identifiers.
	IRobotParamCollection (see page 2187)	Parameter collection.
	IRobotParamSchemaDef (see page 2190)	Definition of the parameter schema.
	IRobotParamSchemaMngr (see page 2194)	Interface for management of all parameter schemata defined in the structure.
	IRobotParamDef (see page 2198)	Complete definition of an external parameter.

(c) 2014 Autodesk Inc. All rights reserved

	IRobotParamSchema (see page 2203)	Parameter schemata have been implemented on order to avoid a collision of names of parameters defined by different external applications. It is recommended that each external application should define its own separate parameter schema.

I IRobotParamServer

Class Hierarchy

RobotOM.IRobotParamServer

C++

```
interface IRobotParamServer : IDispatch;
```

C#

```
public interface IRobotParamServer;
```

Visual Basic

```
Public Interface IRobotParamServer
```

Description

Interface for access to parameters of structure elements of all types. Structure elements are indicated by means of unique identifiers.

Version

Available since version 8.2.

I.1 IRobotParamServer Members

The following tables list the members exposed by IRobotParamServer.

Public Fields

	Name	Description
	Schemas (see page 2182)	External parameter schemata.

Public Methods

	Name	Description
	GetAllParams (see page 2183)	Function enters to the collection all parameters assigned to the given structure element. The number of parameters is returned.
	GetAllParamsForSchema (see page 2183)	Function enters to the collection all parameters from the specified schema assigned to the specified structure element. The number of parameters is returned.
	GetObjectsWithParam (see page 2184)	Function adds to the table unique identifiers of all structure elements which were assigned the determined parameter. Function returns the number of added identifiers.
	GetObjectsWithParamVal (see page 2184)	Function adds to the table unique identifiers of all structure elements which were assigned the determined parameter with the specified value. Function returns the number of added identifiers.
	GetParam (see page 2185)	Function gives access to a value of the specified parameter assigned to a determined structure element. If a given parameter has not been assigned to the structure element, function returns the False value.
	ParamUniqueIdToName (see page 2185)	Function gives access to the name of the parameter and of the schema it belongs to. If the parameter with the specified identifier has not been found, function returns the False value. .

(c) 2014 Autodesk Inc. All rights reserved

	RemoveParam (see page 2185)	Function deletes a determined parameter from the specified structure element.
	RemoveSchemaParams (see page 2186)	Function deletes all parameters belonging to the determined schema from the specified structure element.
	ResetParam (see page 2186)	Function assigns a default value of the determined parameter to a given structure element.
	SetParam (see page 2186)	Function assigns a determined value of the specified parameter to a given structure element.

I.2 IRobotParamServer Fields

The fields of the IRobotParamServer class are listed here.

Public Fields

	Name	Description
	Schemas (see page 2182)	External parameter schemata.

I.2.1 Schemas

C++

```
HRESULT get_Schemas(IRobotParamSchemaMngr**);
```

C#

```
public IRobotParamSchemaMngr Schemas { get; }
```

Visual Basic

```
Public ReadOnly Schemas As IRobotParamSchemaMngr
```

Description

External parameter schemata.

Version

Available since version 8.2.

I.3 IRobotParamServer Methods

The methods of the IRobotParamServer class are listed here.

Public Methods

	Name	Description
	GetAllParams (see page 2183)	Function enters to the collection all parameters assigned to the given structure element. The number of parameters is returned.
	GetAllParamsForSchema (see page 2183)	Function enters to the collection all parameters from the specified schema assigned to the specified structure element. The number of parameters is returned.
	GetObjectsWithParam (see page 2184)	Function adds to the table unique identifiers of all structure elements which were assigned the determined parameter. Function returns the number of added identifiers.
	GetObjectsWithParamVal (see page 2184)	Function adds to the table unique identifiers of all structure elements which were assigned the determined parameter with the specified value. Function returns the number of added identifiers.
	GetParam (see page 2185)	Function gives access to a value of the specified parameter assigned to a determined structure element. If a given parameter has not been assigned to the structure element, function returns the False value.

(c) 2014 Autodesk Inc. All rights reserved

	ParamUniqueIdToName (see page 2185)	Function gives access to the name of the parameter and of the schema it belongs to. If the parameter with the specified identifier has not been found, function returns the False value. .
	RemoveParam (see page 2185)	Function deletes a determined parameter from the specified structure element.
	RemoveSchemaParams (see page 2186)	Function deletes all parameters belonging to the determined schema from the specified structure element.
	ResetParam (see page 2186)	Function assigns a default value of the determined parameter to a given structure element.
	SetParam (see page 2186)	Function assigns a determined value of the specified parameter to a given structure element.

I.3.1 GetAllParams

C++

```
HRESULT GetAllParams(long _unique_obj_id, IRobotParamCollection* _params_col, long* ret);
```

C#

```
public long GetAllParams(long _unique_obj_id, IRobotParamCollection _params_col);
```

Visual Basic

```
Public Function GetAllParams(_unique_obj_id As long, ByRef _params_col As
IRobotParamCollection) As long
```

Description

Function enters to the collection all parameters assigned to the given structure element. The number of parameters is returned.

Version

Available since version 8.2.

I.3.2 GetAllParamsForSchema

C++

```
HRESULT GetAllParamsForSchema(long _unique_obj_id, BSTR _schema_name,
IRobotParamCollection* _params_col, long* ret);
```

C#

```
public long GetAllParamsForSchema(long _unique_obj_id, String _schema_name,
IRobotParamCollection _params_col);
```

Visual Basic

```
Public Function GetAllParamsForSchema(_unique_obj_id As long, _schema_name As String, ByRef
_params_col As IRobotParamCollection) As long
```

Description

Function enters to the collection all parameters from the specified schema assigned to the specified structure element. The number of parameters is returned.

Version

Available since version 8.2.

I.3.3 GetObjectsWithParam

C++

```
HRESULT GetObjectsWithParam(BSTR _param_id, IRobotNumbersArray* _obj_ids, long* ret);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public long GetObjectsWithParam(String _param_id, IRobotNumbersArray _obj_ids);
```

Visual Basic

```
Public Function GetObjectsWithParam(_param_id As String, ByRef _obj_ids As
IRobotNumbersArray) As long
```

Description

Function adds to the table unique identifiers of all structure elements which were assigned the determined parameter. Function returns the number of added identifiers.

Version

Available since version 8.2.

I.3.4 GetObjectsWithParamVal

C++

```
HRESULT GetObjectsWithParamVal(BSTR _param_id, BSTR _param_val, IRobotNumbersArray*
_obj_ids, long* ret);
```

C#

```
public long GetObjectsWithParamVal(String _param_id, String _param_val, IRobotNumbersArray
_obj_ids);
```

Visual Basic

```
Public Function GetObjectsWithParamVal(_param_id As String, _param_val As String, ByRef
_obj_ids As IRobotNumbersArray) As long
```

Description

Function adds to the table unique identifiers of all structure elements which were assigned the determined parameter with the specified value. Function returns the number of added identifiers.

Version

Available since version 8.2.

I.3.5 GetParam

C++

```
HRESULT GetParam(long _unique_obj_id, BSTR _param_id, BSTR _ret_param_val, VARIANT_BOOL*
ret);
```

C#

```
public bool GetParam(long _unique_obj_id, String _param_id, String _ret_param_val);
```

Visual Basic

```
Public Function GetParam(_unique_obj_id As long, _param_id As String, ByRef _ret_param_val
As String) As Boolean
```

Description

Function gives access to a value of the specified parameter assigned to a determined structure element. If a given parameter has not been assigned to the structure element, function returns the False value.

Version

Available since version 8.2.

(c) 2014 Autodesk Inc. All rights reserved

I.3.6 ParamUniqueIdToName

C++

```
HRESULT ParamUniqueIdToName(BSTR _param_id, BSTR _ret_schema_name, BSTR _ret_param_name,
VARIANT_BOOL* ret);
```

C#

```
public bool ParamUniqueIdToName(String _param_id, String _ret_schema_name, String
_ret_param_name);
```

Visual Basic

```
Public Function ParamUniqueIdToName(_param_id As String, ByRef _ret_schema_name As String,
ByRef _ret_param_name As String) As Boolean
```

Description

Function gives access to the name of the parameter and of the schema it belongs to. If the parameter with the specified identifier has not been found, function returns the False value. .

Version

Available since version 8.2.

I.3.7 RemoveParam

C++

```
HRESULT RemoveParam(long _unique_obj_id, BSTR _param_id);
```

C#

```
public void RemoveParam(long _unique_obj_id, String _param_id);
```

Visual Basic

```
Public Sub RemoveParam(_unique_obj_id As long, _param_id As String)
```

Description

Function deletes a determined parameter from the specified structure element.

Version

Available since version 8.2.

I.3.8 RemoveSchemaParams

C++

```
HRESULT RemoveSchemaParams(long _unique_obj_id, BSTR _schema_name);
```

C#

```
public void RemoveSchemaParams(long _unique_obj_id, String _schema_name);
```

Visual Basic

```
Public Sub RemoveSchemaParams(_unique_obj_id As long, _schema_name As String)
```

Description

Function deletes all parameters belonging to the determined schema from the specified structure element.

Version

Available since version 8.2.

(c) 2014 Autodesk Inc. All rights reserved

I.3.9 ResetParam

C++

```
HRESULT ResetParam(long _unique_obj_id, BSTR _param_id);
```

C#

```
public void ResetParam(long _unique_obj_id, String _param_id);
```

Visual Basic

```
Public Sub ResetParam(_unique_obj_id As long, _param_id As String)
```

Description

Function assigns a default value of the determined parameter to a given structure element.

Version

Available since version 8.2.

I.3.10 SetParam

C++

```
HRESULT SetParam(long _unique_obj_id, BSTR _param_id, BSTR _param_val);
```

C#

```
public void SetParam(long _unique_obj_id, String _param_id, String _param_val);
```

Visual Basic

```
Public Sub SetParam(_unique_obj_id As long, _param_id As String, _param_val As String)
```

Description

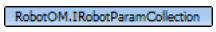
Function assigns a determined value of the specified parameter to a given structure element.

Version

Available since version 8.2.

II IRobotParamCollection

Class Hierarchy



C++

```
interface IRobotParamCollection : IDispatch;
```

C#

```
public interface IRobotParamCollection;
```

Visual Basic

```
Public Interface IRobotParamCollection
```

Description

Parameter collection.

Version

Available since version 8.2.

(c) 2014 Autodesk Inc. All rights reserved

II.1 IRobotParamCollection Members

The following tables list the members exposed by IRobotParamCollection.

Public Fields

	Name	Description
	Count (see page 2187)	Number of parameters in the collection.

Public Methods

	Name	Description
	Find (see page 2188)	Function finds in the collection a parameter with the specified name and returns its index.
	GetFullName (see page 2188)	Function gives access to a name of the parameter and a name of the schema it belongs to.
	GetId (see page 2189)	Function gives access to the unique indentifier of a parameter with the specified index.
	GetName (see page 2189)	Function gives access to the parameter name.
	GetValue (see page 2189)	Function gives access to a value of the parameter with the specified index.

II.2 IRobotParamCollection Fields

The fields of the IRobotParamCollection class are listed here.

Public Fields

	Name	Description
	Count (see page 2187)	Number of parameters in the collection.

II.2.1 Count

C++

```
HRESULT get_Count(long*);
```

C#

```
public long Count { get; }
```

Visual Basic

```
Public ReadOnly Count As long
```

Description

Number of parameters in the collection.

Version

Available since version 8.2.

II.3 IRobotParamCollection Methods

The methods of the IRobotParamCollection class are listed here.

Public Methods

	Name	Description
	Find (see page 2188)	Function finds in the collection a parameter with the specified name and returns its index.

(c) 2014 Autodesk Inc. All rights reserved

	GetFullName (see page 2188)	Function gives access to a name of the parameter and a name of the schema it belongs to.
	GetId (see page 2189)	Function gives access to the unique indentifier of a parameter with the specified index.
	GetName (see page 2189)	Function gives access to the parameter name.
	GetValue (see page 2189)	Function gives access to a value of the parameter with the specified index.

II.3.1 Find

C++

```
HRESULT Find(BSTR _param_name, BSTR _schema_name, long* ret);
```

C#

```
public long Find(String _param_name, String _schema_name);
```

Visual Basic

```
Public Function Find(_param_name As String, _schema_name As String) As long
```

Description

Function finds in the collection a parameter with the specified name and returns its index.

Version

Available since version 8.2.

II.3.2 GetFullName

C++

```
HRESULT GetFullName(long _idx, BSTR _param_name, BSTR _schema_name);
```

C#

```
public void GetFullName(long _idx, String _param_name, String _schema_name);
```

Visual Basic

```
Public Sub GetFullName(_idx As long, ByRef _param_name As String, ByRef _schema_name As
String)
```

Description

Function gives access to a name of the parameter and a name of the schema it belongs to.

Version

Available since version 8.2.

II.3.3 GetId

C++

```
HRESULT GetId(long _idx, BSTR* ret);
```

C#

```
public String GetId(long _idx);
```

Visual Basic

```
Public Function GetId(_idx As long) As String
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Function gives access to the unique indentifier of a parameter with the specified index.

Version

Available since version 8.2.

II.3.4 GetName

C++

```
HRESULT GetName(long _idx, BSTR* ret);
```

C#

```
public String GetName(long _idx);
```

Visual Basic

```
Public Function GetName(_idx As long) As String
```

Description

Function gives access to the parameter name.

Version

Available since version 8.2.

II.3.5 GetValue

C++

```
HRESULT GetValue(long _idx, BSTR* ret);
```

C#

```
public String GetValue(long _idx);
```

Visual Basic

```
Public Function GetValue(_idx As long) As String
```

Description

Function gives access to a value of the parameter with the specified index.

Version

Available since version 8.2.

III IRobotParamSchemaDef

Class Hierarchy

C++

```
interface IRobotParamSchemaDef : IDispatch;
```

C#

```
public interface IRobotParamSchemaDef;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Interface IRobotParamSchemaDef
```

Description

Definition of the parameter schema.

Version

Available since version 8.2.

III.1 IRobotParamSchemaDef Members

The following tables list the members exposed by IRobotParamSchemaDef.

Public Fields

	Name	Description
	Name (see page 2191)	Schema name.
	ParamCount (see page 2191)	Number of parameters defined in the schema.

Public Methods

	Name	Description
	AddParam (see page 2192)	Function defines a new parameter. If the parameter with the specified name already exists in the schema, its definition will be changed. An attempt to change the definition of a parameter which has already been assigned to the structure element fails and function returns the False value.
	AddSimpleParam (see page 2192)	Function adds a simple parameter with the specified name to the schema. If the parameter with the specified name has been earlier defined in the schema, its definition will be changed. An attempt to change the definition of a parameter which has already been assigned to the structure element results in error and function returns the False value.
	GetParam (see page 2193)	Function returns the parameter definition with the specified index in the schema.
	GetParamName (see page 2193)	Function returns the parameter name with the specified index.
	ParamNameToUniqueId (see page 2193)	Function returns a unique identifier of the parameter with the specified name.
	RemoveParam (see page 2194)	Function deletes the parameter with the specified name from the schema. Simultaneously, the parameter will be removed from all structure elements.

III.2 IRobotParamSchemaDef Fields

The fields of the IRobotParamSchemaDef class are listed here.

Public Fields

	Name	Description
	Name (see page 2191)	Schema name.
	ParamCount (see page 2191)	Number of parameters defined in the schema.

III.2.1 Name

C++

```
HRESULT get_Name(BSTR*);
```

C#

```
public String Name { get; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public ReadOnly Name As String
```

Description

Schema name.

Version

Available since version 8.2.

III.2.2 ParamCount

C++

```
HRESULT get_ParamCount(long*);
```

C#

```
public long ParamCount { get; }
```

Visual Basic

```
Public ReadOnly ParamCount As long
```

Description

Number of parameters defined in the schema.

Version

Available since version 8.2.

III.3 IRobotParamSchemaDef Methods

The methods of the IRobotParamSchemaDef class are listed here.

Public Methods

	Name	Description
	AddParam (see page 2192)	Function defines a new parameter. If the parameter with the specified name already exists in the schema, its definition will be changed. An attempt to change the definition of a parameter which has already been assigned to the structure element fails and function returns the False value.
	AddSimpleParam (see page 2192)	Function adds a simple parameter with the specified name to the schema. If the parameter with the specified name has been earlier defined in the schema, its definition will be changed. An attempt to change the definition of a parameter which has already been assigned to the structure element results in error and function returns the False value.
	GetParam (see page 2193)	Function returns the parameter definition with the specified index in the schema.
	GetParamName (see page 2193)	Function returns the parameter name with the specified index.
	ParamNameToUniqueId (see page 2193)	Function returns a unique identifier of the parameter with the specified name.
	RemoveParam (see page 2194)	Function deletes the parameter with the specified name from the schema. Simultaneously, the parameter will be removed from all structure elements.

III.3.1 AddParam

C++

```
HRESULT AddParam(IRobotParamDef* _param_def, VARIANT_BOOL* ret);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public bool AddParam(IRobotParamDef _param_def);
```

Visual Basic

```
Public Function AddParam(ByRef _param_def As IRobotParamDef) As Boolean
```

Description

Function defines a new parameter. If the parameter with the specified name already exists in the schema, its definition will be changed. An attempt to change the definition of a parameter which has already been assigned to the structure element fails and function returns the False value.

Version

Available since version 8.2.

III.3.2 AddSimpleParam

C++

```
HRESULT AddSimpleParam(BSTR _name, BSTR _default_val, VARIANT_BOOL* ret);
```

C#

```
public bool AddSimpleParam(String _name, String _default_val);
```

Visual Basic

```
Public Function AddSimpleParam(_name As String, _default_val As String) As Boolean
```

Description

Function adds a simple parameter with the specified name to the schema. If the parameter with the specified name has been earlier defined in the schema, its definition will be changed. An attempt to change the definition of a parameter which has already been assigned to the structure element results in error and function returns the False value.

Version

Available since version 8.2.

III.3.3 GetParam

C++

```
HRESULT GetParam(long _idx, IRobotParamDef** ret);
```

C#

```
public IRobotParamDef GetParam(long _idx);
```

Visual Basic

```
Public Function GetParam(_idx As long) As IRobotParamDef
```

Description

Function returns the parameter definition with the specified index in the schema.

Version

Available since version 8.2.

III.3.4 GetParamName

C++

```
HRESULT GetParamName(long _idx, BSTR* ret);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public String GetParamName(long _idx);
```

Visual Basic

```
Public Function GetParamName(_idx As long) As String
```

Description

Function returns the parameter name with the specified index.

Version

Available since version 8.2.

III.3.5 ParamNameToUniqueId

C++

```
HRESULT ParamNameToUniqueId(BSTR _name, BSTR* ret);
```

C#

```
public String ParamNameToUniqueId(String _name);
```

Visual Basic

```
Public Function ParamNameToUniqueId(_name As String) As String
```

Description

Function returns a unique identifier of the parameter with the specified name.

Version

Available since version 8.2.

III.3.6 RemoveParam

C++

```
HRESULT RemoveParam(BSTR _param_name);
```

C#

```
public void RemoveParam(String _param_name);
```

Visual Basic

```
Public Sub RemoveParam(_param_name As String)
```

Description

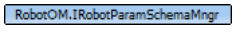
Function deletes the parameter with the specified name from the schema. Simultaneously, the parameter will be removed from all structure elements.

Version

Available since version 8.2.

IV IRobotParamSchemaMngr

Class Hierarchy



(c) 2014 Autodesk Inc. All rights reserved

C++

```
interface IRobotParamSchemaMngr : IDispatch;
```

C#

```
public interface IRobotParamSchemaMngr;
```

Visual Basic

```
Public Interface IRobotParamSchemaMngr
```

Description

Interface for management of all parameter schemata defined in the structure.

Version

Available since version 8.2.

IV.1 IRobotParamSchemaMngr Members

The following tables list the members exposed by IRobotParamSchemaMngr.

Public Fields

	Name	Description
	SchemaCount (see page 2195)	Number of defined schemata.

Public Methods

	Name	Description
	Clear (see page 2196)	Function deletes all parameter schemata defined in the project. It involves simultaneously deletion of all external parameters defined in the project.
	Create (see page 2196)	Function creates a schema with the specified name. If the schema with this name already exists, the function returns its definition.
	Exist (see page 2196)	Function checks if the schema with the specified name exists.
	Get (see page 2197)	Function returns the parameter schema with the specified index.
	GetByName (see page 2197)	Function returns the schema with the specified name.
	Remove (see page 2197)	Function deletes the schema with the specified index.
	RemoveByName (see page 2198)	Function deletes the schema with the specified name.

IV.2 IRobotParamSchemaMngr Fields

The fields of the IRobotParamSchemaMngr class are listed here.

Public Fields

	Name	Description
	SchemaCount (see page 2195)	Number of defined schemata.

IV.2.1 SchemaCount

C++

```
HRESULT get_SchemaCount(long*);
```

C#

```
public long SchemaCount { get; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public ReadOnly SchemaCount As long
```

Description

Number of defined schemata.

Version

Available since version 8.2.

IV.3 IRobotParamSchemaMngr Methods

The methods of the IRobotParamSchemaMngr class are listed here.

Public Methods

	Name	Description
	Clear (see page 2196)	Function deletes all parameter schemata defined in the project. It involves simultaneously deletion of all external parameters defined in the project.
	Create (see page 2196)	Function creates a schema with the specified name. If the schema with this name already exists, the function returns its definition.
	Exist (see page 2196)	Function checks if the schema with the specified name exists.
	Get (see page 2197)	Function returns the parameter schema with the specified index.
	GetByName (see page 2197)	Function returns the schema with the specified name.
	Remove (see page 2197)	Function deletes the schema with the specified index.
	RemoveByName (see page 2198)	Function deletes the schema with the specified name.

IV.3.1 Clear

C++

```
HRESULT Clear();
```

C#

```
public void Clear();
```

Visual Basic

```
Public Sub Clear()
```

Description

Function deletes all parameter schemata defined in the project. It involves simultaneously deletion of all external parameters defined in the project.

Version

Available since version 8.2.

IV.3.2 Create

C++

```
HRESULT Create(BSTR _schema_name, IRobotParamSchema** ret);
```

C#

```
public IRobotParamSchema Create(String _schema_name);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Function Create(_schema_name As String) As IRobotParamSchema
```

Description

Function creates a schema with the specified name. If the schema with this name already exists, the function returns its definition.

Version

Available since version 8.2.

IV.3.3 Exist

C++

```
HRESULT Exist(BSTR _schema_name, VARIANT_BOOL* ret);
```

C#

```
public bool Exist(String _schema_name);
```

Visual Basic

```
Public Function Exist(_schema_name As String) As Boolean
```

Description

Function checks if the schema with the specified name exists.

Version

Available since version 8.2.

IV.3.4 Get

C++

```
HRESULT Get(long _idx, IRobotParamSchema** ret);
```

C#

```
public IRobotParamSchema Get(long _idx);
```

Visual Basic

```
Public Function Get(_idx As long) As IRobotParamSchema
```

Description

Function returns the parameter schema with the specified index.

Version

Available since version 8.2.

IV.3.5 GetByName

C++

```
HRESULT GetByName(BSTR _schema_name, IRobotParamSchema** ret);
```

C#

```
public IRobotParamSchema GetByName(String _schema_name);
```

Visual Basic

```
Public Function GetByName(_schema_name As String) As IRobotParamSchema
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Function returns the schema with the specified name.

Version

Available since version 8.2.

IV.3.6 Remove

C++

```
HRESULT Remove(long _idx);
```

C#

```
public void Remove(long _idx);
```

Visual Basic

```
Public Sub Remove(_idx As long)
```

Description

Function deletes the schema with the specified index.

Version

Available since version 8.2.

IV.3.7 RemoveByName

C++

```
HRESULT RemoveByName(BSTR _schema_name);
```

C#

```
public void RemoveByName(String _schema_name);
```

Visual Basic

```
Public Sub RemoveByName(_schema_name As String)
```

Description

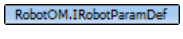
Function deletes the schema with the specified name.

Version

Available since version 8.2.

V IRobotParamDef

Class Hierarchy



C++

```
interface IRobotParamDef : IDispatch;
```

C#

```
public interface IRobotParamDef;
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Interface IRobotParamDef
```

Description

Complete definition of an external parameter.

Version

Available since version 8.2.

V.1 IRobotParamDef Members

The following tables list the members exposed by IRobotParamDef.

Public Fields

	Name	Description
	DefaultValue (see page 2199)	Default parameter value.
	GuiReadOnly (see page 2199)	Flag indicating if a given parameter can be edited in the parameter browser.
	GuiVisible (see page 2200)	Flag indicating if a given parameter should be visible in the parameter browser.
	Name (see page 2200)	Parameter name - should be unique in the schema.
	UniqueId (see page 2200)	Automatically-assigned parameter identifier, unique in the structure.
	ValueList (see page 2201)	List of possible values, if the parameter value is limited only to the specified list.
	ValueType (see page 2201)	Parameter value type.

Public Methods

	Name	Description
	GuiGetName (see page 2202)	Function returns the attribute name in the specified language.
	GuiSetName (see page 2202)	Function sets a language specific attribute name displayed in the attribute browser.

V.2 IRobotParamDef Fields

The fields of the IRobotParamDef class are listed here.

Public Fields

	Name	Description
	DefaultValue (see page 2199)	Default parameter value.
	GuiReadOnly (see page 2199)	Flag indicating if a given parameter can be edited in the parameter browser.
	GuiVisible (see page 2200)	Flag indicating if a given parameter should be visible in the parameter browser.
	Name (see page 2200)	Parameter name - should be unique in the schema.
	UniqueId (see page 2200)	Automatically-assigned parameter identifier, unique in the structure.
	ValueList (see page 2201)	List of possible values, if the parameter value is limited only to the specified list.
	ValueType (see page 2201)	Parameter value type.

V.2.1 DefaultValue

C++

```
HRESULT get_DefaultValue(BSTR*);
HRESULT put_DefaultValue(BSTR);
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public String DefaultValue { get; set; }
```

Visual Basic

```
Public DefaultValue As String
```

Description

Default parameter value.

Version

Available since version 8.2.

V.2.2 GuiReadOnly

C++

```
HRESULT get_GuiReadOnly(VARIANT_BOOL*);
HRESULT put_GuiReadOnly(VARIANT_BOOL);
```

C#

```
public bool GuiReadOnly { get; set; }
```

Visual Basic

```
Public GuiReadOnly As Boolean
```

Description

Flag indicating if a given parameter can be edited in the parameter browser.

Version

Available since version 8.2.

V.2.3 GuiVisible

C++

```
HRESULT get_GuiVisible(VARIANT_BOOL*);
HRESULT put_GuiVisible(VARIANT_BOOL);
```

C#

```
public bool GuiVisible { get; set; }
```

Visual Basic

```
Public GuiVisible As Boolean
```

Description

Flag indicating if a given parameter should be visible in the parameter browser.

Version

Available since version 8.2.

V.2.4 Name

C++

```
HRESULT get_Name(BSTR*);
HRESULT put_Name(BSTR);
```

C#

```
public String Name { get; set; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Name As String
```

Description

Parameter name - should be unique in the schema.

Version

Available since version 8.2.

V.2.5 UniqueId

C++

```
HRESULT get_UniqueId(BSTR*);
```

C#

```
public String UniqueId { get; }
```

Visual Basic

```
Public ReadOnly UniqueId As String
```

Description

Automatically-assigned parameter identifier, unique in the structure.

Version

Available since version 8.2.

V.2.6 ValueList

C++

```
HRESULT get_ValueList(IRobotNamesArray**);
```

C#

```
public IRobotNamesArray ValueList { get; }
```

Visual Basic

```
Public ReadOnly ValueList As IRobotNamesArray
```

Description

List of possible values, if the parameter value is limited only to the specified list.

Version

Available since version 8.2.

V.2.7 ValueType

C++

```
HRESULT get_ValueType(IRobotParamValueType*);
HRESULT put_ValueType(IRobotParamValueType);
```

C#

```
public IRobotParamValueType ValueType { get; set; }
```

Visual Basic

```
Public ValueType As IRobotParamValueType
```

Description

Parameter value type.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 8.2.

V.3 IRobotParamDef Methods

The methods of the IRobotParamDef class are listed here.

Public Methods

	Name	Description
	GuiGetName (see page 2202)	Function returns the attribute name in the specified language.
	GuiSetName (see page 2202)	Function sets a language specific attribute name displayed in the attribute browser.

V.3.1 GuiGetName

C++

```
HRESULT GuiGetName(long _lang_id, BSTR* ret);
```

C#

```
public String GuiGetName(long _lang_id);
```

Visual Basic

```
Public Function GuiGetName(_lang_id As long) As String
```

Description

Function returns the attribute name in the specified language.

Version

Available since version 8.2.

V.3.2 GuiSetName

C++

```
HRESULT GuiSetName(long _lang_id, BSTR _name);
```

C#

```
public void GuiSetName(long _lang_id, String _name);
```

Visual Basic

```
Public Sub GuiSetName(_lang_id As long, _name As String)
```

Description

Function sets a language specific attribute name displayed in the attribute browser.

Version

Available since version 8.2.

VI IRobotParamValueType

C++

```
enum IRobotParamValueType;
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public enum IRobotParamValueType;
```

Visual Basic

```
Public Enum IRobotParamValueType
```

Members

Members	Description
I_PVT_TEXT = 1	Character string. Available since version 8.2.
I_PVT_INTEGER = 2	Integer. Available since version 8.2.
I_PVT_REAL = 3	Real number. Available since version 8.2.

Description

Available types of external parameter values.

Version

Available since version 8.2.

VII IRobotParamSchema

Class Hierarchy

RobotOM.IRobotParamSchema

C++

```
interface IRobotParamSchema : IDispatch;
```

C#

```
public interface IRobotParamSchema;
```

Visual Basic

```
Public Interface IRobotParamSchema
```

Description

Parameter schemata have been implemented on order to avoid a collision of names of parameters defined by different external applications. It is recommended that each external application should define its own separate parameter schema.

Version

Available since version 8.2.

VII.1 IRobotParamSchema Members

The following tables list the members exposed by IRobotParamSchema.

Public Fields

	Name	Description
	Def (see page 2204)	Scheme definition.

(c) 2014 Autodesk Inc. All rights reserved

Public Methods

	Name	Description
	GetAllParams (see page 2205)	Function enters to the collection all parameters from this schema assigned to the given structure element. The number of parameters is returned.
	GetObjectsWithParam (see page 2205)	Function adds to the table unique identifiers of all structure elements which were assigned the determined parameter. Function returns the number of added identifiers.
	GetObjectsWithParamVal (see page 2205)	Function adds to the table unique identifiers of all structure elements which were assigned the determined parameter with the specified value. Function returns the number of added identifiers.
	GetParam (see page 2206)	Function gives access to a value of the specified parameter assigned to a determined structure element. If a given parameter has not been assigned to the structure element, function returns the False value. .
	RemoveAllParams (see page 2206)	Function deletes all parameters belonging to this schema from the specified structure element.
	RemoveParam (see page 2206)	Function deletes a determined parameter from the specified structure element.
	ResetParam (see page 2207)	Function assigns a default value of the determined parameter to the specified structure element.
	SetParam (see page 2207)	Function assigns a determined parameter value to the specified structure element.

VII.2 IRobotParamSchema Fields

The fields of the IRobotParamSchema class are listed here.

Public Fields

	Name	Description
	Def (see page 2204)	Scheme definition.

VII.2.1 Def

C++

```
HRESULT get_Def(IRobotParamSchemaDef**);
```

C#

```
public IRobotParamSchemaDef Def { get; }
```

Visual Basic

```
Public ReadOnly Def As IRobotParamSchemaDef
```

Description

Scheme definition.

Version

Available since version 8.2.

VII.3 IRobotParamSchema Methods

The methods of the IRobotParamSchema class are listed here.

(c) 2014 Autodesk Inc. All rights reserved

Public Methods

	Name	Description
	GetAllParams (see page 2205)	Function enters to the collection all parameters from this schema assigned to the given structure element. The number of parameters is returned.
	GetObjectsWithParam (see page 2205)	Function adds to the table unique identifiers of all structure elements which were assigned the determined parameter. Function returns the number of added identifiers.
	GetObjectsWithParamVal (see page 2205)	Function adds to the table unique identifiers of all structure elements which were assigned the determined parameter with the specified value. Function returns the number of added identifiers.
	GetParam (see page 2206)	Function gives access to a value of the specified parameter assigned to a determined structure element. If a given parameter has not been assigned to the structure element, function returns the False value. .
	RemoveAllParams (see page 2206)	Function deletes all parameters belonging to this schema from the specified structure element.
	RemoveParam (see page 2206)	Function deletes a determined parameter from the specified structure element.
	ResetParam (see page 2207)	Function assigns a default value of the determined parameter to the specified structure element.
	SetParam (see page 2207)	Function assigns a determined parameter value to the specified structure element.

VII.3.1 GetAllParams

C++

```
HRESULT GetAllParams(long _unique_obj_id, IRobotParamCollection* _params_col, long* ret);
```

C#

```
public long GetAllParams(long _unique_obj_id, IRobotParamCollection _params_col);
```

Visual Basic

```
Public Function GetAllParams(_unique_obj_id As long, ByRef _params_col As
IRobotParamCollection) As long
```

Description

Function enters to the collection all parameters from this schema assigned to the given structure element. The number of parameters is returned.

Version

Available since version 8.2.

VII.3.2 GetObjectsWithParam

C++

```
HRESULT GetObjectsWithParam(BSTR _param_name, IRobotNumbersArray* _obj_ids, long* ret);
```

C#

```
public long GetObjectsWithParam(String _param_name, IRobotNumbersArray _obj_ids);
```

Visual Basic

```
Public Function GetObjectsWithParam(_param_name As String, ByRef _obj_ids As
IRobotNumbersArray) As long
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Function adds to the table unique identifiers of all structure elements which were assigned the determined parameter. Function returns the number of added identifiers.

Version

Available since version 8.2.

VII.3.3 GetObjectsWithParamVal

C++

```
HRESULT GetObjectsWithParamVal(BSTR _param_name, BSTR _param_val, IRobotNumbersArray*
_obj_ids, long* ret);
```

C#

```
public long GetObjectsWithParamVal(String _param_name, String _param_val,
IRobotNumbersArray _obj_ids);
```

Visual Basic

```
Public Function GetObjectsWithParamVal(_param_name As String, _param_val As String, ByRef
_obj_ids As IRobotNumbersArray) As long
```

Description

Function adds to the table unique identifiers of all structure elements which were assigned the determined parameter with the specified value. Function returns the number of added identifiers.

Version

Available since version 8.2.

VII.3.4 GetParam

C++

```
HRESULT GetParam(long _unique_obj_id, BSTR _param_name, BSTR _ret_param_val, VARIANT_BOOL*
ret);
```

C#

```
public bool GetParam(long _unique_obj_id, String _param_name, String _ret_param_val);
```

Visual Basic

```
Public Function GetParam(_unique_obj_id As long, _param_name As String, ByRef
_ret_param_val As String) As Boolean
```

Description

Function gives access to a value of the specified parameter assigned to a determined structure element. If a given parameter has not been assigned to the structure element, function returns the False value. .

Version

Available since version 8.2.

VII.3.5 RemoveAllParams

C++

```
HRESULT RemoveAllParams(long _unique_obj_id);
```

C#

```
public void RemoveAllParams(long _unique_obj_id);
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public Sub RemoveAllParams(_unique_obj_id As long)
```

Description

Function deletes all parameters belonging to this schema from the specified structure element.

Version

Available since version 8.2.

VII.3.6 RemoveParam

C++

```
HRESULT RemoveParam(long _unique_obj_id, BSTR _param_name);
```

C#

```
public void RemoveParam(long _unique_obj_id, String _param_name);
```

Visual Basic

```
Public Sub RemoveParam(_unique_obj_id As long, _param_name As String)
```

Description

Function deletes a determined parameter from the specified structure element.

Version

Available since version 8.2.

VII.3.7 ResetParam

C++

```
HRESULT ResetParam(long _unique_obj_id, BSTR _param_name);
```

C#

```
public void ResetParam(long _unique_obj_id, String _param_name);
```

Visual Basic

```
Public Sub ResetParam(_unique_obj_id As long, _param_name As String)
```

Description

Function assigns a default value of the determined parameter to the specified structure element.

Version

Available since version 8.2.

VII.3.8 SetParam

C++

```
HRESULT SetParam(long _unique_obj_id, BSTR _param_name, BSTR _param_val);
```

C#

```
public void SetParam(long _unique_obj_id, String _param_name, String _param_val);
```

Visual Basic

```
Public Sub SetParam(_unique_obj_id As long, _param_name As String, _param_val As String)
```

(c) 2014 Autodesk Inc. All rights reserved

Description

Function assigns a determined parameter value to the specified structure element.

Version

Available since version 8.2.

Robot Kernel

Available since version 2.5.

Interfaces

	Name	Description
	IRobotKernel (see page 2208)	Main interface of the Robot program kernel.
	IRobotKernelPreferences (see page 2215)	Interface describing parameters of the Robot program kernel.

I IRobotKernel

Class Hierarchy

RobotOM.IRobotKernel

C++

```
interface IRobotKernel : IDispatch;
```

C#

```
public interface IRobotKernel;
```

Visual Basic

```
Public Interface IRobotKernel
```

Description

Main interface of the Robot program kernel.

Version

Available since version 2.5.

I.1 IRobotKernel Members

The following tables list the members exposed by IRobotKernel.

Public Fields

	Name	Description
	CalcEngine (see page 2209)	
	CmpntFactory (see page 2210)	
	ConcrReinfEngine (see page 2210)	Calculation module for the concrete reinforcement.
	Preferences (see page 2210)	Settings for the Robot program kernel .
	ProgramName (see page 2210)	Program name.
	ProgramPath (see page 2211)	Full path to the main executable file for process to which RobotKernel has been loaded.

(c) 2014 Autodesk Inc. All rights reserved

	ProgramVersion (see page 2211)	Designation of the program version.
	ProjectPreferences (see page 2211)	Current project settings.
	ProjectUniqueId (see page 2212)	Unique project identifier.
	Structure (see page 2212)	Interface describing the entire structure.
	Version (see page 2212)	Version number.

Public Methods

	Name	Description
	GetExtension (see page 2213)	Function returns the interface for the module extending functionality of the Robot program kernel. If the extension with the indicated name has not been installed, then the function returns zero value (Nothing).
	ProjectNew (see page 2213)	Function creates a new project of the indicated type. .
	ProjectNewFromTemplate (see page 2214)	Function creates a new document based on the template defined by the specified file. .
	ProjectOpen (see page 2214)	Function opens the project saved in the indicated RTD format file. .
	ProjectSave (see page 2214)	Function saves the current project to an RTD format file. .
	ProjectSaveAs (see page 2215)	Function saves the current project to the indicated RTD format file. .

I.2 IRobotKernel Fields

The fields of the IRobotKernel class are listed here.

Public Fields

	Name	Description
	CalcEngine (see page 2209)	
	CmpntFactory (see page 2210)	
	ConcrReinfEngine (see page 2210)	Calculation module for the concrete reinforcement.
	Preferences (see page 2210)	Settings for the Robot program kernel .
	ProgramName (see page 2210)	Program name.
	ProgramPath (see page 2211)	Full path to the main executable file for process to which RobotKernel has been loaded.
	ProgramVersion (see page 2211)	Designation of the program version.
	ProjectPreferences (see page 2211)	Current project settings.
	ProjectUniqueId (see page 2212)	Unique project identifier.
	Structure (see page 2212)	Interface describing the entire structure.
	Version (see page 2212)	Version number.

I.2.1 CalcEngine

C++

```
HRESULT get_CalcEngine(IRobotCalcEngine**);
```

C#

```
public IRobotCalcEngine CalcEngine { get; }
```

Visual Basic

```
Public ReadOnly CalcEngine As IRobotCalcEngine
```

Version

Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

I.2.2 CmpntFactory

C++

```
HRESULT get_CmpntFactory(IRobotComponentFactory**);
```

C#

```
public IRobotComponentFactory CmpntFactory { get; }
```

Visual Basic

```
Public ReadOnly CmpntFactory As IRobotComponentFactory
```

Version

Available since version 2.5.

I.2.3 ConcrReinfEngine

C++

```
HRESULT get_ConcrReinfEngine(IRConcrCalcEngine**);
```

C#

```
public IRConcrCalcEngine ConcrReinfEngine { get; }
```

Visual Basic

```
Public ReadOnly ConcrReinfEngine As IRConcrCalcEngine
```

Description

Calculation module for the concrete reinforcement.

Version

Available since version 5.5.

I.2.4 Preferences

C++

```
HRESULT get_Preferences(IRobotKernelPreferences**);
```

C#

```
public IRobotKernelPreferences Preferences { get; }
```

Visual Basic

```
Public ReadOnly Preferences As IRobotKernelPreferences
```

Description

Settings for the Robot program kernel .

Version

Available since version 2.5.

I.2.5 ProgramName

C++

```
HRESULT get_ProgramName(BSTR*);
```

C#

```
public String ProgramName { get; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public ReadOnly ProgramName As String
```

Description

Program name.

Version

Available since version 5.5.

I.2.6 ProgramPath

C++

```
HRESULT get_ProgramPath(BSTR*);
```

C#

```
public String ProgramPath { get; }
```

Visual Basic

```
Public ReadOnly ProgramPath As String
```

Description

Full path to the main executable file for process to which RobotKernel has been loaded.

Version

Available since version 13.4.

I.2.7 ProgramVersion

C++

```
HRESULT get_ProgramVersion(BSTR*);
```

C#

```
public String ProgramVersion { get; }
```

Visual Basic

```
Public ReadOnly ProgramVersion As String
```

Description

Designation of the program version.

Version

Available since version 5.5.

I.2.8 ProjectPreferences

C++

```
HRESULT get_ProjectPreferences(IRobotProjectPreferences**);
```

C#

```
public IRobotProjectPreferences ProjectPreferences { get; }
```

Visual Basic

```
Public ReadOnly ProjectPreferences As IRobotProjectPreferences
```

Description

Current project settings.

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 2.5.

I.2.9 ProjectUniqueId

C++

```
HRESULT get_ProjectUniqueId(BSTR*);
```

C#

```
public String ProjectUniqueId { get; }
```

Visual Basic

```
Public ReadOnly ProjectUniqueId As String
```

Description

Unique project identifier.

Version

Available since version 2.5.

I.2.10 Structure

C++

```
HRESULT get_Structure(IRobotStructure**);
```

C#

```
public IRobotStructure Structure { get; }
```

Visual Basic

```
Public ReadOnly Structure As IRobotStructure
```

Description

Interface describing the entire structure.

Version

Available since version 2.5.

I.2.11 Version

C++

```
HRESULT get_Version(double*);
```

C#

```
public double Version { get; }
```

Visual Basic

```
Public ReadOnly Version As double
```

Description

Version number.

Version

Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

I.3 IRobotKernel Methods

The methods of the IRobotKernel class are listed here.

Public Methods

	Name	Description
	GetExtension (see page 2213)	Function returns the interface for the module extending functionality of the Robot program kernel. If the extension with the indicated name has not been installed, then the function returns zero value (Nothing).
	ProjectNew (see page 2213)	Function creates a new project of the indicated type. .
	ProjectNewFromTemplate (see page 2214)	Function creates a new document based on the template defined by the specified file. .
	ProjectOpen (see page 2214)	Function opens the project saved in the indicated RTD format file. .
	ProjectSave (see page 2214)	Function saves the current project to an RTD format file. .
	ProjectSaveAs (see page 2215)	Function saves the current project to the indicated RTD format file. .

I.3.1 GetExtension

C++

```
HRESULT GetExtension(BSTR _extmdl_name, IDispatch* ret);
```

C#

```
public IDispatch GetExtension(String _extmdl_name);
```

Visual Basic

```
Public Function GetExtension(_extmdl_name As String) As IDispatch
```

Description

Function returns the interface for the module extending functionality of the Robot program kernel. If the extension with the indicated name has not been installed, then the function returns zero value (Nothing).

Version

Available since version 2.5.

I.3.2 ProjectNew

C++

```
HRESULT ProjectNew(IRobotProjectType _prj_type, VARIANT_BOOL* ret);
```

C#

```
public bool ProjectNew(IRobotProjectType _prj_type);
```

Visual Basic

```
Public Function ProjectNew(_prj_type As IRobotProjectType) As Boolean
```

Description

Function creates a new project of the indicated type. .

Version

Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

I.3.3 ProjectNewFromTemplate

C++

```
HRESULT ProjectNewFromTemplate(BSTR _tmpl_file, VARIANT_BOOL* ret);
```

C#

```
public bool ProjectNewFromTemplate(String _tmpl_file);
```

Visual Basic

```
Public Function ProjectNewFromTemplate(_tmpl_file As String) As Boolean
```

Description

Function creates a new document based on the template defined by the specified file. .

Version

Available since version 3.5.

I.3.4 ProjectOpen

C++

```
HRESULT ProjectOpen(BSTR _file_path, VARIANT_BOOL* ret);
```

C#

```
public bool ProjectOpen(String _file_path);
```

Visual Basic

```
Public Function ProjectOpen(_file_path As String) As Boolean
```

Description

Function opens the project saved in the indicated RTD format file. .

Version

Available since version 2.5.

I.3.5 ProjectSave

C++

```
HRESULT ProjectSave();
```

C#

```
public void ProjectSave();
```

Visual Basic

```
Public Sub ProjectSave()
```

Description

Function saves the current project to an RTD format file. .

Version

Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

I.3.6 ProjectSaveAs

C++

```
HRESULT ProjectSaveAs(BSTR _file_path);
```

C#

```
public void ProjectSaveAs(String _file_path);
```

Visual Basic

```
Public Sub ProjectSaveAs(_file_path As String)
```

Description

Function saves the current project to the indicated RTD format file. .

Version

Available since version 2.5.

II IRobotKernelPreferences

Class Hierarchy

RobotOM.IRobotKernelPreferences

C++

```
interface IRobotKernelPreferences : IDispatch;
```

C#

```
public interface IRobotKernelPreferences;
```

Visual Basic

```
Public Interface IRobotKernelPreferences
```

Description

Interface describing parameters of the Robot program kernel.

Version

Available since version 2.5.

II.1 IRobotKernelPreferences Members

The following tables list the members exposed by IRobotKernelPreferences.

Public Fields

	Name	Description
	Multiprocessing (see page 2216)	Flag activating parallel processing - multiprocessing.

Public Methods

	Name	Description
	GetDirectory (see page 2217)	Function returns the full path to the selected folder.
	GetDirectoryExt (see page 2217)	Function returns folder extension. If for a specified folder no extension has been defined, function returns an empty character string. .
	GetLanguage (see page 2217)	Function returns the current setting of the indicated language.

(c) 2014 Autodesk Inc. All rights reserved

	SetDirectoryExt (see page 2218)	Function defines a folder extension. If an empty character string is given a as a full path to the extension, it will result in deleting the extension. Meaning of the folder extension is best illustrated by the example. If the SetDirectoryExt(I_DE_TEMPLATE, "C:My Templates") function is activated, then from this moment on template files will be searched first in the C:My Templates folder and next, in the standard folder with template files. .
	SetLanguage (see page 2218)	

II.2 IRobotKernelPreferences Fields

The fields of the IRobotKernelPreferences class are listed here.

Public Fields

	Name	Description
	Multiprocessing (see page 2216)	Flag activating parallel processing - multiprocessing.

II.2.1 Multiprocessing

C++

```
HRESULT get_Multiprocessing(VARIANT_BOOL*);
HRESULT put_Multiprocessing(VARIANT_BOOL);
```

C#

```
public bool Multiprocessing { get; set; }
```

Visual Basic

```
Public Multiprocessing As Boolean
```

Description

Flag activating parallel processing - multiprocessing.

Version

Available since version 8.5.

II.3 IRobotKernelPreferences Methods

The methods of the IRobotKernelPreferences class are listed here.

Public Methods

	Name	Description
	GetDirectory (see page 2217)	Function returns the full path to the selected folder.
	GetDirectoryExt (see page 2217)	Function returns folder extension. If for a specified folder no extension has been defined, function returns an empty character string. .
	GetLanguage (see page 2217)	Function returns the current setting of the indicated language.
	SetDirectoryExt (see page 2218)	Function defines a folder extension. If an empty character string is given a as a full path to the extension, it will result in deleting the extension. Meaning of the folder extension is best illustrated by the example. If the SetDirectoryExt(I_DE_TEMPLATE, "C:My Templates") function is activated, then from this moment on template files will be searched first in the C:My Templates folder and next, in the standard folder with template files. .
	SetLanguage (see page 2218)	

(c) 2014 Autodesk Inc. All rights reserved

II.3.1 GetDirectory

C++

```
HRESULT GetDirectory(IRobotDirectory _rob_dir, BSTR* ret);
```

C#

```
public String GetDirectory(IRobotDirectory _rob_dir);
```

Visual Basic

```
Public Function GetDirectory(_rob_dir As IRobotDirectory) As String
```

Description

Function returns the full path to the selected folder.

Version

Available since version 2.5.

II.3.2 GetDirectoryExt

C++

```
HRESULT GetDirectoryExt(IRobotDirectoryExtension _dir_ext, BSTR* ret);
```

C#

```
public String GetDirectoryExt(IRobotDirectoryExtension _dir_ext);
```

Visual Basic

```
Public Function GetDirectoryExt(_dir_ext As IRobotDirectoryExtension) As String
```

Description

Function returns folder extension. If for a specified folder no extension has been defined, function returns an empty character string. .

Version

Available since version 3.5.

II.3.3 GetLanguage

C++

```
HRESULT GetLanguage(IRobotLanguage _lang_id, int* ret);
```

C#

```
public int GetLanguage(IRobotLanguage _lang_id);
```

Visual Basic

```
Public Function GetLanguage(_lang_id As IRobotLanguage) As int
```

Description

Function returns the current setting of the indicated language.

Version

Available since version 2.5.

(c) 2014 Autodesk Inc. All rights reserved

II.3.4 SetDirectoryExt

C++

```
HRESULT SetDirectoryExt(IRobotDirectoryExtension _dir_ext, BSTR _full_path);
```

C#

```
public void SetDirectoryExt(IRobotDirectoryExtension _dir_ext, String _full_path);
```

Visual Basic

```
Public Sub SetDirectoryExt(_dir_ext As IRobotDirectoryExtension, _full_path As String)
```

Description

Function defines a folder extension. If an empty character string is given a as a full path to the extension, it will result in deleting the extension.

Meaning of the folder extension is best illustrated by the example. If the SetDirectoryExt(I_DE_TEMPLATE, "C:My Templates") function is activated, then from this moment on template files will be searched first in the C:My Templates folder and next, in the standard folder with template files. .

Version

Available since version 3.5.

II.3.5 SetLanguage

C++

```
HRESULT SetLanguage(IRobotLanguage _lang_id, int _lang_num);
```

C#

```
public void SetLanguage(IRobotLanguage _lang_id, int _lang_num);
```

Visual Basic

```
Public Sub SetLanguage(_lang_id As IRobotLanguage, _lang_num As int)
```

Version

Available since version 2.5.

STR files analyzer

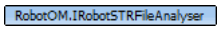
Available since version 2.5.

Interfaces

	Name	Description
	IRobotSTRFileAnalyser (see page 2219)	Analyzer of STR format text files.

I IRobotSTRFileAnalyser

Class Hierarchy



(c) 2014 Autodesk Inc. All rights reserved

C++

```
interface IRobotSTRFileAnalyser : IDispatch;
```

C#

```
public interface IRobotSTRFileAnalyser;
```

Visual Basic

```
Public Interface IRobotSTRFileAnalyser
```

Description

Analyzer of STR format text files.

Version

Available since version 2.5.

I.1 IRobotSTRFileAnalyser Members

The following tables list the members exposed by IRobotSTRFileAnalyser.

Public Fields

	Name	Description
	InsertParams (see page 2220)	Parameters of inserting the structure defined in an STR file to the current project.
	Params (see page 2220)	Parameters defined in the STR file.

Public Methods

	Name	Description
	InsertToProject (see page 2220)	Function inserts definition of the structure described in the indicated STR file to the current project. If the parameters of the inserted structure have been read out and modified earlier, then their new values will be considered during structure insertion. .
	ReadParams (see page 2221)	Function reads in the parameters defined in the indicated STR file. Once read in, the parameters are available by means of the Params (see page 2220) component and may be edited. If read-out of variables fails, then zero value will be returned (False). .

I.2 IRobotSTRFileAnalyser Fields

The fields of the IRobotSTRFileAnalyser class are listed here.

Public Fields

	Name	Description
	InsertParams (see page 2220)	Parameters of inserting the structure defined in an STR file to the current project.
	Params (see page 2220)	Parameters defined in the STR file.

I.2.1 InsertParams

C++

```
HRESULT get_InsertParams(IRobotFileInsertParams**);
```

C#

```
public IRobotFileInsertParams InsertParams { get; }
```

(c) 2014 Autodesk Inc. All rights reserved

Visual Basic

```
Public ReadOnly InsertParams As IRobotFileInsertParams
```

Description

Parameters of inserting the structure defined in an STR file to the current project.

Version

Available since version 2.5.

I.2.2 Params

C++

```
HRESULT get_Params(IRobotSTRParams**);
```

C#

```
public IRobotSTRParams Params { get; }
```

Visual Basic

```
Public ReadOnly Params As IRobotSTRParams
```

Description

Parameters defined in the STR file.

Version

Available since version 2.5.

I.3 IRobotSTRFileAnalyser Methods

The methods of the IRobotSTRFileAnalyser class are listed here.

Public Methods

	Name	Description
	InsertToProject (see page 2220)	Function inserts definition of the structure described in the indicated STR file to the current project. If the parameters of the inserted structure have been read out and modified earlier, then their new values will be considered during structure insertion. .
	ReadParams (see page 2221)	Function reads in the parameters defined in the indicated STR file. Once read in, the parameters are available by means of the Params (see page 2220) component and may be edited. If read-out of variables fails, then zero value will be returned (False). .

I.3.1 InsertToProject

C++

```
HRESULT InsertToProject(BSTR _file_path, VARIANT_BOOL _ignore_warnings, VARIANT_BOOL
_only_geometry, VARIANT_BOOL* ret);
```

C#

```
public bool InsertToProject(String _file_path, bool _ignore_warnings, bool _only_geometry);
```

Visual Basic

```
Public Function InsertToProject(_file_path As String, _ignore_warnings As Boolean,
_only_geometry As Boolean) As Boolean
```

Description

Function inserts definition of the structure described in the indicated STR file to the current project. If the parameters of the

(c) 2014 Autodesk Inc. All rights reserved

inserted structure have been read out and modified earlier, then their new values will be considered during structure insertion. .

Version

Available since version 2.5.

I.3.2 ReadParams

C++

```
HRESULT ReadParams(BSTR _file_path, VARIANT_BOOL* ret);
```

C#

```
public bool ReadParams(String _file_path);
```

Visual Basic

```
Public Function ReadParams(_file_path As String) As Boolean
```

Description

Function reads in the parameters defined in the indicated STR file. Once read in, the parameters are available by means of the Params (see page 2220) component and may be edited. If read-out of variables fails, then zero value will be returned (False). .

Version

Available since version 2.5.

Structure correction

Available since version 3.5.

Enumerations

	Name	Description
	IRobotBarIntersectRelationship (see page 2222)	Relations between bars resulting from bar intersections. .

Interfaces

	Name	Description
	IRobotStructureGeoAnalyser (see page 2222)	Tool component performing anlysis of structure geometry and its automatic correction in order to obtain a correct calculation model.

I IRobotBarIntersectRelationship

C++

```
enum IRobotBarIntersectRelationship;
```

C#

```
public enum IRobotBarIntersectRelationship;
```

Visual Basic

```
Public Enum IRobotBarIntersectRelationship
```

(c) 2014 Autodesk Inc. All rights reserved

Members

Members	Description
I_BIR_UNKNOWN = 0	It is not known if bars remaining in this relation intersect or not. Available since version 3.5.
I_BIR_NOT_INTERSECT = 1	Bars remaining in this relation do not intersect. Available since version 3.5.
I_BIR_INTERSECT_MID_MID = 2	Bars remaining in this relation intersect. Available since version 3.5.
I_BIR_INTERSECT_MID_END = 3	Available since version 3.5.
I_BIR_INTERSECT_END_MID = 4	Available since version 3.5.
I_BIR_INTERSECT_END_END = 5	Available since version 3.5.

Description

Relations between bars resulting from bar intersections. .

Version

Available since version 3.5.

II IRobotStructureGeoAnalyser

Class Hierarchy

RobotOM.IRobotStructureGeoAnalyser

C++

```
interface IRobotStructureGeoAnalyser : IDispatch;
```

C#

```
public interface IRobotStructureGeoAnalyser;
```

Visual Basic

```
Public Interface IRobotStructureGeoAnalyser
```

Description

Tool component performing anlysis of structure geometry and its automatic correction in order to obtain a correct calculation model.

Version

Available since version 3.5.

II.1 IRobotStructureGeoAnalyser Members

The following tables list the members exposed by IRobotStructureGeoAnalyser.

Public Fields

	Name	Description
	CanEliminateIsolatedNodes (see page 2224)	Flag steering with work of the Correct (see page 2225) function; the True value indicates that it is possible to delete isolated nodes in order to perform structure correction.
	CanExtendBars (see page 2224)	Flag managing operation of the Correct (see page 2225) function; True (default) value means that to carry out structure correction bar length may be increased.

(c) 2014 Autodesk Inc. All rights reserved

	CanUseRigidLinks (see page 2224)	Flag managing operation of the Correct (see page 2225) function; True (default) value means that to carry out structure correction rigid links may be defined.
	Precision (see page 2225)	Precision applied by the Correct (see page 2225) function; setting the value (-1) indicates adoption of a default value .

Public Methods

	Name	Description
	Correct (see page 2225)	Function carries out geometry correction for a whole structure or for a structure part based on the parameters set earlier. .
	DefineIntersection (see page 2226)	Function defines a relation of intersection between the specified bars. This information will be used by the function of structure geometry correction. .
	SetEdgeSize (see page 2226)	Function defines the "thickness" of a geometric object edge.
	SetNodeSize (see page 2226)	Function defines the "size" of a node. Structure elements that are located in a distance equal or less than the specified radius from the node will be treated as adjoining to the node.
	StartCollectInfo (see page 2227)	Function runs analysis of operations performed on the structure to make subsequent correction of its geometry easier. .
	StopCollectInfo (see page 2227)	

II.2 IRobotStructureGeoAnalyser Fields

The fields of the IRobotStructureGeoAnalyser class are listed here.

Public Fields

	Name	Description
	CanEliminateIsolatedNodes (see page 2224)	Flag steering with work of the Correct (see page 2225) function; the True value indicates that it is possible to delete isolated nodes in order to perform structure correction.
	CanExtendBars (see page 2224)	Flag managing operation of the Correct (see page 2225) function; True (default) value means that to carry out structure correction bar length may be increased.
	CanUseRigidLinks (see page 2224)	Flag managing operation of the Correct (see page 2225) function; True (default) value means that to carry out structure correction rigid links may be defined.
	Precision (see page 2225)	Precision applied by the Correct (see page 2225) function; setting the value (-1) indicates adoption of a default value .

II.2.1 CanEliminateIsolatedNodes

C++

```
HRESULT get_CanEliminateIsolatedNodes(VARIANT_BOOL*);
HRESULT put_CanEliminateIsolatedNodes(VARIANT_BOOL);
```

C#

```
public bool CanEliminateIsolatedNodes { get; set; }
```

Visual Basic

```
Public CanEliminateIsolatedNodes As Boolean
```

Description

Flag steering with work of the Correct (see page 2225) function; the True value indicates that it is possible to delete isolated nodes in order to perform structure correction.

Version

Available since version 8.5.

(c) 2014 Autodesk Inc. All rights reserved

II.2.2 CanExtendBars

C++

```
HRESULT get_CanExtendBars(VARIANT_BOOL*);
HRESULT put_CanExtendBars(VARIANT_BOOL);
```

C#

```
public bool CanExtendBars { get; set; }
```

Visual Basic

```
Public CanExtendBars As Boolean
```

Description

Flag managing operation of the Correct (see page 2225) function; True (default) value means that to carry out structure correction bar length may be increased.

Version

Available since version 3.5.

II.2.3 CanUseRigidLinks

C++

```
HRESULT get_CanUseRigidLinks(VARIANT_BOOL*);
HRESULT put_CanUseRigidLinks(VARIANT_BOOL);
```

C#

```
public bool CanUseRigidLinks { get; set; }
```

Visual Basic

```
Public CanUseRigidLinks As Boolean
```

Description

Flag managing operation of the Correct (see page 2225) function; True (default) value means that to carry out structure correction rigid links may be defined.

Version

Available since version 3.5.

II.2.4 Precision

C++

```
HRESULT get_Precision(double*);
HRESULT put_Precision(double);
```

C#

```
public double Precision { get; set; }
```

Visual Basic

```
Public Precision As double
```

Description

Precision applied by the Correct (see page 2225) function; setting the value (-1) indicates adoption of a default value .

Version

Available since version 3.5.

(c) 2014 Autodesk Inc. All rights reserved

II.3 IRobotStructureGeoAnalyser Methods

The methods of the IRobotStructureGeoAnalyser class are listed here.

Public Methods

	Name	Description
	Correct (see page 2225)	Function carries out geometry correction for a whole structure or for a structure part based on the parameters set earlier. .
	DefineIntersection (see page 2226)	Function defines a relation of intersection between the specified bars. This information will be used by the function of structure geometry correction. .
	SetEdgeSize (see page 2226)	Function defines the "thickness" of a geometric object edge.
	SetNodeSize (see page 2226)	Function defines the "size" of a node. Structure elements that are located in a distance equal or less than the specified radius from the node will be treated as adjoining to the node.
	StartCollectInfo (see page 2227)	Function runs analysis of operations performed on the structure to make subsequent correction of its geometry easier. .
	StopCollectInfo (see page 2227)	

II.3.1 Correct

C++

```
HRESULT Correct(BSTR _bar_sel = "");
```

C#

```
public void Correct(String _bar_sel = "");
```

Visual Basic

```
Public Sub Correct(Optional _bar_sel As String = "")
```

Description

Function carries out geometry correction for a whole structure or for a structure part based on the parameters set earlier. .

Version

Available since version 3.5.

II.3.2 DefineIntersection

C++

```
HRESULT DefineIntersection(long _bar1, long _bar2, IRobotBarIntersectRelationship
_intersect_rel);
```

C#

```
public void DefineIntersection(long _bar1, long _bar2, IRobotBarIntersectRelationship
_intersect_rel);
```

Visual Basic

```
Public Sub DefineIntersection(_bar1 As long, _bar2 As long, _intersect_rel As
IRobotBarIntersectRelationship)
```

Description

Function defines a relation of intersection between the specified bars. This information will be used by the function of structure geometry correction. .

(c) 2014 Autodesk Inc. All rights reserved

Version

Available since version 3.5.

II.3.3 SetEdgeSize

C++

```
HRESULT SetEdgeSize(long _obj_num, double _radius);
```

C#

```
public void SetEdgeSize(long _obj_num, double _radius);
```

Visual Basic

```
Public Sub SetEdgeSize(_obj_num As long, _radius As double)
```

Description

Function defines the "thickness" of a geometric object edge.

II.3.4 SetNodeSize

C++

```
HRESULT SetNodeSize(long _node_num, double _radius);
```

C#

```
public void SetNodeSize(long _node_num, double _radius);
```

Visual Basic

```
Public Sub SetNodeSize(_node_num As long, _radius As double)
```

Description

Function defines the "size" of a node. Structure elements that are located in a distance equal or less than the specified radius from the node will be treated as adjoining to the node.

II.3.5 StartCollectInfo

C++

```
HRESULT StartCollectInfo();
```

C#

```
public void StartCollectInfo();
```

Visual Basic

```
Public Sub StartCollectInfo()
```

Description

Function runs analysis of operations performed on the structure to make subsequent correction of its geometry easier. .

Version

Available since version 3.5.

II.3.6 StopCollectInfo

C++

```
HRESULT StopCollectInfo();
```

(c) 2014 Autodesk Inc. All rights reserved

C#

```
public void StopCollectInfo();
```

Visual Basic

```
Public Sub StopCollectInfo()
```

Version

Available since version 3.5.

(c) 2014 Autodesk Inc. All rights reserved

Index

_

_VVST_STRUCTURAL_AXIS enumeration member 1406

A

Add-ins Manager 1548

Angle connection 2030

Application - main object of the model 1535

B

Backgrounds 1327

Bars 538

Basic definitions 2

Beam-principal beam connection 2123

Buckling analysis parameters 359

C

Cables 539

Calculation module 1560

Calculation results for bars 869

Calculation results for dynamic analyses 924

Calculation results for finite elements 935

Calculation results for nodes 852

CB71 Code 1853

Claddings 614

Code combinations 148

Column base connections 1970

Compatible nodes 488

Complex attributes of a bar 538

Complex attributes of a node 478

Complex attributes of an object 703

Complex attributes shared by objects of different types 659

Concrete 1665

Concrete column base 1989

Connection module 1938

Continuous Footing 1168

D

Data Server 1

Data structures containing complex calculation results 884

Design of RC members 1788

Direct Analysis Method 1595

Drawing 1266

E

EC3 Code 1808

Edit operations 1048

Elastic ground 542

EPF_HTML enumeration member 1518

EPF_MS_OFFICE enumeration member 1518

EPF_OPEN_OFFICE enumeration member 1518

External parameters 2180

F

FE mesh generator 807

Finite elements 807

Fixed column base 1979

Footfall analysis parameters 398

FRF analysis parameters 395

G

Generating 3D loads 82

Geometric data types 1609

Geometrical imperfections 588

Global data types 1608

Graphical structure views 1379

Grouping of objects 847

Gusset plate connections 2059

H

Harmonic analysis parameters 390

I

I_3PRV_LOCAL_SYSTEM enumeration member 45

I_3PRV_N1 enumeration member 45

I_3PRV_N2 enumeration member 45

I_3PRV_N3 enumeration member 45

I_3PRV_PROJECTION enumeration member 45

I_3PRV_PX1 enumeration member 45

I_3PRV_PX2 enumeration member 45

(c) 2014 Autodesk Inc. All rights reserved

I_3PRV_PX3 enumeration member 45
I_3PRV_PY1 enumeration member 45
I_3PRV_PY2 enumeration member 45
I_3PRV_PY3 enumeration member 45
I_3PRV_PZ1 enumeration member 45
I_3PRV_PZ2 enumeration member 45
I_3PRV_PZ3 enumeration member 45
I_3PRV_TX1 enumeration member 48
I_3PRV_TX2 enumeration member 48
I_3PRV_TX3 enumeration member 48
I_3PRV_TZ1 enumeration member 48
I_3PRV_TZ2 enumeration member 48
I_3PRV_TZ3 enumeration member 48
I_ACI318_BR_10 enumeration member 1750
I_ACI318_BR_11 enumeration member 1750
I_ACI318_BR_14 enumeration member 1750
I_ACI318_BR_18 enumeration member 1750
I_ACI318_BR_3 enumeration member 1750
I_ACI318_BR_4 enumeration member 1750
I_ACI318_BR_5 enumeration member 1750
I_ACI318_BR_6 enumeration member 1750
I_ACI318_BR_7 enumeration member 1750
I_ACI318_BR_8 enumeration member 1750
I_ACI318_BR_9 enumeration member 1750
I_ACI318_LAPT_12 enumeration member 1748
I_ACI318_LAPT_3 enumeration member 1748
I_ACI318_LAPT_6 enumeration member 1748
I_ACI318_LAPT_60 enumeration member 1748
I_ACI318_MBD_10 enumeration member 1751
I_ACI318_MBD_12 enumeration member 1751
I_ACI318_MBD_14 enumeration member 1751
I_ACI318_MBD_16 enumeration member 1751
I_ACI318_MBD_20 enumeration member 1751
I_ACI318_MBD_25 enumeration member 1751
I_ACI318_MBD_32 enumeration member 1751
I_ACI318_MBD_40 enumeration member 1751
I_ACI318_MBD_5 enumeration member 1751
I_ACI318_MBD_6 enumeration member 1751
I_ACI318_MBD_8 enumeration member 1751
I_ACI318_ST_40 enumeration member 1748
I_ACI318_ST_50 enumeration member 1748
I_ACI318_ST_60 enumeration member 1748
I_ACI318_ST_75 enumeration member 1748
I_AMT_DAM enumeration member 1360
I_AMT_MAIN enumeration member 1360
I_AST_COLUMN_CIRCULAR enumeration member 517
I_AST_COLUMN_RECTANGULAR enumeration member 517
I_AST_NODAL enumeration member 517
I_AST_WALL_CONCRETE enumeration member 517
I_AST_WALL_MASONRY enumeration member 517
I_BAM_BLOCK_SUBSPACE_ITERATION enumeration member 363
I_BAM_SUBSPACE_ITERATION enumeration member 363
I_BBT_BIAXIAL enumeration member 1698
I_BBT_FALSE enumeration member 1698
I_BBT_SIMPLE enumeration member 1698
I_BBT_UNIAXIAL enumeration member 1698
I_BCAPT_ELONGATION_DL enumeration member 541
I_BCAPT_ELONGATION_DL_RELATIVE enumeration member 541
I_BCAPT_FORCE_FO enumeration member 541
I_BCAPT_LENGTH_L enumeration member 541
I_BCAPT_STRESSES_SIGMA enumeration member 541
I_BCCT_MEMBER enumeration member 1696
I_BCCT_PLATE enumeration member 1696
I_BCE_BCKL_COLUMN enumeration member 1686
I_BCE_NO_CALC enumeration member 1686
I_BCE_NO_ERRORS enumeration member 1686
I_BCE_SECTION_SHEAR enumeration member 1686
I_BCE_SECTION_TOO_SMALL enumeration member 1686
I_BCE_USER_ERROR enumeration member 1686
I_BCPDDV_CONCRETE_CREEP enumeration member 1678
I_BCPDDV_CONCRETE_FC enumeration member 1678
I_BCPDDV_COVER_SIZE enumeration member 1678
I_BCPDDV_FLT2F enumeration member 1678
I_BCPDDV_LONG_COMPRESSION_BAR_DIAM enumeration member 1678
I_BCPDDV_LONG_STEEL_FE enumeration member 1678
I_BCPDDV_LONG_TENSION_BAR_DIAM enumeration member 1678
I_BCPDDV_MAIN_RNF_COORD_X enumeration member 1678
I_BCPDDV_MAIN_RNF_COORD_Y enumeration member 1678
I_BCPDDV_MAIN_RNF_COORD_Z enumeration member 1678
I_BCPDDV_MAX_BREAK enumeration member 1678

(c) 2014 Autodesk Inc. All rights reserved

I_BCPDDV_MAX_CRACK enumeration member 1678
I_BCPDDV_MAX_DEFLECTION enumeration member 1678
I_BCPDDV_REDISTRIBUTION enumeration member 1678
I_BCPDDV_SAND_CONTENT enumeration member 1678
I_BCPDDV_TRAN_BAR_DIAM enumeration member 1678
I_BCPDDV_TRAN_INCLINATION enumeration member 1678
I_BCPDDV_TRAN_STEEL_FE enumeration member 1678
I_BCPDDV_USER_VALUE enumeration member 1678
I_BCPDIV_ADJUST_DEFLECTION enumeration member 1679
I_BCPDIV_CALC_CRACK enumeration member 1679
I_BCPDIV_CALC_DEFLECTION enumeration member 1679
I_BCPDIV_CONCRETE_BY_CLASS enumeration member 1679
I_BCPDIV_CONCRETE_BY_CONSTRUCTION enumeration member 1679
I_BCPDIV_CONCRETE_LIGHT enumeration member 1679
I_BCPDIV_CRACKING enumeration member 1679
I_BCPDIV_FLT2F enumeration member 1679
I_BCPDIV_LEGS_NUM enumeration member 1679
I_BCPDIV_LONG_BAR_DIAM_THE_SAME enumeration member 1679
I_BCPDIV_LONG_STEEL_BY_CLASS enumeration member 1679
I_BCPDIV_MAIN_RNF_COORD_SYSTEM enumeration member 1679
I_BCPDIV_SEISMICS enumeration member 1679
I_BCPDIV_TRAN_STEEL_BY_CLASS enumeration member 1679
I_BCPDIV_TRAN_TYPE enumeration member 1679
I_BCPDIV_TRAN_ZONES_NUM enumeration member 1679
I_BCPDIV_TRAN_ZONES_OPTIMALIZATION enumeration member 1679
I_BCPDIV_USER_VALUE enumeration member 1679
I_BCPDSV_CONCRETE_CLASS enumeration member 1680
I_BCPDSV_LONG_REINF_CLASS enumeration member 1680
I_BCPDSV_NAME enumeration member 1680
I_BCPDSV_TRAN_REINF_CLASS enumeration member 1680
I_BCPDSV_USER_VALUE enumeration member 1680
I_BCS_CARTESIAN enumeration member 1699
I_BCS_POLAR enumeration member 1699
I_BCT_BAR_SPACING enumeration member 1698
I_BCT_CRACK_WIDTH enumeration member 1698
I_BCT_DIMENSIONING enumeration member 1698
I_BCT_RIGIDITY enumeration member 1698
I_BCW_BREAK_MAX enumeration member 1687
I_BCW_CONCRETE_EXC_QPR enumeration member 1687
I_BCW_CONCRETE_EXC_RAR enumeration member 1687
I_BCW_COT_THETA_CHANGED enumeration member 1687
I_BCW_LAMBDA_MAX enumeration member 1687
I_BCW_NO_WARNINGS enumeration member 1687
I_BCW_RO_MAX enumeration member 1687
I_BCW_RO_MIN enumeration member 1687
I_BCW_SPACE_BY_CODE enumeration member 1687
I_BCW_STEEL_EXC_RAR enumeration member 1687
I_BCW_USER_WARNING enumeration member 1687
I_BD_CB71_CANTILEVER enumeration member 1887
I_BD_CB71_INTERNAL_BRACINGS enumeration member 1887
I_BD_CB71_NONE enumeration member 1887
I_BD_CB71_PINNED_PINNED_1_0 enumeration member 1887
I_BD_CB71_STIFF_PINNED_0_8 enumeration member 1887
I_BD_CB71_STIFF_STIFF_0_65 enumeration member 1887
I_BD_X enumeration member 1698
I_BD_Y enumeration member 1698
I_BD_Z enumeration member 1698
I_BDPDV_AS enumeration member 1698
I_BDPDV_AS_BOTTOM enumeration member 1698
I_BDPDV_AS_LEFT enumeration member 1698
I_BDPDV_AS_RIGHT enumeration member 1698
I_BDPDV_AS_TOP enumeration member 1698
I_BDPDV_CONCRETE_FC enumeration member 1698
I_BDPDV_DIR_ALPHA enumeration member 1698
I_BDPDV_DIR_R enumeration member 1698
I_BDPDV_DIR_THETA enumeration member 1698
I_BDPDV_DIR_X enumeration member 1698
I_BDPDV_DIR_Y enumeration member 1698
I_BDPDV_DIR_Z enumeration member 1698
I_BDPIV_BEND enumeration member 1690
I_BDPIV_BEND_BIAXIAL enumeration member 1690
I_BDPIV_BEND_SIMPLE enumeration member 1690
I_BDPIV_BEND_UNIAXIAL enumeration member 1690
I_BDPIV_CALCULATION_TYPE enumeration member 1690
I_BDPIV_DIR_Y enumeration member 1690
I_BDPIV_DIR_Z enumeration member 1690
I_BDPIV_DIRECTION enumeration member 1690

(c) 2014 Autodesk Inc. All rights reserved

I_BDPIV_LEVEL enumeration member 1690
I_BDPIV_SHEAR enumeration member 1690
I_BDPIV_TORSION enumeration member 1690
I_BDPIV_USER_VALUE enumeration member 1690
I_BDRV_DIL enumeration member 41
I_BDRV_ENTIRE_STRUCTURE enumeration member 44
I_BDRV_REL enumeration member 41
I_BDRV_X enumeration member 44
I_BDRV_Y enumeration member 44
I_BDRV_Z enumeration member 44
I_BEBDV_HEIGHT enumeration member 621
I_BEBDV_LENGTH enumeration member 621
I_BEBDV_THICKNESS_1 enumeration member 621
I_BEBDV_THICKNESS_2 enumeration member 621
I_BEBDV_WIDTH enumeration member 621
I_BEBT_PLATES enumeration member 621
I_BEBT_SECTION enumeration member 621
I_BERV_ELASTIC enumeration member 580
I_BERV_ELASTIC_MINUS enumeration member 580
I_BERV_ELASTIC_PLUS enumeration member 580
I_BERV_ELASTIC_REDUCED enumeration member 580
I_BERV_ELASTIC_REDUCED_MINUS enumeration member 580
I_BERV_ELASTIC_REDUCED_PLUS enumeration member 580
I_BERV_MINUS enumeration member 580
I_BERV_NONE enumeration member 580
I_BERV_NONLINEAR enumeration member 580
I_BERV_PLUS enumeration member 580
I_BERV_STD enumeration member 580
I_BFCRV_ALPHA enumeration member 41
I_BFCRV_BETA enumeration member 41
I_BFCRV_CX enumeration member 41
I_BFCRV_CY enumeration member 41
I_BFCRV_CZ enumeration member 41
I_BFCRV_FX enumeration member 41
I_BFCRV_FY enumeration member 41
I_BFCRV_FZ enumeration member 41
I_BFCRV_GAMMA enumeration member 41
I_BFCRV_GENERATE_CALC_NODE enumeration member 41
I_BFCRV_LOC enumeration member 41
I_BFCRV_OFFSET_Y enumeration member 41
I_BFCRV_OFFSET_Z enumeration member 41
I_BFCRV_REL enumeration member 41
I_BFCRV_X enumeration member 41
I_BFDDV_COEF_LONG enumeration member 1699
I_BFDDV_COEF_LONG2TOT enumeration member 1699
I_BFDIV_SLS_COMB_TYPE enumeration member 1699
I_BFDSLS_COMB_TYPE_FRE enumeration member 1700
I_BFDSLS_COMB_TYPE_NONE enumeration member 1700
I_BFDSLS_COMB_TYPE_QPR enumeration member 1700
I_BFDSLS_COMB_TYPE_RAR enumeration member 1700
I_BGIA_Y enumeration member 592
I_BGIA_Z enumeration member 592
I_BIR_INTERSECT_END_END enumeration member 2222
I_BIR_INTERSECT_END_MID enumeration member 2222
I_BIR_INTERSECT_MID_END enumeration member 2222
I_BIR_INTERSECT_MID_MID enumeration member 2222
I_BIR_NOT_INTERSECT enumeration member 2222
I_BIR_UNKNOWN enumeration member 2222
I_BL_DOWN enumeration member 1699
I_BL_UP enumeration member 1699
I_BLN_BEAM_NATURE_DEAD enumeration member 1165
I_BLN_BEAM_NATURE_LIVE enumeration member 1165
I_BLN_BEAM_NATURE_SEISMIC enumeration member 1165
I_BLN_BEAM_NATURE_SNOW enumeration member 1165
I_BLN_BEAM_NATURE_WIND enumeration member 1165
I_BMDDV_BUCK_COEF_Y enumeration member 1683
I_BMDDV_BUCK_COEF_Z enumeration member 1683
I_BMDDV_LENGTH_Y enumeration member 1683
I_BMDDV_LENGTH_Z enumeration member 1683
I_BMDDV_SUPPORT_WIDTH_1 enumeration member 1683
I_BMDDV_SUPPORT_WIDTH_2 enumeration member 1683
I_BMDDV_USER_VALUE enumeration member 1683
I_BMDIV_CALC_AS_SLENDER_Y enumeration member 1684
I_BMDIV_CALC_AS_SLENDER_Z enumeration member 1684
I_BMDIV_SWAY_Y enumeration member 1684
I_BMDIV_SWAY_Z enumeration member 1684
I_BMDIV_TYPE enumeration member 1684
I_BMDIV_USER_VALUE enumeration member 1684
I_BMDRV_ALPHA enumeration member 42
I_BMDRV_BETA enumeration member 42

(c) 2014 Autodesk Inc. All rights reserved

I_BMDRV_GAMMA enumeration member 42
I_BMDRV_MX enumeration member 42
I_BMDRV_MY enumeration member 42
I_BMDRV_MZ enumeration member 42
I_BMDSV_USER_VALUE enumeration member 1694
I_BMT_BEAM enumeration member 1695
I_BMT_COLUMN enumeration member 1695
I_BMT_PLATE enumeration member 1695
I_BOAP_0_0 enumeration member 623
I_BOAP_0_VPZ enumeration member 623
I_BOAP_0_VZ enumeration member 623
I_BOAP_VPY_0 enumeration member 623
I_BOAP_VPY_VPZ enumeration member 623
I_BOAP_VPY_VZ enumeration member 623
I_BOAP_VY_0 enumeration member 623
I_BOAP_VY_VPZ enumeration member 623
I_BOAP_VY_VZ enumeration member 623
I_BOML_IN_THE_AXIS enumeration member 624
I_BOML_INCREASE_LENGTH enumeration member 624
I_BOML_REDUCE_LENGTH enumeration member 624
I_BRDV_AS_BOTTOM enumeration member 1685
I_BRDV_AS_LEFT enumeration member 1685
I_BRDV_AS_MAX enumeration member 1685
I_BRDV_AS_MIN enumeration member 1685
I_BRDV_AS_MIN_BOTTOM enumeration member 1685
I_BRDV_AS_MIN_TOP enumeration member 1685
I_BRDV_AS_PROVIDED_BOTTOM_MY enumeration member 1685
I_BRDV_AS_PROVIDED_BOTTOM_MZ enumeration member 1685
I_BRDV_AS_PROVIDED_TOP_MY enumeration member 1685
I_BRDV_AS_PROVIDED_TOP_MZ enumeration member 1685
I_BRDV_AS_RIGHT enumeration member 1685
I_BRDV_AS_TOP enumeration member 1685
I_BRDV_BAR_SPACING enumeration member 1685
I_BRDV_CRACK_WIDTH enumeration member 1685
I_BRDV_CRACK_WIDTH_LT_BOTTOM enumeration member 1685
I_BRDV_CRACK_WIDTH_LT_TOP enumeration member 1685
I_BRDV_CRACK_WIDTH_ST_BOTTOM enumeration member 1685
I_BRDV_CRACK_WIDTH_ST_TOP enumeration member 1685
I_BRDV_LAMBDA_Y enumeration member 1685
I_BRDV_LAMBDA_Z enumeration member 1685
I_BRDV_RIGIDITY enumeration member 1685
I_BRDV_RO enumeration member 1685
I_BRDV_RO_MAX enumeration member 1685
I_BRDV_RO_MIN enumeration member 1685
I_BRDV_STIFFNESS_BOTTOM enumeration member 1685
I_BRDV_STIFFNESS_TOP enumeration member 1685
I_BRDV_STIRR_SPACE enumeration member 1685
I_BRDV_STIRR_SPACE_MAX enumeration member 1685
I_BRDV_STIRR_SPACE_MIN enumeration member 1685
I_BRDV_STIRR_SPACE_MOD enumeration member 1685
I_BRDV_USER_VALUE enumeration member 1685
I_BRIV_BEND_CASE_IDX enumeration member 1686
I_BRIV_BEND_ERROR_NUM enumeration member 1686
I_BRIV_BEND_WARNING_NUM enumeration member 1686
I_BRIV_SHEAR_CASE_IDX enumeration member 1686
I_BRIV_SHEAR_ERROR_NUM enumeration member 1686
I_BRIV_SHEAR_WARNING_NUM enumeration member 1686
I_BRIV_TORSION_CASE_IDX enumeration member 1686
I_BRIV_TORSION_ERROR_NUM enumeration member 1686
I_BRIV_TORSION_WARNING_NUM enumeration member 1686
I_BRIV_USER_VALUE enumeration member 1686
I_BRSV_BEND_ERROR_NAME enumeration member 1688
I_BRSV_BEND_WARNING_NAME enumeration member 1688
I_BRSV_SHEAR_ERROR_NAME enumeration member 1688
I_BRSV_SHEAR_WARNING_NAME enumeration member 1688
I_BRSV_TORSION_ERROR_NAME enumeration member 1688
I_BRSV_TORSION_WARNING_NAME enumeration member 1688
I_BRSV_USER_VALUE enumeration member 1688
I_BS8110_CA_1 enumeration member 1758
I_BS8110_CA_28 enumeration member 1758
I_BS8110_CA_3 enumeration member 1758
I_BS8110_CA_365 enumeration member 1758
I_BS8110_CA_7 enumeration member 1758
I_BS8110_CA_90 enumeration member 1758
I_BS8110_CG_AUTO enumeration member 1759
I_BS8110_CG_C12_15 enumeration member 1759

(c) 2014 Autodesk Inc. All rights reserved

I_BS8110_CG_C16_20 enumeration member 1759
I_BS8110_CG_C20_25 enumeration member 1759
I_BS8110_CG_C25_30 enumeration member 1759
I_BS8110_CG_C30_37 enumeration member 1759
I_BS8110_CG_C35_45 enumeration member 1759
I_BS8110_CG_C40_50 enumeration member 1759
I_BS8110_CG_C45_55 enumeration member 1759
I_BS8110_CG_C50_60 enumeration member 1759
I_BS8110_ER_MILD enumeration member 1758
I_BS8110_ER_MODERATE enumeration member 1758
I_BS8110_ER_MOST_SEVERE enumeration member 1758
I_BS8110_ER_SEVERE enumeration member 1758
I_BS8110_ER_VERY_SEVERE enumeration member 1758
I_BS8110_PSF_1985 enumeration member 1760
I_BS8110_PSF_1997 enumeration member 1760
I_BSCCP_LEFT enumeration member 552
I_BSCCP_NONE enumeration member 552
I_BSCCP_RIGHT enumeration member 552
I_BSCDV_BEAM_B enumeration member 551
I_BSCDV_BEAM_BG1 enumeration member 551
I_BSCDV_BEAM_BG2 enumeration member 551
I_BSCDV_BEAM_EL1 enumeration member 551
I_BSCDV_BEAM_EL2 enumeration member 551
I_BSCDV_BEAM_ER1 enumeration member 551
I_BSCDV_BEAM_ER2 enumeration member 551
I_BSCDV_BEAM_H enumeration member 551
I_BSCDV_BEAM_HG1 enumeration member 551
I_BSCDV_BEAM_HG2 enumeration member 551
I_BSCDV_BEAM_I_B enumeration member 551
I_BSCDV_BEAM_I_B1 enumeration member 551
I_BSCDV_BEAM_I_B2 enumeration member 551
I_BSCDV_BEAM_I_H enumeration member 551
I_BSCDV_BEAM_I_HF1 enumeration member 551
I_BSCDV_BEAM_I_HF2 enumeration member 551
I_BSCDV_BEAM_PL enumeration member 551
I_BSCDV_BEAM_PR enumeration member 551
I_BSCDV_BEAM_RECT_B enumeration member 551
I_BSCDV_BEAM_RECT_H enumeration member 551
I_BSCDV_BEAM_RL enumeration member 551
I_BSCDV_BEAM_RR enumeration member 551
I_BSCDV_BEAM_T_B enumeration member 551
I_BSCDV_BEAM_T_BF enumeration member 551
I_BSCDV_BEAM_T_H enumeration member 551
I_BSCDV_BEAM_T_HF enumeration member 551
I_BSCDV_COL_B enumeration member 551
I_BSCDV_COL_DE enumeration member 551
I_BSCDV_COL_H enumeration member 551
I_BSCDV_COL_H1 enumeration member 551
I_BSCDV_COL_H2 enumeration member 551
I_BSCDV_COL_L1 enumeration member 551
I_BSCDV_COL_L2 enumeration member 551
I_BSCDV_COL_N enumeration member 551
I_BSCS_C enumeration member 576
I_BSCS_I enumeration member 576
I_BSCS_L enumeration member 576
I_BSCS_UNDEFINED enumeration member 576
I_BSD_BEAM_SECTION_DIM_B enumeration member 1166
I_BSD_BEAM_SECTION_DIM_BG1 enumeration member 1166
I_BSD_BEAM_SECTION_DIM_BG2 enumeration member 1166
I_BSD_BEAM_SECTION_DIM_BG3 enumeration member 1166
I_BSD_BEAM_SECTION_DIM_BG4 enumeration member 1166
I_BSD_BEAM_SECTION_DIM_CL enumeration member 1166
I_BSD_BEAM_SECTION_DIM_CR enumeration member 1166
I_BSD_BEAM_SECTION_DIM_EL1 enumeration member 1166
I_BSD_BEAM_SECTION_DIM_EL2 enumeration member 1166
I_BSD_BEAM_SECTION_DIM_ER1 enumeration member 1166
I_BSD_BEAM_SECTION_DIM_ER2 enumeration member 1166
I_BSD_BEAM_SECTION_DIM_H1 enumeration member 1166
I_BSD_BEAM_SECTION_DIM_H2 enumeration member 1166
I_BSD_BEAM_SECTION_DIM_HG1 enumeration member 1166
I_BSD_BEAM_SECTION_DIM_HG2 enumeration member 1166
I_BSD_BEAM_SECTION_DIM_HG3 enumeration member 1166
I_BSD_BEAM_SECTION_DIM_HG4 enumeration member 1166
I_BSD_BEAM_SECTION_DIM_L enumeration member 1166

(c) 2014 Autodesk Inc. All rights reserved

I_BSD_BEAM_SECTION_DIM_OFFY1 enumeration member 1166

I_BSD_BEAM_SECTION_DIM_OFFY2 enumeration member 1166

I_BSD_BEAM_SECTION_DIM_OFFZ1 enumeration member 1166

I_BSD_BEAM_SECTION_DIM_OFFZ2 enumeration member 1166

I_BSD_BEAM_SECTION_DIM_PL enumeration member 1166

I_BSD_BEAM_SECTION_DIM_PL_CALC enumeration member 1166

I_BSD_BEAM_SECTION_DIM_PR enumeration member 1166

I_BSD_BEAM_SECTION_DIM_PR_CALC enumeration member 1166

I_BSD_BEAM_SECTION_DIM_RL enumeration member 1166

I_BSD_BEAM_SECTION_DIM_RR enumeration member 1166

I_BSD_CFOOT_SECTION_DIM_B enumeration member 1183

I_BSD_CFOOT_SECTION_DIM_EL1 enumeration member 1183

I_BSD_CFOOT_SECTION_DIM_ER1 enumeration member 1183

I_BSD_CFOOT_SECTION_DIM_H1 enumeration member 1183

I_BSD_CFOOT_SECTION_DIM_H2 enumeration member 1183

I_BSD_CFOOT_SECTION_DIM_L enumeration member 1183

I_BSD_CFOOT_SECTION_DIM_PL enumeration member 1183

I_BSD_CFOOT_SECTION_DIM_PR enumeration member 1183

I_BSD_CFOOT_SECTION_DIM_RL enumeration member 1183

I_BSD_CFOOT_SECTION_DIM_RR enumeration member 1183

I_BSDV_ANGLE1 enumeration member 549

I_BSDV_ANGLE2 enumeration member 549

I_BSDV_AX enumeration member 549

I_BSDV_AY enumeration member 549

I_BSDV_AZ enumeration member 549

I_BSDV_BF enumeration member 549

I_BSDV_BF2 enumeration member 549

I_BSDV_D enumeration member 549

I_BSDV_DIM1 enumeration member 549

I_BSDV_DIM2 enumeration member 549

I_BSDV_DIM3 enumeration member 549

I_BSDV_GAMMA enumeration member 549

I_BSDV_IOMEGA enumeration member 549

I_BSDV_IX enumeration member 549

I_BSDV_IY enumeration member 549

I_BSDV_IZ enumeration member 549

I_BSDV_P1_LENGTH enumeration member 549

I_BSDV_P1_THICKNESS enumeration member 549

I_BSDV_P2_LENGTH enumeration member 549

I_BSDV_P2_THICKNESS enumeration member 549

I_BSDV_P3_LENGTH enumeration member 549

I_BSDV_P3_THICKNESS enumeration member 549

I_BSDV_P4_LENGTH enumeration member 549

I_BSDV_P4_THICKNESS enumeration member 549

I_BSDV_RA enumeration member 549

I_BSDV_RI enumeration member 549

I_BSDV_S enumeration member 549

I_BSDV_SURFACE enumeration member 549

I_BSDV_TF enumeration member 549

I_BSDV_TF2 enumeration member 549

I_BSDV_TW enumeration member 549

I_BSDV_VPY enumeration member 549

I_BSDV_VPZ enumeration member 549

I_BSDV_VY enumeration member 549

I_BSDV_VZ enumeration member 549

I_BSDV_WEIGHT enumeration member 549

I_BSDV_WX enumeration member 549

I_BSDV_WY enumeration member 549

I_BSDV_WZ enumeration member 549

I_BSDV_ZY enumeration member 549

I_BSDV_ZZ enumeration member 549

I_BSM_BEAM_SUPPORT_CONCRETE enumeration member 1168

I_BSM_BEAM_SUPPORT_MASONRY enumeration member 1168

I_BSM_CFOOT_SUPPORT_CONCRETE enumeration member 1184

I_BSM_CFOOT_SUPPORT_MASONRY enumeration member 1184

I_BSNDV_BOX_2_B enumeration member 550

I_BSNDV_BOX_2_B1 enumeration member 550

I_BSNDV_BOX_2_H enumeration member 550

I_BSNDV_BOX_2_TF enumeration member 550

(c) 2014 Autodesk Inc. All rights reserved

I_BSNDV_BOX_2_TW enumeration member 550
I_BSNDV_BOX_3_B enumeration member 550
I_BSNDV_BOX_3_B1 enumeration member 550
I_BSNDV_BOX_3_B2 enumeration member 550
I_BSNDV_BOX_3_H enumeration member 550
I_BSNDV_BOX_3_TF enumeration member 550
I_BSNDV_BOX_3_TF2 enumeration member 550
I_BSNDV_BOX_3_TW enumeration member 550
I_BSNDV_BOX_B enumeration member 550
I_BSNDV_BOX_H enumeration member 550
I_BSNDV_BOX_TF enumeration member 550
I_BSNDV_BOX_TW enumeration member 550
I_BSNDV_C_B enumeration member 550
I_BSNDV_C_H enumeration member 550
I_BSNDV_C_TF enumeration member 550
I_BSNDV_C_TW enumeration member 550
I_BSNDV_CROSS_P1_L enumeration member 550
I_BSNDV_CROSS_P1_T enumeration member 550
I_BSNDV_CROSS_P2_L enumeration member 550
I_BSNDV_CROSS_P2_T enumeration member 550
I_BSNDV_CROSS_P3_L enumeration member 550
I_BSNDV_CROSS_P3_T enumeration member 550
I_BSNDV_CROSS_P4_L enumeration member 550
I_BSNDV_CROSS_P4_T enumeration member 550
I_BSNDV_DRECT_B enumeration member 550
I_BSNDV_DRECT_D enumeration member 550
I_BSNDV_DRECT_H enumeration member 550
I_BSNDV_H_B enumeration member 550
I_BSNDV_H_B1 enumeration member 550
I_BSNDV_H_H enumeration member 550
I_BSNDV_H_TF enumeration member 550
I_BSNDV_H_TW enumeration member 550
I_BSNDV_HOLE_B enumeration member 550
I_BSNDV_HOLE_H enumeration member 550
I_BSNDV_HOLE_X enumeration member 550
I_BSNDV_HOLE_Z enumeration member 550
I_BSNDV_I_B enumeration member 550
I_BSNDV_I_H enumeration member 550
I_BSNDV_I_TF enumeration member 550
I_BSNDV_I_TW enumeration member 550
I_BSNDV_II_B1 enumeration member 550
I_BSNDV_II_B2 enumeration member 550
I_BSNDV_II_H enumeration member 550
I_BSNDV_II_TF1 enumeration member 550
I_BSNDV_II_TF2 enumeration member 550
I_BSNDV_II_TW enumeration member 550
I_BSNDV_L_B enumeration member 550
I_BSNDV_L_H enumeration member 550
I_BSNDV_L_TF enumeration member 550
I_BSNDV_L_TW enumeration member 550
I_BSNDV_POLYGONAL_D enumeration member 550
I_BSNDV_POLYGONAL_D_IS_INT enumeration member 550
I_BSNDV_POLYGONAL_N enumeration member 550
I_BSNDV_POLYGONAL_T enumeration member 550
I_BSNDV_RECT_B enumeration member 550
I_BSNDV_RECT_H enumeration member 550
I_BSNDV_RECT_T enumeration member 550
I_BSNDV_T_B enumeration member 550
I_BSNDV_T_H enumeration member 550
I_BSNDV_T_TF enumeration member 550
I_BSNDV_T_TW enumeration member 550
I_BSNDV_TUBE_D enumeration member 550
I_BSNDV_TUBE_T enumeration member 550
I_BSNDV_XI_B enumeration member 550
I_BSNDV_XI_H enumeration member 550
I_BSNDV_XI_H1 enumeration member 550
I_BSNDV_XI_TF enumeration member 550
I_BSNDV_XI_TW enumeration member 550
I_BSNDV_XT_B enumeration member 550
I_BSNDV_XT_H enumeration member 550
I_BSNDV_XT_H1 enumeration member 550
I_BSNDV_XT_TF enumeration member 550
I_BSNDV_XT_TW enumeration member 550
I_BSNDV_Z_B enumeration member 550
I_BSNDV_Z_H enumeration member 550
I_BSNDV_Z_TF enumeration member 550
I_BSNDV_Z_TW enumeration member 550
I_BSSDV_A enumeration member 579
I_BSSDV_B1 enumeration member 579
I_BSSDV_B2 enumeration member 579
I_BSSDV_BP enumeration member 579
I_BSSDV_C enumeration member 579
I_BSSDV_D enumeration member 579
I_BSSDV_H enumeration member 579

(c) 2014 Autodesk Inc. All rights reserved

I_BSSDV_HS enumeration member 579
I_BSSDV_TF1 enumeration member 579
I_BSSDV_TF2 enumeration member 579
I_BSSDV_TP enumeration member 579
I_BSSDV_TW enumeration member 579
I_BSSDV_W enumeration member 579
I_BSST_CAE enumeration member 546
I_BSST_CAEP enumeration member 546
I_BSST_CAI enumeration member 546
I_BSST_CAIP enumeration member 546
I_BSST_CCL enumeration member 546
I_BSST_CIRC_FILLED enumeration member 546
I_BSST_COLD_C_PLUS enumeration member 546
I_BSST_COLD_L_LIPS enumeration member 546
I_BSST_COLD_SIGMA enumeration member 546
I_BSST_COLD_SIGMA_SL enumeration member 546
I_BSST_COLD_Z enumeration member 546
I_BSST_COLD_Z_ROT enumeration member 546
I_BSST_COMP_2C_BACK enumeration member 546
I_BSST_COMP_2C_BACK_WELD enumeration member 546
I_BSST_COMP_2C_FACE enumeration member 546
I_BSST_COMP_2C_FACE_WELD enumeration member 546
I_BSST_COMP_2I enumeration member 546
I_BSST_COMP_2I_WELD enumeration member 546
I_BSST_COMP_2L_CROSS enumeration member 546
I_BSST_COMP_2L_CROSS_WELD enumeration member 546
I_BSST_COMP_2L_FACE_LONG enumeration member 546
I_BSST_COMP_2L_FACE_LONG_WELD enumeration member 546
I_BSST_COMP_2L_FACE_SHORT enumeration member 546
I_BSST_COMP_2L_FACE_SHORT_WELD enumeration member 546
I_BSST_COMP_2L_LONG enumeration member 546
I_BSST_COMP_2L_LONG_WELD enumeration member 546
I_BSST_COMP_2L_SHORT enumeration member 546
I_BSST_COMP_2L_SHORT_WELD enumeration member 546
I_BSST_COMP_2LI enumeration member 546
I_BSST_COMP_2LI_WELD enumeration member 546
I_BSST_COMP_4L_BACK enumeration member 546
I_BSST_COMP_4L_BACK_WELD enumeration member 546
I_BSST_COMP_4L_FACE enumeration member 546
I_BSST_COMP_4L_FACE_WELD enumeration member 546
I_BSST_COMP_CI enumeration member 546
I_BSST_COMP_CI_BACK enumeration member 546
I_BSST_COMP_CI_BACK_WELD enumeration member 546
I_BSST_COMP_CI_WELD enumeration member 546
I_BSST_CONCR_BEAM enumeration member 546
I_BSST_CONCR_BEAM_I enumeration member 546
I_BSST_CONCR_BEAM_RECT enumeration member 546
I_BSST_CONCR_BEAM_T enumeration member 546
I_BSST_CONCR_COL_C enumeration member 546
I_BSST_CONCR_COL_CH enumeration member 546
I_BSST_CONCR_COL_CQ enumeration member 546
I_BSST_CONCR_COL_L enumeration member 546
I_BSST_CONCR_COL_P enumeration member 546
I_BSST_CONCR_COL_R enumeration member 546
I_BSST_CONCR_COL_T enumeration member 546
I_BSST_CONCR_COL_Z enumeration member 546
I_BSST_CUAP enumeration member 546
I_BSST_DCEC enumeration member 546
I_BSST_DCED enumeration member 546
I_BSST_DCEP enumeration member 546
I_BSST_DCIG enumeration member 546
I_BSST_DCIP enumeration member 546
I_BSST_HEA enumeration member 546
I_BSST_HEAA enumeration member 546
I_BSST_HEB enumeration member 546
I_BSST_HEC enumeration member 546
I_BSST_HEM enumeration member 546
I_BSST_HER enumeration member 546
I_BSST_HHEA enumeration member 546
I_BSST_HHEB enumeration member 546
I_BSST_HHEM enumeration member 546
I_BSST_IIPE enumeration member 546
I_BSST_IPE enumeration member 546
I_BSST_IPEA enumeration member 546
I_BSST_IPEO enumeration member 546
I_BSST_IPER enumeration member 546
I_BSST_IPEV enumeration member 546
I_BSST_IPN enumeration member 546
I_BSST_JOIST_BG enumeration member 546
I_BSST_JOIST_DLH enumeration member 546
I_BSST_JOIST_G enumeration member 546
I_BSST_JOIST_K enumeration member 546

(c) 2014 Autodesk Inc. All rights reserved

I_BSST_JOIST_KCS enumeration member 546
I_BSST_JOIST_LH enumeration member 546
I_BSST_JOIST_SLH enumeration member 546
I_BSST_JOIST_VG enumeration member 546
I_BSST_MHEA enumeration member 546
I_BSST_MHEB enumeration member 546
I_BSST_MHEM enumeration member 546
I_BSST_MIPE enumeration member 546
I_BSST_PRS enumeration member 546
I_BSST_RECT_FILLED enumeration member 546
I_BSST_SPEC_CASTELLATED_WEB_HEXAGONAL_OPENINGS enumeration member 546
I_BSST_SPEC_CASTELLATED_WEB_HEXAGONAL_OPENINGS_SHIFTED enumeration member 546
I_BSST_SPEC_CASTELLATED_WEB_ROUND_OPENINGS enumeration member 546
I_BSST_SPEC_CORRUGATED_WEB enumeration member 546
I_BSST_SPEC_IFBA enumeration member 546
I_BSST_SPEC_IFBB enumeration member 546
I_BSST_SPEC_SFB enumeration member 546
I_BSST_TCAR enumeration member 546
I_BSST_TEAE enumeration member 546
I_BSST_TEAI enumeration member 546
I_BSST_THEX enumeration member 546
I_BSST_TREC enumeration member 546
I_BSST_TRND enumeration member 546
I_BSST_TRON enumeration member 546
I_BSST_UAP enumeration member 546
I_BSST_UNKNOWN enumeration member 546
I_BSST_UPAF enumeration member 546
I_BSST_UPN enumeration member 546
I_BSST_URND enumeration member 546
I_BSST_USER_BOX enumeration member 546
I_BSST_USER_BOX_2 enumeration member 546
I_BSST_USER_BOX_3 enumeration member 546
I_BSST_USER_C_SHAPE enumeration member 546
I_BSST_USER_CIRC_FILLED enumeration member 546
I_BSST_USER_CROSS enumeration member 546
I_BSST_USER_I_BISYM enumeration member 546
I_BSST_USER_I_MONOSYM enumeration member 546
I_BSST_USER_POLYGONAL enumeration member 546
I_BSST_USER_RECT enumeration member 546
I_BSST_USER_T_SHAPE enumeration member 546
I_BSST_USER_TUBE enumeration member 546
I_BSST_UUAP enumeration member 546
I_BSST_UUPN enumeration member 546
I_BSST_WOOD_CIRC enumeration member 546
I_BSST_WOOD_DRECT enumeration member 546
I_BSST_WOOD_RECT enumeration member 546
I_BST_COMPLEX enumeration member 550
I_BST_JOIST enumeration member 550
I_BST_NS_BOX enumeration member 550
I_BST_NS_BOX_2 enumeration member 550
I_BST_NS_BOX_3 enumeration member 550
I_BST_NS_C enumeration member 550
I_BST_NS_CROSS enumeration member 550
I_BST_NS_DRECT enumeration member 550
I_BST_NS_H enumeration member 550
I_BST_NS_HOLE enumeration member 550
I_BST_NS_I enumeration member 550
I_BST_NS_II enumeration member 550
I_BST_NS_L enumeration member 550
I_BST_NS_LP enumeration member 550
I_BST_NS_POLYGONAL enumeration member 550
I_BST_NS_RECT enumeration member 550
I_BST_NS_T enumeration member 550
I_BST_NS_TUBE enumeration member 550
I_BST_NS_XI enumeration member 550
I_BST_NS_XT enumeration member 550
I_BST_NS_Z enumeration member 550
I_BST_NS_ZP enumeration member 550
I_BST_SPECIAL enumeration member 550
I_BST_STANDARD enumeration member 550
I_BTC_COMPRESSION_ONLY enumeration member 651
I_BTC_STANDARD enumeration member 651
I_BTC_TENSION_ONLY enumeration member 651
I_BTRV_ALPHA enumeration member 43
I_BTRV_BETA enumeration member 43
I_BTRV_GAMMA enumeration member 43
I_BTRV_LOCAL enumeration member 43
I_BTRV_PROJECTION enumeration member 43
I_BTRV_PX1 enumeration member 43
I_BTRV_PX2 enumeration member 43
I_BTRV_PY1 enumeration member 43

(c) 2014 Autodesk Inc. All rights reserved

I_BTRV_PY2 enumeration member 43
I_BTRV_PZ1 enumeration member 43
I_BTRV_PZ2 enumeration member 43
I_BTRV_RELATIVE enumeration member 43
I_BTRV_TX enumeration member 43
I_BTRV_TY enumeration member 43
I_BTRV_TZ enumeration member 43
I_BTRV_X1 enumeration member 43
I_BTRV_X2 enumeration member 43
I_BURV_ALPHA enumeration member 42
I_BURV_BETA enumeration member 42
I_BURV_GAMMA enumeration member 42
I_BURV_LOCAL enumeration member 42
I_BURV_OFFSET_Y enumeration member 42
I_BURV_OFFSET_Z enumeration member 42
I_BURV_PROJECTION enumeration member 42
I_BURV_PX enumeration member 42
I_BURV_PY enumeration member 42
I_BURV_PZ enumeration member 42
I_BURV_RELATIVE enumeration member 42
I_BVRT_AUTOMATIC enumeration member 1337
I_BVRT_UNLIMITED enumeration member 1337
I_BVRT_USER_DEFINED enumeration member 1337
I_CAMFT_LIST_OF_MODES enumeration member 437
I_CAMFT_MASS_PERCENTAGE enumeration member 437
I_CAMFT_NON_ACTIVE enumeration member 437
I_CAT_COMB enumeration member 410
I_CAT_COMB_BUCKLING enumeration member 410
I_CAT_COMB_CODE enumeration member 410
I_CAT_COMB_NONLINEAR enumeration member 410
I_CAT_COMB_NONLINEAR_BUCKLING enumeration member 410
I_CAT_DYNAMIC_FOOTFALL enumeration member 410
I_CAT_DYNAMIC_FRF enumeration member 410
I_CAT_DYNAMIC_HARMONIC enumeration member 410
I_CAT_DYNAMIC_MODAL enumeration member 410
I_CAT_DYNAMIC_NONLINEAR_MODAL_WITH_STATIC_FORCE enumeration member 410
I_CAT_DYNAMIC_SEISMIC enumeration member 410
I_CAT_DYNAMIC_SPECTRAL enumeration member 410
I_CAT_MOBILE_MAIN enumeration member 410
I_CAT_PUSH_OVER enumeration member 410
I_CAT_STATIC_BUCKLING enumeration member 410
I_CAT_STATIC_ELF_SEISMIC enumeration member 410
I_CAT_STATIC_LINEAR enumeration member 410
I_CAT_STATIC_LINEAR_AUXILIARY enumeration member 410
I_CAT_STATIC_NONLINEAR enumeration member 410
I_CAT_STATIC_NONLINEAR_BUCKLING enumeration member 410
I_CAT_TIME_HISTORY enumeration member 410
I_CBAELCG_C12_15 enumeration member 1740
I_CBAELCG_C16_20 enumeration member 1740
I_CBAELCG_C20_25 enumeration member 1740
I_CBAELCG_C25_30 enumeration member 1740
I_CBAELCG_C30_37 enumeration member 1740
I_CBAELCG_C35_45 enumeration member 1740
I_CBAELCG_C40_50 enumeration member 1740
I_CBAELCG_C45_55 enumeration member 1740
I_CBAELCG_C50_60 enumeration member 1740
I_CBAELCG_OTHER enumeration member 1740
I_CBAELCT_NOT_SEVERE enumeration member 1739
I_CBAELCT_OTHER enumeration member 1739
I_CBAELCT_SEVERE enumeration member 1739
I_CBAELCT_VERY_SEVERE enumeration member 1739
I_CBAELET_AGGRESSIVE_WATER enumeration member 1739
I_CBAELET_NORMAL_WATER enumeration member 1739
I_CBAELSG_DEFORMED enumeration member 1740
I_CBAELSG_PLAIN enumeration member 1740
I_CBAELWL_EXCEPTIONAL enumeration member 1739
I_CBAELWL_HIGH enumeration member 1739
I_CBAELWL_LOW enumeration member 1739
I_CBC_BEAM_CRACKING_LIMITED enumeration member 1167
I_CBC_BEAM_CRACKING_NOT_PERMISSIBLE enumeration member 1167
I_CBC_BEAM_CRACKING_PERMISSIBLE enumeration member 1167
I_CBC_BEAM_CRACKING_UNDEFINED enumeration member 1167
I_CBCPDT_POINT_COUNT enumeration member 1794
I_CBCPDT_POINT_SPACING enumeration member 1794
I_CBS_BAR_SUBTYPE_ANY enumeration member 1258
I_CBS_BAR_SUBTYPE_BACK enumeration member 1258
I_CBS_BAR_SUBTYPE_DIR_HOR enumeration member 1258

(c) 2014 Autodesk Inc. All rights reserved

I_CBS_BAR_SUBTYPE_DIR_VER enumeration member 1258
I_CBS_BAR_SUBTYPE_DIST enumeration member 1258
I_CBS_BAR_SUBTYPE_DOWN enumeration member 1258
I_CBS_BAR_SUBTYPE_EXT enumeration member 1258
I_CBS_BAR_SUBTYPE_FOOT enumeration member 1258
I_CBS_BAR_SUBTYPE_FRONT enumeration member 1258
I_CBS_BAR_SUBTYPE_GROIN enumeration member 1258
I_CBS_BAR_SUBTYPE_INC_45 enumeration member 1258
I_CBS_BAR_SUBTYPE_INC_LEFT enumeration member 1258
I_CBS_BAR_SUBTYPE_INC_RIGHT enumeration member 1258
I_CBS_BAR_SUBTYPE_INT enumeration member 1258
I_CBS_BAR_SUBTYPE_JOIN enumeration member 1258
I_CBS_BAR_SUBTYPE_LEFT enumeration member 1258
I_CBS_BAR_SUBTYPE_PIN enumeration member 1258
I_CBS_BAR_SUBTYPE_RIGHT enumeration member 1258
I_CBS_BAR_SUBTYPE_SHA enumeration member 1258
I_CBS_BAR_SUBTYPE_SPEC enumeration member 1258
I_CBS_BAR_SUBTYPE_STRAIGHT enumeration member 1258
I_CBS_BAR_SUBTYPE_TABLE enumeration member 1258
I_CBS_BAR_SUBTYPE_UBAR enumeration member 1258
I_CBS_BAR_SUBTYPE_UP enumeration member 1258
I_CBS_BAR_SUBTYPE_UP_RIGHT enumeration member 1258
I_CBS_BAR_SUBTYPE_WALL enumeration member 1258
I_CBS_BEAM_SECTION_LG_BP enumeration member 1165
I_CBS_BEAM_SECTION_LG_LP enumeration member 1165
I_CBS_BEAM_SECTION_LG_NP enumeration member 1165
I_CBS_BEAM_SECTION_LG_RP enumeration member 1165
I_CBS_BEAM_SECTION_NG_BP enumeration member 1165
I_CBS_BEAM_SECTION_NG_LP enumeration member 1165
I_CBS_BEAM_SECTION_NG_NP enumeration member 1165
I_CBS_BEAM_SECTION_NG_RP enumeration member 1165
I_CBS_BEAM_SECTION_RG_BP enumeration member 1165
I_CBS_BEAM_SECTION_RG_LP enumeration member 1165
I_CBS_BEAM_SECTION_RG_NP enumeration member 1165
I_CBS_BEAM_SECTION_RG_RP enumeration member 1165
I_CBS_CFOOT_SECTION_BP enumeration member 1183
I_CBS_CFOOT_SECTION_LP enumeration member 1183
I_CBS_CFOOT_SECTION_NP enumeration member 1183
I_CBS_CFOOT_SECTION_RP enumeration member 1183
I_CBSGT_C enumeration member 1798
I_CBSGT_C2 enumeration member 1798
I_CBSGT_C4 enumeration member 1798
I_CBSGT_I enumeration member 1798
I_CBSGT_L enumeration member 1798
I_CBSGT_P enumeration member 1798
I_CBSGT_R enumeration member 1798
I_CBSGT_T enumeration member 1798
I_CBSGT_TN enumeration member 1798
I_CBSGT_UC enumeration member 1798
I_CBSGT_UC2 enumeration member 1798
I_CBSGT_UC4 enumeration member 1798
I_CBSGT_UL enumeration member 1798
I_CBSGT_UP enumeration member 1798
I_CBSGT_UR enumeration member 1798
I_CBSGT_UT enumeration member 1798
I_CBSGT_UZ enumeration member 1798
I_CBSGT_Z enumeration member 1798
I_CBT_ALS enumeration member 433
I_CBT_BAR_TYPE_LON_BOAT enumeration member 1257
I_CBT_BAR_TYPE_LON_CONS enumeration member 1257
I_CBT_BAR_TYPE_LON_CORN enumeration member 1257
I_CBT_BAR_TYPE_LON_DIST enumeration member 1257
I_CBT_BAR_TYPE_LON_FLAN enumeration member 1257
I_CBT_BAR_TYPE_LON_HOLE enumeration member 1257
I_CBT_BAR_TYPE_LON_JOIN enumeration member 1257
I_CBT_BAR_TYPE_LON_JOIN_STRUT enumeration member 1257
I_CBT_BAR_TYPE_LON_MAIN enumeration member 1257
I_CBT_BAR_TYPE_LON_MAIN_FIRST enumeration member 1257
I_CBT_BAR_TYPE_LON_SPEC enumeration member 1257
I_CBT_BAR_TYPE_LON_SURF enumeration member 1257
I_CBT_BAR_TYPE_LON_TORS enumeration member 1257
I_CBT_BAR_TYPE_TRA_CONS enumeration member 1257
I_CBT_BAR_TYPE_TRA_CORN enumeration member 1257
I_CBT_BAR_TYPE_TRA_HOLE enumeration member 1257
I_CBT_BAR_TYPE_TRA_JOIN enumeration member 1257
I_CBT_BAR_TYPE_TRA_MAIN enumeration member 1257
I_CBT_BAR_TYPE_TRA_SPEC enumeration member 1257
I_CBT_BAR_TYPE_TRA_SURF enumeration member 1257
I_CBT_BAR_TYPE_UNDEFINED enumeration member 1257
I_CBT_SLS enumeration member 433

(c) 2014 Autodesk Inc. All rights reserved

I_CBT_SLS_EC_FRE enumeration member 433
I_CBT_SLS_EC_QPR enumeration member 433
I_CBT_SLS_EC_RAR enumeration member 433
I_CBT_SPC enumeration member 433
I_CBT_ULS enumeration member 433
I_CCACT_GA enumeration member 163
I_CCACT_GS enumeration member 163
I_CCACT_GU_MAX enumeration member 163
I_CCACT_GU_MIN enumeration member 163
I_CCACT_KSI_MAX enumeration member 163
I_CCACT_KSI_MIN enumeration member 163
I_CCACT_PSI0_1 enumeration member 163
I_CCACT_PSI0_2 enumeration member 163
I_CCACT_PSI0_3 enumeration member 163
I_CCACT_PSI0_N enumeration member 163
I_CCACT_PSI1 enumeration member 163
I_CCACT_PSI2_1 enumeration member 163
I_CCACT_PSI2_N enumeration member 163
I_CCACT_PSIK enumeration member 163
I_CCBC_BM_VERIFICATION enumeration member 1806
I_CCCC_CL_VERIFICATION enumeration member 1803
I_CCCPT_ACCIDENTAL enumeration member 167
I_CCCPT_DEAD enumeration member 167
I_CCCPT_LIVE enumeration member 167
I_CCCPT_SEISMIC enumeration member 167
I_CCCT_ACCIDENTAL enumeration member 167
I_CCCT_EXTREMAL enumeration member 167
I_CCCT_FREQUENT enumeration member 167
I_CCCT_FUNDAMENTAL enumeration member 167
I_CCCT_QUASI_PERM enumeration member 167
I_CCCT_RARE enumeration member 167
I_CCCT_SEISMIC enumeration member 167
I_CCCT_SIMPLIFIED enumeration member 167
I_CCCT_STANDARD enumeration member 167
I_CCCT_USER enumeration member 167
I_CCD_COLUMN_DIM_A enumeration member 1201
I_CCD_COLUMN_DIM_B enumeration member 1201
I_CCD_COLUMN_DIM_C enumeration member 1201
I_CCD_COLUMN_DIM_DE enumeration member 1201
I_CCD_COLUMN_DIM_EPD enumeration member 1201
I_CCD_COLUMN_DIM_H1 enumeration member 1201
I_CCD_COLUMN_DIM_H2 enumeration member 1201
I_CCD_COLUMN_DIM_HSD enumeration member 1201
I_CCD_COLUMN_DIM_HSP enumeration member 1201
I_CCD_COLUMN_DIM_HSP_X enumeration member 1201
I_CCD_COLUMN_DIM_HSP_Y enumeration member 1201
I_CCD_COLUMN_DIM_L1 enumeration member 1201
I_CCD_COLUMN_DIM_L2 enumeration member 1201
I_CCD_COLUMN_DIM_N enumeration member 1201
I_CCDVT_DEFLECTION enumeration member 174
I_CCDVT_FX enumeration member 174
I_CCDVT_FXX_MAX enumeration member 174
I_CCDVT_FY enumeration member 174
I_CCDVT_FZ enumeration member 174
I_CCDVT_MX enumeration member 174
I_CCDVT_MXX_MAX enumeration member 174
I_CCDVT_MY enumeration member 174
I_CCDVT_MZ enumeration member 174
I_CCDVT_REACTIONS enumeration member 174
I_CCDVT_SIGMA_X enumeration member 174
I_CCDVT_SXX_MAX enumeration member 174
I_CCDVT_UX enumeration member 174
I_CCDVT_UY enumeration member 174
I_CCDVT_UZ enumeration member 174
I_CCF_MAIN enumeration member 157
I_CCF_MAX enumeration member 157
I_CCF_MIN enumeration member 157
I_CCGT_FULL enumeration member 188
I_CCGT_MANUAL enumeration member 188
I_CCGT_NONE enumeration member 188
I_CCGT_SIMPLIFIED enumeration member 188
I_CCO_AND enumeration member 178
I_CCO_AND_OR enumeration member 178
I_CCO_EXCLUSIVE_OR enumeration member 178
I_CCT_COLUMN_SEC_C18 enumeration member 1202
I_CCT_COLUMN_SEC_C36 enumeration member 1202
I_CCT_COLUMN_SEC_C90 enumeration member 1202
I_CCT_COLUMN_SEC_L enumeration member 1202
I_CCT_COLUMN_SEC_P enumeration member 1202
I_CCT_COLUMN_SEC_R enumeration member 1202
I_CCT_COLUMN_SEC_TC enumeration member 1202
I_CCT_COLUMN_SEC_Z enumeration member 1202
I_CEC2CG_AUTOMATIC enumeration member 1774
I_CEC2CG_C12_15 enumeration member 1774

(c) 2014 Autodesk Inc. All rights reserved

I_CEC2CG_C16_20 enumeration member 1774
I_CEC2CG_C20_25 enumeration member 1774
I_CEC2CG_C25_30 enumeration member 1774
I_CEC2CG_C30_37 enumeration member 1774
I_CEC2CG_C35_45 enumeration member 1774
I_CEC2CG_C40_50 enumeration member 1774
I_CEC2CG_C45_55 enumeration member 1774
I_CEC2CG_C50_60 enumeration member 1774
I_CEC2ER_1 enumeration member 1774
I_CEC2ER_2A enumeration member 1774
I_CEC2ER_2B enumeration member 1774
I_CEC2ER_3 enumeration member 1774
I_CEC2ER_4A enumeration member 1774
I_CEC2ER_4B enumeration member 1774
I_CEC2ER_5A enumeration member 1774
I_CEC2ER_5B enumeration member 1774
I_CEC2ER_5C enumeration member 1774
I_CEC2ITSG_FE_B_22 enumeration member 1774
I_CEC2ITSG_FE_B_32 enumeration member 1774
I_CEC2ITSG_FE_B_38 enumeration member 1774
I_CEC2ITSG_FE_B_44 enumeration member 1774
I_CEC2NAD_BELGIAN enumeration member 1779
I_CEC2NAD_DUTCH enumeration member 1779
I_CEC2NAD_FINNISH enumeration member 1779
I_CEC2NAD_FRENCH enumeration member 1779
I_CEC2NAD_GERMAN enumeration member 1779
I_CEC2NAD_ITALIAN enumeration member 1779
I_CFDT_FOOT_DIM_A enumeration member 1118
I_CFDT_FOOT_DIM_AP enumeration member 1118
I_CFDT_FOOT_DIM_B enumeration member 1118
I_CFDT_FOOT_DIM_BP enumeration member 1118
I_CFDT_FOOT_DIM_COL_A enumeration member 1118
I_CFDT_FOOT_DIM_COL_B enumeration member 1118
I_CFDT_FOOT_DIM_EX enumeration member 1118
I_CFDT_FOOT_DIM_EY enumeration member 1118
I_CFDT_FOOT_DIM_H1 enumeration member 1118
I_CFDT_FOOT_DIM_H2 enumeration member 1118
I_CFDT_FOOT_DIM_H3 enumeration member 1118
I_CFDT_FOOT_DIM_H4 enumeration member 1118
I_CFDT_FOOT_DIM_L enumeration member 1118
I_CFST_FOOT_SHAPE_RECT enumeration member 1118
I_CFST_FOOT_SHAPE_TRAPEZOIDAL enumeration member 1118
I_CFST_FOOT_SHAPE_TWO_CL enumeration member 1118
I_CFT_FOOT_PIER_TYPE_BOLTED enumeration member 1128
I_CFT_FOOT_PIER_TYPE_DOWELED enumeration member 1128
I_CFT_FOOT_PIER_TYPE_HINGE_1 enumeration member 1128
I_CFT_FOOT_PIER_TYPE_HINGE_2 enumeration member 1128
I_CFT_FOOT_PIER_TYPE_PLAIN enumeration member 1128
I_CFT_FOOT_PIER_TYPE_SOCKETED enumeration member 1128
I_CFT_FOOT_TYPE_FOOTING enumeration member 1119
I_CFT_FOOT_TYPE_WALL enumeration member 1119
I_CM_CLOUD enumeration member 1594
I_CM_CLOUD_ASYNCHRONOUS enumeration member 1594
I_CM_LOCAL enumeration member 1594
I_CM_LOCAL_ASYNCHRONOUS enumeration member 1594
I_CM_TRAPEZOIDAL enumeration member 616
I_CMC_COHERENT_FE_MESH enumeration member 1326
I_CMC_USE_OF_KINEMATIC_CONSTRAINTS enumeration member 1326
I_CMRT_FOR_NONZERO_AS enumeration member 1779
I_CMRT_FOR_WHOLE_PANEL enumeration member 1779
I_CMRT_NONE enumeration member 1779
I_CMS_CALCULATIONS enumeration member 1593
I_CMS_VERIFICATION enumeration member 1593
I_CMSL_ERROR enumeration member 1593
I_CMSL_NOTE enumeration member 1593
I_CMSL_WARNING enumeration member 1593
I_CN_ACCIDENTAL enumeration member 410
I_CN_EXPLOATATION enumeration member 410
I_CN_PERMANENT enumeration member 410
I_CN_SEISMIC enumeration member 410
I_CN_SNOW enumeration member 410
I_CN_TEMPERATURE enumeration member 410
I_CN_WIND enumeration member 410
I_CPN84CG_ANOTHER enumeration member 1721
I_CPN84CG_B10 enumeration member 1721
I_CPN84CG_B12_5 enumeration member 1721
I_CPN84CG_B15 enumeration member 1721
I_CPN84CG_B17_5 enumeration member 1721
I_CPN84CG_B20 enumeration member 1721

(c) 2014 Autodesk Inc. All rights reserved

I_CPN84CG_B25 enumeration member 1721

I_CPN84CG_B30 enumeration member 1721

I_CPN84CG_B35 enumeration member 1721

I_CPN84CG_B40 enumeration member 1721

I_CPN84CG_B50 enumeration member 1721

I_CPN84CG_B55 enumeration member 1721

I_CPN84CG_B60 enumeration member 1721

I_CPN84CG_B65 enumeration member 1721

I_CPN84ER_MILD enumeration member 1722

I_CPN84ER_MODERATE enumeration member 1722

I_CPN84ER_SEVERE_OR_LEAKPROOF enumeration member 1722

I_CPN84HT_40 enumeration member 1740

I_CPN84HT_40_75 enumeration member 1740

I_CPN84HT_75 enumeration member 1740

I_CPN84HT_WATER enumeration member 1740

I_CPN99CG_ANOTHER enumeration member 1730

I_CPN99CG_B15 enumeration member 1730

I_CPN99CG_B20 enumeration member 1730

I_CPN99CG_B25 enumeration member 1730

I_CPN99CG_B30 enumeration member 1730

I_CPN99CG_B37 enumeration member 1730

I_CPN99CG_B45 enumeration member 1730

I_CPN99CG_B50 enumeration member 1730

I_CPN99CG_B55 enumeration member 1730

I_CPN99CG_B60 enumeration member 1730

I_CPN99CG_B65 enumeration member 1730

I_CPN99CG_B70 enumeration member 1730

I_CPN99ER_X0 enumeration member 1722

I_CPN99ER_XA1_XA2_XA3 enumeration member 1722

I_CPN99ER_XC1_XC2_XC3_XC4 enumeration member 1722

I_CPN99ER_XD1_XD2_XD3 enumeration member 1722

I_CPN99ER_XF1_XF3 enumeration member 1722

I_CPN99ER_XF2_XF4 enumeration member 1722

I_CPN99ER_XS1_XS2_XS3 enumeration member 1722

I_CPNSG_A_0 enumeration member 1721

I_CPNSG_A_I enumeration member 1721

I_CPNSG_A_II enumeration member 1721

I_CPNSG_A_III enumeration member 1721

I_CPNSG_A_IIIN enumeration member 1721

I_CRCT_BENDING_COMPRESSION_TENSION enumeration member 1765

I_CRCT_COMPRESSION_TENSION enumeration member 1765

I_CRCT_SIMPLE_BENDING enumeration member 1765

I_CRD_ALONG_X enumeration member 1709

I_CRD_ALONG_Y enumeration member 1709

I_CRD_ALONG_Z enumeration member 1709

I_CRD_AUTOMATIC enumeration member 1709

I_CRD_CARTESIAN_ALONG_VECTOR enumeration member 1709

I_CRD_POLAR_POINT enumeration member 1709

I_CRVT_FRF_FREQUENCY enumeration member 446

I_CS_CANCELLED_BY_USER enumeration member 1593

I_CS_COMPLETED enumeration member 1593

I_CS_FAILED_CALCULATION enumeration member 1593

I_CS_FAILED_GENERATION enumeration member 1593

I_CS_FAILED_NO_ENTITLEMENT enumeration member 1593

I_CS_FAILED_VERIFICATION enumeration member 1593

I_CS_IN_PROGRESS enumeration member 1593

I_CSNIPCG_B10 enumeration member 1771

I_CSNIPCG_B12_5 enumeration member 1771

I_CSNIPCG_B15 enumeration member 1771

I_CSNIPCG_B20 enumeration member 1771

I_CSNIPCG_B25 enumeration member 1771

I_CSNIPCG_B30 enumeration member 1771

I_CSNIPCG_B35 enumeration member 1771

I_CSNIPCG_B40 enumeration member 1771

I_CSNIPCG_B45 enumeration member 1771

I_CSNIPCG_B50 enumeration member 1771

I_CSNIPCG_B55 enumeration member 1771

I_CSNIPCG_B60 enumeration member 1771

I_CSNIPCG_B7_5 enumeration member 1771

I_CSNIPCM_AUTOCLAVES enumeration member 1771

I_CSNIPCM_NORMAL enumeration member 1771

I_CSNIPCM_THERMAL_TREATMENT enumeration member 1771

I_CSNIPCT_CELLULAR enumeration member 1771

I_CSNIPCT_FINE_GRAINED_A enumeration member 1771

I_CSNIPCT_FINE_GRAINED_B enumeration member 1771

I_CSNIPCT_FINE_GRAINED_V enumeration member 1771

I_CSNIPCT_HEAVYWEIGHT enumeration member 1771

I_CSNIPCT_LIGHTWEIGHT_NATURAL_FILLER enumeration member 1771

I_CSNIPCT_LIGHTWEIGHT_SYNTHETIC_COARSE_GRAINED

(c) 2014 Autodesk Inc. All rights reserved

enumeration member 1771

I_CSNIPCT_LIGHTWEIGHT_SYNTHETIC_COMPACT enumeration member 1771

I_CSNIPCT_POROUS enumeration member 1771

I_CSNIPE_EXTERNAL enumeration member 1774

I_CSNIPE_GROUND_VARIABLE_WATER_LEVEL enumeration member 1774

I_CSNIPE_INTERNAL enumeration member 1774

I_CSNIPSG_A_I enumeration member 1771

I_CSNIPSG_A_II enumeration member 1771

I_CSNIPSG_A_III enumeration member 1771

I_CSNIPSG_A_IIIB_Y enumeration member 1771

I_CSNIPSG_A_IIIB_YH enumeration member 1771

I_CSNIPSG_A_IV enumeration member 1771

I_CSNIPSG_A_V enumeration member 1771

I_CSNIPSG_A_VI enumeration member 1771

I_CSNIPSG_B_II enumeration member 1771

I_CSNIPSG_BP_I enumeration member 1771

I_CSNIPSG_BP_II enumeration member 1771

I_CSNIPSG_K_19 enumeration member 1771

I_CSNIPSG_K_7 enumeration member 1771

I_CSR_SLAB_RNF_BAR_AND_NET enumeration member 1226

I_CSR_SLAB_RNF_BAR_ONLY enumeration member 1226

I_CSR_SLAB_RNF_NET_ONLY enumeration member 1226

I_CSS_SLAB_SUPPORT_FIXED enumeration member 1227

I_CSS_SLAB_SUPPORT_PINNED enumeration member 1227

I_CST_STEEL_TYPE_LONG enumeration member 1260

I_CST_STEEL_TYPE_SPEC enumeration member 1260

I_CST_STEEL_TYPE_TRAN enumeration member 1260

I_CST_STEEL_TYPE_WIRE enumeration member 1260

I_CT_CALCULATE_IN_CLOUD enumeration member 1548

I_CT_CALCULATE_LOCALLY enumeration member 1548

I_CT_CASE_ANALYSIS_MODES_FILTER enumeration member 1656

I_CT_CODE_COMBINATION enumeration member 433

I_CT_CODE_COMBINATIONS enumeration member 1312

I_CT_COMBINATION enumeration member 433

I_CT_EMITTER enumeration member 1656

I_CT_EXTREME_PARAMS enumeration member 1656

I_CT_FE_EXTREME_PARAMS enumeration member 1656

I_CT_FE_MULTI_RESULT_TYPE enumeration member 1656

I_CT_FE_RESULT_PARAMS enumeration member 1656

I_CT_FOUNDATIONS_DESIGN enumeration member 1312

I_CT_GEO_ARC enumeration member 1656

I_CT_GEO_CIRCLE enumeration member 1656

I_CT_GEO_CONTOUR enumeration member 1656

I_CT_GEO_CURVE_DIV enumeration member 1656

I_CT_GEO_LAYER enumeration member 1656

I_CT_GEO_OBJECT enumeration member 1656

I_CT_GEO_POINT_2D enumeration member 1656

I_CT_GEO_POINT_3D enumeration member 1656

I_CT_GEO_POINT_3D_COLLECTION enumeration member 1656

I_CT_GEO_POLYLINE enumeration member 1656

I_CT_GEO_SEGMENT enumeration member 1656

I_CT_GEO_SEGMENT_ARC enumeration member 1656

I_CT_GEO_SEGMENT_COLLECTION enumeration member 1656

I_CT_GEO_SEGMENT_LINE enumeration member 1656

I_CT_HTML_VIEW enumeration member 1656

I_CT_JOINT_ANGLE_LOAD enumeration member 1656

I_CT_JOINT_FIXED_LOAD enumeration member 1656

I_CT_JOINT_GUSSET_CROSS_LOAD enumeration member 1656

I_CT_JOINT_GUSSET_FLANGE_LOAD enumeration member 1656

I_CT_JOINT_GUSSET_SIMPLE_LOAD enumeration member 1656

I_CT_JOINT_KNEE_LOAD enumeration member 1656

I_CT_JOINT_LOAD enumeration member 1656

I_CT_JOINT_PINNED_LOAD enumeration member 1656

I_CT_JOINT_TUBE_LOAD enumeration member 1656

I_CT_MOBILE enumeration member 433

I_CT_MODIF_EXTRUSION enumeration member 1656

I_CT_MODIF_LATHE enumeration member 1656

I_CT_MODIF_PYRAMID enumeration member 1656

I_CT_MOVING_LOADS enumeration member 1312

I_CT_NAMES_ARRAY enumeration member 1656

I_CT_NUMBERS_ARRAY enumeration member 1656

I_CT_NUMBERS_DICTIONARY enumeration member 1656

I_CT_OBJECTS_ARRAY enumeration member 1656

I_CT_OPER_MESHING enumeration member 1656

I_CT_OPER_ROTATION enumeration member 1656

I_CT_OPER_SCALING enumeration member 1656

I_CT_OPER_TRANSLATION enumeration member 1656

I_CT_PARAM_DEF enumeration member 1656

(c) 2014 Autodesk Inc. All rights reserved

I_CT_POINTS_ARRAY enumeration member 1656

I_CT_RC_REAL_REINF enumeration member 1312

I_CT_RC_THEORETICAL_REINF enumeration member 1312

I_CT_RESULT_QUERY_PARAMS enumeration member 1656

I_CT_RTF_VIEW enumeration member 1656

I_CT_SEISMIC_LOADS enumeration member 1312

I_CT_SIMPLE enumeration member 433

I_CT_SNOW_WIND_LOADS enumeration member 1312

I_CT_SPECTRAL_ANALYSIS_POINTS_COLLECTION enumeration member 1656

I_CT_SPECTRAL_ANALYSIS_SPECTRUM enumeration member 1656

I_CT_STEEL_CONNECTIONS enumeration member 1312

I_CT_STEEL_STRUCTURES enumeration member 1312

I_CT_STRUCTURE_GEO_ANALYSER enumeration member 1656

I_CT_STRUCTURE_MERGE_DATA enumeration member 1656

I_CT_SW_STRUCT3D enumeration member 1656

I_CT_SW_STRUCT3D_ELEMENT enumeration member 1656

I_CT_SW_STRUCT3D_FRAME enumeration member 1656

I_CT_SW_STRUCT3D_GEN_PARAMS enumeration member 1656

I_CT_SW_STRUCT3D_PURLIN_GEN_PARAMS enumeration member 1656

I_CT_TABLE_SCREEN_CAPTURE_PARAMS enumeration member 1656

I_CT_TIMBER_STRUCTURES enumeration member 1312

I_CT_VALUES_ARRAY enumeration member 1656

I_CT_VIEW_SCREEN_CAPTURE_PARAMS enumeration member 1656

I_CT_X enumeration member 616

I_CT_XY enumeration member 616

I_CT_Y enumeration member 616

I_D_CONF enumeration member 1544

I_D_EXE enumeration member 1544

I_D_HELP enumeration member 1544

I_D_MAIN enumeration member 1544

I_D_RES enumeration member 1544

I_D_TEMPLATE enumeration member 1544

I_D_USER_CONF enumeration member 1544

I_D_USER_MAIN enumeration member 1544

I_D_USER_OUTPUT enumeration member 1544

I_D_USER_PROJECTS enumeration member 1544

I_D_USER_TEMPLATE enumeration member 1544

I_DADT_CONSTANT enumeration member 446

I_DADT_MODAL enumeration member 446

I_DADT_NONE enumeration member 446

I_DADT_VARIABLE enumeration member 446

I_DAT_ANSI_AISC_360_10_ASD enumeration member 1597

I_DAT_ANSI_AISC_360_10_LRFD enumeration member 1597

I_DAT_USER_DEFINED enumeration member 1597

I_DBD_EC3_AUTO enumeration member 1818

I_DBD_EC3_CANTILEVER_2_0 enumeration member 1818

I_DBD_EC3_INTERN_ADJBAR_6 enumeration member 1818

I_DBD_EC3_INTERNAL_BRACINGS enumeration member 1818

I_DBD_EC3_NO enumeration member 1818

I_DBD_EC3_PINNED_ADJBAR_1 enumeration member 1818

I_DBD_EC3_PINNED_ADJBAR_3 enumeration member 1818

I_DBD_EC3_PINNED_PINNED_1_0 enumeration member 1818

I_DBD_EC3_PINNED_STIFF_0_7 enumeration member 1818

I_DBD_EC3_STIFF_ADJBAR_1 enumeration member 1818

I_DBD_EC3_STIFF_ADJBAR_3 enumeration member 1818

I_DBD_EC3_STIFF_STIFF_0_5 enumeration member 1818

I_DBD_EC3_TRUSS_CHORD_0_9 enumeration member 1818

I_DBD_EC3_TRUSS_DIAGONAL_0_8 enumeration member 1818

I_DBD_EC3_USER_DEFINED enumeration member 1818

I_DCSFT_CHECKSLEND enumeration member 1915

I_DCSFT_FIRE enumeration member 1915

I_DCSFT_POINTMIDDLE enumeration member 1915

I_DCSPT_CALCMETHOD enumeration member 1915

I_DCSPT_CALCTYPE enumeration member 1915

I_DCSPV_CALCMETHOD_1 enumeration member 1915

I_DCSPV_CALCMETHOD_2 enumeration member 1915

I_DCSPV_CALCMETHOD_3 enumeration member 1915

I_DCSPV_DIMENSIONING enumeration member 1915

I_DCSPV_EFFRATIO enumeration member 1916

I_DCSPV_GROUP_VERIF enumeration member 1915

I_DCSPV_MANUAL_VERIF enumeration member 1915

I_DCSPV_MAXSLEND enumeration member 1916

I_DCSPV_MEMBERS_VERIF enumeration member 1915

I_DCSPV_OPTIMIZATION enumeration member 1915

I_DE_CONF enumeration member 1545

(c) 2014 Autodesk Inc. All rights reserved

I_DE_TEMPLATE enumeration member 1545

I_DE_USER_CONF enumeration member 1545

I_DE_USER_TEMPLATE enumeration member 1545

I_DEDDT_Y enumeration member 1898

I_DEDDT_Z enumeration member 1898

I_DEDIPT_BUCKLING_Y enumeration member 1898

I_DEDIPT_BUCKLING_Z enumeration member 1898

I_DEDIPT_LBUCKLING_L enumeration member 1898

I_DEDIPT_LBUCKLING_U enumeration member 1898

I_DEDPT_ACC enumeration member 1898

I_DEDPT_CASE_NO enumeration member 1898

I_DEDPT_COMP_NO enumeration member 1898

I_DEDPT_ELEM_NO enumeration member 1898

I_DEDPT_LOADCLASS enumeration member 1898

I_DEDPT_POINT_NO enumeration member 1898

I_DEDPT_POINTS_NUM enumeration member 1898

I_DI_BAR_SECTIONS enumeration member 1476

I_DI_PANEL_THICKNESS enumeration member 1476

I_DI_SUPPORTS enumeration member 1476

I_DLBCD_EC3_CANTILEVER_2_0 enumeration member 1818

I_DLBCD_EC3_INTERNAL_BRACINGS enumeration member 1818

I_DLBCD_EC3_NO enumeration member 1818

I_DLBCD_EC3_PINNED_PINNED_1_0 enumeration member 1818

I_DLBCD_EC3_STIFF_STIFF_0_5 enumeration member 1818

I_DLBCD_EC3_USER_DEFINED enumeration member 1818

I_DLBT_EC3_CANTILEVER enumeration member 1817

I_DLBT_EC3_NO enumeration member 1817

I_DLBT_EC3_SYMMETR_LOADED enumeration member 1817

I_DLL_EC3_CENTER_LOADED enumeration member 1817

I_DLL_EC3_LOW_EDGE_LOADED enumeration member 1817

I_DLL_EC3_LOW_SEC_PART_LOADED enumeration member 1817

I_DLL_EC3_UPP_EDGE_LOADED enumeration member 1817

I_DLL_EC3_UPP_SECT_PAR_LOADED enumeration member 1817

I_DLLCT_LATERAL_LOAD_DIR enumeration member 1601

I_DLLCT_ONE_DIR_ONLY enumeration member 1601

I_DLLCT_TWO_DIR_INDEPENDENTLY enumeration member 1601

I_DLT_CB71_FORCE_AT_DISTANCE enumeration member 1888

I_DLT_CB71_MOMENT_AT_END enumeration member 1888

I_DLT_CB71_UNIFORM_LOAD enumeration member 1888

I_DLT_EC3_CONCENTRATED_FORCE enumeration member 1817

I_DLT_EC3_MOMENTS_AT_ENDS enumeration member 1817

I_DLT_EC3_UNIFORM_LOAD enumeration member 1817

I_DLT_EC3_UNIFORM_MOMENT enumeration member 1817

I_DMCBC_Y enumeration member 1919

I_DMCBC_Z enumeration member 1919

I_DMCRV_CORRECT enumeration member 1919

I_DMCRV_INCORRECT enumeration member 1919

I_DMCRV_INCORRECTDATA enumeration member 1919

I_DMCRV_INSTABILITY enumeration member 1919

I_DMDBDT_BUCKLING_U enumeration member 1893

I_DMDBDT_BUCKLING_V enumeration member 1893

I_DMDBDT_BUCKLING_Y enumeration member 1893

I_DMDBDT_BUCKLING_Z enumeration member 1893

I_DMDDDT_DEFL_Y enumeration member 1894

I_DMDDDT_DEFL_Z enumeration member 1894

I_DMDDDT_DISP_X enumeration member 1893

I_DMDDDT_DISP_Y enumeration member 1893

I_DMDDEVT_HUMIDITY enumeration member 1908

I_DMDDEVT_RETRAIT enumeration member 1908

I_DMDLDT_LENGTH_U enumeration member 1893

I_DMDLDT_LENGTH_V enumeration member 1893

I_DMDLDT_LENGTH_Y enumeration member 1893

I_DMDLDT_LENGTH_Z enumeration member 1893

I_DMDLEVT_CATEGORY enumeration member 1908

I_DMDLEVT_NATURE enumeration member 1908

I_DMDLEVT_TIMB_TYPE enumeration member 1908

I_DMDMT_CONCRETE enumeration member 1893

I_DMDMT_STEEL enumeration member 1893

I_DMDMT_TIMBER enumeration member 1893

I_DMDT_ALUMINIUM enumeration member 1908

I_DMDT_BEAM enumeration member 1892

I_DMDT_COLUMN enumeration member 1892

I_DMDT_MEMBER enumeration member 1892

I_DMDT_NONE enumeration member 1908

I_DMDT_STEEL enumeration member 1908

I_DMDT_USER enumeration member 1892

(c) 2014 Autodesk Inc. All rights reserved

I_DMDT_WOOD enumeration member 1908
I_DMDVT_CS enumeration member 1908
I_DMDVT_DAMPCOEF enumeration member 1908
I_DMDVT_E enumeration member 1908
I_DMDVT_E_5 enumeration member 1908
I_DMDVT_FU enumeration member 1908
I_DMDVT_G enumeration member 1908
I_DMDVT_LX enumeration member 1908
I_DMDVT_NU enumeration member 1908
I_DMDVT_PN_E_ADITIONAL enumeration member 1908
I_DMDVT_PN_E_TRANS enumeration member 1908
I_DMDVT_RE enumeration member 1908
I_DMDVT_RE_AX_COMR enumeration member 1908
I_DMDVT_RE_AX_TENS enumeration member 1908
I_DMDVT_RE_BENDING enumeration member 1908
I_DMDVT_RE_SHEAR enumeration member 1908
I_DMDVT_RE_TR_COMPR enumeration member 1908
I_DMDVT_RE_TR_TENS enumeration member 1908
I_DMDVT_RO enumeration member 1908
I_DMDVT_RT enumeration member 1908
I_DMDVT_TRANS enumeration member 1908
I_DMRTC_DESCRIPTION enumeration member 1920
I_DMRTC_NAME enumeration member 1920
I_DMRTC_PARAGRAPH enumeration member 1920
I_DMRTC_UNIT enumeration member 1920
I_DMRTC_VALUE enumeration member 1920
I_DMRTLT_1PARAM enumeration member 1920
I_DMRTLT_4HEADER enumeration member 1920
I_DMRTLT_4PARAM enumeration member 1920
I_DMRTLT_5HEADER enumeration member 1920
I_DMRTLT_5PARAM enumeration member 1920
I_DMRTLT_CAPTION enumeration member 1920
I_DMRTLT_PRINTCAPTION enumeration member 1920
I_DMRTLT_SUBCAPTION enumeration member 1920
I_DOF_RX enumeration member 1657
I_DOF_RY enumeration member 1657
I_DOF_RZ enumeration member 1657
I_DOF_UX enumeration member 1657
I_DOF_UY enumeration member 1657
I_DOF_UZ enumeration member 1657
I_DPDIT_STANDARD enumeration member 1912
I_DPDIT_VAR_BEGEND enumeration member 1912
I_DPDIT_VAR_IN_POINT enumeration member 1912
I_DPDIT_VAR_MIDDLE enumeration member 1912
I_DPDT_CAE enumeration member 1912
I_DPDT_CAEP enumeration member 1912
I_DPDT_CAI enumeration member 1912
I_DPDT_CAIP enumeration member 1912
I_DPDT_CAISSON enumeration member 1912
I_DPDT_CROSS enumeration member 1912
I_DPDT_CUSER enumeration member 1912
I_DPDT_DCEC enumeration member 1912
I_DPDT_DCED enumeration member 1912
I_DPDT_DCEP enumeration member 1912
I_DPDT_DCIG enumeration member 1912
I_DPDT_DCIP enumeration member 1912
I_DPDT_DRECT enumeration member 1912
I_DPDT_FRTG enumeration member 1912
I_DPDT_HEA enumeration member 1912
I_DPDT_HEAA enumeration member 1912
I_DPDT_HEB enumeration member 1912
I_DPDT_HEC enumeration member 1912
I_DPDT_HEM enumeration member 1912
I_DPDT_HER enumeration member 1912
I_DPDT_HHEA enumeration member 1912
I_DPDT_HHEB enumeration member 1912
I_DPDT_HHEM enumeration member 1912
I_DPDT_IIPE enumeration member 1912
I_DPDT_INSYM enumeration member 1912
I_DPDT_IPE enumeration member 1912
I_DPDT_IPEA enumeration member 1912
I_DPDT_IPEO enumeration member 1912
I_DPDT_IPER enumeration member 1912
I_DPDT_IPEV enumeration member 1912
I_DPDT_IPN enumeration member 1912
I_DPDT_ISYM enumeration member 1912
I_DPDT_MHEA enumeration member 1912
I_DPDT_MHEB enumeration member 1912
I_DPDT_MHEM enumeration member 1912
I_DPDT_MIPE enumeration member 1912
I_DPDT_NONE enumeration member 1912
I_DPDT_OSIE enumeration member 1912
I_DPDT_PRS enumeration member 1912
I_DPDT_RECT enumeration member 1912

(c) 2014 Autodesk Inc. All rights reserved

I_DPDT_TCAR enumeration member 1912
I_DPDT_TEAE enumeration member 1912
I_DPDT_TEAI enumeration member 1912
I_DPDT_THEX enumeration member 1912
I_DPDT_TREC enumeration member 1912
I_DPDT_TRON enumeration member 1912
I_DPDT_TUBE enumeration member 1912
I_DPDT_TUSER enumeration member 1912
I_DPDT_UAP enumeration member 1912
I_DPDT_UPN enumeration member 1912
I_DPDT_UUAP enumeration member 1912
I_DPDT_UUPN enumeration member 1912
I_DPDT_X enumeration member 1912
I_DPDVT_B enumeration member 1912
I_DPDVT_B2 enumeration member 1912
I_DPDVT_BD enumeration member 1912
I_DPDVT_BF enumeration member 1912
I_DPDVT_BF2 enumeration member 1912
I_DPDVT_DIS enumeration member 1912
I_DPDVT_EA enumeration member 1912
I_DPDVT_EA2 enumeration member 1912
I_DPDVT_ES enumeration member 1912
I_DPDVT_ES2 enumeration member 1912
I_DPDVT_H enumeration member 1912
I_DPDVT_HD enumeration member 1912
I_DPDVT_HW enumeration member 1912
I_DPDVT_I enumeration member 1912
I_DPDVT_IY enumeration member 1912
I_DPDVT_IZ enumeration member 1912
I_DPDVT_MASSE enumeration member 1912
I_DPDVT_MSY enumeration member 1912
I_DPDVT_MSZ enumeration member 1912
I_DPDVT_R enumeration member 1912
I_DPDVT_R2 enumeration member 1912
I_DPDVT_S enumeration member 1912
I_DPDVT_SY enumeration member 1912
I_DPDVT_SZ enumeration member 1912
I_DPDVT_VPY enumeration member 1912
I_DPDVT_VPZ enumeration member 1912
I_DPDVT_VY enumeration member 1912
I_DPDVT_VZ enumeration member 1912
I_DRV_COEFF enumeration member 51
I_DRV_ENTIRE_STRUCTURE enumeration member 51
I_DRV_X enumeration member 51
I_DRV_Y enumeration member 51
I_DRV_Z enumeration member 51
I_DSMRT_FACTOR_0_8 enumeration member 1605
I_DSMRT_FACTOR_0_8_AND_NL_INCREASE enumeration member 1605
I_DSMRT_FACTOR_0_8_AND_TAU_B enumeration member 1605
I_DST_DOUBLE enumeration member 1889
I_DST_LONG enumeration member 1889
I_DST_TEXT enumeration member 1889
I_DT_ANCHOR_BOLTS enumeration member 1326
I_DT_BOLTS enumeration member 1326
I_DT_BUILDING_SOILS enumeration member 1326
I_DT_REINFORCING_BARS enumeration member 1326
I_DT_SECTIONS enumeration member 1326
I_DT_STANDARD_LOADS enumeration member 1326
I_DT_VEHICLE_LOADS enumeration member 1326
I_DT_WIRE_FABRICS enumeration member 1326
I_DUT_DISPL enumeration member 1935
I_DUT_FORCE enumeration member 1935
I_DUT_LENGTH enumeration member 1935
I_DUT_MOMENT enumeration member 1935
I_DUT_NONE enumeration member 1935
I_DUT_SECDIMEN enumeration member 1935
I_DUT_SECMI enumeration member 1935
I_DUT_SECSUR enumeration member 1935
I_DUT_SECVOL enumeration member 1935
I_DUT_STRESS enumeration member 1935
I_DYST_EC3_AVERAGE enumeration member 1852
I_DYST_EC3_BASIC enumeration member 1852
I_EC8_SAGT_A enumeration member 301
I_EC8_SAGT_B enumeration member 301
I_EC8_SAGT_C enumeration member 301
I_EC8_SAGT_D enumeration member 301
I_EC8_SAGT_E enumeration member 301
I_EC8_SAST_TYPE_1 enumeration member 301
I_EC8_SAST_TYPE_2 enumeration member 301
I_EFF_DXF enumeration member 1274
I_EFF_STR enumeration member 1274
I_ESM_AUTO enumeration member 1567
I_ESM_FRONTAL enumeration member 1567

(c) 2014 Autodesk Inc. All rights reserved

I_ESM_ITERATIVE enumeration member 1567

I_ESM_MULTI_THREADED enumeration member 1567

I_ESM_SKYLINE enumeration member 1567

I_ESM_SPARSE enumeration member 1567

I_ESM_SPARSE_M enumeration member 1567

I_EVT_DEFLECTION_UX enumeration member 915

I_EVT_DEFLECTION_UY enumeration member 915

I_EVT_DEFLECTION_UZ enumeration member 915

I_EVT_DISPLACEMENT_BAR_RX enumeration member 915

I_EVT_DISPLACEMENT_BAR_RY enumeration member 915

I_EVT_DISPLACEMENT_BAR_RZ enumeration member 915

I_EVT_DISPLACEMENT_BAR_UX enumeration member 915

I_EVT_DISPLACEMENT_BAR_UY enumeration member 915

I_EVT_DISPLACEMENT_BAR_UZ enumeration member 915

I_EVT_DISPLACEMENT_NODE_RX enumeration member 915

I_EVT_DISPLACEMENT_NODE_RY enumeration member 915

I_EVT_DISPLACEMENT_NODE_RZ enumeration member 915

I_EVT_DISPLACEMENT_NODE_UX enumeration member 915

I_EVT_DISPLACEMENT_NODE_UY enumeration member 915

I_EVT_DISPLACEMENT_NODE_UZ enumeration member 915

I_EVT_FOOTFALL_A enumeration member 915

I_EVT_FOOTFALL_EXCITATION_NODE enumeration member 915

I_EVT_FOOTFALL_FREQUENCY enumeration member 915

I_EVT_FOOTFALL_RF_OVERALL enumeration member 915

I_EVT_FOOTFALL_RF_RESONANT enumeration member 915

I_EVT_FOOTFALL_RF_TRANSIENT enumeration member 915

I_EVT_FOOTFALL_VRMQ enumeration member 915

I_EVT_FOOTFALL_VRMS enumeration member 915

I_EVT_FORCE_BAR_FX enumeration member 915

I_EVT_FORCE_BAR_FY enumeration member 915

I_EVT_FORCE_BAR_FZ enumeration member 915

I_EVT_FORCE_BAR_MX enumeration member 915

I_EVT_FORCE_BAR_MY enumeration member 915

I_EVT_FORCE_BAR_MZ enumeration member 915

I_EVT_FRF_VX enumeration member 915

I_EVT_FRF_VY enumeration member 915

I_EVT_FRF_VZ enumeration member 915

I_EVT_MODAL_AVERAGE_PARTICIPATION_COEFF enumeration member 915

I_EVT_MODAL_CURMASS_RX enumeration member 915

I_EVT_MODAL_CURMASS_RY enumeration member 915

I_EVT_MODAL_CURMASS_RZ enumeration member 915

I_EVT_MODAL_CURMASS_UX enumeration member 915

I_EVT_MODAL_CURMASS_UY enumeration member 915

I_EVT_MODAL_CURMASS_UZ enumeration member 915

I_EVT_MODAL_DAMPING enumeration member 915

I_EVT_MODAL_EIGENVALUE enumeration member 915

I_EVT_MODAL_EIGENVECTOR_RX enumeration member 915

I_EVT_MODAL_EIGENVECTOR_RX_1 enumeration member 915

I_EVT_MODAL_EIGENVECTOR_RY enumeration member 915

I_EVT_MODAL_EIGENVECTOR_RY_1 enumeration member 915

I_EVT_MODAL_EIGENVECTOR_RZ enumeration member 915

I_EVT_MODAL_EIGENVECTOR_RZ_1 enumeration member 915

I_EVT_MODAL_EIGENVECTOR_UX enumeration member 915

I_EVT_MODAL_EIGENVECTOR_UX_1 enumeration member 915

I_EVT_MODAL_EIGENVECTOR_UY enumeration member 915

I_EVT_MODAL_EIGENVECTOR_UY_1 enumeration member 915

I_EVT_MODAL_EIGENVECTOR_UZ enumeration member 915

I_EVT_MODAL_EIGENVECTOR_UZ_1 enumeration member 915

I_EVT_MODAL_ENERGY enumeration member 915

I_EVT_MODAL_FREQUENCY enumeration member 915

I_EVT_MODAL_MASSES_X enumeration member 915

I_EVT_MODAL_MASSES_Y enumeration member 915

I_EVT_MODAL_MASSES_Z enumeration member 915

I_EVT_MODAL_NODE_MASSES_X enumeration member 915

I_EVT_MODAL_NODE_MASSES_Y enumeration member 915

I_EVT_MODAL_NODE_MASSES_Z enumeration member 915

I_EVT_MODAL_PARTICIPATION_COEFF_RX enumeration member 915

I_EVT_MODAL_PARTICIPATION_COEFF_RY enumeration member 915

(c) 2014 Autodesk Inc. All rights reserved

I_EVT_MODAL_PARTICIPATION_COEFF_RZ enumeration member 915

I_EVT_MODAL_PARTICIPATION_COEFF_UX enumeration member 915

I_EVT_MODAL_PARTICIPATION_COEFF_UY enumeration member 915

I_EVT_MODAL_PARTICIPATION_COEFF_UZ enumeration member 915

I_EVT_MODAL_PERIOD enumeration member 915

I_EVT_MODAL_PRECISION enumeration member 915

I_EVT_MODAL_PULSATION enumeration member 915

I_EVT_MODAL_SUMMASS_RX enumeration member 915

I_EVT_MODAL_SUMMASS_RY enumeration member 915

I_EVT_MODAL_SUMMASS_RZ enumeration member 915

I_EVT_MODAL_SUMMASS_UX enumeration member 915

I_EVT_MODAL_SUMMASS_UY enumeration member 915

I_EVT_MODAL_SUMMASS_UZ enumeration member 915

I_EVT_MODAL_TOTMASS_RX enumeration member 915

I_EVT_MODAL_TOTMASS_RY enumeration member 915

I_EVT_MODAL_TOTMASS_RZ enumeration member 915

I_EVT_MODAL_TOTMASS_UX enumeration member 915

I_EVT_MODAL_TOTMASS_UY enumeration member 915

I_EVT_MODAL_TOTMASS_UZ enumeration member 915

I_EVT_PSEUDOSTATIC_FORCE_FX enumeration member 915

I_EVT_PSEUDOSTATIC_FORCE_FY enumeration member 915

I_EVT_PSEUDOSTATIC_FORCE_FZ enumeration member 915

I_EVT_PSEUDOSTATIC_FORCE_GX enumeration member 915

I_EVT_PSEUDOSTATIC_FORCE_GY enumeration member 915

I_EVT_PSEUDOSTATIC_FORCE_GZ enumeration member 915

I_EVT_PSEUDOSTATIC_FORCE_MX enumeration member 915

I_EVT_PSEUDOSTATIC_FORCE_MY enumeration member 915

I_EVT_PSEUDOSTATIC_FORCE_MZ enumeration member 915

I_EVT_PSEUDOSTATIC_FORCE_TX enumeration member 915

I_EVT_PSEUDOSTATIC_FORCE_TY enumeration member 915

I_EVT_PSEUDOSTATIC_FORCE_TZ enumeration member 915

I_EVT_REACTION_FX enumeration member 915

I_EVT_REACTION_FY enumeration member 915

I_EVT_REACTION_FZ enumeration member 915

I_EVT_REACTION_MX enumeration member 915

I_EVT_REACTION_MY enumeration member 915

I_EVT_REACTION_MZ enumeration member 915

I_EVT_STRESS_BAR_FX_SX enumeration member 915

I_EVT_STRESS_BAR_SMAX enumeration member 915

I_EVT_STRESS_BAR_SMAX_MY enumeration member 915

I_EVT_STRESS_BAR_SMAX_MZ enumeration member 915

I_EVT_STRESS_BAR_SMIN enumeration member 915

I_EVT_STRESS_BAR_SMIN_MY enumeration member 915

I_EVT_STRESS_BAR_SMIN_MZ enumeration member 915

I_EVT_STRESS_BAR_T enumeration member 915

I_EVT_STRESS_BAR_TY enumeration member 915

I_EVT_STRESS_BAR_TZ enumeration member 915

I_EVT_TIME_ARX enumeration member 915

I_EVT_TIME_ARY enumeration member 915

I_EVT_TIME_ARZ enumeration member 915

I_EVT_TIME_AX enumeration member 915

I_EVT_TIME_AY enumeration member 915

I_EVT_TIME_AZ enumeration member 915

I_EVT_TIME_VRX enumeration member 915

I_EVT_TIME_VRY enumeration member 915

I_EVT_TIME_VRZ enumeration member 915

I_EVT_TIME_VX enumeration member 915

I_EVT_TIME_VY enumeration member 915

I_EVT_TIME_VZ enumeration member 915

I_EXT_ANGLE2BEAM enumeration member 2180

I_EXT_NONE enumeration member 2180

I_FAEF_CONCRETE_CENTRE enumeration member 399

I_FAEF_SCI_P354 enumeration member 399

I_FAEM_FULL_EXCITATION enumeration member 398

I_FAEM_SELF_EXCITATION enumeration member 398

I_FANST_ALL_NODES enumeration member 401

I_FANST_NODES_BELONGING_TO_SELECTED_PANELS enumeration member 401

I_FANST_SELECTED_NODES enumeration member 401

I_FET_Q4 enumeration member 835

I_FET_Q8 enumeration member 835

I_FET_T3 enumeration member 835

I_FET_T6 enumeration member 835

I_FET_VOL_B20 enumeration member 835

I_FET_VOL_B8 enumeration member 835

(c) 2014 Autodesk Inc. All rights reserved

I_FET_VOL_T10 enumeration member 835
I_FET_VOL_T4 enumeration member 835
I_FET_VOL_W15 enumeration member 835
I_FET_VOL_W6 enumeration member 835
I_FLT_ARBITRARY enumeration member 937
I_FLT_LOWER enumeration member 937
I_FLT_MIDDLE enumeration member 937
I_FLT_UPPER enumeration member 937
I_FMP_CB71_AUTOMATIC enumeration member 1888
I_FMP_CB71_HORIZONTAL enumeration member 1888
I_FMP_CB71_VERTICAL enumeration member 1888
I_FPT_CB71_LESS enumeration member 1889
I_FPT_CB71_MORE enumeration member 1889
I_FRK_ELEMENT_CENTER enumeration member 986
I_FRK_ELEMENT_NODE enumeration member 986
I_FRK_PANEL_NODE enumeration member 986
I_FRRCP_HORIZONTAL_BOTTOM enumeration member 977
I_FRRCP_HORIZONTAL_MIDDLE enumeration member 977
I_FRRCP_HORIZONTAL_TOP enumeration member 977
I_FRRCP_VERTICAL_LEFT enumeration member 977
I_FRRCP_VERTICAL_MIDDLE enumeration member 977
I_FRRCP_VERTICAL_RIGHT enumeration member 977
I_FRS_GLOBAL_SMOOTHING enumeration member 986
I_FRS_IN_ELEMENT_CENTER enumeration member 986
I_FRS_NO_SMOOTHING enumeration member 986
I_FRS_SMOOTHING_ACCORDING_TO_SELECTION enumeration member 986
I_FRS_SMOOTHING_WITHIN_A_PANEL enumeration member 986
I_FRT_COMPLEX_M_MISES enumeration member 984
I_FRT_COMPLEX_MXX_BOTTOM_NEN enumeration member 984
I_FRT_COMPLEX_MXX_BOTTOM_WA enumeration member 984
I_FRT_COMPLEX_MXX_TOP_NEN enumeration member 984
I_FRT_COMPLEX_MXX_TOP_WA enumeration member 984
I_FRT_COMPLEX_MYY_BOTTOM_NEN enumeration member 984
I_FRT_COMPLEX_MYY_BOTTOM_WA enumeration member 984
I_FRT_COMPLEX_MYY_TOP_NEN enumeration member 984
I_FRT_COMPLEX_MYY_TOP_WA enumeration member 984
I_FRT_COMPLEX_N_MISES enumeration member 984
I_FRT_COMPLEX_S_MISES enumeration member 984
I_FRT_DETAILED_MXX enumeration member 984
I_FRT_DETAILED_MXY enumeration member 984
I_FRT_DETAILED_MYY enumeration member 984
I_FRT_DETAILED_NXX enumeration member 984
I_FRT_DETAILED_NXY enumeration member 984
I_FRT_DETAILED_NYY enumeration member 984
I_FRT_DETAILED_PNORM enumeration member 984
I_FRT_DETAILED_QXX enumeration member 984
I_FRT_DETAILED_QYY enumeration member 984
I_FRT_DETAILED_RNORM enumeration member 984
I_FRT_DETAILED_RXX enumeration member 984
I_FRT_DETAILED_RYY enumeration member 984
I_FRT_DETAILED_SXX enumeration member 984
I_FRT_DETAILED_SXY enumeration member 984
I_FRT_DETAILED_SYY enumeration member 984
I_FRT_DETAILED_TXX enumeration member 984
I_FRT_DETAILED_TYY enumeration member 984
I_FRT_DETAILED_UXX enumeration member 984
I_FRT_DETAILED_UYY enumeration member 984
I_FRT_DETAILED_WNORM enumeration member 984
I_FRT_FOOT_RES_AX enumeration member 1127
I_FRT_FOOT_RES_AY enumeration member 1127
I_FRT_PRINCIPAL_M1 enumeration member 984
I_FRT_PRINCIPAL_M1_2 enumeration member 984
I_FRT_PRINCIPAL_M2 enumeration member 984
I_FRT_PRINCIPAL_MAL enumeration member 984
I_FRT_PRINCIPAL_N1 enumeration member 984
I_FRT_PRINCIPAL_N1_2 enumeration member 984
I_FRT_PRINCIPAL_N2 enumeration member 984
I_FRT_PRINCIPAL_NAL enumeration member 984
I_FRT_PRINCIPAL_Q1_2 enumeration member 984
I_FRT_PRINCIPAL_S1 enumeration member 984
I_FRT_PRINCIPAL_S1_2 enumeration member 984
I_FRT_PRINCIPAL_S2 enumeration member 984
I_FRT_PRINCIPAL_SAL enumeration member 984
I_FRT_PRINCIPAL_T1_2 enumeration member 984
I_FRT_PRINCIPAL_U enumeration member 984
I_FRT_PRINCIPAL_UGX enumeration member 984
I_FRT_PRINCIPAL_UGY enumeration member 984
I_FRT_PRINCIPAL_UGZ enumeration member 984
I_FRT_REDUCED_HEIGHT enumeration member 984

(c) 2014 Autodesk Inc. All rights reserved

I_FRT_REDUCED_LENGTH enumeration member 984
I_FRT_REDUCED_MY enumeration member 984
I_FRT_REDUCED_MZ enumeration member 984
I_FRT_REDUCED_NX enumeration member 984
I_FRT_REDUCED_SE enumeration member 984
I_FRT_REDUCED_SO enumeration member 984
I_FRT_REDUCED_T enumeration member 984
I_FRT_REDUCED_TY enumeration member 984
I_FRT_REDUCED_TZ enumeration member 984
I_FRT_REINF_A_MIN enumeration member 984
I_FRT_REINF_AX enumeration member 984
I_FRT_REINF_AX_BOTTOM enumeration member 984
I_FRT_REINF_AX_TOP enumeration member 984
I_FRT_REINF_AY enumeration member 984
I_FRT_REINF_AY_BOTTOM enumeration member 984
I_FRT_REINF_AY_TOP enumeration member 984
I_FRT_REINF_E_AX_BOTTOM enumeration member 984
I_FRT_REINF_E_AX_TOP enumeration member 984
I_FRT_REINF_E_AY_BOTTOM enumeration member 984
I_FRT_REINF_E_AY_TOP enumeration member 984
I_FRT_REINF_F enumeration member 984
I_GADM_BEGIN_MIDDLE_END enumeration member 1634
I_GADM_CENTER_BEGIN_END enumeration member 1634
I_GCA_NONE enumeration member 1614
I_GCA_OX enumeration member 1614
I_GCA_OY enumeration member 1614
I_GCA_OZ enumeration member 1614
I_GCAS_OX_MINUS enumeration member 1635
I_GCAS_OX_PLUS enumeration member 1635
I_GCAS_OY_MINUS enumeration member 1635
I_GCAS_OY_PLUS enumeration member 1635
I_GCAS_OZ_MINUS enumeration member 1635
I_GCAS_OZ_PLUS enumeration member 1635
I_GCS_GLOBAL enumeration member 1615
I_GCS_LOCAL_AFTER_OFFSET enumeration member 1615
I_GCS_LOCAL_ORIGINAL enumeration member 1615
I_GOT_ARC enumeration member 1616
I_GOT_CIRCLE enumeration member 1616
I_GOT_CONTOUR enumeration member 1616
I_GOT_INTERSECTION enumeration member 1616
I_GOT_NONE enumeration member 1616
I_GOT_POLYLINE enumeration member 1616
I_GP_XY enumeration member 1635
I_GP_XZ enumeration member 1635
I_GP_YZ enumeration member 1635
I_GST_ARC enumeration member 1615
I_GST_LINE enumeration member 1615
I_GST_NONE enumeration member 1615
I_ICRV_LOCAL enumeration member 48
I_ICRV_NPOINTS enumeration member 48
I_ICRV_PROJECTION enumeration member 48
I_ICRV_PX1 enumeration member 48
I_ICRV_PX2 enumeration member 48
I_ICRV_PX3 enumeration member 48
I_ICRV_PY1 enumeration member 48
I_ICRV_PY2 enumeration member 48
I_ICRV_PY3 enumeration member 48
I_ICRV_PZ1 enumeration member 48
I_ICRV_PZ2 enumeration member 48
I_ICRV_PZ3 enumeration member 48
I_IPT_CHOLESKY enumeration member 1574
I_IPT_DIAGONAL enumeration member 1574
I_IPT_GAUSS enumeration member 1574
I_IPT_ICCF enumeration member 1574
I_IS2000_ER_MILD enumeration member 1780
I_IS2000_ER_MODERATE enumeration member 1780
I_IS2000_ER_MOST_SEVERE enumeration member 1780
I_IS2000_ER_SEVERE enumeration member 1780
I_IS2000_ER_VERY_SEVERE enumeration member 1780
I_ISM_0 enumeration member 1578
I_ISM_1 enumeration member 1578
I_ISM_2 enumeration member 1578
I_ISMU_1_2 enumeration member 1578
I_ISMU_1_4 enumeration member 1578
I_ISMU_MAX enumeration member 1578
I_ISMU_MIN enumeration member 1578
I_JAPP_LONG_FLANGE_TO_COLUMN enumeration member 2040
I_JAPP_SHORT_FLANGE_TO_COLUMN enumeration member 2040
I_JAPT_RECT enumeration member 2019
I_JAPT_TUBE enumeration member 2019
I_JAT_PIN enumeration member 2003
I_JAT_PLATE enumeration member 2003
I_JAT_ROD enumeration member 2003

(c) 2014 Autodesk Inc. All rights reserved

I_JAT_SIMPLE enumeration member 2003

I_JAU_DEGREES enumeration member 2179

I_JAU_RADIANS enumeration member 2179

I_JBT_HIGH_TENSION enumeration member 2169

I_JBT_NORMAL enumeration member 2169

I_JCBPCM_ELASTIC enumeration member 1998

I_JCBPCM_PLASTIC enumeration member 1998

I_JCDT_IN_STRUCTURE enumeration member 2156

I_JCDT_STANDALONE enumeration member 2156

I_JCMT_BEAM_ANGLES enumeration member 2180

I_JCMT_BEAM_BRACKETS enumeration member 2180

I_JCMT_BEAM_MAIN enumeration member 2180

I_JCT_ANGLE enumeration member 2156

I_JCT_BEAM_GIRDER enumeration member 2156

I_JCT_BOLTS enumeration member 2152

I_JCT_COL_CONCR enumeration member 2156

I_JCT_COL_FIXED enumeration member 2156

I_JCT_COL_PINNED enumeration member 2156

I_JCT_GUSSET_CROSS enumeration member 2156

I_JCT_GUSSET_FLANGE enumeration member 2156

I_JCT_GUSSET_SIMPLE enumeration member 2156

I_JCT_KNEE_BOLTED enumeration member 2156

I_JCT_KNEE_WELDED enumeration member 2156

I_JCT_TRNS enumeration member 2156

I_JCT_TUBE enumeration member 2156

I_JCT_UNKNOWN enumeration member 2156

I_JCT_WELDS enumeration member 2152

I_JET_L_ANGLES_R_ANGLES_FLANGE enumeration member 2152

I_JET_L_ANGLES_R_ANGLES_WEB enumeration member 2152

I_JET_L_ANGLES_R_PLATES enumeration member 2152

I_JET_L_PLATES_R_ANGLES enumeration member 2152

I_JET_L_PLATES_R_PLATES enumeration member 2152

I_JET_L_SEATS_ANGLES_R_SEATS enumeration member 2152

I_JET_L_SEATS_ANGLES_R_SEATS_ANGLES enumeration member 2152

I_JET_L_SEATS_R_SEATS enumeration member 2152

I_JET_L_SEATS_R_SEATS_ANGLES enumeration member 2152

I_JET_L_STIFF_LONG_R_STIFF_LONG enumeration member 2152

I_JET_L_STIFF_LONG_R_STIFF_SHORT enumeration member 2152

I_JET_L_STIFF_SHORT_R_STIFF_LONG enumeration member 2152

I_JET_L_STIFF_SHORT_R_STIFF_SHORT enumeration member 2152

I_JET_R_ANGLES_FLANGE enumeration member 2152

I_JET_R_ANGLES_WEB enumeration member 2152

I_JET_R_PLATES enumeration member 2152

I_JET_R_SEATS enumeration member 2152

I_JET_R_SEATS_ANGLES enumeration member 2152

I_JET_R_STIFF_LONG enumeration member 2152

I_JET_R_STIFF_SHORT enumeration member 2152

I_JFCBST_COMPLEX enumeration member 1988

I_JFCBST_NONE enumeration member 1988

I_JFCBST_SIMPLE enumeration member 1988

I_JFPT_ANGLE enumeration member 2009

I_JFPT_NONE enumeration member 2009

I_JFPT_WELD enumeration member 2009

I_JFST_1 enumeration member 2029

I_JFST_2 enumeration member 2029

I_JFST_3 enumeration member 2029

I_JFST_4 enumeration member 2029

I_JFST_5 enumeration member 2029

I_JGCPC_ALL_CUTTED enumeration member 2087

I_JGCPC_CONTINUE_LEFTLOWER_RIGHTUPPER enumeration member 2087

I_JGCPC_CONTINUE_LEFTUPPER_RIGHTLOWER enumeration member 2087

I_JGCT_ACUTE enumeration member 2078

I_JGCT_CUT_IN enumeration member 2078

I_JGCT_CUT_OUT enumeration member 2078

I_JGDPT_LONG_FLANGE enumeration member 2122

I_JGDPT_LONG_FLANGE_INV enumeration member 2122

I_JGDPT_SHORT_FLANGE enumeration member 2122

I_JGDPT_SHORT_FLANGE_INV enumeration member 2122

I_JGFPC_CONTINUOUS enumeration member 2102

I_JGFPC_CUTTED enumeration member 2102

I_JGFPRT_IRREGULAR enumeration member 2123

I_JGFPRT_REGULAR enumeration member 2123

I_JGFT_BOLTED enumeration member 2068

I_JGFT_WELDED enumeration member 2068

I_JGSPP_DIAGONAL enumeration member 2068

I_JGSPP_VERTICAL enumeration member 2068

I_JKDST_DOUBLE enumeration member 1965

I_JKDST_LEFT enumeration member 1965

(c) 2014 Autodesk Inc. All rights reserved

I_JKDST_RIGHT enumeration member 1965
I_JKFT_BOLTED enumeration member 1950
I_JKFT_WELDED enumeration member 1950
I_JKM_FLAN_PLATE_LOWER enumeration member 1969
I_JKM_FLAN_PLATE_UPPER enumeration member 1969
I_JKM_PLATE enumeration member 1969
I_JKM_PLATE_LEFT enumeration member 1969
I_JKM_TENSION_PLATE_LOWER enumeration member 1969
I_JKM_TENSION_PLATE_UPPER enumeration member 1969
I_JKM_WEB_BEAM_STIFF_V_LOWER enumeration member 1969
I_JKM_WEB_BEAM_STIFF_V_UPPER enumeration member 1969
I_JKM_WEB_COLUMN_STIFF_H_LOWER enumeration member 1969
I_JKM_WEB_COLUMN_STIFF_H_UPPER enumeration member 1969
I_JKRT_BRACKET enumeration member 1951
I_JKRT_NONE enumeration member 1951
I_JKRT_STIFF enumeration member 1951
I_JKT_BEAM2BEAM enumeration member 1950
I_JKT_BEAM2COLUMN enumeration member 1950
I_JKT_BEAM2COLUMN_CONTINUE enumeration member 1950
I_JKWST_DIAG enumeration member 1965
I_JKWST_NONE enumeration member 1965
I_JKWST_PLATE enumeration member 1965
I_JLT_MANUAL_FORCES enumeration member 2179
I_JLT_ROBOT_FORCES enumeration member 2179
I_JPCBST_LDIAG enumeration member 1971
I_JPCBST_LI enumeration member 1971
I_JPCBST_LRDIAG enumeration member 1971
I_JPCBST_LRI enumeration member 1971
I_JPCBST_NONE enumeration member 1971
I_JPCBST_RDIAG enumeration member 1971
I_JPCBST_RI enumeration member 1971
I_JSFT_ABRUPT_CHANGES_HEIGHT enumeration member 1993
I_JSFT_RECT_CROSS_SECT enumeration member 1993
I_JSFT_TAPERED enumeration member 1993
I_JST_ALONG_BEARER enumeration member 2035
I_JST_ALONG_SECOND enumeration member 2035
I_JTPT_RECT enumeration member 2047
I_JTPT_RECT_RECT enumeration member 2047
I_JTPT_RECT_TUBE enumeration member 2047
I_JTPT_TUBE enumeration member 2047
I_JTPT_TUBE_TUBE enumeration member 2047
I_JTPT_UNKNOWN enumeration member 2047
I_JTT_K enumeration member 2046
I_JTT_KT enumeration member 2046
I_JTT_N enumeration member 2046
I_JTT_NONE enumeration member 2046
I_JTT_T enumeration member 2046
I_JTT_X enumeration member 2046
I_JTT_Y enumeration member 2046
I_JWFRP_FLANGE_LONG enumeration member 2159
I_JWFRP_FLANGE_TRAN enumeration member 2159
I_JWFRP_OTHER enumeration member 2159
I_JWFRP_WEB_LONG enumeration member 2159
I_JWFRP_WEB_TRAN enumeration member 2159
I_JWT_DOUBLE enumeration member 2169
I_JWT_I enumeration member 2013
I_JWT_L enumeration member 2013
I_JWT_NONE enumeration member 2013
I_JWT_SINGLE enumeration member 2169
I_JWT_T enumeration member 2013
I_JWT_X enumeration member 2013
I_L_PRINTOUT enumeration member 1544
I_L_REGIONAL enumeration member 1544
I_L_WORK enumeration member 1544
I_L3DRV_GAMMA enumeration member 47
I_L3DRV_LOCAL enumeration member 47
I_L3DRV_MX1 enumeration member 47
I_L3DRV_MX2 enumeration member 47
I_L3DRV_MY1 enumeration member 47
I_L3DRV_MY2 enumeration member 47
I_L3DRV_MZ1 enumeration member 47
I_L3DRV_MZ2 enumeration member 47
I_L3DRV_PX1 enumeration member 47
I_L3DRV_PX2 enumeration member 47
I_L3DRV_PY1 enumeration member 47
I_L3DRV_PY2 enumeration member 47
I_L3DRV_PZ1 enumeration member 47
I_L3DRV_PZ2 enumeration member 47
I_LBT_CB71_CANTILEVER enumeration member 1887

(c) 2014 Autodesk Inc. All rights reserved

I_LBT_CB71_FIXED_SUPPORTS enumeration member 1887
I_LBT_CB71_NONE enumeration member 1887
I_LBT_CB71_PINNED_SUPPORTS enumeration member 1887
I_LE_CLOUD_SOLVE enumeration member 1547
I_LE_LOCAL_SOLVE enumeration member 1547
I_LES_ENTITLED enumeration member 1547
I_LES_NOT_ENTITLED enumeration member 1547
I_LES_NOT_SIGNED_IN enumeration member 1547
I_LES_UNKNOWN enumeration member 1547
I_LI_DESIGN_CONNECTIONS enumeration member 1475
I_LI_DESIGN_RC_MEMBERS enumeration member 1475
I_LI_DESIGN_STEEL_ALUMINUM enumeration member 1475
I_LI_DESIGN_WOOD enumeration member 1475
I_LI_FOOTING_DEFINITION enumeration member 1475
I_LI_FOUNDATION_DEFINITION enumeration member 1475
I_LI_MODEL_BARS enumeration member 1475
I_LI_MODEL_GEOMETRY enumeration member 1475
I_LI_MODEL_LOADS enumeration member 1475
I_LI_MODEL_NODES enumeration member 1475
I_LI_MODEL_PROPERTIES enumeration member 1475
I_LI_MODEL_SUPPORTS enumeration member 1475
I_LI_RC_BEAM_DEFINITION enumeration member 1475
I_LI_RC_BEAM_WALL_DEFINITION enumeration member 1475
I_LI_RC_COLUMN_DEFINITION enumeration member 1475
I_LI_RC_SLAB_PROVIDED_REINF enumeration member 1475
I_LI_RC_SLAB_REINFORCEMENT enumeration member 1475
I_LI_RC_SLAB_REQUIRED_REINF enumeration member 1475
I_LI_RESULTS_DETAILED enumeration member 1475
I_LI_RESULTS_DIAGRAMS enumeration member 1475
I_LI_RESULTS_MAPS enumeration member 1475
I_LI_RESULTS_STRESS_BARS enumeration member 1475
I_LI_RESULTS_STRESS_STRUCTURE enumeration member 1475
I_LI_TOOL_FINAL_DRAWING enumeration member 1475
I_LI_TOOL_SECTION_DEFINITION enumeration member 1475
I_LI_TOOL_TEXT_FILE enumeration member 1475
I_LL_CB71_BOTTOM enumeration member 1888
I_LL_CB71_MIDDLE enumeration member 1888
I_LL_CB71_UP enumeration member 1888
I_LLT_BEAM_LOAD_CONTINUOUS enumeration member 1164
I_LLT_BEAM_LOAD_SELF_WEIGHT enumeration member 1164
I_LLT_BEAM_LOAD_TRAPEZOIDAL1 enumeration member 1164
I_LLT_BEAM_LOAD_TRAPEZOIDAL2 enumeration member 1164
I_LLT_BEAM_LOAD_TRAPEZOIDAL3 enumeration member 1164
I_LLT_BEAM_LOAD_UNIFORM enumeration member 1164
I_LOERV_GAMMA enumeration member 51
I_LOERV_LOCAL_SYSTEM enumeration member 51
I_LOERV_MX enumeration member 51
I_LOERV_MY enumeration member 51
I_LOERV_MZ enumeration member 51
I_LOERV_PX enumeration member 51
I_LOERV_PY enumeration member 51
I_LOERV_PZ enumeration member 51
I_LRDT_MINUS enumeration member 708
I_LRDT_NONE enumeration member 708
I_LRDT_PLUS enumeration member 708
I_LRDT_RELEASED enumeration member 708
I_LRT_BAR_DILATATION enumeration member 55
I_LRT_BAR_FORCE_CONCENTRATED enumeration member 55
I_LRT_BAR_FORCE_CONCENTRATED_MASS enumeration member 55
I_LRT_BAR_MOMENT_DISTRIBUTED enumeration member 55
I_LRT_BAR_THERMAL enumeration member 55
I_LRT_BAR_TRAPEZOIDALE enumeration member 55
I_LRT_BAR_TRAPEZOIDALE_MASS enumeration member 55
I_LRT_BAR_UNIFORM enumeration member 55
I_LRT_BAR_UNIFORM_MASS enumeration member 55
I_LRT_DEAD enumeration member 55
I_LRT_IN_3_POINTS enumeration member 55
I_LRT_IN_CONTOUR enumeration member 55
I_LRT_LINEAR enumeration member 55
I_LRT_LINEAR_3D enumeration member 55
I_LRT_LINEAR_ON_EDGES enumeration member 55
I_LRT_MASS_ACTIVATION enumeration member 55
I_LRT_MOBILE_DISTRIBUTED enumeration member 55
I_LRT_MOBILE_POINT_FORCE enumeration member 55

(c) 2014 Autodesk Inc. All rights reserved

I_LRT_NODE_ACCELERATION enumeration member 55
I_LRT_NODE_DISPLACEMENT enumeration member 55
I_LRT_NODE_FORCE enumeration member 55
I_LRT_NODE_FORCE_IN_POINT enumeration member 55
I_LRT_NODE_FORCE_MASS enumeration member 55
I_LRT_NODE_VELOCITY enumeration member 55
I_LRT_POINT_AUXILIARY enumeration member 55
I_LRT_PRESSURE enumeration member 55
I_LRT_SPECTRUM_VALUE enumeration member 55
I_LRT_SURFACE_ON_OBJECT enumeration member 55
I_LRT_THERMAL enumeration member 55
I_LRT_THERMAL_IN_3_POINTS enumeration member 55
I_LRT_UNIFORM enumeration member 55
I_LRV_LOCAL enumeration member 46
I_LRV_MX1 enumeration member 46
I_LRV_MX2 enumeration member 46
I_LRV_MY1 enumeration member 46
I_LRV_MY2 enumeration member 46
I_LRV_NODE_1 enumeration member 46
I_LRV_NODE_2 enumeration member 46
I_LRV_PZ1 enumeration member 46
I_LRV_PZ2 enumeration member 46
I_LS_ALS enumeration member 435
I_LS_SLS enumeration member 435
I_LS_SPC enumeration member 435
I_LS_ULS enumeration member 435
I_LT_BAR_CABLE enumeration member 17
I_LT_BAR_ELASTIC_GROUND enumeration member 17
I_LT_BAR_GEO_IMPERFECTIONS enumeration member 17
I_LT_BAR_NONLINEAR_HINGE enumeration member 17
I_LT_BAR_OFFSET enumeration member 17
I_LT_BAR_RELEASE enumeration member 17
I_LT_BAR_SECTION enumeration member 17
I_LT_BAREND_BRACKET enumeration member 17
I_LT_CLADDING enumeration member 17
I_LT_LINEAR_RELEASE enumeration member 17
I_LT_MATERIAL enumeration member 17
I_LT_MEMBER_REINFORCEMENT_PARAMS enumeration member 17
I_LT_MEMBER_TYPE enumeration member 17
I_LT_NODE_COMPATIBILITY enumeration member 17
I_LT_NODE_RELEASE enumeration member 17
I_LT_NODE_RIGID_LINK enumeration member 17
I_LT_PANEL_CALC_MODEL enumeration member 17
I_LT_PANEL_REINFORCEMENT enumeration member 17
I_LT_PANEL_THICKNESS enumeration member 17
I_LT_SOLID_PROPERTIES enumeration member 17
I_LT_SUPPORT enumeration member 17
I_LT_UNKNOWN enumeration member 17
I_LT_VEHICLE enumeration member 17
I_MAA_BASE_REDUCTION enumeration member 193
I_MAA_BLOCK_SUBSPACE_ITERATION enumeration member 193
I_MAA_LANCZOS enumeration member 193
I_MAA_PCG enumeration member 193
I_MAA_PCG_RITZ enumeration member 193
I_MAA_SUBSPACE_ITERATION enumeration member 193
I_MALT_FREQUENCY enumeration member 194
I_MALT_PERIOD enumeration member 194
I_MALT_PULSATION enumeration member 194
I_MAM_MODAL enumeration member 192
I_MAM_SEISMIC enumeration member 192
I_MAM_SEISMIC_PSEUDO enumeration member 192
I_MAMMT_CONSISTENT enumeration member 193
I_MAMMT_LUMPED enumeration member 193
I_MAMMT_LUMPED_WITH_ROTATIONS enumeration member 193
I_MARV_ACTIVATION_DIR enumeration member 46
I_MARV_CASE_NUM enumeration member 46
I_MARV_FACTOR enumeration member 46
I_MARV_INPUT_DIR_X enumeration member 46
I_MARV_INPUT_DIR_Y enumeration member 46
I_MARV_INPUT_DIR_Z enumeration member 46
I_MARV_SIGN enumeration member 46
I_MAT_SURFACE_AND_VOLUMETRIC enumeration member 831
I_MAT_SURFACE_ONLY enumeration member 831
I_MAT_VOLUMETRIC_ONLY enumeration member 831
I_MCAPT_AUTOMATIC enumeration member 343
I_MCAPT_SELECTION enumeration member 343
I_MCF_MAIN enumeration member 335
I_MCF_MAX enumeration member 335
I_MCF_MIN enumeration member 335
I_MCT_10P enumeration member 1040
I_MCT_2SM enumeration member 1040
I_MCT_ALL enumeration member 1040

(c) 2014 Autodesk Inc. All rights reserved

I_MCT_CQC enumeration member 1040
I_MCT_NONE enumeration member 1040
I_MCT_SRSS enumeration member 1040
I_MDRV_DX enumeration member 53
I_MDRV_DY enumeration member 53
I_MDRV_PX enumeration member 53
I_MDRV_PY enumeration member 53
I_MDRV_PZ enumeration member 53
I_MDRV_X enumeration member 53
I_MDRV_Y enumeration member 53
I_MDRV_Z enumeration member 53
I_MDT_DELAUNAY enumeration member 812
I_MDT_DELAUNAY_AND_KANG enumeration member 812
I_MDT_KANG enumeration member 812
I_MET_PERFECTLY_PLASTIC enumeration member 673
I_MET_PLASTIC_WITH_HARDENING enumeration member 673
I_MEUM_DAMAGE enumeration member 676
I_MEUM_ELASTIC enumeration member 676
I_MEUM_MIXED enumeration member 676
I_MEUM_PLASTIC enumeration member 676
I_MFR_ANY enumeration member 809
I_MFR_FORCED enumeration member 809
I_MFR_NONE enumeration member 809
I_MFR_PROPOSED enumeration member 809
I_MFR_RECOMMENDED enumeration member 809
I_MGT_AUTOMATIC enumeration member 813
I_MGT_ELEMENT_SIZE enumeration member 813
I_MGT_USER enumeration member 813
I_MIR_NEVER enumeration member 810
I_MIR_OFTEN enumeration member 810
I_MIR_RARELY enumeration member 810
I_MM_DRUCKER_PRAGER enumeration member 673
I_MM_ELASTIC enumeration member 673
I_MM_HUBER_MISES enumeration member 673
I_MM_MOHR_COULOMB enumeration member 673
I_MM_RANKIN enumeration member 673
I_MMT_COONS enumeration member 812
I_MMT_DELAUNAY enumeration member 812
I_MPDT_SQUARE_IN_RECT enumeration member 810
I_MPDT_TRIANG_AND_SQUARE_IN_TRIANG enumeration member 810
I_MPDT_TRIANG_AND_TRAPEZ_IN_TRIANG enumeration member 810
I_MPDT_TRIANG_IN_RECT enumeration member 810
I_MPDT_TRIANG_IN_TRIANG enumeration member 810
I_MPFRV_FX enumeration member 52
I_MPFRV_FY enumeration member 52
I_MPFRV_FZ enumeration member 52
I_MPFRV_X enumeration member 52
I_MPFRV_Y enumeration member 52
I_MPFRV_Z enumeration member 52
I_MRT_DOUBLE enumeration member 831
I_MRT_SIMPLE enumeration member 831
I_MRT_TRIPLE enumeration member 831
I_MSFET_3NODE_TRIANG enumeration member 811
I_MSFET_4NODE_QUADRIL enumeration member 811
I_MSFET_6NODE_TRIANG enumeration member 811
I_MSFET_8NODE_QUADRIL enumeration member 811
I_MST_ALL enumeration member 1040
I_MST_FIRST_N enumeration member 1040
I_MST_NONE enumeration member 1040
I_MST_SINGLE enumeration member 1040
I_MT_ALL enumeration member 672
I_MT_ALUMINIUM enumeration member 672
I_MT_COARSE enumeration member 808
I_MT_CONCRETE enumeration member 672
I_MT_FINE enumeration member 808
I_MT_FLOORS_COARSE enumeration member 808
I_MT_FLOORS_NORMAL enumeration member 808
I_MT_NORMAL enumeration member 808
I_MT_OTHER enumeration member 672
I_MT_STEEL enumeration member 672
I_MT_TIMBER enumeration member 672
I_MT_USER enumeration member 808
I_MT_WALLS_COARSE enumeration member 808
I_MT_WALLS_NORMAL enumeration member 808
I_MTT_GLUE_LAMINATED enumeration member 672
I_MTT_KERTO_Q enumeration member 672
I_MTT_KERTO_Q_OLD enumeration member 672
I_MTT_KERTO_S enumeration member 672
I_MTT_KERTO_S_OLD enumeration member 672
I_MTT_NORMAL enumeration member 672
I_MVFET_4NODE_TETRAHED enumeration member 811
I_MVFET_8NODE_HEXAHED enumeration member 811

(c) 2014 Autodesk Inc. All rights reserved

I_NAAT_ARC_LENGTH_METHOD enumeration member 317
I_NAAT_DIRECT_ITERATION_METHOD enumeration member 317
I_NAAT_INCREMENTAL_METHOD enumeration member 317
I_NAAT_PREDICTOR_CORRECTOR_METHOD enumeration member 317
I_NACRV_ALPHA enumeration member 54
I_NACRV_BETA enumeration member 54
I_NACRV_GAMMA enumeration member 54
I_NACRV_UX enumeration member 54
I_NACRV_UY enumeration member 54
I_NACRV_UZ enumeration member 54
I_NARV_ALPHA enumeration member 41
I_NARV_BETA enumeration member 41
I_NARV_CX enumeration member 41
I_NARV_CY enumeration member 41
I_NARV_CZ enumeration member 41
I_NARV_FX enumeration member 41
I_NARV_FY enumeration member 41
I_NARV_FZ enumeration member 41
I_NARV_GAMMA enumeration member 41
I_NARV_NR enumeration member 41
I_NDRV_ALPHA enumeration member 40
I_NDRV_BETA enumeration member 40
I_NDRV_GAMMA enumeration member 40
I_NDRV_RX enumeration member 40
I_NDRV_RY enumeration member 40
I_NDRV_RZ enumeration member 40
I_NDRV_UX enumeration member 40
I_NDRV_UY enumeration member 40
I_NDRV_UZ enumeration member 40
I_NFIPRV_ALPHA enumeration member 50
I_NFIPRV_BETA enumeration member 50
I_NFIPRV_FX enumeration member 50
I_NFIPRV_FY enumeration member 50
I_NFIPRV_FZ enumeration member 50
I_NFIPRV_GAMMA enumeration member 50
I_NFIPRV_MX enumeration member 50
I_NFIPRV_MY enumeration member 50
I_NFIPRV_MZ enumeration member 50
I_NFIPRV_POINT_X enumeration member 50
I_NFIPRV_POINT_Y enumeration member 50
I_NFIPRV_POINT_Z enumeration member 50
I_NFRV_ALPHA enumeration member 40
I_NFRV_BETA enumeration member 40
I_NFRV_CX enumeration member 40
I_NFRV_CY enumeration member 40
I_NFRV_CZ enumeration member 40
I_NFRV_FX enumeration member 40
I_NFRV_FY enumeration member 40
I_NFRV_FZ enumeration member 40
I_NFRV_GAMMA enumeration member 40
I_NHCT_FX enumeration member 609
I_NHCT_FY enumeration member 609
I_NHCT_FZ enumeration member 609
I_NHCT_MX enumeration member 609
I_NHCT_MY enumeration member 609
I_NHCT_MZ enumeration member 609
I_NHCT_SX enumeration member 609
I_NHMT_FORCE_DISPLACEMENT enumeration member 603
I_NHMT_MOMENT_ROTATION enumeration member 603
I_NHMT_STRESS_PAIN enumeration member 603
I_NHMUT_DAMAGE enumeration member 606
I_NHMUT_ELASTIC enumeration member 606
I_NHMUT_MIXED enumeration member 606
I_NHMUT_PLASTIC enumeration member 606
I_NLCT_B_LINEAR enumeration member 687
I_NLCT_CUSTOM enumeration member 687
I_NLCT_GAP_HOOK enumeration member 687
I_NLCT_LINEAR enumeration member 687
I_NLCT_PARABOLIC enumeration member 687
I_NLCT_PARABOLIC_EC2 enumeration member 687
I_NLCT_PERFECTLY_PLASTIC enumeration member 687
I_NLCT_PLASTIC_WITH_HARDENING enumeration member 687
I_NLMT_FORCE_DISPLACEMENT enumeration member 699
I_NLMT_MOMENT_ROTATION enumeration member 699
I_NLSAT_ANY enumeration member 699
I_NLSAT_NEGATIVE enumeration member 699
I_NLSAT_POSITIVE enumeration member 699
I_NSFD_RX enumeration member 514
I_NSFD_RY enumeration member 514
I_NSFD_RZ enumeration member 514
I_NSFD_UX enumeration member 514

(c) 2014 Autodesk Inc. All rights reserved

I_NSFD_UY enumeration member 514
I_NSFD_UZ enumeration member 514
I_NSODFT_MINUS enumeration member 514
I_NSODFT_NONE enumeration member 514
I_NSODFT_PLUS enumeration member 514
I_NVRV_ALPHA enumeration member 54
I_NVRV_BETA enumeration member 54
I_NVRV_GAMMA enumeration member 54
I_NVRV_UX enumeration member 54
I_NVRV_UY enumeration member 54
I_NVRV_UZ enumeration member 54
I_OFF_HTML enumeration member 1502
I_OFF_PNG enumeration member 1502
I_OFF_RTF enumeration member 1502
I_OFF_RTF_JPEG enumeration member 1502
I_OFF_TEXT enumeration member 1502
I_OLXDDT_CARTESIAN enumeration member 794
I_OLXDDT_POLAR enumeration member 794
I_OLXDDT_UNDEFINED enumeration member 794
I_OMT_EXTRUSION enumeration member 744
I_OMT_LATHE enumeration member 744
I_OMT_NONE enumeration member 744
I_OMT_PYRAMID enumeration member 744
I_OOT_MESH enumeration member 744
I_OOT_NONE enumeration member 744
I_OOT_ROTATE enumeration member 744
I_OOT_SCALE enumeration member 744
I_OOT_TRANSLATE enumeration member 744
I_OPT_MAIN enumeration member 779
I_OPT_REFERENCE enumeration member 779
I_OPT_SIDE enumeration member 779
I_OST_BEAM enumeration member 850
I_OST_COLUMN enumeration member 850
I_OST_SLAB enumeration member 850
I_OST_UNDEFINED enumeration member 850
I_OST_WALL enumeration member 850
I_OT_BAR enumeration member 36
I_OT_CASE enumeration member 36
I_OT_FAMILY enumeration member 36
I_OT_FINITE_ELEMENT enumeration member 36
I_OT_GEOMETRY enumeration member 36
I_OT_NODE enumeration member 36
I_OT_OBJECT enumeration member 36
I_OT_PANEL enumeration member 36
I_OT_UNDEFINED enumeration member 36
I_OT_VOLUME enumeration member 36
I_PARV_ALPHA enumeration member 44
I_PARV_BETA enumeration member 44
I_PARV_CX enumeration member 44
I_PARV_CY enumeration member 44
I_PARV_CZ enumeration member 44
I_PARV_FX enumeration member 44
I_PARV_FY enumeration member 44
I_PARV_FZ enumeration member 44
I_PARV_GAMMA enumeration member 44
I_PARV_X enumeration member 44
I_PARV_Y enumeration member 44
I_PARV_Z enumeration member 44
I_PCDT_FULL_PLANE enumeration member 984
I_PCDT_LIMITED_PLANE enumeration member 984
I_PCMD_FLEXIBLE enumeration member 720
I_PCMD_NONE enumeration member 720
I_PCMD_RIGID enumeration member 720
I_PCMFET_NONE enumeration member 719
I_PCMFET_SHELL enumeration member 719
I_PCMLT_ANALYTICAL_FINITE_ELEMENTS enumeration member 719
I_PCMLT_SIMPLIFIED_ONE_WAY enumeration member 719
I_PCMLT_SIMPLIFIED_TWO_WAY enumeration member 719
I_PCT_BEAM enumeration member 1269
I_PCT_BWALL enumeration member 1269
I_PCT_COLUMN enumeration member 1269
I_PCT_CONTINUOUS_FOOT enumeration member 1269
I_PCT_DRAWING enumeration member 1269
I_PCT_FOOT enumeration member 1269
I_PCT_JOINT enumeration member 1269
I_PCT_RETAINING_WALL enumeration member 1269
I_PCT_SLAB enumeration member 1269
I_PL_MEMBER_REINFORCEMENT_PARAMS_STANDARD enumeration member 29
I_PL_MEMBER_TYPE_BEAM enumeration member 29
I_PL_MEMBER_TYPE_COLUMN enumeration member 29
I_PL_MEMBER_TYPE_RC_BEAM enumeration member 29
I_PL_MEMBER_TYPE_RC_COLUMN enumeration member 29

(c) 2014 Autodesk Inc. All rights reserved

I_PL_MEMBER_TYPE_SIMPLE_BAR enumeration member 29

I_PL_MEMBER_TYPE_TIMBER_BEAM enumeration member 29

I_PL_MEMBER_TYPE_TIMBER_COLUMN enumeration member 29

I_PL_MEMBER_TYPE_TIMBER_MEMBER enumeration member 29

I_PL_PANEL_CALC_MODEL_CURTAIN_WALL enumeration member 29

I_PL_PANEL_CALC_MODEL_DECK_SLAB enumeration member 29

I_PL_PANEL_CALC_MODEL_SHELL enumeration member 29

I_PL_PANEL_CALC_MODEL_SLAB_FLEXIBLE_DIAPHRAGM
enumeration member 29

I_PL_PANEL_CALC_MODEL_SLAB_RIGID_DIAPHRAGM enumeration member 29

I_PL_PANEL_CALC_MODEL_SLAB_XY_DIAPHRAGM enumeration member 29

I_PL_PANEL_REINFORCEMENT_RC_FLOOR enumeration member 29

I_PL_PANEL_REINFORCEMENT_RC_SHELL enumeration member 29

I_PL_PANEL_REINFORCEMENT_RC_WALL enumeration member 29

I_PLT_BEAM_LOAD_AXIAL_FORCE enumeration member 1165

I_PLT_BEAM_LOAD_FORCE enumeration member 1165

I_PLT_BEAM_LOAD_MOMENT enumeration member 1165

I_PLT_BEAM_LOAD_TORSION_MOMENT enumeration member 1165

I_PLT_BEAM_LOAD_UNIFORM_TORSION_MOMENT enumeration member 1165

I_POD_UX_MINUS enumeration member 395

I_POD_UX_PLUS enumeration member 395

I_POD_UY_MINUS enumeration member 395

I_POD_UY_PLUS enumeration member 395

I_POD_UZ_MINUS enumeration member 395

I_POD_UZ_PLUS enumeration member 395

I_POLDM_ACCELERATION enumeration member 395

I_POLDM_USER_DEFINED enumeration member 395

I_PRV_DIRECTION enumeration member 45

I_PRV_H enumeration member 45

I_PRV_P enumeration member 45

I_PRV_RO enumeration member 45

I_PS_BAR_INACTIVE enumeration member 1041

I_PS_CASE_CODE_COMBINATIONS enumeration member 1041

I_PS_CASE_COMBINATIONS enumeration member 1041

I_PS_CASE_SIMPLE_CASES enumeration member 1041

I_PS_NODE_CALC_NODES enumeration member 1041

I_PS_NODE_SUPPORTED enumeration member 1041

I_PS_NODE_USER_NODES enumeration member 1041

I_PSF_ANF enumeration member 1358

I_PSF_DWG enumeration member 1358

I_PSF_DXF enumeration member 1358

I_PSF_DXF_V14 enumeration member 1358

I_PSF_RTD enumeration member 1358

I_PSF_RTD_NORESULTS enumeration member 1358

I_PSF_S enumeration member 1358

I_PSF_SAT enumeration member 1358

I_PSF_STP_CIM enumeration member 1358

I_PSF_STP_DSTV enumeration member 1358

I_PSF_STR enumeration member 1358

I_PSF_WRL enumeration member 1358

I_PSFT_FRAME enumeration member 1515

I_PSFT_NONE enumeration member 1515

I_PSFT_SEPARATION_LINE enumeration member 1515

I_PSO_LANDSCAPE enumeration member 1514

I_PSO_PORTRAIT enumeration member 1514

I_PSTL_BEGINNING enumeration member 1517

I_PSTL_END enumeration member 1517

I_PT_AXISYMMETRIC enumeration member 1342

I_PT_BUILDING enumeration member 1342

I_PT_CONCRETE_BEAM enumeration member 1342

I_PT_CONCRETE_COLUMN enumeration member 1342

I_PT_CONCRETE_DEEP_BEAM enumeration member 1342

I_PT_FOUNDATION enumeration member 1342

I_PT_FRAME_2D enumeration member 1342

I_PT_FRAME_3D enumeration member 1342

I_PT_GRILLAGE enumeration member 1342

I_PT_PARAMETRIZED enumeration member 1342

I_PT_PLANE_DEFORMATION enumeration member 1342

I_PT_PLANE_STRESS enumeration member 1342

I_PT_PLATE enumeration member 1342

I_PT_SECTION enumeration member 1342

I_PT_SHELL enumeration member 1342

I_PT_STEEL_CONNECTION enumeration member 1342

I_PT_TRUSS_2D enumeration member 1342

(c) 2014 Autodesk Inc. All rights reserved

I_PT_TRUSS_3D enumeration member 1342

I_PT_VOLUMETRIC enumeration member 1342

I_PVT_INTEGER enumeration member 2202

I_PVT_REAL enumeration member 2202

I_PVT_TEXT enumeration member 2202

I_QO_DISCARD_CHANGES enumeration member 1545

I_QO_PROMPT_TO_SAVE_CHANGES enumeration member 1545

I_QO_SAVE_CHANGES enumeration member 1545

I_RBT_D1_BOTTOM enumeration member 1751

I_RBT_D1_TOP enumeration member 1751

I_RBT_D2_BOTTOM enumeration member 1751

I_RBT_D2_TOP enumeration member 1751

I_RCL_LOAD_CASE_ACCIDENTAL enumeration member 1214

I_RCL_LOAD_CASE_DESIGN enumeration member 1214

I_RCL_LOAD_CASE_DESIGN_ACC enumeration member 1214

I_RCL_LOAD_CASE_EXPLOITATION enumeration member 1214

I_RCL_LOAD_CASE_PERMANENT enumeration member 1214

I_RCL_LOAD_CASE_SEISMIC enumeration member 1214

I_RCL_LOAD_CASE_SNOW enumeration member 1214

I_RCL_LOAD_CASE_TEMPERATURE enumeration member 1214

I_RCL_LOAD_CASE_TEST enumeration member 1214

I_RCL_LOAD_CASE_WIND enumeration member 1214

I_RCM_ANALYTICAL enumeration member 956

I_RCM_NEN enumeration member 956

I_RCM_WOOD_ARMER enumeration member 956

I_RIT_BLANK_PAGE enumeration member 1517

I_RIT_PAGE_TEMPLATE enumeration member 1517

I_RIT_VIEW enumeration member 1517

I_RPT_BAR enumeration member 1025

I_RPT_BAR_DIV_COUNT enumeration member 1025

I_RPT_BAR_DIV_POINT enumeration member 1025

I_RPT_BAR_ELEMENT enumeration member 1025

I_RPT_BAR_ELEMENT_DIV_COUNT enumeration member 1025

I_RPT_BAR_ELEMENT_DIV_DISCONTINUITY enumeration member 1025

I_RPT_BAR_ELEMENT_DIV_DISCONTINUITY_OFFSET enumeration member 1025

I_RPT_BAR_ELEMENT_DIV_POINT enumeration member 1025

I_RPT_BAR_RELATIVE_POINT enumeration member 1025

I_RPT_BUFFER_CHUNK_SIZE enumeration member 1025

I_RPT_CALC_POINT enumeration member 1025

I_RPT_DIR_X enumeration member 1025

I_RPT_DIR_X_DEFTYPE enumeration member 1025

I_RPT_ELEMENT enumeration member 1025

I_RPT_ITERATE_LOAD_CASES enumeration member 1025

I_RPT_LAYER enumeration member 1025

I_RPT_LAYER_ARBITRARY_VALUE enumeration member 1025

I_RPT_LINEAR_SUPPORTS enumeration member 1025

I_RPT_LOAD_CASE enumeration member 1025

I_RPT_LOAD_CASE_CMPNT enumeration member 1025

I_RPT_MAX_BUFFER_SIZE enumeration member 1025

I_RPT_MODE enumeration member 1025

I_RPT_MODE_CMB enumeration member 1025

I_RPT_MULTI_THREADS enumeration member 1025

I_RPT_NODE enumeration member 1025

I_RPT_PANEL enumeration member 1025

I_RPT_PANEL_PART enumeration member 1025

I_RPT_REDUCED_CUT_POS enumeration member 1025

I_RPT_REINFORCE_CALC_METHOD enumeration member 1025

I_RPT_RESULT_POINT_COORDINATES enumeration member 1025

I_RPT_SMOOTHING enumeration member 1025

I_RPT_STOREY enumeration member 1025

I_RPT_THREAD_COUNT enumeration member 1025

I_RQRT_DONE enumeration member 1033

I_RQRT_MORE_AVAILABLE enumeration member 1033

I_RST_AVAILABLE enumeration member 1033

I_RST_NONE enumeration member 1033

I_RST_OUT_OF_DATE enumeration member 1033

I_SAAXAT_FREQUENCY enumeration member 316

I_SAAXAT_PERIOD enumeration member 316

I_SAAXAT_PULSATION enumeration member 316

I_SACT_RPA_88_1 enumeration member 264

I_SACT_RPA_88_2 enumeration member 264

I_SACT_RPA_88_3 enumeration member 264

I_SACT_RPA_88_4 enumeration member 264

I_SACT_RPA_88_5 enumeration member 264

I_SACT_RPA_88_6 enumeration member 264

I_SACT_RPA_88_7 enumeration member 264

(c) 2014 Autodesk Inc. All rights reserved

I_SACT_RPA_88_8 enumeration member 264

I_SAD_ITALY_ORDINANZA_HORIZONTAL enumeration member 278

I_SAD_ITALY_ORDINANZA_VERTICAL enumeration member 278

I_SADT_CHINESE_A enumeration member 259

I_SADT_CHINESE_B enumeration member 259

I_SADT_HORIZONTAL enumeration member 261

I_SADT_PS_69_AVERAGE enumeration member 262

I_SADT_PS_69_NORMAL enumeration member 262

I_SADT_PS_69_WEAK enumeration member 262

I_SADT_VERTICAL enumeration member 261

I_SAET_CHINESE_FAR enumeration member 260

I_SAET_CHINESE_NEAR enumeration member 260

I_SAGCT_EAK_2000_ALPHA enumeration member 268

I_SAGCT_EAK_2000_BETA enumeration member 268

I_SAGCT_EAK_2000_DELTA enumeration member 268

I_SAGCT_EAK_2000_GAMMA enumeration member 268

I_SAGT_ARBITRARY enumeration member 1525

I_SAGT_CARTESIAN enumeration member 1525

I_SAGT_CYLINDRICAL enumeration member 1525

I_SAICT_100_92_I enumeration member 262

I_SAICT_100_92_II enumeration member 262

I_SAICT_100_92_III enumeration member 262

I_SAICT_100_92_IV enumeration member 262

I_SAIFT_EAK_2000_SIGMA1 enumeration member 267

I_SAIFT_EAK_2000_SIGMA2 enumeration member 267

I_SAIFT_EAK_2000_SIGMA3 enumeration member 267

I_SAIFT_EAK_2000_SIGMA4 enumeration member 267

I_SAIT_CHINESE_6 enumeration member 259

I_SAIT_CHINESE_7 enumeration member 259

I_SAIT_CHINESE_8 enumeration member 259

I_SAIT_CHINESE_9 enumeration member 259

I_SALT_123 enumeration member 1520

I_SALT_ABC enumeration member 1520

I_SALT_DEFINE enumeration member 1520

I_SALT_VALUE enumeration member 1520

I_SALT_VARIOUS enumeration member 1520

I_SAMPC_SQUARE_ROOT_OF_SUM_SQUARES enumeration member 1579

I_SAMPC_SUM_ABSOLUTE_VALUES enumeration member 1579

I_SAOYAT_ACCELERATION enumeration member 317

I_SAOYAT_EXCITATION enumeration member 317

I_SAOYAT_VELOCITY enumeration member 317

I_SAPCT_DM_16_1_96_1 enumeration member 261

I_SAPCT_DM_16_1_96_12 enumeration member 261

I_SAPCT_DM_16_1_96_14 enumeration member 261

I_SAS_ITALY_ORDINANZA_DESIGN enumeration member 278

I_SAS_ITALY_ORDINANZA_ELASTIC enumeration member 278

I_SASC_RPS_2000_I enumeration member 270

I_SASC_RPS_2000_II enumeration member 270

I_SASCT_IBC_2000_A enumeration member 267

I_SASCT_IBC_2000_B enumeration member 267

I_SASCT_IBC_2000_C enumeration member 267

I_SASCT_IBC_2000_D enumeration member 267

I_SASCT_IBC_2000_E enumeration member 267

I_SASCT_IBC_2000_F enumeration member 267

I_SASCT_IBC_2006_A enumeration member 286

I_SASCT_IBC_2006_B enumeration member 286

I_SASCT_IBC_2006_C enumeration member 286

I_SASCT_IBC_2006_D enumeration member 286

I_SASCT_IBC_2006_E enumeration member 286

I_SASCT_IBC_2006_F enumeration member 286

I_SAST_AFPS_90_A enumeration member 256

I_SAST_AFPS_90_B enumeration member 256

I_SAST_AFPS_90_C enumeration member 256

I_SAST_AFPS_90_S0 enumeration member 257

I_SAST_AFPS_90_S1 enumeration member 257

I_SAST_AFPS_90_S2 enumeration member 257

I_SAST_AFPS_90_S3 enumeration member 257

I_SAST_CHINESE_BRIDGES enumeration member 258

I_SAST_CHINESE_BUILDINGS enumeration member 258

I_SAST_CHINESE_HARBOR_BUILDINGS enumeration member 258

I_SAST_CHINESE_I enumeration member 258

I_SAST_CHINESE_II enumeration member 258

I_SAST_CHINESE_III enumeration member 258

I_SAST_CHINESE_IV enumeration member 258

I_SAST_CHINESE_SPECIAL enumeration member 258

I_SAST_CIRSOC_103_A enumeration member 260

I_SAST_CIRSOC_103_A0 enumeration member 260

I_SAST_CIRSOC_103_B enumeration member 260

I_SAST_CIRSOC_103_I enumeration member 260

I_SAST_CIRSOC_103_II enumeration member 260

(c) 2014 Autodesk Inc. All rights reserved

I_SAST_CIRSOC_103_III enumeration member 260
I_SAST_DIMENSIONING enumeration member 257
I_SAST_ELASITC enumeration member 257
I_SAST_ITALY_ORDINANZA_A enumeration member 278
I_SAST_ITALY_ORDINANZA_B enumeration member 278
I_SAST_ITALY_ORDINANZA_C enumeration member 278
I_SAST_ITALY_ORDINANZA_D enumeration member 278
I_SAST_ITALY_ORDINANZA_E enumeration member 278
I_SAST_PS_69_FLEXIBLE enumeration member 263
I_SAST_PS_69_RIGID enumeration member 263
I_SAST_PS_92_2008_B enumeration member 292
I_SAST_PS_92_2008_C enumeration member 292
I_SAST_PS_92_2008_D enumeration member 292
I_SAST_PS_92_2008_ENVELOPE enumeration member 292
I_SAST_PS_92_2008_S0 enumeration member 292
I_SAST_PS_92_2008_S1 enumeration member 292
I_SAST_PS_92_2008_S2 enumeration member 292
I_SAST_PS_92_2008_S3 enumeration member 292
I_SAST_PS_92_B enumeration member 263
I_SAST_PS_92_C enumeration member 263
I_SAST_PS_92_D enumeration member 263
I_SAST_PS_92_ENVELOPE enumeration member 263
I_SAST_PS_92_S0 enumeration member 263
I_SAST_PS_92_S1 enumeration member 263
I_SAST_PS_92_S2 enumeration member 263
I_SAST_PS_92_S3 enumeration member 263
I_SAST_RPA_2003_S1 enumeration member 273
I_SAST_RPA_2003_S2 enumeration member 273
I_SAST_RPA_2003_S3 enumeration member 273
I_SAST_RPA_2003_S4 enumeration member 273
I_SAST_RPA_88_FLEXIBLE enumeration member 265
I_SAST_RPA_88_RIGID enumeration member 265
I_SAST_RPS_2000_S1 enumeration member 269
I_SAST_RPS_2000_S2 enumeration member 269
I_SAST_RPS_2000_S3 enumeration member 269
I_SAST_TURKISH_23098_Z1 enumeration member 265
I_SAST_TURKISH_23098_Z2 enumeration member 265
I_SAST_TURKISH_23098_Z3 enumeration member 265
I_SAST_TURKISH_23098_Z4 enumeration member 265
I_SAST_UBC_97_A enumeration member 266
I_SAST_UBC_97_B enumeration member 266
I_SAST_UBC_97_C enumeration member 266
I_SAST_UBC_97_Sa enumeration member 266
I_SAST_UBC_97_Sb enumeration member 266
I_SAST_UBC_97_Sc enumeration member 266
I_SAST_UBC_97_Sd enumeration member 266
I_SAST_UBC_97_Se enumeration member 266
I_SAST_UBC_97_Sf enumeration member 266
I_SAUT_RPA_2003_1A enumeration member 273
I_SAUT_RPA_2003_1B enumeration member 273
I_SAUT_RPA_2003_2 enumeration member 273
I_SAUT_RPA_2003_3 enumeration member 273
I_SAUT_RPA_88_1 enumeration member 264
I_SAUT_RPA_88_2 enumeration member 264
I_SAUT_RPA_88_3 enumeration member 264
I_SAVT_ERRORS_AND_WARNINGS enumeration member 1572
I_SAVT_ERRORS_ONLY enumeration member 1572
I_SAVT_NONE enumeration member 1572
I_SAZT_AFPS_90_IA enumeration member 253
I_SAZT_AFPS_90_IB enumeration member 253
I_SAZT_AFPS_90_II enumeration member 253
I_SAZT_AFPS_90_III enumeration member 253
I_SAZT_CIRSOC_103_0 enumeration member 253
I_SAZT_CIRSOC_103_1 enumeration member 253
I_SAZT_CIRSOC_103_2 enumeration member 253
I_SAZT_CIRSOC_103_3 enumeration member 253
I_SAZT_CIRSOC_103_4 enumeration member 253
I_SAZT_EAK_2000_I enumeration member 256
I_SAZT_EAK_2000_II enumeration member 256
I_SAZT_EAK_2000_III enumeration member 256
I_SAZT_EAK_2000_IV enumeration member 256
I_SAZT_ITALY_ORDINANZA_1 enumeration member 278
I_SAZT_ITALY_ORDINANZA_2 enumeration member 278
I_SAZT_ITALY_ORDINANZA_3 enumeration member 278
I_SAZT_ITALY_ORDINANZA_4 enumeration member 278
I_SAZT_P_100_92_A enumeration member 254
I_SAZT_P_100_92_B enumeration member 254
I_SAZT_P_100_92_C enumeration member 254
I_SAZT_P_100_92_D enumeration member 254
I_SAZT_P_100_92_E enumeration member 254
I_SAZT_P_100_92_F enumeration member 254
I_SAZT_PS_92_2008_2 enumeration member 291
I_SAZT_PS_92_2008_3 enumeration member 291
I_SAZT_PS_92_2008_4 enumeration member 291

(c) 2014 Autodesk Inc. All rights reserved

I_SAZT_PS_92_2008_5 enumeration member 291
I_SAZT_PS_92_IA enumeration member 254
I_SAZT_PS_92_IB enumeration member 254
I_SAZT_PS_92_II enumeration member 254
I_SAZT_PS_92_III enumeration member 254
I_SAZT_RPA_2003_I enumeration member 272
I_SAZT_RPA_2003_IIa enumeration member 272
I_SAZT_RPA_2003_IIb enumeration member 272
I_SAZT_RPA_2003_III enumeration member 272
I_SAZT_RPA_88_I enumeration member 255
I_SAZT_RPA_88_II enumeration member 255
I_SAZT_RPA_88_III enumeration member 255
I_SAZT_RPS_2000_1 enumeration member 269
I_SAZT_RPS_2000_2 enumeration member 269
I_SAZT_RPS_2000_3 enumeration member 269
I_SAZT_TURKISH_23098_1 enumeration member 256
I_SAZT_TURKISH_23098_2 enumeration member 256
I_SAZT_TURKISH_23098_3 enumeration member 256
I_SAZT_TURKISH_23098_4 enumeration member 256
I_SAZT_UBC_97_1 enumeration member 255
I_SAZT_UBC_97_2A enumeration member 255
I_SAZT_UBC_97_2B enumeration member 255
I_SAZT_UBC_97_3 enumeration member 255
I_SAZT_UBC_97_4 enumeration member 255
I_SCFT_FIXED enumeration member 526
I_SCFT_PINNED enumeration member 526
I_SCFT_ROLLER enumeration member 526
I_SCUT_COPY_TO_CLIPBOARD enumeration member 1459
I_SCUT_CURRENT_VIEW enumeration member 1459
I_SCUT_UPDATED_UPON_PRINTING enumeration member 1459
I_SCUT_UPDATED_WHOLE_STRUCTURE enumeration member 1459
I_SEET_COLUMN enumeration member 522
I_SEET_WALL enumeration member 522
I_SMSM_MDA enumeration member 1573
I_SMSM_NDM enumeration member 1573
I_SOORV_LOCAL enumeration member 52
I_SOORV_PX enumeration member 52
I_SOORV_PY enumeration member 52
I_SOORV_PZ enumeration member 52
I_SRMDT_LAST_CALC_MODE enumeration member 294
I_SRMDT_LIMIT_FREQUENCY enumeration member 294
I_SRPD_AUTOMATIC enumeration member 1584
I_SRPD_DIR_X enumeration member 1584
I_SRPD_DIR_Y enumeration member 1584
I_SRPD_DIR_Z enumeration member 1584
I_SRS_SLAB_SEG_ENTIRE enumeration member 1227
I_SRS_SLAB_SEG_SINGLE enumeration member 1227
I_SSLDM_AUTOMATIC enumeration member 454
I_SSLDM_BY_STOREY enumeration member 454
I_SSLDM_BY_Z_COORDINATE enumeration member 454
I_SSPT_AS_1170_4_1_100 enumeration member 471
I_SSPT_AS_1170_4_1_1000 enumeration member 471
I_SSPT_AS_1170_4_1_1500 enumeration member 471
I_SSPT_AS_1170_4_1_20 enumeration member 471
I_SSPT_AS_1170_4_1_200 enumeration member 471
I_SSPT_AS_1170_4_1_2000 enumeration member 471
I_SSPT_AS_1170_4_1_25 enumeration member 471
I_SSPT_AS_1170_4_1_250 enumeration member 471
I_SSPT_AS_1170_4_1_2500 enumeration member 471
I_SSPT_AS_1170_4_1_50 enumeration member 471
I_SSPT_AS_1170_4_1_500 enumeration member 471
I_SSPT_AS_1170_4_1_800 enumeration member 471
I_SSSC_AS_1170_4_A enumeration member 470
I_SSSC_AS_1170_4_B enumeration member 470
I_SSSC_AS_1170_4_C enumeration member 470
I_SSSC_AS_1170_4_D enumeration member 470
I_SSSC_AS_1170_4_E enumeration member 470
I_SSSC_EC_8_A enumeration member 467
I_SSSC_EC_8_B enumeration member 467
I_SSSC_EC_8_C enumeration member 467
I_SSSC_EC_8_D enumeration member 467
I_SSSC_EC_8_E enumeration member 467
I_SSSC_EC_8_ENVELOPE enumeration member 467
I_SSSCT_ASCE_7_10_A enumeration member 458
I_SSSCT_ASCE_7_10_B enumeration member 458
I_SSSCT_ASCE_7_10_C enumeration member 458
I_SSSCT_ASCE_7_10_D enumeration member 458
I_SSSCT_ASCE_7_10_E enumeration member 458
I_SSSCT_ASCE_7_10_F enumeration member 458
I_SSST_AS_1170_4_BRACED_STEEL_FRAMES enumeration member 471
I_SSST_AS_1170_4_CONCRETE_FRAMES enumeration member 471
I_SSST_AS_1170_4_OTHER_STRUCTURES enumeration

(c) 2014 Autodesk Inc. All rights reserved

member 471

I_SSST_AS_1170_4_STEEL_FRAMES enumeration member 471

I_SSST_ASCE_7_10_BRACED_STEEL_FRAMES enumeration member 462

I_SSST_ASCE_7_10_CONCRETE_FRAMES enumeration member 462

I_SSST_ASCE_7_10_OTHER_STRUCTURE_SYSTEM enumeration member 462

I_SSST_ASCE_7_10_STEEL_FRAMES enumeration member 462

I_SSST_EC_8_BRACED_STEEL_FRAMES enumeration member 466

I_SSST_EC_8_CONCRETE_FRAMES enumeration member 466

I_SSST_EC_8_OTHER_STRUCTURE_SYSTEM enumeration member 466

I_SSST_EC_8_STEEL_FRAMES enumeration member 466

I_SSST_EC_8_Type1 enumeration member 467

I_SSST_EC_8_Type2 enumeration member 467

I_SST_SURFACE_PLAIN enumeration member 1264

I_SST_SURFACE_RIBBED enumeration member 1264

I_SSTBM_APPROXIMATE_BY_CODE enumeration member 455

I_SSTBM_PRECISE_MODAL enumeration member 455

I_SSTBM_PRECISE_MODAL_WITH_MASS enumeration member 455

I_SSTBM_USER_DEFINED enumeration member 455

I_STR_PT_DOUBLE enumeration member 1281

I_STR_PT_DOUBLE_3 enumeration member 1281

I_STR_PT_GROUP enumeration member 1281

I_STR_PT_INTEGER enumeration member 1281

I_STR_PT_SECTION enumeration member 1281

I_STR_PT_SELECTION enumeration member 1281

I_STR_PT_TEXT enumeration member 1281

I_STR_PT_TEXT_LIST enumeration member 1281

I_SWCECCT_TYPE_I enumeration member 148

I_SWCECCT_TYPE_II enumeration member 148

I_SWCECCT_TYPE_III enumeration member 148

I_SWCECGT_TYPE_I enumeration member 147

I_SWCECGT_TYPE_II enumeration member 147

I_SWCECGT_TYPE_III enumeration member 147

I_SWCECGT_TYPE_IV enumeration member 147

I_SWCECST_TYPE_I enumeration member 147

I_SWCECST_TYPE_II enumeration member 147

I_SWCECST_TYPE_III enumeration member 147

I_SWCECST_TYPE_IV enumeration member 147

I_SWCECST_TYPE_V enumeration member 147

I_SWCECST_TYPE_VI enumeration member 147

I_SWCFRST_ACCIDENTAL enumeration member 130

I_SWCFRST_FOLDED_OR_CORRUGATED enumeration member 130

I_SWCFRST_NORMAL enumeration member 130

I_SWCFRST_NORMAL_AND_ACCIDENTAL enumeration member 130

I_SWCFRST_RIBBED enumeration member 130

I_SWCFRST_SMOOTH_OR_CORRUGATED enumeration member 130

I_SWCFRWS_EXPOSED enumeration member 129

I_SWCFRWS_NORMAL enumeration member 129

I_SWCFRWS_OBSCURED enumeration member 129

I_SWCFRWT_EXTREME enumeration member 129

I_SWCFRWT_NORMAL enumeration member 129

I_SWCPLSZ_I enumeration member 110

I_SWCPLSZ_II enumeration member 110

I_SWCPLSZ_III enumeration member 110

I_SWCPLSZ_IV enumeration member 110

I_SWCPLWPDT_CONSTANT enumeration member 130

I_SWCPLWPDT_VARIABLE enumeration member 130

I_SWCPLWS_A enumeration member 109

I_SWCPLWS_B enumeration member 109

I_SWCPLWS_C enumeration member 109

I_SWCPLWZ_I enumeration member 109

I_SWCPLWZ_II enumeration member 109

I_SWCPLWZ_IIA enumeration member 109

I_SWCPLWZ_IIB enumeration member 109

I_SWCPLWZ_III enumeration member 109

I_TCF_BAR_DEFL_DEFLECTIONS enumeration member 1362

I_TCF_BAR_DEFL_DISPLACEMENTS enumeration member 1362

I_TCF_BAR_DEFL_MAX_DEFLECTIONS enumeration member 1362

I_TCF_CASE_INFO enumeration member 1362

I_TCF_COEXISTENT_VALUES enumeration member 1362

I_TCF_COMB_CODE_COMPONENTS enumeration member 1362

I_TCF_COMB_DEFINITIONS enumeration member 1362

I_TCF_COMB_MOVING_LOADS enumeration member 1362

I_TCF_COMB_TIME_HISTORY_COMPONENTS enumeration member 1362

(c) 2014 Autodesk Inc. All rights reserved

I_TCF_DETAILS enumeration member 1362

I_TCF_EXTREME_COMBINATIONS enumeration member 1362

I_TCF_FE_DIR_X_AXIS enumeration member 1362

I_TCF_FE_DIR_Y_AXIS enumeration member 1362

I_TCF_FE_DIR_Z_AXIS enumeration member 1362

I_TCF_FE_IN_ELEMENT_CENTERS enumeration member 1362

I_TCF_FE_IN_ELENODE enumeration member 1362

I_TCF_FE_IN_NODES enumeration member 1362

I_TCF_FE_LAYER_LOWER enumeration member 1362

I_TCF_FE_LAYER_MAX enumeration member 1362

I_TCF_FE_LAYER_MAX_ABSOLUTE enumeration member 1362

I_TCF_FE_LAYER_MIDDLE enumeration member 1362

I_TCF_FE_LAYER_MIN enumeration member 1362

I_TCF_FE_LAYER_UPPER enumeration member 1362

I_TCF_GLOBAL_ANALYSIS_RESULTS_NPOINTS enumeration member 1362

I_TCF_GLOBAL_ANALYSIS_RESULTS_RELATIVE enumeration member 1362

I_TCF_GLOBAL_COORDINATES enumeration member 1362

I_TCF_LOCAL_COORDINATES enumeration member 1362

I_TCF_LOCALS enumeration member 1362

I_TCF_ORDER_BY_CASE enumeration member 1362

I_TCF_PROP_STD_TAPERED_SECTION enumeration member 1362

I_TCF_REAC_APPLIED_FORCES_SUM enumeration member 1362

I_TCF_REAC_DDC enumeration member 1362

I_TCF_REAC_EQUILIBRIUM_PRECISION enumeration member 1362

I_TCF_REAC_RESIDUUM enumeration member 1362

I_TCF_REAC_VALUES enumeration member 1362

I_TCF_REINF_IN_ELEMENT_CENTERS enumeration member 1362

I_TCF_REINF_IN_NODES enumeration member 1362

I_THAM_HHT enumeration member 369

I_THAM_MODAL_DECOMPOSITION enumeration member 369

I_THAM_NEWMARK enumeration member 369

I_THAM_NEWMARK_ACCELERATION enumeration member 369

I_THT_CONSTANT enumeration member 730

I_THT_VARIABLE_ALONG_LINE enumeration member 730

I_THT_VARIABLE_ON_PLANE enumeration member 730

I_TMV_D_XXXX enumeration member 740

I_TMV_D_XXYY enumeration member 740

I_TMV_D_XYXY enumeration member 740

I_TMV_D_YYYY enumeration member 740

I_TMV_H_XX enumeration member 740

I_TMV_H_YY enumeration member 740

I_TMV_K_XXXX enumeration member 740

I_TMV_K_XXYY enumeration member 740

I_TMV_K_XYXY enumeration member 740

I_TMV_K_YYYY enumeration member 740

I_TO_COPY enumeration member 1048

I_TO_COPY_WITH_DRAG enumeration member 1048

I_TO_MOVE enumeration member 1048

I_TODT_DIR_X enumeration member 730

I_TODT_DIR_Y enumeration member 730

I_TODT_DIR_Z enumeration member 730

I_TODT_VECTOR enumeration member 730

I_TOT_BIDIR_BOX_FLOOR enumeration member 730

I_TOT_DOUBLE_SIDED_UNIDIR_RIBS enumeration member 730

I_TOT_GRILLAGE enumeration member 730

I_TOT_HOLLOW_CORE_SLAB enumeration member 730

I_TOT_MATERIAL enumeration member 730

I_TOT_ONE_SIDED_BIDIR_RIBS enumeration member 730

I_TOT_ONE_SIDED_UNIDIR_RIBS enumeration member 730

I_TOT_SLAB_ON_TRAPEZOID_PLATE enumeration member 730

I_TOT_UNIDIR_BOX_FLOOR enumeration member 730

I_TOT_USER enumeration member 730

I_TRV_GRADIENT_1 enumeration member 50

I_TRV_GRADIENT_2 enumeration member 50

I_TRV_GRADIENT_3 enumeration member 50

I_TRV_T_1 enumeration member 50

I_TRV_T_2 enumeration member 50

I_TRV_T_3 enumeration member 50

I_TSCUT_COPY_TO_CLIPBOARD enumeration member 1379

I_TSCUT_CURRENT_VIEW enumeration member 1379

I_TSCUT_UPDATED_UPON_PRINTING enumeration member 1379

I_TT_HOMOGENEOUS enumeration member 724

I_TT_ORTHOTROPIC enumeration member 724

I_TUT_MINUS enumeration member 724

(c) 2014 Autodesk Inc. All rights reserved

I_TUT_NONE enumeration member 724
I_TUT_PLUS enumeration member 724
I_UD_UY enumeration member 545
I_UD_UZ enumeration member 545
I_UMT_FORCE enumeration member 1285
I_UMT_LENGTH enumeration member 1285
I_UMT_MASS enumeration member 1285
I_URT_UNKNOWN enumeration member 1019
I_URT_VALUE enumeration member 1019
I_URT_VALUE_3D enumeration member 1019
I_URV_LOCAL_SYSTEM enumeration member 49
I_URV_PROJECTED enumeration member 49
I_URV_PX enumeration member 49
I_URV_PY enumeration member 49
I_URV_PZ enumeration member 49
I_URV_RELATIVE enumeration member 49
I_US_MINUS enumeration member 545
I_US_NONE enumeration member 545
I_US_PLUS enumeration member 545
I_UT_ANGLE_ROTATION_DATA enumeration member 1282
I_UT_ANGLE_ROTATION_RESULT enumeration member 1282
I_UT_DIAMETER_RC_BASE enumeration member 1282
I_UT_DIMENSIONLESS_QUALITY enumeration member 1282
I_UT_DISPLACEMENT enumeration member 1282
I_UT_FORCE enumeration member 1282
I_UT_MASS enumeration member 1282
I_UT_MOMENT enumeration member 1282
I_UT_REINFORCEMENT_AREAS enumeration member 1282
I_UT_RULER enumeration member 1282
I_UT_SECTION_DIMENSION enumeration member 1282
I_UT_SECTION_PROPERTIES enumeration member 1282
I_UT_STEEL_CONECTIONS enumeration member 1282
I_UT_STRESS enumeration member 1282
I_UT_STRUCTURE_DIMENSION enumeration member 1282
I_UT_TEMPERATURE enumeration member 1282
I_UT_WEIGHT enumeration member 1282
I_VBMRT_DESIGN_MEMBER_LENGTH enumeration member 1430
I_VBMRT_DESIGN_RATIO enumeration member 1430
I_VBMRT_DESIGN_SLEND_LAY enumeration member 1430
I_VBMRT_DESIGN_SLEND_LAZ enumeration member 1430
I_VBMRT_NOTHING enumeration member 1430
I_VBMRT_NTM_FX enumeration member 1430
I_VBMRT_NTM_FY enumeration member 1430
I_VBMRT_NTM_FZ enumeration member 1430
I_VBMRT_NTM_MX enumeration member 1430
I_VBMRT_NTM_MY enumeration member 1430
I_VBMRT_NTM_MZ enumeration member 1430
I_VBMRT_STRESS_FX_AX enumeration member 1430
I_VBMRT_STRESS_S_MAX enumeration member 1430
I_VBMRT_STRESS_S_MAX_MY enumeration member 1430
I_VBMRT_STRESS_S_MAX_MZ enumeration member 1430
I_VBMRT_STRESS_S_MIN enumeration member 1430
I_VBMRT_STRESS_S_MIN_MY enumeration member 1430
I_VBMRT_STRESS_S_MIN_MZ enumeration member 1430
I_VBMRT_STRESS_SHEAR_TY enumeration member 1430
I_VBMRT_STRESS_SHEAR_TZ enumeration member 1430
I_VBMRT_STRESS_TORSION_T enumeration member 1430
I_VDA_ADVANCED_CABLES enumeration member 1418
I_VDA_ADVANCED_COMPATIBLE_NODES enumeration member 1418
I_VDA_ADVANCED_ELASTIC_FOUNDATION enumeration member 1418
I_VDA_ADVANCED_GEOIMPERFECTIONS enumeration member 1418
I_VDA_ADVANCED_GEOIMPERFECTIONS_NAME enumeration member 1418
I_VDA_ADVANCED_NONLINEAR_HINGES enumeration member 1418
I_VDA_ADVANCED_OFFSETS enumeration member 1418
I_VDA_ADVANCED_RC_BEAM enumeration member 1418
I_VDA_ADVANCED_RC_COLUMN enumeration member 1418
I_VDA_ADVANCED_RC_CONTINOUS_FOOTING enumeration member 1418
I_VDA_ADVANCED_RC_DEEP_BEAM enumeration member 1418
I_VDA_ADVANCED_RC_FOUNDATION enumeration member 1418
I_VDA_ADVANCED_RC_WALL enumeration member 1418
I_VDA_ADVANCED_RELEASE_CODES enumeration member 1418
I_VDA_ADVANCED_RELEASE_SYMBOLS enumeration member 1418
I_VDA_ADVANCED_RIGID_LINK_SURFACE enumeration member 1418
I_VDA_ADVANCED_RIGID_LINKS enumeration member 1418

(c) 2014 Autodesk Inc. All rights reserved

I_VDA_ADVANCED_STEEL_CONNECTION_NAMES enumeration member 1418

I_VDA_ADVANCED_STEEL_CONNECTION_NUMBERS enumeration member 1418

I_VDA_ADVANCED_STEEL_CONNECTION_TYPES enumeration member 1418

I_VDA_ADVANCED_STEEL_CONNECTIONS enumeration member 1418

I_VDA_ADVANCED_STORIES_COLORS enumeration member 1418

I_VDA_ADVANCED_STORIES_RESULTS enumeration member 1418

I_VDA_ADVANCED_TENSION_COMPRESSION enumeration member 1418

I_VDA_FE_AUXILIARY_OBJECTS enumeration member 1418

I_VDA_FE_CHARACTERISTIC_POINTS enumeration member 1418

I_VDA_FE_CLADDING_COLORS enumeration member 1418

I_VDA_FE_CLADDING_INTERIOR enumeration member 1418

I_VDA_FE_COLOR_LEGEND enumeration member 1418

I_VDA_FE_CONTOUR_COMPONENTS enumeration member 1418

I_VDA_FE_EDGE_NUMBERS enumeration member 1418

I_VDA_FE_EMITTERS enumeration member 1418

I_VDA_FE_FE_INTERIOR enumeration member 1418

I_VDA_FE_FINITE_ELEMENT_NUMBERS enumeration member 1418

I_VDA_FE_FINITE_ELEMENTS enumeration member 1418

I_VDA_FE_LOAD_DIRECTION enumeration member 1418

I_VDA_FE_MESH_PREVIEW enumeration member 1418

I_VDA_FE_PANEL_COMPLEX_DESC enumeration member 1418

I_VDA_FE_PANEL_CONTOURS enumeration member 1418

I_VDA_FE_PANEL_INTERIOR enumeration member 1418

I_VDA_FE_PANEL_NAMES enumeration member 1418

I_VDA_FE_PANEL_NUMBERS enumeration member 1418

I_VDA_FE_PANEL_REINFORCEMENT enumeration member 1418

I_VDA_FE_PANEL_THICKNESSES enumeration member 1418

I_VDA_FE_THICKNESS enumeration member 1418

I_VDA_LOADS_AUTOMATICALLY enumeration member 1418

I_VDA_LOADS_DISTRIBUTION_REGIONS enumeration member 1418

I_VDA_LOADS_MOVING_ELEMENTS enumeration member 1418

I_VDA_LOADS_MOVING_ROUTE enumeration member 1418

I_VDA_LOADS_MOVING_VEHICLE enumeration member 1418

I_VDA_LOADS_PRESSURE_MAP enumeration member 1418

I_VDA_LOADS_SYMBOLS_CONCENTRATED enumeration member 1418

I_VDA_LOADS_SYMBOLS_LINEAR enumeration member 1418

I_VDA_LOADS_SYMBOLS_PLANAR enumeration member 1418

I_VDA_LOADS_SYMBOLS_UNIFORM_SIZE enumeration member 1418

I_VDA_LOADS_VALUES enumeration member 1418

I_VDA_OTHER_CALC_ELEM_NUMBERS enumeration member 1418

I_VDA_OTHER_CALC_POINT_NUMBERS enumeration member 1418

I_VDA_OTHER_DIMENSION_LINES enumeration member 1418

I_VDA_OTHER_GRID enumeration member 1418

I_VDA_OTHER_HIDE_INACTIVE enumeration member 1418

I_VDA_OTHER_HIDE_NODES enumeration member 1418

I_VDA_OTHER_OBJECTS_OUT_OF_PLANE enumeration member 1418

I_VDA_OTHER_RULER enumeration member 1418

I_VDA_OTHER_STRUCTURAL_AXIS enumeration member 1418

I_VDA_OTHER_STRUCTURAL_AXIS_DESCRIPTION enumeration member 1418

I_VDA_SECTIONS_CODE_GROUPS enumeration member 1418

I_VDA_SECTIONS_COLORS enumeration member 1418

I_VDA_SECTIONS_MATERIAL enumeration member 1418

I_VDA_SECTIONS_MEMBER_TYPE_LEGEND enumeration member 1418

I_VDA_SECTIONS_MEMBER_TYPE_NAME enumeration member 1418

I_VDA_SECTIONS_NAME enumeration member 1418

I_VDA_SECTIONS_SHAPE enumeration member 1418

I_VDA_SECTIONS_SURFACE enumeration member 1418

I_VDA_SECTIONS_SYMBOLS enumeration member 1418

I_VDA_STRUCTURE_ATTRIBUTE_LABELS enumeration member 1418

I_VDA_STRUCTURE_BAR_NAMES enumeration member 1418

I_VDA_STRUCTURE_BAR_NUMBERS enumeration member 1418

(c) 2014 Autodesk Inc. All rights reserved

I_VDA_STRUCTURE_EXPLODE_BARS enumeration member 1418

I_VDA_STRUCTURE_EXPLODE_FE enumeration member 1418

I_VDA_STRUCTURE_GROUP_COLORS enumeration member 1418

I_VDA_STRUCTURE_LOCAL_SYSTEM_BARS enumeration member 1418

I_VDA_STRUCTURE_LOCAL_SYSTEM_FE enumeration member 1418

I_VDA_STRUCTURE_LOCAL_SYSTEM_PANELS enumeration member 1418

I_VDA_STRUCTURE_NODE_NUMBERS enumeration member 1418

I_VDA_STRUCTURE_ONLY_FOR_SELECTED_OBJECTS enumeration member 1418

I_VDA_STRUCTURE_STRUCTURE enumeration member 1418

I_VDA_STRUCTURE_SUPPORT_CODES enumeration member 1418

I_VDA_STRUCTURE_SUPPORT_DETAILED_SYMBOLS enumeration member 1418

I_VDA_STRUCTURE_SUPPORT_DIRECTION enumeration member 1418

I_VDA_STRUCTURE_SUPPORT_SHAPES enumeration member 1418

I_VDA_STRUCTURE_SUPPORT_SYMBOLS enumeration member 1418

I_VDA_VIEW_COLOR enumeration member 1418

I_VDA_VIEW_HIDDEN_LINES enumeration member 1418

I_VDA_VIEW_NONE enumeration member 1418

I_VDA_VIEW_QUICK_SHADING enumeration member 1418

I_VDA_VIEW_SHADING enumeration member 1418

I_VDA_VIEW_SHADING_EDGES enumeration member 1418

I_VDA_VIEWOGL_ANTYALIASING enumeration member 1418

I_VDA_VIEWOGL_BLACK_EDGES enumeration member 1418

I_VDA_VIEWOGL_COLOR_FACES enumeration member 1418

I_VDA_VIEWOGL_DETAILS enumeration member 1418

I_VDA_VIEWOGL_DRAW_OUT_OF_SCREEN enumeration member 1418

I_VDA_VIEWOGL_HIDE_INVISIBLE_LINES enumeration member 1418

I_VDA_VIEWOGL_HIDE_XY_CUTTING_PLANES enumeration member 1418

I_VDA_VIEWOGL_HIDE_XZ_CUTTING_PLANES enumeration member 1418

I_VDA_VIEWOGL_HIDE_YZ_CUTTING_PLANES enumeration member 1418

I_VDA_VIEWOGL_LIGHT enumeration member 1418

I_VDA_VIEWOGL_LIGHT_ON_MAPS enumeration member 1418

I_VDA_VIEWOGL_REDRAW enumeration member 1418

I_VDA_VIEWOGL_TRANSLUCENT enumeration member 1418

I_VDART_NTM_FX enumeration member 1432

I_VDART_NTM_FY enumeration member 1432

I_VDART_NTM_FZ enumeration member 1432

I_VDART_NTM_MX enumeration member 1432

I_VDART_NTM_MY enumeration member 1432

I_VDART_NTM_MZ enumeration member 1432

I_VDART_NTM_UX enumeration member 1432

I_VDART_NTM_UY enumeration member 1432

I_VDART_NTM_UZ enumeration member 1432

I_VDART_REINFORCE_BOTTOM enumeration member 1432

I_VDART_REINFORCE_BOTTOM_REAL enumeration member 1432

I_VDART_REINFORCE_NUMBER_OF_LOWER_BARS enumeration member 1432

I_VDART_REINFORCE_NUMBER_OF_UPPER_BARS enumeration member 1432

I_VDART_REINFORCE_RATIO enumeration member 1432

I_VDART_REINFORCE_RATIO_REAL enumeration member 1432

I_VDART_REINFORCE_STIRRUP_SPACING enumeration member 1432

I_VDART_REINFORCE_STIRRUP_SPACING_REAL enumeration member 1432

I_VDART_REINFORCE_TOP enumeration member 1432

I_VDART_REINFORCE_TOP_REAL enumeration member 1432

I_VDART_STRESS_AXIAL_FX_AX enumeration member 1432

I_VDART_STRESS_BENDING_S_MAX_MY enumeration member 1432

I_VDART_STRESS_BENDING_S_MAX_MZ enumeration member 1432

I_VDART_STRESS_BENDING_S_MIN_MY enumeration member 1432

I_VDART_STRESS_BENDING_S_MIN_MZ enumeration member 1432

I_VDART_STRESS_S_MAX enumeration member 1432

I_VDART_STRESS_S_MIN enumeration member 1432

I_VDART_STRESS_SHEAR_TY enumeration member 1432

I_VDART_STRESS_SHEAR_TZ enumeration member 1432

(c) 2014 Autodesk Inc. All rights reserved

I_VDART_STRESS_TORSION_T enumeration member 1432
I_VDATT_GLOBAL_EXTREMES enumeration member 1415
I_VDATT_LOCAL_EXTREMES enumeration member 1415
I_VDATT_VALUES enumeration member 1415
I_VDDT_LABELS enumeration member 1427
I_VDDT_NONE enumeration member 1427
I_VDDT_TEXT enumeration member 1427
I_VDFT_FENCE enumeration member 1428
I_VDFT_FILLED enumeration member 1428
I_VDPT_IN_PLANE enumeration member 1447
I_VDPT_NORMAL enumeration member 1447
I_VDRT_DEFORMATION_DEFORMATION enumeration member 1426
I_VDRT_DEFORMATION_EXACT enumeration member 1426
I_VDRT_NTM_FX enumeration member 1426
I_VDRT_NTM_FY enumeration member 1426
I_VDRT_NTM_FZ enumeration member 1426
I_VDRT_NTM_KY enumeration member 1426
I_VDRT_NTM_KZ enumeration member 1426
I_VDRT_NTM_MX enumeration member 1426
I_VDRT_NTM_MY enumeration member 1426
I_VDRT_NTM_MZ enumeration member 1426
I_VDRT_REACTION_FORCES enumeration member 1426
I_VDRT_REACTION_MOMENTS enumeration member 1426
I_VDRT_REACTION_RESIDUAL_FORCES enumeration member 1426
I_VDRT_REACTION_RESIDUAL_MOMENTS enumeration member 1426
I_VDRT_REINFORCE_BOTTOM enumeration member 1426
I_VDRT_REINFORCE_BOTTOM_REAL enumeration member 1426
I_VDRT_REINFORCE_NUMBER_OF_LOWER_BARS enumeration member 1426
I_VDRT_REINFORCE_NUMBER_OF_UPPER_BARS enumeration member 1426
I_VDRT_REINFORCE_RATIO enumeration member 1426
I_VDRT_REINFORCE_RATIO_REAL enumeration member 1426
I_VDRT_REINFORCE_STIRRUP_SPACING enumeration member 1426
I_VDRT_REINFORCE_STIRRUP_SPACING_REAL enumeration member 1426
I_VDRT_REINFORCE_TOP enumeration member 1426
I_VDRT_REINFORCE_TOP_REAL enumeration member 1426
I_VDRT_STRESS_AXIAL_FX_AX enumeration member 1426
I_VDRT_STRESS_BENDING_S_MAX_MY enumeration member 1426
I_VDRT_STRESS_BENDING_S_MAX_MZ enumeration member 1426
I_VDRT_STRESS_BENDING_S_MIN_MY enumeration member 1426
I_VDRT_STRESS_BENDING_S_MIN_MZ enumeration member 1426
I_VDRT_STRESS_S_MAX enumeration member 1426
I_VDRT_STRESS_S_MIN enumeration member 1426
I_VDRT_STRESS_SHEAR_TY enumeration member 1426
I_VDRT_STRESS_SHEAR_TZ enumeration member 1426
I_VDRT_STRESS_TORSION enumeration member 1426
I_VDSDT_DIFFERENTIATED enumeration member 1428
I_VDSDT_UNDIFFERENTIATED enumeration member 1428
I_VDVT_ALL enumeration member 1461
I_VDVT_GLOBAL_EXTREMES enumeration member 1461
I_VDVT_LOCAL_EXTREMES enumeration member 1461
I_VFMCPT_MOMENTS enumeration member 1444
I_VFMCPT_NONE enumeration member 1444
I_VFMCPT_NORMAL_FORCES enumeration member 1444
I_VFMCPT_STRESSES enumeration member 1444
I_VFMLST_AUTOMATIC enumeration member 1443
I_VFMLST_CARTESIAN_ALONG_VECTOR enumeration member 1443
I_VFMLST_CARTESIAN_ALONG_X enumeration member 1443
I_VFMLST_CARTESIAN_ALONG_Y enumeration member 1443
I_VFMLST_CARTESIAN_ALONG_Z enumeration member 1443
I_VFMLST_POLAR_IN_NODE enumeration member 1443
I_VFMLST_POLAR_IN_POINT enumeration member 1443
I_VFMLT_ABSOLUTE_MAXIMUM enumeration member 1444
I_VFMLT_ARBITRARY enumeration member 1444
I_VFMLT_LOWER enumeration member 1444
I_VFMLT_MAXIMUM enumeration member 1444
I_VFMLT_MIDDLE enumeration member 1444
I_VFMLT_MINIMUM enumeration member 1444
I_VFMLT_UPPER enumeration member 1444
I_VFMRT_COMPLEX_MEMBRANE_FORCES enumeration member 1436
I_VFMRT_COMPLEX_MOMENTS enumeration member 1436
I_VFMRT_COMPLEX_REINFORCE_BOTTOM_MXX enumeration member 1436
I_VFMRT_COMPLEX_REINFORCE_BOTTOM_MYY

(c) 2014 Autodesk Inc. All rights reserved

enumeration member 1436

I_VFMRT_COMPLEX_REINFORCE_TOP_MXX enumeration member 1436

I_VFMRT_COMPLEX_REINFORCE_TOP_MYY enumeration member 1436

I_VFMRT_COMPLEX_STRESSES enumeration member 1436

I_VFMRT_DETAILED_DISPLACEMENT_XX enumeration member 1436

I_VFMRT_DETAILED_DISPLACEMENT_YY enumeration member 1436

I_VFMRT_DETAILED_MEMBRANE_FORCE_XX enumeration member 1436

I_VFMRT_DETAILED_MEMBRANE_FORCE_XY enumeration member 1436

I_VFMRT_DETAILED_MEMBRANE_FORCE_YY enumeration member 1436

I_VFMRT_DETAILED_MOMENT_XX enumeration member 1436

I_VFMRT_DETAILED_MOMENT_XY enumeration member 1436

I_VFMRT_DETAILED_MOMENT_YY enumeration member 1436

I_VFMRT_DETAILED_ROTATION_XX enumeration member 1436

I_VFMRT_DETAILED_ROTATION_YY enumeration member 1436

I_VFMRT_DETAILED_ROTATION_Z enumeration member 1436

I_VFMRT_DETAILED_SHEAR_FORCE_XX enumeration member 1436

I_VFMRT_DETAILED_SHEAR_FORCE_YY enumeration member 1436

I_VFMRT_DETAILED_SHEAR_STRESS_XX enumeration member 1436

I_VFMRT_DETAILED_SHEAR_STRESS_YY enumeration member 1436

I_VFMRT_DETAILED_SOIL_REACTION_Z enumeration member 1436

I_VFMRT_DETAILED_STRESS_XX enumeration member 1436

I_VFMRT_DETAILED_STRESS_XY enumeration member 1436

I_VFMRT_DETAILED_STRESS_YY enumeration member 1436

I_VFMRT_DETAILED_STRESS_Z enumeration member 1436

I_VFMRT_GLOBAL_DISPLACEMENT_X enumeration member 1436

I_VFMRT_GLOBAL_DISPLACEMENT_Y enumeration member 1436

I_VFMRT_GLOBAL_DISPLACEMENT_Z enumeration member 1436

I_VFMRT_PRINCIPAL_MEMBRANE_FORCE_1 enumeration member 1436

I_VFMRT_PRINCIPAL_MEMBRANE_FORCE_1_2 enumeration member 1436

I_VFMRT_PRINCIPAL_MEMBRANE_FORCE_2 enumeration member 1436

I_VFMRT_PRINCIPAL_MEMBRANE_FORCE_ANGLE enumeration member 1436

I_VFMRT_PRINCIPAL_MOMENT_1 enumeration member 1436

I_VFMRT_PRINCIPAL_MOMENT_1_2 enumeration member 1436

I_VFMRT_PRINCIPAL_MOMENT_2 enumeration member 1436

I_VFMRT_PRINCIPAL_MOMENT_ANGLE enumeration member 1436

I_VFMRT_PRINCIPAL_SHEAR_FORCE_1_2 enumeration member 1436

I_VFMRT_PRINCIPAL_SHEAR_STRESS_1_2 enumeration member 1436

I_VFMRT_PRINCIPAL_STRESS_1 enumeration member 1436

I_VFMRT_PRINCIPAL_STRESS_1_2 enumeration member 1436

I_VFMRT_PRINCIPAL_STRESS_2 enumeration member 1436

I_VFMRT_PRINCIPAL_STRESS_ANGLE enumeration member 1436

I_VFMRT_TOTAL_DISPLACEMENTS enumeration member 1436

I_VFMST_GLOBAL_SMOOTHING enumeration member 1444

I_VFMST_NO_SMOOTHING enumeration member 1444

I_VFMST_SMOOTHING_ACCORDING_TO_SELECTION enumeration member 1444

I_VFMST_SMOOTHING_WITH_PANEL enumeration member 1444

I_VGAPT_DESIGN_MEMBER_LENGTH enumeration member 1453

I_VGAPT_DESIGN_RATIO enumeration member 1453

I_VGAPT_DESIGN_SLENDERNESS_LAY enumeration member 1453

I_VGAPT_DESIGN_SLENDERNESS_LAZ enumeration member 1453

I_VGAPT_FORCE_FX enumeration member 1453

I_VGAPT_FORCE_FY enumeration member 1453

I_VGAPT_FORCE_FZ enumeration member 1453

I_VGAPT_FORCE_KX enumeration member 1453

(c) 2014 Autodesk Inc. All rights reserved

I_VGAPT_FORCE_KY enumeration member 1453

I_VGAPT_FORCE_MX enumeration member 1453

I_VGAPT_FORCE_MY enumeration member 1453

I_VGAPT_FORCE_MZ enumeration member 1453

I_VGAPT_STRESS_AXIAL_SXFX enumeration member 1453

I_VGAPT_STRESS_BENDING_SMY enumeration member 1453

I_VGAPT_STRESS_BENDING_SMZ enumeration member 1453

I_VGAPT_STRESS_EXTREME enumeration member 1453

I_VGAPT_STRESS_NORMAL enumeration member 1453

I_VGAPT_STRESS_SHEAR enumeration member 1453

I_VGAPT_STRESS_TORSION enumeration member 1453

I_VGART_N_POINTS enumeration member 1452

I_VGART_RELATIVE enumeration member 1452

I_VHLDT_COLOR_SIDES_EDGES enumeration member 1421

I_VHLDT_HIDDEN_LINES enumeration member 1421

I_VHLDT_NONE enumeration member 1421

I_VHLDT_QUICK_SHADING_OF_FACES enumeration member 1421

I_VHLDT_SHADING enumeration member 1421

I_VHLDT_SHADING_EDGES enumeration member 1421

I_VILLST_AUTOMATIC enumeration member 1394

I_VILLST_CARTESIAN_ALONG_VECTOR enumeration member 1394

I_VILLST_CARTESIAN_ALONG_X enumeration member 1394

I_VILLST_CARTESIAN_ALONG_Y enumeration member 1394

I_VILLST_CARTESIAN_ALONG_Z enumeration member 1394

I_VILLST_POLAR_IN_NODE enumeration member 1394

I_VILLST_POLAR_IN_POINT enumeration member 1394

I_VILLT_ARBITRARY enumeration member 1395

I_VILLT_LOWER enumeration member 1395

I_VILLT_MIDDLE enumeration member 1395

I_VILLT_UPPER enumeration member 1395

I_VILRT_COMPLEX_MEMBRANE_FORCE enumeration member 1393

I_VILRT_COMPLEX_MOMENT enumeration member 1393

I_VILRT_COMPLEX_STRESS enumeration member 1393

I_VILRT_DETAILED_DISPLACEMENT_XX enumeration member 1393

I_VILRT_DETAILED_DISPLACEMENT_YY enumeration member 1393

I_VILRT_DETAILED_DISPLACEMENT_Z enumeration member 1393

I_VILRT_DETAILED_MEMBRANE_FORCE_XX enumeration member 1393

I_VILRT_DETAILED_MEMBRANE_FORCE_XY enumeration member 1393

I_VILRT_DETAILED_MEMBRANE_FORCE_YY enumeration member 1393

I_VILRT_DETAILED_MOMENTS_XX enumeration member 1393

I_VILRT_DETAILED_MOMENTS_XY enumeration member 1393

I_VILRT_DETAILED_MOMENTS_YY enumeration member 1393

I_VILRT_DETAILED_ROTATION_XX enumeration member 1393

I_VILRT_DETAILED_ROTATION_YY enumeration member 1393

I_VILRT_DETAILED_ROTATION_Z enumeration member 1393

I_VILRT_DETAILED_SHEAR_FORCE_XX enumeration member 1393

I_VILRT_DETAILED_SHEAR_FORCE_YY enumeration member 1393

I_VILRT_DETAILED_SHEAR_STRESS_XX enumeration member 1393

I_VILRT_DETAILED_SHEAR_STRESS_YY enumeration member 1393

I_VILRT_DETAILED_SOIL_REACTION_Z enumeration member 1393

I_VILRT_DETAILED_STRESS_XX enumeration member 1393

I_VILRT_DETAILED_STRESS_XY enumeration member 1393

I_VILRT_DETAILED_STRESS_YY enumeration member 1393

I_VILRT_DETAILED_STRESS_Z enumeration member 1393

I_VILRT_EXTREME_MEMBRANE_FORCES_1 enumeration member 1393

I_VILRT_EXTREME_MEMBRANE_FORCES_1_2 enumeration member 1393

I_VILRT_EXTREME_MEMBRANE_FORCES_2 enumeration member 1393

I_VILRT_EXTREME_MEMBRANE_FORCES_ANGLE enumeration member 1393

I_VILRT_EXTREME_MOMENTS_1 enumeration member 1393

I_VILRT_EXTREME_MOMENTS_1_2 enumeration member 1393

I_VILRT_EXTREME_MOMENTS_2 enumeration member 1393

(c) 2014 Autodesk Inc. All rights reserved

I_VILRT_EXTREME_MOMENTS_ANGLE enumeration member 1393

I_VILRT_EXTREME_SHEAR_FORCE_1_2 enumeration member 1393

I_VILRT_EXTREME_SHEAR_STRESS_1_2 enumeration member 1393

I_VILRT_EXTREME_STRESS_1 enumeration member 1393

I_VILRT_EXTREME_STRESS_1_2 enumeration member 1393

I_VILRT_EXTREME_STRESS_2 enumeration member 1393

I_VILRT_EXTREME_STRESS_ANGLE enumeration member 1393

I_VILRT_NODES_FX enumeration member 1393

I_VILRT_NODES_FY enumeration member 1393

I_VILRT_NODES_FZ enumeration member 1393

I_VILRT_NODES_MX enumeration member 1393

I_VILRT_NODES_MY enumeration member 1393

I_VILRT_NODES_MZ enumeration member 1393

I_VILRT_NODES_RX enumeration member 1393

I_VILRT_NODES_RY enumeration member 1393

I_VILRT_NODES_RZ enumeration member 1393

I_VILRT_NODES_UX enumeration member 1393

I_VILRT_NODES_UY enumeration member 1393

I_VILRT_NODES_UZ enumeration member 1393

I_VILRT_NTM_FX enumeration member 1393

I_VILRT_NTM_FY enumeration member 1393

I_VILRT_NTM_FZ enumeration member 1393

I_VILRT_NTM_KY enumeration member 1393

I_VILRT_NTM_KZ enumeration member 1393

I_VILRT_NTM_MX enumeration member 1393

I_VILRT_NTM_MY enumeration member 1393

I_VILRT_NTM_MZ enumeration member 1393

I_VILRT_NTM_UX enumeration member 1393

I_VILRT_NTM_UZ enumeration member 1393

I_VLT_CONCENTRATED enumeration member 352

I_VLT_LINEAR enumeration member 352

I_VLT_SURFACE enumeration member 352

I_VP_3DXYZ enumeration member 1405

I_VP_XY enumeration member 1405

I_VP_XY_3D enumeration member 1405

I_VP_XZ enumeration member 1405

I_VP_XZ_3D enumeration member 1405

I_VP_YZ enumeration member 1405

I_VP_YZ_3D enumeration member 1405

I_VPI_ARCHITECT_ADDRESS1 enumeration member 1491

I_VPI_ARCHITECT_ADDRESS2 enumeration member 1491

I_VPI_ARCHITECT_CONTACT enumeration member 1491

I_VPI_ARCHITECT_EMAIL enumeration member 1491

I_VPI_ARCHITECT_FAX enumeration member 1491

I_VPI_ARCHITECT_NAME enumeration member 1491

I_VPI_ARCHITECT_TELEPHONE enumeration member 1491

I_VPI_COMPANY_ADDRESS1 enumeration member 1491

I_VPI_COMPANY_ADDRESS2 enumeration member 1491

I_VPI_COMPANY_CONTACT enumeration member 1491

I_VPI_COMPANY_EMAIL enumeration member 1491

I_VPI_COMPANY_FAX enumeration member 1491

I_VPI_COMPANY_NAME enumeration member 1491

I_VPI_COMPANY_TELEPHONE enumeration member 1491

I_VPI_DATE enumeration member 1491

I_VPI_DESIGNER_ADDRESS1 enumeration member 1491

I_VPI_DESIGNER_ADDRESS2 enumeration member 1491

I_VPI_DESIGNER_CONTACT enumeration member 1491

I_VPI_DESIGNER_EMAIL enumeration member 1491

I_VPI_DESIGNER_FAX enumeration member 1491

I_VPI_DESIGNER_NAME enumeration member 1491

I_VPI_DESIGNER_TELEPHONE enumeration member 1491

I_VPI_INVESTOR_ADDRESS1 enumeration member 1491

I_VPI_INVESTOR_ADDRESS2 enumeration member 1491

I_VPI_INVESTOR_CONTACT enumeration member 1491

I_VPI_INVESTOR_EMAIL enumeration member 1491

I_VPI_INVESTOR_FAX enumeration member 1491

I_VPI_INVESTOR_NAME enumeration member 1491

I_VPI_INVESTOR_TELEPHONE enumeration member 1491

I_VPI_PROJECT_ADDRESS1 enumeration member 1491

I_VPI_PROJECT_ADDRESS2 enumeration member 1491

I_VPI_PROJECT_COMMENT enumeration member 1491

I_VPI_PROJECT_CONTENT enumeration member 1491

I_VPI_PROJECT_CREATED enumeration member 1491

I_VPI_PROJECT_DIRECTORY enumeration member 1491

I_VPI_PROJECT_FILE enumeration member 1491

I_VPI_PROJECT_MODIFIED enumeration member 1491

I_VPI_PROJECT_NAME enumeration member 1491

I_VPI_PROJECT_REFERENCE enumeration member 1491

I_VPI_PROJECT_SIZE enumeration member 1491

I_VPI_PROJECT_STATISTICS enumeration member 1491

I_VPI_PROJECT_STEP enumeration member 1491

(c) 2014 Autodesk Inc. All rights reserved

I_VPI_PROJECT_VERSION enumeration member 1491
I_VPI_REPORT_PAGE_NUMBER enumeration member 1491
I_VPI_ROBOT_APP enumeration member 1491
I_VPI_ROBOT_PROVIDER enumeration member 1491
I_VPI_ROBOT_VER enumeration member 1491
I_VPI_TIME enumeration member 1491
I_VPI_USER_ADDRESS enumeration member 1491
I_VPI_USER_NAME enumeration member 1491
I_VPI_VERIF_ADDRESS1 enumeration member 1491
I_VPI_VERIF_ADDRESS2 enumeration member 1491
I_VPI_VERIF_CONTACT enumeration member 1491
I_VPI_VERIF_EMAIL enumeration member 1491
I_VPI_VERIF_FAX enumeration member 1491
I_VPI_VERIF_NAME enumeration member 1491
I_VPI_VERIF_TELEPHONE enumeration member 1491
I_VRRT_A_MIN enumeration member 1448
I_VRRT_AX enumeration member 1448
I_VRRT_AX_BOTTOM enumeration member 1448
I_VRRT_AX_TOP enumeration member 1448
I_VRRT_AY enumeration member 1448
I_VRRT_AY_BOTTOM enumeration member 1448
I_VRRT_AY_TOP enumeration member 1448
I_VRRT_E_AX_BOTTOM enumeration member 1448
I_VRRT_E_AX_TOP enumeration member 1448
I_VRRT_E_AY_BOTTOM enumeration member 1448
I_VRRT_E_AY_TOP enumeration member 1448
I_VRRT_F enumeration member 1448
I_VSCR_2048 enumeration member 1462
I_VSCR_3072 enumeration member 1462
I_VSCR_4096 enumeration member 1462
I_VSCR_DEFAULT enumeration member 1462
I_VST_DEFORM enumeration member 1412
I_VST_FX enumeration member 1412
I_VST_FY enumeration member 1412
I_VST_FZ enumeration member 1412
I_VST_MX enumeration member 1412
I_VST_MY enumeration member 1412
I_VST_MZ enumeration member 1412
I_VST_SIG enumeration member 1412
I_VST_TAU enumeration member 1412
I_VT_DETAILED_ANALYSIS enumeration member 1395
I_VT_DIAGRAMS enumeration member 1395
I_VT_GLOBAL_ANALYSIS enumeration member 1395
I_VT_INFLUENCE_LINES enumeration member 1395
I_VT_MAPS_ON_BARS enumeration member 1395
I_VT_MAPS_ON_FINITE_ELEMENTS enumeration member 1395
I_VT_PANEL_CUTS enumeration member 1395
I_VT_STANDARD enumeration member 1395
I_VVST_ACTIVE_CASE enumeration member 1406
I_VVST_ACTIVE_MODE enumeration member 1406
I_VVST_ACTIVE_PHASE enumeration member 1406
I_VVST_ACTIVE_QCMB enumeration member 1406
I_VVST_ANALYSIS enumeration member 1406
I_VVST_ANIMATION enumeration member 1406
I_VVST_BAR_LOADS enumeration member 1406
I_VVST_BAR_MAPS enumeration member 1406
I_VVST_BEST enumeration member 1406
I_VVST_BUCKLING enumeration member 1406
I_VVST_COMPATIBILITIES enumeration member 1406
I_VVST_CtrT enumeration member 1406
I_VVST_DEFORMATION enumeration member 1406
I_VVST_Dep enumeration member 1406
I_VVST_DESCRIPTIONS enumeration member 1406
I_VVST_DetDi enumeration member 1406
I_VVST_Dia3D enumeration member 1406
I_VVST_DIM_LINES enumeration member 1406
I_VVST_ELE enumeration member 1406
I_VVST_ENVELOPES enumeration member 1406
I_VVST_EXPLODE enumeration member 1406
I_VVST_EXTREMES enumeration member 1406
I_VVST_FE enumeration member 1406
I_VVST_FE_CUTS enumeration member 1406
I_VVST_FE_DIRECTION enumeration member 1406
I_VVST_FE_LAY enumeration member 1406
I_VVST_FILL enumeration member 1406
I_VVST_FORCES enumeration member 1406
I_VVST_GloDi enumeration member 1406
I_VVST_GRID enumeration member 1406
I_VVST_LOADS enumeration member 1406
I_VVST_LOCAL enumeration member 1406
I_VVST_MAPS enumeration member 1406
I_VVST_MAPS_DEFORMATION enumeration member 1406
I_VVST_MAPS_LAYER enumeration member 1406
I_VVST_MOBILE enumeration member 1406

(c) 2014 Autodesk Inc. All rights reserved

I_VVST_MODE enumeration member 1406
I_VVST_MTC enumeration member 1406
I_VVST_NODE_LOADS enumeration member 1406
I_VVST_NODES enumeration member 1406
I_VVST_NOT_SELECTED enumeration member 1406
I_VVST_OFFSETS enumeration member 1406
I_VVST_PLOT enumeration member 1406
I_VVST_REACTIONS enumeration member 1406
I_VVST_REINF_CROSSES enumeration member 1406
I_VVST_REINF_DIR enumeration member 1406
I_VVST_RELEASES enumeration member 1406
I_VVST_RIGID_LINKS enumeration member 1406
I_VVST_RULER enumeration member 1406
I_VVST_SCALE enumeration member 1406
I_VVST_SECTION_DRAW enumeration member 1406
I_VVST_SECTION_NAMES enumeration member 1406
I_VVST_SECTION_SURF enumeration member 1406
I_VVST_SECTION_SYMB enumeration member 1406
I_VVST_SECTIONS enumeration member 1406
I_VVST_STRESSES enumeration member 1406
I_VVST_STRESSES_GLOBAL enumeration member 1406
I_VVST_STRUCTURE enumeration member 1406
I_VVST_STRUCTURE_3D enumeration member 1406
I_VVST_SUPERPOS enumeration member 1406
I_VVST_SUPPORTS enumeration member 1406
I_VVST_THICKNESS enumeration member 1406
I_VVST_TITLE enumeration member 1406
I_VVSV_ANALYSIS_DET enumeration member 1407
I_VVSV_ANALYSIS_GLO enumeration member 1407
I_VVSV_ANALYSIS_INF enumeration member 1407
I_VVSV_BAR_MAPS_DESIGN_MEMB_LENGTH enumeration member 1407
I_VVSV_BAR_MAPS_DESIGN_RATIO enumeration member 1407
I_VVSV_BAR_MAPS_DESIGN_SLEND_LAY enumeration member 1407
I_VVSV_BAR_MAPS_DESIGN_SLEND_LAZ enumeration member 1407
I_VVSV_BAR_MAPS_FX enumeration member 1407
I_VVSV_BAR_MAPS_FY enumeration member 1407
I_VVSV_BAR_MAPS_FZ enumeration member 1407
I_VVSV_BAR_MAPS_MX enumeration member 1407
I_VVSV_BAR_MAPS_MY enumeration member 1407
I_VVSV_BAR_MAPS_MZ enumeration member 1407
I_VVSV_BAR_MAPS_SHEAR_STRESS_T enumeration member 1407
I_VVSV_BAR_MAPS_SHEAR_STRESS_TY enumeration member 1407
I_VVSV_BAR_MAPS_SHEAR_STRESS_TZ enumeration member 1407
I_VVSV_BAR_MAPS_STRESS_FX_AX enumeration member 1407
I_VVSV_BAR_MAPS_STRESS_S_MAX enumeration member 1407
I_VVSV_BAR_MAPS_STRESS_S_MAX_MY enumeration member 1407
I_VVSV_BAR_MAPS_STRESS_S_MAX_MZ enumeration member 1407
I_VVSV_BAR_MAPS_STRESS_S_MIN enumeration member 1407
I_VVSV_BAR_MAPS_STRESS_S_MIN_MY enumeration member 1407
I_VVSV_BAR_MAPS_STRESS_S_MIN_MZ enumeration member 1407
I_VVSV_DEFORMATION_EXACT enumeration member 1407
I_VVSV_DEFORMATION_STD enumeration member 1407
I_VVSV_DESCRIPTIONS_HORIZONTAL enumeration member 1407
I_VVSV_DESCRIPTIONS_NONE enumeration member 1407
I_VVSV_DESCRIPTIONS_VERTICAL enumeration member 1407
I_VVSV_ELE_BAR_NUM enumeration member 1407
I_VVSV_ELE_CAL_NUM enumeration member 1407
I_VVSV_ELE_FE_NUM enumeration member 1407
I_VVSV_ELE_FE_VIS enumeration member 1407
I_VVSV_ENVELOPES_CMP enumeration member 1407
I_VVSV_ENVELOPES_CUT enumeration member 1407
I_VVSV_ENVELOPES_HST enumeration member 1407
I_VVSV_ENVELOPES_MAX enumeration member 1407
I_VVSV_ENVELOPES_MIN enumeration member 1407
I_VVSV_ENVELOPES_ONE enumeration member 1407
I_VVSV_ENVELOPES_SET enumeration member 1407
I_VVSV_ENVELOPES_SEU enumeration member 1407
I_VVSV_FE_ADV enumeration member 1407
I_VVSV_FE_COE enumeration member 1407
I_VVSV_FE_COI enumeration member 1407
I_VVSV_FE_COL enumeration member 1407
I_VVSV_FE_CON enumeration member 1407
I_VVSV_FE_DSC enumeration member 1407
I_VVSV_FE_DSO enumeration member 1407
I_VVSV_FE_EDG enumeration member 1407

(c) 2014 Autodesk Inc. All rights reserved

I_VVSV_FE_EMI enumeration member 1407

I_VVSV_FE_MES enumeration member 1407

I_VVSV_FE_RES enumeration member 1407

I_VVSV_FILL_AVG enumeration member 1407

I_VVSV_FILL_BOR enumeration member 1407

I_VVSV_FILL_DEF enumeration member 1407

I_VVSV_FILL_FE enumeration member 1407

I_VVSV_FILL_FILL enumeration member 1407

I_VVSV_FILL_ISO enumeration member 1407

I_VVSV_FILL_SMOOTH_CAR enumeration member 1407

I_VVSV_FILL_SMOOTH_GLO enumeration member 1407

I_VVSV_FILL_SMOOTH_LOC enumeration member 1407

I_VVSV_FILL_SMOOTH_NON enumeration member 1407

I_VVSV_FILL_SMOOTH_SEL enumeration member 1407

I_VVSV_FILL_TAN enumeration member 1407

I_VVSV_FILL_VAL enumeration member 1407

I_VVSV_FORCES_BAR_REACT_KY enumeration member 1407

I_VVSV_FORCES_BAR_REACT_KZ enumeration member 1407

I_VVSV_FORCES_DFL enumeration member 1407

I_VVSV_FORCES_FX enumeration member 1407

I_VVSV_FORCES_FXC enumeration member 1407

I_VVSV_FORCES_FY enumeration member 1407

I_VVSV_FORCES_FZ enumeration member 1407

I_VVSV_FORCES_MX enumeration member 1407

I_VVSV_FORCES_MY enumeration member 1407

I_VVSV_FORCES_MZ enumeration member 1407

I_VVSV_FORCES_UX enumeration member 1407

I_VVSV_FORCES_UY enumeration member 1407

I_VVSV_FORCES_UZ enumeration member 1407

I_VVSV_LOADS_SYMBOL enumeration member 1407

I_VVSV_LOADS_VALUE enumeration member 1407

I_VVSV_MAPS_COMPLEX_MEMB_FORCES enumeration member 1407

I_VVSV_MAPS_COMPLEX_MOMENTS enumeration member 1407

I_VVSV_MAPS_COMPLEX_REINFORCEMENT_BOTTOM_MXX
enumeration member 1407

I_VVSV_MAPS_COMPLEX_REINFORCEMENT_BOTTOM_MYY
enumeration member 1407

I_VVSV_MAPS_COMPLEX_REINFORCEMENT_TOP_MXX enumeration member 1407

I_VVSV_MAPS_COMPLEX_REINFORCEMENT_TOP_MYY enumeration member 1407

I_VVSV_MAPS_COMPLEX_STRESSES enumeration member 1407

I_VVSV_MAPS_CROSS_M enumeration member 1407

I_VVSV_MAPS_CROSS_N enumeration member 1407

I_VVSV_MAPS_CROSS_S enumeration member 1407

I_VVSV_MAPS_DESCRIPTION enumeration member 1407

I_VVSV_MAPS_DETAILED_DISPLACEMENTS_XX enumeration member 1407

I_VVSV_MAPS_DETAILED_DISPLACEMENTS_YY enumeration member 1407

I_VVSV_MAPS_DETAILED_DISPLACEMENTS_Z enumeration member 1407

I_VVSV_MAPS_DETAILED_MEMB_FORCES_XX enumeration member 1407

I_VVSV_MAPS_DETAILED_MEMB_FORCES_XY enumeration member 1407

I_VVSV_MAPS_DETAILED_MEMB_FORCES_YY enumeration member 1407

I_VVSV_MAPS_DETAILED_MOMENTS_XX enumeration member 1407

I_VVSV_MAPS_DETAILED_MOMENTS_XY enumeration member 1407

I_VVSV_MAPS_DETAILED_MOMENTS_YY enumeration member 1407

I_VVSV_MAPS_DETAILED_ROTATION_XX enumeration member 1407

I_VVSV_MAPS_DETAILED_ROTATION_YY enumeration member 1407

I_VVSV_MAPS_DETAILED_ROTATION_Z enumeration member 1407

I_VVSV_MAPS_DETAILED_SHEAR_FORCES_XX enumeration member 1407

I_VVSV_MAPS_DETAILED_SHEAR_FORCES_YY enumeration member 1407

I_VVSV_MAPS_DETAILED_SHEAR_STRESSES_XX enumeration member 1407

I_VVSV_MAPS_DETAILED_SHEAR_STRESSES_YY enumeration member 1407

I_VVSV_MAPS_DETAILED_SOIL_REACTIONS enumeration member 1407

I_VVSV_MAPS_DETAILED_STRESSES_XX enumeration member 1407

I_VVSV_MAPS_DETAILED_STRESSES_XY enumeration member 1407

I_VVSV_MAPS_DETAILED_STRESSES_YY enumeration member 1407

I_VVSV_MAPS_DETAILED_STRESSES_Z enumeration member 1407

I_VVSV_MAPS_ISOLINES enumeration member 1407

(c) 2014 Autodesk Inc. All rights reserved

I_VVSV_MAPS_LAYER_ABS_MINIMUM enumeration member 1407

I_VVSV_MAPS_LAYER_ARBITRARY enumeration member 1407

I_VVSV_MAPS_LAYER_LOWER enumeration member 1407

I_VVSV_MAPS_LAYER_MAXIMUM enumeration member 1407

I_VVSV_MAPS_LAYER_MIDDLE enumeration member 1407

I_VVSV_MAPS_LAYER_MINIMUM enumeration member 1407

I_VVSV_MAPS_LAYER_UPPER enumeration member 1407

I_VVSV_MAPS_PRINCIPAL_GLOBAL_DISP_X enumeration member 1407

I_VVSV_MAPS_PRINCIPAL_GLOBAL_DISP_Y enumeration member 1407

I_VVSV_MAPS_PRINCIPAL_GLOBAL_DISP_Z enumeration member 1407

I_VVSV_MAPS_PRINCIPAL_MEMB_FORCES_1 enumeration member 1407

I_VVSV_MAPS_PRINCIPAL_MEMB_FORCES_1_2 enumeration member 1407

I_VVSV_MAPS_PRINCIPAL_MEMB_FORCES_2 enumeration member 1407

I_VVSV_MAPS_PRINCIPAL_MEMB_FORCES_ANGLE enumeration member 1407

I_VVSV_MAPS_PRINCIPAL_MOMENTS_1 enumeration member 1407

I_VVSV_MAPS_PRINCIPAL_MOMENTS_1_2 enumeration member 1407

I_VVSV_MAPS_PRINCIPAL_MOMENTS_2 enumeration member 1407

I_VVSV_MAPS_PRINCIPAL_MOMENTS_ANGLE enumeration member 1407

I_VVSV_MAPS_PRINCIPAL_SHEAR_FORCES enumeration member 1407

I_VVSV_MAPS_PRINCIPAL_SHEAR_STRESSES enumeration member 1407

I_VVSV_MAPS_PRINCIPAL_STRESSES_1 enumeration member 1407

I_VVSV_MAPS_PRINCIPAL_STRESSES_1_2 enumeration member 1407

I_VVSV_MAPS_PRINCIPAL_STRESSES_2 enumeration member 1407

I_VVSV_MAPS_PRINCIPAL_STRESSES_ANGLE enumeration member 1407

I_VVSV_MAPS_PRINCIPAL_TOTAL_DISP enumeration member 1407

I_VVSV_MAPS_SMOOTH_CAR enumeration member 1407

I_VVSV_MAPS_SMOOTH_GLO enumeration member 1407

I_VVSV_MAPS_SMOOTH_LOC enumeration member 1407

I_VVSV_MAPS_SMOOTH_NON enumeration member 1407

I_VVSV_MAPS_SMOOTH_SEL enumeration member 1407

I_VVSV_MOBILE_CAR enumeration member 1407

I_VVSV_MOBILE_ELE enumeration member 1407

I_VVSV_MOBILE_LOADS enumeration member 1407

I_VVSV_MOBILE_ROUTE enumeration member 1407

I_VVSV_MTC_CABLE enumeration member 1407

I_VVSV_MTC_COMPRESSION enumeration member 1407

I_VVSV_MTC_FEM enumeration member 1407

I_VVSV_MTC_MATERIAL enumeration member 1407

I_VVSV_MTC_TENSION enumeration member 1407

I_VVSV_REACTIONS_DIV enumeration member 1407

I_VVSV_REACTIONS_F enumeration member 1407

I_VVSV_REACTIONS_M enumeration member 1407

I_VVSV_REACTIONS_RESID_F enumeration member 1407

I_VVSV_REACTIONS_RESID_M enumeration member 1407

I_VVSV_REACTIONS_ROT enumeration member 1407

I_VVSV_REACTIONS_VAL enumeration member 1407

I_VVSV_REINFORCEMENT_A1 enumeration member 1407

I_VVSV_REINFORCEMENT_A1BARS enumeration member 1407

I_VVSV_REINFORCEMENT_A1R enumeration member 1407

I_VVSV_REINFORCEMENT_A2 enumeration member 1407

I_VVSV_REINFORCEMENT_A2BARS enumeration member 1407

I_VVSV_REINFORCEMENT_A2R enumeration member 1407

I_VVSV_REINFORCEMENT_DISTRIB enumeration member 1407

I_VVSV_REINFORCEMENT_DISTRIBR enumeration member 1407

I_VVSV_REINFORCEMENT_PRC enumeration member 1407

I_VVSV_REINFORCEMENT_PRCR enumeration member 1407

I_VVSV_RIGID_LINKS_DET enumeration member 1407

I_VVSV_RIGID_LINKS_RIG enumeration member 1407

I_VVSV_RIGID_LINKS_SUR enumeration member 1407

I_VVSV_RIGID_LINKS_SYM enumeration member 1407

I_VVSV_SECTION_DRAW_COLOR enumeration member 1407

I_VVSV_SECTION_DRAW_GROUP enumeration member 1407

I_VVSV_SECTION_DRAW_SHAPE enumeration member 1407

I_VVSV_SECTION_DRAW_TOP_BOTTOM enumeration member 1407

(c) 2014 Autodesk Inc. All rights reserved

I_VVSV_SECTION_SYMB_ELA enumeration member 1407
I_VVSV_SECTION_SYMB_SYMB enumeration member 1407
I_VVSV_STRESSES_AXIAL enumeration member 1407
I_VVSV_STRESSES_FLEXMAX_MY enumeration member 1407
I_VVSV_STRESSES_FLEXMAX_MZ enumeration member 1407
I_VVSV_STRESSES_FLEXMIN_MY enumeration member 1407
I_VVSV_STRESSES_FLEXMIN_MZ enumeration member 1407
I_VVSV_STRESSES_GLOBAL_MISES_MAX enumeration member 1407
I_VVSV_STRESSES_GLOBAL_MISES_MIN enumeration member 1407
I_VVSV_STRESSES_GLOBAL_NORMAL_MAX enumeration member 1407
I_VVSV_STRESSES_GLOBAL_NORMAL_MIN enumeration member 1407
I_VVSV_STRESSES_GLOBAL_TAU_MAX enumeration member 1407
I_VVSV_STRESSES_GLOBAL_TAU_MIN enumeration member 1407
I_VVSV_STRESSES_GLOBAL_USER_MAX enumeration member 1407
I_VVSV_STRESSES_GLOBAL_USER_MIN enumeration member 1407
I_VVSV_STRESSES_NORMALMAX enumeration member 1407
I_VVSV_STRESSES_NORMALMIN enumeration member 1407
I_VVSV_STRESSES_SHEAR_Y enumeration member 1407
I_VVSV_STRESSES_SHEAR_Z enumeration member 1407
I_VVSV_STRESSES_TORSION enumeration member 1407
I_VVSV_STRUCTURE_COR enumeration member 1407
I_VVSV_STRUCTURE_HDF enumeration member 1407
I_VVSV_STRUCTURE_HDL enumeration member 1407
I_VVSV_STRUCTURE_HDR enumeration member 1407
I_VVSV_STRUCTURE_HDY enumeration member 1407
I_VVSV_STRUCTURE_HID enumeration member 1407
I_VVSV_STRUCTURE_ON enumeration member 1407
I_VVSV_STRUCTURE_POI enumeration member 1407
I_VVSV_STRUCTURE_SHD enumeration member 1407
I_VVSV_STRUCTURE_SPD enumeration member 1407
I_VVSV_SUPPORTS_COD enumeration member 1407
I_VVSV_SUPPORTS_LBL enumeration member 1407
I_VVSV_SUPPORTS_SYMB enumeration member 1407
I_WS_MAXIMIZE enumeration member 1653
I_WS_MINIMIZE enumeration member 1653
I_WS_NORMAL enumeration member 1653
IBST_BEAM_SUPPORT_FIXED enumeration member 1167
IBST_BEAM_SUPPORT_PINNED enumeration member 1167
IBST_BEAM_SUPPORT_SWAY enumeration member 1167
Influence lines 1385
IRBestBendType 1698
IRBestCalcErrors 1686
IRBestCalcParamsData 1677
about IRBestCalcParamsData 1677
IRBestCalcParamsData fields 1678
IRBestCalcParamsData members 1677
ModularityList 1678
IRBestCalcParamsDataDoubleValue 1678
IRBestCalcParamsDataIntegerValue 1679
IRBestCalcParamsDataList 1680
about IRBestCalcParamsDataList 1680
Create 1682
Delete 1682
Get 1682
IRBestCalcParamsDataList fields 1681
IRBestCalcParamsDataList members 1680
IRBestCalcParamsDataList methods 1682
LabNames 1681
Selected 1681
Store 1683
IRBestCalcParamsDataStringValue 1680
IRBestCalcParamsDlg 1669
about IRBestCalcParamsDlg 1669
DoModal 1669
IRBestCalcParamsDlg members 1669
IRBestCalcParamsDlg methods 1669
SetStandard 1670
IRBestCalculationType 1698
IRBestCalcWarnings 1687
IRBestCodeCalcEngine 1691
about IRBestCodeCalcEngine 1691
Calculate 1692
GetResults 1692
IRBestCodeCalcEngine members 1691
IRBestCodeCalcEngine methods 1692

(c) 2014 Autodesk Inc. All rights reserved

SetForces 1693
SetForcesArray 1693
SetForcesSlender 1693
SetGeometry 1694
SetParams 1694
IRBestCodeCalculationType 1696
IRBestCodeService 1667
about IRBestCodeService 1667
CalcEngine 1668
CalcParamsDlg 1668
IRBestCodeService fields 1668
IRBestCodeService members 1667
MemberDlg 1668
IRBestCodeService2 1700
about IRBestCodeService2 1700
IRBestCodeService2 fields 1701
IRBestCodeService2 members 1701
RobotVersion 1701
ShowBucklingButton 1701
IRBestCodeServiceExt 1696
about IRBestCodeServiceExt 1696
GetCalcParamsDlg 1697
IRBestCodeServiceExt members 1697
IRBestCodeServiceExt methods 1697
IsServed 1697
IRBestCoordSystem 1699
IRBestDimParams 1690
about IRBestDimParams 1690
IRBestDimParamsDoubleValue 1698
IRBestDimParamsIntegerValue 1690
IRBestDirection 1698
IRBestForceData 1688
about IRBestForceData 1688
IRBestForceData fields 1689
IRBestForceData members 1688
LimitState 1690
Values 1690
IRBestForceDataDoubleValue 1699
IRBestForceDataIntegerValue 1699
IRBestForceDataSLSCombType 1700
IRBestLevel 1699
IRBestMemberData 1683
about IRBestMemberData 1683
IRBestMemberDataDoubleValue 1683
IRBestMemberDataIntegerValue 1684
IRBestMemberDataStringValue 1694
IRBestMemberDlg 1670
about IRBestMemberDlg 1670
DoModal 1671
IRBestMemberDlg members 1670
IRBestMemberDlg methods 1670
SetStandard 1671
IRBestMemberType 1695
IRBestParamSet 1671
about IRBestParamSet 1671
Clear 1673
ClearDouble 1673
ClearInteger 1674
ClearString 1674
GetDouble 1674
GetInteger 1674
GetString 1675
IRBestParamSet members 1671
IRBestParamSet methods 1672
IsValidDouble 1675
IsValidInteger 1675
IsValidString 1675
SetDouble 1676
SetInteger 1676
SetString 1676
IRBestPlateCalcParamsDlg 1695
about IRBestPlateCalcParamsDlg 1695
DoModal 1695
IRBestPlateCalcParamsDlg members 1695
IRBestPlateCalcParamsDlg methods 1695
SetStandard 1696
IRBestResults 1684
about IRBestResults 1684
IRBestResultsDoubleValue 1685
IRBestResultsIntegerValue 1686
IRBestResultsStringValue 1688
IRConcr_ACI318_BarDim 1750
IRConcr_ACI318_ConcreteParams 1749
about IRConcr_ACI318_ConcreteParams 1749

(c) 2014 Autodesk Inc. All rights reserved

AsInStructure 1750
IRConcr_ACI318_ConcreteParams fields 1750
IRConcr_ACI318_ConcreteParams members 1749
IRConcr_ACI318_LoadActionPeriodType 1748
IRConcr_ACI318_MetricBarDIm 1751
IRConcr_ACI318_ReinforceData 1741
about IRConcr_ACI318_ReinforceData 1741
GetBarDim 1747
GetMetricBarDim 1747
IRConcr_ACI318_ReinforceData fields 1742
IRConcr_ACI318_ReinforceData members 1741
IRConcr_ACI318_ReinforceData methods 1746
IsMetric 1743
Main 1743
SetBarDim 1747
SetMetricBarDim 1748
SLS_Cracking 1743
SLS_CreepingCoef 1743
SLS_CreepingCoefValue 1744
SLS_Deflection 1744
SLS_DeflectionReinfCorrection 1744
SLS_Factor 1744
SLS_FactorValue 1745
SLS_LoadActionPeriod 1745
SLS_LoadRatio 1745
SLS_MaxCracking 1746
SLS_MaxDeflection 1746
IRConcr_ACI318_SteelGrades 1748
IRConcr_BAEL_ConcreteGrades 1740
IRConcr_BAEL_CrackingType 1739
IRConcr_BAEL_EnvironmentType 1739
IRConcr_BAEL_ReinforceData 1731
about IRConcr_BAEL_ReinforceData 1731
CrackAlpha 1733
CrackAlphaBot 1734
CrackAlphaTop 1734
CrackEnv 1734
CrackEnvBot 1734
CrackEnvTop 1735
CrackExtraParams 1735
CrackExtraParamsBot 1735
CrackExtraParamsTop 1736
Cracking 1736
CrackingBot 1736
CrackingTop 1736
CrackWaterLevel 1737
CrackWaterLevelBot 1737
CrackWaterLevelTop 1737
IRConcr_BAEL_ReinforceData fields 1733
IRConcr_BAEL_ReinforceData members 1731
Main 1737
SLS_Deflection 1738
SLS_DeflectionReinfCorrect 1738
SLS_MaxDeflection 1738
SteelSymbol 1738
IRConcr_BAEL_SteelGrades 1740
IRConcr_BAEL_WaterLevel 1739
IRConcr_BS8110_ConcreteAge 1758
IRConcr_BS8110_ConcreteGrades 1759
IRConcr_BS8110_ExposureRatings 1758
IRConcr_BS8110_PartialSafetyFactors 1760
IRConcr_BS8110_ReinforceData 1752
about IRConcr_BS8110_ReinforceData 1752
IRConcr_BS8110_ReinforceData fields 1753
IRConcr_BS8110_ReinforceData members 1752
Main 1753
PartialSafetyFactors 1754
SLS_ConcreteAge 1754
SLS_Cracking 1754
SLS_CrackingReinfCorection 1755
SLS_CreepingCoef 1755
SLS_CreepingCoefValue 1755
SLS_Deflection 1756
SLS_DeflectionReinfCorection 1756
SLS_EnvHumidity 1756
SLS_Exposure 1757
SLS_LoadRatio 1757
SLS_MaxCracking 1757
SLS_MaxDeflection 1758
IRConcr_EC2_ConcreteGrades 1774
IRConcr_EC2_ExposureRatings 1774
IRConcr_EC2_ITALIAN_SteelGrades 1774
IRConcr_EC2_NAD 1779
IRConcr_EC2_ReinforceData 1775

(c) 2014 Autodesk Inc. All rights reserved

about IRConcr_EC2_ReinforceData 1775
IRConcr_EC2_ReinforceData fields 1776
IRConcr_EC2_ReinforceData members 1775
NAD 1777
SLS_ConcreteAge 1777
SLS_Cracking 1777
SLS_CrackingReinfCorrect 1777
SLS_CreepingCoef 1777
SLS_CreepingCoefValue 1778
SLS_Deflection 1778
SLS_DeflectionReinfCorrect 1778
SLS_EnvHumidityVal 1778
SLS_Exposure 1778
SLS_MaxCracking 1779
SLS_MaxDeflection 1779
IRConcr_IS2000_EnvironmentType 1780
IRConcr_IS2000_ReinforceData 1780
about IRConcr_IS2000_ReinforceData 1780
IRConcr_IS2000_ReinforceData fields 1782
IRConcr_IS2000_ReinforceData members 1780
Main 1782
SLS_ConcreteAge 1783
SLS_Cracking 1783
SLS_CrackingReinfCorrect 1783
SLS_CreepingCoeff 1783
SLS_CreepingCoeffValue 1784
SLS_Deflection 1784
SLS_DeflectionReinfCorrect 1784
SLS_ExposureBot 1784
SLS_ExposureTop 1785
SLS_LoadRatio 1785
SLS_MaxCrackingBot 1785
SLS_MaxCrackingBotEnabled 1785
SLS_MaxCrackingTop 1786
SLS_MaxCrackingTopEnabled 1786
SLS_MaxDeflection 1786
IRConcr_PN_SteelGrades 1721
IRConcr_PN84_ConcreteGrades 1721
IRConcr_PN84_ExposureRatings 1722
IRConcr_PN84_HumidityType 1740
IRConcr_PN84_ReinforceData 1715
about IRConcr_PN84_ReinforceData 1715
CheckConstructionStages 1717
IRConcr_PN84_ReinforceData fields 1716
IRConcr_PN84_ReinforceData members 1715
LongActionAbove100 1717
Main 1718
SingleShortLoad 1718
SLS_ConcreteAge 1718
SLS_Cracking 1718
SLS_CrackingReinfCorrect 1719
SLS_CreepingCoef 1719
SLS_Deflection 1719
SLS_DeflectionReinfCorrect 1719
SLS_EnvHumidity 1720
SLS_Exposure 1720
SLS_LoadsRatio 1720
SLS_MaxCracking 1720
SLS_MaxDeflection 1721
IRConcr_PN99_ConcreteGrades 1730
IRConcr_PN99_ExposureRatings 1722
IRConcr_PN99_ReinforceData 1723
about IRConcr_PN99_ReinforceData 1723
IRConcr_PN99_ReinforceData fields 1724
IRConcr_PN99_ReinforceData members 1723
IsSpecialStructure 1725
Main 1726
SLS_Concrete 1726
SLS_ConcreteAge 1726
SLS_Cracking 1726
SLS_CrackingReinfCorrect 1727
SLS_CreepingCoef 1727
SLS_CreepingCoefValue 1727
SLS_Deflection 1727
SLS_DeflectionReinfCorrect 1728
SLS_EnvHumidityVal 1728
SLS_ExposureBot 1728
SLS_ExposureTop 1728
SLS_LoadRatio 1729
SLS_MaxCrackingBot 1729
SLS_MaxCrackingBotEnabled 1729
SLS_MaxCrackingTop 1729
SLS_MaxCrackingTopEnabled 1730
SLS_MaxDeflection 1730

(c) 2014 Autodesk Inc. All rights reserved

IRConcr_SNIP_ConcreteGrades 1771
IRConcr_SNIP_ConcreteParams 1772
about IRConcr_SNIP_ConcreteParams 1772
ConcretingInLayers 1773
CuringMethod 1773
HighHumidity 1773
IRConcr_SNIP_ConcreteParams fields 1773
IRConcr_SNIP_ConcreteParams members 1772
Type 1773
IRConcr_SNIP_ConcreteTypes 1771
IRConcr_SNIP_CuringMethods 1771
IRConcr_SNIP_Exposure 1774
IRConcr_SNIP_ReinforceData 1769
about IRConcr_SNIP_ReinforceData 1769
IRConcr_SNIP_ReinforceData fields 1770
IRConcr_SNIP_ReinforceData members 1769
SLS_Cracking 1770
SLS_CrackingReinfCorrection 1770
SLS_Exposure 1771
IRConcr_SNIP_SteelGrades 1771
IRConcrBarDiameters 1264
about IRConcrBarDiameters 1264
Add 1266
Count 1265
IRConcrBarDiameters fields 1265
IRConcrBarDiameters members 1265
IRConcrBarDiameters methods 1266
Item 1265
RemoveAll 1266
IRConcrBarSectionData 1794
A 1795
about IRConcrBarSectionData 1794
Ac 1796
Ap 1796
B 1796
Bf 1796
Bfb 1796
H 1797
Hf 1797
Hfb 1797
IRConcrBarSectionData fields 1795
IRConcrBarSectionData members 1795
N 1797
Per 1797
Type 1798
IRConcrBarSectionGeometryType 1798
IRConcrBarSubtype 1258
IRConcrBarType 1257
IRConcrBeam 1129
about IRConcrBeam 1129
Activate 1136
Calculate 1136
CalculationOptions 1131
Concrete 1131
CreateCalculationNoteRtf 1137
CreateFromBars 1137
GenerateSpliceBars 1137
Geometry 1131
ImportOptions 1132
IRConcrBeam fields 1130
IRConcrBeam members 1129
IRConcrBeam methods 1136
IsActive 1132
IsSelected 1132
LinearLoad 1132
LinearLoadsCount 1133
Name 1133
NumberOfElements 1133
PatternOptions 1133
PointLoad 1134
PointLoadsCount 1134
Reinforcement 1134
Save 1137
Steel 1134
StoryOptions 1135
StructureUserNo 1135
StructureUserNoCount 1135
UniqueId 1135
Verify 1138
VerifySpan 1138
IRConcrBeamCalcOptions 1138
about IRConcrBeamCalcOptions 1138
CalcSpanLengthInAxis 1139
CoverBottom 1140

(c) 2014 Autodesk Inc. All rights reserved

CoverBottomFixed 1140
CoverSide 1140
CoverSideFixed 1140
CoverTop 1141
CoverTopFixed 1141
IRConcrBeamCalcOptions fields 1139
IRConcrBeamCalcOptions members 1138
IRConcrBeamCalcPointDefinitionType 1794
IRConcrBeamCrackingType 1167
IRConcrBeamGeometry 1141
about IRConcrBeamGeometry 1141
AutoNameSpans 1142
AutoNameSupports 1143
ConcreteVolume 1143
IRConcrBeamGeometry fields 1142
IRConcrBeamGeometry members 1141
IRConcrBeamGeometry methods 1144
LeftCantilever 1143
RightCantilever 1143
ShutteringArea 1144
Span 1144
SpanNumber 1144
UpdateInternalGeometryData 1145
IRConcrBeamImportOptions 1145
about IRConcrBeamImportOptions 1145
GroupByGeometry 1146
GroupByLevel 1146
ImportCombinations 1146
IRConcrBeamImportOptions fields 1145
IRConcrBeamImportOptions members 1145
LiveLongCoeff 1147
RunCalcAuto 1147
ShowDialog 1147
IRConcrBeamLinearLoad 1147
about IRConcrBeamLinearLoad 1147
Case 1148
IRConcrBeamLinearLoad fields 1148
IRConcrBeamLinearLoad members 1148
Nature 1149
RelativeCoordinates 1149
Spans 1149
Type 1150
Value1 1150
Value2 1150
Value3 1150
X1 1151
X2 1151
X3 1151
X4 1151
IRConcrBeamLinearLoadType 1164
IRConcrBeamLoadNatureType 1165
IRConcrBeamPatternOptions 1152
about IRConcrBeamPatternOptions 1152
IRConcrBeamPatternOptions fields 1152
IRConcrBeamPatternOptions members 1152
SpanBySpan 1152
IRConcrBeamPointLoad 1153
about IRConcrBeamPointLoad 1153
Case 1153
IRConcrBeamPointLoad fields 1153
IRConcrBeamPointLoad members 1153
Nature 1154
RelativeCoordinates 1154
Spans 1154
Type 1154
Value 1155
X1 1155
IRConcrBeamPointLoadType 1165
IRConcrBeamSectionDimType 1166
IRConcrBeamSectionType 1165
IRConcrBeamSegment 1155
about IRConcrBeamSegment 1155
Dim 1156
IRConcrBeamSegment fields 1156
IRConcrBeamSegment members 1156
SectionType 1156
IRConcrBeamSpan 1156
about IRConcrBeamSpan 1156
CalculationLength 1157
HasOpenings 1158
IRConcrBeamSpan fields 1157
IRConcrBeamSpan members 1157
LeftSupport 1158
Length 1158

(c) 2014 Autodesk Inc. All rights reserved

Name 1158
RightSupport 1159
Segment 1159
SegmentNumber 1159
IRConcrBeamSpanNumbers 1159
about IRConcrBeamSpanNumbers 1159
Add 1161
Count 1160
IRConcrBeamSpanNumbers fields 1160
IRConcrBeamSpanNumbers members 1160
IRConcrBeamSpanNumbers methods 1161
Item 1160
RemoveAll 1161
IRConcrBeamStoryOptions 1161
about IRConcrBeamStoryOptions 1161
Cracking 1162
IRConcrBeamStoryOptions fields 1162
IRConcrBeamStoryOptions members 1162
IRConcrBeamSupport 1162
about IRConcrBeamSupport 1162
IRConcrBeamSupport fields 1163
IRConcrBeamSupport members 1163
MaterialType 1163
Name 1163
Type 1164
Width 1164
IRConcrBeamSupportMaterialType 1168
IRConcrBeamSupportType 1167
IRConcrCalcEngine 1798
about IRConcrCalcEngine 1798
IRConcrCalcEngine fields 1799
IRConcrCalcEngine members 1799
MemberRequiredReinf 1799
SlabRequiredReinf 1799
IRConcrCodeBeam 1805
about IRConcrCodeBeam 1805
IRConcrCodeBeam members 1805
IRConcrCodeBeam methods 1805
IsCommandEnabled 1806
Verification 1806
IRConcrCodeBeamCommand 1806
IRConcrCodeColumn 1802
about IRConcrCodeColumn 1802
IRConcrCodeColumn members 1802
IRConcrCodeColumn methods 1802
IsCommandEnabled 1802
Verification 1803
IRConcrCodeColumnCommand 1803
IRConcrCodeReport 1803
about IRConcrCodeReport 1803
AddError 1804
AddMessage 1804
AddWarning 1805
IRConcrCodeReport members 1804
IRConcrCodeReport methods 1804
IRConcrCodeService 1800
about IRConcrCodeService 1800
Beam 1801
Column 1801
IRConcrCodeService fields 1801
IRConcrCodeService members 1800
IRConcrCodeService methods 1801
IsConcrComponentServed 1801
IRConcrColumn 1186
about IRConcrColumn 1186
Activate 1193
BucklingModel 1188
Calculate 1194
CalculationOptions 1188
Concrete 1188
CreateCalculationNoteRtf 1194
CreateFromBars 1194
Geometry 1189
HasUpperColumn 1189
ImportOptions 1189
IRConcrColumn fields 1187
IRConcrColumn members 1186
IRConcrColumn methods 1193
IsActive 1189
IsSelected 1190
Loads 1190
Name 1190
NumberOfElements 1190
PatternOptions 1191

(c) 2014 Autodesk Inc. All rights reserved

Reinforcement 1191
Save 1194
Steel 1191
StructureUserNo 1191
StructureUserNoCount 1192
UniqueId 1192
UpperColumn 1192
UpperColumnDY 1192
UpperColumnDZ 1193
Verify 1195
IRConcrColumnBucklingModel 1202
about IRConcrColumnBucklingModel 1202
IRConcrColumnBucklingModel fields 1203
IRConcrColumnBucklingModel members 1203
IsDirectionYOff 1204
IsDirectionZOff 1204
IsSwayY 1204
IsSwayZ 1204
IsTotalStructureHeight 1205
ky 1205
kz 1205
Ly 1205
Lz 1206
NumberOfStories 1206
TotalStructureHeight 1206
IRConcrColumnCalcOptions 1195
about IRConcrColumnCalcOptions 1195
Cover 1195
IRConcrColumnCalcOptions fields 1195
IRConcrColumnCalcOptions members 1195
IRConcrColumnDimensionType 1201
IRConcrColumnGeometry 1196
about IRConcrColumnGeometry 1196
ConcreteVolume 1197
Dim 1197
DimIsFixed 1197
IRConcrColumnGeometry fields 1196
IRConcrColumnGeometry members 1196
Section 1197
SectionName 1198
ShutteringArea 1198
IRConcrColumnImportOptions 1198
about IRConcrColumnImportOptions 1198
GroupByGeometry 1199
GroupByLevel 1199
ImportCombinations 1200
IRConcrColumnImportOptions fields 1199
IRConcrColumnImportOptions members 1199
RunCalcAuto 1200
ShowDialog 1200
IRConcrColumnLoad 1209
about IRConcrColumnLoad 1209
CaseName 1210
CaseType 1210
Fy 1210
Fz 1211
Gamma 1211
IRConcrColumnLoad fields 1209
IRConcrColumnLoad members 1209
MnsY 1211
MnsZ 1211
MyA 1212
MyB 1212
MyC 1212
MzA 1212
MzB 1213
MzC 1213
N 1213
NdN 1213
IRConcrColumnLoadCaseType 1214
IRConcrColumnLoads 1207
about IRConcrColumnLoads 1207
Add 1208
Count 1207
IRConcrColumnLoads fields 1207
IRConcrColumnLoads members 1207
IRConcrColumnLoads methods 1208
Item 1208
RemoveAll 1208
IRConcrColumnPatternOptions 1200
about IRConcrColumnPatternOptions 1200
IRConcrColumnPatternOptions fields 1201
IRConcrColumnPatternOptions members 1201
TiesToBeam 1201

(c) 2014 Autodesk Inc. All rights reserved

IRConcrColumnSectionType 1202
IRConcrConcrete 1263
about IRConcrConcrete 1263
CharacteristicStrength 1263
IRConcrConcrete fields 1263
IRConcrConcrete members 1263
IRConcrConcreteParams 1709
about IRConcrConcreteParams 1709
Ec 1710
Fc 1711
Fc_calc 1711
Fcj 1711
Ft 1712
Grade 1712
IRConcrConcreteParams fields 1710
IRConcrConcreteParams members 1709
IRConcrContinuousFooting 1168
about IRConcrContinuousFooting 1168
Activate 1174
Calculate 1174
CalculationOptions 1170
Concrete 1170
CreateCalculationNoteRtf 1174
Geometry 1171
ImportOptions 1171
IRConcrContinuousFooting fields 1169
IRConcrContinuousFooting members 1169
IRConcrContinuousFooting methods 1174
IsActive 1171
IsSelected 1171
Name 1172
NumberOfElements 1172
PatternOptions 1172
Reinforcement 1172
Save 1175
Steel 1173
StructureUserNo 1173
StructureUserNoCount 1173
UniqueId 1173
Verify 1175
IRConcrContinuousFootingGeometry 1175
about IRConcrContinuousFootingGeometry 1175
AutoNameSpans 1176
AutoNameSupports 1177
ConcreteVolume 1177
IRConcrContinuousFootingGeometry fields 1176
IRConcrContinuousFootingGeometry members 1175
IRConcrContinuousFootingGeometry methods 1178
LeftCantilever 1177
RightCantilever 1177
ShutteringArea 1178
Span 1178
SpanNumber 1178
UpdateInternalGeometryData 1179
IRConcrContinuousFootingSectionDimType 1183
IRConcrContinuousFootingSectionType 1183
IRConcrContinuousFootingSegment 1179
about IRConcrContinuousFootingSegment 1179
Dim 1179
IRConcrContinuousFootingSegment fields 1179
IRConcrContinuousFootingSegment members 1179
SectionType 1180
IRConcrContinuousFootingSpan 1180
about IRConcrContinuousFootingSpan 1180
CalculationLength 1181
IRConcrContinuousFootingSpan fields 1181
IRConcrContinuousFootingSpan members 1180
LeftSupport 1181
Length 1182
Name 1182
RightSupport 1182
Segment 1182
SegmentNumber 1183
IRConcrContinuousFootingSupportMaterialType 1184
IRConcrDeepBeam 1228
about IRConcrDeepBeam 1228
Activate 1231
Concrete 1229
CreateCalculationNoteRtf 1231
CreateFromObjects 1232
Geometry 1229
IRConcrDeepBeam fields 1229
IRConcrDeepBeam members 1228
IRConcrDeepBeam methods 1231

(c) 2014 Autodesk Inc. All rights reserved

IsActive 1229
IsSelected 1230
Name 1230
Reinforcement 1230
Save 1232
Steel 1230
UniqueId 1231
IRConcrDeepBeamGeometry 1232
about IRConcrDeepBeamGeometry 1232
IRConcrDrawing 1266
about IRConcrDrawing 1266
Group 1267
GroupCount 1267
IRConcrDrawing fields 1267
IRConcrDrawing members 1267
IRConcrDrawingBarGroup 1268
about IRConcrDrawingBarGroup 1268
Bar 1268
BarCount 1269
IRConcrDrawingBarGroup fields 1268
IRConcrDrawingBarGroup members 1268
IRConcrFooting 1106
about IRConcrFooting 1106
Activate 1113
Calculate 1113
CalculationOptions 1108
Concrete 1108
CreateCalculationNoteRtf 1113
CreateFromNodes 1113
CreateReinforcement 1114
DisplayDrawing 1114
FootRotation 1108
Geometry 1108
Ground 1109
IRConcrFooting fields 1107
IRConcrFooting members 1106
IRConcrFooting methods 1112
IsActive 1109
IsSelected 1109
Loads 1110
Name 1110
NumberOfElements 1110
Pattern 1110
Reinforcement 1111
Results 1111
Save 1114
Steel 1111
StructureUserNo 1111
StructureUserNoCount 1112
UniqueId 1112
Verify 1114
IRConcrFootingCalculationOptions 1123
about IRConcrFootingCalculationOptions 1123
Cover 1124
IRConcrFootingCalculationOptions fields 1123
IRConcrFootingCalculationOptions members 1123
NominalCover 1124
NominalPierCover 1124
PierCover 1124
IRConcrFootingDimType 1118
IRConcrFootingGeometry 1115
about IRConcrFootingGeometry 1115
ConcreteVolume 1116
GetDim 1117
IRConcrFootingGeometry fields 1115
IRConcrFootingGeometry members 1115
IRConcrFootingGeometry methods 1117
PierType 1116
SetDim 1117
Shape 1116
ShutteringArea 1116
Type 1117
IRConcrFootingGround 1119
about IRConcrFootingGround 1119
ColumnPierLevel 1120
IRConcrFootingGround fields 1120
IRConcrFootingGround members 1120
SoilLevel 1120
IRConcrFootingLoads 1121
about IRConcrFootingLoads 1121
IRConcrFootingPattern 1125
about IRConcrFootingPattern 1125
Diameter1 1126
Diameter2 1126

(c) 2014 Autodesk Inc. All rights reserved

IRConcrFootingPattern fields 1125
IRConcrFootingPattern members 1125
Spacing1Max 1126
Spacing1Min 1126
Spacing2Max 1127
Spacing2Min 1127
IRConcrFootingPierType 1128
IRConcrFootingResults 1121
about IRConcrFootingResults 1121
Ax 1122
Ay 1122
IRConcrFootingResults fields 1122
IRConcrFootingResults members 1121
Val 1122
IRConcrFootingShapeType 1118
IRConcrFootingType 1119
IRConcrFootResultType 1127
IRConcrMemberRequiredReinfCalcParams 1790
about IRConcrMemberRequiredReinfCalcParams 1790
BeamPointsType 1791
BeamPointsValue 1792
CasesALS 1792
CasesSLS 1792
CasesULS 1792
CombALS 1793
CombSLS 1793
CombULS 1793
IRConcrMemberRequiredReinfCalcParams fields 1791
IRConcrMemberRequiredReinfCalcParams members 1790
Members 1793
IRConcrMemberRequiredReinfEngine 1788
about IRConcrMemberRequiredReinfEngine 1788
Calculate 1789
GetCalculatedMembers 1790
IRConcrMemberRequiredReinfEngine fields 1789
IRConcrMemberRequiredReinfEngine members 1788
IRConcrMemberRequiredReinfEngine methods 1789
Params 1789
IRConcrMinimumReinforcementType 1779
IRConcrPlateCodeService2 1787
about IRConcrPlateCodeService2 1787
IRConcrPlateCodeService2 fields 1787
IRConcrPlateCodeService2 members 1787
RobotVersion 1787
ShowLongTermCracking 1788
IRConcrReinforceBarType 1751
IRConcrReinforceCalcType 1765
IRConcrReinforceData 1704
about IRConcrReinforceData 1704
BarDim_D1_Bot 1705
BarDim_D1_Up 1705
BarDim_D2_Bot 1706
BarDim_D2_Up 1706
CodeName 1706
Concrete 1706
Cover_Bot 1707
Cover_Up 1707
GetMainDirection 1708
IRConcrReinforceData fields 1705
IRConcrReinforceData members 1704
IRConcrReinforceData methods 1707
ReinforcingSteel 1707
SetMainDirection 1708
IRConcrReinforceData2 1765
about IRConcrReinforceData2 1765
IRConcrReinforceData2 fields 1766
IRConcrReinforceData2 members 1766
Main 1766
IRConcrReinforceDataMain 1767
about IRConcrReinforceDataMain 1767
IRConcrReinforceDataMain fields 1768
IRConcrReinforceDataMain members 1767
MembraneReinfInOneLayer 1768
MinimumReinf 1768
ReinfCalcType 1768
UnidirReinf 1769
IRConcrReinforceDirection 1709
IRConcrReinforcement 1245
about IRConcrReinforcement 1245
Bars 1246
Comment 1247
Freeze 1248
GroupGlobally 1247
GroupingMethod 1247

(c) 2014 Autodesk Inc. All rights reserved

IRConcrReinforcement fields 1246
IRConcrReinforcement members 1246
IRConcrReinforcement methods 1248
IsFrozen 1247
LockRefresh 1248
RemoveAll 1248
RemoveParametric 1249
IRConcrReinforcingBar 1249
about IRConcrReinforcingBar 1249
CreateCopy 1254
DirectionX 1250
DirectionY 1251
DirectionZ 1251
GetROBarEditor 1254
GroupNumber 1251
IRConcrReinforcingBar fields 1250
IRConcrReinforcingBar members 1249
IRConcrReinforcingBar methods 1253
IsParametric 1251
Position 1252
ROBarEditorId 1252
SetBarShape 1254
SetDirections 1255
SetROBarEditor 1255
Span 1252
SteelType 1252
Subtype 1253
Type 1253
Weight 1253
IRConcrReinforcingBars 1255
about IRConcrReinforcingBars 1255
Add 1257
Count 1256
IRConcrReinforcingBars fields 1256
IRConcrReinforcingBars members 1255
IRConcrReinforcingBars methods 1256
Item 1256
IRConcrSlab 1215
about IRConcrSlab 1215
Activate 1218
CalculationOptions 1216
Concrete 1216
CreateCalculationNoteRtf 1219
CreateFromObjects 1219
Geometry 1216
IRConcrSlab fields 1216
IRConcrSlab members 1215
IRConcrSlab methods 1218
IsSelected 1217
Name 1217
PatternOptions 1217
Reinforcement 1217
Save 1219
StartWizard 1220
Steel 1218
UniqueId 1218
IRConcrSlabCalculationOptions 1220
about IRConcrSlabCalculationOptions 1220
Cover 1220
CoverLateral 1221
IRConcrSlabCalculationOptions fields 1220
IRConcrSlabCalculationOptions members 1220
IRConcrSlabGeometry 1221
about IRConcrSlabGeometry 1221
ConcreteVolume 1222
IRConcrSlabGeometry fields 1221
IRConcrSlabGeometry members 1221
ShutteringArea 1222
IRConcrSlabPatternOptions 1222
about IRConcrSlabPatternOptions 1222
IRConcrSlabPatternOptions fields 1223
IRConcrSlabPatternOptions members 1222
ReinforcementSegment 1223
ReinforcementType 1223
IRConcrSlabRequiredReinfCalcParams 1760
about IRConcrSlabRequiredReinfCalcParams 1760
CasesACC 1761
CasesSLS 1761
CasesULS 1762
DisplayErrors 1762
ForcesReduction 1762
GloballyAvgDesginForces 1762
IRConcrSlabRequiredReinfCalcParams fields 1761
IRConcrSlabRequiredReinfCalcParams members 1760

(c) 2014 Autodesk Inc. All rights reserved

Method 1763
Panels 1763
IRConcrSlabRequiredReinfEngine 1763
about IRConcrSlabRequiredReinfEngine 1763
Calculate 1765
GetCalculatedPanels 1765
IRConcrSlabRequiredReinfEngine fields 1764
IRConcrSlabRequiredReinfEngine members 1764
IRConcrSlabRequiredReinfEngine methods 1764
Params 1764
IRConcrSlabRnfSegmentType 1227
IRConcrSlabRnfType 1226
IRConcrSlabSupportType 1227
IRConcrSlabWizard 1223
about IRConcrSlabWizard 1223
AddColumnSupport 1225
AddOpening 1225
AddOpeningNode 1225
AddSlabNode 1226
AddWallSupport 1226
Finish 1226
IRConcrSlabWizard fields 1224
IRConcrSlabWizard members 1224
IRConcrSlabWizard methods 1225
Thickness 1224
IRConcrSteel 1260
about IRConcrSteel 1260
AvailableNames 1261
AvailableStrengths 1261
BarDiameters 1262
BarsDatabasePath 1262
CharacteristicStrength 1262
IRConcrSteel fields 1261
IRConcrSteel members 1260
SelectedName 1262
SurfaceType 1262
IRConcrSteelNames 1258
about IRConcrSteelNames 1258
Count 1259
IRConcrSteelNames fields 1259
IRConcrSteelNames members 1259
Item 1259
IRConcrSteelParams 1712
about IRConcrSteelParams 1712
Ey 1713
Ft 1713
Ft_calc 1714
Fy 1714
Fy_calc 1714
Grade 1715
IRConcrSteelParams fields 1713
IRConcrSteelParams members 1713
IRConcrSteelStrengths 1264
about IRConcrSteelStrengths 1264
IRConcrSteelSurfaceType 1264
IRConcrSteelType 1260
IRConcrWall 1233
about IRConcrWall 1233
Activate 1236
Calculate 1236
Concrete 1234
CreateCalculationNoteRtf 1236
CreateFromObjects 1237
Geometry 1234
IRConcrWall fields 1234
IRConcrWall members 1233
IRConcrWall methods 1236
IsSelected 1234
Name 1235
Reinforcement 1235
Save 1237
Steel 1235
UniqueId 1235
Verify 1237
IRConcrWallGeometry 1238
about IRConcrWallGeometry 1238
Height 1239
IRConcrWallGeometry fields 1238
IRConcrWallGeometry members 1238
LeftBoundaryLength 1239
LeftBoundaryThickness 1239
Length 1239
Name 1240
Openings 1240

(c) 2014 Autodesk Inc. All rights reserved

RightBoundaryLength 1240
RightBoundaryThickness 1240
Thickness 1241
IRConcrWallOpening 1241
about IRConcrWallOpening 1241
IRConcrWallOpening fields 1242
IRConcrWallOpening members 1241
Lx 1242
Lz 1242
Name 1242
PosX 1243
PosZ 1243
IRConcrWallOpenings 1243
about IRConcrWallOpenings 1243
Add 1244
Count 1244
IRConcrWallOpenings fields 1244
IRConcrWallOpenings members 1243
IRConcrWallOpenings methods 1244
Item 1244
IRDimBuckDiagramCB71 1887
IRDimBuckDiagramEC3 1818
IRDimCalcState 1916
about IRDimCalcState 1916
GetParam 1917
GetParamValue 1917
IRDimCalcState members 1916
IRDimCalcState methods 1916
IsFlagSet 1917
SetFlag 1917
SetParam 1918
SetParamValue 1918
IRDimCalcStateFlagType 1915
IRDimCalcStateParamType 1915
IRDimCalcStateParamValue 1915
IRDimCalcStateValueType 1916
IRDimClient 1934
about IRDimClient 1934
GetRDimCodeService 1934
IRDimClient members 1934
IRDimClient methods 1934
IRDimCodeResCB71 1863
about IRDimCodeResCB71 1863
BadWidthBfUnderFire 1869
BuckCoefKMax 1869
BuckCoefKy 1869
BuckCoefKz 1870
BuckLengthComposBeamLey 1870
BuckLengthComposBeamLez 1870
BuckLengthLfy 1870
BuckLengthLfz 1871
BuckSlendComposBeamLaym 1871
BuckSlendComposBeamLazm 1871
BuckSLendLay 1871
BuckSlendLaz 1872
CoefDependOnAngle 1872
CoefDependOnHeight 1872
CoefDependOnHumAndBend 1872
CoefDependOnHumAndComp 1872
CoefDependOnWidth 1873
EfficiencyRatio 1873
FireHightInMidSpanH 1873
FireInerMomInMidSpanIy 1873
FireInerMomInMidSpanIz 1874
FireInerMomIx 1874
FireInerMomIy 1874
FireInerMomIz 1874
FireLftEdgeDistVpy 1875
FireLowEdgeDistVpz 1875
FireProtectCoefK2 1875
FireProtection 1875
FireProtectionPosition 1876
FireRedCoefK1b 1876
FireRedCoefK1h 1876
FireResistance 1876
FireRgtEdgeDistVy 1877
FireSecAreaInMidSpanS 1877
FireSecAreaS 1877
FireSeModulWy 1877
FireSeModulWZ 1878
FireShearAreaSY 1878
FireShearAreaSZ 1878
FireUprEdgeDistVz 1878
FireWidthAtMembEndB 1879

(c) 2014 Autodesk Inc. All rights reserved

FireWidthAtMembEndH 1879
FireWidthB 1879
FireWidthH 1879
FireWidthInMidSpanB 1880
IRDimCodeResCB71 fields 1866
IRDimCodeResCB71 members 1864
IsBuckY 1880
IsBuckZ 1880
LatBuckCoefKinst 1880
LatBuckLengthCoef 1881
LatBuckLengthLd 1881
LatBuckRelSlendLam 1881
LatBuckStrsCrit 1881
MatAxCompResist 1882
MatAxTensResist 1882
MatBendResist 1882
MatShearResist 1882
MatTrCompResist 1883
MatTrTensResist 1883
StrsBend 1883
StrsBendInCompEdgeMY 1883
StrsBendInCompEdgeMZ 1884
StrsBendInCurveBeam 1884
StrsComp 1884
StrsFinal 1884
StrsLftEdgeMZ 1885
StrsLowEdgeMY 1885
StrsRgtEdgeMZ 1885
StrsShearY 1885
StrsShearZ 1886
StrsTens 1886
StrsUprEdgeMY 1886
TimberType 1886
IRDimCodeResEC3 1819
about IRDimCodeResEC3 1819
AxForceExcentrEny 1827
AxForcExcentrEnz 1827
BendParamBetaMlty 1827
BendParamBetaMltz 1828
BendParamBetaMz 1828
BendParamBetMy 1828
BuckCoeffMinXi 1828
BuckCoeffXy 1829
BuckCoeffXz 1829
BuckCurveCoeffAlfy 1829
BuckCurveCoeffAlfz 1829
BuckCurveNumbY 1830
BuckCurveNumbZ 1830
BuckParamFiy 1830
BuckParamFiz 1830
BuckParamKy 1831
BuckParamKz 1831
BuckRelSlendLaby 1831
BuckRelSlendLabz 1831
BuckSLendLamy 1832
BuckStrenNbyrd 1832
BuckStrenNbzrd 1832
BuclSlendLamz 1832
ClassOfSect 1833
ClassOfSectElem1 1833
ClassOfSectElem2 1833
ClassOfSectElem3 1833
ClassOfSectElem4 1834
CompParamBetaA 1834
CritMomenMcr 1834
EffectiveMomenMeff 1834
EffSectAreaSeff 1835
EffSectModulWyeff 1835
EffSectModulWzeff 1835
ElastMomStrenMelyrd 1835
ElastMomStrenMelzrd 1836
ElastSectModulWyel 1836
ElastSectModulWzel 1836
FlangeAreaAf 1836
FlangeAreaAw 1837
InteractParamAlfa 1837
InteractParamBeta 1837
InteractParamMilt 1837
InteractParamMiy 1838
InteractParamMiz 1838
IRDimCodeResEC3 fields 1823
IRDimCodeResEC3 members 1819
IsBuckY 1838
IsBuckZ 1838

(c) 2014 Autodesk Inc. All rights reserved

LaodLevel 1839
LatBuckCoeffXlt 1839
LatBuckLengthLd 1839
LatBuckMomStrenmbrd 1839
LatBuckParamBetaW 1840
LatBuckParamC1 1840
LatBuckParamC2 1840
LatBuckParamFilt 1840
LatBuckParamKlt 1841
LatBuckParamLd 1841
LatBuckSlendLamlt 1841
LowFlanSlend1 1841
LowFlanSlend2 1842
MaEffRatio 1842
MaterCoeffGamma0 1842
MaterCoeffGamma1 1842
MaterCoeffGamma2 1843
MaterialCapacityFy 1843
MaxBuckSlend 1843
MomStrengthMpzEuro 1843
OverallBuckStrenNbrd 1844
PartEffRatio1 1844
PartEffRatio2 1844
PartEffRatio3 1844
PartEffRatio4 1845
PartEffRatio5 1845
PlastAxForcStrenNplrd 1845
PlastCompStrenNcrd 1845
PlastMomStrenMplyrd 1846
PlastMomStrenMplzrd 1846
PlastSectModulWypl 1846
PlastSectModulWzpl 1846
PlastTensStrenNtrd 1847
ReducMomStrenMcyrd 1847
ReducMomStrenMczrd 1847
ReducMomStrenMnyrd 1847
ReducMomStrenMnzrd 1848
ReducMomStrenMvyrd 1848
ReducMomStrenMvzrd 1848
ShearStrenTplyrd 1848
ShearStrenTplzrd 1849
StrsComp 1849
StrsLftEdgeMZ 1849
StrsLowEdgeMY 1849
StrsRgtEdgeMZ 1850
StrsShearY 1850
StrsShearZ 1850
StrsTens 1850
StrsUprEdgeMY 1851
TensStrenNurd 1851
TorsMomInertIt 1851
UppFlanSlend 1851
WarpingConstantIw 1852
WebSlend 1852
WebSlend1 1852
IRDimCodeService 1932
about IRDimCodeService 1932
EditMembDef 1933
GetDefaultMembDef 1933
GetMembCalc 1934
IRDimCodeService members 1933
IRDimCodeService methods 1933
IRDimEffDef 1899
about IRDimEffDef 1899
Clear 1902
IRDimEffDef fields 1900
IRDimEffDef members 1899
IRDimEffDef methods 1901
MX 1900
MY 1900
MZ 1901
N 1901
QY 1901
QZ 1901
Read_IntPsEff_M_1P4L 1903
Read_IntPsEff_M_3P4L 1903
Read_IntPsEff_M_MID 1903
Read_IntPsEff_M1 1903
Read_IntPsEff_M12 1903
Read_IntPsEff_M2 1904
Read_IntPsEff_MN_MAX 1904
Read_IntPsEff_MP_MAX 1904
Read_M_1P4L 1904
Read_M_3P4L 1904

(c) 2014 Autodesk Inc. All rights reserved

Read_M_MID 1905
Read_M1 1905
Read_M12 1905
Read_M2 1905
Read_MN_MAX 1905
Read_MP_MAX 1906
ReadParam 1906
WriteForces 1906
WriteIntPsEffSet1 1906
WriteIntPsEffSet2 1907
WriteParam 1907
WriteValuesSet1 1907
WriteValuesSet2 1907
IRDimEffDefDirType 1898
IRDimEffDefIntPsType 1898
IRDimEffDefParamType 1898
IRDimFireMemberPositionsCB71 1888
IRDimFireProtectionTimesCB71 1889
IRDimLatBuckCoeffDiagramEC3 1818
IRDimLaterBuckTypeCB71 1887
IRDimLaterBuckTypeEC3 1817
IRDimLoadLevelCB71 1888
IRDimLoadLevelEC3 1817
IRDimLoadTypeCB71 1888
IRDimLoadTypeEC3 1817
IRDimMatDef 1908
about IRDimMatDef 1908
IRDimMatDef fields 1909
IRDimMatDef members 1909
IRDimMatDef methods 1910
Name 1909
ReadDoubleExtraValue 1910
ReadLongExtraValue 1911
ReadValue 1911
SecondName 1910
SetNames 1911
Type 1910
WriteDoubleExtraValue 1911
WriteLongExtraValue 1911
WriteValue 1912
IRDimMatDefDblExValType 1908
IRDimMatDefLongExValType 1908
IRDimMatDefType 1908
IRDimMatDefValType 1908
IRDimMembCalc 1928
about IRDimMembCalc 1928
CalculBuckling 1930
CalculMember 1931
GetRatio 1931
GetResultsInterface 1931
IRDimMembCalc fields 1929
IRDimMembCalc members 1928
IRDimMembCalc methods 1930
IsExtraMomentsSet1 1929
IsExtraMomentsSet2 1929
IsIntPointsMomentsSet1 1930
IsIntPointsMomentsSet2 1930
SetCalcState 1931
SetEffDef 1931
SetMatDef 1932
SetMembDef 1932
SetProfDef 1932
IRDimMembCalcBuckType 1919
IRDimMembCalcRetValue 1919
IRDimMembDef 1894
about IRDimMembDef 1894
ClientID 1895
DeflYZRelLimit 1896
DisplXYRelLimit 1896
IRDimMembDef fields 1895
IRDimMembDef members 1894
IRDimMembDef methods 1896
IsBuckCoefConst 1897
IsDeflectionYZ 1897
IsDisplacementXY 1897
Length 1895
LengthYZUV 1897
MatType 1895
Name 1895
Retrieve 1897
Store 1898
Type 1896
IRDimMembDefBucklingDataType 1893
IRDimMembDefDeflDataType 1894

(c) 2014 Autodesk Inc. All rights reserved

IRDimMembDefDispDataType 1893
IRDimMembDefLengthDataType 1893
IRDimMembDefMatType 1893
IRDimMembDefType 1892
IRDimMembParamsCB71 1853
about IRDimMembParamsCB71 1853
ArcBeamCheck 1856
ArcRadius 1856
BuckComposBeamCoefCey 1856
BuckComposBeamCoefCez 1857
BuckIsComposedY 1857
BuckIsComposedZ 1857
BuckLengthCoeffZ 1857
BucklingDiagramY 1858
BucklingDiagramZ 1858
BucklLengthCoeffY 1858
FireContinProtForComposBeam 1858
FireLftSideProt 1859
FireLowSideProt 1859
FireMembPosition 1859
FireProtectionTime 1859
FireRequiredResist 1860
FireRgtSideProt 1860
FireUprSideProt 1860
ForceDistFromBeginDistX 1861
HumChangeDeltaH 1861
InflDeflCoefTheta 1861
IRDimMembParamsCB71 fields 1855
IRDimMembParamsCB71 members 1854
LatBuckType 1861
LatCoeffLowerFlangeValue 1862
LatCoeffUpperFlangeValue 1862
LoadLevel 1862
LoadTypeY 1862
TensAreaNetGros 1863
TubeControl 1863
IRDimMembParamsEC3 1809
about IRDimMembParamsEC3 1809
BuckLengthCoeffY 1811
BuckLengthCoeffZ 1811
BucklingDiagramY 1811
BucklingDiagramZ 1812
HotRolledPipes 1812
IRDimMembParamsEC3 fields 1810
IRDimMembParamsEC3 members 1809
LatBuckType 1812
LatCoeffLowerFlange 1812
LatCoeffLowerFlangeValue 1813
LatCoeffUpperFlange 1813
LatCoeffUpperFlangeValue 1813
LoadLevel 1813
LoadLevelValue 1814
LoadTypeY 1814
LoadTypeZ 1814
MaterCoeffGamma0 1815
MaterCoeffGamma1 1815
RelLimitDeflUy 1815
RelLimitdeflUz 1815
TensAreaNetGros 1816
TubeControl 1816
YieldStrengthType 1816
YieldStrengthValue 1816
IRDimMembRes 1920
about IRDimMembRes 1920
BlockCount 1924
CreateResWnd 1924
EffRatio 1922
GetEffDefAccess 1924
GetLineComponent 1925
GetLineType 1925
GetMatDefAccess 1925
GetMaxLineNo 1925
GetMembDefAccess 1926
GetProfDefAccess 1926
IRDimMembRes fields 1921
IRDimMembRes members 1920
IRDimMembRes methods 1923
IsLineActive 1926
IsStatement 1926
Language 1922
Ratio 1922
RecognizedPQ 1926
RefreshUnits 1927
ReplaceMark 1927

(c) 2014 Autodesk Inc. All rights reserved

ResOfCalc 1922
Retrieve 1927
RtfFileName 1922
SetEffDef 1927
SetMatDef 1927
SetMembDef 1928
SetProfDef 1928
SlendY 1923
SlendZ 1923
Units 1923
IRDimMembResTableComp 1920
IRDimMembResTableLineType 1920
IRDimMembSrv 1918
about IRDimMembSrv 1918
CheckLabelName 1919
IRDimMembSrv members 1918
IRDimMembSrv methods 1918
Save 1919
IRDimProfDef 1913
about IRDimProfDef 1913
Clear 1914
IRDimProfDef fields 1913
IRDimProfDef members 1913
IRDimProfDef methods 1914
IsVar 1913
Name 1914
ReadValue 1914
Type 1914
WriteValue 1915
IRDimProfDefItemType 1912
IRDimProfDefType 1912
IRDimProfDefValType 1912
IRDimServer 1937
about IRDimServer 1937
IRDimStream 1889
about IRDimStream 1889
Clear 1890
IRDimStream members 1890
IRDimStream methods 1890
ReadDouble 1891
ReadLong 1891
ReadText 1891
SeekSet 1891
Size 1891
WriteDouble 1892
WriteLong 1892
WriteText 1892
IRDimStreamType 1889
IRDimUnits 1935
about IRDimUnits 1935
AreRobotUnits 1936
Format 1936
IRDimUnits fields 1935
IRDimUnits members 1935
IRDimUnits methods 1936
ReadToUserCoef 1936
ReadUserName 1937
Refresh 1937
IRDimUnitType 1935
IRDimYieldStrengthTypeEC3 1852
IRJointAnchor 2004
about IRJointAnchor 2004
AnchorPlate 2005
Bolts 2005
IRJointAnchor fields 2004
IRJointAnchor members 2004
Tige 2005
Type 2005
IRJointAnchorBolt 2001
about IRJointAnchorBolt 2001
IRJointAnchorBolt fields 2001
IRJointAnchorBolt members 2001
Length1 2001
Length2 2002
Length3 2002
Length4 2002
IRJointAnchorPlate 2002
about IRJointAnchorPlate 2002
IRJointAnchorPlate fields 2003
IRJointAnchorPlate members 2003
Type 2003
IRJointAnchorPlateType 2019
IRJointAnchorType 2003
IRJointAngle 2033

(c) 2014 Autodesk Inc. All rights reserved

about IRJointAngle 2033
DistFromUpperBeamEdge 2034
IRJointAngle fields 2034
IRJointAngle members 2034
Length 2034
Profile 2035
ProfilePosition 2035
Type 2035
IRJointAngleBolts 2030
about IRJointAngleBolts 2030
Area 2031
ClassName 2031
Diameter 2031
DiameterName 2032
DistFromUpperEdge 2032
DistFromWeb 2032
Friction 2033
IRJointAngleBolts fields 2031
IRJointAngleBolts members 2030
Rows 2033
Spacing 2033
IRJointAngleLoad 2041
about IRJointAngleLoad 2041
Fz 2041
IRJointAngleLoad fields 2041
IRJointAngleLoad members 2041
IRJointAngleProfilePosition 2040
IRJointAngleType 2035
IRJointAngleUnit 2179
IRJointBeamCut 2036
about IRJointBeamCut 2036
HeightLower 2037
HeightUpper 2037
IRJointBeamCut fields 2036
IRJointBeamCut members 2036
Length 2037
IRJointBeamGirder 2139
about IRJointBeamGirder 2139
AngleLeft 2141
AngleRight 2141
BeamLeft 2142
BeamRight 2142
DistGirderFlangeToBeamFlangeLeft 2142
DistGirderFlangeToBeamFlangeRight 2143
DistGirderToBeamLeft 2143
DistGirderToBeamRight 2143
ElementType 2143
Girder 2144
IRJointBeamGirder fields 2141
IRJointBeamGirder members 2140
PlateLeft 2144
PlateRight 2144
SeatLeftDown 2144
SeatLeftUp 2145
SeatRightDown 2145
SeatRightUp 2145
StiffenerLeft 2145
StiffenerRight 2146
IRJointBeamGirderAngle 2153
about IRJointBeamGirderAngle 2153
BoltsBeam 2154
BoltsGirder 2154
Element 2154
IRJointBeamGirderAngle fields 2154
IRJointBeamGirderAngle members 2153
IRJointBeamGirderBeam 2155
about IRJointBeamGirderBeam 2155
CutEnd 2155
IRJointBeamGirderBeam fields 2155
IRJointBeamGirderBeam members 2155
Profile 2155
IRJointBeamGirderBolts 2124
about IRJointBeamGirderBolts 2124
Area 2125
ClassName 2125
Cols 2126
Diameter 2126
DiameterName 2126
DistFromUpperBeamEdge 2126
DistFromVertBeamEdge 2127
Friction 2127
IRJointBeamGirderBolts fields 2125
IRJointBeamGirderBolts members 2124
Rows 2127

(c) 2014 Autodesk Inc. All rights reserved

SpacingH 2128
SpacingV 2128
IRJointBeamGirderLoad 2146
about IRJointBeamGirderLoad 2146
IRJointBeamGirderLoad fields 2147
IRJointBeamGirderLoad members 2146
LFx 2147
LFz 2147
LMy 2147
RFx 2148
RFz 2148
RMy 2148
IRJointBeamGirderPlate 2132
about IRJointBeamGirderPlate 2132
BoltsBeam 2133
ConnectorsType 2134
DistFromUpperBeamEdge 2134
IRJointBeamGirderPlate fields 2133
IRJointBeamGirderPlate members 2133
Length 2134
Material 2135
Thick 2135
WeldsBeam 2135
WeldsGirder 2135
Width 2136
IRJointBeamGirderSeat 2136
about IRJointBeamGirderSeat 2136
BoltsBeam 2137
BoltsGirder 2137
ConnectorsToWebType 2138
IRJointBeamGirderSeat fields 2137
IRJointBeamGirderSeat members 2136
Length 2138
Material 2138
Section 2139
WeldsGirder 2139
IRJointBeamGirderSeatBolts 2149
about IRJointBeamGirderSeatBolts 2149
Area 2150
ClassName 2150
Cols 2150
Diameter 2151
DiameterName 2151
DistFromPerpendicularArmEdge 2151
DistFromVertEdge 2151
Friction 2152
IRJointBeamGirderSeatBolts fields 2149
IRJointBeamGirderSeatBolts members 2149
IRJointBeamGirderStiffener 2128
about IRJointBeamGirderStiffener 2128
BoltsBeam 2129
ConnectorsType 2130
CutHeightDown 2130
CutHeightUp 2130
CutLength 2131
IRJointBeamGirderStiffener fields 2129
IRJointBeamGirderStiffener members 2129
Length 2131
Material 2131
Thick 2132
WeldsBeam 2132
IRJointBearingPlate 2013
about IRJointBearingPlate 2013
IRJointBearingPlate fields 2014
IRJointBearingPlate members 2014
ThickBearingBar 2014
IRJointBolts 2166
about IRJointBolts 2166
Area 2167
ClassName 2167
Cols 2167
Diameter 2167
DiameterName 2168
Friction 2168
Height1 2168
IRJointBolts fields 2166
IRJointBolts members 2166
Rows 2168
IRJointBoltType 2169
IRJointColumnBasePlateCalcModel 1998
IRJointColumnBracket 2016
about IRJointColumnBracket 2016
Exist 2019
Height 2017

(c) 2014 Autodesk Inc. All rights reserved

IRJointColumnBracket fields 2017
IRJointColumnBracket members 2017
IRJointColumnBracket methods 2019
Length 2018
Thick 2018
VThick 2018
Width 2018
IRJointColumnSquare 2024
about IRJointColumnSquare 2024
Diameter 2025
IRJointColumnSquare fields 2024
IRJointColumnSquare members 2024
Length 2025
Profile 2025
IRJointComponentType 2180
IRJointConcreteColumn 1990
about IRJointConcreteColumn 1990
Base 1992
CalcModel 1992
Depth 1992
Foundation 1992
IRJointConcreteColumn fields 1991
IRJointConcreteColumn members 1991
Materials 1993
Profile 1993
SpreadFootingType 1993
IRJointConcreteColumnFoundation 1994
about IRJointConcreteColumnFoundation 1994
IRJointConcreteColumnFoundation fields 1994
IRJointConcreteColumnFoundation members 1994
par_a1 1995
par_b 1995
par_b1 1995
par_b2 1995
par_h 1996
par_h1 1996
par_h2 1996
par_l 1996
par_l1 1997
par_l2 1997
par_o1 1997
par_o2 1997
IRJointConcreteColumnLoad 1998
about IRJointConcreteColumnLoad 1998
Fx 1999
FxAssemb 1999
Fy 2000
Fz 2000
IRJointConcreteColumnLoad fields 1999
IRJointConcreteColumnLoad members 1998
My 2000
Mz 2000
IRJointConcreteMaterials 1989
about IRJointConcreteMaterials 1989
CoeffBA 1990
Dosage 1990
IRJointConcreteMaterials fields 1989
IRJointConcreteMaterials members 1989
Sigma 1990
IRJointConnection 2159
about IRJointConnection 2159
GetFromRobot 2161
IRJointConnection fields 2160
IRJointConnection members 2160
IRJointConnection methods 2161
SetToRobot 2161
Type 2160
WFRelPos 2161
IRJointConnectionDefType 2156
IRJointConnectionInfo 2157
about IRJointConnectionInfo 2157
DefType 2158
Elements 2158
IRJointConnectionInfo fields 2157
IRJointConnectionInfo members 2157
Node 2158
Number 2158
Type 2159
UniqueId 2159
IRJointConnectionServer 2169
about IRJointConnectionServer 2169
Calculate 2171
CalculateNote 2171
Count 2170

(c) 2014 Autodesk Inc. All rights reserved

Create 2172
CreateInfo 2172
Delete 2172
Exist 2173
Find 2173
FindWithId 2173
Get 2173
GetAllNumbers 2174
GetInfo 2174
IRJointConnectionServer fields 2170
IRJointConnectionServer members 2170
IRJointConnectionServer methods 2171
IRJointConnectionType 2156
IRJointConnectorsType 2152
IRJointElementType 2152
IRJointExtType 2180
IRJointFixedColumnBase 1982
about IRJointFixedColumnBase 1982
Anchor 1983
Base 1983
BasePlateMaterial 1984
Bolts 1984
ComplexStiff 1984
FootPlate 1984
IRJointFixedColumnBase fields 1983
IRJointFixedColumnBase members 1982
Materials 1985
NodeNumber 1985
Profile 1985
SimpleStiff 1985
StiffType 1986
Washer 1986
Wedge 1986
Welds 1986
IRJointFixedColumnBaseStiffType 1988
IRJointFixedFootWelds 1987
about IRJointFixedFootWelds 1987
FootPlate 1987
IRJointFixedFootWelds fields 1987
IRJointFixedFootWelds members 1987
Stiff 1988
Washer 1988
Wedge 1988
IRJointFixedLoad 1980
about IRJointFixedLoad 1980
Fx 1980
Fy 1981
Fz 1981
IRJointFixedLoad fields 1980
IRJointFixedLoad members 1980
My 1981
Mz 1981
IRJointFootBolts 2006
about IRJointFootBolts 2006
Area 2007
ClassName 2007
Diameter 2007
DiameterName 2007
Distance 2008
Friction 2008
IRJointFootBolts fields 2006
IRJointFootBolts members 2006
Rows 2008
SpacingH 2008
SpacingV 2009
IRJointFootMaterials 2022
about IRJointFootMaterials 2022
CementAmount 2023
ConcrClass 2023
ConcrSteelCoeff 2023
IRJointFootMaterials fields 2022
IRJointFootMaterials members 2022
PlateSigma 2023
PlateYoung 2024
IRJointFootPlate 2009
about IRJointFootPlate 2009
Diameter 2010
IRJointFootPlate fields 2010
IRJointFootPlate members 2010
Type 2010
IRJointFootPlateType 2009
IRJointFootStiffenerComplex 2027
about IRJointFootStiffenerComplex 2027
Height 2028

(c) 2014 Autodesk Inc. All rights reserved

IRJointFootStiffenerComplex fields 2028
IRJointFootStiffenerComplex members 2028
Length 2028
ThickPlateHor 2029
ThickStiff 2029
IRJointFootStiffenerHoriz 2021
about IRJointFootStiffenerHoriz 2021
IRJointFootStiffenerHoriz fields 2021
IRJointFootStiffenerHoriz members 2021
Thick 2021
IRJointFootStiffenerSimple 2025
about IRJointFootStiffenerSimple 2025
Height 2026
IRJointFootStiffenerSimple fields 2026
IRJointFootStiffenerSimple members 2026
Length 2026
Thick 2027
Type 2027
Width 2027
IRJointFootStiffenerVert 2019
about IRJointFootStiffenerVert 2019
IRJointFootStiffenerVert fields 2020
IRJointFootStiffenerVert members 2020
Length 2020
Thick 2020
WidthSpacing 2021
IRJointFootStiffType 2029
IRJointFootWelds 2014
about IRJointFootWelds 2014
Bearing 2015
FootPlate 2015
IRJointFootWelds fields 2015
IRJointFootWelds members 2015
Stiff 2016
Washer 2016
Wedge 2016
IRJointGussetBoltsDiag 2060
about IRJointGussetBoltsDiag 2060
BoltAxisShift 2061
ClassName 2061
Diameter 2062
DistanceH1 2062
Friction 2062
IRJointGussetBoltsDiag fields 2061
IRJointGussetBoltsDiag members 2060
Rows 2062
Spacing 2063
IRJointGussetCornersType 2078
IRJointGussetCross 2096
about IRJointGussetCross 2096
BoltsClassVector 2097
BoltsDimensionNamesVector 2098
BoltsDimensionsVector 2098
DiagLeftLower 2098
DiagLeftUpper 2099
DiagRightLower 2099
DiagRightUpper 2099
FixType_LeftL_RightU 2099
FixType_LeftLower 2100
FixType_LeftU_RightL 2100
FixType_LeftUpper 2100
FixType_RightLower 2101
FixType_RightUpper 2101
GussetPlate 2101
IRJointGussetCross fields 2097
IRJointGussetCross members 2096
ProfileCutting 2101
IRJointGussetCrossLoad 2118
about IRJointGussetCrossLoad 2118
Fx_LeftLower 2119
Fx_LeftUpper 2119
Fx_RightLower 2120
Fx_RightUpper 2120
IRJointGussetCrossLoad fields 2119
IRJointGussetCrossLoad members 2119
IRJointGussetCrossPlate 2087
about IRJointGussetCrossPlate 2087
B1 2089
B2 2089
B3 2089
B4 2089
CornersType 2090
CutH1 2090
CutH2 2090

(c) 2014 Autodesk Inc. All rights reserved

CutH3 2091
CutH4 2091
CutV1 2091
CutV2 2092
CutV3 2092
CutV4 2092
H1 2093
H2 2093
H3 2093
H4 2093
Height 2094
HOffset 2094
HorizontalOffset 2094
IRJointGussetCrossPlate fields 2088
IRJointGussetCrossPlate members 2087
Length 2095
Material 2095
Thick 2095
VOffset 2095
IRJointGussetCrossProfileCutting 2087
IRJointGussetDiagonale 2065
about IRJointGussetDiagonale 2065
BarNumber 2066
BoltsDiag 2066
DistanceEC 2066
Exist 2067
IRJointGussetDiagonale fields 2065
IRJointGussetDiagonale members 2065
Position 2067
Profile 2067
WeldsDiag 2067
IRJointGussetDiagonalePositionType 2122
IRJointGussetFixType 2068
IRJointGussetFlange 2112
about IRJointGussetFlange 2112
BoltsClassVector 2113
BoltsDimensionNamesVector 2114
BoltsDimensionsVector 2114
DiagonalLeft 2114
DiagonalRight 2115
DiagonalUp 2115
FixType_DiagonalLeft 2115
FixType_DiagonalRight 2115
FixType_DiagonalUp 2116
FixType_FlangeLeft 2116
FixType_FlangeRight 2116
FlangeLeft 2116
FlangeRight 2117
GussetPlate 2117
IRJointGussetFlange fields 2113
IRJointGussetFlange members 2112
ProfileCutting 2117
IRJointGussetFlangeLoad 2120
about IRJointGussetFlangeLoad 2120
Fx_FlangeLeft 2121
Fx_FlangeRight 2121
Fx_LeftUpper 2122
Fx_RightUpper 2122
Fx_Upper 2122
IRJointGussetFlangeLoad fields 2121
IRJointGussetFlangeLoad members 2120
IRJointGussetFlangePlate 2102
about IRJointGussetFlangePlate 2102
B1 2104
B2 2104
B3 2104
B4 2104
CornersType 2105
CutH1 2105
CutH2 2105
CutH3 2106
CutH4 2106
CutV1 2106
CutV2 2107
CutV3 2107
CutV4 2107
EH 2108
EV 2108
Geight 2108
H1 2108
H2 2109
H3 2109
H4 2109
HOffset 2109

(c) 2014 Autodesk Inc. All rights reserved

IRJointGussetFlangePlate fields 2103
IRJointGussetFlangePlate members 2102
Length 2110
Material 2110
NewPlateDefinition 2110
Thick 2111
Type 2111
VOffset 2111
IRJointGussetFlangePlateRegularType 2123
IRJointGussetFlangeProfileCutting 2102
IRJointGussetSimple 2068
about IRJointGussetSimple 2068
Attachment 2069
BoltsClassVector 2070
BoltsDimensionNamesVector 2070
BoltsDimensionsVector 2070
Diagonale 2071
FixType 2071
GussetPlate 2071
IRJointGussetSimple fields 2069
IRJointGussetSimple members 2068
ProfilePosition 2071
IRJointGussetSimpleAttachBolts 2080
about IRJointGussetSimpleAttachBolts 2080
ClassName 2081
Diameter 2081
Friction 2081
Horizontal 2081
IRJointGussetSimpleAttachBolts fields 2080
IRJointGussetSimpleAttachBolts members 2080
Vertical 2082
IRJointGussetSimpleAttachBoltsHorizontal 2082
about IRJointGussetSimpleAttachBoltsHorizontal 2082
DistanceEB 2083
DistanceH1 2083
IRJointGussetSimpleAttachBoltsHorizontal fields 2082
IRJointGussetSimpleAttachBoltsHorizontal members 2082
Rows 2083
Spacing 2083
IRJointGussetSimpleAttachBoltsVertical 2084
about IRJointGussetSimpleAttachBoltsVertical 2084
DistanceEB 2084
DistanceH1 2085
IRJointGussetSimpleAttachBoltsVertical fields 2084
IRJointGussetSimpleAttachBoltsVertical members 2084
Rows 2085
Spacing 2085
IRJointGussetSimpleAttachment 2078
about IRJointGussetSimpleAttachment 2078
BoltsAttach 2079
FixType 2079
IRJointGussetSimpleAttachment fields 2078
IRJointGussetSimpleAttachment members 2078
WeldsAttach 2079
WeldType 2080
IRJointGussetSimpleAttachWelds 2085
about IRJointGussetSimpleAttachWelds 2085
IRJointGussetSimpleAttachWelds fields 2086
IRJointGussetSimpleAttachWelds members 2086
ThickEdgeA 2086
ThickEdgeB 2086
IRJointGussetSimpleLoad 2117
about IRJointGussetSimpleLoad 2117
Fx 2118
IRJointGussetSimpleLoad fields 2118
IRJointGussetSimpleLoad members 2118
IRJointGussetSimplePlate 2072
about IRJointGussetSimplePlate 2072
CornersType 2073
DistanceD 2073
DistanceEH 2073
DistanceEV 2074
H1 2074
H2 2074
H3 2075
H4 2075
Height 2075
IRJointGussetSimplePlate fields 2072
IRJointGussetSimplePlate members 2072
Material 2076
Thick 2076
V1 2076
V2 2076
V3 2077

(c) 2014 Autodesk Inc. All rights reserved

V4 2077
Width 2077
IRJointGussetSimpleProfilePosition 2068
IRJointGussetWeldsDiag 2063
about IRJointGussetWeldsDiag 2063
IRJointGussetWeldsDiag fields 2063
IRJointGussetWeldsDiag members 2063
Length1 2064
Length2 2064
ThickCorner 2064
ThickFlange 2065
IRJointKnee 1951
about IRJointKnee 1951
Beam 1954
Bolts 1954
BracketLow 1955
BracketUp 1955
Column 1955
DefComponentMaterial 1955
FixType 1956
FlanPlateLower 1956
FlanPlatePosLower 1956
FlanPlatePosUpper 1957
FlanPlateUpper 1957
IRJointKnee fields 1953
IRJointKnee members 1951
IsBolted 1957
IsDefComponentMaterialSet 1958
KneeType 1958
Material 1958
MaterPlates 1958
Plate 1959
PlatePosition 1959
PlatePositionLow 1959
StiffDiag 1959
StiffLow 1960
StiffTypeLow 1960
StiffTypeUp 1960
StiffUp 1960
TensionPlateLow 1961
TensionPlateUp 1961
VStiffLow 1961
VStiffUp 1961
WebBeamStiffenerVLower 1962
WebBeamStiffenerVUpper 1962
WebColumnStiffenerHLower 1962
WebColumnStiffenerHUpper 1963
WebPlate 1963
WebStiffType 1963
WeldBracketDownFlange 1963
WeldBracketUpFlange 1964
WeldFlange 1964
WeldStiff 1964
WeldWeb 1964
IRJointKneeBolts 1939
about IRJointKneeBolts 1939
ClassName 1940
Cols 1941
DiameterName 1941
EqualSpac 1941
Height1 1942
IRJointKneeBolts fields 1940
IRJointKneeBolts members 1940
Rows 1942
SpacingH 1942
SpacingV 1943
Symmetry 1943
IRJointKneeBracket 1943
about IRJointKneeBracket 1943
Angle 1944
AngleExt 1947
Exist 1944
Height 1945
IRJointKneeBracket fields 1944
IRJointKneeBracket members 1943
IRJointKneeBracket methods 1946
Length 1945
Material 1945
ThickFlange 1946
ThickWeb 1946
Width 1946
IRJointKneeDiagonalStiff 1965
about IRJointKneeDiagonalStiff 1965
Exist 1966

(c) 2014 Autodesk Inc. All rights reserved

IRJointKneeDiagonalStiff fields 1966
IRJointKneeDiagonalStiff members 1965
Material 1966
Thick 1966
Type 1967
IRJointKneeDiagonalStiffType 1965
IRJointKneeFixType 1950
IRJointKneeLoad 1968
about IRJointKneeLoad 1968
IRJointKneeLoad fields 1968
IRJointKneeLoad members 1968
M 1969
N 1969
Q 1969
IRJointKneeMaterials 1969
IRJointKneeReinfType 1951
IRJointKneeStiffColumn 1967
about IRJointKneeStiffColumn 1967
IRJointKneeStiffColumn fields 1967
IRJointKneeStiffColumn members 1967
Thick 1968
IRJointKneeType 1950
IRJointKneeWebPlate 1947
about IRJointKneeWebPlate 1947
Bolts 1948
Height 1948
IRJointKneeWebPlate fields 1948
IRJointKneeWebPlate members 1947
Material 1948
Thick 1949
Type 1949
WeldLong 1949
WeldTran 1949
Width 1950
IRJointKneeWebStiffType 1965
IRJointLoad 2178
about IRJointLoad 2178
Cases 2178
IRJointLoad fields 2178
IRJointLoad members 2178
Type 2179
IRJointLoadType 2179
IRJointPinnedColumnBase 1971
about IRJointPinnedColumnBase 1971
Anchor 1973
Base 1973
BasePlateMaterial 1973
Bearing 1974
BearingPlate 1974
Bolts 1974
FootPlate 1974
IRJointPinnedColumnBase fields 1972
IRJointPinnedColumnBase members 1972
Materials 1975
NodeNumber 1975
Profile 1975
Square 1975
StiffHoriz 1976
StiffType 1976
StiffVert 1976
Washer 1976
Wedge 1977
Welds 1977
IRJointPinnedColumnBaseStiffType 1971
IRJointPinnedLoad 1977
about IRJointPinnedLoad 1977
IRJointPinnedLoad fields 1978
IRJointPinnedLoad members 1977
Nc 1978
Nt 1978
NTy 1978
NTz 1979
Ty 1979
Tz 1979
IRJointPlate 2163
about IRJointPlate 2163
AngleExt 2165
Exist 2164
IRJointPlate fields 2163
IRJointPlate members 2163
IRJointPlate methods 2165
Length 2164
Material 2164
Thick 2165

(c) 2014 Autodesk Inc. All rights reserved

Width 2165
IRJointProfile 2174
about IRJointProfile 2174
Angle 2175
AngleExt 2177
BarNumber 2175
Edit 2177
EditComponent 2177
IRJointProfile fields 2175
IRJointProfile members 2175
IRJointProfile methods 2176
Material 2176
Section 2176
IRJointSpreadFootingType 1993
IRJointTube 2042
about IRJointTube 2042
DiagLeftLower 2043
DiagLeftUpper 2044
DiagRightUpper 2044
Flange 2044
IRJointTube fields 2043
IRJointTube members 2042
PostUpper 2044
StiffHBracketMaterial 2045
StiffHoriz 2045
StiffLateral 2045
TubeType 2045
WeldDiag 2046
WeldStiff 2046
IRJointTubeDiagProfile 2050
about IRJointTubeDiagProfile 2050
Angle 2050
BarNumber 2051
Exist 2051
Gap 2051
IRJointTubeDiagProfile fields 2050
IRJointTubeDiagProfile members 2050
Length 2052
Material 2052
Overlap 2052
Profile 2052
Section 2053
IRJointTubeFlangeProfile 2047
about IRJointTubeFlangeProfile 2047
BarNumber 2048
Excentr 2048
Exist 2048
IRJointTubeFlangeProfile fields 2048
IRJointTubeFlangeProfile members 2047
Material 2049
Profile 2049
Section 2049
IRJointTubeLoad 2055
about IRJointTubeLoad 2055
DiagLeftLowerM 2057
DiagLeftLowerN 2057
DiagLeftUpperM 2057
DiagLeftUpperN 2057
DiagRightUpperM 2058
DiagRightUpperN 2058
FlangeM 2058
FlangeN 2058
FlangeQ 2059
IRJointTubeLoad fields 2056
IRJointTubeLoad members 2056
PostUpperN 2059
IRJointTubePostProfile 2053
about IRJointTubePostProfile 2053
BarNumber 2054
Exist 2054
IRJointTubePostProfile fields 2053
IRJointTubePostProfile members 2053
Length 2054
Material 2055
Profile 2055
Section 2055
IRJointTubeProfileType 2047
IRJointTubeType 2046
IRJointWebFlangeRelativePos 2159
IRJointWebType 2169
IRJointWedge 2011
about IRJointWedge 2011
IRJointWedge fields 2011
IRJointWedge members 2011

(c) 2014 Autodesk Inc. All rights reserved

Length 2011
Profile 2012
Thick 2012
Type 2012
Width 2012
XTypeMaterial 2013
IRJointWedgeType 2013
IRJointWeld 2162
about IRJointWeld 2162
IRJointWeld fields 2162
IRJointWeld members 2162
Strength 2162
Thick 2163
IRJointWithAngles 2037
about IRJointWithAngles 2037
Angle 2039
BeamBolts 2039
BeamCut 2039
BeamProfile 2039
ColumnBolts 2040
ColumnProfile 2040
Distance 2040
IRJointWithAngles fields 2038
IRJointWithAngles members 2038
IRobotActiveModelType 1360
IRobotAddIn 1548
about IRobotAddIn 1548
Connect 1550
Disconnect 1550
DoCommand 1551
GetExpectedVersion 1551
InstallCommands 1551
IRobotAddIn members 1549
IRobotAddIn methods 1549
IRobotAddInMngr 1551
about IRobotAddInMngr 1551
InstallCommand 1552
IRobotAddInMngr members 1552
IRobotAddInMngr methods 1552
IRobotAddInRegistrar 1556
about IRobotAddInRegistrar 1556
Guid 1557
InstallMenu 1559
IRobotAddInRegistrar fields 1557
IRobotAddInRegistrar members 1556
IRobotAddInRegistrar methods 1558
ProductName 1558
ProgId 1558
ProviderName 1558
Register 1559
Unregister 1559
IRobotAdvancedResultServer 924
about IRobotAdvancedResultServer 924
Eigenvalues 925
Eigenvectors 925
FootfallValue 927
FRF 925
IRobotAdvancedResultServer fields 925
IRobotAdvancedResultServer members 924
IRobotAdvancedResultServer methods 927
MassSum 926
SpectralCoeffs 926
TimeHistory 926
IRobotAdvancedSupportData 517
about IRobotAdvancedSupportData 517
B 518
D 518
EquivalentElasticity 519
H 519
IRobotAdvancedSupportData fields 518
IRobotAdvancedSupportData members 518
IsEquivalentElasticity 519
Type 519
IRobotAdvancedSupportType 517
IRobotApplication 1535
about IRobotApplication 1535
CmpntFactory 1537
Interactive 1537
IRobotApplication fields 1536
IRobotApplication members 1536
IRobotApplication methods 1540
Is360 1537
Kernel 1538
LicenseCheckEntitlement 1540

(c) 2014 Autodesk Inc. All rights reserved

Preferences 1538
ProgramVersion 1538
Project 1538
Quit 1540
UserControl 1539
Version 1539
Visible 1539
Window 1540
IRobotBackground 1334
about IRobotBackground 1334
Color 1335
FilePath 1335
InsertParams 1335
IRobotBackground fields 1334
IRobotBackground members 1334
IRobotBackground methods 1337
IsOn 1336
Name 1336
Number 1336
Save 1337
VisibilityRange 1337
IRobotBackgroundInsertParams 1330
about IRobotBackgroundInsertParams 1330
InsertionPoint 1331
IRobotBackgroundInsertParams fields 1331
IRobotBackgroundInsertParams members 1330
Layers 1331
Mirror 1332
Plane 1332
PositionOnNormalAxis 1332
ReferencePoint 1333
RotationAngle 1333
ScalingFactor 1333
Units 1333
IRobotBackgroundLayers 1327
about IRobotBackgroundLayers 1327
Count 1328
FindName 1329
Get 1329
IRobotBackgroundLayers fields 1328
IRobotBackgroundLayers members 1328
IRobotBackgroundLayers methods 1328
IsImported 1329
SetImported 1330
IRobotBackgroundServer 1339
about IRobotBackgroundServer 1339
Create 1340
Get 1340
GetAllNumbers 1341
IRobotBackgroundServer members 1340
IRobotBackgroundServer methods 1340
Remove 1341
RemoveAll 1341
IRobotBackgroundVisibilityRange 1338
about IRobotBackgroundVisibilityRange 1338
FromPos 1338
IRobotBackgroundVisibilityRange fields 1338
IRobotBackgroundVisibilityRange members 1338
ToPos 1339
Type 1339
IRobotBackgroundVisibilityRangeType 1337
IRobotBar 624
about IRobotBar 624
AnalyzeTTMethod 627
CalcGamma 633
ChangeOrientation 634
Elements 627
End 627
EndNode 628
Gamma 628
GetElemsData 634
GetLCS 634
GetSimpleBars 635
InactiveBar 628
IRobotBar fields 626
IRobotBar members 625
IRobotBar methods 633
IsSuperBar 629
Length 629
Name 629
NameTemplate 629
ReversedOffset 630
ReversedRelease 630
ReversedSection 630

(c) 2014 Autodesk Inc. All rights reserved

SetOffset 635
SetOrientation 635
SetSection 636
ShearForces 631
Start 631
StartNode 631
StructuralType 632
TensionCompression 632
TrussBar 632
UniqueId 633
IRobotBarBucklingData 897
about IRobotBarBucklingData 897
BuckLengthY 898
BuckLengthZ 898
CriticalCoef 898
CriticalForce 898
IRobotBarBucklingData fields 897
IRobotBarBucklingData members 897
SlendY 899
SlendZ 899
IRobotBarBucklingServer 881
about IRobotBarBucklingServer 881
CriticalCoef 881
EigenValue 882
IRobotBarBucklingServer members 881
IRobotBarBucklingServer methods 881
IRobotBarCableAssemblingParamType 541
IRobotBarCableData 539
about IRobotBarCableData 539
AssemblingParam 540
AssemblingParamValue 540
IRobotBarCableData fields 539
IRobotBarCableData members 539
MaterialName 540
SectionAX 541
IRobotBarDeadRecordValues 44
IRobotBarDeflectionData 894
about IRobotBarDeflectionData 894
IRobotBarDeflectionData fields 895
IRobotBarDeflectionData members 895
PosUX 895
PosUY 896
PosUZ 896
UX 896
UY 896
UZ 896
IRobotBarDeflectionServer 875
about IRobotBarDeflectionServer 875
DynCombMaxValue 876
DynCombValue 877
DynMaxValue 877
DynValue 877
IRobotBarDeflectionServer members 876
IRobotBarDeflectionServer methods 876
MaxValue 878
MaxValueEx 878
Value 878
ValueEx 878
IRobotBarDilatationRecordValues 41
IRobotBarDisplacementServer 882
about IRobotBarDisplacementServer 882
DynCombValue 883
DynValue 883
IRobotBarDisplacementServer members 882
IRobotBarDisplacementServer methods 883
Value 884
IRobotBarElasticGroundData 542
about IRobotBarElasticGroundData 542
GetOneDir 544
HX 543
IRobotBarElasticGroundData fields 543
IRobotBarElasticGroundData members 542
IRobotBarElasticGroundData methods 544
KY 543
KZ 544
SetOneDir 544
IRobotBarElement 652
about IRobotBarElement 652
EndNode 653
Inactive 653
IRobotBarElement fields 652
IRobotBarElement members 652
IsCalc 653
Number 653

(c) 2014 Autodesk Inc. All rights reserved

StartNode 654
IRobotBarElementData 654
about IRobotBarElementData 654
Bar 655
GetEndNode 656
GetStartNode 656
IRobotBarElementData fields 655
IRobotBarElementData members 654
IRobotBarElementData methods 655
Number 655
IRobotBarElementDataSet 656
about IRobotBarElementDataSet 656
BarCount 657
ElemCount 658
GetBarNumber 658
GetElem 658
GetElemCountForBar 659
GetElemForBar 659
IRobotBarElementDataSet fields 657
IRobotBarElementDataSet members 657
IRobotBarElementDataSet methods 658
IRobotBarEnd 648
about IRobotBarEnd 648
GetLabel 649
GetLabelName 650
GetLabels 650
GetOffsetValue 650
HasLabel 650
IRobotBarEnd fields 649
IRobotBarEnd members 648
IRobotBarEnd methods 649
Node 649
RemoveLabel 651
SetLabel 651
IRobotBarEndBracketData 621
about IRobotBarEndBracketData 621
GetValue 622
IRobotBarEndBracketData fields 622
IRobotBarEndBracketData members 621
IRobotBarEndBracketData methods 622
IsValueRelative 623
SetValue 623
Type 622
IRobotBarEndBracketDataValue 621
IRobotBarEndBracketType 621
IRobotBarEndOffsetData 619
about IRobotBarEndOffsetData 619
IRobotBarEndOffsetData fields 619
IRobotBarEndOffsetData members 619
MemberLength 620
UX 620
UY 620
UZ 620
IRobotBarEndReleaseData 582
about IRobotBarEndReleaseData 582
AX 583
AY 584
AZ 584
BX 584
BY 585
BZ 585
HX 585
HY 585
HZ 586
IRobotBarEndReleaseData fields 583
IRobotBarEndReleaseData members 582
KX 586
KY 586
KZ 586
NonlinearModel 587
RX 587
RY 587
RZ 587
UX 588
UY 588
UZ 588
IRobotBarEndReleaseValue 580
IRobotBarForceConcentrateRecordValues 41
IRobotBarForceData 888
about IRobotBarForceData 888
FX 889
FY 890
FZ 890
IRobotBarForceData fields 889

(c) 2014 Autodesk Inc. All rights reserved

IRobotBarForceData members 889
KY 890
KYAvailable 890
KZ 891
KZAvailable 891
MX 891
MY 891
MZ 891
IRobotBarForceServer 872
about IRobotBarForceServer 872
DynCombValue 873
DynCombValueByNPoints 873
DynValue 873
DynValueByNPoints 874
IRobotBarForceServer members 872
IRobotBarForceServer methods 872
Value 874
ValueByNPoints 874
ValueByNPointsEx 875
ValueEx 875
IRobotBarGeoImperfectionsAxis 592
IRobotBarGeoImperfectionsData 589
about IRobotBarGeoImperfectionsData 589
GetBucklingCoeff 590
GetUser 590
IRobotBarGeoImperfectionsData members 589
IRobotBarGeoImperfectionsData methods 589
IsAutomatic 590
IsMinus 591
SetAutomatic 591
SetMinus 591
SetUser 592
IRobotBarIntersectRelationship 2222
IRobotBarMomentDistributedRecordValues 42
IRobotBarOffsetAutoPosition 623
IRobotBarOffsetData 617
about IRobotBarOffsetData 617
AxisOffset 617
CoordinateSystem 618
End 618
IRobotBarOffsetData fields 617
IRobotBarOffsetData members 617
ObjectNumber 618
Position 618
Start 619
IRobotBarOffsetMemberLength 624
IRobotBarReleaseData 581
about IRobotBarReleaseData 581
EndNode 581
IRobotBarReleaseData fields 581
IRobotBarReleaseData members 581
StartNode 582
IRobotBarResultServer 869
about IRobotBarResultServer 869
Buckling 870
Deflections 870
Displacements 870
Forces 871
GeoImperfections 871
IRobotBarResultServer fields 870
IRobotBarResultServer members 869
IRobotBarResultServer methods 871
Stresses 871
IRobotBarSectionComplexData 572
about IRobotBarSectionComplexData 572
B 573
Count 574
D 574
Get 574
GetShape 575
GetValue 575
IRobotBarSectionComplexData fields 573
IRobotBarSectionComplexData members 573
IRobotBarSectionComplexData methods 574
Set 575
IRobotBarSectionComponentShape 576
IRobotBarSectionConcreteCutsPosition 552
IRobotBarSectionConcreteData 565
about IRobotBarSectionConcreteData 565
BeamCutsPosition 566
CalcGeometry 567
GetReduction 567
GetTapered 568
GetValue 568

(c) 2014 Autodesk Inc. All rights reserved

IRobotBarSectionConcreteData fields 566
IRobotBarSectionConcreteData members 565
IRobotBarSectionConcreteData methods 567
IsBeam 566
IsColumn 566
SetReduction 568
SetTapered 569
SetValue 569
IRobotBarSectionConcreteDataValue 551
IRobotBarSectionData 552
about IRobotBarSectionData 552
AliasCount 555
CalcNonstdGeometry 559
Concrete 555
CreateNonstd 559
DrawSymbol 560
ElasticParams 555
FindAlias 560
FindNonstd 560
GetAlias 561
GetAliasEx 561
GetNonstd 561
GetValue 562
IRobotBarSectionData fields 554
IRobotBarSectionData members 553
IRobotBarSectionData methods 558
IsConcrete 555
IsJoist 556
IsSpecial 556
LoadFromDBase 562
LoadFromDBase2 562
MaterialName 556
Members 557
Name 557
NonstdCount 557
RemoveNonstd 563
SetValue 563
ShapeType 558
Special 558
Type 558
IRobotBarSectionDataValue 549
IRobotBarSectionElasticParams 569
about IRobotBarSectionElasticParams 569
Active 570
IRobotBarSectionElasticParams fields 570
IRobotBarSectionElasticParams members 570
L1 571
L2 571
MaterialModel 571
N 571
N1 572
N2 572
IRobotBarSectionNonstdData 563
about IRobotBarSectionNonstdData 563
GetValue 564
IRobotBarSectionNonstdData fields 564
IRobotBarSectionNonstdData members 563
IRobotBarSectionNonstdData methods 564
Position 564
SetValue 564
IRobotBarSectionNonstdDataValue 550
IRobotBarSectionQuantitySurvey 1082
about IRobotBarSectionQuantitySurvey 1082
Count 1083
GetLength 1083
GetName 1084
GetPaintArea 1084
GetUnitWeight 1084
GetVolume 1085
GetWeight 1085
IRobotBarSectionQuantitySurvey fields 1083
IRobotBarSectionQuantitySurvey members 1082
IRobotBarSectionQuantitySurvey methods 1083
IRobotBarSectionShapeType 546
IRobotBarSectionSpecialData 576
about IRobotBarSectionSpecialData 576
DbFullName 577
DbName 578
GetValue 578
IRobotBarSectionSpecialData fields 577
IRobotBarSectionSpecialData members 577
IRobotBarSectionSpecialData methods 578
Section 578
SetValue 579

(c) 2014 Autodesk Inc. All rights reserved

IRobotBarSectionSpecialDataValue 579
IRobotBarSectionType 550
IRobotBarServer 636
about IRobotBarServer 636
BeginMultiOperation 640
CalcGamma 640
ChangeOrientation 641
Create 641
CreateSuperBar 641
EndMultiOperation 642
FindWithId 642
FreeNumber 638
GetAnalyzeTTMethodEnabled 642
GetElemsData 642
GetLCS 643
GetName 643
GetNameTemplate 643
GetStructuralType 644
GetUniqueId 644
IRobotBarServer fields 638
IRobotBarServer members 636
IRobotBarServer methods 639
IsInactive 644
IsTrussBar 645
NonlinearHingeModels 638
NonlinearHinges 639
SetAnalyzeTTMethod 645
SetInactive 645
SetNameTemplate 646
SetOrientation 646
SetShearForces 646
SetStructuralType 647
SetTensionCompression 647
SetTrussBar 647
Update 647
IRobotBarStressData 892
about IRobotBarStressData 892
FXSX 893
IRobotBarStressData fields 892
IRobotBarStressData members 892
ShearY 893
ShearZ 893
Smax 893
SmaxMY 893
SmaxMZ 894
Smin 894
SminMY 894
SminMZ 894
Torsion 894
IRobotBarStressServer 879
about IRobotBarStressServer 879
DynCombValue 880
DynValue 880
IRobotBarStressServer members 879
IRobotBarStressServer methods 879
Value 880
ValueEx 880
IRobotBarTensionCompression 651
IRobotBarThermalRecordValues 43
IRobotBarTrapezoidaleRecordValues 43
IRobotBarUniformRecordValues 42
IRobotBucklingAnalysisMethod 363
IRobotBucklingAnalysisParams 359
about IRobotBucklingAnalysisParams 359
Increment 361
IRobotBucklingAnalysisParams fields 360
IRobotBucklingAnalysisParams members 360
IsNonlinear 361
IterationsCount 361
Method 361
ModesCount 362
NonlinearParams 362
Shift 362
SturmVerification 363
Tolerance 363
IRobotBucklingDeformationParams 1589
about IRobotBucklingDeformationParams 1589
CaseNumber 1590
GetModeCoeff 1591
IRobotBucklingDeformationParams fields 1589
IRobotBucklingDeformationParams members 1589
IRobotBucklingDeformationParams methods 1591
MaxDisplacement 1590
OmitCaseForDeformations 1590

(c) 2014 Autodesk Inc. All rights reserved

SetModeCoeff 1591
IRobotCalcEngine 1561
about IRobotCalcEngine 1561
AnalysisParams 1562
AutoFreezeResults 1562
AutoGenerateModel 1563
BucklingDeformation 1563
Calculate 1566
CalculateEx 1566
DAM 1563
GenerateModel 1566
GenerationParams 1564
IRobotCalcEngine fields 1562
IRobotCalcEngine members 1561
IRobotCalcEngine methods 1565
SaveResultsInExternalFile 1564
SeismicResultsSaveParams 1564
StatusWindowParent 1565
StopCalculation 1567
UseStatusWindow 1565
IRobotCalcEngineEvents 1591
about IRobotCalcEngineEvents 1591
CalcMessage 1592
IRobotCalcEngineEvents members 1592
IRobotCalcEngineEvents methods 1592
IRobotCalcMessageSeverityLevel 1593
IRobotCalcMessageSource 1593
IRobotCalculationMode 1594
IRobotCalculationModelCoherence 1326
IRobotCalculationResume 1034
about IRobotCalculationResume 1034
DiagonalStiffnessMatrixMax 1035
DiagonalStiffnessMatrixMin 1035
DiagonalStiffnessMatrixPrecision 1035
GetEnergy 1036
GetEnergyPrecision 1036
IRobotCalculationResume fields 1034
IRobotCalculationResume members 1034
IRobotCalculationResume methods 1035
IRobotCalculationStatus 1593
IRobotCalculationsType 1548
IRobotCase 408
about IRobotCase 408
AnalizeType 408
IRobotCase fields 408
IRobotCase members 408
Name 409
Nature 409
Number 409
Type 409
IRobotCaseAnalizeType 410
IRobotCaseAnalysisModesFilter 435
about IRobotCaseAnalysisModesFilter 435
IRobotCaseAnalysisModesFilter fields 436
IRobotCaseAnalysisModesFilter members 436
MassPercentage 436
Modes 436
Type 437
IRobotCaseAnalysisModesFilterType 437
IRobotCaseCollection 435
about IRobotCaseCollection 435
IRobotCaseCombination 426
about IRobotCaseCombination 426
CaseFactors 427
CombinationType 427
GetAnalysisParams 430
IRobotCaseCombination fields 427
IRobotCaseCombination members 426
IRobotCaseCombination methods 429
IsAuxiliary 427
Label 428
NatureName 428
Quadratic 428
SeismicType 429
SetAnalysisParams 430
SetNatureExt 430
Signed 429
UniqueId 429
IRobotCaseFactor 434
about IRobotCaseFactor 434
CaseNumber 434
Factor 434
IRobotCaseFactor fields 434
IRobotCaseFactor members 434

(c) 2014 Autodesk Inc. All rights reserved

IRobotCaseFactorMngr 431
about IRobotCaseFactorMngr 431
Count 431
Delete 432
Get 432
IRobotCaseFactorMngr fields 431
IRobotCaseFactorMngr members 431
IRobotCaseFactorMngr methods 432
New 432
IRobotCaseNature 410
IRobotCaseRelatedValueType 446
IRobotCaseServer 411
about IRobotCaseServer 411
BeginMultiOperation 415
CodeCmbEngine 413
CreateCombination 416
CreateMobile 416
CreateSimple 416
Delete 417
DeleteMany 417
EndMultiOperation 417
Exist 417
FindWithId 418
FreeNumber 413
Get 418
GetAll 418
GetCmpntCount 419
GetMany 419
GetRelatedValue 419
GetUniqueId 420
IRobotCaseServer fields 412
IRobotCaseServer members 411
IRobotCaseServer methods 414
IsAuxiliary 420
QCmbTau 413
SELFSeismicEngine 413
SetAuxiliary 420
SnowWindEngine 414
TimeHistoryFunctions 414
WindLoadsSimulationEngine 414
IRobotCaseType 433
IRobotCladdingData 614
about IRobotCladdingData 614
IRobotCladdingData fields 615
IRobotCladdingData members 615
Method 615
Type 615
IRobotCladdingMethod 616
IRobotCladdingType 616
IRobotCmdInfo 1554
about IRobotCmdInfo 1554
Id 1555
IRobotCmdInfo fields 1555
IRobotCmdInfo members 1554
MenuChecked 1555
MenuEnabled 1556
Name 1556
IRobotCmdList 1552
about IRobotCmdList 1552
Count 1553
Get 1554
IRobotCmdList fields 1553
IRobotCmdList members 1553
IRobotCmdList methods 1553
New 1554
IRobotCodeCmbActionCoeffType 163
IRobotCodeCmbActionServer 160
about IRobotCodeCmbActionServer 160
Count 161
GetCoeff 161
GetName 162
GetNature 162
IRobotCodeCmbActionServer fields 160
IRobotCodeCmbActionServer members 160
IRobotCodeCmbActionServer methods 161
New 162
Remove 162
SetCoeff 163
SetName 163
SetNature 163
IRobotCodeCmbActiveCaseInfo 186
about IRobotCodeCmbActiveCaseInfo 186
CaseNature 187
CaseNumber 187

(c) 2014 Autodesk Inc. All rights reserved

Coefficient 187
GroupNumber 188
IRobotCodeCmbActiveCaseInfo fields 187
IRobotCodeCmbActiveCaseInfo members 186
IsSelected 188
IRobotCodeCmbCombPartType 167
IRobotCodeCmbCombs 164
about IRobotCodeCmbCombs 164
Count 165
Get 166
IRobotCodeCmbCombs fields 165
IRobotCodeCmbCombs members 164
IRobotCodeCmbCombs methods 165
New 166
Remove 166
Set 166
IRobotCodeCmbCombType 167
IRobotCodeCmbComponent 155
about IRobotCodeCmbComponent 155
Count 156
Get 156
IRobotCodeCmbComponent fields 156
IRobotCodeCmbComponent members 156
IRobotCodeCmbComponent methods 156
IRobotCodeCmbComponentMngr 153
about IRobotCodeCmbComponentMngr 153
Count 154
Get 154
IRobotCodeCmbComponentMngr fields 154
IRobotCodeCmbComponentMngr members 153
IRobotCodeCmbComponentMngr methods 154
IRobotCodeCmbDecidingValueType 174
IRobotCodeCmbFactor 154
about IRobotCodeCmbFactor 154
CaseNumber 155
Factor 155
IRobotCodeCmbFactor fields 155
IRobotCodeCmbFactor members 155
IRobotCodeCmbFlag 157
IRobotCodeCmbGenerationParams 168
about IRobotCodeCmbGenerationParams 168
ActiveCases 169
AllBars 169
AllNodes 170
BarSel 170
ExtremalSnowFactor 170
GenType 171
Groups 171
IRobotCodeCmbGenerationParams fields 169
IRobotCodeCmbGenerationParams members 168
IRobotCodeCmbGenerationParams methods 172
IsCombinationSelected 173
IsCombinationTypeSelected 173
IsDecidingValueSelected 173
NodeSel 171
PointsOnBar 171
Regulations 172
Relations 172
SelectCombination 173
SelectCombinationType 174
SelectDecidingValue 174
IRobotCodeCmbGenerationType 188
IRobotCodeCmbGroup 175
about IRobotCodeCmbGroup 175
Add 177
AddAll 177
CaseCount 176
IRobotCodeCmbGroup fields 176
IRobotCodeCmbGroup members 175
IRobotCodeCmbGroup methods 177
IsFull 178
Nature 176
Operator 176
IRobotCodeCmbGroupRelation 180
about IRobotCodeCmbGroupRelation 180
AddGroup 182
Clear 183
GetGroup 183
GetGroupCount 183
GetOperator 184
IRobotCodeCmbGroupRelation fields 181
IRobotCodeCmbGroupRelation members 181
IRobotCodeCmbGroupRelation methods 182
Nature 181

(c) 2014 Autodesk Inc. All rights reserved

NewRow 184
RemoveRow 184
RowCount 182
SetGroup 184
SetOperator 185
IRobotCodeCmbGroupRelationServer 185
about IRobotCodeCmbGroupRelationServer 185
Get 186
IRobotCodeCmbGroupRelationServer members 185
IRobotCodeCmbGroupRelationServer methods 185
Set 186
IRobotCodeCmbGroupServer 178
about IRobotCodeCmbGroupServer 178
FindByCase 179
Get 179
GetCount 180
IRobotCodeCmbGroupServer members 179
IRobotCodeCmbGroupServer methods 179
New 180
IRobotCodeCmbOperator 178
IRobotCodeCmbRegulations 157
about IRobotCodeCmbRegulations 157
Actions 158
CodeName 158
Combinations 158
IRobotCodeCmbRegulations fields 158
IRobotCodeCmbRegulations members 157
IsEuroCode 159
MaterialType 159
Version 159
IRobotCodeCombination 151
about IRobotCodeCombination 151
CombinationType 152
Components 152
FindByFlag 153
Flag 152
IRobotCodeCombination fields 151
IRobotCodeCombination members 151
IRobotCodeCombination methods 153
UniqueId 152
IRobotCodeCombinationEngine 149
about IRobotCodeCombinationEngine 149
Generate 150
IRobotCodeCombinationEngine fields 150
IRobotCodeCombinationEngine members 149
IRobotCodeCombinationEngine methods 150
Params 150
IRobotCodeRegistrar 1646
about IRobotCodeRegistrar 1646
CodeName 1647
CodeType 1648
Country 1648
Guid 1648
IRobotCodeRegistrar fields 1647
IRobotCodeRegistrar members 1647
IRobotCodeRegistrar methods 1649
Manufacturer 1649
ProgId 1649
Register 1649
Unregister 1650
IRobotCodeType 1312
IRobotCollection 9
about IRobotCollection 9
Count 10
Get 10
IRobotCollection fields 10
IRobotCollection members 10
IRobotCollection methods 10
IRobotCombinationType 433
IRobotComponentFactory 1654
about IRobotComponentFactory 1654
Create 1655
CreateExt 1655
IRobotComponentFactory members 1654
IRobotComponentFactory methods 1655
IRobotComponentType 1656
IRobotDAMAnalysisType 1597
IRobotDAMCalcModule 1595
about IRobotDAMCalcModule 1595
DeleteModel 1597
IRobotDAMCalcModule fields 1596
IRobotDAMCalcModule members 1595
IRobotDAMCalcModule methods 1597
IsActive 1596

(c) 2014 Autodesk Inc. All rights reserved

Params 1596
Run 1597
IRobotDAMLateralLoadCombType 1601
IRobotDAMNotionalLoads 1598
about IRobotDAMNotionalLoads 1598
ActiveXN 1599
ActiveXP 1599
ActiveYN 1599
ActiveYP 1600
Coefficient 1600
GravityLoadCombEnabled 1600
IRobotDAMNotionalLoads fields 1598
IRobotDAMNotionalLoads members 1598
LateralLoadCombEnabled 1601
LateralLoadCombType 1601
IRobotDAMParams 1605
about IRobotDAMParams 1605
Analysis 1606
GetNLPDParams 1607
IRobotDAMParams fields 1606
IRobotDAMParams members 1606
IRobotDAMParams methods 1607
NotionalLoads 1607
ReducedStiffness 1607
SetNLPDParams 1608
Update 1608
IRobotDAMReducedStiffness 1602
about IRobotDAMReducedStiffness 1602
IRobotDAMReducedStiffness fields 1603
IRobotDAMReducedStiffness members 1602
RCBeamsValue 1603
RCColumnsAndWallsValue 1603
RCSlabsValue 1604
SteelMembersReductionType 1604
SteelMembersTauBValue 1604
SteelMembersValue 1605
IRobotDAMSteelMembersReductionType 1605
IRobotDatabaseType 1326
IRobotDataObject 30
about IRobotDataObject 30
GetLabel 31
GetLabelName 32
GetLabels 32
HasLabel 32
IRobotDataObject fields 31
IRobotDataObject members 30
IRobotDataObject methods 31
Number 31
RemoveLabel 32
SetLabel 33
IRobotDataObjectServer 33
about IRobotDataObjectServer 33
Delete 34
DeleteMany 34
Exist 35
Get 35
GetAll 35
GetMany 36
IRobotDataObjectServer members 33
IRobotDataObjectServer methods 34
RemoveLabel 36
SetLabel 36
IRobotDeadRecordValues 51
IRobotDegreeOfFreedom 1657
IRobotDialogId 1476
IRobotDirectory 1544
IRobotDirectoryExtension 1545
IRobotDisplacementData 886
about IRobotDisplacementData 886
IRobotDisplacementData fields 887
IRobotDisplacementData members 887
RX 887
RY 887
RZ 888
UX 888
UY 888
UZ 888
IRobotDynamicAnalysisDamping 443
about IRobotDynamicAnalysisDamping 443
ConstValue 444
GetModeDamping 445
GetModes 445
IRobotDynamicAnalysisDamping fields 443
IRobotDynamicAnalysisDamping members 443

(c) 2014 Autodesk Inc. All rights reserved

IRobotDynamicAnalysisDamping methods 444
RemoveModeDamping 445
SetModeDamping 446
Type 444
IRobotDynamicAnalysisDampingType 446
IRobotDynamicAnalysisExcitationDirection 438
about IRobotDynamicAnalysisExcitationDirection 438
CombType 439
Group1 439
Group2 439
Group3 440
IRobotDynamicAnalysisExcitationDirection fields 438
IRobotDynamicAnalysisExcitationDirection members 438
Lambda 440
Mi 440
NormalizedX 440
NormalizedY 440
NormalizedZ 441
QuadraticActive 441
QuadraticSigned 441
ResolutionActive 441
Rx 441
Ry 442
Rz 442
UseNormalized 442
X 442
Y 442
Z 443
IRobotEigenvalues 899
about IRobotEigenvalues 899
AvPartCoeff 900
Damping 900
EigenValue 901
Energy 901
Frequence 901
IRobotEigenvalues fields 900
IRobotEigenvalues members 899
Period 901
Precision 902
Pulsation 902
IRobotEigenvaluesServer 927
about IRobotEigenvaluesServer 927
CombValue 928
IRobotEigenvaluesServer members 927
IRobotEigenvaluesServer methods 928
Value 928
IRobotEigenvectorsServer 932
about IRobotEigenvectorsServer 932
CombValue 933
IRobotEigenvectorsServer members 932
IRobotEigenvectorsServer methods 932
Value 933
IRobotEmitter 515
about IRobotEmitter 515
EstimatedElemNumber 515
H0 516
IRobotEmitter fields 515
IRobotEmitter members 515
R1 516
R2 516
VariableMeshDensityIncrement 516
IRobotEquationSolvingMethod 1567
IRobotEurocodeFactors 1324
about IRobotEurocodeFactors 1324
IRobotEurocodeFactors fields 1325
IRobotEurocodeFactors members 1325
SteelConnections 1325
SteelDesign 1325
IRobotEurocodeSteelConnectionFactors 1319
about IRobotEurocodeSteelConnectionFactors 1319
Gamma0 1320
Gamma1 1321
Gamma2 1321
Gamma3 1321
Gamma3Ser 1322
Gamma4 1322
Gamma5 1322
Gamma6 1323
Gamma7 1323
GammaC 1323
IRobotEurocodeSteelConnectionFactors fields 1320
IRobotEurocodeSteelConnectionFactors members 1319
IRobotEurocodeSteelConnectionFactors methods 1323
LoadFromCode 1324

(c) 2014 Autodesk Inc. All rights reserved

LoadFromCodeNumber 1324
IRobotEurocodeSteelDesignFactors 1316
about IRobotEurocodeSteelDesignFactors 1316
Gamma0 1317
Gamma1 1317
Gamma2 1318
GammaFire 1318
IRobotEurocodeSteelDesignFactors fields 1317
IRobotEurocodeSteelDesignFactors members 1316
IRobotEurocodeSteelDesignFactors methods 1318
LoadFromCode 1318
LoadFromCodeNumber 1319
IRobotExternalFileFormat 1274
IRobotExternalPreviewFormat 1518
IRobotExtremeParams 913
about IRobotExtremeParams 913
BarDivision 914
IRobotExtremeParams fields 914
IRobotExtremeParams members 913
Selection 914
ValueType 915
IRobotExtremeResultServer 1017
about IRobotExtremeResultServer 1017
IRobotExtremeResultServer members 1018
IRobotExtremeResultServer methods 1018
MaxValue 1018
MinValue 1018
IRobotExtremeValue 909
about IRobotExtremeValue 909
Bar 911
Case 911
CaseCmpnt 911
IRobotExtremeValue fields 910
IRobotExtremeValue members 910
IsAvailable 912
ModeCmb 912
Node 912
Position 912
Value 913
IRobotExtremeValueType 915
IRobotFeExtremeParams 987
about IRobotFeExtremeParams 987
CaseCmpnt 988
CaseSel 989
ElementSel 989
GetDirX 992
IRobotFeExtremeParams fields 988
IRobotFeExtremeParams members 987
IRobotFeExtremeParams methods 992
Layer 989
LayerArbitraryValue 990
ModeCmb 990
NodeSel 990
PanelSel 991
ReducedCutPos 991
Reset 992
ResultId 991
SetDirX 993
Smoothing 992
IRobotFeExtremeValue 993
about IRobotFeExtremeValue 993
Case 995
CaseCmpnt 995
Element 995
GetDirX 998
IRobotFeExtremeValue fields 994
IRobotFeExtremeValue members 993
IRobotFeExtremeValue methods 998
IsAvailable 995
Layer 996
LayerArbitraryValue 996
ModeCmb 996
Node 996
Panel 997
ReducedCutPos 997
ResultId 997
Smoothing 997
Value 998
IRobotFeLayerType 937
IRobotFeMultiExtremeValue 1001
about IRobotFeMultiExtremeValue 1001
Count 1001
Get 1002
GetByResType 1002

(c) 2014 Autodesk Inc. All rights reserved

IRobotFeMultiExtremeValue fields 1001
IRobotFeMultiExtremeValue members 1001
IRobotFeMultiExtremeValue methods 1002
IRobotFeMultiResultType 999
about IRobotFeMultiResultType 999
Add 1000
Count 999
Get 1000
IRobotFeMultiResultType fields 999
IRobotFeMultiResultType members 999
IRobotFeMultiResultType methods 1000
Remove 1000
IRobotFeResultComplex 963
about IRobotFeResultComplex 963
IRobotFeResultComplex fields 964
IRobotFeResultComplex members 963
M_MISES 964
MXX_BOTTOM 964
MXX_TOP 965
MYY_BOTTOM 965
MYY_TOP 965
N_MISES 966
S_MISES 966
IRobotFeResultDetailed 943
about IRobotFeResultDetailed 943
IRobotFeResultDetailed fields 944
IRobotFeResultDetailed members 943
MXX 944
MXY 945
MYY 945
NXX 945
NXY 946
NYY 946
PNorm 946
QXX 946
QYY 947
RNorm 947
RXX 947
RYY 948
SXX 948
SXY 948
SYY 949
TXX 949
TYY 949
UXX 949
UYY 950
WNorm 950
IRobotFeResultKind 986
IRobotFeResultParams 937
about IRobotFeResultParams 937
CalcMethod 939
Case 939
CaseCmpnt 939
Element 940
GetDirX 942
IRobotFeResultParams fields 938
IRobotFeResultParams members 938
IRobotFeResultParams methods 942
Layer 940
LayerArbitraryValue 940
ModeCmb 941
Node 941
Panel 941
SetDirX 942
IRobotFeResultPrincipal 956
about IRobotFeResultPrincipal 956
IRobotFeResultPrincipal fields 957
IRobotFeResultPrincipal members 957
M1 958
M1_2 958
M2 958
MAL 958
N1 959
N1_2 959
N2 959
NAL 960
Q1_2 960
S1 960
S1_2 961
S2 961
SAL 961
T1_2 961
U 962
UGX 962

(c) 2014 Autodesk Inc. All rights reserved

UGY 962
UGZ 963
IRobotFeResultReduced 971
about IRobotFeResultReduced 971
CutPos 973
Height 973
IRobotFeResultReduced fields 972
IRobotFeResultReduced members 972
Length 973
MY 974
MZ 974
NodeLeftBottom 974
NodeLeftTop 975
NodeRightBottom 975
NodeRightTop 975
NX 975
SE 976
SO 976
T 976
TY 977
TZ 977
IRobotFeResultReducedCutPosition 977
IRobotFeResultReinforcement 966
A_MIN 968
about IRobotFeResultReinforcement 966
AX 968
AX_BOTTOM 968
AX_TOP 968
AY 969
AY_BOTTOM 969
AY_TOP 969
CalcError 970
E_AX_BOTTOM 970
E_AX_TOP 970
E_AY_BOTTOM 971
E_AY_TOP 971
F 971
IRobotFeResultReinforcement fields 967
IRobotFeResultReinforcement members 967
IRobotFeResultServer 950
about IRobotFeResultServer 950
Complex 952
Detailed 953
IRobotFeResultServer fields 951
IRobotFeResultServer members 951
IRobotFeResultServer methods 952
MaxValue 953
MinValue 953
MultiMaxValue 954
MultiMinValue 954
PanelCuts 952
Principal 954
Reduced 955
ReducedEx 955
Reinforcement 955
IRobotFeResultSmoothing 986
IRobotFeResultType 984
IRobotFileInsertParams 1355
about IRobotFileInsertParams 1355
AsObject 1356
GetInsertionPoint 1357
GetRotation 1357
IRobotFileInsertParams fields 1355
IRobotFileInsertParams members 1355
IRobotFileInsertParams methods 1356
ReferenceNode 1356
ScaleFactor 1356
SetInsertionPoint 1357
SetRotation 1357
IRobotFiniteElement 832
about IRobotFiniteElement 832
CalcArea 835
FeType 833
IRobotFiniteElement fields 833
IRobotFiniteElement members 832
IRobotFiniteElement methods 834
Nodes 833
ObjectNumber 834
ObjectPartIdx 834
UniqueId 834
IRobotFiniteElementData 842
about IRobotFiniteElementData 842
GetNode 844
IRobotFiniteElementData fields 842

(c) 2014 Autodesk Inc. All rights reserved

IRobotFiniteElementData members 842
IRobotFiniteElementData methods 843
NodeCount 843
Number 843
Object 843
IRobotFiniteElementDataSet 844
about IRobotFiniteElementDataSet 844
ElemCount 845
GetElem 846
GetElemCountForObject 846
GetElemForObject 846
GetObjectNumber 847
IRobotFiniteElementDataSet fields 845
IRobotFiniteElementDataSet members 844
IRobotFiniteElementDataSet methods 846
ObjectCount 845
IRobotFiniteElementNodes 836
about IRobotFiniteElementNodes 836
Count 836
Get 837
GetAll 837
IRobotFiniteElementNodes fields 836
IRobotFiniteElementNodes members 836
IRobotFiniteElementNodes methods 837
Set 838
SetAll 838
IRobotFiniteElementServer 838
about IRobotFiniteElementServer 838
CalcArea 840
Create 840
FreeNumber 839
IRobotFiniteElementServer fields 839
IRobotFiniteElementServer members 839
IRobotFiniteElementServer methods 840
MeshConcentrate 841
MeshConsolidate 841
Update 841
IRobotFiniteElementType 835
IRobotFootfallAnalysisExcitationForces 399
IRobotFootfallAnalysisExcitationMethod 398
IRobotFootfallAnalysisModalParams 401
about IRobotFootfallAnalysisModalParams 401
FrequencyLimit 402
IgnoreDensity 402
IncludeMassForDirX 403
IncludeMassForDirY 403
IncludeMassForDirZ 403
IRobotFootfallAnalysisModalParams fields 402
IRobotFootfallAnalysisModalParams members 401
IRobotFootfallAnalysisNodeSelection 399
about IRobotFootfallAnalysisNodeSelection 399
IRobotFootfallAnalysisNodeSelection fields 400
IRobotFootfallAnalysisNodeSelection members 399
SelectedNodes 400
SelectedPanels 400
Type 400
IRobotFootfallAnalysisNodeSelectionType 401
IRobotFootfallAnalysisParams 403
about IRobotFootfallAnalysisParams 403
Damping 405
ExcitationForces 405
ExcitationMethod 405
ExcitationNodes 405
FootstepsNumber 406
IRobotFootfallAnalysisParams fields 404
IRobotFootfallAnalysisParams members 404
MaxWalkingFrequency 406
MinWalkingFrequency 406
ModalParams 407
ResponseNodes 407
WalkersWeight 407
IRobotFootfallResults 920
A 921
about IRobotFootfallResults 920
ExcitationNode 921
Frequency 922
IRobotFootfallResults fields 921
IRobotFootfallResults members 920
RF_Overall 922
RF_Resonant 922
RF_Transient 923
VRMQ 923
VRMS 923
IRobotForcesData 902

(c) 2014 Autodesk Inc. All rights reserved

about IRobotForcesData 902
FX 903
FY 903
FZ 903
IRobotForcesData fields 903
IRobotForcesData members 902
MX 904
MY 904
MZ 904
IRobotFRFAnalysisParams 396
about IRobotFRFAnalysisParams 396
Damping 397
FinalFrequency 397
FrequencyDivision 397
IncludeEigenfrequencies 397
InitialFrequency 398
IRobotFRFAnalysisParams fields 396
IRobotFRFAnalysisParams members 396
IRobotFRFResults 918
about IRobotFRFResults 918
Frequency 919
IRobotFRFResults fields 918
IRobotFRFResults members 918
VX 919
VY 919
VZ 920
IRobotFRFResultServer 934
about IRobotFRFResultServer 934
IRobotFRFResultServer members 935
IRobotFRFResultServer methods 935
Value 935
IRobotGeoArc 1628
about IRobotGeoArc 1628
IRobotGeoArc fields 1629
IRobotGeoArc members 1628
P1 1629
P2 1629
P3 1629
IRobotGeoArcDefinitionMethod 1634
IRobotGeoCircle 1630
about IRobotGeoCircle 1630
IRobotGeoCircle fields 1630
IRobotGeoCircle members 1630
P1 1630
P2 1631
P3 1631
PC 1631
IRobotGeoContour 1626
about IRobotGeoContour 1626
Add 1627
Clear 1628
IRobotGeoContour fields 1627
IRobotGeoContour members 1627
IRobotGeoContour methods 1627
Segments 1627
IRobotGeoCoordinateAxis 1614
IRobotGeoCoordinateAxisSense 1635
IRobotGeoCoordinateSystem 1615
IRobotGeoCurveDiv 1616
about IRobotGeoCurveDiv 1616
IRobotGeoCurveDiv fields 1617
IRobotGeoCurveDiv members 1616
IsAnalytical 1617
Length 1617
Mode 1617
N 1618
IRobotGeoLayer 1632
about IRobotGeoLayer 1632
IRobotGeoLayer fields 1632
IRobotGeoLayer members 1632
IsThickDefined 1633
P1 1633
P2 1633
P3 1634
PDir 1634
Thickness 1634
IRobotGeoObject 1623
about IRobotGeoObject 1623
Initialize 1624
IRobotGeoObject fields 1623
IRobotGeoObject members 1623
IRobotGeoObject methods 1624
Type 1623
IRobotGeoObjectType 1616

(c) 2014 Autodesk Inc. All rights reserved

IRobotGeoPlane 1635
IRobotGeoPoint2D 1610
about IRobotGeoPoint2D 1610
Get 1612
IRobotGeoPoint2D fields 1611
IRobotGeoPoint2D members 1610
IRobotGeoPoint2D methods 1611
Set 1612
X 1611
Y 1611
IRobotGeoPoint3D 1612
about IRobotGeoPoint3D 1612
Get 1614
IRobotGeoPoint3D fields 1613
IRobotGeoPoint3D members 1612
IRobotGeoPoint3D methods 1614
Set 1614
X 1613
Y 1613
Z 1613
IRobotGeoPoint3DCollection 1631
about IRobotGeoPoint3DCollection 1631
IRobotGeoPolyline 1624
about IRobotGeoPolyline 1624
Add 1625
Clear 1626
IRobotGeoPolyline fields 1625
IRobotGeoPolyline members 1624
IRobotGeoPolyline methods 1625
Segments 1625
IRobotGeoSegment 1618
about IRobotGeoSegment 1618
IRobotGeoSegment fields 1618
IRobotGeoSegment members 1618
P1 1619
Type 1619
IRobotGeoSegmentArc 1621
about IRobotGeoSegmentArc 1621
Div 1622
IRobotGeoSegmentArc fields 1622
IRobotGeoSegmentArc members 1621
IRobotGeoSegmentArc methods 1622
P2 1622
Set 1622
IRobotGeoSegmentCollection 1626
about IRobotGeoSegmentCollection 1626
IRobotGeoSegmentLine 1619
about IRobotGeoSegmentLine 1619
Div 1620
IRobotGeoSegmentLine fields 1620
IRobotGeoSegmentLine members 1619
NDiv 1620
Radius 1620
Round 1621
IRobotGeoSegmentType 1615
IRobotGroup 1041
about IRobotGroup 1041
Color 1042
CreateCollection 1044
CreateSelection 1044
IRobotGroup fields 1042
IRobotGroup members 1042
IRobotGroup methods 1043
Name 1042
ObjectType 1043
Read 1044
SelList 1043
Store 1045
IRobotGroupObjectServer 847
about IRobotGroupObjectServer 847
Explode 848
FindFirst 849
FindNext 849
GetContents 849
GroupGiven 850
GroupSelected 850
IRobotGroupObjectServer members 848
IRobotGroupObjectServer methods 848
IRobotGroupServer 1045
about IRobotGroupServer 1045
Create 1046
Delete 1046
Find 1047
Get 1047

(c) 2014 Autodesk Inc. All rights reserved

GetCount 1047
IRobotGroupServer members 1045
IRobotGroupServer methods 1046
IRobotHarmonicAnalysisParams 390
about IRobotHarmonicAnalysisParams 390
Excitation 391
IRobotHarmonicAnalysisParams fields 391
IRobotHarmonicAnalysisParams members 390
MassMatrix 391
IRobotHtmlView 1467
about IRobotHtmlView 1467
IRobotHtmlView fields 1468
IRobotHtmlView members 1468
IRobotHtmlView methods 1468
LoadFromFile 1469
Printable 1468
SaveToFile 1469
IRobotIn3PointsRecordValues 45
IRobotInContourRecordValues 48
IRobotIterativePrediconditionerType 1574
IRobotIterativeSolverMemoryUsage 1578
IRobotIterativeSolverMethod 1578
IRobotIterativeSolverParams 1574
about IRobotIterativeSolverParams 1574
AggregationLevelsCount 1575
AnalyseDiagonale 1575
CalcKMatrix 1576
InternalIterationsCount 1576
IRobotIterativeSolverParams fields 1575
IRobotIterativeSolverParams members 1574
MemoryUsage 1576
Method 1577
Multilevel 1577
PrediconditionerType 1577
Tolerance 1578
IRobotKernel 2208
about IRobotKernel 2208
CalcEngine 2209
CmpntFactory 2210
ConcrReinfEngine 2210
GetExtension 2213
IRobotKernel fields 2209
IRobotKernel members 2208
IRobotKernel methods 2213
Preferences 2210
ProgramName 2210
ProgramPath 2211
ProgramVersion 2211
ProjectNew 2213
ProjectNewFromTemplate 2214
ProjectOpen 2214
ProjectPreferences 2211
ProjectSave 2214
ProjectSaveAs 2215
ProjectUniqueId 2212
Structure 2212
Version 2212
IRobotKernelPreferences 2215
about IRobotKernelPreferences 2215
GetDirectory 2217
GetDirectoryExt 2217
GetLanguage 2217
IRobotKernelPreferences fields 2216
IRobotKernelPreferences members 2215
IRobotKernelPreferences methods 2216
Multiprocessing 2216
SetDirectoryExt 2218
SetLanguage 2218
IRobotLabel 18
about IRobotLabel 18
Data 19
IRobotLabel fields 19
IRobotLabel members 19
Name 20
Type 20
UniqueId 20
IRobotLabelServer 20
about IRobotLabelServer 20
Create 23
CreateLike 23
Delete 24
Exist 24
FindWithId 24
Get 24

(c) 2014 Autodesk Inc. All rights reserved

GetAll 25
GetAvailableNames 25
GetDefault 25
GetMany 26
GetPredefinedName 26
GetUniqueId 26
IRobotLabelServer members 21
IRobotLabelServer methods 22
IsAvailable 27
IsPredefinedName 27
IsUsed 27
SetDefault 28
Store 28
StoreWithName 28
IRobotLabelType 17
IRobotLanguage 1544
IRobotLayoutId 1475
IRobotLicenseEntitlement 1547
IRobotLicenseEntitlementStatus 1547
IRobotLimitState 435
IRobotLinear3DRecordValues 47
IRobotLinearOnEdgesRecordValues 51
IRobotLinearRecordValues 46
IRobotLinearReleaseData 704
about IRobotLinearReleaseData 704
HX 705
IRobotLinearReleaseData fields 705
IRobotLinearReleaseData members 704
KX 705
KY 706
KZ 706
RX 706
UX 707
UY 707
UZ 707
IRobotLinearReleaseDef 713
about IRobotLinearReleaseDef 713
EdgeIdx 714
EdgeObject 715
IRobotLinearReleaseDef fields 714
IRobotLinearReleaseDef members 714
LabelName 715
Object 715
PartIdx 716
IRobotLinearReleaseDefinitionType 708
IRobotLinearReleaseDefList 716
about IRobotLinearReleaseDefList 716
Add 717
Clear 718
Count 717
Get 718
IRobotLinearReleaseDefList fields 717
IRobotLinearReleaseDefList members 716
IRobotLinearReleaseDefList methods 717
Remove 718
IRobotLinearReleaseServer 708
about IRobotLinearReleaseServer 708
Count 709
Find 710
FindEdge 711
FindLabel 711
FindObject 711
Get 712
GetLabel 712
IRobotLinearReleaseServer fields 709
IRobotLinearReleaseServer members 708
IRobotLinearReleaseServer methods 710
Remove 712
Set 713
SetMany 713
IRobotLoadRecord 56
about IRobotLoadRecord 56
Description 57
GetValue 58
IRobotLoadRecord fields 57
IRobotLoadRecord members 56
IRobotLoadRecord methods 58
Objects 57
ObjectType 58
SetValue 59
Type 58
IRobotLoadRecord2 75
about IRobotLoadRecord2 75
GetGeoLimits 76

(c) 2014 Autodesk Inc. All rights reserved

IRobotLoadRecord2 fields 75
IRobotLoadRecord2 members 75
IRobotLoadRecord2 methods 76
SetGeoLimits 76
UniqueId 76
IRobotLoadRecordBarTrapezoidal 77
about IRobotLoadRecordBarTrapezoidal 77
GetPoint 78
IRobotLoadRecordBarTrapezoidal fields 78
IRobotLoadRecordBarTrapezoidal members 77
IRobotLoadRecordBarTrapezoidal methods 78
PointCount 78
SetPoint 79
IRobotLoadRecordCommon 81
about IRobotLoadRecordCommon 81
IRobotLoadRecordCommon fields 81
IRobotLoadRecordCommon members 81
IsAutoGenerated 81
IRobotLoadRecordDead 79
about IRobotLoadRecordDead 79
FiniteElems 80
IRobotLoadRecordDead fields 80
IRobotLoadRecordDead members 79
IRobotLoadRecordIn3Points 63
about IRobotLoadRecordIn3Points 63
GetGeoLimits 65
GetPoint 65
IRobotLoadRecordIn3Points fields 64
IRobotLoadRecordIn3Points members 64
IRobotLoadRecordIn3Points methods 65
SetGeoLimits 65
SetPoint 66
UniqueId 64
IRobotLoadRecordInContour 69
about IRobotLoadRecordInContour 69
GetContourPoint 71
GetGeoLimits 71
GetPoint 71
GetVector 71
IRobotLoadRecordInContour fields 70
IRobotLoadRecordInContour members 69
IRobotLoadRecordInContour methods 70
SetContourPoint 72
SetGeoLimits 72
SetPoint 72
SetVector 72
UniqueId 70
IRobotLoadRecordLinear 73
about IRobotLoadRecordLinear 73
GetPoint 74
IRobotLoadRecordLinear fields 73
IRobotLoadRecordLinear members 73
IRobotLoadRecordLinear methods 74
SetPoint 74
UniqueId 74
IRobotLoadRecordLinear3D 61
about IRobotLoadRecordLinear3D 61
GetPoint 63
IRobotLoadRecordLinear3D fields 62
IRobotLoadRecordLinear3D members 62
IRobotLoadRecordLinear3D methods 63
SetPoint 63
UniqueId 62
IRobotLoadRecordMngr 59
about IRobotLoadRecordMngr 59
Count 60
Create 60
Delete 61
Get 61
IRobotLoadRecordMngr fields 60
IRobotLoadRecordMngr members 59
IRobotLoadRecordMngr methods 60
New 61
IRobotLoadRecordThermalIn3Points 66
about IRobotLoadRecordThermalIn3Points 66
GetGeoLimits 68
GetPoint 68
IRobotLoadRecordThermalIn3Points fields 67
IRobotLoadRecordThermalIn3Points members 66
IRobotLoadRecordThermalIn3Points methods 67
SetGeoLimits 68
SetPoint 68
UniqueId 67
IRobotLoadRecordType 55

(c) 2014 Autodesk Inc. All rights reserved

IRobotMassActivationRecordValues 46
IRobotMassEccentricities 206
about IRobotMassEccentricities 206
IRobotMassEccentricities fields 207
IRobotMassEccentricities members 207
IsDirX 207
IsDirY 208
RelativeValues 208
ValueDirX 208
ValueDirY 209
IRobotMassSumServer 928
about IRobotMassSumServer 928
Current 929
IRobotMassSumServer members 929
IRobotMassSumServer methods 929
PartCoeff 929
Relative 930
Total 930
IRobotMaterialData 660
about IRobotMaterialData 660
CB71_Category 663
CB71_Humidity 663
CB71_Nature 663
CB71_Retreat 663
CS 664
Default 664
DumpCoef 664
E 664
E_5 665
E_Trans 665
EC_Deformation 665
GMean 665
IRobotMaterialData fields 661
IRobotMaterialData members 660
IRobotMaterialData methods 671
Kirchoff 666
LoadFromDBase 671
LX 666
Name 666
NU 666
Nuance 667
PN_Deformation 667
PN_E_Additional 667
PN_E_Trans 667
RE 668
RE_AxCompr 668
RE_AxTens 668
RE_Bending 669
RE_Shear 669
RE_TrCompr 669
RE_TrTens 669
RO 670
RT 670
SaveToDBase 672
SecondName 670
Steel_Thermal 670
Timber_Type 671
Type 671
IRobotMaterialDatabase 1308
about IRobotMaterialDatabase 1308
Get 1310
GetAll 1310
GetDefault 1310
IRobotMaterialDatabase fields 1309
IRobotMaterialDatabase members 1309
IRobotMaterialDatabase methods 1309
Load 1311
LoadFromFile 1311
Name 1309
SetDefault 1311
IRobotMaterialElasticModel 674
about IRobotMaterialElasticModel 674
Coeff 675
IRobotMaterialElasticModel fields 674
IRobotMaterialElasticModel members 674
Model 675
UnloadingCoeff 675
UnloadingMethod 676
IRobotMaterialElasticType 673
IRobotMaterialElasticUnloadingMethod 676
IRobotMaterialModel 673
IRobotMaterialQuantitySurvey 1079
about IRobotMaterialQuantitySurvey 1079
Count 1080

(c) 2014 Autodesk Inc. All rights reserved

GetName 1081
GetType 1081
GetVolume 1081
GetWeight 1082
IRobotMaterialQuantitySurvey fields 1080
IRobotMaterialQuantitySurvey members 1080
IRobotMaterialQuantitySurvey methods 1080
IRobotMaterialTimberType 672
IRobotMaterialType 672
IRobotMeshAccessType 831
IRobotMeshCoonsParams 817
about IRobotMeshCoonsParams 817
ForcingRatio 818
IRobotMeshCoonsParams fields 817
IRobotMeshCoonsParams members 817
PanelDivisionType 818
IRobotMeshDelaunayParams 820
about IRobotMeshDelaunayParams 820
EmittersDefault 821
EmittersSmoothing 822
EmittersUser 822
H_max 822
H0 823
IRobotMeshDelaunayParams fields 821
IRobotMeshDelaunayParams members 820
MeshDensity 823
NumberOfLevels 823
Q 823
RegularMesh 824
Type 824
IRobotMeshDelaunayType 812
IRobotMeshForcingRatio 809
IRobotMeshGeneration 815
about IRobotMeshGeneration 815
Division1 816
Division2 816
ElementSize 816
IRobotMeshGeneration fields 815
IRobotMeshGeneration members 815
Type 817
IRobotMeshGenerationType 813
IRobotMeshImplementDegree 810
IRobotMeshMethod 813
about IRobotMeshMethod 813
ForcingRatio 814
ImplementDegree 814
IRobotMeshMethod fields 814
IRobotMeshMethod members 813
Method 814
IRobotMeshMethodType 812
IRobotMeshPanelDivType 810
IRobotMeshParams 829
about IRobotMeshParams 829
Flag 829
IRobotMeshParams fields 829
IRobotMeshParams members 829
MeshType 830
SurfaceParams 830
VolumeParams 830
IRobotMeshRefinementType 831
IRobotMeshSurfaceFEType 811
IRobotMeshSurfaceFiniteElems 818
about IRobotMeshSurfaceFiniteElems 818
ConversionCoeff 819
ForcingRatio 819
IRobotMeshSurfaceFiniteElems fields 819
IRobotMeshSurfaceFiniteElems members 819
Type 820
IRobotMeshSurfaceParams 824
about IRobotMeshSurfaceParams 824
Coons 825
Delaunay 826
FiniteElems 826
Generation 826
IRobotMeshSurfaceParams fields 825
IRobotMeshSurfaceParams members 825
Method 826
IRobotMeshType 808
IRobotMeshVolumeParams 827
about IRobotMeshVolumeParams 827
AdditionalSurfaceMeshing 828
FiniteElemsType 828
IRobotMeshVolumeParams fields 827
IRobotMeshVolumeParams members 827

(c) 2014 Autodesk Inc. All rights reserved

MeshDensity 828
IRobotMeshVolumetricFEType 811
IRobotMobileCase 327
about IRobotMobileCase 327
ApplicationPlaneBars 328
ApplicationPlaneType 329
Components 329
FindByFlag 331
Flag 329
GetRoute 332
IRobotMobileCase fields 328
IRobotMobileCase members 327
IRobotMobileCase methods 331
IsAuxiliary 330
Label 330
NatureName 330
SetNatureExt 332
SetRoute 332
UniqueId 330
Vehicle 331
IRobotMobileCaseApplicationPlaneType 343
IRobotMobileCaseComponent 334
about IRobotMobileCaseComponent 334
IRobotMobileCaseComponent fields 335
IRobotMobileCaseComponent members 334
Point 335
Records 335
IRobotMobileCaseComponentMngr 333
about IRobotMobileCaseComponentMngr 333
Count 333
Get 334
IRobotMobileCaseComponentMngr fields 333
IRobotMobileCaseComponentMngr members 333
IRobotMobileCaseComponentMngr methods 334
IRobotMobileCaseFlag 335
IRobotMobileCaseRoute 336
about IRobotMobileCaseRoute 336
BeginingRouteLimit 337
EndRouteLimit 337
Geometry 338
GetFactors 339
IRobotMobileCaseRoute fields 337
IRobotMobileCaseRoute members 336
IRobotMobileCaseRoute methods 339
LoadDirection 338
SetFactors 339
Step 338
Tolerance 339
IRobotMobileCaseSegmentFactors 340
about IRobotMobileCaseSegmentFactors 340
Gamma 341
HL 341
HR 341
IRobotMobileCaseSegmentFactors fields 340
IRobotMobileCaseSegmentFactors members 340
LL 342
LR 342
VL 342
VR 343
IRobotMobileDistributedRecordValues 53
IRobotMobilePointForceRecordValues 52
IRobotModalAnalysisAlgorithm 193
IRobotModalAnalysisBase 201
about IRobotModalAnalysisBase 201
Add 201
Count 202
Delete 202
Get 202
IRobotModalAnalysisBase members 201
IRobotModalAnalysisBase methods 201
IRobotModalAnalysisLimits 203
about IRobotModalAnalysisLimits 203
DefineLimits 205
FrequencyLimitValue 204
IRobotModalAnalysisLimits fields 203
IRobotModalAnalysisLimits members 203
IRobotModalAnalysisLimits methods 204
PeriodLimitValue 204
PulsationLimitValue 204
IRobotModalAnalysisLimitType 194
IRobotModalAnalysisMassMatrixType 193
IRobotModalAnalysisMode 192
IRobotModalAnalysisParams 194
about IRobotModalAnalysisParams 194

(c) 2014 Autodesk Inc. All rights reserved

Acceleration 195
Base 196
Damping 196
DisregardDensity 196
IncludeDampingInCalculations 197
IRobotModalAnalysisParams fields 195
IRobotModalAnalysisParams members 194
IterationsCount 197
Limits 197
MassEccentricities 198
MassMatrix 198
MassParticipation 198
Method 199
Mode 199
ModesCount 199
Shifts 200
SturmVerification 200
Tolerance 200
IRobotModalAnalysisShifts 205
about IRobotModalAnalysisShifts 205
IRobotModalAnalysisShifts fields 206
IRobotModalAnalysisShifts members 205
IRobotModalAnalysisShifts methods 206
IterationsCount 206
SetDefault 206
IRobotModalWithStaticForcesAnalysisParams 189
about IRobotModalWithStaticForcesAnalysisParams 189
CreateFromStatic 191
GetStaticStateParams 191
IRobotModalWithStaticForcesAnalysisParams fields 190
IRobotModalWithStaticForcesAnalysisParams members 190
IRobotModalWithStaticForcesAnalysisParams methods 191
Nonlinear 191
SetStaticStateParams 192
IRobotModeCombinationType 1040
IRobotModelGenerationParams 1584
about IRobotModelGenerationParams 1584
AssemblingCase 1586
GenerateNodes_BarsAndFiniteElems 1586
GenerateNodes_DiagonalBars 1586
GenerateNodes_VertHorizBars 1587
IRobotModelGenerationParams fields 1585
IRobotModelGenerationParams members 1585
MaxElementLength 1587
NeglectedBars 1587
NeglectedGeoObjects 1588
ToleranceAutomatic 1588
ToleranceValue 1588
IRobotModeSelection 15
about IRobotModeSelection 15
Combination 16
IRobotModeSelection fields 16
IRobotModeSelection members 15
Mode 16
Type 16
IRobotModeSelectionType 1040
IRobotMultiCollection 13
about IRobotMultiCollection 13
Exist 14
Get 14
IRobotMultiCollection members 14
IRobotMultiCollection methods 14
Set 15
IRobotMultiSelection 11
about IRobotMultiSelection 11
CaseCmpnt 12
Exist 13
Get 13
IRobotMultiSelection fields 11
IRobotMultiSelection members 11
IRobotMultiSelection methods 12
Modes 12
Set 13
IRobotNamesArray 1638
about IRobotNamesArray 1638
Count 1639
Find 1639
Get 1640
IRobotNamesArray fields 1639
IRobotNamesArray members 1638
IRobotNamesArray methods 1639
Set 1640
SetSize 1640

(c) 2014 Autodesk Inc. All rights reserved

IRobotNode 526
about IRobotNode 526
GetCalcSupport 530
GetEmitter 530
HasCalcSupport 528
HasEmitter 528
IRobotNode fields 528
IRobotNode members 527
IRobotNode methods 530
IsCalc 528
RemoveEmitter 531
SetEmitter 531
UniqueId 529
X 529
Y 529
Z 529
IRobotNodeAccelerationRecordValues 54
IRobotNodeAuxiliaryRecordValues 41
IRobotNodeBucklingServer 866
about IRobotNodeBucklingServer 866
EigenVector 867
EigenVectorCmb 867
IRobotNodeBucklingServer members 866
IRobotNodeBucklingServer methods 866
IRobotNodeCompatibilityData 489
about IRobotNodeCompatibilityData 489
Alpha 490
AX 491
AY 491
AZ 491
Beta 492
BX 492
BY 492
BZ 492
Gamma 493
HX 493
HY 493
HZ 494
IRobotNodeCompatibilityData fields 490
IRobotNodeCompatibilityData members 489
KX 494
KY 494
KZ 494
NonlinearModel 495
RX 495
RY 495
RZ 496
UX 496
UY 496
UZ 497
IRobotNodeCompatibilityDef 497
about IRobotNodeCompatibilityDef 497
Bars 498
Compatible 498
IRobotNodeCompatibilityDef fields 498
IRobotNodeCompatibilityDef members 497
LabelName 498
Main 499
IRobotNodeCompatibilityServer 499
about IRobotNodeCompatibilityServer 499
Count 500
Find 501
FindCompatible 501
FindLabel 501
FindMain 502
Get 502
GetLabel 502
IRobotNodeCompatibilityServer fields 500
IRobotNodeCompatibilityServer members 499
IRobotNodeCompatibilityServer methods 500
Remove 503
Set 503
IRobotNodeDisplacementRecordValues 40
IRobotNodeDisplacementServer 864
about IRobotNodeDisplacementServer 864
DynCombValue 865
DynValue 865
IRobotNodeDisplacementServer members 864
IRobotNodeDisplacementServer methods 864
Value 865
ValueEx 866
IRobotNodeForceInPointRecordValues 50
IRobotNodeForceRecordValues 40
IRobotNodeResultServer 852

(c) 2014 Autodesk Inc. All rights reserved

about IRobotNodeResultServer 852
Buckling 853
Displacements 853
IRobotNodeResultServer fields 852
IRobotNodeResultServer members 852
PseudostaticForces 853
Reactions 853
IRobotNodeRigidLinkData 479
about IRobotNodeRigidLinkData 479
IRobotNodeRigidLinkData fields 480
IRobotNodeRigidLinkData members 479
RX 480
RY 480
RZ 481
UX 481
UY 481
UZ 482
IRobotNodeRigidLinkDef 482
about IRobotNodeRigidLinkDef 482
IRobotNodeRigidLinkDef fields 482
IRobotNodeRigidLinkDef members 482
LabelName 483
Master 483
Slaves 483
IRobotNodeRigidLinkServer 484
about IRobotNodeRigidLinkServer 484
Count 485
Find 486
FindLabel 486
FindMaster 486
FindSlave 487
Get 487
GetLabel 487
IRobotNodeRigidLinkServer fields 485
IRobotNodeRigidLinkServer members 484
IRobotNodeRigidLinkServer methods 485
Remove 488
RemoveSlave 488
Set 488
IRobotNodeServer 531
about IRobotNodeServer 531
CompatibleNodes 533
Create 534
FindWithId 535
FreeNumber 533
GetCalcNodes 535
GetCalcSupport 535
GetConnectedBars 535
GetUniqueId 536
GetUserNodes 536
HasCalcSupport 536
IRobotNodeServer fields 532
IRobotNodeServer members 531
IRobotNodeServer methods 534
IsCalc 537
NonlinearLinks 533
RemoveEmitter 537
RigidLinks 533
SetEmitter 537
IRobotNodeSupportData 503
about IRobotNodeSupportData 503
Advanced 506
Alpha 506
AX 506
AY 507
AZ 507
Beta 507
BX 507
BY 507
BZ 508
ElasticLinear 508
ElasticSurface 508
Gamma 509
GetAdvanced 512
GetOneDir 513
GlobalCoordSystem 509
HX 509
HY 509
HZ 510
IRobotNodeSupportData fields 505
IRobotNodeSupportData members 504
IRobotNodeSupportData methods 512
IsFixed 513
KX 510

(c) 2014 Autodesk Inc. All rights reserved

KY 510
KZ 510
NonlinearModel 510
RX 511
RY 511
RZ 511
SetAdvanced 513
SetFixed 513
SetOneDir 514
UX 511
UY 512
UZ 512
IRobotNodeSupportFixingDirection 514
IRobotNodeSupportOneDirectionFixingType 514
IRobotNodeVelocityRecordValues 54
IRobotNonlinearAnalysisAlgorithmType 317
IRobotNonlinearAnalysisParams 318
about IRobotNonlinearAnalysisParams 318
Algorithm 320
DegreeOfFreedom 320
DisplacementsRelativeCodeTolerance 320
GetSettingsFromPreferences 325
IncrementLengthReductionFactor 321
IncrementLengthReductionNumber 321
IRobotNonlinearAnalysisParams fields 319
IRobotNonlinearAnalysisParams members 318
IRobotNonlinearAnalysisParams methods 325
LineSearchMethodFactor 321
LoadIncrementNumber 322
MatrixUpdateAfterEachIteration 322
MatrixUpdateAfterEachSubdivision 322
MaxDisplacement 322
MaximumIterationNumberForOneIncrement 323
MaximumNumberOfBFGSCorrections 323
MaximumNumberOfLineSearches 323
MaxLoadFactor 323
NodeNumber 324
PDelta 324
ResidualForcesRelativeCodeTolerance 324
ResultListEachIteration 325
SaveSettingsInPreferences 326
Stiff 325
IRobotNonlinearHingeComponentType 609
IRobotNonlinearHingeData 606
about IRobotNonlinearHingeData 606
GetModel 608
IRobotNonlinearHingeData fields 607
IRobotNonlinearHingeData members 606
IRobotNonlinearHingeData methods 607
IsActive 608
NormalStress 607
Remove 608
SetModel 609
IRobotNonlinearHingeDef 609
about IRobotNonlinearHingeDef 609
Bar 610
IRobotNonlinearHingeDef fields 610
IRobotNonlinearHingeDef members 610
LabelName 611
Offset 611
Relative 611
IRobotNonlinearHingeModel 595
about IRobotNonlinearHingeModel 595
GetAxisParams 600
GetPoints 600
IRobotNonlinearHingeModel fields 596
IRobotNonlinearHingeModel members 596
IRobotNonlinearHingeModel methods 600
LimitCoordX 597
LimitCoordXValue 597
LimitCoordY 598
LimitCoordYValue 598
MixedUnloadingValue 598
ModelType 599
Name 599
SetAxisParams 601
SetPoints 601
Symetry 599
UnloadingMethod 599
IRobotNonlinearHingeModelAxisParams 601
about IRobotNonlinearHingeModelAxisParams 601
ImmediateOccupancy 602
IRobotNonlinearHingeModelAxisParams fields 602
IRobotNonlinearHingeModelAxisParams members 602

(c) 2014 Autodesk Inc. All rights reserved

LifeSafety 603
StructuralStability 603
IRobotNonlinearHingeModelPoints 604
about IRobotNonlinearHingeModelPoints 604
Count 605
Get 605
IRobotNonlinearHingeModelPoints fields 604
IRobotNonlinearHingeModelPoints members 604
IRobotNonlinearHingeModelPoints methods 605
Set 605
IRobotNonlinearHingeModelServer 593
about IRobotNonlinearHingeModelServer 593
Count 594
Create 594
Delete 595
Find 595
Get 595
IRobotNonlinearHingeModelServer fields 594
IRobotNonlinearHingeModelServer members 593
IRobotNonlinearHingeModelServer methods 594
IRobotNonlinearHingeModelType 603
IRobotNonlinearHingeModelUnloadingType 606
IRobotNonlinearHingeServer 612
about IRobotNonlinearHingeServer 612
Count 613
Get 613
IRobotNonlinearHingeServer members 612
IRobotNonlinearHingeServer methods 612
Remove 613
Set 614
IRobotNonlinearLink 680
about IRobotNonlinearLink 680
GetCurveType 682
GetParams 682
IRobotNonlinearLink fields 680
IRobotNonlinearLink members 680
IRobotNonlinearLink methods 682
ModelType 681
Name 681
SetCurveType 683
SetParams 683
Symetry 681
IRobotNonlinearLinkCurveType 687
IRobotNonlinearLinkMngr 700
about IRobotNonlinearLinkMngr 700
Delete 701
Get 701
IRobotNonlinearLinkMngr members 700
IRobotNonlinearLinkMngr methods 700
IsDefined 701
Set 702
IRobotNonlinearLinkModelType 699
IRobotNonlinearLinkParams 683
about IRobotNonlinearLinkParams 683
CurveType 684
IRobotNonlinearLinkParams fields 684
IRobotNonlinearLinkParams members 684
IRobotNonlinearLinkParamsBLinear 685
about IRobotNonlinearLinkParamsBLinear 685
D1 686
IRobotNonlinearLinkParamsBLinear fields 686
IRobotNonlinearLinkParamsBLinear members 686
K1 686
K2 687
IRobotNonlinearLinkParamsCustom 694
about IRobotNonlinearLinkParamsCustom 694
Count 695
Get 696
IRobotNonlinearLinkParamsCustom fields 695
IRobotNonlinearLinkParamsCustom members 695
IRobotNonlinearLinkParamsCustom methods 696
New 696
Remove 696
Set 697
IRobotNonlinearLinkParamsCustomSegment 697
about IRobotNonlinearLinkParamsCustomSegment 697
Constant 698
Expression 698
IRobotNonlinearLinkParamsCustomSegment fields 698
IRobotNonlinearLinkParamsCustomSegment members 697
OriginPoint 698
IRobotNonlinearLinkParamsGapHook 693
about IRobotNonlinearLinkParamsGapHook 693
D 694

(c) 2014 Autodesk Inc. All rights reserved

IRobotNonlinearLinkParamsGapHook fields 694
IRobotNonlinearLinkParamsGapHook members 694
K 694
IRobotNonlinearLinkParamsLinear 684
about IRobotNonlinearLinkParamsLinear 684
IRobotNonlinearLinkParamsLinear fields 685
IRobotNonlinearLinkParamsLinear members 685
K 685
IRobotNonlinearLinkParamsParabolic 687
about IRobotNonlinearLinkParamsParabolic 687
Dlim 688
Dmax 688
Flim 689
IRobotNonlinearLinkParamsParabolic fields 688
IRobotNonlinearLinkParamsParabolic members 688
K 689
IRobotNonlinearLinkParamsPlastic 689
about IRobotNonlinearLinkParamsPlastic 689
Dlim 690
Flim 690
IRobotNonlinearLinkParamsPlastic fields 690
IRobotNonlinearLinkParamsPlastic members 690
K 691
W 691
IRobotNonlinearLinkParamsPlasticHardening 691
about IRobotNonlinearLinkParamsPlasticHardening 691
Dlim 692
Flim 692
IRobotNonlinearLinkParamsPlasticHardening fields 692
IRobotNonlinearLinkParamsPlasticHardening members 691
K0 692
K1 693
W 693
IRobotNonlinearLinkSemiAxisType 699
IRobotNonlinearLinkServer 677
about IRobotNonlinearLinkServer 677
Count 678
Create 678
Find 679
Get 679
IRobotNonlinearLinkServer fields 678
IRobotNonlinearLinkServer members 678
IRobotNonlinearLinkServer methods 678
Remove 679
IRobotNumbersArray 1641
about IRobotNumbersArray 1641
Count 1641
Get 1642
IRobotNumbersArray fields 1641
IRobotNumbersArray members 1641
IRobotNumbersArray methods 1642
Set 1642
SetSize 1642
IRobotNumbersCollection 1645
about IRobotNumbersCollection 1645
Count 1646
Get 1646
IRobotNumbersCollection fields 1645
IRobotNumbersCollection members 1645
IRobotNumbersCollection methods 1646
IRobotNumbersDictionary 1662
about IRobotNumbersDictionary 1662
Count 1663
Delete 1663
Find 1663
FindGet 1664
Get 1664
IRobotNumbersDictionary fields 1662
IRobotNumbersDictionary members 1662
IRobotNumbersDictionary methods 1663
Set 1664
IRobotObjAttributes 763
about IRobotObjAttributes 763
DirZ 764
GetDirX 765
GetLabelName 766
GetLabels 766
GetLCS 766
GetLCSDisplayPosition 766
HasLabel 767
IRobotObjAttributes fields 764
IRobotObjAttributes members 763
IRobotObjAttributes methods 765
Meshed 764

(c) 2014 Autodesk Inc. All rights reserved

RemoveLabel 767
SetDirX 767
SetLabel 767
IRobotObjectsArray 1643
about IRobotObjectsArray 1643
Count 1644
Get 1644
IRobotObjectsArray fields 1643
IRobotObjectsArray members 1643
IRobotObjectsArray methods 1644
Set 1644
SetSize 1645
IRobotObjectStructuralType 850
IRobotObjectType 36
IRobotObjEdge 759
about IRobotObjEdge 759
Divide 761
DivideByPlane 761
DivideN 761
GetLabel 762
GetLabelName 762
GetLabels 762
HasLabel 762
IRobotObjEdge fields 760
IRobotObjEdge members 759
IRobotObjEdge methods 760
Path 760
RemoveLabel 763
SetLabel 763
IRobotObjEdgeCollection 770
about IRobotObjEdgeCollection 770
IRobotObjLocalXDirDefinitionType 794
IRobotObjMesh 803
about IRobotObjMesh 803
Freeze 804
Generate 805
GetBasePoints 806
GetQuality 806
IRobotObjMesh fields 804
IRobotObjMesh members 804
IRobotObjMesh methods 805
IsGenerated 805
Params 805
Remove 806
SetBasePoints 807
IRobotObjModifExtrusion 749
about IRobotObjModifExtrusion 749
IRobotObjModifExtrusion fields 749
IRobotObjModifExtrusion members 749
Vector 749
IRobotObjModification 745
about IRobotObjModification 745
Add 747
Clear 747
Filled 746
IRobotObjModification fields 746
IRobotObjModification members 745
IRobotObjModification methods 747
NDiv 746
Operations 746
Type 747
IRobotObjModificationCollection 769
about IRobotObjModificationCollection 769
IRobotObjModificationType 744
IRobotObjModifLathe 750
about IRobotObjModifLathe 750
Angle 751
AxsP1 751
AxsP2 751
IRobotObjModifLathe fields 750
IRobotObjModifLathe members 750
IRobotObjModifPyramid 751
about IRobotObjModifPyramid 751
AddPoint 753
ClearPoints 753
Factors 752
IRobotObjModifPyramid fields 752
IRobotObjModifPyramid members 752
IRobotObjModifPyramid methods 753
Points 753
IRobotObjObject 770
about IRobotObjObject 770
AnalyzeTTMethod 772
CalcArea 776

(c) 2014 Autodesk Inc. All rights reserved

CalcVol 776
FiniteElems 772
GetFiniteElemsData 777
GetHostedObjects 777
GetPart 777
GetPartType 778
Host 772
Initialize 778
IRobotObjObject fields 771
IRobotObjObject members 770
IRobotObjObject methods 776
IsVolume 773
Main 773
Mesh 773
Name 774
NameTemplate 774
Nodes 774
PartsCount 774
Reference 775
SetHostedObjects 778
StructuralType 775
UniqueId 775
Update 778
IRobotObjObjectServer 779
about IRobotObjObjectServer 779
AutoRecalcHoles 781
BeginMultiOperation 784
CalcArea 784
CalcVol 785
Create 785
CreateArc 785
CreateCircle 786
CreateCone 786
CreateContour 786
CreateCube 787
CreateCylinder 787
CreateOnFiniteElems 787
CreatePolyline 788
CreateSolid 788
EndMultiOperation 788
FindWithId 789
FreeNumber 782
GetAnalyzeTTMethodEnabled 789
GetFiniteElemsData 789
GetHost 790
GetHostedObjects 790
GetName 790
GetNameTemplate 791
GetStructuralType 791
GetUniqueId 791
IRobotObjObjectServer fields 781
IRobotObjObjectServer members 779
IRobotObjObjectServer methods 783
IsVolume 792
LinearReleases 782
Mesh 782
SetAnalyzeTTMethod 792
SetHost 792
SetHostedObjects 793
SetNameTemplate 793
SetStructuralType 793
IRobotObjOperation 747
about IRobotObjOperation 747
IRobotObjOperation fields 748
IRobotObjOperation members 748
Type 748
IRobotObjOperationCollection 748
about IRobotObjOperationCollection 748
IRobotObjOperationType 744
IRobotObjOperMeshing 757
about IRobotObjOperMeshing 757
Add 759
Clear 759
IRobotObjOperMeshing fields 758
IRobotObjOperMeshing members 757
IRobotObjOperMeshing methods 758
Points 758
Vectors 758
IRobotObjOperRotation 756
about IRobotObjOperRotation 756
Angle 756
AxsP1 757
AxsP2 757
IRobotObjOperRotation fields 756

(c) 2014 Autodesk Inc. All rights reserved

IRobotObjOperRotation members 756
IRobotObjOperScaling 754
about IRobotObjOperScaling 754
Center 755
Factor 755
IRobotObjOperScaling fields 755
IRobotObjOperScaling members 755
IRobotObjOperTranslation 754
about IRobotObjOperTranslation 754
IRobotObjOperTranslation fields 754
IRobotObjOperTranslation members 754
Vector 754
IRobotObjPart 768
about IRobotObjPart 768
Attribs 768
GetGeometry 769
IRobotObjPart fields 768
IRobotObjPart members 768
IRobotObjPart methods 769
Type 769
IRobotObjPart2 801
about IRobotObjPart2 801
CalcArea 803
CalcVol 803
FiniteElems 802
IRobotObjPart2 fields 802
IRobotObjPart2 members 801
IRobotObjPart2 methods 802
Nodes 802
IRobotObjPartMain 794
about IRobotObjPartMain 794
AddModification 798
CalcArea 798
CalcVol 798
ClearModifications 799
CurveDiv 795
DefPoints 796
Edges 796
FiniteElems 796
Geometry 796
IRobotObjPartMain fields 795
IRobotObjPartMain members 794
IRobotObjPartMain methods 797
ModelPoints 797
Modifications 797
Nodes 797
IRobotObjPartReference 799
about IRobotObjPartReference 799
CalcArea 800
CalcVol 801
FiniteElems 800
IRobotObjPartReference fields 799
IRobotObjPartReference members 799
IRobotObjPartReference methods 800
Nodes 800
IRobotObjPartType 779
IRobotOutputFileFormat 1502
IRobotPageSetup 1491
about IRobotPageSetup 1491
Footer 1493
FromEdgeFooter 1494
FromEdgeHeader 1494
GetTemplateName 1500
Gutter 1494
Header 1495
IRobotPageSetup fields 1493
IRobotPageSetup members 1492
IRobotPageSetup methods 1500
IsCurrent 1495
Load 1501
LoadCurrent 1501
MarginBottom 1495
MarginLeft 1496
MarginRight 1496
MarginTop 1496
PageOrientation 1497
PaperHeight 1497
PaperSize 1497
PaperWidth 1498
Save 1501
SaveAs 1502
StartPageNumber 1498
TemplateCount 1498
TemplateName 1499

(c) 2014 Autodesk Inc. All rights reserved

TextFrame 1499
TitlePage 1499
Toc 1499
Variables 1500
IRobotPageSetupFrameType 1515
IRobotPageSetupOrientation 1514
IRobotPageSetupTableOfContents 1515
about IRobotPageSetupTableOfContents 1515
Active 1516
IncludeTitle 1516
IRobotPageSetupTableOfContents fields 1516
IRobotPageSetupTableOfContents members 1515
Location 1516
IRobotPageSetupTocLocation 1517
IRobotPanelCalcModelData 720
about IRobotPanelCalcModelData 720
Diaphragm 721
FinElemType 721
IRobotPanelCalcModelData fields 721
IRobotPanelCalcModelData members 720
LoadTransfer 721
IRobotPanelCalcModelDiaphragm 720
IRobotPanelCalcModelFinElemType 719
IRobotPanelCalcModelLoadTransfer 719
IRobotPanelCut 981
about IRobotPanelCut 981
Active 982
Color 982
DefType 982
IRobotPanelCut fields 982
IRobotPanelCut members 981
Point1 983
Point2 983
Point3 983
IRobotPanelCutDefinitionType 984
IRobotPanelCutMngr 978
about IRobotPanelCutMngr 978
Count 978
Create 979
Find 979
Get 980
GetName 980
IRobotPanelCutMngr fields 978
IRobotPanelCutMngr members 978
IRobotPanelCutMngr methods 979
Remove 980
Store 981
IRobotParamCollection 2187
about IRobotParamCollection 2187
Count 2187
Find 2188
GetFullName 2188
GetId 2189
GetName 2189
GetValue 2189
IRobotParamCollection fields 2187
IRobotParamCollection members 2187
IRobotParamCollection methods 2188
IRobotParamDef 2198
about IRobotParamDef 2198
DefaultValue 2199
GuiGetName 2202
GuiReadOnly 2199
GuiSetName 2202
GuiVisible 2200
IRobotParamDef fields 2199
IRobotParamDef members 2198
IRobotParamDef methods 2201
Name 2200
UniqueId 2200
ValueList 2201
ValueType 2201
IRobotParamSchema 2203
about IRobotParamSchema 2203
Def 2204
GetAllParams 2205
GetObjectsWithParam 2205
GetObjectsWithParamVal 2205
GetParam 2206
IRobotParamSchema fields 2204
IRobotParamSchema members 2203
IRobotParamSchema methods 2204
RemoveAllParams 2206
RemoveParam 2206

(c) 2014 Autodesk Inc. All rights reserved

ResetParam 2207
SetParam 2207
IRobotParamSchemaDef 2190
about IRobotParamSchemaDef 2190
AddParam 2192
AddSimpleParam 2192
GetParam 2193
GetParamName 2193
IRobotParamSchemaDef fields 2191
IRobotParamSchemaDef members 2190
IRobotParamSchemaDef methods 2191
Name 2191
ParamCount 2191
ParamNameToUniqueId 2193
RemoveParam 2194
IRobotParamSchemaMngr 2194
about IRobotParamSchemaMngr 2194
Clear 2196
Create 2196
Exist 2196
Get 2197
GetByName 2197
IRobotParamSchemaMngr fields 2195
IRobotParamSchemaMngr members 2194
IRobotParamSchemaMngr methods 2195
Remove 2197
RemoveByName 2198
SchemaCount 2195
IRobotParamServer 2181
about IRobotParamServer 2181
GetAllParams 2183
GetAllParamsForSchema 2183
GetObjectsWithParam 2184
GetObjectsWithParamVal 2184
GetParam 2185
IRobotParamServer fields 2182
IRobotParamServer members 2181
IRobotParamServer methods 2183
ParamUniqueIdToName 2185
RemoveParam 2185
RemoveSchemaParams 2186
ResetParam 2186
Schemas 2182
SetParam 2186
IRobotParamValueType 2202
IRobotPointAuxiliaryRecordValues 44
IRobotPointsArray 1658
about IRobotPointsArray 1658
Count 1659
Get 1659
IRobotPointsArray fields 1659
IRobotPointsArray members 1658
IRobotPointsArray methods 1659
Set 1660
SetSize 1660
IRobotPredefinedLabel 29
IRobotPredefinedSelection 1041
IRobotPreferences 1541
about IRobotPreferences 1541
CalculationsType 1542
CloudCalculationsEnabled 1542
GetDirectory 1543
GetLanguage 1543
IRobotPreferences fields 1541
IRobotPreferences members 1541
IRobotPreferences methods 1543
Multiprocessing 1542
OpenGL 1542
IRobotPreferencesEvents 1546
about IRobotPreferencesEvents 1546
IRobotPreferencesEvents members 1546
IRobotPreferencesEvents methods 1546
OnDialogOK 1546
IRobotPressureRecordValues 45
IRobotPrintable 1477
about IRobotPrintable 1477
Comment 1478
IncludeDateTime 1478
IRobotPrintable fields 1478
IRobotPrintable members 1478
IRobotPrintable methods 1479
SaveToFile 1479
StartFromNewPage 1479
Title 1479

(c) 2014 Autodesk Inc. All rights reserved

IRobotPrintEngine 1480
about IRobotPrintEngine 1480
AddScToReport 1483
AddTemplateToReport 1483
AddToReport 1484
ClosePreview 1484
CreateReportFromOrganizer 1484
ExternalPreviewReport 1485
IRobotPrintEngine fields 1481
IRobotPrintEngine members 1480
IRobotPrintEngine methods 1482
IsWhilePreview 1481
OrganizerItems 1481
PageSetup 1482
PreviewReport 1485
PrintReport 1485
RemoveFromReport 1485
RemoveScFromReport 1486
ReportTemplates 1482
ResetReport 1486
SaveReportToFile 1486
SaveReportToOrganizer 1486
SaveReportToTemplate 1487
ScreenCaptures 1482
IRobotProject 1342
about IRobotProject 1342
ActiveModel 1345
AxisMngr 1345
Backgrounds 1345
CalcEngine 1346
Close 1351
ComponentMngr 1346
ConcrReinfEngine 1346
Connections 1346
DimServer 1347
ExtFileName 1347
ExtFileParams 1347
FileInsertParams 1348
FileName 1348
InsertExtFile 1351
IRobotProject fields 1344
IRobotProject members 1342
IRobotProject methods 1350
IsActive 1348
Name 1348
New 1352
NewFromTemplate 1352
Open 1352
OpenExtFile 1353
Preferences 1349
PrintEngine 1349
ReadExtFileParams 1353
Save 1353
SaveAs 1354
SaveAsExtFile 1354
SaveToFormat 1354
Structure 1349
Type 1349
UniqueId 1350
ViewMngr 1350
IRobotProjectComponent 1269
about IRobotProjectComponent 1269
Data 1270
IRobotProjectComponent fields 1270
IRobotProjectComponent members 1270
Name 1270
Type 1271
IRobotProjectComponentMngr 1271
about IRobotProjectComponentMngr 1271
Create 1273
Get 1273
GetLevelName 1273
IRobotProjectComponentMngr fields 1272
IRobotProjectComponentMngr members 1271
IRobotProjectComponentMngr methods 1272
LevelCount 1272
StdLevelName 1272
IRobotProjectComponentType 1269
IRobotProjectEvents 1358
about IRobotProjectEvents 1358
IRobotProjectEvents members 1359
IRobotProjectEvents methods 1359
OnClose 1359
OnSave 1359

(c) 2014 Autodesk Inc. All rights reserved

IRobotProjectPreferences 1296
about IRobotProjectPreferences 1296
CalcModelCoherence 1298
EurocodeFactors 1298
GetActiveCode 1302
GetActiveCodeNumber 1303
GetCurrentDatabase 1303
IRobotProjectPreferences fields 1297
IRobotProjectPreferences members 1296
IRobotProjectPreferences methods 1302
KinematicConstraints 1299
Materials 1299
MeshAutoAdjust 1299
MeshAutoAdjustIterationCount 1299
MeshParams 1300
MeshParamsFloors 1300
MeshParamsWalls 1300
Save 1303
SectionsActive 1301
SectionsFound 1301
SetActiveCode 1304
SetActiveCodeNumber 1304
SetCurrentDatabase 1304
Units 1301
VehiclesActive 1301
VehiclesFound 1302
IRobotProjectPreferencesEvents 1315
about IRobotProjectPreferencesEvents 1315
IRobotProjectPreferencesEvents members 1315
IRobotProjectPreferencesEvents methods 1315
OnDialogOK 1316
IRobotProjectSaveFormat 1358
IRobotProjectType 1342
IRobotProtectionInfo 1653
about IRobotProtectionInfo 1653
IRobotProtectionInfo members 1653
IRobotProtectionInfo methods 1654
IsExtensionEnabled 1654
IRobotPseudostaticForceServer 867
about IRobotPseudostaticForceServer 867
CombValue 868
IRobotPseudostaticForceServer members 867
IRobotPseudostaticForceServer methods 868
Value 868
IRobotPushOverAnalysisParams 392
about IRobotPushOverAnalysisParams 392
Direction 393
IRobotPushOverAnalysisParams fields 392
IRobotPushOverAnalysisParams members 392
LoadDefinition 393
MaxDisplacement 393
Node 394
Nonlinear 394
NonlinearParams 394
IRobotPushOverDirection 395
IRobotPushOverLoadDefinitionMethod 395
IRobotQuitOption 1545
IRobotReactionData 885
about IRobotReactionData 885
FX 885
FY 886
FZ 886
IRobotReactionData fields 885
IRobotReactionData members 885
MX 886
MY 886
MZ 886
IRobotReactionServer 854
about IRobotReactionServer 854
DDC 856
DDCEx 856
DDCLocal 857
DDCLocalEx 857
DDCSum 857
DDCSumEx 857
DynCombDDC 858
DynCombDDCLocal 858
DynCombDDCSum 858
DynCombLocal 859
DynCombSum 859
DynCombSumForce 859
DynCombValue 859
DynDDC 860
DynDDCLocal 860

(c) 2014 Autodesk Inc. All rights reserved

DynDDCSum 860
DynLocal 861
DynSum 861
DynSumForce 861
DynValue 861
IRobotReactionServer members 854
IRobotReactionServer methods 855
Local 862
LocalEx 862
Sum 862
SumEx 862
SumForce 863
SumForceEx 863
Value 863
ValueEx 863
IRobotReinforceCalcMethods 956
IRobotReportItem 1509
about IRobotReportItem 1509
CreateView 1513
ExcludeFromReport 1510
GetPageTemplate 1514
HasNoteAfter 1510
HasNoteBefore 1511
IncludeDateTime 1511
IRobotReportItem fields 1510
IRobotReportItem members 1509
IRobotReportItem methods 1513
NoteAfter 1511
NoteBefore 1512
StartFromNewPage 1512
Title 1512
TitleText 1513
Type 1513
IRobotReportItemList 1507
about IRobotReportItemList 1507
Count 1508
Get 1508
IRobotReportItemList fields 1508
IRobotReportItemList members 1508
IRobotReportItemList methods 1508
IRobotReportItemType 1517
IRobotReportStdElementRtf 1505
about IRobotReportStdElementRtf 1505
Active 1505
Frame 1506
IRobotReportStdElementRtf fields 1505
IRobotReportStdElementRtf members 1505
IRobotReportStdElementRtf methods 1506
LoadFromFile 1506
RestoreDefaults 1507
SaveToFile 1507
IRobotReportTemplateMngr 1502
about IRobotReportTemplateMngr 1502
Count 1503
Find 1504
Get 1504
IRobotReportTemplateMngr fields 1503
IRobotReportTemplateMngr members 1503
IRobotReportTemplateMngr methods 1503
Remove 1504
IRobotResultParamType 1025
IRobotResultQueryParams 1029
about IRobotResultQueryParams 1029
GetParam 1031
IRobotResultQueryParams fields 1030
IRobotResultQueryParams members 1030
IRobotResultQueryParams methods 1031
IsParamSet 1032
Reset 1032
ResultIds 1030
Selection 1031
SetParam 1032
IRobotResultQueryReturnType 1033
IRobotResultRow 1023
about IRobotResultRow 1023
GetParam 1024
GetValue 1024
GetValueType 1024
IRobotResultRow members 1023
IRobotResultRow methods 1023
IsAvailable 1025
IRobotResultRowSet 1027
about IRobotResultRowSet 1027
Clear 1029

(c) 2014 Autodesk Inc. All rights reserved

CurrentRow 1028
IRobotResultRowSet fields 1028
IRobotResultRowSet members 1027
IRobotResultRowSet methods 1028
MoveFirst 1029
MoveNext 1029
ResultIds 1028
IRobotResultServer 1002
about IRobotResultServer 1002
Advanced 1004
Any 1004
Bars 1004
CalculationResume 1004
Extremes 1005
FiniteElems 1005
IRobotResultServer fields 1003
IRobotResultServer members 1003
IRobotResultServer methods 1006
Nodes 1005
Query 1007
Status 1006
Storeys 1006
Total 1006
IRobotResultStatusType 1033
IRobotRtfView 1462
about IRobotRtfView 1462
AppendFromFile 1464
Evaluate 1464
IRobotRtfView fields 1463
IRobotRtfView members 1463
IRobotRtfView methods 1463
LoadFromFile 1464
Printable 1463
SaveToFile 1464
IRobotScreenCaptureMngr 1465
about IRobotScreenCaptureMngr 1465
Count 1466
Find 1466
Get 1467
IRobotScreenCaptureMngr fields 1466
IRobotScreenCaptureMngr members 1465
IRobotScreenCaptureMngr methods 1466
Remove 1467
IRobotSectionDatabase 1312
about IRobotSectionDatabase 1312
Description 1313
FullName 1313
GetAll 1314
IRobotSectionDatabase fields 1313
IRobotSectionDatabase members 1312
IRobotSectionDatabase methods 1314
Load 1314
LoadFromFile 1315
Name 1314
IRobotSectionDatabaseList 1305
about IRobotSectionDatabaseList 1305
Add 1306
AddFromFile 1307
ChangeIndex 1307
Count 1306
Find 1307
Get 1307
GetDatabase 1308
IRobotSectionDatabaseList fields 1305
IRobotSectionDatabaseList members 1305
IRobotSectionDatabaseList methods 1306
Remove 1308
IRobotSeismicAnalysis_AFPS_90_Params 212
about IRobotSeismicAnalysis_AFPS_90_Params 212
BehaviorFactor 213
Direction 213
DirectionType 213
Filter 214
IRobotSeismicAnalysis_AFPS_90_Params fields 212
IRobotSeismicAnalysis_AFPS_90_Params members 212
ResidualMode 214
Site 214
SpectrumType 215
StructureType 215
Topography 215
ZoneType 216
IRobotSeismicAnalysis_AFPS_90_SiteType 257
IRobotSeismicAnalysis_AFPS_90_StructureType 256
IRobotSeismicAnalysis_AFPS_90_ZoneType 253

(c) 2014 Autodesk Inc. All rights reserved

IRobotSeismicAnalysis_CHINESE_DesignType 259
IRobotSeismicAnalysis_CHINESE_EarthquakeType 260
IRobotSeismicAnalysis_CHINESE_IntensityType 259
IRobotSeismicAnalysis_CHINESE_Params 216
about IRobotSeismicAnalysis_CHINESE_Params 216
DesignStandard 217
Direction 217
EarthquakeType 218
Factor 218
Filter 218
Intensity 219
IRobotSeismicAnalysis_CHINESE_Params fields 217
IRobotSeismicAnalysis_CHINESE_Params members 216
Site 219
SiteTg 219
StructureType 220
IRobotSeismicAnalysis_CHINESE_SiteType 258
IRobotSeismicAnalysis_CHINESE_StructureType 258
IRobotSeismicAnalysis_CIRSOC_103_Params 220
about IRobotSeismicAnalysis_CIRSOC_103_Params 220
Direction 221
DirectionType 221
Filter 221
IRobotSeismicAnalysis_CIRSOC_103_Params fields 220
IRobotSeismicAnalysis_CIRSOC_103_Params members 220
PlasticityCoeff 222
Soil 222
StructureType 222
ZoneType 223
IRobotSeismicAnalysis_CIRSOC_103_SoilType 260
IRobotSeismicAnalysis_CIRSOC_103_StructureType 260
IRobotSeismicAnalysis_CIRSOC_103_ZoneType 253
IRobotSeismicAnalysis_DM_16_1_96_Params 223
about IRobotSeismicAnalysis_DM_16_1_96_Params 223
Direction 224
Filter 224
IRobotSeismicAnalysis_DM_16_1_96_Params fields 224
IRobotSeismicAnalysis_DM_16_1_96_Params members 223
SeismicCoeff 224
SeismicProtectionCoeff 225
IRobotSeismicAnalysis_DM_16_1_96_ProtectionCoeffType 261
IRobotSeismicAnalysis_EAK_2000_GroundCategoryType 268
IRobotSeismicAnalysis_EAK_2000_ImportanceFactorType 267
IRobotSeismicAnalysis_EAK_2000_Params 248
about IRobotSeismicAnalysis_EAK_2000_Params 248
BehaviorFactor 250
Direction 250
DirectionType 250
Filter 251
FoundationFactor 251
GroundCategory 251
ImportanceFactor 252
IRobotSeismicAnalysis_EAK_2000_Params fields 249
IRobotSeismicAnalysis_EAK_2000_Params members 249
VerticalBehaviorFactor 252
VerticalFoundationFactor 252
ZoneType 252
IRobotSeismicAnalysis_EAK_2000_ZoneType 256
IRobotSeismicAnalysis_EC8_General_Params 296
about IRobotSeismicAnalysis_EC8_General_Params 296
Ag 297
B 298
BehaviorFactor 298
Direction 298
DirectionType 298
ExcitationDir 299
Filter 299
IRobotSeismicAnalysis_EC8_General_Params fields 297
IRobotSeismicAnalysis_EC8_General_Params members 296
ResidualMode 299
S 299
Spectrum 300
Tb 300
Tc 300
Td 300
IRobotSeismicAnalysis_EC8_GroundType 301
IRobotSeismicAnalysis_EC8_Params 302
about IRobotSeismicAnalysis_EC8_Params 302
Ag 303
BehaviorFactor 303

(c) 2014 Autodesk Inc. All rights reserved

Direction 303
DirectionType 303
ExcitationDir 304
Filter 304
GroundType 304
IRobotSeismicAnalysis_EC8_Params fields 302
IRobotSeismicAnalysis_EC8_Params members 302
ResidualMode 305
Spectrum 305
SpectrumType 305
IRobotSeismicAnalysis_EC8_SpectrumType 301
IRobotSeismicAnalysis_IBC_2000_Params 245
about IRobotSeismicAnalysis_IBC_2000_Params 245
BehaviorFactor 246
Direction 246
ExcitationDir 247
Filter 247
Ie 247
IRobotSeismicAnalysis_IBC_2000_Params fields 246
IRobotSeismicAnalysis_IBC_2000_Params members 246
S1 248
SiteClass 248
Ss 248
IRobotSeismicAnalysis_IBC_2000_SiteClassType 267
IRobotSeismicAnalysis_IBC_2006_Params 282
about IRobotSeismicAnalysis_IBC_2006_Params 282
BehaviorFactor 283
Direction 283
ExcitationDir 284
Filter 284
I 284
IRobotSeismicAnalysis_IBC_2006_Params fields 283
IRobotSeismicAnalysis_IBC_2006_Params members 282
S1 284
SiteClass 285
Ss 285
TL 285
IRobotSeismicAnalysis_IBC_2006_SiteClassType 286
IRobotSeismicAnalysis_ITALY_ORDINANZA_Direction 278
IRobotSeismicAnalysis_ITALY_ORDINANZA_Params 276
about IRobotSeismicAnalysis_ITALY_ORDINANZA_Params 276
Direction 276
ExcitationDir 277
FactorQ 277
Filter 277
IRobotSeismicAnalysis_ITALY_ORDINANZA_Params fields 276
IRobotSeismicAnalysis_ITALY_ORDINANZA_Params members 276
Soil 277
Spectrum 277
Zone 278
IRobotSeismicAnalysis_ITALY_ORDINANZA_SoilType 278
IRobotSeismicAnalysis_ITALY_ORDINANZA_Spectrum 278
IRobotSeismicAnalysis_ITALY_ORDINANZA_ZoneType 278
IRobotSeismicAnalysis_P_100_2006_Params 279
about IRobotSeismicAnalysis_P_100_2006_Params 279
Agg 280
B0 280
BehaviorFactor 280
ExcitationDir 280
ImportanceFactor 281
IRobotSeismicAnalysis_P_100_2006_Params fields 279
IRobotSeismicAnalysis_P_100_2006_Params members 279
SpectrumType 281
Tb 281
Tc 281
Td 282
IRobotSeismicAnalysis_P_100_92_ImportanceClassType 262
IRobotSeismicAnalysis_P_100_92_Params 225
about IRobotSeismicAnalysis_P_100_92_Params 225
Direction 226
ExcitationDir 226
Filter 226
ImportanceClass 227
IRobotSeismicAnalysis_P_100_92_Params fields 226
IRobotSeismicAnalysis_P_100_92_Params members 225
Psi 227
Tc 227
ZoneType 228
IRobotSeismicAnalysis_P_100_92_ZoneType 254
IRobotSeismicAnalysis_PS_69_DampingType 262
IRobotSeismicAnalysis_PS_69_Params 228

(c) 2014 Autodesk Inc. All rights reserved

about IRobotSeismicAnalysis_PS_69_Params 228
Alpha 229
Damping 229
Delta 229
Direction 230
Filter 230
IRobotSeismicAnalysis_PS_69_Params fields 229
IRobotSeismicAnalysis_PS_69_Params members 228
Soil 230
IRobotSeismicAnalysis_PS_69_SoilType 263
IRobotSeismicAnalysis_PS_92_2008_Params 286
about IRobotSeismicAnalysis_PS_92_2008_Params 286
Ag 287
BehaviorFactor 288
Direction 288
DirectionType 288
ExcitationDir 289
Filter 289
IRobotSeismicAnalysis_PS_92_2008_Params fields 287
IRobotSeismicAnalysis_PS_92_2008_Params members 287
ResidualMode 289
Site 289
SiteEnvelope 290
SpectrumType 290
StructureType 290
Topography 291
ZoneType 291
IRobotSeismicAnalysis_PS_92_2008_SiteEnvelope 293
about IRobotSeismicAnalysis_PS_92_2008_SiteEnvelope 293
IRobotSeismicAnalysis_PS_92_2008_SiteEnvelope members 293
IRobotSeismicAnalysis_PS_92_2008_SiteEnvelope methods 293
IsActive 293
SetActive 294
IRobotSeismicAnalysis_PS_92_2008_SiteType 292
IRobotSeismicAnalysis_PS_92_2008_StructureType 292
IRobotSeismicAnalysis_PS_92_2008_ZoneType 291
IRobotSeismicAnalysis_PS_92_Params 231
about IRobotSeismicAnalysis_PS_92_Params 231
BehaviorFactor 232
Direction 232
DirectionType 233
ExcitationDir 233
Filter 233
IRobotSeismicAnalysis_PS_92_Params fields 231
IRobotSeismicAnalysis_PS_92_Params members 231
ResidualMode 233
Site 234
SiteEnvelope 234
SpectrumType 234
StructureType 235
Topography 235
ZoneType 235
IRobotSeismicAnalysis_PS_92_SiteEnvelope 268
about IRobotSeismicAnalysis_PS_92_SiteEnvelope 268
IRobotSeismicAnalysis_PS_92_SiteEnvelope members 268
IRobotSeismicAnalysis_PS_92_SiteEnvelope methods 269
IsActive 269
SetActive 269
IRobotSeismicAnalysis_PS_92_SiteType 263
IRobotSeismicAnalysis_PS_92_StructureType 263
IRobotSeismicAnalysis_PS_92_ZoneType 254
IRobotSeismicAnalysis_RPA_2003_Params 273
about IRobotSeismicAnalysis_RPA_2003_Params 273
BehaviorFactor 274
ExcitationDir 274
Filter 274
IRobotSeismicAnalysis_RPA_2003_Params fields 274
IRobotSeismicAnalysis_RPA_2003_Params members 273
QualityCoef 274
ResidualMode 275
Site 275
Usage 275
Zone 275
IRobotSeismicAnalysis_RPA_2003_SiteType 273
IRobotSeismicAnalysis_RPA_2003_UsageType 273
IRobotSeismicAnalysis_RPA_2003_ZoneType 272
IRobotSeismicAnalysis_RPA_88_CategoryType 264
IRobotSeismicAnalysis_RPA_88_Params 236
about IRobotSeismicAnalysis_RPA_88_Params 236
Category 236

(c) 2014 Autodesk Inc. All rights reserved

Direction 237
Filter 237
IRobotSeismicAnalysis_RPA_88_Params fields 236
IRobotSeismicAnalysis_RPA_88_Params members 236
QualityFactor 237
Soil 238
Usage 238
ZoneType 238
IRobotSeismicAnalysis_RPA_88_SoilType 265
IRobotSeismicAnalysis_RPA_88_UsageType 264
IRobotSeismicAnalysis_RPA_88_ZoneType 255
IRobotSeismicAnalysis_RPS_2000_Params 270
about IRobotSeismicAnalysis_RPS_2000_Params 270
BehaviorFactor 271
DirectionType 271
ExcitationDir 271
Filter 271
IRobotSeismicAnalysis_RPS_2000_Params fields 270
IRobotSeismicAnalysis_RPS_2000_Params members 270
ResidualMode 272
Site 272
StructureClass 272
Zone 272
IRobotSeismicAnalysis_RPS_2000_SiteType 269
IRobotSeismicAnalysis_RPS_2000_StructureClass 270
IRobotSeismicAnalysis_RPS_2000_ZoneType 269
IRobotSeismicAnalysis_TURKISH_23098_Params 239
about IRobotSeismicAnalysis_TURKISH_23098_Params 239
BehaviorFactor 240
Direction 240
Filter 240
IRobotSeismicAnalysis_TURKISH_23098_Params fields 239
IRobotSeismicAnalysis_TURKISH_23098_Params members 239
SoilType 241
StructureImportance 241
ZoneType 241
IRobotSeismicAnalysis_TURKISH_23098_SoilType 265
IRobotSeismicAnalysis_TURKISH_23098_ZoneType 256
IRobotSeismicAnalysis_UBC_97_Params 241
about IRobotSeismicAnalysis_UBC_97_Params 241
BehaviorFactor 242
ClosestDistance 243
Direction 243
ExcitationDir 243
Filter 244
I 244
IRobotSeismicAnalysis_UBC_97_Params fields 242
IRobotSeismicAnalysis_UBC_97_Params members 242
Soil 244
Source 245
ZoneType 245
IRobotSeismicAnalysis_UBC_97_SoilType 266
IRobotSeismicAnalysis_UBC_97_SourceType 266
IRobotSeismicAnalysis_UBC_97_ZoneType 255
IRobotSeismicAnalysisDirectionType 261
IRobotSeismicAnalysisSpectrumType 257
IRobotSeismicResidualMode 294
about IRobotSeismicResidualMode 294
AugmentationFactor 295
DefinitionMethod 295
IRobotSeismicResidualMode fields 295
IRobotSeismicResidualMode members 295
IsActive 296
LimitFrequency 296
IRobotSeismicResidualModeDefinitionType 294
IRobotSeismicResultsPanelDirection 1584
IRobotSeismicResultsSaveParams 1579
about IRobotSeismicResultsSaveParams 1579
Displacements 1581
Forces 1581
IRobotSeismicResultsSaveParams fields 1580
IRobotSeismicResultsSaveParams members 1580
LocalDisplacements 1581
NMQ 1581
OnlyQuadraticCombs 1582
PanelsDir 1582
PointNumber 1582
Reactions 1583
Reduced 1583
Save 1583
Stresses 1584
IRobotSelection 3

(c) 2014 Autodesk Inc. All rights reserved

about IRobotSelection 3
Add 5
AddOne 5
AddText 6
And 6
AndText 6
Clear 7
Contains 7
Count 4
Exclude 7
ExcludeOne 8
ExcludeText 8
FromText 8
Get 9
IRobotSelection fields 4
IRobotSelection members 3
IRobotSelection methods 5
ToText 9
Type 4
IRobotSelectionFactory 1037
about IRobotSelectionFactory 1037
Create 1038
CreateByLabel 1038
CreateByStorey 1038
CreateFull 1039
CreateMulti 1039
CreatePredefined 1039
Get 1039
IRobotSelectionFactory members 1037
IRobotSelectionFactory methods 1037
IRobotSELFSeismic_AS_1170_4_Params 467
about IRobotSELFSeismic_AS_1170_4_Params 467
IRobotSELFSeismic_AS_1170_4_Params fields 468
IRobotSELFSeismic_AS_1170_4_Params members 468
Kp 468
Mi 469
Probability 469
SoilCategory 469
Sp 469
StructureTypeX 470
StructureTypeY 470
Z 470
IRobotSELFSeismic_AS_1170_4_ProbabilityType 471
IRobotSELFSeismic_AS_1170_4_SoilCategoryType 470
IRobotSELFSeismic_AS_1170_4_StructureType 471
IRobotSELFSeismic_ASCE_7_10_Params 458
about IRobotSELFSeismic_ASCE_7_10_Params 458
I 459
IRobotSELFSeismic_ASCE_7_10_Params fields 459
IRobotSELFSeismic_ASCE_7_10_Params members 459
R 460
S1 460
SiteClass 460
Ss 460
StructureTypeX 461
StructureTypeY 461
TBaseValueX 461
TBaseValueY 461
TL 462
IRobotSELFSeismic_ASCE_7_10_SiteClassType 458
IRobotSELFSeismic_ASCE_7_10_StructureType 462
IRobotSELFSeismic_EC_8_Params 462
about IRobotSELFSeismic_EC_8_Params 462
Ag 463
Beta 464
IRobotSELFSeismic_EC_8_Params fields 463
IRobotSELFSeismic_EC_8_Params members 463
IRobotSELFSeismic_EC_8_Params methods 465
Q 464
SiteClass 464
SiteClassEnvelopeCheck 466
SiteClassEnvelopeIsChecked 466
SpectrumType 465
StructureTypeX 465
StructureTypeY 465
IRobotSELFSeismic_EC_8_SiteClass 467
IRobotSELFSeismic_EC_8_SpectrumType 467
IRobotSELFSeismic_EC_8_StructureType 466
IRobotSELFSeismicAnalysisParams 455
about IRobotSELFSeismicAnalysisParams 455
CodeName 456
CodeNumber 456
CodeParams 456
Eccentricities 457

(c) 2014 Autodesk Inc. All rights reserved

GetExcitationDir 458
IRobotSELFSeismicAnalysisParams fields 456
IRobotSELFSeismicAnalysisParams members 455
IRobotSELFSeismicAnalysisParams methods 457
StructureParams 457
TBaseMethod 457
IRobotSELFSeismicEngine 448
about IRobotSELFSeismicEngine 448
GenerateLoadCases 449
GenerationParams 448
IRobotSELFSeismicEngine fields 448
IRobotSELFSeismicEngine members 448
IRobotSELFSeismicEngine methods 449
IRobotSELFSeismicGenerationParams 449
about IRobotSELFSeismicGenerationParams 449
CodeName 450
CodeNumber 450
ExcitationDir 451
IRobotSELFSeismicGenerationParams fields 450
IRobotSELFSeismicGenerationParams members 450
ModalCaseParams 451
SeismicParams 451
TBaseMethod 451
IRobotSELFSeismicLevelDefinitionMethod 454
IRobotSELFSeismicStructureParams 452
about IRobotSELFSeismicStructureParams 452
BaseLevelCoordZ 452
BaseLevelDefMethod 453
BaseLevelStorey 453
IRobotSELFSeismicStructureParams fields 452
IRobotSELFSeismicStructureParams members 452
TopLevelCoordZ 453
TopLevelDefMethod 454
TopLevelStorey 454
IRobotSELFSeismicTBaseMethod 455
IRobotSerializable 1660
about IRobotSerializable 1660
IRobotSerializable members 1661
IRobotSerializable methods 1661
Read 1661
Write 1661
IRobotSimpleCase 420
about IRobotSimpleCase 420
GetAnalysisParams 424
GetSeismicCode 425
IRobotSimpleCase fields 421
IRobotSimpleCase members 421
IRobotSimpleCase methods 424
IsAuxiliary 422
Label 422
MainMode 422
ModesCount 423
NatureName 423
Records 423
SetAnalysisParams 425
SetNatureExt 425
TimeStepCount 423
UniqueId 424
IRobotSnowWindEngine 96
about IRobotSnowWindEngine 96
CodeParams 97
Generate 98
Generate3D 98
GenerateStruct3D 98
IRobotSnowWindEngine fields 97
IRobotSnowWindEngine members 96
IRobotSnowWindEngine methods 98
Params 97
ShowParamsDlg 99
IRobotSnowWindParams 93
about IRobotSnowWindParams 93
BaseOnGround 94
BaySpacing 94
Envelope 94
IRobotSnowWindParams fields 93
IRobotSnowWindParams members 93
IsolatedRoofs 95
IsSnow 95
IsWind 95
TotalDepth 95
WithCavities 96
WithParapets 96
IRobotSolidPropertiesData 741
about IRobotSolidPropertiesData 741

(c) 2014 Autodesk Inc. All rights reserved

DampCoef 742
E 742
IRobotSolidPropertiesData fields 741
IRobotSolidPropertiesData members 741
IRobotSolidPropertiesData methods 744
LoadFromDBase 744
LX 742
MaterialModel 743
NU 743
RO 743
IRobotSparseMSolverMethod 1573
IRobotSparseMSolverParams 1572
about IRobotSparseMSolverParams 1572
IRobotSparseMSolverParams fields 1573
IRobotSparseMSolverParams members 1573
Method 1573
IRobotSpectralAnalysisAbscissaXAxisType 316
IRobotSpectralAnalysisOrdinateYAxisType 317
IRobotSpectralAnalysisParams 306
about IRobotSpectralAnalysisParams 306
Direction 307
ExcitationDir 307
Filter 307
IRobotSpectralAnalysisParams fields 306
IRobotSpectralAnalysisParams members 306
Spectrum 307
IRobotSpectralAnalysisPointsCollection 313
about IRobotSpectralAnalysisPointsCollection 313
Add 314
Clear 314
Count 313
Get 315
IRobotSpectralAnalysisPointsCollection fields 313
IRobotSpectralAnalysisPointsCollection members 313
IRobotSpectralAnalysisPointsCollection methods 314
LoadFromFile 315
Remove 315
SaveToFile 316
Set 316
IRobotSpectralAnalysisSpectrum 308
about IRobotSpectralAnalysisSpectrum 308
AbscissaXAxis 309
AbscissaXAxisLogarithmicScale 309
AddFromTimeHistory 311
Average 312
Damping 309
IRobotSpectralAnalysisSpectrum fields 308
IRobotSpectralAnalysisSpectrum members 308
IRobotSpectralAnalysisSpectrum methods 311
LoadFromFile 312
Name 310
OrdinateYAxis 310
OrdinateYAxisLogarithmicScale 310
Points 311
SaveToFile 312
IRobotSpectralCoefficients 930
about IRobotSpectralCoefficients 930
IRobotSpectralCoefficients members 931
IRobotSpectralCoefficients methods 931
ModeCoef 931
PartCoef 931
SpectrCoef 932
IRobotStorey 1052
about IRobotStorey 1052
AutomaticSelection 1053
Color 1054
Ex1 1054
Ey1 1054
Height 1055
IRobotStorey fields 1053
IRobotStorey members 1053
IRobotStorey methods 1056
Lx 1055
Ly 1055
Name 1055
Objects 1056
SetHeight 1056
SetTopLevel 1057
TopLevel 1056
IRobotStoreyDisplacements 1074
about IRobotStoreyDisplacements 1074
DrUX 1075
DrUY 1075
IRobotStoreyDisplacements fields 1074

(c) 2014 Autodesk Inc. All rights reserved

IRobotStoreyDisplacements members 1074
MaxUX 1075
MaxUY 1075
MinUX 1076
MinUY 1076
NodeMaxUX 1076
NodeMaxUY 1076
NodeMinUX 1077
NodeMinUY 1077
IRobotStoreyMngr 1057
about IRobotStoreyMngr 1057
BaseLevel 1058
Count 1059
Create2 1060
Create2Ex 1060
Delete 1060
DeleteAll 1061
DisregardedObjects 1059
FilterStorey 1059
Find 1061
Get 1061
IRobotStoreyMngr fields 1058
IRobotStoreyMngr members 1057
IRobotStoreyMngr methods 1059
IRobotStoreyReducedForces 1069
about IRobotStoreyReducedForces 1069
FX 1071
FX_ToColumns 1071
FX_ToWalls 1071
FY 1071
FY_ToColumns 1072
FY_ToWalls 1072
FZ 1072
FZ_ToColumns 1072
FZ_ToWalls 1073
IRobotStoreyReducedForces fields 1070
IRobotStoreyReducedForces members 1070
MX 1073
MY 1073
MZ 1073
IRobotStoreyResultServer 1077
about IRobotStoreyResultServer 1077
Displacements 1078
IRobotStoreyResultServer members 1078
IRobotStoreyResultServer methods 1078
ReducedForces 1078
Values 1079
IRobotStoreySelection 1062
about IRobotStoreySelection 1062
Add 1063
AddAll 1063
Clear 1063
GetNames 1064
GetNumbers 1064
IRobotStoreySelection members 1062
IRobotStoreySelection methods 1062
Remove 1064
IRobotStoreyValues 1065
about IRobotStoreyValues 1065
Ex0 1066
Ex1 1066
Ex2 1066
Ey0 1067
Ey1 1067
Ey2 1067
G 1068
IRobotStoreyValues fields 1065
IRobotStoreyValues members 1065
Ix 1068
Iy 1068
Iz 1068
Mass 1069
R 1069
IRobotSTRFileAnalyser 2219
about IRobotSTRFileAnalyser 2219
InsertParams 2220
InsertToProject 2220
IRobotSTRFileAnalyser fields 2219
IRobotSTRFileAnalyser members 2219
IRobotSTRFileAnalyser methods 2220
Params 2220
ReadParams 2221
IRobotSTRParameter 1276
about IRobotSTRParameter 1276

(c) 2014 Autodesk Inc. All rights reserved

Description 1277
DoubleVal_1 1278
DoubleVal_2 1278
DoubleVal_3 1278
FilePathVal 1279
IntegerVal 1279
IRobotSTRParameter fields 1277
IRobotSTRParameter members 1276
IsActive 1279
Name 1279
SelectionVal 1280
TextList 1280
TextVal 1280
Type 1280
IRobotSTRParams 1274
about IRobotSTRParams 1274
Count 1275
FindParameter 1276
GetParameter 1276
IRobotSTRParams fields 1275
IRobotSTRParams members 1275
IRobotSTRParams methods 1275
IRobotSTRParamType 1281
IRobotStructuralAxisGrid 1526
about IRobotStructuralAxisGrid 1526
IRobotStructuralAxisGrid fields 1526
IRobotStructuralAxisGrid members 1526
IRobotStructuralAxisGrid methods 1527
Name 1526
Save 1527
Type 1527
IRobotStructuralAxisGridCartesian 1527
about IRobotStructuralAxisGridCartesian 1527
GetRelativeToPoint 1531
IncludeStoreysInZ 1528
IRobotStructuralAxisGridCartesian fields 1528
IRobotStructuralAxisGridCartesian members 1528
IRobotStructuralAxisGridCartesian methods 1530
RotationAngle 1529
RotationAxis 1529
SetRelativeToPoint 1531
StoreysInZ 1529
X 1530
Y 1530
Z 1530
IRobotStructuralAxisGridMngr 1531
about IRobotStructuralAxisGridMngr 1531
Activate 1533
Clear 1533
Count 1532
Create 1533
Delete 1533
FindByName 1534
Get 1534
GetByName 1534
IRobotStructuralAxisGridMngr fields 1532
IRobotStructuralAxisGridMngr members 1532
IRobotStructuralAxisGridMngr methods 1532
IsActive 1535
IRobotStructuralAxisGridType 1525
IRobotStructuralAxisLabelType 1520
IRobotStructuralAxisSequenceList 1520
about IRobotStructuralAxisSequenceList 1520
AddSequence 1523
AxisCount 1522
Clear 1523
DeleteSequence 1523
FindAxisByPos 1524
GetAxis 1524
GetSequence 1524
IRobotStructuralAxisSequenceList fields 1521
IRobotStructuralAxisSequenceList members 1521
IRobotStructuralAxisSequenceList methods 1522
SequenceCount 1522
SetAxisLabel 1524
SetLabelFormat 1525
SingleOutAxis 1525
StartPosition 1522
IRobotStructure 1090
about IRobotStructure 1090
ApplyCache 1097
Bars 1092
Cases 1092
Clear 1097

(c) 2014 Autodesk Inc. All rights reserved

CreateCache 1097
Edit 1092
ExportXml 1098
ExtParams 1092
FiniteElems 1093
GroupObjects 1093
Groups 1093
IRobotStructure fields 1091
IRobotStructure members 1090
IRobotStructure methods 1096
IsCalcModelGenerated 1094
Labels 1094
Merge 1098
Nodes 1094
Objects 1094
QuantitySurvey 1095
Results 1095
ResultsFreeze 1095
Selections 1095
Storeys 1096
Type 1096
IRobotStructureAnalysisModalParticipationCoeff 1579
IRobotStructureAnalysisParams 1567
about IRobotStructureAnalysisParams 1567
AutoBarMerging 1569
AutoVerification 1569
DSCAlgorithm 1569
EquationSolvingMethod 1570
FictitiousRigidityCoeff 1570
IgnoreWarnings 1570
IRobotStructureAnalysisParams fields 1568
IRobotStructureAnalysisParams members 1568
IterativeParams 1571
ModalParticipationCoeff 1571
RLINKElements 1571
SparseMParams 1571
IRobotStructureApplyInfo 1101
about IRobotStructureApplyInfo 1101
Bars 1102
IRobotStructureApplyInfo fields 1101
IRobotStructureApplyInfo members 1101
Nodes 1102
IRobotStructureAutoVerificationType 1572
IRobotStructureCache 1098
about IRobotStructureCache 1098
AddBar 1099
AddNode 1100
EnsureNodeExist 1100
IRobotStructureCache members 1099
IRobotStructureCache methods 1099
SetBarLabel 1100
SetBarName 1101
IRobotStructureEditTools 1048
about IRobotStructureEditTools 1048
DivideBar 1049
IRobotStructureEditTools members 1049
IRobotStructureEditTools methods 1049
SelMirror 1050
SelRotate 1050
SelScale 1050
SelTranslate 1051
TranslateBar 1051
TranslateNode 1052
IRobotStructureGeoAnalyser 2222
about IRobotStructureGeoAnalyser 2222
CanEliminateIsolatedNodes 2224
CanExtendBars 2224
CanUseRigidLinks 2224
Correct 2225
DefineIntersection 2226
IRobotStructureGeoAnalyser fields 2223
IRobotStructureGeoAnalyser members 2223
IRobotStructureGeoAnalyser methods 2225
Precision 2225
SetEdgeSize 2226
SetNodeSize 2226
StartCollectInfo 2227
StopCollectInfo 2227
IRobotStructureMergeData 1102
about IRobotStructureMergeData 1102
CreateStructure 1104
IRobotStructureMergeData fields 1103
IRobotStructureMergeData members 1103
IRobotStructureMergeData methods 1103

(c) 2014 Autodesk Inc. All rights reserved

LoadStructure 1104
Structure 1103
IRobotStructureQuantitySurvey 1088
about IRobotStructureQuantitySurvey 1088
BarSections 1089
IRobotStructureQuantitySurvey fields 1089
IRobotStructureQuantitySurvey members 1089
Materials 1089
PanelThickness 1089
IRobotStructureValues 1019
about IRobotStructureValues 1019
GetEx2 1020
GetEy2 1021
GetG 1021
GetIx 1021
GetIy 1022
GetIz 1022
GetMass 1022
GetT 1023
IRobotStructureValues members 1019
IRobotStructureValues methods 1020
IRobotSupportColumnFixingType 526
IRobotSupportEquivalentColumnElasticity 524
about IRobotSupportEquivalentColumnElasticity 524
Fixing1 525
Fixing2 525
IRobotSupportEquivalentColumnElasticity fields 525
IRobotSupportEquivalentColumnElasticity members 524
L1 525
L2 526
ThroughTwoStories 526
IRobotSupportEquivalentElasticity 520
about IRobotSupportEquivalentElasticity 520
E 521
IRobotSupportEquivalentElasticity fields 520
IRobotSupportEquivalentElasticity members 520
IRobotSupportEquivalentElasticity methods 521
PoissonRatio 521
SetMaterial 522
Type 521
IRobotSupportEquivalentElasticityType 522
IRobotSupportEquivalentWallElasticity 522
about IRobotSupportEquivalentWallElasticity 522
IRobotSupportEquivalentWallElasticity fields 523
IRobotSupportEquivalentWallElasticity members 522
L1 523
L2 523
ThroughTwoStories 524
IRobotSurfaceOnObjectRecordValues 52
IRobotSWCodeECCdType 148
IRobotSWCodeECGroundType 147
IRobotSWCodeECParams 131
about IRobotSWCodeECParams 131
Altitude 134
GlobalCDIR 134
GroundType 134
IRobotSWCodeECParams fields 132
IRobotSWCodeECParams members 131
IRobotSWCodeECParams methods 146
LeftWind2NordAngle 135
NodalLoadsForAllBars 135
NodalLoadsForBarsList 135
PermDoorFront 135
PermDoorFrontPresent 136
PermDoorLeftSide 136
PermDoorLeftSidePresent 136
PermDoorRear 136
PermDoorRearPresent 137
PermDoorRightSide 137
PermDoorRightSidePresent 137
PermFront 137
PermLeftSide 138
PermRear 138
PermRightSide 138
ReferenceLevel 138
SnowBarCoeffGet 146
SnowBarCoeffSet 146
SnowGutterBars 139
SnowPressureExtreme 139
SnowPressureNormal 139
SnowRedistribution 140
StructureAge 140
StructureAgeCode 140
StructureHeight 140

(c) 2014 Autodesk Inc. All rights reserved

StructureP 141
WindAutoCd 141
WindBarCoeffGet 146
WindBarCoeffSet 147
WindCALT 141
WindCd 141
WindCDIR 142
WindCdType 142
WindCt 142
WindCtAuto 142
WindCTEM 143
WindE 143
WindKT 143
WindPressureAutomatic 143
WindQref 144
WindQref0 144
WindQref0p 144
WindSiteType 145
WindVref0 145
WindZ0 145
WindZMin 145
IRobotSWCodeECSiteType 147
IRobotSWCodeFRParams 110
about IRobotSWCodeFRParams 110
Altitude 114
IRobotSWCodeFRParams fields 112
IRobotSWCodeFRParams members 110
IRobotSWCodeFRParams methods 127
IsolatedRoofGetLocation 128
IsolatedRoofs 114
IsolatedRoofSetLocation 128
NodalLoadsForAllBars 114
NodalLoadsForBarsList 115
OpenStructure 115
PermDoorFront 115
PermDoorFrontPresent 115
PermDoorLeftSide 116
PermDoorLeftSidePresent 116
PermDoorRear 116
PermDoorRearPresent 116
PermDoorRightSide 117
PermDoorRightSidePresent 117
PermFront 117
PermLeftSide 117
PermRear 118
PermRightSide 118
ReferenceLevel 118
RiseOfRoof 119
RiseOfRoofAutomatic 119
SnowBarCoeffGet 128
SnowBarCoeffSet 128
SnowGutterBars 119
SnowIsWaterOutflow 119
SnowObstacles 120
SnowPressureExtreme 120
SnowPressureExtremeAutomatic 120
SnowPressureNormal 120
SnowPressureNormalAutomatic 121
SnowRedistribution 121
SnowRegion 121
SnowType 121
SnowWaterOutflow 122
StructureHeight 122
SurfaceLower 122
SurfaceUpper 122
WindBarCoeffGet 129
WindBarCoeffSet 129
WindCoastalArea 123
WindDeltaCoeff 123
WindDeltaCoeffAutomatic 123
WindDynamicAction 124
WindDynamicActionCoeff 124
WindDynamicActionCoeffAutomatic 124
WindDynamicActionPeriod 124
WindDynamicActionSteelStructure 125
WindFacadeOffset 125
WindMultipleRoof 125
WindPressure 125
WindPressureAutomatic 126
WindPressureCeCiMinimum 126
WindPressureVariable 126
WindRegion 126
WindSite 127
WindType 127

(c) 2014 Autodesk Inc. All rights reserved

IRobotSWCodeFRSnowType 130
IRobotSWCodeFRSurfaceType 130
IRobotSWCodeFRWindSite 129
IRobotSWCodeFRWindType 129
IRobotSWCodePLParams 99
about IRobotSWCodePLParams 99
Altitude 101
IRobotSWCodePLParams fields 100
IRobotSWCodePLParams members 99
IRobotSWCodePLParams methods 107
IsolatedRoofGetLocation 107
IsolatedRoofs 101
IsolatedRoofSetLocation 108
NodalLoadsForAllBars 102
NodalLoadsForBarsList 102
PermFront 102
PermLeftSide 102
PermRear 103
PermRightSide 103
ReferenceLevel 103
SnowBarCoeffGet 108
SnowBarCoeffSet 108
SnowPressure 103
SnowPressureAutomatic 104
SnowRedistribution 104
SnowZone 104
StructureHeight 104
WindBarCoeffGet 109
WindBarCoeffSet 109
WindDynamicAction 105
WindDynamicDecrement 105
WindDynamicPeriod 105
WindMultipleRoofs 106
WindPressure 106
WindPressureAutomatic 106
WindPressureDistribOnHeight 106
WindSite 107
WindZone 107
IRobotSWCodePLSnowZone 110
IRobotSWCodePLWindPressDistribType 130
IRobotSWCodePLWindSite 109
IRobotSWCodePLWindZone 109
IRobotSWStruct3D 83
about IRobotSWStruct3D 83
FrameCount 83
FrameElemCount 84
GetFrame 84
IRobotSWStruct3D fields 83
IRobotSWStruct3D members 83
IRobotSWStruct3D methods 84
SetFrame 84
IRobotSWStruct3DElement 85
about IRobotSWStruct3DElement 85
Bar 85
IRobotSWStruct3DElement fields 85
IRobotSWStruct3DElement members 85
IsFacadeLoaded 86
IsFacadeOnly 86
Purlins 86
IRobotSWStruct3DFrame 86
about IRobotSWStruct3DFrame 86
ElemCount 87
GetElem 88
IRobotSWStruct3DFrame fields 87
IRobotSWStruct3DFrame members 87
IRobotSWStruct3DFrame methods 87
SetElem 88
IRobotSWStruct3DGenParams 88
about IRobotSWStruct3DGenParams 88
Bars 89
FacadeLoadedBars 89
FacadeOnlyBars 90
FrameCount 90
GetPurlins 91
IRobotSWStruct3DGenParams fields 89
IRobotSWStruct3DGenParams members 88
IRobotSWStruct3DGenParams methods 91
Offsets 90
SetPurlins 91
Spacings 90
IRobotSWStruct3DPurlinGenParams 91
about IRobotSWStruct3DPurlinGenParams 91
IRobotSWStruct3DPurlinGenParams fields 92
IRobotSWStruct3DPurlinGenParams members 92

(c) 2014 Autodesk Inc. All rights reserved

Locations 92
RelativeLocations 92
SectionName 93
IRobotTable 1364
about IRobotTable 1364
AddColumn 1369
ColCount 1366
Configuration 1366
GetColWidth 1369
GetDataType 1369
GetRowHeight 1370
IRobotTable fields 1365
IRobotTable members 1364
IRobotTable methods 1368
MakeScreenCapture 1370
Printable 1366
RowCount 1367
Select 1370
SelectAllFromStore 1370
SelectFromStore 1371
SelectMode 1371
SelectModeFromStore 1371
SetColWidth 1372
SetRowHeight 1372
StoreModeSelection 1372
StoreSelection 1373
Title 1367
UserControl 1367
Visible 1367
Window 1368
IRobotTableConfig 1363
about IRobotTableConfig 1363
IRobotTableConfig members 1363
IRobotTableConfig methods 1363
SetFlag 1364
SetValue 1364
IRobotTableConfigFlag 1362
IRobotTableConfigValue 1363
IRobotTableDataType 1361
IRobotTableFrame 1373
about IRobotTableFrame 1373
Count 1374
Current 1374
Get 1375
GetName 1375
IRobotTableFrame fields 1374
IRobotTableFrame members 1373
IRobotTableFrame methods 1375
SetName 1375
Window 1374
IRobotTableScreenCaptureParams 1376
about IRobotTableScreenCaptureParams 1376
AddInfoTab 1377
AdjustColumnWidth 1377
Comment 1377
IncludeDateAndTime 1378
IRobotTableScreenCaptureParams fields 1376
IRobotTableScreenCaptureParams members 1376
Name 1378
UpdateType 1378
IRobotTableScreenCaptureUpdateType 1379
IRobotTableType 1361
IRobotThermalIn3PointsRecordValues 48
IRobotThermalRecordValues 50
IRobotThicknessData 722
about IRobotThicknessData 722
Data 723
ElasticFoundation 723
IRobotThicknessData fields 722
IRobotThicknessData members 722
MaterialName 723
ThicknessType 723
Uplift 724
IRobotThicknessHomoData 725
about IRobotThicknessHomoData 725
GetP1 727
GetP2 728
GetP3 728
GetReduction 728
IRobotThicknessHomoData fields 725
IRobotThicknessHomoData members 725
IRobotThicknessHomoData methods 727
SetP1 728
SetP2 729

(c) 2014 Autodesk Inc. All rights reserved

SetP3 729
SetReduction 729
Thick1 726
Thick2 726
Thick3 726
ThickConst 726
Type 727
IRobotThicknessHomoType 730
IRobotThicknessMatrix 739
about IRobotThicknessMatrix 739
GetValue 740
IRobotThicknessMatrix members 739
IRobotThicknessMatrix methods 739
SetValue 740
IRobotThicknessMatrixValue 740
IRobotThicknessOrthoData 731
A 732
A1 733
A2 733
about IRobotThicknessOrthoData 731
B 733
B1 733
DirType 734
DisregardBendStiffDirY 734
GetVector 738
H 734
H0 735
H1 735
H2 735
HA 735
HB 736
HC 736
IRobotThicknessOrthoData fields 732
IRobotThicknessOrthoData members 731
IRobotThicknessOrthoData methods 738
Matrix 736
N1 736
N2 737
SetVector 739
Thick1 737
Thick2 737
Thick3 738
Type 738
IRobotThicknessOrthoDirType 730
IRobotThicknessOrthoType 730
IRobotThicknessQuantitySurvey 1085
about IRobotThicknessQuantitySurvey 1085
Count 1086
GetArea 1087
GetName 1087
GetUnitWeight 1087
GetVolume 1088
GetWeight 1088
IRobotThicknessQuantitySurvey fields 1086
IRobotThicknessQuantitySurvey members 1086
IRobotThicknessQuantitySurvey methods 1086
IRobotThicknessType 724
IRobotThicknessUpliftType 724
IRobotTimeHistoryAnalysisMethod 369
IRobotTimeHistoryAnalysisParams 364
about IRobotTimeHistoryAnalysisParams 364
Count 365
Delete 368
Division 366
End 366
Find 368
Get 369
InitialCase 366
IRobotTimeHistoryAnalysisParams fields 365
IRobotTimeHistoryAnalysisParams members 365
IRobotTimeHistoryAnalysisParams methods 368
Method 367
MethodParams 367
Set 369
TimeStep 367
IRobotTimeHistoryFunctionList 378
about IRobotTimeHistoryFunctionList 378
AddFromFile 380
Count 379
Create 380
CreateSum 381
Delete 381
Find 381
Get 382

(c) 2014 Autodesk Inc. All rights reserved

GetName 382
IRobotTimeHistoryFunctionList fields 379
IRobotTimeHistoryFunctionList members 379
IRobotTimeHistoryFunctionList methods 380
SaveToFile 382
Store 383
IRobotTimeHistoryHHTParams 386
about IRobotTimeHistoryHHTParams 386
Alpha 386
Beta 387
CoeffAlpha 387
IRobotTimeHistoryHHTParams fields 386
IRobotTimeHistoryHHTParams members 386
Nonlinearity 387
NonlinearParams 388
IRobotTimeHistoryModalDecompositionParams 371
about IRobotTimeHistoryModalDecompositionParams 371
Count 372
Delete 373
DeleteMode 373
Get 373
GetDamping 374
IRobotTimeHistoryModalDecompositionParams fields 372
IRobotTimeHistoryModalDecompositionParams members 372
IRobotTimeHistoryModalDecompositionParams methods 373
IsDefined 374
SetDamping 374
IRobotTimeHistoryNewmarkAccelParams 388
about IRobotTimeHistoryNewmarkAccelParams 388
Alpha 389
Beta 389
IRobotTimeHistoryNewmarkAccelParams fields 389
IRobotTimeHistoryNewmarkAccelParams members 388
Nonlinearity 389
NonlinearParams 390
IRobotTimeHistoryNewmarkParams 370
about IRobotTimeHistoryNewmarkParams 370
Alpha 370
Beta 371
IRobotTimeHistoryNewmarkParams fields 370
IRobotTimeHistoryNewmarkParams members 370
MassMatrixType 371
IRobotTimeHistoryNonlinearParams 383
about IRobotTimeHistoryNonlinearParams 383
IRobotTimeHistoryNonlinearParams fields 384
IRobotTimeHistoryNonlinearParams members 383
MatrixUpdateAfterEachIteration 384
MatrixUpdateAfterEachSubdivision 384
MaximumIterationNumberForOneIncrement 385
PDelta 385
ResidualForcesRelativeCodeTolerance 385
IRobotTimeHistoryPointsCollection 375
about IRobotTimeHistoryPointsCollection 375
Add 376
Clear 377
Count 376
Delete 377
Find 377
Get 378
IRobotTimeHistoryPointsCollection fields 375
IRobotTimeHistoryPointsCollection members 375
IRobotTimeHistoryPointsCollection methods 376
LoadFromFile 378
SaveToFile 378
IRobotTimeHistoryResults 904
about IRobotTimeHistoryResults 904
ARX 905
ARY 906
ARZ 906
AX 906
AY 907
AZ 907
IRobotTimeHistoryResults fields 905
IRobotTimeHistoryResults members 905
Time 907
VRX 908
VRY 908
VRZ 908
VX 908
VY 909
VZ 909
IRobotTimeHistoryResultServer 933
about IRobotTimeHistoryResultServer 933

(c) 2014 Autodesk Inc. All rights reserved

IRobotTimeHistoryResultServer members 934
IRobotTimeHistoryResultServer methods 934
Value 934
IRobotTranslateOptions 1048
IRobotUniformRecordValues 49
IRobotUnitComplexData 1290
about IRobotUnitComplexData 1290
IRobotUnitComplexData fields 1290
IRobotUnitComplexData members 1290
Name2 1290
IRobotUnitData 1283
about IRobotUnitData 1283
E 1284
IRobotUnitData fields 1284
IRobotUnitData members 1283
Name 1284
Precision 1285
Type 1285
IRobotUnitEditionData 1294
about IRobotUnitEditionData 1294
Coefficient 1294
IRobotUnitEditionData fields 1294
IRobotUnitEditionData members 1294
Type 1295
Unit 1295
IRobotUnitEditionServer 1291
about IRobotUnitEditionServer 1291
Count 1292
Delete 1292
Find 1292
Get 1293
IRobotUnitEditionServer members 1291
IRobotUnitEditionServer methods 1291
New 1293
Set 1293
IRobotUnitEditionType 1285
IRobotUnitMngr 1286
about IRobotUnitMngr 1286
Count 1288
Get 1288
GetCoeff 1288
GetCoeff2 1289
GetName 1289
IRobotUnitMngr fields 1286
IRobotUnitMngr members 1286
IRobotUnitMngr methods 1287
Refresh 1289
Set 1289
UnitEdition 1287
UseMetricAsDefault 1287
IRobotUnitMngrEvents 1295
about IRobotUnitMngrEvents 1295
IRobotUnitMngrEvents members 1296
IRobotUnitMngrEvents methods 1296
UnitsChanged 1296
IRobotUnitType 1282
IRobotUniversalResultAccess 1007
about IRobotUniversalResultAccess 1007
Available 1009
Bar 1010
CalcPoint 1010
DivCount 1010
DivPoint 1010
Element 1011
GetDirX 1016
IRobotUniversalResultAccess fields 1009
IRobotUniversalResultAccess members 1007
IRobotUniversalResultAccess methods 1016
Layer 1011
LayerArbitraryValue 1011
LinearSupports 1012
LoadCase 1012
LoadCaseCmpnt 1012
Mode 1013
ModeCmb 1013
Node 1013
Panel 1013
ReducedCutPos 1014
ReinforceCalcMethod 1014
RelativePoint 1014
Reset 1016
ResultId 1015
ResultType 1015
ResultValue 1015

(c) 2014 Autodesk Inc. All rights reserved

ResultValue3D 1017
SetDirX 1017
Storey 1015
IRobotUniversalResultType 1019
IRobotUpliftDirection 545
IRobotUpliftSense 545
IRobotValuesArray 1636
about IRobotValuesArray 1636
Count 1637
Get 1637
IRobotValuesArray fields 1636
IRobotValuesArray members 1636
IRobotValuesArray methods 1637
Set 1637
SetSize 1638
IRobotVariableMngr 1487
about IRobotVariableMngr 1487
AddExtension 1488
Delete 1489
Exist 1489
GetPredefinedValue 1489
GetValue 1489
IRobotVariableMngr members 1487
IRobotVariableMngr methods 1488
RemoveExtension 1490
SetPredefinedValue 1490
SetValue 1490
IRobotVariableMngrExtension 1518
about IRobotVariableMngrExtension 1518
GetIndexedValue 1519
GetValue 1519
IRobotVariableMngrExtension members 1518
IRobotVariableMngrExtension methods 1519
IRobotVariablePredefinedId 1491
IRobotVehicleData 343
about IRobotVehicleData 343
b 344
d1 344
d2 345
IRobotVehicleData fields 344
IRobotVehicleData members 344
IRobotVehicleData methods 345
LoadFromDBase 346
Loads 345
StoreToDBase 346
IRobotVehicleDatabase 356
about IRobotVehicleDatabase 356
Description 357
GetAll 358
IRobotVehicleDatabase fields 357
IRobotVehicleDatabase members 357
IRobotVehicleDatabase methods 358
Load 359
LoadFromFile 359
LongName 357
Name 358
IRobotVehicleDatabaseList 352
about IRobotVehicleDatabaseList 352
Add 354
AddFromFile 354
Count 353
Create 355
Default 353
Find 355
Get 355
GetDatabase 356
IRobotVehicleDatabaseList fields 353
IRobotVehicleDatabaseList members 352
IRobotVehicleDatabaseList methods 354
Remove 356
IRobotVehicleLoad 348
about IRobotVehicleLoad 348
DX 350
DY 350
F 350
IRobotVehicleLoad fields 349
IRobotVehicleLoad members 349
S 351
Type 351
X 351
IRobotVehicleLoadMngr 346
about IRobotVehicleLoadMngr 346
Count 347
Delete 347

(c) 2014 Autodesk Inc. All rights reserved

Get 348
IRobotVehicleLoadMngr fields 347
IRobotVehicleLoadMngr members 347
IRobotVehicleLoadMngr methods 347
New 348
IRobotVehicleLoadType 352
IRobotView 1396
about IRobotView 1396
CopyToClipboard 1401
DiagramMagnification 1398
GetRotationPoint 1401
GetScale 1402
GetSize 1402
GetWorkPoint 1402
GetZoom 1402
IRobotView fields 1397
IRobotView members 1396
IRobotView methods 1400
IsLocal 1403
ParamsBarMap 1398
ParamsDiagram 1398
ParamsDisplay 1398
ParamsFeMap 1399
Printable 1399
Projection 1399
Redraw 1403
Rotate 1403
Selection 1399
SetGlobal 1403
SetLocal 1404
SetRotationPoint 1404
SetScale 1404
SetSize 1404
SetWorkPoint 1405
SetZoom 1405
Title 1400
Type 1400
Visible 1400
IRobotView2 1445
about IRobotView2 1445
IRobotView2 fields 1446
IRobotView2 members 1445
ParamsPanelCut 1446
UserControl 1447
Window 1447
IRobotView3 1459
about IRobotView3 1459
IRobotView3 members 1460
IRobotView3 methods 1461
MakeScreenCapture 1461
IRobotViewBarMapParams 1430
about IRobotViewBarMapParams 1430
CurrentResult 1431
Descriptions 1431
IRobotViewBarMapParams fields 1431
IRobotViewBarMapParams members 1431
MapThicknessCoeff 1432
StructureDeformation 1432
IRobotViewBarMapResultType 1430
IRobotViewBarMaps 1429
about IRobotViewBarMaps 1429
IRobotViewDetailedAnalysis 1412
about IRobotViewDetailedAnalysis 1412
CurrentTableTab 1414
IRobotViewDetailedAnalysis fields 1414
IRobotViewDetailedAnalysis members 1413
IRobotViewDetailedAnalysis methods 1415
MakeScreenCapture 1415
ParamsDetailed 1414
UserControl 1414
Window 1415
IRobotViewDetailedAnalysisParams 1433
about IRobotViewDetailedAnalysisParams 1433
Descriptions 1434
Filling 1434
GetColor 1435
IRobotViewDetailedAnalysisParams fields 1434
IRobotViewDetailedAnalysisParams members 1433
IRobotViewDetailedAnalysisParams methods 1435
IsOn 1436
PositiveNegative 1435
ReinforceShowTheoreticAndRealVals 1435
Set 1436
SetColor 1436

(c) 2014 Autodesk Inc. All rights reserved

IRobotViewDetailedAnalysisResultType 1432
IRobotViewDetailedAnalysisTableTab 1415
IRobotViewDiagramDescriptionType 1427
IRobotViewDiagramFillingType 1428
IRobotViewDiagramParams 1422
about IRobotViewDiagramParams 1422
Descriptions 1423
Filling 1423
GetColor 1425
GetScale 1425
IRobotViewDiagramParams fields 1423
IRobotViewDiagramParams members 1422
IRobotViewDiagramParams methods 1424
IsOn 1425
PositiveNegative 1424
ReactionsInLocalSystem 1424
Set 1426
SetColor 1426
SetScale 1426
Values 1424
IRobotViewDiagramPositionType 1447
IRobotViewDiagramResultType 1426
IRobotViewDiagrams 1429
about IRobotViewDiagrams 1429
IRobotViewDiagramSignDifferType 1428
IRobotViewDiagramValueType 1461
IRobotViewDisplayAttributes 1418
IRobotViewDisplayParams 1416
about IRobotViewDisplayParams 1416
HiddenLines 1417
IRobotViewDisplayParams fields 1416
IRobotViewDisplayParams members 1416
IRobotViewDisplayParams methods 1417
IsOn 1417
Set 1418
SymbolSize 1417
IRobotViewFeMapCrossPresentationType 1444
IRobotViewFeMapLayerType 1444
IRobotViewFeMapLocalSystemType 1443
IRobotViewFeMapParams 1438
about IRobotViewFeMapParams 1438
CrossPresentation 1439
CurrentResult 1440
DeformationActive 1440
DeformationConstScale 1440
Direction 1440
DirectionData 1441
DirectionNode 1441
IRobotViewFeMapParams fields 1439
IRobotViewFeMapParams members 1438
IRobotViewFeMapParams methods 1443
Isolines 1441
Layer 1441
LayerArbitraryVal 1442
MapScale 1442
SetDirectionData 1443
Smoothing 1442
WithDescription 1443
IRobotViewFeMapResultType 1436
IRobotViewFeMaps 1429
about IRobotViewFeMaps 1429
IRobotViewFeMapSmoothingType 1444
IRobotViewGlobalAnalysis 1453
about IRobotViewGlobalAnalysis 1453
IRobotViewGlobalAnalysis fields 1455
IRobotViewGlobalAnalysis members 1454
IRobotViewGlobalAnalysis methods 1455
MakeScreenCapture 1455
ParamsGlobalAnalysis 1455
IRobotViewGlobalAnalysisParams 1449
about IRobotViewGlobalAnalysisParams 1449
IRobotViewGlobalAnalysisParams fields 1449
IRobotViewGlobalAnalysisParams members 1449
IRobotViewGlobalAnalysisParams methods 1450
IsOn 1450
Results 1449
Set 1450
IRobotViewGlobalAnalysisParamsType 1453
IRobotViewGlobalAnalysisResultsParams 1450
about IRobotViewGlobalAnalysisResultsParams 1450
IRobotViewGlobalAnalysisResultsParams fields 1451
IRobotViewGlobalAnalysisResultsParams members 1451
NPointsValue 1451
RelativeValue 1452

(c) 2014 Autodesk Inc. All rights reserved

Type 1452
IRobotViewGlobalAnalysisResultsType 1452
IRobotViewHiddenLinesDisplayType 1421
IRobotViewInfluenceLines 1386
about IRobotViewInfluenceLines 1386
IRobotViewInfluenceLines fields 1387
IRobotViewInfluenceLines members 1386
IRobotViewInfluenceLines methods 1387
MakeScreenCapture 1388
ParamsInfluenceLines 1387
IRobotViewInfluenceLinesLayerType 1395
IRobotViewInfluenceLinesLocalSystemType 1394
IRobotViewInfluenceLinesParams 1388
about IRobotViewInfluenceLinesParams 1388
Direction 1389
DirectionData 1389
DirectionNode 1390
Element 1390
IRobotViewInfluenceLinesParams fields 1389
IRobotViewInfluenceLinesParams members 1388
IRobotViewInfluenceLinesParams methods 1392
IsOn 1392
Layer 1390
LayerArbitraryVal 1391
Position 1391
RangeFrom 1391
RangeTo 1392
Set 1392
IRobotViewInfluenceLinesResultType 1393
IRobotViewMngr 1469
about IRobotViewMngr 1469
CreateFromSc 1472
CreateRtfView 1473
CreateTable 1473
CreateView 1473
CurrentLayout 1471
GetTable 1474
GetView 1474
IRobotViewMngr fields 1470
IRobotViewMngr members 1469
IRobotViewMngr methods 1472
NewViewAsTab 1471
RecycleViews 1471
Refresh 1474
ShowDialog 1475
TableCount 1471
ViewCount 1472
IRobotViewPanelCutParams 1381
about IRobotViewPanelCutParams 1381
CurrentReinforcementResult 1382
CurrentResult 1383
Descritpions 1383
Filling 1383
IntegralValue 1384
IRobotViewPanelCutParams fields 1382
IRobotViewPanelCutParams members 1382
Layer 1384
LayerArbritraryValue 1384
Position 1385
Smoothing 1385
IRobotViewPanelCuts 1381
about IRobotViewPanelCuts 1381
IRobotViewProjection 1405
IRobotViewReinforcementResultType 1448
IRobotViewScaleType 1412
IRobotViewScreenCaptureParams 1456
about IRobotViewScreenCaptureParams 1456
Comment 1457
IncludeDateAndTime 1457
IRobotViewScreenCaptureParams fields 1456
IRobotViewScreenCaptureParams members 1456
Name 1457
Orientation 1458
Resolution 1458
ScaleAutomatic 1458
ScaleValue 1459
UpdateType 1459
IRobotViewScreenCaptureResolution 1462
IRobotViewScreenCaptureUpdateType 1459
IRobotViewType 1395
IRobotViewVisibilityStatusType 1406
IRobotViewVisibilityStatusValue 1407
IRobotWindLoadsSimulationEngine 472
about IRobotWindLoadsSimulationEngine 472

(c) 2014 Autodesk Inc. All rights reserved

Generate 473
IRobotWindLoadsSimulationEngine fields 472
IRobotWindLoadsSimulationEngine members 472
IRobotWindLoadsSimulationEngine methods 473
Params 473
IRobotWindLoadsSimulationParams 473
about IRobotWindLoadsSimulationParams 473
DeviationPercent 475
DirectionXNEnabled 475
DirectionXNYNEnabled 475
DirectionXNYPEnabled 475
DirectionXPEnabled 476
DirectionXPYNEnabled 476
DirectionXPYPEnabled 476
DirectionYNEnabled 476
DirectionYPEnabled 477
Elements 477
IRobotWindLoadsSimulationParams fields 474
IRobotWindLoadsSimulationParams members 474
OpeningsClosed 477
TerrainLevel 477
Velocity 478
IRobotWindow 1650
about IRobotWindow 1650
Activate 1652
Caption 1651
Handle 1651
IRobotWindow fields 1651
IRobotWindow members 1650
IRobotWindow methods 1652
IsActive 1651
SendMessage 1652
State 1652
IRobotWindowState 1653

J

Job preferences 1281

K

Knee connection 1939

L

Labels - complex attributes of objects 17
Linear releases 704
Load cases 37
Load records 38

M

Material 659
MaterialType 1184
about MaterialType 1184
IRConcrContinuousFootingSupport fields 1185
IRConcrContinuousFootingSupport members 1185
MaterialType 1185
Name 1185
Width 1185
Member section definition module 1937
Modal analysis parameters 189
Modal analysis parameters recognizing static forces 189
Moving loads 326

N

Nodes 478
Non-linear analysis parameters 317
Non-linear constraints 676
Non-linear hinges 592

O

Objects 702

P

Panel calculation model 719
Panel cuts 1381
Parameters of simplified seismic analysis 447
Pinned column base 1971
Plate and shell reinforcement 1702
Presentation of data 1360
Printouts 1476
Project 1104
Project components 1105
Push-over analysis parameters 391

(c) 2014 Autodesk Inc. All rights reserved

Q

Quantity survey 1079
Querying mechanism 2

R

RC Beam 1128
RC Column 1185
RC Deep Beam 1228
RC Slab 1214
RC Wall 1232
Reinforcement 1245
Releases 580
Results 851
Rigid links 479
Robot Kernel 2207
Robot Object Model 1
RobotOM 1
RobotOM namespace 1
RobotOM.IRBestBendType 1698
RobotOM.IRBestCalcErrors 1686
RobotOM.IRBestCalcParamsData 1677
RobotOM.IRBestCalcParamsData.ModularityList 1678
RobotOM.IRBestCalcParamsDataDoubleValue 1678
RobotOM.IRBestCalcParamsDataIntegerValue 1679
RobotOM.IRBestCalcParamsDataList 1680
RobotOM.IRBestCalcParamsDataList.Create 1682
RobotOM.IRBestCalcParamsDataList.Delete 1682
RobotOM.IRBestCalcParamsDataList.Get 1682
RobotOM.IRBestCalcParamsDataList.LabNames 1681
RobotOM.IRBestCalcParamsDataList.Selected 1681
RobotOM.IRBestCalcParamsDataList.Store 1683
RobotOM.IRBestCalcParamsDataStringValue 1680
RobotOM.IRBestCalcParamsDlg 1669
RobotOM.IRBestCalcParamsDlg.DoModal 1669
RobotOM.IRBestCalcParamsDlg.SetStandard 1670
RobotOM.IRBestCalculationType 1698
RobotOM.IRBestCalcWarnings 1687
RobotOM.IRBestCodeCalcEngine 1691
RobotOM.IRBestCodeCalcEngine.Calculate 1692
RobotOM.IRBestCodeCalcEngine.GetResults 1692
RobotOM.IRBestCodeCalcEngine.SetForces 1693
RobotOM.IRBestCodeCalcEngine.SetForcesArray 1693
RobotOM.IRBestCodeCalcEngine.SetForcesSlender 1693
RobotOM.IRBestCodeCalcEngine.SetGeometry 1694
RobotOM.IRBestCodeCalcEngine.SetParams 1694
RobotOM.IRBestCodeCalculationType 1696
RobotOM.IRBestCodeService 1667
RobotOM.IRBestCodeService.CalcEngine 1668
RobotOM.IRBestCodeService.CalcParamsDlg 1668
RobotOM.IRBestCodeService.MemberDlg 1668
RobotOM.IRBestCodeService2 1700
RobotOM.IRBestCodeService2.RobotVersion 1701
RobotOM.IRBestCodeService2.ShowBucklingButton 1701
RobotOM.IRBestCodeServiceExt 1696
RobotOM.IRBestCodeServiceExt.GetCalcParamsDlg 1697
RobotOM.IRBestCodeServiceExt.IsServed 1697
RobotOM.IRBestCoordSystem 1699
RobotOM.IRBestDimParams 1690
RobotOM.IRBestDimParamsDoubleValue 1698
RobotOM.IRBestDimParamsIntegerValue 1690
RobotOM.IRBestDirection 1698
RobotOM.IRBestForceData 1688
RobotOM.IRBestForceData.LimitState 1690
RobotOM.IRBestForceData.Values 1690
RobotOM.IRBestForceDataDoubleValue 1699
RobotOM.IRBestForceDataIntegerValue 1699
RobotOM.IRBestForceDataSLSCombType 1700
RobotOM.IRBestLevel 1699
RobotOM.IRBestMemberData 1683
RobotOM.IRBestMemberDataDoubleValue 1683
RobotOM.IRBestMemberDataIntegerValue 1684
RobotOM.IRBestMemberDataStringValue 1694
RobotOM.IRBestMemberDlg 1670
RobotOM.IRBestMemberDlg.DoModal 1671
RobotOM.IRBestMemberDlg.SetStandard 1671
RobotOM.IRBestMemberType 1695
RobotOM.IRBestParamSet 1671
RobotOM.IRBestParamSet.Clear 1673
RobotOM.IRBestParamSet.ClearDouble 1673
RobotOM.IRBestParamSet.ClearInteger 1674
RobotOM.IRBestParamSet.ClearString 1674
RobotOM.IRBestParamSet.GetDouble 1674
RobotOM.IRBestParamSet.GetInteger 1674

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRBestParamSet.GetString 1675
RobotOM.IRBestParamSet.IsValidDouble 1675
RobotOM.IRBestParamSet.IsValidInteger 1675
RobotOM.IRBestParamSet.IsValidString 1675
RobotOM.IRBestParamSet.SetDouble 1676
RobotOM.IRBestParamSet.SetInteger 1676
RobotOM.IRBestParamSet.SetString 1676
RobotOM.IRBestPlateCalcParamsDlg 1695
RobotOM.IRBestPlateCalcParamsDlg.DoModal 1695
RobotOM.IRBestPlateCalcParamsDlg.SetStandard 1696
RobotOM.IRBestResults 1684
RobotOM.IRBestResultsDoubleValue 1685
RobotOM.IRBestResultsIntegerValue 1686
RobotOM.IRBestResultsStringValue 1688
RobotOM.IRConcr_ACI318_BarDim 1750
RobotOM.IRConcr_ACI318_ConcreteParams 1749
RobotOM.IRConcr_ACI318_ConcreteParams.AsInStructure 1750
RobotOM.IRConcr_ACI318_LoadActionPeriodType 1748
RobotOM.IRConcr_ACI318_MetricBarDIm 1751
RobotOM.IRConcr_ACI318_ReinforceData 1741
RobotOM.IRConcr_ACI318_ReinforceData.GetBarDim 1747
RobotOM.IRConcr_ACI318_ReinforceData.GetMetricBarDim 1747
RobotOM.IRConcr_ACI318_ReinforceData.IsMetric 1743
RobotOM.IRConcr_ACI318_ReinforceData.Main 1743
RobotOM.IRConcr_ACI318_ReinforceData.SetBarDim 1747
RobotOM.IRConcr_ACI318_ReinforceData.SetMetricBarDim 1748
RobotOM.IRConcr_ACI318_ReinforceData.SLS_Cracking 1743
RobotOM.IRConcr_ACI318_ReinforceData.SLS_CreepingCoef 1743
RobotOM.IRConcr_ACI318_ReinforceData.SLS_CreepingCoefValue 1744
RobotOM.IRConcr_ACI318_ReinforceData.SLS_Deflection 1744
RobotOM.IRConcr_ACI318_ReinforceData.SLS_DeflectionReinfCorrection 1744
RobotOM.IRConcr_ACI318_ReinforceData.SLS_Factor 1744
RobotOM.IRConcr_ACI318_ReinforceData.SLS_FactorValue 1745
RobotOM.IRConcr_ACI318_ReinforceData.SLS_LoadActionPeriod 1745
RobotOM.IRConcr_ACI318_ReinforceData.SLS_LoadRatio 1745
RobotOM.IRConcr_ACI318_ReinforceData.SLS_MaxCracking 1746
RobotOM.IRConcr_ACI318_ReinforceData.SLS_MaxDeflection 1746
RobotOM.IRConcr_ACI318_SteelGrades 1748
RobotOM.IRConcr_BAEL_ConcreteGrades 1740
RobotOM.IRConcr_BAEL_CrackingType 1739
RobotOM.IRConcr_BAEL_EnvironmentType 1739
RobotOM.IRConcr_BAEL_ReinforceData 1731
RobotOM.IRConcr_BAEL_ReinforceData.CrackAlpha 1733
RobotOM.IRConcr_BAEL_ReinforceData.CrackAlphaBot 1734
RobotOM.IRConcr_BAEL_ReinforceData.CrackAlphaTop 1734
RobotOM.IRConcr_BAEL_ReinforceData.CrackEnv 1734
RobotOM.IRConcr_BAEL_ReinforceData.CrackEnvBot 1734
RobotOM.IRConcr_BAEL_ReinforceData.CrackEnvTop 1735
RobotOM.IRConcr_BAEL_ReinforceData.CrackExtraParams 1735
RobotOM.IRConcr_BAEL_ReinforceData.CrackExtraParamsBot 1735
RobotOM.IRConcr_BAEL_ReinforceData.CrackExtraParamsTop 1736
RobotOM.IRConcr_BAEL_ReinforceData.Cracking 1736
RobotOM.IRConcr_BAEL_ReinforceData.CrackingBot 1736
RobotOM.IRConcr_BAEL_ReinforceData.CrackingTop 1736
RobotOM.IRConcr_BAEL_ReinforceData.CrackWaterLevel 1737
RobotOM.IRConcr_BAEL_ReinforceData.CrackWaterLevelBot 1737
RobotOM.IRConcr_BAEL_ReinforceData.CrackWaterLevelTop 1737
RobotOM.IRConcr_BAEL_ReinforceData.Main 1737
RobotOM.IRConcr_BAEL_ReinforceData.SLS_Deflection 1738
RobotOM.IRConcr_BAEL_ReinforceData.SLS_DeflectionReinfCorrect 1738
RobotOM.IRConcr_BAEL_ReinforceData.SLS_MaxDeflection 1738
RobotOM.IRConcr_BAEL_ReinforceData.SteelSymbol 1738

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRConcr_BAEL_SteelGrades 1740

RobotOM.IRConcr_BAEL_WaterLevel 1739

RobotOM.IRConcr_BS8110_ConcreteAge 1758

RobotOM.IRConcr_BS8110_ConcreteGrades 1759

RobotOM.IRConcr_BS8110_ExposureRatings 1758

RobotOM.IRConcr_BS8110_PartialSafetyFactors 1760

RobotOM.IRConcr_BS8110_ReinforceData 1752

RobotOM.IRConcr_BS8110_ReinforceData.Main 1753

RobotOM.IRConcr_BS8110_ReinforceData.PartialSafetyFactors 1754

RobotOM.IRConcr_BS8110_ReinforceData.SLS_ConcreteAge 1754

RobotOM.IRConcr_BS8110_ReinforceData.SLS_Cracking 1754

RobotOM.IRConcr_BS8110_ReinforceData.SLS_CrackingReinfCorection 1755

RobotOM.IRConcr_BS8110_ReinforceData.SLS_CreepingCoef 1755

RobotOM.IRConcr_BS8110_ReinforceData.SLS_CreepingCoefValue 1755

RobotOM.IRConcr_BS8110_ReinforceData.SLS_Deflection 1756

RobotOM.IRConcr_BS8110_ReinforceData.SLS_DeflectionReinfCorection 1756

RobotOM.IRConcr_BS8110_ReinforceData.SLS_EnvHumidity 1756

RobotOM.IRConcr_BS8110_ReinforceData.SLS_Exposure 1757

RobotOM.IRConcr_BS8110_ReinforceData.SLS_LoadRatio 1757

RobotOM.IRConcr_BS8110_ReinforceData.SLS_MaxCracking 1757

RobotOM.IRConcr_BS8110_ReinforceData.SLS_MaxDeflection 1758

RobotOM.IRConcr_EC2_ConcreteGrades 1774

RobotOM.IRConcr_EC2_ExposureRatings 1774

RobotOM.IRConcr_EC2_ITALIAN_SteelGrades 1774

RobotOM.IRConcr_EC2_NAD 1779

RobotOM.IRConcr_EC2_ReinforceData 1775

RobotOM.IRConcr_EC2_ReinforceData.NAD 1777

RobotOM.IRConcr_EC2_ReinforceData.SLS_ConcreteAge 1777

RobotOM.IRConcr_EC2_ReinforceData.SLS_Cracking 1777

RobotOM.IRConcr_EC2_ReinforceData.SLS_CrackingReinfCorrect 1777

RobotOM.IRConcr_EC2_ReinforceData.SLS_CreepingCoef 1777

RobotOM.IRConcr_EC2_ReinforceData.SLS_CreepingCoefValue 1778

RobotOM.IRConcr_EC2_ReinforceData.SLS_Deflection 1778

RobotOM.IRConcr_EC2_ReinforceData.SLS_DeflectionReinfCorrect 1778

RobotOM.IRConcr_EC2_ReinforceData.SLS_EnvHumidityVal 1778

RobotOM.IRConcr_EC2_ReinforceData.SLS_Exposure 1778

RobotOM.IRConcr_EC2_ReinforceData.SLS_MaxCracking 1779

RobotOM.IRConcr_EC2_ReinforceData.SLS_MaxDeflection 1779

RobotOM.IRConcr_IS2000_EnvironmentType 1780

RobotOM.IRConcr_IS2000_ReinforceData 1780

RobotOM.IRConcr_IS2000_ReinforceData.Main 1782

RobotOM.IRConcr_IS2000_ReinforceData.SLS_ConcreteAge 1783

RobotOM.IRConcr_IS2000_ReinforceData.SLS_Cracking 1783

RobotOM.IRConcr_IS2000_ReinforceData.SLS_CrackingReinfCorrect 1783

RobotOM.IRConcr_IS2000_ReinforceData.SLS_CreepingCoeff 1783

RobotOM.IRConcr_IS2000_ReinforceData.SLS_CreepingCoeffValue 1784

RobotOM.IRConcr_IS2000_ReinforceData.SLS_Deflection 1784

RobotOM.IRConcr_IS2000_ReinforceData.SLS_DeflectionReinfCorrect 1784

RobotOM.IRConcr_IS2000_ReinforceData.SLS_ExposureBot 1784

RobotOM.IRConcr_IS2000_ReinforceData.SLS_ExposureTop 1785

RobotOM.IRConcr_IS2000_ReinforceData.SLS_LoadRatio 1785

RobotOM.IRConcr_IS2000_ReinforceData.SLS_MaxCrackingBot 1785

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRConcr_IS2000_ReinforceData.SLS_MaxCrackingBotEnabled 1785

RobotOM.IRConcr_IS2000_ReinforceData.SLS_MaxCrackingTop 1786

RobotOM.IRConcr_IS2000_ReinforceData.SLS_MaxCrackingTopEnabled 1786

RobotOM.IRConcr_IS2000_ReinforceData.SLS_MaxDeflection 1786

RobotOM.IRConcr_PN_SteelGrades 1721

RobotOM.IRConcr_PN84_ConcreteGrades 1721

RobotOM.IRConcr_PN84_ExposureRatings 1722

RobotOM.IRConcr_PN84_HumidityType 1740

RobotOM.IRConcr_PN84_ReinforceData 1715

RobotOM.IRConcr_PN84_ReinforceData.CheckConstructionStages 1717

RobotOM.IRConcr_PN84_ReinforceData.LongActionAbove100 1717

RobotOM.IRConcr_PN84_ReinforceData.Main 1718

RobotOM.IRConcr_PN84_ReinforceData.SingleShortLoad 1718

RobotOM.IRConcr_PN84_ReinforceData.SLS_ConcreteAge 1718

RobotOM.IRConcr_PN84_ReinforceData.SLS_Cracking 1718

RobotOM.IRConcr_PN84_ReinforceData.SLS_CrackingReinfCorrect 1719

RobotOM.IRConcr_PN84_ReinforceData.SLS_CreepingCoef 1719

RobotOM.IRConcr_PN84_ReinforceData.SLS_Deflection 1719

RobotOM.IRConcr_PN84_ReinforceData.SLS_DeflectionReinfCorrect 1719

RobotOM.IRConcr_PN84_ReinforceData.SLS_EnvHumidity 1720

RobotOM.IRConcr_PN84_ReinforceData.SLS_Exposure 1720

RobotOM.IRConcr_PN84_ReinforceData.SLS_LoadsRatio 1720

RobotOM.IRConcr_PN84_ReinforceData.SLS_MaxCracking 1720

RobotOM.IRConcr_PN84_ReinforceData.SLS_MaxDeflection 1721

RobotOM.IRConcr_PN99_ConcreteGrades 1730

RobotOM.IRConcr_PN99_ExposureRatings 1722

RobotOM.IRConcr_PN99_ReinforceData 1723

RobotOM.IRConcr_PN99_ReinforceData.IsSpecialStructure 1725

RobotOM.IRConcr_PN99_ReinforceData.Main 1726

RobotOM.IRConcr_PN99_ReinforceData.SLS_Concrete 1726

RobotOM.IRConcr_PN99_ReinforceData.SLS_ConcreteAge 1726

RobotOM.IRConcr_PN99_ReinforceData.SLS_Cracking 1726

RobotOM.IRConcr_PN99_ReinforceData.SLS_CrackingReinfCorrect 1727

RobotOM.IRConcr_PN99_ReinforceData.SLS_CreepingCoef 1727

RobotOM.IRConcr_PN99_ReinforceData.SLS_CreepingCoefValue 1727

RobotOM.IRConcr_PN99_ReinforceData.SLS_Deflection 1727

RobotOM.IRConcr_PN99_ReinforceData.SLS_DeflectionReinfCorrect 1728

RobotOM.IRConcr_PN99_ReinforceData.SLS_EnvHumidityVal 1728

RobotOM.IRConcr_PN99_ReinforceData.SLS_ExposureBot 1728

RobotOM.IRConcr_PN99_ReinforceData.SLS_ExposureTop 1728

RobotOM.IRConcr_PN99_ReinforceData.SLS_LoadRatio 1729

RobotOM.IRConcr_PN99_ReinforceData.SLS_MaxCrackingBot 1729

RobotOM.IRConcr_PN99_ReinforceData.SLS_MaxCrackingBotEnabled 1729

RobotOM.IRConcr_PN99_ReinforceData.SLS_MaxCrackingTop 1729

RobotOM.IRConcr_PN99_ReinforceData.SLS_MaxCrackingTopEnabled 1730

RobotOM.IRConcr_PN99_ReinforceData.SLS_MaxDeflection 1730

RobotOM.IRConcr_SNIP_ConcreteGrades 1771

RobotOM.IRConcr_SNIP_ConcreteParams 1772

RobotOM.IRConcr_SNIP_ConcreteParams.ConcretingInLayers 1773

RobotOM.IRConcr_SNIP_ConcreteParams.CuringMethod 1773

RobotOM.IRConcr_SNIP_ConcreteParams.HighHumidity 1773

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRConcr_SNIP_ConcreteParams.Type 1773
RobotOM.IRConcr_SNIP_ConcreteTypes 1771
RobotOM.IRConcr_SNIP_CuringMethods 1771
RobotOM.IRConcr_SNIP_Exposure 1774
RobotOM.IRConcr_SNIP_ReinforceData 1769
RobotOM.IRConcr_SNIP_ReinforceData.SLS_Cracking 1770
RobotOM.IRConcr_SNIP_ReinforceData.SLS_CrackingReinf Correction
1770
RobotOM.IRConcr_SNIP_ReinforceData.SLS_Exposure 1771
RobotOM.IRConcr_SNIP_SteelGrades 1771
RobotOM.IRConcrBarDiameters 1264
RobotOM.IRConcrBarDiameters.Add 1266
RobotOM.IRConcrBarDiameters.Count 1265
RobotOM.IRConcrBarDiameters.Item 1265
RobotOM.IRConcrBarDiameters.RemoveAll 1266
RobotOM.IRConcrBarSectionData 1794
RobotOM.IRConcrBarSectionData.A 1795
RobotOM.IRConcrBarSectionData.Ac 1796
RobotOM.IRConcrBarSectionData.Ap 1796
RobotOM.IRConcrBarSectionData.B 1796
RobotOM.IRConcrBarSectionData.Bf 1796
RobotOM.IRConcrBarSectionData.Bfb 1796
RobotOM.IRConcrBarSectionData.H 1797
RobotOM.IRConcrBarSectionData.Hf 1797
RobotOM.IRConcrBarSectionData.Hfb 1797
RobotOM.IRConcrBarSectionData.N 1797
RobotOM.IRConcrBarSectionData.Per 1797
RobotOM.IRConcrBarSectionData.Type 1798
RobotOM.IRConcrBarSectionGeometryType 1798
RobotOM.IRConcrBarSubtype 1258
RobotOM.IRConcrBarType 1257
RobotOM.IRConcrBeam 1129
RobotOM.IRConcrBeam.Activate 1136
RobotOM.IRConcrBeam.Calculate 1136
RobotOM.IRConcrBeam.CalculationOptions 1131
RobotOM.IRConcrBeam.Concrete 1131
RobotOM.IRConcrBeam.CreateCalculationNoteRtf 1137
RobotOM.IRConcrBeam.CreateFromBars 1137
RobotOM.IRConcrBeam.GenerateSpliceBars 1137
RobotOM.IRConcrBeam.Geometry 1131
RobotOM.IRConcrBeam.ImportOptions 1132
RobotOM.IRConcrBeam.IsActive 1132
RobotOM.IRConcrBeam.IsSelected 1132
RobotOM.IRConcrBeam.LinearLoad 1132
RobotOM.IRConcrBeam.LinearLoadsCount 1133
RobotOM.IRConcrBeam.Name 1133
RobotOM.IRConcrBeam.NumberOfElements 1133
RobotOM.IRConcrBeam.PatternOptions 1133
RobotOM.IRConcrBeam.PointLoad 1134
RobotOM.IRConcrBeam.PointLoadsCount 1134
RobotOM.IRConcrBeam.Reinforcement 1134
RobotOM.IRConcrBeam.Save 1137
RobotOM.IRConcrBeam.Steel 1134
RobotOM.IRConcrBeam.StoryOptions 1135
RobotOM.IRConcrBeam.StructureUserNo 1135
RobotOM.IRConcrBeam.StructureUserNoCount 1135
RobotOM.IRConcrBeam.UniqueId 1135
RobotOM.IRConcrBeam.Verify 1138
RobotOM.IRConcrBeam.VerifySpan 1138
RobotOM.IRConcrBeamCalcOptions 1138
RobotOM.IRConcrBeamCalcOptions.CalcSpanLengthInAxis
1139
RobotOM.IRConcrBeamCalcOptions.CoverBottom 1140
RobotOM.IRConcrBeamCalcOptions.CoverBottomFixed 1140
RobotOM.IRConcrBeamCalcOptions.CoverSide 1140
RobotOM.IRConcrBeamCalcOptions.CoverSideFixed 1140
RobotOM.IRConcrBeamCalcOptions.CoverTop 1141
RobotOM.IRConcrBeamCalcOptions.CoverTopFixed 1141
RobotOM.IRConcrBeamCalcPointDefinitionType 1794
RobotOM.IRConcrBeamCrackingType 1167
RobotOM.IRConcrBeamGeometry 1141
RobotOM.IRConcrBeamGeometry.AutoNameSpans 1142
RobotOM.IRConcrBeamGeometry.AutoNameSupports 1143
RobotOM.IRConcrBeamGeometry.ConcreteVolume 1143
RobotOM.IRConcrBeamGeometry.LeftCantilever 1143
RobotOM.IRConcrBeamGeometry.RightCantilever 1143
RobotOM.IRConcrBeamGeometry.ShutteringArea 1144
RobotOM.IRConcrBeamGeometry.Span 1144
RobotOM.IRConcrBeamGeometry.SpanNumber 1144
RobotOM.IRConcrBeamGeometry.UpdateInternalGeometryD
ata
1145
RobotOM.IRConcrBeamImportOptions 1145
RobotOM.IRConcrBeamImportOptions.GroupByGeometry
1146
RobotOM.IRConcrBeamImportOptions.GroupByLevel 1146

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRConcrBeamImportOptions.ImportCombinations 1146
RobotOM.IRConcrBeamImportOptions.LiveLongCoeff 1147
RobotOM.IRConcrBeamImportOptions.RunCalcAuto 1147
RobotOM.IRConcrBeamImportOptions.ShowDialog 1147
RobotOM.IRConcrBeamLinearLoad 1147
RobotOM.IRConcrBeamLinearLoad.Case 1148
RobotOM.IRConcrBeamLinearLoad.Nature 1149
RobotOM.IRConcrBeamLinearLoad.RelativeCoordinates 1149
RobotOM.IRConcrBeamLinearLoad.Spans 1149
RobotOM.IRConcrBeamLinearLoad.Type 1150
RobotOM.IRConcrBeamLinearLoad.Value1 1150
RobotOM.IRConcrBeamLinearLoad.Value2 1150
RobotOM.IRConcrBeamLinearLoad.Value3 1150
RobotOM.IRConcrBeamLinearLoad.X1 1151
RobotOM.IRConcrBeamLinearLoad.X2 1151
RobotOM.IRConcrBeamLinearLoad.X3 1151
RobotOM.IRConcrBeamLinearLoad.X4 1151
RobotOM.IRConcrBeamLinearLoadType 1164
RobotOM.IRConcrBeamLoadNatureType 1165
RobotOM.IRConcrBeamPatternOptions 1152
RobotOM.IRConcrBeamPatternOptions.SpanBySpan 1152
RobotOM.IRConcrBeamPointLoad 1153
RobotOM.IRConcrBeamPointLoad.Case 1153
RobotOM.IRConcrBeamPointLoad.Nature 1154
RobotOM.IRConcrBeamPointLoad.RelativeCoordinates 1154
RobotOM.IRConcrBeamPointLoad.Spans 1154
RobotOM.IRConcrBeamPointLoad.Type 1154
RobotOM.IRConcrBeamPointLoad.Value 1155
RobotOM.IRConcrBeamPointLoad.X1 1155
RobotOM.IRConcrBeamPointLoadType 1165
RobotOM.IRConcrBeamSectionDimType 1166
RobotOM.IRConcrBeamSectionType 1165
RobotOM.IRConcrBeamSegment 1155
RobotOM.IRConcrBeamSegment.Dim 1156
RobotOM.IRConcrBeamSegment.SectionType 1156
RobotOM.IRConcrBeamSpan 1156
RobotOM.IRConcrBeamSpan.CalculationLength 1157
RobotOM.IRConcrBeamSpan.HasOpenings 1158
RobotOM.IRConcrBeamSpan.LeftSupport 1158
RobotOM.IRConcrBeamSpan.Length 1158
RobotOM.IRConcrBeamSpan.Name 1158
RobotOM.IRConcrBeamSpan.RightSupport 1159
RobotOM.IRConcrBeamSpan.Segment 1159
RobotOM.IRConcrBeamSpan.SegmentNumber 1159
RobotOM.IRConcrBeamSpanNumbers 1159
RobotOM.IRConcrBeamSpanNumbers.Add 1161
RobotOM.IRConcrBeamSpanNumbers.Count 1160
RobotOM.IRConcrBeamSpanNumbers.Item 1160
RobotOM.IRConcrBeamSpanNumbers.RemoveAll 1161
RobotOM.IRConcrBeamStoryOptions 1161
RobotOM.IRConcrBeamStoryOptions.Cracking 1162
RobotOM.IRConcrBeamSupport 1162
RobotOM.IRConcrBeamSupport.MaterialType 1163
RobotOM.IRConcrBeamSupport.Name 1163
RobotOM.IRConcrBeamSupport.Type 1164
RobotOM.IRConcrBeamSupport.Width 1164
RobotOM.IRConcrBeamSupportMaterialType 1168
RobotOM.IRConcrBeamSupportType 1167
RobotOM.IRConcrCalcEngine 1798
RobotOM.IRConcrCalcEngine.MemberRequiredReinf 1799
RobotOM.IRConcrCalcEngine.SlabRequiredReinf 1799
RobotOM.IRConcrCodeBeam 1805
RobotOM.IRConcrCodeBeam.IsCommandEnabled 1806
RobotOM.IRConcrCodeBeam.Verification 1806
RobotOM.IRConcrCodeBeamCommand 1806
RobotOM.IRConcrCodeColumn 1802
RobotOM.IRConcrCodeColumn.IsCommandEnabled 1802
RobotOM.IRConcrCodeColumn.Verification 1803
RobotOM.IRConcrCodeColumnCommand 1803
RobotOM.IRConcrCodeReport 1803
RobotOM.IRConcrCodeReport.AddError 1804
RobotOM.IRConcrCodeReport.AddMessage 1804
RobotOM.IRConcrCodeReport.AddWarning 1805
RobotOM.IRConcrCodeService 1800
RobotOM.IRConcrCodeService.Beam 1801
RobotOM.IRConcrCodeService.Column 1801
RobotOM.IRConcrCodeService.IsConcrComponentServed 1801
RobotOM.IRConcrColumn 1186
RobotOM.IRConcrColumn.Activate 1193
RobotOM.IRConcrColumn.BucklingModel 1188
RobotOM.IRConcrColumn.Calculate 1194
RobotOM.IRConcrColumn.CalculationOptions 1188
RobotOM.IRConcrColumn.Concrete 1188
RobotOM.IRConcrColumn.CreateCalculationNoteRtf 1194

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRConcrColumn.CreateFromBars 1194
RobotOM.IRConcrColumn.Geometry 1189
RobotOM.IRConcrColumn.HasUpperColumn 1189
RobotOM.IRConcrColumn.ImportOptions 1189
RobotOM.IRConcrColumn.IsActive 1189
RobotOM.IRConcrColumn.IsSelected 1190
RobotOM.IRConcrColumn.Loads 1190
RobotOM.IRConcrColumn.Name 1190
RobotOM.IRConcrColumn.NumberOfElements 1190
RobotOM.IRConcrColumn.PatternOptions 1191
RobotOM.IRConcrColumn.Reinforcement 1191
RobotOM.IRConcrColumn.Save 1194
RobotOM.IRConcrColumn.Steel 1191
RobotOM.IRConcrColumn.StructureUserNo 1191
RobotOM.IRConcrColumn.StructureUserNoCount 1192
RobotOM.IRConcrColumn.UniqueId 1192
RobotOM.IRConcrColumn.UpperColumn 1192
RobotOM.IRConcrColumn.UpperColumnDY 1192
RobotOM.IRConcrColumn.UpperColumnDZ 1193
RobotOM.IRConcrColumn.Verify 1195
RobotOM.IRConcrColumnBucklingModel 1202
RobotOM.IRConcrColumnBucklingModel.IsDirectionYOff 1204
RobotOM.IRConcrColumnBucklingModel.IsDirectionZOff 1204
RobotOM.IRConcrColumnBucklingModel.IsSwayY 1204
RobotOM.IRConcrColumnBucklingModel.IsSwayZ 1204
RobotOM.IRConcrColumnBucklingModel.IsTotalStructureHeight 1205
RobotOM.IRConcrColumnBucklingModel.ky 1205
RobotOM.IRConcrColumnBucklingModel.kz 1205
RobotOM.IRConcrColumnBucklingModel.Ly 1205
RobotOM.IRConcrColumnBucklingModel.Lz 1206
RobotOM.IRConcrColumnBucklingModel.NumberOfStories 1206
RobotOM.IRConcrColumnBucklingModel.TotalStructureHeight 1206
RobotOM.IRConcrColumnCalcOptions 1195
RobotOM.IRConcrColumnCalcOptions.Cover 1195
RobotOM.IRConcrColumnDimensionType 1201
RobotOM.IRConcrColumnGeometry 1196
RobotOM.IRConcrColumnGeometry.ConcreteVolume 1197
RobotOM.IRConcrColumnGeometry.Dim 1197
RobotOM.IRConcrColumnGeometry.DimIsFixed 1197
RobotOM.IRConcrColumnGeometry.Section 1197
RobotOM.IRConcrColumnGeometry.SectionName 1198
RobotOM.IRConcrColumnGeometry.ShutteringArea 1198
RobotOM.IRConcrColumnImportOptions 1198
RobotOM.IRConcrColumnImportOptions.GroupByGeometry 1199
RobotOM.IRConcrColumnImportOptions.GroupByLevel 1199
RobotOM.IRConcrColumnImportOptions.ImportCombinations 1200
RobotOM.IRConcrColumnImportOptions.RunCalcAuto 1200
RobotOM.IRConcrColumnImportOptions.ShowDialog 1200
RobotOM.IRConcrColumnLoad 1209
RobotOM.IRConcrColumnLoad.CaseName 1210
RobotOM.IRConcrColumnLoad.CaseType 1210
RobotOM.IRConcrColumnLoad.Fy 1210
RobotOM.IRConcrColumnLoad.Fz 1211
RobotOM.IRConcrColumnLoad.Gamma 1211
RobotOM.IRConcrColumnLoad.MnsY 1211
RobotOM.IRConcrColumnLoad.MnsZ 1211
RobotOM.IRConcrColumnLoad.MyA 1212
RobotOM.IRConcrColumnLoad.MyB 1212
RobotOM.IRConcrColumnLoad.MyC 1212
RobotOM.IRConcrColumnLoad.MzA 1212
RobotOM.IRConcrColumnLoad.MzB 1213
RobotOM.IRConcrColumnLoad.MzC 1213
RobotOM.IRConcrColumnLoad.N 1213
RobotOM.IRConcrColumnLoad.NdN 1213
RobotOM.IRConcrColumnLoadCaseType 1214
RobotOM.IRConcrColumnLoads 1207
RobotOM.IRConcrColumnLoads.Add 1208
RobotOM.IRConcrColumnLoads.Count 1207
RobotOM.IRConcrColumnLoads.Item 1208
RobotOM.IRConcrColumnLoads.RemoveAll 1208
RobotOM.IRConcrColumnPatternOptions 1200
RobotOM.IRConcrColumnPatternOptions.TiesToBeam 1201
RobotOM.IRConcrColumnSectionType 1202
RobotOM.IRConcrConcrete 1263
RobotOM.IRConcrConcrete.CharacteristicStrength 1263
RobotOM.IRConcrConcreteParams 1709
RobotOM.IRConcrConcreteParams.Ec 1710
RobotOM.IRConcrConcreteParams.Fc 1711
RobotOM.IRConcrConcreteParams.Fc_calc 1711
RobotOM.IRConcrConcreteParams.Fcj 1711

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRConcrConcreteParams.Ft 1712
RobotOM.IRConcrConcreteParams.Grade 1712
RobotOM.IRConcrContinuousFooting 1168
RobotOM.IRConcrContinuousFooting.Activate 1174
RobotOM.IRConcrContinuousFooting.Calculate 1174
RobotOM.IRConcrContinuousFooting.CalculationOptions 1170
RobotOM.IRConcrContinuousFooting.Concrete 1170
RobotOM.IRConcrContinuousFooting.CreateCalculationNoteRtf 1174
RobotOM.IRConcrContinuousFooting.Geometry 1171
RobotOM.IRConcrContinuousFooting.ImportOptions 1171
RobotOM.IRConcrContinuousFooting.IsActive 1171
RobotOM.IRConcrContinuousFooting.IsSelected 1171
RobotOM.IRConcrContinuousFooting.Name 1172
RobotOM.IRConcrContinuousFooting.NumberOfElements 1172
RobotOM.IRConcrContinuousFooting.PatternOptions 1172
RobotOM.IRConcrContinuousFooting.Reinforcement 1172
RobotOM.IRConcrContinuousFooting.Save 1175
RobotOM.IRConcrContinuousFooting.Steel 1173
RobotOM.IRConcrContinuousFooting.StructureUserNo 1173
RobotOM.IRConcrContinuousFooting.StructureUserNoCount 1173
RobotOM.IRConcrContinuousFooting.UniqueId 1173
RobotOM.IRConcrContinuousFooting.Verify 1175
RobotOM.IRConcrContinuousFootingGeometry 1175
RobotOM.IRConcrContinuousFootingGeometry.AutoNameSpans 1176
RobotOM.IRConcrContinuousFootingGeometry.AutoNameSupports 1177
RobotOM.IRConcrContinuousFootingGeometry.ConcreteVolume 1177
RobotOM.IRConcrContinuousFootingGeometry.LeftCantilever 1177
RobotOM.IRConcrContinuousFootingGeometry.RightCantilever 1177
RobotOM.IRConcrContinuousFootingGeometry.ShutteringArea 1178
RobotOM.IRConcrContinuousFootingGeometry.Span 1178
RobotOM.IRConcrContinuousFootingGeometry.SpanNumber 1178
RobotOM.IRConcrContinuousFootingGeometry.UpdateInternalGeometryData 1179
RobotOM.IRConcrContinuousFootingSectionDimType 1183
RobotOM.IRConcrContinuousFootingSectionType 1183
RobotOM.IRConcrContinuousFootingSegment 1179
RobotOM.IRConcrContinuousFootingSegment.Dim 1179
RobotOM.IRConcrContinuousFootingSegment.SectionType 1180
RobotOM.IRConcrContinuousFootingSpan 1180
RobotOM.IRConcrContinuousFootingSpan.CalculationLength 1181
RobotOM.IRConcrContinuousFootingSpan.LeftSupport 1181
RobotOM.IRConcrContinuousFootingSpan.Length 1182
RobotOM.IRConcrContinuousFootingSpan.Name 1182
RobotOM.IRConcrContinuousFootingSpan.RightSupport 1182
RobotOM.IRConcrContinuousFootingSpan.Segment 1182
RobotOM.IRConcrContinuousFootingSpan.SegmentNumber 1183
RobotOM.IRConcrContinuousFootingSupport 1184
RobotOM.IRConcrContinuousFootingSupport.MaterialType 1185
RobotOM.IRConcrContinuousFootingSupport.Name 1185
RobotOM.IRConcrContinuousFootingSupport.Width 1185
RobotOM.IRConcrContinuousFootingSupportMaterialType 1184
RobotOM.IRConcrDeepBeam 1228
RobotOM.IRConcrDeepBeam.Activate 1231
RobotOM.IRConcrDeepBeam.Concrete 1229
RobotOM.IRConcrDeepBeam.CreateCalculationNoteRtf 1231
RobotOM.IRConcrDeepBeam.CreateFromObjects 1232
RobotOM.IRConcrDeepBeam.Geometry 1229
RobotOM.IRConcrDeepBeam.IsActive 1229
RobotOM.IRConcrDeepBeam.IsSelected 1230
RobotOM.IRConcrDeepBeam.Name 1230
RobotOM.IRConcrDeepBeam.Reinforcement 1230
RobotOM.IRConcrDeepBeam.Save 1232
RobotOM.IRConcrDeepBeam.Steel 1230
RobotOM.IRConcrDeepBeam.UniqueId 1231
RobotOM.IRConcrDeepBeamGeometry 1232
RobotOM.IRConcrDrawing 1266
RobotOM.IRConcrDrawing.Group 1267
RobotOM.IRConcrDrawing.GroupCount 1267
RobotOM.IRConcrDrawingBarGroup 1268
RobotOM.IRConcrDrawingBarGroup.Bar 1268

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRConcrDrawingBarGroup.BarCount 1269
RobotOM.IRConcrFooting 1106
RobotOM.IRConcrFooting.Activate 1113
RobotOM.IRConcrFooting.Calculate 1113
RobotOM.IRConcrFooting.CalculationOptions 1108
RobotOM.IRConcrFooting.Concrete 1108
RobotOM.IRConcrFooting.CreateCalculationNoteRtf 1113
RobotOM.IRConcrFooting.CreateFromNodes 1113
RobotOM.IRConcrFooting.CreateReinforcement 1114
RobotOM.IRConcrFooting.DisplayDrawing 1114
RobotOM.IRConcrFooting.FootRotation 1108
RobotOM.IRConcrFooting.Geometry 1108
RobotOM.IRConcrFooting.Ground 1109
RobotOM.IRConcrFooting.IsActive 1109
RobotOM.IRConcrFooting.IsSelected 1109
RobotOM.IRConcrFooting.Loads 1110
RobotOM.IRConcrFooting.Name 1110
RobotOM.IRConcrFooting.NumberOfElements 1110
RobotOM.IRConcrFooting.Pattern 1110
RobotOM.IRConcrFooting.Reinforcement 1111
RobotOM.IRConcrFooting.Results 1111
RobotOM.IRConcrFooting.Save 1114
RobotOM.IRConcrFooting.Steel 1111
RobotOM.IRConcrFooting.StructureUserNo 1111
RobotOM.IRConcrFooting.StructureUserNoCount 1112
RobotOM.IRConcrFooting.UniqueId 1112
RobotOM.IRConcrFooting.Verify 1114
RobotOM.IRConcrFootingCalculationOptions 1123
RobotOM.IRConcrFootingCalculationOptions.Cover 1124
RobotOM.IRConcrFootingCalculationOptions.NominalCover 1124
RobotOM.IRConcrFootingCalculationOptions.NominalPierCover 1124
RobotOM.IRConcrFootingCalculationOptions.PierCover 1124
RobotOM.IRConcrFootingDimType 1118
RobotOM.IRConcrFootingGeometry 1115
RobotOM.IRConcrFootingGeometry.ConcreteVolume 1116
RobotOM.IRConcrFootingGeometry.GetDim 1117
RobotOM.IRConcrFootingGeometry.PierType 1116
RobotOM.IRConcrFootingGeometry.SetDim 1117
RobotOM.IRConcrFootingGeometry.Shape 1116
RobotOM.IRConcrFootingGeometry.ShutteringArea 1116
RobotOM.IRConcrFootingGeometry.Type 1117
RobotOM.IRConcrFootingGround 1119
RobotOM.IRConcrFootingGround.ColumnPierLevel 1120
RobotOM.IRConcrFootingGround.SoilLevel 1120
RobotOM.IRConcrFootingLoads 1121
RobotOM.IRConcrFootingPattern 1125
RobotOM.IRConcrFootingPattern.Diameter1 1126
RobotOM.IRConcrFootingPattern.Diameter2 1126
RobotOM.IRConcrFootingPattern.Spacing1Max 1126
RobotOM.IRConcrFootingPattern.Spacing1Min 1126
RobotOM.IRConcrFootingPattern.Spacing2Max 1127
RobotOM.IRConcrFootingPattern.Spacing2Min 1127
RobotOM.IRConcrFootingPierType 1128
RobotOM.IRConcrFootingResults 1121
RobotOM.IRConcrFootingResults.Ax 1122
RobotOM.IRConcrFootingResults.Ay 1122
RobotOM.IRConcrFootingResults.Val 1122
RobotOM.IRConcrFootingShapeType 1118
RobotOM.IRConcrFootingType 1119
RobotOM.IRConcrFootResultType 1127
RobotOM.IRConcrMemberRequiredReinfCalcParams 1790
RobotOM.IRConcrMemberRequiredReinfCalcParams.BeamPointsType 1791
RobotOM.IRConcrMemberRequiredReinfCalcParams.BeamPointsValue 1792
RobotOM.IRConcrMemberRequiredReinfCalcParams.CasesALS 1792
RobotOM.IRConcrMemberRequiredReinfCalcParams.CasesSLS 1792
RobotOM.IRConcrMemberRequiredReinfCalcParams.CasesULS 1792
RobotOM.IRConcrMemberRequiredReinfCalcParams.CombALS 1793
RobotOM.IRConcrMemberRequiredReinfCalcParams.CombSLS 1793
RobotOM.IRConcrMemberRequiredReinfCalcParams.CombULS 1793
RobotOM.IRConcrMemberRequiredReinfCalcParams.Members 1793
RobotOM.IRConcrMemberRequiredReinfEngine 1788

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRConcrMemberRequiredReinfEngine.Calculate 1789
RobotOM.IRConcrMemberRequiredReinfEngine.GetCalculatedMembers 1790
RobotOM.IRConcrMemberRequiredReinfEngine.Params 1789
RobotOM.IRConcrMinimumReinforcementType 1779
RobotOM.IRConcrPlateCodeService2 1787
RobotOM.IRConcrPlateCodeService2.RobotVersion 1787
RobotOM.IRConcrPlateCodeService2.ShowLongTermCracking 1788
RobotOM.IRConcrReinforceBarType 1751
RobotOM.IRConcrReinforceCalcType 1765
RobotOM.IRConcrReinforceData 1704
RobotOM.IRConcrReinforceData.BarDim_D1_Bot 1705
RobotOM.IRConcrReinforceData.BarDim_D1_Up 1705
RobotOM.IRConcrReinforceData.BarDim_D2_Bot 1706
RobotOM.IRConcrReinforceData.BarDim_D2_Up 1706
RobotOM.IRConcrReinforceData.CodeName 1706
RobotOM.IRConcrReinforceData.Concrete 1706
RobotOM.IRConcrReinforceData.Cover_Bot 1707
RobotOM.IRConcrReinforceData.Cover_Up 1707
RobotOM.IRConcrReinforceData.GetMainDirection 1708
RobotOM.IRConcrReinforceData.ReinforcingSteel 1707
RobotOM.IRConcrReinforceData.SetMainDirection 1708
RobotOM.IRConcrReinforceData2 1765
RobotOM.IRConcrReinforceData2.Main 1766
RobotOM.IRConcrReinforceDataMain 1767
RobotOM.IRConcrReinforceDataMain.MembraneReinfInOneLayer 1768
RobotOM.IRConcrReinforceDataMain.MinimumReinf 1768
RobotOM.IRConcrReinforceDataMain.ReinfCalcType 1768
RobotOM.IRConcrReinforceDataMain.UnidirReinf 1769
RobotOM.IRConcrReinforceDirection 1709
RobotOM.IRConcrReinforcement 1245
RobotOM.IRConcrReinforcement.Bars 1246
RobotOM.IRConcrReinforcement.Comment 1247
RobotOM.IRConcrReinforcement.Freeze 1248
RobotOM.IRConcrReinforcement.GroupGlobally 1247
RobotOM.IRConcrReinforcement.GroupingMethod 1247
RobotOM.IRConcrReinforcement.IsFrozen 1247
RobotOM.IRConcrReinforcement.LockRefresh 1248
RobotOM.IRConcrReinforcement.RemoveAll 1248
RobotOM.IRConcrReinforcement.RemoveParametric 1249
RobotOM.IRConcrReinforcingBar 1249
RobotOM.IRConcrReinforcingBar.CreateCopy 1254
RobotOM.IRConcrReinforcingBar.DirectionX 1250
RobotOM.IRConcrReinforcingBar.DirectionY 1251
RobotOM.IRConcrReinforcingBar.DirectionZ 1251
RobotOM.IRConcrReinforcingBar.GetROBarEditor 1254
RobotOM.IRConcrReinforcingBar.GroupNumber 1251
RobotOM.IRConcrReinforcingBar.IsParametric 1251
RobotOM.IRConcrReinforcingBar.Position 1252
RobotOM.IRConcrReinforcingBar.ROBarEditorId 1252
RobotOM.IRConcrReinforcingBar.SetBarShape 1254
RobotOM.IRConcrReinforcingBar.SetDirections 1255
RobotOM.IRConcrReinforcingBar.SetROBarEditor 1255
RobotOM.IRConcrReinforcingBar.Span 1252
RobotOM.IRConcrReinforcingBar.SteelType 1252
RobotOM.IRConcrReinforcingBar.Subtype 1253
RobotOM.IRConcrReinforcingBar.Type 1253
RobotOM.IRConcrReinforcingBar.Weight 1253
RobotOM.IRConcrReinforcingBars 1255
RobotOM.IRConcrReinforcingBars.Add 1257
RobotOM.IRConcrReinforcingBars.Count 1256
RobotOM.IRConcrReinforcingBars.Item 1256
RobotOM.IRConcrSlab 1215
RobotOM.IRConcrSlab.Activate 1218
RobotOM.IRConcrSlab.CalculationOptions 1216
RobotOM.IRConcrSlab.Concrete 1216
RobotOM.IRConcrSlab.CreateCalculationNoteRtf 1219
RobotOM.IRConcrSlab.CreateFromObjects 1219
RobotOM.IRConcrSlab.Geometry 1216
RobotOM.IRConcrSlab.IsSelected 1217
RobotOM.IRConcrSlab.Name 1217
RobotOM.IRConcrSlab.PatternOptions 1217
RobotOM.IRConcrSlab.Reinforcement 1217
RobotOM.IRConcrSlab.Save 1219
RobotOM.IRConcrSlab.StartWizard 1220
RobotOM.IRConcrSlab.Steel 1218
RobotOM.IRConcrSlab.UniqueId 1218
RobotOM.IRConcrSlabCalculationOptions 1220
RobotOM.IRConcrSlabCalculationOptions.Cover 1220
RobotOM.IRConcrSlabCalculationOptions.CoverLateral 1221
RobotOM.IRConcrSlabGeometry 1221

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRConcrSlabGeometry.ConcreteVolume 1222
RobotOM.IRConcrSlabGeometry.ShutteringArea 1222
RobotOM.IRConcrSlabPatternOptions 1222
RobotOM.IRConcrSlabPatternOptions.ReinforcementSegment 1223
RobotOM.IRConcrSlabPatternOptions.ReinforcementType 1223
RobotOM.IRConcrSlabRequiredReinfCalcParams 1760
RobotOM.IRConcrSlabRequiredReinfCalcParams.CasesACC 1761
RobotOM.IRConcrSlabRequiredReinfCalcParams.CasesSLS 1761
RobotOM.IRConcrSlabRequiredReinfCalcParams.CasesULS 1762
RobotOM.IRConcrSlabRequiredReinfCalcParams.DisplayErrors 1762
RobotOM.IRConcrSlabRequiredReinfCalcParams.ForcesReduction 1762
RobotOM.IRConcrSlabRequiredReinfCalcParams.GloballyAvgDesginForces 1762
RobotOM.IRConcrSlabRequiredReinfCalcParams.Method 1763
RobotOM.IRConcrSlabRequiredReinfCalcParams.Panels 1763
RobotOM.IRConcrSlabRequiredReinfEngine 1763
RobotOM.IRConcrSlabRequiredReinfEngine.Calculate 1765
RobotOM.IRConcrSlabRequiredReinfEngine.GetCalculatedPanels 1765
RobotOM.IRConcrSlabRequiredReinfEngine.Params 1764
RobotOM.IRConcrSlabRnfSegmentType 1227
RobotOM.IRConcrSlabRnfType 1226
RobotOM.IRConcrSlabSupportType 1227
RobotOM.IRConcrSlabWizard 1223
RobotOM.IRConcrSlabWizard.AddColumnSupport 1225
RobotOM.IRConcrSlabWizard.AddOpening 1225
RobotOM.IRConcrSlabWizard.AddOpeningNode 1225
RobotOM.IRConcrSlabWizard.AddSlabNode 1226
RobotOM.IRConcrSlabWizard.AddWallSupport 1226
RobotOM.IRConcrSlabWizard.Finish 1226
RobotOM.IRConcrSlabWizard.Thickness 1224
RobotOM.IRConcrSteel 1260
RobotOM.IRConcrSteel.AvailableNames 1261
RobotOM.IRConcrSteel.AvailableStrengths 1261
RobotOM.IRConcrSteel.BarDiameters 1262
RobotOM.IRConcrSteel.BarsDatabasePath 1262
RobotOM.IRConcrSteel.CharacteristicStrength 1262
RobotOM.IRConcrSteel.SelectedName 1262
RobotOM.IRConcrSteel.SurfaceType 1262
RobotOM.IRConcrSteelNames 1258
RobotOM.IRConcrSteelNames.Count 1259
RobotOM.IRConcrSteelNames.Item 1259
RobotOM.IRConcrSteelParams 1712
RobotOM.IRConcrSteelParams.Ey 1713
RobotOM.IRConcrSteelParams.Ft 1713
RobotOM.IRConcrSteelParams.Ft_calc 1714
RobotOM.IRConcrSteelParams.Fy 1714
RobotOM.IRConcrSteelParams.Fy_calc 1714
RobotOM.IRConcrSteelParams.Grade 1715
RobotOM.IRConcrSteelStrengths 1264
RobotOM.IRConcrSteelSurfaceType 1264
RobotOM.IRConcrSteelType 1260
RobotOM.IRConcrWall 1233
RobotOM.IRConcrWall.Activate 1236
RobotOM.IRConcrWall.Calculate 1236
RobotOM.IRConcrWall.Concrete 1234
RobotOM.IRConcrWall.CreateCalculationNoteRtf 1236
RobotOM.IRConcrWall.CreateFromObjects 1237
RobotOM.IRConcrWall.Geometry 1234
RobotOM.IRConcrWall.IsSelected 1234
RobotOM.IRConcrWall.Name 1235
RobotOM.IRConcrWall.Reinforcement 1235
RobotOM.IRConcrWall.Save 1237
RobotOM.IRConcrWall.Steel 1235
RobotOM.IRConcrWall.UniqueId 1235
RobotOM.IRConcrWall.Verify 1237
RobotOM.IRConcrWallGeometry 1238
RobotOM.IRConcrWallGeometry.Height 1239
RobotOM.IRConcrWallGeometry.LeftBoundaryLength 1239
RobotOM.IRConcrWallGeometry.LeftBoundaryThickness 1239
RobotOM.IRConcrWallGeometry.Length 1239
RobotOM.IRConcrWallGeometry.Name 1240
RobotOM.IRConcrWallGeometry.Openings 1240
RobotOM.IRConcrWallGeometry.RightBoundaryLength 1240
RobotOM.IRConcrWallGeometry.RightBoundaryThickness 1240

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRConcrWallGeometry.Thickness 1241
RobotOM.IRConcrWallOpening 1241
RobotOM.IRConcrWallOpening.Lx 1242
RobotOM.IRConcrWallOpening.Lz 1242
RobotOM.IRConcrWallOpening.Name 1242
RobotOM.IRConcrWallOpening.PosX 1243
RobotOM.IRConcrWallOpening.PosZ 1243
RobotOM.IRConcrWallOpenings 1243
RobotOM.IRConcrWallOpenings.Add 1244
RobotOM.IRConcrWallOpenings.Count 1244
RobotOM.IRConcrWallOpenings.Item 1244
RobotOM.IRDimBuckDiagramCB71 1887
RobotOM.IRDimBuckDiagramEC3 1818
RobotOM.IRDimCalcState 1916
RobotOM.IRDimCalcState.GetParam 1917
RobotOM.IRDimCalcState.GetParamValue 1917
RobotOM.IRDimCalcState.IsFlagSet 1917
RobotOM.IRDimCalcState.SetFlag 1917
RobotOM.IRDimCalcState.SetParam 1918
RobotOM.IRDimCalcState.SetParamValue 1918
RobotOM.IRDimCalcStateFlagType 1915
RobotOM.IRDimCalcStateParamType 1915
RobotOM.IRDimCalcStateParamValue 1915
RobotOM.IRDimCalcStateValueType 1916
RobotOM.IRDimClient 1934
RobotOM.IRDimClient.GetRDimCodeService 1934
RobotOM.IRDimCodeResCB71 1863
RobotOM.IRDimCodeResCB71.BadWidthBfUnderFire 1869
RobotOM.IRDimCodeResCB71.BuckCoefKMax 1869
RobotOM.IRDimCodeResCB71.BuckCoefKy 1869
RobotOM.IRDimCodeResCB71.BuckCoefKz 1870
RobotOM.IRDimCodeResCB71.BuckLengthComposBeamLey 1870
RobotOM.IRDimCodeResCB71.BuckLengthComposBeamLez 1870
RobotOM.IRDimCodeResCB71.BuckLengthLfy 1870
RobotOM.IRDimCodeResCB71.BuckLengthLfz 1871
RobotOM.IRDimCodeResCB71.BuckSlendComposBeamLaym 1871
RobotOM.IRDimCodeResCB71.BuckSlendComposBeamLazm 1871
RobotOM.IRDimCodeResCB71.BuckSLendLay 1871
RobotOM.IRDimCodeResCB71.BuckSlendLaz 1872
RobotOM.IRDimCodeResCB71.CoefDependOnAngle 1872
RobotOM.IRDimCodeResCB71.CoefDependOnHeight 1872
RobotOM.IRDimCodeResCB71.CoefDependOnHumAndBend 1872
RobotOM.IRDimCodeResCB71.CoefDependOnHumAndComp 1872
RobotOM.IRDimCodeResCB71.CoefDependOnWidth 1873
RobotOM.IRDimCodeResCB71.EfficiencyRatio 1873
RobotOM.IRDimCodeResCB71.FireHightInMidSpanH 1873
RobotOM.IRDimCodeResCB71.FireInerMomInMidSpanIy 1873
RobotOM.IRDimCodeResCB71.FireInerMomInMidSpanIz 1874
RobotOM.IRDimCodeResCB71.FireInerMomIx 1874
RobotOM.IRDimCodeResCB71.FireInerMomIy 1874
RobotOM.IRDimCodeResCB71.FireInerMomIz 1874
RobotOM.IRDimCodeResCB71.FireLftEdgeDistVpy 1875
RobotOM.IRDimCodeResCB71.FireLowEdgeDistVpz 1875
RobotOM.IRDimCodeResCB71.FireProtectCoefK2 1875
RobotOM.IRDimCodeResCB71.FireProtection 1875
RobotOM.IRDimCodeResCB71.FireProtectionPosition 1876
RobotOM.IRDimCodeResCB71.FireRedCoefK1b 1876
RobotOM.IRDimCodeResCB71.FireRedCoefK1h 1876
RobotOM.IRDimCodeResCB71.FireResistance 1876
RobotOM.IRDimCodeResCB71.FireRgtEdgeDistVy 1877
RobotOM.IRDimCodeResCB71.FireSecAreaInMidSpanS 1877
RobotOM.IRDimCodeResCB71.FireSecAreaS 1877
RobotOM.IRDimCodeResCB71.FireSeModulWy 1877
RobotOM.IRDimCodeResCB71.FireSeModulWZ 1878
RobotOM.IRDimCodeResCB71.FireShearAreaSY 1878
RobotOM.IRDimCodeResCB71.FireShearAreaSZ 1878
RobotOM.IRDimCodeResCB71.FireUprEdgeDistVz 1878
RobotOM.IRDimCodeResCB71.FireWidthAtMembEndB 1879
RobotOM.IRDimCodeResCB71.FireWidthAtMembEndH 1879
RobotOM.IRDimCodeResCB71.FireWidthB 1879
RobotOM.IRDimCodeResCB71.FireWidthH 1879
RobotOM.IRDimCodeResCB71.FireWidthInMidSpanB 1880
RobotOM.IRDimCodeResCB71.IsBuckY 1880
RobotOM.IRDimCodeResCB71.IsBuckZ 1880
RobotOM.IRDimCodeResCB71.LatBuckCoefKinst 1880

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRDimCodeResCB71.LatBuckLengthCoef 1881
RobotOM.IRDimCodeResCB71.LatBuckLengthLd 1881
RobotOM.IRDimCodeResCB71.LatBuckRelSlendLam 1881
RobotOM.IRDimCodeResCB71.LatBuckStrsCrit 1881
RobotOM.IRDimCodeResCB71.MatAxCompResist 1882
RobotOM.IRDimCodeResCB71.MatAxTensResist 1882
RobotOM.IRDimCodeResCB71.MatBendResist 1882
RobotOM.IRDimCodeResCB71.MatShearResist 1882
RobotOM.IRDimCodeResCB71.MatTrCompResist 1883
RobotOM.IRDimCodeResCB71.MatTrTensResist 1883
RobotOM.IRDimCodeResCB71.StrsBend 1883
RobotOM.IRDimCodeResCB71.StrsBendInCompEdgeMY 1883
RobotOM.IRDimCodeResCB71.StrsBendInCompEdgeMZ 1884
RobotOM.IRDimCodeResCB71.StrsBendInCurveBeam 1884
RobotOM.IRDimCodeResCB71.StrsComp 1884
RobotOM.IRDimCodeResCB71.StrsFinal 1884
RobotOM.IRDimCodeResCB71.StrsLftEdgeMZ 1885
RobotOM.IRDimCodeResCB71.StrsLowEdgeMY 1885
RobotOM.IRDimCodeResCB71.StrsRgtEdgeMZ 1885
RobotOM.IRDimCodeResCB71.StrsShearY 1885
RobotOM.IRDimCodeResCB71.StrsShearZ 1886
RobotOM.IRDimCodeResCB71.StrsTens 1886
RobotOM.IRDimCodeResCB71.StrsUprEdgeMY 1886
RobotOM.IRDimCodeResCB71.TimberType 1886
RobotOM.IRDimCodeResEC3 1819
RobotOM.IRDimCodeResEC3.AxForceExcentrEny 1827
RobotOM.IRDimCodeResEC3.AxForcExcentrEnz 1827
RobotOM.IRDimCodeResEC3.BendParamBetaMlty 1827
RobotOM.IRDimCodeResEC3.BendParamBetaMltz 1828
RobotOM.IRDimCodeResEC3.BendParamBetaMz 1828
RobotOM.IRDimCodeResEC3.BendParamBetMy 1828
RobotOM.IRDimCodeResEC3.BuckCoeffMinXi 1828
RobotOM.IRDimCodeResEC3.BuckCoeffXy 1829
RobotOM.IRDimCodeResEC3.BuckCoeffXz 1829
RobotOM.IRDimCodeResEC3.BuckCurveCoeffAlfy 1829
RobotOM.IRDimCodeResEC3.BuckCurveCoeffAlfz 1829
RobotOM.IRDimCodeResEC3.BuckCurveNumbY 1830
RobotOM.IRDimCodeResEC3.BuckCurveNumbZ 1830
RobotOM.IRDimCodeResEC3.BuckParamFiy 1830
RobotOM.IRDimCodeResEC3.BuckParamFiz 1830
RobotOM.IRDimCodeResEC3.BuckParamKy 1831
RobotOM.IRDimCodeResEC3.BuckParamKz 1831
RobotOM.IRDimCodeResEC3.BuckRelSlendLaby 1831
RobotOM.IRDimCodeResEC3.BuckRelSlendLabz 1831
RobotOM.IRDimCodeResEC3.BuckSLendLamy 1832
RobotOM.IRDimCodeResEC3.BuckStrenNbyrd 1832
RobotOM.IRDimCodeResEC3.BuckStrenNbzrd 1832
RobotOM.IRDimCodeResEC3.BuclSlendLamz 1832
RobotOM.IRDimCodeResEC3.ClassOfSect 1833
RobotOM.IRDimCodeResEC3.ClassOfSectElem1 1833
RobotOM.IRDimCodeResEC3.ClassOfSectElem2 1833
RobotOM.IRDimCodeResEC3.ClassOfSectElem3 1833
RobotOM.IRDimCodeResEC3.ClassOfSectElem4 1834
RobotOM.IRDimCodeResEC3.CompParamBetaA 1834
RobotOM.IRDimCodeResEC3.CritMomenMcr 1834
RobotOM.IRDimCodeResEC3.EffectiveMomenMeff 1834
RobotOM.IRDimCodeResEC3.EffSectAreaSeff 1835
RobotOM.IRDimCodeResEC3.EffSectModulWyeff 1835
RobotOM.IRDimCodeResEC3.EffSectModulWzeff 1835
RobotOM.IRDimCodeResEC3.ElastMomStrenMelyrd 1835
RobotOM.IRDimCodeResEC3.ElastMomStrenMelzrd 1836
RobotOM.IRDimCodeResEC3.ElastSectModulWyel 1836
RobotOM.IRDimCodeResEC3.ElastSectModulWzel 1836
RobotOM.IRDimCodeResEC3.FlangeAreaAf 1836
RobotOM.IRDimCodeResEC3.FlangeAreaAw 1837
RobotOM.IRDimCodeResEC3.InteractParamAlfa 1837
RobotOM.IRDimCodeResEC3.InteractParamBeta 1837
RobotOM.IRDimCodeResEC3.InteractParamMilt 1837
RobotOM.IRDimCodeResEC3.InteractParamMiy 1838
RobotOM.IRDimCodeResEC3.InteractParamMiz 1838
RobotOM.IRDimCodeResEC3.IsBuckY 1838
RobotOM.IRDimCodeResEC3.IsBuckZ 1838
RobotOM.IRDimCodeResEC3.LaodLevel 1839
RobotOM.IRDimCodeResEC3.LatBuckCoeffXlt 1839
RobotOM.IRDimCodeResEC3.LatBuckLengthLd 1839
RobotOM.IRDimCodeResEC3.LatBuckMomStrenmbrd 1839
RobotOM.IRDimCodeResEC3.LatBuckParamBetaW 1840
RobotOM.IRDimCodeResEC3.LatBuckParamC1 1840
RobotOM.IRDimCodeResEC3.LatBuckParamC2 1840
RobotOM.IRDimCodeResEC3.LatBuckParamFilt 1840
RobotOM.IRDimCodeResEC3.LatBuckParamKlt 1841
RobotOM.IRDimCodeResEC3.LatBuckParamLd 1841
RobotOM.IRDimCodeResEC3.LatBuckSlendLamlt 1841

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRDimCodeResEC3.LowFlanSlend1 1841
RobotOM.IRDimCodeResEC3.LowFlanSlend2 1842
RobotOM.IRDimCodeResEC3.MaEffRatio 1842
RobotOM.IRDimCodeResEC3.MaterCoeffGamma0 1842
RobotOM.IRDimCodeResEC3.MaterCoeffGamma1 1842
RobotOM.IRDimCodeResEC3.MaterCoeffGamma2 1843
RobotOM.IRDimCodeResEC3.MaterialCapacityFy 1843
RobotOM.IRDimCodeResEC3.MaxBuckSlend 1843
RobotOM.IRDimCodeResEC3.MomStrengthMpzEuro 1843
RobotOM.IRDimCodeResEC3.OverallBuckStrenNbrd 1844
RobotOM.IRDimCodeResEC3.PartEffRatio1 1844
RobotOM.IRDimCodeResEC3.PartEffRatio2 1844
RobotOM.IRDimCodeResEC3.PartEffRatio3 1844
RobotOM.IRDimCodeResEC3.PartEffRatio4 1845
RobotOM.IRDimCodeResEC3.PartEffRatio5 1845
RobotOM.IRDimCodeResEC3.PlastAxForcStrenNplrd 1845
RobotOM.IRDimCodeResEC3.PlastCompStrenNcrd 1845
RobotOM.IRDimCodeResEC3.PlastMomStrenMplyrd 1846
RobotOM.IRDimCodeResEC3.PlastMomStrenMplzrd 1846
RobotOM.IRDimCodeResEC3.PlastSectModulWypl 1846
RobotOM.IRDimCodeResEC3.PlastSectModulWzpl 1846
RobotOM.IRDimCodeResEC3.PlastTensStrenNtrd 1847
RobotOM.IRDimCodeResEC3.ReducMomStrenMcyrd 1847
RobotOM.IRDimCodeResEC3.ReducMomStrenMczrd 1847
RobotOM.IRDimCodeResEC3.ReducMomStrenMnyrd 1847
RobotOM.IRDimCodeResEC3.ReducMomStrenMnzrd 1848
RobotOM.IRDimCodeResEC3.ReducMomStrenMvyrd 1848
RobotOM.IRDimCodeResEC3.ReducMomStrenMvzrd 1848
RobotOM.IRDimCodeResEC3.ShearStrenTplyrd 1848
RobotOM.IRDimCodeResEC3.ShearStrenTplzrd 1849
RobotOM.IRDimCodeResEC3.StrsComp 1849
RobotOM.IRDimCodeResEC3.StrsLftEdgeMZ 1849
RobotOM.IRDimCodeResEC3.StrsLowEdgeMY 1849
RobotOM.IRDimCodeResEC3.StrsRgtEdgeMZ 1850
RobotOM.IRDimCodeResEC3.StrsShearY 1850
RobotOM.IRDimCodeResEC3.StrsShearZ 1850
RobotOM.IRDimCodeResEC3.StrsTens 1850
RobotOM.IRDimCodeResEC3.StrsUprEdgeMY 1851
RobotOM.IRDimCodeResEC3.TensStrenNurd 1851
RobotOM.IRDimCodeResEC3.TorsMomInertIt 1851
RobotOM.IRDimCodeResEC3.UppFlanSlend 1851
RobotOM.IRDimCodeResEC3.WarpingConstantIw 1852
RobotOM.IRDimCodeResEC3.WebSlend 1852
RobotOM.IRDimCodeResEC3.WebSlend1 1852
RobotOM.IRDimCodeService 1932
RobotOM.IRDimCodeService.EditMembDef 1933
RobotOM.IRDimCodeService.GetDefaultMembDef 1933
RobotOM.IRDimCodeService.GetMembCalc 1934
RobotOM.IRDimEffDef 1899
RobotOM.IRDimEffDef.Clear 1902
RobotOM.IRDimEffDef.MX 1900
RobotOM.IRDimEffDef.MY 1900
RobotOM.IRDimEffDef.MZ 1901
RobotOM.IRDimEffDef.N 1901
RobotOM.IRDimEffDef.QY 1901
RobotOM.IRDimEffDef.QZ 1901
RobotOM.IRDimEffDef.Read_IntPsEff_M_1P4L 1903
RobotOM.IRDimEffDef.Read_IntPsEff_M_3P4L 1903
RobotOM.IRDimEffDef.Read_IntPsEff_M_MID 1903
RobotOM.IRDimEffDef.Read_IntPsEff_M1 1903
RobotOM.IRDimEffDef.Read_IntPsEff_M12 1903
RobotOM.IRDimEffDef.Read_IntPsEff_M2 1904
RobotOM.IRDimEffDef.Read_IntPsEff_MN_MAX 1904
RobotOM.IRDimEffDef.Read_IntPsEff_MP_MAX 1904
RobotOM.IRDimEffDef.Read_M_1P4L 1904
RobotOM.IRDimEffDef.Read_M_3P4L 1904
RobotOM.IRDimEffDef.Read_M_MID 1905
RobotOM.IRDimEffDef.Read_M1 1905
RobotOM.IRDimEffDef.Read_M12 1905
RobotOM.IRDimEffDef.Read_M2 1905
RobotOM.IRDimEffDef.Read_MN_MAX 1905
RobotOM.IRDimEffDef.Read_MP_MAX 1906
RobotOM.IRDimEffDef.ReadParam 1906
RobotOM.IRDimEffDef.WriteForces 1906
RobotOM.IRDimEffDef.WriteIntPsEffSet1 1906
RobotOM.IRDimEffDef.WriteIntPsEffSet2 1907
RobotOM.IRDimEffDef.WriteParam 1907
RobotOM.IRDimEffDef.WriteValuesSet1 1907
RobotOM.IRDimEffDef.WriteValuesSet2 1907
RobotOM.IRDimEffDefDirType 1898
RobotOM.IRDimEffDefIntPsType 1898
RobotOM.IRDimEffDefParamType 1898
RobotOM.IRDimFireMemberPositionsCB71 1888
RobotOM.IRDimFireProtectionTimesCB71 1889

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRDimLatBuckCoeffDiagramEC3 1818
RobotOM.IRDimLaterBuckTypeCB71 1887
RobotOM.IRDimLaterBuckTypeEC3 1817
RobotOM.IRDimLoadLevelCB71 1888
RobotOM.IRDimLoadLevelEC3 1817
RobotOM.IRDimLoadTypeCB71 1888
RobotOM.IRDimLoadTypeEC3 1817
RobotOM.IRDimMatDef 1908
RobotOM.IRDimMatDef.Name 1909
RobotOM.IRDimMatDef.ReadDoubleExtraValue 1910
RobotOM.IRDimMatDef.ReadLongExtraValue 1911
RobotOM.IRDimMatDef.ReadValue 1911
RobotOM.IRDimMatDef.SecondName 1910
RobotOM.IRDimMatDef.SetNames 1911
RobotOM.IRDimMatDef.Type 1910
RobotOM.IRDimMatDef.WriteDoubleExtraValue 1911
RobotOM.IRDimMatDef.WriteLongExtraValue 1911
RobotOM.IRDimMatDef.WriteValue 1912
RobotOM.IRDimMatDefDblExValType 1908
RobotOM.IRDimMatDefLongExValType 1908
RobotOM.IRDimMatDefType 1908
RobotOM.IRDimMatDefValType 1908
RobotOM.IRDimMembCalc 1928
RobotOM.IRDimMembCalc.CalculBuckling 1930
RobotOM.IRDimMembCalc.CalculMember 1931
RobotOM.IRDimMembCalc.GetRatio 1931
RobotOM.IRDimMembCalc.GetResultsInterface 1931
RobotOM.IRDimMembCalc.IsExtraMomentsSet1 1929
RobotOM.IRDimMembCalc.IsExtraMomentsSet2 1929
RobotOM.IRDimMembCalc.IsIntPointsMomentsSet1 1930
RobotOM.IRDimMembCalc.IsIntPointsMomentsSet2 1930
RobotOM.IRDimMembCalc.SetCalcState 1931
RobotOM.IRDimMembCalc.SetEffDef 1931
RobotOM.IRDimMembCalc.SetMatDef 1932
RobotOM.IRDimMembCalc.SetMembDef 1932
RobotOM.IRDimMembCalc.SetProfDef 1932
RobotOM.IRDimMembCalcBuckType 1919
RobotOM.IRDimMembCalcRetValue 1919
RobotOM.IRDimMembDef 1894
RobotOM.IRDimMembDef.ClientID 1895
RobotOM.IRDimMembDef.DeflYZRelLimit 1896
RobotOM.IRDimMembDef.DisplXYRelLimit 1896
RobotOM.IRDimMembDef.IsBuckCoefConst 1897
RobotOM.IRDimMembDef.IsDeflectionYZ 1897
RobotOM.IRDimMembDef.IsDisplacementXY 1897
RobotOM.IRDimMembDef.Length 1895
RobotOM.IRDimMembDef.LengthYZUV 1897
RobotOM.IRDimMembDef.MatType 1895
RobotOM.IRDimMembDef.Name 1895
RobotOM.IRDimMembDef.Retrieve 1897
RobotOM.IRDimMembDef.Store 1898
RobotOM.IRDimMembDef.Type 1896
RobotOM.IRDimMembDefBucklingDataType 1893
RobotOM.IRDimMembDefDeflDataType 1894
RobotOM.IRDimMembDefDispDataType 1893
RobotOM.IRDimMembDefLengthDataType 1893
RobotOM.IRDimMembDefMatType 1893
RobotOM.IRDimMembDefType 1892
RobotOM.IRDimMembParamsCB71 1853
RobotOM.IRDimMembParamsCB71.ArcBeamCheck 1856
RobotOM.IRDimMembParamsCB71.ArcRadius 1856
RobotOM.IRDimMembParamsCB71.BuckComposBeamCoefCey 1856
RobotOM.IRDimMembParamsCB71.BuckComposBeamCoefCez 1857
RobotOM.IRDimMembParamsCB71.BuckIsComposedY 1857
RobotOM.IRDimMembParamsCB71.BuckIsComposedZ 1857
RobotOM.IRDimMembParamsCB71.BuckLengthCoeffZ 1857
RobotOM.IRDimMembParamsCB71.BucklingDiagramY 1858
RobotOM.IRDimMembParamsCB71.BucklingDiagramZ 1858
RobotOM.IRDimMembParamsCB71.BucklLengthCoeffY 1858
RobotOM.IRDimMembParamsCB71.FireContinProtForComposBeam 1858
RobotOM.IRDimMembParamsCB71.FireLftSideProt 1859
RobotOM.IRDimMembParamsCB71.FireLowSideProt 1859
RobotOM.IRDimMembParamsCB71.FireMembPosition 1859
RobotOM.IRDimMembParamsCB71.FireProtectionTime 1859
RobotOM.IRDimMembParamsCB71.FireRequiredResist 1860
RobotOM.IRDimMembParamsCB71.FireRgtSideProt 1860
RobotOM.IRDimMembParamsCB71.FireUprSideProt 1860
RobotOM.IRDimMembParamsCB71.ForceDistFromBeginDistX 1861
RobotOM.IRDimMembParamsCB71.HumChangeDeltaH 1861

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRDimMembParamsCB71.InflDeflCoefTheta 1861
RobotOM.IRDimMembParamsCB71.LatBuckType 1861
RobotOM.IRDimMembParamsCB71.LatCoeffLowerFlangeValue 1862
RobotOM.IRDimMembParamsCB71.LatCoeffUpperFlangeValue 1862
RobotOM.IRDimMembParamsCB71.LoadLevel 1862
RobotOM.IRDimMembParamsCB71.LoadTypeY 1862
RobotOM.IRDimMembParamsCB71.TensAreaNetGros 1863
RobotOM.IRDimMembParamsCB71.TubeControl 1863
RobotOM.IRDimMembParamsEC3 1809
RobotOM.IRDimMembParamsEC3.BuckLengthCoeffY 1811
RobotOM.IRDimMembParamsEC3.BuckLengthCoeffZ 1811
RobotOM.IRDimMembParamsEC3.BucklingDiagramY 1811
RobotOM.IRDimMembParamsEC3.BucklingDiagramZ 1812
RobotOM.IRDimMembParamsEC3.HotRolledPipes 1812
RobotOM.IRDimMembParamsEC3.LatBuckType 1812
RobotOM.IRDimMembParamsEC3.LatCoeffLowerFlange 1812
RobotOM.IRDimMembParamsEC3.LatCoeffLowerFlangeValue 1813
RobotOM.IRDimMembParamsEC3.LatCoeffUpperFlange 1813
RobotOM.IRDimMembParamsEC3.LatCoeffUpperFlangeValue 1813
RobotOM.IRDimMembParamsEC3.LoadLevel 1813
RobotOM.IRDimMembParamsEC3.LoadLevelValue 1814
RobotOM.IRDimMembParamsEC3.LoadTypeY 1814
RobotOM.IRDimMembParamsEC3.LoadTypeZ 1814
RobotOM.IRDimMembParamsEC3.MaterCoeffGamma0 1815
RobotOM.IRDimMembParamsEC3.MaterCoeffGamma1 1815
RobotOM.IRDimMembParamsEC3.RelLimitDeflUy 1815
RobotOM.IRDimMembParamsEC3.RelLimitdeflUz 1815
RobotOM.IRDimMembParamsEC3.TensAreaNetGros 1816
RobotOM.IRDimMembParamsEC3.TubeControl 1816
RobotOM.IRDimMembParamsEC3.YieldStrengthType 1816
RobotOM.IRDimMembParamsEC3.YieldStrengthValue 1816
RobotOM.IRDimMembRes 1920
RobotOM.IRDimMembRes.BlockCount 1924
RobotOM.IRDimMembRes.CreateResWnd 1924
RobotOM.IRDimMembRes.EffRatio 1922
RobotOM.IRDimMembRes.GetEffDefAccess 1924
RobotOM.IRDimMembRes.GetLineComponent 1925
RobotOM.IRDimMembRes.GetLineType 1925
RobotOM.IRDimMembRes.GetMatDefAccess 1925
RobotOM.IRDimMembRes.GetMaxLineNo 1925
RobotOM.IRDimMembRes.GetMembDefAccess 1926
RobotOM.IRDimMembRes.GetProfDefAccess 1926
RobotOM.IRDimMembRes.IsLineActive 1926
RobotOM.IRDimMembRes.IsStatement 1926
RobotOM.IRDimMembRes.Language 1922
RobotOM.IRDimMembRes.Ratio 1922
RobotOM.IRDimMembRes.RecognizedPQ 1926
RobotOM.IRDimMembRes.RefreshUnits 1927
RobotOM.IRDimMembRes.ReplaceMark 1927
RobotOM.IRDimMembRes.ResOfCalc 1922
RobotOM.IRDimMembRes.Retrieve 1927
RobotOM.IRDimMembRes.RtfFileName 1922
RobotOM.IRDimMembRes.SetEffDef 1927
RobotOM.IRDimMembRes.SetMatDef 1927
RobotOM.IRDimMembRes.SetMembDef 1928
RobotOM.IRDimMembRes.SetProfDef 1928
RobotOM.IRDimMembRes.SlendY 1923
RobotOM.IRDimMembRes.SlendZ 1923
RobotOM.IRDimMembRes.Units 1923
RobotOM.IRDimMembResTableComp 1920
RobotOM.IRDimMembResTableLineType 1920
RobotOM.IRDimMembSrv 1918
RobotOM.IRDimMembSrv.CheckLabelName 1919
RobotOM.IRDimMembSrv.Save 1919
RobotOM.IRDimProfDef 1913
RobotOM.IRDimProfDef.Clear 1914
RobotOM.IRDimProfDef.IsVar 1913
RobotOM.IRDimProfDef.Name 1914
RobotOM.IRDimProfDef.ReadValue 1914
RobotOM.IRDimProfDef.Type 1914
RobotOM.IRDimProfDef.WriteValue 1915
RobotOM.IRDimProfDefItemType 1912
RobotOM.IRDimProfDefType 1912
RobotOM.IRDimProfDefValType 1912
RobotOM.IRDimServer 1937
RobotOM.IRDimStream 1889
RobotOM.IRDimStream.Clear 1890
RobotOM.IRDimStream.ReadDouble 1891

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRDimStream.ReadLong 1891
RobotOM.IRDimStream.ReadText 1891
RobotOM.IRDimStream.SeekSet 1891
RobotOM.IRDimStream.Size 1891
RobotOM.IRDimStream.WriteDouble 1892
RobotOM.IRDimStream.WriteLong 1892
RobotOM.IRDimStream.WriteText 1892
RobotOM.IRDimStreamType 1889
RobotOM.IRDimUnits 1935
RobotOM.IRDimUnits.AreRobotUnits 1936
RobotOM.IRDimUnits.Format 1936
RobotOM.IRDimUnits.ReadToUserCoef 1936
RobotOM.IRDimUnits.ReadUserName 1937
RobotOM.IRDimUnits.Refresh 1937
RobotOM.IRDimUnitType 1935
RobotOM.IRDimYieldStrengthTypeEC3 1852
RobotOM.IRJointAnchor 2004
RobotOM.IRJointAnchor.AnchorPlate 2005
RobotOM.IRJointAnchor.Bolts 2005
RobotOM.IRJointAnchor.Tige 2005
RobotOM.IRJointAnchor.Type 2005
RobotOM.IRJointAnchorBolt 2001
RobotOM.IRJointAnchorBolt.Length1 2001
RobotOM.IRJointAnchorBolt.Length2 2002
RobotOM.IRJointAnchorBolt.Length3 2002
RobotOM.IRJointAnchorBolt.Length4 2002
RobotOM.IRJointAnchorPlate 2002
RobotOM.IRJointAnchorPlate.Type 2003
RobotOM.IRJointAnchorPlateType 2019
RobotOM.IRJointAnchorType 2003
RobotOM.IRJointAngle 2033
RobotOM.IRJointAngle.DistFromUpperBeamEdge 2034
RobotOM.IRJointAngle.Length 2034
RobotOM.IRJointAngle.Profile 2035
RobotOM.IRJointAngle.ProfilePosition 2035
RobotOM.IRJointAngle.Type 2035
RobotOM.IRJointAngleBolts 2030
RobotOM.IRJointAngleBolts.Area 2031
RobotOM.IRJointAngleBolts.ClassName 2031
RobotOM.IRJointAngleBolts.Diameter 2031
RobotOM.IRJointAngleBolts.DiameterName 2032
RobotOM.IRJointAngleBolts.DistFromUpperEdge 2032
RobotOM.IRJointAngleBolts.DistFromWeb 2032
RobotOM.IRJointAngleBolts.Friction 2033
RobotOM.IRJointAngleBolts.Rows 2033
RobotOM.IRJointAngleBolts.Spacing 2033
RobotOM.IRJointAngleLoad 2041
RobotOM.IRJointAngleLoad.Fz 2041
RobotOM.IRJointAngleProfilePosition 2040
RobotOM.IRJointAngleType 2035
RobotOM.IRJointAngleUnit 2179
RobotOM.IRJointBeamCut 2036
RobotOM.IRJointBeamCut.HeightLower 2037
RobotOM.IRJointBeamCut.HeightUpper 2037
RobotOM.IRJointBeamCut.Length 2037
RobotOM.IRJointBeamGirder 2139
RobotOM.IRJointBeamGirder.AngleLeft 2141
RobotOM.IRJointBeamGirder.AngleRight 2141
RobotOM.IRJointBeamGirder.BeamLeft 2142
RobotOM.IRJointBeamGirder.BeamRight 2142
RobotOM.IRJointBeamGirder.DistGirderFlangeToBeamFlangeLeft 2142
RobotOM.IRJointBeamGirder.DistGirderFlangeToBeamFlangeRight 2143
RobotOM.IRJointBeamGirder.DistGirderToBeamLeft 2143
RobotOM.IRJointBeamGirder.DistGirderToBeamRight 2143
RobotOM.IRJointBeamGirder.ElementType 2143
RobotOM.IRJointBeamGirder.Girder 2144
RobotOM.IRJointBeamGirder.PlateLeft 2144
RobotOM.IRJointBeamGirder.PlateRight 2144
RobotOM.IRJointBeamGirder.SeatLeftDown 2144
RobotOM.IRJointBeamGirder.SeatLeftUp 2145
RobotOM.IRJointBeamGirder.SeatRightDown 2145
RobotOM.IRJointBeamGirder.SeatRightUp 2145
RobotOM.IRJointBeamGirder.StiffenerLeft 2145
RobotOM.IRJointBeamGirder.StiffenerRight 2146
RobotOM.IRJointBeamGirderAngle 2153
RobotOM.IRJointBeamGirderAngle.BoltsBeam 2154
RobotOM.IRJointBeamGirderAngle.BoltsGirder 2154
RobotOM.IRJointBeamGirderAngle.Element 2154
RobotOM.IRJointBeamGirderBeam 2155
RobotOM.IRJointBeamGirderBeam.CutEnd 2155
RobotOM.IRJointBeamGirderBeam.Profile 2155
RobotOM.IRJointBeamGirderBolts 2124

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRJointBeamGirderBolts.Area 2125
RobotOM.IRJointBeamGirderBolts.ClassName 2125
RobotOM.IRJointBeamGirderBolts.Cols 2126
RobotOM.IRJointBeamGirderBolts.Diameter 2126
RobotOM.IRJointBeamGirderBolts.DiameterName 2126
RobotOM.IRJointBeamGirderBolts.DistFromUpperBeamEdge 2126
RobotOM.IRJointBeamGirderBolts.DistFromVertBeamEdge 2127
RobotOM.IRJointBeamGirderBolts.Friction 2127
RobotOM.IRJointBeamGirderBolts.Rows 2127
RobotOM.IRJointBeamGirderBolts.SpacingH 2128
RobotOM.IRJointBeamGirderBolts.SpacingV 2128
RobotOM.IRJointBeamGirderLoad 2146
RobotOM.IRJointBeamGirderLoad.LFx 2147
RobotOM.IRJointBeamGirderLoad.LFz 2147
RobotOM.IRJointBeamGirderLoad.LMy 2147
RobotOM.IRJointBeamGirderLoad.RFx 2148
RobotOM.IRJointBeamGirderLoad.RFz 2148
RobotOM.IRJointBeamGirderLoad.RMy 2148
RobotOM.IRJointBeamGirderPlate 2132
RobotOM.IRJointBeamGirderPlate.BoltsBeam 2133
RobotOM.IRJointBeamGirderPlate.ConnectorsType 2134
RobotOM.IRJointBeamGirderPlate.DistFromUpperBeamEdge 2134
RobotOM.IRJointBeamGirderPlate.Length 2134
RobotOM.IRJointBeamGirderPlate.Material 2135
RobotOM.IRJointBeamGirderPlate.Thick 2135
RobotOM.IRJointBeamGirderPlate.WeldsBeam 2135
RobotOM.IRJointBeamGirderPlate.WeldsGirder 2135
RobotOM.IRJointBeamGirderPlate.Width 2136
RobotOM.IRJointBeamGirderSeat 2136
RobotOM.IRJointBeamGirderSeat.BoltsBeam 2137
RobotOM.IRJointBeamGirderSeat.BoltsGirder 2137
RobotOM.IRJointBeamGirderSeat.ConnectorsToWebType 2138
RobotOM.IRJointBeamGirderSeat.Length 2138
RobotOM.IRJointBeamGirderSeat.Material 2138
RobotOM.IRJointBeamGirderSeat.Section 2139
RobotOM.IRJointBeamGirderSeat.WeldsGirder 2139
RobotOM.IRJointBeamGirderSeatBolts 2149
RobotOM.IRJointBeamGirderSeatBolts.Area 2150
RobotOM.IRJointBeamGirderSeatBolts.ClassName 2150
RobotOM.IRJointBeamGirderSeatBolts.Cols 2150
RobotOM.IRJointBeamGirderSeatBolts.Diameter 2151
RobotOM.IRJointBeamGirderSeatBolts.DiameterName 2151
RobotOM.IRJointBeamGirderSeatBolts.DistFromPerpendicularArmEdge 2151
RobotOM.IRJointBeamGirderSeatBolts.DistFromVertEdge 2151
RobotOM.IRJointBeamGirderSeatBolts.Friction 2152
RobotOM.IRJointBeamGirderStiffener 2128
RobotOM.IRJointBeamGirderStiffener.BoltsBeam 2129
RobotOM.IRJointBeamGirderStiffener.ConnectorsType 2130
RobotOM.IRJointBeamGirderStiffener.CutHeightDown 2130
RobotOM.IRJointBeamGirderStiffener.CutHeightUp 2130
RobotOM.IRJointBeamGirderStiffener.CutLength 2131
RobotOM.IRJointBeamGirderStiffener.Length 2131
RobotOM.IRJointBeamGirderStiffener.Material 2131
RobotOM.IRJointBeamGirderStiffener.Thick 2132
RobotOM.IRJointBeamGirderStiffener.WeldsBeam 2132
RobotOM.IRJointBearingPlate 2013
RobotOM.IRJointBearingPlate.ThickBearingBar 2014
RobotOM.IRJointBolts 2166
RobotOM.IRJointBolts.Area 2167
RobotOM.IRJointBolts.ClassName 2167
RobotOM.IRJointBolts.Cols 2167
RobotOM.IRJointBolts.Diameter 2167
RobotOM.IRJointBolts.DiameterName 2168
RobotOM.IRJointBolts.Friction 2168
RobotOM.IRJointBolts.Height1 2168
RobotOM.IRJointBolts.Rows 2168
RobotOM.IRJointBoltType 2169
RobotOM.IRJointColumnBasePlateCalcModel 1998
RobotOM.IRJointColumnBracket 2016
RobotOM.IRJointColumnBracket.Exist 2019
RobotOM.IRJointColumnBracket.Height 2017
RobotOM.IRJointColumnBracket.Length 2018
RobotOM.IRJointColumnBracket.Thick 2018
RobotOM.IRJointColumnBracket.VThick 2018
RobotOM.IRJointColumnBracket.Width 2018
RobotOM.IRJointColumnSquare 2024
RobotOM.IRJointColumnSquare.Diameter 2025
RobotOM.IRJointColumnSquare.Length 2025
RobotOM.IRJointColumnSquare.Profile 2025
RobotOM.IRJointComponentType 2180

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRJointConcreteColumn 1990
RobotOM.IRJointConcreteColumn.Base 1992
RobotOM.IRJointConcreteColumn.CalcModel 1992
RobotOM.IRJointConcreteColumn.Depth 1992
RobotOM.IRJointConcreteColumn.Foundation 1992
RobotOM.IRJointConcreteColumn.Materials 1993
RobotOM.IRJointConcreteColumn.Profile 1993
RobotOM.IRJointConcreteColumn.SpreadFootingType 1993
RobotOM.IRJointConcreteColumnFoundation 1994
RobotOM.IRJointConcreteColumnFoundation.par_a1 1995
RobotOM.IRJointConcreteColumnFoundation.par_b 1995
RobotOM.IRJointConcreteColumnFoundation.par_b1 1995
RobotOM.IRJointConcreteColumnFoundation.par_b2 1995
RobotOM.IRJointConcreteColumnFoundation.par_h 1996
RobotOM.IRJointConcreteColumnFoundation.par_h1 1996
RobotOM.IRJointConcreteColumnFoundation.par_h2 1996
RobotOM.IRJointConcreteColumnFoundation.par_l 1996
RobotOM.IRJointConcreteColumnFoundation.par_l1 1997
RobotOM.IRJointConcreteColumnFoundation.par_l2 1997
RobotOM.IRJointConcreteColumnFoundation.par_o1 1997
RobotOM.IRJointConcreteColumnFoundation.par_o2 1997
RobotOM.IRJointConcreteColumnLoad 1998
RobotOM.IRJointConcreteColumnLoad.Fx 1999
RobotOM.IRJointConcreteColumnLoad.FxAssemb 1999
RobotOM.IRJointConcreteColumnLoad.Fy 2000
RobotOM.IRJointConcreteColumnLoad.Fz 2000
RobotOM.IRJointConcreteColumnLoad.My 2000
RobotOM.IRJointConcreteColumnLoad.Mz 2000
RobotOM.IRJointConcreteMaterials 1989
RobotOM.IRJointConcreteMaterials.CoeffBA 1990
RobotOM.IRJointConcreteMaterials.Dosage 1990
RobotOM.IRJointConcreteMaterials.Sigma 1990
RobotOM.IRJointConnection 2159
RobotOM.IRJointConnection.GetFromRobot 2161
RobotOM.IRJointConnection.SetToRobot 2161
RobotOM.IRJointConnection.Type 2160
RobotOM.IRJointConnection.WFRelPos 2161
RobotOM.IRJointConnectionDefType 2156
RobotOM.IRJointConnectionInfo 2157
RobotOM.IRJointConnectionInfo.DefType 2158
RobotOM.IRJointConnectionInfo.Elements 2158
RobotOM.IRJointConnectionInfo.Node 2158
RobotOM.IRJointConnectionInfo.Number 2158
RobotOM.IRJointConnectionInfo.Type 2159
RobotOM.IRJointConnectionInfo.UniqueId 2159
RobotOM.IRJointConnectionServer 2169
RobotOM.IRJointConnectionServer.Calculate 2171
RobotOM.IRJointConnectionServer.CalculateNote 2171
RobotOM.IRJointConnectionServer.Count 2170
RobotOM.IRJointConnectionServer.Create 2172
RobotOM.IRJointConnectionServer.CreateInfo 2172
RobotOM.IRJointConnectionServer.Delete 2172
RobotOM.IRJointConnectionServer.Exist 2173
RobotOM.IRJointConnectionServer.Find 2173
RobotOM.IRJointConnectionServer.FindWithId 2173
RobotOM.IRJointConnectionServer.Get 2173
RobotOM.IRJointConnectionServer.GetAllNumbers 2174
RobotOM.IRJointConnectionServer.GetInfo 2174
RobotOM.IRJointConnectionType 2156
RobotOM.IRJointConnectorsType 2152
RobotOM.IRJointElementType 2152
RobotOM.IRJointExtType 2180
RobotOM.IRJointFixedColumnBase 1982
RobotOM.IRJointFixedColumnBase.Anchor 1983
RobotOM.IRJointFixedColumnBase.Base 1983
RobotOM.IRJointFixedColumnBase.BasePlateMaterial 1984
RobotOM.IRJointFixedColumnBase.Bolts 1984
RobotOM.IRJointFixedColumnBase.ComplexStiff 1984
RobotOM.IRJointFixedColumnBase.FootPlate 1984
RobotOM.IRJointFixedColumnBase.Materials 1985
RobotOM.IRJointFixedColumnBase.NodeNumber 1985
RobotOM.IRJointFixedColumnBase.Profile 1985
RobotOM.IRJointFixedColumnBase.SimpleStiff 1985
RobotOM.IRJointFixedColumnBase.StiffType 1986
RobotOM.IRJointFixedColumnBase.Washer 1986
RobotOM.IRJointFixedColumnBase.Wedge 1986
RobotOM.IRJointFixedColumnBase.Welds 1986
RobotOM.IRJointFixedColumnBaseStiffType 1988
RobotOM.IRJointFixedFootWelds 1987
RobotOM.IRJointFixedFootWelds.FootPlate 1987
RobotOM.IRJointFixedFootWelds.Stiff 1988
RobotOM.IRJointFixedFootWelds.Washer 1988
RobotOM.IRJointFixedFootWelds.Wedge 1988
RobotOM.IRJointFixedLoad 1980

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRJointFixedLoad.Fx 1980
RobotOM.IRJointFixedLoad.Fy 1981
RobotOM.IRJointFixedLoad.Fz 1981
RobotOM.IRJointFixedLoad.My 1981
RobotOM.IRJointFixedLoad.Mz 1981
RobotOM.IRJointFootBolts 2006
RobotOM.IRJointFootBolts.Area 2007
RobotOM.IRJointFootBolts.ClassName 2007
RobotOM.IRJointFootBolts.Diameter 2007
RobotOM.IRJointFootBolts.DiameterName 2007
RobotOM.IRJointFootBolts.Distance 2008
RobotOM.IRJointFootBolts.Friction 2008
RobotOM.IRJointFootBolts.Rows 2008
RobotOM.IRJointFootBolts.SpacingH 2008
RobotOM.IRJointFootBolts.SpacingV 2009
RobotOM.IRJointFootMaterials 2022
RobotOM.IRJointFootMaterials.CementAmount 2023
RobotOM.IRJointFootMaterials.ConcrClass 2023
RobotOM.IRJointFootMaterials.ConcrSteelCoeff 2023
RobotOM.IRJointFootMaterials.PlateSigma 2023
RobotOM.IRJointFootMaterials.PlateYoung 2024
RobotOM.IRJointFootPlate 2009
RobotOM.IRJointFootPlate.Diameter 2010
RobotOM.IRJointFootPlate.Type 2010
RobotOM.IRJointFootPlateType 2009
RobotOM.IRJointFootStiffenerComplex 2027
RobotOM.IRJointFootStiffenerComplex.Height 2028
RobotOM.IRJointFootStiffenerComplex.Length 2028
RobotOM.IRJointFootStiffenerComplex.ThickPlateHor 2029
RobotOM.IRJointFootStiffenerComplex.ThickStiff 2029
RobotOM.IRJointFootStiffenerHoriz 2021
RobotOM.IRJointFootStiffenerHoriz.Thick 2021
RobotOM.IRJointFootStiffenerSimple 2025
RobotOM.IRJointFootStiffenerSimple.Height 2026
RobotOM.IRJointFootStiffenerSimple.Length 2026
RobotOM.IRJointFootStiffenerSimple.Thick 2027
RobotOM.IRJointFootStiffenerSimple.Type 2027
RobotOM.IRJointFootStiffenerSimple.Width 2027
RobotOM.IRJointFootStiffenerVert 2019
RobotOM.IRJointFootStiffenerVert.Length 2020
RobotOM.IRJointFootStiffenerVert.Thick 2020
RobotOM.IRJointFootStiffenerVert.WidthSpacing 2021
RobotOM.IRJointFootStiffType 2029
RobotOM.IRJointFootWelds 2014
RobotOM.IRJointFootWelds.Bearing 2015
RobotOM.IRJointFootWelds.FootPlate 2015
RobotOM.IRJointFootWelds.Stiff 2016
RobotOM.IRJointFootWelds.Washer 2016
RobotOM.IRJointFootWelds.Wedge 2016
RobotOM.IRJointGussetBoltsDiag 2060
RobotOM.IRJointGussetBoltsDiag.BoltAxisShift 2061
RobotOM.IRJointGussetBoltsDiag.ClassName 2061
RobotOM.IRJointGussetBoltsDiag.Diameter 2062
RobotOM.IRJointGussetBoltsDiag.DistanceH1 2062
RobotOM.IRJointGussetBoltsDiag.Friction 2062
RobotOM.IRJointGussetBoltsDiag.Rows 2062
RobotOM.IRJointGussetBoltsDiag.Spacing 2063
RobotOM.IRJointGussetCornersType 2078
RobotOM.IRJointGussetCross 2096
RobotOM.IRJointGussetCross.BoltsClassVector 2097
RobotOM.IRJointGussetCross.BoltsDimensionNamesVector 2098
RobotOM.IRJointGussetCross.BoltsDimensionsVector 2098
RobotOM.IRJointGussetCross.DiagLeftLower 2098
RobotOM.IRJointGussetCross.DiagLeftUpper 2099
RobotOM.IRJointGussetCross.DiagRightLower 2099
RobotOM.IRJointGussetCross.DiagRightUpper 2099
RobotOM.IRJointGussetCross.FixType_LeftL_RightU 2099
RobotOM.IRJointGussetCross.FixType_LeftLower 2100
RobotOM.IRJointGussetCross.FixType_LeftU_RightL 2100
RobotOM.IRJointGussetCross.FixType_LeftUpper 2100
RobotOM.IRJointGussetCross.FixType_RightLower 2101
RobotOM.IRJointGussetCross.FixType_RightUpper 2101
RobotOM.IRJointGussetCross.GussetPlate 2101
RobotOM.IRJointGussetCross.ProfileCutting 2101
RobotOM.IRJointGussetCrossLoad 2118
RobotOM.IRJointGussetCrossLoad.Fx_LeftLower 2119
RobotOM.IRJointGussetCrossLoad.Fx_LeftUpper 2119
RobotOM.IRJointGussetCrossLoad.Fx_RightLower 2120
RobotOM.IRJointGussetCrossLoad.Fx_RightUpper 2120
RobotOM.IRJointGussetCrossPlate 2087
RobotOM.IRJointGussetCrossPlate.B1 2089
RobotOM.IRJointGussetCrossPlate.B2 2089
RobotOM.IRJointGussetCrossPlate.B3 2089
RobotOM.IRJointGussetCrossPlate.B4 2089

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRJointGussetCrossPlate.CornersType 2090
RobotOM.IRJointGussetCrossPlate.CutH1 2090
RobotOM.IRJointGussetCrossPlate.CutH2 2090
RobotOM.IRJointGussetCrossPlate.CutH3 2091
RobotOM.IRJointGussetCrossPlate.CutH4 2091
RobotOM.IRJointGussetCrossPlate.CutV1 2091
RobotOM.IRJointGussetCrossPlate.CutV2 2092
RobotOM.IRJointGussetCrossPlate.CutV3 2092
RobotOM.IRJointGussetCrossPlate.CutV4 2092
RobotOM.IRJointGussetCrossPlate.H1 2093
RobotOM.IRJointGussetCrossPlate.H2 2093
RobotOM.IRJointGussetCrossPlate.H3 2093
RobotOM.IRJointGussetCrossPlate.H4 2093
RobotOM.IRJointGussetCrossPlate.Height 2094
RobotOM.IRJointGussetCrossPlate.HOffset 2094
RobotOM.IRJointGussetCrossPlate.HorizontalOffset 2094
RobotOM.IRJointGussetCrossPlate.Length 2095
RobotOM.IRJointGussetCrossPlate.Material 2095
RobotOM.IRJointGussetCrossPlate.Thick 2095
RobotOM.IRJointGussetCrossPlate.VOffset 2095
RobotOM.IRJointGussetCrossProfileCutting 2087
RobotOM.IRJointGussetDiagonale 2065
RobotOM.IRJointGussetDiagonale.BarNumber 2066
RobotOM.IRJointGussetDiagonale.BoltsDiag 2066
RobotOM.IRJointGussetDiagonale.DistanceEC 2066
RobotOM.IRJointGussetDiagonale.Exist 2067
RobotOM.IRJointGussetDiagonale.Position 2067
RobotOM.IRJointGussetDiagonale.Profile 2067
RobotOM.IRJointGussetDiagonale.WeldsDiag 2067
RobotOM.IRJointGussetDiagonalePositionType 2122
RobotOM.IRJointGussetFixType 2068
RobotOM.IRJointGussetFlange 2112
RobotOM.IRJointGussetFlange.BoltsClassVector 2113
RobotOM.IRJointGussetFlange.BoltsDimensionNamesVector 2114
RobotOM.IRJointGussetFlange.BoltsDimensionsVector 2114
RobotOM.IRJointGussetFlange.DiagonalLeft 2114
RobotOM.IRJointGussetFlange.DiagonalRight 2115
RobotOM.IRJointGussetFlange.DiagonalUp 2115
RobotOM.IRJointGussetFlange.FixType_DiagonalLeft 2115
RobotOM.IRJointGussetFlange.FixType_DiagonalRight 2115
RobotOM.IRJointGussetFlange.FixType_DiagonalUp 2116
RobotOM.IRJointGussetFlange.FixType_FlangeLeft 2116
RobotOM.IRJointGussetFlange.FixType_FlangeRight 2116
RobotOM.IRJointGussetFlange.FlangeLeft 2116
RobotOM.IRJointGussetFlange.FlangeRight 2117
RobotOM.IRJointGussetFlange.GussetPlate 2117
RobotOM.IRJointGussetFlange.ProfileCutting 2117
RobotOM.IRJointGussetFlangeLoad 2120
RobotOM.IRJointGussetFlangeLoad.Fx_FlangeLeft 2121
RobotOM.IRJointGussetFlangeLoad.Fx_FlangeRight 2121
RobotOM.IRJointGussetFlangeLoad.Fx_LeftUpper 2122
RobotOM.IRJointGussetFlangeLoad.Fx_RightUpper 2122
RobotOM.IRJointGussetFlangeLoad.Fx_Upper 2122
RobotOM.IRJointGussetFlangePlate 2102
RobotOM.IRJointGussetFlangePlate.B1 2104
RobotOM.IRJointGussetFlangePlate.B2 2104
RobotOM.IRJointGussetFlangePlate.B3 2104
RobotOM.IRJointGussetFlangePlate.B4 2104
RobotOM.IRJointGussetFlangePlate.CornersType 2105
RobotOM.IRJointGussetFlangePlate.CutH1 2105
RobotOM.IRJointGussetFlangePlate.CutH2 2105
RobotOM.IRJointGussetFlangePlate.CutH3 2106
RobotOM.IRJointGussetFlangePlate.CutH4 2106
RobotOM.IRJointGussetFlangePlate.CutV1 2106
RobotOM.IRJointGussetFlangePlate.CutV2 2107
RobotOM.IRJointGussetFlangePlate.CutV3 2107
RobotOM.IRJointGussetFlangePlate.CutV4 2107
RobotOM.IRJointGussetFlangePlate.EH 2108
RobotOM.IRJointGussetFlangePlate.EV 2108
RobotOM.IRJointGussetFlangePlate.Geight 2108
RobotOM.IRJointGussetFlangePlate.H1 2108
RobotOM.IRJointGussetFlangePlate.H2 2109
RobotOM.IRJointGussetFlangePlate.H3 2109
RobotOM.IRJointGussetFlangePlate.H4 2109
RobotOM.IRJointGussetFlangePlate.HOffset 2109
RobotOM.IRJointGussetFlangePlate.Length 2110
RobotOM.IRJointGussetFlangePlate.Material 2110
RobotOM.IRJointGussetFlangePlate.NewPlateDefinition 2110
RobotOM.IRJointGussetFlangePlate.Thick 2111
RobotOM.IRJointGussetFlangePlate.Type 2111
RobotOM.IRJointGussetFlangePlate.VOffset 2111
RobotOM.IRJointGussetFlangePlateRegularType 2123
RobotOM.IRJointGussetFlangeProfileCutting 2102
RobotOM.IRJointGussetSimple 2068

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRJointGussetSimple.Attachment 2069
RobotOM.IRJointGussetSimple.BoltsClassVector 2070
RobotOM.IRJointGussetSimple.BoltsDimensionNamesVector 2070
RobotOM.IRJointGussetSimple.BoltsDimensionsVector 2070
RobotOM.IRJointGussetSimple.Diagonale 2071
RobotOM.IRJointGussetSimple.FixType 2071
RobotOM.IRJointGussetSimple.GussetPlate 2071
RobotOM.IRJointGussetSimple.ProfilePosition 2071
RobotOM.IRJointGussetSimpleAttachBolts 2080
RobotOM.IRJointGussetSimpleAttachBolts.ClassName 2081
RobotOM.IRJointGussetSimpleAttachBolts.Diameter 2081
RobotOM.IRJointGussetSimpleAttachBolts.Friction 2081
RobotOM.IRJointGussetSimpleAttachBolts.Horizontal 2081
RobotOM.IRJointGussetSimpleAttachBolts.Vertical 2082
RobotOM.IRJointGussetSimpleAttachBoltsHorizontal 2082
RobotOM.IRJointGussetSimpleAttachBoltsHorizontal.DistanceEB 2083
RobotOM.IRJointGussetSimpleAttachBoltsHorizontal.DistanceH1 2083
RobotOM.IRJointGussetSimpleAttachBoltsHorizontal.Rows 2083
RobotOM.IRJointGussetSimpleAttachBoltsHorizontal.Spacing 2083
RobotOM.IRJointGussetSimpleAttachBoltsVertical 2084
RobotOM.IRJointGussetSimpleAttachBoltsVertical.DistanceEB 2084
RobotOM.IRJointGussetSimpleAttachBoltsVertical.DistanceH1 2085
RobotOM.IRJointGussetSimpleAttachBoltsVertical.Rows 2085
RobotOM.IRJointGussetSimpleAttachBoltsVertical.Spacing 2085
RobotOM.IRJointGussetSimpleAttachment 2078
RobotOM.IRJointGussetSimpleAttachment.BoltsAttach 2079
RobotOM.IRJointGussetSimpleAttachment.FixType 2079
RobotOM.IRJointGussetSimpleAttachment.WeldsAttach 2079
RobotOM.IRJointGussetSimpleAttachment.WeldType 2080
RobotOM.IRJointGussetSimpleAttachWelds 2085
RobotOM.IRJointGussetSimpleAttachWelds.ThickEdgeA 2086
RobotOM.IRJointGussetSimpleAttachWelds.ThickEdgeB 2086
RobotOM.IRJointGussetSimpleLoad 2117
RobotOM.IRJointGussetSimpleLoad.Fx 2118
RobotOM.IRJointGussetSimplePlate 2072
RobotOM.IRJointGussetSimplePlate.CornersType 2073
RobotOM.IRJointGussetSimplePlate.DistanceD 2073
RobotOM.IRJointGussetSimplePlate.DistanceEH 2073
RobotOM.IRJointGussetSimplePlate.DistanceEV 2074
RobotOM.IRJointGussetSimplePlate.H1 2074
RobotOM.IRJointGussetSimplePlate.H2 2074
RobotOM.IRJointGussetSimplePlate.H3 2075
RobotOM.IRJointGussetSimplePlate.H4 2075
RobotOM.IRJointGussetSimplePlate.Height 2075
RobotOM.IRJointGussetSimplePlate.Material 2076
RobotOM.IRJointGussetSimplePlate.Thick 2076
RobotOM.IRJointGussetSimplePlate.V1 2076
RobotOM.IRJointGussetSimplePlate.V2 2076
RobotOM.IRJointGussetSimplePlate.V3 2077
RobotOM.IRJointGussetSimplePlate.V4 2077
RobotOM.IRJointGussetSimplePlate.Width 2077
RobotOM.IRJointGussetSimpleProfilePosition 2068
RobotOM.IRJointGussetWeldsDiag 2063
RobotOM.IRJointGussetWeldsDiag.Length1 2064
RobotOM.IRJointGussetWeldsDiag.Length2 2064
RobotOM.IRJointGussetWeldsDiag.ThickCorner 2064
RobotOM.IRJointGussetWeldsDiag.ThickFlange 2065
RobotOM.IRJointKnee 1951
RobotOM.IRJointKnee.Beam 1954
RobotOM.IRJointKnee.Bolts 1954
RobotOM.IRJointKnee.BracketLow 1955
RobotOM.IRJointKnee.BracketUp 1955
RobotOM.IRJointKnee.Column 1955
RobotOM.IRJointKnee.DefComponentMaterial 1955
RobotOM.IRJointKnee.FixType 1956
RobotOM.IRJointKnee.FlanPlateLower 1956
RobotOM.IRJointKnee.FlanPlatePosLower 1956
RobotOM.IRJointKnee.FlanPlatePosUpper 1957
RobotOM.IRJointKnee.FlanPlateUpper 1957
RobotOM.IRJointKnee.IsBolted 1957
RobotOM.IRJointKnee.IsDefComponentMaterialSet 1958
RobotOM.IRJointKnee.KneeType 1958
RobotOM.IRJointKnee.Material 1958
RobotOM.IRJointKnee.MaterPlates 1958
RobotOM.IRJointKnee.Plate 1959
RobotOM.IRJointKnee.PlatePosition 1959
RobotOM.IRJointKnee.PlatePositionLow 1959

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRJointKnee.StiffDiag 1959
RobotOM.IRJointKnee.StiffLow 1960
RobotOM.IRJointKnee.StiffTypeLow 1960
RobotOM.IRJointKnee.StiffTypeUp 1960
RobotOM.IRJointKnee.StiffUp 1960
RobotOM.IRJointKnee.TensionPlateLow 1961
RobotOM.IRJointKnee.TensionPlateUp 1961
RobotOM.IRJointKnee.VStiffLow 1961
RobotOM.IRJointKnee.VStiffUp 1961
RobotOM.IRJointKnee.WebBeamStiffenerVLower 1962
RobotOM.IRJointKnee.WebBeamStiffenerVUpper 1962
RobotOM.IRJointKnee.WebColumnStiffenerHLower 1962
RobotOM.IRJointKnee.WebColumnStiffenerHUpper 1963
RobotOM.IRJointKnee.WebPlate 1963
RobotOM.IRJointKnee.WebStiffType 1963
RobotOM.IRJointKnee.WeldBracketDownFlange 1963
RobotOM.IRJointKnee.WeldBracketUpFlange 1964
RobotOM.IRJointKnee.WeldFlange 1964
RobotOM.IRJointKnee.WeldStiff 1964
RobotOM.IRJointKnee.WeldWeb 1964
RobotOM.IRJointKneeBolts 1939
RobotOM.IRJointKneeBolts.ClassName 1940
RobotOM.IRJointKneeBolts.Cols 1941
RobotOM.IRJointKneeBolts.DiameterName 1941
RobotOM.IRJointKneeBolts.EqualSpac 1941
RobotOM.IRJointKneeBolts.Height1 1942
RobotOM.IRJointKneeBolts.Rows 1942
RobotOM.IRJointKneeBolts.SpacingH 1942
RobotOM.IRJointKneeBolts.SpacingV 1943
RobotOM.IRJointKneeBolts.Symmetry 1943
RobotOM.IRJointKneeBracket 1943
RobotOM.IRJointKneeBracket.Angle 1944
RobotOM.IRJointKneeBracket.AngleExt 1947
RobotOM.IRJointKneeBracket.Exist 1944
RobotOM.IRJointKneeBracket.Height 1945
RobotOM.IRJointKneeBracket.Length 1945
RobotOM.IRJointKneeBracket.Material 1945
RobotOM.IRJointKneeBracket.ThickFlange 1946
RobotOM.IRJointKneeBracket.ThickWeb 1946
RobotOM.IRJointKneeBracket.Width 1946
RobotOM.IRJointKneeDiagonalStiff 1965
RobotOM.IRJointKneeDiagonalStiff.Exist 1966
RobotOM.IRJointKneeDiagonalStiff.Material 1966
RobotOM.IRJointKneeDiagonalStiff.Thick 1966
RobotOM.IRJointKneeDiagonalStiff.Type 1967
RobotOM.IRJointKneeDiagonalStiffType 1965
RobotOM.IRJointKneeFixType 1950
RobotOM.IRJointKneeLoad 1968
RobotOM.IRJointKneeLoad.M 1969
RobotOM.IRJointKneeLoad.N 1969
RobotOM.IRJointKneeLoad.Q 1969
RobotOM.IRJointKneeMaterials 1969
RobotOM.IRJointKneeReinfType 1951
RobotOM.IRJointKneeStiffColumn 1967
RobotOM.IRJointKneeStiffColumn.Thick 1968
RobotOM.IRJointKneeType 1950
RobotOM.IRJointKneeWebPlate 1947
RobotOM.IRJointKneeWebPlate.Bolts 1948
RobotOM.IRJointKneeWebPlate.Height 1948
RobotOM.IRJointKneeWebPlate.Material 1948
RobotOM.IRJointKneeWebPlate.Thick 1949
RobotOM.IRJointKneeWebPlate.Type 1949
RobotOM.IRJointKneeWebPlate.WeldLong 1949
RobotOM.IRJointKneeWebPlate.WeldTran 1949
RobotOM.IRJointKneeWebPlate.Width 1950
RobotOM.IRJointKneeWebStiffType 1965
RobotOM.IRJointLoad 2178
RobotOM.IRJointLoad.Cases 2178
RobotOM.IRJointLoad.Type 2179
RobotOM.IRJointLoadType 2179
RobotOM.IRJointPinnedColumnBase 1971
RobotOM.IRJointPinnedColumnBase.Anchor 1973
RobotOM.IRJointPinnedColumnBase.Base 1973
RobotOM.IRJointPinnedColumnBase.BasePlateMaterial 1973
RobotOM.IRJointPinnedColumnBase.Bearing 1974
RobotOM.IRJointPinnedColumnBase.BearingPlate 1974
RobotOM.IRJointPinnedColumnBase.Bolts 1974
RobotOM.IRJointPinnedColumnBase.FootPlate 1974
RobotOM.IRJointPinnedColumnBase.Materials 1975
RobotOM.IRJointPinnedColumnBase.NodeNumber 1975
RobotOM.IRJointPinnedColumnBase.Profile 1975
RobotOM.IRJointPinnedColumnBase.Square 1975
RobotOM.IRJointPinnedColumnBase.StiffHoriz 1976
RobotOM.IRJointPinnedColumnBase.StiffType 1976

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRJointPinnedColumnBase.StiffVert 1976
RobotOM.IRJointPinnedColumnBase.Washer 1976
RobotOM.IRJointPinnedColumnBase.Wedge 1977
RobotOM.IRJointPinnedColumnBase.Welds 1977
RobotOM.IRJointPinnedColumnBaseStiffType 1971
RobotOM.IRJointPinnedLoad 1977
RobotOM.IRJointPinnedLoad.Nc 1978
RobotOM.IRJointPinnedLoad.Nt 1978
RobotOM.IRJointPinnedLoad.NTy 1978
RobotOM.IRJointPinnedLoad.NTz 1979
RobotOM.IRJointPinnedLoad.Ty 1979
RobotOM.IRJointPinnedLoad.Tz 1979
RobotOM.IRJointPlate 2163
RobotOM.IRJointPlate.AngleExt 2165
RobotOM.IRJointPlate.Exist 2164
RobotOM.IRJointPlate.Length 2164
RobotOM.IRJointPlate.Material 2164
RobotOM.IRJointPlate.Thick 2165
RobotOM.IRJointPlate.Width 2165
RobotOM.IRJointProfile 2174
RobotOM.IRJointProfile.Angle 2175
RobotOM.IRJointProfile.AngleExt 2177
RobotOM.IRJointProfile.BarNumber 2175
RobotOM.IRJointProfile.Edit 2177
RobotOM.IRJointProfile.EditComponent 2177
RobotOM.IRJointProfile.Material 2176
RobotOM.IRJointProfile.Section 2176
RobotOM.IRJointSpreadFootingType 1993
RobotOM.IRJointTube 2042
RobotOM.IRJointTube.DiagLeftLower 2043
RobotOM.IRJointTube.DiagLeftUpper 2044
RobotOM.IRJointTube.DiagRightUpper 2044
RobotOM.IRJointTube.Flange 2044
RobotOM.IRJointTube.PostUpper 2044
RobotOM.IRJointTube.StiffHBracketMaterial 2045
RobotOM.IRJointTube.StiffHoriz 2045
RobotOM.IRJointTube.StiffLateral 2045
RobotOM.IRJointTube.TubeType 2045
RobotOM.IRJointTube.WeldDiag 2046
RobotOM.IRJointTube.WeldStiff 2046
RobotOM.IRJointTubeDiagProfile 2050
RobotOM.IRJointTubeDiagProfile.Angle 2050
RobotOM.IRJointTubeDiagProfile.BarNumber 2051
RobotOM.IRJointTubeDiagProfile.Exist 2051
RobotOM.IRJointTubeDiagProfile.Gap 2051
RobotOM.IRJointTubeDiagProfile.Length 2052
RobotOM.IRJointTubeDiagProfile.Material 2052
RobotOM.IRJointTubeDiagProfile.Overlap 2052
RobotOM.IRJointTubeDiagProfile.Profile 2052
RobotOM.IRJointTubeDiagProfile.Section 2053
RobotOM.IRJointTubeFlangeProfile 2047
RobotOM.IRJointTubeFlangeProfile.BarNumber 2048
RobotOM.IRJointTubeFlangeProfile.Excentr 2048
RobotOM.IRJointTubeFlangeProfile.Exist 2048
RobotOM.IRJointTubeFlangeProfile.Material 2049
RobotOM.IRJointTubeFlangeProfile.Profile 2049
RobotOM.IRJointTubeFlangeProfile.Section 2049
RobotOM.IRJointTubeLoad 2055
RobotOM.IRJointTubeLoad.DiagLeftLowerM 2057
RobotOM.IRJointTubeLoad.DiagLeftLowerN 2057
RobotOM.IRJointTubeLoad.DiagLeftUpperM 2057
RobotOM.IRJointTubeLoad.DiagLeftUpperN 2057
RobotOM.IRJointTubeLoad.DiagRightUpperM 2058
RobotOM.IRJointTubeLoad.DiagRightUpperN 2058
RobotOM.IRJointTubeLoad.FlangeM 2058
RobotOM.IRJointTubeLoad.FlangeN 2058
RobotOM.IRJointTubeLoad.FlangeQ 2059
RobotOM.IRJointTubeLoad.PostUpperN 2059
RobotOM.IRJointTubePostProfile 2053
RobotOM.IRJointTubePostProfile.BarNumber 2054
RobotOM.IRJointTubePostProfile.Exist 2054
RobotOM.IRJointTubePostProfile.Length 2054
RobotOM.IRJointTubePostProfile.Material 2055
RobotOM.IRJointTubePostProfile.Profile 2055
RobotOM.IRJointTubePostProfile.Section 2055
RobotOM.IRJointTubeProfileType 2047
RobotOM.IRJointTubeType 2046
RobotOM.IRJointWebFlangeRelativePos 2159
RobotOM.IRJointWebType 2169
RobotOM.IRJointWedge 2011
RobotOM.IRJointWedge.Length 2011
RobotOM.IRJointWedge.Profile 2012
RobotOM.IRJointWedge.Thick 2012
RobotOM.IRJointWedge.Type 2012

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRJointWedge.Width 2012
RobotOM.IRJointWedge.XTypeMaterial 2013
RobotOM.IRJointWedgeType 2013
RobotOM.IRJointWeld 2162
RobotOM.IRJointWeld.Strength 2162
RobotOM.IRJointWeld.Thick 2163
RobotOM.IRJointWithAngles 2037
RobotOM.IRJointWithAngles.Angle 2039
RobotOM.IRJointWithAngles.BeamBolts 2039
RobotOM.IRJointWithAngles.BeamCut 2039
RobotOM.IRJointWithAngles.BeamProfile 2039
RobotOM.IRJointWithAngles.ColumnBolts 2040
RobotOM.IRJointWithAngles.ColumnProfile 2040
RobotOM.IRJointWithAngles.Distance 2040
RobotOM.IRobotActiveModelType 1360
RobotOM.IRobotAddIn 1548
RobotOM.IRobotAddIn.Connect 1550
RobotOM.IRobotAddIn.Disconnect 1550
RobotOM.IRobotAddIn.DoCommand 1551
RobotOM.IRobotAddIn.GetExpectedVersion 1551
RobotOM.IRobotAddIn.InstallCommands 1551
RobotOM.IRobotAddInMngr 1551
RobotOM.IRobotAddInMngr.InstallCommand 1552
RobotOM.IRobotAddInRegistrar 1556
RobotOM.IRobotAddInRegistrar.Guid 1557
RobotOM.IRobotAddInRegistrar.InstallMenu 1559
RobotOM.IRobotAddInRegistrar.ProductName 1558
RobotOM.IRobotAddInRegistrar.ProgId 1558
RobotOM.IRobotAddInRegistrar.ProviderName 1558
RobotOM.IRobotAddInRegistrar.Register 1559
RobotOM.IRobotAddInRegistrar.Unregister 1559
RobotOM.IRobotAdvancedResultServer 924
RobotOM.IRobotAdvancedResultServer.Eigenvalues 925
RobotOM.IRobotAdvancedResultServer.Eigenvectors 925
RobotOM.IRobotAdvancedResultServer.FootfallValue 927
RobotOM.IRobotAdvancedResultServer.FRF 925
RobotOM.IRobotAdvancedResultServer.MassSum 926
RobotOM.IRobotAdvancedResultServer.SpectralCoeffs 926
RobotOM.IRobotAdvancedResultServer.TimeHistory 926
RobotOM.IRobotAdvancedSupportData 517
RobotOM.IRobotAdvancedSupportData.B 518
RobotOM.IRobotAdvancedSupportData.D 518
RobotOM.IRobotAdvancedSupportData.EquivalentElasticity 519
RobotOM.IRobotAdvancedSupportData.H 519
RobotOM.IRobotAdvancedSupportData.IsEquivalentElasticity 519
RobotOM.IRobotAdvancedSupportData.Type 519
RobotOM.IRobotAdvancedSupportType 517
RobotOM.IRobotApplication 1535
RobotOM.IRobotApplication.CmpntFactory 1537
RobotOM.IRobotApplication.Interactive 1537
RobotOM.IRobotApplication.Is360 1537
RobotOM.IRobotApplication.Kernel 1538
RobotOM.IRobotApplication.LicenseCheckEntitlement 1540
RobotOM.IRobotApplication.Preferences 1538
RobotOM.IRobotApplication.ProgramVersion 1538
RobotOM.IRobotApplication.Project 1538
RobotOM.IRobotApplication.Quit 1540
RobotOM.IRobotApplication.UserControl 1539
RobotOM.IRobotApplication.Version 1539
RobotOM.IRobotApplication.Visible 1539
RobotOM.IRobotApplication.Window 1540
RobotOM.IRobotBackground 1334
RobotOM.IRobotBackground.Color 1335
RobotOM.IRobotBackground.FilePath 1335
RobotOM.IRobotBackground.InsertParams 1335
RobotOM.IRobotBackground.IsOn 1336
RobotOM.IRobotBackground.Name 1336
RobotOM.IRobotBackground.Number 1336
RobotOM.IRobotBackground.Save 1337
RobotOM.IRobotBackground.VisibilityRange 1337
RobotOM.IRobotBackgroundInsertParams 1330
RobotOM.IRobotBackgroundInsertParams.InsertionPoint 1331
RobotOM.IRobotBackgroundInsertParams.Layers 1331
RobotOM.IRobotBackgroundInsertParams.Mirror 1332
RobotOM.IRobotBackgroundInsertParams.Plane 1332
RobotOM.IRobotBackgroundInsertParams.PositionOnNormal Axis 1332
RobotOM.IRobotBackgroundInsertParams.ReferencePoint 1333
RobotOM.IRobotBackgroundInsertParams.RotationAngle 1333
RobotOM.IRobotBackgroundInsertParams.ScalingFactor 1333
RobotOM.IRobotBackgroundInsertParams.Units 1333
RobotOM.IRobotBackgroundLayers 1327

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRobotBackgroundLayers.Count 1328
RobotOM.IRobotBackgroundLayers.FindName 1329
RobotOM.IRobotBackgroundLayers.Get 1329
RobotOM.IRobotBackgroundLayers.IsImported 1329
RobotOM.IRobotBackgroundLayers.SetImported 1330
RobotOM.IRobotBackgroundServer 1339
RobotOM.IRobotBackgroundServer.Create 1340
RobotOM.IRobotBackgroundServer.Get 1340
RobotOM.IRobotBackgroundServer.GetAllNumbers 1341
RobotOM.IRobotBackgroundServer.Remove 1341
RobotOM.IRobotBackgroundServer.RemoveAll 1341
RobotOM.IRobotBackgroundVisibilityRange 1338
RobotOM.IRobotBackgroundVisibilityRange.FromPos 1338
RobotOM.IRobotBackgroundVisibilityRange.ToPos 1339
RobotOM.IRobotBackgroundVisibilityRange.Type 1339
RobotOM.IRobotBackgroundVisibilityRangeType 1337
RobotOM.IRobotBar 624
RobotOM.IRobotBar.AnalyzeTTMethod 627
RobotOM.IRobotBar.CalcGamma 633
RobotOM.IRobotBar.ChangeOrientation 634
RobotOM.IRobotBar.Elements 627
RobotOM.IRobotBar.End 627
RobotOM.IRobotBar.EndNode 628
RobotOM.IRobotBar.Gamma 628
RobotOM.IRobotBar.GetElemsData 634
RobotOM.IRobotBar.GetLCS 634
RobotOM.IRobotBar.GetSimpleBars 635
RobotOM.IRobotBar.InactiveBar 628
RobotOM.IRobotBar.IsSuperBar 629
RobotOM.IRobotBar.Length 629
RobotOM.IRobotBar.Name 629
RobotOM.IRobotBar.NameTemplate 629
RobotOM.IRobotBar.ReversedOffset 630
RobotOM.IRobotBar.ReversedRelease 630
RobotOM.IRobotBar.ReversedSection 630
RobotOM.IRobotBar.SetOffset 635
RobotOM.IRobotBar.SetOrientation 635
RobotOM.IRobotBar.SetSection 636
RobotOM.IRobotBar.ShearForces 631
RobotOM.IRobotBar.Start 631
RobotOM.IRobotBar.StartNode 631
RobotOM.IRobotBar.StructuralType 632
RobotOM.IRobotBar.TensionCompression 632
RobotOM.IRobotBar.TrussBar 632
RobotOM.IRobotBar.UniqueId 633
RobotOM.IRobotBarBucklingData 897
RobotOM.IRobotBarBucklingData.BuckLengthY 898
RobotOM.IRobotBarBucklingData.BuckLengthZ 898
RobotOM.IRobotBarBucklingData.CriticalCoef 898
RobotOM.IRobotBarBucklingData.CriticalForce 898
RobotOM.IRobotBarBucklingData.SlendY 899
RobotOM.IRobotBarBucklingData.SlendZ 899
RobotOM.IRobotBarBucklingServer 881
RobotOM.IRobotBarBucklingServer.CriticalCoef 881
RobotOM.IRobotBarBucklingServer.EigenValue 882
RobotOM.IRobotBarCableAssemblingParamType 541
RobotOM.IRobotBarCableData 539
RobotOM.IRobotBarCableData.AssemblingParam 540
RobotOM.IRobotBarCableData.AssemblingParamValue 540
RobotOM.IRobotBarCableData.MaterialName 540
RobotOM.IRobotBarCableData.SectionAX 541
RobotOM.IRobotBarDeadRecordValues 44
RobotOM.IRobotBarDeflectionData 894
RobotOM.IRobotBarDeflectionData.PosUX 895
RobotOM.IRobotBarDeflectionData.PosUY 896
RobotOM.IRobotBarDeflectionData.PosUZ 896
RobotOM.IRobotBarDeflectionData.UX 896
RobotOM.IRobotBarDeflectionData.UY 896
RobotOM.IRobotBarDeflectionData.UZ 896
RobotOM.IRobotBarDeflectionServer 875
RobotOM.IRobotBarDeflectionServer.DynCombMaxValue 876
RobotOM.IRobotBarDeflectionServer.DynCombValue 877
RobotOM.IRobotBarDeflectionServer.DynMaxValue 877
RobotOM.IRobotBarDeflectionServer.DynValue 877
RobotOM.IRobotBarDeflectionServer.MaxValue 878
RobotOM.IRobotBarDeflectionServer.MaxValueEx 878
RobotOM.IRobotBarDeflectionServer.Value 878
RobotOM.IRobotBarDeflectionServer.ValueEx 878
RobotOM.IRobotBarDilatationRecordValues 41
RobotOM.IRobotBarDisplacementServer 882
RobotOM.IRobotBarDisplacementServer.DynCombValue 883
RobotOM.IRobotBarDisplacementServer.DynValue 883
RobotOM.IRobotBarDisplacementServer.Value 884
RobotOM.IRobotBarElasticGroundData 542

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRobotBarElasticGroundData.GetOneDir 544
RobotOM.IRobotBarElasticGroundData.HX 543
RobotOM.IRobotBarElasticGroundData.KY 543
RobotOM.IRobotBarElasticGroundData.KZ 544
RobotOM.IRobotBarElasticGroundData.SetOneDir 544
RobotOM.IRobotBarElement 652
RobotOM.IRobotBarElement.EndNode 653
RobotOM.IRobotBarElement.Inactive 653
RobotOM.IRobotBarElement.IsCalc 653
RobotOM.IRobotBarElement.Number 653
RobotOM.IRobotBarElement.StartNode 654
RobotOM.IRobotBarElementData 654
RobotOM.IRobotBarElementData.Bar 655
RobotOM.IRobotBarElementData.GetEndNode 656
RobotOM.IRobotBarElementData.GetStartNode 656
RobotOM.IRobotBarElementData.Number 655
RobotOM.IRobotBarElementDataSet 656
RobotOM.IRobotBarElementDataSet.BarCount 657
RobotOM.IRobotBarElementDataSet.ElemCount 658
RobotOM.IRobotBarElementDataSet.GetBarNumber 658
RobotOM.IRobotBarElementDataSet.GetElem 658
RobotOM.IRobotBarElementDataSet.GetElemCountForBar 659
RobotOM.IRobotBarElementDataSet.GetElemForBar 659
RobotOM.IRobotBarEnd 648
RobotOM.IRobotBarEnd.GetLabel 649
RobotOM.IRobotBarEnd.GetLabelName 650
RobotOM.IRobotBarEnd.GetLabels 650
RobotOM.IRobotBarEnd.GetOffsetValue 650
RobotOM.IRobotBarEnd.HasLabel 650
RobotOM.IRobotBarEnd.Node 649
RobotOM.IRobotBarEnd.RemoveLabel 651
RobotOM.IRobotBarEnd.SetLabel 651
RobotOM.IRobotBarEndBracketData 621
RobotOM.IRobotBarEndBracketData.GetValue 622
RobotOM.IRobotBarEndBracketData.IsValueRelative 623
RobotOM.IRobotBarEndBracketData.SetValue 623
RobotOM.IRobotBarEndBracketData.Type 622
RobotOM.IRobotBarEndBracketDataValue 621
RobotOM.IRobotBarEndBracketType 621
RobotOM.IRobotBarEndOffsetData 619
RobotOM.IRobotBarEndOffsetData.MemberLength 620
RobotOM.IRobotBarEndOffsetData.UX 620
RobotOM.IRobotBarEndOffsetData.UY 620
RobotOM.IRobotBarEndOffsetData.UZ 620
RobotOM.IRobotBarEndReleaseData 582
RobotOM.IRobotBarEndReleaseData.AX 583
RobotOM.IRobotBarEndReleaseData.AY 584
RobotOM.IRobotBarEndReleaseData.AZ 584
RobotOM.IRobotBarEndReleaseData.BX 584
RobotOM.IRobotBarEndReleaseData.BY 585
RobotOM.IRobotBarEndReleaseData.BZ 585
RobotOM.IRobotBarEndReleaseData.HX 585
RobotOM.IRobotBarEndReleaseData.HY 585
RobotOM.IRobotBarEndReleaseData.HZ 586
RobotOM.IRobotBarEndReleaseData.KX 586
RobotOM.IRobotBarEndReleaseData.KY 586
RobotOM.IRobotBarEndReleaseData.KZ 586
RobotOM.IRobotBarEndReleaseData.NonlinearModel 587
RobotOM.IRobotBarEndReleaseData.RX 587
RobotOM.IRobotBarEndReleaseData.RY 587
RobotOM.IRobotBarEndReleaseData.RZ 587
RobotOM.IRobotBarEndReleaseData.UX 588
RobotOM.IRobotBarEndReleaseData.UY 588
RobotOM.IRobotBarEndReleaseData.UZ 588
RobotOM.IRobotBarEndReleaseValue 580
RobotOM.IRobotBarForceConcentrateRecordValues 41
RobotOM.IRobotBarForceData 888
RobotOM.IRobotBarForceData.FX 889
RobotOM.IRobotBarForceData.FY 890
RobotOM.IRobotBarForceData.FZ 890
RobotOM.IRobotBarForceData.KY 890
RobotOM.IRobotBarForceData.KYAvailable 890
RobotOM.IRobotBarForceData.KZ 891
RobotOM.IRobotBarForceData.KZAvailable 891
RobotOM.IRobotBarForceData.MX 891
RobotOM.IRobotBarForceData.MY 891
RobotOM.IRobotBarForceData.MZ 891
RobotOM.IRobotBarForceServer 872
RobotOM.IRobotBarForceServer.DynCombValue 873
RobotOM.IRobotBarForceServer.DynCombValueByNPoints 873
RobotOM.IRobotBarForceServer.DynValue 873
RobotOM.IRobotBarForceServer.DynValueByNPoints 874
RobotOM.IRobotBarForceServer.Value 874
RobotOM.IRobotBarForceServer.ValueByNPoints 874

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRobotBarForceServer.ValueByNPointsEx 875
RobotOM.IRobotBarForceServer.ValueEx 875
RobotOM.IRobotBarGeoImperfectionsAxis 592
RobotOM.IRobotBarGeoImperfectionsData 589
RobotOM.IRobotBarGeoImperfectionsData.GetBucklingCoeff 590
RobotOM.IRobotBarGeoImperfectionsData.GetUser 590
RobotOM.IRobotBarGeoImperfectionsData.IsAutomatic 590
RobotOM.IRobotBarGeoImperfectionsData.IsMinus 591
RobotOM.IRobotBarGeoImperfectionsData.SetAutomatic 591
RobotOM.IRobotBarGeoImperfectionsData.SetMinus 591
RobotOM.IRobotBarGeoImperfectionsData.SetUser 592
RobotOM.IRobotBarIntersectRelationship 2222
RobotOM.IRobotBarMomentDistributedRecordValues 42
RobotOM.IRobotBarOffsetAutoPosition 623
RobotOM.IRobotBarOffsetData 617
RobotOM.IRobotBarOffsetData.AxisOffset 617
RobotOM.IRobotBarOffsetData.CoordinateSystem 618
RobotOM.IRobotBarOffsetData.End 618
RobotOM.IRobotBarOffsetData.ObjectNumber 618
RobotOM.IRobotBarOffsetData.Position 618
RobotOM.IRobotBarOffsetData.Start 619
RobotOM.IRobotBarOffsetMemberLength 624
RobotOM.IRobotBarReleaseData 581
RobotOM.IRobotBarReleaseData.EndNode 581
RobotOM.IRobotBarReleaseData.StartNode 582
RobotOM.IRobotBarResultServer 869
RobotOM.IRobotBarResultServer.Buckling 870
RobotOM.IRobotBarResultServer.Deflections 870
RobotOM.IRobotBarResultServer.Displacements 870
RobotOM.IRobotBarResultServer.Forces 871
RobotOM.IRobotBarResultServer.GeoImperfections 871
RobotOM.IRobotBarResultServer.Stresses 871
RobotOM.IRobotBarSectionComplexData 572
RobotOM.IRobotBarSectionComplexData.B 573
RobotOM.IRobotBarSectionComplexData.Count 574
RobotOM.IRobotBarSectionComplexData.D 574
RobotOM.IRobotBarSectionComplexData.Get 574
RobotOM.IRobotBarSectionComplexData.GetShape 575
RobotOM.IRobotBarSectionComplexData.GetValue 575
RobotOM.IRobotBarSectionComplexData.Set 575
RobotOM.IRobotBarSectionComponentShape 576
RobotOM.IRobotBarSectionConcreteCutsPosition 552
RobotOM.IRobotBarSectionConcreteData 565
RobotOM.IRobotBarSectionConcreteData.BeamCutsPosition 566
RobotOM.IRobotBarSectionConcreteData.CalcGeometry 567
RobotOM.IRobotBarSectionConcreteData.GetReduction 567
RobotOM.IRobotBarSectionConcreteData.GetTapered 568
RobotOM.IRobotBarSectionConcreteData.GetValue 568
RobotOM.IRobotBarSectionConcreteData.IsBeam 566
RobotOM.IRobotBarSectionConcreteData.IsColumn 566
RobotOM.IRobotBarSectionConcreteData.SetReduction 568
RobotOM.IRobotBarSectionConcreteData.SetTapered 569
RobotOM.IRobotBarSectionConcreteData.SetValue 569
RobotOM.IRobotBarSectionConcreteDataValue 551
RobotOM.IRobotBarSectionData 552
RobotOM.IRobotBarSectionData.AliasCount 555
RobotOM.IRobotBarSectionData.CalcNonstdGeometry 559
RobotOM.IRobotBarSectionData.Concrete 555
RobotOM.IRobotBarSectionData.CreateNonstd 559
RobotOM.IRobotBarSectionData.DrawSymbol 560
RobotOM.IRobotBarSectionData.ElasticParams 555
RobotOM.IRobotBarSectionData.FindAlias 560
RobotOM.IRobotBarSectionData.FindNonstd 560
RobotOM.IRobotBarSectionData.GetAlias 561
RobotOM.IRobotBarSectionData.GetAliasEx 561
RobotOM.IRobotBarSectionData.GetNonstd 561
RobotOM.IRobotBarSectionData.GetValue 562
RobotOM.IRobotBarSectionData.IsConcrete 555
RobotOM.IRobotBarSectionData.IsJoist 556
RobotOM.IRobotBarSectionData.IsSpecial 556
RobotOM.IRobotBarSectionData.LoadFromDBase 562
RobotOM.IRobotBarSectionData.LoadFromDBase2 562
RobotOM.IRobotBarSectionData.MaterialName 556
RobotOM.IRobotBarSectionData.Members 557
RobotOM.IRobotBarSectionData.Name 557
RobotOM.IRobotBarSectionData.NonstdCount 557
RobotOM.IRobotBarSectionData.RemoveNonstd 563
RobotOM.IRobotBarSectionData.SetValue 563
RobotOM.IRobotBarSectionData.ShapeType 558
RobotOM.IRobotBarSectionData.Special 558
RobotOM.IRobotBarSectionData.Type 558
RobotOM.IRobotBarSectionDataValue 549
RobotOM.IRobotBarSectionElasticParams 569
RobotOM.IRobotBarSectionElasticParams.Active 570

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRobotBarSectionElasticParams.L1 571
RobotOM.IRobotBarSectionElasticParams.L2 571
RobotOM.IRobotBarSectionElasticParams.MaterialModel 571
RobotOM.IRobotBarSectionElasticParams.N 571
RobotOM.IRobotBarSectionElasticParams.N1 572
RobotOM.IRobotBarSectionElasticParams.N2 572
RobotOM.IRobotBarSectionNonstdData 563
RobotOM.IRobotBarSectionNonstdData.GetValue 564
RobotOM.IRobotBarSectionNonstdData.Position 564
RobotOM.IRobotBarSectionNonstdData.SetValue 564
RobotOM.IRobotBarSectionNonstdDataValue 550
RobotOM.IRobotBarSectionQuantitySurvey 1082
RobotOM.IRobotBarSectionQuantitySurvey.Count 1083
RobotOM.IRobotBarSectionQuantitySurvey.GetLength 1083
RobotOM.IRobotBarSectionQuantitySurvey.GetName 1084
RobotOM.IRobotBarSectionQuantitySurvey.GetPaintArea 1084
RobotOM.IRobotBarSectionQuantitySurvey.GetUnitWeight 1084
RobotOM.IRobotBarSectionQuantitySurvey.GetVolume 1085
RobotOM.IRobotBarSectionQuantitySurvey.GetWeight 1085
RobotOM.IRobotBarSectionShapeType 546
RobotOM.IRobotBarSectionSpecialData 576
RobotOM.IRobotBarSectionSpecialData.DbFullName 577
RobotOM.IRobotBarSectionSpecialData.DbName 578
RobotOM.IRobotBarSectionSpecialData.GetValue 578
RobotOM.IRobotBarSectionSpecialData.Section 578
RobotOM.IRobotBarSectionSpecialData.SetValue 579
RobotOM.IRobotBarSectionSpecialDataValue 579
RobotOM.IRobotBarSectionType 550
RobotOM.IRobotBarServer 636
RobotOM.IRobotBarServer.BeginMultiOperation 640
RobotOM.IRobotBarServer.CalcGamma 640
RobotOM.IRobotBarServer.ChangeOrientation 641
RobotOM.IRobotBarServer.Create 641
RobotOM.IRobotBarServer.CreateSuperBar 641
RobotOM.IRobotBarServer.EndMultiOperation 642
RobotOM.IRobotBarServer.FindWithId 642
RobotOM.IRobotBarServer.FreeNumber 638
RobotOM.IRobotBarServer.GetAnalyzeTTMethodEnabled 642
RobotOM.IRobotBarServer.GetElemsData 642
RobotOM.IRobotBarServer.GetLCS 643
RobotOM.IRobotBarServer.GetName 643
RobotOM.IRobotBarServer.GetNameTemplate 643
RobotOM.IRobotBarServer.GetStructuralType 644
RobotOM.IRobotBarServer.GetUniqueId 644
RobotOM.IRobotBarServer.IsInactive 644
RobotOM.IRobotBarServer.IsTrussBar 645
RobotOM.IRobotBarServer.NonlinearHingeModels 638
RobotOM.IRobotBarServer.NonlinearHinges 639
RobotOM.IRobotBarServer.SetAnalyzeTTMethod 645
RobotOM.IRobotBarServer.SetInactive 645
RobotOM.IRobotBarServer.SetNameTemplate 646
RobotOM.IRobotBarServer.SetOrientation 646
RobotOM.IRobotBarServer.SetShearForces 646
RobotOM.IRobotBarServer.SetStructuralType 647
RobotOM.IRobotBarServer.SetTensionCompression 647
RobotOM.IRobotBarServer.SetTrussBar 647
RobotOM.IRobotBarServer.Update 647
RobotOM.IRobotBarStressData 892
RobotOM.IRobotBarStressData.FXSX 893
RobotOM.IRobotBarStressData.ShearY 893
RobotOM.IRobotBarStressData.ShearZ 893
RobotOM.IRobotBarStressData.Smax 893
RobotOM.IRobotBarStressData.SmaxMY 893
RobotOM.IRobotBarStressData.SmaxMZ 894
RobotOM.IRobotBarStressData.Smin 894
RobotOM.IRobotBarStressData.SminMY 894
RobotOM.IRobotBarStressData.SminMZ 894
RobotOM.IRobotBarStressData.Torsion 894
RobotOM.IRobotBarStressServer 879
RobotOM.IRobotBarStressServer.DynCombValue 880
RobotOM.IRobotBarStressServer.DynValue 880
RobotOM.IRobotBarStressServer.Value 880
RobotOM.IRobotBarStressServer.ValueEx 880
RobotOM.IRobotBarTensionCompression 651
RobotOM.IRobotBarThermalRecordValues 43
RobotOM.IRobotBarTrapezoidaleRecordValues 43
RobotOM.IRobotBarUniformRecordValues 42
RobotOM.IRobotBucklingAnalysisMethod 363
RobotOM.IRobotBucklingAnalysisParams 359
RobotOM.IRobotBucklingAnalysisParams.Increment 361
RobotOM.IRobotBucklingAnalysisParams.IsNonlinear 361
RobotOM.IRobotBucklingAnalysisParams.IterationsCount 361
RobotOM.IRobotBucklingAnalysisParams.Method 361

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRobotBucklingAnalysisParams.ModesCount 362

RobotOM.IRobotBucklingAnalysisParams.NonlinearParams 362

RobotOM.IRobotBucklingAnalysisParams.Shift 362

RobotOM.IRobotBucklingAnalysisParams.SturmVerification 363

RobotOM.IRobotBucklingAnalysisParams.Tolerance 363

RobotOM.IRobotBucklingDeformationParams 1589

RobotOM.IRobotBucklingDeformationParams.CaseNumber 1590

RobotOM.IRobotBucklingDeformationParams.GetModeCoeff 1591

RobotOM.IRobotBucklingDeformationParams.MaxDisplacement 1590

RobotOM.IRobotBucklingDeformationParams.OmitCaseForDeformations 1590

RobotOM.IRobotBucklingDeformationParams.SetModeCoeff 1591

RobotOM.IRobotCalcEngine 1561

RobotOM.IRobotCalcEngine.AnalysisParams 1562

RobotOM.IRobotCalcEngine.AutoFreezeResults 1562

RobotOM.IRobotCalcEngine.AutoGenerateModel 1563

RobotOM.IRobotCalcEngine.BucklingDeformation 1563

RobotOM.IRobotCalcEngine.Calculate 1566

RobotOM.IRobotCalcEngine.CalculateEx 1566

RobotOM.IRobotCalcEngine.DAM 1563

RobotOM.IRobotCalcEngine.GenerateModel 1566

RobotOM.IRobotCalcEngine.GenerationParams 1564

RobotOM.IRobotCalcEngine.SaveResultsInExternalFile 1564

RobotOM.IRobotCalcEngine.SeismicResultsSaveParams 1564

RobotOM.IRobotCalcEngine.StatusWindowParent 1565

RobotOM.IRobotCalcEngine.StopCalculation 1567

RobotOM.IRobotCalcEngine.UseStatusWindow 1565

RobotOM.IRobotCalcEngineEvents 1591

RobotOM.IRobotCalcEngineEvents.CalcMessage 1592

RobotOM.IRobotCalcMessageSeverityLevel 1593

RobotOM.IRobotCalcMessageSource 1593

RobotOM.IRobotCalculationMode 1594

RobotOM.IRobotCalculationModelCoherence 1326

RobotOM.IRobotCalculationResume 1034

RobotOM.IRobotCalculationResume.DiagonalStiffnessMatrixMax 1035

RobotOM.IRobotCalculationResume.DiagonalStiffnessMatrixMin 1035

RobotOM.IRobotCalculationResume.DiagonalStiffnessMatrixPrecision 1035

RobotOM.IRobotCalculationResume.GetEnergy 1036

RobotOM.IRobotCalculationResume.GetEnergyPrecision 1036

RobotOM.IRobotCalculationStatus 1593

RobotOM.IRobotCalculationsType 1548

RobotOM.IRobotCase 408

RobotOM.IRobotCase.AnalizeType 408

RobotOM.IRobotCase.Name 409

RobotOM.IRobotCase.Nature 409

RobotOM.IRobotCase.Number 409

RobotOM.IRobotCase.Type 409

RobotOM.IRobotCaseAnalizeType 410

RobotOM.IRobotCaseAnalysisModesFilter 435

RobotOM.IRobotCaseAnalysisModesFilter.MassPercentage 436

RobotOM.IRobotCaseAnalysisModesFilter.Modes 436

RobotOM.IRobotCaseAnalysisModesFilter.Type 437

RobotOM.IRobotCaseAnalysisModesFilterType 437

RobotOM.IRobotCaseCollection 435

RobotOM.IRobotCaseCombination 426

RobotOM.IRobotCaseCombination.CaseFactors 427

RobotOM.IRobotCaseCombination.CombinationType 427

RobotOM.IRobotCaseCombination.GetAnalysisParams 430

RobotOM.IRobotCaseCombination.IsAuxiliary 427

RobotOM.IRobotCaseCombination.Label 428

RobotOM.IRobotCaseCombination.NatureName 428

RobotOM.IRobotCaseCombination.Quadratic 428

RobotOM.IRobotCaseCombination.SeismicType 429

RobotOM.IRobotCaseCombination.SetAnalysisParams 430

RobotOM.IRobotCaseCombination.SetNatureExt 430

RobotOM.IRobotCaseCombination.Signed 429

RobotOM.IRobotCaseCombination.UniqueId 429

RobotOM.IRobotCaseFactor 434

RobotOM.IRobotCaseFactor.CaseNumber 434

RobotOM.IRobotCaseFactor.Factor 434

RobotOM.IRobotCaseFactorMngr 431

RobotOM.IRobotCaseFactorMngr.Count 431

RobotOM.IRobotCaseFactorMngr.Delete 432

RobotOM.IRobotCaseFactorMngr.Get 432

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRobotCaseFactorMngr.New 432
RobotOM.IRobotCaseNature 410
RobotOM.IRobotCaseRelatedValueType 446
RobotOM.IRobotCaseServer 411
RobotOM.IRobotCaseServer.BeginMultiOperation 415
RobotOM.IRobotCaseServer.CodeCmbEngine 413
RobotOM.IRobotCaseServer.CreateCombination 416
RobotOM.IRobotCaseServer.CreateMobile 416
RobotOM.IRobotCaseServer.CreateSimple 416
RobotOM.IRobotCaseServer.Delete 417
RobotOM.IRobotCaseServer.DeleteMany 417
RobotOM.IRobotCaseServer.EndMultiOperation 417
RobotOM.IRobotCaseServer.Exist 417
RobotOM.IRobotCaseServer.FindWithId 418
RobotOM.IRobotCaseServer.FreeNumber 413
RobotOM.IRobotCaseServer.Get 418
RobotOM.IRobotCaseServer.GetAll 418
RobotOM.IRobotCaseServer.GetCmpntCount 419
RobotOM.IRobotCaseServer.GetMany 419
RobotOM.IRobotCaseServer.GetRelatedValue 419
RobotOM.IRobotCaseServer.GetUniqueId 420
RobotOM.IRobotCaseServer.IsAuxiliary 420
RobotOM.IRobotCaseServer.QCmbTau 413
RobotOM.IRobotCaseServer.SELFSeismicEngine 413
RobotOM.IRobotCaseServer.SetAuxiliary 420
RobotOM.IRobotCaseServer.SnowWindEngine 414
RobotOM.IRobotCaseServer.TimeHistoryFunctions 414
RobotOM.IRobotCaseServer.WindLoadsSimulationEngine 414
RobotOM.IRobotCaseType 433
RobotOM.IRobotCladdingData 614
RobotOM.IRobotCladdingData.Method 615
RobotOM.IRobotCladdingData.Type 615
RobotOM.IRobotCladdingMethod 616
RobotOM.IRobotCladdingType 616
RobotOM.IRobotCmdInfo 1554
RobotOM.IRobotCmdInfo.Id 1555
RobotOM.IRobotCmdInfo.MenuChecked 1555
RobotOM.IRobotCmdInfo.MenuEnabled 1556
RobotOM.IRobotCmdInfo.Name 1556
RobotOM.IRobotCmdList 1552
RobotOM.IRobotCmdList.Count 1553
RobotOM.IRobotCmdList.Get 1554
RobotOM.IRobotCmdList.New 1554
RobotOM.IRobotCodeCmbActionCoeffType 163
RobotOM.IRobotCodeCmbActionServer 160
RobotOM.IRobotCodeCmbActionServer.Count 161
RobotOM.IRobotCodeCmbActionServer.GetCoeff 161
RobotOM.IRobotCodeCmbActionServer.GetName 162
RobotOM.IRobotCodeCmbActionServer.GetNature 162
RobotOM.IRobotCodeCmbActionServer.New 162
RobotOM.IRobotCodeCmbActionServer.Remove 162
RobotOM.IRobotCodeCmbActionServer.SetCoeff 163
RobotOM.IRobotCodeCmbActionServer.SetName 163
RobotOM.IRobotCodeCmbActionServer.SetNature 163
RobotOM.IRobotCodeCmbActiveCaseInfo 186
RobotOM.IRobotCodeCmbActiveCaseInfo.CaseNature 187
RobotOM.IRobotCodeCmbActiveCaseInfo.CaseNumber 187
RobotOM.IRobotCodeCmbActiveCaseInfo.Coefficient 187
RobotOM.IRobotCodeCmbActiveCaseInfo.GroupNumber 188
RobotOM.IRobotCodeCmbActiveCaseInfo.IsSelected 188
RobotOM.IRobotCodeCmbCombPartType 167
RobotOM.IRobotCodeCmbCombs 164
RobotOM.IRobotCodeCmbCombs.Count 165
RobotOM.IRobotCodeCmbCombs.Get 166
RobotOM.IRobotCodeCmbCombs.New 166
RobotOM.IRobotCodeCmbCombs.Remove 166
RobotOM.IRobotCodeCmbCombs.Set 166
RobotOM.IRobotCodeCmbCombType 167
RobotOM.IRobotCodeCmbComponent 155
RobotOM.IRobotCodeCmbComponent.Count 156
RobotOM.IRobotCodeCmbComponent.Get 156
RobotOM.IRobotCodeCmbComponentMngr 153
RobotOM.IRobotCodeCmbComponentMngr.Count 154
RobotOM.IRobotCodeCmbComponentMngr.Get 154
RobotOM.IRobotCodeCmbDecidingValueType 174
RobotOM.IRobotCodeCmbFactor 154
RobotOM.IRobotCodeCmbFactor.CaseNumber 155
RobotOM.IRobotCodeCmbFactor.Factor 155
RobotOM.IRobotCodeCmbFlag 157
RobotOM.IRobotCodeCmbGenerationParams 168
RobotOM.IRobotCodeCmbGenerationParams.ActiveCases 169
RobotOM.IRobotCodeCmbGenerationParams.AllBars 169
RobotOM.IRobotCodeCmbGenerationParams.AllNodes 170
RobotOM.IRobotCodeCmbGenerationParams.BarSel 170

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRobotCodeCmbGenerationParams.ExtremalSnowFactor 170
RobotOM.IRobotCodeCmbGenerationParams.GenType 171
RobotOM.IRobotCodeCmbGenerationParams.Groups 171
RobotOM.IRobotCodeCmbGenerationParams.IsCombinationSelected 173
RobotOM.IRobotCodeCmbGenerationParams.IsCombinationTypeSelected 173
RobotOM.IRobotCodeCmbGenerationParams.IsDecidingValueSelected 173
RobotOM.IRobotCodeCmbGenerationParams.NodeSel 171
RobotOM.IRobotCodeCmbGenerationParams.PointsOnBar 171
RobotOM.IRobotCodeCmbGenerationParams.Regulations 172
RobotOM.IRobotCodeCmbGenerationParams.Relations 172
RobotOM.IRobotCodeCmbGenerationParams.SelectCombination 173
RobotOM.IRobotCodeCmbGenerationParams.SelectCombinationType 174
RobotOM.IRobotCodeCmbGenerationParams.SelectDecidingValue 174
RobotOM.IRobotCodeCmbGenerationType 188
RobotOM.IRobotCodeCmbGroup 175
RobotOM.IRobotCodeCmbGroup.Add 177
RobotOM.IRobotCodeCmbGroup.AddAll 177
RobotOM.IRobotCodeCmbGroup.CaseCount 176
RobotOM.IRobotCodeCmbGroup.IsFull 178
RobotOM.IRobotCodeCmbGroup.Nature 176
RobotOM.IRobotCodeCmbGroup.Operator 176
RobotOM.IRobotCodeCmbGroupRelation 180
RobotOM.IRobotCodeCmbGroupRelation.AddGroup 182
RobotOM.IRobotCodeCmbGroupRelation.Clear 183
RobotOM.IRobotCodeCmbGroupRelation.GetGroup 183
RobotOM.IRobotCodeCmbGroupRelation.GetGroupCount 183
RobotOM.IRobotCodeCmbGroupRelation.GetOperator 184
RobotOM.IRobotCodeCmbGroupRelation.Nature 181
RobotOM.IRobotCodeCmbGroupRelation.NewRow 184
RobotOM.IRobotCodeCmbGroupRelation.RemoveRow 184
RobotOM.IRobotCodeCmbGroupRelation.RowCount 182
RobotOM.IRobotCodeCmbGroupRelation.SetGroup 184
RobotOM.IRobotCodeCmbGroupRelation.SetOperator 185
RobotOM.IRobotCodeCmbGroupRelationServer 185
RobotOM.IRobotCodeCmbGroupRelationServer.Get 186
RobotOM.IRobotCodeCmbGroupRelationServer.Set 186
RobotOM.IRobotCodeCmbGroupServer 178
RobotOM.IRobotCodeCmbGroupServer.FindByCase 179
RobotOM.IRobotCodeCmbGroupServer.Get 179
RobotOM.IRobotCodeCmbGroupServer.GetCount 180
RobotOM.IRobotCodeCmbGroupServer.New 180
RobotOM.IRobotCodeCmbOperator 178
RobotOM.IRobotCodeCmbRegulations 157
RobotOM.IRobotCodeCmbRegulations.Actions 158
RobotOM.IRobotCodeCmbRegulations.CodeName 158
RobotOM.IRobotCodeCmbRegulations.Combinations 158
RobotOM.IRobotCodeCmbRegulations.IsEuroCode 159
RobotOM.IRobotCodeCmbRegulations.MaterialType 159
RobotOM.IRobotCodeCmbRegulations.Version 159
RobotOM.IRobotCodeCombination 151
RobotOM.IRobotCodeCombination.CombinationType 152
RobotOM.IRobotCodeCombination.Components 152
RobotOM.IRobotCodeCombination.FindByFlag 153
RobotOM.IRobotCodeCombination.Flag 152
RobotOM.IRobotCodeCombination.UniqueId 152
RobotOM.IRobotCodeCombinationEngine 149
RobotOM.IRobotCodeCombinationEngine.Generate 150
RobotOM.IRobotCodeCombinationEngine.Params 150
RobotOM.IRobotCodeRegistrar 1646
RobotOM.IRobotCodeRegistrar.CodeName 1647
RobotOM.IRobotCodeRegistrar.CodeType 1648
RobotOM.IRobotCodeRegistrar.Country 1648
RobotOM.IRobotCodeRegistrar.Guid 1648
RobotOM.IRobotCodeRegistrar.Manufacturer 1649
RobotOM.IRobotCodeRegistrar.ProgId 1649
RobotOM.IRobotCodeRegistrar.Register 1649
RobotOM.IRobotCodeRegistrar.Unregister 1650
RobotOM.IRobotCodeType 1312
RobotOM.IRobotCollection 9
RobotOM.IRobotCollection.Count 10
RobotOM.IRobotCollection.Get 10
RobotOM.IRobotCombinationType 433
RobotOM.IRobotComponentFactory 1654

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRobotComponentFactory.Create 1655
RobotOM.IRobotComponentFactory.CreateExt 1655
RobotOM.IRobotComponentType 1656
RobotOM.IRobotDAMAnalysisType 1597
RobotOM.IRobotDAMCalcModule 1595
RobotOM.IRobotDAMCalcModule.DeleteModel 1597
RobotOM.IRobotDAMCalcModule.IsActive 1596
RobotOM.IRobotDAMCalcModule.Params 1596
RobotOM.IRobotDAMCalcModule.Run 1597
RobotOM.IRobotDAMLateralLoadCombType 1601
RobotOM.IRobotDAMNotionalLoads 1598
RobotOM.IRobotDAMNotionalLoads.ActiveXN 1599
RobotOM.IRobotDAMNotionalLoads.ActiveXP 1599
RobotOM.IRobotDAMNotionalLoads.ActiveYN 1599
RobotOM.IRobotDAMNotionalLoads.ActiveYP 1600
RobotOM.IRobotDAMNotionalLoads.Coefficient 1600
RobotOM.IRobotDAMNotionalLoads.GravityLoadCombEnabled 1600
RobotOM.IRobotDAMNotionalLoads.LateralLoadCombEnabled 1601
RobotOM.IRobotDAMNotionalLoads.LateralLoadCombType 1601
RobotOM.IRobotDAMParams 1605
RobotOM.IRobotDAMParams.Analysis 1606
RobotOM.IRobotDAMParams.GetNLPDParams 1607
RobotOM.IRobotDAMParams.NotionalLoads 1607
RobotOM.IRobotDAMParams.ReducedStiffness 1607
RobotOM.IRobotDAMParams.SetNLPDParams 1608
RobotOM.IRobotDAMParams.Update 1608
RobotOM.IRobotDAMReducedStiffness 1602
RobotOM.IRobotDAMReducedStiffness.RCBeamsValue 1603
RobotOM.IRobotDAMReducedStiffness.RCColumnsAndWallsValue 1603
RobotOM.IRobotDAMReducedStiffness.RCSlabsValue 1604
RobotOM.IRobotDAMReducedStiffness.SteelMembersReductionType 1604
RobotOM.IRobotDAMReducedStiffness.SteelMembersTauBValue 1604
RobotOM.IRobotDAMReducedStiffness.SteelMembersValue 1605
RobotOM.IRobotDAMSteelMembersReductionType 1605
RobotOM.IRobotDatabaseType 1326
RobotOM.IRobotDataObject 30
RobotOM.IRobotDataObject.GetLabel 31
RobotOM.IRobotDataObject.GetLabelName 32
RobotOM.IRobotDataObject.GetLabels 32
RobotOM.IRobotDataObject.HasLabel 32
RobotOM.IRobotDataObject.Number 31
RobotOM.IRobotDataObject.RemoveLabel 32
RobotOM.IRobotDataObject.SetLabel 33
RobotOM.IRobotDataObjectServer 33
RobotOM.IRobotDataObjectServer.Delete 34
RobotOM.IRobotDataObjectServer.DeleteMany 34
RobotOM.IRobotDataObjectServer.Exist 35
RobotOM.IRobotDataObjectServer.Get 35
RobotOM.IRobotDataObjectServer.GetAll 35
RobotOM.IRobotDataObjectServer.GetMany 36
RobotOM.IRobotDataObjectServer.RemoveLabel 36
RobotOM.IRobotDataObjectServer.SetLabel 36
RobotOM.IRobotDeadRecordValues 51
RobotOM.IRobotDegreeOfFreedom 1657
RobotOM.IRobotDialogId 1476
RobotOM.IRobotDirectory 1544
RobotOM.IRobotDirectoryExtension 1545
RobotOM.IRobotDisplacementData 886
RobotOM.IRobotDisplacementData.RX 887
RobotOM.IRobotDisplacementData.RY 887
RobotOM.IRobotDisplacementData.RZ 888
RobotOM.IRobotDisplacementData.UX 888
RobotOM.IRobotDisplacementData.UY 888
RobotOM.IRobotDisplacementData.UZ 888
RobotOM.IRobotDynamicAnalysisDamping 443
RobotOM.IRobotDynamicAnalysisDamping.ConstValue 444
RobotOM.IRobotDynamicAnalysisDamping.GetModeDamping 445
RobotOM.IRobotDynamicAnalysisDamping.GetModes 445
RobotOM.IRobotDynamicAnalysisDamping.RemoveModeDamping 445
RobotOM.IRobotDynamicAnalysisDamping.SetModeDamping 446
RobotOM.IRobotDynamicAnalysisDamping.Type 444
RobotOM.IRobotDynamicAnalysisDampingType 446
RobotOM.IRobotDynamicAnalysisExcitationDirection 438
RobotOM.IRobotDynamicAnalysisExcitationDirection.CombType

(c) 2014 Autodesk Inc. All rights reserved

439

RobotOM.IRobotDynamicAnalysisExcitationDirection.Group1 439

RobotOM.IRobotDynamicAnalysisExcitationDirection.Group2 439

RobotOM.IRobotDynamicAnalysisExcitationDirection.Group3 440

RobotOM.IRobotDynamicAnalysisExcitationDirection.Lambda 440

RobotOM.IRobotDynamicAnalysisExcitationDirection.Mi 440

RobotOM.IRobotDynamicAnalysisExcitationDirection.NormalizedX 440

RobotOM.IRobotDynamicAnalysisExcitationDirection.NormalizedY 440

RobotOM.IRobotDynamicAnalysisExcitationDirection.NormalizedZ 441

RobotOM.IRobotDynamicAnalysisExcitationDirection.QuadraticActive 441

RobotOM.IRobotDynamicAnalysisExcitationDirection.QuadraticSigned 441

RobotOM.IRobotDynamicAnalysisExcitationDirection.ResolutionActive 441

RobotOM.IRobotDynamicAnalysisExcitationDirection.Rx 441

RobotOM.IRobotDynamicAnalysisExcitationDirection.Ry 442

RobotOM.IRobotDynamicAnalysisExcitationDirection.Rz 442

RobotOM.IRobotDynamicAnalysisExcitationDirection.UseNormalized 442

RobotOM.IRobotDynamicAnalysisExcitationDirection.X 442

RobotOM.IRobotDynamicAnalysisExcitationDirection.Y 442

RobotOM.IRobotDynamicAnalysisExcitationDirection.Z 443

RobotOM.IRobotEigenvalues 899

RobotOM.IRobotEigenvalues.AvPartCoeff 900

RobotOM.IRobotEigenvalues.Damping 900

RobotOM.IRobotEigenvalues.EigenValue 901

RobotOM.IRobotEigenvalues.Energy 901

RobotOM.IRobotEigenvalues.Frequence 901

RobotOM.IRobotEigenvalues.Period 901

RobotOM.IRobotEigenvalues.Precision 902

RobotOM.IRobotEigenvalues.Pulsation 902

RobotOM.IRobotEigenvaluesServer 927

RobotOM.IRobotEigenvaluesServer.CombValue 928

RobotOM.IRobotEigenvaluesServer.Value 928

RobotOM.IRobotEigenvectorsServer 932

RobotOM.IRobotEigenvectorsServer.CombValue 933

RobotOM.IRobotEigenvectorsServer.Value 933

RobotOM.IRobotEmitter 515

RobotOM.IRobotEmitter.EstimatedElemNumber 515

RobotOM.IRobotEmitter.H0 516

RobotOM.IRobotEmitter.R1 516

RobotOM.IRobotEmitter.R2 516

RobotOM.IRobotEmitter.VariableMeshDensityIncrement 516

RobotOM.IRobotEquationSolvingMethod 1567

RobotOM.IRobotEurocodeFactors 1324

RobotOM.IRobotEurocodeFactors.SteelConnections 1325

RobotOM.IRobotEurocodeFactors.SteelDesign 1325

RobotOM.IRobotEurocodeSteelConnectionFactors 1319

RobotOM.IRobotEurocodeSteelConnectionFactors.Gamma0 1320

RobotOM.IRobotEurocodeSteelConnectionFactors.Gamma1 1321

RobotOM.IRobotEurocodeSteelConnectionFactors.Gamma2 1321

RobotOM.IRobotEurocodeSteelConnectionFactors.Gamma3 1321

RobotOM.IRobotEurocodeSteelConnectionFactors.Gamma3Ser 1322

RobotOM.IRobotEurocodeSteelConnectionFactors.Gamma4 1322

RobotOM.IRobotEurocodeSteelConnectionFactors.Gamma5 1322

RobotOM.IRobotEurocodeSteelConnectionFactors.Gamma6 1323

RobotOM.IRobotEurocodeSteelConnectionFactors.Gamma7 1323

RobotOM.IRobotEurocodeSteelConnectionFactors.GammaC 1323

RobotOM.IRobotEurocodeSteelConnectionFactors.LoadFromCode 1324

RobotOM.IRobotEurocodeSteelConnectionFactors.LoadFromCodeNumber 1324

RobotOM.IRobotEurocodeSteelDesignFactors 1316

RobotOM.IRobotEurocodeSteelDesignFactors.Gamma0 1317

RobotOM.IRobotEurocodeSteelDesignFactors.Gamma1 1317

RobotOM.IRobotEurocodeSteelDesignFactors.Gamma2 1318

RobotOM.IRobotEurocodeSteelDesignFactors.GammaFire 1318

RobotOM.IRobotEurocodeSteelDesignFactors.LoadFromCod

(c) 2014 Autodesk Inc. All rights reserved

e
1318
RobotOM.IRobotEurocodeSteelDesignFactors.LoadFromCodeNumber
1319
RobotOM.IRobotExternalFileFormat 1274
RobotOM.IRobotExternalPreviewFormat 1518
RobotOM.IRobotExtremeParams 913
RobotOM.IRobotExtremeParams.BarDivision 914
RobotOM.IRobotExtremeParams.Selection 914
RobotOM.IRobotExtremeParams.ValueType 915
RobotOM.IRobotExtremeResultServer 1017
RobotOM.IRobotExtremeResultServer.MaxValue 1018
RobotOM.IRobotExtremeResultServer.MinValue 1018
RobotOM.IRobotExtremeValue 909
RobotOM.IRobotExtremeValue.Bar 911
RobotOM.IRobotExtremeValue.Case 911
RobotOM.IRobotExtremeValue.CaseCmpnt 911
RobotOM.IRobotExtremeValue.IsAvailable 912
RobotOM.IRobotExtremeValue.ModeCmb 912
RobotOM.IRobotExtremeValue.Node 912
RobotOM.IRobotExtremeValue.Position 912
RobotOM.IRobotExtremeValue.Value 913
RobotOM.IRobotExtremeValueType 915
RobotOM.IRobotFeExtremeParams 987
RobotOM.IRobotFeExtremeParams.CaseCmpnt 988
RobotOM.IRobotFeExtremeParams.CaseSel 989
RobotOM.IRobotFeExtremeParams.ElementSel 989
RobotOM.IRobotFeExtremeParams.GetDirX 992
RobotOM.IRobotFeExtremeParams.Layer 989
RobotOM.IRobotFeExtremeParams.LayerArbitraryValue 990
RobotOM.IRobotFeExtremeParams.ModeCmb 990
RobotOM.IRobotFeExtremeParams.NodeSel 990
RobotOM.IRobotFeExtremeParams.PanelSel 991
RobotOM.IRobotFeExtremeParams.ReducedCutPos 991
RobotOM.IRobotFeExtremeParams.Reset 992
RobotOM.IRobotFeExtremeParams.ResultId 991
RobotOM.IRobotFeExtremeParams.SetDirX 993
RobotOM.IRobotFeExtremeParams.Smoothing 992
RobotOM.IRobotFeExtremeValue 993
RobotOM.IRobotFeExtremeValue.Case 995
RobotOM.IRobotFeExtremeValue.CaseCmpnt 995
RobotOM.IRobotFeExtremeValue.Element 995
RobotOM.IRobotFeExtremeValue.GetDirX 998
RobotOM.IRobotFeExtremeValue.IsAvailable 995
RobotOM.IRobotFeExtremeValue.Layer 996
RobotOM.IRobotFeExtremeValue.LayerArbitraryValue 996
RobotOM.IRobotFeExtremeValue.ModeCmb 996
RobotOM.IRobotFeExtremeValue.Node 996
RobotOM.IRobotFeExtremeValue.Panel 997
RobotOM.IRobotFeExtremeValue.ReducedCutPos 997
RobotOM.IRobotFeExtremeValue.ResultId 997
RobotOM.IRobotFeExtremeValue.Smoothing 997
RobotOM.IRobotFeExtremeValue.Value 998
RobotOM.IRobotFeLayerType 937
RobotOM.IRobotFeMultiExtremeValue 1001
RobotOM.IRobotFeMultiExtremeValue.Count 1001
RobotOM.IRobotFeMultiExtremeValue.Get 1002
RobotOM.IRobotFeMultiExtremeValue.GetByResType 1002
RobotOM.IRobotFeMultiResultType 999
RobotOM.IRobotFeMultiResultType.Add 1000
RobotOM.IRobotFeMultiResultType.Count 999
RobotOM.IRobotFeMultiResultType.Get 1000
RobotOM.IRobotFeMultiResultType.Remove 1000
RobotOM.IRobotFeResultComplex 963
RobotOM.IRobotFeResultComplex.M_MISES 964
RobotOM.IRobotFeResultComplex.MXX_BOTTOM 964
RobotOM.IRobotFeResultComplex.MXX_TOP 965
RobotOM.IRobotFeResultComplex.MYY_BOTTOM 965
RobotOM.IRobotFeResultComplex.MYY_TOP 965
RobotOM.IRobotFeResultComplex.N_MISES 966
RobotOM.IRobotFeResultComplex.S_MISES 966
RobotOM.IRobotFeResultDetailed 943
RobotOM.IRobotFeResultDetailed.MXX 944
RobotOM.IRobotFeResultDetailed.MXY 945
RobotOM.IRobotFeResultDetailed.MYY 945
RobotOM.IRobotFeResultDetailed.NXX 945
RobotOM.IRobotFeResultDetailed.NXY 946
RobotOM.IRobotFeResultDetailed.NYY 946
RobotOM.IRobotFeResultDetailed.PNorm 946
RobotOM.IRobotFeResultDetailed.QXX 946
RobotOM.IRobotFeResultDetailed.QYY 947
RobotOM.IRobotFeResultDetailed.RNorm 947
RobotOM.IRobotFeResultDetailed.RXX 947
RobotOM.IRobotFeResultDetailed.RYY 948

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRobotFeResultDetailed.SXX 948
RobotOM.IRobotFeResultDetailed.SXY 948
RobotOM.IRobotFeResultDetailed.SYY 949
RobotOM.IRobotFeResultDetailed.TXX 949
RobotOM.IRobotFeResultDetailed.TYY 949
RobotOM.IRobotFeResultDetailed.UXX 949
RobotOM.IRobotFeResultDetailed.UYY 950
RobotOM.IRobotFeResultDetailed.WNorm 950
RobotOM.IRobotFeResultKind 986
RobotOM.IRobotFeResultParams 937
RobotOM.IRobotFeResultParams.CalcMethod 939
RobotOM.IRobotFeResultParams.Case 939
RobotOM.IRobotFeResultParams.CaseCmpnt 939
RobotOM.IRobotFeResultParams.Element 940
RobotOM.IRobotFeResultParams.GetDirX 942
RobotOM.IRobotFeResultParams.Layer 940
RobotOM.IRobotFeResultParams.LayerArbitraryValue 940
RobotOM.IRobotFeResultParams.ModeCmb 941
RobotOM.IRobotFeResultParams.Node 941
RobotOM.IRobotFeResultParams.Panel 941
RobotOM.IRobotFeResultParams.SetDirX 942
RobotOM.IRobotFeResultPrincipal 956
RobotOM.IRobotFeResultPrincipal.M1 958
RobotOM.IRobotFeResultPrincipal.M1_2 958
RobotOM.IRobotFeResultPrincipal.M2 958
RobotOM.IRobotFeResultPrincipal.MAL 958
RobotOM.IRobotFeResultPrincipal.N1 959
RobotOM.IRobotFeResultPrincipal.N1_2 959
RobotOM.IRobotFeResultPrincipal.N2 959
RobotOM.IRobotFeResultPrincipal.NAL 960
RobotOM.IRobotFeResultPrincipal.Q1_2 960
RobotOM.IRobotFeResultPrincipal.S1 960
RobotOM.IRobotFeResultPrincipal.S1_2 961
RobotOM.IRobotFeResultPrincipal.S2 961
RobotOM.IRobotFeResultPrincipal.SAL 961
RobotOM.IRobotFeResultPrincipal.T1_2 961
RobotOM.IRobotFeResultPrincipal.U 962
RobotOM.IRobotFeResultPrincipal.UGX 962
RobotOM.IRobotFeResultPrincipal.UGY 962
RobotOM.IRobotFeResultPrincipal.UGZ 963
RobotOM.IRobotFeResultReduced 971
RobotOM.IRobotFeResultReduced.CutPos 973
RobotOM.IRobotFeResultReduced.Height 973
RobotOM.IRobotFeResultReduced.Length 973
RobotOM.IRobotFeResultReduced.MY 974
RobotOM.IRobotFeResultReduced.MZ 974
RobotOM.IRobotFeResultReduced.NodeLeftBottom 974
RobotOM.IRobotFeResultReduced.NodeLeftTop 975
RobotOM.IRobotFeResultReduced.NodeRightBottom 975
RobotOM.IRobotFeResultReduced.NodeRightTop 975
RobotOM.IRobotFeResultReduced.NX 975
RobotOM.IRobotFeResultReduced.SE 976
RobotOM.IRobotFeResultReduced.SO 976
RobotOM.IRobotFeResultReduced.T 976
RobotOM.IRobotFeResultReduced.TY 977
RobotOM.IRobotFeResultReduced.TZ 977
RobotOM.IRobotFeResultReducedCutPosition 977
RobotOM.IRobotFeResultReinforcement 966
RobotOM.IRobotFeResultReinforcement.A_MIN 968
RobotOM.IRobotFeResultReinforcement.AX 968
RobotOM.IRobotFeResultReinforcement.AX_BOTTOM 968
RobotOM.IRobotFeResultReinforcement.AX_TOP 968
RobotOM.IRobotFeResultReinforcement.AY 969
RobotOM.IRobotFeResultReinforcement.AY_BOTTOM 969
RobotOM.IRobotFeResultReinforcement.AY_TOP 969
RobotOM.IRobotFeResultReinforcement.CalcError 970
RobotOM.IRobotFeResultReinforcement.E_AX_BOTTOM 970
RobotOM.IRobotFeResultReinforcement.E_AX_TOP 970
RobotOM.IRobotFeResultReinforcement.E_AY_BOTTOM 971
RobotOM.IRobotFeResultReinforcement.E_AY_TOP 971
RobotOM.IRobotFeResultReinforcement.F 971
RobotOM.IRobotFeResultServer 950
RobotOM.IRobotFeResultServer.Complex 952
RobotOM.IRobotFeResultServer.Detailed 953
RobotOM.IRobotFeResultServer.MaxValue 953
RobotOM.IRobotFeResultServer.MinValue 953
RobotOM.IRobotFeResultServer.MultiMaxValue 954
RobotOM.IRobotFeResultServer.MultiMinValue 954
RobotOM.IRobotFeResultServer.PanelCuts 952
RobotOM.IRobotFeResultServer.Principal 954
RobotOM.IRobotFeResultServer.Reduced 955
RobotOM.IRobotFeResultServer.ReducedEx 955
RobotOM.IRobotFeResultServer.Reinforcement 955
RobotOM.IRobotFeResultSmoothing 986

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRobotFeResultType 984
RobotOM.IRobotFileInsertParams 1355
RobotOM.IRobotFileInsertParams.AsObject 1356
RobotOM.IRobotFileInsertParams.GetInsertionPoint 1357
RobotOM.IRobotFileInsertParams.GetRotation 1357
RobotOM.IRobotFileInsertParams.ReferenceNode 1356
RobotOM.IRobotFileInsertParams.ScaleFactor 1356
RobotOM.IRobotFileInsertParams.SetInsertionPoint 1357
RobotOM.IRobotFileInsertParams.SetRotation 1357
RobotOM.IRobotFiniteElement 832
RobotOM.IRobotFiniteElement.CalcArea 835
RobotOM.IRobotFiniteElement.FeType 833
RobotOM.IRobotFiniteElement.Nodes 833
RobotOM.IRobotFiniteElement.ObjectNumber 834
RobotOM.IRobotFiniteElement.ObjectPartIdx 834
RobotOM.IRobotFiniteElement.UniqueId 834
RobotOM.IRobotFiniteElementData 842
RobotOM.IRobotFiniteElementData.GetNode 844
RobotOM.IRobotFiniteElementData.NodeCount 843
RobotOM.IRobotFiniteElementData.Number 843
RobotOM.IRobotFiniteElementData.Object 843
RobotOM.IRobotFiniteElementDataSet 844
RobotOM.IRobotFiniteElementDataSet.ElemCount 845
RobotOM.IRobotFiniteElementDataSet.GetElem 846
RobotOM.IRobotFiniteElementDataSet.GetElemCountForObject 846
RobotOM.IRobotFiniteElementDataSet.GetElemForObject 846
RobotOM.IRobotFiniteElementDataSet.GetObjectNumber 847
RobotOM.IRobotFiniteElementDataSet.ObjectCount 845
RobotOM.IRobotFiniteElementNodes 836
RobotOM.IRobotFiniteElementNodes.Count 836
RobotOM.IRobotFiniteElementNodes.Get 837
RobotOM.IRobotFiniteElementNodes.GetAll 837
RobotOM.IRobotFiniteElementNodes.Set 838
RobotOM.IRobotFiniteElementNodes.SetAll 838
RobotOM.IRobotFiniteElementServer 838
RobotOM.IRobotFiniteElementServer.CalcArea 840
RobotOM.IRobotFiniteElementServer.Create 840
RobotOM.IRobotFiniteElementServer.FreeNumber 839
RobotOM.IRobotFiniteElementServer.MeshConcentrate 841
RobotOM.IRobotFiniteElementServer.MeshConsolidate 841
RobotOM.IRobotFiniteElementServer.Update 841
RobotOM.IRobotFiniteElementType 835
RobotOM.IRobotFootfallAnalysisExcitationForces 399
RobotOM.IRobotFootfallAnalysisExcitationMethod 398
RobotOM.IRobotFootfallAnalysisModalParams 401
RobotOM.IRobotFootfallAnalysisModalParams.FrequencyLimit 402
RobotOM.IRobotFootfallAnalysisModalParams.IgnoreDensity 402
RobotOM.IRobotFootfallAnalysisModalParams.IncludeMassForDirX 403
RobotOM.IRobotFootfallAnalysisModalParams.IncludeMassForDirY 403
RobotOM.IRobotFootfallAnalysisModalParams.IncludeMassForDirZ 403
RobotOM.IRobotFootfallAnalysisNodeSelection 399
RobotOM.IRobotFootfallAnalysisNodeSelection.SelectedNodes 400
RobotOM.IRobotFootfallAnalysisNodeSelection.SelectedPanels 400
RobotOM.IRobotFootfallAnalysisNodeSelection.Type 400
RobotOM.IRobotFootfallAnalysisNodeSelectionType 401
RobotOM.IRobotFootfallAnalysisParams 403
RobotOM.IRobotFootfallAnalysisParams.Damping 405
RobotOM.IRobotFootfallAnalysisParams.ExcitationForces 405
RobotOM.IRobotFootfallAnalysisParams.ExcitationMethod 405
RobotOM.IRobotFootfallAnalysisParams.ExcitationNodes 405
RobotOM.IRobotFootfallAnalysisParams.FootstepsNumber 406
RobotOM.IRobotFootfallAnalysisParams.MaxWalkingFrequency 406
RobotOM.IRobotFootfallAnalysisParams.MinWalkingFrequency 406
RobotOM.IRobotFootfallAnalysisParams.ModalParams 407
RobotOM.IRobotFootfallAnalysisParams.ResponseNodes 407
RobotOM.IRobotFootfallAnalysisParams.WalkersWeight 407
RobotOM.IRobotFootfallResults 920
RobotOM.IRobotFootfallResults.A 921
RobotOM.IRobotFootfallResults.ExcitationNode 921
RobotOM.IRobotFootfallResults.Frequency 922

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRobotFootfallResults.RF_Overall 922
RobotOM.IRobotFootfallResults.RF_Resonant 922
RobotOM.IRobotFootfallResults.RF_Transient 923
RobotOM.IRobotFootfallResults.VRMQ 923
RobotOM.IRobotFootfallResults.VRMS 923
RobotOM.IRobotForcesData 902
RobotOM.IRobotForcesData.FX 903
RobotOM.IRobotForcesData.FY 903
RobotOM.IRobotForcesData.FZ 903
RobotOM.IRobotForcesData.MX 904
RobotOM.IRobotForcesData.MY 904
RobotOM.IRobotForcesData.MZ 904
RobotOM.IRobotFRFAnalysisParams 396
RobotOM.IRobotFRFAnalysisParams.Damping 397
RobotOM.IRobotFRFAnalysisParams.FinalFrequency 397
RobotOM.IRobotFRFAnalysisParams.FrequencyDivision 397
RobotOM.IRobotFRFAnalysisParams.IncludeEigenfrequencies 397
RobotOM.IRobotFRFAnalysisParams.InitialFrequency 398
RobotOM.IRobotFRFResults 918
RobotOM.IRobotFRFResults.Frequency 919
RobotOM.IRobotFRFResults.VX 919
RobotOM.IRobotFRFResults.VY 919
RobotOM.IRobotFRFResults.VZ 920
RobotOM.IRobotFRFResultServer 934
RobotOM.IRobotFRFResultServer.Value 935
RobotOM.IRobotGeoArc 1628
RobotOM.IRobotGeoArc.P1 1629
RobotOM.IRobotGeoArc.P2 1629
RobotOM.IRobotGeoArc.P3 1629
RobotOM.IRobotGeoArcDefinitionMethod 1634
RobotOM.IRobotGeoCircle 1630
RobotOM.IRobotGeoCircle.P1 1630
RobotOM.IRobotGeoCircle.P2 1631
RobotOM.IRobotGeoCircle.P3 1631
RobotOM.IRobotGeoCircle.PC 1631
RobotOM.IRobotGeoContour 1626
RobotOM.IRobotGeoContour.Add 1627
RobotOM.IRobotGeoContour.Clear 1628
RobotOM.IRobotGeoContour.Segments 1627
RobotOM.IRobotGeoCoordinateAxis 1614
RobotOM.IRobotGeoCoordinateAxisSense 1635
RobotOM.IRobotGeoCoordinateSystem 1615
RobotOM.IRobotGeoCurveDiv 1616
RobotOM.IRobotGeoCurveDiv.IsAnalytical 1617
RobotOM.IRobotGeoCurveDiv.Length 1617
RobotOM.IRobotGeoCurveDiv.Mode 1617
RobotOM.IRobotGeoCurveDiv.N 1618
RobotOM.IRobotGeoLayer 1632
RobotOM.IRobotGeoLayer.IsThickDefined 1633
RobotOM.IRobotGeoLayer.P1 1633
RobotOM.IRobotGeoLayer.P2 1633
RobotOM.IRobotGeoLayer.P3 1634
RobotOM.IRobotGeoLayer.PDir 1634
RobotOM.IRobotGeoLayer.Thickness 1634
RobotOM.IRobotGeoObject 1623
RobotOM.IRobotGeoObject.Initialize 1624
RobotOM.IRobotGeoObject.Type 1623
RobotOM.IRobotGeoObjectType 1616
RobotOM.IRobotGeoPlane 1635
RobotOM.IRobotGeoPoint2D 1610
RobotOM.IRobotGeoPoint2D.Get 1612
RobotOM.IRobotGeoPoint2D.Set 1612
RobotOM.IRobotGeoPoint2D.X 1611
RobotOM.IRobotGeoPoint2D.Y 1611
RobotOM.IRobotGeoPoint3D 1612
RobotOM.IRobotGeoPoint3D.Get 1614
RobotOM.IRobotGeoPoint3D.Set 1614
RobotOM.IRobotGeoPoint3D.X 1613
RobotOM.IRobotGeoPoint3D.Y 1613
RobotOM.IRobotGeoPoint3D.Z 1613
RobotOM.IRobotGeoPoint3DCollection 1631
RobotOM.IRobotGeoPolyline 1624
RobotOM.IRobotGeoPolyline.Add 1625
RobotOM.IRobotGeoPolyline.Clear 1626
RobotOM.IRobotGeoPolyline.Segments 1625
RobotOM.IRobotGeoSegment 1618
RobotOM.IRobotGeoSegment.P1 1619
RobotOM.IRobotGeoSegment.Type 1619
RobotOM.IRobotGeoSegmentArc 1621
RobotOM.IRobotGeoSegmentArc.Div 1622
RobotOM.IRobotGeoSegmentArc.P2 1622
RobotOM.IRobotGeoSegmentArc.Set 1622
RobotOM.IRobotGeoSegmentCollection 1626

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRobotGeoSegmentLine 1619
RobotOM.IRobotGeoSegmentLine.Div 1620
RobotOM.IRobotGeoSegmentLine.NDiv 1620
RobotOM.IRobotGeoSegmentLine.Radius 1620
RobotOM.IRobotGeoSegmentLine.Round 1621
RobotOM.IRobotGeoSegmentType 1615
RobotOM.IRobotGroup 1041
RobotOM.IRobotGroup.Color 1042
RobotOM.IRobotGroup.CreateCollection 1044
RobotOM.IRobotGroup.CreateSelection 1044
RobotOM.IRobotGroup.Name 1042
RobotOM.IRobotGroup.ObjectType 1043
RobotOM.IRobotGroup.Read 1044
RobotOM.IRobotGroup.SelList 1043
RobotOM.IRobotGroup.Store 1045
RobotOM.IRobotGroupObjectServer 847
RobotOM.IRobotGroupObjectServer.Explode 848
RobotOM.IRobotGroupObjectServer.FindFirst 849
RobotOM.IRobotGroupObjectServer.FindNext 849
RobotOM.IRobotGroupObjectServer.GetContents 849
RobotOM.IRobotGroupObjectServer.GroupGiven 850
RobotOM.IRobotGroupObjectServer.GroupSelected 850
RobotOM.IRobotGroupServer 1045
RobotOM.IRobotGroupServer.Create 1046
RobotOM.IRobotGroupServer.Delete 1046
RobotOM.IRobotGroupServer.Find 1047
RobotOM.IRobotGroupServer.Get 1047
RobotOM.IRobotGroupServer.GetCount 1047
RobotOM.IRobotHarmonicAnalysisParams 390
RobotOM.IRobotHarmonicAnalysisParams.Excitation 391
RobotOM.IRobotHarmonicAnalysisParams.MassMatrix 391
RobotOM.IRobotHtmlView 1467
RobotOM.IRobotHtmlView.LoadFromFile 1469
RobotOM.IRobotHtmlView.Printable 1468
RobotOM.IRobotHtmlView.SaveToFile 1469
RobotOM.IRobotIn3PointsRecordValues 45
RobotOM.IRobotInContourRecordValues 48
RobotOM.IRobotIterativePrediconditionerType 1574
RobotOM.IRobotIterativeSolverMemoryUsage 1578
RobotOM.IRobotIterativeSolverMethod 1578
RobotOM.IRobotIterativeSolverParams 1574
RobotOM.IRobotIterativeSolverParams.AggregationLevelsCount 1575
RobotOM.IRobotIterativeSolverParams.AnalyseDiagonale 1575
RobotOM.IRobotIterativeSolverParams.CalcKMatrix 1576
RobotOM.IRobotIterativeSolverParams.InternalIterationsCount 1576
RobotOM.IRobotIterativeSolverParams.MemoryUsage 1576
RobotOM.IRobotIterativeSolverParams.Method 1577
RobotOM.IRobotIterativeSolverParams.Multilevel 1577
RobotOM.IRobotIterativeSolverParams.PrediconditionerType 1577
RobotOM.IRobotIterativeSolverParams.Tolerance 1578
RobotOM.IRobotKernel 2208
RobotOM.IRobotKernel.CalcEngine 2209
RobotOM.IRobotKernel.CmpntFactory 2210
RobotOM.IRobotKernel.ConcrReinfEngine 2210
RobotOM.IRobotKernel.GetExtension 2213
RobotOM.IRobotKernel.Preferences 2210
RobotOM.IRobotKernel.ProgramName 2210
RobotOM.IRobotKernel.ProgramPath 2211
RobotOM.IRobotKernel.ProgramVersion 2211
RobotOM.IRobotKernel.ProjectNew 2213
RobotOM.IRobotKernel.ProjectNewFromTemplate 2214
RobotOM.IRobotKernel.ProjectOpen 2214
RobotOM.IRobotKernel.ProjectPreferences 2211
RobotOM.IRobotKernel.ProjectSave 2214
RobotOM.IRobotKernel.ProjectSaveAs 2215
RobotOM.IRobotKernel.ProjectUniqueId 2212
RobotOM.IRobotKernel.Structure 2212
RobotOM.IRobotKernel.Version 2212
RobotOM.IRobotKernelPreferences 2215
RobotOM.IRobotKernelPreferences.GetDirectory 2217
RobotOM.IRobotKernelPreferences.GetDirectoryExt 2217
RobotOM.IRobotKernelPreferences.GetLanguage 2217
RobotOM.IRobotKernelPreferences.Multiprocessing 2216
RobotOM.IRobotKernelPreferences.SetDirectoryExt 2218
RobotOM.IRobotKernelPreferences.SetLanguage 2218
RobotOM.IRobotLabel 18
RobotOM.IRobotLabel.Data 19
RobotOM.IRobotLabel.Name 20
RobotOM.IRobotLabel.Type 20
RobotOM.IRobotLabel.UniqueId 20
RobotOM.IRobotLabelServer 20

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRobotLabelServer.Create 23
RobotOM.IRobotLabelServer.CreateLike 23
RobotOM.IRobotLabelServer.Delete 24
RobotOM.IRobotLabelServer.Exist 24
RobotOM.IRobotLabelServer.FindWithId 24
RobotOM.IRobotLabelServer.Get 24
RobotOM.IRobotLabelServer.GetAll 25
RobotOM.IRobotLabelServer.GetAvailableNames 25
RobotOM.IRobotLabelServer.GetDefault 25
RobotOM.IRobotLabelServer.GetMany 26
RobotOM.IRobotLabelServer.GetPredefinedName 26
RobotOM.IRobotLabelServer.GetUniqueId 26
RobotOM.IRobotLabelServer.IsAvailable 27
RobotOM.IRobotLabelServer.IsPredefinedName 27
RobotOM.IRobotLabelServer.IsUsed 27
RobotOM.IRobotLabelServer.SetDefault 28
RobotOM.IRobotLabelServer.Store 28
RobotOM.IRobotLabelServer.StoreWithName 28
RobotOM.IRobotLabelType 17
RobotOM.IRobotLanguage 1544
RobotOM.IRobotLayoutId 1475
RobotOM.IRobotLicenseEntitlement 1547
RobotOM.IRobotLicenseEntitlementStatus 1547
RobotOM.IRobotLimitState 435
RobotOM.IRobotLinear3DRecordValues 47
RobotOM.IRobotLinearOnEdgesRecordValues 51
RobotOM.IRobotLinearRecordValues 46
RobotOM.IRobotLinearReleaseData 704
RobotOM.IRobotLinearReleaseData.HX 705
RobotOM.IRobotLinearReleaseData.KX 705
RobotOM.IRobotLinearReleaseData.KY 706
RobotOM.IRobotLinearReleaseData.KZ 706
RobotOM.IRobotLinearReleaseData.RX 706
RobotOM.IRobotLinearReleaseData.UX 707
RobotOM.IRobotLinearReleaseData.UY 707
RobotOM.IRobotLinearReleaseData.UZ 707
RobotOM.IRobotLinearReleaseDef 713
RobotOM.IRobotLinearReleaseDef.EdgeIdx 714
RobotOM.IRobotLinearReleaseDef.EdgeObject 715
RobotOM.IRobotLinearReleaseDef.LabelName 715
RobotOM.IRobotLinearReleaseDef.Object 715
RobotOM.IRobotLinearReleaseDef.PartIdx 716
RobotOM.IRobotLinearReleaseDefinitionType 708
RobotOM.IRobotLinearReleaseDefList 716
RobotOM.IRobotLinearReleaseDefList.Add 717
RobotOM.IRobotLinearReleaseDefList.Clear 718
RobotOM.IRobotLinearReleaseDefList.Count 717
RobotOM.IRobotLinearReleaseDefList.Get 718
RobotOM.IRobotLinearReleaseDefList.Remove 718
RobotOM.IRobotLinearReleaseServer 708
RobotOM.IRobotLinearReleaseServer.Count 709
RobotOM.IRobotLinearReleaseServer.Find 710
RobotOM.IRobotLinearReleaseServer.FindEdge 711
RobotOM.IRobotLinearReleaseServer.FindLabel 711
RobotOM.IRobotLinearReleaseServer.FindObject 711
RobotOM.IRobotLinearReleaseServer.Get 712
RobotOM.IRobotLinearReleaseServer.GetLabel 712
RobotOM.IRobotLinearReleaseServer.Remove 712
RobotOM.IRobotLinearReleaseServer.Set 713
RobotOM.IRobotLinearReleaseServer.SetMany 713
RobotOM.IRobotLoadRecord 56
RobotOM.IRobotLoadRecord.Description 57
RobotOM.IRobotLoadRecord.GetValue 58
RobotOM.IRobotLoadRecord.Objects 57
RobotOM.IRobotLoadRecord.ObjectType 58
RobotOM.IRobotLoadRecord.SetValue 59
RobotOM.IRobotLoadRecord.Type 58
RobotOM.IRobotLoadRecord2 75
RobotOM.IRobotLoadRecord2.GetGeoLimits 76
RobotOM.IRobotLoadRecord2.SetGeoLimits 76
RobotOM.IRobotLoadRecord2.UniqueId 76
RobotOM.IRobotLoadRecordBarTrapezoidal 77
RobotOM.IRobotLoadRecordBarTrapezoidal.GetPoint 78
RobotOM.IRobotLoadRecordBarTrapezoidal.PointCount 78
RobotOM.IRobotLoadRecordBarTrapezoidal.SetPoint 79
RobotOM.IRobotLoadRecordCommon 81
RobotOM.IRobotLoadRecordCommon.IsAutoGenerated 81
RobotOM.IRobotLoadRecordDead 79
RobotOM.IRobotLoadRecordDead.FiniteElems 80
RobotOM.IRobotLoadRecordIn3Points 63
RobotOM.IRobotLoadRecordIn3Points.GetGeoLimits 65
RobotOM.IRobotLoadRecordIn3Points.GetPoint 65
RobotOM.IRobotLoadRecordIn3Points.SetGeoLimits 65
RobotOM.IRobotLoadRecordIn3Points.SetPoint 66

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRobotLoadRecordIn3Points.UniqueId 64
RobotOM.IRobotLoadRecordInContour 69
RobotOM.IRobotLoadRecordInContour.GetContourPoint 71
RobotOM.IRobotLoadRecordInContour.GetGeoLimits 71
RobotOM.IRobotLoadRecordInContour.GetPoint 71
RobotOM.IRobotLoadRecordInContour.GetVector 71
RobotOM.IRobotLoadRecordInContour.SetContourPoint 72
RobotOM.IRobotLoadRecordInContour.SetGeoLimits 72
RobotOM.IRobotLoadRecordInContour.SetPoint 72
RobotOM.IRobotLoadRecordInContour.SetVector 72
RobotOM.IRobotLoadRecordInContour.UniqueId 70
RobotOM.IRobotLoadRecordLinear 73
RobotOM.IRobotLoadRecordLinear.GetPoint 74
RobotOM.IRobotLoadRecordLinear.SetPoint 74
RobotOM.IRobotLoadRecordLinear.UniqueId 74
RobotOM.IRobotLoadRecordLinear3D 61
RobotOM.IRobotLoadRecordLinear3D.GetPoint 63
RobotOM.IRobotLoadRecordLinear3D.SetPoint 63
RobotOM.IRobotLoadRecordLinear3D.UniqueId 62
RobotOM.IRobotLoadRecordMngr 59
RobotOM.IRobotLoadRecordMngr.Count 60
RobotOM.IRobotLoadRecordMngr.Create 60
RobotOM.IRobotLoadRecordMngr.Delete 61
RobotOM.IRobotLoadRecordMngr.Get 61
RobotOM.IRobotLoadRecordMngr.New 61
RobotOM.IRobotLoadRecordThermalIn3Points 66
RobotOM.IRobotLoadRecordThermalIn3Points.GetGeoLimits 68
RobotOM.IRobotLoadRecordThermalIn3Points.GetPoint 68
RobotOM.IRobotLoadRecordThermalIn3Points.SetGeoLimits 68
RobotOM.IRobotLoadRecordThermalIn3Points.SetPoint 68
RobotOM.IRobotLoadRecordThermalIn3Points.UniqueId 67
RobotOM.IRobotLoadRecordType 55
RobotOM.IRobotMassActivationRecordValues 46
RobotOM.IRobotMassEccentricities 206
RobotOM.IRobotMassEccentricities.IsDirX 207
RobotOM.IRobotMassEccentricities.IsDirY 208
RobotOM.IRobotMassEccentricities.RelativeValues 208
RobotOM.IRobotMassEccentricities.ValueDirX 208
RobotOM.IRobotMassEccentricities.ValueDirY 209
RobotOM.IRobotMassSumServer 928
RobotOM.IRobotMassSumServer.Current 929
RobotOM.IRobotMassSumServer.PartCoeff 929
RobotOM.IRobotMassSumServer.Relative 930
RobotOM.IRobotMassSumServer.Total 930
RobotOM.IRobotMaterialData 660
RobotOM.IRobotMaterialData.CB71_Category 663
RobotOM.IRobotMaterialData.CB71_Humidity 663
RobotOM.IRobotMaterialData.CB71_Nature 663
RobotOM.IRobotMaterialData.CB71_Retreat 663
RobotOM.IRobotMaterialData.CS 664
RobotOM.IRobotMaterialData.Default 664
RobotOM.IRobotMaterialData.DumpCoef 664
RobotOM.IRobotMaterialData.E 664
RobotOM.IRobotMaterialData.E_5 665
RobotOM.IRobotMaterialData.E_Trans 665
RobotOM.IRobotMaterialData.EC_Deformation 665
RobotOM.IRobotMaterialData.GMean 665
RobotOM.IRobotMaterialData.Kirchoff 666
RobotOM.IRobotMaterialData.LoadFromDBase 671
RobotOM.IRobotMaterialData.LX 666
RobotOM.IRobotMaterialData.Name 666
RobotOM.IRobotMaterialData.NU 666
RobotOM.IRobotMaterialData.Nuance 667
RobotOM.IRobotMaterialData.PN_Deformation 667
RobotOM.IRobotMaterialData.PN_E_Additional 667
RobotOM.IRobotMaterialData.PN_E_Trans 667
RobotOM.IRobotMaterialData.RE 668
RobotOM.IRobotMaterialData.RE_AxCompr 668
RobotOM.IRobotMaterialData.RE_AxTens 668
RobotOM.IRobotMaterialData.RE_Bending 669
RobotOM.IRobotMaterialData.RE_Shear 669
RobotOM.IRobotMaterialData.RE_TrCompr 669
RobotOM.IRobotMaterialData.RE_TrTens 669
RobotOM.IRobotMaterialData.RO 670
RobotOM.IRobotMaterialData.RT 670
RobotOM.IRobotMaterialData.SaveToDBase 672
RobotOM.IRobotMaterialData.SecondName 670
RobotOM.IRobotMaterialData.Steel_Thermal 670
RobotOM.IRobotMaterialData.Timber_Type 671
RobotOM.IRobotMaterialData.Type 671
RobotOM.IRobotMaterialDatabase 1308
RobotOM.IRobotMaterialDatabase.Get 1310
RobotOM.IRobotMaterialDatabase.GetAll 1310

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRobotMaterialDatabase.GetDefault 1310
RobotOM.IRobotMaterialDatabase.Load 1311
RobotOM.IRobotMaterialDatabase.LoadFromFile 1311
RobotOM.IRobotMaterialDatabase.Name 1309
RobotOM.IRobotMaterialDatabase.SetDefault 1311
RobotOM.IRobotMaterialElasticModel 674
RobotOM.IRobotMaterialElasticModel.Coeff 675
RobotOM.IRobotMaterialElasticModel.Model 675
RobotOM.IRobotMaterialElasticModel.UnloadingCoeff 675
RobotOM.IRobotMaterialElasticModel.UnloadingMethod 676
RobotOM.IRobotMaterialElasticType 673
RobotOM.IRobotMaterialElasticUnloadingMethod 676
RobotOM.IRobotMaterialModel 673
RobotOM.IRobotMaterialQuantitySurvey 1079
RobotOM.IRobotMaterialQuantitySurvey.Count 1080
RobotOM.IRobotMaterialQuantitySurvey.GetName 1081
RobotOM.IRobotMaterialQuantitySurvey.GetType 1081
RobotOM.IRobotMaterialQuantitySurvey.GetVolume 1081
RobotOM.IRobotMaterialQuantitySurvey.GetWeight 1082
RobotOM.IRobotMaterialTimberType 672
RobotOM.IRobotMaterialType 672
RobotOM.IRobotMeshAccessType 831
RobotOM.IRobotMeshCoonsParams 817
RobotOM.IRobotMeshCoonsParams.ForcingRatio 818
RobotOM.IRobotMeshCoonsParams.PanelDivisionType 818
RobotOM.IRobotMeshDelaunayParams 820
RobotOM.IRobotMeshDelaunayParams.EmittersDefault 821
RobotOM.IRobotMeshDelaunayParams.EmittersSmoothing 822
RobotOM.IRobotMeshDelaunayParams.EmittersUser 822
RobotOM.IRobotMeshDelaunayParams.H_max 822
RobotOM.IRobotMeshDelaunayParams.H0 823
RobotOM.IRobotMeshDelaunayParams.MeshDensity 823
RobotOM.IRobotMeshDelaunayParams.NumberOfLevels 823
RobotOM.IRobotMeshDelaunayParams.Q 823
RobotOM.IRobotMeshDelaunayParams.RegularMesh 824
RobotOM.IRobotMeshDelaunayParams.Type 824
RobotOM.IRobotMeshDelaunayType 812
RobotOM.IRobotMeshForcingRatio 809
RobotOM.IRobotMeshGeneration 815
RobotOM.IRobotMeshGeneration.Division1 816
RobotOM.IRobotMeshGeneration.Division2 816
RobotOM.IRobotMeshGeneration.ElementSize 816
RobotOM.IRobotMeshGeneration.Type 817
RobotOM.IRobotMeshGenerationType 813
RobotOM.IRobotMeshImplementDegree 810
RobotOM.IRobotMeshMethod 813
RobotOM.IRobotMeshMethod.ForcingRatio 814
RobotOM.IRobotMeshMethod.ImplementDegree 814
RobotOM.IRobotMeshMethod.Method 814
RobotOM.IRobotMeshMethodType 812
RobotOM.IRobotMeshPanelDivType 810
RobotOM.IRobotMeshParams 829
RobotOM.IRobotMeshParams.Flag 829
RobotOM.IRobotMeshParams.MeshType 830
RobotOM.IRobotMeshParams.SurfaceParams 830
RobotOM.IRobotMeshParams.VolumeParams 830
RobotOM.IRobotMeshRefinementType 831
RobotOM.IRobotMeshSurfaceFEType 811
RobotOM.IRobotMeshSurfaceFiniteElems 818
RobotOM.IRobotMeshSurfaceFiniteElems.ConversionCoeff 819
RobotOM.IRobotMeshSurfaceFiniteElems.ForcingRatio 819
RobotOM.IRobotMeshSurfaceFiniteElems.Type 820
RobotOM.IRobotMeshSurfaceParams 824
RobotOM.IRobotMeshSurfaceParams.Coons 825
RobotOM.IRobotMeshSurfaceParams.Delaunay 826
RobotOM.IRobotMeshSurfaceParams.FiniteElems 826
RobotOM.IRobotMeshSurfaceParams.Generation 826
RobotOM.IRobotMeshSurfaceParams.Method 826
RobotOM.IRobotMeshType 808
RobotOM.IRobotMeshVolumeParams 827
RobotOM.IRobotMeshVolumeParams.AdditionalSurfaceMeshing 828
RobotOM.IRobotMeshVolumeParams.FiniteElemsType 828
RobotOM.IRobotMeshVolumeParams.MeshDensity 828
RobotOM.IRobotMeshVolumetricFEType 811
RobotOM.IRobotMobileCase 327
RobotOM.IRobotMobileCase.ApplicationPlaneBars 328
RobotOM.IRobotMobileCase.ApplicationPlaneType 329
RobotOM.IRobotMobileCase.Components 329
RobotOM.IRobotMobileCase.FindByFlag 331
RobotOM.IRobotMobileCase.Flag 329
RobotOM.IRobotMobileCase.GetRoute 332
RobotOM.IRobotMobileCase.IsAuxiliary 330

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRobotMobileCase.Label 330
RobotOM.IRobotMobileCase.NatureName 330
RobotOM.IRobotMobileCase.SetNatureExt 332
RobotOM.IRobotMobileCase.SetRoute 332
RobotOM.IRobotMobileCase.UniqueId 330
RobotOM.IRobotMobileCase.Vehicle 331
RobotOM.IRobotMobileCaseApplicationPlaneType 343
RobotOM.IRobotMobileCaseComponent 334
RobotOM.IRobotMobileCaseComponent.Point 335
RobotOM.IRobotMobileCaseComponent.Records 335
RobotOM.IRobotMobileCaseComponentMngr 333
RobotOM.IRobotMobileCaseComponentMngr.Count 333
RobotOM.IRobotMobileCaseComponentMngr.Get 334
RobotOM.IRobotMobileCaseFlag 335
RobotOM.IRobotMobileCaseRoute 336
RobotOM.IRobotMobileCaseRoute.BeginingRouteLimit 337
RobotOM.IRobotMobileCaseRoute.EndRouteLimit 337
RobotOM.IRobotMobileCaseRoute.Geometry 338
RobotOM.IRobotMobileCaseRoute.GetFactors 339
RobotOM.IRobotMobileCaseRoute.LoadDirection 338
RobotOM.IRobotMobileCaseRoute.SetFactors 339
RobotOM.IRobotMobileCaseRoute.Step 338
RobotOM.IRobotMobileCaseRoute.Tolerance 339
RobotOM.IRobotMobileCaseSegmentFactors 340
RobotOM.IRobotMobileCaseSegmentFactors.Gamma 341
RobotOM.IRobotMobileCaseSegmentFactors.HL 341
RobotOM.IRobotMobileCaseSegmentFactors.HR 341
RobotOM.IRobotMobileCaseSegmentFactors.LL 342
RobotOM.IRobotMobileCaseSegmentFactors.LR 342
RobotOM.IRobotMobileCaseSegmentFactors.VL 342
RobotOM.IRobotMobileCaseSegmentFactors.VR 343
RobotOM.IRobotMobileDistributedRecordValues 53
RobotOM.IRobotMobilePointForceRecordValues 52
RobotOM.IRobotModalAnalysisAlgorithm 193
RobotOM.IRobotModalAnalysisBase 201
RobotOM.IRobotModalAnalysisBase.Add 201
RobotOM.IRobotModalAnalysisBase.Count 202
RobotOM.IRobotModalAnalysisBase.Delete 202
RobotOM.IRobotModalAnalysisBase.Get 202
RobotOM.IRobotModalAnalysisLimits 203
RobotOM.IRobotModalAnalysisLimits.DefineLimits 205
RobotOM.IRobotModalAnalysisLimits.FrequencyLimitValue 204
RobotOM.IRobotModalAnalysisLimits.PeriodLimitValue 204
RobotOM.IRobotModalAnalysisLimits.PulsationLimitValue 204
RobotOM.IRobotModalAnalysisLimitType 194
RobotOM.IRobotModalAnalysisMassMatrixType 193
RobotOM.IRobotModalAnalysisMode 192
RobotOM.IRobotModalAnalysisParams 194
RobotOM.IRobotModalAnalysisParams.Acceleration 195
RobotOM.IRobotModalAnalysisParams.Base 196
RobotOM.IRobotModalAnalysisParams.Damping 196
RobotOM.IRobotModalAnalysisParams.DisregardDensity 196
RobotOM.IRobotModalAnalysisParams.IncludeDampingInCalculations 197
RobotOM.IRobotModalAnalysisParams.IterationsCount 197
RobotOM.IRobotModalAnalysisParams.Limits 197
RobotOM.IRobotModalAnalysisParams.MassEccentricities 198
RobotOM.IRobotModalAnalysisParams.MassMatrix 198
RobotOM.IRobotModalAnalysisParams.MassParticipation 198
RobotOM.IRobotModalAnalysisParams.Method 199
RobotOM.IRobotModalAnalysisParams.Mode 199
RobotOM.IRobotModalAnalysisParams.ModesCount 199
RobotOM.IRobotModalAnalysisParams.Shifts 200
RobotOM.IRobotModalAnalysisParams.SturmVerification 200
RobotOM.IRobotModalAnalysisParams.Tolerance 200
RobotOM.IRobotModalAnalysisShifts 205
RobotOM.IRobotModalAnalysisShifts.IterationsCount 206
RobotOM.IRobotModalAnalysisShifts.SetDefault 206
RobotOM.IRobotModalWithStaticForcesAnalysisParams 189
RobotOM.IRobotModalWithStaticForcesAnalysisParams.CreateFromStatic 191
RobotOM.IRobotModalWithStaticForcesAnalysisParams.GetStaticStateParams 191
RobotOM.IRobotModalWithStaticForcesAnalysisParams.Nonlinear 191
RobotOM.IRobotModalWithStaticForcesAnalysisParams.SetStaticStateParams 192
RobotOM.IRobotModeCombinationType 1040
RobotOM.IRobotModelGenerationParams 1584
RobotOM.IRobotModelGenerationParams.AssemblingCase 1586
RobotOM.IRobotModelGenerationParams.GenerateNodes_BarsAndFiniteElems

(c) 2014 Autodesk Inc. All rights reserved

1586

RobotOM.IRobotModelGenerationParams.GenerateNodes_DiagonalBars
1586

RobotOM.IRobotModelGenerationParams.GenerateNodes_VertHorizBars
1587

RobotOM.IRobotModelGenerationParams.MaxElementLength

1587

RobotOM.IRobotModelGenerationParams.NeglectedBars
1587

RobotOM.IRobotModelGenerationParams.NeglectedGeoObjects
1588

RobotOM.IRobotModelGenerationParams.ToleranceAutomatic
1588

RobotOM.IRobotModelGenerationParams.ToleranceValue
1588

RobotOM.IRobotModeSelection 15

RobotOM.IRobotModeSelection.Combination 16

RobotOM.IRobotModeSelection.Mode 16

RobotOM.IRobotModeSelection.Type 16

RobotOM.IRobotModeSelectionType 1040

RobotOM.IRobotMultiCollection 13

RobotOM.IRobotMultiCollection.Exist 14

RobotOM.IRobotMultiCollection.Get 14

RobotOM.IRobotMultiCollection.Set 15

RobotOM.IRobotMultiSelection 11

RobotOM.IRobotMultiSelection.CaseCmpnt 12

RobotOM.IRobotMultiSelection.Exist 13

RobotOM.IRobotMultiSelection.Get 13

RobotOM.IRobotMultiSelection.Modes 12

RobotOM.IRobotMultiSelection.Set 13

RobotOM.IRobotNamesArray 1638

RobotOM.IRobotNamesArray.Count 1639

RobotOM.IRobotNamesArray.Find 1639

RobotOM.IRobotNamesArray.Get 1640

RobotOM.IRobotNamesArray.Set 1640

RobotOM.IRobotNamesArray.SetSize 1640

RobotOM.IRobotNode 526

RobotOM.IRobotNode.GetCalcSupport 530

RobotOM.IRobotNode.GetEmitter 530

RobotOM.IRobotNode.HasCalcSupport 528

RobotOM.IRobotNode.HasEmitter 528

RobotOM.IRobotNode.IsCalc 528

RobotOM.IRobotNode.RemoveEmitter 531

RobotOM.IRobotNode.SetEmitter 531

RobotOM.IRobotNode.UniqueId 529

RobotOM.IRobotNode.X 529

RobotOM.IRobotNode.Y 529

RobotOM.IRobotNode.Z 529

RobotOM.IRobotNodeAccelerationRecordValues 54

RobotOM.IRobotNodeAuxiliaryRecordValues 41

RobotOM.IRobotNodeBucklingServer 866

RobotOM.IRobotNodeBucklingServer.EigenVector 867

RobotOM.IRobotNodeBucklingServer.EigenVectorCmb 867

RobotOM.IRobotNodeCompatibilityData 489

RobotOM.IRobotNodeCompatibilityData.Alpha 490

RobotOM.IRobotNodeCompatibilityData.AX 491

RobotOM.IRobotNodeCompatibilityData.AY 491

RobotOM.IRobotNodeCompatibilityData.AZ 491

RobotOM.IRobotNodeCompatibilityData.Beta 492

RobotOM.IRobotNodeCompatibilityData.BX 492

RobotOM.IRobotNodeCompatibilityData.BY 492

RobotOM.IRobotNodeCompatibilityData.BZ 492

RobotOM.IRobotNodeCompatibilityData.Gamma 493

RobotOM.IRobotNodeCompatibilityData.HX 493

RobotOM.IRobotNodeCompatibilityData.HY 493

RobotOM.IRobotNodeCompatibilityData.HZ 494

RobotOM.IRobotNodeCompatibilityData.KX 494

RobotOM.IRobotNodeCompatibilityData.KY 494

RobotOM.IRobotNodeCompatibilityData.KZ 494

RobotOM.IRobotNodeCompatibilityData.NonlinearModel 495

RobotOM.IRobotNodeCompatibilityData.RX 495

RobotOM.IRobotNodeCompatibilityData.RY 495

RobotOM.IRobotNodeCompatibilityData.RZ 496

RobotOM.IRobotNodeCompatibilityData.UX 496

RobotOM.IRobotNodeCompatibilityData.UY 496

RobotOM.IRobotNodeCompatibilityData.UZ 497

RobotOM.IRobotNodeCompatibilityDef 497

RobotOM.IRobotNodeCompatibilityDef.Bars 498

RobotOM.IRobotNodeCompatibilityDef.Compatible 498

RobotOM.IRobotNodeCompatibilityDef.LabelName 498

RobotOM.IRobotNodeCompatibilityDef.Main 499

RobotOM.IRobotNodeCompatibilityServer 499

RobotOM.IRobotNodeCompatibilityServer.Count 500

RobotOM.IRobotNodeCompatibilityServer.Find 501

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRobotNodeCompatibilityServer.FindCompatible 501
RobotOM.IRobotNodeCompatibilityServer.FindLabel 501
RobotOM.IRobotNodeCompatibilityServer.FindMain 502
RobotOM.IRobotNodeCompatibilityServer.Get 502
RobotOM.IRobotNodeCompatibilityServer.GetLabel 502
RobotOM.IRobotNodeCompatibilityServer.Remove 503
RobotOM.IRobotNodeCompatibilityServer.Set 503
RobotOM.IRobotNodeDisplacementRecordValues 40
RobotOM.IRobotNodeDisplacementServer 864
RobotOM.IRobotNodeDisplacementServer.DynCombValue 865
RobotOM.IRobotNodeDisplacementServer.DynValue 865
RobotOM.IRobotNodeDisplacementServer.Value 865
RobotOM.IRobotNodeDisplacementServer.ValueEx 866
RobotOM.IRobotNodeForceInPointRecordValues 50
RobotOM.IRobotNodeForceRecordValues 40
RobotOM.IRobotNodeResultServer 852
RobotOM.IRobotNodeResultServer.Buckling 853
RobotOM.IRobotNodeResultServer.Displacements 853
RobotOM.IRobotNodeResultServer.PseudostaticForces 853
RobotOM.IRobotNodeResultServer.Reactions 853
RobotOM.IRobotNodeRigidLinkData 479
RobotOM.IRobotNodeRigidLinkData.RX 480
RobotOM.IRobotNodeRigidLinkData.RY 480
RobotOM.IRobotNodeRigidLinkData.RZ 481
RobotOM.IRobotNodeRigidLinkData.UX 481
RobotOM.IRobotNodeRigidLinkData.UY 481
RobotOM.IRobotNodeRigidLinkData.UZ 482
RobotOM.IRobotNodeRigidLinkDef 482
RobotOM.IRobotNodeRigidLinkDef.LabelName 483
RobotOM.IRobotNodeRigidLinkDef.Master 483
RobotOM.IRobotNodeRigidLinkDef.Slaves 483
RobotOM.IRobotNodeRigidLinkServer 484
RobotOM.IRobotNodeRigidLinkServer.Count 485
RobotOM.IRobotNodeRigidLinkServer.Find 486
RobotOM.IRobotNodeRigidLinkServer.FindLabel 486
RobotOM.IRobotNodeRigidLinkServer.FindMaster 486
RobotOM.IRobotNodeRigidLinkServer.FindSlave 487
RobotOM.IRobotNodeRigidLinkServer.Get 487
RobotOM.IRobotNodeRigidLinkServer.GetLabel 487
RobotOM.IRobotNodeRigidLinkServer.Remove 488
RobotOM.IRobotNodeRigidLinkServer.RemoveSlave 488
RobotOM.IRobotNodeRigidLinkServer.Set 488
RobotOM.IRobotNodeServer 531
RobotOM.IRobotNodeServer.CompatibleNodes 533
RobotOM.IRobotNodeServer.Create 534
RobotOM.IRobotNodeServer.FindWithId 535
RobotOM.IRobotNodeServer.FreeNumber 533
RobotOM.IRobotNodeServer.GetCalcNodes 535
RobotOM.IRobotNodeServer.GetCalcSupport 535
RobotOM.IRobotNodeServer.GetConnectedBars 535
RobotOM.IRobotNodeServer.GetUniqueId 536
RobotOM.IRobotNodeServer.GetUserNodes 536
RobotOM.IRobotNodeServer.HasCalcSupport 536
RobotOM.IRobotNodeServer.IsCalc 537
RobotOM.IRobotNodeServer.NonlinearLinks 533
RobotOM.IRobotNodeServer.RemoveEmitter 537
RobotOM.IRobotNodeServer.RigidLinks 533
RobotOM.IRobotNodeServer.SetEmitter 537
RobotOM.IRobotNodeSupportData 503
RobotOM.IRobotNodeSupportData.Advanced 506
RobotOM.IRobotNodeSupportData.Alpha 506
RobotOM.IRobotNodeSupportData.AX 506
RobotOM.IRobotNodeSupportData.AY 507
RobotOM.IRobotNodeSupportData.AZ 507
RobotOM.IRobotNodeSupportData.Beta 507
RobotOM.IRobotNodeSupportData.BX 507
RobotOM.IRobotNodeSupportData.BY 507
RobotOM.IRobotNodeSupportData.BZ 508
RobotOM.IRobotNodeSupportData.ElasticLinear 508
RobotOM.IRobotNodeSupportData.ElasticSurface 508
RobotOM.IRobotNodeSupportData.Gamma 509
RobotOM.IRobotNodeSupportData.GetAdvanced 512
RobotOM.IRobotNodeSupportData.GetOneDir 513
RobotOM.IRobotNodeSupportData.GlobalCoordSystem 509
RobotOM.IRobotNodeSupportData.HX 509
RobotOM.IRobotNodeSupportData.HY 509
RobotOM.IRobotNodeSupportData.HZ 510
RobotOM.IRobotNodeSupportData.IsFixed 513
RobotOM.IRobotNodeSupportData.KX 510
RobotOM.IRobotNodeSupportData.KY 510
RobotOM.IRobotNodeSupportData.KZ 510
RobotOM.IRobotNodeSupportData.NonlinearModel 510
RobotOM.IRobotNodeSupportData.RX 511

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRobotNodeSupportData.RY 511
RobotOM.IRobotNodeSupportData.RZ 511
RobotOM.IRobotNodeSupportData.SetAdvanced 513
RobotOM.IRobotNodeSupportData.SetFixed 513
RobotOM.IRobotNodeSupportData.SetOneDir 514
RobotOM.IRobotNodeSupportData.UX 511
RobotOM.IRobotNodeSupportData.UY 512
RobotOM.IRobotNodeSupportData.UZ 512
RobotOM.IRobotNodeSupportFixingDirection 514
RobotOM.IRobotNodeSupportOneDirectionFixingType 514
RobotOM.IRobotNodeVelocityRecordValues 54
RobotOM.IRobotNonlinearAnalysisAlgorithmType 317
RobotOM.IRobotNonlinearAnalysisParams 318
RobotOM.IRobotNonlinearAnalysisParams.Algorithm 320
RobotOM.IRobotNonlinearAnalysisParams.DegreeOfFreedom 320
RobotOM.IRobotNonlinearAnalysisParams.DisplacementsRelativeCodeTolerance 320
RobotOM.IRobotNonlinearAnalysisParams.GetSettingsFromPreferences 325
RobotOM.IRobotNonlinearAnalysisParams.IncrementLengthReductionFactor 321
RobotOM.IRobotNonlinearAnalysisParams.IncrementLengthReductionNumber 321
RobotOM.IRobotNonlinearAnalysisParams.LineSearchMethodFactor 321
RobotOM.IRobotNonlinearAnalysisParams.LoadIncrementNumber 322
RobotOM.IRobotNonlinearAnalysisParams.MatrixUpdateAfterEachIteration 322
RobotOM.IRobotNonlinearAnalysisParams.MatrixUpdateAfterEachSubdivision 322
RobotOM.IRobotNonlinearAnalysisParams.MaxDisplacement 322
RobotOM.IRobotNonlinearAnalysisParams.MaximumIterationNumberForOneIncrement 323
RobotOM.IRobotNonlinearAnalysisParams.MaximumNumberOfBFGSCorrections 323
RobotOM.IRobotNonlinearAnalysisParams.MaximumNumberOfLineSearches 323
RobotOM.IRobotNonlinearAnalysisParams.MaxLoadFactor 323
RobotOM.IRobotNonlinearAnalysisParams.NodeNumber 324
RobotOM.IRobotNonlinearAnalysisParams.PDelta 324
RobotOM.IRobotNonlinearAnalysisParams.ResidualForcesRelativeCodeTolerance 324
RobotOM.IRobotNonlinearAnalysisParams.ResultListEachIteration 325
RobotOM.IRobotNonlinearAnalysisParams.SaveSettingsInPreferences 326
RobotOM.IRobotNonlinearAnalysisParams.Stiff 325
RobotOM.IRobotNonlinearHingeComponentType 609
RobotOM.IRobotNonlinearHingeData 606
RobotOM.IRobotNonlinearHingeData.GetModel 608
RobotOM.IRobotNonlinearHingeData.IsActive 608
RobotOM.IRobotNonlinearHingeData.NormalStress 607
RobotOM.IRobotNonlinearHingeData.Remove 608
RobotOM.IRobotNonlinearHingeData.SetModel 609
RobotOM.IRobotNonlinearHingeDef 609
RobotOM.IRobotNonlinearHingeDef.Bar 610
RobotOM.IRobotNonlinearHingeDef.LabelName 611
RobotOM.IRobotNonlinearHingeDef.Offset 611
RobotOM.IRobotNonlinearHingeDef.Relative 611
RobotOM.IRobotNonlinearHingeModel 595
RobotOM.IRobotNonlinearHingeModel.GetAxisParams 600
RobotOM.IRobotNonlinearHingeModel.GetPoints 600
RobotOM.IRobotNonlinearHingeModel.LimitCoordX 597
RobotOM.IRobotNonlinearHingeModel.LimitCoordXValue 597
RobotOM.IRobotNonlinearHingeModel.LimitCoordY 598
RobotOM.IRobotNonlinearHingeModel.LimitCoordYValue 598
RobotOM.IRobotNonlinearHingeModel.MixedUnloadingValue 598
RobotOM.IRobotNonlinearHingeModel.ModelType 599
RobotOM.IRobotNonlinearHingeModel.Name 599
RobotOM.IRobotNonlinearHingeModel.SetAxisParams 601
RobotOM.IRobotNonlinearHingeModel.SetPoints 601
RobotOM.IRobotNonlinearHingeModel.Symetry 599
RobotOM.IRobotNonlinearHingeModel.UnloadingMethod 599
RobotOM.IRobotNonlinearHingeModelAxisParams 601
RobotOM.IRobotNonlinearHingeModelAxisParams.ImmediateOccupancy 602

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRobotNonlinearHingeModelAxisParams.LifeSafety 603

RobotOM.IRobotNonlinearHingeModelAxisParams.StructuralStability 603

RobotOM.IRobotNonlinearHingeModelPoints 604

RobotOM.IRobotNonlinearHingeModelPoints.Count 605

RobotOM.IRobotNonlinearHingeModelPoints.Get 605

RobotOM.IRobotNonlinearHingeModelPoints.Set 605

RobotOM.IRobotNonlinearHingeModelServer 593

RobotOM.IRobotNonlinearHingeModelServer.Count 594

RobotOM.IRobotNonlinearHingeModelServer.Create 594

RobotOM.IRobotNonlinearHingeModelServer.Delete 595

RobotOM.IRobotNonlinearHingeModelServer.Find 595

RobotOM.IRobotNonlinearHingeModelServer.Get 595

RobotOM.IRobotNonlinearHingeModelType 603

RobotOM.IRobotNonlinearHingeModelUnloadingType 606

RobotOM.IRobotNonlinearHingeServer 612

RobotOM.IRobotNonlinearHingeServer.Count 613

RobotOM.IRobotNonlinearHingeServer.Get 613

RobotOM.IRobotNonlinearHingeServer.Remove 613

RobotOM.IRobotNonlinearHingeServer.Set 614

RobotOM.IRobotNonlinearLink 680

RobotOM.IRobotNonlinearLink.GetCurveType 682

RobotOM.IRobotNonlinearLink.GetParams 682

RobotOM.IRobotNonlinearLink.ModelType 681

RobotOM.IRobotNonlinearLink.Name 681

RobotOM.IRobotNonlinearLink.SetCurveType 683

RobotOM.IRobotNonlinearLink.SetParams 683

RobotOM.IRobotNonlinearLink.Symetry 681

RobotOM.IRobotNonlinearLinkCurveType 687

RobotOM.IRobotNonlinearLinkMngr 700

RobotOM.IRobotNonlinearLinkMngr.Delete 701

RobotOM.IRobotNonlinearLinkMngr.Get 701

RobotOM.IRobotNonlinearLinkMngr.IsDefined 701

RobotOM.IRobotNonlinearLinkMngr.Set 702

RobotOM.IRobotNonlinearLinkModelType 699

RobotOM.IRobotNonlinearLinkParams 683

RobotOM.IRobotNonlinearLinkParams.CurveType 684

RobotOM.IRobotNonlinearLinkParamsBLinear 685

RobotOM.IRobotNonlinearLinkParamsBLinear.D1 686

RobotOM.IRobotNonlinearLinkParamsBLinear.K1 686

RobotOM.IRobotNonlinearLinkParamsBLinear.K2 687

RobotOM.IRobotNonlinearLinkParamsCustom 694

RobotOM.IRobotNonlinearLinkParamsCustom.Count 695

RobotOM.IRobotNonlinearLinkParamsCustom.Get 696

RobotOM.IRobotNonlinearLinkParamsCustom.New 696

RobotOM.IRobotNonlinearLinkParamsCustom.Remove 696

RobotOM.IRobotNonlinearLinkParamsCustom.Set 697

RobotOM.IRobotNonlinearLinkParamsCustomSegment 697

RobotOM.IRobotNonlinearLinkParamsCustomSegment.Constant 698

RobotOM.IRobotNonlinearLinkParamsCustomSegment.Expression 698

RobotOM.IRobotNonlinearLinkParamsCustomSegment.OriginPoint 698

RobotOM.IRobotNonlinearLinkParamsGapHook 693

RobotOM.IRobotNonlinearLinkParamsGapHook.D 694

RobotOM.IRobotNonlinearLinkParamsGapHook.K 694

RobotOM.IRobotNonlinearLinkParamsLinear 684

RobotOM.IRobotNonlinearLinkParamsLinear.K 685

RobotOM.IRobotNonlinearLinkParamsParabolic 687

RobotOM.IRobotNonlinearLinkParamsParabolic.Dlim 688

RobotOM.IRobotNonlinearLinkParamsParabolic.Dmax 688

RobotOM.IRobotNonlinearLinkParamsParabolic.Flim 689

RobotOM.IRobotNonlinearLinkParamsParabolic.K 689

RobotOM.IRobotNonlinearLinkParamsPlastic 689

RobotOM.IRobotNonlinearLinkParamsPlastic.Dlim 690

RobotOM.IRobotNonlinearLinkParamsPlastic.Flim 690

RobotOM.IRobotNonlinearLinkParamsPlastic.K 691

RobotOM.IRobotNonlinearLinkParamsPlastic.W 691

RobotOM.IRobotNonlinearLinkParamsPlasticHardening 691

RobotOM.IRobotNonlinearLinkParamsPlasticHardening.Dlim 692

RobotOM.IRobotNonlinearLinkParamsPlasticHardening.Flim 692

RobotOM.IRobotNonlinearLinkParamsPlasticHardening.K0 692

RobotOM.IRobotNonlinearLinkParamsPlasticHardening.K1 693

RobotOM.IRobotNonlinearLinkParamsPlasticHardening.W 693

RobotOM.IRobotNonlinearLinkSemiAxisType 699

RobotOM.IRobotNonlinearLinkServer 677

RobotOM.IRobotNonlinearLinkServer.Count 678

RobotOM.IRobotNonlinearLinkServer.Create 678

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRobotNonlinearLinkServer.Find 679
RobotOM.IRobotNonlinearLinkServer.Get 679
RobotOM.IRobotNonlinearLinkServer.Remove 679
RobotOM.IRobotNumbersArray 1641
RobotOM.IRobotNumbersArray.Count 1641
RobotOM.IRobotNumbersArray.Get 1642
RobotOM.IRobotNumbersArray.Set 1642
RobotOM.IRobotNumbersArray.SetSize 1642
RobotOM.IRobotNumbersCollection 1645
RobotOM.IRobotNumbersCollection.Count 1646
RobotOM.IRobotNumbersCollection.Get 1646
RobotOM.IRobotNumbersDictionary 1662
RobotOM.IRobotNumbersDictionary.Count 1663
RobotOM.IRobotNumbersDictionary.Delete 1663
RobotOM.IRobotNumbersDictionary.Find 1663
RobotOM.IRobotNumbersDictionary.FindGet 1664
RobotOM.IRobotNumbersDictionary.Get 1664
RobotOM.IRobotNumbersDictionary.Set 1664
RobotOM.IRobotObjAttributes 763
RobotOM.IRobotObjAttributes.DirZ 764
RobotOM.IRobotObjAttributes.GetDirX 765
RobotOM.IRobotObjAttributes.GetLabelName 766
RobotOM.IRobotObjAttributes.GetLabels 766
RobotOM.IRobotObjAttributes.GetLCS 766
RobotOM.IRobotObjAttributes.GetLCSDisplayPosition 766
RobotOM.IRobotObjAttributes.HasLabel 767
RobotOM.IRobotObjAttributes.Meshed 764
RobotOM.IRobotObjAttributes.RemoveLabel 767
RobotOM.IRobotObjAttributes.SetDirX 767
RobotOM.IRobotObjAttributes.SetLabel 767
RobotOM.IRobotObjectsArray 1643
RobotOM.IRobotObjectsArray.Count 1644
RobotOM.IRobotObjectsArray.Get 1644
RobotOM.IRobotObjectsArray.Set 1644
RobotOM.IRobotObjectsArray.SetSize 1645
RobotOM.IRobotObjectStructuralType 850
RobotOM.IRobotObjectType 36
RobotOM.IRobotObjEdge 759
RobotOM.IRobotObjEdge.Divide 761
RobotOM.IRobotObjEdge.DivideByPlane 761
RobotOM.IRobotObjEdge.DivideN 761
RobotOM.IRobotObjEdge.GetLabel 762
RobotOM.IRobotObjEdge.GetLabelName 762
RobotOM.IRobotObjEdge.GetLabels 762
RobotOM.IRobotObjEdge.HasLabel 762
RobotOM.IRobotObjEdge.Path 760
RobotOM.IRobotObjEdge.RemoveLabel 763
RobotOM.IRobotObjEdge.SetLabel 763
RobotOM.IRobotObjEdgeCollection 770
RobotOM.IRobotObjLocalXDirDefinitionType 794
RobotOM.IRobotObjMesh 803
RobotOM.IRobotObjMesh.Freeze 804
RobotOM.IRobotObjMesh.Generate 805
RobotOM.IRobotObjMesh.GetBasePoints 806
RobotOM.IRobotObjMesh.GetQuality 806
RobotOM.IRobotObjMesh.IsGenerated 805
RobotOM.IRobotObjMesh.Params 805
RobotOM.IRobotObjMesh.Remove 806
RobotOM.IRobotObjMesh.SetBasePoints 807
RobotOM.IRobotObjModifExtrusion 749
RobotOM.IRobotObjModifExtrusion.Vector 749
RobotOM.IRobotObjModification 745
RobotOM.IRobotObjModification.Add 747
RobotOM.IRobotObjModification.Clear 747
RobotOM.IRobotObjModification.Filled 746
RobotOM.IRobotObjModification.NDiv 746
RobotOM.IRobotObjModification.Operations 746
RobotOM.IRobotObjModification.Type 747
RobotOM.IRobotObjModificationCollection 769
RobotOM.IRobotObjModificationType 744
RobotOM.IRobotObjModifLathe 750
RobotOM.IRobotObjModifLathe.Angle 751
RobotOM.IRobotObjModifLathe.AxsP1 751
RobotOM.IRobotObjModifLathe.AxsP2 751
RobotOM.IRobotObjModifPyramid 751
RobotOM.IRobotObjModifPyramid.AddPoint 753
RobotOM.IRobotObjModifPyramid.ClearPoints 753
RobotOM.IRobotObjModifPyramid.Factors 752
RobotOM.IRobotObjModifPyramid.Points 753
RobotOM.IRobotObjObject 770
RobotOM.IRobotObjObject.AnalyzeTTMethod 772
RobotOM.IRobotObjObject.CalcArea 776
RobotOM.IRobotObjObject.CalcVol 776
RobotOM.IRobotObjObject.FiniteElems 772

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRobotObjObject.GetFiniteElemsData 777
RobotOM.IRobotObjObject.GetHostedObjects 777
RobotOM.IRobotObjObject.GetPart 777
RobotOM.IRobotObjObject.GetPartType 778
RobotOM.IRobotObjObject.Host 772
RobotOM.IRobotObjObject.Initialize 778
RobotOM.IRobotObjObject.IsVolume 773
RobotOM.IRobotObjObject.Main 773
RobotOM.IRobotObjObject.Mesh 773
RobotOM.IRobotObjObject.Name 774
RobotOM.IRobotObjObject.NameTemplate 774
RobotOM.IRobotObjObject.Nodes 774
RobotOM.IRobotObjObject.PartsCount 774
RobotOM.IRobotObjObject.Reference 775
RobotOM.IRobotObjObject.SetHostedObjects 778
RobotOM.IRobotObjObject.StructuralType 775
RobotOM.IRobotObjObject.UniqueId 775
RobotOM.IRobotObjObject.Update 778
RobotOM.IRobotObjObjectServer 779
RobotOM.IRobotObjObjectServer.AutoRecalcHoles 781
RobotOM.IRobotObjObjectServer.BeginMultiOperation 784
RobotOM.IRobotObjObjectServer.CalcArea 784
RobotOM.IRobotObjObjectServer.CalcVol 785
RobotOM.IRobotObjObjectServer.Create 785
RobotOM.IRobotObjObjectServer.CreateArc 785
RobotOM.IRobotObjObjectServer.CreateCircle 786
RobotOM.IRobotObjObjectServer.CreateCone 786
RobotOM.IRobotObjObjectServer.CreateContour 786
RobotOM.IRobotObjObjectServer.CreateCube 787
RobotOM.IRobotObjObjectServer.CreateCylinder 787
RobotOM.IRobotObjObjectServer.CreateOnFiniteElems 787
RobotOM.IRobotObjObjectServer.CreatePolyline 788
RobotOM.IRobotObjObjectServer.CreateSolid 788
RobotOM.IRobotObjObjectServer.EndMultiOperation 788
RobotOM.IRobotObjObjectServer.FindWithId 789
RobotOM.IRobotObjObjectServer.FreeNumber 782
RobotOM.IRobotObjObjectServer.GetAnalyzeTTMethodEnabled 789
RobotOM.IRobotObjObjectServer.GetFiniteElemsData 789
RobotOM.IRobotObjObjectServer.GetHost 790
RobotOM.IRobotObjObjectServer.GetHostedObjects 790
RobotOM.IRobotObjObjectServer.GetName 790
RobotOM.IRobotObjObjectServer.GetNameTemplate 791
RobotOM.IRobotObjObjectServer.GetStructuralType 791
RobotOM.IRobotObjObjectServer.GetUniqueId 791
RobotOM.IRobotObjObjectServer.IsVolume 792
RobotOM.IRobotObjObjectServer.LinearReleases 782
RobotOM.IRobotObjObjectServer.Mesh 782
RobotOM.IRobotObjObjectServer.SetAnalyzeTTMethod 792
RobotOM.IRobotObjObjectServer.SetHost 792
RobotOM.IRobotObjObjectServer.SetHostedObjects 793
RobotOM.IRobotObjObjectServer.SetNameTemplate 793
RobotOM.IRobotObjObjectServer.SetStructuralType 793
RobotOM.IRobotObjOperation 747
RobotOM.IRobotObjOperation.Type 748
RobotOM.IRobotObjOperationCollection 748
RobotOM.IRobotObjOperationType 744
RobotOM.IRobotObjOperMeshing 757
RobotOM.IRobotObjOperMeshing.Add 759
RobotOM.IRobotObjOperMeshing.Clear 759
RobotOM.IRobotObjOperMeshing.Points 758
RobotOM.IRobotObjOperMeshing.Vectors 758
RobotOM.IRobotObjOperRotation 756
RobotOM.IRobotObjOperRotation.Angle 756
RobotOM.IRobotObjOperRotation.AxsP1 757
RobotOM.IRobotObjOperRotation.AxsP2 757
RobotOM.IRobotObjOperScaling 754
RobotOM.IRobotObjOperScaling.Center 755
RobotOM.IRobotObjOperScaling.Factor 755
RobotOM.IRobotObjOperTranslation 754
RobotOM.IRobotObjOperTranslation.Vector 754
RobotOM.IRobotObjPart 768
RobotOM.IRobotObjPart.Attribs 768
RobotOM.IRobotObjPart.GetGeometry 769
RobotOM.IRobotObjPart.Type 769
RobotOM.IRobotObjPart2 801
RobotOM.IRobotObjPart2.CalcArea 803
RobotOM.IRobotObjPart2.CalcVol 803
RobotOM.IRobotObjPart2.FiniteElems 802
RobotOM.IRobotObjPart2.Nodes 802
RobotOM.IRobotObjPartMain 794
RobotOM.IRobotObjPartMain.AddModification 798
RobotOM.IRobotObjPartMain.CalcArea 798
RobotOM.IRobotObjPartMain.CalcVol 798

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRobotObjPartMain.ClearModifications 799
RobotOM.IRobotObjPartMain.CurveDiv 795
RobotOM.IRobotObjPartMain.DefPoints 796
RobotOM.IRobotObjPartMain.Edges 796
RobotOM.IRobotObjPartMain.FiniteElems 796
RobotOM.IRobotObjPartMain.Geometry 796
RobotOM.IRobotObjPartMain.ModelPoints 797
RobotOM.IRobotObjPartMain.Modifications 797
RobotOM.IRobotObjPartMain.Nodes 797
RobotOM.IRobotObjPartReference 799
RobotOM.IRobotObjPartReference.CalcArea 800
RobotOM.IRobotObjPartReference.CalcVol 801
RobotOM.IRobotObjPartReference.FiniteElems 800
RobotOM.IRobotObjPartReference.Nodes 800
RobotOM.IRobotObjPartType 779
RobotOM.IRobotOutputFileFormat 1502
RobotOM.IRobotPageSetup 1491
RobotOM.IRobotPageSetup.Footer 1493
RobotOM.IRobotPageSetup.FromEdgeFooter 1494
RobotOM.IRobotPageSetup.FromEdgeHeader 1494
RobotOM.IRobotPageSetup.GetTemplateName 1500
RobotOM.IRobotPageSetup.Gutter 1494
RobotOM.IRobotPageSetup.Header 1495
RobotOM.IRobotPageSetup.IsCurrent 1495
RobotOM.IRobotPageSetup.Load 1501
RobotOM.IRobotPageSetup.LoadCurrent 1501
RobotOM.IRobotPageSetup.MarginBottom 1495
RobotOM.IRobotPageSetup.MarginLeft 1496
RobotOM.IRobotPageSetup.MarginRight 1496
RobotOM.IRobotPageSetup.MarginTop 1496
RobotOM.IRobotPageSetup.PageOrientation 1497
RobotOM.IRobotPageSetup.PaperHeight 1497
RobotOM.IRobotPageSetup.PaperSize 1497
RobotOM.IRobotPageSetup.PaperWidth 1498
RobotOM.IRobotPageSetup.Save 1501
RobotOM.IRobotPageSetup.SaveAs 1502
RobotOM.IRobotPageSetup.StartPageNumber 1498
RobotOM.IRobotPageSetup.TemplateCount 1498
RobotOM.IRobotPageSetup.TemplateName 1499
RobotOM.IRobotPageSetup.TextFrame 1499
RobotOM.IRobotPageSetup.TitlePage 1499
RobotOM.IRobotPageSetup.Toc 1499
RobotOM.IRobotPageSetup.Variables 1500
RobotOM.IRobotPageSetupFrameType 1515
RobotOM.IRobotPageSetupOrientation 1514
RobotOM.IRobotPageSetupTableOfContents 1515
RobotOM.IRobotPageSetupTableOfContents.Active 1516
RobotOM.IRobotPageSetupTableOfContents.IncludeTitle 1516
RobotOM.IRobotPageSetupTableOfContents.Location 1516
RobotOM.IRobotPageSetupTocLocation 1517
RobotOM.IRobotPanelCalcModelData 720
RobotOM.IRobotPanelCalcModelData.Diaphragm 721
RobotOM.IRobotPanelCalcModelData.FinElemType 721
RobotOM.IRobotPanelCalcModelData.LoadTransfer 721
RobotOM.IRobotPanelCalcModelDiaphragm 720
RobotOM.IRobotPanelCalcModelFinElemType 719
RobotOM.IRobotPanelCalcModelLoadTransfer 719
RobotOM.IRobotPanelCut 981
RobotOM.IRobotPanelCut.Active 982
RobotOM.IRobotPanelCut.Color 982
RobotOM.IRobotPanelCut.DefType 982
RobotOM.IRobotPanelCut.Point1 983
RobotOM.IRobotPanelCut.Point2 983
RobotOM.IRobotPanelCut.Point3 983
RobotOM.IRobotPanelCutDefinitionType 984
RobotOM.IRobotPanelCutMngr 978
RobotOM.IRobotPanelCutMngr.Count 978
RobotOM.IRobotPanelCutMngr.Create 979
RobotOM.IRobotPanelCutMngr.Find 979
RobotOM.IRobotPanelCutMngr.Get 980
RobotOM.IRobotPanelCutMngr.GetName 980
RobotOM.IRobotPanelCutMngr.Remove 980
RobotOM.IRobotPanelCutMngr.Store 981
RobotOM.IRobotParamCollection 2187
RobotOM.IRobotParamCollection.Count 2187
RobotOM.IRobotParamCollection.Find 2188
RobotOM.IRobotParamCollection.GetFullName 2188
RobotOM.IRobotParamCollection.GetId 2189
RobotOM.IRobotParamCollection.GetName 2189
RobotOM.IRobotParamCollection.GetValue 2189
RobotOM.IRobotParamDef 2198
RobotOM.IRobotParamDef.DefaultValue 2199
RobotOM.IRobotParamDef.GuiGetName 2202
RobotOM.IRobotParamDef.GuiReadOnly 2199

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRobotParamDef.GuiSetName 2202
RobotOM.IRobotParamDef.GuiVisible 2200
RobotOM.IRobotParamDef.Name 2200
RobotOM.IRobotParamDef.UniqueId 2200
RobotOM.IRobotParamDef.ValueList 2201
RobotOM.IRobotParamDef.ValueType 2201
RobotOM.IRobotParamSchema 2203
RobotOM.IRobotParamSchema.Def 2204
RobotOM.IRobotParamSchema.GetAllParams 2205
RobotOM.IRobotParamSchema.GetObjectsWithParam 2205
RobotOM.IRobotParamSchema.GetObjectsWithParamVal 2205
RobotOM.IRobotParamSchema.GetParam 2206
RobotOM.IRobotParamSchema.RemoveAllParams 2206
RobotOM.IRobotParamSchema.RemoveParam 2206
RobotOM.IRobotParamSchema.ResetParam 2207
RobotOM.IRobotParamSchema.SetParam 2207
RobotOM.IRobotParamSchemaDef 2190
RobotOM.IRobotParamSchemaDef.AddParam 2192
RobotOM.IRobotParamSchemaDef.AddSimpleParam 2192
RobotOM.IRobotParamSchemaDef.GetParam 2193
RobotOM.IRobotParamSchemaDef.GetParamName 2193
RobotOM.IRobotParamSchemaDef.Name 2191
RobotOM.IRobotParamSchemaDef.ParamCount 2191
RobotOM.IRobotParamSchemaDef.ParamNameToUniqueId 2193
RobotOM.IRobotParamSchemaDef.RemoveParam 2194
RobotOM.IRobotParamSchemaMngr 2194
RobotOM.IRobotParamSchemaMngr.Clear 2196
RobotOM.IRobotParamSchemaMngr.Create 2196
RobotOM.IRobotParamSchemaMngr.Exist 2196
RobotOM.IRobotParamSchemaMngr.Get 2197
RobotOM.IRobotParamSchemaMngr.GetByName 2197
RobotOM.IRobotParamSchemaMngr.Remove 2197
RobotOM.IRobotParamSchemaMngr.RemoveByName 2198
RobotOM.IRobotParamSchemaMngr.SchemaCount 2195
RobotOM.IRobotParamServer 2181
RobotOM.IRobotParamServer.GetAllParams 2183
RobotOM.IRobotParamServer.GetAllParamsForSchema 2183
RobotOM.IRobotParamServer.GetObjectsWithParam 2184
RobotOM.IRobotParamServer.GetObjectsWithParamVal 2184
RobotOM.IRobotParamServer.GetParam 2185
RobotOM.IRobotParamServer.ParamUniqueIdToName 2185
RobotOM.IRobotParamServer.RemoveParam 2185
RobotOM.IRobotParamServer.RemoveSchemaParams 2186
RobotOM.IRobotParamServer.ResetParam 2186
RobotOM.IRobotParamServer.Schemas 2182
RobotOM.IRobotParamServer.SetParam 2186
RobotOM.IRobotParamValueType 2202
RobotOM.IRobotPointAuxiliaryRecordValues 44
RobotOM.IRobotPointsArray 1658
RobotOM.IRobotPointsArray.Count 1659
RobotOM.IRobotPointsArray.Get 1659
RobotOM.IRobotPointsArray.Set 1660
RobotOM.IRobotPointsArray.SetSize 1660
RobotOM.IRobotPredefinedLabel 29
RobotOM.IRobotPredefinedSelection 1041
RobotOM.IRobotPreferences 1541
RobotOM.IRobotPreferences.CalculationsType 1542
RobotOM.IRobotPreferences.CloudCalculationsEnabled 1542
RobotOM.IRobotPreferences.GetDirectory 1543
RobotOM.IRobotPreferences.GetLanguage 1543
RobotOM.IRobotPreferences.Multiprocessing 1542
RobotOM.IRobotPreferences.OpenGL 1542
RobotOM.IRobotPreferencesEvents 1546
RobotOM.IRobotPreferencesEvents.OnDialogOK 1546
RobotOM.IRobotPressureRecordValues 45
RobotOM.IRobotPrintable 1477
RobotOM.IRobotPrintable.Comment 1478
RobotOM.IRobotPrintable.IncludeDateTime 1478
RobotOM.IRobotPrintable.SaveToFile 1479
RobotOM.IRobotPrintable.StartFromNewPage 1479
RobotOM.IRobotPrintable.Title 1479
RobotOM.IRobotPrintEngine 1480
RobotOM.IRobotPrintEngine.AddScToReport 1483
RobotOM.IRobotPrintEngine.AddTemplateToReport 1483
RobotOM.IRobotPrintEngine.AddToReport 1484
RobotOM.IRobotPrintEngine.ClosePreview 1484
RobotOM.IRobotPrintEngine.CreateReportFromOrganizer 1484
RobotOM.IRobotPrintEngine.ExternalPreviewReport 1485
RobotOM.IRobotPrintEngine.IsWhilePreview 1481
RobotOM.IRobotPrintEngine.OrganizerItems 1481
RobotOM.IRobotPrintEngine.PageSetup 1482
RobotOM.IRobotPrintEngine.PreviewReport 1485
RobotOM.IRobotPrintEngine.PrintReport 1485

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRobotPrintEngine.RemoveFromReport 1485
RobotOM.IRobotPrintEngine.RemoveScFromReport 1486
RobotOM.IRobotPrintEngine.ReportTemplates 1482
RobotOM.IRobotPrintEngine.ResetReport 1486
RobotOM.IRobotPrintEngine.SaveReportToFile 1486
RobotOM.IRobotPrintEngine.SaveReportToOrganizer 1486
RobotOM.IRobotPrintEngine.SaveReportToTemplate 1487
RobotOM.IRobotPrintEngine.ScreenCaptures 1482
RobotOM.IRobotProject 1342
RobotOM.IRobotProject.ActiveModel 1345
RobotOM.IRobotProject.AxisMngr 1345
RobotOM.IRobotProject.Backgrounds 1345
RobotOM.IRobotProject.CalcEngine 1346
RobotOM.IRobotProject.Close 1351
RobotOM.IRobotProject.ComponentMngr 1346
RobotOM.IRobotProject.ConcrReinfEngine 1346
RobotOM.IRobotProject.Connections 1346
RobotOM.IRobotProject.DimServer 1347
RobotOM.IRobotProject.ExtFileName 1347
RobotOM.IRobotProject.ExtFileParams 1347
RobotOM.IRobotProject.FileInsertParams 1348
RobotOM.IRobotProject.FileName 1348
RobotOM.IRobotProject.InsertExtFile 1351
RobotOM.IRobotProject.IsActive 1348
RobotOM.IRobotProject.Name 1348
RobotOM.IRobotProject.New 1352
RobotOM.IRobotProject.NewFromTemplate 1352
RobotOM.IRobotProject.Open 1352
RobotOM.IRobotProject.OpenExtFile 1353
RobotOM.IRobotProject.Preferences 1349
RobotOM.IRobotProject.PrintEngine 1349
RobotOM.IRobotProject.ReadExtFileParams 1353
RobotOM.IRobotProject.Save 1353
RobotOM.IRobotProject.SaveAs 1354
RobotOM.IRobotProject.SaveAsExtFile 1354
RobotOM.IRobotProject.SaveToFormat 1354
RobotOM.IRobotProject.Structure 1349
RobotOM.IRobotProject.Type 1349
RobotOM.IRobotProject.UniqueId 1350
RobotOM.IRobotProject.ViewMngr 1350
RobotOM.IRobotProjectComponent 1269
RobotOM.IRobotProjectComponent.Data 1270
RobotOM.IRobotProjectComponent.Name 1270
RobotOM.IRobotProjectComponent.Type 1271
RobotOM.IRobotProjectComponentMngr 1271
RobotOM.IRobotProjectComponentMngr.Create 1273
RobotOM.IRobotProjectComponentMngr.Get 1273
RobotOM.IRobotProjectComponentMngr.GetLevelName 1273
RobotOM.IRobotProjectComponentMngr.LevelCount 1272
RobotOM.IRobotProjectComponentMngr.StdLevelName 1272
RobotOM.IRobotProjectComponentType 1269
RobotOM.IRobotProjectEvents 1358
RobotOM.IRobotProjectEvents.OnClose 1359
RobotOM.IRobotProjectEvents.OnSave 1359
RobotOM.IRobotProjectPreferences 1296
RobotOM.IRobotProjectPreferences.CalcModelCoherence 1298
RobotOM.IRobotProjectPreferences.EurocodeFactors 1298
RobotOM.IRobotProjectPreferences.GetActiveCode 1302
RobotOM.IRobotProjectPreferences.GetActiveCodeNumber 1303
RobotOM.IRobotProjectPreferences.GetCurrentDatabase 1303
RobotOM.IRobotProjectPreferences.KinematicConstraints 1299
RobotOM.IRobotProjectPreferences.Materials 1299
RobotOM.IRobotProjectPreferences.MeshAutoAdjust 1299
RobotOM.IRobotProjectPreferences.MeshAutoAdjustIterationCount 1299
RobotOM.IRobotProjectPreferences.MeshParams 1300
RobotOM.IRobotProjectPreferences.MeshParamsFloors 1300
RobotOM.IRobotProjectPreferences.MeshParamsWalls 1300
RobotOM.IRobotProjectPreferences.Save 1303
RobotOM.IRobotProjectPreferences.SectionsActive 1301
RobotOM.IRobotProjectPreferences.SectionsFound 1301
RobotOM.IRobotProjectPreferences.SetActiveCode 1304
RobotOM.IRobotProjectPreferences.SetActiveCodeNumber 1304
RobotOM.IRobotProjectPreferences.SetCurrentDatabase 1304
RobotOM.IRobotProjectPreferences.Units 1301
RobotOM.IRobotProjectPreferences.VehiclesActive 1301
RobotOM.IRobotProjectPreferences.VehiclesFound 1302
RobotOM.IRobotProjectPreferencesEvents 1315
RobotOM.IRobotProjectPreferencesEvents.OnDialogOK 1316
RobotOM.IRobotProjectSaveFormat 1358

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRobotProjectType 1342
RobotOM.IRobotProtectionInfo 1653
RobotOM.IRobotProtectionInfo.IsExtensionEnabled 1654
RobotOM.IRobotPseudostaticForceServer 867
RobotOM.IRobotPseudostaticForceServer.CombValue 868
RobotOM.IRobotPseudostaticForceServer.Value 868
RobotOM.IRobotPushOverAnalysisParams 392
RobotOM.IRobotPushOverAnalysisParams.Direction 393
RobotOM.IRobotPushOverAnalysisParams.LoadDefinition 393
RobotOM.IRobotPushOverAnalysisParams.MaxDisplacement 393
RobotOM.IRobotPushOverAnalysisParams.Node 394
RobotOM.IRobotPushOverAnalysisParams.Nonlinear 394
RobotOM.IRobotPushOverAnalysisParams.NonlinearParams 394
RobotOM.IRobotPushOverDirection 395
RobotOM.IRobotPushOverLoadDefinitionMethod 395
RobotOM.IRobotQuitOption 1545
RobotOM.IRobotReactionData 885
RobotOM.IRobotReactionData.FX 885
RobotOM.IRobotReactionData.FY 886
RobotOM.IRobotReactionData.FZ 886
RobotOM.IRobotReactionData.MX 886
RobotOM.IRobotReactionData.MY 886
RobotOM.IRobotReactionData.MZ 886
RobotOM.IRobotReactionServer 854
RobotOM.IRobotReactionServer.DDC 856
RobotOM.IRobotReactionServer.DDCEx 856
RobotOM.IRobotReactionServer.DDCLocal 857
RobotOM.IRobotReactionServer.DDCLocalEx 857
RobotOM.IRobotReactionServer.DDCSum 857
RobotOM.IRobotReactionServer.DDCSumEx 857
RobotOM.IRobotReactionServer.DynCombDDC 858
RobotOM.IRobotReactionServer.DynCombDDCLocal 858
RobotOM.IRobotReactionServer.DynCombDDCSum 858
RobotOM.IRobotReactionServer.DynCombLocal 859
RobotOM.IRobotReactionServer.DynCombSum 859
RobotOM.IRobotReactionServer.DynCombSumForce 859
RobotOM.IRobotReactionServer.DynCombValue 859
RobotOM.IRobotReactionServer.DynDDC 860
RobotOM.IRobotReactionServer.DynDDCLocal 860
RobotOM.IRobotReactionServer.DynDDCSum 860
RobotOM.IRobotReactionServer.DynLocal 861
RobotOM.IRobotReactionServer.DynSum 861
RobotOM.IRobotReactionServer.DynSumForce 861
RobotOM.IRobotReactionServer.DynValue 861
RobotOM.IRobotReactionServer.Local 862
RobotOM.IRobotReactionServer.LocalEx 862
RobotOM.IRobotReactionServer.Sum 862
RobotOM.IRobotReactionServer.SumEx 862
RobotOM.IRobotReactionServer.SumForce 863
RobotOM.IRobotReactionServer.SumForceEx 863
RobotOM.IRobotReactionServer.Value 863
RobotOM.IRobotReactionServer.ValueEx 863
RobotOM.IRobotReinforceCalcMethods 956
RobotOM.IRobotReportItem 1509
RobotOM.IRobotReportItem.CreateView 1513
RobotOM.IRobotReportItem.ExcludeFromReport 1510
RobotOM.IRobotReportItem.GetPageTemplate 1514
RobotOM.IRobotReportItem.HasNoteAfter 1510
RobotOM.IRobotReportItem.HasNoteBefore 1511
RobotOM.IRobotReportItem.IncludeDateTime 1511
RobotOM.IRobotReportItem.NoteAfter 1511
RobotOM.IRobotReportItem.NoteBefore 1512
RobotOM.IRobotReportItem.StartFromNewPage 1512
RobotOM.IRobotReportItem.Title 1512
RobotOM.IRobotReportItem.TitleText 1513
RobotOM.IRobotReportItem.Type 1513
RobotOM.IRobotReportItemList 1507
RobotOM.IRobotReportItemList.Count 1508
RobotOM.IRobotReportItemList.Get 1508
RobotOM.IRobotReportItemType 1517
RobotOM.IRobotReportStdElementRtf 1505
RobotOM.IRobotReportStdElementRtf.Active 1505
RobotOM.IRobotReportStdElementRtf.Frame 1506
RobotOM.IRobotReportStdElementRtf.LoadFromFile 1506
RobotOM.IRobotReportStdElementRtf.RestoreDefaults 1507
RobotOM.IRobotReportStdElementRtf.SaveToFile 1507
RobotOM.IRobotReportTemplateMngr 1502
RobotOM.IRobotReportTemplateMngr.Count 1503
RobotOM.IRobotReportTemplateMngr.Find 1504
RobotOM.IRobotReportTemplateMngr.Get 1504
RobotOM.IRobotReportTemplateMngr.Remove 1504
RobotOM.IRobotResultParamType 1025

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRobotResultQueryParams 1029
RobotOM.IRobotResultQueryParams.GetParam 1031
RobotOM.IRobotResultQueryParams.IsParamSet 1032
RobotOM.IRobotResultQueryParams.Reset 1032
RobotOM.IRobotResultQueryParams.ResultIds 1030
RobotOM.IRobotResultQueryParams.Selection 1031
RobotOM.IRobotResultQueryParams.SetParam 1032
RobotOM.IRobotResultQueryReturnType 1033
RobotOM.IRobotResultRow 1023
RobotOM.IRobotResultRow.GetParam 1024
RobotOM.IRobotResultRow.GetValue 1024
RobotOM.IRobotResultRow.GetValueType 1024
RobotOM.IRobotResultRow.IsAvailable 1025
RobotOM.IRobotResultRowSet 1027
RobotOM.IRobotResultRowSet.Clear 1029
RobotOM.IRobotResultRowSet.CurrentRow 1028
RobotOM.IRobotResultRowSet.MoveFirst 1029
RobotOM.IRobotResultRowSet.MoveNext 1029
RobotOM.IRobotResultRowSet.ResultIds 1028
RobotOM.IRobotResultServer 1002
RobotOM.IRobotResultServer.Advanced 1004
RobotOM.IRobotResultServer.Any 1004
RobotOM.IRobotResultServer.Bars 1004
RobotOM.IRobotResultServer.CalculationResume 1004
RobotOM.IRobotResultServer.Extremes 1005
RobotOM.IRobotResultServer.FiniteElems 1005
RobotOM.IRobotResultServer.Nodes 1005
RobotOM.IRobotResultServer.Query 1007
RobotOM.IRobotResultServer.Status 1006
RobotOM.IRobotResultServer.Storeys 1006
RobotOM.IRobotResultServer.Total 1006
RobotOM.IRobotResultStatusType 1033
RobotOM.IRobotRtfView 1462
RobotOM.IRobotRtfView.AppendFromFile 1464
RobotOM.IRobotRtfView.Evaluate 1464
RobotOM.IRobotRtfView.LoadFromFile 1464
RobotOM.IRobotRtfView.Printable 1463
RobotOM.IRobotRtfView.SaveToFile 1464
RobotOM.IRobotScreenCaptureMngr 1465
RobotOM.IRobotScreenCaptureMngr.Count 1466
RobotOM.IRobotScreenCaptureMngr.Find 1466
RobotOM.IRobotScreenCaptureMngr.Get 1467
RobotOM.IRobotScreenCaptureMngr.Remove 1467
RobotOM.IRobotSectionDatabase 1312
RobotOM.IRobotSectionDatabase.Description 1313
RobotOM.IRobotSectionDatabase.FullName 1313
RobotOM.IRobotSectionDatabase.GetAll 1314
RobotOM.IRobotSectionDatabase.Load 1314
RobotOM.IRobotSectionDatabase.LoadFromFile 1315
RobotOM.IRobotSectionDatabase.Name 1314
RobotOM.IRobotSectionDatabaseList 1305
RobotOM.IRobotSectionDatabaseList.Add 1306
RobotOM.IRobotSectionDatabaseList.AddFromFile 1307
RobotOM.IRobotSectionDatabaseList.ChangeIndex 1307
RobotOM.IRobotSectionDatabaseList.Count 1306
RobotOM.IRobotSectionDatabaseList.Find 1307
RobotOM.IRobotSectionDatabaseList.Get 1307
RobotOM.IRobotSectionDatabaseList.GetDatabase 1308
RobotOM.IRobotSectionDatabaseList.Remove 1308
RobotOM.IRobotSeismicAnalysis_AFPS_90_Params 212
RobotOM.IRobotSeismicAnalysis_AFPS_90_Params.BehaviorFactor 213
RobotOM.IRobotSeismicAnalysis_AFPS_90_Params.Direction 213
RobotOM.IRobotSeismicAnalysis_AFPS_90_Params.DirectionType 213
RobotOM.IRobotSeismicAnalysis_AFPS_90_Params.Filter 214
RobotOM.IRobotSeismicAnalysis_AFPS_90_Params.ResidualMode 214
RobotOM.IRobotSeismicAnalysis_AFPS_90_Params.Site 214
RobotOM.IRobotSeismicAnalysis_AFPS_90_Params.SpectrumType 215
RobotOM.IRobotSeismicAnalysis_AFPS_90_Params.StructureType 215
RobotOM.IRobotSeismicAnalysis_AFPS_90_Params.Topography 215
RobotOM.IRobotSeismicAnalysis_AFPS_90_Params.ZoneType 216
RobotOM.IRobotSeismicAnalysis_AFPS_90_SiteType 257
RobotOM.IRobotSeismicAnalysis_AFPS_90_StructureType 256
RobotOM.IRobotSeismicAnalysis_AFPS_90_ZoneType 253

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRobotSeismicAnalysis_CHINESE_DesignType 259
RobotOM.IRobotSeismicAnalysis_CHINESE_EarthquakeType 260
RobotOM.IRobotSeismicAnalysis_CHINESE_IntensityType 259
RobotOM.IRobotSeismicAnalysis_CHINESE_Params 216
RobotOM.IRobotSeismicAnalysis_CHINESE_Params.DesignStandard 217
RobotOM.IRobotSeismicAnalysis_CHINESE_Params.Direction 217
RobotOM.IRobotSeismicAnalysis_CHINESE_Params.EarthquakeType 218
RobotOM.IRobotSeismicAnalysis_CHINESE_Params.Factor 218
RobotOM.IRobotSeismicAnalysis_CHINESE_Params.Filter 218
RobotOM.IRobotSeismicAnalysis_CHINESE_Params.Intensity 219
RobotOM.IRobotSeismicAnalysis_CHINESE_Params.Site 219
RobotOM.IRobotSeismicAnalysis_CHINESE_Params.SiteTg 219
RobotOM.IRobotSeismicAnalysis_CHINESE_Params.StructureType 220
RobotOM.IRobotSeismicAnalysis_CHINESE_SiteType 258
RobotOM.IRobotSeismicAnalysis_CHINESE_StructureType 258
RobotOM.IRobotSeismicAnalysis_CIRSOC_103_Params 220
RobotOM.IRobotSeismicAnalysis_CIRSOC_103_Params.Direction 221
RobotOM.IRobotSeismicAnalysis_CIRSOC_103_Params.DirectionType 221
RobotOM.IRobotSeismicAnalysis_CIRSOC_103_Params.Filter 221
RobotOM.IRobotSeismicAnalysis_CIRSOC_103_Params.PlasticityCoeff 222
RobotOM.IRobotSeismicAnalysis_CIRSOC_103_Params.Soil 222
RobotOM.IRobotSeismicAnalysis_CIRSOC_103_Params.StructureType 222
RobotOM.IRobotSeismicAnalysis_CIRSOC_103_Params.ZoneType 223
RobotOM.IRobotSeismicAnalysis_CIRSOC_103_SoilType 260
RobotOM.IRobotSeismicAnalysis_CIRSOC_103_StructureType 260
RobotOM.IRobotSeismicAnalysis_CIRSOC_103_ZoneType 253
RobotOM.IRobotSeismicAnalysis_DM_16_1_96_Params 223
RobotOM.IRobotSeismicAnalysis_DM_16_1_96_Params.Direction 224
RobotOM.IRobotSeismicAnalysis_DM_16_1_96_Params.Filter 224
RobotOM.IRobotSeismicAnalysis_DM_16_1_96_Params.SeismicCoeff 224
RobotOM.IRobotSeismicAnalysis_DM_16_1_96_Params.SeismicProtectionCoeff 225
RobotOM.IRobotSeismicAnalysis_DM_16_1_96_ProtectionCoeffType 261
RobotOM.IRobotSeismicAnalysis_EAK_2000_GroundCategoryType 268
RobotOM.IRobotSeismicAnalysis_EAK_2000_ImportanceFactorType 267
RobotOM.IRobotSeismicAnalysis_EAK_2000_Params 248
RobotOM.IRobotSeismicAnalysis_EAK_2000_Params.BehaviorFactor 250
RobotOM.IRobotSeismicAnalysis_EAK_2000_Params.Direction 250
RobotOM.IRobotSeismicAnalysis_EAK_2000_Params.DirectionType 250
RobotOM.IRobotSeismicAnalysis_EAK_2000_Params.Filter 251
RobotOM.IRobotSeismicAnalysis_EAK_2000_Params.FoundationFactor 251
RobotOM.IRobotSeismicAnalysis_EAK_2000_Params.GroundCategory 251
RobotOM.IRobotSeismicAnalysis_EAK_2000_Params.ImportanceFactor 252
RobotOM.IRobotSeismicAnalysis_EAK_2000_Params.VerticalBehaviorFactor 252
RobotOM.IRobotSeismicAnalysis_EAK_2000_Params.Vertical

(c) 2014 Autodesk Inc. All rights reserved

FoundationFactor
252

RobotOM.IRobotSeismicAnalysis_EAK_2000_Params.ZoneType
252

RobotOM.IRobotSeismicAnalysis_EAK_2000_ZoneType 256

RobotOM.IRobotSeismicAnalysis_EC8_General_Params 296

RobotOM.IRobotSeismicAnalysis_EC8_General_Params.Ag
297

RobotOM.IRobotSeismicAnalysis_EC8_General_Params.B
298

RobotOM.IRobotSeismicAnalysis_EC8_General_Params.BehaviorFactor
298

RobotOM.IRobotSeismicAnalysis_EC8_General_Params.Direction
298

RobotOM.IRobotSeismicAnalysis_EC8_General_Params.DirectionType
298

RobotOM.IRobotSeismicAnalysis_EC8_General_Params.ExcitationDir
299

RobotOM.IRobotSeismicAnalysis_EC8_General_Params.Filter
299

RobotOM.IRobotSeismicAnalysis_EC8_General_Params.ResidualMode
299

RobotOM.IRobotSeismicAnalysis_EC8_General_Params.S
299

RobotOM.IRobotSeismicAnalysis_EC8_General_Params.Spectrum
300

RobotOM.IRobotSeismicAnalysis_EC8_General_Params.Tb
300

RobotOM.IRobotSeismicAnalysis_EC8_General_Params.Tc
300

RobotOM.IRobotSeismicAnalysis_EC8_General_Params.Td
300

RobotOM.IRobotSeismicAnalysis_EC8_GroundType 301

RobotOM.IRobotSeismicAnalysis_EC8_Params 302

RobotOM.IRobotSeismicAnalysis_EC8_Params.Ag 303

RobotOM.IRobotSeismicAnalysis_EC8_Params.BehaviorFactor
303

RobotOM.IRobotSeismicAnalysis_EC8_Params.Direction 303

RobotOM.IRobotSeismicAnalysis_EC8_Params.DirectionType
303

RobotOM.IRobotSeismicAnalysis_EC8_Params.ExcitationDir
304

RobotOM.IRobotSeismicAnalysis_EC8_Params.Filter 304

RobotOM.IRobotSeismicAnalysis_EC8_Params.GroundType
304

RobotOM.IRobotSeismicAnalysis_EC8_Params.ResidualMode
305

RobotOM.IRobotSeismicAnalysis_EC8_Params.Spectrum
305

RobotOM.IRobotSeismicAnalysis_EC8_Params.SpectrumType
305

RobotOM.IRobotSeismicAnalysis_EC8_SpectrumType 301

RobotOM.IRobotSeismicAnalysis_IBC_2000_Params 245

RobotOM.IRobotSeismicAnalysis_IBC_2000_Params.BehaviorFactor
246

RobotOM.IRobotSeismicAnalysis_IBC_2000_Params.Direction
246

RobotOM.IRobotSeismicAnalysis_IBC_2000_Params.ExcitationDir
247

RobotOM.IRobotSeismicAnalysis_IBC_2000_Params.Filter
247

RobotOM.IRobotSeismicAnalysis_IBC_2000_Params.Ie 247

RobotOM.IRobotSeismicAnalysis_IBC_2000_Params.S1 248

RobotOM.IRobotSeismicAnalysis_IBC_2000_Params.SiteClass
248

RobotOM.IRobotSeismicAnalysis_IBC_2000_Params.Ss 248

RobotOM.IRobotSeismicAnalysis_IBC_2000_SiteClassType
267

RobotOM.IRobotSeismicAnalysis_IBC_2006_Params 282

RobotOM.IRobotSeismicAnalysis_IBC_2006_Params.BehaviorFactor
283

RobotOM.IRobotSeismicAnalysis_IBC_2006_Params.Direction
283

RobotOM.IRobotSeismicAnalysis_IBC_2006_Params.ExcitationDir
284

RobotOM.IRobotSeismicAnalysis_IBC_2006_Params.Filter
284

RobotOM.IRobotSeismicAnalysis_IBC_2006_Params.I 284

RobotOM.IRobotSeismicAnalysis_IBC_2006_Params.S1 284

RobotOM.IRobotSeismicAnalysis_IBC_2006_Params.SiteClass
285

RobotOM.IRobotSeismicAnalysis_IBC_2006_Params.Ss 285

RobotOM.IRobotSeismicAnalysis_IBC_2006_Params.TL 285

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRobotSeismicAnalysis_IBC_2006_SiteClassType 286
RobotOM.IRobotSeismicAnalysis_ITALY_ORDINANZA_Direction 278
RobotOM.IRobotSeismicAnalysis_ITALY_ORDINANZA_Params 276
RobotOM.IRobotSeismicAnalysis_ITALY_ORDINANZA_Params.Direction 276
RobotOM.IRobotSeismicAnalysis_ITALY_ORDINANZA_Params.ExcitationDir 277
RobotOM.IRobotSeismicAnalysis_ITALY_ORDINANZA_Params.FactorQ 277
RobotOM.IRobotSeismicAnalysis_ITALY_ORDINANZA_Params.Filter 277
RobotOM.IRobotSeismicAnalysis_ITALY_ORDINANZA_Params.Soil 277
RobotOM.IRobotSeismicAnalysis_ITALY_ORDINANZA_Params.Spectrum 277
RobotOM.IRobotSeismicAnalysis_ITALY_ORDINANZA_Params.Zone 278
RobotOM.IRobotSeismicAnalysis_ITALY_ORDINANZA_SoilType 278
RobotOM.IRobotSeismicAnalysis_ITALY_ORDINANZA_Spectrum 278
RobotOM.IRobotSeismicAnalysis_ITALY_ORDINANZA_ZoneType 278
RobotOM.IRobotSeismicAnalysis_P_100_2006_Params 279
RobotOM.IRobotSeismicAnalysis_P_100_2006_Params.Agg 280
RobotOM.IRobotSeismicAnalysis_P_100_2006_Params.B0 280
RobotOM.IRobotSeismicAnalysis_P_100_2006_Params.BehaviorFactor 280
RobotOM.IRobotSeismicAnalysis_P_100_2006_Params.ExcitationDir 280
RobotOM.IRobotSeismicAnalysis_P_100_2006_Params.ImportanceFactor 281
RobotOM.IRobotSeismicAnalysis_P_100_2006_Params.SpectrumType 281
RobotOM.IRobotSeismicAnalysis_P_100_2006_Params.Tb 281
RobotOM.IRobotSeismicAnalysis_P_100_2006_Params.Tc 281
RobotOM.IRobotSeismicAnalysis_P_100_2006_Params.Td 282
RobotOM.IRobotSeismicAnalysis_P_100_92_ImportanceClassType 262
RobotOM.IRobotSeismicAnalysis_P_100_92_Params 225
RobotOM.IRobotSeismicAnalysis_P_100_92_Params.Direction 226
RobotOM.IRobotSeismicAnalysis_P_100_92_Params.ExcitationDir 226
RobotOM.IRobotSeismicAnalysis_P_100_92_Params.Filter 226
RobotOM.IRobotSeismicAnalysis_P_100_92_Params.ImportanceClass 227
RobotOM.IRobotSeismicAnalysis_P_100_92_Params.Psi 227
RobotOM.IRobotSeismicAnalysis_P_100_92_Params.Tc 227
RobotOM.IRobotSeismicAnalysis_P_100_92_Params.ZoneType 228
RobotOM.IRobotSeismicAnalysis_P_100_92_ZoneType 254
RobotOM.IRobotSeismicAnalysis_PS_69_DampingType 262
RobotOM.IRobotSeismicAnalysis_PS_69_Params 228
RobotOM.IRobotSeismicAnalysis_PS_69_Params.Alpha 229
RobotOM.IRobotSeismicAnalysis_PS_69_Params.Damping 229
RobotOM.IRobotSeismicAnalysis_PS_69_Params.Delta 229
RobotOM.IRobotSeismicAnalysis_PS_69_Params.Direction 230
RobotOM.IRobotSeismicAnalysis_PS_69_Params.Filter 230
RobotOM.IRobotSeismicAnalysis_PS_69_Params.Soil 230
RobotOM.IRobotSeismicAnalysis_PS_69_SoilType 263
RobotOM.IRobotSeismicAnalysis_PS_92_2008_Params 286
RobotOM.IRobotSeismicAnalysis_PS_92_2008_Params.Ag 287
RobotOM.IRobotSeismicAnalysis_PS_92_2008_Params.BehaviorFactor 288
RobotOM.IRobotSeismicAnalysis_PS_92_2008_Params.Direction 288
RobotOM.IRobotSeismicAnalysis_PS_92_2008_Params.DirectionType 288
RobotOM.IRobotSeismicAnalysis_PS_92_2008_Params.Excit

(c) 2014 Autodesk Inc. All rights reserved

ationDir
289
RobotOM.IRobotSeismicAnalysis_PS_92_2008_Params.Filter 289
RobotOM.IRobotSeismicAnalysis_PS_92_2008_Params.ResidualMode
289
RobotOM.IRobotSeismicAnalysis_PS_92_2008_Params.Site 289
RobotOM.IRobotSeismicAnalysis_PS_92_2008_Params.SiteEnvelope
290
RobotOM.IRobotSeismicAnalysis_PS_92_2008_Params.SpectrumType
290
RobotOM.IRobotSeismicAnalysis_PS_92_2008_Params.StructureType
290
RobotOM.IRobotSeismicAnalysis_PS_92_2008_Params.Topography
291
RobotOM.IRobotSeismicAnalysis_PS_92_2008_Params.ZoneType
291
RobotOM.IRobotSeismicAnalysis_PS_92_2008_SiteEnvelope 293
RobotOM.IRobotSeismicAnalysis_PS_92_2008_SiteEnvelope.IsActive
293
RobotOM.IRobotSeismicAnalysis_PS_92_2008_SiteEnvelope.SetActive
294
RobotOM.IRobotSeismicAnalysis_PS_92_2008_SiteType 292
RobotOM.IRobotSeismicAnalysis_PS_92_2008_StructureType
292
RobotOM.IRobotSeismicAnalysis_PS_92_2008_ZoneType
291
RobotOM.IRobotSeismicAnalysis_PS_92_Params 231
RobotOM.IRobotSeismicAnalysis_PS_92_Params.BehaviorFactor
232
RobotOM.IRobotSeismicAnalysis_PS_92_Params.Direction
232
RobotOM.IRobotSeismicAnalysis_PS_92_Params.DirectionType
233
RobotOM.IRobotSeismicAnalysis_PS_92_Params.ExcitationDir
233
RobotOM.IRobotSeismicAnalysis_PS_92_Params.Filter 233
RobotOM.IRobotSeismicAnalysis_PS_92_Params.ResidualMode
233
RobotOM.IRobotSeismicAnalysis_PS_92_Params.Site 234
RobotOM.IRobotSeismicAnalysis_PS_92_Params.SiteEnvelope
234
RobotOM.IRobotSeismicAnalysis_PS_92_Params.SpectrumType
234
RobotOM.IRobotSeismicAnalysis_PS_92_Params.StructureType
235
RobotOM.IRobotSeismicAnalysis_PS_92_Params.Topography
235
RobotOM.IRobotSeismicAnalysis_PS_92_Params.ZoneType 235
RobotOM.IRobotSeismicAnalysis_PS_92_SiteEnvelope 268
RobotOM.IRobotSeismicAnalysis_PS_92_SiteEnvelope.IsActive
269
RobotOM.IRobotSeismicAnalysis_PS_92_SiteEnvelope.SetActive
269
RobotOM.IRobotSeismicAnalysis_PS_92_SiteType 263
RobotOM.IRobotSeismicAnalysis_PS_92_StructureType 263
RobotOM.IRobotSeismicAnalysis_PS_92_ZoneType 254
RobotOM.IRobotSeismicAnalysis_RPA_2003_Params 273
RobotOM.IRobotSeismicAnalysis_RPA_2003_Params.BehaviorFactor
274
RobotOM.IRobotSeismicAnalysis_RPA_2003_Params.ExcitationDir
274
RobotOM.IRobotSeismicAnalysis_RPA_2003_Params.Filter
274
RobotOM.IRobotSeismicAnalysis_RPA_2003_Params.QualityCoef
274
RobotOM.IRobotSeismicAnalysis_RPA_2003_Params.ResidualMode
275
RobotOM.IRobotSeismicAnalysis_RPA_2003_Params.Site
275
RobotOM.IRobotSeismicAnalysis_RPA_2003_Params.Usage
275
RobotOM.IRobotSeismicAnalysis_RPA_2003_Params.Zone
275
RobotOM.IRobotSeismicAnalysis_RPA_2003_SiteType 273
RobotOM.IRobotSeismicAnalysis_RPA_2003_UsageType
273
RobotOM.IRobotSeismicAnalysis_RPA_2003_ZoneType 272

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRobotSeismicAnalysis_RPA_88_CategoryType 264

RobotOM.IRobotSeismicAnalysis_RPA_88_Params 236

RobotOM.IRobotSeismicAnalysis_RPA_88_Params.Category 236

RobotOM.IRobotSeismicAnalysis_RPA_88_Params.Direction 237

RobotOM.IRobotSeismicAnalysis_RPA_88_Params.Filter 237

RobotOM.IRobotSeismicAnalysis_RPA_88_Params.QualityFactor 237

RobotOM.IRobotSeismicAnalysis_RPA_88_Params.Soil 238

RobotOM.IRobotSeismicAnalysis_RPA_88_Params.Usage 238

RobotOM.IRobotSeismicAnalysis_RPA_88_Params.ZoneType 238

RobotOM.IRobotSeismicAnalysis_RPA_88_SoilType 265

RobotOM.IRobotSeismicAnalysis_RPA_88_UsageType 264

RobotOM.IRobotSeismicAnalysis_RPA_88_ZoneType 255

RobotOM.IRobotSeismicAnalysis_RPS_2000_Params 270

RobotOM.IRobotSeismicAnalysis_RPS_2000_Params.BehaviorFactor 271

RobotOM.IRobotSeismicAnalysis_RPS_2000_Params.DirectionType 271

RobotOM.IRobotSeismicAnalysis_RPS_2000_Params.ExcitationDir 271

RobotOM.IRobotSeismicAnalysis_RPS_2000_Params.Filter 271

RobotOM.IRobotSeismicAnalysis_RPS_2000_Params.ResidualMode 272

RobotOM.IRobotSeismicAnalysis_RPS_2000_Params.Site 272

RobotOM.IRobotSeismicAnalysis_RPS_2000_Params.StructureClass 272

RobotOM.IRobotSeismicAnalysis_RPS_2000_Params.Zone 272

RobotOM.IRobotSeismicAnalysis_RPS_2000_SiteType 269

RobotOM.IRobotSeismicAnalysis_RPS_2000_StructureClass 270

RobotOM.IRobotSeismicAnalysis_RPS_2000_ZoneType 269

RobotOM.IRobotSeismicAnalysis_TURKISH_23098_Params 239

RobotOM.IRobotSeismicAnalysis_TURKISH_23098_Params.BehaviorFactor 240

RobotOM.IRobotSeismicAnalysis_TURKISH_23098_Params.Direction 240

RobotOM.IRobotSeismicAnalysis_TURKISH_23098_Params.Filter 240

RobotOM.IRobotSeismicAnalysis_TURKISH_23098_Params.SoilType 241

RobotOM.IRobotSeismicAnalysis_TURKISH_23098_Params.StructureImportance 241

RobotOM.IRobotSeismicAnalysis_TURKISH_23098_Params.ZoneType 241

RobotOM.IRobotSeismicAnalysis_TURKISH_23098_SoilType 265

RobotOM.IRobotSeismicAnalysis_TURKISH_23098_ZoneType 256

RobotOM.IRobotSeismicAnalysis_UBC_97_Params 241

RobotOM.IRobotSeismicAnalysis_UBC_97_Params.BehaviorFactor 242

RobotOM.IRobotSeismicAnalysis_UBC_97_Params.ClosestDistance 243

RobotOM.IRobotSeismicAnalysis_UBC_97_Params.Direction 243

RobotOM.IRobotSeismicAnalysis_UBC_97_Params.ExcitationDir 243

RobotOM.IRobotSeismicAnalysis_UBC_97_Params.Filter 244

RobotOM.IRobotSeismicAnalysis_UBC_97_Params.I 244

RobotOM.IRobotSeismicAnalysis_UBC_97_Params.Soil 244

RobotOM.IRobotSeismicAnalysis_UBC_97_Params.Source 245

RobotOM.IRobotSeismicAnalysis_UBC_97_Params.ZoneType 245

RobotOM.IRobotSeismicAnalysis_UBC_97_SoilType 266

RobotOM.IRobotSeismicAnalysis_UBC_97_SourceType 266

RobotOM.IRobotSeismicAnalysis_UBC_97_ZoneType 255

RobotOM.IRobotSeismicAnalysisDirectionType 261

RobotOM.IRobotSeismicAnalysisSpectrumType 257

RobotOM.IRobotSeismicResidualMode 294

RobotOM.IRobotSeismicResidualMode.AugmentationFactor 295

RobotOM.IRobotSeismicResidualMode.DefinitionMethod 295

RobotOM.IRobotSeismicResidualMode.IsActive 296

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRobotSeismicResidualMode.LimitFrequency 296
RobotOM.IRobotSeismicResidualModeDefinitionType 294
RobotOM.IRobotSeismicResultsPanelDirection 1584
RobotOM.IRobotSeismicResultsSaveParams 1579
RobotOM.IRobotSeismicResultsSaveParams.Displacements 1581
RobotOM.IRobotSeismicResultsSaveParams.Forces 1581
RobotOM.IRobotSeismicResultsSaveParams.LocalDisplacements 1581
RobotOM.IRobotSeismicResultsSaveParams.NMQ 1581
RobotOM.IRobotSeismicResultsSaveParams.OnlyQuadraticCombs 1582
RobotOM.IRobotSeismicResultsSaveParams.PanelsDir 1582
RobotOM.IRobotSeismicResultsSaveParams.PointNumber 1582
RobotOM.IRobotSeismicResultsSaveParams.Reactions 1583
RobotOM.IRobotSeismicResultsSaveParams.Reduced 1583
RobotOM.IRobotSeismicResultsSaveParams.Save 1583
RobotOM.IRobotSeismicResultsSaveParams.Stresses 1584
RobotOM.IRobotSelection 3
RobotOM.IRobotSelection.Add 5
RobotOM.IRobotSelection.AddOne 5
RobotOM.IRobotSelection.AddText 6
RobotOM.IRobotSelection.And 6
RobotOM.IRobotSelection.AndText 6
RobotOM.IRobotSelection.Clear 7
RobotOM.IRobotSelection.Contains 7
RobotOM.IRobotSelection.Count 4
RobotOM.IRobotSelection.Exclude 7
RobotOM.IRobotSelection.ExcludeOne 8
RobotOM.IRobotSelection.ExcludeText 8
RobotOM.IRobotSelection.FromText 8
RobotOM.IRobotSelection.Get 9
RobotOM.IRobotSelection.ToText 9
RobotOM.IRobotSelection.Type 4
RobotOM.IRobotSelectionFactory 1037
RobotOM.IRobotSelectionFactory.Create 1038
RobotOM.IRobotSelectionFactory.CreateByLabel 1038
RobotOM.IRobotSelectionFactory.CreateByStorey 1038
RobotOM.IRobotSelectionFactory.CreateFull 1039
RobotOM.IRobotSelectionFactory.CreateMulti 1039
RobotOM.IRobotSelectionFactory.CreatePredefined 1039
RobotOM.IRobotSelectionFactory.Get 1039
RobotOM.IRobotSELFSeismic_AS_1170_4_Params 467
RobotOM.IRobotSELFSeismic_AS_1170_4_Params.Kp 468
RobotOM.IRobotSELFSeismic_AS_1170_4_Params.Mi 469
RobotOM.IRobotSELFSeismic_AS_1170_4_Params.Probability 469
RobotOM.IRobotSELFSeismic_AS_1170_4_Params.SoilCategory 469
RobotOM.IRobotSELFSeismic_AS_1170_4_Params.Sp 469
RobotOM.IRobotSELFSeismic_AS_1170_4_Params.StructureTypeX 470
RobotOM.IRobotSELFSeismic_AS_1170_4_Params.StructureTypeY 470
RobotOM.IRobotSELFSeismic_AS_1170_4_Params.Z 470
RobotOM.IRobotSELFSeismic_AS_1170_4_ProbabilityType 471
RobotOM.IRobotSELFSeismic_AS_1170_4_SoilCategoryType 470
RobotOM.IRobotSELFSeismic_AS_1170_4_StructureType 471
RobotOM.IRobotSELFSeismic_ASCE_7_10_Params 458
RobotOM.IRobotSELFSeismic_ASCE_7_10_Params.I 459
RobotOM.IRobotSELFSeismic_ASCE_7_10_Params.R 460
RobotOM.IRobotSELFSeismic_ASCE_7_10_Params.S1 460
RobotOM.IRobotSELFSeismic_ASCE_7_10_Params.SiteClass 460
RobotOM.IRobotSELFSeismic_ASCE_7_10_Params.Ss 460
RobotOM.IRobotSELFSeismic_ASCE_7_10_Params.StructureTypeX 461
RobotOM.IRobotSELFSeismic_ASCE_7_10_Params.StructureTypeY 461
RobotOM.IRobotSELFSeismic_ASCE_7_10_Params.TBaseValueX 461
RobotOM.IRobotSELFSeismic_ASCE_7_10_Params.TBaseValueY 461
RobotOM.IRobotSELFSeismic_ASCE_7_10_Params.TL 462
RobotOM.IRobotSELFSeismic_ASCE_7_10_SiteClassType 458
RobotOM.IRobotSELFSeismic_ASCE_7_10_StructureType 462
RobotOM.IRobotSELFSeismic_EC_8_Params 462
RobotOM.IRobotSELFSeismic_EC_8_Params.Ag 463

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRobotSELFSeismic_EC_8_Params.Beta 464

RobotOM.IRobotSELFSeismic_EC_8_Params.Q 464

RobotOM.IRobotSELFSeismic_EC_8_Params.SiteClass 464

RobotOM.IRobotSELFSeismic_EC_8_Params.SiteClassEnvelopeCheck 466

RobotOM.IRobotSELFSeismic_EC_8_Params.SiteClassEnvelopeIsChecked 466

RobotOM.IRobotSELFSeismic_EC_8_Params.SpectrumType 465

RobotOM.IRobotSELFSeismic_EC_8_Params.StructureTypeX 465

RobotOM.IRobotSELFSeismic_EC_8_Params.StructureTypeY 465

RobotOM.IRobotSELFSeismic_EC_8_SiteClass 467

RobotOM.IRobotSELFSeismic_EC_8_SpectrumType 467

RobotOM.IRobotSELFSeismic_EC_8_StructureType 466

RobotOM.IRobotSELFSeismicAnalysisParams 455

RobotOM.IRobotSELFSeismicAnalysisParams.CodeName 456

RobotOM.IRobotSELFSeismicAnalysisParams.CodeNumber 456

RobotOM.IRobotSELFSeismicAnalysisParams.CodeParams 456

RobotOM.IRobotSELFSeismicAnalysisParams.Eccentricities 457

RobotOM.IRobotSELFSeismicAnalysisParams.GetExcitationDir 458

RobotOM.IRobotSELFSeismicAnalysisParams.StructureParams 457

RobotOM.IRobotSELFSeismicAnalysisParams.TBaseMethod 457

RobotOM.IRobotSELFSeismicEngine 448

RobotOM.IRobotSELFSeismicEngine.GenerateLoadCases 449

RobotOM.IRobotSELFSeismicEngine.GenerationParams 448

RobotOM.IRobotSELFSeismicGenerationParams 449

RobotOM.IRobotSELFSeismicGenerationParams.CodeName 450

RobotOM.IRobotSELFSeismicGenerationParams.CodeNumber 450

RobotOM.IRobotSELFSeismicGenerationParams.ExcitationDir 451

RobotOM.IRobotSELFSeismicGenerationParams.ModalCaseParams 451

RobotOM.IRobotSELFSeismicGenerationParams.SeismicParams 451

RobotOM.IRobotSELFSeismicGenerationParams.TBaseMethod 451

RobotOM.IRobotSELFSeismicLevelDefinitionMethod 454

RobotOM.IRobotSELFSeismicStructureParams 452

RobotOM.IRobotSELFSeismicStructureParams.BaseLevelCoordZ 452

RobotOM.IRobotSELFSeismicStructureParams.BaseLevelDefMethod 453

RobotOM.IRobotSELFSeismicStructureParams.BaseLevelStorey 453

RobotOM.IRobotSELFSeismicStructureParams.TopLevelCoordZ 453

RobotOM.IRobotSELFSeismicStructureParams.TopLevelDefMethod 454

RobotOM.IRobotSELFSeismicStructureParams.TopLevelStorey 454

RobotOM.IRobotSELFSeismicTBaseMethod 455

RobotOM.IRobotSerializable 1660

RobotOM.IRobotSerializable.Read 1661

RobotOM.IRobotSerializable.Write 1661

RobotOM.IRobotSimpleCase 420

RobotOM.IRobotSimpleCase.GetAnalysisParams 424

RobotOM.IRobotSimpleCase.GetSeismicCode 425

RobotOM.IRobotSimpleCase.IsAuxiliary 422

RobotOM.IRobotSimpleCase.Label 422

RobotOM.IRobotSimpleCase.MainMode 422

RobotOM.IRobotSimpleCase.ModesCount 423

RobotOM.IRobotSimpleCase.NatureName 423

RobotOM.IRobotSimpleCase.Records 423

RobotOM.IRobotSimpleCase.SetAnalysisParams 425

RobotOM.IRobotSimpleCase.SetNatureExt 425

RobotOM.IRobotSimpleCase.TimeStepCount 423

RobotOM.IRobotSimpleCase.UniqueId 424

RobotOM.IRobotSnowWindEngine 96

RobotOM.IRobotSnowWindEngine.CodeParams 97

RobotOM.IRobotSnowWindEngine.Generate 98

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRobotSnowWindEngine.Generate3D 98
RobotOM.IRobotSnowWindEngine.GenerateStruct3D 98
RobotOM.IRobotSnowWindEngine.Params 97
RobotOM.IRobotSnowWindEngine.ShowParamsDlg 99
RobotOM.IRobotSnowWindParams 93
RobotOM.IRobotSnowWindParams.BaseOnGround 94
RobotOM.IRobotSnowWindParams.BaySpacing 94
RobotOM.IRobotSnowWindParams.Envelope 94
RobotOM.IRobotSnowWindParams.IsolatedRoofs 95
RobotOM.IRobotSnowWindParams.IsSnow 95
RobotOM.IRobotSnowWindParams.IsWind 95
RobotOM.IRobotSnowWindParams.TotalDepth 95
RobotOM.IRobotSnowWindParams.WithCavities 96
RobotOM.IRobotSnowWindParams.WithParapets 96
RobotOM.IRobotSolidPropertiesData 741
RobotOM.IRobotSolidPropertiesData.DampCoef 742
RobotOM.IRobotSolidPropertiesData.E 742
RobotOM.IRobotSolidPropertiesData.LoadFromDBase 744
RobotOM.IRobotSolidPropertiesData.LX 742
RobotOM.IRobotSolidPropertiesData.MaterialModel 743
RobotOM.IRobotSolidPropertiesData.NU 743
RobotOM.IRobotSolidPropertiesData.RO 743
RobotOM.IRobotSparseMSolverMethod 1573
RobotOM.IRobotSparseMSolverParams 1572
RobotOM.IRobotSparseMSolverParams.Method 1573
RobotOM.IRobotSpectralAnalysisAbscissaXAxisType 316
RobotOM.IRobotSpectralAnalysisOrdinateYAxisType 317
RobotOM.IRobotSpectralAnalysisParams 306
RobotOM.IRobotSpectralAnalysisParams.Direction 307
RobotOM.IRobotSpectralAnalysisParams.ExcitationDir 307
RobotOM.IRobotSpectralAnalysisParams.Filter 307
RobotOM.IRobotSpectralAnalysisParams.Spectrum 307
RobotOM.IRobotSpectralAnalysisPointsCollection 313
RobotOM.IRobotSpectralAnalysisPointsCollection.Add 314
RobotOM.IRobotSpectralAnalysisPointsCollection.Clear 314
RobotOM.IRobotSpectralAnalysisPointsCollection.Count 313
RobotOM.IRobotSpectralAnalysisPointsCollection.Get 315
RobotOM.IRobotSpectralAnalysisPointsCollection.LoadFromFile 315
RobotOM.IRobotSpectralAnalysisPointsCollection.Remove 315
RobotOM.IRobotSpectralAnalysisPointsCollection.SaveToFile 316
RobotOM.IRobotSpectralAnalysisPointsCollection.Set 316
RobotOM.IRobotSpectralAnalysisSpectrum 308
RobotOM.IRobotSpectralAnalysisSpectrum.AbscissaXAxis 309
RobotOM.IRobotSpectralAnalysisSpectrum.AbscissaXAxisLogarithmicScale 309
RobotOM.IRobotSpectralAnalysisSpectrum.AddFromTimeHistory 311
RobotOM.IRobotSpectralAnalysisSpectrum.Average 312
RobotOM.IRobotSpectralAnalysisSpectrum.Damping 309
RobotOM.IRobotSpectralAnalysisSpectrum.LoadFromFile 312
RobotOM.IRobotSpectralAnalysisSpectrum.Name 310
RobotOM.IRobotSpectralAnalysisSpectrum.OrdinateYAxis 310
RobotOM.IRobotSpectralAnalysisSpectrum.OrdinateYAxisLogarithmicScale 310
RobotOM.IRobotSpectralAnalysisSpectrum.Points 311
RobotOM.IRobotSpectralAnalysisSpectrum.SaveToFile 312
RobotOM.IRobotSpectralCoefficients 930
RobotOM.IRobotSpectralCoefficients.ModeCoef 931
RobotOM.IRobotSpectralCoefficients.PartCoef 931
RobotOM.IRobotSpectralCoefficients.SpectrCoef 932
RobotOM.IRobotStorey 1052
RobotOM.IRobotStorey.AutomaticSelection 1053
RobotOM.IRobotStorey.Color 1054
RobotOM.IRobotStorey.Ex1 1054
RobotOM.IRobotStorey.Ey1 1054
RobotOM.IRobotStorey.Height 1055
RobotOM.IRobotStorey.Lx 1055
RobotOM.IRobotStorey.Ly 1055
RobotOM.IRobotStorey.Name 1055
RobotOM.IRobotStorey.Objects 1056
RobotOM.IRobotStorey.SetHeight 1056
RobotOM.IRobotStorey.SetTopLevel 1057
RobotOM.IRobotStorey.TopLevel 1056
RobotOM.IRobotStoreyDisplacements 1074
RobotOM.IRobotStoreyDisplacements.DrUX 1075
RobotOM.IRobotStoreyDisplacements.DrUY 1075
RobotOM.IRobotStoreyDisplacements.MaxUX 1075
RobotOM.IRobotStoreyDisplacements.MaxUY 1075
RobotOM.IRobotStoreyDisplacements.MinUX 1076
RobotOM.IRobotStoreyDisplacements.MinUY 1076

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRobotStoreyDisplacements.NodeMaxUX 1076
RobotOM.IRobotStoreyDisplacements.NodeMaxUY 1076
RobotOM.IRobotStoreyDisplacements.NodeMinUX 1077
RobotOM.IRobotStoreyDisplacements.NodeMinUY 1077
RobotOM.IRobotStoreyMngr 1057
RobotOM.IRobotStoreyMngr.BaseLevel 1058
RobotOM.IRobotStoreyMngr.Count 1059
RobotOM.IRobotStoreyMngr.Create2 1060
RobotOM.IRobotStoreyMngr.Create2Ex 1060
RobotOM.IRobotStoreyMngr.Delete 1060
RobotOM.IRobotStoreyMngr.DeleteAll 1061
RobotOM.IRobotStoreyMngr.DisregardedObjects 1059
RobotOM.IRobotStoreyMngr.FilterStorey 1059
RobotOM.IRobotStoreyMngr.Find 1061
RobotOM.IRobotStoreyMngr.Get 1061
RobotOM.IRobotStoreyReducedForces 1069
RobotOM.IRobotStoreyReducedForces.FX 1071
RobotOM.IRobotStoreyReducedForces.FX_ToColumns 1071
RobotOM.IRobotStoreyReducedForces.FX_ToWalls 1071
RobotOM.IRobotStoreyReducedForces.FY 1071
RobotOM.IRobotStoreyReducedForces.FY_ToColumns 1072
RobotOM.IRobotStoreyReducedForces.FY_ToWalls 1072
RobotOM.IRobotStoreyReducedForces.FZ 1072
RobotOM.IRobotStoreyReducedForces.FZ_ToColumns 1072
RobotOM.IRobotStoreyReducedForces.FZ_ToWalls 1073
RobotOM.IRobotStoreyReducedForces.MX 1073
RobotOM.IRobotStoreyReducedForces.MY 1073
RobotOM.IRobotStoreyReducedForces.MZ 1073
RobotOM.IRobotStoreyResultServer 1077
RobotOM.IRobotStoreyResultServer.Displacements 1078
RobotOM.IRobotStoreyResultServer.ReducedForces 1078
RobotOM.IRobotStoreyResultServer.Values 1079
RobotOM.IRobotStoreySelection 1062
RobotOM.IRobotStoreySelection.Add 1063
RobotOM.IRobotStoreySelection.AddAll 1063
RobotOM.IRobotStoreySelection.Clear 1063
RobotOM.IRobotStoreySelection.GetNames 1064
RobotOM.IRobotStoreySelection.GetNumbers 1064
RobotOM.IRobotStoreySelection.Remove 1064
RobotOM.IRobotStoreyValues 1065
RobotOM.IRobotStoreyValues.Ex0 1066
RobotOM.IRobotStoreyValues.Ex1 1066
RobotOM.IRobotStoreyValues.Ex2 1066
RobotOM.IRobotStoreyValues.Ey0 1067
RobotOM.IRobotStoreyValues.Ey1 1067
RobotOM.IRobotStoreyValues.Ey2 1067
RobotOM.IRobotStoreyValues.G 1068
RobotOM.IRobotStoreyValues.Ix 1068
RobotOM.IRobotStoreyValues.Iy 1068
RobotOM.IRobotStoreyValues.Iz 1068
RobotOM.IRobotStoreyValues.Mass 1069
RobotOM.IRobotStoreyValues.R 1069
RobotOM.IRobotSTRFileAnalyser 2219
RobotOM.IRobotSTRFileAnalyser.InsertParams 2220
RobotOM.IRobotSTRFileAnalyser.InsertToProject 2220
RobotOM.IRobotSTRFileAnalyser.Params 2220
RobotOM.IRobotSTRFileAnalyser.ReadParams 2221
RobotOM.IRobotSTRParameter 1276
RobotOM.IRobotSTRParameter.Description 1277
RobotOM.IRobotSTRParameter.DoubleVal_1 1278
RobotOM.IRobotSTRParameter.DoubleVal_2 1278
RobotOM.IRobotSTRParameter.DoubleVal_3 1278
RobotOM.IRobotSTRParameter.FilePathVal 1279
RobotOM.IRobotSTRParameter.IntegerVal 1279
RobotOM.IRobotSTRParameter.IsActive 1279
RobotOM.IRobotSTRParameter.Name 1279
RobotOM.IRobotSTRParameter.SelectionVal 1280
RobotOM.IRobotSTRParameter.TextList 1280
RobotOM.IRobotSTRParameter.TextVal 1280
RobotOM.IRobotSTRParameter.Type 1280
RobotOM.IRobotSTRParams 1274
RobotOM.IRobotSTRParams.Count 1275
RobotOM.IRobotSTRParams.FindParameter 1276
RobotOM.IRobotSTRParams.GetParameter 1276
RobotOM.IRobotSTRParamType 1281
RobotOM.IRobotStructuralAxisGrid 1526
RobotOM.IRobotStructuralAxisGrid.Name 1526
RobotOM.IRobotStructuralAxisGrid.Save 1527
RobotOM.IRobotStructuralAxisGrid.Type 1527
RobotOM.IRobotStructuralAxisGridCartesian 1527
RobotOM.IRobotStructuralAxisGridCartesian.GetRelativeToPoint 1531
RobotOM.IRobotStructuralAxisGridCartesian.IncludeStoreysInZ 1528

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRobotStructuralAxisGridCartesian.RotationAngle 1529

RobotOM.IRobotStructuralAxisGridCartesian.RotationAxis 1529

RobotOM.IRobotStructuralAxisGridCartesian.SetRelativeToPoint 1531

RobotOM.IRobotStructuralAxisGridCartesian.StoreysInZ 1529

RobotOM.IRobotStructuralAxisGridCartesian.X 1530

RobotOM.IRobotStructuralAxisGridCartesian.Y 1530

RobotOM.IRobotStructuralAxisGridCartesian.Z 1530

RobotOM.IRobotStructuralAxisGridMngr 1531

RobotOM.IRobotStructuralAxisGridMngr.Activate 1533

RobotOM.IRobotStructuralAxisGridMngr.Clear 1533

RobotOM.IRobotStructuralAxisGridMngr.Count 1532

RobotOM.IRobotStructuralAxisGridMngr.Create 1533

RobotOM.IRobotStructuralAxisGridMngr.Delete 1533

RobotOM.IRobotStructuralAxisGridMngr.FindByName 1534

RobotOM.IRobotStructuralAxisGridMngr.Get 1534

RobotOM.IRobotStructuralAxisGridMngr.GetByName 1534

RobotOM.IRobotStructuralAxisGridMngr.IsActive 1535

RobotOM.IRobotStructuralAxisGridType 1525

RobotOM.IRobotStructuralAxisLabelType 1520

RobotOM.IRobotStructuralAxisSequenceList 1520

RobotOM.IRobotStructuralAxisSequenceList.AddSequence 1523

RobotOM.IRobotStructuralAxisSequenceList.AxisCount 1522

RobotOM.IRobotStructuralAxisSequenceList.Clear 1523

RobotOM.IRobotStructuralAxisSequenceList.DeleteSequence 1523

RobotOM.IRobotStructuralAxisSequenceList.FindAxisByPos 1524

RobotOM.IRobotStructuralAxisSequenceList.GetAxis 1524

RobotOM.IRobotStructuralAxisSequenceList.GetSequence 1524

RobotOM.IRobotStructuralAxisSequenceList.SequenceCount 1522

RobotOM.IRobotStructuralAxisSequenceList.SetAxisLabel 1524

RobotOM.IRobotStructuralAxisSequenceList.SetLabelFormat 1525

RobotOM.IRobotStructuralAxisSequenceList.SingleOutAxis 1525

RobotOM.IRobotStructuralAxisSequenceList.StartPosition 1522

RobotOM.IRobotStructure 1090

RobotOM.IRobotStructure.ApplyCache 1097

RobotOM.IRobotStructure.Bars 1092

RobotOM.IRobotStructure.Cases 1092

RobotOM.IRobotStructure.Clear 1097

RobotOM.IRobotStructure.CreateCache 1097

RobotOM.IRobotStructure.Edit 1092

RobotOM.IRobotStructure.ExportXml 1098

RobotOM.IRobotStructure.ExtParams 1092

RobotOM.IRobotStructure.FiniteElems 1093

RobotOM.IRobotStructure.GroupObjects 1093

RobotOM.IRobotStructure.Groups 1093

RobotOM.IRobotStructure.IsCalcModelGenerated 1094

RobotOM.IRobotStructure.Labels 1094

RobotOM.IRobotStructure.Merge 1098

RobotOM.IRobotStructure.Nodes 1094

RobotOM.IRobotStructure.Objects 1094

RobotOM.IRobotStructure.QuantitySurvey 1095

RobotOM.IRobotStructure.Results 1095

RobotOM.IRobotStructure.ResultsFreeze 1095

RobotOM.IRobotStructure.Selections 1095

RobotOM.IRobotStructure.Storeys 1096

RobotOM.IRobotStructure.Type 1096

RobotOM.IRobotStructureAnalysisModalParticipationCoeff 1579

RobotOM.IRobotStructureAnalysisParams 1567

RobotOM.IRobotStructureAnalysisParams.AutoBarMerging 1569

RobotOM.IRobotStructureAnalysisParams.AutoVerification 1569

RobotOM.IRobotStructureAnalysisParams.DSCAlgorithm 1569

RobotOM.IRobotStructureAnalysisParams.EquationSolvingMethod 1570

RobotOM.IRobotStructureAnalysisParams.FictitiousRigidityCoeff 1570

RobotOM.IRobotStructureAnalysisParams.IgnoreWarnings 1570

RobotOM.IRobotStructureAnalysisParams.IterativeParams 1571

RobotOM.IRobotStructureAnalysisParams.ModalParticipationCoeff 1571

RobotOM.IRobotStructureAnalysisParams.RLINKElements 1571

RobotOM.IRobotStructureAnalysisParams.SparseMParams 1571

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRobotStructureApplyInfo 1101
RobotOM.IRobotStructureApplyInfo.Bars 1102
RobotOM.IRobotStructureApplyInfo.Nodes 1102
RobotOM.IRobotStructureAutoVerificationType 1572
RobotOM.IRobotStructureCache 1098
RobotOM.IRobotStructureCache.AddBar 1099
RobotOM.IRobotStructureCache.AddNode 1100
RobotOM.IRobotStructureCache.EnsureNodeExist 1100
RobotOM.IRobotStructureCache.SetBarLabel 1100
RobotOM.IRobotStructureCache.SetBarName 1101
RobotOM.IRobotStructureEditTools 1048
RobotOM.IRobotStructureEditTools.DivideBar 1049
RobotOM.IRobotStructureEditTools.SelMirror 1050
RobotOM.IRobotStructureEditTools.SelRotate 1050
RobotOM.IRobotStructureEditTools.SelScale 1050
RobotOM.IRobotStructureEditTools.SelTranslate 1051
RobotOM.IRobotStructureEditTools.TranslateBar 1051
RobotOM.IRobotStructureEditTools.TranslateNode 1052
RobotOM.IRobotStructureGeoAnalyser 2222
RobotOM.IRobotStructureGeoAnalyser.CanEliminateIsolated Nodes
2224
RobotOM.IRobotStructureGeoAnalyser.CanExtendBars 2224
RobotOM.IRobotStructureGeoAnalyser.CanUseRigidLinks 2224
RobotOM.IRobotStructureGeoAnalyser.Correct 2225
RobotOM.IRobotStructureGeoAnalyser.DefineIntersection 2226
RobotOM.IRobotStructureGeoAnalyser.Precision 2225
RobotOM.IRobotStructureGeoAnalyser.SetEdgeSize 2226
RobotOM.IRobotStructureGeoAnalyser.SetNodeSize 2226
RobotOM.IRobotStructureGeoAnalyser.StartCollectInfo 2227
RobotOM.IRobotStructureGeoAnalyser.StopCollectInfo 2227
RobotOM.IRobotStructureMergeData 1102
RobotOM.IRobotStructureMergeData.CreateStructure 1104
RobotOM.IRobotStructureMergeData.LoadStructure 1104
RobotOM.IRobotStructureMergeData.Structure 1103
RobotOM.IRobotStructureQuantitySurvey 1088
RobotOM.IRobotStructureQuantitySurvey.BarSections 1089
RobotOM.IRobotStructureQuantitySurvey.Materials 1089
RobotOM.IRobotStructureQuantitySurvey.PanelThickness 1089
RobotOM.IRobotStructureValues 1019
RobotOM.IRobotStructureValues.GetEx2 1020
RobotOM.IRobotStructureValues.GetEy2 1021
RobotOM.IRobotStructureValues.GetG 1021
RobotOM.IRobotStructureValues.GetIx 1021
RobotOM.IRobotStructureValues.GetIy 1022
RobotOM.IRobotStructureValues.GetIz 1022
RobotOM.IRobotStructureValues.GetMass 1022
RobotOM.IRobotStructureValues.GetT 1023
RobotOM.IRobotSupportColumnFixingType 526
RobotOM.IRobotSupportEquivalentColumnElasticity 524
RobotOM.IRobotSupportEquivalentColumnElasticity.Fixing1 525
RobotOM.IRobotSupportEquivalentColumnElasticity.Fixing2 525
RobotOM.IRobotSupportEquivalentColumnElasticity.L1 525
RobotOM.IRobotSupportEquivalentColumnElasticity.L2 526
RobotOM.IRobotSupportEquivalentColumnElasticity.ThroughTwoStories
526
RobotOM.IRobotSupportEquivalentElasticity 520
RobotOM.IRobotSupportEquivalentElasticity.E 521
RobotOM.IRobotSupportEquivalentElasticity.PoissonRatio 521
RobotOM.IRobotSupportEquivalentElasticity.SetMaterial 522
RobotOM.IRobotSupportEquivalentElasticity.Type 521
RobotOM.IRobotSupportEquivalentElasticityType 522
RobotOM.IRobotSupportEquivalentWallElasticity 522
RobotOM.IRobotSupportEquivalentWallElasticity.L1 523
RobotOM.IRobotSupportEquivalentWallElasticity.L2 523
RobotOM.IRobotSupportEquivalentWallElasticity.ThroughTwo Stories
524
RobotOM.IRobotSurfaceOnObjectRecordValues 52
RobotOM.IRobotSWCodeECCdType 148
RobotOM.IRobotSWCodeECGroundType 147
RobotOM.IRobotSWCodeECParams 131
RobotOM.IRobotSWCodeECParams.Altitude 134
RobotOM.IRobotSWCodeECParams.GlobalCDIR 134
RobotOM.IRobotSWCodeECParams.GroundType 134
RobotOM.IRobotSWCodeECParams.LeftWind2NordAngle 135
RobotOM.IRobotSWCodeECParams.NodalLoadsForAllBars 135
RobotOM.IRobotSWCodeECParams.NodalLoadsForBarsList 135
RobotOM.IRobotSWCodeECParams.PermDoorFront 135
RobotOM.IRobotSWCodeECParams.PermDoorFrontPresent

(c) 2014 Autodesk Inc. All rights reserved

136
RobotOM.IRobotSWCodeECParams.PermDoorLeftSide 136
RobotOM.IRobotSWCodeECParams.PermDoorLeftSidePresent 136
RobotOM.IRobotSWCodeECParams.PermDoorRear 136
RobotOM.IRobotSWCodeECParams.PermDoorRearPresent 137
RobotOM.IRobotSWCodeECParams.PermDoorRightSide 137
RobotOM.IRobotSWCodeECParams.PermDoorRightSidePresent 137
RobotOM.IRobotSWCodeECParams.PermFront 137
RobotOM.IRobotSWCodeECParams.PermLeftSide 138
RobotOM.IRobotSWCodeECParams.PermRear 138
RobotOM.IRobotSWCodeECParams.PermRightSide 138
RobotOM.IRobotSWCodeECParams.ReferenceLevel 138
RobotOM.IRobotSWCodeECParams.SnowBarCoeffGet 146
RobotOM.IRobotSWCodeECParams.SnowBarCoeffSet 146
RobotOM.IRobotSWCodeECParams.SnowGutterBars 139
RobotOM.IRobotSWCodeECParams.SnowPressureExtreme 139
RobotOM.IRobotSWCodeECParams.SnowPressureNormal 139
RobotOM.IRobotSWCodeECParams.SnowRedistribution 140
RobotOM.IRobotSWCodeECParams.StructureAge 140
RobotOM.IRobotSWCodeECParams.StructureAgeCode 140
RobotOM.IRobotSWCodeECParams.StructureHeight 140
RobotOM.IRobotSWCodeECParams.StructureP 141
RobotOM.IRobotSWCodeECParams.WindAutoCd 141
RobotOM.IRobotSWCodeECParams.WindBarCoeffGet 146
RobotOM.IRobotSWCodeECParams.WindBarCoeffSet 147
RobotOM.IRobotSWCodeECParams.WindCALT 141
RobotOM.IRobotSWCodeECParams.WindCd 141
RobotOM.IRobotSWCodeECParams.WindCDIR 142
RobotOM.IRobotSWCodeECParams.WindCdType 142
RobotOM.IRobotSWCodeECParams.WindCt 142
RobotOM.IRobotSWCodeECParams.WindCtAuto 142
RobotOM.IRobotSWCodeECParams.WindCTEM 143
RobotOM.IRobotSWCodeECParams.WindE 143
RobotOM.IRobotSWCodeECParams.WindKT 143
RobotOM.IRobotSWCodeECParams.WindPressureAutomatic 143
RobotOM.IRobotSWCodeECParams.WindQref 144
RobotOM.IRobotSWCodeECParams.WindQref0 144
RobotOM.IRobotSWCodeECParams.WindQref0p 144
RobotOM.IRobotSWCodeECParams.WindSiteType 145
RobotOM.IRobotSWCodeECParams.WindVref0 145
RobotOM.IRobotSWCodeECParams.WindZ0 145
RobotOM.IRobotSWCodeECParams.WindZMin 145
RobotOM.IRobotSWCodeECSiteType 147
RobotOM.IRobotSWCodeFRParams 110
RobotOM.IRobotSWCodeFRParams.Altitude 114
RobotOM.IRobotSWCodeFRParams.IsolatedRoofGetLocation 128
RobotOM.IRobotSWCodeFRParams.IsolatedRoofs 114
RobotOM.IRobotSWCodeFRParams.IsolatedRoofSetLocation 128
RobotOM.IRobotSWCodeFRParams.NodalLoadsForAllBars 114
RobotOM.IRobotSWCodeFRParams.NodalLoadsForBarsList 115
RobotOM.IRobotSWCodeFRParams.OpenStructure 115
RobotOM.IRobotSWCodeFRParams.PermDoorFront 115
RobotOM.IRobotSWCodeFRParams.PermDoorFrontPresent 115
RobotOM.IRobotSWCodeFRParams.PermDoorLeftSide 116
RobotOM.IRobotSWCodeFRParams.PermDoorLeftSidePresent 116
RobotOM.IRobotSWCodeFRParams.PermDoorRear 116
RobotOM.IRobotSWCodeFRParams.PermDoorRearPresent 116
RobotOM.IRobotSWCodeFRParams.PermDoorRightSide 117
RobotOM.IRobotSWCodeFRParams.PermDoorRightSidePresent 117
RobotOM.IRobotSWCodeFRParams.PermFront 117
RobotOM.IRobotSWCodeFRParams.PermLeftSide 117
RobotOM.IRobotSWCodeFRParams.PermRear 118
RobotOM.IRobotSWCodeFRParams.PermRightSide 118
RobotOM.IRobotSWCodeFRParams.ReferenceLevel 118
RobotOM.IRobotSWCodeFRParams.RiseOfRoof 119
RobotOM.IRobotSWCodeFRParams.RiseOfRoofAutomatic 119
RobotOM.IRobotSWCodeFRParams.SnowBarCoeffGet 128
RobotOM.IRobotSWCodeFRParams.SnowBarCoeffSet 128
RobotOM.IRobotSWCodeFRParams.SnowGutterBars 119
RobotOM.IRobotSWCodeFRParams.SnowIsWaterOutflow 119

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRobotSWCodeFRParams.SnowObstacles 120
RobotOM.IRobotSWCodeFRParams.SnowPressureExtreme 120
RobotOM.IRobotSWCodeFRParams.SnowPressureExtremeAutomatic 120
RobotOM.IRobotSWCodeFRParams.SnowPressureNormal 120
RobotOM.IRobotSWCodeFRParams.SnowPressureNormalAutomatic 121
RobotOM.IRobotSWCodeFRParams.SnowRedistribution 121
RobotOM.IRobotSWCodeFRParams.SnowRegion 121
RobotOM.IRobotSWCodeFRParams.SnowType 121
RobotOM.IRobotSWCodeFRParams.SnowWaterOutflow 122
RobotOM.IRobotSWCodeFRParams.StructureHeight 122
RobotOM.IRobotSWCodeFRParams.SurfaceLower 122
RobotOM.IRobotSWCodeFRParams.SurfaceUpper 122
RobotOM.IRobotSWCodeFRParams.WindBarCoeffGet 129
RobotOM.IRobotSWCodeFRParams.WindBarCoeffSet 129
RobotOM.IRobotSWCodeFRParams.WindCoastalArea 123
RobotOM.IRobotSWCodeFRParams.WindDeltaCoeff 123
RobotOM.IRobotSWCodeFRParams.WindDeltaCoeffAutomatic 123
RobotOM.IRobotSWCodeFRParams.WindDynamicAction 124
RobotOM.IRobotSWCodeFRParams.WindDynamicActionCoeff 124
RobotOM.IRobotSWCodeFRParams.WindDynamicActionCoeffAutomatic 124
RobotOM.IRobotSWCodeFRParams.WindDynamicActionPeriod 124
RobotOM.IRobotSWCodeFRParams.WindDynamicActionSteelStructure 125
RobotOM.IRobotSWCodeFRParams.WindFacadeOffset 125
RobotOM.IRobotSWCodeFRParams.WindMultipleRoof 125
RobotOM.IRobotSWCodeFRParams.WindPressure 125
RobotOM.IRobotSWCodeFRParams.WindPressureAutomatic 126
RobotOM.IRobotSWCodeFRParams.WindPressureCeCiMinimum 126
RobotOM.IRobotSWCodeFRParams.WindPressureVariable 126
RobotOM.IRobotSWCodeFRParams.WindRegion 126
RobotOM.IRobotSWCodeFRParams.WindSite 127
RobotOM.IRobotSWCodeFRParams.WindType 127
RobotOM.IRobotSWCodeFRSnowType 130
RobotOM.IRobotSWCodeFRSurfaceType 130
RobotOM.IRobotSWCodeFRWindSite 129
RobotOM.IRobotSWCodeFRWindType 129
RobotOM.IRobotSWCodePLParams 99
RobotOM.IRobotSWCodePLParams.Altitude 101
RobotOM.IRobotSWCodePLParams.IsolatedRoofGetLocation 107
RobotOM.IRobotSWCodePLParams.IsolatedRoofs 101
RobotOM.IRobotSWCodePLParams.IsolatedRoofSetLocation 108
RobotOM.IRobotSWCodePLParams.NodalLoadsForAllBars 102
RobotOM.IRobotSWCodePLParams.NodalLoadsForBarsList 102
RobotOM.IRobotSWCodePLParams.PermFront 102
RobotOM.IRobotSWCodePLParams.PermLeftSide 102
RobotOM.IRobotSWCodePLParams.PermRear 103
RobotOM.IRobotSWCodePLParams.PermRightSide 103
RobotOM.IRobotSWCodePLParams.ReferenceLevel 103
RobotOM.IRobotSWCodePLParams.SnowBarCoeffGet 108
RobotOM.IRobotSWCodePLParams.SnowBarCoeffSet 108
RobotOM.IRobotSWCodePLParams.SnowPressure 103
RobotOM.IRobotSWCodePLParams.SnowPressureAutomatic 104
RobotOM.IRobotSWCodePLParams.SnowRedistribution 104
RobotOM.IRobotSWCodePLParams.SnowZone 104
RobotOM.IRobotSWCodePLParams.StructureHeight 104
RobotOM.IRobotSWCodePLParams.WindBarCoeffGet 109
RobotOM.IRobotSWCodePLParams.WindBarCoeffSet 109
RobotOM.IRobotSWCodePLParams.WindDynamicAction 105
RobotOM.IRobotSWCodePLParams.WindDynamicDecrement 105
RobotOM.IRobotSWCodePLParams.WindDynamicPeriod 105
RobotOM.IRobotSWCodePLParams.WindMultipleRoofs 106
RobotOM.IRobotSWCodePLParams.WindPressure 106
RobotOM.IRobotSWCodePLParams.WindPressureAutomatic 106
RobotOM.IRobotSWCodePLParams.WindPressureDistribOnHeight 106
RobotOM.IRobotSWCodePLParams.WindSite 107

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRobotSWCodePLParams.WindZone 107
RobotOM.IRobotSWCodePLSnowZone 110
RobotOM.IRobotSWCodePLWindPressDistribType 130
RobotOM.IRobotSWCodePLWindSite 109
RobotOM.IRobotSWCodePLWindZone 109
RobotOM.IRobotSWStruct3D 83
RobotOM.IRobotSWStruct3D.FrameCount 83
RobotOM.IRobotSWStruct3D.FrameElemCount 84
RobotOM.IRobotSWStruct3D.GetFrame 84
RobotOM.IRobotSWStruct3D.SetFrame 84
RobotOM.IRobotSWStruct3DElement 85
RobotOM.IRobotSWStruct3DElement.Bar 85
RobotOM.IRobotSWStruct3DElement.IsFacadeLoaded 86
RobotOM.IRobotSWStruct3DElement.IsFacadeOnly 86
RobotOM.IRobotSWStruct3DElement.Purlins 86
RobotOM.IRobotSWStruct3DFrame 86
RobotOM.IRobotSWStruct3DFrame.ElemCount 87
RobotOM.IRobotSWStruct3DFrame.GetElem 88
RobotOM.IRobotSWStruct3DFrame.SetElem 88
RobotOM.IRobotSWStruct3DGenParams 88
RobotOM.IRobotSWStruct3DGenParams.Bars 89
RobotOM.IRobotSWStruct3DGenParams.FacadeLoadedBars 89
RobotOM.IRobotSWStruct3DGenParams.FacadeOnlyBars 90
RobotOM.IRobotSWStruct3DGenParams.FrameCount 90
RobotOM.IRobotSWStruct3DGenParams.GetPurlins 91
RobotOM.IRobotSWStruct3DGenParams.Offsets 90
RobotOM.IRobotSWStruct3DGenParams.SetPurlins 91
RobotOM.IRobotSWStruct3DGenParams.Spacings 90
RobotOM.IRobotSWStruct3DPurlinGenParams 91
RobotOM.IRobotSWStruct3DPurlinGenParams.Locations 92
RobotOM.IRobotSWStruct3DPurlinGenParams.RelativeLocations 92
RobotOM.IRobotSWStruct3DPurlinGenParams.SectionName 93
RobotOM.IRobotTable 1364
RobotOM.IRobotTable.AddColumn 1369
RobotOM.IRobotTable.ColCount 1366
RobotOM.IRobotTable.Configuration 1366
RobotOM.IRobotTable.GetColWidth 1369
RobotOM.IRobotTable.GetDataType 1369
RobotOM.IRobotTable.GetRowHeight 1370
RobotOM.IRobotTable.MakeScreenCapture 1370
RobotOM.IRobotTable.Printable 1366
RobotOM.IRobotTable.RowCount 1367
RobotOM.IRobotTable.Select 1370
RobotOM.IRobotTable.SelectAllFromStore 1370
RobotOM.IRobotTable.SelectFromStore 1371
RobotOM.IRobotTable.SelectMode 1371
RobotOM.IRobotTable.SelectModeFromStore 1371
RobotOM.IRobotTable.SetColWidth 1372
RobotOM.IRobotTable.SetRowHeight 1372
RobotOM.IRobotTable.StoreModeSelection 1372
RobotOM.IRobotTable.StoreSelection 1373
RobotOM.IRobotTable.Title 1367
RobotOM.IRobotTable.UserControl 1367
RobotOM.IRobotTable.Visible 1367
RobotOM.IRobotTable.Window 1368
RobotOM.IRobotTableConfig 1363
RobotOM.IRobotTableConfig.SetFlag 1364
RobotOM.IRobotTableConfig.SetValue 1364
RobotOM.IRobotTableConfigFlag 1362
RobotOM.IRobotTableConfigValue 1363
RobotOM.IRobotTableDataType 1361
RobotOM.IRobotTableFrame 1373
RobotOM.IRobotTableFrame.Count 1374
RobotOM.IRobotTableFrame.Current 1374
RobotOM.IRobotTableFrame.Get 1375
RobotOM.IRobotTableFrame.GetName 1375
RobotOM.IRobotTableFrame.SetName 1375
RobotOM.IRobotTableFrame.Window 1374
RobotOM.IRobotTableScreenCaptureParams 1376
RobotOM.IRobotTableScreenCaptureParams.AddInfoTab 1377
RobotOM.IRobotTableScreenCaptureParams.AdjustColumnWidth 1377
RobotOM.IRobotTableScreenCaptureParams.Comment 1377
RobotOM.IRobotTableScreenCaptureParams.IncludeDateAndTime 1378
RobotOM.IRobotTableScreenCaptureParams.Name 1378
RobotOM.IRobotTableScreenCaptureParams.UpdateType 1378
RobotOM.IRobotTableScreenCaptureUpdateType 1379
RobotOM.IRobotTableType 1361
RobotOM.IRobotThermalIn3PointsRecordValues 48
RobotOM.IRobotThermalRecordValues 50

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRobotThicknessData 722
RobotOM.IRobotThicknessData.Data 723
RobotOM.IRobotThicknessData.ElasticFoundation 723
RobotOM.IRobotThicknessData.MaterialName 723
RobotOM.IRobotThicknessData.ThicknessType 723
RobotOM.IRobotThicknessData.Uplift 724
RobotOM.IRobotThicknessHomoData 725
RobotOM.IRobotThicknessHomoData.GetP1 727
RobotOM.IRobotThicknessHomoData.GetP2 728
RobotOM.IRobotThicknessHomoData.GetP3 728
RobotOM.IRobotThicknessHomoData.GetReduction 728
RobotOM.IRobotThicknessHomoData.SetP1 728
RobotOM.IRobotThicknessHomoData.SetP2 729
RobotOM.IRobotThicknessHomoData.SetP3 729
RobotOM.IRobotThicknessHomoData.SetReduction 729
RobotOM.IRobotThicknessHomoData.Thick1 726
RobotOM.IRobotThicknessHomoData.Thick2 726
RobotOM.IRobotThicknessHomoData.Thick3 726
RobotOM.IRobotThicknessHomoData.ThickConst 726
RobotOM.IRobotThicknessHomoData.Type 727
RobotOM.IRobotThicknessHomoType 730
RobotOM.IRobotThicknessMatrix 739
RobotOM.IRobotThicknessMatrix.GetValue 740
RobotOM.IRobotThicknessMatrix.SetValue 740
RobotOM.IRobotThicknessMatrixValue 740
RobotOM.IRobotThicknessOrthoData 731
RobotOM.IRobotThicknessOrthoData.A 732
RobotOM.IRobotThicknessOrthoData.A1 733
RobotOM.IRobotThicknessOrthoData.A2 733
RobotOM.IRobotThicknessOrthoData.B 733
RobotOM.IRobotThicknessOrthoData.B1 733
RobotOM.IRobotThicknessOrthoData.DirType 734
RobotOM.IRobotThicknessOrthoData.DisregardBendStiffDirY 734
RobotOM.IRobotThicknessOrthoData.GetVector 738
RobotOM.IRobotThicknessOrthoData.H 734
RobotOM.IRobotThicknessOrthoData.H0 735
RobotOM.IRobotThicknessOrthoData.H1 735
RobotOM.IRobotThicknessOrthoData.H2 735
RobotOM.IRobotThicknessOrthoData.HA 735
RobotOM.IRobotThicknessOrthoData.HB 736
RobotOM.IRobotThicknessOrthoData.HC 736
RobotOM.IRobotThicknessOrthoData.Matrix 736
RobotOM.IRobotThicknessOrthoData.N1 736
RobotOM.IRobotThicknessOrthoData.N2 737
RobotOM.IRobotThicknessOrthoData.SetVector 739
RobotOM.IRobotThicknessOrthoData.Thick1 737
RobotOM.IRobotThicknessOrthoData.Thick2 737
RobotOM.IRobotThicknessOrthoData.Thick3 738
RobotOM.IRobotThicknessOrthoData.Type 738
RobotOM.IRobotThicknessOrthoDirType 730
RobotOM.IRobotThicknessOrthoType 730
RobotOM.IRobotThicknessQuantitySurvey 1085
RobotOM.IRobotThicknessQuantitySurvey.Count 1086
RobotOM.IRobotThicknessQuantitySurvey.GetArea 1087
RobotOM.IRobotThicknessQuantitySurvey.GetName 1087
RobotOM.IRobotThicknessQuantitySurvey.GetUnitWeight 1087
RobotOM.IRobotThicknessQuantitySurvey.GetVolume 1088
RobotOM.IRobotThicknessQuantitySurvey.GetWeight 1088
RobotOM.IRobotThicknessType 724
RobotOM.IRobotThicknessUpliftType 724
RobotOM.IRobotTimeHistoryAnalysisMethod 369
RobotOM.IRobotTimeHistoryAnalysisParams 364
RobotOM.IRobotTimeHistoryAnalysisParams.Count 365
RobotOM.IRobotTimeHistoryAnalysisParams.Delete 368
RobotOM.IRobotTimeHistoryAnalysisParams.Division 366
RobotOM.IRobotTimeHistoryAnalysisParams.End 366
RobotOM.IRobotTimeHistoryAnalysisParams.Find 368
RobotOM.IRobotTimeHistoryAnalysisParams.Get 369
RobotOM.IRobotTimeHistoryAnalysisParams.InitialCase 366
RobotOM.IRobotTimeHistoryAnalysisParams.Method 367
RobotOM.IRobotTimeHistoryAnalysisParams.MethodParams 367
RobotOM.IRobotTimeHistoryAnalysisParams.Set 369
RobotOM.IRobotTimeHistoryAnalysisParams.TimeStep 367
RobotOM.IRobotTimeHistoryFunctionList 378
RobotOM.IRobotTimeHistoryFunctionList.AddFromFile 380
RobotOM.IRobotTimeHistoryFunctionList.Count 379
RobotOM.IRobotTimeHistoryFunctionList.Create 380
RobotOM.IRobotTimeHistoryFunctionList.CreateSum 381
RobotOM.IRobotTimeHistoryFunctionList.Delete 381
RobotOM.IRobotTimeHistoryFunctionList.Find 381
RobotOM.IRobotTimeHistoryFunctionList.Get 382
RobotOM.IRobotTimeHistoryFunctionList.GetName 382
RobotOM.IRobotTimeHistoryFunctionList.SaveToFile 382

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRobotTimeHistoryFunctionList.Store 383
RobotOM.IRobotTimeHistoryHHTParams 386
RobotOM.IRobotTimeHistoryHHTParams.Alpha 386
RobotOM.IRobotTimeHistoryHHTParams.Beta 387
RobotOM.IRobotTimeHistoryHHTParams.CoeffAlpha 387
RobotOM.IRobotTimeHistoryHHTParams.Nonlinearity 387
RobotOM.IRobotTimeHistoryHHTParams.NonlinearParams 388
RobotOM.IRobotTimeHistoryModalDecompositionParams 371
RobotOM.IRobotTimeHistoryModalDecompositionParams.Count 372
RobotOM.IRobotTimeHistoryModalDecompositionParams.Delete 373
RobotOM.IRobotTimeHistoryModalDecompositionParams.DeleteMode 373
RobotOM.IRobotTimeHistoryModalDecompositionParams.Get 373
RobotOM.IRobotTimeHistoryModalDecompositionParams.GetDamping 374
RobotOM.IRobotTimeHistoryModalDecompositionParams.IsDefined 374
RobotOM.IRobotTimeHistoryModalDecompositionParams.SetDamping 374
RobotOM.IRobotTimeHistoryNewmarkAccelParams 388
RobotOM.IRobotTimeHistoryNewmarkAccelParams.Alpha 389
RobotOM.IRobotTimeHistoryNewmarkAccelParams.Beta 389
RobotOM.IRobotTimeHistoryNewmarkAccelParams.Nonlinearity 389
RobotOM.IRobotTimeHistoryNewmarkAccelParams.NonlinearParams 390
RobotOM.IRobotTimeHistoryNewmarkParams 370
RobotOM.IRobotTimeHistoryNewmarkParams.Alpha 370
RobotOM.IRobotTimeHistoryNewmarkParams.Beta 371
RobotOM.IRobotTimeHistoryNewmarkParams.MassMatrixType 371
RobotOM.IRobotTimeHistoryNonlinearParams 383
RobotOM.IRobotTimeHistoryNonlinearParams.MatrixUpdateAfterEachIteration 384
RobotOM.IRobotTimeHistoryNonlinearParams.MatrixUpdateAfterEachSubdivision 384
RobotOM.IRobotTimeHistoryNonlinearParams.MaximumIterationNumberForOneIncrement 385
RobotOM.IRobotTimeHistoryNonlinearParams.PDelta 385
RobotOM.IRobotTimeHistoryNonlinearParams.ResidualForcesRelativeCodeTolerance 385
RobotOM.IRobotTimeHistoryPointsCollection 375
RobotOM.IRobotTimeHistoryPointsCollection.Add 376
RobotOM.IRobotTimeHistoryPointsCollection.Clear 377
RobotOM.IRobotTimeHistoryPointsCollection.Count 376
RobotOM.IRobotTimeHistoryPointsCollection.Delete 377
RobotOM.IRobotTimeHistoryPointsCollection.Find 377
RobotOM.IRobotTimeHistoryPointsCollection.Get 378
RobotOM.IRobotTimeHistoryPointsCollection.LoadFromFile 378
RobotOM.IRobotTimeHistoryPointsCollection.SaveToFile 378
RobotOM.IRobotTimeHistoryResults 904
RobotOM.IRobotTimeHistoryResults.ARX 905
RobotOM.IRobotTimeHistoryResults.ARY 906
RobotOM.IRobotTimeHistoryResults.ARZ 906
RobotOM.IRobotTimeHistoryResults.AX 906
RobotOM.IRobotTimeHistoryResults.AY 907
RobotOM.IRobotTimeHistoryResults.AZ 907
RobotOM.IRobotTimeHistoryResults.Time 907
RobotOM.IRobotTimeHistoryResults.VRX 908
RobotOM.IRobotTimeHistoryResults.VRY 908
RobotOM.IRobotTimeHistoryResults.VRZ 908
RobotOM.IRobotTimeHistoryResults.VX 908
RobotOM.IRobotTimeHistoryResults.VY 909
RobotOM.IRobotTimeHistoryResults.VZ 909
RobotOM.IRobotTimeHistoryResultServer 933
RobotOM.IRobotTimeHistoryResultServer.Value 934
RobotOM.IRobotTranslateOptions 1048
RobotOM.IRobotUniformRecordValues 49
RobotOM.IRobotUnitComplexData 1290
RobotOM.IRobotUnitComplexData.Name2 1290
RobotOM.IRobotUnitData 1283
RobotOM.IRobotUnitData.E 1284
RobotOM.IRobotUnitData.Name 1284
RobotOM.IRobotUnitData.Precision 1285
RobotOM.IRobotUnitData.Type 1285
RobotOM.IRobotUnitEditionData 1294

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRobotUnitEditionData.Coefficient 1294
RobotOM.IRobotUnitEditionData.Type 1295
RobotOM.IRobotUnitEditionData.Unit 1295
RobotOM.IRobotUnitEditionServer 1291
RobotOM.IRobotUnitEditionServer.Count 1292
RobotOM.IRobotUnitEditionServer.Delete 1292
RobotOM.IRobotUnitEditionServer.Find 1292
RobotOM.IRobotUnitEditionServer.Get 1293
RobotOM.IRobotUnitEditionServer.New 1293
RobotOM.IRobotUnitEditionServer.Set 1293
RobotOM.IRobotUnitEditionType 1285
RobotOM.IRobotUnitMngr 1286
RobotOM.IRobotUnitMngr.Count 1288
RobotOM.IRobotUnitMngr.Get 1288
RobotOM.IRobotUnitMngr.GetCoeff 1288
RobotOM.IRobotUnitMngr.GetCoeff2 1289
RobotOM.IRobotUnitMngr.GetName 1289
RobotOM.IRobotUnitMngr.Refresh 1289
RobotOM.IRobotUnitMngr.Set 1289
RobotOM.IRobotUnitMngr.UnitEdition 1287
RobotOM.IRobotUnitMngr.UseMetricAsDefault 1287
RobotOM.IRobotUnitMngrEvents 1295
RobotOM.IRobotUnitMngrEvents.UnitsChanged 1296
RobotOM.IRobotUnitType 1282
RobotOM.IRobotUniversalResultAccess 1007
RobotOM.IRobotUniversalResultAccess.Available 1009
RobotOM.IRobotUniversalResultAccess.Bar 1010
RobotOM.IRobotUniversalResultAccess.CalcPoint 1010
RobotOM.IRobotUniversalResultAccess.DivCount 1010
RobotOM.IRobotUniversalResultAccess.DivPoint 1010
RobotOM.IRobotUniversalResultAccess.Element 1011
RobotOM.IRobotUniversalResultAccess.GetDirX 1016
RobotOM.IRobotUniversalResultAccess.Layer 1011
RobotOM.IRobotUniversalResultAccess.LayerArbitraryValue 1011
RobotOM.IRobotUniversalResultAccess.LinearSupports 1012
RobotOM.IRobotUniversalResultAccess.LoadCase 1012
RobotOM.IRobotUniversalResultAccess.LoadCaseCmpnt 1012
RobotOM.IRobotUniversalResultAccess.Mode 1013
RobotOM.IRobotUniversalResultAccess.ModeCmb 1013
RobotOM.IRobotUniversalResultAccess.Node 1013
RobotOM.IRobotUniversalResultAccess.Panel 1013
RobotOM.IRobotUniversalResultAccess.ReducedCutPos 1014
RobotOM.IRobotUniversalResultAccess.ReinforceCalcMethod 1014
RobotOM.IRobotUniversalResultAccess.RelativePoint 1014
RobotOM.IRobotUniversalResultAccess.Reset 1016
RobotOM.IRobotUniversalResultAccess.ResultId 1015
RobotOM.IRobotUniversalResultAccess.ResultType 1015
RobotOM.IRobotUniversalResultAccess.ResultValue 1015
RobotOM.IRobotUniversalResultAccess.ResultValue3D 1017
RobotOM.IRobotUniversalResultAccess.SetDirX 1017
RobotOM.IRobotUniversalResultAccess.Storey 1015
RobotOM.IRobotUniversalResultType 1019
RobotOM.IRobotUpliftDirection 545
RobotOM.IRobotUpliftSense 545
RobotOM.IRobotValuesArray 1636
RobotOM.IRobotValuesArray.Count 1637
RobotOM.IRobotValuesArray.Get 1637
RobotOM.IRobotValuesArray.Set 1637
RobotOM.IRobotValuesArray.SetSize 1638
RobotOM.IRobotVariableMngr 1487
RobotOM.IRobotVariableMngr.AddExtension 1488
RobotOM.IRobotVariableMngr.Delete 1489
RobotOM.IRobotVariableMngr.Exist 1489
RobotOM.IRobotVariableMngr.GetPredefinedValue 1489
RobotOM.IRobotVariableMngr.GetValue 1489
RobotOM.IRobotVariableMngr.RemoveExtension 1490
RobotOM.IRobotVariableMngr.SetPredefinedValue 1490
RobotOM.IRobotVariableMngr.SetValue 1490
RobotOM.IRobotVariableMngrExtension 1518
RobotOM.IRobotVariableMngrExtension.GetIndexedValue 1519
RobotOM.IRobotVariableMngrExtension.GetValue 1519
RobotOM.IRobotVariablePredefinedId 1491
RobotOM.IRobotVehicleData 343
RobotOM.IRobotVehicleData.b 344
RobotOM.IRobotVehicleData.d1 344
RobotOM.IRobotVehicleData.d2 345
RobotOM.IRobotVehicleData.LoadFromDBase 346
RobotOM.IRobotVehicleData.Loads 345
RobotOM.IRobotVehicleData.StoreToDBase 346
RobotOM.IRobotVehicleDatabase 356
RobotOM.IRobotVehicleDatabase.Description 357

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRobotVehicleDatabase.GetAll 358
RobotOM.IRobotVehicleDatabase.Load 359
RobotOM.IRobotVehicleDatabase.LoadFromFile 359
RobotOM.IRobotVehicleDatabase.LongName 357
RobotOM.IRobotVehicleDatabase.Name 358
RobotOM.IRobotVehicleDatabaseList 352
RobotOM.IRobotVehicleDatabaseList.Add 354
RobotOM.IRobotVehicleDatabaseList.AddFromFile 354
RobotOM.IRobotVehicleDatabaseList.Count 353
RobotOM.IRobotVehicleDatabaseList.Create 355
RobotOM.IRobotVehicleDatabaseList.Default 353
RobotOM.IRobotVehicleDatabaseList.Find 355
RobotOM.IRobotVehicleDatabaseList.Get 355
RobotOM.IRobotVehicleDatabaseList.GetDatabase 356
RobotOM.IRobotVehicleDatabaseList.Remove 356
RobotOM.IRobotVehicleLoad 348
RobotOM.IRobotVehicleLoad.DX 350
RobotOM.IRobotVehicleLoad.DY 350
RobotOM.IRobotVehicleLoad.F 350
RobotOM.IRobotVehicleLoad.S 351
RobotOM.IRobotVehicleLoad.Type 351
RobotOM.IRobotVehicleLoad.X 351
RobotOM.IRobotVehicleLoadMngr 346
RobotOM.IRobotVehicleLoadMngr.Count 347
RobotOM.IRobotVehicleLoadMngr.Delete 347
RobotOM.IRobotVehicleLoadMngr.Get 348
RobotOM.IRobotVehicleLoadMngr.New 348
RobotOM.IRobotVehicleLoadType 352
RobotOM.IRobotView 1396
RobotOM.IRobotView.CopyToClipboard 1401
RobotOM.IRobotView.DiagramMagnification 1398
RobotOM.IRobotView.GetRotationPoint 1401
RobotOM.IRobotView.GetScale 1402
RobotOM.IRobotView.GetSize 1402
RobotOM.IRobotView.GetWorkPoint 1402
RobotOM.IRobotView.GetZoom 1402
RobotOM.IRobotView.IsLocal 1403
RobotOM.IRobotView.ParamsBarMap 1398
RobotOM.IRobotView.ParamsDiagram 1398
RobotOM.IRobotView.ParamsDisplay 1398
RobotOM.IRobotView.ParamsFeMap 1399
RobotOM.IRobotView.Printable 1399
RobotOM.IRobotView.Projection 1399
RobotOM.IRobotView.Redraw 1403
RobotOM.IRobotView.Rotate 1403
RobotOM.IRobotView.Selection 1399
RobotOM.IRobotView.SetGlobal 1403
RobotOM.IRobotView.SetLocal 1404
RobotOM.IRobotView.SetRotationPoint 1404
RobotOM.IRobotView.SetScale 1404
RobotOM.IRobotView.SetSize 1404
RobotOM.IRobotView.SetWorkPoint 1405
RobotOM.IRobotView.SetZoom 1405
RobotOM.IRobotView.Title 1400
RobotOM.IRobotView.Type 1400
RobotOM.IRobotView.Visible 1400
RobotOM.IRobotView2 1445
RobotOM.IRobotView2.ParamsPanelCut 1446
RobotOM.IRobotView2.UserControl 1447
RobotOM.IRobotView2.Window 1447
RobotOM.IRobotView3 1459
RobotOM.IRobotView3.MakeScreenCapture 1461
RobotOM.IRobotViewBarMapParams 1430
RobotOM.IRobotViewBarMapParams.CurrentResult 1431
RobotOM.IRobotViewBarMapParams.Descriptions 1431
RobotOM.IRobotViewBarMapParams.MapThicknessCoeff 1432
RobotOM.IRobotViewBarMapParams.StructureDeformation 1432
RobotOM.IRobotViewBarMapResultType 1430
RobotOM.IRobotViewBarMaps 1429
RobotOM.IRobotViewDetailedAnalysis 1412
RobotOM.IRobotViewDetailedAnalysis.CurrentTableTab 1414
RobotOM.IRobotViewDetailedAnalysis.MakeScreenCapture 1415
RobotOM.IRobotViewDetailedAnalysis.ParamsDetailed 1414
RobotOM.IRobotViewDetailedAnalysis.UserControl 1414
RobotOM.IRobotViewDetailedAnalysis.Window 1415
RobotOM.IRobotViewDetailedAnalysisParams 1433
RobotOM.IRobotViewDetailedAnalysisParams.Descriptions 1434
RobotOM.IRobotViewDetailedAnalysisParams.Filling 1434
RobotOM.IRobotViewDetailedAnalysisParams.GetColor 1435
RobotOM.IRobotViewDetailedAnalysisParams.IsOn 1436
RobotOM.IRobotViewDetailedAnalysisParams.PositiveNegative 1435

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRobotViewDetailedAnalysisParams.ReinforceShowTheoreticAndRealVals 1435
RobotOM.IRobotViewDetailedAnalysisParams.Set 1436
RobotOM.IRobotViewDetailedAnalysisParams.SetColor 1436
RobotOM.IRobotViewDetailedAnalysisResultType 1432
RobotOM.IRobotViewDetailedAnalysisTableTab 1415
RobotOM.IRobotViewDiagramDescriptionType 1427
RobotOM.IRobotViewDiagramFillingType 1428
RobotOM.IRobotViewDiagramParams 1422
RobotOM.IRobotViewDiagramParams.Descriptions 1423
RobotOM.IRobotViewDiagramParams.Filling 1423
RobotOM.IRobotViewDiagramParams.GetColor 1425
RobotOM.IRobotViewDiagramParams.GetScale 1425
RobotOM.IRobotViewDiagramParams.IsOn 1425
RobotOM.IRobotViewDiagramParams.PositiveNegative 1424
RobotOM.IRobotViewDiagramParams.ReactionsInLocalSystem 1424
RobotOM.IRobotViewDiagramParams.Set 1426
RobotOM.IRobotViewDiagramParams.SetColor 1426
RobotOM.IRobotViewDiagramParams.SetScale 1426
RobotOM.IRobotViewDiagramParams.Values 1424
RobotOM.IRobotViewDiagramPositionType 1447
RobotOM.IRobotViewDiagramResultType 1426
RobotOM.IRobotViewDiagrams 1429
RobotOM.IRobotViewDiagramSignDifferType 1428
RobotOM.IRobotViewDiagramValueType 1461
RobotOM.IRobotViewDisplayAttributes 1418
RobotOM.IRobotViewDisplayParams 1416
RobotOM.IRobotViewDisplayParams.HiddenLines 1417
RobotOM.IRobotViewDisplayParams.IsOn 1417
RobotOM.IRobotViewDisplayParams.Set 1418
RobotOM.IRobotViewDisplayParams.SymbolSize 1417
RobotOM.IRobotViewFeMapCrossPresentationType 1444
RobotOM.IRobotViewFeMapLayerType 1444
RobotOM.IRobotViewFeMapLocalSystemType 1443
RobotOM.IRobotViewFeMapParams 1438
RobotOM.IRobotViewFeMapParams.CrossPresentation 1439
RobotOM.IRobotViewFeMapParams.CurrentResult 1440
RobotOM.IRobotViewFeMapParams.DeformationActive 1440
RobotOM.IRobotViewFeMapParams.DeformationConstScale 1440
RobotOM.IRobotViewFeMapParams.Direction 1440
RobotOM.IRobotViewFeMapParams.DirectionData 1441
RobotOM.IRobotViewFeMapParams.DirectionNode 1441
RobotOM.IRobotViewFeMapParams.Isolines 1441
RobotOM.IRobotViewFeMapParams.Layer 1441
RobotOM.IRobotViewFeMapParams.LayerArbitraryVal 1442
RobotOM.IRobotViewFeMapParams.MapScale 1442
RobotOM.IRobotViewFeMapParams.SetDirectionData 1443
RobotOM.IRobotViewFeMapParams.Smoothing 1442
RobotOM.IRobotViewFeMapParams.WithDescription 1443
RobotOM.IRobotViewFeMapResultType 1436
RobotOM.IRobotViewFeMaps 1429
RobotOM.IRobotViewFeMapSmoothingType 1444
RobotOM.IRobotViewGlobalAnalysis 1453
RobotOM.IRobotViewGlobalAnalysis.MakeScreenCapture 1455
RobotOM.IRobotViewGlobalAnalysis.ParamsGlobalAnalysis 1455
RobotOM.IRobotViewGlobalAnalysisParams 1449
RobotOM.IRobotViewGlobalAnalysisParams.IsOn 1450
RobotOM.IRobotViewGlobalAnalysisParams.Results 1449
RobotOM.IRobotViewGlobalAnalysisParams.Set 1450
RobotOM.IRobotViewGlobalAnalysisParamsType 1453
RobotOM.IRobotViewGlobalAnalysisResultsParams 1450
RobotOM.IRobotViewGlobalAnalysisResultsParams.NPointsValue 1451
RobotOM.IRobotViewGlobalAnalysisResultsParams.RelativeValue 1452
RobotOM.IRobotViewGlobalAnalysisResultsParams.Type 1452
RobotOM.IRobotViewGlobalAnalysisResultsType 1452
RobotOM.IRobotViewHiddenLinesDisplayType 1421
RobotOM.IRobotViewInfluenceLines 1386
RobotOM.IRobotViewInfluenceLines.MakeScreenCapture 1388
RobotOM.IRobotViewInfluenceLines.ParamsInfluenceLines 1387
RobotOM.IRobotViewInfluenceLinesLayerType 1395
RobotOM.IRobotViewInfluenceLinesLocalSystemType 1394
RobotOM.IRobotViewInfluenceLinesParams 1388
RobotOM.IRobotViewInfluenceLinesParams.Direction 1389
RobotOM.IRobotViewInfluenceLinesParams.DirectionData 1389
RobotOM.IRobotViewInfluenceLinesParams.DirectionNode 1390

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRobotViewInfluenceLinesParams.Element 1390
RobotOM.IRobotViewInfluenceLinesParams.IsOn 1392
RobotOM.IRobotViewInfluenceLinesParams.Layer 1390
RobotOM.IRobotViewInfluenceLinesParams.LayerArbitraryVal 1391
RobotOM.IRobotViewInfluenceLinesParams.Position 1391
RobotOM.IRobotViewInfluenceLinesParams.RangeFrom 1391
RobotOM.IRobotViewInfluenceLinesParams.RangeTo 1392
RobotOM.IRobotViewInfluenceLinesParams.Set 1392
RobotOM.IRobotViewInfluenceLinesResultType 1393
RobotOM.IRobotViewMngr 1469
RobotOM.IRobotViewMngr.CreateFromSc 1472
RobotOM.IRobotViewMngr.CreateRtfView 1473
RobotOM.IRobotViewMngr.CreateTable 1473
RobotOM.IRobotViewMngr.CreateView 1473
RobotOM.IRobotViewMngr.CurrentLayout 1471
RobotOM.IRobotViewMngr.GetTable 1474
RobotOM.IRobotViewMngr.GetView 1474
RobotOM.IRobotViewMngr.NewViewAsTab 1471
RobotOM.IRobotViewMngr.RecycleViews 1471
RobotOM.IRobotViewMngr.Refresh 1474
RobotOM.IRobotViewMngr.ShowDialog 1475
RobotOM.IRobotViewMngr.TableCount 1471
RobotOM.IRobotViewMngr.ViewCount 1472
RobotOM.IRobotViewPanelCutParams 1381
RobotOM.IRobotViewPanelCutParams.CurrentReinforcementResult 1382
RobotOM.IRobotViewPanelCutParams.CurrentResult 1383
RobotOM.IRobotViewPanelCutParams.Descritpions 1383
RobotOM.IRobotViewPanelCutParams.Filling 1383
RobotOM.IRobotViewPanelCutParams.IntegralValue 1384
RobotOM.IRobotViewPanelCutParams.Layer 1384
RobotOM.IRobotViewPanelCutParams.LayerArbritraryValue 1384
RobotOM.IRobotViewPanelCutParams.Position 1385
RobotOM.IRobotViewPanelCutParams.Smoothing 1385
RobotOM.IRobotViewPanelCuts 1381
RobotOM.IRobotViewProjection 1405
RobotOM.IRobotViewReinforcementResultType 1448
RobotOM.IRobotViewScaleType 1412
RobotOM.IRobotViewScreenCaptureParams 1456
RobotOM.IRobotViewScreenCaptureParams.Comment 1457
RobotOM.IRobotViewScreenCaptureParams.IncludeDateAndTime 1457
RobotOM.IRobotViewScreenCaptureParams.Name 1457
RobotOM.IRobotViewScreenCaptureParams.Orientation 1458
RobotOM.IRobotViewScreenCaptureParams.Resolution 1458
RobotOM.IRobotViewScreenCaptureParams.ScaleAutomatic 1458
RobotOM.IRobotViewScreenCaptureParams.ScaleValue 1459
RobotOM.IRobotViewScreenCaptureParams.UpdateType 1459
RobotOM.IRobotViewScreenCaptureResolution 1462
RobotOM.IRobotViewScreenCaptureUpdateType 1459
RobotOM.IRobotViewType 1395
RobotOM.IRobotViewVisibilityStatusType 1406
RobotOM.IRobotViewVisibilityStatusValue 1407
RobotOM.IRobotWindLoadsSimulationEngine 472
RobotOM.IRobotWindLoadsSimulationEngine.Generate 473
RobotOM.IRobotWindLoadsSimulationEngine.Params 473
RobotOM.IRobotWindLoadsSimulationParams 473
RobotOM.IRobotWindLoadsSimulationParams.DeviationPercent 475
RobotOM.IRobotWindLoadsSimulationParams.DirectionXNEnabled 475
RobotOM.IRobotWindLoadsSimulationParams.DirectionXNYNEnabled 475
RobotOM.IRobotWindLoadsSimulationParams.DirectionXNYPEnabled 475
RobotOM.IRobotWindLoadsSimulationParams.DirectionXPEnabled 476
RobotOM.IRobotWindLoadsSimulationParams.DirectionXPYNEnabled 476
RobotOM.IRobotWindLoadsSimulationParams.DirectionXPYPEnabled 476
RobotOM.IRobotWindLoadsSimulationParams.DirectionYNEnabled 476
RobotOM.IRobotWindLoadsSimulationParams.DirectionYPEnabled 477
RobotOM.IRobotWindLoadsSimulationParams.Elements 477
RobotOM.IRobotWindLoadsSimulationParams.OpeningsClosed 477

(c) 2014 Autodesk Inc. All rights reserved

RobotOM.IRobotWindLoadsSimulationParams.TerrainLevel 477

RobotOM.IRobotWindLoadsSimulationParams.Velocity 478

RobotOM.IRobotWindow 1650

RobotOM.IRobotWindow.Activate 1652

RobotOM.IRobotWindow.Caption 1651

RobotOM.IRobotWindow.Handle 1651

RobotOM.IRobotWindow.IsActive 1651

RobotOM.IRobotWindow.SendMessage 1652

RobotOM.IRobotWindow.State 1652

RobotOM.IRobotWindowState 1653

S

Screen captures 1465

Sections 546

Seismic analysis parameters 209

Selections 1036

Snow/wind loads 82

Spectral analysis parameters 305

Spread footing 1105

Steel and timber design 1807

Stories 1052

STR files analyzer 2218

Structural axes 1520

Structure components 37

Structure correction 2221

Supplementing RC modules with new codes 1800

Supplementing the required reinforcement module with new codes 1665

Support of external file formats 1274

T

Tables 1361

Time history analysis parameters 364

Tube connection 2042

U

Uniform access to structure components 29

Units and formats 1282

V

Values describing load records 39

Views and layouts 1360

Views for HTML format files 1467

Views for RTF format files 1462

W

Wind loads simulation 472

(c) 2014 Autodesk Inc. All rights reserved